Local Systems, Equivariant Cohomology and the Integral Hecke Action for $\mathrm{GL}(2)/\mathrm{SL}_2(\mathbb{Z})$

The Binding Mathematical Contract

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Abstract

This is the binding mathematical contract for the implementation: it fixes the objects it constructs and everything asserted about them. It sets out the simplicial model $(K, \mathcal{R})$ of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$, the matrix local systems on it, above all the symmetric powers Sym^n of the standard representation, and the equivariant cochain complex and its cohomology. It defines the integral Hecke datum and the operators T_p, T_{p^ν}, T_N, $T(p,p)$ and U_ℓ in all degrees; the involution ι; and the boundary datum and the mapping cone, which carry the Hecke action to the boundary and to compact supports, with the six groups and their long exact sequence. It states how these are realised and certified: a combinatorial reduction primitive with integer bounds, a certified floating-point filter, numbered correctness obligations and a worked example. Later parts treat the integral line and descent over $\mathbb{Z}$; the Dickson invariant, θ_{L}, the ordinary projector and Hida theory at level one; level N and nebentypus; the $\mathbb{Z}/\ell^K$ Hecke algebra and its saturation; the ℓ-local isotropy model, descent at level N and the Hecke algebra over $\mathbb{Z}$; and the Moerdijk–Pronk model. Code maps say what is implemented. Every assertion carries a status tag; Appendix B indexes every numbered statement, and Appendix E records what each version changed.

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1 Scope, status, and how to read this document

What this chapter fixes. What this document is normative about and what it is not; the vocabulary in which every later statement declares its epistemic status; the rule that makes a `\label` of this document a stable API for the C++ source that cites it; the rules its own authors are bound by; and the version convention under which a statement of the form "as of [C] v$X.Y$" is verifiable.

What this document is

This is the *binding mathematical contract* for the implementation in this repository: the objects it constructs, and everything asserted about them. It fixes the simplicial model $(K, \mathcal{R})$ of $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}$, the matrix local systems on it, the equivariant cochain complex and its cohomology $H^q(K, \mathcal{R}; L)$, the integral Hecke datum and the operators T_p, T_{p^ν}, T_N, $T(p,p)$ and U_ℓ in *all* cohomological degrees, the boundary and cone constructions that carry that action to $H^q(\partial)$ and H^q_c, the involution ι, the combinatorial reduction primitive with its integer bounds, the certified floating-point filter, and the numbered correctness obligations that a conforming implementation must discharge.

It is cited as **[C]**. That short key is not decorative: 53 C++ source files already cite statements of this document in the form `[C, alg:tau]`, at 519 sites across 71 files. The key and every cited label are therefore preserved verbatim (§1.2, Appendix D).

What is normative here, and what is not

The corpus this document consolidates was split along a line that put predictions and plain facts about the objects on the same side. The line is redrawn here:

> **CONTRACT.** **[C]** — this document — contains the objects, their construction, and everything proved about them. Every statement in it is normative.
>
> **[E]** — the companion `hecke_extensions.tex` — is the *programme specification*: what to compute with those objects and how to certify it. It is versioned, expected to churn, and cited from no source file.

Consequently the following moved *into* this document and are no longer sought in [E]: the six groups $X_1, \dots, X_6$ and the long exact sequence relating them, the boundary groups and their characters χ_b, Poincaré duality, the amalgam presentation of Γ and the resulting formula for $\dim H^1$, and the Dickson/Bockstein statement (§16). They are facts about the objects and they are the numerical oracles that Obligation 20.7 reads; they are not a programme. What remains in [E]: the Hecke algebra $\mathbb{T}$ and its local pieces $\mathbb{T}_{\mathfrak{m}}$, the separation and generation bounds, the classification data and the staged algorithms, the scalar extension to $\mathbb{F}_{\ell^f}$, the output schema, and the predictions. Appendix A, entry A.18, records the citation changes this move causes.

Two things this document deliberately does *not* do. It does not assert anything about a coefficient ring beyond the hypotheses written into the individual statements that use one — $R = \mathbb{F}_\ell$, $\mathbb{Z}/\ell^e$ and $\mathbb{Z}/m$ are distinguished at the point of use, never globally (rule (A1)). And it fixes the $\ell = 2$ truncation policy only as a *policy*: 14.24 (OC-1), decided by the owner on 2026-09-16 as (OC-1a) and stated in §14.5 with its alternatives kept reachable, rather than silently assumed away; the stabilisation theorem (OC-1c) that would replace the policy by a proved bound is half proved there (Lemma 14.22) and half open (Remark 14.23).

The superseded documents

- `hecke_action.tex` (**[C]** as it stood; PDF of 2026-07-03) is superseded *by this document*, which keeps its key, its 67 labels and its statements, with the corrections of Appendix A applied.
- `monodromy_v5.tex` (**[v5]**) is superseded and frozen. Its framework — the pair $(K, \mathcal{R})$, the universal cover K^u, the three equivalent categories and the explicit equivariant complex — is re-homed here in full (§§3–6), and its labels come with it. This is not tidiness: five C++ source files cite `[C, thm:explicit_complex_equiv]`, and until this document existed that citation dangled, because the label lived in [v5] (Appendix A, entry A.7). What stays in the frozen [v5] is its §6 Möbius pedagogy and its §12, the latter superseded in full (entries A.10, A.19, A.20).
- The earlier document cited [v5] *by statement number* — Prop. 24, Thm. 26, Def. 28/29, Thm. 31/33, Thm. 35, Def. 39/40, Thm. 44. Most of those statements carry no label and [v5] fixes no version, so those references were not stable. Every one of them is now an internal `\ref` of this document.

How to read this document

Part I (§§3–6) is the framework: the model $(K, \mathcal{R})$, local systems, the equivariant complex, and the isomorphism Ψ through which every later formula is transported onto the fixed complex. Part II (§§7–9) fixes the arithmetic conventions, the coefficient module Sym^n with its contragredient, and the involution. Part III (§§10–13) is the Hecke datum and the primary definition of the operators. Part IV (§§14–17) carries that action to the boundary, the cone and compact supports. Part V (§§18–22.5) is realisation and certification: the reduction primitive, the filter, the obligations, the worked example and the code map. Part VI (§23, since

v1.8.0) is the integral line: what each coefficient ring names, the free model on the unrefined tessellation, and the descent over $\mathbb{Z}$; it is placed last so that no earlier chapter number moves, and continues §17 in substance. Part VII (§§24–25, since v1.11.0-pre.1, §26, since v1.12.0-pre.1, and §27, since v1.13.0-pre.1) is multiplication by the Dickson invariant, the operator θ_{L} and its long exact sequence, the ordinary projector, the foundation of Hida theory at level 1, and its continuation: level structure by Shapiro on the fixed model, the stabilisation to level ℓ, control and the lattice datum; it is placed after Part VI for the same reason. Part VIII (§28, since v1.14.0-pre.1, and §29) is level N for every N — the composite coset table, the cokernel presentation and its Hecke action by Merel's matrices — and nebentypus; Part IX (§§30–32) is the $\mathbb{Z}/\ell^K$ Hecke algebra, the ℓ-local isotropy model with the descent at level N and the Hecke algebra over $\mathbb{Z}$, and the Moerdijk–Pronk model. Chapter 2 is *normative and exhaustive* and binds all nine parts; a reader who skips it will mis-read formulas, because several symbols here mean something different from what they meant in the superseded documents.

1.1 Status vocabulary

Every assertion that is not a definition carries *exactly one* status tag, and the tag is part of the statement.

tag	macro	what it claims
[proved here]	`\PROVED`	proved in this document, from the hypotheses in its own statement
[standard: . . .]	`\STD{...}`	standard; discharged by the *precise* citation in the argument
[conjectural]	`\CONJ`	believed, not proved here and not discharged by a citation
[verified computationally]	`\COMP`	verified computationally, over the stated range, by a named validation
[open]	`\OPENSTAT`	open: the statement is not settled and the document says so

> **CONTRACT.** `\STD` takes a **mandatory** argument, and that argument must be a precise citation — author, work, and the section or theorem number — not a bare pointer to a book. `[standard]` with no pointer does not compile. Where no precise citation exists, the tag is [conjectural] and the reason is written out.

[verified computationally] never upgrades to [proved here]. A computation over a finite grid is evidence about that grid; the range it covers is stated with the tag, and the validation that produces it is named, so the claim can be re-run. Appendix B indexes every numbered statement with its tag.

A sixth marker, not a status tag (v1.17.0-pre.1; the Lean programme's question Q9, ruled by the owner on 2026-10-01). `\LEAN{decl}` and `\LEANlit{decl}{hypotheses}` record that a statement already tagged [proved here] is formally verified in the Lean ground truth [GT] (`lean/blueprint/src/gl2_groundtruth.tex`) — with no literature hypothesis, or modulo the named literature hypotheses of [GT]. The marker stands *beside* [proved here], never in place of it, and never beside [conjectural], [verified computationally] or [open] alone; it does not change the exactly-one-tag rule above, and Appendix B does not count it. *Amended at v1.24.0-rc.2* (the rule as already applied, e.g. at Remark 7.6 and Definition 11.6; the referee `docs/implementation/slot25/RC124_referee.md` F2): the marker stands beside [proved here] or [standard:] — on a [standard:] statement it records that [GT] proves the clause; it neither turns [standard:] into [proved here] nor replaces the citation — and, clause-scoped, on a definition or a displayed equation; never beside [conjectural], [verified computationally] or [open] alone. It was defined at v1.17.0-pre.1; the first markers enter at v1.17.0, the release that records the Lean milestone M1. **The marker is clause-scoped.** It stands at the clause,

or the sentence, that [GT] formalises, never at the head of a statement of which only some clauses are formalised; a parenthesis after it narrows it further where [GT] states a clause in a weaker or different form (for instance "as the injectivity of the canonical surjection"). A clause of a tagged statement that carries no marker is *not* claimed formally verified, whatever its [proved here] tag. A *definition* carries no status tag; on a definition the marker records only that [GT] defines the same objects (with the stated differences of indexing). The list of v1.17.0 is `docs/implementation/slot17/lean_taggable_labels.md` (the slot-17 Lean audit F3's, after its repairs): 42 markers on 20 labels; Corollary 30.6 takes `\LEANlit`, its [GT] proof carrying Lemma 30.4 as a hypothesis (Lemma 30.4 itself is not tagged), and Proposition 30.8(1) takes `\LEANlit` modulo Hida's control theorem. The list of v1.18.0 adds the slot-18 Lean audit F3's §8 (`docs/implementation/slot18/L18-6_F3_audit.md`, milestone M2) with its repairs (`docs/implementation/slot18/L18-6_repairs.md`: Lemma 7.3(2), Definition 7.2's rows, (FD1)): 81 markers on 42 labels in all (75 `\LEAN`, 6 `\LEANlit`); Lemma 30.4 and Proposition 30.7(3), (4)'s second surjection take `\LEANlit` modulo Eichler–Shimura at level one (the [GT] hypotheses `hES`, `hF`); Corollary 30.6's `\LEANlit` stays (its [GT] proof still carries the rank hypothesis, not yet discharged from Lemma 30.4). The list of v1.19.0 adds the slot-19 Lean audit F3's §8 (`docs/implementation/slot19/L19-6_F3_audit.md`, milestone M3) as amended by its repairs (`docs/implementation/slot19/L19-6_repairs.md` §3) and by the pair-hypothesis results of slot-19 L19-4 (`docs/implementation/slot19/L19-4_report.md` §3): 98 markers on 48 labels in all (93 `\LEAN`, 5 `\LEANlit`; the boundary hypotheses (B1)–(B3) and the standing hypotheses of §15 counted under their sections). (H4)(ii) is now tagged in the form this document uses it, the acyclicity (37); Remark 7.6, Lemma 11.3(2), Theorem 11.7(b) and the whole of (c), and the cone chapter's hypotheses on the model are tagged; Corollary 30.6(1) takes `\LEANlit` modulo Eichler–Shimura at level one (the [GT] hypothesis `hES`, the rank hypothesis now discharged), its (2) `\LEAN`. The list of v1.20.0 adds the slot-20 Lean audit F3's §8 (`docs/implementation/slot20/L20-6_F3_audit.md`, milestone M4) as amended by its repairs (`docs/implementation/slot20/L20-6_repairs.md`): 140 markers on 65 labels in all (126 `\LEAN`, 14 `\LEANlit`). The new `\LEANlit` markers (Theorem 10.8, Obligation 20.4, Theorem 29.10) are modulo the Hecke-ring identities of [Shi71, Prop. 3.8, Prop. 3.16, Prop. 3.17, Thm. 3.24, Prop. 3.31, Prop. 3.33, Thm. 3.34, §3.2–3.3] as the [GT] hypotheses `hTp`, `hcomm`, `hcop`, `hTpchi0`, `hTpchi`, `hUchi`. **Since v1.21.0-pre.1 their locators are verified** on the owner's scan of Shimura Ch. 3 (step R20-L, `docs/implementation/slot21/R20-L_report.md`), with the transport of [AS86, §1.1] read on the GDZ page images; R20-L restated the locators, not the clauses: `hcop` is Prop. 3.16 (Thm. 3.24(3) is the full-operator form), `hTp` is Thm. 3.24(5) with Prop. 3.17 (Prop. 3.30, the level-N-to-level-1 homomorphism, is not used), `hcomm` is Prop. 3.8 as applied on p. 56 (Thm. 3.24 does not state it), and the nebentypus identities are Shimura's character-free identities in $R(\Gamma_0(N), \Delta_0(N))$ (Prop. 3.31, Thm. 3.34(3) with Prop. 3.33), χ entering only through $T(p,p)$. Shimura's notation ($T(p^\nu)$ the full sum, $T(1,p^\nu)$ the primitive coset) is confirmed by his Thm. 3.24(1), (2). The full operator is formalised only in the assembled form (19): (18) carries no marker. Theorem 11.10 and the equation (20) carry none either — [GT] proves only a weak form, with cor characterised by $\mathrm{cor} \circ \mathrm{res} = [\Gamma : \Gamma'']$ alone; the maps res, $(\mathrm{conj}_g)_*$ and their compatible pair are formalised (`def:m4-classical`), but (20) carries no marker, not even on them. The level-N and nebentypus markers are on the model and, for the operators, for a fixed table and τ (their independence of both is not formalised). The list of v1.22.0-pre.6 adds §32's markers, clause-scoped per `docs/implementation/slot22/MPL1_report.md` v1.1 §5 ([GT] Part G12, the slot-22 Lean packages MP-L1 / MP-L2): 30 markers on 11 labels, all `\LEAN`, so 170 markers on 76 labels in all (156 `\LEAN`, 14 `\LEANlit`). They cover the descriptor predicates, the MP set, the *unnormalised* MP complex $C^\bullet_{\mathrm{un}}$ and its comparison theorem, discharged in full for M_7 and $\mathsf{R}_{\mathbb{Z}}$; none is on Theorem 32.19, Propositions 32.17, 32.20, 32.21, 32.22 or Remark 32.13 (not formalised), nor on the normalised complex. The list of v1.22.0-pre.7 adds

25 markers of §32, clause-scoped per `docs/implementation/slot22/MPL3_report.md` v1.0 §5 (the slot-22 Lean package MP-L3), all \LEAN, on Definitions 32.18 (4), 32.6 (1), 32.8 (2), Theorems 32.19 (10, and 3 on its [verified computationally]^{r} record), 32.12 (2) and Proposition 32.20 (3), so 195 markers on 79 labels in all (181 \LEAN, 14 \LEANlit). They cover the morphisms with the label factor, naturality and refinement invariance for $C^{\bullet}_{\mathrm{un}}$ (label independence on cohomology), $\varepsilon^{\mathsf{MP}}$ with the "if" of Proposition 32.20(2), and the normalised complex with its equivalence to $C^{\bullet}_{\mathrm{un}}$; none is on Propositions 32.17, 32.21, 32.22 or Remark 32.13. v1.22.0-pre.8 adds no marker (195 on 79 labels); Remark 32.23 (the MP-model Hecke operators) carries none. v1.22.0-pre.9 to pre.11 add none. v1.22.0-pre.12 adds the LR-Y markers (`docs/implementation/slot22/LRY_phase3_report.md` §C as repaired by `LRY3_audit.md`): 7 on Claim 16.2(1),(2), Lemma 14.22(2) and the new Remark 16.3, so 202 markers on 82 labels (188 \LEAN, 14 \LEANlit). v1.22.0-pre.13 adds 4 (LR-Y phase 4, `LRY4_audit.md` §6) on Claim 16.2(2) and Lemma 14.22(2), labels that already carried markers, so 206 on 82 (192 \LEAN). v1.22.0-pre.14 adds 12 (the LV dimension results, worded within `LV2_audit.md` §8) on Lemma 16.9 (its last clause, for fields only), Proposition 16.12(2) (the value of h^0 at the certified cells), Proposition 31.3(4) (for fields only) and the new Remark 16.20 (9), so 218 markers on 86 labels in all (204 \LEAN, 14 \LEANlit); the Lean cells outside [GT]'s census (`L23Cells`, $n > 64$) are cited as data, never by a marker, and Proposition 16.15 carries none. v1.22.0-pre.15 adds none. v1.22.0-pre.16 adds 4 (LR-Y phase 5, `LRY_phase5_report.md` §D as repaired by `LRY5_audit.md`) on Claim 16.2(1),(2), a label that already carried markers, so 222 markers on 86 labels in all (208 \LEAN, 14 \LEANlit). v1.24.0-rc.1 adds 7 (the slot-24 Lean package M6's proposal D5, `docs/implementation/slot24/LEAN_M6_report.md` §4 as corrected by `LEAN_M6_audit.md` F1, F3; the owner's question Q4, approved in principle on 2026-10-18): 3 on Theorem 11.10 and 4 on (20), so 229 markers on 88 labels in all (215 \LEAN, 14 \LEANlit). [GT] Part M6 now proves the group-cohomological form on the model (Θ is the canonical comparison, cor is Brown's transfer, the operator is independent of the resolution and of τ), so the sentence of v1.20.0 above ("carry none either") describes that version. Both statements are tagged [standard: [Bro82, Ch. III.9–10]]; the markers stand beside [standard:] under the rule as amended at v1.24.0-rc.2 above, clause by clause in their parentheses, and move no tag. The third marker of that proposal, on the witness of Proposition 32.20(2), is not applied (it is not part of Q4).

1.2 Label stability is part of the contract

A \label of this document is an API, consumed by source code that this document does not control.

(L1) A label is never renamed and never deleted.

(L2) A label is never retargeted to a statement that asserts something different. If what a statement says changes, the label stays with the statement whose claim it named, and the change is recorded in Appendix A.

(L3) A retired statement keeps its label, attached to the erratum that retires it. Removal is not a way to fix a wrong statement; correction is.

(L4) A new label never reuses a retired string.

(L5) New labels use the scheme `kind:stem` with `kind` one of `sec`, `def`, `thm`, `lem`, `prop`, `cor`, `rmk`, `alg`, `obl`, `cert`, `cl`, `eq`, `app`, `open`, `err`. No global prefix: a prefix would break every citing source file at once, for nothing.

(L6) The policy is machine-checked, not merely declared (§22.5): every label cited from source or from a project document is dereferenced in Appendix D, so a missing or duplicated label fails the build, and a census script fails on a `[C, ...]` citation this document does not define.

1.3 Authoring rules

These bind the authors of this document. They are cheap now and expensive later; the cautionary examples behind each are real and are recorded in Appendix A.

(A1) **Every hypothesis at the point of use.** A statement carries every hypothesis it needs *in its own statement*, even when that hypothesis was declared for the chapter. Chapter-level standing hypotheses may still be declared; a statement that uses one repeats it. (Motivating failure: "X is contractible and two-dimensional" was load-bearing for the uniqueness of Λ and for the existence of the fills, and was unwritten — so a reader could not tell the argument from a wish. A second: entry A.3, a realisation theorem proved over a field and then applied over $\mathbb{Z}/\ell^e$.)

(A2) **No claim rests on "standard".** See §1.1: every [standard:] carries a precise citation, and the tag will not compile without one.

(A3) **Definitions are constructive where the code is constructive.** Where the implementation builds an object by a terminating procedure on explicit data — as Algorithm 18.3 builds τ — the definition *is* that procedure together with the properties it guarantees, and existence and uniqueness are then theorems *about the construction*. "Choose any τ satisfying (T1)–(T3)" with existence deferred to an abstract lemma is not the definition this document gives, because it is not the object the code produces.

(A4) **Every proof obligation is a checkable predicate on emitted data.** An obligation states: the data it reads (which stored table, which matrix, in which basis), the predicate, the ring the predicate is evaluated over, and what happens on failure (abort / record / diagnostic). "The tables must be consistent" is not an obligation; "$\partial\eta + \eta\partial = j\tau_\partial - \tau j$ on every stored representative, both sides evaluated through each table's own (T1) helper, over $\mathbb{Z}$; a failing table aborts" is.

(A5) **One symbol, one meaning, document-wide** — §2, enforced by the retired macros of `gl2_preamble.tex`, which raise a LaTeX error rather than typeset. This is not a style rule: three incompatible meanings of Δ, τ and ι coexisted in the corpus, two of them inside a single document.

(A6) **Diagnostics are never branched on.** Where the implementation records a quantity that could be mistaken for a decision procedure — `cochain_commutes`, `eta_zero`, `solve_regime` — this document says so explicitly. The entrywise identity $r\,\Theta_g = \Theta_g^{\partial}\,r$ holding on the stored data does *not* license $h = 0$, and the text must make that impossible to misread.

1.4 Version marker and changelog

Neither superseded document carried a version marker: both used `\date{\today}`, so every rebuild silently re-dated the PDF and no "as of" statement about either was verifiable (Appendix A, entry A.17). This document fixes a version:

(V1) The date line is `Version` $X.Y$ `--- YYYY-MM-DD` and a status clause, never `\today`. The supersession record — one entry per version — is the first subsection of Appendix E.

(V2) X **increments on any change to a normative statement**; Y increments on prose, on recording an erratum, and on new non-normative material.

(V3) §*Changelog* — Appendix E — carries one entry per version and, within it, **one line per changed statement, naming its label**. "Editorial pass" is not a changelog entry: a contract whose statements move must say which ones moved. The rules under which that appendix is kept, including what counts as one entry and how a statement's provenance is marked, are stated at its head.

(V4) `hecke_action.tex` and `monodromy_v5.tex` each receive a frozen version marker and a superseded-by banner when this document ships. **Done at v1.0.0**: `hecke_action.tex` is frozen at Version 1.0 (2026-07-02) and `monodromy_v5.tex` at Version 5.0 (2026-07-03), each

carrying a banner that names this document as the successor, says that a `[C, <label>]` citation resolves here and not there, and directs the reader to Appendix A. The content of neither was otherwise touched: they are the historical record. [v5]'s banner also says where `thm:explicit_complex_equiv` now lives, that being the label its five C++ citing sites use (entry A.7), and that §12's β is not re-homed (entry A.19).

(V5) **Before completion the version is $0.X.Y$.** While this document is incomplete in the sense below, the leading field is 0 and marks that fact; X and Y are the fields of (V1) and carry (V2)'s bump rules unchanged. The leading field becomes 1, and $0.X.Y$ then reads $1.X.Y$ with no renumbering of the history, when **both** of the following hold:

(i) **no chapter of this document is a stub**; and

(ii) **the label namespace is unambiguous** — (V4)'s two freezes have been done, and no label defined here is also defined by a live companion document.

Criterion (i) was met at v0.19.0 and **(ii) at v1.0.0** (Appendix E). (ii)'s obstruction was concrete rather than ceremonial and was discharged concretely. Nine labels of §16 — `sec:six`, `cl:amalgam`, `cl:dimH1`, `cl:bockstein`, `cl:boundary`, `cl:les`, `eq:les`, `cl:chi`, `cl:pd` — were defined *both* here, by the scope cut of entry A.18, *and* in the live companion `hecke_extensions.tex`, so that a reference `[C, cl:pd]` did not resolve to a unique statement and (L6) — the policy is machine-checked, not merely declared — could not be honoured across the corpus. `hecke_extensions.tex` v1.3 removes its nine copies and cites this document instead, and (V4)'s two freezes are done, so the criterion holds; the evidence is Remark 22.1. The refinement of (V1)'s form that the leading field represents is kept in the numbering, not dropped: "as of [C] v$0.X.Y$" says which normative revision is meant *and* that the document was incomplete when it was made, and 0.19.0 reads as 0.19.0 still. A leading 1 says only that this document is complete and its namespace unambiguous. It does *not* say the programme is finished — two chapters state objects nothing implements (§22.2; three until v1.3.3), and 14.24, decided at v1.3.0, governs a construction none of them has — and §E.1 keeps that list.

2 Notation, fixed once

This chapter is *normative and exhaustive*. A symbol used anywhere in this document has exactly the meaning fixed here, in every chapter, in every proof, and in every displayed formula. A symbol not listed here is either local to the statement that introduces it — in which case that statement introduces it explicitly — or it is a mistake.

The machine-readable form of this chapter is `gl2_preamble.tex`: every macro named below is defined there and nowhere else, and companion documents `\input` that file rather than copying it. The pre-consolidation corpus copied macro blocks between `hecke_action.tex`, `hecke_extensions.tex` and `monodromy_v5.tex`, which is precisely how three incompatible meanings of Δ, τ and ι came to coexist; §2.7 lists the nineteen resulting collisions and their resolution.

2.1 Reading rules

(N1) **One symbol, one meaning, document-wide.** Not "one meaning per chapter": a reader who opens the document at a proof must be able to read every symbol in it without knowing which chapter it is in.

(N2) **Decoration does not create a new symbol.** π_1 and a map called π are a collision, not a disambiguation; so are τ and τ' with different meanings. Where two objects are genuinely two instances of one notion (the transporters $s_1, \dots, s_m$; the vertices $v_0, \dots, v_6$) indices are of course used.

(N3) **Single letters are a scarce resource; operators get names.** Where a map has no

traditional single-letter name, it is written as a named operator in upright type (inc_c, pr_c, $\mathrm{pr}_{\mathcal{R}}$, adj) rather than given a Greek letter. This costs three characters and buys a permanently free letter.

(N4) **Indices are letters, not primes.** p, ℓ and q are reserved (§2.2); a running index is t, j or s. i is never an index: it is reserved for $\sqrt{-1}$ and the elliptic point $v_6 = i$ (§2.3, C15).

(N5) **Retired symbols are errors, not deprecations.** The macros `\K`, `\R`, `\C`, `\M`, `\pR`, `\pu`, `\image` are defined in `gl2_preamble.tex` to raise a LaTeX error. Text re-homed from the old documents therefore cannot carry an old meaning in unnoticed: it fails to build.

2.2 The three arithmetic parameters

This is the single most error-prone point in the corpus, and the one the owner has already decided (OQ1, `hecke_extensions_owner_decisions.md` §OQ1: *adopt the contract's* `ell` / `n_sym` / `p_hecke`, weight $k = n + 2$). It is restated here as the binding rule, extended to the cohomological degree.

CONTRACT.

role	symbol	code identifier	range
coefficient prime (characteristic of R)	ℓ	`ell`	any prime
Hecke prime (index of the operator T_p)	p	`p_hecke`	any prime, $p = \ell$ allowed
symmetric-power degree	n	`n_sym`	$n \geq 0$
cohomological degree	q	`q`	$q \geq 0$
modular weight (derived, never independent)	$k := n + 2$	—	
prime-power exponent in T_{p^ν}, $N = p^\nu$	ν	`r` (code)	$\nu \geq 1$
coefficient-ring exponent in $R = \mathbb{Z}/\ell^e$	e	—	$e \geq 1$
residue degree of $\mathbb{F}_{\ell^f} = \mathbb{T}_{\mathfrak{m}}/\mathfrak{m}$	f	`r` (code)	$f \geq 1$

Consequences, each of which changes text somewhere:

- q is **the cohomological degree and nothing else**. The spec `hecke_extensions.tex` used q for the Hecke prime while also writing H^q and $d_q = \dim H^q$; every T_q, B_q, $\chi_b(q)$, $\Gamma_0(q)$ there becomes T_p, B_p, $\chi_b(p)$, $\Gamma_0(p)$ here.
- p is **the Hecke prime and nothing else**. The owner's notes `hecke_extensions_mathematics.md` and `hecke_extensions_computational_strategy.md` use p for the *coefficient* prime — the exact opposite assignment. A formula copied from them without applying the dictionary is not merely mislabelled, it is wrong.
- n is **the symmetric-power degree and nothing else**. Two other uses are retired: the cone degree (now q, §2.6) and the composite Hecke index T_n (now T_N with $N \in \mathbb{Z}_{>0}$, which `hecke_action.tex` already used at one site and not at two others).
- k is **the modular weight and nothing else**, and it is a *derived* quantity: the document writes k only where the classical weight-k statement is being invoked, and always with $k = n + 2$ visible nearby. The three other uses are retired: $k := \#S$ (the number of Hecke operators; now s), the basis index $u_k = x^{n-k}y^k$ (now u_j), and the chain-term index $n_k^{(i,\sigma)}$ (now $n_t^{(j,\sigma)}$, C15).
- The exponent of a coefficient ring $\mathbb{Z}/\ell^e$ is e, **not** k. This departs from the letter of the owner's OQ18 wording ("$\mathbb{Z}/\ell^k$"), which was written before k was pinned to the weight; the two symbols would otherwise collide in the Sturm chapter, where $\mathbb{Z}/\ell^e$-coefficients and weight-k forms appear in one sentence. **Owner confirmation requested** (§2.9, O1).

2.3 The framework: complexes, the model, the pair

symbol	macro	meaning
K	—	the finite simplicial complex of the model (the 7-vertex complex)
$\mathcal{R}$	\Rel	the *relation set* on K; the model is the pair $(K, \mathcal{R})$
$K/\mathcal{R}$	\KR	the quotient complex
K^u	\Ku	the universal cover complex
$\mathrm{pr}_{\mathcal{R}}$, pr^u	\prR, \pru	the projections $K \to K/\mathcal{R}$, $K^u \to K/\mathcal{R}$
π_1	\piOne	the fundamental group (only with the subscript 1)
σ	—	an *oriented simplex*; $o_q(K)$ is the set of oriented q-simplices
$v_0, \dots, v_6$	—	the seven vertices of the model
Γ	\Ga	$\mathrm{SL}_2(\mathbb{Z})$
Γ''	—	$\Gamma_0(N)$, $N = p^\nu$, the group of the correspondence
Γ_∞	\Gbdry	the boundary group $\langle T \rangle \times \langle -I \rangle$
Δ	\De	the monoid $\{g \in \mathrm{M}_2(\mathbb{Z}) : \det g > 0\}$
$\Delta_{p^\nu}^{\mathrm{prim}}$	\Deprim	the *primitive* double coset
X	\Xbulk	the Γ-tessellation of the truncated upper half plane
∂X	\Xbdry	$\Gamma \cdot i(K_\partial) \subset X$, the Γ-subcomplex of horocycle lines
K_∂	\Kbdry	the circle model: the path $v_0v_1v_2v_3$ with $v_3 = Tv_0$
j	—	the inclusion $\partial X \hookrightarrow X$ (of *spaces/chains*)
r	\rres	the restriction cochain map $r := j^*$
ζ	—	$e^{\pi i/3}$; the elliptic vertices are $v_4 = \zeta^2$, $v_5 = \zeta$, $v_6 = i$
T, S	—	the standard generators $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ of Γ
ι, ι_K	\inv, \invK	the involution $\mathrm{diag}(-1, 1)$ and its simplicial realisation
$\widetilde{\sigma}$	\lift	the distinguished lift of $\sigma \in o_q(K)$ into $o_q(K^u)$, for the strict fundamental domain I

Note on $\mathcal{R}$ and R. The old contract wrote the model as (K, R) and the cohomology as $H^q(K, R; L)$, with R the *relation*; the cone and coefficient-ring material writes R for the *coefficient ring* ($6 \in R^\times$, V_R, $\mathbb{Z}/\ell^e$). Both are indispensable and they cannot share a letter. Here R is the coefficient ring and the relation set is $\mathcal{R}$, so the model is $(K, \mathcal{R})$ and its cohomology $H^q(K, \mathcal{R}; L)$. Individual relations are written as triples $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$, $\gamma \in \Gamma$ (C17), and are given no single-letter name (the old $r \in R_+$ is retired: r is the restriction map). Code identifiers are unaffected: `SimplexRelation` and `EquivariantCohomologyOnK` name the same objects.

2.4 Cochains, coefficients, bases

symbol	macro	meaning
R	\cring	the coefficient ring: $\mathbb{F}_\ell$, $\mathbb{Z}/\ell^e$, or $\mathbb{Z}/m$
V	\Vcoef	$\mathrm{Sym}^n(\mathbb{Z}^2)$, the integral coefficient module
V_R	\VR	$V \otimes R$
M_n, D_n	\Mn, \Dn	$\mathrm{Sym}^n \mathbb{F}_\ell^2$ and its contragredient (divided powers)
ρ_n, $\rho_n^\vee$	\rhon, \rhodual	the symmetric-power action and its contragredient
$u_j = x^{n-j}y^j$	—	the ordered basis of Sym^n, $0 \le j \le n$
d_V	—	$\dim V = n + 1$
M_γ	—	the monodromy matrix $\rho_n(\gamma)$ of a relation
C^q_{eq}, Z^q_{eq}, B^q_{eq}, H^q_{eq}	\Cq…	equivariant cochains, cocycles, coboundaries, cohomology

symbol	macro	meaning
$c_j^{(q)}$, $z_j^{(q)}$, $b_j^{(q)}$	—	the stored bases of C^q_{eq}, Z^q_{eq}, B^q_{eq}
d	—	the *coboundary* $C^q_{\mathrm{eq}} \to C^{q+1}_{\mathrm{eq}}$
∂	—	the *chain boundary* $C_q \to C_{q-1}$
ε	`\aug`	the augmentation $C_0 \to \mathbb{Z}$
P_q	—	the projector $Z^q_{\mathrm{eq}} \to H^q_{\mathrm{eq}}$
f_{12}	`\cf`	the normalised weight-12 cusp form (classically "Δ")

The basis-name scheme $c/z/b$ replaces `monodromy_v5.tex`'s $\beta_j^{(q)}$, $\delta_s^{(q)}$, $\eta_p^{(q)}$, each of which collided with a different reserved symbol — and the last of which was additionally indexed by p.

2.5 The Hecke datum

symbol	macro	meaning
τ	—	the g-twisted chain datum on X (*only* this)
τ_∂	`\taub`	the g-twisted chain datum on ∂X
$s_1, \dots, s_m$	—	the transporters, $\Gamma = \bigsqcup_j \Gamma'' s_j$
δ_j	—	the coset elements $\delta_j = s_j^{-1} g$
β_j, β_∞	—	the upper-triangular *right*-coset representatives
δ_j, δ_∞	—	the upper-triangular *left*-coset representatives (the normative ones)
Ξ_g, Θ_g	`\Xig`, `\Th`	the induced-system map and the fused cochain map $\Theta_g = \omega_\bullet \circ \Xi_g$
α, ω	—	the restriction unit and the corestriction counit
μ	—	the would-be middle map of the induced sequence (*provably does no*
Θ_g^∂	`\Thb`	the boundary cochain map
$A^{(q)}$, $A^{(q)}_\partial$, $A^{(q)}_c$	`\Amat...`	the cohomology-level matrices on H^q, $H^q(\partial)$, H^q_c
T_p, T_{p^ν}, T_N, $T(p,p)$, U_ℓ	—	the operators; $N \in \mathbb{Z}_{>0}$ a composite index
$\mathbb{T}$, $\mathfrak{m}$	`\Talg`, `\midl`	the Hecke algebra and a maximal ideal of it
χ_b	—	a boundary character
$a_N(f_{12})$	—	the N-th Fourier coefficient of f_{12} (classically "$\tau(N)$")

2.6 The pair, the cone, compact supports

symbol	macro	meaning
η	`\etapair`	the *pair homotopy* $C_*(\partial X) \to C_{*+1}(X)$
h	`\hder`	the *derived* cochain homotopy $h := \widetilde{\eta}$ (never freely chosen)
Λ	`\Lam`	the unique $C_0(\partial X) \to C_2(X)$ with $\eta' - \eta = \partial\Lambda - \Lambda\partial$
$C(r)$	`\Cone`	the mapping cone, $C(r)^q = C^{q+1}_{\mathrm{eq}}(K) \oplus C^q_{\mathrm{eq}}(K_\partial)$
(c, c_∂)	—	a cone cochain (bulk component, boundary component)
D	`\Dcone`	the cone differential $D(c, c_\partial) = (-dc,\ dc_\partial + rc)$
Φ_g	`\Phig`	the cone cochain map $\Phi_g(c, c_\partial) = (\Theta_g c,\ \Theta_g^\partial c_\partial + hc)$
inc_c, pr_c	`\coneinc`, `\coneproj`	the cone inclusion and projection
H^q_c	`\Hcpt`	$H^q_c := H^{q-1}(C(r))$
$X_1, \dots, X_6$	—	the six groups of the long exact sequence

Three points where this differs from the sources it is re-homed from:

- **The cone degree is q.** The OQ2 verdict and `monodromy_v5.tex` §8 write $C(r)^n = A^{n+1} \oplus B^n$ and $H^n_c := H^{n-1}(C(r))$ — with n also the symmetric power, in the same notation block. Here: $C(r)^q$, $H^q_c = H^{q-1}(C(r))$.
- **The cone inclusion is inc_c, not ι.** Both the verdict ($\iota_* \circ k_* \circ \pi_*$) and the implementation plan ($\Phi_g \, \iota = \iota \, \Theta^{\partial}_g$) wrote ι for the cone inclusion, 130 lines after the same plan fixed ι as the involution — both inside the text proposed for the contract. Here ι is the involution; the cone maps are named operators.
- **The cone element is (c, c_{∂}), not (a, b).** a is a Fourier coefficient and an exponent in $\mathbb{Z}/\ell^a$; b indexes the boundary characters χ_b. The component names also say which component is which, which the formalization will want.

2.7 The nineteen collisions and their resolution

Eleven collisions were found by the corpus survey of 2026-09-15; three more (C12, C13, C14) were found during this design pass, a fifteenth (C15) during the Chapters 7–8 drafting pass, a sixteenth (C16) during the Chapter 11 drafting pass, and a seventeenth (C17) during the Chapter 13 slot 4 follow-up pass, all 2026-09-15. Two further ones were found while Chapters 14 and 15 were written (2026-09-16) and resolved *locally* there, each with a note saying it was not yet listed here: they are C18 and C19, added 2026-09-16. Each row says which document's usage *changes*.

#	the two (or more) live meanings	resolution
C3	$\iota = \mathrm{diag}(-1, 1)$ (spec (A3); owner decision OQ3; record field `iota_sign`) *vs* ι = the cone inclusion $B \to C(r)$ (`monodromy_v5.tex`:1226; verdict §0, §2; plan §0.4 C-A3)	ι **is the involution**. The cone inclusion becomes inc_c and the projection pr_c. *Changes:* `monodromy_v5.tex` §8, the verdict's and the plan's cone text.
C4	τ = the g-twisted chain datum (contract, 36 code sites via `alg:tau`) *vs* τ = an oriented simplex (`hecke_action.tex`:1012–1015, *inside the proof of* `thm:xi`; `monodromy_v5.tex`:619,628–630) *vs* $\tau(N)$ = Ramanujan's function (spec, six sites)	τ **is the chain datum**. Oriented simplices are σ (already the contract's own convention elsewhere). Ramanujan's function is written $a_N(f_{12})$ and never appears as τ. *Changes:* the proof of `thm:xi`, `monodromy_v5.tex` Thm 44's proof, the spec's $\ell = 691$ section.
C5	$\Delta = \{g \in \mathrm{M}_2(\mathbb{Z}) : \det g > 0\}$ (both documents; "note $\iota \notin \Delta$") *vs* Δ = Ramanujan's cusp form (spec:162, 424–425, 1037, 1327, 1334) — *same macro, same document*	Δ **is the monoid**. The weight-12 cusp form is f_{12}, with $a_N(f_{12})$ its coefficients. *Changes:* the spec.
C6	η = the pair homotopy (verdict §2.3ff; spec:779; plan:346) *vs* $\eta^{(q)}_p$ = a coboundary basis vector (`monodromy_v5.tex`:1182, 1208, 1211, 1325, 1328 — *inside §8, the very cone section*), itself indexed by p	η **is the pair homotopy**. Basis vectors are $c^{(q)}_j$ / $z^{(q)}_j$ / $b^{(q)}_j$ for C^q_{eq} / Z^q_{eq} / B^q_{eq}, indexed by j. *Changes:* `monodromy_v5.tex` §7–§8.

#	the two (or more) live meanings	resolution
C7	$\delta_j = s_j^{-1} g$ (contract, `alg:assembly`) *vs* δ the coboundary $(\delta\varphi)(x) = \varphi(\partial x)$ (`hecke_action.tex`:690, 1014 — same proof as the coset elements) *vs* $\delta_s^{(q)}$ a cocycle basis vector (`monodromy_v5.tex`:1098–1109)	δ **is the coset element**. The coboundary is d throughout (∂ for chains, d for cochains, D for the cone). Basis vectors as in C6. *Changes:* `hecke_action.tex` §§4–5, `monodromy_v5.tex` §7.
C8	$\beta_j = \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}$ right-coset representatives (contract `prop:cosets`) *vs* β the middle map of the induced sequence (`monodromy_v5.tex`:1519, the map proved not to exist) *vs* $\beta_j^{(q)}$ a basis vector (`monodromy_v5.tex`:1098)	β_j **is the right-coset representative**. The non-existent middle map is μ, with a parenthetical "called β in [v5] §12". Basis vectors as in C6. *Changes:* `monodromy_v5.tex` §7, §12; the no-go remark keeps its label `rmk:nogo`.
C9	$\rho = e^{2\pi i/3}$, ρ^2, "the elliptic points i, ρ, ρ^2" (`hecke_action.tex`:194, 222–223, 243–244) *vs* ρ_n, ρ_L the representation (*same subsection*, :245)	ρ **is the representation**. The elliptic points are $\zeta := e^{\pi i/3}$, ζ^2 and i; the vertices are $v_4 = \zeta^2$, $v_5 = \zeta$, $v_6 = i$. (The old naming was nonstandard as well as colliding: $e^{\pi i/3}$ is conventionally $-\bar{\rho}$.) *Changes:* `hecke_action.tex` §2.2.
C2	n = symmetric power (everywhere) *vs* n = the cone degree (verdict:32; plan C-A3(ii.4); `monodromy_v5.tex`:1152) *vs* T_n the composite Hecke index (`hecke_action.tex`:80, 1439, against T_N at :518)	n **is the symmetric power**. Cone degree $\to$ q; composite index $\to$ T_N. *Changes:* the verdict's and the plan's cone text, `monodromy_v5.tex` §8, two sites in `hecke_action.tex`.
C10	$k = n + 2$ the weight (spec:136, 206, 323, 419) *vs* $k := \#S$ (spec:746, *same document*) *vs* $u_k = x^{n-k}y^k$ (contract:294) *vs* $n_k^{(i,\sigma)}$ (contract:977) *vs* k = the symmetric power in [M]/[S]	k **is the weight**, always derived and always with $k = n + 2$ visible. $\#S \to s$; $u_k \to u_j$; $n_k^{(i,\sigma)} \to n_t^{(j,\sigma)}$ (the transporter index by C15); [M]/[S]'s $k \to n$. *Changes:* spec §7, contract §2.3 and §5, both owner notes.
C11	r the exponent in T_{p^r} (contract:496) *vs* r the residue degree of $\mathbb{F}_{\ell^r}$ (spec:40, 345) *vs* r^0, r^1 the restriction maps (spec:238; verdict; plan) *vs* $r \in R_+$ a relation (contract:107, 121) — two of them as superscripts on one line at spec:280	r **is the restriction cochain map** $r = j^*$ (it is the symbol the cone notation is built on: $C(r)$, $\ker r^1$, $\operatorname{coker} r^0$). Exponent $\to \nu$ (T_{p^ν}, $N = p^\nu$); residue degree $\to f$ ($\mathbb{F}_{\ell^f}$) — the standard number-theoretic letter; relations get no letter (see §2.3). *Changes:* contract §3, §2.1; spec §6.
C1	q the Hecke prime (spec:134, and $\sim$ 60 occurrences of T_q) *vs* q the cohomological degree (spec:262, 746, *same Throughout-line*; the whole contract; the code)	q **is the cohomological degree**; the Hecke prime is p (§2.2, owner decision OQ1). *Changes:* the spec throughout.
D3	p the Hecke prime (contract, code) *vs* p the coefficient prime ([M], [S], and the owner's own wording in the decision record)	p **is the Hecke prime**; the coefficient prime is ℓ (owner decision OQ1). *Changes:* [M], [S].

#	the two (or more) live meanings	resolution
C12	$M_\gamma := \rho_n(\gamma)$ a monodromy matrix (contract:245; the code's `SimplexRelation::M`) *vs* $M_n = \mathrm{Sym}^n \mathbb{F}_\ell^2$ and D_n (spec) *vs* $M : C_0(\partial X) \to C_2(X)$ the unique map of the η-uniqueness lemma (verdict Lemma 2.4.1; plan C-A3(ii.3)) *vs* $\mathrm{M}_2(\mathbb{Z})$ (macro `\M`) *vs* M^T, M_T boundary (co)invariants	The coefficient module is V / V_R, with M_n retained as a synonym *defined once* for the classification literature; monodromy matrices keep M_γ (always subscripted); the unique degree-(+2) map is Λ; the matrix algebra is $\mathrm{M}_2(\mathbb{Z})$ (`\Mat`; `\M` is retired to an error); boundary (co)invariants are V_R^T, $(V_R)_T$. *Changes:* the verdict's and plan's cone text, the spec.
C13	R the relation set, in (K, R) and $H^q(K, R; L)$ — *including the title of the contract* — *vs* R the coefficient ring ($6 \in R^\times$, V_R, $R = \mathbb{Z}/\ell^e$) in all of the cone and integrality material	R **is the coefficient ring**; the relation set is $\mathcal{R}$, so the model is $(K, \mathcal{R})$ and its cohomology $H^q(K, \mathcal{R}; L)$. *Changes:* the contract's title and §§1–2, `monodromy_v5.tex` throughout (typographic only; no label and no code identifier moves).
C14	s_i the transporters $\Gamma = \bigsqcup_i \Gamma'' s_i$ (contract) *vs* s an (un)oriented simplex (`monodromy_v5.tex`; the code's flat index $s \cdot d_V + k$) *vs* $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ the generator *vs* S the set of computed operators (spec §7)	s_j **is a transporter** (index j, not i: C15); simplices are σ; S is the generator; the operator set is $\mathcal{S}$ with $s := \#\mathcal{S}$. *Changes:* `monodromy_v5.tex`, spec §7.
C15	(N4)'s reading rule, "a running index is i, j, t or s" *vs* $v_6 = i = \sqrt{-1}$, the elliptic point (§2.3) — the same chapter, the reading rules and the framework table a page apart	i **is** $\sqrt{-1}$ **(and the elliptic point** v_6**) only**. Running indices are t, j or s; (N4) is amended accordingly. The transporters and coset elements are therefore s_j, $\delta_j = s_j^{-1} g$ and the chain-term coefficients $n_t^{(j,\sigma)}$, which is what Chapters 10–11 write. *Changes:* this chapter's own (N4) and its tables (§2.5, C7, C10, C14); Chapters 7–8 already avoided i as an index (found during their drafting); `hecke_action.tex` throughout, where the transporter index is i (crosswalk, §2.8). Owner decision recorded §2.9, 2026-09-15.
C16	$i(\sigma)$ the distinguished lift of a simplex $\sigma \in K$ into the universal cover K^u (the code's `i(sigma)`; `monodromy_v5.tex`'s inclusion $i : K \to K^u$ and `rmk:injectivityofi`; `hecke_action.tex`, the τ-datum and `alg:tau`) *vs* $i = \sqrt{-1}$ and the elliptic point v_6 (C15) — and, in the same formulas, i as the transporter index	i **is** $\sqrt{-1}$ (C15), so the lift gets no letter at all: it is the decoration $\widetilde{\sigma}$ (`\lift`), rendered $\widetilde{\sigma}$, for the strict fundamental domain $K \subseteq K^u$ of (H4)(iii). It is a *map*, not an index, so (N3) does not force an upright operator name. *Changes:* `monodromy_v5.tex` §§1–2 and `hecke_action.tex` §§5–6 wherever $i(\sigma)$ appears; the code identifier `i(sigma)` is unaffected and does *not* move.

#	the two (or more) live meanings	resolution
C17	g the relation element of a triple $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, g) \in \mathcal{R}$ (`def:model7rel`, the seven-vertex model's relation table) *vs* $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$ the Hecke element fixed throughout the Hecke chapters (§2.5) — the two meet in one formula, `thm:xi`(1)	g **is the Hecke element only**. Relation elements are $\gamma \in \Gamma$, so a relation is a triple $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$; this model's relation elements (T, S, ST, TS, $-I$) all lie in Γ, so the rename costs nothing. *Changes:* `def:model7rel` and its contract box (both instances of g), and the one further site where the relation triple's third component was written g (the contragredient local system, `sec:contragredient` in the contract's numbering).
C18	$N = p^{\nu}$ the Hecke level, fixed throughout the Hecke chapters (§2.2; $\Gamma'' = \Gamma_0(N)$, $g = \mathrm{diag}(1, N)$, T_N, $a_N(f_{12})$) *vs* $N := \prod_{c \in \mathbb{F}_\ell}(y + cx) = y^\ell - x^{\ell-1}y$, the degree-$\ell$ boundary invariant of `cl:boundary` and `cl:chi` (`hecke_extensions.tex`; the spelling the boundary chapter inherited) — the two meet in one statement, `prop:bdrychar`, whose eigenvalues are indexed by the *level* p and whose basis is built from the *invariant*	N **is the Hecke level only**. The invariant is Nm — it is the norm, i.e. the $\langle T \rangle$-orbit product, of y — and it gets an upright operator name, not a letter ((N3)). *Changes:* `hecke_extensions.tex` `cl:boundary`, `cl:chi` and every occurrence of the invariant in their neighbourhood; in this document the spelling is already Nm everywhere (`prop:bdrychar` and the V_R^T basis $x^{n-b\ell}\mathrm{Nm}^{\,b}$). Found while Chapter 14 was written; resolved locally there on 2026-09-16 and listed here on the same day.
C19	The **tilde** as a decoration, with four live readings: $\widetilde{\sigma}$, the distinguished lift of a simplex (C16, `\lift`); $\widetilde{F}$, the coset transport of a (T1)-map (`def:cosettransport` — $\widetilde{\tau}$, $\widetilde{H}$, $\widetilde{\Lambda}$, $\widetilde{N_\partial}$, $h = \widetilde{\eta}$); $\widetilde{T}$, the *cochain-level* operator of `def:hecke`; and, in Part I only, $\widetilde{K}$ the saturation and $\widetilde{X}$ the universal cover	**The decoration is read from what it decorates**, and no rename is made: on a *simplex or chain* it is the lift, on a *map with (T1)* it is coset transport. The two never meet in one formula and are typographically distinct (a decorated symbol versus a decorated operator); the point of definition of each says so (§2.3, `def:cosettransport`). $\widetilde{T}$ is *consistent* with the second reading only because $\widetilde{T} = \widetilde{\tau}$ (`def:cosettransport`), T itself being no chain map; it is kept because 53 code sites and both superseded documents spell it so. Part I's two are out of scope of the other three — no lift and no coset transport occurs there — and are not renamed. *Rule:* no **new** tilde decoration may be introduced, in this document or in the code, without a row here. *Changes:* none to any document; this row records the reading rule. Found while Chapter 15 was written; noted there on 2026-09-16 and listed here on the same day.

2.8 Crosswalk: reading the old documents

A reader holding a pre-consolidation document applies this table. It is also the checklist for the re-homing work: a chapter is not done until every row that touches its source has been applied.

here	`hecke_action.tex`	`hecke_extensions.tex`	`monodromy_v5.tex` / [M], [S], [V]
ℓ	ℓ	ℓ	p ([M], [S])
p	p	q	q ([M], [S]); p ([V])
n	n	n	k ([M], [S])
q (degree)	q	q *and* the prime	k, n
$k = n + 2$	—	k, *also* $\#S$	$k + 2$ ([M])
ν	r (in T_{p^r})	r	—
f	—	r (in $\mathbb{F}_{\ell^r}$)	—
e	—	—	k (in $\mathbb{Z}/\ell^k$)
s	—	$k := \#S$	—
t, j, s (running index)	i	i	i
s_j, δ_j, $n_t^{(j,\sigma)}$	s_i, δ_i, $n_k^{(i,\sigma)}$	—	—
$\widetilde{\sigma}$	$i(\sigma)$	—	$i(\sigma)$, $i : K \to K^u$
r	—	r^0, r^1	r
$\mathcal{R}$	R	—	R
γ (relation element)	g_r	—	g_r
R	—	R	R ([V], [X])
σ	σ, *also* τ	—	τ, s
ζ	ρ	—	—
u_j	u_k	—	—
d (coboundary)	δ	—	δ_{simp}
$c_j^{(q)}, z_j^{(q)}, b_j^{(q)}$	—	—	$\beta_j^{(q)}, \delta_s^{(q)}, \eta_p^{(q)}$
$\mathrm{inc}_c, \mathrm{pr}_c$	—	—	ι, π
(c, c_∂)	—	—	(a, b)
Λ	—	—	M ([V])
μ	β (in `rmk:nogo`)	—	β
f_{12}, $a_N(f_{12})$	—	Δ, $\tau(q)$	—
$\mathrm{M}_2(\mathbb{Z})$	$\mathrm{M}_2(\mathbb{Z})$	$\mathrm{M}_2(\mathbb{Z})$	—
T_N	T_n *and* T_N	—	—
Nm (the invariant)	—	N (C18)	—
the tilde	read by what it decorates; no rename (C19)		

2.9 What the owner must confirm

OQ1 fixed the three arithmetic parameters and nothing else. The following go beyond it and are decisions taken in this design pass; each is defensible and each is reversible at zero cost *now* and at the cost of a full re-read *later*. One item that would otherwise belong here does not: C15 (i reserved for $\sqrt{-1}$/the elliptic point, never a running index) was already decided by the owner on 2026-09-15 and needs no confirmation; it is recorded, settled, at its row in §2.7 and in (N4).

(O1) $R = \mathbb{Z}/\ell^e$ rather than $\mathbb{Z}/\ell^k$ (§2.2). Departs from the letter of OQ18's wording; needed because k is the weight.

(O2) R for the coefficient ring and $\mathcal{R}$ for the relation set (C13) — which changes the *title* of the binding contract from $H^*(K, R; L)$ to $H^*(K, \mathcal{R}; L)$. The alternative (keep R for the relation, rename the coefficient ring) was rejected: $6 \in R^\times$, V_R and $\mathbb{Z}/\ell^e$ appear in far more statements than (K, R) does, and no other letter for a commutative ring reads naturally.

(O3) ν for the prime-power exponent and f for the residue degree (C11), so that r can be the restriction map. The alternative (keep T_{p^r}, rename the restriction to res^q) was rejected because res is already the group-cohomology restriction and $C(r)$, $\ker r^1$, $\operatorname{coker} r^0$ are the load-bearing notation of the newest chapter.

(O4) The elliptic points as $\zeta = e^{\pi i/3}$, ζ^2 (C9), rather than keeping ρ and renaming the representation.

(O5) Ramanujan's Δ and $\tau(N)$ written f_{12} and $a_N(f_{12})$ (C5, C4). This is the one resolution that departs from universal mathematical custom; it is taken because Δ and τ are both normative *implementation* objects with code citations, and the cusp form appears only in the classification literature.

(O6) The basis-name scheme $c_j^{(q)}$ / $z_j^{(q)}$ / $b_j^{(q)}$ (C6, C7, C8) replacing three colliding Greek letters at once.

(O7) Cone maps as named operators inc_c, pr_c rather than letters ((N3), C3).

2.10 The integral line

Added at v1.8.0 with §23; every symbol below is used only there and in the statements that point at it.

symbol	macro	meaning
$C^\bullet_{\mathrm{eq}}(R)$	—	$C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$ with fibre V_R and the relations imposed *over* R: the complex the implementation builds for the ring in force. **Never** the base change $C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes R$, which is written so; the two agree when $6 \in R^\times$ (23.2)
$\mathcal{I}_\bullet$, $\mathcal{I}^\partial_\bullet$	`\Iso`, `\Isob`	the isotropy-resolution model on the unrefined $(X, \partial X)$ and its boundary subcomplex (23.4)
$F^\sigma_\bullet$	—	the periodic free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\Gamma_\sigma]$; the analogue, for the stabiliser, of the $F_\bullet$ of 17.1 over $\mathbb{Z}[\Gamma]$ ((N2): the same kind of object for a subgroup, not the same object; worded so at v1.8.1)
Σ_s, Σ^∂_s	—	the orbit representatives of s-simplices of X, ∂X, as in the proof of 17.3
cmp	`\cmpI`	the comparison map $\mathcal{I}_\bullet \to C_\bullet(X)$, a named operator ((N3)); cmp^* its cochain map
$C^\bullet_{\mathcal{I}}(R)$	`\CI`	$\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_\bullet, V_R)$
$\tau_{\mathcal{I}}$, $\tau^\partial_{\mathcal{I}}$	—	the g-twisted chain datum on $\mathcal{I}_\bullet$, $\mathcal{I}^\partial_\bullet$ (23.13); an instance of 11.1's notion, not a new meaning of τ
$\widehat{\mathbb{Z}}_6$, $\widehat{\mathbb{Q}}_6$	`\Zsix`, `\Qsix`	$\mathbb{Z}_2 \times \mathbb{Z}_3$ and $\mathbb{Q}_2 \times \mathbb{Q}_3$
$\mathbb{Z}_\ell$, $\mathbb{Z}_{(\ell)}$	—	the ℓ-adic integers; the localisation of $\mathbb{Z}$ at ℓ
H^q_{fr}	`\Hfr`	$H^q(\Gamma; V)$ modulo its torsion
e	—	the exponent of the ring $\mathbb{Z}/\ell^e$ in force (§2.2); the design notes' "K" for a certified level is this e, since K is the model complex

Reading the design notes. The integral-line brief (`docs/implementation/integral_line_design_brief.md`) and the ℓ-adic investigation write R for the free resolution (here $\mathcal{I}_\bullet$: R is the coefficient ring), ε for the comparison map (here cmp: ε is the augmentation), $\widehat{C}$ for $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(R, V)$ (here $C^\bullet_{\mathcal{I}}$), $\hat{\tau}$ for the lifted datum (here $\tau_{\mathcal{I}}$), R_∂ (here $\mathcal{I}^\partial_\bullet$), Λ^q for the free quotient (here H^q_{fr}: Λ is the η-uniqueness map), $C_7 \otimes \mathbb{Z}/\ell^K$ for the pipeline object (here $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e)$), $\mathbb{Z}/\ell^k$ and K for rings and certified levels (here $\mathbb{Z}/\ell^e$), and e_i for torsion exponents (here a_t, local to the statement that introduces it).

2.11 The Dickson line and the ordinary projector

Added at v1.11.0-pre.1 with §24 and §25. The plan and the code write L, D and θ; this document cannot, because L is the matrix local system (§2.3), D_n the contragredient module and θ_g the translation of §11.1 (rule (A5)).

symbol	macro	meaning
L	`\DkL`	the first Dickson invariant $x^\ell y - xy^\ell \in \mathrm{Sym}^{\ell+1}(\mathbb{F}_\ell^2)$ (24.1); the code's and the plan's L
D	`\DkD`	the second Dickson invariant, of degree $\ell^2 - \ell$ (24.1); the plan's D
θ_{L}	`\thL`	multiplication by L, on fibres, cochains and cohomology (24.9); the literature's and the plan's θ
n^+	`\nplus`	$n + \ell + 1$, the target degree of θ_{L} (the plan's and the code's m; N is the Hecke level only, m the coset count)
Q_n, ρ_{Q_n}	`\Qn`	the quotient $\mathrm{Sym}^{n^+}/\mathsf{L}\cdot\mathrm{Sym}^n$ over $\mathbb{F}_\ell$ and its monoid action (24.7)
$\bar{u}_t$	—	the class of $u_t = x^{n^+-t}y^t$ in Q_n; the complement basis is $(\bar{u}_0, \dots, \bar{u}_{\ell-1}, \bar{u}_{n^+})$
$\mathbf{1}_n$	—	the class $\bar{u}_0 - \bar{u}_{\ell-1} + \bar{u}_{n^+}$, the invariant of Q_n when $(\ell-1) \mid (n+2)$ (24.8)
$h^q(m)$	—	$\dim_{\mathbb{F}_\ell} H^q(\Gamma; M_m)$
e_{ord}	`\eord`	the ordinary projector of an endomorphism of a finite-length module (25.1); *not* the class e of 14.22 nor the exponent e of $\mathbb{Z}/\ell^e$
Added at v1.12.0-pre.1 with §26.		
J_n	—	$\rho_n(S) - 1$ stacked over $\rho_n(T) - 1$, a $2(n+1) \times (n+1)$ integer matrix (26.1)
λ_e	—	$\lambda(H^0(\Gamma; V \otimes \mathbb{Z}/\ell^e))$, the length (26.1)
$e_{\max}$	—	the largest exponent of the ℓ-primary torsion of $H^1(\Gamma; V)$ (0 if none); the code's `e_max`
e'	—	a target precision; [E]'s and the code's k in $K = k + e_{\max}$, which this document cannot write (k is the weight)
$\mathfrak{g}$	—	a number of generators (26.3); the plan's r, which is the restriction map here
$N_{\mathrm{ord}}(\mathfrak{g}, K)$	—	the ordinary exponent $\ell^{K-1+\lceil \log_\ell \mathfrak{g} \rceil} \prod_{f \le \mathfrak{g}} (\ell^f - 1)$ (26.3); N alone is the Hecke level
$h^1_{\mathrm{ord}}(n)$, $\chi^{\mathrm{ord}}_{n,p}$	—	the ordinary rank and the characteristic polynomial of T_p on the ordinary part (26.5)

symbol	macro	meaning
$\mathrm{Tw}_m(\ell; n_0)$	—	the weight tower of level m through n_0 (26.5)
(Hi1)–(Hi6)	—	the predictions of 26.6; the plan's (H1)–(H6), which are the standing hypotheses here
Added at v1.13.0-pre.1 with §27.		
m_N	—	$[\Gamma : \Gamma_0(N)] = \lvert\mathbb{P}^1(\mathbb{Z}/N)\rvert$ (27.1); subscripted, as m alone is a coset count or a tower level
$[A\,g\,B]$	—	$\mathrm{cor}^B_{B\cap gAg^{-1}} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}^A_{A\cap g^{-1}Bg}$ between two groups; $[\Gamma g \Gamma] = T_{\Gamma g \Gamma}$
$\widetilde{[A\,g\,B]}$	—	its block formula on the induced fibre (68), reusing the level-1 τ-table (27.2)
$U_\ell^{(\ell)}$	—	$[\Gamma_0(\ell)\, g\, \Gamma_0(\ell)]$, $g = \mathrm{diag}(1,\ell)$ (27.3); the plan's $U_\ell^{(\ell)}$; $U_\ell = T_\ell$ stays the level-1 operator
U^*	—	$[\Gamma_0(\ell)\,\mathrm{diag}(\ell,1)\,\Gamma_0(\ell)]$, the Atkin–Lehner conjugate, Hida's $T(p)$ at level p (27.10)
dg_1, dg_ℓ	—	the degeneracy maps $\mathrm{res}^\Gamma_{\Gamma_0(\ell)}$ and $[\Gamma\, g\, \Gamma_0(\ell)]$; the plan's α_1, α_ℓ (α is 12.6's restriction unit)
$\alpha^0_{B/A}$, $\omega_{A/B}$	—	the level-change fibre maps, projection and trace (27.2(4)(b)); $\omega_{\Gamma_0(N)/\Gamma} = \omega$ of 12.7
ϖ	—	the block row $(\rho_n(gs_1^{-1}), \dots, \rho_n(gs_\ell^{-1}), 0)$ of (69)
$s_b^{(2)}$	—	$\begin{pmatrix} 1 & 0 \\ \ell b & 1 \end{pmatrix}$, representatives of $\Gamma_0(\ell^2)\backslash\Gamma_0(\ell)$ (27.3)
$\nu_2(N)$, $\nu_3(N)$	—	the numbers of fixed points of S, ST on $\mathbb{P}^1(\mathbb{Z}/N)$ (27.5)
e_1, e_ℓ	—	the ℓ-adic ordinary projectors of T_ℓ at level 1 and of $U_\ell^{(\ell)}$ at level ℓ (27.9)
c_{st}, u_{st}	—	$\omega\, e_\ell\, \mathrm{dg}_\ell$ on $e_1 H(1)$, and the unit root of $u^2 - T_\ell u + \ell^{n+1}$ (27.9)
$M_m(n)$, $\mathbb{T}^{\mathrm{ord}}_{m,\mathcal{P}}(n)$	—	the lattice datum and its Hecke algebra (27.13)
(Hi7)–(Hi12)	—	the predictions of 27.14

Part I — The framework

3 Simplicial complexes with relations, and the universal cover

What this chapter fixes. The combinatorial input of the whole document: a finite simplicial complex K, a *relation set* $\mathcal{R}$ on it, the quotient $K/\mathcal{R}$, and the complex K^u on which a group G acts with K as a strict fundamental domain. It fixes what it *means* for the pair $(K,\mathcal{R})$ to present that quotient (Assumption 3.17) — the property (H4)(iii) of §7 points at — and it proves that the elements occurring in $\mathcal{R}$ generate G (Theorem 3.24). Everything here is stated for an arbitrary finite $(K,\mathcal{R})$: no triangulation is fixed, and a subdivision of one admissible input is another admissible input.

Standing hypotheses. Assumptions 3.1, 3.3, 3.6 and 3.17 are the hypotheses of this Part. Per rule (A1) each statement below repeats the ones it uses; no statement of this document may be read as holding without them merely because they were declared here.

A note on the letter G. Part I is a framework for an arbitrary group acting on an arbitrary complex, and that group is written G. From §7 onwards it is instantiated once and for all as $G = \Gamma = \mathrm{SL}_2(\mathbb{Z})$ acting on $K^u \cong X$ ((H4)), and the letter G is not used again. G is the one symbol of these four chapters that §2 does not tabulate; it is used here in exactly the sense [v5] used it, and nowhere else in this document.

Complexes, orientations, and the quotient by a relation

A finite *simplicial complex* K on an ordered vertex set V_0 is a set of non-empty subsets of V_0 containing every singleton and closed under passing to non-empty subsets; $\dim \sigma = |\sigma| - 1$ and $\dim K = \max_\sigma \dim \sigma$; $|K|$ is the geometric realisation. $o_q(K)$ denotes the set of *oriented* q-simplices, and throughout this document the orientation of a simplex is the one induced by the vertex order — the stored orientation of §7, which is where the storage conventions are fixed and which this chapter does not restate. $\partial_t \sigma$ is the t-th face of σ ($0 \le t \le \dim \sigma$), with its induced orientation; t, j, s are the running indices of this document ((N4)).

Assumption 3.1 (Contractibility of the fundamental domain). K is a finite simplicial complex whose geometric realisation $|K|$ is connected and contractible.

Definition 3.2 (Vertex relation and quotient complex)**.** Let V_0 be the vertex set of K and let $\sim$ be an equivalence relation on V_0, written $[v]$ for the class of v. The *quotient complex* $K/\mathcal{R}$ is the set of subsets $\{[v_0], \dots, [v_q]\}$ of $V_0/\sim$ for which some $\{u_0, \dots, u_q\} \in K$ has $[u_t] = [v_t]$ for all t.

Assumption 3.3 (Non-identification on edges). If $v_0, v_1 \in V_0$ with $v_0 \sim v_1$ and $v_0 \neq v_1$, then $\{v_0, v_1\}$ is not a 1-simplex of K.

Lemma 3.4. *[proved here] Assume 3.3. Then*

1. *for every $\sigma = \{v_0, \dots, v_q\} \in K$ the classes $[v_0], \dots, [v_q]$ are pairwise distinct;*
2. *$K/\mathcal{R}$ is a simplicial complex;*
3. *$\mathrm{pr}_{\mathcal{R}} \colon K \to K/\mathcal{R}$, $\mathrm{pr}_{\mathcal{R}}(\{v_0, \dots, v_q\}) = \{[v_0], \dots, [v_q]\}$, is a surjective simplicial map.*

Proof. (1) If $[v_t] = [v_{t'}]$ with $t \neq t'$ then $v_t \sim v_{t'}$ and $v_t \neq v_{t'}$, while $\{v_t, v_{t'}\}$ is a face of σ and hence a 1-simplex of K — contradicting 3.3. (2) Every singleton $\{[v]\}$ lies in $K/\mathcal{R}$ because $\{v\} \in K$; and if $\sigma' = \{[v_0], \dots, [v_q]\} \in K/\mathcal{R}$ comes from $\sigma = \{u_0, \dots, u_q\} \in K$, any non-empty subset of σ' comes from the corresponding face of σ, which lies in K. (3) Immediate from (1) and Definition 3.2; surjectivity is that definition. □

Definition 3.5 (Boundary and interior)**.** Assume K is *pure* of dimension d, i.e. every maximal simplex has dimension d. The *boundary* ∂K is the subcomplex generated by the non-maximal simplices of K that are contained in a unique maximal simplex. Write $K^\circ := K \setminus \partial K$ for the *interior*; K° is in general not a subcomplex, and we use the convention $|K^\circ| = |K|^\circ$.

Assumption 3.6 (Open interior). K is pure, and for all $\sigma, \sigma' \in K$,

$$\sigma \neq \sigma' \ \wedge \ \mathrm{pr}_{\mathcal{R}}(\sigma) = \mathrm{pr}_{\mathcal{R}}(\sigma') \quad \Longrightarrow \quad \sigma, \sigma' \in \partial K.$$

3.6 says that $K/\mathcal{R}$ is obtained by glueing K to itself *along its boundary only*: no two distinct d-simplices are identified, and the relation is trivial on K°.

Example 3.7 (Simplicial S^1). $K = \{\{0\}, \{1\}, \{2\}, \{3\}, \{0,1\}, \{1,2\}, \{2,3\}\}$ is the interval in three segments, $\partial K = \{\{0\}, \{3\}\}$, and $\sim$ is generated by $0 \sim 3$. Then 3.1, 3.3 and 3.6 all hold and $|K/\mathcal{R}| \cong S^1$. This is the shape of the boundary model K_∂ of §14; it is an example, not a definition, and nothing below specialises to it.

Proposition 3.8. *[proved here] Assume 3.3 and 3.6. Then $\partial K/\sim$ is a simplicial subcomplex of $K/\mathcal{R}$, set-theoretically $|K^\circ| \sqcup |\partial K/\sim| = |K/\mathcal{R}|$, the map $|\mathrm{pr}_\mathcal{R}| : |K| \to |K/\mathcal{R}|$ is injective on $|K|^\circ$, and $\mathrm{pr}_\mathcal{R}$ is injective on d-simplices of K.*

Proof. Purity (3.6) makes ∂K well defined, and 3.3 is inherited by $(\partial K, \sim)$, so Lemma 3.4 applies to it and exhibits $\partial K/\sim$ as a subcomplex. The displayed implication of 3.6 says exactly that the only identifications happen inside ∂K, whence the set-theoretic disjoint decomposition and the injectivity on the interior; injectivity on d-simplices is the case $\sigma, \sigma' \notin \partial K$. □

Definition 3.9 (Saturated complexes)**.** K is *saturated* for $\sim$ if for every $\sigma \in K$ and every subset $\sigma' \subseteq V_0$ such that each vertex of σ is equivalent to exactly one vertex of σ' and conversely, one has $\sigma' \in K$: replacing any number of vertices of a simplex by other representatives of their classes again yields a simplex.

Proposition 3.10. *[proved here] Let K be a finite simplicial complex with an equivalence relation $\sim$ on its vertices. There is a saturated pair $(\widetilde{K}, \widetilde{\sim})$ together with a simplicial map $\iota_{\mathrm{sat}} \colon K \to \widetilde{K}$ descending to $K/\mathcal{R} \to \widetilde{K}/\widetilde{\sim}$, universal among simplicial maps $f \colon K \to L$ that descend to a map $K/\mathcal{R} \to L/\sim_L$: every such f factors as $f = \widetilde{f} \circ \iota_{\mathrm{sat}}$ for a unique simplicial $\widetilde{f} \colon \widetilde{K} \to L$ descending to $\widetilde{K}/\widetilde{\sim} \to L/\sim_L$. Moreover*

(a) for any such $(\widetilde{K}, \widetilde{\sim})$ the induced map $K/\mathcal{R} \to \widetilde{K}/\widetilde{\sim}$ is an isomorphism of simplicial complexes;

(b) $\widetilde{K}$ may be chosen with the same vertex set as K and ι_{sat} induced by the identity on vertices, in which case $\widetilde{\sim} \;= \sim$;

(c) if $(K, \sim)$ satisfies 3.1 (resp. 3.3) then so does $(\widetilde{K}, \widetilde{\sim})$.

Proof. Take $\widetilde{K}$ on the vertex set V_0 to consist of all $\sigma' \subseteq V_0$ obtained from some $\sigma \in K$ by replacing vertices with equivalent ones as in Definition 3.9, and ι_{sat} the identity on vertices. This is a simplicial complex (a subset of such a σ' is obtained the same way from the corresponding face of σ), it is saturated by construction, and it satisfies (b). Any f as in the statement sends equivalent vertices to equivalent vertices, hence sends each replacement σ' of σ to a simplex of L, which forces and defines $\widetilde{f}$ uniquely on vertices; this is the universal property. Since $K/\mathcal{R} = \widetilde{K}/\sim$ canonically — the two complexes have the same simplices by Definition 3.2 — (a) follows, and (c) is immediate from (b), because $\left|\widetilde{K}\right|$ deformation retracts onto $|K|$ along the identity on vertices and $\sim$ is unchanged. □

The universal cover complex

Definition 3.11 (Universal cover complex)**.** Assume 3.1, 3.3 and 3.6. Put $X := |K/\mathcal{R}|$, let $p \colon \widetilde{X} \to X$ be a universal topological cover and let $G := \pi_1(X, x_0)$ act on $\widetilde{X}$ by deck transformations. A *simplicial universal cover* of $(K, \mathcal{R})$ is a (possibly infinite) simplicial complex K^u with:

(a) $|K^u| = \widetilde{X}$;

(b) p is the realisation of a simplicial map $\mathrm{pr}^u \colon K^u \to K/\mathcal{R}$;

(c) there is a simplicial map $K \to K^u$, $\sigma \mapsto \widetilde{\sigma}$, with $\mathrm{pr}_\mathcal{R} = \mathrm{pr}^u \circ \widetilde{(-)}$;

(d) G acts simplicially on K^u, realising the deck action, and pr^u is G-invariant;

(e) K^u is finest with these properties: any G-invariant simplicial complex $K^{u\prime}$ covering $K/\mathcal{R}$ admits a G-equivariant simplicial map $K^{u\prime} \to K^u$ over $K/\mathcal{R}$.

The map of (c) is written $\sigma \mapsto \widetilde{\sigma}$ throughout. [v5] and the code write $i(\sigma)$; this document cannot, because i is $\sqrt{-1}$ and the elliptic vertex v_6 (§2.7, C15–C16). The code identifier `i(sigma)` is unaffected.

Proposition 3.12. *[standard: [Hat02, §1.3, Props. 1.33–1.34]] Assume 3.1, 3.3 and 3.6. Then a simplicial universal cover $(K^u, \mathrm{pr}^u, \widetilde{(-)})$ exists and is unique up to unique isomorphism.*

Proof. Uniqueness is the universal property (e). For existence: $K/\mathcal{R}$ is a simplicial complex by Lemma 3.4, so $X = |K/\mathcal{R}|$ is a connected, locally path-connected, semilocally simply connected CW complex and a universal cover $p\colon \widetilde{X} \to X$ exists [standard: [Hat02, §1.3]]. Take as vertices of K^u the fibre $p^{-1}(V_0/\sim)$, and declare $\widetilde{\sigma} = \{\widetilde{w}_0, \dots, \widetilde{w}_q\}$ a q-simplex whenever $p(\widetilde{\sigma})$ is the vertex set of a q-simplex σ' of $K/\mathcal{R}$ *and* the characteristic map $|\sigma'| \to X$ lifts to $|\sigma'| \to \widetilde{X}$ carrying those vertices to $\widetilde{w}_0, \dots, \widetilde{w}_q$; since $|\sigma'|$ is contractible, such a lift exists and is unique once the image of one vertex is prescribed [standard: [Hat02, Props. 1.33–1.34]]. This gives (a), (b). For (c): $|K|$ is contractible (3.1), so $|\mathrm{pr}_{\mathcal{R}}|$ lifts uniquely to $|K| \to \widetilde{X}$ once a basepoint lift is chosen, and that lift carries vertices to vertices of K^u and simplices to simplices, i.e. it is the realisation of a simplicial $\sigma \mapsto \widetilde{\sigma}$ with $\mathrm{pr}^u \circ \widetilde{(-)} = \mathrm{pr}_{\mathcal{R}}$. (d) holds because deck transformations permute $p^{-1}(V_0/\sim)$ and preserve lifted characteristic maps. (e) holds because any G-invariant simplicial structure on $\widetilde{X}$ over $K/\mathcal{R}$ has the lifted characteristic maps among its cells, so K^u refines it. □

Remark 3.13. Assume 3.1, 3.3 and 3.6. Then $\sigma \mapsto \widetilde{\sigma}$ is injective. Indeed, suppose $\widetilde{y_1} = \widetilde{y_2}$ for $y_1, y_2 \in |K|$, so $\mathrm{pr}_{\mathcal{R}}(y_1) = \mathrm{pr}_{\mathcal{R}}(y_2)$. If both lie in $|K|^\circ$, Proposition 3.8 gives $y_1 = y_2$. Otherwise, if $y_1 \neq y_2$, a path c in $|K|$ from y_1 to y_2 — which exists by 3.1 — projects to a loop $\mathrm{pr}_{\mathcal{R}}(c)$ in X whose lift is the loop $\widetilde{c}$ in the simply connected $\widetilde{X}$; hence $\mathrm{pr}_{\mathcal{R}}(c)$ is nullhomotopic, and the identification of y_1 with y_2 contributes no non-trivial element of G. Injectivity on simplices follows, since a simplex is determined by its vertices. This is the hypothesis Theorem 5.10 needs and the reason 3.6 is assumed.

Definition 3.14 (Ordered relation)**.** With V_0 ordered, put $\mathcal{R}_+ := \{(u,v) \in V_0 \times V_0 : u \sim v,\ u < v\}$ and $\mathcal{R}_- := \{(u,v) : u \sim v,\ u > v\}$; together with the diagonal these generate $\sim$. Set $V_{\mathrm{cl}}^{\pm} := \{u \in V_0 : \exists v,\ (u,v) \in \mathcal{R}_\pm\}$ and let $K_{\mathrm{cl}}^{\pm}$ be the subcomplexes of simplices all of whose vertices lie in $V_{\mathrm{cl}}^{\pm}$. Write $K_{\mathrm{cl}} := K_{\mathrm{cl}}^{-} \cup K_{\mathrm{cl}}^{+}$. Under 3.6, $K_{\mathrm{cl}}^{\pm} \subseteq \partial K$: the relation identifies simplices of K_{cl}^{-} with distinct simplices of K_{cl}^{+}, and it is trivial on the interior.

The relation datum, and what it means to present the quotient

The pair $(K, \mathcal{R})$ of this document carries more than a vertex relation: each identification comes with the group element realising it, and *simplices may be identified with themselves* by a non-trivial element — exactly the case that occurs at the elliptic vertices and, centrally, everywhere (§7.1, Remark 7.7). The following definition is the form of the relation datum the rest of the document uses.

Definition 3.15 (Relations, transporters, the coincidence groupoid)**.** Let a group G act simplicially and *without inversions* on a simplicial complex K^u (i.e. if γ maps a simplex to itself it fixes it pointwise, hence preserves its orientation), and let $K \subseteq K^u$ be a finite subcomplex, with $\widetilde{\sigma} \in o_q(K^u)$ the image of $\sigma \in o_q(K)$. For $\sigma, \sigma' \in o_q(K)$ put

$$\mathrm{Tr}(\sigma, \sigma') \ := \ \{\gamma \in G : \ \gamma \cdot \widetilde{\sigma'} = \widetilde{\sigma} \text{ in } o_q(K^u)\}.$$

A *relation* is a triple $\varrho = (\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma)$ with $\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}} \in o_q(K)$ for a common q (written $\deg \varrho = q$) and $\gamma \in \mathrm{Tr}(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}})$; γ is its *relation element* (C17: relation elements are $\gamma \in G$, never g, which is the Hecke element). The set of all relations is the *coincidence groupoid* $\mathcal{C}(K, K^u, G)$: it contains $(\sigma, \sigma, 1)$ for every σ and is closed under

$$(\sigma, \sigma', \gamma), (\sigma', \sigma'', \gamma') \ \longmapsto \ (\sigma, \sigma'', \gamma\gamma'), \qquad (\sigma, \sigma', \gamma) \ \longmapsto \ (\sigma', \sigma, \gamma^{-1}).$$

A set $\mathcal{R}$ of relations *generates* $\mathcal{C}(K, K^u, G)$ if the smallest subset of $\mathcal{C}(K, K^u, G)$ containing $\mathcal{R}$ and all $(\sigma, \sigma, 1)$ and closed under the two operations above is $\mathcal{C}(K, K^u, G)$ itself. Note $\mathrm{Tr}(\sigma, \sigma) = \mathrm{Stab}_G(\widetilde{\sigma})$ and that $\mathrm{Tr}(\sigma, \sigma')$, when non-empty, is a coset of that stabiliser.

Definition 3.16 (The ineffective kernel N)**.** Let G act simplicially and without inversions on K^u as in Definition 3.15. Write

$$N \;:=\; \ker\big(G \longrightarrow \mathrm{Aut}(K^u)\big) \;=\; \{\gamma \in G :\; \gamma\cdot\widetilde{\tau} = \widetilde{\tau} \text{ for every oriented simplex } \widetilde{\tau} \text{ of } K^u\}$$

for the subgroup acting *trivially* on K^u; the action is *effective* when $N = 1$. N is normal in G, is contained in $\mathrm{Stab}_G(\widetilde{\sigma})$ for every σ, and every non-empty transporter $\mathrm{Tr}(\sigma,\sigma')$ is a union of cosets of N. Both cases occur in this document: $N = 1$ in the free instance of Remark 3.20, while for $G = \Gamma = \mathrm{SL}_2(\mathbb{Z})$ acting on $K^u \cong X$ one has $N = \{\pm I\}$, because $-I$ acts trivially on $\mathcal{H}$ and, by (H4)(iv), no other element of Γ fixes every simplex. A *non-effective* action is therefore an explicit, supported case of this Part, not an unstated exception.

Assumption 3.17 (Fundamental domain; presenting the quotient). Let G, K^u, K, $\mathcal{R}$ be as in Definition 3.15, with $|K^u|$ connected, and let $N \trianglelefteq G$ be the ineffective kernel of Definition 3.16. We say that K **is a strict fundamental domain and** $(K,\mathcal{R})$ **presents the quotient** when all four of the following hold. Write $\widetilde{D} := |K| \setminus |K_{\mathrm{cl}}|$, where $K_{\mathrm{cl}} \subseteq K$ is the subcomplex generated by the simplices that a coincidence *moves*: those $\sigma \in o_\bullet(K)$ admitting a relation $(\sigma,\sigma',\gamma) \in \mathcal{C}(K,K^u,G)$ with $\gamma\cdot\widetilde{\sigma} \neq \widetilde{\sigma}$ — equivalently, with $\sigma' \neq \sigma$, the two conditions agreeing because $\gamma\widetilde{\sigma'} = \widetilde{\sigma}$ and γ is injective on simplices. A simplex merely *fixed* by a non-trivial element of G is therefore *not* in K_{cl}: its isotropy is the business of (FD4), not of K_{cl} (Remark 3.18). In the free instance of Remark 3.20 every stabiliser is trivial and this K_{cl} is the one of Definition 3.14.

(FD1) $|K|$ is connected, $\widetilde{D}$ is connected and non-empty, and $\widetilde{D}$ is dense in $|K|$: $\overline{\widetilde{D}} = |K|$.

(FD2) $G\cdot K = K^u$: every simplex of K^u is $\gamma\cdot\widetilde{\sigma}$ for some $\gamma \in G$ and $\sigma \in K$.

(FD3) $\gamma\widetilde{D} \cap \widetilde{D} = \emptyset$ for every $\gamma \in G \setminus N$, i.e. for every γ acting non-trivially on K^u. Since $\gamma\widetilde{D}$ depends only on the coset γN, this says exactly that the translates $\gamma\widetilde{D}$ are pairwise disjoint as γ ranges over G/N. (What the condition *buys* — the decomposition of $|K^u|$, and that all identifications occur on $|K_{\mathrm{cl}}|$ — is Remark 3.19; it is a consequence, not part of the hypothesis.)

(FD4) **Completeness.** $\mathcal{R}$ generates the coincidence groupoid $\mathcal{C}(K,K^u,G)$ in the sense of Definition 3.15.

Remark 3.18 (Why N is quantified over and never divided out). (FD1) and (FD3) are the two conditions of Assumption 3.17 that speak about *motion*, and both are stated modulo N. Read without N they degenerate the moment the action is non-effective, and the seven-vertex model is exactly such a case: every one of its simplices admits the coincidence $(\sigma,\sigma,-I)$, so a K_{cl} built from "σ admits (σ,σ',γ) with $(\sigma',\gamma) \neq (\sigma,1)$" would be all of K, leaving $\widetilde{D} = \emptyset$ and (FD1) unsatisfiable; and "$\gamma\widetilde{D} \cap \widetilde{D} = \emptyset$ for $\gamma \neq 1$" would fail at $\gamma = -I$ whatever $\widetilde{D}$ is. Quantifying modulo N, and building K_{cl} from the simplices actually moved, is what makes both conditions say what they were meant to say.

What must not be done instead is to replace G by G/N. The action on K^u factors through G/N; **the action on the coefficients does not.** For the model, $-I$ is the identity of $\mathrm{Aut}(K^u)$ but acts on $V = \mathrm{Sym}^n$ as $\rho_n(-I) = (-1)^n\mathrm{id}$ (Definition 8.1), so for odd n it acts as $-\mathrm{id}$; the constraint $f(\sigma) = \rho_n(-I)f(\sigma)$ that (FD4) therefore imposes on *every* simplex is precisely the odd-n vanishing mechanism $H^\bullet_{\mathrm{eq}}(K,\mathcal{R};L) = 0$ of Remark 7.7, and passing to G/N would delete it. Hence G stays $\mathrm{SL}_2(\mathbb{Z})$, N is quantified over in (FD1) and (FD3), and the whole of the isotropy — N included — is carried by (FD4), which is where isotropy belongs (Lemma 3.22(a)).

Remark 3.19 (What (FD1)–(FD3) buy: the decomposition of $|K^u|$). [proved here] Assume (FD1)–(FD3). Then no translate of $\widetilde{D}$ meets $|K_{\mathrm{cl}}|$, and

$$|K^u| \;=\; \Big(\bigsqcup_{\gamma N\in G/N} \gamma\widetilde{D}\Big) \;\sqcup \bigcup_{\gamma N\in G/N} \gamma\,|K_{\mathrm{cl}}|\,, \tag{1}$$

where the first union is disjoint and the second in general is not. Moreover $\overline{\widetilde{D}} = |K|$ maps onto $|K/\mathcal{R}|$ under $|\mathrm{pr}^u|$, and any two distinct points of $|K|$ with the same image both lie in $|K_{\mathrm{cl}}|$.

The form "hence $|K^u| = \bigsqcup_{\gamma\in G} \gamma\widetilde{D}$" would be false, and does not follow from the disjointness in (FD3) — not even for an effective action: the translates of $\widetilde{D}$ omit the translates of $|K_{\mathrm{cl}}|$, which is the whole identified boundary. They can be made to cover only by adding back one representative of each identified boundary simplex, i.e. by passing to a *half-open* fundamental domain. Nothing in this document needs one, and none is constructed.

Proof of Remark 3.19. No translate of $\widetilde{D}$ meets $|K_{\mathrm{cl}}|$. Since K_{cl} is a subcomplex, $|K_{\mathrm{cl}}|$ is the union of the relative interiors of the $\widetilde{\sigma}$, $\sigma \in K_{\mathrm{cl}}$; so let z lie in the relative interior of such a $\widetilde{\sigma}$ and suppose $z \in \gamma\widetilde{D}$. Then $\gamma^{-1}z \in \widetilde{D} \subseteq |K|$. The relative interiors of the simplices of K^u partition $|K^u|$ and γ^{-1} is simplicial, so $\gamma^{-1}z$ lies in the relative interior of $\gamma^{-1}\widetilde{\sigma}$; as it lies in $|K|$ and K is a subcomplex of K^u, that simplex is $\widetilde{\tau}$ for some $\tau \in K$. If $\tau \neq \sigma$ then $(\sigma, \tau, \gamma) \in \mathcal{C}(K, K^u, G)$ moves τ, so $\tau \in K_{\mathrm{cl}}$; if $\tau = \sigma$ then $\tau \in K_{\mathrm{cl}}$ by hypothesis on σ. Either way $\gamma^{-1}z \in |K_{\mathrm{cl}}|$, contradicting $\gamma^{-1}z \in \widetilde{D} = |K| \setminus |K_{\mathrm{cl}}|$. Hence $\gamma\widetilde{D} \cap |K_{\mathrm{cl}}| = \emptyset$ for every $\gamma \in G$, and therefore $\gamma'\widetilde{D} \cap \gamma\,|K_{\mathrm{cl}}| = \emptyset$ for all $\gamma, \gamma' \in G$.

Equation (1). The $\gamma\widetilde{D}$ are pairwise disjoint over G/N by (FD3). Given $z \in |K^u|$, (FD2) supplies γ with $\gamma^{-1}z \in |K| = \widetilde{D} \sqcup |K_{\mathrm{cl}}|$, so z lies in $\gamma\widetilde{D}$ or in $\gamma\,|K_{\mathrm{cl}}|$; the two families are disjoint by the previous paragraph.

Identifications occur on $|K_{\mathrm{cl}}|$. Surjectivity of $\overline{\widetilde{D}} = |K| \to |K/\mathcal{R}|$ is (FD1) together with the definition of $K/\mathcal{R}$. If $z \neq z'$ in $|K|$ have the same image then $z' = \gamma z$ for some $\gamma \in G$, necessarily $\gamma \notin N$. Both in $\widetilde{D}$ contradicts (FD3); one in $\widetilde{D}$ and the other in $|K_{\mathrm{cl}}|$ contradicts the first paragraph. So both lie in $|K_{\mathrm{cl}}|$. □

Remark 3.20 (The two instances that occur in this document). (FD4) is the substantive condition, and it is the one (H4)(iii) of §7 asserts for the seven-vertex model. Two instances occur:

- *The free instance ([v5]'s setting).* K satisfies 3.1, 3.3, 3.6; $G = \pi_1(|K/\mathcal{R}|, x_0)$; K^u is the simplicial universal cover of Definition 3.11. Deck transformations of a universal cover act *freely*, so every stabiliser $\mathrm{Tr}(\sigma, \sigma)$ is trivial — in particular the ineffective kernel is $N = 1$ (Definition 3.16) and (FD3) is the usual condition "for $\gamma \neq 1$" — $\mathcal{R}$ contains no relation (σ, σ, γ) with $\gamma \neq 1$, and $\mathcal{R}$ is the datum $\{(\sigma^-, \sigma^+, \gamma_r)\}$ indexed by $\mathcal{R}_+$ of Definition 3.14. (FD1)–(FD3) then hold with K_{cl} as in Definition 3.14, provided $\widetilde{D}$ is connected, dense in $|K|$ and simply connected.
- *The instance this document computes in.* $G = \Gamma = \mathrm{SL}_2(\mathbb{Z})$ acting on $K^u \cong X$, which has *finite non-trivial* simplex stabilisers — every one of them containing $-I$ ((H4)(iv)), so the action is *not* effective and $N = \{\pm I\}$ (Definition 3.16, Remark 3.18). Here $\mathcal{R}$ necessarily contains self-relations (σ, σ, γ), $\gamma \neq 1$: the three elliptic stabiliser rows and the $|o_\bullet(K)|$ central rows of Definition 7.2. (FD4) is then a statement with content, and omitting the central rows makes it false — which is precisely Remark 7.7, and the reason [v5]'s "issues with torsion, which we ignore" could not be carried over (Appendix A, entry A.8).

Nothing in §§3–6 fixes a triangulation. A subdivision of an admissible $(K, \mathcal{R})$, equivariant for the G-action, is again an admissible input; what is triangulation-indexed (N_q, $|\mathcal{R}|$, the stored simplex order) is recomputed per input and appears only from §7 onwards.

Proposition 3.21 (Relation elements). *[proved here] Let G, K^u, K, $\mathcal{R}$ satisfy 3.17.*

1. *For $\sigma, \sigma' \in o_q(K)$ the transporter $\mathrm{Tr}(\sigma, \sigma')$ is either empty or a left coset $\gamma \cdot \mathrm{Stab}_G(\widetilde{\sigma'})$, and $\mathrm{Tr}(\sigma, \sigma) = \mathrm{Stab}_G(\widetilde{\sigma})$.*
2. *In the free instance of Remark 3.20 — $G = \pi_1(|K/\mathcal{R}|, x_0)$ acting on the simplicial universal cover under 3.1, 3.3, 3.6 — every non-empty transporter is a* singleton. *In particular each pair $(u, v) \in \mathcal{R}_+$ of Definition 3.14, with $\sigma^+ = \{u\} \in K^+_{\mathrm{cl}}$ and $\sigma^- = \{v\} \in K^-_{\mathrm{cl}}$, determines a unique $\gamma_r \in G$ with $\gamma_r \cdot \widetilde{\sigma^+} = \widetilde{\sigma^-}$, and likewise in every degree for the higher-dimensional simplices identified by r.*

Proof. (1) If $\gamma, \gamma' \in \mathrm{Tr}(\sigma, \sigma')$ then $\gamma^{-1}\gamma' \cdot \widetilde{\sigma'} = \widetilde{\sigma'}$, so $\gamma' \in \gamma \,\mathrm{Stab}_G(\widetilde{\sigma'})$; conversely each such element transports. The second claim is the definition. (2) Deck transformations of a universal cover act freely on $\widetilde{X}$ [standard: [Hat02, §1.3]], so every stabiliser is trivial and (1) makes non-empty transporters singletons. That the pair (σ^-, σ^+) has non-empty transporter is (FD2) together with $\mathrm{pr}_{\mathcal{R}}(\sigma^-) = \mathrm{pr}_{\mathcal{R}}(\sigma^+)$ and $\mathrm{pr}^u \circ \widetilde{(-)} = \mathrm{pr}_{\mathcal{R}}$. □

Lemma 3.22 (Criterion for (FD4): stabilisers and orbits)**.** *[proved here] Let G, K^u, K be as in Definition 3.15 (G acting simplicially and without inversions, $K \subseteq K^u$ a finite subcomplex) and let $\mathcal{R} \subseteq \mathcal{C}(K, K^u, G)$ be a set of relations. Write $\langle \mathcal{R} \rangle$ for the sub-groupoid of $\mathcal{C}(K, K^u, G)$ generated by $\mathcal{R}$ in the sense of Definition 3.15. Then $\mathcal{R}$ satisfies (FD4) — that is, $\langle \mathcal{R} \rangle = \mathcal{C}(K, K^u, G)$ — if and only if both of the following hold.*

(a) ***Every stabiliser is reached.*** *For every $\sigma \in o_\bullet(K)$,*

$$\{\gamma \in G : \ (\sigma, \sigma, \gamma) \in \langle \mathcal{R} \rangle\} \ = \ \mathrm{Stab}_G(\widetilde{\sigma}).$$

(b) ***Every coincidence of simplices is reached.*** *For every pair $\sigma, \sigma' \in o_q(K)$ with $\mathrm{Tr}(\sigma, \sigma') \neq \emptyset$ there is* at least one *γ with $(\sigma, \sigma', \gamma) \in \langle \mathcal{R} \rangle$.*

Moreover it suffices, for (a), that for each σ the relation elements of the self-relations *$(\sigma, \sigma, \gamma) \in \mathcal{R}$ generate $\mathrm{Stab}_G(\widetilde{\sigma})$; and for (b), that each such pair $\{\sigma, \sigma'\}$ be joined by a path of members of $\mathcal{R}$, traversed in either direction.*

Remark 3.23 (Why the criterion is the right shape). (a) and (b) split (FD4) into its two independent contents: a *local* one — $\mathcal{R}$ must see each simplex's full isotropy — and a *global* one — $\mathcal{R}$ must connect each G-orbit of simplices of K. Neither implies the other, and the split is exactly the one the seven-vertex model needs: its eight listed rows discharge (b) and the three elliptic stabilisers, while (a) at the remaining twenty simplices is discharged by, and only by, the central rows $(\sigma, \sigma, -I)$ of Definition 7.2 (Remark 7.7, Proposition 7.4).

Both conditions are finite and decidable for finite K with finite stabilisers, which is what makes (FD4) a formalization target: (a) is one equality of subgroups of a finite group per simplex of K, and (b) is a reachability question in the finite graph on $o_\bullet(K)$ whose edges are the members of $\mathcal{R}$, compared against the partition of $o_\bullet(K)$ into G-orbits. No enumeration of K^u, and no enumeration of G, is involved in either.

Proof of Lemma 3.22. Necessity. Both conditions are special cases of $\langle \mathcal{R} \rangle = \mathcal{C}(K, K^u, G)$: the members of $\mathcal{C}(K, K^u, G)$ of the form $(\sigma, \sigma, -)$ are precisely the elements of $\mathrm{Stab}_G(\widetilde{\sigma})$, and $\mathrm{Tr}(\sigma, \sigma') \neq \emptyset$ says that some triple $(\sigma, \sigma', \gamma)$ lies in $\mathcal{C}(K, K^u, G)$.

Sufficiency. Let $(\sigma, \sigma', \gamma) \in \mathcal{C}(K, K^u, G)$. Then $\gamma \in \mathrm{Tr}(\sigma, \sigma')$, which is therefore non-empty, so (b) supplies a γ_0 with $(\sigma, \sigma', \gamma_0) \in \langle \mathcal{R} \rangle$. By Proposition 3.21(1) the transporter is the left coset $\gamma_0 \,\mathrm{Stab}_G(\widetilde{\sigma'})$, so $h := \gamma_0^{-1}\gamma \in \mathrm{Stab}_G(\widetilde{\sigma'})$, and (a) at σ' gives $(\sigma', \sigma', h) \in \langle \mathcal{R} \rangle$. Composing the two,

$$(\sigma, \sigma', \gamma_0) \cdot (\sigma', \sigma', h) \ = \ (\sigma, \sigma', \gamma_0 h) \ = \ (\sigma, \sigma', \gamma) \ \in \ \langle \mathcal{R} \rangle.$$

The two sufficient conditions. $\langle \mathcal{R} \rangle$ is closed under composition and inversion, so for a fixed σ the set $\{\gamma : (\sigma, \sigma, \gamma) \in \langle \mathcal{R} \rangle\}$ is a *subgroup* of $\mathrm{Stab}_G(\widetilde{\sigma})$ — it is contained in the stabiliser because every member of $\mathcal{C}(K, K^u, G)$ is — and it contains the relation element of every self-relation of $\mathcal{R}$ at σ; if those generate the stabiliser, the subgroup is all of it. Likewise a path $\sigma = \sigma_0, \sigma_1, \ldots, \sigma_m = \sigma'$ whose consecutive terms are the two simplices of a member of $\mathcal{R}$ composes, after inverting the members traversed backwards, to a member of $\langle \mathcal{R} \rangle$ over the pair (σ, σ'). □

Theorem 3.24 (Generation by the relation elements)**.** *[proved here] Let G, K^u, K, $\mathcal{R}$ satisfy 3.17, with $|K^u|$ connected and K finite. Then G is generated by the relation elements*

$$\mathcal{E}(\mathcal{R}) \ := \ \{\gamma_\varrho : \varrho \in \mathcal{R}\} \subseteq G.$$

If moreover K satisfies 3.1, 3.3 and 3.6 and $G = \pi_1(X, x_0)$, $X = |K/\mathcal{R}|$, acts on the simplicial universal cover of Definition 3.11, then for $r = (u, v) \in \mathcal{R}_+$ and any path c in $|K|$ from u to v whose interior lies in $|K|^\circ$ and passes through a point x_K with $\mathrm{pr}_{\mathcal{R}}(x_K) = x_0$, the loop $\mathrm{pr}_{\mathcal{R}}(c)$ represents $\gamma_r \in \pi_1(X, x_0)$, independently of the choice of c.

Proof. Put $\Sigma := \{\gamma \in G : \gamma K \cap K \neq \emptyset \text{ as sets of simplices of } K^u\}$ and $H := \langle \Sigma \rangle$.

$\Sigma \subseteq \langle \mathcal{E}(\mathcal{R}) \rangle$. Let $\gamma \in \Sigma$ and let $\widetilde{\sigma} = \gamma \cdot \widetilde{\sigma'}$ with $\sigma, \sigma' \in K$; as G acts without inversions the equality holds for the stored orientations, so $(\sigma, \sigma', \gamma) \in \mathcal{C}(K, K^u, G)$. By (FD4) that triple is obtained from members of $\mathcal{R}$ and identity triples by composition and inversion, so γ is a word in $\mathcal{E}(\mathcal{R})^{\pm 1}$.

$H = G$. Since $|K^u|$ is connected, its 1-skeleton is connected; fix a vertex $\widetilde{v_*}$ with $v_* \in K$. We show by induction on the edge-path distance from $\widetilde{v_*}$ that every vertex $\widetilde{w}$ of K^u lies in ηK for some $\eta \in H$. For distance 0 take $\eta = 1$. If $\widetilde{w}$ is adjacent to a vertex $\widetilde{u} \in \eta K$, $\eta \in H$, then by (FD2) the edge $\{\widetilde{u}, \widetilde{w}\}$ lies in δK for some $\delta \in G$; then $\widetilde{u} \in \eta K \cap \delta K$, so $\eta^{-1}\delta \in \Sigma \subseteq H$, so $\delta \in H$, and $\widetilde{w} \in \delta K$. Now let $\gamma \in G$ be arbitrary and pick a vertex $v \in K$. By the above, $\gamma\widetilde{v} \in \delta K$ for some $\delta \in H$, i.e. $\delta^{-1}\gamma\widetilde{v}$ is a vertex of K; hence $\delta^{-1}\gamma \in \Sigma \subseteq H$ and $\gamma \in H$. Combining with the previous paragraph, $G = H = \langle \mathcal{E}(\mathcal{R}) \rangle$.

The loop description. Assume the further hypotheses. The lift of $\mathrm{pr}_{\mathcal{R}}(c)$ starting at $\widetilde{u}$ is $\widetilde{c}$, which ends at $\widetilde{v}$; by definition of the deck action the loop $\mathrm{pr}_{\mathcal{R}}(c)$, based at $\mathrm{pr}_{\mathcal{R}}(u) = \mathrm{pr}_{\mathcal{R}}(v)$, represents the unique $\gamma \in G$ with $\gamma\widetilde{u} = \widetilde{v}$, which is γ_r (Proposition 3.21(2)). Changing the basepoint from $\mathrm{pr}_{\mathcal{R}}(u)$ to x_0 conjugates by the path $\mathrm{pr}_{\mathcal{R}}(c|_{[u,x_K]})$, whose lift $\widetilde{c}|_{[u,x_K]}$ lies in $|K| = \overline{\widetilde{D}}$; carrying out the conjugation, the lift of the based loop $\mathrm{pr}_{\mathcal{R}}(c|_{[x_K,u]}) \cdot \mathrm{pr}_{\mathcal{R}}(c) \cdot \mathrm{pr}_{\mathcal{R}}(c|_{[u,x_K]})$ starting at $\widetilde{x_K}$ ends at $\gamma_r\widetilde{x_K}$, so that loop represents γ_r in $\pi_1(X, x_0)$. Independence of c: any two admissible paths are homotopic rel endpoints in the contractible $|K|$ (3.1), and the homotopy projects. □

Code map. Definition 3.15: `SimplexRelation` (`LocalSystemSimplicialComplex.h`) is the triple $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma)$, stored with its $\gamma \in \mathrm{M}_2(\mathbb{Z})$ and the derived matrix $\rho_n(\gamma)$; the degree is `q_dim` and the two simplices are `s_lhs_idx_in_Kq`, `s_rhs_idx_in_Kq`. (FD4) for the model: `SL2ZRelationSeeds_7VertexModel` plus `GenerateSL2ZRelationsWithGroupData` (`LocalSystemSL2Z.cpp`), checked by `HeckeTessellationValidation` and `HeckeTessellationSmokeValidati` Definition 3.11(c): `i(sigma)` in the τ-table sources (§18).

Formalization note. Definition 3.15 and (FD4) are the formalization-friendly core: for a finite K and a decidable equality on $o_\bullet(K^u)$, the coincidence groupoid is a decidable relation and "$\mathcal{R}$ generates it" is a reachability statement in a finite graph once the relevant simplices are enumerated. What needs genuine topology is only Proposition 3.12 (existence of the universal cover) and the free-instance half of Proposition 3.21; the instance actually computed in (Remark 3.20, second bullet) takes $K^u \cong X$ and its Γ-action as given by Proposition 7.4 and never constructs a universal cover. Theorem 3.24's first half is a finite induction over an edge path and is the natural first target.

4 Matrix local systems and the three equivalent categories

What this chapter fixes. The coefficient side of the datum: a *matrix local system* is a rule assigning an invertible matrix to each relation, subject to a cocycle condition, and this is the same thing as a representation of G (Proposition 4.3) and — over a field, in the free instance — the same thing as a locally constant sheaf on $|K/\mathcal{R}|$ (Theorems 4.4, 4.5). The chapter exists so that the computational datum $(K, \mathcal{R}, \{M_\varrho\})$ may be used with a clear conscience: it is not an ad hoc encoding but an object of an equivalent category.

Standing hypotheses. R is a commutative ring with 1 and V a free R-module of finite rank d_V; G, K^u, K, $\mathcal{R}$ satisfy 3.17. Where a statement needs R to be a *field* it says so: [v5] worked over a field $\mathbb{F}$ throughout and did not mark where that was used, and $R = \mathbb{Z}/\ell^e$ is in scope here (§2.2).

Definition 4.1 (Matrix local system)**.** A *matrix local system* $L = (V, \{M_\varrho\}_{\varrho\in\mathcal{R}})$ of rank d_V on $(K, \mathcal{R})$ consists of a free R-module V of rank d_V and, for each relation $\varrho \in \mathcal{R}$, an invertible $M_\varrho \in \mathrm{GL}(V)$, subject to the *cocycle condition*: for every word $\gamma_{\varrho_1}^{\epsilon_1} \cdots \gamma_{\varrho_m}^{\epsilon_m} = 1$ in G with $\varrho_t \in \mathcal{R}$ and $\epsilon_t = \pm 1$, one has $M_{\varrho_1}^{\epsilon_1} \cdots M_{\varrho_m}^{\epsilon_m} = \mathrm{id}_V$; and $M_\varrho = M_{\varrho'}$ whenever $\gamma_\varrho = \gamma_{\varrho'}$. A *morphism* $L \to L' = (V', \{M'_\varrho\})$ is an R-linear $\phi\colon V \to V'$ with $\phi \circ M_\varrho = M'_\varrho \circ \phi$ for every $\varrho \in \mathcal{R}$. The resulting category is $\mathbf{LS} = \mathbf{LS}(K, \mathcal{R})$.

Remark 4.2 (The matrices are derived, never entered). In this document $V = \mathrm{Sym}^n(\mathbb{Z}^2) \otimes R$ and $M_\varrho := \rho_n(\gamma_\varrho)$ for the representation ρ_n of Definition 8.1; the cocycle condition is then automatic, because ρ_n is a homomorphism on $\mathrm{SL}_2(\mathbb{Z})$. That is the contract box of §7.1: the list of relations is the single source of truth for the pairs (relation, γ), and the matrix is derived. Definition 4.1 is stated in the general form because the cocycle condition is what makes the general framework work, and because an implementation that entered matrices by hand would need it as a check.

Proposition 4.3. *[proved here] Let $G, K^u, K, \mathcal{R}$ satisfy 3.17 and let V be a free R-module of finite rank. A family $\{M_\varrho\}_{\varrho\in\mathcal{R}}$ of invertible matrices is a matrix local system in the sense of Definition 4.1 if and only if the assignment $\gamma_\varrho \mapsto M_\varrho$ extends to a (necessarily unique) group homomorphism $\rho_L\colon G \to \mathrm{GL}(V)$. This assignment is a bijection, functorial in V.*

Proof. By Theorem 3.24 (whose hypotheses are 3.17 with $|K^u|$ connected and K finite), G is generated by $\mathcal{E}(\mathcal{R}) = \{\gamma_\varrho\}$. A map defined on a generating set extends to a homomorphism exactly when the images satisfy every relation the generators satisfy, which is the cocycle condition; the extension is then unique because the γ_ϱ generate. The consistency requirement $\gamma_\varrho = \gamma_{\varrho'} \Rightarrow M_\varrho = M_{\varrho'}$ of Definition 4.1 is what makes $\gamma_\varrho \mapsto M_\varrho$ a well-defined map on $\mathcal{E}(\mathcal{R})$ in the first place. Functoriality: a morphism ϕ intertwines the M_ϱ iff it intertwines ρ_L on generators iff it intertwines ρ_L. □

Theorem 4.4 (Monodromy equivalence)**.** *[standard: [Hat02, §1.3, Thm. 1.38, and §3.H]] Let X be a connected, locally path-connected and semilocally simply connected topological space with basepoint x_0, let $G = \pi_1(X, x_0)$, and let R be a commutative ring. The category $\mathbf{LCS}(X)$ of locally constant sheaves of finitely generated free R-modules on X is equivalent to the category $\mathbf{Rep}(G)$ of finitely generated free R-modules with a G-action. The equivalence sends a sheaf to its stalk at x_0 with the monodromy action by parallel transport, with quasi-inverse $(V, \rho) \mapsto (\widetilde{X} \times V)/G$.*

$|K/\mathcal{R}|$ satisfies those hypotheses whenever 3.3 holds (Lemma 3.4 makes it the realisation of a simplicial complex, hence a CW complex) and $|K|$ is connected.

Theorem 4.5. *[proved here] Let $G, K^u, K, \mathcal{R}$ satisfy 3.17 with $|K^u|$ connected and K finite, and let R be a commutative ring. Then $\mathbf{Rep}(G)$ and $\mathbf{LS}(K, \mathcal{R})$ are equivalent, by mutually quasi-inverse functors*

$$S\colon (V, \rho) \longmapsto (V, \{\rho(\gamma_\varrho)\}_{\varrho\in\mathcal{R}}), \qquad T\colon (V, \{M_\varrho\}) \longmapsto (V, \rho_L),$$

each the identity on underlying modules and on morphisms.

Proof. S lands in $\mathbf{LS}$: the cocycle condition holds because ρ is a homomorphism, and $\gamma_\varrho = \gamma_{\varrho'}$ forces equal matrices. T is Proposition 4.3. Both are the identity on underlying modules, and a linear map intertwines all $\rho(\gamma_\varrho)$ iff it intertwines ρ on the generating set $\mathcal{E}(\mathcal{R})$ (Theorem 3.24) iff it intertwines ρ; so both are fully faithful, and $TS = \mathrm{id}$, $ST = \mathrm{id}$ on the nose. □

Corollary 4.6. *[proved here] In the free instance of Remark 3.20 — K satisfying 3.1, 3.3, 3.6, $X = |K/\mathcal{R}|$, $G = \pi_1(X, x_0)$, K^u the simplicial universal cover — the three categories* $\mathbf{LCS}(X)$, $\mathbf{Rep}(G)$ *and* $\mathbf{LS}(K, \mathcal{R})$ *are equivalent, over any commutative ring R.*

Proof. Compose Theorems 4.4 and 4.5; the hypotheses of the former hold for $X = |K/\mathcal{R}|$ by the remark after it, those of the latter by Remark 3.20. □

Remark 4.7 (What the equivalence does *not* say in the instance computed in). Corollary 4.6 is stated for the *free* instance only. For $G = \Gamma = \mathrm{SL}_2(\mathbb{Z})$ acting on X the stabilisers are non-trivial, the quotient is an orbifold rather than a space with fundamental group Γ, and $\mathbf{LCS}(|K/\mathcal{R}|)$ is *not* equivalent to $\mathbf{Rep}(\Gamma)$. What survives without any freeness hypothesis is Theorem 4.5, $\mathbf{Rep}(G) \simeq \mathbf{LS}(K, \mathcal{R})$ — which is all the rest of this document uses, since the coefficient datum enters only through ρ_n (Remark 4.2) — together with the isomorphism Ψ of Theorem 5.10, which likewise needs no freeness. The comparison with $H^\bullet(\Gamma; V_R)$ is a separate statement carrying the hypothesis $6 \in R^\times$ at its own point of use (Remark 7.6, (H3)); [v5]'s "up to issues with torsion, which we ignore" is retired (Appendix A, entry A.8).

Code map. Definition 4.1: `LocalSystemSimplicialComplexT<Elt>` carries V as `representation_dim` $= d_V$ and the family $\{M_\varrho\}$ as the `SimplexRelation::M` fields; ρ_L is ρ_n (`ComputeSymmetricPowerMatrixFM`, `FlatMatrix.cpp`). Remark 4.2: `GenerateSL2ZRelationsWithGroupData` derives every `M` from the stored γ and never accepts one by hand; `HeckeTessellationValidation` checks that.

Formalization note. Proposition 4.3 and Theorem 4.5 are pure algebra over a generating set and are formalizable as stated; over $R = \mathbb{F}_\ell$ they are statements about `Matrix.GeneralLinearGroup` and a `MonoidHom`. Theorem 4.4 is the only genuinely topological input of this chapter, is used nowhere downstream (Remark 4.7), and would be the one item to import rather than prove.

5 The explicit equivariant cochain complex

What this chapter fixes. The complex this document computes with, and the isomorphism through which every later formula is transported onto it. Three descriptions of the same cohomology are given — the cochain complex of $K/\mathcal{R}$ with local coefficients, the G-equivariant complex on K^u, and the *explicit* complex $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$ of functions on $o_\bullet(K)$ constrained by $\mathcal{R}$ — and the last two are identified by Ψ (Theorem 5.10). Ψ is what makes a cochain a finite list of vectors indexed by the simplices of the *finite* complex K, and it is the statement six C++ source files cite (Appendix A, entry A.7).

Standing hypotheses. R is a commutative ring, $L = (V, \{M_\varrho\})$ a matrix local system on $(K, \mathcal{R})$ with representation $\rho_L \colon G \to \mathrm{GL}(V)$ (Proposition 4.3), and $G, K^u, K, \mathcal{R}$ satisfy 3.17. Statements that additionally need a *free* action, a *contractible* $|K^u|$, or R a field, say so in their own statement (rule (A1)); Theorem 5.10(1) needs none of the three.

Cochains on the quotient, and on the universal cover

Let L be a matrix local system with representation ρ_L. Fix a total order on the vertices of $K/\mathcal{R}$, orienting its simplices, and for each $\sigma' \in o_q(K/\mathcal{R})$ choose a lift $\widetilde{\sigma'} \in o_q(K^u)$ with $\mathrm{pr}^u(\widetilde{\sigma'}) = \sigma'$. A q-cochain of $K/\mathcal{R}$ with coefficients in L is a function $f \colon o_q(K/\mathcal{R}) \to V$, so

$C^q(K/\mathcal{R};L) = \bigoplus_{\sigma' \in o_q(K/\mathcal{R})} V$, and the coboundary is

$$(d^q f)(\vartheta) \;=\; \sum_{t=0}^{q+1} (-1)^t \, \rho_L\big(\gamma_t(\vartheta)\big)\, f(\partial_t \vartheta), \qquad \vartheta \in o_{q+1}(K/\mathcal{R}), \tag{2}$$

where $\gamma_t(\vartheta) \in G$ is the unique element with $\partial_t \widetilde{\vartheta} = \gamma_t(\vartheta) \cdot \widetilde{(\partial_t \vartheta)}$ when the action is free, and any element of the corresponding transporter otherwise — in which case (2) is well defined only if ρ_L is trivial on the relevant stabilisers, which is one more reason the equivariant description below is the primary one. $H^\bullet(K/\mathcal{R};L)$ denotes the cohomology of (2). The coboundary is written d throughout; δ is a coset element (§2.7, C7).

Remark 5.1 (Equivariant description). [proved here] For a free G-action on K^u and any choice of lifts as above there is a natural isomorphism of cochain complexes

$$C^q(K/\mathcal{R};L) \;\cong\; \mathrm{Hom}_{\mathbb{Z}[G]}\big(C_q(K^u), V\big), \qquad f_\phi(\sigma') := \phi(\widetilde{\sigma'}),$$

where $C_\bullet(K^u)$ is the integral simplicial chain complex of K^u and $\mathrm{Hom}_{\mathbb{Z}[G]}$ means $\phi(\gamma \cdot \widetilde{\sigma}) = \rho_L(\gamma)\phi(\widetilde{\sigma})$. Under it the coboundary (2) becomes $(d\phi)(\widetilde{\vartheta}) = \phi(\partial\widetilde{\vartheta})$, with $\partial = \sum_t (-1)^t \partial_t$ the chain boundary: a G-equivariant ϕ is determined by its values on one lift per orbit, and the $\rho_L(\gamma_t)$ of (2) are exactly the correction factors moving $\partial_t\widetilde{\vartheta}$ to the chosen lift. **The right-hand side is intrinsic: it involves no choice of lifts and no freeness.** It is the definition this document uses, and $H^\bullet(K/\mathcal{R};L)$ is to be read through it.

Theorem 5.2 (Equivalence of cohomologies)**.** *Let $G, K^u, K, \mathcal{R}$ satisfy 3.17, let L be a matrix local system with representation $\rho_L\colon G \to \mathrm{GL}(V)$, and assume in addition that $|K^u|$* ***is contractible and the G-action on K^u is free****. Put $X := |K/\mathcal{R}|$ and let $\mathcal{F}_L$ be the locally constant sheaf on X corresponding to ρ_L under Theorem 4.4. Then there are natural isomorphisms*

$$H^\bullet(K/\mathcal{R};L) \;\cong\; H^\bullet(G;V) \;\cong\; H^\bullet(X;\mathcal{F}_L).$$

Proof. [proved here] from the following two standard inputs. *(1)* K^u is a G-equivariant triangulation of the contractible space $|K^u|$ on which G acts freely, so $C_\bullet(K^u;\mathbb{Z})$ is a complex of free $\mathbb{Z}[G]$-modules — freely permuted basis, trivial stabilisers — and is acyclic with $H_0 = \mathbb{Z}$ via the augmentation ε; i.e. it is a free $\mathbb{Z}[G]$-resolution of $\mathbb{Z}$ [standard: [Bro82, Ch. I.4], where a free action on a complex with contractible realisation is shown to give a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$]. Group cohomology may be computed from any such resolution [standard: [Bro82, Ch. I.7.4] for the independence of the resolution, [Bro82, Ch. III.1] for the definition with coefficients], so $H^\bullet(G;V) = H^\bullet\big(\mathrm{Hom}_{\mathbb{Z}[G]}(C_\bullet(K^u), V)\big)$, which is $H^\bullet(K/\mathcal{R};L)$ by Remark 5.1. *(2)* $X = |K^u|/G$ with $|K^u|$ contractible and G acting freely and properly discontinuously, so X is a $K(G,1)$ and $|K^u|$ is its universal cover; cohomology of X with coefficients in the local system $\mathcal{F}_L$ is by definition the cohomology of $\mathrm{Hom}_{\mathbb{Z}[\pi_1(X)]}(C_\bullet(\widetilde{X}), V)$ [standard: [Hat02, §3.H]], which is the same complex. Naturality in L is naturality of both identifications in V. □

The freeness hypothesis is not decorative. For $G = \Gamma = \mathrm{SL}_2(\mathbb{Z})$ on X the chain groups $C_q(X)$ are sums of $\mathbb{Z}[\Gamma/\Gamma_\sigma]$ with $\Gamma_\sigma \neq 1$, hence not free, and step (1) fails; what replaces it, under the hypothesis $6 \in R^\times$, is the relative-projectivity argument of Remark 7.6, and at $\ell \in \{2,3\}$ nothing replaces it and no comparison is asserted ((H3)). Theorem 5.10 below is the part of this chapter that survives without freeness, and it is the part the implementation uses.

The explicit complex on K

Definition 5.3 (Naive cochains on K)**.** $C^q(K;V)$ is the R-module of functions $f\colon o_q(K) \to V$, i.e. $\bigoplus_{\sigma \in o_q(K)} V$, with the standard coboundary

$$(d^q f)(\sigma) \;=\; \sum_{t=0}^{q+1} (-1)^t f(\partial_t \sigma), \qquad \sigma \in o_{q+1}(K), \tag{3}$$

so $d^q = d_1^q \otimes \mathrm{id}_V$ with d_1^q the coboundary of K with trivial coefficients. The stored orientation and the flat coordinates in which (3) is realised are fixed once in §7 and are not restated here.

$(C^\bullet(K;V), d)$ computes $H^\bullet(K;V)$, which under 3.1 is V in degree 0 and 0 above: it carries no information about L. The local system enters as a *constraint*.

Definition 5.4 (Equivariant cochains)**.** [formally verified: [GT, `def:relcx`]] ([GT] imposes the identity over the full transporter groupoid, Lemma 5.5 under (FD4); the coboundary carries $\rho_L(h_j)$, $= 1$ on the model) $C^q_{\mathrm{eq}}(K, \mathcal{R}; L) \subseteq C^q(K;V)$ is the submodule of those f satisfying, for every relation $\varrho = (\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$ of degree q,

$$f(\sigma_{\mathrm{lhs}}) \;=\; M_\varrho\, f(\sigma_{\mathrm{rhs}}) \;=\; \rho_L(\gamma)\, f(\sigma_{\mathrm{rhs}}). \tag{4}$$

Relations with $\sigma_{\mathrm{lhs}} = \sigma_{\mathrm{rhs}}$ are included and are not vacuous: they read $\big(\mathrm{id} - \rho_L(\gamma)\big) f(\sigma) = 0$ (Remark 7.7).

Lemma 5.5 (Consequences of $\mathcal{R}$ are imposed too)**.** *[proved here] [formally verified: [GT, `def:relcx`]] Let $f \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ and let $(\sigma, \sigma', \gamma)$ lie in the sub-groupoid generated by the degree-q members of $\mathcal{R}$ in the sense of Definition 3.15. Then $f(\sigma) = \rho_L(\gamma) f(\sigma')$. In particular, under (FD4) the identity holds for* every *γ with $\gamma\widetilde{\sigma'} = \widetilde{\sigma}$.*

Proof. The set of triples $(\sigma, \sigma', \gamma)$ for which the identity holds for all $f \in C^q_{\mathrm{eq}}$ contains $\mathcal{R}$ and all $(\sigma, \sigma, 1)$, and is closed under composition ($f(\sigma) = \rho_L(\gamma) f(\sigma') = \rho_L(\gamma)\rho_L(\gamma') f(\sigma'') = \rho_L(\gamma\gamma') f(\sigma'')$) and inversion ($\rho_L(\gamma)$ is invertible). So it contains the generated sub-groupoid. Under (FD4) that is all of $\mathcal{C}(K, K^u, G)$ in degree q, whose degree-q members are exactly the $(\sigma, \sigma', \gamma)$ with $\gamma\widetilde{\sigma'} = \widetilde{\sigma}$. □

Assumption 5.6 (Consistent face identification). $\mathcal{R}$ is *closed under faces*: for every $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$ of degree $q+1$ and every $0 \le t \le q+1$, the triple $(\partial_t\sigma_{\mathrm{lhs}}, \partial_t\sigma_{\mathrm{rhs}}, \gamma)$ lies in the sub-groupoid generated by the degree-q members of $\mathcal{R}$.

Remark 5.7. [proved here] Under 3.17, 5.6 holds automatically as soon as $\mathcal{R}$ contains *all* of $\mathcal{C}(K, K^u, G)$ in each degree, and for a given $\mathcal{R}$ it is a finite check. The geometric reason is one line: $K \subseteq K^u$ is a subcomplex and G acts simplicially, so $\widetilde{\partial_t\sigma} = \partial_t\widetilde{\sigma}$ and

$$\widetilde{\partial_t\sigma_{\mathrm{lhs}}} = \partial_t\widetilde{\sigma_{\mathrm{lhs}}} = \partial_t\big(\gamma \cdot \widetilde{\sigma_{\mathrm{rhs}}}\big) = \gamma \cdot \partial_t\widetilde{\sigma_{\mathrm{rhs}}} = \gamma \cdot \widetilde{\partial_t\sigma_{\mathrm{rhs}}},$$

so the face triple is a coincidence, and (FD4) puts it in the generated groupoid. The seven-vertex model's $\mathcal{R}$ is closed under faces literally, row by row: the T-edge row has the two T-vertex rows as faces, the S-edge row the S-vertex rows, and each central row $(\sigma, \sigma, -I)$ has the central rows of its faces (Definition 7.2).

Lemma 5.8. *[proved here] [formally verified: [GT, `def:relcx`]] Assume 5.6. Then $d^q\big(C^q_{\mathrm{eq}}(K, \mathcal{R}; L)\big) \subseteq C^{q+1}_{\mathrm{eq}}(K, \mathcal{R}; L)$, so $\big(C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L), d\big)$ is a subcomplex of $(C^\bullet(K;V), d)$.*

Proof. Let $f \in C^q_{\mathrm{eq}}$ and $\varrho = (\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$ of degree $q+1$. By 5.6 and Lemma 5.5, $f(\partial_t\sigma_{\mathrm{lhs}}) = \rho_L(\gamma) f(\partial_t\sigma_{\mathrm{rhs}})$ for every t. Hence

$$(d^q f)(\sigma_{\mathrm{lhs}}) = \sum_t (-1)^t f(\partial_t\sigma_{\mathrm{lhs}}) = \rho_L(\gamma) \sum_t (-1)^t f(\partial_t\sigma_{\mathrm{rhs}}) = \rho_L(\gamma)\,(d^q f)(\sigma_{\mathrm{rhs}}).$$ □

Definition 5.9 (Equivariant cohomology of $(K, \mathcal{R}; L)$)**.** $Z^q_{\mathrm{eq}}(K, \mathcal{R}; L) := \ker\big(d^q|_{C^q_{\mathrm{eq}}}\big)$, $B^q_{\mathrm{eq}}(K, \mathcal{R}; L) := d^{q-1}\big(C^{q-1}_{\mathrm{eq}}\big)$ (with $B^0_{\mathrm{eq}} = 0$) and

$$H^q_{\mathrm{eq}}(K, \mathcal{R}; L) \;:=\; Z^q_{\mathrm{eq}}(K, \mathcal{R}; L) \big/ B^q_{\mathrm{eq}}(K, \mathcal{R}; L),$$

also written $H^q(K, \mathcal{R}; L)$. The inclusion $B^q_{\mathrm{eq}} \subseteq Z^q_{\mathrm{eq}}$ is Lemma 5.8 together with $d \circ d = 0$.

Theorem 5.10 (The explicit equivariant complex computes $H^\bullet(K,\mathcal{R};L)$)**.** *Let G act simplicially and without inversions on a simplicial complex K^u, let $K\subseteq K^u$ be a finite subcomplex, let $\mathcal{R}$ be a relation set such that $(K,\mathcal{R})$ presents the quotient (3.17) and is closed under faces (5.6), and let $L=(V,\{M_\varrho\})$ be a matrix local system over a commutative ring R with representation $\rho_L\colon G\to \mathrm{GL}(V)$ (Proposition 4.3). Then:*

1. *[proved here] [formally verified: [GT, `thm:psi`]] (with (5) and the naturality in L; not (2)) The map*

$$\Psi\colon\ \mathrm{Hom}_{\mathbb{Z}[G]}\big(C_q(K^u),V\big)\longrightarrow C^q_{\mathrm{eq}}(K,\mathcal{R};L),\qquad (\Psi\phi)(\sigma):=\phi(\widetilde{\sigma}),$$

is an isomorphism of cochain complexes, natural in L, with inverse $\Phi(f)=\phi_f$ determined by

$$\phi_f\big(\gamma\cdot\widetilde{\sigma}\big)\ =\ \rho_L(\gamma)\,f(\sigma),\qquad \gamma\in G,\ \sigma\in o_q(K). \tag{5}$$

No freeness, contractibility or field hypothesis is used.

2. *[proved here] If in addition $|K^u|$ is contractible and the G-action is free, then $H^\bullet(K/\mathcal{R};L)\cong H^\bullet_{\mathrm{eq}}(K,\mathcal{R};L)$, and by Theorem 5.2 both are $H^\bullet(G;V)$ and $H^\bullet(|K/\mathcal{R}|\,;\mathcal{F}_L)$.*

Proof. (1) *Ψ lands in C^q_{eq}.* Let $\varrho=(\sigma_{\mathrm{lhs}},\sigma_{\mathrm{rhs}},\gamma)\in\mathcal{R}$ of degree q, so $\widetilde{\sigma_{\mathrm{lhs}}}=\gamma\cdot\widetilde{\sigma_{\mathrm{rhs}}}$ by Definition 3.15. Then $(\Psi\phi)(\sigma_{\mathrm{lhs}})=\phi(\gamma\cdot\widetilde{\sigma_{\mathrm{rhs}}})=\rho_L(\gamma)\phi(\widetilde{\sigma_{\mathrm{rhs}}})=M_\varrho(\Psi\phi)(\sigma_{\mathrm{rhs}})$.

Ψ is a chain map. $K\subseteq K^u$ is a subcomplex, so $\widetilde{\partial_t\sigma}=\partial_t\widetilde{\sigma}$ and

$$(d\,\Psi\phi)(\sigma)=\sum_t(-1)^t\phi(\widetilde{\partial_t\sigma})=\phi\Big(\sum_t(-1)^t\partial_t\widetilde{\sigma}\Big)=\phi(\partial\widetilde{\sigma})=(d\phi)(\widetilde{\sigma})=\big(\Psi(d\phi)\big)(\sigma),$$

the fourth equality by Remark 5.1.

Φ is well defined. Every $\widetilde{\vartheta}\in o_q(K^u)$ is $\gamma\cdot\widetilde{\sigma}$ for some $\gamma\in G$, $\sigma\in o_q(K)$, by (FD2). Suppose $\gamma_1\widetilde{\sigma_1}=\gamma_2\widetilde{\sigma_2}$. Then $\gamma_1^{-1}\gamma_2\cdot\widetilde{\sigma_2}=\widetilde{\sigma_1}$, i.e. $(\sigma_1,\sigma_2,\gamma_1^{-1}\gamma_2)\in\mathcal{C}(K,K^u,G)$, so by Lemma 5.5 (using (FD4)) $f(\sigma_1)=\rho_L(\gamma_1^{-1}\gamma_2)f(\sigma_2)$ and therefore $\rho_L(\gamma_1)f(\sigma_1)=\rho_L(\gamma_2)f(\sigma_2)$: the right-hand side of (5) does not depend on the presentation. Extending $\mathbb{Z}$-linearly to $C_q(K^u)$ gives ϕ_f, and $\phi_f(\gamma'\widetilde{\vartheta})=\rho_L(\gamma')\phi_f(\widetilde{\vartheta})$ is immediate from (5), so $\phi_f\in\mathrm{Hom}_{\mathbb{Z}[G]}(C_q(K^u),V)$. *This step is the whole content of the theorem, and (FD4) is exactly what it needs*: the case $\sigma_1=\sigma_2$, $\gamma_1^{-1}\gamma_2\in\mathrm{Stab}_G(\widetilde{\sigma_1})\neq 1$ is the case of a non-trivial stabiliser, and it is handled by the same line — provided $\mathcal{R}$ carries the corresponding self-relations. If it does not, ϕ_f is not well defined and Ψ is not surjective; that is the content of Remark 7.7 and of (H4)(iv).

Ψ and Φ are mutually inverse. $(\Psi\phi_f)(\sigma)=\phi_f(\widetilde{\sigma})=\rho_L(1)f(\sigma)=f(\sigma)$, so $\Psi\Phi=\mathrm{id}$. Conversely, with $f_\phi:=\Psi\phi$, $\phi_{f_\phi}(\gamma\widetilde{\sigma})=\rho_L(\gamma)\phi(\widetilde{\sigma})=\phi(\gamma\widetilde{\sigma})$ by G-equivariance of ϕ, so $\Phi\Psi=\mathrm{id}$. Naturality in L is clear from the formulas. Both maps are R-linear and commute with d, so Ψ is an isomorphism of cochain complexes.

(2) Under the extra hypotheses, Remark 5.1 identifies $C^\bullet(K/\mathcal{R};L)$ with $\mathrm{Hom}_{\mathbb{Z}[G]}(C_\bullet(K^u),V)$, and (1) identifies that with $C^\bullet_{\mathrm{eq}}(K,\mathcal{R};L)$; pass to cohomology and apply Theorem 5.2. □

Remark 5.11 (What the theorem is used for). Part (1) is used, in this form and without part (2), at four places: it is the identification $C^\bullet_{\mathrm{eq}}(K,\mathcal{R};L)\cong\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(K^u),V_R)$ of Remark 7.6; it forces the central relation (7) on every simplex (Remark 7.7); it is the passage $c\mapsto\phi_c$ of (5) on which Ξ_g and Θ_g are defined (Definition 12.10, Theorem 12.11); and it is what lets the induced local system of Definition 12.3 be compared with the group-cohomological Ind. [v5] §12's "Simplicial interpretation" was reaching for part (1) and garbled it; it is not re-homed (Appendix A, entry A.10).

Remark 5.12 (Injectivity of the lift). Theorem 5.10(1) presupposes that $\sigma\mapsto\widetilde{\sigma}$ is injective on $o_q(K)$, which is automatic here because $K\subseteq K^u$ is a subcomplex and the lift is the inclusion. In the free instance, where K is given abstractly and mapped into K^u by Definition 3.11(c), injectivity is Remark 3.13 and rests on 3.6. Without it a single $\widetilde{\vartheta}$ could be the image of several simplices of K and Φ would need a choice; [v5] flagged this and this document removes the issue by taking $K\subseteq K^u$.

Code map. Definition 5.4: the rows of A_{C^q} (§6, `assemble_A_Cq` in `EquivariantCohomologyOnK.cpp`) are exactly (4), one per relation. Definition 5.3: `coboundary_operator_elt` is d_1^q. Definition 5.9: `compute_cohomology()` and the accessors `C_eq_block(q)`, `Z_eq_block(q)`, `B_eq_block(q)`, `H_eq_block(q)`. Theorem 5.10 is cited as `[C, thm:explicit_complex_equiv]` from `LocalSystemSL2Z.h`, `LocalSystemSL2Z.cpp`, `Hecke/InducedLocalSystem.cpp`, `Hecke/HeckeTessellationValidation.h`, `Hecke/HeckeTessellationValidation.cpp` and `Hecke/HeckeTessellationSmokeValidation.cpp`; (5) is the map $c \mapsto \phi_c$ of `HeckeOperatorOnK` and `InducedLocalSystem`.

Formalization note. Theorem 5.10(1) is finite and constructive: Ψ is restriction along a finite inclusion and Φ is defined by (5), whose well-definedness is the single lemma 5.5 plus (FD4). It needs no topology, no field, and no freeness, and is the natural formalization target of Part I — the more so because it is the statement six source files cite. Part (2) and Theorem 5.2 rest on [Bro82, Ch. I.4] and [Hat02, §3.H] and would be imported.

6 Systematic computation: bases, projectors, and the stored objects

What this chapter fixes. *Which* objects the implementation computes and stores for each degree q, in which order, and by which linear algebra — because every matrix of Parts III–V is written in the bases fixed here, and a Hecke matrix $A^{(q)}$ is meaningless without them. Two linear systems, A_{C^q} and A_{Z^q}, and three basis-extraction steps are all that is needed.

Standing hypotheses. R is a field — the elimination-based basis extractions below use division, and the implementation runs over $R = \mathbb{F}_\ell$; the case $R = \mathbb{Z}/\ell^e$ is not claimed here. $(K, \mathcal{R})$ satisfies 3.17 and 5.6, L is a matrix local system of rank d_V, and $N_q := |o_q(K)|$, so $\dim_R C^q(K;V) = N_q d_V$. Cochains are written in the flat coordinates of (8); that convention is fixed in §7 and is not restated.

The two constraint matrices

Write $\mathcal{R}^{(q)}$ for the relations of degree q and $N_{\mathcal{R},q} := \left|\mathcal{R}^{(q)}\right|$.

Proposition 6.1 (The cochain system A_{C^q})**.** *[proved here] Let A_{C^q} be the $N_{\mathcal{R},q} \times N_q$ block matrix over R, with blocks of size $d_V \times d_V$, whose row $\varrho = (\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma)$ is*

$$\begin{cases} \mathrm{id} \text{ at column } \sigma_{\mathrm{lhs}}, \quad -M_\varrho \text{ at column } \sigma_{\mathrm{rhs}}, & \sigma_{\mathrm{lhs}} \neq \sigma_{\mathrm{rhs}}, \\ \mathrm{id} - M_\varrho \text{ at column } \sigma_{\mathrm{lhs}}, & \sigma_{\mathrm{lhs}} = \sigma_{\mathrm{rhs}}, \end{cases}$$

and zero elsewhere. Then $C^q_{\mathrm{eq}}(K, \mathcal{R}; L) = \ker A_{C^q}$ inside $R^{N_q d_V}$. Moreover a relation with $\sigma_{\mathrm{lhs}} = \sigma_{\mathrm{rhs}}$ and $M_\varrho = \mathrm{id}$ contributes the zero row and may be dropped; a relation with $M_\varrho = \alpha\,\mathrm{id}$, $\alpha \neq 1$, contributes the non-zero scalar block $(1-\alpha)\mathrm{id}$ and may not.

Proof. Row ϱ of $A_{C^q} f = 0$ is (4) verbatim, and Definition 5.4 imposes exactly one such equation per relation. The two special cases are the two displayed forms evaluated at $M_\varrho = \mathrm{id}$ and $M_\varrho = \alpha\mathrm{id}$. □

The second sentence is the *vacuous-relation short-circuit*: for even n the central relations of Definition 7.2 have $\rho_n(-I) = \mathrm{id}$ and drop out, for odd n and $\ell \neq 2$ they have $\rho_n(-I) = -\mathrm{id}$ and force $f = 0$ (Remark 7.7). Dropping a zero row changes no value.

Proposition 6.2 (The cocycle system A_{Z^q})**.** *[proved here] Let $Z_1^q := \ker d_1^q \subseteq R^{N_q}$ be the trivial-coefficient cocycle space, with a fixed ordered basis $z_{1,1}^{(q)}, \dots, z_{1,k_q}^{(q)}$, $k_q = \dim Z_1^q$; write $z_{1,j}^{(q)}(\sigma) \in R$ for its component at $\sigma \in o_q(K)$, and*

$$E^{(q)}\colon\ R^{k_q d_V} \longrightarrow R^{N_q d_V}, \qquad (v_1, \dots, v_{k_q}) \longmapsto \sum_{j=1}^{k_q} z_{1,j}^{(q)} \otimes v_j,$$

the $N_q \times k_q$ block matrix with scalar block $z_{1,j}^{(q)}(\sigma)\,\mathrm{id}$ at (σ, j). Let A_{Z^q} be the $N_{\mathcal{R},q} \times k_q$ block matrix whose (ϱ, j) block is the pencil

$$z_{1,j}^{(q)}(\sigma_{\mathrm{lhs}})\,\mathrm{id} \;-\; z_{1,j}^{(q)}(\sigma_{\mathrm{rhs}})\,M_\varrho.$$

Then $E^{(q)}$ is injective and

$$Z^q_{\mathrm{eq}}(K, \mathcal{R}; L) \;=\; E^{(q)}\big(\ker A_{Z^q}\big).$$

Proof. Since $d^q = d_1^q \otimes \mathrm{id}_V$ (Definition 5.3), $\ker d^q = Z_1^q \otimes V$, and $Z^q_{\mathrm{eq}} = (\ker d^q) \cap C^q_{\mathrm{eq}}$ by Lemma 5.8. Every $f \in Z_1^q \otimes V$ is uniquely $f = \sum_j z_{1,j}^{(q)} \otimes v_j$ — uniqueness, hence injectivity of $E^{(q)}$, because the $z_{1,j}^{(q)}$ are linearly independent and V is free. Evaluating at σ gives $f(\sigma) = \sum_j z_{1,j}^{(q)}(\sigma)\, v_j$, so (4) for ϱ reads $\sum_j \big(z_{1,j}^{(q)}(\sigma_{\mathrm{lhs}})\mathrm{id} - z_{1,j}^{(q)}(\sigma_{\mathrm{rhs}})M_\varrho\big)v_j = 0$, which is row ϱ of $A_{Z^q} v = 0$. □

The point of A_{Z^q} is size: it has $k_q d_V$ columns where A_{C^q} has $N_q d_V$, and $k_q \le N_q$ is typically much smaller. A block with both scalars zero is structurally absent; a block with $z_{1,j}^{(q)}(\sigma_{\mathrm{rhs}}) = 0$, or with M_ϱ scalar, is itself scalar and carries no data.

The stored bases, degree by degree

Definition 6.3 (The stored objects)**.** For each degree q the implementation computes and stores

object	what it is	how it is obtained
$c_1^{(q)}, \dots, c_{m_q}^{(q)}$	a basis of C^q_{eq}, $m_q = \dim C^q_{\mathrm{eq}}$	kernel basis of A_{C^q}
$z_1^{(q)}, \dots, z_{d_{Z,q}}^{(q)}$	a basis of Z^q_{eq}	$E^{(q)}$ applied to a kernel basis of A_{Z^q}
$b_1^{(q)}, \dots, b_{d_{B,q}}^{(q)}$	a basis of B^q_{eq}	image basis of d^{q-1} on C^{q-1}_{eq} (Proposition 6.4)
$w_1^{(q)}, \dots, w_{m_q - d_{Z,q}}^{(q)}$	a basis of $W^q := C^q_{\mathrm{eq}}/Z^q_{\mathrm{eq}}$	the $c_j^{(q)}$ at the pivots of $[\, z^{(q)} \,|\, c^{(q)} \,]$
$\mathfrak{h}_q \subseteq \{1, \dots, d_{Z,q}\}$	the indices of a basis of H^q_{eq}	the pivots of $[\, b^{(q)} \,|\, z^{(q)} \,]$
P_q	the projector of Proposition 6.5	from the same elimination

all in the flat coordinates (8). The basis names $c/z/b$ replace [v5]'s $\beta_j^{(q)}$, $\delta_s^{(q)}$, $\eta_p^{(q)}$, each of which collided with a reserved symbol (Appendix A, entry A.11); $w_j^{(q)}$ keeps [v5]'s letter for the quotient W^q. The stored basis of H^q_{eq} is the sub-family $\{[z_j^{(q)}] : j \in \mathfrak{h}_q\}$, re-indexed as $[z_1^{(q)}], \dots, [z_{d_q}^{(q)}]$, $d_q := \dim H^q_{\mathrm{eq}} = |\mathfrak{h}_q|$, wherever a later chapter writes the stored cohomology basis (Algorithm 12.13).

> **CONTRACT. The basis choice is part of the datum.** $A^{(q)}$, the projectors and every matrix of Parts III–V are written in the stored bases of Definition 6.3, and those are fixed by a deterministic elimination rule: leftmost pivot column, topmost

non-zero pivot row, leading 1, elimination above and below; kernel bases are the reduced free-column vectors in ascending order of free column; image bases are the *original* columns at the pivots; a factor basis is the greedy leftmost-independent family of generators modulo the relations. Changing the rule changes the stored bases and hence the matrices, without changing any of the invariants of §20. Cross-checks that compare matrices entry by entry are comparing this convention as well.

Proposition 6.4 (Coboundaries without W^{q-1})**.** *[proved here] Assume 5.6 and $q \geq 1$. Let $C^{(q-1)}$ be the $N_{q-1}d_V \times m_{q-1}$ matrix whose columns are $c_1^{(q-1)}, \dots, c_{m_{q-1}}^{(q-1)}$. Then the column j of $d^{q-1}C^{(q-1)}$ is a pivot column of $d^{q-1}C^{(q-1)}$ if and only if $c_j^{(q-1)}$ is a pivot column of $[\, z^{(q-1)} \,|\, c^{(q-1)} \,]$; consequently*

$$\{\text{pivot columns of } d^{q-1}C^{(q-1)}\} \;=\; \{d^{q-1}w_1^{(q-1)}, \dots, d^{q-1}w_{m_{q-1}-d_{Z,q-1}}^{(q-1)}\}$$

as ordered families, and either side is a basis of B^q_{eq}.

Proof. With the leftmost-pivot rule, column j of a matrix is a pivot column iff it is not in the span of the columns before it. So column j of $d^{q-1}C^{(q-1)}$ is a pivot column iff $d^{q-1}c_j^{(q-1)} \notin \operatorname{span}\{d^{q-1}c_t^{(q-1)} : t < j\}$, i.e. iff $c_j^{(q-1)} \notin \operatorname{span}\{c_t^{(q-1)} : t < j\} + \ker\big(d^{q-1}|_{C^{q-1}_{\mathrm{eq}}}\big)$. By Lemma 5.8 and Definition 5.9 that kernel is Z^{q-1}_{eq}, so the condition is $c_j^{(q-1)} \notin \operatorname{span}\big(Z^{q-1}_{\mathrm{eq}} \cup \{c_t^{(q-1)} : t < j\}\big)$, which is exactly the pivot condition for column j of the c-block of $[\, z^{(q-1)} \,|\, c^{(q-1)} \,]$, i.e. the definition of $w^{(q-1)}$ in Definition 6.3. Both families therefore consist of the same vectors in the same order. They span $B^q_{\mathrm{eq}} = d^{q-1}(C^{q-1}_{\mathrm{eq}})$, and they are independent because they are pivot columns. □

This is the shipped route, and it is why Z^{q-1}_{eq} need not be held while B^q_{eq} is computed: the elimination is $N_q d_V \times m_{q-1}$ instead of $N_{q-1}d_V \times (d_{Z,q-1} + m_{q-1})$. [v5]'s step $(q-1).4$ computed W^{q-1} first and applied d^{q-1} to it; Proposition 6.4 says the two agree *column for column*, which is what makes the change value-preserving.

Proposition 6.5 (The stored projector)**.** *[proved here] Let $\mathfrak{h}_q$ be as in Definition 6.3. There is a unique R-linear $P_q \colon R^{N_q d_V} \to R^{d_q}$ with*

$$P_q\, z_j^{(q)} = e_{\iota(j)} \;(j \in \mathfrak{h}_q), \qquad P_q\, B^q_{\mathrm{eq}} = 0, \qquad P_q\, z_j^{(q)} = \text{the coordinates of } [z_j^{(q)}] \;(j \notin \mathfrak{h}_q),$$

ι the increasing enumeration of $\mathfrak{h}_q$, and P_q vanishes on a fixed complement of Z^q_{eq}. Equivalently, P_q reads off coordinates in the stored basis of H^q_{eq} modulo B^q_{eq}. Its defining certificates are

$$P_q\, H^{(q)} = \mathrm{id}_{d_q}, \qquad P_q\, B^{(q)} = 0, \tag{6}$$

where $H^{(q)}$, $B^{(q)}$ are the matrices whose columns are the stored bases of H^q_{eq} and B^q_{eq}.

Proof. $Z^q_{\mathrm{eq}} = B^q_{\mathrm{eq}} \oplus \operatorname{span}\{z_j^{(q)} : j \in \mathfrak{h}_q\}$ by the pivot property of $\mathfrak{h}_q$ in $[\, b^{(q)} \,|\, z^{(q)} \,]$, so the three displayed requirements prescribe P_q on Z^q_{eq} and on the chosen complement, i.e. on all of $R^{N_q d_V}$; linearity then determines it uniquely. (6) restates the first two requirements. □

Algorithm 6.6 (The cohomology of $(K, \mathcal{R}; L)$ in all degrees)**. Input.** $(K, \mathcal{R})$ satisfying 3.17 and 5.6; a matrix local system L of rank d_V over a field R; the trivial-coefficient data d_1^q and a basis of Z_1^q for each q (independent of L and of d_V, so precomputed once). **Steps.** For $q = 0, \dots, \dim K$:

1. assemble A_{C^q} (Proposition 6.1) and take $c_\bullet^{(q)}$:= its kernel basis; $m_q = \dim C^q_{\mathrm{eq}}$;

2. assemble A_{Z^q} (Proposition 6.2), take a kernel basis, and apply $E^{(q)}$: $z_\bullet^{(q)}$, $d_{Z,q} = \dim Z^q_{\mathrm{eq}}$;
3. for $q \geq 1$, $b_\bullet^{(q)}$:= the image basis of $d^{q-1}C^{(q-1)}$ (Proposition 6.4); $b_\bullet^{(0)}$ is empty;
4. $\mathfrak{h}_q$, the stored basis of H^q_{eq} and P_q from the factor basis of the generators $z_\bullet^{(q)}$ modulo the relations $b_\bullet^{(q)}$ (Proposition 6.5); $d_q = \dim H^q_{\mathrm{eq}} = d_{Z,q} - d_{B,q}$.

Dependencies. Steps 1–2 depend only on q; step 3 in degree q reads only $c_\bullet^{(q-1)}$; step 4 reads only $z_\bullet^{(q)}$ and $b_\bullet^{(q)}$. So the degrees of steps 1–2 are mutually independent, and the chains $3 \to 4$ are mutually independent across q once step 1 has run — which is how the implementation schedules them. **No step depends on the order in which the degrees are run**, so the stored objects are independent of that schedule; this is asserted, and checked, as a determinism property of the implementation.

Remark 6.7 (Cost). With $d_V = n + 1$ the dominant costs are the two kernel extractions: A_{C^q} is $N_{\mathcal{R},q}d_V \times N_q d_V$ and A_{Z^q} is $N_{\mathcal{R},q}d_V \times k_q d_V$, so both are cubic in d_V with a triangulation-dependent constant, and step 4 eliminates a $(d_{Z,q} + d_{B,q}) \times N_q d_V$ system. The d_V-independent data — d_1^q and the basis of Z_1^q — are computed once per $(K, \mathcal{R})$; that separation is the point of Proposition 6.2, and it is the only structural optimisation this chapter asserts. The measured behaviour of the shipped implementation is not part of this contract.

Code map. Proposition 6.1: `assemble_A_Cq`, and the vacuous-relation short-circuit is `relation_is_vacuous` / `live_relations`, gated by `VacuousRelationValidation`. Proposition 6.2: `assemble_A_Zq` and `assemble_beta_expansion` ($E^{(q)}$), with Z_1^q from `get_trivial_cocycle_basis_Z1_elt`. Definition 6.3: the stored `C_eq_block(q)`, `Z_eq_block(q)`, `B_eq_block(q)`, `H_eq_block(q)`, `P_block(q)`, each a multi-vector whose column t is one cochain in the layout (8). Proposition 6.4: `image_basis` applied to `cohom.DC`. Proposition 6.5: `factor_basis`, whose `pivots` are $\mathfrak{h}_q$ and whose `projector` is P_q; (6) is checked in process by `CohomologyBlockModelValidation`. The contract box's elimination rule is the engine's documented basis-choice contract, pinned entrywise by `BlockEliminationValidation` and `GoldenRegressionValidation`.

Formalization note. Everything in this chapter is finite linear algebra over a field, and Propositions 6.1, 6.2, 6.4 and 6.5 are statements about kernels, images and quotients of explicit matrices — the most directly formalizable material in the document, and the natural place to state the stored bases as *definitions by a deterministic elimination* rather than as "choose a basis". The one thing a formalization must not skip is the contract box: the basis is part of the specification, not an implementation detail, and a proof that only fixes the spans would not pin $A^{(q)}$.

Part II — The model and its coefficients

7 Conventions, the two primes, and the 7-vertex model

What this chapter fixes. The two roles a prime plays in this document, and the standing hypotheses (H1)–(H4) under which every later statement is read; the concrete arithmetic datum — the group $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, its generators, the seven-vertex complex K and its relation set $\mathcal{R}$ — on which every later chapter computes; the central $-I$ relation, which is a modelling *requirement* and not a convenience; and the cochain coordinates in which every matrix of this document is written. Nothing here is about the Hecke action: this chapter fixes the object the Hecke action will act on.

Standing hypotheses. (H1)–(H4) below are declared once for Parts II–V and, per rule (A1), are *repeated in every statement that uses one*. No statement of this document may be read as holding for all ℓ merely because (H1) was declared in this chapter.

The two primes ℓ and p

Two distinct roles of primes occur throughout and must never be conflated. The symbols are fixed in §2.2; the box below is the operational form of that fixing, and it is the one the code base implements.

> **CONTRACT.** ℓ is the *residue characteristic of the coefficients*: the local system has fibre V_R over a coefficient ring R of characteristic ℓ, and $\dim_{\mathbb{F}_\ell} M_n = n+1$ for $R = \mathbb{F}_\ell$. p is the *Hecke prime*: T_p is attached to the double coset $\Gamma \operatorname{diag}(1,p)\,\Gamma$ (§10). New code uses the identifiers `ell` (coefficients) and `p_hecke` (Hecke prime); the legacy identifier `p` of the arithmetic layer — `LocalSystemSimplicialComplex::prime`, the `p_mod` arguments — is to be read as ℓ, never as the Hecke prime.

The composite-index operator is written T_N with $N \in \mathbb{Z}_{>0}$, never T_n: n is the symmetric-power degree and nothing else (§2.2; Appendix A, entry A.16).

The standing hypotheses (H1)–(H4)

(H1) **Good case, $p \neq \ell$.** The generic situation. Every statement of §§10–21 applies without restriction.

(H2) **Bad case, $p = \ell$.** The operator defined for $p = \ell$ is U_ℓ. Its *definition* is identical — the construction of §§11–12 never inverts an element of determinant p — but $\rho_n(g)$ for $\det g = \ell$ is *singular* modulo ℓ for $n \geq 1$ (Lemma 8.4), and the Hecke recursion degenerates: $T_{\ell^\nu} = U_\ell^\nu$ over $R = \mathbb{F}_\ell$ (Corollary 10.10).

(H3) **Torsion-of-stabilisers caveat, $\ell \in \{2,3\}$.** The stabilisers in $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ of the elliptic points ζ^2, ζ, i have orders $6, 6, 4$, and every simplex stabiliser contains the centre $\{\pm I\}$; the orders occurring therefore divide 12. If $6 \in R^\times$ — equivalently $\ell \geq 5$, for R of characteristic ℓ — the complex $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$ computes $H^\bullet(\Gamma; V_R)$ (Remark 7.6). If $\ell \in \{2,3\}$ it computes the cohomology of the explicit complex itself, which we call the **orbifold model cohomology** of $(K, \mathcal{R}; L)$; **this may differ from $H^\bullet(\Gamma; V_R)$, and no isomorphism between the two is asserted there, in either direction.** (Since v1.8.0 the orbifold model cohomology is identified, over the ring in force, as a Bredon cohomology independent of the triangulation (Proposition 23.3), and one integral comparison map to group cohomology is constructed, cmp^* of Proposition 23.5(4); it is a quasi-isomorphism only when $6 \in R^\times$. Until v1.7.4 this hypothesis said that no comparison *map* was asserted.) All *constructions* and all *well-definedness* statements of this document hold for every ℓ; the *comparison* statements with $H^\bullet(\Gamma; V_R)$, the Hecke-algebra relations of Theorem 10.8 and the eigenvalue oracles of Obligation 20.7 carry the hypothesis $6 \in R^\times$ in their own statements.

(H4) **Tessellation hypothesis.** Fix a truncation height $Y > 1$ and let X be the Γ-tessellation of the truncated upper half plane: the Γ-invariant simplicial complex whose underlying space is $\mathcal{H}$ with the Γ-translates of the open horoball $\{\operatorname{Im} z > Y\}$ removed, triangulated by the Γ-translates of the five 2-simplices of Definition 7.1. We use, and therefore state, all four of:

- (i) Γ acts simplicially on X, without inversions, with *finite* stabilisers, and $K^u \cong X$ as Γ-complexes;
- (ii) X **is contractible and 2-dimensional**;

(iii) K is a strict fundamental domain for that action, and $(K, \mathcal{R})$ presents the quotient in the sense of §3 (Assumption 3.17);

(iv) the stabiliser of every simplex of X contains $-I$; the stabilisers of the elliptic vertices are $\langle ST\rangle$, $\langle TS\rangle$, $\langle S\rangle$, of orders $6, 6, 4$, at v_4, v_5, v_6 respectively; every other simplex stabiliser is exactly $\{\pm I\}$.

Why (H4)(ii) is written down. It is load-bearing three chapters later — the uniqueness of Λ and the existence of the fills in §15 both use it — and in the superseded corpus it was *never written*, so a reader could not tell the argument from a wish. Rule (A1) exists for exactly this statement.

7.1 The arithmetic group datum and the 7-vertex model

$\Gamma := \mathrm{SL}_2(\mathbb{Z})$ acts on the upper half plane $\mathcal{H}$ by $z \mapsto \frac{az+b}{cz+d}$. The generators used by the model are

$$T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \qquad TS = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix},$$

with $S^2 = (ST)^3 = (TS)^3 = -I$ and $S^4 = (ST)^6 = (TS)^6 = I$. Write $\mathcal{G} := \{T, S, ST, TS\}$ for this generator multiset. The elliptic points are $\zeta := e^{\pi i/3}$, ζ^2 and i; the letter ρ is reserved for the representation (§2.7, C9).

Definition 7.1 (The seven-vertex model)**.** [formally verified: [GT, `def:model7-lean`]] (the simplex list: the seven vertices, the five triangles, the eleven edges, increasing vertex order, lexicographic within each dimension) K is the finite 2-dimensional simplicial complex with

- **vertices** $v_0, \dots, v_6$: v_0, v_1, v_2, v_3 on the horocycle $\{\mathrm{Im}\, z = Y\}$, with Re running from $-\frac{1}{2}$ at v_0 to $+\frac{1}{2}$ at v_3; $v_4 = \zeta^2$ (on $\mathrm{Re}\, z = -\frac{1}{2}$), $v_5 = \zeta$ (on $\mathrm{Re}\, z = +\frac{1}{2}$), $v_6 = i$;
- **2-simplices** $\{v_0v_1v_4\}$, $\{v_1v_2v_6\}$, $\{v_2v_3v_5\}$, $\{v_1v_4v_6\}$, $\{v_2v_5v_6\}$;
- **1-simplices** the eleven edges of those five triangles: v_0v_1, v_0v_4, v_1v_2, v_1v_4, v_1v_6, v_2v_3, v_2v_5, v_2v_6, v_3v_5, v_4v_6, v_5v_6.

So $N_0 = 7$, $N_1 = 11$, $N_2 = 5$ and $\chi(K) = 1$. Every simplex is stored with increasing vertex index, and the stored order within each dimension is lexicographic; that order is part of the datum, because every matrix of this document is written in it. Code: `CreateSL2ZFundamentalDomainTriangulation` (`LocalSystemSL2Z.cpp`), which derives the edge list from the face list exactly as above.

Definition 7.2 (The relation set $\mathcal{R}$ of the model)**.** [formally verified: [GT, `prop:model7coincidences-2`]] (the eight displayed rows: each γ lies in the transporter of its row and carries the stored orientation to the stored orientation) $\mathcal{R}$ consists of the triples $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma)$ below, each read as the equivariance constraint $f(\sigma_{\mathrm{lhs}}) = \rho_n(\gamma)\, f(\sigma_{\mathrm{rhs}})$ — the `SimplexRelation` format $f(s_{\mathrm{lhs}}) = M\, f(s_{\mathrm{rhs}})$ with $M = \rho_n(\gamma)$:

q	σ_{lhs}	σ_{rhs}	γ	code tag
0	$\{v_3\}$	$\{v_0\}$	T	`T_v3_v0`
0	$\{v_5\}$	$\{v_4\}$	T	`T_v5_v4`
1	$\{v_3, v_5\}$	$\{v_0, v_4\}$	T	`T_e(0,4)_e(3,5)`
0	$\{v_4\}$	$\{v_5\}$	S	`S_v4_v5`
0	$\{v_6\}$	$\{v_6\}$	S	`S_v6_stabilizer`
1	$\{v_4, v_6\}$	$\{v_5, v_6\}$	S	`S_e(5,6)_e(4,6)`
0	$\{v_4\}$	$\{v_4\}$	ST	`ST_v4_stabilizer`
0	$\{v_5\}$	$\{v_5\}$	TS	`TS_v5_stabilizer`

and, for *every* simplex σ of K in every dimension, the central relation $(\sigma, \sigma, -I)$ — Remark 7.7, code tags `negI_q0_0`, `negI_q1_0`, ...

The eight listed rows plus the $7 + 11 + 5 = 23$ central relations give $|\mathcal{R}| = 31$. Code: `SL2ZRelationSeeds_7VertexModel` and `GenerateSL2ZRelationsWithGroupData` (`LocalSystemSL2Z.cpp`), in this order.

> **CONTRACT.** The list of `SimplexRelations` is the single source of truth for the pairs (relation, γ). Any relation added to the model carries its $\gamma \in \Gamma$ as exact integer 2×2 data; its local-system matrix is then *derived* as $\rho_n(\gamma)$ (Definition 8.1), never entered by hand. The superseded dictionary "$r \in R_+ \leftrightarrow g_r$" of [v5], Prop. 24 is the fourth column of the table above; individual relations carry no single-letter name here, because r is the restriction cochain map (§2.3).

Lemma 7.3 (The coincidences of the seven-vertex model)**.** *[proved here] Let K be as in Definition 7.1, let $Y > 1$, let X be as in (H4), and let $\widetilde{(-)}$ be the inclusion of K into X as the truncated fundamental domain $D_Y := \{z \in \mathcal{H} : |z| \geq 1,\ |\mathrm{Re}\, z| \leq \frac{1}{2},\ \mathrm{Im}\, z \leq Y\}$. Then:*

1. ***Stabilisers.*** *[formally verified: [GT, `prop:model7h4-iv`]] $\mathrm{Stab}_\Gamma(\widetilde{\sigma}) = \{\pm I\}$ for every simplex σ of K other than the three elliptic vertices, where*

$$\mathrm{Stab}_\Gamma(\widetilde{v_4}) = \langle ST \rangle, \quad \mathrm{Stab}_\Gamma(\widetilde{v_5}) = \langle TS \rangle, \quad \mathrm{Stab}_\Gamma(\widetilde{v_6}) = \langle S \rangle,$$

   of orders 6, 6, 4. In particular every simplex stabiliser is finite, contains $-I$, and has order dividing 12; and no $\gamma \in \Gamma$ permutes the vertices of a simplex of K non-trivially, so Γ acts without inversions.

2. ***Coincidences between distinct simplices.*** *[formally verified: [GT, `prop:model7coincidences-2`]] For $\sigma \neq \sigma'$ in $o_\bullet(K)$ the transporter $\mathrm{Tr}(\sigma, \sigma')$ is non-empty for exactly four unordered pairs $\{\sigma, \sigma'\}$:*

$$\{v_0, v_3\}, \qquad \{v_4, v_5\}, \qquad \big\{\{v_0, v_4\},\ \{v_3, v_5\}\big\}, \qquad \big\{\{v_4, v_6\},\ \{v_5, v_6\}\big\},$$

   realised by $T \in \mathrm{Tr}(v_3, v_0)$, $T \in \mathrm{Tr}(v_5, v_4)$, $T \in \mathrm{Tr}(\{v_3, v_5\}, \{v_0, v_4\})$ and $S \in \mathrm{Tr}(\{v_4, v_6\}, \{v_5, v_6\})$, each carrying the stored (increasing vertex order) orientation to the stored orientation. All five 2-simplices, the three vertices v_1, v_2, v_6 and the seven edges other than the four listed are alone in their Γ-orbit.

Proof. Write $D := \{z \in \mathcal{H} : |z| \geq 1,\ |\mathrm{Re}\, z| \leq \frac{1}{2}\}$, so $|K| = D_Y = D \cap \{\mathrm{Im}\, z \leq Y\}$. Take as given the classical description of the Γ-action on $\mathcal{H}$ [standard: [Ser73, Ch. VII, §1.2, Thms. 1–2]], in the two parts used here:

(A) *Identifications.* If $z, z' \in D$ with $z \neq z'$ and $z = \gamma z'$ for some $\gamma \in \Gamma$, then either $\mathrm{Re}\, z' = \mp\frac{1}{2}$ and $z = z' \pm 1$, with $\gamma = \pm T^{\pm 1}$; or $|z'| = 1$ and $z = -1/z'$, with $\gamma = \pm S$.

(B) *Stabilisers of points.* For $z \in D$, $\mathrm{Stab}_\Gamma(z)$ is $\{\pm I\}$ except at the three points $z = i$, $z = \zeta$, $z = \zeta^2$, where it is $\langle S \rangle$ of order 4, $\langle TS \rangle$ of order 6 and $\langle ST \rangle$ of order 6 respectively. (S fixes i; $ST : z \mapsto -1/(z+1)$ fixes the roots of $z^2 + z + 1$, so ζ^2; $TS : z \mapsto (z-1)/z$ fixes the roots of $z^2 - z + 1$, so ζ. Serre's statement is in $\mathrm{PSL}_2(\mathbb{Z})$, where the three stabilisers have orders $2, 3, 3$; the orders above are those of their preimages in $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, the central $-I$ acting trivially on $\mathcal{H}$.)

Part (2). Let $\sigma \neq \sigma'$ in $o_\bullet(K)$ and $\gamma \in \mathrm{Tr}(\sigma, \sigma')$, so γ maps $|\sigma'|$ onto $|\sigma|$ and hence the relative interior of $|\sigma'|$ onto that of $|\sigma|$. Distinct simplices of a simplicial complex have disjoint relative interiors, so $\gamma z' \neq z'$ for every z' in the relative interior of $|\sigma'|$, while both z' and $\gamma z'$ lie in D. By (A) that relative interior is contained in $L_- \cup L_+ \cup C$, where $L_\mp := \{\mathrm{Re}\, z = \mp\frac{1}{2}\}$ and $C := \{|z| = 1\}$ — three geodesics of $\mathcal{H}$. A 2-simplex has 2-dimensional relative interior and is therefore excluded, so $\dim \sigma' \leq 1$; and if $\dim \sigma' = 1$ its relative interior is an infinite subset of a geodesic segment, so it meets one of the three geodesics in at least two points and therefore lies

in that one, two distinct geodesics of $\mathcal{H}$ meeting in at most one point. Hence $|\sigma'|$, and likewise $|\sigma|$, lies in L_-, in L_+ or in C.

By Definition 7.1 the simplices of K so contained are exactly: in $L_- \cap D_Y$ (the vertical segment from $v_4 = \zeta^2$ to $v_0 = -\frac{1}{2} + iY$) the simplices $\{v_0\}$, $\{v_4\}$, $\{v_0, v_4\}$; in $L_+ \cap D_Y$ (from $v_5 = \zeta$ to $v_3 = \frac{1}{2} + iY$) the simplices $\{v_3\}$, $\{v_5\}$, $\{v_3, v_5\}$; and in $C \cap D_Y$ (the unit arc from ζ^2 through i to ζ) the simplices $\{v_4\}$, $\{v_5\}$, $\{v_6\}$, $\{v_4, v_6\}$, $\{v_5, v_6\}$ — there are no other vertices of K on the two walls or on the arc, and no other edges of K joining two of these vertices along them. Now (A) pins γ and hence σ: if $|\sigma'| \subseteq L_-$ then $\gamma = \pm T$ and $\sigma = T\sigma'$, giving $v_0 \mapsto v_3$, $v_4 \mapsto v_5$, $\{v_0, v_4\} \mapsto \{v_3, v_5\}$; if $|\sigma'| \subseteq L_+$ then $\gamma = \pm T^{-1}$ and $\sigma = T^{-1}\sigma'$, the three inverse assignments; if $|\sigma'| \subseteq C$ then $\gamma = \pm S$ and $\sigma = S\sigma'$, giving $v_4 \leftrightarrow v_5$, $\{v_4, v_6\} \leftrightarrow \{v_5, v_6\}$ and $v_6 \mapsto v_6$, the last excluded by $\sigma \neq \sigma'$. Collecting the unordered pairs gives exactly the four listed, with the stated transporter elements. Each of T and S maps the smaller vertex of the source edge to the smaller vertex of the target edge ($0 \mapsto 3$, $4 \mapsto 5$ for T; $5 \mapsto 4$, $6 \mapsto 6$ for S), so the stored orientations correspond.

Part (1). For a vertex v of K the claim is (B): $v_4 = \zeta^2$, $v_5 = \zeta$, $v_6 = i$ are the three exceptional points, and $v_0, \dots, v_3$ lie on $\operatorname{Im} z = Y > 1$, so are none of them. Let now $\dim \sigma \geq 1$ and $\gamma \in \operatorname{Stab}_\Gamma(\widetilde{\sigma})$; then γ permutes the vertices of σ. If that permutation were non-trivial it would move some vertex v of σ to another vertex w of σ, so $\{v, w\}$ would be one of the two vertex pairs of part (2), namely $\{v_0, v_3\}$ or $\{v_4, v_5\}$ — but neither is an edge of K (Definition 7.1), so neither pair is contained in a common simplex. Hence γ fixes every vertex of σ, i.e. at least two distinct points of $\mathcal{H}$; a non-identity element of $\mathrm{PSL}_2(\mathbb{R})$ fixes at most one point of $\mathcal{H}$, so $\gamma = \pm I$. This also proves the last sentence of part (1): no γ permutes the vertices of a simplex of K non-trivially, and since every simplex of X is a Γ-translate of a simplex of K (that is how X is defined in (H4)), the same holds on X: Γ acts without inversions, and $\operatorname{Stab}_\Gamma(\gamma\widetilde{\sigma}) = \gamma \operatorname{Stab}_\Gamma(\widetilde{\sigma})\gamma^{-1}$ is finite of order dividing 12. □

Proposition 7.4 (The model satisfies (H4))**.** *[proved here] [formally verified: [GT, `prop:model7h4-i`]] ((H4)(i) without the identification $K^u \cong X$, which is the trust boundary) [formally verified: [GT, `prop:model7h4-ii`]] ((H4)(ii) in the form this document uses it, since v1.19.0: the acyclicity of $C_\bullet(X)$ over $\mathbb{Z}$ in the three forms of (37) — $H_0(X) \cong \mathbb{Z}$ by the augmentation, $Z_1 = B_1$, $Z_2 = 0$ — and $\dim X = 2$ ([GT] row `prop:model7h4-ii-dim`); proved combinatorially in [GT] (slot-19 L19-3, `model7_acyclic`), not by the horoball retraction; contractibility as a topological statement is not formalised) [formally verified: [GT, `prop:model7h4-iii`]] ((H4)(iii): (FD1) and (FD2) only) [formally verified: [GT, `prop:model7h4-iv`]] ((H4)(iv) in full) Let K and $\mathcal{R}$ be as in Definitions 7.1 and 7.2, let $Y > 1$, and let X be as in (H4). Then (H4)(i)–(iv) hold. The ineffective kernel of Definition 3.16 is $N = \{\pm I\}$ here, so (FD1) and (FD3) of (H4)(iii) are the conditions modulo $\{\pm I\}$ (Remark 3.18).*

Proof. (i) and (iv). Γ permutes the simplices of X because X is by definition the Γ-translate tessellation of the five 2-simplices of Definition 7.1 and their faces, and $K^u \cong X$ as Γ-complexes is the identification fixed in Remark 3.20 (second bullet) — it is not a claim proved here but the instantiation of the framework of §3. That the action is *without inversions*, that every stabiliser is *finite*, and the whole of (iv) — the stabiliser inventory, and $-I$ in every stabiliser — are Lemma 7.3(1).

(ii). $\dim K = 2$ by Definition 7.1, and X is covered by Γ-translates of $|K|$, so $\dim X = 2$. For contractibility: since $Y > 1$, the open horoballs $\gamma^{-1}\{\operatorname{Im} z > Y\}$, $\gamma \in \Gamma$, are pairwise disjoint or equal; the map $z \mapsto \operatorname{Re} z + i \min(\operatorname{Im} z, Y)$ is a deformation retraction of $\{\operatorname{Im} z > Y\}$ onto the horocycle $\{\operatorname{Im} z = Y\}$, fixing that horocycle pointwise; transporting it by γ^{-1} on each horoball and taking the identity elsewhere is well defined by disjointness, is Γ-equivariant, and is a deformation retraction of $\mathcal{H}$ onto X. As $\mathcal{H}$ is contractible, so is X.

(iii): (FD2). $\Gamma \cdot K = X$ holds by the definition of X just recalled.

(iii): (FD4), the substantive clause. By Lemma 3.22 it suffices to verify its two conditions,

and for each of them the sufficient criterion given there. The stabilisers and the coincidences they refer to are supplied by Lemma 7.3; neither is read off a table of $\mathcal{R}$.

Condition (a) asks that, at each of the $7 + 11 + 5 = 23$ simplices σ of K, the relation elements of the *self*-relations of $\mathcal{R}$ at σ generate $\mathrm{Stab}_\Gamma(\widetilde{\sigma})$. Reading off Definition 7.2 against Lemma 7.3(1):

- $\sigma = \{v_4\}$: the row $(\{v_4\}, \{v_4\}, ST)$ generates $\langle ST\rangle = \mathrm{Stab}_\Gamma(\widetilde{v_4})$;
- $\sigma = \{v_5\}$: the row $(\{v_5\}, \{v_5\}, TS)$ generates $\langle TS\rangle$;
- $\sigma = \{v_6\}$: the row $(\{v_6\}, \{v_6\}, S)$ generates $\langle S\rangle$;
- each of the remaining 20 simplices — v_0, v_1, v_2, v_3, all 11 edges and all 5 triangles — has stabiliser $\{\pm I\}$, generated by the relation element of its central row $(\sigma, \sigma, -I)$.

This is the entire content of the 23 central rows, and it is why omitting them is not an option. Apart from them $\mathcal{R}$ has no self-relation at any of those twenty simplices, so the subgroup reached at each would be trivial while the stabiliser is $\{\pm I\}$: (a) would fail, (FD4) with it, and with (FD4) the surjectivity of Ψ in Theorem 5.10. That is Remark 7.7 and Appendix A, entry A.8, seen as the failure of a stated criterion rather than as a discrepancy in a computed dimension.

Condition (b) asks that each pair of distinct simplices with non-empty transporter be joined by members of $\mathcal{R}$. By Lemma 7.3(2) there are exactly four such pairs, and $\mathcal{R}$ contains one row for each: `T_v3_v0` for $\{v_0, v_3\}$, `T_v5_v4` for $\{v_4, v_5\}$, `T_e(0,4)_e(3,5)` for $\{\{v_0, v_4\}, \{v_3, v_5\}\}$ and `S_e(5,6)_e(4,6)` for $\{\{v_4, v_6\}, \{v_5, v_6\}\}$. This exhausts the four pairs, so (b) holds.

Seven of the eight listed rows have now been used — three for (a) at the elliptic vertices, four for (b). The eighth, $(\{v_4\}, \{v_5\}, S)$ (`S_v4_v5`), is *redundant* for (FD4): inverting `T_v5_v4` gives $(\{v_4\}, \{v_5\}, T^{-1})$, composing it with the self-relation $(\{v_5\}, \{v_5\}, TS)$ gives $(\{v_4\}, \{v_5\}, T^{-1} \cdot TS) = (\{v_4\}, \{v_5\}, S)$, and both factors lie in $\langle\mathcal{R}\rangle$. It is retained because it is the S-wall's own identification and the τ-table path templates of §18 consume it.

(iii): N, (FD1) and (FD3). First $N = \{\pm I\}$: $-I$ acts trivially on $\mathcal{H}$, hence on X; conversely $N \subseteq \mathrm{Stab}_\Gamma(\widetilde{v_1}) = \{\pm I\}$ by Lemma 7.3(1).

By Assumption 3.17, K_{cl} is generated by the simplices a coincidence moves, which by Lemma 7.3(2) are the eight simplices v_0, v_3, v_4, v_5, $\{v_0, v_4\}, \{v_3, v_5\}, \{v_4, v_6\}, \{v_5, v_6\}$; the subcomplex they generate is the two vertical walls and the unit arc together with their vertices (v_6 included, as a face). Hence $\widetilde{D} = D_Y \setminus (L_- \cup L_+ \cup C)$ — the notation $L_\mp$, C for the two vertical walls and the unit circle being that of the proof just given — is the interior of D_Y together with the open horocycle segment from v_0 to v_3: non-empty, connected and dense in $|K| = D_Y$, which with the connectedness of D_Y is (FD1). Note that the three elliptic vertices are *not* put into K_{cl} by their stabilisers — v_4, v_5 are there because T moves them and v_6 only as a face — which is the point of Remark 3.18.

For (FD3), let $\gamma \in \Gamma$ with $\gamma\widetilde{D} \cap \widetilde{D} \neq \emptyset$, say $z, \gamma z \in \widetilde{D}$. Then z lies on none of L_-, L_+, C, so (A) of Lemma 7.3's proof forces $\gamma z = z$, and (B) forces $\gamma \in \{\pm I\} = N$. So $\gamma\widetilde{D} \cap \widetilde{D} = \emptyset$ for every $\gamma \in \Gamma \setminus N$, which is (FD3); the decomposition (1) of $|X|$ and the statement that all identifications occur on $|K_{\mathrm{cl}}|$ then follow from Remark 3.19. □

Remark 7.5 ((FD4) for the model is re-verified as a finite computation). [verified computationally] The two conditions of Lemma 3.22 are precisely what `SL2ZTessellationTables` (`Hecke/SL2ZTessellationTables.cpp`) computes, so the discharge above is re-run on every construction of that class and by the validation named below; it does not rest on the printed argument alone. The class seeds the transporter sets $\mathrm{Tr}(\sigma, \sigma')$ with exactly a generating set of $\langle\mathcal{R}\rangle$ — $\{\pm I\}$ at every simplex (the 23 central rows), the cyclic groups $\langle ST\rangle$, $\langle TS\rangle$, $\langle S\rangle$ at v_4, v_5, v_6 (the three stabiliser rows), and one element per row of `SL2ZRelationSeeds_7VertexModel` (the eight listed rows) — and closes them under composition, inverse and multiplication by $-I$ up to a fixed point, which it verifies is one. Its constructor-time `validate_closure`, and independently `HeckeTessellationValidation` against a hand-derived table committed before the implementation was read, then assert:

- self-transporter orders $2, 2, 2, 2, 6, 6, 4$ at $v_0, \dots, v_6$ and 2 at each of the 11 edges and 5 triangles. Since $\{\gamma : (\sigma, \sigma, \gamma) \in \langle\mathcal{R}\rangle\}$ is a subgroup of $\mathrm{Stab}_\Gamma(\widetilde{\sigma})$ and the orders match those of Lemma 7.3(1), the two groups are equal: this is (a), all 23 simplices of it.
- the orbits of the generated groupoid — $\{v_0, v_3\}$, $\{v_1\}$, $\{v_2\}$, $\{v_4, v_5\}$, $\{v_6\}$ on vertices; $\{\{v_0, v_4\}, \{v_3, v_5\}\}$ and $\{\{v_4, v_6\}, \{v_5, v_6\}\}$ on edges with the other seven singletons; all five triangles singletons — together with $\mathrm{Tr}(v_3, v_0) = \{\pm T\}$, $\mathrm{Tr}(\{v_3, v_5\}, \{v_0, v_4\}) = \{\pm T\}$ and $\mathrm{Tr}(\{v_4, v_6\}, \{v_5, v_6\}) = \{\pm S\}$ by exact set equality. Matched against Lemma 7.3(2), this is (b).
- that $\mathcal{R}$ has exactly $8+23 = 31$ rows: the eight listed ones first, in the order of Definition 7.2 and with relation elements T, T, T, S, S, S, ST, TS, then one central row $(\sigma, \sigma, -I)$ per simplex of every dimension, each with $M_\varrho = \rho_n(-I)$ recomputed independently.

What the computation does *not* supply is Lemma 7.3 itself: the *true* stabilisers and the *true* orbits, against which the generated ones are compared. Those come from [Ser73, Ch. VII, §1.2] through the printed proof, and a hand-derived expected table is what `HeckeTessellationValidation` encodes. Neither half alone discharges (FD4).

Remark 7.6 (The orbifold caveat). Assume (H4). Besides the boundary identifications for T and S, the model imposes the *stabiliser relations* $f(v_4) = \rho_n(ST)f(v_4)$, $f(v_5) = \rho_n(TS)f(v_5)$, $f(v_6) = \rho_n(S)f(v_6)$ (Definition 7.2) and — by Remark 7.7 — the central relation $f(\sigma) = \rho_n(-I)f(\sigma)$ on every simplex. Consequently

$$C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L) \;\cong\; \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(K^u), V_R\big), \qquad K^u \cong X,$$

by Theorem 5.10. By (H4)(i),(iv) each chain group $C_q(X)$ is a direct sum of modules $\mathbb{Z}[\Gamma/\Gamma_\sigma]$ with $|\Gamma_\sigma| \mid 12$ [formally verified: [GT, `lem:xbulk-orient`]] (the direct-sum decomposition, for every X-descriptor, the orientation of each orbit fixed by (H4)(i)); in particular $C_\bullet(X)$ is *not* free over $\mathbb{Z}[\Gamma]$. If moreover $6 \in R^\times$, then $|\Gamma_\sigma| \in R^\times$ for every σ, each $R[\Gamma/\Gamma_\sigma]$ is projective relative to the trivial subgroup, and the standard comparison arguments — the fundamental lemma of relative homological algebra, Shapiro's lemma, transfer — apply verbatim, giving $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L) \cong H^\bullet(\Gamma; V_R)$ [standard: [Ben91, §3.6] for relative projectivity and transfer; [Bro82, Ch. III.6] for Shapiro's lemma; [Bro82, Ch. VII, §7] for the isotropy spectral sequence of a group acting on a complex, whose rows E_1^{*q}, $q > 0$, vanish precisely because $|\Gamma_\sigma| \in R^\times$]. [formally verified: [GT, `thm:xbulk-repaug`]] (the ingredient that $C_\bullet(X) \otimes R \to R$ is a resolution, for every acyclic X-descriptor with a vertex and every R; projective on the model when $6 \in R^\times$) [formally verified: [GT, `thm:model7-rmkorbifold`]] (the isomorphism $H^n_{\mathrm{eq}}(K, \mathcal{R}; L) \cong H^n(\Gamma; V_R)$ for every $n \geq 0$ under $6 \in R^\times$, for every X-descriptor with (H4)(ii) (as acyclicity), a vertex, and stabiliser orders dividing 12 ((H4)(iv)), in particular the model; L a local system with representation ρ_L as in Proposition 4.3, read on Γ (in [GT] a Δ-module, $\Gamma \subseteq \Delta \subseteq G$, through $\rho_L = \rho \circ e^{-1}$ for $e\colon \Gamma \cong \mathrm{SL}_2(\mathbb{Z})$, not only a representation of $\mathrm{SL}_2(\mathbb{Z})$); the statement is tagged, not this argument: [GT] replaces relative projectivity, Shapiro's lemma, transfer and the isotropy spectral sequence by mathlib's `groupCohomologyIso` on the projective resolution; a K outside the descriptor framework is not claimed) This is the precise content of (H3).

For $\ell \in \{2, 3\}$ that vanishing fails and the isomorphism is *not* asserted: the left-hand side computes the orbifold model cohomology of (H3), a well-defined invariant of $(K, \mathcal{R}; L)$ and the object this implementation emits at those primes. Two further points, neither of them settled here. First, π_1 of the quotient *orbifold* is Γ only in the orbifold sense: the torsion may not be ignored, and Remark 7.7 is the place where ignoring it changes an answer (Appendix A, entry A.8). Second, the systematic treatment of the regime $6 \notin R^\times$, by a free Γ-model of which the present one is a specialisation, is §17. The separate question of what happens at $\ell = 2$ *on the boundary*, where $H^\bullet(\Gamma_\infty; V_R)$ is unbounded above for the boundary group Γ_∞ of §14 (Lemma 14.20), is construction item 14.24 (OC-1): the mathematics is §14.5 — including the bulk periodicity of Lemma 14.22 — the *policy* there was decided by the owner on 2026-09-16 ((OC-1a)), and neither is this chapter's.

A design direction, recorded here and not restated elsewhere in this document. The model of Definitions 7.1 and 7.2 is fixed for the regime $6 \in R^\times$. The owner's stated expectation (2026-09-15; `docs/implementation/hecke_extensions_owner_decisions.md`) is that a correct orbifold treatment at $\ell \in \{2,3\}$, and over $\mathbb{Z}$, will require a *refinement* of this triangulation — a subdivision of the present complex, not a different model. This is an expectation about how §17 and 14.24 will be constructed, **not** a theorem and not a claim that the present model is insufficient there; nothing stated in this document depends on it. Its force is on the authors of §17 and §14: the definitions there must be phrased so as to admit a refinement of the 7-vertex complex rather than hard-wiring its seven vertices, eleven edges and five triangles. *Answered at v1.8.0* (owner ruling 6c, 2026-09-24): no refinement is needed. The free model at $\ell \in \{2,3\}$ and over $\mathbb{Z}$ is built on the unrefined tessellation, the missing degrees coming from the stabiliser resolutions (§23, Remark 23.6); and what the present model computes at those primes is identified there (Proposition 23.3).

Remark 7.7 (Modelling requirement: the central $-I$ relation on *every* simplex). Assume (H4). Because $-I \in \Gamma$ is central and acts trivially on $\mathcal{H}$, it is the identity deck transformation of $X \cong K^u$: it fixes the image in X of *every* oriented simplex $\sigma \in o_q(K)$, in every degree q — this is (H4)(iv). The isomorphism of Theorem 5.10 therefore forces the equivariance relation

$$f(\sigma) \;=\; \rho_n(-I)\, f(\sigma) \qquad \text{for } \textit{all } \sigma \in o_q(K),\ \text{all } q, \tag{7}$$

and not merely at the elliptic fixed points v_4, v_5, v_6, where the stabiliser rows of Definition 7.2 already supply $-I = S^2 = (ST)^3 = (TS)^3$ implicitly. Imposing (7) on every free (non-elliptic) simplex as well is a **binding modelling requirement**, stated here because the eight listed rows do not imply it. Its effect, for R of characteristic ℓ, is [proved here] as follows.

- n *even.* $\rho_n(-I) = +I$ and (7) is vacuous; omitting it is a no-op. The implementation emits it anyway and short-circuits it (`VacuousRelationValidation`).
- n *odd,* $\ell \neq 2$. $\rho_n(-I) = -I$, so (7) reads $2f(\sigma) = 0$, i.e. $f(\sigma) = 0$ on every simplex, and the complex vanishes: $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L) = 0$. *Omitting* (7) on the free simplices leaves $H^\bullet$ spuriously non-zero — concretely $\dim H^1 = n+1$ instead of 0 — whereas the correct value is 0. This is correct also on the group side: $-I$ is central, hence acts trivially on $H^\bullet(\Gamma; -)$, while it acts on the coefficients by $\rho_n(-I) = (-1)^n = -1$; so $2H^\bullet(\Gamma; V_R) = 0$ and, 2 being invertible, $H^\bullet(\Gamma; V_R) = 0$. This is the odd-$n$ oracle of Obligation 20.7 and §21.
- $\ell = 2$. $\rho_n(-I) = +I$ for every n, since $-1 = 1$ in R; (7) is again vacuous, and the odd-n vanishing argument above does not apply — it used $2 \in R^\times$.

The implementation imposes (7) globally, as the `FIX-negI` block of `GenerateSL2ZRelationsWithGroupData`.

Cochain conventions and flat coordinates

Orientations. Every q-simplex of K is stored with increasing vertex order, and that stored orientation is the one all signs are taken relative to.

Coboundary. The cochain differential is

$$(d^q f)(\sigma) \;=\; \sum_{t=0}^{q+1} (-1)^t\, f(\partial_t \sigma), \qquad d^q = d_1^q \otimes \mathrm{id}_{V_R},$$

with ∂_t the t-th face map. It is written d for cochains throughout; ∂ is the chain boundary and D the cone differential (§2.4). The superseded documents wrote δ for this map; δ now denotes a coset element (§2.7, C7).

Flat coordinates. A cochain $c \in C^q(K; V_R)$ is the column $\big(c(\sigma_0)^\top, \dots, c(\sigma_{N_q-1})^\top\big)^\top \in R^{N_q d_V}$ in the stored simplex order, $d_V = n+1$: the flat index of the component of $c(\sigma_s)$ along the basis vector u_j of Definition 8.1 is

$$s \cdot d_V + j, \qquad 0 \le s < N_q, \quad 0 \le j \le n. \tag{8}$$

Every matrix in this document — A_{C^q}, A_{Z^q}, $\Theta^{(q)}_{\mathrm{flat}}$, the stored bases $c_j^{(q)}, z_j^{(q)}, b_j^{(q)}$ and the projectors P_q — is written in these coordinates. In the implementation (8) is `engine::BlockMat::to_flat` / `from_flat` under the one-block-column contract, which is its single source of truth, and `H_eq_projectors_from_C_eq_flat` is P_q in it. The stored bases and projectors themselves are the subject of §6.

Code map. (H1), (H2): `ell` / `p_hecke` throughout `Hecke/`; the $p = \ell$ branch is `HeckeOperatorFamily`'s U_ℓ. (H3): gated at every comparison site, e.g. `Hecke/BarCocycleH1.h` ("the denominators $4, 6$ must be invertible"), `Hecke/HeckeRelationsValidation.h`, `ValidationRegistry.cpp`. (H4): `Hecke/SL2ZTessellationTables.cpp`. Definition 7.1: `CreateSL2ZFundamentalDomainTriangulation`. Definition 7.2: `SL2ZRelationSeeds_7VertexModel`, `GenerateSL2ZRelationsWithGroupData`. Remark 7.7: the `FIX-negI` block of `LocalSystemSL2Z.cpp`, checked by `VacuousRelationValidation` and by the odd-n oracle of §21. Equation (8): `engine::BlockMat::to_flat`, gated by `CohomologyBlockModelValidation`.

Formalization note. Definitions 7.1 and 7.2 are finite explicit data — a list of vertex lists per dimension, and a list of $(\text{degree}, \text{index}, \text{index}, \mathrm{M}_2(\mathbb{Z}))$ tuples — so both are `decidable` to state and to check. Proposition 7.4(i),(iii),(iv) is a finite check once the classical fundamental-domain theorem is available; (H4)(ii) is the one hypothesis of this chapter that needs a genuine topological argument, and the proof given reduces it to contractibility of $\mathcal{H}$ plus disjointness of horoballs at $Y > 1$. Remark 7.7's odd-n statement is two lines from centrality and $2 \in R^\times$, and is the natural first formalization target of Part II.

8 The Sym^n convention, its semigroup extension, and the contragredient

What this chapter fixes. The coefficient module and the exact convention by which a 2×2 integer matrix acts on it: ρ_n, defined as the *monoid* homomorphism on all of $\mathrm{M}_2(\mathbb{Z})$ — not only on Γ — because the Hecke coset representatives have determinant p and are not invertible; the extension of that action to the monoid Δ; the determinant of ρ_n, which is where the bad prime $p = \ell$ enters; and the contragredient module D_n with its adjugate action and the perfect pairing $V_R \times D_n \to R$ against which T_p is self-adjoint.

Standing hypotheses. R is a commutative ring with 1; where a characteristic or an invertibility hypothesis is needed it is stated in the statement that needs it (rule (A1)). *No* hypothesis on det is used anywhere in this chapter: every definition and every statement below is valid for singular matrices, which is what makes the bad case (H2) a case of the same construction rather than a different one.

The symmetric power and its action

Definition 8.1 (Integral symmetric power action)**.** Let $W := \mathbb{Z}^2$ with ordered basis (x, y), and let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{M}_2(\mathbb{Z})$ act on W in the *column convention*: $A \cdot x = ax + cy$ and $A \cdot y = bx + dy$, i.e. $A \cdot e_j = \sum_t A_{tj} e_t$ with $x = e_1$, $y = e_2$. Let

$$V := \mathrm{Sym}^n W, \qquad u_j := x^{n-j} y^j \quad (0 \le j \le n), \qquad d_V := \dim V = n + 1,$$

with $(u_0, \dots, u_n)$ as ordered basis. Then $\rho_n(A) := \mathrm{Sym}^n(A) \in \mathrm{End}_{\mathbb{Z}}(V)$ is the multiplicative extension of the above to degree-n monomials; concretely, column j of the matrix of $\rho_n(A)$ in

the basis (u_t) holds the coefficients of

$$A_*(u_j) \;=\; (a\,x + c\,y)^{\,n-j}\,(b\,x + d\,y)^{\,j}, \tag{9}$$

that is, dehomogenising by $x = 1$,

$$\rho_n(A)_{t,j} \;=\; \big[y^t\big]\,(a + cy)^{\,n-j}(b + dy)^{\,j} \;=\; \sum_s \binom{n-j}{s} a^{\,n-j-s} c^{\,s} \binom{j}{t-s} b^{\,j-t+s} d^{\,t-s}.$$

For a coefficient ring R put $V_R := V \otimes_{\mathbb{Z}} R$ and $\rho_n^R := \rho_n \bmod R$; we drop the superscript. For $R = \mathbb{F}_\ell$ we also write $M_n = V_R$, the notation of the classification literature.

This is a constructive definition in the sense of rule (A3): (9) is literally the object the implementation builds (`ComputeSymmetricPowerMatrixFM`, `ModArithmetic/FlatMatrix.cpp`), and $\rho_n(A)$ is therefore a mathematically unique matrix that any correct evaluation route must reproduce entry for entry — which is what `SymmetricPowerValidation` asserts of the eight branches that route actually takes.

Proposition 8.2 (Code convention = standard action; multiplicativity)**.** *[proved here] For every $A \in \mathrm{M}_2(\mathbb{Z})$ and every $n \geq 0$:*

1. $\rho_n(A) \bmod \ell =$ *`ComputeSymmetricPowerMatrixFM`*$(A \bmod \ell,\, n)$*;*
2. $A \mapsto \rho_n(A)$ *is a homomorphism of* monoids $(\mathrm{M}_2(\mathbb{Z}), \cdot) \to (\mathrm{End}_{\mathbb{Z}}(V), \cdot)$*: $\rho_n(A_1A_2) = \rho_n(A_1)\rho_n(A_2)$ and $\rho_n(I) = I$. In particular no transpose, inverse or contragredient is involved; ρ_n restricted to Γ is a group homomorphism, and its values on T, S, ST, TS are exactly the matrices attached to the eight rows of Definition 7.2; [formally verified: [GT, `prop:symconv`]]*
3. *ρ_n is integral, $\rho_n(\mathrm{M}_2(\mathbb{Z})) \subseteq \mathrm{M}_{n+1}(\mathbb{Z})$, and every entry of $\rho_n(A)$ is bounded in absolute value by $(2\max_{t,j}|A_{tj}|)^n$. [formally verified: [GT, `prop:symconv`]]*

Proof. (1) The substitution $x \mapsto ax + cy$, $y \mapsto bx + dy$ is an algebra endomorphism subst_A of $\mathbb{Z}[x, y]$; its matrix on $\mathrm{span}(x, y)$ in the basis (x, y), images as columns, is A itself, i.e. it is the action of Definition 8.1; and the implementation tabulates its matrix on the degree-n part in the basis (u_j), columns being images, which is $\mathrm{Sym}^n(A)$.

(2) $\mathrm{subst}_{A_1} \circ \mathrm{subst}_{A_2} = \mathrm{subst}_{A_1A_2}$: on x, $\mathrm{subst}_{A_1}(a_2x + c_2y) = a_2(a_1x + c_1y) + c_2(b_1x + d_1y) = (a_1a_2 + b_1c_2)x + (c_1a_2 + d_1c_2)y$, and $(A_1A_2)_{11} = a_1a_2 + b_1c_2$, $(A_1A_2)_{21} = c_1a_2 + d_1c_2$; likewise on y; extend multiplicatively to degree n. Composition of matrices in the columns-are-images convention matches composition of maps, and $\mathrm{subst}_I = \mathrm{id}$.

(3) By (9) the sum of the absolute values of the coefficients of $A_*(u_j)$ is at most $(|a| + |c|)^{n-j}(|b| + |d|)^j \leq (2M)^n$ with $M = \max_{t,j}|A_{tj}|$; each single coefficient is bounded by that sum. □

> **CONTRACT.** The action of *any* $g \in \mathrm{M}_2(\mathbb{Z})$ on Sym^n coefficients — Hecke coset representatives, non-invertible modulo ℓ, included — is $\rho_n(g)$ of Definition 8.1, computed as `ComputeSymmetricPowerMatrixFM`$(g \bmod \ell,\, n)$. It acts on cochain values by *left* matrix multiplication, matching the `SimplexRelation` usage $f(\sigma_{\mathrm{lhs}}) = \rho_n(g) f(\sigma_{\mathrm{rhs}})$ of Definition 7.2. **No determinant twist is applied**: the coefficient module is plain $\mathrm{Sym}^n(\mathbb{Z}^2)$, not $\mathrm{Sym}^n \otimes \det^m$. This is the *integral normalisation*, and it is what fixes the eigenvalue normalisation of the oracles in §21.

Definition 8.3 (Semigroup extension)**.** Let

$$\Delta := \mathrm{M}_2(\mathbb{Z}) \cap \mathrm{GL}_2(\mathbb{Q})^+ = \{g \in \mathrm{M}_2(\mathbb{Z}) : \det g > 0\},$$

a multiplicative submonoid of $\mathrm{GL}_2(\mathbb{Q})$ containing Γ. "The representation extends to the subsemigroup of $\mathrm{GL}_2(\mathbb{Q})$ generated by Γ and the chosen coset representatives" means precisely: there is a monoid homomorphism $\rho\colon \Delta \to \mathrm{End}_{\mathbb{Z}}(V)$ with $\rho|_\Gamma = \rho_n|_\Gamma$. For $V = \mathrm{Sym}^n$ this holds with $\rho = \rho_n$, by Proposition 8.2(2), since $\Delta \subseteq \mathrm{M}_2(\mathbb{Z})$ and all coset representatives of Proposition 10.1 lie in Δ.

The sentence just quoted is [v5], §12.1. That section is superseded in full (Appendix A, entries A.19, A.20), but the requirement it states is not among its defects and is met as shown.

Lemma 8.4 (Determinant of the symmetric power)**.** *[proved here] [formally verified: [GT, `lem:detsym`]] (the second sentence for R a field of characteristic ℓ)* $\det \rho_n(A) = (\det A)^{n(n+1)/2}$ *for every $A \in \mathrm{M}_2(\mathbb{Z})$. Hence for $g \in \Delta$ with $\det g = p$ and R of characteristic ℓ: $\rho_n(g)$ is invertible over R if and only if $\ell \neq p$ or $n = 0$.*

Proof. Both sides are polynomials in a, b, c, d with integer coefficients, so it suffices to prove the identity on a Zariski-dense subset of $\mathrm{M}_2(\mathbb{C})$; the matrices with two distinct eigenvalues form such a subset. For $A = P\,\mathrm{diag}(\alpha, \beta)\,P^{-1}$ with P invertible, Proposition 8.2(2) gives $\rho_n(A) = \rho_n(P)\rho_n(\mathrm{diag}(\alpha,\beta))\rho_n(P)^{-1}$ (with $\rho_n(P)^{-1} = \rho_n(P^{-1})$), so $\det \rho_n(A) = \det \rho_n(\mathrm{diag}(\alpha,\beta))$. The latter matrix is diagonal with entries $\alpha^{n-j}\beta^j$, whence $\det = \alpha^{\sum_j (n-j)}\beta^{\sum_j j} = (\alpha\beta)^{n(n+1)/2} = (\det A)^{n(n+1)/2}$. For the second claim: $n(n+1)/2 = 0$ exactly for $n = 0$, and $\ell \mid p^{n(n+1)/2}$ with $n \geq 1$ exactly when $\ell = p$. □

Lemma 8.4 is the whole content of hypothesis (H2): at $p = \ell$ and $n \geq 1$ every coset representative acts by a singular matrix, which is why U_ℓ can be nilpotent on $H^\bullet$ and why the recursion of Corollary 10.10 degenerates — *not* a defect of the construction, which never inverts $\rho_n(g)$.

8.1 The contragredient local system and the pairing

New in this document; owner decision OQ4 (`hecke_extensions_owner_decisions.md` §OQ4) makes D_n a first-class local system *together with* its pairing map, rather than the source of duality predictions only, and requires (architecture rule R2) that both be defined over $\mathbb{Z}$ and the R-versions be their base changes. That is how they are defined here.

Recall the adjugate on 2×2 matrices,

$$\mathrm{adj}\begin{pmatrix} a & b \\ c & d \end{pmatrix} := \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

which satisfies, for all $g, h \in \mathrm{M}_2(\mathbb{Z})$ and by direct computation,

$$g\,\mathrm{adj}\,g = \mathrm{adj}\,g\,g = (\det g)I, \qquad \mathrm{adj}(gh) = (\mathrm{adj}\,h)(\mathrm{adj}\,g), \qquad \mathrm{adj}(\mathrm{adj}\,g) = g, \qquad \mathrm{adj}\,I = I. \tag{10}$$

So adj is an anti-automorphism of the monoid $\mathrm{M}_2(\mathbb{Z})$, restricting to Δ, and $\mathrm{adj}\,\gamma = \gamma^{-1}$ for $\gamma \in \Gamma$.

Definition 8.5 (The contragredient action $\rho_n^\vee(g) = \rho_n(\mathrm{adj}\,g)^\top$)**.** Let R be a commutative ring. Put

$$D_n := \mathrm{Hom}_{\mathbb{Z}}(V, \mathbb{Z}), \qquad D_n \otimes_{\mathbb{Z}} R = \mathrm{Hom}_R(V_R, R),$$

free of rank $d_V = n + 1$ with the basis $(u_0^\vee, \ldots, u_n^\vee)$ dual to $(u_0, \ldots, u_n)$ (the second equality because V is free of finite rank). Define

$$\rho_n^\vee(g) := \rho_n(\mathrm{adj}\,g)^\top \in \mathrm{End}_{\mathbb{Z}}(D_n) \qquad \text{in the dual basis, } g \in \mathrm{M}_2(\mathbb{Z}),$$

equivalently, basis-free,

$$\big(\rho_n^\vee(g)\varphi\big)(v) \;=\; \varphi\big(\rho_n(\mathrm{adj}\,g)\,v\big), \qquad \varphi \in D_n,\ v \in V. \tag{11}$$

[formally verified: [GT, `def:contragredient`]] (the action (11)) The *contragredient local system* on $(K,\mathcal{R})$ is the local system whose fibre is $D_n \otimes R$ and whose relation matrix for the relation $(\sigma_{\mathrm{lhs}},\sigma_{\mathrm{rhs}},\gamma)\in\mathcal{R}$ of Definition 7.2 is $\rho_n^\vee(\gamma)$. Both D_n and $\rho_n^\vee$ are defined over $\mathbb{Z}$; every coefficient ring is a base change of that datum, and the reduction maps $\mathbb{Z}/\ell^{e+1}\to\mathbb{Z}/\ell^e$ commute with $\rho_n^\vee$ entrywise.

Proposition 8.6 (The perfect pairing $V\times D_n\to R$ and self-adjointness of T_p)**.** *[proved here] [formally verified: [GT, `prop:pairing`]] (clauses (1)–(5); (2) as the reflexivity of the evaluation, (5) as two matrix identities) Let R be a commutative ring, $n\ge 0$, and let*

$$\langle\,\cdot\,,\,\cdot\,\rangle\colon V_R\times(D_n\otimes R)\longrightarrow R,\qquad \langle v,\varphi\rangle:=\varphi(v)$$

be the evaluation pairing. Then:

1. *$\rho_n^\vee\colon \mathrm{M}_2(\mathbb{Z})\to\mathrm{End}_{\mathbb{Z}}(D_n)$ is a homomorphism of* monoids*, and $\rho_n^\vee(\gamma)=\rho_n(\gamma^{-1})^{\top}$ for $\gamma\in\Gamma$, i.e. $\rho_n^\vee|_\Gamma$ is the contragredient of $\rho_n|_\Gamma$ in the usual sense.*
2. *The pairing is* perfect *over every R: its Gram matrix in the bases (u_j), $(u_j^\vee)$ is the identity, so the induced map $D_n\otimes R\to\mathrm{Hom}_R(V_R,R)$ is an isomorphism. It is defined over $\mathbb{Z}$ and the R-pairing is its base change; in particular it commutes with the reduction maps $\mathbb{Z}/\ell^{e+1}\to\mathbb{Z}/\ell^e$ on both arguments.*
3. ***Adjointness.*** *For all $g\in\mathrm{M}_2(\mathbb{Z})$, $v\in V_R$, $\varphi\in D_n\otimes R$,*

$$\big\langle \rho_n(g)\,v,\ \varphi\big\rangle \;=\; \big\langle v,\ \rho_n^\vee(\mathrm{adj}\,g)\,\varphi\big\rangle. \tag{12}$$

4. ***Twisted invariance.*** *For all g, $\langle\rho_n(g)v,\rho_n^\vee(g)\varphi\rangle=(\det g)^n\langle v,\varphi\rangle$. In particular the pairing is Γ-invariant, so it is a pairing of local systems on $(K,\mathcal{R})$ in the sense of Definition 8.5; on Δ it is invariant only up to the character $\det^n$.*
5. ***The double coset is self-adjugate.*** *$\mathrm{adj}(\Gamma g\Gamma)=\Gamma\,(\mathrm{adj}\,g)\,\Gamma$ for every g, and for $g=\mathrm{diag}(1,p)$ one has $\mathrm{adj}\,g=\mathrm{diag}(p,1)=SgS^{-1}$, hence $\Gamma(\mathrm{adj}\,g)\Gamma=\Gamma g\Gamma$.*

Proof. (1) By the identities (10) and Proposition 8.2(2), $\rho_n^\vee(gh)=\rho_n(\mathrm{adj}(gh))^{\top}=\big(\rho_n(\mathrm{adj}\,h)\rho_n(\mathrm{adj}\,g)\big)^{\top}=\rho_n(\mathrm{adj}\,g)^{\top}\rho_n(\mathrm{adj}\,h)^{\top}=\rho_n^\vee(g)\rho_n^\vee(h)$, and $\rho_n^\vee(I)=I$. For $\gamma\in\Gamma$, $\mathrm{adj}\,\gamma=(\det\gamma)\gamma^{-1}=\gamma^{-1}$.

(2) $\langle u_t,u_j^\vee\rangle=\delta_{tj}$ by definition of the dual basis, so the Gram matrix is I, which is invertible over every R; everything in sight is the base change of the corresponding $\mathbb{Z}$-datum because V is free of finite rank.

(3) By (11) applied to $\mathrm{adj}\,g$ and $\mathrm{adj}(\mathrm{adj}\,g)=g$: $\langle v,\rho_n^\vee(\mathrm{adj}\,g)\varphi\rangle=\big(\rho_n^\vee(\mathrm{adj}\,g)\varphi\big)(v)=\varphi\big(\rho_n(\mathrm{adj}\,\mathrm{adj}\,g)v\big)=\varphi(\rho_n(g)v)=\langle\rho_n(g)v,\varphi\rangle$.

(4) By (11), Proposition 8.2(2) and the identities (10), $\langle\rho_n(g)v,\rho_n^\vee(g)\varphi\rangle=\varphi\big(\rho_n(\mathrm{adj}\,g)\rho_n(g)v\big)=\varphi\big(\rho_n((\det g)I)\,v\big)=(\det g)^n\varphi(v)$, using $\rho_n(\lambda I)=\lambda^n I$, which is Definition 8.1 for a scalar matrix. For $\gamma\in\Gamma$, $\det\gamma=1$.

(5) $\mathrm{adj}(\gamma_1g\gamma_2)=(\mathrm{adj}\,\gamma_2)(\mathrm{adj}\,g)(\mathrm{adj}\,\gamma_1)=\gamma_2^{-1}(\mathrm{adj}\,g)\gamma_1^{-1}$, and $\gamma\mapsto\gamma^{-1}$ is a bijection of Γ; so $\mathrm{adj}(\Gamma g\Gamma)=\Gamma(\mathrm{adj}\,g)\Gamma$. For $g=\mathrm{diag}(1,p)$, $\mathrm{adj}\,g=\mathrm{diag}(p,1)$, and $S\,\mathrm{diag}(1,p)\,S^{-1}=\begin{pmatrix}0&-1\\1&0\end{pmatrix}\begin{pmatrix}1&0\\0&p\end{pmatrix}\begin{pmatrix}0&1\\-1&0\end{pmatrix}=\mathrm{diag}(p,1)$. ∎

Remark 8.7 (What self-adjointness of T_p means, and where it is proved). Equation (12) is the coefficient-level input to the classical statement that the Hecke operators are self-adjoint for Poincaré duality, and it is exactly what makes the adjoint operator computable: if $\Gamma g\Gamma=\bigsqcup_j\Gamma g_j$ (Proposition 10.1), then the operator adjoint to $T_{\Gamma g\Gamma}$ on the D_n-side is the one built from the representatives $\mathrm{adj}\,g_j$, i.e. $T_{\Gamma(\mathrm{adj}\,g)\Gamma}$; and by Proposition 8.6(5) that is again $T_{\Gamma g\Gamma}$ for $g=\mathrm{diag}(1,p)$ — *no* determinant factor appears, because (12) carries none. The cohomological statement — perfect Hecke-compatible pairings $H_c^t(\Gamma;M_n)\times H^{2-t}(\Gamma;D_n)\to\mathbb{F}_\ell$ and $H^t(\partial;M_n)\times H^{1-t}(\partial;D_n)\to\mathbb{F}_\ell$, under the hypothesis $6\in R^\times$ — is Claim 16.26 of §16, where the compact

supports and the boundary it refers to are constructed. This chapter proves the coefficient-level half and nothing more: Proposition 8.6 says nothing about cohomology, and (12) must not be read as if it did.

Lemma 8.8 (V versus D_n: the integral invariant form)**.** *[proved here] [formally verified: [GT, `lem:symdual`]] (the displayed identities, the equivariance of Φ_n and the invertibility criterion; [GT] defines B_n by its Gram matrix (13), so the permutation-sum formula and its well-definedness are not covered) Define $B_n\colon V\times V\to\mathbb{Z}$ on products of vectors of $W=\mathbb{Z}^2$ by*

$$B_n(v_1\cdots v_n,\, w_1\cdots w_n)\;:=\;\sum_{\pi\in\mathfrak{S}_n}\prod_{t=1}^{n}\det\big[\,v_t\;\big|\;w_{\pi(t)}\,\big].$$

Then B_n is a well-defined symmetric-in-each-argument bilinear form with

$$B_n(u_t,u_j)=(-1)^t\,t!\,(n-t)!\;\delta_{t+j,n},\tag{13}$$

and for all $g\in\mathrm{M}_2(\mathbb{Z})$ and $v,w\in V$

$$B_n\big(\rho_n(g)v,\rho_n(g)w\big)=(\det g)^{\,n}B_n(v,w),\qquad B_n\big(\rho_n(g)v,\,w\big)=B_n\big(v,\,\rho_n(\operatorname{adj}g)w\big).$$

Consequently $\Phi_n\colon V\to D_n$, $\Phi_n(v):=B_n(v,\cdot)$, satisfies $\Phi_n\circ\rho_n(g)=\rho_n^{\vee}(g)\circ\Phi_n$ for every $g\in\mathrm{M}_2(\mathbb{Z})$ — it is a morphism of Δ-modules, hence of local systems on $(K,\mathcal{R})$ — its matrix is antidiagonal with entries (13), and

$$\Phi_n\otimes R\text{ is an isomorphism}\iff\textstyle\prod_{t=0}^{n}t!\,(n-t)!\in R^{\times}\iff n!\in R^{\times},$$

which for R a field of characteristic ℓ means $n\le\ell-1$.

Proof. $\det[\,v\mid w\,]$ is the standard symplectic form on W, and $\det[\,gv\mid gw\,]=(\det g)\det[\,v\mid w\,]$. The displayed sum is unchanged under a permutation of the v_t and under a permutation of the w_t (reindex π), so it descends to $\mathrm{Sym}^nW\times\mathrm{Sym}^nW$ and is bilinear. Multiplying each of the n factors by $\det g$ gives the first equivariance. For (13), take $v_1\cdots v_n=x^{n-t}y^t$ and $w_1\cdots w_n=x^{n-j}y^j$ and use $\det[x\mid x]=\det[y\mid y]=0$, $\det[x\mid y]=1=-\det[y\mid x]$: a permutation contributes only if it matches each of the $n-t$ letters x on the left to a letter y on the right and each of the t letters y on the left to a letter x on the right, which forces $n-t=j$ and $t=n-j$, i.e. $t+j=n$; there are then $(n-t)!\,t!$ such permutations, each contributing $1^{n-t}(-1)^t$.

For the second equivariance, substitute $w\mapsto\rho_n(\operatorname{adj}g)w$ in the first and use $\rho_n(g)\rho_n(\operatorname{adj}g)=\rho_n((\det g)I)=(\det g)^n\mathrm{id}$: $(\det g)^nB_n(\rho_n(g)v,w)=(\det g)^nB_n(v,\rho_n(\operatorname{adj}g)w)$. Both sides are polynomial in the entries of g with integer coefficients, and $\mathbb{Z}[a,b,c,d]$ is a domain in which $(\det g)^n=(ad-bc)^n\neq0$, so the factor cancels there; the identity over any R follows by specialising the resulting polynomial identity. Then, for $w\in V$, $\big(\Phi_n(\rho_n(g)v)\big)(w)=B_n(\rho_n(g)v,w)=B_n(v,\rho_n(\operatorname{adj}g)w)=\big(\Phi_n(v)\big)(\rho_n(\operatorname{adj}g)w)=\big(\rho_n^{\vee}(g)\Phi_n(v)\big)(w)$ by (11).

Finally $\Phi_n(u_t)=(-1)^tt!(n-t)!\,u_{n-t}^{\vee}$, so the matrix of Φ_n is antidiagonal with those entries and $\Phi_n\otimes R$ is bijective iff all of them are units. Since $n!=\binom{n}{t}t!(n-t)!$, each $t!(n-t)!$ divides $n!$, so $n!$ being a unit makes all of them units; conversely the factor at $t=0$ is $0!\,n!=n!$. Over a field of characteristic ℓ, $n!\neq0$ iff $n<\ell$. □

Remark 8.9 (Divided powers, and what is *not* claimed). The classification literature calls D_n the *divided-power* module and records that M_n fails to be self-dual for $n\ge\ell$ ([E, `sec:symdiv`]). What Lemma 8.8 proves is the sharper and weaker statement that the canonical integral comparison map Φ_n fails to be an isomorphism exactly when $n!\notin R^{\times}$. Whether V_R and $D_n\otimes R$ are *abstractly* isomorphic as Γ-modules in that range is a different question; it is not used anywhere in this document and is not asserted here.

Code map. Definition 8.1, Proposition 8.2: `ComputeSymmetricPowerMatrixFM` (`ModArithmetic/FlatMatri`x whose branch table realises (9) in $O(d_V^2)$, against the double-sum reference `ComputeSymmetricPowerMatrixFM_r` gated by `SymmetricPowerValidation` (and its slow-tier twin) entry for entry, and by $\rho_n(A)\rho_n(B) = \rho_n(AB)$ as a structural check. The deprecated `ComputeSymmetricPowerMatrix` returns the same matrix in the retired `Matrix` type. Lemma 8.4: the singular-matrix regime is exercised by the $p = \ell$ arms of `HeckeBadPrimeValidation`. Definition 8.3: the coset representatives of `Hecke/HeckeCosetTables.cpp`, all of determinant $p^\nu > 0$. **Since v1.7.3 (hecke-extensions slot-6 WP-1, phase E7a) §8.1 is implemented**: `Mat22::adj` (`Hecke/ExactInt`, (10)), $\rho_n^\vee = \rho_n(\mathrm{adj}\,\cdot)^T$ as `ComputeContragredientPowerMatrixFM` with the dispatcher `ComputeFibreMatrixFM` (`LocalSystemUtils`), the contragredient relation list `GenerateSL2ZContragredientRelationsWithGroupData` (`LocalSystemSL2Z`), Φ_n of Lemma 8.8 and the pairing of Proposition 8.6 in `Hecke/HeckeContragredient`, and $A(D_n)(T_p)^q$ through `HeckeOperatorOnK`'s `FibreRepresentation` argument (default `SymN`, byte for byte the pre-slot-6 operator). `HeckeDualityValidation` asserts the intertwining *untwisted* (exponent 0, as Lemma 8.8 gives it) and the $p^{\pm n}$-twisted relations to fail. The cohomology-level pairings are `Hecke/HeckePairing` (slot-6 WP-3; §22.1). The v1.7.2 status follows, kept as history. §8.1 **had no implementation at v1.7.2**: adj occurs in the code only as the $\mathrm{adj}(g)/N$ projective trick of `Hecke/HeckeTauTable.cpp` and `Hecke/BarCocycleH1.cpp` (each spelling the 2×2 adjugate out inline; there is no adjugate operation on `Mat22`), and neither $\rho_n^\vee$ nor the pairing is built. They are phase E7 of the extension plan, made unconditional by owner decision OQ4; the perfectness certificate that Proposition 8.6(2) licenses — Gram matrix $= I$ in the stored bases — is the cheapest possible one.

Formalization note. Definition 8.1 is a closed-form matrix over $\mathbb{Z}$ and Proposition 8.2(2) is the statement that Sym^n is a functor on the multiplicative monoid, available in `mathlib4` as the algebra map induced on `MvPolynomial`; the bound in Proposition 8.2(3) is a decidable inequality. Lemma 8.4 is the only statement in this chapter proved by Zariski density; a formalization would more comfortably prove it by induction on n from the closed form (9) for triangular A together with a Bruhat-style factorisation, or simply accept the polynomial-identity argument over $\mathbb{Z}[a, b, c, d]$ used in Lemma 8.8. Definition 8.5 and the whole of Proposition 8.6 are finite linear algebra over a commutative ring and carry no analytic content; (12) is the natural statement to formalise first, as everything else about the duality rests on it.

9 The involution ι

What this chapter fixes. The element $\iota = \mathrm{diag}(-1, 1) \in \mathrm{GL}_2(\mathbb{Z})$, its simplicial realisation ι_K with the orientation sign of every simplex, the involution ι^* it induces on $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$, the exact hypothesis under which ι^* splits $H^\bullet_{\mathrm{eq}}$ into $\pm$ eigenspaces — it is $2 \in R^\times$, and what replaces the splitting at $\ell = 2$ is stated here and not deferred — and the commutation $[\iota^*, T_{\Gamma g \Gamma}] = 0$, with the level at which it holds and the hypothesis each level needs. **Since v1.1.1 every clause of Obligation 9.10 is implemented and has been run**: (i) and (iv) since v0.13.0 — and since v1.1.0 clause (iv) *holds* on the canonical τ and gates it (Remark 9.5) — and (ii), (iii), (v) since v1.1.1, when ι^* was first built as a map of cochain complexes. **Since v1.2.0 the sign of ι^* is fixed by owner decision OQ3 (2026-09-16)**: Definition 9.1(iv) defines $\iota^* := -\iota^*_{\mathrm{geo}}$, the negative of the geometric pullback, so that "+" *is by definition* the eigenspace containing the Eisenstein classes. The v1.1.1 run had found those classes in the (-1)-eigenspace of the geometric involution on every cell of a 77-cell grid, by two independent routes; the owner resolved it by negating the definition, not at a call site, and Remark 9.9 says what the record field `iota_sign` now means. Every predicate of the five clauses is gating and passes, the former "recorded half" of clause (v) included, which is now a definition-consistency oracle; the honest cost of the choice — constants in H^0 are odd, ι^* is the geometric involution twisted by the sign

character det of $\mathrm{GL}_2(\mathbb{Z})/\mathrm{SL}_2(\mathbb{Z})$ rather than a pullback — is stated at Definition 9.1(iv) and in `docs/implementation/oq3_resolution.md`. The code map at the end says which validation discharges what.

Standing hypotheses. (H1)–(H4) of §7, each repeated where it is used (rule (A1)), together with one hypothesis proper to this chapter. It is stated for an arbitrary Γ-tessellation rather than for the seven vertices of Definition 7.1, so that a refinement of that complex is admissible input (the design direction recorded in Remark 7.6).

(I1) **ι-stable tessellation.** The map $z \mapsto -\bar{z}$ of $\mathcal{H}$ carries the complex X of (H4) to itself simplicially — each simplex of X onto a simplex of X — and carries the strict fundamental domain K of (H4)(iii) onto itself.

(I1) is a property of the triangulation alone. It is verified for the seven-vertex model in Lemma 9.2, and an ι-stable refinement of that model satisfies it verbatim.

The element, its symmetric power, and its realisation

Definition 9.1 (The involution ι, its simplicial realisation ι_K, and $\rho_n(\iota)$)**.** 1. **The element.** $\iota := \mathrm{diag}(-1,1) \in \mathrm{GL}_2(\mathbb{Z})$, with $\iota^2 = I$, $\det \iota = -1$ and therefore $\iota \notin \Delta$: it is *not* a Hecke element and $T_{\Gamma\iota\Gamma}$ is not defined. Conjugation $\mathrm{conj}_\iota \colon \gamma \mapsto \iota\gamma\iota^{-1} = \iota\gamma\iota$ is an involutive automorphism of $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ and of the monoid $\mathrm{M}_2(\mathbb{Z})$, and on the generator multiset $\mathcal{G} = \{T, S, ST, TS\}$ of §7.1 and on the centre it is

$$\mathrm{conj}_\iota(T) = T^{-1}, \quad \mathrm{conj}_\iota(S) = S^{-1}, \quad \mathrm{conj}_\iota(ST) = (TS)^{-1}, \quad \mathrm{conj}_\iota(TS) = (ST)^{-1}, \quad \mathrm{conj}_\iota(-I) = -I. \tag{14}$$

On $\mathcal{H}$, ι acts by $z \mapsto -\bar{z}$ (the $\det < 0$ convention $z \mapsto (a\bar{z}+b)/(c\bar{z}+d)$), and $\iota(\gamma z) = \mathrm{conj}_\iota(\gamma)(\iota z)$.

2. **On coefficients.** $\rho_n(\iota) \in \mathrm{GL}(V)$ is the value at ι of the monoid homomorphism ρ_n on all of $\mathrm{M}_2(\mathbb{Z})$ (Definition 8.1, Proposition 8.2(2)) — which is why that chapter defined ρ_n on $\mathrm{M}_2(\mathbb{Z})$ and not on Δ. Explicitly, from (9) with $(a,b,c,d) = (-1,0,0,1)$,

$$\rho_n(\iota)\, u_j = (-1)^{n-j} u_j, \qquad \rho_n(\iota) = \mathrm{diag}\big((-1)^{n-j}\big)_{0\le j\le n}, \qquad \rho_n(\iota)^2 = \rho_n(I) = I. \tag{15}$$

In characteristic 2, $\rho_n(\iota) = I$ for every n.

3. **On the complex.** Assume (H4) and (I1). Write ι_X for the simplicial automorphism of X induced by $z \mapsto -\bar{z}$ and ι_K for its restriction to K, both extended $\mathbb{Z}$-linearly to simplicial chains, where they carry the stored orientations of §7 to stored orientations up to sign; ι_K and ι_X are two realisations of one notion and are indexed accordingly ((N2)). They satisfy $\iota_X(\gamma \cdot x) = \mathrm{conj}_\iota(\gamma) \cdot \iota_X(x)$, $\iota_X \circ \partial = \partial \circ \iota_X$, $\varepsilon \circ \iota_X = \varepsilon$, $\iota_X^2 = \mathrm{id}$, and $\iota_X(\widetilde{\sigma}) = \widetilde{\iota_K \sigma}$ for $\sigma \in o_q(K)$.

4. **On cochains.** For $f \in C^q(K; V_R)$, read as a homomorphism on $C_q(K)$, write ι_{geo}^* for the *geometric* pullback along $(\iota_K, \rho_n(\iota))$ and define ι^* as its negative:

$$(\iota_{\mathrm{geo}}^* f)(\sigma) \;:=\; \rho_n(\iota)\, f\big(\iota_K \sigma\big), \qquad \iota^* \;:=\; -\,\iota_{\mathrm{geo}}^*, \quad \text{i.e.} \quad (\iota^* f)(\sigma) \;=\; -\rho_n(\iota)\, f\big(\iota_K \sigma\big), \qquad \sigma \in o_q(K), \tag{16}$$

in every degree q and for every coefficient module V_R this document uses (D_n of Definition 8.5 included). **ι^*, not ι_{geo}^*, is the involution of this document**: $H^q_{\mathrm{eq},\pm}$, `iota_sign`, Obligation 9.10 and every $\pm$ statement below refer to it. The sign is the owner decision OQ3 of 2026-09-16 (**v1.2.0**; previously $\iota^* = \iota_{\mathrm{geo}}^*$): it is chosen so that "+" is *by definition* the eigenspace containing the Eisenstein classes, Remark 9.9. What the sign is, honestly: $-\rho_n(\iota) = \det(\iota)\,\rho_n(\iota) = (\det \otimes \rho_n)(\iota)$ with $\det|_\Gamma = 1$ — so on every relation element $\gamma \in \Gamma$ the twisted fibre map $(\det \otimes \rho_n)(\gamma)$ equals $\rho_n(\gamma)$ and the relations of $\mathcal{R}$ are preserved — so ι^* is the pullback along $(\iota_K, \det(\iota)\rho_n(\iota))$ — an automorphism of the local system $(K, \mathcal{R}, L)$ in the sense of Proposition 9.4(1)–(2), since the intertwining relation is

linear in the fibre map — but, for n even, *not* the pullback along any lift of $z \mapsto -\bar{z}$ to $\mathrm{GL}_2(\mathbb{Z})$: the two lifts $\pm\iota$ give $\rho_n(\pm\iota) = (\pm 1)^n \rho_n(\iota)$, the same matrix for n even. (For n odd it *is* the pullback along the lift $-\iota$; but for $\ell \neq 2$ and n odd $H^\bullet_{\mathrm{eq}} = 0$, Remark 7.7, and at $\ell = 2$ the sign is invisible, so on every group this document computes with a non-zero answer the twist is not a lift.) It is the geometric involution twisted by the sign character of $\mathrm{GL}_2(\mathbb{Z})/\mathrm{SL}_2(\mathbb{Z})$, the mechanism by which the classical modular-symbol "+" (the side of the boundary symbol [standard: [Cre97, Ch. 2]]) differs from the naive pullback. Consequences: $\iota^* = -\mathrm{id}$ on $H^0(\Gamma; \mathbb{F}_\ell)$ at $n = 0$ (constants are *odd*); at $\ell = 2$ the sign is invisible, $\iota^* = \iota^*_{\mathrm{geo}}$. In the flat coordinates (8) ι^* is a signed block permutation: one block $-\varepsilon_\sigma \rho_n(\iota)$ per simplex σ, at block position $(\sigma, |\iota_K \sigma|)$, where $\varepsilon_\sigma \in \{\pm 1\}$ is the sign of $\iota_K \sigma$ in the table of Lemma 9.2 (the form of the code map below and of `HeckeIotaSignValidation`; ι_K being an involution with $\varepsilon_{\iota_K \sigma} = \varepsilon_\sigma$ and $\rho_n(\iota)$ diagonal, the matrix equals its transpose), and no data of its own beyond the single $d_V \times d_V$ diagonal matrix (15). A third party reproduces any $\pm$ statement of a record from $\iota^* = -\iota^*_{\mathrm{geo}}$ and nothing else.

Lemma 9.2 (The seven-vertex model is ι-stable, with its sign table)**.** *[proved here] Let K be as in Definition 7.1, with the four horocycle vertices placed symmetrically about the imaginary axis, i.e. $\mathrm{Re}\, v_1 = -\mathrm{Re}\, v_2$. (Definition 7.1 fixes $\mathrm{Re}\, v_0 = -\frac{1}{2}$ and $\mathrm{Re}\, v_3 = +\frac{1}{2}$ and leaves $\mathrm{Re}\, v_1, \mathrm{Re}\, v_2$ free, subject only to $|\mathrm{Re}| < \frac{1}{2}$; the symmetric placement is what makes ι simplicial and it is the only use made of it in this document.) Then (I1) holds, and ι_K is the vertex permutation*

$$0 \leftrightarrow 3, \qquad 1 \leftrightarrow 2, \qquad 4 \leftrightarrow 5, \qquad 6 \text{ fixed},$$

an involutive simplicial automorphism of K whose orientation sign is -1 on exactly six of the 23 simplices:

q	$\iota_K\sigma = -\sigma'$ *(sign -1)*	$\iota_K\sigma = +\sigma'$ *(sign $+1$)*
0	—	*all seven vertices*
1	$\{v_0v_1\} \leftrightarrow \{v_2v_3\}$; $\{v_1v_2\}$ *to itself*	*the other eight edges*
2	$\{v_0v_1v_4\} \leftrightarrow \{v_2v_3v_5\}$; $\{v_1v_2v_6\}$ *to itself*	$\{v_1v_4v_6\} \leftrightarrow \{v_2v_5v_6\}$

Proof. $z \mapsto -\bar{z}$ preserves Im and $|z|$ and negates Re, so it fixes setwise the horocycle $\{\mathrm{Im}\, z = Y\}$, the unit circle and the region $D_Y = |K|$ of Lemma 7.3. On vertices it exchanges $v_0 \leftrightarrow v_3$ and, by the symmetric placement, $v_1 \leftrightarrow v_2$; and $v_4 = \zeta^2 = -\frac{1}{2} + \frac{\sqrt{3}}{2} i \mapsto \zeta = v_5$, $v_6 = i \mapsto v_6$. It maps the five faces of Definition 7.1 to $\{v_2v_3v_5\}$, $\{v_1v_2v_6\}$, $\{v_0v_1v_4\}$, $\{v_2v_5v_6\}$, $\{v_1v_4v_6\}$ — the same five — hence the eleven edges to the eleven edges; so ι_K is a simplicial automorphism of K with $\iota_K^2 = \mathrm{id}$. That ι is simplicial on all of X follows from (H4)'s definition of X as the tessellation by Γ-translates of those five simplices, together with $\iota(\gamma \cdot x) = \mathrm{conj}_\iota(\gamma) \cdot \iota(x)$ and $\mathrm{conj}_\iota(\Gamma) = \Gamma$; and $\iota_X(K) = K$ is what was just shown. This is (I1).

The signs are the signs of the permutations sorting the images back into increasing vertex order. Edges: $(0,1) \mapsto (3,2)$, $(1,2) \mapsto (2,1)$ and $(2,3) \mapsto (1,0)$ each need one transposition; the other eight map increasingly — $(0,4) \mapsto (3,5)$, $(1,4) \mapsto (2,5)$, $(1,6) \mapsto (2,6)$, $(2,5) \mapsto (1,4)$, $(2,6) \mapsto (1,6)$, $(3,5) \mapsto (0,4)$, $(4,6) \mapsto (5,6)$, $(5,6) \mapsto (4,6)$. Triangles: $(0,1,4) \mapsto (3,2,5)$, $(1,2,6) \mapsto (2,1,6)$ and $(2,3,5) \mapsto (1,0,4)$ each need one transposition, while $(1,4,6) \mapsto (2,5,6)$ and $(2,5,6) \mapsto (1,4,6)$ are already increasing. □

Remark 9.3 (What the sign table is, and what it is not). The six minus signs are an artefact of the *stored* orientation — increasing vertex index, §7 — which is not a coherent orientation of the surface. They are not a convention and nothing may be chosen about them: they are forced by Definition 7.1's storage rule, and ι^* is simply wrong without them. They are printed because a reader cannot recompute them from the vertex permutation alone without redoing the sort, and because a refinement of K will have a different table, computed the same way.

Automorphism of the local system, and commutation with the Hecke action

Proposition 9.4 ($(\iota_K, \rho_n(\iota))$ is an automorphism of $(K, \mathcal{R}, L)$, and ι commutes with $T_{\Gamma g \Gamma}$)**.** *Assume (H4) and (I1), and let R be a commutative ring.*

1. *[proved here] **(Automorphism of the relation datum; no hypothesis on ℓ.)** For every relation $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$ of Definition 7.2, the triple $\big(\iota_K \sigma_{\mathrm{lhs}}, \iota_K \sigma_{\mathrm{rhs}}, \mathrm{conj}_\iota(\gamma)\big)$ lies in the sub-groupoid $\langle \mathcal{R} \rangle$ of Definition 3.15. It is in fact the* inverse *of a member of the list: each of the six T- and S-rows goes to the inverse of itself, the two elliptic stabiliser rows $(\{v_4\}, \{v_4\}, ST)$ and $(\{v_5\}, \{v_5\}, TS)$ are exchanged, and each central row $(\sigma, \sigma, -I)$ goes to the central row at $|\iota_K \sigma|$. **Up to inversion is the strongest statement available**: $\mathcal{R}$ as a* list *is not preserved, its generated groupoid is. Part of the claim is that in every row the two sides carry the* same *orientation sign, so that the common sign cancels and the image triple is a triple of* stored *simplices, which is what membership in $\langle \mathcal{R} \rangle$ asserts.*
2. *[proved here] **(Involution of the cochain complex; no hypothesis on ℓ.)** ι^* of (16) maps $C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ into itself, commutes with d, and satisfies $(\iota^*)^2 = \mathrm{id}$; so does ι^*_{geo}, and the two differ by the global sign $-\mathrm{id}$, which commutes with everything in sight. Hence ι^* preserves Z^q_{eq} and B^q_{eq} and induces an involution of $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ in every degree q. Under Ψ of Theorem 5.10 it corresponds to $\iota^\# := -\iota^\#_{\mathrm{geo}}$, where $(\iota^\#_{\mathrm{geo}}\phi)(x) := \rho_n(\iota)\, \phi(\iota_X x)$ on $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(K^u), V_R\big)$, i.e. $\Psi \circ \iota^\# = \iota^* \circ \Psi$ (and likewise with the subscript* geo *on both sides).*
3. ***(Hecke.)** Let $g \in \Delta$ with $\mathrm{conj}_\iota(g) = g$ — which holds for* every *operator of §10, all of whose elements $\mathrm{diag}(1, p^\nu)$ and pI are diagonal — let $s_1, \dots, s_m$, $\delta_j = s_j^{-1} g$ and τ be as in Definition 11.6, and put $s_j^\iota := \mathrm{conj}_\iota(s_j)$ and $\tau^\iota := \iota_X \circ \tau \circ \iota_X$. Then:*
   *(3a) [proved here] **no hypothesis on ℓ.** $\{s_j^\iota\}$ represents $\Gamma'' \backslash \Gamma$, τ^ι is a g-twisted chain datum over R (Definition 11.1), and at* cochain *level*

$$\iota^\# \circ \widetilde{T}\big[g, \{s_j\}, \tau\big] \;=\; \widetilde{T}\big[g, \{s_j^\iota\}, \tau^\iota\big] \circ \iota^\#,$$

   where $\widetilde{T}[\,\cdot\,]$ is (21) built from the displayed datum.
   *(3b) [proved here] **no hypothesis on ℓ.** If moreover $\tau^\iota = \tau$, then $\iota^\# \circ \widetilde{T} = \widetilde{T} \circ \iota^\#$ on the nose, hence $\iota^* \Theta_g = \Theta_g \iota^*$ as cochain maps.*
   *(3c) [proved here] **under** $6 \in R^\times$ (hypothesis (H3); for R of characteristic ℓ, $\ell \geq 5$). $[\iota^*, T_{\Gamma g \Gamma}] = 0$ on $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ for every q; equivalently ι^* and $A^{(q)}$ commute as matrices in the stored basis of H^q_{eq}. The same then holds for T_p, T_{p^ν}, T_N, $T(p,p)$ and U_ℓ, and for the primitive operator of §10.1.*

Proof. (1). Read Definition 7.2 against (14) and Lemma 9.2. $(\{v_3\}, \{v_0\}, T) \mapsto (\{v_0\}, \{v_3\}, T^{-1})$, $(\{v_5\}, \{v_4\}, T) \mapsto (\{v_4\}, \{v_5\}, T^{-1})$, $(\{v_4\}, \{v_5\}, S) \mapsto (\{v_5\}, \{v_4\}, S^{-1})$ and $(\{v_6\}, \{v_6\}, S) \mapsto (\{v_6\}, \{v_6\}, S^{-1})$ are the inverses of themselves. The two edge rows likewise, and with sign $+1$ on *both* sides by Lemma 9.2, so no sign survives: $(\{v_3, v_5\}, \{v_0, v_4\}, T) \mapsto (\{v_0, v_4\}, \{v_3, v_5\}, T^{-1})$ and $(\{v_4, v_6\}, \{v_5, v_6\}, S) \mapsto (\{v_5, v_6\}, \{v_4, v_6\}, S^{-1})$. The elliptic stabiliser rows are exchanged: $(\{v_4\}, \{v_4\}, ST) \mapsto (\{v_5\}, \{v_5\}, (TS)^{-1})$, the inverse of $(\{v_5\}, \{v_5\}, TS)$, and symmetrically. A central row $(\sigma, \sigma, -I)$ goes to $(\iota_K\sigma, \iota_K\sigma, -I)$, whose two sides carry the same sign, hence to the central row at $|\iota_K \sigma|$. $\langle \mathcal{R} \rangle$ is closed under inversion, so every image lies in it.

(2). The four properties are proved for ι^*_{geo}; they pass to $\iota^* = -\iota^*_{\mathrm{geo}}$ because the relations of $\mathcal{R}$ and d are linear, $(-\mathrm{id})^2 = \mathrm{id}$, and Ψ is linear, so $\Psi \circ (-\iota^\#_{\mathrm{geo}}) = -\iota^*_{\mathrm{geo}} \circ \Psi$. (Equivalently: $-\rho_n(\iota)$ satisfies the same intertwining relation as $\rho_n(\iota)$ below, so the computation may be read with $-\rho_n(\iota)$ throughout.) *Equivariance.* Let $f \in C^q_{\mathrm{eq}}$ and $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$. From $\iota\, \mathrm{conj}_\iota(\gamma) = \gamma\, \iota$ in $\mathrm{M}_2(\mathbb{Z})$ and Proposition 8.2(2), $\rho_n(\iota)\rho_n(\mathrm{conj}_\iota(\gamma)) = \rho_n(\gamma)\rho_n(\iota)$; so by (1) and Lemma 5.5,

$$(\iota^*_{\mathrm{geo}} f)(\sigma_{\mathrm{lhs}}) = \rho_n(\iota) f(\iota_K \sigma_{\mathrm{lhs}}) = \rho_n(\iota)\rho_n(\mathrm{conj}_\iota(\gamma)) f(\iota_K \sigma_{\mathrm{rhs}}) = \rho_n(\gamma)\, (\iota^*_{\mathrm{geo}} f)(\sigma_{\mathrm{rhs}}).$$

Commutation with d. ι_K is simplicial, so $\iota_K\partial = \partial\iota_K$ on chains, and $\rho_n(\iota)$ is a constant matrix; hence $(\iota_{\mathrm{geo}}^*(df))(\sigma) = \rho_n(\iota)f(\partial\iota_K\sigma) = \rho_n(\iota)f(\iota_K\partial\sigma) = (d(\iota_{\mathrm{geo}}^*f))(\sigma)$. *Involutivity.* $\rho_n(\iota)^2 = I$ by (15) and $\iota_K^2 = \mathrm{id}$ on chains. *Transport.* By (I1), $\iota_X(\widetilde{\sigma}) = \widetilde{\iota_K\sigma}$, so $\Psi(\iota_{\mathrm{geo}}^{\#}\phi)(\sigma) = \rho_n(\iota)\phi(\iota_X\widetilde{\sigma}) = \rho_n(\iota)(\Psi\phi)(\iota_K\sigma) = \iota_{\mathrm{geo}}^*(\Psi\phi)(\sigma)$; that $\iota_{\mathrm{geo}}^{\#}\phi$ is again Γ-equivariant is the displayed computation with arbitrary $\gamma \in \Gamma$ in place of a relation element.

(3a). $\mathrm{conj}_\iota(\Gamma) = \Gamma$ and $\mathrm{conj}_\iota(g) = g$ give $\mathrm{conj}_\iota(\Gamma'') = \mathrm{conj}_\iota(\Gamma \cap g\Gamma g^{-1}) = \Gamma''$, so $\Gamma = \bigsqcup_j \mathrm{conj}_\iota(\Gamma'' s_j) = \bigsqcup_j \Gamma'' s_j^\iota$. Properties (T2) and (T3) of τ^ι are immediate from $\iota_X\partial = \partial\iota_X$, $\varepsilon\iota_X = \varepsilon$ and $\iota_X^2 = \mathrm{id}$. For (T1), let $h \in \Gamma''$; then $\mathrm{conj}_\iota(h) \in \Gamma''$ and

$$\tau^\iota(h\cdot x) = \iota_X\tau\big(\mathrm{conj}_\iota(h)\cdot\iota_X x\big) = \iota_X\big((g^{-1}\,\mathrm{conj}_\iota(h)g)\cdot\tau(\iota_X x)\big) = \mathrm{conj}_\iota\big(g^{-1}\,\mathrm{conj}_\iota(h)g\big)\cdot\tau^\iota(x) = (g^{-1}hg)\cdot\tau^\iota(x),$$

the last step by $\mathrm{conj}_\iota(g) = g$ and $\mathrm{conj}_\iota^2 = \mathrm{id}$. For the intertwining it suffices to treat $\iota_{\mathrm{geo}}^{\#}$: $\iota^{\#} = -\iota_{\mathrm{geo}}^{\#}$ occurs once on each side of the asserted identity, so the signs cancel. Write $\psi := \iota_{\mathrm{geo}}^{\#}\phi$, so that $\phi(\iota_X y) = \rho_n(\iota)\psi(y)$, and note $s_j \cdot \iota_X x = \iota_X(s_j^\iota \cdot x)$ and $\tau \circ \iota_X = \iota_X \circ \tau^\iota$. Then by (21)

$$\big(\iota^{\#}(\widetilde{T}\phi)\big)(x) = \rho_n(\iota)\sum_j \rho_n(\delta_j)\,\phi\big(\iota_X\tau^\iota(s_j^\iota\cdot x)\big) = \sum_j \rho_n\big(\iota\delta_j\iota\big)\,\psi\big(\tau^\iota(s_j^\iota\cdot x)\big),$$

and $\mathrm{conj}_\iota(\delta_j) = \mathrm{conj}_\iota(s_j)^{-1}\,\mathrm{conj}_\iota(g) = (s_j^\iota)^{-1}g$ is exactly the coset element attached to s_j^ι, so the right-hand side is $\big(\widetilde{T}[g,\{s_j^\iota\},\tau^\iota]\,\psi\big)(x)$.

(3b). If $\tau^\iota = \tau$ the two data of (3a) differ only in the representatives, so $s_j^\iota = h_j s_{j'}$ for some $h_j \in \Gamma''$ and some permutation $j \mapsto j'$ of $\{1,\dots,m\}$. With τ held fixed, the computation in the proof of Theorem 11.7(a) shows that each summand of (21) is unchanged under $s \mapsto hs$, $h \in \Gamma''$; permuting the summands does not change the sum; hence $\widetilde{T}[g,\{s_j^\iota\},\tau] = \widetilde{T}[g,\{s_j\},\tau]$ as cochain maps, and (3a) becomes the asserted commutation.

(3c). By (3a) the two sides of $[\iota^{\#},\widetilde{T}]$ are the operators of two admissible data for the *same* g, so by Theorem 11.7(a),(b) they induce the same map on H^q_{eq} — (b) being where $6 \in R^\times$ enters, and the only place. The extension to the operator family is Definition 10.4: every double coset of determinant p^ν has a diagonal representative $\mathrm{diag}(p^a, p^{\nu-a})$, which conj_ι fixes, so each summand of (18) is covered, $T(p,p) = T_{\Gamma pI\Gamma}$ included; conj_ι moreover preserves both the determinant and the content of an integer matrix, hence the primitive double coset $\Delta^{\mathrm{prim}}_{p^\nu}$ of §10.1; and T_N is a $\mathbb{Z}$-combination of the T_{p^ν} by (19) and inherits the commutation. □

Remark 9.5 ((3b) is the only route at $\ell \in \{2,3\}$; its hypothesis was *false* for the canonical τ of v0.13.0–v1.0.0 and is *true* for the canonical τ since v1.1.0). Theorem 11.7(b) is unavailable for $6 \notin R^\times$, and so therefore is Proposition 9.4(3c): at $\ell \in \{2,3\}$ the commutation $[\iota^*, T_{\Gamma g\Gamma}] = 0$ **does not follow** from the general argument. What is available there is (3b), whose hypothesis $\tau^\iota = \tau$ is a finite comparison of two integer tables (Obligation 9.10(iv)).

The finding (v0.13.0; run of 2026-09-16, `HeckeIotaTauSymmetryValidation`*).* [verified computationally] On the canonical τ of Algorithm 18.3 *as it stood before v1.1.0* — kept in the tree as `TauVertexPolicy::GeometricNearestLegacy` and re-run by the same validation, so the number below is reproduced by every run — at $\ell \in \{2,3\}$, over $(p,\nu) \in \{(2,1),(3,1),(5,1),(7,1),(2,2),(3,2)\}$ and $n \in \{0,1,2,3,4,10\}$, 3600 of the 10764 stored table slots satisfy $\iota_X\,\tau\,\iota_X \neq \tau$ — by degree, 456/3276 at $q=0$, 2340/5148 at $q=1$, 804/2340 at $q=2$ — every comparison exact over $\mathbb{Z}$ (no table used the mod-ℓ fallback of Remark 11.5). A minimal witness, sign-free because every vertex carries orientation sign $+1$ in Lemma 9.2's table: at $p=2$, $\nu=1$, $\ell=2$, $n=0$, in the slot $(j,q,\sigma) = (1,0,\{v_0\})$ with $s_1 = I$,

$$\tau(\widetilde{v_0}) = \gamma\cdot\widetilde{v_1}, \qquad (\iota_X\,\tau\,\iota_X)(\widetilde{v_0}) = \gamma\cdot\widetilde{v_2}, \qquad \gamma = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix},$$

the two right-hand sides being distinct as *chains* (the validation canonicalises every term through the public key the τ-table uses). The mechanism is step 1 of Algorithm 18.3: with $g = \mathrm{diag}(1,2)$

and $s_1 = I$ the reduced point of $\mathrm{adj}(g) \cdot \mathrm{pos}(v_0) = -1 + 4i$ is $4i$, **on the mirror** $\mathrm{Re}\, z = 0$, equidistant from v_1 and v_2, which ι_K exchanges; the then tie-break — ascending vertex index — picked v_1. That negative was *certified*: τ^ι satisfies (T1)–(T3) of Definition 11.1 on every stored slot, so it exhibited two admissible data for one g, exactly as Proposition 9.4(3a) predicts, and not a defect.

Why no tie-break of the tied set can repair it. On the mirror the tied set $S = \{v_1, v_2\}$ is ι_K-stable without a fixed point, so $c(S) = \iota_K c(S)$ is impossible for any choice $c(S) \in S$: the repair this remark proposed at v0.13.0 ("among tied vertices prefer the one of smaller index in the orbit of ι_K") cannot work as stated, and an equivariant rule must leave the tied set. The only ι_K-fixed vertex of K is $v_6 = i$, and i is the only vertex of X on the mirror at all ($\gamma \cdot i \in i\mathbb{R}_{>0}$ forces $\gamma \in \{\pm I, \pm S\}$, and no horocycle vertex lies on the axis), so a mirror point *must* be sent to v_6, off-nearest. The tied sets that actually occur on the validation's grid are three: $\{v_1, v_2\}$ on the mirror (288 slots), and the ι-paired $\{v_0, v_1\}$ / $\{v_2, v_3\}$ off it (72 each); no other tie arises.

The repair, taken at v1.1.0 (owner decision of 2026-09-16). Two changes to Algorithm 18.3, neither touching Definition 11.1 or Obligation 20.11: the ι_K-equivariant tie-break of step (1b), and the tree-routed templates of step 2. [verified computationally] **Step 1 alone was not enough**: with the tie-break changed and the templates as before, degree 0 was clean (0/3276) while degrees 1 and 2 still had 1848/5148 and 1068/2340 differing slots. The breadth-first templates of the old step 2, in ascending neighbour order, are not ι_K-equivariant, and an edge with ι_K-sign -1 was built in the *reverse* direction; a 1-chain with prescribed boundary in X is unique only up to the boundary of a 2-chain, so this was a second, genuine obstruction, invisible in the degree-0 witness. Tree routing removes it: in the tree $Y \subset X$ of step 2 a 1-chain with prescribed boundary is *unique*, so the edge chain depends only on its two endpoints and is Γ-equivariant, ι_X-equivariant and antisymmetric; degree 2 then follows, ∂_2 being injective on $C_2(X)$.

[verified computationally] **The verdict since v1.1.0.** $\iota_X \tau \iota_X = \tau$ on every stored slot of every table: 0 of 21528 slots over the 144 tables of the grid ($\ell \in \{2, 3, 5, 691\}$), 0 of 10764 on the binding subgrid, every comparison exact over $\mathbb{Z}$; Obligation 9.10(iv) is *gating* there. The legacy τ reproduces its 3600. Hence Proposition 9.4(3b) applies to the canonical τ: $\iota^\# \circ \widetilde{T} = \widetilde{T} \circ \iota^\#$ on the nose and $\iota^* \Theta_g = \Theta_g \iota^*$ as cochain maps for every ℓ — so at $\ell \in \{2, 3\}$, where (3c) is silent, $[\iota^*, T_{\Gamma g \Gamma}] = 0$ on H^q_{eq} for every operator whose τ-table lies in the grid ($p \in \{2, 3, 5, 7\}$, $\nu \leq 2$). Two limits are part of the statement: the identity is verified table by table by a run, not proved for every (p, ν) — the structural reason above is an argument, not a theorem of this document, and the obligation is the predicate a run discharges; and ι^* as a cochain map is not implemented (clauses (ii), (iii), (v)), so the commutation is a consequence of the proposition, not yet an observed matrix identity.

Two consequences of the tree routing. (a) **The canonical τ vanishes in degree 2:** $\tau(\partial \widetilde{t})$ is a closed loop of Y-paths, hence 0, and the unique fill is 0; so $\Theta^{(2)}_{\mathrm{flat}} = 0$. Admissible — (T1)–(T3) hold — and by Theorem 11.7(b) without effect on H^2_{eq} where $6 \in R^\times$; on the golden grid $\dim H^2_{\mathrm{eq}} = 0$ at every point, $\ell \in \{2, 3\}$ included. (b) **The re-baseline.** 2097 of the 14049 frozen reference values changed — every $\Theta^{(q)}_{\mathrm{flat}}$, its rank, every $\Theta^{(q)}_{\mathrm{eq}}$ and the three JSON records containing them — and *no* dimension, no $A^{(q)}$ characteristic polynomial, rank or trace, no decomposition invariant, no basis and no projector, at any point, $\ell \in \{2, 3\}$ included (Remark 11.9; the object-by-object record is `docs/matrix-engine/rebaseline-2026-09-16-iota-symmetric-tau.md`). That nothing moved on H^q_{eq} at $\ell \in \{2, 3\}$ is an observation, not a theorem: Theorem 11.7(b) promises it only for $6 \in R^\times$.

The $\pm$ decomposition, and what happens at $\ell = 2$

Write $H^q_{\mathrm{eq},\pm}$ for the ± 1 eigenspaces of ι^* on $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$, i.e. $H^q_{\mathrm{eq},\pm} := \ker(\iota^* \mp \mathrm{id})$, and likewise $C^q_{\mathrm{eq},\pm}$, $Z^q_{\mathrm{eq},\pm}$, $B^q_{\mathrm{eq},\pm}$; this is one symbol with one more index, not a new symbol ((N2)). Since $\iota^* = -\iota^*_{\mathrm{geo}}$ (Definition 9.1(iv), v1.2.0), these are the eigenspaces of the geometric involution with the labels exchanged, $H^q_{\mathrm{eq},\pm} = \ker(\iota^*_{\mathrm{geo}} \pm \mathrm{id})$; every $\pm$ statement of this document, the live specification and the validation output is in terms of ι^*. A $\pm$ *computed* before v1.2.0 — the v1.1.1 run record of Remark 9.9 and `docs/implementation/oq3_iota_sign_verification.md` — is the geometric one and must be negated; a $\pm$ *predicted* from owner decision OQ3 before v1.2.0 (the split of Obligation 9.10(v), family (C) of §21.4, the $H^1_\pm$ statements of `hecke_extensions.tex`) was written in the convention v1.2.0 adopted and holds as printed.

Proposition 9.6 (Eigenspace decomposition — and it needs $2 \in R^\times$)**.** *[proved here] Assume (H4) and (I1), let R be a commutative ring and fix a degree q.*

1. ***If*** *$2 \in R^\times$, the maps $\mathrm{pr}_\pm := \frac{1}{2}(\mathrm{id} \pm \iota^*)$ are complementary idempotents commuting with d, and*
$$H^q_{\mathrm{eq}}(K, \mathcal{R}; L) = H^q_{\mathrm{eq},+} \oplus H^q_{\mathrm{eq},-}, \qquad H^q_{\mathrm{eq},\pm} = \mathrm{im}\ \mathrm{pr}_\pm,$$
with the same decomposition of C^q_{eq}, Z^q_{eq} and B^q_{eq}. If in addition $6 \in R^\times$, both summands are stable under every operator of §10 by Proposition 9.4(3c), so $A^{(q)}$ is block diagonal in any basis adapted to the splitting — which is what halves the matrices.
2. ***If*** *$2 \notin R^\times$* ***the decomposition is false in general****, and what holds instead is a filtration. Put $\mathfrak{n} := \iota^* - \mathrm{id}$; then $\mathfrak{n}^2 = -2\mathfrak{n}$, so in characteristic 2, $\mathfrak{n}^2 = 0$ and*
$$0 \subseteq \mathrm{im}\,\mathfrak{n} \subseteq \ker \mathfrak{n} \subseteq H^q_{\mathrm{eq}},$$
a filtration by submodules stable under every operator for which Proposition 9.4(3b) or (3c) applies. Over $R = \mathbb{F}_2$ the module H^q_{eq} over $\mathbb{F}_2[\iota^*] \cong \mathbb{F}_2[x]/(x^2)$ ($x \mapsto \mathfrak{n}$) is a direct sum of a copies of the trivial module and b copies of the free module, with $b = \mathrm{rank}\,\mathfrak{n}$ and $a = \dim \ker \mathfrak{n} - b$; the two eigenspaces coincide, $H^q_{\mathrm{eq},+} = H^q_{\mathrm{eq},-} = \ker \mathfrak{n}$, and they exhaust H^q_{eq} if and only if $b = 0$, i.e. $\iota^* = \mathrm{id}$. ***At $\ell = 2$ no $\pm$ splitting of H_{tot} and no halving of any matrix may be claimed.***

Proof. (1) $(\iota^*)^2 = \mathrm{id}$ (Proposition 9.4(2)) gives $\mathrm{pr}_+\mathrm{pr}_- = \frac{1}{4}\big(\mathrm{id} - (\iota^*)^2\big) = 0$, $\mathrm{pr}_\pm^2 = \mathrm{pr}_\pm$ and $\mathrm{pr}_+ + \mathrm{pr}_- = \mathrm{id}$; each $\mathrm{pr}_\pm$ commutes with d because ι^* does, and $\mathrm{im}\ \mathrm{pr}_\pm = \ker(\iota^* \mp \mathrm{id})$ since $\iota^*\mathrm{pr}_\pm = \pm\mathrm{pr}_\pm$ and $\mathrm{pr}_\pm$ is the identity on that kernel. Stability under the operators is immediate from commutation.

(2) $\mathfrak{n}^2 = (\iota^*)^2 - 2\iota^* + \mathrm{id} = 2\mathrm{id} - 2\iota^* = -2\mathfrak{n}$, which vanishes in characteristic 2; hence $\mathrm{im}\,\mathfrak{n} \subseteq \ker \mathfrak{n}$, and both are submodules stable under anything commuting with ι^*. Over a field a square-zero endomorphism is, in a suitable basis, a direct sum of $b = \mathrm{rank}\,\mathfrak{n}$ Jordan blocks $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $a = \dim \ker \mathfrak{n} - b$ zero blocks; the block $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ is the free $\mathbb{F}_2[x]/(x^2)$-module of rank 1 and the zero block is the trivial module. Finally $-1 = +1$ in characteristic 2, so $\ker(\iota^* - \mathrm{id}) = \ker(\iota^* + \mathrm{id})$, and this is all of H^q_{eq} exactly when $\mathfrak{n} = 0$. □

Remark 9.7 (The three degeneracies at $\ell = 2$, and the odd-n case). Three distinct things happen at $\ell = 2$ and should not be conflated. First $\rho_n(\iota) = I$ by (15), and $-1 = +1$, so $\iota^* = \iota^*_{\mathrm{geo}}$ is the bare simplicial pullback and carries neither coefficient information nor the sign of Definition 9.1(iv) — the v1.2.0 normalisation is invisible at $\ell = 2$. Second the splitting fails (Proposition 9.6(2)). Third, Remark 7.7's odd-n vanishing argument fails as well — it used $2 \in R^\times$ — so $\ell = 2$ is precisely the regime in which $H^\bullet_{\mathrm{eq}}$ can be non-zero for odd n *and* has no ι-splitting. For $\ell \neq 2$ and n odd the question does not arise: $H^\bullet_{\mathrm{eq}} = 0$ there and both eigenspaces are 0. The additional degree bound and truncation policy that $\ell = 2$ forces on the *boundary* are construction item 14.24, whose mathematics is §14.5 and whose policy was decided by the

owner on 2026-09-16 ((OC-1a)); nothing in this chapter depends on either, ι^* being defined degreewise.

Remark 9.8 (ι, the contragredient, and the pairing). [verified computationally]^{r} for the mirror split of H^1_c (v1.6.0, the closing paragraph); the rest is commentary on tagged statements it cites. At coefficient level everything is proved and explicit. Proposition 8.6(4) with $\det \iota = -1$, and $\operatorname{adj} \iota = \operatorname{diag}(1,-1) = -\iota$ with $\rho_n(\iota)$ diagonal, give

$$\big\langle \rho_n(\iota)v,\ \rho_n^\vee(\iota)\varphi \big\rangle = (-1)^n \langle v, \varphi\rangle, \qquad \rho_n^\vee(\iota) = \rho_n(\operatorname{adj}\iota)^\top = (-1)^n \rho_n(\iota).$$

So ι acts on the contragredient module D_n of Definition 8.5 by the same diagonal matrix up to $(-1)^n$, the comparison map Φ_n of Lemma 8.8 intertwines the two actions (that lemma is stated for every $g \in \mathrm{M}_2(\mathbb{Z})$, ι included), and for n even ι is an isometry of the evaluation pairing on the nose. The sign of Definition 9.1(iv) does not disturb this: it applies to *both* coefficient modules, V and D_n, so at cochain level the two involutions carry one factor -1 each and $\langle \iota^* x, \iota^* y\rangle = (-1)(-1)(-1)^n\langle x,y\rangle = (-1)^n\langle x,y\rangle$ — the same statement as for the geometric pair, and every $\pm$ consequence below is invariant under negating both involutions at once.

What this does not establish. Remark 8.7 is explicit that Proposition 8.6 proves the coefficient-level half and says nothing about cohomology; the cohomological pairings are Claim 16.26 of §16, not available here. Neither does this chapter construct the action of ι on $H^\bullet(\partial)$ or on $H^\bullet_c$. That ι normalises $\Gamma_\infty = \langle T\rangle \times \langle -I\rangle$ is immediate from (14), so a boundary involution does exist; but its datum belongs to §14 and the cone to §15. *Contingent* on both, the consequence to expect is this. The coefficient factor $(-1)^n$ is not the whole sign: $z \mapsto -\bar z$ reverses the orientation of $\mathcal{H}$, so ι_X reverses the orientation of the surface and of the pair, and a cohomological pairing through a fundamental class of the pair — which is what the perfect pairings of Claim 16.26(1) are — picks up a further factor -1. The equivariance to expect is therefore $-(-1)^n$: if the pairing of Claim 16.26 satisfies $\langle \iota^* x, \iota^* y\rangle = -(-1)^n\langle x,y\rangle$ and $2 \in R^\times$, then for n even *equal* signs annihilate each other — $\langle x,y\rangle = \varepsilon^2\langle x,y\rangle = \langle \iota^* x, \iota^* y\rangle = -\langle x,y\rangle$ for $x \in H^t_{c,\varepsilon}$, $y \in H^{2-t}_{\mathrm{eq},\varepsilon}$ — so duality would restrict to a perfect pairing of each sign with the *opposite* one, as for modular symbols, where complex conjugation is orientation-reversing and H^1_+ is paired with H^1_- [standard: [Cre97, Ch. 2]]. The dimensions this document predicts agree with that and not with self-pairing: Obligation 9.10(v) puts $(\dim H^1_{\mathrm{eq},+}, \dim H^1_{\mathrm{eq},-}) = (d_{n+2}, d_{n+2}-1)$, and since $H^0(\partial; M_n)$ is one-dimensional and odd under the boundary involution (v1.2.0), H^1_c carries the mirror split $(d_{n+2}-1, d_{n+2})$, so a perfect pairing must match $+$ with $-$. (Until v1.3.2 this remark predicted the self-pairing, having omitted the orientation factor.) That sentence is a prediction about a pairing this document does not have — Claim 16.26 is a stub of §16 — and is asserted here as nothing more.

Its dimension half is no longer contingent (v1.6.0; hecke-extensions slot-4 WP-3, `docs/implementation/slot4/WP-3_report.md` findings 4.1 and 5). The two objects the paragraph above called unavailable now exist: ι^*_∂ on the boundary model and $\iota_c = \iota^* \oplus \iota^*_\partial$ on $C(r)$, with $r^q\iota^* = \iota^*_\partial r^q$ and $D\iota_c = \iota_c D$ *asserted* (`HeckeInvolutionValidation`, tier `standard`, run 2026-09-21: the 77-cell ι grid and a 15-cell cone grid $\ell \in \{2,3,5,7,691\} \times n \in \{0,2,10\}$). **Recorded outcome**: on the **eight** cells of that cone grid with $\ell \in \{3,5,7,691\}$ and $n \in \{2,10\}$ — among them $(n,\ell) = (10,691)$, where $\dim H^1_c = 3$ splits $(1,2)$ against $\dim H^1_{\mathrm{eq}} = 3$'s $(2,1)$; at $n=2$ the split is $(0,1)$ against H^1_{eq}'s $(1,0)$ and at $n=10$ it is $(1,2)$ against $(2,1)$ — H^1_c carries *exactly* the mirror of the split Obligation 9.10(v) puts on H^1_{eq}, which is what the paragraph above predicts and the only prediction this document makes about ι on the cone. (v1.6.1: v1.6.0 wrote "thirteen cells — $(10,691)$ and the twelve cells of the cone grid with $\ell \in \{3,5,7,691\}$", which counts $(10,691)$ twice, that cell being one of the twelve, and folds in the four at $n=0$, where $H^1_c = 0$ and Obligation 9.10(v) makes no prediction at all, its d_k being defined for even $k \geq 4$ and $n=0$ being $k=2$. At $\ell = 3$ that obligation's dimension statement is also outside its own hypothesis $\ell \geq 5$, so the four cells there compare a measured split with a measured split; they are kept in the eight because the mirror relation is what is being recorded, not the

obligation's prediction.) At $\ell = 2$ the recorded pairs are (k, k), Proposition 9.6(2) removing the split and not the involution, and they are correctly not asserted to sum to $\dim H_c^q$. Rule (A6): this is a record, it licenses nothing, no code path may branch on it, and [verified computationally] never upgrades to [proved here] (§1.1) — the *pairing* statement above stays contingent on Claim 16.26, and what would make ι_c commute with $A_c^{(q)}$ is stated, and left open, at the end of §15.3.

Remark 9.9 (The record field `iota_sign`: what is chosen and what is not). Owner decision OQ3 — proposed 2026-09-15 as an assertion, **resolved 2026-09-16 as a definition** — fixes the convention "the Eisenstein class lies in $H^1_{\mathrm{eq},+}$", recorded as `"iota_sign": "eisenstein_in_H1_plus"` in the extension record. Its status has to be stated exactly, because Definition 9.1 leaves *no* sign free.

- The only ambiguity in the *geometric* datum is the lift of $z \mapsto -\bar{z}$ to $\mathrm{GL}_2(\mathbb{Z})$ — ι versus $-\iota = \mathrm{diag}(1, -1)$, both inducing the same map on $\mathcal{H}$ because $-I$ acts trivially there. The two give $\rho_n(-\iota) = (-1)^n \rho_n(\iota)$, hence the same ι^*_{geo} for n even and opposite ι^*_{geo} for n odd. For $\ell \neq 2$ and n odd $H^\bullet_{\mathrm{eq}} = 0$ (Remark 7.7), so on every group this document computes with $6 \in R^\times$ the two lifts induce the same involution, and *no* choice of lift can move the Eisenstein class from one eigenspace to the other.
- **The sign was therefore an assertion about the geometric involution, and it was checked.** At v1.1.1 `HeckeIotaSignValidation` built ι^*_{geo} from Definition 9.1 alone and found the Eisenstein class in $\ker(\iota^*_{\mathrm{geo}} + \mathrm{id})$ on all 74 signed cells of a 77-cell grid, by two independent routes that agree wherever both apply, with $\iota^*_{\mathrm{geo}} = +\mathrm{id}$ on $H^0(\Gamma; \mathbb{F}_\ell)$ pinning the global sign (`docs/implementation/oq3_iota_sign_verification.md`; that report's $\pm$ are geometric). The mechanism is elementary: ι reverses the boundary circle, and the boundary class $[y^n]$ is $\rho_n(\iota)$-fixed, so $\iota^*_{\mathrm{geo}} = -1$ on $H^1(\partial)$ for $\ell > n$; restriction intertwines the two involutions and the Eisenstein line is boundary-visible.
- **Resolution (v1.2.0): the definition was negated, not the record.** Definition 9.1(iv) now reads $\iota^* := -\iota^*_{\mathrm{geo}}$, so `iota_sign = eisenstein_in_H1_plus` is **true by definition**, and the field's meaning changes accordingly: it *names the normalisation* $\iota^* = -\iota^*_{\mathrm{geo}}$ relative to the geometric involution. A third party reproduces any $\pm$ statement of a record by that one formula. The choice was an owner decision on a normative statement with an X bump ((V2)) and a changelog line, exactly as this remark had required; the alternatives and their costs — keeping the geometric involution and renaming its eigenspaces, or a degree-parity twist $(-1)^q$, which anticommutes with d — are weighed in `docs/implementation/oq3_resolution.md`. **It was not flipped at a call site**, which is the failure mode that a recorded convention field invites, and no pipeline value moved: ι^* occurs in no golden object.
- What is *not* chosen: that the geometric sign is uniform across cells. "+ is the side of the Eisenstein classes" is well-posed only because every cell gives the same geometric sign, and that is a fact about the model, not a convention. Obligation 9.10(v) is now the gate on it: a cell whose Eisenstein class lies in $H^1_{\mathrm{eq},-}$ would mean the definition is ill-posed, and the repair would again be a normative one, never a per-cell sign.

Obligation 9.10 (Sign convention and runtime assertions for ι)**.** Assume (H4) and (I1). An implementation of ι emits the signed block permutation of Definition 9.1(iv) in every degree, in the flat coordinates (8), and must discharge the following. Each clause names the stored data it reads, the predicate, the ring it is evaluated over, and the action on failure.

(i) **Simplicial automorphism.** The stored vertex permutation is an involution carrying K to K, and the recomputed orientation sign of every simplex agrees with Lemma 9.2's table. Over $\mathbb{Z}$; a failure **aborts**.

(ii) **Local-system automorphism.** For every row $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma)$ of $\mathcal{R}$, the triple $(\iota_K \sigma_{\mathrm{lhs}}, \iota_K \sigma_{\mathrm{rhs}}, \iota\gamma\iota)$ is the inverse of a row of $\mathcal{R}$ — compared as exact $\mathrm{M}_2(\mathbb{Z})$ data

together with stored simplex indices and their signs — and $\rho_n(\iota)\rho_n(\gamma) = \rho_n(\iota\gamma\iota)\rho_n(\iota)$ entrywise over R. A failure **aborts** (Proposition 9.4(1)).

The two halves, and the complex that has only one of them (v1.7.0; `docs/implementation/slot4/WP-3_re` finding 3, the rider it asked for). The clause has an *exact* half — the triple compared as $\mathrm{M}_2(\mathbb{Z})$ data — and a *fibre* half — the identity in ρ_n. The fibre half names data every local system carries and is asserted **unconditionally, always**: it is what makes ι_C preserve $C^\bullet_{\mathrm{eq}}$, and no implementation is excused from it. The exact half names a row's γ, which a local system presented by its monodromy matrices alone need not store — the shipped $\mathcal{R}$ does not, the exact rows living in the model-specific datum — and it is asserted on **every row for which the complex stores that** γ. This is a statement about the data the clause reads, not a licence: an implementation whose complex carries the exact rows **may not** skip the exact half, and every implementation **records** which rows it ran it on, so that a record says whether the exact comparison happened rather than leaving a reader to assume it. A complex carrying no exact group data discharges the fibre half and records the exact half as not evaluated; it does not report clause (ii) as passed in the exact sense. Rule (A6) governs the record as usual.

(iii) **Cochain involution.** $\iota^*\iota^* = \mathrm{id}$ and $\iota^* d = d\iota^*$ as matrices over R in every degree, and ι^* carries the stored bases of C^q_{eq}, Z^q_{eq}, B^q_{eq} into their own spans. A failure **aborts** (Proposition 9.4(2)).

(iv) **τ-symmetry — a diagnostic, never branched on.** Record whether $\iota_X\,\tau\,\iota_X = \tau$ holds on every stored table entry, compared over $\mathbb{Z}$. This is the hypothesis of Proposition 9.4(3b) and the *only* route to $[\iota^*, T_{\Gamma g\Gamma}] = 0$ when $6 \notin R^\times$. It is **recorded**, and **a true value licenses nothing beyond (3b)** — in particular it may not be used to skip clause (v) (rule (A6); Remark 9.5). On an arbitrary admissible τ a false value is not a failure. **On the canonical τ of Algorithm 18.3 it is, since v1.1.0**: steps (1b) and 2 of that algorithm are specified so that the identity holds, so a differing slot is a defect of the construction and the validation fails. This is not a branch on the diagnostic — nothing downstream reads the verdict — but a regression gate on the construction. [verified computationally] **Discharged — negatively at v0.13.0 on the then-canonical τ, positively since v1.1.0 on the canonical τ**, by `HeckeIotaTauSymmetryValidation` (runs of 2026-09-16): on the canonical τ the recorded value is *true* on every stored slot of every table of the grid (0 of 21528 slots over 144 tables, $\ell \in \{2, 3, 5, 691\}$), every comparison exact over $\mathbb{Z}$; the pre-v1.1.0 τ, kept as the legacy policy and re-run alongside, differs on 3600 of 10764 binding slots (Remark 9.5 for the grid, the witness and the mechanism). The clause remains in force as the predicate a *re-run* on any other τ, on any other vertex policy or on a refined triangulation must evaluate and record. The validation fails on a differing slot of the canonical τ, on clause (i), or if τ^ι is not an admissible datum (Proposition 9.4(3a)).

(v) **Commutation, the sign, and the $\pm$ dimensions.** For $6 \in R^\times$: $\iota^* A^{(q)} = A^{(q)}\iota^*$ over R in every degree and for every computed operator; a failure **aborts** (Proposition 9.4(3c)). **The sign — a definition-consistency oracle, aborting since v1.2.0.** Definition 9.1(iv) defines $\iota^* = -\iota^*_{\mathrm{geo}}$ so that "+" is the eigenspace of the Eisenstein classes; the implementation must exhibit that this is well posed: on every cell (n, ℓ) with $\ell \geq 5$, n even, on which the Eisenstein class of Obligation 20.7 can be located — as the joint T_p-eigenline of eigenvalue $1 + p^{n+1}$ where that eigenspace is a line, or as the class visible under restriction to the cusp where $\operatorname{rank} r^1 = 1$ — it lies in $H^1_{\mathrm{eq},+}$, the two routes agree wherever both apply, and $\iota^* = -\iota^*_{\mathrm{geo}}$ entrywise in every degree. The degree-0 instance is exact and cheap and is what pins the global sign: ι^* on $H^0_{\mathrm{eq}} = V_R^\Gamma$ is $-\rho_n(\iota)$ restricted to the invariants, which for $n = 0$ is $-\mathrm{id}$ (the geometric ι^*_{geo}, which passes every other clause, gives $+\mathrm{id}$ there). A failure **aborts**, and what it would mean is not a wrong sign at a call site but a *non-uniform* geometric sign across cells, i.e. Definition 9.1(iv)'s "+" ill-posed — a defect of the definition, to be repaired normatively (Remark 9.9). **The dimensions**

— **recorded rather than aborting:** for $\ell \geq 5$ off the Dickson locus of Remark 20.8 and n even,

$$\dim H^1_{\mathrm{eq},+} = d_{n+2}, \qquad \dim H^1_{\mathrm{eq},-} = d_{n+2} - 1,$$

where for even $k \geq 4$ the integer $d_k = \lfloor k/12 \rfloor + 1$ if $k \not\equiv 2 \pmod{12}$ and $d_k = \lfloor k/12 \rfloor$ otherwise is the dimension of the space of level-1 modular forms of weight k, its subspace of cusp forms having dimension $d_k - 1$ [standard: [Ser73, Ch. VII, §3]]. Given the gated Eisenstein sign this is the statement that the cuspidal part $\ker r^1$ splits evenly, the mod-ℓ Eichler–Shimura prediction discussed after this obligation, which is why it records.
[verified computationally] **Checked 2026-09-16 (v1.2.0): every predicate of this clause holds.** `HeckeIotaSignValidation` (tier `standard`, 1.4 s, 4 104 267 exact assertions, 0 gating failures) builds both ι^*_{geo} and ι^* from Definition 9.1 alone — the live ρ_n of $\mathrm{diag}(-1, 1)$, ι_K and Lemma 9.2's signs re-derived from the shipped exact coordinates, the global sign of (16) — and evaluates this clause over $n \in \{2, 4, \ldots, 30\}$ even (11 values) and $\ell \in \{5, 7, 11, 13, 101, 134217757, 2^{31} - 1\}$: 77 cells. Commutation $\iota^* A^{(q)} = A^{(q)} \iota^*$ for $p \in \{2, 3\}$ in every degree, $(\iota^*)^2 = \mathrm{id}$ on H^q_{eq}, $\iota^* = -\iota^*_{\mathrm{geo}}$ on cochains and on H^q_{eq}, and the degree-0 anchor $\iota^* = -1$ on $H^0_{\mathrm{eq}} = H^0(\Gamma; \mathbb{F}_\ell)$ at $n = 0$ for $\ell \in \{5, 13, 134217757\}$; the two lifts ι and $-\iota$ of Remark 9.9 induce the same involution on all 77 cells, as n even predicts. On every one of the 74 cells on which a sign is defined (74 by the T_p-eigenline, 53 by the cusp, 53 by both, agreeing) the Eisenstein class lies in $H^1_{\mathrm{eq},+}$; the three cells without a sign are the Dickson cells $(20, 5)$, $(24, 5)$, $(30, 5)$, where the joint eigenspace has dimension 2, 2, 3 (Fermat: $1 + p^{n+1} \equiv 1 + p^{n+1-(\ell-1)}$) and $\mathrm{rank}\, r^1 \neq 1$. The recorded split is $(d_{n+2}, d_{n+2} - 1)$ on all 66 off-Dickson cells, with $\ker r^1$ splitting evenly wherever $\ell > n$. *Under the geometric involution every sign is the opposite*: Eisenstein in $\ker(\iota^*_{\mathrm{geo}} + \mathrm{id})$ on 74/74, the mirror split $(d_{n+2} - 1, d_{n+2})$ on 66/66, $\iota^*_{\mathrm{geo}} = +1$ on $H^0(\Gamma; \mathbb{F}_\ell)$ — the v1.1.1 finding that motivated the decision, recorded by the same run (`IOTA_SIGN eisenstein=plus (geometric=minus) ... h0_anchor=-1 (geometric=+1)`). Evidence: `docs/implementation/oq3_resolution.md` (the decision and its cost) and `docs/implementation/oq3_iota_sign_verification.md` (the 77-cell table, whose $\pm$ are geometric).

The classical statement behind the H^1_{eq} oracle is the Eichler–Shimura decomposition of $H^1(\Gamma; \mathrm{Sym}^n \mathbb{C}^2)$ into a space of weight-$(n + 2)$ modular forms and a conjugate space of cusp forms [standard: [Shi71, Ch. 8]; [Cre97, Ch. 2] for the involution and the $\pm$ parts in the modular-symbol realisation]: each cuspidal eigenform contributes a 2-dimensional ι^*-stable subspace on which ι^* has eigenvalues $+1$ and -1, and the Eisenstein line contributes the one further dimension, which Definition 9.1(iv) places in $H^1_{\mathrm{eq},+}$ by construction. That the mod-ℓ reduction inherits the two dimensions is a *prediction*, not a theorem of this document ([conjectural] as a citation: the Dickson locus of Remark 20.8 is exactly where reduction is known to misbehave) — which is why (v)'s dimension statement records rather than aborts. [verified computationally] The total it must reproduce is already observable without ι: at $\ell = 134217757$ the shipped pipeline gives $\dim H^1_{\mathrm{eq}} = 1, 1, 3, 5, 7$ for $n = 2, 4, 10, 22, 34$ and $\dim H^1_{\mathrm{eq}} = 0$ for $n = 0$, matching $2d_{n+2} - 1$ (respectively 0) in every case (runs of 2026-09-15, `--bench cohom <n> 134217757`).

Code map. All five clauses are implemented, by two validations. `Hecke/HeckeIotaTauSymmetryVali` (added 2026-09-16, registered as `HeckeIotaTauSymmetryValidation`, tier `fast`, 0.5 s, 64699 exact-integer assertions over 432 τ-tables — the canonical, the legacy and the `FixedV6` policy at each of 144 grid points) builds ι_K and ι_X from Definition 9.1(i),(iii) alone — the vertex permutation and the sign of every simplex are *recomputed*, not read from a stored table — and discharges **Obligation 9.10(i)**, which passes, and **Obligation 9.10(iv)**, whose recorded verdict is positive on the canonical τ — gating there since v1.1.0 — and negative on the legacy τ (Remark 9.5). It also verifies Lemma 9.2's unstated hypothesis $\mathrm{Re}\, v_1 = -\mathrm{Re}\, v_2$ exactly in the shipped coordinates ($v_1 = -\frac{1}{6} + 2i$, $v_2 = \frac{1}{6} + 2i$), and

certifies, as a gate on its own construction, that τ^ι satisfies (T1)–(T3) of Definition 11.1 — i.e. Proposition 9.4(3a) on the shipped tables. `Hecke/HeckeIotaSignValidation.{h,cpp}` (added 2026-09-16, registered as `HeckeIotaSignValidation`, tier `standard`, 1.4 s, 4 104 267 assertions, 0 gating failures) covers the remaining three: it builds both ι^*_{geo} and $\iota^* = -\iota^*_{\mathrm{geo}}$ from Definition 9.1 alone as explicit $(N_q d_V) \times (N_q d_V)$ matrices over R — the block $\pm\varepsilon_\sigma \rho_n(\iota)$ at block position $(\sigma, |\iota_K \sigma|)$ in the flat coordinates (8), the two asserted to be negatives of each other — and discharges **Obligation 9.10(ii)** (all 31 relation rows of the model, the 23 central $-I$ rows included, entrywise over R), **Obligation 9.10(iii)** ($\iota^*\iota^* = \mathrm{id}$ and $\iota^* d = d\iota^*$ as full matrices in every degree, and the stored C^q_{eq}, Z^q_{eq}, B^q_{eq} carried into their own spans) and **Obligation 9.10(v)** — commutation, the gating Eisenstein sign ("+" on every signed cell, by both routes) and the degree-0 anchor $\iota^* = -1$, all passing, with the geometric sign recorded alongside; see that clause's [verified computationally] record and Remark 9.9. It reaches $H^\bullet_{\mathrm{eq}}$ through the stored P_q and H_{eq} of Definition 6.3, and the cusp through the shipped `EquivariantCochainComplexMorphismOnK` restriction, on which the same construction is re-run and (16)'s naturality $r\,\iota^* = \iota^*_\partial\, r$ is asserted. **Code map (v1.6.2, the slot-4 close; this replaces the 2026-09-16 census, which found no `HeckeInvolution` translation unit and is now spent).** Three of the 108 registered validations (v1.7.4 count; 106 at v1.7.3, 100 at v1.7.2, 94 at v1.6.2, 89 at v1.5.2, 80 at v1.4.2, 77 at v1.3.3, 68 when that census was taken) touch this chapter: `HeckeIotaTauSymmetryValidation` and `HeckeIotaSignValidation` as before, and `HeckeInvolutionValidation` since hecke-extensions slot-4 WP-3. The *production* objects are `Hecke/HeckeInvolution` — the morphism specified below, whose `compute()` discharges Obligation 9.10(i),(ii),(iii) *blockwise and exactly over* $\mathbb{Z}$ and aborts on failure — and `HeckeExt/HeckeSignSplit`, which carries clause (v) and the $\pm$ decomposition. The acceptance is *entrywise against an independent construction*: WP-3 exports `HeckeIotaSignValidation`'s builders as `hecke::iota_oracle` value-preservingly, and the production object reproduces that validation's reference line field for field. Remark 9.8's closing paragraph and the closing paragraph of §15.3 record the same. The consequence for the rest of the document is unchanged: every [proved here] tag above other than Proposition 9.4(1),(2),(3a) and Lemma 9.2 rests on its printed proof alone. The *specification* recorded here for a *production* ι^* — as opposed to the validation's own matrix, which exists to be checked and not to be depended on — stands unchanged as a *specification* (E3 of the extension programme, gated by amendment C-A1). Since v1.6.2 it has been *built* to this specification, so what follows is what the shipped `Hecke/HeckeInvolution` satisfies and is no longer a description of something absent: ι^* is an `EquivariantCochainComplexMorphismOnK` built through the general constructor, its `FiberBlockTable` holding exactly one `PerSimplexBlock` per simplex — source simplex $|\iota_K \sigma|$, block $\mp\rho_n(\iota)$, the sign of Lemma 9.2 *negated* per Definition 9.1(iv) and folded into the matrix — which is the degenerate single-entry case of the per-simplex sparse block rows that the repair of Remark 12.9 introduced, so no new datum shape is required. That class's documented contravariant naming applies: its `domain` is the *target* of the induced map on $H^\bullet$. Clause (iv) of Obligation 9.10 reads `hecke::HeckeTauTable`'s stored chains and compares them with their ι_X-images, so it needs ι on the simplices of X carried by `hecke::SL2ZTessellationTables`, not only on those of K.

Formalization note. Definition 9.1(i)–(ii) is a 2×2 integer matrix and a closed-form diagonal matrix, and (iii)–(iv) are a vertex permutation with a computable sign; so Lemma 9.2, Proposition 9.4(1),(2) and clauses (i)–(iii) of Obligation 9.10 are `decidable` predicates on finite data and are the natural first targets of this chapter. Proposition 9.4(3a) is a finite calculation with the τ-table and needs no hypothesis on ℓ; (3c) is the only statement here that invokes an abstract input — Theorem 11.7(b), and through it the fundamental lemma of relative homological algebra — and it inherits that theorem's hypothesis $6 \in R^\times$ verbatim. Proposition 9.6(2)'s classification is the Jordan form of a square-zero endomorphism over a field. The statement to

be careful with is (1): its $\frac{1}{2}$ must be carried as an Invertible $(2 : R)$ instance argument and not as a field hypothesis, since $R = \mathbb{Z}/\ell^e$ is in scope.

Part III — The integral Hecke datum

10 The integral Hecke datum for $\mathrm{GL}(2)/\mathrm{SL}_2(\mathbb{Z})$

What this chapter fixes. The group-theoretic datum out of which every Hecke operator of this document is built: the element $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$ with $N = p^\nu$, the two congruence subgroups it cuts out of Γ, the coset spaces of the correspondence with the *explicit* representatives, class map and cocycle the implementation uses, and the operators themselves — T_p, the **full** T_{p^ν}, the **primitive** $T_{\Delta_{p^\nu}^{\mathrm{prim}}}$, the composite T_N, the central $T(p,p)$ and the bad-prime U_ℓ — with the relations among them. Nothing here says what an operator *is* as a map: that is §11. This chapter says which double cosets are fed to that construction, with which auxiliary data, and what is asserted of the output.

Standing hypotheses. (H1)–(H4) of §7. Per rule (A1) each is repeated in every statement that uses it; in particular (H3) ($6 \in R^\times$, i.e. $\ell \geq 5$ for R of characteristic ℓ) is *not* in force by default and is written out wherever it is needed — which is every statement of §10.4 and no statement of §§10.2–10.3.

Notation used throughout. The coset index is $j \in \{1, \dots, m\}$ (never i: §2.7, C15) and the prime-power exponent is ν (never r: C11). The composite-index operator is T_N with $N \in \mathbb{Z}_{>0}$, never T_n (Appendix A, entry A.16).

10.1 Primitive and full: the two operators at a prime power

The single most consequential naming decision of this chapter is made first, because the superseded contract stated two numbered prescriptions that conflated the two objects and were therefore both wrong by $p^n I$ (Appendix A, entry A.1).

> **CONTRACT.** At a prime power p^ν with $\nu \geq 2$ there are **two different operators**, and this document names them differently and never abbreviates either away:
>
> - the **primitive double-coset operator** $T_{\Delta_{p^\nu}^{\mathrm{prim}}} := T_{\Gamma \mathrm{diag}(1,p^\nu)\Gamma}$, attached to the *single* double coset $\Delta_{p^\nu}^{\mathrm{prim}} = \Gamma\,\mathrm{diag}(1,p^\nu)\,\Gamma$ of the *primitive* determinant-p^ν matrices, with $m = p^\nu + p^{\nu-1}$ cosets;
> - the **full operator** T_{p^ν} of (18), the sum over *all* double cosets of determinant p^ν, with $\sigma_1(p^\nu) = 1 + p + \cdots + p^\nu$ right cosets. This is Shimura's $T(p^\nu)$ (his $T(1,p^\nu)$ is the primitive $T_{\Delta_{p^\nu}^{\mathrm{prim}}}$), it is the operator that satisfies the recursion of Theorem 10.8, and **"T_{p^ν}" with no qualifier always means this one.**
>
> They agree at $\nu \in \{0,1\}$ and differ for $\nu \geq 2$, by (19); at $\nu = 2$ the difference is exactly the central term $T(p,p) = p^n I$:
>
> $$T_{p^2} \;=\; T_{\Delta_{p^2}^{\mathrm{prim}}} \;+\; p^n I.$$
>
> Where both appear in one line the emphatic forms $T_{p^\nu}^{\mathrm{full}} := T_{p^\nu}$ and $T_{p^\nu}^{\mathrm{prim}} := T_{\Delta_{p^\nu}^{\mathrm{prim}}}$ may be used. The implementation's comments write T'_{p^r} for the primitive operator;

this document does not, because a prime is a decoration and decorations do not make new symbols (§2.1, (N2)).

Which entry point of the implementation emits which object is tabulated in Appendix A, entry A.1, together with the evidence; the code map at the end of this chapter repeats the table in short form. The arithmetic relations that distinguish them are Theorem 10.8 and its caveat.

10.2 The correspondence datum

Fix a prime p, an exponent $\nu \geq 1$, and put

$$N := p^\nu, \qquad g := \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix} \in \Delta,$$

Δ the monoid of Definition 8.3. Matrices act on column vectors from the left, so Γh is a *right* coset and $h\Gamma$ a *left* coset.

Proposition 10.1 (Coset decompositions)**.** *[proved here] Let p be a prime, $\nu \geq 1$, $N = p^\nu$, $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, and $m := N + p^{\nu-1} = p^\nu + p^{\nu-1}$. Then:*

1. ***Right cosets.*** *$\Gamma g \Gamma$ is the set of all $\delta \in \mathrm{M}_2(\mathbb{Z})$ with $\det \delta = N$ and* gcd *of the entries of δ equal to 1 (*primitive*), and*

$$\Gamma g \Gamma \;=\; \bigsqcup_{d \mid N} \; \bigsqcup_{\substack{0 \leq b < d \\ \gcd(N/d,\, b,\, d) = 1}} \Gamma \begin{pmatrix} N/d & b \\ 0 & d \end{pmatrix},$$

   a disjoint union of exactly m right cosets. For $\nu = 1$ this is $\bigsqcup_{0 \leq b < p} \Gamma \beta_b \sqcup \Gamma \beta_\infty$ with $\beta_b := \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$ and $\beta_\infty := \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$. [formally verified: [GT, `prop:cosets`]] (the first sentence, the count m and the list at $\nu = 1$; not the list for general ν)
2. ***Left cosets.*** *$\Gamma g \Gamma = \bigsqcup \delta \Gamma$ over the transposes of the matrices in (1); there are again m of them, and for $\nu = 1$ they are $\begin{pmatrix} 1 & 0 \\ b & p \end{pmatrix} \Gamma$ $(0 \leq b < p)$ and $\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \Gamma$. [formally verified: [GT, `prop:cosets`]]*
3. ***The two congruence subgroups.***

$$\Gamma' := \Gamma \cap g^{-1} \Gamma g = \Gamma^0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma : \; b \equiv 0 \;\; (N) \right\},$$

$$\Gamma'' := \Gamma \cap g \Gamma g^{-1} = \Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma : \; c \equiv 0 \;\; (N) \right\},$$

   both of index m in Γ, and the conjugation $\mathrm{conj}_g \colon \Gamma' \to \Gamma''$, $h \mapsto ghg^{-1}$, is a group isomorphism with integral values in both directions: $g \begin{pmatrix} a & Nb \\ c & d \end{pmatrix} g^{-1} = \begin{pmatrix} a & b \\ Nc & d \end{pmatrix}$. [formally verified: [GT, `prop:cosets-3-iso`]] (since v1.18.0 including the words "group isomorphism")
4. ***The coset spaces.*** *The bottom-row map $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto (c : d)$ induces a bijection*

$$\Gamma'' \backslash \Gamma \;\xrightarrow{\sim}\; \mathbb{P}^1(\mathbb{Z}/N) := \big\{ (c : d) \in (\mathbb{Z}/N)^2 : \gcd(c, d, N) = 1 \big\} / (\mathbb{Z}/N)^\times,$$

$$\big|\mathbb{P}^1(\mathbb{Z}/N)\big| = m.$$

The right Γ-set $\Gamma\backslash\Gamma g\Gamma$ (cosets $\Gamma\delta$, action $\Gamma\delta\cdot\gamma=\Gamma\delta\gamma$) is isomorphic to $\Gamma'\backslash\Gamma$ via $\Gamma'\gamma\mapsto\Gamma g\gamma$, and the left Γ-set $\Gamma g\Gamma/\Gamma$ is isomorphic to Γ/Γ'' via $\gamma\Gamma''\mapsto\gamma g\Gamma$. [formally verified: [GT, `prop:cosets`]] (the bijection and $\left|\mathbb{P}^1(\mathbb{Z}/N)\right|=m$; not the two isomorphisms of Γ-sets)

5. ***Transporters and coset elements.*** *If $s_1,\dots,s_m\in\Gamma$ represent $\Gamma''\backslash\Gamma$, then*

$$\delta_j:=s_j^{-1}g,\qquad j=1,\dots,m,$$

represent the left cosets: $\Gamma g\Gamma=\bigsqcup_{j=1}^m\delta_j\Gamma$. Every δ_j is integral of determinant N, and no δ_j is ever inverted anywhere in this document. [formally verified: [GT, `prop:cosets-5-any`]] (since v1.18.0 for any transversal, not only the canonical s_j of Definition 10.2) [formally verified: [GT, `lem:m4-cosetdata`]] (since v1.20.0: $\Gamma''=\Gamma\cap g\Gamma g^{-1}$ has finite index, so such s_j exist, for every $g\in\Delta$)

Proof. (1) *Hermite reduction.* Every $\delta\in\mathrm{M}_2(\mathbb{Z})$ with $\det\delta=N>0$ lies in $\Gamma\begin{pmatrix}a&b\\0&d\end{pmatrix}$ for exactly one triple (a,b,d) with $a,d>0$, $ad=N$, $0\le b<d$. Existence: let $(\alpha,\gamma_{21})^\top$ be the first column of δ — not $(0,0)^\top$, since $\det\delta\neq0$ — and $e:=\gcd(\alpha,\gamma_{21})>0$; the row $(-\gamma_{21}/e,\ \alpha/e)$ has coprime entries, so it is the second row of some $\gamma_0\in\Gamma$, and $\gamma_0\delta$ has lower-left entry 0; replacing γ_0 by $-\gamma_0$ if necessary (note $-I\in\Gamma$) makes both diagonal entries positive, and multiplying on the left by $T^{-\lfloor b/d\rfloor}$ moves b into $[0,d)$ without touching a or d. Uniqueness: if two such normal forms are Γ-equivalent, the connecting element is upper triangular with positive diagonal and determinant 1, hence equals T^k, which forces $a=a'$, $d=d'$ and $b\equiv b'\ (d)$, hence $b=b'$. So the listed matrices are pairwise inequivalent and the $\Gamma\begin{pmatrix}a&b\\0&d\end{pmatrix}$ over all admissible triples exhaust $\{\det=N\}$.

Which of them are primitive. The content of a matrix is Γ-biinvariant, so δ is primitive iff its Hermite form is, i.e. iff $\gcd(a,b,d)=\gcd(N/d,b,d)=1$; and the primitive matrices of determinant N form the single double coset $\Gamma\,\mathrm{diag}(1,N)\,\Gamma$, because the elementary divisors $d_1\mid d_2$ of δ — which classify $\Gamma\delta\Gamma$, and which satisfy $d_1=\mathrm{content}(\delta)$, $d_1d_2=\det\delta$ — are then 1 and N [standard: [Shi71, §3.2, pp. 55–57, Lemmas 3.11–3.13]: the Hecke ring of (Γ,Δ), where double cosets are parametrised by elementary divisors; locator verified (R20-L; v1.21.0-pre.1, in place of "Ch. 3, §3.1")]. The count, with $N=p^\nu$ and $d=p^\beta$: for $\beta=0$ the only b is 0 and the condition reads $\gcd(p^\nu,0,1)=1$, giving 1; for $\beta=\nu$ the condition is vacuous, giving p^ν; for $0<\beta<\nu$ it reads $p\nmid b$, giving $p^\beta-p^{\beta-1}$. The total is $1+\sum_{\beta=1}^{\nu-1}(p^\beta-p^{\beta-1})+p^\nu=p^\nu+p^{\nu-1}=m$.

(2) Transposition is an anti-automorphism of $\mathrm{M}_2(\mathbb{Z})$ with $\Gamma^\top=\Gamma$ and $g^\top=g$, so it carries $\Gamma g\Gamma$ onto itself and exchanges left with right cosets.

(3) $g\begin{pmatrix}a&b\\c&d\end{pmatrix}g^{-1}=\begin{pmatrix}a&b/N\\Nc&d\end{pmatrix}$, which is integral iff $N\mid b$ and then has lower-left entry divisible by N; this gives $\Gamma\cap g\Gamma g^{-1}=\Gamma_0(N)$ and simultaneously the stated formula for conj_g. Likewise $g^{-1}\begin{pmatrix}a&b\\c&d\end{pmatrix}g=\begin{pmatrix}a&Nb\\c/N&d\end{pmatrix}$ gives $\Gamma\cap g^{-1}\Gamma g=\Gamma^0(N)$. Both are groups, conj_g is a bijection with inverse $\mathrm{conj}_{g^{-1}}$, and the index statement follows from (4).

(4) The map is well defined into $\mathbb{P}^1(\mathbb{Z}/N)$: $ad-bc=1$ forces $\gcd(c,d,N)=1$. It is constant on Γ''-orbits: for $h=\begin{pmatrix}\alpha&\beta\\N\gamma_0&\epsilon\end{pmatrix}\in\Gamma_0(N)$ the bottom row of $h\gamma$ is $(N\gamma_0a+\epsilon c,\ N\gamma_0b+\epsilon d)\equiv\epsilon\,(c,d)\ (N)$, and $\alpha\epsilon\equiv1\ (N)$ makes ϵ a unit. It is injective on orbits: if $\gamma_1,\gamma_2\in\Gamma$ have bottom rows $(c_1,d_1)\equiv u\,(c_2,d_2)$ with u a unit, then the bottom row of $\gamma_1\gamma_2^{-1}$ is $(c_1d_2-d_1c_2,\ -c_1b_2+d_1a_2)$, whose first entry is $\equiv u(c_2d_2-d_2c_2)=0\ (N)$, so $\gamma_1\gamma_2^{-1}\in\Gamma_0(N)$. It is surjective, with the count m: a class $(c:d)$ with $\gcd(c,d,p)=1$ has either d a unit — normalise to $(k:1)$, $k\in\mathbb{Z}/N$, giving N classes — or $p\mid d$, in which case c is a unit and one normalises to $(1:pc_0)$, $c_0\in\mathbb{Z}/p^{\nu-1}$, giving $p^{\nu-1}$ classes; the two families are disjoint, because $(k:1)=(1:pc_0)$ would

force the unit u with $u \cdot 1 = pc_0$, and pc_0 is not a unit. Each class is hit by the closed-form lift of Definition 10.2, which is in Γ by inspection. The two Γ-set isomorphisms are the orbit–stabiliser bijections for the transitive actions on $\Gamma\backslash\Gamma g\Gamma$ and $\Gamma g\Gamma/\Gamma$, whose stabilisers of Γg and of $g\Gamma$ are Γ' and Γ'' by (3).

(5) The map $\Gamma'' s_j \mapsto s_j^{-1} g\Gamma$ is the composite of the bijection $\Gamma''\backslash\Gamma \xrightarrow{\sim} \Gamma/\Gamma''$, $\Gamma''\gamma \mapsto \gamma^{-1}\Gamma''$, with the bijection $\Gamma/\Gamma'' \xrightarrow{\sim} \Gamma g\Gamma/\Gamma$ of (4); it is therefore a bijection, so the $\delta_j = s_j^{-1} g$ represent the m left cosets. (Directly: if $s' = h\, s_j$ with $h \in \Gamma''$ then $s'^{-1} g = s_j^{-1} g\, (g^{-1} h^{-1} g)$ with $g^{-1}h^{-1}g \in \Gamma' \subseteq \Gamma$, so the left coset does not depend on the representative.) Finally $\det \delta_j = \det g = N$, and δ_j is integral because $s_j^{-1} \in \Gamma$. □

Definition 10.2 (The canonical representatives, the class map and the cocycle)**.** [formally verified: [GT, `def:cosetreps`]] (in [GT] with a 0-based index, the class map an index) Let p, ν, N, m be as in Proposition 10.1. The *canonical transporters* are, in the 1-based index j that this document and the implementation share,

$$s_j := \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \quad \text{with } k = j-1, \quad 1 \le j \le N \qquad (\text{class } (k:1)),$$

$$s_{N+1+c} := \begin{pmatrix} 0 & -1 \\ 1 & pc \end{pmatrix}, \quad 0 \le c < p^{\nu-1} \qquad (\text{class } (1:pc)),$$

each of determinant 1 and with bottom row its assigned class; they exhaust $\Gamma''\backslash\Gamma$ by Proposition 10.1(4). At $\nu = 1$ this is $s_1 = I$, $s_{1+k} = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$ $(1 \le k \le p-1)$, $s_{p+1} = S$. The *coset elements* are $\delta_j := s_j^{-1} g$. The *class map* is

$$\mathrm{cls}\colon \Gamma \to \mathbb{P}^1(\mathbb{Z}/N), \qquad \mathrm{cls} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (c:d),$$

and $j \cdot \gamma$ denotes the index with $\mathrm{cls}(s_j\gamma) = \mathrm{cls}(s_{j\cdot\gamma})$, i.e. the right action of Γ on $\{1, \dots, m\} \cong \Gamma''\backslash\Gamma$; concretely $(c,d) \cdot \gamma$ is the *row* vector product $(c,d)\gamma$ reduced mod N. The *coset cocycle* is

$$h_j(\gamma) \;:=\; s_j\, \gamma\, s_{j\cdot\gamma}^{-1} \;\in\; \Gamma'' = \Gamma_0(N), \qquad \text{equivalently} \qquad s_j\gamma = h_j(\gamma)\, s_{j\cdot\gamma}. \tag{17}$$

Finally, the *upper-triangular transporters* t_j representing $\Gamma^0(N)\backslash\Gamma$ (class map = the *top* row $(a:b)$) are $t_j := \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} = T^k$ with $k = j-1$ for $1 \le j \le N$, and $t_{N+1+c} := \begin{pmatrix} -pc & -1 \\ 1 & 0 \end{pmatrix}$ for $0 \le c < p^{\nu-1}$; at $\nu = 1$, $t_{p+1} = S$. They are used only in §12 (Proposition 12.6).

> **CONTRACT.** Definition 10.2 is **binding as written**, not up to an equally valid choice. The general recipe "lift $(c:d)$ to Γ by an extended-Euclid Bézout solve" does *not* determine the matrix: at $(c,d) = (1,1)$ the textbook iterative extended gcd returns $\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$, which is a correct lift of the class and is *not* $s_2 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. The superseded contract pinned only the $\nu = 1$ table and left $\nu \ge 2$ to the implementation, which therefore chose and documented the closed forms above (`Hecke/HeckeCosetTables.h`); they are recorded here so that the $\nu \ge 2$ tables, the τ-table built on them and every byte-level cross-check have a contractual source. Changing them is a change to this document, not a refactor: the cochain-level matrices depend on the choice (Remark 11.9), only $A^{(q)}$ does not.

Lemma 10.3 (The cocycle is a cocycle)**.** *[proved here] [formally verified: [GT, `lem:cocycle`]] With the notation of Definition 10.2: $j\cdot\gamma$ and $h_j(\gamma)$ are well defined, $(j,\gamma)\mapsto j\cdot\gamma$ is a right action of Γ on $\{1,\dots,m\}$, $h_j(\gamma)\in\Gamma_0(N)$, and*

$$h_j(\gamma_1\gamma_2)=h_j(\gamma_1)\,h_{j\cdot\gamma_1}(\gamma_2),\qquad h_j(1)=I,\qquad \gamma\in\Gamma,\ 1\le j\le m.$$

Moreover $|s_j|\le\max(p^\nu,1)$ entrywise, and the same bound holds for s_j^{-1} and for $\delta_j=s_j^{-1}g$.

Proof. $j\cdot\gamma$ is well defined because cls induces the bijection of Proposition 10.1(4) and the Γ-action on $\Gamma''\backslash\Gamma$ is by right translation; it is an action because cls is Γ-equivariant. $h_j(\gamma)=s_j\gamma s_{j\cdot\gamma}^{-1}$ lies in Γ and has $\mathrm{cls}(h_j(\gamma)s_{j\cdot\gamma})=\mathrm{cls}(s_j\gamma)=\mathrm{cls}(s_{j\cdot\gamma})$, so $h_j(\gamma)\in\Gamma''$ by injectivity in Proposition 10.1(4). For the cocycle identity, expand $s_j\gamma_1\gamma_2=h_j(\gamma_1)s_{j\cdot\gamma_1}\gamma_2=h_j(\gamma_1)h_{j\cdot\gamma_1}(\gamma_2)s_{(j\cdot\gamma_1)\cdot\gamma_2}$ and compare with $s_j(\gamma_1\gamma_2)=h_j(\gamma_1\gamma_2)s_{j\cdot(\gamma_1\gamma_2)}$, using that the s_j are a fixed set of representatives. The bounds are read off Definition 10.2: the entries of s_j lie in $\{0,\pm1,k,pc\}\subseteq[-p^\nu,p^\nu]$, s_j^{-1} negates the off-diagonal and swaps the diagonal, and right multiplication by $g=\mathrm{diag}(1,N)$ scales the second column by $N=p^\nu$ — whence the sharper statement recorded as Lemma 18.5. □

The matrices $\rho_n(h_j(\gamma))$ are the *block transition matrices* of the induced local system; they are assembled in Proposition 12.2 and (22), and the permutation $j\mapsto j\cdot\gamma$ is what makes the induced system a matrix local system on $(K,\mathcal{R})$ (Proposition 12.4). The worked $p=2$ table of $(j\cdot\gamma,h_j(\gamma))$ on the generator multiset $\mathcal{G}$ is §21.

10.3 The operators

Definition 10.4 (Operators)**.** For a double coset $\Gamma\delta\Gamma\subseteq\Delta$ let $T_{\Gamma\delta\Gamma}$ be the operator on $H^q_{\mathrm{eq}}(K,\mathcal{R};L)$ constructed in §11 (Definition 11.6). Write

$$\Delta_{p^\nu}^{\mathrm{prim}}:=\Gamma\,\mathrm{diag}(1,p^\nu)\,\Gamma\qquad(\nu\ge0,\ \Delta_1^{\mathrm{prim}}=\Gamma)$$

for the *primitive* determinant-p^ν double coset, of $m=p^\nu+p^{\nu-1}$ cosets for $\nu\ge1$ and 1 coset for $\nu=0$ (Proposition 10.1(1)). Then:

1. $T_p:=T_{\Delta_p^{\mathrm{prim}}}=T_{\Gamma\mathrm{diag}(1,p)\Gamma}$; [formally verified: [GT, `def:m4-hecke`]]
2. $T(p,p):=T_{\Gamma pI\Gamma}$, the *central* operator; [formally verified: [GT, `def:m4-hecke`]]
3. the **full** operator of degree p^ν is

$$T_{p^\nu}\ :=\sum_{\substack{\Gamma\delta\Gamma\subseteq\Delta\\ \det\delta=p^\nu}}T_{\Gamma\delta\Gamma}\ ,\qquad T_{p^0}:=I,\tag{18}$$

   the sum over *all* double cosets of determinant p^ν — equivalently, the operator attached to the characteristic function of $\{\delta\in\Delta:\det\delta=p^\nu\}$, whose right-coset count is $\sigma_1(p^\nu)=1+p+\cdots+p^\nu$. This is Shimura's $T(p^\nu)$ [standard: [Shi71, §3.2, Thm. 3.24(1), (2), (7), p. 63]]; Shimura's $T(1,p^\nu)$ is the primitive $\Delta_{p^\nu}^{\mathrm{prim}}$ (v1.19.0-pre.5: Shimura's notation corrected — his $T(1,p^\nu)$ is the single double coset $\Delta_{p^\nu}^{\mathrm{prim}}$, his $T(p^\nu)$ the full sum; v1.21.0-pre.1: **verified** on the owner's scan, R20-L — Thm. 3.24(1) $T(m)=\sum_{ad=m,\,a\mid d}T(a,d)$, (2) $T(1,p^k)=T(p^k)-T(p,p)T(p^{k-2})$, (7) $\deg T(m)=\sigma_1(m)$); [formally verified: [GT, `def:m4-hecke`]] (only in the assembled form (19), by which [GT] defines the full operator on the model; (18) itself carries no marker)
4. for $N=\prod_t p_t^{\nu_t}\in\mathbb{Z}_{>0}$ the composite operator is $T_N:=\prod_t T_{p_t^{\nu_t}}$, well defined by Theorem 10.8(1); $T_1=I$;
5. $U_\ell:=T_\ell$, the operator of (1) in the bad case $p=\ell$ of (H2).

Proposition 10.5 (Central operator)**.** *[proved here] [formally verified: [GT, `prop:diamond-model`]] (on the model, for any scalar NI with $\rho(NI)$ the scalar c: with $\tau = \mathrm{id}$ and no hypothesis on ℓ, and for any τ under $6 \in R^\times$) Assume (H4). For every prime p, every $\nu \geq 0$, every commutative coefficient ring R and every degree q* — with no hypothesis on ℓ — *the double coset $\Gamma\, p^\nu I\, \Gamma = p^\nu I \cdot \Gamma$ is a single left and a single right coset, and*

$$T_{\Gamma\, p^\nu I\, \Gamma} \;=\; \rho_n(p^\nu I) \;=\; p^{\nu n} \cdot I \qquad \text{on } H^q_{\mathrm{eq}}(K, \mathcal{R}; L).$$

In particular $T(p,p) = p^n I$. Since $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ has level 1 and hence no non-trivial nebentypus, the classical diamond $\langle p \rangle$ is trivial and $T(p,p) = p^n \langle p \rangle = p^n I$ is the classical normalisation with weight $k = n + 2$: $p \cdot p^n = p^{k-1}$.

Proof. $p^\nu I$ is central in $\mathrm{GL}_2(\mathbb{Q})$, so $\Gamma p^\nu I \Gamma = p^\nu I \Gamma = \Gamma p^\nu I$ is one coset, $m = 1$, $\Gamma' = \Gamma'' = \Gamma$, $s_1 = I$, $\delta_1 = p^\nu I$, and $\mathrm{conj}_{p^\nu I} = \mathrm{id}$. Hence $\tau = \mathrm{id}$ satisfies (T1)–(T3) of Definition 11.1 over every R — (T1) reads $\mathrm{id}(hx) = h\,\mathrm{id}(x)$ — and it is this explicit datum, not Lemma 11.3, that the construction uses, which is why no hypothesis on ℓ is needed. Equation (21) then reads $(\widetilde{T}\phi)(x) = \rho_n(p^\nu I)\phi(x) = p^{\nu n}\phi(x)$ by Definition 8.1 applied to a scalar matrix, and a scalar cochain map induces the same scalar on cohomology. □

Proposition 10.6 (Primitive-to-full assembly)**.** *[proved here] Assume (H4). For every prime p, every $\nu \geq 0$, every commutative coefficient ring R and every degree q — again with* no *hypothesis on ℓ —*

$$T_{p^\nu} \;=\; \sum_{0 \leq 2j \leq \nu} p^{jn}\, T_{\Delta^{\mathrm{prim}}_{p^{\nu-2j}}}, \qquad T_{\Delta^{\mathrm{prim}}_{p^0}} := I. \tag{19}$$

For $\nu = 0, 1$ the sum has one term and $T_{p^\nu} = T_{\Delta^{\mathrm{prim}}_{p^\nu}}$. For $\nu = 2$ it reads $T_{p^2} = T_{\Delta^{\mathrm{prim}}_{p^2}} + p^n I$, the extra term being the central coset pI acting by $T(p,p) = p^n I$ (Proposition 10.5). [formally verified: [GT, `prop:fullassembly-model`]] (the summand only: $T_{\Gamma p^j \mathrm{diag}(1,p^\mu)\Gamma} = p^{jn}\, T_{\Delta^{\mathrm{prim}}_{p^\mu}}$, at cochain level with no hypothesis on ℓ, for an arbitrary τ under $6 \in R^\times$, on the model; not (19) as an identity with (18): the enumeration of the double cosets of determinant p^ν is not formalised, and [GT]'s full operator is defined by the assembled form)

Proof. The index set. Every $\delta \in \mathrm{M}_2(\mathbb{Z})$ with $\det \delta = p^\nu$ has content $\gcd(\delta_{11}, \delta_{12}, \delta_{21}, \delta_{22}) = p^j$ for a unique $j \geq 0$, and $\delta = p^j \delta_0$ with δ_0 primitive of determinant $p^{\nu-2j} \geq 1$, whence $2j \leq \nu$. Content is Γ-biinvariant, so

$$\{\delta \in \Delta : \det \delta = p^\nu\} = \bigsqcup_{0 \leq 2j \leq \nu} p^j I \cdot \Delta^{\mathrm{prim}}_{p^{\nu-2j}},$$

and the double cosets of determinant p^ν are exactly the $p^j I \cdot \Delta^{\mathrm{prim}}_{p^{\nu-2j}}$, one for each admissible j. (Consistency check on right-coset counts: $\sum_{0 \leq 2j \leq \nu} \left(p^{\nu-2j} + p^{\nu-2j-1}\right)$, with the $j = \nu/2$ term read as 1, telescopes to $1 + p + \cdots + p^\nu = \sigma_1(p^\nu)$.)

The scalar. *Within this proof only*, the coset index is written t: the letter j is taken by the content exponent of (19), whose spelling is fixed by Appendix A, entry A.1. Fix j and put $g_0 := \mathrm{diag}(1, p^{\nu-2j})$, $g := p^j g_0$. Since $p^j I$ is central, $\Gamma \cap g\Gamma g^{-1} = \Gamma \cap g_0 \Gamma g_0^{-1}$ and $\mathrm{conj}_g = \mathrm{conj}_{g_0}$, so the *same* transporters s_t and the *same* twisted chain datum τ serve both (Definition 11.1 depends on g only through conj_g), while $\delta_t = s_t^{-1} g = p^j\,(s_t^{-1} g_0)$ and therefore $\rho_n(\delta_t) = p^{jn} \rho_n(s_t^{-1} g_0)$ by Proposition 8.2(2) and $\rho_n(p^j I) = p^{jn} I$. Substituting into (21) gives $T_{p^j I \cdot \Delta^{\mathrm{prim}}_{p^{\nu-2j}}} = p^{jn}\, T_{\Delta^{\mathrm{prim}}_{p^{\nu-2j}}}$ already at cochain level. Summing over j as in (18) gives (19). □

Remark 10.7 (The raw pipeline computes the primitive operator). The coset tables of Definition 10.2 — the $\mathbb{P}^1(\mathbb{Z}/p^\nu)$ representatives — enumerate the *primitive* double coset $\Delta^{\mathrm{prim}}_{p^\nu}$ and nothing else. The pipeline `HeckeCosetTables`(p,ν)→`HeckeTauTable`→`HeckeOperatorOnK`

therefore emits $T_{\Delta_{p^\nu}^{\mathrm{prim}}}$, exposed as `T_p_pow_primitive_direct`. The *standard* operator T_{p^ν} is assembled from it by (19) (`T_p_pow_direct`) or, equivalently, obtained by the recursion of Theorem 10.8 (`T_p_pow`); the two agree by construction and are asserted equal at runtime. This is the whole content of Appendix A, entry A.1: a reader who takes the raw pipeline's output for T_{p^ν} at $\nu \geq 2$ is off by the terms of (19), and two numbered prescriptions of the superseded contract did exactly that.

10.4 The Hecke algebra relations

Theorem 10.8 (Hecke relations, integral normalisation)**.** *Assume (H4) and* $6 \in R^\times$ *— for* R *of characteristic* ℓ, $\ell \geq 5$ *(hypothesis (H3)). Then on* $H^q_{\mathrm{eq}}(K, \mathcal{R}; \mathrm{Sym}^n(R^2))$, *for every degree* q, *all primes* p, p' *and all* $\nu \geq 1$*:*

1. $T_p T_{p'} = T_{p'} T_p$, *[formally verified modulo the commutativity of the Hecke ring, [Shi71, Prop. 3.8, p. 54, applied in §3.2, p. 56] (there for* $\det > 0$*; for* $\det < 0$ *the same argument with a diagonal representative), transported by [AS86, §1.1, p. 194], as the hypothesis* `hcomm`*; locator verified (R20-L): [GT,* `thm:relations-comm-model`*]] (as an instance of the commutativity of all* $T_{\Gamma g \Gamma}$*, Obligation 20.4) and the* full *operators of Definition 10.4(3) satisfy*
$$T_{p^{\nu+1}} = T_p\, T_{p^\nu} - p\, T(p,p)\, T_{p^{\nu-1}} = T_p\, T_{p^\nu} - p^{n+1}\, T_{p^{\nu-1}};$$
equivalently, for $\nu = 1$*:* $(T_p)^2 = T_{p^2} + p^{n+1} I$ *— note that* $(T_p)^2$ *and* T_{p^2} *are different objects and are* not *written alike in this document. [formally verified modulo the primitive products of [Shi71, Thm. 3.24(5), p. 63, with Prop. 3.17, p. 58], transported by [AS86, §1.1, p. 194], as the hypothesis* `hTp`*; locator verified (R20-L): [GT,* `thm:relations-1-model`*]] (on the model, for* τ *over* R*, for the full operators in the assembled form (19), every* $\nu \geq 1$*; not the ring homomorphism of the [standard] note below) [standard: [Shi71, Thm. 3.24(4), p. 63] for the identity in* $\mathcal{H}(\Delta, \Gamma)$ *(verified, R20-L); [AS86, §1.1, p. 194] for the action of* $\mathcal{H}$ *on cohomology (stated there as well known, with a reference to Kuga–Parry–Sah, not proved; read, R20-L) and [Bro82, Ch. III.9–10] for the transfer ([Bro82, Prop. III.9.5, p. 82]), which together make* $\Gamma\delta\Gamma \mapsto T_{\Gamma\delta\Gamma}$ *a ring homomorphism]*
2. $T_M T_N = T_{MN}$ *for* $\gcd(M, N) = 1$*; [formally verified modulo the coprime product of double cosets, [Shi71, Prop. 3.16, p. 57] (Thm. 3.24(3) is its full-operator form), transported by [AS86, §1.1, p. 194], as the hypothesis* `hcop`*; locator verified (R20-L): [GT,* `thm:relations-2-model`*]] (in the primitive form* $T_{\mathrm{diag}(1,M)} T_{\mathrm{diag}(1,N)} = T_{\mathrm{diag}(1,MN)}$*, on the model; not* T_N *of Definition 10.4(4)) with (1) this makes Definition 10.4(4) independent of the order of the factors.*

Caveat, and it is necessary. T_{p^ν} *here is the* ***full*** *operator* $T(p^\nu)$ *of (18) (Shimura's notation),* not *the primitive double-coset operator* $T_{\Delta_{p^\nu}^{\mathrm{prim}}}$*; for* $\nu \geq 2$ *the two differ by the terms of (19), e.g.* $T_{p^2} = T_{\Delta_{p^2}^{\mathrm{prim}}} + p^n I$*. They coincide only at* $\nu \leq 1$*, so* T_p *is unambiguous. Substituting the primitive operator into the recursion falsifies it by* $p^n I$ *(Appendix A, entry A.1). [formally verified modulo the primitive products of [Shi71, Thm. 3.24(5), p. 63, with Prop. 3.17, p. 58], transported by [AS86, §1.1, p. 194], as the hypothesis* `hTp`*; locator verified (R20-L): [GT,* `lem:m4-hTp-hrel`*]] (the* $p^n I$ *difference, derived on the model; its half* $T_{p^2} = T_{\Delta_{p^2}^{\mathrm{prim}}} + p^n I$ *in the assembled form needs no hypothesis) The exponent is pinned by the weight: with* $k = n + 2$, $p \cdot p^n = p^{k-1}$ *is the classical Eichler–Shimura factor (Obligation 20.7).*

Proof sketch. Under $6 \in R^\times$ the assignment $\Gamma\delta\Gamma \mapsto T_{\Gamma\delta\Gamma}$ extends linearly to a ring homomorphism from the abstract Hecke algebra $\mathcal{H}(\Delta, \Gamma)$ (the free $\mathbb{Z}$-module on double cosets with the convolution product) to $\mathrm{End}\big(H^q_{\mathrm{eq}}(K, \mathcal{R}; L)\big)$; this is Obligation 20.4, and the hypothesis $6 \in R^\times$ enters twice — once through Theorem 11.10, which identifies $T_{\Gamma\delta\Gamma}$ with the classical operator (20) on $H^\bullet(\Gamma; V_R)$, and once through Remark 7.6, which identifies the two cohomologies at

all. In $\mathcal{H}(\Delta,\Gamma)$ one has the classical identities $T(p)\,T(p^\nu) = T(p^{\nu+1}) + p\,T(p,p)\,T(p^{\nu-1})$ and commutativity (locally at each prime $\mathcal{H}$ is the polynomial ring on $T(p) = T(1,p)$ and $T(p,p)$, not inverted — v1.21.0-pre.1, R20-L: "$T(p,p)^{\pm1}$" corrected — and $\mathcal{H}$ is commutative) [standard: [Shi71, Thm. 3.20, p. 59; Thm. 3.24(4); Prop. 3.8 with p. 56], verified, R20-L]. Shimura's $T(p^\nu)$ *is* the full operator T_{p^ν} of (18) — both are the characteristic function of $\{\det = p^\nu\}$ — and this identification is exactly what makes the cited relation a statement about T_{p^ν} and not about $T_{\Delta^{\mathrm{prim}}_{p^\nu}}$. Applying the homomorphism and Proposition 10.5 gives (1); (2) is the coprime factorisation $T(M)T(N) = T(MN)$ [standard: [Shi71, Thm. 3.24(3), p. 63]; in the primitive form, the product of the double cosets of coprime determinant, [Shi71, Prop. 3.16, p. 57]; verified, R20-L, in place of "Ch. 3, §3.3"] transported the same way. □

Remark 10.9 (The coefficient of $T_{p^{\nu-1}}$ is p^{n+1}, not $(p+1)p^n$). A working note in this project carried $(T_p)^2 = T_{p^2} + (p+1)p^n$, attributed to a corollary of the superseded contract that says no such thing (Appendix A, entry A.6). The degree count settles it without any convention: T_p has $p+1$ right cosets, $(T_p)^2$ therefore $(p+1)^2$, the full T_{p^2} has $1+p+p^2$, and $(p+1)^2-(1+p+p^2) = p$ — one factor p, multiplying the central $T(p,p) = p^n I$, i.e. p^{n+1}.

Corollary 10.10 ($p = \ell$: the bad prime)**.** *Let $R = \mathbb{F}_\ell$, $n \geq 0$, and let $p = \ell$ (hypothesis (H2)), so that $U_\ell = T_\ell$.*

1. (Assuming in addition $\ell \geq 5$, i.e. (H3).) *Since $p^{n+1} = \ell^{n+1} \equiv 0 \pmod{\ell}$, the recursion of Theorem 10.8(1) collapses to $T_{\ell^{\nu+1}} = U_\ell\, T_{\ell^\nu}$, whence the* full *operators satisfy*

$$T_{\ell^\nu} \;=\; U_\ell^{\nu} \qquad \text{for all } \nu \geq 1.$$

2. (No hypothesis on ℓ.) *For $n \geq 1$, (19) has every correction term multiplied by $\ell^{jn} \equiv 0$ for $j \geq 1$, so*

$$T_{\ell^\nu} \;=\; T_{\Delta^{\mathrm{prim}}_{\ell^\nu}} \quad \text{on } \mathbb{F}_\ell\text{-coefficients}, \qquad n \geq 1,\ \nu \geq 0\,:$$

at the bad prime, and only there, the full and the primitive prime-power operators coincide. *For $n = 0$ they do not: the scalars ℓ^{jn} are then all* 1.

Proof. (1) is Theorem 10.8(1) with $p = \ell$ and an induction on ν from $T_{\ell^0} = I$, $T_{\ell^1} = U_\ell$. (2) is Proposition 10.6, which carries no hypothesis on ℓ, read modulo ℓ. □

Remark 10.11 (Singularity at $p = \ell$ is not a defect of the construction). For $p = \ell$ and $n \geq 1$ every $\rho_n(\delta_j)$ mod ℓ is singular (Lemma 8.4); e.g. $\rho_n(\mathrm{diag}(1,\ell)) = \mathrm{diag}(1,0,\ldots,0)$ mod ℓ kills all of Sym^n except the top monomial $u_0 = x^n$, and U_ℓ may be nilpotent on $H^\bullet_{\mathrm{eq}}$. Nothing in §§11–12 requires invertibility of $\rho_n(\delta)$ for $\delta \notin \Gamma$: the only matrices ever inverted are $\rho_n(\gamma)$ with $\gamma \in \Gamma$, which are unimodular (Obligation 20.3). U_ℓ is therefore defined integrally by the same pipeline, with no branch and no special case.

Code map. Proposition 10.1, Definition 10.2, Lemma 10.3: `Hecke/HeckeCosetTables.h/.cpp` (`s(j)`, `s_inv(j)`, `delta(j)`, `class_index`, `perm(gen,j)` for $j \cdot \gamma$, `cocycle(gen,j)` for $h_j(\gamma)$ and `rho_cocycle(gen,j)` for $\rho_n(h_j(\gamma))$; the index is 1-based, matching this chapter's subscripts verbatim and deliberately so), gated by `HeckeCosetTablesValidation` and `HeckeCosetTablesSelfCheckValidation`. Definition 10.4, Proposition 10.5, Proposition 10.6, Theorem 10.8, Corollary 10.10: `Hecke/HeckeOperatorFamily.h/.cpp`, with

entry point	emits
`T_p(`p`)`	T_p (unambiguous)
`T_p_pow(`p,ν`)`	the **full** T_{p^ν}, by the recursion of Thm. 10.8
`T_p_pow_direct(`p,ν`)`	the **full** T_{p^ν}, by the assembly (19)
`T_p_pow_primitive_direct(`p,ν`)`	the **primitive** $T_{\Delta^{\mathrm{prim}}_{p^\nu}}$ (the raw pipeline)
`T_n(`N`)`	T_N of Def. 10.4(4); the code identifier keeps the retired spelling
`T_pp(`p`)`	$T(p,p) = p^n I$, by the closed formula of Prop. 10.5, with no pipeline run

There is *no* `U_ell` entry point, and that is the contract's own invariant made visible: U_ℓ is `T_p(ell)`, the same pipeline and the same code path, with no branch on $p = \ell$ anywhere (Remark 10.11, Obligation 20.3). [verified computationally] A recorded reach limit of that code path, not a defect of the contract (v1.3.3; `docs/implementation/slot1/WP-2_report.md` §4): at $n = 16$, `--bench hecke 16 <ell> <ell>` (U_ℓ, $\nu = 1$) completes at $\ell \in \{691, 1009, 2003, 3617\}$ and throws `HeckeOverflow` ("exact multiply overflow", `Hecke/ExactInt.h`) in the coset construction at $\ell \in \{5003, 7001, 10007, 43867\}$, independently of n (checked at $n \in \{2, 4, 8, 16\}$), while $\ell = 43867$ with a good prime runs in milliseconds — so the `int64` tier of Theorem 18.7 is exhausted somewhere in $3617 < \ell < 5003$, and a wider tier is what an ℓ beyond that needs (`docs/KNOWN_ISSUES.md` §3).

Repaired and re-measured at v1.5.0 (slot-3 package WP-1, `docs/implementation/slot3/WP-1_report` the repair was planned as slot-2 WP-3 and did not run in slot 2). Two corrections to the record above and one new number. *(a) The site.* The overflow was never in the coset construction: it is in Algorithm 18.3 step 1's fundamental-domain reduction, on the quadratic point Lemma 18.5(6) bounds, and the quantity that overflows is a transient local, not a stored transporter (Remark 18.6). *(b) Lemma 18.5 did not fail*, and neither did Theorem 18.7 on the statements it made: the certified B was inside the tier throughout, and the gap was that no clause covered step 1 — which is clause (6)'s reason for existing and Theorem 18.7(iv)'s. *(c) The new reach*, after the per-operation escalation to the `__int128` tier at that site: [verified computationally] the binding constraint is again Lemma 18.5(5), through clause (e) of the τ-certification, which on this model reduces to $4N^2 \le 2^{31} - 1$, i.e. $\boldsymbol{\ell \le 23170}$ (largest prime 23167); past it the table is **refused by Obligation 20.11's certification**, by the named clause that owns the bound, rather than by an anonymous overflow — the same input now fails where this document says it should. Measured: $(n, \ell) = (16, 5003)$ completes in 73 s / 0.95 GB, $(16, 10007)$ in 612 s / 1.9 GB, and $\ell = 43867$ builds its τ-table with no overflow and is then refused by clause (e). Raising ℓ beyond 23170 is a change to the *stored* type and is deliberately not made: no cell the programme needs is behind it, and the memory admission of U_ℓ refuses those cells first (`HeckeUEllReachValidation`, `HeckeUEllBudgetValidation`). **Lifted at v1.23.0-pre.9** (slot-24 ELL-W) *without* changing the stored type: by Theorem 18.7(4) a table with $B > 2^{31} - 1$ is certified in the stored wide tier, so clause (e) now refuses only when $2NB > 2^{63} - 1$, which with $B = 4N^2$ is $\ell > 1048575$; [verified computationally] `--bench hecke 0 43867 43867` completes (28 s, 2.1 GB) where it was refused, and U_{30011} at $n = 10$ matches $\tau(30011)$ mod 30011 (`HeckeWideTauValidation`). The memory admission of `u_ell = auto` (the frozen model `u_ell_rss/v2-2026-09-17`) is a mathematical setting and is *not* recalibrated here; the module route of the saturation cells does not consult it. The gates are `HeckeOperatorFamilySmokeValidation` — which asserts `T_p_pow` = `T_p_pow_direct` and the decomposition witness `T_p_pow_direct` = `T_p_pow_primitive_direct` $+ p^n I$ at $\nu = 2$ — and `HeckeRelationsValidation` (oracle O9: commutativity, the $\nu = 2$ recursion, and the primitive-versus-full witness). The hypothesis $6 \in R^\times$ of Theorem 10.8 is respected by those gates and not merely by this text: at $\ell \in \{2, 3\}$ the recursion and the U_{ℓ^ν} identities are *recorded and not asserted*, the hard assertions there being only the hypothesis-free ones (pipeline com-

pletes, τ certifies, well-definedness), and the $p = \ell$ recursion is asserted only at $(p, \ell) = (5, 5)$ (`docs/implementation/hecke_action_implementation_plan.md`, amendment A-F1). Remark 10.9 and Remark 10.7 are the prose form of `Hecke/HeckeOperatorFamily.h`:136–173.

Formalization note. Proposition 10.1 is elementary and constructive throughout: (1) is Hermite reduction of a 2×2 integral matrix, (4) is a finite computation once N is fixed and a decidable predicate in general, and Definition 10.2 gives the inverse of the bijection in closed form, so `P1(ZMod N)` together with the two closed-form lifts is a complete formalisation of the coset datum with no choice principle. Lemma 10.3 is then a one-line consequence of uniqueness of coset representatives and is the natural first target. Proposition 10.5 and Proposition 10.6 are proved *at cochain level* from the content decomposition and $\rho_n(\lambda I) = \lambda^n I$, so neither needs cohomology, neither needs a hypothesis on ℓ, and both are formalisable before §11 is. Theorem 10.8 is the one statement of this chapter that is *not* self-contained: it rests on the ring-homomorphism property (Obligation 20.4), which rests on the classical comparison (Theorem 11.10); a formalisation would either import $\mathcal{H}(\Delta, \Gamma)$ and its structure theorem or — the cheaper route for this model — verify the $\nu = 2$ instance directly as a finite identity of cochain maps and induct, which is what `HeckeRelationsValidation` does numerically.

11 The primary definition: $T_{\Gamma g \Gamma}$ in all degrees via a twisted chain map

What this chapter fixes. The *normative* definition of the Hecke operator: a g-twisted chain datum τ on the universal cover, the cochain map $\widetilde{T}$ built from τ and the coset datum of §10, and the operator $T_{\Gamma g \Gamma}$ it induces in *every* cohomological degree at once. Also: what is and is not independent of the choices, at which level; the exact condition under which the shipped τ is integral; and the comparison with the classical operator of the theory of arithmetic groups. Degree-specific recipes — modular symbols, bar cochains — appear in this document only as validation oracles (§13), never as definitions.

Standing hypotheses. (H1)–(H4) of §7, each repeated where it is used (rule (A1)). The reader should note in advance the shape of the dependence, because the superseded document stated it only in passing: *the construction* of $\widetilde{T}$ (Definition 11.6) needs no hypothesis on ℓ at all; *the existence* of an abstract τ needs none either, over $\mathbb{Z}$ and hence over every R (Lemma 11.3(1)); *homotopy uniqueness* of τ needs $6 \in R^\times$ (Lemma 11.3(2)); and *the identification with the classical operator* (Theorem 11.10) needs $6 \in R^\times$ twice over. Remark 11.4 says which downstream statements that apportionment does and does not move.

11.1 The correspondence in cohomology

Topologically the datum of §10 is a correspondence. Write $\mathrm{pr}' \colon \Gamma' \backslash \mathcal{H} \to \Gamma \backslash \mathcal{H}$ and $\mathrm{pr}'' \colon \Gamma'' \backslash \mathcal{H} \to \Gamma \backslash \mathcal{H}$ for the two covering projections (both of degree m) and $\theta_g \colon \Gamma' \backslash \mathcal{H} \to \Gamma'' \backslash \mathcal{H}$, $\Gamma' z \mapsto \Gamma'' g z$, for the translation, well defined because $g \Gamma' g^{-1} = \Gamma''$ (Proposition 10.1(3)); then $T_{\Gamma g \Gamma} = (\mathrm{pr}'' \circ \theta_g)_* \circ (\mathrm{pr}')^*$. (The projections are named operators rather than letters, per (N3): π_1 is the fundamental group and t_j is an upper-triangular transporter, §2.3 and Definition 10.2.) Algebraically — and this is the form every later statement uses —

$$T_{\Gamma g \Gamma} \;=\; \mathrm{cor}^{\Gamma}_{\Gamma''} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}^{\Gamma}_{\Gamma'} \;:\; H^\bullet(\Gamma; V_R) \longrightarrow H^\bullet(\Gamma; V_R), \tag{20}$$

where $(\mathrm{conj}_g)_* \colon H^\bullet(\Gamma'; V_R) \to H^\bullet(\Gamma''; V_R)$ is induced by the compatible pair $\big(\mathrm{conj}_g^{-1} \colon \Gamma'' \to \Gamma',\ \rho_n(g) \colon V_R \to V_R\big)$. That this pair *is* compatible is exactly the semigroup extension of

Definition 8.3: for $h'' \in \Gamma''$ one has $g^{-1}h''g \in \Gamma'$ and, by Proposition 8.2(2) applied to the integral identity $g\,(g^{-1}h''g) = h''g$ in $\mathrm{M}_2(\mathbb{Z})$,

$$\rho_n(g)\,\rho_n(g^{-1}h''g) \;=\; \rho_n(h'')\,\rho_n(g),$$

with every factor integral and *no* inverse of a determinant-N matrix taken (Obligation 20.3). Equation (20) is the standard description of a double-coset operator [standard: [Bro82, Ch. III.9–10] for res, cor and the conjugation action; [AS86, §1.1, p. 194] for the double-coset operator (there in its right-module form on homogeneous cochains; v1.17.0-pre.5, slot-17 R17-4); [Har22, Ch. 2]]; it is recorded here as the object Theorem 11.10 compares with, not as the definition, which is Definition 11.6. [formally verified: [GT, `thm:m6-cor-res`]] [formally verified: [GT, `thm:m6-corH-indep`]] [formally verified: [GT, `thm:m6-cor-res-index`]] [formally verified: [GT, `thm:m6-Tcls-res`]] ($\mathrm{cor}^{\Gamma}_{\Gamma''}$ as Brown's transfer on any projective resolution, independent of the transversal, $\mathrm{cor}\circ\mathrm{res} = [\Gamma:\Gamma'']$, and $\mathrm{cor}\circ(\mathrm{conj}_g)_*\circ\mathrm{res}$ for the pair $(\mathrm{conj}_g^{-1}, \rho_n(g))$, in every degree and over every commutative ring; the topological correspondence form is not formalised; v1.24.0-rc.1)

11.2 The twisted chain datum τ

Throughout this subsection $C_\bullet(K^u)$ is the simplicial chain complex of the Γ-tessellation $K^u \cong X$ of (H4), an augmented complex of $\mathbb{Z}[\Gamma]$-modules, acyclic because X is contractible ((H4)(ii), Proposition 7.4); and $C^\bullet_{\mathrm{eq}}(K,\mathcal{R};L) \cong \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(K^u), V_R\big)$ via the isomorphism Ψ of Theorem 5.10.

Definition 11.1 (g-twisted chain datum)**.** Let R be a commutative ring and $g \in \Delta$, $\Gamma'' = \Gamma \cap g\Gamma g^{-1}$. A *$g$-twisted chain map over R* is a degree-preserving R-linear map

$$\tau\colon C_\bullet(K^u)\otimes_{\mathbb{Z}} R \longrightarrow C_\bullet(K^u)\otimes_{\mathbb{Z}} R$$

such that

(T1) $\tau(h\cdot x) = (g^{-1}hg)\cdot\tau(x)$ for all $h \in \Gamma''$ and all x (the right-hand side is defined because $g^{-1}hg \in \Gamma' \subseteq \Gamma$ by Proposition 10.1(3));
(T2) $\partial\circ\tau = \tau\circ\partial$;
(T3) $\varepsilon(\tau(x)) = \varepsilon(x)$ in degree 0.

A *g-twisted chain datum* is such a τ together with the finite table of its values on a set of Γ''-orbit representatives of the simplices of K^u, from which (T1) determines it everywhere; that table is what Algorithm 18.3 emits and what Obligation 20.11 certifies.

Remark 11.2 (The ring in Definition 11.1, and why it is not fixed to $\mathbb{F}_\ell$). The superseded document stated Definition 11.1 with $R = \mathbb{F}_\ell$ fixed. Nothing in (T1)–(T3) uses that, and two consumers need more: the implementation's table is a table of *integer* chains ($R = \mathbb{Z}$), and the boundary and cone data of §§14–15 are required to be one integral datum whose R-versions are its base changes ("tables once, matrices per R"). The definition is therefore stated over a general R; the label `def:tau` names the same notion and the $\mathbb{F}_\ell$ case is recovered verbatim by $R = \mathbb{F}_\ell$, so this is a widening and not a retargeting (rule (L2)). Every statement below says which ring it is over.

Geometrically τ is a simplicial approximation of $z \mapsto g^{-1}z$ on $\mathcal{H}$, which satisfies (T1) on the nose because $g^{-1}(hz) = (g^{-1}hg)(g^{-1}z)$. That is how Algorithm 18.3 *finds* a candidate; it is not how any statement about τ is proved. Definition, existence, uniqueness and certification are purely algebraic, and the geometry is discharged entirely by Obligation 20.11 — the point of that obligation being precisely that no geometric assertion survives into the correctness of T_p.

Lemma 11.3 (Existence, and homotopy uniqueness)**.** *Assume (H4) and let $g \in \Delta$. The two halves carry* different *hypotheses and are therefore stated separately (rule (A1)); the apportionment is Remark 11.4.*

1. ***(Existence over $\mathbb{Z}$; no hypothesis on ℓ.)*** *[proved here] A g-twisted chain map over $\mathbb{Z}$ exists, and is produced by the terminating procedure (V), (E), (F) of the proof below (rule (A3)). Consequently a g-twisted chain map over R exists for* every *commutative ring R, namely the base change of one over $\mathbb{Z}$.*
2. ***(Homotopy uniqueness; needs $6 \in R^\times$.)*** *[proved here] [formally verified: [GT, `lem:tauexists-2-model`]] (on the model, for R-linear chain maps of the restricted resolutions $C_\bullet(X) \otimes R$ over id_R, including the last sentence: existence over such R, re-proved abstractly; not (1)) Let R be a commutative ring with $6 \in R^\times$ — for R of characteristic ℓ this is $\ell \geq 5$ (hypothesis (H3)). Then any two g-twisted chain maps over R are homotopic* through g-twisted maps*: if τ, τ' both satisfy (T1)–(T3) there is an additive degree-$(+1)$ map H satisfying (T1) with $\tau - \tau' = \partial H + H\partial$. The argument re-proves existence over such an R as well, abstractly; it is (1) that gives existence over $\mathbb{Z}$ and for every ℓ. [formally verified: [GT, `lem:m4-tauR-exists`]] (since v1.20.0: the existence over such R for every $g \in \Delta$, in the form the operator on the model reads, a τ over R; (1), existence over $\mathbb{Z}$, still not)*

Proof of (1). Choose a set of Γ''-orbit representatives of the simplices of K^u: finitely many, because Γ has finitely many orbits ((H4)(iii)) and $[\Gamma : \Gamma''] = m < \infty$ (Proposition 10.1(3)). Define τ on them in ascending dimension and extend by (T1) at each step, $\tau(h \cdot \sigma) := (g^{-1}hg) \cdot \tau(\sigma)$ for $h \in \Gamma''$ — the conjugate lies in Γ by Proposition 10.1(3) — which is well defined exactly when $\tau(\sigma)$ is fixed by $g^{-1}\Gamma''_\sigma g$, Γ''_σ being the Γ''-stabiliser of σ. Since Γ acts without inversions ((H4)(i)) no stabiliser reverses an orientation, so this determines an additive map of the free $\mathbb{Z}$-modules $C_q(K^u)$. *Nothing below inverts a stabiliser order or solves a system over anything but $\mathbb{Z}$.*

(V) **Vertices.** Let v be a representative vertex. By (H4)(i),(iv) Γ_v is cyclic of order 2, 4 or 6 and contains $-I$; so is $\Gamma''_v = \Gamma'' \cap \Gamma_v$ (note $-I \in \Gamma''$, since $-I \in \Gamma$ and $g(-I)g^{-1} = -I$), and so is its conjugate $G := g^{-1}\Gamma''_v g \subseteq \Gamma$. If $G = \{\pm I\}$, put $\tau(v) := w$ for an arbitrary vertex w of K^u; it is fixed, because $-I$ acts trivially ((H4)(iv)). If $|G| \in \{4, 6\}$, a generator γ of G is a torsion element of Γ other than $\pm I$, hence elliptic, hence fixes a unique point $z_0 \in \mathcal{H}$, which $G = \langle\gamma\rangle$ therefore fixes. The stabiliser of z_0 modulo $\pm I$ is non-trivial, so z_0 is Γ-equivalent to i, to ζ or to ζ^2 [standard: [Ser73, Ch. VII, §1.2, Thm. 1(1),(3)]: every point of $\mathcal{H}$ is Γ-equivalent to a point of the standard fundamental domain, and the points there with non-trivial stabiliser modulo $\pm I$ are exactly i, $\zeta^2 = e^{2\pi i/3}$ and $\zeta = e^{\pi i/3}$], i.e. to the vertex v_6, v_5 or v_4 of Definition 7.1; every Γ-translate of those is a vertex of K^u ((H4)(i),(iv), their imaginary parts being $\leq 1 < Y$). Put $\tau(v) :=$ that vertex, which G fixes. In both cases $\tau(v)$ is a *single* vertex, so (T3) holds.

(E) **Edges.** Let e be a representative edge, $\partial e = v'' - v'$. Then $\tau(\partial e) = \tau(v'') - \tau(v')$ has augmentation 0, hence is a boundary, because $\varepsilon\colon H_0(K^u) \to \mathbb{Z}$ is an isomorphism ($K^u \cong X$ is contractible, (H4)(ii); the first of the three consequences collected at (37)). Let $\tau(e)$ be *any* 1-chain with $\partial\tau(e) = \tau(\partial e)$. The (T1)-extension is well defined because $\Gamma''_e \subseteq \Gamma_e = \{\pm I\}$ ((H4)(iv)) and $g^{-1}(-I)g = -I$ acts trivially; and $\partial\tau = \tau\partial$ holds on the whole orbit because both sides satisfy (T1) and agree on e.

(F) **Triangles.** Let σ be a representative triangle. By (T2) in degree 1, $\partial\tau(\partial\sigma) = \tau(\partial\partial\sigma) = 0$, so $\tau(\partial\sigma)$ is a 1-cycle, hence a boundary, and the 2-chain with that boundary is *unique* ($Z_1(K^u) = B_1(K^u)$ and $Z_2(K^u) = 0$, (37)); take it. The (T1)-extension and (T2) are as in (E), the stabiliser of a triangle again being $\{\pm I\}$. There is nothing to do in degrees ≥ 3, K^u being two-dimensional ((H4)(ii)).

The result satisfies (T1)–(T3) over $\mathbb{Z}$, and those are identities between $\mathbb{Z}$-linear maps of free $\mathbb{Z}$-modules, so they survive $\otimes_{\mathbb{Z}} R$.

This is the construction of Proposition 15.13 with its step (B) — which pre-assigns the boundary slots from τ_∂, and is what makes the result a map of *pairs* — omitted; that proposition therefore contains (1), at the cost of the further hypotheses (B1)–(B3). Neither statement uses

(2), so there is no circularity. □

Proof of (2). Write $C_\bullet := C_\bullet(K^u) \otimes_{\mathbb{Z}} R$ and let $C_\bullet^{(c)}$ be the same complex with the Γ''-action $h * x := (g^{-1}hg) \cdot x$, which is defined because $g^{-1}hg \in \Gamma$ (Proposition 10.1(3)) and is an action because conj_g is a group isomorphism. Conditions (T1)–(T3) say precisely that $\tau\colon C_\bullet \to C_\bullet^{(c)}$ is an augmentation-preserving map of complexes of $R[\Gamma'']$-modules.

The source is a complex of relatively $R[\Gamma'']$-projective modules. By (H4)(i),(iv) each $C_q(K^u)$ is a direct sum of permutation modules $\mathbb{Z}[\Gamma/\Gamma_\sigma]$ with $|\Gamma_\sigma|$ dividing 12: the stabilisers occurring are $\{\pm I\}$ of order 2, $\langle S\rangle$ of order 4 at v_6, and $\langle ST\rangle, \langle TS\rangle$ of order 6 at v_4, v_5. This — the *orders 4 and 6 of the elliptic stabilisers* — is where the hypothesis enters, and it is the only place: $6 \in R^\times$ makes $2, 3$ and hence $4, 6, 12$ units, so each $|\Gamma_\sigma|$ is invertible in R and the averaging idempotent $|\Gamma_\sigma|^{-1}\sum_{u\in\Gamma_\sigma} u$ exhibits $R[\Gamma/\Gamma_\sigma]$ as a direct summand of $R[\Gamma]$. Restriction to the finite-index subgroup Γ'' preserves being a summand of a free module.

The target is acyclic and augmented. X is contractible ((H4)(ii), Proposition 7.4), so the augmented complex $C_\bullet(K^u) \to \mathbb{Z}$ is contractible as a complex of abelian groups, hence stays exact after $\otimes_{\mathbb{Z}} R$; the Γ''-action is changed by conj_g but the underlying complex is the same one, so $C_\bullet^{(c)}$ is acyclic over R and augmented over R as well.

Existence and uniqueness up to $R[\Gamma'']$-homotopy now follow from the fundamental lemma of relative homological algebra [standard: [Bro82, Ch. I.7.4]: a map of augmented complexes of relatively projective modules over an acyclic augmented target exists and is unique up to homotopy]. A homotopy over $R[\Gamma'']$ is exactly an additive H with property (T1) — the translation of Γ''-equivariance for the $*$-action into (T1) is the same one made for τ above. □

> **CONTRACT.** For $6 \notin R^\times$ — in particular $\ell \in \{2, 3\}$ — what is available is **Lemma 11.3(1) and not (2)**: a g-twisted chain datum *exists*, over $\mathbb{Z}$ and hence over every R, but **homotopy uniqueness is not asserted**, and this document does not repair it by an abstract argument. Also holding there: the *canonical* τ built by Algorithm 18.3 still exists (that construction never divides by 2 or 3; its only divisions are exact-divisibility checks in the integer fill) and is still certified by Obligation 20.11. Consequently, for $6 \notin R^\times$ the operator is **defined** as the output of the pipeline run with the canonical τ; its independence of τ (Theorem 11.7(b)) and its identification with (20) (Theorem 11.10) are asserted only for $6 \in R^\times$. No cross-τ comparison is made anywhere at $\ell \in \{2, 3\}$, in this document or in the implementation.

Remark 11.4 (The two halves apportioned: what (1) buys, and what it does not). [proved here] (Rule (A6).) Up to and including v0.15.0 Lemma 11.3 carried $6 \in R^\times$ on *both* halves, and the contract box above declared the whole lemma unavailable at $\ell \in \{2, 3\}$. Clause (1) removes the hypothesis from the existence half — by a construction, not by the comparison theorem. Three things must be kept apart from it.

1. **Uniqueness does not follow, and the argument does not transfer.** The proof of (1) may choose a value *and then* impose (T1), because a *vertex* value can be chosen inside the fixed-point set of the twisted stabiliser (Serre's elliptic points) and because edge and triangle stabilisers are central. A twisted homotopy H has no such freedom: $H(v)$ must be a 1-chain with a *prescribed* boundary *and* be fixed by $g^{-1}\Gamma''_v g$, and for $|\Gamma''_v| \in \{4, 6\}$ nothing in the construction produces an invariant fill. Supplying one is exactly what the averaging idempotent $|\Gamma_\sigma|^{-1}\sum_u u$ does in the proof of (2), and it is why (2) keeps $6 \in R^\times$. Whether homotopy uniqueness *holds* at $\ell \in \{2, 3\}$ is [conjectural] either way and is not decided here; what would settle it is a twisted-equivariant fill at the elliptic vertices, or a counterexample — two certified τ at $\ell = 2$ inducing different $A^{(q)}$, which

`HeckeIotaTauSymmetryValidation` already has the two data for (Remark 9.5) — and on the golden grid those two, the legacy and the canonical τ, induce the *same* $A^{(q)}$ at every $\ell \in \{2,3\}$ point (the v1.1.0 re-baseline moved no cohomology-level value), so no counterexample is known there.

2. **No downstream hypothesis is weakened by (1).** Every statement of this document that carries $6 \in R^\times$ "via Lemma 11.3" uses the *uniqueness* half, or relative projectivity of $C_\bullet(K^u) \otimes R$, or both; each was read against its own proof and **none** of them moves: Theorem 11.7(b), and through it (c) for arbitrary admissible choices; Proposition 9.4(3c) (which is (3a) plus Theorem 11.7(a),(b)); Theorem 11.10, which uses the hypothesis twice and whose *first* use — relative projectivity, hence also $H^\bullet_{\mathrm{eq}} \cong H^\bullet(\Gamma; V_R)$ (Remark 7.6, (H3)) — is untouched by (1); Proposition 13.4's agreement clause, which is a comparison of two resolutions; Proposition 15.12(2), and hence Theorem 15.11(c) and (e); and Remark 15.14(i),(ii), and hence Theorem 15.11(c′). **What (1) does buy** is that Definition 11.6 is non-vacuous over $\mathbb{Z}$ for every ℓ — before it, the only datum known to exist at $\ell \in \{2,3\}$ was the output of Algorithm 18.3 — and the corresponding correction inside Remark 11.5.
3. **Existence over $\mathbb{Z}$ is not integrality of the canonical table.** (1) produces *some* datum over $\mathbb{Z}$; it says nothing about whether Algorithm 18.3's triangle fills solve over $\mathbb{Z}$, which is a property of that algorithm's choices, is verified per table, and is governed by owner decision OP2 (Remark 11.5). The two must not be conflated — that conflation is the class of over-claim Appendix A records at entry A.3.

Remark 11.5 (Integrality of the emitted τ-table: the claim and its condition). Algorithm 18.3 emits, for each orbit representative, a chain $\tau(s_j \cdot \widetilde{\sigma}) = \sum_t n_t \gamma_t \cdot \widetilde{\sigma}_t$ with $n_t \in \mathbb{Z}$, $\gamma_t \in \Gamma$. Steps 1 and 2 are integral *unconditionally*, because neither solves a linear system: a vertex value is the single term $1 \cdot \gamma_{\mathrm{red}}^{-1} \widetilde{w}$, and an edge value is a concatenation of stored edge-path templates, hence a $\mathbb{Z}$-combination of oriented edges obtained by addition alone. Step 3, the triangle fill, solves $\partial_2|_{\mathcal{N}} x = \tau(\partial\widetilde{\sigma})$ over $\mathbb{Z}$ on a finite subcomplex $\mathcal{N} \subseteq K^u$, growing $\mathcal{N}$ on failure; that solve is where integrality can be lost, because the algorithm permits, as a fallback, solving the *same* system mod ℓ.

- **What is claimed.** If every fill of a table solved over $\mathbb{Z}$, the table is a g-twisted chain datum over $R = \mathbb{Z}$ in the sense of Definition 11.1, and the datum over any R — in particular every $\mathbb{F}_\ell$, every $\mathbb{Z}/\ell^e$, every $\mathbb{Z}/m$ — is its base change; the certification of Obligation 20.11 is then exact over $\mathbb{Z}$. This is verifiable per table, and it is verified: the regime is recorded per chain and the table-level disjunction is exposed (`used_mod_ell_solve()`).
- **What is *not* claimed.** That the fills always solve over $\mathbb{Z}$. There is no theorem here. Lemma 11.3(1) does give existence of *some* g-twisted chain datum over $\mathbb{Z}$, for every ℓ — but by a different construction, which chooses its vertex values at the elliptic points and its triangle fills anywhere in K^u, whereas Algorithm 18.3 chooses by nearest vertex and fills inside a growing finite neighbourhood $\mathcal{N}$. **Existence of an integral datum is not integrality of *this* datum**: the $\mathbb{Z}$-fill's feasibility on a given $\mathcal{N}$ is a property of the table, not of the model (Remark 11.4(3)). A chain obtained by the mod-ℓ fallback defines τ over $\mathbb{F}_\ell$ only, and Obligation 20.11's certification for every identity touching it is then exact mod ℓ rather than over $\mathbb{Z}$ — which is sound, and is a strictly weaker datum.
- **Policy (owner decision OP2, 2026-09-15).** The mod-ℓ fill fallback is **forbidden by default**. It exists only because of a fixed-width integer tier and never for a mathematical reason, so the remedy is to widen that tier; the fallback stays reachable behind an explicit setting and a datum that uses it is marked `integral = false`. **`integral = false` is a recorded failure to reach $\mathbb{Z}$, not the default escape from a hard fill**, and a datum claiming integrality must have solve regime $\mathbb{Z}$ on every one of its tables.
- [verified computationally] On the shipped grid — $(p,\nu) \in \{(2,1),(3,1),(5,1),(7,1),(2,2)\}$, $\ell \in \{5, 7, 691, 134217757\}$, $n \in \{0, 2, 10\}$, plus the bad-prime points $p = \ell \in \{2,5\}$

at the same n, i.e. 66 points — 0 **of** 66 **used the mod-ℓ triangle solve**: every table is integral, with certified entry bound $B \le 196$ (Lemma 18.5). Validation `HeckeTauTableSelfCheckValidation`, 69/69 points passed, run of 2026-09-15 on the `RelWithDebInfo` build.

11.3 Definition of the operator

For $\sigma \in o_q(K)$ write $\widetilde{\sigma} \in o_q(K^u)$ for its distinguished lift under the strict fundamental domain $K \subseteq K^u$ of (H4)(iii) (macro `\lift`, §2.3); the implementation writes `i(sigma)` for $\widetilde{\sigma}$, a spelling this document cannot use because i is $\sqrt{-1}$ (§2.7, C15, C16).

Definition 11.6 (Normative definition of $T_{\Gamma g \Gamma}$)**.** [formally verified: [GT, `def:heckeP`]] (the definition (21), and $\delta_k = s_k^{-1} g$ for the canonical s_k; since v1.19.0: τ over R is formalised on the model (`TauR`, `heckeTildeR`), [GT]'s operator on a general descriptor takes τ over $\mathbb{Z}$) Let $g \in \Delta$, $\Gamma'' = \Gamma \cap g\Gamma g^{-1}$ of index m, let $s_1, \dots, s_m$ represent $\Gamma''\backslash\Gamma$ and $\delta_j := s_j^{-1} g$ (Proposition 10.1(5); canonically Definition 10.2) [formally verified: [GT, `lem:m4-cosetdata`]] (since v1.20.0: such data exist for every $g \in \Delta$ — Γ'' of finite index, a transversal, $\delta_j \in \Delta$; a non-constructive datum, not the canonical table of Definition 10.2), and let τ be a g-twisted chain datum over R (Definition 11.1). For $\phi \in \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(K^u), V_R\big)$ define

$$(\widetilde{T}\phi)(x) \;:=\; \sum_{j=1}^{m} \rho_n(\delta_j)\, \phi\big(\tau(s_j \cdot x)\big), \qquad x \in C_q(K^u),\ q \ge 0. \tag{21}$$

Then $T_{\Gamma g \Gamma}$ on $C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ is $\Psi \circ \widetilde{T} \circ \Psi^{-1}$ (Theorem 5.10), and on $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ the induced map. The sum is finite, all its factors are integral, and no determinant-N matrix is inverted (Obligation 20.3); the definition is uniform in q — one formula, one pipeline, all degrees — and needs *no* hypothesis on ℓ.

Theorem 11.7 (Well-definedness)**.** *Assume (H4) and let $g, \{s_j\}, \tau, R$ be as in Definition 11.6. [proved here] **(No hypothesis on ℓ.)** [formally verified: [GT, `thm:welldef-lean`]] (equivariance and the chain-map property; the descent to H^q_{eq} is the generic one) $\widetilde{T}$ carries Γ-equivariant cochains to Γ-equivariant cochains and commutes with the coboundary; hence it descends to $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ for every q.*

Moreover the induced map on H^q_{eq} is independent of

- *(a) the choice of representatives s_j — [proved here] **with no hypothesis on** ℓ, and in fact at cochain level,* provided τ is held fixed*; [formally verified: [GT, `thm:welldef-lean`]] (at cochain level, τ fixed; since v1.19.0 for any two choices of representatives) [formally verified: [GT, `prop:fullassembly-model`]] (since v1.20.0 also for τ over R, on the model, at cochain level)*
- *(b) the choice of τ — [proved here] under $6 \in R^\times$ [formally verified: [GT, `thm:welldef-b`]] (on the model, the representatives held fixed; τ, τ' over R, as R-linear maps of $C_q(X) \otimes R$ satisfying (T1)–(T3); on $H^n_{\mathrm{eq}}(K, \mathcal{R}; L)$ itself, every $n \ge 0$) (hypothesis (H3); via the* uniqueness *half Lemma 11.3(2), which is where the hypothesis is used and the only place — existence, Lemma 11.3(1), needs none);*
- *(c) the choice of g inside $\Gamma g \Gamma$ — [proved here] [formally verified: [GT, `thm:welldef-c`]] (since v1.19.0 the whole of (c): the transported half with no hypothesis on ℓ; the general case on the model, under $6 \in R^\times$, with τ_1 over $\mathbb{Z}$) at cochain level for the* transported *choices constructed in the proof, with no hypothesis on ℓ; and for arbitrary admissible choices under $6 \in R^\times$, via (a) and (b).*

None of (a)–(c) is claimed to fix the cochain map $\widetilde{T}$ itself: each is a statement about H^q_{eq}, and Remark 11.9 says exactly what fails at cochain level and what a validation may therefore compare.

Proof. Equivariance. Let $\gamma \in \Gamma$. Write $s_j\gamma = h_j(\gamma)\, s_{j\cdot\gamma}$ with $h_j(\gamma) \in \Gamma''$ as in (17) (Lemma 10.3). Using (T1), Γ-equivariance of ϕ, and $g^{-1}h_j(\gamma)g \in \Gamma' \subseteq \Gamma$:

$$\rho_n(\delta_j)\,\phi\big(\tau(s_j\gamma x)\big) = \rho_n(\delta_j)\,\rho_n\big(g^{-1}h_j(\gamma)g\big)\,\phi\big(\tau(s_{j\cdot\gamma}x)\big) = \rho_n\big(s_j^{-1}h_j(\gamma)g\big)\,\phi\big(\tau(s_{j\cdot\gamma}x)\big),$$

by Proposition 8.2(2) and $\delta_j\, g^{-1}h_j(\gamma)g = s_j^{-1}h_j(\gamma)g$ as an identity in $\mathrm{M}_2(\mathbb{Z})$. Since $s_j^{-1}h_j(\gamma) = \gamma s_{j\cdot\gamma}^{-1}$ by (17), the right-hand side is $\rho_n(\gamma)\,\rho_n(\delta_{j\cdot\gamma})\,\phi\big(\tau(s_{j\cdot\gamma}x)\big)$. Summing over j and using that $j \mapsto j\cdot\gamma$ is a bijection (Lemma 10.3) gives $(\widetilde{T}\phi)(\gamma x) = \rho_n(\gamma)(\widetilde{T}\phi)(x)$.

Chain map. With $(d\phi)(x) = \phi(\partial x)$, property (T2) and additivity give $\widetilde{T}(d\phi) = d(\widetilde{T}\phi)$ term by term.

(a). The general change of admissible representatives is $s_j \mapsto h_j s_j$ with $h_j \in \Gamma''$. Holding τ fixed, the j-th summand of (21) becomes

$$\rho_n\big(s_j^{-1}h_j^{-1}g\big)\,\phi\big(\tau(h_j s_j x)\big) = \rho_n\big(s_j^{-1}h_j^{-1}g\big)\,\rho_n\big(g^{-1}h_j g\big)\,\phi\big(\tau(s_j x)\big) = \rho_n(\delta_j)\,\phi\big(\tau(s_j x)\big),$$

unchanged summand by summand; so for fixed τ the cochain map is literally the same map. **The proviso is not cosmetic.** In an admissible construction τ is built *relative to* the chosen s_j (Algorithm 18.3 stores one chain per pair (j,σ) and derives the rest through (T1)), so a change of representatives rebuilds τ; the summand-by-summand invariance then does not apply and only the induced map on H^q_{eq} is invariant, via (b). This is why (a) is stated at the H^q_{eq} level.

(b). Let $\tau' = \tau + \partial H + H\partial$ with H as in Lemma 11.3(2), and set $(\widetilde{H}\phi)(x) := \sum_j \rho_n(\delta_j)\phi\big(H(s_j x)\big)$. Then $\widetilde{T}' - \widetilde{T} = d\circ\widetilde{H} + \widetilde{H}\circ d$, and $\widetilde{H}\phi$ is Γ-equivariant by the computation above, which used only property (T1) of its argument — which H has. So the two cochain maps are cochain homotopic and induce the same map on $H^\bullet_{\mathrm{eq}}$.

(c). Let $g_1 = \gamma_1 g \gamma_2$ with $\gamma_1, \gamma_2 \in \Gamma$. Then $\Gamma_1'' := \Gamma \cap g_1\Gamma g_1^{-1} = \gamma_1\Gamma''\gamma_1^{-1}$, the elements $\gamma_1 s_j$ represent $\Gamma_1''\backslash\Gamma$ ($\bigsqcup_j \Gamma_1''\gamma_1 s_j = \gamma_1 \bigsqcup_j \Gamma'' s_j = \Gamma$), with $(\gamma_1 s_j)^{-1}g_1 = \delta_j\gamma_2$, and $\tau_1 := \gamma_2^{-1}\cdot\tau(\gamma_1^{-1}\cdot(-))$ is g_1-twisted: for $h_1 \in \Gamma_1''$, using $\gamma_1^{-1}h_1\gamma_1 \in \Gamma''$ and $g_1^{-1}h_1g_1 = \gamma_2^{-1}\big(g^{-1}\gamma_1^{-1}h_1\gamma_1 g\big)\gamma_2$,

$$\tau_1(h_1 x) = \gamma_2^{-1}\cdot\tau\big((\gamma_1^{-1}h_1\gamma_1)\,\gamma_1^{-1}x\big) = \gamma_2^{-1}\big(g^{-1}\gamma_1^{-1}h_1\gamma_1 g\big)\cdot\tau(\gamma_1^{-1}x) = (g_1^{-1}h_1g_1)\cdot\tau_1(x),$$

and (T2), (T3) are inherited. Substituting $(g_1, \{\gamma_1 s_j\}, \tau_1)$ into (21) reproduces the operator for $(g, \{s_j\}, \tau)$ verbatim: $\rho_n(\delta_j\gamma_2)\,\phi\big(\tau_1(\gamma_1 s_j x)\big) = \rho_n(\delta_j)\rho_n(\gamma_2)\,\phi\big(\gamma_2^{-1}\cdot\tau(s_j x)\big) = \rho_n(\delta_j)\,\phi\big(\tau(s_j x)\big)$, γ_1 cancelling inside τ_1's argument and $\rho_n(\gamma_2)$ against Γ-equivariance of ϕ. That identity is unconditional. (**Erratum, v1.18.0**, confirmed by the owner on 2026-10-03; found by slot-18 L18-2, the Lean proof `GL2.Model.thetaG_transported`: through v1.18.0-pre.1 this paragraph took $\{s_j\gamma_1^{-1}\}$ as the representatives of $\Gamma_1''\backslash\Gamma$ — not a transversal in general (its cosets $\Gamma_1'' s_j\gamma_1^{-1} = \gamma_1\big(\Gamma''\gamma_1^{-1}s_j\big)\gamma_1^{-1}$ are distinct iff $\{\gamma_1^{-1}s_j\}$ represents $\Gamma''\backslash\Gamma$, which fails in general), and with τ_1 its summands read $\tau(\gamma_1^{-1}s_j\gamma_1^{-1}x)$, not $\tau(s_j x)$ — and said that $\rho_n(\gamma_1)$ and $\rho_n(\gamma_2)$ both cancel against equivariance. The statement (c) and its conclusion are unchanged. The slot-18 Lean audit F3 confirmed the erratum with a counterexample: $\Gamma'' = \Gamma_0(p)$, $s_j = \left(\begin{smallmatrix}1 & 0\\ j & 1\end{smallmatrix}\right)$ $(j < p)$, $s_p = S$, $\gamma_1 = S$; the class of $\Gamma_0(p)x$ is the bottom row of x in $\mathbb{P}^1(\mathbb{F}_p)$, and $(-1,0)\left(\begin{smallmatrix}1 & 0\\ j & 1\end{smallmatrix}\right) = (-1,0)$ for every j, so the p elements $\gamma_1^{-1}s_j$, $j < p$, lie in one coset.) To pass from it to independence for an *arbitrary* admissible datum at g_1 one must compare that datum with the transported one, which is (a) and (b) — hence the hypothesis $6 \in R^\times$ in the general case. □

Remark 11.8 (What (T3) is for). Property (T3) is used in degree 0 and nowhere else, and it is what pins the classical normalisation there. A 0-cocycle corresponds to ϕ with $\phi(\tilde{v}) = v$ for every vertex $\tilde{v}$ of K^u and a fixed $v \in V_R^\Gamma$; (T3) forces $\phi(\tau(s_j\tilde{v})) = v$, so (21) gives

$$T_{\Gamma g\Gamma}\, v = \sum_{j=1}^{m} \rho_n(\delta_j)\, v,$$

the standard double-coset action on invariants — for $n = 0$, multiplication by m, e.g. $(p+1)v$ for $g = \mathrm{diag}(1, p)$. Without (T3) the degree-0 operator would still be well defined but would not be normalised, and the H^0 eigenvalue oracles of §21 would be off by the augmentation defect.

Remark 11.9 (The level at which each independence holds). All three independence claims of Theorem 11.7 — coset representatives (a), twisted datum (b), double-coset representative (c) — are claims about *cohomology*: the induced matrix $A^{(q)}$ on H^q_{eq} is invariant, bit for bit in every tested setting whenever $6 \in R^\times$. The cochain map Θ_g and its flat matrix $\Theta^{(q)}_{\mathrm{flat}}$ are *not* invariant under any of them: $\Theta^{(q)}_{\mathrm{flat}}$ depends on the chosen τ, and in an admissible construction τ is coupled to the s_j (Algorithm 18.3 builds τ from the representatives), so even a relabelling $s_j \mapsto h_j s_j$ with a non-trivial $h_j \in \Gamma_0(N)$ rebuilds τ and in general changes $\Theta^{(q)}_{\mathrm{flat}}$. The summand-by-summand invariance in the proof of (a) applies only when τ is held fixed, which the coupled construction does not do. Consequently:

- **Variants in s_j, τ or g are compared at the $A^{(q)}$ level, never bit for bit at cochain level.** A validation that compares $\Theta^{(q)}_{\mathrm{flat}}$ across such a variant is testing a statement this document does not make.
- **Cochain-level bit-identity is asserted only between genuinely identical constructions**: the trivial relabelling $h_j = I$ (same s_j, same τ); the two evaluation routes $\omega_\bullet \circ \Xi_g$ and direct (21) on the *same* data (Obligation 20.5); $\omega_\bullet \circ \mu_\bullet \circ \alpha_\bullet$ against Θ_g (Proposition 13.2(3)), because $\mu_\bullet$ and $\alpha_\bullet$ are likewise built from the *same* g, the *same* s_j and the *same* τ-table (Obligation 20.5); and the `HECKE_PURE_INTEGER` on-versus-off determinism check (Obligation 20.9).

This remark is a statement about what may be asserted, not a diagnostic to branch on (rule (A6)).

11.4 Comparison with the classical operator

Theorem 11.10 (Comparison with the classical operator). *Assume (H4) and $6 \in R^\times$ — for R of characteristic ℓ, $\ell \geq 5$ (hypothesis (H3)). Then under the isomorphisms $H^q_{\mathrm{eq}}(K, \mathcal{R}; L) \cong H^q(\Gamma; V_R)$ of Theorem 5.10 and Remark 7.6, the operator of Definition 11.6 equals (20): it is the standard double-coset Hecke operator of the theory of arithmetic groups [standard: [Bro82, Ch. III.9–10] for* res, cor, *the conjugation action and the transfer formula; [AS86, §1] and [Har22, Ch. 2–3] for the double-coset operator on the cohomology of arithmetic groups; [Shi71, Ch. 8] and [Den23] for the classical modular-forms normalisation]. [formally verified: [GT,* `thm:m6-classical-theta`*]] [formally verified: [GT,* `thm:m6-Tcls-res`*]] [formally verified: [GT,* `thm:m6-hecke-res`*]] (on the 7-vertex model, for every commutative ring with $6 \in R^\times$, every representation ρ of Δ (in particular ρ_n), every $g \in \Delta$, coset datum, τ and degree: $\Theta \circ T_{\Gamma g \Gamma} = \mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res} \circ \Theta$ (the right side is M5's* `Tcls`*, identified with $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ by [GT]* `thm:m6-Tcls-res`*) with Θ the comparison of Remark 7.6 (mathlib's* Ext*-built one, equal to the comparison induced by any lift of the identity, [GT]* `thm:m6-theta-ext`*); the operator does not depend on the projective resolution or on τ; the classical modular-forms normalisation of Shimura / Deng is not formalised; v1.24.0-rc.1)*

Proof sketch. The hypothesis $6 \in R^\times$ is used twice, and it is worth separating the uses. First, it makes $C_\bullet(K^u) \otimes R$ a complex of relatively projective $R[\Gamma]$-modules (the computation in the proof of Lemma 11.3(2)), so that this one complex computes $H^\bullet$ of Γ *and* of each of its finite-index subgroups Γ', Γ'', and that $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L) \cong H^\bullet(\Gamma; V_R)$ at all (Remark 7.6). Second, it supplies the homotopy uniqueness of Lemma 11.3(2), which is what makes the middle map below independent of the choice of τ. *Neither* use is met by Lemma 11.3(1), so this theorem does not weaken (Remark 11.4(2)). On that resolution the three constituents of (20) are:

- $\mathrm{res}^\Gamma_{\Gamma'}$ is the inclusion $\mathrm{Hom}_\Gamma(C_\bullet, V_R) \hookrightarrow \mathrm{Hom}_{\Gamma'}(C_\bullet, V_R)$;

- $(\mathrm{conj}_g)_*\colon \mathrm{Hom}_{\Gamma'}(C_\bullet, V_R) \to \mathrm{Hom}_{\Gamma''}(C_\bullet, V_R)$ is $\psi \mapsto \rho_n(g) \circ \psi \circ \tau$: by (T1), τ is exactly a chain map over $\mathrm{conj}_g^{-1}\colon \Gamma'' \to \Gamma'$, so $\rho_n(g)\psi\tau$ is Γ''-equivariant by the computation in the proof of Theorem 11.7, and by Lemma 11.3(2) any two choices of τ give homotopic maps, i.e. the map induced in cohomology is the canonical one;
- $\mathrm{cor}_{\Gamma''}^{\Gamma}$ is the transfer $\psi \mapsto \big(x \mapsto \sum_j \rho_n(s_j)^{-1}\psi(s_j x)\big)$ [standard: [Bro82, Ch. III.9]], well defined and independent of the s_j by the computation in the proof of Theorem 11.7(a) with g replaced by I.

Composing, for Γ-equivariant ϕ,

$$x \;\longmapsto\; \sum_j \rho_n(s_j)^{-1}\rho_n(g)\,\phi\big(\tau(s_j x)\big) \;=\; \sum_j \rho_n(s_j^{-1}g)\,\phi\big(\tau(s_j x)\big) \;=\; (\widetilde{T}\phi)(x),$$

which is (21); the middle equality is Proposition 8.2(2) together with $\rho_n(s_j)^{-1} = \rho_n(s_j^{-1})$, legitimate because $s_j \in \Gamma$ is unimodular. □

Remark 11.11 (Which object of §10 this chapter defines). Definition 11.6 defines $T_{\Gamma g \Gamma}$ for *one* double coset. Fed $g = \mathrm{diag}(1, p^\nu)$ — which is what the coset tables of Definition 10.2 enumerate — it therefore produces the **primitive** operator $T_{\Delta_{p^\nu}^{\mathrm{prim}}}$, and for $\nu \geq 2$ that is *not* T_{p^ν}: the full operator is the sum (18), assembled by (19) (Remark 10.7; Appendix A, entry A.1). At $\nu = 1$, and hence for T_p and U_ℓ, the distinction is empty.

Code map. Definition 11.1, Lemma 11.3, Remark 11.5: `Hecke/HeckeTauTable.h/.cpp` — the stored `ChainTerm` triples $(n_t, \gamma_t, \sigma_t)$, one chain per pair (j, σ), with every Γ''-related slot derived through the single shared (T1) helper `eval_tau_on_translate` used by *both* construction and certification; the solve regime is recorded per chain and reported by `used_mod_ell_solve()`. The constructor certifies (Obligation 20.11) and throws on failure, so an uncertified table cannot exist; this is the implementation's invariant I8, "certification over geometry", and it is the code-level form of the paragraph after Definition 11.1. Gated by `HeckeTauTableValidation`, `HeckeTauTableSelfCheckValidation`. Definition 11.6, (21): `Hecke/HeckeOperatorOnK.h/.cpp`, whose fused route $\Theta_g = \omega_\bullet \circ \Xi_g$ and whose second, independently bracketed route evaluate (21) and are asserted bit-identical (§12, Obligation 20.5); the induced $A^{(q)}$ is `get_H_matrices()`. Theorem 11.7 and Remark 11.9: `HeckeIndependenceValidation`, which varies the s_j (through `HeckeCosetTables`' custom-representative constructor), τ and g and compares at the $A^{(q)}$ level only — exactly as Remark 11.9 requires. Theorem 11.10: cross-checked, not assumed, by `HeckeCrossChecks` and the bar-cocycle route of §13 (Obligation 20.5); the $\ell \in \{2, 3\}$ arms make no cross-τ and no classical-comparison assertion, per the contract box after Lemma 11.3.

Formalization note. Definition 11.1 is three equational conditions on a map of complexes and is directly expressible. The two halves of Lemma 11.3 want different treatments. For (2) the useful reformulation is the one used in its proof — τ is a morphism of augmented complexes of $R[\Gamma'']$-modules $C_\bullet \to C_\bullet^{(c)}$ — since `mathlib4`'s homological algebra then supplies the comparison theorem; there the one hypothesis a formalisation must not lose is the source's relative projectivity, whose entire content here is that 4 and 6 — the orders of the elliptic stabilisers, (H4)(iv) — are units, and stating (2) for a general R with $6 \in R^\times$, rather than for $\mathbb{F}_\ell$ with $\ell \geq 5$, makes that visible in the statement instead of in the proof. Clause (1) needs *none* of that machinery: it is a terminating construction over $\mathbb{Z}$ on a finite set of orbit representatives whose only non-combinatorial inputs are (37) and the elliptic-point classification of [Ser73], and it is therefore the cheaper of the two to formalise and the one to do first. Definition 11.6 is a finite sum over a finite index set and is constructive given the tables of Definition 10.2 and Algorithm 18.3; the equivariance half of Theorem 11.7 is a finite manipulation using only Lemma 10.3 and Proposition 8.2(2), needs no hypothesis on ℓ, and is the natural first target of

Part III. Parts (b) and (c) of Theorem 11.7 and all of Theorem 11.10 depend on the comparison theorem — through Lemma 11.3(2), never (1) — and should be attempted only after it. Obligation 20.11 is already a decidable predicate on emitted finite data, so it needs no idealisation to formalise — which is the shape rule (A4) asks every obligation to have.

12 Induced local systems and the maps α, μ, ω

What this chapter fixes. The same operator as §11, re-organised as a computation on the *fixed* complex $(K, \mathcal{R})$ — which is what makes it executable, since `EquivariantCohomologyOnK` runs on $(K, \mathcal{R})$ and on nothing else. Three things are fixed here: the *induced* matrix local system L_H, which is the Shapiro-theoretic home of the coset bookkeeping; the two honest morphisms of local systems α (restriction unit) and ω (corestriction counit); and the **binding** fused cochain map

$$\Theta_g \;=\; \omega_\bullet \circ \Xi_g,$$

with its closed form (25) and the assembly of the cohomology-level matrix $A^{(q)}$ (Algorithm 12.13). Between the second and the third sits the one place where the superseded framework document was not merely imprecise but wrong: there is *no* middle morphism of local systems, by a Schur argument, and Remark 12.9 states the no-go rather than repairing the formula. The recovered three-map factorisation — which exists, at cochain level, and is a cross-check rather than a definition — is §13.

Standing hypotheses. (H1)–(H4) of §7, each repeated where it is used (rule (A1)). *No construction* of §§12.1–12.4 needs a hypothesis on ℓ: the induced module, its block matrices, α, ω, Ξ_g and Θ_g are built over an arbitrary commutative coefficient ring R and are integral; so is every well-definedness statement about them. The hypothesis $6 \in R^\times$ ((H3)) enters at exactly two points, both flagged in their own statements: the Shapiro identification of Proposition 12.4(4), and the identification of the induced map with $T_{\Gamma g \Gamma}$ in Theorem 12.11(3) — which inherits it from Theorem 11.10 and not from anything in this chapter.

Notation used throughout. The coset index and the induced-module block index are both $j \in \{1, \dots, m\}$ and the chain-term index is t (never i: §2.7, C15); the distinguished lift of $\sigma \in o_q(K)$ into K^u is $\widetilde{\sigma}$, which the implementation spells `i(sigma)` (C16). **A relation of $\mathcal{R}$ is written** $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma)$ **with** $\gamma \in \Gamma$, matching Definition 7.2: in this chapter $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$ is the Hecke element of §10.2, and the two symbols appear in one formula (Theorem 12.11(1)) — §2.7, C17. Every relation element of the model does lie in Γ (Definition 7.2: T, S, ST, TS, $-I$).

Parametric in the triangulation. Every statement of this chapter is stated for an arbitrary finite $(K, \mathcal{R})$ satisfying (H4), never for the seven vertices of Definition 7.1: the input is a simplex set, a relation set whose elements are group elements, and the coset datum of §10, which is a statement about $\mathbb{P}^1(\mathbb{Z}/N)$ and is independent of the triangulation altogether. A refinement of the model (Remark 7.6, owner direction OD-T1) is therefore admissible input here with no change to any definition, proof or algorithm; what a refinement changes is the numbers N_q and the τ-table, not this chapter.

12.1 The induced module and its block matrices

Definition 12.1 (Induced module, ordered basis)**.** Let R be a commutative ring, $H \leq \Gamma$ a subgroup of finite index m, and fix representatives $w_1, \dots, w_m \in \Gamma$ of $H\backslash\Gamma$ with $w_1 = I$. The

induced module is

$$\mathrm{Ind}_H^\Gamma V_R := \big\{ f\colon \Gamma \to V_R \ :\ f(h\gamma) = \rho_n(h) f(\gamma) \ \ \forall h \in H,\ \gamma \in \Gamma \big\}, \qquad (\gamma \cdot f)(x) := f(x\gamma),$$

a left $R[\Gamma]$-module. Restriction to the representatives,

$$f \ \longleftrightarrow\ \big(f(w_1), \dots, f(w_m)\big) \in V_R^{\oplus m},$$

is an isomorphism of R-modules, and the *ordered basis* of $\mathrm{Ind}_H^\Gamma V_R$ is the concatenation, in the order $w_1, \dots, w_m$, of m copies of the ordered basis $(u_0, \dots, u_n)$ of V_R — so the coordinate of block j and in-block index k is $(j-1)(n+1)+k$, and $\dim \mathrm{Ind}_H^\Gamma V_R = m(n+1) = [\Gamma : H] \cdot d_V$. [formally verified: [GT, `def:levelellmodel-model`]] (the module in the coordinates $(f(w_j))_j$, for any table of representatives)

The restriction map is bijective because every $\gamma \in \Gamma$ is uniquely hw_j with $h \in H$, so prescribing $f(w_j)$ prescribes $f(hw_j) = \rho_n(h) f(w_j)$ and no consistency condition remains; $\rho_n(h)$ is invertible for $h \in \Gamma$ by Proposition 8.2(2).

Proposition 12.2 (Block transition matrices)**.** *[proved here] Let R, H, m, (w_j) be as in Definition 12.1. For $\gamma \in \Gamma$ define the index $j \cdot \gamma$ and the element $h_j(\gamma) \in H$ by*

$$w_j\,\gamma \ = \ h_j(\gamma)\, w_{j\cdot\gamma}, \qquad j = 1, \dots, m.$$

Both are well defined; $(j, \gamma) \mapsto j \cdot \gamma$ is a right action of Γ on $\{1, \dots, m\}$, and

$$h_j(I) = I, \qquad h_j(\gamma_1\gamma_2) = h_j(\gamma_1)\, h_{j\cdot\gamma_1}(\gamma_2).$$

In the coordinates of Definition 12.1 the action of γ on $\mathrm{Ind}_H^\Gamma V_R$ is the block matrix

$$M_H(\gamma) \ = \ \sum_{j=1}^{m} E_{j,\, j\cdot\gamma} \otimes \rho_n\big(h_j(\gamma)\big) \qquad \textit{(block row } j\textit{, block column } j\cdot\gamma\textit{)}, \tag{22}$$

i.e. $(\gamma \cdot f)(w_j) = \rho_n(h_j(\gamma))\, f(w_{j\cdot\gamma})$; it has exactly one non-zero block per block row. The map $\gamma \mapsto M_H(\gamma)$ is a group homomorphism $\Gamma \to \mathrm{GL}_{m(n+1)}(R)$, and $M_H(\gamma^{-1}) = M_H(\gamma)^{-1}$. [formally verified: [GT, `def:levelellmodel-model`]] ((22) with $h_j(\gamma) = w_j \gamma w_{j\cdot\gamma}^{-1}$, the two cocycle identities and the homomorphism, for any table; "one non-zero block per block row" in the form of a product of maps)

Proof. Well-definedness and the cocycle identity are the argument of Lemma 10.3 with (Γ'', s_j) replaced by (H, w_j) and the class map replaced by the coset map $\gamma \mapsto H\gamma$: $Hw_j\gamma$ is one of the m cosets, which names $j\cdot\gamma$ uniquely, and $h_j(\gamma) = w_j\gamma w_{j\cdot\gamma}^{-1} \in H$ by construction; expanding $w_j\gamma_1\gamma_2$ in two ways and using uniqueness of the decomposition $\gamma = hw_j$ gives the action property and the cocycle identity. For (22),

$$(\gamma \cdot f)(w_j) = f(w_j\gamma) = f\big(h_j(\gamma) w_{j\cdot\gamma}\big) = \rho_n\big(h_j(\gamma)\big) f(w_{j\cdot\gamma}),$$

which is block row j of the displayed matrix. Homomorphy is then a computation in which the only input is multiplicativity of ρ_n (Proposition 8.2(2)): with $E_{j,a}E_{j',b} = \delta_{aj'}E_{j,b}$,

$$M_H(\gamma_1)M_H(\gamma_2) = \sum_j E_{j,\,(j\cdot\gamma_1)\cdot\gamma_2} \otimes \rho_n\big(h_j(\gamma_1)\big)\rho_n\big(h_{j\cdot\gamma_1}(\gamma_2)\big) = \sum_j E_{j,\,j\cdot(\gamma_1\gamma_2)} \otimes \rho_n\big(h_j(\gamma_1\gamma_2)\big),$$

which is $M_H(\gamma_1\gamma_2)$; and $M_H(I) = I$. Invertibility follows with $\gamma_2 = \gamma_1^{-1}$. □

CONTRACT. For the Hecke operator at p^ν the implementation instantiates Proposition 12.2 twice, and **both instances are the coset datum of §10, not a second construction**:

- $L'' := L_{\Gamma''}$, $\Gamma'' = \Gamma_0(N)$, with $w_j := s_j$ the canonical transporters of Definition 10.2. Then $j \cdot \gamma$ is the bottom-row $\mathbb{P}^1(\mathbb{Z}/N)$ index action and $h_j(\gamma) = s_j \gamma s_{j\cdot\gamma}^{-1}$ is the cocycle (17), both by Lemma 10.3. Only the four generator values $\gamma \in \{T, S, ST, TS\}$ and $\gamma = -I$ are ever needed (Definition 7.2).
- $L' := L_{\Gamma'}$, $\Gamma' = \Gamma^0(N)$, with $w_j := t_j$ the upper-triangular transporters of Definition 10.2 and the *top*-row class map $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \mapsto (a : b)$. This one is used only by α (Proposition 12.6) and by the cross-check of §13; the binding pipeline never constructs it.

Definition 12.3 (Induced matrix local system)**.** Let $L = (V_R, \{\rho_n(\gamma)\}_{(\cdot,\cdot,\gamma)\in\mathcal{R}})$ be the matrix local system of the model (Definition 4.1) and let $H \le \Gamma$ be of finite index m with representatives as in Definition 12.1. Then

$$L_H \;:=\; \Big(\mathrm{Ind}_H^\Gamma V_R,\; \big\{M_H(\gamma)\big\}_{(\cdot,\cdot,\gamma)\in\mathcal{R}}\Big)$$

is the matrix local system on $(K, \mathcal{R})$ obtained from L by replacing the matrix $\rho_n(\gamma)$ of each relation by $M_H(\gamma)$ of (22), **keeping the same relation list, the same simplex pairs** $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}})$ **and the same order**. Its representation dimension is $m(n+1)$.

Proposition 12.4 (L_H is a matrix local system; Shapiro)**.** *[proved here] Assume (H4) and let $H \le \Gamma$ be of finite index m, with L_H as in Definition 12.3 over a commutative ring R. Then:*

1. *every $M_H(\gamma)$ is invertible over R and the assignment $\gamma \mapsto M_H(\gamma)$ is a group homomorphism, so L_H satisfies the cocycle condition of Definition 4.1 and is a matrix local system on $(K, \mathcal{R})$ — with* no *hypothesis on ℓ;*
2. *consequently the data of Definition 12.3 is a valid* `LocalSystemSimplicialComplex` *input (each relation is again of the single-matrix form $f(\sigma_{\mathrm{lhs}}) = M\, f(\sigma_{\mathrm{rhs}})$ over R), and* `EquivariantCohomologyOnK` *computes $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L_H)$* unchanged*;*
3. *if $-I \in H$ — which holds for $H \in \{\Gamma_0(N), \Gamma^0(N)\}$ — then the central relations $(\sigma, \sigma, -I)$ of Definition 7.2 are carried by*

$$M_H(-I) \;=\; I_m \otimes \rho_n(-I) \;=\; (-1)^n\, I_{m(n+1)},$$

   so they impose $f \equiv 0$ for odd n on every block exactly as they do for L (Remark 7.7);
4. *if moreover $6 \in R^\times$ — for R of characteristic ℓ, $\ell \ge 5$ (hypothesis (H3)) — then*

$$H^q_{\mathrm{eq}}(K, \mathcal{R}; L_H) \;\cong\; H^q\big(\Gamma; \mathrm{Ind}_H^\Gamma V_R\big) \;\cong\; H^q(H; V_R) \qquad \text{for every } q,$$

   the first isomorphism by Theorem 5.10 and Remark 7.6 — comparison of the projective $R[H]$-resolution $C_\bullet(X) \otimes R$ with any projective resolution (v1.16.0-pre.3) —, the second by Shapiro's lemma [standard: [Bro82, Ch. III.6.2 and Exercise III.8.2]]. For $6 \notin R^\times$ ***neither isomorphism is asserted*** *and the left-hand side is the orbifold model cohomology of (H3). [formally verified: [GT,* `def:levelellmodel-model`*]] (since v1.20.0: both isomorphisms for $6 \in R^\times$, on the model, for any $H \le \Gamma$ of finite index with a table of representatives, $\Gamma_0(N)$ included; not (1)–(3))*

Proof. (1) is Proposition 12.2. The cocycle condition of Definition 4.1 asks that every relation among the relation elements that holds in the group presented by $(K, \mathcal{R})$ — which is Γ, by Theorem 3.24 and (H4)(iii) — be satisfied by the assigned matrices; a group homomorphism

satisfies every such relation, so there is nothing to check relation by relation. This is the induced-system half of Proposition 4.3. (2) is (1) plus the observation that (22) is a single square matrix of size $m(n+1)$, which is the format `SimplexRelation` accepts; the relation list is unchanged by construction. (3) $-I$ is central in $\mathrm{M}_2(\mathbb{Z})$ and lies in H, so $w_j(-I) = (-I)w_j$ gives $j \cdot (-I) = j$ and $h_j(-I) = w_j(-I)w_j^{-1} = -I$; (22) then reads $M_H(-I) = I_m \otimes \rho_n(-I)$, and $\rho_n(-I)u_k = (-x)^{n-k}(-y)^k = (-1)^n u_k$ by Definition 8.1. (4) The first isomorphism is Theorem 5.10 applied to the local system L_H — whose hypotheses are (H4), not a property of the coefficient module — followed by Remark 7.6, whose *only* use of $6 \in R^\times$ is the invertibility of the simplex-stabiliser orders, again independent of the coefficient module. The second is Shapiro's lemma for the finite-index subgroup H. □

Remark 12.5 (What Proposition 12.4(4) is for, and what it is not). Part (4) is an *oracle*, not a step of the pipeline: it predicts $\dim H^q_{\mathrm{eq}}(K, \mathcal{R}; L_H)$ from classical data about the modular curve $X_0(N)$ and is checked against the computed dimension (Obligation 20.6). The binding construction of Θ_g (Definition 12.10) never builds L' or L'' and never computes the cohomology of either. This is why the whole of §12.1 is, for the pipeline, a *coordinate system* rather than a computation.

12.2 α and ω: two honest morphisms of local systems

Proposition 12.6 (Restriction unit α). *[proved here] Assume (H4), let R be a commutative ring, $N = p^\nu$, $\Gamma' = \Gamma^0(N)$ of index m with the upper-triangular representatives $t_1 = I, \dots, t_m$ of Definition 10.2. Then*

$$\alpha \colon V_R \longrightarrow \mathrm{Ind}_{\Gamma'}^{\Gamma} V_R, \qquad \alpha(v)(\gamma) := \rho_n(\gamma)\, v,$$

is a morphism of $R[\Gamma]$-modules; in the coordinates of Definition 12.1 it is the block column $\big(\rho_n(t_j)\big)_{j=1,\dots,m}$. *Equivalently (Theorem 4.5) it is a morphism of matrix local systems $L \to L'$, i.e. $\alpha\, \rho_n(\gamma) = M_{\Gamma'}(\gamma)\, \alpha$ for every relation element γ of $\mathcal{R}$ — with* no *hypothesis on ℓ. If moreover $6 \in R^\times$, then under the identifications of Proposition 12.4(4) the induced map on cohomology is* $\mathrm{res}^{\Gamma}_{\Gamma'}$ *[standard: [Bro82, Ch. III.6.2 and Exercise III.8.2]].*

Proof. $\alpha(v)$ lies in the induced module: $\alpha(v)(h\gamma) = \rho_n(h\gamma)v = \rho_n(h)\alpha(v)(\gamma)$ by Proposition 8.2(2). Equivariance: $\alpha(\rho_n(\gamma)v)(x) = \rho_n(x\gamma)v = \alpha(v)(x\gamma) = (\gamma \cdot \alpha(v))(x)$, again by Proposition 8.2(2). Coordinates: $\alpha(v)(t_j) = \rho_n(t_j)v$, which is the stated block column. The local-system reformulation is the equivalence **Rep** $\simeq$ **LS** of Theorem 4.5, applicable because every relation element of $\mathcal{R}$ lies in Γ. The Shapiro statement identifies the unit of the restriction–induction adjunction with res. □

Proposition 12.7 (Corestriction counit ω). *[proved here] Assume (H4), let R be a commutative ring, $N = p^\nu$, $\Gamma'' = \Gamma_0(N)$ of index m, and let $s_1, \dots, s_m \in \Gamma$ represent $\Gamma''\backslash\Gamma$ (canonically Definition 10.2). Then the trace map*

$$\omega \colon \mathrm{Ind}_{\Gamma''}^{\Gamma} V_R \longrightarrow V_R, \qquad \omega(f) := \sum_{j=1}^{m} \rho_n(s_j)^{-1} f(s_j) = \sum_{j=1}^{m} \rho_n(s_j^{-1}) f(s_j),$$

is independent of the choice of the s_j and is a morphism of $R[\Gamma]$-modules; in the coordinates of Definition 12.1 it is the block row $\big(\rho_n(s_j^{-1})\big)_{j=1,\dots,m}$. *Equivalently (Theorem 4.5) it is a morphism of matrix local systems $L'' \to L$ — with* no *hypothesis on ℓ. Every s_j lies in Γ, so $\rho_n(s_j^{-1})$ is integral and invertible over every R and* ***no determinant-N matrix is inverted*** *(Obligation 20.3). If moreover $6 \in R^\times$, the induced map on cohomology is the transfer* $\mathrm{cor}^{\Gamma}_{\Gamma''}$ *[standard: [Bro82, Ch. III.9–10]].*

Proof. Independence: replacing s_j by hs_j with $h \in \Gamma''$ replaces $f(s_j)$ by $\rho_n(h)f(s_j)$ and $\rho_n(s_j^{-1})$ by $\rho_n(s_j^{-1})\rho_n(h)^{-1}$, by Proposition 8.2(2), so the summand is unchanged. Equivariance: with $s_j\gamma = h_j(\gamma)s_{j\cdot\gamma}$ and hence $s_j^{-1}h_j(\gamma) = \gamma s_{j\cdot\gamma}^{-1}$ (Lemma 10.3),

$$\omega(\gamma\cdot f) = \sum_j \rho_n(s_j^{-1})f(s_j\gamma) = \sum_j \rho_n(s_j^{-1})\rho_n\big(h_j(\gamma)\big)f(s_{j\cdot\gamma}) = \rho_n(\gamma)\sum_j \rho_n(s_{j\cdot\gamma}^{-1})f(s_{j\cdot\gamma}) = \rho_n(\gamma)\,\omega(f),$$

using Proposition 8.2(2) twice and that $j \mapsto j\cdot\gamma$ is a bijection. The coordinate description and the local-system reformulation are as in Proposition 12.6. The identification with cor is the transfer formula on the resolution $C_\bullet(K^u)\otimes R$, which is the computation already made in the proof of Theorem 11.10. □

Remark 12.8 (The corestriction convention, stated once). The superseded framework document writes the corestriction as $\omega_*(\psi) = \sum_\gamma \rho_L(\gamma)\,\psi(\gamma^{-1})$, the sum over representatives γ of $\Gamma''\backslash\Gamma$ (`monodromy_v5.tex`:1536). That formula and the one in Proposition 12.7 agree, but *only* after the substitution $\gamma = s_j^{-1}$ — which also replaces the index set $\{s_j\}$ of $\Gamma''\backslash\Gamma$ by the set $\{s_j^{-1}\}$ of representatives of Γ/Γ''. The substitution is recorded here because it is unmarked at the source, where a transcriber meets it first (Appendix A, entry A.9). This document uses the form of Proposition 12.7 and no other.

12.3 The middle map: a no-go, not a repair

Remark 12.9 (The middle map μ is not a morphism of local systems). [proved here] The superseded framework document `monodromy_v5.tex` §12.5 proposes a "conjugation" map $\mu_*(\phi)(\gamma) := \rho_L(g)\,\phi(g^{-1}\gamma g)$ — it calls it β; here β_j is a right-coset representative (§2.7, C8) — as a morphism $\mathrm{Ind}_{\Gamma'}^{\Gamma}V_R \to \mathrm{Ind}_{\Gamma''}^{\Gamma}V_R$, and builds the Hecke operator as $\omega\circ\mu\circ\alpha$ (its §§12.6–12.7). This is **not correct as stated**, for two independent reasons, the second of which forecloses every repair.

(i) Ill-typed. An element of $\mathrm{Ind}_{\Gamma'}^{\Gamma}V_R$ is a function on all of Γ, but $g^{-1}\gamma g \in \Gamma$ only for $\gamma \in \Gamma''$ (Proposition 10.1(3)); the formula does not define a function on Γ, and no Γ-equivariant extension of it is specified.

(ii) No morphism of local systems can do the job. Let $R = \mathbb{Q}$ and $V = \mathrm{Sym}^{10}\mathbb{Q}^2$. *Any* composite of three morphisms of $\mathbb{Q}[\Gamma]$-modules

$$V \xrightarrow{\ \alpha\ } \mathrm{Ind}_{\Gamma'}^{\Gamma}V \xrightarrow{\ \mu\ } \mathrm{Ind}_{\Gamma''}^{\Gamma}V \xrightarrow{\ \omega\ } V$$

is an element $u \in \mathrm{End}_\Gamma(V)$, and the map it induces on $H^\bullet(\Gamma;V)$ is u_*. Now $\mathrm{Sym}^{10}\mathbb{Q}^2$ is an irreducible Γ-module, because Γ is Zariski dense in SL_2 [standard: [Ser73, Ch. VII, §1] for the density of $\mathrm{SL}_2(\mathbb{Z})$, and the irreducibility of Sym^n for SL_2 in characteristic 0], so $\mathrm{End}_\Gamma(V) = \mathbb{Q}$ and u_* is a *scalar* in every degree. But T_2 on $H^1(\mathrm{SL}_2(\mathbb{Z});\mathrm{Sym}^{10}\mathbb{Q}^2)$ has eigenvalues $-24, -24, 2^{11}+1 = 2049$ [standard: Eichler–Shimura in weight $k = n+2 = 12$, [Shi71, Ch. 8]; Obligation 20.7], which is not scalar. Hence **no** morphism of the two induced local systems whatsoever — in particular no repair of the displayed formula — realises T_p through $\omega\circ\mu\circ\alpha$.

Resolution adopted by this document. The pipeline does not factor the operator through a third local system at all: it is the fused cochain map $\Theta_g = \omega_\bullet\circ\Xi_g$ of Definition 12.10, which needs neither Γ' nor L' nor any μ. A middle map *does* exist one level down, as a cochain map given by per-simplex block rows rather than by one fibre matrix; it is constructed in Proposition 13.2 and used only as a cross-check. The distinction is exactly the one (ii) makes: a morphism of local systems is a single fibre matrix applied at every simplex, and that is what cannot exist. Recorded as Appendix A, entry A.19.

12.4 The binding cochain map Ξ_g and Θ_g

Throughout this subsection $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, $N = p^\nu$, $\Gamma'' = \Gamma_0(N)$ of index m, $s_1, \dots, s_m$ and $\delta_j = s_j^{-1} g$ are as in Definition 10.2, and τ is a g-twisted chain datum over R (Definition 11.1).

Assume (H4). For $c \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ write ϕ_c for the corresponding Γ-equivariant cochain on K^u under the isomorphism Ψ of Theorem 5.10:

$$\phi_c\big(\gamma \cdot \widetilde{\sigma}\big) \;=\; \rho_n(\gamma)\, c(\sigma), \qquad \gamma \in \Gamma,\ \sigma \in o_q(K), \tag{23}$$

extended R-linearly to $C_q(K^u) \otimes R$. **Two hypotheses make (23) a definition rather than a wish, and both are used below.** First, $K \subseteq K^u$ is a *strict* fundamental domain ((H4)(iii)), so every simplex of K^u is $\gamma \cdot \widetilde{\sigma}$ for a unique σ; second, if $\gamma_1 \widetilde{\sigma} = \gamma_2 \widetilde{\sigma}$ then $\gamma_2^{-1}\gamma_1$ lies in the stabiliser $\Gamma_{\widetilde{\sigma}}$, and the right-hand side is unambiguous *because* c satisfies the stabiliser relations and the central relation of Definition 7.2 — which is precisely what "equivariant" means for $(K, \mathcal{R})$ and is the content of Theorem 5.10. Consistency with the coboundary, $\phi_{dc} = d\,\phi_c$ where $(d\phi)(x) = \phi(\partial x)$, is part of the same theorem.

Definition 12.10 (The map Ξ_g and the fused map Θ_g)**.** Assume (H4). For $c \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ define $\Xi_g(c) \in C^q\big(K; \mathrm{Ind}_{\Gamma''}^{\Gamma} V_R\big)$ blockwise by

$$\big(\Xi_g c\big)(\sigma)_j \;:=\; \rho_n(g)\ \phi_c\Big(\tau\big(s_j \cdot \widetilde{\sigma}\big)\Big) \;\in\; V_R, \qquad j = 1, \dots, m, \quad \sigma \in o_q(K). \tag{24}$$

Explicitly, and this is the form the implementation evaluates: the table of Algorithm 18.3 stores, for each pair (j, σ), the finite chain

$$\tau\big(s_j \cdot \widetilde{\sigma}\big) \;=\; \sum_t n_t^{(j,\sigma)}\ \gamma_t^{(j,\sigma)} \cdot \widetilde{\sigma_t^{(j,\sigma)}}, \qquad n_t \in \mathbb{Z},\ \gamma_t \in \Gamma,\ \sigma_t \in o_q(K),$$

whence, by (23) and Proposition 8.2(2),

$$\big(\Xi_g c\big)(\sigma)_j \;=\; \sum_t n_t^{(j,\sigma)}\ \rho_n\big(g\,\gamma_t^{(j,\sigma)}\big)\ c\big(\sigma_t^{(j,\sigma)}\big). \tag{25}$$

Every matrix occurring in (25) is ρ_n of an *integral* matrix of determinant N (Definition 8.3), precomputed once over R; no determinant-N matrix is inverted (Obligation 20.3). Finally

$$\Theta_g \;:=\; \omega_\bullet \circ \Xi_g, \qquad \big(\Theta_g c\big)(\sigma) \;=\; \sum_{j=1}^{m} \rho_n(s_j^{-1})\,\big(\Xi_g c\big)(\sigma)_j,$$

with $\omega_\bullet$ the simplexwise application of the block row of Proposition 12.7.

Definition 12.10 is constructive in the sense of rule (A3): (25) is a finite sum over a finite table, and it is literally what `HeckeTauTable::xi_terms(j, q, sigma)` stores — one triple $\big(n_t,\ \sigma_t,\ \rho_n(g\gamma_t)\big)$ per term, with the $\rho_n(g\gamma_t)$ shared through a content-addressed pool.

Theorem 12.11 (Ξ_g lands in the induced system; Θ_g realises $T_{\Gamma g \Gamma}$)**.** *Assume (H4), let R be a commutative ring, and let $g, \{s_j\}, \tau$ be as above. Then:*

1. *[proved here] **(No hypothesis on ℓ.)** $\Xi_g(c) \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L'')$ for every $c \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$: for every relation $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$ — the stabiliser and central relations included —*

$$(\Xi_g c)(\sigma_{\mathrm{lhs}}) \;=\; M_{\Gamma''}(\gamma)\,(\Xi_g c)(\sigma_{\mathrm{rhs}}).$$

2. *[proved here] **(No hypothesis on ℓ.)** $\Xi_g \circ d = d \circ \Xi_g$ and $\omega_\bullet \circ d = d \circ \omega_\bullet$; hence $\Theta_g \colon C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L) \to C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$ is a map of cochain complexes.*

3. *[proved here]* $\Theta_g = \Psi \circ \widetilde{T} \circ \Psi^{-1}$ *with* $\widetilde{T}$ *the operator* (21) *of Definition 11.6, as an identity of cochain maps and with* no *hypothesis on* ℓ*. Consequently the map induced on* $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ *is the operator of Definition 11.6 in every degree* q*; and if moreover* $6 \in R^\times$ *(hypothesis (H3)) it is the classical double-coset operator* $T_{\Gamma g \Gamma}$ *of* (20)*, by Theorem 11.10.*

Proof. (1) Let $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$, so that $\widetilde{\sigma_{\mathrm{lhs}}} = \gamma \cdot \widetilde{\sigma_{\mathrm{rhs}}}$ in K^u ((H4)(iii) and Theorem 5.10; for a stabiliser or central relation $\sigma_{\mathrm{lhs}} = \sigma_{\mathrm{rhs}}$ and γ fixes the lift). Write $s_j \gamma = h_j(\gamma) s_{j\cdot\gamma}$ with $h_j(\gamma) \in \Gamma''$ (Lemma 10.3). Then, using (T1) of Definition 11.1 — legitimate because $h_j(\gamma) \in \Gamma''$ — and Γ-equivariance of ϕ_c,

$$\begin{aligned}(\Xi_g c)(\sigma_{\mathrm{lhs}})_j &= \rho_n(g)\,\phi_c\big(\tau(s_j\gamma \cdot \widetilde{\sigma_{\mathrm{rhs}}})\big) = \rho_n(g)\,\phi_c\Big(\tau\big(h_j(\gamma)\, s_{j\cdot\gamma} \cdot \widetilde{\sigma_{\mathrm{rhs}}}\big)\Big)\\ &= \rho_n(g)\,\phi_c\Big(\big(g^{-1}h_j(\gamma)g\big) \cdot \tau\big(s_{j\cdot\gamma} \cdot \widetilde{\sigma_{\mathrm{rhs}}}\big)\Big) = \rho_n(g)\,\rho_n\big(g^{-1}h_j(\gamma)g\big)\,\phi_c\Big(\tau\big(s_{j\cdot\gamma} \cdot \widetilde{\sigma_{\mathrm{rhs}}}\big)\Big)\\ &= \rho_n\big(h_j(\gamma)\big)\,(\Xi_g c)(\sigma_{\mathrm{rhs}})_{j\cdot\gamma},\end{aligned}$$

the last step by Proposition 8.2(2) applied to the identity $g\,(g^{-1}h_j(\gamma)g) = h_j(\gamma)\,g$ in $\mathrm{M}_2(\mathbb{Z})$ — an identity between integral matrices, with no inverse of a determinant-N matrix taken. By (22) the right-hand side is block j of $M_{\Gamma''}(\gamma)\,(\Xi_g c)(\sigma_{\mathrm{rhs}})$.

(2) For Ξ_g: the coboundary of the equivariant complex is $(dc)(\sigma) = c(\partial\sigma)$, and $K \subseteq K^u$ is a simplicial subcomplex ((H4)(iii)), so $\widetilde{\partial\sigma} = \partial\widetilde{\sigma}$. Hence, for $\sigma \in o_{q+1}(K)$,

$$\big(d\,\Xi_g c\big)(\sigma)_j = (\Xi_g c)(\partial\sigma)_j = \rho_n(g)\,\phi_c\big(\tau(s_j \cdot \partial\widetilde{\sigma})\big) = \rho_n(g)\,\phi_c\big(\partial\,\tau(s_j \cdot \widetilde{\sigma})\big) = \rho_n(g)\,(d\phi_c)\big(\tau(s_j \cdot \widetilde{\sigma})\big),$$

using that s_j acts simplicially and (T2) of Definition 11.1; since $\phi_{dc} = d\phi_c$ (Theorem 5.10), the right-hand side is $(\Xi_g\, dc)(\sigma)_j$. For $\omega_\bullet$: it is one fixed fibre matrix applied at every simplex, and the coboundary acts as $d_1 \otimes \mathrm{id}$ on $C^\bullet(K) \otimes V_R$, so the two commute. Composing gives the statement for Θ_g, and (1) gives that Θ_g maps $C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ into itself, since $\omega_\bullet$ carries L''-equivariant cochains to L-equivariant ones (Proposition 12.7).

(3) For $\sigma \in o_q(K)$,

$$(\Theta_g c)(\sigma) = \sum_{j=1}^{m} \rho_n(s_j^{-1})\,\rho_n(g)\,\phi_c\big(\tau(s_j \cdot \widetilde{\sigma})\big) = \sum_{j=1}^{m} \rho_n(\delta_j)\,\phi_c\big(\tau(s_j \cdot \widetilde{\sigma})\big) = (\widetilde{T}\phi_c)\big(\widetilde{\sigma}\big),$$

the middle equality by Proposition 8.2(2) applied to $s_j^{-1} g = \delta_j$ in $\mathrm{M}_2(\mathbb{Z})$ (again integral, and $\rho_n(s_j)^{-1} = \rho_n(s_j^{-1})$ because $s_j \in \Gamma$ is unimodular), and the last by (21). Since Ψ is evaluation at the lifts (23), this says $\Theta_g c = \Psi(\widetilde{T}\phi_c) = \Psi\widetilde{T}\Psi^{-1}c$. The remaining assertions are Definition 11.6 and Theorem 11.10, the latter carrying the hypothesis $6 \in R^\times$ and being the *only* place it is needed. □

Remark 12.12 (Which choices Θ_g depends on). Θ_g depends on g, on the representatives s_j and on the chosen τ, and it is not invariant under changing any of them — only the induced $A^{(q)}$ is, and only under the hypotheses of Theorem 11.7. Remark 11.9 says exactly what may therefore be compared bit for bit and what may not; nothing in this chapter weakens it.

12.5 Assembly of the Hecke matrix on H^q_{eq}

Algorithm 12.13 (Hecke matrix in degree q)**.** Assume (H4). **Input.** The computed equivariant cohomology of L (§6): the stored basis $\big[z_1^{(q)}\big], \dots, \big[z_{d_q}^{(q)}\big]$ of $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ with representative cocycles $z_j^{(q)} \in Z^q_{\mathrm{eq}} \subseteq R^{N_q d_V}$ in the flat layout, and the projector

$$P_q \colon R^{N_q d_V} \longrightarrow R^{d_q}, \qquad P_q\, z_j^{(q)} = e_j, \quad P_q\, B^q_{\mathrm{eq}} = 0,$$

i.e. P_q reads off coordinates in the stored basis modulo B^q_{eq} (this is the defining property of the stored projector, and it is certified as $P_q H_{\mathrm{eq}} = I$, $P_q B_{\mathrm{eq}} = 0$); plus the τ-table of §18 for $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$ and the coset tables of Definition 10.2. **Steps.**

1. For $j = 1, \dots, d_q$ compute $y_j := \Theta_g \, z_j^{(q)} \in R^{N_q d_V}$ by (25) followed by the ω-row of Proposition 12.7.
2. Set column j of $A^{(q)} \in \mathrm{M}_{d_q}(R)$ to $P_q \, y_j$.

Output. $A^{(q)}$, the matrix of the operator of Definition 11.6 on $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ in the stored basis, in the *column* convention

$$T\big[z_j^{(q)}\big] \;=\; \sum_k A^{(q)}_{kj} \, \big[z_k^{(q)}\big].$$

Correctness. $y_j \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ by Theorem 12.11(1) and Proposition 12.7; $d\, y_j = 0$ by Theorem 12.11(2) and $d\, z_j^{(q)} = 0$, so $y_j \in Z^q_{\mathrm{eq}}$ and $P_q y_j$ is the coordinate vector of $[y_j] = T[z_j^{(q)}]$ (Theorem 12.11(3)) modulo B^q_{eq}. **Degenerate degrees.** $d_q = 0$ yields the 0×0 matrix and $\dim C^q_{\mathrm{eq}} = 0$ the 0×0 $\Theta^{(q)}_{\mathrm{eq}}$; nothing is special-cased. This is the normal situation at $q = 2$ for $6 \in R^\times$ and at every q for odd n (Remark 7.7). **No normalisation.** $A^{(q)}$ is neither symmetrised nor rescaled: the integral normalisation of the contract box after Proposition 8.2 is what fixes the eigenvalue normalisation of the oracles (§21).

> **CONTRACT.** The three runtime assertions of Algorithm 12.13 are stated as predicates on emitted data in Obligation 20.1 (for every j: y_j satisfies every relation of $\mathcal{R}$, and $d\, y_j = 0$, over R, a failure aborting the run) and Obligation 20.5 (the two evaluation routes of §13). They are *not* restated here, per rule (A4): an obligation has one home.

Code map. Definitions 12.1, 12.3, Proposition 12.2 and (22): `Hecke/InducedLocalSystem.h/.cpp` — `MakeInducedLocalSystemGamma0` builds L'' from `HeckeCosetTables::perm/rho_cocycle` (bottom-row class map) and `MakeInducedLocalSystemGamma0Upper` builds L' from `perm_upper/rho_cocycle_upper` (top-row class map), both reusing the base relation list of `GenerateSL2ZRelationsWithGroupData` unchanged and replacing only the matrix, which is Definition 12.3 verbatim. (22) is assembled as an `engine::BlockMat` with one block write per block row — the one-non-zero-block-per-row structure of Proposition 12.2 is the storage model, not an optimisation. Proposition 12.4(3) is the code path for the `negI_*` relations, which computes $M_H(-I) = I_m \otimes \rho_n(-I)$ directly rather than through the generator dictionary; (4) is `HeckeShapiroValidation` (Obligation 20.6), which hard-asserts $\dim H^0_{\mathrm{eq}} = \dim H^1_{\mathrm{eq}} = 1$ for $H = \Gamma_0(2), \Gamma_0(3)$ at $n = 0$, $\ell \in \{5, 101\}$ against classical genus and cusp data of $X_0(2)$, $X_0(3)$. Propositions 12.6 and 12.7, and Proposition 13.2: `Hecke/HeckeCrossChecks.h/.cpp`. Definition 12.10, (25), Theorem 12.11 and Algorithm 12.13: `Hecke/HeckeOperatorOnK.h/.cpp` — Θ_g is stored as `Theta_block(q)`, an $N_q \times N_q$ block matrix whose pattern is exactly the set of pairs (σ, σ_t) named by the τ-table (`HeckeThetaBlockValidation` checks that the structural non-zeros are confined to it), the flat form being `get_Theta_flat_matrices()[q]`; the cocycle and relation assertions of Algorithm 12.13 are the compute-time checks (i) and (ii), and $A^{(q)}$ is `get_H_matrices()[q]`. The flat layout used by P_q and by the $z_j^{(q)}$ is simplex-major, fibre-minor — flat index $s \cdot d_V + k$ for the s-th simplex in the stored order and fibre coordinate k — and is the `nbc` $= 1$ block-matrix layout of `BlockMat::to_flat`, which is its single implementation.

Formalization note. Definition 12.1 and Proposition 12.2 are pure finite group theory — a transversal, an index action and a 1-cocycle — and are the easiest part of Part III to formalise;

`mathlib4` supplies induced modules and Shapiro's lemma, so Proposition 12.4(4) is a matter of matching conventions rather than of proof. The load-bearing content of this chapter for a proof assistant is (23): it is the one place where a *choice* (the lift $\widetilde{\sigma}$) is turned into a well-defined map, and its well-definedness is a statement about stabilisers, not about coefficients. A formalisation should therefore take Ψ of Theorem 5.10 as the primitive and define Ξ_g by (24), deriving (25) as a lemma — the reverse of the implementation's order, which computes (25) and never forms ϕ_c. Theorem 12.11(1) and (2) are finite manipulations using only Lemma 10.3, Proposition 8.2(2) and (T1)–(T2), need no hypothesis on ℓ, and are the natural companion target to the equivariance half of Theorem 11.7. Remark 12.9(ii) is a genuinely non-constructive argument (irreducibility plus a numerical non-scalarity) and should be formalised, if at all, as a statement about $\mathrm{End}_\Gamma(\mathrm{Sym}^{10}\mathbb{Q}^2)$ together with an imported eigenvalue fact, not re-derived.

13 Equivalent formulations and cross-checks

What this chapter fixes. Three formulations that compute the *same* operator as Definition 11.6/Definition 12.10, what each is good for, and at which level each is asserted to agree. None of them is a definition; the normative operator is Definition 11.6, realised by Definition 12.10 and Algorithm 12.13. The three are: the equivariant-chain-map picture and its two bracketings (§13.1), which agree *bit for bit* on the same data; the classical three-map factorisation $T_{\Gamma g\Gamma} = \omega_* \circ \mu_* \circ \alpha_*$ (§13.2), whose middle map is a cochain map with per-simplex blocks and *not* a morphism of local systems (Remark 12.9); and the corrected group-cochain formula in degree 1 (§13.3), which is an independent route through the bar resolution and agrees only up to similarity.

Standing hypotheses. (H1)–(H4), repeated at each point of use (rule (A1)). The hypothesis $6 \in R^\times$ ((H3)) is needed by Proposition 13.4 — which computes in $H^1(\Gamma; V_R)$ and uses the amalgam torsion — and by nothing in §§13.1–13.2, whose statements are cochain-level identities over an arbitrary commutative R.

13.1 The equivariant-chain-map picture, and the two bracketings

Definition 11.6 together with Theorem 12.11 *is* the classical "Hecke operator as an equivariant chain map" picture, written out on an explicit resolution. The datum $(\{s_j\}, \tau)$ is a simplicial stand-in for the correspondence of §11.1: $\sum_j s_j(-)$ plays the role of $(\mathrm{pr}')^*$ — unfolding over the m sheets of $\Gamma''\backslash\mathcal{H} \to \Gamma\backslash\mathcal{H}$ — the map τ plays the role of θ_g^*, and the $\rho_n(\delta_j)$-weighted sum plays the role of the trace $(\mathrm{pr}'')_*$. Nothing in this paragraph is used in a proof; it is stated so that a reader arriving from the classical literature can see which factor is which.

What *is* used is that the single formula (21) admits two different bracketings, and that they are required to agree exactly.

Proposition 13.1 (The two evaluation routes agree on the nose)**.** *[proved here] [formally verified: [GT, `prop:tworoutes`]] Assume (H4), let R be a commutative ring, and fix $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, the transporters s_j and the τ-table of Definition 12.10. For $c \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ the two expressions*

$$\underbrace{\sum_{j=1}^{m} \rho_n(s_j^{-1}) \sum_t n_t^{(j,\sigma)}\, \rho_n\big(g\,\gamma_t^{(j,\sigma)}\big)\, c\big(\sigma_t^{(j,\sigma)}\big)}_{\text{fused: } \omega_\bullet \circ \Xi_g,\ \text{Definition 12.10}} \qquad \text{and} \qquad \underbrace{\sum_{j=1}^{m} \sum_t n_t^{(j,\sigma)}\, \rho_n\big(\delta_j\,\gamma_t^{(j,\sigma)}\big)\, c\big(\sigma_t^{(j,\sigma)}\big)}_{\text{route 2: direct (21)}}$$

are equal term by term, for every $\sigma \in o_q(K)$, every j and every t — with no *hypothesis on ℓ, and* as an identity of matrices, not only of their values.

Proof. $\delta_j\gamma_t = s_j^{-1}(g\gamma_t)$ is an identity in $\mathrm{M}_2(\mathbb{Z})$, between integral matrices and with no inverse of a determinant-N matrix taken (Proposition 10.1(5)). Apply ρ_n and Proposition 8.2(2), which is multiplicativity of ρ_n as a monoid homomorphism on all of $\mathrm{M}_2(\mathbb{Z})$ and in particular on the non-invertible elements: $\rho_n(\delta_j\gamma_t) = \rho_n(s_j^{-1})\rho_n(g\gamma_t)$. □

> **CONTRACT.** Proposition 13.1 is the implementer's cross-check, and it is one of **two** places in this document where **bit-identity of cochain-level matrices** is asserted rather than equality in cohomology — the other being Proposition 13.2(3), $\omega_\bullet \circ \mu_\bullet \circ \alpha_\bullet = \Theta_g$ — legitimately, because the two routes consume the *same* g, the *same* s_j and the *same* τ-table, so Remark 11.9's prohibition does not apply. The two sides are formed differently on purpose: the fused route multiplies two *already reduced* stored matrices $\rho_n(s_j^{-1})$ and $\rho_n(g\gamma_t)$ over R, whereas route 2 forms the integer product $\delta_j\gamma_t$ *exactly* in $\mathrm{M}_2(\mathbb{Z})$ and reduces once, at the end. Agreement therefore tests the reduction discipline and the monoid identity of Proposition 8.2(2) simultaneously, and a discrepancy is a bug in one of them and in nothing else. Recorded as Obligation 20.5.

13.2 The three-map factorisation, at cochain level

Remark 12.9 showed that the middle arrow cannot be a morphism of local systems. It *can* be a cochain map, and the following makes it explicit. The price is that it is given by one block matrix per pair of simplices — a `FiberBlockTable` — rather than by a single fibre matrix, which is exactly the generalisation Remark 12.9 says is needed.

Proposition 13.2 (The middle cochain map $\mu_\bullet$)**.** *[proved here] Assume (H4), let R be a commutative ring, and let g, $\{s_j\}$, τ, $L' = L_{\Gamma^0(N)}$ with representatives $t_1 = I, \dots, t_m$, and $L'' = L_{\Gamma_0(N)}$ with representatives $s_1, \dots, s_m$ be as in §12.1. For $c' \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L')$ define*

$$\big(\mu_\bullet c'\big)(\sigma)_j \;:=\; \rho_n(g) \cdot \Big[\, \phi^{L'}_{c'}\big(\tau(s_j \cdot \widetilde{\sigma})\big)\Big]_1, \qquad j = 1, \dots, m,$$

*where $\phi^{L'}_{c'}$ is the $\mathrm{Ind}_{\Gamma'}^{\Gamma} V_R$-valued equivariant cochain attached to c' by (23) applied to the local system L', i.e. $\phi^{L'}_{c'}(\gamma \cdot \widetilde{\sigma}) = M_{\Gamma'}(\gamma)\, c'(\sigma)$, and $[\,\cdot\,]_1$ is evaluation at $1 \in \Gamma$, i.e. block coordinate 1 (legitimate because $t_1 = I$). **The order of the two operations is part of the definition:** $[\,\cdot\,]_1$ lands in V_R and $\rho_n(g)$ is applied there, so no action of $\rho_n(g)$ on $\mathrm{Ind}_{\Gamma'}^{\Gamma} V_R$ is presupposed. Evaluation at 1 is a Γ'-morphism $\mathrm{Ind}_{\Gamma'}^{\Gamma} V_R \to V_R$:*

$$\big[u \cdot f\big]_1 = \rho_n(u)\,\big[f\big]_1 \qquad \text{for all } u \in \Gamma', \tag{26}$$

because $1 \cdot u = 1$ and $h_1(u) = t_1 u t_1^{-1} = u$ in (22). Then:

1. ***Closed form.*** *[formally verified: [GT, `prop:mu-assembled`]] (the three factor clauses, assembled; not (2), (3)) With $\tau(s_j\widetilde{\sigma}) = \sum_t n_t\, \gamma_t \cdot \widetilde{\sigma_t}$ as in Definition 12.10, and with $a_t := 1 \cdot \gamma_t$ the index of the $\Gamma^0(N)$-coset of γ_t,*

$$\big(\mu_\bullet c'\big)(\sigma)_j \;=\; \sum_t n_t\; \rho_n\big(g\,\gamma_t\, t_{a_t}^{-1}\big)\;\big[c'(\sigma_t)\big]_{a_t},$$

   all factors integral of determinant N, with $t_{a_t}^{-1} \in \Gamma$ and no determinant-N matrix inverted.
2. ***It is a cochain map into L''.*** *$\mu_\bullet$ carries $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L')$ into $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L'')$ and commutes with d — with no hypothesis on ℓ.*
3. ***It splits Ξ_g.*** *$\mu_\bullet \circ \alpha_\bullet = \Xi_g$ as maps $C^q(K; V_R) \to C^q(K; \mathrm{Ind}_{\Gamma''}^{\Gamma} V_R)$, entry for entry, where $\alpha_\bullet$ is the simplexwise application of the block column of Proposition 12.6. Consequently $\omega_\bullet \circ \mu_\bullet \circ \alpha_\bullet = \Theta_g$, and if moreover $6 \in R^\times$ then on cohomology $T_{\Gamma g \Gamma} = \omega_* \circ \mu_* \circ \alpha_*$ under the identifications of Proposition 12.4(4).*

Proof. (1) By (23) for L', $\phi^{L'}_{c'}(\tau(s_j\widetilde{\sigma})) = \sum_t n_t\, M_{\Gamma'}(\gamma_t)\, c'(\sigma_t)$. By (22) the block row 1 of $M_{\Gamma'}(\gamma_t)$ has its single non-zero block $\rho_n(h_1(\gamma_t))$ in block column $a_t = 1\cdot\gamma_t$, and $h_1(\gamma_t) = t_1\gamma_t t_{a_t}^{-1} = \gamma_t t_{a_t}^{-1} \in \Gamma^0(N)$ (Proposition 12.2). Hence $\big[\phi^{L'}_{c'}(\tau(s_j\widetilde{\sigma}))\big]_1 = \sum_t n_t\, \rho_n(\gamma_t t_{a_t}^{-1})\big[c'(\sigma_t)\big]_{a_t}$; multiplying by $\rho_n(g)$ and using Proposition 8.2(2) on the integral identity $g\cdot(\gamma_t t_{a_t}^{-1}) = g\gamma_t t_{a_t}^{-1}$ gives the display.

(2) Let $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$ and write $s_j\gamma = h_j(\gamma)s_{j\cdot\gamma}$ with $h_j(\gamma) \in \Gamma''$ (Lemma 10.3), so that $u := g^{-1}h_j(\gamma)g \in \Gamma'$ (Proposition 10.1(3)). By (T1) of Definition 11.1 and Γ-equivariance of $\phi^{L'}_{c'}$,

$$\big(\mu_\bullet c'\big)(\sigma_{\mathrm{lhs}})_j = \rho_n(g)\Big[\,u\cdot\phi^{L'}_{c'}\big(\tau(s_{j\cdot\gamma}\cdot\widetilde{\sigma_{\mathrm{rhs}}})\big)\Big]_1 \overset{(26)}{=} \rho_n(g)\,\rho_n(u)\,\Big[\phi^{L'}_{c'}\big(\tau(s_{j\cdot\gamma}\cdot\widetilde{\sigma_{\mathrm{rhs}}})\big)\Big]_1 = \rho_n\big(h_j(\gamma)\big)\big(\mu_\bullet c'\big)(\sigma_{\mathrm{rhs}})_{j\cdot\gamma},$$

the last step by Proposition 8.2(2) applied to $g\,u = h_j(\gamma)\,g$ in $\mathrm{M}_2(\mathbb{Z})$; by (22) this is block j of $M_{\Gamma''}(\gamma)(\mu_\bullet c')(\sigma_{\mathrm{rhs}})$. Commutation with d is the computation of Theorem 12.11(2) verbatim, which used only (T2), $\widetilde{\partial\sigma} = \partial\widetilde{\sigma}$ and linearity of $[\,\cdot\,]_1$.

(3) By Proposition 12.6, $\big[\alpha_\bullet c(\sigma)\big]_a = \rho_n(t_a)c(\sigma)$. Substituting into (1) and using Proposition 8.2(2) on the integral identity $(g\gamma_t t_{a_t}^{-1})\,t_{a_t} = g\gamma_t$ in $\mathrm{M}_2(\mathbb{Z})$,

$$\big(\mu_\bullet\alpha_\bullet c\big)(\sigma)_j = \sum_t n_t\,\rho_n\big(g\gamma_t t_{a_t}^{-1}\big)\rho_n\big(t_{a_t}\big)c(\sigma_t) = \sum_t n_t\,\rho_n\big(g\gamma_t\big)c(\sigma_t) = \big(\Xi_g c\big)(\sigma)_j$$

by (25). Composing with $\omega_\bullet$ gives Θ_g (Definition 12.10); the cohomological statement is Theorem 12.11(3) together with Propositions 12.6 and 12.7, whose identifications with res and cor carry $6 \in R^\times$ and are the only place it is used. □

> **CONTRACT.** Proposition 13.2 is a **cross-check route only**. It is what makes each of the three classical factors separately testable — $\alpha_\bullet$ and $\omega_\bullet$ through the single-fibre-matrix machinery of §6, $\mu_\bullet$ through per-simplex block rows — and it is asserted against the binding pipeline at three levels of increasing strength: $\mu_\bullet$ alone maps L'-equivariant cochains to L''-equivariant ones; $\mu_\bullet\alpha_\bullet = \Xi_g$ on the full standard basis of $C^q(K; V_R)$, not merely on $\operatorname{im}\alpha_\bullet$; and $\omega_\bullet\mu_\bullet\alpha_\bullet = \Theta_g$ bit-identically as matrices for every q. The binding operator is and remains Θ_g of Definition 12.10, which constructs neither L' nor $\mu_\bullet$ and needs neither. Recorded as Obligation 20.5.

13.3 The corrected group-cochain formula

Remark 13.3 (Deviation/clarification vs. `monodromy_v5.tex` §12.3). [proved here] The superseded framework document states, as its "Algebraic Hecke Operator",

$$(T_g\phi)(\gamma_1,\dots,\gamma_s) = \sum_j \rho_L(g_j)\,\phi\big(g_j^{-1}\gamma_1 g_j,\dots,g_j^{-1}\gamma_s g_j\big).$$

As written this is not well defined on inhomogeneous cochains — $g_j^{-1}\gamma g_j$ need not lie in Γ — and, even read inside the monoid Δ, it **does not commute with the bar differential**, because right multiplication by γ *permutes* the cosets $\delta_j\Gamma$ and the formula holds every j fixed. It is the special case of the correct formula in which every γ_u stabilises every coset. Proposition 13.4 carries the permutation explicitly. Recorded as Appendix A, entry A.20.

Proposition 13.4 (Corrected bar-cochain formula, degree 1)**.** *[proved here] Assume (H4) and let R be a commutative ring. Write $\Gamma g\Gamma = \bigsqcup_{j=1}^{m}\delta_j\Gamma$ for the left-coset decomposition of Proposition 10.1(2),(5), with $\delta_j = s_j^{-1}g$. For $\gamma\in\Gamma$ let $\gamma * j$ be defined by $\gamma\delta_j\Gamma = \delta_{\gamma*j}\Gamma$ and put*

$$\kappa_j(\gamma) \;:=\; \delta_{\gamma*j}^{-1}\,\gamma\,\delta_j \;\in\; \Gamma.$$

Then on inhomogeneous 1*-cochains* $\phi\colon \Gamma \to V_R$ *the assignment*

$$(T_g\phi)(\gamma) \;:=\; \sum_{j=1}^{m} \rho_n\big(\delta_{\gamma * j}\big)\,\phi\big(\kappa_j(\gamma)\big) \tag{27}$$

carries cocycles to cocycles and coboundaries to coboundaries — with no *hypothesis on* ℓ *— and if moreover* $6 \in R^\times$ *(hypothesis (H3)) it induces on* $H^1(\Gamma; V_R)$ *the same operator as Definition 11.6.*

Proof. Chain rule. $\gamma * j$ and $\kappa_j(\gamma)$ are well defined because the $\delta_j\Gamma$ are m disjoint cosets permuted by left multiplication (Proposition 10.1(2)), and $\kappa_j(\gamma) \in \Gamma$ because $\gamma\delta_j$ and $\delta_{\gamma * j}$ lie in the same left coset. From $\gamma_1\gamma_2\delta_j = \gamma_1\delta_{\gamma_2 * j}\kappa_j(\gamma_2) = \delta_{\gamma_1 * (\gamma_2 * j)}\kappa_{\gamma_2 * j}(\gamma_1)\kappa_j(\gamma_2)$ and uniqueness of the coset:

$$(\gamma_1\gamma_2) * j = \gamma_1 * (\gamma_2 * j), \qquad \kappa_j(\gamma_1\gamma_2) = \kappa_{\gamma_2 * j}(\gamma_1)\,\kappa_j(\gamma_2).$$

Cocycles. Let $\phi(ab) = \phi(a) + \rho_n(a)\phi(b)$. Then

$$\begin{aligned}(T_g\phi)(\gamma_1\gamma_2) &= \sum_j \rho_n\big(\delta_{\gamma_1 * (\gamma_2 * j)}\big)\Big[\phi\big(\kappa_{\gamma_2 * j}(\gamma_1)\big) + \rho_n\big(\kappa_{\gamma_2 * j}(\gamma_1)\big)\phi\big(\kappa_j(\gamma_2)\big)\Big] \\ &= (T_g\phi)(\gamma_1) + \sum_j \rho_n\big(\delta_{\gamma_1 * j'}\kappa_{j'}(\gamma_1)\big)\Big|_{j'=\gamma_2 * j}\phi\big(\kappa_j(\gamma_2)\big) = (T_g\phi)(\gamma_1) + \rho_n(\gamma_1)\,(T_g\phi)(\gamma_2),\end{aligned}$$

using $\delta_{\gamma_1 * j'}\kappa_{j'}(\gamma_1) = \gamma_1\delta_{j'}$, then Proposition 8.2(2), and that $j \mapsto \gamma_2 * j$ is a bijection — the reindexing that Remark 13.3 says the superseded formula omits, and without which the first sum is not $(T_g\phi)(\gamma_1)$. *Coboundaries.* For $\phi = dv$, i.e. $\phi(\gamma) = \rho_n(\gamma)v - v$,

$$(T_g\,dv)(\gamma) = \sum_j \Big[\rho_n\big(\delta_{\gamma * j}\kappa_j(\gamma)\big)v - \rho_n\big(\delta_{\gamma * j}\big)v\Big] = \sum_j \Big[\rho_n(\gamma\delta_j)v - \rho_n(\delta_{\gamma * j})v\Big] = \rho_n(\gamma)w - w, \quad w := \sum_j \rho_n(\delta_j)v,$$

using Proposition 8.2(2) and, in the last step, that $j \mapsto \gamma * j$ is a bijection; so $T_g(dv) = dw$. *Agreement.* Both operators compute $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ of (20): (27) is the standard double-coset action on the bar resolution [standard: [Bro82, Ch. III.9–10] for res, cor and the bar resolution; [AS86, §1.1, p. 194] for the double-coset operator (the right-module form on homogeneous cochains; the left-module form by the S^{-1} device on the same page)], with $\{\delta_j\} = \{s_j^{-1}g\}$ the representatives of Proposition 10.1(5), and Definition 11.6 computes the same thing on the resolution $C_\bullet(K^u) \otimes R$ by Theorem 11.10. Both resolutions are complexes of relatively projective $R[\Gamma]$-modules when $6 \in R^\times$ — for $C_\bullet(K^u) \otimes R$ this is the computation in the proof of Lemma 11.3(2), for the bar resolution it is freeness — so the canonical comparison equivalence between them is unique up to homotopy and transports one operator to the other [standard: [Bro82, Ch. I.7.4]]. This is the *only* use of $6 \in R^\times$ in this proposition. □

Remark 13.5 ($\kappa_j(\gamma)$ is computed by an exact division, not by an inversion). The expression $\kappa_j(\gamma) = \delta_{\gamma * j}^{-1}\gamma\delta_j$ displays an inverse of a determinant-N matrix, which Obligation 20.3 forbids in the pipeline. It is not needed: writing $\delta_k = s_k^{-1}g$ and $N\,g^{-1} = \mathrm{adj}(g)$, the identity

$$N \cdot \kappa_j(\gamma) \;=\; \mathrm{adj}(g)\, s_k\, \gamma\, s_j^{-1}\, g, \qquad k = \gamma * j,$$

holds in $\mathrm{M}_2(\mathbb{Z})$, so $\kappa_j(\gamma)$ is obtained by forming an integer product and dividing by N *exactly*; the index k is the unique one for which the division is exact, and the result is verified by the reconstruction identity $\delta_k\,\kappa_j(\gamma) = \gamma\,\delta_j$ over $\mathbb{Z}$. This is what the implementation does (rule (A3)), and it is why an oracle route inherits the integrality discipline of the pipeline rather than relaxing it.

CONTRACT. Proposition 13.4 is a **validation oracle only**: an independent implementation path for H^1 through explicit group cocycles on a presentation of Γ, touching neither the τ-table nor the induced systems nor the simplicial cochain spaces. The normative operator in all degrees is Definition 11.6 / Algorithm 12.13. Because the two routes produce matrices in *different, unrelated bases* of H^1, they are compared by **similarity invariants — characteristic polynomial and minimal polynomial — and not entrywise**; asserting an entrywise identity here would be asserting something this document does not claim (Remark 11.9). Any discrepancy of those invariants is a bug. Recorded as Obligation 20.5.

13.4 Degree-wise specialisation

The pipeline is uniform in q — one formula, one implementation, every degree — and this subsection records what it specialises to, for orientation only. No statement below is used elsewhere.

- $q = 0$. $H^0_{\mathrm{eq}}(K, \mathcal{R}; L) = \{v \in V_R : \rho_n(\gamma)v = v \text{ for every relation element } \gamma \text{ of } \mathcal{R}\}$ and, by Remark 11.8, $T\,v = \sum_{j=1}^{m} \rho_n(\delta_j)v$. For $n = 0$ this is multiplication by m: $T_p = p + 1$, and at the bad prime $U_\ell = \ell + 1 \equiv 1 \pmod{\ell}$ (hypothesis (H2)).
- $q = 1$. The operator matches the classical modular-symbol and group-cocycle computations: Proposition 13.4 is the second route, and the Eichler–Shimura eigenvalues are the oracle (Obligation 20.7).
- $q = 2$. No classical shortcut exists and the pipeline applies unchanged. For $6 \in R^\times$ the model has $H^2_{\mathrm{eq}}(K, \mathcal{R}; L) = 0$ — X is 2-dimensional and Γ has virtual cohomological dimension 1 [standard: [Bro82, Ch. VIII, §§2–3]; [Den23] for this model] — so $A^{(2)}$ is the 0×0 matrix, which is itself a useful consistency assertion. For $\ell \in \{2, 3\}$, H^2_{eq} may be non-zero and Algorithm 12.13 produces its matrix by the same code path, with the caveat of (H3) on what that group is.

Code map. Proposition 13.1: `Hecke/HeckeOperatorOnK.cpp` — the two predefined programs over the τ-pattern (`program_fused` and `program_route2`, the latter in its own block pool so that content-addressed deduplication cannot make the comparison a tautology), compared as mathematical matrices by `blk::equal` over all degrees, with `apply_Theta` and `apply_route2` the vector forms; any mismatch throws, and `HeckeThetaBlockValidation` additionally checks `apply_Theta`$(q, e_b) = \Theta^{(q)}_{\mathrm{flat}}[:, b] =$ `apply_route2`(q, e_b) on the whole standard basis. Proposition 13.2: `Hecke/HeckeCrossChecks.h/.cpp` — $\mu_\bullet$ is the `FiberBlockTable` whose (σ, σ') entry accumulates $\sum_{t:\sigma_t=\sigma',\, a_t=a} n_t\, \rho_n(g\gamma_t t_a^{-1})$ into block (j, a), the product $g\gamma_t t_a^{-1}$ formed exactly in $\mathrm{M}_2(\mathbb{Z})$ and reduced once; the three assertions of the contract box are the class's three checks, gated by `HeckeCrossCheckValidation`. Proposition 13.4 and Remark 13.5: `Hecke/BarCocycleH1.h/.cpp`, which presents $\Gamma = \langle S \rangle *_{\langle S^2 \rangle} \langle ST \rangle$, represents a 1-cocycle by the pair $(\phi(S), \phi(ST))$ subject to the norm and amalgam-matching conditions, evaluates ϕ on an arbitrary $\kappa \in \Gamma$ by reducing κ to a word in T, S and expanding with the cocycle rule (asserting both that the word reconstructs κ exactly and that the accumulated representation matrix equals $\rho_n(\kappa)$), enforces $\ell \geq 5$ at construction, and compares characteristic and minimal polynomials against $A^{(1)}$; the exact division of Remark 13.5 and its reconstruction check are in `hecke_matrix()`.

Formalization note. Proposition 13.1 is a one-line consequence of Proposition 8.2(2) and is the cheapest statement in Part III to formalise; it is also among the most valuable to state separately, because it is one of only **two** claims in the document where a *bit*-level agreement is mathematically forced rather than empirically observed — the other being Proposition 13.2(3),

proved by the same route: an integral identity in $\mathrm{M}_2(\mathbb{Z})$ pushed through the monoid homomorphism ρ_n (Proposition 8.2(2)). Proposition 13.2 is finite linear algebra over the same data and needs nothing beyond (22). Proposition 13.4 splits cleanly for a proof assistant: the cocycle and coboundary computations are elementary and unconditional — the chain rule for $(*, \kappa)$ is the same 1-cocycle structure as Lemma 10.3 and should share its formalisation — whereas the agreement clause is a comparison of two resolutions and should be stated as an instance of the same comparison theorem used by Lemma 11.3(2), not re-proved. The degree-wise specialisations of §13.4 are orientation and carry no formalisation obligation, with one exception worth recording: the vanishing $H^2_{\mathrm{eq}} = 0$ at $6 \in R^\times$ is carried in §13.4 by a citation (rule (A2)) and is *not* derived here, so a formalisation must import it rather than take the emitted 0×0 matrix as a proof of it.

Part IV — The boundary, the cone, and compact supports

14 The boundary datum

What this chapter fixes. The boundary of the model, as a Hecke-equivariant object, and nothing beyond it. The boundary group Γ_∞ and the *definition* $H^\bullet(\partial; V_R) := H^\bullet(\Gamma_\infty; V_R)$; the circle model, and the exact hypothesis — $2 \in R^\times$ — under which it computes that group; the upper-triangular set B_N, its left cosets, and why the right cosets of Proposition 10.1(1) are the wrong ones; the boundary g-twisted chain datum τ_∂ on ∂X, which is the **normative** boundary datum, its existence and homotopy uniqueness over $\mathbb{Z}$ for *every* ℓ, and its unconditional integrality; the boundary operator Θ^∂_g and the boundary characters χ_b; and what $\ell = 2$ costs, where $H^\bullet(\partial)$ is unbounded above in degree and the six-term sequence of §16 is a *window* of a sequence that does not terminate — and where the bulk, unlike the boundary, is *periodic* above degree 2 (Lemma 14.22, Hecke-equivariantly). The $\ell = 2$ truncation policy is construction item 14.24 (**OC-1**), decided by the owner on 2026-09-16 as (OC-1a) and stated in §14.5 with its alternatives kept reachable. **Since v1.3.3 this chapter is implemented**: the certified τ_∂-table, the boundary coset table and the operator Θ^∂_g exist in the tree, and Obligation 14.19 is discharged in full by two registered validations — the code map at the end of the chapter says what was built, which choices the implementation fixed where the statements below leave them open, and what it recorded.

Three errata land here (Appendix A): A.12, a definition and its own normative construction disagreeing inside one document, repaired by Definition 14.2 together with Proposition 14.4; A.4, a uniqueness statement derived from a theorem whose hypothesis the construction exists to avoid, repaired by Lemma 14.10, which never cites that theorem; and the boundary half of A.14, the characters presented in the right-coset convention as though it were the answer, repaired by Proposition 14.16.

What this chapter does not fix. The pair $(X, \partial X)$, the restriction map $r = \jmath^*$, the pair homotopy η, the cone $C(r)$ and the cone action Φ_g are §15; the six groups and the long exact sequence are §16; the free Γ-model that 14.24 will be constructed on is §17. τ_∂ is the boundary *datum*; every statement below that mentions the cone is a forward reference and is marked as one.

Standing hypotheses. (H1)–(H4) of §7, each repeated where it is used (rule (A1)), together with three hypotheses proper to this chapter. Per the design direction recorded in Remark 7.6 they are stated for *the boundary triangulation in force* — an arbitrary Γ-stable boundary subcomplex satisfying (B1)–(B3) — and not for the four horocycle vertices $v_0v_1v_2v_3$ of Defini-

tion 7.1, so that a refinement of that complex is admissible input. Nothing in this chapter is a statement about seven vertices. [formally verified: [GT, `def:model7bd`]] (on the model, for ∂X given by the boundary list $v_0, v_1, v_2, v_3, \{v_0, v_1\}, \{v_1, v_2\}, \{v_2, v_3\}$: (B1) as a Γ-saturated, face-closed subcomplex; of (B2) only $C_q(\partial X) = 0$ for $q \geq 2$ — the component map and the contractibility of the components are not formalised; (B3) as stabilisers of order 2 containing $\pm I$; [GT] numbers these conditions differently)

(B1) **The boundary subcomplex.** $\partial X \subseteq X$ is a Γ-stable full subcomplex of the tessellation of (H4), namely the union of the Γ-translates of the horocycle at ∞ at the truncation height Y. Writing ∂X_∞ for the connected component lying over the cusp ∞, the setwise stabiliser $\{\gamma \in \Gamma : \gamma \partial X_\infty = \partial X_\infty\}$ is the group Γ_∞ of Definition 14.2, and $\partial X = \Gamma \cdot \partial X_\infty$. Moreover Γ has *finitely many* orbits of simplices on ∂X — equivalently Γ_∞ has finitely many on ∂X_∞ — which is what makes the tables below finite.

(B2) **Components.** The map $\pi_0(\partial X) \to \mathbb{P}^1(\mathbb{Q})$ sending a component to the cusp it lies over is a Γ-equivariant bijection, and **every component is a 1-dimensional contractible simplicial complex** — a triangulated line. In particular $Z_1(\partial X_c) = H_1(\partial X_c) = 0$ and $C_q(\partial X) = 0$ for $q \geq 2$.

(B3) **Stabilisers are central.** The Γ-stabiliser of every simplex of ∂X is exactly $\{\pm I\}$, which acts trivially on X ((H4)(iv)). **No elliptic point lies on** ∂X, and no simplex of ∂X has a stabiliser of order 4 or 6.

Remark 14.1 (The shipped triangulation satisfies (B1)–(B3), and so does every equivariant refinement). [proved here] For the model of Definition 7.1, K_∂ is the path $v_0 v_1 v_2 v_3$ on the horocycle $\{\operatorname{Im} z = Y\}$ with $v_3 = T \cdot v_0$, and $\partial X_\infty = \Gamma_\infty \cdot K_\partial$ is the whole horocycle, a line with three Γ_∞-orbits of edges and three of vertices (v_0 and v_3 being in one). (B1) holds because the horocycle at ∞ is a line and its setwise stabiliser in Γ is $\mathrm{Stab}_\Gamma(\infty) = \Gamma_\infty$; (B2) holds because Γ acts transitively on $\mathbb{P}^1(\mathbb{Q})$ with stabiliser Γ_∞ at ∞ and the horoballs removed in (H4) are indexed by the cusps, so $\pi_0(\partial X) \cong \Gamma/\Gamma_\infty \cong \mathbb{P}^1(\mathbb{Q})$ equivariantly; (B3) holds because (H4) fixes $Y > 1$ while $\operatorname{Im}\zeta = \operatorname{Im}\zeta^2 = \sqrt{3}/2$ and $\operatorname{Im} i = 1$, so the three elliptic points of §7.1 lie strictly below the horocycle and off ∂X, and (H4)(iv)'s "every other simplex stabiliser is exactly $\{\pm I\}$" applies to every simplex of ∂X.

If X' is any Γ-equivariant refinement of X and $\partial X'$ the induced subdivision of ∂X, then (B1)–(B3) again hold: subdividing a line gives a line, refinement does not change π_0, and the stabiliser of a simplex of a subdivision is contained in the stabiliser of the simplex containing it, hence still $\{\pm I\}$ (it contains $-I$, which acts trivially). **Every statement of this chapter is therefore inherited by any refinement**, which is what Remark 7.6's design direction asks of it. What is not inherited is triangulation-indexed data: the number of Γ_∞-orbits of boundary simplices, the size of the τ_∂-table, and the bound of Lemma 18.5.

14.1 The boundary group, and what the circle model computes

Definition 14.2 (The boundary group and the boundary cohomology)**.** $\Gamma_\infty := \mathrm{Stab}_\Gamma(\infty) = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \Gamma \right\}$, the subgroup of upper-triangular elements of $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. Since $ad = 1$ with $a, d \in \mathbb{Z}$,

$$\Gamma_\infty = \{\pm T^k : k \in \mathbb{Z}\} = \langle T \rangle \times \langle -I \rangle \;\cong\; \mathbb{Z} \times \mathbb{Z}/2, \tag{28}$$

a direct product because $\langle T \rangle \cap \langle -I \rangle = \{I\}$ and $-I$ is central. For a commutative ring R the **boundary cohomology** is

$$H^\bullet(\partial; V_R) \;:=\; H^\bullet(\Gamma_\infty; V_R),$$

the group cohomology of Γ_∞ with coefficients in $V_R = \mathrm{Sym}^n(\mathbb{Z}^2) \otimes R$ and Γ_∞ acting through ρ_n (Definition 8.1). Owner decision OP5, 2026-09-15; it agrees with §16's own definition of X_2 and X_5, from which it is transported unchanged. It is the half of erratum A.12 that was already right; the other half is Proposition 14.4.

OWNER DECISION. The boundary group is $\Gamma_\infty = \langle T\rangle \times \langle -I\rangle$, *not* $\langle T\rangle$ (OP5, amended and decided 2026-09-15; the verdict gave no recommendation and the owner decided with its argument section, [V] §5.5). The one-dimensional "circle model" on K_∂ is the *specialisation* of this datum at $2 \in R^\times$ (Proposition 14.4), exactly as the 7-vertex complex is the specialisation of the orbifold model at $6 \in R^\times$ (§17). Under (H3) with $\ell \geq 5$ the two coincide and nothing in Parts III–V is affected; at $\ell = 2$ they part company, and substantively — §14.5.

Proposition 14.3 (The cohomology of Γ_∞, in every degree)**.** *[proved here] Let R be a commutative ring and $n \geq 0$. Write $V_R^T := \ker(\rho_n(T) - \mathrm{id})$ and $(V_R)_T := V_R/\operatorname{im}(\rho_n(T) - \mathrm{id})$ (notation C12). Then for every $q \geq 0$ there is a short exact sequence of R-modules*

$$0 \longrightarrow \big(H^{q-1}(\langle -I\rangle; V_R)\big)_T \longrightarrow H^q(\Gamma_\infty; V_R) \longrightarrow \big(H^q(\langle -I\rangle; V_R)\big)^T \longrightarrow 0 \tag{29}$$

(with the left term read as 0 for $q = 0$), natural in V_R. If moreover $\rho_n(-I) = \mathrm{id}$ on V_R — equivalently n is even, or $2 = 0$ in R — then $H^q(\langle -I\rangle; V_R) = V_R$ for $q = 0$, $= V_R[2]$ for q odd and $= V_R/2V_R$ for $q \geq 2$ even.

Proof. Two elementary resolutions, used throughout this chapter. (i) For $\langle T\rangle \cong \mathbb{Z}$,

$$0 \longrightarrow \mathbb{Z}[\langle T\rangle] \xrightarrow{T-1} \mathbb{Z}[\langle T\rangle] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

is a free resolution of length 1; hence for every $\mathbb{Z}[\langle T\rangle]$-module M, $H^0(\langle T\rangle; M) = M^T$, $H^1(\langle T\rangle; M) = M_T$ and $H^s(\langle T\rangle; M) = 0$ for $s \geq 2$, so $\langle T\rangle$ has cohomological dimension ≤ 1. [proved here] (ii) For $\langle -I\rangle \cong \mathbb{Z}/2$ with generator t, the periodic free resolution $\cdots \to \mathbb{Z}[\mathbb{Z}/2] \xrightarrow{1+t} \mathbb{Z}[\mathbb{Z}/2] \xrightarrow{t-1} \mathbb{Z}[\mathbb{Z}/2] \xrightarrow{\varepsilon} \mathbb{Z} \to 0$ gives, for M with *trivial* t-action, the complex $M \xrightarrow{0} M \xrightarrow{2} M \xrightarrow{0} M \xrightarrow{2} \cdots$ and hence $H^0 = M$, $H^q = M[2]$ for q odd, $H^q = M/2M$ for $q \geq 2$ even. [proved here]

Now $\Gamma_\infty = \langle T\rangle \times \langle -I\rangle$ by (28), so the Lyndon–Hochschild–Serre spectral sequence of $1 \to \langle -I\rangle \to \Gamma_\infty \to \langle T\rangle \to 1$ has $E_2^{s,q} = H^s\big(\langle T\rangle; H^q(\langle -I\rangle; V_R)\big)$ [standard: [Bro82, Ch. VII.6]], and the conjugation action of $\langle T\rangle$ on $\langle -I\rangle$ is trivial because $-I$ is central and the extension splits, so the coefficient module is $H^q(\langle -I\rangle; V_R)$ with the T-action induced from V_R. By (i), $E_2^{s,q} = 0$ for $s \geq 2$; the spectral sequence therefore degenerates at E_2 and its two-column filtration in total degree q is (29), the outer terms being $E_2^{1,q-1}$ and $E_2^{0,q}$ read off by (i). The last sentence of the statement is (ii) applied to $M = V_R$, whose t-action is trivial exactly when $\rho_n(-I) = \mathrm{id}$. □

Proposition 14.4 (What the circle model computes, and when it is the boundary cohomology)**.** *[proved here] (Appendix A, entry A.12: the specialisation hypothesis $2 \in R^\times$ is here part of the statement, not of a preamble.) Assume (H4) and (B1)–(B3), let R be a commutative ring and $n \geq 0$, and let $C_\partial^\bullet := \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(\partial X), V_R\big)$ be the equivariant cochain complex of the boundary. For the model of Definition 7.1 let $K_\partial = v_0v_1v_2v_3$ be the boundary path and let $\mathcal{R}_\partial$ be the relation set on K_∂ consisting of the* cusp relation *(v_3, v_0, T), i.e. $f(v_3) = \rho_n(T)f(v_0)$, **together with the $-I$ self-relations** $(\sigma, \sigma, -I)$, i.e. $f(\sigma) = \rho_n(-I)f(\sigma)$, on every simplex σ of K_∂ — the boundary half of (7); the* circle model *is $C_{\mathrm{eq}}^\bullet(K_\partial, \mathcal{R}_\partial; L)$. (Until v1.4.0 this proposition called it "the circle model with its single relation $f(v_3) = \rho_n(T)f(v_0)$", which describes $\mathcal{R}_\partial$ only where the self-relations are vacuous — clause (4); open point (3) of the v1.3.3 changelog.) Then:*

1. ***(Shapiro, unconditionally.)*** *$C_\partial^\bullet \cong \mathrm{Hom}_{\mathbb{Z}[\Gamma_\infty]}\big(C_\bullet(\partial X_\infty), V_R\big) = \mathrm{Hom}_{\mathbb{Z}[\langle T\rangle]}\big(C_\bullet(\partial X_\infty), V_R^{-I}\big)$, where $V_R^{-I} := \ker(\rho_n(-I) - \mathrm{id})$, by restriction of a cochain to the ∞-component.*
2. ***(What it computes.)*** *If n is even, or if $2 = 0$ in R, then $V_R^{-I} = V_R$ and $H^q(C_\partial^\bullet) = H^q(\langle T\rangle; V_R)$ for every q; explicitly $H^0 = V_R^T$, $H^1 = (V_R)_T$ and $H^q = 0$ for $q \geq 2$. If n is odd and $2 \in R^\times$ then $V_R^{-I} = 0$ and $C_\partial^\bullet = 0$.*

3. ***(When it is $H^\bullet(\partial;V_R)$.)*** *Let n be even. If $2\in R^\times$ the inflation map $H^q(C^\bullet_\partial)=H^q(\langle T\rangle;V_R)\to H^q(\Gamma_\infty;V_R)$, along $\Gamma_\infty\twoheadrightarrow\langle T\rangle$ and legitimate because $-I$ acts trivially on V_R, is an isomorphism for every q. If $R=\mathbb{Z}/\ell^e$ with $\ell=2$ and $V_R\neq 0$ it is* not*: $H^2(C^\bullet_\partial)=0$ while $H^2(\Gamma_\infty;V_R)\neq 0$.* ***So the circle model computes $H^\bullet(\partial;V_R)$ exactly when $2\in R^\times$****, and at $\ell=2$ it computes a different, bounded object.*
4. ***(The model computes $C^\bullet_\partial$ — for every n and every R.)*** *For the model of Definition 7.1, $C^\bullet_{\mathrm{eq}}(K_\partial,\mathcal{R}_\partial;L)=C^\bullet_\partial$ as cochain complexes, by restriction of a cochain to the simplices of K_∂, with inverse $f\mapsto\big(\gamma\cdot\widetilde{\sigma}\mapsto\rho_n(\gamma)f(\sigma)\big)$. Consequently the circle model is the single-relation model — the cusp relation alone — with coefficients V_R^{-I} in place of V_R; the self-relations are vacuous, and the identification with the single-relation model on V_R holds literally, exactly when $\rho_n(-I)=\mathrm{id}$, i.e. for n even or $2=0$ in R; for n odd and $2\in R^\times$ the circle model is the zero complex (Remark 14.5). Without the self-relations, the single-relation model on V_R is $\mathrm{Hom}_{\mathbb{Z}[\langle T\rangle]}(C_\bullet(\partial X_\infty),V_R)$, which is* not *$C^\bullet_\partial$ for n odd, $2\in R^\times$.*

Proof. (1) By (B1) and (B2), $C_\bullet(\partial X)=\bigoplus_c C_\bullet(\partial X_c)$ with Γ permuting the summands transitively with stabiliser Γ_∞, i.e. $C_\bullet(\partial X)\cong\mathbb{Z}[\Gamma]\otimes_{\mathbb{Z}[\Gamma_\infty]}C_\bullet(\partial X_\infty)$ as complexes of $\mathbb{Z}[\Gamma]$-modules; the tensor–hom adjunction $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma]\otimes_{\mathbb{Z}[\Gamma_\infty]}P,V_R)\cong\mathrm{Hom}_{\mathbb{Z}[\Gamma_\infty]}(P,V_R)$, $\Phi\mapsto\Phi|_P$, is the first isomorphism [standard: [Bro82, Ch. III.6], where this adjunction is the content of Shapiro's lemma]. For the second: $-I$ acts trivially on X ((H4)(iv)), so a Γ_∞-equivariant φ satisfies $\varphi(x)=\varphi((-I)x)=\rho_n(-I)\varphi(x)$, i.e. it takes values in V_R^{-I}, and conversely.

(2) $\rho_n(-I)=(-1)^n\mathrm{id}$ by Proposition 8.2(2) applied to $-I$, so $V_R^{-I}=V_R$ when n is even and when $2=0$ in R, and $V_R^{-I}=V_R[2]=0$ when n is odd and $2\in R^\times$. By (B2), ∂X_∞ is a contractible 1-dimensional complex on which $\langle T\rangle$ acts freely: by (B3) the Γ-stabiliser of every simplex of ∂X is $\{\pm I\}$, and $T^k\notin\{\pm I\}$ for $k\neq 0$. Hence the augmented complex $C_\bullet(\partial X_\infty)\to\mathbb{Z}$ is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\langle T\rangle]$ — free by the free action, exact by contractibility — and $\mathrm{Hom}_{\mathbb{Z}[\langle T\rangle]}(C_\bullet(\partial X_\infty),V_R)$ computes $H^\bullet(\langle T\rangle;V_R)$ by definition; the displayed values are step (i) of the proof of Proposition 14.3, and they vanish above degree 1 because ∂X_∞ is 1-dimensional.

(3) With n even, $-I$ acts trivially on V_R. If $2\in R^\times$ then $H^q(\langle -I\rangle;V_R)=0$ for $q\geq 1$ (Proposition 14.3, last sentence: $V_R[2]=0$ and $V_R/2V_R=0$), so the E_2-page of the spectral sequence used there is concentrated in the row $q=0$, where it is $H^s(\langle T\rangle;V_R)$; a spectral sequence concentrated in one row degenerates with edge map the inflation, so inflation $H^s(\langle T\rangle;V_R)\to H^s(\Gamma_\infty;V_R)$ is an isomorphism for every s, and $H^s(\langle T\rangle;V_R)=H^s(C^\bullet_\partial)$ by (2). If $R=\mathbb{Z}/\ell^e$, $\ell=2$, then V_R is free of rank $n+1>0$ over R, so $V_R/2V_R\neq 0$; $\rho_n(T)$ is unipotent (Definition 8.1: $T\cdot x=x$, $T\cdot y=x+y$, so $\rho_n(T)u_j=u_j+\sum_{t<j}\binom{j}{t}u_t$ is unitriangular), hence so is its reduction on $V_R/2V_R\neq 0$, hence $\big(V_R/2V_R\big)^T\neq 0$; by (29) at $q=2$, $H^2(\Gamma_\infty;V_R)$ surjects onto it and is non-zero, while $H^2(C^\bullet_\partial)=0$ by (2).

(4) $\partial X=\Gamma\cdot K_\partial$: by Remark 14.1, $\partial X_\infty=\Gamma_\infty\cdot K_\partial$ and $\partial X=\Gamma\cdot\partial X_\infty$ ((B1)). This is (FD2) for the pair $(K_\partial,\partial X)$. The coincidence groupoid $\mathcal{C}(K_\partial,\partial X,\Gamma)$ of Definition 3.15 — the triples (σ,σ',γ) with $\gamma\cdot\widetilde{\sigma'}=\widetilde{\sigma}$ and σ,σ' simplices of K_∂ — is generated by $\mathcal{R}_\partial$: both simplices lie in ∂X_∞, whose setwise stabiliser is Γ_∞ ((B1)), so $\gamma\in\Gamma_\infty=\{\pm T^k\}$; by Remark 14.1 the Γ_∞-orbits of simplices of ∂X_∞ meet K_∂ in $\{v_0,v_3\}$, $\{v_1\}$, $\{v_2\}$ and in each of the three edges separately, and by (B3) every stabiliser is $\{\pm I\}$; hence a triple is $(\sigma,\sigma,\pm I)$, $(v_3,v_0,\pm T)$ or $(v_0,v_3,\pm T^{-1})$, each a composite of the cusp relation, its inverse and the self-relations. This is (FD4) for $(K_\partial,\mathcal{R}_\partial)$. The proof of Theorem 5.10(1) uses exactly (FD2) and (FD4) — it says so at its well-definedness step — and therefore applies verbatim to $(K_\partial,\mathcal{R}_\partial)$ and ∂X: $\Psi\colon\phi\mapsto\phi|_{K_\partial}$ is an isomorphism of cochain complexes $C^\bullet_\partial\to C^\bullet_{\mathrm{eq}}(K_\partial,\mathcal{R}_\partial;L)$ with the stated inverse. (Assumption 3.17 as a whole is not invoked: its standing hypothesis that $|K^u|$ be connected fails for ∂X, and neither it nor (FD1), (FD3) is needed; $\mathcal{R}_\partial$ is closed under faces, Assumption 5.6, the faces of a self-relation being self-relations and the cusp relation a vertex relation.) The self-relations read $(\mathrm{id}-\rho_n(-I))f(\sigma)=0$, i.e. $f(\sigma)\in V_R^{-I}$; keeping the cusp relation and restricting the values to V_R^{-I} is the single-relation model with coefficients V_R^{-I}, which is clause (1)'s second isomorphism

read on K_∂. The case split is $\rho_n(-I) = (-1)^n\mathrm{id}$ (proof of (2)): $V_R^{-I} = V_R$ for n even or $2 = 0$ in R, and $V_R^{-I} = V_R[2] = 0$ for n odd and $2 \in R^\times$. For the last sentence: without the self-relations the well-definedness step of Theorem 5.10(1) fails at the stabiliser $\{\pm I\}$ exactly as Remark 7.7 describes, and for n odd, $2 \in R^\times$, the single-relation model on V_R has non-zero cochains (any value on v_1, say) while $C_\partial^\bullet = 0$. □

Remark 14.5 (Odd n, on the boundary). [proved here] For n odd and $2 \in R^\times$ the boundary complex is the zero complex and $H^\bullet(\partial; V_R) = 0$, for the same reason that $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L) = 0$ there: $-I$ is in every stabiliser and acts by $(-1)^n = -1$ (Remark 7.7). The two vanishings are one phenomenon, and both fail at $\ell = 2$, where $(-1)^n = 1$. Every statement of this chapter that assumes n even says so. Since v1.4.0 the model side is literal: Proposition 14.4(4) identifies the circle model $C^\bullet_{\mathrm{eq}}(K_\partial, \mathcal{R}_\partial; L)$, with the $-I$ self-relations in $\mathcal{R}_\partial$, with $C_\partial^\bullet$ for every n, so the zero complex at odd n, $2 \in R^\times$, is what the implementation computes and not a statement about a different object (code map of this chapter).

14.2 The upper-triangular set B_N and its cosets

Proposition 14.6 (The cosets of B_N, and which decomposition is the right one)**.** *[proved here] Let p be a prime, $\nu \geq 1$, $N = p^\nu$, $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$ and $m = N + p^{\nu-1}$ as in Proposition 10.1, and put*

$$B_N := \{\delta \in \Gamma g \Gamma : \ \delta \text{ upper triangular}\},$$

the elements of the double coset fixing the cusp ∞. Then:

1. ***(Left cosets.)*** *$B_N = \bigsqcup \delta\Gamma_\infty$ over exactly m cosets. For $\nu = 1$ they are represented by*

$$\delta_\infty^\partial := \mathrm{diag}(1, p) = g, \qquad \delta_t^\partial := \begin{pmatrix} p & t \\ 0 & 1 \end{pmatrix} \quad (0 \leq t < p),$$

   and $T\delta_\infty^\partial = \delta_\infty^\partial T^p$, $T\delta_t^\partial = \delta_{t+1}^\partial$ for $t < p-1$, and $T\delta_{p-1}^\partial = \delta_0^\partial T$. [formally verified: [GT, `prop:bdrycosets`]]
2. ***(Bijection with the bulk cosets.)*** *$\delta\Gamma_\infty \mapsto \delta\Gamma$ is a bijection $B_N/\Gamma_\infty \to \Gamma g \Gamma/\Gamma$. Hence for each bulk coset element $\delta_j = s_j^{-1} g$ of Definition 10.2 the set $\{\gamma \in \Gamma : \delta_j \gamma \in B_N\}$ is a single left coset $\gamma_j \Gamma_\infty$, characterised by $\gamma_j \cdot \infty = \delta_j^{-1} \cdot \infty$, and $\{\delta_j \gamma_j\}_{j=1}^m$ is a set of representatives of B_N/Γ_∞. [formally verified: [GT, `prop:bdrycosets-cusp`]] (since v1.18.0 including the characterisation $\gamma_j \cdot \infty = \delta_j^{-1} \cdot \infty$, as a ratio pair)*
3. ***(Two double cosets, not one.)*** *For $\nu = 1$, $B_p = \Gamma_\infty \delta_\infty^\partial \Gamma_\infty \ \sqcup \ \Gamma_\infty \delta_0^\partial \Gamma_\infty$, with $\Gamma_\infty \cap \delta_\infty^\partial \Gamma_\infty (\delta_\infty^\partial)^{-1} = \Gamma_\infty$ (one left coset) and $\Gamma_\infty \cap \delta_0^\partial \Gamma_\infty (\delta_0^\partial)^{-1} = \langle T^p \rangle \times \langle -I \rangle$ of index p (p left cosets). [formally verified: [GT, `prop:bdrycosets`]]*
4. ***(The right cosets are the wrong ones.)*** *The upper-triangular right-coset representatives $\beta_b = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$ $(0 \leq b < p)$ and $\beta_\infty = \mathrm{diag}(p, 1)$ of Proposition 10.1(1) represent $\Gamma_\infty \backslash B_p$, not B_p/Γ_∞. They are **not** to be used: they give the* det*-twisted operator, χ_{b+1} in place of χ_b (Proposition 14.16). [formally verified: [GT, `prop:bdrycosets`]]*

Proof. An upper-triangular $\delta = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ lies in $\Gamma g \Gamma$ iff $ad = N$ and $\gcd(a, b, d) = 1$ (Proposition 10.1(1)), so $a = \pm p^u$, $d = \pm p^{\nu-u}$ with equal signs, $0 \leq u \leq \nu$. Since $\Gamma_\infty = \{\pm T^k\}$ and $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} T^k = \begin{pmatrix} a & b + ka \\ 0 & d \end{pmatrix}$, the left coset $\delta\Gamma_\infty$ is determined by $(u,\ b \bmod a)$ up to the common sign, which $-I \in \Gamma_\infty$ absorbs. Counting admissible b: for $u = 0$ one coset; for $0 < u < \nu$ both a and d are divisible by p so primitivity forces $p \nmid b$, giving $p^u - p^{u-1}$ cosets; for $u = \nu$, $d = \pm 1$ and every $b \bmod N$ occurs, giving N. The total is $1 + \sum_{u=1}^{\nu-1}(p^u - p^{u-1}) +$

$N = p^{\nu-1} + N = m$. At $\nu = 1$ the cosets are $u = 0$, represented by $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \delta_\infty^\partial$, and $u = 1$, represented by $\begin{pmatrix} p & t \\ 0 & 1 \end{pmatrix} = \delta_t^\partial$, $0 \le t < p$; the three displayed identities are the matrix products $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \begin{pmatrix} 1 & p \\ 0 & p \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}\begin{pmatrix} 1 & p \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} p & t \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & t+1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} p & p-1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & p \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

(2) The map is well defined since $\Gamma_\infty \subseteq \Gamma$. It is injective: if $\delta' = \delta\gamma$ with $\delta, \delta' \in B_N$ and $\gamma \in \Gamma$, then $\gamma = \delta^{-1}\delta'$ is upper triangular ($\delta^{-1} = \frac{1}{N}\begin{pmatrix} d & -b \\ 0 & a \end{pmatrix}$, a product of upper-triangular matrices), hence lies in Γ_∞. Both sides have m elements — the left by (1), the right by Proposition 10.1(2) — so it is a bijection. For the last sentence, $\delta_j\gamma$ is upper triangular iff it fixes ∞ iff $\gamma \cdot \infty = \delta_j^{-1} \cdot \infty$; the set of such $\gamma \in \Gamma$ is a left coset of $\mathrm{Stab}_\Gamma(\delta_j^{-1}\infty)$, which for any choice γ_j is $\gamma_j\Gamma_\infty$ (conjugate stabilisers), and $\delta_j\gamma_j\Gamma_\infty$ is then one well-defined element of B_N/Γ_∞. These m elements are distinct because the $\delta_j\Gamma$ are, and the bijection just proved carries one family to the other.

(3) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ conjugated by δ_∞^∂ is $\delta_\infty^\partial T(\delta_\infty^\partial)^{-1} = \begin{pmatrix} 1 & 1/p \\ 0 & 1 \end{pmatrix}$, so $\Gamma_\infty \cap \delta_\infty^\partial\Gamma_\infty(\delta_\infty^\partial)^{-1} = \Gamma_\infty$ and the double coset $\Gamma_\infty\delta_\infty^\partial\Gamma_\infty$ is a single left coset; it is the $u = 0$ coset. Likewise $\delta_0^\partial T(\delta_0^\partial)^{-1} = T^p$, so the intersection is $\langle T^p\rangle \times \langle -I\rangle$ of index p, and by (1)'s identities the p cosets $\delta_t^\partial\Gamma_\infty$ are permuted transitively by Γ_∞; they are the $u = 1$ cosets. The two families exhaust B_p.

(4) $\Gamma_\infty\beta_b = \{\pm\begin{pmatrix} 1 & b+kp \\ 0 & p \end{pmatrix}\}$ exhausts the $u = 0$ family as b runs over $0, \dots, p-1$, and $\Gamma_\infty\beta_\infty = \{\pm\begin{pmatrix} p & k \\ 0 & 1 \end{pmatrix}\}$ is the whole $u = 1$ family; so these $p+1$ sets are the right cosets $\Gamma_\infty\backslash B_p$. They split the two families in the opposite way to (1), which is exactly why the two sums differ; Proposition 14.16(2) computes by how much. □

Remark 14.7 (General ν: what this chapter does not state). Clauses (1) and (2) of Proposition 14.6 hold for every ν; clauses (3) and (4), Proposition 14.16(3) with (31), and the Γ_∞-level datum of the decision box after Definition 14.14 are $\nu = 1$ statements, and **no double-coset decomposition $\Gamma_\infty\backslash B_N/\Gamma_\infty$ for $\nu \ge 2$ is stated or proved in this document**. That is why Obligation 14.19(v) is degree 0 alone at $\nu \ge 2$ (owner confirmation 2 of the slot-2 plan, 2026-09-16: narrow now, add when a phase needs $T_{p^\nu}^\partial$ on H^1; hecke-extensions slot 1, WP-4b, finding 11). The shape a later addition would have to state and prove is recorded so that it is not re-derived from scratch, and is *not asserted here*: writing $\delta_{u,b} := \begin{pmatrix} p^u & b \\ 0 & p^{\nu-u} \end{pmatrix}$ for the representatives of (1), the computation in the proof of (1) gives $T^k\delta_{u,b} = \delta_{u,\,b+kp^{\nu-u}}$ and $\delta_{u,b}T^k = \delta_{u,\,b+kp^u}$, which suggests that the double cosets are indexed by u and $b \bmod p^{\min(u,\nu-u)}$ (with $p \nmid b$ for $0 < u < \nu$), that the double coset of $\delta_{u,b}$ is the union of $p^{\max(0,\,2u-\nu)}$ left cosets, and that the transporters are the corresponding powers of T. The general-ν analogue of (31) on H^1 and a Γ_∞-level twisted datum for route (v.c) at $\nu \ge 2$ would follow from it and are equally not stated. At $\nu = 1$ the shape specialises to clause (3). (v1.12.0-pre.3: an H^1 formula at every ν *is* now stated and proved, Proposition 14.17(2), (32); it is a sum over the left cosets of clause (1), which hold at every ν, and uses none of the double-coset shape above, which stays unasserted, as does the Γ_∞-level datum at $\nu \ge 2$. Obligation 14.19(v) is not extended by it.)

14.3 The boundary twisted chain datum τ_∂

Definition 14.8 (g-twisted chain datum on ∂X)**.** Let R be a commutative ring, $g \in \Delta$ and $\Gamma'' = \Gamma \cap g\Gamma g^{-1}$. A *$g$-twisted chain map on the boundary over R* is a degree-preserving R-linear

map $\tau_\partial\colon C_\bullet(\partial X)\otimes_{\mathbb{Z}} R \to C_\bullet(\partial X)\otimes_{\mathbb{Z}} R$ such that

(T1∂) $\tau_\partial(h\cdot x) = (g^{-1}hg)\cdot\tau_\partial(x)$ for all $h\in\Gamma''$ and all x;

(T2∂) $\partial\circ\tau_\partial = \tau_\partial\circ\partial$;

(T3∂) the induced map on $H_0(\partial X)\otimes R = R[\pi_0(\partial X)]$ is π_* for a map of sets $\pi\colon \pi_0(\partial X)\to\pi_0(\partial X)$: each component is carried into a single component, with total coefficient 1.

A *boundary g-twisted chain datum* τ_∂ is such a map together with the finite table of its values on a set of Γ''-orbit representatives of the simplices of ∂X, from which (T1∂) determines it everywhere.

(T3∂) is the correct analogue of (T3) of Definition 11.1: there $H_0(K^u)=\mathbb{Z}$ and "$\varepsilon\tau=\varepsilon$" says the induced map is the identity; here $H_0(\partial X)$ is free on the cusps and the condition says the induced map is a *permutation-like* map of basis elements, not merely degree-preserving. Both say that τ does not spread a point over several components. On each component separately, (T3∂) is the old (T3).

Lemma 14.9 ((T1∂) and (T3∂) force the component map: it is $c\mapsto g^{-1}c$)**.** *[proved here] Assume (H4) and (B1)–(B3), and let $g=\begin{pmatrix}1&0\\0&N\end{pmatrix}$, $\Gamma''=\Gamma_0(N)$ as in Proposition 10.1. Identify $\pi_0(\partial X)$ with $\mathbb{P}^1(\mathbb{Q})$ by (B2), and let g^{-1} act on $\mathbb{P}^1(\mathbb{Q})$ by Möbius transformations ($\det g = N>0$, so this is defined and $g^{-1}\mathbb{P}^1(\mathbb{Q})=\mathbb{P}^1(\mathbb{Q})$). Then every τ_∂ satisfying (T1∂) and (T3∂) has*

$$\pi(c) \;=\; g^{-1}\cdot c \qquad \text{for every cusp } c.$$

In particular $\pi(\infty)=\infty$, π is a bijection, and π is determined by g alone: ***no choice is made in it****, and it is the same for every admissible τ_∂.*

Proof. Write $\Gamma_c := \mathrm{Stab}_\Gamma(c)$. Γ acts transitively on $\mathbb{P}^1(\mathbb{Q})$ (a primitive column $(a,c)^{\mathsf T}$ completes to a matrix of Γ), so $\Gamma_c=\gamma\Gamma_\infty\gamma^{-1}$ for any γ with $\gamma\infty=c$; hence $\Gamma_c=\langle P_c\rangle\times\langle -I\rangle$ with $P_c=\gamma T\gamma^{-1}$ parabolic, by (28), and Γ_c is abelian. $\Gamma''=\Gamma_0(N)$ has finite index m in Γ (Proposition 10.1(3)), so $\Gamma''\cap\Gamma_c$ has finite index in Γ_c and contains P_c^k for some $k\ge 1$. By Proposition 10.1(3), $\mathrm{conj}_g^{-1}\colon\Gamma''\to\Gamma'$ is an isomorphism with integral values, so $P:=g^{-1}P_c^k g$ lies in Γ; it has the same trace ±2 as P_c^k and is not $\pm I$, hence is a non-trivial parabolic element of Γ and fixes exactly one point of $\mathbb{P}^1(\mathbb{Q})$. That point is $g^{-1}c$, because P_c^k fixes c and $P(g^{-1}c)=g^{-1}P_c^k c=g^{-1}c$.

Now let τ_∂ satisfy (T1∂) and (T3∂), with component map π. For $h\in\Gamma''\cap\Gamma_c$ and x supported in ∂X_c, the chain $h\cdot x$ is again supported in ∂X_c, so comparing the component of each side of (T1∂) — which (T3∂) says is well defined — gives $\pi(c)=(g^{-1}hg)\cdot\pi(c)$. Thus $\pi(c)$ is fixed by $g^{-1}(\Gamma''\cap\Gamma_c)g$, in particular by P, whose unique fixed cusp is $g^{-1}c$. Hence $\pi(c)=g^{-1}c$, and π is a bijection because g^{-1} acts invertibly on $\mathbb{P}^1(\mathbb{Q})$. For $g=\begin{pmatrix}1&0\\0&N\end{pmatrix}$, $g^{-1}\infty=\infty$. □

Lemma 14.10 (Existence, homotopy uniqueness and integrality of τ_∂ — over $\mathbb{Z}$, for every ℓ)**.** *[proved here] (Appendix A, entry A.4: part (3) is proved by the explicit argument below and* ***never*** *by the comparison theorem, whose relative-projectivity hypothesis is exactly what this construction exists to avoid.) Assume (H4) and (B1)–(B3), and let $g=\begin{pmatrix}1&0\\0&N\end{pmatrix}$, $\Gamma''=\Gamma_0(N)$. Then:*

1. ***(Existence over $\mathbb{Z}$, no hypothesis on ℓ.)*** *A boundary g-twisted chain datum over $R=\mathbb{Z}$ exists, and may be built by the following terminating procedure (rule (A3)). For each Γ''-orbit representative vertex v of ∂X, lying in the component ∂X_c, choose any vertex w of the forced target component $\partial X_{g^{-1}c}$ (Lemma 14.9; the canonical choice, the one the implementation makes, is step 2 of Algorithm 14.12) and set $\tau_\partial(v):=w$; for each Γ''-orbit representative edge e of ∂X, let $\tau_\partial(e)$ be the* unique 1*-chain of $\partial X_{g^{-1}c}$ with $\partial\tau_\partial(e)=\tau_\partial(\partial e)$; extend to all of $C_\bullet(\partial X)$ by (T1∂). There is nothing to do in degrees ≥ 2.*

2. ***(Integrality, no hypothesis on ℓ and no linear solve.)*** *Every coefficient of the emitted table lies in $\{0, \pm 1\}$, and no system of equations is solved anywhere in (1). Consequently the table* is *a datum over $\mathbb{Z}$, and the datum over any R is its base change;* ***the mod-ℓ fallback of Algorithm 18.3 step 3 cannot arise on the boundary, for any ℓ, because there is no step 3: ∂X has no 2-simplices ((B2)).***
3. ***(Homotopy uniqueness over $\mathbb{Z}$, no hypothesis on ℓ.)*** *If $\tau_\partial, \tau'_\partial$ both satisfy (T1∂)–(T3∂) over $\mathbb{Z}$ then there is an additive degree-$(+1)$ map $N_\partial \colon C_\bullet(\partial X) \to C_{\bullet+1}(\partial X)$ satisfying (T1∂) with $\tau_\partial - \tau'_\partial = \partial N_\partial + N_\partial \partial$; moreover N_∂ vanishes in degree 1 and is unique in degree 0.*

Proof. (1). By Lemma 14.9 the target component $g^{-1}c$ is forced, and $\partial X_{g^{-1}c} \neq \emptyset$, so a w exists. The (T1∂)-extension $\tau_\partial(h \cdot v) := (g^{-1}hg) \cdot \tau_\partial(v)$, $h \in \Gamma''$, is well defined precisely when $(g^{-1}hg) \cdot w = w$ for every h in the Γ''-stabiliser of v. **By (B3) that stabiliser is** $\Gamma'' \cap \{\pm I\} = \{\pm I\}$, $g^{-1}(-I)g = -I$, and $-I$ acts trivially on X ((H4)(iv)); so the condition is vacuous. *This is the whole of the difference from the abstract route of Lemma 11.3(2)*: there the stabilisers have orders 4 and 6 and the averaging idempotent $|\Gamma_\sigma|^{-1} \sum_u u$ is needed, which is why $6 \in R^\times$ is required there and is not required here. (Lemma 11.3(1) avoids the idempotent in the bulk as well, but by a different device — choosing the vertex value inside the fixed-point set of the twisted stabiliser — and it buys existence only, never uniqueness.) For an edge e of ∂X_c with endpoints $\partial e = w'' - w'$: $\tau_\partial(\partial e) = \tau_\partial(w'') - \tau_\partial(w')$ is a 0-chain of total coefficient 0 in $\partial X_{g^{-1}c}$, hence a boundary, because $\partial X_{g^{-1}c}$ is connected ((B2)); the 1-chain with that boundary is unique because $Z_1(\partial X_{g^{-1}c}) = 0$ ((B2)), and it is the oriented path from $\tau_\partial(w')$ to $\tau_\partial(w'')$. The same stabiliser argument makes the (T1∂)-extension of τ_∂ on edges well defined. (T2∂) holds by construction in degree 1 and is vacuous elsewhere; (T3∂) holds with $\pi(c) = g^{-1}c$ by construction. The procedure terminates because Γ'' has finitely many orbits on the simplices of ∂X: Γ has finitely many by (B1), and $[\Gamma : \Gamma''] = m < \infty$ by Proposition 10.1(3).

(2). A vertex value is a single vertex with coefficient 1; an edge value is an oriented path, i.e. a sum of edges with coefficients ± 1. No other operation occurs.

(3). Both data have the same component map π (Lemma 14.9), so for a Γ''-orbit representative vertex v of ∂X_c the 0-chain $(\tau_\partial - \tau'_\partial)(v)$ has total coefficient 0 and is supported in $\partial X_{g^{-1}c}$; let $N_\partial(v)$ be the unique 1-chain there with $\partial N_\partial(v) = (\tau_\partial - \tau'_\partial)(v)$ (existence by connectedness, uniqueness by $Z_1 = 0$; (B2)), extended by (T1∂), which is well defined by the stabiliser argument of (1). Put $N_\partial := 0$ in degree 1, which is forced since $C_2(\partial X) = 0$. It remains to check $(\tau_\partial - \tau'_\partial)(e) = N_\partial(\partial e)$ for every edge e. The difference $z := (\tau_\partial - \tau'_\partial)(e) - N_\partial(\partial e)$ is a 1-chain of $\partial X_{g^{-1}c}$ with

$$\partial z = (\tau_\partial - \tau'_\partial)(\partial e) - \partial N_\partial(\partial e) = (\tau_\partial - \tau'_\partial)(\partial e) - (\tau_\partial - \tau'_\partial)(\partial e) = 0,$$

using (T2∂) for both data; so $z \in Z_1(\partial X_{g^{-1}c}) = 0$. Uniqueness in degree 0 is the uniqueness of the fill just used. □

Remark 14.11 (Integrality of τ_∂: the claim, its reason, and what it is not). Two distinct facts are in play and have been conflated before, so they are separated here: conflating them is the class of over-claim Appendix A records at entry A.3, a hypothesis that was dropped when a statement was carried into a regime it was not proved in.

- **What is claimed.** τ_∂ exists over $\mathbb{Z}$, with coefficients in $\{0, \pm 1\}$, for *every* ℓ — including $\ell \in \{2, 3\}$ — and is homotopy-unique over $\mathbb{Z}$ for every ℓ (Lemma 14.10). The two reasons are different and both are necessary: (B3) (the stabilisers of boundary simplices are central and act trivially) is what makes the (T1∂)-extension well defined *without inverting a stabiliser order*, and (B2) (each component is a contractible 1-complex) is what makes the fills exist, be unique, and be paths.
- **What is *not* claimed.** That the *uniqueness* half Lemma 11.3(2) holds over $\mathbb{Z}$, or without $6 \in R^\times$, for the *bulk* τ. It is not claimed, and nothing here repairs it: the bulk homotopy

argument needs relative projectivity and therefore needs the elliptic stabiliser orders 4 and 6 to be units. (The *existence* half is a different matter: it does hold over $\mathbb{Z}$ for every ℓ, by a construction, Lemma 11.3(1) — but the device there is fixed points of the twisted stabilisers, not (B3), and it says nothing about uniqueness. See Remark 11.4.) Nor is the abstract comparison theorem used above: Lemma 14.10 is a construction and an explicit uniqueness argument, not an application of [Bro82, Ch. I.7.4].

- **Policy (owner decision OP2, 2026-09-15).** The mod-ℓ fill fallback is forbidden by default and `integral = false` is not the escape from a hard fill (Remark 11.5). **On the boundary the policy is vacuous**: by Lemma 14.10(2) there is no fill to fall back from, so a τ_∂-table always has solve regime $\mathbb{Z}$, and a record in which a τ_∂-table carries `integral = false` is reporting an implementation defect, not a mathematical regime. That is a checkable assertion and it is Obligation 14.19(iv).

Algorithm 14.12 (TAU-BDRY-TABLE: the canonical τ_∂)**.** Assume (H4) and (B1)–(B3), and let $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, $\Gamma'' = \Gamma_0(N)$. **Input:** p, ν, $N = p^\nu$; the boundary subcomplex K_∂ of the model with the exact positions of its vertices (Definition 7.1) and its ∞-line coordinates — the vertices of $\partial X_\infty = \Gamma_\infty \cdot K_\partial$ are $T^k v_i$ with i in the first period, numbered by the *position* $P = k \cdot \mathrm{per} + i$ ($\mathrm{per} = 3$ for the model, $v_3 = Tv_0$ at position 3), with real parts $\mathrm{Re}(T^k v_i) = \mathrm{Re}\, v_i + k$; the representatives $\{s_j\}_{j=1}^m$ of Definition 10.2; and a cusp reduction $\mathrm{red}\colon \mathbb{P}^1(\mathbb{Q}) \to \Gamma$ with $\mathrm{red}(c) \cdot c = \infty$ (the implementation's is the nearest-integer Euclid on the primitive column of (P2); by property (1) below the output does not depend on this choice). **Output:** for every $j \in \{1, \dots, m\}$ and every simplex σ of K_∂ a chain $\tau_\partial(s_j \cdot \widetilde{\sigma})$ in the stored-chain format (56), over $\mathbb{Z}$, with coefficients in $\{0, \pm 1\}$. This is the procedure of Lemma 14.10(1) with its one free choice — "any vertex w of the target component" — fixed; the output is **the canonical** τ_∂, and every statement of this document about "the" τ_∂-table refers to it (rule (A3)). Hecke-extensions slot 1, WP-4a, finding 2; `Hecke/HeckeBoundaryDatum`.

0. **Slots, and the single (T1∂) helper.** As in Algorithm 18.3 step 0. In ascending degree, and within a degree in the model's order of boundary simplices and $j = 1, \dots, m$: a slot (j, σ) is *derived* whenever a stored slot (j', σ') with $s_j \cdot \widetilde{\sigma} = \gamma \cdot \widetilde{\sigma'}$ for a transporter $\gamma = s_j t$, $\gamma = h\, s_{j'}$ with $h \in \Gamma''$, exists — then $\tau_\partial(s_j \sigma) := (g^{-1} h g) \cdot \tau_\partial(s_{j'} \sigma')$ through the one shared (T1∂) helper, with the orientation sign of the transporter tracked — and is *constructed* by steps 1–3 otherwise. Step 4 uses the same helper. By property (1) the order in which the slots are visited does not reach the output.
1. **Vertices: the transported real part.** For a constructed slot (j, v): $z := \big(\mathrm{adj}(g)\, s_j\big) \cdot \mathrm{pos}(v) \in \mathcal{H}$ with $\mathrm{adj}(g) = \mathrm{diag}(N, 1)$, projectively, as in Algorithm 18.3 step 1 (no rational number is formed); $c' := g^{-1} \cdot (s_j \infty)$, the forced target component (Lemma 14.9); $\gamma_{\mathrm{red}} := \mathrm{red}(c')$; and $x := \mathrm{Re}(\gamma_{\mathrm{red}} z) \in \mathbb{Q}$, the real part of the image transported to the ∞-component — its vertical projection onto the horocycle.
2. **The vertex rule.** Let P be the position of the line vertex nearest to x, i.e. with $\big|x - \mathrm{Re}(T^k v_i)\big|$ minimal, decided exactly; **if two vertices are equidistant from x, take the one with the larger real part.** Set $\tau_\partial(s_j \cdot \widetilde{v}) := \gamma_{\mathrm{red}}^{-1} \cdot T^k \widetilde{v_i}$ for $P = k \cdot \mathrm{per} + i$, one term with coefficient 1, so (T3∂) holds by inspection.
3. **Edges.** For a constructed slot (j, e) with $\partial e = v_b - v_a$: the two vertex values $\gamma_a \cdot \widetilde{w_a}$ and $\gamma_b \cdot \widetilde{w_b}$ are already stored (a derived face's value being the one the step-0 helper produced) and lie in one component c' (step 1, Lemma 14.9); with $\gamma_{\mathrm{red}} := \mathrm{red}(c')$ let P_a, P_b be the positions of $\gamma_{\mathrm{red}} \gamma_a \widetilde{w_a}$ and $\gamma_{\mathrm{red}} \gamma_b \widetilde{w_b}$ on the line. Set $\tau_\partial(s_j \cdot \widetilde{e}) := \gamma_{\mathrm{red}}^{-1} \cdot$ (the oriented walk from P_a to P_b), the sum of the line edges between the two positions with coefficient ± 1 according to their orientation — the unique 1-chain of the line with boundary $\gamma_b \widetilde{w_b} - \gamma_a \widetilde{w_a}$ (Lemma 14.10(1)); empty if $P_a = P_b$. No linear solve occurs.
4. **Certification, and the bulk bijection.** Obligation 14.19(i)–(iv) is evaluated on the finished table in the constructor; a failure aborts. The element γ_j of Proposition 14.6(2),

fixed there only up to $\gamma_j\Gamma_\infty$, is normalised as $\gamma_j := \mathrm{red}(g^{-1}s_j\infty)^{-1}$ (WP-4a, finding 5), and the bijection $j \mapsto \delta_j\gamma_j\Gamma_\infty$ is certified.

Properties. [proved here]

1. **(Well defined, and independent of the derivation order.)** Steps 1–2 define a function w on the vertices x of ∂X, $w(x) = \gamma_{\mathrm{red}}^{-1} \cdot \mathrm{nearest}\big(\mathrm{Re}(\gamma_{\mathrm{red}}\, g^{-1}\mathrm{pos}(x))\big)$ with γ_{red} *any* reduction of the cusp of $g^{-1}\mathrm{pos}(x)$, and $w(h \cdot x) = (g^{-1}hg) \cdot w(x)$ for every $h \in \Gamma''$. Hence a slot derived in step 0 carries the chain its direct construction would give, in every degree, and the table is independent of the order of step 0 and of the cusp reduction.
2. **(It is a boundary datum over $\mathbb{Z}$.)** The output satisfies (T1∂)–(T3∂), with coefficients in $\{0, \pm 1\}$ and solve regime $\mathbb{Z}$, for every ℓ: it is an instance of Lemma 14.10(1)–(2).
3. **(Any other admissible τ_∂ is homotopic to it)** over $\mathbb{Z}$, for every ℓ, by Lemma 14.10(3); the vertex rule affects the chains and the size of §15's η, and nothing on cohomology (Proposition 14.15(1)).
4. **(No ℓ and no n in the table.)** Steps 0–4 read neither ℓ nor n: the table is a datum over $\mathbb{Z}$, the same for every coefficient ring and element width, and the datum over R is its base change.

Proof of the properties. (1) For a vertex x of ∂X in the component over the cusp c put $y := g^{-1}\mathrm{pos}(x) \in \mathcal{H}$, which lies over $c' = g^{-1}c$; a reduction γ_{red} is any element of Γ with $\gamma_{\mathrm{red}}c' = \infty$, unique up to left multiplication by $\mathrm{Stab}_\Gamma(\infty) = \Gamma_\infty = \{\pm T^k\}$ (28). Replacing γ_{red} by $\pm T^k\gamma_{\mathrm{red}}$ replaces $x = \mathrm{Re}(\gamma_{\mathrm{red}}y)$ by $x + k$ ($-I$ acts trivially on $\mathcal{H}$), and the nearest position by $P + k \cdot$per: the distances $\big|x - \mathrm{Re}(T^{k'}v_i)\big|$ and the tie rule "larger real part" are both invariant under a common translation. So the vertex becomes $T^k \cdot$(vertex), which $(T^k\gamma_{\mathrm{red}})^{-1} = \gamma_{\mathrm{red}}^{-1}T^{-k}$ carries to the same vertex of $\partial X_{c'}$ as before: $w(x)$ is well defined. For $h \in \Gamma''$ put $\gamma := g^{-1}hg \in \Gamma$ (Proposition 10.1(3)); then $g^{-1}\mathrm{pos}(hx) = \gamma y$ lies over $\gamma c'$, $\gamma_{\mathrm{red}}\gamma^{-1}$ is a reduction of $\gamma c'$, and $w(hx) = (\gamma_{\mathrm{red}}\gamma^{-1})^{-1} \cdot \mathrm{nearest}\big(\mathrm{Re}(\gamma_{\mathrm{red}}\gamma^{-1}\gamma y)\big) = \gamma \cdot w(x)$. A derived vertex slot is therefore $(g^{-1}hg) \cdot \tau_\partial(s_{j'}v') = \gamma\, w(s_{j'}v') = w(hs_{j'}v')$, the directly constructed value; a derived edge slot is the (T1∂)-translate of a walk, and the translate of the unique 1-chain with a given boundary is the unique 1-chain with the translated boundary (Γ acts by chain maps, and $Z_1(\partial X_c) = 0$ by (B2)), again the directly constructed value. (2) is Lemma 14.10(1)–(2) with the choice of w made by step 2: (T1∂) holds on every slot by step 0 and (1), (T2∂) by step 3, (T3∂) by step 2 and Lemma 14.9, and the coefficients are those of a single vertex and of an oriented walk. (3) is Lemma 14.10(3) and Proposition 14.15(1). (4) is read off the steps: the inputs are N, the cosets and the exact geometry of the boundary path. □

Remark 14.13 (τ_∂ is not made ι-symmetric, and no vertex rule can make it so). [proved here] Let ι_X be the involution of §9 (Definition 9.1: $z \mapsto -\bar{z}$, realised by $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z})$, with $\iota(\gamma z) = \mathrm{conj}_\iota(\gamma)\iota(z)$ and $\mathrm{conj}_\iota \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$). It preserves ∂X, each horocycle and the cusp ∞, and acts on ∂X_∞ as ι_K on K_∂ extended by $\iota(T^k x) = T^{-k}\iota(x)$. Hecke-extensions slot 1, WP-4a, finding 3; open point (7) of the v1.3.3 changelog.

1. **(The conjugate datum is admissible, and homotopic.)** $\tau'_\partial := \iota_X\, \tau_\partial\, \iota_X$ is again a boundary g-twisted chain datum over $\mathbb{Z}$ for the same $g = \mathrm{diag}(1, N)$, and $\tau'_\partial \simeq \tau_\partial$ by Lemma 14.10(3): the two induce the same Θ_g^∂ on cohomology.
2. **(No vertex rule is ι-equivariant.)** Let w be *any* rule that assigns to a rational x — the transported real part of Algorithm 14.12 step 1 — a vertex $w(x)$ of ∂X_∞, T-equivariantly ($w(x+1) = T\, w(x)$), with any tie rule whatever or with no reference to distances at all. Then $w(-x) = \iota_X\, w(x)$ fails at $x = 0$ — and, by T-equivariance, at every integer x — because no vertex of ∂X_∞ is ι_X-fixed. An integer transported real part occurs at every $(2, \nu)$ (the vertex v_0 of K_∂ has $x = -N/2$; (3) below), so at $p = 2$ the ι-symmetry of τ_∂ cannot be *arranged by the choice of tie rule* in step 2, nor by any other T-equivariant

vertex rule, in contrast with the bulk, where Algorithm 18.3 step (1b) breaks ties ι_K-equivariantly through the ι_K-fixed vertex v_6 (Remark 9.5). For odd p no transported real part is an integer (proof of (3)), and this clause obstructs nothing there.

3. **(The canonical table is not ι-symmetric at $(p,\nu) = (2,1)$.)** With the shipped coordinates $\operatorname{Re} v_1 = -\operatorname{Re} v_2 = -\frac{1}{6}$ (the ones Lemma 9.2's hypothesis is verified in, §9), the canonical τ_∂ of Algorithm 14.12 at $N = 2$ has $\iota_X\tau_\partial\iota_X \neq \tau_\partial$, at two places on the ∞-line itself: on the vertex v_1 of K_∂ (the slot $s_1 = I$, Definition 10.2) the transported real part is $N\operatorname{Re} v_1 = -\frac{1}{3}$, a tie between v_0 and v_1 resolved to v_1, while on its mirror v_2 it is $+\frac{1}{3}$, a tie between v_2 and v_3 resolved to v_3, and $\iota_K v_3 = v_0 \neq v_1$; and on v_0 it is $N\operatorname{Re} v_0 = -1$, an integer, where (2) obstructs every rule: the tie $\{T^{-1}v_1, T^{-1}v_2\}$ is resolved to $T^{-1}v_2$, on the mirror $v_3 = Tv_0$ the tie $\{Tv_1, Tv_2\}$ to Tv_2, and $\iota_X(T^{-1}v_2) = Tv_1 \neq Tv_2$. (v1.4.0 placed the first exhibit on the slots $S\cdot v_1$, $S\cdot v_2$ over the cusp 0, where the transported real parts are $\mp\frac{1}{12}$ and no tie occurs; corrected in v1.4.1.)
4. **(What is and is not asserted.)** Whether the emitted table satisfies $\iota_X\tau_\partial\iota_X = \tau_\partial$ at a given (p,ν) is a property of that table: it fails at $(2,1)$ by (3) — and at every $(2,\nu)$, under every T-equivariant vertex rule, since $x(v_0) = -2^{\nu-1}$ is an integer — it is asserted nowhere, Obligation 9.10 says nothing about τ_∂, and no validation records it. For odd p the mirror point is never a transported real part (proof of (3)), so (2) obstructs nothing there; whether a *mirror-consistent* T-equivariant tie rule ("ties away from $\operatorname{Re} z = 0$", read modulo 1) would then give an ι-symmetric table at every odd (p,ν) is not settled here and not claimed. If §15 or §9 ever need an ι-symmetric boundary datum, that is a contract addition and is not made here (owner close-out decision (a) of slot 1, 2026-09-16).

Proof. (1) $\operatorname{conj}_\iota$ preserves Γ, preserves $\Gamma'' = \Gamma_0(N)$ ($c \mapsto -c$ keeps $N \mid c$) and fixes $g = \operatorname{diag}(1,N)$. By (I1) ι_X is a simplicial automorphism of X; it preserves $\operatorname{Im} z$, hence the horocycles at height Y and their Γ-translates (a translate $\gamma\cdot\{\operatorname{Im} z = Y\}$ goes to $\operatorname{conj}_\iota(\gamma)\cdot\{\operatorname{Im} z = Y\}$), so it preserves ∂X and acts on $C_\bullet(\partial X)$ as a chain map with $\iota_X(\gamma x) = \operatorname{conj}_\iota(\gamma)\,\iota_X(x)$. For $h \in \Gamma''$ and $\iota := \iota_X$: $\tau'_\partial(hx) = \iota\tau_\partial(\operatorname{conj}_\iota(h)\,\iota x) = \iota\big((g^{-1}\operatorname{conj}_\iota(h)g)\cdot\tau_\partial(\iota x)\big) = \operatorname{conj}_\iota(g^{-1}\operatorname{conj}_\iota(h)g)\cdot\iota\tau_\partial\iota(x) = (g^{-1}hg)\cdot\tau'_\partial(x)$, using $\operatorname{conj}_\iota(g) = g$ and $\operatorname{conj}_\iota^2 = \mathrm{id}$: (T1$\partial$). (T2$\partial$) holds because ι is a chain map. (T3∂): ι permutes the components ($c \mapsto \operatorname{conj}_\iota(\gamma)\infty$ for $c = \gamma\infty$, i.e. $c \mapsto -c$ on $\mathbb{P}^1(\mathbb{Q})$) and carries a single vertex with coefficient 1 to a single vertex with coefficient 1 (vertices carry no orientation sign), so τ'_∂ carries each component into one component with total coefficient 1, with component map $c \mapsto \iota\, g^{-1}\,\iota\, c = g^{-1}c$, as Lemma 14.9 forces in any case. So τ'_∂ is admissible over $\mathbb{Z}$; Lemma 14.10(3) and Proposition 14.15(1) give the rest.

(2) On the horocycle $\{\operatorname{Im} z = Y\}$ the involution is $x \mapsto -x$ on real parts, so ι-equivariance of w at $x = 0$ reads $w(0) = \iota_X w(0)$. The vertices of ∂X_∞ are $T^k v_i$, $i \in \{0,1,2\}$, and $\iota_X(T^k v_i) = T^{-k}\iota_K(v_i)$ with ι_K the permutation $0 \leftrightarrow 3$, $1 \leftrightarrow 2$ of Lemma 9.2: $T^k v_0 \mapsto T^{-k}v_3 = T^{1-k}v_0$, fixed only if $2k = 1$; $T^k v_1 \mapsto T^{-k}v_2$ and $T^k v_2 \mapsto T^{-k}v_1$, never fixed because v_1 and v_2 lie in different Γ_∞-orbits (Remark 14.1). So no vertex is fixed, and no w satisfies the identity at 0. At an integer k the identity reads $w(-k) = \iota_X w(k)$, i.e. $T^{-k}w(0) = T^{-k}\iota_X w(0)$ by T-equivariance and $\iota_X T^k = T^{-k}\iota_X$, which is the identity at 0. (For the nearest-vertex rule the tied set at $x = 0$ is $\{v_1, v_2\}$, one ι_K-orbit; the argument does not need it.)

(3) The vertices of $K_\partial \subset \partial X_\infty$ lie over the cusp ∞ and are the slots $(1,v)$, $s_1 = I$; by property (1) of Algorithm 14.12 the value at any vertex is w of it, whichever slot or (T1∂)-translate the implementation constructs. For $v \in K_\partial$ step 1 gives $z = g^{-1}\mathrm{pos}(v) = N\,\mathrm{pos}(v)$ ($\mathrm{adj}(g) = \operatorname{diag}(N,1)$ acts as $z \mapsto Nz$), $c' = \infty$, $\gamma_{\mathrm{red}} = I$ and $x = N\operatorname{Re} v$. At $N = 2$: for $v = v_1$, $x = -\frac{1}{3}$, equidistant ($\frac{1}{6}$) from $\operatorname{Re} v_0 = -\frac{1}{2}$ and $\operatorname{Re} v_1 = -\frac{1}{6}$ and farther from every other line vertex; the tie rule takes v_1, so $\tau_\partial(v_1) = v_1$. For $v = v_2$, $x = +\frac{1}{3}$, equidistant from v_2 and v_3; the tie rule takes v_3, so $\tau_\partial(v_2) = v_3$. Now $\iota_X v_1 = \iota_K v_1 = v_2$, so ι-symmetry at v_1 would require $\tau_\partial(v_2) = \iota_X\tau_\partial(v_1) = \iota_X v_1 = v_2$; but $\tau_\partial(v_2) = v_3 \neq v_2$. For $v = v_0$, $x = -1$, equidistant ($\frac{1}{6}$) from $T^{-1}v_1$ ($\mathrm{Re} = -\frac{7}{6}$) and $T^{-1}v_2$ ($\mathrm{Re} = -\frac{5}{6}$); the tie rule takes $T^{-1}v_2$, so $\tau_\partial(v_0) = T^{-1}v_2$, and $\tau_\partial(v_3) = \tau_\partial(Tv_0) = T^N\tau_\partial(v_0) = Tv_2$ by (T1∂) with $h = T$,

$g^{-1}Tg = T^N$ (directly: $x(v_3) = +1$, tie $\{Tv_1, Tv_2\}$, resolved to Tv_2). ι-symmetry at v_0 would require $\tau_\partial(v_3) = \iota_X \tau_\partial(v_0) = \iota_X(T^{-1}v_2) = T\,\iota_K v_2 = Tv_1 \neq Tv_2$. This second failure is clause (2)'s obstruction, attained: $x(v_0) = -N/2$ is an integer at every $(2,\nu)$. (On the slots $S \cdot v_1$, $S \cdot v_2$ over the cusp 0, where v1.4.0 had placed the exhibit, $z = g^{-1}S\,\mathrm{pos}(v) = -N/\mathrm{pos}(v)$, $\gamma_{\mathrm{red}} = S$, $\gamma_{\mathrm{red}} z = \mathrm{pos}(v)/N$ and $x = \mathrm{Re}(v)/N = \mp\frac{1}{12}$: no tie, $\tau_\partial(Sv_i) = S^{-1}v_i$ for $i = 1, 2$, which *is* mirror-consistent; v1.4.0 had applied g in place of g^{-1}.)

Which slots have an integer transported real part. For a slot (j, v) the matrix $N\gamma_{\mathrm{red}} g^{-1} s_j = \gamma_{\mathrm{red}}\,\mathrm{adj}(g)\,s_j$ is integral of determinant N and fixes ∞, hence is $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$ with $AD = N$, $A = p^a$, $D = p^{\nu-a}$ after normalising the common sign, and $x = (A\,\mathrm{Re}\,v + B)/D$. An integer x needs $A\,\mathrm{Re}\,v \in \mathbb{Z}$, i.e. $p^a/2 \in \mathbb{Z}$ ($v \in \{v_0, v_3\}$) or $p^a/6 \in \mathbb{Z}$ ($v \in \{v_1, v_2\}$); the second never holds and the first holds exactly for $p = 2$, $a \geq 1$. So for odd p no transported real part is an integer, while at every $(2,\nu)$ the vertex v_0 of K_∂ realises one ($A = N$, $D = 1$, $x = -2^{\nu-1}$), and at $(2,2)$ the slot $\left(s_j = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, v_0\right)$ even has $A = D = 2$, $B = 1$ and $x = 0$. The tie at $x = \pm\frac{1}{3}$ is where $2\,\mathrm{Re}\,v_1$ is the midpoint of $\mathrm{Re}\,v_0$ and $\mathrm{Re}\,v_1$, i.e. exactly at $\mathrm{Re}\,v_1 = -\frac{1}{6}$; with another admissible placement of v_1 the first exhibit may disappear, the second cannot, which is why (3) is stated for the shipped coordinates and (4) claims nothing beyond what (2) proves at odd p. □

14.4 The boundary operator

Definition 14.14 (Θ_g^∂ and T_p^∂)**.** Let $g \in \Delta$, let $s_1, \ldots, s_m$ and $\delta_j = s_j^{-1} g$ be *the same* transporters and coset elements as in Definition 11.6 (Definition 10.2), and let τ_∂ be a boundary g-twisted chain datum over R. For $\varphi \in \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(\partial X), V_R\big) = C_\partial^\bullet$ define

$$(\Theta_g^\partial \varphi)(x) \;:=\; \sum_{j=1}^{m} \rho_n(\delta_j)\,\varphi\big(\tau_\partial(s_j \cdot x)\big), \qquad x \in C_q(\partial X),\ q \geq 0. \tag{30}$$

This is (21) with τ_∂ in place of τ and $C_\bullet(\partial X)$ in place of $C_\bullet(K^u)$, and nothing else changed. T_p^∂ denotes the induced map on $H^\bullet(C_\partial^\bullet)$ for $g = \mathrm{diag}(1, p)$; the operator family of Definition 10.4 is assembled from these exactly as in the bulk.

> **OWNER DECISION. The Γ-level τ_∂ on ∂X is the normative boundary datum** (owner decision OP1, 2026-09-15). The one-dimensional datum on ∂X_∞ obtained by transporting (30) through Proposition 14.4(1), together with the left-coset representatives $\delta_\infty^\partial, \delta_t^\partial$ of Proposition 14.6(1), is its **Shapiro transcription and a validation route only** (Obligation 14.19(v)). The reason is economy of certified objects: with the Γ-level datum, τ_∂ is one more table of the same shape as τ, certified by the same predicates, and the pair homotopy of §15 compares two data on *one* model; with a separately constructed boundary object one would need a fourth certified table, a homotopy between two boundary cochain maps. (v1.12.0-pre.3: the agreement of (30) with the group-level formulas of Obligation 14.19(v.a)/(v.b) on cohomology is now a theorem — Proposition 14.15(2) in degree 0, Proposition 14.17 in degree 1; (v.c)'s Γ_∞-level chain datum remains a checked route. None of this changes the decision: (30) stays the normative object and the transcription a route.)

Proposition 14.15 (Θ_g^∂ is well defined; and what it is in degree 0)**.** *[proved here] Assume (H4), (B1)–(B3), and let $g, \{s_j\}, \tau_\partial, R$ be as in Definition 14.14.*

1. ***(No hypothesis on ℓ.)*** *Θ_g^∂ carries $C_\partial^\bullet$ into itself and commutes with d; hence it descends to $H^q(C_\partial^\bullet)$ for every q. Its independence of $\{s_j\}$, of τ_∂ and of $g \in \Gamma g \Gamma$ on $H^\bullet(C_\partial^\bullet)$ holds as in Theorem 11.7(a),(c),* and (b) holds here over $\mathbb{Z}$ for every ℓ, *by Lemma 14.10(3) in place of Lemma 11.3(2).*

2. ***(Degree* 0*.)*** *$H^0(C^\bullet_\partial) = V_R^{\Gamma_\infty} = V_R^T \cap V_R^{-I}$, and on it*

$$\Theta_g^\partial\, v \;=\; \sum_{\delta \in B_N/\Gamma_\infty} \rho_n(\delta)\, v,$$

the classical double-coset action, the sum being over the representatives of Proposition 14.6(1) and independent of their choice.

Proof. (1) The proof of Theorem 11.7 uses only (T1), (T2), Lemma 10.3 and Proposition 8.2(2); (T1∂) and (T2∂) are the same two conditions on a different $\mathbb{Z}[\Gamma]$-complex, and every step goes through verbatim. Part (b) there invokes Lemma 11.3(2) *only* to produce the twisted homotopy H; Lemma 14.10(3) produces it over $\mathbb{Z}$ with no hypothesis on ℓ, and the rest of the argument — that $(\widetilde{H}\varphi)(x) = \sum_j \rho_n(\delta_j)\varphi(N_\partial(s_j x))$ is Γ-equivariant, and that the two boundary operators built from τ_∂ and $\tau'_\partial = \tau_\partial + \partial N_\partial + N_\partial \partial$ differ by $d\widetilde{H} + \widetilde{H}d$ — uses only (T1∂).

(2) A 0-cochain φ is a cocycle iff $\varphi(\partial e) = 0$ for every edge, i.e. iff φ is constant on each component; Γ-equivariance then makes it determined by $m_0 := \varphi(v)$ for one vertex $v \in \partial X_\infty$, subject to $\rho_n(h)m_0 = m_0$ for every $h \in \Gamma_\infty$ (the setwise stabiliser of ∂X_∞, whose elements move v inside the component on which φ is constant); so $H^0 = V_R^{\Gamma_\infty}$, which is $V_R^T \cap V_R^{-I}$ by (28).

For the formula fix j and put $c := s_j \cdot \infty$, so that $s_j \cdot v$ is a vertex of ∂X_c. By (T3∂), $\tau_\partial(s_j \cdot v)$ is a single vertex w_j with coefficient 1, and by Lemma 14.9 it lies in $\partial X_{c'}$ with $c' = g^{-1}c = g^{-1}s_j\infty = \delta_j^{-1} \cdot \infty$. Let $\gamma_j \in \Gamma$ be as in Proposition 14.6(2), i.e. $\gamma_j \cdot \infty = c'$ and $\delta_j\gamma_j \in B_N$. Then $\gamma_j \cdot \partial X_\infty = \partial X_{c'}$, and since φ is constant on $\partial X_{c'}$, $\varphi(w_j) = \varphi(\gamma_j \cdot v) = \rho_n(\gamma_j)m_0$; by Proposition 8.2(2), $\rho_n(\delta_j)\varphi(w_j) = \rho_n(\delta_j\gamma_j)m_0$. Summing over j and using that $\{\delta_j\gamma_j\}$ represents B_N/Γ_∞ (Proposition 14.6(2)) gives the displayed sum. It is independent of the representatives because $\rho_n(\delta h)m_0 = \rho_n(\delta)\rho_n(h)m_0 = \rho_n(\delta)m_0$ for $h \in \Gamma_\infty$ and $m_0 \in V_R^{\Gamma_\infty}$. □

Proposition 14.16 (The boundary characters, and which coset system gives them)**.** *(Appendix A, entry A.14, boundary half: the characters are stated in the normative left-coset convention, and the right-coset convention's* det*-twist is stated as part of the proposition rather than hedged.) Let R be a commutative ring of characteristic ℓ, let $n \geq 0$, let p be a prime with $p \neq \ell$ ((H1)) —* except in the first sentence of *(1), the identification of V_R^T, which uses no hypothesis on p at all — put $g = \mathrm{diag}(1, p)$, and write*

$$\mathrm{Nm} \;:=\; \prod_{c \in \mathbb{F}_\ell} (y + cx) \;=\; y^\ell - x^{\ell-1}y \;\in\; \mathrm{Sym}^\ell(\mathbb{Z}^2) \otimes R, \qquad \chi_b(p) := p^b + p^{n+1-b}.$$

Notation. *Claim 16.21 writes this invariant N; in this document $N = p^\nu$ is the Hecke level (§2.2), so it is written* Nm *here — it is the norm, i.e. the $\langle T\rangle$-orbit product, of y. This is a genuine collision of the superseded corpus; it was resolved locally in v0.14.0 and is* ***C18*** *of §2.7 since v0.16.0.*

1. *[proved here] $V_R^T = R[x, \mathrm{Nm}]_n = \bigoplus_{0 \leq b \leq \lfloor n/\ell \rfloor} R \cdot x^{n-b\ell}\mathrm{Nm}^b$, of rank $\lfloor n/\ell \rfloor + 1$ [formally verified: [GT, `prop:bdrychar`]] (this first sentence; not the eigenvector sentence, which needs Proposition 14.15(2)); and for n even each $x^{n-b\ell}\mathrm{Nm}^b$ is an eigenvector of Θ_g^∂ on $H^0(C^\bullet_\partial) = V_R^T$ with eigenvalue $\chi_b(p)$.*
2. *[proved here] With the* right*-coset representatives β_b, β_∞ of Proposition 14.6(4) in place of the left ones, the same monomial is an eigenvector with eigenvalue $\chi_{b+1}(p)$. **The two operators are different**, and the left-coset one is the one (30) computes.* What the right-coset sum is *(v1.4.0; open point (6) of the v1.3.3 changelog): $M_\beta := \sum_{0 \leq b < p} \rho_n(\beta_b) + \rho_n(\beta_\infty)$ is a well-defined endomorphism of V_R^T — each summand maps V_R^T into itself, because $T\beta_b = \beta_b T^p$, $T^p\beta_\infty = \beta_\infty T$ and $V_R^{T^p} = V_R^T$ for $p \neq \ell$ — and the monomials $x^{n-b\ell}\mathrm{Nm}^b$ are an eigenbasis for it. It is asserted as an operator on V_R^T **only**: it is not asserted to be induced by any cochain map of $C^\bullet_\partial$, and none is exhibited — a sum over $\Gamma_\infty\backslash B_p$ is not Γ_∞-equivariant as a formula on cochains, unlike the sum over B_p/Γ_∞ of*

Proposition 14.15(2) — so "the right-coset Θ_g^∂" at cochain level is not an object of this document, and a reader should not look for one. The implementation builds M_β as a fibre-level negative control *only (`beta_control_fiber_operator`, never used by the pipeline; code map). [formally verified: [GT, `prop:bdrychar`]] (clause (2) only: the eigenvalue $\chi_{b+1}(p)$ and $M_\beta(V_R^T) \subseteq V_R^T$; that the monomials are an eigenbasis rests on (1), which is not formalised)*

3. *[proved here] Assume in addition $n < \ell$, n even, and that R is a field. Then $H^0(C_\partial^\bullet)$ and $H^1(C_\partial^\bullet)$ are both 1-dimensional, spanned by $[x^n] \in V_R^T$ and $[y^n] \in (V_R)_T$ respectively, and the classical double-coset operator attached to B_p/Γ_∞ acts on both by $\chi_0(p) = 1 + p^{n+1}$ — the Eisenstein eigenvalue of Obligation 20.7. In degree 0 that operator is Θ_g^∂ (Proposition 14.15(2)); in degree 1 its agreement with Θ_g^∂ is the Shapiro transcription,* ***proved since v1.12.0-pre.3*** *as Proposition 14.17(3) — for every n, every R and every p, the coordinate Φ below being an isomorphism whenever $\rho_n(-I) = \mathrm{id}$ (Proposition 14.17(1)) — and re-checked on every run by Obligation 14.19(v.b) (owner decision OP1). (Until v1.12.0-pre.2 this sentence read "which this document does not prove and which Obligation 14.19(v.b) checks".)* The coordinate *(v1.4.0; open point (4) of the v1.3.3 changelog): the identification used here and in (31) is*

$$\Phi\colon H^1(C_\partial^\bullet) \xrightarrow{\sim} (V_R)_T, \qquad [\varphi] \longmapsto \big[\varphi(\langle v_0, Tv_0\rangle)\big],$$

where $\langle v_0, Tv_0\rangle$ is the oriented path from v_0 to $v_3 = Tv_0$ in K_∂, so that $\varphi(\langle v_0, Tv_0\rangle)$ is the signed sum of φ over its three edges; equivalently $[f(T)]$ for the group 1-cocycle $f(\gamma) := \varphi(\langle v_0, \gamma v_0\rangle)$ of $\langle T\rangle$, which satisfies $f(\gamma\gamma') = f(\gamma) + \rho_n(\gamma) f(\gamma')$. Its inverse sends $[m]$ to the class of the equivariant cochain equal to m on the edge $\langle v_0, v_1\rangle$ and to 0 on the other two edges of K_∂. Reversing the orientation of the path replaces Φ by $-\Phi$, which leaves the transported operator unchanged; what a representative map and a lift must share *is that sign — built on opposite orientations, the transport of (31) into the stored basis is off by -1 and does not agree with (30) wherever $2A_\partial^{(1)} \neq 0$. On H^1 the operator is*

$$[m] \;\longmapsto\; \Big[\; \rho_n(\delta_\infty^\partial)\Big(\sum_{a=0}^{p-1} \rho_n(T)^a\Big) m \;+\; \rho_n(\delta_0^\partial)\, m \;\Big]. \tag{31}$$

4. *The statement of (1) and (3) for H^1 and* arbitrary *n — that $(V_R)_T$ carries the same multiset $\{\chi_b\}_{0 \le b \le \lfloor n/\ell \rfloor}$ — together with the analysis of when two characters coincide, is Claim 16.24 of §16 and is not re-proved here. (v1.12.0-pre.3: that the operator on H^1 is, for n even and $p \neq \ell$, diagonalisable over $\mathbb{F}_\ell$, with eigenvalue $\chi_{n-k}(p)$ on a torus weight vector of weight $a^{n-k}d^k$, is Lemma 14.18; which weights occur — the multiset — is not.)*

Proof. (1), the invariants. T acts on $R[x,y]$ by $x \mapsto x$, $y \mapsto x + y$ (Definition 8.1), an automorphism of order ℓ generating an additive $\mathbb{F}_\ell$-action. The identity $\prod_{c \in \mathbb{F}_\ell}(Y + c) = Y^\ell - Y$ in $\mathbb{F}_\ell[Y]$, homogenised by $Y = y/x$ and multiplied by x^ℓ, gives $\prod_{c \in \mathbb{F}_\ell}(y + cx) = y^\ell - x^{\ell-1}y = \mathrm{Nm}$; T permutes the factors, so Nm is T-invariant, as is x. Since Nm is monic of degree ℓ in y over $R[x]$, the elements x, Nm are algebraically independent and $R[x,y]$ is free over $R[x,\mathrm{Nm}]$ with basis $1, y, \dots, y^{\ell-1}$ (division with remainder by a monic polynomial). Write an invariant f as $f = \sum_{t<\ell} a_t(x, \mathrm{Nm})\, y^t$. Then $0 = Tf - f = \sum_t a_t\big[(x+y)^t - y^t\big]$, and $(x+y)^t - y^t$ has y-degree $t-1$ with leading coefficient $t\,x$. If some $a_t \neq 0$ with $t \geq 1$, take the largest such t; the coefficient of y^{t-1} in the sum is $t\,x\,a_t$ (all $t' < t$ contribute y-degree $\leq t' - 1 \leq t - 2$), so $t\,x\,a_t = 0$. Here t is a unit in R for $1 \leq t < \ell$, because R has characteristic ℓ and therefore contains $\mathbb{F}_\ell$; and x is a non-zero-divisor in $R[x,y]$ for *every* commutative R, the monomials being a basis. So $a_t = 0$ — a contradiction. Hence $f = a_0(x, \mathrm{Nm})$, and the degree-n part of $R[x, \mathrm{Nm}]$ is as displayed.

(1), the eigenvalue. By Proposition 14.15(2) the operator on H^0 is $\sum_{\delta \in B_p/\Gamma_\infty} \rho_n(\delta)$, i.e. $\rho_n(\delta_\infty^\partial) + \sum_{t=0}^{p-1} \rho_n(\delta_t^\partial)$. Now $\delta_\infty^\partial = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ acts by $x \mapsto x$, $y \mapsto py$, so $\delta_\infty^\partial \cdot \mathrm{Nm} = p^\ell y^\ell - p\,x^{\ell-1}y =$

$p\,\mathrm{Nm}$ in characteristic ℓ, since $p^\ell \equiv p$ (Fermat); hence $\delta_\infty^\partial \cdot x^{n-b\ell}\mathrm{Nm}^b = p^b x^{n-b\ell}\mathrm{Nm}^b$. And $\delta_t^\partial = \begin{pmatrix} p & t \\ 0 & 1 \end{pmatrix}$ acts by $x \mapsto px$, $y \mapsto tx+y$, so using $u^\ell \equiv u$ and $p^{\ell-1} \equiv 1$ (here $p \neq \ell$, (H1)),

$$\delta_t^\partial \cdot \mathrm{Nm} = (tx+y)^\ell - (px)^{\ell-1}(tx+y) = t\,x^\ell + y^\ell - t\,x^\ell - x^{\ell-1}y = \mathrm{Nm},$$

while $\delta_t^\partial \cdot x^{n-b\ell} = p^{\,n-b\ell}x^{n-b\ell}$ and $p^{\,n-b\ell} \equiv p^{\,n-b}$. Each of the p elements δ_t^∂ therefore contributes $p^{\,n-b}$, and the total is $p^b + p \cdot p^{\,n-b} = \chi_b(p)$.

(2). $\beta_\infty = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$ acts by $x \mapsto px$, $y \mapsto y$, so it fixes Nm (again $p^{\ell-1} \equiv 1$) and scales $x^{n-b\ell}$ by $p^{\,n-b}$; $\beta_b = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$ acts by $x \mapsto x$, $y \mapsto bx+py$, so $\beta_b \cdot \mathrm{Nm} = (bx+py)^\ell - x^{\ell-1}(bx+py) = b\,x^\ell + p\,y^\ell - b\,x^\ell - p\,x^{\ell-1}y = p\,\mathrm{Nm}$ and β_b fixes x. The p elements β_b contribute $p \cdot p^b$ and β_∞ contributes $p^{\,n-b}$, total $p^{\,b+1}+p^{\,n+1-(b+1)} = \chi_{b+1}(p)$. *That M_β preserves V_R^T:* $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} = \begin{pmatrix} 1 & b+p \\ 0 & p \end{pmatrix} = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}\begin{pmatrix} 1 & p \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & p \\ 0 & 1 \end{pmatrix}\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & p \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ are the two identities; for $v \in V_R^T$ they give $\rho_n(T)\rho_n(\beta_b)v = \rho_n(\beta_b)\rho_n(T)^p v = \rho_n(\beta_b)v$ and $\rho_n(T)^p\rho_n(\beta_\infty)v = \rho_n(\beta_\infty)\rho_n(T)v = \rho_n(\beta_\infty)v$, so $\rho_n(\beta_b)v \in V_R^T$ and $\rho_n(\beta_\infty)v \in V_R^{T^p}$. Finally $V_R^{T^p} = V_R^T$: $\rho_n(T)$ is unipotent (Definition 8.1), so in characteristic ℓ, $(\rho_n(T)-\mathrm{id})^{\ell^k} = \rho_n(T)^{\ell^k} - \mathrm{id} = 0$ for $\ell^k \geq n+1$, i.e. $\rho_n(T)$ has order a power of ℓ, and p being prime to ℓ, $\rho_n(T)^p$ generates the same cyclic group as $\rho_n(T)$.

(3). For $n < \ell$ one has $\lfloor n/\ell \rfloor = 0$, so $V_R^T = Rx^n$ is 1-dimensional and, R being a field, so is $(V_R)_T$ (both are the kernel and the cokernel of the same endomorphism $\rho_n(T) - \mathrm{id}$ of a finite-dimensional space). The H^0 eigenvalue is (1) at $b = 0$. For H^1: $H^1(C_\partial^\bullet) = H^1(\langle T\rangle; V_R) = (V_R)_T$ (Proposition 14.4(2)), and the identification is the Φ of the statement: every equivariant 1-cochain φ on ∂X_∞ is a cocycle ($C_2 = 0$), $f(\gamma) := \varphi(\langle v_0, \gamma v_0\rangle)$ satisfies the cocycle rule because $\langle v_0, \gamma\gamma' v_0\rangle = \langle v_0, \gamma v_0\rangle + \gamma\langle v_0, \gamma' v_0\rangle$ as chains and $\varphi(\gamma\, c) = \rho_n(\gamma)\varphi(c)$; a coboundary $d\psi$ has $f(T) = \psi(Tv_0) - \psi(v_0) = (\rho_n(T) - \mathrm{id})\psi(v_0) \in \mathrm{im}(\rho_n(T) - \mathrm{id})$, so Φ is well defined; it is onto because the cochain φ_m of the inverse formula has $\Phi[\varphi_m] = m$; and it is injective because a φ with $\varphi(\langle v_0, Tv_0\rangle) \in \mathrm{im}(\rho_n(T) - \mathrm{id})$, say $= (\rho_n(T) - \mathrm{id})u$, equals $d\psi$ for the equivariant ψ with $\psi(v_0) := u$, $\psi(v_1) := u + \varphi(\langle v_0, v_1\rangle)$, $\psi(v_2) := \psi(v_1) + \varphi(\langle v_1, v_2\rangle)$, which is consistent at $v_3 = Tv_0$ since $\psi(v_2) + \varphi(\langle v_2, v_3\rangle) = u + \varphi(\langle v_0, Tv_0\rangle) = \rho_n(T)u = \rho_n(T)\psi(v_0)$. Reversing the orientation replaces f by $-f$ and Φ by $-\Phi$, and $(-\Phi)\,A\,(-\Phi)^{-1} = \Phi A \Phi^{-1}$ for any operator A. On $(V_R)_T$ the double-coset operator attached to B_p/Γ_∞ is computed one double coset at a time (Proposition 14.6(3)). For $\Gamma_\infty\delta_\infty^\partial\Gamma_\infty$ the index is 1, the operator on a 1-cocycle f of $\langle T\rangle$ is $f \mapsto \rho_n(\delta_\infty^\partial) \circ f \circ \mathrm{conj}_{\delta_\infty^\partial}^{-1}$, and $(\delta_\infty^\partial)^{-1}T\delta_\infty^\partial = T^p$, so $m = f(T) \mapsto \rho_n(\delta_\infty^\partial)f(T^p) = \rho_n(\delta_\infty^\partial)\big(\sum_{a<p}\rho_n(T)^a\big)m$ by the cocycle rule $f(\gamma\gamma') = f(\gamma) + \gamma \cdot f(\gamma')$. For $\Gamma_\infty\delta_0^\partial\Gamma_\infty$ take the transporters T^t, $0 \leq t < p$, of $\langle T^p\rangle \times \langle -I\rangle$ in Γ_∞; then $T^t \cdot T = T^{t+1}$ has trivial cocycle for $t < p-1$ and cocycle T^p for $t = p-1$, so only the last summand survives and gives $m \mapsto \rho_n(T^{-(p-1)}\delta_0^\partial)m$, which is $\rho_n(\delta_0^\partial)m$ modulo $\mathrm{im}(\rho_n(T) - \mathrm{id})$. This is (31). Evaluating at $m = y^n$: $\delta_0^\partial = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$ sends $y \mapsto y$, so the second term is y^n; and $T^a y^n = (ax+y)^n$, $\delta_\infty^\partial(ax+y)^n = (ax+py)^n$, so the first term is $\sum_{a<p}\sum_k \binom{n}{k} a^{\,n-k}p^k x^{\,n-k}y^k$. For $n < \ell$, $\rho_n(T) - \mathrm{id}$ is a single nilpotent Jordan block of rank n whose image is $\mathrm{span}\{x^{\,n-k}y^k : k < n\}$, so every term with $k < n$ dies in $(V_R)_T$ and what is left is $\sum_{a<p} p^n y^n = p^{\,n+1}y^n$. Total $1 + p^{\,n+1} = \chi_0(p)$. □

Corroboration. All three parts were additionally checked by direct polynomial arithmetic in $\mathbb{F}_\ell[x,y]$ — the T-invariance of every $x^{\,n-b\ell}\mathrm{Nm}^{\,b}$, $\dim V_R^T = \lfloor n/\ell \rfloor + 1$ as $\dim\ker(\rho_n(T) - \mathrm{id})$, the χ_b of (1), the χ_{b+1} of (2) and the H^1 statement of (3) — over $\ell \in \{2,3,5,7,11,13\}$, $n \leq 24$ and $p \in \{2,3,5,7,11,13\}$ with $p \neq \ell$: 0 disagreements. This carries **no** [verified computationally] tag,

because it was an *ad hoc* computation and not a registered validation; there is nothing in the tree to re-run. Obligation 14.19(vi) is where it becomes one.

Proposition 14.17 (The Shapiro transcription in degree 1)**.** *[proved here] (v1.12.0-pre.3, hecke-extensions slot-12 WP-16: the degree-*1 *identity that Proposition 14.16(3) stated and left unproved — the degree-*1 *twin of Proposition 14.15(2).) Assume (H4) and (B1)–(B3). Let R be a commutative ring, $n \geq 0$, p a prime, $\nu \geq 1$, $N = p^{\nu}$, $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, and let $\{s_j\}$, $\delta_j = s_j^{-1} g$ and τ_∂ be as in Definition 14.14 —* any *admissible choices. Fix a vertex v_0 of ∂X_∞ (for the model of Definition 7.1, the vertex v_0 of K_∂). For vertices u, w of one component of ∂X write $\langle u, w\rangle$ for the unique* 1*-chain of that component with $\partial\langle u, w\rangle = w - u$ (step 0 of the proof), write $\equiv$ for equality in $(V_R)_T = V_R/\operatorname{im}(\rho_n(T) - \mathrm{id})$, and for $k \in \mathbb{Z}$ put $[k]_X := \sum_{t=0}^{k-1} X^t$ if $k \geq 0$ and $[k]_X := -\sum_{t=k}^{-1} X^t$ if $k < 0$, so that $X^k - 1 = (X-1)[k]_X$ in $\mathbb{Z}[X^{\pm 1}]$.*

1. ***(The coordinate, over every R.)*** *The map $\Phi\colon H^1(C_\partial^\bullet) \to (V_R)_T$, $[\varphi] \mapsto [\varphi(\langle v_0, Tv_0\rangle)]$, is well defined and R-linear, does not depend on v_0, and changes sign with the orientation of the path. If $\rho_n(-I) = \mathrm{id}$ on V_R — n even, or $2 = 0$ in R — it is an isomorphism. For the model it is the Φ of Proposition 14.16(3), which is therefore an isomorphism without that clause's hypotheses $n < \ell$ and "R a field".*
2. ***(Every ν, every p.)*** *Let $\{\beta_C\}$ be any set of representatives of the left cosets $C \in B_N/\Gamma_\infty$ of Proposition 14.6(1), and for each C let $k_C \in \mathbb{Z}$ be defined by $\beta_C^{-1} T \beta_{T^{-1}C} = \pm T^{k_C}$ (the left-hand side lies in Γ_∞, because $T\beta_{T^{-1}C} \in C = \beta_C\Gamma_\infty$). Then for every $\varphi \in C_\partial^1$, with $m := \varphi(\langle v_0, Tv_0\rangle)$,*

$$\Phi\big(\Theta_g^\partial[\varphi]\big) \;=\; \Big[\, \sum_{C \in B_N/\Gamma_\infty} \rho_n(\beta_C)\, [k_C]_{\rho_n(T)}\, m \,\Big]. \tag{32}$$

   Consequently, if $\rho_n(-I) = \mathrm{id}$ on V_R, the right-hand side is a well-defined endomorphism of $(V_R)_T$, independent of the representatives, and is the operator induced by (30) on $H^1(C_\partial^\bullet)$ in the coordinate Φ. (Without $\rho_n(-I) = \mathrm{id}$, over every R: for representatives with positive diagonal, e.g. $\begin{pmatrix} p^u & b \\ 0 & p^{\nu-u} \end{pmatrix}$, every $\beta_C^{-1} T \beta_{T^{-1}C}$ is $+T^{k_C}$, and then $\sum_C \rho_n(\beta_C)[k_C]_{\rho_n(T)}(\rho_n(T) - \mathrm{id}) = \sum_C \big(\rho_n(T\beta_{T^{-1}C}) - \rho_n(\beta_C)\big) = (\rho_n(T) - \mathrm{id}) \sum_C \rho_n(\beta_C)$, so the right-hand side is a well-defined endomorphism of $(V_R)_T$ for such representatives; v1.12.0-pre.3, referee P4.)
3. ***($\nu = 1$: the transcription.)*** *Let $\nu = 1$. Then (31) is a well-defined endomorphism of $(V_R)_T$, and*

$$\Phi \circ T_p^\partial \;=\; (31) \circ \Phi \qquad \text{on } H^1(C_\partial^\bullet).$$

   If moreover $\rho_n(-I) = \mathrm{id}$ on V_R (n even, or $2 = 0$ in R), the operator induced by (30) on $H^1(C_\partial^\bullet)$ ***is*** *(31) under the isomorphism Φ.*

No hypothesis is made on ℓ, on n against ℓ, on p against ℓ (the boundary operator at $p = \ell$ is included), or that R is a field.

Proof. Conventions, read off the definitions. $C_\partial^q = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_q(\partial X), V_R)$ (Proposition 14.4), so $\varphi(\gamma x) = \rho_n(\gamma)\varphi(x)$ for $\gamma \in \Gamma$ — the convention of the proof of Proposition 14.15(2); $(d\psi)(x) = \psi(\partial x)$ (proof of Theorem 11.7); a cochain is extended R-linearly to $C_\bullet(\partial X) \otimes R$, where τ_∂ takes its values (Definition 14.8). By Proposition 14.15(1), $\Theta_g^\partial\varphi \in C_\partial^1$ and $\Theta_g^\partial[\varphi] = [\Theta_g^\partial\varphi]$. Since $C_2(\partial X) = 0$ ((B2)), every 1-cochain is a cocycle. $-I$ fixes every chain ((H4)(iv)).

Step 0: chains on ∂X. By (B2) each component ∂X_c is a contractible 1-dimensional complex, so $0 \to C_1(\partial X_c) \xrightarrow{\partial} C_0(\partial X_c) \xrightarrow{\varepsilon} \mathbb{Z} \to 0$ is an exact sequence of free abelian groups, hence split, and it stays exact after $\otimes R$. Summing over the components: (a) $\partial\colon C_1(\partial X) \otimes R \to C_0(\partial X) \otimes R$ is injective; (b) a 0-chain is a boundary iff its total coefficient on every component is 0, i.e. iff its class in $H_0(\partial X) \otimes R = R[\pi_0(\partial X)]$ vanishes. Hence $\langle u, w\rangle$ exists and is unique;

$\langle u,w\rangle+\langle w,x\rangle=\langle u,x\rangle$; $\gamma\langle u,w\rangle=\langle\gamma u,\gamma w\rangle$ for $\gamma\in\Gamma$, since Γ acts by chain automorphisms permuting the components ((B2)); and $\langle v_0,T^kv_0\rangle=[k]_T\cdot\langle v_0,Tv_0\rangle$ for every $k\in\mathbb{Z}$, both sides having boundary $T^kv_0-v_0$ because $(T-1)[k]_T=T^k-1$.

(1). Write $e:=\langle v_0,Tv_0\rangle$. A coboundary has $(d\psi)(e)=\psi(Tv_0)-\psi(v_0)=(\rho_n(T)-\mathrm{id})\psi(v_0)$, so Φ is well defined; it is clearly R-linear. For another vertex v_0' of ∂X_∞, step 0 gives $\langle v_0',Tv_0'\rangle=\langle v_0',v_0\rangle+e+T\langle v_0,v_0'\rangle$, so the two values differ by $(\rho_n(T)-\mathrm{id})\varphi(\langle v_0,v_0'\rangle)$. Reversing the orientation replaces e by $-e$. Now let $\rho_n(-I)=\mathrm{id}$. A simplicial automorphism of a triangulated line is a shift or a reflection, and a reflection fixes a vertex or an edge setwise; by (B3) (and (H4)(i), no inversions) T fixes no simplex of ∂X_∞, so it is a shift by some $d\neq0$ positions, and the $|d|$ edges of the segment from v_0 to Tv_0 — the support of e, each with coefficient ±1 — sit at $|d|$ consecutive positions and so lie in pairwise distinct $\langle T\rangle$-orbits. *Onto:* given $m\in V_R$, pick an edge ϵ of that segment, with coefficient $c=\pm1$ in e; the Γ_∞-equivariant cochain on ∂X_∞ equal to $\rho_n(T)^kcm$ on $T^k\epsilon$ and to 0 on the other orbits exists ($\langle T\rangle$ acts freely and $-I$ trivially, on the edges and on V_R), extends to a unique $\varphi\in C^1_\partial$ by Proposition 14.4(1), and has $\varphi(e)=m$. *One-to-one:* if $\varphi(e)=(\rho_n(T)-\mathrm{id})u$, put $\psi(x):=u+\varphi(\langle v_0,x\rangle)$ for the vertices x of ∂X_∞. Then $\psi(Tx)=u+\varphi(e)+\rho_n(T)\varphi(\langle v_0,x\rangle)=\rho_n(T)\psi(x)$ and $\psi((-I)x)=\psi(x)=\rho_n(-I)\psi(x)$, so ψ is Γ_∞-equivariant and extends to a 0-cochain of $C^\bullet_\partial$ (Proposition 14.4(1)); on an edge $\langle x,x'\rangle$ of ∂X_∞, $\psi(x')-\psi(x)=\varphi(\langle x,x'\rangle)$, so $d\psi$ and φ agree on ∂X_∞, hence everywhere by equivariance, and $[\varphi]=0$. For the model, the path of K_∂ from v_0 to $v_3=Tv_0$ is a 1-chain with boundary v_3-v_0, hence equals e, and Φ is the map of Proposition 14.16(3).

(2), step 1: the coset attached to each transporter. For each j choose $\gamma_j\in\Gamma$ with $\delta_j\gamma_j\in B_N$ and put $\beta_j:=\delta_j\gamma_j$, $C_j:=\beta_j\Gamma_\infty$. By Proposition 14.6(2) (whose proof uses of δ_j only that $\delta_j\Gamma\in\Gamma g\Gamma/\Gamma$, so it applies to any admissible $\{s_j\}$), $j\mapsto C_j$ is a bijection onto B_N/Γ_∞, the admissible γ_j form the coset $\gamma_j\Gamma_\infty$, and $\gamma_j\cdot\infty=\delta_j^{-1}\cdot\infty$; replacing γ_j within that coset replaces β_j by any element of C_j, so a choice of the γ_j realises the given family $\{\beta_C\}$, and we fix it: $\beta_j=\beta_{C_j}$. For each j let $j\cdot T$ be the index with $\Gamma''s_jT=\Gamma''s_{j\cdot T}$ (right translation on $\Gamma''\backslash\Gamma$, so $j\mapsto j\cdot T$ is a permutation), $h_j:=s_jTs_{j\cdot T}^{-1}\in\Gamma''$ and $h_j':=g^{-1}h_jg\in\Gamma'\subseteq\Gamma$ (Proposition 10.1(3)). In $\mathrm{M}_2(\mathbb{Z})$,

$$\delta_j\,h_j'\;=\;s_j^{-1}h_jg\;=\;T\,s_{j\cdot T}^{-1}g\;=\;T\,\delta_{j\cdot T}. \tag{i}$$

So $\delta_j(h_j'\gamma_{j\cdot T})=T\beta_{j\cdot T}$ is upper triangular and lies in $\Gamma g\Gamma$, i.e. in B_N, and by Proposition 14.6(2) $h_j'\gamma_{j\cdot T}=\gamma_j\varepsilon_j$ with $\varepsilon_j\in\Gamma_\infty$; then $\varepsilon_j=\beta_j^{-1}T\beta_{j\cdot T}$. Hence $T\beta_{j\cdot T}\in C_j$, i.e. $C_{j\cdot T}=T^{-1}C_j$, and $\varepsilon_j=\pm T^{k_{C_j}}$; write $k_j:=k_{C_j}$. (ii)

Step 2: the chains. Put $w_i:=\tau_\partial(s_iv_0)$ and $y_j:=\tau_\partial(s_je)$. As $s_je=\langle s_jv_0,s_jTv_0\rangle$ and $s_jTv_0=h_js_{j\cdot T}v_0$, (T2$\partial$) and (T1$\partial$) give

$$\partial y_j\;=\;\tau_\partial(h_js_{j\cdot T}v_0)-\tau_\partial(s_jv_0)\;=\;h_j'w_{j\cdot T}-w_j.$$

By (T3∂) and Lemma 14.9, the class of w_i in $R[\pi_0(\partial X)]$ is that of the cusp $g^{-1}s_i\infty=\delta_i^{-1}\infty=\gamma_i\infty$ (as s_iv_0 lies in the component over $s_i\infty$: $v_0\in\partial X_\infty$ and the bijection $\pi_0(\partial X)\cong\mathbb{P}^1(\mathbb{Q})$ of (B2) is Γ-equivariant), which is the class of the vertex γ_iv_0 ((B2)); *only this class is used*, not that w_i is a single vertex. By step 0(b), $w_i=\gamma_iv_0+\partial z_i$ for some $z_i\in C_1(\partial X)\otimes R$. The chain $y_j':=\gamma_j\langle v_0,T^{k_j}v_0\rangle+h_j'z_{j\cdot T}-z_j$ has boundary $\gamma_jT^{k_j}v_0-\gamma_jv_0+h_j'(w_{j\cdot T}-\gamma_{j\cdot T}v_0)-(w_j-\gamma_jv_0)=h_j'w_{j\cdot T}-w_j$, because $\gamma_jT^{k_j}v_0=\gamma_j\varepsilon_jv_0=h_j'\gamma_{j\cdot T}v_0$ by (ii); so $y_j=y_j'$ by step 0(a).

Step 3: evaluation. With $m=\varphi(e)$, step 0 and equivariance give $\varphi(\gamma_j\langle v_0,T^{k_j}v_0\rangle)=\rho_n(\gamma_j)[k_j]_{\rho_n(T)}m$. Using (i) and Proposition 8.2(2),

$$\rho_n(\delta_j)\varphi(y_j)\;=\;\rho_n(\beta_j)[k_j]_{\rho_n(T)}m\;+\;\rho_n(T)\rho_n(\delta_{j\cdot T})\varphi(z_{j\cdot T})\;-\;\rho_n(\delta_j)\varphi(z_j).$$

Step 4: the sum. By (30), $(\Theta_g^\partial\varphi)(e)=\sum_j\rho_n(\delta_j)\varphi(y_j)$. Since $j\mapsto j\cdot T$ is a permutation, the last two terms sum over j to $(\rho_n(T)-\mathrm{id})\sum_j\rho_n(\delta_j)\varphi(z_j)\equiv0$: the correction terms telescope. Hence $\Phi(\Theta_g^\partial[\varphi])=[(\Theta_g^\partial\varphi)(e)]=\big[\sum_j\rho_n(\beta_j)[k_j]_{\rho_n(T)}m\big]$, which is (32) since $j\mapsto C_j$ is a bijection

and $\beta_j = \beta_{C_j}$. The family $\{\beta_C\}$ was arbitrary. If $\rho_n(-I) = \mathrm{id}$, Φ is an isomorphism by (1), so the right-hand side equals $\Phi\,\Theta_g^\partial\,\Phi^{-1}[m]$: it depends only on $[m]$ and not on the representatives.

(3). Take the representatives δ_∞^∂, δ_t^∂ of Proposition 14.6(1) and read $T^{-1}C$ and k_C off its identities $T\delta_\infty^\partial = \delta_\infty^\partial T^p$, $T\delta_t^\partial = \delta_{t+1}^\partial$ $(t < p-1)$, $T\delta_{p-1}^\partial = \delta_0^\partial T$: $T^{-1}C_\infty = C_\infty$ with $k = p$; $T^{-1}C_t = C_{t-1}$ with $k = 0$ for $1 \le t < p$; $T^{-1}C_0 = C_{p-1}$ with $k = 1$. So the sum in (32) is $\rho_n(\delta_\infty^\partial)\sum_{a=0}^{p-1}\rho_n(T)^a m + \rho_n(\delta_0^\partial)m$ — (31) on the nose, and (2) gives the displayed identity. That (31) is well defined on $(V_R)_T$, for every R and every p: by the identities just quoted and $\delta_0^\partial T = T^p\delta_0^\partial$, $\rho_n(\delta_\infty^\partial)\big(\sum_{a<p}\rho_n(T)^a\big)(\rho_n(T)-\mathrm{id}) = \rho_n(\delta_\infty^\partial)(\rho_n(T)^p-\mathrm{id}) = (\rho_n(T)-\mathrm{id})\rho_n(\delta_\infty^\partial)$ and $\rho_n(\delta_0^\partial)(\rho_n(T)-\mathrm{id}) = (\rho_n(T)^p-\mathrm{id})\rho_n(\delta_0^\partial)$, both with image in $\mathrm{im}(\rho_n(T)-\mathrm{id})$. The last sentence of (3) is (1). □

What the proof uses, and what it does not (v1.12.0-pre.3). Only (T1∂), (T2∂), (T3∂) — the last only through the H_0-class of $\tau_\partial(s_i v_0)$, so no single-vertex normalisation of the table and no base change from a $\mathbb{Z}$-datum is needed —, (B2), (B3) (for (1)), (H4)(iv), Lemma 14.9, Proposition 14.6(1)–(2), Proposition 10.1(3) and Proposition 8.2(2). The choices of $\{s_j\}$, of τ_∂, of v_0 and of the γ_j drop out, as in Proposition 14.15(2); the independence of τ_∂ of Proposition 14.15(1) is not invoked. No division occurs: the correction terms of step 4 telescope under $j \mapsto j\cdot T$, which is why $p = \ell$ is covered. *At $\nu \ge 2$,* (2) is an H^1 formula over the *left* cosets of Proposition 14.6(1), which hold at every ν; it needs none of the double-coset structure that Remark 14.7 records as not stated, and it does not extend Obligation 14.19(v), which stays (v.a) alone at $\nu \ge 2$. *For n odd and $2 \in R^\times$* both sides of (2) and (3) vanish ($C_\partial^\bullet = 0$, Proposition 14.4(2)) while $(V_R)_T$ need not; that is why the isomorphism clauses carry $\rho_n(-I) = \mathrm{id}$ and nothing else does.

Corroboration (v1.12.0-pre.3). The chain-level operator (30) was evaluated literally on ∂X — with random transporters $\{h_j s_j\}$, $h_j \in \Gamma_0(N)$, a random boundary datum whose vertex values are 0-chains of total coefficient 1 on the forced component plus total-zero chains on it and on a second component, lines with one and with three vertex orbits per period, and random cochains — and compared with (31) and (32) in $(V_R)_T$ over $\mathbb{F}_\ell$, for $\ell \in \{2,3,5,7,11\}$, $p \in \{2,3,5,7\}$ ($p = \ell$ included), even $n \le 22$ at $\nu = 1$, $n \le 12$ at $\nu = 2$, and odd $n \le 7$ at $\ell = 2$: 1488 trials, 0 disagreements (288 of them at $p = \ell$). The negative control that tests part (3)'s coset bookkeeping — the sum $\sum_{a<p}\rho_n(T)^a$ attached to δ_0^∂ instead of δ_∞^∂ — was rejected in 572 of the 1248 trials at $\nu = 1$ (it coincides with the true operator on the rest); a second one, Φ reversed on one side only, is a pure sign control (rejected exactly when $2\,\Theta_g^\partial[\varphi] \ne 0$) and tests no step of the proof. The WP-16 referee's re-run extended the grid ($p = \ell$: 405 trials, 370 with $[m] \ne 0$) and added a control that does target the datum axioms: a table with (T3∂)'s total coefficient broken to $c = 2$ is rejected against the formula wherever $(c-1)\Theta_g^\partial[\varphi] \ne 0$ and matches c times it in every trial (v1.12.0-pre.3, referee P5). Like the corroboration of Proposition 14.16 this is an *ad hoc* computation (`prop_t_check.py`, outside the tree), not a registered validation, and carries no [verified computationally] tag; the proposition is [proved here], and Obligation 14.19(v.b) re-checks it on every run at $\nu = 1$.

Lemma 14.18 (Boundary operators away from ℓ factor through the torus)**.** *[proved here] (v1.12.0-pre.3, hecke-extensions slot-12 WP-16; "Lemma B′" of the slot-12 finality proposal.) Let $R = \mathbb{F}_\ell$, n even, $p \ne \ell$ a prime ((H1)), $g = \mathrm{diag}(1,p)$, $\bar p := p \bmod \ell$, and assume (H4), (B1)–(B3). For $A \in \mathrm{M}_2(\mathbb{F}_\ell)$ write $\rho_n(A)$ for $\rho_n(\tilde A) \bmod \ell$, $\tilde A$ any integral lift (well defined by (9), and multiplicative by Proposition 8.2(2)). Let $U := \{\rho_n(T)^k\}$ and let $\mathbf{T}(\mathbb{F}_\ell)$ be the diagonal torus of $\mathrm{GL}_2(\mathbb{F}_\ell)$; it normalises the unipotent upper-triangular group, so $\rho_n(\mathbf{T}(\mathbb{F}_\ell))$ preserves $V_R^U = V_R^T = H^0(C_\partial^\bullet)$ and $\mathrm{im}(\rho_n(T)-\mathrm{id})$, hence acts on $(V_R)_T \cong H^1(C_\partial^\bullet)$ (the latter through Φ of Proposition 14.17(1)).*

0. *On $H^0(C_\partial^\bullet) = V_R^T$, $T_p^\partial = \rho_n(\mathrm{diag}(1,\bar p)) + \bar p\,\rho_n(\mathrm{diag}(\bar p,1))$.*
1. *On $H^1(C_\partial^\bullet)$, in the coordinate Φ, $T_p^\partial = \bar p\,\rho_n(\mathrm{diag}(1,\bar p)) + \rho_n(\mathrm{diag}(\bar p,1))$.*
2. *Consequently, in both degrees, T_p^∂ lies in the image of $\mathbb{F}_\ell[\mathbf{T}(\mathbb{F}_\ell)]$, is diagonalisable over $\mathbb{F}_\ell$; the T_p^∂ for the primes $p \ne \ell$ commute pairwise; and $T_p^\partial = T_{p'}^\partial$ whenever $p \equiv p' \pmod\ell$.*

3. *On a vector of* $\mathbf{T}(\mathbb{F}_\ell)$*-weight* $\mathrm{diag}(a,d) \mapsto a^{n-k}d^k$ *the eigenvalue of* T_p^∂ *is* $\chi_k(p)$ *in degree* 0 *and* $\chi_{n-k}(p)$ *in degree* 1 *(*$\chi_b(p) = p^b + p^{n+1-b}$ *as in Proposition 14.16).*

Proof. The preliminaries: $\rho_n(T)^\ell = \rho_n(T^\ell) = \rho_n(I)$ as $T^\ell \equiv I \pmod{\ell}$, so U is the image of the unipotent upper-triangular group of $\mathrm{SL}_2(\mathbb{F}_\ell)$; $\mathrm{diag}(a,d)\begin{pmatrix}1 & u\\ 0 & 1\end{pmatrix}\mathrm{diag}(a,d)^{-1} = \begin{pmatrix}1 & au/d\\ 0 & 1\end{pmatrix}$, so for $t \in \mathbf{T}(\mathbb{F}_\ell)$ one has $\rho_n(t)\rho_n(T) = \rho_n(T)^k\rho_n(t)$ for some k, and $\rho_n(T)^k - \mathrm{id} = (\rho_n(T) - \mathrm{id})[k]_{\rho_n(T)}$; this gives both preservation statements, and $V_R^U = V_R^T$. $H^0(C_\partial^\bullet) = V_R^T$ because $\rho_n(-I) = \mathrm{id}$ (Proposition 14.15(2)).

(0) By Proposition 14.15(2) with the representatives of Proposition 14.6(1), $T_p^\partial = \rho_n(\delta_\infty^\partial) + \sum_{t<p}\rho_n(T)^t\rho_n(\delta_0^\partial)$ on V_R^T. From $\delta_0^\partial T = T^p\delta_0^\partial$, $\rho_n(\delta_0^\partial)v \in V_R^{T^p}$ for $v \in V_R^T$, and $V_R^{T^p} = V_R^T$ for $p \neq \ell$ (proof of Proposition 14.16(2)); so the sum is $p\,\rho_n(\delta_0^\partial)$, and reducing the matrices modulo ℓ gives (0).

(1) By Proposition 14.17(3), T_p^∂ is (31) under Φ. On $(V_R)_T$, $\rho_n(T)$ acts as the identity and $\rho_n(\delta_\infty^\partial) = \rho_n(\mathrm{diag}(1,\bar{p}))$ preserves $\mathrm{im}(\rho_n(T) - \mathrm{id})$ (preliminaries), so $\rho_n(\delta_\infty^\partial)\sum_{a<p}\rho_n(T)^a m \equiv \bar{p}\,\rho_n(\mathrm{diag}(1,\bar{p}))m$, and $\rho_n(\delta_0^\partial) = \rho_n(\mathrm{diag}(\bar{p},1))$.

(2) $\mathbf{T}(\mathbb{F}_\ell)$ is abelian and $t^{\ell-1} = 1$ for each of its elements, so each $\rho_n(t)$ is annihilated by $X^{\ell-1} - 1 = \prod_{c\in\mathbb{F}_\ell^\times}(X - c)$, which is split with distinct roots: each $\rho_n(t)$ is diagonalisable over $\mathbb{F}_\ell$, a commuting family of such is simultaneously diagonalisable, and so is every $\mathbb{F}_\ell$-linear combination — in each degree, on the induced actions. The formulas of (0) and (1) depend on p only through $\bar{p}$.

(3) $\mathrm{diag}(1,\bar{p})$ and $\mathrm{diag}(\bar{p},1)$ act on that weight by $\bar{p}^k$ and $\bar{p}^{n-k}$: in degree 0, $p^k + p\cdot p^{n-k} = \chi_k(p)$; in degree 1, $p\cdot p^k + p^{n-k} = p^{n-k} + p^{n+1-(n-k)} = \chi_{n-k}(p)$ (in $\mathbb{F}_\ell$). □

Lemma 14.18(2) is the semisimplicity half of Claim 16.24(2), now proved for $X_5 = H^1(C_\partial^\bullet)$ at n even, $p \neq \ell$; the *multiset* half — which torus weights $(V_R)_T$ carries — does **not** follow and stays [conjectural] there. (3) reads the eigenvalues off a weight; it does not compute the weights. The degree-1 weight rule χ_{n-k}, not χ_k, is consistent with Proposition 14.16(3): for $n < \ell$ the class of y^n $(k = n)$ carries $\chi_0(p)$.

Obligation 14.19 (Certified τ_∂-table and boundary operator)**.** [formally verified: [GT, `chk:taubd`]] [formally verified: [GT, `chk:taubd-p2`]] [formally verified: [GT, `chk:pairgrid`]] (clauses (i)–(iv) as a predicate, verified at $(p,\nu) = (2,1)$ on the golden table and at the five grid points of [GT]; not (v)) Assume (H4) and (B1)–(B3). An implementation of the boundary datum emits, for each $g = \mathrm{diag}(1,N)$ and each Γ''-orbit representative simplex σ of ∂X, a chain $\tau_\partial(s_j\cdot\sigma) = \sum_t n_t^{(j,\sigma)}\,\gamma_t\cdot\sigma_t$ with $n_t^{(j,\sigma)} \in \mathbb{Z}$, $\gamma_t \in \Gamma$, in the stored-chain format of Algorithm 18.3, and must discharge the following. Each clause names the stored data it reads, the predicate, the ring it is evaluated over, and the action on failure.

(i) **Structural.** Every emitted chain is supported in a single component, the component map agrees with $c \mapsto g^{-1}c$ of Lemma 14.9 recomputed from g alone, and in degree 0 each chain is a single simplex with coefficient 1 — i.e. (T3∂). Over $\mathbb{Z}$; a failure **aborts**.

(ii) **(T2∂).** $\partial\tau_\partial(s_j\sigma) = \tau_\partial(\partial(s_j\sigma))$ for every stored (j,σ) in degree 1, both sides evaluated through the *same* (T1∂) helper that the construction uses. Over $\mathbb{Z}$; a failure **aborts**.

(iii) **(T1∂) bookkeeping.** For every stored transporter $h \in \Gamma''$ occurring in the table, $\tau_\partial(h\cdot x) = (g^{-1}hg)\cdot\tau_\partial(x)$ as *chains*, compared after reduction to canonical (n_t,γ_t,σ_t) form. Over $\mathbb{Z}$; a failure **aborts**. This is Obligation 20.11's clause on the boundary complex, with ∂X in place of K^u.

(iv) **Integrality is unconditional here.** Every coefficient $n_t^{(j,\sigma)}$ lies in $\{0,\pm1\}$ and the recorded solve regime of a τ_∂-table is $\mathbb{Z}$, for *every* ℓ. Over $\mathbb{Z}$; a failure **aborts** — unlike Remark 11.5's bulk clause, where a mod-ℓ regime is a recorded weaker datum, a mod-ℓ τ_∂-table is an implementation defect (Lemma 14.10(2), Remark 14.11).

(v) **The Shapiro transcription agrees — a cross-check, and a route, not a definition.** The normative object is (30); per owner decision OP1 each comparison below is a *validation route*, and a disagreement is a defect in the transcription, never a redefinition of Θ_g^∂ (since v1.12.0-pre.3, with (v.a) and (v.b) resting on proved identities — Propositions 14.15(2) and 14.17(3) — a disagreement there is a defect of the implementation). Three comparisons, each between the operator induced by (30) on $H^\bullet(C_\partial^\bullet)$ and a second object, as matrices over R in the stored basis, entrywise; a failure of any **aborts**. (Until v1.4.0 this clause named one comparison where the chapter offers three; open point (1) of the v1.3.3 changelog.)

(v.a) **Degree 0, every ν.** The second object is $\sum_{\delta\in B_N/\Gamma_\infty}\rho_n(\delta)$ on $H^0(C_\partial^\bullet)=V_R^{\Gamma_\infty}$ over the representatives of Proposition 14.6(1)–(2) (Proposition 14.15(2)), transported into the stored basis through Proposition 14.4(1): a 0-cocycle is $(m,m,m,\rho_n(T)m)$ on v_0,v_1,v_2,v_3 with $m\in V_R^{\Gamma_\infty}$.

(v.b) **Degree 1, $\nu=1$.** The second object is (31) on $H^1(C_\partial^\bullet)\cong(V_R)_T$ in the coordinate Φ of Proposition 14.16(3), representative map and lift on the same orientation. *Status note* (v1.12.0-pre.3): the identity this comparison asserts is a theorem, Proposition 14.17(3), for every n, R and p (the coordinate being an isomorphism at $\rho_n(-I)=\mathrm{id}$). (v.b) is therefore **corroboration of a theorem**: what it now tests is the implementation — the τ_∂-table, the stored basis, the transport into it, and that representative map and lift share the orientation — and a failure is an implementation defect. Its predicate, ring and action on failure (abort) are unchanged. Until v1.12.0-pre.2 it was the only support of the degree-1 identity, checked per run on the primes a run builds.

(v.c) **Every computed degree, $\nu=1$: the Γ_∞-level datum.** The second object is the operator of the decision box after Definition 14.14: a Γ_∞-level twisted chain datum on ∂X_∞ built from the line coordinates alone — one δ_∞^∂-twisted and one δ_0^∂-twisted chain map, with the transporters T^t of Proposition 14.6(3) and the left cosets of (1) — each certified before use (twisted equivariance on a window of positions, chain map, preservation of $C_\partial^\bullet$). At cochain level the two operators are not asserted equal, and no chain homotopy between them is claimed — the Γ_∞-level datum is not a boundary g-twisted chain datum in the sense of Definition 14.8, so Lemma 14.10(3) does not speak about it; only the cohomology-level identity is asserted.

At $\nu\geq 2$ clause (v) is (v.a) alone. (v.b) and (v.c) rest on Proposition 14.6(3) and (31), which are $\nu=1$ statements (Remark 14.7); H^1 at $\nu\geq 2$ is covered by (vii) and by the operator's own certificates, Proposition 14.15(1) evaluated in `compute()`. Owner confirmation 2 of the slot-2 plan (2026-09-16): narrowed to what is stated, not extended; open point (2) of the v1.3.3 changelog. (v1.12.0-pre.3: the general-ν formula (32) of Proposition 14.17(2) would give a (v.b) at $\nu\geq 2$; it is *not* added here — no implementation computes it, and adding a clause is the owner's call under the same confirmation.)

(vi) **Oracle — recorded, and gating only where it is proved.** For n even and $p\neq\ell$: the eigenvalues of Θ_g^∂ on $H^0(C_\partial^\bullet)$ in the basis $x^{n-b\ell}\mathrm{Nm}^b$ are $\chi_b(p)$ (Proposition 14.16(1)); a failure **aborts**. For $n<\ell$ the class of y^n in $H^1(C_\partial^\bullet)$ has eigenvalue $1+p^{n+1}$ (Proposition 14.16(3)); a failure **aborts**. The general-n H^1 multiset of Claim 16.24 is **recorded**, not gating, until that claim is proved.

(vii) **Equivariance of the restriction (forward).** $r^q\circ T_{\Gamma g\Gamma}=T_p^\partial\circ r^q$ on cohomology in every computed degree, where $r=j^*$ is the restriction of §15. This clause is the *boundary datum*'s form of a statement expounded in full at Obligation 15.22, whose clause (iii) is the *pair*'s form of it (v1.7.1: through v1.6.2 the two forms had the same content and this sentence read "belongs to the pair and is stated in full at"; v1.7.0 strengthened (iii) alone, so the two are now distinct clauses binding distinct implementations — the paragraph below, and the open point of the v1.7.1 changelog entry); it is named here because it is the only

assertion that ties (30)'s choice of coset system to the bulk, and because it is what Proposition 14.16(2) settles: with the left-coset operator it holds, both sides acting on $r^1(H^1)$ by $\chi_0(p) = 1 + p^{n+1}$ for $n < \ell$; with the right-coset eigenvalue $\chi_1(p)$ of Proposition 14.16(2) in place of $A^{(1)}_\partial = \chi_0(p)$ on the one-dimensional H^1 ($n < \ell$, Proposition 14.16(3); the sum M_β itself is an operator on $V_R^T = H^0$) it fails **whenever $\chi_0(p) \neq \chi_1(p)$ in R and $r^1 \neq 0$** — i.e. $p \not\equiv 1$ and $p^n \not\equiv 1 \pmod{\ell}$; Claim 16.24(3) says when the two coincide as functions of p — and cannot distinguish the two where $\chi_0(p) = \chi_1(p)$. On H^0 the control distinguishes them only on a class whose restriction to V_R^T has a component $x^{n-b\ell}\mathrm{Nm}^b$ with $\chi_b(p) \neq \chi_{b+1}(p)$, which no power of the first Dickson class supplies: the b-th power has $n = b(\ell+1)$ and restricts to the single monomial $x^b\mathrm{Nm}^b$, and $b \equiv n - b \pmod{\ell-1}$, so $\chi_b \equiv \chi_{b+1}$ identically by Claim 16.24(3) (Remark 16.25 is the case $b = 1$). At $6 \in R^\times$ every bulk invariant of degree $n < \ell(\ell-1)$ is such a power (Claim 16.5(2)'s Dickson locus with its second generator absent), so on the model complexes the H^0 control cannot bite below that degree — and did not, on the $n \leq 16$ grid. (Until v1.4.0 the clause said "fails with the right-coset one" without the qualifier; open point (5) of the v1.3.3 changelog.) At $\ell \in \{2, 3\}$ the clause is recorded, not asserted; the record is Remark 15.23. **The carve-out stays here, and stays here alone** (v1.7.0). Since v1.7.0 Obligation 15.22(iii) asserts the *same* identity at *every* ℓ, because an implementation of the *pair* carries the derived homotopy h and that obligation's clause (i) then proves it with no hypothesis on ℓ. This clause is **not** strengthened with it: what *this* obligation requires an implementation to build is τ_∂ and Θ^∂_g, and this clause reads in addition the bulk $T_{\Gamma g \Gamma}$ and the restriction r — which is why it is the only clause here that ties this chapter to the bulk — but it requires no η and no h (§15 builds both), so the proof is unavailable to an implementation of the boundary datum, which is what this obligation binds, and a clause may not rest on data its own obligation does not require. (v1.7.1: through v1.7.0 this sentence read "the data *this* chapter stores are τ_∂ and Θ^∂_g alone", which this clause's own statement falsifies.) An implementation that also discharges Obligation 15.22(i) asserts the identity at every ℓ under that clause, and this carve-out is vacuous for it; the asymmetry, and why the two clauses do not contradict each other, are stated in full at Obligation 15.22(iii).

14.5 What $\ell = 2$ costs, and the truncation policy

Everything above is uniform in ℓ. What is not uniform is the *size* of the answer. This subsection states the degree behaviour at $\ell = 2$ — the boundary is non-zero in every degree (Lemma 14.20), the bulk is 2-periodic above degree 2, Hecke-equivariantly (Lemma 14.22) — and then fixes the policy an implementation follows there: 14.24, **decided by the owner on 2026-09-16**.

Lemma 14.20 (At $\ell = 2$ the boundary cohomology is non-zero in every degree)**.** *[proved here] Let R be a commutative ring with $2 = 0$ in R (in particular R of characteristic $\ell = 2$), let $n \geq 0$ and $V_R \neq 0$. Then*

$$H^0(\partial; V_R) = V_R^T, \qquad 0 \to (V_R)_T \to H^q(\partial; V_R) \to V_R^T \to 0 \quad \textit{for every } q \geq 1,$$

and $V_R^T \neq 0 \neq (V_R)_T$. Hence $H^q(\partial; V_R) \neq 0$ for every *$q \geq 0$: $H^\bullet(\partial; V_R)$ **is unbounded above in degree.** If R is a field then $\dim H^0(\partial) = \dim V_R^T$ and $\dim H^q(\partial) = 2\dim V_R^T$ for every $q \geq 1$; for $R = \mathbb{F}_2$ that is $\lfloor n/2 \rfloor + 1$ and $2(\lfloor n/2 \rfloor + 1)$.*

Proof. With $2 = 0$ in R, $\rho_n(-I) = (-1)^n\mathrm{id} = \mathrm{id}$, so Proposition 14.3's last sentence applies and gives $H^q(\langle -I\rangle; V_R) = V_R$ for every $q \geq 0$: for q odd it is $V_R[2] = V_R$ and for $q \geq 2$ even it is $V_R/2V_R = V_R$, both because $2 = 0$. Substituting into (29) gives the displayed sequences, with the T-action on $H^q(\langle -I\rangle; V_R)$ the natural one on V_R because Γ_∞ is a direct product (proof of Proposition 14.3). Non-vanishing: $\rho_n(T) - \mathrm{id}$ is nilpotent (Definition 8.1; the proof of Proposition 14.4(3)), say $(\rho_n(T) - \mathrm{id})^k = 0$; if $V_R^T = \ker(\rho_n(T) - \mathrm{id})$ were 0 then,

taking the largest i with $(\rho_n(T)-\mathrm{id})^i V_R \neq 0$ — which exists since $V_R \neq 0$ — that module would be a non-zero submodule of the kernel, a contradiction; and if $(\rho_n(T)-\mathrm{id})V_R = V_R$ then $V_R = (\rho_n(T)-\mathrm{id})^k V_R = 0$, also a contradiction. The dimension count over a field is $\dim\ker = \dim\operatorname{coker}$ for an endomorphism of a finite-dimensional space; for $R=\mathbb{F}_2$, $\dim V_R^T = \lfloor n/\ell \rfloor + 1 = \lfloor n/2 \rfloor + 1$ by Proposition 14.16(1). □

Remark 14.21 (Three complexes, and which one the six-term sequence is a window of). At $\ell = 2$ three objects that coincide for $6 \in R^\times$ come apart, and the statement "the sequence is unbounded" is true of exactly one of them. They must not be conflated.

1. **The model complexes.** $C^\bullet_{\mathrm{eq}}(K,\mathcal{R};L)$ vanishes above degree 2 and $C^\bullet_\partial$ above degree 1, for every ℓ, because K is 2-dimensional and K_∂ is 1-dimensional. Their cohomologies, and the cone of r between them, are finite and terminate. **At $\ell \in \{2,3\}$ they do not compute $H^\bullet(\Gamma;V_R)$, $H^\bullet(\partial;V_R)$ or $H^\bullet_c$** ((H3), Remark 7.6, Proposition 14.4(3)); they compute the orbifold model cohomology of (H3) and its boundary analogue, which are well-defined invariants of the triangulation and are what the shipped pipeline emits. Nothing in this document claims more for them, and 14.24 does not apply to them.
2. **The group-cohomological objects.** $H^\bullet(\partial;V_R) = H^\bullet(\Gamma_\infty;V_R)$ of Definition 14.2 is non-zero in every degree (Lemma 14.20). Consequently the long exact sequence of Claim 16.23, read with these objects, **cannot terminate**: for every $q \geq 0$, exactness at $H^q(\partial)$ gives $\operatorname{coker}\big(r^q\colon H^q(\Gamma;V_R) \to H^q(\partial;V_R)\big) \hookrightarrow H^{q+1}_c$, so *at least one* of $H^q(\Gamma;V_R)$ and H^{q+1}_c is non-zero. The six terms $X_1,\dots,X_6$ of §16 are then a **window** of a sequence with a non-zero term in every degree, and the hypotheses that close the window for $6 \in R^\times$ and n even — $H^2(\Gamma;V_R)=0$ and $H^0_c = 0$ — are exactly what fails. **Whether $H^q(\Gamma;V_R)$ is itself non-zero in every degree is not asserted here**, and nothing below needs it: the display already puts a non-zero term in every degree of the sequence. Claim 16.2's twelve-torsion vanishing, which does bound the bulk for $\ell \geq 5$, is unavailable at $\ell = 2$, and no argument in this document bounds $H^\bullet(\Gamma;V_R)$ above there. What *is* proved is that it need not be bounded to be finite information: Lemma 14.22 gives $H^q(\Gamma;V_R) \cong H^{q+2}(\Gamma;V_R)$ for every $q \geq 2$, every R and every n, by cup product with a canonical class $e \in H^2(\Gamma;\mathbb{Z}) \cong \mathbb{Z}/12$ and compatibly with every Hecke operator $T_{\Gamma g\Gamma}$, so the bulk is determined in degrees ≥ 2 by H^2 and H^3 and is bounded above exactly when both vanish. (Farrell cohomology, the packaging in which such periodicity is usually stated, agrees with H^q only *above* the virtual cohomological dimension — 1 for $\mathrm{SL}_2(\mathbb{Z})$ — and is **not used**: the lemma is about ordinary cohomology and is proved from the tree sequence (50) of Claim 16.2(1) — the Mayer–Vietoris sequence of Claim 16.2(2), re-derived in the proof for an arbitrary coefficient module — and the cup product, with no Farrell theory and no vcd bookkeeping.)
3. **The free Γ-model.** The complexes of §17 compute (2) and are infinite-dimensional in the degree direction for every ℓ; they are the only object of the three to which a truncation policy applies. 14.24 is a statement about them.

Lemma 14.22 (Bulk periodicity above the virtual cohomological dimension, and its Hecke equivariance)**.** *Let $\Gamma = \mathrm{SL}_2(\mathbb{Z}) = \langle S\rangle *_{\langle -I\rangle} \langle U\rangle$ as in Claim 16.2, with the Bass–Serre tree $\mathcal{T}$ of its clause (1): vertex set $\Gamma/\langle S\rangle \sqcup \Gamma/\langle U\rangle$, edge set $\Gamma/\langle -I\rangle$, the edge $\gamma\langle -I\rangle$ joining $\gamma\langle S\rangle$ to $\gamma\langle U\rangle$, on which Γ acts by left translation. All cohomology is* ordinary *group cohomology. Write Γ_{fin} for the set of elements of finite order of Γ and, for $h \in \Gamma_{\mathrm{fin}}$,*

$$\zeta_+(h) \in \mathbb{C}^\times \;:=\; \textit{the eigenvalue of } h \textit{ on the vector } (z_0,1)^{\mathsf{T}}, \textit{ where } z_0 \in \mathcal{H} \textit{ is a fixed point of } h \textit{ on } \mathcal{H}$$

($\zeta_+(\pm I) = \pm 1$). For a finite cyclic subgroup $C \leq \Gamma$ let $\kappa_C\colon H^2(C;\mathbb{Z}) \to \mathrm{Hom}(C,\mathbb{Q}/\mathbb{Z})$ be the inverse of the Bockstein of $0\to\mathbb{Z}\to\mathbb{Q}\to\mathbb{Q}/\mathbb{Z}\to 0$, and let $\chi_C\colon C \to \mathbb{Q}/\mathbb{Z}$ be $\zeta_+|_C$ read through $\exp(2\pi i\,\cdot)\colon \mathbb{Q}/\mathbb{Z} \xrightarrow{\sim} \mu_\infty$.

1. *[proved here] (**The class e.**) ζ_+ is well defined on Γ_{fin}, takes values in the roots of unity, and satisfies $\zeta_+(ghg^{-1}) = \zeta_+(h)$ whenever $g \in \mathrm{GL}_2(\mathbb{Q})^+$ and $h, ghg^{-1} \in \Gamma_{\mathrm{fin}}$. On every*

finite cyclic $C \le \Gamma$, χ_C is a well-defined faithful *character, and κ_C is an isomorphism, natural in C. There is a* ***unique*** *class $e \in H^2(\Gamma;\mathbb{Z})$ with*

$$\kappa_{\langle S\rangle}\big(\mathrm{res}^{\Gamma}_{\langle S\rangle} e\big) = \chi_{\langle S\rangle}, \qquad \kappa_{\langle U\rangle}\big(\mathrm{res}^{\Gamma}_{\langle U\rangle} e\big) = \chi_{\langle U\rangle};$$

$H^2(\Gamma;\mathbb{Z}) \cong \mathbb{Z}/12$ and e generates it; and $\kappa_C(\mathrm{res}^{\Gamma}_{C} e) = \chi_C$ for every *finite cyclic $C \le \Gamma$, so that $\mathrm{res}^{\Gamma}_{C} e$ generates $H^2(C;\mathbb{Z}) \cong \mathbb{Z}/|C|$.*

2. *[proved here] [formally verified: [GT, `thm:mp-tree-periodic`]] (only the existence of an isomorphism $H^q(\Gamma;V) \cong H^{q+2}(\Gamma;V)$ for every $q \ge 2$, every commutative ring and every V, from the literal 2-periodicity of $P_\bullet$ (Remark 16.3), natural in V on the cohomology of the explicit complex $V^2 \to V^3 \to \cdots$) [formally verified: [GT, `thm:mp-periodic-natural`]] (a periodicity isomorphism $H^q(\Gamma;-) \cong H^{q+2}(\Gamma;-)$, $q \ge 2$, that is a natural isomorphism of functors on `groupCohomology`, built through a comparison with the bar resolution that is itself natural in V; not shown to coincide with the isomorphism of the preceding marker;* ***not*** *that it is $e \cup -$, not its compatibility with restriction, not the $q = 1$ clause)* ***(Periodicity.)*** *Let V be* any *$\mathbb{Z}[\Gamma]$-module. The cup product $e \cup -\colon H^q(\Gamma;V) \to H^{q+2}(\Gamma;V)$ is an isomorphism for every $q \ge 2$; it is natural in V — hence R-linear for $V = V_R$ — and compatible with restriction to every subgroup. For $q = 1$ it is surjective, with kernel $\partial\big(N_{-I}V\big)$, where $N_{-I}V := (1 + (-I))\,V \subseteq V^{-I}$ and $\partial\colon V^{-I} = H^0(\langle -I\rangle;V) \to H^1(\Gamma;V)$ is the connecting map of the Mayer–Vietoris sequence ((51), in the coefficient form $(*)$ of the proof); it is not injective in general: for $V = V_R$ with $6 \in R^\times$ it is the zero map onto $H^3(\Gamma;V_R) = 0$ (Claim 16.2(3)) on a source that is non-zero for $R = \mathbb{F}_\ell$, $\ell \ge 5$, $n = 2$ (the Euler-characteristic identity of that claim gives $\dim H^1 = \dim H^0 + 3 - 1 - 1 \ge 1$), while for n odd, $N_{-I}V_R = 0$ and it is injective. Consequently*

   $$H^q(\Gamma;V_R) \;\cong\; H^{q+2}(\Gamma;V_R) \qquad \text{for every } q \ge 2, \text{ every commutative ring } R \text{ and every } n \ge 0:$$

   $H^\bullet(\Gamma;V_R)$ is determined in degrees ≥ 2 by $H^2(\Gamma;V_R)$ and $H^3(\Gamma;V_R)$, and vanishes in every degree ≥ 2 if and only if both vanish.
3. *[proved here]* ***(Hecke equivariance.)*** *Let $g \in \Delta$, $\Gamma' = \Gamma \cap g^{-1}\Gamma g$, $\Gamma'' = \Gamma \cap g\Gamma g^{-1}$, and let $T_{\Gamma g\Gamma} = \mathrm{cor}^{\Gamma}_{\Gamma''} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}^{\Gamma}_{\Gamma'}$ be the operator (20) on the ordinary group cohomology $H^\bullet(\Gamma;V_R)$, for any commutative ring R and any $n \ge 0$ — it is defined with no hypothesis on ℓ and with no model. (It is the operator the shipped pipeline computes when $6 \in R^\times$, Theorem 11.10, and the one a free Γ-model of §17 computes for every ℓ, Remark 17.2; nothing in this clause depends on either identification.) Then*
   (a) e is Δ-invariant: $\iota_g^* e = \mathrm{res}^{\Gamma}_{\Gamma''} e$ in $H^2(\Gamma'';\mathbb{Z})$, where $\iota_g\colon \Gamma'' \to \Gamma$, $h \mapsto g^{-1}hg$;
   (b) $T_{\Gamma g\Gamma}(e \cup x) = e \cup T_{\Gamma g\Gamma}(x)$ for every $x \in H^q(\Gamma;V_R)$, $q \ge 0$.
   Consequently, for every $q \ge 2$, $e \cup -\colon H^q(\Gamma;V_R) \to H^{q+2}(\Gamma;V_R)$ is an isomorphism of modules over the R-algebra generated by $\{T_{\Gamma g\Gamma} : g \in \Delta\}$: ***the Hecke module*** *$H^q(\Gamma;V_R)$* ***is 2-periodic in*** *$q \ge 2$, and $A^{(q+2)}(T_{\Gamma g\Gamma})$ is $A^{(q)}(T_{\Gamma g\Gamma})$ written in the transported basis $e \cup$ (basis of H^q) — $A^{(q)}(T_{\Gamma g\Gamma})$ denoting here the matrix of the operator of this clause on $H^q(\Gamma;V_R)$, which at $\ell \in \{2,3\}$ is not the shipped model's matrix (Remark 14.21(1)).*

External inputs. Four standard facts are used in the form (E1)–(E4) below, together with [Bro82, Ch. III.10.2] ($|G|$ annihilates $H^q(G;-)$ for finite G and $q \ge 1$, a numbered statement already used in Claim 16.2(3)) in Steps 0 and 5; the fixed-vertex statement of Step 3 is cited to Serre and proved in place. Each is cited at chapter-and-section level; where the pointer is to a section rather than to a numbered statement it has *not* been verified against the text (rule (A2)). What is then relied on is exactly the displayed form: (E1) and the first two items of (E2) are direct checks on the cochain formulae (33) and $\psi \mapsto f \circ \psi \circ (\alpha \times \cdots \times \alpha)$, which the reader may take as the definitions; the last item of (E2) (an inner automorphism acts trivially), (E3) and (E4) are structural and are used as stated, (E4) being the description of the corestriction $\mathrm{cor}^{\Gamma}_{\Gamma''}$ that (20) means (the transfer formula on a resolution is the form the proof of Theorem 11.10 computes with). (Referee pass, v1.3.1.)

(E1) [standard: [Bro82, Ch. V, §3]: the cup product] For a group G and G-modules A, B there is a product $\cup\colon H^p(G;A)\otimes H^q(G;B)\to H^{p+q}(G;A\otimes_{\mathbb{Z}}B)$, induced on inhomogeneous cochains by

$$(u\cup\varphi)(g_1,\dots,g_{p+q})=u(g_1,\dots,g_p)\otimes(g_1\cdots g_p)\,\varphi(g_{p+1},\dots,g_{p+q}),\tag{33}$$

satisfying the Leibniz rule $d(u\cup\varphi)=du\cup\varphi+(-1)^p u\cup d\varphi$ and natural in A and in B. With $A=\mathbb{Z}$ (trivial action, $\mathbb{Z}\otimes B=B$) it makes $H^\bullet(G;B)$ a module over $H^\bullet(G;\mathbb{Z})$, with $1\in H^0(G;\mathbb{Z})=\mathbb{Z}$ acting as the identity.

(E2) [standard: [Bro82, Ch. III, §8]: compatible pairs] A pair $(\alpha\colon G'\to G,\ f\colon A\to A')$ with $f(\alpha(g')a)=g'f(a)$ induces $(\alpha,f)^*\colon H^\bullet(G;A)\to H^\bullet(G';A')$, $\psi\mapsto f\circ\psi\circ(\alpha\times\cdots\times\alpha)$, functorially in the pair; $\mathrm{res}^G_H=(\mathrm{incl},\mathrm{id})^*$; and for $\gamma\in G$ and trivial A the pair (c_γ,id_A), $c_\gamma(h)=\gamma^{-1}h\gamma$, induces the *identity* of $H^\bullet(G;A)$.

(E3) [standard: [Bro82, Ch. III.6.2 and Exercise III.8.2]: Shapiro's lemma in the form already used at Propositions 12.4 and 12.6] For $H\le G$ and an H-module B put $\mathrm{Coind}^G_H B:=\mathrm{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}[G],B)$ with $(\gamma\varphi)(x)=\varphi(x\gamma)$. Then $\mathrm{sh}\colon H^q(G;\mathrm{Coind}^G_H B)\xrightarrow{\sim}H^q(H;B)$ is res^G_H followed by the map induced by the H-map $\mathrm{ev}\colon\varphi\mapsto\varphi(1)$. In particular $H^q(G;\mathrm{Coind}^G_1 B)=0$ for $q\ge1$.

(E4) [standard: [Bro82, Ch. III, §§5, 9]: induction, coinduction and the transfer] For $[G:H]<\infty$, $\mathrm{Ind}^G_H B=\mathbb{Z}[G]\otimes_{\mathbb{Z}[H]}B\cong\mathrm{Coind}^G_H B$ as G-modules, and $\mathrm{cor}^G_H=\mathrm{tr}_*\circ\mathrm{sh}^{-1}$ for the G-map

$$\mathrm{tr}\colon\mathrm{Coind}^G_H B\to B,\qquad \mathrm{tr}(\varphi)=\sum_{s\in G/H}s\,\varphi(s^{-1})\tag{34}$$

(independent of the representatives s, φ being H-equivariant); on a projective resolution this is the transfer formula used in the proof of Theorem 11.10.

Proof. Step 0: κ_C. For finite C and $q\ge1$, $|C|$ annihilates $H^q(C;\mathbb{Q})$ [standard: [Bro82, Ch. III.10.2], as in the proof of Claim 16.2(3)] and also acts invertibly on it (it is induced by the automorphism $|C|\cdot$ of $\mathbb{Q}$), so $H^q(C;\mathbb{Q})=0$. The long exact sequence of $0\to\mathbb{Z}\to\mathbb{Q}\to\mathbb{Q}/\mathbb{Z}\to0$ therefore gives an isomorphism $\delta\colon H^1(C;\mathbb{Q}/\mathbb{Z})=\mathrm{Hom}(C,\mathbb{Q}/\mathbb{Z})\to H^2(C;\mathbb{Z})$; $\kappa_C:=\delta^{-1}$. (v1.10.0: δ is the snake-lemma connecting map, $\delta[c]=[d\tilde c]$ for a $\mathbb{Q}$-valued cochain $\tilde c$ lifting the $\mathbb{Q}/\mathbb{Z}$-valued cocycle c, with the cochain differential $f\mapsto f\circ\partial$ on a projective resolution carrying no sign — the same convention for C and for Γ. This fixes the sign of the class e of (1): under the opposite convention the class so characterised is $-e$. Nothing in (2) or (3) depends on the sign, since $-e$ also generates; hecke-extensions slot-10 WP-14 review, finding m-6, where the built class is checked against this convention.) Naturality: for a homomorphism $\alpha\colon C'\to C$ the connecting maps commute with $(\alpha,\mathrm{id})^*$ ((E2)), so $\kappa_{C'}(\alpha^*w)=\kappa_C(w)\circ\alpha$. For C cyclic of order m, $\mathrm{Hom}(C,\mathbb{Q}/\mathbb{Z})\cong\mathbb{Z}/m$, generated by any faithful character.

Step 1: ζ_+. Let $h\in\Gamma_{\mathrm{fin}}$ have order m. Then h is diagonalisable over $\mathbb{C}$ (its minimal polynomial divides the separable X^m-1) with eigenvalues ζ,ζ^{-1}, roots of unity. If $h=\pm I$ every $z_0\in\mathcal{H}$ is fixed and every vector is an eigenvector with eigenvalue ±1, so $\zeta_+(h)=\pm1$ is well defined. Otherwise $\zeta\ne\zeta^{-1}$ (a diagonalisable matrix with a single eigenvalue ±1 is $\pm I$), so $\zeta\notin\mathbb{R}$, $t:=\mathrm{tr}\,h=\zeta+\bar\zeta$ satisfies $|t|<2$, and writing $h=\begin{pmatrix}a&b\\c&d\end{pmatrix}$: $4bc=4ad-4<4ad-(a+d)^2=-(a-d)^2\le0$, so $c\ne0$. The fixed points of $z\mapsto(az+b)/(cz+d)$ are the two roots of $cz^2+(d-a)z-b=0$, whose discriminant is $(a+d)^2-4<0$: one root z_0 in $\mathcal{H}$, the other $\bar z_0$. From $h(z_0)=z_0$,

$$h\begin{pmatrix}z_0\\1\end{pmatrix}=\begin{pmatrix}az_0+b\\cz_0+d\end{pmatrix}=(cz_0+d)\begin{pmatrix}h(z_0)\\1\end{pmatrix}=(cz_0+d)\begin{pmatrix}z_0\\1\end{pmatrix},$$

so $(z_0,1)^{\mathsf T}$ is an eigenvector, $\zeta_+(h)=cz_0+d$ is well defined (the fixed point in $\mathcal{H}$ is unique), and $(\bar z_0,1)^{\mathsf T}$ is the eigenvector for the other eigenvalue $\overline{\zeta_+(h)}=\zeta_+(h)^{-1}$; the two form a basis

of $\mathbb{C}^2$ since $z_0 \neq \bar{z}_0$. (Explicitly $\zeta_+(h) = \frac{t}{2} + i\,\mathrm{sgn}(c)\frac{\sqrt{4-t^2}}{2}$; for instance $\zeta_+(S) = i$ and $\zeta_+(U) = \zeta_+(ST) = e^{\pi i/3}$.)

Invariance. Let $g = \begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix} \in \mathrm{GL}_2(\mathbb{Q})^+$ with $h' := ghg^{-1} \in \Gamma_{\mathrm{fin}}$. Since $\det g > 0$, g acts on $\mathcal{H}$ by Möbius transformations, $\mathrm{Im}(g\cdot z) = \det(g)\,\mathrm{Im}(z)/\,|\gamma z+\delta|^2 > 0$, and $g\,(z_0,1)^{\mathsf{T}} = (\gamma z_0+\delta)\,(g\cdot z_0,1)^{\mathsf{T}}$ with $\gamma z_0+\delta \neq 0$ ($z_0 \notin \mathbb{R}$). Now h' fixes $g\cdot z_0 \in \mathcal{H}$, so $\zeta_+(h')$ is the eigenvalue of h' on $(g\cdot z_0,1)^{\mathsf{T}}$, a non-zero multiple of $g(z_0,1)^{\mathsf{T}}$; and $h'\,g(z_0,1)^{\mathsf{T}} = g\,h\,(z_0,1)^{\mathsf{T}} = \zeta_+(h)\,g(z_0,1)^{\mathsf{T}}$. Hence $\zeta_+(h') = \zeta_+(h)$ (trivially so if $h = \pm I$).

Characters. Let $C = \langle h_0\rangle \leq \Gamma$ be finite cyclic. If $h_0 = \pm I$ then $C \subseteq \{\pm I\}$ and $\zeta_+|_C$, being $\pm I \mapsto \pm 1$, is a faithful character. Otherwise let $z_0 \in \mathcal{H}$ be the fixed point of h_0; every $h \in C$ is a power of h_0, fixes z_0, and its eigenvalue on $(z_0,1)^{\mathsf{T}}$ is $\zeta_+(h)$ (for $h \neq \pm I$ because z_0 is then *its* fixed point in $\mathcal{H}$; for $h = \pm I$ trivially). Eigenvalues on a common eigenvector multiply, so $\zeta_+|_C$ is a homomorphism $C \to \mu_\infty$; it is faithful because $\zeta_+(h) = 1$ forces h to act as 1 on both basis vectors $(z_0,1)^{\mathsf{T}}$, $(\bar{z}_0,1)^{\mathsf{T}}$ (the second with eigenvalue $\zeta_+(h)^{-1}$), i.e. $h = I$. Thus χ_C is a faithful character.

Step 2: e, and $H^2(\Gamma;\mathbb{Z}) \cong \mathbb{Z}/12$. Claim 16.2(2) with $R = \mathbb{Z}$, $n = 0$ — $V_R = \mathbb{Z}$ with trivial action — reads, in degrees 1 to 2,

$$H^1(\langle -I\rangle;\mathbb{Z}) \to H^2(\Gamma;\mathbb{Z}) \xrightarrow{(\mathrm{res},\mathrm{res})} H^2(\langle S\rangle;\mathbb{Z}) \oplus H^2(\langle U\rangle;\mathbb{Z}) \xrightarrow{\mathrm{res}-\mathrm{res}} H^2(\langle -I\rangle;\mathbb{Z}),$$

and $H^1(\langle -I\rangle;\mathbb{Z}) = \mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z}) = 0$. So $(\mathrm{res},\mathrm{res})$ is injective with image $\ker(\mathrm{res}-\mathrm{res})$. Through κ (Step 0, naturality) $\mathrm{res}-\mathrm{res}$ is $(\chi,\chi') \mapsto \chi|_{\langle -I\rangle} - \chi'|_{\langle -I\rangle}$ on $\mathrm{Hom}(\langle S\rangle,\mathbb{Q}/\mathbb{Z}) \oplus \mathrm{Hom}(\langle U\rangle,\mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/6$, each restriction being onto $\mathrm{Hom}(\langle -I\rangle,\mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/2$ ($\mathbb{Q}/\mathbb{Z}$ is divisible), so the kernel has order $24/2 = 12$; it contains $(\chi_{\langle S\rangle},\chi_{\langle U\rangle})$, because both restrict to $\zeta_+|_{\langle -I\rangle}$, and that element has order $\mathrm{lcm}(4,6) = 12$ (the two characters are faithful, Step 1). Hence $H^2(\Gamma;\mathbb{Z}) \cong \ker(\mathrm{res}-\mathrm{res}) \cong \mathbb{Z}/12$, and e — the unique preimage of $(\chi_{\langle S\rangle},\chi_{\langle U\rangle})$ — generates it.

Step 3: $\kappa_C(\mathrm{res}^\Gamma_C\, e) = \chi_C$ for every finite cyclic $C \leq \Gamma$. First, *C fixes a vertex of $\mathcal{T}$* [standard: [Ser80, Ch. I, §4.3], a finite group acting without inversion on a tree fixes a vertex; the pointer is not verified against the text (rule (A2)) and the argument is given]: the action is without inversion (the two ends of an edge have different types); pick a vertex v, let $\mathcal{T}_0$ be the finite subtree spanned by the finite orbit Cv, which is C-invariant; the set of leaves of a finite tree is invariant under its automorphisms, and a finite tree with at least three vertices remains a non-empty tree when all its leaves are deleted (the interior vertices of a path have degree ≥ 2); iterating, C preserves a subtree with one vertex, or with two vertices joined by an edge, whose ends it then fixes since it does not invert. So $C \leq \Gamma_v = \gamma D\gamma^{-1}$ for some $\gamma \in \Gamma$ and $D \in \{\langle S\rangle,\langle U\rangle\}$. Put $D' := \gamma D\gamma^{-1}$ and let $c_\gamma\colon \Gamma \to \Gamma$, $h \mapsto \gamma^{-1}h\gamma$, which maps D' isomorphically onto D. By (E2), $c_\gamma^* e = e$ and $\mathrm{res}^\Gamma_{D'} e = \mathrm{res}^\Gamma_{D'}(c_\gamma^* e) = (c_\gamma|_{D'})^*(\mathrm{res}^\Gamma_D e)$, the two composites $D' \to \Gamma$ being the same homomorphism. Applying $\kappa_{D'}$ and Step 0, $\kappa_{D'}(\mathrm{res}^\Gamma_{D'} e) = \chi_D \circ c_\gamma|_{D'} = \zeta_+(\gamma^{-1}(\cdot)\gamma)|_{D'} = \zeta_+|_{D'} = \chi_{D'}$, by Step 2 and the invariance of Step 1 (with $\gamma^{-1} \in \Gamma \subseteq \mathrm{GL}_2(\mathbb{Q})^+$). Restricting further to $C \leq D'$ (Step 0 again) gives $\kappa_C(\mathrm{res}^\Gamma_C e) = \chi_{D'}|_C = \chi_C$. Since χ_C is faithful, $\mathrm{res}^\Gamma_C e$ generates $H^2(C;\mathbb{Z}) \cong \mathbb{Z}/\,|C|$. This proves (1).

Step 4: five formal consequences of (E1)–(E4). Let G be a group, $u \in H^p(G;\mathbb{Z})$.

(P1) For a compatible pair $(\alpha\colon G' \to G, f\colon B \to B')$ and $x \in H^q(G;B)$: $(\alpha,f)^*(u\cup x) = \alpha^* u \cup (\alpha,f)^* x$, where $\alpha^* = (\alpha,\mathrm{id}_\mathbb{Z})^*$. On cochains, by (33) and $f(\alpha(g')b) = g'f(b)$: $f\big(u(\alpha g'_1,\dots,\alpha g'_p)\,\alpha(g'_1\cdots g'_p)\,\varphi(\alpha g'_{p+1},\dots)\big) = u(\alpha g'_1,\dots)\,(g'_1\cdots g'_p)\,f\varphi(\alpha g'_{p+1},\dots)$. In particular $\mathrm{res}^G_H(u\cup x) = \mathrm{res}^G_H u \cup \mathrm{res}^G_H x$.

(P2) *Connecting homomorphisms.* For a short exact sequence $0 \to B' \to B \to B'' \to 0$ of G-modules and $x \in H^q(G;B'')$: $\delta(u\cup x) = (-1)^p u \cup \delta x$. For a short exact sequence $0 \to A' \to A \to A'' \to 0$ of G-modules that stays exact after $\otimes_\mathbb{Z} B$ (e.g. one of free abelian groups), $w \in H^p(G;A'')$ and $x \in H^q(G;B)$: $\delta(w\cup x) = \delta w \cup x$ in $H^{p+q+1}(G;A'\otimes B)$.

Both by the Leibniz rule: lift the cocycle of the class being shifted to a cochain of the middle term, cup it with a cocycle of the other class, and take d; the term with d on the cocycle vanishes, leaving $(-1)^p u \cup d\varphi$, resp. $dw \cup \varphi$.

(P3) *Shapiro.* For $H \le G$, $X \in H^q(G; \mathrm{Coind}_H^G B)$: $\mathrm{sh}(u \cup X) = \mathrm{res}_H^G u \cup \mathrm{sh}(X)$. Indeed $\mathrm{sh}(u \cup X) = \mathrm{ev}_* \mathrm{res}(u \cup X) = \mathrm{ev}_*(\mathrm{res}\, u \cup \mathrm{res}\, X) = \mathrm{res}\, u \cup \mathrm{ev}_* \mathrm{res}\, X$ by (P1) and naturality in B ((E1)).

(P4) *Projection formula.* For $[G : H] < \infty$ and $y \in H^q(H; B)$: $\mathrm{cor}_H^G(\mathrm{res}_H^G u \cup y) = u \cup \mathrm{cor}_H^G y$. Indeed (P3) with $X = \mathrm{sh}^{-1} y$ gives $\mathrm{sh}^{-1}(\mathrm{res}\, u \cup y) = u \cup \mathrm{sh}^{-1} y$, and $\mathrm{cor}(\mathrm{res}\, u \cup y) = \mathrm{tr}_*(u \cup \mathrm{sh}^{-1} y) = u \cup \mathrm{tr}_* \mathrm{sh}^{-1} y = u \cup \mathrm{cor}\, y$ by (E4) and naturality in B.

(P5) For H finite and any H-module M, $H^q(H; \mathbb{Z}[H] \otimes_{\mathbb{Z}} M) = 0$ for $q \ge 1$, the tensor product carrying the diagonal action: $h \otimes m \mapsto h \otimes h^{-1} m$ is an H-isomorphism onto $\mathbb{Z}[H] \otimes M$ with H acting on the left factor only, which is $\mathrm{Ind}_1^H M \cong \mathrm{Coind}_1^H M$ ((E4)), whose cohomology vanishes in positive degrees ((E3)).

Step 5: cyclic periodicity. Let $C = \langle t \rangle$ be cyclic of order m, $N := 1 + t + \cdots + t^{m-1} \in \mathbb{Z}[C]$, $I := \ker(\varepsilon \colon \mathbb{Z}[C] \to \mathbb{Z})$. Then

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1 \mapsto N} \mathbb{Z}[C] \xrightarrow{t-1} \mathbb{Z}[C] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0 \tag{35}$$

is exact: $\ker \varepsilon = I = (t-1)\mathbb{Z}[C]$ since $\mathbb{Z}[C] = \mathbb{Z}[t]/(t^m - 1)$; $\ker(t-1) = \mathbb{Z}[C]^C = \mathbb{Z}N$ (an element $\sum a_j t^j$ fixed by t has all a_j equal); and $1 \mapsto N$ is injective with image $\mathbb{Z}N$. This is the first segment of the periodic resolution of [Bro82, Ch. III.1] used in the proof of Proposition 14.3. Split it as $0 \to I \to \mathbb{Z}[C] \to \mathbb{Z} \to 0$ and $0 \to \mathbb{Z} \to \mathbb{Z}[C] \to I \to 0$, with connecting maps $\delta_1 \colon H^0(C; \mathbb{Z}) \to H^1(C; I)$ and $\delta_2 \colon H^1(C; I) \to H^2(C; \mathbb{Z})$, and put $u_0 := \delta_2 \delta_1(1) \in H^2(C; \mathbb{Z})$. Let M be any C-module. Both short exact sequences consist of free abelian groups, so they stay exact after $\otimes_{\mathbb{Z}} M$; write $\delta_1' \colon H^q(C; M) \to H^{q+1}(C; I \otimes M)$ and $\delta_2' \colon H^{q+1}(C; I \otimes M) \to H^{q+2}(C; M)$ for the connecting maps of the tensored sequences (diagonal actions). By (P2) (first-variable form, twice) and $1 \cup x = x$,

$$u_0 \cup x = \delta_2\big(\delta_1(1)\big) \cup x = \delta_2'\big(\delta_1(1) \cup x\big) = \delta_2' \delta_1'(1 \cup x) = \delta_2' \delta_1'(x) \qquad (x \in H^q(C; M)).$$

By (P5) the terms $H^j(C; \mathbb{Z}[C] \otimes M)$ flanking δ_1' and δ_2' vanish for $j \ge 1$, so δ_2' is an isomorphism for every $q \ge 0$, and δ_1' is an isomorphism for $q \ge 1$ and, for $q = 0$, surjective with kernel the image of $H^0(C; \mathbb{Z}[C] \otimes M) \to H^0(C; M) = M^C$ under $\varepsilon \otimes \mathrm{id}$. That image is $N \cdot M$: an element $\sum_c c \otimes m_c$ is C-invariant iff $m_c = c\, m_1$ for all c, and $\varepsilon \otimes \mathrm{id}$ sends $\sum_c c \otimes c m_1$ to $N m_1$. Hence $u_0 \cup - \colon H^q(C; M) \to H^{q+2}(C; M)$ *is an isomorphism for* $q \ge 1$, *and for* $q = 0$ *a surjection* $M^C \to H^2(C; M)$ *with kernel* NM. Taking $M = \mathbb{Z}$, $q = 0$: $u_0 \cup k = k\, u_0$ and the map is onto, so u_0 generates $H^2(C; \mathbb{Z})$, which is $\cong \mathbb{Z}/m$ by Step 0; thus u_0 has order m. If u is any generator, $u = a\, u_0$ with $\gcd(a, m) = 1$, and since m kills $H^j(C; M)$ for $j \ge 1$ [standard: [Bro82, Ch. III.10.2]] multiplication by a is invertible there; so *cup product with any generator of* $H^2(C; \mathbb{Z})$ *is an isomorphism* $H^q(C; M) \to H^{q+2}(C; M)$ *for every* $q \ge 1$ *and every* M, *and for* $q = 0$ *a surjection with kernel* NM.

Step 6: proof of (2). For $D \in \{\langle S \rangle, \langle U \rangle, \langle -I \rangle\}$ put $P_D := \mathbb{Z}[\Gamma/D]$. Every term of (50) is a free abelian group, so the sequence is $\mathbb{Z}$-split and $\mathrm{Hom}_{\mathbb{Z}}(-, V)$ — with the diagonal action $(\gamma f)(x) = \gamma f(\gamma^{-1} x)$ — turns it into a short exact sequence of $\mathbb{Z}[\Gamma]$-modules

$$0 \to V \to \mathrm{Hom}_{\mathbb{Z}}(P_{\langle S \rangle}, V) \oplus \mathrm{Hom}_{\mathbb{Z}}(P_{\langle U \rangle}, V) \to \mathrm{Hom}_{\mathbb{Z}}(P_{\langle -I \rangle}, V) \to 0, \tag{36}$$

whose long exact cohomology sequence we call $(*)$. For each D, $\Phi_D \colon \mathrm{Hom}_{\mathbb{Z}}(P_D, V) \to \mathrm{Coind}_D^{\Gamma} V$, $\Phi_D(f)(\gamma) := \gamma f(\gamma^{-1} D)$, is a Γ-isomorphism (it is D-equivariant in the argument, Γ-equivariant, and inverted by $\varphi \mapsto (\gamma D \mapsto \gamma \varphi(\gamma^{-1}))$, all by direct substitution), so by (E3) the terms of $(*)$ are $H^q(\Gamma; V)$, $H^q(\langle S \rangle; V) \oplus H^q(\langle U \rangle; V)$ and $H^q(\langle -I \rangle; V)$: those of (51), of which $(*)$ is the coefficient-sequence form (under $\mathrm{Ext}^q_{\mathbb{Z}[\Gamma]}(P, V) \cong H^q(\Gamma; \mathrm{Hom}_{\mathbb{Z}}(P, V))$ for $\mathbb{Z}$-free P it *is* (51); nothing below needs that identification). Cup product with e acts on every term of

$(*)$ and commutes with every map of it: with the maps induced by (36) by naturality in the coefficients ((E1)), with the connecting homomorphism by (P2) with $p=2$, sign $(-1)^2=+1$. On $H^q(\Gamma;\mathrm{Hom}_{\mathbb{Z}}(P_D,V))$ it is transported by Φ_{D*} and sh to $\mathrm{res}^{\Gamma}_{D}\,e\cup-$ on $H^q(D;V)$ ((P3)), where $\mathrm{res}^{\Gamma}_{D}\,e$ generates $H^2(D;\mathbb{Z})$ by (1); so by Step 5 the vertical map is an isomorphism on these terms in every degree $q\geq 1$, and in degree 0 a surjection $V^D\to H^2(D;V)$ with kernel N_DV. The five lemma on

$$H^{q-1}(\oplus)\to H^{q-1}(\langle -I\rangle;V)\to H^q(\Gamma;V)\to H^q(\oplus)\to H^q(\langle -I\rangle;V)$$

(with $H^j(\oplus):=H^j(\langle S\rangle;V)\oplus H^j(\langle U\rangle;V)$), mapped by $e\cup-$ to the same window two degrees up, gives that $e\cup-\colon H^q(\Gamma;V)\to H^{q+2}(\Gamma;V)$ is an isomorphism as soon as the four outer maps are, i.e. for $q-1\geq 1$: **this is where $q\geq 2$ enters**. For $q=1$ the window's left-hand terms are in degree 0, where the vertical maps are only surjective; the four lemma still gives surjectivity of $e\cup-\colon H^1(\Gamma;V)\to H^3(\Gamma;V)$, and a diagram chase gives the kernel: if $e\cup x=0$ then the image of x in $H^1(\oplus)$ dies under an injective map, so $x=\partial y$ with $y\in V^{-I}$; then $\partial(e\cup y)=0$, so $e\cup y=(\mathrm{res}-\mathrm{res})(e\cup z)$ for some $z\in V^S\oplus V^U$ (surjectivity in degree 0), so $y-(\mathrm{res}-\mathrm{res})z\in\ker(e\cup-)=N_IV$ and $x=\partial\big(y-(\mathrm{res}-\mathrm{res})z\big)\in\partial(N_IV)$; conversely $e\cup\partial(N_IV)=\partial(e\cup N_IV)=0$. Naturality in V and compatibility with restriction are (E1) and (P1); R-linearity for $V=V_R$ is naturality applied to the endomorphisms $v\mapsto rv$. The statements about $V=V_R$ follow: $N_IV_R=(1+\rho_n(-I))V_R=(1+(-1)^n)V_R$ (Proposition 8.2(2)), and for $6\in R^\times$, $H^3(\Gamma;V_R)=0$ by Claim 16.2(3) while $\dim H^1(\Gamma;V_R)=\dim H^0(\Gamma;V_R)+\dim V_R-\dim V_R^S-\dim V_R^U=\dim H^0+3-1-1\geq 1$ for $R=\mathbb{F}_\ell$, $\ell\geq 5$, $n=2$ (the eigenvalues of $\rho_n(S)$ are $-1,1,-1$, those of $\rho_n(U)$ are 1 and the two primitive cube roots of unity, both matrices semisimple since $4,6\in\mathbb{F}_\ell^\times$).

Step 7: proof of (3a). $[\Gamma:\Gamma'']<\infty$ (Proposition 10.1). Restricted to $\mathbb{Z}[\Gamma'']$, each P_D of (50) is the finite direct sum, over the Γ''-orbits $\Gamma''x$ in Γ/D, of the permutation modules $\mathbb{Z}[\Gamma''x]\cong\mathbb{Z}[\Gamma''/\Gamma''_x]$ with $\Gamma''_x=\Gamma''\cap\gamma D\gamma^{-1}$ for $x=\gamma D$ — the stabiliser of the vertex or edge x of $\mathcal{T}$ in Γ'', a subgroup of a cyclic group, hence finite cyclic. Applying $\mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma'']}(-,\mathbb{Z})$ to (50) exactly as in the proof of Claim 16.2(2), with Shapiro's lemma ((E3), (E4)) on each summand, gives an exact segment

$$\bigoplus_{\text{edges }x}H^1(\Gamma''_x;\mathbb{Z})\longrightarrow H^2(\Gamma'';\mathbb{Z})\xrightarrow{(\mathrm{res}^{\Gamma''}_{\Gamma''_x})_x}\bigoplus_{\text{vertices }x}H^2(\Gamma''_x;\mathbb{Z}),$$

the middle map being restriction for the same reason the maps of (51) are: Shapiro's isomorphism is restriction followed by evaluation at the base coset ((E3)), and the augmentation $\mathbb{Z}[\Gamma''/\Gamma''_x]\to\mathbb{Z}$ becomes, under Φ, the unit $V\to\mathrm{Coind}\,V$, whose evaluation at the base coset is the identity. Since $H^1(\text{finite};\mathbb{Z})=0$, $H^2(\Gamma'';\mathbb{Z})$ **is detected on the finite cyclic subgroups** Γ''_x. It therefore suffices to show $\mathrm{res}^{\Gamma''}_{C}(\iota_g^*e)=\mathrm{res}^{\Gamma''}_{C}(\mathrm{res}^{\Gamma}_{\Gamma''}e)$ for every finite cyclic $C\leq\Gamma''$. Put $C':=g^{-1}Cg=\iota_g(C)\leq\Gamma$, finite cyclic. By functoriality ((E2)), $\mathrm{res}^{\Gamma''}_{C}\,\iota_g^*e=(\iota_g|_C)^*e=(\iota_g|_C\colon C\to C')^*\,\mathrm{res}^{\Gamma}_{C'}\,e$, and by Step 0 and (1) applied to C', $\kappa_C\big((\iota_g|_C)^*\,\mathrm{res}^{\Gamma}_{C'}\,e\big)=\chi_{C'}\circ\iota_g|_C=\zeta_+(g^{-1}(\cdot)g)|_C=\zeta_+|_C=\chi_C=\kappa_C(\mathrm{res}^{\Gamma}_{C}\,e)$, the middle equality being the invariance of Step 1 for $g^{-1}\in\mathrm{GL}_2(\mathbb{Q})^+$. As κ_C is injective and $\mathrm{res}^{\Gamma}_{C}=\mathrm{res}^{\Gamma''}_{C}\,\mathrm{res}^{\Gamma}_{\Gamma''}$, this is the required equality.

Step 8: proof of (3b) and the consequences. Let $x\in H^q(\Gamma;V_R)$. By (P1) for $\mathrm{res}^{\Gamma}_{\Gamma'}$, $\mathrm{res}^{\Gamma}_{\Gamma'}(e\cup x)=\mathrm{res}^{\Gamma}_{\Gamma'}\,e\cup\mathrm{res}^{\Gamma}_{\Gamma'}\,x$. The map $(\mathrm{conj}_g)_*$ is $(\alpha,f)^*$ for the compatible pair $\alpha=\mathrm{conj}_{g^{-1}}\colon\Gamma''\to\Gamma'$, $f=\rho_n(g)$ of (20), so by (P1) again $(\mathrm{conj}_g)_*(\mathrm{res}^{\Gamma}_{\Gamma'}\,e\cup y)=\alpha^*(\mathrm{res}^{\Gamma}_{\Gamma'}\,e)\cup(\mathrm{conj}_g)_*y=\iota_g^*e\cup(\mathrm{conj}_g)_*y=\mathrm{res}^{\Gamma}_{\Gamma''}\,e\cup(\mathrm{conj}_g)_*y$, using $\mathrm{incl}_{\Gamma'}\circ\alpha=\iota_g$ and (3a). Finally (P4) for $\Gamma''\leq\Gamma$ gives $\mathrm{cor}^{\Gamma}_{\Gamma''}(\mathrm{res}^{\Gamma}_{\Gamma''}\,e\cup z)=e\cup\mathrm{cor}^{\Gamma}_{\Gamma''}\,z$. Composing the three, $T_{\Gamma g\Gamma}(e\cup x)=e\cup T_{\Gamma g\Gamma}x$. For $q\geq 2$, $e\cup-$ is an isomorphism by (2) and commutes with every $T_{\Gamma g\Gamma}$ by (3b), hence is an isomorphism of modules over the algebra they generate; the statement about $A^{(q)}$ is the same sentence in a basis. □

Remark 14.23 (What Lemma 14.22 does and does not settle for (OC-1c)).

1. **Ordinary cohomology, no Farrell theory.** The lemma is a statement about $H^\bullet(\Gamma;-)$. Farrell cohomology $\widehat{H}^\bullet(\Gamma;-)$ ([Bro82, Ch. X]) agrees with it only in degrees *above* $\mathrm{vcd}(\mathrm{SL}_2(\mathbb{Z}))=1$ and is neither used nor asserted about; the periodicity is proved from Claim 16.2(2), the cup product and the five lemma, and is exact about its lower edge: it is an isomorphism from $q=2$, a surjection at $q=1$, and says nothing at $q=0$. The reader may recognise e as the Euler class of the central extension $1\to\mathbb{Z}\to B_3\to\mathrm{SL}_2(\mathbb{Z})\to 1$ (the universal covering of $\mathrm{SL}_2(\mathbb{R})$ restricted to Γ), whose restriction to $\langle S\rangle$ is $0\to\mathbb{Z}\xrightarrow{4}\mathbb{Z}\to\mathbb{Z}/4\to 0$, matching $\zeta_+(S)=i$; this document neither verifies nor uses that identification — e is *defined* in clause (1), and its Δ-invariance is proved there, not inherited from $\mathrm{GL}_2(\mathbb{R})^+$.
2. **What is now a theorem in (OC-1c)'s hypothesis.** Its bulk half, in a form stronger than it asked for: $H^q(\Gamma;V_R)\cong H^{q+2}(\Gamma;V_R)$ for $q\geq 2$, every R and every n, *as Hecke modules* for every $T_{\Gamma g\Gamma}$, $g\in\Delta$, by one canonical isomorphism. On the free Γ-model of §17 the bulk column of (39) in degrees ≥ 4 is therefore a transport of degrees 2 and 3, Hecke matrices included. On the shipped 7-vertex model at $6\in R^\times$ the lemma is vacuous ($H^q=0$ for $q\geq 2$, Claim 16.2(3)).
3. **What remains open in (OC-1c), precisely.** Three things, none of them asserted here.
   (a) *The boundary.* Lemma 14.20 gives the same R-module $H^q(\partial;V_R)$ in every degree $q\geq 1$ at $\ell=2$, and Proposition 14.3 exhibits $H^q(\Gamma_\infty;V_R)$ for every R as an extension of two pieces that are 2-periodic for $q\geq 2$ (Step 5 of the proof); but no *map* $H^q(\partial;V_R)\to H^{q+2}(\partial;V_R)$ is proved to be an isomorphism. The candidate is $\mathrm{res}^\Gamma_{\Gamma_\infty}e\cup-$; its compatibility with $r^q=\mathrm{res}^\Gamma_{\Gamma_\infty}$ is automatic ((P1) of the proof: $r^q(e\cup x)=\mathrm{res}^\Gamma_{\Gamma_\infty}e\cup r^q x$), and its T^∂_p-equivariance would follow by the argument of clause (3) run on Γ_∞ with the boundary coset system of Proposition 14.6, given the boundary analogue of (3a). Nothing of this is proved here.
   (b) *The cone.* Given (a) with an isomorphism from some degree q_0^∂ on, the five lemma applied to (39) would give $H^q_c\cong H^{q+2}_c$ for $q\geq\max(3,q_0^\partial+1)$, *provided* cup product with e acts on $H^\bullet_c=H^\bullet_\Gamma(Y,\partial Y;V_R)$ (Definition 17.1; Theorem 17.3(2) where $6\in R^\times$) compatibly with the maps of that sequence. That is the standard module structure of the long exact sequence of a pair over $H^\bullet(\Gamma;\mathbb{Z})$, and it is not written out in this document.
   (c) *Hecke equivariance on the boundary and on the cone*, by the same projection-formula route as clause (3); not proved here.

   If (a) holds from $q_0^\partial=1$ — the value Lemma 14.20 suggests at $\ell=2$ — then every group of (39) in degree ≥ 5 is a transport of one in degree ≤ 4, which is why the policy of 14.24 recommends $q_{\max}\geq 4$ at $\ell\in\{2,3\}$ where the stabilisation is to be *observed*: a run with $q_{\max}<4$ cannot exhibit even the first bulk isomorphism $H^2(\Gamma;V_R)\cong H^4(\Gamma;V_R)$, and the cone's first, $H^3_c\cong H^5_c$, needs $q_{\max}\geq 5$.

Open construction item 14.24 (OC-1: the $\ell=2$ truncation policy — DECIDED 2026-09-16). **Decided by the owner on 2026-09-16: the policy is (OC-1a) below, with the three riders stated after the box** (owner decision "OC-1: adoption per brief"; the brief is `docs/implementation/oc1_decision_brief.md`, the record is `docs/implementation/hecke_extensions_owner_decisions.md`, entry OC-1). What follows is therefore a *policy* — a definition-like statement, which carries no status tag of its own (§1.1, Appendix B); the truncation lemma inside it keeps its own tag. From v0.14.0 to v1.2.0 this item carried the tag `[open]` and the header "reserved by the owner; a tagged working assumption, not a decision"; the label and the content of the boxed statement are unchanged (rules (L1), (L2)), its status is. At $\ell\in\{2,3\}$ the complexes of §17 are unbounded above (Remark 14.21(3)), so an implementation computes a truncation, and the contract must fix four things: the bound $q_{\max}$, its place in the record schema, the certificate obligations at the top of the truncation, and what "exactness" asserts there. The enabling fact is elementary and

is what makes any truncation legitimate at all:

Truncation lemma. [proved here] For a cochain complex $C^\bullet$ and $q \le q_{\max}$, $H^q(C^\bullet)$ depends only on $C^{\le q_{\max}+1}$: it is $\ker d^q / \operatorname{im} d^{q-1}$, and d^q is defined on C^q with values in C^{q+1}, $q+1 \le q_{\max}+1$. *The same is false for* $q = q_{\max}+1$, where the truncated complex has $\ker d^{q_{\max}+1} = C^{q_{\max}+1}$ and computes a quotient that is in general strictly larger than $H^{q_{\max}+1}$.

The policy taken here (decided 2026-09-16; the box is verbatim the v0.14.0 working assumption, except that its record sentence says "per complex" since v1.4.0 — §22.4 item 5, owner confirmation 1 of the slot-2 plan).

[OC-1: $q_{\max}$ is an explicit, recorded parameter of a run, defaulting to 2.] The free-Γ-model complexes of §17 — bulk, boundary and cone — are built in degrees $\le q_{\max}+1$ and the twisted chain data τ, τ_∂, η are certified in every degree in which they are built. Cohomology is **reported only in degrees** $\le q_{\max}$; degree $q_{\max}+1$ is computed as scaffolding and is never reported as a cohomology group. The long exact sequence is asserted **exact at every spot all three of whose terms are reported**, and asserted to be a *complex* (consecutive composites zero) at the remaining spots inside the computed range. The record carries, *per complex it is about* (v1.4.0), $q_{\max}$, the flag `truncated`, the derived/set marker and that complex's list of degrees; a record with `truncated = true` may not report a vanishing above $q_{\max}$. At $6 \in R^\times$ the model complexes are already finite, `truncated = false`, and the policy is vacuous.

Spelled out, "exact at every spot all three of whose terms are reported" means, for the sequence $\cdots \to H^q_c \to H^q(\Gamma) \xrightarrow{r^q} H^q(\partial) \to H^{q+1}_c \to \cdots$: exactness is asserted at H^q_c and at $H^q(\Gamma)$ for $q \le q_{\max}$, and at $H^q(\partial)$ for $q \le q_{\max}-1$; at $H^{q_{\max}}(\partial)$ only $r^{q_{\max}} \circ (\text{previous}) = 0$ is asserted, because the next term $H^{q_{\max}+1}_c$ is not reported. Likewise $d \circ d = 0$ is checkable in degrees $\le q_{\max}-1$ and the cochain-map identity $d\Phi_g = \Phi_g d$ in degrees $\le q_{\max}$.

Three riders (owner decision, 2026-09-16).

(i) **$q_{\max}$ is derived on a finite model and settable only where `truncated = true`.** On a model whose complexes are finite — the 7-vertex model and every refinement of it at $6 \in R^\times$ — $q_{\max}$ is the top degree of the complex, *not* a choice, and the record marks it as derived; a user may set $q_{\max}$ only on a truncated free model, where it is the bound of the box. $q_{\max}$ is distinct from the *reported-degree list*, which is an independent filter on the degrees a record prints (already present in the campaign specification) and may be a proper subset of $\{0, \dots, q_{\max}\}$; a record with `truncated = true` may still not report a vanishing above $q_{\max}$.

(ii) **The bulk periodicity above the virtual cohomological dimension is a theorem, not a citation**: Lemma 14.22, with its Hecke equivariance, replaces the "standard route, not verified" that Remark 14.21(2) carried until v1.2.0; what it leaves of (OC-1c) is Remark 14.23(3).

(iii) **Obligation 15.22(iv) asserts composites zero only where both terms are reported**, i.e. it carries the box's own qualification ("at $H^{q_{\max}}(\partial)$ only $r^{q_{\max}} \circ (\text{previous}) = 0$") instead of its former blanket "in every computed degree".

Non-normative recommendation. At $\ell \in \{2, 3\}$ a run with $q_{\max} < 4$ cannot exhibit the stabilisation of Lemma 14.22 at all — its first instance, $H^2(\Gamma; V_R) \cong H^4(\Gamma; V_R)$, needs degree 4 reported and degree 5 built, i.e. $q_{\max} \ge 4$; the second, $H^3 \cong H^5$, and the cone's first (Remark 14.23(3)(b), from H^3_c) need $q_{\max} \ge 5$ — so $q_{\max} \ge 4$ is the recommended minimum there. The default stays 2: it is not a knob on the shipped finite models (rider (i)), and at $\ell \ge 5$ the extra degrees are known to vanish (Claim 16.2(3)).

The alternatives: (OC-1a) taken, (OC-1c) kept reachable.

(OC-1a) *Taken (owner decision 2026-09-16):* $q_{\max}$ a recorded run parameter, default 2, with the riders above.

(OC-1b) $q_{\max} = 2$ fixed by the contract. Rejected: it writes the current six-term window into a normative statement, which is precisely what OD-T1 and §17 are being kept open against, and it would have to be undone by a normative amendment the first time anyone asks for H^3.

(OC-1c) $q_{\max}$ determined by a *stabilisation theorem*: by Lemma 14.20 the boundary cohomology is, as an R-module, *the same in every degree* $q \geq 1$ at $\ell = 2$; the bulk *is* periodic above the virtual cohomological dimension, Hecke-equivariantly — Lemma 14.22: $e \cup -\colon H^q(\Gamma; V_R) \xrightarrow{\sim} H^{q+2}(\Gamma; V_R)$ for $q \geq 2$, commuting with every $T_{\Gamma g \Gamma}$, so the bulk half of this hypothesis, Hecke action included, is a theorem since v1.3.0; and if a boundary periodicity isomorphism compatible with r and with T_p^{∂}, and the cone periodicity it implies through (39), are proved as well, then a finite computation determines every degree and $q_{\max}$ is a *theorem*, not a parameter. **What remains unproved is exactly Remark 14.23(3)**: the boundary isomorphism (the candidate $\mathrm{res}^{\Gamma}_{\Gamma_\infty} e \cup -$ is named there), the module structure of (39) over $H^\bullet(\Gamma; \mathbb{Z})$ that the five lemma needs on the cone, and the Hecke equivariance on boundary and cone. This is a strengthening of (OC-1a) and not an alternative to it: adopting (OC-1a) changes nothing that (OC-1c) would later add.

(OC-1d) No truncation: decline to compute the cone at $\ell \in \{2, 3\}$ and report only the model objects of Remark 14.21(1). This is the status quo and remains available; it is a refusal, not a policy, and it forecloses §17's whole purpose.

Decision, and why. (OC-1a), as recommended from v0.14.0 and decided by the owner on 2026-09-16. It is the only one of the four that is correct for every ℓ, degree-generic as §17 and the implementation's data model require, vacuous exactly where the shipped $6 \in R^\times$ pipeline lives, and — the decisive point — **the one that keeps (OC-1c) free**. A truncation bound that is a recorded parameter can be replaced by a proved bound without amending a single statement; a truncation bound written into the contract cannot. The cost of (OC-1a) is that a record must carry three extra fields and that "$H^q = 0$ for $q > 2$" stops being sayable at $\ell \in \{2, 3\}$, which is not a cost but a correction. The decision leaves (OC-1c) exactly as reachable as before, and nearer: its bulk half is Lemma 14.22, so (OC-1b) would have written a normative bound that a theorem already half supersedes.

Where the 14.24 tag is used

So that the policy can be changed mechanically, these are every site in this document that depends on it — items (1)–(6) on its *content*, item (7) on its *status*. Changing 14.24 means changing these and nothing else; the adoption of 2026-09-16 (v1.3.0) changed exactly (1), (3), (6) and (7).

1. 14.24 itself — the boxed policy, the spelled-out exactness clause, the three riders, and the decision.
2. Remark 14.21(3), the sentence naming the free Γ-model as the object a truncation applies to.
3. Obligation 14.19(v.c), (vi)–(vii) and, forward, Obligation 15.22: "in every computed degree" is the phrase that reads $q_{\max}$ (since v1.4.0 clause (v) is split into (v.a)–(v.c), and only (v.c) carries the phrase; (v.a) and (v.b) name their degree), and the exactness assertion of the latter is the one 14.24 narrows. In §15 also Definition 15.9, stated for all $q \in \mathbb{Z}$ and read in degrees $\leq q_{\max}$ on a truncated model, and Remark 15.19, which names these two sites.
4. §17: 17.1 and 17.3 are stated in a degree range, not in degrees $\{0, 1, 2\}$ (written in v0.17.0; 17.1's $q_{\max}$ is this assumption's parameter), and Remark 17.5(4) — the sentence saying that a truncation policy is needed whatever the triangulation — is that chapter's third dependent site.
5. §16: Claim 16.23 and (52) — the six-term presentation is the $q_{\max} = 2$, $6 \in R^\times$ instance,

and Remark 14.21 says so.

6. The record schema of §22: the four per-complex fields $q_{\max}$, `truncated`, the derived/set marker and the complex's degree list (per complex since v1.4.0, §22.4 item 5), and the job-level reported-degree selection.
7. **The sites that assert the item's *status*** rather than read its content — every one of which changed on adoption, and which the list above had omitted until v1.3.0: the abstract ("one open construction item"); the closing paragraph of §1's scope statement ("the single open construction item"); rule (V5) ("14.24 is open"); the paragraph of §7.1 on what happens at $\ell = 2$ on the boundary ("the policy there is a tagged working assumption"); the "What this chapter fixes" paragraph of §14; the opening of §14.5; Remark 9.7's closing sentence and the remark before Obligation 15.22 that names this chapter's two reading sites (both missed at v1.3.0 and repaired at v1.3.2); the "What this appendix does not supply" list of Appendix C ("A decision on 14.24"); Appendix B's row for 14.24 and its counts; and §E.1 ("one open construction item is open"). The historical version lines of the Release record (the first subsection of Appendix E; the date line until v1.25.0-pre.7) and the v0.14.0 changelog entry are *not* status sites and do not change.

Code map. Implemented since v1.3.3 (hecke-extensions slot 1, packages WP-4a and WP-4b, 2026-09-16; `docs/implementation/slot1/WP-4a_report.md`, `WP-4b_report.md`). Until v1.3.2 this paragraph recorded a census of 2026-09-16 that found no τ_∂, no Θ^∂, no B_N, no boundary coset table and no `HeckeBoundary` translation unit in the tree, and only the boundary model without Hecke structure: `TestSL2Z_To_S1Cusp_RestrictionMorphism` (`LocalSystemSL2Z.cpp`), which builds K_∂ as the four-vertex path, its `EquivariantCohomologyOnK`, the inclusion $K_\partial \hookrightarrow K$ and the induced r^q with identity fibre map, pinned by `HeckeMorphismRegressionValida` That demo is unchanged. The specification the census closed with — a unit beside `hecke::HeckeCosetTables`, a τ_∂-table in the `ChainTerm` format of `hecke::HeckeTauTable` with every Γ''-related slot derived through the single shared (T1∂) helper, certified in the constructor, over the boundary subcomplex of `hecke::SL2ZTessellationTables` and degree-generic — is realised, unit by unit:

- `Hecke/SL2ZTessellationTables` (additive): the boundary subcomplex of (B1)–(B3) — the horocycle path $v_0v_1v_2v_3$ at height $Y = 2$ (vertices $\{0,1,2,3\}$ and edges e_{01}, e_{12}, e_{23} of K), the ∞-line coordinates (the position of $T^k v_i$ is $3k + i$), the $\mathbb{P}^1(\mathbb{Q})$ cusp key with the reduction to ∞ by nearest-integer Euclid on the primitive column, all exact and overflow-checked, and a constructor-time `validate_boundary()` asserting (B1)–(B3) on the shipped tables.
- `Hecke/HeckeCosetTables` (additive): `twisted_conjugate`, the one (T1) split-and-conjugate step, which `HeckeTauTable` now also calls — the bulk and the boundary datum literally share the helper, as this paragraph asked.
- `Hecke/HeckeBoundaryDatum` (`hecke::HeckeBoundaryTauTableT<Elt>`): the Γ-level τ_∂ of Definition 14.8, one chain per (j, σ) in `ChainTerm` format, degree-generic (no $q \in \{0,1,2\}$ literal; a boundary with cells above dimension 1 is refused, not truncated), every Γ''-related slot derived through the shared (T1∂) helper `tau_on_translate`; and the B_N/Γ_∞ table of Proposition 14.6 — the canonical representatives $\left(\begin{smallmatrix} p^u & b \\ 0 & p^{\nu-u} \end{smallmatrix}\right)$, $0 \le b < p^u$, and the bijection of (2) — certified in the constructor (invariant I8: codes 11–16 structural, 17 = (iv), 20–22 = (i), 30 = (ii), 40 = (iii), 60–66 the coset table). The table contains no ℓ: it is a datum over $\mathbb{Z}$, built for every ℓ the coset tables are built for, $\ell \in \{2,3\}$ and $p = \ell$ included, and `solve_regime() == "Z"` is a constant of the type, not the outcome of a fill (Remark 14.11).
- `Hecke/HeckeBoundaryOperator` (`hecke::HeckeBoundaryOperatorT<Elt>`): (30) on the circle model $C^\bullet_{\mathrm{eq}}(K_\partial, \mathcal{R}_\partial; L)$ — now a library call, `LocalSystemSL2Z::CreateSL2ZBoundaryPathTriangu` / `GenerateSL2ZBoundaryRelationsElt`, lifted out of the demo — assembled from the

bulk's own s_j, δ_j and τ_∂ in two bracketings asserted byte-identical; runtime assertions that Θ^∂ preserves the relations and the cocycles, the relation closure $\Theta^\partial C^q_{\mathrm{eq}} \subseteq C^q_{\mathrm{eq}}$ and $d\Theta^\partial = \Theta^\partial d$ (Proposition 14.15(1)); $A^{(q)}_\partial = P_q \Theta^\partial z_i$ per Algorithm 12.13; a JSON record `hecke_boundary_operator/v1` carrying the certificate bundle (not a golden object).

Obligation 14.19 is discharged in full, (i)–(vii), by `HeckeBoundaryTauValidation` (`standard`; 138 tables, 52 686 checks; (i)–(iv) re-derived independently of the table's own helpers over $(p,\nu) \in \{(2,1),(3,1),(5,1),(7,1),(11,1),(2,2),(3,2)\}$, $\ell \in \{2,3,5,7,691,134217757\}$ at `uint32_t` and $\ell \in \{2^{61}-1, 2^{32}+15\}$ at `uint64_t`, $n \in \{0,2,10\}$; the ℓ-, n- and width-independence of the chains asserted; seven tampered chains rejected) and `HeckeBoundaryOperatorValidation` (`standard`; 546 cells, 19 669 checks; $(p,\nu) \in \{(2,1),(3,1),(5,1),(7,1),(2,$ $\ell \in \{2,3,5,7,11,13,101,691,134217757,2^{31}-1\}$, even $n \le 16$, plus $2^{61}-1$ at `uint64_t`). Clause by clause: [verified computationally]

- (v) — two routes, both asserted equal to (30) on cohomology, entrywise, inside `compute()` itself (a mismatch aborts): *route A*, the Shapiro transcription of the decision box after Definition 14.14 — a Γ_∞-level twisted chain datum on the ∞-line built from the line coordinates alone, with the canonical left cosets δ_∞, δ_t of Proposition 14.6(1) and the double-coset structure of (3), hence available at $\nu = 1$; and *route B*, the closed form of Proposition 14.15(2) on $H^0 = V_R^T$ (every ν) and (31) on $H^1 = (V_R)_T$ ($\nu = 1$), transported into the stored H_{eq} basis through the coordinates of Proposition 14.4(1) — a 0-cocycle is $(m, m, m, \rho_n(T)m)$ with $m \in V_R^T$, and $H^1 \to (V_R)_T$ is $\varphi \mapsto \varphi([v_0, Tv_0])$, the signed sum over the fundamental path, i.e. the group cocycle $f(T)$ of the proof of Proposition 14.16(3); with the opposite orientation the form would not agree. The validation re-derives route B on its own and re-asserts both on every cell. Since v1.4.0 the clause names the three comparisons the implementation runs: route B on H^0 is (v.a), asserted at every ν of the grid; route B on H^1 is (v.b) and route A is (v.c), both asserted at $\nu = 1$; at $\nu = 2$ the clause is (v.a) alone, and the H^1 operator there is covered by (vii) and by `compute()`'s own certificates, exactly as the clause now says.
- (vi) — on the 330 cells with n even and $p \neq \ell$: every $x^{n-b\ell}\mathrm{Nm}^b$ (own polynomial arithmetic; 575 vectors) has eigenvalue $\chi_b(p)$ on H^0 and charpoly $A^{(0)}_\partial = \prod_b (X - \chi_b)$ — asserted; for $n < \ell$ (232 cells) $[y^n]$ has eigenvalue $1 + p^{n+1}$ on H^1 — asserted; the general-n H^1 multiset of Claim 16.24(2) **recorded**, 330/330 cells agree, the claim still [conjectural]. The β_j right-coset substitution of Proposition 14.6(4) is built as the *negative control* (never used by the pipeline): it acts by $\chi_{b+1}(p)$ on every monomial, as Proposition 14.16(2) proves, differs from $A^{(0)}_\partial$ on 212 cells, and breaks clause (vii) on H^1 on the 170 cells with $\chi_0(p) \neq \chi_1(p)$ and $\operatorname{rank} r^1 > 0$; on H^0 it cannot bite on this model, because the bulk $H^0(\Gamma) \neq 0$ only at $n = 0$, where $\chi_0 = \chi_1$, and on the Dickson cells $n = \ell + 1$, where the invariant $x\,\mathrm{Nm}$ is the $b = 1$ monomial with $\chi_1 \equiv \chi_2$ (Claim 16.24(3)).
- (vii) — $r^q A^{(q)} = A^{(q)}_\partial r^q$ against `HeckeOperatorOnK` through the public restriction morphism, **asserted** on 864 (cell, q) pairs at $6 \in R^\times$, $q \in \{0,1\}$ ($q = 2$ vacuous, $H^2(\partial) = 0$ asserted), $\nu \in \{1,2\}$ and $p = \ell$ included; **recorded** at $\ell \in \{2,3\}$, where the chapter does not assert it: 216/216 pairs hold on the model complexes — offered to §15 as a recorded fact, not as a claim, and recorded there as Remark 15.23 since v1.4.0. $r^1 \neq 0$ on 480 cells. The β_j control's failure to bite on H^0 is what the qualified clause (vii) predicts for $n < \ell(\ell-1)$.

Family (A) of §21.4 has been run with these numbers; its first instance $(n, \ell, p) = (2, 5, 2)$ prints $A^{(0)}_\partial = [4]$, $A^{(1)}_\partial = [4]$, bulk $A^{(1)}(T_2) = [4]$, $\operatorname{rank} r^1 = 1$, as that paragraph predicts.

Choices the implementation fixed where the statements leave them open — recorded as the implementation's choices, consistent with the statements, not as changes to them:

- Lemma 14.10(1) allows "any vertex w of the target component". The canonical τ_∂ takes

the line vertex nearest to $\mathrm{Re}(\gamma_{\mathrm{red}} z)$ (the vertical projection onto the horocycle), ties to the larger real part — a T-equivariant rule, hence independent of which γ_{red} represents the cusp reduction, and $-I$ acts trivially. **Since v1.4.0 this rule is Algorithm 14.12**, the contract's definition of the canonical τ_∂ (`HeckeBoundaryTauTableT::construct_vertex`, `construct_edge`, `SL2ZTessellationTables::boundary_line_nearest`), with the well-definedness of the derivation order as its property (1). Any other admissible choice is homotopic over $\mathbb{Z}$ for every ℓ (Lemma 14.10(3)); the choice matters only for the size of §15's η. The rule is *not* ι-symmetric, and no vertex rule can be: Remark 14.13, which proves it at the level of rules (no line vertex is ι-fixed) and exhibits the failure of the shipped table at $(p, \nu) = (2, 1)$; Obligation 9.10 says nothing about τ_∂.

- Proposition 14.6(2) fixes γ_j only up to $\gamma_j \Gamma_\infty$; the implementation takes $\gamma_j :=$ reduce_to_infinity$(g^{-1} s_j \infty)^{-1}$, deterministic through the (P2) tie rule — Algorithm 14.12 step 4 since v1.4.0.
- Clause (iii)'s "every stored transporter $h \in \Gamma''$ occurring in the table" is read as Obligation 20.11 reads it — the transporter sets $S(\sigma, \sigma')$ of the model between boundary simplices, through $s_j t$ — and the validation covers arbitrary $\gamma \in \Gamma$ by 40 fixed-seed random words per table.
- $\mathcal{R}_\partial$ of the circle model is the cusp relation $f(v_3) = \rho_n(T) f(v_0)$ *and* the $-I$ self-relations $f(\sigma) = \rho_n(-I) f(\sigma)$ on every simplex (the boundary half of Remark 7.7's `FIX-negI`), so that $C^\bullet_{\mathrm{eq}}(K_\partial, \mathcal{R}_\partial; L) = \mathrm{Hom}_{\mathbb{Z}[\Gamma_\infty]}(C_\bullet(\partial X), V_R)$ literally for every n: vacuous for n even or $\ell = 2$, the zero complex for n odd at $2 \in R^\times$ (Remark 14.5, asserted by the odd-n arm). **Since v1.4.0 this is Proposition 14.4(4)**: $\mathcal{R}_\partial$ is defined in the proposition with the self-relations, and the identification is proved for every n; "the circle model with its single relation" survives only as the description of the case $\rho_n(-I) = \mathrm{id}$.

What it records (owner confirmation 11 of the slot-1 plan: compute at every ℓ, never refuse): `boundary_model = "circle"`, `boundary_is_group_cohomology` $= (2 \in R^\times) = (\ell \neq 2)$ — the flag Proposition 14.4(3) asks for, so the code now records which of the two objects it computes — `q_max = 1` with `q_max_source = "derived"` (the top degree of the finite circle model), `truncated = false`, reported degrees $\{0, 1\}$; 14.24 has nothing to truncate on this complex (its rider (i) applies to the free model). At $\ell = 2$ the dimensions are Proposition 14.4(2)'s, $\dim H^0 = \dim H^1 = \dim V_R^T = \lfloor n/2 \rfloor + 1$, and the validation asserts that they are *not* Lemma 14.20's. The bulk object derives $q_{\max} = 2$; since v1.4.0, §22.4 item 5 rules that a record carries these fields *per complex* and never as a maximum over a record (owner confirmation 1 of the slot-2 plan): bulk 2 / `false` / `"derived"` / $[0, 1, 2]$, boundary 1 / `false` / `"derived"` / $[0, 1]$.

The two shape requirements the specification stated hold as built, with one wording corrected: the table is over ∂X — whose simplices *are* simplices of K^u (the Γ-translates of the boundary simplices of K), addressed by the same canonical keys; what differs is the *slot set*, supplied by `SL2ZTessellationTables::boundary_simplices` — and it is degree-generic, not $q \in \{0, 1, 2\}$, because 14.24 indexes the free model by $q_{\max}$. What the pair still lacks — η, the cone and Φ_g — is §15's code map.

Formalization note. Definition 14.2 and (28) are a two-line computation with 2×2 integer matrices. Proposition 14.6 is a finite matrix-arithmetic statement with an explicit counting argument and is decidable for each N; it is the natural first target here. Lemma 14.9 and Lemma 14.10 are the two that carry the content, and both are *constructive*: the second is a terminating procedure plus two uniqueness statements, each of which reduces to $Z_1 = 0$ on a contractible 1-complex, so a formalisation needs simplicial chains and H_1 of a tree, not homological algebra — in particular it needs **neither** the fundamental lemma of relative homological algebra **nor** any invertibility hypothesis, which is the whole point of stating it separately from Lemma 11.3(2). The hypothesis a formalisation must not lose is (B3): it is what makes the

(T1∂)-extension well defined, and it is used three times (existence on vertices, on edges, and in the uniqueness argument). Proposition 14.3 needs the Lyndon–Hochschild–Serre spectral sequence of a direct product with a cyclic factor, which `mathlib4` does not currently carry in this form; the two-column degeneration can be replaced by the explicit $\langle T\rangle$-equivariant periodic resolution of $\mathbb{Z}$ over $\mathbb{Z}[\mathbb{Z}/2]$ tensored with the two-term resolution over $\mathbb{Z}[\mathbb{Z}]$, which is a finite construction and avoids the spectral sequence entirely — the recommended route. Proposition 14.16 is polynomial arithmetic in $R[x,y]$ modulo the Frobenius identities $u^\ell = u$ and $p^{\ell-1} = 1$, both of which are `ZMod` lemmas. Obligation 14.19 is a decidable predicate on emitted finite data in every clause except (vii), which quantifies over cohomology classes and is decidable only through the stored bases.

15 The pair homotopy and the Hecke action on the mapping cone

What this chapter fixes. The Hecke action on compactly supported cohomology, as a definition and not as a slogan. The pair $(X, \partial X)$, the restriction $r = j^*$ and the mapping cone $C(r)$ in the convention of `monodromy_v5.tex` §8; the *pair homotopy* η, a certified chain-level datum one level below the cochains, where τ and τ_∂ live, and the only new table this chapter asks for; the cochain homotopy h, **derived** from η by coset transport and never chosen; the cone cochain map Φ_g and the operator $T^c_{\Gamma g\Gamma}$ it induces on $H^\bullet_c$; and the theorem that makes $H^1_c = X_3$ a Hecke module at all — that the operator is independent of η over $\mathbb{Z}$ for every ℓ, and, for $6 \in R^\times$, of every other choice as well, and coincides with the correspondence on the relative complex. It fixes, too, why the construction has this shape and not a simpler one: a freely chosen cochain homotopy leaves the operator on X_3 undetermined by a $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0)$ that is realised in full over a field and is non-zero in a real case; and the entrywise identity $r\,\Theta_g = \Theta^\partial_g\, r$, however tempting, decides nothing (rule (A6)). Finally: integrality — the three tables are over $\mathbb{Z}$ once and every R-computation is their reduction, which is what makes reductions and Bocksteins equivariant on the nose — and the two obligations that make all of this a checkable property of emitted data.

Five errata land here (Appendix A): A.2 and A.3 in Proposition 15.16, which states the realisation argument with Z^0 and never B^0, and *over a field*, with $\mathbb{Z}/\ell^e$ explicitly not claimed; A.5 in Lemma 15.6, which states the augmentation argument and not the "both are vertices" recipe; A.13 in Lemma 15.8 and Lemma 15.10, where the cochain-level sign is *derived* from the chain-level convention and is therefore asserted in one place only; and A.15, which this whole chapter answers — "cone by functoriality" is not a definition, and Definition 15.9 together with Proposition 15.12 is.

What this chapter does not fix. The boundary datum τ_∂ and the boundary operator Θ^∂_g are §14 and are consumed here, not redefined. The six groups $X_1, \dots, X_6$ and the six-term sequence are §16; this chapter proves the general long exact sequence and says under which hypotheses it closes to that window. The free Γ-model on which all of this is to be repeated at $6 \notin R^\times$ is §17; here $6 \notin R^\times$ is treated as §11 treats it (Remark 15.19).

Standing hypotheses. (H4) of §7 — in particular (H4)(ii), *X is contractible and two-dimensional*, and (H4)(iv), *$-I$ lies in every stabiliser and acts trivially* — and (B1)–(B3) of §14, each repeated where it is used (rule (A1)). [formally verified: [GT, `thm:model7-pairhyp`]] (on the seven-vertex model with its horocycle boundary: [GT]'s pair hypothesis `PairHyp`, the form in which this chapter's lemmas use (H4) and (B1)–(B3), holds with no further hypothesis, for every $g \in \mathrm{GL}_2(\mathbb{Q})$) Per the design direction of Remark 7.6 every statement is made for *the tessellation and boundary triangulation in force* and not for seven vertices; Remark 14.1 is what makes the shipped triangulation, and every equivariant refinement of it, admissible input. The

consequences of (H4)(ii) that are used, and used by name, are

$$H_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \text{ is an isomorphism}, \qquad Z_1(X) = B_1(X), \qquad Z_2(X) = 0, \tag{37}$$

all over $\mathbb{Z}$ [formally verified: [GT, `prop:model7h4-ii`]] (on the model) (and hence, $C_\bullet(X)$ being free, after $\otimes R$ for every R [formally verified: [GT, `thm:xbulk-repaug`]] [formally verified: [GT, `lem:xbulk-contractz`]] (on the model, as the split exactness of the augmented R-complex)): the first two because X is connected with $H_1(X) = 0$, the third because $H_2(X) = 0$ and $C_3(X) = 0$. The OQ2 revision audit (finding F5) found these three facts load-bearing and *unwritten* in every source of this chapter; they are written here once and cited by (37) at each use. Throughout, $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, $\Gamma'' = \Gamma_0(N)$, $s_1, \dots, s_m$ and $\delta_j = s_j^{-1} g$ are the canonical data of Definition 10.2, τ is a g-twisted chain datum over $\mathbb{Z}$ on X (Definition 11.1) and τ_∂ a boundary g-twisted chain datum over $\mathbb{Z}$ on ∂X (Definition 14.8); "(T1)" means the twisted equivariance $F(h \cdot x) = (g^{-1}hg) \cdot F(x)$ for $h \in \Gamma''$, for whichever additive map F is under discussion.

15.1 The pair, the restriction map and the cone

Definition 15.1 (The pair $(X, \partial X)$, and $r := j^*$)**.** [formally verified: [GT, `def:xbdry`]] [formally verified: [GT, `thm:xbdry-pair`]] (∂X as a Γ-saturated, face-closed sub-descriptor; j equivariant, a chain map over the augmentations) Assume (H4) and (B1)–(B3), and let R be a commutative ring. The *pair* is $(X, \partial X)$ with $j\colon \partial X \hookrightarrow X$ the inclusion, a Γ-equivariant simplicial map, written also for the induced chain map $j\colon C_\bullet(\partial X) \to C_\bullet(X)$. As the inclusion of a subcomplex, j commutes with the augmentations: $\varepsilon \circ j = \varepsilon_\partial$, where $\varepsilon_\partial\colon C_0(\partial X) \to \mathbb{Z}$ sends every vertex to 1 (stated explicitly at v1.18.0-pre.2, slot-18 L18-4: the Lean statement `GL2.Cone.eta_exists` needs it as the hypothesis `hjaug`, an abstract j not being an inclusion; Lemma 15.6(V) uses it; no statement changes). Its two equivariant cochain complexes are

$$\begin{aligned} C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L) &\cong \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(X), V_R\big) && \text{(Theorem 5.10, } K^u \cong X\text{)}, \\ C^\bullet_\partial &= \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(\partial X), V_R\big) && \text{(Proposition 14.4)}, \end{aligned}$$

and the *restriction* is the cochain map $r := j^*\colon C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L) \to C^\bullet_\partial$, $\phi \mapsto \phi \circ j$. The *relative complex* is $\ker r$, with the differential restricted from $C^\bullet_{\mathrm{eq}}$. The *mapping cone* of r is the cochain complex

$$C(r)^q := C^{q+1}_{\mathrm{eq}}(K, \mathcal{R}; L) \oplus C^q_\partial, \qquad D(c, c_\partial) := \big(-dc,\ dc_\partial + rc\big), \tag{38}$$

with $\mathrm{inc}_c\colon C^\bullet_\partial \to C(r)$, $c_\partial \mapsto (0, c_\partial)$, and $\mathrm{pr}_c\colon C(r) \to C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)[1]$, $(c, c_\partial) \mapsto c$, where $C^\bullet_{\mathrm{eq}}[1]^q := C^{q+1}_{\mathrm{eq}}$ carries the differential $-d$. **Compactly supported cohomology** is

$$H^q_c \ := \ H^{q-1}\big(C(r)\big), \qquad q \in \mathbb{Z}.$$

This is the convention of `monodromy_v5.tex` §8 (`:1152--1163`) with its (a, b') written (c, c_∂), its cone degree n written q, and its ι, π written $\mathrm{inc}_c, \mathrm{pr}_c$ (§2.6); for the model of Definition 7.1 it is exactly the table entry $C(r)^q = C^{q+1}_{\mathrm{eq}}(K) \oplus C^q_{\mathrm{eq}}(K_\partial)$ of §2.6, since there $C^\bullet_\partial = C^\bullet_{\mathrm{eq}}(K_\partial, \mathcal{R}_\partial; L)$ (Proposition 14.4).

Lemma 15.2 (r is onto; the cone is a complex; $H^\bullet_c$ is relative cohomology)**.** *[proved here] Assume (H4) and (B1)–(B3) and let R be a commutative ring; no hypothesis on ℓ or n.*

1. ***(Surjectivity.)*** *r is surjective in every degree: the* extension by zero $\mathrm{ext}_0\colon C^q_\partial \to C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$, *which gives a cochain the value $\varphi(\sigma)$ on a simplex σ of ∂X and 0 on every other simplex of X, is R-linear, Γ-equivariant, and a section of r. It is* not *a cochain map, and is never used as one.*

2. ***(The cone.)*** *[formally verified: [GT, `lem:conerelative`]]* $D^2 = 0$*; inc_c and pr_c are cochain maps, and $0 \to C_\partial^\bullet \xrightarrow{\mathrm{inc}_c} C(r) \xrightarrow{\mathrm{pr}_c} C_{\mathrm{eq}}^\bullet(K,\mathcal{R};L)[1] \to 0$ is a short exact sequence of complexes of R-modules whose connecting homomorphism $H^q(C_{\mathrm{eq}}^\bullet[1]) = H_{\mathrm{eq}}^{q+1} \to H^{q+1}(C_\partial^\bullet)$ is $+r^{q+1}$. Hence the* ***long exact sequence***

$$\cdots \longrightarrow H_c^q \xrightarrow{\mathrm{pr}_{c*}} H_{\mathrm{eq}}^q(K,\mathcal{R};L) \xrightarrow{r^q} H^q(C_\partial^\bullet) \xrightarrow{\mathrm{inc}_{c*}} H_c^{q+1} \longrightarrow \cdots, \tag{39}$$

in every degree $q \in \mathbb{Z}$.

3. ***(Relative cohomology.)*** *[formally verified: [GT, `lem:conerelative`]] $c \mapsto (c,0)$ is a quasi-isomorphism $(\ker r)[1] \to C(r)$; consequently*

$$H_c^q \;\cong\; H^q(\ker r) \;=\; H^q\Big(\mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(X,\partial X), V_R\big)\Big) \qquad \text{for every } q,$$

where $C_\bullet(X,\partial X) := C_\bullet(X)/j\,C_\bullet(\partial X)$ is the relative chain complex, a complex of $\mathbb{Z}[\Gamma]$-permutation modules on the simplices of X not in ∂X.

4. ***(Vanishing.)*** *$H_c^q = 0$ for $q \le 0$ and for $q \ge 3$. In particular $H_c^0 = 0$* with no hypothesis *on ℓ or n: it is the kernel of $r^0\colon H_{\mathrm{eq}}^0 \to H^0(C_\partial^\bullet)$, and that map is injective.*

5. ***(The window, and the two dimension counts.)*** *[formally verified: [GT, `lem:conerelative`]] (given (4), which is not formalised: a statement about the model) Let $R = \mathbb{F}$ be a field.*

$$\dim H_c^1 = \dim H^0(C_\partial^\bullet) - \dim H_{\mathrm{eq}}^0 + \dim\ker r^1, \qquad \dim H_c^2 = \dim H^1(C_\partial^\bullet) - \operatorname{rank} r^1. \tag{40}$$

The first identity holds with no further hypothesis *(v1.6.0): it uses only the exactness of (39) and $H_c^0 = 0$ from (4), both of which are unconditional.* ***The second identity is asserted under $H_{\mathrm{eq}}^2(K,\mathcal{R};L) = 0$,*** *which is also exactly the hypothesis under which (39) collapses to the six-term exact sequence $0 \to H_{\mathrm{eq}}^0 \to H^0(\partial) \to H_c^1 \to H_{\mathrm{eq}}^1 \to H^1(\partial) \to H_c^2 \to 0$ of Claim 16.23. That hypothesis is Claim 16.2's business (it holds for $6 \in R^\times$) and is* not *proved here. (Through v1.5.2 both identities were stated under it; `docs/implementation/slot3/WP-5_report.md` finding 3.)*

Proof. (1) Γ preserves ∂X ((B1)), so $\gamma\sigma \in \partial X$ iff $\sigma \in \partial X$, and $(\mathrm{ext}_0\varphi)(\gamma\sigma) = \rho_n(\gamma)(\mathrm{ext}_0\varphi)(\sigma)$ holds in both cases; $r \circ \mathrm{ext}_0 = \mathrm{id}$ by construction. That ext_0 is not a cochain map: a boundary edge is a face of an interior triangle, so $d(\mathrm{ext}_0\varphi)$ can be non-zero on a simplex where $\mathrm{ext}_0(d\varphi)$ vanishes.

(2) $D^2(c,c_\partial) = D(-dc, dc_\partial + rc) = \big(d^2c,\ d(dc_\partial + rc) - r\,dc\big) = (0,\ drc - rdc) = 0$, using $rd = dr$. $D\mathrm{inc}_c(c_\partial) = (0,dc_\partial) = \mathrm{inc}_c(dc_\partial)$ and $\mathrm{pr}_c D(c,c_\partial) = -dc$, the differential of the shift. Levelwise exactness is the definition of a direct sum. The long exact sequence is [standard: [Wei94, Thm. 1.3.1]: the long exact sequence of a short exact sequence of complexes, with the connecting map of its proof]. For the connecting map: a class in $H^q(C_{\mathrm{eq}}^\bullet[1])$ is $[c]$ with $c \in Z_{\mathrm{eq}}^{q+1}$; lift to $(c,0) \in C(r)^q$, apply D to get $(0,rc) = \mathrm{inc}_c(rc)$; so the connecting map is $[c] \mapsto [rc] = r^{q+1}[c]$, with the sign $+$ forced by (38). Re-indexing $H^{q-1}(C(r)) = H_c^q$ gives (39).

(3) The map is a cochain map because $D(c,0) = (-dc, rc) = (-dc, 0)$ for $c \in \ker r$, and $-d$ is the differential of the shift. Its cokernel is identified by $(c,c_\partial) \mapsto (rc, c_\partial)$ with the complex $C(\mathrm{id})$ of $\mathrm{id}\colon C_\partial^\bullet \to C_\partial^\bullet$, $C(\mathrm{id})^q = C_\partial^{q+1} \oplus C_\partial^q$, $D_{\mathrm{id}}(b,b') = (-db, db' + b)$: the map is a cochain map ($D_{\mathrm{id}}(rc,c_\partial) = (-drc, dc_\partial + rc)$ is the image of $D(c,c_\partial)$), it is surjective by (1), and its kernel is $\{(c,0) : rc = 0\} = (\ker r)[1]$. $C(\mathrm{id})$ is contractible: with $s(b,b') := (b',0)$ one computes $(D_{\mathrm{id}}s + sD_{\mathrm{id}})(b,b') = (-db', b') + (db' + b, 0) = (b,b')$. So in the long exact sequence of $0 \to (\ker r)[1] \to C(r) \to C(\mathrm{id}) \to 0$ every third term vanishes and the first map is an isomorphism on cohomology [standard: [Wei94, §1.5] for the mapping cone; [Wei94, Thm. 1.3.1] for the sequence]. Finally $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-,V_R)$ is left exact, so applying it to $0 \to C_\bullet(\partial X) \xrightarrow{j} C_\bullet(X) \to C_\bullet(X,\partial X) \to 0$ (exact because j is the inclusion of a subcomplex, (B1)) identifies $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X,\partial X), V_R)$ with $\ker r$.

(4) $\ker r$ vanishes outside degrees $0, 1, 2$ because X is two-dimensional ((H4)(ii)), so $H^q(\ker r) = 0$ for $q < 0$ and $q > 2$; with (3) this is $H_c^q = 0$ for $q \le -1$ and $q \ge 3$. For $q = 0$: $H_c^0 = H^0(\ker r)$ consists of the equivariant 0-cocycles c with $rc = 0$. A 0-cocycle satisfies $c(\partial e) = 0$ for every edge, hence is constant on the vertices of the connected complex X (37); a constant that vanishes on the non-empty set of boundary vertices ((B1)) is 0. Equivalently: r^0 is the inclusion $V_R^\Gamma \hookrightarrow V_R^{\Gamma\infty}$ (Remark 11.8, Proposition 14.15(2)), which is injective.

(5) *The first identity, with no hypothesis.* $H_c^0 = 0$ by (4), so exactness of (39) at H_{eq}^0 makes r^0 injective, and exactness at $H^0(\partial)$, at H_c^1 and at H_{eq}^1 gives the exact sequence

$$0 \longrightarrow H_{\mathrm{eq}}^0 \xrightarrow{r^0} H^0(C_\partial^\bullet) \xrightarrow{\mathrm{inc}_{c*}} H_c^1 \xrightarrow{\mathrm{pr}_{c*}} \ker r^1 \longrightarrow 0,$$

whose last term is $\mathrm{im}(\mathrm{pr}_{c*}\colon H_c^1 \to H_{\mathrm{eq}}^1) = \ker r^1$, again by exactness at H_{eq}^1. Over a field the alternating sum of dimensions of an exact sequence vanishes, which is the first identity. **Nothing here uses** H_{eq}^2, $H^q(C_\partial^\bullet)$ for $q \ge 1$, or any hypothesis on ℓ or n beyond (H4) and (B1)–(B3).

The six-term window and the second identity, under $H_{\mathrm{eq}}^2 = 0$. With that hypothesis the terms H_c^0 (by (4)), H_{eq}^2 (by hypothesis), $H^2(C_\partial^\bullet)$ (∂X is one-dimensional, (B2)) and H_c^3 (by (4)) of (39) all vanish, leaving the displayed six-term sequence; its second half is $0 \to \operatorname{coker} r^1 \to H_c^2 \to 0$, i.e. the second identity. That half is where the hypothesis is used: without $H_{\mathrm{eq}}^2 = 0$ the map $H_c^2 \to H_{\mathrm{eq}}^2$ need not be zero and $\operatorname{coker} r^1 \to H_c^2$ need not be onto. □

Every map built in this chapter is a *coset transport*: the device of Theorem 11.7(b), where it produced $\widetilde{H}$ from a twisted homotopy H, applied to maps between the chain complexes of the pair. It is given a name and four identities once, so that every later proof is a two-line computation.

Definition 15.3 (Coset transport)**.** Let $Y_\bullet, Y'_\bullet$ each be one of $C_\bullet(X)$, $C_\bullet(\partial X)$, and let $F\colon Y_\bullet \to Y'_{\bullet+k}$ be an additive map of degree $k \in \mathbb{Z}$ satisfying (T1). Its *coset transport* is the R-linear map

$$\widetilde{F}\colon \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(Y'_{q+k}, V_R\big) \longrightarrow \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(Y_q, V_R\big), \qquad \big(\widetilde{F}\phi\big)(y) \;:=\; \sum_{j=1}^{m} \rho_n(\delta_j)\,\phi\big(F(s_j \cdot y)\big), \tag{41}$$

of *cochain* degree $-k$. Thus $\widetilde{\tau}$ is the $\widetilde{T}$ of (21), i.e. Θ_g under Ψ (Theorem 12.11(3)); $\widetilde{\tau_\partial}$ is Θ_g^∂ of (30); and for the twisted homotopy H of Lemma 11.3(2), $\widetilde{H}$ is the map of that name in the proof of Theorem 11.7(b). *Notation.* The tilde on a *map* denotes coset transport; the tilde on a *simplex*, $\widetilde{\sigma}$, denotes the lift of §2.3 (C16). The two never meet in one formula, but a reader should know they are unrelated. The decoration has two further readings — the cochain-level operator $\widetilde{T}$ of Definition 11.6, which is $\widetilde{\tau}$ and so agrees with the first, and Part I's saturation $\widetilde{K}$ and universal cover $\widetilde{X}$, which occur nowhere near either. All four are **C19** of §2.7 since v0.16.0: the decoration is read from what it decorates, nothing is renamed, and no new tilde may be introduced without a row there.

Lemma 15.4 (Coset transport: equivariance and four identities)**.** *[proved here] Assume (H4) and (B1), let R be a commutative ring, and let F, G be additive maps with (T1) as in Definition 15.3. Then, with no hypothesis on ℓ:*

1. *$\widetilde{F}\phi$ is Γ-equivariant for every Γ-equivariant ϕ, so (41) is well typed; and $F \mapsto \widetilde{F}$ is additive.*
2. *$\widetilde{\partial \circ F}\,\phi = \widetilde{F}(d\phi)$ and $\widetilde{F \circ \partial}\,\phi = d\big(\widetilde{F}\phi\big)$.*
3. *If $G\colon C_\bullet(\partial X) \to C_{\bullet+k}(\partial X)$, then $\widetilde{j \circ G}\,\phi = \widetilde{G}(r\phi)$ for $\phi \in \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), V_R)$.*
4. *If $G\colon C_\bullet(X) \to C_{\bullet+k}(X)$, then $\widetilde{G \circ j}\,\phi = r\big(\widetilde{G}\phi\big)$.*

Proof. (1) Let $\gamma \in \Gamma$ and write $s_j\gamma = h_j(\gamma)s_{j\cdot\gamma}$ with $h_j(\gamma) \in \Gamma''$ (Lemma 10.3). By (T1) of F, Γ-equivariance of ϕ, and $g^{-1}h_j(\gamma)g \in \Gamma$,

$$\rho_n(\delta_j)\,\phi\big(F(s_j\gamma y)\big) = \rho_n(\delta_j)\,\rho_n\big(g^{-1}h_j(\gamma)g\big)\,\phi\big(F(s_{j\cdot\gamma}y)\big) = \rho_n\big(\gamma\, s_{j\cdot\gamma}^{-1}g\big)\,\phi\big(F(s_{j\cdot\gamma}y)\big),$$

by Proposition 8.2(2) on the integral identity $\delta_j\, g^{-1}h_j(\gamma)g = s_j^{-1}h_j(\gamma)g = \gamma s_{j\cdot\gamma}^{-1}g$ in $\mathrm{M}_2(\mathbb{Z})$ (17). Summing over j, and using that $j \mapsto j\cdot\gamma$ is a bijection, gives $(\widetilde{F}\phi)(\gamma y) = \rho_n(\gamma)(\widetilde{F}\phi)(y)$. This is the equivariance computation of Theorem 11.7 with F in place of τ; it used (T1) of F and nothing else about it. Additivity in F is clear.

(2) $(\widetilde{\partial F}\phi)(y) = \sum_j \rho_n(\delta_j)\phi(\partial F(s_jy)) = \sum_j \rho_n(\delta_j)(d\phi)(F(s_jy)) = (\widetilde{F}d\phi)(y)$, since $(d\phi)(x) = \phi(\partial x)$. And, s_j acting simplicially so that $\partial(s_jy) = s_j\partial y$, $(\widetilde{F\partial}\phi)(y) = \sum_j \rho_n(\delta_j)\phi(F(\partial(s_jy))) = \sum_j \rho_n(\delta_j)\phi(F(s_j\,\partial y)) = (\widetilde{F}\phi)(\partial y) = (d\widetilde{F}\phi)(y)$.

(3) $(\widetilde{jG}\phi)(y) = \sum_j \rho_n(\delta_j)\phi(jG(s_jy)) = \sum_j \rho_n(\delta_j)(\phi\circ j)(G(s_jy)) = (\widetilde{G}(r\phi))(y)$.

(4) $(\widetilde{Gj}\phi)(y) = \sum_j \rho_n(\delta_j)\phi(G(js_jy)) = \sum_j \rho_n(\delta_j)\phi(G(s_j\,jy)) = (\widetilde{G}\phi)(jy) = (r\widetilde{G}\phi)(y)$, using that j is Γ-equivariant. □

15.2 The pair homotopy

Definition 15.5 (Pair homotopy η: $\partial\eta+\eta\partial = j\tau_\partial - \tau j$)**.** [formally verified: [GT, `def:pairhomotopy`]] (the augmentation compatibility is part of Definition 15.1) Assume (H4) and (B1)–(B3), let R be a commutative ring, and let τ, τ_∂ be g-twisted chain data over R on X and ∂X (Definitions 11.1, 14.8), for the same g, the same Γ'' and the same twist $h \mapsto g^{-1}hg$. A *pair homotopy* for (τ, τ_∂) is an additive map of degree $+1$

$$\eta\colon C_\bullet(\partial X)\otimes R \longrightarrow C_{\bullet+1}(X)\otimes R$$

satisfying (T1) and

$$\partial\circ\eta + \eta\circ\partial \;=\; j\circ\tau_\partial - \tau\circ j \qquad \text{as maps } C_\bullet(\partial X)\otimes R \to C_\bullet(X)\otimes R. \tag{42}$$

Both sides of (42) are twisted chain maps $C_\bullet(\partial X) \to C_\bullet(X)$ lifting the same map of components (Lemma 14.9), and η is a twisted chain homotopy between them. An *η-table* is such a map together with the finite table of its values on a set of Γ''-orbit representatives of the simplices of ∂X — a 1-chain of X for each representative vertex, a 2-chain for each representative edge — from which (T1) determines it everywhere. The *Hecke datum of the pair* is the triple $(\tau, \tau_\partial, \eta)$.

> **OWNER DECISION. The sign convention is (42), $\partial\eta + \eta\partial = j\tau_\partial - \tau j$, and the derived homotopy of Definition 15.9 is $h := \widetilde{\eta}$ with no sign** (owner decision OP6, 2026-09-15). The opposite convention would give $h = -\widetilde{\eta}$; it is used nowhere in this document or in the implementation plan. What this decision does *not* move is the cochain identity $dh + hd = \Theta^{\partial}_g r - r\Theta_g$ and the cone map Φ_g, which are forced by D (Lemma 15.8) and are the same under either convention.

Lemma 15.6 (η exists over $\mathbb{Z}$ for every ℓ)**.** *[proved here] [formally verified: [GT, `lem:etaexists`]] [formally verified: [GT, `cor:model7-etaexists`]] (on the seven-vertex model with its horocycle boundary, every $g \in \mathrm{GL}_2(\mathbb{Q})$: (H4) and (B1)–(B3) discharged in [GT]) (Appendix A, entry A.5: the vertex step is an augmentation argument, not "a path between two vertices".) Assume (H4) and (B1)–(B3), and let τ, τ_∂ be g-twisted chain data* over $\mathbb{Z}$. *Then a pair homotopy for (τ, τ_∂) over $\mathbb{Z}$ exists, and one is built by the following terminating procedure (rule (A3)). No hypothesis on ℓ is used, no stabiliser order is inverted, and no linear system is solved over anything but $\mathbb{Z}$.*

(V) ***Vertices.*** *For each Γ''-orbit representative vertex x of ∂X: the 0-chain $(j\tau_\partial - \tau j)(x)$ of X has augmentation $\varepsilon(j\tau_\partial x) - \varepsilon(\tau j x) = 1 - 1 = 0$ — by $\varepsilon \circ j = \varepsilon_\partial$ (Definition 15.1) with (T3∂) for τ_∂, and (T3) for τ — hence is a boundary (37). Choose* any *1-chain $\eta(x) \in C_1(X)$ with $\partial\eta(x) = (j\tau_\partial - \tau j)(x)$, and extend to all boundary vertices by (T1).*

(E) ***Edges.*** *For each Γ''-orbit representative edge e of ∂X: the 1-chain $z_e := (j\tau_\partial - \tau j)(e) - \eta(\partial e)$ of X is a cycle, hence a boundary, and there is a* unique *2-chain $\eta(e)$ with $\partial\eta(e) = z_e$ (37). Extend by (T1).*

(≥ 2) *There is nothing to do: $C_q(\partial X) = 0$ for $q \geq 2$ ((B2)).*

The map so defined satisfies (T1) and (42); the only choice made is the vertex chain in (V), and Lemma 15.7 says exactly what that choice costs. The choice made *(v1.5.0, naming what the implementation does; the statement is unchanged, (V) still admitting* any *1-chain with the stated boundary): the augmentation-0 0-chain $(j\tau_\partial - \tau j)(x)$ is written as a sum of pairs (one vertex with coefficient $+1$, one with -1) and each pair is joined by the* ***pairwise path routing*** *of Algorithm 18.3 step 2 — the stored path template between the two vertices — which is the augmentation argument made constructive, needs no linear solve, and reduces to* the one path joining the two vertices *when $j\tau_\partial x$ and $\tau j x$ are single vertices, as they are on the shipped tables (v1.6.0, corrected from "the single edge path", which could be read as "one edge"; the stored path template between two vertices may be several edges long). §15's code map records it as the implemented choice, and Lemma 15.7 is what makes the naming harmless. The procedure terminates because Γ'' has finitely many orbits of simplices on ∂X ((B1) and $[\Gamma : \Gamma''] = m < \infty$, Proposition 10.1(3)).*

Proof. (T1)-extension is well defined. Both in (V) and in (E) the value on an orbit $\Gamma'' \cdot \sigma$ is set by $\eta(h\sigma) := (g^{-1}hg) \cdot \eta(\sigma)$, which is well defined iff $(g^{-1}hg) \cdot \eta(\sigma) = \eta(\sigma)$ for every h in the Γ''-stabiliser of σ. By (B3) that stabiliser is $\Gamma'' \cap \{\pm I\}$, $g^{-1}(-I)g = -I$, and $-I$ acts trivially on X ((H4)(iv)); so the condition is vacuous. *This is where the difference from the abstract route of Lemma 11.3(2) lies*: no elliptic stabiliser ever occurs on the source, so no averaging idempotent and no invertibility of 4 or 6 is needed — the same remark as in Lemma 14.10(1).

(V) and the identity on vertices. $(j\tau_\partial - \tau j)(x)$ is an integer 0-chain of augmentation 0, and $\varepsilon\colon H_0(X) \to \mathbb{Z}$ is an isomorphism (37), so it is ∂ of an integer 1-chain; this is the statement entry A.5 corrects — $\tau(jx)$ is a 0-chain of augmentation 1 and not, in general, a single vertex, and the argument never needs it to be one. Once $\partial\eta(x) = (j\tau_\partial - \tau j)(x)$ holds on representatives it holds on every boundary vertex, because both sides satisfy (T1) (the right-hand side by (T1) of τ, of τ_∂, and equivariance of j) and agree on representatives. This is (42) in degree 0, where $\eta\partial = 0$.

(E). Using (T2) for τ and τ_∂, that j is a chain map, and the degree-0 identity just proved,

$$\partial z_e = (j\tau_\partial - \tau j)(\partial e) - \partial\eta(\partial e) = (j\tau_\partial - \tau j)(\partial e) - (j\tau_\partial - \tau j)(\partial e) = 0,$$

so $z_e \in Z_1(X) = B_1(X)$ and a 2-chain with boundary z_e exists; it is unique because two such differ by an element of $Z_2(X) = 0$ (37). Both facts hold over $\mathbb{Z}$, so the fill is an integer 2-chain. Then $\partial\eta(e) + \eta(\partial e) = z_e + \eta(\partial e) = (j\tau_\partial - \tau j)(e)$ on representatives, hence on all boundary edges by (T1) of both sides: this is (42) in degree 1. Degrees ≥ 2 are vacuous by (B2), and additivity is by construction. □

Lemma 15.7 (η is unique up to $\partial\Lambda - \Lambda\partial$, Λ unique)**.** *[proved here] [formally verified: [GT, `lem:etaunique`]] [formally verified: [GT, `cor:model7-etaexists`]] (on the seven-vertex model with its horocycle boundary, every $g \in \mathrm{GL}_2(\mathbb{Q})$: (H4) and (B1)–(B3) discharged in [GT]) Assume (H4) and (B1)–(B3), let R be a commutative ring, and let η, η' be two pair homotopies for the same (τ, τ_∂) over R. Then there is a* ***unique*** *additive map of degree $+2$*

$$\Lambda\colon C_\bullet(\partial X) \otimes R \longrightarrow C_{\bullet+2}(X) \otimes R \qquad \text{with} \qquad \eta' - \eta \;=\; \partial\Lambda - \Lambda\partial,$$

and it satisfies (T1) automatically. Explicitly Λ is non-zero only on $C_0(\partial X)$, where $\Lambda(x)$ is the unique 2-chain with $\partial\Lambda(x) = (\eta' - \eta)(x)$; in degree 1 the identity reads $(\eta' - \eta)(e) = -\Lambda(\partial e)$ and is forced. *Over $R = \mathbb{Z}$ this holds with integer chains. The hypotheses used are (H4)(ii) through (37) — for the existence and the uniqueness of the fills — and (B2) for the vanishing of $C_2(\partial X)$; (B3) is* not *needed here, because uniqueness makes (T1) automatic.*

Proof. Put $\xi := \eta' - \eta$. Subtracting the two instances of (42) gives $\partial\xi + \xi\partial = 0$, and ξ has (T1) as a difference of two maps with (T1).

Degree 0. For a boundary vertex x, $\partial\xi(x) = -\xi(\partial x) = 0$, so $\xi(x) \in Z_1(X) = B_1(X)$ (37): there is a 2-chain $\Lambda(x)$ with $\partial\Lambda(x) = \xi(x)$, and it is unique because $Z_2(X) = 0$. Define Λ on *every* boundary vertex this way (not only on representatives) and extend additively to $C_0(\partial X)$. (T1): for $h \in \Gamma''$, $\partial\big((g^{-1}hg)\Lambda(x)\big) = (g^{-1}hg)\xi(x) = \xi(hx) = \partial\Lambda(hx)$, so $(g^{-1}hg)\Lambda(x) = \Lambda(hx)$ by uniqueness. No stabiliser argument is needed.

Degree 1. For a boundary edge e, $\partial\big(\xi(e) + \Lambda(\partial e)\big) = \partial\xi(e) + \xi(\partial e) = 0$, so $\xi(e) + \Lambda(\partial e) \in Z_2(X) = 0$, i.e. $\xi(e) = -\Lambda(\partial e)$. Since $C_3(X) = 0$ the only possible value of Λ on $C_1(\partial X)$ is 0, and with it $(\partial\Lambda - \Lambda\partial)(e) = -\Lambda(\partial e) = \xi(e)$.

Degrees ≥ 2. $C_q(\partial X) = 0$ ((B2)). So $\xi = \partial\Lambda - \Lambda\partial$. *Uniqueness of* Λ. Any Λ' with $\xi = \partial\Lambda' - \Lambda'\partial$ has $\partial\Lambda'(x) = \xi(x)$ on vertices, hence $\Lambda'(x) = \Lambda(x)$ by $Z_2(X) = 0$, and $\Lambda' = 0 = \Lambda$ in degree 1. □

15.3 The derived homotopy and the cone action

The next lemma is pure algebra about the cone and is stated for an *arbitrary* degree-(-1) map h; it is where the sign of the cochain identity comes from, and it is the only place it is asserted.

Lemma 15.8 (The sign of $D\Phi_g - \Phi_g D$)**.** *[proved here] (Appendix A, entry A.13.) Assume (H4), (B1)–(B3), let R be a commutative ring, let $\Theta\colon C^\bullet_{\mathrm{eq}}(K,\mathcal{R};L) \to C^\bullet_{\mathrm{eq}}(K,\mathcal{R};L)$ and $\Theta^\partial\colon C^\bullet_\partial \to C^\bullet_\partial$ be cochain maps, and let $h\colon C^q_{\mathrm{eq}}(K,\mathcal{R};L) \to C^{q-1}_\partial$ $(q \in \mathbb{Z})$ be any R-linear maps. Define $\Phi\colon C(r)^q \to C(r)^q$ by $\Phi(c, c_\partial) := (\Theta c,\ \Theta^\partial c_\partial + hc)$. Then*

$$(D\Phi - \Phi D)(c, c_\partial) \;=\; \big(0,\ (dh + hd + r\Theta - \Theta^\partial r)\, c\big), \tag{43}$$

so Φ is a cochain map ***iff*** *$dh + hd = \Theta^\partial r - r\Theta$. Moreover $\Phi \circ \mathrm{inc}_c = \mathrm{inc}_c \circ \Theta^\partial$ and $\mathrm{pr}_c \circ \Phi = \Theta \circ \mathrm{pr}_c$ hold for every h.*

Proof. $D\Phi(c, c_\partial) = D(\Theta c, \Theta^\partial c_\partial + hc) = \big(-d\Theta c,\ d\Theta^\partial c_\partial + dhc + r\Theta c\big)$ and $\Phi D(c, c_\partial) = \Phi(-dc, dc_\partial + rc) = \big(-\Theta dc,\ \Theta^\partial dc_\partial + \Theta^\partial rc - hdc\big)$. Subtracting, and using $d\Theta = \Theta d$, $d\Theta^\partial = \Theta^\partial d$, gives (43). The two intertwinings are read off the definitions: $\Phi(0, c_\partial) = (0, \Theta^\partial c_\partial)$ and $\mathrm{pr}_c\Phi(c, c_\partial) = \Theta c$. □

This is the sign three independent derivations agree on — the implementation plan v1.2, the OQ2 verdict §2.1 and the OQ2 revision audit B.1 — and it was re-derived for this chapter from (38) alone; the plan audit's $dh + hd = r\Theta - \Theta^\partial r$ is entry A.13, and nothing downstream of it carried the wrong orientation.

Definition 15.9 (The derived cochain homotopy $h := \widetilde{\eta}$, and Φ_g)**.** Assume (H4) and (B1)–(B3), let R be a commutative ring, and let $(\tau, \tau_\partial, \eta)$ be a Hecke datum of the pair (Definition 15.5), with $\Theta_g = \widetilde{\tau}$ and $\Theta_g^\partial = \widetilde{\tau_\partial}$ the cochain maps of Definitions 11.6/12.10 and 14.14 built from the same g, s_j, δ_j. The **derived cochain homotopy** is the coset transport of η,

$$h := \widetilde{\eta}\colon C^q_{\mathrm{eq}}(K,\mathcal{R};L) \longrightarrow C^{q-1}_\partial, \qquad (h\phi)(y) = \sum_{j=1}^{m} \rho_n(\delta_j)\, \phi\big(\eta(s_j \cdot y)\big), \quad y \in C_{q-1}(\partial X), \tag{44}$$

and the **cone cochain map** is

$$\Phi_g\colon C(r)^q \to C(r)^q, \qquad \Phi_g(c, c_\partial) := \big(\Theta_g\, c,\ \Theta_g^\partial\, c_\partial + h\, c\big). \tag{45}$$

The operator $T^c_{\Gamma g\Gamma}$ on $H^q_c = H^{q-1}(C(r))$ is the map induced by Φ_g (Theorem 15.11(a) says it is one), and its matrix $A^{(q)}_c$ in the stored basis of H^q_c is assembled by Algorithm 12.13 run on the complex $C(r)$ in place of $C^\bullet_{\mathrm{eq}}$ — the stored cocycle basis of $Z^{q-1}(C(r))$, the projector modulo $B^{q-1}(C(r))$, and (45) applied columnwise — with the same column convention. For the shipped model h is non-zero in cochain degrees 1 and 2 only: $h^1\colon C^1_{\mathrm{eq}} \to C^0_\partial$ is the transport of η on vertices and $h^2\colon C^2_{\mathrm{eq}} \to C^1_\partial$ of η on edges, and $h^0 = h^3 = 0$ because ∂X is one-dimensional ((B2)). h **is never chosen at cochain level**; the only choices are the chain-level tables, and Proposition 15.12 says what they cost.

Lemma 15.10 (The derived homotopy satisfies the cone identity)**.** *[proved here] Under the hypotheses of Definition 15.9, and with no hypothesis on ℓ:*

$$dh + hd \;=\; \Theta^\partial_g\, r - r\,\Theta_g \qquad \text{on } C^q_{\mathrm{eq}}(K,\mathcal{R};L),\ \text{for every } q.$$

Proof. Apply Lemma 15.4 to (42). By (2), $\widetilde{\partial\eta}\phi = \widetilde{\eta}(d\phi) = hd\phi$ and $\widetilde{\eta\partial}\phi = d(\widetilde{\eta}\phi) = dh\phi$; by (3), $\widetilde{j\tau_\partial}\phi = \widetilde{\tau_\partial}(r\phi) = \Theta^\partial_g r\phi$; by (4), $\widetilde{\tau j}\phi = r(\widetilde{\tau}\phi) = r\Theta_g\phi$. Coset transport is additive in the map (Lemma 15.4(1)), so transporting both sides of $\partial\eta+\eta\partial = j\tau_\partial - \tau j$ gives $hd+dh = \Theta^\partial_g r - r\Theta_g$. □

Theorem 15.11 (Well-definedness of the cone action)**.** *Assume (H4) and (B1)–(B3), let R be a commutative ring, and let $(\tau,\tau_\partial,\eta)$ be a Hecke datum of the pair over $\mathbb{Z}$, with Φ_g as in (45) over R.*

*(a) [proved here] **(No hypothesis on ℓ or n.)** [formally verified: [GT, `thm:conewelldef`]] Φ_g is a cochain map of complexes of R-modules, $\Phi_g\mathrm{inc}_c = \mathrm{inc}_c\Theta^\partial_g$ and $\mathrm{pr}_c\Phi_g = \Theta_g\mathrm{pr}_c$. Hence Φ_g induces $T^c_{\Gamma g\Gamma}$ on H^q_c for every q, and **every map of (39) intertwines** $(T^\partial_{\Gamma g\Gamma}, T^c_{\Gamma g\Gamma}, T_{\Gamma g\Gamma})$: pr_{c*}, inc_{c*} and r^q commute with the three operators. This is not a separate obligation; it is a consequence of (a).*

*(b) [proved here] [formally verified: [GT, `thm:conewelldef`]] [formally verified: [GT, `cor:model7-etaexists`]] (on the seven-vertex model with its horocycle boundary, every $g \in \mathrm{GL}_2(\mathbb{Q})$: (H4) and (B1)–(B3) discharged in [GT]) **(Independence of η — over $\mathbb{Z}$, for every ℓ.)** Two pair homotopies η,η' for the same (τ,τ_∂) give cone-homotopic Φ_g: $\Phi^{\eta'}_g - \Phi^\eta_g = D\Upsilon + \Upsilon D$ for the integral, filtration-preserving homotopy $\Upsilon(c,c_\partial) = (0, -\widetilde{\Lambda}\,c)$ of Proposition 15.12(1). Hence $T^c_{\Gamma g\Gamma}$ on $H^\bullet_c$, and every map it induces on (39), is independent of η.*

*(c) **(For $6 \in R^\times$ — hypothesis (H3).)** [proved here] $T^c_{\Gamma g\Gamma}$ is independent of τ and τ_∂, of the transporters s_j (in the sense of Theorem 11.7(a): at cochain level with the tables held fixed, on cohomology in general) and of g within $\Gamma g\Gamma$ (Proposition 15.12(2)–(4)); and under the identification of Lemma 15.2(3) it is the operator $\mathrm{cor}\circ(\mathrm{conj}_g)_*\circ\mathrm{res}$ on $H^\bullet\big(\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X,\partial X), V_R)\big)$ induced by any* strict *pair (Proposition 15.13), which exists; the identification of* res, cor, conj *on the relative complex is that of Theorem 11.10, on the same external inputs ([AS86, §1], [Bro82, Ch. III.9–10]).*

*(c′) **(For $6 \in R^\times$; the two consequences that reach outside this document, with their status fixed in Remark 15.14.)** The **Hecke relations of Theorem 10.8 hold on $H^\bullet_c$ on the nose**, in the same normalisation and with the same full-versus-primitive caveat — [proved here] through the Steinberg-module form of Remark 15.14(ii), on the same external inputs as Theorem 10.8. The operator of (c) is the classical Hecke correspondence on the relative cohomology of the Borel–Serre pair, i.e. on the compactly supported cohomology of the open orbifold — [standard: [Har87, §§1–2] and [BS73], neither of whose section numbers is verified here (Remark 15.14(iii))]. T_p on H^1_c is adjoint to T_p on $H^1_{\mathrm{eq}}(K,\mathcal{R};D_n)$ under the pairing of Proposition 8.6 — [open] here, being Claim 16.26's (Remark 15.14(iv), Proposition 15.20(3)).*

*(d) [proved here] **(Strictness.)** [formally verified: [GT, `thm:conewelldef`]] $\eta \equiv 0$ is admissible iff $\tau j = j\tau_\partial$ ("τ is a map of pairs"); then $h = 0$ and $\Phi_g = (\Theta_g, \Theta^\partial_g)$. **The cochain identity $r\Theta_g = \Theta^\partial_g r$ alone does not license $h = 0$** (Remark 15.15).*

(e) ***(For $6 \notin R^\times$, on the complexes of this model.)*** *(a), (b) and (d) hold as stated; $T^{\mathrm{c}}_{\Gamma g\Gamma}$ is* defined *as the output of the pipeline run with the canonical $(\tau, \tau_\partial, \eta)$, and (c) is not asserted — exactly as for $T_{\Gamma g\Gamma}$ itself in the contract box after Lemma 11.3 (Remark 15.19).*

Proof. (a) Lemma 15.8 with $\Theta = \Theta_g$, $\Theta^\partial = \Theta^\partial_g$ (cochain maps by Theorem 12.11(2) and Proposition 14.15(1)) and h of (44): the bracket in (43) vanishes by Lemma 15.10, so Φ_g is a cochain map; it is R-linear because Θ_g, Θ^∂_g, h are. The two intertwinings are the last sentence of Lemma 15.8, so $(\Theta^\partial_g, \Phi_g, \Theta_g)$ is a map of the short exact sequence of Lemma 15.2(2) to itself, and a map of short exact sequences of complexes induces a map of long exact sequences [standard: [Wei94, Thm. 1.3.1], naturality of the connecting homomorphism].

(b), (c), (c′) are Proposition 15.12, Proposition 15.13 and Remark 15.14. (d): if $\tau j = j\tau_\partial$ then $\eta = 0$ satisfies (42) and (T1); conversely if $\eta = 0$ is a pair homotopy then (42) reads $0 = j\tau_\partial - \tau j$. With $\eta = 0$, (44) gives $h = 0$. (e): (a), (b), (d) used no hypothesis on ℓ; (c) used Lemma 11.3(2), which is unavailable there. □

Proposition 15.12 (Independence of η over $\mathbb{Z}$; of $\tau, \tau_\partial, s_j, g$ for $6 \in R^\times$)**.** *Assume (H4) and (B1)–(B3), and let R be a commutative ring.*

1. *[proved here] (η; **over $\mathbb{Z}$, every ℓ.**) [formally verified: [GT, `prop:coneindep`]] Let η, η' be pair homotopies for (τ, τ_∂) and Λ the unique map of Lemma 15.7, so $\eta' - \eta = \partial\Lambda - \Lambda\partial$. Then $h' - h = \widetilde{\Lambda}\, d - d\,\widetilde{\Lambda}$, where $\widetilde{\Lambda}\colon C^q_{\mathrm{eq}} \to C^{q-2}_\partial$ is the coset transport of Λ, and with $\Upsilon\colon C(r)^q \to C(r)^{q-1}$, $\Upsilon(c, c_\partial) := (0, -\widetilde{\Lambda}c)$,*

$$\Phi^{\eta'}_g - \Phi^{\eta}_g \;=\; D\Upsilon + \Upsilon D. \tag{46}$$

   Υ is R-linear, is defined over $\mathbb{Z}$ when the tables are, and is non-zero only on $C(r)^1$.
2. *[proved here] (τ **and** τ_∂**; needs $6 \in R^\times$ for the bulk half only.**) Let $\tau' = \tau + \partial H + H\partial$ with H a degree-$(+1)$ map with (T1) (Lemma 11.3(2): any two τ are so related when $6 \in R^\times$), and $\tau'_\partial = \tau_\partial + \partial N_\partial + N_\partial\partial$ with N_∂ as in Lemma 14.10(3) (any two τ_∂ are so related over $\mathbb{Z}$, for every ℓ). Then $\eta' := \eta + jN_\partial - Hj$ is a pair homotopy for (τ', τ'_∂), and with $\Upsilon(c, c_\partial) := (-\widetilde{H}c,\ \widetilde{N_\partial}c_\partial)$,*

$$\Phi^{(\tau', \tau'_\partial, \eta')}_g - \Phi^{(\tau, \tau_\partial, \eta)}_g \;=\; D\Upsilon + \Upsilon D. \tag{47}$$

   Any other pair homotopy for (τ', τ'_∂) is then covered by (1). In particular the induced operator is independent of τ_∂ over $\mathbb{Z}$ for every ℓ, and of τ for $6 \in R^\times$.
3. *[proved here] (s_j**; no hypothesis on ℓ when the tables are held fixed.**) Replacing s_j by $h_j s_j$ with $h_j \in \Gamma''$ and keeping $(\tau, \tau_\partial, \eta)$ fixed leaves Θ_g, Θ^∂_g, h and hence Φ_g unchanged, summand by summand. In an admissible construction the tables are built relative to the s_j and are rebuilt with them (Theorem 11.7(a), the proviso); the induced operator is then invariant by (2), hence for $6 \in R^\times$.*
4. *[proved here] (g **within** $\Gamma g\Gamma$.) For $g_1 = \gamma_1 g \gamma_2$, $\gamma_1, \gamma_2 \in \Gamma$, the transported triple $\big(\gamma_2^{-1}\tau(\gamma_1^{-1}\cdot),\ \gamma_2^{-1}\tau_\partial(\gamma_1^{-1}\cdot),\ \gamma_2^{-1}\eta(\gamma_1^{-1}\cdot)\big)$ with transporters $s_j\gamma_1^{-1}$ is a Hecke datum of the pair for g_1 and reproduces Φ_g verbatim, with no hypothesis on ℓ; an arbitrary admissible datum at g_1 is compared with the transported one by (2) and (3), hence under $6 \in R^\times$.*

Proof. (1) By Lemma 15.4(1),(2) applied to $\partial\Lambda - \Lambda\partial$: $h' - h = \widetilde{\eta' - \eta} = \widetilde{\partial}\widetilde{\Lambda} - \widetilde{\Lambda}\widetilde{\partial} = \widetilde{\Lambda}d - d\widetilde{\Lambda}$. Now, for $(c, c_\partial) \in C(r)^q$, using Θ_g, Θ^∂_g unchanged,

$$D\Upsilon(c, c_\partial) = D(0, -\widetilde{\Lambda}c) = (0, -d\widetilde{\Lambda}c), \qquad \Upsilon D(c, c_\partial) = \Upsilon(-dc, dc_\partial + rc) = (0, \widetilde{\Lambda}dc),$$

whose sum is $(0, (\widetilde{\Lambda}d - d\widetilde{\Lambda})c) = (0, (h' - h)c) = (\Phi^{\eta'}_g - \Phi^{\eta}_g)(c, c_\partial)$. Degrees: $c \in C^{q+1}_{\mathrm{eq}}$, $\widetilde{\Lambda}c \in C^{q-1}_\partial$, so Υ lands in $C(r)^{q-1}$; Λ is non-zero only on $C_0(\partial X)$, so $\widetilde{\Lambda}$ is non-zero only on C^2_{eq} and Υ only

on $C(r)^1$. Integrality: Λ is an integer table when η, η' are (Lemma 15.7), and (41) has integral coefficients.

(2) η' has (T1) as a sum of maps with (T1) (j is equivariant). And $\partial\eta' + \eta'\partial = (j\tau_\partial - \tau j) + j(\partial N_\partial + N_\partial\partial) - (\partial H + H\partial)j = j\tau'_\partial - \tau' j$, using that j is a chain map. By Lemma 15.4, $\Theta'_g = \Theta_g + d\widetilde{H} + \widetilde{H}d$ (this is Theorem 11.7(b)), $\Theta^{\partial\prime}_g = \Theta^\partial_g + d\widetilde{N_\partial} + \widetilde{N_\partial}d$ (Proposition 14.15(1)), and by (3),(4) of that lemma $h' = h + \widetilde{N_\partial}r - r\widetilde{H}$. Then

$$D\Upsilon(c, c_\partial) = \big(d\widetilde{H}c,\ d\widetilde{N_\partial}c_\partial - r\widetilde{H}c\big), \qquad \Upsilon D(c, c_\partial) = \big(\widetilde{H}dc,\ \widetilde{N_\partial}dc_\partial + \widetilde{N_\partial}rc\big),$$

and the sum is $\big((d\widetilde{H} + \widetilde{H}d)c,\ (d\widetilde{N_\partial} + \widetilde{N_\partial}d)c_\partial + (\widetilde{N_\partial}r - r\widetilde{H})c\big)$, which is the difference of the two cone maps. Two pair homotopies for (τ', τ'_∂) are covered by (1), and cone homotopies compose. The hypothesis $6 \in R^\times$ enters only through the existence of H (Lemma 11.3(2)); N_∂ exists over $\mathbb{Z}$ for every ℓ (Lemma 14.10(3)).

(3) The computation in the proof of Theorem 11.7(a) shows that the j-th summand of (41) is unchanged under $s_j \mapsto h_j s_j$ for any F with (T1); apply it to $F = \tau, \tau_\partial, \eta$. The proviso is that of Theorem 11.7(a) and Remark 11.9, unchanged.

(4) As in the proof of Theorem 11.7(c): the transported $\tau_1, \tau_{\partial 1}$ are g_1-twisted (that computation, verbatim, on each complex), and for the transported η_1 one has $\partial\eta_1 + \eta_1\partial = \gamma_2^{-1}(j\tau_\partial - \tau j)(\gamma_1^{-1}\cdot) = j\tau_{\partial 1} - \tau_1 j$ because j is equivariant. Substituting into (41) with transporters $s_j\gamma_1^{-1}$ reproduces each of Θ_g, Θ^∂_g, h, the factors $\rho_n(\gamma_1)$, $\rho_n(\gamma_2)$ cancelling against equivariance of ϕ. □

Proposition 15.13 (Strict pairs exist over $\mathbb{Z}$, for every ℓ)**.** *[proved here] Assume (H4) and (B1)–(B3), and let τ_∂ be a boundary g-twisted chain datum over $\mathbb{Z}$ (Lemma 14.10(1)). Then there is a g-twisted chain datum τ^s over $\mathbb{Z}$ on X with*

$$\tau^s \circ j \ = \ j \circ \tau_\partial,$$

so that $\eta \equiv 0$ is a pair homotopy for (τ^s, τ_∂) and $\Phi_g = (\Theta^s_g, \Theta^\partial_g)$ preserves $(\ker r)[1] \subseteq C(r)$. It is built by the following terminating procedure, which never inverts a stabiliser order and never solves a system over anything but $\mathbb{Z}$; the only external fact it uses is that every finite subgroup of Γ of order 4 or 6 fixes a vertex of X [standard: [Ser73, Ch. VII, §1.2, Thm. 1(1),(3)]: a point of $\mathcal{H}$ with non-trivial stabiliser modulo $\pm I$ is Γ-equivalent to i, to ζ^2 or to ζ; these are the vertices v_6, v_4, v_5 of Definition 7.1, whose Γ-translates all lie in X since $\mathrm{Im} \le 1 < Y$. The pointer read "Thm. 2" until v0.16.0; Serre's Theorem 2 is the generation by S and T and is not what is used here. That such a subgroup has a fixed point at all is the step spelled out in the proof of Lemma 11.3(1): it is cyclic, and a generator is a torsion element of Γ other than $\pm I$, hence elliptic].

- *(B)* ***Boundary simplices.*** *$\tau^s(j\sigma) := j\tau_\partial(\sigma)$ for every simplex σ of ∂X. This has (T1) and (T2) because τ_∂ does and j is an equivariant chain map.*
- *(V)* ***Interior vertices.*** *For each Γ''-orbit representative vertex v of X not in ∂X: the twisted stabiliser $g^{-1}\Gamma''_v g$ is a finite subgroup of Γ (Proposition 10.1(3)); choose $\tau^s(v) := w$, a vertex of X fixed by it — any vertex if the group is $\{\pm I\}$, the fixed vertex supplied by the cited fact otherwise. Extend by (T1).*
- *(E)* ***Interior edges.*** *For each Γ''-orbit representative edge e of X not in ∂X: $\tau^s(\partial e)$ is a 0-chain of augmentation 0, hence a boundary (37); choose any 1-chain with that boundary. Extend by (T1); this is well defined because the stabiliser of an edge is $\{\pm I\}$ ((H4)(iv)).*
- *(F)* ***Triangles.*** *For each Γ''-orbit representative triangle σ: $\tau^s(\partial\sigma)$ is a 1-cycle, hence the boundary of a* unique *2-chain (37); take it. Extend by (T1), well defined since the stabiliser is $\{\pm I\}$.*

Proof. (B) is well defined on the Γ''-stable subcomplex ∂X and gives (T1), (T2), (T3) there. (V): the (T1)-extension $\tau^s(hv) := (g^{-1}hg)w$ is well defined iff w is fixed by $g^{-1}\Gamma''_v g$, which is how w was chosen; if Γ''_v has order 4 or 6 then so does its conjugate $g^{-1}\Gamma''_v g \subseteq \Gamma$, which by the cited fact fixes a point of $\Gamma \cdot i \cup \Gamma \cdot \zeta$, a vertex of X by (H4)(i),(iv). (T3) holds since $\tau^s(v)$ is

one vertex. (E), (F): (T2) is imposed on representatives and propagates to orbits because both sides have (T1); existence of the fills is (37) over $\mathbb{Z}$, exactly as in Lemma 15.6, and uniqueness in (F) is $Z_2(X) = 0$. (T1) for the extension needs the stabiliser condition, vacuous for $\{\pm I\}$ by (H4)(iv). Termination: finitely many Γ''-orbits ((H4)(iii), $[\Gamma : \Gamma''] < \infty$). The consequences: $\eta = 0$ satisfies (42) for (τ^s, τ_∂); $h = 0$; and for $c \in \ker r$, $r\Theta_g^s c = \widetilde{\tau^s} j\, c = \widetilde{j\tau_\partial}\, c = \Theta_g^\partial r c = 0$ by Lemma 15.4(3),(4), so Θ_g^s preserves $\ker r$ and $\Phi_g(c, 0) = (\Theta_g^s c, 0)$. □

Proposition 15.13 contains, as its bulk half, exactly the statement Lemma 11.3(1) makes: existence of a g-twisted chain datum over $\mathbb{Z}$ with no hypothesis on ℓ, by a construction. (Historically it came first; that steps (V), (E), (F) prove existence on their own, with step (B) dropped and hence without (B1)–(B3), is what Lemma 11.3(1) records. Neither proof uses the other, so there is no circularity.) What this proposition adds is *strictness* — the boundary slots preassigned from τ_∂, so that $\eta \equiv 0$ is admissible. What neither does is repair the *uniqueness* half Lemma 11.3(2), which needs $6 \in R^\times$ and is used in Proposition 15.12(2); see Remark 11.4. And the shipped τ of Algorithm 18.3 is *not* built this way (Remark 15.15).

Remark 15.14 (The relative complex, the Steinberg module, and where the Hecke relations come from). This remark fixes the status of the assertions of Theorem 15.11(c′) that reach outside this document, and gives the algebraic form under which one of them becomes provable on inputs this document already uses. Assume (H4), (B1)–(B3) and $6 \in R^\times$ throughout.

(i) The operator on the relative complex is canonical. [proved here] For a strict pair (τ^s, τ_∂) the cone operator is, by Lemma 15.2(3) and Proposition 15.13, the map induced by $\Theta_g^s|_{\ker r}$ on $H^\bullet(\ker r) = H^\bullet(\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X, \partial X), V_R))$, and $\Theta_g^s|_{\ker r}$ is $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ on that complex with $(\mathrm{conj}_g)_*$ realised by $\psi \mapsto \rho_n(g) \circ \psi \circ \tau^s_{\mathrm{rel}}$, τ^s_{rel} the map induced on $C_\bullet(X, \partial X)$ — the proof sketch of Theorem 11.10 verbatim, on the relative complex, which for $6 \in R^\times$ is again a complex of relatively projective $R[\Gamma]$-modules (its terms are the permutation modules on the interior simplices, with the same stabilisers). Two strict pairs induce *homotopic* τ_{rel} over Γ'': with H, N_∂ as in Proposition 15.12(2), the map $Hj - jN_\partial$ satisfies $\partial(Hj - jN_\partial) + (Hj - jN_\partial)\partial = 0$, so by Lemma 15.7 (applied to the two pair homotopies 0 and $jN_\partial - Hj$ for $(\tau^{s\prime}, \tau'_\partial)$) it equals $\partial\Lambda - \Lambda\partial$ for a unique Λ; extending Λ by zero on interior vertices to $\widehat{\Lambda}\colon C_0(X) \to C_2(X)$ (a (T1)-map, the boundary vertex set being Γ''-stable) and replacing H by $H' := H - (\partial\widehat{\Lambda} - \widehat{\Lambda}\partial)$ gives a twisted homotopy $\tau^{s\prime} - \tau^s = \partial H' + H'\partial$ with $H'j = jN_\partial$, i.e. a homotopy *of pairs*, which descends to the relative complex. So the homotopy class of τ_{rel}, and with it $(\mathrm{conj}_g)_*$, does not depend on the strict pair; and by Proposition 15.12 every admissible triple gives the same operator on $H_c^\bullet$ as every strict pair.

(ii) The Steinberg form. [proved here] (sketch; the derived-category bookkeeping is routine and is the part a formalisation would have to supply). X is contractible and ∂X is a disjoint union of contractible components indexed by $\mathbb{P}^1(\mathbb{Q})$ ((H4)(ii), (B2)), so the long exact homology sequence of the pair gives $H_q(X, \partial X) = 0$ for $q \neq 1$ and $H_1(X, \partial X) \cong \ker\big(\varepsilon\colon \mathbb{Z}[\mathbb{P}^1(\mathbb{Q})] \to \mathbb{Z}\big) =: \mathrm{St}$, the Steinberg module of $\mathrm{GL}_2(\mathbb{Q})$, as $\mathbb{Z}[\Gamma]$-modules; hence $C_\bullet(X, \partial X) \simeq \mathrm{St}[1]$ in the derived category of $\mathbb{Z}[\Gamma]$-modules. Since $C_\bullet(X, \partial X) \otimes R$ is a bounded complex of relatively projective modules and St is $\mathbb{Z}$-free,

$$H_c^q \;\cong\; H^q\big(\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X, \partial X), V_R)\big) \;\cong\; \mathrm{Ext}^q_{\mathbb{Z}[\Gamma]}\big(\mathrm{St}[1], V_R\big) \;\cong\; H^{q-1}\big(\Gamma; \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R)\big) \qquad (6 \in R^\times),$$

[standard: [Bro82, Ch. VII.7] for equivariant cohomology of a Γ-complex with invertible stabiliser orders being computed by $\mathrm{Hom}_\Gamma(C_\bullet, -)$, as in Remark 7.6; [Wei94, §§10.7, 5.7] for hypercohomology and Ext of a complex]. The monoid Δ acts on $\mathbb{P}^1(\mathbb{Q})$ by Möbius transformations and hence on St, and $\mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R)$ is a Δ-module via $(\delta \cdot f)(s) := \rho_n(\delta) f(\delta^{-1} s)$, δ^{-1} acting through $\mathrm{GL}_2(\mathbb{Q})$. Under the displayed isomorphisms, a twisted chain map of pairs induces on $H_1 = \mathrm{St}$ the map $s \mapsto g^{-1}s$ (Lemma 14.9: the component map of τ_∂ is $c \mapsto g^{-1}c$), so $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ on the relative complex corresponds to the double-coset operator of $\Gamma g \Gamma$ on $H^\bullet(\Gamma; \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R))$ in the sense of [AS86, §1]. **Consequently the Hecke relations of Theorem 10.8 hold for**

$T^c_{\Gamma g\Gamma}$ **on** $H^\bullet_c$, by the same argument and on the same external inputs as that theorem [standard: [Shi71, Thm. 3.24]; [AS86, §1]], since those inputs concern $H^\bullet(\Gamma; M)$ for an *arbitrary* Δ-module M and $\mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R)$ is one. This is the sense in which "the Hecke relations hold on $H^\bullet_c$ on the nose" is a theorem here and not a citation of geometry.

(iii) The geometric identification. That $H^\bullet(\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X, \partial X), V_R))$ is the cohomology of the Borel–Serre pair $(\Gamma\backslash\overline{\mathcal{H}}, \partial)$ with coefficients in the local system of V_R, that this is the compactly supported cohomology of the open orbifold $\Gamma\backslash\mathcal{H}$, and that the operator of (i) is the action of the Hecke correspondence on that pair, is [standard: [Har87, §§1–2], where the long exact sequence $H^\bullet_c \to H^\bullet \to H^\bullet(\partial)$ and the Hecke action on all three terms are set up for GL_2 over a number field, the case here being $\mathbb{Q}$; [BS73] for the bordification] — *with the caveat*, per rule (A2), that this document has not verified the section numbers of [BS73] — nor those of [Har87] — against the text, and cites both works as a whole; where a precise pointer is required this half is to be read as [conjectural] *as a citation*. (The *bibliographic* data of [Har87] *have* been checked, in v0.16.0: the entry had carried a wrong title until then.) Nothing in this document depends on (iii): every statement made about $H^\bullet_c$ is proved on the complexes of Definition 15.1, and (iii) only names what they compute.

(iv) Poincaré duality. The adjointness of T_p on $H^1_c(V_R)$ and on $H^1_{\mathrm{eq}}(K, \mathcal{R}; D_n)$ under a perfect pairing $H^1_c \times H^1_{\mathrm{eq}}(D_n) \to H^2_c(R) \cong R$ is Claim 16.26, a stub of §16 at this version; its consequence for this chapter, Proposition 15.20(3), is stated conditionally on it and carries [open] until it is written.

Remark 15.15 ($\eta \equiv 0$ iff $\tau j = j\tau_\partial$; the entrywise identity decides nothing). [proved here] (Rule (A6).) Theorem 15.11(d) says when $h = 0$ is admissible: exactly when τ is a map of pairs. Two things must be kept apart from it.

1. **The shipped τ is not a map of pairs.** [verified computationally] Verified for this chapter on the emitted table `docs/implementation/tau_table_p2.txt` ($p = 2$, $g = \mathrm{diag}(1, 2)$, $m = 3$, every fill over $\mathbb{Z}$), reading its stored slots against the vertex and edge lists of Definition 7.1. *On the canonical τ since v1.1.0* (policy `GeometricNearest`, the ι-symmetric re-baseline; the counts re-read at v1.4.2 by slot-2 WP-7's cusp-support walk, `hecke::cusp_support_walk`, and confirmed by hand on the table): of the 12 stored slots at boundary vertices $v_0, \ldots, v_3$, 10 map to an interior vertex — every j at v_0 and v_3 (all to $v_6 = i$; e.g. $\tau(s_1\widetilde{v_0}) = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix} \cdot \widetilde{v_6}$, `:13`) and $j \in \{2, 3\}$ at v_1, v_2 (to $v_4 = \zeta^2$ and v_6), the two $j = 1$ slots at v_1, v_2 staying on ∂X (both to v_0); of the 9 stored slots at boundary edges, 5 carry interior edges (`:34`, `:35`, `:40`, `:49`, `:50` — the $j = 1$ slots at all three boundary edges carry $\{v_0, v_4\}$ and $\{v_4, v_6\}$, the $j = 2$ slots at $\{v_0, v_1\}$ and $\{v_2, v_3\}$ carry $\{v_4, v_6\}$), the other four being the zero chain. (v1.3.3–v1.4.1 gave 8 of 12 and 2 of 9 with "the $j = 1$ slots all stay on ∂X": the pre-v1.1.0 table.) A different statistic on the same table is how many slots carry a *non-zero* η once the η-table exists (v1.4.2, slot-2 WP-8): 11 of the 12 vertex slots and 6 of the 9 edge slots — more than leave the boundary, because $\eta(s_j\sigma) \neq 0$ also where $\tau(s_j\sigma)$ stays on ∂X but differs from $j\tau_\partial(s_j\sigma)$. Both counts are ℓ- and n-independent. So $\tau j \neq j\tau_\partial$ for *every* τ_∂, $\eta \equiv 0$ is unavailable, and the η-table is genuinely needed. This is a property of Algorithm 18.3's nearest-vertex policy (step 1), not of the model: Proposition 15.13 builds a strict τ^s on the same model.
2. **$r\Theta_g = \Theta^\partial_g r$ on cochains does not license $h = 0$.** Suppose the identity holds entrywise for some (τ, τ_∂) with $\tau j \neq j\tau_\partial$. Then $h = \widetilde{\eta}$ satisfies $dh + hd = 0$ (Lemma 15.10), i.e. h is a degree-(-1) cochain map, and by the computation of Proposition 15.16(1), $\Phi^*_g - (\Theta_g, \Theta^\partial_g)_* = \mathrm{inc}_{c*} \circ h_* \circ \mathrm{pr}_{c*}$ on $H^\bullet_c$, an element of $\mathrm{Hom}(\ker r^{q+1}, \mathrm{coker}\, r^q)$ that *need not vanish*. Whether it vanishes for some actual (τ, τ_∂) with the identity is [conjectural] either way and is not decided here; what is [proved here] is that the identity does not imply it. The correct rule is (d): $h = 0$ is admissible iff the η-table is identically zero, i.e. iff $\tau j = j\tau_\partial$ *exactly*. The quantities `eta_zero` := ($\eta \equiv 0$) and `cochain_commutes`

$:= (r\Theta_g = \Theta_g^{\partial} r$ entrywise) are therefore **recorded diagnostics and gate nothing**; `eta_zero` implies `cochain_commutes`, not conversely.

Proposition 15.16 (Why a free cochain-level h will not do: the $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0)$ ambiguity)**.** *(Appendix A, entries A.2 and A.3: the realisation is stated with Z^0, never B^0, and* over a field*.) Assume (H4), (B1)–(B3), let Θ_g, Θ_g^{∂} be as in Definition 15.9, and let $h, h' \colon C^{\bullet}_{\mathrm{eq}} \to C^{\bullet-1}_{\partial}$ be* any *two R-linear maps with $dh + hd = dh' + h'd = \Theta_g^{\partial} r - r\Theta_g$, so that both $\Phi := (\Theta_g, \Theta_g^{\partial} + h)$ and $\Phi' := (\Theta_g, \Theta_g^{\partial} + h')$ are cochain maps of $C(r)$ (Lemma 15.8).*

1. *[proved here] **(The ambiguity, over any commutative R.)** $\xi := h' - h$ is a cochain map of degree -1 ($d\xi + \xi d = 0$), and on $H_c^{q+1} = H^q(C(r))$*

$$\Phi'_* - \Phi_* \;=\; \mathrm{inc}_{c*} \circ \xi_* \circ \mathrm{pr}_{c*} \;:\; H_c^{q+1} \twoheadrightarrow \ker r^{q+1} \xrightarrow{\xi_*} H^q(C^{\bullet}_{\partial}) \twoheadrightarrow \mathrm{coker}\, r^q \hookrightarrow H_c^{q+1},$$

   an element of $\mathrm{Hom}_R(\ker r^{q+1}, \mathrm{coker}\, r^q)$; for $H_c^1 = X_3$ this is $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0)$.
2. *[proved here] **(Realisation — over a field only, and inside $C^{\bullet}_{\mathrm{eq}}$.)** [formally verified: [GT, `prop:coneambiguity`]] (over a field; not over $\mathbb{Z}/\ell^e$) Let $R = \mathbb{F}$ be a* field*, and read every cochain space of the construction below as the **equivariant** one: $C^1 := C^1_{\mathrm{eq}}(K, \mathcal{R}; L)$, $B^1 := d\,C^0_{\mathrm{eq}}$, $Z^1 := Z^1_{\mathrm{eq}}$ (v1.6.0, stated explicitly; `docs/implementation/slot3/WP-6_report.md` finding 3 — on the contractible X the flat reading is* vacuous*, the flat complex $C^{\bullet}(K; V_R)$ being acyclic in degree 1 (37), so that the flat B^1 would already contain every flat 1-cocycle and a ξ annihilating it would be zero on every cocycle). For every $\mathbb{F}$-linear $\lambda \colon H^1_{\mathrm{eq}}(K, \mathcal{R}; L) \to Z^0(C^{\bullet}_{\partial}) = H^0(C^{\bullet}_{\partial})$ there is a degree-(-1) cochain map ξ of $C^{\bullet}_{\mathrm{eq}}$ with $\xi_* = \lambda$; hence for every $\mu \in \mathrm{Hom}_{\mathbb{F}}(\ker r^1, \mathrm{coker}\, r^0)$ there is a h' satisfying the cone identity with $\Phi'_* - \Phi_* = \mu$ on H_c^1. **The statement is not claimed over $\mathbb{Z}/\ell^e$ or $\mathbb{Z}/m$,** where the splittings used below need not exist (entry A.3).*
3. ***(Consequence.)** For a freely chosen cochain-level h the operator on X_3 is undetermined to the full extent of $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0)$, which is non-zero in real cases (Remark 15.17); X_3 would not be a Hecke module. For the* derived *$h = \widetilde{\eta}$ the ambiguity vanishes identically, over $\mathbb{Z}$, for every ℓ (Proposition 15.12(1)): the difference of two derived homotopies is not merely a chain map but a cone* homotopy*. That is the whole reason the datum is a chain-level η and not a cochain-level h.*

Proof. (1) $d\xi + \xi d = 0$ by subtraction. Let $(c, c_{\partial}) \in Z^q(C(r))$, i.e. $dc = 0$ and $dc_{\partial} = -rc$; then $[c] \in \ker r^{q+1}$ (since rc is a coboundary) and $\mathrm{pr}_{c*}[(c, c_{\partial})] = [c]$; the image of pr_{c*} is $\ker r^{q+1}$ and the kernel of inc_{c*} is $\mathrm{im}\, r^q$ by exactness of (39). Now $(\Phi' - \Phi)(c, c_{\partial}) = (0, \xi c)$, and $d(\xi c) = -\xi(dc) = 0$, so $[(0, \xi c)] = \mathrm{inc}_{c*}[\xi c] = \mathrm{inc}_{c*}\xi_*[c]$.

(2) Over a field choose complements *inside the equivariant complex*: $C^1_{\mathrm{eq}} = B^1_{\mathrm{eq}} \oplus \widetilde{H}^1 \oplus W^1$ with $\widetilde{H}^1 \subseteq Z^1_{\mathrm{eq}}$ mapping isomorphically onto H^1_{eq} and $d \colon W^1 \to B^2_{\mathrm{eq}}$ an isomorphism. Define ξ on C^1_{eq} by $\xi := \lambda$ on $\widetilde{H}^1$ (through $\widetilde{H}^1 \cong H^1_{\mathrm{eq}}$) and $\xi := 0$ on $B^1_{\mathrm{eq}} \oplus W^1$; define $\xi := 0$ on C^q_{eq} for $q \neq 1$. Then $d\xi = 0$ because ξ takes values in $Z^0(C^{\bullet}_{\partial})$ — note $Z^0 \subseteq C^0_{\partial}$ is the kernel of d and is in general a proper subspace (entry A.2), and it equals $H^0(C^{\bullet}_{\partial})$ because there are no 0-coboundaries; and $\xi d = 0$ because $d\,C^0_{\mathrm{eq}} = B^1_{\mathrm{eq}} \subseteq \ker \xi$ and ξ vanishes on C^2_{eq}. So $d\xi + \xi d = 0$ and $\xi_* = \lambda$ on H^1_{eq}. Given μ, extend it to $\lambda \colon H^1_{\mathrm{eq}} \to H^0(C^{\bullet}_{\partial})$ by choosing a complement of $\ker r^1$ in H^1_{eq} and a lift $\mathrm{coker}\, r^0 \to H^0(C^{\bullet}_{\partial})$ (a field again); then put $h' := h + \xi$ and apply (1).

(3) is (1), (2) and Proposition 15.12(1). □

Remark 15.17 (The ambiguity is non-zero at $(n, \ell) = (10, 691)$)*.* [verified computationally]^{r} for the dimensions, [proved here] for X_2, and [open] *in this document* for one input — the reason r^1 has the rank it has, which is a claim of the companion [E] and is not re-derived here. The arithmetic is recorded so that Proposition 15.16(3) has a witness. **Rewritten at v1.5.0 from the measured values** (Appendix E, v1.4.2, deferred item 1; slot-2 WP-7 and slot-3 WP-5): through v1.4.2 it was computed under the assumption $r^1 = 0$, which is false here, and the old numbers are quoted at the end.

At $(n,\ell) = (10, 691)$, $R = \mathbb{F}_{691}$, p any good prime:

- $X_1 = H^0_{\mathrm{eq}} = 0$: 10 is not in the Dickson semigroup $\langle \ell+1, \ell^2-\ell\rangle$ (Claim 16.5(2), Remark 20.8), so there is no invariant at all;
- $X_2 = H^0(C^\bullet_\partial) = V_R^T \cong \mathbb{F}_{691}$ — [proved here], Proposition 14.16(1) with $n < \ell$;
- $\dim X_4 = \dim H^1_{\mathrm{eq}} = 3 = 2d_{12} - 1$ (Claim 16.5(1));
- $\operatorname{rank} r^0 = 0$, since its source X_1 is 0; hence $\operatorname{coker} r^0 = X_2$ is 1-dimensional;
- $\operatorname{rank} r^1 = 1$, hence $\dim \ker r^1 = \dim H^1_! = 2$. This is the *measured* value (`HeckeBoundaryParabolicValidation`, slot-2 WP-7; `HeckeConeValidation`, slot-3 WP-5, which finds it on all ten Kummer cells of [E, Claim 1.8], with $\dim H^1_! = 2\dim S_{n+2}$), and it is where v1.4.2 and earlier were wrong.

So by (40), $\dim X_3 = \dim X_2 - \dim X_1 + \dim \ker r^1 = 1 - 0 + 2 = 3$ — which `HeckeConeValidation` computes directly as $\dim H^1_c = 3$, the two routes agreeing — and $\dim H^2_c = \dim H^1(C^\bullet_\partial) - \operatorname{rank} r^1 = 1 - 1 = 0$. The ambiguity is $\operatorname{Hom}_{\mathbb{F}}(\ker r^1, \operatorname{coker} r^0) \cong \mathbb{F}_{691}^{2\times 1} = \mathbb{F}_{691}^2 \neq 0$: **a free cochain-level h leaves the T_p-action on this 3-dimensional X_3 undetermined by a 2-dimensional space, and the derived one does not** — Proposition 15.16(3)'s witness, which the repair does not touch. Only the two dimensions move: the conclusion is the same, and is the reason this remark exists.

Why $r^1 \neq 0$, and what "the Eisenstein class" means mod ℓ. 691 divides the numerator of $B_{12}/12$, which is what makes $(10, 691)$ a Kummer cell; the inference drawn from it through v1.4.2 — that the Eisenstein class restricts to zero, so $r^1 = 0$ — does not hold mod ℓ, because there is no single Eisenstein class there. The joint $1 + p^{n+1}$-eigenspace E of $\{T_2, T_3\}$ on H^1_{eq} is *two*-dimensional at every Kummer cell (the integral Eisenstein and one cuspidal eigenline reduce into the same socle of a non-split block; [E] v1.4.0 `cl:harder`), r^1 vanishes on all of E and is non-zero on the top of the block, so $\operatorname{rank} r^1 = 1$. Wherever this document speaks of "the Eisenstein class" modulo ℓ — Obligation 20.7's last factor above all — read the joint eigenspace E. Obligation 9.10(v) already carries the qualification this needs, locating the Eisenstein class only *where that eigenspace is a line*, or through restriction to the cusp where $\operatorname{rank} r^1 = 1$; what a Kummer cell removes is the first of those two routes, E being two-dimensional there. This is the input that keeps the [open] tag: [E]'s claim is not proved in this document.

What v1.4.2 and earlier said, kept so that the repair is visible and a reader of the old PDF is not left guessing: "$r^1 = 0 \ldots$ so $\ker r^1 = X_4$, $\operatorname{Hom}(\ker r^1, \operatorname{coker} r^0) \cong \mathbb{F}_{691}^3$, and $\dim X_3 = 1 - 0 + 3 = 4$". The 4 propagated to [X] §6.2 E5 acceptance (c), which slot 3 corrected ([X] v1.7). The measured $\dim H^1_c = 3$ at $(10, 691)$ is `HeckeConeValidation`'s (v1.5.2).

What this chapter does *not* say about ι and the cone action (v1.6.0; an open point, recorded). §9 builds ι^* on $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$ and, on the boundary model, ι^*_∂; $r^q\iota^* = \iota^*_\partial r^q$ is an identity of the stored matrices, and with it $\iota_c := \iota^* \oplus \iota^*_\partial$ satisfies $D\iota_c = \iota_c D$ and $\iota_c^2 = \mathrm{id}$, so ι_c is an involution of $C(r)$ and acts on $H^\bullet_c$. **That ι_c commutes with $A_c^{(q)}$ does not follow from any statement of this document.** Φ_g is $(\Theta_g, \Theta_g^\partial + h)$ by Definition 15.9, and $\iota_c\Phi_g = \Phi_g\iota_c$ needs, beyond the two bulk and boundary commutations of Proposition 9.4 and Obligation 14.19, the third identity $\iota^*_\partial h = h\,\iota^*$ on the stored $C^\bullet_{\mathrm{eq}}$ basis — i.e. that the pair homotopy η intertwines ι_X and $\iota_{\partial X}$, up to the homotopy Lemma 15.7 allows. Neither Lemma 15.6's construction nor Algorithm 14.12's vertex policy is built to respect ι — the ι-symmetry that *is* imposed is on τ alone (Obligation 9.10(iv), the policy of Algorithm 18.3 step 1 since v1.1.0) — so the identity is neither proved nor expected on a general choice of η. *Measured* (`HeckeInvolutionValidation`, hecke-extensions slot-4 WP-3, run 2026-09-21): at $(n, \ell) = (10, 691)$ the commutator $[\iota_c, A_c^{(1)}(T_2)]$ vanishes, on the one degree with $H^q_c \neq 0$; and on the 15-cell cone grid $\ell \in \{2, 3, 5, 7, 691\} \times n \in \{0, 2, 10\}$ both $r^q\iota^* = \iota^*_\partial r^q$ and $D\iota_c = \iota_c D$ are *asserted* and hold, while $\dim H^q_{c,\pm}$ is **recorded** (v1.6.1: that grid's arm builds ι_c, D and the $\pm$ dimensions and no cone Hecke action at all, so the commutator is recorded at $(10, 691)$ *alone*, as the next sentence says and as v1.6.0's "$\dim H^q_{c,\pm}$ and the commutator are recorded" did not).

One cell is not evidence, and rule (A6) applies: the recorded commutator licenses nothing, and no code path may branch on it; the tagged record of this measurement is Remark 9.8. **For the next** X: either a clause of Obligation 15.21 requiring $\iota_\partial^* h = h\,\iota^*$ where an ι-equivariant η has been constructed, or a counterexample cell. Until then $[\iota_c, A_c^{(q)}] = 0$ is asserted nowhere in this document and must be asserted nowhere in an implementation.

15.4 Integrality: one integral cochain map, and the Bocksteins

Proposition 15.18 (Integrality: one integral cochain map $\Phi_{g,\mathbb{Z}}$; Bockstein equivariance)**.** *Assume (H4) and (B1)–(B3), and let $(\tau, \tau_\partial, \eta)$ be a Hecke datum of the pair* over $\mathbb{Z}$ *— τ with every fill over $\mathbb{Z}$ (Remark 11.5), τ_∂ and η unconditionally (Lemmas 14.10, 15.6).*

1. *[proved here]* ***(Tables once.)*** *τ, τ_∂, η are tables of integer chains, independent of R and of n. Writing $C(r)_\mathbb{Z}$ for the cone over $R = \mathbb{Z}$, its terms $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_q(X), V) \cong \bigoplus_\sigma V^{\Gamma_\sigma}$ and $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_q(\partial X), V) \cong \bigoplus_\sigma V^{\{\pm I\}}$ (sums over Γ-orbit representatives, finitely many by (H4)(iii), (B1)) are finitely generated* free $\mathbb{Z}$*-modules, each summand being a saturated sublattice of $V = \mathbb{Z}^{n+1}$; and $\Phi_{g,\mathbb{Z}} := \Phi_g$ over $\mathbb{Z}$ is a single cochain map of $C(r)_\mathbb{Z}$ with integer matrix entries.*
2. *[proved here]* ***(Matrices per R.)*** *For every commutative R the reduction $C(r)_\mathbb{Z} \otimes R \to C(r)_R$ is a map of complexes carrying $\Phi_{g,\mathbb{Z}} \otimes \mathrm{id}$ to Φ_g over R; it is an isomorphism when $6 \in R^\times$. The matrices Θ_g, Θ_g^∂, h over $\mathbb{Z}$ have entries of size $\sim N^n$ and are never stored; over $R = \mathbb{Z}/m$ they are formed from the tables and the reduced $\rho_n(g\gamma_t) \bmod m$, exactly as (25) does for Θ_g.*
3. *[proved here]* ***(Reductions and Bocksteins.)*** *For $a, b \geq 1$ the reduction maps $C(r)_\mathbb{Z} \otimes \mathbb{Z}/\ell^{a+b} \to C(r)_\mathbb{Z} \otimes \mathbb{Z}/\ell^a$ and the short exact sequences of complexes $0 \to C(r)_\mathbb{Z} \otimes \mathbb{Z}/\ell^a \to C(r)_\mathbb{Z} \otimes \mathbb{Z}/\ell^{a+b} \to C(r)_\mathbb{Z} \otimes \mathbb{Z}/\ell^b \to 0$ are $\Phi_{g,\mathbb{Z}}$-equivariant; hence so are the induced maps on cohomology and the connecting (Bockstein) homomorphisms $H^q(C(r)_\mathbb{Z} \otimes \mathbb{Z}/\ell^b) \to H^{q+1}(C(r)_\mathbb{Z} \otimes \mathbb{Z}/\ell^a)$.* On the nose*, not up to homotopy.*
4. *[proved here]* ***(Independence, on every page.)*** *Two Hecke data of the pair over $\mathbb{Z}$ give $\Phi'_{g,\mathbb{Z}} = \Phi_{g,\mathbb{Z}} + D\Upsilon + \Upsilon D$ with Υ* integral *and preserving the filtration $\ell^k C(r)_\mathbb{Z}$, provided $6 \in R^\times$ for the τ-half (for two η and two τ_∂ with the same τ, unconditionally). Consequently the two induce the same maps on every page E_r, $r \geq 1$, of the spectral sequence of the ℓ-adically filtered complex $C(r)_\mathbb{Z}$ [standard: [Wei94, §5.4] for the spectral sequence of a filtered complex], so the pages carry a well-defined $T^c_{\Gamma g \Gamma}$-action.*
5. ***(What a mod-m fill would cost — and why it cannot occur here.)*** *A table solved only modulo m defines Φ_g over $\mathbb{Z}/m$ only, and a record built on it must say so. For τ_∂ there is no fill (Lemma 14.10(2)); for η the edge fills are* unique *over $\mathbb{Z}$ and over every $\mathbb{Z}/m$ (37), so a mod-m solve returns the reduction of the integer fill and the fallback is never mathematically necessary; the only way a non-integral η-table can arise is a fixed-width integer tier that fails to find the integer fill.* ***Owner decision OP2 (2026-09-15) applies**: the fallback is forbidden by default, the tier is widened instead, and a record claiming integrality must have solve regime $\mathbb{Z}$ on all three tables (Remark 11.5, Obligation 15.21(iv)).*

Proof. (1) The tables are integer chains by hypothesis and involve neither R nor n; the coefficients $\rho_n(\delta_j)$, $\rho_n(g\gamma_t)$ are integral (Definition 8.3). An equivariant cochain is determined by its values on orbit representatives, each value constrained to $V^{\Gamma_\sigma} = \bigcap_{u \in \Gamma_\sigma} \ker(\rho_n(u) - \mathrm{id})$, a kernel of a map between free $\mathbb{Z}$-modules of finite rank, hence a saturated sublattice and free. (2) The map $V^{\Gamma_\sigma} \otimes R \to (V_R)^{\Gamma_\sigma}$ is injective because a saturated sublattice is a direct summand, and surjective when $|\Gamma_\sigma| \in R^\times$: for $f \in (V_R)^{\Gamma_\sigma}$ with any lift $\hat{f} \in V$, $\sum_{u \in \Gamma_\sigma} \rho_n(u)\hat{f} \in V^{\Gamma_\sigma}$ reduces to $|\Gamma_\sigma|\, f$. With $|\Gamma_\sigma| \mid 12$ ((H4)(iv), (B3)) this is $6 \in R^\times$. Compatibility with Φ_g is that (41) is a formula with integer coefficients. (3) Freeness of the terms (1) makes $- \otimes_\mathbb{Z} C(r)_\mathbb{Z}$ preserve the exactness of $0 \to \mathbb{Z}/\ell^a \to \mathbb{Z}/\ell^{a+b} \to \mathbb{Z}/\ell^b \to 0$; $\Phi_{g,\mathbb{Z}} \otimes \mathrm{id}$ commutes with maps of the form

id $\otimes$ (ring map); the connecting homomorphism is natural [standard: [Wei94, Thm. 1.3.1]]. (4) Proposition 15.12(1),(2) with integer tables give integer Υ; an integral map preserves $\ell^k C(r)_{\mathbb{Z}}$. A filtration-preserving homotopy induces, on each graded piece $\ell^k C(r)_{\mathbb{Z}}/\ell^{k+1}C(r)_{\mathbb{Z}}$, a homotopy between the two induced maps, so they agree on $E_1 = H(\mathrm{gr})$; the maps on $E_{r+1} = H(E_r, d_r)$ are induced from those on E_r, so they agree on every page. (5) is Lemma 14.10(2), (37) over $\mathbb{Z}/m$ (the complex $C_\bullet(X)\otimes\mathbb{Z}/m$ is still acyclic, being the base change of a contractible complex of free modules), and the cited policy. □

Remark 15.19 (6 $\notin R^\times$ on this model, and the truncation policy). At $\ell \in \{2,3\}$ the complexes of Definition 15.1 are the *model* complexes of Remark 14.21(1): finite, terminating, and computing the orbifold model cohomology of (H3) and its boundary and cone analogues — *not* $H^\bullet(\Gamma; V_R)$, $H^\bullet(\Gamma_\infty; V_R)$ or the compactly supported cohomology of the orbifold. On them, Theorem 15.11(a),(b),(d), Lemma 15.2, Proposition 15.18(1),(3) and the η/τ_∂-halves of Proposition 15.12 hold verbatim, none having used a hypothesis on ℓ; $T^c_{\Gamma g\Gamma}$ is then *defined* as the output of the pipeline with the canonical triple, and its independence of τ, its identification with any correspondence, and the Hecke relations are not asserted — the contract box after Lemma 11.3, applied to the cone. The systematic treatment is §17, where the same definitions are to be repeated on a free Γ-model of the pair; there the complexes are unbounded above at $\ell = 2$ (Remark 14.21(2),(3)) and the truncation policy of construction item 14.24, decided by the owner on 2026-09-16 ((OC-1a)), governs. **This chapter reads that tag at two sites**, listed in §14.5: Obligation 15.22, whose "every computed degree" and whose exactness clause are narrowed by 14.24; and Definition 15.9, which is stated for all $q \in \mathbb{Z}$ and, on a truncated model, is to be read in degrees $\le q_{\max}$ with $C(r)$ built to $q_{\max}+1$. On the model complexes of this chapter the tag is vacuous (`truncated = false`).

15.5 The two cross-routes and the Poincaré-duality gate

Proposition 15.20 (The two cross-routes and the Poincaré-duality gate)**.** *Assume (H4) and (B1)–(B3), let R be a commutative ring and $(\tau, \tau_\partial, \eta)$ a Hecke datum of the pair over $\mathbb{Z}$. The following are* cross-checks *of Definition 15.9, never definitions; the cone of (38) remains normative. Both routes (1) and (2) are committed validation arms, and (3) is the acceptance gate for X_3 at $\ell \ge 5$ — owner decision OP3, 2026-09-15, which* amended *the verdict's "one of the two" to both.*

1. *[proved here] **(Strict-pair route; needs $6 \in R^\times$.)** Let τ^s be a strict datum for τ_∂ (Proposition 15.13) and $\Phi^s_g := (\Theta^s_g, \Theta^\partial_g)$ with $\eta = 0$. Then Φ^s_g and Φ_g induce the same $T^c_{\Gamma g\Gamma}$, hence the same $A^{(q)}_c$ in the same stored basis, for every q — by Proposition 15.12(2), which is where $6 \in R^\times$ enters; at $\ell \in \{2,3\}$ agreement is* recorded, *not asserted. On $(\ker r)[1]$ the operator is $\Theta^s_g|_{\ker r}$, so this route computes $H^\bullet_c$ as relative cohomology with the bulk pipeline run on $\ker r$.*
2. *[proved here] **(Relative-complex route, "route 3"; no hypothesis on ℓ.)** With ext_0 of Lemma 15.2(1) put $E := \mathrm{ext}_0 \circ h \colon C^q_{\mathrm{eq}} \to C^{q-1}_{\mathrm{eq}}$ and*

$$\Theta'_g := \Theta_g + dE + Ed. \tag{48}$$

Then Θ'_g is a cochain map, cochain-homotopic to Θ_g (so it induces the same $A^{(q)}$), satisfies $r\Theta'_g = \Theta^\partial_g r$ on the nose, hence preserves $\ker r$; and $\Phi'_g := (\Theta'_g, \Theta^\partial_g)$ with $h = 0$ is cone-homotopic to Φ_g via $\Upsilon(c, c_\partial) := (Ec, 0)$: $\Phi'_g - \Phi_g = -(D\Upsilon + \Upsilon D)$. Consequently $T^c_{\Gamma g\Gamma}$ is the map induced by $\Theta'_g|_{\ker r}$ on $H^\bullet(\ker r) \cong H^\bullet_c$, for the same *triple, over $\mathbb{Z}$ and for every ℓ; and the $A^{(q)}_c$ so computed equals the normative one.*
3. ***(Poincaré-duality gate; $6 \in R^\times$, conditional on Claim 16.26.)** [open] If the pairing of Proposition 8.6 induces a perfect pairing $H^1_c(K, \mathcal{R}; V_R) \times H^1_{\mathrm{eq}}(K, \mathcal{R}; D_n) \to R$ under which T_p is self-adjoint (Claim 16.26, with $\mathrm{adj}(g) = \mathrm{diag}(N, 1) \in \Gamma g \Gamma$), then $\langle A^{(1)}_c x, y\rangle =$*

$\langle x, A^{(1)}(D_n)\, y\rangle$ for all x, y, and $A_c^{(1)}$ is determined by $A^{(1)}(D_n)$ — an operator the bulk pipeline already computes. This is a complete independent pin *of $A_c^{(1)}$ and the acceptance gate once the pairing exists at cochain level (owner decision OP3). It is an oracle: a disagreement is a defect, never a redefinition. Its status here is that of Claim 16.26, a stub at this version.*

Proof. (1) Proposition 15.13 gives τ^s; Proposition 15.12(2) with $\tau' = \tau^s$ (and $\tau'_\partial = \tau_\partial$, $N_\partial = 0$) gives the cone homotopy, whose existence needs H with $\tau^s - \tau = \partial H + H\partial$, i.e. Lemma 11.3(2) — the uniqueness half, which is why this route keeps $6 \in R^\times$ even though both τ^s and τ exist over $\mathbb{Z}$; the last sentence is Lemma 15.2(3) and the end of the proof of Proposition 15.13.

(2) E is R-linear and equivariant (Lemma 15.2(1)), and $rE = r\,\mathrm{ext}_0\, h = h$. Then $r\Theta'_g = r\Theta_g + d(rE) + (rE)d = r\Theta_g + dh + hd = \Theta_g^\partial r$ by Lemma 15.10, using $rd = dr$. Θ'_g is a cochain map because $dE + Ed$ is one ($d(dE + Ed) = dEd = (dE + Ed)d$), and $\Theta'_g - \Theta_g = dE + Ed$ is a cochain homotopy, so $A^{(q)}$ is unchanged. For $c \in \ker r$, $r\Theta'_g c = \Theta_g^\partial rc = 0$. Finally $D\Upsilon(c, c_\partial) = D(Ec, 0) = (-dEc, rEc) = (-dEc, hc)$ and $\Upsilon D(c, c_\partial) = \Upsilon(-dc, \cdot) = (-Edc, 0)$, so $D\Upsilon + \Upsilon D = \big(-(dE + Ed)c,\ hc\big) = -(\Phi'_g - \Phi_g)(c, c_\partial)$. Hence Φ'_g and Φ_g induce the same map on $H^\bullet(C(r))$; Φ'_g preserves $(\ker r)[1]$ and acts there by Θ'_g, and the quasi-isomorphism of Lemma 15.2(3) transports the induced maps. No hypothesis on ℓ was used. (3) is the stated conditional, the adjointness identity being the definition of "self-adjoint" read in the stored bases. □

15.6 Obligations

The four certificates of the extension programme's data model, and what each one is *for*. The first two are the content; the last two are consequences re-checked.

certificate	stores / `verify()` re-derives	ring	what it pins
`TauBoundary-Certificate`	the τ_∂-table; Obligation 14.19(i)–(iv)	$\mathbb{Z}$	Θ_g^∂ up to homotopy (Lemma 14.10(3))
`PairHomotopy-Certificate`	the η-table; Obligation 15.21(i)–(iv)	$\mathbb{Z}$	**the operator on** $H_c^\bullet$ (Prop. 15.12(1))
`ConeAction-Certificate`	Θ_g^∂, h, Φ_g, $A_c^{(q)}$ over R; Obligation 15.21(v) and 15.22(i)–(iii)	R	that the stored h *is* the transport of the stored η
`ConeCertificate`	D, the maps of (39), $A^{(q)}, A_\partial^{(q)}, A_c^{(q)}$; Obligation 15.22(iv)–(vi)	R	the *sequence*, not the module: any Φ of Prop. 15.16 passes it

Obligation 15.21 (Certified η-table)**.** [formally verified: [GT, `chk:eta`]] [formally verified: [GT, `chk:eta-p2`]] [formally verified: [GT, `chk:pairgrid`]] (clauses (i)–(iv) as a predicate, verified at the same (p, ν); not (v), (vi); the link to Definition 15.5: [formally verified: [GT, `thm:etaspec-pairhomotopy`]] [formally verified: [GT, `cor:etap2-pairhomotopy`]] (the latter at $(p, \nu) = (2, 1)$) — an accepted table is a pair homotopy for every τ, τ_∂, η realising the three tables on the generators $s_j \cdot \sigma$; that such realisations exist is not formalised) Assume (H4) and (B1)–(B3). An implementation of the cone action emits, for each $g = \mathrm{diag}(1, N)$ and each Γ''-orbit representative simplex σ of ∂X, the chain $\eta(s_j \cdot \sigma) = \sum_t n_t^{(j,\sigma)}\, \gamma_t \cdot \sigma_t$ with $n_t^{(j,\sigma)} \in \mathbb{Z}$, $\gamma_t \in \Gamma$, σ_t a simplex of X of dimension $\dim \sigma + 1$, in the stored-chain format of Algorithm 18.3 and in canonical form, together with the **table hashes** of the stored τ- and τ_∂-tables it was built against and the solve regime of every fill. It must discharge the following; each clause names the data read, the predicate, the ring, and the action on failure.

The three hashes (v1.5.0; §22.4): what identifies "the tables it was built against" in a record is a hash of each stored table — `table_fnv64()` on the η-table itself and the same function on each

base table, so that the three are one function and not three (`Hecke/HeckePairHomotopy.h`: `table_fnv64()`, `tau_table_fnv64()`, `tau_boundary_table_fnv64()`; the byte-wise FNV-1a-64 of `ModArithmetic/Fnv.h`). They are *identifying and never certifying*: two admissible η for the same (τ, τ_∂) differ in the first hash while $A_c^{(q)}$ does not (Lemma 15.7, Proposition 15.12(1)), so a hash may be compared and recorded but not branched on (rule (A6)), and clause (i) — not a hash — is what ties the table to its certified inputs.

(i) **Structural.** $\det \gamma_t = 1$; every σ_t is a simplex of X of the right dimension; and the tables it refers to are the certified τ (Obligation 20.11) and τ_∂ (Obligation 14.19) of the same g, Γ'' and s_j. Over $\mathbb{Z}$; a failure **aborts**.

(ii) **The homotopy identity** (42)**.** $\partial\eta(s_j\sigma) = (j\tau_\partial - \tau j)(s_j\sigma)$ for every stored vertex slot, and $\partial\eta(s_j\sigma) + \eta(\partial(s_j\sigma)) = (j\tau_\partial - \tau j)(s_j\sigma)$ for every stored edge slot, **both sides evaluated only through each table's own (T1) helper, all built on the one (P1) split** from the stored τ-, τ_∂- and η-tables — the same helpers the construction uses, so that construction and certification cannot disagree on how a non-representative slot is derived — and compared after reduction to canonical $(n_t, \gamma_t, \sigma_t)$ form. Over $\mathbb{Z}$; a failure **aborts**. (v1.5.0, corrected from "the shared (T1) helper — the same helper the construction uses": there is no one helper shared by the three tables. Each has its own — `eval_tau_on_translate`, `tau_on_translate`, `eta_on_translate` — and all three route through the one conjugation primitive of (P1), `HeckeCosetTables::twisted_conjugate`, which is what the clause is about: that construction and certification share the derivation, not that one function serves three tables. On the shipped tables a τ-value on a translate is never asked for, (T1) of the defect supplying it.)

(iii) **(T1) bookkeeping.** For every stored transporter $h \in \Gamma''$ occurring in the table, $\eta(h\cdot x) = (g^{-1}hg)\cdot\eta(x)$ as chains in canonical form — Obligation 20.11's clause with η in place of τ; and the entry bound of Lemma 18.5 recounted on the η-chains, so that the arithmetic tier of Theorem 18.7 covers the derived h. Over $\mathbb{Z}$; a failure **aborts**.

(iv) **Solve regime (owner decision OP2).** The recorded regime of every edge fill is $\mathbb{Z}$; a fill recorded as solved modulo m marks the table `integral = false`, is **forbidden by default** and **aborts** under the default policy; where an implementation offers a mod-m path at all it is behind an explicit setting, and then the record carries the flag, may not claim integrality, and states that Φ_g is defined over $\mathbb{Z}/m$ only (Proposition 15.18(5)). Since the fills are unique (37), a mod-m regime is a tier failure and never a mathematical one — and therefore **an implementation MAY omit the mod-m regime altogether** (v1.5.0): it then has no such path and registers no such setting, a tier failure aborts unconditionally, and the recorded regime is $\mathbb{Z}$ by construction. The shipped implementation is of that kind (`Hecke/HeckePairHomotopy`, `Hecke/HeckeIntegerSolve`; `HeckeSettings` registers no key for it), and the absence of the setting is conformance, not a gap.

(v) **The derived h is the transport of the stored η.** The stored matrix h over R equals (44) recomputed from the η-table and $\rho_n(\delta_j\gamma_t)$ mod m — the analogue of the `xi_terms` recomputation for Θ_g — entrywise, in every computed degree. Over R; a failure **aborts**. **This clause, and not any identity among matrices, is what pins** $A_c^{(q)}$: a tampered $h' = h + \xi$ with ξ a degree-(-1) cochain map passes every clause of Obligation 15.22 and fails this one (Proposition 15.16).

(vi) **Diagnostics — recorded, never branched on (rule (A6)).** `eta_zero` := (every stored η-chain is 0); `cochain_commutes` := ($r\Theta_g = \Theta_g^\partial r$ entrywise over R); the three solve regimes. The implication `eta_zero` $\Rightarrow$ `cochain_commutes` is asserted (its failure **aborts**, being a contradiction in the stored data); nothing else about them is asserted, and no code path may select $h = 0$ on the strength of `cochain_commutes` (Remark 15.15(2)).

Obligation 15.22 (Cone exactness and intertwining, re-derived over R)**.** Assume (H4) and (B1)–(B3), and let the stored data be: the bases of $Z^q(C(r))$, $B^q(C(r))$, $H^q(C(r))$ and the

projector, in the flat layout of Algorithm 12.13 on $C(r)$; the matrices D, r, Θ_g, Θ_g^{∂}, h, Φ_g over R; $A^{(q)}$, $A_{\partial}^{(q)}$, $A_c^{(q)}$; and the **three map families** $\mathrm{pr}_{c*}^{\,q}$, r^q, $\mathrm{inc}_{c*}^{\,q}$ of (39), stored per degree in the stored bases (v1.6.0, corrected from "the six maps of (39)": the sequence is degree-indexed and the code stores the three families per degree, while "six" counted the six-term window of Lemma 15.2(5) — on the model complexes there are eight non-trivial map instances, $\mathrm{pr}_{c*}^0, r^0, \mathrm{inc}_{c*}^0, \mathrm{pr}_{c*}^1, r^1, \mathrm{inc}_{c*}^1, \mathrm{pr}_{c*}^2, r^2$, of which two are zero-dimensional and inc_{c*}^2 lands in $H_c^3 = 0$; `docs/implementation/slot3/WP-5_report.md` finding 11). "Every computed degree" means every q in the reported range, which is all of $\{0,1,2\}$ on the model complexes and is read from $q_{\max}$ under 14.24 on a truncated free model (§14.5).

(i) **The cone identity.** $dh + hd = \Theta_g^{\partial} r - r\Theta_g$ and $D\Phi_g = \Phi_g D$ **as matrices on the stored** $C_{\mathrm{eq}}^{\bullet}$ **basis** — equivalently, the bracket of Lemma 15.8 annihilates the stored equivariant cochains — in every computed degree; and $D^2 = 0$, which involves only d and r and holds on the flat spaces as well. Over R; a failure **aborts**.

Why the basis is named, and what is recorded instead (v1.6.0; `docs/implementation/slot3/WP-6_report` finding 1, the reading implemented). The coset transport of Definition 15.3 is defined on $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, V_R)$, and the step that turns $\varphi(\gamma_t \sigma_t)$ into $\rho_n(\gamma_t)\varphi(\sigma_t)$ — the step that makes Θ_g, Θ_g^{∂} and h finite matrices at all — *is* the Γ-equivariance of φ. The stored matrices are honest linear maps of the flat cochain spaces and agree with the matrix-free routes on every flat vector, but the two identities above are consequences of the transport only on the equivariant subspace, and **the same two identities on the full flat cochain space are recorded, never asserted**: on the shipped grid the bracket vanishes on $C_{\mathrm{eq}}^{\bullet}$ in 424 of 424 computed degrees and on the full flat space in 29 of them, and $D\Phi_g = \Phi_g D$ holds flat on 1 of 212 operators. A record that reports the cone carries the two as the diagnostics `phi_flat_identity` and `residual_flat_zero` (§22.4 item 6(e)), under rule (A6): a `false` there is conformance and not a defect, and no code path may branch on either. The same qualifier governs the tampered h' of clause (viii)(b), which is required to satisfy this clause on $C_{\mathrm{eq}}^{\bullet}$ and not flat. (Through v1.5.2 the clause read "as matrices, in every computed degree", which an implementer can — and the first one did — read as a statement about the stored flat matrices, and which then aborts on correct data.)

(ii) **Assembly.** Φ_g maps every stored cocycle of $Z^{q-1}(C(r))$ into $Z^{q-1}(C(r))$ (checked as $D\Phi_g z = 0$), and $A_c^{(q)}$ is the projection of those images (Algorithm 12.13 step 2, recomputed). Over R; a failure **aborts**.

(iii) **Intertwining.** Each map of (39) in the stored bases commutes with the corresponding pair among $(A_{\partial}^{(q)}, A_c^{(q)}, A^{(q)})$ (Theorem 15.11(a)); in particular $r^q A^{(q)} = A_{\partial}^{(q)} r^q$, which is the pair's form of Obligation 14.19(vii). Over R **at every** ℓ**, the** r**-half included**; a failure **aborts** (v1.7.0; through v1.6.2 the r-half carried an $\ell \in \{2,3\}$ carve-out, asserted for $6 \in R^{\times}$ and *recorded* otherwise, which clause (i) already proved away).

Why the r*-half needs no hypothesis on* ℓ here (v1.7.0; `docs/implementation/slot5/WP-4_report.md` §4, which re-checked the hypothesis this proof uses against the shipped code rather than against prose; owner ruling of 2026-09-22, `[OD]` slot-4 close-out 3). An implementation that reaches this clause has already discharged clause (i), which carries no hypothesis on ℓ and whose failure aborts: $dh + hd = \Theta_g^{\partial} r - r\Theta_g$ on the stored $C_{\mathrm{eq}}^{\bullet}$ basis, in every computed degree. Let $z \in Z_{\mathrm{eq}}^q$ be a stored *equivariant* cocycle — an element of the span of the stored $C_{\mathrm{eq}}^{\bullet}$ basis, which is where clause (i) holds and which is what Algorithm 12.13 step 2 reads — so that $dz = 0$. Then

$$\Theta_g^{\partial} r z - r\Theta_g z = (dh + hd)z = d(hz),$$

a coboundary of the boundary complex — so $r\Theta_g z$ and $\Theta_g^{\partial} r z$ have the same class, and since r is a cochain map it carries B_{eq}^q into the boundary coboundaries, whence $r^q A^{(q)} = A_{\partial}^{(q)} r^q$ on cohomology in the stored bases, both sides assembled by Algorithm 12.13 step 2 as

clause (ii) requires. Nothing in the argument inverts 2, 3 or 6: h is the coset transport (44) of an η-table built over $\mathbb{Z}$ for every ℓ (Obligation 15.21(iv)), clause (i) is asserted at every ℓ, and the projection is linear over R. The hypothesis is the *existence of the stored* h, not the invertibility of 6.

What this clause does not say. It is an assertion about **the objects this document's implementation computes** — $H^\bullet_{\mathrm{eq}}(K,\mathcal{R};L)$, its boundary analogue $H^\bullet(C^\bullet_\partial)$, and the stored $r^\bullet$, $A^{(q)}$, $A^{(q)}_\partial$ — and **nothing more**. At $\ell \in \{2,3\}$ those groups are the orbifold model cohomology and its boundary analogue and are *not* the group cohomology of Γ and of Γ_∞ (Remark 15.19, 14.24); no statement of this document identifies the two there, and asserting the identity above at $\ell \in \{2,3\}$ does not assert, imply or license the corresponding statement for $H^\bullet(\Gamma;M_n)$ and its boundary restriction. A reader who transports it there is reading a clause about the computed complexes as a clause about group cohomology, which is exactly the step Remark 15.19 forbids. Recorded outcome, unchanged by the strengthening: 424/424 (cell,q) pairs, the 74 operators at $\ell \in \{2,3\}$ included (Remark 15.23).

The asymmetry with Obligation 14.19(vii), which keeps its carve-out (v1.7.0). (vii) states the same identity in the boundary chapter, whose stored data are τ_∂ and Θ^∂_g alone: an implementation of the boundary datum has no η (Definition 15.5) and no h (44), so the proof above is *not available to it*, and (vii) therefore continues to assert the identity for $6 \in R^\times$ and to *record* it at $\ell \in \{2,3\}$. The two clauses do not contradict each other because they bind different implementations: (vii) binds an implementation of the boundary datum, (iii) binds an implementation of the *pair*, which by clause (i) carries h. For an implementation that discharges both — the shipped one — the identity is asserted at every ℓ under (iii), and (vii)'s carve-out is vacuous for it. This is the only place in this document where a strictly stronger statement is available to the richer datum, and it is stated here rather than moved into (vii) so that the boundary chapter stays readable, and implementable, on its own. The other maps of (39) were, and are, asserted at every ℓ.

(iv) **Exactness — where it is asserted.** Each composite of consecutive maps of (39) *both of whose end terms are reported* is zero: on the model complexes that is every composite, and on a truncated free model it is every composite in degrees $\le q_{\max}$ except the outgoing one at $H^{q_{\max}}(\partial)$, whose target $H_c^{q_{\max}+1}$ is scaffolding with no stored basis — there only the incoming composite $r^{q_{\max}} \circ (H_c^{q_{\max}} \to H^{q_{\max}}_{\mathrm{eq}}) = 0$ is asserted (14.24, the spelled-out clause; rider (iii)). *What "reported" means, per complex* (v1.5.0; `docs/implementation/slot3/WP-5_report.md` finding 1, the reading implemented): a term $H^q(F)$ of one of the three complexes F — bulk, boundary, cone — is **reported** iff $q < 0$ (vacuously: the term is 0), or $q \le q_{\max}(F)$, or F is not truncated; an untruncated complex of finite length *does* report its vanishing above its top degree, the box of 14.24 forbidding that only where `truncated = true`. The three $q_{\max}$ are the per-complex fields of §22.4 item 5, and the sentence before this one is exactly this rule in the case $q_{\max}(\mathrm{bulk}) = q_{\max}(\mathrm{boundary}) = q_{\max}$ that 14.24's spelled-out clause states. On the shipped models nothing changes: all three complexes are finite and untruncated, so every term is reported and every composite and every spot is asserted. At every spot all three of whose terms are reported, ker = im: over a field by the rank identity, over $\mathbb{Z}/\ell^e$ or $\mathbb{Z}/m$ by a solve exhibiting each kernel generator as an image (a non-field elimination, [OD OQ18]; for a ring over which no such solve is built the non-field case is **recorded** as unchecked, never reported as passed). A failure at an asserted spot **aborts**. *The non-field case, as built* (v1.9.0; through v1.8.1 the parenthesis read "until it exists the non-field case is recorded as unchecked", stale since the elimination exists): over $R = \mathbb{Z}/\ell^e$, $\ell \ge 5$, $\ell^e < 2^{64}$, the non-field elimination is `src/ModArithmetic/ENGINE_API.md` §13.6 (hecke-extensions slot 8), and the check is **made**: at every spot, every composite of consecutive maps is shown zero by a solve into the target's relations, and every Howell generator of the kernel

of the outgoing map is exhibited as the image of the incoming map plus a relation, the witnesses stored and re-derived (`ConeModuleCertificate` item 14, hecke-extensions slot-9 WP-4; measured by `HeckeReductionValidation`, arm (4b-C), at five of the seven points of [E, `sec:ring`] (e) and at the two word-regime cells, the point $(60, 5)$ at both levels $k \in \{2, 3\}$ running in `HeckeReductionCoverageSlowValidation` since hecke-extensions slot-9 WP-4t (v1.9.1), which also covers the extension stage's case list at $\ell \geq 5$). Beside it the length identity is asserted at every term — the length of the term is the sum of the lengths of the two images — and with the composite-zero check it implies exactness on its own, a submodule of a finite module of the same length being the whole; so the witness list is a second, independent discharge, not the only one. At $\ell \in \{2, 3\}$ and for $\mathbb{Z}/m$ no cone over the ring is built ([E, `sec:ring`] (b1)). $\operatorname{im}(\mathrm{pr}_{c*}^{q}) = \ker r^q$ is checked **explicitly** — both inclusions, by solves, and not only through the rank identity — at **every** bulk spot of (39) all three of whose terms are reported, and not at $q = 1$ alone (v1.6.0, read degree-generically as implemented; `docs/implementation/slot3/WP-5_report.md` finding 5: through v1.5.2 the clause named the single instance $\operatorname{im}(H_c^1 \to H_{\mathrm{eq}}^1) = \ker r^1$, a degree-1 literal in a degree-generic obligation. On the model complexes that is $q \in \{0, 1, 2\}$; on a truncated free model every $q \leq q_{\max}$ whose spot is asserted. A ring that is not a field, and over which no such solve is built, records the check unchecked, as the rank identity above does, and never reports it as passed; over $\mathbb{Z}/\ell^e$, $\ell \geq 5$, both inclusions are made by solves since v1.9.0, as above). At $H^{q_{\max}}(\partial)$ on a truncated model the rank of $r^{q_{\max}}$ is **recorded**, and with it the consequence of exactness that needs no basis of the unreported term, $\operatorname{coker} r^{q_{\max}} \hookrightarrow H_c^{q_{\max}+1}$ (Remark 14.21(2)); at $\ell = 2$, where $H^q(\partial; V_R)$ is known in closed form in every degree (Lemma 14.20), that cokernel's dimension is a lower bound for $\dim H_c^{q_{\max}+1}$ and is recorded as such, never as a reported group. 14.24 narrows the set of asserted spots on a truncated model and nothing else.

(v) **Bookkeeping, where the window closes.** $H_c^0 = 0$ is asserted unconditionally (Lemma 15.2(4)); $H_{\mathrm{eq}}^2 = 0$ itself is asserted for $6 \in R^\times$ (Claim 16.2) and **recorded** otherwise. Of the two identities of (40), over a field and wherever the terms they name are reported: **the first is asserted without consulting H_{eq}^2 at all** (v1.6.0) — Lemma 15.2(5) proves it from the exactness of (39) and $H_c^0 = 0$ alone, so an implementation must assert it on *every* computed cell and may not make the assertion conditional on a stored $H_{\mathrm{eq}}^2 = 0$; **the second is asserted where $H_{\mathrm{eq}}^2 = 0$ holds in the stored data** and **recorded** otherwise. A failure of an asserted item **aborts** — the stored object's own certificate for **both** identities, each under its own minimal guard, *and* the validation arm that carries the first, in the sense of clause (viii) (v1.7.0, restated: through v1.6.2 this sentence assigned the second identity to the certificate and the unconditional first one to "the validation arm that carries it" alone, which was the division of labour the shipped certificate then had; the certificate now carries the first identity too, so the clause names both and neither is the other's substitute).

What the strengthening does and does not say (v1.6.0; `docs/implementation/slot3/WP-5_report.md` finding 3; through v1.5.2 both identities were gated on $H_{\mathrm{eq}}^2 = 0$). It is a statement about the *dependence*, not about a known counterexample: on the shipped model complexes $H_{\mathrm{eq}}^2 = 0$ holds on every computed cell — including all 18 at $\ell \in \{2, 3\}$, where Claim 16.2 does not apply and it is therefore *recorded* rather than asserted (`HeckeConeValidation`: 74 cells, $\ell \in \{2, 3\}$ cells with $H_{\mathrm{eq}}^2 \neq 0$: none) — so no cell currently separates the two readings. What the strengthening buys is that a cell which did would still be covered, and that a conforming implementation may not silently skip the count where its own H_{eq}^2 is non-zero or unreported.

Carried by the certificate since v1.7.0 (hecke-extensions slot-5 WP-4, phase OI-2, owner ruling `[OD]` slot-4 close-out 2; `docs/implementation/slot5/WP-4_report.md` §1, §3). `ConeCertificate`'s item C11 is split into two guards under the one item code: the first

identity is evaluated without consulting the stored $H^2_{\mathrm{eq}} = 0$, and without the two further reported-term conjuncts that belong to the second identity alone; the second keeps them. The split is not vacuous — a *constructed* cell in which H^2_{eq} is unreported (a bulk complex declared truncated at $q_{\max} = 1$, so that the flag is structurally false with no stored dimension corrupted) and r^1 is perturbed entrywise makes the first identity fire and leaves the second silent, where the shared guard would have skipped both. Value-preserving on the shipped grid, as the paragraph above predicts: $A_c^{(q)}$ is bit-identical and both recorded collector digests are unchanged.

(vi) **Consistency assertions — theorems re-checked.** For $6 \in R^\times$: $[A_c^{(q)}(T_p), A_c^{(q)}(T_{p'})] = 0$ and the $\nu = 2$ recursion of Theorem 10.8(1) on H_c^q, with the *full* T_{p^2} and the coefficient p^{n+1} (entries A.6, A.14), in every computed degree; a failure **aborts**. For $6 \notin R^\times$ the same quantities are **recorded**, not asserted (Remark 15.19). *Where the full operator comes from* (v1.6.0; `docs/implementation/slot3/WP-6_report.md` finding 4): the cone pipeline run at $g = \mathrm{diag}(1, p^\nu)$ emits the *primitive* operator $T_{\Delta^{\mathrm{prim}}_{p^\nu}}$, exactly as the bulk pipeline does; the full T_{p^ν} this clause requires is assembled from the primitive ones by (19) (Proposition 10.6), at $\nu = 2$ by adding the central $p^n I$ (§10.1, entry A.1). An implementation that checks the recursion against the primitive operator checks a false statement. *Over $\mathbb{Z}/\ell^e$, as built* (v1.9.0, status; hecke-extensions slot-9 WP-4 and its review repair): the commutator $[A_c^{(q)}(T_p), A_c^{(q)}(T_{p'})] = 0$, and likewise $[A_\partial^{(q)}(T_p), A_\partial^{(q)}(T_{p'})] = 0$, is asserted at every $\ell \geq 5$ cell over the ring for every pair of computed primes (`HeckeReductionValidation`, arm (4b-H)); **the $\nu = 2$ recursion is not computed over the ring in this build**, so this clause is **partial** there (§22.2).

(vii) **Cross-routes (owner decision OP3).** The strict-pair route and route 3 of Proposition 15.20(1),(2) each produce $A_c^{(q)}$ in the stored basis equal to the normative one, entrywise, in every computed degree; route 3's agreement **aborts** on failure for every ℓ; the strict-pair route's agreement **aborts** for $6 \in R^\times$ and is **recorded** otherwise. Once Claim 16.26 and the cochain-level pairing exist, the adjointness of Proposition 15.20(3) is asserted for $\ell \geq 5$ and is the acceptance gate for X_3; until then it is **recorded** as unavailable. *Over $\mathbb{Z}/\ell^e$, as built* (v1.9.0–v1.9.1, status; owner ruling 3 (iii) (A) of the slot-9 decisions, 2026-09-24): only the normative route, route 1 of Proposition 15.20, was built over the ring. The strict-pair route, route 3, and the relative sub-complex constructor that route 3 needs refused a composite ring by name, each with a validation arm; so the agreement this clause asserts was an $\mathbb{F}_\ell$ statement in that build, and its composite half was open (stale from hecke-extensions slot-10 WP-3's merge, `c9684e9`, until v1.10.0). *Over $\mathbb{Z}/\ell^e$, since v1.10.0* (status; owner ruling 3 (A) of the slot-10 decisions, 2026-09-25; hecke-extensions slot-10 WP-3): the strict-pair route, route 3 and the relative sub-complex constructor are built over the ring at $\ell \geq 5$ (the constructor on the module route of the bulk, refusing at $\ell \in \{2, 3\}$ by name), and **the clause's first two sentences are discharged over the ring at $\ell^e < 2^{32}$ — at every point of [E, `sec:ring`] (e)**: both cross-routes give $A_c^{(q)}$ entrywise equal to route 1's in its cyclic-summand basis, in every computed degree, and both agreements abort on failure (only $6 \in R^\times$ reaches them). Route 3's transport uses no inverse matrix: the correspondence $M\colon [z] \mapsto [(z, 0)]$ and the explicit $N\colon [(c, c_\partial)] \mapsto [c + d\,\mathrm{ext}_0 c_\partial]$, ext_0 an equivariant section of r, are asserted mutually inverse. The cross-routes follow the companion's `uint32_t` regime rule: above 2^{32} they refuse by the ring context's named refusal, and route 1 alone runs there. **The third sentence over the ring is recorded as unavailable**, by its own rule: the cochain-level pairing and the contragredient refuse a composite ring (`Hecke/HeckePairing`, `Hecke/HeckeContragredient`), so the adjointness of Proposition 15.20(3) is not evaluated there (the slot's independent review, `docs/implementation/slot10/WP-14_review.md`, finding M-1). Measured

by `HeckeConeCrossRouteRingValidation` (386 checks) and the changed rows of `HeckeReductionValidation`'s arms (X) and (X4b) (§22.2).

(viii) **Two arms the validation suite must contain.** (a) *Independence*: two η-tables built with different vertex-chain policies in Lemma 15.6(V), and two τ_∂-tables, must give *bit-identical* $A_c^{(q)}$, bit-identical cone bases and D (which involve only d and r), while h and the table hashes may move; a difference in $A_c^{(q)}$ **aborts** (over $\mathbb{Z}$ for the η/τ_∂ arm, by Proposition 15.12(1),(2)). (b) *Negative control*: a tampered $h' = h + \xi$ with ξ built as in the proof of Proposition 15.16(2) must *pass* clauses (i)–(vi) and *fail* Obligation 15.21(v), and must change $A_c^{(1)}$ at a cell where $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0) \neq 0$ (Remark 15.17); the arm's failure to behave so **aborts** the validation. Arm (b) is the executable form of the statement that clauses (i)–(vi) certify the sequence and not the module.

Remark 15.23 (Restriction equivariance at $\ell \in \{2,3\}$ on the model complexes: the measurement that preceded the proof). [verified computationally] **Since v1.7.0 this remark is load-bearing for no clause of §15**: clause (iii) above asserts $r^q A^{(q)} = A_\partial^{(q)} r^q$ at *every* ℓ, on the strength of clause (i), and what follows is the record of the outcome *measured before* that implication was written into the document — kept because [verified computationally] never upgrades and never disappears (§1.1), and because it is still the record that discharges Obligation 14.19(vii) at $\ell \in \{2,3\}$, where the boundary chapter has no h and the proof of clause (iii) is unavailable (clause (iii)'s asymmetry paragraph). Read as written, for that clause: Obligation 14.19(vii) — $r^q T_{\Gamma g \Gamma} = T_p^\partial r^q$ — is asserted at $6 \in R^\times$ and is *recorded* at $\ell \in \{2,3\}$, where the model complexes compute the orbifold model cohomology and its boundary analogue rather than group cohomology (Remark 15.19) and no statement of this document asserts it. Recorded outcome (`HeckeBoundaryOperatorValidation`, tier `standard`, run 2026-09-16, hecke-extensions slot 1 WP-4b, finding 15): on every (cell, q) pair of its grid at $\ell \in \{2,3\}$ — $(p,\nu) \in \{(2,1),(3,1),(5,1),(7,1),(2,2),(3,2)\}$, even $n \leq 16$, $q \in \{0,1\}$ ($q = 2$ vacuous), the bad cases $p = \ell$ and $\nu = 2$ included — the identity holds entrywise over $\mathbb{F}_\ell$ in the stored bases: 216/216 pairs. The observation is consistent with what a pair homotopy η over $\mathbb{Z}$ (Definition 15.5, Lemma 15.6) gives once Φ_g exists — the cohomology-level identity is a consequence of Theorem 15.11(a) under its hypotheses and is not proved without them — and it is offered to the phase that builds this chapter (E5) as a recorded fact and as an outcome its $\ell \in \{2,3\}$ arm may expect, **not** as a claim: a failure at another (n,p,ν) or on another triangulation would contradict nothing in this document. Rule (A6): the record licenses nothing, and no code path may branch on it. [verified computationally] never upgrades to [proved here] (§1.1). *Re-recorded through the cone* (v1.5.2; `HeckeConeActionValidation`, hecke-extensions slot-3 WP-6, run 2026-09-17): clause (iii)'s r-half holds on 424/424 (cell, q) pairs of that grid ($\ell \in \{2,3,5,7,691,134217757,2^{31}-1\}$, even $n \leq 16$, 74 of its 212 operators at $\ell \in \{2,3\}$). *Superseded as the clause's authority* (v1.7.0): the same 424/424 are now **asserted** by clause (iii) at every ℓ — not on the strength of this record, which licenses nothing, but on the strength of clause (i), which proves the identity with no hypothesis on ℓ for any implementation carrying h. The sentence above about Theorem 15.11(a)'s hypotheses remains correct for the boundary datum alone, which is the case Obligation 14.19(vii) still covers, and is no longer the last word for the pair. (Open point (8) of the v1.3.3 changelog; carried non-normatively in v1.4.0.)

Code map. Implemented, except clause (vii), since v1.5.2 (hecke-extensions slot 3, packages WP-5 and WP-6, 2026-09-17; `docs/implementation/slot3/WP-5_report.md`, `WP-6_report.md`). *The cone complex* (WP-5, phase E5-2): `EquivariantMappingCone` (`EquivariantMappingCone`, generic, no Hecke include) — $C(r)$ of Definition 15.1 / (38) on the flat layout of Algorithm 12.13, its $Z/B/H$ bases and projector through the block model with the pipeline's basis rules, the three map families of (39) in stored bases per degree, the per-complex degree fields of §22.4 item 5 derived, and `ConeCertificate` (Obligation 15.22(iv),(v)

and $D^2 = 0$, with a re-deriving `verify()`) — built on `Hecke/SL2ZBoundaryPair`, the restriction $r^\bullet$ lifted pipeline-side and value-preservingly out of `HeckeExt/HeckeBoundary`; `HeckeConeValidation` (tier `standard`, 74 cells: every spot exact, $H_c^0 = 0$ at every ℓ and n, $\dim H_c^1 = 3$ at $(10, 691)$ and $1 + 2\dim S_{n+2}$ with $H_c^2 = 0$ at the ten Kummer cells). *The Hecke action* (WP-6, phase E5-3): `Hecke/HeckeConeAction` (`HeckeConeAction`) — h by the coset transport (44) of the stored η-table in two bracketings asserted bit-identical, never chosen; Φ_g of (45); the residual of Lemma 15.8; $A_c^{(q)}$ by Algorithm 12.13 on $C(r)$; `ConeActionCertificate` (Obligation 15.21(v),(vi) and Obligation 15.22(i)–(iii)) and `ConeRelationsCertificate` (Obligation 15.22(vi), the *full* T_{p^2} of §10.1); the value-preserving vertex policies `EtaVertexPolicy` (`Hecke/HeckePairHomotopy`) and `TauBoundaryVertexPolicy` (`Hecke/HeckeBoundaryDatum`, default the canonical Algorithm 14.12, the second an admissible non-tie choice of Lemma 14.10(1)) with `table_fnv64()` on all three tables, which is what Obligation 15.22(viii)(a) runs on; a standalone record `hecke_cone_action/v1`; `HeckeConeActionValidation` (tier `standard`, 63 cells, 212 operators). Measured there: the identities of clause (i) hold on the stored $C_{\mathrm{eq}}^\bullet$ basis in every degree and on the full flat cochain space only in 29 of 424 degrees, which is what the coset transport's use of Γ-equivariance predicts and is recorded, not asserted (a wording point for the next X). The golden collectors ($\dim H_c^q$, the tier-A invariants of $A_c^{(q)}$, h, Θ_g^∂, the cone bases, the table hashes) are *defined and never emitted* — called from the two validations only; their emission is phase E2's. **Since v1.7.3 clause (vii) is implemented** — `HeckeConeStrictPairRoute` (slot-6 WP-4), `HeckeConeRelativeRoute` (WP-5), both in `Hecke/HeckeConeAction`, and the PD gate of `Hecke/HeckePairing` (WP-3); the rest of this sentence is the v1.7.2 status. **Not implemented** (v1.7.2): clause (vii) of Obligation 15.22 — the strict-pair route and route 3, phase E5's cross-route arms (slot 6 of the programme) — and PD adjointness (recorded unavailable until Claim 16.26); the cone's integration into `HeckeReport` and the campaign record (E2). The paragraph below is the v1.4.2 status, kept as the record of the slot-2 half.

Implemented in part since v1.4.2 (hecke-extensions slot 2, package WP-8, 2026-09-16; `docs/implementation/slot2/WP-8_report.md`): the η-table exists — `Hecke/HeckePairHomotopy` (`hecke::HeckePairHomotopyT<Elt>`), built by Lemma 15.6 for every ℓ ((V) by the pairwise path routing of Algorithm 18.3 step 2, (E) by the unique fill through the one integer solver `Hecke/HeckeIntegerSolve`, moved verbatim out of `Hecke/HeckeTauTable.cpp`), certified in its constructor against Obligation 15.21(i)–(iv) with no mod-m path (a tier failure aborts; owner decision OP2), `eta_zero` recorded ((vi)), `table_fnv64()` and a JSON record `hecke_pair_homotopy/v1` carrying the per-complex fields of §22.4 item 5 in the boundary degrees $[0, 1]$; `HeckePairHomotopyValidation` (tier `standard`) re-derives (i)–(iv) independently on 152 tables, rejects 27 tampered chains, exhibits Lemma 15.7 on two vertex-chain policies, and finds `eta_zero` false on every table (Remark 15.15(1)). **Not implemented** at v1.4.2: the cone $C(r)$, Φ_g, the derived h, $A_c^{(q)}$, clause (v) and Obligation 15.22 — slot 3 of the extension programme, which built them (above). Until v1.4.1 this paragraph recorded that nothing in the chapter was implemented; the census it rested on is kept below. A census of the tree on 2026-09-16 — `grep` over `src/` for `eta`, `PairHomotopy`, `ConeAction`, `HeckeCone`, `EquivariantMappingCone`, `HeckePairDatum`, `TauBoundary` and `Phi_g` — found no η-table, no equivariant cone complex, no Φ_g, no $A_c^{(q)}$ and no `HeckeConeValidation`; none of the 80 registered validations (v1.4.2 count) computed $H_c^\bullet$, and the validation of the η-table was `HeckePairHomotopyValidation`, the cone's to be a second name (since v1.5.2 two: `HeckeConeValidation` and `HeckeConeActionValidation`). (The τ_∂-table the same census also missed exists since v1.3.3 — `Hecke/HeckeBoundaryDatum`, §14's code map — and is this chapter's second input; the pair homotopy η that joins it to the bulk τ is built since v1.4.2.) What exists is (a) the *non-equivariant* algebraic cone of a `CochainComplexMorphism`, `CochainComplexMorphism::ConeComplex` (`SimplicialCohomology.h`:227, `SimplicialCohomology.cpp`:832), the implementation of

`monodromy_v5.tex` §8 on `SimplicialComplex` without a local system — the model for the equivariant cone of (38), not an instance of it; (b) the restriction morphism r^q of the circle model as an `EquivariantCochainComplexMorphismOnK` (§14, code map); and (c) a comment in `EquivariantCohomologyOnK.h`:115 that names "the Cone class" as the intended user of a protected constructor — a class that does not exist. (Since v1.5.2 a cone class exists, `EquivariantMappingCone`, and it does not use that constructor; the comment of (c) still stands in the header.) The *specification* for the phase that builds this (E5 of the extension programme, gated by amendment C-A3): an η-table builder reusing `hecke::HeckeTauTable`'s step-2/3 machinery, (T1) helper and constructor-time certification (invariant I8) with ∂X as source and X as target; a `HeckePairDatum` $\{\tau, \tau_\partial, \eta;$ solve regimes; `integral`$\}$ as tables of `ChainTerm`s over $\mathbb{Z}$, degree-generic per 14.24; an `EquivariantMappingCone` realising (38) on `EquivariantCohomologyOnK` objects; the derived h and Φ_g of (44)–(45) assembled through the block model of §6; the four certificates of the table above with `verify()` re-deriving and never re-running a search; and, per owner decision OP7, golden tiers — $\dim H_c^q$ and the char-poly/rank/trace of $A_c^{(q)}$ for $\ell \geq 5$ in tier A, and Θ_g^∂, h, the cone bases and the three table hashes in tier B, because they depend on the choices Proposition 15.12 shows to be immaterial.

Formalization note. This is the chapter a formalisation should attack first, because it is the one whose statements have no external corroboration: every one of Lemmas 15.2–15.10, Theorem 15.11(a),(b),(d), Propositions 15.12(1),(2)-boundary-half, 15.13, 15.16 and 15.18(1)–(4) is proved here from (37), (B1)–(B3), (H4)(iv), Lemma 10.3 and Proposition 8.2(2) by finite manipulations, and none of them cites anything a proof assistant would have to import beyond the long exact sequence of a short exact sequence of complexes. The recommended order: Lemma 15.4 first — it is four one-line identities and its equivariance clause is the equivariance half of Theorem 11.7 with a parameter — then Lemma 15.8 (pure algebra on a direct sum), Lemma 15.10 (a substitution), Lemma 15.7 (two uses of $Z_2 = 0$, one of $Z_1 = B_1$), and Proposition 15.12(1), which is the theorem that matters and is one displayed computation. The constructive content is Lemmas 15.6 and Proposition 15.13: both are terminating procedures whose correctness reduces to (37) on a 2-dimensional complex and a stabiliser condition, so they need simplicial chains and the homology of a contractible 2-complex, not homological algebra; the hypothesis a formalisation must not lose is (B3) in Lemma 15.6 (three uses: the (T1)-extension on vertices, on edges, and in the strict-pair construction), and (H4)(ii) everywhere. Two things are genuinely harder. Proposition 15.16(2) chooses complements and is therefore classical linear algebra over a field, correctly so restricted (entry A.3). Theorem 15.11(c) and Remark 15.14 depend on the comparison theorem through Lemma 11.3(2) and, for the Steinberg form, on derived-category bookkeeping ($C_\bullet(X, \partial X) \simeq \mathrm{St}[1]$, Ext of a bounded complex of relative projectives) that `mathlib4` has in pieces; they should be attempted after Part III's comparison theorem, and a formalisation may take (iii) of that remark as an axiom about names rather than a theorem, since nothing in the document depends on it. Obligations 15.21 and 15.22 are decidable predicates on emitted finite data in every clause except 15.22(iv) over a non-field ring, which is decidable through a Howell or Smith form once one is available.

16 The six groups and the long exact sequence

What this chapter fixes. The six cohomology groups this project computes or predicts, the exact sequence that relates them, the characters by which the Hecke algebra acts on the two boundary terms, and the duality that pins the two compactly supported ones. Everything here is a *fact about the objects* constructed in Parts I–IV; nothing here is a programme. That is why these nine labels live in this document and not in the companion specification (§1; Appendix A, entry A.18), and it is why Obligation 20.7 may read them.

Standing hypotheses. (H1)–(H4) and (B1)–(B3) are in force, and per rule (A1) every statement repeats the ones it uses. **Three hypotheses recur and are never suppressed**: $6 \in R^\times$ (equivalently $\ell \geq 5$ for R of characteristic ℓ), which is (H3) and is what makes the model complexes compute group cohomology (Remark 7.6); $2 \in R^\times$, which is what makes the circle model compute the boundary cohomology (Proposition 14.4(3)); and n even, without which every group below vanishes for $2 \in R^\times$ (Remark 7.7, Remark 14.5).

16.1 The six groups

CONTRACT. Let R be a commutative ring, $n \geq 0$ and $V_R = \mathrm{Sym}^n(\mathbb{Z}^2) \otimes R$ (at $R = \mathbb{F}_\ell$ also written M_n). The **six groups** are

$$\begin{aligned} X_1 &= H^0(\Gamma; V_R), \quad X_2 = H^0(\partial; V_R), \quad X_3 = H^1_c, \\ X_4 &= H^1(\Gamma; V_R), \quad X_5 = H^1(\partial; V_R), \quad X_6 = H^2_c, \end{aligned} \tag{49}$$

with $H^\bullet(\partial; V_R) := H^\bullet(\Gamma_\infty; V_R)$ (Definition 14.2) and $H^q_c := H^{q-1}(C(r))$ (Definition 15.1, Lemma 15.2(3)). Write $H_{\mathrm{tot}} := \bigoplus_{t=1}^{6} X_t$ and, over a field, $d := \dim H_{\mathrm{tot}}$.

X_1 and X_4 are the groups the shipped pipeline emits, as H^0_{eq} and H^1_{eq} of $(K, \mathcal{R}; L)$; the identification with group cohomology is Remark 7.6 and holds for $6 \in R^\times$. X_2 and X_5 are computed today as the cohomology of the circle model K_∂, which is the $2 \in R^\times$ specialisation of (49) (Proposition 14.4); **no Hecke action on them is implemented** — the operator is Definition 14.14 and the code map of §14.5 says what is missing. X_3 and X_6 are not computable today at all: the cone of Definition 15.1 exists for the non-equivariant `CochainComplex` only. Their Hecke action is *defined* — Definition 15.9, Theorem 15.11 — and that definition is what §15 exists to supply.

Remark 16.1 (What this chapter claims, and two hypotheses it does not inherit). The nine statements below were written in `hecke_extensions.tex`:186–338 under a blanket "$\ell \geq 5$ throughout, n even" declared once at :157–160. Re-homed here under rule (A1), each carries its own hypotheses, and two of them are strictly weaker than the blanket:

1. $H^0_c = 0$ **is unconditional.** Lemma 15.2(4) proves it with no hypothesis on ℓ and none on n: r^0 is the injection $V_R^\Gamma \hookrightarrow V_R^{\Gamma_\infty}$. Claim 16.23 therefore does *not* list it as a hypothesis, where [E] did.
2. n **even is not needed for exactness.** It is needed for the six groups to be non-zero, not for the sequence; Claim 16.23 asks only for the two vanishings it actually uses.

The third hypothesis is not removable. $H^2(\Gamma; V_R) = 0$ genuinely requires an invertibility assumption: Lemma 16.4(2) exhibits $\ell = 2$, $n = 0$ with $H^q(\Gamma; \mathbb{F}_2) \neq 0$ for *every* $q \geq 0$. Claim 16.23 keeps it.

16.2 The amalgam, and what it gives in each degree

Claim 16.2 (Amalgam presentation). *Let R be a commutative ring and $n \geq 0$; put $U := ST$, so that $\langle S\rangle$, $\langle U\rangle$, $\langle -I\rangle$ have orders 4, 6, 2.*

1. *[standard: [Ser80, Ch. I, §4]: the amalgam decomposition of $\mathrm{SL}_2(\mathbb{Z})$ and the correspondence between amalgams and group actions on trees. The pointer is given at chapter-and-section level and is* **not** *verified against the text (rule (A2)): no copy is in `docs/literature/`. (50) is proved independently of this pointer (the markers below), so nothing in this document rests on it; the amalgam decomposition itself is used nowhere else (v1.22.0-pre.12, LR-Y)] $\Gamma = \langle S\rangle *_{\langle -I\rangle} \langle U\rangle$, and the augmented simplicial chain complex of the associated Bass–Serre tree is a short exact sequence of $\mathbb{Z}[\Gamma]$-modules*

$$0 \to \mathbb{Z}[\Gamma/\langle -I\rangle] \to \mathbb{Z}[\Gamma/\langle S\rangle] \oplus \mathbb{Z}[\Gamma/\langle U\rangle] \to \mathbb{Z} \to 0. \tag{50}$$

The vertices of Y (the Γ-triangulation of $\mathcal{H}$ behind M_7, Definition 7.1) lying in the Γ-orbits of ζ^2 and i, with the edges of Y between them, form the tree ($\mathcal{T}$ in Lemma 14.22, $\mathcal{Y}$ in §28): vertex set $\mathcal{T}_0 = \Gamma/\langle S\rangle \sqcup \Gamma/\langle U\rangle$, edge set $\mathcal{T}_1 = \Gamma/\langle -I\rangle$, the edge $\gamma\langle -I\rangle$ joining $\gamma\langle S\rangle$ to $\gamma\langle U\rangle$. [proved here] [formally verified: [GT, `thm:mp-treeses-exact`]] *(exactness of the augmented chain complex of the tree, over every commutative ring, without Serre: the full subcomplex of Y on the Γ-orbits of ζ^2 and i is a retract of the acyclic $C_\bullet(Y)$ of M_7)* [formally verified: [GT, `thm:mp-model7-tree-orbits`]] *(the terms: $\mathcal{T}_1 \cong \Gamma/\langle -I\rangle$, $\mathcal{T}_0 \cong \Gamma/\langle U\rangle \sqcup \Gamma/\langle S\rangle$ as Γ-sets and as permutation modules over every ring, $k[\Gamma/\langle U\rangle \sqcup \Gamma/\langle S\rangle] = k[\Gamma/\langle S\rangle] \oplus k[\Gamma/\langle U\rangle]$ up to* `Finsupp.sumFinsuppLEquivProdFinsupp` *and the order of the summands; the differential is the simplicial boundary, Γ-equivariant)* [formally verified: [GT, `thm:mp-treeses-rep`]] *(the sequence assembled as one short exact sequence in $\operatorname{Rep} k\,\Gamma$ over every commutative ring k, $\gamma\langle -I\rangle \mapsto \gamma\langle S\rangle - \gamma\langle U\rangle$ and every coset $\mapsto 1$, with middle term the permutation module $k[\Gamma/\langle S\rangle \sqcup \Gamma/\langle U\rangle]$, which is $k[\Gamma/\langle S\rangle] \oplus k[\Gamma/\langle U\rangle]$ up to* `Finsupp.sumFinsuppLEquivProdFinsupp`*; exactness transported from [GT]* `thm:mp-treeses-exact` *along the orbit–stabiliser bijections).*

2. [proved here] [formally verified: [GT, `thm:mp-tree-mv`]] *(the long exact sequence of (50) for every R, every $R[\Gamma]$-module V (in particular V_R) and every q, built on the periodic cochain complexes $V^2 \to V^2 \to \cdots$ ($(1-S, 1-U)$, (N_S, N_U) alternating) and $V \to V \to \cdots$ ($1-(-I)$, $1+(-I)$ alternating, shifted up by one); the middle map is induced by φ^* of Remark 16.3 on cocycles; the maps into and out of $H^q(\Gamma; V)$ go through the comparison $H^q(\Gamma; V) \cong H^q(\operatorname{Hom}_\Gamma(P_\bullet, V))$ built on* Ext *(mathlib's* `groupCohomologyIso`*; [GT]* `thm:mp-tree-cochains`*))* [formally verified: [GT, `thm:mp-mv-terms`]] *(the terms: these periodic complexes compute $H^q(\langle S\rangle; V) \oplus H^q(\langle U\rangle; V)$ and $H^q(\langle -I\rangle; V)$ as mathlib's* `groupCohomology` *of the subgroups, every q, through evaluation at the generators and a fixed homotopy equivalence with the bar resolution; the identification of the edge term is the identity on cocycles)* [formally verified: [GT, `thm:mp-mv-resres`]] *(under those identifications the middle map is $\mathrm{res}_S - \mathrm{res}_U$, mathlib's restriction along $\langle -I\rangle \subset \langle S\rangle$ and $\langle -I\rangle \subset \langle U\rangle$, with the sign as displayed, every q; on the periodic cochains restriction is $x \mapsto x$ in even and $x \mapsto (1+S)x$, resp. $(1+U+U^2)x$, in odd degrees)* [formally verified: [GT, `thm:mp-mv-low`]] *(the beginning $0 \to V^\Gamma \to V^S \oplus V^U \to V^{-I}$, $a \mapsto (a,a)$, $(x,y) \mapsto x - y$, is exact and $V^\Gamma = V^S \cap V^U$, proved directly, not as the degree-0 case of the sequence; restriction on H^0 is the inclusion of invariants under mathlib's $H^0(G;V) \cong V^G$)* [formally verified: [GT, `thm:mp-mv-res`]] *(through the comparison $H^q(\Gamma; V) \cong H^q(\operatorname{Hom}_\Gamma(P_\bullet, V))$ built on a fixed homotopy equivalence with the bar resolution (not the* Ext*-built one of the first marker) and the identifications of [GT]* `thm:mp-mv-terms`*, the map $H^q(\Gamma; V) \to H^q(\langle S\rangle; V) \oplus H^q(\langle U\rangle; V)$ of the sequence is $(\mathrm{res}_S, \mathrm{res}_U)$, mathlib's restriction, every q, $q = 0$ included; on cochains the projections $((x,y),z) \mapsto x$, y)* [formally verified: [GT, `thm:mp-mv-les`]] *(the whole sequence (51) in* `groupCohomology` *terms, under the same comparison and the identifications of [GT]* `thm:mp-mv-terms`*, exact at every term, $q = 0$ included ($H^0(\Gamma; V) \to H^0(\langle S\rangle; V) \oplus H^0(\langle U\rangle; V)$ injective), with maps $(\mathrm{res}_S, \mathrm{res}_U)$, $\mathrm{res}_S - \mathrm{res}_U$ and a connecting map $H^q(\langle -I\rangle; V) \to H^{q+1}(\Gamma; V)$ defined by transport of the inclusion of the edge part of $\operatorname{Hom}_\Gamma(P_\bullet, V)$ and characterised on cocycles ($[z] \mapsto [((0,0),z)]$); the connecting map is not identified with that of mathlib's* `groupCohomology` *sequence of a coefficient sequence, and the agreement of the bar-built and the* Ext*-built comparisons is not proved)* [formally verified: [GT, `thm:mp-mv-natural`]] *(naturality in V: each of the three maps commutes with the maps induced by a morphism $V \to V'$ of $R[\Gamma]$-modules and its restrictions to $\langle S\rangle$, $\langle U\rangle$, $\langle -I\rangle$, every q)* ***(Mayer–Vietoris, every R, every n, every degree.)*** *(50) yields a long exact sequence, natural in V_R,*

$$\cdots \to H^q(\Gamma; V_R) \to H^q(\langle S\rangle; V_R) \oplus H^q(\langle U\rangle; V_R) \xrightarrow{\mathrm{res} - \mathrm{res}} H^q(\langle -I\rangle; V_R) \to H^{q+1}(\Gamma; V_R) \to \cdots \tag{51}$$

beginning $0 \to H^0(\Gamma;V_R) \to V_R^S \oplus V_R^U \to V_R^{-I}$.

3. *[proved here]* ***(Consequences for*** $6 \in R^\times$***.)*** *If* $6 \in R^\times$ *then* $H^q(\Gamma;V_R) = 0$ *for every* $q \geq 2$*;* $H^0(\Gamma;V_R) = V_R^S \cap V_R^U$*; and, for* n *even,*

$$H^1(\Gamma;V_R) \;=\; V_R/\big((1+\rho_n(S))V_R + (1+\rho_n(U)+\rho_n(U)^2)V_R\big).$$

If moreover R *is a field, then for* n *even* $\dim H^0(\Gamma;V_R) - \dim H^1(\Gamma;V_R) = \dim V_R^S + \dim V_R^U - \dim V_R$*. (The letter* χ *is* not *used for this Euler characteristic: it is the boundary character of Claim 16.24, §2.5.)*

4. *[proved here]* ***(Torsion bound, every*** R***, every*** n***.)*** $24 \cdot H^q(\Gamma;V_R) = 0$ *for every* $q \geq 2$*; the sharp classical constant is* 12*,* Γ *having a free subgroup of index* 12 *[standard: [Ser80, Ch. I, §4] for free subgroups of a free product of finite groups, with the same rule (A2) caveat; the commutator subgroup of* $\mathrm{SL}_2(\mathbb{Z})$ *is the classical witness,* $\mathrm{SL}_2(\mathbb{Z})^{\mathrm{ab}} = \mathbb{Z}/12$*]. Either constant gives (3)'s vanishing, since both are invertible when* 6 *is.*

Proof. (2) For each of the three subgroups $\Gamma_0 \in \{\langle S\rangle, \langle U\rangle, \langle -I\rangle\}$, $\mathbb{Z}[\Gamma/\Gamma_0] = \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_0]} \mathbb{Z}$ is an induced module, so $\mathrm{Ext}^q_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma/\Gamma_0], V_R) \cong H^q(\Gamma_0;V_R)$ [standard: [Bro82, Ch. III.6], Shapiro's lemma, as in the proof of Proposition 14.4(1)]; applying $\mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(-,V_R)$ to (50) and using $\mathrm{Ext}^q_{\mathbb{Z}[\Gamma]}(\mathbb{Z},V_R) = H^q(\Gamma;V_R)$ gives (51), with the connecting maps of the Ext sequence, and naturality in V_R is naturality of that sequence. The initial segment is $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma/\Gamma_0], V_R) = V_R^{\Gamma_0}$ together with $H^0(\Gamma;V_R) = V_R^\Gamma$.

(3) For a finite subgroup $\Gamma_0 \leq \Gamma$ and $q \geq 1$, $|\Gamma_0| \cdot H^q(\Gamma_0;V_R) = 0$ [standard: [Bro82, Ch. III.10.2] (Corollary 10.2, p. 84), from $\mathrm{cor} \circ \mathrm{res} = [\Gamma_0 : 1]$ ([Bro82, Prop. III.9.5(ii), p. 82]) on $H^q(\Gamma_0;-)$ for $q \geq 1$, where $H^q(1;-) = 0$]; with $4, 6, 2 \in R^\times$ every term of (51) in a positive degree vanishes, so $H^q(\Gamma;V_R) = 0$ for $q \geq 2$ and $H^1(\Gamma;V_R) = \mathrm{coker}(V_R^S \oplus V_R^U \to V_R^{-I})$. For n even $\rho_n(-I) = \mathrm{id}$ (Proposition 8.2(2)), so $V_R^{-I} = V_R$; and $\rho_n(S)^2 = \rho_n(U)^3 = \rho_n(-I) = \mathrm{id}$, so the averaging idempotents of $\langle S\rangle$ and $\langle U\rangle$ are $\frac{1}{2}(1+\rho_n(S))$ and $\frac{1}{3}(1+\rho_n(U)+\rho_n(U)^2)$, whence $V_R^S = (1+\rho_n(S))V_R$ and $V_R^U = (1+\rho_n(U)+\rho_n(U)^2)V_R$; the image of the difference map is their sum. $H^0(\Gamma;V_R) = V_R^S \cap V_R^U$ is exactness at the second term, S and U generating Γ. The Euler characteristic is the alternating count on $0 \to H^0 \to V_R^S \oplus V_R^U \to V_R \to H^1 \to 0$.

(4) In (51) at $q \geq 2$, exactness gives $0 \to C \to H^q(\Gamma;V_R) \to A \to 0$ with A a submodule of $H^q(\langle S\rangle;V_R) \oplus H^q(\langle U\rangle;V_R)$ and C a quotient of $H^{q-1}(\langle -I\rangle;V_R)$. By the annihilator cited in (3), $\mathrm{lcm}(4,6) = 12$ kills A and 2 kills C (note $q-1 \geq 1$), so 24 kills $H^q(\Gamma;V_R)$. For the sharp constant: if $F \leq \Gamma$ is free of index 12 then $H^q(F;V_R) = 0$ for $q \geq 2$ (F has cohomological dimension ≤ 1, its Cayley graph being a contractible free F-complex of dimension 1), and $\mathrm{cor} \circ \mathrm{res} = 12$ on $H^q(\Gamma;V_R)$ [standard: [Bro82, Ch. III.9–10]]. □

Code map. The presentation of Claim 16.2(3) is the independent H^1 route of `Hecke/BarCocycleH1.cpp`: it presents $\Gamma = \langle S\rangle *_{\langle S^2\rangle} \langle ST\rangle$, represents a 1-cocycle by the pair $(\phi(S), \phi(ST))$ subject to the norm and amalgam-matching conditions, enforces $\ell \geq 5$ at construction — which is exactly hypothesis (3) — and compares characteristic and minimal polynomials against $A^{(1)}$ (Proposition 13.4, §13.4).

Remark 16.3 (The tree resolution over every ring). [proved here] [formally verified: [GT, `thm:mp-model7-cone-res`]] (the free resolution $P_\bullet$, acyclic in every degree) [formally verified: [GT, `thm:mp-tree-mp-comparison`]] ($P_\bullet$ is a projective resolution; $H^q(\Gamma;V) \cong H^q(\mathrm{Hom}_\Gamma(P_\bullet,V)) \cong H^q(C^\bullet_{\mathrm{un}}(\mathsf{S}(\mathsf{M}_7);V))$; the homotopy equivalence with the MP resolution is the abstract one; the normalised $C^\bullet$ through Theorem 32.12(4)) [formally verified: [GT, `thm:mp-tree-cochains`]] ($\mathrm{Hom}_\Gamma(P_\bullet,V) \cong (V^2 \to V^3 \to V^3 \to \cdots)$ by evaluation at the generators, hence $H^q(\Gamma;V) \cong H^q(V^2 \to V^3 \to \cdots)$, every commutative ring, every V, every q, $-I$ included) Let $P^{\langle S\rangle}$, $P^{\langle U\rangle}$, $P^{\langle -I\rangle}$ be the 2-periodic free resolutions of $\mathbb{Z}[\Gamma/\langle S\rangle]$, $\mathbb{Z}[\Gamma/\langle U\rangle]$, $\mathbb{Z}[\Gamma/\langle -I\rangle]$ induced from the cyclic groups ($\cdots \xrightarrow{N} \mathbb{Z}[\Gamma] \xrightarrow{1-t} \mathbb{Z}[\Gamma] \to \mathbb{Z}[\Gamma/\langle t\rangle]$, right multiplication). The boundary of (50) is oriented as $\mathfrak{e} \mapsto \mathfrak{v}_S - \mathfrak{v}_U$ (the edge $[\zeta^2, i]$ runs from ζ^2 to i). This is

the **negative** of the convention $\partial(\gamma\mathfrak{e}) = \gamma\mathfrak{v}_U - \gamma\mathfrak{v}_S$ of the tree $\mathcal{Y}$ of §28 (its notation paragraph), which moves no statement. The boundary lifts to $\varphi\colon P^{\langle -I\rangle} \to P^{\langle S\rangle} \oplus P^{\langle U\rangle}$, with $\varphi = (1, -1)$ in even and $(1+S, -(1+U+U^2))$ in odd degrees, in the order $(\langle S\rangle, \langle U\rangle)$. The mapping cone of φ, augmented to $\mathbb{Z}$, is a free resolution $P_\bullet\colon \mathbb{Z}[\Gamma]^2 \leftarrow \mathbb{Z}[\Gamma]^3 \leftarrow \mathbb{Z}[\Gamma]^3 \leftarrow \cdots$ of $\mathbb{Z}$ over $\mathbb{Z}[\Gamma]$. Its terms are 2-periodic from degree 1 and its differentials from degree 2: $P_{n+2} = P_n$ for $n \geq 1$ and $\partial_{n+2} = \partial_n$ for $n \geq 2$; dually $\mathrm{Hom}_\Gamma(P_\bullet, V)$ has $C^{n+2} = C^n$ and $d^{n+2} = d^n$ for every $n \geq 1$. For every commutative ring R the same holds over $R[\Gamma]$. So for every $R[\Gamma]$-module V and every q, $H^q(\Gamma; V) \cong H^q(\mathrm{Hom}_\Gamma(P_\bullet, V)) \cong H^q(C^\bullet(\mathsf{S}(\mathsf{M}_7); V))$ (Theorem 32.12). With $N_S = \sum_{j<4} S^j$, $N_U = \sum_{j<6} U^j$, and S, U, $-I$ acting on V through its structure map, $\mathrm{Hom}_\Gamma(P_\bullet, V)$ is the cochain complex $V^2 \to V^3 \to V^3 \to \cdots$ with $d^0(x,y) = ((1-S)x, (1-U)y, x-y)$ and $d^{m+1}((x,y),z) = (v_{m+1}(x,y), \varphi^*_{m+1}(x,y) - e_m z)$, where $v_j = (1-S, 1-U)$ and $\varphi^*_j(x,y) = x - y$ for even j, $v_j = (N_S, N_U)$ and $\varphi^*_j(x,y) = (1+S)x - (1+U+U^2)y$ for odd j, and $e_m = 1 \mp (-I)$ for m even/odd. $P_\bullet$ and $R[\mathfrak{F}'_\bullet]$ are homotopy equivalent, as any two projective resolutions of R are; the equivalence is not computed. When $6 \in R^\times$, Lemma 28.4's length-1 resolution $C_\bullet(\mathcal{Y}) \otimes R$ is a second projective resolution. $P_\bullet$ needs no unit: its terms are free.

Lemma 16.4 ($H^2(\Gamma; V_R) = 0$ needs its hypothesis)**.** *1. [proved here] If $6 \in R^\times$ then $H^q(\Gamma; V_R) = 0$ for every $q \geq 2$ — Claim 16.2(3). This is the hypothesis under which §13.4 reports $A^{(2)}$ as the 0×0 matrix, and under which $H^2_{\mathrm{eq}}(K, \mathcal{R}; L) = 0$.*

*2. [proved here] **It is not removable.** Let $R = \mathbb{F}_2$ and $n = 0$, so $V_R = \mathbb{F}_2$ with trivial Γ-action. Then $H^q(\Gamma; \mathbb{F}_2) \neq 0$ for every $q \geq 0$; in particular $H^2(\Gamma; \mathbb{F}_2) \neq 0$, and the hypothesis $H^2(\Gamma; M) = 0$ of Claim 16.23 fails at $(\ell, n) = (2, 0)$, a pair with n even.*

Proof. (2) With trivial coefficients $\mathbb{F}_2$: $H^q(\langle -I\rangle; \mathbb{F}_2) = \mathbb{F}_2$ for every $q \geq 0$ (the periodic resolution of step (ii) in the proof of Proposition 14.3, with $2 = 0$ in $\mathbb{F}_2$, gives $\mathbb{F}_2 \xrightarrow{0} \mathbb{F}_2 \xrightarrow{0} \cdots$); $H^q(\langle S\rangle; \mathbb{F}_2) = \mathbb{F}_2$ for every $q \geq 0$ by the analogous periodic resolution of a cyclic group of order 4, whose two maps are $g - 1 = 0$ and the norm $4 = 0$; and $H^q(\langle U\rangle; \mathbb{F}_2) = H^q(\langle -I\rangle; \mathbb{F}_2) = \mathbb{F}_2$, because $\langle U\rangle \cong \mathbb{Z}/2 \times \mathbb{Z}/3$ with $3 \in \mathbb{F}_2^\times$, so the Lyndon–Hochschild–Serre sequence of that product collapses to $H^q(\mathbb{Z}/2; \mathbb{F}_2^{\mathbb{Z}/3}) = H^q(\mathbb{Z}/2; \mathbb{F}_2)$ [standard: [Bro82, Ch. VII.6], as in the proof of Proposition 14.3]. Now read (51): for every $q \geq 0$ the map $H^q(\langle S\rangle; \mathbb{F}_2) \oplus H^q(\langle U\rangle; \mathbb{F}_2) \to H^q(\langle -I\rangle; \mathbb{F}_2)$ has a 2-dimensional source and a 1-dimensional target, so its kernel is non-zero; by exactness $H^q(\Gamma; \mathbb{F}_2)$ surjects onto that kernel, and is therefore non-zero. □

This settles, for $n = 0$, the question Remark 14.21(2) left open. That remark records that no argument in this document bounds $H^\bullet(\Gamma; V_R)$ above at $\ell = 2$ and does not assert non-vanishing in every degree; Lemma 16.4(2) asserts it for the trivial module, by the amalgam alone and with no appeal to Farrell–Tate cohomology. The general-n statement at $\ell = 2$ is *not* claimed here: the same count needs $\dim V_R^S + \dim V_R^U > \dim V_R$, which holds at $n = 0$ and is not automatic.

16.3 The dimension of H^1, and the Dickson correction

Claim 16.5 (Dimension of H^1)**.** *Let $\ell \geq 5$, $R = \mathbb{F}_\ell$, $n > 0$ even, $k := n + 2$, and let*

$$d_k := \begin{cases} \lfloor k/12 \rfloor & k \equiv 2 \pmod{12}, \\ \lfloor k/12 \rfloor + 1 & \text{otherwise}, \end{cases} \qquad k \text{ even},\ k \geq 4,$$

be the dimension of the space of level-1 modular forms of weight k, its subspace of cusp forms having dimension $d_k - 1$ [standard: [Ser73, Ch. VII, §3]] — the same d_k as in Obligation 9.10(v). Then

1. [standard: [Shi71, Ch. 8] for Eichler–Shimura over $\mathbb{C}$; [Cre97, Ch. 2] for its modular-symbol form; [Den23] for the mod-ℓ statement with the correction term]

$$\dim_{\mathbb{F}_\ell} H^1(\Gamma; M_n) \;=\; \underbrace{d_k}_{\text{modular forms}} \;+\; \underbrace{(d_k - 1)}_{\text{cusp forms}} \;+\; \dim_{\mathbb{F}_\ell} H^0(\Gamma; M_n) \;=\; 2d_k - 1 + \dim_{\mathbb{F}_\ell} H^0(\Gamma; M_n).$$

The first two terms are the Eichler–Shimura decomposition of $H^1(\Gamma;\mathrm{Sym}^n\mathbb{C}^2)$ into the weight-k level-1 modular forms and a conjugate copy of its cusp subspace, of dimensions d_k and d_k-1 (Obligation 9.10(v) uses the same two numbers); the third is the Dickson correction of (2), invisible over $\mathbb{C}$.

2. *[proved here] (v1.23.0-pre.15, proved in place, and since v1.23.0-pre.16 as Lemma 16.6, which holds at every prime ℓ; until v1.23.0-pre.14 this clause was [standard:] on [Wil83], a pointer never verified against the text, rule (A2); Dickson's theorem, [Dic11], [Wil83], is the classical source, and nothing here rests on either pointer)* $\mathbb{F}_\ell[x,y]^\Gamma=\mathbb{F}_\ell[x,y]^{\mathrm{SL}_2(\mathbb{F}_\ell)}$ *— the reduction $\Gamma\to\mathrm{SL}_2(\mathbb{F}_\ell)$ being onto, since $\mathrm{SL}_2(\mathbb{F}_\ell)$ is generated by the elementary matrices $\left(\begin{smallmatrix}1&c\\0&1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}1&0\\c&1\end{smallmatrix}\right)$, which lift — is a polynomial ring on two homogeneous invariants of degrees $\ell+1$ and $\ell^2-\ell$, so*

$$\dim_{\mathbb{F}_\ell} H^0(\Gamma;M_n)=\#\big\{(a,b)\in\mathbb{Z}^2_{\geq 0}:a(\ell+1)+b(\ell^2-\ell)=n\big\},$$

*which, for $\ell\geq 3$ (the standing hypothesis $\ell\geq 5$ is more than needed), is 0 for $0<n<\ell+1$ and first non-zero at $n=\ell+1$, where the invariant is $x^\ell y-xy^\ell$ (at $\ell=2$ it is non-zero already at $n=2$, the invariant $Q=d_2$: Lemma 16.6). The locus of (n,ℓ) on which it is non-zero is the **Dickson locus**; Remark 20.8 is where its small-ℓ exceptional behaviour is recorded.*

Proof of (2). Lemma 16.6 below, stated and proved for every prime ℓ (v1.23.0-pre.16, referee D-1; in v1.23.0-pre.15 this proof stood here, and the three sites that used it at $\ell=2$ or at every ℓ cited it). $\mathbb{F}_\ell[x,y]^\Gamma=S^G$ because Γ acts through its image $\mathrm{SL}_2(\mathbb{F}_\ell)$, onto as stated, and the count is that of the basis $\{L^aQ^b\}$ of the degree-n part. □

Lemma 16.6 (The Dickson invariants of $\mathrm{SL}_2(\mathbb{F}_\ell)$, every prime ℓ)**.** *[proved here] For* every *prime ℓ (no hypothesis $\ell\geq 5$, none on n), with $L:=x^\ell y-xy^\ell$ and $Q:=(x^{\ell^2}y-xy^{\ell^2})/L$,*

$$\mathbb{F}_\ell[x,y]^{\mathrm{SL}_2(\mathbb{F}_\ell)}=\mathbb{F}_\ell[L,Q]\quad(\textit{these are }\mathsf{L},\mathsf{D}\textit{ of Definition 24.1}),$$

a polynomial ring on homogeneous generators of degrees $\ell+1$ and $\ell^2-\ell$, so $\dim_{\mathbb{F}_\ell}\big(\mathbb{F}_\ell[x,y]^{\mathrm{SL}_2(\mathbb{F}_\ell)}\big)_n=\#\{(a,b)\in\mathbb{Z}^2_{\geq 0}:a(\ell+1)+b(\ell^2-\ell)=n\}$ for every $n\geq 0$; and $\mathbb{F}_\ell[x,y]^{\mathrm{GL}_2(\mathbb{F}_\ell)}=\mathbb{F}_\ell[L^{\ell-1},Q]$. (The count vanishes for $0<n<\ell+1$ only when $\ell\geq 3$; at $\ell=2$ the degrees are 3 and 2 and the count is 1 at $n=2$.)

Proof. (v1.23.0-pre.15, in place of the pointer [Wil83]; finding F1 of `docs/implementation/slot24/C-D1_refer` made a lemma at v1.23.0-pre.16, referee D-1.) Nothing below uses $\ell\geq 5$: it holds for every prime ℓ. By Definition 8.1, M_n is the degree-n part of $S:=\mathbb{F}_\ell[x,y]$, γ acting by the $\mathbb{F}_\ell$-algebra automorphism with $(\gamma x,\gamma y)=(x,y)\gamma$, i.e. $\gamma x=ax+cy$, $\gamma y=bx+dy$. Γ acts through its image $\mathrm{SL}_2(\mathbb{F}_\ell)$ (onto, as stated), so $\mathbb{F}_\ell[x,y]^\Gamma=S^G$, where G is the group of the automorphisms g_A, $(g_Ax,g_Ay)=(x,y)A$, $A\in\mathrm{SL}_2(\mathbb{F}_\ell)$; A is read off from g_Ax, g_Ay, so $|G|=|\mathrm{SL}_2(\mathbb{F}_\ell)|=\ell(\ell^2-1)$. *Step 1 (the invariant L).* Put $L:=x^\ell y-xy^\ell=-\det\left(\begin{smallmatrix}x&y\\x^\ell&y^\ell\end{smallmatrix}\right)$. As $a^\ell=a$ on $\mathbb{F}_\ell$, $(g_Ax^\ell,g_Ay^\ell)=(x^\ell,y^\ell)A$, so the matrix $\left(\begin{smallmatrix}g_Ax&g_Ay\\g_Ax^\ell&g_Ay^\ell\end{smallmatrix}\right)$ is $\left(\begin{smallmatrix}x&y\\x^\ell&y^\ell\end{smallmatrix}\right)A$, and $g_AL=\det(A)\,L=L$. *Step 2 (the invariant Q; integrality).* Let $V:=\mathbb{F}_\ell x+\mathbb{F}_\ell y\subset S$ (ℓ^2 elements) and $F(T):=\prod_{v\in V}(T-v)\in S[T]$. Each g_A restricts to an $\mathbb{F}_\ell$-linear bijection of V, so it permutes the factors, and the coefficients of F lie in S^G. Let $\Delta(T):=\det\left(\begin{smallmatrix}x&y&T\\x^\ell&y^\ell&T^\ell\\x^{\ell^2}&y^{\ell^2}&T^{\ell^2}\end{smallmatrix}\right)\in S[T]$. For $v=\alpha x+\beta y\in V$, $v^{\ell^i}=\alpha x^{\ell^i}+\beta y^{\ell^i}$, so the last column at $T=v$ is a combination of the first two and $\Delta(v)=0$. The $v\in V$ are pairwise distinct and S is a domain, so dividing by

the monic $T - v$ one root at a time gives $F \mid \Delta$ in $S[T]$; Δ has degree ℓ^2 in T with leading coefficient $\det \begin{pmatrix} x & y \\ x^\ell & y^\ell \end{pmatrix} = -L \neq 0$, so $\Delta = -L\,F$. Expanding Δ along its last column, $\Delta = (-L)^\ell\, T + L_2\, T^\ell - L\, T^{\ell^2}$ with $L_2 := x^{\ell^2} y - x y^{\ell^2}$; dividing by $-L$ (signs are immaterial at $\ell = 2$),

$$F(T) \;=\; T^{\ell^2} - Q\, T^\ell + L^{\ell-1}\, T, \qquad Q := L_2/L \in S,$$

Q homogeneous of degree $\ell^2 - \ell$ and G-invariant, as a coefficient of F. (L, Q are L, D of Definition 24.1, Lemma 24.2(2).) Put $\mathcal{D} := \mathbb{F}_\ell[L, Q] \subseteq S^G$. x and y are roots of the monic $F \in \mathcal{D}[T]$, so $S = \mathcal{D}[x, y]$ is integral over $\mathcal{D}$. *Step 3 (the fraction fields).* Let $\mathcal{K} := \mathbb{F}_\ell(x, y)$, on which G acts by field automorphisms, and $k := \operatorname{Frac} \mathcal{D} \subseteq \mathcal{K}^G$. *Upper bound.* $x \neq 0$ is a root of $F(T)/T = T^{\ell^2-1} - Q\, T^{\ell-1} + L^{\ell-1} \in \mathcal{D}[T]$, so $[k(x) : k] \leq \ell^2 - 1$; and $L = -x\,(y^\ell - x^{\ell-1} y)$, so y is a root of $T^\ell - x^{\ell-1} T + L/x \in k(x)[T]$ and $[\mathcal{K} : k(x)] \leq \ell$, as $\mathcal{K} = k(x, y)$. Hence $[\mathcal{K} : k] \leq \ell(\ell^2 - 1) = |G|$. *Lower bound.* $\theta := x + y^2 \in \mathcal{K}$ has trivial stabiliser in G: $g_A \theta = \theta$ for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ reads $(ax + cy) + (bx + dy)^2 = x + y^2$ in S; comparing the coefficients of x, y (degree 1) and of x^2, xy, y^2 (degree 2) gives $a = 1$, $c = 0$, $b^2 = 0$, $2bd = 0$, $d^2 = 1$, so $b = 0$ (at $\ell = 2$ the xy-condition is empty, but $b^2 = 0$ already forces it), and $\det A = d = 1$. So the $|G|$ elements $g\theta$ are pairwise distinct. By the upper bound $\mathcal{K}$ is algebraic over $k \subseteq \mathcal{K}^G$, so θ has a minimal polynomial over $\mathcal{K}^G$, whose coefficients every g fixes, so each $g\theta$ is one of its roots; hence $[\mathcal{K} : \mathcal{K}^G] \geq |G|$. From $k \subseteq \mathcal{K}^G \subseteq \mathcal{K}$: $|G| \geq [\mathcal{K} : k] = [\mathcal{K} : \mathcal{K}^G]\,[\mathcal{K}^G : k] \geq |G|\,[\mathcal{K}^G : k]$, so $\mathcal{K}^G = k$. Moreover $\mathcal{K}$, of transcendence degree 2 over $\mathbb{F}_\ell$, is algebraic over $k = \mathbb{F}_\ell(L, Q)$, so L and Q are algebraically independent: $\mathcal{D}$ is a polynomial ring in L, Q. *Step 4 (conclusion).* $S^G \subseteq \mathcal{K}^G = k$, and every $f \in S^G$ is integral over $\mathcal{D}$ (Step 2). The polynomial ring $\mathcal{D}$ is factorial, hence integrally closed: if $f = a/b$ with $a, b \in \mathcal{D}$ coprime and $f^m + c_{m-1} f^{m-1} + \cdots + c_0 = 0$ ($c_i \in \mathcal{D}$), then $a^m = -b\,(c_{m-1} a^{m-1} + \cdots + c_0 b^{m-1})$, so every prime factor of b divides a, and b is a unit. Hence $S^G = \mathcal{D} = \mathbb{F}_\ell[L, Q]$, and since L, Q are algebraically independent and homogeneous of degrees $\ell + 1$, $\ell^2 - \ell$, the monomials $L^a Q^b$ with $a(\ell + 1) + b(\ell^2 - \ell) = n$ are a basis of $(S^G)_n = H^0(\Gamma; M_n)$: the displayed count. At $n = \ell + 1$ (for $\ell \geq 3$) the basis is $\{L\}$. *The* $\mathrm{GL}_2(\mathbb{F}_\ell)$ *statement* (v1.23.0-pre.16, referee D-3). Steps 1–2 hold verbatim for every $A \in \mathrm{GL}_2(\mathbb{F}_\ell)$, as $a^\ell = a^{\ell^2} = a$: $g_A L = \det(A)\, L$ and $g_A L_2 = \det(A)\, L_2$, so $g_A Q = Q$ and $g_A(L^a Q^b) = \det(A)^a L^a Q^b$. The $L^a Q^b$ are a basis of $S^{\mathrm{SL}_2(\mathbb{F}_\ell)}$, so they are eigenvectors of $\mathrm{GL}_2(\mathbb{F}_\ell)$ and the invariants are spanned by those with $\det(A)^a = 1$ for all A, i.e. $(\ell - 1) \mid a$ (det is onto the cyclic group $\mathbb{F}_\ell^\times$ of order $\ell - 1$): $S^{\mathrm{GL}_2(\mathbb{F}_\ell)} = \mathbb{F}_\ell[L^{\ell-1}, Q]$. (An exact check of Steps 1–3 — the identity for F, the invariance of L and Q, the trivial stabiliser of θ, at $\ell \in \{2, 3, 5, 7\}$ — is `docs/implementation/slot24/cd1fix_dickson_check.py`; the referee's 800-cell check of the count is `C-D1_referee.md` §2. Neither is a tag.) □

[verified computationally] **Corroboration.** Both clauses were checked against the shipped campaign table `docs/matrix-engine/results/results_table.tsv` (2026-09-16): over its 2118 rows with $\ell \geq 5$ and $n > 0$ — 710 distinct pairs (n, ℓ) with $n \leq 1000$ and $\ell \in \{5, 7, 11, 13, 691, 10007, 134217757, 2^{61} - 1, 9223372036854775783\}$, of which 203 lie on the Dickson locus — the emitted $\dim H^0_{\mathrm{eq}}$ equals the count of (2) and the emitted $\dim H^1_{\mathrm{eq}}$ equals $2d_k - 1 + \dim H^0_{\mathrm{eq}}$ in *every* row, with 0 disagreements, and $\dim H^2_{\mathrm{eq}} = 0$ throughout. This is a re-reading of an existing artefact, not a new registered validation; $\dim H^\bullet_{\mathrm{eq}} = \dim H^\bullet(\Gamma; M_n)$ uses Remark 7.6 and therefore $\ell \geq 5$.

Claim 16.7 (Torsion origin of the Dickson classes)**.** *[proved here] Let $n > 0$, $\ell \geq 5$, $V = \mathrm{Sym}^n(\mathbb{Z}^2)$ and $M_n = V \otimes \mathbb{F}_\ell$.*

1. *$H^q(\Gamma; V)$ is annihilated by 24 for every $q \geq 2$ (Claim 16.2(4)), hence has no ℓ-torsion; and $H^0(\Gamma; V) = 0$, an integral $\mathrm{SL}_2(\mathbb{Z})$-invariant of positive degree being 0.*
2. *Consequently the long exact cohomology sequence of $0 \to V \xrightarrow{\ell} V \to M_n \to 0$ gives $H^1(\Gamma; M_n) = H^1(\Gamma; V) \otimes \mathbb{F}_\ell$ and $H^1(\Gamma; V)[\ell] \cong H^0(\Gamma; M_n)$, the second isomorphism being*

the Bockstein. ***The Dickson classes are therefore reductions of ℓ-torsion:*** *invisible to characteristic-zero Eichler–Shimura, which is why they appear in Claim 16.5(1) as a correction term and not as forms.*

3. *Both isomorphisms are equivariant for every double-coset operator* $\mathrm{cor}\circ(\mathrm{conj}_g)_*\circ\mathrm{res}$, $g\in\Delta$, *hence — under* $6\in R^\times$, *by Theorem 11.10 — for* $T_{\Gamma g\Gamma}$ *and every operator of §10 built from it.*

Reference for the classical form of (2): [Den23].

Proof. (1) The first half is Claim 16.2(4) with $R=\mathbb{Z}$; 24 is prime to $\ell\geq 5$. For the second, let $f\in V^\Gamma$, viewed as a binary form of degree n over $\mathbb{Q}$. Invariance under T reads $f(x,y)=f(x,x+y)$ (Definition 8.1), hence $f(x,y)=f(x,y+tx)$ for every $t\in\mathbb{Z}$; for fixed $x\neq 0$ the polynomial $f(x,-)$ of degree $\leq n$ takes one value at infinitely many points, so it is constant and $f\in\mathbb{Q}[x]$, i.e. $f=c\,x^n$. Invariance under S then gives $c\,x^n=\rho_n(S)(c\,x^n)=c\,y^n$, so $c=0$ for $n>0$.

(2) No cochain complex is needed: $0\to V\xrightarrow{\ell}V\to M_n\to 0$ is a short exact sequence of $\mathbb{Z}[\Delta]$-modules (V is $\mathbb{Z}$-free, and multiplication by ℓ is Δ-equivariant), and $H^\bullet(\Gamma;-)$ is a δ-functor, so there is a long exact sequence

$$\cdots\to H^q(\Gamma;V)\xrightarrow{\ell}H^q(\Gamma;V)\to H^q(\Gamma;M_n)\xrightarrow{\beta}H^{q+1}(\Gamma;V)\xrightarrow{\ell}\cdots,$$

β its connecting map, i.e. the Bockstein [standard: [Bro82, Ch. III, §§0–1]: the long exact cohomology sequence of a short exact sequence of coefficient modules]. It gives $0\to H^q(\Gamma;V)/\ell\to H^q(\Gamma;M_n)\to H^{q+1}(\Gamma;V)[\ell]\to 0$ in every degree. At $q=1$ the right-hand term is $H^2(\Gamma;V)[\ell]=0$ by (1), giving $H^1(\Gamma;M_n)=H^1(\Gamma;V)/\ell=H^1(\Gamma;V)\otimes\mathbb{F}_\ell$; at $q=0$ the left-hand term is $H^0(\Gamma;V)/\ell=0$ by (1), giving $H^0(\Gamma;M_n)\cong H^1(\Gamma;V)[\ell]$ through β.

(3) Each of res, $(\mathrm{conj}_g)_*$ and cor is a morphism of δ-functors in the coefficient module, so the composite commutes with the connecting maps of the sequence in (2); the sequence is one of Δ-modules, which is what makes $(\mathrm{conj}_g)_*$ act on all three terms at once. □

Remark 16.8 (The exponents of the Dickson torsion: ℓ^2-torsion at $n=\ell m$, unbounded along $n=\ell^j m$). Claim 16.7 fixes the *number* of ℓ-primary cyclic summands of $H^1(\Gamma;V)$ — $\dim H^0(\Gamma;M_n)$, the Dickson count — and says nothing about their *exponents*. This remark records what is known about them (finding F-A of `docs/implementation/e8_ladic_investigation.md`, [L]; owner ruling 10 of 2026-09-24); **it changes no claim**. Let $n>0$ and ℓ be a prime. Clauses (1) and (3) assume $\ell\geq 5$, where Claim 16.7 is; clauses (2) and (4) hold at **every** prime ℓ, 2 and 3 included (clause (2) since v1.9.0 — through v1.8.1 the remark assumed $\ell\geq 5$ throughout, clause (2) included, although its proof never used it, slot-8 WP-9 finding 6, owner ruling 10 (A) of the slot-9 decisions, 2026-09-24; clause (4) since v1.24.0-rc.1). For $\ell\geq 5$ write the ℓ-primary torsion of $H^1(\Gamma;V)$ as $\bigoplus_{t=1}^{s}\mathbb{Z}/\ell^{a_t}$, $s=\dim H^0(\Gamma;M_n)$.

1. [proved here] **(The Bockstein counts the exponent-one summands.)** The mod-ℓ Bockstein $\beta^0\colon H^0(\Gamma;M_n)\to H^1(\Gamma;M_n)$, the connecting map of $0\to M_n\xrightarrow{\ell}V\otimes\mathbb{Z}/\ell^2\to M_n\to 0$, has rank $\#\{t:a_t=1\}$. So $\dim H^0(\Gamma;M_n)-\mathrm{rank}\,\beta^0=\#\{t:a_t\geq 2\}$, and β^0 is injective iff $H^1(\Gamma;V)$ has no ℓ^2-torsion.
2. [proved here] **(The Frobenius lower bound, at every prime.)** Let ℓ be any prime. If $m>0$ and $H^0(\Gamma;M_m)\neq 0$, $M_m=\mathrm{Sym}^m\mathbb{Z}^2\otimes\mathbb{F}_\ell$, then for every $j\geq 0$ the group $H^1(\Gamma;\mathrm{Sym}^{\ell^j m}\mathbb{Z}^2)$ contains an element of order ℓ^{j+1}. In particular H^1 has ℓ^2-torsion at $n=\ell m$, and the largest ℓ-primary exponent is unbounded along $n=\ell^j m$. *The instances at* $\ell\in\{2,3\}$ (v1.9.0; the integral-line design brief's finding F-4): at $\ell=2$, $m=2$ and the invariant x^2+xy+y^2 of M_2 (Proposition 23.2(2)) give an element of order 2^{j+1} at $n=2^{j+1}$; at $\ell=3$, $m=4$ and $m=6$ and the $\mathrm{SL}_2(\mathbb{F}_3)$-invariants x^3y-xy^3 and $x^6+x^4y^2+x^2y^4+y^6$ (the reduction $\Gamma\to\mathrm{SL}_2(\mathbb{F}_3)$ being onto) give an element of order

9 at $n = 12$ and at $n = 18$. (Through v1.8.1 this clause read "the argument uses no hypothesis on ℓ; it is stated here where Claim 16.7 is", and was stated for $\ell \geq 5$ only.)

3. [verified computationally][a] **(The probe.)** [L] §2.5 and §5.4 record, from the exact out-of-tree probe `docs/implementation/e8_ladic_probe.py` (not a registered validation; certified at $\mathbb{Z}/\ell^8$), on its sweep $\ell = 5$, $n \leq 130$; $\ell = 7$, $n \leq 114$; $\ell \in \{11, 13\}$, $n \leq 98$; and at $(150, 5)$, $(180, 5)$, $(200, 5)$: $\mathbb{Z}/25$ at $(30, 5)$ with $\operatorname{rank}\beta^0 = 0$ of one Dickson class; $\mathbb{Z}/25 \oplus \mathbb{Z}/5$ at $(60, 5)$, $\operatorname{rank}\beta^0 = 1$ of two; $\mathbb{Z}/125 \oplus (\mathbb{Z}/5)^2$ at $(150, 5)$, $\operatorname{rank}\beta^0 = 2$ of three; $\mathbb{Z}/49$ at $(56, 7)$ and $(112, 7)$; one $\mathbb{Z}/25$ among the classes at $(90, 5)$, $(100, 5)$, $(120, 5)$, $(130, 5)$, $(180, 5)$, $(200, 5)$; and exponent 1 on every other torsion cell. On all of them the largest exponent *equals* the lower bound of (2) — $j+1$ at the largest j with $n = \ell^j m$, m a Dickson degree, and 1 where n has no such form — and (1)'s count holds. (v1.9.0, beside this clause and not part of its scope $\ell \geq 5$: at $\ell \in \{2, 3\}$ the integral-line probe's table, Appendix A of `docs/implementation/integral_line_design_brief.md`, $n \leq 20$, shows $\mathbb{Z}/2^j$ at $n = 2^j$, $1 \leq j \leq 4$, and $\mathbb{Z}/9$ at $n = 12$ and $n = 18$ — the instances of (2) named there, each at its bound; the table is re-derived in the suite by `HeckeFreeModelValidation` (hecke-extensions slot-9 WP-11), which asserts the groups and not the equality with the bound.) *The $\mathbb{Z}/\ell^2$ depth window* (added at v1.14.0-pre.1, taken out at pre.2, restored at v1.14.0; the first cluster run, published 2026-09-28 on the owner's sign-off, [S14] Q2, `docs/matrix-engine/results/noctua2_run1/depth_window_l2.tsv`; [EX] `nc:depth`): over $\mathbb{Z}/\ell^2$, at even $66 \leq n \leq 300$ for $\ell \in \{7, 11, 13\}$ and $66 \leq n \leq 200$ for $\ell \in \{5, 691\}$, ℓ^2-torsion occurs at exactly 17 cells, which are exactly the loci of (2) in that window — $n = \ell m$ with $H^0(\Gamma; M_m) \neq 0$ —, each with exactly one summand of exponent ≥ 2 (so (1)'s $\dim H^0(\Gamma; M_n) - \operatorname{rank}\beta^0 = 1$), and exactly where the $K = 2$ mode is uncertified. (2) with $j = 2$, $m = 6$ forces an element of order 5^3 at $(150, 5) = (5^2 \cdot 6, 5)$, which $\mathbb{Z}/25$ does not read. That there is no ℓ^2-torsion off the loci is observed on this window only; this document does not prove it (item (4) cites it from the literature). *Since v1.24.0-rc.1* every cell of this clause, and of the $\ell \in \{2, 3\}$ table beside it, is an instance of the theorem cited in (4); the clause is a computational confirmation of that theorem, not an observation beyond it.

4. [standard: [CCS13, Thm. 4.5 (= Thm. 3.7(2))], numbering of arXiv:1204.5390, read verbatim by the phenomenology literature search (`docs/phenomenology/literature_search_B.md` §1), not by a referee of this document; VERIFY] **(The exponents, from the literature; v1.24.0-rc.1, replacing "not asserted" for the first two items of the old list.)** For every prime ℓ and every $n > 0$ the ℓ-primary torsion of $H^1(\Gamma; V)$ is $\bigoplus \mathbb{Z}/\ell^{\,v_\ell(\gcd(k,h))+1}$, one summand for each Dickson monomial $P_\ell^k Q_\ell^h$ of degree n, i.e. each (k, h) with $k(\ell+1) + h\,\ell(\ell-1) = n$, where $P_\ell = xy(x^{\ell-1} - y^{\ell-1})$ and $Q_\ell = P_{\ell^2}/P_\ell$ are the Dickson invariants. Consequences: the number of summands is $\dim H^0(\Gamma; M_n)$ (the Dickson monomials of degree n are a basis of the $\mathrm{SL}_2(\mathbb{F}_\ell)$-invariants, onto which Γ maps), at every prime; and the largest exponent *equals* the lower bound of (2) — some (k, h) has $\ell^j \mid \gcd(k, h)$ iff $n = \ell^j m$ with $m = (k/\ell^j)(\ell+1) + (h/\ell^j)\ell(\ell-1)$ a Dickson degree — so it is $j+1$ at the largest such j, and 1 where n has no such form with $j \geq 1$ (and ℓ-torsion at all). *Convention.* The module of [CCS13] is $\mathbb{Z}[x, y]$ graded with $\deg x = \deg y = 2$ (v1.24.0-rc.2), so their degree-$2n$ part is our V (polynomial degree n) and their condition $2k(\ell+1) + 2h\,\ell(\ell-1) = 2n$ is the one above; it carries the substitutions $\sigma_1 \colon x \mapsto x - y$, $\sigma_2 \colon y \mapsto x + y$ (the other variable fixed), acting through $B_3 \to \mathrm{SL}_2(\mathbb{Z})$. In the convention of Proposition 8.2(1) these are ρ_n of $\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$ and of $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, two matrices that generate Γ; so their module is (V, ρ_n) up to an automorphism of Γ, which does not change H^1. The five integral Smith forms of the phenomenology catalogue (PH-07, $n = 4, 10, 12, 16, 22$) agree with their multiset exactly, in polynomial degree (read literally, without the factor 2, they would not: at $n = 12$, $\ell = 2$ the exponents are $\{1, 2, 3\}$, not $\{1, 2\}$).

5. **Not asserted**: [L]'s estimate $1+\log_\ell\big(n/(\ell+1)\big)$ for the growth of the largest exponent (through v1.24.0-pre.1 this item also listed "that the largest exponent is exactly $j+1$ in general" and "that it is 1 off the loci of (2)", now (4)). Claim 16.7 is unchanged. (1) is the *meaning* the companion specification gives its $\mathbb{Z}/\ell^e$ bookkeeping identity, whose count $\#\{t : a_t \geq 2\}$ is asserted where the exponent is certified ([E, `sec:ring`]); what (2) adds for those computations is that no fixed e reads every exponent correctly, which is Proposition 23.17's subject.

Proof. (1) Reduction mod ℓ, mod ℓ^2 and the identity form a map of short exact sequences from $0 \to V \xrightarrow{\ell} V \to M_n \to 0$ to $0 \to M_n \xrightarrow{\ell} V \otimes \mathbb{Z}/\ell^2 \to M_n \to 0$, so by naturality of connecting maps [standard: [Wei94, Thm. 1.3.1]] $\beta^0 = \rho \circ \beta$, with β the integral Bockstein, an isomorphism $H^0(\Gamma; M_n) \to H^1(\Gamma; V)[\ell]$ (Claim 16.7(2)), and $\rho\colon H^1(\Gamma; V) \to H^1(\Gamma; M_n)$ the reduction, which factors as $H^1(\Gamma; V) \to H^1(\Gamma; V)/\ell \hookrightarrow H^1(\Gamma; M_n)$ (the same claim's sequence at $q = 1$). So $\operatorname{rank} \beta^0$ is the rank of $G[\ell] \to G/\ell G$ for $G = H^1(\Gamma; V)$, whose kernel is $G[\ell] \cap \ell G$: the free and prime-to-ℓ parts contribute nothing to $G[\ell]$, and the generator $\ell^{a_t - 1}$ of $(\mathbb{Z}/\ell^{a_t})[\ell]$ lies in ℓG iff $a_t \geq 2$.

(2) Lift a non-zero Γ-invariant of M_m to $f \in \operatorname{Sym}^m \mathbb{Z}^2 \subset \mathbb{Z}[x, y]$, so $\rho_m(\gamma) f - f \in \ell \operatorname{Sym}^m \mathbb{Z}^2$ for every $\gamma \in \Gamma$. Each $\rho_\bullet(\gamma) = \operatorname{Sym}^\bullet(\gamma)$ is a ring endomorphism of $\mathbb{Z}[x, y]$ (Definition 8.1: the multiplicative extension), so for $h_i := f^{\ell^i}$ induction gives $\rho(\gamma) h_i - h_i \in \ell^{i+1} \mathbb{Z}[x, y]$: if $\rho(\gamma) h_i = h_i + \ell^{i+1} u$ then $\rho(\gamma) h_{i+1} = (h_i + \ell^{i+1} u)^\ell$, and every term but h_i^ℓ is divisible by $\binom{\ell}{k} \ell^{k(i+1)}$ with $1 \leq k \leq \ell - 1$, hence by ℓ^{i+2}, or is $\ell^{\ell(i+1)} u^\ell$, again divisible by ℓ^{i+2}. So h_j is a Γ-invariant of $V \otimes \mathbb{Z}/\ell^{j+1}$ for $n = \ell^j m$, of additive order exactly ℓ^{j+1}, since $h_j \bmod \ell = (f \bmod \ell)^{\ell^j} \neq 0$ in the domain $\mathbb{F}_\ell[x, y]$. The long exact sequence of $0 \to V \xrightarrow{\ell^{j+1}} V \to V \otimes \mathbb{Z}/\ell^{j+1} \to 0$ and $H^0(\Gamma; V) = 0$ (Claim 16.7(1), $n > 0$; that half of its proof is about $\mathbb{Z}$-coefficients and uses no hypothesis on ℓ) give an injection $H^0(\Gamma; V \otimes \mathbb{Z}/\ell^{j+1}) \hookrightarrow H^1(\Gamma; V)$ — the argument of Claim 16.7(2) with ℓ^{j+1} in place of ℓ — and the image of h_j has order ℓ^{j+1}. □

16.4 Odd n at $\ell = 2$, and the prime-to-ℓ Hecke action at $\ell \in \{2, 3\}$

What this subsection fixes. The classes of $H^\bullet(\Gamma; \operatorname{Sym}^n)$ at the primes 2 and 3 that the rest of this section leaves open: the complete integral cohomology for n odd (it is an $\mathbb{F}_2$-vector space in every degree, of closed-form dimension, with $H^1 \neq 0$ for $n \geq 3$), its $\mathbb{Z}/2^k$ reductions, and the prime-to-ℓ Hecke action on $H^q(\Gamma; \operatorname{Sym}^n \mathbb{F}_\ell^2)$, $\ell \in \{2, 3\}$, in every degree, which is the scalar $1 + p$. Throughout $\Gamma = \langle S \rangle *_{\langle -I \rangle} \langle U \rangle$, $U := ST$, as in Claim 16.2; $A := \langle S \rangle$, $B := \langle U \rangle$, $C := \langle -I \rangle$ (orders 4, 6, 2); $V = \operatorname{Sym}^n \mathbb{Z}^2$ with ρ_n of Definition 8.1; for a prime p, $g_p := \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ and $T_p := T_{\Gamma g_p \Gamma} = \operatorname{cor}^{\Gamma}_{\Gamma''} \circ (\operatorname{conj}_{g_p})_* \circ \operatorname{res}^{\Gamma}_{\Gamma'}$, the operator of Lemma 14.22(3) on ordinary group cohomology, defined for every $\mathbb{Z}[\Delta]$-module (here $\Gamma' = \Gamma^0(p)$, $\Gamma'' = \Gamma_0(p)$). All cohomology is ordinary group cohomology. The census and plan documents that motivated this subsection are not cited as authority: every statement below carries its own proof.

Lemma 16.9 (A central element acting by -1 forces exponent 2)**.** *[proved here] Let X be a $\mathbb{Z}[\Gamma]$-module on which $-I$ acts as -1. Then $2 \cdot H^q(\Gamma; X) = 0$ for every $q \geq 0$. This applies to $X = V$, $V/2^k V$ ($k \geq 1$) and $V \otimes \mathbb{Z}_{(2)}$ for n odd. Consequently, for n odd and $k \geq 1$, every $H^q(\Gamma; V)$ and every $H^q(\Gamma; V/2^k)$ is an $\mathbb{F}_2$-vector space, the long exact sequence of $0 \to V \xrightarrow{2^k} V \to V/2^k \to 0$ splits into short exact sequences*

$$0 \to H^q(\Gamma; V) \xrightarrow{\mathrm{red}_k} H^q(\Gamma; V/2^k) \xrightarrow{\delta_k} H^{q+1}(\Gamma; V) \to 0 \qquad (q \geq 0),$$

*and every $H^q(\Gamma; V_R)$ vanishes when $2 \in R^\times$ and n is odd. [formally verified: [GT, `thm:lv3-tree-negone`]] (only this last clause, and only **for a field**: if K is a field with $2 \neq 0$ and X a finite-dimensional $K[\Gamma]$-module on which $-I$ acts as -1, then $H^q(\Gamma; X) = 0$ for every q, mathlib's*

*`groupCohomology`, proved on the complex of Remark 16.3; instantiated at $X = \mathrm{Sym}^n \mathbb{F}_3^2$, n odd, by `V3_dims_odd` ([GT] `thm:lv3-v3-dims`, marked in Remark 16.20(2)); **not** for a ring R with $2 \in R^\times$ as stated here, and not $2 \cdot H^q(\Gamma; X) = 0$, the $\mathbb{F}_2$-structure or the split sequences). (This generalises Proposition 23.16(2), which is the case $X = V$.)*

Proof. $z := -I$ is central, so $(c_z,\ x \mapsto zx)$, $c_z(h) = z^{-1}hz = h$, is a compatible pair, and it induces the identity of $H^\bullet(\Gamma; X)$ [standard: [Bro82, Ch. III, §8]: an inner automorphism, paired with the action of the conjugating element on the module, acts trivially on cohomology (the form used in the proof of Proposition 23.10(1); the pointer is at section level and is not verified against the text, rule (A2))]. As $c_z = \mathrm{id}_\Gamma$, the induced map is the one induced by the module automorphism $x \mapsto zx = -x$, i.e. -1. So $-1 = 1$ on $H^q(\Gamma; X)$. For $X = V$ etc. with n odd, $\rho_n(-I) = (-1)^n = -1$ (Definition 8.1). The short exact sequences follow because multiplication by 2^k is 0 on $H^\bullet(\Gamma; V)$; and if 2 is a unit of R, multiplication by 2 is invertible on $H^q(\Gamma; V_R)$ and zero, so the group vanishes. □

Lemma 16.10 (The degree-0 Bockstein, at every prime)**.** *[proved here] Let ℓ be any prime (including 2 and 3) and $n > 0$. The connecting map β of $0 \to V \xrightarrow{\ell} V \to M_n \to 0$ is an isomorphism $H^0(\Gamma; M_n) \xrightarrow{\sim} H^1(\Gamma; V)[\ell]$, equivariant for every operator $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$, $g \in \Delta$. The reduction $H^1(\Gamma; V) \otimes \mathbb{F}_\ell \to H^1(\Gamma; M_n)$ is an isomorphism if and only if $H^2(\Gamma; V)[\ell] = 0$ — which holds for $\ell \geq 5$ (Claim 16.7(2)) and **fails at $\ell = 2$ for every odd n** (Proposition 16.12(3): $H^2(\Gamma; V) \neq 0$ is 2-torsion). So Claim 16.7(2)'s second isomorphism holds at every prime, its first one only where $H^2(\Gamma; V)$ has no ℓ-torsion.*

Proof. The long exact sequence [standard: [Bro82, Ch. III, §§0–1], as in the proof of Claim 16.7(2)] gives, in every degree, $0 \to H^q(\Gamma; V)/\ell \to H^q(\Gamma; M_n) \to H^{q+1}(\Gamma; V)[\ell] \to 0$. At $q = 0$ the left term is 0 because $H^0(\Gamma; V) = 0$ for $n > 0$ (the argument of Claim 16.7(1), which is about $\mathbb{Z}$-coefficients and uses no hypothesis on ℓ); at $q = 1$ the right term is $H^2(\Gamma; V)[\ell]$. Equivariance is the argument of Claim 16.7(3), which uses no hypothesis on ℓ: res, $(\mathrm{conj}_g)_*$ and cor are morphisms of δ-functors and the sequence is one of $\mathbb{Z}[\Delta]$-modules. □

Lemma 16.11 (Cyclic groups: cohomology and restriction)**.** *[proved here] Let $G = \langle t \rangle$ be cyclic of order m, X a $\mathbb{Z}[G]$-module, $N_t := 1 + t + \cdots + t^{m-1}$, and let $d \mid m$, $h := t^d$. Then $H^0(G; X) = X^G$, $H^{2j+1}(G; X) = \ker N_t/(t-1)X$ and $H^{2j+2}(G; X) = X^G/N_t X$ for $j \geq 0$, and $\mathrm{res}^G_{\langle h \rangle}$ is induced in odd degrees by $x \mapsto (1 + t + \cdots + t^{d-1})x$ and in even degrees ≥ 2 by $x \mapsto x$.*

Proof. The cohomology is that of $\mathrm{Hom}_G(-, X)$ on the periodic resolution $\cdots \xrightarrow{N_t} \mathbb{Z}[G] \xrightarrow{t-1} \mathbb{Z}[G] \to \mathbb{Z}$ [standard: [Bro82, Ch. I.6 and Ch. III.1]: the periodic resolution of a finite cyclic group; section-level pointer, not verified against the text, rule (A2)]. A chain map from the periodic resolution of $\langle h \rangle$ (differentials $h - 1$, N_h) to the restriction of that of G lifting the identity of $\mathbb{Z}$ is $f_{2j} = 1$, $f_{2j+1} = 1 + t + \cdots + t^{d-1}$: indeed $(t-1)(1 + \cdots + t^{d-1}) = h - 1$ and $N_t = (1 + \cdots + t^{d-1})N_h$. Applying $\mathrm{Hom}(-, X)$ gives the stated maps. □

Proposition 16.12 (Odd n: the integral cohomology is $\mathbb{F}_2$-elementary, in closed form)**.** *[proved here] Let n be odd, $M_n = \mathrm{Sym}^n \mathbb{F}_2^2$, $h^0 := \dim_{\mathbb{F}_2} M_n^\Gamma$, $c_U := \dim_{\mathbb{F}_2} M_n^U$.*

1. ***(Mayer–Vietoris collapses.)*** *$H^0(\Gamma; V) = 0$ and, for* every *$j \geq 0$, (51) reduces to the exact sequence*

$$0 \to H^{2j+1}(\Gamma; V) \to V/(S-1)V \ \oplus\ V/(U-1)V \xrightarrow{r} M_n \to H^{2j+2}(\Gamma; V) \to 0,$$

$r(a, b) = (1+S)a + (1+U+U^2)b \bmod 2$, with $V/(S-1)V \cong \mathbb{F}_2^{(n+1)/2}$, $V/(U-1)V \cong M_n^U$, and both summands of r injective, with images M_n^S and M_n^U.

2. ***(The invariants.)*** *$M_n^S = (1+S)M_n$ has dimension $(n+1)/2$; $c_U = 2h^0$; and*

$$h^0 = \lfloor (\lfloor n/3 \rfloor + 1)/2 \rfloor,$$

which is 0 for $n = 1$ and ≥ 1 for $n \geq 3$. [formally verified: [GT, `chk:lv-cells`]] (only the value *of h^0, as $\dim_{\mathbb{F}_2} H^0(\Gamma; M_n)$ (mathlib's `groupCohomology`, which is M_n^Γ), at the odd $n \leq 63$ and at $n = 101, 129, 255$, each by a kernel decision of two ranks; not the formula for general n, not the Dickson description; the same cells give $\dim_{\mathbb{F}_2} H^q(\Gamma; M_n) = (n+1)/2$ for every $q \geq 1$, the value that (3) and the split sequences of Lemma 16.9 predict; the integral groups $H^q(\Gamma; V)$, and (1), (3), (4), are **not** formalised; the other odd $n \leq 699$: Remark 16.20) $M_n^\Gamma = M_n^{\mathrm{GL}_2(\mathbb{F}_2)}$ is the degree-n part of the Dickson algebra $\mathbb{F}_2[d_2, d_3]$, $d_2 = x^2 + xy + y^2$, $d_3 = x^2y + xy^2$ (Lemma 16.6 at $\ell = 2$, with $L = d_3$, $Q = d_2$ and $\mathrm{GL}_2(\mathbb{F}_2) = \mathrm{SL}_2(\mathbb{F}_2)$, proved in place since v1.23.0-pre.15; no [standard:] tag: [Wil83] is provenance only, and the dimension formula above does not depend on it), spanned by the $d_3^a d_2^b$ with $3a + 2b = n$ (a odd).*

3. ***(The groups.)*** *For every $j \geq 0$,*

$$H^{2j+1}(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2) \cong \mathbb{F}_2^{h^0} \ \text{(zero iff } n = 1), \qquad H^{2j+2}(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2) \cong \mathbb{F}_2^{(n+1)/2 - h^0} \ \text{(never zero)},$$

and $H^0 = 0$. So $H^\bullet(\Gamma; V)$ is 2-periodic from degree 1 on, not only from degree 2 as Lemma 14.22(2) gives in general: $e \cup -\colon H^q(\Gamma; V) \to H^{q+2}(\Gamma; V)$ is a Hecke-equivariant isomorphism for every $q \geq 1$.

4. ***(The odd-degree classes are Dickson classes.)*** *For every $j \geq 0$ the composite $e^j \cup \beta\colon H^0(\Gamma; M_n) \to H^{2j+1}(\Gamma; V)$ (β of Lemma 16.10) is an isomorphism, equivariant for every $T_{\Gamma g \Gamma}$, $g \in \Delta$.*

Proof. (1) $V = V$ is torsion-free and $-I = S^2 = U^3$ acts on it by -1, so $V^S = V^U = V^C \subseteq V^{-I} = 0$ and $N_S = (1+S)(1+S^2) = 0$, $N_U = (1+U+U^2)(1+U^3) = 0$, $N_{-I} = 1 + (-I) = 0$ on V. By Lemma 16.11 every even-degree group of A, B, C (degree 0 included) vanishes, and $H^{2j+1}(A; V) = V/(S-1)V$, $H^{2j+1}(B; V) = V/(U-1)V$, $H^{2j+1}(C; V) = V/2V = M_n$, with restriction maps to $C = \langle S^2 \rangle = \langle U^3 \rangle$ induced by $1+S$ and $1+U+U^2$ (the sign of the difference map in (51) is immaterial mod 2). Exactness of (51) at the vanishing even-degree terms gives the displayed sequences and $H^0(\Gamma; V) \subseteq V^S = 0$; at $j = 0$ the left end uses $H^0(C; V) = 0$. Through $S \mapsto i$ (as $S^2 = -1$ on V), V is a torsion-free, hence free, $\mathbb{Z}[i]$-module of rank $(n+1)/2$, so $V/(S-1)V \cong (\mathbb{Z}[i]/(i-1))^{(n+1)/2} = \mathbb{F}_2^{(n+1)/2}$. As $(U-1)(1+U+U^2) = U^3 - 1 = -2$ on V, $2V \subseteq (U-1)V$ and $V/(U-1)V = M_n/(U-1)M_n$. On M_n, U has order 3 ($U^3 = -I$ acts trivially mod 2), so $\pi_U := 1 + U + U^2$ is idempotent ($\pi_U^2 = 3\pi_U = \pi_U$), $\pi_U M_n = M_n^U$, and $M_n = \pi_U M_n \oplus (1 - \pi_U) M_n$ with $U - 1$ invertible on $(1 - \pi_U)M_n$ (no fixed vector there) and zero on $\pi_U M_n$; hence $(U-1)M_n = (1-\pi_U)M_n$ and $M_n/(U-1)M_n \cong M_n^U$. The images of the two summands of r are $(1+S)M_n$ and $\pi_U M_n = M_n^U$; $(1+S)M_n = M_n^S$ has dimension $(n+1)/2$ by (2), so both summands are surjections onto their images between spaces of equal dimension, i.e. injective.

(2) $\rho_n(S)$ mod 2 substitutes $x \mapsto y$, $y \mapsto x$ (Definition 8.1), so it swaps $x^a y^{n-a}$ and $x^{n-a} y^a$, which differ for n odd: M_n is a free $\mathbb{F}_2[\langle \bar{S} \rangle]$-module of rank $(n+1)/2$, and $M_n^S = (1+S)M_n$. The image of U in $\mathrm{SL}_2(\mathbb{F}_2) \cong S_3$ generates the normal subgroup A_3, so SUS^{-1} acts on M_n as U or U^2 and S commutes with π_U: $M_n^U = \pi_U M_n$ is an $\mathbb{F}_2[\langle \bar{S} \rangle]$-direct summand of the free module M_n, hence projective, hence free ($\mathbb{F}_2[\mathbb{Z}/2]$ is local). Its S-invariants are therefore half of it, and they are $M_n^{\langle S, U \rangle} = M_n^\Gamma$: $c_U = 2h^0$. Over $\mathbb{F}_4$, $\bar{U} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ has the eigenvalues ω, ω^2 ($\omega^2 + \omega + 1 = 0$), so on Sym^n its eigenvalues are $\omega^a \omega^{2(n-a)} = \omega^{2n-a}$, $0 \leq a \leq n$, and $c_U = \#\{0 \leq a \leq n : \ a \equiv 2n \pmod 3\}$. With $n = 3m + s$, $s \in \{0, 1, 2\}$: $c_U = m + 1$ for $s \in \{0, 2\}$ and $c_U = m$ for $s = 1$; n odd forces m odd in the first case and even in the second, so $h^0 = c_U/2 = \lfloor (m+1)/2 \rfloor$ in

both. For the Dickson description: $M_n^{\Gamma} = M_n^{\mathrm{SL}_2(\mathbb{F}_2)}$ since $\Gamma \to \mathrm{SL}_2(\mathbb{F}_2)$ is onto, and the count $\#\{(a,b) : 3a+2b=n\}$ of the cited theorem agrees with h^0 (for n odd, a runs over the odd $a \le \lfloor n/3 \rfloor$).

(3) By (1), $H^{2j+1} \cong \ker r$ and $H^{2j+2} \cong \operatorname{coker} r$; r restricted to each summand is injective, so $\ker r \cong M_n^S \cap M_n^U = M_n^{\Gamma}$, of dimension h^0, and $\dim \operatorname{coker} r = (n+1) - \dim(M_n^S + M_n^U) = (n+1) - \frac{n+1}{2} - c_U + h^0 = \frac{n+1}{2} - h^0$, which is ≥ 1 since $h^0 \le (m+1)/2 < (n+1)/2$. The groups are $\mathbb{F}_2$-vector spaces by Lemma 16.9. For $e \cup -$: Lemma 14.22(2) gives an isomorphism for $q \ge 2$, and for $q = 1$ a surjection with kernel $\partial(N_{-I}V) = \partial(0) = 0$; equivariance is Lemma 14.22(3)(b).

(4) $H^1(\Gamma; V)[2] = H^1(\Gamma; V)$ by Lemma 16.9, so β is an equivariant isomorphism onto $H^1(\Gamma; V)$ (Lemma 16.10); then apply (3). □

Lemma 16.13 (The $\mathbb{Z}/2^k$ reductions split, Hecke-equivariantly)**.** *[proved here] Let n be odd, $k \ge 2$ and $q \ge 0$. The short exact sequence $0 \to H^q(\Gamma; V) \to H^q(\Gamma; V/2^k) \to H^{q+1}(\Gamma; V) \to 0$ of Lemma 16.9 splits as a sequence of modules over the algebra generated by all $T_{\Gamma g \Gamma}$, $g \in \Delta$ (T_2 included):*

$$H^q(\Gamma; V/2^k) \;\cong\; H^q(\Gamma; V) \oplus H^{q+1}(\Gamma; V).$$

A retraction is $\varsigma := \mathrm{red}_1^{-1} \circ \mathrm{red}_{k,1}$, $\mathrm{red}_{k,1} \colon H^q(\Gamma; V/2^k) \to H^q(\Gamma; V/2)$ the reduction. Consequently the $\mathbb{Z}/2^k$-cohomology, $k \ge 2$, carries no information beyond $H^q(\Gamma; V)$ and $H^{q+1}(\Gamma; V)$ with their Hecke actions. For $k = 1$ no equivariant splitting is asserted.

Proof. Multiplication by 2^{k-1} on the left term, the identity in the middle and reduction on the right form a morphism of short exact sequences from $0 \to V \xrightarrow{2^k} V \to V/2^k \to 0$ to $0 \to V \xrightarrow{2} V \to V/2 \to 0$, so by naturality of connecting maps [standard: [Wei94, Thm. 1.3.1], as in the proof of Remark 16.8(1)] $\delta_1 \circ \mathrm{red}_{k,1} = (2^{k-1})_* \circ \delta_k = 0$ for $k \ge 2$ (Lemma 16.9). Hence $\mathrm{red}_{k,1}$ lands in $\ker \delta_1 = \mathrm{red}_1 H^q(\Gamma; V)$, and $\mathrm{red}_{k,1} \circ \mathrm{red}_k = \mathrm{red}_1$; so ς is defined and $\varsigma \circ \mathrm{red}_k = \mathrm{id}$. All of $\mathrm{red}_{k,1}$, red_k, red_1 commute with every $T_{\Gamma g \Gamma}$ (maps of $\mathbb{Z}[\Delta]$-modules; the argument of Claim 16.7(3)), so $\ker \varsigma$ is a Hecke-stable complement, carried isomorphically onto $H^{q+1}(\Gamma; V)$ by δ_k. □

Lemma 16.14 (T_ℓ has rank at most 2 at $\ell \in \{2,3\}$)**.** *[proved here] Let $\ell \in \{2,3\}$, $n \ge 0$, and $T_\ell = T_{\Gamma g_\ell \Gamma}$ on $H^q(\Gamma; M_n)$. If $\ell = 2$, or $\ell = 3$ and n is even, T_ℓ factors through $H^q(\Gamma_0(\ell); \mathbb{F}_\ell)$, whose dimension is*

$$\dim_{\mathbb{F}_2} H^q(\Gamma_0(2); \mathbb{F}_2) = 1, 2, 2, 2, \ldots, \qquad \dim_{\mathbb{F}_3} H^q(\Gamma_0(3); \mathbb{F}_3) = 1, 2, 1, 1, \ldots \qquad (q = 0, 1, 2, \ldots).$$

So $\operatorname{rank} T_2 \le 2$ on every H^q, $q \ge 1$; $\operatorname{rank} T_3 \le 2$ on H^1 and ≤ 1 on H^q, $q \ne 1$; and $\operatorname{rank} T_\ell \le 1$ on H^0. (For $\ell = 3$, n odd, $H^\bullet(\Gamma; M_n) = 0$ by Lemma 16.9.) Whether T_ℓ is a rank-one idempotent on H^1 is not asserted.

Proof. $\Gamma' = \Gamma \cap g_\ell^{-1} \Gamma g_\ell = \Gamma^0(\ell)$, $\Gamma'' = \Gamma \cap g_\ell \Gamma g_\ell^{-1} = \Gamma_0(\ell)$, and $(\mathrm{conj}_{g_\ell})_*$ is induced by the pair $(\alpha \colon \Gamma'' \to \Gamma',\ h \mapsto g_\ell^{-1} h g_\ell;\ m \mapsto \rho_n(g_\ell)m)$. Mod ℓ, $\rho_n(g_\ell)$ substitutes $x \mapsto x$, $y \mapsto 0$, so $\rho_n(g_\ell)m = \varpi(m) x^n$ with $\varpi(m)$ the coefficient of x^n. For $h = \begin{pmatrix} a & b \\ \ell c & d \end{pmatrix} \in \Gamma_0(\ell)$, $\alpha(h) = \begin{pmatrix} a & \ell b \\ c & d \end{pmatrix}$ substitutes $x \mapsto ax + cy$, $y \mapsto dy$ mod ℓ, so $\varpi(\rho_n(\alpha(h))m) = a^n \varpi(m)$; and $\rho_n(h) x^n = a^n x^n$. Under the hypothesis $a^n = 1$ ($a \equiv 1 \bmod 2$; $a \equiv \pm 1 \bmod 3$ and n even), the pair factors as $(\alpha, \varpi) \colon (\Gamma', M_n) \to (\Gamma'', \mathbb{F}_\ell)$ followed by $(\mathrm{id},\ 1 \mapsto x^n) \colon (\Gamma'', \mathbb{F}_\ell) \to (\Gamma'', M_n)$, both compatible, so $(\mathrm{conj}_{g_\ell})_*$, and with it T_ℓ, factors through $H^q(\Gamma_0(\ell); \mathbb{F}_\ell)$ ((E2), functoriality in the pair).

For the dimensions, restrict (50) to $\Gamma_0(\ell)$: each $\mathbb{Z}[\Gamma/D]$, $D \in \{A, B, C\}$, becomes $\bigoplus \mathbb{Z}[\Gamma_0(\ell)/\mathrm{Stab}]$ over the $\Gamma_0(\ell)$-orbits on Γ/D, i.e. over the D-orbits on $\Gamma_0(\ell)\backslash\Gamma \cong \mathbb{P}^1(\mathbb{F}_\ell)$ (rows $(c:d)$, Γ acting on the right; Definition 27.1), with stabilisers conjugate to the stabilisers in D of the points; Shapiro's lemma [standard: [Bro82, Ch. III.6], as in Claim 16.2(2)] gives a long exact sequence $\cdots \to H^q(\Gamma_0(\ell)) \to \bigoplus_{A\text{- and }B\text{-orbits}} H^q(\mathrm{Stab}) \xrightarrow{r_q} \bigoplus_{\text{points}} H^q(C) \to H^{q+1}(\Gamma_0(\ell)) \to \cdots$

(C acts trivially on $\mathbb{P}^1$; r_q is restriction, up to conjugations that act trivially on $H^\bullet(C)$). $\bar{S}\colon (c:d)\mapsto(d:-c)$ and $\bar{U}\colon (c:d)\mapsto(d:d-c)$. $\ell=2$, points $(0{:}1),(1{:}0),(1{:}1)$: A fixes $(1{:}1)$ (stabiliser A) and swaps the other two (stabiliser C); B is transitive (stabiliser C). With $\mathbb{F}_2$-coefficients every term is $\mathbb{F}_2$; by Lemma 16.11 res^A_C is 0 in odd and the identity in even degrees, res^B_C the identity ($1+U+U^2=3=1$) in odd and even degrees. In the basis (vertices $A\cdot(1{:}1)$, $A\cdot(0{:}1)$, $B\cdot(0{:}1)$; edges $(1{:}1),(0{:}1),(1{:}0)$) r_q has columns $(\epsilon,0,0)$, $(0,1,1)$, $(1,1,1)$ with $\epsilon=0$ for q odd and 1 for q even: rank 2 in every degree, so $\ker r_q$ and $\operatorname{coker} r_q$ are lines and $\dim H^q(\Gamma_0(2);\mathbb{F}_2)=\dim\operatorname{coker} r_{q-1}+\dim\ker r_q=2$ for $q\geq 1$, 1 for $q=0$. $\ell=3$, points $(0{:}1),(1{:}0),(1{:}1),(1{:}2)$: A has the two orbits $\{(0{:}1),(1{:}0)\}$, $\{(1{:}1),(1{:}2)\}$ (stabilisers C); B fixes $(1{:}2)$ (stabiliser B) and permutes the other three cyclically (stabiliser C). With $\mathbb{F}_3$-coefficients $H^q(C)=H^q(A)=0$ and $H^q(B)=\mathbb{F}_3$ for $q\geq 1$ (Lemma 16.11); so $r_q=0$ on a line for $q\geq 1$, and r_0 is the incidence map of a connected graph with four vertices and four edges (rank 3). Hence $\dim H^q(\Gamma_0(3);\mathbb{F}_3)=1,\ 1+1,\ 0+1,\dots$. □

Proposition 16.15 ($T_p=1+p$ as a matrix, at $\ell\in\{2,3\}$)**.** *Let $\ell\in\{2,3\}$, $n\geq 0$ and $p\neq\ell$ a prime. Then*

$$T_p\ =\ (1+p)\cdot\mathrm{id}\qquad \text{on } H^q(\Gamma;\mathrm{Sym}^n\mathbb{F}_\ell^2) \text{ for every } q\geq 0,$$

[proved here] given the inputs (F1), (I1) *below; in particular $T_p=0$ for every odd p at $\ell=2$; every T_p, $p\neq\ell$, is a scalar on $H^q(\Gamma;M_n)$, so the algebra generated by all T_p (p prime) acts on $H^q(\Gamma;M_n)$ through $\mathbb{F}_\ell[T_\ell]$ (v1.22.0-pre.11, F-L23 R2). The statement is the matrix identity, not an eigenvalue statement.*

(F1) [standard: [Ser80, Ch. I, §4] for the free subgroups of $\mathrm{SL}_2(\mathbb{Z})$ and [Shi71, §1.6] for the cusps of $\Gamma(N)$; section-level pointers, **not** *verified against the text (rule (A2))] With $\Gamma(\ell):=\ker(\Gamma\to\mathrm{SL}_2(\mathbb{F}_\ell))$ and $T':=\begin{pmatrix}1&0\\1&1\end{pmatrix}$: $\Gamma(2)=\{\pm I\}\times F$ with F free on T^2, T'^2; $\Gamma(3)$ is free of rank 3; and $\Gamma(3)\backslash\mathcal{H}^*$ has genus 0 with four cusps.* Derived from these *(v1.22.0-pre.11, F-L23 R4): $\Gamma(3)^{\mathrm{ab}}=\mathbb{Z}[\text{cusps}]/(\sum)$ on the parabolic classes, which Γ permutes by conjugation as $\mathrm{SL}_2(\mathbb{F}_3)$ permutes $\mathbb{P}^1(\mathbb{F}_3)$; hence any three of them form a basis — in particular the class of T'^3 (the parabolic at the cusp 0) is non-zero in $\Gamma(3)^{\mathrm{ab}}\otimes\mathbb{F}_3$ — and $H^1(\Gamma(3);\mathbb{F}_3)=\mathrm{Hom}(\Gamma(3),\mathbb{F}_3)$ is the augmentation-zero submodule of $\mathbb{F}_3[\mathbb{P}^1(\mathbb{F}_3)]=\mathbb{F}_3\oplus\mathsf{St}_3$, $\mathsf{St}_3=\mathrm{Sym}^2\mathbb{F}_3^2$ (the Steinberg module of $\mathrm{SL}_2(\mathbb{F}_3)$ is projective, so the permutation module splits); so $H^1(\Gamma(3);\mathbb{F}_3)\cong\mathrm{Sym}^2\mathbb{F}_3^2$ as $\mathbb{F}_3[\mathrm{SL}_2(\mathbb{F}_3)]$-modules ($\Gamma$ acting by conjugation).* Origin *(v1.24.0-rc.1): the mechanism — $\Gamma(\ell)$ acts trivially on $\mathrm{Sym}^n\mathbb{F}_\ell^2$, $X(\ell)$ has genus 0, and T_p acts on the cusps by $1+p$ for $p\equiv\pm1 \bmod \ell$ — is [CE, Lemma 3.1] ($\ell=2,3,5$, on $W_k/\ell\otimes H^1(Y(\ell),\mathbb{Z}/\ell)$), which its authors credit to Hatada (their ref. [11]; [Rus] quotes Hatada, Math. Ann. 239 (1979); that the two are the same paper is not checked; not seen); at $\ell\in\{2,3\}$ every prime $p\neq\ell$ is $\equiv\pm1 \bmod \ell$. The level-one, every-degree form with (I1) is this document's own.*

(I1) [standard: [Bro82, Ch. III, §9] (the transfer on inhomogeneous cochains, (E4)); section-level pointer, not verified (rule (A2))] For every $\mathbb{Z}[\Delta]$-module X, $T_{\Gamma g\Gamma}$ on $H^1(\Gamma;X)$ is induced by the coset formula (27) of Proposition 13.4, for any system of left-coset representatives δ_j of $\Gamma g\Gamma/\Gamma$. That the class of $T_g\phi$ does not depend on the representatives is a cochain identity (v1.22.0-pre.11, F-L23 R3): replacing δ_j by $\delta_j b_j$, $b_j\in\Gamma$, replaces $\kappa_j(\gamma)$ by $b^{-1}_{\gamma*j}\kappa_j(\gamma)b_j$, and the cocycle rule with $\phi(b^{-1})=-\rho_n(b^{-1})\phi(b)$ gives $(T'_g\phi)(\gamma)=(T_g\phi)(\gamma)+\rho_n(\gamma)w-w$, $w:=\sum_j\rho_n(\delta_j)\phi(b_j)$ (the terms $-\rho_n(\delta_{\gamma*j})\phi(b_{\gamma*j})$ re-index by the bijection $j\mapsto\gamma*j$). So (I1) is needed only for the representatives $\delta_j=s_jg$, s_j running over Γ/Γ'' (those of Proposition 10.1(5) with s_j^{-1} in place of s_j), where it is the transfer formula on inhomogeneous 1-cocycles, $(\mathrm{cor}\,\psi)(\gamma)=\sum_i s_{\gamma\cdot i}\,\psi(h_i(\gamma))$ for $\gamma s_i=s_{\gamma\cdot i}h_i(\gamma)$, composed with $(\mathrm{conj}_g)_*$ and res *[standard: [Bro82, Ch. III.9–10] and [AS86, §1.1, p. 194], as in the proof of Proposition 13.4]. The clause is used: Step 0(c)'s $\delta_\infty=\pm\begin{pmatrix}p&0\\0&1\end{pmatrix}$ is not of the shape s_jg_p.*

Proof. Step 0: two general facts. (a) H^0: T_p on $H^0(\Gamma;X)=X^\Gamma$ is $m\mapsto\sum_j\rho_n(\delta_j)m$ over left-coset representatives of $\Gamma g_p\Gamma/\Gamma$ (by (E4): cor on H^0 is $\sum_{s\in\Gamma/\Gamma''}s$, and $\{sg_p\}$ represents $\Gamma g_p\Gamma/\Gamma$). (b) *Inflated classes.* Let X be an $\mathbb{F}_\ell[\mathrm{GL}_2(\mathbb{F}_\ell)]$-module, Δ acting through reduction (on M_n this is ρ_n, Definition 8.1), $d_p:=g_p \bmod \ell$. For $\bar x\in H^q(\mathrm{SL}_2(\mathbb{F}_\ell);X)$,

$$T_p\inf(\bar x)=(p+1)\inf\big((c_{d_p},\,d_p)_*\bar x\big),$$

$(c_{d_p},d_p)_*$ the automorphism of $H^q(\mathrm{SL}_2(\mathbb{F}_\ell);X)$ induced by the pair $(u\mapsto d_p^{-1}ud_p,\ x\mapsto d_px)$: Γ' and Γ'' map onto $\mathrm{SL}_2(\mathbb{F}_\ell)$ (they contain $\Gamma(p)$, and $\mathrm{SL}_2(\mathbb{Z})\to\mathrm{SL}_2(\mathbb{Z}/p\ell)=\mathrm{SL}_2(\mathbb{F}_p)\times\mathrm{SL}_2(\mathbb{F}_\ell)$ is onto), res and $(\mathrm{conj}_{g_p})_*$ commute with inflation by functoriality in the pair ((E2); $g_p^{-1}hg_p\equiv d_p^{-1}\bar h d_p$), and $\mathrm{cor}^\Gamma_{\Gamma''}\circ\inf=\mathrm{cor}^\Gamma_{\Gamma''}\circ\mathrm{res}^\Gamma_{\Gamma''}\circ\inf=[\Gamma:\Gamma'']\inf$ [standard: [Bro82, Prop. III.9.5(ii), p. 82]]. For $\ell=2$, and for $\ell=3$ with $p\equiv1$, $d_p=1$ and the pair is the identity; for $\ell=3$, $p\equiv2$ the factor is $p+1\equiv0$. So $T_p=1+p$ on $\inf H^q(\mathrm{SL}_2(\mathbb{F}_\ell);X)$ in all cases. (c) *The level-$\Gamma(\ell)$ operator.* Choose the left-coset representatives $\delta_c=\begin{pmatrix}1&0\\ \ell c&p\end{pmatrix}$ $(0\le c<p)$ and $\delta_\infty=\pm\begin{pmatrix}p&0\\0&1\end{pmatrix}$, the sign making $\delta_\infty\equiv d_p \bmod\ell$ (they represent the $p+1$ column lattices of index p; $c\mapsto\ell c$ is a bijection mod p). All $\delta_j\equiv d_p$, so $\kappa_j(\gamma)=\delta_{\gamma*j}^{-1}\gamma\delta_j\in\Gamma(\ell)$ for $\gamma\in\Gamma(\ell)$: restricted to $\Gamma(\ell)$, (27) only reads ϕ on $\Gamma(\ell)$, and defines an operator $T^{(\ell)}$ on $H^1(\Gamma(\ell);X)$ (the proof of Proposition 13.4 applies verbatim) with $\mathrm{res}^\Gamma_{\Gamma(\ell)}\circ T_p=T^{(\ell)}\circ\mathrm{res}^\Gamma_{\Gamma(\ell)}$ on H^1, by (I1). Since $\Gamma(\ell)$ acts trivially on X, $H^1(\Gamma(\ell);X)=\mathrm{Hom}(\Gamma(\ell),\mathbb{F}_\ell)\otimes X$ and $T^{(\ell)}=T_0^{(\ell)}\otimes d_p$, $T_0^{(\ell)}$ the operator for trivial coefficients. A change of representatives within the same classes mod ℓ ($\delta_j\mapsto\delta_ju_j$, $u_j\in\Gamma(\ell)$) changes $T_0^{(\ell)}\phi$ by $\sum_j(\phi(u_j)-\phi(u_{\gamma*j}))=0$; using the representatives $\gamma_1^{-1}\delta_j\gamma_0$ ($\gamma_0,\gamma_1\in\Gamma$, $\gamma_1\equiv d_p\gamma_0d_p^{-1}$) one finds $T_0^{(\ell)}\circ\gamma_0=\gamma_1\circ T_0^{(\ell)}$ for the conjugation action $(\gamma\phi)(h)=\phi(\gamma^{-1}h\gamma)$. Finally, for $\gamma=T'^\ell$: $T'^\ell\delta_c=\delta_{c+1}$ for $c<p-1$, $T'^\ell\delta_{p-1}=\delta_0T'^\ell$, $T'^\ell\delta_\infty=\delta_\infty T'^{\ell p}$, so

$$(T_0^{(\ell)}\phi)(T'^\ell)=\phi(T'^\ell)+\phi(T'^{\ell p})=(1+p)\,\phi(T'^\ell)\qquad(\phi\in\mathrm{Hom}(\Gamma(\ell),\mathbb{F}_\ell)).$$

$\ell=3$. $T_0^{(3)}=(1+p)\mathrm{id}$ on $H^1(\Gamma(3);\mathbb{F}_3)$: if $p\equiv1$, $d_p=1$ and $T_0^{(3)}$ is $\mathrm{SL}_2(\mathbb{F}_3)$-equivariant, hence (by (F1) and $\mathrm{End}_{\mathrm{SL}_2(\mathbb{F}_3)}(\mathrm{Sym}^2\mathbb{F}_3^2)=\mathbb{F}_3$: an endomorphism commutes with $\bar T$, whose only Jordan block on Sym^2 has size 3, so it is $a+bN+cN^2$, $N=\bar T-1$, and it preserves the $\bar T'$-fixed line $\mathbb{F}_3y^2$, which forces $b=c=0$) a scalar, read off at T'^3 as $1+p$; if $p\equiv2$, $\sigma_0:=\mathrm{diag}(1,-1)\in\mathrm{GL}_2(\mathbb{Z})$ normalises $\Gamma(3)$, has $\bar\sigma_0=d_p$, and $T_0^{(3)}\circ\sigma_0^{-1}$ is equivariant, so $T_0^{(3)}=\lambda\sigma_0$, and $(1+p)\phi(T'^3)=\lambda\phi(\sigma_0^{-1}T'^3\sigma_0)=-\lambda\phi(T'^3)$ forces $\lambda=-(1+p)=0$. So $T^{(3)}=(1+p)\mathrm{id}$ on $H^1(\Gamma(3);X)$. Now $\Gamma(3)$ is free, so the Lyndon–Hochschild–Serre spectral sequence of $\Gamma(3)\to\Gamma\to\mathrm{SL}_2(\mathbb{F}_3)$ [standard: [Bro82, Ch. VII.6]] has two rows, $E_2^{a,1}=H^a(\mathrm{SL}_2(\mathbb{F}_3);H^1(\Gamma(3);\mathbb{F}_3)\otimes X)$, and these vanish for $a\ge1$: restriction to the Sylow subgroup $\langle\bar T\rangle$ is injective (index 8, [Bro82, Prop. III.9.5(ii)]), and $\mathrm{Sym}^2\mathbb{F}_3^2\otimes X$ is free over $\mathbb{F}_3[\langle\bar T\rangle]$ (a free module tensored with any module is free). Hence inflation $H^q(\mathrm{SL}_2(\mathbb{F}_3);X)\to H^q(\Gamma;X)$ is onto for $q\ge2$ and bijective for $q=0$, and for $q=1$ there is an exact $0\to\inf H^1(\mathrm{SL}_2(\mathbb{F}_3);X)\to H^1(\Gamma;X)\xrightarrow{\mathrm{res}}H^1(\Gamma(3);X)$. By Step 0(b) $T_p=1+p$ in degrees $q\ne1$ and on $\mathcal{I}:=\inf H^1(\mathrm{SL}_2(\mathbb{F}_3);X)$. Put $D_p:=T_p-(1+p)$ on H^1: $\mathrm{res}\circ D_p=(T^{(3)}-(1+p))\circ\mathrm{res}=0$, so $D_p(H^1)\subseteq\mathcal{I}$; and $e\cup D_p(x)=D_p(e\cup x)=0$ (Lemma 14.22(3)(b); $T_p=1+p$ on H^3). $e\cup-$ is injective on $\mathcal{I}$: res^Γ_B is injective on $\mathcal{I}$ ($B\to\mathrm{SL}_2(\mathbb{F}_3)$ is injective with image of index 4), $\mathrm{res}^\Gamma_B e$ generates $H^2(B;\mathbb{Z})$ (Lemma 14.22(1)), and cup product with a generator is injective on $H^1(B;X)$ (Step 5 of the proof of Lemma 14.22). So $D_p=0$.

$\ell=2$. Here $d_p=1$ and Δ_{odd} acts on M_n through $\mathrm{SL}_2(\mathbb{F}_2)\cong S_3$. Put $\pi_U:=1+U+U^2$ (central idempotent of $\mathbb{F}_2[S_3]$, A_3 normal). Then $M_n=\pi_UM_n\oplus(1+\pi_U)M_n$; π_UM_n is an $\mathbb{F}_2[S_3/A_3]$-module, so $\pi_UM_n\cong\mathbb{F}_2^t\oplus\mathsf{P}^i$ with $\mathsf{P}:=\mathbb{F}_2[S_3/A_3]$; and $(1+\pi_U)M_n$ is a module over $\mathbb{F}_2[S_3](1+\pi_U)$, which the action on $\mathsf{St}_2:=\mathrm{Sym}^1\mathbb{F}_2^2$ identifies with $\mathrm{M}_2(\mathbb{F}_2)$ (it is onto, as $\mathrm{GL}_2(\mathbb{F}_2)$ spans $\mathrm{M}_2(\mathbb{F}_2)$, and both have dimension 4), so $(1+\pi_U)M_n\cong\mathsf{St}_2^s$. These are isomorphisms of $\mathbb{Z}[\Delta_{\mathrm{odd}}]$-modules, and T_p is additive, so it suffices that $T_p=0$ (p odd) on

$H^q(\Gamma;X)$ for $X \in \{\mathbb{F}_2, \mathsf{P}, \mathsf{St}_2\}$. Two Mayer–Vietoris counts ((51) and Lemma 16.11; res^B_C is bijective on $H^\bullet(-;\mathbb{F}_2)$ for trivial coefficients, $1+U+U^2$ acting as $3=1$): $\dim H^q(\Gamma;\mathbb{F}_2)=1$ and $\dim H^q(\Gamma;\mathsf{P})=1$ for all q ($\mathsf{P}|_A = \mathrm{Ind}_C^A\mathbb{F}_2$, $\mathsf{P}|_B$, $\mathsf{P}|_C$ trivial). $X=\mathbb{F}_2$: $T_p e = T_p(e\cup 1) = e\cup T_p 1 = (p+1)e$ in $H^2(\Gamma;\mathbb{Z})=\mathbb{Z}/12$ (Lemma 14.22(1),(3)), so $T_p\bar{e}=0$ for the reduction $\bar{e}\neq 0$ ($e\notin 2H^2$), which spans $H^2(\Gamma;\mathbb{F}_2)$; $H^1(\Gamma;\mathbb{F}_2)\cong H^2(\Gamma;\mathbb{Z})[2]=\langle 6e\rangle$ equivariantly (Bockstein, $H^1(\Gamma;\mathbb{Z})=0$), and $T_p(6e)=6(p+1)e=0$; H^0: $T_p=p+1=0$; and $e\cup -$ is an equivariant isomorphism $H^q\to H^{q+2}$ for $q\geq 1$ ($N_{-I}\mathbb{F}_2=0$). $X=\mathsf{P}$: in $0\to\mathbb{F}_2\to\mathsf{P}\to\mathbb{F}_2\to 0$ (maps of $\mathbb{Z}[\Delta_{\mathrm{odd}}]$-modules), $H^q(\Gamma;\mathsf{P})$ is a line, so either it injects into $H^q(\Gamma;\mathbb{F}_2)$ or it is the image of $H^q(\Gamma;\mathbb{F}_2)$; in both cases $T_p=0$ on it. $X=\mathsf{St}_2$: $H^0=0$. $q=1$: inflation–restriction [standard: [Bro82, Ch. VII.6]] gives $0\to H^1(S_3;\mathsf{St}_2)\to H^1(\Gamma;\mathsf{St}_2)\xrightarrow{\mathrm{res}} H^1(\Gamma(2);\mathsf{St}_2)^{S_3}$, and $H^1(S_3;\mathsf{St}_2)=0$ (restriction to the Sylow $\langle\bar{S}\rangle$ is injective, and St_2 is free over it). By (F1) $\mathcal{W}:=\mathrm{Hom}(\Gamma(2),\mathbb{F}_2)$ is 3-dimensional with $\phi\mapsto(\phi(-I),\phi(T^2),\phi(T'^2))$ bijective, and $T_0^{(2)}$ is S_3-equivariant (Step 0(c), $d_p=1$), so it preserves $\mathcal{W}_{\mathsf{St}}:=(1+\pi_U)\mathcal{W}$. $(\mathcal{W}\otimes\mathsf{St}_2)^{S_3}\subseteq\mathcal{W}_{\mathsf{St}}\otimes\mathsf{St}_2$ (on $\pi_U\mathcal{W}\otimes\mathsf{St}_2$ already A_3 has no invariant), and $\mathcal{W}_{\mathsf{St}}\neq 0$ because $H^1(\Gamma;\mathsf{St}_2)\supseteq\mathrm{coker}(\mathsf{St}_2^S\oplus\mathsf{St}_2^U\to\mathsf{St}_2)\neq 0$ ((51); $\mathsf{St}_2^U=0$, $\dim\mathsf{St}_2^S=1$) injects there. So $\mathcal{W}_{\mathsf{St}}\cong\mathsf{St}_2$ (a sum of copies, of dimension ≤ 3), with $\mathrm{End}_{S_3}(\mathsf{St}_2)=\mathbb{F}_2$ ($\bar{U}$ makes St_2 a line over $\mathbb{F}_2[\bar{U}]=\mathbb{F}_4$, on which $\bar{S}$ is Frobenius-semilinear); $T_0^{(2)}=\lambda$ on it. If every $\phi\in\mathcal{W}_{\mathsf{St}}$ vanished at T'^2, then also at $ST'^2S^{-1}=T^{-2}$, so $\mathcal{W}_{\mathsf{St}}\subseteq\{\phi:\phi(T'^2)=\phi(T^2)=0\}$, a line — impossible; so Step 0(c) gives $\lambda=1+p=0$, $T^{(2)}=0$ on $\mathcal{W}_{\mathsf{St}}\otimes\mathsf{St}_2$, and $T_p=0$ on $H^1(\Gamma;\mathsf{St}_2)$ since res is injective. $q=2$: $\mathsf{St}_2=\mathrm{Sym}^1\mathbb{F}_2^2$, and for $n=1$ Proposition 16.12(3) gives $H^1(\Gamma;\mathrm{Sym}^1\mathbb{Z}^2)=H^3=0$, $H^2=\mathbb{F}_2$, so Lemma 16.9's sequences give equivariant isomorphisms $H^1(\Gamma;\mathsf{St}_2)\xrightarrow{\delta_1}H^2(\Gamma;\mathrm{Sym}^1\mathbb{Z}^2)\xrightarrow{\mathrm{red}_1}H^2(\Gamma;\mathsf{St}_2)$. $q\geq 3$: $e\cup -$ is an equivariant isomorphism for $q\geq 1$ ($N_{-I}\mathsf{St}_2=0$). □

Proposition 16.16 (Odd n at $\ell=2$: the Hecke action)**.** *Let n be odd and $k\geq 1$.*

1. *[proved here] (given Proposition 16.15's inputs) For every odd prime p, $T_p=0$ on $H^q(\Gamma;V)$ and on $H^q(\Gamma;V/2^k)$, every $q\geq 0$.*
2. *[proved here] $T_2=0$ on $H^0(\Gamma;M_n)$ and on $H^{2j+1}(\Gamma;V)$ for every $j\geq 0$.*
3. *[proved here] $\mathrm{rank}\,T_2\leq 2$ on every $H^q(\Gamma;V)$; and for $k\geq 2$, T_2 on $H^q(\Gamma;V/2^k)$ is $T_2|_{H^q}\oplus T_2|_{H^{q+1}}$ under Lemma 16.13.*
4. *[conjectural] T_2 is a rank-one idempotent on $H^{2j+2}(\Gamma;V)$ for every $j\geq 0$ — equivalently, by (2), the sequence of Lemma 16.9 at $q=1$ and the periodicity of Proposition 16.12(3): T_2 on $H^1(\Gamma;M_n)$, which kills $\mathrm{red}_1 H^1(\Gamma;V)$, induces a rank-one idempotent on the quotient $H^2(\Gamma;V)$. The stronger form observed in the census classification review (`docs/implementation/slot22/census_l23_classification_review.md`, Theorem B(2)) and by `oddn_l2.py` — T_2 itself a rank-one idempotent on $H^1(\Gamma;M_n)$ whose image is not contained in $\mathrm{red}_1 H^1(\Gamma;V)$ — implies it (the image line meets the sub trivially) and is not asserted to be equivalent to it: Lemma 16.14 allows $\mathrm{rank}\,T_2=2$ on $H^1(\Gamma;M_n)$ with a rank-one induced map (v1.22.0-pre.11, F-L23 R1). Evidence: out-of-tree computations for every odd $n\leq 101$ (the L23 master plan's `oddn_l2.py`); not a registered validation. If (4) holds, the Hecke algebra is $\mathbb{F}_2$ on odd degrees and $\mathbb{F}_2\times\mathbb{F}_2$ on even degrees with $(n+1)/2-h^0\geq 2$; that it is $\mathbb{F}_2$ on odd degrees is (1)–(2).*

Proof. (1) $\mathrm{red}_1\colon H^q(\Gamma;V)\hookrightarrow H^q(\Gamma;M_n)$ is equivariant (Lemma 16.9), and $T_p=1+p=0$ on $H^q(\Gamma;M_n)$ (Proposition 16.15); for $k\geq 2$ use Lemma 16.13, for $k=1$ it is the cited proposition. (2) By Step 0(a) of the proof of Proposition 16.15, $T_2 m=\sum_j\rho_n(\delta_j)m$ over left-coset representatives of determinant 2, each of rank 1 mod 2, $\delta_j\equiv u_j w_j^{\mathsf{T}}$; by Definition 8.1 $\rho_n(\delta_j)$ substitutes $(x,y)\mapsto\lambda_j(x,y)\,w_j^{\mathsf{T}}$ with λ_j a linear form, so $\rho_n(\delta_j)m=m(w_j)\,\lambda_j^n$. For $m\in M_n^\Gamma$ the value $m(w)$ is the same for the three non-zero $w\in\mathbb{F}_2^2$ (Γ permutes them transitively and $(\rho_n(\gamma)m)(w)=m(w\bar{\gamma})$), say c_m; and $m\in M_n^S=(1+S)M_n$ (Proposition 16.12(2)), $m=f+\rho_n(S)f$, so $3c_m=\sum_{w\neq 0}m(w)=\sum_{w\neq 0}f(w)+\sum_{w\neq 0}f(w\bar{S})=0$. Hence $T_2=0$ on $H^0(\Gamma;M_n)$, and on $H^{2j+1}(\Gamma;V)$ by the equivariant isomorphism of Proposition 16.12(4). (3)

red_1 is an equivariant injection into $H^q(\Gamma; M_n)$, where Lemma 16.14 applies; the second half is Lemma 16.13. □

Lemma 16.17 (The minimal divisor of $T_p-(1+p)$ is not an independent datum)**.** *[proved here] Let ℓ be a prime, $\mathcal{N}$ a free $\mathbb{Z}_\ell$-module of finite rank and, for $Y \in \mathrm{End}(\mathcal{N})$, $\nu(Y) := \sup\{m : Y\mathcal{N} \subseteq \ell^m\mathcal{N}\} \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, the least exponent among the elementary divisors of Y (∞ iff $Y = 0$).*

1. *If $Y \in \mathrm{End}(\mathcal{N})$ kills a primitive vector e_0 and $c \in \mathbb{Z}_\ell$, then $\nu(Y + c) = \min\big(\nu(Y), v_\ell(c)\big)$.*
2. *Let $n > 0$ be even, $p \neq \ell$ a prime, $\mathcal{N} := \big(H^1(\Gamma; V)/\mathrm{tors}\big) \otimes \mathbb{Z}_\ell$ with its T_p, and assume $\mathcal{N} \otimes \mathbb{Q}_\ell$ contains a T_p-eigenvector of eigenvalue $1 + p^{n+1}$ (the Eisenstein class [standard: [Har22, Ch. 2–3], the pointer of Obligation 9.10]). Then*

$$\nu\big(T_p - (1+p)\big) \;=\; \min\Big(\nu\big(T_p - (1+p^{n+1})\big),\; v_\ell(p^n - 1)\Big).$$

Proof. (1) If $Y \equiv 0$ and $c \equiv 0 \bmod \ell^m$ then $Y + c \equiv 0$. Conversely, if $(Y + c)\mathcal{N} \subseteq \ell^m\mathcal{N}$ then $ce_0 = (Y + c)e_0 \in \ell^m\mathcal{N}$, so $v_\ell(c) \geq m$ (e_0 primitive), and then $Y\mathcal{N} \subseteq \ell^m\mathcal{N}$. (2) The torsion of $H^1(\Gamma; V)$ is stable under every endomorphism, so $\mathcal{N}$ carries T_p; the saturation of the eigenline in $\mathcal{N}$ is generated by a primitive e_0 with $(T_p - (1 + p^{n+1}))e_0 = 0$. Apply (1) with $Y = T_p - (1 + p^{n+1})$ and $c = p^{n+1} - p = p(p^n - 1)$, $v_\ell(c) = v_\ell(p^n - 1)$. □

Remark 16.18 (The standard twist: odd n inside even n). [proved here] Let n be even and $R = \mathbb{Z}_{(2)}$. The Koszul sequence $0 \to \mathrm{Sym}^{n-1} \otimes \det \to \mathrm{Sym}^n \otimes \mathrm{Sym}^1 \to \mathrm{Sym}^{n+1} \to 0$ (multiplication on the right, $f \otimes (x \wedge y) \mapsto fx \otimes y - fy \otimes x$ on the left) is exact and natural for every $A \in \mathrm{M}_2(\mathbb{Z})$, and the comultiplication $\mathrm{Sym}^{n+1} \to \mathrm{Sym}^n \otimes \mathrm{Sym}^1$, $f \mapsto \partial_x f \otimes x + \partial_y f \otimes y$, composed with multiplication is $(n + 1)\mathrm{id}$; as $n + 1$ is a unit,

$$\mathrm{Sym}^n R^2 \otimes \mathrm{Sym}^1 R^2 \;\cong\; \mathrm{Sym}^{n+1} R^2 \;\oplus\; \mathrm{Sym}^{n-1} R^2 \otimes \det$$

as $R[\mathrm{M}_2(\mathbb{Z})]$-modules, so $H^q(\Gamma; \mathrm{Sym}^n \otimes \mathrm{Sym}^1)$ is the sum of $H^q(\Gamma; \mathrm{Sym}^{n+1})$ and $H^q(\Gamma; \mathrm{Sym}^{n-1})$ over R. On the second summand $T_{\Gamma g \Gamma}$ acts as $\det(g) \cdot T_{\Gamma g \Gamma}$ (the pair $(\mathrm{conj}_g)_*$ carries the scalar $\det g$). For odd p this factor is invisible mod 2; **for $p = 2$ it is not**: the operator induced by T_2 on the det-summand is $2\,T_2^{(n-1)}$, which is 0, since $H^q(\Gamma; \mathrm{Sym}^{n-1} R^2)$ has exponent 2 (Lemma 16.9).

Remark 16.19 (An erratum to a project review, and the scope of this subsection). The census classification review `docs/implementation/slot22/census_l23_classification_review.md`, Theorem B(6), stated $H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2) = 0$ and $(\mathbb{Z}/2)^{(n+1)/2}$ in every even degree ≥ 2, for n odd. By Proposition 16.12(3) that is true only for $n = 1$: for every odd $n \geq 3$, $H^1 \cong \mathbb{F}_2^{h^0} \neq 0$ ($H^1(\Gamma; \mathrm{Sym}^3 \mathbb{Z}^2) = \mathbb{Z}/2$, generated by the Bockstein of $d_3 = xy(x + y)$) and the even degrees carry $(n+1)/2 - h^0$. Its universal-coefficient recursion was started at $\dim H^0(\Gamma; M_n) = (n+1)/2$ instead of h^0; that review carries a dated erratum. No statement of this document depended on it. Not stated here, because they are not proved: T_ℓ a rank-one idempotent on $H^1(\Gamma; M_n)$ for even n (the analogue of Proposition 16.16(4)); a growth law for the torsion exponents at 2 and 3 (cf. Remark 16.8(4)); the Gouvêa bound on slopes; the $\ell = 3$ comparison of the $\pm$ halves of the lattice $H^1(\Gamma; V)/\mathrm{tors} \otimes \mathbb{Z}_3$; and the Hodge–Newton comparison of elementary divisors and slopes, which will be written with its Eichler–Shimura input in a later version.

Remark 16.20 (The dimensions at $\ell \in \{2, 3\}$, $n \leq 700$, formalised). [proved here] Let $\ell \in \{2, 3\}$, $M_n = \mathrm{Sym}^n \mathbb{F}_\ell^2$ with $g \mapsto \rho_n(g) \bmod \ell$ ((9)), $d = n + 1$, and $h^q := \dim_{\mathbb{F}_\ell} H^q(\Gamma; M_n)$. On the complex $V^2 \to V^3 \to V^3 \to \cdots$ of Remark 16.3 the h^q are ranks of explicit matrices over $\mathbb{F}_\ell$, by rank–nullity. Write $s = \mathrm{rank}(1 - S, 1 - U) \colon M_n \to M_n^2$.

1. **($\ell = 2$.)** $-I$ acts trivially and $N_S = N_U = e_m = 0$; with $a = \mathrm{rank}\,\psi$, $\psi(x, y) = (1 + S)x - (1 + U + U^2)y$,

$$h^0 = d - s, \qquad h^q = 2d - a - s \quad \text{for every } q \geq 1.$$

[formally verified: [GT, `thm:lv-tree-f2-dims`]] (for every finite-dimensional $\mathbb{F}_2[\Gamma]$-module on which $-I$ acts trivially, with dim in place of d; the degree-0 identity for every finite-dimensional $\mathbb{F}_2[\Gamma]$-module, the others when $-I$ acts trivially) [formally verified: [GT, `thm:lv-f1`]] (the matrices of $1+S$, $1+U+U^2$, $1-S$, $1-U$ on M_n, bit-packed, are those of (9) reduced mod 2, every n) [formally verified: [GT, `thm:lv-v2-dims`]] (the formula for $\mathrm{Sym}^n\mathbb{F}_2^2$, every n, a and s computed by `rankF2`, proved equal to the rank over $\mathbb{F}_2$)

2. **($\ell = 3$.)** For n odd, $h^q = 0$ for every q (Lemma 16.9's last clause and Proposition 31.3(4), at $\mathbb{F}_3$). For n even $-I$ acts trivially; with $r_S^+ = \mathrm{rank}(1+S)$, $r_3 = \mathrm{rank}(1+U+U^2)$, $r_U^- = \mathrm{rank}(1-U)$,

$$h^0 = d - s, \qquad h^1 = 2d - s - r_S^+ - r_3, \qquad h^q = d - r_3 - r_U^- \quad \text{for every } q \geq 2.$$

[formally verified: [GT, `thm:lv3-tree-triv`]] (for every field K with $2 \neq 0$ and every finite-dimensional $K[\Gamma]$-module on which $-I$ acts trivially; proved on the complex of Remark 16.3, the resolution of the amalgam of Claim 16.2; over $\mathbb{F}_3$ these are the dimensions of the terms of Proposition 31.3(2) — a count, not a formalisation of its isomorphisms or of its sequence) [formally verified: [GT, `thm:lv3-rankf3`]] (`rankF3` is the rank over $\mathbb{F}_3$) [formally verified: [GT, `thm:lv3-f1`]] (the matrices of $1 \pm S$, $1-U$, $1+U+U^2$ on M_n, as two bit-planes, are those of (9) reduced mod 3, every n) [formally verified: [GT, `thm:lv3-v3-dims`]] (the vanishing for every odd n and the formula for every even n, for $\mathrm{Sym}^n\mathbb{F}_3^2$)

3. **(The cells.)** [formally verified: [GT, `chk:lv-cells`]] ($\ell = 2$: h^0 and h^q, $q \geq 1$, as numbers, for $n \leq 64$ and $n = 100, 101, 128, 129, 255, 256$) [formally verified: [GT, `chk:lv3-cells`]] ($\ell = 3$: the same for every even $n \leq 64$) For every other $n \leq 700$ at $\ell = 2$, and every other even $n \leq 700$ at $\ell = 3$, the same kernel decision (`decide +kernel`, no `native_decide`) is made in the Lean library `L23Cells`, which is outside [GT]'s census; those cells are **data, not \LEAN evidence**: values, modules, ledger commits and replay logs are `docs/implementation/slot22/lv/lean_dims_l2.tsv`, `lean_dims_l3.tsv` and the ledgers `lv1_ledger.tsv`, `lv2_ledger.tsv`, `lv3_ledger.tsv` there. So $h^0, \ldots, h^3$ are formalised for $\ell = 2, 3$ and every $n \leq 700$ (701 values of n at each ℓ; at $\ell = 3$, n odd, by the theorem), as Lean theorems on mathlib's `groupCohomology` with the coefficient module of (9), standard axioms only.

This is a **different model** from the census's: the Lean values are computed through the tree complex, the C++ values on the isotropy-resolution model $C_{\mathcal{I}}^\bullet$ of Proposition 23.5. They agree wherever both exist — read at the census snapshot of 2026-10-11, 22:47, recorded in `lean_dims_l2.tsv` / `lean_dims_l3.tsv` at commit `9e07e1a0`: at $\ell = 2$, 264 computed records, all at even $n \leq 528$; at $\ell = 3$, 61 (31 even $n \leq 60$, 30 odd); no mismatch — so the agreement is a cross-model check, not a re-run; later census passes add records, and a `census_mismatch` on any of them would be a contradiction between the kernel and a C++ record, to be reported. At $\ell = 2$, n odd, the census rows are closed forms, not computations: there the Lean cell is the first computation of the number, and a per-cell check of Proposition 16.12's values (h^0, and $h^q = (n+1)/2$, $q \geq 1$; all 350 odd $n \leq 699$ agree with the closed forms, same snapshot). Nothing here concerns the integral groups $H^q(\Gamma; V)$, the Hecke operators (Proposition 16.15 carries no marker), or any census column other than $h^0, \ldots, h^3$; and no clause of this document is superseded.

16.5 The boundary groups, and the long exact sequence

Claim 16.21 (Boundary groups)**.** *[proved here] Let R be a field of characteristic ℓ, n even, and $2 \in R^\times$ (i.e. $\ell \neq 2$). Then*

$$H^0(\partial; V_R) = V_R^T, \qquad H^1(\partial; V_R) = (V_R)_T, \qquad H^q(\partial; V_R) = 0 \quad (q \geq 2),$$

both non-zero groups of dimension $\lfloor n/\ell \rfloor + 1$*, and*

$$V_R^T = R[x, \mathrm{Nm}]_n = \bigoplus_{0 \le b \le \lfloor n/\ell \rfloor} R \cdot x^{\,n-b\ell} \mathrm{Nm}^{\,b}, \qquad \mathrm{Nm} = y^\ell - x^{\ell-1} y.$$

Proof. $\Gamma_\infty = \langle T \rangle \times \langle -I \rangle$ and $\rho_n(-I) = \mathrm{id}$ for n even, so with $2 \in R^\times$ Proposition 14.4(3) identifies $H^q(\Gamma_\infty; V_R)$ with $H^q(\langle T \rangle; V_R)$ for every q, and Proposition 14.4(2) evaluates the latter as V_R^T, $(V_R)_T$, $0, 0, \dots$. The description of V_R^T and its rank are Proposition 14.16(1); $\dim (V_R)_T = \dim V_R^T$ because both are the kernel and the cokernel of one endomorphism of a finite-dimensional space; non-vanishing is the nilpotence argument in the proof of Lemma 14.20, which used no hypothesis on ℓ. □

Remark 16.22 (Two repairs in Claim 16.21). **(i)** [E] states "$H^i(\partial; M) = 0$ for $i \ge 2$" under its blanket $\ell \ge 5$. As a statement about $H^\bullet(\Gamma_\infty; V_R)$ that is *false at* $\ell = 2$, where every degree is non-zero (Lemma 14.20); the hypothesis that carries it is $2 \in R^\times$, and it is now in the statement. **(ii)** The invariant [E] writes N is written Nm here: $N = p^\nu$ is the Hecke level (§2.2), and the two meet in Proposition 14.16. This is collision **C18** of §2.7.

Claim 16.23 (The six-term sequence)**.** *Assume (H4) and (B1)–(B3) and let* R *be a commutative ring.*

1. *[proved here]* ***(The sequence, in all degrees, with no hypothesis on*** ℓ ***or*** n***.)*** *The short exact sequence of complexes* $0 \to C_\partial^\bullet \xrightarrow{\mathrm{inc}_c} C(r) \xrightarrow{\mathrm{pr}_c} C_{\mathrm{eq}}^\bullet(K, \mathcal{R}; L)[1] \to 0$ *of Lemma 15.2(2) gives the long exact sequence (39),*

$$\cdots \to H_c^q \to H_{\mathrm{eq}}^q(K, \mathcal{R}; L) \xrightarrow{r^q} H^q(C_\partial^\bullet) \to H_c^{q+1} \to \cdots,$$

and ***every map in it intertwines*** $(T^\partial_{\Gamma g \Gamma}, T^c_{\Gamma g \Gamma}, T_{\Gamma g \Gamma})$*, by Theorem 15.11(a) — which needs only that* Φ_g *is a cochain map, and that is a theorem here, not a hypothesis.*

2. *[proved here]* ***(Collapse to six terms.)*** *[formally verified: [GT,* `cl:les`*]] (the injectivity of* r_*^0 *and the Euler identity, under* $H^2_{\mathrm{eq}} = 0$*; exactness is Lemma 15.2(2)) $H_c^q = 0$ for $q \le 0$ and $q \ge 3$* unconditionally*, and $H^q(C_\partial^\bullet) = 0$ for $q \ge 2$ because ∂X is one-dimensional ((B2)). If in addition $H^2_{\mathrm{eq}}(K, \mathcal{R}; L) = 0$, the sequence of (1) collapses to*

$$0 \to X_1 \xrightarrow{r^0} X_2 \to X_3 \to X_4 \xrightarrow{r^1} X_5 \to X_6 \to 0, \tag{52}$$

with r^0 the inclusion $V_R^\Gamma \hookrightarrow V_R^{\Gamma_\infty}$, and over a field

$$\dim X_3 = \dim X_2 - \dim X_1 + \dim \ker r^1, \qquad \dim X_6 = \dim X_5 - \operatorname{rank} r^1. \tag{53}$$

Write $H^1_! := \ker r^1 = \mathrm{im}(X_3 \to X_4)$.

3. ***(When*** (52) ***is a statement about the six groups*** (49)***.)*** *[proved here] Assume $6 \in R^\times$ and n even. Then $H^q_{\mathrm{eq}} \cong H^q(\Gamma; V_R)$ (Remark 7.6), $H^q(C_\partial^\bullet) \cong H^q(\partial; V_R)$ (Proposition 14.4(3), which uses n even and $2 \in R^\times$), and $H^2_{\mathrm{eq}} = 0$ (Lemma 16.4(1)); so (52) is exact with the entries (49).* ***The hypothesis*** *$H^2(\Gamma; V_R) = 0$* ***is not removable*** *(Lemma 16.4(2)); the hypothesis $H_c^0 = 0$ that [E] listed beside it* ***is****, being unconditional by (2). At $\ell \in \{2, 3\}$ the three identifications fail and (52) is a* window *of a sequence with a non-zero term in every degree: Remark 14.21, and 14.24 for the policy. Read on the model complexes, (1) and (2) hold at every ℓ.*

4. *[conjectural]* as a citation. *In the presentation of Claim 16.2(3), r^1 is induced by*

$$r^1[m] \;=\; \big[\rho_n(S)^{-1}\big(\rho_n(U) - \mathrm{id}\big)m\big] \;\in\; (V_R)_T, \qquad U = ST. \tag{54}$$

The map is [proved here] well defined — it kills both generators of the denominator of Claim 16.2(3): $(\rho_n(U) - \mathrm{id})(1 + \rho_n(U) + \rho_n(U)^2) = \rho_n(U)^3 - \mathrm{id} = \rho_n(-I) - \mathrm{id} = 0$ *for*

n even, and $\rho_n(S)^{-1}(\rho_n(U)-\mathrm{id})(1+\rho_n(S)) = (\rho_n(T)-\mathrm{id}) + (\rho_n(T)-\mathrm{id})\rho_n(S)$ lies in $\mathrm{im}(\rho_n(T)-\mathrm{id})$, using $\rho_n(S)^{-1}\rho_n(ST) = \rho_n(T)$, $\rho_n(S)^{-1} = \rho_n(S)$ for n even, and $\rho_n(TS) - \rho_n(S) = (\rho_n(T)-\mathrm{id})\rho_n(S)$. ***That it is the restriction map*** *is* not *proved here: its only source is the working note* `hecke_extensions_mathematics.md`*, which Appendix A, entry A.14, declares not a source of record. Nothing below (54) depends on it: (53) is proved from (2) alone.*

Proof. (1) is Lemma 15.2(2) and Theorem 15.11(a) verbatim. (2) The two vanishings are Lemma 15.2(4) and (5); the collapse and (53) are Lemma 15.2(5), whose hypothesis is exactly $H^2_{\mathrm{eq}} = 0$; r^0 is identified in the proof of Lemma 15.2(4). (3) collects the three cited identifications. (4) the displayed computations, using ρ_n's multiplicativity (Proposition 8.2(2)) and $\rho_n(-I) = \mathrm{id}$. □

16.6 The boundary characters

Claim 16.24 (Boundary eigenvalues)**.** *Let R be a field of characteristic ℓ, n even, $2 \in R^\times$ — so that X_2 and X_5 are the two groups of Claim 16.21; clauses (1) and (2) hold verbatim for $H^\bullet(C^\bullet_\partial)$ at every ℓ — let $p \neq \ell$ be a prime ((H1)), $g = \mathrm{diag}(1,p)$, and $\chi_b(p) := p^b + p^{n+1-b}$ for $0 \le b \le \lfloor n/\ell \rfloor$.*

1. *[proved here] On $X_2 = H^0(\partial; V_R) = V_R^T$ the monomials $x^{n-b\ell}\mathrm{Nm}^b$ are eigenvectors of Θ^∂_g with eigenvalues $\chi_b(p)$. This is Proposition 14.16(1), and the right-coset convention gives χ_{b+1} instead — Proposition 14.16(2), a* different operator*. For $n < \ell$ both X_2 and X_5 are 1-dimensional and carry $\chi_0(p) = 1 + p^{n+1}$, the Eisenstein eigenvalue of Obligation 20.7 (Proposition 14.16(3)).*
2. *[conjectural] $X_5 = H^1(\partial; V_R) = (V_R)_T$ carries the* same *multiset $\{\chi_b\}_{0\le b\le\lfloor n/\ell\rfloor}$, with the class of y^n carrying χ_0. Proposition 14.16(3) proves the case $n < \ell$; the general case is not proved here and is what Obligation 14.19(vi)* records *rather than gates. [verified computationally] It was checked for every even $n \le 24$, $\ell \in \{2,3,5,7,11,13\}$ and $p \in \{2,3,5,7,11,13,17\} \setminus \{\ell\}$ — 468 cases — by forming ρ_n over $\mathbb{F}_\ell$, the operator (31) on $(V_R)_T$ and $\sum_\delta \rho_n(\delta)$ on V_R^T, and factoring both characteristic polynomials: in every case both multisets are exactly $\{\chi_b\}$ and both polynomials split.* Ad hoc*, not a registered validation.* Status note *(v1.12.0-pre.3): the clause has two halves, and they now differ. The* semisimplicity *half — that Θ^∂_g on X_5 is diagonalisable over $\mathbb{F}_\ell$, the operators for $p \neq \ell$ commuting and depending on $p \bmod \ell$ only, with eigenvalue $\chi_{n-k}(p)$ on a torus weight vector of weight $a^{n-k}d^k$ — is* ***proved****: Lemma 14.18(1)–(3) at $R = \mathbb{F}_\ell$, and over any field of characteristic ℓ by extension of scalars, which rests on Proposition 14.17 (the operator of (30) on X_5 is (31)). The* multiset *half — which torus weights $(V_R)_T$ carries, i.e. that the eigenvalues are exactly $\{\chi_b\}_{0\le b\le\lfloor n/\ell\rfloor}$ — does not follow and* ***stays*** **[conjectural]***; the tag above is that half's. The clause's last part, "with the class of y^n carrying χ_0",* is *proved for every even n and $p \neq \ell$: $y^n \notin \mathrm{im}(\rho_n(T)-\mathrm{id})$ (every element of that image has y-degree $< n$, as $(\rho_n(T)-\mathrm{id})u_j = \sum_{t<j}\binom{j}{t}u_t$), and y^n has torus weight d^n, so Lemma 14.18(3) with $k = n$ gives $\chi_0(p)$ (v1.12.0-pre.3, referee P3). Later evidence for it, all ad hoc or recorded, none gating:* 2080 *group-form cases ($\ell = 5$, even $n \le 200$; $\ell \in \{7,11,13\}$, even $n \le 120$; slot-12 WP-13) and* 814/814 *shipped operators on X_5 (slot-12 WP-14,* `HeckeBoundaryFinalityValidation` *arm (L), recorded).*
3. *[proved here] **(Coincidences.)** As functions of p on primes $p \neq \ell$, $\chi_a \equiv \chi_{a'}$ if and only if $a \equiv a'$ or $a \equiv n+1-a' \pmod{\ell-1}$. In particular $\chi_0 \equiv \chi_1$ iff $(\ell-1) \mid n$, and $\chi_1 \equiv \chi_2$ iff $(\ell-1) \mid (n-2)$.*

Proof. (3) $\chi_a(p)$ depends on p only through $p \bmod \ell$, and $p \mapsto p \bmod \ell$ is onto $\mathbb{F}_\ell^\times$ as p runs over the primes $\neq \ell$ [standard: [Ser73, Ch. VI, §4]: Dirichlet's theorem on primes in arithmetic progressions; the section pointer is not verified against the text (rule (A2)), and the statement is

used only through this surjectivity, which is why (3) is really a statement about functions on $\mathbb{F}_\ell^\times$]. So $\chi_a \equiv \chi_{a'}$ on primes iff $t^a + t^{n+1-a} = t^{a'} + t^{n+1-a'}$ for every $t \in \mathbb{F}_\ell^\times$.

Put $m := \ell - 1$ and fix a generator g of the cyclic group $\mathbb{F}_\ell^\times$. A function $\mathbb{F}_\ell^\times \to \mathbb{F}_\ell$ is the vector of its values at $g^0, \dots, g^{m-1}$, and $t \mapsto t^c$ has value vector $(g^{cs})_{s<m}$, which depends only on $c \bmod m$. The $m \times m$ matrix $(g^{cs})_{c,s\in\mathbb{Z}/m}$ is a Vandermonde matrix in the m *distinct* elements $g^0, \dots, g^{m-1}$ of $\mathbb{F}_\ell$, hence invertible; so the m functions $t \mapsto t^c$, $c \in \mathbb{Z}/m$, are linearly independent over $\mathbb{F}_\ell$. The displayed equality is a vanishing $\mathbb{F}_\ell$-linear combination of them with integer coefficients in $\{0, \pm 1, \pm 2\}$, all of which are non-zero in $\mathbb{F}_\ell$ when they are non-zero in $\mathbb{Z}$, provided $\ell > 2$. Hence for $\ell > 2$ the multisets $\{a, n+1-a\}$ and $\{a', n+1-a'\}$ agree modulo m, which is the stated disjunction: if $a \equiv a'$ then $n+1-a \equiv n+1-a'$, and if $a \equiv n+1-a'$ then $n+1-a \equiv a'$. For $\ell = 2$, $m = 1$ and both sides of the equivalence hold for all a, a'. Specialising $(a, a') = (0, 1)$ gives $0 \equiv 1$ or $0 \equiv n$, and $(1, 2)$ gives $1 \equiv 2$ or $1 \equiv n - 1$; for $\ell > 2$ the first alternative fails in both. □

Remark 16.25 (Which Dickson classes fail to separate the two indexings). (Appendix A, entry A.21.) [E] concludes from Claim 16.24(3) that "$\chi_1 \equiv \chi_2$ at every Dickson locus, so Dickson classes cannot discriminate the two indexings". **The first half is false.** On the Dickson locus $n = a(\ell+1) + b(\ell^2 - \ell)$ of Claim 16.5(2) one has $n \equiv 2a \pmod{\ell - 1}$, since $\ell + 1 \equiv 2$ and $\ell^2 - \ell \equiv 0$; so $\chi_1 \equiv \chi_2$ iff $(\ell - 1) \mid 2(a - 1)$, **not** always. It does hold at the *first* Dickson class $n = \ell + 1$ ($a = 1$, $b = 0$), which is [E]'s example $(n, \ell) = (6, 5)$: there $n - 2 = \ell - 1$ and $\chi_1(2) = 2 + 2^6 \equiv 1 \equiv \chi_2(2)$. It fails already at the next one, $(n, \ell) = (12, 5)$ ($a = 2$): $\chi_1(2) = 2 + 2^{12} \equiv 3$ and $\chi_2(2) = 2^2 + 2^{11} \equiv 2$ in $\mathbb{F}_5$. [verified computationally] Checked for $\ell \in \{5, 7, 11, 13\}$ and every Dickson locus with $n < 60$: the criterion $(\ell - 1) \mid (n - 2)$ agrees with direct evaluation in every case, and the criterion of Claim 16.24(3) was checked exhaustively for $\ell \le 19$, $n < 40$ and all $0 \le a, a' \le n + 1$ — 190 560 cases, 0 disagreements. **What survives** is the conclusion [E] drew it for: the first Dickson class never separates the left- and right-coset indexings, so the separation must come from the Eisenstein eigenvalue $1 + p^{n+1}$ instead — which is how Proposition 14.16(3) and Obligation 14.19(vii) settle the convention.

16.7 Poincaré duality

Claim 16.26 (Poincaré duality). ***[open] This is owner Open Question 4 and is not proved in this document****; its coefficient-level input is proved (Proposition 8.6) and its cohomological content is not. Let $\ell \ge 5$, $R = \mathbb{F}_\ell$, n even, $M_n = \mathrm{Sym}^n \mathbb{F}_\ell^2$ and D_n its contragredient (Definition 8.5).*

1. ***(Perfect pairings.)*** *The evaluation pairing of Proposition 8.6 induces perfect pairings*

$$H_c^t(K, \mathcal{R}; M_n) \times H_{\mathrm{eq}}^{2-t}(K, \mathcal{R}; D_n) \to \mathbb{F}_\ell, \qquad H^t(\partial; M_n) \times H^{1-t}(\partial; D_n) \to \mathbb{F}_\ell,$$

   equivalently, under $6 \in R^\times$ (Remark 7.6, Lemma 15.2(3)), $H_c^t(\Gamma; M_n) \times H^{2-t}(\Gamma; D_n) \to \mathbb{F}_\ell$ — which is how [E] and Claim 16.23 name the same groups.
2. ***(Self-adjointness — in the adjointness form, and only there.)*** *Under (1), for every $g \in \Delta$ the operator $T_{\Gamma g \Gamma}$ on the M_n-side is adjoint to $T_{\Gamma(\mathrm{adj}\, g)\Gamma}$ on the D_n-side:*

$$\langle T_{\Gamma g \Gamma}\, x,\ y \rangle = \langle x,\ T_{\Gamma(\mathrm{adj}\, g)\Gamma}\, y \rangle,$$

   with no determinant factor*; and for $g = \mathrm{diag}(1, N)$ one has $\Gamma(\mathrm{adj}\, g)\Gamma = \Gamma g \Gamma$ (Proposition 8.6(5)), so T_p, T_{p^ν} and T_N are* self-*adjoint. This is the form in which Proposition 15.20(3) reads this claim.*
3. ***(Consequences.)*** *For a maximal ideal $\mathfrak{m}$ of the Hecke algebra $\mathbb{T}$, $H_c^t(M_n)_\mathfrak{m}$ and $H^{2-t}(D_n)_\mathfrak{m}$ have the same characteristic polynomials, and the* Loewy *layers of one are the* socle *layers of the other in reverse order — $\mathrm{rad}^i(V^\vee) = (\mathrm{soc}^i V)^\perp$ for a module V over the local ring $\mathbb{T}_\mathfrak{m}$. The two series coincide only for rigid modules. The classification consequence drawn from this is [E, `pr:dual`] and stays there.*

Remark 16.27 (What Claim 16.26 rests on, and the p^n twist it must not inherit). (Appendix A, entries A.14 and A.18.)

1. **The adjointness form is twist-free; the diagonal one is not.** Proposition 8.6 proves two different identities. (3), $\langle \rho_n(g)v, \varphi\rangle = \langle v, \rho_n^\vee(\operatorname{adj} g)\varphi\rangle$, is exact, integral and carries *no* determinant factor. (4), $\langle \rho_n(g)v, \rho_n^\vee(g)\varphi\rangle = (\det g)^n\langle v, \varphi\rangle$, carries the twist $(\det g)^n = p^n$ for $g = \operatorname{diag}(1, p)$. [E]'s wording "$T_q$ self-adjoint once D_n carries the adjugate Δ-structure $g \mapsto \det(g)g^{-1}$" is true of the *adjointness* form and false of the diagonal one, and Claim 16.26(2) is stated in the former. Note $\det(g)g^{-1} = \operatorname{adj} g$, so the structure [E] names is Definition 8.5's.
2. **What it depends on.** (i) Proposition 8.6(2),(3),(5) — the perfectness of the coefficient pairing over every R, the adjointness identity, and the self-adjugacy of the double coset; (ii) the construction of $H_c^\bullet$ as relative cohomology, Lemma 15.2(3), and of its Hecke action, Definition 15.9; (iii) the boundary cohomology Definition 14.2 and its operator Definition 14.14; (iv) the hypothesis $6 \in R^\times$, through Remark 7.6 and Lemma 16.4(1). **Not** yet available: a cochain-level pairing on the stored bases. Until it exists, (1) and (2) are assertions with no certificate, and Proposition 15.20(3) — the acceptance gate for X_3 — inherits their [open] status.
3. **Socle, not Loewy.** Entry A.14 of Appendix A records the working note's statement of (3) for the *Loewy* series. It is the socle series that reverses; Claim 16.26(3) says so.
4. M_n **is not self-dual for** $n \geq \ell$, so (1) genuinely relates two different local systems; the integral comparison map Φ_n of Lemma 8.8 is an isomorphism exactly when $n! \in R^\times$, i.e. $n \leq \ell - 1$ (Remark 8.9). This is why the pipeline, which computes M_n only, cannot check (1) without building D_n — phase E7 of the extension programme, made unconditional by owner decision OQ4.

Code map. Two of the six groups are computed with their Hecke action, two without, two not at all. $X_1 = H^0_{\mathrm{eq}}$ and $X_4 = H^1_{\mathrm{eq}}$, their bases, projectors and every matrix $A^{(q)}(T_p)$, $q \in \{0, 1, 2\}$: `EquivariantCohomologyOnK` and `HeckeOperatorOnK`, with $A^{(2)}$ emitted as the 0×0 matrix (§13.4, Lemma 16.4(1)). X_2, X_5 and the restriction maps r^0, r^1: the circle model of `LocalSystemPresets::CreateS1Generic` and the morphism `TestSL2Z_To_S1Cusp_RestrictionMorphism` (`LocalSystemSL2Z.cpp`:843–922), pinned by `HeckeMorphismRegressionValidation`; **no boundary Hecke operator exists** (census of 2026-09-16, §14.5 code map). X_3, X_6: **not computable**; the mapping cone exists only for the non-equivariant `CochainComplex` (`ConeComplex()`, `SimplicialCohomology.h`), not for `EquivariantCochainComplexMorphismOnK`. Claim 16.5 is corroborated by `docs/matrix-engine/results/results_table.tsv` as recorded above; Claim 16.2(3) is the presentation `BarCocycleH1` implements; Claim 16.26 has **no implementation**, $\rho_n^\vee$ and the pairing not being built (§8.1 code map).

Formalization note. Claim 16.2(2) is the cheapest target in this chapter and the most reusable: it is $\operatorname{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(-, V_R)$ applied to one short exact sequence, and `mathlib4` has both the long exact sequence of Ext and Shapiro's lemma in the form $\operatorname{Ext}^\bullet_R(\operatorname{Ind} M, N)$. Its input (50) is the one genuinely external object — the Bass–Serre tree — and a formalisation may take (50) itself as the definition of the amalgam datum rather than derive it. Claim 16.2(3),(4), Lemma 16.4 and Claim 16.21 are then finite consequences, the first needing only $|\Gamma_0|\, H^q(\Gamma_0; -) = 0$ for a finite subgroup Γ_0. Claim 16.24(3) is character theory of a cyclic group and is fully elementary; only its appeal to Dirichlet is external, and that can be avoided by restating the claim over $\mathbb{F}_\ell^\times$, which is what the proof actually uses. Claim 16.5 and Claim 16.26 are *not* formalisation targets at this stage: the first imports Eichler–Shimura and Dickson, the second is open.

17 The general orbifold model, and the torsion primes

What this chapter fixes. Definitions only; the construction is deferred — owner decision OP4, 2026-09-15. What is fixed here is *which object* the words "the orbifold cohomology of the pair" name when $6 \notin R^\times$, and in what shape its Hecke datum is to be written, so that the deferred construction cannot later be taken to define something else. Three things are stated: the Borel construction of the pair and its free Γ-model (Definition 17.1); the sense in which the 7-vertex complex of Definition 7.1 is *a* specialisation of it at $6 \in R^\times$ (Theorem 17.3); and what Moerdijk–Pronk's simplicial complex is and is not cited for (Remark 17.4). What is *not* here: any construction of a free model, of its τ-tables, or of the $\ell = 2$ truncation — that is one piece of work with 14.24, and it is deferred with it. (Since v1.8.0 the construction, on the unrefined $(X, \partial X)$ and as existence, is §23.)

Two standing instructions on how these definitions are phrased.

1. **Degree-generic.** Every definition below is stated in a degree range $\{0, \dots, q_{\max}\}$ with $q_{\max}$ a parameter, never in $\{0, 1, 2\}$. This is 14.24's site (4) (§14.5), and it is what makes the truncation policy expressible at all: at $\ell \in \{2, 3\}$ the complexes below are unbounded above (Remark 14.21(3)).
2. **Refinement-parametric** (owner direction OD-T1, 2026-09-15, recorded at Remark 7.6). The input is an *arbitrary* Γ-pair satisfying (G1)–(G4) of Definition 17.1 — which are (H4)(i),(ii) and (B1),(B2) over an arbitrary input, *without* (H4)(iv)'s stabiliser inventory — not the 7-vertex one; a Γ-equivariant refinement of the 7-vertex complex is then admissible input rather than an exception. Nothing below names a vertex, an edge or a triangle of Definition 7.1, and nothing below may be read as "the 7-vertex complex plus corrections". Remark 17.5 says what a refinement would and would not change.

Definition 17.1 (The Borel construction of the pair, and its twisted chain data)**.** Let R be a commutative ring and V_R a $\mathbb{Z}[\Delta]$-module, free and finitely generated over $\mathbb{Z}$ (Definition 8.1: $V_R = \mathrm{Sym}^n(\mathbb{Z}^2) \otimes R$). Let $(Y, \partial Y)$ be a pair of simplicial Γ-complexes satisfying:

(G1) Γ acts simplicially on Y without inversions, with finite stabilisers, and $\partial Y \subseteq Y$ is a Γ-subcomplex;
(G2) Y is contractible and finite-dimensional;
(G3) Y has finitely many Γ-orbits of simplices in each degree, and so does ∂Y;
(G4) ∂Y is a disjoint union of contractible components permuted transitively by Γ, the stabiliser of a distinguished component ∂Y_∞ being Γ_∞.

((G1)–(G4) are *rule tags*, in the style of (H1)–(H4) and (B1)–(B3); they do not reintroduce the letter G of §3 as a symbol, and no statement of this chapter uses that letter.) (G1) and (G2) are (H4)(i),(ii), (G4) is (B1),(B2), and (G3) is what the finiteness of K supplies in (H4)(iii) — all four with the 7-vertex complex replaced by an arbitrary input. (H4)(iv)'s *inventory* of stabilisers is deliberately **not** assumed, only their finiteness; (H4)(iii) itself, the existence of a strict fundamental domain, is assumed only where Theorem 17.3 needs it. Fix a bound $q_{\max} \geq 0$ (14.24) and a free resolution $F_\bullet \to \mathbb{Z}$ of the trivial module over $\mathbb{Z}[\Gamma]$.

1. **(The free Γ-model.)** Put

$$\widehat{C}_\bullet(Y) := F_\bullet \otimes_{\mathbb{Z}} C_\bullet(Y), \qquad \widehat{C}_\bullet(Y, \partial Y) := F_\bullet \otimes_{\mathbb{Z}} \big(C_\bullet(Y)/C_\bullet(\partial Y)\big), \tag{55}$$

each with the diagonal Γ-action and the total differential. Both are complexes of **free** $\mathbb{Z}[\Gamma]$-modules, because $F_\bullet$ is free and the simplicial chain modules are free over $\mathbb{Z}$; and $\widehat{C}_\bullet(Y) \to \mathbb{Z}$ is a free resolution, because Y is contractible ((G2)). This is the simplicial-chain incarnation of the Borel construction $E\Gamma \times_\Gamma Y$.

2. **(The orbifold cochain complexes and their cohomology.)** For $0 \leq q \leq q_{\max} + 1$ set

$$\widehat{C}^{\,q}(Y; V_R) := \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(\widehat{C}_q(Y), V_R\big),$$

and likewise for the pair and for ∂Y, with the cone $C(r)$ of the restriction $r := j^*$ formed exactly as in Definition 15.1. The **orbifold cohomology of the pair** is the cohomology of these complexes, reported in degrees $\leq q_{\max}$:

$$H^q_\Gamma(Y;V_R), \qquad H^q_\Gamma(\partial Y;V_R), \qquad H^q_\Gamma(Y,\partial Y;V_R).$$

[proved here] By (1), $\widehat{C}_\bullet(Y) \to \mathbb{Z}$ is a free resolution over $\mathbb{Z}[\Gamma]$, so $H^q_\Gamma(Y;V_R) = H^q(\Gamma;V_R)$ by the definition of group cohomology. On the boundary, (G4) gives $C_\bullet(\partial Y) \cong \mathbb{Z}[\Gamma]\otimes_{\mathbb{Z}[\Gamma_\infty]} C_\bullet(\partial Y_\infty)$, hence $\widehat{C}_\bullet(\partial Y) \cong \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\infty]} \big(F_\bullet \otimes_{\mathbb{Z}} C_\bullet(\partial Y_\infty)\big)$ by the projection formula, and the right-hand factor is a free $\mathbb{Z}[\Gamma_\infty]$-resolution of $\mathbb{Z}$ ($F_\bullet$ restricted to Γ_∞ is still free, ∂Y_∞ is contractible); so by the adjunction of Proposition 14.4(1), $H^q_\Gamma(\partial Y;V_R) = H^q(\Gamma_\infty;V_R) = H^q(\partial;V_R)$ of Definition 14.2. Both hold for every R, every n and every reported degree $q \leq q_{\max}$ — with *no* invertibility hypothesis, which is the point of the construction. (Degree $q_{\max}+1$ is scaffolding and is not reported: 14.24's truncation lemma.)

3. **(Twisted chain data on the free model.)** A **Hecke datum of the pair on** $(Y,\partial Y)$ is a triple $(\tau,\tau_\partial,\eta)$ of additive maps

$$\tau\colon \widehat{C}_\bullet(Y) \to \widehat{C}_\bullet(Y), \quad \tau_\partial\colon \widehat{C}_\bullet(\partial Y) \to \widehat{C}_\bullet(\partial Y), \quad \eta\colon \widehat{C}_\bullet(\partial Y) \to \widehat{C}_{\bullet+1}(Y),$$

defined in every degree $\leq q_{\max}+1$, satisfying (T1)–(T3) of Definition 11.1 and (T1∂)–(T3∂) of Definition 14.8 verbatim — with $\widehat{C}_\bullet$ in place of $C_\bullet(K^u)$ and $C_\bullet(\partial X)$ — together with the pair identity $\partial\eta + \eta\partial = j\tau_\partial - \tau j$ of Definition 15.5. The cochain maps Θ_g, Θ^∂_g, the derived homotopy h and the cone map Φ_g are their coset transports (Definition 15.3, Definition 15.9), unchanged.

Remark 17.2 (Why the free model buys what it buys). [proved here] On $\widehat{C}_\bullet$ the source of every comparison argument is *free* over $\mathbb{Z}[\Gamma]$ and the target is acyclic, so the fundamental lemma of homological algebra applies with no relative projectivity and no invertibility hypothesis [standard: [Wei94, Thm. 2.2.6]; [Bro82, Ch. I.7.4]]. Consequently, on a model satisfying (G1)–(G4), *both* clauses of Lemma 11.3 hold over $\mathbb{Z}$ for every ℓ — existence, which Lemma 11.3(1) already gives on the 7-vertex model, *and* homotopy uniqueness, which on that model needs $6 \in R^\times$ (Lemma 11.3(2), Remark 11.4). The same applies to τ_∂ (Lemma 14.10, which is already unconditional) and to η (Lemma 15.6, Lemma 15.7). Hence Theorem 15.11(c) — independence of τ, τ_∂, s_j and g, and the identification with $\mathrm{cor}\circ(\mathrm{conj}_g)_*\circ\mathrm{res}$ — holds on this model **for every** ℓ, where on the 7-vertex model it is asserted only for $6 \in R^\times$ (Theorem 15.11(e)). **That is the entire reason the free model is wanted**, and it is a statement about the model, not about a construction of one: no such model is built in this document.

Theorem 17.3 (The 7-vertex complex is *a* specialisation at $6 \in R^\times$)**.** *Let $(Y,\partial Y)$ satisfy (G1)–(G4), let R be a commutative ring with $6 \in R^\times$ — more precisely, with $|\Gamma_\sigma| \in R^\times$ for every simplex σ of Y and of ∂Y — let $q_{\max} \geq 0$, and let $(K_Y,\mathcal{R}_Y)$ present the quotient of Y in the sense of §3 (Assumption 3.17). Then for every q with $0 \leq q \leq q_{\max}$:*

1. *[proved here] $H^q_\Gamma(Y;V_R) \cong H^q_{\mathrm{eq}}(K_Y,\mathcal{R}_Y;L)$ and $H^q_\Gamma(\partial Y;V_R) \cong H^q\big(\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(\partial Y),V_R)\big)$ — the latter being $H^q(C^\bullet_\partial)$ when $\partial Y = \partial X$ — naturally; i.e. the equivariant cochain complex of §5 is the $t=0$ row of the isotropy spectral sequence of the free model, and the rows $t>0$ vanish.*
2. *[proved here] $H^q_\Gamma(Y,\partial Y;V_R) \cong H^q_c$ of Lemma 15.2(3), compatibly with (1) and with the long exact sequence (39).*
3. *[proved here] The two Hecke actions agree: the operator induced on $H^\bullet_\Gamma(Y,\partial Y;V_R)$ by a Hecke datum of the pair on $(Y,\partial Y)$ (Definition 17.1(3)) corresponds under (2) to $T^c_{\Gamma g\Gamma}$ of Definition 15.9, and likewise in the bulk and on the boundary.* Both *are $\mathrm{cor}\circ(\mathrm{conj}_g)_*\circ\mathrm{res}$ (Theorem 11.10, Theorem 15.11(c), Remark 17.2), which determines them.*

4. *Taking* $(Y, \partial Y) = (X, \partial X)$ *and* $(K_Y, \mathcal{R}_Y) = (K, \mathcal{R})$ *of Definitions 7.1 and 7.2, which satisfies (G1)–(G4) by Proposition 7.4 and (B1)–(B2), recovers the whole shipped pipeline.* ***It is one admissible input among others****: any* Γ*-equivariant refinement of* $(K, \mathcal{R})$ *is another, with the* same *left-hand sides in (1)–(3) and different tables (Remark 17.5).*

Not asserted, at any ℓ*:* *that a model satisfying (G1)–(G4) with* ∂Y *as in (G4) exists for a* refinement *of* $(K, \mathcal{R})$*, or that one can be built effectively. That is the deferred construction.* Since v1.8.0 *a finite-rank free model on* $(X, \partial X)$ *itself — which (4) already admits with* $F \otimes C_\bullet(X)$ *— is constructed, with its comparison map and a Hecke datum of the pair on it, as existence (Definition 23.4, Proposition 23.5, Proposition 23.13); for a refinement nothing is asserted and nothing is needed (Remark 23.6). (Worded so at v1.8.1: v1.8.0's sentence read as if the unrefined case had been deferred.)*

Proof. (1) $\widehat{C}_\bullet(Y) = F_\bullet \otimes C_\bullet(Y)$ is a double complex of $\mathbb{Z}[\Gamma]$-modules; its second spectral sequence is the isotropy spectral sequence, with $E_1^{s,t} = \bigoplus_{\sigma \in \Sigma_s} H^t(\Gamma_\sigma; V_R)$, Σ_s a set of representatives of the Γ-orbits of s-simplices (finite by (G3)), converging to $H_\Gamma^{s+t}(Y; V_R)$ [standard: [Bro82, Ch. VII, §7, (7.10)], with the same rule (A2) caveat on the pointer as in Remark 7.6, which is the same citation]. Each $H^t(\Gamma_\sigma; V_R)$ with $t \geq 1$ is annihilated by $|\Gamma_\sigma|$ [standard: [Bro82, Ch. III.10.2]], which is a unit, so those rows vanish; the surviving row $t = 0$ is $s \mapsto \bigoplus_{\sigma \in \Sigma_s} V_R^{\Gamma_\sigma}$, which is $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(Y), V_R)$, and that is $C_{\mathrm{eq}}^\bullet(K_Y, \mathcal{R}_Y; L)$ by Theorem 5.10. The boundary statement is the same argument on ∂Y. This is Remark 7.6 with $(K, \mathcal{R})$ replaced by $(K_Y, \mathcal{R}_Y)$; no step of it used the 7-vertex complex.

(2) The relative version of (1) gives $H_\Gamma^q(Y, \partial Y; V_R) \cong H^q(\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(Y, \partial Y), V_R))$, and Lemma 15.2(3) identifies the right-hand side with H_c^q; the compatibility with (39) is the naturality of both identifications in the short exact sequence of Lemma 15.2(2), whose three terms are the images of the three complexes of (1) and (2).

(3) Both operators are $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ on the same group cohomology: for the free model by Remark 17.2, for the 7-vertex model by Theorem 15.11(c), whose hypothesis $6 \in R^\times$ is in force. A map determined by a formula on both sides agrees on both sides.

(4) Proposition 7.4 gives (H4)(i)–(iv), hence (G1) and (G2), and (G3) because K is finite; (G4) is (B1)–(B2). □

Remark 17.4 (Moerdijk–Pronk as a computational model, and the effectivity caveat). Moerdijk and Pronk [MP99] give a simplicial set $S(\mathcal{T})$ attached to a triangulated orbifold and prove $H^\bullet(\mathcal{M}, \mathcal{A}) \cong H^\bullet(S, A)$. Line numbers below are of the LaTeX source `simplicial.tex` in `docs/literature/orbifolds/`, and every quotation in this remark **was checked against that source on 2026-09-16**. *Inside the quotation marks the symbols are theirs, not this document's* — in particular their G is a chart's isotropy group, their X is the space acted on and their S_k is the degree-k level of a simplicial set, where §3 retired G after Part I, X is the tessellation and $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

1. **What they are cited for.** (i) The identification of the orbifold cohomology of a global quotient with the cohomology of the Borel construction, in their own words: "If the orbifold is given as the orbit space X/G of a group action as above, this cohomology is the equivariant sheaf cohomology of the group action. It agrees with the (ordinary) cohomology of the Borel construction $EG \times_G X$" (:72–75). That sentence is the warrant for Definition 17.1 being *the* definition and not one option among several. (ii) $S(\mathcal{T})$ as *one computational model* of it, for an *adapted* triangulation — one in which every top simplex lies in a chart and every simplex has a vertex of maximal isotropy (:520–528) — with S_0 the top simplices and, for $k \geq 1$, $S_k = \sum_{\sigma_0 \cap \cdots \cap \sigma_k \neq \emptyset} G^k_{v(\sigma_0 \cap \cdots \cap \sigma_k)}$ (:589–592), and the comparison theorem at :678–684 (Theorem 2.1 of the published version, a *number* this document has not checked against the journal text).

2. **What they are *not* cited for: the normative definition.** Their charts are effective by construction — a chart is "a finite group G of C^∞-automorphisms of $\tilde{U}$" (:125), i.e. a subgroup of $\mathrm{Aut}(\tilde{U})$ — and their construction of the face maps μ rests on "there exists a unique $h \in H$ such that $\mu = h \circ \lambda$" (:148), which is exactly what fails when a subgroup acts trivially. $[\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}]$ **is not effective**: $-I$ acts trivially on $\mathcal{H}$ ((H4)(iv), Remark 7.7), and the effective quotient $[\mathrm{PSL}_2(\mathbb{Z})\backslash\mathcal{H}]$ has isotropy $\mathbb{Z}/2, \mathbb{Z}/3$ where this document needs $\mathbb{Z}/4, \mathbb{Z}/6$ — the distinction that Remark 7.7 shows is not cosmetic. So *as written* their construction computes the $\mathrm{PSL}_2(\mathbb{Z})$-orbifold. That it nevertheless "does the job in general" is plausible — their proof (:1063–1440) runs through the translation groupoid and never uses the uniqueness of h — but it is [conjectural] *as a citation*, and nothing in this document rests on it.
3. **A second reason not to make it normative.** They state the independence of μ from its choices twice without proof: "It can be shown that this definition is independent of the choices made above" (:772) and "It can be shown that this definition of μ is again independent of the various choices. ... We omit the details" (:811–815). A normative definition of a *contract* may not rest on an omitted proof.
4. **The verdict.** Definition 17.1 is normative; $S(\mathcal{T})$, and equally the isotropy-resolution model (one periodic resolution per elliptic orbit, assembled along faces), are computational models of it, to be compared with it when either is built. Both are infinite-dimensional in the degree direction, which is 14.24's subject. (Since v1.8.0 the isotropy-resolution model is Definition 23.4, compared with Definition 17.1 by Proposition 23.5(5); it is specified, not built. Since v1.9.0 it is built, on the unrefined tessellation, by hecke-extensions slot-9 WP-10, §22.1; since v1.10.0 with its lifted Hecke datum for T_2, T_3, T_5, Remark 23.14.)

Addendum (v1.22.0-pre.5, §32). Items (2) and (3) are *resolved*, not withdrawn. The normative object stays Definition 17.1. The Moerdijk–Pronk-type set is now defined choice-free (Definition 32.6: no charts, no μ-maps, no effectivity, no adaptedness), and its comparison theorem (Theorem 32.12) is proved in this document, independently of their argument, for the non-effective $[\mathrm{SL}_2(\mathbb{Z})\backslash X]$, in every degree and over every ring. It is their own $S(\mathcal{T})$ exactly when the action is effective, the triangulation adapted and the quotient simplicial (Remark 32.7(3)); for $\mathrm{SL}_2(\mathbb{Z})$ it is a non-effective generalisation of it. The [conjectural] of item (2) concerns their own construction and stays as stated; nothing rests on it.

Remark 17.5 (What a refinement of the 7-vertex complex would and would not change). Owner direction OD-T1 (Remark 7.6) expects the correct treatment at $\ell \in \{2, 3\}$ and over $\mathbb{Z}$ to need a refinement — a subdivision — of Definition 7.1's complex. This remark records the consequence for what is already written, so that the deferred construction is not mistaken for a revision.

1. **Survives verbatim in content.** A Γ-equivariant subdivision of a contractible 2-dimensional Γ-complex is again contractible and 2-dimensional, and it preserves the stabiliser of every point, hence the stabiliser inventory. So Proposition 7.4(i)–(iv) holds of the refinement, as do (G1)–(G4), and Theorem 17.3 applies to it unchanged. Definition 10.2 is untouched: it is a statement about $\mathbb{P}^1(\mathbb{Z}/N)$ with no reference to any triangulation.
2. **Must be recomputed, per refinement.** Everything indexed by the simplices: the counts N_0, N_1, N_2 and the Euler characteristic of Definition 7.1; the relation table of Definition 7.2 and its size; the stored simplex order, in which every matrix of this document is written; Algorithm 18.3's orbit representatives and the bounds of Lemma 18.5; and hence the golden references of the implementation.
3. **Is not a revision of anything.** OD-T1 is the owner's stated expectation, a design direction and not a theorem (Remark 7.6, last paragraph); no statement of this document depends on it, and none is weakened by it. Its whole force is that (G1)–(G4) and Theorem 17.3 are phrased over an arbitrary input, which is the instruction discharged above.

4. **What a refinement does *not* by itself buy.** It does not make the complexes of Definition 17.1 finite: those are unbounded above in degree at $\ell \in \{2,3\}$ for *every* triangulation, the elliptic and central stabilisers contributing in every degree (Remark 14.21(2),(3), Lemma 14.20, Lemma 16.4(2)). A truncation policy is therefore needed whatever the triangulation, which is why 14.24 is stated over "the boundary triangulation in force" and not over $v_0v_1v_2v_3$.

(v1.8.0: on these grounds OD-T1 is answered "not needed" and the free model is built on the unrefined complex; Remark 23.6.) (v1.22.0-pre.5, owner ruling Q2 of 2026-10-09: for Moerdijk–Pronk's rule (ii) on the actual isotropy the 7-vertex complex is *not* adapted, at every ℓ and over $\mathbb{Z}$, and the refinement $\mathsf{R}_{\mathbb{Z}}$ — one subdivision point on v_4v_6 and its S-image — is the smallest listed adapted one, the same at $\ell = 2$, $\ell = 3$ and over $\mathbb{Z}$ (Definition 32.4, Table 9, Remark 32.5); so OD-T1's expectation is borne out *for their construction*, while every model proved in §32 already holds on the unrefined complex, which stays the economical model. What distinguishes $\ell = 2$, $\ell = 3$ and $\mathbb{Z}$ is the isotropy datum and the coefficient system (Definition 32.3), not the triangulation.)

Code map. Nothing in this chapter is implemented, and that is the point of OP4: the definitions exist so that what is built later cannot silently be a different object. Two shape requirements reach the code and are recorded at §14.5's code map as well: the Hecke datum must be typed as tables of twisted chains on an *abstract* Γ-pair — orbit representatives, stabilisers, transporters and the (T1) bookkeeping — rather than on the seven vertices, and it must be **degree-generic**, not the $q \in \{0,1,2\}$ that `hecke::HeckeTauTable` hard-codes today, because 14.24 indexes it by $q_{\max}$. Both are properties of a data model, checkable by inspection, and neither requires the deferred construction.

Formalization note. Definition 17.1(1) is the one statement here a proof assistant would find easy and useful: $F \otimes_{\mathbb{Z}} M$ with the diagonal action is free over $\mathbb{Z}[\Gamma]$ when F is free and M is $\mathbb{Z}$-free, and `mathlib4` has the ingredients. It is worth doing early precisely because Remark 17.2 — the removal of every invertibility hypothesis — is a one-line consequence of it plus the fundamental lemma, which `mathlib4` also has. Theorem 17.3(1) needs the isotropy spectral sequence, which it does not, and should be deferred; its statement is however the right interface, since (2) and (3) are formal consequences of (1) together with Lemma 15.2(3) and Theorem 15.11(c). Remark 17.4 is bibliography, not mathematics, and carries no obligation.

Part V — Realisation, bounds, and certification

18 The combinatorial reduction primitive and integer bounds

What this chapter fixes. Everything in §§11–13 consumes exactly one combinatorial primitive: *exact bookkeeping of* Γ- *and* Γ''*-translates of simplices and cosets.* This chapter specifies that primitive as exact arithmetic on bounded integers (Definition 18.1), gives the terminating construction of the τ-table whose output Definition 11.1 consumes and Obligation 20.11 certifies (Algorithm 18.3), proves the magnitude bounds the construction obeys (Lemma 18.5), and fixes the arithmetic backend that those bounds select (Theorem 18.7). Nothing here is a statement about H^q_{eq}: the chapter's entire output is a certified finite table of integer chains and a certified integer bound B.

Standing hypotheses. (H4) throughout; p a prime, $\nu \geq 1$, $N = p^\nu$, $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, $\Gamma'' = \Gamma_0(N)$, $m = N + p^{\nu-1}$, and the transporters s_j, coset elements $\delta_j = s_j^{-1} g$, class map cls, index action $j \cdot \gamma$ and cocycle $h_j(\gamma)$ of Definition 10.2 and Lemma 10.3. Per rule (A1) each statement repeats what it uses. **No statement of this chapter carries a hypothesis on** ℓ: the chapter is integral, and the only place ℓ enters is the recorded mod-ℓ fallback of Algorithm 18.3(3) and the mod-ℓ backend box of Theorem 18.7.

Notation local to this chapter. Four symbols are used here and nowhere else in this document, and are declared once rather than added to chapter 2 (rule (A5) governs *document-wide* symbols; a chapter-local declaration is the device already used inside the proof of Proposition 10.6): $\|A\| := \max_{j,t} |A_{jt}|$, the entrywise sup-norm on $\mathrm{M}_2(\mathbb{Z})$; $\mathcal{F} := \{z \in \mathcal{H} : |\mathrm{Re}\, z| \leq \frac{1}{2},\ |z| \geq 1\}$, the standard closed fundamental domain of Γ; $\mathrm{pos}(v)$, the exact position in $\mathcal{H}$ of a vertex v of K (Definition 7.1); and $\mathcal{N}$, a finite subcomplex of K^u. The generator multiset of the model is $\mathcal{G} = \{T, S, ST, TS\}$ of §7.1; its elements and $-I$ all have entries in $\{0, \pm 1\}$, so $\|\gamma\| \leq 1$ for $\gamma \in \mathcal{G} \cup \{-I\}$.

18.1 The sub-primitives

Definition 18.1 (Reduction primitives)**.** The four primitives below are the *only* places where exact unbounded-in-principle integer arithmetic occurs. Each is a terminating procedure on explicit data (rule (A3)); each is stated with its output specification, and the properties it guarantees are Lemma 18.5 and the certification of Obligation 20.11, not assumptions.

(P1) **Coset class map, index action and cocycle.** Given $\gamma \in \Gamma$ and a coset index j: compute $\mathrm{cls}(s_j\gamma) = (c, d)\gamma \bmod N$ as a *row*-vector product of the bottom row (c, d) of s_j with γ (never transposed), look the class up in the table of Definition 10.2 to get $j \cdot \gamma$, and form the cocycle $h_j(\gamma) = s_j \gamma s_{j\cdot\gamma}^{-1} \in \Gamma''$ by two exact 2×2 products and one inversion of a unimodular matrix. The class lookup is arithmetic modulo N; only the cocycle needs exact integers.

(P2) **Word reduction in** Γ**.** Given $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, produce a word

$$\gamma \;=\; \pm T^{a_1} S\, T^{a_2} S \cdots T^{a_k}$$

by the *nearest-integer* Euclidean algorithm on the first column: while $c \neq 0$, choose the integer q with $|a - qc| \leq |c|/2$ (ties resolved by rounding the residue up, so that q is a function of (a, c) alone and the word is deterministic), replace M by $S^{-1}T^{-q}M$, and emit the letters T^q (omitted when $q = 0$) and S. The loop terminates because $|c|$ strictly decreases; on exit $c = 0$ and $\det M = 1$ force $M = \pm T^{\pm b}$, which supplies the optional leading $-I$ and the trailing T-power. Intermediate magnitudes obey Lemma 18.5(4), and $k = O(\log\|\gamma\|)$. Used to express the relative position of two translates in step 2 of Algorithm 18.3; *not* used in the hot path, where $\rho_n(\gamma)$ is evaluated directly from $\gamma \bmod \ell$.

(P3) **Hermite reduction at determinant** N**.** Given $\delta \in \mathrm{M}_2(\mathbb{Z})$ with $\det \delta = N$, produce $\gamma \in \Gamma$ and the normal form $\gamma\delta = \begin{pmatrix} N/d & b \\ 0 & d \end{pmatrix}$ of Proposition 10.1(1): Euclid on the first column, then column normalisation. Intermediate entries are bounded by $2\max(\|\delta\|, N)$ by the same argument as (P2). **This primitive is specified and has no live call site** — the closed forms of Definition 10.2 produce s_j, and hence $\delta_j = s_j^{-1} g$, directly, so the pipeline never needs to reduce an arbitrary determinant-N matrix. It is retained because the *proof* of Proposition 10.1(1) is this procedure, and because any future entry point taking an arbitrary δ needs it.

(P4) **Translate bookkeeping, and the canonical key.** A q-simplex of K^u is represented as a pair $(\gamma, \sigma) \in \Gamma \times o_q(K)$ standing for $\gamma \cdot \widetilde{\sigma}$. Two pairs represent the same oriented

simplex iff they differ by a stored transporter, so a *canonical key* is computed by ranging over the finite transporter set $\mathrm{Tr}(\sigma, \widehat{\sigma})$ of the model, taking the lexicographically least (a, b, c, d) among the candidates $\gamma \cdot$ (transporter), and recording the parity of the induced permutation of the vertex slots as a sign ± 1. The key quotients by $-I$ (the transporter set is closed under negation and $-I$ acts trivially on K^u by (H4)(iv)); it is idempotent, and it is the *only* equality test on simplices of K^u used anywhere in this document. Every group element arising is a product of at most two stored elements.

Remark 18.2 (Why (P1)–(P4) and nothing else). The Hecke pipeline of §§11–12 touches $\mathbb{Z}$ in exactly four places: building the coset tables (P1), building the τ-table (Algorithm 18.3, which uses (P2), (P4) and the predicates of §19), comparing simplices of K^u (P4), and reducing a group element modulo ℓ or modulo N for a consumer (Theorem 18.7, early reduction). Everything downstream — ρ_n, Θ_g, $\Theta_{\mathrm{eq}}^{(q)}$, $A^{(q)}$, the eliminations — is linear algebra over R. That is the content of the contract box at the end of this chapter, and it is what makes Theorem 18.7 a statement about a *finite* arithmetic workload rather than about the whole computation.

18.2 Construction of the τ-table

Algorithm 18.3 (TAU-TABLE). Assume (H4). **Input:** p, ν, $N = p^\nu$; the model $(K, \mathcal{R})$ with its relation and transporter tables and the exact vertex positions of Definition 7.1; the representatives $\{s_j\}_{j=1}^m$ of Definition 10.2. **Output:** for every $j \in \{1, \dots, m\}$ and every $\sigma \in o_q(K)$, $q = 0, 1, 2$, an integer chain

$$\tau\big(s_j \cdot \widetilde{\sigma}\big) \;=\; \sum_t n_t^{(j,\sigma)}\, \gamma_t \cdot \widetilde{\sigma}_t, \qquad n_t^{(j,\sigma)} \in \mathbb{Z},\ \gamma_t \in \Gamma,\ \sigma_t \in o_q(K), \tag{56}$$

stored as triples $(n_t^{(j,\sigma)}, \gamma_t, \sigma_t)$ in the canonical form of (P4), satisfying (T1)–(T3) of Definition 11.1 by construction and by step 4. By (T1) it suffices to store τ on the finitely many slots (j, σ); that is one slot per simplex of $\Gamma''\backslash K^u$ *up to the derivation of step 0.* At $p = 2$, $\nu = 1$ $(m = 3)$ the table is $3 \cdot (7 + 11 + 5) = 69$ chains.

0. Slots, and the single (T1) helper. The pairs (j, σ) are *not* pairwise Γ''-inequivalent: $o_q(K)$ is not a section of the Γ-groupoid orbits (already $\widetilde{v_3} = T \cdot \widetilde{v_0}$, so $s_j \cdot \widetilde{v_3} = h_j(T)\, s_{j \cdot T} \cdot \widetilde{v_0}$ with $h_j(T) \in \Gamma''$). Exactly one slot per Γ''-orbit is therefore *constructed* by steps 1–3, and every other slot is *derived* from it through one shared helper implementing (T1), $\tau(h \cdot x) = (g^{-1} h g) \cdot \tau(x)$, with the sign of (P4) tracked. The *same* helper is used by step 4, so construction and certification cannot disagree about how a derived slot is obtained.

1. Vertices. For each constructed slot (j, v): form the integral matrix $\mathrm{adj}(g)\, s_j$ with $\mathrm{adj}(g) = \mathrm{diag}(N, 1) = N\, g^{-1}$ and let $z := \big(\mathrm{adj}(g) s_j\big) \cdot \mathrm{pos}(v) \in \mathcal{H}$, where $\mathrm{pos}(v)$ is the exact quadratic point of Definition 7.1. (The Möbius action is projective and $\det(\mathrm{adj}(g) s_j) = N > 0$, so $\mathrm{adj}(g) s_j$ acts exactly as $g^{-1} s_j$; **no rational number is ever formed**.) Reduce z to $\mathcal{F}$ by the standard alternation — shift by T^{-q} with q the nearest-integer quotient of the cleared real part, then apply S while $|z| < 1$, repeat — obtaining $\gamma_{\mathrm{red}} \in \Gamma$ with $\gamma_{\mathrm{red}} z \in \mathcal{F}$. Choose a vertex w of K:

(1a) if $\gamma_{\mathrm{red}} z$ is *exactly* the position of a model vertex, take that vertex;

(1b) otherwise let S be the set of vertices at minimal euclidean distance from $\gamma_{\mathrm{red}} z$, decided exactly. If $|S| = 1$ take its element. Otherwise the tie is broken ι_K*-equivariantly* (the vertex involution of Lemma 9.2, read off the exact coordinates): if S contains a full ι_K-orbit — a swapped pair $\{v, \iota_K v\}$, or v_6 — take v_6, the unique ι_K-fixed vertex of K; otherwise S meets every orbit in at most one vertex, and take the one in the orbit of smallest index ($\{v_0, v_3\} < \{v_1, v_2\} < \{v_4, v_5\} < \{v_6\}$). The first branch is forced: a reduced point on the mirror $\mathrm{Re}\, z = 0$ has an ι_K-stable tied set, and no choice *within* a swapped pair commutes with ι_K (Remark 9.5). Before v1.1.0 ties were broken by ascending vertex index, which is not ι_K-equivariant.

Set $\tau(s_j \cdot \widetilde{v}) := \gamma_{\mathrm{red}}^{-1} \cdot \widetilde{w}$, a single term with coefficient 1, so (T3) holds by inspection. All branch decisions of the reduction and of (1b) are the certified predicates of §19.

2. **Edges.** For a constructed slot (j, e) with $e = (v_0, v_1)$: the two faces of $s_j \cdot \widetilde{e}$ are themselves slots, so their τ-values $\gamma_a \cdot \widetilde{w_a}$ and $\gamma_b \cdot \widetilde{w_b}$ are already stored (fetched through the step-0 helper where the face is derived). If the two canonical keys coincide, set $\tau(s_j \cdot \widetilde{e}) := 0$: the empty 1-chain has exactly the required boundary. Otherwise reduce $\gamma_a^{-1}\gamma_b$ to a word in $T^{\pm 1}, S$ by (P2), drop the $-I$ letter (it acts trivially on K^u), and concatenate, for each remaining letter L, the stored *path template* from $\widetilde{w}$ to $L \cdot \widetilde{w'}$ translated by the running prefix. A path template is a 1-chain of K^u supported in $\widetilde{K} \cup L \cdot \widetilde{K}$, computed once from the model and *certified at construction*: its boundary equals $L \cdot \widetilde{w'} - \widetilde{w}$ exactly over $\mathbb{Z}$. **Since v1.1.0 the template is tree-routed.** Let $Y \subset K^u$ be the Γ-orbit of the two edges $\{v_4v_6\}$, $\{v_5v_6\}$ — a tree, the Bass–Serre tree of $\mathrm{PSL}_2(\mathbb{Z})$ with the elliptic points as its vertices — and let t be the spoke attachment $t(v_0) = v_4$, $t(v_1) = t(v_2) = v_6$, $t(v_3) = v_5$, $t(v) = v$ on $\{v_4, v_5, v_6\}$, a table that commutes with ι_K and with $\widetilde{v_3} = T\widetilde{v_0}$ ($t(v_3) = T\,t(v_0)$). The template is $[\widetilde{w}, \widetilde{t(w)}] + Y\text{-path}\big(\widetilde{t(w)}, L\widetilde{t(w')}\big) - L[\widetilde{w'}, \widetilde{t(w')}]$, the Y-path being the unique 1-chain of Y with that boundary. Concatenated along *any* word the templates telescope, because Y-paths compose to the unique Y-path, to $[A, t(A)] + Y\text{-path}(t(A), t(B)) - [B, t(B)]$ for the two endpoints $A = \gamma_a\widetilde{w_a}$, $B = \gamma_b\widetilde{w_b}$: a chain of the endpoints alone, Γ-equivariant, ι_X-equivariant and antisymmetric in $A \leftrightarrow B$, so neither the word's tie rule nor any traversal order reaches the output. (Before v1.1.0 the templates were breadth-first vertex paths in K in ascending neighbour order, which are neither ι_K-equivariant nor reversal-symmetric; Remark 9.5.) If the word is empty while the two keys differ, use the net-identity detour T, T^{-1}, whose boundary still telescopes. Set $\tau(s_j \cdot \widetilde{e}) :=$ the accumulated 1-chain. (This is the Manin/continued-fraction device [Man72, Cre97] in general position.)
3. **Triangles.** For a constructed slot (j, t): $c := \tau(\partial(s_j \cdot \widetilde{t}))$ is a 1-cycle of K^u, by step 2 and (T2) in lower degree. Solve $\partial x = c$ for a 2-chain x supported on a finite subcomplex $\mathcal{N} \subseteq K^u$: initialise $\mathcal{N}$ as the union of the translates occurring in $\operatorname{supp} c$, take as columns the canonical keys of the 2-simplices of $\mathcal{N}$ and as rows the canonical keys of $\operatorname{supp} c$ together with all column faces, and solve the resulting sparse integer system by column-echelon elimination with exact divisibility checks. On failure, grow $\mathcal{N}$ by one ring of adjacent translates (T, T^{-1}, S) and retry. **Termination:** $\varinjlim \mathcal{N} = K^u$ is acyclic ((H4)(ii)), so every 1-cycle bounds at a finite stage. Set $\tau(s_j \cdot \widetilde{t}) := x$. *Solve regime (recorded).* The integer solve is attempted first and is the default; solving the same system modulo ℓ instead is permitted only behind the explicit setting of owner decision OP2, marks the affected chains and the table `integral = false`, and is recorded — see Remark 11.5, which also records that 0 of 66 shipped points used it. Both regimes produce a valid τ over their ring, and they agree on cohomology by Theorem 11.7(b) where that theorem's hypothesis holds. No Smith/Hermite normal-form library and no arbitrary-precision arithmetic is used; the entry growth of the integer solve is bounded by the fixed-width tier of Theorem 18.7 and overflow-checked, and an overflow is a tier failure that aborts ([HM97] is the classical reference for controlling that growth in general; the systems here are small enough that the checked tier suffices).
4. **Certification (mandatory).** Verify *exactly* — over $\mathbb{Z}$, or over $\mathbb{F}_\ell$ for identities involving a chain that step 3 solved modulo ℓ — the full predicate list of Obligation 20.11: structure and canonicity, (T3) in degree 0, (T2) in degrees 1 and 2 with the right-hand side evaluated *only* through the step-0 helper, the (T1) bookkeeping over every stored transporter, the magnitude bound of Lemma 18.5(5), and the consumer table of (25). **A table failing any clause is rejected**: the datum never exists in an uncertified state.

Remark 18.4 (Certification over geometry). Step 4 converts every heuristic and geometric choice of steps 1–3 — the nearest vertex, the tie-break, the path templates, the neighbourhood growth,

the FP-filtered reduction — into a *proof* that the emitted table is a g-twisted chain datum in the sense of Definition 11.1. Correctness of the resulting Hecke operator then depends only on Theorems 11.7 and 12.11, and on no geometric assertion whatever. Two consequences, both used elsewhere in this document and neither optional:

1. **The choices in steps 1–3 are free, and changing one changes the table.** The tie-break of (1b) and the templates of step 2 are, since v1.1.0, ι_K-equivariant, and Remark 9.5 records the computation showing that the canonical τ is consequently ι-symmetric — and that the pre-v1.1.0 choices (ascending-index tie-break, breadth-first templates) were not, each on its own. A different admissible choice is a different admissible τ; by Theorem 11.7(b) it moves nothing on H^q_{eq} where $6 \in R^\times$, and by Remark 11.9 it moves $\Theta^{(q)}_{\mathrm{flat}}$ — which is why the v1.1.0 change re-baselined every frozen cochain-level reference value and no cohomology-level one.
2. **Elliptic slots are forced, not chosen.** If the slot $s_j \cdot \widetilde{v}$ has a non-trivial stabiliser in Γ'' modulo $\pm I$, then (T1) forces the assigned vertex, the reduced point $\gamma_{\mathrm{red}} z$ is exactly an elliptic fixed point, and branch (1a) returns it. This is an explanation of why the construction succeeds, not a step of any proof: step 4 is what accepts the table.

18.3 Integer bounds

Lemma 18.5 (Entry bounds)**.** *[proved here] Assume (H4), and let* $\|A\| = \max_{j,t} |A_{jt}|$ *for* $A \in \mathrm{M}_2(\mathbb{Z})$*. Then, with* $N = p^\nu$ *and no hypothesis on* ℓ*:*

1. $\|AB\| \le 2\|A\|\|B\|$ *for* $A, B \in \mathrm{M}_2(\mathbb{Z})$*; hence a product of* k *matrices is bounded by* $2^{k-1}\prod_t \|A_t\|$*. [formally verified: [GT, `lem:bounds`]]*
2. ***Fixed input data.*** $\|s_j\|$, $\|s_j^{-1}\|$, $\|\delta_j\|$, $\|g\| \le \max(N, 1)$ *for every* j *(Lemma 10.3) [formally verified: [GT, `lem:bounds`]] (this clause), and* $\|\gamma\| \le 1$ *for* $\gamma \in \mathcal{G} \cup \{-I\}$*.*
3. ***Cocycles.*** $\|h_j(\gamma)\| = \|s_j \gamma s_{j\cdot\gamma}^{-1}\| \le 4N^2\|\gamma\| \le 4N^2$ *for* $\gamma \in \mathcal{G} \cup \{-I\}$*. [formally verified: [GT, `lem:bounds`]] (the first inequality)*
4. ***Euclidean reduction.*** *Every intermediate matrix of (P2) on input* γ *satisfies* $\|M\| \le \max(\|\gamma\|, 2)$*, and in particular* $\|M\| \le 2\max(\|\gamma\|, 1)$ *[formally verified: [GT, `lem:bounds`]] (for (P2), on an abstract step relation); the same bound holds for (P3) with* $\max(\|\delta\|, N)$ *in place of* $\|\gamma\|$*.*
5. ***τ-table elements.*** *Every* γ_t *stored by Algorithm 18.3 satisfies* $\|\gamma_t\| \le C_K N^2$ *for a constant* C_K *of the fixed model, independent of* p *and* ν*; and the implementation must* certify *the actual attained bound* $B_{\mathrm{tab}} := \max_{j,\sigma,t}\|\gamma_t\|$ *at build time, by a full recount over the stored table in overflow-checked arithmetic (Obligation 20.11, Obligation 20.10).*
6. ***Reduction intermediates — which (1)–(5) do* not *bound*** *(v1.5.0; `docs/implementation/slot3/W` finding 1). Clauses (1)–(5) bound the matrices the pipeline* stores*. The fundamental-domain reduction of Algorithm 18.3 step 1 also carries a quantity that is not one of them: the quadratic point* $z = (u + v\sqrt{D}\,\mathrm{i})/w$ *on cleared denominators* $(u, v, w \in \mathbb{Z}$*,* $w > 0$*,* $v > 0$*,* $D \in \{1, 3\}$*,* $\gcd(u, v, w) = 1$ *— the exact representation on which (P2) and the predicates of §19 act), with* $\|z\| := \max(|u|, |v|, w)$*. For it:*
   (a) One Möbius step. *For* $\gamma \in \mathrm{M}_2(\mathbb{Z})$ *with* $\det\gamma > 0$*, every integer formed in evaluating* $\gamma \cdot z$ *— the three coordinates and the products that build them — has magnitude at most* $7\|\gamma\|^2\|z\|^2$*, and hence* $\|\gamma \cdot z\| \le 7\|\gamma\|^2\|z\|^2$ *before the* gcd *normalisation.*
   (b) The point step 1 starts from. *With* $C := \max_v \|\mathrm{pos}(v)\|$ *the largest cleared coordinate of a vertex position of the fixed model (*$C = 12$ *on the 7-vertex model of §7.1), the initial point* $z_0 = \big(\mathrm{adj}(g)s_j\big) \cdot \mathrm{pos}(v)$ *of step 1 satisfies* $\|z_0\| \le 7C^2N^2$*, and the order* C^2N^2 *is attained.*
   (c) Consequence for the backend. $\|z_0\|$ *is of order* N^2 *with a* model *constant in front of it, not of order* B*; so the arithmetic of step 1 is bounded by* $7\|z\|^2$ *and* not *by the* $2B^2$ *of Theorem 18.7(i), and a tier chosen against* B *alone does not cover it. The tier*

criterion for step 1 is therefore $7\max_z\|z\|^2$*, the maximum taken over the points the reduction actually passes through; (6b) bounds the first of them, and that the others stay at that order is* measured *and not asserted here (Remark 18.6; v1.5.1, which struck v1.5.0's unproved "with* $\max_z\|z\|$ *of order* C^2N^2*" from the clause and left it where the measurement is recorded). In particular the width must be chosen against at least* $\max\bigl(B,\,7C^2N^2\bigr)$ *and its square, and an implementation that cannot bound the reduction a priori must be overflow-checked per operation, which Theorem 18.7(3) requires of it anyway.*

The gap this clause closes was a real one and not a hypothetical: the measurement that found it, and the reach it fixes, are recorded at Remark 18.6 and in §10's code map.

7. ***The wide tier — two stored-class operands in*** 128 ***bits*** *(v1.23.0-pre.9; slot-24 package ELL-W,* `docs/implementation/slot24/ELLW_report.md`*). Let* $N \geq 2$*, let* $B = \max(B_{\mathrm{tab}}, 4N^2)$ *be the certified bound of Theorem 18.7, and assume*

$$2NB \;\leq\; 2^{63} - 1.$$

Then $B < 2^{61}$*, and:*

(a) Storage. *Every stored* γ_t*, the inverse of every stored* γ_t*, and every matrix of norm* $\leq 2B$ *— in particular every intermediate of (P2) on an input of norm* $\leq B$*, by (4) — has entries of magnitude* $< 2^{62}$*, so signed* 64*-bit integers hold them.*

(b) Two stored-class operands. *For* $A, A' \in \mathrm{M}_2(\mathbb{Z})$ *with* $\|A\|, \|A'\| \leq B$*, each of the two products in an entry of* AA'*, and each of the products* ad*,* bc *of* $\det A$*, has magnitude* $\leq B^2 < 2^{122}$*, and each sum of two of them has magnitude* $< 2^{123}$*; so* AA' *and* $\det A$ *are evaluated exactly in signed* 128*-bit integers. Clause (b) bounds the* intermediates*; the* result *is bounded only by* $2B^2$ *(by (1)), which need not be below* 2^{63}*, and is returned to the* 64*-bit tier by a checked narrowing, which aborts rather than wraps (Obligation 20.10(iii)).*

(c) A fixed datum times a stored matrix. *For* $F \in \{g, \mathrm{adj}(g), s_j, s_j^{-1}, \delta_j\}$ *(*$\|F\| \leq \max(N,1) = N$ *by (2)) and* $\|\gamma\| \leq B$*:* $\|F\gamma\|, \|\gamma F\| \leq 2NB \leq 2^{63} - 1$*, and every intermediate is one of the two products (*$\leq NB$*) or their sum; exact in signed* 64*-bit integers. These are the products the level-one consumers of* τ *form before the early reduction (Theorem 18.7,* Early-reduction scheme*); other consumers (the level structures, e.g. the three-fold product* $v\,\gamma\,w^{-1}$ *of the level-*ℓ *route) form checked products not covered by (c), and abort beyond* 64 *bits.*

(d) Reach. *When* $B = 4N^2$ *(i.e.* $B_{\mathrm{tab}} \leq 4N^2$*) the hypothesis reads* $8N^3 \leq 2^{63} - 1$*, i.e.* $N \leq 2^{20} - 1 = 1048575$ *(largest prime* 1048573*).*

That the result of each product of (b) the pipeline actually forms is below 2^{63} *is [verified computationally]* measured *and not asserted here (Remark 18.6): the clause makes no value depend on it, because an exceeding result aborts.* ***Not bounded a priori by (a)–(c)****, and formed in overflow-checked arithmetic whose completion is likewise [verified computationally] measured (v1.23.0-pre.9, audit F1): the input* $\mathrm{rel} = \gamma_a^{-1}\gamma_b$ *of (P2) in Algorithm 18.3 step 2, a product of two stored matrices (*$\|\mathrm{rel}\| \leq 2B^2$*, so its (P2) intermediates are* $\leq 4B^2$ *by (4)), and the running edge prefix* $P = \gamma_a L_1 \cdots L_j$ *along its word (a priori of order* B^3*; the word is expanded letter by letter); the (T1) conjugate* $g^{-1}hg$ *of step 0 (*$\|\cdot\|$ *up to* $2N^2B$*, by (1), (2) — the gap Remark 18.2 leaves, and the same in the narrow tier), whose product with a stored* γ_t *goes through the* 128*-bit form, itself checked, so it can only abort; and the consumers named at the end of (c).*

Proof. (1) Each entry of AB is a sum of two products of entries.

(2) is Lemma 10.3: the entries of s_j lie in $\{0, \pm 1, k, pc\}$ with $0 \leq k < N$ and $0 \leq pc < N$; s_j^{-1} swaps the diagonal and negates the off-diagonal; $\delta_j = s_j^{-1}g$ scales the second column by N, which does not increase the bound $\max(N,1)$ because the second column of s_j^{-1} has entries in $\{0, \pm 1\}$ for both branches of Definition 10.2. For $\mathcal{G} \cup \{-I\}$, read off §7.1.

(3) Apply (1) twice to (2): $\|s_j\gamma s_{j\cdot\gamma}^{-1}\| \le 2\|s_j\| \cdot 2\|\gamma\|\|s_{j\cdot\gamma}^{-1}\| \le 4N^2\|\gamma\|$.

(4) Write $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad-bc = 1$ and $c \neq 0$, and let q be the nearest-integer quotient, so $a' := a - qc$ satisfies $|a'| \le |c|/2$. Put $b' := b - qd$. From $a'd - b'c = \det M = 1$ we get $b' = (a'd-1)/c$, whence

$$|b'| \;\le\; \frac{|a'|\,|d|+1}{|c|} \;\le\; \frac{|c|\,|d|/2+1}{|c|} \;=\; \frac{|d|}{2} + \frac{1}{|c|} \;\le\; \frac{|d|}{2} + 1.$$

The intermediate matrix $T^{-q}M = \begin{pmatrix} a' & b' \\ c & d \end{pmatrix}$ therefore has $\|T^{-q}M\| \le \max\big(|c|, |d|, |c|/2, |d|/2+1\big) \le \max(\|M\|, 2)$, and $S^{-1}T^{-q}M$ has the same entries up to sign and order. Induction on the loop from $\|M_0\| = \|\gamma\|$ gives $\|M_k\| \le \max(\|\gamma\|, 2)$ for every k. The exit matrix $\pm T^{\pm b}$ is one of the M_k. For (P3) the same computation applies to the first column of δ with $\det\delta = N$ in place of 1, giving the extra additive $N/|c| \le N$.

(5) The reduction word of step 1 for $\big(\mathrm{adj}(g)s_j\big)\cdot\mathrm{pos}(v)$ has convergents bounded by the entries of the integralised input $\mathrm{adj}(g)s_j$, whose norm is at most $2N \cdot N = 2N^2$ by (1) and (2); by (4) the intermediates never exceed twice that. The path templates of step 2 and the neighbourhood of step 3 contribute a factor depending on the fixed model only, and products of at most three such elements occur; both are absorbed into C_K. **The a-priori constant is not what makes the pipeline sound**: the build-time recount of B_{tab} is, and it is what Theorem 18.7 consumes.

(6a) On cleared denominators the Möbius action of $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\Delta := \det\gamma > 0$ on $z = (u + v\sqrt{D}\,\mathrm{i})/w$ is

$$u' = (au+bw)(cu+dw) + D\,ac\,v^2, \qquad v' = vw\Delta, \qquad w' = (cu+dw)^2 + D\,c^2v^2,$$

which is the only formula evaluated. Put $G := \|\gamma\|$, $Z := \|z\|$. Then $|au+bw| \le 2GZ$ and $|cu+dw| \le 2GZ$, so the first summand of u' and of w' is $\le 4G^2Z^2$ and the second is $\le DG^2Z^2 \le 3G^2Z^2$; and $|\Delta| \le 2G^2$ gives $|v'| \le 2G^2Z^2$. Every intermediate the formula forms is one of these products or their sum, hence $\le 7G^2Z^2$, and so is each of $|u'|, |v'|, w'$. The gcd normalisation only divides. *The two shapes step 1 applies* (v1.5.1; the bound Theorem 18.7(iv) quotes is $7\|z\|^2$, with no G, and G is not 1 for a translation by a large quotient, so the specialisation is part of the proof and not a corollary of the displayed inequality). The reduction applies only $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, where $G = 1$ and (6a) gives $7Z^2$ outright; and the shift $T^{-q} = \begin{pmatrix} 1 & -q \\ 0 & 1 \end{pmatrix}$ by the *nearest*-integer quotient q of u by w, where $c = 0$ and $d = \Delta = 1$ collapse the formulas to $u' = (u-qw)w$, $v' = vw$, $w' = w^2$ with $|u-qw| \le w/2$, so that qw, $u-qw$ and each of $|u'|, |v'|, w'$ is $\le 2Z^2$. Both are therefore $\le 7Z^2$ as (iv) states, although (6a) applied with $G = |q|$ would give only $7q^2Z^2$. (6b) Apply (6a) with $z = \mathrm{pos}(v)$, so $Z \le C$, and $\gamma = \mathrm{adj}(g)s_j$, whose norm is at most N: by Definition 10.2 and $\mathrm{adj}(g) = \mathrm{diag}(N, 1)$ the product is $\begin{pmatrix} N & 0 \\ k & 1 \end{pmatrix}$ or $\begin{pmatrix} 0 & -N \\ 1 & pc \end{pmatrix}$ with $0 \le k, pc < N$, so $G \le N$ and $\|z_0\| \le 7C^2N^2$. Attainment (v1.5.1, which names the coordinate that attains it; v1.5.0 cited $v' = vw\Delta$, which for these matrices is of order C^2N and not C^2N^2): it is the *third* coordinate, $w' = (cu+dw)^2 + D\,c^2v^2$, which for $\gamma = \begin{pmatrix} N & 0 \\ k & 1 \end{pmatrix}$ with k of order N is of order $k^2\|\mathrm{pos}(v)\|^2 \asymp C^2N^2$ — its second summand $D\,k^2v^2$ already is — so $\|z_0\| \asymp C^2N^2$, measured as $\approx 144.8\,N^2$ against $C^2 = 144$ with $C = 12$ on the shipped model (Remark 18.6). (6c) is (6a), (6b) and the definition of $B = \max(B_{\mathrm{tab}}, 4N^2)$: B is a bound on stored matrices, $\|z_0\|$ is not bounded by it, and $7\|z\|^2$ is what an overflow-checked step of (P2) must fit.

(7) From $N \geq 2$: $B \leq (2^{63}-1)/(2N) \leq (2^{63}-1)/4 < 2^{61}$. (7a) The stored γ_t have norm $\leq B_{\mathrm{tab}} \leq B$; the inverse of a unimodular matrix has the same entries up to sign and order; and (4) bounds the (P2) intermediates on an input of norm $\leq B$ by $\max(B, 2) \leq 2B < 2^{62}$. (7b) Each product is of two entries of magnitude $\leq B$, hence $\leq B^2 < 2^{122}$, and a sum of two such is $< 2^{123} < 2^{127}$; the result bound $2B^2$ is (1). (7c) Each entry of $F\gamma$ (or γF) is a sum of two products of an entry of F ($\leq N$) and one of γ ($\leq B$), so each product is $\leq NB$ and the entry $\leq 2NB$, which is $\leq 2^{63}-1$ by hypothesis. (7d) $8N^3 \leq 2^{63}-1$ iff $N^3 \leq (2^{63}-1)/8$, i.e. $N^3 < 2^{60}$, i.e. $N < 2^{20}$; and $N = 2^{20}-1$ gives $8N^3 < 2^{63}$. □

Remark 18.6 (C_K is never evaluated, and B_{tab} is small). [verified computationally] Lemma 18.5(5) is stated with an unevaluated model constant on purpose: no statement of this document depends on its value, because every consumer reads the certified B_{tab} instead. Over the shipped grid $(p, \nu) \in \{(2,1),(3,1),(5,1),(7,1),(2,2)\}$ at $\ell \in \{5, 7, 691, 134217757\}$ and $n \in \{0, 2, 10\}$, plus the bad-prime points $p = \ell \in \{2, 5\}$ — 66 points — the certified whole-pipeline bound is $B \leq 196$, far inside the first tier of Theorem 18.7(1) (`HeckeTauTableSelfCheckValidation`, run of 2026-09-15; the same figure is recorded at Remark 11.5).

And B is not the whole story (v1.5.0; slot-3 package WP-1, `docs/implementation/slot3/WP-1_report` `HeckeUEllReachValidation`). The one place where the certified B does *not* bound the arithmetic is Lemma 18.5(6)'s reduction point, and it was found the hard way. At $N = \ell = 5003$ — the bad prime, U_ℓ — $B = \max(B_{\mathrm{tab}}, 4N^2) = 4N^2 \approx 1.0 \cdot 10^8$, so $2B^2 \approx 2 \cdot 10^{16}$ and clause (e) of Obligation 20.11's certification passes with four orders of magnitude to spare, while Algorithm 18.3 step 1 overflows the 64-bit tier: the reduction point was measured at $\|z_0\| \approx 144.8\, N^2$, i.e. $\asymp C^2 N^2$ with $C^2 = 144$, $C = 12$ (v1.5.1: v1.5.0 wrote the two as an equality), attained at the *first* Möbius step, with $\max\|z\|$ constant along the reduction (the gcd normalisation returns each step to the entry size, so the growth is one squaring and not a cascade) — measured at $\ell \in \{11, 101, 691, 1009, 2003, 3617\}$, with $\|z\|/N^2$ rising to 144.8 and stable there, and independent of n, step 1 never seeing n. The int64 tier therefore dies at $(144.8\, N^2)^2 \geq 2^{63}$, i.e. $N \geq 4580$, which is exactly the bracket $3617 < N < 5003$ the slot-1 record had measured and attributed to the coset construction: the attribution was wrong, the site is step 1, and the quantity is the one clause (6) now bounds. The repair escalates that site to the `__int128` tier *per operation* (Theorem 18.7(2),(iv)), value-preservingly, and moves the binding constraint back onto clause (5) — see §10's code map for the resulting reach.

Past $N = 23170$ (v1.23.0-pre.9; slot-24 ELL-W, `docs/implementation/slot24/ELLW_report.md`). For U_ℓ ($N = \ell$, $\nu = 1$, $n = 0$) the certified B_{tab} is far below $4N^2$ — $B_{\mathrm{tab}} = 4415$ at $\ell = 5003$, 20442 at 23167, 20447 at 23173, 26480 at 30011, i.e. $B_{\mathrm{tab}} \approx 0.88\, N$ — so $B = 4N^2$, the cocycle bound of clause (3), and clause (7)'s hypothesis is its (7d). On every table built past 23170 (the primes 23173, 30011, 43867 at $n = 0$ and 30011 at $n = 10$, U_ℓ and T_{30011}) the 64-bit form of every product of clause (7)(b) the pipeline formed *completed*: the wide tier escalated 0 times, so every result was below 2^{63} with a margin of many orders of magnitude, and the tier is a certified guarantee there rather than a path taken. Measured, not asserted (clause (7), last sentence).

18.4 The arithmetic backend

Theorem 18.7 (Arithmetic backend requirements)**.** *[proved here] Assume (H4), and let*

$$B \;:=\; \max\bigl(B_{\mathrm{tab}},\, 4N^2\bigr)$$

be the certified bound of Lemma 18.5(5) and (3). All exact integer arithmetic required by the entire Hecke pipeline consists of

(i) 2×2 *matrix products of at most two stored matrices at a time, of magnitude* $\leq 2B^2$*, and unimodular inversions, of magnitude* $\leq B$*;*

(ii) *Euclidean steps (P2), (P3), of magnitude* $\leq 2B$ *by Lemma 18.5(4);*

(iii) *the sign predicates of §19, of magnitude* $\leq 2B^2$ *on their own inputs;*

(iv) ***(v1.5.0)*** *the Möbius steps of the fundamental-domain reduction of Algorithm 18.3 step 1, of magnitude* $\leq 7B_{\text{red}}^2$ *(Lemma 18.5(6a) and its proof, which bounds the two shapes step 1 applies), where* $B_{\text{red}} := \max_z \|z\|$ *over the quadratic points* z *that reduction passes through. Lemma 18.5(6b) bounds the* entry *point,* $\|z_0\| \leq 7C^2N^2$*, and that* B_{red} *is of the same order — i.e. that the reduction's own steps do not exceed the norm they start from — is [verified computationally]* measured *and not proved here (Remark 18.6); nothing below needs more, the criteria being stated in* B_{red} *itself and the implementation being overflow-checked per operation (v1.5.1, replacing v1.5.0's "*$B_{\text{red}} \leq 7C^2N^2$ *by Lemma 18.5(6)", which that lemma proves for* z_0 *only).* B_{red} ***is not bounded by*** B*: it is of order* C^2N^2 *with the model constant* C *in front, where* B *is of order* N^2 *with the constant* 4 *— and it is squared, so the two criteria below are genuinely different.*

Consequently:

1. ***Signed*** 64***-bit suffices*** *for (i)–(iii) — the* stored*-matrix arithmetic — iff* $2B^2 < 2^{63}$*, i.e.* $B \leq 2^{31}$*, which holds for every* N *whose certified* B_{tab} *stays under* 2^{31} *and, since* $B = \max(B_{\text{tab}}, 4N^2)$*, for every* N *with* $4N^2 \leq 2^{31} - 1$*, i.e.* $N \leq 23170$*, with such a table (v1.5.1, correcting v1.5.0's "*$N \leq 2^{15}$*": at* $N = 2^{15}$ *already* $4N^2 = 2^{32} > 2^{31}$*, and* $N \leq 23170$ *is exactly the reach §10 records, clause (e) of Obligation 20.11's certification being this criterion). It suffices for (iv) iff* $7B_{\text{red}}^2 < 2^{63}$*, i.e.* $B_{\text{red}} \leq 2^{30}$*, which on the shipped model (*$B_{\text{red}} \approx C^2N^2 \approx 144N^2$*) means* $N \lesssim 2700$ a priori*; the* measured *first overflow is later, at* $N \approx 4580$*, because the largest intermediate actually formed is* $\approx B_{\text{red}}^2$ *and not* $7B_{\text{red}}^2$ *(Remark 18.6, where that threshold is computed; v1.5.1, which separated the a-priori criterion from the measured threshold — v1.5.0 quoted the measured* 4580 *as though it followed from the criterion, while clause (2) below derives its figure from the criterion).* ***The second criterion does not follow from the first and is the binding one first****: through v1.4.2 this clause read "with the early-reduction scheme below this covers every* $N < 2^{15}$ *outright by Lemma 18.5(5)", which is false — at* $N = 5003 < 2^{15}$ *the certified* $B = 4N^2$ *leaves* $2B^2$ *four orders of magnitude inside the tier while step 1's reduction leaves it (Lemma 18.5(6), Remark 18.6).*
2. 128***-bit integers suffice*** *for (i)–(iii) iff* $B \leq 2^{63}$*, and for (iv) iff* $7B_{\text{red}}^2 < 2^{127}$*, i.e. on the shipped model* $N \lesssim 1.8 \cdot 10^8$*.*
3. ***Arbitrary precision is never required.*** *For the target regimes* $N < 2^{32}$ *and* $N < 2^{64}$*, fixed-width* 192*-bit, respectively* 320*-bit, integers bound all magnitudes a priori (*$2B^2 \leq 2^{1+2(2\log_2 N + \log_2 C_K)}$*); and the implementation must be overflow-checked, so that a violation of the certified bound* ***aborts rather than corrupts****. Arbitrary-precision arithmetic must not be linked into the pipeline: by this clause it is unreachable, and its absence is itself a guard against a silent performance cliff.*
4. ***The stored wide tier*** *(v1.23.0-pre.9; slot-24 ELL-W). If* $B > 2^{31} - 1$*, clause (1) no longer covers (i) a priori. The pipeline then keeps the* stored *matrices in signed* 64*-bit integers and forms each product of (i) on two stored-class operands, and each determinant of a stored matrix, in* 128*-bit integers — attempting the* 64*-bit evaluation first and escalating only after its overflow check fires, so that every value the* 64*-bit tier computes is computed by the same operations — and narrows the result back with a check. Under* $2NB \leq 2^{63} - 1$ *(Lemma 18.5(7)) the intermediates of (i) on two stored-class operands are exact in* 128 *bits (7b), (ii) on inputs of norm* $\leq B$ *and the unimodular inversions are* 64*-bit (7a), the level-one consumers' products of a fixed datum with a stored matrix are* 64*-bit (7c), (iii) is unchanged — its inputs are step-1 quadratic points, bounded through (iv) — and (iv) is clause (2) unchanged. This is a sufficient condition, not a characterisation: the results of (7b), the (P2) input* $\gamma_a^{-1}\gamma_b$ *and the edge prefix of step 2, the (T1) conjugate, and the level-structure consumers are formed in overflow-checked arithmetic whose completion is*

[verified computationally] measured and not proved here (Lemma 18.5(7), last paragraph; Remark 18.6) — as for (iv), an exceeding operation aborts, it never wraps. On this model, with $B = 4N^2$, that is $N \le 2^{20}-1$, below (iv)'s $N \lesssim 1.8\cdot 10^8$, so it is the binding constraint of the τ-certification. A stored 128*-bit type would not extend it: a* 128*-bit product of two stored matrices needs $B < 2^{63}$, which* 64*-bit storage already holds; the next step past $N = 2^{20}$ is* 128*-bit consumer products, not a wider storage.*

One predicate needs the wide tier, and says so. *The nearest-vertex comparison of Algorithm 18.3(1b) compares two squared* Euclidean *distances in the coordinates of Lemma 19.1(Q1), on cleared denominators — a heuristic choice, which is why the metric it uses is immaterial and only Algorithm 18.3(4) disposes; the comparison is formed in the* 128*-bit tier, reduced by the* gcd *of its two coefficients, and only then handed to the* 64*-bit predicate of §19. If it still exceeds the* 64*-bit tier after that reduction the implementation* ***aborts****. This is the one place where (iii)'s bound $2B^2$ is on the* predicate's own *inputs rather than on the pipeline's stored matrices, and it is the reason clause (iii) is phrased that way.*

Early-reduction scheme (mandatory). *The pipeline needs exact integers only inside (P2), (P3) and the predicates. All* consumers *of a group element need it only modulo ℓ (for ρ_n) or modulo N (for the class map (P1)). The implementation must therefore carry each group element as the triple*

$$\big(\textit{exact } \mathrm{M}_2(\mathbb{Z}) \textit{ (only while needed)},\quad \gamma \bmod \ell,\quad \gamma \bmod N\big)$$

and reduce eagerly, dropping the exact component at its last use; all recurring arithmetic is then single-word.

Mod-ℓ backend. *A modular product $(ab) \bmod \ell$ computed through a* 64*-bit product is exact iff $(\ell-1)^2 \le 2^{64}-1$, i.e. for every $\ell < 2^{32}$; no floating-point and no probabilistic arithmetic is involved. For $2^{32} \le \ell < 2^{64}$ this document requires either* 128*-bit products or a Barrett/Montgomery reduction with proved parameters. Fermat exponent reduction* $\exp \bmod (\ell-1)$ *remains valid for prime ℓ with bases reduced mod ℓ; note that it sends* $\exp \equiv 0$ *with base* $\not\equiv 0$ *to* 1*, which is correct.*

Proof. (i)–(iii) are the closure statement of Remark 18.2 together with Lemma 18.5: every stored matrix has norm $\le B$ by Lemma 18.5(2),(3),(5) and the definition of B; a product of two of them has magnitude $\le 2B^2$ by Lemma 18.5(1); and the predicates of §19 evaluate $a_1b_1 \pm a_2b_2$ on inputs of magnitude $\le B$, hence of magnitude $\le 2B^2$. Clauses (1)–(3) are then the representable ranges of the respective fixed-width signed types. The early-reduction scheme is well defined because ρ_n factors through $\gamma \bmod \ell$ (Definition 8.1 applied to a matrix with entries reduced mod ℓ) and the class map through $\gamma \bmod N$ (Definition 10.2); no other consumer of a group element exists, by Remark 18.2. The mod-ℓ clause is the exactness condition for a product of two residues in a 64-bit word. Clause (4) is Lemma 18.5(7) together with the order of evaluation: the 128-bit form is reached only after the 64-bit form's overflow check fired, and both are exact, so wherever the 64-bit form completes the result is that form's. □

CONTRACT. Separation of concerns (binding). Floating point may be used *only* inside the geometric predicates of §19, and only through the certified filter. All linear algebra over R — cochains, elimination, projectors, Hecke matrices — and all coset, word and translate bookkeeping over $\mathbb{Z}$ and $\mathbb{Z}/N$ is exact integer arithmetic. A build setting must exist that disables the floating-point fast path entirely (*pure-integer mode*) for differential testing, and both settings must produce bit-identical results (Obligation 20.9).

Code map. Definition 18.1: (P1) `Hecke/HeckeCosetTables.h/.cpp` (`class_index`/`class_index_of`, the bottom-row class map; `perm`, the index action $j \cdot \gamma$; `cocycle`, $h_j(\gamma)$, with `rho_cocycle` its eagerly reduced consumer table); (P2) `Hecke/SL2ZTessellationTables.cpp` `word_in_T_S` / `word_in_T_S_with_bound`, with `evaluate_word` as the inverse and the Lemma 18.5(4) bound checked at every intermediate step; (P3) — no call site, by the clause above; (P4) `SL2ZTessellationTables::canonical_key`. Algorithm 18.3: `Hecke/HeckeTauTable.h/.cpp` — step 0 `eval_tau_on_translate` / `try_derive`, step 1 `construct_vertex`, `reduce_to_fundamental_domain`, `choose_vertex` (the (1b) tie-break) and `compare_vertex_distances`, step 2 `construct_edge` with `SL2ZTessellationTables::path_template` (the tree-routed `Canonical` family and the pre-v1.1.0 `LegacyBfs` family, both certified in its own constructor), step 3 `solve_triangle_system` with `used_mod_ell_solve()` as the recorded regime, step 4 `certify()`, called by the constructor, which throws on a non-zero result. The exact quadratic points of step 1 are `Hecke/QuadPoint.h/.cpp`, and the exact vertex positions $\mathrm{pos}(v)$ — with the truncation height of (H4) fixed to $Y = 2$ — are `SL2ZTessellationTables::vertex_position`. Lemma 18.5(5) and Theorem 18.7: `Hecke/ExactInt.h/.cpp` (`Mat22`, `Mat22W`, `checked_mul`/`checked_add`/`chec` `HeckeOverflow`, and `GroupElt`, which *is* the early-reduction triple), with `HeckeTauTable::certified_bound_B` and `certified_bound_B()` the certified B_{tab} and B, and clause (e) of `certify()` the recount and the tier assertion $2B^2 < 2^{63}$. The mod-ℓ backend is `ModArithmetic::mod_mul` and, since the engine work, `engine::ModCtx`. Validations: `HeckeTessellationValidation` and `HeckeTessellationSmokeValidation` ((P2), (P4), the path templates), `HeckeCosetTablesValidation` and `HeckeCosetTablesSelfCheckValidation` ((P1)), `HeckeTauTableValidation` and `HeckeTauTableSelfCheckValidation` (Algorithm 18.3 and the bounds), `HeckeExactIntSmokeValidation` and `HeckeExtendedRegimeValidation` (Theorem 18.7). Since v1.5.2 (slot-3 WP-1): `Hecke/QuadPoint.h/.cpp` is `QuadPointT<I>` with $I \in \{$`int64_t`, `__int128`$\}$ (`QuadPoint`, `QuadPointW`) and `HeckeTauTable::construct_vertex` runs Algorithm 18.3 step 1 through `hecke::escalate_on_overflow` — the `int64` tier first, widened per operation only after a `HeckeOverflow` (Lemma 18.5(6), Theorem 18.7(2),(iv)), `hecke::ScopedExactTier` forcing either tier; `HeckeUEllReachValidation` (the forced-wide arm bit-identical to the `int64` tier wherever that tier completes, and the forced-narrow negative control) and `HeckeUEllReachSlow` (the coset tables at $\ell = 43867$, and the refusal there by clause (e), code 51). Since v1.23.0-pre.9 (slot-24 ELL-W): Theorem 18.7(4) is `hecke::mul_tiered`, `det_tiered` and `inv_unimodular_tiered` (`Hecke/ExactInt.h/.cpp`; the `int64` `Mat22` operation first, the `Mat22W` form after a `HeckeOverflow`, the result narrowed with a check; `ScopedExactTier` forcing either tier), used at the stored-class sites of `Hecke/HeckeTauTable.cpp` (the (T1) translate of step 0, the vertex relation of step 2, clause (a)'s determinant), `HeckeCosetTables::twisted_conjugate` and `SL2ZTessellationTables::word_in_T_S`; clause (e) of `certify()` records the tier (`wide_tier()`) and asserts $2NB \le 2^{63}-1$ there (code 52); **one rollback lever, value-neutral**: `HECKE_TAU_WIDE=0` (line `EFFECTIVE HeckeTauWide`) restores the refusal with code 51 above $B = 2^{31} - 1$ and never changes a value at or below it. Gates: `HeckeWideTauValidation` (the helpers with fault injection, forced-wide = narrow on $\ell \le 5003$, the first wide prime 23173, U_{30011} at $n = 10$ against $\tau(30011)$ from the q-expansion of Δ) and `HeckeWideTauSlowValidation` (forced-wide = narrow at $\ell = 23167$, T_{30011} modulo 13); `HeckeUEllReachSlow`'s arms (9), (10) follow the lever. The boundary datum and η (`Hecke/HeckeBoundaryDatum.cpp`, `Hecke/HeckePairHomotopy.cpp`) keep the narrow assertion (code 51 above 23170): the cone route is not widened.

Formalization note. Definition 18.1 and Algorithm 18.3 are constructive as written (rule (A3)): (P2) is structural recursion on $|c|$ with the decrease exhibited in the proof of Lemma 18.5(4); step 3's termination is the only non-structural argument, and it rests on acyclicity of K^u ((H4)(ii)) plus the finiteness of the support of a chain. Lemma 18.5(1)–(4) are arithmetic inequalities over $\mathbb{Z}$ and need no library beyond `Int.lt` and `abs`; (5) is a statement

about an emitted table and is best formalised as a predicate on it, not as a theorem about C_K — which is how Obligation 20.10 already states it. Theorem 18.7(1),(2) are representability facts about fixed-width integer types; a formalization would carry them as hypotheses on the arithmetic model, and (3) would become the statement that the abstract algorithm never leaves a stated interval.

19 The certified floating-point filter

What this chapter fixes. The reduction steps of Algorithm 18.3 make branch decisions on integer data. This chapter fixes the *shape* of every one of them, proves a threshold below which a floating-point evaluation of that shape certifies its own sign, proves the two regimes in which the exact fallback is provably unreachable, and fixes the priority order and the admissible constants as a binding contract. Nothing in this chapter is a statement about K^u, τ or H^q_{eq}: a predicate returns the sign of an integer, and by Remark 18.4 even a *wrong* sign could not survive Algorithm 18.3(4). The filter is therefore about speed and about the *absence* of unverified floating point, not about correctness of the Hecke operator.

Standing hypotheses. None on ℓ, none on p, none on the model. The two theorems below are statements about IEEE-754 arithmetic on integers, and they carry their floating-point hypotheses in their own statements (rule (A1)), as (FP1)–(FP3).

19.1 The predicate inventory

Lemma 19.1 (Every branch decision is a two-term integer dot product)**.** *[proved here] Assume (H4). Every branch decision taken by Algorithm 18.3 and by the primitives of Definition 18.1 is the sign of an integer of the form*

$$x \;=\; a_1 b_1 \;+\; s\, a_2 b_2, \qquad a_1, b_1, a_2, b_2 \in \mathbb{Z}, \quad s \in \{+1, -1\}, \tag{57}$$

and there are exactly four such decisions:

(D) Orientation / determinant. $\operatorname{sign}(ad - bc)$ *for* $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{M}_2(\mathbb{Z})$*, i.e. (57) with* $s = -1$*.*

(Q1) $|\operatorname{Re} z|$ ***against*** $\tfrac{1}{2}$***.*** *For a point* $z = (u + v\sqrt{-D})/w$ *of* $\mathcal{H}$ *with* $u, v, w \in \mathbb{Z}$*,* $v, w > 0$ *and* $D \in \{1, 3\}$ *— the exact quadratic points in which the model's vertices and all their translates live — the test* $|\operatorname{Re} z| \lesseqgtr \tfrac{1}{2}$ *is* $\operatorname{sign}(2\,|u| - w)$*, i.e. (57) with* $(a_1, b_1, a_2, b_2, s) = (2, |u|, 1, w, -1)$*.*

(Q2) $|z|$ ***against*** 1***.*** *The test* $|z| \lesseqgtr 1$ *is* $\operatorname{sign}(u^2 + Dv^2 - w^2) = \operatorname{sign}\big((u - w)(u + w) + (Dv)\,v\big)$*, i.e. (57) with* $s = +1$*.*

(Q3) The nearest-vertex comparison. *The difference of two squared Euclidean distances on cleared denominators —* $|z - P|^2\,(ww')^2$ *for* $z = (u + v\sqrt{-D})/w$ *and* $P = (u' + v'\sqrt{-D'})/w'$ *— is rational when the two points share their* D *and has the form* $A + B\sqrt{3}$ *with* $A, B \in \mathbb{Z}$ *when they do not* ($\{D, D'\} = \{1, 3\}$)*. Its sign is decided by the signs of* A *and* B *when they agree or one vanishes, and otherwise by* $\operatorname{sign}(A^2 - 3B^2)$*, i.e. (57) with* $(a_1, b_1, a_2, b_2, s) = (A, A, 3B, B, -1)$*; the case* $A^2 = 3B^2$ *with* $(A, B) \neq (0, 0)$ *does not occur,* $\sqrt{3}$ *being irrational.*

No other branch decision on integer data occurs: the class map (P1) is arithmetic modulo N*, the quotient choice in (P2) and in the reduction is an integer division, and the linear solves of Algorithm 18.3(3) are exact eliminations with divisibility tests. In particular,* ***no comparison anywhere in this document is made between floating-point numbers that are not of the form*** *(57).*

Proof. (D), (Q1), (Q2) are the displayed identities, valid over $\mathbb{Z}$; for (Q2) expand $(u - w)(u + w) + Dv^2 = u^2 - w^2 + Dv^2$. (Q1) and (Q2) are the two containment tests defining $\mathcal{F}$, cleared

of denominators; the class of points is closed under the Möbius action of a matrix of positive determinant, since for $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\det M > 0$ and $r := au + bw$, $t := cu + dw$ one has

$$M \cdot z = \frac{\big(rt + D\,ac\,v^2\big) + \big(vw \det M\big)\sqrt{-D}}{t^2 + D\,c^2 v^2},$$

with integer numerator and positive denominator (the denominator vanishes only if $t = cv = 0$, which forces $\det M = 0$). For (Q3), the squared Euclidean distance from z to a point $P = (u' + v'\sqrt{-D'})/w'$, cleared by $(ww')^2$, is $(uw' - u'w)^2 + Dv^2w'^2 + D'v'^2w^2 - 2vv'ww'\sqrt{DD'}$; subtracting the corresponding quantity for a second vertex, after scaling both by the other's w^2, leaves $A + B\sqrt{DD'}$ with $A, B \in \mathbb{Z}$, and $DD' \in \{1, 3, 9\}$, so $\sqrt{DD'} \in \{1, 3, \sqrt{3}\}$ and the only irrational case is $\{D, D'\} = \{1, 3\}$. If A, B have the same strict sign the sign is that sign; if one vanishes it is the other's; otherwise $\operatorname{sign}(A + B\sqrt{3}) = \operatorname{sign}(A) \cdot \operatorname{sign}(A^2 - 3B^2)$. The enumeration is exhaustive by inspection of Algorithm 18.3 and Definition 18.1. □

19.2 The filter

Theorem 19.2 (FP64 filter)**.** *[proved here] Let $a_1, b_1, a_2, b_2 \in \mathbb{Z}$ with $|a_1|, |b_1|, |a_2|, |b_2| \le B$ for an integer $B \le 2^{53}$, let $s \in \{+1, -1\}$, and let $x = a_1b_1 + s\,a_2b_2$, so $x \in \mathbb{Z}$ and $|x| \le 2B^2$. Let $\hat{x}$ be computed in IEEE-754 binary64 under the hypotheses*

(FP1) the four inputs are converted to binary64 exactly — which holds because $B \le 2^{53}$;

(FP2) the rounding attribute is round-to-nearest, so that each elementary operation satisfies the standard model $\mathrm{fl}(\alpha \odot \beta) = (\alpha \odot \beta)(1 + \theta)$ with $|\theta| \le 2^{-53}$;

(FP3) the evaluation order is exactly $\hat{x} = \mathrm{fl}\big(\mathrm{fl}(a_1b_1) + s\,\mathrm{fl}(a_2b_2)\big)$, with one rounding per step — no fused multiply-add, no contraction of the two operations into one, and no wider intermediate precision.

Then

$$|\hat{x} - x| \;\le\; 4B^2 \cdot 2^{-53}\big(1 + 2^{-53}\big) \;<\; E(B) \qquad \text{where} \qquad E(B) := 2^{-50}B^2, \tag{58}$$

and in fact $|\hat{x} - x| < \frac{1}{2}E(B)\big(1 + 2^{-53}\big)$. Consequently:

1. *if $|\hat{x}| > E(B)$ then $\operatorname{sign}(\hat{x}) = \operatorname{sign}(x)$, so the floating-point verdict is* certified *and no exact evaluation is needed;*
2. *otherwise nothing is claimed, and the sign must be obtained by the exact bounded-integer evaluation of Theorem 18.7.*

Proof. By (FP1) the products a_1b_1 and a_2b_2 are products of exactly represented operands, so by (FP2) $\mathrm{fl}(a_tb_t) = a_tb_t(1 + \theta_t)$ with $|\theta_t| \le 2^{-53}$ for $t = 1, 2$, and by (FP3) the sum rounds once, $\hat{x} = \big(a_1b_1(1 + \theta_1) + s\,a_2b_2(1 + \theta_2)\big)(1 + \theta_3)$ with $|\theta_3| \le 2^{-53}$. Write $\eta := a_1b_1\theta_1 + s\,a_2b_2\theta_2$, so that $\hat{x} = (x + \eta)(1 + \theta_3)$ and $|\eta| \le 2B^2 \cdot 2^{-53}$. Then

$$|\hat{x} - x| \;=\; |\eta + (x + \eta)\theta_3| \;\le\; |\eta| + \big(|x| + |\eta|\big)2^{-53} \;\le\; 2B^2 2^{-53} + 2B^2 2^{-53} + 2B^2 2^{-106} \;=\; 4B^2 2^{-53}\big(1 + 2^{-53}\big),$$

which is (58); and $4B^2 2^{-53} = \frac{1}{2} \cdot 2^{-50}B^2 = \frac{1}{2}E(B)$, giving the sharper form. For (1): if $|\hat{x}| > E(B)$ then $|\hat{x}| > E(B) > |\hat{x} - x|$, so x and $\hat{x}$ cannot have opposite signs and $x \neq 0$; hence $\operatorname{sign}(x) = \operatorname{sign}(\hat{x})$. (The design — a cheap filter with a proved threshold and an exact fallback — is Shewchuk's [She97]; the bound above is elementary and is proved here rather than cited.) □

Remark 19.3 (The threshold is itself computed in floating point, and that is sound). [proved here] An implementation cannot form $E(B) = 2^{-50}B^2$ exactly for every $B \le 2^{53}$; it computes $\widehat{E} := \mathrm{fl}\big(\mathrm{fl}(B \cdot B) \cdot 2^{-50}\big)$. Conversion of B is exact, the scaling by a power of two is exact (and cannot underflow for $B \ge 1$), so $\widehat{E} = E(B)(1 + \theta)$ with $|\theta| \le 2^{-53}$. The certified test $|\hat{x}| > \widehat{E}$ is therefore still sound: $\widehat{E} \ge E(B)(1 - 2^{-53}) > \frac{1}{2}E(B)(1 + 2^{-53}) \ge |\hat{x} - x|$ by (58), which is

all the proof of Theorem 19.2(1) used. **The factor 2 of slack between the proved error bound and the normative constant $E(B)$ exists exactly for this reason**, and is why $E(B) = 2^{-50}B^2$ — and not the sharper $2^{-51}B^2$ — is the binding constant.

Corollary 19.4 (Totality threshold)**.** *[proved here] In the situation of Theorem 19.2, suppose in addition that*

$$E(B) + |\hat{x} - x| \;<\; 1, \tag{59}$$

*for which each of the following is sufficient: (i) $E(B) \le \frac{1}{2}$ — **not** $E(B) < \frac{1}{2}$, see Appendix A, entry A.22; (ii) $2B^2 \le 2^{53}$ and $E(B) < 1$. Then the filter is* total*: $|\hat{x}| > E(B)$ certifies $\operatorname{sign}(x) = \operatorname{sign}(\hat{x}) \neq 0$ as before, and $|\hat{x}| \le E(B)$ certifies $x = 0$. The exact fallback is then provably unreachable. Both sufficient conditions hold for*

$$B \;\le\; 2^{24} \qquad (\text{then } E(B) \le 2^{-2} \text{ and } 2B^2 \le 2^{49}),$$

which is the threshold this document fixes and an implementation must use. Under (ii) one has moreover $\hat{x} = x$ exactly.

Proof. $x \in \mathbb{Z}$, so $x \neq 0$ implies $|x| \ge 1$ and hence $|\hat{x}| \ge |x| - |\hat{x} - x| \ge 1 - |\hat{x} - x| > E(B)$ by (59); contrapositively $|\hat{x}| \le E(B)$ forces $x = 0$. (i) implies (59) because $|\hat{x} - x| \le \frac{1}{2}E(B)(1 + 2^{-53})$ by (58), so the left side is at most $E(B)\big(\frac{3}{2} + 2^{-54}\big)$, which for $E(B) \le \frac{1}{2}$ is at most $\frac{3}{4} + 2^{-55} < 1$. For (ii): $2B^2 \le 2^{53}$ makes $|a_t b_t| \le B^2 \le 2^{52}$ and $|x| \le 2B^2 \le 2^{53}$ integers exactly representable in binary64, so under (FP1)–(FP3) every rounding is exact and $\hat{x} = x$, whence $|\hat{x} - x| = 0$ and (59) reduces to $E(B) < 1$. At $B = 2^{24}$, $E(B) = 2^{-50}2^{48} = 2^{-2}$ and $2B^2 = 2^{49}$. □

Corollary 19.5 (FP32 analogue)**.** *[proved here] The statements of Theorem 19.2 and Corollary 19.4 hold verbatim in IEEE-754 binary32, with 2^{-53} replaced by 2^{-24} in (FP2), with (FP1) requiring $B \le 2^{24}$, with the constant*

$$E_{32}(B) := 2^{-21}B^2, \qquad |\hat{x} - x| \le 4B^2 \cdot 2^{-24}(1 + 2^{-24}) = \tfrac{1}{2}E_{32}(B)(1 + 2^{-24}),$$

and with the totality threshold

$$B \;\le\; 2^{10} \qquad (\text{then } E_{32}(B) = \tfrac{1}{2} < 1 \text{ and } 2B^2 = 2^{21} \le 2^{24}),$$

which is certified by Corollary 19.4(i) in its corrected, non-strict form $E_{32}(B) \le \frac{1}{2}$, and independently by Corollary 19.4(ii) — the evaluation is exact *there. It is **not** certified by the* strict *criterion "$E(B) < \frac{1}{2}$" of the superseded contract, which $B = 2^{10}$ fails with equality; that document nevertheless derived this threshold from it, and so did not prove it (Appendix A, entry A.22). The binary32 path is provided for future device inner loops on the fixed model constants and is otherwise not used: $B \le 2^{10}$ covers only very small data, and Theorem 19.2 is preferred everywhere else.*

Proof. Identical to the proofs above with 2^{-24} for 2^{-53}; $4B^2 2^{-24} = \frac{1}{2} \cdot 2^{-21}B^2$. For the threshold, $B = 2^{10}$ gives $|a_t b_t| \le 2^{20}$ and $|x| \le 2^{21}$, all integers of magnitude $\le 2^{24}$ and therefore exactly representable in binary32, so $\hat{x} = x$ and Corollary 19.4(ii) applies with $E_{32}(2^{10}) = \frac{1}{2} < 1$; criterion (i) applies as well, with equality. □

CONTRACT. Priority order (binding). (1) the FP64 filtered predicate of Theorem 19.2; (2) exact bounded-integer evaluation, 64-bit then 128-bit per Theorem 18.7; (3) fixed-width wide integers. **Arbitrary-precision arithmetic must not be linked into the pipeline**; by Theorem 18.7(3) it is unreachable in the target regime, and its absence is itself a guard against a silent performance cliff. **The only admissible thresholds** are $E(B) = 2^{-50}B^2$ (binary64) and $E_{32}(B) = 2^{-21}B^2$

(binary32), evaluated with the *certified* B of Lemma 18.5(5), with the *strict* inequality $|\hat{x}| > E(B)$, and with the evaluation order (FP3). Each of those four — constant, certified B, strict inequality, evaluation order — is load-bearing in the proof above, and none may be changed as an optimisation. **A preconditions violation aborts.** $s \notin \{\pm 1\}$, B above the exact-conversion limit, or an input of magnitude exceeding B leaves the verdict uncertified; the implementation must throw, never return (Theorem 18.7(3), Obligation 20.10). **A pure-integer build must exist** in which the floating-point path is compiled out entirely, and the two builds must produce bit-identical results (Obligation 20.9).

Code map. Theorem 19.2: `Hecke/Predicates.h/.cpp` `sign_dot2`, whose signature $(a_1, b_1, a_2, b_2, s, B)$ is (57) verbatim; (FP3) is enforced by `volatile` intermediates, (FP1) and the magnitude preconditions by `check_dot2_preconditions`, which throws `HeckeOverflow`; the threshold is `std::ldexp(dB*dB, -50)`, which is Remark 19.3's $\widehat{E}$; the exact fallback is `sign_dot2_exact`, the `int64`→`__int128` ladder of Theorem 18.7. Corollary 19.4: the branch `B <= (1 << 24)`. Corollary 19.5: `sign_dot2_f32` with constant 2^{-21} and branch `B <= (1 << 10)`. Lemma 19.1: (D) is `sign_det2`; (Q1) is `QuadPoint::cmp_abs_re_half`; (Q2) is `QuadPoint::cmp_norm_one`; (Q3) is `QuadPoint::sign_lin_sqrt3`, fed by `HeckeTauTable::compare_vertex_di` which forms A, B in the 128-bit tier, divides by $\gcd(A, B)$ and aborts if the result still exceeds the 64-bit tier (Theorem 18.7, the wide-tier clause). Counters `fp_hits` / `exact_fallbacks` (`PredicateStats`) are the *diagnostic* that Corollary 19.4 is observed, and are never branched on (rule (A6)). The build setting is the CMake option `HECKE_PURE_INTEGER`, which compiles out both floating-point paths by `#ifndef`, so that every call becomes an `exact_fallback`. Validation: `HeckeFpFilterValidation` (Obligation 20.9).

Formalization note. Lemma 19.1 is a finite case analysis over $\mathbb{Z}$ and needs no floating-point model at all; it is the statement worth formalising first, because it is what makes the rest of the chapter a self-contained lemma about IEEE-754 rather than a claim about the pipeline. Theorem 19.2 and its corollaries are statements in the standard model (FP2), which a formalization carries as a hypothesis on an abstract rounding function rather than as a fact about hardware; mathlib4's `FloatingPoint` development is not needed, since (FP2) plus exact representability is all the proofs use. Corollary 19.4(ii) is the clause a formalization should prefer: it replaces an error estimate by the statement that every intermediate is an integer of bounded magnitude, which is decidable.

20 Correctness obligations

What this chapter fixes. The implementation is entitled to rely on the numbered statements of this document; in exchange it must *discharge* the obligations below. This chapter is the complete roster: the eleven obligations of the superseded contract, re-homed and repaired, the remark on the exceptional loci that two of them are gated by, and — integrated here, not merely listed — the four obligations that Chapters 9, 14 and 15 added. It closes with the table (§20.3) that says, for each one, what ring it is evaluated over, whether a failure **aborts** or is **recorded**, and whether a *registered* validation stands behind it.

How to read an obligation (rule (A4)). An obligation is a checkable predicate on emitted data, not a wish. Each clause names *(i)* the stored data it reads, *(ii)* the predicate, *(iii)* the ring the predicate is evaluated over, and *(iv)* what happens on failure. Two failure actions exist and they are not interchangeable:

- **aborts** — the computation stops and emits nothing. An obligation clause that aborts is an assertion this document makes and stakes the run on.
- **recorded** — the value is written into the emitted record and the run continues. A recorded quantity is *never* branched on (rule (A6)): it may not select a code path, skip another clause, or be reported as a pass.

A clause is **gated** when it aborts under a stated hypothesis and is recorded otherwise; every gate in this chapter is $6 \in R^\times$ ((H3)), $p \neq \ell$ ((H1)) or "off the exceptional locus of Remark 20.8", and each is written into its own clause.

Standing hypotheses. (H4) throughout, and the datum $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, $N = p^\nu$, $\Gamma'' = \Gamma_0(N)$, $\{s_j\}$, τ of §§10–11. Per rule (A1) each obligation repeats the hypotheses it needs; in particular **no obligation is under a blanket** $\ell \geq 5$, and the three that need it say so in the clause that needs it.

20.1 The core obligations

Obligation 20.1 (Well-definedness in all degrees)**.** Assume (H4); **no hypothesis on** ℓ. Reading the stored τ-table, the coset tables, and the stored bases and projector of §6, an implementation of Θ_g (Definition 12.10) must discharge:

(i) **Equivariance on emitted data.** For every stored cohomology-basis cocycle $z_j^{(q)}$, the image $y_j := \Theta_g z_j^{(q)}$ satisfies *every* relation of $\mathcal{R}$: $y_j(\sigma_{\mathrm{lhs}}) = \rho_n(\gamma)\, y_j(\sigma_{\mathrm{rhs}})$ for every row $(\sigma_{\mathrm{lhs}}, \sigma_{\mathrm{rhs}}, \gamma) \in \mathcal{R}$, i.e. $y_j \in C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$. Over R; a failure **aborts**.

(ii) **Cocycle.** $d\, y_j = 0$ with d the coboundary of §5. Over R; a failure **aborts**.

(iii) **Restriction to C^q_{eq} is exactly solvable.** Each Θ_g-image of a stored C^q_{eq}-basis vector lies in the span of that basis, the solve that defines $\Theta^{(q)}_{\mathrm{eq}}$ succeeds, and its residual, recomputed independently, is identically zero. Over R; a failure **aborts**.

Clauses (i)–(iii) are the runtime form of Theorems 11.7 and 12.11: Θ_g maps $C^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ to itself, commutes with d, and induces $T_{\Gamma g \Gamma}$ on $H^q_{\mathrm{eq}}(K, \mathcal{R}; L)$ for $q = 0, 1, 2$ *simultaneously*. They are the three assertions that Algorithm 12.13's contract box delegates here.

Obligation 20.2 (Independence of all auxiliary choices)**.** Assume (H4). The induced map on H^q_{eq} is independent of the choices listed in Theorem 11.7(a)–(c) and Proposition 12.7. **Every comparison is made on $A^{(q)}$, never on $\Theta^{(q)}_{\mathrm{flat}}$** (Remark 11.9: a validation comparing cochain matrices across a varied choice is testing a statement this document does not make).

(i) **Coset representatives.** Rebuild the whole chain — coset tables, τ-table, operator — against $s'_j = h_j s_j$ with $h_j \in \Gamma''$, and compare $A^{(q)}$ entrywise in every degree. Over R; a failure **aborts**. With $h_j = I$ for every j the rebuild must additionally reproduce $\Theta^{(q)}_{\mathrm{flat}}$ bit for bit — that is the same construction, not a varied choice, and a difference is a defect in the rebuild machinery. With non-trivial h_j the cochain-level agreement is **recorded**, never asserted, because rebuilding the chain rebuilds τ and this clause then exercises Theorem 11.7(a) and (b) together.

(ii) **The twisted chain datum.** Rebuild τ under a different admissible vertex policy of Algorithm 18.3(1) against the *same* cosets and compare $A^{(q)}$ entrywise. Over R; a failure **aborts** *when* $6 \in R^\times$, and is **recorded** otherwise — the uniqueness half Lemma 11.3(2) is exactly where that hypothesis is used, and no cross-τ assertion is made at $\ell \in \{2, 3\}$.

(iii) **The double-coset representative g.** Theorem 11.7(c). **This clause is not reachable through the present interface** and is discharged by its proof alone: the coset tables fix $g = \mathrm{diag}(1, N)$ and $\Gamma'' = \Gamma_0(N)$, and admit only representatives in the same Γ''-classes, so a genuinely different g, or a conjugate subgroup $\gamma_1 \Gamma_0(N) \gamma_1^{-1}$ with $\gamma_1 \notin \Gamma_0(N)$, cannot be built. What *is* run — replacing every s_j by γs_j for one fixed $\gamma \in \Gamma''$ — is the constant-h_j

case of clause (i) and must be reported as such, not as an instance of (c). Making (iii) checkable requires an entry point parameterised by an arbitrary $g \in \Delta$ and an arbitrary finite-index $\Gamma'' \leq \Gamma$.

(iv) **Reduction words.** The words produced by (P2)/(P3) are functions of their input alone (Definition 18.1), so this clause is discharged by determinism of the primitive and needs no run; a non-deterministic quotient rule would be a defect in (P2), not an independence failure.

Obligation 20.3 (Integrality and the two primes)**.** Assume (H4); **no hypothesis on** ℓ.

(i) **Every matrix is integral.** Every matrix entering the pipeline is $\rho_n(\delta) \bmod \ell$ for an *integral* $\delta \in \Delta$ with $\det \delta \in \{1, N\}$, read from the stored τ- and coset tables. Over $\mathbb{Z}$ for the integrality, over R for the reduction; a failure **aborts**.

(ii) **No determinant-N matrix is ever inverted.** Only unimodular inversion occurs, and only on s_j, $s_{j\cdot\gamma}$ and γ_{red}. Over $\mathbb{Z}$; a failure **aborts**. (Lemma 8.4: for $p = \ell$ and $n \geq 1$ every $\rho_n(\delta_j) \bmod \ell$ is singular, so an inversion would not merely be unjustified, it would fail.)

(iii) **The normalisation is the integral one.** $A^{(0)}(T_p) = 1 + p$ on the trivial coefficient system, and the recursion coefficient of Theorem 10.8(1) is $p^{n+1} = p^{k-1}$ with $k = n + 2$ — neither symmetrised nor rescaled (Algorithm 12.13, "no normalisation"). Over R; a failure **aborts**.

(iv) **The bad prime uses the same pipeline.** For $p = \ell$ ((H2)) the operator U_ℓ is produced by the identical construction with no special case, and Corollary 10.10(2)'s coincidence of the full and primitive prime-power operators at $n \geq 1$ holds in the emitted matrices. Over R; a failure **aborts**. The recursion $T_{\ell^\nu} = U_\ell^\nu$ of Corollary 10.10(1) carries $6 \in R^\times$ and is **recorded**, not asserted, at $\ell \in \{2, 3\}$.

Obligation 20.4 (Hecke algebra structure)**.** Assume (H4) and $6 \in R^\times$ — for R of characteristic ℓ, $\ell \geq 5$ (hypothesis (H3)). Then $\Gamma\delta\Gamma \mapsto T_{\Gamma\delta\Gamma}$ extends to a ring homomorphism $\mathcal{H}(\Delta, \Gamma) \to \mathrm{End}\big(H^q_{\mathrm{eq}}(K, \mathcal{R}; L)\big)$ for each q (Theorem 10.8; [standard: [Shi71, Thm. 3.24, §3.2–3.3] for the identities in $\mathcal{H}(\Delta, \Gamma)$ (verified, R20-L), [AS86, §1.1, p. 194] (the action, stated as well known) and [Bro82, Ch. III.9–10] (the transfer) for the computation that makes the assignment a ring homomorphism]), whence the $\{T_N\}_N$ commute [formally verified modulo the commutativity of the Hecke ring, [Shi71, Prop. 3.8, p. 54, applied in §3.2, p. 56] (there for $\det > 0$; for $\det < 0$ the same argument with a diagonal representative), transported by [AS86, §1.1, p. 194], as the hypothesis `hcomm`; locator verified (R20-L): [GT, `obl:algebra-model`]] (all $T_{\Gamma g \Gamma}$, $g \in \Delta$, commute and the R-algebra they generate is commutative, on the model; not the ring homomorphism) and satisfy $T_{p^{\nu+1}} = T_p T_{p^\nu} - p^{n+1} T_{p^{\nu-1}}$ and $T_M T_{M'} = T_{MM'}$ for $\gcd(M, M') = 1$.

The runtime tests, in the normalisation in which each is true. Everything below is over R, entrywise on $A^{(q)}$, in every computed degree; a failure **aborts** under $6 \in R^\times$ and is **recorded** otherwise.

(i) **Commutation.** $[A^{(q)}(T_p), A^{(q)}(T_{p'})] = 0$ for distinct primes p, p' (the shipped instance is $p, p' = 2, 3$).

(ii) **The recursion at $\nu = 2$, for the *full* operator.**

$$A^{(q)}\big(T_{p^2}\big) \;=\; A^{(q)}(T_p)^2 \;-\; p^{n+1} I,$$

with T_{p^2} the **full** operator of (18) obtained *independently of the recursion* — by the double-coset assembly (19), not by the identity being tested, which would make the check vacuous. [formally verified modulo the primitive products of [Shi71, Thm. 3.24(5), p. 63, with Prop. 3.17, p. 58], transported by [AS86, §1.1, p. 194], as the hypothesis `hTp`; locator verified (R20-L): [GT, `lem:m4-hTp-hrel`]] (the identity as algebra on the model, the full operator in the assembled form; the runtime test itself is the C++)

(iii) **The primitive operator, and the exact size of the difference.** The raw pipeline — coset tables, τ-table, operator — run at $\nu = 2$ emits the **primitive** double-coset operator $T_{\Delta_{p^2}^{\mathrm{prim}}}$, whose $p^2 + p$ cosets are what Definition 10.2 enumerates, and it satisfies

$$A^{(q)}\big(T_{\Delta_{p^2}^{\mathrm{prim}}}\big) \;=\; A^{(q)}(T_p)^2 - p^{n+1}I - p^n I, \qquad \text{equivalently} \qquad A^{(q)}\big(T_{p^2}\big) = A^{(q)}\big(T_{\Delta_{p^2}^{\mathrm{prim}}}\big) + p^n I,$$

the difference being the diamond scalar $T(p,p) = p^n I$ of Proposition 10.5. [formally verified modulo the primitive products of [Shi71, Thm. 3.24(5), p. 63, with Prop. 3.17, p. 58], transported by [AS86, §1.1, p. 194], as the hypothesis `hTp`; locator verified (R20-L): [GT, `lem:m4-hTp-hrel`]] (both identities as algebra on the model, the second with no hypothesis; the runtime test itself is the C++) **Both identities are asserted, and the emitting entry point of each is named in the record**; the two are *not* interchangeable and substituting either operator into the other's identity falsifies it by $p^n I$.

(iv) **Diamond.** $A^{(q)}(T(p,p)) = p^n I$ (Proposition 10.5, which carries no hypothesis on ℓ, so this clause aborts for every ℓ).

(v) **Coprime multiplicativity.** $A^{(q)}(T_{MM'}) = A^{(q)}(T_M)A^{(q)}(T_{M'})$ for $\gcd(M, M') = 1$ (the shipped instance is $T_6 = T_2 T_3$).

ERRATUM. Clauses (ii) and (iii) are **erratum E1** (Appendix A, entry A.1). The superseded contract stated this obligation's runtime test as "compute $T_{\Gamma\,\mathrm{diag}(1,p^2)\,\Gamma}$ by the same pipeline with its $p^2 + p$ cosets and compare with $T_p^2 - p^{n+1}$": a $p^2 + p$-coset operator is the *primitive* one and $T_p^2 - p^{n+1}I$ is the *full* one, so the prescription as written fails against a correct implementation, by exactly $p^n I$. §10.1 names the two operators and (19) relates them; entry A.1 carries the evidence, the entry-point table and the runs, and is not restated here.

Obligation 20.5 (Agreement of the three formulations)**.** Assume (H4); **no hypothesis on** ℓ. The three routes of §§12–13 must agree, and the level at which each agreement is asserted is part of the obligation (Remark 11.9).

(i) **Bit-identically on cochains, on the same datum.** The fused route $\Theta_g = \omega_\bullet \circ \Xi_g$ and the direct evaluation of (21) ("route 2"), built from the *same* g, the *same* s_j and the *same* τ-table, agree on every entry of $\Theta_{\mathrm{flat}}^{(q)}$ in every degree. Over R; a failure **aborts**. Likewise $\omega_\bullet \circ \mu_\bullet \circ \alpha_\bullet$ against Θ_g (Proposition 13.2(3)), same data, same level.

(ii) **Through the public evaluation interface.** Applying the operator to each standard basis vector e_b of $C^q(K;V)$ reproduces column b of $\Theta_{\mathrm{flat}}^{(q)}$ exactly, by both routes. Over R; a failure **aborts**. This clause exists because (i) can be discharged against private state; (ii) cannot.

(iii) **On H^1, the bar-cochain route.** The corrected degree-1 group-cochain formula of Proposition 13.4 — *with* the coset permutation $j \mapsto j \cdot \gamma$, whose omission is entry A.20 — induces the same map on H^1_{eq}. Over R, compared at the $A^{(1)}$ level; a failure **aborts**. It is an independent route and not a reformulation, so the comparison is on cohomology, not on cochains.

(iv) **Across the two arithmetic builds.** The pure-integer build and the filtered build produce bit-identical stored objects; see Obligation 20.9, which owns that predicate.

Obligation 20.6 (Shapiro consistency of L_H)**.** Assume (H4). $\dim_R H^q_{\mathrm{eq}}(K, \mathcal{R}; L_{\Gamma''})$, computed by the equivariant cohomology of the *induced* local system of Definition 12.3, equals $\dim_R H^q_{\mathrm{eq}}$ of the $\Gamma_0(N)$-model with coefficients V_R, in every computed degree (Proposition 12.4). Over R; a failure **aborts**. For $p = 2$, $n = 0$, $6 \in R^\times$ the values are $\dim H^0_{\mathrm{eq}} = 1$ and $\dim H^1_{\mathrm{eq}} = 1$, the latter from $\Gamma_0(2)^{\mathrm{ab}}$, the model being the Borel–Serre-type truncation with both cusps of $X_0(2)$

on the boundary circle covering the cusp circle of the base. The identification of either side with $H^\bullet(\Gamma_0(N);V_R)$ needs $6\in R^\times$ (Remark 7.6); the *equality of the two computed dimensions* does not, and that is what this obligation asserts.

Obligation 20.7 (Classical eigenvalue oracle)**.** Assume (H4), $6\in R^\times$ (so $\ell\geq 5$ for $R=\mathbb{F}_\ell$), $p\neq\ell$ ((H1)), n even, and (n,ℓ) **off** the exceptional locus of Remark 20.8. Then, under Eichler–Shimura and reduction modulo ℓ [standard: [Shi71, Ch. 8] for the Eichler–Shimura isomorphism over $\mathbb{C}$; [Har22, Ch. 2–3] for the Eisenstein class and its eigenvalue; [Den23] for the mod-ℓ statement], the characteristic polynomial of $A^{(1)}(T_p)$ on $H^1_{\mathrm{eq}}(K,\mathcal{R};\mathrm{Sym}^n\mathbb{F}_\ell^2)$ is

$$\Big(\prod_f \big(X-a_p(f)\big)^2\Big)\cdot\big(X-(1+p^{n+1})\big)\pmod{\ell},\tag{60}$$

the product running over the Galois orbit of normalised cuspidal eigenforms f of level 1 and weight $k=n+2$ — each with multiplicity *two*, from the holomorphic and antiholomorphic summands of the Eichler–Shimura decomposition (Claim 16.5(1)) — and the last factor the Eisenstein class, of boundary eigenvalue $1+p^{n+1}=1+p^{k-1}$. In particular $\deg=2(d_k-1)+1=2d_k-1$, which is $\dim H^1_{\mathrm{eq}}$ off the exceptional locus.

"The Eisenstein class" modulo ℓ is a space, not a line, at a Kummer cell (v1.5.0; Appendix E, v1.4.2 deferred item 1). The *factor* $X-(1+p^{n+1})$ of (60) is well defined wherever this obligation applies, and nothing above changes. The *class* named by the second standard pointer of this obligation — [Har22, Ch. 2–3], the Eisenstein class and its eigenvalue, a characteristic-0 statement — need not survive reduction as a line: where ℓ divides the numerator of the relevant Bernoulli quotient, the integral Eisenstein eigenline and a cuspidal one reduce into the same socle of a non-split block of the local Hecke algebra, and the joint $1+p^{n+1}$-eigenspace E on H^1_{eq} is two-dimensional ([E] v1.4.0 `cl:harder`; measured at all ten Kummer cells, Remark 15.17). **Wherever this document, or a record it governs, speaks of *the* Eisenstein class modulo ℓ, read the joint eigenspace E**; (60) singles out no line inside it, and Obligation 9.10(v) already states the two conditions — the eigenspace a line, or restriction to the cusp with $\operatorname{rank} r^1=1$ — under which a class, rather than a space, is what is being located. Off the Kummer loci E is a line — measured, not proved here: on every off-Dickson non-Kummer cell with $k\leq 26$ (43 cells, slot-2 WP-7; the statement in general is [E]'s `cl:harder`) — and the two readings then coincide. *No assertion of this obligation moves in either case*: what (i)–(iii) below assert are dimensions, the characteristic polynomial (60) and the full/primitive distinction, none of which names a class inside E (v1.5.1, which states that reason rather than resting on the line case).

What is asserted and what is recorded. Over R, in the stored basis:

(i) **Asserted, aborting:** the dimension $\dim H^1_{\mathrm{eq}}=2d_k-1$ and the characteristic polynomial (60) at those (n,ℓ,p) for which the constants $a_p(f)$ have been independently verified — the shipped set being weights $k\in\{12,16,24\}$, the Eisenstein factor at every admissible (n,ℓ,p), and the Ramanujan congruence at $\ell=691$; $A^{(0)}(T_p)=1+p$ at $n=0$; and $H^\bullet_{\mathrm{eq}}=0$ for odd n at every $\ell\neq 2$ (Remark 7.7).

(ii) **Recorded, never asserted:** the same quantities at weights whose $a_p(f)$ this document has not independently verified (the shipped set being $k\in\{18,20,22,26\}$); every (n,ℓ) on the exceptional locus of Remark 20.8; and every $\ell\in\{2,3\}$, where $6\notin R^\times$ and the identification of $H^\bullet_{\mathrm{eq}}$ with $H^\bullet(\Gamma;V_R)$ is not claimed in either direction ((H3)).

(iii) **The full/primitive distinction applies here too.** An oracle stated for a prime-power index must name which operator it is about; the identity $A^{(q)}(T_{p^2})=A^{(q)}(T_p)^2-p^{n+1}I$ is about the *full* operator, and the corresponding primitive statement carries the extra $-p^nI$ (Obligation 20.4(ii),(iii); Appendix A, entry A.1). **The worked oracles of §21 are bound by this clause**, oracle 6 in particular.

The specialisations used as unit tests are listed in §21; the boundary analogue is Obligation 14.19(vi), and the ±-eigenspace refinement is Obligation 9.10(v). *Provenance of the pointers* (rule (A2)): all three are carried unchanged from the places this document already cites them — [Shi71, Ch. 8] and [Den23] from Claim 16.5(1), [Har22, Ch. 2–3] from Theorem 11.10 — and were not independently re-verified against the texts here; [Den23] carries no section pointer in either place, which is the weakest of the three.

Remark 20.8 (Small-ℓ exceptional loci: where the classical oracle does not apply). [proved here] Let $\ell \geq 5$, $R = \mathbb{F}_\ell$, $n > 0$. The dimension and characteristic-polynomial oracles of Obligation 20.7 equal the characteristic-0 values *except* where Sym^n acquires genuine mod-ℓ $\mathrm{SL}_2(\mathbb{F}_\ell)$-invariants, i.e. where $H^0_{\mathrm{eq}}(K, \mathcal{R}; M_n) = (M_n)^{\mathrm{SL}_2(\mathbb{F}_\ell)} \neq 0$. By Claim 16.5(2) — which is where the invariant-theoretic input and its citation live, and which is not restated here — that happens exactly when n lies in the numerical semigroup

$$\langle \ell+1,\ \ell^2-\ell \rangle \ = \ \{a(\ell+1)+b(\ell^2-\ell)\ :\ a,b \in \mathbb{Z}_{\geq 0}\},$$

generated by the degrees of the two Dickson generators; the first instance is $n = \ell+1$, where the invariant is $x^\ell y - xy^\ell$. Call this the **exceptional locus**. On it, $H^0_{\mathrm{eq}} \neq 0$ for $n > 0$ and $\dim H^\bullet_{\mathrm{eq}}$ legitimately *exceeds* the characteristic-0 value, by Claim 16.5(1); the classical oracle does not apply and must be replaced by a directly computed and recorded baseline (Obligation 20.7(ii)).

Both generators matter, and a test on the first alone is unsound. The predicate is membership in the semigroup, *not* divisibility by $\ell+1$: at $(n,\ell) = (20,5)$ one has $20 = 1\cdot(\ell^2-\ell)$ with $\ell+1 = 6 \nmid 20$, so a $(\ell+1) \mid n$ test would assert a characteristic-0 dimension at a genuine exceptional point. [verified computationally] The implemented predicate is the semigroup membership (`HeckeEndToEndValidation.cpp`, `IsDicksonExceptional`), and $(20,5)$ is in the shipped grid.

The oracle settings of §21 avoid the locus, and it must be checked rather than assumed, since $n \geq \ell$ there: for $n = 10$, $10 \notin \langle 6, 20\rangle$ ($\ell = 5$) and $10 \notin \langle 8, 42\rangle$ ($\ell = 7$), so $\dim H^1_{\mathrm{eq}} = 3 = 2d_{12} - 1$ holds at both.

Obligation 20.9 (FP-filter soundness)**.** Assume (H4); **no hypothesis on** ℓ.

(i) **No uncertified sign.** Every verdict returned by the filtered path satisfies the hypothesis of Theorem 19.2(1) or Corollary 19.4/19.5; equivalently, for every call the returned sign equals the sign computed by an *independent* exact evaluation of (57). Over $\mathbb{Z}$; a failure **aborts**. The exact reference must not be the production fallback: this clause is a differential test, and sharing the routine would make it vacuous.

(ii) **Dead code below the totality threshold.** For $B \leq 2^{24}$ (binary64) and $B \leq 2^{10}$ (binary32) the exact fallback is never entered. Over $\mathbb{Z}$; a failure **aborts**. The counters that witness it are a diagnostic and are never branched on (rule (A6)).

(iii) **Preconditions abort.** $s \notin \{\pm 1\}$, an input of magnitude exceeding B, or B above the exact-conversion limit (2^{53}, resp. 2^{24}) throws; it never returns a value. Over $\mathbb{Z}$; by construction.

(iv) **The two builds are bit-identical.** The build with the floating-point path compiled out and the build with it enabled produce *bit-identical* stored objects — τ-tables, $\Theta^{(q)}_{\mathrm{flat}}$, $\Theta^{(q)}_{\mathrm{eq}}$, $A^{(q)}$ and every hashed golden object. Over R and $\mathbb{Z}$; a difference **aborts**. **This clause is a two-build protocol, not a predicate a single run can evaluate**, and the roster of §20.3 records it as such: a single-process validation can establish (i)–(iii) and can establish that *its own* output is build-independent, but (iv) as stated is discharged only by building twice and comparing.

Obligation 20.10 (Bounded-integer soundness)**.** Assume (H4); **no hypothesis on** ℓ.

(i) **The bound is certified, not assumed.** At table-build time $B_{\mathrm{tab}} = \max\|\gamma_t\|$ is recomputed by a full recount over every stored chain term and compared with the stored value, and $B = \max(B_{\mathrm{tab}}, 4N^2)$ is recomputed from it (Lemma 18.5(3),(5)). Over $\mathbb{Z}$; a failure **aborts**.

(ii) **The selected tier covers B.** The assertion for the 64-bit tier is $2B^2 < 2^{63}$, i.e. $B \le 2^{31} - 1$ (Theorem 18.7(1)); an implementation using a wider tier asserts that tier's inequality instead. Over $\mathbb{Z}$; a failure **aborts**. *Since v1.23.0-pre.9* the wider tier is the stored wide tier of Theorem 18.7(4): a τ-table with $B > 2^{31} - 1$ asserts $2NB \le 2^{63} - 1$ (Lemma 18.5(7); code 52 if it fails), and is refused with code 51, as before, when the wide tier may not run (the rollback `HECKE_TAU_WIDE=0`, or a build forced narrow).

(iii) **Overflow aborts, never wraps.** Every exact product, sum, difference and negation is overflow-checked, including the cases whose naive form is undefined behaviour (negating the most negative value, taking its magnitude). Over $\mathbb{Z}$; a failure **aborts**.

(iv) **Arbitrary precision is absent.** No arbitrary-precision arithmetic is linked into the pipeline (Theorem 18.7(3)); the escalation ladder is 64-bit → 128-bit → fixed-width wide, and the wide tier is unreachable in the target regime. This is a property of the build, **recorded** in the implementation map of §22 rather than asserted at runtime. *Scope of the exclusion* (v1.11.0-pre.2; worded at v1.11.0 to the route as built, `ENGINE_API.md` §13.11.2 and §13.11.8; owner ruling (ii) of 2026-09-25, hecke-extensions slot 11): a value of $\mathbb{Z}/\ell^K$ carried as $L = \lceil K/m \rceil$ digits in base ℓ^m of a *fixed* word width $w \in \{32, 64, 128\}$, m the largest exponent with $\ell^m < 2^{w-1}$ — the ℓ-adic digit induction of `ENGINE_API.md` §13.11, in which L is fixed per cell at construction from the working precision K (on the pipeline, $K = k + e_{\max}$, fixed before the digit run from a certified bound, and capped by the grammar of [E, `sec:ring`]: $k \le 4096$, $K\lceil \log_2 \ell \rceil \le 8192$ bits), is never grown during a computation, and in which every primitive is a single-word operation or one product into a *fixed* double word of $2w$ bits (for $w = 128$ the fixed two-limb type of `ENGINE_API.md` §13.12.2), with every intermediate bounded by a constant stated at its definition — is **not** the arbitrary-precision arithmetic this clause excludes: no digit count is chosen by the data, no allocation follows a carry, and no arbitrary-precision library is linked. The escalation ladder above governs the exact-integer tables of Theorem 18.7 and is unchanged; the digit route is not a rung of it. A route whose digit count grows at run time, or whose digits are not native words, remains excluded. (v1.11.0-pre.2 read "every operation on the digits is a native w-bit operation of the mod-ℓ^m backend of (v)"; as built, a digit product goes through the double word, and (v) is the mod-ℓ backend.)

(v) **The mod-ℓ backend is exact.** A modular product is computed exactly for the ℓ in force — through a 64-bit product for $\ell < 2^{32}$, and through a 128-bit product or a Barrett/Montgomery reduction with proved parameters for $2^{32} \le \ell < 2^{64}$ (Theorem 18.7, mod-ℓ box). Over $\mathbb{F}_\ell$; a failure **aborts**.

Obligation 20.11 (Certified τ-table)**.** Assume (H4); **no hypothesis on** ℓ. An implementation of Algorithm 18.3 emits, for each g and each slot (j, σ), the chain (56) in canonical form, together with the recorded solve regime of every triangle fill, and must discharge the following. Everything is evaluated *exactly*, over $\mathbb{Z}$, or over $\mathbb{F}_\ell$ for an identity involving a chain that Algorithm 18.3(3) solved modulo ℓ; **every clause aborts**, and the certification runs in the constructor, so that an uncertified table cannot exist (Remark 18.4).

(i) **Structure.** Every stored γ_t has $\det \gamma_t = 1$; every term is in the canonical form of Definition 18.1(P4), with sign $+1$ and non-zero coefficient; the canonicalisation is idempotent.

(ii) **(T3), augmentation.** $\varepsilon\big(\tau(s_j \cdot \widetilde{v})\big) = 1$ for every vertex slot.

(iii) **(T2), chain map.** $\partial\, \tau(x) = \tau(\partial x)$ for every stored representative x of degree 1 and 2, the right-hand side evaluated *only* through the shared (T1) helper of Algorithm 18.3(0) and compared after reduction to canonical form — so that construction and certification cannot disagree about how a derived slot is obtained.

(iv) **(T1), twisted equivariance bookkeeping.** For every stored transporter h occurring in the table and every j, the slot (j, σ) equals the sign-tracked (T1)-translate of the slot it is related to; this includes the self-constraints imposed by the elliptic stabilisers, and it is what makes the non-inequivalent slot set of Algorithm 18.3(0) consistent.

(v) **Bounds.** Obligation 20.10(i),(ii) on this table.

(vi) **The consumer table.** The eagerly reduced table of (25) matches a recomputation of $\rho_n\big((g\gamma_t) \bmod \ell\big)$ *from each term's own stored exact* γ_t, entrywise — so that a term pointing at the wrong cached matrix is caught. Over R.

(vii) **The solve regime is recorded.** Whether any fill used the mod-ℓ fallback is written into the record; a table using it is marked `integral = false`, may not claim integrality, and is forbidden by default (owner decision OP2, Remark 11.5). **Recorded**, and the default-policy violation **aborts**.

Given (i)–(vii), no geometric statement enters the correctness of $T_{\Gamma g \Gamma}$: that is the entire purpose of this obligation, and it is why Algorithm 18.3's steps 1–3 are free to choose.

20.2 The obligations the later chapters added

Four further obligations are stated in full where their objects are defined, and are *not* restated here (rule (A4): an obligation has one home). What this section adds is the integration the roster needs — what each is over, and what its status is.

- **Obligation 9.10** (§9), the involution. Clauses (i)–(iii) and (v) are over $\mathbb{Z}$ and R and **abort**; clause (iv), the τ-symmetry $\iota_X \tau \iota_X = \tau$, is over $\mathbb{Z}$ and is **recorded**, because it is the hypothesis of Proposition 9.4(3b) and not a decision procedure (rule (A6)). **Clause (iv) was discharged negatively at v0.13.0 and is discharged positively since v1.1.0**: on the then-canonical τ the recorded value was *false*, 3600 of 10764 slots differing over the shipped $\ell \in \{2, 3\}$ grid, with an exhibited witness and mechanism — the ascending-index tie-break of Algorithm 18.3(1b) on a mirror point, structurally incompatible with ι_K, and, behind it, the breadth-first templates of step 2; on the canonical τ since v1.1.0 the value is *true* on every slot of every table of the grid, and the validation gates it (Remark 9.5; Remark 18.4(1)). The consequence is that at $\ell \in \{2, 3\}$ this document now has a route to $[\iota^*, T_{\Gamma g \Gamma}] = 0$ for the canonical τ: Proposition 9.4(3b). The clause stays in force as the predicate any other τ, vertex policy or refined triangulation must re-evaluate. **Clauses (ii), (iii) and (v) were implemented and run at v1.1.1** (`HeckeIotaSignValidation`), so the obligation is evaluated in full. At v1.1.1 their aborting content passed while the then-*recorded* sign half of (v) came out false for the geometric involution (Eisenstein class in $\ker(\iota_{\mathrm{geo}}^* + \mathrm{id})$, mirror split, on every off-Dickson cell of a 77-cell grid). **At v1.2.0 the owner resolved it by negating Definition 9.1(iv)**, $\iota^* := -\iota_{\mathrm{geo}}^*$, so that "+" is by definition the side of the Eisenstein classes; clause (v)'s sign statement is now a gating definition-consistency oracle (a failure would mean a non-uniform geometric sign, i.e. an ill-posed definition) and holds on every signed cell, the degree-0 anchor is $\iota^* = -1$ on $H^0(\Gamma; \mathbb{F}_\ell)$, and the $\pm$ split $(d_{n+2}, d_{n+2} - 1)$ is recorded as predicted on 66/66 cells (Remark 9.9; `docs/implementation/oq3_resolution.md`). Nothing downstream changed: no pipeline value contains ι^*.
- **Obligation 14.19** (§14), the boundary datum. Clauses (i)–(v) and (vii) are over $\mathbb{Z}$ or R and **abort**; clause (iv) additionally makes integrality unconditional on the boundary, unlike the bulk; clause (vi)'s general-n H^1 multiset is **recorded** until Claim 16.24 is proved. **Unimplemented** (§14, code map).
- **Obligation 15.21** (§15), the pair homotopy. All clauses are over $\mathbb{Z}$ or R and **abort**, except clause (vi)'s three diagnostics — `eta_zero`, `cochain_commutes` and the solve regimes — which are **recorded** and may never select $h = 0$. Clause (v), that the stored h is the transport of the stored η, is what pins $A_c^{(q)}$; no identity among matrices does.

Unimplemented (§15, code map).

- **Obligation 15.22** (§15), cone exactness and intertwining. Over R; clauses (i)–(iii) and (viii) **abort** unconditionally, (iv)'s non-field spots and (v)'s $H^2_{\mathrm{eq}} = 0$ at $6 \notin R^\times$ are **recorded**, (vi) aborts for $6 \in R^\times$ and records otherwise, and (vii) is split by route. **Unimplemented** (§15, code map). Note that (vi) re-checks Theorem 10.8(1) on H^q_c *with the full* T_{p^2} *and the coefficient* p^{n+1}: erratum E1 binds there exactly as it binds Obligation 20.4(ii).

20.3 The roster

> **CONTRACT. A printed proof and a run are different epistemic positions, and this table keeps them apart.** An obligation with a registered validation behind it has been evaluated, on a stated grid, by something that can be re-run; an obligation without one rests on the proof of the statement it discharges, plus whatever *ad hoc* computation is recorded at that statement. Neither is worthless and neither substitutes for the other, and [verified computationally] never upgrades to [proved here] (§1.1). "Registered" below means: named in the validation registry of the shipped tree, hence runnable by name and included in a full suite run.

Of the 94 registered validations (v1.6.2; 89 at v1.5.2, 80 at v1.4.2, 77 at v1.3.3, 68 at v1.3.2), the column below names those that evaluate the obligation's own predicate. Status is one of **discharged** (every clause evaluated, or proved where the clause is not a run), **partial** (some clause is not evaluable through the present interface, and the obligation says which), **negative** (the predicate was evaluated and is false, and the document records the consequence), and **unimplemented** (the objects the obligation reads do not exist in the tree).

obligation	over	failure	status	registered validations
20.1	R	aborts	discharged	`HeckeOperatorSmoke-Validation`, `HeckeIndependence-Validation`, `HeckeEndToEnd-Validation`
20.2	R	aborts; (i) partly records	**partial**: (iii) not reachable	`HeckeIndependenceValidation`
20.3	$\mathbb{Z}$, R	aborts	discharged	`HeckeEndToEndValidation`, `HeckeCosetTablesValidation`, `HeckeBadPrimeValidation`
20.4	R	aborts; records at $\ell \in \{2,3\}$	discharged	`HeckeRelationsValidation`, `HeckeOperatorFamily-SmokeValidation`, `HeckeEndToEndValidation`
20.5	R	aborts	discharged	`HeckeCrossCheckValidation`, `HeckeIndependence-Validation`, `HeckeEndToEnd-Validation`; also in `compute()`
20.6	R	aborts	discharged	`HeckeShapiroValidation`, `HeckeInducedLocal-SystemSmokeValidation`
20.7	R	aborts / records per (i),(ii)	discharged on the shipped grid	`HeckeOracleValidation`, `HeckeEndToEndValidation`, `HeckeBlockFormValidation`
20.8	$\mathbb{F}_\ell$	gates 20.7	discharged	`HeckeEndToEndValidation` (`hecke::IsDicksonExceptionalCount`, `Hecke/DicksonCount.h` since v1.3.3 — one count shared by the three former file-local copies)
20.9	$\mathbb{Z}$, R	aborts	discharged; (iv) by build protocol	`HeckeFpFilterValidation`; (iv) needs two builds
20.10	$\mathbb{Z}$, $\mathbb{F}_\ell$	aborts; (iv) records	discharged	`HeckeExactIntSmoke-Validation`, `HeckeTauTableValidation`, `HeckeExtendedRegime-Validation`
20.11	$\mathbb{Z}$ / $\mathbb{F}_\ell$	aborts; (vii) records	discharged; (vi) vacuous on a `ChainsOnly` table (no consumer table; code map)	`HeckeTauTableValidation`, `HeckeTauTableSelf-CheckValidation`, `HeckeModuleStreamValidation` (`ChainsOnly`); also in the constructor
9.10	$\mathbb{Z}$, R	(i)–(iii),(v) abort; (iv) records; (v)'s $\pm$ dimensions record	(i),(ii),(iii) discharged; (iv) discharged (**negative** at v0.13.0, positive and gating since v1.1.0); (v) discharged — the sign gating and **true by definition** since v1.2.0 (owner decision OQ3, Definition 9.1(iv) negated), the recorded split as predicted	`HeckeIotaTauSymmetry-Validation`, `HeckeIotaSign-Validation`
14.19	$\mathbb{Z}$, R	aborts; (vi) partly records; (vii) records at $\ell \in \{2,3\}$	discharged since v1.3.3, and re-read against the v1.4.0 clauses with no arm changed: (i)–(iv) over $\mathbb{Z}$ on every table, $\ell \in \{2,3\}$ included; (v.a) route B on H^0 at ev-	`HeckeBoundaryTau-Validation` ((i)–(iv)), `HeckeBoundaryOperator-Validation` ((v.a)–(v.c), (vi), (vii)); (i)–(iv) also in the constructor, (v.a)–(v.c) also in `compute()`

Reading the table. Twelve of the fifteen rows are discharged by a registered validation — ten at v1.3.2, Obligation 14.19 since v1.3.3, whose two validations and the clause-by-clause record are in §14's code map, and Obligation 15.21 since v1.5.2 (slot-3 WP-6 discharged its clause (v)). One — Obligation 20.2 — is partial for a stated structural reason, and the missing clause is the one that needs an entry point this tree does not have. One — Obligation 9.10 — is evaluated in full and carries *two* negative records in its history rather than a hope: clause (iv) was discharged negatively at v0.13.0 before the v1.1.0 repair made it hold and gate, and clause (v)'s sign prediction was false for the geometric involution at v1.1.1 before the owner resolved it at v1.2.0 by negating Definition 9.1(iv), after which the clause gates and holds. One — Obligation 15.22 — was partial from v1.5.2 to v1.7.2: every clause but (vii) was evaluated (slot-3 WP-5 and WP-6), and (vii)'s cross-routes were a scheduled phase, not a missing entry point; **since v1.7.3 (slot-6 WP-3, WP-4, WP-5) clause (vii) is evaluated too**, all three of its sentences; until v1.5.1 it and Obligation 15.21 were *unimplemented*, their clauses a specification written as predicates on emitted data precisely so that they could be handed to the phase that builds them unchanged (which is what happened to Obligation 14.19: written at v1.0.0 as such a specification, handed to the phase unchanged, and discharged at v1.3.3 with no clause moved — and to Obligations 15.21 and 15.22 in slots 2–3, likewise with no clause moved). One further clause, Obligation 20.9(iv), is a *protocol* rather than a predicate and is recorded as such. [verified computationally] That protocol has been executed: on 2026-09-16 the tree was configured and built twice out of source, with the floating-point filter enabled and with `HECKE_PURE_INTEGER=ON`, and in both builds `GOLDEN_STRICT=1 ./cohomology_app GoldenRegressionValidation` reported tier A 9446 checked, 0 failing and tier B 4603 checked, 0 mismatching against the same frozen reference — so all 14049 hashed objects, $\Theta^{(q)}_{\mathrm{flat}}$, $A^{(q)}$ and the τ-derived data included, agree bit for bit between the two builds — and `HeckeFpFilterValidation`, `HeckeExactIntSmokeValidation`, `HeckeTauTableValidation`, `HeckeEndToEndValidation` and `HeckeOracleValidation` each exited 0 in both (`docs/matrix-engine/tools/cross_config.sh`). This is a recorded run of an existing script, not a registered validation, and the table says so.

Code map. Obligations 20.1 and 20.5(i) are asserted inside `HeckeOperatorOnK::compute()`, which throws `std::runtime_error` and leaves the object in the not-computed state; the executed counts and the route bit-identity flag are carried in the `certificate` object of `to_json()` and in `certificates()`. Obligation 20.11 is `hecke::HeckeTauTable::certify()`, called by the constructor, clause by clause: (i)–(iv) its items (a)–(d), (v) its item (e), (vi) its item (f), (vii) `used_mod_ell_solve()`. **At v1.23.0-pre.5** (slot-23 G2-F, value-preserving): a table built `hecke::TauTableContents::ChainsOnly` — the module route's table, built by `hecke_ext::RunModuleT` on the stream route — keeps the chain records only and carries *no consumer table* (no table of (25), no ρ_n pool), so on it (vi) has no object and its certification is vacuous (item (f) and the length match of (a) are skipped); (i)–(v) and (vii) run unchanged in the constructor. The module route reads no consumer table on any path: it evaluates $\rho_n \bmod \ell^k$ from each term's exact γ_t (the base change of the integral datum, Remark 11.5). Every other consumer builds the table `Full`, with (vi) as written (`docs/implementation/slot23/G2F_report.md`, `G2F_audit.md`). Obligation 20.10 is `Hecke/ExactInt.h` (`checked_*`, `HeckeOverflow`, `uabs64`) together with that item (e). Obligation 20.9 is `Hecke/Predicates.cpp` and the `HECKE_PURE_INTEGER` option. Obligation 20.4 is `HeckeOperatorFamily::T_p_pow` (recursion), `T_p_pow_direct` (the full operator by (19)) and `T_p_pow_primitive_direct` (the raw primitive operator) — the three entry points of entry A.1's table, and the reason clause (iii) can name which identity belongs to which. Obligation 20.7's gating predicate is `IsDicksonExceptional`. The aggregated gate is `HeckeEndToEndValidation`, which prints an obligation-to-test-identifier coverage table for the nine it covers; Obligation 20.2 is covered by `HeckeIndependenceValidation`

alone, and Obligation 9.10 by the two validations of Chapter 9's code map — (i),(iv) by `HeckeIotaTauSymmetryValidation` and (ii),(iii),(v) by `HeckeIotaSignValidation` — neither of which is part of that aggregate gate.

v1.23.0-pre.4 (slot-23 G2: the module route over $\mathbb{Z}/\ell^k$, a value-preserving re-implementation; no clause relaxed). On the module route (`HeckeOperatorOnKT::compute_module()`, `Hecke/HeckeOperatorOnK.{h,cpp}`, and `EquivariantCohomologyOnKT::compute_cohomology_module()`, `EquivariantCohomologyOnK.{h,cpp}`) Obligations 20.1 and 20.5(i) are discharged as written, in every degree $q = 0, 1, 2$, by a re-implementation (`docs/implementation/slot23/G2_plan.md`, `G2AB_report.md`, `G2CDE_report.md`, `G2G_report.md` and their adversarial audits; merged at `c5af90e9`, `cb967760`, `b33e6fb7`): (G2-A) the solve that defines $\Theta_{\mathrm{eq}}^{(q)}$ (20.1(iii)), the solve for X_q in the cohomology and the first solve of `H_class_coordinates()` are read off the unit Howell pivots of the C_{eq} each solves against (C_{eq}^{q} for $\Theta_{\mathrm{eq}}^{(q)}$, C_{eq}^{q+1} for X_q; `ModuleDegree::piv`, $C[\mathrm{piv},:] = I$ checked, so C is injective and the read-off is the unique solution) by `engine::elim::solve_unit_pivot_readoff` (`src/ModArithmetic/DenseElim.{h,cpp}`, `ENGINE_API.md` §13.6), with the residual recomputed exactly over every row; a non-unit pivot (Hida's lattice), a failed identity check or a non-zero residual falls back to the Howell solve `solve_multi_rhs_module`, with its error texts; (G2-B) the $d \circ d$ certificate $X_q X_{q-1} = 0$ is decided by one exact GEMM (the scalar loop `engine::elim::product_is_zero` is the rollback's), and `elim::cyclic_decomposition`'s column clear runs with the loops interchanged; (G2-C) `compute_module_stream()` assembles $\Theta^{(q)}$ coset by coset ($\Omega_i = \rho_n(s_i^{-1})$ evaluated once per degree and coset, memos scoped to one degree and coset), and on T_ℓ's low-rank path sums one coset's terms per block by one GEMM (`hecke::lowrank::sum_of_low_rank`, `Hecke/HeckeLowRank.h`); (G2-D) route 2 is built in σ-row chunks and compared with $\Theta^{(q)}$ entrywise on every entry of every degree before it is dropped (20.5(i) unchanged, the mismatch text unchanged), the scan of `factor_low_rank` taking its valuation by `FastValuation` (equal to `ModCtx::valuation` on every residue); (G2-E) $\Theta_{\mathrm{eq}}^{(q)}$ is read off the product $\Theta^{(q)}C$ (`engine::elim::solve_unit_pivot_readoff_product`, only the non-pivot rows formed, the residual over every row), and the sat / Loewy runner `RunModuleT` (`HeckeExt/HeckeRingHeckeAlgebra.cpp`), alone and on the stream route, sets `ModuleRetention::Lean` on its operators and calls `release_module_intermediates()` on its cohomology, so that $\Theta^{(q)}$, $\Theta_{\mathrm{eq}}^{(q)}$ and the cohomology's intermediates X_q, `Zc` are freed after their last certificate and their guarded accessors (`Theta_module`, `Theta_eq_module`, `module_X`, `module_Zc`) throw a named error; (G2-G) `elim::cyclic_decomposition` and `elim::invariant_factors` keep lean work buffers (one Howell form of $[\Gamma^{\mathsf{T}} \mid I]$ for the kernel and the coordinates, `kernel_from_augmented`, `solve_from_augmented`). No certificate item is sampled, relaxed or moved out of `compute_module()`. **One rollback lever, value-neutral**: `HECKE_MODULE_STREAM=0` (read once per process by `engine::elim::module_stream_enabled()`; any value other than `0` or `1` warns and keeps the default; scoped as `ScopedModuleStream`) runs `compute_module_legacy()`, the Howell solves, the scalar $d \circ d$ loop, the original loop order and buffers and full retention, i.e. the pre-G2 route (the lever also selects the legacy elimination buffers in every caller, cohomology-only runs and Integral subquotients included); when the variable is set, one `EFFECTIVE HECKE_MODULE_STREAM` line is printed at process exit with the counters of `module_stream_totals()`. It selects a route, never a value: every entry computed is the residue of an exact sum or the unique solution of an injective system, so order and grouping cannot move a byte; G2-B's loop interchange is neutral because its column operations are independent (none writes the pivot column, and the pivot row's quotients are taken first), and G2-G's shared buffer because the Howell form is canonical; and the gate is `HeckeModuleStreamValidation` (`standard`) and `HeckeModuleStreamSlowValidation` (`slow`), registered in `Registry/Slot23/G2.cpp` (source `HeckeExt/HeckeModuleStreamValidation.cpp`): stream against legacy entry-wise on every object, byte-equal `SAT_DEFECT` lines, negative controls on the residual, on

route 2 and on a released object, and non-vacuity counters. The golden reference and the digests are unchanged across these packages. **At v1.23.0-pre.7** (slot-23 G2-T, value-preserving; `docs/implementation/slot23/G2T_report.md`, `G2T_audit.md`): on the stream route only, `compute_module_stream()` evaluates each $\rho_n(g)$ mod ℓ^k straight into row-major buffers (`ComputeSymmetricPowerRowsRing`, `src/ModArithmetic/FlatMatrix.{h,cpp}`: one Pascal row per operator, the column steps in a wavefront, the inverse of the divisor folded into one reduction per entry where the accumulator cannot overflow) and factors T_ℓ's terms by `factor_low_rank(..., fast_scan)` (a divisibility scan by ℓ^v, a valuation floor, a rank-one GEMM update; the same pivot sequence); **one rollback lever, value-neutral**: `HECKE_MODULE_EVAL=0` (`engine::elim::module_eval_enabled()`, scoped as `ScopedModuleEval`; inert under `HECKE_MODULE_STREAM=0`; when set, one `EFFECTIVE HECKE_MODULE_EVAL` line at exit) restores the G2-F evaluation; both compute the same canonical residues, so the lever selects a route, never a value (gate: `HeckeModuleStreamValidation` (V)). **At v1.23.0-pre.8** (slot-23 G2-T2 and G2-P, value-preserving; `docs/implementation/slot23/G2T2_report.md`, `G2T2_audit.md`, `G2P_report.md`, `G2P_audit.md`). G2-T2: `ComputeSymmetricPowerRowsRing`'s column steps run on AVX2 / AVX-512F lanes where the modulus allows Shoup products ($\ell^k \le 2^{31}$; otherwise scalar), each lane computing the scalar step's canonical residue; and route 2 of a degree is built in one chunk with one ρ_n memo per coset shared across the σ-rows (the memo is keyed by the entries mod ℓ^k, of which ρ_n is a function); two **rollback levers, value-neutral**: `HECKE_RHO_SIMD=0` (the scalar fill; `RhoFillPath`, scoped as `ScopedRhoFillPath`) and `HECKE_ROUTE2_SHARED_MEMO=0` (G2-D's σ-row chunks; `ScopedRoute2SharedMemo`), each with an `EFFECTIVE` line at exit when set (gate: `HeckeModuleStreamValidation` (V)(d)–(g)). G2-P: when the consumer pool (`ENGINE_THREADS`) has more than one worker, `compute_module_stream()` assembles $\Theta^{(q)}$ and route 2 in coset batches (the distinct keys listed serially, evaluated by the pool, each σ-row's block accumulated by one task in the serial term order and grouping), and `howell_rows` and the product read-off split their disjoint row updates over the elimination hook; **the values do not depend on the thread count or the schedule** — every entry is the canonical residue of an exact sum formed from the same operands, the Howell form and the read-off are unique, and `ENGINE_THREADS=1` is the serial route itself; **one rollback lever, value-neutral**: `HECKE_MODULE_PARALLEL=0` (`engine::elim::module_parallel_enabled()`, scoped as `ScopedModuleParallel`; the `EFFECTIVE HECKE_MODULE_PARALLEL` line at exit when set, also naming `ENGINE_HOST_ELIM_HOOK=0`, which keeps the Howell and read-off splits serial) restores the serial route at any thread count (gate: `HeckeModuleStreamValidation` (P): every object at consumer pools of 1, 2, 7, 16 workers against pool 1, the rollback, the negative controls of the parallel read-off on a pool worker; `HeckeModuleStreamSlowValidation` (P)). **At v1.23.0-pre.10** (slot-24 MT-1, value-preserving; `docs/implementation/slot24/MT1_report.md`, `MT1_audit.md`, MERGE after fixes F1–F3). (0) At every thread count, `elim::cyclic_decomposition`'s pivot search on the lean module route visits the trailing entries in the same column-major order with the same strict comparison and stops at the first unit: the serial search keeps the first entry of least valuation, valuation 0 is the least possible, so the held candidate is the one the full scan returns, and everything after the search is unchanged. (a)–(e) When `HECKE_MODULE_PARALLEL` is on and the consumer pool has more than one worker, the module route also splits over the pool (all but the assembly's split over the elimination hook, so under `ENGINE_HOST_ELIM_HOOK=0` only the assembly splits, as G2-P's Howell and read-off splits stay serial there): the remaining pivot search (tiles over the columns, folded in tile order with the same strict comparison, i.e. the first minimum of the concatenation) and `cyclic_decomposition`'s disjoint row operations; the products $Y = \Theta H$, DC and $C \cdot$ basis in row or column blocks (each entry the same exact dot product modulo ℓ^k); the runtime assertions of Obligation 20.1(i), (ii) checked per column by pure reads, the serial loops run verbatim, with their first message, if any column fails (so the outcome, the message and

the executed counts are the serial ones); the assembly as one task per (σ, σ_k) block, in the serial term order; and the residual of the read-off behind Obligation 20.1(iii) and the right-hand sides of `solve_from_augmented` in blocks, the smallest failing index's message thrown. So Obligations 20.1 and 20.5(i) are discharged as written and **the values do not depend on the thread count or the schedule**; **one rollback lever, value-neutral**: `HECKE_MODULE_MT=0` (`engine::elim::module_mt_enabled()`, scoped as `ScopedModuleMt`; any other value warns and keeps the default; the `EFFECTIVE HECKE_MODULE_MT` line at exit when set, with the counter `pivot_exit`; inert under `HECKE_MODULE_STREAM=0`) restores G2-P's route verbatim, the full scan included (gate: `HeckeModuleMtValidation`, `standard`, `Registry/Slot24/MT1.cpp`: `cyclic_decomposition` at consumer pools of 1, 2, 3, 7, 13, 16 workers, the read-off and the solves at pools of 1, 2, 7, 16, every object of a set of cells and whole `SAT_DEFECT` cells at pools of 1, 2, 7, 16, each against the lever off at pool 1). **At v1.23.0-pre.11** (slot-24 G2-H, value-preserving; `docs/implementation/slot24/G2H_report.md`, `G2H_audit.md`, MERGE after fixes F1, F2). On the stream route, under the lean retention of the sat / Loewy runner `RunModuleT` only, Obligation 20.1(iii) is **certified without forming** $\Theta^{(q)}_{\mathrm{eq}}$: `engine::elim::unit_pivot_residual_product` makes the read-off's checks that $C^q_{\mathrm{eq}}[\mathrm{piv},:] = I$, in the same order, and then checks $(\Theta^{(q)}[\mathrm{np},:] - C^q_{\mathrm{eq}}[\mathrm{np},:]\,\Theta^{(q)}[\mathrm{piv},:])\,C^q_{\mathrm{eq}} = 0$ entrywise, which by associativity and distributivity over $\mathbb{Z}/\ell^k$ (no unit used) is exactly the vanishing of the read-off's residual $\Theta^{(q)}C - C\,(\Theta^{(q)}C)[\mathrm{piv},:]$; with $C[\mathrm{piv},:] = I$ the read-off $(\Theta^{(q)}C)[\mathrm{piv},:]$ is the unique solution of the solve that defines $\Theta^{(q)}_{\mathrm{eq}}$, so the certificate is the read-off's, item for item, and Obligation 20.1(iii) is discharged as written; any other outcome takes the Howell solve with its texts. There, route 2 is subtracted from $\Theta^{(q)}$ in place after $Y = \Theta^{(q)}H$ and (i)–(iii), and Obligation 20.5(i) is checked as $\Theta^{(q)} - (\text{route 2}) = 0$ entrywise (the same exact sums, the same mismatch text); the cohomology's X_q and `Zc` are freed inside `compute_cohomology_module()` after their last reader (`set_module_early_release`). On the stream route in every caller, `howell_rows` adopts its store as H (the same rows in the same order) when no closure row was appended and every row of the store is a pivot row, and the lean `cyclic_decomposition` / `invariant_factors` take the length from its pivot loop. No certificate item is sampled, relaxed or moved out of `compute_module()`. **One rollback lever, value-neutral**: `HECKE_MODULE_LEAN_PEAK=0` (`engine::elim::module_lean_peak_enabled()`, scoped as `ScopedModuleLeanPeak`; any other value warns and keeps the default; inert under `HECKE_MODULE_STREAM=0`; when set, one `EFFECTIVE HECKE_MODULE_LEAN_PEAK` line at exit) restores the route of before (gate: `HeckeModuleStreamValidation` (H)).

Formalization note. Every obligation here is already a decidable predicate on a finite emitted table (rule (A4)), which is the shape a formalization wants: the statement to prove is "if the predicate holds on the emitted data then the stored $A^{(q)}$ is the matrix of $T_{\Gamma g \Gamma}$", and the predicate itself needs no proof, only evaluation. Three shapes recur and are worth factoring once: *entrywise equality of two matrices over R* (Obligations 20.1(iii), 20.5, 20.4), *equality of two integer chains after canonicalisation* (Obligation 20.11(iii),(iv), and identically in Obligations 14.19 and 15.21), and *a magnitude recount against a stored bound* (Obligation 20.10(i)). The one obligation that is not of this shape is 20.9(iv), which quantifies over *builds*; a formalization would express it as the statement that the pipeline's output is a function of the abstract sign predicate alone — which is exactly what Lemma 19.1 plus Theorem 19.2 give, and is a better target than the two-build diff.

21 Worked micro-example ($p = 2$) and the numerical oracles

What this chapter fixes. Two things, and they are different in kind. First, the *worked datum*: the coset tables, the cocycle table and the induced block matrices for $g = \mathrm{diag}(1, 2)$, printed in full, so that §§10–12 have an instance small enough to check by hand and so that

§11's forward reference has a target. Second, the *oracles*: the nine numerical unit tests of the superseded contract, re-homed with erratum A.1 applied to oracle 6 and erratum A.23 to oracle 7, and the three new families the boundary, the cone and the involution require. An oracle is not an obligation: Obligation 20.7 is the normative statement and this chapter is the list of instances it reads, together with the registered validation that evaluates each. **Nothing here may be cited as a licence to weaken an obligation**, and Obligation 20.7(iii) binds oracle 6 specifically.

Standing hypotheses. (H4) throughout, and the datum $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, $N = p^\nu$, $\Gamma'' = \Gamma_0(N)$, $\{s_j\}$ of Definition 10.2, τ of Algorithm 18.3. Per rule (A1) every oracle below repeats the hypotheses it needs; in particular the eigenvalue oracles carry $6 \in R^\times$, $p \neq \ell$ ((H1)), n even, and "off the exceptional locus of Remark 20.8", and the two oracles that carry none say so.

21.1 The datum for T_2

Take $p = 2$, $\nu = 1$, $N = 2$, $g = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, $m = p + 1 = 3$ (Proposition 10.1(1)). The canonical transporters of Definition 10.2, their classes in $\mathbb{P}^1(\mathbb{Z}/2)$, the coset elements $\delta_j = s_j^{-1} g$ and the upper-triangular transporters t_j are

j	s_j	$\mathrm{cls}(s_j)$	$\delta_j = s_j^{-1} g$	t_j
1	I	(0:1)	$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$	I
2	$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$	(1:1)	$\begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$	T
3	$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	(1:0)	$\begin{pmatrix} 0 & 2 \\ -1 & 0 \end{pmatrix}$	S

(61)

Each δ_j has determinant 2 and is integral; the right coset representatives of Proposition 10.1(1) are $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$. The t_j are T^k for $k = j - 1$ up to $j = N = 2$ and $t_{N+1} = S$, with top-row classes (1:0), (1:1), (0:1).

Proposition 21.1 (The $p = 2$ cocycle table)**.** *[proved here] With $\mathcal{G} = \{T, S, ST, TS\}$ the generator multiset of §7.1 and the s_j of (61), the permutation $j \cdot \gamma$ and the cocycle $h_j(\gamma) = s_j \gamma s_{j \cdot \gamma}^{-1}$ of Definition 10.2 are*

γ	$1 \cdot \gamma$	$2 \cdot \gamma$	$3 \cdot \gamma$	$h_1(\gamma)$	$h_2(\gamma)$	$h_3(\gamma)$
T	1	3	2	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 1 \\ -2 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$
S	3	2	1	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
ST	2	3	1	$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & -1 \\ 0 & -1 \end{pmatrix}$
TS	3	1	2	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 2 & -1 \end{pmatrix}$

(62)

Every entry has determinant 1 and even lower-left entry, i.e. lies in $\Gamma_0(2)$; the three permutations are the transpositions/3-cycles shown, and Lemma 10.3's two identities hold on the table: $j \cdot (TS) = (j \cdot T) \cdot S$ and $h_j(TS) = h_j(T)\, h_{j \cdot T}(S)$ for $j = 1, 2, 3$.

Proof. Twelve products of 2×2 integer matrices, each displayed above; the class $\mathrm{cls}(s_j \gamma)$ determines $j \cdot \gamma$ by Proposition 10.1(4), and $h_j(\gamma) := s_j \gamma s_{j \cdot \gamma}^{-1}$ is then read off. The two identities are

four more products each. [verified computationally] The same table is recomputed from p alone and compared entry by entry — representatives, classes, permutations and all twelve cocycles — by `HeckeCosetTablesValidation` (oracle `O16`), which reads it from this chapter and not from `Hecke/HeckeCosetTables.cpp`. □

The induced system. By (22) the induced local system $L'' = L_{\Gamma_0(2)}$ of Definition 12.3 is the relation table of §7.1 with each $M_\gamma = \rho_n(\gamma)$ replaced by the $3(n{+}1) \times 3(n{+}1)$ block matrix $M_{\Gamma''}(\gamma) = \sum_j E_{j,\,j\cdot\gamma} \otimes \rho_n(h_j(\gamma))$ read off (62) — one non-zero block per block row. Because $\gamma \mapsto M_{\Gamma''}(\gamma)$ is a homomorphism (Proposition 12.2) and $S^2 = (ST)^3 = -I$ in Γ, the assembled table must satisfy

$$M_{\Gamma''}(S)^4 = M_{\Gamma''}(ST)^6 = I, \qquad M_{\Gamma''}(S)^2 = M_{\Gamma''}(ST)^3 = M_{\Gamma''}(-I), \tag{63}$$

and these are assertions about the *assembly*, not about Γ: they fail if a permutation or a block is transcribed wrongly. They are checked at $n \in \{0, 2, 10\}$, $\ell \in \{5, 691\}$ by `HeckeCosetTablesValidation`.

α and ω. With the t_j and s_j of (61), Propositions 12.6 and 12.7 give the block column and the block row

$$\alpha = \begin{pmatrix} \rho_n(I) \\ \rho_n(T) \\ \rho_n(S) \end{pmatrix} : V_R \to V_R^{\oplus 3}, \qquad \omega = \Big(\rho_n(I),\ \rho_n\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix},\ \rho_n\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\Big) : V_R^{\oplus 3} \to V_R,$$

the entries of ω being $\rho_n(s_j^{-1})$, all integral and invertible over every R (no determinant-N matrix is inverted; Obligation 20.3). The middle map is *not* of this shape (Remark 12.9): it is the per-simplex block map Ξ_g of Definition 12.10, determined by the certified τ-table.

Example 21.2 (The τ-table at $p = 2$, as emitted). [verified computationally] Algorithm 18.3 at $p = 2$, $\nu = 1$ stores one chain per pair (j, σ) with $1 \le j \le 3$ and $\sigma \in o_q(K)$, i.e. $3{\cdot}(N_0{+}N_1{+}N_2) = 3{\cdot}(7{+}11{+}5) = 69$ chains — small enough to read in a code review, which is why the superseded contract printed the count. Run of 2026-09-16 (`HeckeTauTableSelfCheckValidation`, which writes the table to `docs/implementation/tau_table_p2.txt`): 69 stored chains, 15 triangle fills, **all** 15 **solved over** $\mathbb{Z}$ **and none modulo** ℓ, $B_{\mathrm{tab}} = 2$ and hence the certified bound $B = \max(B_{\mathrm{tab}}, 4N^2) = 16$ of Lemma 18.5; every chain coefficient in $\{0, \pm 1\}$; certification (Obligation 20.11) passed at all 69 grid points, and 0 of the 66 grid points of the wider self-check used the mod-ℓ solve. **The** 69 **slots are not** 69 **independent constructions.** Step 0 of Algorithm 18.3 builds one slot per Γ''-orbit and derives the rest through the shared (T1) helper; the emitted table marks each derived slot, and $\widetilde{v_3} = T \cdot \widetilde{v_0}$ is the first place where the superseded step list's "one representative per orbit, since $\{(j, \sigma)\} \leftrightarrow$ simplices of $\Gamma''\backslash K^u$" is false (Appendix E, v0.18.0, under `alg:tau`).

21.2 A fully worked instance: $(n, \ell, p) = (2, 5, 2)$

At $n = 2$ the coefficient module is $V_R = \mathrm{Sym}^2(\mathbb{F}_\ell^2)$, $d_V = 3$, basis $(u_0, u_1, u_2) = (x^2, xy, y^2)$. By Definition 8.1 the three blocks of α and the three of ω are, over $\mathbb{Z}$,

$$\rho_n(I) = I_3, \quad \rho_n(T) = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}, \quad \rho_n(S) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$\rho_n(s_1^{-1}) = I_3, \quad \rho_n(s_2^{-1}) = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}, \quad \rho_n(s_3^{-1}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

and the three coset elements act by $\rho_n(\delta_1) = \mathrm{diag}(1,2,4)$, $\rho_n(\delta_2) = \left(\begin{smallmatrix} 1 & 0 & 0 \\ -2 & 2 & 0 \\ 1 & -2 & 4 \end{smallmatrix}\right)$, $\rho_n(\delta_3) = \left(\begin{smallmatrix} 0 & 0 & 4 \\ 0 & -2 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right)$. All of these are reduced modulo $\ell = 5$ before use, never after (§18). [verified computationally] The pipeline on this instance emits $\dim H^0_{\mathrm{eq}} = \dim H^2_{\mathrm{eq}} = 0$, $\dim H^1_{\mathrm{eq}} = 1$ and

$$A^{(1)}(T_2) = (9 \bmod 5) = (4) = \left(1 + 2^{n+1}\right) \bmod 5,$$

the Eisenstein eigenvalue of Obligation 20.7; and at $n = 0$, $\ell = 5$, $A^{(0)}(T_2) = (3) = (p+1)$. Both are asserted, with the two evaluation routes bit-identical and the runtime assertions of Obligation 20.1 executed, by `HeckeOperatorSmokeValidation` (run of 2026-09-16, exit code 0). **The cochain-level matrices are not printed here and must not be**: at $n = 2$ they are 21×21, 33×33 and 15×15, they depend on the choice of s_j and of τ (Remark 11.9), and only $A^{(q)}$ is an invariant of the datum.

21.3 The nine oracles

Let $A^{(q)}(T_p)$ be the matrix of Algorithm 12.13, and write $\mathrm{Eis}(p,n) := 1 + p^{n+1}$. Each oracle names its hypotheses, its assertion, and the registered validation that evaluates it; the numbering 1–9 is the superseded contract's and is preserved, because Obligation 20.7(iii) cites *oracle 6* by number. Whether a given instance **aborts** or is **recorded** is decided by Obligation 20.7(i),(ii) and not here.

1. **Trivial coefficients.** $n = 0$, $R = \mathbb{F}_\ell$ with $\ell \geq 5$, any p: $\dim H^0_{\mathrm{eq}} = 1$, $\dim H^1_{\mathrm{eq}} = \dim H^2_{\mathrm{eq}} = 0$, and $A^{(0)}(T_p) = (p+1)$ — the normalisation (T3) pins this (Remark 11.8). [proved here] for $A^{(0)}$: Remark 11.8's computation $T_{\Gamma g \Gamma} v = \sum_j \rho_n(\delta_j) v = mv$ on invariants, with $m = p+1$. `HeckeOracleValidation` (`O1`).
2. **Odd n.** n odd, $2 \in R^\times$ (so any $\ell \neq 2$), any p: $H^q_{\mathrm{eq}} = 0$ for every q and the pipeline returns 0×0 matrices in every degree. [proved here]: Remark 7.7 — $-I$ is central, lies in the stabiliser relation at v_6, and $\rho_n(-I) = (-1)^n I = -I$. `HeckeOracleValidation` (`O2`). This oracle carries **no** hypothesis $\ell \geq 5$ and is the one small-ℓ statement of the list.
3. **Weight 4.** $n = 2$, $\ell = 5$ (so $6 \in R^\times$), $p = 2$: $\dim H^1_{\mathrm{eq}} = 1$, $A^{(1)}(T_2) = (\mathrm{Eis}(2,2) \bmod 5) = (4)$, $\dim H^2_{\mathrm{eq}} = 0$; and $A^{(1)}(T_3) = (28 \bmod 5) = (3)$. There are no cusp forms of weight 4, so the whole of H^1_{eq} is Eisenstein. `HeckeOracleValidation` (`O3`); §21.2 is this oracle worked out.
4. **Weight 10.** $n = 8$, $\ell \in \{5, 7, 101\}$, $p = 2$: $\dim H^1_{\mathrm{eq}} = 1$ and $A^{(1)}(T_2) = (513 \bmod \ell) = (\mathrm{Eis}(2,8) \bmod \ell)$; $\dim S_{10} = 0$. At $(n,\ell) = (8,7)$ the pair is *on* the exceptional locus of Remark 20.8 ($8 = \ell+1$) and is therefore **recorded**, not asserted. `HeckeOracleValidation` (`O4`).
5. **Weight 12 — the cusp form.** $n = 10$, $\ell \in \{5, 7, 13, 101, 10007\}$ (all off the locus: $10 \notin \langle 6, 20\rangle$, $10 \notin \langle 8, 42\rangle$; Remark 20.8), $\ell \neq 691$, n even, $p \neq \ell$: $\dim H^1_{\mathrm{eq}} = 3$ and

$$\det\left(X - A^{(1)}(T_p)\right) \equiv \left(X - a_p(f_{12})\right)^2 \left(X - \mathrm{Eis}(p,10)\right) \pmod{\ell}, \tag{64}$$

   with f_{12} the normalised weight-12 cusp form (LMFDB `1.12.a.a`), the multiplicity 2 coming from the two Eichler–Shimura summands (Obligation 20.7, Claim 16.5(1)). The embedded constants are $a_2(f_{12}) = -24$, $a_3(f_{12}) = 252$, $a_5(f_{12}) = 4830$, $a_7(f_{12}) = -16744$, and $\mathrm{Eis}(p,10) = 1 + p^{11}$, so at $p = 2$ the polynomial is $(X+24)^2(X-2049)$. `HeckeOracleValidation` (`O5`), `HeckeEndToEndValidation`.
6. **Hecke relations — with erratum A.1.** $6 \in R^\times$, $n \in \{0, 2, 10\}$, $\ell \in \{5, 101\}$, in every degree q:
   (a) $[A^{(q)}(T_2), A^{(q)}(T_3)] = 0$;
   (b) $A^{(q)}\left(T_4^{\mathrm{full}}\right) = A^{(q)}(T_2)^2 - 2^{n+1} I$, with T_4^{full} obtained independently of the recursion by the double-coset assembly (19) (`HeckeOperatorFamily::T_p_pow_direct`);

(c) $A^{(q)}\big(T_{\Delta_4^{\mathrm{prim}}}\big) = A^{(q)}(T_2)^2 - 2^{n+1}I - 2^nI$, with $T_{\Delta_4^{\mathrm{prim}}}$ the raw 6-coset pipeline (`HeckeOperatorFamily::T_p_pow_primitive_direct`);
(d) $A^{(q)}(T(p,p)) = p^nI$ and $A^{(q)}(T_6) = A^{(q)}(T_2)A^{(q)}(T_3)$.
`HeckeRelationsValidation` (O9), `HeckeOperatorFamilySmokeValidation`.

> **ERRATUM. This is the site erratum A.1 was found in.** The superseded oracle 6 read "$A^{(q)}(T_4) = A^{(q)}(T_2)^2 - 2^{n+1}I$ with T_4 computed independently from $\Gamma\,\mathrm{diag}(1,4)\,\Gamma$ (6 cosets)": $6 = p^2 + p$ cosets is the *primitive* operator and $T_2^2 - 2^{n+1}I$ is the *full* one, so the prescription as written fails against a correct implementation by exactly 2^nI. Clauses (b) and (c) are the two correct statements, each with its emitting entry point named. **Obligation 20.7(iii) binds this oracle**: an oracle stated for a prime-power index must name which operator it is about, and this one does, in both directions. Entry A.1 carries the evidence and the entry-point table and is not restated here.

7. **Bad prime $p = \ell$ — with erratum A.23.** $p = \ell$ (hypothesis (H2)), $n \in \{0, 2, 10\}$. *Asserted for every ℓ:* the pipeline runs to completion and certifies, i.e. Obligations 20.1, 20.3 and 20.11 are discharged, and $U_\ell := T_\ell$ is well defined. *Asserted for $\ell \geq 5$ only* (the shipped instance is $p = \ell = 5$, $n = 10$): $T_{\ell^2} = U_\ell^2$ for the *full* operator, which is Corollary 10.10(1) and needs $\ell \geq 5$; and the ordinarity stabilisation rank $A^{(q)}(U_\ell^r)$ non-increasing and stationary for $r \geq \dim H^q_{\mathrm{eq}}$. *Recorded, never asserted, at $\ell \in \{2, 3\}$:* the same two quantities, and every eigenvalue statement, because $6 \notin R^\times$ ((H3)). `HeckeRelationsValidation` and `HeckeBadPrimeValidation` (O10).

> **ERRATUM.** The superseded oracle 7 asserted $U_4 = U_2^2$ at $p = \ell = 2$ "(Corollary `cor:ulrecursion`)". Corollary 10.10(1) carries $\ell \geq 5$, inherited from Theorem 10.8, and clause (2) says something else entirely (full = primitive for $n \geq 1$); neither licenses the assertion at $\ell = 2$. Entry A.23; the implementation already records rather than asserts it there (amendment A-F1, `HeckeRelationsValidation`:273–330), so the defect is in the contract text alone.

8. **Ramanujan congruence and non-semisimplicity.** $n = 10$, $\ell = 691$, $p = 2$. Since $2049 + 24 = 3 \cdot 691$, (64) degenerates to $(X + 24)^3$ mod 691; the assertion with content is that $A^{(1)}(T_2) \neq -24\,I$, i.e. the minimal polynomial is not $X + 24$ — the Eisenstein/cuspidal extension is non-split. This is the torsion-sensitive part of the pipeline; cf. [Den23]. The Jordan partition (expected $(2, 1)$) is **recorded**, not asserted. `HeckeOracleValidation` (O6), `HeckeBlockFormValidation`.
9. **Internal agreement, and restriction compatibility.** (a) Cross-route agreement and pure-integer-versus-filtered agreement are Obligations 20.5 and 20.9; they are *bit*-identity statements, not eigenvalue statements, and are listed here only so that the oracle roster is complete (`HeckeIndependenceValidation`, `HeckeFpFilterValidation`, `HeckeCrossCheckValidation`). (b) *Restriction compatibility:* for $\ell \geq 5$, $n \in \{2, 10\}$, $p = 2$, and only when the Eisenstein and cuspidal T_2 eigenvalues are distinct modulo ℓ, the composite $H^1_{\mathrm{eq}}(K, \mathcal{R}; L) \to H^1(K_\partial; L|_{K_\partial})$ has rank 1: the eigenvector of eigenvalue $\mathrm{Eis}(p, n)$ maps to a non-zero class and the cuspidal generalised eigenspace maps to 0 [standard: [Har22, Ch. 2–3] for the Eisenstein class and its boundary eigenvalue]. `HeckeTessellationValidation`, `HeckeMorphismRegressionValidation` (O15). **What this clause is about.** The object on the right is the *circle model* K_∂ of Proposition 14.4, i.e. the $2 \in R^\times$ specialisation of Definition 14.2, and the composite is r^1 of §15 *without* any Hecke structure on the target. It is therefore *not* the boundary oracle of 14.19(vi), and it is not evidence for Obligation 15.22(iii).

Remark 21.3 (Two implementations of the Dickson predicate). [verified computationally] The gating predicate of Remark 20.8 occurs twice in the tree under the same name. `HeckeEndToEndValidation.cpp`'s `IsDicksonExceptional` is the semigroup membership $n \in \langle \ell{+}1, \ell^2{-}\ell \rangle$, which is the predicate Remark 20.8 states and pins; `HeckeOracleValidation.cpp`'s copy tests only the first generator, $(\ell+1) \mid n$. On the grid that second copy is used on — $n \in \{0,2,4,6,8,10,12,14,22\}$ against $\ell \in \{5,7,13,101,691,10007,134217757\}$ — the two predicates **agree at every point**, checked exhaustively on 2026-09-16; the exceptional points there are $(6,5)$, $(12,5)$ and $(8,7)$, all caught by either test. The weaker copy is therefore sound where it is used and unsound in general (Remark 20.8's counterexample $(20,5)$ is outside its grid and inside the other's). Recorded so that extending `HeckeOracleValidation`'s grid is known to require replacing it.

21.4 The new oracle families: ∂, the cone, and ι

One of these three families is a specification, not a run; the other two have been run. Chapter 15 is implemented only in its η-half — the η-table exists since v1.4.2, no Φ_g and no $A_c^{(q)}$ does (its code map says so) — so family (B) below is what the phase that builds it must evaluate, written as numbers so that they can be evaluated rather than argued about, and it may not be read as evidence that its obligation is discharged. Family (A), the boundary oracles, **was run at v1.3.3** (2026-09-16, `HeckeBoundaryOperatorValidation`; until v1.3.2 it, too, was a specification, chapter 14 being unimplemented) and its own paragraph records the outcome. Family (C), the involution oracles, **was run at v1.1.1 and again at v1.2.0** and its own paragraph records both outcomes — the second under the normalisation $\iota^* = -\iota^*_{\mathrm{geo}}$ of Definition 9.1(iv) that the first motivated; Obligation 9.10 is evaluated in full and holds. Each family names the obligation it serves.

(A) **Boundary oracles** (Obligation 14.19(vi)). For n even, $p \neq \ell$: on $H^0(C^\bullet_\partial)$ in the basis $x^{n-b\ell}\mathrm{Nm}^{\,b}$ the eigenvalues of Θ^∂_g are the characters $\chi_b(p)$ of Proposition 14.16(1) — *asserted*; for $n < \ell$ the class of y^n in $H^1(C^\bullet_\partial)$ has eigenvalue $\mathrm{Eis}(p,n) = 1 + p^{n+1}$ (Proposition 14.16(3)) — *asserted*; the general-n H^1 multiset of Claim 16.24(2) — *recorded* until that claim is proved. **The rank is $\lfloor n/\ell \rfloor + 1$, not** 1, so an oracle written for $n < \ell$ does not extend by inspection: at $(n,\ell) = (10,5)$ the space V_R^T is *three*-dimensional, with the three eigenvalues χ_0, χ_1, χ_2 of Proposition 14.16(1). The first instance to run is $(n,\ell,p) = (2,5,2)$, where $n < \ell$ and n is even, so Proposition 14.16(3) applies: H^0 and H^1 are both one-dimensional, spanned by $[x^2]$ and $[y^2]$, and both carry the eigenvalue $\chi_0(2) = 1 + 2^3 = 9 \equiv 4 \pmod 5$. That number is the point: it is the same 4 as $A^{(1)}(T_2)$ of §21.2, and Obligation 15.22(iii) — $r^1 A^{(1)} = A^{(1)}_\partial r^1$ — is what forces the agreement. [verified computationally] **Run at v1.3.3** (`HeckeBoundaryOperatorValidation`, 546 cells): every asserted number above holds — $\chi_b(p)$ on each $x^{n-b\ell}\mathrm{Nm}^{\,b}$ (575 vectors on 330 cells, the rank $\lfloor n/\ell \rfloor + 1$ realised, three eigenvalues at $(10,5)$), $1 + p^{n+1}$ on $[y^n]$ for $n < \ell$ (232 cells) — the first instance $(2,5,2)$ prints $A^{(0)}_\partial = [4]$, $A^{(1)}_\partial = [4]$, bulk $A^{(1)}(T_2) = [4]$, $\operatorname{rank} r^1 = 1$, and the general-$n$ H^1 multiset of Claim 16.24(2) agrees on 330/330 cells — recorded, the claim's tag unchanged. The right-coset substitution of Proposition 14.6(4), run as the negative control, returns χ_{b+1} and breaks Obligation 15.22(iii) on 170 cells, as Proposition 14.16(2) says it must (§14, code map).

(B) **Cone oracles** (Obligation 15.22(iv),(v)). At $6 \in R^\times$, n even, off the Dickson locus, the six-term sequence (52) with $H^2_{\mathrm{eq}} = 0$ gives, by (53), a dimension book the implementation must reproduce exactly. The first instance to run is again $(n,\ell,p) = (2,5,2)$, where every term is known independently: $\dim X_1 = \dim H^0_{\mathrm{eq}} = 0$ (Remark 20.8: $2 \notin \langle 6, 20 \rangle$), $\dim X_2 = \dim H^0(C^\bullet_\partial) = 1$ and $\dim X_5 = \dim H^1(C^\bullet_\partial) = 1$ by Proposition 14.16(3), $\dim X_4 = \dim H^1_{\mathrm{eq}} = 1$ (§21.2), and $\operatorname{rank} r^1 = 1$ with $\ker r^1 = 0$ (oracle 9(b), and $S_4 = 0$, so all of H^1_{eq} is Eisenstein); whence (53) forces $\dim X_3 = \dim H^1_c = 1 - 0 + 0 = 1$

and $\dim X_6 = \dim H^2_c = 1 - 1 = 0$. At $n = 10$ the same book is to be run with $\dim X_2 = \dim X_5 = \lfloor 10/\ell \rfloor + 1$, which is 3 at $\ell = 5$ and 1 at $\ell \in \{13, 101\}$: the $\ell \le n$ rows are where a boundary implementation is actually exercised, and they are not $n < \ell$ rows in disguise. In addition $H^0_c = 0$ unconditionally (Lemma 15.2(4)), every composite of consecutive maps vanishes, and $A^{(q)}_c$ satisfies the relations of Obligation 15.22(vi) — including clause (b) of oracle 6 above with $A^{(q)}$ replaced by $A^{(q)}_c$. The negative control of Obligation 15.22(viii)(b) is part of this family and is the one arm that must *fail* a clause to pass.

(C) **Involution oracles** (Obligation 9.10(v)). For $\ell \ge 5$ off the Dickson locus and n even, $\dim H^1_{\mathrm{eq},+} = d_{n+2}$ and $\dim H^1_{\mathrm{eq},-} = d_{n+2} - 1$ with d_k the dimension of level-1 weight-k modular forms; at $n = 10$, $k = 12$, $d_{12} = 2$, so the split of $\dim H^1_{\mathrm{eq}} = 3$ is $2+1$, and the T_p-eigenvector of eigenvalue $\mathrm{Eis}(p, 10)$ lies in the $+$ part. At $n = 2$, $k = 4$, $d_4 = 1$, so the split of $\dim H^1_{\mathrm{eq}} = 1$ is $1+0$. The dimensions are **recorded**, not asserted (Obligation 9.10(v)); *asserted* there are $\iota^* A^{(q)} = A^{(q)} \iota^*$ for $6 \in R^\times$ and, since v1.2.0, the sign itself — the T_p-eigenvector of eigenvalue $\mathrm{Eis}(p, n)$ lies in the $+$ part on every cell on which it can be located (the Dickson cells carry no sign), which under $\iota^* = -\iota^*_{\mathrm{geo}}$ (Definition 9.1(iv)) is a consistency statement about the definition. The degree-0 instance is exact and needs no cohomology: ι^* on $H^0_{\mathrm{eq}} = V_R^\Gamma$ is $-\rho_n(\iota)$ restricted to the invariants, which at $n = 0$ is $-\mathrm{id}$. [verified computationally] **Run at v1.1.1 and at v1.2.0 by `HeckeIotaSignValidation`.** At v1.1.1, for the then-current involution ι^*_{geo}, the asserted part held and the recorded prediction came out as the exact mirror — the Eisenstein eigenvector in the $-$ part, the split $(d_{n+2} - 1, d_{n+2})$: $1 + 2$ at $n = 10$, $0 + 1$ at $n = 2$ — on every cell of a 77-cell grid where a sign is defined, with $\iota^*_{\mathrm{geo}} = +1$ at $n = 0$ pinning the global sign. That finding is what owner decision OQ3 of 2026-09-16 acted on (Remark 9.9). **At v1.2.0, under the normalised ι^*, every number above holds as printed**: the Eisenstein eigenvector is in the $+$ part on all 74 signed cells (gating), the split is $(d_{n+2}, d_{n+2} - 1)$ on all 66 off-Dickson cells — $2 + 1$ at $n = 10$, $1 + 0$ at $n = 2$ — and $\iota^* = -1$ at $n = 0$ (gating). The two runs are one computation read through $\iota^* = -\iota^*_{\mathrm{geo}}$; the validation prints both. *The diagnostic of Obligation 9.10(iv) had been run before this family and was false on the then-canonical τ; on the canonical τ it holds and gates since v1.1.0* (Remark 9.5). It is not part of this family and may not be used to skip clause (v) — and was not: (v) was evaluated independently.

Code map. The worked datum is `hecke::HeckeCosetTables` (representatives, class map, permutation and cocycle, `Hecke/HeckeCosetTables.h`:21 citing this section) and `hecke::HeckeTauTable`; the printed table of Example 21.2 is written by `HeckeTauTableSelfCheckValidation`. Of the oracles, 1–5 and 8 are `HeckeOracleValidation` (its families `O1`–`O8`, `O14`), 6 and 7 are `HeckeRelationsValidation` together with `HeckeOperatorFamilySmokeValidation` and `HeckeBadPrimeValidation`, 9(a) is `HeckeIndependenceValidation` / `HeckeCrossCheckValidation` / `HeckeFpFilterValidation` and 9(b) is `HeckeTessellationValidation` / `HeckeMorphismRegressionValida` `HeckeEndToEndValidation` re-runs the whole matrix and prints the obligation-to-test coverage table. The three families of §21.4 were, until v1.3.3, two without an implementation and one with; since v1.3.3 family (A) is run (`HeckeBoundaryOperatorValidation`), since v1.4.2 the η-table of family (B) exists while no cone assembly does, and family (C) is `HeckeIotaSignValidation`, which since v1.1.1 discharges Obligation 9.10(ii),(iii),(v). `sec:example` is cited from 16 sites in 8 source files (§22.5, census of 2026-09-16), so this section's numbering is load-bearing: renaming oracle 6 would silently retarget them.

Formalization note. Proposition 21.1 and (63) are finite computations in $\mathrm{M}_2(\mathbb{Z})$ and $\mathrm{GL}_{3(n+1)}$, decidable as stated, and are the cheapest thing in this chapter to formalise — but also the least valuable, since they instantiate Lemma 10.3 and Proposition 12.2, which are the

statements worth having. Oracles 1 and 2 are the two worth formalising: each is a *theorem* of this document (Remark 11.8, Remark 7.7) with a two-line proof, not an imported constant, and together they pin the normalisation and the odd-n vanishing that every other oracle is read against. Oracles 3–5, 8 and 9(b) depend on Eichler–Shimura and on embedded $a_p(f_{12})$; a formalisation should carry the constants as hypotheses and prove only the *shape* of (64), which is Obligation 20.7. Oracle 6 is finite linear algebra over R once $A^{(q)}$ exists, and clauses (b),(c) are the pair whose *difference* $p^n I$ is the content (Proposition 10.5); that difference, not either identity alone, is what a formalisation should state, because it is what entry A.1 exists to stop anyone getting wrong again.

22 Implementation notes: mapping to the code base

What this chapter fixes. Three things the per-chapter code maps do not: the *aggregate* map from this document to the tree and back, in one table; the *configuration surface* — which switches exist, and the standing rule that none of them may change a computed value; and the *record schema*, i.e. what an emitted run must contain for the obligations of §20 to be checkable after the fact. It closes with §22.5, the guards on this document itself.

> **CONTRACT. No switch in §22.3 may change a computed value.** Every build option, environment variable and command-line flag listed there selects a *route*, a *backend*, a *schedule* or an *amount of certification* — never an answer. The two that select a route are exactly the two rollbacks (`HECKE_PURE_INTEGER`, Obligation 20.9(iv); `ENGINE_BACKEND`), and for both the contract is bit-identity of every stored object, not agreement to within anything. The one that changes what is *emitted* rather than what is computed is the certification level (`--lean`, `certify`), and §22.4 requires the record to say which level ran.

22.1 The code map

Chapter by chapter, the translation units that realise it. "—" means the chapter has no implementation; §22.2 says what that costs.

chapter	translation units (paths relative to the repository root)
3–6 (Part I)	`src/hacs/`: `SimplicialCohomology`, `LocalSystemSimplicialComplex` (the $(K, \mathcal{R})$ datum, `SimplexRelation`), `EquivariantCohomologyOnK` (the $C/Z/B/H$ bases and P_q of 6.3, 6.1, 6.2, 6.5, 6.6), `EquivariantCochainComplexMorphismOnK` (`FiberBlockTable`; the A.19 fix)
7 (model)	`LocalSystemSL2Z` (`SL2ZRelationDatum`, `GenerateSL2ZRelationsWithGroupData`); `Hecke/SL2ZTessellationTables` (the closure and the canonical key of 18.1(P4)); `Hecke/HeckeModelDescriptor` (the 7/11/5 literal is now a descriptor)
8 (Sym^n)	`ComputeSymmetricPowerMatrixFM` (`src/ModArithmetic/FlatMatrix.cpp`) — (9) literally; `LocalSystemUtils`. **Since v1.7.3** (slot-6 WP-1, phase E7a) the contragredient: `Mat22::adj` (`Hecke/ExactInt`), `ComputeContragredientPowerMatrixFM` and `ComputeFibreMatrixFM` (`LocalSystemUtils`), `GenerateSL2ZContragredientRelationsWithGroupData` (`LocalSystemSL2Z`), `Hecke/HeckeContragredient` (Φ_n of 8.8, the pairing of 8.6), the `FibreRepresentation` argument of `Hecke/HeckeOperatorOnK` ($A(D_n)(T_p)^q$); since v1.7.3 (slot-6 WP-3, phase E7b) `Hecke/HeckePairing` — the relative fundamental 2-chain, the cone-level and circle pairings, `PairingCertificate`, `PairingAdjoint`, i.e. 16.26(1)–(3) tested and the PD gate of 15.22(vii); `HeckeDualityValidation`. **Since v1.7.4** (slot-7 WP-1) phase E7 is **accepted**: 16.26(1)–(2) tested on all 54 cells of the extension stage's case list at $\ell \geq 5$ (`HeckeDualityValidation` arm A16 on 34 of them, the `slow` `HeckeDualityCoverageSlowValidation` on three, two more under an environment switch), failing on none, the twisted controls rejected; 16.26 stays open — a test, not a proof; the E7 objects are golden since the stage's add-only emission of slot 7 (651 entries). (At v1.7.2: not built, scheduled as phase E7.) **Since v1.8.1** (slot-8 WP-3, phase E8, status) ρ_n over $\mathbb{Z}/\ell^k$: `ComputeSymmetricPowerMatrixRing` (`src/ModArithmetic/FlatMatrix.cpp`) — (9)'s integer entries reduced modulo ℓ^k, by the unit recurrences where the divisor is a unit, a division-free Pascal route otherwise, and the split $\mathrm{diag}(\ell, 1) \cdot g'$ for U_ℓ; the $\mathbb{F}_\ell$ builder is unchanged
9 (ι)	`Hecke/HeckeIotaTauSymmetryValidation` — ι_K, its signs and τ^ι, for Obligation 9.10(i),(iv), (iv) gating on the canonical τ since v1.1.0 — and, since v1.1.1, `Hecke/HeckeIotaSignValidation` — ι^*_{geo} and $\iota^* = -\iota^*_{\mathrm{geo}}$ on cochains and on $H^\bullet_{\mathrm{eq}}$, the $\pm$ split and the Eisenstein sign (gating and true by definition since v1.2.0, the geometric sign recorded), for Obligation 9.10(ii),(iii),(v). Both are validations. **Since v1.6.2 (slot-4 WP-3, phase E3) the chapter has production code**: `Hecke/HeckeInvolution` — the `EquivariantCochainComplexMorphismOnK` this chapter's code map specifies, one $-\varepsilon_\sigma \rho_n(\iota)$ block per simplex, `compute()` discharging Obligation 9.10(i),(ii),(iii) blockwise and exactly over $\mathbb{Z}$ and aborting on failure — and `HeckeExt/HeckeSignSplit` — clause (v), $A^{(q)}_{\pm}$, the per-sign decomposition and the Eisenstein line — gated entrywise against the two validations' own construction by `HeckeInvolutionValidation`. **Since v1.7.2 (slot-5 WP-1 and WP-12a, phase E2) the chapter's objects are in the record**: `hecke_ext/v2` carries the per-sign objects with `iota_sign`, and, per the owner's ruling of 2026-09-22, the record-level `iota_exact_comparison` — Obligation 9.10(ii)'s rider as an emitted object (`ran` / `relation_rows` / `rows_with_exact` / `rows_compared`), refused by `HeckeExtRecord::verify()` code 23 if it misreports. `HeckeExtRecordValidation`. **Planned at v1.20.0-pre.1** (slot-20 WP-LV3(d), [E, `rmk:lviotak`], [E, `chk:lvtriangle`]): `Hecke/HeckeInvolution` and the console honour ι^* over $R = \mathbb{Z}/\ell^k$ (the console's exit 5 removed); Proposition 9.6 ($2 \in R^\times$) and Proposition 9.4 already cover it, so no statement of this chapter changes. **Built at v1.20.0-pre.2** (WP-LV3, merged `c1fda051`): `hecke::BuildRingInvolution` (record `hecke_involution_ring/v1`; $\iota^2 = \mathrm{id}$ and the length split checked; $\ell \in \{2, 3\}$ refused), the console's `iota` on a $\mathbb{Z}/\ell^k$ handle ([E, `sec:ring`] (h)), script `049_lv3_iota_ring.oc`; gated in `HeckeLValueValidation` (arm `LV3-I`).
10, 11	`Hecke/HeckeCosetTables` (10.2, 10.3), `Hecke/HeckeTauTable` (11.1, 18.3, 20.11, certified in the constructor), `Hecke/HeckeOperatorFamily` (10.4, 10.5, 10.8, 10.10; the three entry points of A.1); since v1.4.2 `Hecke/HeckeIntegerSolve`, 18.3 step 3's integer solver, moved verbatim out of `HeckeTauTable.cpp` by slot-2 WP-8 so that the η fills use the one solver (value-preserving: golden and `tau_table_p2.txt` unchanged). Since v1.5.2 (slot-3 WP-1, the U_ℓ reach repair recorded in this chapter's code map): 18.3 step 1 runs through `hecke::escalate_on_overflow` on `QuadPointT<I>` (`Hecke/QuadPoint`,

The arithmetic substrate below all of this — `ModArithmetic/` and the engine (`engine::kernels`, `engine::blk`, `engine::tape`, `engine::cuda`) — realises no statement of this document. It is bound only by Theorem 18.7 and by the invariant that $\mathbb{F}_\ell$-values are integers: the engine's one floating-point step is a quotient *estimate* corrected by an integer compare-and-subtract, so it computes an exact residue, and the device kernels are exact-integer twins of the CPU ones. Consequently no value in this document depends on the thread count, the batching, the layout, the arena or the device (`EngineDeterminismValidation`, `EngineCrossDeviceValidation`, `CudaBitIdentityValidation`).

22.2 What is not implemented

CONTRACT. v1.20.0-pre.1 (hecke-extensions slot-20 WP-2). Slot 20's code is written against text outside this document, contract first (rule 21), and merges only after that text is at pre.1 (rule 22). **WP-LV3** (the L-value route at depth 2, [E, `sec:lvdepth2`]): a generic $\mathfrak{m}_{\mathrm{Eis}}$-block reader over $\mathcal{O} = \mathbb{Z}_\ell[x]/(P)$, the cross-kind gate ([E, `cl:lvtriangle`], [E, `chk:lvtriangle`]), the change-of-lattice constant ([E, `def:lvlattice`]), ι^* over $\mathbb{Z}/\ell^k$ (the §9 row; this document's part is Definition 9.1 and Propositions 9.4, 9.6), the twisted rung $(37; 62, 6)$ on §29's character model. **WP-MEAS** (the measure-valued linear evaluation, [E, `sec:lvmeasure`]): the Mazur–Tate–Teitelbaum measure on this document's model through the cone of §15 (its H^1_c) and the modular symbols of [E, `rmk:lvshapes`] (B); `HeckeLValueMeasureValidation` (`standard`). **NB-4** (the atlas sidecar and `--serve-atlas`): a transport of the console and the atlas store, specified by [P]; it realises no statement of this document, and the console's values, transcripts and digests are invariants of it. **WP-JET** (the build on an `aarch64` Jetson Orin, `sm_87`): every patch is value-neutral and golden-clean, gated by `--all` and `cross_config.sh`. None of them realises a statement of this document; nothing of this box is built at this version. The milestone-M4 `\LEAN` markers are not in this pre-release: they follow slot-20 L20-6 (the Lean audit F3) and its repairs. **v1.20.0-pre.2: built** (each merged after its contract text, rule 22; zero golden movement and the pinned digests unchanged at each merge). WP-MEAS (`f2897f74`): `ModularSymbolMeasure`, `LValueRoute::measure()`, `HeckeLValueMeasureValidation`; extended by slot-20 S1 / S2 (`af3bf133`: the record `lfunction/v1` and the `read_only` verb `lfunction`, [E, `sec:lvlfunction`]). NB-4 (`6218ebc6`): the atlas sidecar `oc-atlas-index/1`, `--serve-atlas`, `ConsoleServeAtlasValidation`. WP-LV3 (`c1fda051`): the generic $\mathfrak{m}_{\mathrm{Eis}}$-block reader (tame, totally ramified orders; [E, `chk:lvtriangle`]), the cross-kind gate, c_f^{ε} in the normalisation record ([E, `prop:lvlattice`]), ι^* over $\mathbb{Z}/\ell^k$ (the §9 row), the twisted rung $(37; 62, 6)$ on §29's character model. WP-JET: value-neutral, as stated. Still none realises a statement of this document.

CONTRACT. v1.19.0-pre.5 (hecke-extensions slot-19 WP-2 continued). Built and merged since the v1.19.0-pre.4 box: the bilinear and linear instances of [E, `def:lvroute`] (slot-19 WP-BIL, `HidaPairingRoute`, `HeckeExt/LValue/HidaPairing`; slot-19 WP-LIN, `BirchModularSymbolRoute`, `HeckeExt/LValue/BirchModularSymbol`; arms of `HeckeLValueValidation`; [E, `rmk:lvshapes`], [E, `rmk:lvlegc`]). They read this document's model, the cone of §15 and the pairing of Proposition 8.6 (`Hecke/HeckePairing.h`); their gates observe, consistently with Definition 9.1, that this document's T_p on H^1_c is the modular-symbol T_p (no adjoint, no scalar) and that a geometric-even symbol lies

in the minus part, and that the cone-level cup form is -1 times Hida's pairing on the stored bases. None realises a statement of this document; no statement, tag or count of this document changes.

CONTRACT. v1.19.0-pre.4 (hecke-extensions slot-19 WP-2 continued). Built and merged since the v1.19.0-pre.3 box: WP-LV's twisted instance (slot-19 WP-LV2, `HeckeExt/LValue/`, `HeckeLValueValidation`; [E, `cl:lvtwist`] on §29's character model) and WP-HL (the truncated Λ-adic arm, `HeckeExt/HeckeHidaLambda`, `HeckeExt/HeckeHidaLambdaForms`, `HeckeLambdaValidation`; [U] `sec:u-lambda`). So every slot-19 package of the v1.19.0-pre.1 box is built. None realises a statement of this document; no statement, tag or count of this document changes.

CONTRACT. v1.19.0-pre.3 (hecke-extensions slot-19 WP-2 continued). Of the v1.19.0-pre.1 box below, **built and merged**: WP-LC (`HeckeExt/HeckeLatticeModule`, `HeckeLatticeModuleValidation`, [E, `chk:latcmp`] held); WP-LV's untwisted instance (`HeckeExt/LValue/`, `HeckeLValueValidation`); WP-NF (the seventh static target `NumberField/`, `NumberFieldValidation`); NB-2 (`--serve`, `ConsoleReadOnlyTagValidation`, `ConsoleServeValidation`; compiled only with `-DOC_BUILD_NOTEBOOK=ON`). **Not built**: WP-HL (the truncated Λ-adic arm; its research prototype, slot-19 R19-2, ran and is read in [U] 0.3.0-pre.3) and WP-LV's twisted instance ([E, `cl:lvtwist`], refused by the route). As the box below says, none of them realises a statement of this document; no statement, tag or count of this document changes.

CONTRACT. v1.19.0-pre.1 (hecke-extensions slot-19 WP-2). Slot 19's code is written against text outside this document, contract first (rule 21), and merges only after that text is at pre.1 (rule 22). **WP-LC** (the lattice comparison): $H^1(\Gamma; M_n \otimes \mathbb{Z}/\ell^M)_{\mathfrak{m}}$ as a module over $\mathbb{T}_{\mathfrak{m}}$ on the §30 route, gated by [E, `chk:latcmp`] at $(258, 37)$, $M \leq 4$; it uses Proposition 30.7 and Corollary 30.6 and proves nothing new here. **WP-LV** (the L-value route, [E, `sec:lvalue`]): the Eisenstein congruence module on the level-one model (its ordinary block, §26) and, twisted, of §29's character model at $N = \ell$; this document's part is the model and Corollary 30.6. **WP-HL** (the truncated Λ-adic arm, [U] `sec:u-lambda`): an arm of the Hida tower code of §26–§27, upper core, built only on the referee F1's ACCEPT of [U]'s part 3. **WP-NF** (the number-field core): a seventh static target `NumberField/`, STL only, guarded like `HeckeCore`; like the engine it realises *no* statement of this document. **NB-2** (`--serve`, `PrintJson`, the `read_only` verb tags): a transport of the console, specified by [P] v3.10 (slot-19 NB-1); it realises no statement of this document, and the console's values, transcripts and digests are invariants of it. Nothing of this box is built at this version.

CONTRACT. v1.14.0-pre.1 (hecke-extensions slot-14 WP-4). §28 is written *before* its implementation (rule 21), its predictions (LN1)–(LN6) before the code that tests them (rule 20): the composite-N table of Definition 28.1, the orbit-wise fixed spaces and the Heilbronn–Merel operator of Proposition 28.9 are slot-14 WP-12's, gated on the referee of the chapter (WP-5) and on the owner's [S14] Q4; the sparse elimination at large N is slot 15's. Nothing of §28 is built at this version.

CONTRACT. v1.15.0-pre.1 (hecke-extensions slot-15 WP-3). §29 (nebentypus: $\Gamma_1(N)$ and $(\Gamma_0(N), \chi)$ over $\mathbb{F}_q$ and $W_K(\mathbb{F}_q)$) is written before its implementation (rule 21), its predictions (NB1)–(NB6) before the code (rule 20); the code is slot 16's (owner ruling Q3). §30 turns the readings of slot-14's `HeckeExt/HeckeRingHeckeAlgebra` into statements; its code change (the certified verdict) is slot-15 WP-5's, gated on the referee. **At v1.16.0** §29 is built (slot-16 WP-10, WP-11; its code map).

CONTRACT. v1.15.0-pre.3 (hecke-extensions slot-15 WP-17). One sentence in §28's scope paragraph, proposed by slot-15 WP-8: a Γ-module map between induced modules induces the map on the cokernel presentations of Proposition 27.5(2), so the presentation-level fibre maps α^0, ω of `Hecke/HeckeLevelNSparse` rest on a statement of this document; the degeneracy maps α_d, $d > 1$, are asserted on the model only (Proposition 27.2(4)(c)). The code map of §28 records the sparse elimination (WP-8) and R3 at scale (WP-9) as built. No statement added or removed, no label renamed, no tag moved.

CONTRACT. v1.16.0-pre.1 (hecke-extensions slot-16 WP-12). §28.6 (the model-to-presentation map, α_d on the presentation, higher weight at composite N; (LN7)–(LN9)) and §31 (the ℓ-local isotropy model and the closed form at $\ell = 3$; (IL1)–(IL4)) are written before their implementation (rule 21), their predictions before the code (rule 20). The first is slot-16 WP-13's, gated on the referee (WP-14); the second is not scheduled. v1.15.0-pre.3's sentence below is superseded by Proposition 28.13(3) and Proposition 28.14. **At v1.16.0** (the release) both are built: the first by slot-16 WP-13, the second at $\ell = 3$ by the stretch package WP-16, with (IL3)'s level-N clause and the pair and cone half of Proposition 31.2(5) not run (the chapters' code maps).

CONTRACT. v1.17.0-pre.1 (hecke-extensions slot-17 WP-2). Contract first for slot 17 (rule 21), no statement of the three chapters changed: in §28 the table's reach beyond composite $N \approx 5.5 \cdot 10^4$ (WP-5); in §29 the sparse cokernel with a character and $T_{p^\nu}^{(N,\chi)}$, $\nu \geq 2$ (WP-6; the Galois ring at $d(\chi) \geq 2$ a stretch); in §31 (IL3) at level N, (IL4) over $\mathbb{Z}/3^e$, the pair and cone half of Proposition 31.2(5) and a synthetic refined descriptor (WP-7). None is built or run at this version; each chapter's code map says what the package must keep. **At v1.17.0-pre.3** (slot-17 WP-3 referee F1's m-W3): WP-5 (§28: the coset products in `int64` then `__int128`) and WP-6 (§29: the sparse cokernel with a character over $\mathbb{F}_q$; $T_{p^\nu}^{(N,\chi)}$ from Mer_{p^ν}, tested, not proved) are built, merged and audited, as their chapters' code maps record; still **not built**: the sparse route over $W_K(\mathbb{F}_q)$ and the Galois ring at $d(\chi) \geq 2$, the model-route double coset $[\Gamma_0(N)\,\mathrm{diag}(1, p^\nu)\,\Gamma_0(N)]_\chi$; WP-7's items as the code map of §31 says. **At v1.17.0-pre.5**: WP-7 (§31) is run, merged and audited (ACCEPT): (IL3) at level N over $\mathbb{F}_3$, (IL4) over $\mathbb{Z}/3^e$, the pair and cone, and the synthetic refined descriptor on cohomology; still **not built**: a strict lift on a refined descriptor and the Moerdijk–Pronk simplicial set itself.

CONTRACT. v1.18.0-pre.1 (hecke-extensions slot-18 WP-2). Contract first for slot 18 (rule 21), no statement of the three chapters changed, one checkable item added ((IL5), before the code, rule 20): in §28 the structure-aware fill model as a cost statement (Remark 28.16(4)), a value-preserving elimination order gated by bit-identity, the measurements at $N = 99330$ ($k = 2$) and 90090 ($k = 4$), and the wide-integer route for the $\Gamma_1(N)$ membership and the Heilbronn action with a character (WP-5); in §29 cohomology over $W_K(\mathbb{F}_q)$ at $d(\chi) \geq 2$ and the model-route $[\Gamma_0(N)\,\mathrm{diag}(1, p^2)\,\Gamma_0(N)]_\chi$ (WP-6); in §31 a strict lift on a synthetic refined descriptor ((IL5)) and (IL3) over $\mathbb{F}_9$, $\mathbb{Z}/9$ (WP-7). None is built or run at this version; each chapter's code map says what the package must keep. **At v1.18.0-pre.2** (the WP-3 repairs): WP-5, WP-6 and WP-7 are built and merged, as their chapters' code maps record — WP-5's generator order, `contains_wide` and `HeilbronnChi`'s wide route (its structure-aware fill model failed its rule, a finding; the two window measurements are **not run**); WP-6's cohomology over $W_K(\mathbb{F}_q)$ at $d(\chi) \geq 2$ and the model-route T_{p^2}; WP-7's strict lift on a synthetic refined descriptor ((IL5)) and (IL3) over $\mathbb{F}_9$, $\mathbb{Z}/9$. Still **not built**: a Moerdijk–Pronk $S(\mathcal{T})$ descriptor (Remark 17.4).

CONTRACT. v1.13.0-pre.1 (hecke-extensions slot-13 WP-2). §27 is written *before* its implementation (rule 21), its predictions (Hi7)–(Hi12) before the code that tests them (rule 20): the level-ℓ operator, the degeneracy maps and (Hi10) are slot-13 WP-6's, the lattice datum and (Hi7)–(Hi9) WP-5's, (Hi11), (Hi12) WP-10's (owner-gated). Nothing of §27 is built at this version. v1.13.0-pre.2 (WP-3's repairs) adds, also unbuilt: the variant (H) of Definition 27.13 on Hida's lattice (WP-5), and, not scheduled, the Hecke action on the cokernel presentation (Remark 27.7, (O2)). **At v1.13.0** (the release) what slot 13 built is recorded in §27's code map; still not built: the monomial and orbit-wise routes, the Heilbronn-matrix action and the composite-N tables ((O2), (O3)); (Hi12) not run. **At v1.14.0** (Hi12) has been run once on the first cluster and held (its [verified computationally][a] sentence); **at v1.14.1** a second cluster run added (Hi11) at $n = 4150$ and (Hi12) at $(4830, 2411)$, both held; the level-N interface is specified in §28.

CONTRACT. v1.12.0-pre.1 (hecke-extensions slot-12 WP-2). §26 is written *before* its implementation (the slot plan's rule 21, "contract first"), its predictions (Hi1)–(Hi6) before the code that tests them (rule 20): the torsion pre-pass and the projector exponent are slot-12 WP-4's, the towers and $(10, 2411)$ WP-5's. Nothing of §26 is built at this version; what is built will be recorded at v1.12.0.

CONTRACT. v1.11.1 (the slot-11 close, status). Unchanged in substance from v1.11.0: the WP-15 referee's repairs are a provenance sentence, a new [proved here] remark (Remark 25.4, which narrows 25.6 to its per-cell bound; the item stays open) and wording. One code change is recorded: `DicksonThetaCell` refuses $\ell \in \{2, 3\}$ as the precondition of Remark 24.12 and Lemma 16.4 (a `std::logic_error`, exit 3), as [E, `sec:thetaobj`] (h) specifies, where through v1.11.0 it threw `NotImplementedInThisBuild`. The counts are re-derived from `--list` at the close: 141 registered validations (55 `fast`, 74 `standard`, 12 `slow`), 82 `Hecke*`.

CONTRACT. v1.11.0 (hecke-extensions slot-11 WP-15). What slot 11 built, against the v1.11.0-pre.1 box below. **§24 is built and tested**: Q_n as a third fibre representation, the six terms and five maps of (67) with δ by the snake construction, the Hecke operators on all three terms (slot-11 WP-5, WP-6; the code map of Part VII); every checkable item of Remark 24.14 held (its last paragraph). **23.23's lifts are built** for $\mathrm{diag}(1,p^2)$, $\mathrm{diag}(1,p^3)$ and pI and its recursion checked on the free model (slot-11 WP-11; the note after its proof). **The ordinary projector** of 25.1 is built for U_ℓ on the free block of H^1 over $\mathbb{Z}/\ell^k$, as a certified Fitting idempotent, on the ℓ-adic digit route (slot-11 WP-18c, [E, `sec:ring`]); on $\mathbb{F}_\ell$ and on the word-sized composite route it is not built, and 25.6 stays open (its status note). **The digit route** (`ENGINE_API.md` §13.11, slot-11 WP-18a–c) is what Obligation 20.10(iv)'s scope sentence reads; nothing else of this document moves with it. Not built: θ_{L} over $\mathbb{Z}/\ell^e$, θ_{L} at the console, the Hida chapter (slot 12).

CONTRACT. v1.11.0-pre.1 (hecke-extensions slot-11 WP-2). Part VII is written *before* its implementation (the slot plan's rule 21, "contract first"): §24's Q_n, the six terms and five maps of (67) and the Hecke operators on them are slot-11 WP-5 and WP-6's; §25's projector has no implementation in slot 11. 23.23 needs lifts for $\mathrm{diag}(1,p^2)$ and pI, which are slot-11 WP-11's (M-3). What was built is recorded at v1.11.0.

CONTRACT. v1.10.0 (hecke-extensions slot-10 WP-15). What slot 10 built, against the v1.9.1 text below (§22.1). **Obligation 15.22(vii) over $\mathbb{Z}/\ell^e$, $\ell \geq 5$**: its first two sentences — the strict-pair route and route 3, each entrywise equal to the normative $A_c^{(q)}$ — are **discharged at $\ell^e < 2^{32}$, every E8 point** (slot-10 WP-3), the cross-routes refusing above 2^{32} by the `uint32_t` regime rule; its third sentence, the Poincaré-duality adjointness gate, is **recorded as unavailable over the ring**, the pairing and the contragredient refusing a composite ring. So over the ring Obligation 15.22 is partial in (vi) (the $\nu = 2$ recursion not computed) and in (vii)'s third sentence only. Phase E8 is complete: items 1–4b as specified, and (vii) over the ring as just stated ([E, `sec:ring`]). At $\ell \in \{2,3\}$ the Hecke operator, the reductions, the boundary and the cone over $\mathbb{Z}/\ell^e$ keep refusing, now for the reason of Remark 23.15. §23: $\tau_{\mathcal{I}}$ **and** $\Theta_{\mathcal{I},g}$ **are built** for T_2, T_3, T_5 (line 2's L2-1b, slot-10 WP-10; Remark 23.14), **the class** e **and** E_q (L2-1b, WP-13), and **the integral Hecke action read basis-free** at $n \in \{10,12,16\}$ (M1(a), slot-10 WP-11) — with $T_pT_{p'} = T_{p'}T_p$ tested on $H^{\leq 5}$ and the recursion of Theorem 10.8(1) **not** tested there, since it needs a lifted datum for a double coset of determinant p^2, which is not built. Not built: the descent assembly, the value $\mathbb{Z}$ of the ring, integer Hecke matrices in a basis (line 2's M1(b)), and §17's general free model. The other partial item is still Obligation 20.2(iii); Claim 16.26 is still tested and not proved. The registered-validation counts of §22.1 are the slot close's. **v1.9.0 (hecke-extensions slot-9 WP-15).** What slot 9 built, against the v1.8.1 text below (§22.1). **Phase E8 item 4b is built**: the boundary and the cone over $\mathbb{Z}/\ell^k$, $\ell \geq 5$ ([E, `sec:ring`] (f), slot-9 WP-4), so **the non-field half of Obligation 15.22(iv) is checked** — by solves at every spot of (39), with the witnesses re-derived — and its 18 objects per point are golden. Over the ring, Obligation 15.22 is **partial** in two clauses: **(vi)**, whose commutator is asserted and whose $\nu = 2$ recursion is not computed over the ring; and **(vii)**, by owner ruling 3 (iii) (A) of 2026-09-24 — **route 1 only**: the strict-pair route, route 3 and the

relative sub-complex constructor refuse the ring, each with an arm, and **(vii)'s composite half is open**. Phase E8 is therefore complete as specified: items 1–4b as specified, 4b over $\mathbb{Z}/\ell^k$ by route 1, and (vii)'s composite half open. At $\ell \in \{2,3\}$ the third complex over $\mathbb{Z}/\ell^k$ is built as the Bredon object it is (Definition 23.20(2), lengths and invariant factors only; slot-9 WP-3), and its K-mode, the Hecke operator, the reductions, the boundary and the cone keep refusing there. §23 is **partly built**: $\mathcal{I}_\bullet$, $\mathcal{I}_\bullet^\partial$ and cmp (line 2's phase L2-1a, slot-9 WP-10) and the integral groups for $n \le 20$, $q \le 5$ (line 2's milestone M0, slot-9 WP-11); $\tau_\mathcal{I}$ and the descent assembly are not built, and §17's general free model is still unbuilt. The other partial item is still Obligation 20.2(iii); Claim 16.26 is still tested and not proved. The registered-validation counts of §22.1 are the slot close's. **v1.9.1 (the slot-9 close, status).** Unchanged in substance from v1.9.0: the referee's repairs are wording, and the counts are re-derived from `--list` (125 registered, 71 `Hecke*`). Since hecke-extensions slot-9 WP-4t the point $(60, 5)$ of item 4b's ring arms runs in the `slow` tier (`HeckeReductionCoverageSlowValidation`); its golden objects stay in every run of the suite. **v1.8.1 (the slot-8 close).** Unchanged in substance from v1.8.0. Two chapters have no implementation of their free models: §17 and §23 ($\mathcal{I}_\bullet$, cmp, $\tau_\mathcal{I}$, the descent). What slot 8 built is the complex over the ring in force at $\mathbb{Z}/\ell^k$, $\ell \ge 5$ (Definition 23.1's third complex; phase E8 items 2, 3 and 4a), with its reductions and Bockstein, on the model complexes; it is refused at $\ell \in \{2,3\}$. **The boundary and the cone over $\mathbb{Z}/\ell^k$ are not built** (phase E8 item 4b, the next slot's), so the non-field half of Obligation 15.22(iv) stays unchecked. The only partial item is still Obligation 20.2(iii); Claim 16.26 is still tested and not proved. **v1.8.0 (slot-8 WP-9).** The chapter added in this version, §23, has no implementation beyond two validations that test statements of it (§22.1): the free model $\mathcal{I}_\bullet$, the comparison map cmp, the lifted datum $\tau_\mathcal{I}$ and the descent assembly are specified and not built. The *prerequisite* the v1.7.4 sentence below calls deferred — the concrete refinement — is re-scoped away: C-A5 is that chapter, on the unrefined tessellation (Remark 23.6). Everything else in the v1.7.4 text stands, and follows. **v1.7.4 (the slot-7 close).** Unchanged from v1.7.3 in substance: one chapter has no implementation (Chapter 17; its contract prerequisite, the concrete refinement, is deferred to slot 8), Chapter 15 is implemented in full, the only partial item is Obligation 20.2(iii), and Claim 16.26 is tested — since v1.7.4 on the whole case list at $\ell \ge 5$ — and not proved. The v1.7.3 text follows. **v1.7.3 (the slot-6 close).** One chapter has no implementation: Chapter 17. **Chapter 15 is implemented in full**: Obligation 15.22(vii)'s strict-pair route, route 3 and PD adjointness gate are evaluated since v1.7.3 (slot-6 WP-4, WP-5, WP-3; §22.1), the last being a *test* of Claim 16.26, which this document still does not prove; and §8.1 is implemented since v1.7.3 (slot-6 WP-1). The only partial item left is Obligation 20.2(iii), below. The rest of this box is the v1.7.2 text, kept as history. One chapter has no implementation and two have a partial one. Chapter 17 (the general orbifold model) is **unimplemented**: there is no free Γ-model anywhere in the tree. Chapter 15 (the pair homotopy and the cone) is **implemented except one clause since v1.5.2** (partial since v1.4.2, when only the certified η-table existed): the equivariant cone (`EquivariantMappingCone`), h, Φ_g and $A_c^{(q)}$ (`Hecke/HeckeConeAction`) exist, Obligation 15.21 is discharged in full and every clause of Obligation 15.22 but (vii) is evaluated (`HeckeConeValidation`, `HeckeConeActionValidation`; two of the 108 registered validations compute $H_c^\bullet$). What it still lacks is clause (vii)'s two cross-routes (the strict pair and route 3, phase E5's slot-6 arms), PD adjointness (which waits on Claim 16.26). **The cone's integration into the record landed in v1.7.2** (slot-5 WP-1, phase E2): `hecke_ext/v2`'s `pair` object carries the com-

plex, `les_maps` per reported spot and the two flags v1.6.2 reserved. **Chapter 14 is no longer among them**: since v1.3.3 the certified Γ-level τ_∂, the boundary coset table B_N/Γ_∞ and the operator Θ_g^∂ exist (`Hecke/HeckeBoundaryDatum`, `Hecke/HeckeBoundaryOperator`), and every clause of Obligation 14.19 is implemented and run, (i)–(iv) by `HeckeBoundaryTauValidation` and (v)–(vii) by `HeckeBoundaryOperatorValidation`; since v1.4.2 the parabolic cohomology $H^1_! = \ker r^1$, its Hecke action and Harder's criterion are computed as well (`HeckeExt/HeckeBoundary`, slot-2 WP-7). What the chapter's objects still lack is their integration into the report and the campaign record (phase E2), and what the *pair* lacks is §15's cone. **Chapter 9 is not among them either**: since v1.1.1 every clause of Obligation 9.10 is implemented and run, (i),(iv) by `HeckeIotaTauSymmetryValidation` and (ii),(iii),(v) by `HeckeIotaSignValidation`. What that chapter lacked up to v1.5.2 was a *production* ι^* — the `EquivariantCochainComplexMorphismOnK` its code map specifies; the two validations built their own. **It exists** (v1.6.2, the slot-4 close; built by hecke-extensions slot-4 WP-3 to exactly the specification of §9): `Hecke/HeckeInvolution` is that morphism and `HeckeExt/HeckeSignSplit` the sign split of clause (v), gated by `HeckeInvolutionValidation` (**standard**, 2954 checks), which compares it *entrywise* against `HeckeIotaSignValidation`'s independently built one on that validation's 77-cell grid and reproduces its reference line field for field. See Remark 9.8, §15.3 and §22.1's row. It has no open item: clause (v)'s sign question was resolved at v1.2.0 by owner decision OQ3 (Definition 9.1(iv) negated, the sign gating and holding). Obligation 20.2 is **partial**: clause (iii) is not reachable through the present interface, and that chapter says which. Their obligations are written as predicates on emitted data precisely so that they can be handed to the phase that builds them unchanged; they are a specification, and nothing in this section may be read as a claim that they hold.

22.3 The build and configuration surface

Six library/executable targets (`ModArithmetic`, `HeckeCore`, `MatrixEngine`, `EngineCuda`, `ConsoleCore` — the console's language core, STL-only, realising no statement of this document — and `cohomology_app`); the reference toolchain is recorded in `CLAUDE.md` and is not part of this contract. What *is* part of it:

- **The two rollbacks.** `cmake -DHECKE_PURE_INTEGER=ON` disables the certified floating-point filter of §19 and forces exact-integer predicates; `ENGINE_BACKEND=legacy` (runtime, no rebuild) sends every retrofitted path back to the pre-engine loop. **Both must produce bit-identical stored objects**, which is Obligation 20.9(iv) for the first and an engine invariant for the second; the first is a two-build *protocol*, not a predicate, and §20.3 records its executed run.
- **The two shadow builds.** `-DENGINE_SHADOW_LEGACY=ON` runs both arms of every retrofitted call and throws on the first differing entry; `-DENGINE_VERIFY_DEVICE=ON` re-executes every GPU node on the CPU inside a real run and compares every written window. Both are diagnostics (rule (A6)): a green shadow build is evidence, not a discharge.
- **The execution switches** — `ENGINE_THREADS`, `ENGINE_BATCH`, `ENGINE_DEVICE` (`cpu` is the default and the golden path), `ENGINE_ARENA`, `ENGINE_LAYOUT`, `ENGINE_CSE`, `ENGINE_FUSE`, `ENGINE_KSPLIT` and the CUDA and tracing families. None can change a value; that is asserted, not assumed (`EngineDeterminismValidation`: identical digests across thread counts, batching, arena, layout, CSE/fusion and device).

- **The four entry points.** `./cohomology_app <Name>` runs one registered validation (exit code = its own); `--all` runs the fast and standard tiers and `--all-full` adds the slow tier — **the Hecke acceptance gate needs `--all-full`**, because `HeckeBenchmarkSmoke` is slow-tier; `--bench <mode>` drives the shipped pipeline through the public API only; and `--campaign <spec>` enumerates an (n, ℓ, p, ν, q) grid and writes one record per job. All four use the same code path, which is what makes a validation's verdict and a campaign's record statements about the same object.
- **The certification level.** `--lean` and a spec's `certify = full | sampled(...) | none` reduce the *external* cross-checks only. The in-process certificates — $\partial\tau = \tau\partial$ in the τ-table constructor, the two-route entrywise comparison and the exact Θ_{eq} residual inside `compute()`, and the level-(c) certification — run on every job in every mode, `certify = none` included (`CampaignCertificateValidation`). They are Obligations 20.11, 20.5(i) and 20.1(iii), and they are not sampleable.

22.4 The record schema

An emitted record is what makes the obligations of §20 checkable *after* the run, by someone who did not make it. The shipped writer is `hecke::BuildHeckeReportJson` (`Hecke/HeckeReport.cpp`, schema string `hecke_report/v2` since v1.4.2 — hecke-extensions phase E0b, slot-2 package WP-2, the stage's one golden re-baseline; `v1` until then) wrapped, per job, by the campaign driver. Binding content:

1. **The parameters**: ℓ, n, p, ν, the model identifier, and the reduction regime actually used — in particular the τ-table's recorded solve regime ($\mathbb{Z}$ or mod ℓ; Obligation 20.11(vii), Remark 11.5) and the certified bound B of Lemma 18.5.
2. **The conventions, in band**: polynomial coefficients low-to-high degree, matrices as arrays of rows, entries canonical residues in $[0, \ell)$. A record that does not say this is not self-describing, and the two orders are exactly the pair a consumer gets wrong.
3. **The three levels**: (a) $\Theta^{(q)}_{\mathrm{flat}}$, optional and size-gated — it is $(N_q d_V)^2$; (b) $A^{(q)}(T_p)$ for every computed p and q; (c) the primary decomposition, with, per block, *either* a proved `residue_degree` *or* a `residue_degree_lower_bound` — never both — a machine-readable `residue_degree_certified` flag, and the certifying linear combination. This is rule (A6) in the schema: a lower bound may not be read as a degree.
4. **Which certificates ran**, and the level they ran at, with counts: the τ-certification verdict, the two-route entrywise comparison, the Θ_{eq} residual (and `"theta_eq_certified": false` when `--lean` turned it off), the decomposition certification, and the external cross-checks together with the sampling mode and stride. A record in which a certificate is absent and one in which it ran and passed must be distinguishable.
5. **The reported degrees** — and here the schema carries 14.24, decided 2026-09-16. **Per complex** (v1.4.0; owner confirmation 1 of the slot-2 plan, 2026-09-16; open point (9) of the v1.3.3 changelog): every complex a record is about — the bulk $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$, the boundary $C^\bullet_\partial$, the cone — carries its own four fields, adopted with the policy: `q_max` (an integer, the top degree of *that* complex), `truncated` (a boolean), `q_max_source` $\in$ {`"derived"`, `"set"`} — rider (i) of 14.24: derived on a finite model, and `q_max` may be set only where `truncated = true` — and `degrees`, the complex's own degree list $[0, \ldots, q_{\max}]$, derived. **Never a maximum over the complexes of a record.** A record about one complex (the per-degree `hecke_report` record, which is about the bulk) carries the four at record level; a record about several (the extension record of phase E2, about bulk, boundary and pair) carries them inside each complex's object. The *reported-degree selection* of a campaign job — rider (i)'s independent filter, the campaign's `point.degrees`, which may be a proper subset — is a job-level field distinct from any complex's `degrees` and keeps its meaning. A record's identity — its output path and its resume key — in-

cludes each complex's tuple and the reported selection, so that two jobs differing only in them are distinct jobs (the campaign job identity carries the reported degree list since 2026-09-16, `Job::id()`, pinned by `CampaignSmokeValidation`). **On the model complexes every field is a constant**: bulk 2 / `false` / `"derived"` / $[0,1,2]$, boundary 1 / `false` / `"derived"` / $[0,1]$ (the boundary object derives its own $q_{\max} = 1$; Proposition 14.4, code map of §14); on a truncated free model they are the bound of the box, and "$H^q = 0$ for $q > 2$" is not sayable. **Implemented at** `hecke_report/v2` (v1.4.2; hecke-extensions phase E0b, slot-2 package WP-2, 2026-09-16, commits `6ced10c` / `70c37de`; until v1.4.0 this instruction was addressed to E5, and v1.4.0–v1.4.1 recorded it as scheduled): the per-degree writer `Hecke/HeckeReport.{h,cpp}` moved from `hecke_report/v1` to `v2` and carries the four fields at record level beside its per-degree `degree`, *per complex* and derived — from `HeckeReportInput::num_degrees`, the degree count of the complex the record is about, which the writer now *requires* (a zero count throws), so that no caller can obtain the fields as a maximum over anything; the `"set"` value of `q_max_source` is reserved for the truncated free model of phase E9 and is produced by no shipped caller; the campaign writer `Campaign/CampaignDriver.cpp`, which embeds one such report per degree and emits the job-level `"degrees":[...]` selection, emits `q_max`, `truncated` and `q_max_source` beside it, un-hashed. The schema move was the stage's one golden re-baseline — exactly the 384 `report_canonical`^{q} / `report_full`^{q} lines of `Golden/GoldenReference.inc` predicted, 0 elsewhere, every changed key path allow-listed (`docs/matrix-engine/rebaseline-2026-09-16-hecke-report-v2.md`); the extension record of E2, about several complexes, is item 6 below. This is site (6) of §14.5, and it is the only place in this chapter that depends on the policy: changing 14.24 changes these fields and nothing else here.

Maps, and complexes built from complexes (v1.5.0; deferred item 5 of the v1.4.2 changelog entry — `docs/implementation/slot2/WP-8_report.md` finding 4 and `docs/implementation/slot3/WP-5_report.md` finding 2). The four fields are defined for a *complex*; two objects a record is about are not complexes, or not primitive ones, and the rule for them is stated here once rather than per object.

- **A map** — the restriction r (degree 0), the pair homotopy η (a chain map of degree $+1$ from the boundary chains to the bulk chains, Definition 15.5), the derived h (a cochain map of degree -1, (44)) — carries in its record the four fields of its **source** complex, and no others. **The degree shift is a constant of the schema**, fixed once per map kind by the definition that introduces the map; it is not a field, and a record may not vary it. So the η-record's fields are the boundary's — on the shipped model $q_{\max} = 1$, `truncated = false`, `"derived"`, `degrees` $[0,1]$ — with $\eta(q)$ landing in bulk degree $q{+}1$ (implemented that way at `hecke_pair_homotopy/v1`; §15's code map). On a truncated free model (phase E9) a map is built, and certified, in every source degree in which the box of 14.24 builds its source complex, i.e. $q \le q_{\max}(\text{source}) + 1$, and *reported* in the source degrees $q \le q_{\max}(\text{source})$; its target degrees are the shift of those, and the box's bound on the target complex must cover them — a condition on the pair of $q_{\max}$, not a further field.
- **A complex built from others** — the cone $C(r)$ — carries `q_max_source = "derived"` *unconditionally*, together with the names of the complexes it is derived from, and its $q_{\max}$ and `truncated` are functions of theirs: $q_{\max} = \max\big(q_{\max}(\text{bulk}),\, q_{\max}(\text{boundary}) + 1\big)$ with `truncated = false` when both inputs are finite and untruncated, and $q_{\max} = \min\big(q_{\max}(\text{bulk}),\, q_{\max}(\text{boundary}) + 1\big)$ with `truncated = true` otherwise. `"set"` is legal on a *source* complex only: a derived complex's $q_{\max}$ is never a user's to choose, which is rider (i) of 14.24 applied one level up. On the shipped model this gives the cone 2 / `false` / `"derived"` / $[0,1,2]$, from bulk 2 and boundary 1.

And the names are the record's own derived flag, so that the rule is checkable from the record (v1.7.0; `docs/implementation/slot5/WP-1_report.md` finding 3). The clause above has been a rule about *conforming records* since v1.5.0, but a verifier could not apply it, because nothing in a record said which of its complexes were derived: a hand-assembled record could declare `q_max_source = "set"` on the cone and be accepted. It is therefore binding that **the list of names is present and non-empty on a derived complex and absent or empty on a source complex** — it is not decoration beside `"derived"`, it is the predicate — and consequently that a record declaring `q_max_source = "set"` on a complex whose list of names is non-empty is **non-conforming**, and that a verifier which checks a record's internal consistency **refuses it**. This costs no field: v1.5.0 already required the names. What it buys is the property this section exists for, that an obligation be checkable "*after* the run, by someone who did not make it": the alternative reading — that the rule binds only the builder — leaves a conforming-looking record that claims a user chose a degree bound no user may choose, and leaves it undetectable by construction. The builder's own guard (a derived object asserting its fields as it emits them) stays; the two are complementary, and neither is the other's substitute. *Implementation note*: the shipped extension record already carries the names (`derived_from` on its `pair` object, empty elsewhere) and the cone builder already guards its own fields; what this rule adds is the record-level refusal, which `HeckeExtRecord::verify()` makes (code 22) and which `HeckeExtRecordValidation` *asserts*, with a positive control on a source complex, since hecke-extensions slot-5 WP-11a (v1.7.1; at v1.7.0 the refusal had not yet landed and that validation pinned the gap in its *negative* form).

6. **The extension record**, `hecke_ext/v1` (v1.6.0), `hecke_ext/v2` **since v1.7.0**. A second writer, *not* an extension of the per-degree one: one JSON per (n, ℓ) cell, about several complexes, whose field inventory is [E, §7] and whose rules binding on this document are the nine below — (a)–(f) since v1.6.0, (g)–(i) with `v2`. It is a *separate* writer by construction, because the per-degree writer's text is a golden object and a new key in it is a re-baseline (§E, (CL5)); the two share the conventions of item 2 and the degree rule of item 5 and nothing else.
   (a) **Settings are part of the record, and part of its hash input.** The record carries every mathematical setting in force — its *value* and its `source` $\in$ {`default`, `spec`, `console`} — together with the **fingerprint** of the *non-default* entries, which is 0 at the defaults. The fingerprint enters every hash the record reports and the identity of the job that wrote it, so that only the default configuration is golden and a golden mismatch can never be explained by a setting. This is the schema half of owner decision R1: the mathematical settings are the family of switches each of which may change a published value, as no `ENGINE_*` variable may. *How* a value was set is therefore itself a value-bearing field: a record whose `source` says `console` and one whose `source` says `default` are different records even at the same value.

   The block names what it carries, and the exclusion is by family (v1.7.0; the first half of the item v1.6.0's changelog deferred: whether the settings block should name the engine settings it excludes. *It should not*, and the reason is a rule rather than a taste). A setting belongs in the block iff it may change a published value — that is the definition of the family, owner decision R1 — and the `ENGINE_*` variables are excluded *as a family*, by the rule and not by an enumeration. Enumerating them would make the settings block a function of the build: a record would acquire a key whenever the engine gained a knob, two builds at identical mathematical settings would emit different blocks, and a reader could not tell a new engine knob from a

new mathematical one. One sentence naming the criterion and the excluded family is the whole content; the list of mathematical keys in force is [E, §7]'s, and grows with its consumers.

And a resumed run writes the settings block a *fresh* run at the same effective values would write (v1.6.0; slot-4 WP-2 finding 4). A resume is the second place a value enters the block, and the rule is not implied by the first: an implementation that installs the budget it was resumed *to* makes a resume to the default budget produce a different settings text, and therefore a different fingerprint and a different hash, from a fresh run at the default — although the two runs computed the same thing. The binding rule is the equality: *a resume to a value equal to the default installs the default's canonical text*, any other value installs its own, and "the resumed record equals a fresh default run byte for byte, fingerprint included" is then a statement an implementation can and must assert. Without it two conforming implementations differ in a published hash.

(b) **A budgeted item carries a `resumable` object at its own site**, with its kind, whether it is `complete`, the budget, the point reached and the budget history; the record carries the count of incomplete items. Inline while the state is small; otherwise a **sidecar** file named by the record layer — the only layer that knows the record's stem — and referenced by **a path relative to the directory holding the record** (in practice its basename, so that record and sidecar move together) together with its FNV-1a-64 from the one byte-wise FNV of `ModArithmetic/Fnv.h`. **The name must be injective**: a record may carry several oversize states of the *same* kind — one per ideal — so the first of a kind may take the bare `<stem>.state.<kind>.json` and every further one is disambiguated by an index. A writer with no stem, which cannot name a file, inlines the state and says so with a flag rather than emitting a path that names nothing. **The record's count of incomplete items counts record *entries***, not distinct decisions (v1.6.0; slot-4 WP-2 finding 5): where a schema attaches a verdict that is a property of the whole run — a memory-budget refusal, say — to each of several sub-objects, one refusal is reported once per sub-object and the count rises with them. That is a legal schema, and it is what the shipped extension record does; what is *not* legal is leaving it unsaid, because a reader who takes the count for a number of decisions will be wrong. A schema that carries a run-level verdict per sub-object states that it does.

And in the extension family, since `hecke_ext/v2`, it may not do it (v1.7.0; owner ruling `[OD]` slot-4 close-out 5 of 2026-09-22). A verdict that is a property of the *whole run* — the memory-budget refusal of U_ℓ is the instance — is carried **at record level, once, with a single `resumable`**, and the record's count of incomplete items is then a count of **decisions**, entries and decisions coinciding. The paragraph above stands as the general rule for a schema that does replicate such a verdict, and it is why this repair is a *version bump* and not an amendment in place: the value of an existing key changes, so a reader must be able to tell the two shapes apart by the schema string alone ((CL5)'s discipline, as `hecke_report/v1` → `v2` set it in slot 2). The bump is made **before the first emission** of an extension record as a golden object, which is what makes it free.

(c) **An unavailable complex is a placeholder, never a silent omission.** A complex a record is about but has not computed is emitted as {`"status": "unavailable"`} and nothing else — distinguishable, as item 4 requires of a certificate, from a complex that was computed. The four degree fields of item 5 are emitted *only* for an available complex, because they would otherwise be a claim about a complex that does not exist in this record.

One level down, the same rule for a scalar and for a flag (v1.7.0;

`docs/implementation/slot5/WP-1_report.md` findings 2 and 4). A key a record's schema requires it to carry but which *this* record has not computed is emitted as `null` — never as a default value, and never dropped. Dropping it makes a closed key list a function of the cell, which destroys the property that made the list a rule; emitting a default conflates two different facts, and item 4's "absent versus ran and passed must be distinguishable" is the same requirement one level up. So a dimension not defined until a complex this record does not carry is `null` and not 0; and a *diagnostic* this record never evaluated is `null` and not `false` — under rule (A6) a `false` on a diagnostic means "evaluated, and it did not hold", which is conformance, so a record writing `false` for "nobody asked" has destroyed the distinction the diagnostic exists to carry.

(d) **Reasons and phases are named, not numbered ad hoc.** A verdict that is not final carries a reason from the fixed list of [E, §7]; the entry that names a failed certificate is written as `certificate_failed:<code or certificate name>`, where a *code* is a non-zero clause index of the `verify()` that failed and a *name* is one of the closed list of certificate names of that section — a certificate route may report "the sweep ran and its answer was no" without any `verify()` having failed, and the vocabulary has to be able to say so. *Where the list lives* (v1.7.0; the second half of the item v1.6.0's changelog deferred). The closed list of certificate names is [E, §7]'s, with the rest of the field inventory, and is not restated here: this document binds that the list is **closed**, that the two payload forms — a `verify()` clause index and a certificate name — are distinguishable by a reader who has only the record, and that a payload outside the list makes the record non-conforming. A name added to the list is an [E] revision and not a [C] one. Where a routine reports the phase it reached, the record carries the **per-item** value and the run's **high-water mark** under separate keys: the two are different facts, and a phase belonging to a block that a split has since removed is the run's, not that block's.

(e) **The two recorded cone flags.** `phi_flat_identity` and `residual_flat_zero` are the names under which a record reports the flat readings of Obligation 15.22(i), inside the object of the complex they are about. They are diagnostics under rule (A6): recorded, never branched on, and a `false` is conformance. The names are reserved here so that the value they will carry has one spelling from the start.

(f) **What this document binds and what it does not.** The field inventory, the example and the per-block content of the extension record are [E, §7]'s; this item binds only (a)–(e) and (g)–(i), which are schema rules, and item 2's conventions and item 5's degree rule, which apply to every record of the programme.

(g) **A record about several complexes reports the maps between them** (v1.7.0, with `hecke_ext/v2`; the objects are [E, §7]'s). Where a record carries two complexes joined by a long exact sequence — the pair of §15 is the instance — the object of the *derived* complex carries one entry per spot of that sequence, and each entry says, per spot, what Obligation 15.22(iv) makes of it: the ranks, whether the composite was found zero, and whether exactness was *asserted* there. Item 5's reported-degree rule governs which spots exist, and rule (c)'s `null` governs the rest: a spot whose composite was not built, and a spot at which the sequence's exactness was not asserted, carry `null` and not `false`, because clause (iv) asserts exactness only where all three terms are reported and a record may not report a verdict the obligation did not reach. A Hecke-equivariance flag on such a sequence is the *action*'s claim and not the complex's: a record carrying the cone but no cone action has not evaluated it, so it is `null`.

(h) **A $\pm$-split record is two further records of the same cell** (v1.7.0, with `hecke_ext/v2`). Where an implementation carries the involution's splitting of

Proposition 9.6(1), each record — the unsplit one and the two per-sign ones — names its sign, every per-block object inside a record carries the *same* sign as that record, and the two per-sign records' dimensions sum, degree by degree, to the unsplit one's. The sum is **asserted** and not recorded: it is Proposition 9.6(1) itself, and a record family that fails it is not a report about one cell. At $2 \notin R^\times$ there is no split and the sign field says so, which is item 4's "absent and ran" distinction again (Obligation 9.10(v)).

(i) **A matching a record performed is a matching it reports** (v1.7.0, with `hecke_ext/v2`). Where a record's objects are indexed by one thing but identified across another — the extension record's maximal ideals, which a run matches across the ambient groups it computed — the record carries, per object, the list of the ambient objects it was matched *against*, the verdict per entry, and the **witness** the test used, including the identity entry for the object's own ambient group, which is what makes two such lists comparable. A *count* of matches is not a report of the matching: it cannot be re-checked by a reader who has only the record, which is what item 4 requires of a certificate and this section of everything. A non-match carries the reason it failed, by the same rule as (d).

Implemented at `hecke_ext/v1` (hecke-extensions slot 4, package WP-1, phase E2a, 2026-09-21; `docs/implementation/slot4/WP-1_report.md`): `HeckeExt/HeckeExtRecord.h` and `HeckeExt/HeckeExtReport`, with `HeckeExtRecordValidation`; `Hecke/HeckeReport.{h,cpp}` is untouched. **At** `hecke_ext/v2` **since v1.7.0** (slot 5, package WP-1, phase E2 *cont.*; `docs/implementation/slot5/WP-1_report.md`), in the same two files and with `HeckeExt/HeckeExtComplexRecords` the one translation layer between the computed objects and the record: (b)'s record-level verdict with its single `resumable`, (g)'s `boundary` and `pair` objects — computed, not (c)'s placeholder, so (e)'s two flags are emitted — (h)'s per-sign records and (i)'s matching list. `Hecke/HeckeReport.{h,cpp}` is still untouched, which is what "a separate writer" means operationally. *Status at v1.6.2 (the slot-4 close)*: the record **is written**, for the **bulk** complex, end to end from a cell — slot-4 WP-2 added `HeckeExt/HeckeExtDriver`, the Stage 1–5 orchestration behind it, gated by `HeckeExtRamanujanValidation` on [E, Claim 8.1] at $(10, 691)$. The boundary, cone and per-sign objects, and the campaign's own `ext` columns, are slot 5's. *Status at v1.7.0*: the boundary, cone and per-sign objects are written (slot-5 WP-1, above); the campaign's `ext` columns and the *emission* of an extension record as a golden object are the slot's other packages, and the schema was bumped before that emission for exactly the reason (b) gives. The fuller status sentences are the slot close's. *Status at v1.7.4 (the slot-7 close)*: **the record is** `hecke_ext/v3` (slot-7 WP-3, stage 3), bumped because the default record gained nine keys, all of them [E]'s field inventory under (f) — none of the nine rules above moves. Rule (a) holds as written: the record carries every setting with its value and `source`, and the fingerprint enters every hash the record reports; the golden reference hashes the record text with the `settings` block reduced to that fingerprint (slot-7 stage 1), so that registering a key at its default moves no golden line. The first ext campaign publishes 505 `v3` records.

7. **The integral descent record**, `integral_descent_record/v1` (v1.23.0-pre.37; plan milestone D6 of `docs/implementation/slot24/DESCENT_plan.md` v0.4, the carried L2-4; [IB] OD-9 (a), OD-10 (a) ([X] S8-6d; v1.23.0-pre.38, referee T2); `docs/implementation/slot24/DESCENT6_report.md`). The record of Theorem 31.16 for one cell (N, n) and a finite set $\mathcal{P}$ of primes $p \nmid N$ (default: the first three of $2, 3, 5, 7, 11, 13$ not dividing N), written under the setting `coefficient_ring = Z` and under no other. Binding content:

   (a) **The ring and the object.** `coefficient_ring = "Z"` and `object = "group"`: the object is $H^q(\Gamma_0(N); V)$, group cohomology, Definition 23.20(3) — never the

Bredon object, never $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$. The setting selects the record and is never implied by it: a builder asked for the record under $\mathbb{F}_\ell$ or $\mathbb{Z}/\ell^e$ refuses (the console verb with a runtime error), and the descent's own tools, which build $\mathbb{F}_{\ell'}$- and $\mathbb{Z}/\ell^K$-objects of their own, run with the ring reset to $\mathbb{F}_\ell$ for the computation only. $\mathbb{Z}$ **is un-reserved for the descent only**: every other consumer of the ring — `EquivariantCohomologyOnK`, `HeckeOperatorOnK`, the console verbs `symmetric_power`, `cohomology`, `hecke`, `iota`, the field job loop of the campaign — **refuses $\mathbb{Z}$ by name** (`hecke::NotImplementedInThisBuild` naming the site; console exit 5), by two mechanisms (v1.23.0-pre.38, referee m1): (i) every construction site reads the exponent through the one reader `CoefficientRingK(who)`, which refuses $\mathbb{Z}$ naming `who`; (ii) the console verbs that compute over $\mathbb{F}_\ell$ without reading the ring are wrapped at registration by `console::bind::GateVerbsOnField`, an *opt-in list* of verb names (`Console/Bindings/RegisterPipelineVerbs.cpp`, `HeckeExt/RegisterHeckeExtensionVerbs.cpp`), each wrapper calling the same reader; the list is kept by `IntegralDescentRecordValidation` arm (F), which probes a fixed list of calls. **Residual risk**: a future $\mathbb{F}_\ell$-only verb that neither reads the ring nor is added to the list would run under $\mathbb{Z}$ as over $\mathbb{F}_\ell$; a registration-time default (refuse unless a verb is tagged ring-aware) would close it and is a code follow-up, not a clause of this version. The general `Z/<m>` stays reserved and refused when set ([E, `sec:ring`] (a)).

(b) **The degrees.** `q_max = 3`, `truncated = true`, `q_max_source = "derived"`, `degrees` $[0,1,2,3]$, `periodic_from = 2` — item 5's four fields at record level, which report the group cohomology of (1)–(3), the bulk, an infinite complex. The bulk occurs a second time, as the `bulk` column of the pair complex of (f), with that object's own four fields ($q_{\max} = 4$, no `periodic_from`): the two are the same complex reported to two ranges for two purposes, and they are compared in degrees ≤ 3 (the `ENGINE_CHECK` of (e)); a verifier applying item 5 reads each object's fields for that object (v1.23.0-pre.38, referee m2). $q_{\max} = 3$ is *derived*, not set: by Theorem 31.16(6) the data of (1)–(4) in degree $q+2$ are those of degree q for every $q \geq 2$ (cup product with $\mathrm{res}^\Gamma_A e$, Proposition 31.7(5)), so degrees $0, 1$ and one period $2, 3$ determine every degree, and 3 is the least $q_{\max}$ that does — degrees 0 and 1 are not periodic (H^0 and the free part of H^1 have no counterpart in degrees 2, 3), and below 3 the period is incomplete. `truncated = true` because the complex is not bounded; `periodic_from` is what licenses a reader to continue the list, and nothing else does: in particular no vanishing above $q_{\max}$ is reported or implied.

(c) **The groups and the bad primes.** `groups`: $H^0, \ldots, H^3$ as abelian groups, Theorem 31.16(1); `bad_primes`: $\mathcal{B}(M_N) \cap [5, \infty)$ with its counts, Proposition 31.8(1); `torsion_primes`: the primes whose torsion is read, $\{2\} \cup \{3 : n \text{ even}\} \cup \mathcal{B}(M_N)$. K **per prime**: for each torsion prime the route of the groups' run (`tree` or `dense` at 2, 3; `shortcut` at 3 with $\nu_3(N) = 0$; `sparse` or `bredon` at $\ell \geq 5$), the certified precision K, the widest ring width W read ($W > K$ when a 2-adic differential escalated) and the certificate's verdict. K and the route are *provenance*: recorded, not hashed (item (h)).

(d) **The characteristic polynomials.** Per $p \in \mathcal{P}$, $\chi^{(N)}_{p,1} = \det(x - T_p \mid H^1_{\mathrm{fr}}(N))$ by Proposition 31.14's output rule (5), Theorem 31.16(2): its degree $r = b_1$, its coefficients (an integer; beyond 2^{63} a sign and a digit count, **and since v1.24.0-pre.1 the exact value** in the entry's list `big`, item (l); `?` when not certified), `determined`, the number undetermined, and **the digest** $\mathrm{char}_{\mathbb{Z}}$ (the JSON key `digest`; the golden object `charZ_T<p>`; v1.23.0-pre.38, referee T1). **Coefficient order** (v1.23.0-pre.38, referee M1): the list `coefficients` is $c_0, c_1, \ldots, c_r$ of Proposition 31.14(1), c_i the coefficient of x^{r-i} — **from the leading coefficient down**, $c_0 = 1$ first, the

constant term last (`CrtCoefficient::i`, `Integral/Descent/DescentHecke.h`; `CharpolyOverZ`, `DescentHeckeCrt.cpp`); at $(11, 0)$, T_2 reads `1,1,-8,-12`, i.e. $x^3 + x^2 - 8x - 12 = (x + 2)^2(x - 3)$. This is an **explicit exception to item 2**'s low-to-high rule, which governs `hecke_report` and the console's `charpoly` objects (constant term first) and not this record. `v1` carries no in-band key for the order; one outside the hashed body (e.g. `"coefficient_order":"high_to_low"`) is an open code item, and until it exists this clause is the convention's only statement. The digest is recomputable from this text (v1.23.0-pre.38, referee m7; `CharpolyZ::digest()`, `DescentHeckeCrt.cpp`, is the implementation): FNV-1a-64 (`ModArithmetic/Fnv.h`, offset basis `0xcbf29ce484222325`) fed a sequence of **unsigned 64-bit words**, each as its 8 bytes least significant first: r; then, per $i = 0, \dots, r$ in this order, i, followed — for a certified coefficient — by the word $1 + [c_i < 0]$ (so 1 or 2), $c_i \bmod 2^{64}$ and $c_i \bmod 3^{40}$ (canonical residues in $[0, 2^{64})$, $[0, 3^{40})$), and — for one not certified — by the word 0, the number of residue primes, and the residues in the order of the prime list. On a determined polynomial the digest is a function of the integers c_i alone, hence of (N, n, p), and **independent of every choice** (Theorem 31.16); it is a fingerprint, not a certificate: $c_i \bmod 2^{64}3^{40}$ with the sign does not recover a coefficient beyond about 2^{126}, and a reader who needs the integers recomputes them. On an undetermined coefficient the digest identifies the residue data of the build's fixed rule for the residue primes (descending from 2^{30}), not an integer; two such digests are comparable under that rule only. That case is **not reachable from the record** (v1.23.0-pre.38, referee N1): the record's options carry no prime limit, D3's `CrtOptions::max_primes = 0` takes as many residue primes as $b_{p,r} + 1$ needs, so every coefficient of an `ok` entry is certified and `determined = true`; it is reachable only through D3's `CrtOptions` directly. $\chi^{(N)}_{p,q}$ for $q \neq 1$ is not recorded: for $q \geq 2$ because $H^q_{\mathrm{fr}} = 0$; for $q = 0$ *by choice* (H^0_{fr} is $\mathbb{Z}$ at $n = 0$ with $T_p = p + 1$, and 0 otherwise — nothing a record needs to carry; v1.23.0-pre.38, referee N3).

(e) **The torsion Hecke modules.** Per (q, ℓ) with ℓ in (c)'s `torsion_primes` (v1.23.0-pre.38, referee m3: for n odd that is $\{2\}$, so the record has the three entries $(q, 2)$ and no entry at 3 or at $\ell \geq 5$, $-I$ acting by -1 and the cohomology being 2-primary; an entry absent for that reason is not a zero group in the sense of (g)(iii)) — $1 \leq q \leq 3$ at $\ell \in \{2, 3\}$, $q = 1$ at $\ell \in \mathcal{B}(M_N)$ (no other (q, ℓ) carries torsion, Theorem 31.16(1); H^0 is torsion-free) — the $\mathbb{Z}[x_p : p \in \mathcal{P}]$-module $(\mathrm{tors}\, H^q(A; V)_{(\ell)}, (T_p))$ of Theorem 31.16(3) by its choice-free invariants (Proposition 31.15(1)): the exponents, `module_certified` ($2\sigma_{\max} \leq K$), and per p one canonical text — the type, the layer polynomials P_j, the layer blocks' polynomials modulo ℓ^{e_j}, the Fitting types, and, when the module is certified, the types of $\ker(T_p - c)$, $\mathrm{coker}(T_p - c)$ — in one system of generators, with `commute`. The ker / coker types are read, and the canonical text carries its `ker=` / `cok=` parts, exactly when (v1.23.0-pre.38, referee m4) the module is certified, **and** the number r of cyclic summands is at most 400 (`max_full_summands`), **and** $K_{\max}(\ell) \geq a_{\max} + 2$, where $K_{\max}(\ell)$ is the largest K with $\ell^K < 2^{31}$ (30 at $\ell = 2$) and $a_{\max}$ the largest exponent (`Invariants`, `Integral/Descent/DescentHeckeTorsion.cpp`); otherwise the text omits them, the entry stays `module_certified = true` and unrefused, and no flag says so. The read's working width is a ladder (v1.23.0-pre.39, item (j)): $k_0 = \min(K_{\max}(\ell), 2a_{\max} + 3)$, then $K_{\max}(\ell)$ when the read falls short at k_0 by a precision shortfall; the width is not in the text and is not one of these caps. These caps enter the hashed canonical text, so they are record content: changing one is a golden event (h)(1), not a rollback. A hashed flag for the omission would be a schema revision (`v2`). Read on the tree model at 2, 3 (every N) and on the cokernel presentation at $\ell \geq 5$, as Theorem 31.16(3)

prescribes. `module_certified = false` is *not* a refusal: it is the narrower certified content of Theorem 31.16(3) (the layer polynomials and the Fitting types at every certified K), and the canonical text then omits what needs $2\sigma_{\max} \le K$. A certified read whose exponents differ from the groups' ℓ-primary part, or a pair whose bulk differs from the groups in a degree ≤ 3, is a **defect** (`ENGINE_CHECK`), never a refusal and never a record.

(f) **The boundary pair.** The pair columns that Theorem 31.16(7)(c) assigns to D6: H^q, $H^q(\partial)$, H^q_c of §31.4 (the tree route of §31.5 by default) for $0 \le q \le 4$, **each complex with its own four fields** (item 5: never a maximum over the complexes of a record) — `q_max = 4`, `truncated = true`, `"derived"`, $[0, \dots, 4]$ — and the primes S and the 2-adic route as provenance. 4 is the least top degree the pair builds admit (§31.4's code map; Proposition 31.25(6) refuses $q_{\max} \notin \{4, 5\}$), and it carries the segment $H^3_c \to H^3 \to H^3(\partial) \to H^4_c$ of (79), which closes the bulk's reported range $q \le 3$. No periodicity of the pair columns is asserted here — Theorem 31.16(6) is a statement about the bulk — so nothing above 4 is reported or implied. The compact-support complex is the cone, a *derived* complex: by item 5 its object must carry the non-empty list of the complexes it is derived from (bulk, boundary), and its fields are those of item 5's rule, $q_{\max} = \min(4, 4+1) = 4$, `truncated = true`. **Met since** `19a969de` (v1.23.0-pre.38, referee M2): the `compact` object carries `"derived_from":["bulk","boundary"]` (`kCompactDerivedFrom`, `Integral/Descent/DescentRecord.h`), asserted by arm (J); the list is outside the hashed body, so it moved no hash and no golden line. (v1.23.0-pre.37: "not met by `v1`", the record as built on `538cf881`.)

(g) **Refusals.** `status` $\in$ {`"ok"`, `"refused"`} with `refusal`, the first refused part's line; each χ entry, each torsion entry and the pair carry their own `refused` and `refusal` (v1.23.0-pre.38, referee m5, in place of "every part"): the groups carry none of their own — a refused groups assembly shows as an empty `groups` list together with the record-level `status` and `refusal` — and the per-prime K entries carry `certified` only. A part's `refusal` may be **non-empty on a part that is not refused**: a `[ModulePrecision]` torsion entry is `"refused":false` (by (e)) with `refusal` naming the narrowed certificate; a reader keys on `refused`, never on a non-empty `refusal`. A refusal is a legitimate limit of the build (a cap, a precision, a degree range, rule 19's RSS ceiling), named by its code. **A refused record is never a valid empty answer**, by three rules: (i) a refused groups assembly reports **no** groups — the list is empty, never partial — whereas an `ok` record reports exactly $q_{\max}+1$ groups, the zero group included ("0", not an absent entry), so an empty list occurs only together with `status = "refused"`; (ii) likewise for the pair's three lists ($q_{\max}+1 = 5$ entries each, or none and `refused`); (iii) a torsion entry with no exponents is the zero group only when it is not `refused`. A reader reads `status` and the part's flag first; `status = "refused"` with parts present means that the parts present are certified and the refused part is absent, as at $(1, 200)$ below. A defect is never a refusal: it throws (`hecke::HeckeInternalError`, `ENGINE_CHECK`) and no record is written. A record of this schema is built with every part (v1.23.0-pre.38, referee N2): the builder's internal options `RecordOptions{hecke = false}` and `{pair = false}` give an `ok` record without χ and torsion, or with `"pair":{"ran":false}`, which is **not** a conforming `v1` record; no caller uses them (dropping them, or recording their state, is a code follow-up).

(h) **The hash: what is hashed, what is not.** `record_fnv1a64` is the one FNV-1a-64 (`ModArithmetic/Fnv.h`) of a fixed-order text, `DescentRecordZ::body()` (`Integral/Descent/DescentRecord.cpp`), **normative for the order**; the partition is exhaustive (v1.23.0-pre.38, referee m6). **Hashed**: the schema, $(N, n, \mathcal{P})$ with

$\mathcal{P}$ ascending, ring and object, (b)'s fields, the groups, the bad primes with their counts and the torsion primes, per p the degree r, `determined`, the digest of (d) and the `refused` marker, (since v1.24.0-pre.1) the exact value of every coefficient beyond 2^{63} (item (l)), per (q, ℓ) the exponents, `module_certified`, the canonical texts of (e) and the `refused` marker, the pair's $q_{\max}$ with its `refused` marker and its groups, `status`, and the settings' canonical text with their fingerprint. **Not hashed**: K, W, the per-prime certificate verdicts and every route (of the groups, of the torsion reads, of the pair), the pair's prime set S, the coefficient strings of (d) and `undetermined`, `commute`, the pair objects' four fields and `derived_from`, the settings' JSON (values with their sources), the refusal texts, timings and RSS. `commute` is record content (e) without being hashed: since `19a969de` the golden `DESCENT` points `ENGINE_CHECK` it (with `module_certified`), and the hash does not see it. The reason is that every hashed field is an *invariant*: by Theorem 31.16 the data of (1)–(3) are independent of every choice made in computing them, models, comparisons, Smith transforms and primes included, and by Remark 31.17(i), (ii) the torsion data are certified exactly, so a record's invariants have one value per cell and setting. The routes and the precisions are such choices, and the descent's rollbacks — `HECKE_DESCENT_TREE=0` (§31.2's code map) and `HECKE_DESCENT_TREE_PAIR=0` (§31.5's code map, obligation (c)) — are **read-only and value-preserving**, the preserved values being exactly "the record's invariants (the group types and the choice-free Hecke invariants), not Smith generators or coordinates". A field that a rollback may change is therefore a field the hash must not contain, and conversely: with K and the routes outside the body, **a rollback cannot change the hash of a record that is `ok` on both routes**. Two qualifications, both by design. (1) The certification flags (`module_certified`, `determined`) are hashed; they are route-independent because the torsion read is on the tree at $\ell \in \{2, 3\}$ and on the cokernel at $\ell \geq 5$ under every rollback (Theorem 31.16(3)), at the build's fixed read width $\ell^K < 2^{31}$, and χ is read on the O-piece alone (Remark 31.17(iv)); changing a fixed width, a cap or a canonical text is a change of the record's content, gated as a golden event, not a rollback. (2) A refusal is not a value: a cap reached on one route and not on the other gives different `status`, hence different hashes, and that is correct. A non-default mathematical setting changes the settings text, hence the hash and the campaign job id (`locality = strict` changes `record_fnv1a64` at $(1, 12)$ from `0x6c841fb033fbd3d1` to `0x0be335f8f71d98cc`, the invariants unchanged; these two values are **illustrative**, as printed by arm (J) at `724b9aaa` (`DESCENT6_report.md` §4), which asserts only that they differ; no validation pins them, and pinning them in (J) is a code follow-up; v1.23.0-pre.38, referee m8).

(i) **What is not in the record.** Theorem 31.16(4) (the invariant factors of $T_p - c$ on the free quotient $H^1_{\mathrm{fr}}(N) \otimes \mathbb{Z}_{(\ell)}$) and (5) (level pairs $\Gamma_0(MN) \subset \Gamma_0(N)$ with T_p) are **not recorded and not built** in the record — (4) exists at $\ell \geq 5$ as D3's `FreeQuotientFactors` and is not called, (5) is read by no built code; (7)'s non-assertions stay non-assertions, and integer Hecke matrices in a basis are not named by the ring (Definition 23.20(3)).

(j) **The refusal at $(1, 200)$, $\ell = 2$, and its name. Up to v1.23.0-pre.38** (this paragraph is that version's text, kept as history; the paragraph "The width ladder" below is the current statement) at $(N, n) = (1, 200)$ the record was `refused` at $(q, \ell) = (1, 2)$ with `REFUSED [KerCokerPrecision] ker / coker of phi - 0: kernel: I in L not certified (30 < 34)` (since `19a969de`, `Integral/Descent/DescentHeckeTorsion.cpp`; v1.23.0-pre.37 quoted the code name of `538cf881`, `[NotEndomorphism]`), the groups, χ and the other parts present. What fails is the read of the types of $\ker(T_p - c)$, $\operatorname{coker}(T_p - c)$ (Propo-

sition 31.15(1)(c)) on the certified module: the code computes them on the presentation $\bigoplus_i \mathbb{Z}/2^{a_i}$ over $\mathbb{Z}/2^k$, $k = \min(30, 2a_{\max} + 3)$ (pre.37 and pre.38 called this "the ring width capped at 2^{30}"; at $(1, 200)$ it is $k = 11$, see below), and at $(1, 200)$ the Smith form of the image lattice certifies 30 of the 34 pivots its rational rank requires — $30 < 34$ counts pivots, not bits. It is a **precision shortfall of the build**, not a statement about T_p: by Theorem 31.16(3) T_p is an endomorphism of the certified module, which is what `NotEndomorphism` would deny. **The taxonomy of `538cf881` misnamed it**, and v1.23.0-pre.37 fixed the name: a rank shortfall or an exhausted precision inside the ker / coker read is `[KerCokerPrecision]`, a precision refusal of the family of `[ModulePrecision]` (the module and the other data certified, the ker / coker types not read at this width); `[NotEndomorphism]` stays reserved for a failed endomorphism safeguard (a relation not carried into the relations, a non-zero free coordinate, an image off the lattice), which by Proposition 28.9(1) and Theorem 31.16(3) signals a defect of the code rather than a limit. The code emits `[KerCokerPrecision]` since `19a969de` (`D3Refusal::KerCokerPrecision`, `Integral/Descent/DescentHecke.h`). (v1.23.0-pre.37's reading rule for a record of `538cf881` — its `NotEndomorphism` with "not certified" or "precision exhausted" read as `KerCokerPrecision` — is vestigial: no merged build writes such a record.) The renaming moves no hash (refusal texts are unhashed) and no golden line (no golden point refuses). Keeping the module's certified parts at such an entry — as `[ModulePrecision]` does — would change the hashed body and is a schema revision (`v2`), not part of the fix. (Pre.38 ended: "a wider ring at 2 for this read ($W = 61$, as the groups' escalation) would lift the cell"; the ladder below lifts it at $k = 30$, without $W = 61$.) **The width ladder** (v1.23.0-pre.39, ICENSUS-P1; `docs/implementation/slot24/ICP1_acceptance.md`). The width of the paragraph above is $k_0 = \min(30, 2a_{\max} + 3)$, at $(1, 200)$ $k_0 = 11$ ($a_{\max} = 4$), not the word bound. Since v1.23.0-pre.39 a read that falls short at k_0 by a *precision shortfall* (a rank over $\mathbb{Z}/\ell^k$ below the rational rank, or a precision exhausted inside the read; `MapRes::precision`) is redone at $k = K_{\max}(\ell)$ (30 at 2, 19 at 3, and so on: the word bound $\ell^K < 2^{31}$ of this read), at every ℓ, not only at 2; only a shortfall there refuses `[KerCokerPrecision]`. $W = 61$ is not needed. A failed safeguard inside the read (a relation not carried into the relations, I not contained in L, the rank input) is not escalated and refuses `[NotEndomorphism]`, a defect by the paragraph above. The read stays certified: each width certifies its own read (rank over $\mathbb{Z}/\ell^k$ equal to the rational rank, the precision checks), and a certified type of $\ker(T_p - c)$, $\operatorname{coker}(T_p - c)$ is choice-free (Proposition 31.15(1)), so it does not depend on k. An entry that reads at k_0 is computed as before, so its text and hash are unchanged; an entry that used to refuse now reads, and its `status` and hash change, as (h)(2) says they should. No golden point refuses, so no golden line moves. The six level-1 cells $n \in \{48, 64, 72, 80, 120, 200\}$, $\mathcal{P} = \{2, 3, 5\}$, where the ICENSUS probes refused at $(q, \ell) = (1, 2)$, now read (`refused=0`). Arm (L) of `IntegralDescentHeckeValidation` checks $n \in \{48, 64, 80, 200\}$: the module against the groups' 2-primary part; each escalated read refuses when forced back to k_0, and each other read gives the same text when forced to $k = 30$; every type of $\ker(T_p - c)$, $\operatorname{coker}(T_p - c)$ against an independent 2-adic Smith form (of $[F \mid \operatorname{diag} 2^{a_i}]$ for the cokernel, of the Pontryagin dual for the kernel); and that each cell escalates at least one read. The canonical text at $(1, 48)$ also equals that of the Python oracle `docs/implementation/slot24/d3_hecke_oracle.py`.

(k) **Parts not run** (v1.23.0-pre.40). A campaign may compute only a prefix of the record's parts (the spec key `parts` $\in$ {`all`, `groups,hecke`, `groups`}, default `all`): a part not run is marked `"ran":false` — the pair already by item (f)'s `pair`, the χ

and torsion entries by `"hecke":{"ran":false}`, written only then — and its arrays are empty *by absence of the part, not by result*, never a zero group. The hashed body carries the line `hecke=not_run` when the Hecke part was not run, and omits the pair lines when the pair was not run (item (f)); either way a partial record never has the hash of a full one. A full record is unchanged byte for byte, and a non-default `parts` enters the cell id (`_partsGH`, `_partsG`); the default adds nothing. A `groups` record also skips the cross-check of the torsion read against the groups' ℓ-primary part (item (e)), which lives in the Hecke part: its groups are certified by their own assembly (Theorem 31.16(1)), but not cross-checked as in a full record.

(l) **The faithful coefficients** (v1.24.0-pre.1; plan item FMTMP, owner request of 2026-11-03, `docs/implementation/slot24/FMTMP_acceptance.md`). A χ entry of (d) carries **every certified coefficient exactly**: one with $|c_i| < 2^{63}$ as before by its decimal integer in `coefficients`, one with $|c_i| \geq 2^{63}$ — whose `coefficients` token stays `[±,`D` digits]` (D the number of residue primes used, unchanged) — **also by its value**, in the entry's list `"big":[{"i":`i`,"hex":"`h`"},...]` (i the index into `coefficients`, ascending), written only when the entry has such a coefficient. **The encoding** h is the canonical signed hexadecimal text of the integer c_i: `-` iff $c_i < 0$, then `0x`, then $|c_i|$ in lowercase hexadecimal digits without a leading zero (the regular expression `^-?0x(0|[1-9a-f][0-9a-f]*)$`, `-0x0` excluded); it is a function of the integer alone — not of the residue primes, their number or the mixed-radix digits of Proposition 31.14(4) from which the builder computes it — hence **choice-free** in the sense of (h), and unique. It is produced with word arithmetic only (Horner on a base-2^{32} limb vector; no multi-precision library: the owner's ruling [R-Q2], no bignum, stands), and **checked at build time**: an independent decode of h (Horner on its hexadecimal digits with the word `mulmod` of Proposition 31.14(4)) must give D3's c_i mod 2^{64}, c_i mod 3^{40} and sign, else the builder stops (a defect, never a refusal). **Hashed**: the body of (h) gains, directly after the line of the χ entry, one line `charpoly p=`p `i=`i `exact=`h per such coefficient, so `record_fnv1a64` covers the values; a record *without* a coefficient beyond 2^{63} has no such line and no `big` key, and keeps its bytes and its hash (`0x6c841fb033fbd3d1` at $(1, 12)$ stays). For a coefficient below 2^{63} the digest of (d) already determines it (its residue modulo 2^{64} with its sign), so with the new lines the body determines every coefficient of a determined entry. The digest of (d), its text and the golden objects `charZ_T<p>` do not change. A reader decodes h by one call (Python `int(h, 16)`; a decoder without sign handling strips the `-` first) and re-verifies the entry with the digest of (d), recomputed from the integers; the reference decoder is `docs/integral_census/icensus_exact.py`. **Old cells.** A cell written before v1.24.0-pre.1 has tokens without `big`: its large coefficients are not in the file, and readers report them as unavailable, never guessed. The campaign's cell ids and resume key do not change (the mathematical settings are unchanged; the precedent of (f)'s `derived_from` and (k)): a resumed campaign directory accepts a complete old-format file, so a re-run that is to carry the values writes to a fresh directory. **Size**: the text grows with $\sum_i \log_{16} |c_i|$; the bound of Proposition 31.14(3) caps it at about $\frac{1}{4}\sum_i b_{p,i}$ characters per p, and a cell file can reach several MB — at $(N, n) = (90, 20)$, $r = 756$, the cell file measures 7.95 MB (2259 values, the largest of 29 415 bits; 72 kB before), against the bound's ≤ 17.5 MB (`FMTMP_acceptance.md` §4); no compression is applied.

Code map (v1.23.0-pre.37, **restated at v1.23.0-pre.38**, referee M2: built on branch `s24-d6impl`, tip `538cf881`; **audited** — `docs/implementation/slot24/D6_code_audit.md`, 2 M / 3 m / 8 N, the fixes at `19a969de`, N5 included — and **merged** into `hecke-extensions` at `2aa61397`; `docs/implementation/slot24/DESCENT6_report.md` §§2–8, §8 the audit's fixes). The setting in `Hecke/HeckeSettings` (`ParseCoefficientRing("Z")`

stores $k = 0$; `CoefficientRingK(who)` and `RequireFieldCoefficientRing(who)` refuse $\mathbb{Z}$ by name; `CoefficientRingIsZ()`); the record in `Integral/Descent/DescentRecord` (`BuildDescentRecord`: D1's `AssembleDescentN`, D3's `RunDescentHecke`, D4/D5c's `AssembleDescentPair`); the console verb `descent(N, n, primes)` returning `DescentRecord` (`Console/Bindings/Descent`), `read_only` with its `--serve` twin, bounded by $N \le 300$, $n \le 60$, $\psi(N)(n+1) \le 300$ and (the audit's m2) $1 \le |\mathcal{P}| \le 6$, every $p \le 97$, $\sum_{p\in\mathcal{P}} p \le 100$ (measured worst cases at the bound $\approx$ 40 s and $\approx$ 1.4 GiB; the default primes sum to at most 31), $\mathcal{P}$ sorted ascending; the campaign's descent cells (`Campaign/CampaignDescent`, `docs/matrix-engine/campaign.md` §17; one cell per $(\texttt{level}, n)$, `integral_descent_cell/v1`); `IntegralDescentRecordValidation` (`standard`; arms (S), (F), (R) against the four oracle tables outside the build — `descentN_oracle.tsv`, `descent_oracle_n128.tsv`, `d3_hecke_oracle.tsv`, `d4_pair_oracle.tsv` — on 13 cells (v1.23.0-pre.38, referee N4, in place of "three oracles"), (J), (K), (B), (G)). **The golden event R1 has landed** at `6958a5d2`: nine points `DESCENT N=<N> n=<n>`, 156 tier-A additions, 0 changed, 0 removed (18117 → 18273), predicted before any build and dry-run into scratch (`docs/implementation/slot24/d6golden/`, ledger `d6golden/ledger.md`). **The four code items of v1.23.0-pre.37 are closed** at `19a969de` (and `8fb4ab2c` for (4)): (1) the ker / coker refusal is `[KerCokerPrecision]`, (j); (2) the `compact` object carries `derived_from`, (f); (3) arm (B) compares `record_fnv1a64` and the hashed body under `HECKE_DESCENT_TREE=0` and `HECKE_DESCENT_TREE_PAIR=0` with the default, in process and through the real environment variables in a child, so (h)'s "a rollback cannot change the hash" is pinned on the arm's cells; (4) arm (G) checks the points present in the reference with equal hashes and pins the 156 values (`d6golden/values.tsv`). **Open code items** (v1.23.0-pre.38; none is a clause, none moves a hash): an in-band coefficient-order key outside the hash, (d); pinning the two $(1, 12)$ hashes of (h) in (J); the `RecordOptions hecke/pair` switches, (g); a registration-time ring default for console verbs, (a).

8. **The integral Hecke structure record**, `integral_hecke_structure/v1` (v1.25.0-pre.0 on the branch `s24-wip-heckez-z0`, refereed A7 / A7b; in the main line from v1.25.0-pre.1, 2026-11-06, a pre-release, not owner-released; the HeckeZ plan `docs/implementation/slot24/heckez/04-record-census.md` v0.4 §1, owner rulings Q1(a), Q2(a), Q3(a), Q12, Q17 of 2026-11-04). The record of §31.6 for one cell (N, n) and one algebra. **Not built** (the plan's package HZ-REC, slot 25); this item fixes what it must contain. (v1.25.0-pre.4, 2026-11-06: built by HZ-REC on `s25-wip-hzrec`, not merged; its schema audit `docs/implementation/slot25/HZREC_schema_audit.md` led to the clauses marked "pre.4" below, binding changes **not yet seen by the owner**. v1.25.0-pre.5: its re-audit `docs/implementation/slot25/HZREC_schema_reaudit.md` led to the clauses marked "pre.5", likewise **not yet seen by the owner**.) Binding content:
   (a) **The ring, the object and the key.** `coefficient_ring = "Z"`, `object = "group"`, as item 7 (a): the object is $H^q(\Gamma_0(N); V)$ and the algebras of Definition 31.27; a builder asked for the record under any other ring refuses. The key is $(N, n, \texttt{algebra})$, $\texttt{algebra} \in \{\texttt{full}, \texttt{anemic}\}$, only `full` at $N = 1$. The record is **a new record beside** `integral_descent_record/v1` (Q2(a)): it references its descent cell by `descent_ref` = {`record_fnv1a64`, `cell_sha256`, `primes`} and copies nothing item 7 already holds, so integral census edition 1 and the golden entries of item 7 stay byte-identical. It is selected by the campaign key `structure = hecke` (default `none`: the descent campaign, byte-identical), which chooses the record schema and is *not* a twelfth mathematical setting (Q17(v)).
   (b) **The two algebras and the rank.** `d` is the a-priori rank of Proposition 31.29: $\dim M_k(\Gamma_0(N))$ for `full` (`d_source = "hES_N"`), d^{an} of (3′) for `anemic` (`d_source = "hES_N_an"`), with `b1` and `e_k`; never the computed $\deg f_\vartheta$. The `operators` actually

used (T_p, $p \nmid N$; U_p, $p \mid N$, `full` only), ascending.

(c) **Statuses: one closed list, each final or pending; no exact value where only a bound is proved.** Every structural field carries one of: `certified`, `upper_bound_certified`, `non_reduced_certified`, `bounds` (with both values), `zero`, `distinct_certified`, `common_certified`, `agrees_on_P` (not a certificate), `partial(unfactored)`, `open(sandwich)`, `open(cap)`, `open(precision)`, `refused(<reason>)` with `<reason>` ∈ {`by_theory`, `non_reduced`, `word_modulus`, `no_lift`, `precision`}, `not_run` — all **final** (they change only when the mathematics or the settings change; two readings apart (v1.26.0-pre.8, confirm-referee D1): the derived `maximal_ideals = not_run` of item (d′)(ii), on any record (v1.26.0-pre.6, referee-light R1; v1.26.0-pre.7, confirm-referee C3), and the aggregate `G3.C_T.status` while a `C_T` read or `G3.C_T.lower` it depends on is pending, which is rewritten when those move (no byte of a final record moves: the record is `pending` by the pending field beside it)) — and `blocked(export)`, `blocked(U_p)`, `refused(export_word)`, `refused(k1_route)`, `refused(word_prime_cap)` [v1.25.0-pre.7, below], the only **pending** statuses (a named package moves them: EXPORT resp. UP-Z the first two (`HZ-EISUP` for `G1.components` and the `maximal_ideals` of item (d′)(ii); v1.26.0-pre.7, confirm-referee W-A), EXPORT-Z step 2 `export_word`, EXPORT-Z or a slot-26 $K = 1$ route `k1_route`; v1.25.0-pre.2, Z0 referee F3, lead decision on O1; a later version with a larger count cap `word_prime_cap`), or a value of the closed enumerations of the plan's §1.2. The overall `status` is `pending` iff some field is pending (a pending refusal reason included), `refused` iff the cell is refused, else `final`. `by_theory` is (C3) at 2, 3, level 1, $d \geq 4$ (Proposition 31.39(6)); `non_reduced` (v1.26.0-pre.3, re-referee W1) is the full (C1) refuted by Theorem 31.45(1) at a `full` record, $N > 1$, whose `G1.reduced_Q` is `non_reduced_certified`, written in `G1.C1.status` only (item (d′)(ii)); `word_modulus` a read whose modulus ℓ^K is $\geq 2^{63}$ (tested first, before any route is asked; $K = 1$ for (C3), $K \geq 2$ for the at-root read, so the latter from $\ell \geq 2^{31.5}$; Proposition 31.48); `no_lift` the named refusal of a cochain lift by EXPORT or UP-Z once landed; `precision` as item 7 (i), (j). (v1.25.0-pre.1, 2026-11-06, as built by HZ-LOC; restated in v1.25.0-pre.2, Z0 referee F3, F4:) `export_word` a read served by EXPORT (level $N > 1$; at level 1 only a torsion prime of the cell, $\ell \leq n + 1$ in practice) with $\ell^K < 2^{63}$ but K above what EXPORT serves: $\ell^K \geq 2^{31}$, the bound of EXPORT's word ring, or $K > K_{\max} = K_{\mathrm{run}} - \sigma_{\max} - 2\sigma^{(1)}_{\max}$, the precision that ring loses to the ℓ-torsion (v1.25.0-pre.3, Z0 re-check; for $\ell < 2^{31}$, K_{run} is the largest K with $\ell^K < 2^{31}$, so there the first case is contained in the second) — that is the binding rule. As built (`Integral/HeckeZLocal.cpp`, `LatticeFromExport`) the code writes it at $\ell \geq 2^{31}$ for *every* K, $K = 1$ included (so at level $N > 1$, $\ell \in [2^{31}, 2^{63})$, (C3) is refused as well; for $\ell \in [46341, 2^{31})$ EXPORT serves $K = 1$ only, (C3) runs and the at-root read is refused), and at $\ell \geq 46341$ whenever K exceeds EXPORT's $K_{\max}$; at $\ell < 46341$ with K above EXPORT's $K_{\max}$ it writes `open(precision)`, although that is the same limit of EXPORT's ring (the word bound when $\ell^K \geq 2^{31}$, e.g. $K \geq 3$ at $\ell \geq 1291$, else its torsion precision), and HZ-REC writes that case as `export_word`. `export_word` is **pending**: EXPORT-Z step 2 (owner ruling 2026-11-06; the free block F_p over $\mathbb{Z}$, of which L/ℓ^K is a reduction for every $\ell^K < 2^{63}$) widens the ring and moves these reads (`docs/implementation/slot25/EXPORT_report.md`, "For HZ-LOC / HZ-TOR", O1). `k1_route` (v1.25.0-pre.2, lead decision on O1): (C3) at $K = 1$ at level 1 for $\ell \in [2^{31.5}, 2^{63})$, which the level-one module route does not serve (it runs at $K \geq 2$); the code writes `refused(word_modulus)` there although $\ell < 2^{63}$, and HZ-REC writes `refused(k1_route)`; pending, an EXPORT-Z / slot-26 item; no weight of the first golden set is affected. `open(precision)`

(v1.25.0-pre.2, F4): the at-root read has not closed at the step cap of the settings ((j), count-based; not EXPORT's $K_{\max}$, which is `export_word`), so it is final; it is written into `root` and `C_T` of the read. (v1.25.0-pre.4, HZ-REC schema audit F7, F8; not yet seen by the owner:) No other pending word exists [v1.25.0-pre.7: besides `refused(word_prime_cap)`, below] (the transient `pending(HZ-TOR)` of the HZ-REC brief is retired: a Tor read the export refuses is `blocked(export)`). A refusal of the export is mapped *by its name* (`ExportRefusalName`): `[Lift]` with a certificate code [v1.25.0-pre.5, R6, below] → `refused(no_lift)` (final), `[ModulePrecision]`, and a Tor read whose $K_{\max}$ lies below $e = \sigma_{\max}$, → `refused(export_word)` (pending), every other name → `blocked(export)`; wherever (e)–(h) list `blocked(export)`, these two words may stand instead. (v1.25.0-pre.5, HZ-REC re-audit R6; not yet seen by the owner:) `[Lift]` is `refused(no_lift)` *only when a certificate refuses the lift*: the pair-lift code it carries ("the strict lift is refused [`<Code>`]") is one of `LiftCertificate`, `IsobClosure`, `StrictIdentity`, `InputTableCertificate` — then no lift exists on the route, which is final. A `[Lift]` that carries a cap of the lift (`RssCeiling`, `IntegerCap`, `DomainCap`) or no pair code (a refusal of the cochain-level operator that the export files under `[Lift]`) is a limit of the route and is written `blocked(export)` (pending). (v1.25.0-pre.7, HZ-RAT-PERF and its audit F2, F6, `docs/implementation/slot25/HZRATPERF_audit.md`; not yet seen by the owner:) `refused(word_prime_cap)`: the word-prime list of the rational step (Proposition 31.29) or of a U_p tag (`full`, $p \mid N$) would exceed its count cap `word_prime_cap` $= 2^{21}$ primes. The list is the `CrtPrimes` order (the odd primes downward from $2^{30} - 1$, all above 2^{29}) without the divisors of N, and it is prefix-stable: the rational step sizes it from its largest per-ϑ certified target (the bound of Proposition 31.29 with the routes' slack) and grows it on demand, the U_p tag starts at min(2048, `word_prime_cap`) primes and doubles; both take primes in that order until their certified cover is reached, so neither the initial length nor the cap changes which primes are used or any value. The cap counts primes, so only a larger cap (a later version) lifts the word: it is **pending**. A refused record contains G0 (`schema`, N, n, `algebra`, `descent_ref.record_fnv1a64`, `status = refused`), `G1.C2_status = refused(word_prime_cap)` and the provenance line `G10.rat.settings`, and no other field (the record's JSON envelope also carries the non-field `refusal` text, `"refused(word_prime_cap): ..."`): everything computed before the refusal is discarded, no partial value is written; its overall `status` is `refused` although the word is of the pending class (as for a refused descent). The word is not in the schema frozen at `heckez-schema-v1-frozen`; it enters as an amendment after the freeze (`docs/implementation/slot25/HZREC_report.md` §11, amendment A1): hash-neutral, since the cap is outside the hashed settings text ((i)) and no record of the default campaign reaches it (the largest predicted need, 354 276 primes at $(1, 702)$ (HZ-KIT's prediction, not a measurement), is below 2^{21}); a scanner pinned to the frozen list rejects a record that carries it.

(d) **G1, the rational structure.** `reduced_Q`, the result of (C1) (`deg_f` $= d$), the primitive ϑ and the operator set (both functions of cell and settings, hence hashed), the components (type, degree, multiplicity on H^1, the minimal polynomial of $\lambda_i(X_{p_0})$ for the smallest p_0 whose eigenvalue generates the coefficient field), the candidate set S_{cand} with its reasons, and the discriminant by its digest, bit count, the factorisation of its square part $\{q, e,$ `prime_proof`$\}$, the cofactor (exact; as built, v1.25.0-pre.2, Z0 referee F5: of the gcd over the primitive ϑ_j that (C2) uses, see §31.6, *The cofactor test*) with `cofactor_test` $\in \{$`one`, `squarefree_by_size`, `unfactored`, `upper_bound`$\}$ (v1.25.0-pre.1, 2026-11-06, HZ-RAT audit F4: `upper_bound` when a cofactor above 2^{64} cannot be certified to divide every D_j by the residue route, or a certified

prime divides it; it is then only a bound and never `complete`) and `C2_status` $\in$ {`complete`, `partial(unfactored)`} (Theorem 31.45(C2): `S` is the set of *all* certified q_i) or, as built (v1.25.0-pre.7, HZ-RAT-PERF audit F6), one of the pending words `blocked(U_p)` (written, until UP-Z's (d′)(ii), for every `full` record at $N > 1$: (C2) for the full algebra needs U_p at the word primes; v1.26.0-pre.3, re-referee W2) and `refused(word_prime_cap)` ((c)); and (v1.26.0-pre.3, re-referee W1, amendment A2) the final `not_run`, written only on the non-reduced branch of (d′)(ii), where no ϑ satisfies (C1) and (C2) is not asked; the closed list of `C2` is these five. *Reading* (v1.26.0-pre.3, re-referee W1): `partial(unfactored)` beside a `G1.C1.status` other than `certified` certifies nothing — Theorem 31.45(2) presupposes (C1) — and says only that no factorisation was attempted; this is the reading of the words an `anemic` record with (C1) `open(cap)` carries (the code path `Integral/HeckeZRecord.cpp`, l. 1076, `else` branch; v1.26.0-pre.4, confirm-referee W5: no record of edition 1 reaches it — every `anemic` record of edition 1 has (C1) `certified`, and its `partial(unfactored)` is the genuine (C2) outcome of Theorem 31.45(2)), so no byte of any record moves. `prime_proof` $\in$ {`small`, `pratt`, `probable`}, `pratt` meaning that the C++ build has verified the Pratt certificate itself; a factor whose certificate was not found or not verified is `probable` and counts as part of the cofactor. (v1.25.0-pre.4, F5, F10; not yet seen by the owner:) *The components as v1 writes them.* `components[0]` is the rational Eisenstein component (`type = eis`, degree 1, multiplicity e_k [v1.25.0-pre.5, R1; pre.4 said 1], p_0 the smallest prime of $\mathcal{P}$, minimal polynomial $x - (1 + p_0^{k-1})$; the eigenvalue of ϑ beside it as `lambda_theta`). `components[1]` (`type = cusp`) is the *whole* cuspidal part as one block, of degree $d - 1$, with `minpoly` $= c_\vartheta$ and `irreducible` $\in$ {`true`, `open(cap)`}: `true` when the degree is 1 or c_ϑ is irreducible of full degree modulo some prime $q \geq 3$ (a deterministic certificate; the first 64 [v1.25.0-pre.5: 256] odd primes are tried in order, a count cap of the settings text, (j); the prime in G9); `open(cap)` is final and says that the block may be a union of components (v1 has no factoriser over $\mathbb{Q}$). $p_0 \in$ {`T_p`, `theta`}: `T_p` when the block is irreducible and $\vartheta = T_p$ (the search tries the T_p in ascending order and takes the first with $\deg f = d$, so p is the smallest prime whose eigenvalue generates the coefficient field), else `theta` (the minimal polynomial is that of $\lambda(\vartheta)$). At $d = 1$ [v1.25.0-pre.5, R3; pre.4: $d = e_k$] (at level 1: $d = e_k$, no cusp forms) `components[1]` is not written. `type` $\in$ {`eis`, `cusp`}. (v1.25.0-pre.5, HZ-REC re-audit R1, R3, R4, R8; not yet seen by the owner:) *Corrections.* (1) The multiplicity of `components[0]` on H^1 is e_k, not 1: a rational Eisenstein system means one anemic Eisenstein system, which all e_k Eisenstein classes carry ($b_1 = 2 \dim S_k + e_k$, and no cuspidal system equals it); $e_k = 1$ at level 1, e_k the number of cusps of $\Gamma_0(N)$ (-1 at $k = 2$) otherwise; e.g. $(2, 6)$ `anemic`, $\chi_L(T_3) = (x - 2188)^2(x - 12)^2$, multiplicity 2. (2) The criterion is $d = 1$: `components[1]` is written iff $d \geq 2$ (at level 1 this is $d = e_k$, no cusp forms; at $N > 1$ `anemic`, $d = e_k = 2$ can hold with a cusp form, and $d = 1$ with $e_k = 3$). (3) At $N > 1$ v1 does not compute the multiplicity of the cuspidal block and writes `components[1].mult = not_run` (final). (4) The irreducibility cap is 256 odd primes (the settings line `heckez.irreducible`); all 256 primes cost 0.19 s at degree 58 ($n \approx 700$), and at most 14 ms on every cell measured (a count cap, never time-based).

(d′) **The full-N reading of the caps text; frozen with v1** (v1.26.0-pre.1, 2026-11-08, slot 26, UP-Z, route (i); `docs/implementation/slot26/UPZ_design.md` §1.4 and its critical audit `docs/implementation/slot26/UPZ_design_audit.md`, ACCEPT route (i) with conditions C1–C7; refereed ACCEPT-with-conditions, `docs/implementation/slot26/C126pre1_referee.md`, its conditions and fixes at

v1.26.0-pre.2; re-refereed ACCEPT-with-conditions, `docs/implementation/slot26/C126pre2_rere` its conditions R1, R2 and should-fix W1, W2 at v1.26.0-pre.3; confirm-refereed ACCEPT, `docs/implementation/slot26/C126pre3_confirm.md`, its should-fix W3, W4, W5 and note N1 at v1.26.0-pre.4; aligned with the audited Z1 code (`docs/implementation/slot26/Z1_code_audit.md`, should-fix S1 (and S2's key)) at v1.26.0-pre.5; referee-light-checked ACCEPT-with-conditions, `docs/implementation/slot26/C126pre5_referee_light.md`, its conditions R1–R3, should-fix W1, W2 and nits N1–N3 at v1.26.0-pre.6; confirm-refereed ACCEPT-with-conditions, `docs/implementation/slot26/C126pre6_confirm_referee.md`, its conditions C1–C3, should-fix W-A, W-B, W-D and nits N-a, N-b at v1.26.0-pre.7; confirm-refereed ACCEPT-with-conditions, `docs/implementation/slot26/C126pre7_confirm_ref` its conditions D1, D2 at v1.26.0-pre.8; extended by Z2's G3 words at v1.26.0-pre.9, `docs/implementation/slot26/C126pre9_disposition.md`, refereed with v1.26.0-pre.8 ACCEPT-with-conditions, `docs/implementation/slot26/C126pre9_referee.md`, its condition E1, should-fix W-1 and nits N-1–N-3 at v1.26.0-pre.10; confirm-refereed with v1.26.0-pre.10 ACCEPT-with-conditions, `docs/implementation/slot26/C126pre10_confirm` its condition C1 and should-fix W-A, W-B at v1.26.0-pre.11; **a binding change not yet seen by the owner**). At level $N > 1$ with `algebra = full` the phrases of the settings text that speak of $\mathcal{P}$ — `heckez.P_rule` ("enlarged … until (C1) is certified", the read set "`P u {T_ell}`"), `heckez.maximal_ideals` ("the first of theta_search's order over P") — and HZ-RAT's `theta_search` read as follows. At $N = 1$ (no prime divides N) and for `anemic` records nothing changes (but for the reading of the derived word of (ii), v1.26.0-pre.7, confirm-referee C3, which moves no byte): no hashed byte of a level-1 or `anemic` record moves, and the settings text, its fingerprint `0x8ffc29730c4f9672`, every record id and the schema string stay.

(i) *The operator list.* ϑ's operator list is `G1.P`'s T_p in ascending order, then U_p for the primes $p \mid N$ in ascending order — **not** `G0.operators`, which also holds every T_ℓ with `t_ell = added` ((j)). Every "over P" of the settings text reads "over the operator list", and "`P u {T_ell}`" reads "the operator list $\cup\{T_\ell\}$" (T_ℓ last). (v1.26.0-pre.2, referee E6.) These are exactly three phrases: the read set "`P u {T_ell}`" of `heckez.P_rule`; "the first of theta_search's order over P" of `heckez.maximal_ideals`; and that line's "(theta_search = P u {T_ell}) … over the read set P u {T_ell}". Nothing else is re-read: $g_{\mathcal{P}}$ of `heckez.S_rule` keeps its meaning (the bound over `G1.P`; the U_p bounds are `S_rule`'s own parenthesis), and `heckez.K_read` names no list.

(ii) *$\mathcal{P}$ and (C1).* $\mathcal{P}$ is selected by `P_rule` exactly as for the `anemic` record of the same cell: `P_rule`'s "(C1)" is the **anemic** (C1) (Theorem 31.45(C1) with $d = d^{\mathrm{an}}$), and U_p never enlarges $\mathcal{P}$. The full (C1) is the degree condition $\deg f_\vartheta = d$ over the operator list, $d =$ `G1.d` $= \dim M_k(\Gamma_0(N))$ ((b)). (v1.26.0-pre.2, referee E5.) $\mathcal{P}$ is meant in the record's sense throughout: `G1.P` is the $\mathcal{P}$ of Theorem 31.45 for the *anemic* algebra, while that theorem's $\mathcal{P}$ for the full algebra is the operator list of (i) (its X_p, $p \mid N$, are U_p by Definition 31.27). *The non-reduced cells* (v1.26.0-pre.2, referee S4). If the record's U_p tag certifies `G1.reduced_Q = non_reduced_certified` (the counting criterion of §31.6, "Two forms, and what is reachable today", asserted for N a power of p only; e.g. $(8, 10)$, where U_2 has a nilpotent block), then $\mathbb{T}_{\mathbb{Q}}$ is not reduced, so $\deg f_\vartheta < d$ for *every* ϑ (Theorem 31.45(1): (C1) implies that $\mathbb{T}_{\mathbb{Q}}$ is reduced) and the full (C1) is false. The record then runs **no** ϑ search (no `theta_cap` is spent) and writes (v1.26.0-pre.3, re-referee W1, R1(a): the accurate words, fixed before any record carries them): `G1.C1.status = refused(non_reduced)`, **final** ((C1) is refuted, not undecided: it can never become `certified`, for any ϑ, cap,

presentation, ℓ or U_p convention; the hashed `G1.reduced_Q` beside it is the reason), with no `G1.C1.deg_f`; `G1.theta = not_run` (final; no search ran); `G1.components = not_run`; no `G1.disc.*` and no `G1.discriminants_used`; `G1.C2_status = not_run` (final; (C2) presupposes (C1), so nothing is certified and nothing is asked); `G1.theta_ops` as in (vi); `G3.partition = U_p_tags` with the U_p-tag fields of `G3.C_T`; no ϑ-derived reason in S_{cand}; and the G2 words explicitly: for every $\ell \in \mathsf{S}$ with $\ell \geq 5$, `G2.ell=<ell>.generation.status = open(sandwich)` (final, no read runs, as for a record whose (C1) is not certified), and for $\ell \in \mathsf{S} \cap \{2,3\}$ `G2.ell=<ell>.generation.status = blocked(U_p)` (pending, by (v)) until Z2 lands, Z2 then writing `open(sandwich)` there (or, if Z2-D's negative is recorded and audited, `refused(no_lift)` by (v)). The words `refused(non_reduced)` in `C1` and `not_run` in `C2` enter the frozen schema as an amendment after the freeze (A2, of the kind of `docs/implementation/slot25/HZREC_report.md` §11, A1): hash-neutral, since no record of edition 1 carries them, the settings text and its fingerprint do not move, and every `full` record at $N > 1$ that will carry them is pending in edition 1 and moves anyway ((ix)); `not_run` in `G1.theta` is already a word of that field. *The verifier* (`docs/integral_census/hz_dataset_verify.py`, its `FINAL` set, l. 54–56, copied from `StatusTable()`) gains `refused(non_reduced)` as **final**, in the same commit as Z1's `StatusTable()` row and the closed lists `C1` (`+refused(non_reduced)`) and `C2` (`+not_run`); no word changes class, and `PENDING` is unchanged. *Otherwise* the search runs as for the `anemic` record: `G1.C1.status` $\in \{$`certified`, `open(cap)`$\}$, `open(cap)` meaning `theta_cap` exhausted, final; `G1.reduced_Q` is `true` when (C1) is certified, else `true` when the U_p tag certifies it (split degree), else `undecided` (final). **These final words only where the search ran** (v1.26.0-pre.3, re-referee R2): `G1.C1.status` $\in \{$`certified`, `open(cap)`$\}$, and the final values of `G1.reduced_Q`, `G1.theta`, `G1.components` and `G1.C2_status` beside it, are written only when the ϑ search over the operator list ran with U_p read at the word primes by (v). Where Z1's word-prime U_p route did not run — a predicted RSS above the ceiling, a refused Heilbronn build or another refused route, a cell outside Z1's coverage — these five ℓ-independent fields (v1.26.0-pre.4, confirm-referee W4: `G1.components` is the fifth, ϑ-dependent as the others and `blocked(U_p)` on every `full` record at $N > 1$ of edition 1, never the final `not_run` of the non-reduced branch here) keep `blocked(U_p)`, or `blocked(export)` for the RSS ceiling (pending, (c)); never `open(cap)`, which would be false-final (a refusal by `word_prime_cap` refuses the whole record, (c)). Subject to that, `blocked(U_p)` is no longer written in `G1.C1.status`, `G1.reduced_Q`, `G1.theta` or `G1.C2_status` (the other four ℓ-independent fields). *The components and the maximal ideals where the search ran* (v1.26.0-pre.5, Z1 code audit S1(a), `docs/implementation/slot26/Z1_code_audit.md`; the text aligned with the audited code, replacing pre.4's inclusion of `G1.components` in the list above): where the search ran and (C1) is `certified`, `G1.components` = `blocked(U_p)`, and at every ℓ of G2 with `generation.status` = `certified`, `G2.ell=<`ℓ`>.maximal_ideals` = `blocked(U_p)` — both **pending** ((c)): the components and the decomposition of (d) need the Eisenstein U_p systems, which Z1 does not build; they stay `blocked(U_p)` until the package that builds them writes their values, a move pending → value, never final → final. (v1.26.0-pre.6, referee-light W1; restated at v1.26.0-pre.7, confirm-referee W-D:) that package is **not** Z2, whose card in `docs/implementation/slot26/00-slot26-plan.md` (v0.6) is U_2, U_3 at $\ell \in \{2,3\}$; it is `HZ-EISUP`, the card of that plan (§2.2,

sequenced right after Z2 by the owner's ruling of 2026-11-08) that builds the Eisenstein U_p systems. Z2 is therefore necessary and not sufficient: a record carrying one of these fields stays `pending` after Z2 until `HZ-EISUP` writes it. Where the search ran and (C1) is `open(cap)`, `G1.components` = `not_run` (final; as for the `anemic` record, no component is computed without (C1); v1.26.0-pre.6, referee-light W2), and where `generation.status` is not `certified`, `maximal_ideals` = `not_run` by the rule of (d) (*The maximal ideals as v1 writes them*); v1.26.0-pre.5 introduces no final word. *The derived word* (v1.26.0-pre.6, referee-light R1; v1.26.0-pre.7, confirm-referee C3: the scope "`full` at $N > 1$" dropped). On any record, `maximal_ideals` = `not_run` beside a `generation.status` of the pending class is a *derived* word: it records that the decomposition was not attempted because generation is pending, it asserts nothing final, and it is rewritten when `generation.status` moves. The rule "stays `blocked(U_p)`" of (v) does not apply to this field. `not_run` beside a final `generation.status` (`open(sandwich)`, `open(cap)`, ...) remains final. The word stays of the final class in (c) and in `StatusTable()`, so no byte and no record status moves: the pending `generation.status` beside it makes the record `pending` by (c). This paragraph alone of (d′) reaches beyond `full` records at $N > 1$ (the word of (d) stands also beside `refused(export_word)` on an `anemic` record and beside `refused(k1_route)` on a `full` record at $N = 1$): only the reading of the word changes, and no byte of a level-1 or `anemic` record moves. Every `full` record at $N > 1$ of this paragraph is pending in any case, by the pending `generation.status` beside the word, and before Z2 also through `G3.C_T.lower` (the rule below). *No full record at $N > 1$ is final before Z2* (v1.26.0-pre.3, re-referee R1(b); the forecast F10 of `docs/implementation/slot26/UPZ_design_audit.md` made a rule): on **every** `full` record at $N > 1$, whatever its $\mathsf{S}_{\mathrm{cand}}$, until Z2 lands (in the sense of *The G3 words from Z2 on*, below; v1.26.0-pre.10, referee N-1) or Z2-D's negative is recorded in `docs/implementation/slot26/UPZ_design.md` and upheld by its audit, the aggregate fields of G3 are written as follows (v1.26.0-pre.4, confirm-referee W3: the list replaces pre.3's open-ended "every other aggregate field whose value may need U_p at $\ell \in \{2,3\}$"; it is exhaustive for this rule): `G3.C_T.lower` = `blocked(U_p)` (pending); `G3.C_T.status` = `upper_bound_certified` where every U_p tag of the record certifies a bound, else `blocked(U_p)` — `upper_bound_certified` is a bound's status, of the final class, and **is** the planned move of (ix): it may be sharpened to `certified` only together with `G3.C_T.lower` (lower = upper at every ℓ, item (d), G3), never alone (no package of this version does so: *The G3 words from Z2 on*, below); and `G3.C_T.exact` is absent. So every such record stays `pending` ((c)) through `G3.C_T.lower` and none is golden-eligible before then (nor, after Z2 alone, any record carrying one of the `blocked(U_p)` fields of *The components and the maximal ideals where the search ran* above; v1.26.0-pre.6, referee-light W1; v1.26.0-pre.7, confirm-referee C1). Those are the records with `G1.C1.status` = `certified` (67 of the 85 records of Z1's recompute). The other 18 (9 with `open(cap)`, 9 with `refused(non_reduced)`) carry no field waiting on `HZ-EISUP`: their pending fields are `generation.status` at $\ell \in \{2,3\}$ and G3's `C_T` fields, which the words of *The G3 words from Z2 on* below make final. The per-ℓ words at $\ell \in \mathsf{S} \cap \{2,3\}$ (G2's `generation.status`) are governed by (v) and by the non-reduced branch above, not by this rule. No level-1 or `anemic` byte moves. *G3 on the non-reduced branch* (v1.26.0-pre.4, confirm-referee N1). There `G3.C_T.status` is `blocked(U_p)`, never

`upper_bound_certified`: as built (`Integral/HeckeZUTag.cpp`) a U_p tag certifies a bound only when $\chi_{\mathbb{Q}}(U_p)$ splits with $m_0 = 0$ and `distinct` $= d$, i.e. (C1) for $\vartheta = U_p$, which Theorem 31.45(1) excludes when $\mathbb{T}_{\mathbb{Q}}$ is not reduced (e.g. the pin $(8, 10, \texttt{full})$ reads `C_T=blocked(U_p)`); and `G3.C_T.lower` is `blocked(U_p)` by the rule above. These are the words before Z2; the final words of these two fields on this branch are those of *The G3 words from Z2 on* (v1.26.0-pre.9). *The G3 words from Z2 on* (v1.26.0-pre.9, UP-Z (b), contract first; `docs/implementation/slot26/C126pre9_disposition.md`; the one place of these words). Z2 has landed for a record iff the binary that wrote it carries the route of (v) at $\ell \in \{2, 3\}$, which the record states by `G10.contract` ((vii)); a binary carrying that route writes the words of this paragraph, so by (vii) its records state `1.26.0-pre.9` or a later version that moves a word it writes (v1.26.0-pre.10, referee N-3). On every `full` record at $N > 1$ of such a binary on which the U_p tag step ran at every $p \mid N$ (v1.26.0-pre.10, referee W-1; as built this is every record such a binary writes), on either branch of (ii), the words of `G3` are: `G3.C_T[U_p].status` = `upper_bound_certified` where that U_p tag certifies its bound, else `not_run`; the rows `G3.C_T[U_p:a,b].status` and `.upper` exist only where the tag certifies its bound, with status `upper_bound_certified`; `G3.C_T.status` = `upper_bound_certified` where every U_p tag of the record certifies its bound, else `not_run`; `G3.C_T.lower` = `not_run`; `G3.C_T.exact` absent. All are **final** ((c)). Where the tag step did not run at some $p \mid N$ (no record of this version: a refusal by `word_prime_cap` refuses the whole record, (c)), `G3.C_T[U_p].status` of that p, `G3.C_T.status` and `G3.C_T.lower` keep `blocked(U_p)` (pending, (c)), as the five fields of (ii) do where the search did not run; never `not_run`, which would be false-final. (`blocked(U_p)` also at an RSS ceiling: it is the only pending word of the closed list `utag`.) `not_run` in the two status fields says that this route ran and certifies no bound, nothing about $C_{\mathbb{T}}$ itself. *Why `lower` is `not_run`* (v1.26.0-pre.10, referee E1, N-2): the rows of `G3.partition` = `U_p_tags` are pairs of the U_p-eigen-systems. For $d \geq 3$ a pair is not all components, and η_L, with it the `lower` $= \exp \eta_L$ of G3, is defined by Definition 31.47 only when $\mathcal{S} \sqcup \mathcal{S}'$ is all components, the hypothesis of every clause of Proposition 31.48. For $d = 2$ the pair of a certifying tag is all components and η_L is defined (e.g. $(N, n) = (2, 4)$, the pair `eis(1),eis(32)`, bound 31); its read is Proposition 31.48(5) with $X_p = U_p$, which no route of this version performs on a `U_p_tags` row and for which the partition `U_p_tags` has no field (v1.26.0-pre.11, confirm-referee W-B). On the non-reduced branch that Definition does not apply at all (Remark 31.46: it assumes $\mathbb{T}_{\mathbb{Q}}$ reduced), so the absence of a ϑ is not the reason and the question left open at v1.26.0-pre.4 is answered by that scope. So no statement of this document prescribes a read of a lower bound on these rows, `G3.C_T.lower` is not a derived word and does not wait on `generation.status`, and on such a record no field is sharpened to `certified`: the rule "only together with `G3.C_T.lower`" above is not exercised, and the reading of the aggregate in (c) does not apply, since `G3.C_T.lower` is not pending. Where no tag certifies there is no row (77 of Z1's 85 records, a count at this version; v1.26.0-pre.11, confirm-referee W-A), and the partition `U_p_tags` carries only the fields of this paragraph (item (d), G3), so `G3.C_T.lower` is `not_run` there too: on the non-reduced branch and on the `open(cap)` records, which are outside the scope of Definition 31.47 ((C1) does not pass), and also where η_L is defined and Proposition 31.48(5) could read it with a T_p: e.g. $(N, n) = (11, 0)$, $d = 2$, one Eisenstein system, no certifying tag, whose `anemic` record reads `G3.C_T.exact` = `0x5`. A `full` record of this

version does not perform that read. A later value of these fields is a settings step or a schema step ((viii)), never a package filling a pending field: the read of Proposition 31.48(5) for one rational Eisenstein system against the rest (with $X_p = T_p$ where a T_p separates, U_p where only a U_p does; v1.26.0-pre.11, confirm-referee W-A) would be a new partition row with its own statement, and `HZ-EISUP`, which writes `G1.components` and `maximal_ideals`, does not move these fields. At $\ell \in \mathsf{S} \cap \{2,3\}$ on a record whose `G1.C1.status` is `open(cap)`, `G2.ell=<ell>.generation.status` = `open(sandwich)` (final; no read without (C1), as at $\ell \geq 5$ and on the non-reduced branch), `maximal_ideals` then being the final `not_run` of (d). *Amendment A3*: `not_run` joins the closed list `utag` of the U_p tag fields, after the freeze, of the kind of A1 and A2: hash-neutral, since that list is not hashed, no record of edition 1 carries `not_run` in such a field, every `full` record at $N > 1$ is pending in edition 1 and moves anyway, and `not_run` is already in the verifier's `FINAL` set (no class moves). The rules of this item for a binary without that route stay in force for what such a binary writes, so no record it wrote is invalidated; such a record stays `pending` until regenerated. On Z1's recompute of the 85 records (a count at this version, not a rule) these words leave the 18 records of (ii) (`open(cap)` and `refused(non_reduced)`) with no pending field, hence `final`, and the 67 with (C1) `certified` pending through `HZ-EISUP`'s fields, 3 of them also through `refused(export_word)` at $\ell \geq 5$.

(iii) *The ϑ search.* HZ-RAT's order on the operator list: the units first, in list order, then the combinations, within `theta_cap` and `theta_coeff_max` = $\binom{d}{2}$ with $d =$ `G1.d` of the record — the full d, not d^{an}.

(iv) *Which U_p.* $U_p = [A \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} A]$, $A = \Gamma_0(N)$: the $U_p^{(N)}$ of Proposition 27.2(4)(d) and of Proposition 31.13(9), on p-old spaces the **adjoint** U_p^* of the classical operator (Proposition 29.34(3)(d)). The convention is named here and in the header of every U_p certificate dump; the $\mathbb{Z}_{(\ell)}$-algebras generated with U_p and with U_p^* are not asserted to be isomorphic at $\ell \mid N$, so a comparison with another convention (the literature, the Lean side) names it, citing Remark 27.10(1), (2) (U^* and $U_\ell^{(\ell)}$ are W_ℓ-conjugate and not interchangeable over $\mathbb{Z}_\ell$; Hida's $T(p)$ is U^*) and [U]'s *Remark on the orientation* (`docs/latex/hida_hecke_algebra.tex`, after `prop:u-towerorient`). (v1.26.0-pre.2, referee S3.) [U]'s "[C]'s U_ℓ is the classical U" is a statement about Hecke *systems* on the χ-parts of the $\Gamma_1(\ell^r)$ model at tame level 1: it fixes the dictionary $U \mapsto a_\ell$, $\langle d \rangle \mapsto \chi(d)$, an orientation that remark decides by computation through the coupled swap $U \leftrightarrow U^*$, $\langle d \rangle \leftrightarrow \langle d \rangle^{-1}$. Proposition 29.34(3)(d) is a statement about the operator against the inclusions dg_1, dg_p of old classes at $\Gamma_0(N)$, $\chi = 1$. The two concern different objects and do not contradict each other; neither is read by this record, which carries no character and no $\Gamma_1(\ell^r)$ level and names its operator by the double coset $[A \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} A]$. That they are one operator-level convention on a common model is not asserted.

(v) *Where U_p is read.* At $\ell \geq 5$ and at the word primes $q \nmid 6N$: the operator $H_{N,p}$ on the cokernel presentation (Proposition 28.9(1), (2), stated for $p \mid N$). For the free block over $\mathbb{Z}$ and every field at $\ell \in \{2,3\}$: $\widetilde{U}_{\mathcal{I},p}^{(N)}$ of Proposition 31.13(9) and its transports. Until that route is built, every U_p-dependent field at $\ell \in \{2,3\}$ stays `blocked(U_p)` (pending, (c)), except `maximal_ideals`, which beside a pending `generation.status` carries the derived `not_run` of (ii) (v1.26.0-pre.6, referee-light R1), and after it is built the fields of G3 follow *The G3 words from Z2 on* of (ii) (v1.26.0-pre.9); `refused(no_lift)` is written only

for a per-cell certificate refusal ((c)), never on the ground that a lift at $p \mid N$ is missing (Proposition 31.13(9) constructs one); a prediction above the RSS ceiling is `blocked(export)`. The digests of the U_p matrices are certificate keys `G9.Up.p=<p>.mat_sha256` (v1.26.0-pre.5, Z1 code audit S1(b); v1.26.0-pre.6, referee-light R2): unhashed (G9, outside `record_fnv1a64`; inside `cert_sha256` as every G9 key), one key per $p \mid N$, present only on `full` records at $N > 1$ whose *record build* serves the U_p matrix over $\mathbb{Z}$ (its τ and M included), and written by the package that serves it there; **no record of Z1 carries it**. Running the U_p dump (`HECKE_DUMP_HZ_CERT`, `WriteUpMatrixFiles`) does not create the key, so `cert_sha256` does not depend on whether a dump ran. That package also adds the key's schema row (the table has none at this version) and fixes which digest the key means. The key is absent elsewhere and carries no status word; no level-1 or `anemic` record carries one, so neither of their digests moves.

(vi) *Self-description (required).* Every `full` record at $N > 1$ built under this reading carries the hashed G1 field `G1.theta_ops` (free text, with its own schema row): the operator list of (i), in order, comma-separated (e.g. `T_2,T_3,T_5,U_2`), by which the coefficient vector `G1.theta` is indexed. Level-1 and `anemic` records do not carry it (their index list is `G1.P`), so no byte of theirs moves.

(vii) *Provenance (required).* Every `full` record at $N > 1$ names in G10 (unhashed) the version of this document whose (d′) it implements, as `G10.contract = gl2_contract <version>, item 8 (d')`. (v1.26.0-pre.5, Z1 code audit S2:) the key is exactly `G10.contract` (not `G10.full.contract`); G10 is outside `record_fnv1a64` and `cert_sha256`, so its text moves no digest. (v1.26.0-pre.6, referee-light R3:) `<version>` is the last version of this document, among those the binary that wrote the record implements, whose change to (d′) moved a word that binary writes, that is (v1.26.0-pre.7, confirm-referee C2), changed, for some record and some written key or field other than `G10.contract` itself, the word or value that (d′) prescribes, or the key's presence; supplying a reason, a class or a pointer, or resolving a conflict between two sentences of (d′) in favour of a word one of them already prescribed, moves no written word; naming for the first time the word of a field that (d′) had left open moves a written word for a binary that writes it (v1.26.0-pre.8, confirm-referee D2). A *written* key or field is one that binary writes, so a change to the prescription of a key no binary has written (v1.26.0-pre.6's for `G9.Up.*`) moves no written word. It makes no difference whether the code was conformed to the text or, as at v1.26.0-pre.5, the text to the code; a later version that moves no written word, and a signature as such, does not change it. At this version `<version>` is `1.26.0-pre.5` for a binary that does not write the words of *The G3 words from Z2 on* in (ii) (none of v1.26.0-pre.6, v1.26.0-pre.7 and v1.26.0-pre.8 moves a written word by this test, v1.26.0-pre.9 prescribes for that binary what it wrote before, and neither v1.26.0-pre.10 nor v1.26.0-pre.11 moves one), so its value is `gl2_contract 1.26.0-pre.5, item 8 (d')`; it is `1.26.0-pre.9` for a binary that writes those words (v1.26.0-pre.9 names words (d′) had left open and changes others, so by this test it moves a written word for that binary; v1.26.0-pre.10 supplies a reason and pointers and adds a proviso for records no binary writes, and v1.26.0-pre.11 supplies status pointers and a reason, so neither moves one), so its value is `gl2_contract 1.26.0-pre.9, item 8 (d')`. A written value of this key is a historical fact and is never rewritten, at a signature or otherwise. The manifest of the integral Hecke dataset edition 2 states both the version of this document under which it is issued (1.26.0 once signed) and the `G10.contract` value(s) its `full` records carry, and asserts, by the test of this item, that no version after

each such value up to the issuing version moved a written word of a record carrying that value (v1.26.0-pre.8, confirm-referee D2); if one did, the affected records are regenerated, never edited.

(viii) *Frozen with v1.* This reading is part of `integral_hecke_structure/v1`. Any later change of it — another order of the list, U_p^* for U_p, another list, another (C1) for $\mathcal{P}$ — is a settings change (a new caps line, hence a new fingerprint and new record ids) or a schema step to `integral_hecke_structure/v2`; **never an edit of this item in place**.

(ix) *Why this is not a re-baseline.* The `blocked(U_p)` fields are the move the schema planned ((c): "a named package moves them"); every `full` record at $N > 1$ of edition 1 carries `blocked(U_p)` in exactly the status fields this reading fills (v1.26.0-pre.7, confirm-referee W-B) — apart from `G2.ell=<ℓ>.maximal_ideals`, which edition 1 writes `not_run` beside `generation.status` = `blocked(U_p)` (v1.26.0-pre.6, referee-light R1): that is the derived word of (ii), which asserts nothing final, and its move `not_run` → `blocked(U_p)` where `generation.status` becomes `certified` is the rewriting (ii) prescribes, not a move from a final word — so no final value under another reading exists; and the index list it refers to is hashed (`G0.operators`, `G1.P` and, by (vi), `G1.theta_ops`). Beside the status fields, non-status hashed values of these records also move between edition 1 and Z1: `G0.operators` (the enlargement of (i), on 64 of the 85 records) and `G1.S_cand.ell=<ℓ>.reason` and `.v_disc` (123 fields each, the discriminant reasons now available). None is a status word and every record concerned was `pending`, so the conclusion stands.

(e) **G2, per prime $\ell \in \mathsf{S}$ and per maximal ideal** (sorted by residue degree and the monic residue factor of $\chi(X_{p_0})$ mod ℓ): `generation` with its `status` and `route` $\in$ {`saturation`, `eis_single`, `rational`, `cyclic`, `commutant`, `AK_unsaturated`, `mazur`} (Propositions 31.32, 31.49, Theorem 31.45, Propositions 31.29(5)(c), 31.34; `mazur` is local at the Eisenstein ideal of prime level and carries the good prime q it used together with $v_\ell(\mathsf{F}_q(1+q))$ and $v_\ell(\nu)$; it is `certified` **only when the two valuations agree**, the check of Proposition 31.49(3) — goodness alone does not exclude a non-Eisenstein block in the Hensel factor (v1.25.0-pre.0b, A7-M1)); the maximal ideals with their residue data and `support` $\in$ {`free`, `torsion`, `both`} (at $\ell \in \{2,3\}$ the degree attribution of Proposition 31.37(4)); `components_through` (Proposition 31.36). **Local types only under certified generation**: `local_type` (`kind` $\in$ {`etale`, `monogenic`, `non_monogenic`}, rank, embedding dimension, `gorenstein` $\in$ {`true`, `false`} from the socle of $\mathbb{T}_{\mathfrak{m}}/\ell$, length, Hilbert function, Loewy length) and `pair_kinds` (depth and kind, `certified` for a generating ϑ, else `upper_bound_certified`) are written only when `generation.status = certified`; otherwise the mod-ℓ invariants of A_1' appear labelled as a statement about the computed subalgebra (Remark 31.46(4)). **Generation is never `certified` from a read of $H^1(\Gamma; V/\ell^K)$ at a torsion prime of the cell** (§31.6, "Two forms, and what is reachable today"). (v1.25.0-pre.4, F6, F12; not yet seen by the owner:) *The maximal ideals as v1 writes them*: decomposed (`maximal_ideals = certified`, by the factorisation of f_ϑ mod ℓ) iff `generation` is `certified` by `saturation` or `rational` and some ϑ of `theta_search`'s order over $\mathcal{P}$ (at most `theta_cap` candidates) has $\bar\vartheta$ generating A_1' ($1, \bar\vartheta, \dots, \bar\vartheta^{d-1}$ independent in $\mathrm{End}(L/\ell)$, $\dim A_1' = d$; then $\mathbb{T} \otimes \mathbb{Z}_\ell = \mathbb{Z}_\ell[\vartheta]$ by (C3) and Nakayama, and f_ϑ mod ℓ is the Krylov relation of $\bar\vartheta^d$); otherwise, and under `eis_single`, `maximal_ideals = not_run` (final, except for the derived word of item (d′)(ii) beside a pending `generation.status`, v1.26.0-pre.6, on any record since v1.26.0-pre.7, confirm-referee C3; the rule is in the settings text, so a wider rule is a new fingerprint). `support` of a maximal ideal

is `both` when a common torsion point of (g) has the same ϑ-residue, `not_run` when that residue is not determined (a non-unit ϑ with a residue of degree > 1). v1 does not compute the degree attribution at $\ell \in \{2, 3\}$ and writes `degree_attr = not_run` (final). (v1.25.0-pre.5, HZ-REC re-audit R4, R9; not yet seen by the owner:) *The ϑ search* is widened to the read set: the read's own generating ϑ, else the first of `theta_search`'s order over $\mathcal{P}$, else — when `t_ell = added`, $\ell \le 23170$, (j) — the first of `theta_search`'s order over the read set $\mathcal{P} \cup \{T_\ell\}$ (T_ℓ last; its Eisenstein eigenvalue is $1 + \ell^{k-1} \equiv 1$); where generation is certified by saturation only with T_ℓ (no ϑ over $\mathcal{P}$ generates A_1', e.g. $(1, 26)$ at 7) the maximal ideals are then decomposed (there by $\vartheta = T_5 + T_7$), and `decomposed_by` names the operators (`theta=(0 0 1 1) on T_2,T_3,T_5,T_7`). *`components_through.<i>`*: `certified` when the component passes through $\mathfrak{m}$, and `not_run` when it is *certified not to pass* ($\pi \nmid f_i \bmod \ell$ at the generating ϑ, Proposition 31.36); under `maximal_ideals = certified` the word `not_run` here therefore means "excluded, certified", not "unknown" (v1 has no separate word for it). *`pair_kinds[eis,cusp].kind`* is `meeting` when the cuspidal block has degree 1 and `not_run` (final) when it has degree ≥ 2: v1 does not decide the kind of a pair with a block that may be a union of components.

(f) **G3, congruence modules: exact, upper, lower.** Per partition treated (all Eisenstein components against the rest; each pair of components meeting somewhere): `C_T` with `upper` $= g_{\mathcal{P}}$ (`upper_bound_certified`), written as the word-size factorisation of $g_{\mathcal{P}}$ — each $v_\ell(g_{\mathcal{P}})$ by residues modulo ℓ^K — plus a cofactor (as built, v1.25.0-pre.2, F5: that of the gcd over the primitive unit $\vartheta = T_p$, the smallest base cofactor certified by residues to divide every $c_{\mathcal{S}',p}(\lambda(X_p))$, else `cofactor_test = upper_bound` as in (d)), with the ℓ-wise form of Proposition 31.48(4) (no big gcd, Q1(a); A7-m4), `lower` $= \exp \eta_L$ per $\ell \mid g_{\mathcal{P}}$, and `exact only` when lower $=$ upper at every $\ell \mid g_{\mathcal{P}}$ (`certified`); `partial(unfactored)` when $g_{\mathcal{P}}$ keeps a cofactor; an ℓ-part whose at-root read is `refused(word_modulus)` stays `upper_bound_certified`; `eta_L` (the Smith exponents of the at-root read, r, the ι-sign); `fibre_product` $\in \{$`certified`, `not_certified`$\}$ (Propositions 31.48(4), 31.49(1)); `second_route` — the Bernoulli valuation $v_\ell(B_k/k)$ and whether it agrees (level 1), Mazur's $v_\ell(\mathrm{num}((N-1)/12))$ (prime level, weight 2) — which never lifts a status (`agrees = false` is a defect, not a status); `general_case` $\in \{$`bounds`, `from_certified_T`$\}$ for non-rational Eisenstein components and cusp–cusp partitions. (v1.25.0-pre.4, F5, F10, O5, O6; not yet seen by the owner:) `partition` $\in \{$`eisenstein` (`S` the Eisenstein component, `S_prime` the cuspidal block), `U_p_tags` (full algebra at $N > 1$: the bounds of the U_p tags; its fields are those of (d′)(ii)), `general` (no rational Eisenstein system), `none(C1_open)`, `none(single_component)`$\}$; at $d = 1$ [v1.25.0-pre.5; pre.4: $d = e_k$] the partition is `none(single_component)` and G3 carries no congruence module. `second_route.agrees` $\in \{$`true`, `false`, `not_read`$\}$, `not_read` when the read it would be compared with has not closed. `eta_L.sign` is `+` at level 1 and `both` at $N > 1$ (the read is on L, not split by ι). (v1.25.0-pre.5, HZ-REC re-audit R2, R3; not yet seen by the owner:) The partition is `none(single_component)` iff $d = 1$ (at level 1: $d = e_k$), (d). *The second route has word caps*, in the settings text (`heckez.second_route`, (j)): $v_\ell(B_k/k)$ is read modulo ℓ^{e+2}, $e = v_\ell(g_{\mathcal{P}})$, at level 1, $\ell \ge 5$, $(\ell - 1) \nmid k$, by the tangent-number route of `HeckeExt/HeckeHidaLambda`, and is served iff $\ell^{e+5} < 2^{128}$, $\ell^{e+2} + \ell \le 2^{84}$ and $\ell^{e+2+V} + \ell \le 2^{110}$ ($V = v_\ell(4^{k/2} - 1)$) and the read gives $v < e + 2$; otherwise `v_Bk = not_run` (final). `agrees = not_read` stands where the at-root read has not closed *or* `v_Bk = not_run`; e.g. $(1, 28)$ at $\ell = 1001259881$ and $(1, 30)$ at 305065927 ($e = 1$, $\ell^6 \ge 2^{128}$), whose at-root reads are `certified`.

(g) **G4, the glue J: `zero` or `bounds`, never `exact`.** Per $(\ell, \mathfrak{m})$ with `support =`

`both`: `zero` (separated supports, or upper bound of order 1), `bounds` with `lower` $= \log_\ell |J \cap \mathbb{T}'|$ and `upper` $= \log_\ell |\mathrm{Hom}_{\mathbb{T}'}(L/\ell^{e_\ell}, \mathrm{tors}_\ell)|$, or `blocked(export)` (Proposition 31.41(3)); an exact value would need Remark 31.43, which no package proves. (v1.25.0-pre.4, F3, F12; not yet seen by the owner:) *The keys as v1 writes them*: the *torsion points* at ℓ are the joint $\mathbb{T}'$-blocks of H^1/ℓ^e ($\mathbb{T}' = \mathbb{Z}[T_p : p$ in the read set], $e = \sigma_{\max}$) with non-zero torsion, keyed `pt[i]` in the order of their residual systems; a point with free rank > 0 is *common* (`support = both`). G4 has one row group per common point, no aggregate per ℓ; `lower` is 0, a valid lower bound (equal to $\log_\ell |J \cap \mathbb{T}'|$ where the read's $j' = 1$ certifies $J \cap \mathbb{T}' = 0$), `upper` the Hom bound of that point. Where $\mathcal{P}$ does not generate on H^1/ℓ^K a torsion point is a point of $\mathbb{T}'$, not necessarily one maximal ideal of $\mathbb{T}$ (Remark 31.43).

(h) **G5–G7, torsion points, Bockstein data, compact support, each with its `theory` tag.** G5: per torsion maximal ideal, its system, `relation_to_free` $\in$ {`distinct_certified` (with the witness T_p), `common_certified` (route `T_ell_level1` or `nonsplit`), `agrees_on_P`} (Theorem 31.44(5)) and `generation_tors`. G6: per common point `split` $\in$ {`split_by_support`, `split`, `non_split`, `blocked(export)`} (Propositions 31.38, 31.41(2)) — the solver runs at every block size, a Lean cap governs only the certificate layer — and the bound `hom_order`; the universal-coefficient types are derived from item 7's groups, not stored (Proposition 31.37). G7: the primes $\ell \geq 5$ with $H^2_c[\ell] \neq 0$, `theory = H_c`, and `hecke = not_run`: no route computes a Hecke action on H^2_c-torsion, and the record asserts none. Every mod-ℓ^K field of G2–G6 has `theory = H`. (v1.25.0-pre.4, F1, F3, F12, F13; not yet seen by the owner:) G5 and G6 are written per torsion point of (g), `pt[i]`, G5 carrying its system (the residue π_p per operator); a point that is not common is `distinct_certified` with the witness `T_p` (the first operator whose residue differs from that of every point with a free part) or `joint`, and its G6 `split` is `split_by_support`. `split_on_P` (final, a statement about $\mathbb{T}'$ like `agrees_on_P`) is added to G6's list: the extension splits $\mathbb{T}'$-equivariantly at the point (Proposition 31.38). It is *not* `split` for $\mathbb{T}$: Proposition 31.41(2)(b) needs $\mathcal{P}$ to generate the image of $\mathbb{T}$ on H^1_ℓ/ℓ^K, torsion included, the mixed generation of Remark 31.43 ([conjectural]), and G2's generation (of $\mathbb{T}^{\mathrm{fr}}$ on L) does not give it. v1 writes `split_on_P` for every $\mathbb{T}'$-split point and never `split`, which is kept for a later theorem; `non_split` and `split_by_support` are statements about $\mathbb{T}$. v1 computes neither `generation_tors` nor the Tor terms of H^2: it writes `generation_tors = not_run` and, at $q = 2$, G5 / G6 `not_run` per ℓ (final). The body has G4, then G5, then G6. (v1.25.0-pre.5, HZ-REC re-audit R4, R11; not yet seen by the owner:) `G7.pair = not_run` (final): the descent cell of `descent_ref` runs `parts = groups,hecke`, without the pair, so v1 writes no pair datum. *Where the `theory` tag is written*: once per $\ell \in$ S (`G2.ell=<l>.theory = H`) and once on G7 (`G7.theory = H_c`); the rows of G4–G6 carry no tag of their own and are `H`. **G8, the Hida block per maximal ideal** (v1.25.0-pre.6, the slot-25 H1 merge after its audit `docs/implementation/slot25/H1_audit.md`; **not yet seen by the owner**). Written *only* under the non-default settings value `heckez.G8 = hida` ((i), (j)); a default record has no G8 field, and any value other than `off`, `hida` is refused. The paths: `G8.scope` $\in$ {`level1`, `not_run(level_N)`} (tame level 1 only); per prime $\ell \geq 5$ of S, `G8.ell=<l>.blocks` $\in$ {`read`, `not_run(no_decomposition)` (G2 did not decompose the maximal ideals at ℓ), `not_run(class_zero)` ($(\ell - 1) \mid k$), `not_run(forms_cap)`}; and per maximal ideal `m[i]` of G2 at ℓ (the same index i) `G8.ell=<l>.m[i].ordinary` $\in$ {`true`, `false`} (the Eisenstein ideal is ordinary by theorem and the forms side must agree; another ideal is ordinary iff the residue of ϑ at $\mathfrak{m}$ divides $\chi(\vartheta \mid e\,M_k)$ mod ℓ, with $T_\ell = U$ on $e\,M_k$; a multiplicity other than 0 or the full one is a defect, never a value). At an ordinary Eisenstein

ideal through the cuspidal component, in addition: `chain_key` $= (1, \ell, k \bmod (\ell-1))$ (three decimals); `g`, the rank g_a of the ordinary Eisenstein block of $e\,S_k$ (e the ordinary projector) (Fitting split by the full residual system, U included); `L_k`, G3's certified C_T ℓ-length, else `not_run`; `v_Bk` $= v_\ell(B_k/k)$ read modulo ℓ^W, else `not_run`; `structure` $\in$ {`fibre_LxL`, `regular_deg_g`, `open`}: `fibre_LxL` iff $g = 1$ (the fibre product $\Lambda \times_{\Lambda/(G_a)} \Lambda$, [U, `prop:u-H11`]), `regular_deg_g` iff $g \geq 2$, some $T_q - (1+q^{k-1})$ generates the block modulo ℓ with $v_\ell \det(T_q - (1+q^{k-1}) \mid \text{block}) =$ `v_Bk`, and $v\big(G(x_k) - G(x_{k+\ell-1})\big) = 1$ ($\lambda = 1$; then $\mathbf{h}(a)_{\mathfrak{m}} \cong \mathbb{Z}_\ell[[Y]]$, [U, `prop:u-H12`](1), (2) and [U, `rmk:u-H12general`]), else `open` (the conservative word); `I0_generator` $\in$ {`T_<q>`, `none_found`}, the first such T_q (q ascending, $q \neq \ell$). *Caps*: the forms side is read modulo ℓ^W, $W = \max(2, \text{upper}_v + 2)$, and served iff $\ell^W < 2^{84}$, $k - 1 \geq W$ and the q-expansion length $\max(\ell, k/12+1, 7,$ the operators of $\vartheta) \cdot (\dim S_k + 2) + 1 \leq 2^{18}$, else `blocks = not_run(forms_cap)`; count-based record content, in the hashed `heckez.G8 = hida` line ((j)). Outside the body: G9 carries K9's integers `G9.K9.ell=<l>` (the certificate layer; `absent(not_applicable)` under `hida` where no ordinary Eisenstein ideal meets a cusp form, `absent(H1)` under the default), G10 the read's W, time and refusal as `G10.hida.ell=<l>` (provenance).

(i) **The hash partition.** `record_fnv1a64`, the one FNV-1a-64 (`ModArithmetic/Fnv.h`) of a fixed-order body — schema, key, operators, `descent_ref.record_fnv1a64`, G1 to G8 in order, every list sorted by its stated invariant key, every big integer as its canonical hexadecimal text (item 7 (l)), every status, then the settings' canonical text with its fingerprint. **A hashed field is determined by (cell, algebra, settings) and independent of the lattice basis and of algorithm internals.** The *certificate layer* G9 (Bézout pairs, witness positions, dump hashes, the K used, Pratt trees, the Lean caps; the fields of `07-lean-plan.md` §11.1) has its own `cert_sha256`, outside `record_fnv1a64`; *provenance* (K and route of each read, word primes, effort, timings, RSS, binary hash) is in neither; the *Lean status* of each certificate lives in a side manifest keyed by (cell, checker), outside both (Q17(i)), so a Lean run never moves a record. (v1.25.0-pre.4, F2, F11; not yet seen by the owner:) **Default v1 records carry no G8 field**: G8 (the Hida block per maximal ideal) is written only under a non-default value of the settings line `heckez.G8` (default `off`), by the package H1, so no default record moves when H1 lands (v1.25.0-pre.6: that value is `hida`; its paths, closed lists and caps are the G8 paragraph of (h)). The word primes that carry the residues (`CrtPrimes`, below 2^{30}) are provenance and are not in the settings text. (v1.25.0-pre.7, HZ-RAT-PERF; not yet seen by the owner:) Their count cap is printed in the same provenance key, `word_primes=CrtPrimes(<2^30,sized_from_bound,cap=2^21)`, the last key of HZ-RAT's settings text, which the record erases from its hashed caps text (so `G10.rat.settings` and `G10.rat.word_primes` show it); it moves no record id, hash, fingerprint or pin. (v1.25.0-pre.5, HZ-REC re-audit R10; not yet seen by the owner:) One use of that sequence is not provenance: (C1)'s primitivity probe runs at its first `c1_probe_primes` primes (the count is hashed, 3 by default), so `primitive` could be a false negative if all of them divided a discriminant (never a false positive; the chance is far below 10^{-15}). The sequence is fixed in v1; a build that changed it would change the settings text (a new fingerprint).

(j) **Caps are record content.** `P_rule`, `theta_search`, `factor_budget` (trial division to 10^6; rho, ECM and the Pratt search by fixed counts), `bezout_record_cap` and $K_{\max}$ per read are in the settings text and count-based, **never time-based**, so a record is the same on every machine (Q17(ii)); a non-default value is a new fingerprint, cell id and hash input (item 7 (h)). *The read set at ℓ* (v1.25.0-pre.1, 2026-11-06; the slot-25 amendment of the plan's `04-record-census.md` §1.5, lead

decision of 2026-11-06 on the HZ-LOC audit's F1). At each read prime ℓ the operator set of the (C3) and at-root reads is $\mathcal{P}$ plus T_ℓ when $\ell \nmid N$ and $\ell \le 23170$ (T_ℓ appended, so the primitive ϑ stays on $\mathcal{P}$; the cap is count-based, T_ℓ having $\ell + 1$ cosets, and is the narrow tier of Lemma 18.5), and v_ℓ of the upper bound is taken over $\mathcal{P} \cup \{\ell\}$: $v_\ell(g_{\mathcal{P}\cup\{\ell\}}) = \min(v_\ell g_{\mathcal{P}}, v_\ell c_\ell(\lambda_\ell))$ (Definition 31.47 and Proposition 31.48 hold for every operator set). Otherwise $\mathcal{P}$ is unchanged. The per-read field `t_ell` takes its value in a closed list (v1.25.0-pre.2, Z0 referee F6): `in_P` ($\ell \in \mathcal{P}$ already), `added`, `omitted(cap:ell>23170)`, `omitted(ell|N:U_ell_not_exported)` (EXPORT exports no U_p, $p \mid N$), `omitted(export:<Name>)` with `<Name>` the export's refusal *name* (`ExportRefusalName`) only, never its free-text message. The cap belongs to the settings text: `P_rule`'s default text carries "$\mathcal{P} \cup \{T_\ell\}$, $\ell \nmid N$, $\ell \le 23170$". `t_ell` is a function of (cell, settings, ℓ) and is hashed; the word regime of a read (`u32`, `u64`, ...) is provenance, as (i). Effect on the plan's cells: $(1, 42)$ at 11 becomes (C3) `certified` with the 11-part of $C_{\mathbb{T}}$ exactly 1 (on $\mathcal{P} = \{2, 3, 5, 7\}$ it was `open(sandwich)` and `upper_bound_certified`), $(1, 40)$ at 3 closes with 3-part 1 (`docs/implementation/slot25/HZLOC_report.md`, "Audit fixes"). *The factor budget as built* (v1.25.0-pre.1, HZ-RAT, `docs/implementation/slot25/HZRAT_report.md` §4): trial division to 10^6, the word factoriser on cofactors below 2^{64}, the Pratt search and verification, and the cell's pool of certified primes; no multi-limb rho or ECM yet, so the C++ frontier lies below the measured one of §31.6 (level 1: (C2) complete for $k \le 34$ and $k = 38$; $g_{\mathcal{P}}$ `partial(unfactored)` from $k = 36$ on, $k = 44$ excepted). The first golden set of (k) lies inside it. Adding rho / ECM later changes `factor_budget`, hence the fingerprint (as above). The Lean caps (`k1_cap`: $f \le 8$, $b_1 \le 40$; `k7_cap`: block rank ≤ 6, $\dim \mathrm{tors}_{\mathfrak{m}} \le 6$; Q12) sit in G9's header and decide only the Lean status. (v1.25.0-pre.4, F2, F11, F16 and the Z0 referee's F10; not yet seen by the owner:) The default caps text has the lines `heckez.P_rule`, `heckez.theta_search+factor_budget` (HZ-RAT's text without its `word_primes` key, then `pool=fixed_point;multilimb_rho=none;ecm=none`: one `rho=` key, the word factoriser), `heckez.S_rule` (every certified prime of $\gcd_j D_j$, $v_{\mathrm{disc}} = 1$ included — no LC-3b drop rule in v1, so the reason list carries no word for it), `heckez.K_read` ((C3) at $K = 1$ when the K_0 read is not served), `heckez.maximal_ideals` ((e)), `heckez.irreducible` ((d)), `heckez.G8 = off` ((i)) and `heckez.bezout_record_cap`. (v1.25.0-pre.5, HZ-REC re-audit R2, R8, R9; not yet seen by the owner:) `heckez.maximal_ideals` names the widened search over $\mathcal{P} \cup \{T_\ell\}$ ((e)), `heckez.irreducible` the cap of 256 primes ((d)), and the new line `heckez.second_route` the three word caps of the second route ((f)); the second route is one of the widenings below. **Defaults never change**: every later widening (rho / ECM, a wider decomposition rule, a factoriser over $\mathbb{Q}$, a drop rule for S, H1's G8) enters as a non-default value with a new fingerprint, so the default records and the golden objects of (k) never move.

(k) **Golden: records with no pending field only, add-only** (Q3(a)). A record becomes a golden object iff its overall status is `final` *and it is built under the default settings only* (v1.25.0-pre.6, H1 audit F1, not yet seen by the owner: a final record under a non-default value, e.g. `heckez.G8 = hida`, is never golden; the record's `golden_eligible` flag, outside the hash, says so); final statuses that are bounds or partial (`bounds` for J, `upper_bound_certified`, `partial(unfactored)`, `open(...)`, `agrees_on_P`, `split_on_P` (v1.25.0-pre.4), `not_run`, `refused(...)` with a final reason) do not prevent it. One tier-A object `HZ_N<N>_n<n>_<alg>` = `record_fnv1a64` per cell; the event is add-only, landed at the end of a slot (the first planned at the slot-25 close: $(1, 10)$, $(1, 22)$ and $(1, n)$, $n \in \{2, 4, 6, 8, 12, 14, 16, 18, 20, 24, 26, 28, 30\}$, 15 objects under Q1(a)).

(l) **Big integers and the literature routes** (Q1(a), *residue-only*). Every integer of any size in the record is produced from residues modulo word primes with a certified bound and written by item 7 (l)'s limb route, the cofactor included (its residues are $D\prod q_i^{-e_i}$); the C++ build certifies factor multiplicities of word-size primes (by residues, or by the cofactor route of §31.6), the cofactor, Pratt certificates below 2^{64} and the cofactor test; a factor above 2^{64} is `probable`; no multi-precision division, gcd, modular power or square root enters the certified build, and the factor finder is untrusted. The routes `mazur` and `second_route` rest on the literature hypotheses named in Proposition 31.49(2), (3) ([U, `pr:u-U1`]; [Maz77]), and the Lean certificates name every C++ output they trust (the dumped operator matrices, the descent's $\chi_{\mathbb{Z}}$, $B_k/k \bmod \ell^2$; Q10).

Code map (v1.25.0-pre.1, 2026-11-06): the record writer HZ-REC is not built (`06-work-packages.md` §2.3). Its inputs are, in the main line: EXPORT, `Integral/Descent/DescentMod` (the module $H^1 \otimes \mathbb{Z}/\ell^K$ with every T_p, $p \nmid N$; `HeckeZExportValidation`); HZ-RAT and HZ-UTAG, `Integral/HeckeZRational.{h,cpp}`, `Integral/HeckeZFactorFinder.{h,cpp}` (untrusted), `Integral/HeckeZUTag.{h,cpp}` (G1, $g_{\mathcal{P}}$, the two full-algebra tags; `HeckeZRationalValidation`, `HeckeZUTagValidation`); HZ-LOC, `Integral/HeckeZLocal.{h,cpp}` and `HeckeExt/HeckeZLocalSource.{h,cpp}` ((C3), the local types, the at-root read; `HeckeZLocalValidation`, `HeckeZLocalCensusXcheck`). Not yet in the main line: EXPORT-Z step 1 (the integral blocks τ_p, M_p of T_p on H^1 over $\mathbb{Z}$, the free block pending its step 2), which this item does not use.

What a record is not. It is not a proof, and no field of it may be branched on by the pipeline that wrote it (rule (A6)). `cochain_commutes`, `eta_zero`, the ι-τ symmetry flag of Obligation 9.10(iv) and the recorded solve regime are diagnostics: they are written so that a later reader can tell what happened, and a true value licenses nothing beyond the statement that names it.

22.5 Document machinery and its guards

This document has four guards, of which three exist and one does not.

1. **The build gate** (`docs/latex/build_gl2_contract.sh`). Runs `pdflatex`, `bibtex`, `pdflatex`, `pdflatex` and **fails** on any undefined reference, undefined citation or multiply-defined label. The label namespace is part of the contract ((L6)), so a dangling `\ref` is an error and not a warning. Three appendices are written so as to be caught by it: Appendix D dereferences every externally cited label, Appendix E dereferences every label it names into a discarded box, and Appendix B prints `\ref` for every statement it indexes — so an index entry naming a label this document does not define is a build failure, exactly as (L6) requires.
2. **The citation census.** The converse check — that every label the *code* cites as `[C, ...]` is defined here — is not gated by the LaTeX build, because the citing sites are C++ comments. It is specified as `label_census.sh` (architecture §8.4). Until v1.5.1 it **was not written**; had it existed it would have caught entry A.7 years earlier, and Remark 22.1 records the runs of it done by hand. **Since v1.5.2 it exists and is registered** (slot-3 WP-10): `docs/matrix-engine/tools/label_census.sh` with its parser `label_census.py` and an allow-list with a reason per entry, run inside the suite by `ContractCensusValidation`, regenerating `docs/latex/label_census.tsv` on every run and failing on any unresolved `[C, <label>]` or `\HA{<label>}` citation; the site count is never gated.
3. **The two migration guards** (`docs/matrix-engine/tools/`). `raw_mod_calls.sh` and `check_matrix_includes.sh` each own an allow-list with a reason per entry and are run inside the validation suite by `ConsumerMigrationValidation` and `MatrixRetirementValidation`, so they cannot rot. They guard the substrate, not this

document; they are named here because they are the pattern item 2 should follow — a script with an allow-list, registered as a validation.

4. **The version and changelog convention**, §1.4 ((V1)–(V5)), whose record is Appendix E and whose rules (CL1)–(CL6) are stated there. (V4)'s two supersession steps — freezing `hecke_action.tex` and `monodromy_v5.tex` with a banner — have not been taken.

Remark 22.1 (The citation census, run by hand). [verified computationally] On 2026-09-16 every occurrence of the citation form `[C, <label>]` was extracted from the whole repository outside `docs/latex/` and intersected with the \label set of `gl2_contract.tex` and `gl2_notation.tex`. Result: 64 **distinct labels cited,** 761 **citation sites in** 78 **files, and** 0 **cited labels that this document does not define** — of which 53 distinct labels and 447 sites in 68 files are in C++ translation units (`.h`, `.cpp`, `.cu`, `.inc`), the rest in project documents. The most-cited are `alg:tau` (37 C++ sites), `thm:backend` (30), `obl:tau` (18), `eq:xiexplicit` (17), `cor:fp32` (17) and `sec:example` (16, in 8 files). Two apparent misses were rejected on inspection: both are prose in `docs/implementation/math_corpus_survey.md` that writes the citation *form* rather than a citation. **This is a recorded run of an ad-hoc script, not a registered validation**, and the figures differ in detail from the architecture's own census of 2026-09-15 (519 sites in 71 files) because the two used different extraction rules; the load-bearing number — zero dangling citations — is the one that does not depend on the rule. Until item 2 above exists, this check is re-run by hand or not at all.

[verified computationally] **Re-run at v1.0.0**, after `hecke_extensions.tex` v1.3, by the same rule: 63 **distinct labels cited,** 778 **citation sites in** 79 **files, and** 0 **cited labels that this document does not define**; the sole non-resolving match is again the prose `[C, <label>]` of `docs/implementation/math_corpus_survey.md`, which writes the citation form rather than a citation. Two further checks were run for (V5)(ii), which the census alone does not settle, because it tests only that a *cited* label resolves: **(i)** the \label set of this document together with `gl2_notation.tex` was intersected with that of `hecke_extensions.tex` — the only companion still live — and the intersection is **empty**; **(ii)** in particular none of the nine labels of §16 is defined there any longer. Check (i) found, and the companion's v1.3 removed, one collision the nine did not include: `hecke_extensions.tex` used `A1`, `A2`, `A3` for its three required additions while this document uses them for authoring rules (A1)–(A3). No citation anywhere used either reading, and the companion renamed its three; but a census of citations would never have found it, which is the argument for making item 2's script compare label *sets* and not only citations.

[verified computationally] **Registered since v1.5.2** (hecke-extensions slot 3, package WP-10, `docs/implementation/slot3/WP-10_report.md`): item 2's script exists, so this check is no longer re-run by hand. `ContractCensusValidation`, run in the slot-3 close's `--all` on 2026-09-17: 429 **labels defined,** 1408 **citation sites in** 121 **of** 430 **scanned files,** 141 **distinct labels cited,** 4 **allow-listed (prose writing the citation form with the placeholder word),** 0 **unresolved**; entry A.7's historical dangling citation (`thm:explicit_complex_equiv`) resolves. The registered script compares citations against the label set; the label-*set* comparison with the companion argued for above is not part of it and stays a check run by hand.

[verified computationally] **Content-keyed since v1.6.2** (hecke-extensions slot 4, package WP-6, `docs/implementation/slot4/WP-6_report.md`). The allow-list is no longer keyed by `relpath:line` but by `relpath|<verbatim fragment of the site's source line>`, so that inserting or deleting lines above a cited passage cannot silently move an entry off its site — which is what makes the format safe for a revision of this document. An entry matching *zero* sites is flagged STALE and *fails* the census, by the same "removal is the unit of progress" rule the other allow-lists use, and the retired `relpath:line` form is rejected with a named error rather than silently reinterpreted. `ContractCensusValidation` gained four arms for exactly these properties, all against scratch roots and never the real allow-list. Run in the slot-4 close's `--all` on 2026-09-22: 429 **labels defined,** 1575 **citation sites in** 128 **of** 471 **scanned files,** 152 **distinct labels cited,** 4 **allow-listed,** 0 **unresolved,** 0 **stale**. The site count is never

gated — it grows every slot; only the unresolved and stale counts are.

Formalization note. This chapter states no mathematics and is not a formalization target. One thing in it is worth carrying across, though: the record schema of §22.4 is the *interface* at which every obligation of §20 becomes decidable, and a formalization that models the pipeline as a function from parameters to a record, with the obligations as predicates on that record, gets the whole of (A4) for free. The one field that resists this treatment is the certification level, because it changes which predicates are evaluated rather than what they say — which is why §22.3 requires the record to carry it.

Part VI — The integral line

23 The integral line: coefficient rings, the free model on the unrefined tessellation, and descent over $\mathbb{Z}$

What this chapter fixes. The mathematics of *integral coefficients* on the model, which until v1.7.4 this document fixed only in pieces (Proposition 15.18, Claims 16.2 and 16.7, Lemma 14.22):

1. which object the shipped complex is *over the ring in force* — three objects live on the model at $\ell \in \{2,3\}$, and the pipeline computes the third (Definition 23.1, Proposition 23.2); what that object is, independently of the triangulation (Proposition 23.3);
2. a free Γ-model on the *unrefined* tessellation $(X, \partial X)$ — the isotropy-resolution model $\mathcal{I}_\bullet$ named in Remark 17.4(4), with its boundary subcomplex $\mathcal{I}_\bullet^\partial$ and one integral comparison map cmp (Definition 23.4, Proposition 23.5); this is the contract item C-A5 of the extension plan, **re-scoped** by owner ruling 6c of 2026-09-24 (Remark 23.6);
3. the descent: the arithmetic fracture square, cohomology-level glueing, the vanishing of the one coherence term the bulk could carry, and the realisation of the owner's descent by that one integral map (Propositions 23.7–23.10);
4. the Hecke datum on the free model: a strict lift of the strict pair datum of Proposition 15.13 — *existence* only (Proposition 23.13);
5. the bad primes (Proposition 23.16), the certified-exponent rule, stated so that it holds at $\ell \in \{2,3\}$ (Proposition 23.17), and the residue identity behind the characteristic polynomial over $\mathbb{Z}$ (Proposition 23.19);
6. what each coefficient ring *names* (Definition 23.20) — in particular what H^q over $\mathbb{Z}$ means on the model and what the integral line will compute. The setting that selects the ring, its spelling, default, the reservation of $\mathbb{Z}$ and the refusal rule are the companion specification's, [E, `sec:ring`] (single-editor rule); this chapter never restates them.

What is not here: any implementation of $\mathcal{I}_\bullet$, cmp or the lifted datum, and in particular their *effective, certified* construction in degrees $\le q_{\max}+1$, which is the key risk of the integral line and a code matter (Remark 23.22) (v1.9.0: $\mathcal{I}_\bullet$, $\mathcal{I}_\bullet^\partial$ and cmp are built since hecke-extensions slot 9, with the integral groups for $n \le 20$; the lifted datum is not — §22.1, Remark 23.21; v1.10.0: the lifted datum is built for $g = \mathrm{diag}(1,p)$, $p \in \{2,3,5\}$, in degrees $\le q_{\max}+1$ by slot 10, Remark 23.14, and the proposition itself still claims existence only); arbitrary-precision arithmetic, which Theorem 18.7(3), the contract box of §19 and Obligation 20.10(iv) still forbid in the build, unchanged; integer Hecke matrices in a basis; the divided-power lattice D_n; families.
Placement. The chapter is a Part of its own, after Part V, so that no chapter number of v1.7.4 moves; mathematically it continues §17.

Standing hypotheses. (H4) throughout, and (B1)–(B3) wherever the boundary occurs; per rule (A1) every statement repeats the ones it uses. $V = \mathrm{Sym}^n(\mathbb{Z}^2)$ is the monomial lattice of Definition 8.1 and nothing else (owner decision OD-4 (a), confirmed 2026-09-24); $U := ST$, as in Claim 16.2. For a commutative ring R, $C^\bullet_{\mathrm{eq}}(R)$ is short for $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$ with fibre V_R *and the relations imposed over* R — the complex the implementation builds for that ring; by Theorem 5.10 and Remark 7.6, $C^\bullet_{\mathrm{eq}}(R) \cong \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), V_R)$, and this identification is used without further mention.

23.1 Coefficient rings, and the three objects on the model

Definition 23.1 (The three complexes on the model, and the notation rule)**.** Assume (H4), let $n \geq 0$ and let R be a commutative ring.

1. The **integral model complex** is $C^\bullet_{\mathrm{eq}}(\mathbb{Z}) = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), V) \cong \bigoplus_\sigma V^{\Gamma_\sigma}$ (sums over Γ-orbit representatives in each degree), a complex of finitely generated free $\mathbb{Z}$-modules (Proposition 15.18(1)).
2. Its **base change** is $C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes_{\mathbb{Z}} R$.
3. The **model complex over the ring in force** is $C^\bullet_{\mathrm{eq}}(R) = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), V_R) \cong \bigoplus_\sigma (V_R)^{\Gamma_\sigma}$.

The reduction map $C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes R \to C^\bullet_{\mathrm{eq}}(R)$ is the map of Proposition 15.18(2). **Notation rule** (binding in this document from v1.8.0; [IB]'s risk K12): $C^\bullet_{\mathrm{eq}}(R)$ *always* denotes (3), the object the implementation computes; the base change is always written $C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes R$; and no statement may silently identify them — the identification needs $6 \in R^\times$ (Proposition 23.2(1)).

Proposition 23.2 (The reduction map, and where the three objects differ)**.** *[proved here] Assume (H4), let $n \geq 0$ and let R be a commutative ring.*

1. *The reduction map $C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes R \to C^\bullet_{\mathrm{eq}}(R)$ is an injective map of complexes in every degree, and an isomorphism of complexes when $6 \in R^\times$. Consequently, for $\ell \geq 5$ and every $e \geq 1$, $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e) = C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes \mathbb{Z}/\ell^e$.*
2. ***At $\ell = 2$ it is not a quasi-isomorphism in general.*** *For $n = 2$ and $R = \mathbb{F}_2$ the map $H^0(C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes \mathbb{F}_2) \to H^0(C^\bullet_{\mathrm{eq}}(\mathbb{F}_2))$ is not onto: the class of the Dickson form $x^2 + xy + y^2$ lies in $H^0(C^\bullet_{\mathrm{eq}}(\mathbb{F}_2))$ and not in the image. Already in degree 0 of the complexes, at $\sigma = v_6$, the reduction $V^{\langle S\rangle} \otimes \mathbb{F}_2 \to (V_R)^{\langle S\rangle}$ has image of rank 1 in a target of rank 2.*
3. ***Odd n.*** *For n odd, $C^\bullet_{\mathrm{eq}}(\mathbb{Z}) = 0$; for every R with $2 \in R^\times$, $H^\bullet(C^\bullet_{\mathrm{eq}}(R)) = 0$; but $\dim_{\mathbb{F}_2} H^1(C^\bullet_{\mathrm{eq}}(\mathbb{F}_2)) = 1$ at $n = 1$, and $H^2(\Gamma; \mathrm{Sym}^1\mathbb{Z}^2) \neq 0$. So at odd n the integral model complex, its reductions, the model complex over $\mathbb{F}_2$ and group cohomology over $\mathbb{Z}$ are four different things.*

Proof. (1) is Proposition 15.18(2) and its proof (a saturated sublattice is a direct summand; the averaging argument needs $|\Gamma_\sigma| \in R^\times$ and $|\Gamma_\sigma| \mid 12$), applied to the bulk complex alone; $\mathbb{Z}/\ell^e$ has $6 \in (\mathbb{Z}/\ell^e)^\times$ for $\ell \geq 5$.

(2) S acts on Sym^2 by a linear substitution sending $x \mapsto \pm y$, $y \mapsto \mp x$ (Definition 8.1), so $x^2 \leftrightarrow y^2$ and $xy \mapsto -xy$. Hence $V^{\langle S\rangle} = \mathbb{Z}(x^2 + y^2)$, while over $\mathbb{F}_2$, where $-xy = xy$, $(V_R)^{\langle S\rangle} = \mathbb{F}_2(x^2+y^2) \oplus \mathbb{F}_2\, xy$: the image of the reduction is the first summand. Now H^0 of either complex is its space of 0-cocycles, and the cochain map is injective by (1), so $H^0(C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes \mathbb{F}_2) \to H^0(C^\bullet_{\mathrm{eq}}(\mathbb{F}_2))$ is injective and its image consists of cocycles whose value at v_6 lies in $\mathbb{F}_2(x^2 + y^2)$. A 0-cocycle of $C^\bullet_{\mathrm{eq}}(\mathbb{F}_2)$ is a Γ-invariant vector $f \in V_R$ placed at every vertex (X is connected); the Dickson form $f = x^2 + xy + y^2$ is invariant under $\mathrm{GL}_2(\mathbb{F}_2)$ (it is invariant under $x \leftrightarrow y$ and under $(x, y) \mapsto (y, x + y)$, which generate), hence under Γ, whose action on V_R factors through the reduction $\Gamma \to \mathrm{SL}_2(\mathbb{F}_2) = \mathrm{GL}_2(\mathbb{F}_2)$. Its value at v_6 is $f \notin \mathbb{F}_2(x^2 + y^2)$.

(3) For n odd the central relation $(\sigma, \sigma, -I)$ reads $f(\sigma) = \rho_n(-I) f(\sigma) = -f(\sigma)$ (Remark 7.7), so $2f(\sigma) = 0$ and $f(\sigma) = 0$ in the torsion-free V; over R with $2 \in R^\times$ the same relation kills every cochain. At $n = 1$, $R = \mathbb{F}_2$: by Proposition 23.3(2) below, $\dim H^1 = \dim V_R^{-I} - \dim(V_R^S + V_R^U)$;

$-I$ acts trivially, S acts on $\mathbb{F}_2^2$ as the swap, with fixed line $\mathbb{F}_2(x+y)$, and $U-\mathrm{id}$ has determinant 1 over $\mathbb{F}_2$ (for any $A \in \mathrm{SL}_2$, $\det(A-\mathrm{id}) = 2-\operatorname{tr}A$, and $\operatorname{tr}U = 1$), so $V_R^U = 0$; hence $\dim H^1 = 2-1 = 1$. For the last clause read (51) over $R = \mathbb{Z}$ with $V = \mathbb{Z}^2$, on which $-I$ acts by -1. For a cyclic group $\langle g\rangle$ of order m the periodic free resolution $\cdots \xrightarrow{N} \mathbb{Z}[\langle g\rangle] \xrightarrow{g-1} \mathbb{Z}[\langle g\rangle] \to \mathbb{Z} \to 0$, $N = 1+g+\cdots+g^{m-1}$, is exact ($\ker(g-1) = \mathbb{Z}N = \operatorname{im}N$ and $\ker N = \operatorname{im}(g-1)$ in $\mathbb{Z}[\langle g\rangle]$; for $m = 2$ it is the resolution (ii) of the proof of Proposition 14.3), so $H^1(\langle g\rangle; M) = \ker N/\operatorname{im}(g-1)$ and $H^2(\langle g\rangle; M) = M^g/NM$. On $\mathbb{Z}^2$: for $\langle -I\rangle$, the norm is 0 and $\operatorname{im}(-I-\mathrm{id}) = 2\mathbb{Z}^2$, so $H^1 = (\mathbb{Z}/2)^2$; for $\langle S\rangle$, the norm $(1+S)(1+S^2)$ is 0 and $|\det(S-\mathrm{id})| = 2$, so $H^1 = \mathbb{Z}/2$, and $(\mathbb{Z}^2)^S = 0$ gives $H^2 = 0$; for $\langle U\rangle$, $\det(U-\mathrm{id}) = 1$, so $H^1 = 0$ and $H^2 = 0$. The segment $H^1(\langle S\rangle) \oplus H^1(\langle U\rangle) \to H^1(\langle -I\rangle) \to H^2(\Gamma) \to H^2(\langle S\rangle) \oplus H^2(\langle U\rangle) = 0$ of (51) then gives $H^2(\Gamma; \mathbb{Z}^2) \cong \operatorname{coker}\big(\mathbb{Z}/2 \oplus 0 \to (\mathbb{Z}/2)^2\big)$, which is non-zero because $(\mathbb{Z}/2)^2$ has four elements and the image at most two. □

Proposition 23.3 (The shipped complex is Bredon cohomology over the ring in force)**.** *Assume (H4). Let $\mathcal{T}$ be the Bass–Serre tree of Lemma 14.22 (vertex set $\Gamma/\langle S\rangle \sqcup \Gamma/\langle U\rangle$, edge set $\Gamma/\langle -I\rangle$, the edge $\gamma\langle -I\rangle$ oriented from $\gamma\langle S\rangle$ to $\gamma\langle U\rangle$), whose augmented chain complex is (50) (Claim 16.2(1)). Let $\mathcal{D}$ be the family of subgroups of Γ contained in a conjugate of $\langle S\rangle$ or of $\langle U\rangle$. ($\mathcal{D}$ is in fact the family of all finite subgroups of Γ, the family for which Bredon cohomology of proper actions is usually stated; this document does not use that.)*

1. *[proved here] **(Fixed sets.)** For every $H \in \mathcal{D}$: if $H \subseteq \{\pm I\}$ then $X^H = X$ and $\mathcal{T}^H = \mathcal{T}$; otherwise X^H and $\mathcal{T}^H$ are each a single vertex. In particular every fixed set is acyclic, and every simplex stabiliser of X and of $\mathcal{T}$ lies in $\mathcal{D}$.*
2. *[proved here] **(The comparison.)** $C_\bullet(X)$ and $C_\bullet(\mathcal{T})$ are chain homotopy equivalent as complexes of $\mathbb{Z}[\Gamma]$-modules, by augmentation-preserving maps. Hence, for* every *$\mathbb{Z}[\Gamma]$-module M — in particular $M = V_R$ for every commutative R and every n — the complex $\operatorname{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), M)$ is chain homotopy equivalent to the tree complex*

$$M^{\langle S\rangle} \oplus M^{\langle U\rangle} \longrightarrow M^{\langle -I\rangle}, \qquad (a,b) \longmapsto b-a, \tag{65}$$

in degrees $0, 1$. So, for $M = V_R$: $H^0(C^\bullet_{\mathrm{eq}}(R)) = V_R^S \cap V_R^U = V_R^\Gamma$, $H^1(C^\bullet_{\mathrm{eq}}(R)) = V_R^{-I}/(V_R^S + V_R^U)$ and $H^q(C^\bullet_{\mathrm{eq}}(R)) = 0$ for $q \geq 2$; over a field, $\dim H^1 = \dim V_R^{-I} - \dim V_R^S - \dim V_R^U + \dim H^0$.
3. *[proved here] **(What it is.)** $H^\bullet(C^\bullet_{\mathrm{eq}}(R))$ is $\mathrm{Ext}^\bullet$, over the category of contravariant functors from the orbit category of $\mathcal{D}$ to abelian groups, from the constant functor $\mathbb{Z}$ to the fixed-point functor $\Gamma/H \mapsto V_R^H$: the **Bredon cohomology** of Γ for the family $\mathcal{D}$ with the fixed-point coefficient system. This is the invariant (H3) calls the orbifold model cohomology; it is **independent of the triangulation** — any simplicial Γ-complex Y acting without inversions, with stabilisers in $\mathcal{D}$ and every Y^H ($H \in \mathcal{D}$) acyclic, gives a complex chain homotopy equivalent to it by the same argument, a Γ-equivariant subdivision of X among them — and it agrees with $H^\bullet(\Gamma; V_R)$ when $6 \in R^\times$ (Remark 7.6) but* not in general *otherwise: at $\ell = 2$, $n = 0$ group cohomology is non-zero in every degree (Lemma 16.4(2)), while (2) vanishes above degree 1.*

Proof. (1) $-I$ is central and lies in every simplex stabiliser of X ((H4)(iv)) and of $\mathcal{T}$ ($-I = S^2 = U^3$, and the edge stabilisers are $\gamma\langle -I\rangle\gamma^{-1} = \langle -I\rangle$), so it acts trivially on both, which is the case $H \subseteq \{\pm I\}$. Let $H \in \mathcal{D}$ with $H \not\subseteq \{\pm I\}$, say $H \subseteq \gamma\langle S\rangle\gamma^{-1} = \mathrm{Stab}_\Gamma(\gamma v_6)$ or $H \subseteq \gamma\langle U\rangle\gamma^{-1} = \mathrm{Stab}_\Gamma(\gamma v_4)$ (Lemma 7.3(1): $\langle S\rangle$ and $\langle ST\rangle$ are the stabilisers of $v_6 = i$ and $v_4 = \zeta^2$). *On X:* H fixes the vertex γv_6 (resp. γv_4); a non-central $h \in H$ fixes at most one point of $\mathcal{H}$ (the last step of the proof of Lemma 7.3), so H fixes no other point of X, and no simplex of positive dimension — a fixed simplex is fixed pointwise, the action being without inversions ((H4)(i)). *On $\mathcal{T}$:* no edge is fixed, the edge stabilisers being $\{\pm I\}$. The map $\varphi(\gamma\langle S\rangle) := \gamma v_6$, $\varphi(\gamma\langle U\rangle) := \gamma v_4$ from the vertices of $\mathcal{T}$ to $\mathcal{H}$ is well defined and Γ-equivariant,

with $\mathrm{Stab}_\Gamma(x) = \mathrm{Stab}_\Gamma(\varphi(x))$ for every vertex x; it is injective, because $\gamma v_6 = \gamma' v_6$ forces $\gamma^{-1}\gamma' \in \langle S\rangle$, likewise for v_4, and v_6, v_4 lie in different Γ-orbits (Lemma 7.3(2)). So H fixes the vertex x iff it fixes the point $\varphi(x)$, which happens for exactly one x by the argument on $\mathcal{H}$ just given. The stabilisers of X are $\{\pm I\}$, $\langle S\rangle$, $\langle ST\rangle = \langle U\rangle$ and $\langle TS\rangle = T\langle U\rangle T^{-1}$ and their conjugates ((H4)(iv)); those of $\mathcal{T}$ are the conjugates of $\langle S\rangle$, $\langle U\rangle$, $\langle -I\rangle$. All lie in $\mathcal{D}$. Acyclicity: X is contractible ((H4)(ii)), $\mathcal{T}$ is acyclic by (50), and a point is acyclic.

(2) Let $\mathcal{O}$ be the category whose objects are the Γ-sets Γ/H, $H \in \mathcal{D}$, and whose morphisms are the Γ-maps, and let $\mathcal{A}$ be the abelian category of contravariant functors $\mathcal{O} \to \mathrm{Ab}$. For a simplicial Γ-complex Y acting without inversions with stabilisers in $\mathcal{D}$, orient each simplex Γ-invariantly (possible because a stabiliser fixes its simplex pointwise) and let $\underline{C}_q(Y) \in \mathcal{A}$ be $\Gamma/H \mapsto C_q(Y^H)$, the chains on the fixed subcomplex, with the maps induced by Γ-maps $\Gamma/H \to \Gamma/H'$ in the evident way. Since $\Sigma^H \cong \mathrm{Map}_\Gamma(\Gamma/H, \Sigma)$ naturally for a Γ-set Σ, $\underline{C}_q(Y) \cong \bigoplus_\sigma \mathbb{Z}[\mathrm{Mor}_{\mathcal{O}}(-, \Gamma/\Gamma_\sigma)]$, a sum over orbit representatives of q-simplices of representable functors, each of which is projective in $\mathcal{A}$ by the Yoneda lemma ($\mathrm{Nat}(\mathbb{Z}[\mathrm{Mor}(-,c)], N) \cong N(c)$ is exact in N). The augmented complex $\underline{C}_\bullet(Y) \to \underline{\mathbb{Z}}$ is exact at the object Γ/H iff Y^H is acyclic. By (1), $\underline{C}_\bullet(X)$ and $\underline{C}_\bullet(\mathcal{T})$ are therefore two projective resolutions of $\underline{\mathbb{Z}}$ in $\mathcal{A}$, and the comparison theorem [standard: [Wei94, Thm. 2.2.6]: a map of a projective complex over M into a resolution of N lifting $M \to N$ exists and is unique up to chain homotopy] gives chain maps over $\mathrm{id}_{\underline{\mathbb{Z}}}$ in both directions whose composites are chain homotopic to the identities. Evaluating at the object $\Gamma/\{1\}$ — whose automorphism group in $\mathcal{O}$ is Γ, so that naturality there is Γ-equivariance — gives the $\mathbb{Z}[\Gamma]$-chain homotopy equivalence $C_\bullet(X) \simeq C_\bullet(\mathcal{T})$ with $\mathbb{Z}[\Gamma]$-linear homotopies, and applying the additive functor $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, M)$ preserves it. Finally $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma/\Gamma_0], M) = M^{\Gamma_0}$, and the cochain differential of $\mathcal{T}$ is evaluation on $\partial(\gamma\langle -I\rangle) = \gamma\langle U\rangle - \gamma\langle S\rangle$, which is (65) (note $M^S, M^U \subseteq M^{-I}$, since $-I = S^2 = U^3$). $H^0 = M^S \cap M^U = M^\Gamma$ because S and U generate Γ; the dimension formula is the rank count on the two-term complex.

(3) By Yoneda, $\mathrm{Nat}(\underline{C}_q(X), \underline{M}) \cong \bigoplus_\sigma M^{\Gamma_\sigma} \cong \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_q(X), M)$ for the fixed-point functor $\underline{M}(\Gamma/H) = M^H$, compatibly with the differentials; since $\underline{C}_\bullet(X)$ is a projective resolution of $\underline{\mathbb{Z}}$, this computes $\mathrm{Ext}^\bullet_{\mathcal{A}}(\underline{\mathbb{Z}}, \underline{M})$. The independence statement is (2) with Y in place of $\mathcal{T}$; a Γ-equivariant subdivision of X has the same fixed *point sets* as X (subdivided), hence the same acyclicity, and the same stabilisers. The last clause: for $6 \in R^\times$ it is Remark 7.6; for $\ell = 2$, $n = 0$, (2) gives $H^q = 0$ for $q \geq 2$ while $H^q(\Gamma; \mathbb{F}_2) \neq 0$ for every q (Lemma 16.4(2)). □

Evidence. [verified computationally]^{r} is recorded in Remark 23.21: the tree complex (65) over $\mathbb{F}_\ell$ reproduces the shipped pipeline's $\dim H^q_{\mathrm{eq}}$ at $\ell \in \{2,3\}$ on every $n \leq 40$, odd n included, and the sixteen golden cells there. Proposition 23.3(2) is also the *oracle* for the model complex over any ring in force, $R = \mathbb{Z}/\ell^e$ at $\ell \in \{2,3\}$ included (then with lengths and invariant factors of (65) in place of dimensions); **it is not** the universal-coefficient reduction of the integral model complex, which is a different object there by Proposition 23.2(2).

23.2 The free model on the unrefined tessellation

Definition 23.4 (The isotropy-resolution model $\mathcal{I}_\bullet$, its boundary subcomplex, and the comparison map)**.** Assume (H4) and (B1)–(B3). For $s \in \{0,1,2\}$ let Σ_s be the set of s-simplices of K that represent the Γ-orbits of s-simplices of X — by Lemma 7.3(2), $\Sigma_0 = \{v_0, v_1, v_2, v_4, v_6\}$, Σ_1 the nine edges of K other than v_3v_5 and v_5v_6, and Σ_2 the five triangles, lifted into X and stored-oriented — and let $\Sigma_0^\partial = \{v_0, v_1, v_2\}$, $\Sigma_1^\partial = \{v_0v_1, v_1v_2, v_2v_3\}$ be the representatives lying in ∂X. Each Γ_σ is cyclic ((H4)(iv)); fix the generator $g_\sigma = ST$ at v_4, S at v_6, $-I$ elsewhere, of order $m_\sigma \in \{6,4,2\}$, and let $F^\sigma_\bullet$ be the periodic free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\Gamma_\sigma]$ ($F^\sigma_t = \mathbb{Z}[\Gamma_\sigma]$, differential $g_\sigma - 1$ from odd t and the norm $\sum_{u\in\Gamma_\sigma} u$ from even $t \geq 2$). Put $G_{s,t} := \bigoplus_{\sigma\in\Sigma_s} \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\sigma]} F^\sigma_t$. An **isotropy-resolution model** of $(X, \partial X)$ is a complex of

$\mathbb{Z}[\Gamma]$-modules $\mathcal{I}_\bullet$ with

$$\mathcal{I}_q = \bigoplus_{s+t=q,\ 0\le s\le 2} G_{s,t}, \qquad d = d^{(0)} + d^{(1)} + d^{(2)},$$

where $d^{(0)} = (-1)^s(\mathrm{id}\otimes\partial^F)$ on $G_{s,\bullet}$ and $d^{(r)}$ maps $G_{s,t}$ to $G_{s-r,t+r-1}$, together with the map $\mathrm{cmp}\colon \mathcal{I}_\bullet \to C_\bullet(X)$ that is the augmentation $\mathbb{Z}[\Gamma]\otimes_{\mathbb{Z}[\Gamma_\sigma]}\mathbb{Z}[\Gamma_\sigma] \to \mathbb{Z}[\Gamma/\Gamma_\sigma] \cong \mathbb{Z}[\Gamma\cdot\sigma]$ on $G_{s,0}$ and 0 on $G_{s,t}$, $t\ge 1$, such that:

(IM1) $d\circ d = 0$ and cmp is a chain map;

(IM2) the sub-sum $\mathcal{I}^\partial_\bullet$ over the representatives in $\Sigma_0^\partial \sqcup \Sigma_1^\partial$ is a subcomplex, and $\mathrm{cmp}(\mathcal{I}^\partial_\bullet) \subseteq C_\bullet(\partial X)$.

Write $C^\bullet_{\mathcal{I}}(R) := \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_\bullet, V_R)$ and likewise $C^\bullet_{\mathcal{I}^\partial}(R)$; and $\mathrm{cmp}^* := \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathrm{cmp}, V_R)\colon C^\bullet_{\mathrm{eq}}(R) \to C^\bullet_{\mathcal{I}}(R)$. ($d^{(1)}$ lifts the simplicial boundary through the transporters, and $d^{(2)}$ is the perturbation correction; $d^{(r)}$ for $r\ge 3$ would leave the columns $0\le s\le 2$, X being 2-dimensional.)

Proposition 23.5 (The isotropy-resolution model: existence, what it computes, and the one integral map)**.** *[proved here] Assume (H4) and (B1)–(B3), and let $n\ge 0$ and R be a commutative ring.*

1. ***(Existence.)*** *An isotropy-resolution model exists; $d^{(1)}$ and $d^{(2)}$ are produced degree by degree by a terminating procedure (lifts along $\mathbb{Z}$-linear contracting homotopies of the augmented F^σ), boundary summands first.*
2. ***(A free resolution.)*** *$\mathcal{I}_\bullet$ is a complex of free $\mathbb{Z}[\Gamma]$-modules of finite rank, $\mathcal{I}_q \cong \mathbb{Z}[\Gamma]^{w_q}$ with $w_0 = 5$, $w_1 = 14$ and $w_q = 19$ for $q\ge 2$; cmp is surjective in every degree and a quasi-isomorphism; hence $\varepsilon\circ\mathrm{cmp}\colon \mathcal{I}_\bullet \to \mathbb{Z}$ is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\Gamma]$. Likewise $\mathcal{I}^\partial_q \cong \mathbb{Z}[\Gamma]^{w^\partial_q}$ with $w^\partial_0 = 3$ and $w^\partial_q = 6$ for $q\ge 1$, and $\mathrm{cmp}\colon \mathcal{I}^\partial_\bullet \to C_\bullet(\partial X)$ is a surjective quasi-isomorphism.*
3. ***(What it computes, over every ring, in every degree.)*** *$C^\bullet_{\mathcal{I}}(R) \cong V_R^{\,w_q}$ in degree q, $C^\bullet_{\mathcal{I}}(\mathbb{Z})\otimes R = C^\bullet_{\mathcal{I}}(R)$, and $H^q(C^\bullet_{\mathcal{I}}(R)) = H^q(\Gamma; V_R)$ for every $q\ge 0$; likewise $H^q(C^\bullet_{\mathcal{I}^\partial}(R)) = H^q(\Gamma_\infty; V_R) = H^q(\partial; V_R)$ of Definition 14.2. No invertibility hypothesis is used.*
4. ***(The comparison map.)*** *$\mathrm{cmp}^*\colon C^\bullet_{\mathrm{eq}}(R) \to C^\bullet_{\mathcal{I}}(R)$ is an injective cochain map in every degree, defined for every R and compatible with base change from $\mathbb{Z}$: cmp^* over R, composed with the reduction $C^\bullet_{\mathrm{eq}}(\mathbb{Z})\otimes R \to C^\bullet_{\mathrm{eq}}(R)$ of Definition 23.1, is cmp^* over $\mathbb{Z}$ tensored with R (as maps into $C^\bullet_{\mathcal{I}}(\mathbb{Z})\otimes R = C^\bullet_{\mathcal{I}}(R)$). The cohomology of its cone is annihilated by 48; so cmp^* is a quasi-isomorphism whenever $6\in R^\times$, and its cone is 6-power torsion over $\mathbb{Z}$. The same holds on the boundary with 4 in place of 48, so there $2\in R^\times$ suffices.* The integral line uses cmp^* over $\mathbb{Z}[1/6]$ only, where the three objects of Definition 23.1 coincide.
5. ***(Relation to Definition 17.1.)*** *$\mathcal{I}_\bullet$ and $F_\bullet\otimes_{\mathbb{Z}} C_\bullet(X)$, for any free resolution $F_\bullet$ of $\mathbb{Z}$ over $\mathbb{Z}[\Gamma]$, are $\mathbb{Z}[\Gamma]$-chain homotopy equivalent by maps over $C_\bullet(X)$ (compatibly with cmp and with $\varepsilon\otimes\mathrm{id}$). So $\mathcal{I}_\bullet$ is the isotropy-resolution model that Remark 17.4(4) names as a computational model of Definition 17.1, compared with it here; and Remark 17.2, whose argument uses only that the source is free and the target acyclic, applies to it verbatim.*

Proof. The columns. For a finite cyclic $\langle g\rangle$ of order m the periodic complex $F^\sigma \to \mathbb{Z}$ is exact: $\ker(g-1) = \mathbb{Z}\cdot N = \mathrm{im}\, N$ and $\ker N = \mathrm{im}(g-1)$ (the augmentation ideal) in $\mathbb{Z}[\langle g\rangle]$, N the norm. Being a bounded-below exact complex of free abelian groups it is split exact, so it has a $\mathbb{Z}$-linear contracting homotopy; closed formulas for one are standard. Since $\mathbb{Z}[\Gamma]$ is free as a right $\mathbb{Z}[\Gamma_\sigma]$-module, $G_{s,\bullet} \to C_s(X) = \bigoplus_{\sigma\in\Sigma_s}\mathbb{Z}[\Gamma/\Gamma_\sigma]$ (the augmentation cmp) is a free resolution of the $\mathbb{Z}[\Gamma]$-module $C_s(X)$, and so is its sub-sum over Σ_s^∂ of $C_s(\partial X)$.

(1) $d^{(1)}$. By the comparison theorem [standard: [Wei94, Thm. 2.2.6]], the boundary $\partial\colon C_s(X) \to C_{s-1}(X)$ lifts to a chain map $d^{(1)}\colon G_{s,\bullet} \to G_{s-1,\bullet}$ over ∂; lifting first on the summands of Σ_1^∂ into the resolution of $C_0(\partial X)$ — possible since $\partial C_1(\partial X) \subseteq C_0(\partial X)$ — makes

$d^{(1)}$ carry boundary summands to boundary summands. The lift is computed on the free generators by the contracting homotopies, so it is a terminating procedure in each degree. $d^{(2)}$. $d^{(1)}d^{(1)}\colon G_{2,\bullet} \to G_{0,\bullet}$ is a chain map over $\partial\partial = 0$, hence null-homotopic by the uniqueness half of the same theorem: there is $d^{(2)}\colon G_{2,t} \to G_{0,t+1}$ with $\partial^F d^{(2)} + d^{(2)}\partial^F = -d^{(1)}d^{(1)}$, again computed by the contracting homotopies. $d^2 = 0$. Sorting $d \circ d$ by the drop in s: drop 0 is $(d^{(0)})^2 = 0$; drop 1 is $d^{(0)}d^{(1)} + d^{(1)}d^{(0)} = 0$ because $d^{(1)}$ commutes with ∂^F and $d^{(0)}$ carries the sign $(-1)^s$; drop 2 occurs only from $s = 2$, where $d^{(0)} = +\partial^F$ on both columns 2 and 0, and is $d^{(1)}d^{(1)} + \partial^F d^{(2)} + d^{(2)}\partial^F = 0$; drops 3 and 4 leave the columns. cmp *is a chain map*: on $G_{s,0}$, $\mathrm{cmp}\, d^{(1)} = \partial\, \mathrm{cmp}$ is the lifting property, $d^{(0)}$ vanishes and $d^{(2)}$ lands in $t = 1$ where cmp $= 0$; on $G_{s,1}$, $\mathrm{cmp}\, d^{(0)} = \pm\varepsilon\, \partial^F = 0$; elsewhere both sides vanish. *The boundary*: $d^{(2)}$ is defined on column 2 only, which has no boundary summand, and $d^{(0)}$, $d^{(1)}$ preserve boundary summands by construction; this is (IM2).

(2) Each $\mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\sigma]} F_t^\sigma \cong \mathbb{Z}[\Gamma]$, so w_q counts the pairs (s, σ) with $\sigma \in \Sigma_s$, $s \le \min(q, 2)$: $|\Sigma_0| = 5$, $|\Sigma_1| = 9$, $|\Sigma_2| = 5$ by Lemma 7.3(2), and 3, 3 on the boundary (Remark 14.1). Surjectivity of cmp is that of the augmentations. For the quasi-isomorphism, filter both sides by s — $F_s\mathcal{I} := \bigoplus_{s' \le s} G_{s',\bullet}$ is a subcomplex since no component of d raises s, and $F_sC := C_{\le s}(X)$ — and note that cmp respects the filtrations and that on the graded pieces it is the augmentation $G_{s,\bullet}[s] \to C_s(X)[s]$, a quasi-isomorphism. The filtrations have length three, so induction on s with the long exact sequences of $0 \to F_{s-1} \to F_s \to \mathrm{gr}_s \to 0$ and the five lemma gives the claim; the boundary is the same argument on $s \le 1$. X is contractible ((H4)(ii)), so $\varepsilon \circ$ cmp is a resolution.

(3) $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\sigma]} \mathbb{Z}[\Gamma_\sigma], V_R) \cong V_R$, and $\mathcal{I}_q$ being finite free, $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_q, V) \otimes R = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_q, V_R)$. By (2) the bulk complex computes $\mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(\mathbb{Z}, V_R) = H^\bullet(\Gamma; V_R)$. On the boundary, ε on each component of ∂X gives a quasi-isomorphism $C_\bullet(\partial X) \to \mathbb{Z}[\Gamma/\Gamma_\infty]$ ((B1), (B2): the components are contractible and permuted transitively, with stabiliser Γ_∞), so $\mathcal{I}^\partial_\bullet$ is a free resolution of $\mathbb{Z}[\Gamma/\Gamma_\infty] = \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\infty]} \mathbb{Z}$ and computes $H^\bullet(\Gamma_\infty; V_R)$ [standard: [Bro82, Ch. III.6], Shapiro's lemma, as in the proof of Claim 16.2(2)].

(4) Injectivity in each degree: cmp is surjective and $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, V_R)$ is left exact. Base change: cmp is a map of $\mathbb{Z}[\Gamma]$-complexes and the reduction of Definition 23.1 is induced by $V \to V_R$. The cone: $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, V_R)$ applied to the filtrations of (2) — which are split in each degree as graded modules — gives filtrations of $C^\bullet_{\mathrm{eq}}(R)$ and $C^\bullet_{\mathcal{I}}(R)$ of length three, respected by cmp*, whose graded map is, summand by summand, $V_R^{\Gamma_\sigma} \to \mathrm{Hom}_{\mathbb{Z}[\Gamma_\sigma]}(F^\sigma_\bullet, V_R)$, the inclusion of $H^0(\Gamma_\sigma; V_R)$ into a complex computing $H^\bullet(\Gamma_\sigma; V_R)$ [standard: [Bro82, Ch. III.6]]. Its cone therefore has cohomology $H^t(\Gamma_\sigma; V_R)$, $t \ge 1$, which $|\Gamma_\sigma|$ annihilates [standard: [Bro82, Ch. III.10.2]]: 12 on column 0, 2 on columns 1 and 2. A three-step filtration of a complex whose graded pieces have cohomology annihilated by a_0, a_1, a_2 has cohomology annihilated by $a_0a_1a_2 = 48$ (long exact sequences). The boundary has two columns, both with stabilisers $\{\pm I\}$ ((B3)): $2 \cdot 2 = 4$.

(5) $F_\bullet \otimes_{\mathbb{Z}} C_\bullet(X) \to C_\bullet(X)$, $\varepsilon \otimes \mathrm{id}$, is surjective, and a quasi-isomorphism because $F_\bullet \to \mathbb{Z}$ is a chain homotopy equivalence of complexes of free abelian groups; its source is free (Definition 17.1(1)). Both it and cmp are surjective quasi-isomorphisms from bounded-below free complexes onto $C_\bullet(X)$, so by Lemma 23.12 below each lifts through the other strictly, over $C_\bullet(X)$. Each composite minus the identity is a chain map into the kernel of the corresponding surjection, which is acyclic; by the comparison theorem [standard: [Wei94, Thm. 2.2.6]] it is null-homotopic through a homotopy with values in that kernel, so the composites are homotopic to the identities over $C_\bullet(X)$. □

Remark 23.6 (C-A5, re-scoped; OD-T1 answered "not needed"). The extension plan's contract item C-A5 was "the concrete *refinement* of the 7-vertex triangulation (owner direction OD-T1) and a free Γ-model satisfying (G1)–(G4), its boundary, its stabiliser inventory, its $\tau/\tau_\partial/\eta$ tables' existence in every degree $\le q_{\max} + 1$, and the truncation instance of 14.24 on it". By owner ruling 6c of 2026-09-24 (the recommendation of the integral-line brief's OD-8 (a)) it is

re-scoped to **the free model on the unrefined** X, and is discharged by this chapter item by item: the model and its boundary are Definition 23.4 and Proposition 23.5; (G1)–(G4) hold for $(X, \partial X)$ by Theorem 17.3(4); the stabiliser inventory is (H4)(iv); the Hecke datum of the pair, in every degree, is Proposition 23.13, with $\eta \equiv 0$; the truncation instance is Remark 23.11. **OD-T1 is answered "not needed"**, for the reasons already written into this document: a Γ-equivariant subdivision stays 2-dimensional, preserves every stabiliser and buys nothing at $\ell \in \{2, 3\}$ (Remark 17.5(1),(4)), the degrees above 2 come from the stabiliser resolutions F^σ and not from any triangulation, and $(X, \partial X)$ already satisfies (G1)–(G4) (Theorem 17.3(4)). [proved here] Should a refinement Y' ever be wanted, it compares integrally: the subdivision is covered by Proposition 23.3(3), so $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), M) \simeq \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(Y'), M)$ for every $\mathbb{Z}[\Gamma]$-module M, and an isotropy-resolution model on Y' is related to $\mathcal{I}_\bullet$ by Proposition 23.5(5)'s argument. The owner's direction is thereby not overruled but answered: it was a design direction (Remark 7.6), and the definitions of §17 were phrased to admit a refinement; none is required.

23.3 The descent: the fracture square and glueing over $\mathbb{Z}$

Write $\widehat{\mathbb{Z}}_6 := \mathbb{Z}_2 \times \mathbb{Z}_3$ and $\widehat{\mathbb{Q}}_6 := \widehat{\mathbb{Z}}_6[1/6] = \mathbb{Q}_2 \times \mathbb{Q}_3$. The object the integral line computes is the group cohomology $H^\bullet(\Gamma; V)$, i.e. the cohomology of $C^\bullet_{\mathcal{I}}(\mathbb{Z})$, a complex of finitely generated free $\mathbb{Z}$-modules, bounded below and *not* bounded above (Remark 23.11).

Proposition 23.7 (The arithmetic fracture square)**.** *[proved here] The sequence of abelian groups*

$$0 \longrightarrow \mathbb{Z} \xrightarrow{a \mapsto (a,a)} \mathbb{Z}[1/6] \oplus \widehat{\mathbb{Z}}_6 \xrightarrow{(a,b) \mapsto a-b} \widehat{\mathbb{Q}}_6 \longrightarrow 0$$

is exact. Consequently, for every cochain complex M of abelian groups, $0 \to M \to M[1/6] \oplus (M \otimes \widehat{\mathbb{Z}}_6) \to M \otimes \widehat{\mathbb{Q}}_6 \to 0$ is an exact sequence of complexes, and M is quasi-isomorphic to the homotopy fibre product (the mapping cocone) of $M[1/6] \to M \otimes \widehat{\mathbb{Q}}_6 \leftarrow M \otimes \widehat{\mathbb{Z}}_6$ by the canonical map: the square is a homotopy pullback.

Proof. Injectivity is clear. The middle: $a = b$ in $\mathbb{Q}_2$ and in $\mathbb{Q}_3$ with $a \in \mathbb{Z}[1/6]$, $b \in \mathbb{Z}_2 \times \mathbb{Z}_3$ forces $a \in \mathbb{Z}[1/6] \cap \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)} = \mathbb{Z}$. Surjectivity: given $(x, y) \in \mathbb{Q}_2 \times \mathbb{Q}_3$, let $\{x\}_2 \in \mathbb{Z}[1/2]$ and $\{y\}_3 \in \mathbb{Z}[1/3]$ be their principal parts (so $x - \{x\}_2 \in \mathbb{Z}_2$, $y - \{y\}_3 \in \mathbb{Z}_3$) and put $a := \{x\}_2 + \{y\}_3 \in \mathbb{Z}[1/6]$; then $a - x = \{y\}_3 - (x - \{x\}_2) \in \mathbb{Z}_2$, since $\mathbb{Z}[1/3] \subset \mathbb{Z}_2$, and symmetrically $a - y \in \mathbb{Z}_3$, so $(a, (a - x, a - y)) \mapsto (x, y)$. $\widehat{\mathbb{Q}}_6$ is flat over $\mathbb{Z}$ (torsion-free over a principal ideal domain), so $\mathrm{Tor}_1(M^q, \widehat{\mathbb{Q}}_6) = 0$ and the sequence stays exact after $M^q \otimes -$. For a short exact sequence of complexes $0 \to M \to A \to C \to 0$ the canonical map from M to the cocone of $A \to C$ is a quasi-isomorphism [standard: [Wei94, §1.5]: the cone of a map and the long exact sequence]. □

Proposition 23.8 (Glueing at the cohomology level is exact over $\mathbb{Z}$)**.** *[proved here] Let M be a cochain complex of abelian groups. For every q the sequence*

$$0 \to H^q(M) \to H^q(M)[1/6] \oplus \big(H^q(M) \otimes \widehat{\mathbb{Z}}_6\big) \to H^q(M) \otimes \widehat{\mathbb{Q}}_6 \to 0$$

is exact, the middle terms being $H^q(M[1/6])$ and $H^q(M \otimes \widehat{\mathbb{Z}}_6)$; so $H^q(M)$ is the fibre product of $H^q(M[1/6])$ and $H^q(M \otimes \widehat{\mathbb{Z}}_6)$ over $H^q(M \otimes \widehat{\mathbb{Q}}_6)$. The connecting maps of the long exact sequence of Proposition 23.7 vanish. For a chain map $f\colon M \to N$, $H^q(f)$ is determined by $H^q(f[1/6])$ and $H^q(f \otimes \widehat{\mathbb{Z}}_6)$.

Proof. $\mathbb{Z}[1/6]$, $\widehat{\mathbb{Z}}_6$ and $\widehat{\mathbb{Q}}_6$ are flat, so $H^q(M \otimes A) = H^q(M) \otimes A$ for each. Tensoring the exact sequence of Proposition 23.7 with any abelian group G stays exact on the left, because $\mathrm{Tor}_1(G, \widehat{\mathbb{Q}}_6) = 0$; with $G = H^q(M)$ this is the displayed sequence, and it shows that the map $H^q(M) \to H^q(M[1/6]) \oplus H^q(M \otimes \widehat{\mathbb{Z}}_6)$ in the long exact sequence is injective, so every connecting map vanishes. A short exact sequence $0 \to G \to A \oplus B \xrightarrow{(a,b) \mapsto a-b} C \to 0$ exhibits G as $A \times_C B$. For $f\colon H^q(N) \to H^q(N[1/6]) \oplus H^q(N \otimes \widehat{\mathbb{Z}}_6)$ is injective and $H^q(f)$ commutes with it. □

Proposition 23.8 is [IB]'s Prop. 2.4 with a hypothesis removed: the proof uses flatness only. Finite generation of $H^q(M)$ is what identifies $M\otimes\widehat{\mathbb{Z}}_6$ with the 6-adic completion and $H^q(M)\otimes\mathbb{Z}_\ell$ with $\varprojlim_e H^q(M\otimes\mathbb{Z}/\ell^e)$, which is how the local runs reach the $\widehat{\mathbb{Z}}_6$-corner (Remark 23.11); the exactness does not need it.

Proposition 23.9 (The owner's descent, realised by one integral map)**.** *[proved here] Assume (H4) and (B1)–(B3), let $n\geq 0$, and put*

$$A := C^\bullet_{\mathrm{eq}}(\mathbb{Z})[1/6] = C^\bullet_{\mathrm{eq}}(\mathbb{Z}[1/6]), \qquad B := C^\bullet_{\mathcal{I}}(\mathbb{Z})\otimes\widehat{\mathbb{Z}}_6, \qquad C := C^\bullet_{\mathcal{I}}(\mathbb{Z})\otimes\widehat{\mathbb{Q}}_6,$$

with $\alpha\colon A\to C$ the composite of $\mathrm{cmp}^*[1/6]$ and the canonical $C^\bullet_{\mathcal{I}}(\mathbb{Z})[1/6]\to C$, and $\beta\colon B\to C$ the canonical map.

1. *The mapping cocone of $\alpha-\beta\colon A\oplus B\to C$ is quasi-isomorphic to $C^\bullet_{\mathcal{I}}(\mathbb{Z})$, hence computes $H^\bullet(\Gamma;V)$.*
2. *For every q, $H^q(\Gamma;V)\cong H^q(A)\times_{H^q(C)}H^q(B)$, the fibre product being taken along $H^q(\alpha)=H^q(\mathrm{cmp}^*)\otimes\widehat{\mathbb{Q}}_6$ and $H^q(\beta)$.*

The only glueing datum is cmp^**: the comparison isomorphisms over $\mathbb{Q}_2$ and $\mathbb{Q}_3$ are its base changes, and nothing further is chosen or certified beyond* cmp^* *being a cochain map (Proposition 23.5(4)).*

Proof. $C^\bullet_{\mathrm{eq}}(\mathbb{Z})[1/6]=C^\bullet_{\mathrm{eq}}(\mathbb{Z}[1/6])$ is Proposition 23.2(1). By Proposition 23.5(4) over $\mathbb{Z}[1/6]$, $\mathrm{cmp}^*[1/6]\colon A\to C^\bullet_{\mathcal{I}}(\mathbb{Z})[1/6]$ is a quasi-isomorphism, so $(\mathrm{cmp}^*[1/6],\mathrm{id},\mathrm{id})$ is a map from the diagram $(A\to C\leftarrow B)$ to $(C^\bullet_{\mathcal{I}}(\mathbb{Z})[1/6]\to C\leftarrow B)$ that is a quasi-isomorphism on each corner; the induced map of cocones is a quasi-isomorphism by the five lemma on their long exact sequences. The second cocone is quasi-isomorphic to $C^\bullet_{\mathcal{I}}(\mathbb{Z})$ by Proposition 23.7, and $H^\bullet(C^\bullet_{\mathcal{I}}(\mathbb{Z}))=H^\bullet(\Gamma;V)$ by Proposition 23.5(3). (2) is (1) together with Proposition 23.8 applied to $C^\bullet_{\mathcal{I}}(\mathbb{Z})$, transported along the quasi-isomorphism $\mathrm{cmp}^*[1/6]$. □

Proposition 23.10 (The bulk carries no coherence datum: the term Hom^{-1} vanishes)**.** *[proved here] Assume (H4) and (B1)–(B3), let $n\geq 0$ and put $N:=C^\bullet_{\mathcal{I}}(\mathbb{Z})\otimes\widehat{\mathbb{Q}}_6$, a complex over the product of fields $\widehat{\mathbb{Q}}_6$.*

1. *$H^\bullet(N)=H^\bullet(\Gamma;V)\otimes\widehat{\mathbb{Q}}_6$ is concentrated in one degree: in degree 1 for $n>0$ even, in degree 0 for $n=0$, and it is zero for n odd.*
2. *Hence $\mathrm{Hom}_{D(\widehat{\mathbb{Q}}_6)}(N,N[-i])=0$ for every $i\geq 1$.*

Proof. (1) $\widehat{\mathbb{Q}}_6$ is flat, so $H^q(N)=H^q(\Gamma;V)\otimes\widehat{\mathbb{Q}}_6$, which vanishes when $H^q(\Gamma;V)$ is torsion: for $q\geq 2$ it is annihilated by 24 (Claim 16.2(4) with $R=\mathbb{Z}$); for n odd it is annihilated by 2 in every degree ($-I$ is central, so conjugation by it acts trivially on $H^\bullet(\Gamma;V)$ [standard: [Bro82, Ch. III, §8]: an inner automorphism acts trivially on cohomology], while it acts by -1 on V — the argument of Remark 7.7); for $n>0$, $H^0(\Gamma;V)=0$ (the argument of Claim 16.7(1), which uses no hypothesis on ℓ); for $n=0$, $H^1(\Gamma;\mathbb{Z})=\mathrm{Hom}(\Gamma,\mathbb{Z})=0$, Γ being generated by the elements S, U of finite order. (2) A complex whose cohomology sits in a single degree d is isomorphic in the derived category to that cohomology placed in degree d (by the two truncations at d), so $\mathrm{Hom}_{D(\widehat{\mathbb{Q}}_6)}(N,N[-i])=\mathrm{Ext}^{-i}_{\widehat{\mathbb{Q}}_6}(H^d,H^d)=0$ for $i\geq 1$ [standard: [Wei94, §10.7]: $\mathrm{Hom}_{D(A)}(X,Y[i])=\mathrm{Ext}^i_A(X,Y)$ for modules X,Y, which is 0 for $i<0$]; for n odd, N is acyclic. □

Reading. (2) is the vanishing of the one term, Hom^{-1} over $\widehat{\mathbb{Q}}_6$, in which [IB] §2.3(c) locates the coherence datum a descent could carry: with it zero, a morphism of glued objects is determined by its two local components and relations among local Hecke operators hold globally. At the cohomology level, which is all the bulk needs, no derived category is required: that is Proposition 23.8's morphism clause, since $H^q(\Gamma;V)$ injects into the product of the two local cohomologies. **The datum is real for the pair**: there the corresponding term contains

$\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0) \otimes \mathbb{Q} \neq 0$, which is Proposition 15.16's ambiguity, present in every model and removed on the free model by the strict datum (Proposition 23.13).

Remark 23.11 (The truncation is a computation device, and the object is not perfect). [proved here] **What is glued is the untruncated** $C^\bullet_{\mathcal{I}}(\mathbb{Z})$, in Propositions 23.7–23.10; none of them needs a bounded complex. $C^\bullet_{\mathcal{I}}(\mathbb{Z})$ is not perfect over $\mathbb{Z}$ in general: at $n = 0$, $H^{2j}(\Gamma; \mathbb{Z}) \cong \mathbb{Z}/12$ for every $j \geq 1$ (Lemma 14.22(1),(2)), so its cohomology is unbounded. **What a local run builds is the brutal truncation** $\sigma^{\leq q_{\max}+1} C^\bullet_{\mathcal{I}}$ — the instance, on $\mathcal{I}_\bullet$, of 14.24's box, which is the "truncation instance" C-A5 asked for: built in degrees $\leq q_{\max} + 1$, reported in degrees $\leq q_{\max}$ by the truncation lemma there, `truncated = true`, $q_{\max}$ a set parameter (rider (i)). Brutal truncation commutes strictly with base change, so the runs over $\mathbb{Z}/\ell^e$ are reductions of one integral truncated complex, and by Lemma 14.22(2),(3) nothing is lost for $q_{\max} \geq 3$: every degree ≥ 2 is H^2 or H^3, Hecke-equivariantly (14.24's non-normative recommendation $q_{\max} \geq 4$, and ≥ 5 to see both periodicity instances, stands). For the $\widehat{\mathbb{Z}}_6$-corner: $H^q(C^\bullet_{\mathcal{I}}(\mathbb{Z}))$ is finitely generated, so $H^q(C^\bullet_{\mathcal{I}}(\mathbb{Z}) \otimes \mathbb{Z}_\ell) = H^q(\Gamma; V) \otimes \mathbb{Z}_\ell$ is determined by the $\mathbb{Z}_{(\ell)}$-type of $H^q(\Gamma; V)$, which one certified run over $\mathbb{Z}/\ell^e$ yields (Proposition 23.17); no inverse limit is computed.

23.4 The Hecke datum on the free model: a strict lift

Lemma 23.12 (Strict lifting along a surjective quasi-isomorphism). *[proved here] Let G be a group, $P_\bullet$ a complex of free $\mathbb{Z}[G]$-modules with $P_q = 0$ for $q < 0$, $p\colon E_\bullet \to C_\bullet$ a surjective quasi-isomorphism of complexes of $\mathbb{Z}[G]$-modules, $P' \subseteq P$ a subcomplex spanned by part of a basis in each degree, $E' \subseteq E$ a subcomplex, and $f\colon P \to C$ a chain map. Given a chain map $f'\colon P' \to E'$ with $pf' = f|_{P'}$, there is a chain map $\hat{f}\colon P \to E$ with $p\hat{f} = f$ and $\hat{f}|_{P'} = f'$ — a* strict *lift, not one up to homotopy.*

Proof. $\ker p$ is acyclic, by the long exact sequence of $0 \to \ker p \to E \to C \to 0$. Define $\hat{f}$ on the basis elements outside P' by induction on the degree. For such an x in degree q, choose $y_0 \in E_q$ with $p(y_0) = f(x)$ (p is onto). Then $z := \hat{f}(\partial x) - \partial y_0$ has $p(z) = f(\partial x) - \partial f(x) = 0$ and $\partial z = 0$, so z is a cycle of the acyclic $\ker p$ and $z = \partial k$ with $k \in (\ker p)_q$ (for $q = 0$ the same step applies; in the applications below $E_{-1} = 0$ and $z = 0$). Put $\hat{f}(x) := y_0 + k$; then $p\hat{f}(x) = f(x)$ and $\partial \hat{f}(x) = \hat{f}(\partial x)$. Extend G-linearly. □

Proposition 23.13 (The strict lift of the strict pair datum). *[proved here] Assume (H4) and (B1)–(B3); let $g \in \Delta$, $\Gamma'' = \Gamma \cap g\Gamma g^{-1}$, τ_∂ a boundary g-twisted chain datum over $\mathbb{Z}$ (Lemma 14.10(1)), τ^s the strict datum of Proposition 15.13 for it, and $(\mathcal{I}_\bullet, \mathrm{cmp})$ an isotropy-resolution model.*

1. *There is an additive map $\tau_{\mathcal{I}}\colon \mathcal{I}_\bullet \to \mathcal{I}_\bullet$, defined in every degree, satisfying (T1)–(T3) of Definition 11.1 with $\mathcal{I}_\bullet$ in place of $C_\bullet(K^u)$ and $\varepsilon \circ \mathrm{cmp}$ as augmentation, with*

$$\mathrm{cmp} \circ \tau_{\mathcal{I}} = \tau^s \circ \mathrm{cmp} \quad \textit{strictly}, \qquad \tau_{\mathcal{I}}(\mathcal{I}^\partial_\bullet) \subseteq \mathcal{I}^\partial_\bullet.$$

   Its restriction $\tau^\partial_{\mathcal{I}}$ to $\mathcal{I}^\partial_\bullet$ satisfies $\mathrm{cmp} \circ \tau^\partial_{\mathcal{I}} = \tau_\partial \circ \mathrm{cmp}$, and $\eta \equiv 0$ is a pair homotopy for $(\tau_{\mathcal{I}}, \tau^\partial_{\mathcal{I}})$: this is a Hecke datum of the pair on the free model $\mathcal{I}_\bullet$ — Definition 17.1(3) with $\mathcal{I}_\bullet$, $\mathcal{I}^\partial_\bullet$ in place of $\widehat{C}_\bullet(Y)$, $\widehat{C}_\bullet(\partial Y)$ — in every degree.
2. *Two such lifts are homotopic through g-twisted maps over $\mathbb{Z}$, for every ℓ (Remark 17.2, whose argument applies by Proposition 23.5(5)).*
3. *Writing Θ^s_g for the coset transport of τ^s on $C^\bullet_{\mathrm{eq}}(R)$ and $\Theta_{\mathcal{I},g}$ for that of $\tau_{\mathcal{I}}$ on $C^\bullet_{\mathcal{I}}(R)$ — formula (41) verbatim — one has, over every commutative R,*

$$\mathrm{cmp}^* \circ \Theta^s_g = \Theta_{\mathcal{I},g} \circ \mathrm{cmp}^* \quad \textit{on the nose},$$

   and $\Theta_{\mathcal{I},g}$ induces $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ on $H^\bullet(\Gamma; V_R)$ for every R, in every degree.

4. *On cohomology over any R with $6 \in R^\times$, Θ_g^s and the shipped Θ_g (the coset transport of the canonical τ of Algorithm 18.3) induce the same operator.* ***At $\ell \in \{2,3\}$ they are not compared****: on the Bredon object $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e)$ they may differ, and no statement of this chapter reads them there.*

Existence only. *The effective, certified construction of $\tau_{\mathcal{I}}$ in degrees $\le q_{\max}+1$ — with a (T1)-bookkeeping over Γ'' and the strict identity as certificates — is the integral line's key risk (K1 of [IB]) and a code matter; nothing here bounds its integer sizes or asserts it is built. (v1.10.0: the statement is unchanged; what hecke-extensions slot 10 built and certified, for $g = \mathrm{diag}(1,p)$, $p \in \{2,3,5\}$, in degrees $\le q_{\max}+1 = 6$, and the integer sizes it measured, are Remark 23.14, which is evidence and bounds nothing either.)*

Proof. Let $\mathcal{I}^{(c)}$, $C_\bullet^{(c)}(X)$ be the same complexes with the twisted Γ''-action $h * x := (g^{-1}hg)\cdot x$ ($g^{-1}hg \in \Gamma$ by Proposition 10.1(3)); (T1) for a map $\mathcal{I} \to \mathcal{I}$ is $\mathbb{Z}[\Gamma'']$-linearity $\mathcal{I} \to \mathcal{I}^{(c)}$. As $[\Gamma : \Gamma''] < \infty$, $\mathcal{I}_q$ is free over $\mathbb{Z}[\Gamma'']$ on a basis that restricts to one of $\mathcal{I}_q^\partial$, and $\mathrm{cmp}\colon \mathcal{I}^{(c)} \to C^{(c)}(X)$ is still a Γ''-equivariant surjective quasi-isomorphism, as is its boundary restriction. (1) First apply Lemma 23.12 to $P = \mathcal{I}^\partial$, $E = (\mathcal{I}^\partial)^{(c)}$, $C = C^{(c)}(\partial X)$, $f = \tau_\partial \circ \mathrm{cmp}$ (with $P' = 0$), obtaining $\tau_{\mathcal{I}}^\partial$; then to $P = \mathcal{I}$, $E = \mathcal{I}^{(c)}$, $C = C^{(c)}(X)$, $f = \tau^s \circ \mathrm{cmp}$, $P' = \mathcal{I}^\partial$, $E' = (\mathcal{I}^\partial)^{(c)}$ and $f' = \tau_{\mathcal{I}}^\partial$, which is admissible because $\mathrm{cmp}\tau_{\mathcal{I}}^\partial = \tau_\partial \mathrm{cmp} = \tau^s \mathrm{cmp}$ on $\mathcal{I}^\partial$ ($\tau^s \circ j = j \circ \tau_\partial$). (T2) holds because the lift is a chain map; (T3): $\varepsilon \mathrm{cmp}\tau_{\mathcal{I}} = \varepsilon\tau^s\mathrm{cmp} = \varepsilon\mathrm{cmp}$ by (T3) for τ^s. The pair identity with $\eta = 0$ is $\tau_{\mathcal{I}} \circ j = j \circ \tau_{\mathcal{I}}^\partial$, which holds by construction. (2) The difference of two lifts is a (T1)-chain map from the free $\mathbb{Z}[\Gamma'']$-complex $\mathcal{I}$ into $\ker \mathrm{cmp}$, which is acyclic; the fundamental lemma [standard: [Wei94, Thm. 2.2.6]] makes it null-homotopic through (T1)-maps. (3) For $\phi \in \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), V_R)$ and $x \in \mathcal{I}$, $(\Theta_{\mathcal{I},g}\mathrm{cmp}^*\phi)(x) = \sum_j \rho_n(\delta_j)\,\phi(\mathrm{cmp}\,\tau_{\mathcal{I}}(s_j x)) = \sum_j \rho_n(\delta_j)\,\phi(\tau^s(s_j\,\mathrm{cmp}x)) = (\mathrm{cmp}^*\Theta_g^s\phi)(x)$, using $\mathrm{cmp}\tau_{\mathcal{I}} = \tau^s\mathrm{cmp}$ and the Γ-equivariance of cmp. The coset transport of Lemma 15.4(1),(2) applies verbatim on $\mathcal{I}$ (its proof used (T1) and $\partial(s_j y) = s_j \partial y$ only), and on the free resolution $\mathcal{I}$ the operator is $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ by Remark 17.2 (with Proposition 23.5(5)). (4) τ^s and the canonical τ are g-twisted chain maps over $\mathbb{Z}$, homotopic through twisted maps when $6 \in R^\times$ (Lemma 11.3(2)), so they induce the same operator on $H^\bullet_{\mathrm{eq}}$ (Theorem 11.7(b)). □

Lifting the *strict* datum, boundary summands first, is what makes the lifted pair homotopy vanish; lifting the shipped τ instead would work the same way for the bulk and leave a boundary datum to certify ([IB] §3.2 item 2, the fallback).

Remark 23.14 (The strict lift as built (hecke-extensions slot 10)). [verified computationally]^{r} (v1.10.0.) For $g = \mathrm{diag}(1,p)$, $p \in \{2,3,5\}$ — $\Gamma'' = \Gamma_0(p)$, $m = p+1$ right cosets $\Gamma'' s_j$ with $s_1 = I$ — the map $\tau_{\mathcal{I}}$ of Proposition 23.13(1) is built in degrees $0, \dots, q_{\max}+1 = 6$ on the model $\mathcal{I}_\bullet$ of Definition 23.4, and certified exactly over $\mathbb{Z}[\Gamma'']$ (`Integral/Model/StrictLift`, hecke-extensions slot-10 WP-10; registered `HeckeStrictLiftValidation`, `standard`, 1824 checks, 0 failing; `docs/implementation/slot10/WP-10_report.md`). The fallback of the paragraph above was not needed.

1. *Construction* — Lemma 23.12's induction, degree by degree, boundary summands first, on the free $\mathbb{Z}[\Gamma'']$-basis $\{s_j e_c\}$, extended by (T1). The preimage y_0 is *read off*, not solved for: at a cell with $t = 0$, cmp is the augmentation of free generators, so the stored chain $\tau^s(s_j\,\mathrm{cmp}x)$ is rewritten term by term as an element of $\mathcal{I}_q$; at $t \ge 1$, $y_0 = 0$. The cycle z of $\ker \mathrm{cmp}$ is contracted *by descent through the columns*: $\ker \mathrm{cmp}$ is the sum of the augmented periodic columns $F_\bullet^\sigma$, whose closed-form $\mathbb{Z}$-linear contracting homotopy h (the one Definition 23.4's differentials are built from) gives $k_s = (-1)^s h(z_s)$ on the top column, and the rest is pushed into lower columns. No linear system is solved; the result is deterministic.
2. *Certified*, re-derived from the stored values and trusting nothing from the construction: (T1) on the Schreier generators of Γ'' (on a free $\mathbb{Z}[\Gamma'']$-basis (T1) is the *defining* property of the extension, so this certifies its bookkeeping); (T2) and (T3); the strict identity

$\mathrm{cmp} \circ \tau_{\mathcal{I}} = \tau^s \circ \mathrm{cmp}$, compared as canonical chains of X through a routine independent of the construction's conversion; $\tau_{\mathcal{I}}(\mathcal{I}^{\partial}_{\bullet}) \subseteq \mathcal{I}^{\partial}_{\bullet}$; and $\mathrm{cmp} \circ \tau^{\partial}_{\mathcal{I}} = \tau_{\partial} \circ \mathrm{cmp}$ against the boundary table's own chains — so $\eta \equiv 0$ is a pair homotopy, as (1) states. A perturbed entry is refused.

3. *(3) in use.* $\Theta_{\mathcal{I},g}$ is a cochain map over $\mathbb{F}_\ell$, $\mathbb{Z}/\ell^k$ and at the 64-bit word $(\mathbb{Z}/2^{40}, \mathbb{Z}/3^{30})$, on 165 and 24 cells (Lemma 15.4(2) on the model). The identity $\mathrm{cmp}^* \circ \Theta^s_g = \Theta_{\mathcal{I},g} \circ \mathrm{cmp}^*$ holds entrywise on $C^{\bullet}_{\mathrm{eq}}(R)$ in degrees $q \le 2$, $n \le 20$, on eight rings among those with a shipped Θ^s_g (v1.10.1): $\mathbb{F}_\ell$, $\ell \in \{2,3,5,7\}$, and $\mathbb{Z}/\ell^k$, $\ell \in \{5,7\}$, $k \in \{2,3\}$ (359 cells; v1.10.1, in place of v1.10.0's arithmetic slip "349") — a consistency check of the transport code, since the identity is a theorem. Over $\mathbb{Z}/\ell^e$ at $\ell \in \{2,3\}$ no Θ^s_g exists in the build (Remark 23.15) and the identity is not separately checked; it holds there by the proof of (3), a formal computation valid over every commutative R, whose three inputs are certified over $\mathbb{Z}$ in item 2 or hold by construction: the strict identity, (T1) of $\tau_{\mathcal{I}}$ (which makes the coset transport well typed, Lemma 15.4(1)) and the Γ-equivariance of cmp.
4. *(4) in use.* At $\ell \ge 5$ the operator $\Theta_{\mathcal{I},g}$ induces on H^1, transported by cmp^*, equals the shipped $A^{(1)}(T_p)$ — built on the *canonical* τ of Algorithm 18.3, not on τ^s — entrywise on 26 cells, so (4) is exercised through Lemma 11.3(2); and $\mathrm{char}(\Theta_{\mathcal{I},g} \,|\, H^1)$, with H^1 taken on the model alone, equals the Eichler–Shimura values of the design brief (its oracle O5) modulo ℓ at $n \in \{10, \dots, 20\}$ (21 cells).
5. *Integer sizes* — the first measurement of [IB]'s key risk K1: largest coefficient 3, 2, 2 and largest matrix entry p for $p = 2, 3, 5$; at most 26 terms in one value; 20, 29 and 64 distinct group elements; the term counts stationary with period 2 from degree 2 on, as the periodic columns suggest. Nothing approaches the word; no `__int128` and no big integer occurs (Remark 23.22(1)).

What this remark does not say. It bounds nothing: Proposition 23.13's "existence only" stands as a statement, and no effectivity is claimed for any other g — in particular not for the double cosets of determinant p^2 that T_{p^2} and $T(p,p)$ need (Theorem 10.8) — nor in degrees above 6 (Remark 23.22(2)). *Later builds* (v1.23.0-pre.28; this remark stays the record of what slot 10 built): hecke-extensions slot 24, milestone D4, builds the same lift for $g = \mathrm{diag}(1,p)$ at **every prime** p requested, on demand, in degrees ≤ 6, by item 1's procedure, and certifies it by item 2's checks, the boundary items included, at run time before first use (Proposition 31.21(1), its code map (f)); measured to $p = 127$ (largest coefficient ≤ 3, largest matrix entry p; `docs/implementation/slot24/DESCENT4_report.md` §4). That too is evidence and bounds nothing; effectivity at a given p is the run-time certificate, refused on failure or at the cap (the full list of refusals: Proposition 31.21(5), v1.23.0-pre.29).

Remark 23.15 (The Hecke operator on the Bredon object at $\ell \in \{2,3\}$ is not compared). (v1.10.0; owner ruling 4 (A) of the hecke-extensions slot-10 decisions, 2026-09-25.) Assume (H4), $\ell \in \{2,3\}$ and $R = \mathbb{Z}/\ell^e$, $e \ge 1$. On the Bredon object $C^{\bullet}_{\mathrm{eq}}(R)$ of Definition 23.1(3) two Hecke objects could be read: *(a)* the coset transport (41) of an admissible integral τ — the canonical one of Algorithm 18.3, or the strict τ^s — a cochain map on $C^{\bullet}_{\mathrm{eq}}(R)$ (v1.10.1: the canonical τ is integral in practice, not by Algorithm 18.3's construction, which records a mod-ℓ triangle-fill fallback in its step (3); since v1.1.0 it vanishes in degree 2, Remark 9.5, consequence (a), so no fill is solved); and *(b)* the operator it induces on $H^{\bullet}_{\mathrm{eq}}$. This document says nothing about (b) beyond Proposition 23.13(4): homotopy uniqueness of admissible τ (Lemma 11.3(2)) needs $6 \in R^\times$, which fails here, so the operator in (b) is **not shown to be independent of the choice of** τ, and this document **compares none of these operators there** — neither Θ^s_g with the canonical Θ_g, nor either with $\Theta_{\mathcal{I},g}$, whose cohomology is $H^{\bullet}(\Gamma; V_R)$ (Proposition 23.5(3)), a different object at $\ell \in \{2,3\}$ (Proposition 23.2). (v1.23.0-pre.25, the C-D4 referee's M2: at level N with $\nu_3(N) = 0$ and over $\mathbb{Z}_{(3)}$-algebras the Bredon object does compute $H^{\bullet}(\Gamma_0(N); V_R)$, and Proposition 31.13(6)(b) identifies the operator of a datum over $\mathbb{Z}$ certified by (T1)–(T3); the shipped canonical τ is not so certified, and §31.3 reads neither

— its 3-local route is Proposition 31.13(5). The level-1 statement of this remark is untouched: $\mathrm{SL}_2(\mathbb{Z})$ has elements of order 3.) (v1.10.1: the one comparison it does record there is of another pair, the legacy and the canonical τ, which induce the same $A^{(q)}$ at every $\ell \in \{2,3\}$ golden point — an observation, not a theorem, Remark 11.4(1).) This is the **absence of a proof of independence, not a proof of dependence**: no two admissible τ with different operators on the Bredon object are known, and none is claimed. What the build does accordingly: at $e = 1$ the shipped $\mathbb{F}_\ell$ pipeline computes (a) and (b) for the *canonical* τ — the canonical τ of Algorithm 18.3 since v1.1.0 (Remark 9.5; v1.10.1, in place of v1.10.0's "unchanged since v1.0.0", which is false of τ and of (a)); its operator (b) did not move in the v1.1.0 re-baseline and equals the legacy τ's at every $\ell \in \{2,3\}$ golden point, Remark 11.4(1) — an observation, not a theorem. It is a definite object, the canonical datum's operator, which no statement of this document identifies with anything τ-independent at $\ell \in \{2,3\}$; at $e \geq 2$ neither is built, by the companion specification's rule ([E, `sec:ring`] (b3)) — (a) is not needed, since the integral line reads T_p through $\Theta_{\mathcal{I},g}$ on $C^\bullet_{\mathcal{I}}(R)$ (Remark 23.14), and (b) is refused by name. Should two admissible τ with different operators in (b) ever be exhibited, a later version records that here in place of this remark's reading.

23.5 Bad primes, the certified exponent, and the characteristic polynomial over $\mathbb{Z}$

Proposition 23.16 (The bad primes form a finite, closed-form list)**.** *[proved here] Let $n \geq 0$ and $V = \mathrm{Sym}^n(\mathbb{Z}^2)$.*

1. *For $n > 0$ even, every prime ℓ at which $H^\bullet(\Gamma; V)$ has ℓ-torsion lies in*

$$\{2,3\} \cup \big\{\, 5 \leq \ell \leq n-1 : \ \#\{(a,b) \in \mathbb{Z}^2_{\geq 0} : a(\ell+1) + b(\ell^2-\ell) = n\} > 0 \,\big\},$$

   and at each $\ell \geq 5$ of the list the ℓ-primary torsion lies in H^1 and has exactly that many cyclic summands.
2. *For n odd, $2 \cdot H^q(\Gamma; V) = 0$ for every q; for $n = 0$ the torsion primes are 2 and 3.*

Proof. (1) For $q \geq 2$, $24 \cdot H^q(\Gamma; V) = 0$ (Claim 16.2(4)); $H^0 = 0$ (Claim 16.7(1)). For $\ell \geq 5$, $H^1(\Gamma; V)[\ell] \cong H^0(\Gamma; M_n)$ (Claim 16.7(2)), whose dimension is the displayed count (Claim 16.5(2), proved in place since v1.23.0-pre.15); the number of cyclic ℓ-primary summands of a finitely generated group is the dimension of its ℓ-torsion over $\mathbb{F}_\ell$, and the count vanishes for $n < \ell + 1$. (2) The centrality argument in the proof of Proposition 23.10(1); at $n = 0$, $H^1(\Gamma; \mathbb{Z}) = 0$ and Claim 16.2(4) confine the torsion to 2 and 3, and $H^2(\Gamma; \mathbb{Z}) \cong \mathbb{Z}/12$ (Lemma 14.22(1)) exhibits both. □

Exponents are *not* bounded by this list. At $\ell \geq 5$ they grow along $n = \ell^j m$ (Remark 16.8, whose lower-bound argument uses no hypothesis on ℓ); at 2 and 3 the integral-line probe (a computation outside the tree, [verified computationally][a], [IB] Appendix A) finds $\mathbb{Z}/2^j$ in H^1 at $n = 2^j$, $j \leq 4$, and $\mathbb{Z}/9$ at $n = 12, 18$. No fixed exponent is safe at any bad prime, which is what the next proposition is for.

Proposition 23.17 (The certified-exponent rule, at every prime, 2 and 3 included)**.** *[proved here] Let $C^\bullet$ be a complex of finitely generated free $\mathbb{Z}$-modules, $q_{\max} \geq 0$, ℓ a prime and $e \geq 1$. For an integer matrix D let $\ell^{a_1}, \ldots, \ell^{a_r}$ ($r = \mathrm{rank}_{\mathbb{Q}} D$) be its non-zero elementary divisors over $\mathbb{Z}_{(\ell)}$, and let $r_e(D)$ be the number of diagonal entries ℓ^a with $a < e$ in the Smith form of D mod ℓ^e over $\mathbb{Z}/\ell^e$.*

1. *$r_e(D) = \#\{t : a_t < e\} \leq \mathrm{rank}_{\mathbb{Q}} D$, with equality iff every $a_t < e$.*
2. *For each q, $\mathrm{tors}\, H^q(C)_{(\ell)} \cong \bigoplus_t \mathbb{Z}/\ell^{a_t}$ over the non-unit elementary divisors ℓ^{a_t} of d^{q-1}, and $\mathrm{rank}\, H^q(C) = \mathrm{rank}\, C^q - \mathrm{rank}_{\mathbb{Q}}\, d^q - \mathrm{rank}_{\mathbb{Q}}\, d^{q-1}$.*

3. ***The rule.*** *If $r_e(d^q) = \operatorname{rank}_{\mathbb{Q}} d^q$ for every $q \le q_{\max}$ (with $d^{-1} = 0$), then for every $q \le q_{\max}$ the $\mathbb{Z}_{(\ell)}$-type of $H^q(C)$ is read off exactly from the Smith forms over $\mathbb{Z}/\ell^e$, every exponent being $< e$, and $H^q(C \otimes \mathbb{Z}/\ell^f) \cong H^q(C) \otimes \mathbb{Z}/\ell^f \oplus \operatorname{Tor}(H^{q+1}(C), \mathbb{Z}/\ell^f)$ is thereby known for* every *$f \ge 1$. If the equality fails for some $q \le q_{\max}$, e is too small: a summand $\mathbb{Z}/\ell^a$ with $a \ge e$ is indistinguishable from a free one mod ℓ^e.*
4. ***Where*** $\operatorname{rank}_{\mathbb{Q}}$ ***comes from.*** *$\operatorname{rank}_{\mathbb{Q}} d^q = \operatorname{rank}_{\mathbb{F}_{\ell'}} d^q$ at any prime ℓ' not dividing an elementary divisor of d^q, i.e. at which $H^{q+1}(C)$ has no ℓ'-torsion. For $C = C^\bullet_{\mathcal{I}}(\mathbb{Z})$ and for $C = C^\bullet_{\mathrm{eq}}(\mathbb{Z})$, every prime $\ell' \ge 5$ with $\ell' > n+1$ qualifies (Proposition 23.16, and Proposition 23.5(4) for $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$, whose cohomology agrees with group cohomology after inverting 6).*
5. ***Scope.*** *The rule applies to $\sigma^{\le q_{\max}+1} C^\bullet_{\mathcal{I}}(\mathbb{Z})$ at every ℓ (its differentials d^q, $q \le q_{\max}$, are those of $C^\bullet_{\mathcal{I}}(\mathbb{Z})$), and to $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$ at every ℓ — at $\ell \ge 5$, where $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e) = C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes \mathbb{Z}/\ell^e$, it is [L]'s stopping rule. It does* ***not*** *apply to the model complex $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e)$ at $\ell \in \{2,3\}$: that complex is not the reduction of an integral complex (Proposition 23.2(2)), carries no integral lattice to certify, and is checked against the tree complex (65) over the same ring instead.*

Proof. (1) Over the local principal ideal domain $\mathbb{Z}_{(\ell)}$, $D = U \operatorname{diag}(\ell^{a_t}, 0)\, V$ with U, V invertible; reducing mod ℓ^e keeps U, V invertible, so the Smith form of D mod ℓ^e has diagonal $\ell^{\min(a_t, e)}$, where $\ell^e = 0$. (2) If $x \in C^q$ and $mx \in \operatorname{im} d^{q-1}$ for some $m \ne 0$, then $m\, d^q x = 0$, so $d^q x = 0$ because C^{q+1} is torsion-free; hence $\operatorname{tors} H^q(C) = \operatorname{tors} \operatorname{coker} d^{q-1}$, whose ℓ-part is read from the Smith form. The rank count is additivity over $\mathbb{Q}$. (3) By (1), the hypothesis says every elementary divisor of d^{q-1} and of d^q is $< \ell^e$ for $q \le q_{\max}$, so (2) gives the types of H^q and H^{q+1}'s torsion; the displayed decomposition is the universal coefficient theorem for a complex of free modules over a principal ideal domain [standard: [Wei94, §3.6]]. If $a \ge e$ for some ℓ^a, (1) counts it as 0. (4) Reduction mod ℓ' preserves the rank exactly when ℓ' divides no elementary divisor, and those are the torsion orders of $\operatorname{coker} d^q$, i.e. of $H^{q+1}(C)$, by (2). For $C^\bullet_{\mathcal{I}}(\mathbb{Z})$ the torsion of $H^\bullet(\Gamma; V)$ lies over the primes of Proposition 23.16, all $\le \max(3, n-1)$; for $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$ the cone of cmp^* is 6-power torsion, so its cohomology has the same torsion primes ≥ 5. (5) is (1)–(3) read against Proposition 23.2. □

Remark 23.18 (The rule, differential by differential). [proved here] (v1.9.0.) In the setting of Proposition 23.17, the hypothesis of (3) may be checked one differential at a time, each at its own level. If $r_{e_q}(d^q) = \operatorname{rank}_{\mathbb{Q}} d^q$ for some $e_q \ge 1$, then every elementary divisor ℓ^{a_t} of d^q has $a_t < e_q$, and $\operatorname{tors} H^{q+1}(C)_{(\ell)}$ is read off exactly from the Smith form of d^q mod ℓ^{e_q}. The $\mathbb{Z}_{(\ell)}$-type of $H^q(C)$ needs e_{q-1} (its torsion comes from d^{q-1}) and the two ranks $\operatorname{rank}_{\mathbb{Q}} d^{q-1}$, $\operatorname{rank}_{\mathbb{Q}} d^q$; the displayed decomposition of $H^q(C \otimes \mathbb{Z}/\ell^f)$ in (3) needs e_{q-1} and e_q. Clause (3) with one e is the uniform case $e = \max_{q \le q_{\max}} e_q$. *Proof.* (1) applied to $D = d^q$ at $e = e_q$ gives $a_t < e_q$ for every t; (2) reads $\operatorname{tors} H^{q+1}(C)_{(\ell)}$ from the non-unit elementary divisors of d^q alone, and H^q's rank from the two ranks over $\mathbb{Q}$; the universal coefficient decomposition involves $H^q(C)$ and $\operatorname{Tor}(H^{q+1}(C), \mathbb{Z}/\ell^f)$ and nothing else. No step couples d^q to $d^{q'}$ for $q' \ne q$. *In use*: the integral groups of line 2's milestone M0 are assembled with K certified per differential (`HeckeFreeModelValidation`, hecke-extensions slot-9 WP-11, which also checks that every level below each certified one is refused, and the displayed decomposition at one level above the largest); Definition 23.20(3)(b)'s "e certified per degree" is this reading. A clarification of (3), not a correction: no statement or proof of the proposition changes (`docs/implementation/slot9/WP-11_report.md` finding 2, confirmed by the slot's independent review, WP-14, `docs/implementation/slot9/WP-14_review.md`).

Proposition 23.19 (The characteristic polynomial over $\mathbb{Z}$, by residues)**.** *[proved here] Let $n \ge 0$, p a prime, $g = \operatorname{diag}(1, p)$, and write $H^1_{\mathrm{fr}} := H^1(\Gamma; V)/\mathrm{tors}$, on which the double-coset operator $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ acts (it preserves torsion); let $\chi_p \in \mathbb{Z}[x]$ be its characteristic polynomial. Let*

$\ell' \geq 5$ be a prime, $\ell' \neq p$, at which $H^1(\Gamma; V)$ has no ℓ'-torsion — every $\ell' > n+1$ qualifies (Proposition 23.16).

1. *Reduction $V \to M_n$ induces an isomorphism, equivariant for that operator, $H^1_{\mathrm{fr}} \otimes \mathbb{F}_{\ell'} \cong H^1(\Gamma; M_n)$ over $\mathbb{F}_{\ell'}$, and $\chi_p \bmod \ell'$ is the characteristic polynomial of the shipped matrix $A^{(1)}(T_p)$ over $\mathbb{F}_{\ell'}$.*
2. *Hence, if $P \in \mathbb{Z}[x]$ is monic of the same degree, $\mathcal{S}$ a finite set of such primes, and $\prod_{\ell' \in \mathcal{S}} \ell' > 2B$ where B bounds the absolute values of the coefficients of both P and χ_p, then $P = \chi_p$ iff $P \equiv \operatorname{charpoly} A^{(1)}(T_p) \pmod{\ell'}$ for every $\ell' \in \mathcal{S}$. No integer beyond a word is formed: the residues of P are reduced digit-wise.*

The bound B for χ_p is an input, not a statement of this document; the one the integral line uses is Deligne's bound on the eigenvalues through Eichler–Shimura (Remark 23.21).

Proof. (1) The long exact sequence of $0 \to V \xrightarrow{\ell'} V \to M_n \to 0$ gives $0 \to H^1(\Gamma; V) \otimes \mathbb{F}_{\ell'} \to H^1(\Gamma; M_n) \to H^2(\Gamma; V)[\ell'] \to 0$ (the argument of Claim 16.7(2)), and the right-hand term is 0 because 24 kills H^2 (Claim 16.2(4)) and $\ell' \geq 5$; without ℓ'-torsion, $H^1(\Gamma; V) \otimes \mathbb{F}_{\ell'} = H^1_{\mathrm{fr}} \otimes \mathbb{F}_{\ell'}$. Every map is induced by a map of $\mathbb{Z}[\Delta]$-modules, so it commutes with $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ (Claim 16.7(3)'s argument). At $6 \in \mathbb{F}_{\ell'}^\times$ the shipped matrix is that operator on $H^1_{\mathrm{eq}} \cong H^1(\Gamma; M_n)$ (Remark 7.6, Theorem 11.10, (H1) as $p \neq \ell'$). (2) $P - \chi_p$ has coefficients of absolute value $\leq 2B < \prod \ell'$ and is divisible by every $\ell' \in \mathcal{S}$. □

23.6 What each coefficient ring names

Definition 23.20 (The mathematical meaning of the coefficient ring)**.** Assume (H4). The setting that selects the coefficient ring — its spelling, default, reserved values and refusal rule — is [E, `sec:ring`]'s; the exponent it writes k is this document's e (§2.2, (O1)). For a cell (n, ℓ), the ring in force names the following object.

1. $\mathbb{F}_\ell$ (the default). $C^\bullet_{\mathrm{eq}}(\mathbb{F}_\ell)$ and its cohomology — $H^\bullet(\Gamma; M_n)$ when $\ell \geq 5$ (Remark 7.6), the Bredon cohomology of Proposition 23.3 over $\mathbb{F}_\ell$ when $\ell \in \{2, 3\}$. Unchanged since v1.0.0.
2. $\mathbb{Z}/\ell^e$, $e \geq 2$. $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e)$, whose cohomology is reported as a $\mathbb{Z}/\ell^e$-module (lengths and invariant factors), never as a dimension. At $\ell \geq 5$ it is $C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes \mathbb{Z}/\ell^e$ and computes $H^\bullet(\Gamma; V \otimes \mathbb{Z}/\ell^e)$ (Proposition 23.2(1), Remark 7.6 with $6 \in R^\times$), with the Hecke action on the nose from the integral tables (Proposition 15.18(2),(3)). At $\ell \in \{2, 3\}$ it names the Bredon object over $\mathbb{Z}/\ell^e$ — which is *not* a reduction of $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$ and not group cohomology. *What this build computes of it there* (v1.9.0; [E, `sec:ring`] (b1), (b3), (c1′)): its **lengths and invariant factors** in degrees $q = 0, 1, 2$, through a flat-coordinate module route — the cochain groups of $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e)$ are not free there (the invariants computation of Proposition 23.2(2)'s proof, read over $\mathbb{Z}/2^e$; [E, `sec:ring`] (c1′) gives it; v1.9.1, the citation through v1.9.0 named Proposition 23.2(2),(3), which do not state it), so no basis of them is used — measured by `HeckeBredonTreeRingValidation` against the tree complex (65) over the same ring (Proposition 23.3(2)); the certified K-mode, which is undefined there (Proposition 23.17(5)), the Hecke operator and the reductions are **refused** by name, as build capabilities (`hecke::NotImplementedInThisBuild`). (Through v1.8.1 this clause ended "and the implementation of this build refuses it there", stale from hecke-extensions slot-9 WP-3's merge, `b53836f`, until this version.)
3. $\mathbb{Z}$ (reserved through v1.23.0-pre.36; **un-reserved for the integral descent only** at v1.23.0-pre.37, item 7 of §22.4: the value is honoured by the descent record `integral_descent_record/v1`, its console verb and its campaign cells, and **refused by name** by every other reader of the ring — a build capability, `hecke::NotImplementedInThisBuild`, never a silent fall-back to $\mathbb{F}_\ell$; the general $\mathbb{Z}/m$ stays reserved). The object is **group cohomology** $H^q(\Gamma; V)$ in degrees $q \leq q_{\max}$, i.e. $H^q(C^\bullet_{\mathcal{I}}(\mathbb{Z}))$ — never the integral model complex $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$, which at $\ell \in \{2, 3\}$ is a different object (Proposition 23.2) and at odd n is zero while group cohomology is not

(Proposition 23.2(3)). It decomposes as $H^q(\Gamma;V) \cong \mathbb{Z}^{r_q} \oplus \bigoplus_\ell T_\ell^q$ over the primes of Proposition 23.16 (at level N: $H^q(\Gamma_0(N);V)$ over the primes of Proposition 31.8, the object of the descent record, item 7 of §22.4; v1.23.0-pre.38, referee m10). What the integral line computes, and from what:

(a) $\mathbb{Z}[1/6]$-part and ranks: from $C^\bullet_{\mathrm{eq}}$ runs at good primes and at the Dickson primes $\ell \geq 5$ over $\mathbb{Z}/\ell^e$, with e certified (Propositions 23.2(1), 23.17); the ranks equally from $C^\bullet_{\mathcal{I}}(\mathbb{Z})$ at a good prime, which Proposition 23.17(4) covers as well (v1.9.0: line 2's milestone M0 takes them there and cross-checks the $C^\bullet_{\mathrm{eq}}$ runs, `HeckeFreeModelValidation`);

(b) 2- and 3-parts: from $\sigma^{\leq q_{\max}+1}C^\bullet_{\mathcal{I}}(\mathbb{Z}) \otimes \mathbb{Z}/\ell^e$ with e certified per degree (Proposition 23.17(5), Remark 23.11; "per degree" read differential by differential, Remark 23.18);

(c) the group, glued by Proposition 23.9, the only glueing datum being cmp*;

(d) the Hecke action: by $\Theta_{\mathcal{I},g}$ (Proposition 23.13); on each T_ℓ^q from the local run (the ℓ-primary part is canonical and stable under $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$); on H^1_{fr} as the characteristic polynomial over $\mathbb{Z}$ by residues (Proposition 23.19), **basis-free** (v1.10.0: $\Theta_{\mathcal{I},g}$ is built for T_2, T_3, T_5 and read on $H^q(C^\bullet_{\mathcal{I}}(R))$ basis-free at small n, Remarks 23.14 and 23.21; the glued action over $\mathbb{Z}$ waits for the descent assembly; v1.23.0-pre.38: built at level N, §31.3, merged at `dc8261a3`, and carried by the record of item 7 of §22.4).

Not named by this ring: integer Hecke matrices on H^1_{fr} in any basis, and the global lattice class of H^1_{fr} as a Hecke module — owner decision OD-11 (b), confirmed 2026-09-24, puts the integral line basis-free first; a coordinate rule is a later revision's (OD-11 (a), small n only). Nor the $\pm$ split at $\ell = 2$ (it needs $2 \in R^\times$, Obligation 9.10), nor Poincaré duality, which over $\mathbb{Z}$ holds only after inverting 6.

Which record carries which of these, and under which setting, is [E]'s: under OD-10 (a) (confirmed 2026-09-24) the record keys that say "$\mathbb{Z}$", the truncation and which object is reported are emitted only under the value naming $\mathbb{Z}$, so that no golden record moves. (v1.23.0-pre.37: so it is built — the record's keys `coefficient_ring = "Z"` and `object = "group"` exist only in `integral_descent_record/v1`, item 7 of §22.4.)

Remark 23.21 (What is tested today (slots 8 to 11)). [verified computationally]^{r} Nine registered validations test statements of this chapter on the shipped tree — two since slot 8, three since slot 9, three since slot 10, one since slot 11 (v1.11.0; the title read "(slots 8, 9 and 10)" and the count "eight" at v1.10.0–v1.11.0-pre.2, "(slots 8 and 9)" and "five" at v1.9.0–v1.9.1, "(slot 8)" and "two" through v1.8.1); none upgrades any of them.

- `HeckeBredonTreeValidation` (`Integral/HeckeBredonTreeValidation.cpp`, slot-8 WP-8, `fast`): Proposition 23.3(2) at $R = \mathbb{F}_\ell$ — $\dim H^q_{\mathrm{eq}}$ of the shipped pipeline equals the tree complex (65), computed by an independent ρ_n and rank routine, on 82 cells ($\ell \in \{2,3\}$, $0 \leq n \leq 40$, odd n included), after first checking the tree complex against the sixteen golden cells at $\ell \in \{2,3\}$; 0 mismatches (`docs/implementation/slot8/WP-8_report.md`).
- `HeckeIntegralCharpolyValidation` (`Integral/`, slot-8 WP-7, `standard`): Proposition 23.19 in use — χ_2 for $n \in \{10,12,14,16,18,20\}$ and χ_3 at $n = 10$ are certified equal over $\mathbb{Z}$ to the Eichler–Shimura values of the integral-line brief (its oracle O5, e.g. $(x-2049)(x+24)^2$ at $n = 10$, $p = 2$) with no integer beyond a word, each cell with one held-out prime; the bound B is the validation's input (Deligne's, as a bit length); at $n \in \{64, 128\}$ the residue vectors and the bound are recorded, never compared with an oracle ($n = 128$ under the read-only switch `HECKE_INTEGRAL_CAMPAIGN=1`); 322 checks, 0 failing (`docs/implementation/slot8/WP-7_report.md`).
- `HeckeBredonTreeRingValidation` (`Integral/`, slot-9 WP-3, `standard`): Proposition 23.3(2) at $R = \mathbb{Z}/\ell^k$, $\ell \in \{2,3\}$, $k \in \{2,3,4\}$ — the lengths and invariant factors of H^q_{eq} computed by the shipped pipeline over the ring (Definition 23.20(2)) equal the tree complex (65) over the same ring, frozen from the design brief's probe

(lattices over $\mathbb{Z}$ and the probe's own Smith form, no modular elimination), on 96 cells (even $n \le 20$, odd $n \le 9$, $H^2 = 0$ throughout), with controls at $k = 1$ against the shipped $\mathbb{F}_\ell$ route and at $\ell = 5$ against the $\ell \ge 5$ module route; 1394 checks, 0 failing (`docs/implementation/slot9/WP-3_report.md`).

- `HeckeIsotropyModelValidation` (`Integral/`, slot-9 WP-10, `standard`): Definition 23.4 and Proposition 23.5(1)–(3) as built — the ranks $5, 14, 19, \ldots$ and $3, 6, 6, \ldots$, (IM1) and (IM2) exactly over $\mathbb{Z}[\Gamma]$, cmp a chain map; cmp* a cochain map, injective, and an isomorphism on H^0 and H^1 against the shipped pipeline at $\ell \ge 5$ (Proposition 23.5(4), where $6 \in R^\times$); $\dim H^q(C^\bullet_{\mathcal{I}}(\mathbb{F}_\ell))$ and the lengths over $\mathbb{Z}/\ell^k$ at $\ell \in \{2, 3\}$ against the design brief's Appendix A through universal coefficients; the boundary model against $H^\bullet(\Gamma_\infty)$; 1032 checks, 0 failing (`docs/implementation/slot9/WP-10_report.md`).
- `HeckeFreeModelValidation` (`Integral/`, slot-9 WP-11, `standard`): line 2's milestone M0 — $H^q(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2)$ for $n \le 20$, $q \le 5$, assembled as Definition 23.20(3)(a)–(b) says (ranks at the good prime 10007 by Proposition 23.17(4); the 2- and 3-parts on $C^\bullet_{\mathcal{I}}$ by Proposition 23.17 per differential, Remark 23.18; the Dickson-prime parts from the shipped K-mode, cross-checked on $C^\bullet_{\mathcal{I}}$), equals Appendix A exactly on 126 of 126 (n, q) cells; the cone of cmp$^* \otimes \mathbb{Z}/\ell^K$ is acyclic through degree 5 at the ten Dickson cells; $H^2 \cong H^4$, $H^3 \cong H^5$ (14.22); at odd n the integral model complex vanishes while the group does not (Proposition 23.2(3)); 803 checks, 0 failing (`docs/implementation/slot9/WP-11_report.md`; v1.10.1: 801 since slot-10 WP-8, whose arm (R) cites the two registered names it used to re-run in process, and whose test (6) refuses on the 7-vertex complex's own number of degrees). (v1.10.0: the design brief's tests (4) and (5) for it, which it declared no slot for, are hosted in `HeckeIntegralActionValidation` below, and its test (3) in `HeckePeriodicityClassValidation`.)
- `HeckeStrictLiftValidation` (`Integral/`, slot-10 WP-10, `standard`): Proposition 23.13(1), (3) and (4) as built, for T_2, T_3, T_5 in degrees ≤ 6 — the certificates, the strict compatibility on the rings with a shipped Θ^s_g, the operator on H^1 against the shipped canonical-τ operator at $\ell \ge 5$, O5 on the model, and a negative control; Remark 23.14; 1824 checks, 0 failing (`docs/implementation/slot10/WP-10_report.md`).
- `HeckePeriodicityClassValidation` (`Integral/`, slot-10 WP-13, `standard`): Lemma 14.22(1)–(2) as built. (1): e is a 2-cocycle in $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_2, \mathbb{Z})$, the Bockstein of the character ζ_+ in the convention of the lemma's Step 0; $\kappa_C(\mathrm{res}_C\, e) = \chi_C$ on the stabilisers, with $\chi = \zeta_+$ re-checked on 224 elements of finite order (v1.10.1: v1.10.0 attributed the 224-element check to the first identity); $[e]$ has order 12 and generates $H^2(\Gamma; \mathbb{Z}) \cong \mathbb{Z}/12$ ($12e$ a coboundary, $6e$ and $4e$ not). $E = e \cup -$ is a chain map $\mathcal{I}_{\bullet+2} \to \mathcal{I}_\bullet$ with $\varepsilon \circ E_0 = e$, certified exactly over $\mathbb{Z}[\Gamma]$. (2), clause by clause, on 126 ring cells ($n \le 20$ over $\mathbb{F}_\ell$ and $\mathbb{Z}/\ell^K$ at $\ell \in \{2, 3\}$, and over $\mathbb{F}_5$, $\mathbb{F}_7$): an isomorphism for $q \ge 2$; for $q = 1$ surjective, with kernel equal to $\partial(N_{-I}V)$ as submodules (inclusion plus equal lengths), injective at odd n; no $q = 0$ clause. The connecting map is realised through the Bass–Serre tree as the first-variable connecting map of (50), which agrees with the coefficient form $(*)$ of the lemma's proof up to one global sign under the identification that proof names; the kernel clause reads only its image, which the sign does not move, and an independent Mayer–Vietoris length computed in V alone, $\mathrm{length}\big((N_{-I}V + V^S + V^U)/(V^S + V^U)\big)$, agrees on all 126 cells (the slot's review, finding m-7); 2482 checks, 0 failing (v1.10.1: 2483 since slot-10 WP-8's oriented-disc arm, in which the fill of x reaches a second round) (`docs/implementation/slot10/WP-13_report.md`).
- `HeckeIntegralActionValidation` (`Integral/`, slot-10 WP-11, `standard`): line 2's milestone M1(a) — T_p as $\Theta_{\mathcal{I},g}$, $p \in \{2, 3, 5\}$, read on $H^q(C^\bullet_{\mathcal{I}}(R))$ with no integral basis, at $n \in \{10, 12, 16\}$ (arms on torsion and on universal coefficients also for every $n \le 20$). The Eichler–Shimura identities of the design brief (O6) on $H^1(\mathbb{F}_\ell) = H^1_{\mathrm{fr}} \otimes \mathbb{F}_\ell$ at eight good primes, with O5's at $n = 12, 16$; the 691-congruence module of $H^1_+ \otimes \mathbb{Z}_{691}$ at $n = 10$ is

$\mathbb{Z}/691$ on *both* lines (the E8 route over $\mathbb{Z}/691^k$ and $\Theta_{\mathcal{I},g}$ on $C^\bullet_{\mathcal{I}} \otimes \mathbb{Z}/691^k$, $k \in \{2,3\}$), by two readings that assume T_2 diagonalisable over $\mathbb{Q}_{691}$ on H^1_{fr} — Eichler–Shimura, of which the identity O6 checked modulo 691^k is the computed evidence (the slot's review, finding m-15); on the 2- and 3-primary torsion in degrees ≤ 5, Lemma 14.22(3)(b) for $q = 0, \ldots, 3$, $T_p T_{p'} = T_{p'} T_p$ on cohomology (non-vacuous: the cochain maps do not commute), $\operatorname{char} A^{(3)}(T_p) \mid \operatorname{char} A^{(1)}(T_p)$, and the torsion modules read twice (by reduction and by the Bockstein) with exponents equal to M0's; **not** tested: the recursion of Theorem 10.8(1), which needs a lifted datum for determinant p^2 (Remark 23.14, last paragraph; the review's finding M-3); the local type of $H^1_{\mathrm{fr}} \otimes \mathbb{Z}_\ell$ at $\ell \in \{2,3\}$ recorded (split at $n = 10$ and 16; at $(16,3)$ with no T_3 oracle, finding m-17), with O5 and O6 holding on $H^1_{\mathrm{fr}} \otimes \mathbb{Z}/\ell^4$ there; the universal-coefficient lengths at every precision below the run's (a consistency check); 4601 checks, 0 failing (`docs/implementation/slot10/WP-11_report.md`).

- `HeckeStrictLiftRelationsValidation` (`Integral/`, slot-11 WP-11, `standard`; v1.11.0): Proposition 23.13(1) built and certified exactly over $\mathbb{Z}[\Gamma'']$ for $g = \operatorname{diag}(1, p^2)$, $\operatorname{diag}(1, p^3)$ and pI ($p \in \{2,3\}$; $p = 5$ for $\operatorname{diag}(1,25)$ and $5I$), with $\tau = \mathrm{id}$ of Proposition 10.5's proof as the datum at pI; the transports are cochain maps over fields, $\mathbb{Z}/\ell^k$ and 64-bit $\mathbb{Z}/\ell^K$; $\Theta_{\mathcal{I},pI}$ induces p^n on every $H^q(C^\bullet_{\mathcal{I}}(R))$, $q \leq 5$, $\ell \in \{2,3\}$ and $p = \ell$ included; and **Proposition 23.23(2) for** $\nu \in \{1,2\}$ holds on H^q, $q \leq 5$, basis-free, over $\mathbb{Z}/2^8$, $\mathbb{Z}/3^5$, $\mathbb{F}_2$, $\mathbb{F}_3$ ($p = \ell$ included), $\mathbb{F}_5$, $\mathbb{F}_7$, $\mathbb{F}_{691}$, with the full operators assembled as in the proof of (19) ($T_{p^2} = \Theta_{\operatorname{diag}(1,p^2)} + \Theta_{pI}$, $T_{p^3} = \Theta_{\operatorname{diag}(1,p^3)} + p^n \Theta_p$); non-vacuous (the identities fail on cochains in 684 of 684 cases), with A.1's primitive operator and 10.9's coefficient as negative controls, refused exactly where p^n acts non-trivially; the $\ell \geq 5$ consistency with the shipped primitive T_{p^ν} through cmp* ($\nu \in \{2,3\}$, 23.13(4)); 3619 checks, 0 failing (`docs/implementation/slot11/WP-11_report.md`). The item "**not** tested: the recursion of Theorem 10.8(1)" of the bullet above is thereby answered on the free model; on $C^\bullet_{\mathrm{eq}}$ at $\ell \in \{2,3\}$ nothing is compared (Remark 23.15).

Remark 23.22 (What this chapter does not say). Five things a reader might expect here and will not find.

1. **No arbitrary precision.** Owner decision OD-3 (c) (confirmed 2026-09-24): Theorem 18.7(3), the box of §19 and Obligation 20.10(iv) are unchanged. Integer outputs of the integral line are residue vectors with a certified bound (Proposition 23.19); a decimal or rational reconstruction happens outside the build. Re-scoping the ban is a separate amendment that this version does not make.
2. **No effectivity claim for** $\tau_{\mathcal{I}}$, and no bound on its integer sizes (Proposition 23.13, last paragraph). $\mathcal{I}_\bullet$ itself is produced by a terminating procedure (Proposition 23.5(1)). (v1.10.0: for $g = \operatorname{diag}(1,p)$, $p \in \{2,3,5\}$, in degrees ≤ 6, $\tau_{\mathcal{I}}$ is built and its sizes are measured, Remark 23.14; that is evidence, not a bound, and for every other g and degree this item stands as written. v1.23.0-pre.24: the bulk lift for $g = \operatorname{diag}(1,p)$, every p, is *produced* by a terminating procedure, Proposition 31.13(2); its sizes stay unbounded. v1.23.0-pre.28: the pair lift for $g = \operatorname{diag}(1,p)$ is built on demand at every prime p requested, degrees ≤ 6, and certified at run time, refused on a failed certificate or at the cap, Proposition 31.21(1) (every refusal: its (5), v1.23.0-pre.29); measured to $p = 127$, still no bound.)
3. **No record schema.** The keys are [E]'s (Definition 23.20, last paragraph); §22.4 is not amended here.
4. **No higher level and no families**, and no divided powers: V is $\mathrm{Sym}^n(\mathbb{Z}^2)$ (OD-4 (a)); D_n and levels via Shapiro need their own chapter. (v1.23.0-pre.14, the level half of the plan's C-D7: levels via Shapiro, for the groups and the descent, are Propositions 31.7 and 31.8; families, divided powers and the level-N Hecke action over $\mathbb{Z}$ stand as written. v1.23.0-pre.18: restriction, trace, conjugation, the level cone and its long exact sequence over $\mathbb{Z}$, and the Hecke squares on cohomology, are Proposition 31.10. v1.23.0-pre.24: the

level-N operator T_p, $p \nmid N$, over $\mathbb{Z}$ at cochain level, and its invariants, are §31.3. v1.23.0-pre.26: the pair and its cone at level N over $\mathbb{Z}$, the boundary by cusps, and T_p on them, are §31.4.)

5. **Nothing about the pair and the cone over** $\mathbb{Z}$ beyond Proposition 23.13(1)'s datum; the cone over $\widehat{\mathbb{Z}}_6$ under 14.24 is a later revision's. (v1.23.0-pre.26, the pair half of the plan's C-D7: the pair models over $\mathbb{Z}$, O_ℓ and $\mathbb{Z}[1/6]$, the long exact sequence over $\mathbb{Z}$ glued from them, the boundary in closed form, compact support, the certified exponent under 14.24's truncation, and the cone with the strict pair datum and T_p, $p \nmid N$, are §31.4; the tree cusp pair stands as written, the plan's D5c. v1.23.0-pre.33: the tree cusp pair — the cusp model, the axis map, the tree cone, its comparisons with $(\mathcal{I}, \mathcal{I}^\partial)$ and T_p by transport — is §31.5, contract first; v1.23.0-pre.36: built, audited and merged at `4064e164`.)

Code map. Implemented: `Integral/HeckeBredonTreeValidation` (Proposition 23.3(2) at $R = \mathbb{F}_\ell$) and `Integral/IntegralCharpolyCrt` with `HeckeIntegralCharpolyValidation` (Proposition 23.19), both slot 8. **Not implemented**: $\mathcal{I}_\bullet$, $\mathcal{I}^\partial_\bullet$, cmp, $\tau_\mathcal{I}$, the descent assembly and the value $\mathbb{Z}$ of the ring (§22.2). The model complex over $\mathbb{Z}/\ell^e$ at $\ell \geq 5$ is E8's ([E, `sec:ring`]). (v1.10.0: this paragraph was the v1.8.0 text and stale from v1.9.0 on; §22.1's row is the current one. Implemented since v1.9.0: `Integral/Model/IsotropyModel` ($\mathcal{I}_\bullet$, $\mathcal{I}^\partial_\bullet$, cmp) and `Integral/Model/IntegralGroups` (the groups at $n \leq 20$); since v1.10.0: `Integral/Model/StrictLift` ($\tau_\mathcal{I}$, $\Theta_{\mathcal{I},g}$; Remark 23.14), `Integral/Model/PeriodicityClass` (e, E_q; Lemma 14.22) and `Integral/Model/IntegralHecke` (the Hecke action on $H^\bullet(C^\bullet_\mathcal{I}(R))$, basis-free). Not implemented: the descent assembly and the value $\mathbb{Z}$ of the ring.)

Formalization note. Propositions 23.7 and 23.8 are elementary facts about flat abelian groups and are cheap and worth formalizing first: they are what makes the descent exact. Lemma 23.12 is a finite induction. Proposition 23.3 needs a functor category and the comparison theorem there, and Proposition 23.5(1) the assembly of resolutions over a complex; both are standard homological algebra with no arithmetic.

23.7 The Hecke relations on the free model

New at v1.11.0-pre.1 (hecke-extensions slot-11 WP-2; the slot-10 review's finding 6, and the precondition of line 2's milestone M-3). Theorem 10.8 is a statement about $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$ and carries $6 \in R^\times$, because on the shipped complex the operator is identified with the classical one only then (Theorem 11.10, Remark 7.6). On the free model $C^\bullet_\mathcal{I}(R)$ that identification holds over *every* ring (Proposition 23.13(3)), and the relations follow there with no hypothesis on ℓ. This subsection states that. It changes nothing about Theorem 10.8, whose hypothesis stays, and nothing about the Bredon object $C^\bullet_{\mathrm{eq}}(R)$ at $\ell \in \{2, 3\}$ (Remark 23.15).

Proposition 23.23 (The Hecke relations on $H^\bullet(C^\bullet_\mathcal{I}(R))$, at every prime)**.** *Assume (H4) and (B1)–(B3); let R be* any *commutative ring — no hypothesis on 6, so $R = \mathbb{F}_\ell$ and $R = \mathbb{Z}/\ell^e$ with $\ell \in \{2, 3\}$ are included — let $n \geq 0$, $V_R = \mathrm{Sym}^n(\mathbb{Z}^2) \otimes R$, and let $(\mathcal{I}_\bullet, \mathrm{cmp})$ be an isotropy-resolution model (Definition 23.4). For $\delta \in \Delta$ choose a strict lift $\tau_\mathcal{I}$ for $g = \delta$ (Proposition 23.13(1)) and write $T^\mathcal{I}_{\Gamma\delta\Gamma}$ for the map that its coset transport $\Theta_{\mathcal{I},\delta}$ induces on $H^q(C^\bullet_\mathcal{I}(R))$. Put, for a prime p and $\nu \geq 0$,*

$$T^\mathcal{I}_{p^\nu} := \sum_{\substack{\Gamma\delta\Gamma \subseteq \Delta \\ \det \delta = p^\nu}} T^\mathcal{I}_{\Gamma\delta\Gamma}, \qquad T^\mathcal{I}_{p^0} := \mathrm{id},$$

the full operator of (18) read on the free model, and $T^\mathcal{I}_p := T^\mathcal{I}_{p^1}$.

1. *[proved here]* $T^{\mathcal{I}}_{\Gamma\delta\Gamma}$ *does not depend on the lift* $\tau_{\mathcal{I}}$*, on the coset representatives, or on* δ *inside* $\Gamma\delta\Gamma$*, and it equals the classical double-coset operator* $\mathrm{cor}\circ(\mathrm{conj}_\delta)_*\circ\mathrm{res}$ *of* (20) *on* $H^q(\Gamma;V_R)$*, in every degree* q*. In particular* $T^{\mathcal{I}}_{\Gamma\,pI\,\Gamma}=p^n\mathrm{id}$.
2. *[proved here] (given the citations below) For every degree* q*, all primes* p,p' *and all* $\nu\geq 1$*, on* $H^q(C^\bullet_{\mathcal{I}}(R))$*:*
$$T^{\mathcal{I}}_pT^{\mathcal{I}}_{p'}=T^{\mathcal{I}}_{p'}T^{\mathcal{I}}_p,\qquad T^{\mathcal{I}}_{p^{\nu+1}}=T^{\mathcal{I}}_p\,T^{\mathcal{I}}_{p^\nu}-p^{n+1}\,T^{\mathcal{I}}_{p^{\nu-1}},$$
and $T^{\mathcal{I}}_{p^2}=T^{\mathcal{I}}_{\Gamma\,\mathrm{diag}(1,p^2)\,\Gamma}+p^n\mathrm{id}$*. With [standard: [Shi71, Thm. 3.24 and §3.2] for the identities in the abstract Hecke ring* $\mathcal{H}(\Delta,\Gamma)$*; [RW70, Thm. 1] and [AS86, §1.1, p. 194] for the fact that* $\Gamma\delta\Gamma\mapsto\mathrm{cor}\circ(\mathrm{conj}_\delta)_*\circ\mathrm{res}$ *is a ring homomorphism* $\mathcal{H}(\Delta,\Gamma)\to\mathrm{End}H^q(\Gamma;W)$ *for* every $\mathbb{Z}[\Delta]$*-module* W *and every* q*, with no invertibility hypothesis. The locator into [RW70] is not verified against the text (rule (A2)): no copy is in* `docs/literature/`*; [AS86, §1.1, p. 194] is verified (slot-17 R17-4, v1.17.0-pre.5) — it* states *the action as "well known" for right modules, left modules by the* S^{-1} *device, without proof and without an invertibility hypothesis].*
3. ***What this does not say.*** *Nothing about* $H^\bullet_{\mathrm{eq}}$ *at* $\ell\in\{2,3\}$*, where Theorem 10.8 keeps its hypothesis and the operator of the Bredon object is not compared with anything (Remark 23.15); and nothing at cochain level, where the recursion does not hold (the cochain maps of different double cosets are built from different lifts). It says nothing about effectivity either: a lift for* $\mathrm{diag}(1,p^2)$ *and* pI *is asserted to exist (Proposition 23.13(1)), not to be built.*

Proof. (1) By Proposition 23.13(3), $\Theta_{\mathcal{I},\delta}$ induces $\mathrm{cor}\circ(\mathrm{conj}_\delta)_*\circ\mathrm{res}$ on $H^\bullet(\Gamma;V_R)=H^\bullet(C^\bullet_{\mathcal{I}}(R))$ for every R, in every degree — the identification of the cohomology of the free resolution with group cohomology needs no hypothesis on R (Proposition 23.5(3)). The classical operator depends on none of the three choices [standard: [Bro82, Ch. III.9–10], as cited in (20)]; so neither does $T^{\mathcal{I}}_{\Gamma\delta\Gamma}$. For $\delta=pI$: $\Gamma''=\Gamma$, $\mathrm{conj}_{pI}=\mathrm{id}$, $\mathrm{cor}=\mathrm{res}=\mathrm{id}$, and the coefficient map is $\rho_n(pI)=p^n$ (proof of Proposition 10.5).

(2) By (1) the assignment $\Gamma\delta\Gamma\mapsto T^{\mathcal{I}}_{\Gamma\delta\Gamma}$ is the classical one, which extends to a ring homomorphism from $\mathcal{H}(\Delta,\Gamma)$ for the $\mathbb{Z}[\Delta]$-module V_R (the monoid Δ acts through ρ_n, Definition 8.3), by the cited theorem. In $\mathcal{H}(\Delta,\Gamma)$, Shimura's $T(p^\nu)$ is the sum of all double cosets of determinant p^ν (v1.19.0-pre.5; earlier written $T(1,p^\nu)$, which in his notation is the primitive coset), the local algebras at distinct primes commute, and $T(p)T(p^\nu)=T(p^{\nu+1})+p\,T(p,p)T(p^{\nu-1})$. Apply the homomorphism: $T(p^\nu)$ goes to $T^{\mathcal{I}}_{p^\nu}$ by definition and $T(p,p)=\Gamma\,pI\,\Gamma$ to $p^n\mathrm{id}$ by (1). The last identity is the $\nu=2$ case of the content decomposition in the proof of Proposition 10.6: the double cosets of determinant p^2 are $\Gamma\,\mathrm{diag}(1,p^2)\,\Gamma$ and $\Gamma\,pI\,\Gamma$. □

For line 2 (slot-11 WP-11). Clause (2) is the statement the X7 recursion checks: at $\ell\in\{2,3\}$, $\nu\leq 2$, on $H^q(C^\bullet_{\mathcal{I}}(R))$, $q\leq 5$, entrywise on cohomology (a basis-free comparison such as the inclusion $P(Z^q)\subseteq B^q$ of Remark 23.21), with $T^{\mathcal{I}}_{p^2}$ assembled from the two lifts for $\mathrm{diag}(1,p^2)$ and pI. It is a theorem there, so a failure is a defect of a lift or of its transport, not a counterexample. At $\ell\geq 5$ the same identity on $C^\bullet_{\mathrm{eq}}(R)$ is Theorem 10.8, and the comparison of the two routes through cmp* is Proposition 23.13(4). *Built at v1.11.0* (slot-11 WP-11, `docs/implementation/slot11/WP-11_report.md`): the lifts that (3) calls "asserted to exist, not to be built" are built for $\mathrm{diag}(1,p^2)$, $\mathrm{diag}(1,p^3)$ and pI at $p\in\{2,3\}$ (and at $p=5$ for $\mathrm{diag}(1,25)$ and $5I$), certified exactly over $\mathbb{Z}[\Gamma'']$; at pI the datum is $\tau=\mathrm{id}$ of Proposition 10.5's proof, and the induction returns a lift homotopic to id (either is a strict lift). Clause (2) holds for $\nu\in\{1,2\}$ on $H^q(C^\bullet_{\mathcal{I}}(R))$, $q\leq 5$, on every cell tried, at $\ell\in\{2,3\}$ with $p=\ell$ included; the evidence is Remark 23.21's last bullet. It is evidence for the statement, not a replacement for its proof, and nothing about effectivity for other δ follows.

Part VII — The Dickson invariant, θ, and the ordinary projector

24 The Dickson invariant and the operator θ_{L}

What this chapter fixes. Multiplication by the first Dickson invariant $\mathsf{L} = x^\ell y - xy^\ell$ as a map of local systems $\mathrm{Sym}^n \to \mathrm{Sym}^{n^+}$, $n^+ = n+\ell+1$, twisted by the determinant; the operator θ_{L} it induces on the equivariant complex and on cohomology, with its Hecke equivariance *in the code's normalisation* (Definition 8.1, (H2)): $T_p\theta_{\mathsf{L}} = p\,\theta_{\mathsf{L}}T_p$, and $U_\ell\theta_{\mathsf{L}} = 0$ over $\mathbb{F}_\ell$; the quotient Q_n with a *fixed* complement basis; the long exact sequence of $0 \to \mathrm{Sym}^n \to \mathrm{Sym}^{n^+} \to Q_n \to 0$ and its termination; $H^0 = \mathbb{F}_\ell[\mathsf{L},\mathsf{D}]$ with its Hecke action; and which statements of the literature on θ are checkable on the model at level 1. It is the contract of hecke-extensions slot 11's Dickson/θ line (WP-5, WP-6): every claim a validation asserts about θ_{L} is a statement of this section.

Standing hypotheses. (H4) throughout; ℓ is a prime and the ring is $\mathbb{F}_\ell$ unless a statement says otherwise. Where $6 \in \mathbb{F}_\ell^\times$ ($\ell \geq 5$) is needed it is written into the statement (rule (A1)); Remark 24.12 says what survives at $\ell \in \{2,3\}$. Throughout, $u_t = x^{m-t}y^t$ is the basis of $M_m = \mathrm{Sym}^m(\mathbb{F}_\ell^2)$ of Definition 8.1, and $\rho_m(A)f = f(ax+cy,\, bx+dy)$ for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is that definition's action, reduced mod ℓ; it is defined for every $A \in \mathrm{M}_2(\mathbb{Z})$, singular modulo ℓ or not, and $A \mapsto \rho_\bullet(A)$ is a monoid homomorphism into the *ring* endomorphisms of $\mathbb{F}_\ell[x,y]$ (Proposition 8.2(1),(2) and its proof).

24.1 The two Dickson invariants

Definition 24.1 (The Dickson invariants)**.**

$$\mathsf{L} := x^\ell y - xy^\ell \in M_{\ell+1}, \qquad \mathsf{D} := \sum_{i=0}^{\ell} x^{\,i(\ell-1)}\, y^{\,(\ell-i)(\ell-1)} \in M_{\ell^2-\ell}.$$

Write $M_{\mathsf{L}}\colon M_m \to M_{m+\ell+1}$ for multiplication by L; in the bases (u_t) it sends $u_i \mapsto u_{i+1} - u_{i+\ell}$, which is the matrix `HeckeExt/HeckeDicksonMultiplication` builds.

Lemma 24.2 (The determinant twist, and what singular matrices do)**.** *[proved here] Let ℓ be a prime and $A \in \mathrm{M}_2(\mathbb{Z})$, read modulo ℓ.*

1. $\rho_{\ell+1}(A)\mathsf{L} = \det(A)\,\mathsf{L}$. *Hence, for every $m \geq 0$,*

$$\rho_{m+\ell+1}(A)\, M_{\mathsf{L}} \;=\; \det(A)\, M_{\mathsf{L}}\, \rho_m(A) \qquad \text{for every } A \in \mathrm{M}_2(\mathbb{Z}), \text{ singular included.} \tag{66}$$

2. $\mathsf{L} = xy\prod_{a\in\mathbb{F}_\ell^\times}(x-ay)$ *is, up to a unit, the product of one linear form for each of the $\ell+1$ points of $\mathbb{P}^1(\mathbb{F}_\ell)$; and* $\mathsf{L}\cdot\mathsf{D} = x^{\ell^2}y - xy^{\ell^2}$.
3. *If A is invertible modulo ℓ then $\rho(A)\mathsf{D} = \mathsf{D}$. If A is singular modulo ℓ then $\rho(A)\mathsf{L} = 0$, and D need not be fixed:* $\rho(\mathrm{diag}(1,0))\mathsf{D} = x^{\ell^2-\ell}$.
4. *Evaluation:* $\mathsf{L}(v) = 0$ *and* $\mathsf{D}(v) = 1$ *for every $v \in \mathbb{F}_\ell^2 \smallsetminus \{0\}$.*
5. *If $f \in M_m$, $m \geq 1$, vanishes at every point of $\mathbb{F}_\ell^2$, then $\mathsf{L} \mid f$.*

Proof. (1) Modulo ℓ, $(ax+cy)^\ell = ax^\ell + cy^\ell$ (Frobenius, and $a^\ell = a$ in $\mathbb{F}_\ell$). So $\rho(A)\mathsf{L} = (ax^\ell + cy^\ell)(bx+dy) - (ax+cy)(bx^\ell+dy^\ell) = (ad-bc)(x^\ell y - xy^\ell)$, the $x^{\ell+1}$ and $y^{\ell+1}$ terms cancelling. Since $\rho_\bullet(A)$ is a ring endomorphism, $\rho_{m+\ell+1}(A)(\mathsf{L}f) = \rho_{\ell+1}(A)\mathsf{L}\cdot\rho_m(A)f = \det(A)\,\mathsf{L}\,\rho_m(A)f$, which is (66). No inverse and no invertibility is used. (2) $x^{\ell-1} - y^{\ell-1} = \prod_{a\in\mathbb{F}_\ell^\times}(x-ay)$; the

forms x, y and $x-ay$ vanish on the $\ell+1$ lines of $\mathbb{F}_\ell^2$. With $X=x^{\ell-1}$, $Y=y^{\ell-1}$: $\mathsf{L}=xy(X-Y)$, $x^{\ell^2}y-xy^{\ell^2}=xy(X^{\ell+1}-Y^{\ell+1})$ and $\mathsf{D}=\sum_{i=0}^{\ell}X^iY^{\ell-i}$. (3) The computation of (1) with ℓ^2 in place of ℓ gives $\rho(A)(x^{\ell^2}y-xy^{\ell^2})=\det(A)(x^{\ell^2}y-xy^{\ell^2})$. For $\det A\neq 0$ in $\mathbb{F}_\ell$, apply $\rho(A)$ to (2): $\det(A)\mathsf{L}\cdot\rho(A)\mathsf{D}=\det(A)\mathsf{L}\mathsf{D}$, and cancel in the domain $\mathbb{F}_\ell[x,y]$. For singular A, (1) gives $\rho(A)\mathsf{L}=0$; $\mathrm{diag}(1,0)$ substitutes $y\mapsto 0$, which leaves the $i=\ell$ term of D. (4) $\mathsf{L}(v)=0$ by (2). For $v=(v_1,v_2)\neq 0$: $X(v),Y(v)\in\{0,1\}$ and not both 0; if both are 1, $\mathsf{D}(v)=\ell+1=1$; if exactly one is, exactly one term of D survives, with value 1. (5) A homogeneous f with $f(a,b)=0$, $(a,b)\neq 0$, is divisible by $bx-ay$ (dehomogenise). The $\ell+1$ forms of (2) are pairwise non-associate primes of the factorial ring $\mathbb{F}_\ell[x,y]$, so their product divides f. □

Proposition 24.3 (H^0 is the Dickson ring)**.** *Let ℓ be a prime and $m\geq 0$.*

1. *[standard: [Dic11, §§2–3] for the invariants of $\mathrm{SL}_2(\mathbb{F}_q)$ and $\mathrm{GL}_2(\mathbb{F}_q)$ acting on $\mathbb{F}_q[x,y]$; [Wil83, §1] for the modern statement. Neither locator is verified against the text (rule (A2))]* $\mathbb{F}_\ell[x,y]^{\mathrm{SL}_2(\mathbb{F}_\ell)}=\mathbb{F}_\ell[\mathsf{L},\mathsf{D}]$, *a polynomial ring on the two generators of Definition 24.1, and* $\mathbb{F}_\ell[x,y]^{\mathrm{GL}_2(\mathbb{F}_\ell)}=\mathbb{F}_\ell[\mathsf{L}^{\ell-1},\mathsf{D}]$. *(v1.23.0-pre.16: both statements are proved in place, for every prime ℓ, by Lemma 16.6; the citations are provenance only.) (The action here is by substitution, $f\mapsto f\circ A^{\top}$ on points; a group closed under transposition has the same invariants either way.)*
2. *[proved here] (by the SL_2 half of (1), Lemma 16.6)* $H^0(\Gamma;M_m)=H^0_{\mathrm{eq}}(K,\mathcal{R};M_m)=M_m^{\mathrm{SL}_2(\mathbb{F}_\ell)}$ *has basis*

$$\{\mathsf{L}^a\mathsf{D}^b\ :\ a,b\geq 0,\ a(\ell+1)+b(\ell^2-\ell)=m\},$$

   so $h^0(m):=\dim H^0(\Gamma;M_m)$ is the count of Lemma 16.6 (Claim 16.5(2)), whose two generators these are. The $-I$ parity: *for ℓ odd both degrees $\ell+1$ and $\ell^2-\ell$ are even, so $H^0=0$ for m odd, as $\rho_m(-I)=(-1)^m$ requires (Remark 7.7); at $\ell=2$, $\deg\mathsf{L}=3$ is odd and $-I\equiv I$.*
3. *[proved here] $M_{\mathsf{L}}\colon H^0(\Gamma;M_m)\to H^0(\Gamma;M_{m+\ell+1})$ is injective, and its cokernel has dimension $[\,\ell(\ell-1)\mid m+\ell+1\,]$, spanned by the class of $\mathsf{D}^{(m+\ell+1)/(\ell^2-\ell)}$ when that exponent is an integer. Equivalently $h^0(m+\ell+1)-h^0(m)=[\,\ell(\ell-1)\mid m+\ell+1\,]$.*

Proof. (2) $H^0_{\mathrm{eq}}=M_m^{\Gamma}$ by Proposition 23.3(2) (every $\mathbb{Z}[\Gamma]$-module, every ℓ), and Γ acts through the surjection $\Gamma\to\mathrm{SL}_2(\mathbb{F}_\ell)$ of Claim 16.5(2); by Lemma 16.6 the invariants are $\mathbb{F}_\ell[\mathsf{L},\mathsf{D}]$. A polynomial ring on homogeneous generators of degrees $\ell+1$ and $\ell^2-\ell$ has the monomials $\mathsf{L}^a\mathsf{D}^b$ of degree m as a basis of its degree-m part. (3) Multiplication by $\mathsf{L}\neq 0$ is injective in a domain. The monomials of degree $m+\ell+1$ with $a\geq 1$ are L times those of degree m; the only other one is D^b with $b(\ell^2-\ell)=m+\ell+1$. □

24.2 The Hecke action on H^0

The operator of Definition 11.6 on H^0 is, by Remark 11.8 (which uses (T3) only, satisfied by every datum, and no hypothesis on ℓ), $T_{\Gamma g\Gamma}v=\sum_{j=1}^{m}\rho(\delta_j)v$ for $v\in M^{\Gamma}$, with $\delta_j=s_j^{-1}g$ the coset elements of Definition 10.2. For $g=\mathrm{diag}(1,p)$ they are $\delta_{1+k}=\begin{pmatrix}1&0\\-k&p\end{pmatrix}$, $0\leq k<p$, and $\delta_{p+1}=\begin{pmatrix}0&p\\-1&0\end{pmatrix}$.

Proposition 24.4 (Hecke operators on $H^0(\Gamma;M_m)$)**.** *[proved here] Assume (H4). Let ℓ be a prime, $m\geq 1$, and $R=\mathbb{F}_\ell$; (1) and (2) need no hypothesis on ℓ beyond primality, and none on the datum; (3) is stated for ℓ odd, and (4) holds at every prime ($\ell=2$ by the direct computation at the end of the proof).*

1. ***(Good primes.)*** *If $g \in \Delta$ with $\ell \nmid \det g$ and $\Gamma g \Gamma$ has m_g left cosets, then $T_{\Gamma g\Gamma}(\mathsf{L}^a\mathsf{D}^b) = m_g\,(\det g)^a\,\mathsf{L}^a\mathsf{D}^b$. In particular, for $p \neq \ell$,*

$$T_p(\mathsf{L}^a\mathsf{D}^b) = (1+p)\,p^a\,\mathsf{L}^a\mathsf{D}^b, \qquad T(p,p)(\mathsf{L}^a\mathsf{D}^b) = p^{2a}\,\mathsf{L}^a\mathsf{D}^b.$$

(The second agrees with Proposition 10.5: $p^m = p^{2a}$ in $\mathbb{F}_\ell$, since $m = a(\ell+1)+b(\ell^2-\ell) \equiv 2a \pmod{\ell-1}$.)

2. ***(The bad prime: rank one, square zero.)*** *For every $v \in M_m^\Gamma$,*

$$U_\ell\, v \;=\; v(1,0)\cdot G_m, \qquad G_m := \sum_{k\in\mathbb{F}_\ell} (x-ky)^m + (-y)^m \in M_m,$$

where $v(1,0)$ is the coefficient of $u_0 = x^m$ in v. $G_m(1,0) = \ell = 0$, so $U_\ell^2 = 0$ and $\operatorname{rank} U_\ell \le 1$ on $H^0(\Gamma; M_m)$.

3. ***(Which monomials U_ℓ kills.)*** *$U_\ell(\mathsf{L}^a\mathsf{D}^b) = 0$ for $a \ge 1$. For $a = 0$, ℓ odd and $M := b(\ell-1)$,*

$$U_\ell(\mathsf{D}^b) = \big(G'_M\big)^\ell, \qquad G'_M := \sum_{k\in\mathbb{F}_\ell} (x-ky)^M + y^M = -\sum_{s=1}^{b-1} \binom{M}{s(\ell-1)} x^{\,M-s(\ell-1)} y^{\,s(\ell-1)},$$

*so $U_\ell(\mathsf{D}^b) = 0$ if and only if $\ell \mid \binom{b(\ell-1)}{s(\ell-1)}$ for every $0 < s < b$. This holds for every $b \le \ell$ and **fails for** $b = \ell+1$.*

4. ***(Consequence.)*** *$U_\ell = 0$ on $H^0(\Gamma; M_m)$ whenever $m < \ell^3 - \ell$, and $U_\ell \neq 0$ on $H^0(\Gamma; M_{\ell^3-\ell})$. The first instance is $(m,\ell) = (120,5)$, where H^0 has basis $\mathsf{L}^{20}, \mathsf{L}^{10}\mathsf{D}^3, \mathsf{D}^6$ and*

$$U_5\,\mathsf{D}^6 = -\mathsf{L}^{20}, \qquad U_5\,\mathsf{L}^{20} = U_5\,\mathsf{L}^{10}\mathsf{D}^3 = 0.$$

Within $\ell \in \{5,7,11,13\}$, $m \le 200$, it is the only one.

Proof. (1) Each δ_j is invertible modulo ℓ, so by Lemma 24.2(1),(3) and multiplicativity $\rho(\delta_j)(\mathsf{L}^a\mathsf{D}^b) = \det(\delta_j)^a\mathsf{L}^a\mathsf{D}^b$, and $\det\delta_j = \det g$; sum over the m_g cosets ($p+1$ for T_p, one for $\Gamma\,pI\,\Gamma$). (2) Modulo ℓ, $\delta_{1+k} \equiv \begin{pmatrix} 1 & 0 \\ -k & 0 \end{pmatrix}$ substitutes $x \mapsto x - ky$, $y \mapsto 0$, and $\delta_{\ell+1} \equiv \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}$ substitutes $x \mapsto -y$, $y \mapsto 0$. For v homogeneous of degree m, $v(X,0) = v(1,0)X^m$, so $\sum_j \rho(\delta_j)v = v(1,0)G_m$. At $(1,0)$ every $(x-ky)^m$ is 1 and $(-y)^m$ is 0 ($m \ge 1$), so $G_m(1,0) = \ell\cdot 1 = 0$; hence $U_\ell(U_\ell v) = v(1,0)\,G_m(1,0)\,G_m = 0$. (That $G_m \in M_m^\Gamma$ whenever some invariant has $v(1,0) \neq 0$ follows from U_ℓ preserving M_m^Γ, Theorem 11.7.) (3) $\mathsf{L}(1,0) = 0$ and $\mathsf{D}(1,0) = 1$, so $v(1,0) = [a = 0]$. For $a = 0$: $\mathsf{D}(X,0) = X^{\ell(\ell-1)}$, so $U_\ell\mathsf{D}^b = \sum_k (x-ky)^{\ell M} + y^{\ell M}$ (ℓM is even), which is $(G'_M)^\ell$ by Frobenius. Expanding, $\sum_{k\in\mathbb{F}_\ell} k^t = -1$ if $t \ge 1$ and $(\ell-1) \mid t$, and 0 otherwise ($t = 0$ gives $\ell = 0$); with $(-1)^{s(\ell-1)} = 1$ this leaves $-\sum_{s=1}^{b}\binom{M}{s(\ell-1)}x^{M-s(\ell-1)}y^{s(\ell-1)}$, whose $s = b$ term $-y^M$ cancels the added y^M. The surviving monomials are distinct, and $\mathbb{F}_\ell[x,y]$ is a domain, so $(G'_M)^\ell = 0$ iff every coefficient vanishes. For $1 \le s < b \le \ell$, the base-ℓ digits of $b(\ell-1)$ are $(b-1, \ell-b)$ and those of $s(\ell-1)$ are $(s-1, \ell-s)$, so by Lucas's theorem [standard: [Luc78, §XXI]: $\binom{A}{B} \equiv \prod_i \binom{A_i}{B_i} \pmod\ell$ on base-ℓ digits — the rule `mod_nCr` implements; locator not verified against the text] the binomial is $\binom{b-1}{s-1}\binom{\ell-b}{\ell-s} \equiv 0$, since $\ell - s > \ell - b$. For $b = \ell+1$, $M = \ell^2-1$ has digits $(\ell-1, \ell-1)$ and $s = 1$ gives $\binom{\ell-1}{0}\binom{\ell-1}{\ell-1} = 1$. (4) By (3) a non-zero U_ℓ needs a basis monomial D^b with $b \ge \ell+1$, i.e. $m = b(\ell^2-\ell) \ge \ell^3-\ell$; at $m = \ell^3-\ell$, $b = \ell+1$ gives $U_\ell\mathsf{D}^{\ell+1} \neq 0$. At $\ell = 5$, $b = 6$: every $\binom{24}{4s}$, $1 \le s \le 5$, is $\equiv 1 \pmod 5$ (digits $(4,4)$ against $(s-1, 5-s)$), so $G'_{24} = -\sum_{s=1}^{5} x^{24-4s}y^{4s} = -(xy)^4(x^4-y^4)^4 = -\mathsf{L}^4$, using $(x^4-y^4)^5 = x^{20}-y^{20}$; hence $U_5\mathsf{D}^6 = (-\mathsf{L}^4)^5 = -\mathsf{L}^{20}$. The next values of m at which some D^b with $b \ge \ell+1$ occurs are $m = 140, \dots, 200$ at $\ell = 5$ ($b = 7, \dots, 10$), where every $\binom{4b}{4s}$, $0 < s < b$, vanishes modulo 5

by Lucas's theorem (for $b = 7$: 28 has base-5 digits $(1,0,3)$, and each $4s \le 24$ has a digit exceeding the corresponding one); at $\ell \ge 7$ the first such m is $(\ell+1)(\ell^2-\ell) \ge 336$. [verified computationally][a] An independent script (slot-11 WP-2 report) lists the $b < 3\ell$ with $U_\ell \mathsf{D}^b \ne 0$ as $\{6,11,12\}$, $\{8,15,16\}$, $\{12,23,24\}$, $\{14,27,28\}$ at $\ell = 5,7,11,13$, and confirms $U_5\mathsf{D}^6 = -\mathsf{L}^{20}$. *At* $\ell = 2$ (v1.11.0-pre.2; the WP-3 referee's m-1), where (3)'s expansion is not used: by (2), $U_2 v = v(1,0)\big(x^m + (x+y)^m + y^m\big)$; the basis monomials with $v(1,0) \ne 0$ are the D^b, $m = 2b$, and $x^m + (x+y)^m + y^m$ vanishes for $m \in \{2,4\}$ and equals $x^4y^2 + x^2y^4 = \mathsf{L}^2$ at $m = 6 = \ell^3 - \ell$. So $U_2\mathsf{D} = U_2\mathsf{D}^2 = 0$ and $U_2\mathsf{D}^3 = \mathsf{L}^2 \ne 0$: (4) holds at $\ell = 2$, with $+\mathsf{L}^2$ in place of the odd-ℓ sign. □

Remark 24.5 (The plan's statement, and what replaces it). The slot-11 plan (and its audit's finding M-1, which corrected an earlier "$(1+\ell)\ell^a$") stated "$U_\ell = 0$ on all of $H^0(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell)$". That is true for $n < \ell^3 - \ell$ and false at $n = \ell^3 - \ell$ (Proposition 24.4(4)); the audit's check covered D and D^2 only. What holds at every $n \ge 1$ is $U_\ell^2 = 0$ on H^0, which is what "every Dickson class is non-ordinary" needs (Corollary 25.3). At $n = 0$, $U_\ell = \ell + 1 = 1$ on $H^0(\Gamma; \mathbb{F}_\ell)$: the constant class is ordinary.

24.3 The quotient Q_n and the operator θ_{L}

Lemma 24.6 (Definition 11.6 with any monoid fibre)**.** *[proved here] Assume (H4). Let W be an R-module with a monoid homomorphism $\rho_W : \mathrm{M}_2(\mathbb{Z}) \to \mathrm{End}_R(W)$ (for instance M_m, Q_n, via reduction mod ℓ). Then (21) with ρ_W in place of ρ_n defines $T_{\Gamma g \Gamma}$ on $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), W)$ for every $g \in \Delta$ and every datum $(\{s_j\}, \tau)$; the hypothesis-free half of Theorem 11.7 (equivariance and the chain-map property) and Remark 11.8 hold verbatim; and a $\mathbb{Z}[\Delta]$-linear map $\phi : W \to W'$ induces a cochain map that commutes with $T_{\Gamma g \Gamma}$.* The datum $(\{s_j\}, \tau)$ does not involve W *(Definition 11.1): one datum serves every fibre at once.*

Proof. The proof of Theorem 11.7's first half uses of ρ_n only $\rho_n(A_1A_2) = \rho_n(A_1)\rho_n(A_2)$ on the integral identity $\delta_j\, g^{-1} h_j(\gamma) g = s_j^{-1} h_j(\gamma) g$; that of Remark 11.8 uses (T3) only. For ϕ: $\phi\rho_W(\delta_j) = \rho_{W'}(\delta_j)\phi$, term by term in (21). □

Definition 24.7 (The quotient Q_n, its fixed basis, and its monoid action)**.** Let ℓ be a prime, $n \ge 0$ and $n^+ := n + \ell + 1$. Put $Q_n := M_{n^+}/\mathsf{L} \cdot M_n$ over $\mathbb{F}_\ell$, with $\pi : M_{n^+} \to Q_n$ the projection, and for $A \in \mathrm{M}_2(\mathbb{Z})$ let $\rho_{Q_n}(A)$ be the map induced by $\rho_{n^+}(A)$ — well defined for *every* A, singular included, because $\rho_{n^+}(A)(\mathsf{L} \cdot M_n) \subseteq \mathsf{L} \cdot M_n$ by (66). Then:

1. $\dim Q_n = \ell + 1$ for every $n \ge 0$, and

$$\mathcal{B}_n := (\bar{u}_0, \bar{u}_1, \dots, \bar{u}_{\ell-1}, \bar{u}_{n^+})$$

   is a basis — **the** basis in which ρ_{Q_n} is written, fixed here. With the class map $c(0) = 0$, $c(n^+) = n^+$ and $c(j) = 1 + \big((j-1) \bmod (\ell-1)\big)$ for $1 \le j \le n^+ - 1$, one has $\pi(u_j) = \bar{u}_{c(j)}$ for every j, and hence

$$\rho_{Q_n}(A)_{s,t} \;=\; \sum_{j:\ c(j)=s} \rho_{n^+}(A)_{j,t} \qquad (s,t \in \{0,1,\dots,\ell-1,n^+\}).$$

   Equivalently: reduce $u_r \mapsto u_{r-\ell+1}$ for $r = n^+ - 1$ down to ℓ, then read the coordinates at $0,\dots,\ell-1,n^+$.
2. $A \mapsto \rho_{Q_n}(A)$ is a monoid homomorphism $\mathrm{M}_2(\mathbb{Z}) \to \mathrm{End}_{\mathbb{F}_\ell}(Q_n)$, and $\rho_{Q_n}(A)\pi = \pi\rho_{n^+}(A)$ — *untwisted.*
3. The local system L_{Q_n} on $(K, \mathcal{R})$ is that of Definition 7.2 with ρ_{Q_n} in place of ρ_n on every relation, the central relation included ($\rho_{Q_n}(-I) = (-1)^{n^+} = (-1)^n$ for ℓ odd); its complex is $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; Q_n) \cong \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), Q_n)$ by Theorem 5.10.

(1) is a statement, not only a definition, and its proof is one line: $\mathsf{L}u_i = u_{i+1} - u_{i+\ell}$, $0 \le i \le n$, so $\mathsf{L} \cdot M_n$ is spanned by the $n+1$ relations $u_j \equiv u_{j+\ell-1}$, $1 \le j \le n+1$, which are independent (their largest indices $j+\ell-1$ are distinct); every u_r with $\ell \le r \le n^+ - 1$ reduces to $u_{r-(\ell-1)}$, u_0 and u_{n^+} occur in no relation, and $\dim Q_n = (n^+ + 1) - (n+1) = \ell + 1 = |\mathcal{B}_n|$. (2) holds because ρ_{n^+} is multiplicative and preserves the subspace. This is the basis `FibreRepresentation::QuotientL` implements (slot-11 WP-5); the alternative normal form $(\bar{u}_0, \bar{u}_{n+2}, \dots, \bar{u}_{n^+})$ is *not* this document's.

Lemma 24.8 (Q_n as homogeneous functions, and its invariants)**.** *[proved here] Let ℓ be a prime and $n \ge 0$.*

1. *Evaluation $f \mapsto (v \mapsto f(v))$ induces an isomorphism of Q_n onto the space $\mathcal{F}_{n^+}$ of functions $\phi\colon \mathbb{F}_\ell^2 \to \mathbb{F}_\ell$ with $\phi(0) = 0$ and $\phi(\lambda v) = \lambda^{n^+}\phi(v)$ for $\lambda \in \mathbb{F}_\ell^\times$, which intertwines $\rho_{Q_n}(A)$ with $(A\star\phi)(v) := \phi(A^\top v)$ for every $A \in \mathrm{M}_2(\mathbb{F}_\ell)$. So Q_n depends, as a $\mathrm{M}_2(\mathbb{F}_\ell)$-module, only on $n \bmod (\ell-1)$. (As a $\mathrm{GL}_2(\mathbb{F}_\ell)$-module it is the principal series of [E, `cl:glover`](2).)*
2. *$H^0(\Gamma; Q_n) = H^0_{\mathrm{eq}}(K, \mathcal{R}; Q_n) = Q_n^{\mathrm{SL}_2(\mathbb{F}_\ell)}$ has dimension $[\,(\ell-1) \mid n+2\,]$. When it is non-zero it is spanned by*
$$\mathbf{1}_n := \bar{u}_0 - \bar{u}_{\ell-1} + \bar{u}_{n^+},$$
the class whose function is 1 on $\mathbb{F}_\ell^2 \smallsetminus \{0\}$. The image of $H^0(\Gamma; M_{n^+})$ in it is spanned by $\pi(\mathsf{D}^b) = \mathbf{1}_n$ when $b = n^+/(\ell^2-\ell)$ is an integer, and is 0 otherwise.
3. *On $\mathbf{1}_n$: $T_{\Gamma g \Gamma}\mathbf{1}_n = m_g\,\mathbf{1}_n$ if $\ell \nmid \det g$ (so $T_p\mathbf{1}_n = (p+1)\mathbf{1}_n$), and $U_\ell\,\mathbf{1}_n = 0$.*

Proof. (1) $(\rho(A)f)(v) = f(av_1 + cv_2, bv_1 + dv_2) = f(A^\top v)$. The kernel of evaluation on M_{n^+} ($n^+ \ge 1$) is $\mathsf{L} \cdot M_n$ by Lemma 24.2(5) and (4); the image lies in $\mathcal{F}_{n^+}$, which has dimension $\ell+1$ (one value per line), and so does Q_n. The space $\mathcal{F}_{n^+}$ and the $\star$-action depend on n^+ only through $\lambda \mapsto \lambda^{n^+}$, i.e. on $n^+ \bmod (\ell-1)$, and $n^+ \equiv n+2$. (2) $H^0_{\mathrm{eq}} = Q_n^\Gamma$ by Proposition 23.3(2), and Γ acts through $\mathrm{SL}_2(\mathbb{F}_\ell)$, which — like its transpose — acts transitively on $\mathbb{F}_\ell^2 \smallsetminus \{0\}$; so an invariant function is a constant c there, and a non-zero constant lies in $\mathcal{F}_{n^+}$ iff $\lambda^{n^+} = 1$ for all $\lambda \in \mathbb{F}_\ell^\times$, iff $(\ell-1) \mid n^+$, iff $(\ell-1) \mid n+2$. The representative $x^{n^+} - x^{n^+-\ell+1}y^{\ell-1} + y^{n^+}$ of $\mathbf{1}_n$ takes the value $1-1+1$, 1, 1 at points with both, only the first, only the second coordinate non-zero (all exponents are positive multiples of $\ell-1$). $\pi(\mathsf{D}^b)$ has function $\mathsf{D}(v)^b = 1$ by Lemma 24.2(4); the other basis monomials of $H^0(M_{n^+})$ are multiples of L and map to 0. (3) By Remark 11.8 (with ρ_{Q_n}, Lemma 24.6), $T\mathbf{1}_n = \sum_j \delta_j \star \mathbf{1}$ and $(A \star \mathbf{1})(v) = [A^\top v \ne 0]$. If $\ell \nmid \det \delta_j$ every term is $\mathbf{1}$. For U_ℓ, with the δ_j of Proposition 24.4(2): $\delta_{1+k}^\top v = (v_1 - kv_2, 0)$ and $\delta_{\ell+1}^\top v = (-v_2, 0)$ modulo ℓ; at $v_2 \ne 0$ the count of non-zero images is $(\ell-1)+1 = \ell$, at $v_2 = 0$ it is $\ell + 0 = \ell$, so the sum is the function $\ell = 0$. □

Definition 24.9 (The operator θ_{L})**.** For $n \ge 0$ let $\theta_{\mathsf{L}}\colon C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; M_n) \to C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; M_{n^+})$ be M_{L} applied to every cochain value, and $\pi_*\colon C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; M_{n^+}) \to C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; Q_n)$ likewise. Both are cochain maps: $M_{\mathsf{L}}\rho_n(\gamma) = \rho_{n^+}(\gamma)M_{\mathsf{L}}$ for $\gamma \in \Gamma$ by (66) with $\det\gamma = 1$, so θ_{L} carries the relations of M_n to those of M_{n^+}, and it commutes with d because it acts on values. The same symbol denotes the induced maps $H^q(\Gamma; M_n) \to H^q(\Gamma; M_{n^+})$. This is the classical θ in the cohomological model (§24.5); [E, `cl:Tlnil`]'s "$L\cdot$".

Proposition 24.10 (Twisted Hecke equivariance of θ_{L}; $U_\ell\theta_{\mathsf{L}} = 0$)**.** *[proved here] Assume (H4); ℓ any prime, $R = \mathbb{F}_\ell$. Let $g \in \Delta$ and let $(\{s_j\}, \tau)$ be one g-twisted datum; write $\widetilde{T}^{(W)}$ for (21) with fibre W (Lemma 24.6). Then,* ***at cochain level****,*

$$\widetilde{T}^{(M_{n^+})} \circ \theta_{\mathsf{L}} \;=\; \det(g) \cdot \theta_{\mathsf{L}} \circ \widetilde{T}^{(M_n)}, \qquad \pi_* \circ \widetilde{T}^{(M_{n^+})} \;=\; \widetilde{T}^{(Q_n)} \circ \pi_*.$$

Consequently, on H^q for every q and every prime p:

$$T_p\,\theta_{\mathsf{L}} = p\,\theta_{\mathsf{L}}\,T_p, \qquad T_{p^\nu}\,\theta_{\mathsf{L}} = p^\nu\,\theta_{\mathsf{L}}\,T_{p^\nu}, \qquad T(p,p)\,\theta_{\mathsf{L}} = p^2\,\theta_{\mathsf{L}}\,T(p,p),$$

the left operator on M_{n^+}, the right one on M_n; and at $p=\ell$, $U_\ell\circ\theta_{\mathsf{L}}=0$ already on cochains. ***The twist is p, not p^{-1}:*** *the $\delta_j=s_j^{-1}g$ have determinant $\det g$, are never inverted ((H2), Obligation 20.3), and $\rho_\bullet$ is covariant (Proposition 8.2(2)). The $T(p,p)$ line is consistent with Proposition 10.5: $p^{n^+}=p^n\,p^{\ell+1}=p^2p^n$ in $\mathbb{F}_\ell$.*

Proof. $(\widetilde{T}^{(M_{n^+})}\theta_{\mathsf{L}}\phi)(x)=\sum_j\rho_{n^+}(\delta_j)M_{\mathsf{L}}\phi(\tau(s_jx))=\sum_j\det(\delta_j)\,M_{\mathsf{L}}\rho_n(\delta_j)\phi(\tau(s_jx))=\det(g)\,(\theta_{\mathsf{L}}\widetilde{T}^{(M_n)}\phi)(x)$ by (66), with the same τ and s_j on both sides (Lemma 24.6). The second identity is Lemma 24.6 for π, which is $\mathbb{Z}[\Delta]$-linear by Definition 24.7(2). Summing over the double cosets of (18), each of determinant p^ν, gives the T_{p^ν} line; $T(p,p)$ is $\Gamma\,pI\,\Gamma$, of determinant p^2. At $p=\ell$, $\det(g)=\ell=0$ in $\mathbb{F}_\ell$. □

On cohomology the datum does not matter when $\ell\geq5$ (Theorem 11.7(b)); at $\ell\in\{2,3\}$ the identities hold for the operators built from one datum, which is what the shipped pipeline does (the canonical τ of Algorithm 18.3 is a datum on $C_\bullet(K^u)$, not on a fibre). This discharges the claim [E, `cl:Tlnil`] marks [standard:]: "U_ℓ kills θ-images" is Proposition 24.10, in the stronger form $U_\ell\circ\theta_{\mathsf{L}}=0$ on cochains.

24.4 The long exact sequence

Theorem 24.11 (The θ_{L}-sequence)**.** *Assume (H4) and $\ell\geq5$ (so $6\in\mathbb{F}_\ell^\times$); let $n\geq0$ and write $h^q(m)=\dim H^q(\Gamma;M_m)$.*

1. *[proved here]* $0\to C^\bullet_{\mathrm{eq}}(K,\mathcal{R};M_n)\xrightarrow{\theta_{\mathsf{L}}}C^\bullet_{\mathrm{eq}}(K,\mathcal{R};M_{n^+})\xrightarrow{\pi_*}C^\bullet_{\mathrm{eq}}(K,\mathcal{R};Q_n)\to0$ *is an exact sequence of cochain complexes, and its long exact sequence is*
$$0\to H^0(M_n)\xrightarrow{\theta_{\mathsf{L}}}H^0(M_{n^+})\xrightarrow{\pi_*}H^0(Q_n)\xrightarrow{\delta}H^1(M_n)\xrightarrow{\theta_{\mathsf{L}}}H^1(M_{n^+})\xrightarrow{\pi_*}H^1(Q_n)\to0,\tag{67}$$
every group being $H^\bullet_{\mathrm{eq}}(K,\mathcal{R};-)$, which is $H^\bullet(\Gamma;-)$ for the two Sym *terms (Remark 7.6) and is written so for all three; $H^q=0$ for $q\geq2$ on all three terms, which is why it stops. δ is the connecting map of the snake construction on cochains.*
2. *[proved here]* ***(Hecke compatibility.)*** *For $g\in\Delta$, the triple $(\det(g)\,T^{(M_n)}_{\Gamma g\Gamma},\,T^{(M_{n^+})}_{\Gamma g\Gamma},\,T^{(Q_n)}_{\Gamma g\Gamma})$ is an endomorphism of (67): besides Proposition 24.10, $\delta\circ T^{(Q_n)}_{\Gamma g\Gamma}=\det(g)\,T^{(M_n)}_{\Gamma g\Gamma}\circ\delta$. At $p=\ell$ this reads $\delta\circ U^{(Q_n)}_\ell=0$ and says nothing about U_ℓ on $\operatorname{im}\delta$.*
3. *[proved here]* ***($\ker\theta_{\mathsf{L}}=\operatorname{im}\delta$, and what it is.)*** *On H^1,*
$$\dim\ker\bigl(\theta_{\mathsf{L}}|_{H^1(M_n)}\bigr)=\operatorname{rank}\delta=[\,(\ell-1)\mid n+2\,]-[\,\ell(\ell-1)\mid n+\ell+1\,]\in\{0,1\}.$$
*When it is 1, $\ker\theta_{\mathsf{L}}=\mathbb{F}_\ell\,\delta(\mathbf{1}_n)$ and $T_p\,\delta(\mathbf{1}_n)=(1+p^{-1})\,\delta(\mathbf{1}_n)$ for every $p\neq\ell$; since $(\ell-1)\mid k:=n+2$, $1+p^{-1}=1+p^{k-1}$ in $\mathbb{F}_\ell$ — **the kernel of θ_{L} on H^1 is at most the Eisenstein line, and it is exactly that line when $(\ell-1)\mid k$ and $\ell(\ell-1)\nmid k+\ell-1$.***
4. *[proved here] (given Claim 16.5(1), which is [standard: [Shi71, Ch. 8], [Den23], as cited there])* ***(Closed forms.)*** *For n even, with d_k as in Claim 16.5 (so $d_2=0$) and $h^1(0)=0$:*
$$\begin{aligned}\dim H^0(Q_n)&=[\,(\ell-1)\mid n+2\,],\\ \operatorname{rank}\theta_{\mathsf{L}}|_{H^0}&=h^0(n),\qquad\dim\operatorname{coker}\theta_{\mathsf{L}}|_{H^0}=[\,\ell(\ell-1)\mid n+\ell+1\,],\\ \operatorname{rank}\theta_{\mathsf{L}}|_{H^1}&=h^1(n)-\operatorname{rank}\delta,\\ \dim H^1(Q_n)&=2\bigl(d_{n+\ell+3}-d_{n+2}\bigr)+[\,(\ell-1)\mid n+2\,],\end{aligned}$$
so $\dim H^1(Q_n)-\dim H^0(Q_n)=2(d_{n+\ell+3}-d_{n+2})$. For n odd all six groups are 0.
5. *[proved here]* ***(Multiplicities.)*** *Let $a=(a_p)_{p\neq\ell}$ be a system of eigenvalues in $\bar{\mathbb{F}}_\ell$ and $E_a(H)$ its generalised eigenspace in $H\otimes\bar{\mathbb{F}}_\ell$ for the commuting family $\{T_p\}_{p\neq\ell}$; write $pa:=(p\,a_p)_p$. Then θ_{L} maps $E_a(H^1(M_n))$ into $E_{pa}(H^1(M_{n^+}))$, injectively unless $a_p=1+p^{-1}$ for all p; so*
$$\dim E_{pa}\bigl(H^1(M_{n^+})\bigr)\;\geq\;\dim E_a\bigl(H^1(M_n)\bigr)-[\,a=(1+p^{-1})_p\,]\cdot\operatorname{rank}\delta.$$

Proof. (1) $C^\bullet_{\mathrm{eq}}(K,\mathcal{R};W) \cong \bigoplus_\sigma W^{\Gamma_\sigma}$ over orbit representatives (Definition 23.1(3) with W for V_R; (H4)(i),(iv)), and each $|\Gamma_\sigma| \in \{2,4,6\}$ is a unit of $\mathbb{F}_\ell$ for $\ell \geq 5$, so $W \mapsto W^{\Gamma_\sigma}$ is exact (averaging). The sequence of fibres $0 \to M_n \to M_{n^+} \to Q_n \to 0$ is exact and Γ-equivariant (Definition 24.7, (66) with $\det\gamma = 1$), so the sequence of complexes is exact in each degree. Its long exact sequence has $H^q = 0$ for $q \geq 2$ on every term by Proposition 23.3(2), which holds for every $\mathbb{Z}[\Gamma]$-module and every ℓ. (2) $T_{\Gamma g \Gamma}$ on the three terms, twisted by $\det g$ on the first, commute with θ_{L} and π_* by Proposition 24.10, so they form an endomorphism of the short exact sequence of complexes, and connecting maps are natural [standard: [Wei94, Thm. 1.3.1]]. (3) By exactness $\ker\theta_{\mathsf{L}}|_{H^1} = \operatorname{im}\delta \cong H^0(Q_n)/\pi_* H^0(M_{n^+})$; by Lemma 24.8(2) and Proposition 24.3(3) the two dimensions are $[(\ell-1) \mid n+2]$ and $[\ell(\ell-1) \mid n^+]$, the second implying the first. When $\operatorname{rank}\delta = 1$, $\delta(\mathbf{1}_n) \neq 0$ spans the image and, by (2) with $g = \operatorname{diag}(1,p)$ and Lemma 24.8(3), $p\,T_p\,\delta(\mathbf{1}_n) = \delta(T_p\mathbf{1}_n) = (p+1)\,\delta(\mathbf{1}_n)$. Finally $p^k = 1$ for $(\ell-1) \mid k$ gives $p^{-1} = p^{k-1}$. (4) The first two lines are Lemma 24.8(2) and Proposition 24.3(3); the third is exactness at $H^1(M_n)$; for the fourth, exactness at the last two spots gives $\dim H^1(Q_n) = h^1(n^+) - h^1(n) + \operatorname{rank}\delta$, and Claim 16.5(1), $h^1(m) = 2d_{m+2} - 1 + h^0(m)$ for even $m > 0$ — which also holds at $m = 0$, where $H^1(\Gamma;\mathbb{F}_\ell) = \operatorname{Hom}(\Gamma,\mathbb{F}_\ell) = 0$ ($\Gamma^{\mathrm{ab}} = \mathbb{Z}/12$, Claim 16.2(4)'s citation) and $d_2 = 0$ by the formula — together with Proposition 24.3(3), gives $2(d_{n^+2} - d_{n+2}) + [\ell(\ell-1) \mid n^+] + \operatorname{rank}\delta$, which is the displayed value. For n odd, n^+ is odd too, $-I$ acts by -1 on all three fibres, and every $W^{\Gamma_\sigma} = 0$ ($-I \in \Gamma_\sigma$, $2 \in \mathbb{F}_\ell^\times$). (5) $T_p\theta_{\mathsf{L}} = p\,\theta_{\mathsf{L}}T_p$ (Proposition 24.10) maps generalised eigenspaces as stated, and $\ker\theta_{\mathsf{L}} = \operatorname{im}\delta$ lies in the eigenspace of $a_p = 1 + p^{-1}$ by (3). □

Re-derivation note (slot-11 WP-2, not a registered validation, and not a status tag of Theorem 24.11; v1.11.0-pre.2, the WP-3 referee's m-3). (3)'s rank formula, the three dimension lines of (4) and $\dim H^0(Q_n)$ were re-derived independently of the pipeline, from the presentation of Claim 16.2(3) applied to M_n, M_{n^+} and Q_n with the basis of Definition 24.7, at $\ell \in \{5,7,11,13\}$ and every even $n \leq 60$ (0 disagreements); ρ_{Q_n}'s well-definedness and multiplicativity on random, singular included, A at $n < 40$; Proposition 24.4(1)–(3) at $\ell \in \{5,7\}$, $n \leq 220$. The registered check is slot-11 WP-5 / WP-6's `HeckeDicksonThetaValidation`. *v1.11.0*: and `HeckeDicksonThetaHeckeValidation`, each with a `slow` companion; their results are the last paragraph of Remark 24.14.

Remark 24.12 (At $\ell \in \{2,3\}$ the sequence is not available on the model). Definition 24.9, Proposition 24.10, Lemma 24.8 and Proposition 24.4 hold at every prime. Theorem 24.11 does not: at $\ell \in \{2,3\}$ the invariants functor $W \mapsto W^{\Gamma_\sigma}$ is not exact for a stabiliser whose order ℓ divides — $\langle U\rangle$ (order 6) at $\ell = 3$, where the order-4 $\langle S\rangle$ is harmless, and every stabiliser at $\ell = 2$ ($-I$ lies in each) — so the sequence of *equivariant cochain complexes* in (1) need not be exact, and the model computes the Bredon object of Proposition 23.3, not $H^\bullet(\Gamma;-)$ ((H3); the plan's audit, finding m-12). Group cohomology has its own long exact sequence for $0 \to M_n \to M_{n^+} \to Q_n \to 0$, but it does not terminate — $H^2(\Gamma;\mathbb{F}_2) \neq 0$ (Lemma 16.4(2)) — and the model does not compute it. Nothing in this section is asserted about either at $\ell \in \{2,3\}$.

24.5 θ in the literature, and what is checkable on the model

The operator θ_{L} is the cohomological counterpart of Serre's and Swinnerton-Dyer's $\theta = q\,d/dq$ on modular forms modulo ℓ. **None of the pointers below is verified against the text**: no copy of the cited works is in `docs/literature/` (rule (A2)); each is at section or theorem level from the standard literature and is flagged *VERIFY* in the slot-11 WP-2 report for the referee.

Remark 24.13 (The classical statements). Let $\ell \geq 5$, and let $\widetilde{M}_k$ be the space of reductions modulo ℓ of level-1 modular forms of weight k with ℓ-integral q-expansion; for $f \neq 0$ let $w(f)$, the *filtration*, be the least k with $f \in \widetilde{M}_k$.

1. [standard: [Ser72, §§1–2], [SD73, §3], and for general level [Kat77, its main theorem]; locators not verified; v1.21.0-pre.2: [Kat77, Theorem (2), p. 55] read (slot-21 literature pass,

`docs/implementation/slot21/missing_literature.md` §7): $\ell \nmid w(f)$ implies exact filtration $w(f)+\ell+1$, for $N \geq 3$ (level one is Serre and Swinnerton-Dyer's case); the "iff" is [Joc82b, Lemma 4.1(b), p. 257], read] $\theta\colon \widetilde{M}_k \to \widetilde{M}_{k+\ell+1}$, $a_n(\theta f) = n\,a_n(f)$; on eigenforms $a_p(\theta f) = p\,a_p(f)$ — Proposition 24.10's $T_p\theta_{\mathsf{L}} = p\theta_{\mathsf{L}}T_p$ is this, in the coefficients of cohomology; and $w(\theta f) \leq w(f)+\ell+1$ with **equality iff** $w(f) \not\equiv 0 \pmod{\ell}$ (the *filtration law*); $E_{\ell-1} \equiv 1$, so $f \in \widetilde{M}_k$ implies $f \in \widetilde{M}_{k+\ell-1}$.

2. [standard: [Joc82a, Thm. 4.1, p. 276], finiteness [Joc82a, Thm. 2.2, p. 272]; verified, slot-17 R17-4] every level-1 system of Hecke eigenvalues modulo ℓ away from ℓ is a θ-twist $(p^i a_p)_p$ of one occurring in weight $\leq \ell+1$; there are finitely many ([E, `cl:sigma`]).
3. [standard: [AS86, Lemma 2.1, p. 204] for the inclusion $\subseteq$ (verified, slot-17 R17-4); the equality and the $\det^j$ / p^j bookkeeping are not in [AS86] and rest on [E, `cl:glover`] and Proposition 24.10] the systems of eigenvalues in $H^1(\Gamma; M_n)$ are those of the Jordan–Hölder factors $\mathrm{Sym}^r \otimes \det^j$ $(0 \leq r \leq \ell-1)$, twisted by p^j, and they are the systems of mod-ℓ eigenforms of weight $k = n+2$ up to that bookkeeping — the cohomological form of (2). The factors are those of [E, `cl:glover`]; the twist $\det^j$ is realised by θ_{L}^j (Proposition 24.10).
4. [standard: [Edi92, §3] (Tate's θ-cycles) and [Joc82b]; locators not verified; v1.21.0-pre.2: [Joc82b, §7, pp. 261–264] (θ-cycles) read and in agreement (slot-21 literature pass); Edixhoven's not] **θ-cycles.** For an eigenform f with $\theta f \neq 0$ the sequence $w(\theta f), w(\theta^2 f), \ldots$ is periodic of period dividing $\ell-1$ ($\theta^\ell = \theta$ on q-expansions); by the filtration law each step raises w by $\ell+1$ except at a *drop*, which occurs exactly after a filtration $\equiv 0 \pmod{\ell}$ and lowers it to $w+\ell+1-s(\ell-1)$ for some $s \geq 1$.
5. [standard: [Gro90, the main theorem, as stated in its introduction], re-proved in [CV92]; locators not verified, and whether the latter's range includes $k = \ell-1$ is not verified either] **Companion forms** (Gross): for an ordinary eigenform f of weight $2 \leq k \leq \ell-1$ with $\bar{\rho}_f$ irreducible, there is an eigenform g of weight $k' = \ell+1-k$ with $\bar{\rho}_g \cong \bar{\rho}_f \otimes \omega^{1-k}$ (ω the mod-ℓ cyclotomic character) if and only if $\bar{\rho}_f|_{G_\ell}$ is split (tamely ramified), under Gross's further hypotheses, which are not restated here. In q-expansions $\theta g = \theta^{\ell-k+1} f$: the companion is the $\theta^{\ell-k}$-twist point of the θ-cycle, and at level 1 its existence is the statement that the cycle's filtration drops there. **Δ modulo 11 is outside this range**: $k = 12 = \ell+1$ (it is the level-1 avatar of the weight-2 newform of level 11). The endpoint $k = \ell-1$, where the companion has weight 2, is the case (C8) of Remark 24.14 uses; its hypotheses there are not restated (v1.11.0-pre.2).

Remark 24.14 (What is checkable on the model at level 1, $n \leq 200$: the prediction list). The list slot-11 WP-6 predicts from, in writing, before it computes (rule 20 of the slot plan). Each item says what it rests on. $\tau(p)$ is Ramanujan's function ($a_p(f_{12})$, §2.5).

(C1) [proved here] **On every cell** with $\ell \geq 5$: $T_p\theta_{\mathsf{L}} = p\theta_{\mathsf{L}}T_p$ entrywise on H^0 and H^1 for $p \in \{2,3,5,7\} \smallsetminus \{\ell\}$; $U_\ell\theta_{\mathsf{L}} = 0$; exactness of (67) at all six spots; the closed forms of Theorem 24.11(4); $\ker\theta_{\mathsf{L}} = \mathrm{im}\,\delta$ with $T_p = 1+p^{-1}$ on it when non-zero; the multiplicity inequality (5).

(C2) [proved here] H^0: T_p by $(1+p)p^a$ on $\mathsf{L}^a\mathsf{D}^b$; $U_\ell = 0$ on $H^0(M_n)$ at every grid cell **except** $(n,\ell) = (120,5)$, where U_5 has rank 1 and $U_5\mathsf{D}^6 = -\mathsf{L}^{20}$; $U_\ell^2 = 0$ everywhere $(n \geq 1)$; $T_p = p+1$ and $U_\ell = 0$ on $H^0(Q_n)$.

(C3) [proved here] (given Claim 16.5(1)) **Ramanujan mod 5.** At $\ell = 5$, $H^1(M_4)$ is one-dimensional (weight 6, no Dickson class), with $T_p = 1+p^5 = 1+p$ (the Eisenstein class of weight 6, by the Eichler–Shimura identification that claim cites), and $\mathrm{rank}\,\delta = 0$ $(4 \nmid 6)$, so θ_{L} embeds it in $H^1(M_{10})$ (dimension 3) as a line with $T_p = p+p^2$. [standard: the congruence $\tau(n) \equiv n\sigma_1(n) \pmod 5$, [SD73, §1]] identifies $p+p^2$ with $\tau(p)$ mod 5: the line lies in the generalised eigenspace of Δ's system, which is 2-dimensional (the $\pm$ pair).

(C4) [proved here] (same basis) **Ramanujan mod 7.** At $\ell = 7$, θ_{L} embeds the Eisenstein line of $H^1(M_2)$ ($T_p = 1+p^3$; $\mathrm{rank}\,\delta = 0$, $6 \nmid 4$) in $H^1(M_{10})$ with $T_p = p+p^4$, which is $\tau(p)$ mod 7 by $\tau(n) \equiv n\sigma_3(n) \pmod 7$ (same citation).

(C5) [proved here] (same basis) Δ **mod** 11. $\operatorname{rank}\delta = 0$ at $(n,\ell) = (10,11)$ ($10 \nmid 12$), so θ_{L} embeds $H^1(M_{10})$ (dimension 3: Eisenstein $1+p^{11} = 1+p$ and the Δ pair) in $H^1(M_{22})$ (dimension 5), with systems $p+p^2$ (once) and $p\,\tau(p)$ (twice); $\dim H^1(Q_{10}) = 2$. *The whole of* $H^1(M_{22})$ (v1.11.0-pre.2; the WP-3 referee's m-2, read on the shipped pipeline): $\theta_{\mathsf{L}}H^1(M_{10})$ (dimension 3) $\oplus$ the weight-24 Eisenstein class ($1+p^{23} = 1+p^3$) $\oplus$ a second class with system $p+p^2$ (the $\pm$ twin of θE_{12} mod 11, a cusp form of filtration 24). So the generalised eigenspace of $p+p^2$ in $H^1(M_{22})$ is **two**-dimensional and contains the θ_{L}-image as a line; "once" above is the image, not the target.

(C6) [proved here] (same basis) Δ **mod** 13. At $(10,13)$, $(\ell-1) = 12 \mid n+2$ and $156 \nmid 24$, so $\operatorname{rank}\delta = 1$: θ_{L} **kills exactly the Eisenstein line** of $H^1(M_{10})$ ($T_p = 1+p^{11} = 1+p^{-1}$; consistently $E_{12} \equiv 1 \pmod{13}$) and embeds the Δ pair in $H^1(M_{24})$ (dimension 3) with system $p\,\tau(p)$; the third class there, $T_p = 1+p$, maps onto $H^1(Q_{10})$ (dimension 1).

(C7) [conjectural] (it rests on Remark 24.13(1),(3) and on reading "the least n at which a system occurs in $H^1(\Gamma; M_n)$" as $w-2$, a dictionary this document does not prove; the Dickson torsion of Claim 16.7 is where it could fail) **Filtration.** A system with $\det\bar\rho = \omega^{k-1}$ occurs only in weights $\equiv k \pmod{\ell-1}$. At $\ell = 11$: $w(\Delta) = 12 \not\equiv 0$, so $w(\theta\Delta) = 24$ and $p\,\tau(p)$ is absent from $H^1(M_n)$ for $n \in \{2,12\}$ and present at $n = 22$. At $\ell = 13$: $w(\theta\Delta) = 26$, so $p\,\tau(p)$ is absent at $n = 12$ and present at $n = 24$; $26 \equiv 0 \pmod{13}$, so the next step drops: $p^2\tau(p)$ first occurs at $n = 14$ or $n = 26$ (filtration 16 or 28; 4 is excluded, the system being irreducible), and the model decides which. *Recorded at v1.11.0-pre.2* (the WP-3 referee's m-9, so that the answer is on record before WP-6 reports): the shipped pipeline gives $T_2|H^1(M_{14})$ at $\ell = 13$ the eigenvalues $9^1, 8^2, 6^1$ and T_3 the eigenvalues $12^1, 6^2, 2^1$, with $8 = 2^2\tau(2)$ and $6 = 3^2\tau(3)$ modulo 13: $p^2\tau(p)$ first occurs at $n = 14$ (filtration 16, the $s = 1$ drop), with multiplicity 2. The statement above stays the disjunction; this is the model's answer to it.

(C8) [conjectural] (it rests on Remark 24.13(5), whose hypotheses are not all checked here) **Gross at** 13. Δ mod 13 is ordinary ($\tau(13) = -577738 \equiv 8$) of weight $12 = \ell-1$; its companion would have weight 2, and $\widetilde{M}_2 = 0$ at level 1. So Gross's criterion, if it applies, says $\bar\rho_\Delta|_{G_{13}}$ is *not* split. On the model this is (C7)'s statement that $p\,\tau(p)$ first occurs at $n = 24$, not earlier; nothing further is checkable. At $\ell = 11$ Gross's theorem says nothing ($k = \ell+1$).

(C9) [proved here] **The Glover multiplicities at** $\ell = 5$, $n = 34,\dots,44$: whether the multiplicity-4 ideals of the census at the cell n (the *target*) are θ_{L}-images of multiplicity-≥ 4 systems at $n' = n-6$ (the *source*) is decided by (5) for $\theta_{n'}$ and by $\operatorname{rank}\delta$ of that map, i.e. at the source: $\operatorname{rank}\delta(n') = 1$ exactly at $n' \in \{30,38\}$ (the targets $n = 36$, 44), where $4 \mid n'+2$ and $20 \nmid n'+6$; the explanation by [E, `cl:glover`]'s factor count is then confirmed or refuted cell by cell. *Indexing* (v1.11.0; slot-11 WP-6's finding 1): through v1.11.0-pre.2 the parenthesis read "$\operatorname{rank}\delta = 1$ exactly at $n = 38$ and $n = 42$, where $4 \mid n+2$ and $20 \nmid n+6$" — $\operatorname{rank}\delta$ of θ_n *out of* the cell, in Definition 24.9's source indexing. That is true, but it is not the map the question is about; both readings are now named.

The ordinary-part predictions at $\ell = 691$ and $\ell = 2411$ belong to the Hida chapter, which slot 11 does not write (owner ruling on decision 2, 2026-09-25).

[verified computationally]^{r} *Checked at v1.11.0* (slot-11 WP-5, WP-6. *Provenance, v1.11.1* (the slot-11 WP-15 referee's M-1): WP-6's predictions (its report's §§1–2 skeleton) were committed before their runs; WP-5 records one deviation — the two closed forms $\dim H^0(Q_n)$ and $\operatorname{rank}\delta$ were checked on their first run, the predictions in hand but not committed before it — and WP-6's §1a re-aligned (C5) and (C7) to v1.11.0-pre.2 after the first runs, (C5)'s assertion tightened from "$\geq$" to "$=$"; every item held as finally stated. Through v1.11.0 this parenthesis read "each prediction was committed in writing before its run, and none was edited after it", which the reports do not support.) `HeckeDicksonThetaValidation` (`standard`, 108 cells, 979 checks) and `HeckeDicksonThetaCoverageSlowValidation` (`slow`,

404 cells, $\ell \in \{5, 7, 11, 13\}$, even $n \le 200$, 3590 checks): (C1)'s exactness at all six spots as subspaces, $H^2 = 0$ on the three terms, the closed forms of Theorem 24.11(3),(4), and H^0 spanned by the $\mathsf{L}^a\mathsf{D}^b$. `HeckeDicksonThetaHeckeValidation` (`standard`, 5482 checks; 34 337 under the read-only interlock `HECKE_THETA_IMAGE_ALL=1`, on the 384 census cells) and its `slow` companion (3836 checks): Proposition 24.10 at cochain level and on cohomology (the τ datum identical at n and n^+, Lemma 24.6), the rest of (C1), and (C2) with $(120, 5)$'s exception entrywise. **Every item held**: (C3)–(C6) as stated, each matched to one system of $\Sigma(\ell)$ and its twist ([E, `sec:atlas`]); (C7) at $\ell = 11$ and $\ell = 13$ as stated, the open alternative decided by the model ($p^2\tau(p)$ first at $n = 14$, as recorded at v1.11.0-pre.2); (C8) consistent; (C9) the inequality on every block, and cell by cell the dimension-4 ideals at $n = 34, 38, 42$ are partial images ($d_\theta = 3$) and those at $n = 40, 44$ full ($d_\theta = 4$). The composite $\theta_{\mathsf{L}}^{\ell-1}$ on the Δ block at $n_0 = 10$ has rank 2 after every step at every ℓ (at $\ell = 5$ the possible loss at $n = 22$ does not occur). This is evidence; it changes no [proved here] or [conjectural] tag above.

25 The ordinary projector

What this chapter fixes. The ordinary projector of an endomorphism of a finite-length module over $\mathbb{F}_\ell$ or $\mathbb{Z}/\ell^e$, as a basis-free object, with the properties the θ line needs ("θ-images are non-ordinary") and the ones the Hida line of slot 12 will need (functoriality, exactness, compatibility with reduction). The precision question — which e certifies the ordinary part over $\mathbb{Z}_\ell$ — is stated as open. No hypothesis on ℓ.

Definition 25.1 (The ordinary projector)**.** Let ℓ be a prime, $e \ge 1$, M a finitely generated $\mathbb{Z}/\ell^e$-module (so of finite length $\lambda(M)$ and finite as a set), and $U \in \mathrm{End}(M)$. Put

$$e_{\mathrm{ord}} = e_{\mathrm{ord}}(U) := \lim_{j \to \infty} U^{j!},$$

the limit in the discrete topology — the sequence is eventually constant (Proposition 25.2(1)). $e_{\mathrm{ord}}M$ is the **ordinary part** and $(1 - e_{\mathrm{ord}})M$ the **non-ordinary part** of (M, U); an element or submodule is *non-ordinary* if e_{ord} kills it. For H^q of the model with $U = U_\ell$ this is the ordinary part in Hida's sense.

Proposition 25.2 (Existence, Fitting, and naturality)**.** *[proved here] With the notation of Definition 25.1 and $\lambda = \lambda(M)$:*

1. ***(Fitting.)*** *$M = \mathrm{im}\, U^\lambda \oplus \ker U^\lambda$, both summands U-stable, U bijective on the first and $U^\lambda = 0$ on the second. For every $j \ge j_0 := \max(\lambda, r)$, where r is the order of $U|_{\mathrm{im}\, U^\lambda}$ in the finite group $\mathrm{Aut}(\mathrm{im}\, U^\lambda)$, $U^{j!}$ is the projection onto $\mathrm{im}\, U^\lambda$ along $\ker U^\lambda$. So e_{ord} exists, is idempotent, and $e_{\mathrm{ord}}M = \mathrm{im}\, U^\lambda$, $(1 - e_{\mathrm{ord}})M = \ker U^\lambda = \{x : U^i x = 0 \text{ for some } i\}$.*
2. ***(Canonical.)*** *e_{ord} is a polynomial in U (namely $U^{j_0!}$); it depends on no basis and commutes with every endomorphism of M commuting with U. $e_{\mathrm{ord}}M$ is the largest U-stable submodule on which U is bijective, and $(1 - e_{\mathrm{ord}})M$ the largest on which U is nilpotent.*
3. ***(Functorial and exact.)*** *If $f\colon M \to M'$ is linear with $fU = U'f$, then $f\, e_{\mathrm{ord}}(U) = e_{\mathrm{ord}}(U')\, f$. A short exact sequence of such pairs stays exact after applying e_{ord} (and after applying $1 - e_{\mathrm{ord}}$).*
4. ***(Reduction.)*** *If M_{e+1} is a $\mathbb{Z}/\ell^{e+1}$-module with U_{e+1}, and $\mathrm{red}\colon M_{e+1} \to M_e$ is a U-equivariant map to a $\mathbb{Z}/\ell^e$-module with U_e (for instance $M_e = M_{e+1}/\ell^e M_{e+1}$, or the reduction $H^q(\mathbb{Z}/\ell^{e+1}) \to H^q(\mathbb{Z}/\ell^e)$ of [E, `sec:ring`] with $U = U_\ell$), then $\mathrm{red} \circ e_{\mathrm{ord}}(U_{e+1}) = e_{\mathrm{ord}}(U_e) \circ \mathrm{red}$.*

Proof. (1) The chains $\mathrm{im}\, U^i$ and $\ker U^i$ are monotone and stabilise by $i = \lambda$ (each strict step changes a length by at least one). For $x \in M$, $U^\lambda x = U^{2\lambda} y$ for some y, so $x - U^\lambda y \in \ker U^\lambda$;

and $\operatorname{im} U^\lambda \cap \ker U^\lambda = 0$ since U^λ is injective on $\operatorname{im} U^\lambda = \operatorname{im} U^{2\lambda}$. U is surjective, hence bijective, on the finite set $\operatorname{im} U^\lambda$. For $j \geq j_0$, $r \mid j!$ and $j! \geq \lambda$, so $U^{j!}$ is the identity on the first summand and 0 on the second. (2) is read off (1): a U-stable submodule on which U is bijective lies in $\bigcap_i \operatorname{im} U^i = \operatorname{im} U^\lambda$, one on which U is nilpotent in $\ker U^\lambda$; a commuting endomorphism commutes with $U^{j_0!}$. (3) $fU^{j!} = U'^{j!}f$ for every j; take j large for both. For a short exact sequence $0 \to M' \to M \to M'' \to 0$ of pairs, e_{ord} is a natural idempotent on it, and the image of a natural idempotent on an exact sequence is a direct summand of it as a complex, hence exact. (4) is (3) for $f = \mathrm{red}$, a map of $\mathbb{Z}/\ell^{e+1}$-modules with M_e viewed as one. □

Corollary 25.3 (θ-images and Dickson classes are non-ordinary)**.** *[proved here] Assume (H4); $R = \mathbb{F}_\ell$, $U = U_\ell$.*

1. *$e_{\mathrm{ord}}(\operatorname{im}\theta_{\mathsf{L}}) = 0$ in $H^q(\Gamma; M_{n^+})$ for every q and $n \geq 0$ (Proposition 24.10; $\operatorname{im}\theta_{\mathsf{L}} \subseteq \ker U_\ell$ and e_{ord} kills $\ker U_\ell$).*
2. *$e_{\mathrm{ord}} = 0$ on $H^0(\Gamma; M_m)$ for every $m \geq 1$ (Proposition 24.4(2): $U_\ell^2 = 0$), and on $H^0(\Gamma; Q_n)$ (Lemma 24.8(3)). At $m = 0$, $e_{\mathrm{ord}} = \mathrm{id}$ on $H^0(\Gamma; \mathbb{F}_\ell)$.*
3. *If $\ell \geq 5$ and $m \geq 1$: U_ℓ is nilpotent on the ℓ-torsion $H^1(\Gamma; \mathrm{Sym}^m \mathbb{Z}^2)[\ell]$, so this subgroup is non-ordinary — by the U_ℓ-equivariant Bockstein $H^0(\Gamma; M_m) \cong H^1(\Gamma; \mathrm{Sym}^m \mathbb{Z}^2)[\ell]$ of Claim 16.7(2),(3) and (2) here. U_ℓ is zero on it for $m < \ell^3 - \ell$ and not zero at $m = \ell^3 - \ell$.*

Remark 25.4 (The ordinary part is torsion-free at level 1). [proved here] (v1.11.1; the slot-11 WP-15 referee's ruling on the author's finding F-4.) Let $\ell \geq 5$, $n \geq 1$, $H := H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2) \otimes \mathbb{Z}_\ell$ with $U = U_\ell$, and $T := H_{\mathrm{tors}}$, a finite U-stable ℓ-group. On H put $e_{\mathrm{ord}} := \varprojlim_e e_{\mathrm{ord}}(U \bmod \ell^e)$ on the $H/\ell^e H$ (compatible by Proposition 25.2(4)); on T, a module of finite length, Definition 25.1 applies directly, and the two agree on T (Proposition 25.2(3) for $T \hookrightarrow H/\ell^e H$, $\ell^e T = 0$, which is injective since $T \cap \ell^e H = 0$ for such e). Then:

1. $e_{\mathrm{ord}}(T) = 0$: the torsion of H is non-ordinary.
2. $e_{\mathrm{ord}} H$ is torsion-free, and the quotient map $H \to H/T$ restricts to an isomorphism $e_{\mathrm{ord}} H \cong e_{\mathrm{ord}}(H/T)$ commuting with every T_p: **the ordinary part of H is the ordinary part of its free quotient**, so its rank and the characteristic polynomials of the T_p on it are read on the free quotient.

Proof. (1) Put $S = e_{\mathrm{ord}}(T)$. By Proposition 25.2(1),(2) U is bijective on S, hence on the U-stable subgroup $S[\ell]$, which is finite. By Corollary 25.3(3) U is nilpotent on $T[\ell] = H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2)[\ell] \otimes \mathbb{Z}_\ell \supseteq S[\ell]$. An endomorphism that is both bijective and nilpotent exists only on the zero group, so $S[\ell] = 0$, and $S = 0$, since a non-zero finite ℓ-group has non-zero ℓ-torsion. (2) e_{ord} is $\mathbb{Z}_\ell$-linear, so $e_{\mathrm{ord}}(T) \subseteq T$ and $(e_{\mathrm{ord}} H)_{\mathrm{tors}} = e_{\mathrm{ord}} H \cap T = e_{\mathrm{ord}}(T) = 0$ (an $x = e_{\mathrm{ord}} x \in T$ lies in $e_{\mathrm{ord}}(T)$). For e with $\ell^e T = 0$ the sequence $0 \to T \to H/\ell^e H \to (H/T)/\ell^e \to 0$ is exact and U-equivariant, so e_{ord} keeps it exact (Proposition 25.2(3)): $e_{\mathrm{ord}}(H/\ell^e H) \to e_{\mathrm{ord}}((H/T)/\ell^e)$ is an isomorphism, its kernel being $e_{\mathrm{ord}}(T) = 0$ by (1). Passing to the limit over e gives $e_{\mathrm{ord}} H \cong e_{\mathrm{ord}}(H/T)$; the T_p commute with U, hence with e_{ord} (Proposition 25.2(2)). □

What this does and does not settle. Question 25.6 asks two things: whether the ordinary part is torsion-free, and which precision, computable from the cell before the run, determines it. This remark answers the first at $\ell \geq 5$, $n \geq 1$, from statements of this document alone; the second — a per-cell bound known before the run — stays open (v1.12.0-pre.1: answered at $\ell \geq 5$, $n \geq 1$ by Corollary 26.2, the item's status pending the release). Slot-11 WP-18c's idempotent on the free block (Question 25.6's v1.11.0 status note) is therefore the whole ordinary part of $H^1 \otimes \mathbb{Z}/\ell^k$'s free reading, not a part of it. At $\ell \in \{2, 3\}$ nothing is asserted (Corollary 25.3(3) is stated at $\ell \geq 5$).

Remark 25.5 (The Eisenstein line one level up is a theorem). [proved here] Let $\ell \geq 5$, $e \geq 1$ and let $p \neq \ell$ be a prime whose class generates $(\mathbb{Z}/\ell^{e+1})^\times$ — for instance a primitive root modulo ℓ^2, which then generates modulo every ℓ^{e+1}. Then for weights k, k', $1 + p^{k-1} \equiv 1 + p^{k'-1} \pmod{\ell^{e+1}}$ if and only if $k \equiv k' \pmod{(\ell-1)\ell^e}$: the Eisenstein eigenvalues of two weights congruent modulo $(\ell-1)\ell^{e-1}$ but not modulo $(\ell-1)\ell^e$ *do* differ modulo ℓ^{e+1} at such a p. (The

order of p is $(\ell-1)\ell^e$.) On a cuspidal ordinary part the corresponding failure is an observation, not a consequence of this remark (the plan's audit, finding m-13); nothing here predicts it.

Open construction item 25.6 (Certified precision for the ordinary part — ANSWERED 2026-09-26). [open] For a cell (n,ℓ), which exponent e — computable from the cell, before the run — suffices for $e_{\mathrm{ord}}(U_\ell)$ on $H^1(\Gamma;\mathrm{Sym}^n\mathbb{Z}^2\otimes\mathbb{Z}/\ell^e)$ to determine the ordinary part of $H^1(\Gamma;\mathrm{Sym}^n\mathbb{Z}^2\otimes\mathbb{Z}_\ell)$ (its rank, its torsion, and the characteristic polynomials of the T_p on it modulo a target $\ell^{e'}$)? What is known here: Proposition 25.2(4) makes the projectors compatible along reductions; Corollary 25.3(3) puts the socle of the Dickson torsion in the non-ordinary part; Remark 16.8(2) shows the torsion exponents are unbounded along $n=\ell^j m$, so no e independent of n reads every exponent (Proposition 23.17). Whether the ordinary part is torsion-free at level 1, and a certified per-cell bound, are **open**; they are the Hida chapter's (slot 12), together with the engine's per-cell precision rule (`src/ModArithmetic/ENGINE_API.md` §13.11.11; the pointer read §13.12, the 128-bit draft, through v1.11.0-pre.2). *Status at v1.11.0* (slot-11 WP-18c, [E, `sec:ring`], the digit route): a per-cell certified precision bound for the *free* part is built — $K=k+e_{\max}$, with $e_{\max}$ read from the certified K-mode at word size (Proposition 23.17(3)) and the stopping rule re-evaluated at K — so that the characteristic polynomials of the T_p on $H^1_{\mathrm{fr}}\otimes\mathbb{Z}/\ell^k$ are certified to precision k; and at $p=\ell$ the idempotent of Definition 25.1 is built on that free block as a Fitting idempotent with four certificates. $e_{\max}$ is found by a run at the cell, not "computable from the cell, before the run", and nothing is said about the torsion's ordinary part; the question above is therefore **not** answered, and the item stays [open]. *Status at v1.11.1* (the slot-11 WP-15 referee's ruling on F-4): the first half — whether the ordinary part is torsion-free at level 1 — is answered at $\ell\geq 5$, $n\geq 1$ by Remark 25.4: it is, and it equals the ordinary part of the free quotient. The sentence "What is known here" above stopped one step short of that corollary. **What stays open** is the second half alone: an exponent computable from the cell *before* the run ($e_{\max}$ is read from a run). The item stays [open], narrowed to that bound; its question text is kept as written (one version, one text). *Status at v1.12.0-pre.1* (slot-12 WP-2, §26): **answered at $\ell\geq 5$, $n\geq 1$, pending release.** By Corollary 26.2(1) the exponent asked for is the target itself, $e=e'$ (the rank already at $e=1$; v1.12.0-pre.2, R3): the torsion being non-ordinary (Remark 25.4), $e_{\mathrm{ord}}(U_\ell)$ on $H^1(\Gamma;\mathrm{Sym}^n\mathbb{Z}^2\otimes\mathbb{Z}/\ell^{e'})$ is the ordinary part of $H^1\otimes\mathbb{Z}_\ell$ reduced modulo $\ell^{e'}$ — its rank (read already over $\mathbb{F}_\ell$), its torsion (0) and the characteristic polynomials of the T_p modulo $\ell^{e'}$. For the whole free quotient (slot-11 WP-18c's free block) the working precision is $K=e'+e_{\max}$ (Corollary 26.2(2)–(4)), with $e_{\max}$ read before the run from $\rho_n(S)$, $\rho_n(T)$ by Proposition 26.1. **The status macro stays [open] until the owner-signed release v1.12.0** (slot-12 plan §5 ruling L3, §6 item 1): changing a binding item's status is a release decision. At $\ell\in\{2,3\}$ nothing is asserted. *Status at v1.12.0* (owner-signed release, 2026-09-26; slot-12 plan §5 ruling L3, §6 item 1): **the item is answered, and the tag moves [open] → [proved here].** Corollary 26.2 (with Proposition 26.1 and Remark 25.4) answers the question in full at $\ell\geq 5$, $n\geq 1$: $e=e'$ for the ordinary part, $K=e'+e_{\max}$ for the free quotient, with $e_{\max}$ computable before the run from $\rho_n(S)$, $\rho_n(T)$ alone. At $\ell\in\{2,3\}$ nothing is asserted. The question text and the boxed statement above are unchanged (rules (L1), (L2)); only the status tag moves, as 14.24's did at v1.3.0.

Code map. Not built at v1.11.0-pre.1. L· and (66)'s certificate are `HeckeExt/HeckeDicksonMultiplication` (slot 5). Definition 24.7, Theorem 24.11 and Proposition 24.4(1) are slot-11 WP-5's (`FibreRepresentation::QuotientL`, the six terms and five maps, `HeckeDicksonThetaValidation`); Proposition 24.10, Proposition 24.4(2)–(4) and Remark 24.14 WP-6's. Definition 25.1 has no implementation in slot 11 (decision 2 not taken) (through v1.11.0-pre.2; v1.11.1, the WP-15 referee's m-3: superseded by the "Built at v1.11.0" paragraph below). *Built at v1.11.0.* `LocalSystemUtils` (`FibreRepresentation::QuotientL`: ρ_{Q_n} on Definition 24.7's basis for every 2×2 matrix, singular included), `LocalSystemSL2Z` (the relation list

of Q_n), `Hecke/HeckeOperatorOnK` (its `QuotientL` arm, on the coset and τ tables of Sym^n), `HeckeExt/HeckeDicksonTheta` (`DicksonThetaCell`: the six terms, the five maps in the stored bases, δ by the snake construction, the Dickson span; slot-11 WP-5) and `HeckeExt/HeckeDicksonThetaHecke` (the Hecke operators on the three terms and the checks of Proposition 24.10, Theorem 24.11(2),(3),(5), Proposition 24.4(2)–(4) and Remark 24.14; slot-11 WP-6); validations as in Remark 24.14's last paragraph. $\ell \in \{2,3\}$ is refused by name (Remark 24.12), and so is a non-default coefficient ring. (v1.11.1: since slot-11 WP-17 the $\ell \in \{2,3\}$ refusal is the mathematical precondition [E, `sec:thetaobj`] (h) specifies — `ENGINE_CHECK_MSG` naming Lemma 16.4, a `std::logic_error` (console exit 3) — and no longer `hecke::NotImplementedInThisBuild` (exit 5); the WP-15 referee's ruling on F-2.) Definition 25.1: built at $p = \ell$ on the free block of H^1 over $\mathbb{Z}/\ell^k$, $\ell^k \geq 2^{64}$, by `Hecke/LadicCell` (slot-11 WP-18c; [E, `sec:ring`], the digit route, whose record key is `hida`); `HeckeLadicCellValidation` certifies it at $(10, 691)$, where U_{691} is ordinary on the whole free block, and at $(30, 5)$, a genuine Fitting split (U_5 with a nilpotent part of dimension 4, $\mathrm{rank}_{\mathbb{F}_5} e = 1$).

Formalization note. Lemma 24.2, Proposition 24.4 and Lemma 24.8 are polynomial identities over $\mathbb{F}_\ell$ and finite counts; Proposition 25.2 is Fitting's lemma. Theorem 24.11 needs the long exact sequence of a short exact sequence of complexes and nothing about Γ beyond Proposition 23.3(2). All are cheap; Dickson's theorem (Proposition 24.3(1)) is the one import.

26 Hida theory at level one: foundation

What this chapter fixes. The three statements every later Hida step needs, and the predictions its first computations test. (i) The ℓ-primary torsion of $H^1(\Gamma; V)$ is read *before the run* from the two matrices $\rho_n(S)$, $\rho_n(T)$ alone (Proposition 26.1); with Remark 25.4 this answers Question 25.6 at $\ell \geq 5$, $n \geq 1$ (Corollary 26.2). (ii) The ordinary projector of Definition 25.1 is one explicit power of U (Lemma 26.3). (iii) The weight towers (Definition 26.5) and what is predicted on them (Remark 26.6). Throughout $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $V = \mathrm{Sym}^n \mathbb{Z}^2$ with the action of Definition 8.1, $n \geq 1$, and $H^\bullet(\Gamma; -)$ is *group* cohomology; the model enters only through Remark 7.6 at $\ell \geq 5$. The exponent of a coefficient ring is e (§2.2); a *target* precision is written e', as in Question 25.6 — it is [E, `sec:ring`]'s k, which this document may not write, $k = n+2$ being the weight. The plan and [X] write the predictions (H1)–(H6) and the generator count of Lemma 26.3 r; here they are (Hi1)–(Hi6) and $\mathfrak{g}$, since (H1)–(H4) are the standing hypotheses and r is the restriction map (rule (A5)).

Proposition 26.1 (The torsion pre-pass)**.** *[proved here] Let ℓ be a prime and $n \geq 1$. Let $J_n \in \mathrm{M}_{2(n+1)\times(n+1)}(\mathbb{Z})$ be the matrix $\rho_n(S) - 1$ stacked over $\rho_n(T) - 1$ (in the basis (u_j) of Definition 8.1). Write the ℓ-primary torsion of $H^1(\Gamma; V)$ as $\bigoplus_{t=1}^{s} \mathbb{Z}/\ell^{a_t}$, $a_t \geq 1$, put $e_{\max} := \max_t a_t$ ($e_{\max} := 0$ if $s = 0$), and for $e \geq 0$*

$$\lambda_e := \lambda\big(H^0(\Gamma; V \otimes \mathbb{Z}/\ell^e)\big) \qquad (\lambda_0 = 0),$$

λ the length. Then:

1. ***(The e-fold Bockstein.)*** *For every $e \geq 1$ the connecting map of $0 \to V \xrightarrow{\ell^e} V \to V \otimes \mathbb{Z}/\ell^e \to 0$ is an isomorphism*

$$H^0(\Gamma; V \otimes \mathbb{Z}/\ell^e) \xrightarrow{\sim} H^1(\Gamma; V)[\ell^e], \qquad \text{hence} \qquad \lambda_e = \sum_{t=1}^{s} \min(a_t, e).$$

At $e = 1$ and $\ell \geq 5$ this is the second isomorphism of Claim 16.7(2), which it extends: the argument is the same, and it needs only $H^0(\Gamma; V) = 0$.

2. ***(Two matrices suffice.)*** *$H^0(\Gamma; V \otimes \mathbb{Z}/\ell^e) = \ker(J_n \bmod \ell^e)$, and $\operatorname{rank}_{\mathbb{Q}} J_n = n+1$. If $\ell^{b_1}, \dots, \ell^{b_{n+1}}$ are the elementary divisors of J_n over $\mathbb{Z}_{(\ell)}$, the multiset of the positive b_i is the multiset of the a_t. In particular $s = \#\{i : b_i > 0\}$ and $e_{\max} = \max_i b_i$; at $\ell \geq 5$, $s = \dim_{\mathbb{F}_\ell} H^0(\Gamma; M_n)$ is the Dickson count of Claim 16.5(2), so $e_{\max} = 0$ whenever $n < \ell + 1$ (the hypothesis $\ell \geq 5$ is needed here: at $\ell = 2$, $n = 2$ has $\mathbb{Z}/2$, the invariant $x^2 + xy + y^2$ of Remark 16.8(2), as $\ell^2 - \ell < \ell + 1$; v1.12.0-pre.2, R2).*
3. ***(The stopping rule.)*** *$\lambda_{e+1} - \lambda_e = \#\{t : a_t > e\} = (n+1) - r_{e+1}(J_n)$, with r_e as in Proposition 23.17; it is non-increasing in e and vanishes exactly for $e \geq e_{\max}$. Hence*

$$e_{\max} = \min\{e \geq 0 : \lambda_{e+1} = \lambda_e\} = \min\{e \geq 0 : r_{e+1}(J_n) = n+1\},$$

the minimum exists, and the number of summands $\mathbb{Z}/\ell^j$ is $(\lambda_j - \lambda_{j-1}) - (\lambda_{j+1} - \lambda_j)$ for $j \geq 1$. The rule reads J_n over $\mathbb{Z}/\ell^e$ for $e = 1, \dots, e_{\max} + 1$ and needs no cochain complex.
4. ***(Hecke.)*** *The isomorphism of (1) is equivariant for every double-coset operator $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$, $g \in \Delta$ — U_ℓ's $g = \mathrm{diag}(1, \ell)$ included — and, under $6 \in R^\times$, for $T_{\Gamma g \Gamma}$ (Theorem 11.10) and every operator of §10 built from it.*
5. ***(Integrally.)*** *The torsion subgroup of $H^1(\Gamma; V)$ is isomorphic to $\bigoplus_i \mathbb{Z}/d_i$, $d_1 \mid \cdots \mid d_{n+1}$ the elementary divisors of J_n over $\mathbb{Z}$.*
6. ***(Scope.)*** *(1)–(5) are statements about group cohomology and hold at* every *prime,* 2 *and* 3 *included. What needs $\ell \geq 5$ is their* use *as a precision for the model: there H^1 of $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e)$ is $H^1(\Gamma; V \otimes \mathbb{Z}/\ell^e)$ (Remark 7.6, Definition 23.20(2)), and $H^1(C^\bullet_{\mathrm{eq}}(\mathbb{Z}))$ agrees with $H^1(\Gamma; V)$ after inverting* 6 *(Proposition 23.5(4)), so its $\mathbb{Z}_{(\ell)}$-type at $\ell \geq 5$ is that of $H^1(\Gamma; V)$ (v1.12.0-pre.2, the WP-3 referee's R1: through pre.1 this read "H^1 of $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$ and of $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^e)$ is group cohomology"); hence $e_{\max}$ is the largest exponent of the $\mathbb{Z}_{(\ell)}$-type of H^1 that Proposition 23.17(3)'s K-mode reads from a run; and $H^2(\Gamma; V)$ has no ℓ-torsion (Claim 16.7(1)), so $H^1(\Gamma; V \otimes \mathbb{Z}/\ell^e) = H^1(\Gamma; V) \otimes \mathbb{Z}/\ell^e$. At $\ell \in \{2, 3\}$ neither holds — the model computes the Bredon object (Proposition 23.3; Proposition 23.17(5)), and $H^2(\Gamma; V)$ is killed by* 24 *(Claim 16.2(4)) but not asserted to vanish — so (1)–(5) give the* 2- *and* 3*-primary torsion of $H^1(\Gamma; V)$ and* ***no*** *precision for the model, and Corollary 26.2 is not asserted there.*

Proof. (1) $H^0(\Gamma; V) = V^\Gamma = 0$ for $n \geq 1$ by the argument of Claim 16.7(1), which takes place over $\mathbb{Q}$ and uses no hypothesis on ℓ. The sequence $0 \to V \xrightarrow{\ell^e} V \to V \otimes \mathbb{Z}/\ell^e \to 0$ is a short exact sequence of $\mathbb{Z}[\Delta]$-modules (V is $\mathbb{Z}$-free and ℓ^e is central), so it has a long exact cohomology sequence [standard: [Bro82, Ch. III, §§0–1]], whose first terms

$$0 = H^0(\Gamma; V) \to H^0(\Gamma; V \otimes \mathbb{Z}/\ell^e) \xrightarrow{\delta} H^1(\Gamma; V) \xrightarrow{\ell^e} H^1(\Gamma; V)$$

say that δ is injective with image $\ker \ell^e = H^1(\Gamma; V)[\ell^e]$. $H^1(\Gamma; V)$ is finitely generated: Γ is generated by S and T (Claim 16.2(3) presents it on S and ST), so a 1-cocycle is determined by its values at S and T and Z^1 embeds in $V \oplus V$. Write $H^1(\Gamma; V) \cong \mathbb{Z}^h \oplus F \oplus \bigoplus_t \mathbb{Z}/\ell^{a_t}$ with F finite of order prime to ℓ; the ℓ^e-torsion of $\mathbb{Z}/\ell^a$ is $\ell^{a - \min(a,e)}\mathbb{Z}/\ell^a$, of length $\min(a, e)$, and $\mathbb{Z}^h$, F have none. Length is additive.

(2) Since S and T generate Γ, an element of $V \otimes \mathbb{Z}/\ell^e$ is Γ-invariant iff $\rho_n(S)$ and $\rho_n(T)$ fix it, i.e. iff it lies in $\ker(J_n \bmod \ell^e)$. Over $\mathbb{Q}$ the kernel of J_n is $V_{\mathbb{Q}}^\Gamma = 0$ (the argument of Claim 16.7(1)), so $\operatorname{rank}_{\mathbb{Q}} J_n = n+1$. Over the local principal ideal domain $\mathbb{Z}_{(\ell)}$, $J_n = P \left(\begin{smallmatrix} \Lambda \\ 0 \end{smallmatrix}\right) Q$ with P, Q invertible and $\Lambda = \mathrm{diag}(\ell^{b_1}, \dots, \ell^{b_{n+1}})$, every b_i finite as the rank is full. Reduction modulo ℓ^e keeps P, Q invertible, so $\ker(J_n \bmod \ell^e) \cong \bigoplus_i \ker(\ell^{b_i} \colon \mathbb{Z}/\ell^e \to \mathbb{Z}/\ell^e)$, of length $\sum_i \min(b_i, e)$. With (1): $\sum_t \min(a_t, e) = \sum_i \min(b_i, e)$ for every $e \geq 0$; the difference between e and $e-1$ gives $\#\{t : a_t \geq e\} = \#\{i : b_i \geq e\}$ for every $e \geq 1$, and these counts determine both multisets of positive entries. At $\ell \geq 5$, $s = \dim_{\mathbb{F}_\ell} H^1(\Gamma; V)[\ell] = \lambda_1 = \dim_{\mathbb{F}_\ell} H^0(\Gamma; M_n)$, which is Claim 16.5(2)'s count, zero for $0 < n < \ell + 1$.

(3) By (1), $\lambda_{e+1}-\lambda_e=\sum_t\big(\min(a_t,e+1)-\min(a_t,e)\big)=\#\{t:a_t\ge e+1\}$. By (2) and Proposition 23.17(1) for $D=J_n$ (whose proof uses only the Smith form over $\mathbb{Z}_{(\ell)}$), $r_{e+1}(J_n)=\#\{i:b_i<e+1\}$, so $(n+1)-r_{e+1}(J_n)=\#\{i:b_i\ge e+1\}$, the same count. It is non-increasing and zero iff $e\ge\max_t a_t$; the multiplicity of j among the a_t is $\#\{a_t\ge j\}-\#\{a_t\ge j+1\}$.

(4) As in Claim 16.7(3): the sequence is one of $\mathbb{Z}[\Delta]$-modules, and res, $(\mathrm{conj}_g)_*$ and cor are morphisms of δ-functors in the coefficient module, so their composite commutes with δ. Theorem 11.10 identifies the composite with $T_{\Gamma g\Gamma}$ under $6\in R^\times$.

(5) The torsion subgroup of the finitely generated group $H^1(\Gamma;V)$ is finite and is the product of its ℓ-primary parts; the Smith form of J_n over $\mathbb{Z}$ localises to the one over $\mathbb{Z}_{(\ell)}$, with elementary divisors $\ell^{v_\ell(d_i)}$, so by (2) the ℓ-primary part is $\bigoplus_i\mathbb{Z}/\ell^{v_\ell(d_i)}$, the ℓ-primary part of $\bigoplus_i\mathbb{Z}/d_i$, for every ℓ.

(6) is read off the cited statements; the identification of the model's $H^1(\mathbb{Z}/\ell^e)$ with $H^1(\Gamma;V)\otimes\mathbb{Z}/\ell^e$ at $\ell\ge5$ is the degree-1 half of the sequence in (1), whose next term is $H^2(\Gamma;V)[\ell^e]=0$. □

Re-derivation note (not a tag; an out-of-tree script, recorded in slot-12 WP-2's report). The elementary divisors of J_n over $\mathbb{Z}_{(\ell)}$, computed exactly modulo ℓ^8, reproduce Remark 16.8(3)'s table — $\mathbb{Z}/25$ at $(30,5)$, $\mathbb{Z}/25\oplus\mathbb{Z}/5$ at $(60,5)$, $\mathbb{Z}/125\oplus(\mathbb{Z}/5)^2$ at $(150,5)$, $\mathbb{Z}/49$ at $(56,7)$, one $\mathbb{Z}/25$ among the classes at $(90,5)$, $(100,5)$, $(120,5)$ — and, by clause (5) at $\ell\in\{2,3\}$, the 2- and 3-primary torsion of H^1 in [IB] Appendix A at $n\in\{2,4,6,8,12,16,18,20\}$. This is evidence for the statement, not part of its proof; the registered check is slot-12 WP-4's.

Corollary 26.2 (The certified precision, before the run)**.** *[proved here] Let $\ell\ge5$ and $n\ge1$. Let $H:=H^1(\Gamma;V)\otimes\mathbb{Z}_\ell$ with $U=U_\ell$ and e_{ord} as in Remark 25.4, $h:=\mathrm{rank}_{\mathbb{Z}_\ell}H$ — which is $2d_k-1$ for even n (Claim 16.5(1)) and 0 for odd n, so it too is known before the run; (3) below reads it off a cyclic decomposition as a check, not as an input (v1.12.0-pre.2, R4) — and let $e_{\max}$ be Proposition 26.1's — computed from $\rho_n(S)$, $\rho_n(T)$ alone, before any cochain complex is built. For $e\ge1$ put $H_e:=H^1(\Gamma;V\otimes\mathbb{Z}/\ell^e)$; then $H_e=H/\ell^eH$, compatibly with every double-coset operator (Proposition 26.1(4),(6)), and H_e is what the model computes over $\mathbb{Z}/\ell^e$ (Definition 23.20(2)). Fix a target precision $e'\ge1$.*

1. ***(The ordinary part needs no precision beyond the target.)*** *The image of $e_{\mathrm{ord}}(U|_{H_{e'}})$ is the image of $e_{\mathrm{ord}}H$ in $H_{e'}$, and $e_{\mathrm{ord}}(U|_{H_{e'}})H_{e'}\cong e_{\mathrm{ord}}H/\ell^{e'}e_{\mathrm{ord}}H$, compatibly with every T_p: a free $\mathbb{Z}/\ell^{e'}$-module of rank $h_{\mathrm{ord}}:=\mathrm{rank}_{\mathbb{Z}_\ell}e_{\mathrm{ord}}H$ on which T_p acts by the reduction of its action on $e_{\mathrm{ord}}H$. So the rank of the ordinary part is $\dim_{\mathbb{F}_\ell}e_{\mathrm{ord}}(U_\ell)H^1(\Gamma;M_n)$ (at $e'=1$), its torsion is 0 (Remark 25.4), and the characteristic polynomial of T_p on it modulo $\ell^{e'}$ is that of T_p on $e_{\mathrm{ord}}(U|_{H_{e'}})H_{e'}$. **The exponent Question 25.6 asks for is** $e=e'$ (the rank and the torsion already at $e=1$); it depends on the cell through nothing. (v1.12.0-pre.2, R3: through pre.1 this read "$e=\max(1,e')$", vacuous under $e'\ge1$.)*
2. ***(The free quotient needs $K=e'+e_{\max}$.)*** *Put $K:=e'+e_{\max}$. The quotient $H\to H/H_{\mathrm{tors}}$ induces a surjection $H_K\to(H/H_{\mathrm{tors}})/\ell^{e'}(H/H_{\mathrm{tors}})$, compatible with every T_p, whose kernel is $H_K[\ell^{e_{\max}}]$; and multiplication by $\ell^{e_{\max}}$ gives $H_K/H_K[\ell^{e_{\max}}]\cong\ell^{e_{\max}}H_K$. So*
$$\ell^{e_{\max}}H_K\ \cong\ H_K/H_K[\ell^{e_{\max}}]\ \cong\ H^1_{\mathrm{fr}}\otimes\mathbb{Z}/\ell^{e'},$$
free of rank h over $\mathbb{Z}/\ell^{e'}$, compatibly with every T_p (U_ℓ included): the characteristic polynomials of the T_p on the free quotient are read modulo $\ell^{e'}$ on H_K, with K fixed before the run.
3. ***(Any cyclic decomposition.)*** *Every decomposition of H_K into cyclic summands has exactly h summands of order ℓ^K and the others of order $\le\ell^{e_{\max}}$; let F' be the sum of the former, T' of the latter, $\pi_{F'}$ the projection along T'. Then $F'/\ell^{e'}F'\to H_K/H_K[\ell^{e_{\max}}]$ is an isomorphism, and for Hecke operators A,B with F'-blocks $A_{F'}:=\pi_{F'}A|_{F'}$: $A_{F'}$ mod $\ell^{e'}$ is*

the operator (2) induces, and $(AB)_{F'} \equiv A_{F'}B_{F'} \pmod{\ell^{e'}}$*. This is the free-block reading of slot-11 WP-18c (Question 25.6's v1.11.0 note), whose* $e_{\max}$ *comes from a run; (2) makes the same* K *available before it.*

4. ***(The two readings agree.)*** e_{ord} *of the operator* U *induces on the module of (2) is the reduction of* $e_{\mathrm{ord}}(U|_{H/H_{\mathrm{tors}}})$*, and its image is isomorphic to* $e_{\mathrm{ord}}H/\ell^{e'}e_{\mathrm{ord}}H$*, compatibly with every* T_p *— the ordinary part of (1).*

Proof. $H_e = H/\ell^e H$: the degree-1 part of the long exact sequence of Proposition 26.1(1) is $0 \to H^1(\Gamma;V)/\ell^e \to H_e \to H^2(\Gamma;V)[\ell^e] \to 0$ and the last term is 0 (Claim 16.7(1)); tensoring the finitely generated $H^1(\Gamma;V)$ with $\mathbb{Z}_\ell$ does not change it modulo ℓ^e. Equivariance is Proposition 26.1(4)'s argument.

(1) Write $\epsilon := e_{\mathrm{ord}}$ on H. It is Hida's projector: $\epsilon = \lim_j U^{j!}$ in the ℓ-adic topology of $\mathrm{End}_{\mathbb{Z}_\ell}(H)$, since modulo ℓ^i the sequence $U^{j!}$ is eventually $e_{\mathrm{ord}}(U \bmod \ell^i)$ (Proposition 25.2(1)) and these are compatible (Proposition 25.2(4)), H being finitely generated, hence ℓ-adically complete; so "the ordinary part of $H^1 \otimes \mathbb{Z}_\ell$" of Question 25.6 is the object ϵH (v1.12.0-pre.2, R3). By its definition in Remark 25.4 as the limit of the $e_{\mathrm{ord}}(U \bmod \ell^j)$, ϵ is a $\mathbb{Z}_\ell$-linear idempotent commuting with U and every T_p (Proposition 25.2(2)), whose reduction modulo $\ell^{e'}$ is $e_{\mathrm{ord}}(U|_{H_{e'}})$. So $H = \epsilon H \oplus (1-\epsilon)H$ and $H_{e'} = \epsilon H/\ell^{e'}\epsilon H \oplus (1-\epsilon)H/\ell^{e'}(1-\epsilon)H$, with $e_{\mathrm{ord}}(U|_{H_{e'}})$ the projection onto the first summand. ϵH is finitely generated and torsion-free (Remark 25.4(2)), hence free of rank h_{ord}, so the first summand is free of rank h_{ord} over $\mathbb{Z}/\ell^{e'}$; each T_p preserves ϵH and the characteristic polynomial of the reduction of an endomorphism of a free module is the reduction of its characteristic polynomial. At $e' = 1$, $H_1 = H^1(\Gamma; M_n)$.

(2) By Proposition 26.1(1), $H_{\mathrm{tors}} \cong \bigoplus_t \mathbb{Z}/\ell^{a_t}$ with every $a_t \le e_{\max} < K$. H_{tors} is stable under every T_p (a torsion element maps to one), so $H \to H/H_{\mathrm{tors}}$ and hence $H_K = H/\ell^K H \to (H/H_{\mathrm{tors}})/\ell^K \to (H/H_{\mathrm{tors}})/\ell^{e'}$ are compatible with every T_p; they are onto. To compute the kernel choose a $\mathbb{Z}_\ell$-module splitting $H = F_0 \oplus H_{\mathrm{tors}}$ (not Hecke-stable; used only to count): $H_K = F_0/\ell^K F_0 \oplus H_{\mathrm{tors}}$, the map is the reduction $F_0/\ell^K \to F_0/\ell^{e'}$ on the first summand and 0 on H_{tors}, so its kernel is $\ell^{e'}F_0/\ell^K F_0 \oplus H_{\mathrm{tors}}$; and $H_K[\ell^{e_{\max}}] = (F_0/\ell^K)[\ell^{e_{\max}}] \oplus H_{\mathrm{tors}}[\ell^{e_{\max}}] = \ell^{K-e_{\max}}F_0/\ell^K F_0 \oplus H_{\mathrm{tors}}$, the same module since $K - e_{\max} = e'$. Multiplication by $\ell^{e_{\max}}$ is central, with kernel $H_K[\ell^{e_{\max}}]$ and image $\ell^{e_{\max}}H_K$.

(3) $H_K \cong (\mathbb{Z}/\ell^K)^h \oplus \bigoplus_t \mathbb{Z}/\ell^{a_t}$ with $a_t \le e_{\max} < K$ (as $e' \ge 1$), and the orders of the summands of a cyclic decomposition of a finite abelian ℓ-group are unique; so $F' \cong (\mathbb{Z}/\ell^K)^h$, $\ell^{e_{\max}}T' = 0$, and $H_K[\ell^{e_{\max}}] = F'[\ell^{e_{\max}}] \oplus T' = \ell^{e'}F' \oplus T'$. Hence $F' \cap H_K[\ell^{e_{\max}}] = \ell^{e'}F'$ and $F' + H_K[\ell^{e_{\max}}] = H_K$, which is the isomorphism. For $x \in F'$, $Ax = A_{F'}x + \pi_{T'}Ax$ with $\pi_{T'}Ax \in T' \subseteq H_K[\ell^{e_{\max}}]$, so $A_{F'}$ induces the operator of (2). $(AB)_{F'} - A_{F'}B_{F'} = \pi_{F'}A\,\pi_{T'}B|_{F'}$, and $A(T') \subseteq A(H_K[\ell^{e_{\max}}]) \subseteq H_K[\ell^{e_{\max}}]$, whose image under $\pi_{F'}$ is $\ell^{e'}F'$.

(4) By Remark 25.4(2) the quotient map restricts to $e_{\mathrm{ord}}H \cong e_{\mathrm{ord}}(H/H_{\mathrm{tors}})$, compatibly with every T_p; apply the argument of (1) to the free module H/H_{tors} and the projector $e_{\mathrm{ord}}(U|_{H/H_{\mathrm{tors}}})$ (the limit of the $e_{\mathrm{ord}}(U \bmod \ell^j)$ on $(H/H_{\mathrm{tors}})/\ell^j$, Proposition 25.2(4)). □

Lemma 26.3 (The ordinary projector as one power of U)**.** *[proved here] Let* ℓ *be a prime,* $K \ge 1$*,* $\mathfrak{g} \ge 1$*,* M *a* $\mathbb{Z}/\ell^K$*-module generated by at most* $\mathfrak{g}$ *elements, and* $U \in \mathrm{End}(M)$*. Let* $c := \lceil \log_\ell \mathfrak{g} \rceil$*, the least* $c \ge 0$ *with* $\ell^c \ge \mathfrak{g}$*, and put*

$$N_{\mathrm{ord}}(\mathfrak{g},K) := \ell^{\,K-1+c}\prod_{f=1}^{\mathfrak{g}}(\ell^f-1), \qquad N^{\mathrm{lcm}}_{\mathrm{ord}}(\mathfrak{g},K) := \ell^{\,K-1+c}\operatorname*{lcm}_{1\le f\le \mathfrak{g}}(\ell^f-1).$$

1. $N^{\mathrm{lcm}}_{\mathrm{ord}}(\mathfrak{g},K)$ *divides* $N_{\mathrm{ord}}(\mathfrak{g},K)$*, and* $N^{\mathrm{lcm}}_{\mathrm{ord}}(\mathfrak{g},K) \ge \mathfrak{g}K \ge \lambda(M)$*.*
2. *Every element of* $\mathrm{GL}_{\mathfrak{g}}(\mathbb{Z}/\ell^K)$ *has order dividing* $N^{\mathrm{lcm}}_{\mathrm{ord}}(\mathfrak{g},K)$*.*
3. $U^N = e_{\mathrm{ord}}(U)$ *for every positive multiple* N *of* $N^{\mathrm{lcm}}_{\mathrm{ord}}(\mathfrak{g},K)$ *— in particular for* $N = N_{\mathrm{ord}}(\mathfrak{g},K)$*, and for* $N = N_{\mathrm{ord}}(\mathfrak{g}',K')$ *whenever* $\mathfrak{g}' \ge \mathfrak{g}$ *and* $K' \ge K$*.*

No hypothesis on ℓ*.*

Proof. (1) The divisibility is that of the product by the lcm. $\ell^{K-1} \geq K$ for $\ell \geq 2$, $K \geq 1$ (induction: $\ell^K \geq 2\ell^{K-1} \geq 2K \geq K+1$), and $\ell^c \geq \mathfrak{g}$, so $N_{\mathrm{ord}}^{\mathrm{lcm}} \geq \ell^{K-1+c} \geq K\mathfrak{g}$; and M is a quotient of $(\mathbb{Z}/\ell^K)^{\mathfrak{g}}$, of length $\mathfrak{g}K$.

(2) Let $g \in \mathrm{GL}_{\mathfrak{g}}(\mathbb{Z}/\ell^K)$, $\bar{g}$ its reduction modulo ℓ, and $\chi = \prod_i p_i^{c_i} \in \mathbb{F}_\ell[x]$ the characteristic polynomial of $\bar{g}$, the p_i distinct monic irreducibles of degrees $f_i \leq \mathfrak{g}$, and $c_i \leq \mathfrak{g}$. No p_i is x, since $\chi(0) = \pm\det\bar{g} \neq 0$; so the class of x in the field $\mathbb{F}_\ell[x]/(p_i)$, of order ℓ^{f_i}, is a unit, and $p_i \mid x^{\ell^{f_i}-1} - 1 \mid x^m - 1$, $m := \mathrm{lcm}_{f\leq\mathfrak{g}}(\ell^f - 1)$. The p_i being distinct irreducibles, $\prod_i p_i \mid x^m - 1$, and since $c_i \leq \mathfrak{g} \leq \ell^c$,

$$\chi \;\Big|\; \Big(\prod_i p_i\Big)^{\ell^c} \;\Big|\; (x^m-1)^{\ell^c} = x^{m\ell^c} - 1$$

(Frobenius in characteristic ℓ). By Cayley–Hamilton $\bar{g}^{m\ell^c} = 1$, i.e. $g^{m\ell^c} \in 1 + \ell\mathrm{M}_{\mathfrak{g}}(\mathbb{Z}/\ell^K)$. For $j \geq 1$ and $Y \in \mathrm{M}_{\mathfrak{g}}(\mathbb{Z}/\ell^K)$, $(1+\ell^j Y)^\ell = 1 + \sum_{i=1}^{\ell-1}\binom{\ell}{i}\ell^{ij}Y^i + \ell^{\ell j}Y^\ell \in 1 + \ell^{j+1}\mathrm{M}_{\mathfrak{g}}$, since $\ell \mid \binom{\ell}{i}$ for $0 < i < \ell$ and $\ell j \geq 2j \geq j+1$ (this holds at $\ell = 2$ too). By induction $g^{m\ell^c\ell^{K-1}} \in 1+\ell^K\mathrm{M}_{\mathfrak{g}} = \{1\}$.

(3) By Proposition 25.2(1), $M = M_o \oplus M_n$ with $M_o = \mathrm{im}\,U^\lambda$, $M_n = \ker U^\lambda$, $\lambda = \lambda(M)$, U bijective on M_o, $U^\lambda = 0$ on M_n, and $e_{\mathrm{ord}}(U)$ the projection onto M_o along M_n. On M_n, $U^N = 0$ as $N \geq \lambda$ by (1). M_o is a direct summand, hence a quotient, of M; put $\mathfrak{g}_o := \dim_{\mathbb{F}_\ell} M_o/\ell M_o \leq \mathfrak{g}$ and choose $x_1, \ldots, x_{\mathfrak{g}_o} \in M_o$ lifting a basis of $M_o/\ell M_o$; they generate M_o (Nakayama over the local ring $\mathbb{Z}/\ell^K$). Let $\pi\colon P := (\mathbb{Z}/\ell^K)^{\mathfrak{g}_o} \to M_o$ send the standard basis to the x_i; it is onto, and an isomorphism modulo ℓ. As P is free there is $\tilde{U} \in \mathrm{End}(P)$ with $\pi\tilde{U} = U\pi$ (lift each Ux_i). Modulo ℓ, $\tilde{U}$ is the endomorphism U induces on $M_o/\ell M_o$, which is onto since U is onto M_o, hence bijective; so $\det\tilde{U}$ is a unit and $\tilde{U} \in \mathrm{GL}_{\mathfrak{g}_o}(\mathbb{Z}/\ell^K)$. Since $N_{\mathrm{ord}}^{\mathrm{lcm}}(\mathfrak{g}_o, K) \mid N_{\mathrm{ord}}^{\mathrm{lcm}}(\mathfrak{g}, K) \mid N$, (2) gives $\tilde{U}^N = 1$, so $U^N\pi = \pi\tilde{U}^N = \pi$ and $U^N = 1$ on M_o (π is onto). Thus $U^N = e_{\mathrm{ord}}(U)$. The last sentence: $N_{\mathrm{ord}}^{\mathrm{lcm}}(\mathfrak{g}, K)$ divides $N_{\mathrm{ord}}^{\mathrm{lcm}}(\mathfrak{g}', K')$, hence $N_{\mathrm{ord}}(\mathfrak{g}', K')$. □

Remark 26.4 (The terms of the exponent are needed, and where the lemma is applied). [proved here] (1) *No term can be dropped.* At $K = 1$ a unipotent Jordan block $1 + E$ of size $\mathfrak{g}$ over $\mathbb{F}_\ell$ has $(1+E)^{\ell^j} = 1 + E^{\ell^j}$, which is 1 iff $\ell^j \geq \mathfrak{g}$: its order is exactly ℓ^c. At $\mathfrak{g} = 1$ and ℓ odd, $1 + \ell \in (\mathbb{Z}/\ell^K)^\times$ has order ℓ^{K-1} and a generator of $(\mathbb{Z}/\ell^K)^\times$ has order $\ell^{K-1}(\ell-1) = N_{\mathrm{ord}}(1, K)$. A companion matrix of an irreducible polynomial of degree f whose root generates $\mathbb{F}_{\ell^f}^\times$ has order $\ell^f - 1$. [verified computationally][a] (v1.12.0-pre.2, R5; the slot-12 WP-3 referee's out-of-tree `exponent_check.py`, not a registered validation.) In every case tested — $\mathrm{GL}_{\mathfrak{g}}(\mathbb{Z}/\ell^K)$ exhaustively for $(\ell, \mathfrak{g}, K) \in \{(2,2,1),(2,2,2),(2,2,3),(2,3,1),(2,3,2),(3,2,1),(3,2,2),(3,3,1),(5,2,1),(7,2,1)\}$ and by sampling for $\mathrm{GL}_2(\mathbb{Z}/25)$ — the exponent of the group *equals* $N_{\mathrm{ord}}^{\mathrm{lcm}}(\mathfrak{g}, K)$ (e.g. 168 for $\mathrm{GL}_3(\mathbb{Z}/4)$, 600 for $\mathrm{GL}_2(\mathbb{Z}/25)$); equality in general is not claimed, the examples above showing only that each factor is attained. (The corrected exponent of [X] v1.15 §8 — slot-11 WP-14 pass 2's ruling — is $N_{\mathrm{ord}}(r, K)$; an exponent without the $\lceil\log_\ell r\rceil$ term fails on the first example as soon as $\mathfrak{g} > \ell^0 = 1$ and $K = 1$.) (2) *Where it is applied* (with Corollary 26.2; $h^1(n) := \dim_{\mathbb{F}_\ell} H^1(\Gamma; M_n)$, which is $2d_k - 1 + \lambda_1$ for even n by Claim 16.5(1), and $h = \mathrm{rank}\,H$). On $H_{e'}$ — generated by $h^1(n)$ elements, since $H_{e'}/\ell H_{e'} = H/\ell H = H^1(\Gamma; M_n)$ — $e_{\mathrm{ord}} = U^N$ with $N = N_{\mathrm{ord}}(h^1(n), e')$; on H_K likewise with $N_{\mathrm{ord}}(h^1(n), K)$; on the free module of Corollary 26.2(2), with $N_{\mathrm{ord}}(h, e')$. On an F'-block of Corollary 26.2(3) over $\mathbb{Z}/\ell^K$, the pair $(F', U_{F'})$ is a $\mathbb{Z}/\ell^K$-module generated by h elements with an endomorphism, so $(U_{F'})^N = e_{\mathrm{ord}}(U_{F'})$ for every multiple N of $N_{\mathrm{ord}}^{\mathrm{lcm}}(h, K)$; but the pair depends on the chosen decomposition and $U_{F'}$ is not the operator U induces on any Hecke module over $\mathbb{Z}/\ell^K$, so $e_{\mathrm{ord}}(U_{F'})$ means nothing beyond its reduction modulo $\ell^{e'}$, which is canonical: $U_{F'}$ mod $\ell^{e'}$ *is* the induced operator of Corollary 26.2(2),(3), and $(U_{F'})^N$ mod $\ell^{e'}$ is its e_{ord} for every multiple N of $N_{\mathrm{ord}}^{\mathrm{lcm}}(h, e')$ — [X]'s $N(r, K) = N_{\mathrm{ord}}(h, K)$ among them. Slot-11 WP-18c's idempotent is $e_{\mathrm{ord}}(U_{F'})$ of such a pair. (v1.12.0-pre.2, R6: through pre.1 this sentence said the block power "is *not* an idempotent of anything over $\mathbb{Z}/\ell^K$", which was too strong.) (3) *Size.* $\log_2 N_{\mathrm{ord}}(\mathfrak{g}, K) < (K + c + \mathfrak{g}(\mathfrak{g}+1)/2)\log_2\ell$, so U^N costs

$O\big((K+c+\mathfrak{g}^2)\log\ell\big)$ products by repeated squaring; N is never needed as an integer, only its binary digits (or the factors, one power at a time).

Definition 26.5 (Weight towers at level 1)**.** Let $\ell \geq 5$, $m \geq 1$ and $n_0 \geq 2$ even. The **tower of level m through n_0** is the set of cells

$$\mathrm{Tw}_m(\ell; n_0) := \big\{(n,\ell) :\ n \geq 2 \text{ even},\ n \equiv n_0 \pmod{(\ell-1)\ell^{m-1}}\big\}.$$

($\ell-1$ is even, so a class is parity-constant; odd n are left out because $H^1(\Gamma;V)\otimes\mathbb{Z}_\ell = 0$ for odd n at $\ell \geq 5$, 2 killing $H^\bullet(\Gamma;V)$ (Proposition 23.16(2)). $n=0$ is left out as outside Corollary 26.2.) $\mathrm{Tw}_{m+1}(\ell;n_0) \subseteq \mathrm{Tw}_m(\ell;n_0)$, and the towers of level 1 are the classes of the weight $k = n+2$ modulo $\ell-1$. To a cell attach, with $H(n) := H^1(\Gamma;\mathrm{Sym}^n\mathbb{Z}^2)\otimes\mathbb{Z}_\ell$ and $U = U_\ell$: its **ordinary rank** $h^1_{\mathrm{ord}}(n) := \mathrm{rank}_{\mathbb{Z}_\ell} e_{\mathrm{ord}}H(n)$, and for each prime $p \neq \ell$ its **ordinary characteristic polynomial** $\chi^{\mathrm{ord}}_{n,p} \in \mathbb{Z}_\ell[X]$, that of T_p on the free module $e_{\mathrm{ord}}H(n)$ (Remark 25.4). The **datum of the tower at level** m is $\big(h^1_{\mathrm{ord}}(n),\ \chi^{\mathrm{ord}}_{n,p} \bmod \ell^m\big)_n$. By Corollary 26.2(1) it is computed on $H^1(\Gamma; V\otimes\mathbb{Z}/\ell^m)$ — over the ring $\mathbb{Z}/\ell^m$, with no extra precision — and the rank already over $\mathbb{F}_\ell$; equivalently, by Corollary 26.2(2)–(4), on H_K at $K = m + e_{\max}(n,\ell)$.

Remark 26.6 (What is checkable on the towers, and at $(10, 2411)$: the prediction list). Committed at v1.12.0-pre.1, before the code that tests them (slot-12 WP-4 and WP-5; the plan's rule 20). Each item carries its own tag. A failure of a [proved here] item is a defect of the implementation or of this document and stops the package; a failure of a [conjectural] item is a finding, recorded and not repaired.

(Hi1) [proved here] **Pre-pass versus run.** At every cell (n,ℓ), $\ell \geq 5$, $n \geq 1$, where the word-size K-mode of Proposition 23.17(3) certifies, the pre-pass $e_{\max}$ of Proposition 26.1(3) equals the largest torsion exponent of the certified $\mathbb{Z}_{(\ell)}$-type of $H^1(C^\bullet_{\mathrm{eq}}(\mathbb{Z}))$ (slot-11 WP-18c's $e_{\max}$), and $\lambda_e = \sum_t \min(a_t, e)$ for its invariant factors ℓ^{a_t} at every e; the isomorphism of Proposition 26.1(1) intertwines the T_p on H^0 with those on $H^1[\ell^e]$. (Proposition 26.1(1), (2),(4),(6).)

(Hi2) [proved here] **The power is the idempotent**, on named modules (v1.12.0-pre.2, R7). (i) On $F'/\ell^{e'}F'$ (a free block of Corollary 26.2(3)): $(U_{F'})^N \bmod \ell^{e'}$, N any multiple of $N^{\mathrm{lcm}}_{\mathrm{ord}}(h,e')$, equals slot-11 WP-18c's certified idempotent $e_{\mathrm{ord}}(U_{F'})$ reduced modulo $\ell^{e'}$, and both are e_{ord} of the induced operator (Remark 26.4(2)). (ii) On $H_{e'}$: $U^N = e_{\mathrm{ord}}(U|_{H_{e'}})$ for $N = N_{\mathrm{ord}}(h^1(n), e')$, idempotent and commuting with T_2, T_3. And the ordinary part read on $H_{e'}$ (Corollary 26.2(1)) has the same rank and the same characteristic polynomials modulo $\ell^{e'}$ as the one read on the free block at $K = e' + e_{\max}$ (Corollary 26.2(4)). (Lemma 26.3; uniqueness of the Fitting projection.)

(Hi3) [conjectural] **Ordinary ranks are constant on level-1 towers.** $h^1_{\mathrm{ord}}(n) = h^1_{\mathrm{ord}}(n')$ for $n, n' \geq 2$ even with $n \equiv n' \pmod{\ell-1}$, and it is read over $\mathbb{F}_\ell$ (Corollary 26.2(1)). *Source*: Hida's control theorem — the rank of the ordinary part of weight-k cusp forms on $\Gamma_0(\ell)$, $\ell \geq 5$, depends only on k modulo $\ell-1$ for $k \geq 2$ [Hid86b, Hid86a], [Hid93, Ch. 7] (locators **not verified**, rule (A2); none of the three is in `docs/literature/`). The constancy is invoked *from* $k=2$: the class $k \equiv 2 \pmod{\ell-1}$ takes its value from $S_2(\Gamma_0(\ell))$, all of which is ℓ-new with $a_\ell = \pm 1$ and therefore ordinary — this is why the genus of $X_0(\ell)$ enters — while the other classes take theirs at a weight $k \geq 3$ through the identification that follows (v1.12.0-pre.2, R8). (v1.13.0-pre.1 note, no tag moved: the locators of [Hid86b, Hid86a] are now verified, and the statements used are Theorems 27.11, 27.12; the level-ℓ identification that follows is Proposition 27.9. The sentence "not verified" above is superseded for those two papers, not for [Hid93].) For $k \geq 3$ the U_ℓ-ordinary part at level ℓ has the dimension of the level-1 one (a level-1 eigenform contributes one ordinary ℓ-stabilisation iff a_ℓ is a unit; an ℓ-new eigenform none, as $a_\ell^2 = \ell^{k-2}$), and $e_{\mathrm{ord}}H(n)\otimes\mathbb{Q}_\ell$ is the Eisenstein line plus two copies of the ordinary cusp forms (Obligation 20.7's Eichler–Shimura decomposition; the Eisenstein class has U_ℓ-eigenvalue $1+\ell^{n+1}$, a unit). Hence

the predicted *values*, from $S_2(\Gamma_0(\ell))$ (genus of $X_0(\ell)$: 0 at $5,7,13$; 1 at 11, the curve $11a$ with $a_{11}=1$) and the small weights (d_k of Claim 16.5): $h^1_{\mathrm{ord}}(n)=1$ for every even $n\geq 2$ at $\ell\in\{5,7\}$ (no ordinary cusp form of level 1 — consistent with $\tau(5)\equiv 0 \bmod 5$, $\tau(7)\equiv 0 \bmod 7$); at $\ell=11$, $h^1_{\mathrm{ord}}(n)=3$ for $n\equiv 0 \pmod{10}$ and 1 otherwise; at $\ell=13$, 3 for $n\equiv 10 \pmod{12}$ and 1 otherwise ($\tau(11)\equiv 1 \bmod 11$ — the classical congruence $\Delta\equiv f_{11a} \pmod{11}$ — and $\tau(13)\equiv 8 \bmod 13$). [verified computationally][a] evidence on the classical side, not a model computation (the slot-12 WP-3 referee's out-of-tree `hida_check.py`: T_ℓ on $S_k(\mathrm{SL}_2(\mathbb{Z}))$ from q-expansions, exactly): for every even $12\leq k\leq 72$ and $\ell\in\{5,7,11,13\}$ the number of ℓ-ordinary cusp eigenforms is exactly the table's, on all 120 pairs.

(Hi4) [conjectural] **Ordinary characteristic polynomials are congruent along towers.** For $m\in\{1,2\}$, $p\in\{2,3\}$ and two cells n,n' of one $\mathrm{Tw}_m(\ell;n_0)$ (so of equal ordinary rank, (Hi3)), $\chi^{\mathrm{ord}}_{n,p}\equiv\chi^{\mathrm{ord}}_{n',p} \pmod{\ell^m}$. *Source*: the Λ-adic form of Hida's theorem (the ordinary Hecke module is free over $\Lambda=\mathbb{Z}_\ell[[X]]$ with specialisations the weight-k ordinary parts; weights $k\equiv k' \pmod{(\ell-1)\ell^{m-1}}$ give points $u^k-1\equiv u^{k'}-1 \pmod{\ell^m}$, $u=1+\ell$) [Hid86a], [Hid93, Ch. 7] (locators not verified; v1.13.0-pre.1 note, no tag moved: for [Hid86a] and [Hid86b] they are now, Theorem 27.11). The Eisenstein factor $X-(1+p^{n+1})$ obeys it unconditionally: $p^{n+1}\equiv p^{n'+1} \pmod{\ell^m}$ as $(\ell-1)\ell^{m-1}$ is the order of $(\mathbb{Z}/\ell^m)^\times$.

(Hi5) [proved here] [standard: [Har22, Ch. 2–3], the Eisenstein class of weight k and its eigenvalues $1+p^{k-1}$ for every prime p, $p=\ell$ included, as in Obligation 20.7] **The Eisenstein root, and where the congruence stops.** At every cell, $1+p^{n+1}$ is a root of $\chi^{\mathrm{ord}}_{n,p}$ in $\mathbb{Z}_\ell$ (the Eisenstein eigenline of $H(n)\otimes\mathbb{Q}_\ell$ is a *simultaneous* eigenline of U_ℓ and of every T_p (v1.12.0-pre.2, R9); it lies in $e_{\mathrm{ord}}H(n)\otimes\mathbb{Q}_\ell$: U_ℓ is invertible on $e_{\mathrm{ord}}H(n)$ and nilpotent modulo ℓ on $(1-e_{\mathrm{ord}})H(n)$ (Proposition 25.2(1) on $H(n)/\ell H(n)$), so the eigenvalues on the second summand are non-units, while the Eisenstein eigenvalue $1+\ell^{n+1}$ is a unit; and $X-(1+p^{n+1})$ divides the monic $\chi^{\mathrm{ord}}_{n,p}$ over $\mathbb{Q}_\ell$, hence over $\mathbb{Z}_\ell$); so $\chi^{\mathrm{ord}}_{n,p}(1+p^{n+1})\equiv 0 \pmod{\ell^{e'}}$ at every precision. If p is a primitive root modulo ℓ^2 and n,n' lie in one $\mathrm{Tw}_m(\ell;n_0)$ but $n\not\equiv n' \pmod{(\ell-1)\ell^m}$, then $1+p^{n+1}\not\equiv 1+p^{n'+1} \pmod{\ell^{m+1}}$ (Remark 25.5, $k-1=n+1$): the Eisenstein roots of the two cells agree modulo ℓ^m and not modulo ℓ^{m+1}. **Not predicted**: that $\chi^{\mathrm{ord}}_{n,p}\not\equiv\chi^{\mathrm{ord}}_{n',p} \pmod{\ell^{m+1}}$ as polynomials — a cuspidal root may be congruent to the other cell's Eisenstein root (Remark 25.5's last sentence). *Addendum* (v1.21.0-pre.3; no statement or tag of this item changes): for $n<\ell-1$, [U]'s (U17)(a) of `pr:u-msym` — the ordinary $\mathfrak{m}_{\mathrm{Eis}}$-block of H^+ is free of rank one over its Hecke algebra — is equivalent to Harder's denominator [E, `cl:lveis`](2), $[H^+ : H^+_! \oplus \mathbb{Z}_\ell E] = \ell^{v(B_k/k)}$, on that block, as read in [E] after the proof of [E, `cl:lveis`] (the slot-21 referee F4 of [U], M3); for $n\geq\ell-1$ both stay conjectural.

(Hi6) [proved here] [standard: [Shi71, Ch. 8], the Eichler–Shimura isomorphism, for the characteristic polynomial of T_{2411} on $H^1(\Gamma;\mathrm{Sym}^{10}\mathbb{Q}^2)$] [verified computationally][a] **The non-ordinary example** $(10,2411)$**.** $e_{\max}(10,2411)=0$ ($n<\ell+1$, Proposition 26.1(2)), $h=h^1(10)=3$, and $T_{2411}=U_{2411}$ has characteristic polynomial $(X-\tau(2411))^2\big(X-(1+2411^{11})\big)$ on $H(10)$. Since $\tau(2411)\equiv 0 \pmod{2411}$ — indeed $\tau(2411)\equiv 150\cdot 2411 \pmod{2411^2}$, an ad hoc computation from the q-expansion of Δ ([verified computationally][a]; slot-12 WP-2's report, re-derived independently by the WP-3 referee: $\tau(2411)=4\,542\,041\,100\,095\,889\,012$), which the model is to confirm — the monic factors $f_1=(X-\tau(2411))^2$ and $f_2=X-(1+2411^{11})$ are coprime modulo 2411, so their resultant is a unit and $af_1+bf_2=1$ in $\mathbb{Z}_{2411}[X]$ (Bezout), whence $H(10)=\ker f_1(U)\oplus\ker f_2(U)$ as $\mathbb{Z}_{2411}$-modules (v1.12.0-pre.2, R10), U-stable of ranks 2 and 1, and Δ's rank-2 summand is non-ordinary ($U^2=2\tau(2411)U-\tau(2411)^2\in 2411\mathrm{End}$ on it by Cayley–Hamilton, so $U^{2m}\equiv 0$ modulo 2411^m), and over $\mathbb{Z}/2411^m$ for every $m\geq 1$ the ordinary part is the Eisenstein line: $h^1_{\mathrm{ord}}(10)=1$, the Eisenstein count, and $\chi^{\mathrm{ord}}_{10,p}=X-(1+p^{11})$.

Scope: what this chapter does not do. It proves no part of Hida theory: (Hi3) and (Hi4) are predictions from the literature, tagged [conjectural], and the model can only confirm them on a finite grid. Slot 13's (the slot-12 plan's ruling L1): control and weight independence as statements of this document; the $\Gamma_0(\ell)$-stabilisation and the level-ℓ model (at level 1, U_ℓ is the operator of (H2), $T_{\Gamma g\Gamma}$ for $g = \mathrm{diag}(1,\ell)$); the Λ-adic lattice and Hida's Hecke algebra; towers beyond ℓ^2; the cells $(700, 691)$ (beyond the 40 GB rule) and $(2420, 2411)$. At $\ell \in \{2, 3\}$ nothing of Corollary 26.2, Definition 26.5 or Remark 26.6 is asserted (Proposition 26.1(6)).

Code map. Built (slot-12 WP-4 and WP-5; recorded at v1.12.1 — until then this paragraph read "Not built at v1.12.0-pre.1 (contract first)", the plan below being its text).

- `Hecke/HeckeOrdinaryPart` (`hecke::ordinary`, WP-4): Proposition 26.1's pre-pass — $J_n = [\rho_n(S)-1; \rho_n(T)-1]$ over $\mathbb{Z}/\ell^e$, the lengths λ_e of its kernel, stopped at the first e with $\lambda_{e+1} = \lambda_e$, giving $e_{\max}$ and the torsion multiset before any run — and Lemma 26.3 (the ordinary projector as one power of U, the exponent N_{ord}); the precision use at $\ell \in \{2, 3\}$ is refused by name (Proposition 26.1(6)). `Hecke/LadicCell` takes the pre-pass as an opt-in precision source (Corollary 26.2). `HeckeOrdinaryPrecisionValidation` (`standard`): (Hi1) (pre-pass = the certified word-size K-mode on 20 cells) and (Hi2) (the projector against the digit route's certified idempotent, and on the torsion-carrying modules).
- `HeckeExt/HeckeHidaTower` (`hecke_ext::hida`, WP-5): Definition 26.5 — per cell $K = m_{\max} + e_{\max}$ from the pre-pass, U_ℓ, T_2, T_3 on H_K by the shipped module route, the ordinary part read on $H_{e'}$ — and (Hi2)–(Hi6) across the weight towers Tw_1, Tw_2, with the non-ordinary cell $(10, 2411)$. `HeckeHidaTowerValidation` (`standard`, 163 cells) and `HeckeHidaTowerSlowValidation` (`slow`, 280 cells).

Nothing of the chapter's "not asserted" list (the Λ-adic lattice, towers beyond ℓ^2, the cells $(700, 691)$, $(2420, 2411)$) is built.

Formalization note. Proposition 26.1 needs the long exact sequence of a coefficient sequence, finite generation of Γ, and the Smith form over a discrete valuation ring; Lemma 26.3 needs Fitting's lemma, Cayley–Hamilton and the binomial theorem. Both are cheap. Corollary 26.2 adds Remark 25.4. Hida's theorems ((Hi3), (Hi4)) are not formalization targets of this document.

27 Hida theory at level one: control, level ℓ, the lattice

What this chapter fixes. Everything slot 13 implements on the Hida line, and the predictions it tests. (i) **Level structure by Shapiro** (§27.1): the cohomology of $\Gamma_0(N)$ is computed on the *fixed* model $(K, \mathcal{R})$ with the induced local system of Definition 12.3, never on a triangulation of $\Gamma_0(N)\backslash\mathcal{H}$ (the owner's direction of 2026-09-26); every level-N Hecke operator, degeneracy map and level-change map is one block formula on the induced fibre that reuses the *level*-1 τ-table (Proposition 27.2); at $N = \ell$ the operator $U_\ell^{(\ell)}$ is written out (Proposition 27.3) together with the relations of the old space (Proposition 27.4); the fibre's monomial structure and the orbit decomposition over $\mathbb{P}^1(\mathbb{Z}/N)$ give H^1 as a cokernel of fixed-point spaces (Proposition 27.5), and §27.2 is the cost model, with the two regimes it serves and the obstacles to scaling (Remark 27.7). (ii) **Stabilisation and control** (§27.3): the ordinary part at level ℓ is the ordinary part at level 1, integrally (Lemma 27.8, Proposition 27.9); Hida's control and weight-independence theorems are stated as literature with verified locators (Theorems 27.11, 27.12). (iii) **The finite-level lattice datum** (Definition 27.13) and (iv) **the predictions** (Hi7)–(Hi12) (Remark 27.14).

Notation and standing conventions. $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $V = \mathrm{Sym}^n\mathbb{Z}^2$ with ρ_n of Definition 8.1, Δ the monoid of Definition 8.3, on which ρ_n is multiplicative (Proposition 8.2(2)). $g := \begin{pmatrix} 1 & 0 \\ 0 & \ell \end{pmatrix}$

throughout §§27.1–27.3 except in Proposition 27.2, where $g \in \Delta$ is arbitrary. For $N \geq 1$, $m_N := [\Gamma : \Gamma_0(N)] = \left|\mathbb{P}^1(\mathbb{Z}/N)\right|$ (the index m of §12 for $H = \Gamma_0(N)$; the subscript keeps it apart from the tower level m of Definition 26.5). For finite-index $A, B \leq \Gamma$ and $g \in \Delta$ write

$$[A\,g\,B] \;:=\; \mathrm{cor}^{B}_{B\cap gAg^{-1}} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}^{A}_{A\cap g^{-1}Bg} \colon H^\bullet(A;V_R) \to H^\bullet(B;V_R),$$

the form of (20) between two groups; $[\Gamma g \Gamma] = T_{\Gamma g \Gamma}$ (Theorem 11.10). The **level-ℓ operator** is $U_\ell^{(\ell)} := [\Gamma_0(\ell)\, g\, \Gamma_0(\ell)]$ (the plan's $U_\ell^{(\ell)}$); the level-1 operator $U_\ell = T_\ell = T_{\Gamma g \Gamma}$ keeps its name. The two **degeneracy maps** are $\mathrm{dg}_1 := [\Gamma\, 1\, \Gamma_0(\ell)] = \mathrm{res}^{\Gamma}_{\Gamma_0(\ell)}$ and $\mathrm{dg}_\ell := [\Gamma\, g\, \Gamma_0(\ell)]$ (the plan's α_1, α_ℓ; the letter α is Proposition 12.6's restriction unit into $\mathrm{Ind}_{\Gamma^0(N)}$, rule (A5)). $s_1, \ldots, s_{\ell+1}$ are the canonical transporters of Definition 10.2 at $N = \ell$ — $s_{1+b} = \left(\begin{smallmatrix} 1 & 0 \\ b & 1 \end{smallmatrix}\right)$ $(0 \leq b < \ell)$, $s_{\ell+1} = S$ — and $s_b^{(2)} := \left(\begin{smallmatrix} 1 & 0 \\ \ell b & 1 \end{smallmatrix}\right)$ $(0 \leq b < \ell)$. τ is the level-1 g-twisted chain datum of Definition 11.1 for $g = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \ell \end{smallmatrix}\right)$ — the table the pipeline already builds for U_ℓ — and $\Gamma_g'' := \Gamma \cap g\Gamma g^{-1} = \Gamma_0(\ell)$ (Proposition 10.1(3)). Group cohomology is written $H^\bullet(A;-)$; the model enters through Proposition 12.4(4), which needs $6 \in R^\times$.

27.1 Level structure by Shapiro on the fixed model

Definition 27.1 (The level-N induced model; built at $N = \ell$)**.** Let $N \geq 1$ and let $w_1 = I, w_2, \ldots, w_{m_N} \in \Gamma$ be a fixed table of representatives of $\Gamma_0(N)\backslash\Gamma$, one per class of $\mathbb{P}^1(\mathbb{Z}/N)$ under the bottom-row map of Proposition 10.1(4). (Its proof is written for $N = p^\nu$; for every N, well-definedness and injectivity hold verbatim, and surjectivity with the count $\left|\mathbb{P}^1(\mathbb{Z}/N)\right| = N\prod_{p\mid N}(1+p^{-1})$ follow from the Chinese remainder theorem and the surjectivity of $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)$; v1.13.0-pre.2, WP-3 referee m2.) At $N = p^\nu$ the table is Definition 10.2's; for composite N it is part of the interface and fixed once, like Definition 10.2, since the cochain-level matrices depend on it (Remark 11.9) (fixed for every N by Definition 28.1, v1.14.0-pre.1). The **level-N model** is the induced matrix local system $L_{\Gamma_0(N)}$ of Definition 12.3 on $(K, \mathcal{R})$: fibre $\mathrm{Ind}^{\Gamma}_{\Gamma_0(N)} V_R$ of rank $m_N(n+1)$, relation matrices $M_{\Gamma_0(N)}(\gamma)$ of (22). Each of these is **monomial**: the permutation $j \mapsto j \cdot \gamma$ of $\mathbb{P}^1(\mathbb{Z}/N)$ (the row-vector action $(c:d) \mapsto (c:d)\gamma$) with the m_N blocks $\rho_n(h_j(\gamma))$, $h_j(\gamma) = w_j \gamma w_{j\cdot\gamma}^{-1} \in \Gamma_0(N)$ — Proposition 12.2. Its cohomology $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L_{\Gamma_0(N)})$ is *the level-N cohomology on the model*; for $6 \in R^\times$ it is $H^\bullet(\Gamma_0(N); V_R)$ (Proposition 12.4(4)) [formally verified: [GT, `def:levelellmodel-model`]] (this sentence, for any table of $\Gamma_0(N)\backslash\Gamma$; not the fixed table nor the monomial matrices as C++ objects), and for $6 \notin R^\times$ nothing is asserted. Slot 13 builds $N = \ell$ (fibre rank $(\ell+1)(n+1)$) on the shipped `Hecke/InducedLocalSystem`; general N is future work, and every statement below is written for the pair of groups it needs so that nothing precludes it.

Proposition 27.2 (Level-N correspondences on the induced fibre)**.** *[proved here] Assume (H4). Let $A, B \leq \Gamma$ be of finite index, $g \in \Delta$, and fix representatives w_a of $A\backslash\Gamma$ and v_i of $B\backslash\Gamma$ ($w_1 = v_1 = I$), r_t of $B''\backslash B$ where $B'' := B \cap gAg^{-1}$, and s_j of $\Gamma_g''\backslash\Gamma$ with a g-twisted chain datum τ over R (Definition 11.1: the level-1 datum of $T_{\Gamma g\Gamma}$). For each pair (i,t) write*

$$r_t v_i = h\, s_j \quad (h \in \Gamma_g''), \qquad k := g^{-1}hg, \qquad a := \text{the index with } Ak = Aw_a, \qquad \delta_{i,t} := v_i\, s_j^{-1}\, g\, w_a^{-1},$$

so $j = j(i,t)$, $a = a(i,t)$. Define, for $\phi \in \mathrm{Hom}_{\mathbb{Z}[\Gamma]}\big(C_\bullet(K^u), \mathrm{Ind}_A^\Gamma V_R\big)$ and $x \in C_q(K^u)$, block by block in the coordinates of Definition 12.1,

$$\big(\widetilde{[A\,g\,B]}\,\phi\big)(x)_i \;:=\; \sum_t \rho_n(\delta_{i,t})\,\phi\big(\tau(s_{j(i,t)}\,x)\big)_{a(i,t)}. \tag{68}$$

Then:

1. ***(Integral, no new τ-table.)*** *$B'' \subseteq \Gamma''_g$; $k \in \Gamma$; and $\delta_{i,t} = r_t^{-1}\, g\,(k w_a^{-1})$ with $k w_a^{-1} \in A$, so $\delta_{i,t} \in \Delta$ is integral of determinant $\det g$. Formula (68) reads only the level-1 τ-table of $T_{\Gamma g \Gamma}$ (on the $s_j \widetilde{\sigma}$), the class maps of $A\backslash\Gamma$, $B\backslash\Gamma$, $\Gamma''_g\backslash\Gamma$, and ρ_n of integral matrices; no determinant-$\det g$ matrix is inverted (Obligation 20.3).*
2. ***(A cochain map, every R.)*** *$\widetilde{[A\,g\,B]}$ maps Γ-equivariant cochains to Γ-equivariant cochains and commutes with the coboundary; under Ψ (Theorem 5.10) it is a cochain map $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L_A) \to C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L_B)$, with* no *hypothesis on ℓ. [formally verified: [GT, `prop:levelcorrespondence-model`]] (in the reciprocity form of (3), on the model)*
3. ***(What it computes.)*** *Under the cochain-level Frobenius reciprocity $\mathrm{Hom}_\Gamma(C_\bullet, \mathrm{Ind}_A^\Gamma V_R) \cong \mathrm{Hom}_A(C_\bullet, V_R)$, $\phi \mapsto \phi(\cdot)_1$, it is the map $\psi \mapsto \big(y \mapsto \sum_t \rho_n(r_t^{-1} g)\, \psi(\tau(r_t y))\big)$ [formally verified: [GT, `prop:levelcorrespondence-model`]] (this formula, and its factorisation $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ at cochain level and on H^q, on the model; not the rest of (3): that it is $[A\,g\,B]$ on $H^\bullet(A; V_R)$, independent of τ and of every representative); if $6 \in R^\times$ the map it induces on $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L_A) \cong H^\bullet(A; V_R)$ is $[A\,g\,B]$, independent of τ and of every choice of representatives [standard: [Bro82, Ch. III.9–10], as in Theorem 11.10].*
4. ***(The instances.)*** *(a) $A = B = \Gamma$: (68) is (21). (b)* ***Level change as fibre maps*** *($g = I$, so $\Gamma''_I = \Gamma$, $s_1 = I$, $\tau = \mathrm{id}$): for $A \subseteq B$, $[A\,1\,B] = \mathrm{cor}^B_A$ is the* trace *$\omega_{A/B}\colon f \mapsto \big(\sum_{a:\, A w_a \subseteq B v_i} \rho_n(v_i w_a^{-1}) f_a\big)_i$; for $B \subseteq A$, $[A\,1\,B] = \mathrm{res}^A_B$ is the* projection *(pull-back along $B\backslash\Gamma \to A\backslash\Gamma$) $\alpha^0_{B/A}\colon f \mapsto \big(\rho_n(v_i w_{a(i)}^{-1}) f_{a(i)}\big)_i$, $A v_i = A w_{a(i)}$. [formally verified: [GT, `prop:levelcorrespondence-model`]] (*cor *and* res *as the factors of (3), at cochain level) Both are morphisms of matrix local systems — one monomial fibre matrix applied at every simplex — so the level change between $\Gamma_0(N') \subseteq \Gamma_0(N)$ ($N \mid N'$) is a fibre map in both directions; $\omega_{\Gamma_0(N)/\Gamma}$ is the ω of Proposition 12.7. (c)* ***Degeneracy maps as Hecke-type correspondences****: $A = \Gamma_0(N)$, $B = \Gamma_0(N')$, $g = \left(\begin{smallmatrix} 1 & 0 \\ 0 & d \end{smallmatrix}\right)$, $dN \mid N'$: then $B'' = B$ (one t) and $[A\,g\,B] = (\mathrm{conj}_g)_* \circ \mathrm{res}$ [formally verified: [GT, `prop:levelcorrespondence-model`]] (as the factor $(\mathrm{conj}_g)_* \circ \mathrm{res}$ of (3), at cochain level) — (68) with one term per block, reading, for $d = p^\nu$ a prime power, the level-1 τ-table of the primitive double coset $\Gamma \left(\begin{smallmatrix} 1 & 0 \\ 0 & d \end{smallmatrix}\right) \Gamma$ (Definition 10.2). For composite $d = d_1 d_2$ the degeneracy map is the composite of prime-power ones through $C := \Gamma_0(N d_1)$: $[A\,g\,B] = [C\,g_2\,B] \circ [A\,g_1\,C]$, $g_i = \left(\begin{smallmatrix} 1 & 0 \\ 0 & d_i \end{smallmatrix}\right)$, on cohomology for $6 \in R^\times$ (each factor has one term, and $(\mathrm{conj}_{g_2})_* \mathrm{res} \circ (\mathrm{conj}_{g_1})_* \mathrm{res} = (\mathrm{conj}_{g_1 g_2})_* \mathrm{res}$ by functoriality); at cochain level the composite of the two block formulas is the formula used, so no table for composite d is needed (v1.13.0-pre.2, WP-3 referee m2). (d)* ***Hecke operators at level*** *N: $A = B = \Gamma_0(N)$, $g = \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$ — $T_p^{(N)}$ for $p \nmid N$, $U_p^{(N)}$ for $p \mid N$ — reading the level-1 table of T_p (of U_ℓ at $p = \ell$).*

Proof. (1) $B'' = B \cap gAg^{-1} \subseteq \Gamma \cap g\Gamma g^{-1} = \Gamma''_g$, as $A, B \subseteq \Gamma$. $r_t v_i \in \Gamma$ lies in exactly one coset $\Gamma''_g s_j$, which defines j and h; $g^{-1}\Gamma''_g g \subseteq \Gamma$ by definition of Γ''_g, so $k \in \Gamma$ and a is defined. From $r_t v_i = h s_j$: $v_i s_j^{-1} = r_t^{-1} h$, so $\delta_{i,t} = r_t^{-1} h g w_a^{-1} = r_t^{-1} g\,(k w_a^{-1})$, and $k w_a^{-1} \in A$ because $Ak = Aw_a$. The rest is read off (68).

(3) The reciprocity $\phi \mapsto \phi_1 := \phi(\cdot)_1$ is a bijection with inverse $\psi \mapsto \big(x \mapsto (\psi(w_a x))_a\big)$, by $\phi(w_a x)_1 = (w_a \cdot \phi(x))(w_1) = \phi(x)(w_a)$ (Definition 12.1); it commutes with precomposition by any chain map, hence with the coboundary. Put $\psi = \phi_1$ and $(T\psi)(y) := \sum_t \rho_n(r_t^{-1} g) \psi(\tau(r_t y))$. Then $(T\psi)(v_i x) = \sum_t \rho_n(r_t^{-1} g)\, \phi(\tau(h s_j x))_1$, and by (T1) for $h \in \Gamma''_g$, $\tau(h s_j x) = k\, \tau(s_j x)$; by (22), $\phi(kz)_1 = (M_A(k)\phi(z))_1 = \rho_n(k w_a^{-1}) \phi(z)_a$ (block row 1 of $M_A(k)$ is the single block $\rho_n(h_1(k))$, $h_1(k) = w_1 k w_a^{-1}$, in column $a = 1 \cdot k$). With Proposition 8.2(2), $\rho_n(r_t^{-1} g)\rho_n(k w_a^{-1}) = \rho_n(\delta_{i,t})$, which is (68): so $\widetilde{[A\,g\,B]}\phi$ is the transport of $T\psi$. *$T\psi$ is B-equivariant*: for $b \in$

B write $r_t b = b_t r_{t\cdot b}$ with $b_t \in B''$; then $\tau(r_t b y) = (g^{-1}b_t g)\tau(r_{t\cdot b}y)$ with $g^{-1}b_t g \in A$, so $\rho_n(r_t^{-1}g)\psi(\tau(r_t b y)) = \rho_n(r_t^{-1}b_t g)\psi(\tau(r_{t\cdot b}y)) = \rho_n(b)\rho_n(r_{t\cdot b}^{-1}g)\psi(\tau(r_{t\cdot b}y))$, and $t \mapsto t\cdot b$ is a bijection — the computation of Theorem 11.7. T commutes with the coboundary because τ and each r_t are chain maps. This proves (2) by transport. If $6 \in R^\times$, $C_\bullet(K^u)\otimes R$ is a resolution of R by modules that are projective over $R[A]$ and over $R[B]$ (the stabiliser orders divide 12; the proof of Lemma 11.3(2), restriction to a finite-index subgroup keeping summands of free modules), and the three factors of $[A\,g\,B]$ are, on it, the inclusion, $\psi \mapsto \rho_n(g)\psi\circ\tau$ and the transfer $\psi \mapsto \sum_t \rho_n(r_t^{-1})\psi(r_t\,\cdot)$ — the proof of Theorem 11.10 with $(\Gamma,\Gamma',\Gamma'')$ replaced by $(A, A\cap g^{-1}Bg, B'')$; τ is a chain map over $\mathrm{conj}_g^{-1}\colon B'' \to A\cap g^{-1}Bg$ lifting the identity of R, and any two such are homotopic by the argument of Lemma 11.3(2). Composing gives T.

(4) (a) $v_1 = w_1 = I$, $B'' = \Gamma_g''$, $r_t = s_t$, so $j = t$, $h = I$, $k = I$, $a = 1$ and $\delta = s_t^{-1}g$. (b) $g = I$: $\Gamma_I'' = \Gamma$, one j with $s_1 = I$, $h = r_t v_i$, $k = h$; for $A \subseteq B$, $B'' = A$ and the $r_t v_i$ run through representatives of $A\backslash\Gamma$ inside Bv_i, giving $\rho_n(v_i w_a^{-1})$ as a runs over the A-cosets in Bv_i; for $B \subseteq A$, $B'' = B$, one t ($r_1 = I$), $a = a(i)$. A formula with $\tau = \mathrm{id}$ and one fixed matrix per block pair is a morphism of local systems (Theorem 4.5). (c) $gAg^{-1}\cap\Gamma = \Gamma_0(dN)$ by the computation of Proposition 10.1(3), which contains $B = \Gamma_0(N')$; so $B'' = B$. (d) is the definition. □

Proposition 27.3 (The level-ℓ operator as a block formula)**.** *[proved here] Assume (H4); R any commutative ring; ℓ any prime; $H := \Gamma_0(\ell)$ with $w_i = v_i = s_i$. Put*

$$\varpi \;:=\; \big(\rho_n(gs_1^{-1}),\ \rho_n(gs_2^{-1}),\ \dots,\ \rho_n(gs_\ell^{-1}),\ 0\big)\colon \mathrm{Ind}_H^\Gamma V_R \to V_R, \qquad gs_{1+b}^{-1} = \begin{pmatrix} 1 & 0 \\ -\ell b & \ell \end{pmatrix},$$

the block row whose last block (the class $(1:0)$, $s_{\ell+1} = S$) is zero. Then the instances of Proposition 27.2 at $N = \ell$ are:

$$\big(\widetilde{U}_\ell^{(\ell)}\phi\big)(x)_i = \varpi\,\phi\big(\tau(s_i x)\big), \qquad (\widetilde{\mathrm{dg}_\ell}\phi)(x)_i = \rho_n(g)\,\phi\big(\tau(s_i x)\big), \qquad (\widetilde{\mathrm{dg}_1}\phi)(x)_i = \rho_n(s_i)\,\phi(x), \tag{69}$$

$i = 1,\dots,\ell+1$, and the trace $\omega\colon f \mapsto \sum_i \rho_n(s_i^{-1})f_i$ of Proposition 12.7. On the model, with $\tau(s_i\widetilde{\sigma}) = \sum c\,\gamma\,\widetilde{\sigma'}$ the stored chain,

$$\big(\widetilde{U}_\ell^{(\ell)}\phi\big)(\widetilde{\sigma})_i \;=\; \sum_{c\,\gamma\,\widetilde{\sigma'}}\; c \sum_{j\neq(\ell+1)\cdot\gamma} \rho_n\big(g\,\gamma\,s_j^{-1}\big)\,\phi(\widetilde{\sigma'})_j, \tag{70}$$

*every $g\gamma s_j^{-1}$ ($j \neq (\ell+1)\cdot\gamma$) integral of determinant ℓ. In particular $U_\ell^{(\ell)}$, dg_ℓ and dg_1 need **no coset table and no τ-table beyond level 1's** U_ℓ: the ℓ cosets $s_b^{(2)}$ of $\Gamma_0(\ell^2)\backslash\Gamma_0(\ell)$ enter only through the closed form $s_b^{(2)-1}g = gs_{1+b}^{-1}$. By Proposition 27.2(2),(3) each is a cochain map over every R, and at $6 \in R^\times$ induces $U_\ell^{(\ell)}$, dg_ℓ, dg_1 on $H^\bullet(\Gamma_0(\ell); V_R)$.*

Proof. $U_\ell^{(\ell)}$: $B'' = H\cap gHg^{-1} = \Gamma_0(\ell^2)$ (the computation of Proposition 10.1(3)), with representatives $r_b = s_b^{(2)}$ of $\Gamma_0(\ell^2)\backslash\Gamma_0(\ell)$: their bottom rows $(\ell b, 1)$ are the ℓ classes $(c:d) \in \mathbb{P}^1(\mathbb{Z}/\ell^2)$ with $\ell \mid c$, which is $\Gamma_0(\ell^2)\backslash\Gamma_0(\ell)$ by Proposition 10.1(4) at $N = \ell^2$. Since $s_b^{(2)} \in \Gamma_0(\ell) = \Gamma_g''$, $s_b^{(2)}s_i \in \Gamma_g'' s_i$: $j = i$, $h = s_b^{(2)}$, $k = g^{-1}s_b^{(2)}g = \begin{pmatrix} 1 & 0 \\ b & 1\end{pmatrix} = s_{1+b}$, $a = 1+b$, and $\delta_{i,b} = s_i s_i^{-1} g s_{1+b}^{-1} = g s_{1+b}^{-1}$, whose value is displayed; it equals $s_b^{(2)-1}g$ by multiplying out. dg_ℓ: $A = \Gamma$ (so $\mathrm{Ind}_\Gamma^\Gamma V_R = V_R$, $a = 1$), $B = H$, $B'' = H\cap g\Gamma g^{-1} = H$, one t, $j = i$, $\delta = g$. dg_1 is Proposition 27.2(4)(b) with $A = \Gamma \supseteq B = H$. For (70): ϕ is Γ-equivariant, so $\phi(\gamma\widetilde{\sigma'}) = M_H(\gamma)\phi(\widetilde{\sigma'})$, and $\varpi M_H(\gamma)f = \sum_b \rho_n(gs_{1+b}^{-1})\rho_n(h_{1+b}(\gamma))f_{(1+b)\cdot\gamma}$ with $s_{1+b}^{-1}h_{1+b}(\gamma) = \gamma s_{(1+b)\cdot\gamma}^{-1}$ (Lemma 10.3); as b runs over $0,\dots,\ell-1$ the index $j = (1+b)\cdot\gamma$ runs over all indices but $(\ell+1)\cdot\gamma$, γ acting bijectively. Integrality: $g\gamma s_j^{-1} = gs_{1+b}^{-1}\,h_{1+b}(\gamma)$, a product of integral matrices. □

Proposition 27.4 (The old space at level ℓ). *[proved here] [verified computationally]*[a] *Assume (H4); R any commutative ring; notation of Proposition 27.3, with $T_\ell = T_{\Gamma g \Gamma}$ of level 1.*

1. $\widetilde{U}_\ell^{(\ell)} \circ \widetilde{\mathrm{dg}_1} = \ell \cdot \widetilde{\mathrm{dg}_\ell}$ as cochain maps.
2. $\omega \circ \widetilde{\mathrm{dg}_1} = (\ell+1)\cdot \mathrm{id}$ *and* $\omega \circ \widetilde{\mathrm{dg}_\ell} = \widetilde{T}_\ell$ *(the cochain map of (21))* as cochain maps.
3. *If* $6 \in R^\times$: $U_\ell^{(\ell)} \circ \mathrm{dg}_\ell = \mathrm{dg}_\ell \circ T_\ell - \ell^n \mathrm{dg}_1$ *on cohomology.*
4. *Hence, for* $6 \in R^\times$, *the map* $\Phi\colon H^\bullet(\Gamma; V_R)^{\oplus 2} \to H^\bullet(\Gamma_0(\ell); V_R)$, $(x,y) \mapsto \mathrm{dg}_1 x + \mathrm{dg}_\ell y$, *satisfies* $U_\ell^{(\ell)}\Phi = \Phi\,\mathsf{M}$ *with*

$$\mathsf{M} := \begin{pmatrix} 0 & -\ell^n \\ \ell & T_\ell \end{pmatrix}, \qquad \det(X - \mathsf{M}) = X^2 - T_\ell X + \ell^{n+1} = X^2 - T_\ell X + \ell^{k-1}.$$

[verified computationally][a] *evidence (slot-13 WP-2's out-of-tree* `levelell_check.py`*, not a registered validation; group cohomology by 1-cocycles on S, T, the operators in the form $[A\,g\,B]$ with the representatives above): (1)–(3) hold on cohomology classes over $\mathbb{F}_{1000003}$ at $(n,\ell) \in \{(2,5),(4,5),(10,5),(2,7),(6,7),(10,7),(0,5),(0,11)\}$. The script has no τ and works on cohomology classes only: the cochain-level forms of (1), (2), and the block formula (69) itself, have no numerical evidence until slot-13 WP-6 evaluates them on the model (v1.13.0-pre.2, WP-3 referee m6, m10).*

Proof. (1) By (69), block i of $\widetilde{U}_\ell^{(\ell)}\widetilde{\mathrm{dg}_1}\phi$ at x is $\varpi\big((\rho_n(s_j)\phi(\tau(s_i x)))_j\big) = \sum_{b<\ell} \rho_n(g s_{1+b}^{-1})\rho_n(s_{1+b})\phi(\tau(s_i x)) = \ell\,\rho_n(g)\phi(\tau(s_i x))$. (2) $\sum_i \rho_n(s_i^{-1})\rho_n(s_i) f = (\ell+1)f$, and $\sum_i \rho_n(s_i^{-1})\rho_n(g)\phi(\tau(s_i x)) = \sum_i \rho_n(s_i^{-1} g)\phi(\tau(s_i x))$ is (21). (3) Block i of $\widetilde{\mathrm{dg}_\ell}\widetilde{T}_\ell\phi$ is $\sum_{j=1}^{\ell+1} \rho_n(g s_j^{-1} g)\phi(\tau(s_j \tau(s_i x)))$, and block i of $\widetilde{U}_\ell^{(\ell)}\widetilde{\mathrm{dg}_\ell}\phi$ is $\sum_{b<\ell} \rho_n(g s_{1+b}^{-1})\rho_n(g)\phi(\tau(s_{1+b}\tau(s_i x)))$, the terms $j \le \ell$ of the former. The difference is the term $j = \ell+1$, $s_{\ell+1} = S$: since $gS^{-1}g = \ell S^{-1}$ and $\rho_n(\ell I) = \ell^n$ (Proposition 10.5), it is $\psi(x)_i = \ell^n \rho_n(S^{-1})\phi(\hat\tau(s_i x))$ with $\hat\tau := \tau \circ S \circ \tau$. Through the reciprocity of Proposition 27.2(3) ψ is the H-cochain map $\phi \mapsto \ell^n \rho_n(S^{-1})\phi\circ\hat\tau$. Now $\hat\tau$ is an augmentation-preserving chain map with $\hat\tau(hy) = (ShS^{-1})\hat\tau(y)$ for $h \in H$: $\tau(hy) = (g^{-1}hg)\tau(y)$, $S(g^{-1}hg)S^{-1} \in \Gamma_0(\ell)$ as $g^{-1}hg \in \Gamma^0(\ell)$, so $\tau(S(g^{-1}hg)\tau y) = g^{-1}Sg^{-1}hgS^{-1}g\,\hat\tau(y)$, and $g^{-1}Sg^{-1} = \ell^{-1}S$ gives ShS^{-1}. The map $y \mapsto Sy$ has the same three properties. For $6 \in R^\times$ the two are therefore homotopic through such maps (the argument of Lemma 11.3(2) for the subgroup H and the homomorphism $\mathrm{conj}_S\colon H \to \Gamma$), so ψ is cochain-homotopic to $\phi \mapsto \ell^n\rho_n(S^{-1})\phi(S\,\cdot) = \ell^n\phi$, i.e. to $\ell^n\widetilde{\mathrm{dg}_1}$ in the reciprocity. (4) $U_\ell^{(\ell)}(\mathrm{dg}_1 x + \mathrm{dg}_\ell y) = \ell\,\mathrm{dg}_\ell x + \mathrm{dg}_\ell T_\ell y - \ell^n \mathrm{dg}_1 y$ by (1), (3); $\det(X - \mathsf{M}) = X(X - T_\ell) + \ell^{n+1}$, and $k - 1 = n + 1$. □

Proposition 27.5 (Orbit-wise fixed points; H^1 as a cokernel). *[proved here] Let $N \ge 1$, R a commutative ring, $M := \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V_R$ with the monomial action of Definition 27.1, and $F \le \Gamma$ a finite subgroup.*

1. ***(Mackey, orbit-wise.)*** *$M^F = \bigoplus_O M_O^F$ over the F-orbits O on $\mathbb{P}^1(\mathbb{Z}/N)$, M_O the sum of the blocks $j \in O$. For $j_O \in O$, with F_{j_O} its stabiliser acting on V_R by $\gamma \mapsto \rho_n(h_{j_O}(\gamma))$ ($h_{j_O}(\gamma) \in \Gamma_0(N)$), restriction to block j_O is an isomorphism $M_O^F \to V_R^{F_{j_O}}$, inverse $f_{j_O\cdot\gamma} := \rho_n(h_{j_O}(\gamma))^{-1} f_{j_O}$. A free orbit modulo $F \cap \{\pm I\}$ contributes a copy of $V_R^{F\cap\{\pm I\}}$ and* ***no solve****.*
2. ***(The cokernel.)*** *If $6 \in R^\times$ and n is even, then*

$$H^1(\Gamma_0(N); V_R) \;\cong\; H^1(\Gamma; M) \;=\; M \big/ \big(M^{\langle S\rangle} + M^{\langle ST\rangle}\big), \qquad M^{\langle S\rangle} = \tfrac{1}{2}(1+M(S))M, \quad M^{\langle ST\rangle} = \tfrac{1}{3}\big(1+M(S$$

and $H^0 = M^{\langle S\rangle} \cap M^{\langle ST\rangle}$; both invariant spaces are computed orbit-wise by (1). If n is odd and $2 \in R^\times$, $H^\bullet(\Gamma_0(N); V_R) = 0$.
3. ***(Dimensions.)*** *If R is a field in which 6 is invertible and n is even,*

$$\dim H^1 - \dim H^0 \;=\; m_N(n+1) - \Big(\tfrac{m_N - \nu_2}{2}(n+1) + \nu_2\, d_S\Big) - \Big(\tfrac{m_N - \nu_3}{3}(n+1) + \nu_3\, d_{ST}\Big),$$

where $\nu_2 = \nu_2(N)$, $\nu_3 = \nu_3(N)$ *are the numbers of fixed points of* S, ST *on* $\mathbb{P}^1(\mathbb{Z}/N)$ *and* $d_S := \dim V_R^{\langle S\rangle}$, $d_{ST} := \dim V_R^{\langle ST\rangle}$ *(independent of* N*). At* $N = \ell \geq 5$*:* $m_\ell = \ell + 1$, $\nu_2(\ell) = 1 + \left(\frac{-1}{\ell}\right)$, $\nu_3(\ell) = 1 + \left(\frac{-3}{\ell}\right)$.

[verified computationally][a] *evidence (the same script): (3) equals the cocycle dimension of* $H^1(\Gamma; M)$ *over* $\mathbb{F}_{1000003}$ *at the eight cells of Proposition 27.4.*

Proof. (1) $(\gamma \cdot f)_j = \rho_n(h_j(\gamma))f_{j\cdot\gamma}$ ((22)), so F preserves each M_O, and $f \in M_O$ is fixed iff $f_j = \rho_n(h_j(\gamma))f_{j\cdot\gamma}$ for all $j \in O$, $\gamma \in F$. Taking $j = j_O$: f is determined by f_{j_O}, which must be fixed by $\rho_n(h_{j_O}(\gamma))$ for $\gamma \in F_{j_O}$; conversely, for such f_{j_O} the displayed inverse is well defined (if $j_O \cdot \gamma = j_O \cdot \gamma'$ then $\gamma'\gamma^{-1} \in F_{j_O}$ and $h_{j_O}(\gamma') = h_{j_O}(\gamma'\gamma^{-1})h_{j_O}(\gamma)$ by Lemma 10.3's cocycle identity) and fixed, by the same identity. On a free orbit $F_{j_O} = F \cap \{\pm I\}$ ($-I$ fixes every point of $\mathbb{P}^1$). (2) Claim 16.2(2),(3) are proved from (50) alone, which is a sequence of $\mathbb{Z}[\Gamma]$-modules; applying $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, M)$ instead of $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, V_R)$ gives the same statements for M: $H^q(\langle S\rangle; M) = H^q(\langle ST\rangle; M) = 0$ for $q \geq 1$ as the orders $4, 6$ are units, and $M^{-I} = M$ for n even ($M(-I) = (-1)^n$, Proposition 12.4(3)). The averaging idempotents are the displayed ones because $M(S)^2 = M(-I) = 1$ and $M(ST)^3 = M(-I) = 1$. The first isomorphism is Shapiro's lemma [standard: [Bro82, Ch. III.6.2]]. For n odd, $-I \in \Gamma_0(N)$ is central and acts by -1, so 2 kills $H^\bullet$ (the argument of Proposition 23.16(2)). (3) From (2), over a field, $\dim H^1 - \dim H^0 = \dim M - \dim M^{\langle S\rangle} - \dim M^{\langle ST\rangle}$ (the exact sequence $0 \to H^0 \to M^S \oplus M^{ST} \to M \to H^1 \to 0$). S and ST act on $\mathbb{P}^1(\mathbb{Z}/N)$ through $\Gamma/\{\pm I\}$ with orders 2 and 3, so every orbit is free or a fixed point; a fixed point j has $F_j = F$ acting through $h_j(\gamma) = w_j\gamma w_j^{-1}$, conjugate to F, whence $\dim V_R^{F_j} = d_F$ by (1). At $N = \ell$: $(c : d) \cdot S = (d : -c)$ is fixed iff $d = \lambda c$, $-c = \lambda d$ for a unit λ, i.e. $\lambda^2 = -1$ ($c = 0$ gives $(0 : 1) \mapsto (1 : 0)$), two points or none; $(c : d) \cdot ST = (d : d - c)$ is fixed iff $\lambda^2 - \lambda + 1 = 0$, of discriminant -3, two roots or none ($\ell \neq 3$). □

27.2 The cost model

Remark 27.6 (Dense versus monomial versus orbit-wise; in m_N, k and ℓ^l). A model of operation counts, not an assertion about running times; $d := n + 1 = k - 1$, $D_N := m_N\, d$ the fibre rank, $m_N = N\prod_{p|N}(1 + p^{-1})$ (Proposition 10.1(4) for prime powers, the Chinese remainder theorem for the rest). All counts are in operations of R.

1. **Dense** (the shipped `InducedLocalSystem` + `EquivariantCohomologyOnK`, which is what slot 13 builds at $N = \ell$, $D_\ell = (\ell + 1)d \leq 574$ on WP-6's grid): each relation matrix costs D_N^2 to store and $O(D_N^3)$ to take a fixed space; the equivariant cochain spaces have dimension $\Theta(N_q D_N)$ and their eliminations cost $O((N_q D_N)^3)$ — **cubic in** $m_N(k - 1)$. A Hecke *matrix* on H^1 on this route costs, beyond the elimination, the dense projector to cohomology, of size $\Theta((N_q D_N)^2)$, applied to each of the $\dim H^1$ image columns of the cochain map of (5) — so the first level-N Hecke matrix this programme computes is cubic in $m_N(k - 1)$, whatever (5) says about the operator (v1.13.0-pre.2, WP-3 referee M2).
2. **Monomial**: $M_{\Gamma_0(N)}(\gamma)$ is a permutation of $\mathbb{P}^1(\mathbb{Z}/N)$ and m_N blocks: $m_N d^2$ to store, $O(m_N d^2)$ to apply (against $D_N^2 = m_N^2 d^2$).
3. **Orbit-wise** (Proposition 27.5(1)): the fixed spaces of the stabilisers cost nothing on free orbits and one $d \times d$ solve per fixed point, $O((\nu_2 + \nu_3)d^3)$ in all — and cheaper still, since $V_R^{F_j} = \rho_n(w_j)V_R^{\langle S\rangle}$ resp. $\rho_n(w_j)V_R^{\langle ST\rangle}$ at a fixed point j: one $d \times d$ solve for each of S, ST once, then $O(d^2)$ per fixed point. The counts are exact for every N: a fixed point $(c : d)$ of S or ST has c a unit (at a prime $p \mid N$ with $p \mid c$, d is a unit modulo p and the fixed-point equation of the proof of Proposition 27.5(3) forces $p \nmid c$), so $\nu_2(N) = \#\{\lambda \in \mathbb{Z}/N : \lambda^2 + 1 \equiv 0\}$ and $\nu_3(N) = \#\{\lambda \in \mathbb{Z}/N : \lambda^2 - \lambda + 1 \equiv 0\}$; by the Chinese remainder theorem both are $\leq 2^{\omega(N)}$ for every N, $\nu_2(N) = 0$ if $4 \mid N$ or a prime $\equiv 3 \pmod 4$ divides N, and $\nu_3(N) = 0$ if $9 \mid N$ or a prime $\equiv 2 \pmod 3$ divides N (v1.13.0-pre.2, WP-3 referee m3; the counts checked numerically for $N \leq 30$ and six

composites by the referee's script; checked against Definition 28.1's table for every $N \le 60$ and eleven $N \le 2310$, slot-14 WP-12).

4. **The cokernel** (Proposition 27.5(2),(3)): after the orbit-local quotient by $M^{\langle S\rangle}$ (on a free S-orbit $\{j, j\cdot S\}$, $f \mapsto f_j - \rho_n(h_j(S))f_{j\cdot S}$ up to the factor $\frac{1}{2}$), what remains is one sparse elimination against $M^{\langle ST\rangle}$: about $D_N/2$ rows, $D_N/3$ columns, at most three $d\times d$ blocks per block column, with answer of dimension about $D_N/6$ ($\dim H^1 \approx 2\dim S_k(\Gamma_0(N))$). This elimination is the only super-linear step; its worst case is cubic in D_N, and this document asserts no better bound (the block sparsity is what modular-symbol implementations exploit).
5. **Hecke operators and level change** (Proposition 27.2): per simplex and per pair (i,t), one level-1 τ-chain and one $d\times d$ block per term — $O([\Gamma : B'']\,\bar{c}\,d^2)$, $\bar{c}$ the mean number of terms of a level-1 chain; for $U_\ell^{(\ell)}$, $\ell+1$ output blocks each costing ℓ blocks per term by (70). No $D_N \times D_N$ matrix is formed, and the level-1 τ-table — the dominant memory of U_ℓ (slot-13 WP-4's lever) — is shared, not replicated per level. The fibre maps of Proposition 27.2(4)(b) cost $O(m_{N'}d^2)$. So *applying* a level-N Hecke operator is linear in m_N and quadratic in d, and needs only the level-1 τ-tables of the primes in use (composite degeneracies compose, Proposition 27.2(4)(c)). **The count is for the cochain map.** Reading it on H^1 costs, on the dense route (1), the $\Theta((N_qD_N)^2)$ projector per image column — cubic in $m_N(k-1)$ in all — and on the cokernel presentation (4) it needs a Hecke action *on the presentation*, which the formula of Proposition 27.2 is not: that action is the Manin–Merel Heilbronn-matrix formula, specified by pointer and not built (Remark 27.7, (O2)).
6. **The ring $\mathbb{Z}/\ell^l$.** The monomial and orbit structures are ring-independent (a permutation and integral 2×2 matrices); the orbit-wise averaging needs only $6 \in R^\times$, so $\ell \ge 5$. Each count above is multiplied by the cost of one ring operation: a word operation for $\ell^l < 2^{64}$, and beyond it the digit route of `ENGINE_API.md` §13.11, whose per-operation cost grows with the number of digit planes. None of (2)–(5) is built at this version.

The regimes this model serves, and the obstacles to scaling, are Remark 27.7.

Remark 27.7 (The two regimes, and the obstacles to scaling in N, k and ℓ^l). [standard: [Man72] (Manin symbols and their relations); [Mer94, §2] (the Hecke operators on Manin symbols by sets of integral matrices of determinant p, the Heilbronn–Merel matrices); [Ste07, Ch. 3 and Ch. 8] and [Cre97, Ch. 2] (the algorithmic form); pointers at section and chapter level, **not** verified against the texts (rule (A2)): no copy is available to this document] Added at v1.13.0-pre.2 (WP-3 referee M2 and its scalability section §7), so that the owner's requirement — the level-N route must scale in large k, large N and large ℓ^l — is tracked in the contract and not only in a report. Notation of Remark 27.6.

The two regimes. (R-N) **Large N at small k** — the modular-symbol regime: the cokernel presentation of Proposition 27.5(2) is, for $6 \in R^\times$ and n even, the presentation of the modular symbols of weight k for $\Gamma_0(N)$ by Manin symbols $\mathbb{P}^1(\mathbb{Z}/N)\times V_R$ modulo the relations $x+xS=0$, $x+xU+xU^2=0$ ($U=ST$; the tree of (50) is the Farey tree), and the cost is the block-sparse elimination of Remark 27.6(4) plus a Hecke action on the presentation (O2). (R-k) **Large k at level 1 or ℓ** — the Hida regime of §26 and this chapter: $m_1=1$, $m_\ell=\ell+1$, the dense route of Remark 27.6(1) on the fixed model, dominated by the level-1 U_ℓ τ-table (O5). **Large N and large k together are outside every route of Remark 27.6**: every route has dense $d\times d$ blocks (O4).

The obstacles, in order of weight:

(O1) **The only built route is dense**: $O((N_qm_Nd)^3)$ per cohomology computation and a $\Theta((N_qm_Nd)^2)$ projector per Hecke column (Remark 27.6(1), (5)). Adequate on slot-13 WP-6's grid ($D_\ell \le 574$, $N_qD_\ell \approx 10^4$), not beyond.

(O2) **The cokernel presentation has no Hecke action in this document.** The action is classical: T_p on Manin symbols is a fixed finite sum over a set of integral matrices

of determinant p (Merel's Heilbronn-type matrices), acting on $\mathbb{P}^1(\mathbb{Z}/N) \times V_R$ without any τ-search — a closed-form Hecke action on the tree of (50). It is **specified here by pointer only**: transcribing it into this document's conventions (the column convention of Definition 8.1, the right action on $\mathbb{P}^1(\mathbb{Z}/N)$, the form $[A\,g\,B]$ of (20)) and proving it equal to Proposition 27.2(4)(d) on cohomology is a contract task of the size of Definition 10.2, not a search like Algorithm 18.3. Not scheduled; not built. **Status at v1.14.0-pre.1**: transcribed and proved in §28 (Proposition 28.9, for T_p, $p \nmid N$, and U_p, $p \mid N$, with its cost in Remark 28.11); not built. **At v1.14.2**: still not built (slot 15); the τ-route of Proposition 27.2(4)(d) at composite N is built (WP-12).

(O3) **Composite-N coset tables are not pinned**: the representatives of $\Gamma_0(N)\backslash\Gamma$ and $B''\backslash B$ with their class maps are a binding datum (Remark 11.9) fixed by Definition 10.2 for prime powers only (Definition 27.1). A table, not a search; no new τ-table is needed for any N (Proposition 27.2(1), (4)(c)). **Status at v1.14.0-pre.1**: pinned for every N by Definition 28.1; not built. **At v1.14.2**: built (slot-14 WP-12, `Gamma0LevelN`); (O3) is closed.

(O4) **Large k and large N together**: the cokernel elimination is at best $(m_N/6)$-block-sparse with d^3 per block operation; the Hida regime ($n \approx (\ell-1)\ell^{m-1}$) at a level $N \geq \ell \geq 691$ is out of reach of every route above.

(O5) **The level-1 U_ℓ τ-table at large ℓ** is shared by every level, not replicated (Remark 27.6(5)) — but it must exist, and it is the bottleneck already at level 1. Slot-13 WP-4's opt-in rank-$\leq k$ lever (`docs/implementation/slot13/WP-4_report.md`) makes it about five times smaller: its fitted model puts $(700, 691)$ over $\mathbb{F}_{691}$ at $\approx$ 14.7 GiB (upper band 16.9 GiB) and $(2420, 2411)$ at ≈ 540 GiB (band 647 GiB), the latter beyond a single box; the remaining floor is the coset tables' full-rank ρ_n of unimodular matrices (a streaming lever, unscheduled). A model of memory, not an assertion; over $\mathbb{Z}/\ell^2$ at these n no prediction exists.

(O6) $\ell \in \{2, 3\}$ **and $\mathbb{Z}$**: the orbit-wise averaging and the cokernel need $6 \in R^\times$, so only the dense route survives there (it does survive: the cochain-level statements of the chapter hold over every ring, Scope). $\mathbb{Z}/\ell^l$, $\ell \geq 5$: the cokernel route survives, by Howell elimination over $\mathbb{Z}/\ell^l$ and the digit route beyond 2^{64} (Remark 27.6(6)).

Verdict recorded (the referee's, adopted): Shapiro on the fixed model is the right route — the induced model and a triangulation of $\Gamma_0(N)\backslash\mathcal{H}$ give the same cochain complex, the triangulation saves no dimension and would need a new certified τ-table per (N, g), and the scalable presentation for large N (Manin symbols on the Farey tree) is itself the Shapiro route on the tree; (O1) and (O2) decide whether large N is ever reached.

27.3 Stabilisation and control

Lemma 27.8 (Level ℓ: the torsion is non-ordinary)**.** *[proved here] Let ℓ be a prime, $n \geq 1$, $H := \Gamma_0(\ell)$, $U := U_\ell^{(\ell)}$.*

1. *For every $e \geq 1$, U acts on $H^0(H; V \otimes \mathbb{Z}/\ell^e) = (V/\ell^e)^H$ by $v \mapsto \sum_{b<\ell} \rho_n(g s_{1+b}^{-1})v$, over every ring and on the model as well; $U = 0$ at $e = 1$ and $U^e = 0$ in general.*
2. *$H^0(H; V \otimes \mathbb{Q}) = 0$, so the connecting map is a U-equivariant isomorphism $H^0(H; V/\ell^e) \cong H^1(H; V)[\ell^e]$; U is nilpotent on the ℓ-primary torsion T_ℓ of $H(\ell) := H^1(H; V) \otimes \mathbb{Z}_\ell$.*
3. *Consequently $e_{\mathrm{ord}}(T_\ell) = 0$, $e_{\mathrm{ord}}H(\ell)$ is free of finite rank, and $e_{\mathrm{ord}}H(\ell) \cong e_{\mathrm{ord}}(H(\ell)/T_\ell)$ (the argument of Remark 25.4, with (2) in place of Corollary 25.3(3)).*
4. *If $\ell \geq 5$: $H^1(H; V \otimes \mathbb{Z}/\ell^e) = H(\ell)/\ell^e H(\ell)$, compatibly with U and every operator of Proposition 27.2(4)(d); so Corollary 26.2(1) holds at level ℓ verbatim: the ordinary part is read over $\mathbb{Z}/\ell^{e'}$ at the target precision itself, its rank over $\mathbb{F}_\ell$.*

Proof. (1) A 0-cocycle of $\mathrm{Hom}_H(C_\bullet(K^u), M)$ is constant, K^u being connected, with value $v \in M^H$; by (T3) $\psi(\tau(y)) = v$ for every vertex y, so the formula of Proposition 27.2(3) gives

$\sum_b \rho_n(s_b^{(2)-1}g)v = \sum_b \rho_n(gs_{1+b}^{-1})v$ with no hypothesis on R. Every $gs_{1+b}^{-1} = \begin{pmatrix} 1 & 0 \\ -\ell b & \ell \end{pmatrix}$ is $\equiv \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \pmod{\ell}$, so modulo ℓ the ℓ terms are equal and their sum is $\ell\,\rho_n(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix})v \equiv 0$: $U = 0$ at $e = 1$, and in general $Uv \in \ell(V/\ell^e)$. Filter $F_i := (V/\ell^e)^H \cap \ell^i(V/\ell^e)$. For $v = \ell^i w \in F_i$, w is H-invariant modulo ℓ^{e-i} (multiplication by ℓ^i is injective $V/\ell^{e-i} \to V/\ell^e$), so $Uv = \ell^i Uw \in \ell^{i+1}(V/\ell^e)$, and Uv is H-invariant: $UF_i \subseteq F_{i+1}$, $F_e = 0$. (2) $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ \ell & 1 \end{pmatrix}$ lie in H; over $\mathbb{Q}$ the fixed space of $\rho_n(T)$ is $\mathbb{Q}u_0$, $u_0 = x^n$ (Definition 8.1: $T \cdot x = x$, $T \cdot y = x+y$), and $\begin{pmatrix} 1 & 0 \\ \ell & 1 \end{pmatrix}$ sends x^n to $(x+\ell y)^n \neq x^n$ for $n \geq 1$. The Bockstein is the argument of Proposition 26.1(1) for the group H (which is finitely generated, of finite index in Γ), and its U-equivariance that of Proposition 26.1(4). With (1), U^e kills $H^1(H;V)[\ell^e]$. (3) is Remark 25.4's proof verbatim. (4) The next term of the sequence is $H^2(H;V)[\ell^e] = H^2(\Gamma;\mathrm{Ind}_H^\Gamma V)[\ell^e]$ (Shapiro, [standard: [Bro82, Ch. III.6.2]]), which 24 kills by the argument of Claim 16.2(4) (it uses (50) alone); so it vanishes for $\ell \geq 5$. Then the proof of Corollary 26.2(1) applies with (3) in place of Remark 25.4. □

Proposition 27.9 (Stabilisation from level 1 to level ℓ)**.** *Tags per item (v1.13.0-pre.2, WP-3 referee m1): (1), (2) [proved here]; (3) [proved here] except the T_p-intertwining, [standard]; (4) the rank [standard], the isomorphism then [proved here] from (3); (5) [standard]. Let $\ell \geq 5$, $n \geq 1$; $H(1) := H^1(\Gamma;V) \otimes \mathbb{Z}_\ell$ with $e_1 := e_{\mathrm{ord}}(T_\ell)$ and $H(\ell) := H^1(\Gamma_0(\ell);V) \otimes \mathbb{Z}_\ell$ with $e_\ell := e_{\mathrm{ord}}(U_\ell^{(\ell)})$, both the ℓ-adic projectors of Remark 25.4 (resp. Lemma 27.8(3)); $h^1_{\mathrm{ord}}(n) = \mathrm{rank}\, e_1 H(1)$ (Definition 26.5).*

1. *[proved here] $e_\ell \,\mathrm{dg}_1 = \ell\,(U_\ell^{(\ell)})^{-1} e_\ell \,\mathrm{dg}_\ell$ ($U_\ell^{(\ell)}$ is invertible on $e_\ell H(\ell)$). In particular $e_\ell \circ \mathrm{dg}_1 \equiv 0 \pmod{\ell}$: **in this document's convention the restriction is not the integral stabilisation** (Remark 27.10).*
2. *[proved here] $c_{\mathrm{st}} := \omega \circ e_\ell \circ \mathrm{dg}_\ell$ restricted to $e_1 H(1)$ is an automorphism of the free $\mathbb{Z}_\ell$-module $e_1 H(1)$; it lies in the closure of $\mathbb{Z}_\ell[T_\ell]$ and is $\equiv T_\ell$ modulo ℓ there.*
3. *[proved here] (the last clause [standard]) So $e_\ell \,\mathrm{dg}_\ell$: $e_1 H(1) \to e_\ell H(\ell)$ is a split injection of free $\mathbb{Z}_\ell$-modules (left inverse $c_{\mathrm{st}}^{-1}\omega$) whose image is $U_\ell^{(\ell)}$-stable; $U_\ell^{(\ell)}$ acts on it as u_{st}, the root of $u^2 - T_\ell u + \ell^{n+1} = 0$ that is invertible (the unit root), and it intertwines T_p with $T_p^{(\ell)}$ for every prime $p \neq \ell$ [standard: [Shi71, §3.1 and Ch. 8]: the Hecke rings act on cohomology as rings, and the degeneracy correspondences commute with T_p for p prime to the level; pointers at section level, **not** verified (rule (A2)) — slot-13 WP-6 tests it].*
4. ***(Ranks.)** $\mathrm{rank}_{\mathbb{Z}_\ell} e_\ell H(\ell) = h^1_{\mathrm{ord}}(n)$ ([standard], the citation at the end of the item; ranks do not depend on the lattice, Remark 27.10(3)). [proved here] from this and (3): $e_\ell \mathrm{dg}_\ell$ is an isomorphism of $\mathbb{Z}_\ell[T_p : p \neq \ell]$-modules $e_1 H(1) \cong e_\ell H(\ell)$, and the tower datum of Definition 26.5 (rank, $\chi^{\mathrm{ord}}_{n,p} \bmod \ell^m$, $p \neq \ell$) is the same at level 1 and at level ℓ [standard: [Hid86a, Prop. 4.7, p. 572, with $N = 1$, $r = 1$; and Lemma 4.6, p. 571], locators **verified** against the PDF (slot-13 WP-2), transported by Remark 27.10; equivalently Eichler–Shimura for $\Gamma_0(\ell)$ [Shi71, Thm. 8.4, p. 234] (verified, R20-L: $S_{n+2}(\Gamma,\Psi) \cong H^1_P(\Gamma, X_n^\Psi)$, the parabolic part, $\mathbb{R}$-linearly) with the newform theory of [AL70] ($a_\ell(f)^2 = \ell^{k-2}$ for an ℓ-new eigenform of weight k and trivial character), the latter pointer not verified].*
5. ***($k = 2$.)** [standard] (the citations in the item) At $n = 0$: $H(1) = 0$, while $H(\ell) = H^1(\Gamma_0(\ell);\mathbb{Z}_\ell)$ is free of rank $2g(X_0(\ell))+1$ and $e_\ell = \mathrm{id}$ on it (the weight-2 Eisenstein class has $U_\ell^{(\ell)}$-eigenvalue 1, an ℓ-new form $a_\ell = \pm 1$) [standard: [Shi71, Thm. 8.4, p. 234] for the parabolic part of rank $2g(X_0(\ell))$ (verified, R20-L; the Eisenstein rank 1 is not part of Thm. 8.4); [AL70], not verified]; and $H^1(\Gamma_0(\ell);\mathbb{F}_\ell) = H(\ell)/\ell$. The genus is 0 at $\ell \in \{5, 7, 13\}$ and 1 at $\ell = 11$.*

[verified computationally][a] evidence (the same script, over $\mathbb{F}_\ell$): $\dim e_{\mathrm{ord}}H^1(\Gamma_0(\ell);M_n) = \dim e_{\mathrm{ord}}H^1(\Gamma;M_n)$ *at every even* $2\le n\le 24$ *for* $\ell\in\{5,7\}$ *and every even* $2\le n\le 12$ *for* $\ell=11$ *(*3 *at* $n=10$*, else* 1*); at* $n=0$ *the level-*ℓ *values are* $1,1,3$ *at* $\ell=5,7,11$*; and* $e_\ell\,\mathrm{dg}_\ell\,e_1$ *has rank* $h^1_{\mathrm{ord}}(n)$ *while* $e_\ell\,\mathrm{dg}_1\,e_1=0$ *over* $\mathbb{F}_\ell$ *at* $(n,\ell)\in\{(10,5),(12,5),(16,7)\}$ *— (1) and (3) modulo* ℓ*. At* $\ell=13$ *(the WP-3 referee's re-run of the same script): the level-*13 *value is* 1 *at* $n=0$*, and* $1=1$ *at* $n=2$*.*

Proof. (1) Proposition 27.4(1) on cohomology, then e_ℓ (which commutes with $U:=U_\ell^{(\ell)}$) and U invertible on $e_\ell H(\ell)$ (Proposition 25.2(1), in the limit).

(2) Let M act on $H(1)^{\oplus 2}$ as in Proposition 27.4(4), a finitely generated $\mathbb{Z}_\ell$-module; $U\Phi=\Phi\mathsf{M}$, so $e_\ell\Phi=\Phi\,e_{\mathrm{ord}}(\mathsf{M})$ (Proposition 25.2(3) modulo each ℓ^j, then the limit), and $\omega\Phi(x,y)=(\ell+1)x+T_\ell y$ by Proposition 27.4(2). Hence $c_{\mathrm{st}}=(\ell+1,\ T_\ell)\,e_{\mathrm{ord}}(\mathsf{M})\binom{0}{1}$, built from T_ℓ by ring operations and limits: it commutes with T_ℓ, hence preserves $e_1H(1)$, and lies in the closure of $\mathbb{Z}_\ell[T_\ell]$. Modulo ℓ (Proposition 25.2(4)), $n\ge 1$ gives $\bar{\mathsf{M}}=\mathrm{diag}(0,\bar T_\ell)$ on $(e_1H(1)/\ell)^{\oplus 2}$, where $\bar T_\ell$ is invertible; so $e_{\mathrm{ord}}(\bar{\mathsf{M}})=\mathrm{diag}(0,1)$ and $\bar c_{\mathrm{st}}=\bar T_\ell$, invertible. By Nakayama c_{st} is onto the finitely generated $e_1H(1)$, hence bijective; $e_1H(1)$ is free by Remark 25.4(2).

(3) Split injectivity is (2). dg_ℓ and ω are $\mathbb{Z}_\ell$-linear and $e_\ell H(\ell)$ is free (Lemma 27.8(3)), so the image Y is a direct summand. By (1) and Proposition 27.4(3), on $e_1H(1)$: $Ue_\ell\mathrm{dg}_\ell=e_\ell\mathrm{dg}_\ell T_\ell-\ell^{n+1}U^{-1}e_\ell\mathrm{dg}_\ell$, i.e. $(U^2-UT'+\ell^{n+1})e_\ell\mathrm{dg}_\ell=0$ with T' the transport of T_ℓ. Over $\mathbb{Q}_\ell$, $Y\otimes\mathbb{Q}_\ell$ is the image of $e_{\mathrm{ord}}(\mathsf{M})$ on the pairs, which is U-stable; a direct summand whose rationalisation is U-stable is U-stable. The transported U satisfies $u^2-T_\ell u+\ell^{n+1}=0$ and is invertible; it is the unit root because the other root is $\ell^{n+1}u^{-1}\in\ell\mathrm{End}$. The T_p-equivariance is the cited standard fact.

(4) $e_\ell H(\ell)$ and $e(U^*)H(\ell)$ have the same rank (Remark 27.10(1)), and Hida's Proposition 4.7 with $N=1$, $r=1$ identifies the latter with the level-1 ordinary part (whose rank is $h^1_{\mathrm{ord}}(n)$, the level-1 $T(p)$ being T_ℓ up to the same convention). A split injection of free $\mathbb{Z}_\ell$-modules of equal rank is an isomorphism. Through the standard route instead: $H(\ell)\otimes\mathbb{C}$ is two copies of $S_k(\Gamma_0(\ell))$ plus two Eisenstein classes; $S_k(\Gamma_0(\ell))$ is the old space, image of level 1 under the two degeneracies, plus the ℓ-new space, on which $U^2=\ell^{k-2}=\ell^n$, non-ordinary for $n\ge 1$; so the ordinary part is the ordinary part of the old space, of rank $h^1_{\mathrm{ord}}(n)$ by (3).

(5) is read off the cited statements: at $n=0$ the ordinary part is everything. □

Remark 27.10 (Two conventions for the level-ℓ operator). [proved here] (1) Put $\beta_\ell:=\begin{pmatrix}\ell&0\\0&1\end{pmatrix}$ and $U^*:=[\Gamma_0(\ell)\,\beta_\ell\,\Gamma_0(\ell)]$. The Atkin–Lehner element $W_\ell:=\begin{pmatrix}0&-1\\\ell&0\end{pmatrix}\in\Delta$ normalises $\Gamma_0(\ell)$ and $W_\ell g W_\ell^{-1}=\beta_\ell$; so the compatible pair $(\mathrm{conj}_{W_\ell},\rho_n(W_\ell))$ induces an endomorphism w of $H^\bullet(\Gamma_0(\ell);V_R)$ with $wU_\ell^{(\ell)}=U^*w$, and $w^2=\rho_n(-\ell I)=\ell^n$ (Proposition 10.5; n even). Over $\mathbb{Q}_\ell$, w is invertible: $U_\ell^{(\ell)}$ and U^* are conjugate, and their ordinary parts have equal rank. Over $\mathbb{Z}_\ell$ they are not interchangeable, as Proposition 27.9(1) shows. (2) Hida's $T(p)$ at level p^r is the double coset $[\Phi\begin{pmatrix}1&0\\0&p\end{pmatrix}\Phi]$ acting through the main involution $\alpha\mapsto\alpha^\iota=\mathrm{adj}\,\alpha$ [Hid86a, (4.6a) and p. 563]; in degree 0 it is $x\mapsto\sum_{i<p}\begin{pmatrix}1&-i\\0&p\end{pmatrix}^\iota x$ [Hid86a, proof of Lemma 4.6, p. 571], i.e. a sum over matrices $\begin{pmatrix}p&i\\0&1\end{pmatrix}\in\Gamma_0(p)\beta_p\Gamma_0(p)$ — in this document's convention U^*, not $U_\ell^{(\ell)}$ (compare Lemma 27.8(1): here $U=0$ on H^0 modulo ℓ, there $T(p)\neq 0$ but $T(p)^2=0$, as Hida states). Hida's isomorphism "$e\circ\mathrm{res}$" is therefore $e(U^*)\circ\mathrm{dg}_1$ here, and this document's integral stabilisation is $e_\ell\circ\mathrm{dg}_\ell$ (Proposition 27.9(3)). (3) **Hida's lattice is not** V (v1.13.0-pre.2, WP-3 referee M1; the pre.1 sentence "up to the order of the basis" was false for $n\ge\ell$). Hida's $L_n(\mathbb{Z})=\mathbb{Z}^{n+1}$ is the column vector ${}^t(x^n,x^{n-1}y,\dots,y^n)$ of monomials in a variable vector

${}^t(x,y)$, on which $A=\begin{pmatrix}a&b\\c&d\end{pmatrix}\in \mathrm{M}_2(\mathbb{Z})$ acts by ${}^t(x,y)\mapsto {}^t(ax+by,cx+dy)$ [Hid86a, p. 566]. Its matrix in that basis is

$$\rho_n^{\mathrm{H}}(A)\;:=\;\rho_n(A^T)^T\;=\;D^{-1}\rho_n(A)\,D,\qquad D:=\operatorname{diag}\Big(\tbinom{n}{0},\tbinom{n}{1},\dots,\tbinom{n}{n}\Big).$$

The first equality is (9) read for A^T: row j of $\rho_n^{\mathrm{H}}(A)$ holds the coefficients of $(ax+by)^{n-j}(cx+dy)^j$ in the u_t, which is column j of $\rho_n(A^T)$. For the second, the rational symmetric form $\langle u_j,u_t\rangle:=\delta_{jt}\binom{n}{j}^{-1}$ satisfies $\langle v^n,w^n\rangle=\langle v,w\rangle^n$ for $v,w\in W\otimes\mathbb{Q}$ (the standard form on W; check on $v=x+sy$, $w=x+ty$), the v^n span $V\otimes\mathbb{Q}$, and $\langle Av,w\rangle=\langle v,A^Tw\rangle$; so $\rho_n(A^T)$ is the adjoint of $\rho_n(A)$ for a form of Gram matrix D^{-1}, $\rho_n(A^T)=D\rho_n(A)^TD^{-1}$; transpose. (Checked symbolically at $n\le 5$ by the referee and by the author.) So ρ_n^{H} is multiplicative on $\mathrm{M}_2(\mathbb{Z})$, and $f\mapsto Df$ is an injective $\mathrm{M}_2(\mathbb{Z})$-map $L_n(\mathbb{Z})\to V$ with image $\bigoplus_j\binom{n}{j}\mathbb{Z}\,u_j$, of index $\prod_j\binom{n}{j}$ — the image of the divided powers $\Gamma^n\mathbb{Z}^2$ under the canonical map to $\mathrm{Sym}^n\mathbb{Z}^2$. Hence $L_n(\mathbb{Z}_\ell)=V\otimes\mathbb{Z}_\ell$ (through D) iff $\ell\nmid\binom{n}{j}$ for every j, iff (Lucas) $n+1=c\,\ell^s$ with $1\le c\le\ell-1$ — true for $n<\ell$ (so at $(10,691)$, $(10,2411)$), false at $(10,5)$, $(700,691)$ and every other tower cell with $n\ge\ell$ not of that form. *Lattice-independent* (unaffected): every $\mathbb{Z}_\ell$-rank of an ordinary part and every characteristic polynomial of a Hecke operator on it (both live on $eH\otimes\mathbb{Q}_\ell$, and $D_*\colon H^1(-;L_n)\to H^1(-;V)$ is a Hecke-equivariant isomorphism after $\otimes\mathbb{Q}$) — so Proposition 27.9(4), (Hi3), (Hi4), (Hi7), (Hi9)–(Hi12) and Definition 26.5's datum do not see the difference, and Proposition 27.9(1)–(3) are proved for V and need nothing from Hida. *Lattice-dependent*: the isomorphism class of a finite-level module, i.e. Definition 27.13's datum and (Hi8). *Addendum* (v1.21.0-pre.2; not an erratum: "lattice-dependent" here means "not shown independent", and no sentence of this remark is false): since [U] v0.5.0-pre.2 (`docs/latex/hida_hecke_algebra.tex`, `lem:u-latticeord`, [proved here] there, elementary; slot-21 referee F4, `docs/implementation/slot21/P8_F4_referee_report.md`, ACCEPT WITH REPAIRS) the finite-level datum is lattice-*independent* on the ordinary part: $D_*\colon e_1H^q(\Gamma;L_n(\mathbb{Z}_\ell))\cong e_1H^q(\Gamma;\mathrm{Sym}^n\otimes\mathbb{Z}_\ell)$ for every q, also on ∂ and at level ℓ; so Definition 27.13's $M_m(n)$ and its variant (H) are isomorphic Hecke modules, (Hi8)'s arms (a) and (b) coincide, and its report (c) (45/45 isomorphisms, slot-13 WP-5) is a theorem. (4) **In Hida's convention the trace is not an integral left inverse** (v1.13.0-pre.2, WP-3 referee m8). From $w\,\mathrm{dg}_1=\mathrm{dg}_\ell$, $w\,\mathrm{dg}_\ell=\ell^n\mathrm{dg}_1$ (from $W_\ell=gS$, conj_S inner on Γ, and $w^2=\ell^n$), $U^*=wUw^{-1}$ and Proposition 27.4(1),(3): $U^*\Phi=\Phi\,\mathsf{M}^*$ with $\mathsf{M}^*=\begin{pmatrix}T_\ell&\ell^{n+1}\\-1&0\end{pmatrix}$, the textbook old-space matrix, of the same characteristic polynomial as M. On $e_1H(1)$, $n\ge1$, with u the unit root and $u'=\ell^{n+1}u^{-1}$ the other: the unit-root eigenspace of M^* is $\{(-uy,y)\}$, and $\omega\circ e(U^*)\circ\mathrm{dg}_1=\big((\ell+1)u-T_\ell\big)(u-u')^{-1}=\ell\,(u-\ell^nu^{-1})(u-u')^{-1}$, which is ℓ times a unit. So the plain trace ω is *not* a left inverse of $e(U^*)\circ\mathrm{dg}_1$ over $\mathbb{Z}_\ell$ (how Hida's "inverse induced by $e_0\circ\mathrm{Tr}$" is normalised is not transcribed here; his proof goes through [Hid86a, Cor. 4.5], per the WP-3 referee's reading); slot-13 WP-6 tests the pair $(e_\ell\circ\mathrm{dg}_\ell,\ c_{\mathrm{st}}^{-1}\omega)$ of Proposition 27.9(3), never a trace inverse in the U^* convention. *Proof of (1)*: $W_\ell\begin{pmatrix}a&b\\\ell c&d\end{pmatrix}W_\ell^{-1}=\begin{pmatrix}d&-c\\-\ell b&a\end{pmatrix}$; $W_\ell gW_\ell^{-1}$ by multiplying out; conjugating each factor of $[A\,g\,B]$ by W_ℓ gives the intertwining, and $W_\ell^2=-\ell I$.

Theorem 27.11 (Hida: Λ-freeness and control)**.** *[standard: [Hid86a, Thm. 3.1, p. 562; Thm. 6.6, p. 590]; [Hid86b, Thm. 3.1 and Cor. 3.2, p. 250]; locators* **verified** *against the PDFs in the owner's local library (papers/h/Hida/) (slot-13 WP-2)] Let $p=\ell\ge5$, Hida's N equal to 1 (so his $\Phi_r=\Gamma_1(N)\cap\Gamma_0(p^r)$ is $\Gamma_0(\ell^r)$ and his $N_0=1$), $\Lambda=\mathbb{Z}_\ell[[1+\ell\mathbb{Z}_\ell]]$, and e Hida's ordinary idempotent (Remark 27.10(2)).* Hypotheses*: Hida assumes throughout that $p\ge5$ [Hid86a, p. 545] (read from the page image; the extracted text misreads it as "$p>5$"), and admits every N prime to p, so $N=1$, $\Phi_1=\Gamma_0(\ell)$, $\ell=5$ are inside them; his (4.4) (no finite*

subgroups) is used only for the sheaf comparison (4.5), and at $N = 1$ the $\delta \equiv S \pmod p$ of Lemma 4.6 is S itself (v1.13.0-pre.2, WP-3 referee m5). Lattice*: $L_n(A)$ is Hida's lattice, $L_n(\mathbb{Z}_\ell) \cong \bigoplus_j \binom{n}{j}\mathbb{Z}_\ell\, u_j \subseteq V \otimes \mathbb{Z}_\ell$ (Remark 27.10(3)), equal to $V \otimes \mathbb{Z}_\ell$ for $n < \ell$ and in general* not *for $n \geq \ell$; the statements below are about it, and are transcribed faithfully (v1.13.0-pre.2, WP-3 referee M1).*

1. (Invent. Thm. 3.1.) *The Pontryagin duals V^0, W^0 of the ordinary parts of $\varinjlim_r H^1(X_r, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ and $\varinjlim_r H^1(Y_r, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$, $Y_r = \Gamma_1(\ell^r)\backslash\mathcal{H}$ and X_r its compactification, are free of finite rank over Λ; and for every $r > 0$ restriction identifies the level-r ordinary part with the invariants of $1 + \ell^r\mathbb{Z}_\ell$.*
2. (Invent. Thm. 6.6.) *For every $m > 0$, $eH^1_P(\Phi_1, L_n(\mathbb{Z}_\ell)) \otimes \mathbb{Z}/\ell^m \cong eH^1_P(\Phi_1, L_n(\mathbb{Z}/\ell^m))$ (H^1_P the parabolic cohomology); and for $n \equiv a \pmod{\ell-1}$, $eH^1_P(\Phi_1, L_n(\mathbb{Z}_\ell)) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \cong eH^1_P(\Phi_1, L_n(\mathbb{Q}_\ell/\mathbb{Z}_\ell)) \cong \mathcal{V}^0(a)[P_n]$ as modules over Hida's universal Hecke algebra, $\mathcal{V}^0$ the (discrete) ordinary part of $\varinjlim_r H^1(X_r, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ — so V^0 is its Pontryagin dual — $\mathcal{V}^0(a)$ its a-component and P_n Hida's weight-n prime of Λ (defined in [Hid86a, §6]; its normalisation is not restated here — v1.21.0-pre.2: it is [Hid86a, Introduction, p. 545], read on the page by the slot-21 referee F4 (M2): P_{k-2} is the kernel of the evaluation $X \mapsto u^k - 1$, $u = 1+\ell$ (trivial character), so Hida's P_n is $(1+X) - (1+\ell)^{n+2}$, the P_k of (3) with $k = n+2$; no twist of the Λ-structure intervenes).*
3. (ENS Thm. 3.1, Cor. 3.2; level $N = p$, $N_0 = 1$.) *The ordinary parts of the universal Hecke algebras (of modular forms, of cusp forms) are free of finite rank over Λ; with $P_k = (1+X) - (1+\ell)^k$, the cuspidal one satisfies $h^0(N,a)/P_k h^0(N,a) \cong h^0_k(\Gamma_1(N_0\ell), a)$ for $k \geq 2$, and the rank of $h^0(N,a)$ over Λ equals the $\mathbb{Z}_\ell$-rank of the ordinary weight-k cusp forms for $k \geq 3$, $k \equiv a \pmod{\ell-1}$.*

To Theorem 27.11(2), after "no twist of the Λ-structure intervenes (placed after the theorem, so that its statement is unchanged). *Addendum* (v1.22.0-pre.1, slot-22 P10-C / F-U5): "no twist intervenes" is relative to Hida's own Λ-structure on $\mathcal{V}^0$, the one induced by $f|z = z_p^k\,(f|_k\sigma_z)$ [Hid86a, p. 549, p. 552], i.e. $\iota(l) = l^2T(l,l)$; on the weight-2 tower $\mathcal{V}^0$ this is $1 + X \mapsto u^2\langle u\rangle$ with $\langle u\rangle$ the natural diamond ("the twist by $z \mapsto z_p^2$", [Hid86a, p. 593], read on the page for [U] part 5), and the weight-k classes are those with $\langle u\rangle = u^{k-2}$ ([Hid86a, p. 589]: $H^1(\Gamma_1(Np^r), n; \cdot) = \{v : v|z = z^n v\}$, $n = k-2$), which is why they sit at $X = u^k - 1$ — relative to the natural action alone they would sit at $u^{k-2} - 1$. The index a of $\mathcal{V}^0(a)$ above is the *natural* one ([Hid86a, p. 589]: $\mathcal{V}^0(a) = \{v : v|\zeta = \zeta^a v\}$, $a \equiv n = k-2$): the weight-k classes of the class $k \bmod (\ell-1)$ land in the component of nebentypus ω^{k-2} — Hida's "$h^0_2(a)$ acts on $\mathcal{V}^0(a-2)$" (p. 593) with a there the Hecke class. Verified on the page images by the slot-22 referee F-U5 (`docs/implementation/slot22/P10C_F4_referee_report.md`, item 7).

Theorem 27.12 (Hida: weight independence of the ordinary rank, and the level-p comparison)**.** *[standard: [Hid86b, Thm. 3.1, p. 250]; [Hid86a, Lemma 4.6, p. 571; Prop. 4.7, p. 572]; locators* **verified** *(slot-13 WP-2)] Let $\ell \geq 5$. (1) For weights $k \geq 3$ in one class a modulo $\ell-1$, the $\mathbb{Z}_\ell$-rank of the ordinary cusp forms of weight k on $\Gamma_0(\ell)$ (trivial character, as $k \equiv a$) is the Λ-rank of $h^0(\ell, a)$, independent of k; by the control of Theorem 27.11(3) at $k = 2$ the same holds from $k = 2$ for the cusp forms. (2) For every $n \geq 0$ the ordinary parts $e_0H^1(\mathrm{SL}_2(\mathbb{Z}), L_n(\mathbb{Z}_\ell))$ and $eH^1(\Gamma_0(\ell^r), L_n(\mathbb{Z}_\ell))$, $r \geq 1$, are $\mathbb{Z}_\ell$-free. (3) For $n > 0$, $r > 0$: $e \circ \mathrm{res}\colon e_0H^1(\mathrm{SL}_2(\mathbb{Z}), L_n(\mathbb{Z}_\ell)) \to eH^1(\Gamma_0(\ell^r), L_n(\mathbb{Z}_\ell))$ is an isomorphism, with inverse induced by $e_0 \circ \mathrm{Tr}$, and likewise for the parabolic cohomology — in Hida's convention (Remark 27.10). L_n in (2), (3) is Hida's lattice, not V for $n \geq \ell$ (Remark 27.10(3)); the consequences below use ranks only, which do not depend on the lattice.* Consequences in this document*: Proposition 27.9(4); and with Eichler–Shimura (Obligation 20.7), $h^1_{\mathrm{ord}}(n) = 2\,\mathrm{rank}\, S^{\mathrm{ord}}_{n+2} + 1$ is constant along Tw_1 — the source of (Hi3), whose locators this theorem supplies (its tag is not moved at this version). Algebra-level consequences (residual ordinary systems and h^{ord}_k/ℓ constant along a class) do* not *follow from (1); they are Proposition 30.8, from Theorem 27.11(3) and Proposition 27.9 (pointer added at v1.15.0-pre.1, P18; no tag moved).*

27.4 The finite-level lattice datum

Definition 27.13 (The lattice datum of a tower cell)**.** Let $\ell \geq 5$, $n \geq 2$ even, $m \geq 1$, and $\mathcal{P}$ a finite set of primes $p \neq \ell$ (default $\{2,3\}$). The lattice is **the contract's** $V = \mathrm{Sym}^n\mathbb{Z}^2$ of Definition 8.1; Hida's L_n is a different $\mathbb{Z}_\ell$-lattice for $n \geq \ell$ (Remark 27.10(3)), and its datum is the variant (H) below. With $H(n) := H^1(\Gamma; \mathrm{Sym}^n\mathbb{Z}^2) \otimes \mathbb{Z}_\ell$ and $U = U_\ell$ as in Definition 26.5, put

$$M_m(n) := e_{\mathrm{ord}} H(n)/\ell^m e_{\mathrm{ord}} H(n),$$

a free $\mathbb{Z}/\ell^m$-module of rank $h^1_{\mathrm{ord}}(n)$ (Corollary 26.2(1): it *is* $e_{\mathrm{ord}}(U|_{H_m})H_m$, $H_m = H^1(\Gamma; \mathrm{Sym}^n\mathbb{Z}^2 \otimes \mathbb{Z}/\ell^m)$, computed over $\mathbb{Z}/\ell^m$), with the operators T_p, $p \in \mathcal{P}$, and its Hecke algebra $\mathbb{T}^{\mathrm{ord}}_{m,\mathcal{P}}(n)$, the $\mathbb{Z}/\ell^m$-subalgebra of $\mathrm{End}(M_m(n))$ generated by them. The **lattice datum of the cell at level** m is the isomorphism class of the $\mathbb{Z}/\ell^m[X_p : p \in \mathcal{P}]$-module $(M_m(n), (T_p)_{p\in\mathcal{P}})$ — equivalently the orbit of the tuple of matrices $(A_p)_{p\in\mathcal{P}}$ in a basis under simultaneous conjugation by $\mathrm{GL}_h(\mathbb{Z}/\ell^m)$, $h = h^1_{\mathrm{ord}}(n)$. By Proposition 27.9(4) the same datum is read at level ℓ on $e_\ell H^1(\Gamma_0(\ell); \mathrm{Sym}^n\mathbb{Z}^2 \otimes \mathbb{Z}/\ell^m)$ through $e_\ell \mathrm{dg}_\ell$. U_ℓ is *not* part of the datum: the level-1 $U_\ell = T_\ell$ transports to $u_{\mathrm{st}} + \ell^{n+1}u_{\mathrm{st}}^{-1}$, not to Hida's U (Proposition 27.9(3)). What is *not* computed: Hida's Λ-adic module and algebra themselves; what is computable is this finite-level datum along Tw_m.

Variant (H): the datum on Hida's lattice (v1.13.0-pre.2, WP-3 referee M1(v)). Replace ρ_n by $\rho_n^{\mathrm{H}}(A) = \rho_n(A^T)^T$ of Remark 27.10(3) *everywhere* ρ_n enters the model: the relation matrices of $(K, \mathcal{R})$, the coefficients of the τ-chains and the blocks $\rho_n(\delta)$ of (21). ρ_n^{H} is integral and multiplicative on $\mathrm{M}_2(\mathbb{Z})$ with $\rho_n^{\mathrm{H}}(\lambda I) = \lambda^n$, and D-conjugate to ρ_n over $\mathbb{Q}$, so the constructions of §11 apply to it unchanged, and T_p is (21) with ρ_n^{H} (at level 1 Hida's $T(p) = [\Phi \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Phi]$ acting through ι is the double coset $\Gamma \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) \Gamma = \Gamma \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma$, so no convention question arises there). Put $H^{\mathrm{H}}(n) := H^1(\Gamma; L_n(\mathbb{Z})) \otimes \mathbb{Z}_\ell$ and $M_m^{\mathrm{H}}(n) := e_{\mathrm{ord}} H^{\mathrm{H}}(n)/\ell^m$: free of rank $h^1_{\mathrm{ord}}(n)$ (Theorem 27.12(2) for the freeness, the rank being lattice-independent), and computed over $\mathbb{Z}/\ell^m$ as $e_{\mathrm{ord}}(U|_{H_m^{\mathrm{H}}})H_m^{\mathrm{H}}$, $H_m^{\mathrm{H}} = H^1(\Gamma; L_n(\mathbb{Z}/\ell^m))$, by the argument of Corollary 26.2(1) (the H^2 term is killed by 24, Claim 16.2(4), for any coefficient module). Its datum is the class of $(M_m^{\mathrm{H}}(n), (T_p)_{p\in\mathcal{P}})$ as above. *Implementation*: one `FibreRepresentation` variant, like the contragredient and `QuotientL` — the shipped `ComputeSymmetricPowerMatrixFM` with the input matrix transposed and the output transposed. (Through v1.13.0-pre.2 this sentence ended "nothing else changes"; amended at v1.13.0, as the slot-13 WP-13 code audit's A3 required.) *One further step changes, on the module route over* $\mathbb{Z}/\ell^m$ (slot-13 WP-5): the equivariant cochain module $C^q_{\mathrm{eq}} = \ker A_{C^q}$ is free on L_n as on V (a direct summand, the stabilisers having order prime to ℓ), but its Howell form may carry a non-unit pivot, a property of the coordinate order and not of freeness (on V every pivot is a unit, Proposition 15.18(2), and the step never runs). The route then takes a *Nakayama basis* — the Howell rows independent modulo ℓ, chosen greedily — and checks it free by $\mathrm{length}(\ker A_{C^q}) = m \cdot \mathrm{rank}$, refusing by name otherwise. Every object computable before is computed on the unchanged unit-pivot branch, so the step is value-preserving there. The map $D_*\colon M_m^{\mathrm{H}}(n) \to M_m(n)$ induced by $f \mapsto Df$ is Hecke-equivariant; whether it is an isomorphism (iff it is surjective modulo ℓ) is a lattice comparison, recorded per cell.

Remark 27.14 (What is checkable at level ℓ, along Tw_3, and at the heavy cells: (Hi7)–(Hi12)). [proved here] [conjectural] [verified computationally][a] Committed at v1.13.0-pre.1, before the code that tests them (slot-13 WP-5, WP-6, WP-10; rule 20); each item carries its own tag, with the convention of Remark 26.6: a failing [proved here] item stops the package, a failing [conjectural] item is a finding. The [conjectural] items' sources are Theorems 27.11, 27.12 (locators verified); they stay [conjectural] because the chain from those statements to this document's objects — Hida's duality between Hecke algebras and forms, his parabolic cohomology against the full H^1,

his convention against this one (Remark 27.10(2)), his lattice L_n against V (Remark 27.10(3); it matters for (Hi8) only) — is not carried out here.

(Hi7) [conjectural] **Congruences modulo ℓ^3.** For $p \in \{2,3\}$ and n, n' in one $\mathrm{Tw}_3(\ell; n_0)$, $\chi^{\mathrm{ord}}_{n,p} \equiv \chi^{\mathrm{ord}}_{n',p} \pmod{\ell^3}$, read over $\mathbb{Z}/\ell^3$ (Corollary 26.2(1)). The period is $(\ell-1)\ell^2$: 100 at $\ell = 5$ ($n = 10, 110, 210$), 294 at $\ell = 7$ ($n = 10, 304$, the slow arm, sized by WP-5 before it runs). The Eisenstein factor obeys it unconditionally ((Hi4)'s argument with $m = 3$).

(Hi8) [conjectural] **The lattice datum is constant on towers.** For $m \in \{1,2\}$, $\mathcal{P} = \{2,3\}$ and n, n' in one $\mathrm{Tw}_m(\ell; n_0)$: $(M_m(n), T_2, T_3) \cong (M_m(n'), T_2, T_3)$ (Definition 27.13) — strictly stronger than (Hi4). Two arms: **(a)** on the contract's lattice V (Definition 27.13); **(b)** on Hida's lattice, $(M^{\mathrm{H}}_m(n), T_2, T_3) \cong (M^{\mathrm{H}}_m(n'), T_2, T_3)$ (Definition 27.13, variant (H)); and, reported per cell and *not* a prediction, (c) whether $D_*\colon M^{\mathrm{H}}_m(n) \to M_m(n)$ is an isomorphism. *Source, stated honestly* (v1.13.0-pre.2, WP-3 referee M1(iv)): Theorem 27.11(1),(2) — the weight-n *parabolic* module on *Hida's lattice* is the quotient of a Λ-free module by (P_n, ℓ^m), and $(P_n, \ell^m) = (P_{n'}, \ell^m)$ in the same a-component when $n \equiv n' \pmod{(\ell-1)\ell^{m-1}}$. Arm (b) extends that from the parabolic part to the full H^1 (the Eisenstein line added); arm (a) is in addition **this document's extrapolation** to a different lattice, on which no cited statement bears — the isomorphism class of a finite-level module is lattice-dependent (Remark 27.10(3)); for $n, n' < \ell$ the two arms coincide. *The test is a decision procedure* (v1.13.0-pre.2, WP-3 referee m4): the solution module $\mathcal{H} := \{P \in \mathrm{M}_h(\mathbb{Z}/\ell^m) : PA_p = A'_pP,\ p \in \mathcal{P}\}$ ($= \mathrm{Hom}_{\mathbb{T}}(M, M')$) is computed by a module solve over $\mathbb{Z}/\ell^m$; an isomorphism exists iff some $P \in \mathcal{H}$ is invertible, iff (Nakayama) some element of the reduction $\bar{\mathcal{H}} \subseteq \mathrm{M}_h(\mathbb{F}_\ell)$ is invertible, iff $\det$ restricted to the $\mathbb{F}_\ell$-space $\bar{\mathcal{H}}$ (of dimension $\le h^2$) is not the zero polynomial. It has degree h; for $h < \ell$ it is non-zero iff it is non-zero at some point of the grid $\{0, \dots, h\}^{\dim \bar{\mathcal{H}}}$ (a non-zero polynomial of degree $\le h$ in each variable does not vanish on a grid of side $h+1$), $\le 4^9$ evaluations at the grid's ranks $h \le 3$; for $h \ge \ell$ all of $\bar{\mathcal{H}}$ is finite. Outcomes: *isomorphic*, with the certificate P printed, or *not isomorphic*, by the vanishing; no "inconclusive". The invariants of pre.1 (the lengths of $\mathrm{Hom}(M,M)$, $\mathrm{Hom}(M,M')$, $\mathrm{Hom}(M',M')$, the Fitting ideals of $A_p - c$) stay as cheap necessary conditions run first.

(Hi9) [conjectural] **(Hi4) at** $p \in \{5,7\}$, $p \neq \ell$, $m \in \{1,2\}$; same source.

(Hi10) [proved here] (Proposition 27.9 and Lemma 27.8, with their [standard] pointers) **Level ℓ against level 1.** At $\ell \in \{5,7,11,13\}$ and even $n \ge 2$ (WP-6's grid: $n \le 40$): $\dim_{\mathbb{F}_\ell} e_{\mathrm{ord}} H^1(\Gamma_0(\ell); M_n) = \mathrm{rank}\, e_\ell H(\ell) = h^1_{\mathrm{ord}}(n)$, the value of (Hi3); at odd n both are 0; at $n = 0$ the level-ℓ value is $2g(X_0(\ell)) + 1 = 1, 1, 3, 1$ against level 1's 0 — so the cells $(0, \ell)$ enter at level ℓ only. Also checkable entrywise on the model: Proposition 27.4(1),(2) at cochain level, (3) on cohomology, and $U^{(\ell)}_\ell$ commuting with $T^{(\ell)}_2$, $T^{(\ell)}_3$ — the commutativity of the Hecke ring of $\Gamma_0(\ell)$ at primes prime to each other, [standard: [Shi71, Ch. 3]; chapter level, **not** verified (rule (A2)); v1.13.0-pre.2, WP-3 referee m9], not proved in this document. [verified computationally][a]: WP-2's script, recorded in Proposition 27.9.

(Hi11) [conjectural] (the Eisenstein half [proved here] by (Hi5)) $(700, 691)$ **lies in** $\mathrm{Tw}_1(691; 10)$ ($700 - 10 = 690 = \ell - 1$): $h^1_{\mathrm{ord}}(700) = h^1_{\mathrm{ord}}(10) = 3$ and $\chi^{\mathrm{ord}}_{700,p} \equiv \chi^{\mathrm{ord}}_{10,p} \equiv \big(X - (1 + p^{11})\big)^3 \pmod{691}$ for $p \in \{2,3\}$ — at $n = 10$, Δ is ordinary at 691 and $\tau(p) \equiv 1 + p^{11}$ by the Ramanujan congruence, which the model shows at $(10, 691)$ (Obligation 20.7(i)). Proved: $1 + p^{701} \equiv 1 + p^{11} \pmod{691}$ is a root of $\chi^{\mathrm{ord}}_{700,p}$ modulo 691, and $h^1_{\mathrm{ord}}(700) \ge 1$. Run only behind the U_ℓ memory lever and the owner's rule 19 (slot-13 WP-10). [verified computationally][a] (added at v1.14.0-pre.1, taken out at pre.2, restored at v1.14.0; the first cluster run, jobs `34550469`, `34550468`, `34550470` at commit `07f413f`, `docs/matrix-engine/results/noctua2_run1/`, published 2026-09-28 on the owner's sign-off, [S14] Q2; [EX] `nc:hi11`): **also held at** $n = 1390$**,** 2080 **and** 2770, the next three members of $\mathrm{Tw}_1(691; 10)$ (k up to 2772, $\dim H^1 = 233, 347, 463$): $h^1_{\mathrm{ord}} = 3$ and the same three polynomials, $\big(X - (1 + p^{11})\big)^3$ for $p \in \{2,3\}$ and $(X-1)^3$ for U_{691} — a

comparison of printed values, the probe gating (Hi11) at $n = 700$ only; and at $m = 2$ the $(700, 691)$ cell reproduces on a second machine, χ^{ord} modulo 691^2 equal to slot 13's, with the probe's gated `HI11 ... verdict=HELD`. [verified computationally][a] (added at v1.14.1; *after the second cluster run*, second-cluster job `3467349` at commit `e85e25a`, with the slot-14 levers on; data signed off by the owner 2026-09-29, [S14] Q-B; [EX] `nc:hi11`, `s14:sept`; the WP-C10 report of the September run, §1): **also held at** $n = 4150$, the seventh member of $\mathrm{Tw}_1(691; 10)$ ($k = 4152$, $\dim H^1 = 693 = 2d_k - 1$, $d_k = 347$): $h^1_{\mathrm{ord}} = 3$ and the same three polynomials, and the probe's own gated verdict line `HI11 n=4150 ell=691 h_ord=3 predicted_h_ord=3 verdict=HELD` — from this build the probe prints the (Hi11) verdict at the cell it is given, so "at $n = 700$ only" above is the first run's build, a dated record. The cell took 711.8 s (the case 715 s) and 49.9 GiB (the probe's own peak; the campaign's sampler read the same). The sixth member, $n = 3460 = 5(\ell + 1)$, is not served: $e_{\max} = 1$ there and the probe refuses it by name (second-cluster job `3464192`), so the computed members are $n = 10$, 700, 1390, 2080, 2770, 4150. *A route to a proof* (v1.24.0-rc.1; the phenomenology catalogue, PH-32): the ordinary rank 3 and the three polynomials along $n \equiv 10 \bmod 690$ would follow from Hida's control theorem modulo ℓ at level one ($\ell \geq 5$, weight ≥ 3) together with $n = 10$; the module-level form of that control has not been checked against Hida's text, so the tag stays [conjectural] and the [verified computationally][a] records above stay the evidence.

(Hi12) [conjectural] (the Eisenstein half [proved here]) $(2420, 2411)$ **lies in** $\mathrm{Tw}_1(2411; 10)$ ($2420 - 10 = 2410 = \ell - 1$): $h^1_{\mathrm{ord}}(2420) = h^1_{\mathrm{ord}}(10) = 1$ — the Eisenstein line alone, as (Hi6) proves at $n = 10$ — and $\chi^{\mathrm{ord}}_{2420,p} = X - (1 + p^{2421}) \equiv X - (1 + p^{11}) \pmod{2411}$. Proved: $h^1_{\mathrm{ord}}(2420) \geq 1$ and the Eisenstein root. [verified computationally][a] (added at v1.14.0-pre.1, taken out at pre.2, restored at v1.14.0; first-cluster job `34550471` at commit `07f413f`, published 2026-09-28 on the owner's sign-off, [S14] Q2; [EX] `nc:hi12`): **run once at** $(2420, 2411)$**, and the prediction held**: $h^1_{\mathrm{ord}} = 1$, $\chi^{\mathrm{ord}}(T_2) = X + 362 = X - (1 + 2^{11})$, $\chi^{\mathrm{ord}}(T_3) = X + 1266 = X - (1 + 3^{11})$ and $\chi^{\mathrm{ord}}(U) = X - 1$ modulo 2411, equal to $n = 10$'s. The Eisenstein half is gated by `HeckeHidaHeavyCellProbe`; the equality with $n = 10$ is a comparison of printed values (the first-cluster build printed no (Hi12) verdict; the build `e85e25a` prints `HI12 ... verdict=HELD`, comparing h^1_{ord} with the prediction only; v1.14.1). [verified computationally][a] (added at v1.14.1; *after the second cluster run*, second-cluster job `3467349` at commit `e85e25a`, levers on; data signed off by the owner 2026-09-29, [S14] Q-B; [EX] `nc:hi12`, `s14:sept`; the WP-C10 report of the September run, §1): **also held at** $(4830, 2411)$, the third computed member of $\mathrm{Tw}_1(2411; 10)$ ($4830 = 10 + 2 \cdot 2410$, $k = 4832$, $\dim H^1 = 805 = 2d_k - 1$, $d_k = 403$): $h^1_{\mathrm{ord}} = 1$ and $\chi^{\mathrm{ord}}(T_2) = X + 362$, $\chi^{\mathrm{ord}}(T_3) = X + 1266$, $\chi^{\mathrm{ord}}(U) = X - 1$ modulo 2411, equal to $n = 10$'s, with `HI12 n=4830 ell=2411 h_ord=1 predicted_h_ord=1 verdict=HELD`; the cell took 1222.4 s and 68.4 GiB (the probe's own peak; the campaign's sampler read 65.3 GiB). $(2420, 2411)$ was recomputed with the levers and gave the same values in 189.3 s and 19.2 GiB (probe; sampler 17.5 GiB), against 5137 s and 493.3 GiB on the first cluster, and the same run's $(1210, 2411)$ reproduces the degree-203 polynomials $\chi^{\mathrm{ord}}(T_2)$, $\chi^{\mathrm{ord}}(T_3)$, $\chi^{\mathrm{ord}}(U)$ of the first-cluster run coefficient for coefficient.

Scope: what this chapter does not do. It proves no part of Hida theory: Theorems 27.11 and 27.12 are literature, with verified locators; (Hi7)–(Hi9), (Hi11), (Hi12) are predictions. What is asserted: the level-N correspondences on the induced model for every N (Proposition 27.2), the level-ℓ operator and its old space, the orbit and cokernel structure, and the integral stabilisation at $\ell \geq 5$, $n \geq 1$. *At* $\ell \in \{2, 3\}$ (v1.13.0-pre.2, WP-3 referee m7): the cochain-level statements hold over every ring — the reciprocity of Proposition 27.2(3) is stated over every R, so the induced model computes the $\Gamma_0(N)$-equivariant (Bredon-type) cohomology of the same tessellation over $\mathbb{Z}/2^k$, $\mathbb{Z}/3^k$, $\mathbb{Z}$, and Proposition 27.2(1),(2), Proposition 27.3,

Proposition 27.4(1),(2) and Lemma 27.8(1)–(3) hold there; what is not asserted there is the cohomology-level independence of τ, the comparison with $H^\bullet(\Gamma_0(N); V_R)$ (Proposition 12.4(4) needs $6 \in R^\times$), the orbit-wise averaging and the cokernel of Proposition 27.5(2),(3), and everything of §27.3 beyond Lemma 27.8(1)–(3) — as at level 1. Not asserted either: Hida's Λ-adic module and Hecke algebra as computed objects; level ℓ^r, $r \geq 2$, and general N beyond the interface of Proposition 27.2 (composite-N coset tables, (O3)); a Hecke action on the cokernel presentation of Proposition 27.5(2) — which is the Manin–Merel Heilbronn-matrix formula, specified by pointer in Remark 27.7, (O2), and not built. (v1.14.0-pre.1: both are now stated and proved in §28, Definition 28.1 and Proposition 28.9; neither is built.) (v1.14.2: Definition 28.1 built, Proposition 28.9 not.)

Code map. Built at v1.13.0 (slot 13; not built at v1.13.0-pre.1 and pre.2, contract first). `Hecke/HeckeLevelEll` (`namespace hecke::level`, slot-13 WP-6): one routine for (68) — $U_\ell^{(\ell)}$, $\widetilde{\mathrm{dg}}_\ell$, $T_p^{(\ell)}$, T_ℓ at level 1 and, with the identity datum, the fibre maps $\widetilde{\mathrm{dg}}_1$, ω — reading only the level-1 τ-table; (70) as a second model route; the operators as block terms applied without a $D_N \times D_N$ matrix, the determinant-ℓ blocks stored factored; the induced relations over $\mathbb{Z}/\ell^k$; the reading on H^q between two local systems. `HeckeHidaLevelEllValidation`: (70) = (68) entrywise as cochain maps, both against an independent group-cocycle route (Shapiro on $\langle S, T\rangle$, no τ) by characteristic polynomials; Proposition 27.4(1), (2) as cochain maps; Proposition 27.5(3); Lemma 27.8(1); Proposition 27.9 as the pair $(e_\ell \circ \widetilde{\mathrm{dg}}_\ell, c_{\mathrm{st}}^{-1}\omega)$; (Hi10) on 92 cells. `HeckeExt/HeckeHidaTower` part 2 (WP-5): towers modulo ℓ^3, T_5, T_7, Definition 27.13 and its variant (H) (`FibreRepresentation::HidaLatticeH`, with the module route's Nakayama basis of the amended sentence), (Hi8)'s decision procedure and arms (a), (b), (c), (Hi7), (Hi9); `HeckeHidaControlValidation`. The heavy cells (WP-10): `HeckeHidaHeavyCellProbe` (`slow`, interlocked), behind the U_ℓ memory lever (`HeckeOperatorOptions::ScopedUEllLowRank`, WP-4, value-preserving; its rank claim is Lemma 27.15, v1.14.1): (Hi11) held at $(700, 691)$ over $\mathbb{F}_{691}$ and $\mathbb{Z}/691^2$; (Hi12) not run (predicted beyond the box; deferred to the cluster roadmap). Every [proved here] and [conjectural] item tested held. Not built: the monomial and orbit-wise routes of §27.2(2)–(5), the Heilbronn-matrix action of (O2) and the composite-N tables of (O3) — the interface for general N, not scheduled (v1.14.0-pre.1: both specified in §28, not built). **At v1.14.2**: the monomial and orbit-wise routes and the composite-N tables are built (slot-14 WP-12, §28's code map); the Heilbronn-matrix action is not. *After the release* (v1.14.0's record; the first cluster run, `docs/matrix-engine/results/noctua2_run1/`, published 2026-09-28 on the owner's sign-off, [S14] Q2): the same probe ran (Hi11) at $n = 1390, 2080, 2770$ and $(700, 691)$ at $m = 2$, and (Hi12) at $(2420, 2411)$ (the items' [verified computationally][a] sentences). The latter needed 493.3 GiB on a 2 TiB node, 0.912 of the U_ℓ-lever fit (541.2 GiB) of (O5) — the regime the cost model placed it in. (v1.14.0-pre.2 kept only this provenance line, under [S14] Q2's default; v1.14.0 restores the rest.) A second run (v1.14.1's record; second-cluster job `3467349`, commit `e85e25a`, the slot-14 levers on by default — coset streaming, the serial phase on the pool and the rank-one closed form of Lemma 27.15 beside WP-4's factored blocks; data signed off by the owner 2026-09-29, [S14] Q-B) added (Hi11) at $n = 4150$ and (Hi12) at $(4830, 2411)$, both held, and recomputed $(2420, 2411)$ in 19.2 GiB (the probe's own peaks 49.9, 68.4, 19.2 GiB; the campaign's sampler 49.9, 65.3, 17.5 GiB). The campaign's admission predictor over-predicted the three cases (84, 182, 46 GiB) and was refitted (`549200e`); the fit of (O5) predates the slot-14 levers and is not restated — it is not a model of a levered run.

Lemma 27.15 (The rank of ρ_n at a matrix singular modulo ℓ: the U_ℓ lever's rank claim)**.** *[proved here] (v1.14.1.) Let ℓ be a prime, $k \geq 1$, $n \geq 0$ and $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{M}_2(\mathbb{Z})$ with $\ell \mid \det g$.*

1. *Over $\mathbb{Z}/\ell^k$ the matrix $\rho_n(g)$ of Definition 8.1 has at most $\min(k, n+1)$ non-zero invariant factors; hence it is a sum of at most $\min(k, n+1)$ outer products $u\,v^{\mathsf{T}}$, $u, v \in (\mathbb{Z}/\ell^k)^{n+1}$.*

2. *Over $\mathbb{F}_\ell$: if $(a,c) \not\equiv (0,0)$, then $(b,d) \equiv \lambda(a,c) \pmod{\ell}$ for a unique $\lambda \in \mathbb{F}_\ell$ and $\rho_n(g) \equiv w\, v^{\mathsf{T}}$ with $w_t = \binom{n}{t} a^{n-t} c^t$ (the coefficients of $(a+cy)^n$) and $v_j = \lambda^j$ ($0^0 = 1$); if $(a,c) \equiv (0,0)$, then $\rho_n(g) \equiv w\, e_n^{\mathsf{T}}$ with w the coefficients of $(b+dy)^n$. In particular $\operatorname{rank}_{\mathbb{F}_\ell} \rho_n(g) \le 1$.*

Proof. (1) By the Smith normal form over $\mathbb{Z}_\ell$, $g = P\,\mathrm{diag}(\ell^\alpha, \ell^\beta)\,Q$ with $P, Q \in \mathrm{GL}_2(\mathbb{Z}_\ell)$ and $0 \le \alpha \le \beta \le \infty$ ($\ell^\infty := 0$, $\infty \cdot 0 := 0$), $\alpha + \beta = v_\ell(\det g) \ge 1$. The formula of Definition 8.1 is polynomial in the entries, so ρ_n extends to $\mathrm{M}_2(\mathbb{Z}_\ell)$ and stays multiplicative ($\rho_n = \mathrm{Sym}^n$); $\rho_n(P)$, $\rho_n(Q)$ are invertible (with inverses $\rho_n(P^{-1})$, $\rho_n(Q^{-1})$), and $\rho_n(\mathrm{diag}(s,t))$ is diagonal with entries $s^{n-j}t^j$ by (9). So the invariant factors of $\rho_n(g)$ over $\mathbb{Z}_\ell$, and after reduction over $\mathbb{Z}/\ell^k$, are ℓ^{e_j} with $e_j = \alpha(n-j) + \beta j$, $0 \le j \le n$, and the non-zero ones modulo ℓ^k are those with $e_j < k$. If $\alpha = 0$, then $\beta \ge 1$, $e_j \ge j$, and at most $\min(k, n+1)$ indices have $e_j < k$. If $\alpha \ge 1$, then $e_j = \alpha n + (\beta - \alpha) j$: for $\beta = \alpha$ all $e_j = \alpha n$, and $e_j < k$ forces $n+1 \le \alpha n + 1 \le k$; for $\beta > \alpha$, $e_j < k$ forces $j < k - \alpha n \le k$. A matrix $P'DQ'$ with D diagonal with r non-zero entries is a sum of r outer products. (2) Column j of $\rho_n(g)$ holds the coefficients of $(a+cy)^{n-j}(b+dy)^j$ (9). If $(a,c) \not\equiv 0$, $\det g \equiv 0$ makes (b,d) proportional to it, so the column is $\lambda^j(a+cy)^n$; if $(a,c) \equiv 0$, every column $j < n$ vanishes and column n is $(b+dy)^n$. □

Code (not a status tag; slot-13 WP-4, slot-14 G0 `a249a6d`, `714e553`): `Hecke/HeckeLowRank.h`. `hecke::lowrank::factor_low_rank` stores a block $\rho_n(g)$, g singular modulo ℓ, as UV by minimum-valuation pivoting, with exactly as many outer products as non-zero invariant factors, and `ENGINE_CHECK`s their number against $\min(k, n+1)$ at every block; `rank_one_rho_factors` is (2) over $\mathbb{F}_\ell$ and `ENGINE_CHECK`s $ad \equiv bc$ per key. By the lemma neither check can fire: they guard the implementation, and the lever cannot change a value (`HeckeUEllLowRankValidation`, `HeckeUEllStreamValidation`).

Formalization note. Propositions 27.2–27.5 and Lemma 27.8 need Frobenius reciprocity, the comparison of projective resolutions, and the Mayer–Vietoris sequence of an amalgam; Proposition 27.9(1)–(3) adds Fitting's lemma and Nakayama. Its part (4) and Theorems 27.11, 27.12 rest on the theory of modular forms and are not formalization targets.

Part VIII — Level and character

28 Level N: the composite coset table, the cokernel presentation and its Hecke action

What this chapter fixes. (i) **The binding coset table for every N** (Definition 28.1, obstacle (O3)): the representatives of $\Gamma_0(N)\backslash\Gamma$, their index order and class map, by the Chinese remainder theorem over the prime-power factors; Definition 10.2 verbatim at $N = p^\nu$; and the representatives of $\Gamma_0(N')\backslash\Gamma_0(N)$ read off the level-N' table. A table, not a search; no new τ-table. (ii) **The level-N model** for every N and every coefficient ring (Definition 28.3), with $\ell \mid N$ (the Hida setting), $\ell \nmid N$, and the setting $\mathbb{F}_p$, $p \nmid N$, of Mazur's Eisenstein ideal. (iii) **The cokernel presentation pinned**: the tree resolution (Lemma 28.4), Manin's exact sequence (Lemma 28.5), the isomorphism $\varkappa_N$ of Proposition 27.5(2) made explicit (Definition 28.6), and a duality with cusp-divisor coinvariants that commutes with restriction, corestriction and conjugation (Lemma 28.7). (iv) **The Hecke action on the presentation** (obstacle (O2)): Merel's set of integral matrices of determinant p, its combinatorial lemma (Lemma 28.8), and the transcription into this document's conventions, *proved* to induce the operator of Proposition 27.2(4)(d) on H^1 — for $T_p^{(N)}$, $p \nmid N$, and $U_p^{(N)}$, $p \mid N$, by one formula (Proposition 28.9). (v) **The cost of**

that route (Remark 28.11) and **the predictions** (LN1)–(LN6) (Remark 28.12), among them route R3 of the slot-14 Loewy note: mod-p Loewy length $e+1$ at Mazur's Eisenstein ideal.

Notation and standing conventions. Those of §27; $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $U := ST = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ (the U of Claim 16.2), ρ_n of Definition 8.1, adj the adjugate of (10). For a prime p, $\Delta_p := \{\delta \in \mathrm{M}_2(\mathbb{Z}) : \det\delta = p\}$. **Cusps**: $\mathbb{P}^1(\mathbb{Q})$ with the *column* action $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \cdot (x : y) := (ax + by : cx + dy)$ of every $\delta \in \Delta$; scalars act trivially, so δ and $\mathrm{adj}\,\delta = (\det\delta)\delta^{-1}$ act as mutually inverse bijections. $[\infty] := (1:0)$, $[0] := (0:1)$, $[-1] := (-1:1)$; $\mathrm{Div} := \mathbb{Z}[\mathbb{P}^1(\mathbb{Q})]$, Div^0 the kernel of the degree, $D_0 := [\infty] - [0] \in \mathrm{Div}^0$; $\mathrm{Div}_R := \mathrm{Div} \otimes R$, likewise Div^0_R. **Level**: $\mathbb{P}^1(\mathbb{Z}/N)$ with the *row* action $(c:d)\cdot\delta := \big((c,d)\delta \bmod N\big)$ of any integral δ for which the result is primitive ($\gcd(c',d',N) = 1$) — always when $\gcd(\det\delta, N) = 1$. **The tree**: $\mathcal{Y}$ is the Bass–Serre tree of Claim 16.2(1); its chain complex is $C_1(\mathcal{Y}) = \mathbb{Z}[\Gamma/\langle -I\rangle]$ with the generator $\mathfrak{e} := \langle -I\rangle$, $C_0(\mathcal{Y}) = \mathbb{Z}[\Gamma/\langle S\rangle] \oplus \mathbb{Z}[\Gamma/\langle U\rangle]$ with the generators $\mathfrak{v}_S$, $\mathfrak{v}_U$, $\partial(\gamma\mathfrak{e}) = \gamma\mathfrak{v}_U - \gamma\mathfrak{v}_S$, and augmentation 1 on every vertex — the sequence (50), whose maps that claim leaves unnamed (a sign convention, which moves no statement). The *edges* are $\mathcal{Y}_1 := \Gamma\mathfrak{e}$, the *vertices* $\mathcal{Y}_0 := \Gamma\mathfrak{v}_S \sqcup \Gamma\mathfrak{v}_U$; $-I$ fixes every edge and vertex. For a, b vertices, $\mathrm{path}(a \to b)$ is the unique 1-chain with boundary $b - a$ (unique: ∂ is injective on $C_1(\mathcal{Y})$, (50)). $\mathfrak{e}, \mathfrak{v}_S, \mathfrak{v}_U, \mathcal{Y}, \varkappa_N, \Pi_A, \mathfrak{b}, \mathrm{Mer}_p, \mathrm{Mer}_n, W_A, W_{N,p}, \mu_N, \zeta_A, H_{N,p}, \mathrm{Div}, D_0$ are new symbols of this chapter and mean nothing else (rule (A5)); Div is not Δ, $\mathcal{Y}$ is not the tessellation X, W_A and $W_{N,p}$ are not Definition 8.1's W, and μ_N is not Proposition 13.2's $\mu_\bullet$ (v1.14.0-pre.2, WP-5 referee F6).

28.1 The coset table for every N

Definition 28.1 (The canonical table of $\Gamma_0(N)\backslash\Gamma$ for every N)**.** Let $N \geq 1$ with prime factorisation $N = \prod_{i=1}^{r} q_i^{\nu_i}$, $q_1 < \cdots < q_r$, $Q_i := q_i^{\nu_i}$, $m_i := Q_i + Q_i/q_i$, $m_N := \prod_i m_i$ ($r = 0$, $m_1 = 1$ at $N = 1$).

1. **Component index.** For $(c,d) \in \mathbb{Z}^2$ primitive modulo q_i: $j_i(c,d) := 1 + (cd^{-1} \bmod Q_i)$ if $q_i \nmid d$, and $j_i(c,d) := Q_i + 1 + (dc^{-1} \bmod Q_i)/q_i$ otherwise — the index of Definition 10.2 at Q_i. The *component pair* of j_i is $(j_i - 1, 1)$ for $j_i \leq Q_i$ and $(1, q_i(j_i - Q_i - 1))$ otherwise.
2. **Index** (mixed radix, primes ascending): for (c,d) primitive modulo N, $j(c,d) := 1 + \sum_{i=1}^{r} \big(j_i(c,d) - 1\big) \prod_{i'<i} m_{i'} \in \{1, \ldots, m_N\}$.
3. **Canonical pair** $(c_j, d_j) \in \{0, \ldots, N-1\}^2$ of an index j: the Chinese-remainder combination of the component pairs of its digits j_i. At $N = 1$: $j = 1$, $(c_1, d_1) := (0,1)$, $w_1 = I$ (v1.14.0-pre.2, WP-5 referee F4).
4. **Representative** $w_j \in \Gamma$, by the first case that applies: (i) $\gcd(d_j, N) = 1$: $w_j := \left(\begin{smallmatrix} 1 & 0 \\ k & 1 \end{smallmatrix}\right)$, $k := c_j d_j^{-1} \bmod N$; (ii) $\gcd(c_j, N) = 1$: $w_j := \left(\begin{smallmatrix} 0 & -1 \\ 1 & k' \end{smallmatrix}\right)$, $k' := d_j c_j^{-1} \bmod N$; (iii) otherwise (a *mixed* class, possible only if $r \geq 2$): t the least integer ≥ 0 with $\gcd(c_j, d_j + tN) = 1$, $d' := d_j + tN$, $a := d'^{-1} \bmod c_j \in [0, c_j)$, $b := (ad' - 1)/c_j$, $w_j := \left(\begin{smallmatrix} a & b \\ c_j & d' \end{smallmatrix}\right)$.
5. **Class map and action**: $\mathrm{cls}_N(\gamma) := j(c,d)$ for γ with bottom row (c,d); for integral δ with $(c_j, d_j)\delta$ primitive modulo N, $j \cdot \delta := j\big((c_j, d_j)\delta\big)$ and $h_j(\delta) := w_j\, \delta\, w_{j\cdot\delta}^{-1}$ (for $\gamma \in \Gamma$ the cocycle of (17)).
6. **Representatives of** $\Gamma_0(N')\backslash\Gamma_0(N)$, $N \mid N'$: the $w_j^{(N')}$ of the level-N' table over the indices j whose canonical pair has $N \mid c_j$, in increasing j. For $A = B = \Gamma_0(N)$ and $g = \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$ they are the r_t of Proposition 27.2, $B'' = \Gamma_0(Np)$.

Lemma 28.2 (The table is well defined, and is Definition 10.2 at prime powers)**.** *[proved here] With the notation of Definition 28.1:*

1. *j is constant on $(\mathbb{Z}/N)^\times$-classes and induces a bijection $\mathbb{P}^1(\mathbb{Z}/N) \to \{1,\dots,m_N\}$; $m_N = N\prod_{q\mid N}(1+q^{-1})$.*
2. *Every w_j lies in Γ and has bottom row in the class j; in case (iii) $t < c_j$. $w_1 = I$. The w_j represent $\Gamma_0(N)\backslash\Gamma$.*
3. *At $N = p^\nu$ index, canonical pairs and representatives are those of Definition 10.2.*
4. *$(j,\gamma) \mapsto j\cdot\gamma$ is a right action of Γ, computed digit by digit (the digit i of $j\cdot\gamma$ is the level-Q_i action on the digit j_i), and $h_j(\gamma) \in \Gamma_0(N)$. For integral δ with $(c_j,d_j)\delta$ primitive modulo N, $h_j(\delta)$ is integral, $\det h_j(\delta) = \det\delta$, and its lower left entry is $\equiv 0 \pmod N$. [formally verified: [GT, `prop:heilbronn-model`]] (this last sentence, for any table of matrices of Γ with pairwise inequivalent bottom rows)*
5. *The set of item (6) represents $\Gamma_0(N')\backslash\Gamma_0(N)$, lies in $\Gamma_0(N)$, and starts with I; it has $m_{N'}/m_N$ elements ($p+1$ for $N' = Np$, $p\nmid N$; p for $p\mid N$). At $N = 1$ it is Definition 10.2 at level N'.*

Proof. (1) Reduction modulo the Q_i is a bijection $\mathbb{P}^1(\mathbb{Z}/N) \to \prod_i \mathbb{P}^1(\mathbb{Z}/Q_i)$: $(\mathbb{Z}/N)^\times = \prod_i(\mathbb{Z}/Q_i)^\times$, and (c,d) is primitive modulo N iff it is modulo every q_i (Chinese remainder theorem). Each j_i is the bijection $\mathbb{P}^1(\mathbb{Z}/Q_i) \to \{1,\dots,m_i\}$ of Proposition 10.1(4) and Definition 10.2, and the mixed radix is a bijection $\prod_i\{1,\dots,m_i\} \to \{1,\dots,m_N\}$.

(2) The canonical pair reduces modulo Q_i to the component pair, which is primitive and of class j_i; so (c_j,d_j) is primitive modulo N and of class j. Case (i): the bottom row $(k,1) \equiv d_j^{-1}(c_j,d_j)$; case (ii): $(1,k') \equiv c_j^{-1}(c_j,d_j)$; both matrices have determinant 1. Case (iii): $c_j \neq 0$ (else d_j is a unit, case (i)) and $c_j \neq 1$ (case (ii)), so $2 \le c_j < N$. For a prime $s \mid c_j$: if $s \mid N$, then $s\nmid d_j$ by primitivity, so $s\nmid d_j + tN$ for every t; if $s\nmid N$, exactly one residue of t modulo s has $s \mid d_j + tN$. By the Chinese remainder theorem some t modulo $\prod_{s\mid c_j,\, s\nmid N} s \le c_j$ avoids all of them, so $t < c_j$ and $\gcd(c_j,d') = 1$; then $ad' \equiv 1 \pmod{c_j}$, $b\in\mathbb{Z}$, $\det w_j = ad' - bc_j = 1$, and the bottom row $(c_j,d') \equiv (c_j,d_j) \pmod N$. $j = 1$ has all digits 1, component pairs $(0,1)$, canonical pair $(0,1)$, case (i) with $k = 0$: $w_1 = I$. The last sentence is the bottom-row bijection $\Gamma_0(N)\backslash\Gamma \to \mathbb{P}^1(\mathbb{Z}/N)$ (Definition 27.1's parenthetical, for every N).

(3) $r = 1$: $j = j_1$ is Definition 10.2's index, the canonical pair is the component pair; $(k,1)$ has $d = 1$ a unit and case (i) gives $\begin{pmatrix}1 & 0\\ k & 1\end{pmatrix} = s_{1+k}$; $(1,pc)$ has $c = 1$ a unit and $d = pc$ not, and case (ii) gives $\begin{pmatrix}0 & -1\\ 1 & pc\end{pmatrix} = s_{N+1+c}$.

(4) The class of $(c_j,d_j)\gamma_1$ is $j\cdot\gamma_1$, so $(c_{j\cdot\gamma_1},d_{j\cdot\gamma_1}) \equiv u\,(c_j,d_j)\gamma_1$ with u a unit, and $(j\cdot\gamma_1)\cdot\gamma_2$ is the class of $u(c_j,d_j)\gamma_1\gamma_2$, i.e. $j\cdot(\gamma_1\gamma_2)$; $j\cdot I = j$. The class modulo Q_i of $(c,d)\gamma$ depends only on that of (c,d), whence the digitwise computation. $h_j(\gamma) \in \Gamma_0(N)$ by the argument of Lemma 10.3. For δ: $w_j, w_{j\cdot\delta} \in \Gamma$, so $h_j(\delta)$ is integral of determinant $\det\delta$; its bottom row is $\mathrm{row}_2(w_j)\,\delta\,w_{j\cdot\delta}^{-1}$, and $\mathrm{row}_2(w_j)\delta \equiv u'\,\mathrm{row}_2(w_{j\cdot\delta}) \pmod N$ for a unit u' (both primitive, same class), while $\mathrm{row}_2(w_{j\cdot\delta})\,w_{j\cdot\delta}^{-1} = (0,1)$.

(5) $\Gamma_0(N')\gamma \subseteq \Gamma_0(N)$ iff N divides the lower left entry of γ, a condition on the class of the bottom row modulo N' ($N \mid N'$); so the classes with $N \mid c_j$ are the image of $\Gamma_0(N')\backslash\Gamma_0(N)$ in $\Gamma_0(N')\backslash\Gamma$. Their lifts lie in $\Gamma_0(N)$: at a prime $q_i \mid N$, $q_i^{\nu_i} \mid c$ forces $q_i\nmid d$, so the component pair modulo Q_i' is $(k,1)$ with $k \equiv cd^{-1} \equiv 0 \pmod{Q_i}$; hence $c_j \equiv 0 \pmod N$, and the lift keeps the lower left entry $\equiv c_j$ in case (i) ($k = c_jd_j^{-1}$), exactly in case (iii), and case (ii) needs c_j a unit modulo N', so $N = 1$. The class of I has index 1. The count is $[\Gamma_0(N) : \Gamma_0(N')] = m_{N'}/m_N$ by (1) and Proposition 10.1(4). □

CONTRACT. Definition 28.1 is **binding as written** (Remark 11.9: the cochain-level matrices of (68) depend on it, $H^\bullet$ does not), and it closes obstacle (O3).

It is **a table, not a search**: cases (i), (ii) are closed forms, case (iii) the least $t < c_j$ with $\gcd(c_j, d_j + tN) = 1$ (a bounded loop, at most c_j gcd's, in practice a few; table-building time, never per operator) and one inverse modulo c_j; the class map and the action are digitwise, $O(\omega(N))$ operations modulo the Q_i. **No new τ-table** is needed for any N: every level-N operator of Proposition 27.2 reads the level-1 table of $T_{\Gamma g \Gamma}$ (its item (1)), and the Hecke action of Proposition 28.9 reads no τ-table at all. Any other transversal of $B''\backslash B$ gives the same operator on $H^\bullet$ (Proposition 27.2(3), $6 \in R^\times$) and, in general, different cochain-level matrices; slot 13's `Gamma0SubTransversal` ($\begin{pmatrix} 1 & 0 \\ Nt & 1 \end{pmatrix}$, $t < p$, and $\begin{pmatrix} x & y \\ N & p \end{pmatrix}$ for $p \nmid N$) is such an alternative, used by `HeckeHidaLevelEllValidation` at $N = \ell$ and compared with item (6)'s on H^1 by `HeckeLevelNValidation` (arms (H), (E)); it is not the binding datum, and no shipped record, report or golden text reads either.

28.2 The level-N model and the cokernel presentation

Definition 28.3 (The level-N model for every N; its coefficient settings)**.** Let $N \geq 1$ and R a commutative ring. $M_N := \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V_R$ in the coordinates of Definition 12.1 with the w_j of Definition 28.1, and $L_N := L_{\Gamma_0(N)}$ (Definition 12.3): Definition 27.1 with the table now fixed for every N. Its relation matrices are the monomial $M_N(\gamma) = \sum_j E_{j,\,j\cdot\gamma} \otimes \rho_n(h_j(\gamma))$ ((22)); the orbit structure is Proposition 27.5, stated there for every N, with $\nu_2(N)$, $\nu_3(N)$ of Remark 27.6(3) multiplicative over the Q_i and the S-, U-orbits on $\mathbb{P}^1(\mathbb{Z}/N)$ computed digitwise. The **coefficient settings**:

(a) every R at cochain level (Proposition 27.2(1),(2));

(b) $6 \in R^\times$ for the comparison with $H^\bullet(\Gamma_0(N); V_R)$ — $R = \mathbb{F}_\ell$, $\mathbb{Z}/\ell^k$ ($\ell \geq 5$), $\mathbb{Q}$, $\mathbb{Z}[1/6]$ — in either relative position: $\ell \mid N$ (the Hida setting; $N = \ell$ is slot 13's) or $\ell \nmid N$;

(c) the **R3 setting** (route R3 of `docs/implementation/slot14/WP-M1_loewy_note.md` §6): $R = \mathbb{F}_p$ (or $\mathbb{Z}/p^k$), $p \geq 5$, $n = 0$ (weight 2), N prime with $p \mid \mathrm{num}\left(\frac{N-1}{12}\right)$, so $p \nmid N$ — Mazur's Eisenstein ideal (Remark 28.12, (LN4)–(LN6)). *Since v1.21.0-pre.4*, at $n = 0$ only, also $p \in \{2, 3\}$ on the **curve object**: $R = \mathbb{F}_p$ (or $\mathbb{Z}$), every prime p, the fixed spaces $M_N^{\langle S \rangle}$, $M_N^{\langle U \rangle}$ taken as the spans of the orbit sums; the cokernel is then $H^1(Y_0(N); \mathbb{F}_p)$, the cohomology of the open coarse curve, which at $p \in \{2, 3\}$ is **not** $H^1(\Gamma_0(N); \mathbb{F}_p)$ (Proposition 28.18, Remark 28.19).

The level-N operators are those of Proposition 27.2 with $A = B = \Gamma_0(N)$, the r_t of Definition 28.1(6), and the s_j and τ of Definition 10.2 for $T_{\Gamma g \Gamma}$ at level 1.

Lemma 28.4 (The tree is a projective resolution)**.** *[proved here] Let $6 \in R^\times$ and $A_0 \leq \Gamma$ any subgroup. Then $C_\bullet(\mathcal{Y}) \otimes R \to R$ is a projective resolution of length 1 of R over $R[A_0]$. Hence $H^q(A_0; X) = H^q\big(\mathrm{Hom}_{A_0}(C_\bullet(\mathcal{Y}), X)\big)$ and $H_q(A_0; X) = H_q\big((C_\bullet(\mathcal{Y}) \otimes X)_{A_0}\big)$ for every $R[A_0]$-module X, both zero for $q \geq 2$.*

Proof. (50) is an exact sequence of free abelian groups, hence split over $\mathbb{Z}$, so it stays exact after $\otimes R$. As an $R[A_0]$-module, $R[\Gamma/F]$ ($F \in \{\langle -I \rangle, \langle S \rangle, \langle U \rangle\}$, of orders $2, 4, 6$) is $\bigoplus R[A_0/H']$ over the double cosets $A_0 \gamma F$, $H' = A_0 \cap \gamma F \gamma^{-1}$ of order dividing 4 or 6, a unit of R; $e := |H'|^{-1} \sum_{h \in H'} h$ is an idempotent with $R[A_0]e \cong R[A_0/H']$, a direct summand of $R[A_0]$. □

Lemma 28.5 (Manin's exact sequence)**.** *[proved here] [standard: [Man72] for the statement (Manin symbols and their relations); pointer* **not** *verified against the text (rule (A2)); the proof below is complete] Define Γ-maps*

$$\partial^t \colon C_0(\mathcal{Y}) \to C_1(\mathcal{Y}), \quad \gamma\mathfrak{v}_S \mapsto \gamma\mathfrak{e} + \gamma S\mathfrak{e}, \quad \gamma\mathfrak{v}_U \mapsto \gamma\mathfrak{e} + \gamma U\mathfrak{e} + \gamma U^2\mathfrak{e}; \qquad \mathfrak{b} \colon C_1(\mathcal{Y}) \to \mathrm{Div}, \quad \gamma\mathfrak{e} \mapsto \gamma \cdot D_0 = [\gamma\infty] - [$$

Then $0 \to C_0(\mathcal{Y}) \xrightarrow{\partial^t} C_1(\mathcal{Y}) \xrightarrow{\mathfrak{b}} \mathrm{Div}^0 \to 0$ is exact.

Proof. The maps are well defined: $S^2 = U^3 = -I$ fixes $\mathfrak{e}$ and every cusp, so $\gamma S\mathfrak{v}_S = \gamma\mathfrak{v}_S$ and $\gamma U\mathfrak{v}_U = \gamma\mathfrak{v}_U$ give the same sums. $\mathsf{b}\,\partial^t = 0$: S exchanges $[\infty]$ and $[0]$, and U permutes $[\infty] \mapsto [0] \mapsto [-1] \mapsto [\infty]$, so $\mathsf{b}(\mathfrak{e}+S\mathfrak{e}) = 0$ and $\mathsf{b}(\mathfrak{e}+U\mathfrak{e}+U^2\mathfrak{e}) = ([\infty]-[0])+([0]-[-1])+([-1]-[\infty]) = 0$.

∂^t *is injective.* The edge $\gamma\mathfrak{e}$ has exactly the two end points $\gamma\mathfrak{v}_S$, $\gamma\mathfrak{v}_U$; if $\partial^t(\sum_v a_v v) = 0$ then $a_v + a_w = 0$ for every edge $\{v, w\}$, so $|a_v|$ is constant on the tree, which is connected (exactness of (50) at $\mathbb{Z}$) and infinite; a finite sum forces $a = 0$.

The Farey complex. Let F be the simplicial complex with vertex set $\mathbb{P}^1(\mathbb{Q})$, edges $\{\gamma\infty, \gamma 0\}$ and triangles $\{\gamma\infty, \gamma 0, \gamma(-1)\}$, $\gamma \in \Gamma$. (F1) For coprime pairs, $\{(a : c), (b : d)\}$ is an edge iff $ad - bc = \pm 1$ ($\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ or $\left(\begin{smallmatrix} a & -b \\ c & -d \end{smallmatrix}\right)$ lies in Γ). (F2) The triangles are exactly the triples of pairwise adjacent cusps: the sides of $\{\gamma\infty, \gamma 0, \gamma(-1)\}$ are $\gamma\{\infty, 0\}$, $\gamma U\{\infty, 0\}$, $\gamma U^2\{\infty, 0\}$; conversely, move two of the three to $\infty, 0$ by (F1), and the third, adjacent to both, is ± 1, with $\{\infty, 0, 1\} = S\{\infty, 0, -1\}$. *Height*: $\mathrm{ht}(a : c) := |c|$ for coprime (a, c); F_h is the full subcomplex on the cusps of height $\le h$. (F3) A cusp $x = (a : c)$ with $c \ge 2$ has exactly two neighbours of smaller height, $y = (b_1 : d_1)$ with $ad_1 - b_1 c = 1$, $d_1 \in [1, c)$ ($d_1 \equiv a^{-1} \bmod c$), and $z = (a - b_1 : c - d_1)$; $\{y, z\}$ is an edge ($b_1(c - d_1) - (a - b_1)d_1 = -1$); no neighbour has height c ($c \mid ad - bc$ there); and by (F2) the only triangle of F_c containing x is $\{x, y, z\}$. *Elementary move*: if $L' = L \cup \{x, \{x, y\}, \{x, z\}, \{x, y, z\}\}$ with $x \notin L$, $\{y, z\} \in L$, the relative chain complex $C_\bullet(L', L)$ is $\mathbb{Z}\{x, y, z\} \to \mathbb{Z}\{x, y\} \oplus \mathbb{Z}\{x, z\} \to \mathbb{Z}x$ with the maps $\{x, y, z\} \mapsto \{x, z\} - \{x, y\}$ up to orientation signs and $\{x, y\}, \{x, z\} \mapsto \pm x$, which is acyclic; so $L \subseteq L'$ is a homology isomorphism. F_1 (cusps ∞ and $\mathbb{Z}$, triangles $\{\infty, k, k+1\} = T^{k+1}\{\infty, 0, -1\}$) is the increasing union of the fans on $\{\infty, -k, \dots, k\}$, each obtained from the edge $\{\infty, 0\}$ by elementary moves; F_h is obtained from F_{h-1} by one elementary move per cusp of height $h \ge 2$ (two cusps of height h are not adjacent), and $\mathsf{F} = \bigcup_h \mathsf{F}_h$. Homology commutes with increasing unions of subcomplexes, so $\tilde{H}_\bullet(\mathsf{F}; \mathbb{Z}) = 0$.

Conclusion. Let $q\colon C_1(\mathcal{Y}) \to C_1(\mathsf{F})$ (oriented 1-chains, $[x, y] = -[y, x]$), $q(\gamma\mathfrak{e}) := [\gamma 0, \gamma\infty]$: well defined, onto by (F1), with $\mathsf{b} = \partial_1^{\mathsf{F}} \circ q$. $q(\gamma\mathfrak{e}) = \pm q(\gamma'\mathfrak{e})$ iff $\gamma' \in \pm\gamma$ (sign $+$) or $\gamma' \in \pm\gamma S$ (sign $-$), the stabiliser of an ordered Farey pair in Γ being $\{\pm I\}$ (it fixes $(1 : 0)$ and $(0 : 1)$); so $\ker q = \partial^t\mathbb{Z}[\Gamma/\langle S\rangle]$. And $q\,\partial^t(\gamma\mathfrak{v}_U) = [\gamma 0, \gamma\infty] + [\gamma(-1), \gamma 0] + [\gamma\infty, \gamma(-1)] = \partial[\gamma(-1), \gamma 0, \gamma\infty]$, so $q\,\partial^t\mathbb{Z}[\Gamma/\langle U\rangle] = \partial C_2(\mathsf{F})$. Then $\operatorname{im}\mathsf{b} = \operatorname{im}\partial_1^{\mathsf{F}} = \mathrm{Div}^0$ ($\tilde{H}_0 = 0$), and $\ker\mathsf{b} = q^{-1}(\partial C_2(\mathsf{F})) = \partial^t\mathbb{Z}[\Gamma/\langle U\rangle] + \ker q = \operatorname{im}\partial^t$ ($H_1 = 0$). □

Definition 28.6 (The pinned cokernel isomorphism $\varkappa_N$)**.** [proved here] (the claims; proof below) Let $6 \in R^\times$, n even, $M := M_N$ and $\mathrm{Rel}_N := M^{\langle S\rangle} + M^{\langle U\rangle} = (1 + M_N(S))M + (1 + M_N(U) + M_N(U)^2)M$ (Proposition 27.5(2)). For $x \in M$ let $\psi_x \in \mathrm{Hom}_{\Gamma_0(N)}(C_1(\mathcal{Y}), V_R)$, $\psi_x(\gamma\mathfrak{e}) := \big(M_N(\gamma)x\big)_1 = \rho_n(h_1(\gamma))\,x_{1\cdot\gamma}$, and let $\varkappa_N(x) \in H^1(\Gamma_0(N); V_R)$ be its class (Lemma 28.4). Then $\varkappa_N$ induces an isomorphism $M/\mathrm{Rel}_N \to H^1(\Gamma_0(N); V_R)$ — **the** isomorphism of Proposition 27.5(2), which that proposition did not pin. In group cocycles: the inhomogeneous cocycle of ψ_x transported to Γ is $C_x(\gamma) := \phi_x(\mathrm{path}(\mathfrak{v}_U \to \gamma\mathfrak{v}_U))$, $\phi_x(\gamma\mathfrak{e}) := M_N(\gamma)x$, so $C_x(U) = 0$ and $C_x(S) = (M_N(S) - 1)x$.

Proof of the claims. The reciprocity $\phi \mapsto \phi(\cdot)_1$ of Proposition 27.2(3) is a cochain isomorphism $\mathrm{Hom}_\Gamma(C_\bullet(\mathcal{Y}), M) \to \mathrm{Hom}_{\Gamma_0(N)}(C_\bullet(\mathcal{Y}), V_R)$ (its proof uses only that the domain is a $\mathbb{Z}[\Gamma]$-module). $\mathrm{Hom}_\Gamma(C_1, M) = M^{-I} = M$ by $\phi \mapsto \phi(\mathfrak{e})$ ($M_N(-I) = 1$, n even), $\mathrm{Hom}_\Gamma(C_0, M) = M^{\langle S\rangle} \oplus M^{\langle U\rangle}$, and the coboundary is $(a, b) \mapsto b - a$; its image is Rel_N, the equality $M^{\langle S\rangle} = (1 + M_N(S))M$ holding because $\frac{1}{2}(1 + M_N(S))$ is the averaging idempotent ($M_N(S)^2 = M_N(-I) = 1$), likewise for U. The cocycle: $f_0(\gamma) := \gamma\mathfrak{v}_U$, $f_1(\gamma_0, \gamma_1) := \mathrm{path}(\gamma_0\mathfrak{v}_U \to \gamma_1\mathfrak{v}_U)$ is a Γ-chain map from the standard resolution to $C_\bullet(\mathcal{Y})$ over the identity (the boundary of a path is its end minus its start), so f^* is the comparison isomorphism, and the inhomogeneous cocycle is $\gamma \mapsto \phi(f_1(1, \gamma))$; $\mathrm{path}(\mathfrak{v}_U \to S\mathfrak{v}_U) = -\mathfrak{e} + S\mathfrak{e}$ and $U\mathfrak{v}_U = \mathfrak{v}_U$. □

Lemma 28.7 (Cokernel duality, and its naturality)**.** *[proved here] Let $6 \in R^\times$, n even, and let $A \le \Gamma$ be of finite index with $-I \in A$. Put $W_A := H_0(A; \mathrm{Div}^0_R \otimes V_R) = (\mathrm{Div}^0_R \otimes V_R)_A$, A acting*

diagonally ($\gamma\cdot(D\otimes v)=\gamma D\otimes\rho_n(\gamma)v$)*, and for* $\psi\in\mathrm{Hom}_A(C_1(\mathcal{Y}),V_R)$

$$\Pi_A(\psi):=\sum_{\xi\in A\backslash\mathcal{Y}_1}\big[\mathsf{b}(\xi)\otimes\psi(\xi)\big]_A,$$

the sum over any set of representatives of the A*-orbits of edges (finitely many; the summand depends only on the orbit, the stabiliser* $\{\pm I\}$ *of an edge acting trivially on* Div *and on* V_R*).*

1. Π_A *vanishes on coboundaries and induces* $\Pi_A\colon H^1(A;V_R)\to W_A$*.*
2. ***(Coordinates.)*** *For* $A=\Gamma_0(N)$*:* $\Pi_A(\varkappa_N(x))=\mu_N(x):=\sum_j[w_jD_0\otimes x_j]_A$*; the* Manin map $\mu_N\colon M_N\to W_A$ *is onto with kernel* Rel_N*; hence* Π_A *is an isomorphism.*
3. ***(Restriction.)*** *For* $A'\le A$ *of finite index with* $-I\in A'$*:* $\Pi_{A'}\circ\mathrm{res}^A_{A'}=\mathrm{tr}\circ\Pi_A$*,* $\mathrm{tr}[y]_A:=\sum_{r\in A'\backslash A}[r\cdot y]_{A'}$*.*
4. ***(Corestriction.)*** *Likewise* $\Pi_A\circ\mathrm{cor}^A_{A'}=\mathrm{pr}\circ\Pi_{A'}$*,* $\mathrm{pr}\colon W_{A'}\to W_A$ *the projection.*
5. ***(Conjugation.)*** *For* $g\in\Delta$ *and* $A'\le\Gamma$ *with* $A'':=gA'g^{-1}\le\Gamma$ *(both of finite index, containing* $-I$*):* $\Pi_{A''}\circ(\mathrm{conj}_g)_*=\mathrm{conj}_g\circ\Pi_{A'}$*, where* $(\mathrm{conj}_g)_*\colon H^1(A';V_R)\to H^1(A'';V_R)$ *is induced by the compatible pair* $(\mathrm{conj}_g^{-1},\rho_n(g))$ *of* (20) *and* $\mathrm{conj}_g[y]_{A'}:=[g\cdot y]_{A''}$*,* g *acting on* Div^0 *by the column action and on* V_R *by* $\rho_n(g)$*.*
6. ***(Hecke operators.)*** *Consequently, for* $g\in\Delta$ *with* $A'':=A\cap gAg^{-1}$ *of finite index,* $\Pi_A\circ[A\,g\,A]=[A\,g\,A]_*\circ\Pi_A$*, where* $[A\,g\,A]$ *is the operator of §27's notation and* $[A\,g\,A]_*[y]_A:=\sum_{A\beta\in A\backslash AgA}[\beta\cdot y]_A$*.*

Proof. Star sums. For an A-invariant function F on edges with values in an abelian group in which 6 is invertible, $\sum_{\xi\in A\backslash\mathcal{Y}_1}F(\xi)=\sum_v k_v^{-1}\sum_{\xi\colon\mathrm{head}(\xi)=v}F(\xi)$, the outer sum over representatives v of the A-orbits of U-vertices, $\mathrm{head}(\gamma\mathfrak{e}):=\gamma\mathfrak{v}_U$, and $k_v:=|\mathrm{Stab}_A(v)/\{\pm I\}|\in\{1,3\}$: $\mathrm{Stab}_A(v)/\{\pm I\}$ acts freely on the three edges at v, so each A-orbit of edges meets the star of exactly one representative v, in k_v edges. The same holds with tails and S-vertices ($k_v\in\{1,2\}$).

(1) For $\psi=\delta f$, $f\in\mathrm{Hom}_A(C_0(\mathcal{Y}),V_R)$, $\psi(\xi)=f(\mathrm{head}\,\xi)-f(\mathrm{tail}\,\xi)$; the star sum turns $\sum_\xi[\mathsf{b}(\xi)\otimes f(\mathrm{head}\,\xi)]$ into $\sum_v k_v^{-1}[\mathsf{b}(\partial^t v)\otimes f(v)]=0$ by Lemma 28.5, and likewise for tails.

(2) $A\backslash\mathcal{Y}_1\cong A\backslash\Gamma$ ($\pm I\subseteq A$ central), with representatives $w_j\mathfrak{e}$, and $\psi_x(w_j\mathfrak{e})=\rho_n(h_1(w_j))x_{1\cdot w_j}=x_j$ since $1\cdot w_j=j$ and $h_1(w_j)=I$; so $\Pi_A(\varkappa_N(x))=\mu_N(x)$. Tensor the sequence of Lemma 28.5 (split over $\mathbb{Z}$) with V_R and take A-coinvariants (right exact): $(C_0\otimes V_R)_A\to(C_1\otimes V_R)_A\to W_A\to 0$ is exact. $x\mapsto\sum_j[w_j\mathfrak{e}\otimes x_j]$ identifies M with $(C_1\otimes V_R)_A$ (each A-orbit of edges once), with inverse $[hw_i\mathfrak{e}\otimes v]\mapsto$ the vector with block i equal to $\rho_n(h)^{-1}v$ ($h\in A$). Under it, $[w_i\mathfrak{v}_S\otimes v]$ maps to $e_i\otimes v+e_{i\cdot S}\otimes\rho_n(h_i(S))^{-1}v=(1+M_N(S)^{-1})(e_i\otimes v)$, and $M_N(S)^{-1}=M_N(S)M_N(-I)=M_N(S)$; the U-vertices give $(1+M_N(U)+M_N(U)^2)(e_i\otimes v)$ likewise ($U^{-1}=U^2\cdot(-I)$). So the image of $(C_0\otimes V_R)_A$ is Rel_N, and the composite $M\to W_A$ is μ_N. With Definition 28.6, Π_A is an isomorphism.

(3) $\{r\xi\}$, $r\in A'\backslash A$, $\xi\in A\backslash\mathcal{Y}_1$, represents $A'\backslash\mathcal{Y}_1$ (edge stabilisers are $\{\pm I\}\subseteq A'$), and $[r\cdot(\mathsf{b}(\xi)\otimes\psi(\xi))]_{A'}=[\mathsf{b}(r\xi)\otimes\psi(r\xi)]_{A'}$.

(4) The corestriction on the resolution of Lemma 28.4 is the transfer $\psi'\mapsto\sum_r\rho_n(r^{-1})\psi'(r\cdot)$ (as in the proof of Proposition 27.2(3)); then $[\mathsf{b}(\xi)\otimes\rho_n(r^{-1})\psi'(r\xi)]_A=[r^{-1}\cdot(\mathsf{b}(r\xi)\otimes\psi'(r\xi))]_A$, and summing over ξ, r gives $\mathrm{pr}\,\Pi_{A'}(\psi')$ by (3)'s bijection.

(5) *A pairing.* For A as in the statement and any $R[A]$-module X, put $\psi\frown z:=\sum_i[\mathsf{x}_i\otimes\psi(y_i)]_A\in(X\otimes V_R)_A$ for $\psi\in\mathrm{Hom}_A(P_1,V_R)$ and a 1-cycle $z=[\sum_i y_i\otimes\mathsf{x}_i]$ of $(P_1\otimes X)_A$, P any projective resolution of length 1 (such as Lemma 28.4's). It is well defined on $H^1\times H_1$: on δf it is the image of $(\partial\otimes 1)z=0$ under $[y\otimes\mathsf{x}]\mapsto[\mathsf{x}\otimes f(y)]$, and there are no 1-boundaries. For a chain map $F\colon P\to P'$ of such resolutions over id_R, $(F^*\psi)\frown z=\psi\frown(F_*z)$ at chain level, F^*, F_* being the comparison isomorphisms: so $\frown$ is a pairing $H^1(A;V_R)\times H_1(A;X)\to(X\otimes V_R)_A$ independent of P. By definition $\Pi_A(\psi)=\psi\frown\zeta_A$ with $\zeta_A:=\sum_{\xi\in A\backslash\mathcal{Y}_1}[\xi\otimes\mathsf{b}(\xi)]$, a cycle of $(C_1(\mathcal{Y})\otimes\mathrm{Div}^0_R)_A$ by the star sums and Lemma 28.5. *Transport.* Let P^g be $C_\bullet(\mathcal{Y})\otimes R$ with $h''\in A''$ acting as $g^{-1}h''g\in A'$: a length-1 projective resolution over $R[A'']$. The maps of the compatible pairs $(\mathrm{conj}_g^{-1},\rho_n(g))$ on H^1 and (conj_g,g) on $H_1(-;\mathrm{Div}^0)$ and $H_0(-;\mathrm{Div}^0\otimes V_R)$ are

computed on P and P^g by $\psi \mapsto \rho_n(g)\circ\psi$ and $y\otimes \mathsf{x} \mapsto y\otimes g\mathsf{x}$ [standard: [Bro82, Ch. III.8], the map of a compatible pair by a chain map over it — here the identity of the underlying modules; **not** verified (rule (A2)); the dependence is definitional (the meaning of $(\mathrm{conj}_g)_*$ in (20)), no theorem of the source is used; v1.14.0-pre.2, WP-5 referee F7], and at chain level $(\rho_n(g)\psi) \frown (y\otimes g\mathsf{x}) = \mathrm{conj}_g\big(\psi \frown (y\otimes \mathsf{x})\big)$. So (5) follows from $(\mathrm{conj}_g)_*[\zeta_{A'}] = [\zeta_{A''}]$ in $H_1(A''; \mathrm{Div}_R^0)$. *The class* $[\zeta_A]$. $H_2(A; R) = 0$ (Lemma 28.4), so $0 \to \mathrm{Div}_R^0 \to \mathrm{Div}_R \to R \to 0$ makes $H_1(A; \mathrm{Div}_R^0) \to H_1(A; \mathrm{Div}_R)$ injective, and $\mathrm{Div}_R = \bigoplus_{c\in A\backslash\mathbb{P}^1(\mathbb{Q})} R[A/A_c]$, A_c the stabiliser, gives $H_1(A; \mathrm{Div}_R) \cong \bigoplus_c H_1(A_c; R)$ by $[x\otimes aA_c] \mapsto [a^{-1}x]$ on $(C_1(\mathcal{Y})\otimes R[A/A_c])_A \cong (C_1(\mathcal{Y})\otimes R)_{A_c}$ (and P is a resolution over $R[A_c]$, Lemma 28.4). For $c = \sigma\infty$, $\sigma\in\Gamma$: $A_c = \pm\langle\gamma_c^+\rangle$ with $\gamma_c^+ := \sigma T^{h_c}\sigma^{-1}$, $h_c \geq 1$ (independent of σ). *Translation classes*: for a tree $\mathcal{X}$ with an action of A_c making $C_\bullet(\mathcal{X})\otimes R$ a resolution as above and a vertex x, $[\mathrm{path}(x\to\gamma_c^+ x)] \in H_1(A_c; R)$ is a cycle independent of x ($\mathrm{path}(x\to\gamma x) - \mathrm{path}(y\to\gamma y) = \mathrm{path}(x\to y) - \gamma\,\mathrm{path}(x\to y)$) and of $\mathcal{X}$: for a chain map F over id to another such tree, $F_1\mathrm{path}(x\to\gamma x)$ and $\sum_i a_i\mathrm{path}(y_i\to\gamma y_i)$ ($F_0 x = \sum a_i y_i$, $\sum a_i = 1$) have the same boundary, hence are equal (∂ injective). Call it $[\gamma_c^+]$. The c-component of ζ_A: at $c=\infty$ the edges with b-coefficient at $[\infty]$ are $\pm T^k\mathfrak{e}$ (head) and $\pm T^k S\mathfrak{e}$ (tail), and the component is $\sum_{k<h_\infty}(T^k\mathfrak{e} - T^k S\mathfrak{e})$, whose boundary is $\sum_{k<h_\infty}(T^k\mathfrak{v}_U - T^{k+1}\mathfrak{v}_U) = \mathfrak{v}_U - T^{h_\infty}\mathfrak{v}_U$ because $S\mathfrak{v}_U = T\mathfrak{v}_U$ ($T^{-1}S = U^2$); so it is $-\mathrm{path}(\mathfrak{v}_U \to \gamma_\infty^+\mathfrak{v}_U)$, of class $-[\gamma_\infty^+]$; at $c=\sigma\infty$ translate by σ. So $[\zeta_A]$ is **the unique class whose component at every cusp c is $-[\gamma_c^+]$**. Finally conj_g maps A'_c onto A''_{gc} and γ_c^+ to γ_{gc}^+: writing $g\sigma = \tau\begin{pmatrix} a & b\\ 0 & d\end{pmatrix}$ ($\tau\in\Gamma$, $a,d>0$, Hermite), $g\gamma_c^+ g^{-1} = \tau\begin{pmatrix} 1 & ah_c/d\\ 0 & 1\end{pmatrix}\tau^{-1} \in A''$ with $ah_c/d > 0$ a positive translation conjugate, which generates A''_{gc} modulo $\pm I$; transported to P^g the translation class of γ_c^+ is that of γ_{gc}^+. Hence both classes have the same image, and are equal.

(6) $[A\,g\,A] = \mathrm{cor}^A_{A''}\circ(\mathrm{conj}_g)_*\circ\mathrm{res}^A_{A'}$, $A' := A\cap g^{-1}Ag$, $A'' = gA'g^{-1}$ (both contain $-I$), so by (3)–(5) $\Pi_A[A\,g\,A]\psi = \mathrm{pr}\,\mathrm{conj}_g\,\mathrm{tr}\,\Pi_A\psi$, i.e. $[y]_A \mapsto \sum_{r\in A'\backslash A}[gr\cdot y]_A$, and the gr represent $A\backslash AgA$ ($Agr = Agr'$ iff $rr'^{-1}\in A\cap g^{-1}Ag$). □

28.3 The Hecke action on the presentation: Merel's set

Lemma 28.8 (Merel's set and its path condition)**.** *[proved here] [standard: [Mer94, §2] for the set and the condition; pointer* **not** *verified (rule (A2)); the proof below is complete] Let $n\geq 1$ and*

$$\mathrm{Mer}_n := \left\{\begin{pmatrix} a & b\\ c & d\end{pmatrix}\in \mathrm{M}_2(\mathbb{Z}) :\ ad-bc=n,\ a>b\geq 0,\ d>c\geq 0\right\},$$

finite ($n = ad-bc \geq a+d-1$). For every left coset $\eta\Gamma$ of determinant-n matrices,

$$\sum_{\epsilon\in\mathrm{Mer}_n\cap\eta\Gamma}\epsilon\cdot D_0 \;=\; D_0 \qquad \text{in } \mathrm{Div}^0,$$

[formally verified: [GT, `lem:merelpath-model`]] (the set, its finiteness and the path condition for every $\eta\Gamma$; not the bound that follows) and $|\mathrm{Mer}_n\cap\eta\Gamma| \leq x_0(\eta\mathbb{Z}^2)$, $x_0(L) := \min\{x>0 : (x,0)\in L\}$.

Proof. $\epsilon\Gamma = \eta\Gamma$ iff $\epsilon\mathbb{Z}^2 = \eta\mathbb{Z}^2 =: L$ (column lattices), so $B_L := \mathrm{Mer}_n\cap\eta\Gamma$ is the set of positively oriented bases $(v_1\,|\,v_2)$ of L, $v_1 = (a,c)^\top$, $v_2 = (b,d)^\top$, with $a>b\geq 0$, $d>c\geq 0$; and $\epsilon\cdot D_0 = [v_1]-[v_2]$, $[v]$ the cusp of v ("tail" $[v_1]$, "head" $[v_2]$). Both vectors lie in the closed first quadrant, and $\det(v_1|v_2)>0$ makes the argument strictly increase from v_1 to v_2. (i) *The tail determines ϵ*: v_1 is the primitive vector of L on its ray; $v_2'-v_2\in L\cap\mathbb{Q}v_1 = \mathbb{Z}v_1$, and $0\leq b,b'<a$ forces $v_2' = v_2$. Likewise the head, by $0\leq c<d$. (ii) *Successor*: if $b>0$, then $(v_2\,|\,mv_2-v_1)\in B_L$ for $m := \lceil a/b\rceil \geq 2$: the determinant is n, the new entries are $a' = b > b' = mb-a\in[0,b)$ and $c' = d < d' = md-c$ ($m\geq 2$, $d>c$). (iii) *Predecessor*: if $c>0$, then $(mv_1-v_2\,|\,v_1)\in B_L$ for

$m := \lceil d/c \rceil \geq 2$: $a'' = ma - b \geq 2a - b > a$, $c'' = mc - d \in [0, c)$, $d'' = c$. (iv) *Start*: exactly one $\epsilon \in B_L$ has tail $[\infty]$, namely $v_1 = (x_0, 0)$, $v_2 = (b, n/x_0)$ with $b \in [0, x_0)$ the unique such that $v_2 \in L$ ($L/(L \cap \mathbb{Z} \times 0) \cong y_1\mathbb{Z}$, $x_0 y_1 = [\mathbb{Z}^2 : L] = n$). The head is $[0]$ iff $b = 0$. By (i) the successor of (ii) is the only element whose tail is the given head, and the predecessor of (iii) the only element whose head is the given tail; arguments increase, B_L is finite, so iterating the predecessor from any ϵ ends at the start of (iv), and iterating the successor from the start ends at an element with head $[0]$. So B_L is one chain $\epsilon_0, \ldots, \epsilon_r$ with $\mathrm{head}(\epsilon_k) = \mathrm{tail}(\epsilon_{k+1})$, from $[\infty]$ to $[0]$, and the sum telescopes to D_0. The first coordinates of the tails strictly decrease from x_0 ($a > b$), so $r + 1 \leq x_0$. □

Proposition 28.9 (The Hecke action on the cokernel presentation: Merel's matrices in this document's conventions)**.** *[proved here] Let R be a commutative ring with $6 \in R^\times$, $n \geq 0$ even, $N \geq 1$, p a prime, $A := \Gamma_0(N)$, $g := \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$. Define $H_{N,p} \colon M_N \to M_N$ by*

$$H_{N,p} := \sum_{\epsilon \in \mathrm{Mer}_p} \sum_{j \,:\, (c_j, d_j)\epsilon \text{ primitive mod } N} E_{j\cdot\epsilon,\, j} \otimes \rho_n\big(\mathrm{adj}\, h_j(\epsilon)\big), \qquad \mathrm{adj}\, h_j(\epsilon) = w_{j\cdot\epsilon}\,(\mathrm{adj}\,\epsilon)\,w_j^{-1}, \tag{71}$$

i.e. $(H_{N,p}x)_i = \sum \rho_n(\mathrm{adj}\, h_j(\epsilon))\, x_j$ over the pairs (ϵ, j) with $j \cdot \epsilon = i$. For $p \nmid N$ every j is admissible; for $p \mid N$ exactly those with $(c_j, d_j)\epsilon \not\equiv 0 \pmod p$. Every $\mathrm{adj}\, h_j(\epsilon)$ is integral of determinant p. [formally verified: [GT, `prop:heilbronn-model`]] (the definition, as an executable specification over $\mathbb{Z}/q$, and the integrality and determinant of every $\mathrm{adj}\, h_j(\epsilon)$, with lower left entry $\equiv 0 \pmod N$)) [formally verified: [GT, `chk:merel-n5`]] (oracle note: the C++ $H_{5,p}$ at $n = 2$ over $\mathbb{F}_{1000003}$, $p \in \{2, 5\}$, is this operator on its dumped table, kernel-checked)

1. *$H_{N,p}(\mathrm{Rel}_N) \subseteq \mathrm{Rel}_N$.*
2. *$\varkappa_N \circ [H_{N,p}] = [A\, g\, A] \circ \varkappa_N$ on M_N/Rel_N; $[A\, g\, A]$ is $T_p^{(N)}$ for $p \nmid N$ and $U_p^{(N)}$ for $p \mid N$ — **the operator that Proposition 27.2(4)(d) induces on** H^1 (its item (3), with Proposition 12.4(4) and Shapiro's lemma for the model), independent of τ and of every table.*
3. *At $N = 1$: T_p on $V_R/(V_R^S + V_R^U) = H^1(\Gamma; V_R)$ is $x \mapsto \sum_{\epsilon \in \mathrm{Mer}_p} \rho_n(\mathrm{adj}\,\epsilon)x$.*
4. *$H_{N,p}$ reads Definition 28.1's table, Mer_p and ρ_n of integral matrices; no τ-table, no coset search, and no matrix outside Γ is inverted (Obligation 20.3). [formally verified: [GT, `prop:heilbronn-model`]] (not (1)–(3))*

Proof. By Lemma 28.7(2),(6), $\Pi_A \varkappa_N = \mu_N$ and $\Pi_A[A\, g\, A] = [A\, g\, A]_* \Pi_A$, with Π_A an isomorphism and $\ker \mu_N = \mathrm{Rel}_N$. It therefore suffices to prove

$$(*) \qquad [A\, g\, A]_*\, \mu_N(x) = \mu_N(H_{N,p}x) \qquad (x \in M_N)\,:$$

then (1) follows ($\mu_N(H_{N,p}r) = [A\, g\, A]_* \mu_N(r) = 0$ for $r \in \mathrm{Rel}_N$), and (2) from $\Pi_A \varkappa_N(H_{N,p}x) = [A\, g\, A]_* \Pi_A \varkappa_N(x) = \Pi_A[A\, g\, A]\varkappa_N(x)$. (3) is $N = 1$, (4) is read off (71); integrality and the determinant by Lemma 28.2(4) and (10).

Step 1: the right cosets. $A \cap g^{-1}Ag = \{a \in A : p \mid a_{12}\}$, and $a \mapsto (a_{11} : a_{12}) \bmod p$ identifies its right cosets in A with a subset of $\mathbb{P}^1(\mathbb{F}_p)$ (h in it multiplies the top row by the unit h_{11}; proportional top rows give $a'a^{-1}$ with $(a'a^{-1})_{12} \equiv 0$). If $p \nmid N$ the image is all of $\mathbb{P}^1(\mathbb{F}_p)$ ($T^y \in A$ gives $(1 : y)$; $\begin{pmatrix} p & u \\ N & v \end{pmatrix} \in A$, $pv - uN = 1$, gives $(0 : 1)$), so $|A\backslash AgA| = p + 1$; if $p \mid N$ it is $\{(1 : y)\}$ ($a_{11}a_{22} \equiv 1 \bmod p$), so $A\backslash AgA$ is represented by $\beta_b := gT^b = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$, $0 \leq b < p$. *Claim*: for each j the right cosets $\Gamma\beta w_j$, $A\beta \in A\backslash AgA$, are pairwise distinct and are all the $p+1$ right cosets of Γ in Δ_p (Proposition 10.1(1)) if $p \nmid N$, all but $\Gamma \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} w_j$ if $p \mid N$. For $p \mid N$:

the $\Gamma\beta_b$ are distinct Hermite forms and the missing form is $\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$. For $p \nmid N$: every $\beta \in AgA$ is upper triangular modulo N with unit diagonal (A is, and $g \equiv \mathrm{diag}(1,p)$), so $\Gamma\beta = \Gamma\beta'$ gives $\gamma = \beta' \operatorname{adj}\beta/p \in \Gamma$ with $\gamma_{21} \equiv 0 \pmod N$, i.e. $A\beta' = A\beta$; $p+1$ distinct cosets exhaust $\Gamma\backslash\Delta_p$.

Step 2: Merel's decomposition. For $\eta \in \Delta_p$, apply Lemma 28.8 at the left coset $\operatorname{adj}(\eta)\Gamma$ and then η: every $\epsilon \in \mathrm{Mer}_p \cap \operatorname{adj}(\eta)\Gamma$ has $\eta\epsilon = p\gamma_\epsilon$ with $\gamma_\epsilon \in \Gamma$, and

$$\eta \cdot D_0 \;=\; \sum_{\epsilon \in \mathrm{Mer}_p \cap \operatorname{adj}(\eta)\Gamma} \gamma_\epsilon \cdot D_0.$$

Step 3: the computation. $[A\,g\,A]_* \mu_N(x) = \sum_j \sum_{A\beta} [\beta w_j D_0 \otimes \rho_n(\beta) x_j]_A$. Apply Step 2 to $\eta = \beta w_j$: $\gamma_\epsilon = p^{-1}\beta w_j \epsilon$. Write $\gamma_\epsilon = h w_i$, $h \in A$, $i = \mathrm{cls}_N(\gamma_\epsilon)$: $[\gamma_\epsilon D_0 \otimes \rho_n(\beta)x_j]_A = [w_i D_0 \otimes \rho_n(w_i\gamma_\epsilon^{-1}\beta)x_j]_A$ and $w_i\gamma_\epsilon^{-1}\beta = w_i(\operatorname{adj}\epsilon)w_j^{-1}$. The index: for $p \mid N$ and $\beta = \beta_b$, $\mathrm{row}_2(\gamma_\epsilon) = \mathrm{row}_2(w_j)\epsilon$ exactly; for $p \nmid N$, $\mathrm{row}_2(\beta) \equiv (0,u) \pmod N$ with u a unit and $\mathrm{row}_2(\gamma_\epsilon) \equiv (u/p)\,\mathrm{row}_2(w_j)\epsilon$; either way $i = j \cdot \epsilon$, and the term is $[w_{j\cdot\epsilon} D_0 \otimes \rho_n(\operatorname{adj} h_j(\epsilon)) x_j]_A$. The result does not depend on the representative β of $A\beta$ (γ_ϵ moves by A on the left). Finally, for fixed j, the left cosets $\operatorname{adj}(\beta w_j)\Gamma = \operatorname{adj}(\Gamma\beta w_j)$ are by Step 1 pairwise distinct and are all left cosets of Δ_p ($p \nmid N$), respectively all but $\operatorname{adj}(\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} w_j)\Gamma = w_j^{-1} g \Gamma$; and $\epsilon \in w_j^{-1} g\Gamma$ iff $w_j\epsilon \in g\Gamma = \{\delta \in \Delta_p : \mathrm{row}_2(\delta) \equiv 0 \bmod p\}$ iff $(c_j, d_j)\epsilon \equiv 0 \pmod p$ — exactly the inadmissible pairs, as $\det\epsilon = p$ is a unit modulo the other primes of N. Every $\epsilon \in \mathrm{Mer}_p$ lies in one left coset, so the double sum is over the admissible pairs, which is $\mu_N(H_{N,p}x)$. □

Remark 28.10 (The transcription, and the trap in it). [proved here] [verified computationally][a] (1) **Against Merel's and Stein's conventions** (pointers of Remark 27.7, not verified): a Manin symbol $[P,(c:d)]$ with the right action $P|\delta$ is, here, block j with the class $(c:d)$ and a value in V_R; the index moves by the *same* row action $j \mapsto j\cdot\epsilon$, and the value by $\rho_n(\operatorname{adj} h_j(\epsilon))$ — the column convention's form of a right action, $\rho_n(\operatorname{adj}\cdot)$ being an anti-homomorphism. Terms with $j\cdot\epsilon$ not primitive are dropped, which is Merel's rule at $p \mid N$. (2) **The trap**: what Step 2 of the proof uses is the *right-coset* condition for $\operatorname{adj}(\mathrm{Mer}_p)$, which Lemma 28.8 gives through adj; Mer_p itself does *not* satisfy it — at $p=2$ the right coset $\Gamma\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ contains no element of $\mathrm{Mer}_2 = \{\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}\}$. Two plausible mis-transcriptions — (a) $\rho_n(h_j(\epsilon))$ in place of $\rho_n(\operatorname{adj} h_j(\epsilon))$; (b) Mer_p itself as the right-coset set, i.e. index $i = j\cdot\operatorname{adj}\epsilon$ and value $\rho_n(w_i\epsilon w_j^{-1})$ — do not even preserve Rel_N at $(N,n,p) = (1,10,2), (5,2,2), (5,2,5), (6,2,5)$ ([verified computationally][a], slot-14 WP-4's `docs/implementation/slot14/wp4/variants.py`); at weight 2, where ρ_n is trivial, (a) coincides with (71) and only (b) is caught, which is why (LN3) tests weights above 2. (3) **Two routes, one operator.** Proposition 27.2(4)(d) is a cochain map on the fixed model; $H_{N,p}$ is a map on M_N that preserves Rel_N but is not the restriction of a cochain map of the model. They agree on H^1, under the canonical identifications, and are compared there — by characteristic polynomials, or through any Γ-chain map between $C_\bullet(K^u)$ and $C_\bullet(\mathcal{Y})$ over the identity. (4) **Composite** n: Lemma 28.8 holds for every n, but the double-coset bookkeeping of Step 1 is done for primes only; T_{p^r} and T_m follow from T_p, U_p by Theorem 10.8(1) and its recursion.

28.4 The cost of the route

Remark 28.11 (The Heilbronn route's cost; scalability in N, k and ℓ^l). [verified computationally][a] A model of operation counts, in the notation of Remark 27.6 ($d = n+1$, $D_N = m_N d$); not an assertion about running times.

1. **Size of** Mer_p**.** Proved: $|\mathrm{Mer}_p| \le \sum_L x_0(L) = p^2+1$ (Lemma 28.8, over the $p+1$ lattices of index p: $x_0 = 1$ for $\{y \equiv 0\ (p)\}$ and $x_0 = p$ for the p lattices $\{x \equiv by\ (p)\}$). Counted

(slot-14 WP-4's `docs/implementation/slot14/wp4/merel_count.py`): $4, 7, 15, 25, 49, 63$ at $p = 2, 3, 5, 7, 11, 13$; 1097, 13045, 48653, 259245 at $p =$ 101, 691, 2003, 8009 — 0.40–$0.51\, p \log^2 p$ on $101 \le p \le 8009$, the ratio decreasing, while $|\mathrm{Mer}_p| / (p \log p)$ *increases* (2.35 at 101 to 3.60 at 8009): Merel's set is not $O(p \log p)$ on this range. The smaller set of Heilbronn matrices built from continued fractions [standard: [Cre97, Ch. 2]; **not** verified, rule (A2)] ($O(p \log p)$ elements by the cited source) is not specified here; any finite set Y with $\sum_{\delta \in Y \cap \Gamma\eta} \mathrm{adj}(\delta) \cdot D_0 = D_0$ for every right coset $\Gamma\eta \subseteq \Delta_p$ can replace $\mathrm{adj}(\mathrm{Mer}_p)$ in the proof without change.

2. **Applying** $H_{N,p}$ to $x \in M_N$: per pair (ϵ, j) the index $j \cdot \epsilon$ ($O(\omega(N))$ operations modulo the Q_i), the 2×2 product $\mathrm{adj}\, h_j(\epsilon)$, the $d \times d$ matrix $\rho_n(\mathrm{adj}\, h_j(\epsilon))$ ($O(d^2)$ by `ComputeSymmetricPowerMatrixFM`) and one block times a vector ($O(d^2)$): $O(|\mathrm{Mer}_p|\, m_N\, d^2)$ ring operations, memory $O(D_N + d^2)$. **No** $D_N \times D_N$ **object**, no τ-table, no coset search.
3. **A Hecke matrix on** H^1 ($\dim H^1 \approx D_N/6$): on a Manin-symbol basis (one block, one basis vector of V_R) $H_{N,p}$ costs $O(|\mathrm{Mer}_p|\, d^2)$ and touches $\le |\mathrm{Mer}_p|$ blocks; expressing the image in the basis of M_N/Rel_N costs $O(|\mathrm{Mer}_p|\, d \cdot \dim H^1)$ through the normal-form table of Rel_N ($D_N \times \dim H^1$, the output of (4); memory $\Theta(D_N \cdot \dim H^1)$), or one sparse solve per column. In all $O(|\mathrm{Mer}_p| \cdot d \cdot (\dim H^1)^2)$ for the matrix, **quadratic in** m_N, against the dense route's $\Theta((N_q D_N)^2)$ projector per column (Remark 27.6(1),(5)). What obstacle (O1) loses for Hecke operators is the N_q^2 factor and the dense model projector, not the quadratic order (v1.14.0-pre.2, WP-5 referee F1).
4. **The elimination** (Remark 27.6(4)): the orbit-local quotient by $M^{\langle S \rangle}$, then one block-sparse elimination against $M^{\langle U \rangle}$ — about $D_N/2$ rows, $D_N/3$ columns, ≤ 3 blocks of size $d \times d$ per block column; worst case cubic in D_N, and **no better bound is asserted**. At $d = 1$ (weight 2) it is the classical Manin-symbol elimination, with ≤ 3 entries per relation. Nothing in Proposition 28.9 forces a dense $D_N \times D_N$ object, so the sparse route at $N \sim 10^4$, $k = 2$ is admitted: $m_N \sim 10^4$, $|\mathrm{Mer}_5| = 15$, about $1.5 \cdot 10^5$ block operations per application of $H_{N,5}$. The normal form of Rel_N is the one $D_N \times \dim H^1$ object of the route; (2)'s "no $D_N \times D_N$ object" is about one application (v1.14.0-pre.2, WP-5 referee F1).
5. **The three axes.** Large N at small k (the regime (R-N) of Remark 27.7): linear in m_N per operator application, quadratic per Hecke matrix on H^1 (3), the elimination as in (4). Large k: d^2 per term, d^3 once for the fixed spaces of $\rho_n(S)$, $\rho_n(U)$ (Remark 27.6(3)); large k and large N together stay (O4) (the fill of (4) with dense $d \times d$ blocks). $R = \mathbb{Z}/\ell^l$: every count times one ring operation (a word, or the digit route beyond 2^{64}), Mer_p independent of R, N, k; the orbit-wise averaging and the cokernel need $6 \in R^\times$ ((O6)).

[verified computationally][a] here means the counts of (1), from an out-of-tree script; every other item is a model.

28.5 The predictions

Remark 28.12 (What is checkable at level N: (LN1)–(LN6)). [proved here] [standard: [Shi71, Maz77, Mer96, WWE20]] [conjectural] [verified computationally][a] Committed at v1.14.0-pre.1, before slot-14 WP-12's code (rule 20), with the convention of Remark 26.6: a failing [proved here] item stops the package, a failing [conjectural] item is a finding. [verified computationally][a] evidence: slot-14 WP-4's scripts in `docs/implementation/slot14/wp4/` — `levelN.py`, `one_cell.py`, `n1_dims.py`, `eisenstein_scan.py` (tracked, read-only, not registered validations); their independent route is *not* the τ-route and *not* the Heilbronn route — $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ on inhomogeneous $\Gamma_0(N)$-cocycles, carried to Γ by the cochain-level Shapiro map and normalised as in Definition 28.6.

(LN1) [standard: [Shi71, Ch. 2, Ch. 8], the dimension formulas and Eichler–Shimura; chapter level, **not** verified] **Dimensions.** For n even and a field R with $6 \in R^\times$, the right-hand side of

Proposition 27.5(3) (which is $\dim H^1 - \dim H^0$, independent of the characteristic) equals $2\dim S_{n+2}(\Gamma_0(N)) + c(N) - 2\delta_{n,0}$, $c(N) = \sum_{t|N} |(\mathbb{Z}/\gcd(t, N/t))^\times|$ the number of cusps; over $\mathbb{Q}$, where $\dim H^0 = \delta_{n,0}$, this is $\dim H^1 = 2\dim S_{n+2} + c(N) - \delta_{n,0}$. [verified computationally][a]: equal for all $N < 200$ and even $n \le 20$ (`n1_dims.py`, 0 mismatches). The code's check: $\dim H^1 - \dim H^0$ computed by either route against the formula at composite N (over $\mathbb{F}_\ell$ both dimensions can exceed the characteristic-zero ones by the same amount).

(LN2) [proved here] (Proposition 27.5(1)) **Orbit-wise = dense**: the fixed spaces $M_N^{\langle S\rangle}$, $M_N^{\langle U\rangle}$ computed orbit by orbit equal the dense ones as subspaces, so both routes give the same Rel_N and the same cokernel, entrywise in the coordinates of Definition 28.3.

(LN3) [proved here] (Proposition 28.9) **Heilbronn = τ-route on H^1**: at every tested (N, n, p, R) — $p \nmid N$ and $p \mid N$, prime and composite N including mixed classes, $R = \mathbb{F}_\ell$ with $\ell \mid N$ and $\ell \nmid N$ — $H_{N,p}(\mathrm{Rel}_N) \subseteq \mathrm{Rel}_N$ entrywise, and the characteristic polynomial of $[H_{N,p}]$ on M_N/Rel_N equals that of Proposition 27.2(4)(d) on $H^1_{\mathrm{eq}}(K, \mathcal{R}; L_N)$. [verified computationally][a]: equal to the independent cocycle route on every basis vector, and $H_{N,p}(\mathrm{Rel}_N) \subseteq \mathrm{Rel}_N$, at the cells listed in `docs/implementation/slot14/WP-4_report.md` (among them $(N, n, p) = (1, 10, 2)$, $(11, 0, 2)$, $(11, 0, 3)$, $(11, 0, 11)$, $(5, 2, 2)$, $(5, 2, 5)$, $(6, 2, 2)$, $(6, 2, 3)$, $(6, 2, 5)$, $(35, 0, 2)$, $(35, 0, 5)$, $(35, 0, 7)$ over $\mathbb{F}_{1000003}$), with the classical eigenvalues where they are known: $(X+24)^2(X-2049)$ for T_2 at weight 12, level 1; $(X+2)^2(X-3)$, $(X+1)^2(X-4)$, $(X-1)^3$ for T_2, T_3, U_{11} at weight 2, level 11; $(X+4)^2(X-9)^2$ and $(X+5)^2(X-1)(X-125)$ for T_2, U_5 at weight 4, level 5; $(X+2)^2(X-1)^2(X-8)^2$ for U_2 at weight 4, level 6; $X^2(X-3)^3(X^2+X-4)^2$ for T_2 at weight 2, level 35; and, re-run by the slot-14 WP-5 referee (`docs/implementation/slot14/WP-5_referee.md` §3.2), $(13, 2, 2)$, $(15, 2, 5)$, $(4, 4, 2)$, $(20, 2, 5)$, $(22, 2, 11)$, $(8, 2, 2)$ over $\mathbb{F}_7$, and $(30, 0, 2)$. That independent route tests Proposition 28.9(2), $H_{N,p} = [A\,g\,A]$; the equality with the τ-route of Proposition 27.2(4)(d) on the model — this item's other half — is WP-12's (v1.14.0-pre.2, WP-5 referee F8). [verified computationally][a] (v1.14.2, WP-12): the τ-route half — the characteristic polynomial of Proposition 27.2(4)(d) on $H^1_{\mathrm{eq}}(K, \mathcal{R}; L_N)$ equals the independent cocycle route's on 20 cells (among them the mixed-class levels $6, 10, 12, 15, 21, 30, 35$ and $n = 2, 4$) and the classical values listed above; the Heilbronn half awaits $H_{N,p}$. [verified computationally][a] (v1.15.0-pre.2, slot-15 WP-7): the Heilbronn half — $H_{N,p}$ of (71) built (`Hecke/HeckeLevelNHeilbronn`), Rel_N preserved and its characteristic polynomials equal to the τ-route's, to the independent cocycle route's and to the classical values above, the two mis-transcriptions of Remark 28.10(2) caught; `HeckeLevelNHeilbronnValidation` (`standard`, 636 checks; `docs/implementation/slot15/WP-7_report.md`). (LN3) is tested on both halves.

(LN4) [standard: [Maz77, Ch. II]; chapter level, **not** verified] [conjectural] **Mazur's Eisenstein ideal, in cohomology.** Let N be prime, $p \ge 5$ prime, $p \mid \mathrm{num}\left(\frac{N-1}{12}\right)$, and $e := \mathrm{rank}_{\mathbb{Z}_p} \mathbb{T}^0_{\mathfrak{m}}$ the rank of Mazur's cuspidal Hecke algebra completed at the Eisenstein maximal ideal $\mathfrak{m}$ (it is ≥ 1 by the cited source). Prediction: the joint generalised eigenspace $W_{N,p} \subseteq H^1(\Gamma_0(N); \mathbb{F}_p)$ of $(T_q)_{q \ne N}$ at the Eisenstein eigenvalues $1 + q$ has $\dim W_{N,p} = 2e + 1 \ge 3$. (The evidence of (LN6) computes $W_{N,p}$ jointly only for $q \le 7$, $q \ne N$; that space contains this $W_{N,p}$, and their equality is part of the prediction; v1.14.0-pre.2, WP-5 referee F2.) [conjectural] because the chain from Mazur's statements about $J_0(N)$ and its Hecke algebra to this document's $H^1(\Gamma_0(N); \mathbb{F}_p)$ is not carried out here.

(LN5) [standard: [Mer96, Théorème 2, p. 72] (verified on the GDZ page images, slot-17 R17-4, v1.17.0-pre.5; in Merel's notation the prime level is p and the residue characteristic q), restated as [WWE20, Thm. 1.5.1, p. 5] (verified on the arXiv text, slot-16 WP-8)] [conjectural] **Merel's criterion**: under Merel's hypotheses — $p > 3$ (here $p \ge 5$) and $p \mid (N-1)/\gcd(N-1, 12)$, which for $p \ge 5$ is (LN4)'s $p \mid \mathrm{num}\left(\frac{N-1}{12}\right)$ — $e \ge 2$ iff $\prod_{i=1}^{(N-1)/2} i^i$ is a p-th power in $(\mathbb{Z}/N)^\times$; so, with (LN4), $\dim W_{N,p} \ge 5$ exactly at those cells. The criterion is evaluated

as $\sum_{i\le(N-1)/2} i\log i \equiv 0 \pmod p$, log the discrete logarithm to any primitive root modulo N.

(LN6) [standard: [WWE20, p. 8 and Cor. 7.2.2, p. 30] (verified on the arXiv text, $p>3$, slot-16 WP-8)] [conjectural] **Mod-p cohomological Loewy length $e+1$.** The Hecke algebra acting on $W_{N,p}$ has Loewy length $e+1$ — the cuspidal part contributing e (by the cited source $\mathbb{T}^0_{\mathfrak{m}}/p \cong \mathbb{F}_p[x]/(x^e)$) and the Eisenstein class one more step. This is route R3: level N gives mod-p Loewy length ≥ 3 in cohomology, which level 1 shows only on classes that are not congruence-free, the classes [E] [E, `pr:latticesplit`] (R) excludes: on cohomology at Kummer classes ([E, `pr:kfonset`]; (608, 547)), and on the forms side also at irreducible congruence classes (slot-17 R17-2 §7) (v1.17.0-pre.3, slot-17 referee F1's m-C3: through v1.17.0-pre.2 this read "which the census at level 1 never shows", false since slot 15's (258, 37); final wording v1.17.0-pre.4, after the owner's ruling on the referee's B-1 restated (R) on congruence-free classes). [verified computationally][a] (`eisenstein_scan.py`; Loewy length by the radical series): $W_{N,p}$ is computed as the joint generalised eigenspace of T_2, T_3, T_5, T_7 ($q \ne N$) only, which contains (LN4)'s $W_{N,p}$ — equality is part of the prediction — and e is not computed independently but read off $\dim W_{N,p}$ (v1.14.0-pre.2, WP-5 referee F2). On all 168 cells with $N \le 2000$, $p \in \{5,7,11,13\}$, $\dim W_{N,p} \in \{3,5,7\}$, $\dim W_{N,p} \ge 5$ exactly at the Merel cells, and the Loewy length is $(\dim W_{N,p}+1)/2$ throughout: $\dim W_{N,p}=3$ at 137 cells, 5 at 28, 7 at 3 — under (LN4) $e=1$, 2, 3; the three $\dim W_{N,p}=7$ cells are $(N,p)=(181,5)$, $(1321,11)$, $(1571,5)$, **Loewy length** 4. [verified computationally][a] (v1.14.2, slot-14 WP-12, `HeckeLevelNValidation` arm (R3)): a second, independent route — the τ-route on the fixed model, $H^1(\Gamma_0(N);\mathbb{F}_p)$ by (68) with Definition 28.1's table, against the Manin-symbol cokernel of `levelN.py` — reproduces $\dim H^1$, $\dim W_{N,p}$, the Loewy length and Merel's criterion (recomputed in C++) on all 26 cells with $N \le 211$, and gives $\dim W_{N,p}=7$, radical layers $7,5,3,1$, Loewy length 4 at $(181,5)$, $(1321,11)$, $(1571,5)$; on every cell the layers are $2e+1, 2e-1, \dots, 1$. Both routes compute $W_{N,p}$ jointly for $q \le 7$, $q \ne N$, only, and read e off $\dim W_{N,p}$ (the WP-5 referee F2 caveat stands); the Loewy length is that of the subalgebra generated by those T_q — a single q can under-generate (at $(151,5)$ the nilpotency index of T_2-3 is 1 against Loewy length 2), so the joint $q \le 7$ is the least admissible generating set. The scan to $N \le 10^4$ (Merel's criterion only) has 733 cells, 118 of them Merel cells (78, 22, 11, 7 at $p=5,7,11,13$). **Named for WP-12**: $(181,5)$ ($m_N=182$, weight 2: $D_N=182$, within the dense route's reach), then $(1321,11)$ and $(1571,5)$. *Addendum* (v1.21.0-pre.4; the extension of (LN4) and (LN6) to $p \in \{2,3\}$, $n=0$, on the curve object of Proposition 28.18, Remark 28.19; [conjectural] as here): the same F2 caveat applies there — the code reads the joint generalised eigenspace $W_{\mathcal{F}}$ of a finite family $\mathcal{F}$ of T_q ($q \le 13$, $q \ne N$, by default), which contains $W_{N,p}$, and a radical series at most $e+1$ long; equality of both is part of the prediction, and a family can fail it by generation (no Mazur good prime in $\mathcal{F}$) or by separation (Remark 28.19(1),(2)).

28.6 Degeneracy maps on the presentation, the model-to-presentation map, and higher weight at scale

This subsection does three things. It reads $\varkappa_N$ of Definition 28.6 on the fixed model, so that the model's cocycles and the presentation can be compared *entrywise* (Proposition 28.13). It puts the degeneracy maps $\alpha_d = [\Gamma_0(M)\begin{pmatrix}1&0\\0&d\end{pmatrix}\Gamma_0(N)]$, $d>1$, on the cokernel presentation by Merel's set of determinant d, proved equal to Proposition 27.2(4)(c) on H^1 (Proposition 28.14). And it records the reach of the chapter in N and k together (Remark 28.16) with the checks (LN7)–(LN9) (Remark 28.17). κ_N^{mod}, e_{46}, $D_{M,N,d}$, $\mathrm{R3}_k$ are new symbols of this subsection and mean nothing else (rule (A5)); $D_{M,N,d}$ is neither the matrix D of Remark 27.10(3) nor D_N of Remark 27.6.

Proposition 28.13 (The model-to-presentation map: $\varkappa_N$ read on the fixed model)**.** *[proved here] Assume (H4). Let R be a commutative ring with $6 \in R^\times$, $N \geq 1$, and let M be the fibre of an induced matrix local system L on $(K, \mathcal{R})$ in the sense of Lemma 29.5 — $M = M_N$ with n even (Definition 28.3), or $M = M_{N,\chi}$ with $\chi(-1) = (-1)^n$ (Definition 29.7) — so that $-I$ acts trivially on M, and write* $\mathrm{Rel} := M^{\langle S\rangle} + M^{\langle U\rangle}$ *(Rel_N of Definition 28.6 for $M = M_N$). Let $e_{46} := \{v_4, v_6\}$ be the stored edge of Definition 7.1, so $\partial\widetilde{e_{46}} = \widetilde{v_6} - \widetilde{v_4}$, and put*

$$\kappa_N^{\mathrm{mod}} \colon C^1_{\mathrm{eq}}(K, \mathcal{R}; L) \to M, \quad f \mapsto -f(e_{46}); \qquad \kappa_N^{\mathrm{mod},0} \colon C^0_{\mathrm{eq}}(K, \mathcal{R}; L) \to M, \quad g \mapsto g(v_4).$$

On a Γ-equivariant refinement of X (Remark 17.5(1)), $\widetilde{e_{46}}$ is replaced by the 1-chain $\mathrm{path}(\widetilde{v_4} \to \widetilde{v_6})$ of the refined edges and $\kappa_N^{\mathrm{mod}}(f) := -f(\mathrm{path}(\widetilde{v_4} \to \widetilde{v_6}))$; every item holds verbatim, $F_1(\mathfrak{e}) := -\mathrm{path}(\widetilde{v_4} \to \widetilde{v_6})$ being a Γ-chain map by the same computation (v1.16.0-pre.3, WP-14 referee C-2).

1. ***(It computes the cokernel.)*** *κ_N^{mod} maps B^1_{eq} into* Rel *and induces an isomorphism $H^1_{\mathrm{eq}}(K, \mathcal{R}; L) \to M/\mathrm{Rel}$; $\kappa_N^{\mathrm{mod},0}$ induces an isomorphism $H^0_{\mathrm{eq}}(K, \mathcal{R}; L) \to M^{\langle S\rangle} \cap M^{\langle U\rangle}$.*
2. ***(It is the pinned isomorphism.)*** *For $M = M_N$, $\varkappa_N \circ [\kappa_N^{\mathrm{mod}}]$ is the identification $H^1_{\mathrm{eq}}(K, \mathcal{R}; L_N) \cong H^1(\Gamma_0(N); V_R)$ of Proposition 12.4(4); likewise with χ (Proposition 29.11(3)). So $[\kappa_N^{\mathrm{mod}}]$ is the inverse of Definition 28.6's isomorphism read on the model, and* ***no choice*** *enters beyond the binding table of Definition 28.1 and the stored orientation of e_{46}.*
3. ***(Naturality: fibre maps.)*** *Let $\phi \colon M \to M'$ be a map of $R[\Gamma]$-modules between two such fibres. Then $\phi(\mathrm{Rel}) \subseteq \mathrm{Rel}'$, the induced map $M/\mathrm{Rel} \to M'/\mathrm{Rel}'$ is $H^1(\Gamma; \phi)$ under the isomorphisms of (1), and $\kappa_N^{\mathrm{mod}}(\phi \circ f) = \phi\big(\kappa_N^{\mathrm{mod}}(f)\big)$* exactly*, with no relation used. In particular the fibre maps α^0 and ω of Proposition 27.2(4)(b) and the diamonds $\langle d\rangle$ of Proposition 29.8(1) act on the presentation by their own block matrices. (This is the sentence v1.15.0-pre.3 added to this chapter's scope paragraph, now a numbered statement.)*
4. ***(Entrywise comparison of operators.)*** *For every cocycle $f \in Z^1_{\mathrm{eq}}(K, \mathcal{R}; L_N)$, every prime p and $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$,*

$$\big(\widetilde{[A\, g\, A]}\, f\big)(e_{46}) - H_{N,p}\, f(e_{46}) \in \mathrm{Rel}_N \qquad (A = \Gamma_0(N)),$$

with (68) on the left and (71) on the right; for $dM \mid N$ and $f \in Z^1_{\mathrm{eq}}(K, \mathcal{R}; L_M)$, $\big(\widetilde{[\Gamma_0(M) \begin{pmatrix} 1 & 0 \\ 0 & d \end{pmatrix} \Gamma_0(N)]}\, f\big)(e_{46}) - D_{M,N,d}\, f(e_{46}) \in \mathrm{Rel}_N$ with (72); and with a character, the same with $M_{N,\chi}$, (74) and $H_{N,p,\chi}$ of Remark 29.12.

Proof. *A Γ-chain map from the tree to the model.* Define $F_0 \colon C_0(\mathcal{Y}) \to C_0(X)$ by $F_0(\gamma\mathfrak{v}_S) := \gamma\widetilde{v_6}$, $F_0(\gamma\mathfrak{v}_U) := \gamma\widetilde{v_4}$, and $F_1 \colon C_1(\mathcal{Y}) \to C_1(X)$ by $F_1(\gamma\mathfrak{e}) := -\gamma\widetilde{e_{46}}$. They are well defined because $\langle S\rangle = \mathrm{Stab}_\Gamma(\widetilde{v_6})$, $\langle U\rangle = \langle ST\rangle = \mathrm{Stab}_\Gamma(\widetilde{v_4})$ (Lemma 7.3(1)) and $-I$ fixes $\widetilde{e_{46}}$ with its orientation (it acts trivially on X, (H4)(iv)); they are Γ-linear by construction, and $\partial F_1(\mathfrak{e}) = -(\widetilde{v_6} - \widetilde{v_4}) = F_0(\mathfrak{v}_U - \mathfrak{v}_S) = F_0(\partial\mathfrak{e})$, so F is a chain map, and it preserves the augmentation (vertices to vertices). For $6 \in R^\times$ both $C_\bullet(\mathcal{Y}) \otimes R$ (Lemma 28.4) and $C_\bullet(X) \otimes R$ (the proof of Proposition 27.2(3): the stabiliser orders divide 12) are projective resolutions of R over $R[\Gamma]$, and over $R[A]$ for every finite-index $A \leq \Gamma$; so F is a comparison map and $F^* = \mathrm{Hom}_\Gamma(F, M)$ induces **the** comparison isomorphism on cohomology, for every $R[\Gamma]$-module M [standard: [Wei94, Thm. 2.2.6]]. *(1)* By Theorem 5.10(1) a cochain f is ϕ_f with $\phi_f(\widetilde{\sigma}) = f(\sigma)$, so $F^*\phi_f(\mathfrak{e}) = -f(e_{46})$ and, in degree 0, $F^*\phi_g = (g(v_6), g(v_4)) \in M^{\langle S\rangle} \oplus M^{\langle U\rangle}$. In the tree complex $\mathrm{Hom}_\Gamma(C_1(\mathcal{Y}), M) = M^{\langle -I\rangle} = M$ ($\phi \mapsto \phi(\mathfrak{e})$) and the coboundary is $(a, b) \mapsto b - a$ (the proof of Definition 28.6's claims; Proposition 29.11(3) with a character), so $H^1 = M/\mathrm{Rel}$ and $H^0 = \{(a, b) : a = b\} \cong M^{\langle S\rangle} \cap M^{\langle U\rangle}$ via b; F^* is a quasi-isomorphism. Directly: $(dg)(e_{46}) = g(v_6) - g(v_4)$, so $\kappa_N^{\mathrm{mod}}(dg) = g(v_4) - g(v_6) \in M^{\langle U\rangle} + M^{\langle S\rangle}$ ($g(v_6)$ is fixed by

S, $g(v_4)$ by ST: the stabiliser rows of Definition 7.2). *(2)* The reciprocity $\phi \mapsto \phi(\cdot)_1$ commutes with precomposition by any Γ-chain map (the proof of Proposition 27.2(3)). So $\varkappa_N \circ [\kappa_N^{\mathrm{mod}}]$ is the map $\mathrm{Hom}_A(F, V_R)$ on A-cochains, $A = \Gamma_0(N)$, under the identifications of both sides with $H^1(A; V_R)$ by comparison with a fixed projective resolution; F restricted to A is a chain map over id_R between projective $R[A]$-resolutions, so it induces the identity of $H^1(A; V_R)$ under those identifications — which are the ones Proposition 12.4(4) and Definition 28.6 use (Theorem 5.10, Remark 7.6, Shapiro's lemma, Lemma 28.4): the identification of Proposition 12.4(4) is comparison of the projective $R[A]$-resolution $C_\bullet(X) \otimes R$ with any projective resolution, then Shapiro, so "no choice enters" is checkable from that item alone (v1.16.0-pre.3, WP-14 referee C-3). With χ the same, with $\rho_{n,\chi}$ (Lemma 29.5). *(3)* ϕ commutes with $M(S)$ and $M(U)$, so it maps $M^{\langle S\rangle}$, $M^{\langle U\rangle}$ into $M'^{\langle S\rangle}$, $M'^{\langle U\rangle}$; the tree complex is natural in the module, and F^* commutes with ϕ_*. Exactness: $(\phi \circ f)(e_{46}) = \phi(f(e_{46}))$. A morphism of matrix local systems is one fibre matrix at every simplex (Theorem 4.5), which is what (4)(b) of Proposition 27.2 and (1) of Proposition 29.8 provide. *(4)* By Proposition 27.2(3) the model operator induces $[A\,g\,B]$ under the identification of Proposition 12.4(4); by Proposition 28.9(2), Proposition 28.14(2) and Remark 29.12 the presentation operators induce the same map under $\varkappa$; by (2) the two identifications differ by $[\kappa_N^{\mathrm{mod}}]$. So both sides of each display have the same class in M/Rel for every cocycle f. □

Proposition 28.14 (The degeneracy maps α_d on the cokernel presentation)**.** *[proved here] Let R be a commutative ring with $6 \in R^\times$, n even, $M, N, d \geq 1$ with $dM \mid N$, $A := \Gamma_0(M)$, $B := \Gamma_0(N)$, $g := \begin{pmatrix} 1 & 0 \\ 0 & d \end{pmatrix}$; let w'_j $(1 \leq j \leq m_M)$ and w_i $(1 \leq i \leq m_N)$ be the tables of Definition 28.1 at levels M and N, $\mathrm{Mer}_1 := \{I\}$, and for $\epsilon \in \mathrm{Mer}_d$ (Lemma 28.8) call (i, ϵ)* admissible *if $\mathrm{row}_2(w_i)\ \mathrm{adj}\,\epsilon \equiv 0 \pmod d$, and then put $j(i,\epsilon) := \mathrm{cls}_M\big(\mathrm{row}_2(w_i)\ \mathrm{adj}\,\epsilon/d\big)$. Define $D_{M,N,d}\colon M_M \to M_N$ by*

$$D_{M,N,d} := \sum_{\epsilon \in \mathrm{Mer}_d} \sum_{i\,:\,(i,\epsilon)\ admissible} E_{i,\,j(i,\epsilon)} \otimes \rho_n\big(w_i\,(\mathrm{adj}\,\epsilon)\,w'^{-1}_{j(i,\epsilon)}\big), \tag{72}$$

i.e. $(D_{M,N,d}\,x)_i = \sum_\epsilon \rho_n(w_i(\mathrm{adj}\,\epsilon)w'^{-1}_{j(i,\epsilon)})\,x_{j(i,\epsilon)}$ over the ϵ admissible with i. Every $w_i(\mathrm{adj}\,\epsilon)w'^{-1}_{j(i,\epsilon)}$ equals $g\,a$ with $a \in A$, so it is integral of determinant d; $\mathrm{row}_2(w_i)\,\mathrm{adj}\,\epsilon/d$ is primitive, and admissibility and $j(i,\epsilon)$ depend only on the class i (the canonical pair (c_i, d_i) may replace $\mathrm{row}_2(w_i)$).

1. *$D_{M,N,d}(\mathrm{Rel}_M) \subseteq \mathrm{Rel}_N$.*
2. *$\varkappa_N \circ [D_{M,N,d}] = \alpha_d \circ \varkappa_M$ on M_M/Rel_M, where $\alpha_d := [A\,g\,B] = (\mathrm{conj}_g)_* \circ \mathrm{res}^A_{g^{-1}Bg}$ is* ***the operator Proposition 27.2(4)(c) induces on*** *H^1, independent of τ and of every table.*
3. ***($d = 1$.)*** *$D_{M,N,1}$ is, entry for entry, the fibre map α^0 of Proposition 27.2(4)(b) for $B \subseteq A$ ($\epsilon = I$, every i admissible, $j(i, I) = \mathrm{cls}_M(w_i)$).*
4. ***(Composite d.)*** *For $d = d_1 d_2$, $[D_{M,N,d}] = [D_{Md_1,N,d_2}] \circ [D_{M,Md_1,d_1}]$ on M_M/Rel_M (by (2) and the factorisation of Proposition 27.2(4)(c)); either side may be used, and they differ at cochain level by an element of Rel_N.*
5. ***(The trace.)*** *If $N = dM$, then $[\omega_{B/A}] \circ [D_{M,N,d}] = [A\,g\,A]$ on M_M/Rel_M, $\omega_{B/A}$ the trace of Proposition 27.2(4)(b) acting by its block matrix (Proposition 28.13(3)): T_p for $d = p \nmid M$, U_p for $d = p \mid M$, and at $M = 1$, $d = p^2$, the primitive $T(1, p^2) = T_{p^2} - p^n = T_p^2 - p^n(p+1)$ (T_{p^2} the full operator of §A.1).*
6. ***(With a character.)*** *If $\chi\colon G_M \to R^\times$, $\chi(-1) = (-1)^n$, and $M_{N,\chi}$ is formed with χ composed with $G_N \to G_M$, then (1), (2) hold for $D_{M,N,d,\chi}$, (72) with each block multiplied by $\chi\big(a_{w_i(\mathrm{adj}\,\epsilon)w'^{-1}_j}\big)$, on $M_{M,\chi}/\mathrm{Rel} \to M_{N,\chi}/\mathrm{Rel}$.*
7. *$D_{M,N,d}$ reads only the two tables, Mer_d and ρ_n of integral matrices: no τ-table, no coset search, and no matrix outside Γ is inverted (Obligation 20.3); $|\mathrm{Mer}_d| \leq \sigma_2(d) = \sum_{a|d} a^2$*

(Lemma 28.8, summed over the lattices of index d, $x_0 = a$ for the a lattices with Hermite diagonal $(a, d/a)$); and at most $2m_N/m_d$ admissible i per ϵ (the admissible bottom rows modulo d lie in the kernel of $v \mapsto v \operatorname{adj}\epsilon$ on $(\mathbb{Z}/d)^2$, of order d, with at most two primitive projective points), so $D_{M,N,d}$ costs $O(|\mathrm{Mer}_d|\ m_M\, d_V^2)$ when the admissible classes are enumerated per ϵ rather than scanned (v1.16.0-pre.3, WP-14 referee C-5).

Proof. *The groups.* $g^{-1}Bg = \{\left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right) \in \Gamma : d \mid b,\ (N/d) \mid c\} \subseteq A$ (as $M \mid N/d$), and $B \subseteq gAg^{-1}$, so in Proposition 27.2(4)(c)'s notation $B'' = B$, $A' := A \cap g^{-1}Bg = g^{-1}Bg$, and $[A\,g\,B] = (\mathrm{conj}_g)_* \circ \mathrm{res}^A_{A'}$; $-I \in A'$, $gA'g^{-1} = B$. By Lemma 28.7(3),(5),

$$\Pi_B \circ [A\,g\,B] = [A\,g\,B]_* \circ \Pi_A, \qquad [A\,g\,B]_*[y]_A := \sum_{r \in A'\backslash A} [gr \cdot y]_B.$$

As in the proof of Proposition 28.9, (1) and (2) follow from $\Pi_A \varkappa_M = \mu_M$, $\Pi_B \varkappa_N = \mu_N$ (Lemma 28.7(2), $\ker\mu = \mathrm{Rel}$, Π an isomorphism) and

$$(*) \qquad [A\,g\,B]_*\,\mu_M(x) = \mu_N(D_{M,N,d}\,x) \qquad (x \in M_M).$$

Step 1. $[A\,g\,B]_*\mu_M(x) = \sum_j \sum_{r \in A'\backslash A} [grw'_j D_0 \otimes \rho_n(gr)x_j]_B$. As r runs over $A'\backslash A$ and j over the table, $u := rw'_j$ runs once over $A'\backslash\Gamma$, with $j = \mathrm{cls}_M(u)$ and $\rho_n(gr) = \rho_n(guw_j'^{-1})$; the summand $[guD_0 \otimes \rho_n(guw_j'^{-1})x_j]_B$ depends only on $A'u$ ($ga'g^{-1} \in B$ for $a' \in A'$). Put $\eta := gu \in g\Gamma$; $A'u \mapsto B\eta$ is a bijection $A'\backslash\Gamma \to B\backslash g\Gamma$ ($Bg = gA'$). *Step 2 (Merel at determinant d).* As in Step 2 of the proof of Proposition 28.9, Lemma 28.8 at the left coset $\mathrm{adj}(\eta)\Gamma$ gives $\eta \cdot D_0 = \sum_{\epsilon \in \mathrm{Mer}_d \cap \mathrm{adj}(\eta)\Gamma} \gamma_\epsilon \cdot D_0$ with $\gamma_\epsilon := d^{-1}\eta\epsilon \in \Gamma$. Write $\gamma_\epsilon = hw_i$, $h \in B$, $i = \mathrm{cls}_N(\gamma_\epsilon)$: the term is $[w_iD_0 \otimes \rho_n(w_i\gamma_\epsilon^{-1}\eta\, w_j'^{-1})x_j]_B$, and $\gamma_\epsilon^{-1}\eta = d\,\epsilon^{-1} = \mathrm{adj}\,\epsilon$. *Step 3 (re-indexing).* The pairs $(B\eta, \epsilon)$ with $\eta \in g\Gamma$, $\epsilon \in \mathrm{Mer}_d \cap \mathrm{adj}(\eta)\Gamma$ correspond bijectively to the pairs $(B\gamma, \epsilon)$ with $\gamma \in \Gamma$, $\epsilon \in \mathrm{Mer}_d$, $\gamma\,\mathrm{adj}\,\epsilon \in g\Gamma$, by $\gamma = \gamma_\epsilon$, $\eta = \gamma\,\mathrm{adj}\,\epsilon$ ($\epsilon \in \mathrm{adj}(\gamma\,\mathrm{adj}\,\epsilon)\Gamma = \epsilon\gamma^{-1}\Gamma$ always; $b \in B$ multiplies η and γ on the left together). Taking $\gamma = w_i$: $w_i\,\mathrm{adj}\,\epsilon \in g\Gamma = \{\delta \in \mathrm{M}_2(\mathbb{Z}) : \det\delta = d,\ \mathrm{row}_2(\delta) \equiv 0\ (d)\}$ is admissibility, and then $u = g^{-1}w_i\,\mathrm{adj}\,\epsilon \in \Gamma$ has bottom row $\mathrm{row}_2(w_i)\,\mathrm{adj}\,\epsilon/d$, so $j = j(i,\epsilon)$ and $w_i(\mathrm{adj}\,\epsilon)w_j'^{-1} = g\,(uw_j'^{-1})$ with $uw_j'^{-1} \in A$. The double sum is therefore $\sum_{(i,\epsilon)\ \text{admissible}} [w_iD_0 \otimes \rho_n(w_i(\mathrm{adj}\,\epsilon)w_{j(i,\epsilon)}'^{-1})x_{j(i,\epsilon)}]_B = \mu_N(D_{M,N,d}x)$, which is $(*)$. *Dependence on the class i.* $\mathrm{row}_2(w_i) \equiv \lambda(c_i, d_i)$ (mod N) for a unit λ (Lemma 28.2(2)); $d \mid N$ and $M \mid N/d$. *(3)* $\mathrm{Mer}_1 = \{I\}$ (the only matrix with $ad - bc = 1$, $a > b \geq 0$, $d > c \geq 0$), and the block is $\rho_n(w_iw'^{-1}_{\mathrm{cls}_M(w_i)})$, which is Proposition 27.2(4)(b)'s $\alpha^0_{B/A}$ with $v_i = w_i$, $w_{a(i)} = w'_{\mathrm{cls}_M(w_i)}$. *(4)* is (2) for the three pairs and Proposition 27.2(4)(c)'s factorisation on cohomology. *(5)* $A \cap gAg^{-1} = \Gamma_0(dM) = B$ (the computation of Proposition 10.1(3)) and $A \cap g^{-1}Ag = g^{-1}Bg = A'$, so $[A\,g\,A] = \mathrm{cor}^A_B \circ (\mathrm{conj}_g)_* \circ \mathrm{res}^A_{A'} = [B\,1\,A] \circ [A\,g\,B]$; $[B\,1\,A] = \omega_{B/A}$ (Proposition 27.2(4)(b)), and Proposition 28.13(3) carries it to the presentation. The double coset $A\left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)A$ is T_p resp. U_p (Proposition 27.2(4)(d)); at $M = 1$, $\Gamma\left(\begin{smallmatrix} 1 & 0 \\ 0 & p^2 \end{smallmatrix}\right)\Gamma$ is the primitive double coset, $T_{p^2} = T(1,p^2) + T(p,p)$ with $T(p,p) = p^n$ (Proposition 10.5) and $T_{p^2} = T_p^2 - p^{n+1}$ (Theorem 10.8). *(6)* As in Remark 29.12: the proofs above use of ρ_n only multiplicativity on the matrices they evaluate — elements of A and B, gr with $r \in A$, and $w_i\gamma_\epsilon^{-1}\eta w_j'^{-1} = h^{-1}gr$ — all in $\Delta_0(M)$ ($a_g = 1$, $c_g = 0$), where $\rho_{n,\chi}$ is multiplicative (Lemma 29.2(2) at level M); $V_R(\chi)$ restricted to B is the level-N module for χ composed with $G_N \to G_M$; and Lemma 28.7(5) is used for $g \in \Delta_0(M)$ only. *(7)* is read off (72); the count is Lemma 28.8's bound summed over the column lattices of index d, which in column Hermite form $\left(\begin{smallmatrix} a & b \\ 0 & d/a \end{smallmatrix}\right)$, $0 \leq b < a$, are a in number for each $a \mid d$, each with $x_0 = a$. □

Remark 28.15 (The transcription of α_d, its traps, and its evidence). [proved here] [verified computationally][a] (1) **The shape.** $D_{M,N,d}$ has the shape of (71) read from the target: the source block j is computed from the target class i by the row action of $\operatorname{adj}\epsilon/d$, and the value is ρ_n of $w_i(\operatorname{adj}\epsilon)w_j'^{-1}$, of the form $g \cdot A$. At $d = p$ prime and $M = 1$, $N = p$ it is dg_p of Proposition 27.3 read on the presentation. (2) **The traps**, as in Remark 28.10(2): (a) ϵ in place of $\operatorname{adj}\epsilon$ (Merel's set itself as the right-coset set) and (b) the adjugate of the block in place of the block do *not* preserve Rel at $(M, N, d, n) = (1, 2, 2, 10)$, $(1, 6, 3, 4)$, $(2, 12, 3, 2)$, $(1, 4, 4, 6)$, and disagree with the independent route there (`docs/implementation/slot16/wp12/negctl.py`). (3) **Evidence** (`docs/implementation/slot16/wp12/degen.py`, over $\mathbb{F}_{1000003}$, importing slot-14 WP-4's `levelN.py`; the independent route is $(\mathrm{conj}_g)_* \circ \mathrm{res}$ on inhomogeneous $\Gamma_0(M)$-cocycles, carried to Γ by the cochain-level Shapiro map at level N and normalised as in Definition 28.6 — no Merel set, no presentation formula): (1), (2) and Hecke equivariance $H_{N,p}D = DH_{M,p}$ modulo Rel_N ($p \in \{5, 7\}$, $p \nmid N$) on 28 cells (M, N, d, n) — among them $d = 4, 6, 9$, mixed-class levels $N = 6, 12, 30, 35$, $M > 1$ with $\gcd(d, M) > 1$ ($(2, 4, 2)$, $(3, 9, 3)$, $(2, 8, 4)$), and $d = 1$ entrywise against α^0; (5) on 8 cells (four T_p, two at $M > 1$, two $T(1, p^2)$); (4) on 6 composites; and the old-space relations $U_p D_{M,Mp,1} = p\, D_{M,Mp,p}$ and $U_p D_{M,Mp,p} = D_{M,Mp,p} T_p - p^n D_{M,Mp,1}$ modulo Rel ($p \nmid M$; the analogue of Proposition 27.4(1),(3) with $\Gamma_0(M) \to \Gamma_0(Mp)$) on 7 cells. All held; the run of record is in `docs/implementation/slot16/WP-12_report.md` §3.

Remark 28.16 (Higher weight and composite N at scale: the reach of this chapter). A scope statement and a model of operation counts and memory, in the notation of Remarks 27.6 and 28.11 ($d_V := n + 1 = k - 1$, $D_N = m_N d_V$, $h := \dim H^1$); not an assertion about running times.

1. **Scope.** Every statement of this chapter — Definitions 28.1, 28.3, 28.6, Lemmas 28.4–28.8, Propositions 28.9, 28.13, 28.14 — holds for *every* $N \geq 1$ and every even $n \geq 0$ with $6 \in R^\times$. "$N \leq 10^4$ at weight 2" is the reach of slot 15's runs ((LN6)'s scan), not a hypothesis; higher weight $k > 2$ at composite N needs no new statement, only the costs below.
2. **Fixed spaces.** One $d_V \times d_V$ solve for each of S, U, then $O(d_V^2)$ per fixed point (Remark 27.6(3)); at most $2^{\omega(N)+1}$ fixed points.
3. **The cokernel at $k > 2$.** After the orbit-local quotient by $M^{\langle S \rangle}$ the elimination against $M^{\langle U \rangle}$ has ≤ 3 blocks of size $d_V \times d_V$ per block column (Remark 27.6(4)); a block operation costs $O(d_V^3)$. Worst case cubic in D_N, no better bound asserted; (O4) stands: at large k *and* large N the fill is dense $d_V \times d_V$ blocks.
4. **Memory of the normal form.** The normal-form table of Rel_N (Remark 28.11(3)) has $D_N \times h$ entries with $h \approx D_N/6$ ((LN1)): about $m_N^2(k-1)^2/6$ words, i.e. $\frac{2}{3}m_N^2(k-1)^2$ bytes in 32-bit words — $\approx$ 67 MB at $(m_N, k) = (10^4, 2)$, $\approx$ 0.6 GB at $(10^4, 4)$, $\approx$ 6.7 GB at $(10^4, 11)$ — an upper bound for a dense table; a sparse normal form stores less (the shipped `SparseCokernel` of `Hecke/HeckeLevelNSparse`: `final_nnz` entries plus the pivots, `HeckeLevelNSparseValidation`'s cost line — the figure WP-13's RSS prediction should use). With the rule-19 prediction this is the RSS model for slot-16 WP-13's ladder. *Caveat at composite N, $k > 2$* (v1.17.0-pre.1; slot-16 WP-13's finding F-3, `docs/implementation/slot16/WP-13_report.md` §4; a statement about the cost model only, no value depends on it): [verified computationally][a] the fitted fill law of `CokernelCostModel` (fitted on prime N) *underestimates* the measured `final_nnz` at composite N and $k > 2$, by a factor of 1.47 at $(N, k) = (30030, 4)$, 5.70 at $(4620, 6)$ and 6.80 at $(2310, 12)$ (WP-13's ladder rungs `30030:2`, `4620:4`, `2310:10` in its $N{:}n$ notation); the dense bound above stays an upper bound there (3.9 GB against 2.4 GB measured at $(2310, 12)$). A prediction at composite N, $k > 2$ therefore scales the measured `final_nnz` of the last rung run by $m_N^{1.335} d_V^2$ rather than reading the fitted law. *Sharpened at v1.17.0-pre.3* (slot-17 WP-5's finding W5-F1, `docs/implementation/slot17/WP-5_report.md` §3): that scaling, too, fails at composite N with a large prime factor — at $N = 99330 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 43$, $k = 2$, it predicted

42 M non-zeros and 1.4 GB, and the sparse presentation grew past 8 GB, where the run was stopped by its watchdog (rule 19) before it finished; this is an open measurement, not a result (no `final_nnz` was recorded), and it is recorded as a failed prediction. Along the primorial rungs 30030, 60060, 90090 the measured-to-fitted fill ratio rose (3.08, 3.79, 4.01). No value depends on it. *The structure-aware fill model* (v1.18.0-pre.1, contract first for slot-18 WP-5; **not built, not run** at this version; a statement about the cost model only). The fill of the sparse elimination is to be modelled through the structure the Shapiro presentation already has — the factorisation of N (the number $\omega(N)$ of prime factors and the largest one, $P(N)$), the orbit and block sizes of the permutation $\otimes\,\rho_n$ form (22) (blocks of size d_V permuted by the coset action), and $d_V = k - 1$ — rather than through m_N alone; fitted on the logged `SPARSE_CELL` and `DEGEN_CELL` rungs of slot-17 WP-5 and WP-6 *before* any new run (rule 20), and the same model replaces `CokernelCostModel`'s fitted law (slot-17 WP-6's finding F-1: final fill 4–5× the law at $m_N \approx 10^4$, $d(\chi) \geq 5$). It replaces the rule-19 RSS hook (`hook_rss/v1`) only if it predicts every logged rung within a factor 1.5 and from above; otherwise it is recorded as a finding and `hook_rss/v1` stays. The two open measurements, $N = 99330$ at $k = 2$ ($m_N = D_N = 304\,128$, 10 991 MiB predicted by `hook_rss/v1`) and $N = 90090$ at $k = 4$ ($m_N = 290\,304$, $D_N = 870\,912$, 9622 MiB predicted by `hook_rss/v1`), both admitted by `HECKE_LEVEL_N_HOOK_RSS_CAP_GIB=12` (v1.18.0-pre.2, the slot-18 referee F1's m-C1: pre.1 gave the figures of 90090 at $k = 4$ only, attached to both cells), are each predicted by the model first and then run once in an exclusive memory window; their `final_nnz` and peak RSS enter this item as [verified computationally][a] at the pre-release that records them, a missed prediction as a failed prediction. A change of the elimination order (column order by block, as the permutation structure suggests) is admissible only as a choice that changes no value: the presentation and every Hecke matrix of this chapter must stay bit-identical on every logged rung (the canonical form is the natural reduced row echelon form in the shipped column order), checked as a gate; if the blow-up is not ordering-driven, nothing is changed and the finding is recorded. *Outcome at v1.18.0-pre.2* (slot-18 WP-5, `docs/implementation/slot18/WP-5_report.md`; predictions committed first, `docs/implementation/slot18/WP-5_predictions.md`). [verified computationally][a] **The structure-aware model failed its rule**: fitted on the 57 logged cokernels of slots 15–17 ($\log\lambda$ against $\log m_N$, $\omega(N)$, $\log P(N)$ and $[d_V > 1]$, λ the fill per free column), it leaves an rms log-residual of 0.43 and a maximum of 0.81, beyond the factor 1.5; the fill is not even monotone in $P(N)$ (at $N = 2310\,P$, $k = 2$, a 3.8-fold spread over $P = 13, \ldots, 43$). So `hook_rss/v1` and `CokernelCostModel`'s fitted law **stay**; this is a finding, recorded as the rule above prescribes, and no cost statement of this item is replaced. *W5-F1's root cause is the order of the generators, not the fill*: `final_nnz` at $N = 99330$, $k = 2$, is 54 450 319 (computed by WP-5's scratch prototype, which reproduces nine logged `final_nnz` exactly; close to 90090's 50.7 M), and the slot-15–17 consumption order (all S-generators, then all ST-generators) made the back-substitutions explode at composite N with a large prime factor (> 16 min and `bad_alloc` at 5.8 GB in the prototype), while the generators sorted by their largest column, descending, eliminate in 5.7 s. The shipped `SparseCokernel` now consumes them in that order (rollback `HECKE_LEVEL_N_GEN_ORDER=as_built`); the choice changes no value, the stored form being the unique natural reduced row echelon form, and the gate held: presentation hash and `final_nnz` equal under both orders on all 21 logged rungs (the column order is *not* changed). At $d_V > 1$ with $P(N) = 13$ the sorted order costs up to +16 % peak RSS (60060 at $k = 4$); `hook_rss/v1` stays an upper bound on every rung. The two window measurements above are **pending** (deferred in slot 18 behind the LAT37 runs and the disk emergency of 2026-10-03); WP-5 predicted 54.45 M non-zeros and ≈ 1.9 GB at $(99330, k = 2)$, and 216 M (216–370 M) and ≈ 5.5 GB (5.5–9.6 GB) at $(90090, k = 4)$.

5. **Operators.** $H_{N,p}$ costs $O(|\mathrm{Mer}_p|\ m_N d_V^2)$ per application, $D_{M,N,d}$ $O(|\mathrm{Mer}_d|\ m_M d_V^2)$ with the admissible classes enumerated per ϵ (at most $2m_N/m_d$ per ϵ; scanning every i costs $O(|\mathrm{Mer}_d|\ m_N)$ checks, $\approx 4 \cdot 10^9$ at $M = 1$, $d = N \approx 10^4$) (Proposition 28.14(7)), and a Hecke matrix on H^1 $O(|\mathrm{Mer}_p|\ d_V\, h^2)$ (Remark 28.11(3)); every fibre map $O(m_N d_V^2)$. None forms a $D_N \times D_N$ object.
6. **The ring.** Over $\mathbb{Z}/\ell^l$, $\ell \geq 5$, every count times one ring operation, and the elimination by Howell forms (Remark 27.7, (O6)); with a character, over $\mathbb{F}_q$ or $W_K(\mathbb{F}_q)$ (Remark 29.13).

Remark 28.17 (What is checkable at level N, continued: (LN7)–(LN9)). [proved here] [standard: [AL70], [BM16]] [conjectural] [verified computationally][a] Committed at v1.16.0-pre.1, before slot-16 WP-13's code (rule 20), with the convention of Remark 26.6: a failing [proved here] item stops the package, a failing [conjectural] item is a finding.

(LN7) [proved here] (Proposition 28.13) **Model = presentation, entrywise.** At every tested (N, n, p, R), for a basis of $Z^1_{\mathrm{eq}}(K, \mathcal{R}; L_N)$ modulo B^1_{eq}: the vectors $-f(e_{46})$ span M_N/Rel_N and are independent there; $(\widetilde{[A\,g\,A]}f)(e_{46}) - H_{N,p}f(e_{46}) \in \mathrm{Rel}_N$ for $p \nmid N$ and $p \mid N$; α^0, ω (and $\langle d\rangle$ with a character) commute with $f \mapsto f(e_{46})$ *exactly*. This replaces (LN3)'s comparison of characteristic polynomials by a comparison of classes.

(LN8) [proved here] (Proposition 28.14) α_d **on the presentation.** $D_{M,N,d}(\mathrm{Rel}_M) \subseteq \mathrm{Rel}_N$; $(\widetilde{[A\,g\,B]}f)(e_{46}) - D_{M,N,d}f(e_{46}) \in \mathrm{Rel}_N$ for $f \in Z^1_{\mathrm{eq}}(K, \mathcal{R}; L_M)$ (the model's α_d, Proposition 27.2(4)(c), against the presentation's); $d = 1$ is α^0 entrywise; the trace (5) and the composite (4); among WP-13's cells at least one with $d = p \mid M$, where (5) gives U_p (proved, so far untested; WP-14 referee). [standard: [AL70]; not verified (rule (A2))] The old-space relations $U_p\alpha_1 = p\,\alpha_p$, $U_p\alpha_p = \alpha_p T_p - p^n\alpha_1$ at $M \to Mp$, $p \nmid M$ (the analogue of Proposition 27.4(1),(3) with $\Gamma_0(M) \to \Gamma_0(Mp)$; [standard: [AL70], not verified]; [verified computationally] in Remark 28.15(3)), and Hecke equivariance $T_p\alpha_d = \alpha_d T_p$ for $p \nmid N$. [verified computationally][a]: Remark 28.15(3).

(LN9) [standard: [BM16]; **not** verified (rule (A2))] [conjectural] R3$_k$**: the Eisenstein reading at weight** $k \geq 4$**.** For N prime, $p \geq 5$ prime, $p \nmid N$, $p > k+1$, $k = n+2 \geq 4$ even, let $W^{(k)}_{N,p} \subseteq H^1(\Gamma_0(N); \mathrm{Sym}^n\mathbb{F}_p^2)$ be the joint generalised eigenspace of $(T_q)_{q\leq 7,\ q\nmid Np}$ at $1 + q^{k-1}$ (the code's reading: $\dim W^{(k)}_{N,p}$, the Loewy length and radical layers of the algebra generated by those T_q on it, reported beside $v_p(N^k - 1)$, $v_p(N^{k-2} - 1)$ and $v_p(B_k/k)$). Prediction: $\dim W^{(k)}_{N,p} \geq 2$ (the two cusps), and $\dim W^{(k)}_{N,p} \geq 3$ *only if* p divides $(N^k - 1)(N^{k-2} - 1)\,\mathrm{num}(B_k/k)$. Reason (not a proof here; it uses that the mod-p joint generalised eigenspace counts characteristic-0 eigensystems with multiplicity, i.e. that $H^1(\Gamma_0(N); \mathrm{Sym}^{k-2}\mathbb{Z}_p^2)$ is torsion-free, which holds since $H^0(\Gamma_0(N); \mathrm{Sym}^{k-2}\mathbb{F}_p^2) = 0$ for $0 < k-2 \leq p-3$ — exactly the hypothesis $p > k+1$, which is why it is the right one; WP-14 referee C-4): a cuspidal eigensystem congruent to the Eisenstein one is old from level 1 — possible only if $p \mid \mathrm{num}(B_k/k)$ (Herbrand–Ribet, $p > k+1$) — or new of level N, where the Steinberg shape of $\rho_f|_{G_{\mathbb{Q}_N}}$ with $a_N = \pm N^{k/2-1}$ forces the Frobenius eigenvalues $\{1, N^{k-1}\}$ of $1 \oplus \bar{\chi}_p^{k-1}$ to be $\{\pm N^{k/2}, \pm N^{k/2-1}\}$ modulo p, i.e. $N^k \equiv 1$ or $N^{k-2} \equiv 1$. The cited source treats the converse; it is not stated here. At $k = 2$, (LN4)–(LN6) apply instead ($N^{k-2} - 1 = 0$). **Reported, not gated** (slot-16 plan, WP-13).

Scope: what this chapter does not do. It asserts, for every N, the coset table, the tree resolution, Manin's sequence, the cokernel isomorphism and its duality, and the Hecke action of T_p ($p \nmid N$) and U_p ($p \mid N$) on the cokernel presentation, and — since v1.16.0-pre.1, §28.6 — the model-to-presentation map with the fibre maps carried by naturality (Proposition 28.13) and the degeneracy maps α_d, $d \geq 1$, on the presentation (Proposition 28.14), all [proved here] for $6 \in R^\times$, n even. Not asserted: anything at $\ell \in \{2, 3\}$ beyond the cochain level of Definition 28.3(a) (the tree resolution and the cokernel need $6 \in R^\times$), except, since v1.21.0-pre.4, the weight-two

curve object of Proposition 28.18 (which is not group cohomology there); nebentypus ($\Gamma_1(N)$; since v1.15.0-pre.1 §29); composite n in Proposition 28.9's bookkeeping (Remark 28.10(4)); any bound on the elimination better than cubic; (LN4)–(LN6), (LN9), which are literature and predictions. (v1.16.0-pre.1: the sentence v1.15.0-pre.3 placed here — a Γ-module map between induced modules carries Rel to Rel and induces $H^1(\Gamma;\phi)$, so the fibre maps α^0, ω act on the presentation, while α_d, $d>1$, was asserted on the model only — was released *unrefereed* at v1.15.0; it is superseded by Proposition 28.13(3), which proves it, and Proposition 28.14, which puts α_d on the presentation.)

Code map. Built at v1.14.2 (slot-14 WP-12, `7a5158d`; not built at v1.14.0, the release, and at v1.14.1 — contract first, rule 21; refereed by slot-14 WP-5, ACCEPT WITH REPAIRS, repaired at pre.2). `Hecke/HeckeLevelN` (`namespace hecke::level`): Definition 28.1(1)–(5) as `Gamma0LevelN`, a `CosetLevel` for every N (at $N=p^\nu$ the shipped table entrywise, Lemma 28.2(3)), item (6) as `Gamma0SubTransversalN`; Definition 28.3 as `InducedRelationsLevel` on any `CosetLevel` over any word-sized ring (the monomial $M_N(g_r)$ of (22); `Hecke/InducedLocalSystem` is unchanged and stays at prime powers); Proposition 27.5(1),(2) orbit-wise (`OrbitWiseFixedSpace`, `CokernelFromFixedSpaces`); the operators by (68) on the level-1 τ-table (`Hecke/HeckeLevelEll`, unchanged), the r_t of item (6). `HeckeLevelNValidation`: (LN1), (LN2), the τ-route half of (LN3) (against an independent group-cocycle route at composite N and the classical characteristic polynomials), (LN4)–(LN6) on the 26 scan cells $N\le 211$ and, under `HECKE_LEVEL_N_R3_CELLS`, the named cells $(1321,11)$, $(1571,5)$. **Built at v1.15.0-pre.2** (slot-15 WP-7): $H_{N,p}$ of (71), `Hecke/HeckeLevelNHeilbronn` (`HeilbronnOperator`), tested by `HeckeLevelNHeilbronnValidation` ((LN3) on both halves). **Built at v1.15.0-pre.3** (slot-15 WP-8, WP-9): the sparse cokernel elimination of Remark 27.6(4) (`Hecke/HeckeLevelNSparse`: `SparseCokernel`, entrywise equal to the dense route, the cost model and its route choice, the fibre maps α^0, ω on the presentation), the degeneracy maps α_d at composite N on the model (Proposition 27.2(4)(c)), `HeckeLevelNSparseValidation`; (LN6) at scale over the Merel candidates $N\le 10^4$ (`Hecke/HeckeLevelNR3`, `HeckeLevelNValidation` arm (R3s)). **Built at v1.16.0** (slot-16 WP-13, `docs/implementation/slot16/WP-13_report.md`; cross-model audit `docs/implementation/slot16/WP-13_audit.md`, ACCEPT WITH REPAIRS, repaired by WP-17; not built at v1.16.0-pre.1 — contract first, rule 21): §28.6 — `Hecke/HeckeLevelNDegen`: κ_N^{mod} of Proposition 28.13 (`KappaModelPoints`, which finds e_{46} by its vertex set and checks its orientation against the model's coboundary, `KappaModel`, `KappaModel0`) and $D_{M,N,d}$ of (72) on the sparse presentation (`DegeneracyOperator`, the admissible classes enumerated per ϵ as in Proposition 28.14(7) and, as a check, by a full scan); the R3$_k$ reading (LN9) (`ReadEisensteinR3k` in `Hecke/HeckeLevelNSparse`); arms of `HeckeLevelNSparseValidation` (no new name): (LN7) on 17 cells, (LN8) against the model on 40 cells (the 28 cells of `docs/implementation/slot16/wp12/degen.py` and 12 new, among them $d=p\mid M$) with 4 negative-control cells, the trace (5) on 20 cells including U_p at $d=p\mid M$ on 12, α_d at scale on 4 cells; (LN9) reported and not gated (145 readings, no finding); weight $k>2$ at composite N on the sparse route up to $N=30030$ at $k=4$ (Remark 28.16; the ladder of the report). **Built at v1.17.0-pre.3** (slot-17 WP-5, `docs/implementation/slot17/WP-5_report.md`, merged; cross-model audit `docs/implementation/slot17/WP-5_audit.md`, ACCEPT WITH REPAIRS, text only; contract first at v1.17.0-pre.1, rule 21; slot-16 WP-13's finding F-1): **the reach of the table at composite N.** Definition 28.1 and its stored table are unchanged (entries $O(N^2)$, in `int64`). Every coset product $h=XYZ$ behind a block $\rho_n(h)$ of (22) (`MonomialOf`, the orbit-wise fixed spaces and generators of Proposition 27.5(1), the level-change maps of Proposition 27.2(4)(b)) is formed in `int64` first and, only on an overflow, in `__int128`, with $h\in\Gamma_0(N)$ and $\det h=1$ checked exactly over $\mathbb{Z}$ and the block keyed on $h \bmod \ell^k$ (`SubgroupProductForRho`; attempt-then-escalate as in Theorem 18.7).

Hence the table and every Hecke matrix of the chapter are bit-identical wherever the old width completed (for $N \le 30030$ the escalation never fires: `HeckeLevelNSparseValidation` arm (W5b), a gate), and the reach is $N < 2^{31}$ for the products. [verified computationally][a] on the ladder: $N = 60060$ and 90090 at $k = 2$, $N = 60060$ at $k = 4$ ($D_N = 580\,608$), (LN3), (LN8) (for $\alpha_2 \colon 30030 \to 60060$, $\alpha_3 \colon 30030 \to 90090$) and the trace (5) held; the old width's refusal at $N = 60060$ is the negative control (W5n). The expected *verdicts* of WP-5's committed predictions (`docs/implementation/slot17/WP-5_predictions.md`) held; its *numeric* predictions did not all hold: `final_nnz` was under-predicted by up to 30 % (90090, $k = 2$: 50.7 M against 39 M) and over-predicted at 60060, $k = 4$ (118.8 M against 165 M), and the $N = 99330$ prediction failed (Remark 28.16(4)). **Built at v1.18.0-pre.2** (slot-18 WP-5, `docs/implementation/slot18/WP-5_report.md`, merged): the sparse elimination consumes the generators of Rel_N sorted by their largest column, descending (`GeneratorOrder::kLastColumnDescending`, stable; rollback `HECKE_LEVEL_N_GEN_ORDER=as_built`) — a value-free choice, because the stored form is the unique natural reduced row echelon form (all 21 logged rungs bit-identical under both orders, a gate; arm (W18o) repeats the comparison entry for entry on 12 cells in `--all`, and the Hecke matrices on 4). Finding W5-F1 is that order's cost: the fill at $N = 99330$, $k = 2$, is 54.45 M, not a fill-law excess. The structure-aware fill model of Remark 28.16(4) **fails** its factor-1.5 rule (the fill is not monotone in $P(N)$), so `hook_rss/v1` stays; this is a finding (the remark's outcome paragraph). `Gamma1LevelN::contains_wide` (exact in `__int128`) and `HeilbronnChi` (its product formed by `Gamma0LevelN::cocycle_wide`) are on the `int64`-then-`__int128` route, arm (W18g). Item (c), the two window measurements, is **pending**. The plan, as committed at v1.18.0-pre.1 (contract first, rule 21; "not built" at that version; kept for the record; no statement of this chapter changes): (a) the structure-aware fill model of Remark 28.16(4), with its replacement rule for the RSS hook and for `CokernelCostModel`'s law; (b) at most one value-preserving change of the elimination order, merged only if every logged slot-16 / slot-17 rung stays bit-identical (a gate, not a report); (c) the measurements $N = 99330$ ($k = 2$) and $N = 90090$ ($k = 4$) in an exclusive window, each predicted by (a) first, under a VmHWM watchdog; (d) `Gamma1LevelN::contains_wide` and `HeilbronnChi`'s products on the `int64`-then-`__int128` route of the v1.17.0-pre.3 paragraph above (the slot-17 WP-6 `__int128` fallback becomes the one route), bit-identical wherever `int64` completes, checked by forcing the wide route on cells where both run. Arms of `HeckeLevelNSparseValidation` only (its `SPARSE_CELL` and `DEGEN_CELL` arms; no new name; the slot-18 plan's "`HeckeLevelNDegenValidation`" is the translation unit of those arms, `Hecke/HeckeLevelNDegenValidation`, not a registered name); its `EFFECTIVE` lines name the RSS model in use; golden unchanged. The Moerdijk–Pronk line: none of (a)–(d) uses the 7-vertex model beyond §12's parametric statements.

Formalization note. Lemma 28.5 needs the Farey complex and a filtration argument; Lemma 28.8 is elementary lattice combinatorics; Lemma 28.7 needs projective resolutions of length 1, Shapiro's lemma for homology and the map of a compatible pair; Proposition 28.9 adds only coset counting. None depends on the theory of modular forms; (LN1), (LN4)–(LN6) do.

28.7 The weight-two presentation at every prime: the curve object

Proposition 28.18 (The $n = 0$ cokernel presentation at every prime is the cohomology of the open modular curve)**.** *[proved here] Let $N \ge 1$, $A := \Gamma_0(N)$, $E := A\backslash\Gamma \cong \mathbb{P}^1(\mathbb{Z}/N)$ with the right action of Γ (Definition 28.1), $D := \mathbb{Z}[E] = M_N$ at $n = 0$ (Definition 28.3), and, for a finite subgroup $F \le \Gamma$, D^F the fixed submodule, which is the $\mathbb{Z}$-span of the F-orbit sums $\sum_{x\in O} e_x$. Let $\mathcal{T}$ be the Bass–Serre tree of Claim 16.2(1) (vertices $\Gamma/\langle S\rangle \sqcup \Gamma/\langle U\rangle$, edges $\Gamma/\langle -I\rangle$, oriented from $\gamma\langle S\rangle$ to $\gamma\langle U\rangle$), $U = ST$, and $G := A\backslash\mathcal{T}$: a finite graph with edge set E ($-I \in A$ is the edge stabiliser), vertex set the $\langle S\rangle$-orbits V_2 and the $\langle U\rangle$-orbits V_3 on E, the edge x joining*

the two orbits containing it. A acts without inversions (the orientation is preserved), so G is a graph and $\mathrm{Hom}_A(C_\bullet(\mathcal{T}),X)=C^\bullet_{\mathrm{cell}}(G;X)$ *for every abelian group X with trivial action. Let $Y_0(N):=A\backslash\mathfrak{H}$ and $X_0(N)$ its compactification, both as coarse spaces.*

1. ***(Free, and what it is.)*** $D/(D^{\langle S\rangle}+D^{\langle U\rangle})=H^1(G;\mathbb{Z})\cong H^1(Y_0(N);\mathbb{Z})=\mathrm{Hom}(\Gamma_0(N),\mathbb{Z})$*, a free $\mathbb{Z}$-module of rank*

$$b_1(G)\;=\;|E|-|V_2|-|V_3|+1\;=\;1+\tfrac{m_N}{6}-\tfrac{\nu_2(N)}{2}-\tfrac{2\nu_3(N)}{3}\;=\;2g(X_0(N))+c(N)-1$$

(ν_2,ν_3 as in Remark 27.6(3), $c(N)$ the number of cusps, g the genus); and $H^1(Y_0(N);\mathbb{Z})\cong H_1(X_0(N),\mathrm{cusps};\mathbb{Z})$ by Poincaré–Lefschetz duality. The quotient by the smaller lattice spanned by $e_x+e_{x\cdot S}$ and $e_x+e_{x\cdot U}+e_{x\cdot U^2}$ alone has torsion whenever $\nu_2(N)+\nu_3(N)>0$ ([verified computationally][a]: $\mathbb{Z}/2$ at $N=5,17,41$, $\mathbb{Z}/3$ at $N=43$, $\mathbb{Z}/6$ at $N=13,37,73$); the fixed points contribute e_x itself, not $2e_x$ or $3e_x$.

2. ***(Every characteristic.)*** *For every prime p and $M:=\mathbb{F}_p[E]$, with $M^{\langle S\rangle}$, $M^{\langle U\rangle}$ the fixed subspaces (spanned by the orbit sums — no $\frac12$, no $\frac13$),*

$$M/(M^{\langle S\rangle}+M^{\langle U\rangle})\;=\;H^1(G;\mathbb{F}_p)\;=\;H^1(Y_0(N);\mathbb{F}_p)\;\cong\;\big(D/(D^{\langle S\rangle}+D^{\langle U\rangle})\big)\otimes\mathbb{F}_p,$$

of dimension $2g+c-1$ for every p, and $M^{\langle S\rangle}\cap M^{\langle U\rangle}=H^0(G;\mathbb{F}_p)=\mathbb{F}_p$. For $p\ge5$ this is $H^1(\Gamma_0(N);\mathbb{F}_p)$ of Proposition 27.5(2) (the same cochain complex); for $p\in\{2,3\}$ it is ***not*** *group cohomology ($\dim H^1(\Gamma_0(N);\mathbb{F}_3)=2g+c-1+\nu_3$, $\dim H^1(\Gamma_0(N);\mathbb{F}_2)=2g+c-1+\nu_2+\delta_{\nu_2,0}$), but the Bredon object of hypothesis (H3) and Proposition 23.3(3) with the permutation coefficients $\mathbb{Z}[E]$: the cohomology of the coarse quotient $Y_0(N)$.*

3. ***(Hecke.)*** *For a prime $q\nmid N$ the operator $H_{N,q}$ of (71) at $n=0$ (every block $\rho_n=1$: $(H_{N,q}x)_i=\sum_{(\epsilon,j):\,j\cdot\epsilon=i}x_j$) is defined over $\mathbb{Z}$, preserves $D^{\langle S\rangle}+D^{\langle U\rangle}$, and induces on $H^1(Y_0(N);\mathbb{Z})$ the Hecke operator T_q, i.e. the endomorphism of the lattice $H^1(Y_0(N);\mathbb{Z})\subset H^1(Y_0(N);\mathbb{Q})$ that Proposition 28.9(2) identifies over $\mathbb{Q}$; its reduction is T_q on (2) for every p, $p\in\{2,3\}$ included. Under the duality of (1) it is T_q on $H_1(X_0(N),\mathrm{cusps};\mathbb{Z})$: for $q\nmid N$ the Hecke correspondence equals its transpose, and w_N commutes with T_q.*

Proof. (1) *The quotient graph.* The edge $\gamma\langle -I\rangle$ has end points $\gamma\langle S\rangle$ and $\gamma\langle U\rangle$; its A-orbit is the coset $A\gamma\in E$, and the A-orbits of vertices are the $\langle S\rangle$- and $\langle U\rangle$-orbits on E ($A\backslash\Gamma/\langle S\rangle$, $A\backslash\Gamma/\langle U\rangle$). A vertex of G coming from a fixed point x (i.e. $w_xSw_x^{-1}\in A$, an elliptic point of A) has the single edge x at it; a free orbit gives a vertex of valence 2 (resp. 3). The cellular coboundary $\delta\colon\mathbb{Z}^V\to\mathbb{Z}^E$ sends 1_v to $\pm$ the indicator of the orbit v (sign $-$ on V_2, $+$ on V_3): *once* per edge, since each edge has one tail and one head in G. So $\operatorname{im}\delta=D^{\langle S\rangle}+D^{\langle U\rangle}$ and $H^1(G;\mathbb{Z})=\operatorname{coker}\delta=D/(D^{\langle S\rangle}+D^{\langle U\rangle})$. (By contrast the chain map ∂^t of Lemma 28.5 on *coinvariants* counts a fixed edge 2 or 3 times: that is the torsion of the last sentence of (1).) H^1 of a finite connected graph is free of rank $|E|-|V|+1$; $|V_2|=(m_N+\nu_2)/2$, $|V_3|=(m_N+2\nu_3)/3$, and $12g=12+m_N-3\nu_2-4\nu_3-6c$ (the Hurwitz formula for $X_0(N)\to X_0(1)$) give the displayed rank. Alternatively, the incidence matrix of a graph is totally unimodular, so every invariant factor of δ is 1 and $\operatorname{coker}\delta$ is free: the rank over $\mathbb{F}_p$ equals the rank over $\mathbb{Q}$ for every p ([verified computationally][a]: an integer elimination with ±1 pivots only on every prime $N\le3000$ tested, `docs/implementation/slot22/ln23ref/`).

G and $Y_0(N)$. $\mathcal{T}$ embeds in $\mathfrak{H}$ as the union of the geodesic arcs $\gamma[i,\rho]$ ($\gamma\in\Gamma$, $\rho=e^{2\pi i/3}$), the dual tree of the Farey tessellation; $\mathfrak{H}\setminus\mathcal{T}$ is the disjoint union, over the cusps $c\in\mathbb{P}^1(\mathbb{Q})$, of open discs R_c bounded by $\mathcal{T}$, and flowing each R_c along the geodesics ending at c is a Γ-equivariant strong deformation retraction of $\mathfrak{H}$ onto $\mathcal{T}$ [standard: [Ser80, Ch. I, §4.2] for the tree and its embedding; the retraction is the classical *spine* statement; locator **not** verified (rule (A2))]. It descends to a deformation retraction of $Y_0(N)=A\backslash\mathfrak{H}$ onto $G=A\backslash\mathcal{T}$, so $H^\bullet(Y_0(N);X)=H^\bullet(G;X)$ for every coefficient group X. (A reader who prefers not to rely on the retraction may use Proposition 23.3(2) instead: $C_\bullet(X)$ and $C_\bullet(\mathcal{T})$ are chain homotopy equivalent over

$\mathbb{Z}[\Gamma]$, hence so are $\mathrm{Hom}_A(C_\bullet(X),X)$ and $\mathrm{Hom}_A(C_\bullet(\mathcal{T}),X)=C^\bullet(G;X)$, and $\mathrm{Hom}_A(C_\bullet(X),X)$ is the cellular cochain complex of $A\backslash X$, which is $Y_0(N)$ with a small open disc removed at every cusp, a homotopy equivalence.) $H^1(Y_0(N);\mathbb{Z})=\mathrm{Hom}(\pi_1(Y_0(N)),\mathbb{Z})=\mathrm{Hom}(A,\mathbb{Z})$ because π_1 of the coarse quotient is A modulo the normal closure of its elliptic elements (torsion, invisible to $\mathrm{Hom}(-,\mathbb{Z})$). Finally $X_0(N)$ is a compact orientable surface and $Y_0(N)=X_0(N)\setminus\text{cusps}$, so Poincaré–Lefschetz duality gives $H^1(Y_0(N);\mathbb{Z})\cong H_1(X_0(N),\text{cusps};\mathbb{Z})$, free of rank $2g+c-1$ by the long exact sequence of the pair.

(2) The cokernel of a map of free $\mathbb{Z}$-modules commutes with $\otimes\mathbb{F}_p$ (right exactness), and the fixed space of a permutation module over any ring is the span of the orbit sums (a fixed vector is constant on orbits); so the $\mathbb{F}_p$-presentation is $\delta\otimes\mathbb{F}_p$ and its cokernel is $H^1(G;\mathbb{F}_p)=H^1(G;\mathbb{Z})\otimes\mathbb{F}_p$ ($H^2(G)=0$). $H^0(G;\mathbb{F}_p)=\mathbb{F}_p$ as G is connected ($\mathcal{T}$ is). For $p\geq 5$, $C^\bullet(G;\mathbb{F}_p)=\mathrm{Hom}_A(C_\bullet(\mathcal{T}),\mathbb{F}_p)$ computes $H^\bullet(A;\mathbb{F}_p)$ by Lemma 28.4, and this is the identification of Proposition 27.5(2) and Definition 28.6 (the same complex). The dimensions of $H^1(A;\mathbb{F}_p)=\mathrm{Hom}(A,\mathbb{F}_p)$ at $p\in\{2,3\}$ follow from A^{ab}: the elliptic elements of order 3 (resp. 4) modulo $\pm I$ contribute the extra classes ([verified computationally][a]: arm (IL3N) of `Integral/HeckeIsotropyModelEllOpenHalves`, Proposition 31.3).

(3) Proposition 28.9(1),(2) hold for the ring $R=\mathbb{Z}[1/6]$: $H_{N,q}$ preserves $\mathrm{Rel}_{\mathbb{Z}[1/6]}=(D^{\langle S\rangle}+D^{\langle U\rangle})[1/6]$ and induces $[A\,g\,A]=T_q$ on $H^1(A;\mathbb{Z}[1/6])=H^1(G;\mathbb{Z}[1/6])$. Since $D/(D^{\langle S\rangle}+D^{\langle U\rangle})$ is torsion-free by (1), $(D^{\langle S\rangle}+D^{\langle U\rangle})[1/6]\cap D=D^{\langle S\rangle}+D^{\langle U\rangle}$; hence $H_{N,q}(D^{\langle S\rangle}+D^{\langle U\rangle})\subseteq D^{\langle S\rangle}+D^{\langle U\rangle}$ and the induced endomorphism of the lattice $H^1(G;\mathbb{Z})$ is the restriction of T_q on $H^1(G;\mathbb{Q})$. Reduction mod p commutes with the induced map by right exactness, which gives (2)'s T_q for every p. For the duality: the transpose of the correspondence $T_q=\pi_{2*}\pi_1^*$ on $X_0(N)$ ($q\nmid N$) is $\pi_{1*}\pi_2^*=T_q$ itself, because $\pi_2=\pi_1\circ w_q$ with w_q an involution of $X_0(Nq)$; and w_N commutes with T_q for $q\nmid N$. □

Remark 28.19 (Reading Mazur's Eisenstein ideal on the curve object; the Hecke family). [standard: [Maz77, II, §9 (9.7), §16 Def. p. 124, Prop. (16.6), Cor. (16.2), §17 Prop. (17.1)], read on the journal scan (slot-22 F4); [CE05, §1 (Thm. 1.1), §6] (LIT-30, held)] [conjectural] Let N be prime and $p\mid \mathrm{num}\left(\frac{N-1}{12}\right)$ — at $p=2$: $N\equiv 1 \pmod 8$, exactly iff $N\equiv 9 \pmod{16}$; at $p=3$: $N\equiv 1 \pmod 9$, exactly iff $N\equiv 10,19 \pmod{27}$ — so that Mazur's Eisenstein maximal ideal $\mathfrak{m}=(p,\,T_q-1-q\ (q\neq N),\,T_N-1)$ of the Hecke algebra $\mathbb{T}$ of $M_2(\Gamma_0(N))$ has $\mathbb{T}^0_{\mathfrak{m}}\neq 0$. On the curve object $H:=H_1(X_0(N),\text{cusps};\mathbb{Z})$ of Proposition 28.18, $H\otimes\mathbb{Q}\cong\mathbb{T}_{\mathbb{Q}}\oplus\mathbb{T}^0_{\mathbb{Q}}$ as a Hecke module, so the $\mathfrak{m}$-generalised eigenspace $W_{\mathfrak{m}}\subseteq H\otimes\mathbb{F}_p$ has $\dim W_{\mathfrak{m}}=2e_p+1$ when $p\,\|\,\mathrm{num}$ ($e_p+1=\mathrm{rank}_{\mathbb{Z}_p}\mathbb{T}_{\mathfrak{m}}$, Calegari–Emerton's e_p) and $2g_p+1$ in general; with $\mathbb{T}_{\mathfrak{m}}/p\cong\mathbb{F}_p[x]/x^{e_p+1}$ the Loewy length of $W_{\mathfrak{m}}$ is e_p+1 iff $\mathbb{T}_{\mathfrak{m}}/p$ acts faithfully ((LN4), (LN6) extended to $p\in\{2,3\}$). What the code reads is the joint generalised eigenspace $W_{\mathcal{F}}$ and the radical series of the ideal generated by $\eta_q:=T_q-1-q$, $q\in\mathcal{F}$, for a finite family $\mathcal{F}$ of primes. Two distinct things can fail, and only the second is a matter of a bound:

1. **Generation.** The radical series of $(\eta_q:q\in\mathcal{F})$ on $W_{\mathfrak{m}}$ has length e_p+1 iff the ideal is $\mathfrak{m}\mathbb{T}_{\mathfrak{m}}$, iff some $\eta_q\notin\mathfrak{m}^2$ (membership in $\mathfrak{m}^2$ is read modulo p throughout, i.e. in $\mathbb{T}_{\mathfrak{m}}/p$). By Mazur this holds *if* $\mathcal{F}$ contains a *good* prime: $\ell\neq N$ with ℓ not a p-th power modulo N and $\frac{\ell-1}{2}\not\equiv 0 \pmod p$ (for $p=2$: $\ell\equiv 3 \pmod 4$ a non-residue; $\ell=2$ is *excluded* by Prop. (17.1), although Mazur's definition admits it when 2 is not a quartic residue). For p odd and $\ell=2$, $\frac{\ell-1}{2}=\frac{1}{2}$ is a p-adic unit, so 2 is good iff it is not a p-th power modulo N (at $(1747,3)$, $\{2\}$ alone reads Loewy $4=e_3+1$). The condition is *sufficient, not necessary*, at $p=2$: at $(2473,2)$ ($e_2=1$, 2 a quartic residue modulo 2473, no good prime ≤ 13) η_2 generates $\mathfrak{m}\mathbb{T}_{\mathfrak{m}}/2$, while $\eta_3,\eta_5,\eta_7,\eta_{11}$ do not ([verified computationally]); at $(41,2)$ (2 not a quartic residue) $\eta_2\in\mathfrak{m}^2$, which is why Prop. (17.1) excludes $\ell=2$ although Mazur's definition admits it. A family without a good prime may therefore read the Loewy length too short (and $\dim W$ right): $(2137,2)$ with $q\leq 13$ reads 15 / Loewy 2; the smallest good prime is 23, and $\{23\}$ alone reads 15 / Loewy $8=e_2+1$. At $(41,2)$ the good primes 3 or

7 alone give $7,5,3,1$; η_2 and η_5 (not good) give $7,3$.

2. **Separation.** $W_{\mathcal{F}} \supseteq W_{\mathfrak{m}}$, with equality iff no other maximal ideal $\mathfrak{m}' \ni p$ of $\mathbb{T}$ contains every η_q, $q \in \mathcal{F}$. A family that does not separate reads $\dim W$ too large and a non-curvilinear series ($(2953, 2)$ with $q \le 13$: 39 / $39, 33, 23, 13, 8, 4$; with 19 added: 7 / $7, 5, 3, 1$). Every maximal ideal of $\mathbb{T}^0/p$ is determined by the eigenvalues of T_n, $n \le B := \lfloor (N+1)/6 \rfloor$, because $\mathbb{T}^0 = \mathrm{Hom}_{\mathbb{Z}}(S_2(\Gamma_0(N); \mathbb{Z}), \mathbb{Z})$ under $(t, f) \mapsto a_1(tf)$ and a cusp form with $a_n \equiv 0 \pmod p$ for $n \le B$ vanishes mod p [standard: Sturm, [Stu87, Thm. 1, p. 276], held (LIT-13), every residue characteristic]; so $\mathcal{F} = \{q \le B\}$ separates, and also contains a good prime when $e_p \ge 2$ (the T_n, $n \le B$, span $\mathbb{T}^0_{\mathfrak{m}}/p$, and $T_n - \sigma(n) \in \mathfrak{m}^2$ for all $n \le B$ would put that span inside $\mathbb{F}_p + \mathfrak{m}^2$). In practice a few primes beyond 13 suffice; the Sturm bound is a proof device, not a cost.

A reading with a family containing a good prime and reading a curvilinear series $2e+1, 2e-1, \ldots, 1$ of length $e+1$ is consistent with $e_p = e$; it is a statement only with a separating family. [verified computationally][a]: on every cell of `HeckeLevelNCurveValidation`'s default grids ($N \le 2000$ at $p = 2$, $N \le 2500$ at $p = 3$) the family $q \le 13$ agrees with Calegari–Emerton; on the Thm 1.1 grid to $3 \cdot 10^4$ it fails at 28 of 397 cells in exactly the two ways above, all repaired by $q \le 100$ (`docs/implementation/slot22/LN23_report.md` §5, the F4 report). The heavy oracle of that validation, $(21929, 2)$ with $e_2 = 127$ ($\dim W = 255$, Loewy length 128, $h(-4 \cdot 21929) = 256$), is Calegari–Emerton's own example [CE05, Thm. 1.1, §6] (v1.24.0-rc.1; the phenomenology catalogue, PH-39). Reported, not asserted here (v1.24.0-rc.2): the validation's heavy arm reproduces it on the curve object (`docs/implementation/slot22/LN23_report.md` Q6), printing as `layers=` the radical-series dimensions $\dim \mathfrak{m}^i W = 255 - 2i$, $i = 0, \ldots, 127$ (so the Loewy layers have dimension 2, the last dimension 1). Their §6 sentence that 2143 is the smallest prime $N \equiv 1 \bmod 9$ with an element of order 9 in the class group of their sextic field is reported false by an exact class-number computation out of tree (1297 is smaller; PH-40, `docs/implementation/slot22/NFQ1_report.md` §0, §1); this document does not assert it, and nothing here uses either sentence.

29 Nebentypus: $\Gamma_1(N)$ and $(\Gamma_0(N), \chi)$ on the fixed model by Shapiro

What this chapter fixes. (i) **The groups, the characters and the coefficient rings** (Definition 29.1): $\Gamma_1(N) \le \Gamma_0(N)$, a Dirichlet character $\chi\colon (\mathbb{Z}/N)^\times \to \mathbb{F}_q^\times$ for a finite field $\mathbb{F}_q$ of characteristic ℓ, $q = \ell^d$ arbitrary, and its Teichmüller lift to the Galois ring $W_K(\mathbb{F}_q) = W(\mathbb{F}_q)/\ell^K$ (Lemma 29.3); the coefficient module $V_R(\chi)$ of $\Gamma_0(N)$ and its extension to the semigroup $\Delta_0(N)$. (ii) **The two models by Shapiro** (Definition 29.7): the *character model* $L_{N,\chi}$ on the table of Definition 28.1, whose relation matrices are the monomial matrices of Definition 28.3 with every block scaled by a value of χ (Lemma 29.5), and the $\Gamma_1(N)$ *model* on a table fibred over it (Lemma 29.6). (iii) **The diamond operators and the χ-decomposition** (Proposition 29.8): $\langle d \rangle$ as a fibre map, the character model as the χ-eigenspace of the diamonds inside the $\Gamma_1(N)$ model, the decomposition over a ring holding the values of every character, and $\chi = 1$ as Definition 28.3 bit for bit. (iv) **The Hecke operators with character** (Proposition 29.9, Theorem 29.10): T_p for $p \nmid N$, U_p for $p \mid N$, $\langle p \rangle$ and $T(p, p)$, all read off the shipped block formula (68) with a scalar per block, on the level-1 τ-table; the relations with the $\chi(p)$ factor. (v) **The orbit structure and the cokernel** with a character (Proposition 29.11), the Heilbronn action with a character (Remark 29.12), the cost (Remark 29.13), the refusal rules (the box after Remark 29.13) and the predictions (NB1)–(NB6) (Remark 29.14).

Notation and standing conventions. Those of §28. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{M}_2(\mathbb{Z})$ write $a_\gamma := a$, $d_\gamma := d$. $N \geq 1$; $(\mathbb{Z}/N)^\times$ is written G_N, of order $\varphi(N)$ and exponent $\lambda(N)$ (Carmichael). ℓ is a prime, $q = \ell^d$ with $d \geq 1$, and $\mathbb{F}_q$ is *the* finite field of Remark 29.4 (a fixed model, so that matrices over it are data). **Rule (A5)**: in this chapter an undecorated χ, and χ_{-3}, χ_{-4}, χ_{-7}, denote Dirichlet characters; the decorated symbols χ_b, $\chi_{n,p}^{\mathrm{ord}}$, χ_C keep their meanings elsewhere and do not occur here. ς_N, $\Delta_0(N)$, $\rho_{n,\chi}$, $V_R(\chi)$, $L_{N,\chi}$, $L_{\Gamma_1(N)}$, σ_u, R_σ, $\langle d \rangle$, $W_K(\mathbb{F}_q)$, $\mathfrak{t}$ (the Teichmüller lift) are new symbols of this chapter and mean nothing else; $\langle d \rangle$ is not the cyclic group $\langle S \rangle$ (the argument of a diamond is always a residue class, never a matrix).

29.1 The groups, the characters, the coefficient rings

Definition 29.1 (Groups, semigroup, characters, coefficients)**.**

1. $\varsigma_N \colon \Gamma_0(N) \to G_N$, $\gamma \mapsto d_\gamma \bmod N$, and $\Gamma_1(N) := \ker \varsigma_N = \{\gamma \in \Gamma_0(N) : d_\gamma \equiv 1 \ (N)\}$.
2. $\Delta_0(N) := \{\delta \in \mathrm{M}_2(\mathbb{Z}) : \det \delta > 0,\ N \mid c_\delta,\ \gcd(a_\delta, N) = 1\}$, a monoid containing $\Gamma_0(N)$ [Shi71, §3.3, (3.3.3), p. 68] (his Δ' with $t = 1$ and $\mathfrak{h} = (\mathbb{Z}/N\mathbb{Z})^\times$; his Γ' of (3.3.2), p. 67, is then $\Gamma_0(N)$; verified, R20-L). [formally verified: [GT, `def:nebdata-model`]] (without the condition $\det \delta > 0$)
3. A *character* is a homomorphism $\chi \colon G_N \to R^\times$, extended to $\mathbb{Z}$ by $\chi(a) := \chi(a \bmod N)$ for $\gcd(a, N) = 1$. Its *parity* is $\chi(-1) \in \{\pm 1\}$. [formally verified: [GT, `def:nebdata-model`]] (over any commutative R)
4. The **coefficient rings** of this chapter are $R = \mathbb{F}_q$ and $R = W_K(\mathbb{F}_q)$ (Lemma 29.3), $K \geq 1$; the characters are $\chi \colon G_N \to \mathbb{F}_q^\times$ — so $\operatorname{ord} \chi \mid q - 1$ and $\ell \nmid \operatorname{ord} \chi$ automatically — and, over $W_K(\mathbb{F}_q)$, their Teichmüller lifts $\mathfrak{t} \circ \chi$. The least q that carries χ is $\ell^{d(\chi)}$, $d(\chi)$ the order of ℓ in $(\mathbb{Z}/\operatorname{ord} \chi)^\times$; any $q = \ell^d$ with $d(\chi) \mid d$ may be used.
5. $V_R(\chi) := V_R$ with $\Gamma_0(N)$ acting by $\rho_{n,\chi}(\gamma) := \chi(a_\gamma)\, \rho_n(\gamma)$, and $\Delta_0(N)$ acting by the same formula. [formally verified: [GT, `def:nebdata-model`]]

Lemma 29.2 (The definition is consistent)**.** *[proved here] (1) ς_N is a surjective homomorphism, so $\Gamma_0(N)/\Gamma_1(N) \cong G_N$, and $a_\gamma \equiv d_\gamma^{-1} \pmod N$ on $\Gamma_0(N)$ [formally verified: [GT, `def:nebdata-model`]] (the congruence $a_\gamma d_\gamma \equiv 1$ only; not the surjectivity of ς_N). (2) $\delta \mapsto a_\delta \bmod N$ is a homomorphism of monoids $\Delta_0(N) \to G_N$; hence $\rho_{n,\chi}$ is multiplicative on $\Delta_0(N)$ and $V_R(\chi)$ is a $R[\Delta_0(N)]$-module; on $\Gamma_0(N)$, $\chi(a_\gamma) = \chi(d_\gamma)^{-1} = \chi^{-1}(\varsigma_N(\gamma))$ [formally verified: [GT, `def:nebdata-model`]]. (3) $-I \in \Gamma_0(N)$ acts on $V_R(\chi)$ by the scalar $\chi(-1)(-1)^n$. (4) $\Gamma_1(N)$ acts on $V_R(\chi)$ through ρ_n: $V_R(\chi)|_{\Gamma_1(N)} = V_R|_{\Gamma_1(N)}$. [formally verified: [GT, `def:nebdata-model`]] (over any commutative R)*

Proof. (1),(2) For γ, γ' with $N \mid c, c'$: $(\gamma\gamma')_{22} = cb' + dd' \equiv dd'$ and $(\gamma\gamma')_{11} = aa' + bc' \equiv aa' \pmod N$; $\det \gamma = ad - bc \equiv ad$, so $ad \equiv 1$ on $\Gamma_0(N)$. A unit u modulo N is d_γ for a lift $\gamma \in \Gamma$ of $\mathrm{diag}(u^{-1}, u) \in \mathrm{SL}_2(\mathbb{Z}/N)$ (surjectivity of $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)$, as in Definition 27.1), and such a lift lies in $\Gamma_0(N)$. $\Delta_0(N)$ is closed under products by the same congruences, and $\gcd(aa', N) = 1$. Multiplicativity of $\rho_{n,\chi}$ is that of ρ_n (Proposition 8.2(2)) times that of $\chi \circ a$. (3) $a_{-I} = -1$ and $\rho_n(-I) = (-1)^n$. (4) $a_\gamma \equiv d_\gamma^{-1} \equiv 1$ on $\Gamma_1(N)$. □

Lemma 29.3 (The Galois ring and the Teichmüller lift)**.** *[proved here] Let $C \in \mathbb{F}_\ell[Y]$ be monic irreducible of degree d (the project-canonical $C_{\ell,d}$ of [E, `def:canon`], which fixes $\mathbb{F}_q = \mathbb{F}_\ell[Y]/(C)$), and $\widetilde{C} \in \mathbb{Z}[Y]$ its coefficientwise lift with coefficients in $[0, \ell)$. Put $W_K(\mathbb{F}_q) := (\mathbb{Z}/\ell^K)[Y]/(\widetilde{C})$ (any monic lift of C gives an isomorphic ring, the Galois ring $\mathrm{GR}(\ell^K, d)$; the coefficientwise lift is fixed so that matrices over it are data, as in Remark 29.4(1)).*

1. *$W_K(\mathbb{F}_q)$ is a local ring, free of rank d over $\mathbb{Z}/\ell^K$, with maximal ideal (ℓ) and residue field $\mathbb{F}_q$; every element is $\ell^j u$ with u a unit, and it is a quotient of $W_{K+1}(\mathbb{F}_q)$ by ℓ^K.*

2. *For each* $x \in \mathbb{F}_q^\times$ *there is exactly one* $\mathfrak{t}(x) \in W_K(\mathbb{F}_q)$ *with* $\mathfrak{t}(x)^{q-1} = 1$ *and* $\mathfrak{t}(x) \equiv x \pmod{\ell}$*;* $\mathfrak{t}\colon \mathbb{F}_q^\times \to W_K(\mathbb{F}_q)^\times$ *is an injective homomorphism, and for* every *lift* $\tilde{x}$ *of* x*,* $\mathfrak{t}(x) = \tilde{x}^{\,q^{K-1}}$*.*
3. $W_K(\mathbb{F}_q)^\times = \mathfrak{t}(\mathbb{F}_q^\times) \times (1 + \ell W_K(\mathbb{F}_q))$*, the second factor an* ℓ*-group. Consequently a homomorphism* $G_N \to W_K(\mathbb{F}_q)^\times$ *whose order is prime to* ℓ *is* $\mathfrak{t}\circ\chi$ *for exactly one* $\chi\colon G_N \to \mathbb{F}_q^\times$*.*

Proof. (1) $\widetilde{C}$ is monic, so the quotient is free of rank d; modulo ℓ it is the field $\mathbb{F}_q$, so (ℓ) is the unique maximal ideal (ℓ is nilpotent), and an element not in (ℓ) is a unit. Writing an element as $\ell^j u$: take j maximal with the element in (ℓ^j). (2) The polynomial $Y^{q-1} - 1$ has the simple root x in $\mathbb{F}_q$ (its derivative $(q-1)x^{q-2}$ is a unit), so by Hensel's lemma over the complete (here: ℓ-adically nilpotent) local ring it has exactly one root lifting x. For a lift $\tilde{x}$ write $\tilde{x} = \mathfrak{t}(x)(1+\ell y)$; then $\tilde{x}^{\,q^{K-1}} = \mathfrak{t}(x)^{q^{K-1}}(1+\ell y)^{q^{K-1}}$, $\mathfrak{t}(x)^q = \mathfrak{t}(x)$ gives $\mathfrak{t}(x)^{q^{K-1}} = \mathfrak{t}(x)$, and $(1+\ell y)^{\ell^m} \equiv 1 \pmod{\ell^{m+1}}$ by induction on m (binomial theorem, $\ell \geq 2$), with $m = d(K-1) \geq K-1$. Uniqueness makes $\mathfrak{t}$ multiplicative; it is injective because it is a section of reduction. (3) Reduction $W_K(\mathbb{F}_q)^\times \to \mathbb{F}_q^\times$ is onto with kernel $1 + \ell W_K$, of order q^{K-1}, and $\mathfrak{t}$ splits it. A homomorphism of order prime to ℓ has trivial component in the ℓ-group $1 + \ell W_K$. □

Remark 29.4 (Which field, and which characters). (1) **The field is data.** $\mathbb{F}_q$ is $\mathbb{F}_\ell[Y]/(C_{\ell,d})$ of [E, `def:canon`] — the field the extension layer already uses for residue fields of degree d — so a character is a table of elements of that field, and the cochain-level matrices of this chapter are well defined. Another choice of C changes those matrices by a field isomorphism and changes no dimension, rank or characteristic polynomial up to that isomorphism. (2) **Characters of ℓ-power order** (possible when $\ell \mid \varphi(N)$, e.g. $\ell^2 \mid N$ or $\ell \mid p-1$ for a prime $p \mid N$) take values in μ_{ℓ^j}, which lives in no $W_K(\mathbb{F}_q)$ outside the trivial group and reduces to 1 in every $\mathbb{F}_q$; they need a *ramified* coefficient ring $\mathbb{Z}_\ell[\zeta_{\ell^j}]/\ell^K$ and are **refused by name** (`character_wild`; the box after Remark 29.13). Over $\mathbb{F}_q$ such a character is trivial, so it is not a restriction of the owner's scope (Q2: $\operatorname{ord}\chi \mid q-1$). Since v1.22.0-pre.1 such characters are admitted over the ramified chain rings $\mathcal{O}_{j,e}$ of Definition 29.15 (§29.7); over $\mathbb{F}_q$ and $W_K(\mathbb{F}_q)$ the refusal stands. (3) The Galois group of $\mathbb{F}_q/\mathbb{F}_\ell$ acts on characters by $\chi \mapsto \chi^\ell$ and on everything below by conjugating matrix entries; the models of χ and χ^ℓ are Frobenius conjugates, so one character per Frobenius orbit suffices for dimensions and characteristic polynomials (up to conjugating their coefficients).

29.2 The two models

Lemma 29.5 (Induced modules of an arbitrary representation; the character model is monomial)**.** *[proved here] Let* R *be a commutative ring,* $H \leq \Gamma$ *of finite index* m *with representatives* $w_1 = I, \dots, w_m$ *of* $H\backslash\Gamma$*, and* $\eta\colon H \to \mathrm{GL}_r(R)$ *a homomorphism,* $X = R^r$*. Definition 12.1 and Proposition 12.2 hold verbatim for* $\mathrm{Ind}_H^\Gamma X$ *with* η *in place of* $\rho_n|_H$*:*

$$M_H^\eta(\gamma) = \sum_j E_{j,\,j\cdot\gamma} \otimes \eta\big(h_j(\gamma)\big),$$

a homomorphism $\Gamma \to \mathrm{GL}_{mr}(R)$ *with one non-zero block per block row; Definition 12.3 and Proposition 12.4(1),(2),(4) hold for the induced matrix local system* L_H^η *(in (4),* $H^q(H; X)$ *with the action* η*). In particular, for* $H = \Gamma_0(N)$ *with the table of Definition 28.1 and* $\eta = \rho_{n,\chi}$*,*

$$M_{N,\chi}(\gamma) \;=\; \sum_{j=1}^{m_N} E_{j,\,j\cdot\gamma} \otimes \chi\big(a_{h_j(\gamma)}\big)\,\rho_n\big(h_j(\gamma)\big), \tag{73}$$

the matrix $M_N(\gamma)$ *of Definition 28.3 with block* j *multiplied by the scalar* $\chi(a_{h_j(\gamma)}) \in \mu_{\operatorname{ord}\chi}(R)$*:* ***same permutation, same blocks up to a root of unity.***

Proof. The proofs cited use of ρ_n only that it is a homomorphism on H (the bijectivity of restriction to the representatives, the cocycle identity, and homomorphy of M_H), and of Proposition 12.4(4) only Theorem 5.10, Remark 7.6 (hypotheses on the model and on $6 \in R^\times$, not on the module) and Shapiro's lemma, which holds for every $R[H]$-module. (73) is the displayed formula with $\eta = \rho_{n,\chi}$ and Definition 29.1(5). □

Lemma 29.6 (The table of $\Gamma_1(N)\backslash\Gamma$)**.** *[proved here] For $u \in G_N$ let $\sigma_1 := I$ and, for $u \neq 1$, $\sigma_u := \left(\begin{smallmatrix}\alpha & \beta\\ N & u'\end{smallmatrix}\right)$ with $u' \in [1,N)$ the representative of u, $\alpha := u'^{-1} \bmod N \in [1,N)$ and $\beta := (\alpha u' - 1)/N$. Order G_N by the representatives u'.*

1. *$\sigma_u \in \Gamma_0(N)$ and $\varsigma_N(\sigma_u) = u$; the σ_u represent $\Gamma_1(N)\backslash\Gamma_0(N)$.*
2. *The elements $w_{(j,u)} := \sigma_u w_j$, $1 \le j \le m_N$, $u \in G_N$ (w_j of Definition 28.1), represent $\Gamma_1(N)\backslash\Gamma$, with $w_{(1,1)} = I$; ordered j-major, u-minor, index $J(j,u) := (j-1)\varphi(N) + \mathrm{rk}(u)$. For $N \ge 3$ there are $m_N\varphi(N) = N^2\prod_{p\mid N}(1-p^{-2})$ of them ($[\Gamma : \Gamma_1(N)]$, as $-I \notin \Gamma_1(N)$); for $N \le 2$, $\Gamma_1(N) = \Gamma_0(N)$ and the table is Definition 28.1's.*
3. *For $\gamma \in \Gamma$: $(j,u)\cdot\gamma = \big(j\cdot\gamma,\ u\,\varsigma_N(h_j(\gamma))\big)$ and $h_{(j,u)}(\gamma) = \sigma_u\, h_j(\gamma)\, \sigma^{-1}_{u\varsigma_N(h_j(\gamma))} \in \Gamma_1(N)$. The model $L_{\Gamma_1(N)} := L^{\rho_n}_{\Gamma_1(N)}$ (Lemma 29.5 with $H = \Gamma_1(N)$, $\eta = \rho_n|_{\Gamma_1(N)}$) of Definition 12.3 on this table has fibre rank $m_N\varphi(N)(n+1)$ and monomial relation matrices whose block permutation is fibred over that of M_N.*

Proof. (1) $\alpha u' - \beta N = 1$ and $N \mid \alpha u' - 1$, so $\beta \in \mathbb{Z}$ and $\det\sigma_u = 1$; $c = N$, $d = u'$. Cosets of $\ker\varsigma_N$ are the fibres of ς_N. (2) $\Gamma = \bigsqcup_j \Gamma_0(N) w_j$ and $\Gamma_0(N) = \bigsqcup_u \Gamma_1(N)\sigma_u$. The count is $m_N \cdot |G_N|$, and $m_N\varphi(N) = N\prod(1+p^{-1}) \cdot N\prod(1-p^{-1})$. (3) $w_{(j,u)}\gamma = \sigma_u w_j \gamma = \sigma_u h_j(\gamma) w_{j\cdot\gamma} = \big[\sigma_u h_j(\gamma)\sigma^{-1}_{u'}\big]\sigma_{u'} w_{j\cdot\gamma}$ with $u' = u\,\varsigma_N(h_j(\gamma))$; the bracket lies in $\Gamma_0(N)$ with $\varsigma_N = u \cdot \varsigma_N(h_j(\gamma)) \cdot u'^{-1} = 1$. Uniqueness of the decomposition gives the action and the cocycle. □

Definition 29.7 (The level-N models with nebentypus)**.** Let $N \ge 1$, $R \in \{\mathbb{F}_q, W_K(\mathbb{F}_q)\}$ (Definition 29.1(4)) — or any commutative ring at cochain level — and $\chi\colon G_N \to R^\times$.

1. The **character model** $L_{N,\chi} := L^{\rho_{n,\chi}}_{\Gamma_0(N)}$: fibre $\mathrm{Ind}^\Gamma_{\Gamma_0(N)} V_R(\chi)$ of rank $m_N(n+1)$, coordinates of Definition 12.1 with the w_j of Definition 28.1, relation matrices (73). For $6 \in R^\times$ its cohomology is $H^\bullet(\Gamma_0(N); V_R(\chi))$ (Lemma 29.5).
2. The $\Gamma_1(N)$ **model** $L_{\Gamma_1(N)}$ of Lemma 29.6(3): for $6 \in R^\times$ its cohomology is $H^\bullet(\Gamma_1(N); V_R)$.

The character model is **the** model of this chapter: its size is that of the $\Gamma_0(N)$ model, whatever χ. The $\Gamma_1(N)$ model is $\varphi(N)$ times larger and is specified as the oracle of (NB3) and for $\ell \mid \varphi(N)$, where no character decomposition exists (Proposition 29.8(3)).

Proposition 29.8 (Diamond operators, and the character model inside the $\Gamma_1(N)$ model)**.** *[proved here] Let R be a commutative ring and view $\mathrm{Ind}^\Gamma_{\Gamma_1(N)} V_R$ as functions $f\colon \Gamma \to V_R$ with $f(h\gamma) = \rho_n(h)f(\gamma)$, $h \in \Gamma_1(N)$ (Definition 12.1).*

1. ***(Diamonds as fibre maps.)*** *For $\sigma \in \Gamma_0(N)$, $(R_\sigma f)(x) := \rho_n(\sigma) f(\sigma^{-1}x)$ is an automorphism of $R[\Gamma]$-modules of $\mathrm{Ind}^\Gamma_{\Gamma_1(N)} V_R$ depending only on $\varsigma_N(\sigma)$, and $\sigma \mapsto R_\sigma$ is a homomorphism. $\langle d\rangle := R_{\sigma_d}$ ($d \in G_N$) is therefore a* fibre map *of $L_{\Gamma_1(N)}$ — one matrix at every simplex, a morphism of matrix local systems — and for $6 \in R^\times$ it induces on $H^\bullet(\Gamma_1(N); V_R)$ the operator $[\Gamma_1(N)\,\sigma_d\,\Gamma_1(N)] = (\mathrm{conj}_{\sigma_d})_*$ of §27's notation. $\langle -1\rangle$ is $(-1)^n\cdot$ the action of $-I$, and $\langle d\rangle\langle d'\rangle = \langle dd'\rangle$.*
2. ***(The character model is the χ-eigenspace.)*** *The inclusion of functions identifies $\mathrm{Ind}^\Gamma_{\Gamma_0(N)} V_R(\chi)$ with the $R[\Gamma]$-submodule $\{f : \langle d\rangle f = \chi(d) f\ \forall d\}$ of $\mathrm{Ind}^\Gamma_{\Gamma_1(N)} V_R$, and $P_\chi f := \sum_{u\in G_N} \chi(u)^{-1}\langle u\rangle f$ is a $R[\Gamma]$-map onto a submodule of it with $P_\chi| = \varphi(N)\cdot\mathrm{id}$ there. No division is used.*
3. ***(Decomposition, (NB3).)*** *If $\varphi(N) \in R^\times$ and R is a domain or a local ring containing a primitive $\lambda(N)$-th root of unity whose powers have unit differences (e.g. $\mathbb{F}_q$ or $W_K(\mathbb{F}_q)$*

with $\lambda(N) \mid q-1$), then $e_\chi := \varphi(N)^{-1}P_\chi$ are orthogonal idempotents with $\sum_\chi e_\chi = 1$ over the $\varphi(N)$ characters $G_N \to R^\times$, and

$$\mathrm{Ind}_{\Gamma_1(N)}^{\Gamma} V_R = \bigoplus_\chi \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V_R(\chi), \qquad H^\bullet(\Gamma_1(N); V_R) = \bigoplus_\chi H^\bullet(\Gamma_0(N); V_R(\chi)) \quad (6 \in R^\times),$$

$\langle d\rangle$ acting on the χ-summand by $\chi(d)$, and compatibly with every operator of Proposition 29.9.

4. *($\chi = 1$, **(NB2).**) For $\chi = 1$, (73) is $M_N(\gamma)$ entry for entry and $L_{N,1}$ is Definition 28.3's L_N; every operator of Proposition 29.9 at $\chi = 1$ is the operator of Definition 28.3 on the same cochains.*

Proof. (1) For $h \in \Gamma_1(N)$: $(R_\sigma f)(hx) = \rho_n(\sigma) f(\sigma^{-1}h\sigma \cdot \sigma^{-1}x) = \rho_n(\sigma)\rho_n(\sigma^{-1}h\sigma) f(\sigma^{-1}x) = \rho_n(h)(R_\sigma f)(x)$, using that $\Gamma_1(N) = \ker \varsigma_N$ is normal in $\Gamma_0(N)$. Γ-equivariance: $(\gamma \cdot R_\sigma f)(x) = \rho_n(\sigma) f(\sigma^{-1}x\gamma) = R_\sigma(\gamma \cdot f)(x)$. For $h \in \Gamma_1(N)$: $R_{\sigma h} f(x) = \rho_n(\sigma)\rho_n(h) f(h^{-1}\sigma^{-1}x) = R_\sigma f(x)$. Homomorphy: $R_\sigma R_{\sigma'} = R_{\sigma\sigma'}$ by expanding. A $R[\Gamma]$-endomorphism of the fibre acts at every simplex by the same matrix (Theorem 4.5). The induced map: under the reciprocity $\phi \mapsto \psi := \phi(\cdot)(I)$ of Proposition 27.2(3), $(R_\sigma \circ \phi)(y)(I) = \rho_n(\sigma)\phi(y)(\sigma^{-1}) = \rho_n(\sigma)\phi(\sigma^{-1}y)(I) = \rho_n(\sigma)\psi(\sigma^{-1}y)$ (Γ-equivariance of ϕ), which is Proposition 27.2(3)'s formula for $A = B = \Gamma_1(N)$, $g = \sigma$ (one coset, $\tau = \sigma^{-1}\cdot$, a valid σ-twisted datum since $\sigma \in \Gamma$). $-I$: $R_{-I} f(x) = (-1)^n f(-x) = (-1)^n((-I)\cdot f)(x)$ as $-I$ is central. (2) $F \in \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V_R(\chi)$ means $F(\sigma x) = \chi(a_\sigma)\rho_n(\sigma)F(x)$ for $\sigma \in \Gamma_0(N)$; restricted to $\Gamma_1(N)$ this is the defining condition of the larger module (Lemma 29.2(4)), and then $R_\sigma F(x) = \rho_n(\sigma)\chi(a_{\sigma^{-1}})\rho_n(\sigma^{-1})F(x) = \chi(\varsigma_N(\sigma))F(x)$ (Lemma 29.2(2)); conversely this identity for all σ is the defining condition. P_χ commutes with Γ by (1); $\langle d\rangle P_\chi f = \sum_u \chi(u)^{-1}\langle du\rangle f = \chi(d)P_\chi f$; on the eigenspace P_χ is $\sum_u \chi(u)^{-1}\chi(u) = \varphi(N)$. (3) Character orthogonality $\sum_\chi \chi(u) = \varphi(N)[u = 1]$ and $\sum_u \chi(u)\chi'(u)^{-1} = \varphi(N)[\chi = \chi']$ hold over such a ring (the characters of the finite abelian group G_N with values in $\mu_{\lambda(N)}(R)$ are $\varphi(N)$ in number and the standard proof needs only that $1-\zeta$ is a non-zero-divisor for $\zeta \neq 1$ in $\mu_{\lambda(N)}$). So the e_χ are orthogonal idempotents summing to 1, each a $R[\Gamma]$-map with image the χ-eigenspace, i.e. the character model by (2). Cohomology commutes with finite direct sums; the Hecke compatibility is Proposition 29.9(4). (4) $\chi(\cdot) = 1$ in (73); the formulas of Proposition 29.9 at $\chi = 1$ are (68). □

29.3 The Hecke operators with a character

Proposition 29.9 (Level-N correspondences with a character)**.** *[proved here] Let R be a commutative ring and $\chi\colon G_N \to R^\times$. Let $A, B \leq \Gamma_0(N)$ be of finite index, $g \in \Delta_0(N) \cap \Delta$, and fix w_a, v_i, r_t, s_j, τ exactly as in Proposition 27.2, with τ the level-1 g-twisted chain datum of $T_{\Gamma g \Gamma}$ (Definition 11.1) — **no new τ-table**. Define*

$$\big(\widetilde{[A\,g\,B]}_\chi\,\phi\big)(x)_i \;:=\; \sum_t \chi\big(a_{\delta_{i,t}}\big)\,\rho_n(\delta_{i,t})\,\phi\big(\tau(s_{j(i,t)}\,x)\big)_{a(i,t)}, \tag{74}$$

(68) with one scalar per term, for cochains with values in $\mathrm{Ind}_A^\Gamma V_R(\chi)$ (in the coordinates of Lemma 29.5). [formally verified: [GT, `prop:nebcorr-model`]] (in the reciprocity form of Proposition 27.2(3), on the model, for a fixed table and τ)

1. *Every $\delta_{i,t}$ lies in $\Delta_0(N)$, so the scalar $\chi(a_{\delta_{i,t}})$ is defined; [formally verified: [GT, `prop:nebcorr-model`]] (this sentence; not the congruence that follows) $a_{\delta_{i,t}} \equiv a_{r_t}^{-1}\, a_g\, a_{kw_a^{-1}} \pmod N$.*
2. *(74) is a cochain map $C_{\mathrm{eq}}^\bullet(K, \mathcal{R}; L_A^\chi) \to C_{\mathrm{eq}}^\bullet(K, \mathcal{R}; L_B^\chi)$ over every R, and for $6 \in R^\times$ it induces $[A\,g\,B]\colon H^\bullet(A; V_R(\chi)) \to H^\bullet(B; V_R(\chi))$ — the map $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ of the compatible pair $(\mathrm{conj}_g^{-1}, \rho_{n,\chi}(g))$ — independent of τ and of every choice of representatives. [formally verified: [GT, `prop:nebcorr-model`]] (the cochain map over every R and the factorisation $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$, on the model; not the independence of τ and of the representatives)*

3. ***(The operators.)*** *With $A = B = \Gamma_0(N)$ and the r_t of Definition 28.1(6): $T_p^{(N,\chi)} := [A \left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right) A]$ for $p \nmid N$; $U_p^{(N,\chi)} := [A \left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right) A]$ for $p \mid N$; for $p \nmid N$, $T^{(N,\chi)}(p,p) := [A\,pI\,A] = \rho_{n,\chi}(pI) = \chi(p)\,p^n$ (one coset, $\tau = \mathrm{id}$, every R, as in Proposition 10.5). [formally verified: [GT, `prop:nebcorr-model`]] ($T^{(N,\chi)}(p,p) = \chi(p)\,c$, c the scalar of $\rho(pI)$; not the $\Gamma_1(N)$ model) On the $\Gamma_1(N)$ model ($A = B = \Gamma_1(N)$, χ trivial, $g \in \Delta_0(N)$): $T_p^{(\Gamma_1)}$, $U_p^{(\Gamma_1)}$ likewise, and $T^{(\Gamma_1)}(p,p) := [\Gamma_1(N)\,p\sigma_p\,\Gamma_1(N)] = p^n\langle p\rangle$.*
4. ***(Compatibility.)*** *Under the inclusion of Proposition 29.8(2) the cochain maps of (3) on the character model are the restrictions of those on the $\Gamma_1(N)$ model, for $g \in \{\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right), pI\}$ ($T(p,p)$ with σ_p), and they commute with every $\langle d\rangle$. [formally verified: [GT, `prop:nebcorr-model`]] (term-wise only: for $a \equiv 1 \pmod N$ the scalar $\chi(a)$ of a term is 1; not the restriction form on the $\Gamma_1(N)$ model, nor the commutation with $\langle d\rangle$)*

Proof. (1) $\delta_{i,t} = r_t^{-1}\,g\,(kw_a^{-1})$ by Proposition 27.2(1), with $r_t \in B \subseteq \Gamma_0(N)$, $kw_a^{-1} \in A \subseteq \Gamma_0(N)$ and $g \in \Delta_0(N)$; $\Delta_0(N)$ is a monoid (Lemma 29.2(2)), and $r_t^{-1} \in \Gamma_0(N)$. (2) The proof of Proposition 27.2(2),(3) goes through with ρ_n replaced by $\rho_{n,\chi}$ on $\Delta_0(N)$, because every matrix at which it evaluates ρ_n in the transport and in the B-equivariance step — $r_t^{-1}g$, $kw_a^{-1} = h_1(k)$, $r_t^{-1}b_t g = b\,r_{t\cdot b}^{-1}g$ with $b \in B$, $b_t \in B''$ — lies in $\Delta_0(N)$ by (1), and it uses of ρ_n only multiplicativity on the matrices it multiplies (Lemma 29.2(2)) and (22), here (73) (Lemma 29.5). In particular $\rho_{n,\chi}(r_t^{-1}g)\rho_{n,\chi}(kw_a^{-1}) = \rho_{n,\chi}(\delta_{i,t}) = \chi(a_{\delta_{i,t}})\rho_n(\delta_{i,t})$, which is (74). The classical identification is the proof of Theorem 11.10 for the pair $(\mathrm{conj}_g^{-1}, \rho_{n,\chi}(g))$, compatible because $\rho_{n,\chi}(ghg^{-1})\rho_{n,\chi}(g) = \rho_{n,\chi}(g)\rho_{n,\chi}(h)$ for $h \in A \cap g^{-1}Bg$ (all in $\Delta_0(N)$), with [standard: [Bro82, Ch. III.9–10]] as there. (3) pI is central, $\Gamma_0(N)\,pI\,\Gamma_0(N) = pI\,\Gamma_0(N)$, and $a_{pI} = p$ is a unit modulo N iff $p \nmid N$. On $\Gamma_1(N)$: $p\sigma_p$ has $a \equiv p\cdot p^{-1} = 1$, lies in $\Delta_0(N)$, normalises $\Gamma_1(N)$ and acts as $\rho_n(p\sigma_p) = p^n\rho_n(\sigma_p)$, which is $p^n R_{\sigma_p}$ by the computation in the proof of Proposition 29.8(1). (4) Let $g \in \{\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right),\ p\sigma_p\}$, so $a_g \equiv 1 \pmod N$, and put $A_0'' := \Gamma_0(N) \cap g\Gamma_0(N)g^{-1}$, $A_1'' := \Gamma_1(N) \cap g\Gamma_1(N)g^{-1} = A_0'' \cap \Gamma_1(N)$ (for $g = \left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)$: $g^{-1}\gamma' g \in \Gamma_0(N)$ iff $Np \mid c_{\gamma'}$). *The map $A_1''\backslash\Gamma_1(N) \to A_0''\backslash\Gamma_0(N)$ is a bijection.* Injective: $r, r' \in \Gamma_1(N)$ with $rr'^{-1} \in A_0''$ have $rr'^{-1} \in A_0'' \cap \Gamma_1(N) = A_1''$. Surjective: $\varsigma_N(A_0'') = G_N$, since for $u \in G_N$ a lift $\gamma \in \Gamma$ of $\mathrm{diag}(u^{-1}, u) \in \mathrm{SL}_2(\mathbb{Z}/Np)$ (which exists: u is a unit modulo Np when $p \mid N$, and by the Chinese remainder theorem with $\gamma \equiv I \pmod p$ when $p \nmid N$) has $Np \mid c_\gamma$ and $d_\gamma \equiv u$; hence $A_0''\Gamma_1(N) = \Gamma_0(N)$ and every coset $A_0''r$ meets $\Gamma_1(N)$. Choose the r_t in $\Gamma_1(N)$, for both models. In the function picture of Proposition 27.2(3), both operators read $\psi \mapsto \sum_t \rho(r_t^{-1}g)\,\psi(\tau(r_t\,\cdot))$ with the same r_t and the same level-1 τ; on $\psi \in \mathrm{Hom}_{\Gamma_0(N)}(C_\bullet, V_R(\chi)) \subseteq \mathrm{Hom}_{\Gamma_1(N)}(C_\bullet, V_R)$ (the inclusion of Proposition 29.8(2) under reciprocity, $V_R(\chi)|_{\Gamma_1(N)} = V_R$) one has $\rho_{n,\chi}(r_t^{-1}g) = \chi(a_{r_t^{-1}})\chi(a_g)\rho_n(r_t^{-1}g) = \rho_n(r_t^{-1}g)$, as $a_{r_t^{-1}} = d_{r_t} \equiv 1$ and $a_g \equiv 1$. So the two cochain maps agree on the character model, and Proposition 27.2(3)'s independence of the representatives makes the choice $r_t \in \Gamma_1(N)$ harmless. Commutation with $\langle d\rangle$: on a sum of χ-eigenspaces it follows, since each operator preserves each character model and $\langle d\rangle$ is the scalar $\chi(d)$ there (Proposition 29.8(3)'s situation); in general it is the statement $\sigma_d g \sigma_d^{-1} \in \Gamma_1(N)g\Gamma_1(N)$, i.e. that the $\Gamma_1(N)$-double cosets in $\Gamma_0(N)g\Gamma_0(N)$ are indexed by $a_\delta \bmod N$ [standard: [Shi71, Lemma 3.29(1), (2), p. 66] (printed as a Lemma; for $\det\delta$ prime to N) and [Shi71, Prop. 3.32(1), Prop. 3.33, pp. 68–70] (with $\mathfrak{h} = \{1\}$, $t = 1$, i.e. $\Gamma' = \Gamma_1(N)$, for general $\det\delta$); verified, R20-L, v1.21.0-pre.1, in place of "Prop. 3.29 and §3.3"] (v1.15.0-pre.2, WP-4 referee A1). □

Theorem 29.10 (Hecke relations with a character)**.** *Assume (H4), $6 \in R^\times$, and $\chi\colon G_N \to R^\times$. On $H^\bullet(\Gamma_0(N); V_R(\chi))$, computed by the character model:*

1. *the operators $T_p^{(N,\chi)}$ ($p \nmid N$), $U_p^{(N,\chi)}$ ($p \mid N$) and $T^{(N,\chi)}(p,p)$ commute;*

2. *for $p \nmid N$ and $\nu \geq 1$ the full operators satisfy $T_{p^{\nu+1}} = T_p T_{p^\nu} - p\,T(p,p)\,T_{p^{\nu-1}} = T_p T_{p^\nu} - \chi(p)\,p^{n+1}\,T_{p^{\nu-1}}$; [formally verified modulo the primitive products of [Shi71, Thm. 3.24(5), Prop. 3.17] carried to the Hecke ring of $(\Gamma_0(N), \Delta_0(N))$ by [Shi71, Prop. 3.31 and p. 71, before Thm. 3.35] ($p \nmid N$; Shimura's ring carries no character, χ enters only through $T(p,p) \mapsto \chi(p)c$), as the hypotheses `hTpchi0`, `hTpchi`; locator verified (R20-L): [GT, `thm:nebrelations-model`]] (on the model, for a fixed table and τ, the full operators in the assembled form $\sum_j (\chi(p)c)^j\, T_{\Delta^{\mathrm{prim}}}$)*
3. *for $p \mid N$, $U_{p^\nu} = (U_p)^\nu$. [formally verified modulo the products of the U-double cosets, [Shi71, Thm. 3.34(3) with Prop. 3.33, pp. 70–71] ($p \mid N$; $T'(p^\mu)$ is the single double coset of $\mathrm{diag}(1, p^\mu)$), as the hypothesis `hUchi`; locator verified (R20-L): [GT, `thm:nebrelations-model`]] (on the model, for a fixed table and τ)*

[standard: [Shi71, Thm. 3.34, Thm. 3.35 and §3.3] for the identities in the Hecke ring $\mathcal{H}(\Gamma_0(N), \Delta_0(N))$ — for $p \nmid N$ Thm. 3.24 carried over by Prop. 3.31 (p. 71, before Thm. 3.35), for $p \mid N$ Thm. 3.34(3) with Prop. 3.33; Shimura's ring has no character, χ enters through $T(p,p)$ (Proposition 29.9(3)) — and [Bro82, Ch. III.9–10] with Obligation 20.4 for the ring homomorphism (the character twist: [AS86, §1.1, Lemma 1.1.5, p. 196]); the Shimura locators are **verified** *on the owner's scan (R20-L, v1.21.0-pre.1)]*

Proof sketch. As for Theorem 10.8: the double-coset action of $\mathcal{H}(\Gamma_0(N), \Delta_0(N))$ on $H^\bullet(\Gamma_0(N); M)$ for a $R[\Delta_0(N)]$-module M (here $V_R(\chi)$, Lemma 29.2(2)) is a ring homomorphism (Obligation 20.4's argument, which uses only that the coefficients form a $\Delta_0(N)$-module), the cited identities hold in that ring, and Proposition 29.9(3) evaluates $T(p,p)$ as the scalar $\chi(p)p^n$. With $k = n+2$, $\chi(p)p^{n+1} = \chi(p)p^{k-1}$, the classical factor. □

29.4 Orbit structure, the cokernel, and the Heilbronn action

Proposition 29.11 (Orbit-wise fixed points and the cokernel with a character)**.** *[proved here] Let R be a commutative ring, $\chi\colon G_N \to R^\times$, $M := \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V_R(\chi)$ with (73), and $F \leq \Gamma$ finite.*

1. ***(Mackey, orbit-wise.)*** *Proposition 27.5(1) holds with $\rho_n(h_{j_O}(\gamma))$ replaced by $\rho_{n,\chi}(h_{j_O}(\gamma))$: restriction to block j_O is an isomorphism $M_O^F \to V_R^{F_{j_O}}$ for the* twisted *action, with inverse $f_{j_O\cdot\gamma} := \rho_{n,\chi}(h_{j_O}(\gamma))^{-1} f_{j_O}$; a free orbit modulo $F \cap \{\pm I\}$ contributes $V_R^{\pm I}$, which is V_R if $\chi(-1) = (-1)^n$ and 0 otherwise, with* ***no solve.***
2. ***(Parity.)*** *If $2 \in R^\times$ and $\chi(-1) \neq (-1)^n$ then $H^\bullet(\Gamma_0(N); V_R(\chi)) = 0$.*
3. ***(The cokernel.)*** *If $6 \in R^\times$ and $\chi(-1) = (-1)^n$ then*

$$H^1(\Gamma_0(N); V_R(\chi)) \cong M \big/ \big(M^{\langle S\rangle} + M^{\langle U\rangle}\big), \qquad H^0 = M^{\langle S\rangle} \cap M^{\langle U\rangle},$$

with the averaging idempotents of Proposition 27.5(2) ($M_{N,\chi}(S)^2 = M_{N,\chi}(-I) = 1$); the isomorphism is Definition 28.6's $\varkappa_N$ with $M_{N,\chi}$ in place of M_N. More generally, for any finite-index $H \leq \Gamma$ and $R[H]$-module X free of finite rank, $H^1(H; X) \cong M^{\langle -I\rangle}/(M^{\langle S\rangle} + M^{\langle U\rangle})$, $M = \mathrm{Ind}_H^\Gamma X$ — the form needed for the $\Gamma_1(N)$ model, $-I \notin \Gamma_1(N)$ for $N \geq 3$.

4. ***(Dimensions, (NB1)'s proved half.)*** *If R is a field with $6 \in R^\times$ containing a primitive 12-th root of unity ζ_{12} (else pass to an extension, which changes no dimension), and $\chi(-1) = (-1)^n$:*

$$\dim H^1 - \dim H^0 = m_N(n+1) - \Big(\tfrac{m_N - \nu_2}{2}(n+1) + \sum_{\lambda} d_S(\chi(-\lambda))\Big) - \Big(\tfrac{m_N - \nu_3}{3}(n+1) + \sum_{\mu} d_U(\chi(\mu^{-1}))\Big),$$

the first sum over the $\nu_2 = \nu_2(N)$ roots λ of $\lambda^2 + 1 \equiv 0 \pmod N$, the second over the ν_3 roots μ of $\mu^2 - \mu + 1 \equiv 0 \pmod N$ (the fixed points of S, U on $\mathbb{P}^1(\mathbb{Z}/N)$, $(c:d)S = \lambda(c:d)$, $(c:d)U = \mu(c:d)$), and $d_S(z) := \#\{0 \leq m \leq n : z\,i^{n-2m} = 1\}$, $d_U(z) := \#\{0 \leq m \leq n : z\,\zeta_6^{n-2m} = 1\}$ with $i := \zeta_{12}^3$, $\zeta_6 := \zeta_{12}^2$, the eigenvalue labels fixed by $\rho_n(S) \sim \mathrm{diag}(i^{n-2m})_m$, $\rho_n(U) \sim \mathrm{diag}(\zeta_6^{n-2m})_m$. At $\chi = 1$ this is Proposition 27.5(3). The right-hand side depends on χ only through the values at the $\nu_2 + \nu_3$ fixed points.

Proof. (1) The proof of Proposition 27.5(1) uses only the cocycle identity and the block formula, both valid for $\rho_{n,\chi}$ (Lemma 29.5); on a free orbit the stabiliser is $F \cap \{\pm I\}$, acting through $\rho_{n,\chi}(-I) = \chi(-1)(-1)^n$ (Lemma 29.2(3)). (2) $-I$ is central in $\Gamma_0(N)$ and acts on $V_R(\chi)$ by the scalar $\varepsilon := \chi(-1)(-1)^n$; conjugation by an element of the group acts trivially on its cohomology, and it acts by ε through the coefficients, so $(\varepsilon - 1)$ kills $H^\bullet$; $\varepsilon - 1 = -2$ is a unit. (3) Lemma 28.4 for H and Shapiro: $H^\bullet(H; X) = H^\bullet(\mathrm{Hom}_\Gamma(C_\bullet(\mathcal{Y}), M))$ with $\mathrm{Hom}_\Gamma(C_1, M) = M^{\langle -I\rangle}$ (the stabiliser of $\mathfrak{e}$), $\mathrm{Hom}_\Gamma(C_0, M) = M^{\langle S\rangle} \oplus M^{\langle U\rangle}$ and coboundary $(a, b) \mapsto b - a$, exactly as in the proof of Definition 28.6's claims; for $H = \Gamma_0(N)$ and $\chi(-1) = (-1)^n$, $M^{\langle -I\rangle} = M$. (4) The exact sequence $0 \to H^0 \to M^{\langle S\rangle} \oplus M^{\langle U\rangle} \to M \to H^1 \to 0$ and (1). S and U have orders 4, 6 with $S^2 = U^3 = -I$ acting trivially on $\mathbb{P}^1(\mathbb{Z}/N)$, so orbits are free modulo $\pm I$ (sizes 2, 3) or fixed points. At a fixed point j of S, $h := h_j(S) = w_j S w_j^{-1} \in \Gamma_0(N)$ and its bottom row is $\mathrm{row}_2(w_j) S w_j^{-1} \equiv \lambda\, \mathrm{row}_2(w_j) w_j^{-1} = \lambda(0, 1)$, so $d_h \equiv \lambda$ and $a_h \equiv \lambda^{-1} = -\lambda$; h is Γ-conjugate to S, so $\rho_n(h)$ has the eigenvalues of $\rho_n(S)$ (semisimple, its order 4 being a unit), and the fixed space of $\chi(-\lambda)\rho_n(h)$ has dimension $d_S(\chi(-\lambda))$. Likewise for U with $a_h \equiv \mu^{-1}$. The fixed points of S are the classes $(1 : \lambda)$ with $\lambda^2 \equiv -1$, of U the $(1 : \mu)$ with $\mu^2 - \mu + 1 \equiv 0$, one each per root (the computation in the proof of Proposition 27.5(3), done modulo N by the Chinese remainder theorem; a class $(c : d)$ with c a non-unit is not fixed, as then λ would be a non-unit). $\rho_n(S)$ acts on $x^{n-m}y^m$ with eigenvalues $i^{\,n-m}(-i)^m = i^{\,n-2m}$ after diagonalising $S \sim \mathrm{diag}(i, -i)$, and $\rho_n(U)$ with ζ_6^{n-2m} ($U \sim \mathrm{diag}(\zeta_6, \zeta_6^{-1})$). □

Remark 29.12 (The Heilbronn action with a character). [proved here] Proposition 28.9 holds for the character model with

$$H_{N,p,\chi} \;:=\; \sum_{\epsilon \in \mathrm{Mer}_p} \;\sum_{j \text{ admissible}} E_{j\cdot\epsilon,\, j} \otimes \chi\big(d_{h_j(\epsilon)}\big)\, \rho_n\big(\mathrm{adj}\, h_j(\epsilon)\big)$$

in place of (71) ($a_{\mathrm{adj}\, h} = d_h$), on $M_{N,\chi}/\mathrm{Rel}_{N,\chi}$, $\chi(-1) = (-1)^n$, inducing $T_p^{(N,\chi)}$ for $p \nmid N$ and $U_p^{(N,\chi)}$ for $p \mid N$. *Proof.* Lemma 28.7 and the proof of Proposition 28.9 use of ρ_n only its multiplicativity on the matrices they evaluate, and those are: elements of $A = \Gamma_0(N)$; the coset representatives $\beta \in AgA \subseteq \Delta_0(N)$; and, in Step 3, $w_i \gamma_\epsilon^{-1} \beta = \mathrm{adj}\, h_j(\epsilon)$, where $\gamma_\epsilon = h w_i$ with $h \in A$ gives $w_i \gamma_\epsilon^{-1} \beta = h^{-1}\beta \in \Delta_0(N)$ — so every one lies in $\Delta_0(N)$, where $\rho_{n,\chi}$ is multiplicative, and the conjugation of Lemma 28.7(5) is used only for $g \in \Delta_0(N)$. With $\rho_{n,\chi}$ for ρ_n throughout, the proofs are unchanged (the hypotheses "n even, $-I \in A$" of Lemma 28.7 are used there only through $M_N(-I) = 1$, which is $\chi(-1) = (-1)^n$ here; v1.15.0-pre.2, WP-4 referee A2). In particular $d_{h_j(\epsilon)}$ is a unit modulo N at every admissible pair. □

29.5 The cost of the route

Remark 29.13 (Cost; scalability in N, k, $\mathrm{ord}\,\chi$ and q). A model of operation counts, in the notation of Remarks 27.6 and 28.11; not an assertion about running times. (1) **The character costs nothing structurally**: the character model has the fibre rank $D_N = m_N(n+1)$ of the $\Gamma_0(N)$ model, the same block permutation and the same number of non-zero blocks; each block is $\rho_n(h)$ times a table-lookup scalar $\chi(a_h)$ (χ is a table of $\varphi(N)$ entries). Orbit-wise, the free orbits need no solve (Proposition 29.11(1)), the $\nu_2 + \nu_3 \le 2^{\omega(N)+1}$ fixed points each one $(n+1)$-dimensional fixed space (the $\chi(a_h)^{-1}$-eigenspace of $\rho_n(h)$, one $d \times d$ solve per distinct scalar, then $O(d^2)$ per fixed point as in Remark 27.6(3)). Every count of Remarks 27.6 and 28.11 carries over with ℓ-operations replaced by $\mathbb{F}_q$-operations. (2) **The field**: an $\mathbb{F}_q$ operation is $O(d^2)$ $\mathbb{F}_\ell$-operations naively ($O(d \log d)$ with fast multiplication); take $q = \ell^{d(\chi)}$, the least field of Definition 29.1(4); $d(\chi) = 1$ iff $\mathrm{ord}\,\chi \mid \ell - 1$ (every ω^i at $N = \ell$, and every quadratic character). One character per Frobenius orbit (Remark 29.4(3)). (3) **All characters at once** costs $\varphi(N)$ character-model runs, which is exactly the size of one $\Gamma_1(N)$-model run ($D_{\Gamma_1} = \varphi(N) D_N$) but with the elimination cubic in D_N instead of $\varphi(N) D_N$: a gain of $\varphi(N)^2$ on the dense route.

The $\Gamma_1(N)$ model is therefore specified as an oracle ((NB3)), not as a production route. (4) Over $W_K(\mathbb{F}_q)$ the module route of §28's code map applies unchanged with $\mathbb{Z}/\ell^K$ replaced by the Galois ring, whose Howell forms exist because $W_K(\mathbb{F}_q)$ is a chain ring with the same ideal lattice (ℓ^j) (Lemma 29.3(1)); **built at v1.18.0-pre.2** (slot-18 WP-6, `Hecke/HeckeNebentypusGalois`, the code map's item (iv), dense and sparse, $K \le 3$, $d \le 4$; v1.18.0-pre.1 read "specified, not built").

CONTRACT. Refusal rules of §29, each by name. (i) $\ell \in \{2, 3\}$: cochain level only (Proposition 29.9(2)'s first half, every R); no comparison with group cohomology, no cokernel, no dimension formula is asserted, and nothing defers to the Bredon object (`nebentypus_small_ell`). (ii) A character of ℓ-power order into $\mathbb{F}_q$ or $W_K(\mathbb{F}_q)$ (`character_wild`, Remark 29.4(2)): still refused. Over the wild rings $\mathcal{O}_{j,e}$ of Definition 29.15 a character $\chi_{i,t}$ is admitted; refused there by name are a character whose wild order does not divide ℓ^j (`wild_ring_order`), a requested π-adic precision above the ring's e (`wild_precision`), and a tame level $N \neq \ell^r$ (`wild_tame_level`: tame level $N > 1$ with wild characters is slot-23 P11, not this text). (iii) A character value outside the declared $\mathbb{F}_q$ (the field must contain $\mu_{\operatorname{ord}\chi}$, `character_field`). (iv) **Not a refusal**: $\chi(-1) \neq (-1)^n$ gives 0 by Proposition 29.11(2) — a theorem, reported as such; and n odd with χ odd is *allowed* (odd weight), unlike level N with trivial character. (v) **The MP constraint** (owner, 2026-09-28): nothing in this chapter uses the 7-vertex model, a trivial central action or Bredon semantics — every statement is for an arbitrary $(K, \mathcal{R})$ satisfying (H4) (§12's "parametric in the triangulation"), the central element $-I$ enters only through its scalar $\chi(-1)(-1)^n$ on the coefficients, and the refined Moerdijk–Pronk model with a non-trivial central action is admissible input.

29.6 The predictions

Remark 29.14 (What is checkable with nebentypus: (NB1)–(NB6)). Committed before slot 16's code (rule 20), with the convention of Remark 26.6: a failing [proved here] item stops the package, a failing [conjectural] item is a finding. Cells are (N, χ, n) with $k = n + 2$, over $\mathbb{F}_q$, $\ell \ge 5$, $\ell \nmid N$ unless said.

(NB1) [proved here] (Proposition 29.11(4)) the Euler characteristic of the character model; and for $n > 0$ [standard: [Shi71, Ch. 8] for Eichler–Shimura with a character; [CO77] for $\dim S_k(\Gamma_0(N), \chi)$; both **not** verified; v1.21.0-pre.2: [CO77, Théorème 1, pp. 71–72] read ($\dim S_k - \dim M_{2-k}$; the Eisenstein dimension is not in that paper), and [Shi71, Thm. 8.4] covers the cuspidal part only (R20-L)] the prediction $\dim H^1 - \dim H^0 = 2\dim S_k(\Gamma_0(N), \chi^{\pm 1}) + \dim \mathcal{E}_k(\Gamma_0(N), \chi^{\pm 1})$ ($\mathcal{E}_k$ the Eisenstein space), the sign of the exponent immaterial for dimensions (complex conjugation). Reason: over a field of characteristic $\ell \ge 5$ the right-hand side of Proposition 29.11(4) equals the characteristic-0 one for $\mathsf{t} \circ \chi$ (the fixed-space dimensions of the order-4 and order-6 elements are Brauer-character multiplicities, invariant under reduction), and over $\mathbb{C}$, $H^0 = 0$ for $n > 0$.

(NB2) [proved here] (Proposition 29.8(4)) $\chi = 1$ reproduces Definition 28.3's level-N model and operators *bit for bit*: relation matrices, cochain maps, h^0, h^1 and every Hecke matrix.

(NB3) [proved here] (Proposition 29.8(3)) with $\lambda(N) \mid q - 1$ and $\ell \nmid \varphi(N)$: $h^\bullet(\Gamma_1(N)) = \sum_\chi h^\bullet(\Gamma_0(N), \chi)$, $\langle d \rangle = \chi(d)$ on the χ-part, and the characteristic polynomial of T_p on the $\Gamma_1(N)$ model is the product over χ of those on the character models. Cells: $N \in \{5, 7, 13\}$, $n \le 10$, $\ell \in \{5, 7, 11, 13, 29\} \setminus \{N\}$ with $\lambda(N) \mid q - 1$.

(NB4) [conjectural] **The weight-shift lemma, on cohomology.** For $N = \ell \ge 5$, $\omega\colon G_\ell = \mathbb{F}_\ell^\times \to \mathbb{F}_\ell^\times$ the identity (so $q = \ell$), $0 \le i \le \ell - 2$, $n \ge 0$ with $(-1)^i = (-1)^n$:

every system of T_p-eigenvalues ($p \neq \ell$) in $H^1(\Gamma_0(\ell); V_{\mathbb{F}_\ell}(\omega^i)) \otimes \bar{\mathbb{F}}_\ell$ is of the form $(p^e b_p)_p$ with $(b_p)_p$ a system in $H^1(\Gamma; \mathrm{Sym}^m \bar{\mathbb{F}}_\ell^2)$ at level 1 for some m and e; the prediction to be tested is the sharp form *at $n = 0$: the systems of weight $i+2$ at level 1 ($m = i$, $e = 0$) and p^i· the systems of weight $\ell + 1 - i$ ($m = \ell - 1 - i$, $e = i$), the two constituents* Sym^i *and* $\mathrm{Sym}^{\ell-1-i} \otimes \det^i$ *of* $\mathrm{Ind}_B^{\mathrm{SL}_2(\mathbb{F}_\ell)} \omega^i$ *as Δ-modules; for $n > 0$, the same with* $\mathrm{Sym}^n \otimes$ *each constituent in place of it, whose own constituents are left to the first run* (v1.15.0-pre.2, WP-4 referee A4) (the weight "$k + i$" of `docs/implementation/slot14/WP-C16_dihedral_over_extension_fields_plan.md` M8, in its normalisation, and the other constituent of $\mathrm{Ind}_{B(\mathbb{F}_\ell)}^{\mathrm{SL}_2(\mathbb{F}_\ell)} \omega^i$ with the θ-twist $e > 0$). Mechanism: $\Gamma_0(\ell) \to B(\mathbb{F}_\ell)$ and the Jordan–Hölder filtration of $\mathrm{Sym}^n \otimes \mathrm{Ind}_B^{\mathrm{SL}_2(\mathbb{F}_\ell)} \omega^i$ (the long exact sequence of Lemma 28.4's length-1 resolution makes every system of H^1 of the middle term a system of H^1 of a constituent, or of an H^0). *Not proved here*: the constituents with their det-twists as Δ-modules for $n > 0$ (at $n = 0$ they are derived in (c) below, v1.17.0-pre.3, and the sharp form at $n = 0$ is [proved here] there given one [standard:] input, v1.17.0-pre.4, whose locator [AS86, Lemmas 1.1.1, 1.1.4] is verified since v1.17.0-pre.5). At $n = 0$ the prediction reads: weight 2, level ℓ, character $\omega^i \leftrightarrow$ level 1, weight $i' + 2$.

(v1.17.0-pre.1; slot-16 WP-11, `docs/implementation/slot16/WP-11_report.md` §1, §4; proposed for the referee, slot-17 WP-3.) **(a) The sharp form at $n = 0$ as data.** [verified computationally]^{r} (`HeckeNebentypusValidation`, arm (X4), gated, every run; the cokernel route at level ℓ against the level-1 cokernel, T_p for $p \in \{2, 3, 5, 7\} \setminus \{\ell\}$) For every $\ell \in \{5, 7, \ldots, 43\}$ and every even $i \leq \ell - 3$ (132 cells), with ω^i acting through $\chi(a_\gamma)$ (Definition 29.1(5)),

$$\mathrm{charpoly}\big(T_p \mid H^1(\Gamma_0(\ell); \mathbb{F}_\ell(\omega^i))\big) = \mathrm{charpoly}\big(T_p \mid H^1(\Gamma; \mathrm{Sym}^i \mathbb{F}_\ell^2)\big) \cdot \mathrm{charpoly}\big(p^i\, T_p \mid H^1(\Gamma; \mathrm{Sym}^{\ell-1-i} \mathbb{F}_\ell^2)\big),$$

held on 132 of 132. **The orientation it pins**: under the $\chi(a_\gamma)$ convention Sym^i is the *untwisted* constituent and $\mathrm{Sym}^{\ell-1-i} \otimes \det^i$ the twisted one; the swapped assignment ($\mathrm{Sym}^{\ell-1-i}$ untwisted, $\mathrm{Sym}^i \otimes \det^{\ell-1-i}$) holds only on the 18 cells where the two coincide. **(b) The reading at $n > 0$.** [verified computationally]^{r} (the same arm; *reported, not gated*) At $n \in \{1, 2\}$ (240 cells, $\ell \leq 43$) the constituents are those of Clebsch–Gordan, $\mathrm{Sym}^n \otimes \mathrm{Sym}^i = \bigoplus_j \mathrm{Sym}^{n+i-2j} \otimes \det^j$ and $\mathrm{Sym}^n \otimes \mathrm{Sym}^{\ell-1-i} \otimes \det^i = \bigoplus_j \mathrm{Sym}^{n+\ell-1-i-2j} \otimes \det^{i+j}$, where it holds over $\mathbb{F}_\ell$ ($n \leq i \leq \ell - 1 - n$); the product of their H^1 characteristic polynomials (each twisted constituent's T_p multiplied by p^{twist}) matched on 216 cells as it stands and on the 24 cells $i = n$ after removing one factor $X - p^n(1 + p)$: the constituent $\mathrm{Sym}^0 \otimes \det^n$ carries $H^0 = \mathbb{F}_\ell$ with $T_p = p^n(1 + p)$, and the connecting map of the long exact sequence removes that class (240 of 240). This is the "or of an H^0" clause of the mechanism above; its form was found by the run. **(c) A proof of (a), given one standard input.** [proved here] (given the [standard] input named; owner ruling of 2026-10-02 on the slot-17 referee F1's M-1, v1.17.0-pre.4; the input's locator **verified** at v1.17.0-pre.5, slot-17 R17-4) *(Proposed at v1.17.0-pre.1; completed at v1.17.0-pre.3 by the referee F1's repair M-1, `docs/implementation/slot17/WP-3_referee_F1.md` §4: the orientation is now derived, not pinned by the data.)* Given [standard: [AS86, §1.1]: the long exact sequence of Γ-cohomology of an exact sequence of modules over the Hecke monoid, and Shapiro's isomorphism $H^\bullet(\Gamma_0(\ell); \mathbb{F}_\ell(\omega^i)) \cong H^\bullet(\Gamma; \mathrm{Ind}_{\Gamma_0(\ell)}^{\Gamma} \mathbb{F}_\ell(\omega^i))$ (Lemma 29.5, Proposition 29.9), commute with the double-coset operators of a compatible pair — [AS86, Lemma 1.1.1, p. 194] (the long exact sequence, for right or left modules over the monoid) and [AS86, Def. 1.1.2, p. 194; Lemma 1.1.4, p. 195] (Shapiro, for a compatible pair with $[\Gamma : \Gamma_0] < \infty$; its proof uses their Lemma 1.1.3); **verified** on the GDZ page images, slot-17 R17-4 (`docs/implementation/slot17/R17-4_report.md` §1a, rows 1–2; v1.17.0-pre.5)], used for the pair $(\Gamma_0(\ell), \Delta_0(\ell)) \subset (\Gamma, \{\delta \in \mathrm{M}_2(\mathbb{Z}) : \det \delta > 0,\ \ell \nmid \det \delta\})$, whose induced

module is the module I below. The pair is compatible in the sense of [AS86, Def. 1.1.2]: (b) in their orientation, $\Gamma\Delta_0(\ell)$ is the whole of the larger monoid, since $\mathrm{GL}_2(\mathbb{F}_\ell) = \mathrm{SL}_2(\mathbb{F}_\ell) \cdot B(\mathbb{F}_\ell)$ (the other orientation, $\Delta_0(\ell)\Gamma$, holds as well); (c) $\Gamma \cap \Delta_0(\ell)\Delta_0(\ell)^{-1} = \Gamma_0(\ell)$, since $B(\mathbb{F}_\ell)B(\mathbb{F}_\ell)^{-1} = B(\mathbb{F}_\ell)$.

The constituents, in this document's conventions. By Proposition 8.2, $\rho_m(b)$ for $b = \begin{pmatrix} a & b' \\ 0 & d \end{pmatrix}$ substitutes $x \mapsto ax$, $y \mapsto b'x+dy$: so $\rho_m(b)x^m = a^m x^m$, the span of the monomials divisible by x is B-stable, and $\varphi_m :=$ "the coefficient of y^m" satisfies $\varphi_m(\rho_m(b)v) = d^m\varphi_m(v)$. Let $\psi(b) := a^i$, $I := \{f\colon \mathrm{GL}_2(\mathbb{F}_\ell) \to \mathbb{F}_\ell : \ f(bg) = \psi(b)f(g)\ \forall b \in B(\mathbb{F}_\ell)\}$ with $(hf)(g) := f(gh)$ (Definition 12.1's convention, with the $\chi(a_\delta)$ of Definition 29.1(5)), $0 < i < \ell - 1$, and $m := \ell - 1 - i$; on $\mathrm{Sym}^m \otimes \det^i$ an element g acts by $\det(g)^i\rho_m(g)$. (α) $\mathrm{Sym}^m \otimes \det^i \to I$, $v \mapsto (g \mapsto \varphi_m(g \cdot v))$, is well defined ($d^m(ad)^i = a^i d^{\ell-1} = a^i$ on B), equivariant, and injective: $\varphi_m(g \cdot v) = \det(g)^i\, v(c_g, d_g)$, so if it vanishes for all g then $v(c,1) = 0$ for every $c \in \mathbb{F}_\ell$ and the coefficient of x^m vanishes; then $v(x,1)$ has degree $< m \le \ell - 1$ and ℓ roots, so $v = 0$. (β) $I \to \mathrm{Sym}^i$, $f \mapsto \sum_{Bg \in B\backslash \mathrm{GL}_2(\mathbb{F}_\ell)} f(g)\, \rho_i(g)^{-1}x^i$, is well defined ($\rho_i(b)^{-1}x^i = a^{-i}x^i$), equivariant, and onto: its image contains x^i, y^i and every $(x+cy)^i$, which span Sym^i as $i < \ell$. ($\beta \circ \alpha = 0$) An equivariant map $\mathrm{Sym}^m \otimes \det^i \to \mathrm{Sym}^i$ sends x^m to a vector fixed by the upper unipotent group, hence to a multiple of x^i; the torus acts on $x^m \otimes \det^i$ by $a^{m+i}d^i = d^i$ and on x^i by a^i, which differ for $0 < i < \ell - 1$; so $x^m \mapsto 0$, and x^m generates Sym^m. Dimensions: $(\ell - i) + (i + 1) = \ell + 1 = \dim I$. Hence

$$0 \to \mathrm{Sym}^{\ell-1-i} \otimes \det^i \to I \to \mathrm{Sym}^i \to 0$$

is exact: Sym^i is the *untwisted* constituent, as (a) found. For $i = 0$ the constant functions split off ($\ell + 1 \equiv 1$ is a unit) and (α) embeds $\mathrm{Sym}^{\ell-1}$ as a complement (its image contains no non-zero constant, as $(\mathrm{Sym}^{\ell-1})^{\mathrm{SL}_2(\mathbb{F}_\ell)} = 0$). Had the coefficient character been $\chi^{-1}(d_\delta)$ (equal to $\chi(a_\delta)$ on $\Gamma_0(\ell)$, not on $\Delta_0(\ell)$), the same computation would give $\mathrm{Sym}^{\ell-1-i}$ untwisted — the swapped reading that (a) rejects on 114 distinguishing cells.

The long exact sequence. $\mathrm{Ind}_{\Gamma_0(\ell)}^{\Gamma}\mathbb{F}_\ell(\omega^i)$ is I restricted along $\Gamma \to \mathrm{SL}_2(\mathbb{F}_\ell)$, with the Hecke monoid acting through $\mathrm{GL}_2(\mathbb{F}_\ell)$; let $0 \to A \to \mathrm{Ind} \to Q \to 0$ be the sequence just proved. $H^2(\Gamma; -) = 0$ on $\mathbb{F}_\ell[\Gamma]$-modules (Claim 16.2(3), whose argument uses nothing about the module), so the long exact sequence ends $\to H^1(\Gamma; Q) \to 0$, and in the Grothendieck group of $\mathbb{F}_\ell[T_p : p \neq \ell]$-modules $[H^1(\mathrm{Ind})] = [H^1(A)] + [H^1(Q)] - [H^0(A)] + [H^0(\mathrm{Ind})] - [H^0(Q)]$. For $0 < i \le \ell - 3$ every H^0 vanishes: $H^0(\Gamma; \mathrm{Sym}^a\mathbb{F}_\ell^2) = (\mathrm{Sym}^a)^{\mathrm{SL}_2(\mathbb{F}_\ell)} = 0$ for $0 < a \le \ell - 1$, and $H^0(\Gamma_0(\ell); \mathbb{F}_\ell(\omega^i)) = 0$ since a_γ takes every value of $\mathbb{F}_\ell^\times$ on $\Gamma_0(\ell)$. For $i = 0$, $[H^0(\mathrm{Ind})] = [H^0(\mathbb{F}_\ell)]$ cancels the constituent $\mathbb{F}_\ell$'s H^0 and $H^1(\Gamma; \mathbb{F}_\ell) = 0$. Since $H^1(\Gamma; \mathrm{Sym}^a \otimes \det^e) = H^1(\Gamma; \mathrm{Sym}^a)$ with T_p multiplied by p^e, the characteristic polynomials multiply as displayed in (a).

Tag at this version. The sharp form at $n = 0$ is [proved here] by (c), given the one [standard:] input above, and [verified computationally]^{r} by (a) (v1.17.0-pre.4, owner ruling of 2026-10-02 on the referee F1's M-1: the tag the referee proposed, with the locator then flagged **not verified**; at v1.17.0-pre.3 the referee's alternative, [verified computationally]^{r} only until the locator is read, was applied instead). *v1.17.0-pre.5:* the locator is read and **verified** on the GDZ page images (slot-17 R17-4, `docs/implementation/slot17/R17-4_report.md` §1a): the long exact sequence is [AS86, Lemma 1.1.1, p. 194] (not Lemma 1.1.3, which is restriction / corestriction), and Shapiro is [AS86, Lemma 1.1.4, p. 195] for the compatible pairs of their Def. 1.1.2 (p. 194), checked for this pair above. The "not verified" flag is dropped; the [verified computationally]^{r} fallback is not triggered. The general statement of (NB4) (every n, every system) stays [conjectural], its mechanism being the same long exact sequence over the constituents of $\mathrm{Sym}^n \otimes \mathrm{Ind}$ plus the identification of the H^0 systems $p^e(1+p)$ with a level-1 H^1 system, which (b) shows is needed and which this version does not write out.

(NB5) [standard: LMFDB newform data; **not** verified, nothing transcribed at this version] [conjectural] Classical characteristic polynomials: for $(N,\chi,k) \in \{(5,\chi_5,4),(13,\chi_{13},4),(7,\chi_{-7},3),(4,\chi_{-4},5),(3,$ (χ_5, χ_{13} the quadratic characters, even as $5 \equiv 13 \equiv 1 \pmod 4$), the characteristic polynomials of T_2, T_3 on the character model's H^1 equal, modulo ℓ, the product over the newforms $f \in S_k(\Gamma_0(N),\chi)$ of $(X - a_p(f))^2$ times the Eisenstein factors — each Eichler–Shimura half contributes once; for real χ this is the form $(X - a_p(f))(X - \overline{a_p(f)})$ written through v1.16.0-pre.1, which for non-real χ is **wrong** (it is the characteristic polynomial on the holomorphic halves of the χ- *and* $\bar\chi$-parts, since $\overline{a_p(f)} = a_p(f^\rho)$ with $f^\rho \in S_k(\Gamma_0(N),\bar\chi)$; v1.16.0-pre.2, slot-16 WP-9 audit `docs/implementation/slot16/WP-9_audit.md` §3: the written form fails in 20 of 24 non-real cases, the squared form holds in 24 of 24). [proved here] *(the squaring, in characteristic 0 through the Teichmüller lift; the identification of $S_k(\Gamma_0(N),\chi)$ rather than $S_k(\Gamma_0(N),\bar\chi)$ as the space whose eigenvalues occur is the eigenvalue dictionary, [proved here] modulo the named literature hypotheses (L1)–(L5) in [GT, `bridge:nebentypus-dictionary`] (`lean/blueprint/src/gl2_groundtruth.tex`; Lean, `proved_mod_lit`, audited by `docs/implementation/slot16/L16-2_fable_audit.md`), and [conjectural] only for those literature inputs (Eichler–Shimura with a character, [Shi71, Ch. 8], VERIFY); the squared form holds whichever way it falls; v1.16.0-pre.3, WP-14 referee C-9 at the owner's ruling.)* Let $\varepsilon := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Then $\varepsilon\Gamma_0(N)\varepsilon = \Gamma_0(N)$ and $a_{\varepsilon\gamma\varepsilon} = a_\gamma$, so $\rho_{n,\chi}(\varepsilon\gamma\varepsilon) = \rho_n(\varepsilon)\rho_{n,\chi}(\gamma)\rho_n(\varepsilon)$ and the compatible pair $(\mathrm{conj}_\varepsilon, \rho_n(\varepsilon))$ defines an involution ι_ε of $H^1(\Gamma_0(N); V_R(\chi))$, of the χ-part itself; it commutes with T_p and U_p because $\varepsilon \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \varepsilon = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ and the double coset is ε-stable. By Eichler–Shimura with a character [standard: [Shi71, Ch. 8]], $H^1_{\mathrm{par}} \otimes \mathbb{C} = H \oplus H^{\mathrm{anti}}$ with H the Hecke-stable image of the holomorphic forms and H^{anti} the antiholomorphic half — for non-real χ *not* the complex conjugate of H, which lives in $V_R(\bar\chi)$ (WP-14 referee C-8) — and ι_ε maps H onto H^{anti} (the holomorphic class of f to the antiholomorphic class of $f^\rho(z) = \overline{f(-\bar z)}$), $H \cap \iota_\varepsilon H = 0$, $H \oplus \iota_\varepsilon H = H^1_{\mathrm{par}}$. So $h \mapsto h \pm \iota_\varepsilon h$ are Hecke-equivariant isomorphisms of H onto the ±1-eigenspaces of ι_ε on H^1_{par}, and $\mathrm{charpoly}(T \mid H^1_{\mathrm{par}}) = \mathrm{charpoly}(T \mid H)^2$, with H carrying the eigenvalues of a single space of forms of character χ or $\bar\chi$. The passage from characteristic 0 (through the Teichmüller lift) to $\mathbb{F}_q$ uses $H^0(\Gamma_0(N); V_{\mathbb{F}_q}(\chi)) = 0$, true on the (NB5) cells ($n \le 5$; WP-9 audit §1); the [proved here] covers the squaring only. **The reduction rule**: a non-real character given by a label (Conrey $N.m$) is reduced modulo ℓ by its value at the least primitive root g modulo N ($g = 3,2,2$ at $N = 7,9,13$): $\psi(g) := s$, the *least* primitive $\mathrm{ord}\,\chi$-th root of unity in $\mathbb{F}_\ell$ (in $[1,\ell)$), $\bar\psi$ its inverse — a rule that presumes $d(\chi) = 1$ ($\mathrm{ord}\,\chi \mid \ell - 1$, Definition 29.1(4)), which holds at the (NB5) cells (C-8). Without this rule the mod-ℓ reading is ambiguous, since choosing s^{-1} is the $\chi \leftrightarrow \bar\chi$ swap itself. The values were transcribed with LMFDB labels by slot-16 WP-9 / WP-9b (`docs/implementation/slot16/WP-9_audit.md`) before slot 16's code package ran.

(NB6) [standard: [Rib77] for CM forms with nebentypus; LMFDB for dim = 1; **not** verified] [conjectural] **CM cells.** At $(4,\chi_{-4},5)$ (CM by $\mathbb{Q}(i)$), $(3,\chi_{-3},7)$ (by $\mathbb{Q}(\sqrt{-3})$), $(7,\chi_{-7},3)$ (by $\mathbb{Q}(\sqrt{-7})$), where the cusp space is predicted to be one CM newform: for every prime $p \nmid N\ell$ *inert* in the CM field, $T_p^{(N,\chi)}$ acts on the cuspidal part of H^1 with characteristic polynomial X^2 ($a_p = 0$), and for p split the two eigenvalues are $\pi^{k-1} + \bar\pi^{k-1}$, $p = \pi\bar\pi$, reduced modulo ℓ. These are the sanity cells of slot 16's [U] part 2 (CM Hida families need exactly such level and character).

Scope: what this chapter does not do. It asserts, for every N, every finite field $\mathbb{F}_q$ of characteristic ℓ and every $\chi\colon G_N \to \mathbb{F}_q^\times$ (and its Teichmüller lift over $W_K(\mathbb{F}_q)$): the two models,

the diamonds and the χ-decomposition, the Hecke operators T_p, U_p, $T(p,p)$ on the level-1 τ-table, the orbit-wise fixed spaces, the cokernel and the Euler characteristic, all [proved here] for $6 \in R^\times$ (cochain level for every R); the relations are standard (Theorem 29.10). Not asserted: anything at $\ell \in \{2,3\}$ beyond the cochain level; ramified coefficient rings; the degeneracy maps between levels with characters (Proposition 27.2(4)(b),(c) with $\rho_{n,\chi}$ applies verbatim when $N' \mid N$ and χ factors through $G_{N'}$, not written out); Atkin–Lehner operators (they move χ to $\bar\chi$ and need Δ-matrices outside $\Delta_0(N)$); the weight-shift lemma ((NB4) is a prediction; since v1.17.0-pre.1 its sharp form at $n = 0$ is data, [verified computationally]^{r}, and since v1.17.0-pre.4 [proved here] given one [standard:] input, whose locator is verified since v1.17.0-pre.5, (NB4)(a), (c)); the comparison with classical forms beyond (NB1), (NB5), (NB6).

Code map. **Built at v1.16.0** (slot-16 WP-10 and WP-11, `docs/implementation/slot16/WP-10_report.md`, `docs/implementation/slot16/WP-11_report.md`; cross-model audit `docs/implementation/slot16/WP-10_W` repaired; not built at v1.15.0-pre.1 or at v1.15.0 — contract first, rule 21, owner ruling Q3 of 2026-09-30). `Hecke/HeckeNebentypus` (`hecke::neb`): an $\mathbb{F}_q$ context with its linear algebra, a Galois-ring context with the Teichmüller lift, Dirichlet characters and the reduction rule of (NB5), the character model as a cokernel with $H_{N,p,\chi}$ (Remark 29.12), (73) and (74) on the fixed model, the $\Gamma_1(N)$ table and the diamonds (Lemma 29.6, Proposition 29.8(1)); no coset or τ-table is added. `HeckeNebentypusValidation` (`standard`) carries (NB1)–(NB6): Proposition 29.11(4) on every χ modulo $N \le 30$, $n \le 5$; (NB2) bit for bit; (NB3) on the $\Gamma_1(N)$ model; (NB5) against the committed table `docs/implementation/slot16/WP-9_nb_predictions.tsv`, every non-real distinguishing case reading χ, not $\bar\chi$; (NB6) on its three cells. (NB4) (WP-11): the sharp form at $n = 0$ is **gated** and held on 132 cells, $\ell \le 43$ (under the $\chi(a_\gamma)$ convention Sym^i is the untwisted constituent); for $n = 1, 2$ the reading is recorded, not gated. Also WP-11, on level 1 blocks (readings of §30, not statements of this chapter): $m = 2$ and $H^1_{\mathfrak{m}} \cong A^2_W$ on the non-ordinary dihedral blocks of slot-14 WP-C16 at $(72, 47)$, $(118, 47)$, $(156, 103)$, with $A_W \cong \mathbb{F}_q[\varepsilon]/(\varepsilon^2)$ the block's local algebra (certified over $\mathbb{Z}_\ell$ as $\Lambda_{\mathfrak{m}} \cong \mathbb{T}'^2_{\mathfrak{m}}$ by Lemma 30.4 and Corollary 30.6); at the F1 hidden blocks $(220, 17)$, $(246, 19)$ $m = 2$ but $H^1_{\mathfrak{m}} \not\cong \mathbb{T}^2_{\mathfrak{m}}$ over the full Hecke algebra (a generation certificate by the T_p, $p \le \dim S_k$). The tags of (NB1)–(NB6) are unchanged at v1.16.0. **Built at v1.17.0-pre.3** (slot-17 WP-6, merged and audited — `docs/implementation/slot17/WP-6_report.md`, cross-model audit `docs/implementation/slot17/WP-6_audit.md`, ACCEPT; contract first at v1.17.0-pre.1, rule 21; the text below is WP-6's proposal with the slot-17 referee F1's repairs M-2, m-W1–m-W3, `docs/implementation/slot17/WP-3_referee_F1.md` §4): (i) **the sparse cokernel with a character** — `Hecke/HeckeNebentypusSparse` (`NebSparseCokernel`): the elimination of Remark 27.6(4) on the generators of Proposition 29.11(1) over $\mathbb{F}_q$, the pivot of every reduced generator its smallest column, so that the result is the natural reduced row echelon form of $\mathrm{Rel}_{N,\chi}$, hence equal to the dense route entrywise wherever both run (checked: pivots and free columns, every row, every $H_{N,p,\chi}$ for $p \le 7$, $p \ne \ell$, and T_4 / U_4, on 13 cells and one parity-zero cell with $d(\chi) \in \{1, 2\}$; the full (NB3) grid of 132 cells reads the same verdicts on the sparse route). It lifts the cap $D \le 4096$ (reached: $N = 30030$, $k = 4$, $D_N = 290\,304$, over $\mathbb{F}_{17}$; $d(\chi) = 2$ at $N = 2003$, $k = 12$). Over $W_K(\mathbb{F}_q)$ it is **not built** (with (iii)). The route is a choice, never a value: `route=auto` is the dense route at $D \le 4096$. (ii) $T^{(N,\chi)}_{p^\nu}$, $\nu \ge 2$: the operator $H_{N,p^\nu,\chi}$ defined by Remark 29.12's formula with Mer_{p^ν} (Lemma 28.8 holds for every determinant) in place of Mer_p, every element of determinant p^ν included. That it induces the full T_{p^ν} is Merel's theorem at composite determinant and is **not** a statement of this document (Proposition 28.9 is for a prime); it is **tested, not proved**: it agrees with the recursion of Theorem 29.10(2) for T_{p^2}, T_{p^3} at $p \in \{2, 3\}$ and T_{25}, and $U_{p^\nu} = (U_p)^\nu$ (Theorem 29.10(3)), on six cells; the recursion without $\chi(p)$ fails wherever $\chi(p) \ne 1$. The model-route double coset $[\Gamma_0(N)\,\mathrm{diag}(1, p^\nu)\,\Gamma_0(N)]_\chi$ of Proposition 29.9 is **not built**. (iii) Galois-ring cohomology at $d(\chi) \ge 2$ (Remark 29.13(4)): **not built**. No coset or τ-table is added; no statement, tag or label changes (Theorem 29.10(2)'s

test moves from "untested" to "tested on six cells"). **Built at v1.18.0-pre.2** (slot-18 WP-6, `docs/implementation/slot18/WP-6_report.md`, merged; contract first at v1.18.0-pre.1, rule 21): (iv) **built**: `Hecke/HeckeNebentypusGalois` — cohomology over $W_K(\mathbb{F}_q)$, $K \le 3$, $d \le 4$, on the dense and the sparse Howell route (equal entrywise on 8 cells, $d \in \{2,3,4\}$); $K = 1$ equals the $\mathbb{F}_q$ route entrywise; at $\chi = 1$, $d = 1$ it equals the fixed model over $\mathbb{Z}/\ell^K$ (5 cells, 3 with torsion); the $\Gamma_1(N)$ model over $\mathbb{Z}/\ell^K$ equals the union over the characters (Proposition 29.8(3)) at $N = 5, 7$. (v) **built**: the model-route T_{p^2} through the shipped `Gamma0LevelN` table on 6 cells / 11 pairs, equal to the recursion $T_p^2 - \chi(p)p^{n+1}$ entrywise and to (ii)'s Mer_{p^2} operator by characteristic polynomial. (vi) **done**: the $D = 193\,536$ ladder cell on a second route (entrywise), and a natural `__int128` escalation in the default run. No statement of this chapter changes; (ii)'s operator stays *tested, not proved*. The plan, as committed at v1.18.0-pre.1 (`docs/implementation/slot18/00-slot18-plan.md`; contract first, rule 21; "not built" at that version; kept for the record; no statement of this chapter changes): (iv) **cohomology over $W_K(\mathbb{F}_q)$ at $d(\chi) \ge 2$, $K \le 3$** (Remark 29.13(4); scope (i)'s $W_K(\mathbb{F}_q)$ half), first on the dense route, then on the sparse route of (i). What the package must keep: every statement used is already one of this chapter over a commutative ring ($R = W_K(\mathbb{F}_q)$, χ valued in its Teichmüller lift, Lemma 29.3); the elimination is by Howell forms over the chain ring (Remark 29.13(4)), and the result is read as lengths and invariant factors, never as a dimension; two checks are gated where both routes run — the reduction $K = 1$ equals the $\mathbb{F}_q$ route entrywise, and at $\chi = 1$ (so $d = 1$, $W_K(\mathbb{F}_\ell) = \mathbb{Z}/\ell^K$) the $\Gamma_0(N)$ character model equals §28's model over $\mathbb{Z}/\ell^K$ ((NB2) over that ring; and, at $N = 1$, `EquivariantCohomologyOnK` over $\mathbb{Z}/\ell^K$) bit for bit; and the sparse route over $W_K(\mathbb{F}_q)$ equals the dense one entrywise on every cell where both run. A disagreement is a finding, never a re-pin. (v) **The model route for T_{p^2}**: the double coset $[\Gamma_0(N)\,\mathrm{diag}(1,p^2)\,\Gamma_0(N)]_\chi$ of Proposition 29.9 on the fixed model, built from the shipped `Gamma0LevelN` table (`BuildLevelCorrespondenceChi` on a $\mathrm{diag}(1,p^2)$ datum) — **never a second coset or τ construction**. Since $\mathcal{H}(\Gamma_0(N),\Delta_0(N)) \ni T_{p^2} = [\mathrm{diag}(1,p^2)] + [\mathrm{diag}(p,p)]$ for $p \nmid N$ and $[\mathrm{diag}(p,p)] = T(p,p)$ acts as $\chi(p)p^n$ (Proposition 29.9(3)), the model route gives the full operator as $[\mathrm{diag}(1,p^2)]_\chi + \chi(p)p^n$; it must equal the recursion $T_p^2 - \chi(p)p^{n+1}$ of Theorem 29.10(2) (a [proved here] identity given that theorem's [standard:] input: a failure stops the package) and is compared with (ii)'s $H_{N,p^2,\chi}$ from Mer_{p^2} on at most six cells, which turns (ii)'s test from "against the recursion" into "against a second route through this chapter's own double coset"; (ii)'s operator stays *tested, not proved*. (vi) The slot-17 WP-6 audit's two minors: a second route at one ladder cell with $D \ge 10^5$ (the $\mathbb{F}_q$ reduction of a W_K run, or (NB6)'s cocycle route at a reduced prime), and one synthetic `__int128`-overflow case outside the ladder. Arms of `HeckeNebentypusValidation` only (no new name); its default run stays within its tier target; the slot-18 golden event's $d = 2$ cell (an $\mathbb{F}_q$-route point) must read the same after (iv).

Formalization note. Lemmas 29.2, 29.3, 29.5, 29.6 and Propositions 29.8, 29.11 are elementary group theory over a commutative ring; Proposition 29.9(2) inherits Proposition 27.2's dependence on Brown's double-coset formalism, and (4) the Γ_1/Γ_0 comparison of Hecke rings; Theorem 29.10 needs the Hecke ring of $(\Gamma_0(N),\Delta_0(N))$. (NB1), (NB4)–(NB6) need modular forms.

29.7 Wild characters, the groups $\Gamma_{\mathsf{H}}(N)$, and the tower at ℓ

What this subsection fixes. (i) The **ramified coefficient rings** $\mathcal{O}_{j,e} = \mathbb{Z}_\ell[\zeta_{\ell^j}]/(\pi^e)$ and the **wild characters** $\chi_{i,t}$ of $(\mathbb{Z}/\ell^r)^\times$ with values in them (Definition 29.15, Lemma 29.16) — lifting refusal (ii) for these rings only. (ii) The **intermediate groups** $\Gamma_0(N) \supseteq \Gamma_{\mathsf{H}}(N) \supseteq \Gamma_1(N)$, $\mathsf{H} \le G_N$, with a character on H, their models, diamonds and operators (Definition 29.17, Propositions 29.18, 29.19) — the level structure of slot-22 WAKE-1 (`Hecke/HeckeGammaH`) and of the ℓ-adic tower of [U] part 5, where H is the Teichmüller subgroup of $(\mathbb{Z}/\ell^r)^\times$. (iii) **Specialisation**

from $\Gamma_{\mathsf H}(N)$ to $\Gamma_0(N)$ with a character, and **coinvariants** along the tower, without dividing by $|G_N/\mathsf H|$ (Lemma 29.20) — the replacement of Proposition 29.8(3), which needs $\varphi(N)\in R^\times$ and fails at $\ell\mid\varphi(N)$. (iv) The **torsion** of the wild character models (Lemma 29.21) and the **elimination over a chain ring** (Lemma 29.22). (v) The predictions (NB7)–(NB10) (Remark 29.23) and the shared `CosetLevel` interface (Remark 29.24). *Notation*: as in §29; $\ell\ge 5$ throughout this subsection; $u:=1+\ell$; for $d\in\mathbb Z$ prime to ℓ, $\omega(d)\in\mu_{\ell-1}(\mathbb Z_\ell)$ is the Teichmüller representative of d mod ℓ and $\langle d\rangle_\ell:=d\,\omega(d)^{-1}\in 1+\ell\mathbb Z_\ell$ (the subscript keeps it apart from the diamond $\langle d\rangle$); the sans-serif $\mathsf H$ is a subgroup of G_N, *not* the $H\le\Gamma$ of Lemma 29.5.

Definition 29.15 (Wild coefficient rings and wild characters)**.** Let $j\ge 1$, $\varphi_j:=\varphi(\ell^j)=\ell^{j-1}(\ell-1)$, $\mathcal O_j:=\mathbb Z_\ell[z]/(\Phi_{\ell^j}(z))$, ζ the image of z, $\pi:=1-\zeta$, and for $e\ge1$, $\mathcal O_{j,e}:=\mathcal O_j/\pi^e\mathcal O_j$. **As data**: for $e=K\varphi_j$, $\mathcal O_{j,e}=(\mathbb Z/\ell^K)[z]/(\Phi_{\ell^j}(z))$ on the power basis $1,z,\dots,z^{\varphi_j-1}$ with coefficients in $[0,\ell^K)$ (the representation of [E, `rmk:lvwild`] and of `nf::CycloModRing`); for other e, the quotient of $\mathcal O_{j,\lceil e/\varphi_j\rceil\varphi_j}$ by π^e, an element stored as its reduction against the Howell basis of $\pi^e\mathcal O$ (Lemma 29.16(2)). For $r\ge1$, $N=\ell^r$, $i\in\mathbb Z/(\ell-1)$ and $t\in\mathbb Z$ with $\ell^j\mid t\,\ell^{r-1}$, put $L(d)\in\mathbb Z/\ell^{r-1}$ by $\langle d\rangle_\ell\equiv u^{L(d)}\pmod{\ell^r}$ and

$$\chi_{i,t}(d):=\omega(d)^i\,\zeta^{\,t\,L(d)}\in\mathcal O_j^\times\qquad(d\in G_{\ell^r}),\qquad\psi_t:=\chi_{0,t}.$$

$\chi_{i,t}$ is the **wild character** of tame part ω^i and wild part ψ_t; a character of G_{ℓ^r} into $\mathcal O_{j,e}^\times$ is used only in this form (P4's $\chi_{i,t}$ of [E, `rmk:lvwild`], with its $m=r$).

Lemma 29.16 (The wild rings)**.** *[proved here] (1) $\mathcal O_j$ is a complete discrete valuation ring, totally ramified of degree φ_j over $\mathbb Z_\ell$, with uniformiser π, $\ell\mathcal O_j=\pi^{\varphi_j}\mathcal O_j$ and residue field $\mathbb F_\ell$ [standard: [Was97, Lemma 1.4, Prop. 2.1], cyclotomic fields of prime-power conductor; VERIFY]. Hence $\mathcal O_{j,e}$ is a* chain ring*: its ideals are $\pi^s\mathcal O_{j,e}$, $0\le s\le e$; it is local with residue field $\mathbb F_\ell$, and $\mathcal O_{j,K\varphi_j}=\mathcal O_j/\ell^K$ is free of rank φ_j over $\mathbb Z/\ell^K$; $\mathcal O_{j,e}\cong\mathbb F_\ell[\pi]/(\pi^e)$ for $e\le\varphi_j$. (2) Every matrix over $\mathcal O_{j,e}$ is equivalent to* $\operatorname{diag}(\pi^{s_1},\dots,\pi^{s_k},0,\dots)$ *(Smith), and every submodule of $\mathcal O_{j,e}^D$ has a unique Howell form (reduced row echelon, pivots π^s, closed under multiplication by π^{e-s}); a finitely generated module is $\bigoplus\mathcal O/\pi^{e_i}$. (3) $\chi_{i,t}$ is a homomorphism $G_{\ell^r}\to\mathcal O_j^\times$, $\chi_{i,t}\equiv\omega^i\pmod\pi$ and $\psi_t\equiv1\pmod\pi$. For $m\in\mathbb Z$ with $\ell^j\nmid m$, $v_\pi(\zeta^m-1)=\ell^{v_\ell(m)}$; so $c(\psi_t):=\min_d v_\pi(\psi_t(d)-1)=\ell^{v_\ell(t)}$ if $\psi_t\ne1$. ψ_t has order $\ell^{j-v_\ell(t)}$ and, if non-trivial, conductor $\ell^{j-v_\ell(t)+1}$. (4) $\sigma_s\colon\zeta\mapsto\zeta^s$ $(\ell\nmid s)$ is a ring automorphism of every $\mathcal O_{j,e}$ carrying the $\chi_{i,t}$-model (Definition 29.7) entrywise onto the $\chi_{i,st}$-model; one t per orbit suffices for lengths and characteristic polynomials, as in Remark 29.4(3).*

Proof. (1) The ring statements are the cited ones; a quotient of a discrete valuation ring by a power of the maximal ideal is a chain ring. (2) A chain ring is a local principal ideal ring: elimination with a pivot of least valuation in the remaining block (row and column operations) gives Smith, rows only (and the closure condition) Howell; uniqueness of the reduced form as for $\mathbb Z/\ell^K$ ([C, `rmk:nebcost`](4) for $W_K(\mathbb F_q)$, the same argument). (3) $d\mapsto\omega(d)$ and $d\mapsto L(d)$ are homomorphisms ($\langle\cdot\rangle_\ell$ is), and $\zeta^{tL(d)}$ is well defined because $\zeta^{t\ell^{r-1}}=1$. ζ^m is a primitive $\ell^{j-v(m)}$-th root of unity, and $1-\zeta'$ for a primitive ℓ^s-th root ζ' has $v_\ell=1/\varphi(\ell^s)$, i.e. $v_\pi=\varphi_j/\varphi(\ell^s)=\ell^{j-s}$. The order of ψ_t is the order of ζ^t (L is onto $\mathbb Z/\ell^{r-1}$); a character of $(\mathbb Z/\ell^r)^\times$ of exact order $\ell^s\ge\ell$, trivial on $\mu_{\ell-1}$, has conductor ℓ^{s+1}. (4) σ_s preserves $\pi\mathcal O$, and the model's matrices are polynomials in the character values. □

Definition 29.17 (The groups $\Gamma_{\mathsf H}(N)$ and their models)**.** Let $N\ge1$, $\mathsf H\le G_N$ a subgroup and $\chi\colon G_N\to R^\times$ a character (only $\chi|_{\mathsf H}$ enters the module; the extension to G_N fixes the diamonds, item (4) below). Put

$$\Gamma_{\mathsf H}(N):=\varsigma_N^{-1}(\mathsf H)=\{\gamma\in\Gamma_0(N):d_\gamma\bmod N\in\mathsf H\},$$

so $\Gamma_{\{1\}}(N)=\Gamma_1(N)$, $\Gamma_{G_N}(N)=\Gamma_0(N)$. Let $\mathsf{U}\subseteq G_N$ be the set of the least representatives $u'\in[1,N)$ of the cosets G_N/H (so $1\in\mathsf{U}$), ordered by u', and $w_{(j,u)}:=\sigma_u\,w_j$ (w_j of Definition 28.1, σ_u of Lemma 29.6), ordered j-major, u-minor. The **Γ_{H}-model with character** is $L^{\chi}_{\Gamma_{\mathsf{H}}(N)}:=L^{\rho_{n,\chi}}_{\Gamma_{\mathsf{H}}(N)}$ of Lemma 29.5 on this table, $\rho_{n,\chi}(\gamma)=\chi(a_\gamma)\rho_n(\gamma)$ (Definition 29.1(5)), and for $\sigma\in\Gamma_0(N)$, $R_\sigma f(x):=\rho_{n,\chi}(\sigma)f(\sigma^{-1}x)$ on $\mathrm{Ind}^{\Gamma}_{\Gamma_{\mathsf{H}}(N)}V_R(\chi)$; the **diamond** of $d\in G_N$ is $\langle d\rangle:=R_{\sigma_d}$.

Proposition 29.18 ($\Gamma_{\mathsf{H}}(N)$: table, model, diamonds, operators)**.** *[proved here] In the situation of Definition 29.17:*

1. $\Gamma_{\mathsf{H}}(N)\trianglelefteq\Gamma_0(N)$, $\Gamma_0(N)/\Gamma_{\mathsf{H}}(N)\cong G_N/\mathsf{H}$ *through* ς_N; $-I\in\Gamma_{\mathsf{H}}(N)$ *iff* $-1\in\mathsf{H}$.
2. *The* $w_{(j,u)}$ *represent* $\Gamma_{\mathsf{H}}(N)\backslash\Gamma$*; there are* $m_N\,[G_N:\mathsf{H}]$ *of them;* $w_{(1,1)}=I$*. The bottom row of* $w_{(j,u)}$ *is* $\equiv u'(c_j,d_j)\pmod N$*, and the class of* $\gamma\in\Gamma$ *is the* H*-orbit of its bottom row modulo* N*. For* $\gamma\in\Gamma$*:* $(j,u)\cdot\gamma=(j\cdot\gamma,[u\,\varsigma_N(h_j(\gamma))])$*,* $[x]$ *the element of* U *in* $x\mathsf{H}$*, and* $h_{(j,u)}(\gamma)=\sigma_u\,h_j(\gamma)\,\sigma^{-1}_{[u\varsigma_N(h_j(\gamma))]}\in\Gamma_{\mathsf{H}}(N)$*. At* $\mathsf{H}=\{1\}$ *this is Lemma 29.6; at* $\mathsf{H}=G_N$ *it is Definition 28.1's table* entry for entry*.*
3. *The relation matrices are monomial,* $M^{\chi}_{\mathsf{H}}(\gamma)=\sum_J E_{J,J\cdot\gamma}\otimes\chi(a_{h_J(\gamma)})\rho_n(h_J(\gamma))$*, and for* $6\in R^\times$ *the model computes* $H^\bullet(\Gamma_{\mathsf{H}}(N);V_R(\chi))$*; Proposition 29.11(1),(3) hold for it ((3) in its general form, with* $M^{\langle -I\rangle}$ *when* $-1\notin\mathsf{H}$*).*
4. R_σ *is an automorphism of* $R[\Gamma]$*-modules depending only on* $\sigma\Gamma_{\mathsf{H}}(N)$*,* $R_\sigma R_{\sigma'}=R_{\sigma\sigma'}$*; so* G_N/H *acts by fibre maps* $\langle d\rangle$*, and* $\langle h\rangle=1$ *for* $h\in\mathsf{H}$*. For a character* $\psi\colon G_N/\mathsf{H}\to R^\times$ *the inclusion of functions identifies* $\mathrm{Ind}^{\Gamma}_{\Gamma_0(N)}V_R(\chi\psi)$ *with* $\{f:\langle d\rangle f=\psi(d)f\ \forall d\}$ *(no division).*
5. *The operators of Proposition 29.9(3) with* $A=B=\Gamma_{\mathsf{H}}(N)$ *and the transversals* r_t *chosen in* $\Gamma_{\mathsf{H}}(N)$ *—* T_p *(*$p\nmid N$*),* U_p *(*$p\mid N$*), and* $[\Gamma_{\mathsf{H}}\,p\sigma_p\,\Gamma_{\mathsf{H}}]=p^n\chi(p)\langle p\rangle$ *— commute with every* $\langle d\rangle$ *and restrict, under (4), to the operators of the character model of* $\chi\psi$ *(Proposition 29.9(4)'s statement with* $\Gamma_{\mathsf{H}}(N)$ *for* $\Gamma_1(N)$*).*
6. *((NB10)'s oracle) If* $|\mathsf{H}|\in R^\times$ *and* R *holds the values of the characters of* H *with unit differences, then* $e_\chi:=|\mathsf{H}|^{-1}\sum_{h\in\mathsf{H}}\chi(h)^{-1}\langle h\rangle_{\Gamma_1}$ *on the* $\Gamma_1(N)$ *model is an idempotent with image the* Γ_{H}*-model of* χ*, compatibly with every operator of (5).*

Proof. (1) $\Gamma_{\mathsf{H}}(N)$ is the preimage of a subgroup under the homomorphism ς_N (Lemma 29.2(1)); $\varsigma_N(-I)=-1$. (2) $\Gamma=\bigsqcup_j\Gamma_0(N)w_j$ and $\Gamma_0(N)=\bigsqcup_{u\in\mathsf{U}}\Gamma_{\mathsf{H}}(N)\sigma_u$ ($\varsigma_N(\sigma_u)=u$); the bottom row of σ_u is (N,u'), so that of $\sigma_u w_j$ is $\equiv u'(c_j,d_j)$; two elements of $\Gamma_0(N)w_j$ lie in one Γ_{H}-coset iff their ς_N-values agree modulo H, i.e. iff their bottom rows agree up to H. The action and the cocycle are the computation of Lemma 29.6(3) with $\varsigma_N(\cdot)\in\mathsf{H}$ in place of $=1$. At $\mathsf{H}=G_N$, $\mathsf{U}=\{1\}$, $\sigma_1=I$. (3) Lemma 29.5 with $H=\Gamma_{\mathsf{H}}(N)$, $\eta=\rho_{n,\chi}|_{\Gamma_{\mathsf{H}}}$; Proposition 29.11(1),(3) use only the cocycle identity and the block formula. (4) The proof of Proposition 29.8(1),(2), with $\rho_{n,\chi}$ (multiplicative on $\Gamma_0(N)$) for ρ_n and $\Gamma_{\mathsf{H}}(N)$ (normal) for $\Gamma_1(N)$: for $h\in\Gamma_{\mathsf{H}}$, $R_hf(x)=\rho_{n,\chi}(h)f(h^{-1}x)=f(x)$; and for $F\in\mathrm{Ind}_{\Gamma_0}V_R(\chi\psi)$, $R_\sigma F=\chi(a_\sigma)\chi\psi(a_\sigma)^{-1}F=\psi(d_\sigma)F$. (5) The proof of Proposition 29.9(3),(4) with $\Gamma_{\mathsf{H}}(N)$ for $\Gamma_1(N)$: the transversal exists because $\varsigma_N(A_0'')=G_N$ there, and $a_g\equiv1\pmod N$ for $g\in\{\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right),p\sigma_p\}$; $\rho_{n,\chi}(p\sigma_p)=\chi(1)p^n\rho_n(\sigma_p)=p^n\chi(p)\rho_{n,\chi}(\sigma_p)$ as $a_{\sigma_p}\equiv p^{-1}$. (6) Proposition 29.8(3)'s argument for the finite abelian group H. □

Proposition 29.19 (The Heilbronn action on $\Gamma_{\mathsf{H}}(N)$)**.** *[proved here] Proposition 28.9 and Remark 29.12 hold for* $A=\Gamma_{\mathsf{H}}(N)$*,* $-1\in\mathsf{H}$*,* n *even with* $\chi(-1)=1$*, on the table of Proposition 29.18(2): with* $J\cdot\epsilon$ *the class of the* H*-orbit of* $\mathrm{row}_2(w_J)\epsilon$ *and* $h_J(\epsilon):=w_J\,\epsilon\,w^{-1}_{J\cdot\epsilon}$*,*

$$H^{\mathsf{H}}_{N,p,\chi}:=\sum_{\epsilon\in\mathrm{Mer}_p}\ \sum_{J\ \textit{admissible}}E_{J\cdot\epsilon,\,J}\otimes\chi\big(d_{h_J(\epsilon)}\big)\,\rho_n\big(\mathrm{adj}\,h_J(\epsilon)\big)$$

preserves Rel *and induces* T_p *(*$p\nmid N$*),* U_p *(*$p\mid N$*) of Proposition 29.18(5) on* $H^1(\Gamma_{\mathsf{H}}(N);V_R(\chi))$*,* $6\in R^\times$*.*

Proof. The proof of Proposition 28.9 with two sentences changed; the rest (Lemma 28.7 for A of finite index with $-I \in A$, Merel's decomposition, the left-coset bookkeeping) does not use $A = \Gamma_0(N)$. *Step 1*: for $p \nmid N$, every $\beta \in AgA$ is upper triangular modulo N with $a_\beta d_\beta \equiv p$ and $d_\beta \in p\mathsf{H}$ (d is multiplicative modulo N on $\Gamma_0(N)$-type matrices, $d_g = p$); if $\Gamma\beta = \Gamma\beta'$, then $\gamma = \beta' \operatorname{adj}\beta/p \in \Gamma$ has $\gamma_{21} \equiv 0$ and $d_\gamma \equiv d_{\beta'} a_\beta/p \equiv d_{\beta'} d_\beta^{-1} \in \mathsf{H}$, so $A\beta = A\beta'$; the count $|A\backslash AgA| = p+1$ (needed for the exhaustion) holds because $\Gamma_1(N) \subseteq A$ and $\Gamma_1(N) \to \mathrm{SL}_2(\mathbb{F}_p)$ is onto for $p \nmid N$ (Chinese remainder theorem), so $(0{:}1) \in \mathbb{P}^1(\mathbb{F}_p)$ is attained inside A (Proposition 28.9's element $\left(\begin{smallmatrix} p & u \\ N & v \end{smallmatrix}\right)$ with v arbitrary is *not* available in A; v0.6.0-pre.2, F-U5 m6); for $p \mid N$ the representatives gT^b are unchanged and the image is $\{(1{:}y)\}$ as before. *Step 3*: for $p \nmid N$, $\mathrm{row}_2(\gamma_\epsilon) \equiv (d_\beta/p)\,\mathrm{row}_2(w_J)\epsilon$ with $d_\beta/p \in \mathsf{H}$, so the class is $J \cdot \epsilon$; for $p \mid N$ it is exact. The character scalar is Remark 29.12's, $a_{\operatorname{adj} h} = d_h$ a unit modulo N at every admissible pair; moreover $\mathrm{row}_2(w_J)\epsilon \equiv \eta\,\mathrm{row}_2(w_{J\cdot\epsilon}) \pmod N$ with $\eta \in \mathsf{H}$ by the definition of $J \cdot \epsilon$, so $d_{h_J(\epsilon)} \equiv \eta \in \mathsf{H}$: *only $\chi|_{\mathsf{H}}$ enters the operator*, as it must (the extension of χ to G_N normalises the diamonds, not the Hecke operators; F-U5 m7). □

Lemma 29.20 (Specialisation and coinvariants without division)**.** *[proved here] Let $\Gamma'' \trianglelefteq \Gamma' \leq \Gamma$ be of finite index, $C := \Gamma'/\Gamma''$ abelian, R commutative with $6 \in R^\times$, and X an $R[\Gamma']$-module free of finite rank over R. Let C act on $H^1(\Gamma''; X)$ by conjugation (the diamonds of Proposition 29.18(4) when $\Gamma'' = \Gamma_{\mathsf{H}}(N)$, $\Gamma' = \Gamma_0(N)$). Then for every finitely presented $R[C]$-module P, Γ' acting on P through $\Gamma' \to C$, $\gamma \mapsto (\gamma\Gamma'')^{-1}$ — **the inverse of the quotient map** (v0.6.0-pre.2, F-U5 M1: under Shapiro's isomorphism $\mathrm{Ind}_{\Gamma''}^{\Gamma'} X \cong R[C] \otimes_R X$, $f \mapsto \sum_{u \in C} u \otimes u^{-1} f(u)$, the conjugation R_c becomes multiplication by c on the group-ring factor while Γ' acts there by $\bar\gamma^{-1} \otimes \rho(\gamma)$; for $\Gamma'' = \Gamma_{\mathsf{H}}(N)$ the map is $\gamma \mapsto a_\gamma \bmod \mathsf{H}$),*

$$H^1(\Gamma''; X) \otimes_{R[C]} P \;\cong\; H^1(\Gamma'; X \otimes_R P),$$

naturally in P; in particular (a) $P = R_\psi$ for a character ψ of C: $H^1(\Gamma''; X) \otimes_{R[C]} R_\psi \cong H^1(\Gamma'; X(\psi))$, where $X(\psi)$ is X with γ acting by $\psi(a_\gamma)\,\rho(\gamma)$ — for $X = V_R(\chi)$ this is $V_R(\chi\psi)$ of Definition 29.17, i.e. the ψ-coinvariants of the diamonds ($\langle d\rangle \sim \psi(d)$) are the $\chi\psi$-model, *in agreement with the eigenspace statement of Proposition 29.18(4) (checked on the model: $q(f) := \sum_u \psi(u)\rho_{n,\chi}(\sigma_u)^{-1} f(\sigma_u)$ has $q(R_\sigma f) = \psi(d_\sigma) q(f)$ and $q(\gamma f) = \psi(d_\gamma)^{-1}\chi(a_\gamma)\rho_n(\gamma) q(f) = \rho_{n,\chi\psi}(\gamma) q(f)$); (b) $P = R[C/C']$ for a subgroup $C' = \Gamma'''/\Gamma''$: the C'-coinvariants of $H^1(\Gamma''; X)$ are $H^1(\Gamma'''; X)$. The isomorphisms commute with every operator $[\Gamma' g \Gamma'] \,/\, [\Gamma'' g \Gamma'']$ of Proposition 29.18(5) whose transversal is chosen in Γ'', and with the ordinary projector e of U_p when both sides carry it.*

Proof. Shapiro: $H^1(\Gamma''; X) \cong H^1(\Gamma'; R[C] \otimes_R X)$, C acting on $H^1(\Gamma''; X)$ by conjugation corresponding to the right regular action on $R[C]$ (Lemma 29.5, Proposition 29.8(1)'s computation). $H^2(\Gamma'; -) = 0$ on $R[\Gamma']$-modules ($6 \in R^\times$, Lemma 28.4: a projective resolution of length one), so $H^1(\Gamma'; -)$ is right exact and additive; choosing a presentation $R[C]^a \to R[C]^b \to P \to 0$ and tensoring $R[C] \otimes X$ over $R[C]$ with it gives the isomorphism (the five lemma on the two right-exact rows). The operators: on both sides they are given by the same $r_t \in \Gamma''$ and the same level-one τ (Proposition 29.9(4)'s choice), so Shapiro's isomorphism intertwines them term by term; e is a limit of powers of U_p in the finite (or profinite) commutative algebra they generate, and $e(M \otimes P) = eM \otimes P$. □

Lemma 29.21 (The wild character models: torsion and lengths)**.** *[proved here] Let $N = \ell^r$, $n = 0$, $\chi = \chi_{i,t}$ with values in $\mathcal{O}_j$, χ even (i even), and $EC(\chi)$ the right-hand side of Proposition 29.11(4) for χ (over $\operatorname{Frac}\mathcal{O}_j$ with ζ_{12} adjoined; it depends on χ only through ω^i on $\mu_4, \mu_6 \subset \mu_{\ell-1}$). Put $c := 0$ if $i \not\equiv 0 \pmod{\ell-1}$, $c := c(\psi_t)$ of Lemma 29.16(3) if $i \equiv 0$ and $\psi_t \neq 1$.*

1. *If* $\chi \neq 1$*:* $H^1(\Gamma_0(\ell^r);\mathcal{O}_j(\chi)) \cong \mathcal{O}_j^{EC(\chi)} \oplus \mathcal{O}_j/\pi^c$ *and, for every* e, $H^1(\Gamma_0(\ell^r);\mathcal{O}_{j,e}(\chi)) \cong \mathcal{O}_{j,e}^{EC(\chi)} \oplus \mathcal{O}/\pi^{\min(c,e)}$, $H^0 \cong \mathcal{O}/\pi^{\min(c,e)}$.
2. *The ordinary projector* e *of* U_ℓ *kills the torsion summand:* $eH^1(\Gamma_0(\ell^r);\mathcal{O}_{j,e}(\chi))$ *is free over* $\mathcal{O}_{j,e}$ *and equals* $eH^1(\Gamma_0(\ell^r);\mathcal{O}_j(\chi))/\pi^e$.

Proof. (1) $H^2 = 0$ (6 invertible), so the sequences of $0 \to \mathcal{O} \xrightarrow{\pi^e} \mathcal{O} \to \mathcal{O}/\pi^e \to 0$ and $0 \to \mathcal{O} \to \mathcal{K} \to \mathcal{K}/\mathcal{O} \to 0$ ($\mathcal{K}$ the fraction field) give $H^1(\mathcal{O}/\pi^e) = H^1(\mathcal{O})/\pi^e$ and $H^1(\mathcal{O})_{\text{tors}} \cong H^0(\mathcal{K}/\mathcal{O}(\chi))/\operatorname{im} H^0(\mathcal{K}(\chi))$. For $\chi \neq 1$, $H^0(\mathcal{K}(\chi)) = 0$ (some a_γ has $\chi(a_\gamma) \neq 1$), and $H^0(\mathcal{K}/\mathcal{O}(\chi)) = \{x : (\chi(a)-1)x = 0 \ \forall a \in G_{\ell^r}\} = \pi^{-c}\mathcal{O}/\mathcal{O}$ with $c = \min_a v_\pi(\chi(a)-1)$, which is 0 when $\omega^i \neq 1$ (a unit difference) and $c(\psi_t)$ otherwise. The free rank is the dimension over $\mathcal{K}$, which is $EC(\chi)$ (Proposition 29.11(4), $H^0 = 0$). $H^0(\mathcal{O}/\pi^e(\chi))$ is the $\pi^{\min(c,e)}$-torsion of $\mathcal{O}/\pi^e$. (2) The torsion is the image of the connecting map from $H^0(\Gamma_0(\ell^r);\mathcal{K}/\mathcal{O}(\chi))$, which is Hecke-equivariant ([standard: [AS86, Lemma 1.1.1, p. 194]], as in [C, NB4](c)); on H^0, U_ℓ is the sum over $b < \ell$ of $\chi(a_{\delta_b}) = \chi(1)$, i.e. multiplication by ℓ, nilpotent on a finite module, so e kills it; then e commutes with $/\pi^e$ as in (1). □

Lemma 29.22 (Reduced echelon form over a chain ring with free quotient)**.** *[proved here] Let* R *be a chain ring with maximal ideal* (π) *and residue field* k *($\mathbb{Z}/\ell^K$, $W_K(\mathbb{F}_q)$, $\mathcal{O}_{j,e}$), and* Rel $\subseteq R^D$ *a submodule generated by* $g_1, \dots, g_s$ *with* $R^D/$Rel *free. Then* Rel *is a free direct summand; its image in* k^D *is spanned by the* $\bar{g}_i$ *and has the same rank; and* Rel *has a unique reduced row echelon basis* with unit pivots*, whose pivot columns are those of the reduced row echelon form of* $\langle \bar{g}_i \rangle \subseteq k^D$*. The free columns index a basis of the cokernel.*

Proof. A free quotient splits, so Rel is projective, hence free over the local ring, and $\operatorname{Tor}_1(R^D/\text{Rel}, k) = 0$ makes Rel $\otimes\, k \to k^D$ injective with image $\langle \bar{g}_i \rangle$. Let P be the pivot columns over k. The projection Rel $\to R^P$ is an isomorphism modulo π, between free modules of rank $|P|$, hence an isomorphism (Nakayama); the preimages of the unit vectors are the unique reduced basis with unit pivots on P. □

Applicability. For the cokernel of Proposition 29.11(3) the quotient is H^1, free over $\mathbb{Z}/\ell^K$ for the $\Gamma_{\mathsf{H}}(\ell^r)$-models of [U] part 5 (their $H^1(\mathbb{Z}_\ell)$ is torsion-free: Lemma 29.21's argument with χ tame, where $c = 0$, or $\chi = 1$, where $H^0(\mathbb{Q}_\ell) \to H^0(\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ is onto). So the existing sparse elimination over $\mathbb{F}_\ell$ (`Hecke/HeckeLevelNSparse`, `Hecke/HeckeNebentypusSparse`) fixes the pivots and the free columns, and the lift to $\mathbb{Z}/\ell^K$ is a back-substitution with unit pivots. Over $\mathcal{O}_{j,e}$ at $i \equiv 0$ the cokernel has the torsion of Lemma 29.21(1) and the lemma does *not* apply: there the elimination is a genuine Howell form (Lemma 29.16(2)).

Remark 29.23 (What is checkable: (NB7)–(NB10)). Committed before slot-22 P10's code (rule 20), convention of Remark 26.6. Cells: $\ell \in \{5, 7, 11, 13\}$, $r \leq 3$, $n = 0$; numbers in `docs/implementation/slot22/P10_spec.md` §7.

(NB7) [proved here] (Lemma 29.16(3), Proposition 29.11(3)) **Reduction modulo π.** The relation matrices and every operator of the $\chi_{i,t}$-model over $\mathcal{O}_{j,1} = \mathbb{F}_\ell$ are entrywise those of the ω^i-model over $\mathbb{F}_\ell$ ([C, NB2]'s bit-for-bit standard), so the cokernel and the operators on it agree.

(NB8) [proved here] (Lemma 29.21) **Lengths.** length $H^1(\Gamma_0(\ell^r);\mathcal{O}_{j,e}(\chi_{i,t})) = e \cdot EC + \min(c,e)$, length $H^0 = \min(c,e)$, and the ordinary part is free over $\mathcal{O}_{j,e}$; its rank is $2d(a)+1$ ($a \equiv i+2$, [U] Theorem `thm:u-towercontrol`(5)) — the rank clause [proved here] *given* [U] Proposition `prop:u-towerorient` ([verified computationally] at $r = 1$, gated by (U18)(o); F-U5 M3).

(NB9) [proved here] (Lemma 29.20(a)) **Wild specialisation.** $eH^1(\Gamma_{\mathsf{H}_r}(\ell^r);\mathbb{Z}/\ell^K(\omega^i)) \otimes_{(\mathbb{Z}/\ell^K)[G_r]} \mathcal{O}_{j,e}$, $\langle u \rangle \mapsto \zeta^t$, is $eH^1(\Gamma_0(\ell^r);\mathcal{O}_{j,e}(\chi_{i,t}))$ ($e \leq K\varphi_j$, H_r the Teichmüller subgroup, $G_r = G_{\ell^r}/\mathsf{H}_r$), with equal T_p and U_ℓ; computed by the two independent routes (the Γ_0 character model over $\mathcal{O}_{j,e}$ with a Howell elimination; the Γ_{H} model over $\mathbb{Z}/\ell^K$ tensored down).

(NB10) [proved here] (Proposition 29.18(2),(6)) **The Γ_{H} model against its two neighbours.** At $\mathsf{H} = G_N$ it is the character model entry for entry; at $\mathsf{H} = \mathsf{H}_r$, $r \le 2$, its H^1, T_p, U_ℓ and diamonds equal those of e_{ω^i} applied to the $\Gamma_1(\ell^r)$ model.

Remark 29.24 (One level interface: $\Gamma_{\mathsf{H}}(N)$ for WAKE-1 and for the tower). (Not a statement; the code contract.) `hecke::level::CosetLevel` (`Hecke/HeckeLevelEll.h`: `N`, `m`, `rep`, `class_of`, `contains`, `contains_wide`, `name`) is the one interface. Slot-22 WAKE-1 adds `GammaHLevelN` (`Hecke/HeckeGammaH`; H by generators) with the table of Proposition 29.18(2), its sub-transversal (the analogue of `Gamma1SubTransversal`) and its diamonds; P10 *reuses* it with $N = \ell^r$ and $\mathsf{H} = \mathsf{H}_r$ generated by the Teichmüller lift of a primitive root modulo ℓ, and adds only what is level-independent: the tame scalar $\chi(a_\delta)$ (`neb::BlockScalar`, already generic in δ), the chain-ring elimination (Lemma 29.22) and Proposition 29.19's operator. `Gamma1LevelN` is the case $\mathsf{H} = \{1\}$ and stays the oracle ((NB10)); `Gamma0LevelN` the case $\mathsf{H} = G_N$. A second Γ_{H} class is not to be written (the slot-22 runbook: "reuse WAKE-1's `CosetLevel` additions, do not fork them"); if WAKE-1's class lacks the diamonds of a quotient G_N/H of order prime to 6 but divisible by ℓ, P10 extends that class.

Code map of §29.7 **Planned** (slot-22 P10, after F-U5's ACCEPT): `Hecke/HeckeChainRing.h` (header: $\mathbb{Z}/\ell^K$ and $\mathcal{O}_{j,e}$ element arithmetic, v_π, unit inverse, Howell / Smith), `Hecke/HeckeNebentypusWild.{h` (the $\chi_{i,t}$ character model over $\mathcal{O}_{j,e}$: Definition 29.15, Lemmas 29.16, 29.21), WAKE-1's `Hecke/HeckeGammaH` (Proposition 29.18) extended by P10 with Proposition 29.19 and Lemma 29.22; validation `HeckeNebentypusWildValidation` (`standard`) gating (NB7)–(NB10).

29.8 The level-change cone and inflation–restriction

What this subsection fixes. (i) For a pair $B \le A$ of finite-index subgroups of Γ with compatible characters, the **restriction** and the **trace** as fibre maps of the induced models (Definition 29.25, Lemmas 29.26, 29.28). (ii) The **mapping cone** of the restriction and its **six-term exact sequence** on the fixed model, at every ring and every ℓ (Definition 29.29, Lemma 29.30); what it computes (Proposition 29.31); when the model computes group cohomology at $\ell \in \{2,3\}$ (Proposition 29.32); the split case (Proposition 29.33). (iii) The **Hecke operators** on the sequence (Proposition 29.34). (iv) The **normal case**: the cone sequence is inflation–restriction plus two pieces (Definition 29.35, Proposition 29.36), made explicit for a cyclic quotient of order p (Proposition 29.37, Lemmas 29.38, 29.39). (v) **Which pairs are normal**: $\Gamma_0(MN) \trianglelefteq \Gamma_0(N)$ iff $M \mid \gcd(N, 24)$ (Proposition 29.40), the steps of the p-tower (Proposition 29.41), level one (Remark 29.42). (vi) The obligations (Obligation 29.43), the predictions (LES1)–(LES9) (Remark 29.44), the deferred old-space cone (Remark 29.45) and the code map (planned). *Scope* (slot-22 LES-0, owner questions Q2–Q5 at the recommendation): coefficients $\mathbb{F}_\ell$ (and $\mathbb{F}_q$) and $\mathbb{Z}/\ell^K$; at $\ell \in \{2,3\}$ the sequence is the model's (Bredon) sequence, the group-cohomology sequence in all degrees is deferred to the free model of §23 (stated over $\mathbb{Z}$ at v1.23.0-pre.18: Proposition 31.10); the old-space U_p cone is deferred; no console verb. *Notation*: as in §29; $B \le A \le \Gamma$ (the names of Proposition 27.2), $m_A := [\Gamma : A]$, $m_B := [\Gamma : B]$, $U := ST$; $V_R = \mathrm{Sym}^n$ over R with ρ_n. $\mathsf{G} := A/B$ in the normal case (sans-serif, *not* the $G_N = (\mathbb{Z}/N)^\times$ of §29); Nm_{G} its norm element; $\alpha_{B/A}$, $\omega_{B/A}$, $Q_{A/B}$, G, Γ^2, Γ^3 are new symbols of this subsection and mean nothing else.

Definition 29.25 (Level pair with characters)**.** A *character datum* on a finite-index $A \le \Gamma$ is a level $N_A \ge 1$ with $A \subseteq \Gamma_0(N_A)$ and a character $\chi\colon G_{N_A} \to R^\times$ (Definition 29.1(3); $N_A = 1$, $\chi = 1$ is allowed for every A); $V_R(\chi)$ is V_R with A acting by $\rho_{n,\chi}(\gamma) = \chi(a_\gamma)\rho_n(\gamma)$, a homomorphism on $\Gamma_0(N_A)$ (Lemma 29.2(2)). A *level pair* is $(A,\chi) \supseteq (B,\chi')$: $B \le A$ of finite index, with character data (N_A,χ) on A and (N_B,χ') on B. The levels in use are $\Gamma_{\mathsf{H}}(N)$ (Definition 29.17; $\Gamma_0(N)$, $\Gamma_1(N)$ among them), their intersections $\Gamma_{\mathsf{H}}(N) \cap \Gamma_0(N')$, and preimages of subgroups of $\mathrm{SL}_2(\mathbb{Z}/M)$ ($\Gamma(M)$, and Γ^2, Γ^3 of Remark 29.42), the last with $\chi = 1$.

$\mathrm{Ind}_A^\Gamma V_R(\chi)$ and its model L_A^χ are those of Lemma 29.5, on fixed tables $w_1 = I, \dots, w_{m_A}$ of $A\backslash\Gamma$ and $v_1 = I, \dots, v_{m_B}$ of $B\backslash\Gamma$; $a(i)$ is the index with $Av_i = Aw_{a(i)}$ and $h_i := v_i w_{a(i)}^{-1} \in A$.

Lemma 29.26 (Compatibility)**.** *[proved here] In a level pair, $V_R(\chi)|_B = V_R(\chi')$ as $R[B]$-modules iff $\chi(a_\gamma) = \chi'(a_\gamma)$ for every $\gamma \in B$. For $B = \Gamma_{\mathsf{H}'}(N') \subseteq A = \Gamma_{\mathsf{H}}(N)$ $(N \mid N')$ this is $\chi'(u) = \chi(u \bmod N)$ for every $u \in \mathsf{H}'$; in particular, for $\Gamma_0(MN) \subseteq \Gamma_0(N)$ it is $\chi' = \chi \circ \mathrm{red}_{MN\to N}$.*

Proof. Both actions are $\gamma \mapsto (\text{scalar}) \cdot \rho_n(\gamma)$ and $\rho_n(\gamma)$ is invertible, so they agree iff the scalars agree. On $\Gamma_{\mathsf{H}'}(N')$ the values $a_\gamma \bmod N'$ are exactly H': $a_\gamma \equiv d_\gamma^{-1}$ (Lemma 29.2(1)), $d_\gamma \bmod N'$ ranges over H' by the surjectivity of $\varsigma_{N'}$, and H' is a group. □

A level pair whose characters are *not* compatible has no restriction map of the kind below; the code refuses it by name (`les_incompatible_character`). This is the hypothesis of the owner's request ("compatible Nebentypes").

Definition 29.27 (The restriction and the trace on the fibre)**.** For a compatible level pair, view $\mathrm{Ind}_A^\Gamma V_R(\chi) = \{f\colon \Gamma \to V_R : f(ax) = \rho_{n,\chi}(a) f(x),\ a \in A\}$ (Definition 12.1) and let $r_1 = I, \dots, r_{[A:B]}$ represent $B\backslash A$. Put

$$\alpha_{B/A} f := f \ \text{ (the inclusion of functions)}, \qquad (\omega_{B/A} g)(x) := \sum_t \rho_{n,\chi}(r_t)^{-1}\, g(r_t x),$$

$\alpha_{B/A}\colon \mathrm{Ind}_A^\Gamma V_R(\chi) \to \mathrm{Ind}_B^\Gamma V_R(\chi')$ and $\omega_{B/A}\colon \mathrm{Ind}_B^\Gamma V_R(\chi') \to \mathrm{Ind}_A^\Gamma V_R(\chi)$. In the coordinates of Definition 12.1,

$$(\alpha_{B/A} f)_i = \chi(a_{h_i})\, \rho_n(h_i)\, f_{a(i)}, \qquad (\omega_{B/A} g)_a = \sum_{i:\, a(i)=a} \chi(a_{h_i})^{-1} \rho_n(h_i)^{-1} g_i. \tag{75}$$

Lemma 29.28 (Restriction and trace are fibre maps)**.** *[proved here] For a compatible level pair over any commutative R, no hypothesis on ℓ or n:*

1. *$\alpha_{B/A}$ and $\omega_{B/A}$ are well defined, independent of the r_t, morphisms of $R[\Gamma]$-modules given by (75), hence (one monomial fibre matrix at every simplex) morphisms of matrix local systems $L_A^\chi \rightleftarrows L_B^{\chi'}$ (Theorem 4.5); $\alpha_{B/A}$ is injective and split over R, $\omega_{B/A}$ is surjective.*
2. *$\omega_{B/A} \circ \alpha_{B/A} = [A : B] \cdot \mathrm{id}$.*
3. *Under the reciprocity $\phi \mapsto \phi(\cdot)(I)$ of Proposition 27.2(3), $(\alpha_{B/A})_*$ is $\psi \mapsto \psi$ (an A-equivariant cochain read as a B-equivariant one) and $(\omega_{B/A})_*$ is the transfer $\psi \mapsto \sum_t \rho_{n,\chi}(r_t)^{-1} \psi(r_t\,\cdot)$; both are cochain maps $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L_A^\chi) \rightleftarrows C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L_B^{\chi'})$, and for $6 \in R^\times$ they induce res_B^A and cor_B^A on $H^\bullet(A; V_R(\chi))$, $H^\bullet(B; V_R(\chi'))$ [standard: [Bro82, Ch. III.9–10], as in Proposition 12.7; locator read at an earlier R20-L pass ([Bro82, Prop. III.9.5, p. 82])].*
4. *At $\chi = \chi' = 1$, (75) is Proposition 27.2(4)(b)'s $\alpha^0_{B/A}$ and trace, entry for entry on the same tables.*

Proof. (1) For $b \in B$, $f(bx) = \rho_{n,\chi}(b) f(x) = \rho_{n,\chi'}(b) f(x)$ (Lemma 29.26), so $\alpha_{B/A} f \in \mathrm{Ind}_B$; right translation commutes with the inclusion. For $\omega_{B/A}$: replacing r_t by br_t ($b \in B$) changes the term by $\rho(r_t)^{-1}\rho(b)^{-1}\rho(b)$, i.e. not at all ($\rho := \rho_{n,\chi}$, multiplicative on A); for $a \in A$ write $r_t a = b_t r_{t\cdot a}$ ($b_t \in B$, $t \mapsto t \cdot a$ a permutation), then $(\omega g)(ax) = \sum_t \rho(r_t)^{-1}\rho(b_t) g(r_{t\cdot a} x) = \sum_t \rho(a r_{t\cdot a}^{-1}) g(r_{t\cdot a} x) = \rho(a)(\omega g)(x)$; right translation commutes. Coordinates: $(\alpha f)_i = f(v_i) = f(h_i w_{a(i)}) = \rho(h_i) f_{a(i)}$; for ω, the $r_t w_a$ represent the B-cosets in Aw_a, i.e. $r_t w_a = b v_i$ with $a(i) = a$, and then $r_t = b h_i$ and $\rho(r_t)^{-1} g(r_t w_a) = \rho(h_i)^{-1} g(v_i)$. Each block row of α has one non-zero block, an invertible one, and every a occurs as some $a(i)$, so α is injective; $f \mapsto (f_{a(i_a)})_a$ for one i_a with $a(i_a) = a$, corrected by $\rho(h_{i_a})^{-1}$, is a retraction over R. Given $f \in \mathrm{Ind}_A$, the $g \in \mathrm{Ind}_B$ with $g_{i_a} := \chi(a_{h_{i_a}}) \rho(h_{i_a}) f_a$ and $g_i := 0$ otherwise has $\omega g = f$. (2) $(\omega\alpha f)(x) = \sum_t \rho(r_t)^{-1}\rho(r_t) f(x)$.

(3) $(\omega\circ\phi)(y)(I)=\sum_t\rho(r_t)^{-1}\phi(y)(r_t)=\sum_t\rho(r_t)^{-1}\phi(r_ty)(I)$ by the Γ-equivariance of ϕ; the statement for α is immediate. Cochain maps: both are post-composition with module maps. For $6\in R^\times$, $C_\bullet(X)\otimes R$ is a projective resolution of R over $R[A]$ and over $R[B]$ (Remark 7.6, Lemma 28.4), on which restriction of equivariance and the displayed transfer compute res and cor (the cited section). (4) Compare with Proposition 27.2(4)(b): $\rho_n(v_iw_{a(i)}^{-1})=\rho_n(h_i)$, and $\rho_n(h_i)^{-1}=\rho_n(w_av_i^{-1})$ is its trace block. □

Definition 29.29 (The level cone and relative cohomology)**.** For a compatible level pair, $C(\alpha_{B/A})^q:=C^{q+1}_{\mathrm{eq}}(K,\mathcal{R};L^\chi_A)\oplus C^q_{\mathrm{eq}}(K,\mathcal{R};L^{\chi'}_B)$ with $D(c,c'):=(-dc,\ dc'+(\alpha_{B/A})_*c)$ — (38) with r replaced by $(\alpha_{B/A})_*$ — and the *relative cohomology of the pair* $H^q(A,B):=H^{q-1}(C(\alpha_{B/A}))$ (the indexing of H^q_c in Definition 15.1). $H^q(A)$, $H^q(B)$ abbreviate $H^q_{\mathrm{eq}}(K,\mathcal{R};L^\chi_A)$, $H^q_{\mathrm{eq}}(K,\mathcal{R};L^{\chi'}_B)$, and $\mathrm{res}^q:=(\alpha_{B/A})_*$ on H^q, $\mathrm{cor}^q:=(\omega_{B/A})_*$.

Lemma 29.30 (The six-term sequence on the model)**.** *[proved here] For a compatible level pair, over every commutative R, at every ℓ and every n:*

1. *$D^2=0$, and $0\to C^\bullet_{\mathrm{eq}}(L^{\chi'}_B)\to C(\alpha_{B/A})\to C^\bullet_{\mathrm{eq}}(L^\chi_A)[1]\to 0$ is a short exact sequence of complexes whose connecting map is $+\mathrm{res}^{q+1}$.*
2. *$H^q(A)=H^q(B)=0$ for $q\geq 2$, and res^0 is injective.*
3. *Hence $H^q(A,B)=0$ for $q\notin\{1,2\}$ and the long exact sequence is the **six-term exact sequence***

$$0\to H^0(A)\xrightarrow{\mathrm{res}^0}H^0(B)\to H^1(A,B)\to H^1(A)\xrightarrow{\mathrm{res}^1}H^1(B)\to H^2(A,B)\to 0;\qquad(76)$$

over a field $\dim H^1(A,B)=\dim H^0(B)-\dim H^0(A)+\dim\ker\mathrm{res}^1$ and $\dim H^2(A,B)=\dim H^1(B)-\mathrm{rank}\,\mathrm{res}^1$.

Proof. (1) is the proof of Lemma 15.2(2) verbatim: it uses of r only that it is a cochain map, which $(\alpha_{B/A})_*$ is (Lemma 29.28(3)). (2) By Theorem 5.10 the model complex of L^χ_A is $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X),M)$ with $M=\mathrm{Ind}_A^\Gamma V_R(\chi)$, and Proposition 23.3(2) — which holds for *every* $\mathbb{Z}[\Gamma]$-module M — makes it chain homotopy equivalent to the two-term complex (65); so $H^q=0$ for $q\geq 2$ and $H^0=M^{\langle S\rangle}\cap M^{\langle U\rangle}=M^\Gamma$, likewise for B. The equivalence is a chain map on the chain side, so it is natural in M and res^q is the map induced by $\alpha_{B/A}$ on the tree complexes; on H^0 this is the restriction of the injective $\alpha_{B/A}$ to M^Γ. (3) Exactness of the long sequence and the two vanishings; $H^3(A,B)$ sits between $H^2(B)=0$ and $H^3(A)=0$. □

Contrast with the boundary cone of §15. There r is onto and $H^2_{\mathrm{eq}}=0$ is a hypothesis (Lemma 15.2(5)); here $\alpha_{B/A}$ is injective and the vanishing is unconditional, because both ends are cohomologies of the same model with different coefficient modules. The analogue of $\ker r$ is $\mathrm{coker}\,\alpha_{B/A}$ (Proposition 29.31).

Proposition 29.31 (What the cone computes)**.** *[proved here] For a compatible level pair over a commutative R:*

1. *$Q_{A/B}:=\mathrm{Ind}_B^A V_R(\chi')/V_R(\chi)$ (the cokernel of the unit $v\mapsto(x\mapsto\rho_{n,\chi}(x)v)$) is an $R[A]$-module, free over R of rank $([A:B]-1)(n+1)$, and $\mathrm{coker}\,\alpha_{B/A}\cong\mathrm{Ind}_A^\Gamma Q_{A/B}$.*
2. *If $6\in R^\times$, then $C(\alpha_{B/A})\to C^\bullet_{\mathrm{eq}}(K,\mathcal{R};L_{\mathrm{coker}\,\alpha})$, $(c,c')\mapsto[c']$, is a quasi-isomorphism, and $H^q(A,B)\cong H^{q-1}(A;Q_{A/B})$ for every q (Shapiro, Lemma 29.5).*
3. *At $\ell\in\{2,3\}$ (2) is* not *asserted: the sequence (76) is a sequence of Bredon groups (Proposition 23.3(3)), and an end is group cohomology exactly when Proposition 29.32 applies to it.*

(For orientation only, not used: with this shift $H^\bullet(A;Q_{A/B})$ is the relative cohomology of the group pair (A,B) in the sense of [Ada54] and of [BE78], $\mathrm{Hom}(\Delta,V)=Q_{A/B}$ for $\Delta=\ker(\mathbb{Z}[A/B]\to\mathbb{Z})$; neither paper is held, no locator is claimed, VERIFY.)

Proof. (1) Shapiro coordinates $f \mapsto (f(r_t))_t$ identify $\mathrm{Ind}_B^A V_R(\chi')$ with $V_R^{[A:B]}$ over R; the unit is $v \mapsto (\rho(r_t)v)_t$, whose image meets $\{f : f(r_1) = 0\}$ trivially and spans with it ($r_1 = I$), so the quotient is free of the stated rank. Induction from A to Γ is exact (it is a finite direct sum over R of copies of the module, Definition 12.1) and transitive, $\mathrm{Ind}_A^\Gamma \mathrm{Ind}_B^A = \mathrm{Ind}_B^\Gamma$, with $\mathrm{Ind}_A^\Gamma(\mathrm{unit}) = \alpha_{B/A}$. (2) For $6 \in R^\times$ the fixed-point functors $M \mapsto M^{\Gamma_\sigma}$ ($|\Gamma_\sigma| \mid 12$) are exact (averaging), so $0 \to C_{\mathrm{eq}}^\bullet(L_A) \to C_{\mathrm{eq}}^\bullet(L_B) \to C_{\mathrm{eq}}^\bullet(L_{\mathrm{coker}\,\alpha}) \to 0$ is exact (Theorem 5.10: the model complex is $\bigoplus M^{\Gamma_\sigma}$ in each degree). The map of the statement is a cochain map ($D(c, c') \mapsto [dc']$), surjective, with kernel $\{(c, \alpha_* c'') : c \in C_A^{q+1}, c'' \in C_A^q\} \cong C(\mathrm{id}_{C_A})$, which is acyclic; so it is a quasi-isomorphism. Then $H^{q-1}(C_{\mathrm{eq}}^\bullet(L_{\mathrm{Ind}_A Q})) \cong H^{q-1}(\Gamma; \mathrm{Ind}_A^\Gamma Q_{A/B}) \cong H^{q-1}(A; Q_{A/B})$ by Lemma 29.5 (Proposition 12.4(4) for an arbitrary representation). (3) is Proposition 23.3(3) applied to both ends. □

Proposition 29.32 (When the model computes group cohomology at $\ell \in \{2, 3\}$). *[proved here] Let $B \le \Gamma$ have finite index, X an $R[B]$-module, and suppose that* every finite subgroup of B has order in $R^\times$. *Then $H_{\mathrm{eq}}^q(K, \mathcal{R}; L_B^X) \cong H^q(B; X)$ for every q (so $H^q(B; X) = 0$ for $q \ge 2$), whatever ℓ. The finite subgroups of Γ have orders $1, 2, 3, 4, 6$, and every subgroup of even order contains $-I$; so at $\ell = 2$ the hypothesis is $-I \notin B$, and at $\ell = 3$ it is "B contains no element of order 3". It holds at every ℓ for torsion-free B: $\Gamma_1(N)$ for $N \ge 4$, and every subgroup of it (among them $\Gamma(N)$, $N \ge 4$, and $\Gamma_1(p^r) \cap \Gamma_0(p^{r+1})$, $p^r \ge 4$), and $\Gamma(3)$. $\Gamma_0(N)$ contains an element of order 3 iff $x^2 + x + 1 \equiv 0 \pmod N$ is solvable (not for $2 \mid N$ or $9 \mid N$; yes for $N = 3$; the full characterisation — iff $9 \nmid N$ and no prime $p \equiv 2 \pmod 3$ divides N — is Proposition 31.9, v1.23.0-pre.18, referee m2).*

Proof. By Proposition 23.3(2) the model complex is chain homotopy equivalent to $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(\mathcal{T}), \mathrm{Ind}_B^\Gamma X) \cong \mathrm{Hom}_{\mathbb{Z}[B]}(C_\bullet(\mathcal{T}), X)$ (Frobenius reciprocity, $\phi \mapsto \phi(\cdot)(I)$, as in Proposition 27.2(3)). The proof of Lemma 28.4 shows that $C_\bullet(\mathcal{T}) \otimes R \to R$ is a projective resolution over $R[B]$ as soon as the orders of the groups $B \cap \gamma F \gamma^{-1}$ ($F \in \{\langle -I\rangle, \langle S\rangle, \langle U\rangle\}$) are units — these are finite subgroups of B, which is our hypothesis. So the complex computes $H^\bullet(B; X)$. Finite subgroups: they are cyclic, contained in a conjugate of $\langle S\rangle$ ($\cong \mathbb{Z}/4$) or of $\langle U\rangle$ ($\cong \mathbb{Z}/6$) (Proposition 23.3: $\mathcal{D}$ is the family of finite subgroups), and the unique element of order 2 in either is $-I$. Torsion: a torsion element $\gamma \ne \pm I$ of Γ has eigenvalues primitive k-th roots of unity, $k \in \{3, 4, 6\}$, so $\mathrm{tr}\,\gamma \in \{-1, 0, 1\}$; on $\Gamma_1(N)$, $\mathrm{tr}\,\gamma \equiv 2 \pmod N$, which is $\not\equiv -1, 0, 1$ for $N \ge 4$, and $-I \notin \Gamma_1(N)$ for $N \ge 3$. $\Gamma(3)$: $-I \notin \Gamma(3)$; order 4 needs trace $0 \not\equiv 2 \pmod 3$; an element of order 6 has a square of order 3; and $\gamma = I + 3X$ of order 3 would satisfy $\gamma^2 + \gamma + I = 0$ (Cayley–Hamilton, trace -1), but $\gamma^2 + \gamma + I = 3I + 9(X + X^2) \equiv 3I \pmod 9$. Order 3 in $\Gamma_0(N)$: $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of order 3 has $a + d = -1$, and $N \mid c$ gives $ad \equiv 1$, i.e. $a^2 + a + 1 \equiv 0 \pmod N$; conversely, for such an a, $\begin{pmatrix} a & -(a^2 + a + 1)/N \\ N & -1 - a \end{pmatrix}$ has determinant 1 and trace -1. $a^2 + a + 1$ is odd, and takes no value $\equiv 0 \pmod 9$ (check a mod 9); $a = 1$ works for $N = 3$. □

Proposition 29.33 (Index prime to ℓ: the split case). *[proved here] If $[A : B] \in R^\times$, then $e := [A : B]^{-1} \alpha_{B/A} \omega_{B/A}$ is an idempotent $R[\Gamma]$-endomorphism of $\mathrm{Ind}_B^\Gamma V_R(\chi')$ with image $\alpha_{B/A}(\mathrm{Ind}_A^\Gamma V_R(\chi))$ and kernel $\ker \omega_{B/A}$, so $\mathrm{Ind}_B^\Gamma V_R(\chi') = \alpha_{B/A}(\mathrm{Ind}_A) \oplus \ker \omega_{B/A}$ on the fibre, at every ℓ. Hence every res^q is split injective, $H^q(B) = \mathrm{res}^q H^q(A) \oplus \ker \mathrm{cor}^q$, $H^1(A, B) \cong \mathrm{coker}\,\mathrm{res}^0 \cong \ker \mathrm{cor}^0$ and $H^2(A, B) \cong \mathrm{coker}\,\mathrm{res}^1 \cong \ker \mathrm{cor}^1$. Instances: $\Gamma_0(p) \subset \Gamma$ at $\ell \nmid p + 1$; the first step $\Gamma_0(N) \cap \Gamma_1(p) \trianglelefteq \Gamma_0(pN)$ (index $p - 1$) at $\ell = p$.*

Proof. $e^2 = [A : B]^{-2} \alpha(\omega\alpha)\omega = e$ by Lemma 29.28(2), $e\alpha = \alpha$ and $\omega e = \omega$, so $\mathrm{im}\, e = \mathrm{im}\,\alpha$ and $\ker e = \ker \omega$. The model complex is additive in the fibre module (Theorem 5.10), so $C_{\mathrm{eq}}^\bullet(L_B) = \alpha_* C_{\mathrm{eq}}^\bullet(L_A) \oplus C_{\mathrm{eq}}^\bullet(L_{\ker \omega})$ as complexes; $\mathrm{cor} \circ \mathrm{res} = [A : B]$ is invertible, so res is split injective and $\ker \mathrm{cor}^q = H^q(L_{\ker \omega})$. Insert in (76). □

Proposition 29.34 (Hecke operators on the six-term sequence)**.** *Let $(A,\chi)\supseteq(B,\chi')$ be a compatible level pair with $A=\Gamma_{\mathsf H}(N_A)\cap\Gamma_0(N'_A)$, $B=\Gamma_{\mathsf H'}(N_B)\cap\Gamma_0(N'_B)$ with $N_A\mid N'_A$, $N_B\mid N'_B$ (any of the levels of Definition 29.25 of this form; so $\Gamma_1(N'_A)\subseteq A$, $\Gamma_1(N'_B)\subseteq B$), $g=\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)$, and the operators of Propositions 27.2, 29.9 in the reciprocity form $\psi\mapsto\sum_t\rho_{n,\chi}(r_t^{-1}g)\,\psi(\tau(r_t\,\cdot))$, r_t a transversal of $A''\backslash A$, $A''=A\cap gAg^{-1}$, with the level-1 τ of $T_{\Gamma g\Gamma}$.*

1. *[proved here] **(The criterion.)** If $B\cap A''=B''=B\cap gBg^{-1}$ and $A=A''B$, then a transversal r_t of $B''\backslash B$ is one of $A''\backslash A$, and with this common transversal $T_g^B\circ(\alpha_{B/A})_*=(\alpha_{B/A})_*\circ T_g^A$ **as cochain maps**, over every R; so $\Phi_g:=(T_g^A,T_g^B)$ is a cochain map of $C(\alpha_{B/A})$ commuting with the two maps of Lemma 29.30(1), and the whole sequence (76) is equivariant — **no pair homotopy** (contrast Definition 15.5).*
2. *[proved here] **(Instances.)** The criterion holds for T_p, $p\nmid N_B$ (any pair), with $r_t\in B$ representing the $p+1$ classes of $\mathbb P^1(\mathbb F_p)$ — and likewise for $A=\Gamma$ and B the preimage of a subgroup of $\mathrm{SL}_2(\mathbb Z/M)$ normalised by $\mathrm{diag}(1,p)$, $p\nmid M$ (the level-one cells Γ^2, Γ^3, $\Gamma(M)$ of (LES4)); for U_p, $p\mid N_A$, when A and B have the* same Γ_0*-level ($N'_A=N'_B$, the $\Gamma_{\mathsf H}$/diamond direction), with $r_t=\left(\begin{smallmatrix}1&0\\tN'_A&1\end{smallmatrix}\right)\in B$, $t<p$; and the diamonds R_σ ($\sigma\in A\cap\Gamma_0(N'_B)$, Definition 29.17) commute with $\alpha_{B/A}$ on the fibre. So the shipped sub-transversals of `Hecke/HeckeGammaH` (`Gamma1SubTransversal`, `GammaHSubTransversal`, inside $\Gamma_1\subseteq B$) are the ones to use: both levels' operators are built from the **smaller** group's transversal.*
3. *[proved here] [verified computationally][a] **(Level-raising U_p: the square does not commute in general.)** If $p\mid N'_A$ and $pN'_A\mid N'_B$, then $B\subseteq A''$, so $A''B=A''\neq A$ and the criterion fails. Let $\mathrm{dg}_g:=[A\,g\,B]=(\mathrm{conj}_g)_*\circ\mathrm{res}$, $\psi\mapsto\rho_n(g)\,\psi\circ\tau$, the one-term cochain map of Proposition 27.2(4)(c) ($B\cap gAg^{-1}=B$, $r_1=I$).*
   (a) *$T_g^B\circ(\alpha_{B/A})_*=[B:B'']\cdot\mathrm{dg}_g=p\cdot\mathrm{dg}_g$ **as cochain maps**, over every R. In particular $U_p^B\circ(\alpha_{B/A})_*=0$ on cochains when $p=0$ in R (e.g. $R=\mathbb F_p$, $\ell=p$).*
   (b) *$T_g^A=(\omega_{A''/A})_*\circ[A\,g\,A'']$ as cochain maps (the factorisation $\mathrm{cor}\circ(\mathrm{conj}_g)_*\circ\mathrm{res}$ of Proposition 27.2(3), read at cochain level), so $(\alpha_{B/A})_*\circ T_g^A=(\alpha_{B/A})_*(\omega_{A''/A})_*\circ[A\,g\,A'']$.*
   (c) *Let $B=A''$ and $6,p\in R^\times$. On H^q, $\mathrm{res}\circ U_p^A-U_p^B\circ\mathrm{res}=p\,(e-1)\,\mathrm{dg}_g$ with the idempotent e of Proposition 29.33; so the square commutes on H^q **iff** $\mathrm{dg}_g(H^q(A))\subseteq\mathrm{res}\,H^q(A)$. This **fails** on H^1 at $(\Gamma_0(25)\subset\Gamma_0(5)$, $\ell=7$, $n=0$, $\chi=1)$.*
   (d) *On p-old spaces this document's $U_p^{(N)}$ is the* adjoint *$U_p^*=W_pU_pW_p^{-1}$ of the classical operator: the relations $U\circ\mathrm{dg}_1=p\,\mathrm{dg}_p$, $U\circ\mathrm{dg}_p=\mathrm{dg}_p\circ T_p-p^n\mathrm{dg}_1$ of Proposition 27.4(1),(3) are those of U_p^* (the classical U_p has $U_p\circ\mathrm{dg}_p=\mathrm{dg}_1$, $U_p\circ\mathrm{dg}_1=\mathrm{dg}_1\circ T_p-p^{k-1}\mathrm{dg}_p$), and* res *is the inclusion of old forms under any Eichler–Shimura normalisation, so the identification is convention-free. The classical U_p, which commutes with the inclusion of old forms, is $[A\,g'A]$ with $g':=\left(\begin{smallmatrix}p&0\\0&1\end{smallmatrix}\right)=SgS^{-1}$ and the twisted datum $\tau':=S\circ\tau\circ S^{-1}$ (it satisfies (T1)–(T3) of Definition 11.1 for g', on the same level-1 table); at $\chi=1$ the criterion (1) holds for it on* every *level-raising pair, with the common transversal $T^b\in B$, $b<p$ ($A\cap g'Ag'^{-1}=A\cap\Gamma^0(p)$, $B\cap g'Bg'^{-1}=B\cap(A\cap\Gamma^0(p))$, $A=(A\cap\Gamma^0(p))B$), so its square is strict. That construction (and the character convention for g', whose a-entry is p) is not needed in slot 22 and stays deferred (Remark 29.45).*

Obligation 29.43(vi) gates (a) and records the square on H^1. [verified computationally][a] evidence: `docs/implementation/slot22/les0/levelraise_up_check.py` (H^1 at $n=0$ as $\mathrm{Hom}(K,\mathbb F_\ell)$, independent of the tree complex and of τ): (a) on H^1, the failure of (c), and the commuting classical square at $(\Gamma_0(25)\subset\Gamma_0(5)$, $\ell\in\{5,7,11\})$, $(\Gamma_0(49)\subset\Gamma_0(7)$, $\ell=$

5), ($\Gamma_0(121) \subset \Gamma_0(11)$, $\ell = 7$), *output* `les0/out_levelraise_up.txt`, 0 *mismatches.*

4. *[proved here]* ***(Independence of the transversal.)*** *Let ψ be A-equivariant and $b \in A'' \subseteq \Gamma_g''$. By (T1) of Definition 11.1, $\tau(b\,r_t\,y) = (g^{-1}bg)\,\tau(r_t\,y)$ with $g^{-1}bg \in A$, hence $\rho_{n,\chi}\big((br_t)^{-1}g\big)\psi(\tau(br_ty)) = \rho_{n,\chi}(r_t^{-1}g)\,\psi(\tau(r_ty))$ (Proposition 8.2(2); the scalars agree by multiplicativity of a modulo N_A on $\Delta_0(N_A)$, Lemma 29.2(2), equivalently $a_{xg} = a_x$ for $g = \mathrm{diag}(1,p)$ together with multiplicativity on $\Gamma_0(N_A)$, and $a_{g^{-1}bg} = a_b$). So each term of the reciprocity form depends only on the coset $A''r_t$, and T_g^A on $C^\bullet_{\mathrm{eq}}(K,\mathrm{Rel};L_A^\chi)$ does* not *depend on the transversal. Consequently, under the criterion of (1), $T_g^B \circ (\alpha_{B/A})_* = (\alpha_{B/A})_* \circ T_g^A$ as cochain maps for* ***every*** *pair of transversals of $A''\backslash A$ and $B''\backslash B$ — the common transversal of (1) is a device of the proof, not a requirement — and no homotopy h and no ambiguity of Proposition 15.16(1) arise. (The operator of (68) as implemented is transversal-independent even on the whole flat cochain space $C^q(K;L_A^\chi)$: it uses r_t only through the indices $j(i,t)$, $a(i,t)$ of (68) (the class of r_tv_i in $\Gamma_g''\backslash\Gamma$ and the class in $A\backslash\Gamma$ of the conjugate $g^{-1}r_tv_is_j^{-1}g$, read off* `BuildLevelCorrespondence` *and* `BuildLevelCorrespondenceChi`*), both functions of the coset $A''r_t$; so the independence is not specific to equivariant cochains in the implementation, and no record on that formula separates it from (T1).) When the criterion of (1) fails the square need not commute even on cohomology, (3)(c).* ***Rule*** *(as in (2)): LES code builds both operators from the* ***smaller*** *group's transversal, for definiteness; no certificate depends on that choice.*

Proof. (1) $B \cap A'' = B''$ makes $B''\backslash B \to A''\backslash A$ injective, and $A = A''B$ makes it surjective, so a transversal of the former is one of the latter. In the reciprocity form both operators are $\psi \mapsto \sum_t \rho_{n,\chi}(r_t^{-1}g)\psi(\tau(r_t\,\cdot))$ (the scalar $\chi(a_{r_t^{-1}g}) = \chi(a_{r_t})^{-1}$ equals $\chi'(a_{r_t})^{-1}$, as $r_t \in B$ and $a_g = 1$; Lemma 29.26), with the same τ, and $(\alpha_{B/A})_*$ is $\psi \mapsto \psi$ (Lemma 29.28(3)); so the two composites are the same formula. $D\Phi_g = \Phi_g D$ is then the computation of Lemma 15.8 with $h = 0$. (2) $gAg^{-1} \cap \Gamma = \{\begin{pmatrix} a & b' \\ c' & d \end{pmatrix} : c' = pc,\ \begin{pmatrix} a & pb' \\ c & d \end{pmatrix} \in A\}$, so $A'' = A \cap \Gamma_0(pN'_A)$ for $p \mid N'_A$ and $A'' = A \cap \Gamma_0(p)$ for $p \nmid N_AN'_A$ (the a-, d-conditions are unchanged), and likewise for B. T_p, $p \nmid N_B$: $B \cap A'' = B \cap \Gamma_0(p) = B''$; $A''\backslash A$ is the set of the $p+1$ points of $\mathbb{P}^1(\mathbb{F}_p)$ (bottom rows modulo p), and $B \to \mathrm{SL}_2(\mathbb{F}_p)$ is onto ($B \supseteq \Gamma_1(\mathrm{lcm})$, Chinese remainder theorem, p prime to the levels of B), so $A = A''B$. The same argument applies to $A = \Gamma$ ($A'' = \Gamma_0(p)$) and B the preimage of a subgroup $\bar{B} \le \mathrm{SL}_2(\mathbb{Z}/M)$ normalised by $D := \mathrm{diag}(1,p) \bmod M$, $p \nmid M$: $B \supseteq \Gamma(M)$ and $\Gamma(M) \to \mathrm{SL}_2(\mathbb{F}_p)$ is onto, and $x \in \Gamma$ lies in gBg^{-1} iff $p \mid c_x$ and $D^{-1}xD \in \bar{B}$ modulo M, so $B \cap gBg^{-1} = B \cap \Gamma_0(p)$. This covers $\Gamma(M)$ and Γ^2, Γ^3 (preimages of the characteristic subgroups A_3, Q_8), i.e. the level-one cells of Remark 29.42. *U_p, same level* $N' := N'_A = N'_B$, $p \mid N'$: here $N_A \mid N'_A$ and $N_B \mid N'_B$ (the levels of $\Gamma_{\mathsf{H}}(N) \cap \Gamma_0(N')$ are written with $N \mid N'$), so $\Gamma_1(N') \subseteq B$; $B \cap A'' = B \cap \Gamma_0(pN') = B''$; the right cosets of $A'' = A \cap \Gamma_0(pN')$ in A are the classes of the invariant $a_\gamma c_\gamma/N' \bmod p$ (the map of Proposition 29.40, here with $M = p$; it is a coset invariant because $(h\gamma)_{11}(h\gamma)_{21}/N' \equiv a_\gamma c_\gamma/N' \pmod p$ for $h \in A''$), and $\begin{pmatrix} 1 & 0 \\ tN' & 1 \end{pmatrix}$ ($t < p$) lies in $\Gamma_1(N') \subseteq B$, has invariant t, and so meets each class. *Diamonds*: $\alpha_{B/A}$ is the inclusion of functions and $R_\sigma f(x) = \rho_{n,\chi}(\sigma)f(\sigma^{-1}x)$ is the same formula on both modules.

(3) For $\gamma \in B$, $pN'_A \mid c_\gamma$, so $B \subseteq A \cap \Gamma_0(pN'_A) = A''$ by the computation of (2). (a) $B'' = B \cap \Gamma_0(pN'_B)$ by (2), of index p in B (the invariant $a_\gamma c_\gamma/N'_B \bmod p$, as in (2)); let r_t ($t < p$) be a transversal. For an A-equivariant ψ and $r_t \in B \subseteq A'' \subseteq \Gamma_g''$, (T1) of Definition 11.1 gives $\tau(r_ty) = (g^{-1}r_tg)\,\tau(y)$ with $g^{-1}r_tg \in A$ (as $r_t \in gAg^{-1}$), so $\psi(\tau(r_ty)) = \rho_{n,\chi}(g^{-1}r_tg)\psi(\tau(y))$ and the t-th term of $T_g^B(\alpha_{B/A})_*\psi$ is $\rho_{n,\chi'}(r_t^{-1}g)\rho_{n,\chi}(g^{-1}r_tg)\psi(\tau(y)) = \rho_n(g)\psi(\tau(y))$: the ρ_n-factors multiply to $\rho_n(g)$ (Proposition 8.2), and the scalars $\chi'(a_{r_t})^{-1}\chi(a_{g^{-1}r_tg})$ cancel, since $a_{g^{-1}r_tg} = a_{r_t}$ and $\chi'(a_{r_t}) = \chi(a_{r_t})$ (Lemma 29.26). All p terms equal $\mathrm{dg}_g\psi$. This is the proof of Proposition 27.4(1) with $(\Gamma,\Gamma_0(\ell))$ replaced by (A,B). (b) With r_t a transversal of $A''\backslash A$ and $a_g = 1$,

$\rho_{n,\chi}(r_t^{-1}g)\psi(\tau(r_ty)) = \rho_{n,\chi}(r_t)^{-1}\big(\rho_n(g)\psi(\tau(r_ty))\big)$; the bracket is $([A\,g\,A'']\psi)(r_ty)$ (Proposition 27.2(4)(c): one term) and the sum over t is the transfer $(\omega_{A''/A})_*$ of Lemma 29.28(3). (c) By (a), (b) and $B = A''$, on H^q: $\mathrm{res}\circ U_p^A = \mathrm{res}\circ\mathrm{cor}\circ\mathrm{dg}_g$ and $U_p^B\circ\mathrm{res} = p\,\mathrm{dg}_g$. As $[A : A''] = p \in R^\times$ (the invariant of (2)), $\mathrm{res}\circ\mathrm{cor} = (\alpha_{B/A}\omega_{B/A})_* = pe$ with e the idempotent of Proposition 29.33, whose image on $H^q(B)$ is $\mathrm{res}\,H^q(A)$ [standard: [Bro82, Ch. III.9–10], through Lemma 29.28(3); locator read at an earlier R20-L pass, Prop. III.9.5, p. 82]. So the defect is $p(e-1)\mathrm{dg}_g$, zero iff $\mathrm{dg}_g(H^q(A)) \subseteq \ker(e-1) = \mathrm{res}\,H^q(A)$. *The cell.* At $n = 0$, $\chi = 1$, $R = \mathbb{F}_7$ the coefficients are trivial, so $H^1(K;\mathbb{F}_7) = \mathrm{Hom}(K,\mathbb{F}_7)$ for $K \in \{A,B\}$. $A = \Gamma_0(5)$ is generated by $-I$, T, $e_1 := \begin{pmatrix}2&-1\\5&-2\end{pmatrix}$, $e_2 := \begin{pmatrix}-2&-1\\5&2\end{pmatrix}$ (both of order 4), and $P := \begin{pmatrix}1&0\\5&1\end{pmatrix} = e_1Te_2$ (generation: `levelraise_up_check.py` reduces every Schreier generator of $\Gamma_0(5)$ to $\pm I$ by these). As $4 \in \mathbb{F}_7^\times$, every $E \in \mathrm{Hom}(A,\mathbb{F}_7)$ kills $-I, e_1, e_2$, so $\dim H^1(A;\mathbb{F}_7) \le 1$, and $= 1$ ((LES3), arm (i) $p = 5$, $n = 0$, read from $B = \Gamma_0(5)$ there); for $E \neq 0$ normalise $E(T) = 1$; then $E(P) = E(e_1) + E(T) + E(e_2) = 1$. Then $\mathrm{dg}_gE(\gamma) = E(g^{-1}\gamma g) = E\begin{pmatrix}a&5b\\c/5&d\end{pmatrix}$, so $\mathrm{dg}_gE(T) = E(T^5) = 5$ and $\mathrm{dg}_gE\begin{pmatrix}1&0\\25&1\end{pmatrix} = E(P) = 1$, while $\mathrm{res}\,E(T) = 1$ and $\mathrm{res}\,E\begin{pmatrix}1&0\\25&1\end{pmatrix} = E(P^5) = 5$. $\mathrm{dg}_gE = \lambda\,\mathrm{res}\,E$ would need $\lambda = 5$ and $5\lambda = 1$, i.e. $25 \equiv 1 \pmod 7$, which is false; so $\mathrm{dg}_gE \notin \mathrm{res}\,H^1(A)$ and the square fails on H^1. (The argument needs only $E(P) \neq 0$ and $p^2 \not\equiv 1 \pmod\ell$.) (d) The two sets of relations are read off Proposition 27.4(1),(3) and the classical q-expansion identities $U_p(f(pz)) = f(z)$, $U_pf = T_pf - p^{k-1}f(pz)$, $k = n+2$. For g': $SgS^{-1} = \begin{pmatrix}p&0\\0&1\end{pmatrix}$; τ' is a chain map preserving the augmentation, and for $h \in \Gamma\cap g'\Gamma g'^{-1} = S\Gamma_g''S^{-1}$, $\tau'(hx) = S\tau\big((S^{-1}hS)S^{-1}x\big) = (g'^{-1}hg')\tau'(x)$. $g'\begin{pmatrix}a&b\\c&d\end{pmatrix}g'^{-1} = \begin{pmatrix}a&pb\\c/p&d\end{pmatrix}$, so $g'Ag'^{-1}\cap\Gamma \supseteq A\cap\Gamma^0(p)$ and $A\cap g'Ag'^{-1} = A\cap\Gamma^0(p)$ (intersecting with A restores the c-condition; the a-, d-conditions are unchanged), likewise for B; for $\gamma \in A \subseteq \Gamma_0(p)$ the right coset $(A\cap\Gamma^0(p))\gamma$ is determined by $b_\gamma a_\gamma^{-1} \bmod p$ (left multiplication by $h \in \Gamma^0(p)\cap\Gamma_0(p)$ scales (a,b) by a_h modulo p, and $\gamma_1\gamma_2^{-1} \in \Gamma^0(p)$ iff $b_1a_2 \equiv a_1b_2$), and T^t has invariant t, so the $T^b \in \Gamma_1(N_B') \subseteq B$ represent $(A\cap\Gamma^0(p))\backslash A$ and $(B\cap\Gamma^0(p))\backslash B$ alike. (4) Replacing r_t by br_t, $b \in A''$, the displayed computation of the statement uses (T1) for $A'' \subseteq \Gamma_g''$, $g^{-1}bg \in A$ (as $b \in gAg^{-1}$), multiplicativity of ρ_n on Δ (both factors $\rho_n((br_t)^{-1}g)$ and $\rho_n(g^{-1}bg)$ being integral with product $\rho_n(r_t^{-1}g)$), and multiplicativity of a modulo N_A on $\Gamma_0(N_A)$ (Lemma 29.2(2)); this is the well-definedness of the double-coset sum, performed for $r_t \in B \subseteq A''$ in the proof of Proposition 27.2(3)(a). Under the criterion, pick a common transversal c; then for any transversals a, b, $T^B(b)\alpha_*\psi = T^B(c)\alpha_*\psi = \alpha_*T^A(c)\psi = \alpha_*T^A(a)\psi$, using only that α_* maps $C_{\mathrm{eq}}^\bullet(L_A)$ into $C_{\mathrm{eq}}^\bullet(L_B)$ (Lemma 29.28(3)). The parenthetical remark is read off `BuildLevelCorrespondence`. □

Definition 29.35 (Normal pair)**.** A compatible level pair is *normal* if $B \trianglelefteq A$; put $\mathsf{G} := A/B$, finite, and $\mathrm{Nm}_\mathsf{G} := \sum_{g\in\mathsf{G}} R_g$. For $\sigma \in A$ the *conjugation* R_σ on $\mathrm{Ind}_B^\Gamma V_R(\chi')$ is $(R_\sigma f)(x) := \rho_{n,\chi}(\sigma)f(\sigma^{-1}x)$ — Proposition 29.8(1), Definition 29.17 for an arbitrary normal B; for $B = \Gamma_{\mathsf{H}'}(N) \trianglelefteq A = \Gamma_0(N)$ it is the diamond. In coordinates, $(R_\sigma f)_J = \rho_{n,\chi}(\sigma h)f_{J'}$ where $\sigma^{-1}v_J = hv_{J'}$, $h \in B$.

Proposition 29.36 (The normal case is inflation–restriction)**.** *Let the pair be normal,* $\mathsf{G} = A/B$.

1. *[proved here] **(Fibre identities, every** R**, every** ℓ**.)** R_σ is a $R[\Gamma]$-automorphism of $\mathrm{Ind}_B^\Gamma V_R(\chi')$ depending only on σB, with $R_\sigma R_{\sigma'} = R_{\sigma\sigma'}$; $\alpha_{B/A}(\mathrm{Ind}_A) = (\mathrm{Ind}_B)^\mathsf{G}$; $\omega_{B/A}\alpha_{B/A} = [A : B]$ and $\alpha_{B/A}\omega_{B/A} = \mathrm{Nm}_\mathsf{G}$; $\varphi(x) := \rho_{n,\chi}(x)^{-1}f(x)$ ($x \in A$) iden-*

tifies $\mathrm{Ind}_B^A V_R(\chi')$ *with* $\mathrm{Map}(\mathsf{G}, V_R)$, R_σ *becoming left translation and* $\omega_{B/A}$ *the sum, so* $\ker \omega_{B/A} = \sum_g (R_g - 1)\mathrm{Ind}_B$. *For* $6 \in R^\times$, R_σ *induces the conjugation action of* σ *on* $H^\bullet(B; V_R(\chi'))$, *and* res^q *lands in* $H^q(B)^{\mathsf{G}}$.

2. *[standard: [Bro82, Ch. VII, §6, Thm. 6.3 and Cor. 6.4], the Hochschild–Serre spectral sequence and its five-term sequence; VERIFY]* ***(Two rows.)*** *For* $6 \in R^\times$: $E_2^{a,b} = H^a(\mathsf{G}; H^b(B; V_R(\chi'))) \Rightarrow H^{a+b}(A; V_R(\chi))$ *has only the rows* $b \in \{0, 1\}$ *and abuts to zero in total degree* ≥ 2. *Hence* $d_2 \colon H^a(\mathsf{G}; H^1(B)) \to H^{a+2}(\mathsf{G}; H^0(B))$ *is an isomorphism for every* $a \geq 1$, *and*

$$0 \to H^1(\mathsf{G}; H^0(B)) \xrightarrow{\inf} H^1(A) \xrightarrow{\mathrm{res}} H^1(B)^{\mathsf{G}} \xrightarrow{\mathrm{tg}} H^2(\mathsf{G}; H^0(B)) \to 0 \qquad (77)$$

is exact ($H^0(B) = V_R(\chi')^B$).

3. *[proved here]* ***(The cone sequence is*** (77) ***plus two pieces.)*** *For* $6 \in R^\times$, (76) *gives* $0 \to H^0(B)/H^0(A) \to H^1(A, B) \to \ker \mathrm{res}^1 \to 0$ *with* $\ker \mathrm{res}^1 = \inf H^1(\mathsf{G}; H^0(B)) \cong H^1(\mathsf{G}; H^0(B))$, *and* $H^2(A, B) = \mathrm{coker}\, \mathrm{res}^1 \supseteq H^1(B)^{\mathsf{G}}/\mathrm{res}\, H^1(A) \cong H^2(\mathsf{G}; H^0(B))$ *with quotient* $H^1(B)/H^1(B)^{\mathsf{G}}$: *inflation–restriction is the part of the cone sequence inside the* G*-invariants, and* $H^0(B)/H^0(A)$, $H^1(B)/H^1(B)^{\mathsf{G}}$ *are the two pieces it leaves out.*

4. *[proved here] If* $|\mathsf{G}| \in R^\times$: $H^{\geq 1}(\mathsf{G}; -) = 0$, $H^q(A) \cong H^q(B)^{\mathsf{G}}$, *and Proposition 29.33 applies; for* $B = \Gamma_{\mathsf{H}}(N) \trianglelefteq A = \Gamma_0(N)$ *and* R *holding the values of the characters of* G_N/H, $H^q(B)$ *is the sum of the character models (Propositions 29.8(3), 29.18(6)).*

5. *[proved here]* ***(Coinvariants.)*** *For* $6 \in R^\times$, cor^1 *induces* $H^1(B)_{\mathsf{G}} \xrightarrow{\sim} H^1(A)$, *and* $\mathrm{res}^1 \circ \mathrm{cor}^1$ *is the norm map* $H^1(B)_{\mathsf{G}} \to H^1(B)^{\mathsf{G}}$ *of the* G*-module* $H^1(B)$.

Proof. (1) R_σ: the proof of Proposition 29.8(1) with B for $\Gamma_1(N)$ and $\rho_{n,\chi}$ for ρ_n: it uses only $\sigma^{-1} B \sigma = B$, the multiplicativity of $\rho_{n,\chi}$ on A and $\rho_{n,\chi}|_B = \rho_{n,\chi'}$ (Lemma 29.26); it also gives the induced action on cohomology. Coordinates: $\sigma^{-1} v_J = h v_{J'}$ gives $f(\sigma^{-1} v_J) = \rho(h) f_{J'}$. f is G-fixed iff $f(\sigma x) = \rho(\sigma) f(x)$ for all $\sigma \in A$, i.e. iff $f \in \mathrm{Ind}_A$. With $B \backslash A = \mathsf{G}$ the r_t of Definition 29.27 run over G, and $(\alpha \omega f)(x) = \sum_t \rho(r_t^{-1}) f(r_t x) = \sum_t R_{r_t^{-1}} f(x)$. $\varphi(bx) = \rho(x)^{-1}\rho(b)^{-1}\rho(b) f(x) = \varphi(x)$, so φ is a function on $B \backslash A = \mathsf{G}$, arbitrary; $(R_\sigma f)$ has $\varphi'(x) = \rho(x)^{-1}\rho(\sigma) f(\sigma^{-1} x) = \varphi(\sigma^{-1} x)$, and $\omega f(I) = \sum_t \varphi(r_t)$. The kernel of the sum on $\mathrm{Map}(\mathsf{G}, V_R)$ is spanned by the $(\delta_g - \delta_1)v = (R_g - 1)(\delta_1 v)$. Since $R_\sigma \alpha = \alpha$ (fixed points), $R_{\sigma *} \mathrm{res}^q = \mathrm{res}^q$. (2) The spectral sequence of the extension $1 \to B \to A \to \mathsf{G} \to 1$ (the citation); for $6 \in R^\times$, $H^b(B; -) = 0$ for $b \geq 2$ and $H^{a+b}(A; -) = 0$ for $a + b \geq 2$ (Lemma 28.4). With two rows, $E_3 = E_\infty$, and $E_\infty^{a,1} = \ker d_2$, $E_\infty^{a+2,0} = \mathrm{coker}\, d_2$ vanish for $a \geq 1$; for $a = 0$, $E_\infty^{2,0} = 0$ makes $\mathrm{tg} = d_2$ onto. (3) Read (76) with (2); $H^0(B)/H^0(A) = \mathrm{coker}\, \mathrm{res}^0$, and res lands in the invariants by (1). (4) $H^{\geq 1}(\mathsf{G}; M)$ is killed by $|\mathsf{G}|$ [standard: [Bro82, Ch. III, Cor. 10.2]; VERIFY]. (5) $H^1(A; -)$ is right exact and additive ($H^2(A; -) = 0$), so it preserves the cokernel $\mathrm{Ind}_B^A V_R \to (\mathrm{Ind}_B^A V_R)_{\mathsf{G}} = \mathrm{Ind}_B^A V_R / \ker \omega \cong V_R$ of (1); by Shapiro $H^1(A; \mathrm{Ind}_B^A V_R) = H^1(B; V_R)$, the map induced by ω being cor (Lemma 29.28(3)) and R_g the conjugation. $\mathrm{res} \circ \mathrm{cor} = \mathrm{Nm}_{\mathsf{G}}$ by (1). □

Proposition 29.37 (A cyclic quotient of order p)**.** *Let* $\mathsf{G} = \langle \sigma \rangle \cong \mathbb{Z}/p$ *and* $N := \sum_{i<p} \sigma^i$.

1. *[standard: [Bro82, Ch. III, §1, Example 2], the periodic resolution of a cyclic group; VERIFY] For an* $R[\mathsf{G}]$*-module* M: $H^0 = \ker(\sigma - 1)$, $H^{2k-1} = \ker N/(\sigma - 1)M$, $H^{2k} = \ker(\sigma - 1)/NM$ $(k \geq 1)$.

2. *[proved here] Over a field* $\mathbb{F}$ *of characteristic* p: $\mathbb{F}[\mathsf{G}] = \mathbb{F}[t]/(t^p)$, $t = \sigma - 1$, $N = t^{p-1}$; *a finite-dimensional module is* $\bigoplus J_k$ *(Jordan blocks of* σ, $1 \leq k \leq p$*),* J_p *the free module;* $\dim H^a(\mathsf{G}; J_k) = 1$ *for* $k < p$, $a \geq 1$, *and* $= 0$ *for* $k = p$. *So* $\dim H^a(\mathsf{G}; M)$ $(a \geq 1)$ *is the number of* non-free *blocks, and the Jordan type is read from* $\mathrm{rank}\, t^k$, $k = 1, \ldots, p$.

3. *[proved here]* ***(The level step.)*** *Let the pair be normal with* $\mathsf{G} \cong \mathbb{Z}/p$ *and* $R = \mathbb{F}$ *of characteristic* $p \geq 5$. *Then*

   (a) the number of non-free Jordan blocks of σ *on* $H^1(B)$ *equals that on* $H^0(B) = V_R(\chi')^B$;

(b) *if* $H^0(B)=0$, *then* $H^1(B)$ *is a* ***free*** $\mathbb{F}[\mathsf{G}]$*-module, and* $H^1(A)\cong H^1(B)^{\mathsf{G}}\cong H^1(B)_{\mathsf{G}}$ *(one* $\mathbb{Z}/p$ *step of the freeness of [U,* `prop:u-towerfree`*], for the whole of* H^1*);*

(c) *if* A *is torsion-free and* $H^0(B)=H^0(A)$*, of dimension* c*, then with* $h:=\dim H^1(A)$*:* $\dim H^1(B)=ph-(p-1)c$*, the Jordan type of* σ *on* $H^1(B)$ *is* $J_p^{h-c}\oplus J_1^c$*,* $\dim\ker\mathrm{res}^1=c$*,* $\operatorname{rank}\mathrm{res}^1=h-c$*,* $\dim H^1(A,B)=c$*,* $\dim H^2(A,B)=(p-1)h-(p-2)c$*, and* $\dim H^1(B)^{\mathsf{G}}=\dim H^1(B)_{\mathsf{G}}=h$*;*

(d) *if* $H^0(B)=H^0(A)=\mathbb{F}$ *(trivial action) and* A *is arbitrary: exactly one non-free block, of size* $s=\dim H^1(B)-p(h-1)$*, and* $h-1$ *free blocks.*

Proof. (2) $\sigma^p-1=(\sigma-1)^p$ in characteristic p, so $N=(\sigma^p-1)/(\sigma-1)=t^{p-1}$; $\mathbb{F}[t]/(t^p)$ is a principal ideal ring with ideals (t^k), so a finitely generated module is a sum of cyclic ones $J_k=\mathbb{F}[t]/(t^k)$. On J_k, $k<p$: $N=0$, $\ker N/tJ_k$ and $\ker t/NJ_k=\ker t$ are one-dimensional; on J_p: $\ker N=tJ_p$, $\ker t=t^{p-1}J_p=NJ_p$. The number of blocks of size $\ge k$ is $\operatorname{rank}t^{k-1}-\operatorname{rank}t^k$. (3)(a) By Proposition 29.36(2) ($6\in\mathbb{F}^\times$), $H^1(\mathsf{G};H^1(B))\cong H^3(\mathsf{G};H^0(B))$, and $\dim H^3=\dim H^1$ for cyclic G by (1); apply (2) to both. (b) No non-free block; with $H^0(B)=0$ (77) gives $H^1(A)\cong H^1(B)^{\mathsf{G}}$, and Proposition 29.36(5) the coinvariants. (c) G acts trivially on $H^0(B)=H^0(A)=H^0(B)^{\mathsf{G}}$, so $H^0(B)\cong J_1^c$ and $\dim H^a(\mathsf{G};H^0(B))=c$ ($a=1,2$). Lemma 29.38 for the torsion-free A and its subgroup B of index p gives $\dim H^1(B)-c=p(h-c)$. By (a) $H^1(B)$ has exactly c non-free blocks; by (77) the number of all blocks, $\dim H^1(B)^{\mathsf{G}}$, is $\operatorname{rank}\mathrm{res}^1+c=(h-c)+c=h$, since $\ker\mathrm{res}^1=\inf H^1(\mathsf{G};H^0(B))$ has dimension c. So $h-c$ blocks are free, and the c non-free blocks have total size $\dim H^1(B)-p(h-c)=c$: each has size 1. The cone dimensions are Lemma 29.30(3); the coinvariants Proposition 29.36(5). (d) The same count with $c=1$, without Lemma 29.38: one non-free block by (a), h blocks in all, so $h-1$ free ones and the rest has dimension $\dim H^1(B)-p(h-1)$. □

Lemma 29.38 (Torsion-free groups: the Euler characteristic)**.** *[proved here] If* $A\le\Gamma$ *has finite index and is torsion-free, then over every field, at every* ℓ *and for every character datum,* $\dim H^1(A)-\dim H^0(A)=m_A(n+1)/12$ *on the model. In particular it is multiplicative along torsion-free pairs:* $\dim H^1(B)-\dim H^0(B)=[A:B](\dim H^1(A)-\dim H^0(A))$.

Proof. Let $M=\mathrm{Ind}_A^\Gamma V_R(\chi)$, of dimension $m_A(n+1)$. For $F\in\{\langle -I\rangle,\langle S\rangle,\langle U\rangle\}$, of order $f\in\{2,4,6\}$, Mackey decomposes $M|_F$ over the double cosets $A\gamma F$ into modules induced from $F\cap\gamma^{-1}A\gamma=\{1\}$; so $M|_F$ is free over $\mathbb{F}[F]$ and $\dim M^F=\dim M/f$ (one fixed line per free rank-one summand, spanned by its norm). Proposition 23.3(2): $\dim H^1-\dim H^0=\dim M^{-I}-\dim M^S-\dim M^U=\dim M\,(\frac12-\frac14-\frac16)$. □

Lemma 29.39 (Invariants of a unipotent image)**.** *[proved here] Let* $\ell=p$*,* $R=\mathbb{F}_p$*,* $\chi'=1$*, and let* $B\le\Gamma$ *contain* T *and lie in* $\Gamma_1(p)$ *(e.g.* $\Gamma_1(p^r)\cap\Gamma_0(N)$*,* $\Gamma_1(p^r)\cap\Gamma_0(p^{r+1}N)$*,* $r\ge1$*). Then* $\dim V_R^B=\lfloor n/p\rfloor+1$*. If* $A\supseteq B$ *also has these two properties, then* $V_R^B=V_R^A$*, with trivial action of* A/B*. For the character* $\chi=\omega^{a-2}$ *(*$n=0$*) on* $B=\Gamma_{\mathsf{H}}(p^r)$ *with* $\mathsf{H}\supseteq\mu_{p-1}$ *(the tower of §29.7):* $\mathbb{F}_p(\chi)^B=\mathbb{F}_p$ *if* $a\equiv2\pmod{p-1}$ *and* 0 *otherwise.*

Proof. $V_R=\mathrm{Sym}^n\mathbb{F}_p^2$ is a representation of $\mathrm{SL}_2(\mathbb{F}_p)$, and the image of B there is $\{\left(\begin{smallmatrix}1&b\\0&1\end{smallmatrix}\right)\}$ ($c\equiv0$, $d\equiv1$ force $a\equiv1$), generated by T. $\rho_n(T)$ is the substitution $x\mapsto x$, $y\mapsto y+x$ (Proposition 8.2). Put $\mathcal{N}:=\prod_{u\in\mathbb{F}_p}(y+ux)=y^p-x^{p-1}y$, which is invariant; $\mathbb{F}_p[x,y]$ is free over $\mathbb{F}_p[x,\mathcal{N}]$ on $1,y,\dots,y^{p-1}$ ($\mathcal{N}$ is monic of degree p in y). For an invariant $P=\sum_{k<p}c_ky^k$, $c_k\in\mathbb{F}_p[x,\mathcal{N}]$, $0=P(x,y+x)-P(x,y)=\sum_k c_k((y+x)^k-y^k)$, and if $k_0\ge1$ is the largest k with $c_k\ne0$, the coefficient of y^{k_0-1} in this basis is $k_0xc_{k_0}\ne0$ ($k_0<p$). So the invariants are $\mathbb{F}_p[x,\mathcal{N}]$, whose degree-$n$ part has the basis $x^{n-pj}\mathcal{N}^j$, $0\le j\le\lfloor n/p\rfloor$. The second sentence: both images are the same group. The third: $\gamma\in B$ acts by $\omega(a_\gamma)^{a-2}$, and a_γ mod p runs over $\mathbb{F}_p^\times$ (d_γ mod p runs over the reduction of H, which contains μ_{p-1}). □

Proposition 29.40 (Which Γ_0-steps are normal)**.** *[proved here] [verified computationally]*[a] *Let $N \geq 1$, $M \geq 2$. Then $\Gamma_0(MN) \trianglelefteq \Gamma_0(N)$ **iff** $M \mid \gcd(N, 24)$, and then $\gamma \mapsto a_\gamma c_\gamma / N \bmod M$ is an isomorphism $\Gamma_0(N)/\Gamma_0(MN) \cong \mathbb{Z}/M$. [verified computationally]*[a] *evidence: `docs/implementation/slot22/les0/normality_check.py` (all 150 pairs $N \leq 24$, $MN \leq 48$, by brute force in $\mathrm{SL}_2(\mathbb{Z}/MN)$) and `les_predict.py` (all 1380 pairs $N \leq 60$, $2 \leq M \leq 24$, by conjugating generators), 0 mismatches.*

Proof. $\Gamma_0(MN)$ is normal iff it acts trivially, by right multiplication, on $\Gamma_0(MN)\backslash\Gamma_0(N)$. By Definition 27.1 (bottom rows), this set is $\mathcal{F} := \{(c : d) \in \mathbb{P}^1(\mathbb{Z}/MN) : N \mid c\}$ (every such row lifts to $\Gamma_0(N)$, as $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/MN)$ is onto), with the right action of $\Gamma_0(N)$ on rows; $\Gamma_0(MN)$ acts through its image, the upper triangular group of $\mathrm{SL}_2(\mathbb{Z}/MN)$, generated by T and $D_u := \mathrm{diag}(u^{-1}, u)$, $u \in (\mathbb{Z}/MN)^\times$. By the Chinese remainder theorem $\mathcal{F} = \prod_q \mathcal{F}_q$ over the primes $q \mid MN$, the generators act componentwise and the D_u realise every tuple of local units. Fix q, $r := v_q(N)$, $s := v_q(M)$. If $s = 0$, $\mathcal{F}_q$ is the single point $(0 : 1)$. If $s \geq 1$, $r = 0$: $\mathcal{F}_q = \mathbb{P}^1(\mathbb{Z}/q^s)$ and T moves $(1 : 0)$ to $(1 : 1)$. If $s, r \geq 1$: $\mathcal{F}_q = \{(q^r x : 1) : x \in \mathbb{Z}/q^s\}$, T sends x to $x(1 + q^r x)^{-1}$, i.e. moves it by $q^r x^2 (1 + q^r x)^{-1}$ in $\mathbb{Z}/q^s$ — trivial for all x iff $s \leq r$ — and D_u sends x to $u^{-2}x$, trivial for all x iff $u^2 \equiv 1 \pmod{q^s}$ for every unit u, i.e. iff $q^s \in \{2, 3, 4, 8\}$. So normality holds iff $q^s \mid q^r$ and $q^s \mid 24$ for every $q \mid M$, i.e. iff $M \mid \gcd(N, 24)$. The map: for $\gamma, \gamma' \in \Gamma_0(N)$, $(\gamma\gamma')_{21}/N = (c/N)a' + d(c'/N)$ and $(\gamma\gamma')_{11} \equiv aa' \pmod N$, so $\phi(\gamma\gamma') \equiv aa'^2(c/N) + ada'(c'/N) \equiv \phi(\gamma) + \phi(\gamma') \pmod M$, using $ad \equiv 1$ and $a'^2 \equiv 1$ modulo $M \mid 24$; $\phi\begin{pmatrix} 1 & 0 \\ N & 1 \end{pmatrix} = 1$, and $\phi(\gamma) = 0$ iff $MN \mid c$ (a is a unit modulo M). □

Proposition 29.41 (The steps of the p-tower)**.** *[proved here] [verified computationally]*[a] *Let $p \geq 5$ be prime, $p \nmid N$, $r \geq 1$, all groups intersected with $\Gamma_0(N)$, $\mu := \mu_{p-1} \subset (\mathbb{Z}/p^k)^\times$ the Teichmüller subgroup, $\Gamma_r^t := \Gamma_\mu(p^r)$ ([U] part 5's tame tower) and $\mathsf{H}_{r+1,r} := \{u \in (\mathbb{Z}/p^{r+1})^\times : u \bmod p^r \in \mu\}$.*

step	*index*	*normal*	*quotient*
$\Gamma_0(pN) \supset \Gamma_0(N) \cap \Gamma_1(p)$	$p-1$	*yes*	$(\mathbb{Z}/p)^\times$, *cyclic of order* $p-1$
$\Gamma_1(p^r) \supset \Gamma_1(p^r) \cap \Gamma_0(p^{r+1})$	p	*yes*	$\mathbb{Z}/p$, *via* $c/p^r \bmod p$
$\Gamma_1(p^r) \cap \Gamma_0(p^{r+1}) \supset \Gamma_1(p^{r+1})$	p	*yes*	$\mathbb{Z}/p$, *via* $d \bmod p^{r+1}$ ($d \equiv 1 \bmod p^r$)
$\Gamma_1(p^r) \supset \Gamma_1(p^{r+1})$	p^2	***no***	—
$\Gamma_0(p^r) \supset \Gamma_r^t$	p^{r-1}	*yes*	$(1 + p\mathbb{Z})/(1 + p^r\mathbb{Z}) \cong \mathbb{Z}/p^{r-1}$, *filtered by* $\mathbb{Z}/p$ *steps*
$\Gamma_r^t \supset \Gamma_r^t \cap \Gamma_0(p^{r+1}) = \Gamma_{\mathsf{H}_{r+1,r}}(p^{r+1})$	p	***no***	—
$\Gamma_{\mathsf{H}_{r+1,r}}(p^{r+1}) \supset \Gamma_{r+1}^t$	p	*yes*	$\mathsf{H}_{r+1,r}/\mu \cong \mathbb{Z}/p$
$\Gamma_0(p^r) \supset \Gamma_0(p^{r+1})$	p	***no***	—

More generally, for $M \mid N'$: $\Gamma_1(N') \cap \Gamma_0(MN') \trianglelefteq \Gamma_1(N')$ with quotient $\mathbb{Z}/M$ ($c/N' \bmod M$), $\Gamma_1(MN') \trianglelefteq \Gamma_1(N') \cap \Gamma_0(MN')$ with quotient $\{d \equiv 1 \bmod N'\} \cong \mathbb{Z}/M$, and $\Gamma_1(MN') \not\trianglelefteq \Gamma_1(N')$ for $M > 1$ prime to N. So raising the d-condition (the diamonds, the Λ-direction) is normal with quotient $\mathbb{Z}/p$, raising the c-condition (Γ_0, Γ^t) is not, for $p \geq 5$; $\Gamma_1^t = \Gamma_0(p)$, and the first step has quotient $(\mathbb{Z}/p)^\times$, not $\mathbb{Z}/p$. [verified computationally][a] *evidence: `les_predict.py`, $p \in \{5, 7\}$, $r \leq 3$, $N \in \{1, 2, 3\}$, and the Γ_1 steps for all $M \mid N' \leq 24$; 0 mismatches.*

Proof. Row 1: $\Gamma_0(N) \cap \Gamma_1(p) = \varsigma_{pN}^{-1}(\ker((\mathbb{Z}/pN)^\times \to (\mathbb{Z}/p)^\times))$, normal with that quotient (Proposition 29.18(1)). The general Γ_1 statements: for $\gamma, \gamma' \in \Gamma_1(N')$, $(\gamma\gamma')_{21}/N' = (c/N')a' + d(c'/N') \equiv c/N' + c'/N' \pmod M$ ($a' \equiv d \equiv 1 \bmod N'$, $M \mid N'$), a homomorphism onto $\mathbb{Z}/M$ ($\begin{pmatrix} 1 & 0 \\ N' & 1 \end{pmatrix} \mapsto 1$) with kernel $\Gamma_1(N') \cap \Gamma_0(MN')$; $\varsigma_{MN'}$ restricted to $\Gamma_1(N') \cap \Gamma_0(MN')$ is a homomorphism onto $\{d \equiv 1 \bmod N'\} = \{1 + N'k\}$, $(1 + N'k)(1 + N'k') \equiv 1 + N'(k + k') \pmod{MN'}$, with kernel $\Gamma_1(MN')$; and with $x := NN'$ and $\sigma = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} \in \Gamma_1(N') \cap$

$\Gamma_0(N)$ (inside $\Gamma_0(N)$, as the proposition requires), $\sigma T\sigma^{-1} = \begin{pmatrix} 1-x & 1 \\ -x^2 & 1+x \end{pmatrix} \notin \Gamma_1(MN')$ ($1+NN' \not\equiv 1 \bmod MN'$, as $\gcd(N,M)=1$ and $M>1$; in particular for $M=p\nmid N$) while $T \in \Gamma_1(MN')$. Rows 2–4 are the case $N'=p^r$, $M=p$. Row 5: Proposition 29.18(1) with $(\mathbb{Z}/p^r)^\times = \mu\times(1+p\mathbb{Z}/p^r)$, the second factor cyclic of order p^{r-1} (p odd), filtered by $\Gamma_{\mu(1+p^s)}(p^r)$, $s=r,\dots,1$. Row 6: $\gamma\in\Gamma_0(p^{r+1})$ has $d \bmod p^r \in \mu$ iff $d \bmod p^{r+1} \in \mathsf{H}_{r+1,r}$. The cosets are $x\in\mathbb{Z}/p$ (bottom rows $(p^r x:1)$), and $\gamma\in\Gamma_r^t$ acts by $x\mapsto a^2x+ac/p^r$ modulo p (the local computation of Proposition 29.40's proof with $s=1\le r$); the stabiliser of $x=0$ contains a lift of $\mathrm{diag}(\omega(g)^{-1},\omega(g))$, g a primitive root modulo p, which sends $x=1$ to $g^{-2}\neq 1$ ($p\ge 5$). Row 7: both groups are $\varsigma_{p^{r+1}}^{-1}$ of subgroups of $(\mathbb{Z}/p^{r+1})^\times$, hence normal in $\Gamma_0(p^{r+1})$. Row 8: Proposition 29.40 with $M=p\nmid 24$. □

Remark 29.42 (Level one). [proved here] (Given Claim 16.2(1).) $\Gamma^{\mathrm{ab}}\cong\mathbb{Z}/12$: the abelianisation of $\langle S\rangle *_{\langle -I\rangle}\langle U\rangle$ is $(\mathbb{Z}/4\oplus\mathbb{Z}/6)/\langle(2,-3)\rangle\cong\mathbb{Z}/12$, $S\mapsto 3$, $U\mapsto 2$. So a normal subgroup of Γ with quotient $\mathbb{Z}/p$ exists only for $p\in\{2,3\}$, and is unique: Γ^2, the preimage of $A_3\subset \mathrm{SL}_2(\mathbb{F}_2)\cong S_3$ ($S\mapsto 1$, $U\mapsto 0$: it contains $-I$, U and every element of order 3, not S), and Γ^3, the preimage of the quaternion group $Q_8\trianglelefteq \mathrm{SL}_2(\mathbb{F}_3)$ ($S\mapsto 0$, $U\mapsto 2$: it contains $-I$ and S, no element of order 3 or 6). At $\ell=3$, Γ^3 satisfies Proposition 29.32 while Γ does not: the cell where one end of (76) is group cohomology and the other Bredon; at $\ell=2$ both are Bredon. Further normal pairs at level one: $\Gamma^4,\Gamma^6,\Gamma^{12}=[\Gamma,\Gamma]$ (not built in slot 22) and $\Gamma(M)\trianglelefteq\Gamma$, $\mathsf{G}=\mathrm{SL}_2(\mathbb{Z}/M)$, torsion-free for $M\ge 3$ (Proposition 29.32). At $M=\ell\ge 5$, $\Gamma(\ell)$ acts trivially on $\mathrm{Sym}^n\mathbb{F}_\ell$, so $H^0(B)=V_R$ and $\ker\mathrm{res}^1\cong H^1(\mathrm{SL}_2(\mathbb{F}_\ell);\mathrm{Sym}^n\mathbb{F}_\ell)$ (Proposition 29.36(3)) — computable from the finite group alone, the oracle O3 of (LES8).

Proof of the identifications. The pushout of $\mathbb{Z}/4\leftarrow\mathbb{Z}/2\to\mathbb{Z}/6$ has order 12 and is generated by the image of $(1,1)$; a surjection $\Gamma\to\mathbb{Z}/p$ factors through $\mathbb{Z}/12$. The reductions $\Gamma\to\mathrm{SL}_2(\mathbb{F}_2)\to\{\pm1\}$ (sign; $S \bmod 2$ is a transposition) and $\Gamma\to\mathrm{SL}_2(\mathbb{F}_3)\to\mathrm{SL}_2(\mathbb{F}_3)/Q_8\cong\mathbb{Z}/3$ ($U \bmod 3$ has order 6, so $U\notin Q_8$, whose elements have order dividing 4) are onto, so their kernels are Γ^2, Γ^3 by uniqueness. Membership is read off $S\mapsto 3$, $U\mapsto 2$: an element of order 3 is conjugate to $U^{\pm2}\mapsto\pm4$, of order 4 to $S^{\pm1}\mapsto\pm3$, of order 6 to $U^{\pm1}\mapsto\pm2$. □

Obligation 29.43 (Level-cone certificates, re-derived over R)**.** Data: the fibre matrices $M_A(\gamma)$, $M_B(\gamma)$ ($\gamma\in\{S,T,-I\}$, (73)), $\alpha_{B/A}$, $\omega_{B/A}$ of (75) and, in the normal case, R_σ for a generator set of G; the stored bases of $C^\bullet_{\mathrm{eq}}$, Z^q, B^q, H^q of both models and of $C(\alpha_{B/A})$ (Route A, the model) or the tree complexes (65) (Route B, $6\in R^\times$); the operators of Proposition 29.34(1). Ring: R ($\mathbb{F}_\ell$ by ranks; $\mathbb{Z}/\ell^K$ by Howell lengths). Each failure **aborts** unless marked.

(i) **Fibre.** $\alpha M_A(\gamma)=M_B(\gamma)\alpha$, $M_A(\gamma)\omega=\omega M_B(\gamma)$, $\omega\alpha=[A:B]$; normal case: $\alpha\omega=\mathrm{Nm}_{\mathsf{G}}$, $R_\sigma M_B(\gamma)=M_B(\gamma)R_\sigma$, $R_\sigma^{\mathrm{ord}\,\sigma}=1$; at $\chi=1$ on Γ_0-pairs, α equals `LevelChangeMap` entry for entry.

(ii) **Cone.** $D^2=0$ and $d\alpha_*=\alpha_* d$ on the stored equivariant bases.

(iii) **Exactness** at the six terms of (76) (image = kernel, injectivity of res^0, surjectivity onto $H^2(A,B)$), $H^0(A,B)=H^3(A,B)=0$, and the connecting map re-derived by the zig-zag equal to $+\mathrm{res}^1$.

(iv) **Route A = Route B** ($\ell\ge 5$): the six dimensions and $\mathrm{rank}\,\mathrm{res}^1$ at every overlapping cell.

(v) **Hecke.** Φ_g of Proposition 29.34(1) commutes with D *as matrices* on the stored equivariant basis, for T_p ($p\nmid N_B$) and for same-level U_p, and the three map families intertwine it.

(vi) **Level-raising** U_p (Proposition 29.34(3)): **gate** the strict identity $U_p^B\circ(\alpha_{B/A})_*=p\cdot\mathrm{dg}_g$ *as matrices* on the stored equivariant bases (0 failing entries; $U_p^B\circ(\alpha_{B/A})_*=0$ at $\ell=p$); the square $U_p^B\,\mathrm{res}^1$ versus $\mathrm{res}^1\,U_p^A$ on H^1 is computed and **recorded** (its defect, Proposition 29.34(3)(c)), not gated (rule (A6): no code path branches on it).

(vii) **Normal case.** $\mathrm{im}\,\mathrm{res}^q\subseteq H^q(B)^{\mathsf{G}}$; $\dim\ker\mathrm{res}^1=\dim H^1(\mathsf{G};H^0(B))$; $\dim H^1(B)^{\mathsf{G}}-\mathrm{rank}\,\mathrm{res}^1=\dim H^2(\mathsf{G};H^0(B))$; for $\mathsf{G}\cong\mathbb{Z}/p$ the Jordan type of σ on $H^1(B)$ and the

equalities $\dim H^a(\mathsf{G}; H^1(B)) = \dim H^{a+2}(\mathsf{G}; H^0(B))$, $a \in \{1,2\}$ (Proposition 29.37); cor^1 an isomorphism on $H^1(B)_{\mathsf{G}}$.

(viii) **Model tags.** Every term at $\ell \in \{2,3\}$ carries `model=group` or `model=bredon` by Proposition 29.32; no $\ell \in \{2,3\}$ dimension is compared with a group-cohomology formula.

(ix) **Negative controls**, each caught: χ replaced by χ^{-1} in α (unless χ is real), $\rho_n(h_i)$ replaced by I, the sign of D flipped, an incompatible χ' (refused, `les_incompatible_character`), R_σ for σ not normalising B (refused, `les_not_normal`), and the **criterion-fails control**: at a pair where the criterion of Proposition 29.34(1) fails (level-raising, (3)), $U_p^B(\alpha_{B/A})_* \neq (\alpha_{B/A})_* U_p^A$ as matrices on the stored equivariant bases, $q = 0,1,2$, at $(\Gamma_0(25) \subset \Gamma_0(5),\ \ell = 7,\ n = 0,\ \chi = 1)$ — caught. (The former control "operators on independently fixed transversals: the strict square fails at some cell" is **retired**: by Proposition 29.34(4) it can never fire. Operators built on two transversals are **gated** equal on $C^\bullet_{\mathrm{eq}}$ (0 differing entries, together with the strict square on A's own transversal) at a cell with $m_A > 1$, e.g. $(\Gamma_0(25) \subset \Gamma_0(5),\ T_2,\ \ell = 7,\ n = 2)$; the comparison on the flat cochains is only *recorded* (0, by construction of (68)); neither is a negative control, since the identity is a theorem.)

Remark 29.44 (What is checkable: (LES1)–(LES9)). Committed before any LES code (rule 20) in `docs/implementation/slot22/LES_predictions.md` with the numbers of `docs/implementation/slot22/les0/out_predict.txt` (`les_predict.py`, `les_model.py`: the tree complex of Proposition 23.3(2) in exact arithmetic, and the oracle O3, an independent crossed-homomorphism computation on $\mathrm{SL}_2(\mathbb{F}_\ell)$); convention of Remark 26.6.

(LES1) [proved here] (Propositions 29.40, 29.41, Remark 29.42) **Normality**: the table of Proposition 29.41, $\Gamma_0(MN) \trianglelefteq \Gamma_0(N)$ iff $M \mid \gcd(N, 24)$ on $N \le 60$, $M \le 24$, the content of Γ^2, Γ^3, and the fibre identities of Obligation 29.43(i) at every cell below.

(LES2) [proved here] (Lemma 29.30) **Exactness** of (76) at every cell, every ℓ.

(LES3) [proved here] (Proposition 27.5(3), Proposition 29.11(4), Proposition 29.33) **Dimensions** at $\ell \ge 5$: the ends by the closed forms, the split cells by $\dim H^1(A,B) = \dim H^0(B) - \dim H^0(A)$, $\dim H^2(A,B) = \dim H^1(B) - \dim H^1(A)$; the non-split cells [verified computationally][a].

(LES4) [proved here] (Proposition 29.34(1),(2)) **Strict Hecke squares**: 0 failing entries.

(LES5) [proved here] (Proposition 23.3(2)) **Route A = Route B** at every overlapping cell.

(LES6) [proved here] (Lemma 29.30, Proposition 29.32) $\ell \in \{2,3\}$: exactness and the `model=` tags; the model dimensions [verified computationally][a] (tree complex), never compared with O1.

(LES7) [proved here] (Proposition 29.37(3)(c), Lemmas 29.38, 29.39) **The Γ_1-type $\mathbb{Z}/p$ steps** at $\ell = p$: $c = \lfloor n/p \rfloor + 1$, $\dim H^1(B) = p \dim H^1(A) - (p-1)c$, Jordan type $J_p^{h-c} \oplus J_1^c$, and the cone dimensions; [verified computationally][a] at the smaller cells.

(LES8) [proved here] (Proposition 29.37(3)(b),(d), Lemma 29.39, Remark 29.42) **The tower cells** $\Gamma_2^t \trianglelefteq \Gamma_0(p^2)$, $n = 0$, $\chi = \omega^{a-2}$: $H^1(\Gamma_2^t)$ free over $\mathbb{F}_p[\mathbb{Z}/p]$ for $a \not\equiv 2$, exactly one non-free block at $a \equiv 2$ (its size [verified computationally][a]); and $\Gamma(\ell) \trianglelefteq \Gamma$: $\dim \ker \mathrm{res}^1 = \dim H^1(\mathrm{SL}_2(\mathbb{F}_\ell); \mathrm{Sym}^n \mathbb{F}_\ell)$ (O3, [verified computationally][a]).

(LES9) [proved here] (Proposition 29.34(2),(3)) **Hecke squares on H^1** by Route B: for T_p and same-level U_p they commute; for level-raising U_p the strict identity $U_p^B \circ \mathrm{res} = p \cdot \mathrm{dg}_g$ holds with 0 failing entries (= 0 at $\ell = p$), and the square's defect is **non-zero** at $(\Gamma_0(25) \subset \Gamma_0(5),\ \ell = 7,\ n = 0,\ \chi = 1)$ ([verified computationally][a] `les0/out_levelraise_up.txt`, also at $\ell \in \{5, 11\}$, $(\Gamma_0(49) \subset \Gamma_0(7),\ \ell = 5)$, $(\Gamma_0(121) \subset \Gamma_0(11),\ \ell = 7)$); other level-raising cells recorded.

Remark 29.45 (The old-space cone, deferred). (Not a statement.) For $p \mid N_B$, $p \nmid N_A$ the restriction does not intertwine T_p^A and U_p^B (Proposition 27.4(1),(3)); the U_p-equivariant object is the cone of $\Phi = (\mathrm{dg}_1, \mathrm{dg}_p)\colon C^\bullet_{\mathrm{eq}}(L_A)^{\oplus 2} \to C^\bullet_{\mathrm{eq}}(L_B)$ with U_p acting on the source by M of Proposition 27.4(4), whose H^1-cokernel is the p-new quotient. Since Proposition 27.4(3) holds only up to homotopy, its cone action needs an η-type certificate; deferred (slot-22 LES-0, owner question Q4). The alternative is the classical $U_p = [A\,\mathrm{diag}(p,1)A]$ of Proposition 29.34(3)(d),

whose square with res is strict by Proposition 29.34(1); this document's $U_p^{(N)}$ is its adjoint U_p^* on old spaces.

Code map of §29.8 **Planned** (slot-22 LES-1..3, after the referee F-LES): `CongruenceLevel(M,K)` (for $\Gamma(M)$, Γ^2, Γ^3, only for levels not of Γ_{H} type), `RestrictionFibreMap`, `ConjugationFibreMap` and the trace (Definition 29.27, Lemma 29.28) in `Hecke/`; the cone on the model (an audit of `EquivariantMappingCone`, or a sibling `LevelPairCone`); the tree-complex route over $\mathbb{F}_\ell$ and $\mathbb{Z}/\ell^K$ reusing `OnCokernelLevelChange` and Lemma 29.22; validations `HeckeLevelNormalityValidation` (`fast`, (LES1)), `HeckeLevelConeValidation` (`standard`, (LES2)–(LES6), (LES9)) and `HeckeLevelInfResValidation` (`standard`, (LES7), (LES8)).

Part IX — Integral and local structure

30 The $\mathbb{Z}/\ell^K$ Hecke algebra: saturation, the defect, and the comparison with modular forms

What this chapter fixes. (i) **Saturation** (Definition 30.1, Proposition 30.2, P15): for a commutative order $\mathbb{T}$ acting faithfully on a free $\mathbb{Z}_\ell$-lattice, the image A_K of $\mathbb{T}$ on the lattice modulo ℓ^K is $\mathbb{T}/(\mathbb{T}\cap\ell^K\mathbb{T}^{\mathrm{sat}})$; reducing the *module* first loses exactly $(\mathbb{T}\cap\ell\mathbb{T}^{\mathrm{sat}})/\ell\mathbb{T}$; and the stopping rule "$\dim \mathsf{A}_K/\ell = \operatorname{rank}\mathbb{T}$" is *equivalent* to $\mathsf{A}_K/\ell\cong\mathbb{T}/\ell$ — which makes the `stable` verdict of `HeckeExt/HeckeRingHeckeAlgebra` a theorem, not a reading. (ii) **The defect** (Proposition 30.3, P17): $\varepsilon_K^{\mathrm{sat}} = \operatorname{length}\big((\mathbb{T}^{\mathrm{sat}}/\mathbb{T})[\ell^K]\big)$. (iii) **The level-1 inputs** (Lemma 30.4, Lemma 30.5): the rank $\lceil f/2\rceil$ that the stopping rule compares with, and the free part of a torsion block. (iv) **The certified reading** (Corollary 30.6), with its generation caveat. (v) **The comparison of the cohomological and the forms Hecke algebra**, torsion-aware (Proposition 30.7). (vi) **P18**: the algebra-level constancy along a Hida class that Theorem 27.12(1), a statement about ranks, does not give (Proposition 30.8). (vii) **P20**: the forms-side θ and its Frobenius kernel, against Theorem 24.11(3) (Remark 30.9). (viii) (v1.17.0-pre.1) **The defect as a Tor term** (§30.6, Remark 30.10). (ix) (v1.23.0-pre.3) **One module run per cell** (§30.7, Lemma 30.11, Obligation 30.12): the levels K the route reads are reductions of the module run at the largest of them, which licenses one run per cell in place of one per level.

Notation. ℓ a prime; $\mathbb{Z}_\ell$-modules are finitely generated. For a finite $\mathbb{Z}_\ell$-module G, $G[\ell^K]$ is its ℓ^K-torsion and $\operatorname{length}(G)$ its length ($=\log_\ell|G|$). $\mathbb{T}$ (§2's symbol for a Hecke algebra) is decorated here: $\mathbb{T}^{\mathrm{sat}}$, $\mathbb{T}^{\mathrm{coh}}$, $\mathbb{T}^{\mathbb{L}}$, $\mathbb{T}^{\mathrm{f}}$. $\mathbb{L} := H^1(\Gamma; V\otimes\mathbb{Z}_\ell)$ and $\mathbb{L}' := \mathbb{L}/\mathbb{L}_{\mathrm{tors}}$; A_K, $\varepsilon_K^{\mathrm{sat}}$, δ^{sat}, c^{sat} are new symbols of this chapter and mean nothing else (rule (A5); $\mathbb{L}$ is not Hida's Λ nor Λ, A_K not $A^{(q)}$, and $\varepsilon_K^{\mathrm{sat}}$ not the augmentation ε).

30.1 Saturation and the defect

Definition 30.1 (Saturation data)**.** [formally verified: [GT, `def:satdata`]] Let L be a free $\mathbb{Z}_\ell$-module of finite rank and $\mathbb{T}\subseteq\operatorname{End}_{\mathbb{Z}_\ell}(L)$ a commutative $\mathbb{Z}_\ell$-subalgebra (so $\mathbb{T}$ is free of finite rank $r := \operatorname{rank}_{\mathbb{Z}_\ell}\mathbb{T}$ and acts faithfully). Put

$$\mathbb{T}^{\mathrm{sat}} := (\mathbb{T}\otimes\mathbb{Q})\cap\operatorname{End}_{\mathbb{Z}_\ell}(L)\quad(\text{inside }\operatorname{End}_{\mathbb{Q}_\ell}(L\otimes\mathbb{Q})),\qquad \mathsf{A}_K := \operatorname{im}\big(\mathbb{T}\to\operatorname{End}(L/\ell^KL)\big)\ (K\geq 1),$$

$\delta^{\mathrm{sat}} := \operatorname{length}(\mathbb{T}^{\mathrm{sat}}/\mathbb{T})$, c^{sat} the least $c\geq 0$ with $\ell^c\mathbb{T}^{\mathrm{sat}}\subseteq\mathbb{T}$, and $\varepsilon_K^{\mathrm{sat}} := K\,r - \operatorname{length}(\mathsf{A}_K)$. $\mathbb{T}$ is *saturated* if $\mathbb{T}=\mathbb{T}^{\mathrm{sat}}$.

$\mathbb{T}^{\mathrm{sat}}$ is a commutative $\mathbb{Z}_\ell$-algebra containing $\mathbb{T}$, free of rank r (a saturated submodule of $\operatorname{End}(L)$ with $\mathbb{T}^{\mathrm{sat}}\otimes\mathbb{Q}=\mathbb{T}\otimes\mathbb{Q}$), so $\mathbb{T}^{\mathrm{sat}}/\mathbb{T}$ is finite, δ^{sat} and c^{sat} are finite, and c^{sat} is the exponent of $\mathbb{T}^{\mathrm{sat}}/\mathbb{T}$. L is a $\mathbb{T}^{\mathrm{sat}}$-module.

Proposition 30.2 (Saturation: what reducing the module loses (P15)). *[proved here] In the notation of Definition 30.1:*

1. $\ker\big(\mathbb{T} \to \mathrm{End}(L/\ell^K)\big) = \mathbb{T} \cap \ell^K \mathrm{End}(L) = \mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}}$, *so* $\mathsf{A}_K = \mathbb{T}/(\mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}})$; *in particular* $\ker(\mathbb{T}/\ell\mathbb{T} \to \mathsf{A}_1) = (\mathbb{T} \cap \ell\mathbb{T}^{\mathrm{sat}})/\ell\mathbb{T}$. *[formally verified: [GT, `prop:saturation`]]*
2. *The following are equivalent: (a)* $\mathbb{T}/\ell\mathbb{T} \to \mathsf{A}_1$ *is an isomorphism; (b)* $L/\ell L$ *is a faithful* $\mathbb{T}/\ell\mathbb{T}$*-module; (c)* $\mathbb{T}$ *is saturated. They hold if* L *contains a* $\mathbb{T}$*-submodule* $N \cong \mathbb{T}$ *(free of rank one) with* L/N *torsion-free. [formally verified: [GT, `prop:saturation`]] ((a) as the injectivity of the canonical surjection)*
3. ***(The stopping rule.)*** *For every* $K \geq 1$, $\dim_{\mathbb{F}_\ell} \mathsf{A}_K/\ell\mathsf{A}_K \leq r$, *and the following are equivalent: (a)* $\dim_{\mathbb{F}_\ell} \mathsf{A}_K/\ell\mathsf{A}_K = r$; *(b) the canonical surjection* $\mathbb{T}/\ell\mathbb{T} \to \mathsf{A}_K/\ell\mathsf{A}_K$ *is an isomorphism (of* $\mathbb{F}_\ell$*-algebras); (c)* $\mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}} \subseteq \ell\mathbb{T}$. *If they hold at* K, *they hold at every* $K' \geq K$; *they hold for every* $K \geq c^{\mathrm{sat}} + 1$. *[formally verified: [GT, `prop:saturation`]] ((b) as the injectivity of the canonical surjection)*
4. *For* $K \geq c^{\mathrm{sat}}$: $\mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}} = \ell^K \mathbb{T}^{\mathrm{sat}}$ *and* $\mathrm{length}(\mathsf{A}_K) = K r - \delta^{\mathrm{sat}}$. *[formally verified: [GT, `prop:saturation`]]*
5. *Every maximal ideal* $\mathfrak{m}$ *of* $\mathbb{T}$ *contains* ℓ *[formally verified: [GT, `prop:saturation`]] (this sentence only); in (3)(b) the isomorphism carries the* $\mathfrak{m}$*-adic filtration to the* $\mathfrak{m}$*-adic filtration, so the Hilbert function and the Loewy length of* $\mathsf{A}_K/\ell\mathsf{A}_K$ *at* $\mathfrak{m}$ *are those of* $\mathbb{T}/\ell\mathbb{T}$. *All statements localise: for* $\mathbb{T} = \prod_{\mathfrak{m}} \mathbb{T}_{\mathfrak{m}}$, $L = \bigoplus_{\mathfrak{m}} L_{\mathfrak{m}}$, *they hold for each pair* $(\mathbb{T}_{\mathfrak{m}}, L_{\mathfrak{m}})$, *with* $(\mathbb{T}^{\mathrm{sat}})_{\mathfrak{m}} = (\mathbb{T}_{\mathfrak{m}})^{\mathrm{sat}}$.

Proof. (1) $t \in \mathbb{T}$ acts as 0 on $L/\ell^K L$ iff $tL \subseteq \ell^K L$ iff $t \in \ell^K \mathrm{End}(L)$ (L free: the matrix entries). If $t = \ell^K e$ with $e \in \mathrm{End}(L)$, then $e = \ell^{-K} t \in \mathbb{T} \otimes \mathbb{Q}$, so $e \in \mathbb{T}^{\mathrm{sat}}$; the converse inclusion holds as $\mathbb{T}^{\mathrm{sat}} \subseteq \mathrm{End}(L)$. (2) (a)⇔(b): both say $\mathbb{T} \cap \ell\mathrm{End}(L) = \ell\mathbb{T}$, by (1). (c)⇒(a): $\mathbb{T} \cap \ell\mathbb{T} = \ell\mathbb{T}$. (a)⇒(c): let $x \in \mathbb{T}^{\mathrm{sat}} \setminus \mathbb{T}$ and $j \geq 1$ least with $\ell^j x \in \mathbb{T}$ (exists, $\mathbb{T}^{\mathrm{sat}}/\mathbb{T}$ being finite); then $\ell^j x \in \mathbb{T} \cap \ell\mathbb{T}^{\mathrm{sat}} = \ell\mathbb{T}$ by (a) and (1), so $\ell^{j-1} x \in \mathbb{T}$ ($\mathbb{T}$ is torsion-free), against minimality. The sufficient condition: let $N = \mathbb{T}v$ with $t \mapsto tv$ injective, and $x \in \mathbb{T}^{\mathrm{sat}}$. Then $xv \in L$ and $xv \in (\mathbb{T} \otimes \mathbb{Q})v = N \otimes \mathbb{Q}$; as L/N is torsion-free, $L \cap (N \otimes \mathbb{Q}) = N$, so $xv = sv$ with $s \in \mathbb{T}$. Then $(x - s)v = 0$ with $x - s \in \mathbb{T} \otimes \mathbb{Q}$, and $t \mapsto tv$ is injective on $\mathbb{T} \otimes \mathbb{Q}$ ($N \otimes \mathbb{Q} \cong \mathbb{T} \otimes \mathbb{Q}$), so $x = s \in \mathbb{T}$. (3) By (1), $\mathsf{A}_K/\ell\mathsf{A}_K = \mathbb{T}/(\ell\mathbb{T} + \mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}})$, a quotient of the r-dimensional $\mathbb{T}/\ell\mathbb{T}$; it has dimension r iff $\mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}} \subseteq \ell\mathbb{T}$. The map is induced by a ring map, hence is one of algebras. Monotonicity: $\mathbb{T} \cap \ell^{K'} \mathbb{T}^{\mathrm{sat}} \subseteq \mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}}$ for $K' \geq K$. For $K \geq c^{\mathrm{sat}} + 1$: $\ell^K \mathbb{T}^{\mathrm{sat}} = \ell^{K - c^{\mathrm{sat}}} \ell^{c^{\mathrm{sat}}} \mathbb{T}^{\mathrm{sat}} \subseteq \ell^{K - c^{\mathrm{sat}}} \mathbb{T} \subseteq \ell\mathbb{T}$. (4) $\ell^K \mathbb{T}^{\mathrm{sat}} \subseteq \ell^{K - c^{\mathrm{sat}}} \mathbb{T} \subseteq \mathbb{T}$, so the intersection is $\ell^K \mathbb{T}^{\mathrm{sat}}$, and $\mathrm{length}(\mathbb{T}/\ell^K \mathbb{T}^{\mathrm{sat}}) = \mathrm{length}(\mathbb{T}^{\mathrm{sat}}/\ell^K \mathbb{T}^{\mathrm{sat}}) - \mathrm{length}(\mathbb{T}^{\mathrm{sat}}/\mathbb{T}) = Kr - \delta^{\mathrm{sat}}$. (5) $\mathbb{T}$ is finite over the local ring $\mathbb{Z}_\ell$, so ℓ lies in every maximal ideal. A surjective algebra map carries $\mathfrak{m}^i$ onto the i-th power of the image of $\mathfrak{m}$; an isomorphism preserves the filtration. $\mathbb{T}$ is semilocal and ℓ-adically complete, hence the product of its localisations, and L decomposes by the corresponding idempotents; each idempotent lies in $\mathbb{T}$, so saturation, intersections and images are computed factorwise. □

Proposition 30.3 (The defect (P17)). *[proved here] In the notation of Definition 30.1, for every* $K \geq 1$:

1. $\varepsilon_K^{\mathrm{sat}} = \mathrm{length}\big((\mathbb{T}^{\mathrm{sat}}/\mathbb{T})[\ell^K]\big)$; *[formally verified: [GT, `prop:satdefect`]]*
2. *hence* $0 \leq \varepsilon_K^{\mathrm{sat}} \leq \varepsilon_{K+1}^{\mathrm{sat}} \leq \delta^{\mathrm{sat}}$, $\varepsilon_K^{\mathrm{sat}} = \delta^{\mathrm{sat}}$ *for* $K \geq c^{\mathrm{sat}}$, *and* $\varepsilon_K^{\mathrm{sat}} = \varepsilon_{K+1}^{\mathrm{sat}}$ *implies* $\varepsilon_K^{\mathrm{sat}} = \delta^{\mathrm{sat}}$ *(a plateau is final); [formally verified: [GT, `prop:satdefect`]]*
3. $\delta^{\mathrm{sat}} = 0$ *iff* $\mathbb{T}$ *is saturated iff* $\varepsilon_1^{\mathrm{sat}} = 0$. *[formally verified: [GT, `prop:satdefect`]]*

The quantity $K \dim(\mathsf{A}_K/\ell) - \mathrm{length}(\mathsf{A}_K)$ *equals* $\varepsilon_K^{\mathrm{sat}}$ *exactly when* $\dim \mathsf{A}_K/\ell = r$, *and not in general.*

Proof. (1) $\ell^K \mathbb{T} \subseteq \mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}}$, so by Proposition 30.2(1) $\mathrm{length}(\mathsf{A}_K) = \mathrm{length}(\mathbb{T}/\ell^K \mathbb{T}) - \mathrm{length}\big((\mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}})/\ell^K \mathbb{T}\big)$, and $\mathbb{T}$ free of rank r gives $\mathrm{length}(\mathbb{T}/\ell^K \mathbb{T}) = Kr$. Multiplication by ℓ^{-K} inside $\mathbb{T} \otimes \mathbb{Q}$ is an isomorphism $(\mathbb{T} \cap \ell^K \mathbb{T}^{\mathrm{sat}})/\ell^K \mathbb{T} \to (\ell^{-K} \mathbb{T} \cap \mathbb{T}^{\mathrm{sat}})/\mathbb{T} = \{x \in \mathbb{T}^{\mathrm{sat}}/\mathbb{T} : \ell^K x = 0\}$. (2) $G[\ell^K] \subseteq G[\ell^{K+1}] \subseteq G$ for $G = \mathbb{T}^{\mathrm{sat}}/\mathbb{T}$, and $G[\ell^K] = G$ for $K \geq c^{\mathrm{sat}}$, the exponent. If

$G[\ell^K] = G[\ell^{K+1}]$ and $x \in G$ had order ℓ^j, $j > K$, then $\ell^{j-K-1}x$ would lie in $G[\ell^{K+1}]\backslash G[\ell^K]$. (3) $G = 0$ iff $G[\ell] = 0$. The last sentence: $K \dim(\mathsf{A}_K/\ell) - \mathrm{length}(\mathsf{A}_K) = \varepsilon_K^{\mathrm{sat}} - K(r - \dim \mathsf{A}_K/\ell)$. □

30.2 The level-1 inputs

Setting. $\ell \geq 5$, $n > 0$ even, $k := n + 2$, $\mathbb{L} = H^1(\Gamma; V \otimes \mathbb{Z}_\ell)$, $\mathbb{L}_{\mathrm{tors}}$ its torsion, $\mathbb{L}' = \mathbb{L}/\mathbb{L}_{\mathrm{tors}}$. $\mathbb{T}^{\mathrm{coh}} \subseteq \mathrm{End}(\mathbb{L})$ is the $\mathbb{Z}_\ell$-algebra generated by all T_p, p prime, with $T_\ell = T_{\Gamma g \Gamma}$ of level 1 ($g = \mathrm{diag}(1, \ell)$; the pipeline's U_ℓ, Proposition 27.4's T_ℓ); it preserves $\mathbb{L}_{\mathrm{tors}}$, and $\mathbb{T}^{\mathbb{L}} \subseteq \mathrm{End}(\mathbb{L}')$ is its image. $\mathbb{T}^{\mathrm{f}} \subseteq \mathrm{End}(M_k(\mathbb{Z}_\ell))$ is the $\mathbb{Z}_\ell$-algebra generated by the classical T_p on the level-1 modular forms with q-expansion coefficients in $\mathbb{Z}_\ell$. For a maximal ideal $\mathfrak{m}$ of $\mathbb{T}^{\mathrm{coh}}$ (it lies over one of $\mathbb{T}^{\mathbb{L}}$ when $\mathbb{L}'_{\mathfrak{m}} \neq 0$), subscripts $\mathfrak{m}$ denote localisation; $f_{\mathfrak{m}} := \mathrm{rank}\, \mathbb{L}'_{\mathfrak{m}}$.

Lemma 30.4 (The rank of the free Hecke algebra of a block)**.** *[proved here] (given the [standard] inputs named in the proof) [formally verified modulo Eichler–Shimura at level one ([DI95, Thm. 12.2.2, Prop. 12.4.10]), as the hypothesis `hES`: [GT, `lem:heckerankone`]] (over a coefficient field $E \supseteq \mathbb{Q}_\ell$) In the Setting, let $e \in \mathbb{T}^{\mathbb{L}}$ be any idempotent (e.g. the idempotent of a maximal ideal of $\mathbb{T}^{\mathbb{L}}$, or of a maximal ideal $\mathfrak{m}'$ of a subalgebra $\mathbb{T}' \subseteq \mathbb{T}^{\mathbb{L}}$, whose block $e\mathbb{L}'$ may be a sum of several blocks of $\mathbb{T}^{\mathbb{L}}$). Then $\mathrm{rank}_{\mathbb{Z}_\ell} e\mathbb{T}^{\mathbb{L}} = \lceil \mathrm{rank}(e\mathbb{L}')/2 \rceil$; in particular $\mathrm{rank}_{\mathbb{Z}_\ell} \mathbb{T}^{\mathbb{L}}_{\mathfrak{m}} = \lceil f_{\mathfrak{m}}/2 \rceil$ (v1.15.0-pre.2, WP-4 referee B1). And $\mathbb{T}^{\mathbb{L}} \otimes \mathbb{Q}_\ell$ is a product of fields, one factor per $\mathrm{Gal}(\bar{\mathbb{Q}}_\ell/\mathbb{Q}_\ell)$-orbit of systems of Hecke eigenvalues of normalised eigenforms in M_k.*

Proof. Eichler–Shimura [standard: [DI95, Thm. 12.2.2, p. 108] (the isomorphism $\mathcal{M}_k(\Gamma) \oplus \bar{\mathcal{S}}_k(\Gamma) \to H^1(\Gamma, V_k)$, for every congruence subgroup Γ, $k \geq 2$) with its Hecke equivariance [DI95, Prop. 12.4.10, p. 117] (compatibility with the action of $R(\Gamma, \Delta)$), both **verified** on page images of the library's copy (slot-17 L17-5 repairs, the Lean entry `DiamondImEichlerShimura`; v1.18.0-pre.1, the slot-17 referee F1's m-C4); [Shi71, Ch. 8, Prop. 8.5, pp. 237–238], which Diamond–Im cite for the equivariance, kept as a secondary pointer, **verified** for the cuspidal part ($S_k \to H^1_P$; R20-L, v1.21.0-pre.1); the pipeline's T_ℓ is the classical T_ℓ by Theorem 11.10; Claim 16.5(1) and Obligation 20.7 state the same decomposition] gives $\mathbb{L}' \otimes \mathbb{C} \cong M_k(\mathbb{C}) \oplus \overline{S_k(\mathbb{C})}$ as modules over the polynomial ring in the T_p (after fixing $\bar{\mathbb{Q}}_\ell \cong \mathbb{C}$). $M_k(\mathbb{C})$ has a basis of normalised eigenforms $E_k, f_1, \ldots, f_s$ [standard: two inputs, cited separately (v1.17.0-pre.1, slot-17 L17-5's finding, and v1.17.0-pre.2, the citation repair of the WP-3 brief's addendum: the earlier pointer to [Ser73, Ch. VII, §5] alone does not give the existence). *The eigenbasis*: the T_m commute and are self-adjoint for the Petersson inner product on $S_k(\mathbb{C})$, so $S_k(\mathbb{C})$ has a basis of simultaneous eigenvectors, [DI95, §3.6, p. 55] (the T_n, $(n, N) = 1$, self-adjoint for the Petersson product, and $S_k(N, \varepsilon)$ an orthogonal direct sum of $\mathbb{T}^{(N)}$-eigenspaces) with [DI95, §3.5, p. 54] (every eigenspace at most one-dimensional); level 1 is the case $N = 1$ there; **verified** on the page images of the library's copy (slot-17 L17-5 audit and R17-4, v1.17.0-pre.5). Secondary pointers: [Shi71, Thm. 3.41, Thm. 3.42 and the paragraph after it, p. 77] (**verified**, R20-L, v1.21.0-pre.1: at $N = 1$ the $[\Gamma\alpha\Gamma]_k$ on $S_k(\Gamma)$ commute and are self-adjoint, hence simultaneously diagonalisable with real eigenvalues), [DS05, §5.5, Thm. 5.5.4] (**not verified**, rule (A2): not in the library); and E_k spans the Eisenstein complement and is an eigenform. *Multiplicity one*: a non-zero eigenform has $a_1 \neq 0$ and, normalised, $a_m = \lambda_m$, [Ser73, Ch. VII, §5, Thm. 7, p. 102], with [Ser73, Ch. VII, §5, Cor. 1 of Thm. 7, p. 102; Prop. 11, p. 101], verified on page images (slot-17 L17-5); Serre's §5 gives this distinctness half only, not the existence of the basis] whose eigenvalue systems are pairwise distinct: for two normalised eigenforms with the same T_m-eigenvalues for all $m \geq 1$ (the T_m are polynomials in the T_p by Theorem 10.8), $a_m = \lambda_m$ for $m \geq 1$, so their difference is a constant modular form of weight $k \geq 4$, hence 0, and a cusp form cannot coincide with E_k, whose constant term is $\neq 0$. So the T_p act semisimply on $\mathbb{L}' \otimes \bar{\mathbb{Q}}_\ell$ with each cuspidal system twice and the Eisenstein system once, and $\mathbb{T}^{\mathbb{L}} \otimes \bar{\mathbb{Q}}_\ell \cong \bar{\mathbb{Q}}_\ell^{s+1}$ (an algebra generated by commuting semisimple operators is the product over its distinct systems). Localising at $\mathfrak{m}$ keeps the systems reducing to $\mathfrak{m}$: $f_{\mathfrak{m}} = 2s_{\mathfrak{m}} + e_{\mathfrak{m}}$, $e_{\mathfrak{m}} \in \{0, 1\}$ the number of Eisenstein systems in $\mathfrak{m}$, and $\mathrm{rank}\, \mathbb{T}^{\mathbb{L}}_{\mathfrak{m}} = s_{\mathfrak{m}} + e_{\mathfrak{m}} = \lceil f_{\mathfrak{m}}/2 \rceil$. For an idempotent e the same count reads $\mathrm{rank}(e\mathbb{L}') = 2s + \epsilon$

with $\epsilon \in \{0,1\}$ the number of Eisenstein systems in the block — at most one, M_k having one Eisenstein series — so rank $e\mathbb{T}^{\mathbb{L}} = s + \epsilon = \lceil \mathrm{rank}(e\mathbb{L}')/2 \rceil$ for every e. □

Lemma 30.5 (The free part of a torsion block (P-WP6-2)). *[proved here] In the Setting, let $t \geq 0$ with $\ell^t \mathbb{L}_{\mathrm{tors},\mathfrak{m}} = 0$ and $K > t$. Put $N := \mathbb{L}_{\mathfrak{m}}/\ell^K\mathbb{L}_{\mathfrak{m}} = H^1(\Gamma; V \otimes \mathbb{Z}/\ell^K)_{\mathfrak{m}}$ and $\mathsf{A}_K := \mathrm{im}(\mathbb{T}^{\mathrm{coh}}_{\mathfrak{m}} \to \mathrm{End}N)$.*

1. *Multiplication by ℓ^t induces a Hecke-equivariant isomorphism $\mathbb{L}'_{\mathfrak{m}}/\ell^{K-t}\mathbb{L}'_{\mathfrak{m}} \cong \ell^t N$. [formally verified: [GT, `lem:freepart`]]*
2. *The image of $\mathbb{T}^{\mathrm{coh}}_{\mathfrak{m}}$ in $\mathrm{End}(\ell^t N)$ is $\mathsf{A}_K/\mathsf{A}_K[\ell^t]$, and $a \mapsto \ell^t a$ is an isomorphism of $\mathbb{Z}_\ell$-modules $\mathsf{A}_K/\mathsf{A}_K[\ell^t] \to \ell^t\mathsf{A}_K \subseteq \mathrm{End}N$; by (1) this image is the image A'_{K-t} of $\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}$ in $\mathrm{End}(\mathbb{L}'_{\mathfrak{m}}/\ell^{K-t})$, to which Proposition 30.2 applies with $(L,\mathbb{T}) = (\mathbb{L}'_{\mathfrak{m}}, \mathbb{T}^{\mathbb{L}}_{\mathfrak{m}})$ at level $K-t$. [formally verified: [GT, `lem:freepart`]] (in full since v1.18.0)*
3. *If $\mathbb{L}'_{\mathfrak{m}} = 0$ (a pure torsion block) and $K \geq t$, then $N = \mathbb{L}_{\mathfrak{m}}$ and $\mathsf{A}_K = \mathbb{T}^{\mathrm{coh}}_{\mathfrak{m}}$. [formally verified: [GT, `lem:freepart`]]*

The identification $H^1(\Gamma; V \otimes \mathbb{Z}/\ell^K) = \mathbb{L}/\ell^K$ is Claim 16.7's argument with ℓ^K in place of ℓ ($H^2(\Gamma; V)$ has no ℓ-torsion).

Proof. (1) $\mathbb{L}_{\mathfrak{m}} \cong \mathbb{L}'_{\mathfrak{m}} \oplus \mathbb{L}_{\mathrm{tors},\mathfrak{m}}$ as $\mathbb{Z}_\ell$-modules, so $\mathbb{L}_{\mathfrak{m}}[\ell^t] = \mathbb{L}_{\mathrm{tors},\mathfrak{m}}$ and $\ell^t\colon \mathbb{L}_{\mathfrak{m}} \to \ell^t\mathbb{L}_{\mathfrak{m}}$ has kernel $\mathbb{L}_{\mathrm{tors},\mathfrak{m}}$, inducing $\mathbb{L}'_{\mathfrak{m}} \cong \ell^t\mathbb{L}_{\mathfrak{m}}$. Since $K \geq t$, $\ell^t N = \ell^t\mathbb{L}_{\mathfrak{m}}/\ell^K\mathbb{L}_{\mathfrak{m}}$, and the preimage of $\ell^K\mathbb{L}_{\mathfrak{m}} = \ell^t(\ell^{K-t}\mathbb{L}_{\mathfrak{m}})$ in $\mathbb{L}'_{\mathfrak{m}}$ is $\ell^{K-t}\mathbb{L}'_{\mathfrak{m}}$. Multiplication by ℓ^t commutes with every endomorphism. (2) $a \in \mathsf{A}_K$ kills $\ell^t N$ iff $\ell^t a = 0$ in $\mathrm{End}N$; and $\mathbb{T}^{\mathrm{coh}}_{\mathfrak{m}} \to \mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}$ is onto. (3) $N = \mathbb{L}_{\mathfrak{m}}/\ell^K\mathbb{L}_{\mathfrak{m}} = \mathbb{L}_{\mathfrak{m}}$, and $\mathbb{T}^{\mathrm{coh}}_{\mathfrak{m}}$ acts faithfully on $\mathbb{L}_{\mathfrak{m}}$ by definition. □

Corollary 30.6 (The certified reading of the $\mathbb{Z}/\ell^K$ Hecke-algebra route). *[proved here] (given Lemma 30.4) In the Setting, let S be a finite set of primes, $\mathbb{T}' := \mathbb{Z}_\ell[T_p : p \in S] \subseteq \mathbb{T}^{\mathbb{L}}$, $\mathfrak{m}'$ a maximal ideal of $\mathbb{T}'$ (the code's joint system), e its idempotent, $f := \mathrm{rank}(e\mathbb{L}')$, and A'_K the image of $\mathbb{T}'_{\mathfrak{m}'}$ in $\mathrm{End}(e\mathbb{L}'/\ell^K)$ (v1.15.0-pre.2, WP-4 referee B1: the block of the code is one of $\mathbb{T}'$, which without the certificate of (2) may be a sum of several blocks of $\mathbb{T}^{\mathbb{L}}$).*

1. *If $\dim_{\mathbb{F}_\ell} \mathsf{A}'_K/\ell\mathsf{A}'_K = \lceil f/2 \rceil$ at some K, then $\mathrm{rank}\,\mathbb{T}'_{\mathfrak{m}'} = \mathrm{rank}\; e\mathbb{T}^{\mathbb{L}}$ and $\mathsf{A}'_K/\ell\mathsf{A}'_K \cong \mathbb{T}'_{\mathfrak{m}'}/\ell\mathbb{T}'_{\mathfrak{m}'}$ as $\mathbb{F}_\ell$-algebras (Hilbert function and Loewy length included), at this and every larger K; such a K exists iff $\mathrm{rank}\,\mathbb{T}'_{\mathfrak{m}'} = \mathrm{rank}\; e\mathbb{T}^{\mathbb{L}}$, and then $K \leq c^{\mathrm{sat}}(\mathbb{T}'_{\mathfrak{m}'}) + 1$. [formally verified modulo Eichler–Shimura at level one ([DI95, Thm. 12.2.2, Prop. 12.4.10]), as the hypothesis `hES`: [GT, `cor:stablecert`]] (since v1.19.0 at $T_e = e\,T^{\mathbb{L}}$ on $e\mathbb{L}'$, for every $\mathbb{T}' \hookrightarrow T_e$; the rank hypothesis is discharged from Lemma 30.4)*
2. *$\mathbb{T}'_{\mathfrak{m}'} = e\mathbb{T}^{\mathbb{L}}$ (then $e\mathbb{L}'$ is one block of $\mathbb{T}^{\mathbb{L}}$) iff the images of the T_p, $p \in S$, generate $e\mathbb{T}^{\mathbb{L}}/\ell$ as an $\mathbb{F}_\ell$-algebra (Nakayama). [formally verified: [GT, `cor:stablecert-uncond`]] (this sentence) This* generation certificate *is a separate check; without it, (1) certifies $\mathbb{T}'_{\mathfrak{m}'}/\ell$, not $e\mathbb{T}^{\mathbb{L}}/\ell$, and a proper suborder of full rank can have a different, even a larger, Loewy length ($\{(a,b,c) \in \mathbb{Z}_\ell^3 : a \equiv b \equiv c\ (\ell)\}$ has reduction of Loewy length 2, its suborder $\mathbb{Z}_\ell[(0,\ell,\ell^2)]$ one of Loewy length 3).*
3. *Applied to a torsion-free block ($\mathbb{L}_{\mathfrak{m}} = \mathbb{L}'_{\mathfrak{m}}$) this is the `stable` verdict; applied through Lemma 30.5(2) to the free part of a torsion block, the `torsion_free_stable` verdict; Lemma 30.5(3) is the `torsion_stable` verdict. The full algebra of a torsion block with $f > 0$ is* not *covered: $\mathbb{T}^{\mathrm{coh}}_{\mathfrak{m}}$ may have $\mathbb{Z}_\ell$-torsion, and no stopping rule is asserted for it.*

Proof. (1) $\dim \mathsf{A}'_K/\ell \leq \mathrm{rank}\,\mathbb{T}'_{\mathfrak{m}'} \leq \mathrm{rank}\; e\mathbb{T}^{\mathbb{L}} = \lceil f/2 \rceil$ (Proposition 30.2(3) for $(e\mathbb{L}', \mathbb{T}'_{\mathfrak{m}'})$; Lemma 30.4 applied to e; $\mathbb{T}'_{\mathfrak{m}'} = e\mathbb{T}' \subseteq e\mathbb{T}^{\mathbb{L}}$); equality forces both equalities, and Proposition 30.2(3),(5) give the rest. (2) $\mathbb{T}'_{\mathfrak{m}'} \subseteq e\mathbb{T}^{\mathbb{L}}$ are finite $\mathbb{Z}_\ell$-modules; $\mathbb{T}'_{\mathfrak{m}'} + \ell e\mathbb{T}^{\mathbb{L}} = e\mathbb{T}^{\mathbb{L}}$ implies equality by Nakayama, and the image of $\mathbb{T}'_{\mathfrak{m}'}$ in $e\mathbb{T}^{\mathbb{L}}/\ell$ is the subalgebra generated by the images of the T_p. The example: the reduction of the first order is $\mathbb{F}_\ell[x,y]/(x,y)^2$, and $t = (0,\ell,\ell^2)$ satisfies $t(t-\ell)(t-\ell^2) = 0$ with $\mathbb{Z}_\ell[t]/\ell = \mathbb{F}_\ell[t]/t^3$. (3) is the dictionary of `HeckeExt/HeckeRingHeckeAlgebra.h` ((ii)–(iv) and its stopping rule): its block $e_{\mathfrak{m}}M_K$ is $e\mathbb{L}/\ell^K$ for the idempotent e of its joint system $\mathfrak{m}'$ of $\mathbb{T}'$ (Lemma 30.5 read with e in place of

the localisation at a maximal ideal of $\mathbb{T}^{\mathrm{coh}}$, which changes nothing in its proof), its A_K is the algebra generated by T_p, $p \in S$, and its $\lceil f/2 \rceil$ is Lemma 30.4 for e. □

30.3 Cohomology against modular forms, with torsion

Proposition 30.7 (The cohomological and the forms Hecke algebra)**.** *In the Setting:*

1. *[proved here]* $H^1(\Gamma; M_n) = \mathbb{L}/\ell\mathbb{L}$ *Hecke-equivariantly, an extension* $0 \to \mathbb{L}_{\mathrm{tors}}/\ell \to H^1(\Gamma; M_n) \to \mathbb{L}'/\ell \to 0$; $\mathbb{L}_{\mathrm{tors}}/\ell \cong \mathbb{L}[\ell] \cong H^0(\Gamma; M_n)$ *as* $\mathbb{F}_\ell$*-spaces (Claim 16.7). [formally verified: [GT,* `prop:cohforms`*]] (the exact sequence and the equality of lengths; not the identification* $H^1(\Gamma; M_n) = \mathbb{L}/\ell\mathbb{L}$*, an interface of [GT])*
2. *[proved here] Restriction* $\mathbb{T}^{\mathrm{coh}} \to \mathbb{T}^{\mathbb{L}}$ *is onto, with kernel the operators mapping* $\mathbb{L}$ *into* $\mathbb{L}_{\mathrm{tors}}$*. [formally verified: [GT,* `prop:cohforms`*]]*
3. *[proved here] (given the [standard] inputs of Lemma 30.4)* $T_p \mapsto T_p$ *extends to an isomorphism* $\mathbb{T}^{\mathbb{L}} \cong \mathbb{T}^{\mathrm{f}}$*:* ***the natural map runs from cohomology to forms****,* $\mathbb{T}^{\mathrm{coh}} \twoheadrightarrow \mathbb{T}^{\mathbb{L}} \cong \mathbb{T}^{\mathrm{f}}$*. [formally verified modulo Eichler–Shimura at level one and the forms-side Hecke algebra, as the hypotheses* `hES`*,* `hF`*: [GT,* `prop:cohforms`*]]*
4. *[proved here] Modulo* ℓ*: with* $\mathsf{A} := \mathrm{im}(\mathbb{T}^{\mathrm{coh}} \to \mathrm{End}H^1(\Gamma; M_n))$ *(the census algebra) and* $\mathsf{A}^{\mathbb{L}} := \mathrm{im}(\mathbb{T}^{\mathrm{coh}} \to \mathrm{End}(\mathbb{L}'/\ell))$*, there are natural surjections* $\mathsf{A} \twoheadrightarrow \mathsf{A}^{\mathbb{L}}$ *and* $\mathbb{T}^{\mathrm{f}}/\ell \cong \mathbb{T}^{\mathbb{L}}/\ell \twoheadrightarrow \mathsf{A}^{\mathbb{L}}$ *[formally verified: [GT,* `prop:cohforms`*]] (the first surjection* $\mathsf{A} \twoheadrightarrow \mathsf{A}^{\mathbb{L}}$*) [formally verified modulo the hypotheses* `hES`*,* `hF`*: [GT,* `prop:cohforms`*]] (the second surjection, since v1.18.0).* ***No natural map between*** A ***and*** $\mathbb{T}^{\mathrm{f}}/\ell$ ***is asserted****: both are quotients of* $\mathbb{T}^{\mathrm{coh}}/\ell$*, with kernels* $(\mathbb{T}^{\mathrm{coh}} \cap \ell\mathrm{End}\mathbb{L})/\ell$ *and* $(\ker(\mathbb{T}^{\mathrm{coh}} \to \mathbb{T}^{\mathbb{L}}) + \ell\mathbb{T}^{\mathrm{coh}})/\ell$*, and neither contains the other in general — an operator may vanish on* $\mathbb{L}'$ *without vanishing on a torsion class, and conversely.*
5. *[proved here] If* $\mathbb{L}_{\mathfrak{m}}$ *is torsion-free:* $\mathsf{A}_{\mathfrak{m}} = \mathsf{A}^{\mathbb{L}}_{\mathfrak{m}} = \mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}/(\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}} \cap \ell\mathbb{T}^{\mathbb{L},\mathrm{sat}}_{\mathfrak{m}})$*, a quotient of* $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ *with kernel* $(\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}} \cap \ell\mathbb{T}^{\mathbb{L},\mathrm{sat}}_{\mathfrak{m}})/\ell\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}$*, and* $\mathsf{A}_{\mathfrak{m}} \cong \mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ *iff* $\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}$ *is saturated in* $\mathrm{End}(\mathbb{L}_{\mathfrak{m}})$*. This holds if* $\mathbb{L}_{\mathfrak{m}}$ *contains a* $\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}$*-submodule* $N \cong \mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}$ *with torsion-free quotient — e.g.* $\mathbb{L}_{\mathfrak{m}}$ *free over* $\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}$*, or one of the* $\pm$*-parts of* ι^* *(§9) free of rank one over it (multiplicity one).*
6. *[proved here] On every block,* $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ *is recovered from cohomology as* $\mathsf{A}_K/\ell\mathsf{A}_K$ *of Corollary 30.6 (torsion-free blocks) or of its free part (Lemma 30.5), with the generation caveat of Corollary 30.6(2).*

Proof. (1) Claim 16.7(2) with $\mathbb{Z}_\ell$-coefficients ($\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$); the extension is $\mathbb{L}/\ell$ modulo the image of $\mathbb{L}_{\mathrm{tors}}$, which is $\mathbb{L}_{\mathrm{tors}}/(\mathbb{L}_{\mathrm{tors}} \cap \ell\mathbb{L}) = \mathbb{L}_{\mathrm{tors}}/\ell\mathbb{L}_{\mathrm{tors}}$, the torsion being a direct summand as a $\mathbb{Z}_\ell$-module; its dimension is the number of cyclic summands, $\dim \mathbb{L}[\ell]$. Hecke operators preserve torsion, so the sequence is equivariant. (2) Every $\mathbb{Z}_\ell$-linear endomorphism preserves $\mathbb{L}_{\mathrm{tors}}$; the image of the generators generates the image. (3) By Lemma 30.4's proof, $\mathbb{L}' \otimes \bar{\mathbb{Q}}_\ell$ and $M_k(\bar{\mathbb{Q}}_\ell)$ carry the same set of systems, each semisimply; so a polynomial in the T_p with $\mathbb{Q}_\ell$-coefficients vanishes on one iff it vanishes on the other (both iff it vanishes at every system). $\mathbb{L}'$ and $M_k(\mathbb{Z}_\ell)$ are torsion-free, so the kernels of $\mathbb{Z}_\ell[X_p] \to \mathrm{End}(\mathbb{L}')$ and $\to \mathrm{End}(M_k(\mathbb{Z}_\ell))$ are the $\mathbb{Q}_\ell$-kernels intersected with $\mathbb{Z}_\ell[X_p]$, hence equal. (4) An endomorphism of $\mathbb{L}/\ell$ induced by $\mathbb{T}^{\mathrm{coh}}$ preserves $\mathbb{L}_{\mathrm{tors}}/\ell$ and induces one of the quotient $\mathbb{L}'/\ell$, zero if the first is zero; the second surjection is $\mathbb{T}^{\mathbb{L}} \to \mathrm{End}(\mathbb{L}'/\ell)$ reduced modulo ℓ. The kernels are read off Proposition 30.2(1) and (2). (5) $\mathbb{L}_{\mathfrak{m}} = \mathbb{L}'_{\mathfrak{m}}$; Proposition 30.2(1),(2) for $(\mathbb{L}_{\mathfrak{m}}, \mathbb{T}^{\mathbb{L}}_{\mathfrak{m}})$, and (3). (6) Corollary 30.6(1) and (3). □

Where the torsion lives. $\mathbb{L}_{\mathrm{tors}} \neq 0$ only on the Dickson locus ($\dim H^0(\Gamma; M_n) > 0$, Claim 16.5(2)); the blocks it meets are those of the systems of Proposition 24.4, and it is non-ordinary (Remark 25.4). On a block without such a system, (5) applies. The census rows of slot-13 WP-9 were torsion-free on all 654 non-reduced rows tested ([verified computationally][a], `docs/implementation/slot13/WP-9_census_lattice.tsv`); that is data, not a statement of this document.

30.4 Hida's control at the level of algebras (P18)

Proposition 30.8 (The ordinary cuspidal Hecke algebra modulo ℓ is constant along a class)**.** *Let $\ell \geq 5$, $a \bmod \ell - 1$, and $k, k' \geq 2$ with $k \equiv k' \equiv a \pmod{\ell - 1}$. Write $h^0_k := h^0_k(\Gamma_1(\ell), a)$ and $h^0 := h^0(\ell, a)$ in the notation of Theorem 27.11(3).*

1. *[proved here] (from Theorem 27.11(3), [standard]) $(\ell, P_k) = (\ell, X)$ in $\Lambda = \mathbb{Z}_\ell[[X]]$, so $h^0_k/\ell h^0_k \cong h^0/(\ell, X)h^0 \cong h^0_{k'}/\ell h^0_{k'}$ as $\mathbb{F}_\ell$-algebras, each Hecke operator of the universal algebra going to itself. [formally verified modulo Hida's control theorem, Theorem 27.11(3): [GT, `prop:hidaalgconst`]] (as an isomorphism of rings $h^0_k/\ell h^0_k \cong h^0_{k'}/\ell h^0_{k'}$; not "each Hecke operator going to itself")*
2. *[proved here] (from (1)) The maximal ideals of h^0_k with their residual systems of Hecke eigenvalues, and the Hilbert functions and Loewy lengths of the local factors of h^0_k/ℓ, do not depend on k in the class.*
3. *[standard] (the level-1 reading; the citations of Proposition 27.9(4)) For $k \geq 3$ the ordinary cuspidal Hecke algebra of level 1 and weight k (on $e_1 S_k(\mathbb{Z}_\ell)$, T_ℓ the level-one operator) is isomorphic to h^0_k, with $T_p \mapsto T_p$ ($p \neq \ell$) and $T_\ell \mapsto U_\ell + \ell^{k-1}U_\ell^{-1}$; so modulo ℓ, $T_\ell \mapsto U_\ell$, and (2) holds for the level-1 ordinary cuspidal algebras for $k, k' \geq 3$.*

Theorem 27.12(1) is a statement about ranks*; every algebra-level consequence used in this project — that the residual ordinary systems and h^{ord}_k/ℓ are constant along a class — rests on this proposition, i.e. on Theorem 27.11(3) with Proposition 27.9, not on Theorem 27.12(1). The statements are on the forms side (Hecke algebras of cusp forms), where Hida's lattice $L_n \neq V$ (Remark 27.10(3)) plays no role; they reach the cohomology of V through Proposition 30.7(3), for the free part only.*

Proof. (1) $P_k = (1+X) - (1+\ell)^k = X - \big((1+\ell)^k - 1\big)$ and $(1+\ell)^k - 1 \in \ell\mathbb{Z}_\ell$, so $(\ell, P_k) = (\ell, X)$. Theorem 27.11(3) gives $h^0/P_k h^0 \cong h^0_k$; reducing modulo ℓ, $h^0_k/\ell \cong h^0/(\ell, P_k) = h^0/(\ell, X)$, and the same for k'. The isomorphisms are of Λ-algebras sending the universal $T(m)$ to $T(m)$. (2) An isomorphism of algebras preserves maximal ideals, their residue fields, the images of each $T(m)$ and the $\mathfrak{m}$-adic filtrations. (3) For $k \geq 3$ the ordinary cusp forms of weight k on $\Gamma_0(\ell)$ are the ordinary ℓ-stabilisations of the ordinary level-1 forms (an ℓ-new form has $a_\ell^2 = \ell^{k-2}$, not a unit): the stabilisation $f \mapsto f(z) - \beta f(\ell z)$ with $\beta = \ell^{k-1}/u$ has U_ℓ-eigenvalue the unit root u of $X^2 - a_\ell X + \ell^{k-1}$ and the same a_p, $p \neq \ell$ (the sources cited in Proposition 27.9(4)). u lies in $\mathbb{Z}_\ell[T_\ell]$ (Hensel: $X^2 - T_\ell X + \ell^{k-1} \equiv X(X - T_\ell)$ with T_ℓ a unit on the ordinary part), and $T_\ell = u + \ell^{k-1}u^{-1}$, so the two algebras are generated by corresponding operators; $\ell^{k-1}u^{-1} \equiv 0 \pmod{\ell}$. □

30.5 The forms-side θ (P20)

Remark 30.9 (The θ of modular forms modulo ℓ has a Frobenius kernel; the cohomological θ_{L} does not see it). (1) [proved here] On q-expansions, $\theta := q\frac{d}{dq}$ maps $\sum a_m q^m \in \mathbb{F}_\ell[[q]]$ to $\sum m a_m q^m$, so $\ker\theta = \mathbb{F}_\ell[[q^\ell]]$. If $g \in M_{k_0}(\mathbb{F}_\ell)$ then $V(g) := g(q^\ell) = g^\ell$ (Frobenius on coefficients in $\mathbb{F}_\ell$) is a mod-ℓ modular form of weight ℓk_0, and $E_{\ell-1}^j V(g)$, $E_{\ell-1} \equiv 1$, one of weight $\ell k_0 + j(\ell-1)$ with the same q-expansion; it lies in $\ker\theta$. So the forms-side θ on $M_k(\mathbb{F}_\ell)_{\mathfrak{m}}$ is *not* injective whenever such a $V(g)$ lies in the block — although Theorem 24.11(3) makes the cohomological θ_{L} injective on $H^1(\Gamma; M_n)$ away from the Eisenstein line. [formally verified: [GT, `rmk:thetaforms`]] (the two power-series facts $\ker\theta = \mathbb{F}_\ell[[q^\ell]]$ and $g(q^\ell) = g^\ell$ only, nothing about modular forms) (2) [verified computationally][a] (slot-14 WP-C9 §3.3, `theta_check.py` on q-expansions; the cluster run of that report is not published, the forms-side check is local and re-runnable) At $k = 176$, $\ell = 7$ (the cell $(174, 7)$), θ has rank 6 on the 7-dimensional block W_{176} of the system $(a_2, a_3) \equiv (1, 0)$, its kernel spanned by $V(g)$ with g the weight-20 eigenform in that system; likewise at $(406, 11)$ (rank 6 of 7); at $(144, 7)$ rank 5 of 6 with no change of Hilbert function. Along the cohomologically bijective θ_{L}-steps $(174 \to 182, 7)$ and $(406 \to 418, 11)$ (equal dim, A_1 preserved)

the forms-side $\mathbb{T}_{\mathfrak{m}}/\ell$ changes Hilbert function $(1,3,2,1) \to (1,2,2,2)$ at equal dimension and Loewy length. (3) This is consistent with Proposition 30.7(4),(5): $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ is not a function of $H^1(\Gamma; M_n)_{\mathfrak{m}}$ (the saturation defect intervenes), so a map bijective on $H^1(\Gamma; M_n)$ need not preserve it. No statement relating the forms-side θ and θ_{L} as maps is made here.

30.6 The saturation defect as a Tor term

Remark 30.10 (The saturation defect as a Tor term). [proved here] (v1.17.0-pre.1; proposed by slot-16 WP-5, `docs/implementation/slot16/WP-5_tor_note.md` §5) In the notation of Definition 30.1 put $G := \mathbb{T}^{\mathrm{sat}}/\mathbb{T}$ and $I := \mathrm{Tor}_1^{\mathbb{Z}_\ell}(G, \mathbb{F}_\ell)$.

1. $I = G[\ell] \cong (\mathbb{T} \cap \ell\mathbb{T}^{\mathrm{sat}})/\ell\mathbb{T} = \ker\big(\mathbb{T}/\ell\mathbb{T} \to \mathrm{End}(L/\ell L)\big)$, canonically; the last two are ideals of $\mathbb{T}/\ell\mathbb{T}$, and $(\mathbb{T}/\ell\mathbb{T})/I \cong \mathsf{A}_1$ as $\mathbb{F}_\ell$-algebras.
2. $\dim_{\mathbb{F}_\ell} I = \varepsilon_1^{\mathrm{sat}} = r - \dim_{\mathbb{F}_\ell} \mathsf{A}_1$, the number of cyclic summands of G; and $I = 0$ iff $\mathbb{T}$ is saturated. More generally the number of summands $\mathbb{Z}/\ell^e$ of G with $e \geq j$ is $\varepsilon_j^{\mathrm{sat}} - \varepsilon_{j-1}^{\mathrm{sat}}$ ($\varepsilon_0^{\mathrm{sat}} := 0$).
3. At every K at which the equivalent conditions of Proposition 30.2(3) hold, I is the kernel of the composite $\mathsf{A}_K/\ell\mathsf{A}_K \to \mathrm{End}\big((L/\ell^K L)/\ell\big) = \mathrm{End}(L/\ell L)$, read through the isomorphism $\mathbb{T}/\ell\mathbb{T} \cong \mathsf{A}_K/\ell\mathsf{A}_K$ of Proposition 30.2(3)(b).
4. If L contains a $\mathbb{T}$-submodule $N \cong \mathbb{T}$ with L/N torsion-free, then $I = 0$ (Proposition 30.2(2)); the converse fails: a saturated $\mathbb{T}$ need not have such an N.
5. I is not $\mathrm{Tor}_1^{\mathbb{T}}(L, \mathbb{T}/\mathfrak{m})$: for a maximal ideal $\mathfrak{m}$ of a local $\mathbb{T}$ the latter vanishes iff L is free over $\mathbb{T}$ (the multiplicity-one question), so by (4) $I \neq 0$ implies that L is not free over $\mathbb{T}$, while $I = 0$ says nothing about freeness.

Proof. (1) Tensoring $0 \to \mathbb{Z}_\ell \xrightarrow{\ell} \mathbb{Z}_\ell \to \mathbb{F}_\ell \to 0$ with G gives $\mathrm{Tor}_1^{\mathbb{Z}_\ell}(G, \mathbb{F}_\ell) = \ker(\ell\colon G \to G) = G[\ell]$ ($\mathbb{Z}_\ell$ is free). Inside $\mathbb{T} \otimes \mathbb{Q}$, $G[\ell] = (\ell^{-1}\mathbb{T} \cap \mathbb{T}^{\mathrm{sat}})/\mathbb{T}$, and multiplication by ℓ is a $\mathbb{T}$-linear isomorphism $(\ell^{-1}\mathbb{T} \cap \mathbb{T}^{\mathrm{sat}})/\mathbb{T} \to (\mathbb{T} \cap \ell\mathbb{T}^{\mathrm{sat}})/\ell\mathbb{T}$ ($\mathbb{T} \otimes \mathbb{Q}$ is torsion-free). By Proposition 30.2(1) at $K = 1$, $\ker(\mathbb{T} \to \mathrm{End}(L/\ell L)) = \mathbb{T} \cap \ell\mathbb{T}^{\mathrm{sat}} \supseteq \ell\mathbb{T}$, whose image in $\mathbb{T}/\ell\mathbb{T}$ is the displayed kernel; a kernel of a ring map is an ideal, and the quotient is the image A_1. (2) $\dim_{\mathbb{F}_\ell} G[\ell] = \mathrm{length}(G[\ell]) = \varepsilon_1^{\mathrm{sat}}$ by Proposition 30.3(1), and $\varepsilon_1^{\mathrm{sat}} = r - \mathrm{length}(\mathsf{A}_1)$ by definition; for a finite $\mathbb{Z}_\ell$-module $G \cong \bigoplus_i \mathbb{Z}/\ell^{e_i}$, $\mathrm{length}(G[\ell^j]) = \sum_i \min(e_i, j)$, so $\mathrm{length}(G[\ell^j]) - \mathrm{length}(G[\ell^{j-1}]) = \#\{i : e_i \geq j\}$; at $j = 1$ this is the number of summands. $I = 0$ iff $G = 0$ (Proposition 30.3(3)). (3) The composite $\mathbb{T} \to \mathsf{A}_K \to \mathrm{End}(L/\ell^K L) \to \mathrm{End}(L/\ell L)$ is the reduction map of (1), and $\ell\mathsf{A}_K$ acts as 0 on $L/\ell L$, so it factors through $\mathsf{A}_K/\ell\mathsf{A}_K$; under (3)(b) its kernel there is the kernel of $\mathbb{T}/\ell\mathbb{T} \to \mathrm{End}(L/\ell L)$, which is I by (1). (4) The first sentence is Proposition 30.2(2) with (2). For the second, take the order $\mathbb{T} := \{(a,b,c) \in \mathbb{Z}_\ell^3 : a \equiv b \equiv c \pmod{\ell}\}$ of Corollary 30.6(2), with $\mathbb{T}/\ell\mathbb{T} \cong \mathbb{F}_\ell[x,y]/(x,y)^2$, and $L := \mathbb{T}^\vee = \mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{T}, \mathbb{Z}_\ell)$ with $(t\lambda)(s) := \lambda(ts)$; $\mathbb{T}$ acts faithfully ($t\lambda = 0$ for all λ gives $\lambda(t) = 0$ for all λ). If $x \in \mathbb{T} \otimes \mathbb{Q}$ preserves L (the action extended $\mathbb{Q}$-linearly), then $\lambda(x) = (x\lambda)(1) \in \mathbb{Z}_\ell$ for every $\lambda \in \mathbb{T}^\vee$, so $x \in \mathbb{T}^{\vee\vee} = \mathbb{T}$ ($\mathbb{T}$ is free): $\mathbb{T}$ is saturated (the argument uses nothing about this $\mathbb{T}$: every order is saturated on its $\mathbb{Z}_\ell$-dual). A submodule $N \cong \mathbb{T}$ has rank $3 = \mathrm{rank}\, L$, so L/N torsion-free forces $N = L$, i.e. $\mathbb{T}^\vee \cong \mathbb{T}$; then $\mathbb{T}^\vee/\ell \cong \mathbb{T}/\ell$ would be a free $\mathbb{T}/\ell$-module of rank one, so $\mathbb{T}/\ell \cong (\mathbb{T}/\ell)^\vee$ would be Gorenstein, but $\mathbb{F}_\ell[x,y]/(x,y)^2$ has a two-dimensional socle. (5) For L finitely generated over the local Noetherian ring $\mathbb{T}$, $\mathrm{Tor}_1^{\mathbb{T}}(L, \mathbb{T}/\mathfrak{m}) = 0$ iff L is free (Nakayama and a minimal presentation); a free $L \neq 0$ contains $N \cong \mathbb{T}$ as a direct summand with free, hence torsion-free, quotient, so (4) applies. □

Scope: what this chapter does not do. It proves the saturation identities for any commutative order on a free lattice, the certified reading of the $\mathbb{Z}/\ell^K$ route on free and torsion blocks, and the comparison of cohomological and forms Hecke algebras at level 1, $\ell \geq 5$, given Eichler–Shimura. Not asserted: why A has Loewy length ≤ 2 on the non-reduced blocks where it has ([E] [E, `pr:latticesplit`], [conjectural] — refuted as stated at $\ell = 37$ (slot-15

WP-12 (a), `docs/implementation/slot15/WP-4_referee.md` §8), its v1.16.0 form refuted as stated at $(608, 547)$ and contradicted on the forms side (slot-17 R17-2; v1.17.0-pre.3, m-C3), its v1.17.0-pre.2 Kummer-free form contradicted on the forms side at irreducible congruence classes, the regular $\ell = 139$ among them, and (R) now asserted on congruence-free classes only (v1.17.0-pre.4, owner ruling on the referee F1's B-1); no bound on the Loewy length of A is asserted here); any bound on c^{sat} or δ^{sat} in terms of (n, ℓ); a stopping rule for $\mathbb{T}^{\mathrm{coh}}_{\mathfrak{m}}$ on a torsion block with $f_{\mathfrak{m}} > 0$; (the faithfulness of $\mathbb{T}^{\mathrm{f}}/\ell$ on $M_k(\mathbb{F}_\ell)$ at $\ell \geq 5$ is [E, `cl:mkfaithful`], slot-15 WP-19, not restated here; at $\ell \in \{2, 3\}$ nothing is asserted); levels other than 1.

Code map. `HeckeExt/HeckeRingHeckeAlgebra` (slot-14 WP-6) computes A_K, its length, the Hilbert function of A_K/ℓ, $\varepsilon_K^{\mathrm{sat}}$ (its `eps`), the free part of a torsion block on ℓ^t-scaled generators (Lemma 30.5(2)), and the verdicts `stable`, `torsion_free_stable`, `torsion_stable` of Corollary 30.6(3) with $\mathbb{T}'$ the algebra of its generator set S. At v1.15.0-pre.1 these verdicts were **readings**; since slot-15 WP-5 (built at v1.15.0; WP-4's ACCEPT of Propositions 30.2, 30.3, owner ruling Q4) they are *certified by Corollary 30.6* in its B1 form (for $\mathbb{T}'_{\mathfrak{m}'}$; for $\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}$ only with a generation certificate, which is not run), printed as `stable_cert=certified` (Fable audit ACCEPT WITH REPAIRS, repaired). The digit route for $\ell^K \geq 2^{64}$ is built (WP-5). **Built at v1.16.0, recorded at v1.17.0-pre.1** (slot-16 WP-5, `docs/implementation/slot16/WP-5_report.md`; cross-model audit `docs/implementation/slot16/WP-3_WP-5_audit.md`, repaired): `HeckeExt/HeckeSaturationDefect` computes Remark 30.10's I on a torsion-free block at a stable K by two routes compared per block — the kernel of (3) in the regular representation of $\mathsf{A}_K/\ell\mathsf{A}_K$, and the length formula of (2) from the route's own A_1 — with its minimal generators, radical layers, Loewy length, socle and annihilator, and the filtration $(\varepsilon_j^{\mathrm{sat}})_j$ with the plateau of Proposition 30.3(2) as `final`; torsion blocks are refused by name (Corollary 30.6(3) asserts no stopping rule there); `HeckeSaturationDefectValidation` (`standard`). **At v1.23.0-pre.4** (slot 23, value-preserving, no statement moves): the route's levels come from one module run per cell in the common case (G1; §30.7's code map), and the module route under it is the G2 re-implementation mapped in the code map of §20. The forms side of Proposition 30.7 (the sat route's forms cross-route, `BuildForms` / `RunForms` in `HeckeExt/HeckeSaturationDefect.cpp`, and every q-series product of `HeckeExt/HeckeFormsHeckeAlgebra.{h,cpp}`) multiplies through `forms::MulSeries`, which since slot-23 G2-X (`docs/implementation/slot23/G2X_report.md`, `G2X_audit.md` with its fixes; merged at `1ba46525`) is `MulSeriesFast`: the exact integer convolution of the residues by NTT over the primes $5 \cdot 2^{25} + 1$ and $7 \cdot 2^{26} + 1$ and CRT, reduced modulo ℓ, inside its checked exactness bounds; short operands take `MulSeriesSchoolbook` (delayed reduction), products past the transform length `MulSeriesChunked` (blocks of 2^{24}, each an NTT product within both bounds), and $\ell \notin [2, 2^{16})$ the naive product; `EisensteinSeries` takes its divisor sums by a sieve. **The rollback lever is value-neutral**: `HECKE_FORMS_FASTMUL=0` restores the $O(N^2)$ product `MulSeriesNaive` and the trial-division loop (any other value warns and keeps the default; scoped as `ScopedFormsFastMul`; its `EFFECTIVE` line `FormsFastMulEffectiveLine()`, counters `FormsFastMulCountersNow()`), and since the truncated product modulo ℓ and the divisor sums are unique, no route can move a byte; gate `HeckeFormsFastMulValidation` (`standard`; `Registry/Slot23/G2X.cpp`, source `HeckeExt/HeckeFormsFastMulValidation.cpp`): every route against the naive oracle byte for byte, the identities $E_4^2 = E_8$ and $E_4^3 - E_6^2 = 1728\Delta$ beyond the oracle's reach, and whole `SAT_DEFECT` cells. **At v1.23.0-pre.10** (slot-24 FNTT, value-preserving; `docs/implementation/slot24/FNTT_report.md`, `FNTT_audit.md` with its fixes): `BuildCuspSpace` / `BuildModularSpace` compute the same basis series in the transform domain of the same two primes (the spectra of E_6^2 and Δ cached, S_k and the extra basis series of M_k in one pass, no power tables), and `HeckeOnCusp` reduces modulo ℓ once per accumulated sum (whenever both levers `HECKE_FORMS_FASTMUL`, `HECKE_FORMS_NTT_CACHE` are on

and $(m+1)\ell^2 < 2^{64}$, m the number of columns, at any N); the transform-domain route applies when the fast product is on, $N \geq 1024$, $2N-1 \leq 2^{25}$ and $N(\ell-1)^2 < (5\cdot 2^{25}+1)(7\cdot 2^{26}+1)$ (`FnttApplicable`; every operand is a residue vector of length $\leq N$, so every coefficient of the exact integer convolution lies in $[0, N(\ell-1)^2]$ and the CRT recovers it), the earlier route elsewhere. Since the truncated product modulo ℓ is unique, no route moves a byte; **the rollback lever is value-neutral**: `HECKE_FORMS_NTT_CACHE=0` restores the G2-X route verbatim (any other value warns and keeps the default; scoped as `ScopedFormsNttCache`; its `EFFECTIVE` line `FormsNttEffectiveLine()`). **The forms ℓ-range**: the sentence above on $\ell \notin [2, 2^{16})$ stays true of `MulSeriesFast`, but no longer bounds the forms cross-route. A forms space is built (`forms::FormsSpaceSupported`(ℓ, N)) for every $5 \leq \ell < 2^{16}$, and for $2^{16} \leq \ell < 2^{27}$ only when $N < 1024$ or the route above applies and is exact; any other space is refused by `BuildCuspSpace` / `BuildModularSpace` (an `ENGINE_CHECK` naming the bounds), and a cell whose forms space would be refused is refused by `ComputeSaturationDefectCell` and the `HECKE_SAT_DEFECT_CELLS` parser before the module run, so no forms product at $\ell \geq 2^{16}$, $N \geq 1024$ runs by the $O(N^2)$ product. At the sat route's precision $N = 2(k/12+1)p_{\max}+2$ (`forms::SatFormsPrecision`) the condition is roughly $2(k/12+1)\ell^3 < 7.9\cdot 10^{16}$: $\ell = 100003$ is admitted for $n \leq 464$, the largest admitted integer ℓ is 114868 at $n = 300$ and 77702 at $n = 1000$ (the largest admitted primes, the only ℓ a cell takes, 114859 and 77699; the audit's derivation, `FNTT_audit.md` F5). The gate is `HeckeFormsFastMulValidation` (arms (N1)–(N3), (N5), (N6): the transform-domain route against the schoolbook, the naive product and the G2-X route byte for byte, at ℓ up to 1000003, and the bound at its edge).

Formalization note. Propositions 30.2, 30.3, Lemma 30.5, Corollary 30.6 and Remark 30.10 (v1.17.0-pre.1) are commutative algebra over $\mathbb{Z}_\ell$ (finite modules, Nakayama, Tor over a DVR) and a good first target; Lemma 30.4 and Proposition 30.7(3) need Eichler–Shimura, Proposition 30.8 Hida theory. (v1.23.0-pre.3) Lemma 30.11 is homological algebra over $\mathbb{Z}_\ell$ (one long exact sequence and its naturality) on top of Proposition 23.2(1), Remark 7.6 (with its naturality in the ring), Claim 16.2(3) and Proposition 15.18(1),(2), Proposition 15.12(2)–(4) with Lemma 11.3(2) over $\mathbb{Z}_\ell$, applied to the bulk complex alone, and Theorem 10.8(1) (the T_p commute on M_K; used by (2), (3) and the module structure in (1); its proof uses Theorem 11.10, and its Lean marker is a `\LEANlit` with the hypothesis `hcomm`); the compatibility of its (1) with the classical T_p on $\mathbb{L}$ (as opposed to the T_p transported along Remark 7.6) needs Theorem 11.10 as well, which is unformalized; it carries no `\LEAN` marker.

30.7 One module run per cell: the precision levels as reductions

Why this section exists. The route of Corollary 30.6 reads each block at the levels $K = 1$ (through M_K/ℓ), $K = 2, 3, \ldots$ up to the first stable, final level, and at $K_{\mathrm{cls}} := \max(2, e_{\max}+1)$, $e_{\max}$ of Proposition 26.1, never above $K_{\mathrm{need}} := \max(\mathrm{KMAX}, K_{\mathrm{cls}})$. Here KMAX is the route's level cap, a run setting, not a mathematical quantity (the largest precision read for the stopping rule: the code's `LoewyOptions::kmax` and `SatOptions::kmax`, default 6; a block not stable by it reads `not_stable_at_KMAX`), and K_{stab} is the level at which the stopping rule (read from $K = 2$; on a torsion block decided only from K_{cls}) first reads stable in the sense of Corollary 30.6(1), i.e. $\dim_{\mathbb{F}_\ell} \mathsf{A}'_K/\ell\mathsf{A}'_K = \lceil f/2 \rceil$ (on a torsion block, for its free part, Corollary 30.6(3)); so a block stable already at $K = 1$ reads 2. It is the code's `LoewyBlock::K_stab`, printed as `K_stab=`, on a stable verdict read at $K \geq 2$ (on a not-stable block the field holds the last level read, on `reduced_rank1`, read at $K = 1$ only, it holds 0). Through v1.23.0-pre.2 the implementation ran the whole module route once per level read: at $(n, \ell) = (300, 7)$ five runs, $K = 2, \ldots, 6$, 377.1 of the cell's 378.0 s (`ELL_scout.md` §1.3). Lemma 30.11 says that one run at the largest level read determines every lower level up to isomorphism of Hecke modules, which is all the route reads; Obligation 30.12 says how the implementation may use that. Nothing in this section changes a value.

Lemma 30.11 (The precision levels of the module route are reductions of one run)**.** *[proved here] Assume (H4), $\ell \geq 5$ and $n \geq 0$. Let S be a finite set of primes and, for $p \in S$, let T_p act on $C^{\bullet}_{\mathrm{eq}}(\mathbb{Z}/\ell^K)$ through Θ_g (Definition 12.10; $T_\ell = T_{\Gamma g \Gamma}$ with $g = \mathrm{diag}(1,\ell)$, as in the Setting of §30.2) formed, at every level K, from a τ-table with every fill over $\mathbb{Z}$ (Remark 11.5; the table may be computed afresh at each level, each one integral) and the reduced $\rho_n(g\gamma)$. For $K \geq 1$ put $M_K := H^1_{\mathrm{eq}}(C^{\bullet}_{\mathrm{eq}}(\mathbb{Z}/\ell^K))$ with the induced operators T_p, and for $1 \leq K \leq K'$ let $r_{K',K} \colon M_{K'} \to M_K$ be the map induced by the reduction $\mathbb{Z}/\ell^{K'} \to \mathbb{Z}/\ell^K$.*

1. ***(Reduction.)*** *$r_{K',K}$ is surjective, commutes with every T_p, and has kernel $\ell^K M_{K'}$. Hence it induces an isomorphism $M_{K'}/\ell^K M_{K'} \cong M_K$ of $\mathbb{Z}[X_p : p \in S]$-modules ($X_p$ acting as T_p). With $\mathbb{L} = H^1(\Gamma; V \otimes \mathbb{Z}_\ell)$ there are isomorphisms $M_K \cong \mathbb{L}/\ell^K\mathbb{L}$, natural in K and carrying $r_{K',K}$ to the canonical projection, compatible with the identification $H^1(\Gamma; V \otimes \mathbb{Z}/\ell^K) = \mathbb{L}/\ell^K$ of Lemma 30.5 through Remark 7.6 (whose comparison is natural in the ring, being induced by one complex $C_\bullet(K^u)$ for every R). They commute with the T_p when the T_p on $\mathbb{L}$ are transported along Remark 7.6 over $\mathbb{Z}_\ell$ (the pipeline's operators, as in the Setting); that these are the classical double-coset operators on $\mathbb{L}$ is Theorem 11.10, which this sentence then also rests on. The first sentence of (1), and (2)–(5), use neither directly; but (2) and (3), and the module structure over the polynomial ring $\mathbb{Z}[X_p : p \in S]$ in (1), use that the T_p commute on M_K, Theorem 10.8(1), whose proof uses Theorem 11.10 (through Remark 7.6); (4) and (5) use no Hecke relation.*
2. ***(Presentations.)*** *If $M_{K'} = \bigoplus_{i=1}^{D} (\mathbb{Z}/\ell^{a_i})g_i$, $1 \leq a_i \leq K'$, with $T_p g_j = \sum_i A_{ij}^{(p)} g_i$, then $M_K = \bigoplus_{i=1}^{D} (\mathbb{Z}/\ell^{\min(a_i,K)})\bar{g}_i$, $\bar{g}_i := r_{K',K}(g_i)$, with $T_p \bar{g}_j = \sum_i \big(A_{ij}^{(p)} \bmod \ell^{\min(a_i,K)}\big)\bar{g}_i$. No summand vanishes, so $D = \dim_{\mathbb{F}_\ell} M_1$ at every level. Write $M_{K'}{\downarrow}K$ for this presentation (the* derived module*). If e' is an idempotent of the image $\mathsf{B}_{K'}$ of $\mathbb{Z}_\ell[X_p : p \in S]$ in $\mathrm{End}(M_{K'})$, its image e in B_K is an idempotent, $r_{K',K}$ restricts to an isomorphism $e'M_{K'}/\ell^K \cong eM_K$, and e is the unique idempotent of B_K with the same image in B_1; so the block of a joint system at level K is the reduction of its block at level K'.*
3. ***(Readings.)*** *Let Φ be any function of a finite $\mathbb{Z}_\ell$-module with commuting endomorphisms indexed by S that is constant on isomorphism classes of $\mathbb{Z}_\ell[X_p : p \in S]$-modules. Then $\Phi(M_{K'}{\downarrow}K) = \Phi(M_K)$, and likewise on blocks; and, by (1), likewise for every reading of the tower $(M_K, r_{K',K})_{K \leq K'}$ that is constant on isomorphism classes of towers of $\mathbb{Z}_\ell[X_p : p \in S]$-modules (one that uses the maps between levels: the derived tower $(M_{K'}{\downarrow}K)_K$ with the reductions of (2) is isomorphic to it, compatibly with the $r_{K',K}$). In particular the joint $\mathbb{F}_\ell$-rational systems of M_1 and their dimensions, the block eM_K and its cyclic exponents (at K_{cls}: the free rank f and the torsion exponents), the algebra A_K generated by the T_p in $\mathrm{End}(eM_K)$ with its length, cyclic exponents and the Hilbert function of $\mathsf{A}_K/\ell\mathsf{A}_K$ at $\mathfrak{m}$, the free-part reading of Lemma 30.5(2), $\varepsilon_K^{\mathrm{sat}}$, the ideal I of Remark 30.10(3) with its radical layers, socle and annihilator, and the hypotheses and conclusions of Corollary 30.6(1),(3) are the same read on $M_{K'}{\downarrow}K$ as on M_K; so are the monomial basis of A_K/ℓ chosen greedily in a fixed order of the monomials in the T_p and the generators of I as the reduced echelon form of I in that basis (the reduced echelon form of a subspace in fixed coordinates is unique, and the monomial coordinates are fixed by $(\mathsf{A}_K, (T_p)_p)$), hence the* `SAT_DEFECT` *fields* `generators`*,* `monomial_basis`*,* `quotient_hilbert`*,* `dim_image`*,* `tor_routes` *and* `sat_structure`*, and the* `forms=(...)` *reading, which sees only the $K = 1$ block through the dimension of a joint algebra.*
4. ***(Downward only.)*** *Nothing here determines a level above the levels run: the $\mathbb{Z}_\ell$-lattices with one endomorphism $(\mathbb{Z}_\ell, 0)$ and $(\mathbb{Z}_\ell, \ell^K)$ have isomorphic reductions modulo ℓ^K and non-isomorphic ones modulo ℓ^{K+1}.*
5. ***(The hypothesis is the absence of ℓ-torsion one degree up, and it is needed.)*** *Let C be a complex of free $\mathbb{Z}_\ell$-modules of finite rank with a cochain map φ, $1 \leq K \leq K'$,*

q any degree. There is an exact sequence, natural in K and φ-equivariant,

$$0 \to H^q(C)/\ell^K \to H^q(C/\ell^K) \xrightarrow{\delta_K} H^{q+1}(C)[\ell^K] \to 0,$$

under which $r_{K',K}$ induces the reduction on the left and multiplication by $\ell^{K'-K}$ on the right. So $r_{K',K}$ is onto with kernel $\ell^K H^q(C/\ell^{K'})$ for every $K \le K'$ if and only if $H^{q+1}(C)[\ell] = 0$; if $H^{q+1}(C)$ has a cyclic summand $\mathbb{Z}/\ell^a$, $a \ge 1$, then $r_{K+1,K}$ is not onto for every $K \ge a$. Example: $C = (\mathbb{Z}_\ell \xrightarrow{\ell} \mathbb{Z}_\ell)$ in degrees $q, q+1$ has $H^q(C/\ell^K) = \ell^{K-1}\mathbb{Z}/\ell^K \cong \mathbb{Z}/\ell$ for every K and $r_{K',K} = 0$ for $K' > K$, although the groups are isomorphic. On the model this excludes H^0_{eq} on the Dickson locus, where $\mathbb{L}$ has ℓ-torsion: there $H^0_{\mathrm{eq}}(C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^K)) \cong \mathbb{L}[\ell^K]$ and $r_{K',K}$ is multiplication by $\ell^{K'-K}$. Nothing is asserted at $\ell \in \{2,3\}$ (where $C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^K)$ is not a base change, Proposition 23.2(2)), for the cone, or at other levels.

Proof. The integral complex. Put $C := C^\bullet_{\mathrm{eq}}(\mathbb{Z}_\ell)$. By Proposition 23.2(1) ($6 \in \mathbb{Z}_\ell^\times$) $C = C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes \mathbb{Z}_\ell$, a complex of free $\mathbb{Z}_\ell$-modules of finite rank (Definition 23.1(1)), and $C/\ell^K C = C^\bullet_{\mathrm{eq}}(\mathbb{Z}) \otimes \mathbb{Z}/\ell^K = C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^K)$ for every $K \ge 1$. Fix one integral table. It makes Θ_g the base change of one integral cochain map $\Theta_{g,\mathbb{Z}}$ of $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$ (Proposition 15.18(1),(2), applied to the bulk complex alone: (41) has integer coefficients); write φ for $\Theta_{g,\mathbb{Z}} \otimes \mathbb{Z}_\ell$. The table used at level K may be another one, but it is integral too, and two integral tables give cochain maps of $C = C^\bullet_{\mathrm{eq}}(\mathbb{Z}_\ell)$ (each the base change of its integral one, as above) differing by $d\Upsilon + \Upsilon d$ with Υ $\mathbb{Z}_\ell$-linear (Proposition 15.12(2) with Lemma 11.3(2) over $R = \mathbb{Z}_\ell$, $6 \in \mathbb{Z}_\ell^\times$: the two τ differ by $\partial H + H\partial$ with H additive with (T1); replacing H by the $\mathbb{Z}_\ell$-linear extension of its restriction to $C_\bullet(K^u)\otimes 1$ if necessary, H is $\mathbb{Z}_\ell$-linear, keeps (T1), and still gives $\tau - \tau' = \partial H + H\partial$, both sides being $\mathbb{Z}_\ell$-linear, agreeing on $C_\bullet(K^u) \otimes 1$, which spans, and ∂ preserving it; and $\Theta'_g = \Theta_g + d\widetilde{H} + \widetilde{H}d$, the bulk component of (47) (Theorem 11.7(b)), so $\Upsilon := \widetilde{H}$; Proposition 15.12(3),(4), again under $6 \in \mathbb{Z}_\ell^\times$, if the transporters s_j or $g \in \Gamma g \Gamma$ are rechosen; applied to the bulk complex alone); reduced modulo ℓ^K the homotopy stays a homotopy ($C/\ell^K C = C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^K)$), so the operator the level-K table induces on $H^\bullet(C/\ell^K)$ is the one $\varphi \otimes \mathbb{Z}/\ell^K$ induces. That map commutes with the reductions $C/\ell^{K'} \to C/\ell^K$; the matrices of T_p on M_K are those it induces (Algorithm 12.13). By Remark 7.6 over $R = \mathbb{Z}_\ell$ and Claim 16.2(3), $H^2(C) \cong H^2(\Gamma; V \otimes \mathbb{Z}_\ell) = 0$ and $H^1(C) \cong \mathbb{L}$; the comparison of Remark 7.6 is induced by the one complex $C_\bullet(K^u)$ for every R, hence natural along $\mathbb{Z}_\ell \to \mathbb{Z}/\ell^K$.

(5) C is torsion-free, so $0 \to C \xrightarrow{\ell^K} C \to C/\ell^K \to 0$ is a short exact sequence of complexes, and its long exact sequence gives the displayed sequence [standard: [Wei94, Thm. 1.3.1]]. The triple $(\ell^{K'-K}, \mathrm{id}, \mathrm{reduction})$ is a map from the sequence at K' to the one at K (left square: $\ell^K \circ \ell^{K'-K} = \mathrm{id} \circ \ell^{K'}$), so naturality of the connecting map gives $\delta_K \circ r_{K',K} = \ell^{K'-K} \circ \delta_{K'}$, and on the left $r_{K',K}$ is induced by id_C, i.e. it is the reduction $H^q(C)/\ell^{K'} \to H^q(C)/\ell^K$. φ commutes with all vertical maps and with ℓ^K, hence with every map in sight. If $H^{q+1}(C)[\ell] = 0$, then $H^{q+1}(C)[\ell^K] = 0$ for all K, the left maps are isomorphisms, and $r_{K',K}$ is the reduction $H^q(C)/\ell^{K'} \to H^q(C)/\ell^K$: onto, with kernel $\ell^K(H^q(C)/\ell^{K'})$, which corresponds to $\ell^K H^q(C/\ell^{K'})$. Conversely, write the torsion of $H^{q+1}(C)$ as $\bigoplus_j \mathbb{Z}/\ell^{a_j}$ and let $K \ge a = a_{j_0}$. Multiplication by ℓ from $H^{q+1}(C)[\ell^{K+1}]$ to $H^{q+1}(C)[\ell^K]$ respects the decomposition, and on the summand j_0, which lies in both, it is $\ell\colon \mathbb{Z}/\ell^a \to \mathbb{Z}/\ell^a$, not onto. So $\delta_K \circ r_{K+1,K} = \ell \circ \delta_{K+1}$ is not onto, and neither is $r_{K+1,K}$, δ_K being onto. The example: $Z^q(C/\ell^K) = \ker(\ell\colon \mathbb{Z}/\ell^K \to \mathbb{Z}/\ell^K) = \ell^{K-1}\mathbb{Z}/\ell^K$, $B^q = 0$, and the reduction carries $\ell^{K'-1}$ to 0 modulo ℓ^K when $K' > K$. The Dickson sentence is the case $q = 0$ on C (the Dickson locus has $n > 0$): $H^0(C) = H^0(\Gamma; V \otimes \mathbb{Z}_\ell) = 0$ (Claim 16.7(1), $\mathbb{Z}_\ell$ flat), so δ_K is an isomorphism onto $H^1(C)[\ell^K] = \mathbb{L}[\ell^K]$.

(1) is (5) at $q = 1$, $H^2(C) = 0$; the isomorphism $M_K \cong H^1(C)/\ell^K \cong \mathbb{L}/\ell^K$ is the left map of (5), and $r_{K',K}$ corresponds to the reduction; every T_p is induced by a cochain map of the form φ, so all of this is T_p-equivariant. The identification with $\mathbb{L}/\ell^K$ and its compatibility with

Lemma 30.5 come from the naturality of Remark 7.6 along $\mathbb{Z}_\ell \to \mathbb{Z}/\ell^K$; it is T_p-equivariant for the transported T_p on $\mathbb{L}$ by construction, and for the classical ones by Theorem 11.10 over $\mathbb{Z}_\ell$ ($6 \in \mathbb{Z}_\ell^\times$).

(2) $\ell^K M_{K'} = \bigoplus_i \ell^K(\mathbb{Z}/\ell^{a_i})g_i$, so $M_{K'}/\ell^K M_{K'} = \bigoplus_i(\mathbb{Z}/\ell^{\min(a_i,K)})\bar{g}_i$ with the reduced matrices, and (1) carries it isomorphically and equivariantly onto M_K; $\min(a_i, K) \geq 1$. The map $\mathbb{Z}_\ell[X_p] \to \mathrm{End}(M_{K'}) \to \mathrm{End}(M_{K'}/\ell^K) \cong \mathrm{End}(M_K)$ shows that $\mathsf{B}_{K'} \to \mathrm{End}(M_K)$ has image B_K, a ring map, so e is idempotent. e' commutes with ℓ^K, so $e'M_{K'} \cap \ell^K M_{K'} = \ell^K e' M_{K'}$ and $e'M_{K'}/\ell^K$ maps isomorphically onto $e(M_{K'}/\ell^K) = eM_K$. Uniqueness: B_K is commutative (the T_p commute on cohomology, Theorem 10.8(1)), and the kernel of $\mathsf{B}_K \to \mathsf{B}_1$ consists of operators x with $xM_K \subseteq \ell M_K$, hence $x^K = 0$; in a commutative ring two idempotents congruent modulo a nil ideal are equal ($(e_1 - e_2)^3 = e_1 - e_2$ for commuting idempotents, and a nilpotent y with $y^3 = y$ is 0).

(3) By (2) the derived module and M_K are isomorphic $\mathbb{Z}_\ell[X_p]$-modules, and blocks correspond by (2). Every listed reading is defined from the module and the operators alone: the systems and dimensions from $M_1 = M_K/\ell$; the block from its idempotent; A_K as an abstract algebra with its generators T_p is carried by any equivariant isomorphism ϕ to the corresponding one ($a \mapsto \phi a \phi^{-1}$), hence so are its length, exponents, $\mathfrak{m}$-adic filtration and I; the hypotheses and conclusions of Corollary 30.6(1),(3) are statements about these readings. Readings across levels: for $K \leq K'' \leq K'$ the isomorphisms $M_{K'}/\ell^K M_{K'} \cong M_K$ of (1) are induced by the $r_{K',K}$, and $r_{K'',K} \circ r_{K',K''} = r_{K',K}$, so they carry the canonical projections $M_{K'}/\ell^{K''} \to M_{K'}/\ell^K$ (on the derived modules: the reduction of the coordinates) to $r_{K'',K}$; the derived tower and $(M_K, r_{K'',K})$ are isomorphic towers.

(4) Both reductions modulo ℓ^K are $(\mathbb{Z}/\ell^K, 0)$; modulo ℓ^{K+1} the endomorphisms 0 and $\ell^K \neq 0$ are not conjugate. □

Obligation 30.12 (One module run per cell; the levels below it derived)**.** Data: per cell (n, ℓ) of the route of Corollary 30.6 (`HeckeExt/HeckeRingHeckeAlgebra`, `HeckeExt/HeckeSaturationDefect`): the set $\mathrm{Lv}_{\mathrm{read}}$ of levels at which a module is read (the per-K route's `K_run`), the set $\mathrm{Lv}_{\mathrm{run}}$ of levels at which the module route is run, the run modules M_{K^*}, the derived modules $M_{K^*}{\downarrow}K$ of Lemma 30.11(2), and the per-K route (one run per level read, the route through v1.23.0-pre.2) as the oracle. Each failure **aborts** unless marked.

(i) **Scope.** A level is derived only in the situation of Lemma 30.11(1): $\ell \geq 5$, level 1, degree 1, the module route at the run level (native; on the digit route the derived module keeps the lowest $\min(a_i, K)$ base-ℓ digits of row i and is reshaped to K digits per entry), at every run level a τ-table with every fill over $\mathbb{Z}$ (`used_mod_ell_solve()` false; the module route refuses a mod-ℓ table at every run level, `Hecke/HeckeOperatorOnK.cpp`, `RefuseCoefficientRing`, so this is enforced, not only checked). Anything else — $\ell \in \{2, 3\}$, H^0_{eq}, the cone, level $N > 1$, a table with a mod-ℓ fill — takes the per-K route, by name. So does every failure of a run of this route, the intended run or a restart of (ii): a refusal, an exception (`ENGINE_CHECK`, a certificate of the digit route), or a resource cap (the memory cap, a word or digit-route limit). A run at K'' can meet such a failure where the per-K route, never running above $\max \mathrm{Lv}_{\mathrm{read}}$ of the block, would not: a crossing from the 32- to the 64-bit word, or into the digit route when $\ell^{K''} \geq 2^{64}$. The levels still unread are then taken per K; the readings already taken are kept.

(ii) **Downward only; the restart rule.** A level K is read from $M_{K^*}{\downarrow}K$ with $K \leq K^* := \max \mathrm{Lv}_{\mathrm{run}}$ so far ($K = K^*$: the run module itself). A level $K > K^*$ is never derived (Lemma 30.11(4)): each such event triggers a new run at some K'' with $K \leq K'' \leq K_{\mathrm{need}}$, which becomes K^* (so a later level above K'' triggers a further run); the readings taken before the restart are kept (by Lemma 30.11(1) they agree with those the new run would give). The choice of run levels — $K^* = K_{\mathrm{need}}$ in advance, or K_{cls} first and a restart at K_{need} — is free and moves no printed value; it can change the run's word regime, its memory peak and whether it fails, and a failure falls under (i). $K_{\mathrm{cls}} \leq K^*$ always (the

free-rank reading needs it).

(iii) **The derived module, exactly.** Exponents $\min(a_i, K)$; entry (i, j) reduced modulo $\ell^{\min(a_i,K)}$ (the row-exponent convention of `OpModule`); checked: the number of summands is D, every exponent lies in $[1, K]$, the T_p commute, and at $K = 1$ the result equals `ReduceModEll` entry for entry.

(iv) **Invariance of what is printed.** Every field printed or recorded from a derived level (the `LOEWY_RING` and `SAT_DEFECT` lines, the verbs' records) is a reading in the sense of Lemma 30.11(3). The shipped readings are: the joint systems and their dimensions; lengths, cyclic exponents, $\mathbb{F}_\ell$-ranks and Hilbert functions (from Howell spans); $\varepsilon_K^{\mathrm{sat}}$, the stopping verdicts and `stable_cert`; the monomial basis, chosen greedily in a fixed monomial order by $\mathbb{F}_\ell$-linear independence in A_K/ℓ (`ReadDefect`); the generators of I as the reduced echelon form of I in that basis, filtered against $\mathfrak{m}I$; the `SAT_DEFECT` fields `generators`, `monomial_basis`, `quotient_hilbert`, `dim_image`, `tor_routes`, `sat_structure`; `forms=(...)`, which reads the $K = 1$ block only; and the `LOEWY_RING` fields computed from these (`D`, `rank_T`, `free_level`, `free_hilbert`, `hidden`, `cf9`, `cert`) (the reason each is invariant: Lemma 30.11(3)). The presentation-ordered `exps_K2` is stored and printed nowhere, and may not be printed from a derived level. A new field read from a module must be shown to be a reading before it is printed from a derived level. **Recorded** (a review item, not a run).

(v) **Oracle, rollback and gate.** The per-K route stays in the tree as the oracle and the rollback arm, behind a read-only environment switch that prints its `EFFECTIVE` line. A registered validation compares the whole printed output of the two routes byte for byte, timings and the fields of (vi) excepted, on a grid containing at least: a cell with $K_{\mathrm{stab}} \geq 3$ and a plateau ($(300, 7)$, `eps=[9,16,21,25,27,27]`); a cell with two levels ($(300, 11)$); a Dickson-locus cell with a torsion block (`torsion_free_stable`; `refused:torsion_block`); a cell with only `reduced_rank1` blocks at $\ell \geq 37$; a `not_stable_at_KMAX` block (KMAX lowered); a restart; run levels with $\ell^{K^*} < 2^{32}$ and $\geq 2^{32}$; a cell run on the digit route (`ModuleRoute::Digit` with `native_range`, as `HeckeRingLoewyValidation`'s (C3) does), derived to levels with $\ell^K < 2^{64}$ and compared with the per-K route's native reading there. Negative controls, each of which must fire the named check (a control that passes is a failure of the gate):

- a derived module keeping a_i in place of $\min(a_i, K)$, at a level $K < \max_i a_i$: caught by (iii)'s exponent check (an exponent above K);
- one operator entry (i, j) of T_p on a derived module at a level $K \geq 2$ changed by ℓ^{K-1}, chosen in a row with $\min(a_i, K) = K$, inside a block of dimension ≥ 2, at an entry where $\ell^{K-1}[E_{ij}, T_q] \neq 0$ on the derived module for some $q \in S \setminus \{p\}$: caught by (iii)'s commutation check.

Tier. The cell $(300, 7)$ costs $\approx$ 378 s single-thread on the per-K oracle alone (`ELL_scout.md` §1.3), so an arm carrying it is `slow`-tier; a `standard`-tier arm uses a cheaper cell with $K_{\mathrm{stab}} \geq 3$ and a plateau in its place, and keeps every other grid item. Each new registered name moves the roster, `docs/cluster/noctua2/EXPECT_NAMES` and the manual's tables in the same commit.

(vi) **Record honesty** (rule (A6)). `K_run` (of the `hecke_ring_algebra` record, and `SatCell::K_run`) keeps the per-K route's list, i.e. $\mathrm{Lv}_{\mathrm{read}}$, so the record stays byte-stable; its documented meaning changes from "the precisions actually run" to "the levels read", and the header comments (`HeckeExt/HeckeRingHeckeAlgebra.h`, `HeckeExt/HeckeSaturationDefect.h`) and the `hecke_ring_algebra` verb help (`HeckeExt/RegisterHeckeExtensionVerbs.cpp`) say so in the commit that lands the route. $\mathrm{Lv}_{\mathrm{run}}$ is reported in a new field of both cells (the `hecke_ring_algebra` record and `SatCell`), and `module_seconds` counts the runs only. **Recorded**.

(vii) **Certificates.** `CertifyStable` runs on the modules the route read, derived or run, un-

changed; on a derived module its `certified` rests on Lemma 30.11(1) as well as on Corollary 30.6, and the run level it was derived from is recorded beside it (not in `stable_cert`'s value).

Code map of §30.7. Built at v1.23.0-pre.4 (slot-23 lever G1, owner decision 2026-10-16; `docs/implementation/slot23/G1_clause_note.md`; implemented and audited, `docs/implementation/slot23/G1_report.md`, `G1_impl_audit.md` with its fixes; merged at `e4a5b2b3`; planned through v1.23.0-pre.3). One class, `ringalg::LevelCache` (`HeckeExt/HeckeRingHeckeAlgebra.{h,cpp}`), is the level cache of both cell drivers, `ComputeRingLoewyCell` (`HeckeExt/HeckeRingHeckeAlgebra.cpp`) and `ComputeSaturationDefectCell` (`HeckeExt/HeckeSaturationDefect.cpp`): its `get(K)` returns the module read at level K, derived from the latest run M_{K^*} when $K \le K^*$ and otherwise after a (re)run through `RunModule` (`RunModuleT`, or `RunModuleDigit` on the digit route) — one run of the module route per cell in the common case. Obligation 30.12 maps clause by clause: (i) the scope test and the fallback in `LevelCache` ($\ell < 5$ and `ModuleRoute::Hida` take the per-K route by name; a failed run above the request drops the cell to the per-K route for the levels still unread, with one `WARNING HeckeRingHeckeAlgebra: precision reuse fell back to per_K` line; the validation-only `ScopedInjectRunFailure` exercises it); (ii) `RunLevelPolicy` (`Auto`, `AtRequest`, `Lazy`, `SameWord`, `Max`; `Auto` looks ahead to K_{need} within the word for $\ell \le$ `kEagerEllMax`, $= 11$ since slot-23 G2-A, 13 before, a choice that moves only $\mathrm{Lv}_{\mathrm{run}}$; the word test `FitsU32Word`); (iii) `ReduceToLevel` (Lemma 30.11(2), native and digit forms; it never derives upwards) with `CheckLevelModule`, `CheckDerivedModule` and `CheckRunModule` (the checks `summands`, `exponent`, `commute`, `range`, `ReduceModEll`); (v) the rollback arm `HECKE_PRECISION_REUSE=0` (exactly `0` selects the per-K route, one run per level read; any other value warns and keeps the default), scoped as `ScopedPrecisionReuse`, its `EFFECTIVE` line `PrecisionReuseEffectiveLine()`, and the gate `HeckePrecisionReuseValidation` (`standard`) and `HeckePrecisionReuseSlowValidation` (`slow`; $(300, 7)$, $(300, 11)$), registered in `Registry/Slot23/G1.cpp` (source `HeckeExt/HeckePrecisionReuseValidation.cpp`); (vi) the fields `Lv_run` and `level_mode` of `LoewyCell` and `SatCell` and of the `hecke_ring_algebra` record (`HeckeExt/RegisterHeckeExtensionVerbs.cpp`), `K_run` keeping $\mathrm{Lv}_{\mathrm{read}}$; (vii) `LoewyBlock::cert_from_cls` / `cert_from_K`. **The lever is value-neutral**: `HECKE_PRECISION_REUSE` selects a route, never a value; the byte-identity of the two routes' printed output, timings and the run list excepted, is the gate of (v). G1 does not change `compute_module()` (`Hecke/HeckeOperatorOnK.cpp`); its slot-23 G2 re-implementation is mapped in the code map of §20. Through v1.23.0-pre.2 every level read was a run, and the implementation already used the case $K = 1$ of Lemma 30.11(2) (`ReduceModEll`; "$M_K/\ell = H^1(\mathbb{F}_\ell)$ at every K"). The module route builds a fresh τ-table per run and per p (`RunModuleT`, `hida::RunCellOn`) and refuses a table with a mod-ℓ fill at every run level $K \ge 2$ (`Hecke/HeckeOperatorOnK.cpp`, `RefuseCoefficientRing`; `RunModuleT` requires $K \ge 2$), so Obligation 30.12(i)'s τ condition is enforced at every run level.

31 The ℓ-local isotropy model, and the closed form at $\ell = 3$

What this chapter fixes. Over a $\mathbb{Z}_{(\ell)}$-algebra the prime-to-ℓ part of every simplex stabiliser is a unit, so its column in the isotropy-resolution model of Definition 23.4 can be replaced by its invariants. (i) **The ℓ-local model $\mathcal{I}_\bullet^{(\ell)}$** (Definition 31.1): Definition 23.4 with the column of σ the inflation of the periodic resolution of the ℓ-Sylow subgroup, projective and not free, and truncated to degree 0 where that subgroup is trivial. (ii) **What it computes and how it compares** (Proposition 31.2): group cohomology in every degree, cochain groups M^{Q_σ}, comparison maps with $\mathcal{I}_\bullet \otimes R$ in both directions, the Hecke action transported, level N by Shapiro. (iii) **At $\ell = 3$** (Proposition 31.3): one non-trivial column (the orbit of ζ^2), the closed form

$H^q(\Gamma;M)\cong H^q(\mathbb{Z}/3;M^{-I})$ for $q\geq 2$, and the exact comparison with the shipped Bredon route: *that route computes H^0 and a submodule of H^1, and misses exactly $H^{\geq 1}(\mathbb{Z}/3;M^{-I})$.* (iv) Its place beside the Moerdijk–Pronk refined model (Remark 31.4) and the checks (IL1)–(IL4) (Remark 31.5). (v) (v1.23.0-pre.12) **The certified-exponent rule on it** (Proposition 31.6): one complex over $\mathbb{Z}[1/\mathsf{q}_\ell]$ whose base changes are the complexes built over $\mathbb{Z}/\ell^K$ and $\mathbb{F}_{\ell'}$, Proposition 23.17 over $\mathbb{Z}_{(\ell)}$, and the Smith-valuation read-out. **Notation**: those of §23.2; R is a commutative $\mathbb{Z}_{(\ell)}$-algebra ($\mathbb{Z}_{(\ell)}$, $\mathbb{Z}/\ell^e$, $\mathbb{F}_\ell$, …), M an $R[\Gamma]$-module. P_σ, Q_σ, $\bar g_\sigma$, F'^σ, $G'_{s,t}$, $\mathcal{I}'^{(\ell)}$, cmp', Φ_ℓ, Ψ_ℓ are new symbols of this chapter and mean nothing else (rule (A5)); Φ_ℓ is not the cone map Φ_g, nor Ψ_ℓ the map Ψ of Theorem 5.10. q_ℓ, O_ℓ, $C^\bullet_\ell$, F^q, P_q, δ^q are new symbols of Proposition 31.6 (v1.23.0-pre.12), and so are b_j and s (v1.23.0-pre.13); they mean nothing else there.

Definition 31.1 (The ℓ-local isotropy model)**.** Assume (H4) and (B1)–(B3); let ℓ be a prime and R a $\mathbb{Z}_{(\ell)}$-algebra. For $\sigma\in\Sigma_s$ (Definition 23.4) the stabiliser $\Gamma_\sigma=\langle g_\sigma\rangle$ is cyclic of order m_σ; let P_σ be its ℓ-Sylow subgroup, Q_σ its subgroup of order prime to ℓ (so $\Gamma_\sigma=P_\sigma\times Q_\sigma$ and $|Q_\sigma|\in R^\times$), $\bar g_\sigma$ the image of g_σ in $\Gamma_\sigma/Q_\sigma\cong P_\sigma$, and $e_\sigma:=|Q_\sigma|^{-1}\sum_{w\in Q_\sigma}w$. The **column** $F'^\sigma_\bullet$ is: if $P_\sigma\neq 1$, the periodic complex $F'^\sigma_t:=R[\Gamma_\sigma/Q_\sigma]=R[\Gamma_\sigma]e_\sigma$ $(t\geq 0)$ with differential $\bar g_\sigma-1$ from odd t and the norm $\sum_{x\in\Gamma_\sigma/Q_\sigma}x$ from even $t\geq 2$; if $P_\sigma=1$, $F'^\sigma:=R$ in degree 0 alone. Put $G'_{s,t}:=\bigoplus_{\sigma\in\Sigma_s}R[\Gamma]\otimes_{R[\Gamma_\sigma]}F'^\sigma_t$. An **$\ell$-local isotropy model** is a complex of $R[\Gamma]$-modules $\mathcal{I}'^{(\ell)}_\bullet$ with $\mathcal{I}'^{(\ell)}_q=\bigoplus_{s+t=q,\ 0\leq s\leq 2}G'_{s,t}$ and $d=d'^{(0)}+d'^{(1)}+d'^{(2)}$ of the shape of Definition 23.4 ($d'^{(0)}=(-1)^s(\mathrm{id}\otimes\partial^{F'})$, $d'^{(r)}\colon G'_{s,t}\to G'_{s-r,t+r-1}$), with the augmentation $\mathrm{cmp}'\colon\mathcal{I}'^{(\ell)}_\bullet\to C_\bullet(X)\otimes R$ on $G'_{s,0}$ ($R[\Gamma]\otimes_{R[\Gamma_\sigma]}F'^\sigma_0\to R[\Gamma/\Gamma_\sigma]$) and 0 on $t\geq 1$, satisfying (IM1) and (IM2) (the sub-sum over $\Sigma_0^\partial\sqcup\Sigma_1^\partial$ a subcomplex mapping into $C_\bullet(\partial X)\otimes R$). Write $C^\bullet_{\mathcal{I}'^{(\ell)}}(M):=\mathrm{Hom}_{R[\Gamma]}(\mathcal{I}'^{(\ell)}_\bullet,M)$.

Proposition 31.2 (The ℓ-local model: existence, what it computes, the comparison, the Hecke action)**.** *[proved here] In the setting of Definition 31.1:*

1. ***(Existence.)*** *$\mathcal{I}'^{(\ell)}$ exists; $d'^{(1)}$, $d'^{(2)}$ are produced degree by degree by lifts along R-linear contracting homotopies of the augmented columns, boundary summands first — the procedure of Proposition 23.5(1).*
2. ***(A projective resolution.)*** *Each $G'_{s,t}\cong\bigoplus_\sigma R[\Gamma/Q_\sigma]$ is a projective, in general not free, $R[\Gamma]$-module;* cmp' *is a surjective quasi-isomorphism, so $\varepsilon\circ\mathrm{cmp}'\colon\mathcal{I}'^{(\ell)}_\bullet\to R$ is a projective resolution of R over $R[\Gamma]$; likewise on the boundary.*
3. ***(What it computes.)*** *$C^q_{\mathcal{I}'^{(\ell)}}(M)=\bigoplus_{s+t=q}\bigoplus_{\sigma\in\Sigma_s,\ (t=0\text{ or }P_\sigma\neq 1)}M^{Q_\sigma}$, and $H^q(C^\bullet_{\mathcal{I}'^{(\ell)}}(M))=H^q(\Gamma;M)$ for **every** $q\geq 0$ and every $R[\Gamma]$-module M; on the boundary $H^q(\Gamma_\infty;M)$. In particular, for $M=\mathrm{Ind}_{\Gamma_0(N)}^\Gamma V_R$ (Definition 28.3) or $\mathrm{Ind}_{\Gamma_0(N)}^\Gamma V_R(\chi)$ (Definition 29.7) it computes $H^q(\Gamma_0(N);V_R)$, resp. $H^q(\Gamma_0(N);V_R(\chi))$, in every degree (Shapiro's lemma, [standard: [Bro82, Ch. III.6.2]]).*
4. ***(Comparison with the free model.)*** *There are $R[\Gamma]$-chain maps $\Phi_\ell\colon\mathcal{I}_\bullet\otimes R\to\mathcal{I}'^{(\ell)}_\bullet$ and $\Psi_\ell\colon\mathcal{I}'^{(\ell)}_\bullet\to\mathcal{I}_\bullet\otimes R$ with $\mathrm{cmp}'\Phi_\ell=\mathrm{cmp}\otimes R$ and $(\mathrm{cmp}\otimes R)\Psi_\ell=\mathrm{cmp}'$ (strict lifts over $C_\bullet(X)\otimes R$, boundary to boundary), unique up to homotopy over $C_\bullet(X)\otimes R$, mutually inverse up to homotopy; so $\Phi_\ell^*\colon C^\bullet_{\mathcal{I}'^{(\ell)}}(M)\to C^\bullet_{\mathcal{I}}(M)$ and Ψ_ℓ^* are mutually inverse quasi-isomorphisms, $C^\bullet_{\mathcal{I}}(M):=\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_\bullet,M)$. Column by column, the map $\varphi^\sigma_t:=|Q_\sigma|^{\lfloor t/2\rfloor}\pi_\sigma\colon R[\Gamma_\sigma]\to R[\Gamma_\sigma/Q_\sigma]$ (π_σ the projection; 0 for $t\geq 1$ when $P_\sigma=1$) is a chain map $F^\sigma\otimes R\to F'^\sigma$ over id_R, defined over $\mathbb{Z}$ already (no inverse of $|Q_\sigma|$).*
5. ***(The Hecke action.)*** *For $g\in\Delta$ and $T_{\Gamma g\Gamma}$ any cochain-level Hecke operator on $C^\bullet_{\mathcal{I}}(M)$ inducing $[\Gamma g\Gamma]$ — the strict lift of Proposition 23.13, $p\in\{2,3,5\}$, and its extensions in Proposition 23.23 — the cochain map $T'_{\Gamma g\Gamma}:=\Psi_\ell^*\circ T_{\Gamma g\Gamma}\circ\Phi_\ell^*$ induces $[\Gamma g\Gamma]$ on $H^\bullet(\Gamma;M)$, independently of the choices of Φ_ℓ, Ψ_ℓ; likewise for the pair and the cone (the boundary stabilisers are $\{\pm I\}$, (B3)), and U_ℓ needs nothing new (the Δ-module structure of M is unchanged, (H2)).*

Proof. The columns. e_σ is an idempotent of $R[\Gamma_\sigma]$ ($|Q_\sigma| \in R^\times$) with $R[\Gamma_\sigma]e_\sigma \cong R[\Gamma_\sigma/Q_\sigma]$, so each F'^{σ}_t is a projective $R[\Gamma_\sigma]$-module. $F'^{\sigma} \to R$ is exact: for $P_\sigma \neq 1$ it is the periodic resolution of R over the cyclic group Γ_σ/Q_σ (the argument of Proposition 23.5's proof, over R), and for $P_\sigma = 1$ it is $R \to R$. Bounded below and consisting of free R-modules, it is split exact over R, so it has an R-linear contracting homotopy. $R[\Gamma]$ is free as a right $R[\Gamma_\sigma]$-module, so $R[\Gamma] \otimes_{R[\Gamma_\sigma]} -$ is exact and preserves projectives, and $G'_{s,\bullet} \to C_s(X) \otimes R$ is a projective resolution, with $R[\Gamma] \otimes_{R[\Gamma_\sigma]} R[\Gamma_\sigma/Q_\sigma] \cong R[\Gamma/Q_\sigma]$. (1) The proof of Proposition 23.5(1) uses only that the column resolutions are projective with R-linear contracting homotopies and the comparison theorem [standard: [Wei94, Thm. 2.2.6]], which needs projective sources; it applies with one change: the generator $1 \otimes e_\sigma$ of $R[\Gamma]e_\sigma$ is Q_σ-fixed, so every lift produced by the R-linear contracting homotopies is replaced by its Q_σ-average, $y \mapsto e_\sigma y$ ($|Q_\sigma| \in R^\times$). The average still lifts the same element, because the element lifted is the image of a Q_σ-fixed element under an $R[\Gamma]$-map and the differentials commute with e_σ; this is the identity $\mathrm{Hom}_{R[\Gamma]}(R[\Gamma]e_\sigma, Y) = e_\sigma Y$ (v1.16.0-pre.3, WP-14 referee C-1). The average is kept in this procedure for uniformity and as the general statement; on X (and on any Γ-equivariant refinement of it) it changes nothing on d itself: a lift of $d'^{(1)}$ or $d'^{(2)}$ starts at a source of dimension $s \geq 1$, whose stabiliser fixes a segment of $\mathcal{H}$ pointwise (no inversions, (H4)(i)) and is therefore $\{\pm I\}$, a non-central element of Γ fixing only isolated points of $\mathcal{H}$; so $Q_\sigma \subseteq \{\pm I\}$ ($= \{\pm I\}$ at $\ell = 3$, trivial at $\ell = 2$) is central and contained in every target's Q_τ, and $e_\sigma y = y$ for every $y \in R[\Gamma]e_\tau$. Where the average is load-bearing is the comparison of (4): the source generators of Ψ_ℓ are Q_x-fixed while the free target $\mathcal{I} \otimes R$ is not (v1.16.0, slot-16 WP-16 audit F-1). (2) The filtration and five-lemma argument of Proposition 23.5(2), with $C_\bullet(X) \otimes R$ exact over R (split over $\mathbb{Z}$, (H4)(ii)). (3) $\mathrm{Hom}_{R[\Gamma]}(R[\Gamma/Q_\sigma], M) = M^{Q_\sigma}$, and $H^q(\mathrm{Hom}_{R[\Gamma]}(P_\bullet, M)) = \mathrm{Ext}^q_{R[\Gamma]}(R, M) = H^q(\Gamma; M)$ for any projective resolution P ($R \otimes_{\mathbb{Z}}$ of a free $\mathbb{Z}[\Gamma]$-resolution is a free $R[\Gamma]$-resolution). The boundary as in Proposition 23.5(3). (4) $\mathcal{I} \otimes R$ is free and $\mathcal{I}^{(\ell)}$ projective over $R[\Gamma]$, both bounded below, and both augmentations onto $C_\bullet(X) \otimes R$ are surjective quasi-isomorphisms; Lemma 23.12's proof gives the strict lifts — with a projective source, the step "choose y_0" and the step "$z = \partial k$" are lifts of a map out of a projective module through a surjection — constructively, $y_0 := e_\sigma y'_0$ and $k := e_\sigma k'$ for any y'_0, k' produced as in Lemma 23.12's proof (C-1) —, onto ($\ker p$)'s boundaries in the second case ($\ker p$ is acyclic) — boundary first, as there. Uniqueness and inverse up to homotopy: Proposition 23.5(5)'s argument (a chain map into an acyclic kernel from a projective complex is null-homotopic through that kernel). The column maps: on odd t, $\pi_\sigma(g_\sigma - 1) = (\bar{g}_\sigma - 1)\pi_\sigma$, so $\varphi_{t-1} = \varphi_t$ is needed and holds ($\lfloor (t-1)/2 \rfloor = \lfloor t/2 \rfloor$ for odd t); on even $t \geq 2$, $\pi_\sigma(\sum_{u \in \Gamma_\sigma} u) = |Q_\sigma| \sum_{x \in \Gamma_\sigma/Q_\sigma} x$, so $\varphi_{t-1} \circ N = N' \circ \varphi_t$ needs $\varphi_t = |Q_\sigma|\, \varphi_{t-1}$, which is the exponent; augmentations agree in degree 0. For $P_\sigma = 1$ the target has $\bar{g}_\sigma = 1$, and the degree-0 map composed with $g_\sigma - 1$ is 0. (5) The Hecke operator on $H^\bullet(\Gamma; M)$ is $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$, defined independently of the projective resolution [standard: [Bro82, Ch. III.9–10]]; Φ^*_ℓ and Ψ^*_ℓ are the comparison isomorphisms between the resolutions, and conjugating a cochain map inducing it by them induces it again. The pair and the cone: (4) holds on the boundary sub-sum, so the same conjugation applies to the pair datum of Proposition 23.13. □

Proposition 31.3 (At $\ell = 3$: one column, the closed form, and the comparison with the shipped Bredon route)**.** *[proved here] Let $\ell = 3$, R a $\mathbb{Z}_{(3)}$-algebra, M an $R[\Gamma]$-module, and $\mathcal{I}^{(3)}$ on the model of Definition 7.1.*

1. ***(One column.)*** *$P_\sigma \neq 1$ for exactly one representative, $\sigma = v_4$ ($\Gamma_{v_4} = \langle ST \rangle$, $P_{v_4} = \langle (ST)^2 \rangle \cong \mathbb{Z}/3$, $Q_{v_4} = \{\pm I\}$); every other column is $R[\Gamma/\Gamma_\sigma]$ in degree 0 ($\Gamma_{v_6} = \langle S \rangle$ has order 4, the rest $\{\pm I\}$). So $C^q_{\mathcal{I}^{(3)}}(M)$ differs from the shipped (Bredon) cochain group C^q_{eq} (Proposition 23.3) only at v_4: degree 0 is $3 \cdot M^{-I} \oplus M^{\langle S \rangle} \oplus M^{-I}$ (M^{-I} at v_4 in place of $M^{\langle U \rangle}$), degree 1 is $9 \cdot M^{-I} \oplus M^{-I}$, degree 2 is $5 \cdot M^{-I} \oplus M^{-I}$, and every degree $q \geq 3$ is one M^{-I}. The boundary sub-model is the Bredon boundary complex ($P_\sigma = 1$ on ∂X, (B3)).*

2. ***(The closed form.)*** *With $U = ST$ and $\langle U\rangle = \{\pm I\} \times \langle U^2\rangle$,*

$$H^q(\Gamma; M) \;\cong\; H^q(\langle U\rangle; M) \;=\; H^q(\langle U^2\rangle; M^{-I}) \qquad (q \geq 2),$$

which is $(M^{-I})^{U^2}/N(M^{-I})$ for even q and $\ker N/(U^2-1)M^{-I}$ for odd q, $N = 1+U^2+U^4$; and there is an exact sequence

$$0 \to M^{-I}/\big(M^{\langle S\rangle} + M^{\langle U\rangle}\big) \xrightarrow{\iota} H^1(\Gamma; M) \to H^1(\langle U^2\rangle; M^{-I}) \to 0,$$

with $H^0(\Gamma; M) = M^{\langle S\rangle} \cap M^{\langle U\rangle} = M^\Gamma$.

3. ***(The shipped Bredon route.)*** *The complex the shipped pipeline computes over R, $C^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$ with fibre M (Proposition 23.3(2),(3): Bredon cohomology for the finite subgroups with fixed-point coefficients), has $H^0 = M^\Gamma$, $H^1 = M^{-I}/(M^{\langle S\rangle} + M^{\langle U\rangle})$ and $H^q = 0$ for $q \geq 2$; the map $\mathrm{cmp}^*\colon C^\bullet_{\mathrm{eq}} \to C^\bullet_{\mathcal{I}}(M)$ of Proposition 23.5(4) induces the identity on H^0 and, on H^1, an injection whose cokernel is $H^1(\langle U^2\rangle; M^{-I})$ (the same terms as (2)'s sequence; that its image is the image of ι is not asserted). So at $\ell = 3$* ***the Bredon route is exact in degree*** *0,* ***injective in degree*** *1* ***with cokernel*** *$H^1(\mathbb{Z}/3; M^{-I})$,* ***and misses*** *$H^q(\mathbb{Z}/3; M^{-I})$* ***in every degree*** *$q \geq 2$ — nothing else.*

4. ***(Parity.)*** *[formally verified: [GT, `thm:lv3-tree-negone`]] (only the vanishing, and only for $R = K$ a field with $2 \neq 0$ and M finite-dimensional: $H^q(\Gamma; M) = 0$ for every q, proved on the complex of Remark 16.3, not through the terms of (1); instantiated at $M = \mathrm{Sym}^n \mathbb{F}_3^2$, n odd) If $-I$ acts on M by -1 (e.g. V_R with n odd, or $V_R(\chi)$ with $\chi(-1) \neq (-1)^n$), every term of (1) vanishes and $H^\bullet(\Gamma; M) = 0$; this needs only $2 \in R^\times$ (Proposition 29.11(2)'s argument), not 6.*

Proof. (1) The stabilisers are those of (H4)(iv) and Lemma 7.3(1) on the representatives Σ_s of Definition 23.4 (v_5 is not a representative); $\langle ST\rangle$ has order $6 = 2 \cdot 3$, $\langle S\rangle$ order 4, $\{\pm I\}$ order 2. The ranks are read off Proposition 31.2(3) with $M^{Q_\sigma} = M^{\Gamma_\sigma}$ where $P_\sigma = 1$, and $M^{Q_{v_4}} = M^{-I}$. (2) Claim 16.2(2), (51), with $H^q(\langle S\rangle; M) = H^q(\langle -I\rangle; M) = 0$ for $q \geq 1$ (4 and 2 are units; [standard: [Bro82, Ch. III.10.2]]): for $q \geq 2$ the sequence gives $H^q(\Gamma; M) \cong H^q(\langle U\rangle; M)$, and at $q = 1$ it reads $M^S \oplus M^U \to M^{-I} \to H^1(\Gamma; M) \to H^1(\langle U\rangle; M) \to 0$. The Lyndon–Hochschild–Serre sequence of $\{\pm I\} \trianglelefteq \langle U\rangle$ collapses (2 a unit, [standard: [Bro82, Ch. VII.6]]): $H^q(\langle U\rangle; M) = H^q(\langle U^2\rangle; M^{-I})$; the cyclic group $\langle U^2\rangle$ of order 3 has the periodic resolution. (3) The Bredon complex is Proposition 23.3(2). The map: filter $C^\bullet_{\mathcal{I}}(M)$ and $C^\bullet_{\mathrm{eq}}$ by s as in the proof of Proposition 23.5(4); the E_1-page of $C^\bullet_{\mathcal{I}}(M)$ is $E_1^{s,t} = \bigoplus_{\sigma \in \Sigma_s} H^t(\Gamma_\sigma; M)$, its row $t = 0$ is $C^\bullet_{\mathrm{eq}}$, and cmp^* is the inclusion of that row, so on cohomology it is the edge map $E_2^{q,0} \to H^q$. Over a $\mathbb{Z}_{(3)}$-algebra $E_1^{s,t} = 0$ for $t \geq 1$ except $E_1^{0,t} = H^t(\langle U\rangle; M)$ at $\sigma = v_4$. The differentials out of $E_r^{0,t}$, $t \geq 1$, land in $E_r^{r,t-r+1}$, which is 0 for $t - r + 1 \geq 1$ (no column $s \geq 1$ has higher cohomology) and is $E_2^{2,0} = H^2(C^\bullet_{\mathrm{eq}}) = 0$ for $r = t + 1 = 2$ (Proposition 23.3(2)), and $E_r^{r,t-r+1} = 0$ for $r \geq 3$ ($s \leq 2$). So the sequence degenerates at E_2: $0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \to 0$, $H^0 = E_2^{0,0}$, $H^q = E_2^{0,q}$ for $q \geq 2$; $E_2^{1,0} = H^1(C^\bullet_{\mathrm{eq}})$, the edge map $E_2^{1,0} \to H^1$ is injective, and $E_2^{0,1} = H^1(\langle U\rangle; M)$. (4) $-I$ is central and in every Γ_σ; $M^{-I} = 0$ and $M^{Q_\sigma} \subseteq M^{-I} = 0$ when $-I$ acts by -1 and 2 is a unit. □

Remark 31.4 (Its place beside the Bredon route, the free model, and the Moerdijk–Pronk refined model). [proved here] (1) **The Bredon route is not replaced.** What $\mathbb{F}_\ell$ and $\mathbb{Z}/\ell^e$ *name* at $\ell \in \{2, 3\}$ is Definition 23.20(1) and [E, `sec:ring`]: the shipped route computes the Bredon object, and switching the default object to group cohomology would change published values (golden-gated, an owner decision). $\mathcal{I}^{(\ell)}$ is a second, smaller computational model of the *group* cohomology that $\mathcal{I}$ already computes (Proposition 23.5(3)); $\mathcal{I}$ is its oracle in every degree. Proposition 31.3(3) is the exact statement of the difference at $\ell = 3$. (2) **Size.** At $\ell = 3$ the degrees $q \geq 3$ cost one copy of M^{-I} instead of 19 copies of M (Proposition 23.5(2));

at $\ell = 2$ only the corner column shrinks ($P_{v_4} = \{\pm I\}$, $Q_{v_4} = \langle U^2 \rangle$, M^Q of rank about $\frac{1}{3}$), every simplex keeping a $\mathbb{Z}/2$ column — the generic central isotropy, which no 2-local reduction removes ($\Gamma \to \mathrm{PSL}_2(\mathbb{Z})$ does not split, Lemma 14.22's class of order 12). So $\ell = 2$ is specified by Definition 31.1 and gains little; $\ell = 3$ is where it pays. (3) **The MP constraint** (owner, 2026-09-28). Definition 31.1 and Proposition 31.2 use of the model only (H4), (B1)–(B3) and the stabiliser inventory: no vertex count, no edge list, no special role of v_4 beyond "the representatives whose stabiliser has order divisible by ℓ". A Γ-equivariant subdivision Y' of X — in particular the Moerdijk–Pronk-conformant refinement the owner's direction names (subdividing the edge from ζ^2 to i and the edge from ζ^2 towards the cusp) — preserves every point stabiliser (Remark 17.5(1)); the new vertices have stabiliser $\{\pm I\}$, so the simplices with $|P_\sigma| > 2$ (i.e. $P_\sigma \supsetneq \{\pm I\}$) are again the orbit of ζ^2 at $\ell = 3$, and of i at $\ell = 2$ (at $\ell = 2$ every simplex has $P_\sigma \supseteq \{\pm I\}$; WP-14 referee C-7), and the model on Y' is homotopy equivalent to the one on X over R (Proposition 31.2(4)'s argument with $C_\bullet(Y') \simeq C_\bullet(X)$, Remark 23.6). The centre is **not** trivialised: $-I$ lies in every Γ_σ, enters Q_σ at $\ell \neq 2$ as a unit-order factor whose invariants are taken (*not* deleted), and at $\ell = 2$ keeps its column; a coefficient module on which $-I$ acts by a scalar other than 1 (odd n, a character with $\chi(-1) = -1$) is admissible input, which is the "non-trivial action of the centre everywhere" the refined model is to carry. Nothing here is Bredon-only: the Bredon object appears only as the $t = 0$ row and in the comparison (3). (v1.22.0-pre.5: two refinements are to be kept apart. Slot 17's synthetic descriptor, written R_ζ in §32, subdivides all three edge orbits at ζ^2; the Moerdijk–Pronk-conformant refinement of the owner's ruling Q2 is $\mathsf{R}_\mathbb{Z}$, which subdivides only the edge orbit of v_4v_6 (Definition 32.4). Both are adapted; the edge from ζ^2 towards the cusp is forced by none of Moerdijk–Pronk's rules, and goodness forces, among others, the *other* edges from ζ^2 and ζ towards the horocycle, v_1v_4 and v_2v_5 (Remark 32.5, Table 9).)

Remark 31.5 (What is checkable: (IL1)–(IL4)). [proved here] [verified computationally][a] Committed at v1.16.0-pre.1, before any code (rule 20); a failing [proved here] item stops the package. Cells over $\mathbb{Z}/3^e$ and $\mathbb{F}_3$, $n \le 200$, $q \le q_{\max}$.

(IL1) [proved here] (Proposition 31.2(1),(3),(4)) the built $\mathcal{I}^{(3)}$ — every lift averaged by e_σ (C-1) — satisfies $d \circ d = 0$, (IM1) and (IM2); and $\mathcal{I}^{(3)}$ and $\mathcal{I}$ give the same lengths and invariant factors of $H^q(\Gamma; \mathrm{Sym}^n(\mathbb{Z}/3^e)^2)$ for $q \le q_{\max}$, and those of [IB] Appendix A for $n \le 20$, $q \le 5$ (`HeckeFreeModelValidation`'s oracle).

(IL2) [proved here] (Proposition 31.3(2)) the closed form: for $q \ge 2$, $H^q(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2_{(3)}) \cong H^q(\mathbb{Z}/3; \mathrm{Sym}^n \mathbb{Z}^2_{(3)})$ (n even), 2-periodic in q. [verified computationally][a] (slot-14 WP-Q1's `z3check.py`, kept in the slot-14 scratch, not tracked; `docs/implementation/slot14/WP-Q1_small_ell_o` §3.3): the 3-parts of H^2 and H^3 in [IB] Appendix A agree with H^2, H^3 of $\mathbb{Z}/3$ on 22 / 22 cells $n = 0, 2, \ldots, 20$ ($\mathbb{Z}/3$ in H^2 exactly at $n \equiv 0 \pmod 6$, in H^3 exactly at $n \equiv 4 \pmod 6$).

(IL3) [proved here] (Proposition 31.3(3)) over $\mathbb{F}_3$: $\dim H^1(\Gamma; \mathrm{Sym}^n \mathbb{F}_3^2) = \dim H^1_{\mathrm{Bredon}} + \dim H^1(\mathbb{Z}/3; \mathrm{Sym}^n \mathbb{F}_3^2)$ for n even, the first summand being the shipped pipeline's $\dim H^1_{\mathrm{eq}}$ (`HeckeBredonTreeValidation`'s cells), and H^0 equal; the same at every level N with $M = \mathrm{Ind}_{\Gamma_0(N)}^\Gamma V_R$. [verified computationally][a]: [IB] Appendix A's H^1 defect (group minus model) has 3-part $\mathbb{Z}/3$ exactly at $n = 4, 10, 16$, which is $H^1(\mathbb{Z}/3; M)$ (the same note, §3.3). [verified computationally][a] *partial* (v1.16.0 at level 1, slot-16 WP-16; v1.17.0-pre.5 at level N, slot-17 WP-7, `HeckeIsotropyModelEllValidation` arm (IL3N)): held at level 1 over $\mathbb{F}_3$, $n \le 40$, with the map-level comparison over $\mathbb{F}_3$ and $\mathbb{Z}/9$; and at level N with $M = \mathrm{Ind}_{\Gamma_0(N)}^\Gamma V_R$ built from the shipped `GammaOLevelN` table (§28; no second coset construction), *against the shipped level-N model's H^1_{eq} over $\mathbb{F}_3$ only* (Definition 28.3(a), the cochain level), $N \in \{1, 2, 3, 4, 5, 7, 11, 13\}$, $n \in \{0, 1, 2, 4, 6\}$ ($N \le 16$, $n \le 6$ under `HECKE_ISOMODEL_ELL_FULL=1`); over $\mathbb{Z}/9$ and $\mathbb{Z}/27$ the level-N cells compare $\mathcal{I}^{(3)}$ with $\mathcal{I}$ (lengths) and the closed form, not the pipeline, which is mod p; the $\mathbb{Z}/3$ summand is non-zero at $N = 7$. [verified computationally][a] (v1.18.0-pre.2, slot-18 WP-7, arm (IL3Q),

code map item (vi)): over $\mathbb{F}_9$ and, in lengths, over $\mathbb{Z}/9$ against the shipped level-N model's H^1_{eq} over $\mathbb{Z}/9$, 12 cells. **Tag moved** from [verified computationally]^{a} *partial* to [verified computationally]^{a} at v1.18.0-pre.2 (pre.1 through v1.17.0: "*partial* until the next referee rules"), on the slot-18 referee F1's ruling C-4 ("move both at the close on WP-7's (IL5) + (vi) pass", `docs/implementation/slot18/WP-3_F1_referee_report.md`) and WP-7's pass (`docs/implementation/slot18/WP-7_report.md`; `--all` 156/156).

(IL4) [proved here] (Proposition 31.2(5)) the transported T_2', T_5' (and $T_3' = U_3$) on $\mathcal{I}^{(3)}$ have the characteristic polynomials of T_2, T_5, T_3 on $\mathcal{I}$ (Proposition 23.13, `HeckeIntegralActionValidation`), degree by degree, $q \le q_{\max}$. [verified computationally]^{a} *partial* (v1.16.0 over $\mathbb{F}_3$, slot-16 WP-16; v1.17.0-pre.5, slot-17 WP-7, arms (IL4e), (PC)): held over $\mathbb{F}_3$, even $n \le 12$, $q \le 5$, $p \in \{2,3,5\}$; over $\mathbb{Z}/3^e$, $e \in \{2,3\}$, even $n \le 12$ resp. $n \le 8$ ($n \le 20$ / 16 under the full grid), in lengths and invariant factors of H^q and of $\mathrm{im}(T_p' - a)$, $a \in \{0,1,-1\}$ (basis-free), and in characteristic polynomials where H^q is free; and the pair and cone half of Proposition 31.2(5): Φ_ℓ^*, Ψ_ℓ^* and T_p commute with the restriction to the boundary sub-model, and the transported T_p' on the mapping cone of the restriction and on the boundary has the characteristic polynomials (over $\mathbb{F}_3$) and image invariant factors (over $\mathbb{Z}/9$) of T_p on $\mathcal{I}$. **Tag moved** from [verified computationally]^{a} *partial* to [verified computationally]^{a} at v1.18.0-pre.2, with (IL3) and on the same ground (the slot-18 referee F1's C-4, WP-7's pass of (IL5), which re-checks the transported T_p' against T_p on the free model over $\mathbb{Z}/\ell^e$ and $\mathbb{F}_\ell$, and of (vi)).

(IL5) [proved here] (Proposition 23.13(3)'s argument with Proposition 31.2(4),(5) in place of Remark 17.2 — the source is projective, not free — given that $\mathcal{I}^{(\ell)}(Y')$ is an ℓ-local isotropy model over Y' and that the built map satisfies (T1)–(T3); committed at v1.18.0-pre.1, before slot-18 WP-7's code, rule 20; route and hypothesis repaired at v1.18.0-pre.2, the slot-18 referee F1's M-1: pre.1 read "Proposition 23.13(3)'s argument and Proposition 31.2(5)", whose last step passes through Remark 17.2, a statement about a model *free* over $\mathbb{Z}[\Gamma]$) **The strict action on a refined descriptor.** Let Y' be a synthetic refined descriptor as in Remark 31.4(3) (a Γ-equivariant subdivision of X, given as data), and suppose $\mathcal{I}^{(\ell)}(Y')$ is an ℓ-local isotropy model over Y' in the sense of Definition 31.1 with Y' in place of X (it exists by Proposition 31.2(1), Remark 31.4(3): projective, in general not free, over $R[\Gamma]$, with its augmentation onto $C_\bullet(Y') \otimes R$ a surjective quasi-isomorphism). Let $g = \mathrm{diag}(1,p)$, $\Gamma'' = \Gamma \cap g\Gamma g^{-1}$, and τ' an additive map on the resolution side over Y' satisfying (T1) (over Γ''), (T2) and (T3) of Definition 11.1 with that model in place of $C_\bullet(K^u)$. (How slot-18 WP-7 obtains τ' — a strict lift in the sense of Lemma 23.12, certified by the strict identity relative to the comparison with X as in Remark 23.14 — is a certificate of the built map, not a hypothesis of this item.) Then its coset transport $T_p^{Y'}$ and the transported T_p' of Proposition 31.2(5) (through the comparison of the models over Y' and X) induce the same operator $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ on $H^q(\Gamma; V_R)$ (*proof*: by (T1), τ' is an $R[\Gamma'']$-chain map between projective resolutions (restriction to Γ'' keeps projectivity, $R[\Gamma]$ being free over $R[\Gamma'']$), unique up to Γ''-homotopy by the fundamental lemma; (T3) fixes the augmentation; Lemma 15.4(1),(2) apply verbatim (they use (T1) and $\partial(s_j y) = s_j \partial y$ only); so the coset transport induces $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$, by Proposition 31.2(4),(5) (Brown III.9–10: independent of the projective resolution), and so does the transported T_p' by (5)); checked degree by degree, $q \le q_{\max}$: equal characteristic polynomials over $\mathbb{F}_3$ (and $\mathbb{F}_2$), and equal lengths and invariant factors of H^q and of $\mathrm{im}(T - a)$, $a \in \{0,1,-1\}$, over $\mathbb{Z}/3^e$, $e \le 3$, at $\ell \in \{2,3\}$, $p \in \{2,3,5\}$, on every refined descriptor the package builds, with one control cell with $-I \mapsto -1$ (every H^q zero on both sides). A failure stops the package. (v1.18.0-pre.2, the referee F1's m-C2: at $\ell = 2$, where $\mathcal{I}^{(2)}$ "gains little" (Remark 31.4(2)), the checks over $\mathbb{F}_2$ and $\mathbb{Z}/2^e$ are a check of the code path — the projective repair of the lift, the periodic column at v_4 — not new mathematics, as (vi) is for $\mathbb{F}_9$.) [verified computationally]^{a} (v1.18.0-pre.2, slot-18

WP-7, `HeckeIsotropyModelEllValidation` arm (IL5)): held on the two synthetic refined descriptors of the code map's item (iv), $\ell \in \{3,2\}$, $p \in \{2,3,5\}$: the hypothesis certified on each model ($d \circ d = 0$, well defined, (IM1), (IM2)); $T_p^{Y'}$ and the transported T'_p equal on H^q, $q \le 5$ (their difference maps cocycles to coboundaries), with equal image invariant factors of $T - a$ over $\mathbb{Z}/\ell^e$, $e \le 3$, and equal characteristic polynomials over $\mathbb{F}_\ell$, both also equal to T_p on $\mathcal{I}$; the control $n = 1$ at $\ell = 3$; 684 (1080 under the full grid) degree comparisons. The [proved here] tag is unchanged.

Proposition 31.6 (The certified-exponent rule on the ℓ-local model: one complex over $\mathbb{Z}[1/\mathsf{q}_\ell]$)**.** *[proved here] (v1.23.0-pre.12; refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D0_referee.md`; its fixes F1–F7 applied at v1.23.0-pre.13, not re-refereed.) Assume (H4) and (B1)–(B3); let ℓ be a prime, Σ_s, Γ_σ, P_σ, Q_σ and e_σ as in Definition 31.1, and put*

$$\mathsf{q}_\ell := \operatorname{lcm}\{|Q_\sigma| : \sigma \in \Sigma_0 \sqcup \Sigma_1 \sqcup \Sigma_2\}, \qquad \mathsf{O}_\ell := \mathbb{Z}[1/\mathsf{q}_\ell] \subset \mathbb{Z}_{(\ell)}.$$

On the model of Definition 7.1: at $\ell = 2$, $|Q_\sigma| \in \{1,3\}$ ($Q_{v_4} = \langle U^2 \rangle$, $U = ST$, the only non-trivial one), so $\mathsf{q}_2 = 3$ and $\mathsf{O}_2 = \mathbb{Z}[1/3]$; at $\ell = 3$, $|Q_\sigma| \in \{2,4\}$ ($Q_{v_6} = \Gamma_{v_6} = \langle S \rangle$ of order 4, every other $Q_\sigma = \{\pm I\}$), so $\mathsf{q}_3 = 4$ and $\mathsf{O}_3 = \mathbb{Z}[1/2]$; at $\ell \ge 5$, $Q_\sigma = \Gamma_\sigma$, $\mathsf{q}_\ell = 12$ and $\mathsf{O}_\ell = \mathbb{Z}[1/6]$. Let M be a $\mathbb{Z}[\Gamma]$-module that is finitely generated and free over $\mathbb{Z}$ ($V = \operatorname{Sym}^n \mathbb{Z}^2$, $\operatorname{Ind}_{\Gamma_0(N)}^{\Gamma} V$, ...), $M_S := M \otimes_{\mathbb{Z}} S$ for a ring S, and $q_{\max} \ge 0$.

1. ***(One complex, and its base changes.)*** *Definition 31.1 and Proposition 31.2(1)–(3) hold verbatim with R any commutative ring in which q_ℓ is a unit, not necessarily a $\mathbb{Z}_{(\ell)}$-algebra. Fix one such model $\mathcal{I}^{(\ell)}_{\mathsf{O}_\ell}$ over $R = \mathsf{O}_\ell$. For every O_ℓ-algebra S, the complex $\mathcal{I}^{(\ell)}_{\mathsf{O}_\ell} \otimes_{\mathsf{O}_\ell} S$ with the base-changed d and cmp' is a model in this sense over S — for S a $\mathbb{Z}_{(\ell)}$-algebra an ℓ-local isotropy model of Definition 31.1 — and so computes $H^q(\Gamma; M_S)$ for every q (Proposition 31.2(3)). In particular the complexes over $S = \mathbb{Z}/\ell^K$ (every $K \ge 1$) and over $S = \mathbb{F}_{\ell'}$ (ℓ' a prime, $\ell' \nmid \mathsf{q}_\ell$; every $\ell' \ge 5$) are base changes of **one** complex: no lift $d'^{(1)}$, $d'^{(2)}$ is chosen per ring.*
2. ***(The integral cochains.)*** *$C^\bullet_\ell := \operatorname{Hom}_{\mathsf{O}_\ell[\Gamma]}(\mathcal{I}^{(\ell)}_{\mathsf{O}_\ell}, M_{\mathsf{O}_\ell}) = \bigoplus_{s+t=q} \bigoplus_\sigma M_{\mathsf{O}_\ell}^{Q_\sigma}$ (the index set of Proposition 31.2(3)) is a complex of finitely generated free O_ℓ-modules; $C^\bullet_\ell \otimes_{\mathsf{O}_\ell} S = C^\bullet_{\mathcal{I}^{(\ell)} \otimes S}(M_S)$ for every O_ℓ-algebra S; $H^q(C^\bullet_\ell) \cong H^q(\Gamma; M) \otimes \mathsf{O}_\ell$ and $H^q(C^\bullet_\ell \otimes \mathbb{Z}_{(\ell)}) \cong H^q(\Gamma; M) \otimes \mathbb{Z}_{(\ell)}$ for every q.*
3. ***(The rule over the local principal ideal domain $\mathbb{Z}_{(\ell)}$.)*** *Proposition 23.17(1)–(3) and Remark 23.18 hold for $C := C^\bullet_\ell \otimes_{\mathsf{O}_\ell} \mathbb{Z}_{(\ell)}$, a complex of finitely generated free $\mathbb{Z}_{(\ell)}$-modules, in place of a complex of free $\mathbb{Z}$-modules, at the prime ℓ: the elementary divisors ℓ^{a_t} are those of d^q over $\mathbb{Z}_{(\ell)}$, $r_e(d^q)$ is read on $C \otimes \mathbb{Z}/\ell^e$, which is the complex of (1) over $\mathbb{Z}/\ell^e$, and what is read off is the $\mathbb{Z}_{(\ell)}$-type of $H^q(\Gamma; M) \otimes \mathbb{Z}_{(\ell)}$, together with $H^q(\Gamma; M \otimes \mathbb{Z}/\ell^f) \cong H^q(\Gamma; M) \otimes \mathbb{Z}/\ell^f \oplus \operatorname{Tor}(H^{q+1}(\Gamma; M), \mathbb{Z}/\ell^f)$ for every $f \ge 1$. As in Proposition 23.17(5), it applies to the truncation $\sigma^{\le q_{\max}+1}$ (its d^q, $q \le q_{\max}$, are those of the full complex).*
4. ***(Where $\operatorname{rank}_{\mathbb{Q}}$ comes from.)*** *$\operatorname{rank}_{\mathbb{Q}} d^q = \operatorname{rank}_{\mathbb{F}_{\ell'}}(d^q \otimes \mathbb{F}_{\ell'})$ for every prime $\ell' \nmid \mathsf{q}_\ell$ at which $H^{q+1}(\Gamma; M)$ has no ℓ'-torsion (such an ℓ' qualifies for d^q), and $\operatorname{rank} C^q_\ell = \dim_{\mathbb{F}_{\ell'}} C^q_\ell \otimes \mathbb{F}_{\ell'}$ for every $\ell' \nmid \mathsf{q}_\ell$. Put $b_j := \dim_{\mathbb{Q}} H^j(\Gamma; M \otimes \mathbb{Q})$. (v1.23.0-pre.13, the C-D0 referee's F1, F2:) for every M as above,*
   (a) ***($q \ge 1$.)*** *$24 \cdot H^j(\Gamma; M) = 0$ for every $j \ge 2$, so every prime $\ell' \ge 5$ qualifies for d^q, $q \ge 1$ (the case $q = 1$, where $H^{q+1} = H^2$, included); $\ell' \ge 5$ divides neither 24 nor $\mathsf{q}_\ell \mid 12$. Moreover $b_j = 0$ for $j \ge 2$.*
   (b) ***($q = 0$.)*** *A prime $\ell' \nmid \mathsf{q}_\ell$ qualifies for d^0 iff $\dim_{\mathbb{F}_{\ell'}} H^0(\Gamma; M \otimes \mathbb{F}_{\ell'}) = b_0$, the left side measured on this complex as $\operatorname{rank} P_0 - \operatorname{rank} \delta^0$ over $\mathbb{F}_{\ell'}$ (with (5)); and $\operatorname{rank}_{\mathbb{Q}} d^0 = \operatorname{rank} C^0_\ell - b_0$ in any case. b_0 is an input stated per M: for $M = V$ and for $M = \operatorname{Ind}_{\Gamma_0(N)}^{\Gamma} V$ ($N \ge 1$), $b_0 = 0$ if $n > 0$ and $b_0 = 1$ if $n = 0$. For $M = V$ every prime $\ell' \ge 5$ with $\ell' > n + 1$ qualifies for every d^q (Proposition 23.16), the set of Proposition 23.17(4).*

(c) ***(The admissible sources.)*** $\operatorname{rank}_{\mathbb{Q}} d^q$ *may be taken from exactly two sources: (i)* $\operatorname{rank}_{\mathbb{F}_{\ell'}}$ *of the reduction mod* ℓ' *of* ***this*** O_ℓ*-complex* $C_\ell^\bullet$ *(equivalently of* δ^q*, by (5)) — the same model* $\mathcal{I}^{(\ell)}_{\mathsf{O}_\ell}$ *reduced mod* ℓ' *— at an* ℓ' *that qualifies for* d^q *by (a) or (b); (ii) the rational Betti numbers, through* $\operatorname{rank}_{\mathbb{Q}} d^q = \operatorname{rank} C_\ell^q - b_q - \operatorname{rank}_{\mathbb{Q}} d^{q-1}$ *(*$d^{-1} := 0$*;* $b_j = 0$ *for* $j \geq 2$ *by (a)), each* b_j *used taken from a certified source. The rank of a differential of any* ***other*** *complex is* ***not*** *admissible — not a census rank, not the shipped* C_7 *(Bredon) pipeline's rank at* ℓ'*, and not the rank of the* ℓ'*-local model* $\mathcal{I}^{(\ell')}$ *over* $\mathbb{F}_{\ell'}$*, whose columns and differentials are not those of* $\mathcal{I}^{(\ell)}$ *although its cohomology is the same; a computation on another complex enters only through (ii), as a certified value of some* b_j*.*

5. ***(The full layout.)*** *Let* $F^q := \bigoplus_x M_{\mathsf{O}_\ell}$ *over the cells* x *of degree* q*,* $P_q := \operatorname{diag}_x \rho_M(e_x)$*, and* $\delta^q \colon F^q \to F^{q+1}$ *the matrix whose block at* (y,x) *is* $\rho_M(a_{yx})$*,* a_{yx} *the coefficient of* d *from the cell* y *of degree* $q+1$ *to the cell* x *of degree* q *(so* $a_{yx} = e_y\, a_{yx}\, e_x$*, its well-definedness as a map* $R[\Gamma]e_y \to R[\Gamma]e_x$*). Then* P_q *is an idempotent over* O_ℓ *with image* C_ℓ^q*,* $\delta^q = P_{q+1}\delta^q P_q$*, and there are* U, V *invertible over* O_ℓ *with* $U\delta^q V = \operatorname{diag}(d^q, 0)$*. Hence* δ^q *and* d^q *have the same rank over* $\mathbb{Q}$ *and over every* $\mathbb{F}_{\ell'}$*,* $\ell' \nmid \mathfrak{q}_\ell$*, and the same non-zero elementary divisors over* $\mathbb{Z}_{(\ell)}$*;* $\operatorname{rank} C_\ell^q = \operatorname{rank} P_q$ *over* $\mathbb{Q}$ *and over every* $\mathbb{F}_{\ell'}$*; and (3), (4) and (6) may be applied to* δ^q*,* P_q *in place of* d^q*,* C_ℓ^q*.*
6. ***(The valuation read-out is the invariant-factor count.)*** *Let* A *be* d^q *or* δ^q *reduced to* $\mathbb{Z}/\ell^K$*,* $K \geq 1$ *(at* $\ell = 2$ *possibly through* $\mathbb{Z}/2^W$*,* $W \geq K$*, and then read mod* 2^K*). The Smith valuations of* A *over* $\mathbb{Z}/\ell^K$ *form the* ***list, with multiplicity****,* $(\sigma_1, \ldots, \sigma_s)$*,* $\sigma_1 \leq \cdots \leq \sigma_s < K$*, of the exponents of the non-zero diagonal entries* ℓ^{σ_i} *of its Smith form,* s *the number of those entries (counted with multiplicity; repeated values,* $\sigma_i = 0$ *above all, are the rule) — equivalently the invariant factors* $\ell^{K-\sigma_i}$ *of its column span. They are exactly the exponents* $a_t < K$*, with multiplicity, of the elementary divisors of* d^q *over* $\mathbb{Z}_{(\ell)}$*, and* $s = r_K(d^q)$ *of Proposition 23.17(1). Hence, if* $s = \operatorname{rank}_{\mathbb{Q}} d^q$*, read from a source admissible by (4)(c), then every* $a_t < K$*,* $(\sigma_1, \ldots, \sigma_s)$ *is the full list of exponents of the non-zero elementary divisors of* d^q *over* $\mathbb{Z}_{(\ell)}$*, and* $\operatorname{tors} H^{q+1}(\Gamma; M) \otimes \mathbb{Z}_{(\ell)} \cong \bigoplus_{\sigma_i > 0} \mathbb{Z}/\ell^{\sigma_i}$ *exactly (Remark 23.18 with* $e_q = K$*); if* $s < \operatorname{rank}_{\mathbb{Q}} d^q$*, then* K *is too small and nothing about* d^q*'s divisors* ℓ^a*,* $a \geq K$*, is read off. (v1.23.0-pre.13, the referee's F3, F4:)* ***A-priori exponents.*** *For* $q \geq 1$ *every* $a_t \leq v_\ell(24)$*:* $a_t \leq 3$ *at* $\ell = 2$*,* $a_t \leq 1$ *at* $\ell = 3$*,* $a_t = 0$ *at* $\ell \geq 5$*; so for* $q \geq 1$ *every* $K \geq v_\ell(24)+1$ *certifies, and* $\sigma_i \leq v_\ell(24)$ *is a consistency check of the read-out. For* $q = 0$ *(the* ℓ*-primary torsion of* H^1*) no a-priori bound is asserted here (Proposition 23.16's closing note, Remark 16.8).* ***The stopping rule:*** *if* $s < \operatorname{rank}_{\mathbb{Q}} d^q$*, raise* K *and rerun, up to the cap of the ring in use (the code map); at the cap, refuse — never report a group from* $s < \operatorname{rank}_{\mathbb{Q}} d^q$*. By the a-priori bound a cap* $\geq v_\ell(24) + 1$ *binds only at* d^0*.*

Proof. (1) The proof of Proposition 31.2(1)–(3) uses of R only that $|Q_\sigma| \in R^\times$ for every σ: e_σ is then an idempotent of $R[\Gamma_\sigma]$ and $F_t'^\sigma = R[\Gamma_\sigma]e_\sigma$ is projective; for $P_\sigma = 1$, $Q_\sigma = \Gamma_\sigma$ and the column $R = R[\Gamma_\sigma]e_\sigma$ is projective; the averaging $y \mapsto e_\sigma y$ is defined. The exactness of $F'^\sigma \to R$ holds over every ring (the periodic resolution of a finite cyclic group, resp. $R \to R$), the columns consist of free R-modules, so the contracting homotopies are R-linear; the comparison theorem, the filtration argument and $\operatorname{Ext}^q_{R[\Gamma]}(R, M) = H^q(\Gamma; M)$ are ring-independent. Nowhere is ℓ inverted or localised at, so the hypothesis "$\mathbb{Z}_{(\ell)}$-algebra" may be replaced by "$\mathfrak{q}_\ell \in R^\times$"; O_ℓ is such a ring, and Proposition 31.2(1) produces $\mathcal{I}^{(\ell)}_{\mathsf{O}_\ell}$. *Base change.* $G'_{s,t} \otimes_{\mathsf{O}_\ell} S = \bigoplus_\sigma S[\Gamma/Q_\sigma]$ and $F'^\sigma \otimes S$ is the column over S (the same group-ring coefficients), so the base-changed complex has the terms and the shape $d = d'^{(0)} + d'^{(1)} + d'^{(2)}$ of Definition 31.1 over S; $d \circ d = 0$, (IM1) and (IM2) are identities between finitely many group-ring coefficients and survive every ring map. The augmentation cmp′ over O_ℓ is a surjective quasi-isomorphism (Proposition 31.2(2)); its mapping cone is a bounded-below exact complex of free O_ℓ-modules ($\mathsf{O}_\ell[\Gamma/Q_\sigma]$ and $C_s(X) \otimes \mathsf{O}_\ell$

are free over O_ℓ), hence contractible over O_ℓ, and $-\otimes_{\mathsf{O}_\ell} S$ keeps it contractible; so $\mathrm{cmp}'\otimes S$ is a quasi-isomorphism, surjective by right exactness. Hence $\mathcal{I}^{(\ell)}_{\mathsf{O}_\ell}\otimes S$ satisfies Definition 31.1 over S (with the replaced hypothesis), and Proposition 31.2(3) applies. A model produced over S directly by the lifting procedure would be another model, homotopy equivalent to this one (Proposition 31.2(4)'s uniqueness argument with both sides projective); that gives the same cohomology but *not* a common integral form, which is what (3) needs and why (1) fixes one complex.

(2) $\mathrm{Hom}_{\mathsf{O}_\ell[\Gamma]}(\mathsf{O}_\ell[\Gamma/Q_\sigma], M_{\mathsf{O}_\ell}) = M_{\mathsf{O}_\ell}^{Q_\sigma} = \rho_M(e_\sigma)M_{\mathsf{O}_\ell}$, a direct summand of the finitely generated free O_ℓ-module M_{O_ℓ}, hence finitely generated projective over the principal ideal domain O_ℓ (a localisation of $\mathbb{Z}$), hence free. Taking the image of an idempotent commutes with $-\otimes S$, so $\rho_M(e_\sigma)M_{\mathsf{O}_\ell}\otimes S = \rho_M(e_\sigma)M_S = M_S^{Q_\sigma}$, and the cochain differential, given by the coefficients of d acting through ρ_M, is base-changed with them. By Proposition 31.2(3) over O_ℓ, $H^q(C^\bullet_\ell) = H^q(\Gamma; M_{\mathsf{O}_\ell})$. Since $\mathcal{I}_q$ is finitely generated free over $\mathbb{Z}[\Gamma]$ (Proposition 23.5(2)), $C^\bullet_{\mathcal{I}}(M_{\mathsf{O}_\ell}) = C^\bullet_{\mathcal{I}}(M)\otimes\mathsf{O}_\ell$, and O_ℓ is flat over $\mathbb{Z}$, so $H^q(\Gamma; M_{\mathsf{O}_\ell}) = H^q(\Gamma; M)\otimes\mathsf{O}_\ell$; likewise for the flat O_ℓ-algebra $\mathbb{Z}_{(\ell)}$, with $C^\bullet_\ell\otimes\mathbb{Z}_{(\ell)} = C^\bullet_{\mathcal{I}^{(\ell)}\otimes\mathbb{Z}_{(\ell)}}(M_{\mathbb{Z}_{(\ell)}})$ by the first part.

(3) The proof of Proposition 23.17(1)–(3) uses of $\mathbb{Z}$ only: the Smith form over the local principal ideal domain $\mathbb{Z}_{(\ell)}$ of a matrix with entries in $\mathbb{Z}_{(\ell)}$ (an integer matrix is one case of it); that C^{q+1} is torsion-free, and additivity of ranks over the fraction field $\mathbb{Q}$; and the universal coefficient theorem for a complex of free modules over a principal ideal domain [standard: [Wei94, §3.6]]. All three hold for C, which is a complex of finitely generated free $\mathbb{Z}_{(\ell)}$-modules by (2). $C\otimes_{\mathbb{Z}_{(\ell)}}\mathbb{Z}/\ell^e = C^\bullet_\ell\otimes_{\mathsf{O}_\ell}\mathbb{Z}/\ell^e$, which by (1), (2) is the complex built over $\mathbb{Z}/\ell^e$ and computes $H^q(\Gamma; M\otimes\mathbb{Z}/\ell^e)$; and $H^q(C) = H^q(\Gamma; M)\otimes\mathbb{Z}_{(\ell)}$ by (2), so its torsion is the ℓ-primary torsion of $H^q(\Gamma; M)$. Remark 23.18's proof is (1), (2) and the universal coefficient theorem again, one differential at a time. The truncation statement is Proposition 23.17(5)'s observation, unchanged.

(4) Over the principal ideal domain O_ℓ, $d^q = U\,\mathrm{diag}(c_1,\dots,c_r,0)\,V$ with U, V invertible and $c_i\neq 0$, $r = \mathrm{rank}_{\mathbb{Q}}\,d^q$. The proof of Proposition 23.17(2), over O_ℓ, gives $\mathrm{tors}\,\mathrm{coker}\,d^q = \mathrm{tors}\,H^{q+1}(C^\bullet_\ell) = \mathrm{tors}\,H^{q+1}(\Gamma; M)\otimes\mathsf{O}_\ell$, whose order ideal is generated by $\prod c_i$. A prime $\ell'\nmid\mathsf{q}_\ell$ is a prime of O_ℓ, with residue field $\mathbb{F}_{\ell'}$; the reduction keeps the rank r iff ℓ' divides no c_i, i.e. iff $H^{q+1}(\Gamma; M)\otimes\mathsf{O}_\ell$, equivalently (as ℓ' is not inverted) $H^{q+1}(\Gamma; M)$, has no ℓ'-torsion. A free module of rank m reduces to an $\mathbb{F}_{\ell'}$-space of dimension m. For $M = V$: by Proposition 23.16, the torsion primes of $H^\bullet(\Gamma; V)$ are 2, 3 and primes $\le n-1$ ($n>0$ even), 2 (n odd), 2, 3 ($n=0$); every $\ell'\ge 5$, $\ell' > n+1$, avoids them and does not divide $\mathsf{q}_\ell\mid 12$.

(4)(a) (v1.23.0-pre.13). Apply $\mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(-, M)$ to the short exact sequence (50) of Claim 16.2(1). For a subgroup $H\le\Gamma$, $\mathbb{Z}[\Gamma/H] = \mathbb{Z}[\Gamma]\otimes_{\mathbb{Z}[H]}\mathbb{Z}$ and $\mathbb{Z}[\Gamma]$ is free over $\mathbb{Z}[H]$, so $\mathrm{Ext}^j_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma/H], M)\cong\mathrm{Ext}^j_{\mathbb{Z}[H]}(\mathbb{Z}, M) = H^j(H; M)$ (Shapiro). This is the Mayer–Vietoris sequence (51) for the arbitrary $\mathbb{Z}[\Gamma]$-module M (Claim 16.2(2), whose argument uses nothing about the module). For $j\ge 2$ it gives $0\to C\to H^j(\Gamma; M)\to A\to 0$ with C a quotient of $H^{j-1}(\langle -I\rangle; M)$ and A a submodule of $H^j(\langle S\rangle; M)\oplus H^j(\langle U\rangle; M)$. Since $j-1\ge 1$ and $|H|$ kills $H^i(H; -)$ for $i\ge 1$ and H finite [standard: [Bro82, Ch. III.10]], 2 kills C and $\mathrm{lcm}(4,6) = 12$ kills A, so 24 kills $H^j(\Gamma; M)$ — the argument of Claim 16.2(4), which uses nothing about V. For $q\ge 1$, $j = q+1\ge 2$ (at $q=1$, $j=2$ and $j-1=1\ge 1$), so $H^{q+1}(\Gamma; M)$ has no ℓ'-torsion for $\ell'\ge 5$, and $\ell'\nmid\mathsf{q}_\ell$ as $\mathsf{q}_\ell\mid 12$. As $H^j(\Gamma; M\otimes\mathbb{Q}) = H^j(\Gamma; M)\otimes\mathbb{Q}$ (flatness, as in (2)), $b_j = 0$ for $j\ge 2$. A level-N module $\mathrm{Ind}_{\Gamma_0(N)}^{\Gamma}V$ is a $\mathbb{Z}[\Gamma]$-module, so the bound holds for it with no further argument.

(4)(b). $C^\bullet_\ell$ is a complex of finitely generated free modules over the principal ideal domain O_ℓ (by (2)), so the universal coefficient theorem of (3) gives $H^0(C^\bullet_\ell\otimes\mathbb{F}_{\ell'})\cong H^0(C^\bullet_\ell)\otimes\mathbb{F}_{\ell'}\oplus\mathrm{Tor}_1^{\mathsf{O}_\ell}(H^1(C^\bullet_\ell), \mathbb{F}_{\ell'})$. Here $H^0(C^\bullet_\ell) = \ker d^0$ is a submodule of the free C^0_ℓ, hence free, of rank $\dim_{\mathbb{Q}} H^0(C^\bullet_\ell)\otimes\mathbb{Q} = b_0$ (by (2) and flatness), and $\mathrm{Tor}_1(H^1(C^\bullet_\ell), \mathbb{F}_{\ell'}) = H^1(C^\bullet_\ell)[\ell']$, which vanishes iff $H^1(\Gamma; M)\otimes\mathsf{O}_\ell$, equivalently $H^1(\Gamma; M)$, has no ℓ'-torsion. The left side is $H^0(\Gamma; M\otimes\mathbb{F}_{\ell'})$ by (1),

(2), of dimension $\operatorname{rank} C^0_\ell - \operatorname{rank}_{\mathbb{F}_{\ell'}} d^0 = \operatorname{rank} P_0 - \operatorname{rank}_{\mathbb{F}_{\ell'}} \delta^0$ by (5). Over $\mathbb{Q}$, rank–nullity gives $\operatorname{rank}_{\mathbb{Q}} d^0 = \operatorname{rank} C^0_\ell - b_0$. *The values of* b_0. By Shapiro, $H^0(\Gamma; \operatorname{Ind}_{\Gamma_0(N)}^{\Gamma} V_{\mathbb{Q}}) = H^0(\Gamma_0(N); V_{\mathbb{Q}})$ ($N = 1$ is $M = V$; induction and coinduction agree at finite index). For $n = 0$ this is $\mathbb{Q}$. For $n > 0$, $\Gamma_0(N)$ contains $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $T_N := \begin{pmatrix} 1 & 0 \\ N & 1 \end{pmatrix}$. In $V \otimes \mathbb{Q} = \operatorname{Sym}^n \mathbb{Q}^2$, homogeneous polynomials of degree n in x, y, one of the two acts by a substitution $f(x, y) \mapsto f(x, y + cx)$ and the other by $f(x, y) \mapsto f(x + c'y, y)$ ($c, c' \in \{\pm 1, \pm N\}$; which is which, and the signs, depend on the convention of Definition 8.1, not on the argument). If $f(x, y + cx) = f(x, y)$, then $f(x, y + mcx) = f(x, y)$ for every $m \in \mathbb{Z}$, so $t \mapsto f(x, y + tx) - f(x, y)$, a polynomial in t with infinitely many zeros, vanishes; hence $x\,\partial_y f = 0$, and $f \in \mathbb{Q}x^n$. Likewise the other fixes only $\mathbb{Q}y^n$. As $\mathbb{Q}x^n \cap \mathbb{Q}y^n = 0$ for $n > 0$, $b_0 = 0$.

(4)(c). (i) is (4) at a qualifying ℓ'. (ii) is rank–nullity over $\mathbb{Q}$, $b_q = \dim \ker(d^q \otimes \mathbb{Q}) - \operatorname{rank}_{\mathbb{Q}} d^{q-1} = \operatorname{rank} C^q_\ell - \operatorname{rank}_{\mathbb{Q}} d^q - \operatorname{rank}_{\mathbb{Q}} d^{q-1}$, with $b_q = H^q(C^\bullet_\ell) \otimes \mathbb{Q}$'s dimension by (2). *Why nothing else.* (6)'s certificate compares $s = r_K(d^q)$, which never exceeds $\operatorname{rank}_{\mathbb{Q}} d^q$, with a number that must *equal* $\operatorname{rank}_{\mathbb{Q}} d^q$ of this complex; a smaller number (an $\mathbb{F}_{\ell'}$ rank at a non-qualifying ℓ') can certify a too-small K, and the rank of another complex's differential is a different number in general — cohomology is invariant under the choice of model, the ranks of the differentials are not (they depend on the ranks of the cochain modules, which differ from model to model). Betti numbers are invariants, which is why (ii) admits them from any certified source.

(5) e_x is an idempotent of $\mathsf{O}_\ell[Q_x]$ and ρ_M is multiplicative, so P_q is an idempotent with image $\bigoplus_x \rho_M(e_x) M_{\mathsf{O}_\ell} = C^q_\ell$ (by (2), cell by cell). By Proposition 31.2(3)'s identification $\operatorname{Hom}(R[\Gamma]e_x, M) = e_x M$, the cochain differential d^q of $C^\bullet_\ell$ sends the component at x to the component at y by $\rho_M(a_{yx})$; this is δ^q restricted to C^q_ℓ, and $a_{yx} = e_y a_{yx} e_x$ gives $\rho_M(a_{yx}) = \rho_M(e_y)\rho_M(a_{yx})\rho_M(e_x)$, i.e. $\delta^q = P_{q+1}\delta^q P_q$: δ^q vanishes on $\ker P_q$ and lands in $\operatorname{im} P_{q+1}$. $F^q = \operatorname{im} P_q \oplus \ker P_q$ with both summands free over O_ℓ (direct summands of a finitely generated free module over a principal ideal domain); in adapted bases of F^q and F^{q+1}, $\delta^q = \operatorname{diag}(d^q, 0)$, and the base-change matrices U, V are invertible over O_ℓ, hence over every O_ℓ-algebra ($\mathbb{Z}_{(\ell)}$, $\mathbb{Z}/\ell^K$, $\mathbb{F}_{\ell'}$, $\mathbb{Q}$). Ranks and elementary divisors are invariant under such U, V, and the zero block adds none. $\operatorname{rank} P_q = \operatorname{rank} \operatorname{im} P_q$ over every field $\mathsf{O}_\ell \to k$, the image of an idempotent commuting with base change.

(6) By (5), A and the reduction of d^q have the same Smith form up to zero rows and columns, so it suffices to treat d^q. Over $\mathbb{Z}_{(\ell)}$, $d^q = U \operatorname{diag}(\ell^{a_t}, 0)\, V$; reducing mod ℓ^K keeps U, V invertible, and the Smith form over the local ring $\mathbb{Z}/\ell^K$ is unique, so it is $\operatorname{diag}(\ell^{\min(a_t, K)}, 0)$ with $\ell^K = 0$: its non-zero diagonal entries are the ℓ^{a_t} with $a_t < K$ (this is Proposition 23.17(1)'s proof). At $\ell = 2$ the composite $\mathsf{O}_2 \to \mathbb{Z}/2^W \to \mathbb{Z}/2^K$ is the reduction map $\mathsf{O}_2 \to \mathbb{Z}/2^K$, so a matrix formed mod 2^W and read mod 2^K is the reduction to $\mathbb{Z}/2^K$. The count, with multiplicity, is $s = \#\{t : a_t < K\} = r_K(d^q)$; by (4)(c) the number compared with it is $\operatorname{rank}_{\mathbb{Q}} d^q$; and Proposition 23.17(1) with (3) and Remark 23.18 give the rest. *A-priori exponents.* By (3) and the proof of (4), $\bigoplus_t \mathbb{Z}/\ell^{a_t} \cong \operatorname{tors} \operatorname{coker}(d^q \otimes \mathbb{Z}_{(\ell)}) = \operatorname{tors} H^{q+1}(\Gamma; M) \otimes \mathbb{Z}_{(\ell)}$, which for $q \geq 1$ is killed by $\ell^{v_\ell(24)}$ by (4)(a); so $a_t \leq v_\ell(24)$, and with $K \geq v_\ell(24) + 1$ every $a_t < K$, i.e. $s = \operatorname{rank}_{\mathbb{Q}} d^q$. The stopping rule only restates the two cases of the certificate. That a given routine returns the Smith valuations of its input over $\mathbb{Z}/\ell^K$, with multiplicity, is that routine's specification, not part of this proposition. □

Scope: what this chapter does not do. It asserts the ℓ-local model for every prime ℓ over every $\mathbb{Z}_{(\ell)}$-algebra, its comparisons and Hecke transport, and the closed form and the Bredon comparison at $\ell = 3$, all [proved here]. Not asserted: anything over $\mathbb{Z}$ (the model exists only ℓ-locally — at best over $\mathbb{Z}[1/\mathsf{q}_\ell]$, Proposition 31.6 (v1.23.0-pre.12); the glueing is §23.3's); a strict lift on $\mathcal{I}^{(\ell)}$ of X itself (the transport (5) is a conjugation; v1.18.0-pre.2 builds one on a synthetic refinement Y', item (v) of the code map); the Moerdijk–Pronk simplicial set $S(\mathcal{T})$ as a normative object (Remark 17.4); a change of what the default coefficient ring names at

$\ell \in \{2, 3\}$.

Code map. Built at v1.16.0 at $\ell = 3$ (slot-16 WP-16, the stretch package, audited by a cross-model audit `docs/implementation/slot16/WP-16_audit.md`, ACCEPT WITH REPAIRS, repaired by WP-17; not built at v1.16.0-pre.1 — contract first, rule 21). `Integral/Model/IsotropyModelEll` (`integral_model::LocalModel`) on `Integral/Model/IsotropyModel`'s plug point, with a parameter ℓ that builds the columns F'^{σ} and the cochain groups M^{Q_σ} ([Q1n]: `docs/implementation/slot14/WP-Q1_small_ell_orbifold_note.md` §4 item 2), exact over $\mathbb{Q}[\Gamma]$ with denominators prime to ℓ; the strict comparisons Φ_ℓ, Ψ_ℓ of Proposition 31.2(4). `HeckeIsotropyModelEllValidation` (`standard`; `docs/implementation/slot16/WP-16_report` runs (IL1)–(IL4) against $\mathcal{I}$: (IL1) on $\mathbb{Z}/3^e$, $e \le 3$, $n \le 20$ against $\mathcal{I}$ and [IB] Appendix A, and $\mathcal{I}^{(3)} = \mathcal{I}$ on the $\mathbb{F}_3$ cells $20 < n \le 200$; (IL2) on 103 cells up to $n = 200$; (IL3) at level 1 and (v1.17.0-pre.5, slot-17 WP-7) at level N against the pipeline over $\mathbb{F}_3$, (IL4) over $\mathbb{F}_3$ and (v1.17.0-pre.5) over $\mathbb{Z}/3^e$ with the pair and cone (both [verified computationally][a] partial, above); the case $\ell = 2$ through the same code (the MP arm). The lifting procedure averages every lift by e_σ (Proposition 31.2(1)); (IL1) tests $d \circ d = 0$ and (IM1), (IM2) on the built complex (v1.16.0-pre.3, WP-14 referee C-1), where the average is a no-op on d (the reason is in the proof of (1)). **Where the average is tested** is the comparison map Ψ_ℓ's well-definedness check $e_x \Psi_\ell(x) = \Psi_\ell(x)$ in (IL4): without the average Ψ_3 still passes the chain-map identity and strictness but fails well-definedness on 22 of its 25 source generators, the validation's negative control (v1.16.0, WP-16 audit F-1). **Run at v1.17.0-pre.5** (slot-17 WP-7, merged `ea3a102`, audited ACCEPT, `docs/implementation/slot17/WP-7_report.md`, `docs/implementation/slot17/WP-7_audit.md`): (i) (IL3)'s level-N clause, (ii) (IL4) over $\mathbb{Z}/3^e$, $e \le 3$, (iii) the pair and cone half of Proposition 31.2(5), as stated in (IL3), (IL4); (iv) Remark 31.4(3)'s refinement half on a synthetic refined descriptor. The descriptor is data (`Integral/Model/IsotropyModel`'s `ModelData`: representatives, stabiliser inventory, transports, horocycle, and the `Hecke/HeckeModelDescriptor` header, whose N_q are checked against the data), read by the one constructor that also reads the shipped descriptor; `SubdivideEdgeOrbit` subdivides every edge orbit at ζ^2 (and, in a second descriptor, a horocycle edge), the new vertices have stabiliser $\{\pm I\}$ (the code gives a midpoint its edge's stabiliser, correct for the pointwise stabilisers of the shipped data), $-I$ lies in every stabiliser (checked at load), and a module with $-I \mapsto -1$ is admitted (cohomology 0). The comparison is on cohomology only: lengths and invariant factors of H^q, $q \le 5$, of $\mathcal{I}^{(3)}$ on Y' against $\mathcal{I}^{(3)}$ on X and against $\mathcal{I}$ on both, over $\mathbb{Z}/3^e$, $e \le 3$, on the boundary, with a level-N module, and at $\ell = 2$ through the same code; the orbifold Euler characteristic $-1/12$ is unchanged. That the data reproduce the pre-slot-17 constructor is guarded indirectly: arm (REF) is a *descriptor round trip* (the shipped model is now built from the same data, so it compares one code path with itself), and the real guard is that `HeckeIsotropyModelValidation`, `HeckeStrictLiftValidation`, `HeckeFreeModelValidation`, `IsotropyModelDeltaSquaredValidation`, the arm (IL0), (IL1)–(IL4) and the golden and determinism gates pass unchanged (WP-7 audit MINOR-1). Not built: a strict lift on a refined descriptor, and the Moerdijk–Pronk simplicial set itself (Remark 17.4). *The plan of the runs, as committed at v1.17.0-pre.1 (contract first, rule 21; kept for the record):* (i) (IL3)'s clause at level N: $M = \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V_R$ through the shipped `GammaOLevelN` table (§28) — no second coset construction — read as (IL3) states it, its first summand the model's H^1_{eq} at level N over $\mathbb{F}_3$ (Definition 28.3(a), the cochain level, which is all §28 asserts at $\ell = 3$; its tree resolution and cokernel need $6 \in R^\times$ and are not used); (ii) (IL4) over $\mathbb{Z}/3^e$, $e \le 3$, with the T'_p of Proposition 31.2(5) against $\mathcal{I}$ in lengths and invariant factors; (iii) the pair and cone half of Proposition 31.2(5); (iv) Remark 31.4(3)'s refinement half, on a **synthetic refined descriptor**. What the MP constraint requires of (iv): the descriptor is a Γ-equivariant subdivision Y' of X given as data (vertices, simplices, stabiliser inventory, relations), read by the same code path as the shipped descriptor (`Hecke/HeckeModelDescriptor`),

so that no vertex count, edge list or special role of v_4 is hard-wired; it subdivides at least one edge at ζ^2 (the Moerdijk–Pronk-conformant refinement of Remark 31.4(3)), so the new vertices have stabiliser $\{\pm I\}$; the centre is kept ($-I$ acts on the coefficients by its scalar, a module with $-I \mapsto -1$ is admissible input); and the comparison is on cohomology (lengths and invariant factors of H^q, $q \le q_{\max}$, with $\mathcal{I}^{(3)}$ on Y' against $\mathcal{I}^{(3)}$ on X and against $\mathcal{I}$), never on cochains; and one control cell with $-I \mapsto -1$ (odd n), on which every H^q is 0 on Y' and on X, so that "the centre is kept" is run, not only stated (v1.17.0-pre.3, slot-17 referee F1's m-C5). A pass moves (IL3) and (IL4) from [verified computationally]^{a} partial to [verified computationally]^{a}; the tag move is recorded as a proposal for this document's next referee (the runs passed at v1.17.0-pre.5; the tags are not moved here). **Run at v1.18.0-pre.2** (slot-18 WP-7, `docs/implementation/slot18/WP-7_report.md`; contract first, rule 21): (v) **a strict lift on a synthetic refined descriptor** (`Integral/Model/RefinedStrictLift`): the descriptor carries a subdivision record (`ModelData`'s `collapse`, `refines`, written by `SubdivideEdgeOrbit`), from which the collapse $c\colon C_\bullet(Y') \to C_\bullet(X)$ and the subdivision sd are built and certified ($dc = cd$, $d\,\mathrm{sd} = \mathrm{sd}\,d$, $c\,\mathrm{sd} = \mathrm{id}$); τ' is Lemma 23.12's induction on $\mathcal{I}^{(\ell)}(Y')$ for the base datum $\mathrm{sd} \circ \tau^s \circ c$ (τ^s the shipped strict pair datum, read through the shipped lift's strict identity; the shipped coset tables; no new coset or τ-table), made well defined on the projective generators by the Q_σ-average of the twisted extension to the free cover (load-bearing at $\ell = 2$, where $Q_{v_4} = \langle U^2 \rangle$ has a periodic column; a no-op at $\ell = 3$); certified exactly: (T1) on the Schreier generators of $\Gamma_0(p)$, (T2), (T3) and the strict identity over Y'. The comparison of the models over Y' and X is the strict lift of sd, resp. c (Lemma 23.12 along the surjective quasi-isomorphisms onto the chain complexes of Y' and X). Checked by (IL5) (above). (vi) **(IL3) beyond** $\mathbb{F}_3$: over $\mathbb{F}_9$ with $\mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V_R$ through the shipped twisted-monomial route (trivial character), $N \in \{1, 2, 5, 7\}$, $n \le 4$ ($N \le 13$, $n \le 6$ under the full grid): dimensions equal the $\mathbb{F}_3$ ones and $\mathcal{I}$'s, and $\dim H^1 = \dim H^1_{\mathrm{eq}} + \dim H^1(\mathbb{Z}/3; M^{-I})$ with the first summand the shipped pipeline's over $\mathbb{F}_3$ (it does not run over $\mathbb{F}_9$); over $\mathbb{Z}/9$ in lengths against the shipped level-N model's H^1_{eq} *over* $\mathbb{Z}/9$ (its flat-coordinate route, Definition 28.3(a)), 12 cells (32). The MP items of (v) hold (the report's §4 names what a Moerdijk–Pronk $S(\mathcal{T})$ descriptor would still need). On the pass of (IL5) with (vi) (IL3) and (IL4) are moved to [verified computationally]^{a} at v1.18.0-pre.2 (above; the slot-18 referee F1's C-4); (IL5) keeps its [proved here] tag and gains the [verified computationally]^{a} line. *The plan of (v), (vi), as committed at v1.18.0-pre.1 (contract first, rule 21; kept for the record; "not built" at that version):* (v) **a strict lift on a synthetic refined descriptor**, so that T_p acts on the model over Y' directly and not only by transport: τ' as in (IL5), built from the descriptor's data by the code path that reads the shipped descriptor; it may use the shipped strict pair datum τ^s (Proposition 23.13) through a Γ-equivariant comparison between the models over Y' and over X (Lemma 23.12 lifts along any surjective quasi-isomorphism), but builds no new coset or τ-table. Checked by (IL5); the integer sizes are recorded as in Remark 23.14 (evidence, bounding nothing). (vi) **(IL3) beyond** $\mathbb{F}_3$: Proposition 31.3(2),(3) hold over every $\mathbb{Z}_{(3)}$-algebra, so (IL3)'s identity is read over $\mathbb{F}_9$ (flat over $\mathbb{F}_3$: the dimensions must equal the $\mathbb{F}_3$ ones, a check of the $\mathbb{F}_q$ code path, not new mathematics) and over $\mathbb{Z}/9$ in lengths ($\mathrm{length}\, H^1 = \mathrm{length}\, H^1_{\mathrm{Bredon}} + \mathrm{length}\, H^1(\mathbb{Z}/3; M^{-I})$, from (3)'s exact sequence), against the shipped level-N model's H^1_{eq} over that ring where the shipped route runs over it (Definition 28.3(a), the cochain level), and against $\mathcal{I}$ otherwise, as at v1.17.0-pre.5. **What the MP constraint requires of (v)**, beyond item (iv) above: the refined descriptor stays out of `Hecke/HeckeModelDescriptor`'s registry and out of the golden harness; nothing in (v) hard-wires the 7-vertex model, a trivial centre or Bredon-only semantics (the centre $-I$ lies in every stabiliser of Y' and acts on the coefficients by its scalar); and the package report names what a Moerdijk–Pronk $S(\mathcal{T})$ descriptor would still need beyond (v) (Remark 17.4). Arms of `HeckeIsotropyModelEllValidation` only (no new name), its default run within the `standard` tier target. A pass of (IL5) on every cell, with (vi), is proposed to the next referee as the ground for moving (IL3), (IL4) to [verified computationally]^{a}; the tags are

not moved here. (End of the pre.1 plan.)

Code map of Proposition 31.6. Stated at v1.23.0-pre.12, before D0's code (contract first, rule 21; slot-24 milestone D0 of `docs/implementation/slot24/DESCENT_plan.md` v0.2, clause C-D0; audit finding M3 of `DESCENT_plan_audit.md`); refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D0_referee.md`), the fixes applied at v1.23.0-pre.13 (F2, F5–F7 in this code map). What the shipped code already does, read against the clauses: (1)–(2) **hold of `Integral/Model/IsotropyModelEll` as built.** `LocalModel` builds d once, in its constructor, exactly over $\mathbb{Q}[\Gamma]$ (`Rat`, overflow-checked `int64`; an overflow throws and refuses) before any coefficient ring is named, and `coboundary`, `projector` and `Pullback` reduce the stored coefficients entrywise into the `ModCtx` they are given (`reduce_rat`); so the complexes over $\mathbb{Z}/\ell^K$ and over $\mathbb{F}_{\ell'}$ are base changes of one complex, as (1) requires. The only denominators are the coefficients $1/|Q_\sigma|$ of e_σ (`build_columns`); every stored coefficient therefore lies in $\mathsf{O}_\ell[\Gamma]$, which is what the proof of (1) gives. The code does *not* certify this at build time: a denominator that is not a unit of the target ring is refused at reduction (`inverse_mod`'s `ENGINE_CHECK`), never computed; a stray denominator prime to both ℓ and ℓ' would pass silently. **D0 obligation, the denominator check** (v1.23.0-pre.13, referee F5; not built at this version): at build time, every `Rat` denominator in `diff_` and in every e_σ divides a power of q_ℓ (the per-term denominators; `QGroupRingElt::max_den` exists), else the build refuses. (5) is the code's layout: cochains on the full $\bigoplus_x M$ with δ^q and $P_q = \operatorname{diag}\rho_M(e_x)$, $\delta^q = P_{q+1}\delta^q P_q$ resting on `certify()`'s `well_defined` check ($e_{\mathrm{col}}\, a\, e_{\mathrm{row}} = a$ in $\mathbb{Q}[\Gamma]$, i.e. $a_{yx} = e_y a_{yx} e_x$); `cohomology_lengths` returns $\operatorname{len} P_q - \operatorname{len}\operatorname{im}\delta^q - \operatorname{len}\operatorname{im}\delta^{q-1}$. **Not built** (D0's, `Integral/Descent/DescentAssembly` as planned): (i) the read-out of (3), (6) on $\mathcal{I}^{(\ell)}$ — `LocalModel` returns lengths, not Smith valuations, and no code compares the count s with $\operatorname{rank}_{\mathbb{Q}}\delta^q$ on $\mathcal{I}^{(\ell)}$ (on $C^\bullet_{\mathcal{I}}$ it is `HeckeFreeModelValidation`'s). The $\mathbb{F}_{\ell'}$ rank of (4)(c)(i) is that of the **same** `LocalModel`(desc, ℓ) object reduced into `ModCtx`(ℓ'), $\ell' \nmid \mathsf{q}_\ell$; nothing reads it yet. **D0 requirement** (v1.23.0-pre.13, referee F2; the header's audit note F-5: nothing ties `ell` to the `ModCtx` later passed in, so a `LocalModel`(desc, ℓ') over `ModCtx`(ℓ') runs without complaint and is $\mathcal{I}^{(\ell')}$, inadmissible by (4)(c)): the read-out guards that the model's `ell()` is the ℓ whose K is being certified and that the $\mathbb{F}_{\ell'}$ rank is read on that same object, never on another `LocalModel`, the census or the shipped C_7 route; D0's validation prints, per cell, (4)(b)'s comparison $\operatorname{rank} P_0 - \operatorname{rank}\delta^0$ over $\mathbb{F}_{\ell'}$ against b_0, and asserts $\sigma_i \le v_\ell(24)$ for $q \ge 1$ ((6)'s a-priori bound). (i′) (v1.23.0-pre.13, referee F7) The hand-off from `LocalModel`'s `engine::DenseBuf<Elt>` over a `ModCtx` to `ring2k::Mat<uint64_t>` and to `Ring3k`'s canonical `uint32_t` input is not written. *Degrees:* `LocalModel` inherits the descriptor's construction parameter $q_{\max}$ (`Integral/Model/IsotropyModel.h`, default 5, built to $q_{\mathrm{top}} = q_{\max} + 1 = 6$; d^q exists for $q \le q_{\max}$); the plan's (P) arm at degree 6 needs $q_{\max} \ge 5$, i.e. at least the default, never the production value 3. (ii) At $\ell = 2$ (v1.23.0-pre.13, referee F6), the one-run read-out on `engine::ring2k` (rung `uint64_t`, `src/ModArithmetic/Binary/Ring2k.h`) through $\mathbb{Z}/2^W$: $W \le 61$ **is reachable now** — `LocalModel` is instantiated at `Elt = uint64_t`, `ModCtx<uint64_t>::prime_power(2, W)` exists, `reduce_rat` accepts moduli $< 2^{62}$, and `smith(A, k)` reads its input by masking (`Ring2k.h`: "nothing ever reduces"), so these residues are valid input at $k = W$ and one run certifies every exponent $< W$, as (6)'s "formed mod 2^W, read mod 2^K" allows; $W \in [62, 64]$ **is not built**: `reduce_rat` refuses a modulus $\ge 2^{62}$, and the reduction $\mathsf{O}_2 \to \mathbb{Z}/2^W$ (the odd denominators 3^j inverted to width W) is to be written. (iii) At $\ell = 3$, `engine::ring3k` carries $K \le 19$ only ($3^{19} < 2^{31}$, `Ring3k.h`); a 3-adic exponent ≥ 19 is then a refusal under (6) ("K too small"), and no wider rung at 3 exists. Both engines' Smith routines are specified in `src/ModArithmetic/ENGINE_API.md` §§8.2, 8.4 and gated by `Ring2kLadderValidation` and `OrbifoldMod3kValidation`; their correctness is the hypothesis of (6)'s last sentence, not a consequence of it. By (6)'s a-priori bound both caps ($K \le 19$ at 3; $W \le 61$, later 64, at 2) bind only at d^0; reaching one is a refusal by the stopping rule. (iv)

The plan's "$|Q| \in \{1,2\}$ at $\ell = 3$" (`DESCENT_plan.md` §2.7, row C-D0) is corrected here to $|Q_\sigma| \in \{2,4\}$ ($Q_{v_6} = \langle S \rangle$; `build_columns` computes Q_{v_6} of order 4); the ring $\mathsf{O}_3 = \mathbb{Z}[1/2]$ is the same. The plan row is amended to match at v1.23.0-pre.13 (referee F8, `DESCENT_plan.md` v0.3).

Formalization note. Definition 31.1 and Proposition 31.2 are homological algebra over $R[\Gamma]$ (projective resolutions, the comparison theorem, Shapiro's lemma); Proposition 31.3 adds the amalgam's Mayer–Vietoris sequence and the cohomology of cyclic groups. None depends on modular forms. Proposition 31.6 (v1.23.0-pre.12) is base change and Smith forms over the principal ideal domains $\mathbb{Z}[1/\mathfrak{q}_\ell]$ and $\mathbb{Z}_{(\ell)}$; it carries **no** `\LEAN` marker, since none of it is formalised.

31.1 The descent at level N, and its bad primes

(v1.23.0-pre.14; milestone D1 of `docs/implementation/slot24/DESCENT_plan.md` v0.3, clauses C-D1 and C-D2 of its §2.7, the audit findings M8, m3, m4, m7 of `DESCENT_plan_audit.md`.) §23.3 glues $H^\bullet(\Gamma; V)$ from one integral map; nothing in its proofs uses that the coefficient module is V rather than any $\mathbb{Z}[\Gamma]$-module that is finitely generated and free over $\mathbb{Z}$. This subsection states the descent in that generality — so for $\mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V$ and its nebentypus twists, by Shapiro's lemma the level-N groups — with the two places where the generality changes a statement: the coherence term of Proposition 23.10, which survives only when $b_0 b_1 = 0$, and the bad primes, which at level N are not bounded by $n-1$. (It sits in §31's section, after that section's last statement, only so that no statement number moves; it uses that section's Proposition 31.6, and nothing in it is specific to the ℓ-local model. Referee F7, v1.23.0-pre.15.)

Notation. For a $\mathbb{Z}[\Gamma]$-module M and a ring S, $M_S := M \otimes_{\mathbb{Z}} S$; $C^\bullet_{\mathcal{I}}(M) := \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_\bullet, M)$ and $C^\bullet_{\mathrm{eq}}(M) := \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X), M)$, so that $C^\bullet_{\mathcal{I}}(R)$, $C^\bullet_{\mathrm{eq}}(R)$ of §23 are $C^\bullet_{\mathcal{I}}(V_R)$, $C^\bullet_{\mathrm{eq}}(V_R)$ (the model complex with fibre M, Theorem 5.10(1)); $\mathrm{T}(M)$ is the tree complex (65); $b_j(M) := \dim_{\mathbb{Q}} H^j(\Gamma; M_{\mathbb{Q}})$; $\mathsf{O} := \mathbb{Z}[1/6]$. $M_N := \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V$ is Definition 28.3's fibre over $\mathbb{Z}$. For a character $\chi\colon G_N \to \mathbb{C}^\times$ of order m_χ, $\mathsf{O}_\chi := \mathbb{Z}[\zeta_{m_\chi}]$ and χ is read with values in $\mu_{m_\chi} \subset \mathsf{O}_\chi^\times$; $V_{\mathsf{O}_\chi}(\chi)$ is Definition 29.1(5) over $R = \mathsf{O}_\chi$ (Lemma 29.2's proof is ring-independent), and $M_N(\chi) := \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V_{\mathsf{O}_\chi}(\chi)$ in the coordinates (73), free of rank $\varphi(m_\chi) m_N (n+1)$ over $\mathbb{Z}$; $M_N(1) = M_N$. This is an integral form of §29's character model, whose coefficient rings are $\mathbb{F}_q$ and $W_K(\mathbb{F}_q)$ (item (8) below).

Proposition 31.7 (The descent for every finite free coefficient module; level N and nebentypus)**.** *[proved here] (v1.23.0-pre.14; refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D1_refere` the fixes applied at v1.23.0-pre.15, not re-refereed.) Assume (H4) and (B1)–(B3), and let M be a $\mathbb{Z}[\Gamma]$-module that is finitely generated and free over $\mathbb{Z}$ (M_N, $M_N(\chi)$, ...).*

1. ***(The three complexes.)*** *$C^\bullet_{\mathcal{I}}(M)$ is a complex of finitely generated free $\mathbb{Z}$-modules with $C^\bullet_{\mathcal{I}}(M) \otimes S = C^\bullet_{\mathcal{I}}(M_S)$ and $H^q(C^\bullet_{\mathcal{I}}(M_S)) = H^q(\Gamma; M_S)$ for every ring S and every q. $C^\bullet_{\mathrm{eq}}(M)$ is a complex of finitely generated free $\mathbb{Z}$-modules in degrees $0, 1, 2$ with $H^2(C^\bullet_{\mathrm{eq}}(M)) = 0$, and $\mathrm{cmp}^*\colon C^\bullet_{\mathrm{eq}}(M) \to C^\bullet_{\mathcal{I}}(M)$ is an injective cochain map whose cone has cohomology annihilated by 48. Over O: $C^\bullet_{\mathrm{eq}}(M)[1/6] = C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}})$ and $\mathrm{T}(M)[1/6] = \mathrm{T}(M_{\mathsf{O}})$ are complexes of finitely generated free O-modules, and for every O-algebra S — $\mathbb{Z}/\ell^K$ and $\mathbb{F}_{\ell'}$ for primes $\ell, \ell' \geq 5$ among them — $C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}}) \otimes_{\mathsf{O}} S = C^\bullet_{\mathrm{eq}}(M_S)$, $\mathrm{T}(M_{\mathsf{O}}) \otimes_{\mathsf{O}} S = \mathrm{T}(M_S)$, and both compute $H^q(\Gamma; M_S)$ for every q (0 for $q \geq 2$).*
2. ***(The descent.)*** *With $A := C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}})$, $B := C^\bullet_{\mathcal{I}}(M) \otimes \widehat{\mathbb{Z}}_6$, $C := C^\bullet_{\mathcal{I}}(M) \otimes \widehat{\mathbb{Q}}_6$ and α, β as in Proposition 23.9, the mapping cocone of $\alpha - \beta$ is quasi-isomorphic to $C^\bullet_{\mathcal{I}}(M)$, and $H^q(\Gamma; M) \cong H^q(A) \times_{H^q(C)} H^q(B)$ for every q. The only glueing datum is cmp^*. By Shapiro's lemma $H^q(\Gamma; M_N) = H^q(\Gamma_0(N); V)$ and $H^q(\Gamma; M_N(\chi)) = H^q(\Gamma_0(N); V_{\mathsf{O}_\chi}(\chi))$.*
3. ***(Maps and Hecke operators are determined locally.)*** *For a chain map f between complexes of abelian groups (those of (1), (2) among them), $H^q(f)$ is determined by $H^q(f[1/6])$ and $H^q(f \otimes \widehat{\mathbb{Z}}_6)$ (Proposition 23.8's morphism clause). An endomorphism*

*ϕ of a finitely generated abelian group G is determined by $\phi\otimes\mathbb{Z}[1/6]$ and $\phi\otimes\widehat{\mathbb{Z}}_6$; so every Hecke operator on $H^q(\Gamma_0(N);V)$ or $H^q(\Gamma_0(N);V_{\mathsf{O}_\chi}(\chi))$, on whatever model it is computed, is determined by its two local components. A $\mathbb{Z}[\Gamma]$-map $u\colon M\to M'$ (the fibre maps of restriction, trace and R_σ; the O_χ-action on $M_N(\chi)$) induces $\mathrm{Hom}(\mathcal{I}_\bullet,u)$ and $\mathrm{Hom}(C_\bullet(X),u)$, which commute strictly with cmp^*. **No coherence datum is used**: neither (2) nor (3) needs Proposition 23.10 or (4).*

4. ***(The term Hom^{-1}, for M.)** Put $\mathcal{N}_M:=C^\bullet_{\mathcal{I}}(M)\otimes\widehat{\mathbb{Q}}_6$. Then $\mathrm{Hom}_{D(\widehat{\mathbb{Q}}_6)}(\mathcal{N}_M,\mathcal{N}_M[-1])\cong\mathrm{Hom}_{\widehat{\mathbb{Q}}_6}\big(H^1(\mathcal{N}_M),H^0(\mathcal{N}_M)\big)\cong\widehat{\mathbb{Q}}_6^{\,b_0(M)b_1(M)}$ and $\mathrm{Hom}_{D(\widehat{\mathbb{Q}}_6)}(\mathcal{N}_M,\mathcal{N}_M[-i])=0$ for $i\geq 2$; so Proposition 23.10(2) holds for M **iff** $b_0(M)\,b_1(M)=0$. For $M=M_N(\chi)$: $b_0=1$ if $n=0$ and $\chi=1$, and $b_0=0$ otherwise; so the vanishing holds whenever $n>0$ or $\chi\neq 1$ — for $n>0$ with $\chi(-1)=(-1)^n$, and for $\chi(-1)\neq(-1)^n$, where $\mathcal{N}_M$ is acyclic — and **fails** at $n=0$, $\chi=1$ as soon as $b_1>0$: at $N=2$, Proposition 27.5(3) over $\mathbb{Q}$ gives $b_1=1$ (the audit's finding M8), and for every $N>1$, at $n=0$, $\chi=1$, $6\,b_1=6+m_N-3\nu_2(N)-4\nu_3(N)\geq 6$, so $b_1\geq 1$ and the vanishing fails at every level $N>1$ (proved below, v1.23.0-pre.15; $b_1=1$ exactly at $N\in\{2,3,5,7,13\}$; nothing else in this proposition uses it).*
5. ***(Truncation and periodicity.)** Remark 23.11 holds for M verbatim: what is glued is the untruncated $C^\bullet_{\mathcal{I}}(M)$, a local run builds $\sigma^{\leq q_{\max}+1}$, and brutal truncation commutes with base change. For $q\geq 2$, $e\cup-\colon H^q(\Gamma;M)\to H^{q+2}(\Gamma;M)$ is an isomorphism, natural in M (Lemma 14.22(2)). **At chain level** (v1.23.0-pre.15): if $E\colon\mathcal{I}_{\bullet+2}\to\mathcal{I}_\bullet$ is a $\mathbb{Z}[\Gamma]$-chain map with $\varepsilon\circ E_0$ a cocycle representing e (`Integral/Model/PeriodicityClass`'s E, Remark 23.21), then for every M, $\mathrm{Hom}(E,M)\colon C^\bullet_{\mathcal{I}}(M)^\bullet\to C^\bullet_{\mathcal{I}}(M)^{\bullet+2}$ induces $e\cup-$ in every degree, and so, through Shapiro's isomorphism below, $\mathrm{res}^\Gamma_A e\cup-$ on $H^\bullet(A;X)$ for $M=\mathrm{Ind}^\Gamma_A X$. Let $A\leq\Gamma$ have finite index, X a $\mathbb{Z}[A]$-module and $\mathrm{sh}\colon H^\bullet(\Gamma;\mathrm{Ind}^\Gamma_A X)\to H^\bullet(A;X)$ Shapiro's isomorphism; then $\mathrm{sh}(e\cup y)=\mathrm{res}^\Gamma_A e\cup\mathrm{sh}(y)$. Hence $\mathrm{res}^\Gamma_A e\cup-\colon H^q(A;X)\to H^{q+2}(A;X)$ **is an isomorphism for every** $q\geq 2$, and $\mathrm{res}^\Gamma_C e$ generates $H^2(C;\mathbb{Z})$ for every finite cyclic $C\leq A$ (Lemma 14.22(1)). **Hecke equivariance**: let $g\in\Delta$, $A'':=A\cap gAg^{-1}$, $A':=A\cap g^{-1}Ag$, both of finite index, $\iota_g\colon A''\to A'$, $h\mapsto g^{-1}hg$, and $f\colon X\to X$ additive with $f(g^{-1}hg\cdot x)=h\cdot f(x)$ for $h\in A''$ (a compatible pair; $f=\rho_n(g)$ for $X=V$, $f=\rho_{n,\chi}(g)$, $g\in\Delta_0(N)$, for $X=V_{\mathsf{O}_\chi}(\chi)$). For $[AgA]:=\mathrm{cor}^A_{A''}\circ(\iota_g,f)^*\circ\mathrm{res}^A_{A'}$,*

$$[AgA]\big(\mathrm{res}^\Gamma_A e\cup x\big)=\mathrm{res}^\Gamma_A e\cup[AgA](x)\qquad(x\in H^q(A;X),\ q\geq 0).$$

In particular, for $A=\Gamma_0(N)$, T_p ($p\nmid N$) on $H^\bullet(\Gamma_0(N);V)$ and on $H^\bullet(\Gamma_0(N);V_{\mathsf{O}_\chi}(\chi))$ commutes with $\mathrm{res}^\Gamma_A e\cup-$, which for $q\geq 2$ is an isomorphism of Hecke modules; $q_{\max}=3$ determines every degree ≥ 2 at level N, Hecke-equivariantly. The argument does not use $p\nmid N$: it covers $U_p:=[\Gamma_0(N)\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right)\Gamma_0(N)]$, $p\mid N$, with $f=\rho_n(\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right))$, resp. $\rho_{n,\chi}(\left(\begin{smallmatrix}1&0\\0&p\end{smallmatrix}\right))$ — the double coset of Proposition 27.2, resp. 29.9, at $A=B=\Gamma_0(N)$ — on cohomology as well (its cochain operator is deferred, [R-Q4] of the plan; lifted for the free model by Proposition 31.13(9), v1.26.0-pre.1).
6. ***(The good prime.)** A prime $\ell'\geq 5$ qualifies for d^q if $H^{q+1}(\Gamma;M)$ has no ℓ'-torsion; then $\mathrm{rank}_{\mathbb{Q}}\,d^q=\mathrm{rank}_{\mathbb{F}_{\ell'}}\,d^q$ on each of $C^\bullet_{\mathcal{I}}(M)$, $C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}})$, $\mathrm{T}(M_{\mathsf{O}})$ and $C^\bullet_\ell$ of Proposition 31.6 (every $\ell'\geq 5$ qualifies for d^q, $q\geq 1$, by Proposition 31.6(4)(a)). **For** M_N ($\chi=1$, every N, every n) every prime $\ell'\geq 5$ with $\ell'>n+1$ qualifies for every d^q (Proposition 31.8(2)); the plan's rule "$\ell'>n+1$ and $\ell'\nmid 6N$" is a subset, its condition $\ell'\nmid N$ not needed for M_N. **For** $M_N(\chi)$, $\chi\neq 1$, every prime $\ell'\geq 5$ with $\ell'\nmid m_\chi N$ qualifies for every d^q (Proposition 31.8(3)(a)), and $\ell'\nmid N$ is needed: $\ell'\mid N$, $\ell'>n+1$ may carry torsion (the example of Proposition 31.8(3)). One rule valid for every χ: $\ell'\geq 5$, $\ell'>n+1$, $\ell'\nmid m_\chi N$.*
7. ***(The certified exponent on the O-piece at $\ell\geq 5$.)** Let $\ell\geq 5$ and $K\geq 1$.*

(a) ***Direct Smith form on the Bredon complex.*** *Proposition 23.17(1)–(3) and Remark 23.18 hold for* $C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}}) \otimes_{\mathsf{O}} \mathbb{Z}_{(\ell)}$ *in place of a complex of free* $\mathbb{Z}$*-modules, at the prime* ℓ*; its reduction mod* ℓ^K *is* $C^\bullet_{\mathrm{eq}}(M_{\mathbb{Z}/\ell^K})$*, its target* C^1 *written in a basis over* $\mathbb{Z}/\ell^K$ *of the free modules* $(M_{\mathbb{Z}/\ell^K})^{\Gamma_\sigma}$*, its source in a basis or by any generating family, as in (b) (only the image of* d^0 *enters).* d^1 *is onto (its non-zero elementary divisors are units), so all* ℓ*-torsion is read from* d^0*: if the number* s *of non-zero diagonal entries* ℓ^{σ_i} *of the Smith form of* $d^0 \bmod \ell^K$ *equals* $\operatorname{rank}_{\mathbb{Q}} d^0 = \operatorname{rank} C^0_{\mathrm{eq}}(M_{\mathsf{O}}) - b_0(M)$*, then* $\operatorname{tors} H^1(\Gamma; M) \otimes \mathbb{Z}_{(\ell)} \cong \bigoplus_{\sigma_i>0} \mathbb{Z}/\ell^{\sigma_i}$ *exactly; if* $s < \operatorname{rank}_{\mathbb{Q}} d^0$*,* K *is too small, and the stopping rule of Proposition 31.6(6) applies (raise* K*; at the cap of the ring, refuse). No a-priori bound on the exponents of* d^0 *is asserted (Remark 16.8).*

(b) ***The sparse cokernel route at level*** N ***([m7]).*** *Let* $\bar{R}$ *be any matrix over* $\mathbb{Z}/\ell^K$*, written in a basis of the free module* $(M_{\mathbb{Z}/\ell^K})^{\langle -I\rangle}$*, whose rows* generate *the submodule* $(M_{\mathbb{Z}/\ell^K})^{\langle S\rangle} + (M_{\mathbb{Z}/\ell^K})^{\langle U\rangle}$ *— for instance the columns of the averaging idempotents, or the orbit-wise generators of Propositions 27.5(1), 29.11(1) provided that at each fixed point they generate* $(V_{\mathbb{Z}/\ell^K})^{F_j}$ *over* $\mathbb{Z}/\ell^K$ *(a kernel basis computed over* $\mathbb{F}_\ell$ *need not) — and* s *the number of non-zero diagonal entries* ℓ^{σ_i} *of its Smith form. Then* $s = r_K(d^0)$ *for* d^0 *of* $\mathrm{T}(M_{\mathsf{O}})$*; if* $s = \operatorname{rank} \mathrm{T}^1 - b_1(M)$ $\big(= \operatorname{rank} \mathrm{T}^0 - b_0(M)\big)$*, then* $\operatorname{tors} H^1(\Gamma; M) \otimes \mathbb{Z}_{(\ell)} \cong \bigoplus_{\sigma_i>0} \mathbb{Z}/\ell^{\sigma_i}$ *exactly; otherwise* K *is too small, with the same stopping rule. For* M_N*,* n *even:* $\operatorname{rank} \mathrm{T}^1 = m_N(n+1)$ *and* $b_1 = b_0 +$ *the right-hand side of Proposition 27.5(3) over* $\mathbb{Q}$*; the comparison value is this closed form, from Proposition 27.5(1)'s orbit counts, never the number of rows of* $\bar{R}$*.* The certificate that the rows generate *(v1.23.0-pre.15, referee F4): the averaging-idempotent columns need none (*$e_F x$ *is fixed, and* $x = e_F x$ *for fixed* x*); any other family of vectors in block* j *generates* $(V_{\mathbb{Z}/\ell^K})^{F_j}$ *iff (i) each is fixed mod* ℓ^K *by a generator of the cyclic group* F_j*, and (ii) their reductions mod* ℓ *span* $(V_{\mathbb{F}_\ell})^{F_j}$*, of dimension* d_{F_j}*.*

(c) ***Proposition 31.6(4) at level*** N***.*** *(4)(a) holds for every* M *as stated. (4)(b)'s values extend:* $b_0(M_N(\chi)) = 1$ *if* $n = 0$ *and* $\chi = 1$*, and* 0 *otherwise; a prime* $\ell' \geq 5$ *qualifies for* d^0 *iff* $\ell' \notin \mathcal{B}(M)$ *of Proposition 31.8(1). (4)(c)'s two admissible sources apply on* $C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}})$ *and on* $\mathrm{T}(M_{\mathsf{O}})$ *with "this complex" read as the complex whose differential is being certified; the rank of one of* $C^\bullet_{\mathcal{I}}$*,* $C^\bullet_{\mathrm{eq}}$*,* T*,* $C^\bullet_\ell$ *is not admissible for another.*

(At $\ell = 3$*, (7)(b) holds verbatim for* M_N *when* $\nu_3(N) = 0$*: Proposition 31.9(3), v1.23.0-pre.17.)*

8. ***(Nebentypus,*** ℓ***-locally.)*** *Let* ℓ *be a prime,* $\ell \nmid 6m_\chi$*, and* $K \geq 1$*. Then* $M_N(\chi) \otimes \mathbb{Z}/\ell^K \cong \bigoplus_{\lambda|\ell} \operatorname{Ind}_{\Gamma_0(N)}^{\Gamma} V_{W_\lambda}(\mathfrak{t} \circ \bar{\chi}_\lambda)$ *as* $\mathbb{Z}/\ell^K[\Gamma]$*-modules, the sum over the primes* λ *of* O_χ *above* ℓ*, with* $W_\lambda \cong W_K(\mathbb{F}_{q_\lambda})$*,* $q_\lambda := |\mathsf{O}_\chi/\lambda|$*, and* $\bar{\chi}_\lambda := \chi \bmod \lambda \colon G_N \to \mathbb{F}_{q_\lambda}^\times$*. So the* ℓ*-primary part of* $H^\bullet(\Gamma_0(N); V_{\mathsf{O}_\chi}(\chi))$ *is read on §29's character models over* $W_K(\mathbb{F}_{q_\lambda})$*, one per* λ*, at the characters* $\bar{\chi}_\lambda$*.*

Proof. (1) $\mathcal{I}_q$ is finitely generated free over $\mathbb{Z}[\Gamma]$ (Proposition 23.5(2)), so $C^\bullet_{\mathcal{I}}(M) \cong M^{w_q}$ in degree q, $C^\bullet_{\mathcal{I}}(M) \otimes S = C^\bullet_{\mathcal{I}}(M_S)$, and $H^q(C^\bullet_{\mathcal{I}}(M_S)) = \operatorname{Ext}^q_{\mathbb{Z}[\Gamma]}(\mathbb{Z}, M_S) = H^q(\Gamma; M_S)$ — the proof of Proposition 23.5(3), which uses nothing about V_R. $C^\bullet_{\mathrm{eq}}(M) = \bigoplus_\sigma M^{\Gamma_\sigma}$, a submodule of a finitely generated free $\mathbb{Z}$-module in each degree, hence free; $H^2(C^\bullet_{\mathrm{eq}}(M)) = 0$ by Proposition 23.3(2), which is stated for every $\mathbb{Z}[\Gamma]$-module. The injectivity and the bound 48 are the proof of Proposition 23.5(4) with M for V_R: the graded pieces of the cone have cohomology $H^t(\Gamma_\sigma; M)$, $t \geq 1$, annihilated by $|\Gamma_\sigma|$ for every module [standard: [Bro82, Ch. III.10.2]]. Over O: invariants are a kernel and O is flat, so $M^{\Gamma_\sigma}[1/6] = M_{\mathsf{O}}^{\Gamma_\sigma}$; as $|\Gamma_\sigma| \mid 12$ is a unit, $M_{\mathsf{O}}^{\Gamma_\sigma} = e_\sigma M_{\mathsf{O}}$ for the averaging idempotent e_σ, a direct summand of a finitely generated free O-module, hence free over the principal ideal domain O, and $e_\sigma M_{\mathsf{O}} \otimes S = e_\sigma M_S = M_S^{\Gamma_\sigma}$ (the image of an idempotent commutes with base change); the differentials, given by group-ring coefficients acting on M, are base-changed with them. The same holds for T, whose terms are $M^{\langle S\rangle} \oplus M^{\langle U\rangle}$ and $M^{\langle -I\rangle}$.

$C^\bullet_{\mathrm{eq}}(M_S)$ computes $H^q(\Gamma;M_S)$ because the cone of cmp* over S is annihilated by $48\in S^\times$; $\mathrm{T}(M_S)$ does by Lemma 28.4 with $A_0=\Gamma$ and $R=S$; and $H^q(\Gamma;M_S)=0$ for $q\ge 2$ by the same lemma.

(2) The proof of Proposition 23.9 with (1) in place of Proposition 23.5(3),(4): cmp$^*[1/6]\colon A\to C^\bullet_{\mathcal{I}}(M)[1/6]$ is a quasi-isomorphism, the induced map of cocones is one by the five lemma, and Propositions 23.7 and 23.8 apply to $C^\bullet_{\mathcal{I}}(M)$, a complex of abelian groups. Shapiro: [standard: [Bro82, Ch. III.6.2]], induction and coinduction agreeing at finite index, as in the proof of Proposition 27.5(2).

(3) The first sentence is Proposition 23.8. For the second apply Proposition 23.8 to G placed in degree 0: $G\to G[1/6]\oplus(G\otimes\widehat{\mathbb{Z}}_6)$ is injective, and ϕ commutes with it. A Hecke operator is an endomorphism of the finitely generated group $H^q(\Gamma;M)$ (finitely generated by (1)). cmp$^*=\mathrm{Hom}(\mathrm{cmp},M)$, and $\mathrm{Hom}(-,u)\circ\mathrm{Hom}(\mathrm{cmp},M)=\mathrm{Hom}(\mathrm{cmp},M')\circ\mathrm{Hom}(-,u)$ is functoriality of Hom in two variables.

(4) $\widehat{\mathbb{Q}}_6=\mathbb{Q}_2\times\mathbb{Q}_3$ is a product of two fields, so every $\widehat{\mathbb{Q}}_6$-module is projective and injective, and a complex X of $\widehat{\mathbb{Q}}_6$-modules is isomorphic in $D(\widehat{\mathbb{Q}}_6)$ to $\bigoplus_q H^q(X)[-q]$ (split $X^q=Z^q\oplus X^q/Z^q$ and $Z^q=B^q\oplus H^q$). Hence $\mathrm{Hom}_D(X,X[-i])=\prod_{q,p}\mathrm{Ext}^{q-p-i}_{\widehat{\mathbb{Q}}_6}(H^q,H^p)=\prod_q\mathrm{Hom}_{\widehat{\mathbb{Q}}_6}(H^q,H^{q-i})$, since Ext^j vanishes for $j\ne 0$ over a semisimple ring [standard: [Wei94, §10.7]]. For $X=\mathcal{N}_M$, $H^q(\mathcal{N}_M)=H^q(\Gamma;M)\otimes\widehat{\mathbb{Q}}_6\cong\widehat{\mathbb{Q}}_6^{\,b_q}$ (flatness; H^q finitely generated), and $b_q=0$ for $q\ge 2$ (Proposition 31.6(4)(a)); so only $q=1$, $i=1$ survives, and $\mathrm{Hom}_{\widehat{\mathbb{Q}}_6}(\widehat{\mathbb{Q}}_6^{b_1},\widehat{\mathbb{Q}}_6^{b_0})\cong\widehat{\mathbb{Q}}_6^{b_0b_1}$. *The values of* b_0. $H^0(\Gamma;M_N(\chi)_{\mathbb{Q}})=H^0(\Gamma_0(N);V(\chi)_{\mathbb{Q}})$ (Shapiro in degree 0). For $n>0$, $\Gamma_1(N)\subseteq\Gamma_0(N)$ acts on $V(\chi)$ through ρ_n (Lemma 29.2(4)) and contains $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1&0\\N&1\end{smallmatrix}\right)$; the argument of Proposition 31.6(4)(b), over the field $\mathbb{Q}(\zeta_{m_\chi})$ of characteristic 0, leaves no non-zero fixed vector. For $n=0$, $V(\chi)=\mathsf{O}_\chi$ with γ acting by $\chi(a_\gamma)$, and $\gamma\mapsto a_\gamma$ mod N is onto G_N (Lemma 29.2(1)), so the invariants vanish unless $\chi=1$, where they are $\mathbb{Q}$. If $\chi(-1)\ne(-1)^n$, $-I$ acts on the coefficients by -1 and trivially on cohomology, so 2 kills $H^\bullet(\Gamma_0(N);V_{\mathsf{O}_\chi}(\chi))$ (the proof of Proposition 29.11(2), over $\mathbb{Z}$) and every $b_j=0$. At $N=2$, $n=0$: $m_2=3$, $\nu_2(2)=1$ ($(1:1)$ is the fixed point of S), $\nu_3(2)=0$, and Proposition 27.5(3) gives $\dim H^1-\dim H^0=3-2-1=0$, so $b_1=b_0=1$. *Every* $N>1$ (v1.23.0-pre.15, referee F2). At $n=0$, $d_S=d_{ST}=1$ and $b_0=1$, so Proposition 27.5(3) over $\mathbb{Q}$ gives $b_1=1+m_N/6-\nu_2/2-2\nu_3/3$, and $b_1\ge 1$ iff $m_N\ge 3\nu_2+4\nu_3$. By the Chinese remainder theorem $\mathbb{P}^1(\mathbb{Z}/N)=\prod_{p^e\|N}\mathbb{P}^1(\mathbb{Z}/p^e)$ compatibly with Γ, so m_N, ν_2, ν_3 are multiplicative, $m_{p^e}=p^{e-1}(p+1)$. Let $g\in\{S,ST\}$, or its transpose (whichever side (22)'s action uses; g and g^{T} have the same characteristic polynomial χ_g, $\chi_S=X^2+1$, $\chi_{ST}=X^2-X+1$, and each has an off-diagonal entry ±1). A point of $\mathbb{P}^1(\mathbb{Z}/p^e)$ fixed by g is a line spanned by a primitive v with $gv=uv$; then $\chi_g(u)v=\mathrm{adj}(g-u)(g-u)v=0$ and v has a unit coordinate, so $\chi_g(u)=0$. Conversely, for a root u, $g-u$ has determinant 0 and an entry ±1, so its kernel is free of rank 1, spanned by a primitive vector (solve the row with the entry ±1; the other row is then automatic); and the line determines u. So $\nu_2(p^e)$, $\nu_3(p^e)$ are the numbers of roots of X^2+1, X^2-X+1 in $\mathbb{Z}/p^e$. At a root the derivative is a unit when $p\ne 2$, resp. $p\ne 3$ ($(2u)^2=-4$, $(2u-1)^2=-3$), so Hensel's lemma gives $\nu_2(p^e)=1+\left(\frac{-1}{p}\right)$ for p odd and $\nu_3(p^e)=1+\left(\frac{-3}{p}\right)$ for $p\ne 2,3$; directly, $\nu_2(2)=1$, $\nu_2(2^e)=0$ for $e\ge 2$ ($u^2\not\equiv -1 \bmod 4$), $\nu_3(2^e)=0$ (X^2+X+1 has no root mod 2), $\nu_3(3)=1$ and $\nu_3(3^e)=0$ for $e\ge 2$ (the one root -1 mod 3, and $\chi_{ST}(-1+3t)\equiv 3 \bmod 9$). Let $k\ge 1$ be the number of primes dividing N. If $\nu_2\nu_3\ne 0$, every local factor of both is non-zero, which excludes $p=2,3$ and forces $p\equiv 1 \pmod{12}$: $\nu_2=\nu_3=2^k$ and $m_N\ge 14^k\ge 7\cdot 2^k=3\nu_2+4\nu_3$. If $\nu_2\ne 0=\nu_3$, every $p^e\|N$ is 2 or has $p\equiv 1 \pmod 4$, and $m_{p^e}\ge 3\nu_2(p^e)$ ($3=3\cdot 1$, resp. $p+1\ge 6=3\cdot 2$), so $m_N\ge 3^k\nu_2\ge 3\nu_2$. If $\nu_3\ne 0=\nu_2$, every $p^e\|N$ is 3 or has $p\equiv 1 \pmod 3$, and $m_{p^e}\ge 4\nu_3(p^e)$ ($4=4\cdot 1$, resp. $p+1\ge 8=4\cdot 2$), so $m_N\ge 4\nu_3$. If $\nu_2=\nu_3=0$ there is nothing to prove. (Equality, $b_1=1$, *exactly* at $N=2,3,5,7,13$ (v1.23.0-pre.16, referee D-5: in each case above

equality forces $k = 1$ and equality in the one local factor; the referee's direct count to $N = 2000$ agrees); at $N = 1$, $6b_1 = 6 + 1 - 3 - 4 = 0$. The referee's numerical check, $2 \le N \le 20000$, minimum $6b_1 = 6$: `C-D1_referee.md` §2; the local counts against direct fixed points, $N \le 120$: `docs/implementation/slot24/cd1fix_f2_check.py`. Not tags.)

(5) The first part is Remark 23.11's argument, which uses of $C^\bullet_{\mathcal{I}}(\mathbb{Z})$ only that it is a complex of free modules, and Lemma 14.22(2), stated for every $\mathbb{Z}[\Gamma]$-module. *Chain level.* $E^* := \operatorname{Hom}(E, -)$ is a chain map $C^\bullet_{\mathcal{I}}(M) \to C^\bullet_{\mathcal{I}}(M)[2]$, natural in M; as each $\mathcal{I}_q$ is free, $\operatorname{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_q, -)$ is exact, so a short exact sequence of modules gives one of complexes, E^* maps it to the shifted one, and the induced maps commute with the connecting maps. $e \cup -$ is natural in M and commutes with the connecting maps as well ((E1): the Leibniz rule, e a cocycle of even degree, [standard: [Bro82, Ch. V, §3], section level, not verified against the text, rule (A2)]). In degree 0 they agree: $x \in M^\Gamma$ is the cocycle $x \circ \varepsilon$, and $x \circ \varepsilon \circ E_0$ is the image of e under $x_* \colon \mathbb{Z} \to M$, $1 \mapsto x$, which is $e \cup x$ ((E1)'s naturality in the second variable and its unit). Dimension shifting: for $0 \to M \to \operatorname{Coind} M \to M'' \to 0$ with $H^q(\Gamma; \operatorname{Coind} M) = 0$, $q \ge 1$, the connecting map $H^q(\Gamma; M'') \to H^{q+1}(\Gamma; M)$ is onto for every $q \ge 0$, so two such families that agree in degree q on every module agree in degree $q+1$. Shapiro then transports $e \cup -$ to $\operatorname{res}^\Gamma_A e \cup -$ by the identity below. The Shapiro identity is (P3) of that lemma's proof, at $G = \Gamma$, $H = A$, $u = e$ (Ind = Coind at finite index, (E4)). So $\operatorname{res}^\Gamma_A e \cup - = \mathrm{sh} \circ (e \cup -) \circ \mathrm{sh}^{-1}$, an isomorphism for $q \ge 2$. The generator statement is Lemma 14.22(1) for $C \le A \le \Gamma$. *Equivariance.* $\iota_g(A'') \subseteq A'$: for $h \in A''$, $g^{-1}hg \in A$ as $h \in gAg^{-1}$, and $g^{-1}hg \in g^{-1}Ag$ as $h \in A$. Since $A'' \subseteq \Gamma'' := \Gamma \cap g\Gamma g^{-1}$, functoriality of pull-backs and Lemma 14.22(3)(a) give $\iota_g^* \operatorname{res}^\Gamma_{A'} e = \operatorname{res}^{\Gamma''}_{A''} \iota_g^* e = \operatorname{res}^{\Gamma''}_{A''} \operatorname{res}^\Gamma_{\Gamma''} e = \operatorname{res}^\Gamma_{A''} e$. Then, by (P1) (twice: for $\operatorname{res}^A_{A'}$ and for the compatible pair (ι_g, f)) and the projection formula (P4), $[AgA](\operatorname{res}^\Gamma_A e \cup x) = \operatorname{cor}^A_{A''}\big(\iota_g^* \operatorname{res}^\Gamma_{A'} e \cup (\iota_g, f)^* \operatorname{res}^A_{A'} x\big) = \operatorname{cor}^A_{A''}\big(\operatorname{res}^A_{A''}(\operatorname{res}^\Gamma_A e) \cup y\big) = \operatorname{res}^\Gamma_A e \cup \operatorname{cor}^A_{A''} y$ with $y := (\iota_g, f)^* \operatorname{res}^A_{A'} x$; this is the proof of Lemma 14.22(3)(b) with A for Γ and $\operatorname{res}^\Gamma_A e$ for e. T_p at level N is $[AgA]$ with $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ and $f = \rho_n(g)$, resp. $\rho_{n,\chi}(g)$ (Propositions 27.2, 29.9; $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \in \Delta_0(N)$ for every p), and nothing used $p \nmid N$. The consequences for $q_{\max}$ are those of Remark 23.11.

(6) Proposition 31.6(4) and its proof, over $\mathbb{Z}$ for $C^\bullet_{\mathcal{I}}(M)$ and over O for $C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}})$, $\mathrm{T}(M_{\mathsf{O}})$: a free complex over a principal ideal domain reduces mod a prime ℓ' (not inverted) with the same ranks iff ℓ' divides no elementary divisor, i.e. iff the torsion of H^{q+1} of the complex has no ℓ'-part; for $C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}})$ and $\mathrm{T}(M_{\mathsf{O}})$ that cohomology is $H^{q+1}(\Gamma; M) \otimes \mathsf{O}$ by (1). The rest is Proposition 31.8: for M_N, every $\ell' \ge 5$ outside $\mathcal{B}(n, N)$ carries no torsion and every element of $\mathcal{B}(n, N) \cap [5, \infty)$ is $\le n + 1$ ((2) there, and $n = 0$, n odd); for $\chi \ne 1$, (3)(a) there gives $s_{\ell'} = 0$ for $\ell' \nmid 6m_\chi N$, and the example in (3) has $\ell' = 5 \mid N$, $\ell' > n + 1 = 3$.

(7)(a) $C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}}) \otimes \mathbb{Z}_{(\ell)}$ is a complex of finitely generated free $\mathbb{Z}_{(\ell)}$-modules (by (1), $\mathbb{Z}_{(\ell)}$ an O-algebra for $\ell \ge 5$) with cohomology $H^q(\Gamma; M) \otimes \mathbb{Z}_{(\ell)}$; the proof of Proposition 23.17(1)–(3) uses of $\mathbb{Z}$ only Smith forms over $\mathbb{Z}_{(\ell)}$, torsion-freeness of the terms and the universal coefficient theorem, exactly as in the proof of Proposition 31.6(3). Its reduction mod ℓ^K is $C^\bullet_{\mathrm{eq}}(M_{\mathbb{Z}/\ell^K})$ by (1). A basis over $\mathbb{Z}/\ell^K$ of the free module $(M_{\mathbb{Z}/\ell^K})^{\Gamma_\sigma} = e_\sigma M_{\mathbb{Z}_{(\ell)}} \otimes \mathbb{Z}/\ell^K$ lifts to a basis of $e_\sigma M_{\mathbb{Z}_{(\ell)}}$ (Nakayama over the local ring $\mathbb{Z}_{(\ell)}$, the lifts taken in $e_\sigma M_{\mathbb{Z}_{(\ell)}}$, i.e. after applying e_σ), so the matrix of d^0 mod ℓ^K in such bases is the reduction of the matrix of d^0 in bases over $\mathbb{Z}_{(\ell)}$, and its Smith form over $\mathbb{Z}/\ell^K$ is $\operatorname{diag}(\ell^{\min(a_t, K)}, 0)$ (the proof of Proposition 23.17(1)). $H^2(C^\bullet_{\mathrm{eq}}(M_{\mathsf{O}})) = 0$ means d^1 is onto, so $\operatorname{coker} d^1$ has no torsion and every non-zero elementary divisor of d^1 is a unit; $\operatorname{tors} H^1$ is $\operatorname{tors} \operatorname{coker} d^0$ (Proposition 23.17(2)). For a generating family of the source the image of d^0 mod ℓ^K, so its cokernel, is unchanged, and (b)'s comparison of multiplicities applies verbatim. $\operatorname{rank}_{\mathbb{Q}} d^0 = \operatorname{rank} C^0_{\mathrm{eq}} - b_0$ is rank–nullity over $\mathbb{Q}$, the kernel of $d^0 \otimes \mathbb{Q}$ being $H^0(\Gamma; M_{\mathbb{Q}})$ by (1); it is source (ii) of Proposition 31.6(4)(c). The certificate and the stopping rule are Proposition 23.17(1),(3) with Remark 23.18 at $e_0 = K$, and Proposition 31.6(6)'s last paragraph; the absence of an a-priori bound at d^0 is Proposition 23.16's closing note, which

applies to $M = V$, a case of M.

(7)(b) By (1), $\mathrm{T}(M_{\mathsf{O}}) \otimes \mathbb{Z}/\ell^K = \mathrm{T}(M_{\mathbb{Z}/\ell^K})$, so the image of $d^0 \bmod \ell^K$ is the submodule $\mathsf{R} := (M_{\mathbb{Z}/\ell^K})^{\langle S \rangle} + (M_{\mathbb{Z}/\ell^K})^{\langle U \rangle}$, and $\operatorname{coker} \bar{R} = (M_{\mathbb{Z}/\ell^K})^{\langle -I \rangle}/\mathsf{R} = \operatorname{coker}(d^0) \otimes \mathbb{Z}/\ell^K$ (right exactness). Over $\mathbb{Z}_{(\ell)}$, $\operatorname{coker} d^0 \cong \mathbb{Z}_{(\ell)}^{D-r} \oplus \bigoplus_t \mathbb{Z}/\ell^{a_t}$ ($D := \operatorname{rank} \mathrm{T}^1$, $r := \operatorname{rank}_{\mathbb{Q}} d^0$, the $a_t \geq 1$ the non-unit elementary divisors), so $\operatorname{coker} \bar{R} \cong (\mathbb{Z}/\ell^K)^{D-r+\#\{a_t \geq K\}} \oplus \bigoplus_{a_t < K} \mathbb{Z}/\ell^{a_t}$. A Smith form of $\bar{R}$ with s non-zero diagonal entries ℓ^{σ_i} gives $\operatorname{coker} \bar{R} \cong (\mathbb{Z}/\ell^K)^{D-s} \oplus \bigoplus_i \mathbb{Z}/\ell^{\sigma_i}$. Over the local principal ideal ring $\mathbb{Z}/\ell^K$ a finitely generated module is a direct sum of cyclic modules $\mathbb{Z}/\ell^c$, $1 \leq c \leq K$, with uniquely determined multiplicities; comparing the multiplicity of $c = K$ gives $s = r - \#\{a_t \geq K\} = r_K(d^0)$, and comparing $c < K$ gives $\{\sigma_i > 0\} = \{a_t < K\}$ as multisets. So $s = r$ iff every $a_t < K$, and then the torsion is read off. $r = \operatorname{rank} \mathrm{T}^0 - b_0 = \operatorname{rank} \mathrm{T}^1 - b_1$ is rank–nullity over $\mathbb{Q}$ for the two-term complex $\mathrm{T}(M_{\mathbb{Q}})$, whose cohomology is b_0, b_1 by (1). For M_N with n even, $M_N(-I) = 1$, so $\mathrm{T}^1 = M_N$, and Proposition 27.5(3) over the field $\mathbb{Q}$ gives $b_1 - b_0$. The orbit-wise generators generate R iff they generate each orbit's fixed module, which on a free orbit holds by Proposition 27.5(1) with no solve, and at a fixed point is the stated condition. *The certificate.* As $\ell \nmid |F_j|$ ($|F_j| \mid 12$), $W := (V_{\mathbb{Z}/\ell^K})^{F_j} = e_{F_j} V_{\mathbb{Z}/\ell^K}$ is free with $W/\ell W = e_{F_j} V_{\mathbb{F}_\ell} = (V_{\mathbb{F}_\ell})^{F_j}$ (the image of an idempotent commutes with base change, as in (1)). A vector fixed by a generator of F_j lies in W; if the reductions of a family in W span $W/\ell W$, the family generates W by Nakayama's lemma over the local ring $\mathbb{Z}/\ell^K$; conversely a generating family of W lies in W and its reductions span $W/\ell W$.

(7)(c) (4)(a) of Proposition 31.6 is stated for every M. The values of b_0 are proved in (4) above. A prime $\ell' \geq 5$ qualifies for d^0 iff $H^1(\Gamma; M)$ has no ℓ'-torsion, i.e. iff $s_{\ell'}(M) = 0$ (Proposition 31.8(1)). The admissibility statement is Proposition 31.6(4)(c)'s reason: ranks of differentials depend on the complex, Betti numbers do not.

(8) $\mathsf{O}_\chi = \mathbb{Z}[Y]/(\Phi_{m_\chi})$. As $\ell \nmid m_\chi$, $\Phi_{m_\chi} \bmod \ell$ divides the separable $Y^{m_\chi} - 1$, so it is a product of pairwise distinct monic irreducibles $\bar{g}_\lambda$, one per $\lambda \mid \ell$, of degree f_λ with $\ell^{f_\lambda} = q_\lambda$; by Hensel's lemma $\Phi_{m_\chi} \equiv \prod_\lambda g_\lambda \pmod{\ell^K}$ with monic, pairwise comaximal lifts g_λ, and the Chinese remainder theorem gives $\mathsf{O}_\chi/\ell^K \cong \prod_\lambda W_\lambda$, $W_\lambda := (\mathbb{Z}/\ell^K)[Y]/(g_\lambda)$, a local ring free of rank f_λ over $\mathbb{Z}/\ell^K$ with residue field $\mathbb{F}_{q_\lambda}$ (as in Lemma 29.3(1)). $W_\lambda \cong W_K(\mathbb{F}_{q_\lambda})$: $\bar{g}_\lambda$ has a simple root in $\mathbb{F}_{q_\lambda}$, which lifts by Hensel's lemma to a root of g_λ in $W_K(\mathbb{F}_{q_\lambda})$; the induced ring map $W_\lambda \to W_K(\mathbb{F}_{q_\lambda})$ is an isomorphism modulo ℓ (an injective map of fields of the same size), hence onto by Nakayama's lemma, hence bijective (free modules of the same rank). Γ acts O_χ-linearly on $M_N(\chi)$, so $M_N(\chi) \otimes \mathbb{Z}/\ell^K = M_N(\chi) \otimes_{\mathsf{O}_\chi} (\mathsf{O}_\chi/\ell^K) = \bigoplus_\lambda M_N(\chi) \otimes_{\mathsf{O}_\chi} W_\lambda$, and the coordinates (73) show $M_N(\chi) \otimes_{\mathsf{O}_\chi} W_\lambda = \mathrm{Ind} V_{W_\lambda}(\chi_\lambda)$, χ_λ the composite $G_N \to \mathsf{O}_\chi^\times \to W_\lambda^\times$. Its order divides m_χ, prime to ℓ, so $\chi_\lambda = \mathfrak{t} \circ \bar{\chi}_\lambda$ by Lemma 29.3(3). Cohomology is additive in the coefficients. □

Reading. (2) and (3) are the level-N descent with the Hecke action determined locally, by Proposition 23.8 alone; (4) records why the plan's C-D1 may not extend Proposition 23.10 unrestricted: at weight 2 and trivial character a level-N group has rational cohomology in two degrees. Nothing in (2) or (3) depends on it. (7)(a) is the route the D0 report recommends for D1 at $\ell \geq 5$ in place of the shipped C_7 K-mode (`docs/implementation/slot24/DESCENT0_report.md` §1), and (7)(b) the sparse cokernel route at level N; both certify with the closed-form b_0 and need no second prime.

Proposition 31.8 (The bad primes at level N: a closed-form list, and the one degree of torsion at $\ell \geq 5$). *[proved here] (v1.23.0-pre.14; refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D1_referee.md`, the fixes applied at v1.23.0-pre.15, not re-refereed.) For a prime $\ell \geq 5$, $n \geq 0$ and $t \in \mathbb{Z}/(\ell - 1)$ put*

$$\delta_\ell(n) := \#\{(a,b) \in \mathbb{Z}_{\geq 0}^2 : a(\ell+1) + b(\ell^2 - \ell) = n\}, \qquad \beta_\ell(n,t) := \#\{\, j \in \mathbb{Z} : 0 \leq j \leq n/\ell,\ n - 2j + t \equiv 0 \pmod{\ell}$$

and $\beta_\ell(n) := \beta_\ell(n, 0)$; $\delta_\ell(n)$ is the count of Proposition 23.16(1).

1. ***(Every finite free M.)*** *Let M be a $\mathbb{Z}[\Gamma]$-module, finitely generated and free over $\mathbb{Z}$, and ℓ a prime. Then $H^1(\Gamma;M)[\ell] \cong H^0(\Gamma;M/\ell M)/\big(H^0(\Gamma;M)/\ell\big)$, of dimension $s_\ell(M) := \dim_{\mathbb{F}_\ell} H^0(\Gamma;M/\ell M) - b_0(M)$. If $\ell \geq 5$, $H^q(\Gamma;M)$ has no ℓ-torsion for $q \neq 1$, and the ℓ-primary torsion of $H^1(\Gamma;M)$ has exactly $s_\ell(M)$ cyclic summands. Put $\mathcal{B}(M) := \{\ell \geq 5 : s_\ell(M) > 0\}$: the primes ≥ 5 at which $H^\bullet(\Gamma;M)$ has torsion.*
2. ***(Level N, trivial character: the list.)*** *Let $N \geq 1$, $n \geq 0$, and $\ell \geq 5$. If $n > 0$:*

$$s_\ell(M_N) = \delta_\ell(n) \quad (\ell \nmid N), \qquad s_\ell(M_N) = \beta_\ell(n) \quad (\ell \mid N);$$

if $n = 0$, $s_\ell(M_N) = 0$. Hence, for $n > 0$ even, every prime at which $H^\bullet(\Gamma_0(N);\mathrm{Sym}^n\mathbb{Z}^2)$ has torsion lies in

$$\mathcal{B}(n,N) := \{2,3\} \cup \{\,5 \leq \ell \leq n-1 :\ \ell \nmid N,\ \delta_\ell(n) > 0\,\} \cup \{\,5 \leq \ell \leq n+1 :\ \ell \mid N,\ \beta_\ell(n) > 0\,\};$$

*at each $\ell \geq 5$ of $\mathcal{B}(n,N)$ the ℓ-primary torsion lies in H^1 and has exactly $s_\ell(M_N)$ cyclic summands, so $\mathcal{B}(n,N) \cap [5,\infty) = \mathcal{B}(M_N)$; **every prime $\ell \geq 5$ outside $\mathcal{B}(n,N)$ — in particular every $\ell > n+1$ — carries no torsion in any degree**, by this proof and not by omission (for $\ell > n$ by Claims A and B of the proof alone; at $5 \leq \ell \leq n$, $\ell \nmid N$, by Claim 16.5(2), proved in place since v1.23.0-pre.15). A prime $\ell = n+1 \geq 5$ lies in $\mathcal{B}(n,N)$ iff $\ell \mid N$, with $s_\ell = 1$; e.g. $n = 4$, $\ell = N = 5$, the fixed vector x^4 (the audit's finding m3). For n odd, $2 \cdot H^q(\Gamma_0(N);\mathrm{Sym}^n\mathbb{Z}^2) = 0$ for every q; for $n = 0$ the torsion primes lie in $\{2,3\}$. At $N = 1$ this is Proposition 23.16.*
3. ***(Nebentypus.)*** *Let χ, m_χ, O_χ and $M_N(\chi)$ be as in Proposition 31.7, $(n,\chi) \neq (0,1)$, and $\ell \geq 5$ a prime with $\ell \nmid m_\chi$. Write $N = \ell^k N'$ with $\ell \nmid N'$, $\chi = \chi_\ell \chi'$ along $G_N = (\mathbb{Z}/\ell^k)^\times \times (\mathbb{Z}/N')^\times$, and for a prime $\lambda \mid \ell$ of O_χ let $\mathbb{F}_\lambda := \mathsf{O}_\chi/\lambda \supseteq \mathbb{F}_\ell$ and $\bar\chi_\lambda := \chi \bmod \lambda$. Then $s_\ell(M_N(\chi)) = \sum_{\lambda\mid\ell}[\mathbb{F}_\lambda : \mathbb{F}_\ell]\, s_\lambda$, $s_\lambda := \dim_{\mathbb{F}_\lambda} H^0(\Gamma_0(N);V_{\mathbb{F}_\lambda}(\bar\chi_\lambda))$, and*
   (a) if $\ell \nmid N$: $s_\lambda = \delta_\ell(n)$ if $\chi = 1$, and $s_\lambda = 0$ if $\chi \neq 1$;
   (b) if $\ell \mid N$: $s_\lambda = 0$ if $\chi' \neq 1$; if $\chi' = 1$, then χ factors through $(\mathbb{Z}/\ell)^\times$, $\bar\chi_\lambda(a) = (a \bmod \ell)^{t_\lambda}$ for a unique $t_\lambda \in \mathbb{Z}/(\ell-1)$, and $s_\lambda = \beta_\ell(n,t_\lambda)$.

   So for $\chi \neq 1$ the torsion at primes $\ell \geq 5$, $\ell \nmid m_\chi$, lies in H^1 and occurs only at $\ell \mid N$ with $\chi' = 1$ (the conductor of χ a power of ℓ); such an ℓ need not be $\leq n+1$: for $N = 5$, $\chi = \left(\frac{\cdot}{5}\right)$ (even, $t = 2$) and $n = 2$, $s_5 = \beta_5(2,2) = 1$. The primes $\ell \mid 6m_\chi$ are not treated by this clause (at $\ell \mid m_\chi$, O_χ is ramified over ℓ); they are bad primes until a later clause says otherwise.

Proof. (1) M is free over $\mathbb{Z}$, so $0 \to M \xrightarrow{\ell} M \to M/\ell M \to 0$ is exact, and its long exact sequence $0 \to H^0(M) \xrightarrow{\ell} H^0(M) \to H^0(M/\ell M) \to H^1(M) \xrightarrow{\ell} H^1(M)$ gives the displayed isomorphism. $H^0(\Gamma;M) = M^\Gamma$ is a submodule of the finitely generated free $\mathbb{Z}$-module M, hence free, of rank $b_0(M)$ (flatness of $\mathbb{Q}$), and $H^0(M)/\ell \to H^0(M/\ell M)$ is injective by the sequence; so $\dim H^1[\ell] = s_\ell(M)$. The number of cyclic ℓ-primary summands of a finitely generated abelian group G is $\dim_{\mathbb{F}_\ell} G[\ell]$. For $\ell \geq 5$: H^0 is torsion-free, and $24 \cdot H^q(\Gamma;M) = 0$ for $q \geq 2$ (Proposition 31.6(4)(a), proved there for every $\mathbb{Z}[\Gamma]$-module).

(2) $M_N/\ell M_N = \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} M_n$ (the coordinates (22) commute with reduction), so $H^0(\Gamma;M_N/\ell) = H^0(\Gamma_0(N);M_n)$ by Frobenius reciprocity in degree 0; $b_0(M_N) = 0$ for $n > 0$ and 1 for $n = 0$ (Proposition 31.6(4)(b)). For $n = 0$, $H^0(\Gamma_0(N);\mathbb{F}_\ell) = \mathbb{F}_\ell$ and $s_\ell = 0$. Let $n > 0$. ρ_n has entries that are integer polynomials in those of γ ((9)), so $\Gamma_0(N)$ acts on M_n through its image $\bar\Gamma$ in $\mathrm{SL}_2(\mathbb{F}_\ell)$, and $H^0(\Gamma_0(N);M_n) = M_n^{\bar\Gamma}$. *The image* (with (3)'s refinement). $\Gamma_0(N)$ is the preimage of the upper-triangular group $B(\mathbb{Z}/N)$ under $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)$, and $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/M')$ is onto for every $M' \geq 1$ (as in the proof of Lemma 29.2). If $\ell \nmid N$, $\mathrm{SL}_2(\mathbb{Z}/N\ell) = \mathrm{SL}_2(\mathbb{Z}/N) \times \mathrm{SL}_2(\mathbb{F}_\ell)$ (Chinese remainder theorem), so $\Gamma_0(N)$, the preimage of $B(\mathbb{Z}/N) \times \mathrm{SL}_2(\mathbb{F}_\ell)$, maps onto $B(\mathbb{Z}/N) \times \mathrm{SL}_2(\mathbb{F}_\ell)$ and $\bar\Gamma = \mathrm{SL}_2(\mathbb{F}_\ell)$. If $\ell \mid N$, $\Gamma_0(N) \to B(\mathbb{Z}/N) = B(\mathbb{Z}/\ell^k) \times B(\mathbb{Z}/N')$ is onto, and so is $B(\mathbb{Z}/\ell^k) \to B(\mathbb{F}_\ell)$ (lift the diagonal

unit a to a unit $\tilde{a}$ and take $\begin{pmatrix}\tilde{a} & \tilde{b}\\ 0 & \tilde{a}^{-1}\end{pmatrix}$); so $\bar{\Gamma} = B(\mathbb{F}_\ell)$. *Case* $\ell \nmid N$. $M_n^{\mathrm{SL}_2(\mathbb{F}_\ell)} = H^0(\Gamma; M_n)$, as $\Gamma \to \mathrm{SL}_2(\mathbb{F}_\ell)$ is onto as well, and its dimension is $\delta_\ell(n)$ by the proof of Proposition 23.16(1) (Claim 16.5(2), proved in place since v1.23.0-pre.15 — the same input as at level 1). For $\ell > n$ the count needs no Claim 16.5(2): see *the list* below. *Case* $\ell \mid N$. By Definition 8.1, M_n is the degree-n part of $\mathbb{F}_\ell[x,y]$, γ acting by the algebra automorphism with $\gamma \cdot x = ax + cy$, $\gamma \cdot y = bx + dy$. $B(\mathbb{F}_\ell)$ is generated by $u_b := \begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}$ and $h_a := \mathrm{diag}(a, a^{-1})$; u_b fixes x and sends $y \mapsto y + bx$, and h_a sends $x \mapsto ax$, $y \mapsto a^{-1}y$. Put $P := \prod_{c\in\mathbb{F}_\ell}(y + cx) = y^\ell - x^{\ell-1}y$. *Claim A: the u_b-invariants ($b \in \mathbb{F}_\ell$) of degree n are spanned by the x^iP^j, $i + \ell j = n$, and these are linearly independent.* u_b permutes the factors of P ($c \mapsto c + b$), so x and P are invariant. Independence: x^iP^j has y-degree ℓj. Spanning, by induction on n: let f be invariant of degree n; divide by P, monic of degree ℓ in y over $\mathbb{F}_\ell[x]$: $f = gP + r$ with $\deg_y r < \ell$, uniquely, g and r homogeneous. u_b preserves y-degrees and fixes f and P, so by uniqueness $u_bg = g$ and $u_br = r$; $g \in \mathbb{F}_\ell[x, P]$ by induction. Let $e := \deg_y r$ and suppose $e \geq 1$, $r = \sum_{k\leq e} r_k(x)y^k$ with $r_e \neq 0$. The polynomial $R(t) := r(x, y + tx) - r(x, y) \in \mathbb{F}_\ell[x, y][t]$ has degree $\leq e < \ell$ in t, its t^e-coefficient $r_e(x)x^e \neq 0$, and the ℓ distinct roots $t \in \mathbb{F}_\ell$ — impossible over the domain $\mathbb{F}_\ell[x, y]$. So $e = 0$ and $r \in \mathbb{F}_\ell x^n$. *Claim B: $h_a(x^iP^j) = a^{i-j}x^iP^j$:* $h_aP = \prod_c(a^{-1}y + cax) = a^{-\ell}\prod_c(y + ca^2x) = a^{-\ell}P$ ($c \mapsto ca^2$ is a bijection of $\mathbb{F}_\ell$) and $a^\ell = a$. So the torus $\{h_a\}$ acts diagonally on the basis of Claim A, the $B(\mathbb{F}_\ell)$-invariants are spanned by the x^iP^j with $a^{i-j} = 1$ for all $a \in \mathbb{F}_\ell^\times$, i.e. $(\ell - 1) \mid i - j$, and with $i = n - \ell j$, $i - j = n - (\ell + 1)j \equiv n - 2j \pmod{\ell - 1}$: the dimension is $\beta_\ell(n)$. *The list.* $\delta_\ell(n) > 0$, $n > 0$, needs $n \geq \ell + 1$ (the least positive value of $a(\ell+1)+b(\ell^2-\ell)$, as $\ell^2 - \ell \geq \ell + 1$); $\beta_\ell(n) > 0$ needs a j with $(\ell - 1) \mid n - 2j$: $j = 0$ forces $\ell - 1 \leq n$, $j \geq 1$ forces $\ell \leq n$. At $\ell = n + 1$, $j \leq n/\ell < 1$ leaves $j = 0$, and $(\ell - 1) = n$ divides n: $\beta_\ell(n) = 1$; $\delta_{n+1}(n) = 0$. *The good primes without Claim 16.5(2)* (v1.23.0-pre.15, referee F1(a)): for $\ell > n > 0$ and $\ell \nmid N$, Claim A leaves only x^n among the u_b-invariants of degree n, since $\deg_y P = \ell > n$; the lower unipotent $\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix} \in \mathrm{SL}_2(\mathbb{F}_\ell) = \bar{\Gamma}$ sends $x \mapsto x + y$, and $(x + y)^n \neq x^n$ (its coefficient of $x^{n-1}y$ is $n \not\equiv 0$), so x^n is not invariant and $s_\ell(M_N) = 0$. So the rule "every $\ell > n + 1$ is good" rests on Claims A and B alone. The rest of the list statement is (1). *n odd:* $M_N(-I) = (-1)^n = -1$ (the proof of Proposition 27.5(2)), $-I$ is central and acts trivially on cohomology, so 2 kills $H^\bullet$ (Proposition 23.16(2)'s argument). *$n = 0$:* $s_\ell = 0$ for $\ell \geq 5$, so by (1) there is no ℓ-torsion for $\ell \geq 5$. At $N = 1$, $\ell \nmid N$ always and the list is Proposition 23.16(1)'s.

(3) $b_0(M_N(\chi)) = 0$ by Proposition 31.7(4). $M_N(\chi)/\ell = \mathrm{Ind}V_{\mathsf{O}_\chi/\ell}(\chi \bmod \ell)$ and $\mathsf{O}_\chi/\ell = \prod_\lambda \mathbb{F}_\lambda$ (the proof of Proposition 31.7(8) at $K = 1$), so $H^0(\Gamma; M_N(\chi)/\ell) = \bigoplus_\lambda H^0(\Gamma_0(N); V_{\mathbb{F}_\lambda}(\bar{\chi}_\lambda))$, of $\mathbb{F}_\ell$-dimension $\sum_\lambda[\mathbb{F}_\lambda : \mathbb{F}_\ell]s_\lambda$; then apply (1). $\gamma \in \Gamma_0(N)$ acts on $V_{\mathbb{F}_\lambda}(\bar{\chi}_\lambda)$ by $\bar{\chi}_\lambda(a_\gamma)\,\rho_n(\gamma \bmod \ell)$. $\bar{\chi}_\lambda$ has order prime to ℓ (its values lie in $\mathbb{F}_\lambda^\times$), so its $(\mathbb{Z}/\ell^k)^\times$-component is trivial on the ℓ-group $\ker\big((\mathbb{Z}/\ell^k)^\times \to \mathbb{F}_\ell^\times\big)$, $k \geq 1$. Moreover $\mu_{m_\chi} \to \mathbb{F}_\lambda^\times$ is injective ($Y^{m_\chi} - 1$ is separable mod ℓ), so $\bar{\chi}_\lambda = 1$ iff $\chi = 1$, and $\bar{\chi}'_\lambda = 1$ iff $\chi' = 1$. (a) $\ell \nmid N$: the action factors through $\phi\colon \Gamma_0(N) \to \mathrm{SL}_2(\mathbb{F}_\ell) \times G_N$, $\gamma \mapsto (\gamma \bmod \ell, a_\gamma \bmod N)$, which is onto by the computation in (2) ($\Gamma_0(N) \to B(\mathbb{Z}/N) \times \mathrm{SL}_2(\mathbb{F}_\ell)$ is onto, and so is $B(\mathbb{Z}/N) \to G_N$, $\beta \mapsto a_\beta$). The fixed space of $\mathrm{SL}_2(\mathbb{F}_\ell) \times G_N$ on $\mathrm{Sym}^n\mathbb{F}_\lambda^2 \otimes \mathbb{F}_\lambda(\bar{\chi}_\lambda)$ is $(\mathrm{Sym}^n\mathbb{F}_\lambda^2)^{\mathrm{SL}_2(\mathbb{F}_\ell)}$ if $\bar{\chi}_\lambda = 1$ and 0 otherwise; the former has dimension $\delta_\ell(n)$ (a fixed space of matrices over $\mathbb{F}_\ell$ commutes with the extension $\mathbb{F}_\ell \subseteq \mathbb{F}_\lambda$). (b) $\ell \mid N$: the action factors through $\Gamma_0(N) \to B(\mathbb{F}_\ell) \times (\mathbb{Z}/N')^\times$, $\gamma \mapsto (\gamma \bmod \ell, a_\gamma \bmod N')$, onto by (2) ($B(\mathbb{Z}/N') \to (\mathbb{Z}/N')^\times$ onto). The second factor acts by $\bar{\chi}'_\lambda$, so the fixed space is 0 unless $\chi' = 1$. If $\chi' = 1$, $\chi = \chi_\ell$ has order prime to ℓ, so it factors through $(\mathbb{Z}/\ell)^\times = \mathbb{F}_\ell^\times$; $\bar{\chi}_\lambda$ maps the cyclic group $\mathbb{F}_\ell^\times$ into $\mu_{\ell-1}(\mathbb{F}_\lambda) = \mathbb{F}_\ell^\times$, hence is $a \mapsto a^{t_\lambda}$ for a unique $t_\lambda \bmod \ell - 1$. Claim A holds over $\mathbb{F}_\lambda$ verbatim (P and the u_b, $b \in \mathbb{F}_\ell$, are unchanged; $\mathbb{F}_\lambda[x, y]$ is a domain), and u_b has $a_{u_b} = 1$, so the twist does not change the u_b-invariants; h_a has $a_{h_a} = a$ and acts on x^iP^j by $a^{t_\lambda}a^{i-j}$. So the fixed space has dimension $\#\{j : (\ell - 1) \mid n - 2j + t_\lambda\} = \beta_\ell(n, t_\lambda)$. *The example.* $\chi = \left(\frac{\cdot}{5}\right)$ has $\chi(-1) = 1$, $m_\chi = 2$,

$\mathsf{O}_\chi = \mathbb{Z}$, $\bar\chi(a) = a^2$ in $\mathbb{F}_5$ (Euler's criterion), so $t = 2$; $\beta_5(2,2) = \#\{j = 0 : 4 \mid 4\} = 1$. The parity is built in: if $\chi(-1) \neq (-1)^n$, $n + t_\lambda$ is odd ($\bar\chi_\lambda(-1) = (-1)^{t_\lambda}$) and $\beta_\ell(n, t_\lambda) = 0$, as Proposition 29.11(2) requires. □

Reading. At $\ell \geq 5$ the torsion of any finite free coefficient module sits in H^1 and is counted by H^0 modulo ℓ, so the bad-prime list is the list of primes at which the reduction has more invariants than the lattice; at level N the reduction of $\Gamma_0(N)$ is $\mathrm{SL}_2(\mathbb{F}_\ell)$ for $\ell \nmid N$ (the level-1 Dickson count) and the Borel subgroup for $\ell \mid N$ (the count β_ℓ, which reaches $\ell = n+1$). The descent's $\mathbb{Z}[1/6]$-piece at level N therefore needs certified $\mathbb{Z}/\ell^K$ runs at the primes of $\mathcal{B}(n, N) \cap [5, \infty)$ only, in H^1 only (Proposition 31.7(7)); every other prime ≥ 5 is torsion-free by Proposition 31.8, not by omission.

Proposition 31.9 (The 3-adic shortcut at level N: no element of order 3)**.** *[proved here] (v1.23.0-pre.17; **not refereed**; the D1 code audit's finding M1,* `docs/implementation/slot24/D1_code_audit` *Assume (H4) and (B1)–(B3). Let $N \geq 1$ with $\nu_3(N) = 0$, $n \geq 0$, $M := M_N = \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V$, $U := ST$, and $\mathrm{T}(M_S)$ the tree complex (65) with fibre M_S. By Proposition 29.32 and the local counts in the proof of Proposition 31.7(4), $\nu_3(N) = 0$ **iff** $\Gamma_0(N)$ contains no element of order 3 **iff** $9 \mid N$ or N has a prime factor $p \equiv 2 \pmod 3$ ($p = 2$ included).*

1. ***(The stabilisers.)** Every finite subgroup of $\Gamma_0(N)$ has order 2 or 4 (or 1); for every $\gamma \in \Gamma$, $\Gamma_0(N) \cap \gamma\langle U\rangle\gamma^{-1} = \{\pm I\}$, and $\Gamma_0(N) \cap \gamma\langle S\rangle\gamma^{-1}$ is $\{\pm I\}$ or $\gamma\langle S\rangle\gamma^{-1}$. So 3 **is prime to the order of every stabiliser of** $\Gamma_0(N)$ on $\mathcal{T}$ and on X. Equivalently, U has no fixed point on $\mathbb{P}^1(\mathbb{Z}/N)$ (the $\langle U\rangle$-stabiliser of a point contains $U^3 = -I$, so it is $\{\pm I\}$ or $\langle U\rangle$; v1.23.0-pre.18, referee m1), every $\langle U\rangle$-orbit is free modulo $\{\pm I\}$ (of length 3), and M restricted to $\langle U^2\rangle \cong \mathbb{Z}/3$ is induced from the trivial subgroup, so $H^t(\langle U^2\rangle; M_S) = 0$ for every $t \geq 1$ and every ring S.*
2. ***(What the tree complex computes 3-locally.)** For every $\mathbb{Z}_{(3)}$-algebra S: $\mathrm{T}(M_{\mathbb{Z}_{(3)}})$ is a complex of finitely generated free $\mathbb{Z}_{(3)}$-modules in degrees $0, 1$, $\mathrm{T}(M_{\mathbb{Z}_{(3)}}) \otimes_{\mathbb{Z}_{(3)}} S = \mathrm{T}(M_S)$, and*

$$H^q(\mathrm{T}(M_S)) \;\cong\; H^q(C^\bullet_{\mathrm{eq}}(M_S)) \;\cong\; H^q(\Gamma; M_S) \;=\; H^q(\Gamma_0(N); \mathrm{Sym}^n S^2) \qquad \text{for every } q;$$

   *moreover $\mathrm{cmp}^* \otimes \mathbb{Z}_{(3)} \colon C^\bullet_{\mathrm{eq}}(M_{\mathbb{Z}_{(3)}}) \to C^\bullet_{\mathcal{I}}(M) \otimes \mathbb{Z}_{(3)}$ is a quasi-isomorphism. Hence $H^q(\Gamma_0(N); V)$ **has no 3-torsion for** $q \geq 2$, and $\mathrm{tors}\, H^1(\Gamma_0(N); V) \otimes \mathbb{Z}_{(3)} = \mathrm{tors}\,\mathrm{coker}(d^0)$ for d^0 of $\mathrm{T}(M_{\mathbb{Z}_{(3)}})$. This is Proposition 31.7(1)'s statement for T (for $C^\bullet_{\mathrm{eq}}$: its cohomology) with O replaced by $\mathbb{Z}_{(3)}$, for this M; it is false for M_N when $\nu_3(N) > 0$ (item (5)).*
3. ***(The certified exponent at $\ell = 3$.)** Let n be even and $K \geq 1$. Proposition 31.7(7)(b) holds at $\ell = 3$ verbatim: let $\bar R$ be any matrix over $\mathbb{Z}/3^K$, written in a basis of $M_{\mathbb{Z}/3^K} = (M_{\mathbb{Z}/3^K})^{\langle -I\rangle}$, whose rows generate $\mathsf{R} := (M_{\mathbb{Z}/3^K})^{\langle S\rangle} + (M_{\mathbb{Z}/3^K})^{\langle U\rangle}$, and s the number of non-zero diagonal entries 3^{σ_i} of its Smith form. Then $s = r_K(d^0) \leq r := \mathrm{rank}\,\mathrm{T}^0 - b_0 = m_N(n+1) - b_1$, with the closed forms $\mathrm{rank}\,\mathrm{T}^0 = \frac{m_N - \nu_2}{2}(n+1) + \nu_2\, d_S + \frac{m_N}{3}(n+1)$ (Proposition 27.5(1) at $\nu_3 = 0$), $b_0 = [n = 0]$ and $b_1 = b_0 +$ the right-hand side of Proposition 27.5(3) over $\mathbb{Q}$ — never the number of rows of $\bar R$. If $s = r$, then $\mathrm{tors}\, H^1(\Gamma_0(N); V) \otimes \mathbb{Z}_{(3)} \cong \bigoplus_{\sigma_i > 0} \mathbb{Z}/3^{\sigma_i}$ exactly; if $s < r$, K is too small, and the stopping rule of Proposition 31.6(6) applies (raise K; at the cap of the ring, refuse). No a-priori bound on the exponents of d^0 is asserted.* The generators and their certificate. *(a) On a free orbit — every $\langle U\rangle$-orbit, and the free $\langle S\rangle$-orbits — the orbit-wise generators $\sum_{i<o} M(g)^i e_{(j_0,a)}$, $a = 0, \ldots, n$ ($g = U$, $o = 3$; $g = S$, $o = 2$), are a basis of the orbit's fixed module over every ring (Proposition 27.5(1): no solve, no division). (b) At each of the $\nu_2(N)$ fixed points j of S, $F_j = \langle S\rangle$ has order 4, prime to 3: the columns of the averaging idempotent $e_{F_j} = \frac{1}{2}\big(1 + \rho_n(h_j(S))\big)$ (n even) need no certificate, and any other family generates $(V_{\mathbb{Z}/3^K})^{F_j}$ iff (i) each vector is fixed mod 3^K by a generator of F_j and (ii) the reductions mod 3 span $(V_{\mathbb{F}_3})^{F_j}$, of dimension d_S — (7)(b)'s certificate (referee F4), whose only use*

of $\ell \geq 5$ is $\ell \nmid |F_j|$. (c) $\langle U\rangle$ has ***no*** *fixed points, so the case $|F_j| = 6$, where $3 \mid |F_j|$ and (7)(b)'s certificate would fail, does not occur. So the rows built by (a) and (b) are admissible with no certificate.*

4. *(n* ***odd.)*** *2 kills $H^\bullet(\Gamma_0(N); V)$ (Proposition 31.8(2)), so its 3-primary part is 0 in every degree, at every N; nothing is computed.*
5. ***(Scope: the hypothesis is needed, and nothing else is.)*** *$\nu_2(N)$ plays no role (the stabilisers $\{\pm I\}$, $\langle S\rangle$ have orders 2, 4), and $9 \mid N$ is a case of the proposition ($\nu_3(9m) = 0$), not an exception. If $\nu_3(N) > 0$, $\Gamma_0(N)$ contains a conjugate of $\langle U\rangle$ (Proposition 29.32), $H^q(\Gamma_0(N); V)$ may carry 3-torsion in degrees $q \geq 2$, which T and $C^\bullet_{\mathrm{eq}}$ cannot see ($H^q = 0$ for $q \geq 2$ on both; Proposition 31.3(3) at $N = 1$), and the 3-primary part needs the ℓ-local model $\mathcal{I}^{(3)}$ (Proposition 31.6 at $\ell = 3$). Instances (the independent oracle* `docs/implementation/slot24/descentN_oracle.tsv`*): $H^2(\mathrm{SL}_2(\mathbb{Z}); \mathbb{Z}) \supseteq \mathbb{Z}/3$ ($N = 1$, $n = 0$), $H^2(\Gamma_0(7); \mathbb{Z}) \supseteq (\mathbb{Z}/3)^2$, $H^3(\Gamma_0(3); \mathrm{Sym}^4\mathbb{Z}^2) \supseteq \mathbb{Z}/3$.*

In the descent (Proposition 31.7(2)) the 3-primary part of $H^q(\Gamma; M) \otimes \widehat{\mathbb{Z}}_6$ may therefore be read on $\mathrm{T}(M_{\mathbb{Z}_{(3)}})$ in place of $\mathcal{I}^{(3)}$ when $\nu_3(N) = 0$: by (2) the identification is induced by cmp^*, the descent's only glueing datum, and the fibre maps u of Proposition 31.7(3) (restriction, trace, R_σ) commute strictly with it, so their 3-local components may be read on $C^\bullet_{\mathrm{eq}}(M_{\mathbb{Z}_{(3)}})$ as well. No cochain-level Hecke operator on T at level N is asserted here.

Proof. (1) The finite subgroups of Γ are cyclic of order $1, 2, 3, 4, 6$, each contained in a conjugate of $\langle S\rangle \cong \mathbb{Z}/4$ or of $\langle U\rangle \cong \mathbb{Z}/6$, and every one of even order contains $-I$ (Proposition 29.32 and its proof). A subgroup of order 3 or 6 contains an element of order 3, which $\Gamma_0(N)$ lacks: by Proposition 29.32, $\Gamma_0(N)$ has one iff $a^2 + a + 1 \equiv 0 \pmod N$ is solvable, and $a \mapsto -a$ maps the roots of $X^2 + X + 1$ bijectively onto those of $\chi_{ST} = X^2 - X + 1$, whose number modulo N is $\nu_3(N)$ (the proof of Proposition 31.7(4)); that proof's local counts give $\nu_3(p^e) = 0$ exactly for $p \equiv 2 \pmod 3$ ($p = 2$ included) and for $p^e = 3^e$, $e \geq 2$, and ν_3 is multiplicative, whence the stated equivalence. As $-I \in \Gamma_0(N)$ is central, $\Gamma_0(N) \cap \gamma\langle U\rangle\gamma^{-1}$ is a subgroup of the cyclic group $\gamma\langle U\rangle\gamma^{-1}$ of order 6 containing $-I$ and no element of order 3, i.e. $\{\pm I\}$; likewise $\Gamma_0(N) \cap \gamma\langle S\rangle\gamma^{-1} \supseteq \{\pm I\}$ has order 2 or 4. The stabilisers of $\Gamma_0(N)$ on cells of $\mathcal{T}$ (conjugates of $\langle -I\rangle$, $\langle S\rangle$, $\langle U\rangle$ intersected with $\Gamma_0(N)$) and of X (finite subgroups of Γ, Proposition 23.3(1), intersected with $\Gamma_0(N)$) are finite subgroups of $\Gamma_0(N)$. The stabiliser in $\langle U\rangle$ of the coset $\Gamma_0(N)w$ is $\langle U\rangle \cap w^{-1}\Gamma_0(N)w = w^{-1}\big(\Gamma_0(N) \cap w\langle U\rangle w^{-1}\big)w = \{\pm I\}$, so $\langle U\rangle/\{\pm I\} \cong \langle U^2\rangle$ acts freely on $\Gamma_0(N)\backslash\Gamma = \mathbb{P}^1(\mathbb{Z}/N)$ (the block indices $j \mapsto j \cdot \gamma$ of (22), Definition 28.3), with orbits of length 3. $M = \bigoplus_j V_j$ is the sum of its blocks (one per point of $\mathbb{P}^1(\mathbb{Z}/N)$), permuted by Γ as the points are; on an orbit $O = \{j, U^2 j, U^4 j\}$ the block sum M_O is $\bigoplus_{i<3} U^{2i}V_j$, so $M_O \otimes S \cong S[\langle U^2\rangle] \otimes_S (V_j \otimes S)$ is induced from the trivial subgroup. For a cyclic group $C = \langle c\rangle$ of order 3 and such an $X = \bigoplus_{i<3} c^i X_0$, the periodic resolution gives $H^{2k}(C; X) = X^C/\mathcal{N}X$ and $H^{2k+1}(C; X) = \ker\mathcal{N}/(c-1)X$ ($k \geq 1$, resp. $k \geq 0$; $\mathcal{N} = 1 + c + c^2$); $x \in X^C$ is $x = \sum_i c^i x_0 = \mathcal{N}x_0$ with x_0 its X_0-component, and $y = \sum_i c^i y_i \in \ker\mathcal{N}$ (each $y_i \in X_0$) has $y_0 + y_1 + y_2 = 0$ (the X_0-component of $\mathcal{N}y$), so $y = (c-1)\big(cy_2 - y_0\big)$ — indeed $(c-1)(cy_2 - y_0) = c^2y_2 - cy_2 - cy_0 + y_0 = c^2y_2 + cy_1 + y_0$. So $H^t(C; X) = 0$ for $t \geq 1$, over every ring.

(2) Let $A_0 := \Gamma_0(N)$ and S a $\mathbb{Z}_{(3)}$-algebra; 2 and 4 are units in S. *The tree.* The proof of Lemma 28.4 uses of $6 \in R^\times$ only that the orders of the groups $A_0 \cap \gamma F\gamma^{-1}$ are units (as noted in the proof of Proposition 29.32); by (1) they are 2 or 4, so $C_\bullet(\mathcal{T}) \otimes S \to S$ is a projective resolution over $S[A_0]$ and $H^q(A_0; V_S) = H^q(\mathrm{Hom}_{A_0}(C_\bullet(\mathcal{T}), V_S))$, 0 for $q \geq 2$. Frobenius reciprocity (as in the proof of Proposition 29.32) identifies $\mathrm{Hom}_{A_0}(C_\bullet(\mathcal{T}), V_S)$ with $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(\mathcal{T}), \mathrm{Ind}_{A_0}^\Gamma V_S) = \mathrm{T}(M_S)$ (the derivation of (65); $\mathrm{Ind}_{A_0}^\Gamma V_S = M_S$, induction commuting with $\otimes S$), and Shapiro's lemma [standard: [Bro82, Ch. III.6.2]] gives $H^q(A_0; V_S) = H^q(\Gamma; M_S)$. *The Bredon complex.* Proposition 29.32 with $B = A_0$, $X = V_S$, $R = S$ (its hypothesis is (1)) gives $H^q(C^\bullet_{\mathrm{eq}}(M_S)) \cong H^q(A_0; V_S)$. *The comparison.* $C_s(X) \otimes S = \bigoplus_\sigma S[\Gamma/\Gamma_\sigma]$ restricts over $S[A_0]$ to a sum of $S[A_0/H']$, $H' = A_0 \cap \gamma\Gamma_\sigma\gamma^{-1}$ over the double cosets, of order

a unit by (1), hence projective (the idempotent of Lemma 28.4's proof); $C_\bullet(X)\otimes S\to S$ is exact ((H4)(ii), split over $\mathbb{Z}$). So $C_\bullet(X)\otimes S$ and $\mathcal{I}_\bullet\otimes S$ (free over $S[\Gamma]$, hence over $S[A_0]$) are projective resolutions of S over $S[A_0]$, and $\mathrm{cmp}\otimes S$ is a chain map between them over id_S, hence a homotopy equivalence over $S[A_0]$ (the comparison theorem, [standard: [Wei94, Thm. 2.2.6]]). $\mathrm{Hom}_{A_0}(-,V_S)$ preserves homotopy equivalences, and Frobenius reciprocity is natural in the first variable, so $\mathrm{cmp}^*\colon \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X),M_S)\to \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_\bullet,M_S)$ is a quasi-isomorphism; at $S=\mathbb{Z}_{(3)}$ the target is $C^\bullet_{\mathcal{I}}(M)\otimes\mathbb{Z}_{(3)}$ (Proposition 31.7(1)), the source $C^\bullet_{\mathrm{eq}}(M_{\mathbb{Z}_{(3)}})$, and the cohomology is $H^q(\Gamma;M)\otimes\mathbb{Z}_{(3)}$ ($\mathbb{Z}_{(3)}$ flat). *Freeness and base change of* T. $\mathrm{T}^1=M_S^{\langle -I\rangle}=e_{-I}M_S$ and $M_S^{\langle S\rangle}=e_{\langle S\rangle}M_S$ for the averaging idempotents (2, 4 units): direct summands of finitely generated free modules, free over the principal ideal domain $\mathbb{Z}_{(3)}$, and $e(M_{\mathbb{Z}_{(3)}})\otimes S=eM_S$ (the image of an idempotent commutes with base change). $\langle U\rangle$ has order 6, not a unit, but by (1) every $\langle U\rangle$-orbit is free modulo $\{\pm I\}=\langle U\rangle\cap\{\pm I\}$, so by Proposition 27.5(1), over every ring, restriction to the blocks j_O is an isomorphism $M_S^{\langle U\rangle}\to\bigoplus_O V_S^{\{\pm I\}}=\bigoplus_O e_{-I}V_S$, with the inverse given by the transported formula with coefficients in $\mathbb{Z}$; so $M^{\langle U\rangle}_{\mathbb{Z}_{(3)}}$ is free and $M^{\langle U\rangle}_{\mathbb{Z}_{(3)}}\otimes S\to M_S^{\langle U\rangle}$ is an isomorphism. The differential $(a,b)\mapsto b-a$ is base-changed with them. *The consequences.* $H^q(\Gamma_0(N);V)\otimes\mathbb{Z}_{(3)}=0$ for $q\geq 2$, a finitely generated group, so it has no 3-torsion; $H^1(\mathrm{T}(M_{\mathbb{Z}_{(3)}}))=\mathrm{coker}\,d^0$. *Against Proposition 31.3.* With $R=S$ there, (2) of that proposition gives $H^q(\Gamma;M_S)\cong H^q(\langle U^2\rangle;M_S^{-I})$ for $q\geq 2$, and (3) gives that the Bredon route is exact in degree 0 and injective in degree 1 with cokernel $H^1(\langle U^2\rangle;M_S^{-I})$. $M_S^{-I}=e_{-I}M_S$ is a $\langle U^2\rangle$-summand of M_S ($-I$ is central), so its H^t, $t\geq 1$, vanish by (1): the two statements agree. At $N=1$ ($\nu_3(1)=1$) neither term vanishes in general: $H^2(\mathrm{SL}_2(\mathbb{Z});\mathbb{Z})\supseteq\mathbb{Z}/3$.

(3) The proof of Proposition 31.7(7)(b), with (2) in place of (1) there: $\mathrm{T}(M_{\mathbb{Z}_{(3)}})\otimes\mathbb{Z}/3^K=\mathrm{T}(M_{\mathbb{Z}/3^K})$, so the image of d^0 mod 3^K is R and $\mathrm{coker}\,\bar R=\mathrm{coker}(d^0)\otimes\mathbb{Z}/3^K$; over $\mathbb{Z}_{(3)}$, $\mathrm{coker}\,d^0\cong\mathbb{Z}_{(3)}^{D-r}\oplus\bigoplus_t\mathbb{Z}/3^{a_t}$ ($D=m_N(n+1)$, the $a_t\geq 1$), and comparing multiplicities of the cyclic summands $\mathbb{Z}/3^c$ over the local principal ideal ring $\mathbb{Z}/3^K$ gives $s=r-\#\{a_t\geq K\}=r_K(d^0)$ and $\{\sigma_i>0\}=\{a_t<K\}$. By (2), $\mathrm{tors}\,\mathrm{coker}\,d^0=\mathrm{tors}\,H^1(\Gamma;M)\otimes\mathbb{Z}_{(3)}$, and $H^1(\Gamma;M)=H^1(\Gamma_0(N);V)$ (Proposition 31.7(2)). $r=\mathrm{rank}\,\mathrm{T}^0-b_0$ is rank–nullity for $\mathrm{T}(M_{\mathbb{Q}})$, whose H^0 is $M_{\mathbb{Q}}^{\langle S\rangle}\cap M_{\mathbb{Q}}^{\langle U\rangle}=M_{\mathbb{Q}}^{\Gamma}$, of dimension b_0 (Proposition 31.7(4)); $\mathrm{rank}\,\mathrm{T}^0=\dim M_{\mathbb{Q}}^{\langle S\rangle}+\dim M_{\mathbb{Q}}^{\langle U\rangle}$ by (2), and Proposition 27.5(1) over $\mathbb{Q}$ gives the free orbits' $(n+1)$ each and the fixed points' d_S; $r=m_N(n+1)-b_1$ is rank–nullity in degree 1 ($\mathrm{T}^1=M$ for n even, $M(-I)=1$). *The generators.* The rows generate R iff they generate each orbit's fixed module. (a) On an orbit free modulo $\{\pm I\}$ the restriction to block j_0 of $\sum_{i<o}M(g)^ie_{(j_0,a)}$ is e_a (the other terms lie in other blocks), so by Proposition 27.5(1) they map to a basis of $V^{\{\pm I\}}=V$ (n even) under an isomorphism defined over every ring. (b) The proof of (7)(b)'s certificate, with $\ell=3\nmid 4=|F_j|$: $W:=(V_{\mathbb{Z}/3^K})^{F_j}=e_{F_j}V_{\mathbb{Z}/3^K}$ is free with $W/3W=(V_{\mathbb{F}_3})^{F_j}$, the idempotent's columns generate W, and Nakayama's lemma over $\mathbb{Z}/3^K$ gives the iff. $e_{F_j}=\frac14\sum_{h\in F_j}\rho_n(h)=\frac12(1+\rho_n(h_j(S)))$ since $\rho_n(-I)=1$ for n even. (c) is (1). The stopping rule and the absence of an a-priori bound at d^0 are Proposition 31.6(6) (its bound $a_t\leq v_\ell(24)$ is for $q\geq 1$ only).

(4) Proposition 31.8(2)'s proof ($M_N(-I)=-1$, $-I$ central acting trivially on cohomology), for every N.

(5) The orders are (1). If $\nu_3(N)>0$, $\Gamma_0(N)$ contains an element γ of order 3, so $\langle -I,\gamma\rangle$, a cyclic group of order 6, is a conjugate of $\langle U\rangle$ (the classification in (1)). $H^q=0$ for $q\geq 2$ on T and $C^\bullet_{\mathrm{eq}}$: T has length 1, and $H^2(C^\bullet_{\mathrm{eq}}(M))=0$ (Proposition 31.7(1)) with $C^\bullet_{\mathrm{eq}}$ in degrees ≤ 2. The instances are rows of the oracle table: $(N,n,q)=(1,0,2)$: $\mathbb{Z}/2^2{+}\mathbb{Z}/3$; $(7,0,2)$: $\mathbb{Z}/2{+}\mathbb{Z}/3{+}\mathbb{Z}/3$; $(3,4,3)$: $\mathbb{Z}/2{+}\mathbb{Z}/2{+}\mathbb{Z}/2{+}\mathbb{Z}/2{+}\mathbb{Z}/2{+}\mathbb{Z}/3$; $\nu_3(1)=\nu_3(3)=1$, $\nu_3(7)=2$. The closing paragraph: Proposition 31.7(2)'s $H^q(B)=H^q(\Gamma;M)\otimes\widehat{\mathbb{Z}}_6$, a finitely generated $\widehat{\mathbb{Z}}_6$-module, is the sum of its 2- and 3-primary torsion and its free part; its 3-primary part is $\mathrm{tors}\,H^q(\Gamma;M)\otimes\mathbb{Z}_{(3)}$, which (2) computes through cmp^*, the descent's only glueing datum. *A check outside the build, not a tag* (v1.23.0-pre.17): `docs/implementation/slot24/cd1b3_three_shortcut_check.py` computes,

independently of `cohomology_app`, the 3-primary torsion of $\operatorname{coker}\big(\frac{1}{2}(1+M(S)) \mid 1+M(U)+M(U)^2\big)$ over $\mathbb{Z}/3^6$, certified by the exact-trace value of r, and compares it with the oracle's 3-part of H^1, and the absence of 3-torsion in the oracle's H^q, $q=2,\dots,5$, on every even-n cell with $\nu_3(N)=0$ of `descentN_oracle.tsv` (80 cells, $N\in\{2,4,5,6,8,9,10,11,12,14,15,16,25\}$, $9\mid N$ at $N=9$) and on $(18,4)$, $(27,2)$, $(27,4)$, $(45,2)$ (the oracle run for this check): $84=80+4$ cells in two runs, 0 mismatches; over the grid's 148 cells with $\nu_3(N)=0$ (n odd included) the oracle has no 3-torsion in $H^{\geq 2}$, and 22 of its 32 even-n cells with $\nu_3(N)>0$ have some (v1.23.0-pre.18, referee m3: the header of `cd1b3_three_shortcut_check.out` records one command per figure). The script imports the oracle's construction of M_N, so a convention error there would cancel (referee m4); the referee's own script, with its own M_N and its own resolution (`docs/implementation/slot24/C-D1_three_referee.md` §4), closes that dependence: 162 cells with $\nu_3(N)=0$ and 20 control cells, 0 mismatches. □

Proposition 31.10 (Maps over $\mathbb{Z}$: fibre maps, the level cone and its long exact sequence, Hecke squares, the normal case)**.** *[proved here] [standard: [Bro82, Ch. III.6.2, III.8–10, VII.6], [Wei94, Thm. 2.2.6]; VERIFY] (v1.23.0-pre.18; milestone D2 of `docs/implementation/slot24/DESCENT_plan.md` v0.3, clause C-D3 of its §2.7, the owner's LES-0 request of slot 22; refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D3_referee.md`; its minors m1–m5 at v1.23.0-pre.19, among them the new item (3)(b′), **not re-refereed**; item (2)(g) and (2)(b)'s signs at v1.23.0-pre.22, from the D2 code audit, refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D3g_referee.md`); its fixes M1, m1–m5 at v1.23.0-pre.23, **not re-refereed**.) Assume (H4) and (B1)–(B3).* Models. *For a commutative ring S, an S-*model *is a complex $\mathsf{P}_\bullet$ of finitely generated projective $S[\Gamma]$-modules, $\mathsf{P}_q=0$ for $q<0$, with an augmentation $\mathsf{P}_0\to S$ making it a resolution of S; $\mathsf{P}(M):=\operatorname{Hom}_{S[\Gamma]}(\mathsf{P}_\bullet,M_S)$. A* comparison *is an $S[\Gamma]$-chain map $\varphi\colon\mathsf{P}\to\mathsf{P}'$ over id_S; one exists between any two S-models, unique up to homotopy [standard: [Wei94, Thm. 2.2.6]], and $\varphi^*\colon\mathsf{P}'(M)\to\mathsf{P}(M)$ is a quasi-isomorphism compatible with the identifications of both with $H^\bullet(\Gamma;M_S)=\operatorname{Ext}^\bullet_{S[\Gamma]}(S,M_S)$. The models in use: $\mathcal{I}_\bullet\otimes S$ (every S); the free tree resolution of Remark 16.3 (every S); $C_\bullet(X)\otimes S$ and $C_\bullet(\mathcal{T})\otimes S$ for $6\in S^\times$ (Remark 7.6, Lemma 28.4); $\mathcal{I}^{(\ell)}\otimes S$ for O_ℓ-algebras S (Proposition 31.6(1)); and the ℓ-local tree models of the plan's later clause C-D6.* cmp *(whenever $C_\bullet(X)\otimes S$ is a model), Φ_ℓ, Ψ_ℓ and* cmp′ *are comparisons.* Fibres. *Let $(A,\chi)\supseteq(B,\chi')$ be a compatible level pair (Definition 29.25, Lemma 29.26) whose characters take values in $\mathsf{O}:=\mathsf{O}_m=\mathbb{Z}[\zeta_m]$ for some m ($\mathsf{O}=\mathbb{Z}$ if $\chi=\chi'=1$), $V(\chi):=V_{\mathsf{O}}(\chi)$, $M_A:=\operatorname{Ind}_A^\Gamma V(\chi)$, $M_B:=\operatorname{Ind}_B^\Gamma V(\chi')$, $\alpha:=\alpha_{B/A}$, $\omega:=\omega_{B/A}$ (Definition 29.27, over O), and $Q:=\operatorname{coker}\alpha\cong\operatorname{Ind}_A^\Gamma Q_{A/B}$ (Proposition 29.31(1)); M_A, M_B, Q are finitely generated and free over $\mathbb{Z}$. $H^q(A):=H^q(A;V(\chi)_S)$, $H^q(B):=H^q(B;V(\chi')_S)$, the ring S read from the context; $\beta_j:=b_j(M_B)-b_j(M_A)$ for $j\in\{0,1\}$ and $\beta_j:=0$ otherwise (the b_j of §31.1).*

1. ***(Fibre maps commute strictly; they glue to maps over $\mathbb{Z}$.)*** *Let $u\colon M\to M'$ be a $\mathbb{Z}[\Gamma]$-map of finitely generated $\mathbb{Z}$-free modules and $u_{\mathsf{P}}:=\operatorname{Hom}(\mathsf{P}_\bullet,u_S)\colon\mathsf{P}(M)\to\mathsf{P}(M')$.*
   - *(a) $\varphi^*\circ u_{\mathsf{P}'}=u_{\mathsf{P}}\circ\varphi^*$ **as cochain maps**, for every comparison $\varphi\colon\mathsf{P}\to\mathsf{P}'$, every S and every degree — in particular for* cmp* *($=\varepsilon^*$ of the plan), Φ_ℓ^*, Ψ_ℓ^*,* cmp′**, and for the chain homotopy equivalence $C^\bullet_{\mathrm{eq}}(M_S)\simeq\mathrm{T}(M_S)$ of Proposition 23.3(2).*
   - *(b) If $\mathsf{P}_q=\bigoplus_x S[\Gamma/Q_x]$ ($|Q_x|\in S^\times$; every model in use), then $\mathsf{P}(M)^q=\bigoplus_x M_S^{Q_x}$ and u_{P} is u restricted block by block: no lift, no solve.*
   - *(c) $H^q(u_{\mathsf{P}})=H^q(\Gamma;u_S)$ under these identifications, so $H^q(\Gamma;u)\colon H^q(\Gamma;M)\to H^q(\Gamma;M')$ is the map of fibre products of Proposition 31.7(2) induced by u on the O-piece $C^\bullet_{\mathrm{eq}}(M_{\mathbb{Z}[1/6]})$ and on the ℓ-local pieces, $\ell\in\{2,3\}$,* each read on any $\mathbb{Z}_{(\ell)}$- or O_ℓ-model*: u maps the descent diagram of M to that of M', hence the mapping cocones, compatibly with the quasi-isomorphisms of Proposition 31.7(2). No coherence datum enters.*
   - *(d)* ***(Instances, in every degree, over every ring.)*** *Under Shapiro's isomorphism, $H^q(\Gamma;\alpha)=\mathrm{res}^A_B$, $H^q(\Gamma;\omega)=\mathrm{cor}^A_B$ and, for a normal pair (Definition 29.35),*

$H^q(\Gamma;R_\sigma)=c_\sigma$, *the conjugation action of* $\sigma\in A$ *on* $H^q(B)$ *— for every* q *and every* S*, with no hypothesis on* 6 *(contrast Lemma 29.28(3) and Proposition 29.36(1), stated on the model for* $6\in S^\times$*). Hence over* $\mathbb{Z}$*:* $\operatorname{cor}\circ\operatorname{res}=[A:B]$*,* $c_\sigma\circ\operatorname{res}=\operatorname{res}$*,* $\operatorname{res}\circ\operatorname{cor}=\operatorname{Nm}_{\mathsf{G}}$ *(normal pair), in every degree; and if* $[A:B]\in S^\times$*,* $H^q(B)=\operatorname{res}H^q(A)\oplus\ker\operatorname{cor}^q$ *for* ***every*** q *(Proposition 29.33 in all degrees). The diamonds* R_σ *of Proposition 29.34(2) and the* O*-action on* M_A*,* M_B *are fibre maps as well.*

2. ***(The level cone on every model; the long exact sequence over*** $\mathbb{Z}$***.)***

   (a) *For every* S*-model* P*,* $0\to\mathsf{P}(M_A)\xrightarrow{\alpha_\mathsf{P}}\mathsf{P}(M_B)\to\mathsf{P}(Q)\to 0$ *is exact, and* $\pi_\mathsf{P}\colon C(\alpha_\mathsf{P})\to\mathsf{P}(Q)$*,* $(c,c')\mapsto[c']$*, is a surjective quasi-isomorphism, where* $C(\alpha_\mathsf{P})^q=\mathsf{P}(M_A)^{q+1}\oplus\mathsf{P}(M_B)^q$ *(*$q\geq-1$*),* $D(c,c')=(-dc,\,dc'+\alpha_\mathsf{P}c)$*, as in Definition 29.29. For a comparison* φ*,* $(\varphi^*[1],\varphi^*)$ *is a cochain map of cones commuting with* π *and with the two maps of the cone's short exact sequence, and a quasi-isomorphism.*

   (b) $H^q(A,B):=H^{q-1}(C(\alpha_\mathsf{P}))\cong H^{q-1}(\Gamma;Q_S)\cong H^{q-1}(A;Q_{A/B}\otimes S)$*, independently of* P*, and the long sequence*

$$0\to H^0(A)\xrightarrow{\text{res}}H^0(B)\to H^1(A,B)\to H^1(A)\xrightarrow{\text{res}}H^1(B)\to H^2(A,B)\to H^2(A)\xrightarrow{\text{res}}H^2(B)\to H^3 \tag{78}$$

   is exact ***in every degree*** *over every* S*; over* $\mathbb{Z}$ *(*$\mathsf{P}=\mathcal{I}$*) it is the long exact sequence of* $0\to V(\chi)\to\operatorname{Ind}_B^A V(\chi')\to Q_{A/B}\to 0$ *over* A*, and over* $\mathbb{Z}_{(\ell)}$ *its localisation.* $24\cdot H^q(A,B)=0$ *for* $q\geq 3$*, and* $\operatorname{res}^\Gamma_A e\cup-$ *is an isomorphism* $H^q(A,B)\to H^{q+2}(A,B)$ *for* $q\geq 3$ *(so* $q_{\max}=4$ *determines every degree* ≥ 3*). Over a* $\mathbb{Z}[1/6]$*-algebra* (78) *is the six-term sequence* (76)*, read as group cohomology (Proposition 29.31(2)); at* $\ell\in\{2,3\}$ *it is* ***not*** *six-term in general: e.g.* $H^3(\Gamma_0(4),\Gamma_0(8))=\mathbb{Z}/2$ *at* $n=0$*, and* $H^3(\Gamma,\Gamma_0(11))=\mathbb{Z}/3$ *at* $n=4$ *(on every one of the 74 cells of the check below some* $H^q(A,B)$*,* $q\geq 3$*, is non-zero).* ***Signs*** *(v1.23.0-pre.22): the maps* $H^q(B)\to H^{q+1}(A,B)$ *and* $H^q(A,B)\to H^q(A)$ *of* (78) *are induced by the two maps* $c'\mapsto(0,c')$ *and* $(c,c')\mapsto c$ *of the cone's short exact sequence; under* π_P *the second is* $-\delta$*,* $\delta\colon H^{q-1}(\Gamma;Q_S)\to H^q(\Gamma;M_{A,S})$ *the connecting map of* $0\to M_A\to M_B\to Q\to 0$*, with the convention* $\delta[x]=[d\tilde{x}]$ *(*$\tilde{x}$ *a lift, the cochain differential carrying no sign, as for the snake-lemma connecting maps elsewhere in this document); the "*$-$*" depends on that convention. Immaterial for groups, kernels, images, cokernels and exactness, and for equivariance (*$T\circ\delta=\delta\circ T$ *is unchanged under* $\delta\mapsto-\delta$*, so (3)(c) is unaffected); it matters only when the map is compared entrywise with an independently computed* δ*, or composed with maps defined through* δ *(LES-1's predictions, a Bockstein compatibility). (v1.23.0-pre.23, referee m5.)*

   (c) ***(Gluing.)*** $C(\alpha_\mathcal{I})$ *is a complex of finitely generated free* $\mathbb{Z}$*-modules, and Proposition 23.8 applies to it:* $H^q(A,B)$ *over* $\mathbb{Z}$ *is the fibre product of the cohomology of* $C(\alpha_{C_\bullet(X)})$ *over* $\mathbb{Z}[1/6]$ *(the* O*-piece; the only glueing datum is* $(\mathrm{cmp}^*[1],\mathrm{cmp}^*)$*) and of the cones on any* $\mathbb{Z}_{(2)}$*- and* $\mathbb{Z}_{(3)}$*-models, over* $H^q(C(\alpha_\mathcal{I})\otimes\widehat{\mathbb{Q}}_6)$*. The maps of* (78) *(both maps of the cone's short exact sequence and* res*) commute strictly with every comparison, so the sequence glued from the local sequences is* (78) *over* $\mathbb{Z}$*, and every local sequence is exact.*

   (d) ***(The certified exponent on the cone.)*** *On every* $\mathbb{Z}_{(\ell)}$*-model whose cochains are finitely generated free* $\mathbb{Z}_{(\ell)}$*-modules (*$\mathcal{I}\otimes\mathbb{Z}_{(\ell)}$*,* $\mathcal{I}^{(\ell)}\otimes\mathbb{Z}_{(\ell)}$*, and the models of (f)),* $C(\alpha_\mathsf{P})$ *is a bounded below complex of finitely generated free* $\mathbb{Z}_{(\ell)}$*-modules, and Proposition 23.17(1)–(3), Remark 23.18 and Proposition 31.6(6) (the Smith-valuation readout and the stopping rule) hold for its differentials* D^q*,* $q\geq-1$*, verbatim.* $\operatorname{rank}_\mathbb{Q}D^q$ *is admissible from exactly two sources: (i) the rank over* $\mathbb{F}_{\ell'}$ *of the reduction mod* ℓ' *of* ***this*** *cone on the model's integral form —* $\mathcal{I}$ *over* $\mathbb{Z}$*,* $\mathcal{I}^{(\ell)}$ *over* O_ℓ *with* $\ell'\nmid\mathsf{q}_\ell$ *(as in Proposition 31.6(4)(c); a* $\mathbb{Z}_{(\ell)}$*-module reduced mod* $\ell'\neq\ell$ *is* 0*), the tree and*

Bredon complexes of (f) over $\mathbb{Z}[1/6]$ with $\ell' \geq 5$ (v1.23.0-pre.19, referee m4) —, at an ℓ' that qualifies for D^q, i.e. at which $H^{q+1}(\Gamma; Q)$ has no ℓ'-torsion — every prime $\ell' \geq 5$ when $q \neq 0$, and for $q = 0$ iff $s_{\ell'}(Q) = 0$; (ii) the Betti recursion $\operatorname{rank}_{\mathbb{Q}} D^q = \operatorname{rank} C(\alpha_{\mathsf{P}})^q - \beta_q - \operatorname{rank}_{\mathbb{Q}} D^{q-1}$ ($D^{-2} := 0$), with $b_j(Q) = \beta_j$. The rank of a differential of $\mathsf{P}(M_A)$ or $\mathsf{P}(M_B)$ alone is not admissible for the cone (Proposition 31.6(4)(c)). Equivalently the rule may be run on the smaller complex $\mathsf{P}(Q)$ (Proposition 31.7 for $M = Q$): the groups are the same by (a); the maps of (78) need the cone.

(e) ***(Bad primes of the cone.)*** *For $\ell \geq 5$, $H^q(A, B)$ has no ℓ-torsion for $q \neq 2$, and the ℓ-primary torsion of $H^2(A, B) = H^1(\Gamma; Q)$ has exactly $s_\ell(Q) = \dim_{\mathbb{F}_\ell} H^0(\Gamma; Q/\ell) - \beta_0$ cyclic summands (Proposition 31.8(1) for $M = Q$). If moreover $\ell \nmid [A : B]$, then $s_\ell(Q) = s_\ell(M_B) - s_\ell(M_A)$; for $\Gamma_0(MN) \subseteq \Gamma_0(N)$, $\chi = 1$, these are the closed forms of Proposition 31.8(2).*

(f) ***(The cheap routes.)*** *At $\ell \geq 5$ $C_\bullet(\mathcal{T}) \otimes \mathbb{Z}_{(\ell)}$ and $C_\bullet(X) \otimes \mathbb{Z}_{(\ell)}$ are $\mathbb{Z}_{(\ell)}$-models, so (a)–(d) hold for the tree and the Bredon cones; $\operatorname{tors} H^2(A, B) \otimes \mathbb{Z}_{(\ell)} = \operatorname{tors} \operatorname{coker} d^0$ for d^0 of $\mathrm{T}(Q_{\mathbb{Z}_{(\ell)}})$, read by the sparse cokernel route of Proposition 31.7(7)(b) for $M = Q$: the images under $M_B \to Q$ of generators of $(M_B)^{\langle S \rangle} + (M_B)^{\langle U \rangle}$ over $\mathbb{Z}/\ell^K$ generate $Q^{\langle S \rangle} + Q^{\langle U \rangle}$, and the comparison value is $r = \operatorname{rank} \mathrm{T}^0(Q) - \beta_0$ with $\operatorname{rank} \mathrm{T}^j(Q) = \operatorname{rank} \mathrm{T}^j(M_B) - \operatorname{rank} \mathrm{T}^j(M_A)$. At $\ell = 3$, for $A = \Gamma_0(N)$, $B = \Gamma_0(MN)$, $\chi = 1$ and $\nu_3(N) = 0$ (so $\nu_3(MN) = 0$): the same holds over $\mathbb{Z}_{(3)}$ — T and $C^\bullet_{\mathrm{eq}}$ compute $H^\bullet(A; Q_{A/B} \otimes S)$ for every $\mathbb{Z}_{(3)}$-algebra S, $\mathrm{cmp}^* \otimes \mathbb{Z}_{(3)}$ is a quasi-isomorphism of cones, $H^q(A, B) \otimes \mathbb{Z}_{(3)} = 0$ for $q \geq 3$, the 3-local sequence (78) is six-term, and $\operatorname{tors} H^2(A, B) \otimes \mathbb{Z}_{(3)}$ is read as in Proposition 31.9(3) for $M = Q$ (the images of (3)(a), (3)(b)'s generators of M_B; $r = \operatorname{rank} \mathrm{T}^0(Q) - \beta_0$, $\operatorname{rank} \mathrm{T}^0(Q)$ the difference of the closed forms of Proposition 31.9(3) at MN and at N). At $\ell = 2$, and at $\ell = 3$ when $\nu_3(N) > 0$, the cone is read on $\mathcal{I}^{(\ell)}$ or another $\mathbb{Z}_{(\ell)}$-model.*

(g) ***(Generators, coordinates and maps from a certified Smith form: the precision rule.)*** *(v1.23.0-pre.22; the D2 code audit's F1; refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D3g_referee.md`, its fixes at v1.23.0-pre.23.) (d) and Proposition 31.6(6) certify Smith* valuations*, i.e. group types; the maps of (1) and (78) are read from the* transforms*, and this item says how far those are exact. Let A be an $r \times c$ matrix over $\mathbb{Z}_{(\ell)}$, $K \geq 1$, and $LAR = \Sigma$ over $\mathbb{Z}/\ell^K$ with $L \in \mathrm{GL}_r(\mathbb{Z}/\ell^K)$, $R \in \mathrm{GL}_c(\mathbb{Z}/\ell^K)$, Σ zero but for the diagonal entries $u_1\ell^{\sigma_1}, \ldots, u_s\ell^{\sigma_s}$ in its first s places, u_i units (the code does not normalise its pivots; v1.23.0-pre.23, referee m4), every $\sigma_i < K$; put $D := \operatorname{diag}(u_i\ell^{\sigma_i})$. Suppose the form is **certified**: $s = \operatorname{rank}_{\mathbb{Q}} A$. Put $\sigma_{\max} := \max_i \sigma_i$ (0 if $s = 0$) and $K' := K - \sigma_{\max} \geq 1$. Then:*

*(i) There are $\tilde{L} \in \mathrm{GL}_r(\mathbb{Z}_{(\ell)})$, $\tilde{R} \in \mathrm{GL}_c(\mathbb{Z}_{(\ell)})$ with $\tilde{L}A\tilde{R} = \operatorname{diag}(D, 0)$ **exactly**, and $\tilde{L} \equiv L$, $\tilde{L}^{-1} \equiv L^{-1}$, $\tilde{R} \equiv R$, $\tilde{R}^{-1} \equiv R^{-1} \pmod{\ell^{K'}}$. **Moreover** (v1.23.0-pre.23, referee M1) the first s rows of $\tilde{L}$, the last $r - s$ columns of $\tilde{L}^{-1}$ and the last $c - s$ rows of $\tilde{R}^{-1}$ are $\equiv$ those of L, L^{-1}, R^{-1} **modulo** ℓ^K; only the other blocks (the last $r - s$ rows of L, the first s columns of L^{-1}, all of R, the first s rows of R^{-1}) are exact merely modulo $\ell^{K'}$.*

*(ii) Hence, modulo $\ell^{K'}$ and for one exact choice: the last $c - s$ columns of R are a basis of $\ker A$ (saturated in $\mathbb{Z}^c_{(\ell)}$), the matching rows of R^{-1} its coordinates; the columns of L^{-1} are generators of $\operatorname{coker} A \cong \bigoplus_{\sigma_i > 0} \mathbb{Z}/\ell^{\sigma_i} \oplus \mathbb{Z}^{r-s}_{(\ell)}$, the i-th of order ℓ^{σ_i} ($i \leq s$), and the rows of L the coordinate functionals: the class of y is $((Ly)_i \bmod \ell^{\sigma_i})_{i \leq s}$, $((Ly)_i)_{i > s}$. By the "moreover" of (i), the torsion coordinates $(Ly)_i \bmod \ell^{\sigma_i}$, $i \leq s$, are exact for **every** $\sigma_i < K$; the torsion generators (the first s columns of L^{-1}) and the free part (coordinates and generators) are exact modulo $\ell^{K'}$. A read-out that refuses unless $2\sigma_{\max} \leq K$ (the code's `BuildPres`) applies a **conservative policy**, sufficient and not necessary (v1.23.0-pre.23, ref-*

eree m1). (This is about reading exact vectors. A read that applies a map to the torsion generators, Proposition 31.15(3), ***does*** *need $2\sigma_{\max} \le K$ for the module; v1.23.0-pre.25, the C-D4 referee's m1.)* ***Lattice coordinates.*** *For y in the lattice $A\mathbb{Z}^c_{(\ell)}$ spanned by the columns of A, its coordinates in the basis $(\tilde{L}^{-1}De_i)_{i\le s}$ of that lattice are $D^{-1}(\tilde{L}y)_{\le s}$, known modulo $\ell^{K-\sigma_i}$ from $(Ly)_{\le s}$, and $(Ly)_{>s} \equiv 0 \pmod{\ell^K}$ (v1.23.0-pre.23, referee M1; the code's* `LatticeQuotient`*).*

(iii) ***(Composition.)*** *A matrix formed by sums and products from data so read is known modulo $\ell^{K'}$ and is the reduction of the same formula in $\tilde{L}, \tilde{R}$ over $\mathbb{Z}_{(\ell)}$; once its own $\operatorname{rank}_{\mathbb{Q}}$ is certified, (i), (ii) apply to it with K' in place of K.* ***Other operations*** *(v1.23.0-pre.23, referee m2): an exact division by ℓ^{σ_i} costs σ_i digits (a datum known modulo ℓ^k and divisible by ℓ^{σ_i} gives a quotient known modulo $\ell^{k-\sigma_i}$); reducing a torsion row modulo ℓ^{a_i} costs nothing (its entries are defined only modulo ℓ^{a_i}, and any lift is admissible when the relation matrix Δ is adjoined); data from several forms combine at the* ***minimum*** *of their precisions.* ***Certifying ranks*** *(v1.23.0-pre.23, referee m3): since the exact matrix of a derived form depends on the unknown $\tilde{L}$, $\tilde{R}$, its certifying $\operatorname{rank}_{\mathbb{Q}}$ must be an invariant of the map, not of a matrix; the admissible ones are, besides (d)'s two sources for $\operatorname{rank}_{\mathbb{Q}} D^q_{\mathrm{cone}}$: for an induced map $f\colon X \to Y$ with presentation relations Δ_Y (t_Y torsion generators), $\operatorname{rank}_{\mathbb{Q}}[F \mid \Delta_Y] = t_Y + \operatorname{rank}_{\mathbb{Q}} f$; for the kernel lattice $L_f = \{x : Fx \in \operatorname{im}\Delta_Y\}$, $\operatorname{rank}_{\mathbb{Q}} L_f = g_X - \operatorname{rank}_{\mathbb{Q}} f$ (g_X generators of X); for the coordinate matrix C of a sublattice $I \subseteq L_f$, $\operatorname{rank}_{\mathbb{Q}} I$: t_X for $I = \operatorname{im}\Delta_X$ (the kernel of f), $t_X + \operatorname{rank}_{\mathbb{Q}} g$ for $I = F_g\mathbb{Z}^{g_W}_{(\ell)} + \operatorname{im}\Delta_X$ (the comparison $\operatorname{im} g = \ker f$); and $\operatorname{rank}_{\mathbb{Q}} f$ of the maps themselves: res^q, cor^q: $b_q(A)$; $j^q\colon H^q(B) \to H^{q+1}(A,B)$: β_q; $(c,c') \mapsto c$: 0; c_σ: $b_q(B)$; for a normal pair the invariants map $x \mapsto (c_\sigma x - x)_\sigma$ on $H^1(B)$: $b_1(B) - b_1(A)$ (its rational kernel is $H^1(B)^{\mathsf{G}}_{\mathbb{Q}} = \operatorname{res} H^1(A)_{\mathbb{Q}}$). These hold because rationally $\mathrm{cor}\circ\mathrm{res} = [A:B]$ ((1)(d)), so res is injective and the rational long sequence splits into short exact sequences $0 \to H^q(A)_{\mathbb{Q}} \to H^q(B)_{\mathbb{Q}} \to H^{q+1}(A,B)_{\mathbb{Q}} \to 0$. Along a chain of certified Smith forms the precisions therefore decrease additively, $K \ge K' \ge K'' \ge \cdots$, and kernels, images and cokernels of induced maps, as well as the subgroup comparisons "$\operatorname{im} = \ker$", computed so are exact* ***provided*** *every exponent read stays below the precision left at its step; otherwise the read-out is refused (the stopping rule of Proposition 31.6(6)), never reported.*

Without the certification nothing is asserted: if $s < \operatorname{rank}_{\mathbb{Q}} A$, (L,Σ,R) need not be the reduction of any exact Smith form.

3. ***(Hecke squares.)*** *Let $A = \Gamma_0(N) \supseteq B = \Gamma_0(MN)$, $\chi' = \chi \circ \mathrm{red}_{MN\to N}$, $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$, and $[AgA]$, $[BgB]$ the operators of Proposition 31.7(5) on $H^\bullet(A)$, $H^\bullet(B)$ ($f = \rho_{n,\chi}(g)$, resp. $\rho_{n,\chi'}(g)$).*

   (a) *If $p \nmid M$, then $\mathrm{res}\circ[AgA] = [BgB]\circ\mathrm{res}$ and $\mathrm{cor}\circ[BgB] = [AgA]\circ\mathrm{cor}$ on H^q for* ***every*** *q, over every ring. For $p \nmid MN$ these are T_p at both levels; for $p \mid N$, $p \nmid M$, they are U_p at both levels —* ***this document's*** *$U_p = [AgA]$, which on p-old spaces is the adjoint U_p^* of the classical operator (Proposition 29.34(3)(d)). Over $\mathbb{Z}$ the squares hold on the glued groups and are read on any of their local components ((1)(c), Proposition 31.7(3)); on the model over every ring they hold as cochain maps by the criterion Proposition 29.34(1), whose two hypotheses $B \cap A'' = B''$ and $A = A''B$ the proof verifies (its instance (2) for U_p asks for equal Γ_0-levels, which $\Gamma_0(MN) \subset \Gamma_0(N)$, $M > 1$, does not have; v1.23.0-pre.19, referee m2).*

   (b) *If $p \mid M$ (level raising at p) the square with res fails over $\mathbb{Z}$ in general* ***for this document's operator*** *$[AgA]$, $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$: at $(\Gamma_0(25) \subset \Gamma_0(5), n = 0, \chi = 1)$ on*

H^1 *(Proposition 29.34(3)(c) at $\ell = 7$). For $p \mid N$ this is the convention, not level raising:* $[AgA]$ *is the adjoint* U_p^* *on p-old spaces, and the* ***classical*** $U_p = [A\,g'A]$, $g' = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$, *satisfies the criterion Proposition 29.34(1) on every level-raising pair at* $\chi = 1$, *so its square with* res *commutes, as cochain maps (Proposition 29.34(3)(d)). For* $p \nmid N$, $p \mid M$, $[AgA]$ *is* T_p *at level* N *and* U_p *at level* MN, *and both squares fail in general: at* $(\Gamma_0(2) \subset \Gamma,\ n = 2,\ p = 2,\ \chi = 1)$ *on* H^1 *over* $\mathbb{Z}$ *(v1.23.0-pre.19, referee m1).*

(b′) ***(Corestriction and*** U_p***, every*** M***.)*** *If* $p \mid N$, *then*

$$\mathrm{cor}_B^A \circ [BgB] = [AgA] \circ \mathrm{cor}_B^A \quad \text{on } H^q \text{ for } \textbf{every } q,$$

every M *($p \mid M$ included) and over every ring, and over* $\mathbb{Z}$ *on the glued groups ((1)(c)); here* $[AgA]$, $[BgB]$ *are this document's* U_p *at levels* N *and* MN. *In particular, on the Hida p-tower* $\Gamma_0(p^{r+1}) \subset \Gamma_0(p^r)$, $r \geq 1$, ***corestriction is*** U_p***-equivariant over*** $\mathbb{Z}$ ***in every degree****, and restriction is not, for this* U_p *((b)). If* $p \nmid N$ *and* $p \mid M$, *the square with* cor *fails in general, at the cell of (b) (v1.23.0-pre.19, referee m1).*

(c) ***(The relative term.)*** *If* $p \nmid M$, *then over* $\mathbb{Z}[1/6]$*-algebras the pair* (T_g^A, T_g^B) *of Proposition 29.34(1) is a cochain endomorphism of the cone on the model commuting with its short exact sequence, so (78)*$\otimes\mathbb{Z}[1/6]$ *is Hecke-equivariant. At* $\ell \in \{2, 3\}$ *a Hecke operator on* $H^q(A, B) \otimes \mathbb{Z}_{(\ell)}$ *and the equivariance of the connecting map are* ***not asserted****: they need cochain-level operators* T^A, T^B *on an* ℓ*-local model with* $T^B \alpha_{\mathsf{P}} = \alpha_{\mathsf{P}} T^A$ strictly — *the level-N operators of the plan's C-D4. Given such operators on* $\mathcal{I}$, *their transports* $\Psi^* T \Phi^*$ *to any other model satisfy the same strict identity by (1)(a), and* $(T^A[1], T^B)$ *makes (78)*$\otimes\mathbb{Z}_{(\ell)}$ *equivariant. (v1.23.0-pre.24: such operators are Proposition 31.13(3), with the strict identity in its (7).)*

4. ***(The normal case: inflation–restriction over*** $\mathbb{Z}$***.)*** *Let the pair be normal,* $\mathsf{G} := A/B$.

(a) *Over every ring and in every degree,* G *acts on* $H^q(B)$ *through* $c_\sigma = H^q(\Gamma; R_\sigma)$, *glued strictly by (1);* res *lands in* $H^q(B)^{\mathsf{G}}$, $\mathrm{res} \circ \mathrm{cor} = \mathrm{Nm}_{\mathsf{G}}$, $\mathrm{cor} \circ \mathrm{res} = |\mathsf{G}|$; *invariants commute with the flat base changes of the descent, so* $H^q(B)^{\mathsf{G}}$ *over* $\mathbb{Z}$ *is glued from its local components.*

(b) *[standard: [Bro82, Ch. VII, §6]; VERIFY] The Hochschild–Serre spectral sequence* $E_2^{a,b} = H^a(\mathsf{G}; H^b(B)) \Rightarrow H^{a+b}(A)$ *exists over every ring, with the five-term exact sequence* $0 \to H^1(\mathsf{G}; H^0(B)) \xrightarrow{\inf} H^1(A) \xrightarrow{\mathrm{res}} H^1(B)^{\mathsf{G}} \xrightarrow{\mathrm{tg}} H^2(\mathsf{G}; H^0(B)) \xrightarrow{\inf} H^2(A)$. *Over* $\mathbb{Z}[1/6]$*-algebras its last map is* 0 *and the rest is Proposition 29.36(2)–(5); over* $\mathbb{Z}_{(2)}$, $\mathbb{Z}_{(3)}$ *the rows* $b \geq 2$ *need not vanish when B has torsion (*$\Gamma_0(MN) \trianglelefteq \Gamma_0(N)$, $M \mid \gcd(N, 24)$, *contains* $-I$*), and neither the surjectivity of* tg *nor Proposition 29.36(2), (3)'s second half, (5) is asserted there.*

(c) ***(Torsion-free*** B***: two rows over every ring.)*** *If* B *is torsion-free (*$\Gamma_1(N)$, $N \geq 4$, *and its subgroups,* $\Gamma(3)$*: Proposition 29.32), then* $H^b(B; X) = 0$ *for* $b \geq 2$, *every* X *and every ring, and over every* S

$$0 \to H^1(\mathsf{G}; H^0(B)) \to H^1(A) \to H^0(\mathsf{G}; H^1(B)) \xrightarrow{d_2} \cdots \to H^a(\mathsf{G}; H^0(B)) \xrightarrow{\inf} H^a(A) \to H^{a-1}(\mathsf{G}; H^1($$

is exact. If $n > 0$ *and* S *is* $\mathbb{Z}$*-torsion-free (*$\mathbb{Z}$, $\mathbb{Z}_{(\ell)}$, $\mathbb{Z}[1/6]$, …*), then* $H^0(B) = 0$ *and*

$$H^q(A; V(\chi)_S) \;\cong\; H^{q-1}\big(\mathsf{G}; H^1(B; V(\chi')_S)\big) \qquad \text{for every } q \geq 1$$

($q = 1$*:* res *is an isomorphism onto* $H^1(B)^{\mathsf{G}}$*). Hence* $|\mathsf{G}| \cdot H^q(A; V(\chi)) = 0$ *for* $q \geq 2$; *for* $\ell \nmid |\mathsf{G}|$, $H^q(A; V(\chi)) \otimes \mathbb{Z}_{(\ell)} = 0$ *for* $q \geq 2$ *and* $H^1(A) \otimes \mathbb{Z}_{(\ell)} = (H^1(B) \otimes \mathbb{Z}_{(\ell)})^{\mathsf{G}}$; *for* G *cyclic,* $H^q(A) \cong H^{q+2}(A)$ *for* $q \geq 2$ *by the periodic resolution of Proposition 29.37(1).* ***The primary instance*** *is the bottom of the Hida p-tower:* $A = \Gamma_0(N)$, $B = \Gamma_1(N)$,

$N \geq 4$, $\mathsf{G} = (\mathbb{Z}/N)^\times$, *in particular* $N = p^r \geq 4$ *and* $n > 0$*:* $H^q(\Gamma_0(p^r); \mathrm{Sym}^n \mathbb{Z}^2) \cong H^{q-1}\big((\mathbb{Z}/p^r)^\times; H^1(\Gamma_1(p^r); \mathrm{Sym}^n \mathbb{Z}^2)\big)$ *for* $q \geq 2$*, all of it* 2*- and* 3*-primary (*24 *kills* $H^{\geq 2}(A)$*, Proposition 31.6(4)(a)); its* ℓ*-primary part vanishes for* $\ell \in \{2, 3\}$*,* $\ell \nmid p^{r-1}(p-1)$*, and* $H^{\geq 1}\big((\mathbb{Z}/p^r)^\times; H^1(\Gamma_1(p^r); \mathrm{Sym}^n \mathbb{Z}^2)\big)$ *has no* ℓ*-torsion for* $\ell \geq 5$*, even when* $\ell \mid p^{r-1}(p-1)$ *(v1.23.0-pre.19, referee m5). If* A *is torsion-free as well (the steps* $\Gamma_1(p^{r+1}) \trianglelefteq \Gamma_1(p^r) \cap \Gamma_0(p^{r+1}) \trianglelefteq \Gamma_1(p^r)$*,* $p^r \geq 4$*, of Proposition 29.41), the model computes group cohomology over every ring (Proposition 29.32), and Lemma 29.30 and Proposition 29.36(2)–(5) hold over* $\mathbb{Z}$ *verbatim.*

5. ***(What the tree must supply.)*** *For an* ℓ*-local tree model* $\mathsf{P}_\mathcal{T}$ *of the plan's D5 (clause C-D6),* $\ell \in \{2, 3\}$*, to carry items (1), (2) and (4) it suffices that (T1)* $\mathsf{P}_\mathcal{T}$ *is an* S*-model for* $S = \mathsf{O}'$ *a localisation of* $\mathbb{Z}$ *in which* ℓ *is not inverted, or* $S = \mathbb{Z}_{(\ell)}$*; (T2)* $\mathsf{P}_{\mathcal{T},q} = \bigoplus_x S[\Gamma/Q_x]$ *with* $|Q_x| \in S^\times$*, so that* $\mathsf{P}_\mathcal{T}(M)^q = \bigoplus_x M_S^{Q_x}$ *is finitely generated and free over* $\mathbb{Z}_{(\ell)}$ *after base change and fibre maps act block by block ((1)(b), (2)(d)); (T3) a comparison* $\Phi_\mathcal{T} \colon \mathcal{I} \otimes S \to \mathsf{P}_\mathcal{T}$ *(or* $\Psi_\mathcal{T}$ *the other way), given explicitly when classes are to be transported or the model is to be checked against* $\mathcal{I}$ *at a test cell. Nothing else: no strictness over* $C_\bullet(X)$*, no Hecke datum, no boundary. Item (3)(c) at* ℓ *needs in addition the operators of C-D4, transported as there.*

Proof. (1)(a) $\varphi^*(u_{\mathsf{P}'} c) = u \circ c \circ \varphi = u_\mathsf{P}(\varphi^* c)$: $\mathrm{Hom}(-,-)$ is a bifunctor, and nothing is assumed of S, φ or the degree. $\mathrm{cmp} \colon \mathcal{I} \to C_\bullet(X)$, Φ_ℓ, Ψ_ℓ, cmp' are $S[\Gamma]$-chain maps over id_S (Propositions 23.5(4), 31.2(4)), and the equivalence of Proposition 23.3(2) is induced by chain maps on the chain side (as used in the proof of Lemma 29.30(2)). (b) $\mathrm{Hom}_{S[\Gamma]}(S[\Gamma/Q], M_S) \to M_S^Q$, $c \mapsto c(Q)$, is an isomorphism, and $(u \circ c)(Q) = u(c(Q))$. (c) $H^q(\mathsf{P}(M)) \cong \mathrm{Ext}^q_{S[\Gamma]}(S, M_S)$ through comparisons, naturally in M [standard: [Wei94, Thm. 2.2.6]]; Ext is a functor in the second variable, so $H^q(u_\mathsf{P})$ is $H^q(\Gamma; u_S)$. In Proposition 31.7(2) the corners $C^\bullet_{\mathrm{eq}}(M_\mathsf{O})$, $C^\bullet_\mathcal{I}(M) \otimes \widehat{\mathbb{Z}}_6$, $C^\bullet_\mathcal{I}(M) \otimes \widehat{\mathbb{Q}}_6$ and the maps α (base change after $\mathrm{cmp}^*[1/6]$) and β are natural in M by (a), so u induces a morphism of diagrams and of their mapping cocones; the quasi-isomorphisms of Proposition 23.7 ($M \to$ cocone) and of the proof of Proposition 23.9 (the map of cocones induced by $\mathrm{cmp}^*[1/6]$) are natural as well. By Proposition 23.8's morphism clause $H^q(\Gamma; u)$ is determined by its two components, and the ℓ-primary part of the $\widehat{\mathbb{Z}}_6$-component may be read through any comparison by (a). (d) Take $\mathsf{P} = \mathcal{I} \otimes S$: it is free over $S[\Gamma]$, hence over $S[A]$ and $S[B]$, a projective resolution of S over both, so the reciprocity $\phi \mapsto \phi(\cdot)(I)$ of Proposition 27.2(3) identifies $\mathrm{Hom}_{S[\Gamma]}(\mathsf{P}, \mathrm{Ind}_A^\Gamma X)$ with $\mathrm{Hom}_{S[A]}(\mathsf{P}, X)$ and computes Shapiro's isomorphism [standard: [Bro82, Ch. III.6.2]]. The computation in the proof of Lemma 29.28(3) is algebraic and uses only the Γ-equivariance of ϕ, so it holds for P verbatim: α_P is "read an A-equivariant cochain as a B-equivariant one", which induces res, and ω_P is the transfer formula, which induces cor [standard: [Bro82, Ch. III.9]]; the role of $6 \in S^\times$ there was only to make $C_\bullet(X) \otimes S$ a projective resolution, which $\mathcal{I} \otimes S$ is over every S. For R_σ: $\phi(\sigma^{-1} y)(I) = (\sigma^{-1}\phi(y))(I) = \phi(y)(\sigma^{-1})$, so $(R_\sigma \circ \phi)(y)(I) = \rho_{n,\chi}(\sigma)\, \phi(y)(\sigma^{-1})$ and the reciprocity carries $R_{\sigma,\mathsf{P}}$ to $\psi \mapsto \rho_{n,\chi}(\sigma) \psi(\sigma^{-1} \cdot)$ on $\mathrm{Hom}_{S[B]}(\mathsf{P}, V(\chi'))$, the conjugation c_σ computed on a projective resolution over the bigger group A [standard: [Bro82, Ch. III.8]]. The identities are the fibre identities of Lemma 29.28(2) and Proposition 29.36(1), which hold over every ring, pushed through $H^q(\Gamma; -)$. For $[A : B] \in S^\times$ the idempotent e of Proposition 29.33 is a Γ-endomorphism of $M_{B,S}$, and $\mathsf{P}(-)$ is additive, so $\mathsf{P}(M_B) = \alpha_\mathsf{P} \mathsf{P}(M_A) \oplus \mathsf{P}(\ker \omega)$ as complexes in every degree, and $\mathrm{cor} \circ \mathrm{res} = [A : B]$ is invertible. The diamonds commute with α on the fibre (Proposition 29.34(2)); the O-action commutes with the Γ-action by definition.

(2)(a) α is injective and split over O (Lemma 29.28(1)), hence over $\mathbb{Z}$ (O is free over $\mathbb{Z}$), so $0 \to M_{A,S} \to M_{B,S} \to Q_S \to 0$ is exact for every S, with $Q_S = Q \otimes S$; each P_q is projective, so $\mathrm{Hom}_{S[\Gamma]}(\mathsf{P}_q, -)$ is exact. The rest is the proof of Proposition 29.31(2) with this exactness in place of the averaging over $|\Gamma_\sigma| \mid 12$: π_P is a surjective cochain map whose kernel is $\{(c, \alpha_\mathsf{P} c'')\} \cong C(\mathrm{id})$, acyclic. For a comparison, φ^* commutes with α by (1)(a), so $(\varphi^*[1], \varphi^*)$

commutes with D and with both maps of $0 \to \mathsf{P}(M_B) \to C(\alpha_{\mathsf{P}}) \to \mathsf{P}(M_A)[1] \to 0$; it is a quasi-isomorphism by the five lemma on the two long exact sequences. (b) The first isomorphism is (a); the second is Shapiro's lemma for $Q = \mathrm{Ind}_A^\Gamma Q_{A/B}$ [standard: [Bro82, Ch. III.6.2]]. Independence of P is (a) for a comparison. The long sequence of the short exact sequence of complexes in (a) is that of $0 \to M_A \to M_B \to Q \to 0$ in $H^\bullet(\Gamma; -)$, whose connecting map is that of the cone (Lemma 29.30(1)); Shapiro's isomorphism is natural, and Ind_A^Γ is exact and carries $0 \to V(\chi) \to \mathrm{Ind}_B^A V(\chi') \to Q_{A/B} \to 0$ to $0 \to M_A \to M_B \to Q \to 0$ (Proposition 29.31(1)). Localisation is flat. $24 \cdot H^j(\Gamma; Q) = 0$ for $j \geq 2$ is Proposition 31.6(4)(a) for $M = Q$, and the periodicity is Proposition 31.7(5) for $M = Q$ (natural in M, so it is compatible with Shapiro's isomorphism). Over a $\mathbb{Z}[1/6]$-algebra $H^q(\Gamma; -) = 0$ for $q \geq 2$, which leaves six terms. The examples are rows of the check below. (c) $C(\alpha_{\mathcal{I}})$ is built from the finitely generated free $\mathbb{Z}$-modules $C^\bullet_{\mathcal{I}}(M_A)$, $C^\bullet_{\mathcal{I}}(M_B)$. Proposition 23.8 for it, with $C(\alpha_{\mathcal{I}})[1/6] = C(\alpha_{\mathcal{I}\otimes\mathbb{Z}[1/6]})$ and (a) for the comparison cmp over $\mathbb{Z}[1/6]$ (where $C_\bullet(X) \otimes \mathbb{Z}[1/6]$ is a model) and for any comparison $\mathcal{I} \otimes \mathbb{Z}_{(\ell)} \to \mathsf{P}$ at $\ell \in \{2, 3\}$ ($H^q(C() \otimes \widehat{\mathbb{Z}}_6) = \bigoplus_{\ell\in\{2,3\}} H^q(C() \otimes \mathbb{Z}_{(\ell)}) \otimes \mathbb{Z}_\ell$). The maps: (a) and (1)(a). (d) The proofs of Proposition 23.17(1)–(3) and Remark 23.18 use only that C is a complex of finitely generated free modules over $\mathbb{Z}$, and Proposition 31.6(3), (6) carry them to $\mathbb{Z}_{(\ell)}$; $C(\alpha_{\mathsf{P}})^q$ is a sum of two such modules. (i): by Proposition 23.17(2), $\mathrm{rank}_{\mathbb{F}_{\ell'}} D^q = \mathrm{rank}_{\mathbb{Q}} D^q$ iff no elementary divisor of D^q is divisible by ℓ', iff $H^{q+1}(C(\alpha_{\mathsf{P}})) = H^{q+1}(\Gamma; Q)$ has no ℓ'-torsion; for $q + 1 \geq 2$ the group is killed by 24, for $q + 1 = 1$ the condition is $s_{\ell'}(Q) = 0$ by Proposition 31.8(1), and for $q + 1 = 0$ it is a subgroup of the free group Q. (ii): rank–nullity over $\mathbb{Q}$ in each degree, with $\dim_{\mathbb{Q}} H^j(C() \otimes \mathbb{Q}) = b_j(Q)$; over $\mathbb{Q}$ the idempotent e splits $M_{B,\mathbb{Q}} = \alpha M_{A,\mathbb{Q}} \oplus \ker \omega_{\mathbb{Q}}$ Γ-equivariantly and $\ker \omega_{\mathbb{Q}} \cong Q_{\mathbb{Q}}$, so $b_j(Q) = b_j(M_B) - b_j(M_A)$, which is 0 for $j \geq 2$ (Proposition 31.6(4)(a)) and for $j = -1$. The last sentence is (a). (e) Proposition 31.8(1) for the finitely generated free module Q, with $b_0(Q) = \beta_0$ by (d). If $\ell \nmid [A : B]$, e splits the fibre sequence over $\mathbb{Z}_{(\ell)}$ and over $\mathbb{F}_\ell$, so $H^0(\Gamma; M_B/\ell) = H^0(\Gamma; M_A/\ell) \oplus H^0(\Gamma; Q/\ell)$; subtract $b_0(M_B) = b_0(M_A) + \beta_0$. (f) At $\ell \geq 5$ the stabilisers have orders dividing 12, units in $\mathbb{Z}_{(\ell)}$: $C_\bullet(\mathcal{T}) \otimes \mathbb{Z}_{(\ell)}$ is a model by Lemma 28.4, and $C_\bullet(X) \otimes \mathbb{Z}_{(\ell)}$ by Remark 7.6; (a) applies. Exactness of $\mathrm{T}(M_B) \to \mathrm{T}(Q) \to 0$ in degree 0 is (a), i.e. $(M_B)^F \to Q^F$ is onto over $\mathbb{Z}/\ell^K$ for $F \in \{\langle S\rangle, \langle U\rangle\}$, so images of generators generate; the ranks are additive by the same exactness. Proposition 31.7(7)(b) is stated for every finitely generated free M, here $M = Q$, with $b_0(Q) = \beta_0$. At $\ell = 3$: by Proposition 31.9(1) every finite subgroup of $A = \Gamma_0(N)$ has order 1, 2 or 4, so for every $\mathbb{Z}_{(3)}$-algebra S, $C_\bullet(\mathcal{T}) \otimes S$ and $C_\bullet(X) \otimes S$ are projective resolutions of S over $S[A]$ (the proof of Proposition 31.9(2)), as is $\mathcal{I} \otimes S$, and $\mathrm{cmp} \otimes S$ is a comparison over $S[A]$. Frobenius reciprocity identifies $\mathrm{T}(\mathrm{Ind}_A^\Gamma X_S)$ with $\mathrm{Hom}_{S[A]}(C_\bullet(\mathcal{T}) \otimes S, X_S)$, naturally in the A-module X; so (a) and (b) hold verbatim with $S[A]$-models in place of $S[\Gamma]$-models and the A-sequence $0 \to V \to \mathrm{Ind}_B^A V \to Q_{A/B} \to 0$ in place of the fibre sequence (the middle term is $\mathrm{Hom}_{S[B]}(C_\bullet(\mathcal{T}) \otimes S, V_S) = \mathrm{T}(M_B)$ by transitivity of induction, and $B \subseteq A$ has no element of order 3 either). T has length 1, so $H^j(A; Q_{A/B} \otimes \mathbb{Z}_{(3)}) = 0$ for $j \geq 2$, i.e. $H^q(A, B) \otimes \mathbb{Z}_{(3)} = 0$ for $q \geq 3$. The generators: $C_0(\mathcal{T}) \otimes S$ restricted to $S[A]$ is projective, so $\mathrm{T}^0(M_B) \to \mathrm{T}^0(Q)$ is onto, and the images of the generators of Proposition 31.9(3)(a), (b) for M_B generate $Q^{\langle S\rangle} + Q^{\langle U\rangle}$; the proof of Proposition 31.9(3) applies to $M = Q$ with (a) here in place of its (2), and $\mathrm{rank}\, \mathrm{T}^0$ is additive by the same surjectivity and the injectivity of α. (b), signs. A cocycle (c, c') of $C(\alpha_{\mathsf{P}})^{q-1}$ has $dc' = -\alpha_{\mathsf{P}} c$. δ lifts $\pi_{\mathsf{P}}(c, c') = [c']$ to c', and $dc' = \alpha_{\mathsf{P}}(-c)$, so $\delta[c'] = [-c]$. (g) (v1.23.0-pre.22; the D2 code audit's Appendix B, made integral over $\mathbb{Z}_{(\ell)}$.) (i) Lift L, R to L_0, R_0 over $\mathbb{Z}_{(\ell)}$; their determinants are units mod ℓ, so they are invertible over the local ring $\mathbb{Z}_{(\ell)}$. Then $L_0 A R_0 = \Sigma_0 + \ell^K E$ over $\mathbb{Z}_{(\ell)}$, Σ_0 the lift of Σ with entries $u_i \ell^{\sigma_i}$, $u_i \in \mathbb{Z}_{(\ell)}^\times$ lifting the units (all that follows is verbatim with the units; v1.23.0-pre.23, referee m4); in blocks of sizes s and $r - s$, resp. s and $c - s$,

$$L_0 A R_0 = \begin{pmatrix} P & \ell^K E_{12} \\ \ell^K E_{21} & \ell^K E_{22} \end{pmatrix}, \qquad P = D + \ell^K E_{11} = DU, \quad D = \mathrm{diag}(u_i \ell^{\sigma_i}), \quad U = 1 + D^{-1} \ell^K E_{11}.$$

$D^{-1}\ell^K$ has entries $u_i^{-1}\ell^{K-\sigma_i}$, all divisible by $\ell^{K'}$, $K' \geq 1$; so $U \equiv 1 \pmod{\ell^{K'}}$, $U \in \mathrm{GL}_s(\mathbb{Z}_{(\ell)})$, and $P^{-1} = U^{-1}D^{-1}$. The matrices $X := -P^{-1}\ell^K E_{12} = -U^{-1}(\ell^K D^{-1})E_{12}$ and $Y := -\ell^K E_{21}P^{-1} = -E_{21}U^{-1}(\ell^K D^{-1})$ (ℓ^K is a scalar) are integral over $\mathbb{Z}_{(\ell)}$ and $\equiv 0 \pmod{\ell^{K'}}$. Then

$$\begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} L_0 A R_0 \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} P & 0 \\ 0 & \Sigma' \end{pmatrix}, \qquad \Sigma' = \ell^K\big(E_{22} - \ell^K E_{21}P^{-1}E_{12}\big).$$

The left side has $\mathrm{rank}_{\mathbb{Q}} = \mathrm{rank}_{\mathbb{Q}} A = s = \mathrm{rank}_{\mathbb{Q}} P$, so $\Sigma' = 0$ — **this is where the certification enters**. Right-multiplying by $\mathrm{diag}(U^{-1}, 1)$ turns P into D. So $\tilde{L} := \left(\begin{smallmatrix} 1 & 0 \\ Y & 1 \end{smallmatrix}\right) L_0$ and $\tilde{R} := R_0 \left(\begin{smallmatrix} 1 & X \\ 0 & 1 \end{smallmatrix}\right) \mathrm{diag}(U^{-1}, 1)$ satisfy $\tilde{L}A\tilde{R} = \mathrm{diag}(D, 0)$; each factor besides L_0, R_0 is $\equiv 1 \pmod{\ell^{K'}}$, hence so is its inverse, which gives the four congruences. *Moreover* (v1.23.0-pre.23, referee M1): $\tilde{L} = \left(\begin{smallmatrix} 1 & 0 \\ Y & 1 \end{smallmatrix}\right) L_0$ leaves the first s rows of L_0 unchanged; $\tilde{L}^{-1} = L_0^{-1}\left(\begin{smallmatrix} 1 & 0 \\ -Y & 1 \end{smallmatrix}\right)$ leaves the last $r - s$ columns of L_0^{-1} unchanged; $\tilde{R}^{-1} = \mathrm{diag}(U, 1)\left(\begin{smallmatrix} 1 & -X \\ 0 & 1 \end{smallmatrix}\right) R_0^{-1}$ leaves the last $c - s$ rows of R_0^{-1} unchanged; and L_0, L_0^{-1}, R_0^{-1} reduce to L, L^{-1}, R^{-1} modulo ℓ^K. (ii) For an exact Smith form $\tilde{L}A\tilde{R} = \mathrm{diag}(D, 0)$ over the principal ideal domain $\mathbb{Z}_{(\ell)}$, $\ker A = \tilde{R}(0 \oplus \mathbb{Z}_{(\ell)}^{c-s})$, a direct summand, and $y \mapsto \tilde{L}y$ identifies $\mathrm{coker}\, A$ with $\bigoplus_{i \leq s} \mathbb{Z}_{(\ell)}/\ell^{\sigma_i} \oplus \mathbb{Z}_{(\ell)}^{r-s}$, with inverse given by the columns of $\tilde{L}^{-1}$; reduce by (i). A torsion coordinate $(\tilde{L}y)_i$, $i \leq s$, is a row of the first block of $\tilde{L}$, hence $\equiv (Ly)_i \pmod{\ell^K}$ by the "moreover" of (i), and $\sigma_i < K$; the generators (columns of $\tilde{L}^{-1}$, first block) and the free coordinates (last rows of $\tilde{L}$) are exact modulo $\ell^{K'}$ only, which is why $2\sigma_{\max} \leq K$ is sufficient (all torsion exponents $\leq K'$) and not necessary (v1.23.0-pre.23, referee m1). Lattice coordinates: for $y = Ax$, $\tilde{L}y = \mathrm{diag}(D, 0)\tilde{R}^{-1}x$, so $A\mathbb{Z}_{(\ell)}^c = \tilde{L}^{-1}(D\mathbb{Z}_{(\ell)}^s \oplus 0)$ with basis $(\tilde{L}^{-1}De_i)_{i \leq s}$, and the coordinates of y are $D^{-1}(\tilde{L}y)_{\leq s} = D^{-1}(L_0 y)_{\leq s}$; $(L_0 y)_{\leq s}$ is known modulo ℓ^K and its i-th entry is divisible by ℓ^{σ_i}, so the i-th coordinate is known modulo $\ell^{K-\sigma_i}$. The lower block of $\tilde{L}y$ vanishes: $Y(L_0y)_{\leq s} + (L_0y)_{>s} = 0$, so $(L_0y)_{>s} = -Y(L_0y)_{\leq s} = \ell^K E_{21}U^{-1} \cdot D^{-1}(L_0y)_{\leq s} \equiv 0 \pmod{\ell^K}$, and $(L_0y)_{>s} \equiv (Ly)_{>s}$. (iii) Sums and products commute with reduction mod $\ell^{K'}$; a matrix known modulo $\ell^{K'}$ that is the reduction of a $\mathbb{Z}_{(\ell)}$-matrix A' is all (i) needs, with K' in place of K and $\mathrm{rank}_{\mathbb{Q}} A'$ in place of $\mathrm{rank}_{\mathbb{Q}} A$. The other operations: if $x \equiv x_0 \pmod{\ell^k}$ and $\ell^\sigma \mid x, x_0$, then $x/\ell^\sigma \equiv x_0/\ell^\sigma \pmod{\ell^{k-\sigma}}$; a torsion row is an element of $\mathbb{Z}/\ell^{a_i}$, so its reduction loses nothing; a formula in data known modulo ℓ^{k_1} and ℓ^{k_2} is known modulo $\ell^{\min(k_1,k_2)}$. The certifying ranks: rationally $\mathrm{cor} \circ \mathrm{res} = [A : B]$ ((1)(d)) makes res injective, so the rational long sequence breaks into the short exact sequences stated, which give the ranks of res, j, the connecting map (0), cor (surjective, as $\mathrm{cor} \circ \mathrm{res}$ is invertible) and c_σ (an automorphism); for a normal pair $H^1(B)^{\mathsf{G}}_{\mathbb{Q}} = \mathrm{res}\, H^1(A)_{\mathbb{Q}}$ ($\mathrm{res} \circ \mathrm{cor} = \sum_\sigma c_\sigma$). The columns $\ell^{a_i}e_i$ of Δ span the torsion coordinates rationally, and on the free coordinates F is $f \otimes \mathbb{Q}$; this gives $t_Y + \mathrm{rank}_{\mathbb{Q}} f$ for $[F \mid \Delta_Y]$, $g_X - \mathrm{rank}_{\mathbb{Q}} f$ for L_f (rank–nullity), and t_X, resp. $t_X + \mathrm{rank}_{\mathbb{Q}} g$, for the sublattices I. Kernels, images and cokernels of an induced map, and a subgroup comparison, are lattice computations (kernels and quotients of the matrices so formed), each a certified Smith form read under (ii). The last sentence: $A = (\ell^K)$, $L = R = 1$, $s = 0 < 1 = \mathrm{rank}_{\mathbb{Q}} A$; then $\Sigma = 0$ over $\mathbb{Z}/\ell^K$, but every exact Smith form of A is (ℓ^K).

(3)(a) Put $A'' := A \cap gAg^{-1}$, $A' := A \cap g^{-1}Ag$, B'', B' likewise, $\iota_g \colon A'' \to A'$, $h \mapsto g^{-1}hg$, so that $[AgA] = \mathrm{cor}^A_{A''} \circ (\iota_g, f)^* \circ \mathrm{res}^A_{A'}$ (Proposition 31.7(5)). From $g \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) g^{-1} = \left(\begin{smallmatrix} a & b/p \\ pc & d \end{smallmatrix}\right)$ and $g^{-1} \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) g = \left(\begin{smallmatrix} a & pb \\ c/p & d \end{smallmatrix}\right)$: $A'' = \Gamma_0(pN)$, $A' = \Gamma_0(N) \cap \Gamma^0(p)$, $B'' = \Gamma_0(pMN)$, $B' = \Gamma_0(MN) \cap \Gamma^0(p)$. If $p \nmid M$, then $B \cap A'' = \Gamma_0(\mathrm{lcm}(MN, pN)) = B''$ and $B \cap A' = B'$, and $[A : A''] = [A : A'] = [B : B''] = [B : B']$ ($= p + 1$ if $p \nmid N$, $= p$ if $p \mid N$); so $[A''B : A''] = [B : B \cap A''] = [A : A'']$, i.e. $A = A''B$, and likewise $A = A'B$. The double coset formula [standard: [Bro82, Ch. III.9], Mackey] with a single double coset gives $\mathrm{res}^A_B \mathrm{cor}^A_{A''} = \mathrm{cor}^B_{B''} \mathrm{res}^{A''}_{B''}$ and $\mathrm{res}^A_{A'} \mathrm{cor}^A_B = \mathrm{cor}^{A'}_{B'} \mathrm{res}^B_{B'}$. With the naturality of res and cor under the isomorphism of pairs (ι_g, f), which maps B'' onto

B': $\operatorname{res}^A_B[AgA] = \operatorname{cor}^B_{B''}(\iota_g,f)^*\operatorname{res}^{A'}_{B'}\operatorname{res}^A_{A'} = \operatorname{cor}^B_{B''}(\iota_g,f)^*\operatorname{res}^B_{B'}\operatorname{res}^A_B = [BgB]\operatorname{res}^A_B$, and $[AgA]\operatorname{cor}^A_B = \operatorname{cor}^A_{A''}(\iota_g,f)^*\operatorname{cor}^{A'}_{B'}\operatorname{res}^B_{B'} = \operatorname{cor}^A_{A''}\operatorname{cor}^{A''}_{B''}(\iota_g,f)^*\operatorname{res}^B_{B'} = \operatorname{cor}^A_B[BgB]$. The character is harmless: $a_{g^{-1}hg} = a_h$, so $f = \rho_{n,\chi}(g)$ is compatible on every subgroup, and $\chi'|_B = \chi|_B$. Over $\mathbb{Z}$: (1)(c). The cochain statement is the criterion Proposition 29.34(1), whose hypotheses $B\cap A'' = B''$ and $A = A''B$ were just verified. (b) $H^1(\cdot;\mathbb{Z})\otimes\mathbb{F}_7 \to H^1(\cdot;\mathbb{F}_7)$ is an isomorphism, natural in the group and compatible with the operators, at both levels: its cokernel $\operatorname{Tor}(H^2,\mathbb{F}_7)$ vanishes, since 24 kills H^2. A square commuting over $\mathbb{Z}$ would commute over $\mathbb{F}_7$. The classical U_p: Proposition 29.34(3)(d), whose hypotheses $p \mid N'_A = N$ and $pN'_A \mid N'_B = MN$ are $p \mid N$, $p \mid M$. The cell $(\Gamma_0(2)\subset\Gamma, n=2, p=2)$ is a row of the referee's check (`C-D3_referee.md` §2.2). (b′) Let $p \mid N$ and M arbitrary; $A' = A\cap g^{-1}Ag = \Gamma_0(N)\cap\Gamma^0(p)$, $B' = B\cap g^{-1}Bg = \Gamma_0(MN)\cap\Gamma^0(p)$ and $B'' = B\cap gBg^{-1}$ as in (a), so $B\cap A' = \Gamma_0(MN)\cap\Gamma^0(p) = B'$ for every M. Let $\gamma = \begin{pmatrix} a & b \\ c & d\end{pmatrix} \in A$. As $p \mid N \mid c$, a is a unit mod p; for $j \equiv ba^{-1} \pmod p$, $\gamma T^{-j} = \begin{pmatrix} a & b-ja \\ c & d-jc\end{pmatrix}$ $(T = \begin{pmatrix} 1 & 1 \\ 0 & 1\end{pmatrix})$ lies in $\Gamma_0(N)\cap\Gamma^0(p) = A'$, so $\gamma \in A'T^j \subseteq A'B$: $A = A'B$, whatever M is. The double coset formula [standard: [Bro82, Ch. III.9], Mackey] with the single double coset $A'\cdot 1\cdot B$ gives $\operatorname{res}^A_{A'}\operatorname{cor}^A_B = \operatorname{cor}^{A'}_{A'\cap B}\operatorname{res}^B_{A'\cap B} = \operatorname{cor}^{A'}_{B'}\operatorname{res}^B_{B'}$. The isomorphism of pairs (ι_g,f), $\iota_g\colon A''\to A'$, carries B'' onto $g^{-1}B''g = g^{-1}Bg\cap B = B'$ (for every M), and cor is natural under it: $(\iota_g,f)^*\operatorname{cor}^{A'}_{B'} = \operatorname{cor}^{A''}_{B''}(\iota_g,f)^*$. Since $B''\subseteq A''\cap B$, transitivity gives $\operatorname{cor}^A_{A''}\operatorname{cor}^{A''}_{B''} = \operatorname{cor}^A_{B''} = \operatorname{cor}^A_B\operatorname{cor}^B_{B''}$. Hence $[AgA]\operatorname{cor}^A_B = \operatorname{cor}^A_{A''}(\iota_g,f)^*\operatorname{res}^A_{A'}\operatorname{cor}^A_B = \operatorname{cor}^A_{A''}(\iota_g,f)^*\operatorname{cor}^{A'}_{B'}\operatorname{res}^B_{B'} = \operatorname{cor}^A_{A''}\operatorname{cor}^{A''}_{B''}(\iota_g,f)^*\operatorname{res}^B_{B'} = \operatorname{cor}^A_B\operatorname{cor}^B_{B''}(\iota_g,f)^*\operatorname{res}^B_{B'} = \operatorname{cor}^A_B[BgB]$, the last step Proposition 31.7(5) at level MN. The character is harmless as in (a). Nothing used $p\nmid M$: the condition $B\cap A'' = B''$ of (a), which fails for $p\mid M$ ($B\subseteq A''$), enters only the square with res. Over $\mathbb{Z}$: (1)(c). If $p\nmid N$, $p\mid M$, then $[A:A'] = p+1$ while $[B:B\cap A'] = [B:B'] = p$, so $A'B\neq A$, the double coset formula has a second term, and the referee's cell shows it is non-zero. (c) The first sentence is Proposition 29.34(1) with Proposition 29.31(2); the transport: $\Psi^*T^B\Phi^*\alpha = \Psi^*T^B\alpha\Phi^* = \Psi^*\alpha T^A\Phi^* = \alpha\Psi^*T^A\Phi^*$ by (1)(a) twice; a cochain endomorphism of the cone commuting with its short exact sequence makes the long sequence equivariant.

(4)(a) is (1)(d) and (1)(c); $(-)^{\mathsf{G}}$ is a kernel, and $\mathbb{Z}[1/6]$, $\mathbb{Z}_{(\ell)}$, $\widehat{\mathbb{Z}}_6$ are flat. (b) The citation; over a $\mathbb{Z}[1/6]$-algebra $H^2(A) = 0$. (c) Torsion-free B: every finite subgroup has order 1, so Proposition 29.32 gives $H^b(B;X) = 0$ for $b\geq 2$ over every ring. With two rows $d_r = 0$ for $r\geq 3$, and $H^m(A)$ has the filtration $0\to E_\infty^{m,0}\to H^m(A)\to E_\infty^{m-1,1}\to 0$ with $E_\infty^{m,0} = \operatorname{coker}(d_2\colon E_2^{m-2,1}\to E_2^{m,0})$ and $E_\infty^{m-1,1} = \ker(d_2\colon E_2^{m-1,1}\to E_2^{m+1,0})$; splicing gives the sequence, the first map being the edge map inf and, in degree 1, the second the restriction to $E_2^{0,1} = H^1(B)^{\mathsf{G}}$. $H^0(B) = V(\chi')^B_S = 0$: B has finite index, so it contains T^k and $\begin{pmatrix} 1 & 0 \\ k & 1\end{pmatrix}$ for some $k\geq 1$, both with $a=1$, so they act untwisted; over the $\mathbb{Q}$-algebra $S\otimes\mathbb{Q}$ the T^k-invariants of Sym^n $(x\mapsto x,\ y\mapsto y+kx)$ are $S_{\mathbb{Q}}x^n$, and $\begin{pmatrix} 1 & 0 \\ k & 1\end{pmatrix}$ sends x^n to $(x+ky)^n$, whose coefficient of $x^{n-1}y$ is $nk\in(S\otimes\mathbb{Q})^\times$; so the invariants vanish over $S\otimes\mathbb{Q}$, and over $S\subseteq S\otimes\mathbb{Q}$. Then $E_2^{\bullet,0} = 0$, and the sequence gives $H^q(A)\cong E_2^{q-1,1}$. $|\mathsf{G}|$ kills $H^{\geq 1}(\mathsf{G};-)$ [standard: [Bro82, Ch. III.10.2]], which gives the next two consequences, and the periodic resolution the last. $\Gamma_1(N)\trianglelefteq\Gamma_0(N)$ is the kernel of $\gamma\mapsto d_\gamma \bmod N$, onto $(\mathbb{Z}/N)^\times$; $|(\mathbb{Z}/p^r)^\times| = p^{r-1}(p-1)$. If A is torsion-free too, Proposition 29.32 applies to both ends at every ℓ, and the proofs of Lemma 29.30 and Proposition 29.36(2)–(5) use $6\in R^\times$ only through that and through $H^{\geq 2}(A) = H^{\geq 2}(B) = 0$.

(5) Items (1), (2) and (4) use of a model only that it is an S-model and that comparisons exist ((T1); they exist by the comparison theorem, and (T3) makes them explicit), and (2)(d) uses that its cochains are finitely generated free over $\mathbb{Z}_{(\ell)}$: under (T2), $M_S^{Q_x} = e_{Q_x}M_S$ is a direct summand of a finitely generated free module, free over the local principal ideal domain

$\mathbb{Z}_{(\ell)}$, and (1)(b) applies. (3)(c)'s transport is its last sentence.

A check outside the build, not a tag (v1.23.0-pre.18): `docs/implementation/slot24/cd3_descentmaps_che` independent of `cohomology_app` and of `descentN_oracle.py` (own coset tables, induced modules, the free tree resolution of $\mathbb{Z}$ over $\mathbb{Z}[\Gamma]$ in every degree, exact integer Smith forms; the oracle table is read as data only), output `cd3_descentmaps_check.out` (its header records the command). On 74 cells $\Gamma_0(MN) \subseteq \Gamma_0(N)$ (16 pairs, $N \le 6$, $MN \le 16$, even $n \le 8$, $\psi(MN)(n+1) \le 140$; 29 with $\nu_3(N) = 0$, 21 normal): the fibre identities of (1)(d) (and of (4)(a) on the normal pairs), $H^j(C(\alpha)) = H^j(\Gamma; Q)$ as groups for $-1 \le j \le 4$ ((2)(a)), the Betti recursion of (2)(d)(ii) for every D^q, the bad-prime count of (2)(e) at every $\ell \ge 5$ that occurs and at $5, 7, 11, 13$, the sparse cokernel of (2)(f) against $H^1(\Gamma; Q)$ at every $\ell \ge 5$ and at $\ell = 3$ when $\nu_3(N) = 0$, and no 3-torsion in $H^{\ge 3}(A, B)$ when $\nu_3(N) = 0$; the ends against the oracle on 888 rows. On 12 cells $\Gamma_1(N) \trianglelefteq \Gamma_0(N)$ ($N \in \{4, 5, 7, 9\}$, $1 \le n \le 4$), $H^q(\Gamma_0(N); \mathrm{Sym}^n\mathbb{Z}^2) = H^{q-1}((\mathbb{Z}/N)^\times; H^1(\Gamma_1(N); \mathrm{Sym}^n\mathbb{Z}^2))$ for $q = 1, \dots, 5$ ((4)(c); $H^1(\Gamma_1(N))$ on the tree complex of the torsion-free $\Gamma_1(N)$, checked against the free resolution), e.g. $H^3(\Gamma_0(7); \mathrm{Sym}^4\mathbb{Z}^2) \supseteq (\mathbb{Z}/3)^2$, while $3 \nmid |(\mathbb{Z}/5)^\times|$ leaves no 3-torsion in $H^{\ge 2}(\Gamma_0(5); \mathrm{Sym}^n\mathbb{Z}^2)$. 0 mismatches (`CD3_CHECK ... mismatch=0 ... verdict=OK`). Item (3) is not covered by that script; the referee's independent script (`docs/implementation/slot24/C-D3_referee.md` §2.2: H^1 over $\mathbb{Z}$ as crossed homomorphisms, $[AgA]$ through Shapiro, torsion included) checks both squares on 32 cells, and at v1.23.0-pre.19 on 10 more: the square with cor commutes on all 15 cells with $p \mid N$, $p \mid M$ (e.g. $\Gamma_0(8) \subset \Gamma_0(2)$ and $\Gamma_0(8) \subset \Gamma_0(4)$ at $p = 2$, $n \le 6$; $\Gamma_0(18) \subset \Gamma_0(6)$, $\Gamma_0(9) \subset \Gamma_0(3)$ at $p = 3$; $\Gamma_0(25) \subset \Gamma_0(5)$), where the square with res fails on every one, and fails on all 5 cells with $p \nmid N$, $p \mid M$, $n > 0$ ((b), (b′)). □

What is deferred, and why (v1.23.0-pre.18). (i) The **second degeneracy map** $\mathrm{dg}_p = [A\, g\, B]$ and every Hecke-type map: they are not fibre maps (conjugation by $g \notin \Gamma$ is not an automorphism of Γ), so (1) does not reach them; on cohomology they are determined locally (Proposition 31.7(3)), and their cochain lifts on $\mathcal{I}$ or $\mathcal{I}^{(\ell)}$ at level N are C-D4's, as is the Hecke action on the relative term at $\ell \in \{2, 3\}$ ((3)(c)). (ii) The **Hochschild–Serre spectral sequence over $\mathbb{Z}_{(2)}$, $\mathbb{Z}_{(3)}$ when B has torsion**, beyond the five-term sequence (4)(b): its higher differentials are neither computed nor needed, because $H^q(A)$ is computed directly; Proposition 29.36(5) (coinvariants) is not asserted there. (iii) The **old-space cone** (Remark 29.45) stays deferred: its U_p-action needs an η-type certificate. (iv) The **boundary pair and its cone over $\mathbb{Z}$** are the plan's D4, not this clause. (v) Integral nebentypus modules $M_N(\chi)$ over O_χ are not built (Proposition 31.7(8)); (1)–(2) and (4) are stated for them, (2)(f) at $\ell = 3$ and (3)(b)'s example for $\chi = 1$ only. (vi) The tree models (5) are the plan's D5, gated by G-T.

Code map of Propositions 31.7, 31.8, 31.9 and 31.10. Stated at v1.23.0-pre.14, before D1's code (contract first, rule 21). **Built** (and used by D0 at level 1; `Integral/Descent/DescentAssembly`, `docs/implementation/slot24/DESCENT0_report.md`): the $\widehat{\mathbb{Z}}_6$-corner on $\mathcal{I}^{(2)}$, $\mathcal{I}^{(3)}$ for any `CoefficientModule`, under Proposition 31.6; the Smith-valuation hand-off `smith_valuations`; the shipped C_7 K-mode (`EquivariantCohomologyOnK` over a composite context, linear K search) at level 1; the tree Bredon complex $\mathrm{T}(M)$ over $\mathbb{Z}/\ell^K$ and $\mathbb{F}_{\ell'}$ as a *measured comparator* (D0's (c′), feeding no result), and `RunLevelNProbe` (the level-N probe, `MonomialOf`, which checks b_0 against (7)(c)'s closed form at the good prime); `Integral/Model/PeriodicityClass`'s chain map E on $\mathcal{I}_\bullet$, which is natural in the coefficient module. **Not built**: (7)(a), the direct Smith form of d^0 of $C^\bullet_{\mathrm{eq}}(M_{\mathbb{Z}/\ell^K})$ as a production route at any level; (7)(b), the sparse cokernel over $\mathbb{Z}/\ell^K$ — `Hecke/HeckeLevelNSparse`'s `SparseCokernel` eliminates over $\mathbb{F}_p$ by RREF and is gated to a prime field (`ctx.is_field()`, `InvP(order, p)`, the constructor's `IsPrimeU64(p)`); its generators (`OrbitWiseFixedGenerators`: transported blocks on free orbits, the columns of the averaging idempotent $\frac{1}{|F_j|}\sum_{h \in F_j} \rho_n(h)$ at fixed points, orbit sums at $n = 0$)

are (7)(b)-admissible over $\mathbb{Z}/\ell^K$ once $1/|F_j|$ is taken mod ℓ^K; what is not built is the elimination over $\mathbb{Z}/\ell^K$ (minimal-valuation pivoting, the Smith read-out) (v1.23.0-pre.15, referee F3: pre.14 described those generators as kernel bases over $\mathbb{F}_p$, which they are not); the level-N descent assembly (D1's `IntegralDescentLevelNValidation`, with its arm (B), Proposition 31.8's list against $\mathbb{F}_\ell$ runs at every prime $\leq n+2$ including an $\ell = N = n+1$ cell); the matrix of E on $C^\bullet_{\mathcal{I}}(M_N)$; (8)'s integral form $M_N(\chi)$ (the build carries characters over $\mathbb{F}_q$ and $W_K(\mathbb{F}_q)$ only); every level-N Hecke operator over $\mathbb{Z}$ (C-D4). **A check outside the build, not a tag**: a script in exact modular arithmetic (`docs/implementation/slot24/cd2_badprimes_check.py`) compares s_ℓ with $\mathrm{rank}_{\mathbb{Q}} - \mathrm{rank}_{\mathbb{F}_\ell}$ ($\mathrm{rank}_{\mathbb{Q}}$ taken as the common rank at two primes $> 10^6$) of the cokernel presentation over $\mathbb{Z}$ on 280 cells ($N \in \{1,2,3,4,5,6,7,10,11,13\}$, even $n \leq 12$, $\ell \in \{5,7,11,13\}$) and, for the Legendre characters modulo $5, 7, 11, 13$, on 112 cells, and the twisted Borel count of (3)(b) with direct fixed spaces on 984 cells: 0 mismatches. **Proposition 31.9** (v1.23.0-pre.17, stated *after* D1's code, finding M1 of `docs/implementation/slot24/D1_code_audit.md`; on branch `s24-d1impl`, `726c3a67` and `87e8cf32`, not merged at pre.17). `Integral/Descent/DescentLevelN`: `RunSparseCokernelSmith`$(N, n, 3)$ is (3) — `FixedGenerators` builds (3)(a)'s orbit sums (S with $o = 2$, ST with $o = 3$, no division) and (3)(b)'s idempotent columns $\frac{1}{2}(1+B)$ at the S-fixed points (an `ENGINE_CHECK` that $\gcd(o, 3^K) = 1$ at a fixed point, and that the fixed-point counts are ν_2, ν_3), the elimination is sparse with unit pivots and the residual goes to `SmithLevelsEll`; $K =$ `ring3k::kMaxK` $= 19$ at once (no K search: $s < r$ is a refusal, the stopping rule at the cap), $r =$ `rank_T0` $- b_0$ from `LevelNClosedForm` (the closed forms of (3)); `AssembleDescentN` takes it when $\nu_3 = 0$ and puts nothing 3-primary in $H^{\geq 2}$, as (2) says, and runs $\mathcal{I}^{(3)}$ otherwise (item (5)); `force_dense_three` is the comparison arm (S) of `IntegralDescentLevelNValidation`, at $(N, n) = (5,4), (11,6), (2,12), (35,4), (8,6)$. n odd: only the 2-adic piece runs, item (4). (D1's code was merged into `hecke-extensions` after v1.23.0-pre.17, merge `1cc9a6d3`.) **Proposition 31.10** (v1.23.0-pre.18, **stated before D2's code**, contract first, rule 21; plan milestone D2). **Built** and used: the fibre maps on the 7-vertex model over $\mathbb{F}_\ell$, $\mathbb{Z}/\ell^K$ (§29.8's code), the $\widehat{\mathbb{Z}}_6$-corner on $\mathcal{I}^{(2)}$, $\mathcal{I}^{(3)}$ and the $\nu_3 = 0$ shortcut of D1 for one module at a time. **Not built**: (1) α, ω, R_σ on $C^\bullet_{\mathcal{I}}(-)$ and on $C^\bullet_{\mathcal{I}^{(\ell)}}(-)$ (block by block, (1)(b); a `FibreMap` applied to the column layout of Proposition 31.6(5)); (2) the level cone $C(\alpha_{\mathsf{P}})$ on $\mathcal{I}$, $\mathcal{I}^{(2)}$, $\mathcal{I}^{(3)}$ and on T at $\ell \geq 5$ and at $\ell = 3$, $\nu_3(N) = 0$, its certified-K read-out with the Betti recursion (2)(d)(ii), the sparse cokernel of Q over $\mathbb{Z}/\ell^K$ ((2)(f); the same missing elimination as for (7)(b)), and the glued sequence (78) to $q_{\max}$; (3) the Hecke squares (3)(a) over $\mathbb{Z}$ need the level-N operators of C-D4 (only their local components on the model over $\mathbb{F}_\ell$, $\ell \geq 5$, exist, §29.8); (4) the two-row read-out (4)(c) for $\Gamma_1(N) \trianglelefteq \Gamma_0(N)$. The plan's validation `IntegralDescentLESValidation` (arms (E) exactness, (S) split case, (I) $\mathrm{cor} \circ \mathrm{res} = [A : B]$, (L) against LES-1's predictions after $\otimes\mathbb{Z}[1/6]$, (2) the 2-adic pieces at $\Gamma_0(4) \trianglelefteq \Gamma_0(2)$, $\Gamma_0(9) \trianglelefteq \Gamma_0(3)$) is **not built**; the check `docs/implementation/slot24/cd3_descentmaps_check.py` (outside the build) is its oracle. (v1.23.0-pre.22:) **Built at D2** (`Integral/Descent/DescentMaps`, `IntegralDescentMapsValidation` and its slow companion, the plan's `IntegralDescentLESValidation` under another name): (1) block by block on $\mathcal{I}^{(2)}$, $\mathcal{I}^{(3)}$ and on M for the cokernel route; (2) the cone on $\mathcal{I}^{(2)}$, on $\mathcal{I}^{(3)}$ for $\nu_3(N) > 0$, the cokernel routes (2)(f) at $\ell \geq 5$ and at $\ell = 3$, $\nu_3(N) = 0$, the certified read-out (2)(d)(ii), the glued sequence to $q = 4$ (degree 6 in one arm), exactness as subgroups; every generator, coordinate and induced map read under (2)(g), its precisions tracked and refused when exhausted; (1)(d)'s identities and (4)(a), (4)(b) in degree 1 on normal pairs. Still not built: (3), (4)(c), (5). Oracles: `d2_maps_oracle.py` and, independent of it, the D2 code audit's `d2audit/d2audit_oracle.py`; report `DESCENT2_report.md`.

Formalization note. Propositions 31.7 and 31.8 (v1.23.0-pre.14), 31.9 (v1.23.0-pre.17) and 31.10 (v1.23.0-pre.18) are base change, Shapiro's lemma, the comparison theorem for projective resolutions, the semisimple derived category of $\widehat{\mathbb{Q}}_6$, invariant theory of $B(\mathbb{F}_\ell)$ on

$\mathbb{F}_\ell[x,y]$, the double coset formula and a two-row spectral sequence; they carry **no** \LEAN marker.

31.2 The tree models: the ℓ-local resolutions on the Bass–Serre tree, bulk only

(v1.23.0-pre.20; milestone D5 of `docs/implementation/slot24/DESCENT_plan.md` v0.4, clause C-D6 of its §2.7; the owner's ruling at gate G-T, 2026-10-25, on the evidence of `DESCENT1_report.md`.) §31 replaces the columns of the free model $\mathcal{I}_\bullet$ on X by their ℓ-local forms; this subsection does the same on the Bass–Serre tree $\mathcal{T}$ of Claim 16.2(1), whose free resolution is Remark 16.3. It fixes (i) **the ℓ-local tree models** $\mathcal{I}^{(\ell)}_{\mathcal{T}}$ (Definition 31.11): the cone of a lift of the tree's boundary between ℓ-local columns, with closed-form lifts at $\ell=2$, $\ell=3$ and $\ell\geq 5$; (ii) (Proposition 31.12) what they compute, in every degree and for every coefficient module, in particular $\mathrm{Ind}_{\Gamma_0(N)}^{\Gamma}$ of §31.1; their integral form over O_ℓ and the certified-exponent rule on it; their cochains in closed form; closed-form comparisons with the free tree resolution and *effective* comparisons with $\mathcal{I}$ over $\mathbb{Z}$, i.e. over the augmentation, constructed through the inclusion of the tree in X and an explicit retraction back (v1.23.0-pre.21); the fibre maps and the level cone of Proposition 31.10 on them; and the Hecke action by transport. **Bulk only**: nothing here concerns ∂X, $\mathcal{I}^{\partial}_{\bullet}$, the pair (Γ,Γ_∞), its long exact sequence or its cone — the tree cusp pair is the plan's clause C-D8, not stated in this document. *Notation.* $\mathbf{c}:=-I$ *as an element of* Γ (so $1-\mathbf{c}\neq 0$ in $\mathbb{Z}[\Gamma]$; it acts on a module M by $\rho_M(-I)$); $U=ST$ as in Claim 16.2; $N_S=\sum_{j<4}S^j$, $N_U=\sum_{j<6}U^j$ as in Remark 16.3; for $a\in R[\Gamma]$, r_a is right multiplication $y\mapsto ya$, a left $R[\Gamma]$-module map; for a subgroup H of finite order invertible in R, $e_H:=|H|^{-1}\sum_{h\in H}h$ and $M^H=\rho_M(e_H)M$. The symbols $x\in\{S,U,\mathfrak{e}\}$, Γ_x, g_x, P_x, Q_x, e_x, N'_x, F^x, φ', $\mathcal{I}_{\mathcal{T}}$, $\mathcal{I}^{(\ell)}_{\mathcal{T}}$, $C^\bullet_{\mathcal{T}}$, $C^\bullet_{\mathcal{T},\ell}$, $\Phi^{\mathcal{T}}_\ell$, $\Psi^{\mathcal{T}}_\ell$, $\Phi^{\mathbb{Z}}_{\mathcal{T}}$, $\Psi^{\mathbb{Z}}_{\mathcal{T}}$, $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$, k_j, κ_t, κ'_t, w, $T'_{\mathcal{T}}$ and (v1.23.0-pre.21) $\mathsf{i}_{\mathcal{T}}$, $\mathsf{r}_{\mathcal{T}}$, $\mathsf{G}_{\mathcal{T}}$, R_σ, h, h_p, (L), (L′), $\mathsf{H}_{\mathcal{T}}$, H_X, H_ℓ, K, K' are new symbols of this subsection and mean nothing else (rule (A5)); P_x, Q_x, e_x are those of Definition 31.1 for the three tree stabilisers, $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$ are the plan's names, and q_ℓ, O_ℓ are Proposition 31.6's.

Definition 31.11 (The free and the ℓ-local tree models)**.** Let x run over the two vertices and the edge of the tree, $x\in\{S,U,\mathfrak{e}\}$ for $\mathfrak{v}_S$, $\mathfrak{v}_U$, $\mathfrak{e}$ of Remark 16.3, with stabilisers $\Gamma_x=\langle g_x\rangle$, $g_S=S$, $g_U=U$, $g_{\mathfrak{e}}=\mathbf{c}$, of orders $4,6,2$. For a prime ℓ let P_x be the ℓ-Sylow subgroup of Γ_x, Q_x its subgroup of order prime to ℓ, and $e_x:=e_{Q_x}$; let R be a commutative ring in which every $|Q_x|$ is a unit, i.e. an O_ℓ-algebra, $\mathsf{O}_\ell=\mathbb{Z}[1/\mathsf{q}_\ell]$, $\mathsf{q}_\ell:=\mathrm{lcm}_x\,|Q_x|$:

	Q_S, P_S	Q_U, P_U	$Q_{\mathfrak{e}}$, $P_{\mathfrak{e}}$	q_ℓ
$\ell=2$	1, $\langle S\rangle$	$\langle U^2\rangle$, $\langle U^3\rangle=\langle\mathbf{c}\rangle$	1, $\langle\mathbf{c}\rangle$	3
$\ell=3$	$\langle S\rangle$, 1	$\langle\mathbf{c}\rangle$, $\langle U^2\rangle$	$\langle\mathbf{c}\rangle$, 1	4
$\ell\geq 5$	$\langle S\rangle$, 1	$\langle U\rangle$, 1	$\langle\mathbf{c}\rangle$, 1	12

(the values of q_ℓ of Proposition 31.6).

1. **(Columns.)** $F^x_t:=R[\Gamma]e_x$ for every $t\geq 0$ if $P_x\neq 1$, and for $t=0$ only if $P_x=1$; its differential is $\partial^x_t:=r_{(1-g_x)e_x}$ for odd t and $r_{N'_x}$ for even $t\geq 2$, $N'_x:=\sum_{j<|P_x|}g_x^j e_x$; its augmentation is the restriction to $F^x_0=R[\Gamma]e_x$ of the projection $R[\Gamma]\to R[\Gamma/\Gamma_x]$, $u\mapsto u\Gamma_x$, so $ye_x\mapsto y\Gamma_x$.
2. **(The model.)** An **ℓ-local tree model** is the mapping cone $\mathcal{I}^{(\ell)}_{\mathcal{T}}$ of a $R[\Gamma]$-chain map $\varphi'\colon F^{\mathfrak{e}}_\bullet\to F^S_\bullet\oplus F^U_\bullet$ that lifts the boundary of (50) in the orientation of Remark 16.3, $\gamma\langle\mathbf{c}\rangle\mapsto\gamma\langle S\rangle-\gamma\langle U\rangle$, through the augmentations:

$$\mathcal{I}^{(\ell)}_{\mathcal{T},q}=F^S_q\oplus F^U_q\oplus F^{\mathfrak{e}}_{q-1},\qquad \partial(a,z)=\big(\partial a+\varphi'_{q-1}z,\ -\partial^{\mathfrak{e}}z\big),$$

augmented by $F^S_0\oplus F^U_0\to R$, $(u,u')\mapsto\varepsilon(u)+\varepsilon(u')$, ε the augmentation of $R[\Gamma]$ (so $ye_x\mapsto\varepsilon(y)$). For a $R[\Gamma]$-module M put $C^\bullet_{\mathcal{T},\ell}(M):=\mathrm{Hom}_{R[\Gamma]}(\mathcal{I}^{(\ell)}_{\mathcal{T}},M)$.

3. **(The canonical lifts.)** Writing $\varphi'_t = (r_{a_t}, r_{b_t})$ (components into F_t^S, F_t^U):
   (a) $\ell = 2$: $e_S = e_{\mathfrak{e}} = 1$, $e_U = \frac{1}{3}(1+U^2+U^4)$; $(a_t, b_t) = (1, -e_U)$ for even t and $(1+S, -e_U)$ for odd t.
   (b) $\ell = 3$: $e_S = \frac{1}{4}N_S$, $e_U = e_{\mathfrak{e}} = \frac{1}{2}(1+\mathbf{c})$; $(a_0, b_0) = (e_S, -e_{\mathfrak{e}})$, and $F_t^{\mathfrak{e}} = 0$ for $t \geq 1$.
   (c) $\ell \geq 5$: $e_S = \frac{1}{4}N_S$, $e_U = \frac{1}{6}N_U$, $e_{\mathfrak{e}} = \frac{1}{2}(1+\mathbf{c})$; $(a_0, b_0) = (e_S, -e_U)$, and $F_t^x = 0$ for $t \geq 1$.

   The ℓ-local tree model $\mathcal{I}_{\mathcal{T}}^{(\ell)}$ is the cone of the canonical lift.
4. **(The free tree model.)** $\mathcal{I}_{\mathcal{T}}$ is the same construction with $Q_x = 1$ for every x (every column free and periodic, $e_x = 1$, $N'_x = \sum_{u \in \Gamma_x} u$), over $\mathbb{Z}$ or any R, with the lift $(a_t, b_t) = (1, -1)$ for even and $(1+S, -(1+U+U^2))$ for odd t: the resolution $P_\bullet$ of Remark 16.3; $C^\bullet_{\mathcal{T}}(M) := \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_{\mathcal{T}}, M)$.

Proposition 31.12 (The tree models: what they compute, closed forms, the comparisons with $\mathcal{I}$ over $\mathbb{Z}$, fibre maps, Hecke by transport)**.** *[proved here] [standard: [Wei94, Thm. 2.2.6], [Bro82, Ch. III.6.2]] (v1.23.0-pre.20; milestone D5 of* `docs/implementation/slot24/DESCENT_plan.md` *v0.4, clause C-D6; refereed ACCEPT-with-fixes,* `docs/implementation/slot24/C-D6_referee.md`*; its fixes M1, m1–m5 applied at v1.23.0-pre.21 — (4) made effective —, not re-refereed.) Assume (H4) and (B1)–(B3); let ℓ be a prime, R an O_ℓ-algebra and M a $R[\Gamma]$-module; for the integral statements let M be a $\mathbb{Z}[\Gamma]$-module finitely generated and free over $\mathbb{Z}$ (as in Proposition 31.7: V, M_N, $M_N(\chi)$) and $M_R := M \otimes R$.*

1. ***(What the models compute.)***
   (a) *The canonical lifts of Definition 31.11(3) are well defined ($a_t \in e_{\mathfrak{e}}R[\Gamma]e_S$, $b_t \in e_{\mathfrak{e}}R[\Gamma]e_U$) and are chain maps lifting the boundary; so is the lift of (4). Any two lifts are homotopic, and their cones are isomorphic by an isomorphism that is the identity on $F^S \oplus F^U$ and on the quotient $F^{\mathfrak{e}}[-1]$; so every ℓ-local tree model is isomorphic to* the *one.*
   (b) *$\mathcal{I}_{\mathcal{T}}^{(\ell)} \to R$ is a resolution of R by finitely generated projective $R[\Gamma]$-modules, $R[\Gamma]e_x \cong R[\Gamma/Q_x]$ — an R-model in the sense of Proposition 31.10 — and, restricted, a projective resolution of R over $R[A]$ for every subgroup $A \leq \Gamma$. $\mathcal{I}_{\mathcal{T}} \to \mathbb{Z}$ is a free resolution over $\mathbb{Z}[\Gamma]$ (Remark 16.3).*
   (c) *$C^q_{\mathcal{T},\ell}(M) = \bigoplus_x M^{Q_x}$, the sum over the cells x of degree q (the columns F_q^S, F_q^U, $F_{q-1}^{\mathfrak{e}}$ that are non-zero), and $H^q(C^\bullet_{\mathcal{T},\ell}(M)) = H^q(\Gamma; M)$ for* ***every*** *$q \geq 0$ and every $R[\Gamma]$-module M; for $M = \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V$ this is $H^q(\Gamma_0(N); V)$ (Shapiro's lemma, [standard: [Bro82, Ch. III.6.2]]), in particular for $M_{N,R}$ and $M_N(\chi)_R$ of §31.1. If R is a principal ideal domain of characteristic 0 and M is finitely generated and free over R, each M^{Q_x} is free of rank $\frac{1}{|Q_x|}\sum_{w \in Q_x} \mathrm{tr}\, \rho_M(w)$.*
   (d) ***(One integral form; the certified-exponent rule.)*** *The canonical models have all coefficients in $\mathsf{O}_\ell[\Gamma]$ (the free one in $\mathbb{Z}[\Gamma]$). For every O_ℓ-algebra R, $\mathcal{I}^{(\ell)}_{\mathcal{T},\mathsf{O}_\ell} \otimes_{\mathsf{O}_\ell} R$ is the canonical model over R; no lift is chosen per ring. Proposition 31.6(2)–(6) hold verbatim with $C^\bullet_{\mathcal{T},\ell}(M_{\mathsf{O}_\ell})$ in place of $C^\bullet_\ell$, with the same $\mathfrak{q}_\ell$: the integral cochains are free over O_ℓ with $H^q = H^q(\Gamma; M) \otimes \mathsf{O}_\ell$; the rule over $\mathbb{Z}_{(\ell)}$; $\mathrm{rank}_{\mathbb{Q}}\, d^q$ from exactly the two sources of (4)(c) — the reduction mod a qualifying $\ell' \nmid \mathfrak{q}_\ell$ of* ***this*** *complex, or the Betti recursion — the rank of a differential of $\mathcal{I}^{(\ell)}$, of $\mathcal{I}$ or of the free tree being* ***not*** *admissible for it, nor conversely; the full layout; the Smith-valuation read-out with the a-priori bound $\sigma_i \leq v_\ell(24)$ for $q \geq 1$ and the stopping rule.*
2. ***(The cochains in closed form.)*** *For every $R[\Gamma]$-module M, with $x \in M$ or M^{Q_S}, $y \in M^{Q_U}$, $z \in M^{Q_{\mathfrak{e}}}$ the components on F_q^S, F_q^U, $F_{q-1}^{\mathfrak{e}}$:*

(a) $\ell=2$: $C^0=M\oplus M^{U^2}$, $C^q=M\oplus M^{U^2}\oplus M$ $(q\ge 1)$, and

$$\begin{aligned} d^0(x,y)&=\big((1-S)x,\ (1-\mathbf{c})y,\ x-y\big),\\ d^q(x,y,z)&=\big(N_S x,\ (1+\mathbf{c})y,\ (1+S)x-y-(1-\mathbf{c})z\big)\qquad (q\ \text{odd}),\\ d^q(x,y,z)&=\big((1-S)x,\ (1-\mathbf{c})y,\ x-y-(1+\mathbf{c})z\big)\qquad (q\ge 2\ \text{even}).\end{aligned}$$

(b) $\ell=3$: $C^0=M^{\langle S\rangle}\oplus M^{-I}$, $C^1=M^{-I}\oplus M^{-I}$ (the U-column in degree 1 and the edge), $C^q=M^{-I}$ $(q\ge 2)$, and

$$d^0(x,y)=\big((1-U)y,\ x-y\big),\qquad d^1(y,z)=(1+U+U^2)y,$$
$$d^q=1-U\quad (q\ge 2\ \text{even}),\qquad d^q=1+U+U^2\quad (q\ge 3\ \text{odd}).$$

In degrees ≥ 2 this is the periodic complex of the cyclic group $\langle U^2\rangle$ acting on M^{-I} (on which U acts as U^{-2}), whence $H^q(\Gamma;M)\cong H^q(\langle U^2\rangle;M^{-I})$ for $q\ge 2$ — Proposition 31.3(2), read off the cochains.

(c) $\ell\ge 5$: $C^0=M^{\langle S\rangle}\oplus M^{\langle U\rangle}$, $C^1=M^{-I}$, $d^0(x,y)=x-y$, $C^q=0$ $(q\ge 2)$: the tree complex (65) up to the sign of d^0.

(d) ***(Periodicity.)*** At $\ell=2$, $C^{q+2}=C^q$ and $d^{q+2}=d^q$ for $q\ge 1$; at $\ell=3$ for $q\ge 2$. In both cases $H^{q+2}(\Gamma;M)=H^q(\Gamma;M)$ as groups for $q\ge 2$, read on the same cochains, so a run built to degree $q_{\max}+1$ with $q_{\max}=3$ yields every degree (that this identification is $e\cup -$ of Proposition 31.7(5) is **not** asserted).

(e) ***(Ranks.)*** With $r_H:=\operatorname{rank} M^H$ (for M as in (1)(c); $r_1=\operatorname{rank} M$): $\operatorname{rank} C^q$ is $r_1+r_{U^2}$, $2r_1+r_{U^2}$ $(q\ge 1)$ at $\ell=2$; $r_{\langle S\rangle}+r_{-I}$, $2r_{-I}$, r_{-I} $(q\ge 2)$ at $\ell=3$; $r_{\langle S\rangle}+r_{\langle U\rangle}$, r_{-I} at $\ell\ge 5$; and $2r_1$, $3r_1$ for the free tree.

3. ***(Comparison with the free tree, in closed form.)*** Let ℓ be a prime and R an O_ℓ-algebra. Define $\Phi_\ell^{\mathcal{T}}:\mathcal{I}_{\mathcal{T}}\otimes R\to\mathcal{I}_{\mathcal{T}}^{(\ell)}$ and $\Psi_\ell^{\mathcal{T}}:\mathcal{I}_{\mathcal{T}}^{(\ell)}\to\mathcal{I}_{\mathcal{T}}\otimes R$ by

$$\Phi_\ell^{\mathcal{T}}(a,z)=\big(f\,a+k\,z,\ f\,z\big),\qquad \Psi_\ell^{\mathcal{T}}(a,z)=\big(g\,a+k'z,\ g\,z\big),$$

where on each column $f_t^x:=r_{|Q_x|^{\lfloor t/2\rfloor}e_x}$ and g_t^x is $|Q_x|^{-\lfloor t/2\rfloor}$ times the inclusion $R[\Gamma]e_x\subseteq R[\Gamma]$ (both 0 where the ℓ-local column vanishes), and the corrections $k_t\colon F_t^{\mathfrak{e}}\to F_{t+1}^U$ (free source, ℓ-local target), k_t' from the ℓ-local $F_t^{\mathfrak{e}}$ to the free vertex columns of degree $t+1$, are:

(a) $\ell=2$: $k_t=r_{-\mathsf{k}_{\lfloor t/2\rfloor}e_U}$ with $\mathsf{k}_j:=\frac{3^j-1}{2}$; $k_t'=r_{\kappa_t'}$ into F_{t+1}^U, with $\kappa_t':=\frac{1-3^{-\lceil t/2\rceil}}{2}\,e_U+[t\ \text{even}]\,\mathsf{w}$ and $\mathsf{w}:=\frac{1}{3}(2+2U+U^2+U^3)-(1+\mathbf{c})e_U$;

(b) $\ell=3$: $k_t=r_{-2^{(t-1)/2}e_{\mathfrak{e}}}$ for odd t, $k_t=0$ for even t; $k_0'=r_{-\frac12 e_{\mathfrak{e}}}$ into F_1^S (and $k_t'=0$, $t\ge 1$, its source being 0);

(c) $\ell\ge 5$: $k=0$; $k_0'=\big(r_{-\frac12 e_{\mathfrak{e}}},\ r_{\frac13 e_{\mathfrak{e}}(2+2U+U^2+U^3)}\big)$ into $F_1^S\oplus F_1^U$.

Then $\Phi_\ell^{\mathcal{T}}$ and $\Psi_\ell^{\mathcal{T}}$ are $R[\Gamma]$-chain maps over id_R, defined over O_ℓ, with

$$\Phi_\ell^{\mathcal{T}}\circ\Psi_\ell^{\mathcal{T}}=\mathrm{id}\quad\text{strictly},\qquad \Psi_\ell^{\mathcal{T}}\circ\Phi_\ell^{\mathcal{T}}\simeq\mathrm{id}.$$

Hence $\Psi_\ell^{\mathcal{T}*}\Phi_\ell^{\mathcal{T}*}=\mathrm{id}$ on $C^\bullet_{\mathcal{T},\ell}(M)$ and $\Phi_\ell^{\mathcal{T}*}\Psi_\ell^{\mathcal{T}*}\simeq\mathrm{id}$ on $C^\bullet_{\mathcal{T}}(M)$: $C^\bullet_{\mathcal{T},\ell}(M)$ ***is a retract of*** $C^\bullet_{\mathcal{T}}(M)$ ***with*** $\Psi_\ell^{\mathcal{T}*}\Phi_\ell^{\mathcal{T}*}=\mathrm{id}$ ***strictly and*** $\Phi_\ell^{\mathcal{T}*}\Psi_\ell^{\mathcal{T}*}\simeq\mathrm{id}$, the homotopy being H_ℓ^* for the effective H_ℓ of (4)(c) (no side conditions such as $\mathsf{h}\Phi^*=0$, $\Psi^*\mathsf{h}=0$, $\mathsf{h}^2=0$ are asserted), and $\Phi_\ell^{\mathcal{T}*}$, $\Psi_\ell^{\mathcal{T}*}$ are mutually inverse on cohomology. On cochains, with $m:=\lfloor q/2\rfloor$ and the components of (2): at $\ell=2$, $\Phi_2^{\mathcal{T}*}(x,y,z)=(x,\,3^m y,\,z-\mathsf{k}_m y)$ and $\Psi_2^{\mathcal{T}*}(x,y,z)=(x,\,3^{-m}e_U y,\,z+\kappa'_{q-1}y)$ for $q\ge 1$, the identity and $(x,y)\mapsto(x,e_U y)$ for $q=0$; at $\ell=3$, $\Phi_3^{\mathcal{T}*}$ is $(x,y)\mapsto(x,y)$, $(y,z)\mapsto(0,y,z)$ and, for $q\ge 2$, $y\mapsto(0,\,2^m y,\,-[q\ \text{even}]\,2^{m-1}y)$, and $\Psi_3^{\mathcal{T}*}$ is $(x,y)\mapsto(e_S x,e_{\mathfrak{e}}y)$, $(x,y,z)\mapsto\big(e_{\mathfrak{e}}y,\,e_{\mathfrak{e}}(z-\frac12 x)\big)$ and, for $q\ge 2$, $(x,y,z)\mapsto 2^{-m}e_{\mathfrak{e}}y$.

4. ***(Comparison with $\mathcal{I}$, over $\mathbb{Z}$: effective.)*** *(v1.23.0-pre.21; the referee's M1.) Simplices of K stand for their lifts into X (Lemma 7.3), Σ_s, Γ_σ, g_σ, m_σ, $G_{s,t}$, $d^{(r)}$ and* cmp *are those of Definition 23.4, and $\mathcal{I}_\bullet$ is any isotropy-resolution model (e.g. the one of Proposition 23.5(1)'s procedure, which the code builds).*

   (a) ***(The tree in X, a retraction, a homotopy.)*** *The vertices in the Γ-orbits of $v_6 = i$ and $v_4 = \zeta^2$ and the edges in the orbit of $\{v_4, v_6\}$ form a Γ-subcomplex of X, and $\mathsf{i}_\mathcal{T}\colon C_\bullet(\mathcal{T}) \to C_\bullet(X)$, $\gamma\mathfrak{v}_S \mapsto \gamma v_6$, $\gamma\mathfrak{v}_U \mapsto \gamma v_4$, $\gamma\mathfrak{e} \mapsto \gamma\{v_4, v_6\}$ (stored orientation), is an injective $\mathbb{Z}[\Gamma]$-chain map preserving augmentations. Define $\mathsf{r}_\mathcal{T}\colon C_\bullet(X) \to C_\bullet(\mathcal{T})$ and $\mathsf{G}_\mathcal{T}\colon C_s(X) \to C_{s+1}(X)$ $\mathbb{Z}[\Gamma]$-linearly by their values on the representatives:*

$\sigma \in \Sigma_s$	$\mathsf{r}_\mathcal{T}(\sigma)$	$\mathsf{G}_\mathcal{T}(\sigma)$
v_0	$\mathfrak{v}_U$	$-\{v_0, v_4\}$
v_1; v_2	$\mathfrak{v}_S$; $\mathfrak{v}_S$	$-\{v_1, v_6\}$; $-\{v_2, v_6\}$
v_4; v_6	$\mathfrak{v}_U$; $\mathfrak{v}_S$	0; 0
$\{v_0, v_1\}$	$\mathfrak{e}$	$\{v_0, v_1, v_4\} - \{v_1, v_4, v_6\}$
$\{v_1, v_2\}$; $\{v_1, v_4\}$	0; $-\mathfrak{e}$	$\{v_1, v_2, v_6\}$; $\{v_1, v_4, v_6\}$
$\{v_2, v_3\}$	$-S\mathfrak{e}$	$\{v_2, v_3, v_5\} + \{v_2, v_5, v_6\}$
$\{v_2, v_5\}$	$-S\mathfrak{e}$	$\{v_2, v_5, v_6\}$
$\{v_4, v_6\}$	$\mathfrak{e}$	0
$\{v_0, v_4\}$, $\{v_1, v_6\}$, $\{v_2, v_6\}$	0	0
the five triangles	0	0

   Then $\mathsf{r}_\mathcal{T}$ and $\mathsf{G}_\mathcal{T}$ are well defined, $\mathsf{r}_\mathcal{T}$ is a $\mathbb{Z}[\Gamma]$-chain map preserving augmentations, $\mathsf{r}_\mathcal{T} \circ \mathsf{i}_\mathcal{T} = \mathrm{id}$, and $\mathrm{id} - \mathsf{i}_\mathcal{T} \circ \mathsf{r}_\mathcal{T} = \partial\mathsf{G}_\mathcal{T} + \mathsf{G}_\mathcal{T}\partial$ on $C_\bullet(X)$: the equivalence of Proposition 23.3(2), in closed form.

   (b) ***(Contractions and the lifting formula.)*** *Fix for each $\sigma \in \Sigma_0 \sqcup \Sigma_1 \sqcup \Sigma_2 \sqcup \{S, U, \mathfrak{e}\}$ a set $\mathsf{R}_\sigma \subset \Gamma$ of representatives of Γ/Γ_σ (any choice; the code's is* `decompose()` *of* `Integral/Model/PeriodicColumn.h`*, shared by* `IsotropyModel`*,* `IsotropyModelEll`*,* `StrictLift`*,* `RefinedStrictLift` *and* `PeriodicityClass` *(v1.23.0-pre.23, referee t2) and documented at* `GKey` *in* `IsotropyModel.h`*: the least matrix of the coset in the* ***signed*** *lexicographic order on $(a, b, c, d) \in \mathbb{Z}^4$, and j taken relative to the* ***fixed*** *generator g_σ — v1.23.0-pre.22, re-referee N1); write $\gamma = \rho g_\sigma^j$ with $\rho \in \mathsf{R}_\sigma$, $0 \le j < m_\sigma$. On a column $\mathbb{Z}[\Gamma]$ with differentials r_{g-1} (odd) and r_N (even, N the norm of $\langle g\rangle$), as in Definition 23.4, put*

$$\mathsf{h}(\rho g^j) := \rho \textstyle\sum_{i<j} g^i \text{ (from even degrees)}, \qquad \mathsf{h}(\rho g^j) := [j = m{-}1]\,\rho \text{ (from odd degrees)},$$

   and $\mathsf{h}_{-1}(\gamma\Gamma_\sigma) := \rho$; for the tree's columns (r_{1-g}, r_N) negate h from even degrees; on a column that enters a total complex with a global sign ϵ ($G_{s,\bullet}$ with $\epsilon = (-1)^s$; the edge column of a cone, $\epsilon = -1$) use $\epsilon\mathsf{h}$. Let p be $\mathrm{cmp}\colon \mathcal{I}_\bullet \to C_\bullet(X)$ or the column augmentation $\varepsilon_\mathcal{T}\colon \mathcal{I}_\mathcal{T} \to C_\bullet(\mathcal{T})$ (vertex columns $s = 0$, edge column $s = 1$; a surjective quasi-isomorphism, (1)(b)'s proof). For a cycle z of $\ker p$ define $\mathsf{h}_p(z)$ by descending induction on the column index s: if z_s is the component of z on the largest s, put $k_s := \epsilon\mathsf{h}(z_s)$ column by column, replace z by $z - \partial k_s$ and continue; after at most three steps $z = 0$, and $\mathsf{h}_p(z) := \sum k_s$ satisfies $\partial\mathsf{h}_p(z) = z$ with $\mathsf{h}_p(z)$ in column degrees ≥ 1, so $p\,\mathsf{h}_p(z) = 0$. Let P' be a bounded-below complex of free $\mathbb{Z}[\Gamma]$-modules with given bases and $f\colon P' \to C$ a chain map into the target of p. For a basis element x, with y_0 any preimage of $f(x)$ under p (e.g. $\rho \otimes 1$ in $G_{s,0}$, resp. F_0^x, for the basis element $\rho\sigma$),

$$(L) \qquad \hat{f}(x) := y_0 + \mathsf{h}_p\big(\hat{f}(\partial x) - \partial y_0\big),$$

   extended $\mathbb{Z}[\Gamma]$-linearly, is a chain map with $p\hat{f} = f$: Lemma 23.12, made effective. Likewise, for chain maps $A, B\colon P' \to E$ (E the source of p) with $pA = a\,p'$, $pB = b\,p'$

for a $\mathbb{Z}[\Gamma]$-chain map $p'\colon P'\to C'$ (no surjectivity is needed; v1.23.0-pre.22, re-referee N3) and $a-b=\partial g+g\partial$ for a $\mathbb{Z}[\Gamma]$-map $g\colon C'_\bullet\to C_{\bullet+1}$, and y_1 a preimage of $g\,p'(x)$,

$$(L')\qquad H(x):=y_1+\mathsf{h}_p\big(Ax-Bx-H(\partial x)-\partial y_1\big)$$

*gives $\partial H+H\partial=A-B$ and $pH=g\,p'$. (L) and (L′) are finite in each degree, but the maps they produce are **not periodic** in the degree: with the code's representatives the number of terms per generator is bounded, but the integer coefficients grow linearly in q (so (2)'s periodicity applies to the cochains, not to the comparisons); they are built to the degree in use — to $q+1$ for $\Phi^{\mathbb{Z}}_{\mathcal{T}}$, $\Psi^{\mathbb{Z}}_{\mathcal{T}}$ and to q for the homotopies when an operator is transported in degree q (v1.23.0-pre.22, re-referee N2).*

(c) ***(The integral pair and its homotopies.)*** *$\Phi^{\mathbb{Z}}_{\mathcal{T}}\colon\mathcal{I}_\bullet\to\mathcal{I}_{\mathcal{T}}$ is (L) for $f=\mathsf{r}_{\mathcal{T}}\circ\mathrm{cmp}$ along $p=\varepsilon_{\mathcal{T}}$, and $\Psi^{\mathbb{Z}}_{\mathcal{T}}\colon\mathcal{I}_{\mathcal{T}}\to\mathcal{I}_\bullet$ is (L) for $f=\mathsf{i}_{\mathcal{T}}\circ\varepsilon_{\mathcal{T}}$ along $p=\mathrm{cmp}$: $\mathbb{Z}[\Gamma]$-chain maps over $\mathsf{r}_{\mathcal{T}}$, resp. $\mathsf{i}_{\mathcal{T}}$, hence over $\mathrm{id}_{\mathbb{Z}}$ (compatible with $\varepsilon\circ\mathrm{cmp}$ and the augmentation of $\mathcal{I}_{\mathcal{T}}$). By (L′): $\mathsf{H}_{\mathcal{T}}$ ($A=\Phi^{\mathbb{Z}}_{\mathcal{T}}\Psi^{\mathbb{Z}}_{\mathcal{T}}$, $B=\mathrm{id}$, along $\varepsilon_{\mathcal{T}}$, $g=0$ as $\mathsf{r}_{\mathcal{T}}\mathsf{i}_{\mathcal{T}}=\mathrm{id}$) with $\partial\mathsf{H}_{\mathcal{T}}+\mathsf{H}_{\mathcal{T}}\partial=\Phi^{\mathbb{Z}}_{\mathcal{T}}\Psi^{\mathbb{Z}}_{\mathcal{T}}-\mathrm{id}$; H_X ($A=\mathrm{id}$, $B=\Psi^{\mathbb{Z}}_{\mathcal{T}}\Phi^{\mathbb{Z}}_{\mathcal{T}}$, along cmp, $g=\mathsf{G}_{\mathcal{T}}$) with $\partial\mathsf{H}_X+\mathsf{H}_X\partial=\mathrm{id}-\Psi^{\mathbb{Z}}_{\mathcal{T}}\Phi^{\mathbb{Z}}_{\mathcal{T}}$; and, for each ℓ and R, H_ℓ ($A=\Psi^{\mathcal{T}}_\ell\Phi^{\mathcal{T}}_\ell$, $B=\mathrm{id}$ on $\mathcal{I}_{\mathcal{T}}\otimes R$, along $\varepsilon_{\mathcal{T}}\otimes R$, $g=0$) with $\partial\mathsf{H}_\ell+\mathsf{H}_\ell\partial=\Psi^{\mathcal{T}}_\ell\Phi^{\mathcal{T}}_\ell-\mathrm{id}$. $\Phi^{\mathbb{Z}}_{\mathcal{T}}$, $\Psi^{\mathbb{Z}}_{\mathcal{T}}$ are built once over $\mathbb{Z}$, independently of ℓ; another choice of $\mathcal{I}_\bullet$, of the R_σ or of the preimages changes them by a homotopy; as comparisons over $\mathrm{id}_{\mathbb{Z}}$ they are unique up to homotopy.*

(d) ***(Over R.)*** *Put*

$$\Phi_{\mathcal{T}}:=\Phi^{\mathcal{T}}_\ell\circ(\Phi^{\mathbb{Z}}_{\mathcal{T}}\otimes R)\colon\mathcal{I}_\bullet\otimes R\to\mathcal{I}^{(\ell)}_{\mathcal{T}},\qquad\Psi_{\mathcal{T}}:=(\Psi^{\mathbb{Z}}_{\mathcal{T}}\otimes R)\circ\Psi^{\mathcal{T}}_\ell\colon\mathcal{I}^{(\ell)}_{\mathcal{T}}\to\mathcal{I}_\bullet\otimes R.$$

*Then $\Phi_{\mathcal{T}}\Psi_{\mathcal{T}}-\mathrm{id}=\partial\mathsf{K}+\mathsf{K}\partial$ with $\mathsf{K}:=\Phi^{\mathcal{T}}_\ell\mathsf{H}_{\mathcal{T}}\Psi^{\mathcal{T}}_\ell$, and $\mathrm{id}-\Psi_{\mathcal{T}}\Phi_{\mathcal{T}}=\partial\mathsf{K}'+\mathsf{K}'\partial$ with $\mathsf{K}':=\mathsf{H}_X-\Psi^{\mathbb{Z}}_{\mathcal{T}}\mathsf{H}_\ell\Phi^{\mathbb{Z}}_{\mathcal{T}}$ (all $\otimes R$). So $\Phi^*_{\mathcal{T}}\colon C^\bullet_{\mathcal{T},\ell}(M)\to C^\bullet_{\mathcal{I}}(M):=\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_\bullet,M)$ and $\Psi^*_{\mathcal{T}}$ are mutually inverse cochain homotopy equivalences, with the explicit homotopies K^*, K'^*, compatible with the identifications of both sides with $H^\bullet(\Gamma;M)$; every other comparison between the two (e.g. a strict lift through $\mathcal{I}^{(\ell)}_{\mathcal{T}}$ directly, or one through $\mathcal{I}^{(\ell)}$ and Φ_ℓ, Ψ_ℓ — Proposition 31.2(4), whose proof uses of R only that every $|Q_\sigma|$ is a unit, so it holds over every O_ℓ-algebra) is homotopic to these. The comparisons are maps **over the augmentation to** $\mathbb{Z}$ (resp. R); (4) needs no chain map between $\mathcal{I}^{(\ell)}_{\mathcal{T}}$ and $C_\bullet(X)$: the effective construction passes through $C_\bullet(\mathcal{T})\rightleftarrows C_\bullet(X)$, at the integral level only.*

5. ***(Fibre maps and the level cone: (T1)–(T3) hold.)*** *For $\ell\in\{2,3\}$, $\mathcal{I}^{(\ell)}_{\mathcal{T}}$ satisfies (T1)–(T3) of Proposition 31.10(5): it is an S-model for $S=\mathsf{O}_\ell$, a localisation of $\mathbb{Z}$ in which ℓ is not inverted, and for $S=\mathbb{Z}_{(\ell)}$ ((1)(b), (1)(d)); its terms are $\bigoplus_x S[\Gamma/Q_x]$ with $|Q_x|\in S^\times$ ((1)(c)); and (4) gives the comparisons and the homotopies between their composites explicitly, the integral pair $\Phi^{\mathbb{Z}}_{\mathcal{T}}$, $\Psi^{\mathbb{Z}}_{\mathcal{T}}$ included (v1.23.0-pre.21). Hence Proposition 31.10(1), (2) and (4) hold on it verbatim: a $\mathbb{Z}[\Gamma]$-map u of fibres (restriction α, trace ω, R_σ, the diamonds, the O_χ-action) acts on $C^\bullet_{\mathcal{T},\ell}$ block by block, on each M^{Q_x}, and commutes **strictly** with $\Phi^*_{\mathcal{T}}$, $\Psi^*_{\mathcal{T}}$, $\Phi^{\mathcal{T}*}_\ell$, $\Psi^{\mathcal{T}*}_\ell$; the level cone $C(\alpha_{\mathcal{T}})$ on $\mathcal{I}^{(\ell)}_{\mathcal{T}}$ is exact in the sense of (2)(a) there, computes $H^{\bullet-1}(A;Q_{A/B}\otimes S)$, carries the long exact sequence (78) over $\mathbb{Z}_{(\ell)}$ in every degree, and every comparison above induces a quasi-isomorphism of cones commuting with both maps of the cone's short exact sequence; the certified-exponent rule of (2)(d) there holds for D^q on it, with source (i) read on the reduction mod a qualifying $\ell'\nmid\mathsf{q}_\ell$ of **this** cone on $\mathcal{I}^{(\ell)}_{\mathcal{T},\mathsf{O}_\ell}$. So the ℓ-local pieces of Proposition 31.7(2) and of Proposition 31.10(1)(c), (2)(c) may be read on the tree models at $\ell=2$ and $\ell=3$, with no change to the glueing datum cmp^* of the $\mathbb{Z}[1/6]$-piece. This is a statement about the groups and the maps induced by fibre maps; any other map between the glued groups (an explicit fibre-product isomorphism, a Hecke operator through Proposition 31.7(3)) enters the ℓ-local piece through $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$ of (4)(d).*

6. ***(Hecke by transport.)*** *Let T be* any *cochain endomorphism of $C^\bullet_{\mathcal{I}}(M_R)$ inducing an endomorphism t of $H^\bullet(\Gamma;M_R)$ — e.g. $\Theta_{\mathcal{I},g}$ of Proposition 23.13(3), $g\in\Delta$, at level 1; (a)–(d) are stated for every such T, U_p-type cochain operators included once they exist, and are instantiated at level N by the cochain-level T_p, $p\nmid N$, of the plan's clause C-D4 once it is stated (the plan's ruling [R-Q4]: U_p deferred) — and put*

$$T'_{\mathcal{T}} := \Psi^*_{\mathcal{T}}\circ T\circ\Phi^*_{\mathcal{T}}\colon C^\bullet_{\mathcal{T},\ell}(M_R)\to C^\bullet_{\mathcal{T},\ell}(M_R).$$

   (a) *$T'_{\mathcal{T}}$ is a cochain map inducing t under the identification of (1)(c), in every degree; up to cochain homotopy it does not depend on the choice of $\Phi^{\mathbb{Z}}_{\mathcal{T}}$, $\Psi^{\mathbb{Z}}_{\mathcal{T}}$ (nor of any comparison homotopic to $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$).*

   (b) ***(It equals the operator on $\mathcal{I}^{(\ell)}$.)*** *For $T'' := \Psi^*_\ell\circ T\circ\Phi^*_\ell$ on $C^\bullet_{\mathcal{I}^{(\ell)}}(M_R)$ (Proposition 31.2(5), which holds over every O_ℓ-algebra R: its proof, and that of (4) there, use of R only that every $|Q_\sigma|$ is a unit) and every comparison $\vartheta\colon\mathcal{I}^{(\ell)}_{\mathcal{T}}\to\mathcal{I}^{(\ell)}$ over id_R, $H(\vartheta^*)\circ H(T'')=H(T'_{\mathcal{T}})\circ H(\vartheta^*)$: the two transports are the same operator on $H^\bullet(\Gamma;M_R)$. An effective such comparison is $\vartheta:=\Phi_\ell\circ\Psi_{\mathcal{T}}$ (Φ_ℓ of Proposition 31.2(4), $\Psi_{\mathcal{T}}$ of (4)(d); built from constructed pieces only), the one a numerical check uses (v1.23.0-pre.22, re-referee N4; wording v1.23.0-pre.23, referee t1).*

   (c) *For two such T_1, T_2, $T'_{1,\mathcal{T}}T'_{2,\mathcal{T}}\simeq(T_1T_2)'_{\mathcal{T}}$; so every relation that holds among the T_i up to cochain homotopy holds among their transports up to homotopy, and on cohomology exactly.*

   (d) *If T commutes strictly with $\mathrm{Hom}(\mathcal{I}_\bullet,u)$ for a fibre map u, then $T'_{\mathcal{T}}$ commutes strictly with u on $C^\bullet_{\mathcal{T},\ell}$; given level-A, level-B operators with $T^B\alpha_{\mathcal{I}}=\alpha_{\mathcal{I}}T^A$ strictly (the plan's C-D4), $(T'^A_{\mathcal{T}}[1],T'^B_{\mathcal{T}})$ is a cochain endomorphism of $C(\alpha_{\mathcal{T}})$ commuting with its short exact sequence, so Proposition 31.10(3)(c)'s equivariance holds on the tree.*

   No τ-type table on the tree is used: the Hecke datum stays on $\mathcal{I}$, and applying $T'_{\mathcal{T}}$ composes cochain maps, all of them effective by (4).

Scope. *Bulk only. Nothing here is stated about the boundary ∂X, $\mathcal{I}^\partial_\bullet$, the pair (Γ,Γ_∞), its long exact sequence, its cone, or a Hecke action on any of them (the plan's C-D8, tree cusp pair; v1.23.0-pre.33: §31.5); nothing about the Bredon object $C^\bullet_{\mathrm{eq}}$ at $\ell\in\{2,3\}$; and no native Hecke datum on the tree.*

Proof. A cone lemma (∗). Let $\varphi\colon E\to V$, $\varphi'\colon E'\to V'$ be chain maps, $f^V\colon V\to V'$, $f^E\colon E\to E'$ chain maps and $k\colon E_t\to V'_{t+1}$ maps with $\partial k+k\partial=f^V\varphi-\varphi' f^E$. Then $F(a,z):=(f^Va+kz,\ f^Ez)$ is a chain map $C(\varphi)\to C(\varphi')$ (cones as in Definition 31.11(2)): $\partial F(a,z)=(\partial f^Va+\partial kz+\varphi' f^Ez,\ -\partial f^Ez)$ and $F\partial(a,z)=(f^V\partial a+f^V\varphi z-k\partial z,\ -f^E\partial z)$ agree exactly when the displayed identity holds. With $V=V'$, $E=E'$, $f=\mathrm{id}$ and $\varphi_2-\varphi_1=\partial h+h\partial$ it gives the isomorphism $(a,z)\mapsto(a+hz,z)$ from $C(\varphi_2)$ to $C(\varphi_1)$, inverse $(a,z)\mapsto(a-hz,z)$. *Conventions.* Every map below between the modules $R[\Gamma]e_x$ is r_a for an $a\in e_yR[\Gamma]e_x$ (well defined on $R[\Gamma]e_y$ exactly then); $r_b\circ r_a=r_{ab}$; all elements that occur lie in the commutative rings $R[\langle S\rangle]$ or $R[\langle U\rangle]$, which contain the central $\mathbf{c}=S^2=U^3$, and in them e_U, $e_{\mathfrak{e}}$, e_S commute with everything. Repeatedly used: $Ue_U=\mathbf{c}e_U$ at $\ell=2$ ($U=\mathbf{c}U^{-2}$ and $U^{-2}e_U=e_U$), so $(1-U)e_U=(1-\mathbf{c})e_U$, $(1+U+U^2)e_U=(2+\mathbf{c})e_U$ and $N_Ue_U=3(1+\mathbf{c})e_U$; $(1+S)(1-S)=1-\mathbf{c}$, $(1+S)(1+\mathbf{c})=N_S$, $(1+U+U^2)(1-U)=1-\mathbf{c}$, $(1+U+U^2)(1+\mathbf{c})=N_U$; and $N_xe_x=|Q_x|\,N'_x$, since $\Gamma_x\to\Gamma_x/Q_x$ has fibres of size $|Q_x|$ and $\{g_x^j\}_{j<|P_x|}$ maps onto Γ_x/Q_x, which is cyclic of order $|P_x|$ generated by the image of g_x.

(1)(a) *Well defined*: at $\ell=2$, $e_{\mathfrak{e}}=e_S=1$ and $e_U^2=e_U$; at $\ell=3$ and $\ell\geq5$, $\mathbf{c}\in\langle S\rangle$ and $\mathbf{c}\in\langle U\rangle$, so $e_{\mathfrak{e}}e_S=e_S$, $e_{\mathfrak{e}}e_U=e_U$. *Chain maps*: the condition is $a_t\,\delta^S_t=\varepsilon_t\,a_{t-1}$ and $b_t\,\delta^U_t=\varepsilon_t\,b_{t-1}$ for $t\geq1$, where δ^x_t, ε_t are the coefficients of ∂^x_t, $\partial^{\mathfrak{e}}_t$. At $\ell=2$: for odd t, $(1+S)(1-S)=1-\mathbf{c}=\varepsilon_t\cdot1$ and $-e_U(1-U)e_U=-(1-\mathbf{c})e_U=\varepsilon_t(-e_U)$; for even $t\geq2$, $1\cdot N_S=(1+\mathbf{c})(1+S)$ and $-e_U\cdot N'_U=-(1+U)e_U=-(1+\mathbf{c})e_U=\varepsilon_t(-e_U)$. At $\ell=3$ and $\ell\geq5$ the edge column vanishes in degrees ≥1, so the condition is empty. The free lift

is Remark 16.3's, checked by the same identities with $1+U+U^2$ in place of e_U. *Lifting*: in degree 0, (a_0, b_0) applied to $e_{\mathfrak{e}}$ augments to $\varepsilon(a_0)\langle S\rangle - \varepsilon(b_0)\langle U\rangle = \langle S\rangle - \langle U\rangle$, the image of $\langle \mathbf{c}\rangle$. *Homotopic*: two lifts of the same map from the projective complex $F^{\mathfrak{e}}$ to the resolution $F^S \oplus F^U$ of $C_0(\mathcal{T}) \otimes R$ are homotopic [standard: [Wei94, Thm. 2.2.6]] (projectivity: (b)); the cone lemma gives the isomorphism.

(1)(b) e_x is an idempotent of $R[\Gamma_x]$ ($|Q_x| \in R^\times$), $R[\Gamma]e_x \oplus R[\Gamma](1-e_x) = R[\Gamma]$, and $ye_x \mapsto yQ_x$ identifies $R[\Gamma]e_x$ with $R[\Gamma] \otimes_{R[Q_x]} R = R[\Gamma/Q_x]$. Under $R[\Gamma_x]e_x \cong R[\Gamma_x/Q_x]$, $g_x e_x \mapsto \bar{g}_x$ and $N'_x \mapsto \sum_{y\in\Gamma_x/Q_x} y$, so the column is $R[\Gamma]\otimes_{R[\Gamma_x]}$ of the periodic resolution of R over the cyclic group $\Gamma_x/Q_x \cong P_x$ inflated to Γ_x (for $P_x \neq 1$; for $P_x = 1$, of $R \to R$), with the sign of the odd differentials reversed relative to Definition 31.1's $\bar{g}-1$ (Remark 16.3 already uses $1-t$; an isomorphic complex: multiply degree t by $(-1)^{\lceil t/2\rceil}$). That complex is exact over every ring (the proof of Proposition 31.2, "the columns", and Proposition 31.6(1)), and $R[\Gamma]\otimes_{R[\Gamma_x]} -$ is exact ($R[\Gamma]$ is free over $R[\Gamma_x]$), so $F^x \to R[\Gamma/\Gamma_x]$ is a projective resolution. The column augmentations give a map of cones from $\mathcal{I}_{\mathcal{T}}^{(\ell)}$ to the cone of the boundary $C_1(\mathcal{T})\otimes R \to C_0(\mathcal{T})\otimes R$, i.e. to $C_\bullet(\mathcal{T})\otimes R$; it is a quasi-isomorphism by the five lemma on the long sequences of $0 \to F^S \oplus F^U \to C(\varphi') \to F^{\mathfrak{e}}[-1] \to 0$ and of its target, the outer maps being quasi-isomorphisms. $C_\bullet(\mathcal{T})\otimes R \to R$ is a quasi-isomorphism because (50) is split exact over $\mathbb{Z}$ (the proof of Lemma 28.4). The terms are finitely generated projective. For $A \leq \Gamma$, $R[\Gamma]$ is free over $R[A]$, so projective $R[\Gamma]$-modules are projective over $R[A]$. The free case is Remark 16.3.

(1)(c) $\mathrm{Hom}_{R[\Gamma]}(R[\Gamma]e_x, M) \to e_xM = M^{Q_x}$, $c \mapsto c(e_x)$, is an isomorphism, and $H^q(\mathrm{Hom}(\mathsf{P}, M)) = \mathrm{Ext}^q_{R[\Gamma]}(R, M) = H^q(\Gamma; M)$ for every projective resolution P (Proposition 31.2(3)'s argument). Shapiro: [standard: [Bro82, Ch. III.6.2]], induction and coinduction agreeing at finite index. Ranks: $\rho_M(e_x)$ is an idempotent endomorphism of the finitely generated free module M over the principal ideal domain R; its image is a direct summand, hence free, of rank dim of the image over the fraction field, which for an idempotent in characteristic 0 is its trace, $|Q_x|^{-1}\sum_w \mathrm{tr}\,\rho_M(w)$.

(1)(d) The denominators of the canonical lifts, of the e_x and of N'_x are divisors of $|Q_x|$, which divide q_ℓ. Base change replaces each coefficient by its image and keeps the shape; the augmented model over O_ℓ is a bounded-below exact complex of free O_ℓ-modules ($\mathsf{O}_\ell[\Gamma]e_x \cong \mathsf{O}_\ell[\Gamma/Q_x]$), hence split exact, and stays exact after $-\otimes_{\mathsf{O}_\ell} R$ (the argument of Proposition 31.6(1)'s proof). The proofs of Proposition 31.6(2)–(6) use of $C_\ell^\bullet$ only: (i) $C^q = \bigoplus_x M_{\mathsf{O}_\ell}^{Q_x}$, images of idempotents, free over the principal ideal domain O_ℓ and commuting with base change — (1)(c) here; (ii) $H^q(C_\ell^\bullet) = H^q(\Gamma; M)\otimes \mathsf{O}_\ell$, which there comes from a free model and flatness — here (1)(c) over O_ℓ, and $C_{\mathcal{T}}^\bullet(M)\otimes\mathsf{O}_\ell = C_{\mathcal{T}}^\bullet(M_{\mathsf{O}_\ell})$ ($\mathcal{I}_{\mathcal{T}}$ finitely generated free) with O_ℓ flat; (iii) the statements (4)(a), (4)(b) about $H^\bullet(\Gamma; M)$, which do not see the model; (iv) for (5), $a_{yx} = e_y a_{yx} e_x$ for every coefficient: for the lifts a_t, b_t this is (1)(a), and for the column differentials $(1-g_x)e_x$, N'_x and the edge's $-\varepsilon_t$ it holds because e_x is central in the commutative ring $R[\Gamma_x]$ and idempotent; (v) Smith forms over O_ℓ and $\mathbb{Z}_{(\ell)}$ and the universal coefficient theorem. All hold for the tree model, and (4)(c)'s exclusion of other complexes is by its own argument (the rank of a differential is a property of that differential).

(2) Dualise (1)(c): for $a \in e_y R[\Gamma]e_x$, $\mathrm{Hom}(r_a, M)\colon M^{Q_x} \to M^{Q_y}$ is $m \mapsto am$ ($c(e_y a) = c(ae_x) = a\,c(e_x)$). So $d^q(u,w)$ is $\delta^x_{q+1}u_x$ on F^x_{q+1} and $a_q u_S + b_q u_U - \varepsilon_q w$ on $F^{\mathfrak{e}}_q$ (the sign of $-\partial^{\mathfrak{e}}$). Substituting the coefficients and $e_U y = y$ for $y \in M^{U^2}$, $e_{\mathfrak{e}} y = y$ for $y \in M^{-I}$, $e_S x = x$ for $x \in M^{\langle S\rangle}$ gives (a)–(c); at $\ell = 3$ the columns F^S, $F^{\mathfrak{e}}$ live in degree 0 only, which removes every term but those displayed. On M^{-I}, U and $U^{-2} = \mathbf{c}^{-1}U$ agree, U^{-2} generates $\langle U^2\rangle$, and $1+U+U^2$ is its norm; $\mathrm{im}\,d^1 = (1+U+U^2)M^{-I}$, so the cohomology in degrees ≥ 2 is that of the periodic resolution of $\langle U^2\rangle$ [standard: [Bro82, Ch. III.6.2]] on M^{-I} (the case $q=2$ included, its incoming image being the norm's). (d) is inspection of (a), (b): H^q uses d^q and d^{q-1}, both periodic once $q-1 \geq 1$ at $\ell = 2$; at $\ell = 3$ once $q-1 \geq 2$, and H^2 has $\mathrm{im}\,d^1 = \mathrm{im}\,d^3$. (e) is (1)(c) read on (a)–(c).

(3) *Column maps.* f^x: for odd t, $(1-g_x)\cdot e_x = e_x \cdot (1-g_x)e_x$ with equal scalars ($\lfloor t/2\rfloor =$

$\lfloor (t-1)/2 \rfloor$); for even t, $N_x \cdot |Q_x|^{t/2-1} e_x = |Q_x|^{t/2} e_x \cdot N'_x$ by $N_x e_x = |Q_x|\, N'_x$; where the ℓ-local column vanishes in degree $t \geq 1$, $f^x_{t-1}\partial_t = r_{(1-g_x)e_x} = 0$ as $g_x \in Q_x$. g^x: the same identities with the inverse scalars ($|Q_x|^{-1} \in R$). Both preserve the augmentations. $f^x g^x = \mathrm{id}$ column by column ($e_x^2 = e_x$, the scalars cancel). *The corrections.* By $(*)$ $\Phi^{\mathcal{T}}_\ell$ is a chain map iff $\kappa_t \delta^U_{t+1} + \varepsilon_t \kappa_{t-1} = D_t := b_t\, f^U_t - f^{\mathfrak{e}}_t\, b'_t$ (coefficients in F^U_t; the S-components of D vanish: at $\ell = 2$ because $f^S = f^{\mathfrak{e}} = \mathrm{id}$ and the a_t agree, at $\ell \geq 3$ because $f^S_t = 0$ for $t \geq 1$ and $e_{\mathfrak{e}} e_S = e_S$ at $t = 0$), b_t the free and b'_t the local lift, $\kappa_{-1} := 0$. At $\ell = 2$: $D_t = (1 - 3^{t/2})e_U$ (t even) and $(1 - 3^{(t-1)/2}(2 + \mathbf{c}))e_U$ (t odd); with $\kappa_{2j-1} = \kappa_{2j} = -\mathsf{k}_j e_U$, $t = 2j+1$ gives $-\mathsf{k}_{j+1}(1+\mathbf{c}) - \mathsf{k}_j(1-\mathbf{c}) = 1 - 3^j(2+\mathbf{c})$ (as $\mathsf{k}_{j+1} + \mathsf{k}_j = 2 \cdot 3^j - 1$, $\mathsf{k}_{j+1} - \mathsf{k}_j = 3^j$), and $t = 2j+2$ gives $-\mathsf{k}_{j+1}(1-\mathbf{c}) - \mathsf{k}_{j+1}(1+\mathbf{c}) = 1 - 3^{j+1}$, both times e_U; $t = 0$: $D_0 = 0 = \kappa_0$. At $\ell = 3$: $D_0 = 0$, $D_t = -2^{(t-1)/2}(1 + U + U^2)e_{\mathfrak{e}}$ (t odd), $-2^{t/2} e_{\mathfrak{e}}$ ($t \geq 2$ even), and δ^U_{t+1} is $(1 + U + U^2)e_{\mathfrak{e}}$, resp. $(1-U)e_{\mathfrak{e}}$; for odd t, $\kappa_{t-1} = 0$ and $\kappa_t(1 + U + U^2)e_{\mathfrak{e}} = D_t$; for even $t \geq 2$, $\kappa_t = 0$ and $\varepsilon_t \kappa_{t-1} = -(1+\mathbf{c})2^{(t-2)/2} e_{\mathfrak{e}} = -2^{t/2} e_{\mathfrak{e}} = D_t$. At $\ell \geq 5$, $D = 0$ and $k = 0$. For $\Psi^{\mathcal{T}}_\ell$ the identity is $\kappa'_t \delta_{t+1} + \varepsilon_t \kappa'_{t-1} = D'_t := b'_t\, g_t - g^{\mathfrak{e}}_t\, b_t$ with free target coefficients δ. At $\ell = 2$, D'_t has S-component 0 and U-component $1 - 3^{-t/2} e_U$ (t even), $(1 + U + U^2) - 3^{-(t-1)/2} e_U$ (t odd); split $R[\langle U \rangle] = R[\langle U \rangle] e_U \times R[\langle U \rangle](1 - e_U)$. On the second factor $N_U = 3(1+\mathbf{c})e_U$ vanishes and $\mathsf{w}(1-U) = 1 - e_U$ (as $(1-U)(2 + 2U + U^2 + U^3) = 2 - U^2 - U^4 = 3(1 - e_U)$ and $(1-U)(1+\mathbf{c})e_U = (1-\mathbf{c})(1+\mathbf{c})e_U = 0$), so the identity reads $\mathsf{w}(1-U) = 1 - e_U$ for even t and $(1-\mathbf{c})\mathsf{w} = (1 + U + U^2)(1-U)\mathsf{w} = (1 + U + U^2)(1 - e_U)$ for odd t; on the first factor, with $\kappa'_t = \lambda_t e_U$, $\lambda_{2j-1} = \lambda_{2j} = \frac{1 - 3^{-j}}{2}$, it reads $3(1+\mathbf{c})\lambda_{2j-1} + (1-\mathbf{c})\lambda_{2j-2} = 2 + \mathbf{c} - 3^{-(j-1)}$ and $(1-\mathbf{c})\lambda_{2j} + (1+\mathbf{c})\lambda_{2j-1} = 1 - 3^{-j}$, both direct. At $\ell = 3$ and $\ell \geq 5$ the only condition is in degree $t = 0$, $\kappa'_0(1 - S) = e_S - e_{\mathfrak{e}}$ and (at $\ell \geq 5$) $\kappa'_{0,U}(1-U) = e_{\mathfrak{e}} - e_U$, and $-\frac{1}{2} e_{\mathfrak{e}}(1 - S) = -\frac{1}{4}(1 - S + S^2 - S^3) = e_S - e_{\mathfrak{e}}$, $\frac{1}{3} e_{\mathfrak{e}}(2 + 2U + U^2 + U^3)(1 - U) = e_{\mathfrak{e}}(1 - e_{\langle U^2 \rangle}) = e_{\mathfrak{e}} - e_U$. $\Phi\Psi = \mathrm{id}$. $\Phi^{\mathcal{T}}_\ell \Psi^{\mathcal{T}}_\ell (a, z) = (fga + (fk' + kg)z,\, fgz) = (a + (fk' + kg)z,\, z)$. At $\ell = 2$, $f^U_{t+1} k'_t = r_{3^{\lceil t/2 \rceil} \kappa'_t e_U} = r_{\mathsf{k}_{\lceil t/2 \rceil} e_U}$ ($\mathsf{w} e_U = 0$) and $k_t g^{\mathfrak{e}}_t = r_{-\mathsf{k}_{\lceil t/2 \rceil} e_U}$; at $\ell = 3$, k' lands in F^S_1, where $f^S_1 = 0$, and $k_0 = 0$; at $\ell \geq 5$, $f_1 = 0$ and $k = 0$. $\Psi\Phi \simeq \mathrm{id}$: $\Psi^{\mathcal{T}}_\ell \Phi^{\mathcal{T}}_\ell$ and id are chain maps over id_R from the projective (free) complex $\mathcal{I}_{\mathcal{T}} \otimes R$ to the resolution $\mathcal{I}_{\mathcal{T}} \otimes R$, hence homotopic [standard: [Wei94, Thm. 2.2.6]]; an explicit homotopy is H_ℓ of (4)(c), since $\Psi^{\mathcal{T}}_\ell \Phi^{\mathcal{T}}_\ell$ is over the identity of $C_\bullet(\mathcal{T}) \otimes R$ (the column maps preserve the augmentations and k, k' land in column degrees ≥ 1). The cochain statements are $\mathrm{Hom}(-, M)$ of these; the cochain formulas are the dual of (2)'s kind, $(F^* c)(y) = \sum_x b_{yx} c(x)$.

(4) (a) *Well defined*: by Lemma 7.3(1) the stabilisers are $\Gamma_{v_6} = \langle S \rangle$, fixing $\mathfrak{v}_S$, $\Gamma_{v_4} = \langle ST \rangle = \langle U \rangle$, fixing $\mathfrak{v}_U$, and $\{\pm I\}$ for every other representative, acting trivially on $C_\bullet(\mathcal{T})$, on $C_\bullet(X)$'s orientations and on the tables; so $\mathsf{i}_{\mathcal{T}}$, $\mathsf{r}_{\mathcal{T}}$, $\mathsf{G}_{\mathcal{T}}$ extend $\mathbb{Z}[\Gamma]$-linearly. The subcomplex is Claim 16.2(1)'s tree, whose vertices and edges lie on the unit arc of $D_Y \subset X$, and $\Gamma_{\{v_4, v_6\}} = \{\pm I\}$; so $\mathsf{i}_{\mathcal{T}}$ is a bijection of the bases onto that subcomplex's, and a chain map since $\partial\{v_4, v_6\} = v_6 - v_4$. The faces of the representatives that are not representatives are $v_3 = T v_0$, $v_5 = T v_4$, $\{v_3, v_5\} = T\{v_0, v_4\}$ and $\{v_5, v_6\} = S^{-1}\{v_4, v_6\}$ (Lemma 7.3(2)), so $\mathsf{r}_{\mathcal{T}}(v_3) = \mathsf{r}_{\mathcal{T}}(v_5) = T\mathfrak{v}_U = S\mathfrak{v}_U$ ($S^{-1}T = -U \in \langle U \rangle$), $\mathsf{r}_{\mathcal{T}}\{v_3, v_5\} = 0$, $\mathsf{r}_{\mathcal{T}}\{v_5, v_6\} = S\mathfrak{e}$, $\mathsf{G}_{\mathcal{T}}(v_3) = -\{v_3, v_5\}$, $\mathsf{G}_{\mathcal{T}}(v_5) = 0$, $\mathsf{G}_{\mathcal{T}}\{v_3, v_5\} = \mathsf{G}_{\mathcal{T}}\{v_5, v_6\} = 0$. *Chain map*: on edges $\partial \mathsf{r}_{\mathcal{T}}(\{a, b\}) = \mathsf{r}_{\mathcal{T}}(b) - \mathsf{r}_{\mathcal{T}}(a)$ is read off the table (e.g. $\partial(-S\mathfrak{e}) = S\mathfrak{v}_U - \mathfrak{v}_S = \mathsf{r}_{\mathcal{T}}(v_3) - \mathsf{r}_{\mathcal{T}}(v_2)$); on triangles $\mathsf{r}_{\mathcal{T}}(\partial\Delta) = 0$, e.g. $\partial\{v_2, v_3, v_5\} \mapsto 0 - (-S\mathfrak{e}) + (-S\mathfrak{e})$, $\partial\{v_0, v_1, v_4\} \mapsto -\mathfrak{e} - 0 + \mathfrak{e}$. $\mathsf{r}_{\mathcal{T}}\mathsf{i}_{\mathcal{T}} = \mathrm{id}$ on $\mathfrak{v}_S$, $\mathfrak{v}_U$, $\mathfrak{e}$ by the table. *Homotopy*: on vertices $\partial \mathsf{G}_{\mathcal{T}}(v) = v - \mathsf{i}_{\mathcal{T}}\mathsf{r}_{\mathcal{T}}(v)$ (e.g. $\partial(-\{v_0, v_4\}) = v_0 - v_4$); on edges, e.g. $\partial(\{v_0, v_1, v_4\} - \{v_1, v_4, v_6\}) + \mathsf{G}_{\mathcal{T}}(v_1) - \mathsf{G}_{\mathcal{T}}(v_0) = \{v_0, v_1\} - \{v_4, v_6\}$ and $\partial(\{v_2, v_3, v_5\} + \{v_2, v_5, v_6\}) + \mathsf{G}_{\mathcal{T}}(v_3) - \mathsf{G}_{\mathcal{T}}(v_2) = \{v_2, v_3\} + \{v_5, v_6\}$, i.e. $x - \mathsf{i}_{\mathcal{T}}\mathsf{r}_{\mathcal{T}}(x)$; on triangles ($C_3 = 0$) $\mathsf{G}_{\mathcal{T}}(\partial\Delta) = \Delta$, e.g. $\mathsf{G}_{\mathcal{T}}(\partial\{v_2, v_3, v_5\}) = 0 - \{v_2, v_5, v_6\} + (\{v_2, v_3, v_5\} + \{v_2, v_5, v_6\})$. The other cases are the same bookkeeping (all 19 are checked by `cd6_iso_check.py`). (b) $\mathbb{Z}[\Gamma]$ is free as a right $\mathbb{Z}[\langle g \rangle]$-module on R_σ, so the column is $\bigoplus_\rho \rho \otimes F$ over $\mathbb{Z}$, and h is the standard contraction of each summand: $(g-1)\sum_{i<j} g^i = g^j - 1$ gives $\partial\mathsf{h} + \mathsf{h}_{-1}\varepsilon = \mathrm{id}$ in degree 0; in odd degrees $N[j = m-1] + \mathsf{h}((g-1)g^j) = g^j$ (for $j < m-1$ the second term is g^j, for $j = m-1$ it

is $-\sum_{i<m-1} g^i = g^{m-1} - N$); in even degrees ≥ 2, $(g-1)\sum_{i<j} g^i + \mathsf{h}(Ng^j) = g^j - 1 + 1$. Negating the even h for r_{1-g}, and multiplying by ϵ for a global sign, preserves these identities. h_p: no component of ∂ raises s (Definition 23.4; the cone), so the top component z_s of a cycle is a cycle of its columns, in column degree 0 in the kernel of their augmentation (since $pz = 0$ and p is the direct sum of the column augmentations), whence $z_s = \partial^{\mathrm{col}}\epsilon\mathsf{h}(z_s)$ and $z - \partial k_s$ is a cycle of $\ker p$ without component on s or above. (L) is the proof of Lemma 23.12 with $k := \mathsf{h}_p(z)$. (L′): by induction $pH = g\,p'$ in lower degrees, so with $w := Ax - Bx - H(\partial x)$, $p(w - \partial y_1) = (a-b)p'x - g\,p'\partial x - \partial g\,p'x = 0$, and $\partial(w - \partial y_1) = (A-B)\partial x - \partial H\partial x = H\partial\partial x = 0$ by induction; then $\partial H(x) = w$, which is $\partial H + H\partial = A - B$ on x, and $pH(x) = py_1 = g\,p'x$. (c) $\varepsilon_{\mathcal{T}}\Phi^{\mathbb{Z}}_{\mathcal{T}}\Psi^{\mathbb{Z}}_{\mathcal{T}} = \mathsf{r}_{\mathcal{T}}\mathrm{cmp}\Psi^{\mathbb{Z}}_{\mathcal{T}} = \mathsf{r}_{\mathcal{T}}\mathsf{i}_{\mathcal{T}}\varepsilon_{\mathcal{T}} = \varepsilon_{\mathcal{T}}$ and $\mathrm{cmp}\Psi^{\mathbb{Z}}_{\mathcal{T}}\Phi^{\mathbb{Z}}_{\mathcal{T}} = \mathsf{i}_{\mathcal{T}}\mathsf{r}_{\mathcal{T}}\mathrm{cmp}$, so (L′) applies as stated, with $g = 0$, resp. $g = \mathsf{G}_{\mathcal{T}}$ by (a); $\Psi^{\mathcal{T}}_{\ell}\Phi^{\mathcal{T}}_{\ell}$ is over the identity of $C_\bullet(\mathcal{T}) \otimes R$ (the proof of (3)). Both $\mathcal{I}_\bullet$ and $\mathcal{I}_{\mathcal{T}}$ are bounded-below free resolutions of $\mathbb{Z}$ (Proposition 23.5(2), Remark 16.3); uniqueness up to homotopy of comparisons over $\mathrm{id}_{\mathbb{Z}}$: the argument of Proposition 23.5(5) (a chain map from a projective bounded-below complex into the acyclic kernel of an augmentation is null-homotopic). (d) $\Phi_{\mathcal{T}}\Psi_{\mathcal{T}} - \mathrm{id} = \Phi^{\mathcal{T}}_{\ell}(\Phi^{\mathbb{Z}}_{\mathcal{T}}\Psi^{\mathbb{Z}}_{\mathcal{T}} - \mathrm{id})\Psi^{\mathcal{T}}_{\ell} + (\Phi^{\mathcal{T}}_{\ell}\Psi^{\mathcal{T}}_{\ell} - \mathrm{id})$, the second term 0 by (3) and the first $\partial\mathsf{K} + \mathsf{K}\partial$ as $\Phi^{\mathcal{T}}_{\ell}$, $\Psi^{\mathcal{T}}_{\ell}$ are chain maps; $\mathrm{id} - \Psi_{\mathcal{T}}\Phi_{\mathcal{T}} = (\mathrm{id} - \Psi^{\mathbb{Z}}_{\mathcal{T}}\Phi^{\mathbb{Z}}_{\mathcal{T}}) - \Psi^{\mathbb{Z}}_{\mathcal{T}}(\Psi^{\mathcal{T}}_{\ell}\Phi^{\mathcal{T}}_{\ell} - \mathrm{id})\Phi^{\mathbb{Z}}_{\mathcal{T}} = \partial\mathsf{K}' + \mathsf{K}'\partial$. The cochain statements are $\mathrm{Hom}(-, M)$ of these; each comparison is compatible with the identification $H^q(\mathrm{Hom}(\mathsf{P}, M)) = \mathrm{Ext}^q(R, M)$, which is itself computed through comparisons. A comparison through $\mathcal{I}^{(\ell)}$ (Φ_ℓ, Ψ_ℓ of Proposition 31.2(4), with any comparison $\mathcal{I}^{(\ell)} \rightleftarrows \mathcal{I}^{(\ell)}_{\mathcal{T}}$ between projective resolutions) is a comparison $\mathcal{I}_\bullet \otimes R \rightleftarrows \mathcal{I}^{(\ell)}_{\mathcal{T}}$ again, so homotopic to $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$; Proposition 31.2(4)'s proof (strict lifts along R-linear contractions, e_σ-averaged) needs of R only $|Q_\sigma| \in R^\times$ and the exactness of $C_\bullet(X) \otimes R$.

(5) (T1): $\mathsf{q}_2 = 3$, $\mathsf{q}_3 = 4$, so $\ell \nmid \mathsf{q}_\ell$ and $\mathsf{O}_\ell \subset \mathbb{Z}_{(\ell)}$; (1)(b), (1)(d). (T2): (1)(b), (1)(c), with $M^{Q_x}_{\mathbb{Z}_{(\ell)}}$ free over the local principal ideal domain $\mathbb{Z}_{(\ell)}$. (T3): (4). Proposition 31.10(5) then gives (1), (2), (4) there, whose proofs use nothing else of a model. The source (i) of (2)(d) there is read on a model's integral form; here that form is $\mathcal{I}^{(\ell)}_{\mathcal{T},\mathsf{O}_\ell}$ by (1)(d), and a prime $\ell' \nmid \mathsf{q}_\ell$ is a prime of O_ℓ, as in Proposition 31.6(4). The last sentence is Proposition 31.10(1)(c), (2)(c), which allow any $\mathbb{Z}_{(\ell)}$-model for the ℓ-local pieces.

(6) (a) $\Phi^*_{\mathcal{T}}$, $\Psi^*_{\mathcal{T}}$ are cochain maps inducing the comparison isomorphisms, mutually inverse on cohomology (4); conjugating a cochain map that induces t by them induces t again (the argument of Proposition 31.2(5)). Homotopic comparisons give homotopic Φ^*, Ψ^*, and composition respects homotopy. (b) $H(\vartheta^*)$ is the comparison isomorphism between the two identifications with $H^\bullet(\Gamma; M_R)$, and both T'' and $T'_{\mathcal{T}}$ induce t through them, by (a) and Proposition 31.2(5) (over every O_ℓ-algebra: its proof uses the resolution-independence of $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ and the comparisons of (4) there, nothing about ℓ). (c) $T'_1T'_2 = \Psi^*T_1(\Phi^*\Psi^*)T_2\Phi^*$ and $\Phi^*_{\mathcal{T}}\Psi^*_{\mathcal{T}} = (\Psi_{\mathcal{T}}\Phi_{\mathcal{T}})^* \simeq \mathrm{id}$ (4). (d) Proposition 31.10(1)(a) for $\Phi_{\mathcal{T}}$ and $\Psi_{\mathcal{T}}$: $uT'_{\mathcal{T}} = \Psi^*uT\Phi^* = \Psi^*Tu\Phi^* = T'_{\mathcal{T}}u$; the cone is the computation in the proof of Proposition 31.10(3)(c).

A check outside the build, not a tag (v1.23.0-pre.20): `docs/implementation/slot24/cd6_tree_check.py`, independent of `cohomology_app`, of `descentN_oracle.py` and of `integral_line_probe.py` (its own group ring $\mathbb{Q}[\Gamma]$, its own *induced* module M_N on left cosets by first column — the oracle's is coinduced on right cosets by bottom row —, the four models built from Definition 31.11 as chain data, ℓ-adic Smith forms mod ℓ^K certified by exact ranks from the Betti recursion; the oracle tables are read as data only), output `cd6_tree_check.out` (its header records the command). It checks, exactly in $\mathbb{Q}[\Gamma]$ to degree 8, (1)(a) and (3)'s chain-map identities and $\Phi^{\mathcal{T}}_{\ell}\Psi^{\mathcal{T}}_{\ell} = \mathrm{id}$ at $\ell = 2, 3, \geq 5$; and on 145 cells — level 1, $0 \leq n \leq 30$ and $n \in \{40, 56\}$, and $\Gamma_0(N)$ for $2 \leq N \leq 16$, $N \neq 15$, $n \leq 8$, $\psi(N)(n+1) \leq 120$, and $(N, n) \in \{(25, 2), (49, 2)\}$ — at $\ell = 2$ and 3: (2)'s closed forms against the chain data entry by entry, (2)(e)'s ranks, the groups H^q, $q \leq 5$, of the ℓ-local and of the free tree model against the ℓ-parts of `descentN_oracle.tsv` and `descent_oracle_n128.tsv`, the certification and the a-priori bound of (1)(d), $\Psi^{\mathcal{T}*}_{\ell}\Phi^{\mathcal{T}*}_{\ell} = \mathrm{id}$ on cochains, and $\Phi^{\mathcal{T}*}_{\ell}\Psi^{\mathcal{T}*}_{\ell}$ the identity on $H^q(- \otimes \mathbb{Z}/\ell^K)$ (on every

cell by that left inverse and equal finite lengths, and directly on generators of the cocycles, $1 \le q \le 4$, where $3\psi(N)(n+1) \le 90$); at $\ell \in \{5,7\}$ the groups on the Bredon tree; and on 18 level pairs $\Gamma_0(MN) \subset \Gamma_0(N)$ the strict commutation of the fibre maps α, ω with $\Phi_\ell^{\mathcal{T}*}$, $\Psi_\ell^{\mathcal{T}*}$ and the equality of the level-cone groups (degrees -1 to 4) on the ℓ-local and the free tree model: 49 303 checks, 0 mismatches (`CD6_CHECK ... mismatch=0 ... verdict=OK`). It does **not** build $\mathcal{I}$. Its companion (v1.23.0-pre.21) `docs/implementation/slot24/cd6_iso_check.py` (output `cd6_iso_check.out`) does: it builds an $\mathcal{I}_\bullet$ to degree 9 by Proposition 23.5(1)'s procedure with the representatives of (4)(b), and checks exactly in $\mathbb{Z}[\Gamma]$, resp. $\mathbb{Q}[\Gamma]$, to degree 8: $d \circ d = 0$, cmp a chain map, $w_q = 5, 14, 19, \ldots$; (4)(a) on all 19 representatives; $\Phi_{\mathcal{T}}^{\mathbb{Z}}$, $\Psi_{\mathcal{T}}^{\mathbb{Z}}$ chain maps over $\mathsf{r}_{\mathcal{T}}$, $\mathsf{i}_{\mathcal{T}}$; the identities of $\mathsf{H}_{\mathcal{T}}$, H_X, and of H_ℓ, K, K' for $\ell \in \{2,3,5\}$; and on 10 cells ($\mathrm{Ind}_{\Gamma_0(N)}^{\Gamma}\mathrm{Sym}^n$, $N \le 5$, $n \le 6$) mod $p \in \{2,3,5\}$, degrees ≤ 6, equal dimensions of $H^q(C_{\mathcal{I}}^\bullet)$ and $H^q(C_{\mathcal{T}}^\bullet)$ and $\Phi_{\mathcal{T}}^{\mathbb{Z}*}$, $\Psi_{\mathcal{T}}^{\mathbb{Z}*}$ mutually inverse on them: 12 870 checks, 0 mismatches (`CD6_ISO_CHECK ... mismatch=0 ... verdict=OK`); four mutations of the tables (a sign of $\mathsf{r}_{\mathcal{T}}$, a value of $\mathsf{r}_{\mathcal{T}}$, of $\mathsf{G}_{\mathcal{T}}$, of $\mathsf{i}_{\mathcal{T}}$) are each caught. (6) is not exercised (no cochain Hecke operator on $\mathcal{I}$ is built there). □

Code map (v1.23.0-pre.20, **stated before D5's code**, contract first, rule 21; **updated at v1.23.0-pre.30**: D5 is **built and merged**, merge `5eb7e9de`; source `docs/implementation/slot24/DESCENT5` including its §8). **Built.** The files are `Integral/Model/TreeCochains.{h,cpp}` (the ℓ-local tree cochains $C_{\mathcal{T},\ell}^\bullet(M)$ of (2) in closed form, block by block, for any module given by its block-monomial action), `Integral/Model/IsotropyModelTree.{h,cpp}` (the free tree model and the canonical ℓ-local models of Definition 31.11; $\Phi_\ell^{\mathcal{T}}$, $\Psi_\ell^{\mathcal{T}}$ of (3); and (4) made effective on the code's $\mathcal{I}$: the tables of (4)(a), the solvers h_p, $\Phi_{\mathcal{T}}^{\mathbb{Z}}$, $\Psi_{\mathcal{T}}^{\mathbb{Z}}$ by (L), $\mathsf{H}_{\mathcal{T}}$, H_X by (L′), H_ℓ, K, K'), `Integral/Descent/DescentTree.{h,cpp}` (`RunTreeLocalPiece`: the ℓ-primary torsion of $C_{\mathcal{T},\ell}^\bullet(M) \otimes \mathbb{Z}/\ell^K$ by the certified-exponent rule of (1)(d)) and `Integral/Descent/TreeComparisonArms.{h,cpp}` (the validations' arms); the validations are `HeckeIsotropyModelTreeValidation` (standard) and `HeckeIsotropyModelTreeSlowValidation` (slow), registered in `Registry/Slot24/DESCENT5.cpp`, sources `cmake/slot24/DESCENT5.cmake`. **The tree is the default route for the dense ℓ-local bulk pieces**: D0's 2- and 3-adic pieces, D1's 2-adic piece and its dense 3-adic piece at $\nu_3(N) > 0$, and D2's dense pair pieces at 2 and (for $\nu_3 > 0$) at 3; **rollback** `HECKE_DESCENT_TREE=0` (read-only, value-preserving; read once per call) returns to the dense route. Unchanged: the $\nu_3(N) = 0$ shortcut at 3 (Proposition 31.9) and every $\ell \ge 5$ route (the sparse route is cheaper than a dense tree there); the Bredon tree at level 1, $\ell \ge 5$, is built and checked against the direct Smith form of C_{eq} at every Dickson prime, but is not the production route. $\Phi_{\mathcal{T}}^{\mathbb{Z}}$, $\Psi_{\mathcal{T}}^{\mathbb{Z}}$ and the homotopies are (4)(b)–(c) on the code's $\mathcal{I}$ (`Integral/Model/IsotropyModel`, whose column contractions and signed-lexicographically least coset representatives, `decompose()` and `column_homotopy` of `PeriodicColumn.h`, are those of (4)(b)). Both routes agree with the oracles on every cell where both run, and D0's, D1's and D2's validations pass unchanged on both settings of the switch. **Hecke by transport** ((6)): the hook `TransportToTree` ($\mathsf{T}'_{\mathcal{T}} = \Psi_{\mathcal{T}}^* T \Phi_{\mathcal{T}}^*$ on any cochain endomorphism of C_{I}) **exists** and is smoke-tested; its **real arm**, "the transported T'_p equals the one on $\mathcal{I}^{(2)}$" ((6)(b)), is **pending D3**'s level-N cochain operators (C-D4). (v1.23.0-pre.31: that arm is built with D3 on branch `s24-d3impl`, pending merge and audit — the code map of §31.3; v1.23.0-pre.33: audited and merged at `dc8261a3`.) **Not built here**: the tree cusp pair (C-D8, D5c; contract first at v1.23.0-pre.33, §31.5; v1.23.0-pre.36: built, audited and merged at `4064e164`, §31.5's code map); at v1.23.0-pre.30 D4's cone was the pair $(\mathcal{I}, \mathcal{I}^\partial)$ and the tree had no boundary, so the tree bulk did not move D4's wall (since v1.23.0-pre.36 the tree cusp pair is the default 2- and 3-adic pair route, `Integral/Descent/DescentTreePair`; v1.23.0-pre.38, referee m9). The free tree resolution in closed form remains D0's and D1's comparator (c) (`DESCENT0_report.md`, `DESCENT1_report.md`).

Formalization note. Definition 31.11 and Proposition 31.12 are homological algebra over $R[\Gamma]$ (mapping cones, the comparison theorem, Shapiro's lemma) and identities in the group rings of $\mathbb{Z}/4$ and $\mathbb{Z}/6$; the free tree resolution they build on carries three \LEAN markers (Remark 16.3). They carry **no** \LEAN marker.

31.3 Hecke operators T_p, $p \nmid N$, on the descent over $\mathbb{Z}$: the operator at level N, the characteristic polynomial, the torsion

(v1.23.0-pre.24; milestone D3 of `docs/implementation/slot24/DESCENT_plan.md` v0.4, its clauses C-D4 and C-D5 as restricted by the audit's M5; the owner's rulings of 2026-10-24: residues and a certified bound, no big integer [R-Q2]; Hecke output as invariants only — characteristic polynomials over $\mathbb{Z}$ and data on certified groups, integer Hecke matrices later (D8) [R-Q3]; T_p for $p \nmid N$ only, every U_p deferred [R-Q4] (lifted for the free model by Proposition 31.13(9), v1.26.0-pre.1); and of 2026-10-25: the tree is the 2-adic route [R-GT].) §31.1 glues the *groups* $H^q(\Gamma_0(N); V)$ and the maps induced by fibre maps; Proposition 31.7(3) says that a Hecke operator on them is determined by its two local components, but no cochain operator at level N exists on a model that computes the 2- and 3-adic parts, and Proposition 31.10(3)(c) leaves the 2- and 3-adic level sequence without a Hecke action for that reason. This subsection fixes: (i) the level-N operator T_p, $p \nmid N$, at cochain level on the free model $\mathcal{I}_\bullet$, built from a bulk strict lift that exists *and is constructed* for every p, its compatibility with the glueing map cmp^* ($= \varepsilon^*$ of the plan), its transports to $\mathcal{I}^{(\ell)}$ and to the tree models — the route at 2 —, the cheap routes at $\ell \geq 5$ and at 3 (the latter for an integral datum only, and not used by the record; v1.23.0-pre.25), and its strict commutation with the restriction of D2 (Proposition 31.13); (ii) the characteristic polynomial over $\mathbb{Z}$ at level N from residues and a certified bound, with the output rule for a coefficient beyond the word (Proposition 31.14); (iii) which invariants of an endomorphism of a finite ℓ-group are choice-free, and how they are read exactly from a local run (Proposition 31.15); (iv) the record those make certified (Theorem 31.16, the plan's C-D5), and what it does not assert.

What is used. The Hecke clauses of this document that the subsection rests on, and nothing else of the Hecke chapters: Definition 11.1 (the g-twisted chain datum), Proposition 15.13 with Lemma 14.10(1) (an integral datum on X for every $g \in \Delta$), Lemma 23.12 and the bulk half of Proposition 23.13 and of its proof, Remark 23.14(1), (2) (the construction and its certificates, read for every p), Lemma 15.4, Proposition 27.2 (formula (68), items (1)–(4)), Proposition 29.34(1), (2), (4) (the criterion, its instance for T_p, $p \nmid N_B$, and the independence of the transversal; (4) with $\tau_{\mathcal{I}}$ in place of τ, its proof using (T1) only, v1.23.0-pre.25, referee t3), Proposition 23.23 (the level-1 relations on the free model, and its citations), Propositions 31.2(4), (5) and 31.12(3), (4), (6) (comparisons and transport), Propositions 31.7(3), (5) (local determination, periodicity) and 31.10(1), (2)(g), (3) (fibre maps, the precision rule, the Hecke squares), and Proposition 23.19 (level 1); at v1.23.0-pre.31 also Propositions 27.5(1), (2) and 28.9(1), (2) with Definition 28.6 (the cokernel presentation at $\ell \geq 5$ and at the residue primes, with 6 inverted) and Proposition 31.21(1) (the pair lift, one admissible bulk lift); at v1.23.0-pre.32 (referee N6), for completeness of D3's proofs, Proposition 23.5(2) (cmp onto, the check through the lift), Proposition 31.7(1), (7)(b) (flat base change and b_1, the certifying rank) and Lemma 28.7 (through Proposition 28.9); at v1.26.0-pre.1 (item (9) of Proposition 31.13; v1.26.0-pre.2, referee S2) also Proposition 29.34(3)(d) (the adjoint convention) and Proposition 31.10(3)(b′) (within its (3) above), and Proposition 31.28(0) is named by (9)(d); Definition 31.27 and Proposition 31.28, of a later chapter that rests on this one, are cited by item (9) only to *identify* the operator and to point at a relation, not used. *Not used*: Theorems 10.8, 11.10 at $\ell \in \{2, 3\}$, the shipped operator on the Bredon object at $\ell \in \{2, 3\}$ (Remark 23.15: never read here — Proposition 31.13(6)(b) is a statement about an integral datum and is read by no item of Theorem 31.16, whose 3-local piece is the transport of Proposition 31.13(5) for every N; v1.23.0-pre.25, referee M2), Corollary 30.6, and any τ-type table on the tree.

Notation. $A := \Gamma_0(N)$, $g := \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$, $\Gamma''_g := \Gamma \cap g\Gamma g^{-1}$, $M_N := \mathrm{Ind}_A^\Gamma V$ and $M_{N,R} := M_N \otimes R$ (§31.1), $\mathsf{O} := \mathbb{Z}[1/6]$; $H^q_{\mathrm{fr}}(N) := H^q(\Gamma_0(N); V)/\mathrm{tors}$ (so $H^q_{\mathrm{fr}}(1) = H^q_{\mathrm{fr}}$). The symbols τ_X, $\tau_{\mathcal{I}}$ (here: an admissible bulk lift of Proposition 31.13(2)), $\widetilde{T}^{(N)}_{X,p}$, $\widetilde{T}^{(N)}_{\mathcal{I},p}$, $T_p^{\prime(\ell)}$, $T'_{\mathcal{T},p}$, $\chi^{(N)}_{p,q}$, r_N, B_p, $B_{p,i}$, $\mathsf{b}_{p,i}$, $\mathcal{S}$, $V_j(G)$, ϕ_j, $P_j(\phi)$, d_j, e_j, $G_{[c]}$, $G_\circ$, len and $\tilde{x}_j$, $\tilde{\Phi}$ are new symbols of this subsection and mean nothing else (rule (A5)). T_p is always $[AgA]$, $p \nmid N$; $U_p := [AgA]$, $p \mid N$, is defined and used only in item (9) of Proposition 31.13 and its proof (v1.26.0-pre.1; item (1) names it, and U_p^*, through Proposition 29.34(3)(d); v1.26.0-pre.2, referee E1), where $\widetilde{U}^{(N)}_{\mathcal{I},p}$, $\widetilde{U}^{(N)}_{X,p}$, $U_p^{\prime(\ell)}$, $U'_{\mathcal{T},p}$ and U_p^{cl} (the classical operator) are new symbols as well.

Proposition 31.13 (The level-N operator T_p on the free model, every $p \nmid N$ (and U_p, $p \mid N$, item (9)): the bulk lift, the induced-fibre form, the glueing map, the transports)**.** *[proved here] [standard: [Bro82, Ch. III.8–10], [Wei94, Thm. 2.2.6]] (v1.23.0-pre.24; milestone D3, clause C-D4 of the plan; refereed ACCEPT-with-fixes,* `docs/implementation/slot24/C-D4_referee.md`*, its fixes at v1.23.0-pre.25.) Assume (H4) and (B1)–(B3). Let $N \geq 1$, $n \geq 0$, p a prime with $p \nmid N$ (in item (9): $p \mid N$), R a commutative ring, and $(\mathcal{I}_\bullet, \mathrm{cmp})$ an isotropy-resolution model.*

1. ***(The double coset; no adjoint.)*** *With $u, b \in \mathbb{Z}$, $up - bN = 1$, $\gamma_1 := \begin{pmatrix} p & -b \\ -N & u \end{pmatrix}$ and $\gamma_2 := \begin{pmatrix} pu & b \\ N & 1 \end{pmatrix}$ lie in A and $\gamma_1 g \gamma_2 = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$. So $AgA = A\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}A$, and $T_p := [AgA] = \mathrm{cor}^A_{A''} \circ (\iota_g, \rho_n(g))^* \circ \mathrm{res}^A_{A'}$ on $H^\bullet(A; V_R)$ (Proposition 31.7(5)) is the double-coset operator of $\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$ as well: for $p \nmid N$ the distinction between this document's operator and the classical one, which separates U_p from U_p^* when $p \mid N$ (Proposition 29.34(3)(d)), does not arise.*
2. ***(A bulk strict lift for every p, constructed.)*** *Let τ_X be a g-twisted chain datum over $\mathbb{Z}$ on $C_\bullet(X)$ (Definition 11.1 with X for K^u). The construction below and items (3)–(8) use of τ_X nothing else (v1.23.0-pre.25, referee m3). One exists for every p: the strict datum τ^s of Proposition 15.13, with the boundary datum of Lemma 14.10(1), both built by terminating procedures; and the shipped canonical τ of Algorithm 18.3 qualifies* once its stored table is certified integral and checked against (T1)–(T3) over $\mathbb{Z}$ *on $C_\bullet(X)$ (it is integral in practice, not by construction, Remark 23.15), in which case no boundary datum is built at all — the cheaper input.* ***Code obligation**: whichever τ_X is read, its table is checked integral and against (T1)–(T3) over $\mathbb{Z}$ before the lift is used, and the lift is refused otherwise.* ***The check may be made through the lift** (v1.23.0-pre.31, D3's contract note 2): let the table read be a family of* integral *chains $t_{j,c} \in C_s(X)$, one for each element $s_j e_c$ of the basis below with $e_c \in G_{s,0}$, and let $\tau_{\mathcal{I}}$ be additive and certified by (T1) on Schreier generators of Γ''_g, (T2), (T3) and the strict identity on* ***every*** *basis element — $\mathrm{cmp}\tau_{\mathcal{I}}(s_j e_c) = t_{j,c}$ for $e_c \in G_{s,0}$, and $\mathrm{cmp}\tau_{\mathcal{I}}(s_j e_c) = 0$ for $e_c \in G_{s,t}$, $t \geq 1$, where $\mathrm{cmp}(s_j e_c) = 0$. Then the table* ***is*** *the table of a g-twisted chain datum over $\mathbb{Z}$ on $C_\bullet(X)$: there is exactly one additive τ'_X with $\tau'_X \circ \mathrm{cmp} = \mathrm{cmp} \circ \tau_{\mathcal{I}}$, it satisfies (T1)–(T3) over $\mathbb{Z}$, and $\tau'_X(s_j\,\mathrm{cmp} e_c) = t_{j,c}$ (the proof,* the check through the lift*). So this certificate discharges the obligation for $\tau_X := \tau'_X$, no separate check of the table on X is needed, and a table that is not the table of any such datum fails the certificate for* every *$\tau_{\mathcal{I}}$. Then there is an additive $\tau_{\mathcal{I}} \colon \mathcal{I}_\bullet \to \mathcal{I}_\bullet$ with (T1)–(T3) of Definition 11.1 over $\mathbb{Z}$ ($\mathcal{I}_\bullet$ in place of $C_\bullet(K^u)$, augmentation $\varepsilon \circ \mathrm{cmp}$) and $\mathrm{cmp} \circ \tau_{\mathcal{I}} = \tau_X \circ \mathrm{cmp}$ strictly; two such are homotopic through (T1)-maps over $\mathbb{Z}$.* ***Every** such $\tau_{\mathcal{I}}$ is admissible below (v1.23.0-pre.31, D3's contract note 3): items (3)–(8) read of it only (T1)–(T3) over $\mathbb{Z}$ and the strict identity, and, for the cochain-level statements of (3)(c) and (7), that it is fixed independently of the transversal r_t and of N; the bulk component of a pair lift (Proposition 31.21(1), built with the boundary step first) is one, and it is the one the code reads. Every read made from*

*the induced operator is independent of the choice (*independence of the lift*, below). One is* ***produced****, in every degree q, by the procedure of Remark 23.14(1) with no boundary step: on the free $\mathbb{Z}[\Gamma_g'']$-basis $\{s_j e_c\}$ of $\mathcal{I}_q$, the preimage of $\tau_X(s_j \mathrm{cmpe}_c)$ is read off (at $t=0$; it is 0 at $t\geq 1$), and the cycle of* $\ker \mathrm{cmp}$ *left over is contracted by descent through the columns. The procedure is deterministic, solves no linear system, terminates, and is uniform in p; its input τ_X and the coset table of $\Gamma_g''\backslash\Gamma$ ($p+1$ cosets) depend on p. Its output is certified by Remark 23.14(2)'s checks without the boundary items ((T1) on Schreier generators of Γ_g'', (T2), (T3), and the strict identity).* ***No bound on its integer sizes is asserted*** *(Remark 23.22(2)); a coefficient that leaves the word is refused, never truncated (Theorem 18.7(3)).* ***Independence of the lift*** *(v1.23.0-pre.31). Let $\tau_{\mathcal{I}}$, $\tau'_{\mathcal{I}}$ be admissible (for the same τ_X, or for two data τ_X, τ'_X as above), and $\widetilde{T}$, $\widetilde{T}'$ the operators of (3) built from them with the same tables. Then $\widetilde{T}-\widetilde{T}' = d_{\mathrm{I}}\mathsf{H}+\mathsf{H}d_{\mathrm{I}}$, d_{I} the coboundary of $C^\bullet_{\mathcal{I}}(M_{N,R})$, with $\mathsf{H}=\Theta_{h_{\mathcal{I}}}$ (formula (68) with $h_{\mathcal{I}}$ for τ), integral, of degree -1, for a (T1)-homotopy $h_{\mathcal{I}}$ from $\tau_{\mathcal{I}}$ to $\tau'_{\mathcal{I}}$ (constructible by the comparison procedure; v1.23.0-pre.32, referee m1, N4), over every R; so $\widetilde{T}$ and $\widetilde{T}'$, and their transports (5) (conjugated by the same comparisons), induce the same endomorphism T_p of $H^q(A;V_R)$ in every degree. Hence every datum that the record of Theorem 31.16 reads from the induced operator — the torsion modules and every invariant of Proposition 31.15(1), the invariant factors of its (4) — is the same for both lifts; for a fixed certified Smith form, even the matrix $\tilde{\Phi}$ of Proposition 31.15(3) is the same when $2\sigma_{\max}\leq K$, and agrees modulo $\ell^{\min(\sigma_i,K')}$ in row i otherwise. (The characteristic polynomial of Proposition 31.14 reads no lift at all.)*

3. ***(The induced-fibre form.)*** *Let $\widetilde{T}^{(N)}_{\mathcal{I},p}$ be formula (68) with $A=B=\Gamma_0(N)$ ($w_a=v_i$ the table of Definition 28.3, r_t a transversal of $A''\backslash A$, $A''=\Gamma_0(pN)$), with $\tau_{\mathcal{I}}$ in place of τ and x running over $\mathcal{I}_\bullet$: for $\phi\in\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathcal{I}_q,M_{N,R})$,*

$$\big(\widetilde{T}^{(N)}_{\mathcal{I},p}\phi\big)(x)_i=\sum_t \rho_n(\delta_{i,t})\,\phi\big(\tau_{\mathcal{I}}(s_{j(i,t)}x)\big)_{a(i,t)}.$$

   (a) *It is integral and reads only the level-1 table of $\tau_{\mathcal{I}}$ (its values on the $s_j e_c$), the coset tables, and ρ_n of integral matrices of determinant p (Proposition 27.2(1)); no level-N table of any lift is needed.*
   (b) *It is a cochain map $C^\bullet_{\mathcal{I}}(M_{N,R})\to C^\bullet_{\mathcal{I}}(M_{N,R})$ over* ***every*** *R.*
   (c) *Under Shapiro's isomorphism (the reciprocity $\phi\mapsto\phi(\cdot)_1$) it induces T_p on $H^q(A;V_R)$ for* ***every*** *q and every R, with no hypothesis on 6; on cohomology it is independent of τ_X, $\tau_{\mathcal{I}}$, the s_j, the r_t and the table, and at cochain level it is independent of the transversal r_t ($\tau_{\mathcal{I}}$ fixed). The $\tau_{\mathcal{I}}$ produced in (2)* ***is*** *fixed in this sense: it is built once, on $\{s_j e_c\}$, independently of r_t and of N (a level-1 object), which is what (7) needs at both levels; so is the bulk component of a pair lift (Proposition 31.21(1), built once per p; v1.23.0-pre.31). A datum built from the transversal (as the natural one of the bar model) changes $\widetilde{T}^{(N)}_{\mathcal{I},p}$ by a coboundary when r_t changes; then only the cohomology statement survives (v1.23.0-pre.25, referee m4).*
   (d) *At $N=1$ it is the coset transport $\Theta_{\mathcal{I},g}$ of Proposition 23.13(3).*

4. ***(The glueing map.)*** *Let $\widetilde{T}^{(N)}_{X,p}$ be (68) on the shipped level-N model with the datum τ_X of (2). Then, over every R,*

$$\mathrm{cmp}^*\circ\widetilde{T}^{(N)}_{X,p}=\widetilde{T}^{(N)}_{\mathcal{I},p}\circ\mathrm{cmp}^* \qquad \textit{on the nose.}$$

If the shipped model reads another admissible datum (the canonical τ of Algorithm 18.3), its operator and $\widetilde{T}^{(N)}_{X,p}$ agree on cohomology whenever $6\in R^\times$ (Proposition 27.2(3)). Hence T_p on $H^q(A;V)$ over $\mathbb{Z}$ is the endomorphism glued (Proposition 31.7(3)) from the shipped operator on the O-piece $C^\bullet_{\mathrm{eq}}(M_{N,\mathsf{O}})$ and from $\widetilde{T}^{(N)}_{\mathcal{I},p}\otimes\mathbb{Z}_{(\ell)}$, or any of its transports (5), on the ℓ-local pieces, $\ell\in\{2,3\}$; no coherence datum enters.

5. ***(The transports; the route at 2.)*** *Let ℓ be a prime and R an O_ℓ-algebra. Then*

$$T_p'^{(\ell)} := \Psi_\ell^* \circ \widetilde{T}_{\mathcal{I},p}^{(N)} \circ \Phi_\ell^* \text{ on } C^\bullet_{\mathcal{I}^{(\ell)}}(M_{N,R}), \qquad T'_{\mathcal{T},p} := \Psi_{\mathcal{T}}^* \circ \widetilde{T}_{\mathcal{I},p}^{(N)} \circ \Phi_{\mathcal{T}}^* \text{ on } C^\bullet_{\mathcal{T},\ell}(M_{N,R})$$

are cochain maps defined over O_ℓ (Propositions 31.2(4), 31.12(3), (4)) that induce T_p on $H^q(A;V_R)$ in every degree, independently of the comparisons up to cochain homotopy, and that agree with each other on cohomology (Proposition 31.12(6)(a), (b) with $T := \widetilde{T}_{\mathcal{I},p}^{(N)}$). At $\ell = 2$ the route is $T'_{\mathcal{T},p}$ on $\mathcal{I}_{\mathcal{T}}^{(2)}$ ([R-GT]); every factor is constructed, and applying it to a cochain composes three cochain maps — the Hecke datum stays on $\mathcal{I}_\bullet$, no τ-type table on the tree is used.

6. ***(The cheap routes.)*** *(a) If $6 \in R^\times$ — the O-piece, and its reductions modulo ℓ^K, $\ell \geq 5$ — the shipped operator induces T_p on $H^q(C^\bullet_{\mathrm{eq}}(M_{N,R})) = H^q(A;V_R)$, and so does $\mathsf{i}_{\mathcal{T}}^* \circ \widetilde{T}_{X,p}^{(N)} \circ \mathsf{r}_{\mathcal{T}}^*$ on the tree complex $\mathrm{T}(M_{N,R})$ (65), with the integral chain maps $\mathsf{i}_{\mathcal{T}}$, $\mathsf{r}_{\mathcal{T}}$ of Proposition 31.12(4)(a). (b) If $\nu_3(N) = 0$ (Proposition 31.9: $\Gamma_0(N)$ has no element of order 3) and R is a $\mathbb{Z}_{(3)}$-algebra, the same two statements hold **for an integral datum**: (68) with a g-twisted chain datum over $\mathbb{Z}$ on X, certified by (T1)–(T3) over $\mathbb{Z}$, induces T_p on $H^q(A;V_R)$ for every q. The shipped canonical τ is such a datum only once so certified (Remark 23.15: integral in practice, not by construction). **This item is read by no statement of this subsection** (v1.23.0-pre.25, referee M2, resolution (a)): the 3-local piece of Theorem 31.16 is the transport $T_p^{\prime(3)}$ (or $T'_{\mathcal{T},p}$) of (5) for* every *N, $\nu_3(N) = 0$ included, so the shipped operator on the Bredon object is never read at $\ell \in \{2,3\}$.*

7. ***(Strict squares with the restriction; the level sequence.)*** *Let $B := \Gamma_0(MN) \subseteq A$ with $p \nmid MN$, and build $\widetilde{T}_{\mathcal{I},p}^{(N)}$ and $\widetilde{T}_{\mathcal{I},p}^{(MN)}$ from the same $\tau_{\mathcal{I}}$ and a common transversal $r_t \in B$ of $\Gamma_0(pMN)\backslash B$ (which is one of $A''\backslash A$, Proposition 29.34(2)). Then $\widetilde{T}_{\mathcal{I},p}^{(MN)} \circ \alpha_{\mathcal{I}} = \alpha_{\mathcal{I}} \circ \widetilde{T}_{\mathcal{I},p}^{(N)}$ **as cochain maps**, over every R; so $(\widetilde{T}_{\mathcal{I},p}^{(N)}[1], \widetilde{T}_{\mathcal{I},p}^{(MN)})$ is a cochain endomorphism of $C(\alpha_{\mathcal{I}})$ commuting with its short exact sequence, its transports to every model of Proposition 31.10 satisfy the same strict identity, and the long exact sequence (78) is T_p-equivariant **over $\mathbb{Z}$ and over every $\mathbb{Z}_{(\ell)}$, $\ell \in \{2,3\}$ included**, in every degree — which supplies what Proposition 31.10(3)(c) left unasserted at $\ell \in \{2,3\}$. The square with* cor *commutes on cohomology (Proposition 31.10(3)(a)); no cochain-level statement is made for it.*

8. ***(Relations.)*** *For primes $p, p' \nmid N$, $T_p T_{p'} = T_{p'} T_p$ on $H^q(A;V_R)$, every q, every R.*

9. ***($p \mid N$: the operator U_p on the free model.)*** *(v1.26.0-pre.1, slot 26, UP-Z; `docs/implementation/slot26/UPZ_design.md` §7.1 and its critical audit `docs/implementation/slot26/UPZ_design_audit.md` §2.3, condition C5; refereed ACCEPT-with-conditions, `docs/implementation/slot26/C126pre1_referee.md`, its conditions and fixes at v1.26.0-pre.2; **not yet seen by the owner**.) In this item p is a prime with $p \mid N$, in place of $p \nmid N$ in the hypothesis above; everything else is as there, $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$. Put $U_p := [AgA] = \mathrm{cor}_{A''}^{A} \circ (\iota_g, \rho_n(g))^* \circ \mathrm{res}_{A'}^{A}$ on $H^\bullet(A;V_R)$.*

   (a) ***(The coset data.)*** *$A'' = A \cap gAg^{-1} = \Gamma_0(pN)$, of index p in A, with the transversal $r_t := \begin{pmatrix} 1 & 0 \\ tN & 1 \end{pmatrix}$, $0 \leq t < p$, of $A''\backslash A$; and $A'' \subseteq \Gamma_g'' = \Gamma_0(p)$. (For $p \nmid N$ the index is $p+1$, the transversal of (3).)*

   (b) ***(Which U_p.)*** *U_p is this document's $U_p^{(N)}$ of Proposition 27.2(4)(d), the $[A\,g\,A]$ of Proposition 28.9(2) at $p \mid N$, and the operator to which Definition 31.27 sends X_p (through Proposition 31.10(3)(b′)). On p-old spaces it is the* adjoint *$U_p^* = W_p U_p^{\mathrm{cl}} W_p^{-1}$ of the classical operator U_p^{cl} (Proposition 29.34(3)(d); v1.26.0-pre.2, referee E2); item (1) has no analogue at $p \mid N$: $A \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} A \neq AgA$ as double cosets, since by*

Step 1 of the proof of Proposition 28.9 the Hermite form $\left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$ *is the one right coset of* Δ_p *absent from* AgA *at* $p \mid N$ *(v1.26.0-pre.2, referee E9), and* $[A \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) A]$ *is not asserted to equal* U_p*. Every statement and record of this document that reads* U_p*,* $p \mid N$*, reads this operator.*

(c) ***(Items (2)–(6) hold, their references to (7), (8) dropped.)*** *Write* $\widetilde{U}^{(N)}_{\mathcal{I},p}$ *for formula (68) as in (3) with* $A'' = \Gamma_0(pN)$*, the transversal of (a) and* ***the same*** $\tau_{\mathcal{I}}$ *of (2) — the level-1 lift, built and certified once per* p*, independently of* N *(the bulk component of a pair lift, Proposition 31.21(1), is one; the coset table of* $\Gamma''_g\backslash\Gamma$ *still has* $p+1$ *cosets, the level-*N *transversal has* p*) —* $\widetilde{U}^{(N)}_{X,p}$ *for the shipped operator of (4),* $U'^{(\ell)}_p$*,* $U'_{\mathcal{T},p}$ *for the transports of (5). Then items (2)–(6), and the independence of the lift stated in (2), hold (with their references to (7) and (8) dropped: item (2)'s "items (3)–(8)" reads "items (3)–(6)", its "(3)(c) and (7)" reads "(3)(c)"; v1.26.0-pre.2, referee E7) with* U_p *for* T_p *and these operators for* $\widetilde{T}^{(N)}_{\mathcal{I},p}$*,* $\widetilde{T}^{(N)}_{X,p}$*,* $T'^{(\ell)}_p$*,* $T'_{\mathcal{T},p}$*; (3)(d) is vacuous (no prime divides* 1*). In particular* $\widetilde{U}^{(N)}_{\mathcal{I},p}$ *is integral and reads only the level-*1 *table of* $\tau_{\mathcal{I}}$*, is a cochain map over every* R*, induces* U_p *on* $H^q(A; V_R)$ *for* ***every*** q *and every* R *with no hypothesis on* 6*, and is independent of* r_t *at cochain level;* U_p *over* $\mathbb{Z}$ *is glued from the* O*-piece and the* ℓ*-local pieces,* $\ell \in \{2,3\}$ *((4)); at* $\ell \in \{2,3\}$ *the route is the transport (5), at* $\ell = 2$ *on the tree. Every datum the record reads from the induced operator (Proposition 31.15) is independent of the lift.* ***Code obligation****: the level-*N *coset count is* $[A : A''] = p$ *at* $p \mid N$ *(not* $p+1$*), and the lift is the certified level-*1 *one of (2); a lift refused by a certificate is a per-cell refusal, never evidence that no lift exists.*

(d) ***(Not asserted at*** $p \mid N$***.)*** *Items (1), (7) and (8). Nothing in this subsection, or in the record of item 8 of §22.4, reads them for* U_p*. The commutativity of* U_p *with the* T_p *that the record's* ϑ *needs is Proposition 31.28(0), not item (8) (v1.26.0-pre.2, referee E3).*

Proof. (1) $\det\gamma_1 = pu - bN = 1$, $\det\gamma_2 = pu - bN = 1$, the lower-left entries are $\mp N$, so both lie in A; and $\gamma_1 g = \left(\begin{smallmatrix} p & -bp \\ -N & up \end{smallmatrix}\right)$, $\gamma_1 g \gamma_2 = \left(\begin{smallmatrix} p^2u - bpN & pb - bp \\ -Npu + upN & -Nb + up \end{smallmatrix}\right) = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$. The operator $[AgA]$ depends only on the double coset: replacing g by $\gamma_1 g \gamma_2$ composes the three factors with conjugations by γ_1, $\gamma_2 \in A$, which act trivially on $H^\bullet(A; -)$ [standard: [Bro82, Ch. III, §8]: an inner automorphism acts trivially on cohomology], together with the compatible coefficient maps $\rho_n(\gamma_i)$ (the computation of Theorem 11.7(c); the compatibility is the multiplicativity of ρ_n on $\mathrm{M}_2(\mathbb{Z})$, Proposition 8.2(2), v1.23.0-pre.25, referee t2).

(2) Proposition 15.13 builds τ^s for every $g \in \Delta$ from a boundary datum, and Lemma 14.10(1) builds that for $g = \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right)$; a certified canonical table is a g-twisted chain datum over $\mathbb{Z}$ by definition of the certificate. Nothing below reads more of τ_X than (T1)–(T3) over $\mathbb{Z}$. Then the proof of Proposition 23.13(1) with $P' = 0$ (no boundary step): Γ''_g has finite index in Γ, so $\mathcal{I}_q$ is free over $\mathbb{Z}[\Gamma''_g]$ on $\{s_j e_c\}$, and $\mathrm{cmp}\colon \mathcal{I}^{(c)} \to C^{(c)}_\bullet(X)$ (twisted actions) is a Γ''_g-equivariant surjective quasi-isomorphism; Lemma 23.12 with $f = \tau_X \circ \mathrm{cmp}$ gives $\tau_{\mathcal{I}}$, (T2) because it is a chain map, (T3) from (T3) of τ_X; uniqueness up to (T1)-homotopy is Proposition 23.13(2)'s argument. *The procedure.* The induction of Lemma 23.12 needs a preimage y_0 of $f(x)$ and a contraction of the cycle z of the acyclic $\ker\mathrm{cmp}$. At $t = 0$, cmp is the augmentation of free generators, so a chain $\sum c\,\gamma\,\sigma'$ of X is the image of $\sum c\,\gamma\,(1 \otimes 1)_{\sigma'} \in G_{\bullet,0}$; at $t \geq 1$ or in degrees ≥ 3, $f(x) = 0$ and $y_0 = 0$. For the contraction filter $\ker\mathrm{cmp}$ by s: if s is the largest column on which z has a component z_s, then $d^{(0)} z_s = 0$ (it is the top-column part of $dz = 0$) and z_s lies in the augmented kernel of $G_{s,\bullet} \to C_s(X)$, whose closed-form $\mathbb{Z}$-linear contracting homotopy

h (Proposition 23.5's proof, the columns) gives $k_s := (-1)^s h(z_s)$ with $d^{(0)}k_s = z_s$; k_s lies in $t \geq 1$ or in the augmentation kernel, so in $\ker \mathrm{cmp}$, and $z - dk_s$ has no component on columns $\geq s$. Descending on $s \leq 2$ contracts z. This is the formula (L) of Proposition 31.12(4)(b) with cmp as its surjection, $f = \tau_X \circ \mathrm{cmp}$, the basis $\{s_j e_c\}$ and the twisted $\mathbb{Z}[\Gamma''_g]$-linear extension (T1) in place of the $\mathbb{Z}[\Gamma]$-linear one — (L) uses of the extension only that h_p is $\mathbb{Z}$-linear with $\partial \mathsf{h}_p(z) = z$ and $\mathrm{cmp}\, \mathsf{h}_p = 0$. No step solves a system or depends on p except through the coset table. The certificates are those of Remark 23.14(2), which re-derive (T1)–(T3) and the strict identity from the stored values; their boundary items concern $\tau^{\partial}_{\mathcal{I}}$, not built here.

The check through the lift (v1.23.0-pre.31). Put $T := \mathrm{cmp} \circ \tau_{\mathcal{I}} \colon \mathcal{I}_\bullet \to C_\bullet(X)$, additive. On $\mathcal{I}$, (T1) is the defining property of the extension from the free $\mathbb{Z}[\Gamma''_g]$-basis (the Schreier check certifies its bookkeeping), and cmp is Γ-equivariant, so $T(hy) = (g^{-1}hg)\,T(y)$ for $h \in \Gamma''_g$. *Step 1: $T = 0$ on $G_{s,t}$, $t \geq 1$.* $G_{s,t} = \bigoplus_\sigma \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\sigma]} F^\sigma_t$ with $F^\sigma_t = \mathbb{Z}[\Gamma_\sigma]$ is free over $\mathbb{Z}[\Gamma]$ on its e_c, hence spanned over $\mathbb{Z}[\Gamma'']$ by the $s_j e_c$ in it, on which T vanishes by hypothesis; twisted linearity gives the rest. *Step 2: $T = 0$ on $\ker \mathrm{cmp} \cap \bigoplus_s G_{s,0}$.* On $G_{s,0} = \bigoplus_\sigma \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\sigma]} \mathbb{Z}[\Gamma_\sigma]$, cmp is $\mathrm{id} \otimes \varepsilon$ (Definition 23.4); $\mathbb{Z}[\Gamma]$ is free, hence flat, over $\mathbb{Z}[\Gamma_\sigma]$, so its kernel is $\bigoplus_\sigma \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\sigma]} I_\sigma$, I_σ the augmentation ideal of the cyclic group Γ_σ, which is $(g_\sigma - 1)\mathbb{Z}[\Gamma_\sigma] = \partial^F(F^\sigma_1)$. So such an x is $x = d^{(0)}y$ for some $y \in G_{s,1}$ (the sign $(-1)^s$ of $d^{(0)}$ is absorbed into y). Now $dy = d^{(0)}y + d^{(1)}y + d^{(2)}y$ with $d^{(r)}y \in G_{s-r,r}$, at $t = r \geq 1$ for $r \geq 1$; by Step 1, (T2) of $\tau_{\mathcal{I}}$ and the chain map cmp, $T(x) = T(dy) = \mathrm{cmp}\, \tau_{\mathcal{I}}(dy) = \mathrm{cmp}\, d\, \tau_{\mathcal{I}}(y) = \partial\, \mathrm{cmp}\, \tau_{\mathcal{I}}(y) = \partial\, T(y) = 0$. *Step 3.* cmp vanishes on $t \geq 1$, so $\ker \mathrm{cmp} = (\ker \mathrm{cmp} \cap \bigoplus_s G_{s,0}) \oplus \bigoplus_{t \geq 1} G_{s,t}$, and $T(\ker \mathrm{cmp}) = 0$ by Steps 1–2. As cmp is onto in every degree (Proposition 23.5(2)), $\tau'_X(\mathrm{cmp}x) := T(x)$ defines an additive map on $C_\bullet(X)$, the only one with $\tau'_X \mathrm{cmp} = \mathrm{cmp}\tau_{\mathcal{I}}$; it is integral (cmp and $\tau_{\mathcal{I}}$ are). For $y = \mathrm{cmp}x$ and $h \in \Gamma''_g$: (T1) $\tau'_X(hy) = T(hx) = (g^{-1}hg)T(x) = (g^{-1}hg)\tau'_X(y)$; (T2) $\partial\tau'_X(y) = \partial \mathrm{cmp}\tau_{\mathcal{I}}x = \mathrm{cmp}\tau_{\mathcal{I}}dx = \tau'_X(\mathrm{cmp}dx) = \tau'_X(\partial y)$; (T3) $\varepsilon\tau'_X(y) = (\varepsilon \circ \mathrm{cmp})(\tau_{\mathcal{I}}x) = (\varepsilon \circ \mathrm{cmp})(x) = \varepsilon(y)$, by (T3) of $\tau_{\mathcal{I}}$ with the augmentation $\varepsilon \circ \mathrm{cmp}$. Finally $\tau'_X(s_j \mathrm{cmp} e_c) = T(s_j e_c) = t_{j,c}$; as $\Gamma = \bigsqcup_j \Gamma''_g s_j$ and every simplex of X is $\gamma\, \mathrm{cmp} e_c$, $\gamma \in \Gamma$ (the stored-oriented representatives Σ_s), the simplices $s_j \mathrm{cmp} e_c$ contain a set of Γ''_g-orbit representatives, and the (T1)-extension of the table from any such set is τ'_X: the table is the table of the datum τ'_X. Conversely, if the table is not the table of any g-twisted chain datum over $\mathbb{Z}$, no additive $\tau_{\mathcal{I}}$ passes the certificate, since one that did would produce such a datum.

Independence of the lift. Both $\tau_{\mathcal{I}}$ and $\tau'_{\mathcal{I}}$ are chain maps $\mathcal{I}_\bullet \to \mathcal{I}_\bullet$ that are $\mathbb{Z}[\Gamma''_g]$-linear for the twisted action $h \mapsto g^{-1}hg$ on the target (T1) and lift $\mathrm{id}_{\mathbb{Z}}$ (T3); the source is free over $\mathbb{Z}[\Gamma''_g]$ and the target, restricted along $h \mapsto g^{-1}hg$, is an acyclic augmented complex, so the fundamental lemma [standard: [Wei94, Thm. 2.2.6]] gives a $\mathbb{Z}[\Gamma''_g]$-linear $h_{\mathcal{I}}$ of degree $+1$, i.e. with (T1), and $\tau_{\mathcal{I}} - \tau'_{\mathcal{I}} = \partial h_{\mathcal{I}} + h_{\mathcal{I}}\partial$ — whether or not the two lifts start from the same τ_X. For any additive u of degree k with (T1) put $(\Theta_u \phi)(x)_i := \sum_t \rho_n(\delta_{i,t})\, \phi(u(s_{j(i,t)}x))_{a(i,t)}$, the formula (68) with u for τ; the proof of Proposition 27.2(3) shows, with (T1) of u only, that Θ_u maps Γ-equivariant cochains of degree $q+k$ to Γ-equivariant cochains of degree q. Θ_u is additive in u, $\Theta_{\tau_{\mathcal{I}}} = \widetilde{T}$, and, because $\partial(s_j x) = s_j \partial x$ and the indices $j(i,t)$, $a(i,t)$, $\delta_{i,t}$ do not depend on x, $\Theta_{\partial \circ h_{\mathcal{I}}}\phi = \Theta_{h_{\mathcal{I}}}(\phi \circ \partial)$ and $\Theta_{h_{\mathcal{I}} \circ \partial}\phi = (\Theta_{h_{\mathcal{I}}}\phi) \circ \partial$. With $(d_{\mathrm{I}}\phi)(x) = \phi(\partial x)$, unsigned (the convention of Lemma 15.4(2); v1.23.0-pre.32, referee m1), $\mathsf{H} := \Theta_{h_{\mathcal{I}}}$ gives $\widetilde{T} - \widetilde{T}' = \Theta_{\partial h_{\mathcal{I}}} + \Theta_{h_{\mathcal{I}}\partial} = \mathsf{H} d_{\mathrm{I}} + d_{\mathrm{I}} \mathsf{H}$ in every degree, with no sign; these are the two identities of Lemma 15.4(2) in the level-N form. (Under a signed convention $d_{\mathrm{I}} = \varepsilon_q(\,\cdot \circ \partial)$ on C^q one would take $\mathsf{H} := \varepsilon_{q-1}\Theta_{h_{\mathcal{I}}}$ on C^q; not the convention here.) It is integral because $h_{\mathcal{I}}$ and the $\delta_{i,t}$ are. For the transports, $\Psi^*(\widetilde{T} - \widetilde{T}')\Phi^* = d(\Psi^* \mathsf{H} \Phi^*) + (\Psi^* \mathsf{H} \Phi^*)d$, d the coboundary of the target model, the comparisons being cochain maps. So the induced endomorphisms of H^q coincide (both are T_p by (3)(c), (5)). The torsion module (G, ϕ) and the lattice (Λ, ψ) of Proposition 31.15(3), (4) are defined from the induced endomorphism, so their invariants agree; for a fixed certified Smith form, $\tilde{\Phi}_{ij}$ is the exact coordinate of $\phi[x_j]$ when $2\sigma_{\max} \leq K$ and agrees with it modulo $\ell^{\min(\sigma_i, K')}$ otherwise (ibid. (3) and its proof), which gives the last sentence.

(3) (a) is Proposition 27.2(1), whose proof reads nothing of the complex. (b) is Propo-

sition 27.2(2)'s proof with $\mathcal{I}_\bullet$ for $C_\bullet(K^u)$: it uses (T1), (T2) of τ and $\partial(s_j x) = s_j \partial x$, both true on $\mathcal{I}_\bullet$ (Lemma 15.4). (c) The proof of Proposition 27.2(3) assumes $6 \in R^\times$ only to make $C_\bullet(K^u) \otimes R$ a projective resolution of R over $R[A]$, over $R[A']$ and over $R[A'']$; $\mathcal{I}_\bullet \otimes R$ is a *free* $R[\Gamma]$-resolution (Proposition 23.5(2)), hence free over every subgroup of finite index. On it the inclusion computes $\mathrm{res}^A_{A'}$, $\psi \mapsto \rho_n(g)\psi \circ \tau_{\mathcal{I}}$ computes $(\iota_g, \rho_n(g))^*$ — $\tau_{\mathcal{I}}$ is, by (T1) and (T3), a chain map along $\iota_g \colon A'' \to A'$ lifting id_R, unique up to homotopy by the fundamental lemma, the source being free over $R[A'']$ and the target acyclic [standard: [Wei94, Thm. 2.2.6]] — and the transfer computes $\mathrm{cor}^A_{A''}$ [standard: [Bro82, Ch. III.9–10]]; the composite is the reciprocity form, which equals (68) by Proposition 27.2(3)'s computation. Independence on cohomology: the double-coset operator does not depend on these choices (Brown, loc. cit.). At cochain level, the independence of the transversal is Proposition 29.34(4), whose proof uses (T1) of τ only, for a τ that does not depend on r_t; the $\tau_{\mathcal{I}}$ of (2) is built on $\{s_j e_c\}$ from τ_X alone. (d) is Proposition 27.2(4)(a).

(4) For ϕ on $C_q(X)$ and $x \in \mathcal{I}_q$, using $\mathrm{cmp}\,\tau_{\mathcal{I}} = \tau_X\,\mathrm{cmp}$ and the Γ-equivariance of cmp ($s_j \in \Gamma$): $(\widetilde{T}^{(N)}_{\mathcal{I},p}\mathrm{cmp}^*\phi)(x)_i = \sum_t \rho_n(\delta_{i,t})\phi(\mathrm{cmp}\,\tau_{\mathcal{I}}(s_j x))_a = \sum_t \rho_n(\delta_{i,t})\phi(\tau_X(s_j\,\mathrm{cmp}x))_a = (\mathrm{cmp}^*\widetilde{T}^{(N)}_{X,p}\phi)(x)_i$ — the proof of Proposition 23.13(3), blockwise. For $6 \in R^\times$ both shipped operators induce $[AgA]$ by Proposition 27.2(3), and $H(\mathrm{cmp}^*)$ is an isomorphism intertwining $\widetilde{T}^{(N)}_{X,p}$ with $\widetilde{T}^{(N)}_{\mathcal{I},p}$. So on $H^q(A;V) \otimes \mathsf{O}$ and on $H^q(A;V) \otimes \mathbb{Z}_{(\ell)}$ the local operators are $T_p \otimes \mathsf{O}$, $T_p \otimes \mathbb{Z}_{(\ell)}$, and Proposition 31.7(3) glues them to T_p.

(5) Proposition 31.12(6)(a)–(c) is stated for any cochain endomorphism T of $C^\bullet_{\mathcal{I}}(M_R)$ inducing an endomorphism t of $H^\bullet(\Gamma; M_R)$; take $T = \widetilde{T}^{(N)}_{\mathcal{I},p} \otimes R$ and t the transport of T_p under Shapiro's isomorphism, which is natural and independent of the resolution. The same argument with Φ_ℓ, Ψ_ℓ (Proposition 31.2(4), (5): conjugating a cochain map that induces t by comparisons induces t) gives $T'^{(\ell)}_p$. The coefficients lie in O_ℓ because Φ_ℓ, Ψ_ℓ, $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$ do and $\widetilde{T}^{(N)}_{\mathcal{I},p}$ is integral.

(6) (a) For $6 \in R^\times$ the first statement is Proposition 27.2(3); $C_\bullet(X) \otimes R$ and $C_\bullet(\mathcal{T}) \otimes R$ are projective resolutions of R (Remark 7.6, Lemma 28.4) and $\mathsf{i}_{\mathcal{T}}$, $\mathsf{r}_{\mathcal{T}}$ are augmentation-preserving $\mathbb{Z}[\Gamma]$-chain maps between them (Proposition 31.12(4)(a)), i.e. comparisons; conjugating by them preserves the induced operator. (b) By Proposition 31.9(1) every stabiliser in A of a simplex of X, or of a cell of $\mathcal{T}$, has order 1, 2 or 4, a unit in R; so $C_\bullet(X) \otimes R$ and $C_\bullet(\mathcal{T}) \otimes R$ are projective resolutions of R over $R[A]$ and over its subgroups A', A'' (the proof of Proposition 31.9(2)), which, with a datum over $\mathbb{Z}$ satisfying (T1)–(T3), is all the proof of Proposition 27.2(3) uses; a datum known only modulo ℓ, or not checked, is outside the hypothesis.

(7) The proof of Proposition 29.34(1) reads, for both operators, the reciprocity form $\psi \mapsto \sum_t \rho_n(r_t^{-1}g)\psi(\tau(r_t\,\cdot))$ with the same τ and the same r_t, and $(\alpha_{B/A})_*$ as $\psi \mapsto \psi$; it uses nothing of the complex, so it holds with $\mathcal{I}_\bullet$, $\tau_{\mathcal{I}}$. Its hypotheses $B \cap A'' = B''$ and $A = A''B$ for T_p, $p \nmid MN$, are verified in Proposition 29.34(2). The cone: the computation of Lemma 15.8 with $h = 0$, as there. Transport: Proposition 31.10(1)(a) (α commutes strictly with every comparison), so $T'^B\alpha = \Psi^* T^B \Phi^* \alpha = \Psi^* T^B \alpha \Phi^* = \Psi^* \alpha T^A \Phi^* = \alpha T'^A$; on the tree this is Proposition 31.12(6)(d). Over $\mathbb{Z}$ the cone $C(\alpha_{\mathcal{I}})$ itself carries the operator; Proposition 31.10(2)(c) glues its cohomology from the local cones, compatibly by the strict identities.

(8) V is a module over the semigroup $\Delta_0(N)$ of integral matrices with lower-left entry $\equiv 0 \pmod N$ and upper-left entry prime to N (ρ_n is multiplicative on $\mathrm{M}_2(\mathbb{Z})$), so the Hecke ring $\mathcal{H}(\Gamma_0(N), \Delta_0(N))$ acts on $H^q(A; V_R)$ through $AgA \mapsto [AgA]$, a ring homomorphism for every module and every q (the citations of Proposition 23.23(2): [RW70, Thm. 1], [AS86, §1.1, p. 194]); and $T(1,p)$, $T(1,p')$ commute in that ring [standard: [Shi71, Thm. 3.34, Thm. 3.35 and §3.3], as cited in Theorem 29.10; commutativity among the $T(1,p)$, $p \nmid N$, VERIFY].

(9) (v1.26.0-pre.1.) (a) For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in A$, $g\gamma g^{-1} = \begin{pmatrix} a & b/p \\ pc & d \end{pmatrix}$; so the integral elements of gAg^{-1} are the $\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \in \Gamma$ with $c' = pc$, $N \mid c$, i.e. $pN \mid c'$, and $A'' = A \cap gAg^{-1} = \Gamma_0(pN)$;

with Γ in place of A the same computation gives $\Gamma_g'' = \Gamma_0(p) \supseteq \Gamma_0(pN)$. As $[\Gamma : \Gamma_0(M)] = M\prod_{q\mid M}(1+1/q)$, $[A : A''] = p$ when $p \mid N$. The r_t lie in A (lower-left entry tN), and $r_s r_t^{-1} = \begin{pmatrix} 1 & 0 \\ (s-t)N & 1 \end{pmatrix} \in \Gamma_0(pN)$ iff $p \mid s-t$; so the p matrices r_t represent distinct classes of $A''\backslash A$, hence all of them. (b) Proposition 27.2(4)(d) is (68) with $A = B = \Gamma_0(N)$ and this g, named $U_p^{(N)}$ for $p \mid N$; Proposition 28.9(2) and Definition 31.27 (through Proposition 31.10(3)(b′)) use the double coset of the same g; the adjointness is Proposition 29.34(3)(d). The proof of item (1) needs u, b with $up - bN = 1$, which do not exist when $p \mid N$. (c) Item by item, the hypothesis $p \nmid N$ is not used. (2): the statement and its proof read τ_X, Γ_g'' and the coset table of $\Gamma_g''\backslash\Gamma$ only — no N. The lift the code reads, the bulk component of a pair lift, is admissible at $p \mid N$ as well (v1.26.0-pre.2, referee S1): Proposition 31.21(1)'s construction, certificate and homotopy uniqueness read $\Gamma_g'' = \Gamma_0(p)$, g, τ_∂, τ^s and $\mathcal{I}^\partial$ only, nothing of N (the datum is a level-1 object, built once per p); the hypothesis $p \nmid N$ of that proposition is the scope of its other items and is not used by its item (1). (3)(a), (b): Proposition 27.2(1), (2) and their proofs hold for any finite-index A, B and any $g \in \Delta$; Lemma 15.4 reads nothing of N. (3)(c): the proof uses that $\mathcal{I}_\bullet \otimes R$ is free over every subgroup of finite index, that $\tau_{\mathcal{I}}$ is a chain map along $\iota_g\colon A'' \to A'$ lifting id_R (by (T1) on $\Gamma_g'' \supseteq A''$, (a), and (T3)), the fundamental lemma and the transfer; and, for the independence of r_t, Proposition 29.34(4), whose proof uses (T1) and $A'' \subseteq \Gamma_g''$ only. (4): the displayed identity is the computation of its proof, which uses the strict identity $\mathrm{cmp}\tau_{\mathcal{I}} = \tau_X \mathrm{cmp}$ and the Γ-equivariance of cmp; the comparison for $6 \in R^\times$ is Proposition 27.2(3), for any $g \in \Delta$; the glueing is Proposition 31.7(3), with the same argument as there. (5): Propositions 31.12(6) and 31.2(4), (5) hold for any cochain endomorphism that induces an endomorphism of $H^\bullet$. (6): (a) is Proposition 27.2(3) with Proposition 31.12(4)(a), (b) is Proposition 31.9 (orders of stabilisers); neither uses $p \nmid N$. The independence of the lift uses the fundamental lemma over $\mathbb{Z}[\Gamma_g'']$ and the formal identities of Θ_u, which read the indices $j(i,t)$, $a(i,t)$, $\delta_{i,t}$ only. (d) Item (1) is excluded by (b); item (7) rests on Proposition 29.34(2)'s instance for T_p, $p \nmid MN$, and item (8) on the Hecke-ring relations among the $T(1,p)$, $p \nmid N$; neither is claimed for U_p here. □

Proposition 31.14 (The characteristic polynomial over $\mathbb{Z}$ at level N, by residues; the certified bound; the output rule)**.** *[proved here] [standard: [DI95, Thm. 12.2.2, p. 108, Prop. 12.4.10, p. 117], [Del74, Thm. 8.2]VERIFY, [DS05, §4.5–4.6, §5.2]VERIFY] (given the eigenvalue bound of (3) as the input, as in Proposition 23.19; v1.23.0-pre.24; clause C-D4 of the plan; refereed ACCEPT-with-fixes, its fixes at v1.23.0-pre.25.) Let $N \geq 1$, $n \geq 0$, $p \nmid N$ a prime, $T_p = [AgA]$ (Proposition 31.13(1)), and $\chi_{p,q}^{(N)} := \det\big(x - T_p \mid H^q_{\mathrm{fr}}(N)\big) \in \mathbb{Z}[x]$ (T_p preserves torsion).*

1. ***(Only degree 1 carries anything.)*** *$H^q_{\mathrm{fr}}(N) = 0$ for $q \geq 2$; $H^0_{\mathrm{fr}}(N) = H^0(A;V)$ is 0 for $n > 0$ and $\mathbb{Z}$ for $n = 0$, where $T_p = p+1$. So $\chi_{p,q}^{(N)} = 1$ for $q \geq 2$, $\chi_{p,0}^{(N)} = x - (p+1)$ if $n = 0$ and 1 otherwise, and $\chi_{p,1}^{(N)}$ is monic of degree $r_N := b_1(M_N)$, the rank of Proposition 31.7(7)(b).*
2. ***(Residues on the O-piece.)*** *Let $\ell' \geq 5$ be a prime with $s_{\ell'}(M_N) = 0$ (Proposition 31.8(1), (2); every $\ell' > n+1$ qualifies; $\ell' = p$ is admissible — neither the argument nor the model needs $\ell' \neq p$ — and the code may exclude it, v1.23.0-pre.25, referee n1). Reduction $V \to V \otimes \mathbb{F}_{\ell'}$ induces a T_p-equivariant isomorphism $H^1_{\mathrm{fr}}(N) \otimes \mathbb{F}_{\ell'} \cong H^1(A; V \otimes \mathbb{F}_{\ell'})$, so $\chi_{p,1}^{(N)} \bmod \ell'$ is the characteristic polynomial of T_p on $H^1(C^\bullet_{\mathrm{eq}}(M_{N,\mathbb{F}_{\ell'}}))$ — the shipped level-N operator, or its transport to the tree complex $\mathrm{T}(M_{N,\mathbb{F}_{\ell'}})$ (Proposition 31.13(6)(a)), or, for n even, the endomorphism $[H_{N,p}]$ induced on the **cokernel presentation** $M_{N,\mathbb{F}_{\ell'}}/\mathrm{Rel}_N$ (Proposition 28.9(1), (2), through $\varkappa_N$ of Definition 28.6; v1.23.0-pre.31, D3's contract note 1: the route the code reads); for n odd, $r_N = 0$ and nothing is read. **No 2- or 3-adic input enters $\chi_{p,1}^{(N)}$.***
3. ***(The bound — the input.)*** *Every complex eigenvalue λ of T_p on $H^1_{\mathrm{fr}}(N) \otimes \mathbb{C}$ satisfies*

$|\lambda| \le \mathsf{B}_p := 1 + p^{n+1}$. *Hence the coefficient* c_i *of* $x^{r_N - i}$ *in* $\chi_{p,1}^{(N)}$ *satisfies* $|c_i| \le B_{p,i} := \binom{r_N}{i}\mathsf{B}_p^i \le 2^{\mathsf{b}_{p,i}}$ *with the word-size integer* $\mathsf{b}_{p,i} := r_N + i\,(n+1)\,\mathrm{bitlen}(p)$, $\mathrm{bitlen}(p) = \lfloor \log_2 p \rfloor + 1$. *The bound is attained, by the Eisenstein eigenvalue* $1 + p^{n+1}$, *whenever* $r_N > 0$, *i.e.* n *even and* $(N, n) \ne (1, 0)$ *(*$r_N = 0$ *for* n *odd; v1.23.0-pre.25, referee t1).*

4. ***(Recovery in words.)*** *Let* $\mathcal{S}$ *be a finite set of primes as in (2),* $P := \prod_{\ell' \in \mathcal{S}} \ell'$, *and* $\rho_{\ell'} \in \mathbb{F}_{\ell'}[x]$ *the residues of (2). If* $\sum_{\ell' \in \mathcal{S}} (\mathrm{bitlen}(\ell') - 1) \ge \mathsf{b}_{p,i} + 1$, *then* $P > 2B_{p,i}$ *and* c_i *is the unique integer in* $(-P/2, P/2]$ *with* $c_i \equiv \rho_{\ell',i}$ *for every* $\ell' \in \mathcal{S}$. *It is computed without a big integer,* ***provided every*** $\ell' \in \mathcal{S}$ ***is*** $< 2^{32}$ *(v1.23.0-pre.25, referee m2): Garner's mixed-radix digits* $d_k < \ell'_k$ *of the representative* $c^+ \in [0, P)$, *each digit a word operation (its products are* $< \ell'^2_k < 2^{64}$*); the sign by comparing the digits, from the top, with those of* $(P-1)/2$, *which are* $(\ell'_k - 1)/2$*;* $|c_i|$ *in mixed radix by digit-wise subtraction from* P*; and* $c_i \bmod Q$, *for any word-size* $Q \le 2^{64}$ *(for instance* 2^{64}, 3^{40}, ℓ^K*), by Horner's rule on the digits modulo* Q, ***with a word mulmod****: the product of an accumulator* $< Q$ *and a radix* $\ell'_k < 2^{32}$ *exceeds the word for* $Q > 2^{32}$ *(about* 2^{91} *at* $Q = 3^{40}$*), so it is formed by shift-and-add or Montgomery multiplication modulo* Q, *never by a double-word integer (Remark 23.22(1)).* ***The negative case****: if* $c_i = c^+ - P < 0$, *then* $c_i \bmod Q = (c^+ \bmod Q - P \bmod Q) \bmod Q$, *with* $P \bmod Q$ *by the same Horner rule on the radices (equivalently: Horner on the digits of* $|c_i|$*, then negated modulo* Q*).*
5. ***(The output rule, [R-Q2].)*** *Per coefficient:* certified *(the inequality of (4) holds) and* $|c_i| < 2^{63}$*: the integer;* certified *and* $|c_i| \ge 2^{63}$ *(decided on the digits): the sign, the mixed-radix digits and their primes, and* $\mathsf{b}_{p,i}$ *— determined, not printed in decimal;* not certified*: the residue vector* $(\rho_{\ell',i})_{\ell' \in \mathcal{S}}$ *and* $\mathsf{b}_{p,i}$*, flagged undetermined,* ***never*** *a lifted integer.* $\chi_{p,1}^{(N)}$ *is reported as determined iff every coefficient is certified. A held-out prime is a consistency check, not a certificate.*
6. ***(Other moduli.)*** *Residues of* $\chi_{p,1}^{(N)}$ *modulo* ℓ^k, ℓ *any prime, read on an* ℓ*-local lattice known modulo* ℓ^k *(Proposition 31.15(4)), may join* $\mathcal{S}$ *as further pairwise coprime moduli (at* $\ell = 2$ *with the sign test adapted to an even* P*); this is admissible and never needed.*

At $N = 1$ *this is Proposition 23.19 with its bound made explicit.*

Proof. (1) $24 \cdot H^q(\Gamma; M_N) = 0$ for $q \ge 2$ (Proposition 31.6(4)(a)), and Shapiro. $H^0(A; V) \subseteq V$ is torsion-free, of rank $b_0(M_N)$, which is 1 for $n = 0$ and 0 for $n > 0$ (Proposition 31.7(4), (7)(c)). For $n = 0$ the coefficients are trivial and $\rho_0(g) = 1$, so on H^0 res and $(\iota_g, 1)^*$ are the identity of $\mathbb{Z}$ and $\mathrm{cor}^A_{A''}$ is multiplication by $[A : A''] = \left|\mathbb{P}^1(\mathbb{F}_p)\right| = p + 1$ [standard: [Bro82, Ch. III.9]].

(2) The long exact sequence of $0 \to V \xrightarrow{\ell'} V \to V \otimes \mathbb{F}_{\ell'} \to 0$ over A gives $0 \to H^1(A; V) \otimes \mathbb{F}_{\ell'} \to H^1(A; V \otimes \mathbb{F}_{\ell'}) \to H^2(A; V)[\ell'] \to 0$; the right-hand term vanishes (24 kills it, $\ell' \ge 5$), and $H^1(A; V)$ has no ℓ'-torsion when $s_{\ell'}(M_N) = 0$ (Proposition 31.8(1)), so the left-hand term is $H^1_{\mathrm{fr}}(N) \otimes \mathbb{F}_{\ell'}$. The map is induced by the coefficient map, which commutes with $\rho_n(g)$, so it commutes with each factor of $[AgA]$ (naturality of res, cor and of $(\iota_g, f)^*$ in compatible pairs, [standard: [Bro82, Ch. III.8–9]]). The model statements are Proposition 31.13(6)(a) at $R = \mathbb{F}_{\ell'}$. The presentation (v1.23.0-pre.31): for n even and $R = \mathbb{F}_{\ell'}$, $6 \in R^\times$ as $\ell' \ge 5$, so $\varkappa_N \colon M_{N,R}/\mathrm{Rel}_N \to H^1(A; V_R)$ is an isomorphism (Definition 28.6), $H_{N,p}$ preserves Rel_N and $\varkappa_N \circ [H_{N,p}] = [A\,g\,A] \circ \varkappa_N$ (Proposition 28.9(1), (2)), with $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ there as here and $[A\,g\,A] = T_p$ for $p \nmid N$; conjugate endomorphisms have the same characteristic polynomial. For n odd, $-I \in A$ is central and acts on V by -1, so 2 kills $H^\bullet(A; V)$ (the argument of Proposition 27.5(2)), $r_N = 0$ and $\chi_{p,1}^{(N)} = 1$.

(3) By Eichler–Shimura [standard: [DI95, Thm. 12.2.2, p. 108], verified (Lemma 30.4)] $H^1(A; V \otimes \mathbb{C}) \cong M_k(\Gamma_0(N)) \oplus \bar{S}_k(\Gamma_0(N))$, $k = n + 2$, for every $n \ge 0$, compatibly with the Hecke ring [standard: [DI95, Prop. 12.4.10, p. 117], verified], T_p acting on $\bar{S}_k$ through complex conjugation; and $H^1_{\mathrm{fr}}(N) \otimes \mathbb{C} = H^1(A; V \otimes \mathbb{C})$. For $p \nmid N$: on $S_k(\Gamma_0(N))$ the eigenvalues of T_p are the a_p of normalised newforms of levels dividing N [standard: [DS05, §5.6–5.8], Atkin–Lehner–Li,

VERIFY], and $|a_p| \le 2p^{(k-1)/2}$ [standard: [Del74, Thm. 8.2], the Ramanujan–Petersson bound, VERIFY]; on the Eisenstein subspace they are $\psi(p) + \varphi(p)p^{k-1}$ with Dirichlet characters ψ, φ (the eigenforms $E_k^{\psi,\varphi,t}$, and for $k = 2$ the differences of E_2, eigenvalue $1 + p$) [standard: [DS05, §4.5–4.6, §5.2], VERIFY], of absolute value $\le 1 + p^{k-1}$. As $2p^{(n+1)/2} \le 1 + p^{n+1}$, every $|\lambda| \le \mathsf{B}_p$. The coefficient c_i is $\pm$ the i-th elementary symmetric function of the r_N eigenvalues, a sum of $\binom{r_N}{i}$ products of i of them. Bits: $\binom{r_N}{i} \le 2^{r_N}$, and $p \le 2^{\mathrm{bitlen}(p)} - 1$ gives $1 + p^{n+1} \le 2^{(n+1)\mathrm{bitlen}(p)}$. Sharpness: E_k ($k \ge 4$), resp. a difference of E_2 at $N > 1$, contributes $1 + p^{n+1}$.

(4) $\mathrm{bitlen}(\ell') - 1 = \lfloor \log_2 \ell' \rfloor$, so $P \ge 2^{\sum \lfloor \log_2 \ell' \rfloor} \ge 2^{\mathsf{b}_{p,i}+1} > 2B_{p,i}$; two integers in $(-P/2, P/2]$ congruent modulo every ℓ' are equal (Chinese remainder theorem), and c_i is one of them. Garner: $c^+ := \sum_k d_k \prod_{j<k} \ell'_j$ with $d_k := (\rho_{\ell'_k,i} - \sum_{m<k} d_m \prod_{j<m} \ell'_j) \cdot (\prod_{j<k} \ell'_j)^{-1} \bmod \ell'_k$, every quantity reduced modulo ℓ'_k before it is formed, so no intermediate exceeds ℓ'^2_k. Mixed-radix representations with digits $d_k < \ell'_k$ are unique and ordered lexicographically from the top, and $(P-1)/2 = \sum_k \frac{\ell'_k - 1}{2} \prod_{j<k} \ell'_j$ (all ℓ'_k odd; the sum telescopes); $c_i = c^+$ if $c^+ \le (P-1)/2$ and $c^+ - P$ otherwise. Horner modulo Q forms only residues modulo Q and products of two of them, which the word mulmod reduces without leaving the word; $c^+ - P \equiv c^+ \bmod Q - P \bmod Q$.

(5) is the reading rule of (4); it forms no integer beyond a word (Theorem 18.7(3), Remark 23.22(1)). (6): $\chi \bmod \ell^k = \det(x - X) \bmod \ell^k$ for any matrix X of T_p on $H^1_{\mathrm{fr}}(N) \otimes \mathbb{Z}_{(\ell)}$ (base change of a characteristic polynomial), and the Chinese remainder theorem holds for pairwise coprime moduli. □

Proposition 31.15 (Hecke on finite ℓ-groups: what is choice-free, and how it is read exactly from a local run)**.** *[proved here] (v1.23.0-pre.24; clause C-D5 of the plan; refereed ACCEPT-with-fixes, its fixes M1, m1 at v1.23.0-pre.25.) Let ℓ be a prime, G a finite abelian ℓ-group, $\mathrm{len}(G) := \log_\ell |G|$, and $\phi \in \mathrm{End}(G)$. Write $G = \bigoplus_i \mathbb{Z}/\ell^{a_i} e_i$ and $\phi e_k = \sum_i \phi_{ik} e_i$, $\phi_{ik} \in \mathbb{Z}/\ell^{a_i}$ (so $\ell^{a_k}\phi_{ik} = 0$ in $\mathbb{Z}/\ell^{a_i}$).*

1. ***(Choice-free invariants.)*** *Each of the following depends only on the isomorphism class of the $\mathbb{Z}[x]$-module (G, ϕ):*
   - (a) ***the layers**: for $j \ge 1$, $V_j(G) := (\ell^{j-1}G)[\ell]/(\ell^j G)[\ell]$, an $\mathbb{F}_\ell$-space of dimension $\#\{i : a_i = j\}$, with the endomorphism ϕ_j induced by ϕ, and $P_j(\phi) := \det(x - \phi_j) \in \mathbb{F}_\ell[x]$. In the coordinates above, ϕ_j is the block $(\phi_{ik} \bmod \ell)_{a_i = a_k = j}$;*
   - (b) ***the Fitting components**: for $c \in \mathbb{Z}$, $G_{[c]} := \ker(\phi - c)^{\mathrm{len}(G)}$, which depends only on $c \bmod \ell$; $G = \bigoplus_{\bar{c} \in \mathbb{F}_\ell} G_{[c]} \oplus G_\circ$, $G_\circ := \bigcap_c \mathrm{im}(\phi - c)^{\mathrm{len}(G)}$, all summands ϕ-stable; and $G_{[c]}$ has exactly $\mathrm{mult}_{x-\bar{c}} P_j(\phi)$ cyclic summands of order ℓ^j, so its type is read off the P_j;*
   - (c) *the types of $\ker(\phi - c)$ and $\mathrm{coker}(\phi - c)$, $c \in \mathbb{Z}$ (they depend on c modulo the exponent of G);*
   - (d) *for commuting $\phi_1, \dots, \phi_m$, the isomorphism class of the $\mathbb{Z}[x_1, \dots, x_m]$-module G, and every invariant derived from it;*
   - (e) ***the layer blocks modulo** ℓ^{e_j} (v1.23.0-pre.25, referee M1): with $d_j := \min\{|a_i - j| : a_i \ne j\}$ ($d_j := \infty$ if G has one layer) and $e_j := \min(j, d_j)$, the characteristic polynomial modulo ℓ^{e_j} of the diagonal block $(\phi_{ik})_{a_i = a_k = j}$; its reduction modulo ℓ is $P_j(\phi)$.*
2. ***(Choice-free only modulo** ℓ^{e_j}**.)** The characteristic polynomial over $\mathbb{Z}/\ell^j$ of the diagonal block $(\phi_{ik})_{a_i = a_k = j}$ — "the characteristic polynomial on the Smith summands of order ℓ^j" — depends on the decomposition: on $G = \mathbb{Z}/\ell\, e_1 \oplus \mathbb{Z}/\ell^2 e_2$, $\phi e_1 = \ell e_2$, $\phi e_2 = 0$, it is x, and in the basis e_1, $e_1 + e_2$ it is $x - \ell$. It is an invariant modulo ℓ^{e_j} (1)(e), and in general no finer (v1.23.0-pre.25, referee M1, replacing "only its reduction modulo ℓ"): for every $1 \le d < j$, on $G = \mathbb{Z}/\ell^{j-d} e_1 \oplus \mathbb{Z}/\ell^j e_2$ with $\phi e_1 = \ell^d e_2$, $\phi e_2 = 0$ the layer-j block is (0) in the basis (e_1, e_2) and (ℓ^d) in the basis $(e_1, e_1 + e_2)$, and on $G = \mathbb{Z}/\ell^{j+d} e_1 \oplus \mathbb{Z}/\ell^j e_2$ with $\phi e_2 = \ell^d e_1$, $\phi e_1 = 0$ it is (0) and $(-\ell^d)$ in the basis $(e_1 + e_2, e_2)$; in both $d_j = d = e_j$, and the two polynomials differ modulo ℓ^{d+1} (the example above is $j = 2$, $d = 1$). If $d_j \ge j$*

the invariance modulo ℓ^j is the whole block's. Modulo ℓ it is $P_j(\phi)$; if G has one layer (G free over $\mathbb{Z}/\ell^a$), the characteristic polynomial over $\mathbb{Z}/\ell^a$ is an invariant. (For instance $G = \mathbb{Z}/2 \oplus \mathbb{Z}/8$: the layer-3 polynomial is an invariant modulo 4, not modulo 8.)

3. ***(The exact read from a local run.)*** *Let $C^\bullet$ be a complex of finitely generated free $\mathbb{Z}_{(\ell)}$-modules, $q \geq 1$, F a cochain endomorphism of $C^\bullet$ (over $\mathbb{Z}_{(\ell)}$), $G := \operatorname{tors} H^q(C)$ and $\phi := H^q(F)|_G$. Let $L\,(d^{q-1} \bmod \ell^K)\,R = \Sigma$ be a Smith form over $\mathbb{Z}/\ell^K$ with pivots $u_i\ell^{\sigma_i}$, certified ($s = \operatorname{rank}_{\mathbb{Q}} d^{q-1}$, Proposition 31.10(2)(g)), $J := \{i \leq s : \sigma_i > 0\}$, $\sigma_{\max} := \max_J \sigma_i$, $\tilde{x}_j$ the j-th column of L^{-1} ($j \in J$), and $\tilde{\Phi}_{ij} := (L\,F\,\tilde{x}_j)_i \bmod \ell^{\sigma_i}$ ($i, j \in J$). If $2\sigma_{\max} \leq K$ (i.e. $\sigma_i \leq K - \sigma_{\max}$ for every $i \in J$), then $G = \bigoplus_{j\in J} \mathbb{Z}/\ell^{\sigma_j}[x_j]$ for cocycles $x_j \equiv \tilde{x}_j \pmod{\ell^{K-\sigma_{\max}}}$, and $\phi[x_j] = \sum_{i\in J} \tilde{\Phi}_{ij}[x_i]$ exactly: $(G, \phi) \cong (\bigoplus_J \mathbb{Z}/\ell^{\sigma_i}, \tilde{\Phi})$, and every invariant of (1) computed from $(\sigma, \tilde{\Phi})$ is certified. Otherwise the read is refused (or K raised; at the cap of the ring, refused). It applies F to the $|J|$ vectors $\tilde{x}_j$ only.* ***The condition is necessary*** *for the module (v1.23.0-pre.25, referee m1): for every $\sigma \geq 1$ and $\sigma < K < 2\sigma$ there is a certified form whose read is wrong (the proof), and on real data $K = 2\sigma_{\max} - 1$ gives a wrong module in 33 (cell, p, ℓ) cases of the referee's check. It is* not *the conservative policy of Proposition 31.10(2)(g)(ii), whose "exact for every $\sigma_i < K$" concerns the coordinates of an* exact *vector: here the vector is $F\tilde{x}_j$, and $\tilde{x}_j$ is exact only modulo $\ell^{K'}$, $K' := K - \sigma_{\max}$; a read may not be relaxed by analogy with* `BuildPres`. ***What needs only a certified form*** *($K > \sigma_{\max}$, so $K' \geq 1$): $\tilde{\Phi}_{ij}$ agrees with the true coordinate modulo $\ell^{\min(\sigma_i, K')}$, so the layer polynomials P_j, through (1)(b) the types of the Fitting components $G_{[c]}$, and the layer blocks' polynomials modulo $\ell^{\min(e_j, K')}$ are certified at* ***every*** *certified K; the module, (1)(c), (d) and (1)(e) beyond $\ell^{K'}$ need $2\sigma_{\max} \leq K$. The read uses of the complex only d^{q-1} and F on C^q.* ***(3′) A presentation in place of a complex*** *(v1.23.0-pre.31, D3's contract note 1). Let Δ be an $r \times g$ matrix over $\mathbb{Z}_{(\ell)}$, $Q := \operatorname{coker} \Delta = \mathbb{Z}_{(\ell)}^r / \operatorname{im} \Delta$, and $F \in \operatorname{End}(\mathbb{Z}_{(\ell)}^r)$ with $F(\operatorname{im}\Delta) \subseteq \operatorname{im}\Delta$; put $G := \operatorname{tors} Q$, ϕ the endomorphism of G induced by F, $\Lambda := Q/G$ and ψ the endomorphism of Λ induced by F. Then everything in (3) holds verbatim with $d^{q-1} := \Delta$, $C^q := \mathbb{Z}_{(\ell)}^r$ and the certifying rank $s = \operatorname{rank}_{\mathbb{Q}} \Delta$: the same read $L(\Delta \bmod \ell^K)R = \Sigma$, $\tilde{\Phi}_{ij} = (LF\tilde{x}_j)_i \bmod \ell^{\sigma_i}$, the same precisions, the same condition $2\sigma_{\max} \leq K$ for the module.* ***Moreover the free part is read****: with $\tilde{x}_j$ the j-th column of L^{-1}, $s < j \leq r$, the classes of* any *exact lifts $x_j \equiv \tilde{x}_j \pmod{\ell^K}$ form a basis of Λ (one such is $x_j := \tilde{L}^{-1}e_j$, the proof), and $X_{ij} := (LF\tilde{x}_j)_i \bmod \ell^{K'}$, $s < i, j \leq r$, is the matrix of ψ in that basis modulo $\ell^{K'}$, at every certified K — an input of (4) with $k = K'$ (v1.23.0-pre.32, referee m3). The precision K' is* ***sharp*** *(referee N3): the rows $> s$ of $\tilde{L}$ are exact only modulo $\ell^{K'}$, and on the referee's certified instances the read differs from the exact matrix at exactly $\ell^{K'}$; unlike the torsion coordinates of Proposition 31.10(2)(g)(ii), it may not be relaxed.* ***The read is presentation-independent****: if (Δ', F') is another such datum and $\theta\colon Q \to Q'$ an isomorphism with $\theta \circ [F] = [F'] \circ \theta$ (for instance two presentations of one module with one operator, or a presentation and a complex $C^\bullet$ as in (3) with $H^q(C) \cong Q$ compatibly with the operators), then θ restricts to an isomorphism $(G, \phi) \cong (G', \phi')$ and induces $(\Lambda, \psi) \cong (\Lambda', \psi')$, so every invariant of (1) and the invariant factors of (4) agree, each computed from a read that certifies it — the module and (1)(c), (d), (e) from a read with $2\sigma_{\max} \leq K$, the rest as in (3), the factors of (4) when its pivot condition holds (v1.23.0-pre.32, referee N1).*

4. ***(The free quotient: the invariant factors of $\phi - c$.)*** *Let Λ be a free $\mathbb{Z}_{(\ell)}$-module of rank r with an endomorphism ψ, $\chi_\psi \in \mathbb{Z}_{(\ell)}[x]$ its characteristic polynomial, X its matrix in some basis, known modulo ℓ^k (e.g. $\Lambda = H^1_{\mathrm{fr}}(N) \otimes \mathbb{Z}_{(\ell)}$, $\psi = T_p$, X read on a local run with the precision left by Proposition 31.10(2)(g)(iii)), and $c \in \mathbb{Z}$. If the Smith form of $(X - c) \bmod \ell^k$ has r pivots of valuation $< k$, these valuations are the exponents of the invariant factors of $\psi - c$ on Λ, i.e. the type of the finite $\mathbb{Z}_{(\ell)}[x]$-module $\operatorname{coker}(\psi - c)$*

— *choice-free. This happens iff k exceeds the largest exponent; the exponents sum to $v_\ell(\chi_\psi(c))$, so $k > v_\ell(\chi_\psi(c))$ suffices when $\chi_\psi(c) \neq 0$. If $\chi_\psi(c) = 0$ it never happens, and nothing is asserted.*

Proof. (1)(a) $\ell^m G$ and $G[\ell]$ are ϕ-stable, so ϕ induces ϕ_j, and an isomorphism of $\mathbb{Z}[x]$-modules induces isomorphisms of these subquotients intertwining the ϕ_j. In coordinates, $(\ell^{j-1}G)[\ell] = \bigoplus_{a_i \geq j} \mathbb{Z}\,\ell^{a_i-1}e_i$, so the classes of $\ell^{j-1}e_k$, $a_k = j$, form a basis of $V_j(G)$, and $\phi(\ell^{j-1}e_k) = \sum_i \ell^{j-1}\phi_{ik}e_i$: the terms with $a_i < j$ vanish; for $a_i > j$, $\ell^{a_k}\phi_{ik} = 0$ in $\mathbb{Z}/\ell^{a_i}$ gives $v_\ell(\phi_{ik}) \geq a_i - j$, so $\ell^{j-1}\phi_{ik}e_i \in (\ell^{a_i-1}G)[\ell] \subseteq (\ell^j G)[\ell]$ vanishes in V_j; for $a_i = j$ it is $(\phi_{ik} \bmod \ell)\,\ell^{j-1}e_i$. (b) *Fitting's lemma.* For $\psi \in \mathrm{End}(G)$ the chains $\ker\psi^m$, $\operatorname{im}\psi^m$ are monotone in the finite group G, and each strict step changes the order by a factor $\geq \ell$, so both are stationary from $m = \mathrm{len}(G)$ on; then $G = \ker\psi^L \oplus \operatorname{im}\psi^L$ ($L := \mathrm{len}(G)$): if $y = \psi^L x$ and $\psi^L y = 0$ then $x \in \ker\psi^{2L} = \ker\psi^L$, so $y = 0$, and $\left|\ker\psi^L\right|\left|\operatorname{im}\psi^L\right| = |G|$. ψ is nilpotent on the first summand and bijective on the second. Take $\psi = \phi - c$. If $c' \equiv c \pmod{\ell}$, $\phi - c' = (\phi - c) - (c' - c)$ with $c' - c$ nilpotent on the ℓ-group G and central, so $\phi - c'$ is nilpotent on $G_{[c]}$ and $G_{[c]} \subseteq G_{[c']}$, and conversely. If $c' \not\equiv c$, $c - c'$ is a unit on G and $\phi - c' = (\phi - c) + (c - c')$ is bijective on $G_{[c]}$, so $G_{[c]} \cap G_{[c']} = 0$ and $G_{[c]} \subseteq \operatorname{im}(\phi - c')^L$. Applying Fitting's lemma to $\phi - c_2$ on $\operatorname{im}(\phi - c_1)^L$, and so on over the classes of $\mathbb{F}_\ell$, gives the decomposition, with $G_\circ$ the last image. The functor V_j is additive on ϕ-stable direct sums; on $V_j(G_{[c]})$, $\phi_j - \bar{c}$ is nilpotent, and on $V_j(G_{[c']})$, $c' \not\equiv c$, and on $V_j(G_\circ)$ it is induced by an automorphism, hence bijective (an automorphism maps $(\ell^m G)[\ell]$ onto itself). So $\mathrm{mult}_{x-\bar{c}}\, P_j(\phi) = \dim V_j(G_{[c]})$, the number of summands of order ℓ^j of $G_{[c]}$ by (a). (c), (d) Kernels, cokernels and module structure are carried by isomorphisms of modules. $\phi - c$ depends on c modulo the exponent. (e) A homomorphism $\mathbb{Z}/\ell^b \to \mathbb{Z}/\ell^{b'}$ is multiplication by an integer m, defined modulo $\ell^{b'}$, with $\ell^b m \equiv 0 \pmod{\ell^{b'}}$, so $v_\ell(m) \geq \max(0, b' - b)$; composites multiply. A second decomposition of the same type, ordered alike, is $e'_k = he_k$ for an automorphism h, and the matrix of ϕ in it is $h^{-1}\phi h$ (entries of the i-th row reduced modulo ℓ^{a_i}). An entry of the layer-j block of a product of such matrices is a sum over paths $j = b_0 \to b_1 \to \cdots \to b_m = j$ of layers; a term is divisible by ℓ to the sum of the ascents $\max(0, b_{t+1} - b_t)$, and a closed path from j that visits a layer $a \neq j$ ascends by at least $|a - j| \geq d_j$ (for $a > j$ on the way up, for $a < j$ on the way back). Hence, writing X_j for the layer-j block and reading in $\mathbb{Z}/\ell^j$, $(h^{-1}\phi h)_j \equiv (h^{-1})_j\,\phi_j\,h_j$ and $(h^{-1})_j h_j \equiv (h^{-1}h)_j = 1 \pmod{\ell^{e_j}}$: h_j is invertible modulo ℓ^{e_j} and the block changes by a conjugation modulo ℓ^{e_j}, which leaves its characteristic polynomial unchanged. Its reduction modulo ℓ is $\det(x - \phi_j) = P_j(\phi)$ by (a).

(2) $\phi(e_1 + e_2) = \ell e_2 = \ell(e_1 + e_2)$, as $\ell e_1 = 0$; so in the basis $(e_1, e_1 + e_2)$ — orders ℓ, ℓ^2, a basis of the same type — the $\mathbb{Z}/\ell^2$-block is (ℓ), against (0) before. The general pairs: in $\mathbb{Z}/\ell^{j-d}e_1 \oplus \mathbb{Z}/\ell^j e_2$ the basis $(e_1, e_1 + e_2)$ has the same type (orders ℓ^{j-d}, ℓ^j; it generates), and $\phi(e_1 + e_2) = \ell^d e_2 = \ell^d(e_1 + e_2) - \ell^d e_1$, whose coefficient on the layer-j generator is ℓ^d; in $\mathbb{Z}/\ell^{j+d}e_1 \oplus \mathbb{Z}/\ell^j e_2$ the basis $(e_1 + e_2, e_2)$ has orders ℓ^{j+d}, ℓ^j, and $\phi e_2 = \ell^d e_1 = \ell^d(e_1 + e_2) - \ell^d e_2$, coefficient $-\ell^d$ (both ϕ are endomorphisms: $\ell^{j-d} \cdot \ell^d = \ell^j$, $\ell^j \cdot \ell^d = \ell^{j+d}$). As $d < j$, x and $x \mp \ell^d$ differ modulo ℓ^{d+1}. One layer: a change of decomposition is then a change of basis of a free $\mathbb{Z}/\ell^a$-module, under which the characteristic polynomial is invariant.

(3) The certified form gives $\mathrm{tors}\,\mathrm{coker}\, d^{q-1} \cong \bigoplus_J \mathbb{Z}/\ell^{\sigma_i}$, with exact transforms $\tilde{L}$, $\tilde{R}$ over $\mathbb{Z}_{(\ell)}$, the first s rows of $\tilde{L}$ congruent to those of L modulo ℓ^K and the first s columns of $\tilde{L}^{-1}$ congruent to those of L^{-1} modulo $\ell^{K'}$, $K' := K - \sigma_{\max}$ (Proposition 31.10(2)(g)(i)); the classes of $x_j := \tilde{L}^{-1}e_j$ generate, with coordinates $y \mapsto (\tilde{L}y)_i \bmod \ell^{\sigma_i}$ (ibid. (ii)). As C^{q+1} is torsion-free, $\mathrm{tors}\, H^q(C) = \mathrm{tors}\,\mathrm{coker}\, d^{q-1}$ and the x_j are cocycles (the proof of Proposition 23.17(2)). F is $\mathbb{Z}_{(\ell)}$-linear and preserves torsion classes, and

$$(L\,F\,\tilde{x}_j)_i - (\tilde{L}\,F\,x_j)_i = \big((L - \tilde{L})F\tilde{x}_j\big)_i + \big(\tilde{L}\,F(\tilde{x}_j - x_j)\big)_i \equiv 0 \pmod{\ell^{K'}}$$

for $i \leq s$ (the first term is $\equiv 0 \bmod \ell^K$, the second $\bmod \ell^{K'}$). If $\sigma_i \leq K'$, this is 0 modulo ℓ^{σ_i}, so $\tilde{\Phi}_{ij}$ is the i-th coordinate of $\phi[x_j] = [Fx_j]$. For every i, the same congruence gives $\tilde{\Phi}_{ij} \equiv$

the true coordinate modulo $\ell^{\min(\sigma_i,K')}$, with $K'\ge 1$ for a certified form; so the layer blocks are read exactly modulo $\ell^{\min(j,K')}$, which gives the P_j (modulo ℓ), the Fitting types by (1)(b), and the block polynomials modulo $\ell^{\min(e_j,K')}$ by (1)(e). *The condition is necessary* (v1.23.0-pre.25, referee m1). Proposition 31.10(2)(g)(ii)'s exactness of torsion coordinates for every $\sigma_i<K$ is for an exact vector y; here $y=F\tilde{x}_j$ and the error $\ell^{K'}w$ of $\tilde{x}_j$ passes through F. The minimal instance: $C^{q-1}=C^q=\mathbb{Z}_{(\ell)}^2$, $C^{q+1}=0$, $d^{q-1}=\operatorname{diag}(\ell^\sigma,0)$, $F^{q-1}=0$ and $F^qe_1=0$, $F^qe_2=e_1$ (a cochain map), so $G=\mathbb{Z}/\ell^\sigma[e_1]$ and $\phi=0$; for $\sigma<K$ the form with $L=\begin{pmatrix}1&0\\ \ell^{K-\sigma}&1\end{pmatrix}$, $R=1$ is certified ($Ld^{q-1}=\begin{pmatrix}\ell^\sigma&0\\ \ell^K&0\end{pmatrix}\equiv d^{q-1} \bmod \ell^K$), $\tilde{x}_1=(1,-\ell^{K-\sigma})^T$, and $\tilde{\Phi}_{11}=(LF\tilde{x}_1)_1=-\ell^{K-\sigma}$, which is $\ne 0$ modulo ℓ^σ exactly when $K<2\sigma$.

(3′) (v1.23.0-pre.31.) *Reduction to (3).* Let $C^\bullet$ be the two-term complex $C^{q-1}:=\mathbb{Z}_{(\ell)}^g\xrightarrow{\Delta}C^q:=\mathbb{Z}_{(\ell)}^r$, all other terms 0. Then $H^q(C)=Q$, and $C^{q+1}=0$ is torsion-free. Put $F^q:=F$ and $F^m:=0$ for $m\ne q-1,q$; for each basis vector e_k of $\mathbb{Z}_{(\ell)}^g$, $F\Delta e_k\in\operatorname{im}\Delta$, so choose $F^{q-1}e_k$ with $\Delta F^{q-1}e_k=F\Delta e_k$ and extend linearly ($\mathbb{Z}_{(\ell)}^g$ is free). Then $\Delta F^{q-1}=F\Delta$, and $d^q=0$, so (F^m) is a cochain endomorphism of $C^\bullet$ over $\mathbb{Z}_{(\ell)}$, and $H^q(F^\bullet)$ is the endomorphism $[F]$ of Q; its restriction to $G=\operatorname{tors}H^q(C)$ is ϕ. The read of (3) uses only $d^{q-1}=\Delta$ and $F^q=F$, never the chosen F^{q-1}; so it is the read of (3) for this complex, and (3), with its proof, applies verbatim (the x_j are cocycles trivially, $d^q=0$). *The free part.* By Proposition 31.10(2)(g)(i), there are exact $\tilde{L}$, $\tilde{R}$ with $\tilde{L}\Delta\tilde{R}=\operatorname{diag}(D,0)$, the last $r-s$ columns of $\tilde{L}^{-1}$ congruent to those of L^{-1} modulo ℓ^K (its "moreover") and all of $\tilde{L}$ congruent to L modulo $\ell^{K'}$. By (2)(g)(ii), $y\mapsto\tilde{L}y$ identifies Q with $\bigoplus_{i\le s}\mathbb{Z}_{(\ell)}/\ell^{\sigma_i}\oplus\mathbb{Z}_{(\ell)}^{r-s}$; the first summand is G, so $y\mapsto((\tilde{L}y)_i)_{i>s}$ identifies $\Lambda=Q/G$ with $\mathbb{Z}_{(\ell)}^{r-s}$, and $x_j:=\tilde{L}^{-1}e_j$, $j>s$, map to the standard basis. F induces ψ because it preserves $\operatorname{im}\Delta$ and every endomorphism of Q preserves G; the matrix of ψ in the basis $([x_j])_{j>s}$ is $((\tilde{L}Fx_j)_i)_{i,j>s}$. For $i,j>s$,

$$(LF\tilde{x}_j)_i-(\tilde{L}Fx_j)_i=\big((L-\tilde{L})F\tilde{x}_j\big)_i+\big(\tilde{L}F(\tilde{x}_j-x_j)\big)_i\equiv 0\pmod{\ell^{K'}},$$

the first term because $L\equiv\tilde{L}\pmod{\ell^{K'}}$, the second because $\tilde{x}_j\equiv x_j\pmod{\ell^K}$. So X is the reduction modulo $\ell^{K'}$ of the exact matrix of ψ, which is the input of (4) with $k=K'$. *Any exact lifts* (v1.23.0-pre.32, referee m3). Let x'_j, $s<j\le r$, be exact with $x'_j\equiv\tilde{x}_j\pmod{\ell^K}$. As $x_j\equiv\tilde{x}_j\pmod{\ell^K}$ too, $x'_j=x_j+\ell^Kw_j$ with $w_j\in\mathbb{Z}_{(\ell)}^r$, so the coordinates of $[x'_j]$ in the basis $([x_j])_{j>s}$ of Λ form the matrix $P=I+\ell^KM$, $M:=((\tilde{L}w_j)_i)_{i,j>s}$, which is invertible over $\mathbb{Z}_{(\ell)}$ ($\det P\equiv 1 \bmod \ell$); so $([x'_j])$ is a basis, the matrix of ψ in it is $P^{-1}X_{\mathrm{ex}}P$, X_{ex} the exact matrix in the basis $([x_j])$, and $P^{-1}X_{\mathrm{ex}}P\equiv X_{\mathrm{ex}}\pmod{\ell^K}$ ($P\equiv P^{-1}\equiv I$); in particular X is its reduction modulo $\ell^{K'}$ as well. *Independence.* θ maps torsion onto torsion, so it restricts to $G\to G'$ and induces $\Lambda\to\Lambda'$, isomorphisms intertwining the induced maps because $\theta[F]=[F']\theta$; the invariants of (1) depend only on the isomorphism class of (G,ϕ) (by (1)), and those of (4) only on that of (Λ,ψ) (the type of $\operatorname{coker}(\psi-c)$ is not changed by an isomorphism).

(4) Proposition 23.17(1) for the exact matrix $X-c$ over $\mathbb{Z}_{(\ell)}$, whose reduction modulo ℓ^k is $(X \bmod \ell^k)-c$: the number of pivots of valuation $<k$ is $\#\{t:a_t<k\}\le\operatorname{rank}_{\mathbb{Q}}(X-c)\le r$; equality with r forces $\operatorname{rank}_{\mathbb{Q}}(X-c)=r$ and every $a_t<k$, and then the pivots are the elementary divisors up to units. Their product is $\det(X-c)=\pm\chi_\psi(c)$ up to a unit, so $\sum a_t=v_\ell(\chi_\psi(c))$. If $\chi_\psi(c)=0$, $\operatorname{rank}_{\mathbb{Q}}(X-c)<r$. The type of $\operatorname{coker}(X-c)$ does not depend on the basis. □

Theorem 31.16 (The certified descent record: groups, characteristic polynomials over $\mathbb{Z}$, Hecke data on certified groups)**.** *[proved here] (v1.23.0-pre.24; clause C-D5 of the plan, restricted by the audit's M5 and corrected as in Remark 31.17; refereed ACCEPT-with-fixes, its fixes M2, m5 at v1.23.0-pre.25.) Assume (H4) and (B1)–(B3). Let $N\ge 1$, $n\ge 0$, $q_{\max}\ge 3$, $\mathcal{P}$ a finite set of primes not dividing N, and $A=\Gamma_0(N)$. For every $q\le q_{\max}$ the following are determined*

by the descent pieces — the O*-piece* ($C^\bullet_{\mathrm{eq}}(M_{N,\mathsf{O}})$ *with the shipped operator or its tree transport, or the sparse route: the cokernel presentation* $M_{N,\mathsf{O}}/\mathrm{Rel}_N$ *of Proposition 27.5(2) with* $[H_{N,p}]$ *of Proposition 28.9, v1.23.0-pre.31) and the* ℓ*-local pieces,* $\ell \in \{2,3\}$ *(*$\mathcal{I}^{(\ell)}$ *or the tree models, with the transported operators) —, certified by the rules cited, and* ***independent of every choice*** *made in computing them (models, comparisons,* τ_X*,* $\tau_{\mathcal{I}}$*, tables, transversals, Smith transforms, primes):*

1. ***(Groups.)*** $H^q(A;V) \cong \mathbb{Z}^{b_q(M_N)} \oplus \bigoplus_{\ell\in\{2,3\}\cup\mathcal{B}(M_N)} \operatorname{tors} H^q(A;V)_{(\ell)}$*, with the certified exponents of Propositions 31.7(7), 31.6, 31.12(1)(d) and 31.9; no prime* $\ell \geq 5$ *outside* $\mathcal{B}(M_N)$ *(the closed form* $\mathcal{B}(n,N)\cap[5,\infty)$ *for* $n>0$ *even, empty for* $n=0$ *and for* n *odd) contributes, by Proposition 31.8(1), (2); at* $\ell \geq 5$ *only* $q=1$ *carries torsion.*
2. ***(Characteristic polynomials over*** $\mathbb{Z}$***.)*** *For* $p \in \mathcal{P}$*,* $\chi^{(N)}_{p,q}$ *as in Proposition 31.14(1), with* $\chi^{(N)}_{p,1}$ *reported by its output rule (5) from the* O*-piece alone.*
3. ***(Torsion.)*** *For each* ℓ *with* $\operatorname{tors} H^q(A;V)_{(\ell)} \neq 0$*, the isomorphism class of the* $\mathbb{Z}[x_p : p\in\mathcal{P}]$*-module* $\big(\operatorname{tors} H^q(A;V)_{(\ell)}, (T_p)_{p\in\mathcal{P}}\big)$ *— in particular the layer polynomials* $P_j(T_p)$*, the types of the Fitting components and of* $\ker(T_p-c)$*,* $\operatorname{coker}(T_p-c)$*, and the layer blocks' polynomials modulo* ℓ^{e_j} *(Proposition 31.15(1)(e)) — read by Proposition 31.15(3) (the layer polynomials and the Fitting types at every certified* K*, the rest at* $2\sigma_{\max}\leq K$*): at* $\ell\geq 5$ *(*$q=1$*) on the* O*-piece modulo* ℓ^K *— the shipped operator on* $C^\bullet_{\mathrm{eq}}$ *or* T *(Proposition 31.13(6)(a)),* ***or*** $[H_{N,p}]$ *on the cokernel presentation* $M_{N,\mathbb{Z}_{(\ell)}}/\mathrm{Rel}_N$ *read by Proposition 31.15(3′) (Proposition 28.9(1), (2); v1.23.0-pre.31, D3's contract note 1: the route the code reads); at* $\ell=3$ *on* $\mathcal{I}^{(3)}$ *with* $T_p'^{(3)}$ *or on the tree model with* $T'_{\mathcal{T},p}$*, for* ***every*** N*,* $\nu_3(N)=0$ *included — never on* $C^\bullet_{\mathrm{eq}}$ *or* T *at* $\ell=3$ *(v1.23.0-pre.25, referee M2, resolution (a)); at* $\ell=2$ *on* $\mathcal{I}^{(2)}_{\mathcal{T}}$ *with* $T'_{\mathcal{T},p}$ *(Proposition 31.13(5)).*
4. ***(The free quotient.)*** *For* $p\in\mathcal{P}$*, a prime* ℓ *and* $c\in\mathbb{Z}$ *with* $\chi^{(N)}_{p,1}(c)\neq 0$*: the invariant factors of* T_p-c *on* $H^1_{\mathrm{fr}}(N)\otimes\mathbb{Z}_{(\ell)}$*, by Proposition 31.15(4) (the lattice* $H^1_{\mathrm{fr}}(N)\otimes\mathbb{Z}_{(\ell)}$ *read on the* ℓ*-piece at* $\ell\in\{2,3\}$*, on the* O*-piece otherwise — at* $\ell\geq 5$ *on the cokernel presentation by Proposition 31.15(3′), with* $k=K'$*; v1.23.0-pre.31) — so* $K > \sigma_{\max} + v_\ell(\chi^{(N)}_{p,1}(c))$ *suffices there,* $\sigma_{\max}$ *that of the certified form of* Δ *(*$\sigma_{\max}=0$ *and* $K'=K$ *when* $\operatorname{tors} H^1(A;V)_{(\ell)}=0$*, e.g.* $\ell\notin\mathcal{B}(M_N)$*); a read that falls short of (4)'s pivot condition is refused, never misread (v1.23.0-pre.32, referee m2).*
5. ***(D2's maps.)*** *For* $B=\Gamma_0(MN)$ *and* $p\in\mathcal{P}$ *with* $p\nmid M$*:* res *and* cor *commute with* T_p *on* H^q *over* $\mathbb{Z}$*, the sequence (78) is* T_p*-equivariant over* $\mathbb{Z}$ *in every degree (Proposition 31.13(7)), and*

$$\chi^{(MN)}_{p,1} = \chi^{(N)}_{p,1}\cdot\det\big(x-T_p \mid H^2(A,B)/\mathrm{tors}\big), \qquad \text{in particular } \chi^{(N)}_{p,1}\mid\chi^{(MN)}_{p,1} \text{ in } \mathbb{Z}[x].$$

   For $p\mid M$ *nothing is asserted (the squares fail in general, Proposition 31.10(3)(b)); for* $p\mid N$*, nothing is asserted here of* U_p *([R-Q4]; on cohomology, Proposition 31.10(3)(b′); its lift on the free model is Proposition 31.13(9), v1.26.0-pre.1, whether a binary serves it over* $\mathbb{Z}$ *is item 8 (d′)(v) of §22.4, v1.26.0-pre.11, confirm-referee C1).*
6. ***(Every degree.)*** *For* $q\geq 2$ *the data of (1)–(4) in degree* $q+2$ *are those of degree* q*, through* $\mathrm{res}^\Gamma_A\, e\cup -$*, an isomorphism of Hecke modules (Proposition 31.7(5));* $q_{\max}=3$ *determines every degree.*
7. ***(Not asserted.)*** *(a) the* $\mathbb{Z}_{(\ell)}[T_p : p\in\mathcal{P}]$*-module structure of* $H^1_{\mathrm{fr}}(N)\otimes\mathbb{Z}_{(\ell)}$ *beyond (4) — in particular any invariant of* T_p-c *at a root* c *of* $\chi^{(N)}_{p,1}$ *— and the class of the extension* $0\to\mathrm{tors}\to H^1\to H^1_{\mathrm{fr}}(N)\to 0$ *as Hecke modules (the plan's D-δ7 and Q-R1): no certified read is specified here. When* $\chi^{(N)}_{p,1}$ *is squarefree over* $\mathbb{Q}$ *the* $\mathbb{Z}_{(\ell)}$*-similarity class should be determined by* T_p *modulo a finite power of* ℓ *(finite determinacy for a separable characteristic polynomial), but no bound is given and none is asserted; for repeated roots — which occur at level* N *whenever there are old forms,* $(\chi^{(1)}_{p,1})^2 \mid \chi^{(N)}_{p,1}$ *— no finite read is known*

to suffice (v1.23.0-pre.25, referee m5, in place of "open mathematics"). ***The read at a root*** *(v1.25.0-pre.0; wording v1.25.0-pre.0b, A7-m3): the invariant at a root that §31.6 uses — the lattice congruence module η_L at the root $c = \lambda(T_p)$ of a rational Eisenstein component, read as one Smith form of $c_{\mathcal{S}',p}(T_p)$ modulo ℓ^K, $K > v_\ell c_{\mathcal{S}',p}(\lambda(T_p))$ — is specified and proved by Proposition 31.48(5) (with the minimal polynomial on the complement, never $T_p - c$ alone); the extension class is determined by finite data, ε_K for any $K \geq e_\ell$ (Proposition 31.41(2)(a)), and from computed operators a non-split verdict is certified, a split one only under Remark 31.43 (Proposition 31.41(2)(b)); both are statements of §31.6, recorded by item 8 of §22.4,* ***not*** *data of this record, which still asserts neither; (b) integer Hecke matrices in a basis (tier B; the plan's D8); (c) U_p, nebentypus $\chi \neq 1$, groups other than $\Gamma_0(N)$, the boundary pair and its cone (D4, D5c). (v1.23.0-pre.26: the pair and its cone are §31.4, a separate statement; this record is unchanged, and the record's pair columns are D6's. v1.23.0-pre.37: the record is* `integral_descent_record/v1`*, item 7 of §22.4, which fixes its degrees, its hashed and unhashed fields and its refusals; (4) and (5) are not in it; v1.23.0-pre.38: built, audited and merged at* `2aa61397`*, its golden points landed at* `6958a5d2`*.)*

Proof. (1) is the cited statements. (2) is Proposition 31.14, whose residues are read on the O-piece (its (2)). (3) Each local operator named in (3) — Proposition 31.13(5) at $\ell \in \{2,3\}$, (6)(a) at $\ell \geq 5$; (6)(b) is not used — is a cochain endomorphism, over $\mathsf{O}_\ell \subset \mathbb{Z}_{(\ell)}$, of a complex of finitely generated free $\mathbb{Z}_{(\ell)}$-modules (Propositions 31.6(1), 31.12(1)(d)), inducing $T_p \otimes \mathbb{Z}_{(\ell)}$ under the comparison with $H^q(A;V) \otimes \mathbb{Z}_{(\ell)}$ (flat base change, Proposition 31.7(1)); Proposition 31.15(3) reads (G, T_p) exactly, one p at a time, in one system of generators, so the joint module is read as well, and the parts that need only a certified form are those of Proposition 31.15(3). Two computations differ by an isomorphism of $\mathbb{Z}_{(\ell)}$-modules that intertwines the operators (both induce the same T_p), so their invariants agree (Proposition 31.15(1)). *The cokernel presentation at $\ell \geq 5$* (v1.23.0-pre.31). For n odd, 2 kills $H^\bullet(A;V)$ (the proof of Proposition 31.14(2)), so there is no ℓ-torsion and nothing is read. For n even put $R := \mathbb{Z}_{(\ell)}$; $6 \in R^\times$. $\mathrm{Rel}_N \subseteq M_{N,R}$ is spanned by finitely many vectors with entries in $\mathbb{Z}[1/6] \subseteq R$, the columns of a matrix Δ (the generators of $M^{\langle S\rangle}$, $M^{\langle U\rangle}$ ($U = ST$, written $M^{\langle ST\rangle}$ there; v1.23.0-pre.32, referee N2) of Proposition 27.5(1), (2), orbit by orbit, or the columns of $1 + M_N(S)$ and $1 + M_N(U) + M_N(U)^2$); $H_{N,p}$ is an endomorphism of $M_{N,R}$ with integral entries (ρ_n of integral matrices) and $H_{N,p}(\mathrm{Rel}_N) \subseteq \mathrm{Rel}_N$ (Proposition 28.9(1)). The certifying rank is $\mathrm{rank}_{\mathbb{Q}} \Delta = m_N(n+1) - b_1(M_N)$: $\mathrm{coker}(\Delta \otimes \mathbb{Q}) = M_{N,\mathbb{Q}}/\mathrm{Rel}_N \cong H^1(A; V \otimes \mathbb{Q})$ by $\varkappa_N$ over $\mathbb{Q}$, of dimension $b_1(M_N)$ (Proposition 31.7(7)(b) and flat base change, Proposition 31.7(1)). Proposition 31.15(3′) therefore reads $(\mathrm{tors}\,\mathrm{coker}\,\Delta, [H_{N,p}])$, for all $p \in \mathcal{P}$ in the one system of generators of one certified Smith form; and $\varkappa_N$ over R (Definition 28.6) is an isomorphism $\mathrm{coker}\,\Delta \to H^1(A; V \otimes R)$ with $\varkappa_N[H_{N,p}] = T_p \varkappa_N$ (Proposition 28.9(2), $[A\, g\, A] = T_p$ as $p \nmid N$), so by the independence clause of (3′) the invariants read are those of $(\mathrm{tors}\, H^1(A; V \otimes \mathbb{Z}_{(\ell)}), T_p) = (\mathrm{tors}\, H^1(A;V)_{(\ell)}, T_p)$ (flat base change). (4) Proposition 31.15(4) with $\psi = T_p$, whose characteristic polynomial is $\chi_{p,1}^{(N)}$ by base change; $\chi_{p,1}^{(N)}(c) \bmod \ell^K$ is read in words by Proposition 31.14(4). At $\ell \geq 5$ on the presentation: $\varkappa_N$ induces an isomorphism of $\mathrm{coker}\,\Delta/\mathrm{tors}$ with $H^1(A; V \otimes \mathbb{Z}_{(\ell)})/\mathrm{tors} = H^1_{\mathrm{fr}}(N) \otimes \mathbb{Z}_{(\ell)}$ intertwining $[H_{N,p}]$ with T_p, and Proposition 31.15(3′) gives the matrix of T_p on it modulo $\ell^{K'}$ in the basis of its free part. *The precision* (v1.23.0-pre.32, referee m2): Proposition 31.15(4) with $k = K' = K - \sigma_{\max}$ certifies once K' exceeds the largest exponent, and the exponents sum to $v_\ell(\chi_{p,1}^{(N)}(c))$, so $K > \sigma_{\max} + v_\ell(\chi_{p,1}^{(N)}(c))$ suffices; $\sigma_{\max}$ is over *all* pivots of Δ's certified form, and the word read of $\chi_{p,1}^{(N)}(c) \bmod \ell^K$ above only locates $v_\ell(\chi_{p,1}^{(N)}(c))$, it is not the precision of the read. If $\mathrm{tors}\, H^1(A;V)_{(\ell)} = 0$ then $\mathrm{tors}\,\mathrm{coker}\,\Delta = 0$ (through $\varkappa_N$), every pivot is a unit, $\sigma_{\max} = 0$ and $K' = K$. The bound is sufficient, not necessary; a read with fewer than $r - s$ pivots of valuation $< K'$ is refused by (4), never wrong. (5) The squares: Proposition 31.10(3)(a) on cohomology, and Proposition 31.13(7) for

the sequence. Rationally res is injective ($\mathrm{cor} \circ \mathrm{res} = [A : B]$) and the sequence splits into $0 \to H^1(A)_{\mathbb{Q}} \to H^1(B)_{\mathbb{Q}} \to H^2(A,B)_{\mathbb{Q}} \to 0$ (Proposition 31.10(2)(g)(iii)), T_p-equivariantly, so the characteristic polynomials multiply; the quotient of two monic integer polynomials that is monic in $\mathbb{Q}[x]$ lies in $\mathbb{Z}[x]$ (Gauss). (6) Proposition 31.7(5). (7) states scope. □

Remark 31.17 (What the plan stated, and what holds). [proved here] (v1.23.0-pre.24; (i)–(iv) amended at v1.23.0-pre.25 after the referee's M1, M2, m1, m5.) The plan's §2.3 and its clause rows C-D4, C-D5 are right in substance; four points are corrected here. (i) **Torsion is not open.** The plan lists "a certified $\mathbb{Z}/\ell^K[T]$-module type … on torsion" among the things not asserted. On a *certified finite* group the operators are read exactly once $K \geq \sigma_i + \sigma_{\max}$ (Proposition 31.15(3)), so the whole module structure under any finite set of T_p is certified (Theorem 31.16(3)); what stays unasserted of D-δ7 is the *lattice* $H^1_{\mathrm{fr}}(N) \otimes \mathbb{Z}_{(\ell)}$ under T at the roots of χ, and the extension class — not asserted because no certified read is specified, and for repeated roots none is known to suffice (Theorem 31.16(7)(a)). (ii) **The bound.** "K above the group's exponent plus a certified bound on the congruence exponent" becomes: for torsion, $K \geq 2\sigma_{\max}$ for the module (necessary, Proposition 31.15(3)), while the layer polynomials P_j and the Fitting types need only a certified form ($K > \sigma_{\max}$), and no congruence bound; for the lattice, $k > v_\ell(\chi(c))$ at a non-root c, with χ known exactly — or simply the self-certifying count of Proposition 31.15(4). (iii) **"The characteristic polynomial over $\mathbb{Z}/\ell^k$ on each Smith summand"** is not well defined (Proposition 31.15(2)); its well-defined replacement is the layer block's characteristic polynomial modulo ℓ^{e_j}, $e_j = \min(j, d_j)$ (Proposition 31.15(1)(e); in general no finer), whose reduction modulo ℓ is P_j. (iv) **The lever of D-δ2 is a theorem**: $\chi^{(N)}_{p,1}$ needs no 2- or 3-adic input at all (Proposition 31.14(2)); the local pieces at 2 and 3 enter only (3) and (4). The plan's C-D4 "strict lift for $\mathrm{diag}(1,p)$, every p" is existence already (Proposition 23.13(1)); what is new is the bulk lift without a boundary step, its construction for every p, and its level-N form (Proposition 31.13(2), (3)). C-D4′ (a strict lift directly on $\mathcal{I}^{(3)}$) is not drafted; it is a cost lever. The cheap route of Proposition 31.13(6)(b) at $\ell = 3$, $\nu_3(N) = 0$, holds for an integral datum certified over $\mathbb{Z}$ only, and the shipped canonical τ is integral in practice, not by construction; so the record does not use it, and its 3-local piece is the transport $T_p'^{(3)}$ for every N (v1.23.0-pre.25, referee M2, resolution (a): one route at $\ell = 3$, and the shipped Bredon operator never read at $\ell \in \{2,3\}$).

A check outside the build, not a tag (v1.23.0-pre.24): `docs/implementation/slot24/cd4_hecke_check.py` (`python3 -I`, pure integers, no computer algebra system), output `cd4_hecke_check.out`. It reads no line of `cohomology_app` and imports nothing from the tree: it rebuilds the cosets of $\Gamma_0(N)$, the coinduced module M_N, the group 1-cocycles of $\Gamma = \langle S, U \mid S^4, S^2U^{-3}\rangle$ with Shapiro, T_p as $\mathrm{cor} \circ (\iota_g, \rho_n(g))^* \circ \mathrm{res}$ on cocycles of $\Gamma_0(N)$, res and cor between levels, exact Smith forms over $\mathbb{Z}$ and over $\mathbb{Z}/\ell^K$. On 21 cells (level 1, $n \in \{1,2,3,4,6,8,10,12,16\}$; $\Gamma_0(N)$, $N \in \{2,3,4,5,7,11\}$, $n = 0$; $(N,n) \in \{(2,2),(2,10),(3,4),(5,4),(7,2),(11,2)\}$) and $p \leq 7$, $p \nmid N$, it checks, in degree 1: $\chi^{(N)}_{p,1}$ over $\mathbb{Z}$ against *independent* oracles (level 1: q-expansions of $E_4^a E_6^b$, e.g. $(x - 2049)(x + 24)^2$ at $n = 10$, $p = 2$; $\Gamma_0(11)$, $n = 0$: point counts on $X_0(11)$; genus-0 levels at $n = 0$: $(x - 1 - p)^{c-1}$); (2) on the Bredon quotient $M^{-I}/(M^S + M^U)$ over $\mathbb{F}_{\ell'}$, with equal dimensions; (3)'s bound coefficientwise; (4)'s recovery in 64-bit words by Garner's digits, the sign test and $\chi(7)$ mod 2^{64}, mod3^{40} from the digits, including coefficients reported as residues and bound; Proposition 31.15(1) under random automorphisms and (3)'s read on the local Smith form at every $K \geq 2\sigma_{\max}$ with admissibly perturbed lifts (every torsion prime: 2, 3, 5, 7, 11); (4) at $\ell \in \{691, 3\}$, $n = 10$, and $\ell = 5$, $N = 11$; Theorem 31.16(5) on eight level pairs over $\mathbb{Z}$ (torsion coordinates included), the square with cor and U_p, $p \mid N$ (Proposition 31.10(3)(b′)), the failing square at $p \mid M$ as a control, and $(\chi^{(1)}_{p,1})^2 \mid \chi^{(N)}_{p,1}$ at prime N; at $n = 10$ the relations $(T_2 - 2049)(T_2 + 24) = 0$, $3(T_3 - 252) = 256(T_2 + 24)$ on H^1_{fr} and $T_2T_3 = T_3T_2$ on H^1 over $\mathbb{Z}$. Planted faults, each detected: a dropped coset, a single CRT prime, a bound without the Eisenstein term, a perturbed residue (held-out prime), the torsion read

below $2\sigma_{\max}$, and a per-summand characteristic polynomial. 931 checks, 0 failing (`CD4_CHECK ... failing=0 verdict=OK`). It covers H^1 only (the cocycle model has no higher degrees) and does not build $\mathcal{I}_\bullet$, $\mathcal{I}^{(\ell)}$ or a tree model: it checks the statements, not their models. *The referee's check* (v1.23.0-pre.25), `docs/implementation/slot24/cd4referee/cd4referee_check.py`, shares no code with it (Reidemeister–Schreier on $\Gamma_0(N)$, H^2 through the Bockstein, $[AgA]$ for $\begin{pmatrix}1&0\\0&p\end{pmatrix}$ and $\begin{pmatrix}p&0\\0&1\end{pmatrix}$ separately, adversarial certified Smith transforms, 18 cells and the heavy cell $(11,10)$, 14 level pairs); at pre.25 the author added an arm `M`: the sharpness pairs of Proposition 31.15(2) for every $1 \le d < j \le 6$ ($\ell = 2$), ≤ 4 ($\ell = 3$), on both sides, the referee's $\mathbb{Z}/2 \oplus \mathbb{Z}/8$, the minimal example of (3) for $\sigma \le 4$, $K \in (\sigma, 2\sigma+1]$, and on every real torsion read the block polynomials modulo $\ell^{\min(e_j,K')}$ (`CD4R_CHECK ... failing=0 verdict=OK`, output `cd4referee_check.out`).

Code map (v1.23.0-pre.24, **stated before D3's code**, contract first, rule 21). **Not built.** The plan's D3 builds: `Integral/Model/StrictLift` parametrised in p, bulk only, through degree $q_{\max} = 3$ (Proposition 31.13(2)); the level-N operator $\widetilde{T}^{(N)}_{\mathcal{I},p}$ on the code's $\mathcal{I}$ with `Hecke/Gamma0LevelN`'s table, applied to columns, and its transports to $\mathcal{I}^{(\ell)}$ and to `Integral/Model/IsotropyModelTree` (D5) by the code's comparisons; the characteristic polynomial with the output rule (Proposition 31.14(4), (5)), on the existing `Integral/IntegralCharpolyCrt`; the torsion read (Proposition 31.15(3)) on `Integral/Descent/DescentMaps`'s presentations; and `IntegralDescentHeckeValidation` with the plan's arms (R), (C), (E), (O), (M), (L), (K). The existing `HeckeIntegralCharpolyValidation` is level 1, $N = 1$ of Proposition 31.14. **Obligations of the code** (v1.23.0-pre.25, from the referee's M2, m2, m3, m4): (a) the 3-local piece — torsion read and lattice — is computed with $T'^{(3)}_p = \Psi_3^* \widetilde{T}^{(N)}_{\mathcal{I},p} \Phi_3^*$ on $\mathcal{I}^{(3)}$ (or $T'_{\mathcal{T},p}$) for **every** N; the code does not branch on $\nu_3(N)$, and the shipped operator on $C^\bullet_{\mathrm{eq}}$ or T is read only with 6 inverted (the O-piece, and its reductions modulo ℓ^K, $\ell \ge 5$), never at $\ell \in \{2,3\}$; (b) the datum τ_X fed to the lift, strict or canonical, is checked integral and against (T1)–(T3) over $\mathbb{Z}$ first, and the lift refused otherwise; (c) $\tau_{\mathcal{I}}$ is built once per p, on $\{s_j e_c\}$, independently of r_t and of N; (d) the residue primes are $< 2^{32}$, and every Horner step modulo $Q > 2^{32}$ uses a word mulmod; a negative c_i is reduced as $c^+ \bmod Q - P \bmod Q$.

Code map, status at v1.23.0-pre.31. **Built on branch `s24-d3impl`, pending merge and audit** (not merged into `hecke-extensions`; tip `a1375159`; source `docs/implementation/slot24/DESCENT3_repo` read at its §§2, 3, 6). The files, all new, in `Integral/Descent/`: `DescentHecke.h` (the API and the refusal taxonomy); `DescentHeckeCrt.cpp` (Proposition 31.14: residues of $[H_{N,p}]$ on the shipped sparse cokernel presentation over $\mathbb{F}_{\ell'}$ — the presentation form of its (2) —, residue primes descending from 2^{30}, the bound of (3), Garner's digits, the sign test, $c_i \bmod Q$ with a shift-and-add word mulmod and the negative case, the output rule (5), a held-out prime); `DescentHeckeOps.cpp` ($\widetilde{T}^{(N)}_{\mathcal{I},p}$ of Proposition 31.13(3) with the bulk component of D4's pair lift as $\tau_{\mathcal{I}}$ — admissible by (2) —, certified by `StrictLift::certify()`, built once per p on $\{s_j e_c\}$, applied to cochains block by block; and the **transport at $\ell = 2$ and at $\ell = 3$**: $T'_{\mathcal{T},p} = \Psi_{\mathcal{T}}^* \widetilde{T}^{(N)}_{\mathcal{I},p} \Phi_{\mathcal{T}}^*$ on the ℓ-local tree model with D5's certified comparisons, for every N, with no branch on $\nu_3(N)$, the shipped Bredon τ never read at 2 or 3); `DescentHeckeTorsion.cpp` (the read of Proposition 31.15(3): `TorsionOnTree` at $\ell \in \{2,3\}$, $q \le 3$, production; `TorsionOnCokernel` at $\ell \ge 5$, $q = 1$, production, the presentation form (3′) with Δ the generators of Rel_N and the certifying rank $m_N(n+1) - b_1(M_N)$; `TorsionOnFreeIso`, `TorsionOnIsoEll`, oracles; `FreeQuotientFactors`, Theorem 31.16(4) at $\ell \ge 5$ on the presentation with $k = K'$; `RunDescentHecke`, the record of Theorem 31.16(2), (3)); the validations `IntegralDescentHeckeValidation` (`standard`) and `IntegralDescentHeckeSlowValidation` (`slow`), registered in `Registry/Slot24/DESCENT3.cpp`, sources `cmake/slot24/DESCENT3.cmake`; an oracle outside the build, `docs/implementation/slot24/d3_hecke_oracle.py`. The obligations: (a) met (the 3-local piece is the tree transport for every N); (b) met **through the**

lift (Proposition 31.13(2), the check through the lift): the table read is the strict datum τ^s's integral table, and `StrictLift::certify()` checks (T1), (T2), (T3) and the strict identity on every basis element, those at $t \geq 1$ included, which makes that table a g-twisted chain datum over $\mathbb{Z}$; τ^s is not re-checked separately on X, and need not be. (v1.23.0-pre.32, referee N5: `certify()` checks the strict identity for $q \leq 2$ only; for $q \geq 3$ every summand of $\mathcal{I}_q$ has $t \geq 1$ ($s \leq 2$) and $C_q(X) = 0$, so the identity holds trivially there, and the code checks exactly what the clause requires. $H_{N,p}(\mathrm{Rel}_N) \subseteq \mathrm{Rel}_N$ is a theorem, Proposition 28.9(1), re-checked at two residue primes only; `TorsionOnCokernel`'s `NotEndomorphism` tests are partial safeguards, not a certificate.) (c) met (the pair lift, once per p, independent of r_t and N); (d) met. **Not built**: Theorem 31.16(4) at $\ell \in \{2, 3\}$ (the lattice $H^1_{\mathrm{fr}}(N) \otimes \mathbb{Z}_{(\ell)}$ on the ℓ-piece; it needs the free part of $H^1 = \ker d^1 / \operatorname{im} d^0$ in the tree's coordinates, which the presentation form (3′) does not supply, $d^1 \neq 0$ there), and Theorem 31.16(5) (no level pair is read with T_p by D3). Theorem 31.16(6) is exercised by an arm of the standard validation (H^4 against H^2 with T_p on the tree). Whatever the branch builds is recorded here as built only on that branch; the merge and the audit are separate events. (v1.23.0-pre.33: **audited and merged** into `hecke-extensions` at `dc8261a3` — the audit `docs/implementation/slot24/D3_code_audit.md` (`73a00c9b`), its fixes M1, m1–m3 at `51a4b295`, `DESCENT3_report.md` §9.) The $\ell \geq 5$ cokernel torsion read (`TorsionOnCokernel`) refuses `[RssCeiling]` on its predicted RSS **before building** anything (rule 19). `HeldOutMismatch` and a residue-prime rank defect (`[ResidueRank]`: the presentation's $\dim H^1 \neq b_1$ at a residue prime $\ell' > n + 1$) throw `hecke::HeckeInternalError`: they are **failures of the code, not refusals**, and propagate. **Open code item** (v1.23.0-pre.34, the C-D8 referee's N7): the enumerator `HeldOutMismatch` survives in `enum class D3Refusal` (`DescentHecke.h`) although it is never a refusal; it is to be removed or marked vestigial, so that counts of refusal codes stay honest. Not a statement of this document. (v1.23.0-pre.35: **closed** on branch `s24-d5cimpl` — the enumerator is removed; `[HeldOutMismatch]` is still thrown as a `hecke::HeckeInternalError`; §31.5's code map; v1.23.0-pre.36: audited and merged at `4064e164`.) D5c extends this section's code in two places, with no value changed (v1.23.0-pre.36): the comparison of Proposition 31.12(4) on the code's $\mathcal{I}$ (now `SharedTreeComparison()`) and `TreeTransport` are built to degree 5 (`kTreeComparisonTop`; the maps in degrees ≤ 4 are those of the degree-4 build, the construction being degree by degree), and `IsoHeckeN::apply_boundary` applies $\widetilde{T}^{(N)}_{\mathcal{I},p}$ on $\mathcal{I}^\partial$'s rows, reporting a term of $\tau_{\mathcal{I}}(\mathcal{I}^\partial)$ off $\mathcal{I}^\partial$ (the closure, `[IsobClosure]`). (v1.23.0-pre.37, D6: the record that carries this section's data is item 7 of §22.4; its (j) renames the refusal of the ker / coker read at the ring width — `TorsionOnTree`'s `Invariants`, reported as `[NotEndomorphism]` — to `[KerCokerPrecision]`, an open code item of this section's `D3Refusal` at v1.23.0-pre.37, closed at `19a969de`: `D3Refusal::KerCokerPrecision`, merged at `2aa61397`; v1.23.0-pre.38.)

Formalization note. Propositions 31.15 and 31.14(1), (4), (5) are finite algebra (Fitting's lemma, the Chinese remainder theorem, mixed radix) and cheap to formalise; Proposition 31.13 is the comparison theorem and the double-coset formalism; 31.14(3) rests on Eichler–Shimura and Deligne. No `\LEAN` marker.

31.4 The boundary pair and the cone over $\mathbb{Z}$: the Borel–Serre pair on the descent

(v1.23.0-pre.26; milestone D4 of `docs/implementation/slot24/DESCENT_plan.md` v0.4, clause C-D7 of its §2.7, its §2.2 row "Boundary LES" and its D-δ6; `DESCENT_plan_audit.md` on D-δ6.) §31.1–§31.3 glue the *bulk* $H^\bullet(\Gamma_0(N); V)$, its maps and its Hecke operators; §15 builds the pair and its cone on the cellular model, where they compute the Borel–Serre long exact sequence only with 6 inverted (Lemma 15.2, Remark 15.14). This subsection fixes, over $\mathbb{Z}$: (i) the *pair models* — $(\mathcal{I}_\bullet, \mathcal{I}^\partial_\bullet)$ over $\mathbb{Z}$, $(\mathcal{I}^{(\ell)}, \mathcal{I}^{\partial(\ell)})$ over O_ℓ (a free boundary at $\ell = 2$, the Bredon boundary

at $\ell \geq 3$), the cellular pair over $\mathbb{Z}[1/6]$ — and the comparisons of pairs, which exist and are unique up to homotopy of pairs (Definition 31.18, Lemma 31.19); (ii) what they compute, the long exact sequence $H_c^q \to H^q(\Gamma) \to H^q(\Gamma_\infty) \to H_c^{q+1}$ over $\mathbb{Z}$ glued from the three pieces, the boundary in closed form at every level, compact support (H_c^1 torsion-free, the rational Betti numbers, exponent bounds, periodicity), and the certified exponent on the boundary and the cone, the infinite 2-adic boundary included (Proposition 31.20); (iii) the cone with the strict pair datum ($h = 0$) and T_p, $p \nmid N$, on it, with exactly what is certified at which p, its relation to §31.3's bulk operator, and D2's maps on the pair (Proposition 31.21); (iv) where the plan's wording is corrected (Remark 31.22). **Not here**: the tree cusp pair (the plan's §5, D5c, clause C-D8; v1.23.0-pre.33: §31.5), U_p ([R-Q4]), Hecke operators with nebentypus.

What is used. Definition 15.1, (38), Lemma 15.2, Definition 14.2, Propositions 14.3, 14.4(4), Lemmas 14.20, 14.9, 14.10(1), 28.5, Propositions 15.13, 15.12, Theorem 15.11, §23's 23.4–23.14, 14.22, and Propositions 31.2, 31.6, 31.7, 31.10, 31.13. *Notation.* $\mathbb{P}^1 := \mathbb{P}^1(\mathbb{Q})$ with Γ, and Δ, acting by Möbius transformations; $\mathbb{Z}[\mathbb{P}^1] = \mathbb{Z}[\Gamma/\Gamma_\infty]$ ((B2)); $\mathrm{St} := \ker(\varepsilon\colon \mathbb{Z}[\mathbb{P}^1] \to \mathbb{Z})$, a $\mathbb{Z}$-free $\mathbb{Z}[\Gamma]$-module (= Div^0 of Lemma 28.5), $\mathrm{St}_S := \mathrm{St} \otimes S$; $c(N)$ is the number of cusps of $\Gamma_0(N)$, as in §28. M is a $\mathbb{Z}[\Gamma]$-module, finitely generated and free over $\mathbb{Z}$, $d := \mathrm{rank}_{\mathbb{Z}} M$, $M_S := M \otimes S$, $M^\vee := \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z})$ with the contragredient action; $b_j(M) := \dim_{\mathbb{Q}} H^j(\Gamma; M_{\mathbb{Q}})$ and $b_j(\partial; M) := \dim_{\mathbb{Q}} H^j(\Gamma_\infty; M_{\mathbb{Q}})$. When $-I$ acts on M by a sign we write it ε_M ($\varepsilon_{M_N(\chi)} = \chi(-1)(-1)^n$). For a $\mathbb{Z}[\Gamma_\infty]$-module X, X^T, X_T are the invariants and coinvariants of T (notation C12). P, P^∂, r_{P}, $H_c^q(\mathsf{P}; M)$, c (a cusp), w_c, $\Gamma_\infty^{(w)}$, $F_{n,w}$, ε_M, St, Υ_k, j_{St}, $R_\bullet$, ε_R, ε^∂ (v1.23.0-pre.27), τ^{rel}, $\widetilde{T}^{(N)}_{\mathcal{I}^\partial, p}$, $\Phi_p^{(N)}$ are new symbols of this subsection and mean nothing else (rule (A5)); P is *not* the P of Lemma 23.12.

Definition 31.18 (Pair models, their cochains and cones; comparisons and homotopies of pairs)**.** Assume (H4) and (B1)–(B3), and let S be a commutative ring. An **S-pair model** is an S-model $\mathsf{P}_\bullet$ in the sense of Proposition 31.10 (finitely generated projective $S[\Gamma]$-modules, $\mathsf{P}_q = 0$ for $q < 0$, an augmentation making it a resolution of S) together with a subcomplex $\mathsf{P}^\partial_\bullet \subseteq \mathsf{P}_\bullet$ whose term in each degree is a direct summand with projective complement, and an augmentation $\mathsf{P}_0^\partial \to S[\mathbb{P}^1]$ making $\mathsf{P}^\partial_\bullet$ a resolution of $S[\mathbb{P}^1]$, such that $\mathsf{P}_0^\partial \to \mathsf{P}_0 \to S$ equals $\mathsf{P}_0^\partial \to S[\mathbb{P}^1] \xrightarrow{\varepsilon} S$. Put $\mathsf{P}(M) := \mathrm{Hom}_{S[\Gamma]}(\mathsf{P}_\bullet, M_S)$, $\mathsf{P}^\partial(M)$ likewise, $r_{\mathsf{P}}\colon \mathsf{P}(M) \to \mathsf{P}^\partial(M)$ the restriction, $C(r_{\mathsf{P}})$ by (38) (with $\mathsf{P}(M)$, $\mathsf{P}^\partial(M)$ for $C^\bullet_{\mathrm{eq}}$, $C^\bullet_\partial$), inc_c, pr_c as there, and

$$H_c^q(\mathsf{P}; M) := H^{q-1}\big(C(r_{\mathsf{P}})\big), \qquad q \in \mathbb{Z}.$$

A **comparison of pairs** $\varphi\colon (\mathsf{P}, \mathsf{P}^\partial) \to (\mathsf{P}', \mathsf{P}'^\partial)$ is an $S[\Gamma]$-chain map $\mathsf{P} \to \mathsf{P}'$ over id_S with $\varphi(\mathsf{P}^\partial) \subseteq \mathsf{P}'^\partial$, its restriction over $\mathrm{id}_{S[\mathbb{P}^1]}$. Two comparisons are **homotopic as pairs** if their difference is $\partial k + k\partial$ with $k(\mathsf{P}^\partial) \subseteq \mathsf{P}'^\partial$. The pair models in use:

(a) $(\mathcal{I}_\bullet, \mathcal{I}^\partial_\bullet) \otimes S$, every S (Definition 23.4, (IM2): $\mathcal{I}^\partial$ is the sub-sum over $\Sigma_0^\partial \sqcup \Sigma_1^\partial$; its augmentation $\mathcal{I}_0^\partial \xrightarrow{\mathrm{cmp}} C_0(\partial X) \to \mathbb{Z}[\pi_0 \partial X] = \mathbb{Z}[\mathbb{P}^1]$, (B2); Proposition 23.5(2));

(b) $(\mathcal{I}^{(\ell)}, \mathcal{I}^{\partial(\ell)}) \otimes_{\mathsf{O}_\ell} S$ for O_ℓ-algebras S, $\mathcal{I}^{\partial(\ell)}$ the sub-sum of the summands $G'_{s,t}$ over $\Sigma_0^\partial \sqcup \Sigma_1^\partial$ (Definition 31.1, (IM2)), the model fixed once over O_ℓ, boundary summands first (Propositions 31.2(1), 31.6(1));

(c) $(C_\bullet(X), C_\bullet(\partial X)) \otimes S$ for $6 \in S^\times$ (Remark 7.6; $C_s(\partial X) \otimes S = S[\Gamma/\{\pm I\}]^3$ is projective once $2 \in S^\times$, (B3), and resolves $S[\mathbb{P}^1]$ because the components of ∂X are lines, (B1), (B2)); here $C(r_{\mathsf{P}})$ is Definition 15.1's cone with fibre M.

Lemma 31.19 (Comparisons of pairs exist, boundary first, and are unique up to homotopy of pairs)**.** *[proved here] [standard: [Wei94, Thm. 2.2.6]] Let $(\mathsf{P}, \mathsf{P}^\partial)$, $(\mathsf{P}', \mathsf{P}'^\partial)$ be S-pair models.*

1. *Every $S[\Gamma]$-chain map $\varphi^\partial\colon \mathsf{P}^\partial \to \mathsf{P}'^\partial$ over $\mathrm{id}_{S[\mathbb{P}^1]}$ (one exists) extends to a comparison of pairs φ with $\varphi|_{\mathsf{P}^\partial} = \varphi^\partial$.*
2. *Any two comparisons of pairs φ, φ' are homotopic as pairs; a homotopy k^∂ between $\varphi|_{\mathsf{P}^\partial}$ and $\varphi'|_{\mathsf{P}^\partial}$ with values in P'^∂ exists and extends to a homotopy k of pairs.*

3. *For a comparison φ: φ^* and $\varphi^{\partial *}$ are quasi-isomorphisms; $r_{\mathsf{P}}\varphi^* = \varphi^{\partial *} r_{\mathsf{P}'}$ **strictly**; $(\varphi^*[1], \varphi^{\partial *})\colon C(r_{\mathsf{P}'}) \to C(r_{\mathsf{P}})$ is a cochain map commuting with inc_c and pr_c, and a quasi-isomorphism. A homotopy of pairs k between φ, φ' gives the cone homotopy $\Upsilon_k(c, c_\partial) := (-k^* c,\ k^{\partial *} c_\partial)$: $(\varphi^*[1], \varphi^{\partial *}) - (\varphi'^*[1], \varphi'^{\partial *}) = D\Upsilon_k + \Upsilon_k D$. So the maps induced on the three terms of the long exact sequence, and on the sequence, do not depend on the comparison.*

Proof. (1) P^∂ is a bounded below complex of projectives and P'^∂ a resolution of $S[\mathbb{P}^1]$, so φ^∂ exists [standard: [Wei94, Thm. 2.2.6]]. Apply Lemma 23.12 with $P := \mathsf{P}$, $P' := \mathsf{P}^\partial$, $E := \mathsf{P}'$, $C := S$ in degree 0, p the augmentation (a surjective quasi-isomorphism), $E' := \mathsf{P}'^\partial$, $f := \varepsilon$ and $f' := \varphi^\partial$; $pf' = f|_{\mathsf{P}^\partial}$ is the compatibility of the augmentations. The lemma is stated for a free source with P' spanned by part of a basis; with a projective source and a projective complement its two steps are lifts of maps out of a projective module through surjections, exactly as in the proof of Proposition 31.2(4). (2) $u := \varphi - \varphi'$ is a chain map with $\varepsilon u = 0$, and $u|_{\mathsf{P}^\partial}$ is a chain map into P'^∂ over 0: $S[\mathbb{P}^1] \to S[\mathbb{P}^1]$, hence $u|_{\mathsf{P}^\partial} = \partial k^\partial + k^\partial \partial$ with k^∂ into P'^∂ (the uniqueness half of the comparison theorem). Extend k by induction on the degree over a projective complement Q_q of P^∂_q: for $x \in Q_q$ the element $u(x) - k(\partial x)$ is a cycle ($\partial u(x) - \partial k \partial x = u\partial x - (u\partial x - k\partial\partial x) = 0$ by induction), of augmentation 0 if $q = 0$; P' is acyclic in positive degrees and $\ker\varepsilon = \operatorname{im}\partial$ in degree 0, so the map $x \mapsto u(x) - k(\partial x)$ out of the projective Q_q lifts through $\partial\colon \mathsf{P}'_{q+1} \to \operatorname{im}\partial$; call the lift k. (3) Both complexes compute $\mathrm{Ext}^\bullet_{S[\Gamma]}(S, M_S)$, resp. $\mathrm{Ext}^\bullet_{S[\Gamma]}(S[\mathbb{P}^1], M_S)$, and φ^*, $\varphi^{\partial *}$ are the comparison maps; both sides of the strict identity send c to $c \circ \varphi|_{\mathsf{P}^\partial}$. The cone: $D(\varphi^* c, \varphi^{\partial *} c_\partial) = (-\varphi^* dc,\ \varphi^{\partial *} dc_\partial + r\varphi^* c) = (\varphi^*[1], \varphi^{\partial *}) D(c, c_\partial)$, and the five lemma on the two long exact sequences gives the quasi-isomorphism. The homotopy: $\varphi^* - \varphi'^* = dk^* + k^* d$, likewise on the boundary, and $D\Upsilon_k + \Upsilon_k D$ has first component $(dk^* + k^* d)c$, second component $(dk^{\partial *} + k^{\partial *} d)c_\partial + (k^{\partial *} r - r k^*)c$, and the last term vanishes because $k(\mathsf{P}^\partial) \subseteq \mathsf{P}'^\partial$. □

Proposition 31.20 (The pair over $\mathbb{Z}$: what it computes, the glued long exact sequence, the boundary in closed form, compact support, the certified exponent)**.** *[proved here] [standard: [Bro82, Ch. III.2, III.5–6, III.8–9, V.1–3, VI.9], [Wei94, Thm. 1.3.1, Thm. 2.2.6]; VERIFY] (v1.23.0-pre.26; milestone D4, clause C-D7 of the plan; refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D7_referee.md`, its fixes M1, M2, m1–m5 at v1.23.0-pre.27; the pre.26 locator VII.6 is withdrawn, referee m1.) Assume (H4) and (B1)–(B3); M as in the notation.*

1. ***(What a pair model computes, over every ring.)*** *Let $(\mathsf{P}, \mathsf{P}^\partial)$ be an S-pair model.*
   (a) *$H^q(\mathsf{P}(M)) = H^q(\Gamma; M_S)$, $H^q(\mathsf{P}^\partial(M)) = H^q(\Gamma_\infty; M_S)$ (Shapiro's lemma), and $H^q(r_{\mathsf{P}}) = \mathrm{res}^\Gamma_{\Gamma_\infty}$, for every q.*
   (b) *r_{P} is surjective in every degree (split by extension by zero along the complement); with Lemma 15.2(2)'s argument the sequence*
   $$\cdots \to H^q_c(\mathsf{P}; M) \xrightarrow{\mathrm{pr}_{c*}} H^q(\Gamma; M_S) \xrightarrow{\mathrm{res}} H^q(\Gamma_\infty; M_S) \xrightarrow{\mathrm{inc}_{c*}} H^{q+1}_c(\mathsf{P}; M) \to \cdots \tag{79}$$
   is exact in every degree $q \in \mathbb{Z}$.
   (c) *$H^q_c(\mathsf{P}; M) \cong H^q\big(\mathrm{Hom}_{S[\Gamma]}(\mathsf{P}/\mathsf{P}^\partial, M_S)\big) \cong \mathrm{Ext}^{q-1}_{S[\Gamma]}(\mathrm{St}_S, M_S) \cong H^{q-1}\big(\Gamma; \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, M_S)\big)$, independently of the model, and under these identifications (79) is the long exact* Ext*-sequence of $0 \to \mathrm{St} \xrightarrow{j_{\mathrm{St}}} \mathbb{Z}[\mathbb{P}^1] \xrightarrow{\varepsilon} \mathbb{Z} \to 0$ in the first variable, **with these signs** (v1.23.0-pre.27, referee m4, in the manner of Proposition 31.10(2)(b)): identify through $R_\bullet := (\mathsf{P}/\mathsf{P}^\partial)_{\bullet+1}$ with differential $-\partial$ and augmentation $\varepsilon_R[x] := \varepsilon^\partial(\partial x) \in \mathrm{St}$ ($\varepsilon^\partial\colon \mathsf{P}^\partial_0 \to S[\mathbb{P}^1]$ the augmentation; the connecting map of $0 \to \mathsf{P}^\partial \to \mathsf{P} \to \mathsf{P}/\mathsf{P}^\partial \to 0$ on H_1, no sign), the cochains $\mathrm{Hom}(\mathsf{P}/\mathsf{P}^\partial, M_S)^q = \mathrm{Hom}(R, M_S)^{q-1}$ identified identically. Then res is induced by ε; $\mathrm{inc}_{c*} = j^*_{\mathrm{St}}$; and $\mathrm{pr}_{c*} = -\delta$, $\delta\colon \mathrm{Ext}^{q-1}(\mathrm{St}, M_S) \to \mathrm{Ext}^q(S, M_S)$ the connecting map with the convention $\delta[x] = [d\tilde{x}]$ on a resolution of the short exact sequence ($\tilde{x}$ a lift, the cochain differential carrying no sign). The "$-$" depends on the convention: identifying*

$H_0(R)$ *with* St *through* $-\varepsilon_R$ *moves it to* inc_{c*}. *Immaterial for groups, kernels, images, exactness and* T_p*-equivariance; it matters only when a map is compared entrywise with an independently computed* Ext *map. Hence* $H^q_c = 0$ *for* $q \le 0$ *over every* S*; over* $S = \mathbb{Z}$, $H^1_c(\Gamma; M) = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathrm{St}, M)$ *is* ***torsion-free****; and for the model (c) this is Lemma 15.2's* $H^\bullet_c$ *with fibre* M.

(d) ***(The models and their integral forms.)*** *On (a),* $C(r_{\mathcal{I}})$ *is a complex of finitely generated free* $\mathbb{Z}$*-modules of ranks* $5d$, $17d$ *and* $25d$ *in degrees* -1, 0 *and* ≥ 1, *with* $C(r_{\mathcal{I}}) \otimes S$ *the cone over* S. *On (b), the cochains are finitely generated free* O_ℓ*-modules (Proposition 31.6(2)) with every base change;* ***at*** $\ell = 2$ *every boundary stabiliser is its own 2-Sylow (*$P_\sigma = \Gamma_\sigma = \{\pm I\}$*, (B3)), so* $\mathcal{I}^{\partial(2)}_q \cong \mathsf{O}_2[\Gamma]^{w^\partial_q}$ *(*$w^\partial_q = 3, 6, 6, \dots$*) is free and computes* $H^\bullet(\Gamma_\infty; M_{\mathsf{O}_2})$*, non-zero in infinitely many degrees by (4);* ***at*** $\ell \ge 3$ *every boundary column is* O_ℓ *in degree* 0 *(*$P_\sigma = 1$, $Q_\sigma = \{\pm I\}$*),* cmp′ *restricts to an* ***isomorphism of complexes*** $\mathcal{I}^{\partial(\ell)} \to C_\bullet(\partial X) \otimes \mathsf{O}_\ell$*, and* $\mathcal{I}^{\partial(\ell)}(M)$ *is the circle model with fibre* M_{O_ℓ} *(Proposition 14.4(4)):* ***on*** $\mathcal{I}^{(3)}$ ***the boundary is Bredon****. On (c) over* $\mathbb{Z}[1/6]$*, at level* 1 *with* $M = V$*, the cone tensored with* $\mathbb{Z}/\ell^K$, $\ell \ge 5$*, is the shipped E8 item-4b pair (Definition 15.1 over* $R = \mathbb{Z}/\ell^K$*; Proposition 23.2(1)) — an identity of complexes.*

2. ***(The glued long exact sequence over*** $\mathbb{Z}$***.)*** cmp *is a comparison of* $\mathbb{Z}[1/6]$*-pair models* $(\mathcal{I}, \mathcal{I}^\partial) \otimes \mathbb{Z}[1/6] \to (C_\bullet(X), C_\bullet(\partial X)) \otimes \mathbb{Z}[1/6]$ *((IM2)), and* $\alpha_c := (\mathrm{cmp}^*[1], \mathrm{cmp}^{\partial *})$ *is a quasi-isomorphism of cones over* $\mathbb{Z}[1/6]$ *commuting strictly with* r, inc_c, pr_c. *With* $A_c := C(r_{C(X)})$ *over* $\mathbb{Z}[1/6]$, $B_c := C(r_{\mathcal{I}}) \otimes \widehat{\mathbb{Z}}_6$, $C_c := C(r_{\mathcal{I}}) \otimes \widehat{\mathbb{Q}}_6$*:*
   (a) *the mapping cocone of* $A_c \oplus B_c \to C_c$ *(*α_c *followed by base change, minus the canonical map) is quasi-isomorphic to* $C(r_{\mathcal{I}})$*, and* $H^q_c(\Gamma; M) \cong H^{q-1}(A_c) \times_{H^{q-1}(C_c)} H^{q-1}(B_c)$ *for every* q*;* ***the only glueing datum is the pair map*** cmp*;*
   (b) *every map of (79) commutes strictly with every comparison of pairs (Lemma 31.19(3)) and with base change, so the sequence over* $\mathbb{Z}$ *is determined by its localisations: the* ℓ*-primary parts of its terms, and the kernels, images and cokernels of its maps, at* $\ell \ge 5$ *on the* $\mathbb{Z}[1/6]$*-piece (model (c), or (a)), at* $\ell \in \{2, 3\}$ *on model (b) or (a) over* $\mathbb{Z}_{(\ell)}$*, the free parts on any piece. It is exact over* $\mathbb{Z}$*; and since a finitely generated abelian group vanishes iff it vanishes after* $\otimes\mathbb{Z}[1/6]$, $\otimes\mathbb{Z}_{(2)}$ *and* $\otimes\mathbb{Z}_{(3)}$*,* ***exactness at a spot of the glued sequence is equivalent to exactness of the three local sequences there****;*
   (c) *the maps are read on each piece by Proposition 31.10(2)(g)'s precision rule, from Smith forms certified by (6); kernels, images and the comparisons "*$\mathrm{im} = \ker$*" so computed are exact provided every exponent read stays below the precision left at its step, and are refused otherwise.*

   No coherence datum enters (a)–(c): every map is induced by a comparison or by the inclusion $\mathsf{P}^\partial \subseteq \mathsf{P}$*. Proposition 15.16's ambiguity concerns a cochain-level Hecke homotopy, not these maps; it is removed for the Hecke operator in Proposition 31.21.*

3. ***(Level*** N***; the cusps.)*** *For* $A \le \Gamma$ *of finite index and* X *a* $\mathbb{Z}[A]$*-module, finitely generated and free over* $\mathbb{Z}$*, Shapiro's lemma (in the second variable; induced and coinduced modules agree in finite index, Definition 12.1) gives* $H^q(\Gamma_\infty; \mathrm{Ind}^\Gamma_A X) \cong \mathrm{Ext}^q_{\mathbb{Z}[A]}(\mathbb{Z}[\mathbb{P}^1], X) \cong \bigoplus_{c \in A\backslash\mathbb{P}^1} H^q(A_c; X)$*,* A_c *the stabiliser of the cusp* c*, and* $H^q_c(\Gamma; \mathrm{Ind}^\Gamma_A X) \cong \mathrm{Ext}^{q-1}_{\mathbb{Z}[A]}(\mathrm{St}, X)$*: the group-theoretic pair of* A*,* ***block-diagonal by cusps on the boundary****. For* $A = \Gamma_0(N)$ *and the coordinates of Definition 28.1: the cusps are the* $\langle T\rangle$*-orbits on* $\Gamma_0(N)\backslash\Gamma \cong \mathbb{P}^1(\mathbb{Z}/N)$ *(bottom rows, right action),* $c(N)$ *their number; the orbit of the coset of* w *has length* w_c *(the* width*),* $\sum_c w_c = m_N$*; its stabiliser in* Γ_∞ *is* $\Gamma_\infty^{(w_c)} := \langle T^{w_c}\rangle \times \langle -I\rangle$*; and* $\mathrm{Res}_{\Gamma_\infty} M_N \cong \bigoplus_c \mathrm{Ind}^{\Gamma_\infty}_{\Gamma_\infty^{(w_c)}} V$ *with* T^{w_c} *acting on* V *by* $\rho_n(T^{w_c})$ *and* $-I$ *by* $(-1)^n$ *(for* $M_N(\chi)$ *by* $\chi(-1)(-1)^n$ *on the* $-I$ *part; the* T^{w_c}*-action is then twisted by the value of* χ *on* $wT^{w_c}w^{-1}$*). So* $H^q(\Gamma_\infty; M_N) \cong \bigoplus_c H^q(\Gamma_\infty^{(w_c)}; V)$.

4. ***(The boundary in closed form.)*** *Let X be a $\mathbb{Z}[\Gamma_\infty^{(w)}]$-module, finitely generated and free over $\mathbb{Z}$, on which $-I$ acts by $\varepsilon=\pm1$, T^w by t. Then, with X^t, X_t the invariants and coinvariants of t,*

$$\varepsilon=+1:\ H^0=X^t,\ H^1=X_t,\ H^{2j}=(X/2)^t,\ H^{2j+1}=(X/2)_t\ (j\ge1);\qquad \varepsilon=-1:\ H^0=0,\ H^{2j-1}=(X$$

*with $\dim_{\mathbb{F}_2}(X/2)^t=\dim_{\mathbb{F}_2}(X/2)_t$. Hence $2\,H^q(\Gamma_\infty^{(w)};X)=0$ for $q\ge2$ (for $q\ge1$ if $\varepsilon=-1$), $H^q\otimes\mathbb{Z}[1/2]=0$ for $q\ge2$ (**the boundary is bounded away from** 2), and $H^q\otimes\mathbb{Z}_{(2)}\neq0$ for every $q\ge2$ as soon as $X\neq0$ and t is unipotent modulo 2 (as for $\mathrm{Res}_{\Gamma_\infty}M_N$, (3)): then $(X/2)^t\neq0$. For $X=V$ and $t=\rho_n(T^w)$, $w\ge1$:*

$$V^{T^w}=\mathbb{Z}\,x^n,\qquad (V)_{T^w}\cong\mathbb{Z}\oplus F_{n,w},\quad |F_{n,w}|=n!\,w^n,\qquad \dim_{\mathbb{F}_2}(V/2)^{T^w}=\begin{cases}n+1, & w\ \text{even},\\ \lfloor n/2\rfloor+1, & w\ \text{odd}.\end{cases}$$

*So for M_N, n even: $H^0(\Gamma_\infty;M_N)=\mathbb{Z}^{c(N)}$, $H^1=\bigoplus_c(\mathbb{Z}\oplus F_{n,w_c})$, and H^q, $q\ge2$, is elementary abelian of $\mathbb{F}_2$-rank $\sum_c\dim(V/2)^{T^{w_c}}$; n odd: $H^0=0$ and H^q, $q\ge1$, elementary abelian of that rank. At $N=1$ this is Proposition 14.3 and Lemma 14.20, with H^1 explicit: for n even, $\mathrm{tors}\,H^1(\Gamma_\infty;V)$ has order $n!$; for n odd, $H^1(\Gamma_\infty;V)=(V/2)^T$, of order $2^{\lfloor n/2\rfloor+1}$ (v1.23.0-pre.27, referee m3). **Periodicity**: $\mathrm{res}_{\Gamma_\infty^{(w)}}e\cup-\colon H^q\to H^{q+2}$ is an isomorphism for $q\ge2$ ($q\ge1$ if $\varepsilon=-1$).*

5. ***(Compact support.)***
   (a) ***(Low degrees.)*** *r^0 induces the inclusion $M_S^\Gamma\subseteq M_S^{\Gamma_\infty}$ over every S, so $H^0_c=0$ and D^{-1} is injective over every field: on $C(r_{\mathcal{I}})$ over $\mathbb{Z}$ every elementary divisor of D^{-1} is a unit, $\mathrm{rank}_{\mathbb{Q}}D^{-1}=5d$, and $\mathrm{tors}\,H^1_c=0$ (the integral form of (1)(c)).*
   (b) ***(Rational ranks.)*** *Let F be a field with $6\in F^\times$. Then $H^q_c(\Gamma;M_F)=0$ for $q\ge3$, $\dim_F H^2_c(\Gamma;M_F)=\dim_F(M_F^\vee)^\Gamma$, and*

$$\dim_F H^1_c=\dim H^1-\dim H^0+\dim(M_F^\vee)^\Gamma,\qquad \mathrm{rank}\big(\mathrm{res}^1\colon H^1(\Gamma;M_F)\to H^1(\Gamma_\infty;M_F)\big)=\dim H^1(\Gamma_\infty$$

using $\dim H^0(\Gamma_\infty;M_F)=\dim H^1(\Gamma_\infty;M_F)$. For $M=M_N(\chi)$ over $\mathbb{Q}$: $(M_{\mathbb{Q}}^\vee)^\Gamma\neq0$ iff $n=0$ and $\chi=1$, so $b_0(M^\vee)=b_0(M)$, $\dim H^1_c=b_1(M)$, $\dim H^2_c=b_0(M)$; for M_N, n even, $\mathrm{rank}\,\mathrm{res}^1=c(N)-b_0(M_N)$.
   (c) ***(Exponents and the primes.)*** *For $q\ge3$, $48\,H^q_c(\Gamma;M)=0$. At $\ell\ge5$, $H^q_c\otimes\mathbb{Z}_{(\ell)}=0$ for $q\ge3$ and the ℓ-local sequence (79) is six-term, $0\to H^0\to H^0(\Gamma_\infty)\to H^1_c\to H^1\to H^1(\Gamma_\infty)\to H^2_c\to0$. At $\ell=3$, $\mathrm{pr}_{c*}\colon H^q_c\otimes\mathbb{Z}_{(3)}\to H^q(\Gamma;M)\otimes\mathbb{Z}_{(3)}$ is an isomorphism for $q\ge3$ (at level 1: $H^q(\mathbb{Z}/3;M^{-I})$, Proposition 31.3(2), non-zero e.g. for q even at $n=0$). At $\ell=2$ the boundary and, in general, the cone are non-zero in infinitely many degrees (D-δ6). (Observed on every cell of the two checks below: the exponent of H^q_c, $q\ge3$, divides 12; not asserted, v1.23.0-pre.27, referee m5.)*
   (d) ***(Periodicity.)*** *$e\cup-$, $e\in H^2(\Gamma;\mathbb{Z})$ of Lemma 14.22, acts on the three terms of (79) in their coefficient form $H^\bullet(\Gamma;X)$, $X\in\{M,\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[\mathbb{P}^1],M),\mathrm{Hom}_{\mathbb{Z}}(\mathrm{St},M)\}$ (on the boundary term it is $\mathrm{res}_{\Gamma_\infty}e\cup-$, the periodicity of (4)), and commutes with its maps; it is an isomorphism on $H^q(\Gamma;M)$ and on $H^q(\Gamma_\infty;M)$ for $q\ge2$, and on H^q_c for $q\ge3$. Degrees ≤4 of the pair determine every degree: $q_{\max}\ge4$ **for the pair**, one more than the bulk's 3.*
   (e) ***(H^2_c at $\ell\ge5$; v1.23.0-pre.27, referee M2.)*** *For every prime $\ell\ge5$,*

$$H^2_c(\Gamma;M)\otimes\mathbb{Z}_{(\ell)}\ \cong\ H_0(\Gamma;M)\otimes\mathbb{Z}_{(\ell)}=M_\Gamma\otimes\mathbb{Z}_{(\ell)}.$$

*For $M=M_N$, $M_\Gamma\cong V_A$ ($A=\Gamma_0(N)$), which is $\mathbb{Z}$ for $n=0$ and is killed by $\otimes\mathbb{Z}_{(\ell)}$ for every $\ell>n+1$ if $n>0$, every N: so $\mathrm{tors}\,H^2_c(\Gamma;M_N)$ has **no ℓ-torsion for $\ell\ge5$**, $\ell>n+1$, and its ℓ-part for $\ell\ge5$ is that of $\mathrm{tors}\,V_A$, a cokernel of a $d\times2d$ integer matrix. (5)(b)'s $\dim H^2_c=\dim(M_F^\vee)^\Gamma$ is the $\otimes\mathbb{Q}$ shadow ($\dim M_{\Gamma,\mathbb{Q}}=\dim(M_{\mathbb{Q}}^\vee)^\Gamma$).*

6. ***(The certified exponent on the pair; D-δ6.)*** *Let* P *be a* $\mathbb{Z}_{(\ell)}$*-pair model whose cochains are finitely generated free* $\mathbb{Z}_{(\ell)}$*-modules ((a) or (b) over* $\mathbb{Z}_{(\ell)}$*). Propositions 23.17(1)–(3), Remark 23.18 and Proposition 31.6(6) (the Smith-valuation read-out and the stopping rule) hold verbatim for the differentials* $D^{\partial,q}$ *of* $\mathsf{P}^{\partial}(M)$ *and* D^q *of* $C(r_{\mathsf{P}})$ *(*$q \geq -1$*), read on the brutal truncations* $\sigma^{\leq q_{\max}+1}$ *of bulk and boundary; the cone of the truncated restriction equals the cone in degrees* $\leq q_{\max}$*, so* H^q_c *is read for* $q \leq q_{\max}$*.* ***Admissible sources for*** $\operatorname{rank}_{\mathbb{Q}}$*, besides the Betti recursion* $\operatorname{rank}_{\mathbb{Q}} D^q = \operatorname{rank} C^q - b_q - \operatorname{rank}_{\mathbb{Q}} D^{q-1}$ *on the complex in question:*
   (a) *boundary: the closed form,* $b_0(\partial; M) = b_1(\partial; M) = \operatorname{rank} M^{\Gamma_\infty}$ *(*$= c(N)$ *for* M_N*,* n *even;* 0 *for* n *odd),* $b_q(\partial; M) = 0$ *for* $q \geq 2$*; or the same boundary complex modulo a prime* ℓ' *(not dividing* q_ℓ *on (b)) at which* $H^{q+1}(\Gamma_\infty; M)$ *has no* ℓ'*-torsion — every* $\ell' \geq 3$ *for* $q \geq 1$*;*
   (b) *cone,* D^{-1}*:* $\operatorname{rank}_{\mathbb{Q}} = \operatorname{rank} \mathsf{P}(M)^0$*, no source needed ((5)(a));*
   (c) *cone,* D^0*: the Betti recursion with* $b_0(C(r)) = \dim_{\mathbb{Q}} H^1_c$ *of (5)(b), its* $b_j(M)$ *from a source admissible by Propositions 31.6(4), 31.7(7);* ***or*** *the same cone modulo a prime* $\ell' \geq 5$ *(not dividing* q_ℓ *on (b)) at which* $H_0(\Gamma; M)$ *has no* ℓ'*-torsion, which by (5)(e) is exactly when* H^2_c *has none —* ***for*** M_N***, every*** $\ell' \geq 5$ ***with*** $\ell' > n+1$*, the rule of Proposition 31.7(7) for the bulk; for* $M_N(\chi)$*,* $\chi \neq 1$*, the computed* $\operatorname{tors} H_0(\Gamma; M)$ *decides (no closed-form rule is asserted). At* $\ell' \in \{2, 3\}$ *there is no such rule, and a prime dividing* $\operatorname{tors} H^2_c$ *gives a deficient rank (v1.23.0-pre.27, referee M2, replacing "not known a priori");*
   (d) *cone,* D^q*,* $q \geq 1$*: the same cone modulo any prime* $\ell' \geq 5$ *(not dividing* q_ℓ*;* H^{q+2}_c *is killed by* 48*), or the recursion with* $b_1(C(r)) = \dim H^2_c$*,* $b_q(C(r)) = 0$ *for* $q \geq 2$*.*

   Not admissible: the rank of a differential of $\mathsf{P}(M)$ *or* $\mathsf{P}^{\partial}(M)$ *alone for the cone, of another model, or of the cone at a non-qualifying prime.* ***A-priori exponents*** *(consistency checks, and caps):* $a_t \leq v_\ell(2)$ *on* $D^{\partial,q}$*,* $q \geq 1$*; on* $D^{\partial,0}$*,* $\sum_t a_t = v_\ell \left|\operatorname{tors} H^1(\Gamma_\infty; M)\right|$*, for* M_N*,* n *even,* $\sum_c v_\ell(n!\, w_c^n)$*;* $a_t = 0$ *on* D^{-1}*;* $a_t \leq v_\ell(48)$ *on* D^q*,* $q \geq 1$*; on* D^0 *(*$\operatorname{tors} H^2_c$*), at* $\ell \geq 5$*,* $\sum_t a_t = v_\ell \left|\operatorname{tors} H_0(\Gamma; M)\right|$ *((5)(e);* 0 *for* M_N *when* $\ell > n+1$*),* ***no cap at*** $\ell \in \{2, 3\}$ *(v1.23.0-pre.27, referee M2).* ***The truncation*** *is 14.24's box:* $H^q(\Gamma_\infty)$ *and* H^q_c *are reported for* $q \leq q_{\max}$*, (79) is asserted exact at every spot whose three terms are reported; by (4) and (5)(d) nothing is lost for* $q_{\max} \geq 4$ *—* ***the infinite 2-adic boundary is a closed form in every degree, and the truncation reports a finite, periodic part of it****.*

Proof. (1)(a) P resolves S, and P^{∂} resolves $S[\mathbb{P}^1] = S[\Gamma] \otimes_{S[\Gamma_\infty]} S$, so $H^\bullet(\mathsf{P}^{\partial}(M)) = \operatorname{Ext}^\bullet_{S[\Gamma]}(S[\Gamma/\Gamma_\infty], M_S) = H^\bullet(\Gamma_\infty; M_S)$ [standard: [Bro82, Ch. III.6]]; the inclusion $\mathsf{P}^{\partial} \subseteq \mathsf{P}$ lies over $\varepsilon\colon S[\mathbb{P}^1] \to S$, which induces the restriction [standard: [Bro82, Ch. III.8–9]]. (b) Restriction to a direct summand is split surjective. Lemma 15.2(2)'s proof of the cone sequence and its connecting map uses only that r is a cochain map. (c) $\ker r_{\mathsf{P}} = \operatorname{Hom}_{S[\Gamma]}(\mathsf{P}/\mathsf{P}^{\partial}, M_S)$, and $c \mapsto (c, 0)$ is a quasi-isomorphism $(\ker r)[1] \to C(r)$ by Lemma 15.2(3)'s argument (surjectivity of r). The long exact homology sequence of $0 \to \mathsf{P}^{\partial} \to \mathsf{P} \to \mathsf{P}/\mathsf{P}^{\partial} \to 0$, with $H_0(\mathsf{P}^{\partial}) = S[\mathbb{P}^1] \xrightarrow{\varepsilon} S = H_0(\mathsf{P})$ onto and all other homology zero, shows that $\mathsf{P}/\mathsf{P}^{\partial}$, a bounded below complex of projectives, has homology St_S in degree 1 only (St is $\mathbb{Z}$-free, so $\mathrm{St} \otimes S = \ker(S[\mathbb{P}^1] \to S)$). Splitting off the contractible part in degrees ≤ 1 ($H_0 = 0$, ∂_1 onto the projective degree-0 term) leaves a projective resolution of St_S shifted by one, so the Hom complex computes $\operatorname{Ext}^{\bullet-1}_{S[\Gamma]}(\mathrm{St}_S, M_S)$; and $\operatorname{Ext}_{S[\Gamma]}(\mathrm{St}_S, M_S) = H^\bullet(\Gamma; \operatorname{Hom}_{\mathbb{Z}}(\mathrm{St}, M_S))$ because St is $\mathbb{Z}$-free [standard: [Bro82, Ch. III, Prop. 2.2]]. The identification with the Ext-sequence: $\mathsf{P}^{\partial} \to \mathsf{P} \to \mathsf{P}/\mathsf{P}^{\partial}$ is a short exact sequence of projective resolutions over $0 \to \mathrm{St} \to \mathbb{Z}[\mathbb{P}^1] \to \mathbb{Z} \to 0$ shifted, and $\operatorname{Hom}(-, M_S)$ of it is the cone sequence [standard: [Wei94, Thm. 1.3.1, §1.5]]. *The signs* (v1.23.0-pre.27, referee m4). Write $\mathsf{P} = \mathsf{P}^{\partial} \oplus C$ as graded modules, C the projective complement, and $\partial = \begin{pmatrix} \partial^{\partial} & \beta \\ 0 & \partial^{C} \end{pmatrix}$; $\partial^2 = 0$

gives $\partial^{\partial}\beta = -\beta\partial^{C}$, and $C \cong \mathsf{P}/\mathsf{P}^{\partial}$, so $R_q = C_{q+1}$ with $\partial^{R} = -\partial^{C}$ (a complex of projectives with homology St_S in degree 0 only; $R_{-1} = C_0$ splits off, $H_0(\mathsf{P}/\mathsf{P}^{\partial}) = 0$). (i) $\beta\colon R \to \mathsf{P}^{\partial}$ is a chain map ($\partial^{\partial}\beta = \beta\partial^{R}$) with $\varepsilon^{\partial}\beta = \varepsilon_R$, a lift of j_{St}, so $j_{\mathrm{St}}^{*}[x] = [x \circ \beta]$. Under $c \mapsto (c, 0)$, $\mathrm{inc}_{c*}[c_{\partial}] = [(0, c_{\partial})] = [(d\tilde{c}, 0)]$ ($\tilde{c}$ the extension by zero: $(0, c_{\partial}) - D(\tilde{c}, 0) = (d\tilde{c}, 0)$), and $d\tilde{c} = \tilde{c} \circ \partial$ vanishes on P^{∂} and is $c_{\partial} \circ \beta$ on C: $\mathrm{inc}_{c*} = j_{\mathrm{St}}^{*}$. (ii) $E := R \oplus \mathsf{P}$ with $\partial^{E}(r, y) := (\partial^{R} r + \pi_C y,\ \partial y)$ (π_C the projection, a chain map $\mathsf{P} \to C$) is a complex ($\partial^{R}\pi_C + \pi_C\partial = 0$) in a short exact sequence $0 \to R \to E \to \mathsf{P} \to 0$ of bounded below complexes of projectives, each with homology in degree 0 only; $\kappa(r, a, c) := a - \beta r$ ($a \in \mathsf{P}^{\partial}$, $c \in C$) is a chain map $E \to \mathsf{P}^{\partial}$, and $\varepsilon^{\partial}\kappa$ makes $H_0(E) \cong S[\mathbb{P}^1]$ with $R \to E$ over $-j_{\mathrm{St}}$ and $E \to \mathsf{P}$ over ε (on a 0-cycle, $c = \partial^{C} r$, the two augmentations differ by $\varepsilon(\partial r) = 0$). For this resolution of $0 \to \mathrm{St} \xrightarrow{-j_{\mathrm{St}}} S[\mathbb{P}^1] \to S \to 0$, $\tilde{x}(r, y) := x(r)$ lifts a cocycle x of $\mathrm{Hom}(R, M_S)$ and $d\tilde{x} = x \circ \pi_C$, the extension by zero, which is $\mathrm{pr}_{c*}[x]$. The isomorphism $(-1, \mathrm{id}, \mathrm{id})$ from the sequence with j_{St} to the one with $-j_{\mathrm{St}}$ and the naturality of δ give $\mathrm{pr}_{c*} = -\delta$. $H_c^1 = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathrm{St}, M) \subseteq \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, M)$, torsion-free as M is. For (c) of Definition 31.18 the cone is Definition 15.1's. (d) The ranks: Proposition 23.5(2), $C(r_{\mathcal{I}})^q = C_{\mathcal{I}}^{q+1}(M) \oplus C_{\mathcal{I}\partial}^{q}(M)$. At $\ell = 2$: $Q_\sigma = 1$ on the boundary, so $F'^{\sigma} = F^{\sigma} \otimes \mathsf{O}_2$, free. At $\ell \geq 3$: $G'_{s,0} = \mathsf{O}_\ell[\Gamma] \otimes_{\mathsf{O}_\ell[\Gamma_\sigma]} \mathsf{O}_\ell = \mathsf{O}_\ell[\Gamma/\{\pm I\}]$ for $\sigma \in \Sigma^{\partial}$, and cmp' on it is the identification with $C_s(\partial X) \otimes \mathsf{O}_\ell$; there is no $t \geq 1$ on the boundary, so $d'|_{\mathcal{I}^{\partial(\ell)}} = d'^{(1)}$, and $\mathrm{cmp}' d'^{(1)} = \partial \mathrm{cmp}'$ ((IM1)) with cmp' bijective forces $d'^{(1)} = \partial$ under it. Then $\mathcal{I}^{\partial(\ell)}(M) = \mathrm{Hom}_{\Gamma}(C_\bullet(\partial X), M_{\mathsf{O}_\ell})$, which Proposition 14.4(4) identifies with the circle model. The E8 pair: the shipped item-4b complexes are $C_{\mathrm{eq}}^{\bullet}(\mathbb{Z}/\ell^K)$ and $C_{\mathrm{eq}}^{\bullet}(K_\partial, \mathcal{R}_\partial; L)$ over $\mathbb{Z}/\ell^K$, which are Definition 15.1's with $R = \mathbb{Z}/\ell^K$ (Proposition 14.4(4)), and $C_{\mathrm{eq}}^{\bullet}(\mathbb{Z}[1/6]) \otimes \mathbb{Z}/\ell^K = C_{\mathrm{eq}}^{\bullet}(\mathbb{Z}/\ell^K)$ (Proposition 23.2(1)), the same on the boundary.

(2) $\mathrm{cmp}(\mathcal{I}^{\partial}) \subseteq C_\bullet(\partial X)$ is (IM2); on $\mathcal{I}_0^{\partial}$ the augmentations to $\mathbb{Z}[\mathbb{P}^1]$ agree by definition; so cmp is a comparison of pairs once both are $\mathbb{Z}[1/6]$-pair models, and Lemma 31.19(3) applies. (a) Proposition 23.9's proof with the cones in place of $C_{\mathrm{eq}}^{\bullet}$, $C_{\mathcal{I}}^{\bullet}$: Proposition 23.7 for the complex of free $\mathbb{Z}$-modules $C(r_{\mathcal{I}})$, α_c a quasi-isomorphism over $\mathbb{Z}[1/6]$, the five lemma on the cocones, and Proposition 23.8. (b) The strict identities are Lemma 31.19(3) and cmp^{*}'s base-change compatibility (Proposition 23.5(4)); Proposition 23.8's morphism clause determines each map by its two local components; the localisation $G_{(\ell)}$ of a finitely generated abelian group at $\ell \geq 5$ is $(G[1/6])_{(\ell)}$, and $G = 0$ iff $G_{(\ell)} = 0$ for every ℓ; exactness at a spot is the vanishing of a subquotient, and localisation is exact. (c) is Proposition 31.10(2)(g) applied to the differentials and maps of the pieces.

(3) $\mathrm{Ext}^{\bullet}_{\mathbb{Z}[\Gamma]}(Y, \mathrm{Ind}_A^{\Gamma} X) \cong \mathrm{Ext}^{\bullet}_{\mathbb{Z}[A]}(\mathrm{Res}\, Y, X)$ for every $\mathbb{Z}[\Gamma]$-module Y (Shapiro, [standard: [Bro82, Ch. III.6]]: Ind = CoInd in finite index), with $Y = \mathbb{Z}[\mathbb{P}^1]$ and $Y = \mathrm{St}$; as an A-set $\mathbb{P}^1 = \bigsqcup_c A/A_c$, so $\mathbb{Z}[\mathbb{P}^1] = \bigoplus_c \mathrm{Ind}_{A_c}^{A} \mathbb{Z}$ and Shapiro again. For $A = \Gamma_0(N)$: $A\backslash\Gamma/\Gamma_\infty \to A\backslash\mathbb{P}^1$, $A\gamma\Gamma_\infty \mapsto A\gamma\infty$, is a bijection, and $A\backslash\Gamma$ is $\mathbb{P}^1(\mathbb{Z}/N)$ through bottom rows (Lemma 28.2(1)) with Γ_∞ acting on the right through T ($-I$ acts trivially on $\mathbb{P}^1(\mathbb{Z}/N)$); the stabiliser of Aw in Γ_∞ is $\Gamma_\infty \cap w^{-1}Aw \ni -I$, of index w_c, hence $\langle T^{w_c}\rangle \times \langle -I\rangle$. In the coordinates of (22) the orbit's blocks span a Γ_∞-submodule permuted transitively with block stabiliser $\Gamma_\infty^{(w_c)}$, acting on the block of w by $\rho_n(h_w(\gamma))$, $h_w(T^{w_c}) = wT^{w_c}w^{-1} \in \Gamma_0(N)$; a module whose summands are permuted transitively is induced from a block stabiliser [standard: [Bro82, Ch. III, Prop. 5.3]], and $\rho_n(w)$ intertwines $\rho_n(wT^{w_c}w^{-1})$ with $\rho_n(T^{w_c})$ (ρ_n is multiplicative, Proposition 8.2(2)). With nebentypus the character value on $wT^{w_c}w^{-1}$ multiplies the block.

(4) $\Gamma_\infty^{(w)} = \langle T^w\rangle \times \langle -I\rangle$ and Proposition 14.3's proof (the Lyndon–Hochschild–Serre sequence with two columns) applies with T^w for T: $0 \to (H^{q-1}(\langle -I\rangle; X))_t \to H^q \to (H^q(\langle -I\rangle; X))^t \to 0$. For X $\mathbb{Z}$-free, the periodic complex $X \xrightarrow{\varepsilon-1} X \xrightarrow{1+\varepsilon} X \xrightarrow{\varepsilon-1} \cdots$ gives $H^0(\langle -I\rangle; X) = X$, $H^{\mathrm{odd}} = 0$, $H^{\mathrm{even}\geq 2} = X/2$ for $\varepsilon = 1$, and $H^0 = 0$, $H^{\mathrm{odd}} = X/2$, $H^{\mathrm{even}} = 0$ for $\varepsilon = -1$; in each degree one outer term vanishes, which gives the display without an extension problem. $(X/2)^t$ and $(X/2)_t$ are the kernel and cokernel of an endomorphism of a finite-dimensional $\mathbb{F}_2$-space. The vanishing statements follow. For V: dehomogenise $u_j = x^{n-j}y^j \mapsto y^j$, $V \cong \mathbb{Z}[y]_{\leq n}$; by (9)

with $T^w = \begin{pmatrix} 1 & w \\ 0 & 1 \end{pmatrix}$, $\rho_n(T^w)u_j = x^{n-j}(wx+y)^j \mapsto (y+w)^j$, so $\rho_n(T^w) - 1$ is $\Delta_w\colon P(y) \mapsto P(y+w) - P(y)$. $\Delta_w P = 0$ forces P constant ($P(y+kw) = P(y)$ for all k), so $V^{T^w} = \mathbb{Z}x^n$ ($u_0 = x^n$); Δ_w maps $\mathbb{Z}[y]_{\le n}$ into $\mathbb{Z}[y]_{\le n-1}$, and in the bases $(y^1, \dots, y^n)$, $(y^0, \dots, y^{n-1})$ its matrix is triangular with diagonal $(jw)_{j=1}^n$ ($\Delta_w y^j = jwy^{j-1} + \cdots$); so $\mathbb{Z}[y]_{\le n-1}/\Delta_w\mathbb{Z}[y]_{\le n}$ is finite of order $\prod_j jw = n!\, w^n$, and $(V)_{T^w} = \mathbb{Z}y^n\oplus$ it. Modulo 2: if w is even, every coefficient $\binom{j}{i}w^{j-i}$, $i < j$, of Δ_w is even and $\Delta_w \equiv 0$; if w is odd, $P(y+w) = P(y+1)$ over $\mathbb{F}_2$, and the kernel of Δ_1 on $\mathbb{F}_2[y]_{\le n}$ is $\mathbb{F}_2[y^2+y]_{\le n}$ (a polynomial invariant under $y \mapsto y+1$ is a polynomial in y^2+y, by division with remainder), of dimension $\lfloor n/2 \rfloor + 1$. M_N by (3). *Periodicity*: $H^2(\Gamma_\infty^{(w)}; \mathbb{Z}) \cong H^2(\langle -I\rangle; \mathbb{Z}) = \mathbb{Z}/2$ by restriction (the case $X = \mathbb{Z}$, $\varepsilon = 1$, $t = 1$ of the display: $(X/2)^t = \mathbb{Z}/2$), and $\operatorname{res} e$ restricts to a generator on $\langle -I\rangle$ (Lemma 14.22(1)); so $\operatorname{res}_{\Gamma_\infty^{(w)}} e = \mathrm{pr}^* u_0$, u_0 the generator of $H^2(\langle -I\rangle; \mathbb{Z})$, pr the projection onto $\langle -I\rangle$. (v1.23.0-pre.27, referee m1: the tensor resolution, no multiplicativity of a spectral sequence.) Let F be $\mathbb{Z}[\langle T^w\rangle] \xrightarrow{T^w-1} \mathbb{Z}[\langle T^w\rangle]$ and F' the periodic resolution of $\mathbb{Z}$ over $\langle -I\rangle$ ($F'_b = \mathbb{Z}[\langle -I\rangle]$, differentials $\sigma - 1$, $\sigma + 1$ alternately, $\sigma = -I$); $F \otimes F'$ is a free resolution of $\mathbb{Z}$ over $\Gamma_\infty^{(w)}$ [standard: [Bro82, Ch. V.1]]. With $K := \operatorname{Hom}_{\langle -I\rangle}(F', X)$, $\operatorname{Hom}_{\Gamma_\infty^{(w)}}(F \otimes F', X)$ is the cocone of $t - 1\colon K \to K$, whose long exact sequence $H^{q-1}(K) \xrightarrow{t-1} H^{q-1}(K) \to H^q \to H^q(K) \xrightarrow{t-1} H^q(K)$ is the two-column sequence above. The shift $s\colon F'_{b+2} \to F'_b$ (the identity of $\mathbb{Z}[\langle -I\rangle]$) is a chain map; s^* is cup product with the generator of $H^2(\langle -I\rangle; \mathbb{Z}) = \mathbb{Z}/2$ [standard: [Bro82, Ch. VI.9]] (a unique generator, so no sign), and $(\mathrm{id} \otimes s)^*$ is cup product with $\mathrm{pr}^* u_0$ (the cross product with u_0, [standard: [Bro82, Ch. V.2–3]]). It commutes with $t - 1$, so maps the sequence of degree q to that of degree $q + 2$; on $H^b(K) = H^b(\langle -I\rangle; X)$ it is an isomorphism for $b \ge 1$, and for $b = 0$ when $\varepsilon = -1$ (both sides vanish). The five lemma gives the claim for $q \ge 2$, and for $q \ge 1$ if $\varepsilon = -1$.

(5)(a) Under (1)(a), r^0 on H^0 is the inclusion of invariants; $H^{-1}(\Gamma_\infty) = 0$. So $H_c^0 = 0$ over every S, i.e. $\ker(D^{-1} \otimes F) = H^{-1}(C(r) \otimes F) = 0$ for every field F; an integer matrix of full column rank modulo every prime has unit elementary divisors. (b) With $6 \in F^\times$, Lemma 28.5's sequence $0 \to F[\Gamma/\langle S\rangle] \oplus F[\Gamma/\langle U\rangle] \to F[\Gamma/\{\pm I\}] \to \mathrm{St}_F \to 0$ is a projective resolution of length 1 (the stabilisers have orders 4, 6, 2), so by (1)(c) $H_c^q = \operatorname{Ext}^{q-1}(\mathrm{St}_F, M_F) = 0$ for $q \ge 3$ and $H_c^2 = \operatorname{coker}\big(\mu\colon M_F^{\{\pm I\}} \to M_F^{\langle S\rangle} \oplus M_F^{\langle U\rangle}\big)$, $\mu(v) = ((1+S)v, (1+U+U^2)v)$ (pull back along $\gamma\mathfrak{v}_S \mapsto \gamma\mathfrak{e} + \gamma S\mathfrak{e}$, $\gamma\mathfrak{v}_U \mapsto \gamma\mathfrak{e} + \gamma U\mathfrak{e} + \gamma U^2\mathfrak{e}$). For a finite subgroup H the invariants M_F^H and $(M_F^\vee)^H$ are in perfect duality (the averaging idempotent is self-adjoint), and under it the adjoint of $v \mapsto (1+S)v$, $M^{\{\pm I\}} \to M^{\langle S\rangle}$, is $\lambda \mapsto \lambda + S^{-1}\lambda = 2\lambda$, $(M^\vee)^{\langle S\rangle} \to (M^\vee)^{\{\pm I\}}$, that of $1 + U + U^2$ is 3λ. So $\dim \operatorname{coker} \mu = \dim \ker\big((\lambda, \lambda') \mapsto 2\lambda + 3\lambda'\big)$, and $2\lambda = -3\lambda'$ lies in $(M^\vee)^{\langle S\rangle} \cap (M^\vee)^{\langle U\rangle} = (M^\vee)^\Gamma$ (S, U generate Γ), the kernel being isomorphic to it. The two other formulas are the exactness of the rational sequence $0 \to H^0 \to H^0(\Gamma_\infty) \to H_c^1 \to H^1 \to H^1(\Gamma_\infty) \to H_c^2 \to H^2 = 0$ (Proposition 31.6(4)(a): $24H^2(\Gamma; M) = 24H^3(\Gamma; M) = 0$, $24 \in F^\times$), and $\dim H^0(\Gamma_\infty; M_F) = \dim H^1(\Gamma_\infty; M_F)$ (kernel and cokernel of $t - 1$ on M_F^{-I}, (4) with $2 \in F^\times$). For $M_N(\chi)$: $(M_{\mathbb{Q}}^\vee)^\Gamma = ((V(\chi)^\vee)_{\mathbb{Q}})^{\Gamma_0(N)}$ (Shapiro in degree 0), and $V(\chi)_{\mathbb{Q}}^\vee \cong \mathrm{Sym}^n \otimes \chi^{-1}$ has invariants iff $n = 0$, $\chi = 1$ (the argument of Proposition 31.7(4)); for M_N, n even, $\dim H^1(\Gamma_\infty; M_{N,\mathbb{Q}}) = c(N)$ by (4). (c) For $q \ge 3$: $H^{q-1}(\Gamma_\infty)$ is killed by 2 ((4)) and $H^q(\Gamma; M)$ by 24 (Proposition 31.6(4)(a)), and H_c^q is an extension of a subgroup of the second by a quotient of the first. At $\ell \ge 5$ both vanish for $q \ge 2$, resp. $q - 1 \ge 2$, and $H^2(\Gamma; M) \otimes \mathbb{Z}_{(\ell)} = 0$; at $\ell = 3$ the boundary terms vanish for $q - 1 \ge 2$ ((4)), so pr_{c*} is an isomorphism for $q \ge 3$. (d) (v1.23.0-pre.27, referee m1 and n1: no extension of [Bro82, Ch. V.4] to $\operatorname{Ext}(Y, -)$ is used, and the dangling "Lemma 14.22(4)" of pre.26 is gone.) By [standard: [Bro82, Ch. III, Prop. 2.2]] the three terms are $H^\bullet(\Gamma; X)$ for $X = M$, $X = \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}[\mathbb{P}^1], M) = \operatorname{Coind}_{\Gamma_\infty}^\Gamma M$ and $X = \operatorname{Hom}_{\mathbb{Z}}(\mathrm{St}, M)$ (the last in degree $q - 1$), and (79) is, up to the signs of (1)(c), the long exact sequence of the coefficient sequence $0 \to M \to \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}[\mathbb{P}^1], M) \to \operatorname{Hom}_{\mathbb{Z}}(\mathrm{St}, M) \to 0$, which is $\mathbb{Z}$-split (St is $\mathbb{Z}$-free). $e \cup -$ on each is Lemma 14.22(2)'s (any $\mathbb{Z}[\Gamma]$-module); it is

induced by a cochain map natural in the coefficients (cup product with a cocycle ϵ representing e through a diagonal approximation, [standard: [Bro82, Ch. V.1–3]]), so it commutes with the coefficient maps and, the sequence of cochain complexes being exact, with the connecting maps ($d(\epsilon \cup \tilde{x}) = \epsilon \cup d\tilde{x}$: $\deg \epsilon = 2$, $d\epsilon = 0$); a sign of (1)(c) does not affect commutation. The Shapiro isomorphism $H^\bullet(\Gamma; \operatorname{Coind} M) \cong H^\bullet(\Gamma_\infty; M)$ is $\operatorname{res}^\Gamma_{\Gamma_\infty}$ followed by the coefficient map "evaluation at 1" [standard: [Bro82, Ch. III.6]], so it carries $e\cup -$ to $\operatorname{res}_{\Gamma_\infty} e\cup -$ (Lemma 14.22(2): compatible with restriction, natural in the coefficients) — the Shapiro–cup compatibility for Coind, here stated once; it holds at infinite index. The isomorphisms: Lemma 14.22(2) for $X = M$ and $X = \operatorname{Coind} M$ in degrees $q \geq 2$ (for the latter, equivalently item (4) above), and for $X = \operatorname{Hom}_\mathbb{Z}(\mathrm{St}, M)$ in degrees $q - 1 \geq 2$, i.e. on H^q_c, $q \geq 3$, directly.

(5)(e) Over $\mathbb{Z}_{(\ell)}$, $\ell \geq 5$, 6 is a unit, and the cone over $\mathbb{Z}_{(\ell)}$ is the cone over $\mathbb{Z}$ tensored with the flat $\mathbb{Z}_{(\ell)}$; so by (1)(c) and Lemma 28.5's resolution, as in (b), $H^2_c \otimes \mathbb{Z}_{(\ell)} = \operatorname{coker} \mu$ over $\mathbb{Z}_{(\ell)}$, $\mu(v) = ((1+S)v, (1+U+U^2)v)$, $M^{\{\pm I\}} \to M^{\langle S\rangle} \oplus M^{\langle U\rangle}$. For a finite $H \leq \Gamma$, $|H| \in \{2, 4, 6\}$ is a unit, and the norm $N_H \colon M_H \to M^H$ is an isomorphism (its composites with the canonical $M^H \to M_H$ are multiplication by $|H|$). Since $S^2 = U^3 = -I$, $N_{\langle S\rangle} = (1+S)N_{\{\pm I\}}$ and $N_{\langle U\rangle} = (1+U+U^2)N_{\{\pm I\}}$, so $\mu \circ N_{\{\pm I\}} = (N_{\langle S\rangle} \oplus N_{\langle U\rangle}) \circ \nu$ with $\nu \colon M_{\{\pm I\}} \to M_{\langle S\rangle} \oplus M_{\langle U\rangle}$, $[v] \mapsto ([v], [v])$, and $\operatorname{coker} \mu \cong \operatorname{coker} \nu$. And $\operatorname{coker} \nu \cong M_\Gamma$ through $([a], [b]) \mapsto [a - b]$: it is onto, and if $a - b = (S-1)x + (U-1)y$ (which is all of $\sum_g (g-1)M$, as S, U generate Γ: $gh - 1 = g(h-1) + (g-1)$) then $([a], [b]) = \nu[v]$ with $v = a - (S-1)x = b + (U-1)y$. For M_N: $(\operatorname{Ind}^\Gamma_A V)_\Gamma = \mathbb{Z} \otimes_{\mathbb{Z}[A]} V = V_A$. Modulo ℓ, $V_A \otimes \mathbb{F}_\ell = (V/\ell)_A$ is a quotient of $(V/\ell)_B$, B the upper triangular subgroup of $\mathrm{SL}_2(\mathbb{F}_\ell)$, which lies in the image of A for every N ($\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\ell)$ is onto, and the upper triangular matrices modulo $N\ell$ lift into $\Gamma_0(N)$). The unipotent coinvariants are one line for $n < \ell$ (by (4)'s computation, Δ_1 is triangular with diagonal $1, \ldots, n$, units modulo ℓ), spanned by the class of u_n, on which $\operatorname{diag}(a, a^{-1})$ acts by $a^{\pm n}$; so $(V/\ell)_B = 0$ unless $(\ell - 1) \mid n$, in particular for $0 < n < \ell - 1$. By Nakayama's lemma for the finitely generated V_A, $V_A \otimes \mathbb{Z}_{(\ell)} = 0$. The $\otimes\mathbb{Q}$ shadow: coinvariants and invariants of the dual are dual over $\mathbb{Q}$. (6) The cone is a bounded below complex of finitely generated free $\mathbb{Z}_{(\ell)}$-modules, so the cited statements apply as in Proposition 31.10(2)(d). The sources: (a) by (4); (b) by (5)(a); (c) by (5)(b) and the Betti recursion, and its prime by (5)(e) with the criterion that follows; a reduction at ℓ' has the rational rank iff ℓ' divides no elementary divisor, i.e. iff $\operatorname{tors} H^{q+1}(C(r)) = \operatorname{tors} H^{q+2}_c$ has no ℓ'-torsion (Proposition 23.17(4)); (d) by (5)(c). The exponents: the elementary divisors of D^q are the exponents of $\operatorname{tors} H^{q+2}_c$, those of $D^{\partial,q}$ of $\operatorname{tors} H^{q+1}(\Gamma_\infty)$; then (4), (5)(a), (5)(c). The truncation lemma of 14.24 on bulk and boundary, and $C(r_\mathsf{P})^q = \mathsf{P}(M)^{q+1} \oplus \mathsf{P}^\partial(M)^q$. □

Proposition 31.21 (The cone over $\mathbb{Z}$ with the strict pair datum, and T_p, $p \nmid N$, on the pair; D2's maps)**.** *[proved here] [standard: [Bro82, Ch. III.2, III.8–10], [Wei94, Thm. 2.2.6]; VERIFY] (v1.23.0-pre.26; milestone D4, clause C-D7 of the plan; refereed ACCEPT-with-fixes, its fix M3 at v1.23.0-pre.27; (1) and (5) amended at v1.23.0-pre.28 for the owner's scope addition of 2026-10-25: the pair lift for every $p \nmid N$ on demand, certified at run time; pre.28 refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D7p_referee.md`, its fixes M1, m1–m3, N1 at v1.23.0-pre.29.) Assume (H4) and (B1)–(B3). Let $N \geq 1$, $n \geq 0$, $p \nmid N$ prime, $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$, $\Gamma''_g = \Gamma \cap g\Gamma g^{-1} = \Gamma_0(p)$, $A = \Gamma_0(N)$, M_N as in §31.1.*

1. ***(The pair lift: for every p, produced; built on demand and certified at run time; v1.23.0-pre.28.)*** *Let τ_∂ be the boundary datum of Lemma 14.10(1) for g (integral, coefficients in $\{0, \pm 1\}$, by a terminating procedure, every p), and τ^s the strict datum of Proposition 15.13 for it (terminating, every p; $\tau^s j = j\tau_\partial$). A **pair lift** is a $\tau_\mathcal{I}$ as in Proposition 23.13(1): (T1)–(T3) over $\mathbb{Z}$, $\mathrm{cmp}\tau_\mathcal{I} = \tau^s \mathrm{cmp}$, $\tau_\mathcal{I}(\mathcal{I}^\partial) \subseteq \mathcal{I}^\partial$; $\tau^\partial_\mathcal{I}$ its restriction. For **every** p one is **produced**, in every degree, by Remark 23.14(1)'s procedure (Proposition 31.13(2)'s, with the boundary step first): deterministic, no system solved, terminating; it is certified by Remark 23.14(2)'s checks, the boundary items included. Two pair lifts — for any admissible τ_∂, τ^s — are homotopic as pairs through (T1)-maps*

over $\mathbb{Z}$. ***Built and certified*** *(v1.23.0-pre.28, the owner's scope addition of 2026-10-25, replacing pre.27's "$p \in \{2,3,5\}$; for every other p ... not built"): for* ***every prime*** p *requested, the lift is built* on demand *by that procedure in degrees* ≤ 6*, so the pair is served for* $q_{\max} \le 5$ *(*$q_{\max}+1 \le 6$*; obligation (g) of the code map asks* $q_{\max} \ge 4$*); a request for cochains above degree* 6 *(*$q_{\top} > 6$*, or* $q_{\max} > 5$ *for the pair) is* ***refused****, and higher degrees come from (7),* ***never from extending the lift*** *(v1.23.0-pre.29, referee m2). It is certified by* ***all*** *of Remark 23.14(2)'s checks — (T1) on the Schreier generators of* $\Gamma_0(p)$*, (T2), (T3), the strict identity* $\mathrm{cmp}\tau_{\mathcal{I}} = \tau^s \mathrm{cmp}$*,* $\tau_{\mathcal{I}}(\mathcal{I}^{\partial}) \subseteq \mathcal{I}^{\partial}$ *and* $\mathrm{cmp}\tau^{\partial}_{\mathcal{I}} = \tau_{\partial}\mathrm{cmp}$ *— before its first use, and kept for later requests. A lift is* ***refused****, never truncated or used (Theorem 18.7(3)), when (v1.23.0-pre.29, referee M1; pre.28 named only a failed certificate and an int64 product) its* ***predicted RSS*** *exceeds the ceiling (4 GiB; rule 19, decided before anything is built), when its construction* ***does not complete within the code's checked bounds*** *—* the cap*: an exact product outside* 64*-bit signed integers or a* τ*-table chain coefficient outside* 32*-bit signed integers (the exact-integer layer's overflow exception),* p *(or* p^r*) outside the coset tables'* 32*-bit unsigned domain, or a failed mandatory self-certification of an input table (Obligation 20.11) — or when its certificate fails.* ***Any other exception*** *during the construction (an internal check, an internal error) is a* ***failure of the code****, not a refusal: it propagates, and nothing is cached.* ***The certificate suffices on its own****, whatever* p*: the operator of (2) and the proof of (3) read of the lift only (T1)–(T3) over* $\mathbb{Z}$ *and* $\tau_{\mathcal{I}}(\mathcal{I}^{\partial}) \subseteq \mathcal{I}^{\partial}$*, each re-derived by the certificate from the stored values, and these* ***force*** *the component map of* $\tau^{\partial}_{\mathcal{I}}$ *on* $H_0(\mathcal{I}^{\partial}) = \mathbb{Z}[\mathbb{P}^1]$ *to be* $c \mapsto g^{-1}c$ *(the proof of (1), without the strict identities; v1.23.0-pre.29, referee m1). The strict identities with* τ^s*,* τ_{∂} *tie the lift to its inputs, so that an input built wrongly at a new* p *is refused, and are what the cellular comparisons of (4) read. Since the datum is a level-*1 *object, built once per* p*, independently of* N *and of the transversal (Proposition 31.13(3)(c)), that is all level* N *needs at* $p \nmid N$*.* ***No bound on its sizes is asserted*** *(Remark 23.22(2)); what was measured (evidence, bounding nothing): largest coefficient* ≤ 3 *and largest matrix entry* p *for* $p \le 127$*, certification time* $0.12\,s$ *at* $p = 13$*,* $26\,s$ *at* $p = 127$*, growing* $\approx p^{2.4}$ *(`docs/implementation/slot24/DESCENT4_report.md` §4). These sizes are a property of the* shipped model*, not of the procedure (v1.23.0-pre.29, referee N1): on another free model of the shape of Definition 23.4 the same procedure gives largest coefficients* $1, 1, 3, 4, 9, 13, 27, 49, 81, 157, 243, 481$ *in degrees* 0–11 *at* $p = 7$*, growing* $\approx \times 3$ *per two degrees (`docs/implementation/slot24/C-D7p_referee.md`, N1), and the shipped model's stationary term counts (Remark 23.14(5)) are model-specific; hence (7), not a longer lift, is the route above degree* 6*. Which* p *a run has certified is a run-time fact, not a statement of this document.*

2. ***(The level-*N *operator on the pair.)*** *For a pair lift,* $\widetilde{T}^{(N)}_{\mathcal{I},p}$ *is Proposition 31.13(3)'s operator on* $C^{\bullet}_{\mathcal{I}}(M_{N,R})$*, and* $\widetilde{T}^{(N)}_{\mathcal{I}^{\partial},p}$ *the same formula (68) with* $\tau^{\partial}_{\mathcal{I}}$ *and* x *running over* $\mathcal{I}^{\partial}$*, on* $C^{\bullet}_{\mathcal{I}^{\partial}}(M_{N,R})$*. Over every* R *both are integral cochain maps,* $r \circ \widetilde{T}^{(N)}_{\mathcal{I},p} = \widetilde{T}^{(N)}_{\mathcal{I}^{\partial},p} \circ r$ ***on the nose****, and*
$$\Phi^{(N)}_p := \big(\widetilde{T}^{(N)}_{\mathcal{I},p}[1],\ \widetilde{T}^{(N)}_{\mathcal{I}^{\partial},p}\big)$$
is a cochain endomorphism of $C(r_{\mathcal{I}})(M_{N,R})$ *with* $h = 0$*, commuting with* inc_c *and* pr_c *and preserving* $(\ker r)[1]$*. At* $N = 1$ *it is the cone operator of the Hecke datum of the pair* $(\tau_{\mathcal{I}}, \tau^{\partial}_{\mathcal{I}}, 0)$ *(Proposition 23.13(1)).*

3. ***(What it induces; independence; the long exact sequence.)*** *On* $H^q(A; V_R)$*,* pr_c*-compatibly,* $\Phi^{(N)}_p$ *induces* T_p *—* ***the operator of §31.3****: the bulk component of a pair lift is a bulk lift in the sense of Proposition 31.13(2) (with* $\tau_X = \tau^s$*), and the induced operator does not depend on the lift (Proposition 31.13(3)(c)). On* $H^q_c(A; V_R) = \mathrm{Ext}^{q-1}_{\mathbb{Z}[A]}(\mathrm{St}, V_R) = H^{q-1}(A; \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R))$ *it induces the double-coset operator* $[AgA] = \mathrm{cor}^{A}_{A''} \circ (\iota_g, f)^* \circ \mathrm{res}^{A}_{A'}$ *of Proposition 31.7(5) for the compatible pair* $(\mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R), f)$*,* $f(\phi) := \rho_n(g) \circ$

$\phi \circ g^{-1}$ (g^{-1} *acting on* $\mathrm{St} \subseteq \mathbb{Z}[\mathbb{P}^1]$ *by Möbius transformations); on* $H^q(\Gamma_\infty; M_N) = \mathrm{Ext}^q_{\mathbb{Z}[A]}(\mathbb{Z}[\mathbb{P}^1], V_R)$ *the same with* $\mathbb{Z}[\mathbb{P}^1]$ *for* St. *The induced operators are independent of the pair lift,* τ_∂, τ^s, *the* s_j *and the transversal,* ***over*** $\mathbb{Z}$***, for every*** ℓ*; so* (79) *for* A *is* T_p*-equivariant over* $\mathbb{Z}$ *and over every* $\mathbb{Z}_{(\ell)}$, $\ell \in \{2, 3\}$ ***included****, in every degree, and* $T_p T_{p'} = T_{p'} T_p$ *on its three terms.*

4. ***(The pieces.)*** *For an* O_ℓ*-algebra* S *and comparisons of pairs* $\Phi_\ell \colon (\mathcal{I}, \mathcal{I}^\partial) \otimes S \to (\mathcal{I}^{(\ell)}, \mathcal{I}^{\partial(\ell)}) \otimes S$, Ψ_ℓ *back (Lemma 31.19; Proposition 31.2(4)), the transport* $(\Psi_\ell^*[1], \Psi_\ell^{\partial *}) \circ \Phi_p^{(N)} \circ (\Phi_\ell^*[1], \Phi_\ell^{\partial *})$ *is a cochain endomorphism of the* ℓ*-local cone inducing the same operator, independently of the comparisons. On the* $\mathbb{Z}[1/6]$*-piece: at* $N = 1$ *the cone operator of Definition 15.9 with the shipped canonical* $(\tau, \tau_\partial, \eta)$ *and* $h = \widetilde{\eta}$ *(where built, §22.1) induces the same operator on* $H_c^\bullet \otimes \mathbb{Z}[1/6]$*, for* ***every*** p *(Theorem 15.11(b), (c), Proposition 15.12, Lemma 31.19(3) for* cmp*); at level* N *the Bredon pair carries* (68) *with the strict cellular pair* (τ^s, τ_∂)*, and* α_c *of Proposition 31.20(2) intertwines it with* $\Phi_p^{(N)}$ ***on the nose*** *(Proposition 31.13(4)'s computation on bulk and boundary).* T_p *on the glued sequence over* $\mathbb{Z}$ *is determined by these local components (Proposition 23.8's morphism clause on the cone).*

5. ***(What is certified at which*** p***.)*** *The groups and maps of* (79)*: no* p *enters.* T_p *on the bulk: every* $p \nmid N$ *(§31.3).* T_p *on the boundary and the cone (v1.23.0-pre.27, referee M3: weakened from pre.26's "at* $\ell \geq 5$*, every* $p \nmid N$*", so that statement and code map agree; v1.23.0-pre.28, the owner's scope addition: the pair lift is built on demand, (1)):* ***certified, at every*** ℓ ***and every*** N***, for every prime*** $p \nmid N$ ***for which a pair lift has been built and its certificate passed (at run time)*** *((1), (2) over every* R*), with the strict identity* $r \circ \widetilde{T}^{(N)}_{\mathcal{I},p} = \widetilde{T}^{(N)}_{\mathcal{I}^\partial,p} \circ r$ *checked entrywise in every degree used (a theorem by (2), checked as a consistency check of the code; a failure refuses);* ***at*** $\ell \geq 5$ ***and level*** 1***, also at every*** p ***at which the shipped cone operator of Definition 15.9 is built*** *(the* $\mathbb{Z}[1/6]$*-piece of (4), §22.1).* ***Refused exactly when*** *(v1.23.0-pre.29, referee M1; each refusal names its cause, by one of the code's eleven refusal codes, listed as the code enumerates them): (a)* p *is not prime; (b)* $p \mid N$ *(*U_p *deferred; outside this proposition); (c) the bulk-only lift is requested for the cone (below); (d) cochains above degree* 6 *are requested (*$q_{\mathrm{top}} > 6$*; for the pair,* $q_{\max} > 5$*; (1)); (e) the pair lift's predicted RSS exceeds the ceiling, decided before building ((1)); (f)–(h) the cap of (1), during the lift's construction* or the operator's assembly *— (f) an exact product outside* 64*-bit or a* τ*-table coefficient outside* 32*-bit signed integers (and, sharing that exception type in the code, an internal guard of the exact integer geometry: the code map's (f)), (g)* p *(*p^r*) outside the coset tables'* 32*-bit unsigned domain, (h) a failed self-certification of an input table; (i) the pair lift's certificate fails; (j) on the operator's layout a term of* $\tau_{\mathcal{I}}(\mathcal{I}^\partial)$ *lies off* $\mathcal{I}^\partial$*; (k) the strict identity* $r \circ \widetilde{T}^{(N)}_{\mathcal{I},p} = \widetilde{T}^{(N)}_{\mathcal{I}^\partial,p} \circ r$ *fails entrywise,* $\widetilde{T}^{(N)}_{\mathcal{I}^\partial,p}$ *assembled separately from* $\tau^\partial_{\mathcal{I}}$*. (i)–(k) cannot occur for a correct implementation ((j) is implied by (i)'s item* $\tau_{\mathcal{I}}(\mathcal{I}^\partial) \subseteq \mathcal{I}^\partial$*, and (k) is a theorem by (2)); they are consistency checks of the code.* ***No other condition refuses****: any other exception is a failure of the code, never a refusal, and no refused value is truncated or used.* ***In cases (d)–(k)*** *the operator still* ***exists and is determined*** *((1)–(4), and (7) above degree* 6*; at* $\ell \geq 5$ *it is also computable from the strict cellular pair* (τ^s, τ_∂) *at* p*, both terminating, with* (68) *on the level-*N *Bredon pair, (4)) but is* ***not asserted as computed****: that cellular route is not built (the code map's (f)). The bulk-only lift of Proposition 31.13(2), in particular one from the canonical* τ*, need not preserve* $\mathcal{I}^\partial$ *(Remark 15.15); it is* ***not*** *a pair datum, and is never used on the cone (that would need a pair homotopy on* $\mathcal{I}$*, not constructed here).*

6. ***(D2's maps on the pair.)*** *Let* $u \colon M \to M'$ *be a* $\mathbb{Z}[\Gamma]$*-map of finitely generated* $\mathbb{Z}$*-free modules (*$\alpha_{B/A}$*,* $\omega_{B/A}$*,* R_σ *of Proposition 31.10). On every pair model* $(u_{\mathsf{P}}[1], u_{\mathsf{P}^\partial})$ *is a cochain map of cones commuting with* r*,* inc_c*,* pr_c *and with every comparison of pairs,*

strictly; so res^A_B, cor^A_B *and* c_σ *act on* (79) *over* $\mathbb{Z}$ *as a commuting ladder of long exact sequences, glued from the pieces, in every degree (on the boundary:* res, cor *between the cusp blocks of* A *and of* B*). For* $B = \Gamma_0(MN)$ *with* $p \nmid MN$ *and a common transversal (Proposition 31.13(7)):* $\widetilde{T}^{(MN)}_{\mathcal{I}^\partial,p}\alpha = \alpha\,\widetilde{T}^{(N)}_{\mathcal{I}^\partial,p}$ *as well, so the ladder for* res *is* T_p*-equivariant, over* $\mathbb{Z}$ *and every* $\mathbb{Z}_{(\ell)}$*, at the* p *of (5).* $\mathrm{cor}\circ T_p = T_p \circ \mathrm{cor}$ *on the three terms on cohomology (Proposition 31.10(3)(a)'s argument, which is formal in the coefficient module). For* $p \mid M$ *nothing is asserted (Proposition 31.10(3)(b)); of* U_p*,* $p \mid N$*, nothing is asserted here (its lift on the free model: Proposition 31.13(9), v1.26.0-pre.1; v1.26.0-pre.11, confirm-referee C1).*

7. ***(Every degree, Hecke-equivariantly.)*** *For* $q \geq 3$*,* $e \cup -\colon H^q_c \to H^{q+2}_c$ *commutes with* T_p *(Proposition 31.7(5) for the compatible pair of (3)); so* $q_{\max} = 4$ *determines the Hecke modules* H^q_c *in every degree, and* $q_{\max} = 3$ *those of* $H^q(\Gamma_\infty)$*,* $q \geq 2$*, likewise.*

Proof. (1) Existence and the procedure: Proposition 23.13(1)'s proof (Lemma 23.12, first on $\mathcal{I}^\partial$ with $f = \tau_\partial \circ \mathrm{cmp}$, then on $\mathcal{I}$ relative to it) for $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$, whose inputs τ_∂ (Lemma 14.10(1), stated for $\begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$, every N) and τ^s (Proposition 15.13, every $g \in \Delta$) exist for every p; the procedure is Remark 23.14(1)'s, whose read-off of y_0 and contraction by descent through the columns use nothing of p but the coset table of $\Gamma''_g\backslash\Gamma$, exactly as in Proposition 31.13(2)'s proof. The boundary step is that procedure on $\mathcal{I}^\partial$ (v1.23.0-pre.28, spelled out for general p): $\mathcal{I}^\partial$ is the sub-sum of the columns over $\Sigma^\partial_0 \sqcup \Sigma^\partial_1$ (Definition 31.18(a)), a subcomplex on which cmp is a surjective quasi-isomorphism onto $C_\bullet(\partial X)$; the read-off of y_0 at $t = 0$ lands in a boundary column, and the column homotopy h maps each column to itself, so the descent contracts a cycle of $\ker \mathrm{cmp} \cap \mathcal{I}^\partial$ inside $\mathcal{I}^\partial$ — whence $\tau_{\mathcal{I}}(\mathcal{I}^\partial) \subseteq \mathcal{I}^\partial$ by construction. Termination (finitely many Γ''_g-orbits of basis elements, $p+1$ cosets; finitely many columns in each degree) and integrality (the inputs τ_∂, τ^s are integral by Lemma 14.10(2) and Proposition 15.13, h is $\mathbb{Z}$-linear, (T1) multiplies by elements of Γ) hold with no restriction on p: **no step of the proof was verified only at** $p \in \{2, 3, 5\}$; what was, before v1.23.0-pre.28, is the *implementation* (Remark 23.14), and at a new p the run-time certificate of (1) stands in for it. *The certificate suffices; the component map is forced* (v1.23.0-pre.29, referee m1: in pre.28 only a parenthesis of the statement, and the proof read the component map off the strict identity, i.e. off (B-ε), which the statement says is not needed). Let $\tau_{\mathcal{I}}$ satisfy (T1)–(T3) over $\mathbb{Z}$ and $\tau_{\mathcal{I}}(\mathcal{I}^\partial) \subseteq \mathcal{I}^\partial$, and nothing else. (i) By (T2) and $\tau_{\mathcal{I}}(\mathcal{I}^\partial) \subseteq \mathcal{I}^\partial$, $\tau^\partial_{\mathcal{I}}$ is a chain map on $\mathcal{I}^\partial$; by (T1) it is twisted-equivariant along ι_g; so it induces a map ϕ on $H_0(\mathcal{I}^\partial) = \mathbb{Z}[\mathbb{P}^1]$, twisted along ι_g. (ii) Let c be a cusp, P_c a generator of its stabiliser in Γ modulo $\pm I$, and $k \geq 1$ with $P_c^k \in \Gamma_0(p)$. By (i), $\phi([c])$ is a finite formal sum of cusps fixed by $P := g^{-1}P_c^k g \in \Gamma$; P is parabolic, fixes the single cusp $g^{-1}c$ and has infinite orbits on every other point of $\mathbb{P}^1(\mathbb{Q})$, so a finitely supported P-fixed sum is $a\,[g^{-1}c]$, $a \in \mathbb{Z}$. (iii) Definition 31.18 makes the augmentation of $\mathcal{I}^\partial$ (onto $\mathbb{Z}[\mathbb{P}^1]$, followed by $\mathbb{Z}[\mathbb{P}^1] \to \mathbb{Z}$) the restriction of that of $\mathcal{I}$; as $\tau_{\mathcal{I}}$ extends $\tau^\partial_{\mathcal{I}}$ and preserves the augmentation by (T3), $a = \varepsilon(\phi[c]) = \varepsilon([c]) = 1$. So $\phi(c) = g^{-1}c$ for every lift with these properties; no step uses (S) or (B-ε), nor $p \in \{2, 3, 5\}$. *Second proof,* under the strict identities: $\mathrm{cmp}\tau^\partial_{\mathcal{I}} = \tau_\partial \mathrm{cmp}$ and Lemma 14.9 for τ_∂. Homotopy of pairs: both $\tau^\partial_{\mathcal{I}}$ induce on $H_0(\mathcal{I}^\partial) = \mathbb{Z}[\mathbb{P}^1]$ the component map $c \mapsto g^{-1}c$ (by (i)–(iii)); $\mathcal{I}$ is free over $\mathbb{Z}[\Gamma''_g]$ on a basis restricting to one of $\mathcal{I}^\partial$, and the twisted targets $\mathcal{I}^{(c)}$, $\mathcal{I}^{\partial(c)}$ are Γ''_g-resolutions of $\mathbb{Z}$, $\mathbb{Z}[\mathbb{P}^1]^{(c)}$; the proof of Lemma 31.19(2) over $\mathbb{Z}[\Gamma''_g]$, with (T1) as $\mathbb{Z}[\Gamma''_g]$-linearity into the twisted complexes. (2) Integrality, cochain maps and the $N = 1$ case are Proposition 31.13(3)(a), (b), (d), whose proofs read of the complex only (T1), (T2) and $\partial(s_j x) = s_j \partial x$, true on $\mathcal{I}^\partial$. The strict identity: for $x \in \mathcal{I}^\partial$, $(\widetilde{T}^{(N)}_{\mathcal{I},p}\phi)(x)_i = \sum_t \rho_n(\delta_{i,t})\phi(\tau_{\mathcal{I}}(s_j x))_a$ and $\tau_{\mathcal{I}}(s_j x) = \tau^\partial_{\mathcal{I}}(s_j x) \in \mathcal{I}^\partial$, so it equals $(\widetilde{T}^{(N)}_{\mathcal{I}^\partial,p} r\phi)(x)_i$ — Lemma 15.4(3), (4) blockwise. The cone: Lemma 15.8's computation with $h = 0$. (3) The bulk: Proposition 31.13(3)(c). The cone: $\Phi^{(N)}_p$ preserves $(\ker r)[1]$ and

$c \mapsto (c, 0)$ is a quasi-isomorphism (Proposition 31.20(1)(c)), so the operator on $H_c^\bullet$ is that of the coset transport of τ^{rel}, the map induced by $\tau_{\mathcal{I}}$ on $\mathcal{I}/\mathcal{I}^\partial$, on $\mathrm{Hom}(\mathcal{I}/\mathcal{I}^\partial, M_N)$. $\mathcal{I}/\mathcal{I}^\partial$ is a complex of free $\mathbb{Z}[\Gamma]$-modules, free over every subgroup of finite index, with homology St in degree 1, and τ^{rel} is a chain map along ι_g (by (T1), (T3)) lifting $g^{-1}\colon \mathrm{St} \to \mathrm{St}$ (the component map on $H_0(\mathcal{I}^\partial)$, restricted to the kernel of the augmentation); unique up to homotopy by the fundamental lemma. Proposition 31.13(3)(c)'s proof applies verbatim with St in place of $\mathbb{Z}$ — the inclusion computes res, $\psi \mapsto \rho_n(g)\psi\circ\tau^{\mathrm{rel}}$ computes $(\iota_g, f)^*$ on $\mathrm{Ext}(\mathrm{St}, -)$, the transfer computes cor [standard: [Bro82, Ch. III.9–10]] — and $\mathrm{Ext}_{\mathbb{Z}[A]}(\mathrm{St}, V_R) = H^\bullet(A; \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R))$ [standard: [Bro82, Ch. III, Prop. 2.2]] carries these maps to those of the compatible pair $(\mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R), f)$; Shapiro for the level. The boundary is the same argument on $\mathcal{I}^\partial$ with $\mathbb{Z}[\mathbb{P}^1]$. Independence: two pair lifts differ by a homotopy of pairs k through (T1)-maps ((1)); its coset transport gives the cone homotopy Υ_k of Lemma 31.19(3) (with Lemma 15.4(2)), and the transversal by Proposition 29.34(4) on both summands. Equivariance of (79): $\Phi_p^{(N)}$ commutes with inc_c, pr_c, and the identities hold over every R. Commutativity: the three terms are $H^\bullet(A; X)$ for $X \in \{V_R, \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[\mathbb{P}^1], V_R), \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R)\}$, Hecke modules over the same Hecke ring, and Proposition 31.13(8)'s argument is formal in X. (4) Lemma 31.19(3) and Proposition 31.2(4), (5). At $N = 1$ over $\mathbb{Z}[1/6]$: Theorem 15.11(b), (c) and Proposition 15.12 say the shipped operator does not depend on $(\tau, \tau_\partial, \eta)$ when 6 is inverted, and the strict datum $(\tau^s, \tau_\partial, 0)$ is one admissible choice; Proposition 23.13(3)'s identity $\mathrm{cmp}^*\Theta_g^s = \Theta_{\mathcal{I},g}\mathrm{cmp}^*$, and its boundary analogue from $\mathrm{cmp}\tau_{\mathcal{I}}^\partial = \tau_\partial\mathrm{cmp}$, make α_c intertwine on the nose. At level N: Proposition 31.13(4)'s computation with $\mathrm{cmp}\tau_{\mathcal{I}} = \tau^s\mathrm{cmp}$ and $\mathrm{cmp}\tau_{\mathcal{I}}^\partial = \tau_\partial\mathrm{cmp}$. (5) collects (1)–(4) and Proposition 31.13; the bulk-only lift: Remark 15.15. (6) u_{P} is $\mathrm{Hom}(\mathsf{P}, u_S)$, which commutes with restriction to P^∂ and with precomposition by any comparison; the cone identity is immediate. The square with α: Proposition 31.13(7)'s proof (Proposition 29.34(1)'s reciprocity computation reads nothing of the complex) on $\mathcal{I}^\partial$ with $\tau_{\mathcal{I}}^\partial$. cor: Proposition 31.10(3)(a). (7) Proposition 31.7(5) is stated for any compatible pair (X, f); by (3) and Proposition 31.20(5)(d) the cup product on $H_c^\bullet$ is $\mathrm{res}_A^\Gamma e \cup -$ on $H^{\bullet-1}(A; \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R))$. □

Remark 31.22 (What the plan stated, and what holds). [proved here] (v1.23.0-pre.26.) The plan's D4 and its rows are right in substance; six points are corrected or sharpened. (i) **C-D7 is a clause, not an amendment.** The plan's row "(amend 23.22(4), (5))" undersells it: the pair needs Definition 31.18, Lemma 31.19 and Propositions 31.20, 31.21; the amendment of Remark 23.22(4), (5) is made as well. (ii) **D-δ6, sharpened.** The boundary is unbounded *only* at 2: at 3 and at $\ell \geq 5$ it vanishes above degree 1 (Proposition 31.20(4)); the cone is, in general, unbounded at 2 ((5)(c)), and **can be** unbounded at 3 as well, **exactly when** $H^q(A; V)\otimes\mathbb{Z}_{(3)} \neq 0$ for some $q \geq 2$ ($H_c^q \otimes \mathbb{Z}_{(3)} \cong H^q \otimes \mathbb{Z}_{(3)}$ for $q \geq 3$, (5)(c), and the bulk is 2-periodic for $q \geq 2$): e.g. level 1, $n \in \{0, 4, 6, 10, 12\}$; not for n odd, nor for $n \in \{2, 8\}$ at level 1 (v1.23.0-pre.27, referee m2; pre.26 said "unbounded at 2 and at 3"). Both are periodic from degree 3 with closed-form boundary terms, so the truncation loses nothing at $q_{\max} \geq 4$ ((5)(d)) — the pair needs one degree more than the bulk. "With a certified exponent": every exponent is bounded a priori ((6)), that of $\mathrm{tors}\, H_c^2$ at $\ell \geq 5$ only (by $\mathrm{tors}\, H_0(\Gamma; M)$, (5)(e)), not at $\ell \in \{2, 3\}$; its differential D^0 is certified by the Betti recursion **or** by a prime $\ell' \geq 5$ at which $H_0(\Gamma; M)$ has no torsion — for M_N every $\ell' \geq 5$, $\ell' > n+1$, as in the bulk (v1.23.0-pre.27, referee M2; pre.26 said "not by a prime"). (iii) **"Built for $p \in \{2, 3, 5\}$"** is right for *built*; the pair lift *exists and is produced* for every p (Proposition 31.21(1)), and what is *asserted* at other p is Proposition 31.21(5): existence and determinacy for every p; computed only at the p with a built pair lift, and at $\ell \geq 5$, level 1, where the shipped cone operator is built (v1.23.0-pre.27, referee M3). *Since v1.23.0-pre.28* (the owner's scope addition of 2026-10-25) the pair lift is built on demand for every prime $p \nmid N$ and certified at run time (Proposition 31.21(1)), so T_p on the boundary and the cone is computed, at every ℓ, level 1 and level N, at every such p whose lift passes its certificate within the cap; refused only on the eleven conditions of Proposition 31.21(5) (v1.23.0-pre.29, referee M1; pre.28

listed four). The plan's arm (S), $\tau_{\mathcal{I}}(\mathcal{I}^{\partial}) \subseteq \mathcal{I}^{\partial}$, is Remark 23.14(2)'s existing certificate. (iv) **Arm (Q)**, "the $\mathbb{Z}[1/6]$ part equals the shipped E8 4b pair after $\otimes\mathbb{Z}/\ell^K$", holds at level 1 as an identity of complexes (Proposition 31.20(1)(d)); at level N no shipped pair is the $\mathbb{Z}[1/6]$-piece (D1's O-piece at level N is the sparse cokernel route, which carries no cone), and the Bredon pair with fibre M_N is a new object of D4 (the code map). (v) **Coherence.** "The cone is where coherence is real" (the plan's §2.2) is right for the *Hecke operator* on the cone and wrong for its groups and maps, which glue with no datum (Proposition 31.20(2)); on the free model the strict datum makes the operator choice-free over $\mathbb{Z}$ (Proposition 31.21(3)). (vi) **Level N needs no new geometry**: the boundary is block-diagonal by the cusps of $\Gamma_0(N)$, read from the T-orbits on $\mathbb{P}^1(\mathbb{Z}/N)$, and its groups depend only on n and the widths (Proposition 31.20(3), (4)) — the same statement the plan's §5.5 makes for the tree.

A check outside the build, not a tag (v1.23.0-pre.26): `docs/implementation/slot24/cd7_pair_check.py` (`python3 -I`, pure integers, no computer algebra system), output `cd7_pair_check.out`. It reads no line of `cohomology_app`, imports nothing from the tree and builds none of $\mathcal{I}$, $\mathcal{I}^{\partial}$, $\mathcal{I}^{(\ell)}$: its pair model is the free tree resolution of $\mathbb{Z}/4 *_{\mathbb{Z}/2} \mathbb{Z}/6$ (ranks $2, 3, 3, \dots$) with the minimal resolution of Γ_∞ (ranks $1, 2, 2, \dots$) and a comparison φ *computed* by solving $\partial y = \varphi(\partial x)$ over $\mathbb{Z}$ on balls of Γ (Hermite forms) and certified ($\partial\varphi = \varphi\partial$ through degree 8). A second complex, Lemma 28.5's sequence resolved column by column, computes $\mathrm{Ext}^{\bullet}(\mathrm{St}, M)$; a third, the tree Bredon complex with the axis of T, is the $\mathbb{Z}[1/6]$-piece. On 24 cells (level 1, $n \in \{0, 1, 2, 3, 4, 6, 8, 10, 12\}$; $\Gamma_0(N)$, $N \in \{2, \dots, 9, 11, 13\}$, $n = 0$; $(N, n) \in \{(2,1), (2,2), (2,3), (2,6), (3,1), (3,2), (4,2), (5,2), (7,2)\}$), degrees ≤ 6, it checks: the boundary groups against the closed form read from the cusp widths alone, $|F_{n,w}| = n!\, w^n$ and $2H^q(\Gamma_\infty) = 0$ (arm B); $H^q_c = \mathrm{Ext}^{q-1}(\mathrm{St}, M)$ on the two independent complexes (X); exactness of (79) over $\mathbb{Z}$ at every spot as equalities of lattices (E); $H^0_c = 0$, H^1_c torsion-free, $\operatorname{rank} H^1_c = b_1$, $\operatorname{rank} H^2_c = b_0$, $\operatorname{rank}\mathrm{res}^1 = b_1(\partial) - b_0$, $48H^q_c = 0$, the periodicities and the 3- and $\ell \geq 5$-local statements of (5)(c) (C); the certified local reads at 2 and 3 with the sources of (6) and the exponent caps (L); the Bredon pair against the free one at every $\ell \geq 5$ dividing a term, and the 3-local boundary as the Bredon one (Q); model independence under other comparisons $T\varphi$, $(-I)\varphi$ (I). Planted faults, each detected: the circle boundary at $\ell = 2$; the Bredon bulk at $\ell = 3$ for H^q_c, $q \geq 3$; a perturbed entry of r; the truncated top degree reported; K too small (refused); a prime dividing $\operatorname{tors} H^2_c$ as rank source for D^0; the closed form without the widths; the bulk-only rank for the cone. Arm R (added at v1.23.0-pre.27, referee M2): $\operatorname{tors} H^2_c \otimes \mathbb{Z}_{(\ell)} = H_0(\Gamma; M) \otimes \mathbb{Z}_{(\ell)}$ for every $\ell \geq 5$ with H_0 the cokernel of $[S - 1 \mid T - 1]$, the free ranks equal, every such $\ell \leq n + 1$, and D^0 modulo $\ell' \in \{5, \dots, 31\}$ of rational rank iff $\ell' \nmid \operatorname{tors} H_0$ (Proposition 31.20(5)(e), (6)(c)). `CD7_CHECK checks=2528 ... R=308 ... failing=0 verdict=OK` at pre.27 (2220 at pre.26, arm R new). *Corrected at pre.27* (referee M1): pre.26 said here that on these cells H^2_c has no torsion at $\ell \geq 5$; the script's own output refutes it — $H^2_c = \mathbb{Z}/4 \oplus \mathbb{Z}/60$ at $(N, n) = (1, 8)$ and $\mathbb{Z}/2 \oplus \mathbb{Z}/12 \oplus \mathbb{Z}/1260$ at $(1, 12)$. By Proposition 31.20(5)(e) the $\ell \geq 5$ torsion of H^2_c is that of $H_0(A; V)$, zero for $\ell > n + 1$: on these cells $\mathbb{Z}/5$ at $(1, 8)$, $\mathbb{Z}/5 \oplus \mathbb{Z}/7$ at $(1, 12)$, none elsewhere. It is not an Eisenstein-congruence phenomenon; 691 is absent at $n = 10$ because $691 > n + 1$. The script checks no Hecke operator; (S) is the code's certificate. The referee's independent check, `docs/implementation/slot24/cd7referee/cd7referee_check.py` (a Γ-CW pair, the mapping cylinder of the horocycle lines into the Bass–Serre tree; 32 cells, Hecke on H^1_c), rerun at pre.27: `CD7R_CHECK checks=2457 ... failing=0 verdict=OK`.

Code map (v1.23.0-pre.26, amended at v1.23.0-pre.27 in (c) and (f); **stated before D4's code**, contract first, rule 21; (f) restated and the build recorded at v1.23.0-pre.28; (f), (g) and the validation's controls updated at v1.23.0-pre.29). **Built** (v1.23.0-pre.28) on branch `s24-d4impl` (`d82c7761`); the D4 code audit's fixes (`docs/implementation/slot24/D4_code_audit.md`, `46fc0c73`) and the code side of the pre.28 referee's report (`C-D7p_referee.md`, `6daf1e03`) applied there, and **merged** into `hecke-extensions` at `3991a2f1` (v1.23.0-pre.29):

`Integral/Descent/DescentPair.{h,cpp}` (the pieces — `Iso2`, `Iso3`, the shipped E8 pair at level 1, the Bredon pair at level N, the free `IsoFree` arm —, the cone, its certification, the closed-form boundary `BoundaryClosedForm` from `CuspWidths`, the long exact sequence by D2's `ExactAt`), `Integral/Descent/DescentPairHecke.cpp` (`BuildConeHecke`: T_p on the boundary and the cone from the pair lift, built on demand, certified and cached per p; the refusal codes `PairRefusal`, printed `REFUSED [<code>]`; `PredictPairLiftRss`), `Integral/Model/StrictLift.{h,cpp}` (the validation hook `StrictLift::with_homotopy`), sources in `cmake/slot24/DESCENT4.cmake`, registered in `Registry/Slot24/DESCENT4.cpp`: `IntegralDescentPairValidation` (`standard`) and `IntegralDescentPairSlowValidation` (`slow`); report `docs/implementation/slot24/DESCENT4_report.md`. **Obligations of the code** (plan D4, `IntegralDescentPairValidation`): (a) the pair on the code's $\mathcal{I}$: $\mathcal{I}^\partial(M_N)$ as the block selection of the boundary summands (no lift: r is a restriction), the cone of (38) over $\mathbb{Z}/\ell^K$ and over $\mathbb{F}_{\ell'}$, built to degree $q_{\max}+1 \geq 5$; on $\mathcal{I}^{(\ell)}$ the boundary sub-sum of the *same* O_ℓ-model, at $\ell = 3$ checked to be the circle model (Proposition 31.20(1)(d)), at $\ell = 2$ free; (b) the $\mathbb{Z}[1/6]$-piece: level 1, the shipped E8 pair (arm (Q)); level N, the Bredon pair with fibre M_N (dense $C^\bullet_{\mathrm{eq}}(M_N)$ in degrees ≤ 2 and the circle model), a new object — the sparse cokernel route carries no cone; (c) certification by Proposition 31.20(6): D^{-1} by $\operatorname{rank}\mathsf{P}(M)^0$; D^0 by the Betti recursion with $\dim H^1_c = b_1$ from a certified source, **or** by the same cone modulo a prime $\ell' \geq 5$, $\ell' > n+1$ (for M_N; for $M_N(\chi)$, $\chi \neq 1$, only at an $\ell' \geq 5$ checked not to divide the computed $\operatorname{tors} H_0(\Gamma; M)$; never at $\ell' \in \{2,3\}$; v1.23.0-pre.27, referee M2); D^q, $q \geq 1$, by the same cone modulo $\ell' \geq 5$; the boundary by its closed form; the a-priori exponents as consistency checks, at $\ell \geq 5$ on D^0 the ℓ-part of $\operatorname{tors} H_0(\Gamma; M)$; refusal at the cap; (d) arm (B): the boundary against the closed form from the cusp widths on $\mathbb{P}^1(\mathbb{Z}/N)$, including $\left|\operatorname{tors} H^1(\Gamma_\infty; M_N)\right| = \prod_c n!\, w_c^n$; (e) arm (E): exactness of (79) at every reported spot, by Proposition 31.10(2)(g)'s precision rule, on each piece; (f) Hecke on the boundary and the cone (v1.23.0-pre.28, the owner's scope addition of 2026-10-25, replacing pre.27's "… for $p \in \{2,3,5\}$ …; not a D4 obligation: the level-N boundary and cone operator at $p \notin \{2,3,5\}$ … is refused"): for **every prime** $p \nmid N$, level 1 and level N, the pair lift of `Integral/Model/StrictLift` is built on the first request (boundary step first, degrees ≤ 6), certified by Remark 23.14(2)'s checks with the boundary items before first use, and cached per p; $\widetilde{T}^{(N)}_{\mathcal{I},p}$ and $\widetilde{T}^{(N)}_{\mathcal{I}^\partial,p}$ by (68) with $\tau_{\mathcal{I}}$, $\tau^\partial_{\mathcal{I}}$; the strict identity $r\widetilde{T}_{\mathcal{I}} = \widetilde{T}_{\mathcal{I}^\partial} r$ checked entrywise in every degree used, on every call, $\widetilde{T}_{\mathcal{I}^\partial}$ assembled separately on $\mathcal{I}^\partial$'s own layout from $\tau^\partial_{\mathcal{I}}$; **refused only** on the eleven conditions of Proposition 31.21(5), each with its `PairRefusal` code (v1.23.0-pre.29, referee M1, replacing pre.28's four): `NotPrime`, `PDividesN`, `BulkOnly` (never the bulk-only lift on the cone), `DegreeRange` ($q_{\mathrm{top}} > 6$ in `BuildConeHecke`; $q_{\max} > 5$ in `AssembleDescentPair`, where $q_{\max} < 4$ is a programming error and throws), `RssCeiling` (`PredictPairLiftRss`$(p) > 4\,$GiB, before building), `IntegerCap`, `DomainCap`, `InputTableCertificate` (the cap: `ClassifyPairException` maps `hecke::HeckeOverflow` to `IntegerCap`, `std::overflow_error` to `DomainCap`, a `HeckeTauTable` "certification FAILED" to `InputTableCertificate`, while the lift is built *and* while the operator is assembled, referee m3), `LiftCertificate`, `IsobClosure`, `StrictIdentity`; every other exception (an `ENGINE_CHECK`, `solve_regime() ≠ "Z"`, an internal error) **propagates** as a failure and leaves the per-p slot empty. *Known width of `IntegerCap`* (v1.23.0-pre.29, **resolved in the code at v1.23.0-pre.30**, commit `bc128cb7`, merged at `5eb7e9de`): at pre.29 `hecke::HeckeOverflow` was also thrown by internal guards of the exact integer geometry on the lift's path, which were therefore refused as `IntegerCap`. The new type `hecke::HeckeInternalError` (`Hecke/ExactInt.h`, a subclass of `HeckeOverflow`, so every existing handler is unchanged) is now thrown at the non-cap sites (`CuspKey`: $(0:0)$, $\det \neq 1$, singular, `reduce_to_infinity`'s termination, primitivity and internal checks; `word_in_T_S`'s determinant and internal error; `evaluate_word`; `boundary_line_nearest`'s denominator; `QuadPoint`'s domain checks; the τ-table reduction's termination; the integer solvers' termination and self-verification), and

`ClassifyPairException` returns "no refusal" for it: a genuine internal error **propagates** as a failure; **only a real overflow** (`HeckeOverflow` proper) is refused as `IntegerCap`. The classification check of `IntegralDescentPairValidation` covers the new type. The cellular route of Proposition 31.21(4), (5) at $\ell \geq 5$ is not built (not needed: the pair lift serves every ℓ); at level 1, $\ell \geq 5$, the shipped cone operator is read only where built (not read by D4); (g) $4 \leq q_{\max} \leq 5$ for the pair (the lift's degree 6; v1.23.0-pre.29, referee m2); (h) DESCENT4's report prices the pair on $\mathcal{I}^{\partial}$, $\mathcal{I}^{(2)}$ (the third input to gate G-T); (i) *the negative controls of (f)* in `IntegralDescentPairValidation` (v1.23.0-pre.29, referee m4 and the D4 code audit): a perturbed lift value (breaks (T2)) refused as `LiftCertificate`; **homotopy faults** $\tau' = \tau_{\mathcal{I}} + dK + Kd$, $K(s_1 e_v) = e_e$ extended by (T1) (`StrictLift::with_homotopy`; v the first $t = 0$ boundary 0-cell, e a $t = 0$ 1-cell in $\mathcal{I}^{\partial}$ or in the bulk), which keep (T1)–(T3) — asserted — and are refused as `LiftCertificate` by (S) and the boundary items alone at $p = 2, 7$ (flags printed: into $\mathcal{I}^{\partial}$, (S) and (B-ε) fail; into the bulk, (S), (B)-preserved and (B-ε) fail — the referee's Arm F); an off-$\mathcal{I}^{\partial}$ term planted with the closure test ignored, refused by the strict identity alone (`StrictIdentity`) at $(1, 10)$, $p = 2, 7$, and $(11, 0)$, $p = 13$; $q_{\text{top}} = 7$ and $q_{\max} = 6$ (`DegreeRange`); $p = 100003$ (`RssCeiling`, before building); the bulk-only lift (`BulkOnly`); and the exception classification on six synthetic exceptions (three caps refuse; a logic, an internal and a range error propagate).

Formalization note. Lemma 31.19 is the comparison theorem relative to a subcomplex; Proposition 31.20(4) and (5)(a), (b) are finite algebra and cheap; (1)(c) and Proposition 31.21(3) need Ext with Shapiro and the double-coset formalism. No \LEAN marker.

31.5 The tree cusp pair: the boundary pair and the cone over $\mathbb{Z}$ on the tree models

(v1.23.0-pre.33; milestone D5c of `docs/implementation/slot24/DESCENT_plan.md` v0.4, clause C-D8 of its §2.7 and its §5; the owner's ruling at gate G-T.) §31.4 computes the Borel–Serre pair of $\Gamma_0(N)$ over $\mathbb{Z}$ on $(\mathcal{I}, \mathcal{I}^{\partial})$ and its ℓ-local forms; its 2-adic piece meets a memory wall at $D := \operatorname{rank}_{\mathbb{Z}} M_N \approx 1060$–$1240$ (`DESCENT4_report.md` §4), the dense bulk $\mathcal{I}^{(2)}$ inside the cone. §31.2 moved the bulk to the tree but carries no boundary: the Γ-translates of the axis of T cover $\mathcal{T}$, so the tree has no subcomplex indexed by the cusps (the plan's §5.1). This subsection fixes, over every O_ℓ-algebra: (i) a *cusp model* $\mathsf{B}_{\mathcal{T}}^{(\ell)}$, a projective resolution of $R[\mathbb{P}^1]$ built from two copies of the tree's edge column, and an explicit chain map $\mathsf{a}^{(\ell)} \colon \mathsf{B}_{\mathcal{T}}^{(\ell)} \to \mathcal{I}_{\mathcal{T}}^{(\ell)}$ over $R[\mathbb{P}^1] \xrightarrow{\varepsilon} R$ (the *axis map*), whose cochain cone $C(\mathsf{a}^{(\ell)*})$ is the computed object (Definition 31.23); (ii) the mapping cylinder, which makes it a pair model in the sense of Definition 31.18, and the strict deformation of that pair's cone onto $C(\mathsf{a}^{(\ell)*})$ (Lemma 31.24); (iii) what it computes, in every degree, with the signs of Proposition 31.20(1)(c), its cochains in closed form, the certified exponent and the precision rule, explicit comparisons with $(\mathcal{I}, \mathcal{I}^{\partial})$ made effective as in Proposition 31.12(4), and T_p by transport, with the off-diagonal term it acquires on $C(\mathsf{a}^{(\ell)*})$ (Proposition 31.25; v1.23.0-pre.34, referee m1); (iv) where the plan's §5 is corrected (Remark 31.26). *What is used.* Definitions 31.18, 31.11, Lemma 31.19, Propositions 31.20, 31.21, 31.12, 31.10(2)(g), 31.15(3), Lemma 7.3, Definition 23.4, (B1)–(B3), (38), Definition 15.9 (the shape of a cone datum) and Proposition 15.16 (why the five lemma does not suffice). *Notation.* As in §31.2 and §31.4; R is an O_ℓ-algebra, ℓ a prime or $\ell = 0$ for the free models over every ring ($\mathsf{O}_0 := \mathbb{Z}$); $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $S^{-1} = -S$, $U = ST$, so $S^{-1}U = T$; a map written r_a is right multiplication as in §31.2, and on cochains $\operatorname{Hom}(r_a, M)$ is $m \mapsto am$. $\mathcal{L}$, $\mathfrak{v}_\infty$, $\mathfrak{e}_\infty$, $\varepsilon_{\mathcal{L}}$, $\mathsf{B}_{\mathcal{T}}$, $\mathsf{B}_{\mathcal{T}}^{(\ell)}$, e_t, z_t, v_t^x, ε_{B}, $\varepsilon_{\mathcal{T}}^{(\ell)}$, $\varepsilon_{\mathsf{B}}^{(\ell)}$, a, $\mathsf{a}^{(\ell)}$, f_{a}, i_∂, r_∂, G_∂, $\mathsf{g}_\square$, g', Cyl, π_{Cyl}, ϖ, ς, ς', h_{Cyl}, $\Phi_\partial^{\mathbb{Z}}$, $\Psi_\partial^{\mathbb{Z}}$, Φ_ℓ^{∂}, Ψ_ℓ^{∂}, Φ_∂, Ψ_∂, $\mathsf{H}_\partial^{\mathsf{B}}$, H_∂^{X}, y_ℓ, y'_ℓ, Y_Φ, Y_Ψ, j_Φ, j_Ψ, Θ_Φ, Θ_Ψ, $\operatorname{pr}_\partial$, $T_{\mathcal{T}}^{\mathrm{c}}$, T'_∂, $\eta_{\mathcal{T}}$ are new symbols of this subsection and mean nothing else (rule (A5)); ϖ, ς are not the symbols of

other chapters.

Definition 31.23 (The cusp model, the axis map, the tree cone; the geometric tables)**.** Assume (H4) and (B1)–(B3). Let ℓ be a prime, R an O_ℓ-algebra, and $F^{\mathfrak{e}}_\bullet$, $e_{\mathfrak{e}}$, e_S, e_U, F^x, (a_t, b_t) as in Definition 31.11 (so $F^{\mathfrak{e}}_t = R[\Gamma]e_{\mathfrak{e}}$ for every t at $\ell = 2$ and for the free model, for $t = 0$ only at $\ell \geq 3$; ε_t the coefficient of its differential: $(1-\mathbf{c})e_{\mathfrak{e}}$ for odd t, $(1+\mathbf{c})e_{\mathfrak{e}}$ for even $t \geq 2$).

1. **(The cusp model.)** $\mathsf{B}^{(\ell)}_{\mathcal{T}}$ is the mapping cone of $r_{(1-T)e_{\mathfrak{e}}} \colon F^{\mathfrak{e}}_\bullet \to F^{\mathfrak{e}}_\bullet$:

$$\mathsf{B}^{(\ell)}_{\mathcal{T},q} = F^{\mathfrak{e}}_q \oplus F^{\mathfrak{e}}_{q-1}, \qquad \partial \mathsf{e}_t = \varepsilon_t\, \mathsf{e}_{t-1}, \qquad \partial \mathsf{z}_t = (1-T)\,\mathsf{e}_t - \varepsilon_t\, \mathsf{z}_{t-1},$$

   e_t the generator $e_{\mathfrak{e}}$ of the first summand in degree t, z_t that of the second in degree $t+1$ (absent where $F^{\mathfrak{e}}_t = 0$; $\varepsilon_0 \mathsf{z}_{-1} := 0$); augmented by $\mathsf{B}^{(\ell)}_{\mathcal{T},0} = R[\Gamma]e_{\mathfrak{e}} \to R[\mathbb{P}^1]$, $ye_{\mathfrak{e}} \mapsto y \cdot \infty$. Ranks over $R[\Gamma]e_{\mathfrak{e}}$: $1, 2, 2, \ldots$ at $\ell = 2$ (free) and for the free model $\mathsf{B}_{\mathcal{T}}$ ($e_{\mathfrak{e}} = 1$, every R); $1, 1, 0, \ldots$ at $\ell \geq 3$.

2. **(The axis map.)** $\mathsf{a}^{(\ell)} \colon \mathsf{B}^{(\ell)}_{\mathcal{T}} \to \mathcal{I}^{(\ell)}_{\mathcal{T}}$ is $R[\Gamma]$-linear with

$$\mathsf{a}^{(\ell)}(\mathsf{e}_t) = a_t\, \mathsf{v}^S_t, \qquad \mathsf{a}^{(\ell)}(\mathsf{z}_t) = -[t \text{ even}]\, S^{-1} e_S\, \mathsf{v}^S_{t+1} + [t \text{ even}]\, S^{-1} e_U\, \mathsf{v}^U_{t+1} + (S^{-1} - T) e_{\mathfrak{e}}\, \mathsf{v}^{\mathfrak{e}}_t,$$

   $\mathsf{v}^x_t := e_x$ the generator of F^x_t in $\mathcal{I}^{(\ell)}_{\mathcal{T}}$ ($\mathsf{v}^{\mathfrak{e}}_t$ in total degree $t+1$) and a_t the S-component of the canonical lift, every term dropped whose target column vanishes; $\mathsf{a} := \mathsf{a}^{(0)}$ on the free models, with $e_x = 1$. At $\ell \geq 3$ **the choice of S^{-1} rather than S is immaterial** (v1.23.0-pre.34, referee N8): only z_0 occurs, every surviving term of $\mathsf{a}^{(\ell)}(\mathsf{z}_0)$ carries $e_{\mathfrak{e}} = \frac{1}{2}(1+\mathbf{c})$ on the right ($e_U = e_{\mathfrak{e}}$ at $\ell = 3$; the v^S_1-term is dropped, $F^S_1 = 0$), and $S^{-1} = \mathbf{c}S$ with $\mathbf{c}$ central gives $Se_{\mathfrak{e}} = S^{-1}e_{\mathfrak{e}}$; writing S there yields the same map. At $\ell \in \{0, 2\}$ it does not as a chain map, but **on cochains it is immaterial wherever $\mathbf{c}$ acts trivially** (v1.23.0-pre.35): S^{-1} enters $\mathsf{a}^{(\ell)}$ only as the left factor of the coefficients $S^{-1}e_x$ ($x \in \{S, U, \mathfrak{e}\}$), and $(S - S^{-1})e_x = (1-\mathbf{c})Se_x$, so on $\mathrm{Hom}_{R[\Gamma]}(-, M)$ (where $\mathrm{Hom}(r_b, M)$ is $m \mapsto bm$) the two maps differ by $m \mapsto Se_x(1-\mathbf{c})m$. For $M_N = \mathrm{Ind}^{\Gamma}_{\Gamma_0(N)} \mathrm{Sym}^n$ (any M induced from a subgroup containing $-I$), $\mathbf{c}(g \otimes v) = g \otimes (-I)v$ gives $\rho_{M_N}(\mathbf{c}) = (-1)^n$: for n **even** the two axis maps have the **same cochains** at every ℓ (every group, map and operator computed from these cochains agrees; the chain-level constructions of Proposition 31.25(3) presuppose a chain map and use $\mathsf{a}^{(\ell)}$; v1.23.0-pre.36, the C-D8b referee's N6); for n odd they differ by $2Se_x$, which vanishes modulo 2.

3. **(The tree cone.)** For a $R[\Gamma]$-module M put $\mathsf{B}^\bullet_\ell(M) := \mathrm{Hom}_{R[\Gamma]}(\mathsf{B}^{(\ell)}_{\mathcal{T}}, M)$ and

$$C(\mathsf{a}^{(\ell)*})^q := C^{q+1}_{\mathcal{T},\ell}(M) \oplus \mathsf{B}^q_\ell(M), \qquad D(\alpha, c) := \big(-d\alpha,\ dc + \mathsf{a}^{(\ell)*}\alpha\big),$$

   (38) with $\mathsf{a}^{(\ell)*}$ for r, with $\mathrm{inc}_c(c) = (0, c)$, $\mathrm{pr}_c(\alpha, c) = \alpha$, and $H^q_c(\mathcal{T}_\ell; M) := H^{q-1}(C(\mathsf{a}^{(\ell)*}))$.

4. **(The geometric tables.)** $C_\bullet(\mathcal{L})$ is the complex $\mathbb{Z}[\Gamma/\langle\mathbf{c}\rangle]\mathfrak{e}_\infty \to \mathbb{Z}[\Gamma/\langle\mathbf{c}\rangle]\mathfrak{v}_\infty$, $\partial\mathfrak{e}_\infty = (1-T)\mathfrak{v}_\infty$, with $\varepsilon_{\mathcal{L}}(\gamma\mathfrak{v}_\infty) = \gamma \cdot \infty$ (the cellular chains of $\Gamma \times_{\Gamma_\infty} \mathbb{R}$, one vertex and one edge per period); $\varepsilon_{\mathsf{B}} \colon \mathsf{B}_{\mathcal{T}} \to C_\bullet(\mathcal{L})$ is $\mathsf{e}_0 \mapsto \mathfrak{v}_\infty$, $\mathsf{z}_0 \mapsto \mathfrak{e}_\infty$, 0 on $\mathsf{e}_t, \mathsf{z}_t$, $t \geq 1$. **The ℓ-local column augmentations** (v1.23.0-pre.34, referee m4): $\varepsilon^{(\ell)}_{\mathcal{T}} \colon \mathcal{I}^{(\ell)}_{\mathcal{T}} \to C_\bullet(\mathcal{T}) \otimes R$ is $ye_x \mapsto y\Gamma_x$ ($= y\mathfrak{v}_x$) on F^x_0, $x \in \{S, U\}$, and $ye_{\mathfrak{e}} \mapsto y\langle\mathbf{c}\rangle$ ($= y\mathfrak{e}$) on $F^{\mathfrak{e}}_0$ (total degree 1), 0 on every column degree $t \geq 1$ (the vertex part is Definition 31.11(1)'s augmentation); $\varepsilon^{(\ell)}_{\mathsf{B}} \colon \mathsf{B}^{(\ell)}_{\mathcal{T}} \to C_\bullet(\mathcal{L}) \otimes R$ is $ye_{\mathfrak{e}} \mapsto y\mathfrak{v}_\infty$ on e_0, $ye_{\mathfrak{e}} \mapsto y\mathfrak{e}_\infty$ on z_0, 0 on $\mathsf{e}_t, \mathsf{z}_t$, $t \geq 1$. Both are well defined ($Q_x \subseteq \Gamma_x$, $Q_{\mathfrak{e}} \subseteq \langle\mathbf{c}\rangle$ fixes $\mathfrak{v}_\infty$, $\mathfrak{e}_\infty$); $\varepsilon^{(0)}_{\mathcal{T}} = \varepsilon_{\mathcal{T}}$ of Proposition 31.12(4)(b), $\varepsilon^{(0)}_{\mathsf{B}} = \varepsilon_{\mathsf{B}}$, and at $\ell = 2$ $\varepsilon^{(2)}_{\mathsf{B}} = \varepsilon_{\mathsf{B}} \otimes R$. With the boundary simplices v_0, v_1, v_2, v_0v_1, v_1v_2, v_2v_3 of Σ^∂ (Definition 23.4; $v_3 = Tv_0$), the edges and triangles of Σ_1, Σ_2, and $\mathfrak{v}_S$, $\mathfrak{v}_U$, $\mathfrak{e}$ of the tree, define $\mathbb{Z}[\Gamma]$-linearly on representatives:

$$
\begin{array}{ll}
\mathsf{i}_\partial\colon C_\bullet(\mathcal{L})\to C_\bullet(\partial X) & \mathfrak{v}_\infty\mapsto v_0,\ \mathfrak{e}_\infty\mapsto -(v_0v_1+v_1v_2+v_2v_3)\\
\mathsf{r}_\partial\colon C_\bullet(\partial X)\to C_\bullet(\mathcal{L}) & v_0,v_1,v_2\mapsto \mathfrak{v}_\infty;\ v_0v_1,v_1v_2\mapsto 0,\ v_2v_3\mapsto -\mathfrak{e}_\infty\\
\mathsf{G}_\partial\colon C_0(\partial X)\to C_1(\partial X) & v_0\mapsto 0,\ v_1\mapsto v_0v_1,\ v_2\mapsto v_0v_1+v_1v_2\\
\mathsf{f}_\mathsf{a}\colon C_\bullet(\mathcal{L})\to C_\bullet(\mathcal{T}) & \mathfrak{v}_\infty\mapsto \mathfrak{v}_S,\ \mathfrak{e}_\infty\mapsto (S^{-1}-T)\mathfrak{e}\\
\mathsf{g}_\square\colon C_0(\partial X)\to C_1(\mathcal{T}) & v_0\mapsto -\mathfrak{e},\ v_1,v_2\mapsto 0\\
\mathsf{g}'\colon C_s(\mathcal{L})\to C_{s+1}(X) & \mathsf{g}':=-\big(\mathsf{G}_\mathcal{T}\,j_X\,\mathsf{i}_\partial+\mathsf{i}_\mathcal{T}\,\mathsf{g}_\square\,\mathsf{i}_\partial\big)\colon \mathfrak{v}_\infty\mapsto v_0v_4+v_4v_6,\\
 & \mathfrak{e}_\infty\mapsto (v_0v_1v_4-v_1v_4v_6)+v_1v_2v_6+(v_2v_3v_5+v_2v_5v_6)
\end{array}
$$

(j_X the inclusion $C_\bullet(\partial X)\subseteq C_\bullet(X)$; $\mathsf{i}_\mathcal{T}$, $\mathsf{r}_\mathcal{T}$, $\mathsf{G}_\mathcal{T}$ those of Proposition 31.12(4)(a); $\mathsf{g}_\square$ is 0 on edges, G_∂ on edges likewise, $C_2(\mathcal{T})=C_2(\partial X)=0$).

Lemma 31.24 (The cylinder: a pair model, and its cone deformed onto the tree cone)**.** *[proved here] [standard: [Wei94, §1.5]] Let* $\mathsf{a}\colon \mathsf{B}\to\mathsf{P}$ *be a chain map from a complex of* ***finitely generated*** *projective* $R[\Gamma]$*-modules,* ***zero in negative degrees****, resolving* $R[\mathbb{P}^1]$ *(as Definition 31.18 requires;* $\mathsf{B}^{(\ell)}_\mathcal{T}$ *satisfies this, Proposition 31.25(1)(a); v1.23.0-pre.34, referee N1) to an* R*-model* P *(Proposition 31.10), over* $\varepsilon\colon R[\mathbb{P}^1]\to R$*. Put* $\mathrm{Cyl}_q:=\mathsf{P}_q\oplus\mathsf{B}_q\oplus\mathsf{B}_{q-1}$, $\partial(x,b,u):=(\partial x-\mathsf{a}u,\ \partial b+u,\ -\partial u)$*, with* B *the middle summand, and* $\pi_{\mathrm{Cyl}}(x,b,u):=x+\mathsf{a}b$*, augmented by* $\varepsilon\circ\pi_{\mathrm{Cyl}}$*.*

1. $(\mathrm{Cyl},\mathsf{B})$ *is an* R*-pair model (Definition 31.18);* π_{Cyl} *and the inclusion* i_P *of* P *are mutually inverse comparisons over* id_R *(*$\pi_{\mathrm{Cyl}}i_\mathsf{P}=\mathrm{id}$*); on cochains* $r_{\mathrm{Cyl}}\circ\pi^*_{\mathrm{Cyl}}=\mathsf{a}^*$*. Its cochains are* $\mathrm{Cyl}(M)^q=\mathsf{P}(M)^q\oplus\mathsf{B}(M)^q\oplus\mathsf{B}(M)^{q-1}$ *with* $d(\alpha,\beta,\gamma)=(d\alpha,\ d\beta,\ -\mathsf{a}^*\alpha+\beta-d\gamma)$*, and* $r_{\mathrm{Cyl}}(\alpha,\beta,\gamma)=\beta$*.*
2. *On* $C(r_{\mathrm{Cyl}})^q=\mathrm{Cyl}(M)^{q+1}\oplus\mathsf{B}(M)^q$ *put* $\varpi(\alpha,\beta,\gamma,c):=(\alpha,\ c+\gamma)$, $\varsigma(\alpha,c):=(\alpha,0,c,0)$, $\varsigma'(\alpha,c):=(\alpha,\mathsf{a}^*\alpha,0,c)$, $\mathsf{h}_{\mathrm{Cyl}}(\alpha,\beta,\gamma,c):=(0,c,0,0)$*. Then* $\varpi\colon C(r_{\mathrm{Cyl}})\to C(\mathsf{a}^*)$*,* ς*,* ς' *are cochain maps,* $\varpi\varsigma=\varpi\varsigma'=\mathrm{id}$*,* $\mathrm{id}-\varsigma\varpi=D\mathsf{h}_{\mathrm{Cyl}}+\mathsf{h}_{\mathrm{Cyl}}D$*, and* $\varsigma'-\varsigma=D(\mathsf{h}_{\mathrm{Cyl}}\varsigma')+(\mathsf{h}_{\mathrm{Cyl}}\varsigma')D$*.* ***Strictly****,* $\varpi\circ\mathrm{inc}_c=\mathrm{inc}_c$ *and* $\mathrm{pr}_c\circ\varpi=i^*_\mathsf{P}\circ\mathrm{pr}_c$*.*
3. *Hence Proposition 31.20(1)(a)–(c) hold for* $C(\mathsf{a}^*)$*:* $H^q(\mathsf{B}(M))=H^q(\Gamma_\infty;M)$*,* $H^q(\mathsf{a}^*)=\mathrm{res}^\Gamma_{\Gamma_\infty}$*,* $H^{q-1}(C(\mathsf{a}^*))=H^q_c\cong\mathrm{Ext}^{q-1}_{R[\Gamma]}(\mathrm{St}_R,M)$*, and its long exact sequence is (79), exact in every degree, with* $\mathrm{inc}_{c*}=j^*_{\mathrm{St}}$ *and* $\mathrm{pr}_{c*}=-\delta$ *under the identifications through* $H(\varpi)$ ***with the signs of Proposition 31.20(1)(c)****; its connecting map* $H^q(\Gamma;M)\to H^q(\Gamma_\infty;M)$ *is* $H(\mathsf{a}^*)$*, with no sign.*

Proof. (1) $\partial^2=0$: the first component of $\partial^2(x,b,u)$ is $\partial(\partial x-\mathsf{a}u)-\mathsf{a}(-\partial u)=0$, the second $\partial(\partial b+u)-\partial u=0$. The middle summand is a subcomplex with projective complement $\mathsf{P}\oplus\mathsf{B}[-1]$. $\pi_{\mathrm{Cyl}}\partial(x,b,u)=\partial x-\mathsf{a}u+\mathsf{a}\partial b+\mathsf{a}u=\partial\pi_{\mathrm{Cyl}}(x,b,u)$, and with $s(x,b,u):=(0,0,b)$, $\partial s+s\partial=\mathrm{id}-i_\mathsf{P}\pi_{\mathrm{Cyl}}$ (both send (x,b,u) to $(-\mathsf{a}b,b,u)$), so $\mathrm{Cyl}\simeq\mathsf{P}$ resolves R; on B_0 the two augmentations agree because a lies over ε. The cochain formulas are $\mathrm{Hom}(-,M)$ of ∂, the dual of $(x,b,u)\mapsto\alpha x+\beta b+\gamma u$. (2) With $D(\alpha,\beta,\gamma,c)=(-d\alpha,-d\beta,\mathsf{a}^*\alpha-\beta+d\gamma,dc+\beta)$: $\varpi D=(-d\alpha,\ dc+d\gamma+\mathsf{a}^*\alpha)=D\varpi$; $D\varsigma(\alpha,c)=(-d\alpha,0,\mathsf{a}^*\alpha+dc,0)=\varsigma D(\alpha,c)$; $D\varsigma'(\alpha,c)=(-d\alpha,-\mathsf{a}^*d\alpha,0,dc+\mathsf{a}^*\alpha)=\varsigma' D(\alpha,c)$; $D\mathsf{h}_{\mathrm{Cyl}}+\mathsf{h}_{\mathrm{Cyl}}D$ sends (α,β,γ,c) to $(0,-dc,-c,c)+(0,dc+\beta,0,0)=(0,\beta,-c,c)=(\mathrm{id}-\varsigma\varpi)(\alpha,\beta,\gamma,c)$; and $\varsigma'-\varsigma=(\mathrm{id}-\varsigma\varpi)\varsigma'$. The strict identities are read off. (3) By (1) Proposition 31.20(1) applies to $(\mathrm{Cyl},\mathsf{B})$; ϖ is a quasi-isomorphism by (2), commuting with inc_c and, through the comparison i_P over id_R (which induces the identity under the identifications with $H^\bullet(\Gamma;M)$), with pr_c. The connecting map of $0\to\mathsf{B}(M)\to C(\mathsf{a}^*)\to\mathsf{P}(M)[1]\to 0$: a cocycle α lifts to $(\alpha,0)$, and $D(\alpha,0)=(0,\mathsf{a}^*\alpha)$; a^* is the comparison along ε, which induces res (the proof of Proposition 31.20(1)(a)). □

Proposition 31.25 (The tree cusp pair: what it computes, closed forms, comparisons with $(\mathcal{I},\mathcal{I}^\partial)$ over $\mathbb{Z}$, Hecke by transport, precision)**.** *[proved here] [standard: [Wei94, Thm. 2.2.6, §1.5], [Bro82, Ch. III.6, III.8–9]] (v1.23.0-pre.33; milestone D5c, clause C-D8 of the plan; refereed ACCEPT after fixes,* `docs/implementation/slot24/C-D8_referee.md`*; the fixes at v1.23.0-pre.34, not re-refereed.) Assume (H4) and (B1)–(B3);* ℓ*,* R *as in Definition 31.23,* M *a* $R[\Gamma]$*-module; for the integral statements* M *as in §31.4 (*$\mathbb{Z}$*-free of finite rank) and* $M_R:=M\otimes R$*.*

1. ***(What it computes, over every ring.)***
   (a) $\mathsf{B}^{(\ell)}_{\mathcal{T}} \to R[\mathbb{P}^1]$ *is a resolution by finitely generated projective* $R[\Gamma]$*-modules* ($R[\Gamma]e_{\mathfrak{c}} \cong R[\Gamma/Q_{\mathfrak{c}}]$)*, free at* $\ell = 2$ *and on the free model;* $\varepsilon^{(\ell)}_{\mathsf{B}}$ *is a surjective quasi-isomorphism onto* $C_\bullet(\mathcal{L}) \otimes R$*, which resolves* $R[\mathbb{P}^1]$ *through* $\varepsilon_{\mathcal{L}}$*.*
   (b) $\mathsf{a}^{(\ell)}$ *is well defined (every coefficient lies in* $e_{\mathfrak{c}}R[\Gamma]e_y$*), a chain map, and lies over* ε*; on column degrees* 0 *it lies over* $\mathsf{f_a}$ *(*$\varepsilon^{(\ell)}_{\mathcal{T}}\mathsf{a}^{(\ell)} = \mathsf{f_a}\varepsilon^{(\ell)}_{\mathsf{B}}$*, the* ℓ*-local column augmentations of Definition 31.23(4); v1.23.0-pre.34, referee m4).*
   (c) *Lemma 31.24 applies to* $\mathsf{a}^{(\ell)}$*:* $(\mathrm{Cyl}(\mathsf{a}^{(\ell)}), \mathsf{B}^{(\ell)}_{\mathcal{T}})$ *is an* R*-pair model, and* $C(\mathsf{a}^{(\ell)*})$ *computes, in every degree and for every* M*,* $H^\bullet(\Gamma; M)$*,* $H^\bullet(\Gamma_\infty; M)$*,* $H^\bullet_c(\Gamma; M)$ *and (79), with the signs of Proposition 31.20(1)(c). For* $M = \mathrm{Ind}^{\Gamma}_{\Gamma_0(N)}V$ *(Shapiro) these are the groups of* $\Gamma_0(N)$*, and Proposition 31.20(3)–(5) hold verbatim (they concern the groups, not a model).*
2. ***(The cochains in closed form.)*** *With the components of Proposition 31.12(2) on* $C^\bullet_{\mathcal{T},\ell}(M)$ *(*x *on* F^S_q*,* y *on* F^U_q*,* z *on* $F^{\mathfrak{c}}_{q-1}$*) and* (u, w) *on* $(\mathsf{e}_q, \mathsf{z}_{q-1})$*:*
   (a) $\ell = 2$ *(and the free model,* $y \in M$*):* $\mathsf{B}^0 = M$*,* $\mathsf{B}^q = M \oplus M$ *(*$q \geq 1$*),* $d^0u = ((1-\mathbf{c})u, (1-T)u)$*,* $d^q(u, w) = (\varepsilon_{q+1}u,\ (1-T)u - \varepsilon_q w)$*; and* $\mathsf{a}^{(2)*}(x, y) = x$ *in degree* 0*,*

$$\mathsf{a}^{(2)*}(x, y, z) = \big(a_q\, x,\ [q \text{ odd}]\, S^{-1}(y-x) + (S^{-1}-T)z\big) \qquad (q \geq 1;\ a_q = 1 \text{ for even, } 1+S \text{ for odd } q).$$

   (b) $\ell = 3$*:* $\mathsf{B}^0 = \mathsf{B}^1 = M^{-I}$*,* $d^0 = 1 - T$*,* $\mathsf{B}^q = 0$ *(*$q \geq 2$*);* $\mathsf{a}^{(3)*}(x, y) = x$ *(*$x \in M^{\langle S\rangle} \subseteq M^{-I}$*),* $\mathsf{a}^{(3)*}(y, z) = S^{-1}y + (S^{-1} - T)z$ *on* $C^1 = M^{-I} \oplus M^{-I}$*, and* 0 *above.*
   (c) $\ell \geq 5$*:* B *as at* $\ell = 3$*;* $\mathsf{a}^*(x, y) = x$*,* $\mathsf{a}^*z = (S^{-1} - T)z$ *on* $C^1 = M^{-I}$*; the tree cone is* $M^{\langle S\rangle} \oplus M^{\langle U\rangle} \to M^{-I} \oplus M^{-I} \to M^{-I}$*, and* $H^2_c = M^{-I}/\big((1-T)M^{-I} + (1-S)M^{-I}\big)$*, the coinvariants of Proposition 31.20(5)(e).*
   (d) ***(Ranks.)*** $\operatorname{rank} C(\mathsf{a}^{(\ell)*})^q = \operatorname{rank} C^{q+1}_{\mathcal{T},\ell} + \operatorname{rank} \mathsf{B}^q_\ell$*; at* $\ell = 2$*, with* $d = \operatorname{rank} M$ *and* $r := \operatorname{rank} M^{U^2} = \frac{1}{3}(d + 2\operatorname{tr}\rho_M(U^2))$*:* $d + r$*,* $3d + r$*,* $4d + r$*,* $4d + r, \ldots$ *in degrees* $-1, 0, 1, 2, \ldots$ *— with* $r \approx d/3$*,* $\approx \frac{4}{3}d, \frac{10}{3}d, \frac{13}{3}d, \ldots$*, against* $\approx 16\frac{1}{3}d, 24\frac{1}{3}d, \ldots$ *in degrees* $0, 1, \ldots$ *on* $(\mathcal{I}^{(2)}, \mathcal{I}^{\partial(2)})$ *(`DESCENT_plan.md` §5.3).*
3. ***(Comparisons with*** $(\mathcal{I}, \mathcal{I}^\partial)$ ***over*** $\mathbb{Z}$***: effective.)***
   (a) ***(The tables.)*** i_∂*,* r_∂*,* $\mathsf{f_a}$*,* ε_{B} *are chain maps over* $\mathbb{Z}[\mathbb{P}^1]$*, resp. (for* $\mathsf{f_a}$*) over* ε*;* $\mathsf{r}_\partial\mathsf{i}_\partial = \mathrm{id}$*;* $\mathrm{id} - \mathsf{i}_\partial\mathsf{r}_\partial = \partial\mathsf{G}_\partial + \mathsf{G}_\partial\partial$ *on* $C_\bullet(\partial X)$*;* $\mathsf{r}_{\mathcal{T}}j_X - \mathsf{f_a}\mathsf{r}_\partial = \partial\mathsf{g}_\square + \mathsf{g}_\square\partial$ *on* $C_\bullet(\partial X)$*;* $\mathsf{i}_{\mathcal{T}}\mathsf{f_a} - j_X\mathsf{i}_\partial = \partial\mathsf{g}' + \mathsf{g}'\partial$ *on* $C_\bullet(\mathcal{L})$*.*
   (b) ***(The boundary comparisons.)*** The solver, generalised *(v1.23.0-pre.34, referee m2). Let* E *be a complex of* $\mathbb{Z}[\Gamma]$*-modules filtered by finitely many columns* $E = \bigoplus_{s \leq s_{\max}} E_{s,\bullet}$ *(total degree* $s + t$*), each column free and periodic as in Proposition 31.12(4)(b) (generators* $\mathbb{Z}[\Gamma/\Gamma_\sigma]$*-free, differentials* r_{g-1}*,* r_N *or the tree's* r_{1-g}*,* r_N*), entering with a global sign* ϵ_s*, the differential lowering* s *off the columns, and let* $p\colon E \to C$ *be its* column augmentation *(the augmentations of the columns* s *in column degree* 0*, a surjective quasi-isomorphism onto a complex* C *with* C_s *the column's* H_0*). Then* h_p *of loc. cit. — descending in* s*,* $\epsilon_s\mathsf{h}$ *column by column (sign rule:* $\epsilon = -1$ *on a cone's shifted column; the tree's columns with even-degree* h *negated), at most* $s_{\max} + 1$ *steps — satisfies* $\partial\mathsf{h}_p(z) = z$ *and* $p\,\mathsf{h}_p(z) = 0$ *on cycles* z *of* $\ker p$*, and (L), (L′) hold verbatim along* p*; everything is* $\mathbb{Z}$*-linear, so all of it holds over every ring* R *after* $\otimes R$*, for* $R[\Gamma]$*-maps* A*,* B*,* f *with coefficients in* $R[\Gamma]$ *(they may carry the denominators of* O_ℓ *and the idempotents* e_x*) and* P' *free over* $R[\Gamma]$ *with given bases. Loc. cit. is the case* $p \in \{\mathrm{cmp}, \varepsilon_{\mathcal{T}}\}$*; here it is used for* $p = \varepsilon_{\mathsf{B}}$ *(two columns,* e *(*$s = 0$*) and* z *(*$s = 1$*, on top,* $\epsilon = -1$*), the tree's sign convention* r_{1-g}*), for* $p = \mathrm{cmp}|_{\mathcal{I}^\partial}$ *(the columns* $s \leq 1$ *over* Σ^∂*; each* $\epsilon_s\mathsf{h}$ *maps a column into itself, so the descent stays in* $\mathcal{I}^\partial$*), and, over* R*, for* $\varepsilon_{\mathcal{T}} \otimes R$ *and* $\mathrm{cmp} \otimes R$ *in (c). Put* $\Phi^{\mathbb{Z}}_\partial :=$ *(L) for* $f = \mathsf{r}_\partial \circ \mathrm{cmp}$ *along* ε_{B}*,* $\Psi^{\mathbb{Z}}_\partial :=$ *(L) for* $f = \mathsf{i}_\partial \circ \varepsilon_{\mathsf{B}}$ *along* $\mathrm{cmp}|_{\mathcal{I}^\partial}$*: chain maps* $\mathcal{I}^\partial \rightleftarrows \mathsf{B}_{\mathcal{T}}$ *over* $\mathrm{id}_{\mathbb{Z}[\mathbb{P}^1]}$*,*

over r_∂*, resp.* i_∂*; by (L′),* $\mathsf{H}^{\mathsf{B}}_\partial\colon \Phi^{\mathbb{Z}}_\partial\Psi^{\mathbb{Z}}_\partial \simeq \mathrm{id}$ *(*$g=0$*) and* $\mathsf{H}^{X}_\partial\colon \mathrm{id} \simeq \Psi^{\mathbb{Z}}_\partial\Phi^{\mathbb{Z}}_\partial$ *(*$g=\mathsf{G}_\partial$*). ℓ-locally,* $\Phi^\partial_\ell\colon \mathsf{B}_\mathcal{T}\otimes R \to \mathsf{B}^{(\ell)}_\mathcal{T}$ *is* $r_{|Q_\mathfrak{e}|^{\lfloor t/2\rfloor}e_\mathfrak{e}}$ *on both columns (0 where the ℓ-local column vanishes) and* Ψ^∂_ℓ *the inclusions with the inverse scalars — the scalars are* 1 *in every degree that occurs (*$Q_\mathfrak{e}=1$ *at* $\ell=2$*; only* $t=0$ *at* $\ell\geq 3$*; v1.23.0-pre.34, referee N3) —; both lie over the identity of* $C_\bullet(\mathcal{L})\otimes R$ *(*$\varepsilon^{(\ell)}_{\mathsf{B}}\Phi^\partial_\ell = \varepsilon_{\mathsf{B}}\otimes R$*,* $(\varepsilon_{\mathsf{B}}\otimes R)\Psi^\partial_\ell = \varepsilon^{(\ell)}_{\mathsf{B}}$*): chain maps over* id*,* $\Phi^\partial_\ell\Psi^\partial_\ell = \mathrm{id}$ ***strictly****, no correction terms (*$r_{(1-T)e}$ *commutes with the column maps); at* $\ell=2$ *both are the identity.* $\Phi_\partial := \Phi^\partial_\ell\Phi^{\mathbb{Z}}_\partial$*,* $\Psi_\partial := \Psi^{\mathbb{Z}}_\partial\Psi^\partial_\ell$*.*

(c) ***(The two squares.)*** *With* $\Phi_\mathcal{T}$*,* $\Psi_\mathcal{T}$*,* $\Phi^\mathcal{T}_\ell$*,* $\Psi^\mathcal{T}_\ell$ *of Proposition 31.12(3), (4) and* $j\colon \mathcal{I}^\partial \subseteq \mathcal{I}$*:* $\mathsf{y}_\ell :=$ *(L′) along* $\varepsilon_\mathcal{T}\otimes R$ *on* $\mathcal{I}^\partial\otimes R$ *with* $A = \Phi^{\mathbb{Z}}_\mathcal{T} j$*,* $B = \Psi^\mathcal{T}_\ell \mathsf{a}^{(\ell)}\Phi_\partial$*,* $p' = \mathrm{cmp}|_{\mathcal{I}^\partial}$*,* $g = \mathsf{g}_\square$*;* $\mathsf{y}'_\ell :=$ *(L′) along* $\mathrm{cmp}\otimes R$ *on* $\mathsf{B}_\mathcal{T}\otimes R$ *with* $A = \Psi_\mathcal{T}\mathsf{a}^{(\ell)}\Phi^\partial_\ell$*,* $B = j\Psi^{\mathbb{Z}}_\partial$*,* $p' = \varepsilon_{\mathsf{B}}$*,* $g = \mathsf{g}'$*. Put* $\mathsf{Y}_\Phi := \Phi^\mathcal{T}_\ell\mathsf{y}_\ell$*,* $\mathsf{Y}_\Psi := \mathsf{y}'_\ell\Psi^\partial_\ell$*. (*$\Phi^\mathcal{T}_\ell$*,* $\Psi^\mathcal{T}_\ell$ *lie over the identity of* $C_\bullet(\mathcal{T})\otimes R$ *—* $\varepsilon^{(\ell)}_\mathcal{T}\Phi^\mathcal{T}_\ell = \varepsilon_\mathcal{T}\otimes R$*,* $(\varepsilon_\mathcal{T}\otimes R)\Psi^\mathcal{T}_\ell = \varepsilon^{(\ell)}_\mathcal{T}$*: in column degree* 0 *they are* r_{e_x}*, resp. the inclusion,* $ye_x\Gamma_x = y\Gamma_x$*, and the corrections* k*,* k' *land in column degree* 1 *— and* Φ^∂_ℓ*,* Ψ^∂_ℓ *over that of* $C_\bullet(\mathcal{L})\otimes R$*, (b); v1.23.0-pre.34, referee m4.) Then*

$$\Phi_\mathcal{T}\, j - \mathsf{a}^{(\ell)}\Phi_\partial = \partial\mathsf{Y}_\Phi + \mathsf{Y}_\Phi\partial, \qquad \Psi_\mathcal{T}\,\mathsf{a}^{(\ell)} - j\,\Psi_\partial = \partial\mathsf{Y}_\Psi + \mathsf{Y}_\Psi\partial.$$

(d) ***(The cone maps.)***

$$\mathsf{j}_\Phi(\alpha,c) := \big(\Phi^*_\mathcal{T}\alpha,\ \Phi^*_\partial c - \mathsf{Y}^*_\Phi\alpha\big), \qquad \mathsf{j}_\Psi(\alpha',c') := \big(\Psi^*_\mathcal{T}\alpha',\ \Psi^*_\partial c' - \mathsf{Y}^*_\Psi\alpha'\big)$$

are cochain maps $C(\mathsf{a}^{(\ell)*}) \rightleftarrows C(r_\mathcal{I})(M_R)$*, and* ***strictly*** $\mathsf{j}_\Phi\mathrm{inc}_c = \mathrm{inc}_c\Phi^*_\partial$*,* $\mathrm{pr}_c\mathsf{j}_\Phi = \Phi^*_\mathcal{T}\mathrm{pr}_c$*, likewise for* j_Ψ*. They are quasi-isomorphisms,* ***mutually inverse on cohomology****,* $H(\mathsf{j}_\Phi)$ *is the comparison isomorphism of Lemma 31.19(3) (through* ϖ *and a comparison of pairs, the proof), and* $(H(\Phi^*_\mathcal{T}), H(\Phi^*_\partial), H(\mathsf{j}_\Phi))$ *is an isomorphism of long exact sequences from the tree's (79) to that of* $(\mathcal{I},\mathcal{I}^\partial)$*. On cohomology none of these depends on the choices (representatives, preimages,* $\mathcal{I}$*, the solvers' order).*

4. ***(Hecke by transport.)*** *Let* $T = (T_\mathrm{b}[1], T_\partial)$ *be a cochain endomorphism of* $C(r_\mathcal{I})(M_R)$ *with* $r\,T_\mathrm{b} = T_\partial\, r$ *strictly — e.g.* $\Phi^{(N)}_p$ *of Proposition 31.21(2) for* $M = M_N$*,* $p\nmid N$ *— and put* $T^\mathrm{c}_\mathcal{T} := \mathsf{j}_\Psi\circ T\circ\mathsf{j}_\Phi$*.*

   (a) $T^\mathrm{c}_\mathcal{T}(\alpha,c) = \big(T'_\mathcal{T}\alpha,\ T'_\partial c + \eta_\mathcal{T}\alpha\big)$ *with* $T'_\mathcal{T} = \Psi^*_\mathcal{T}T_\mathrm{b}\Phi^*_\mathcal{T}$ *(the operator of Proposition 31.12(6)),* $T'_\partial := \Psi^*_\partial T_\partial\Phi^*_\partial$ *and*

$$\eta_\mathcal{T} := -\Psi^*_\partial T_\partial\mathsf{Y}^*_\Phi - \mathsf{Y}^*_\Psi T_\mathrm{b}\Phi^*_\mathcal{T}, \qquad d\,\eta_\mathcal{T} + \eta_\mathcal{T}\,d = T'_\partial\,\mathsf{a}^{(\ell)*} - \mathsf{a}^{(\ell)*}\,T'_\mathcal{T}.$$

   $(T'_\mathcal{T}, T'_\partial, \eta_\mathcal{T})$ has the shape of Definition 15.9's datum. ***Where*** $\eta_\mathcal{T}$ ***comes from*** *(v1.23.0-pre.34, referee m1):* $C(\mathsf{a}^{(\ell)*})$ *is not the cone of a restriction to a subcomplex. On the pair model* $(\mathrm{Cyl}(\mathsf{a}^{(\ell)}), \mathsf{B}^{(\ell)}_\mathcal{T})$ *of (1)(c) the transported datum is* ***strict****:* $T^\mathrm{Cyl} := (\Theta^*_\Psi[1], \Psi^*_\partial)\circ T\circ(\Theta^*_\Phi[1], \Phi^*_\partial)$ *is diagonal on* $C(r_\mathrm{Cyl})$ *(*Θ_Φ*,* Θ_Ψ *the comparisons of pairs of the proof of (3)(d)), and* $T^\mathrm{c}_\mathcal{T} \simeq \varpi\circ T^\mathrm{Cyl}\circ\varsigma'$*; the off-diagonal term enters through* ϖ *and* ς'*. The diagonal* $(T'_\mathcal{T}[1], T'_\partial)$ ***need not*** *be a cochain map and* $\eta_\mathcal{T}$ ***need not*** *vanish* — observed*, not part of the proved statement: for* $T = \theta^*$ *in the check below and for the C-D8 referee's* T_ε *(*`docs/implementation/slot24/cd8referee/`*); for fibre maps* $\eta_\mathcal{T}$ *may be non-zero while the diagonal is a cochain map (the check below).* $\eta_\mathcal{T}$ *is* ***determined*** *by* T *and the comparisons; the ambiguity of Proposition 15.16 does not arise.*

   (b) $T^\mathrm{c}_\mathcal{T}$ *is a cochain endomorphism of* $C(\mathsf{a}^{(\ell)*})$*, commuting with* inc_c *through* T'_∂ *and with* pr_c *through* $T'_\mathcal{T}$*, and under* $H(\mathsf{j}_\Phi)$ *it induces* $H(T)$ *on* $H^\bullet_c$*;* T'_∂*,* $T'_\mathcal{T}$ *induce* $H(T_\partial)$*,* $H(T_\mathrm{b})$*. For* $T = \Phi^{(N)}_p$*:* T_p *of Proposition 31.21(3) on the three terms of (79), over every* R*, in every degree* $\leq q_\mathrm{max}$ *for which the pair lift is built; the tree's sequence is* T_p*-equivariant. By (e), the comparisons are needed only in chain degrees* $\leq q_\mathrm{max}$ *for this.*

(c) *On cohomology $T^{\mathrm{c}}_{\mathcal{T}}$ does not depend on the comparisons; $H(T^{\mathrm{c}}_{1,\mathcal{T}})H(T^{\mathrm{c}}_{2,\mathcal{T}}) = H((T_1T_2)^{\mathrm{c}}_{\mathcal{T}})$. (No cochain homotopy between $T^{\mathrm{c}}_1T^{\mathrm{c}}_2$ and $(T_1T_2)^{\mathrm{c}}$ is constructed here.)*

(d) *A $\mathbb{Z}[\Gamma]$-map u of fibres acts on every term blockwise and commutes **strictly** with $\mathsf{a}^{(\ell)*}$, j_Φ, j_Ψ; so Proposition 31.21(6) (D2's maps on the pair, the T_p-equivariant ladder for* res*) holds on the tree pair verbatim, and the level ladder's signs (Proposition 31.10(2)(b)) are unchanged.*

(e) ***(Truncation: what the read of H^q_c uses; v1.23.0-pre.35.)** Let $m \geq 1$. Suppose given, over R, only: $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$, Φ_∂, Ψ_∂ in chain degrees $\leq m$, commuting with ∂ there and lying over the identities of (3)(b), (c); Y_Φ, Y_Ψ in chain degrees $\leq m-1$ (into chain degrees $\leq m$) with the two identities of (3)(c) in chain degrees $\leq m-1$; and T as in (4), or the truncation of one to cochain degrees $\leq m$ — the maps below use T_{b} on $C_{\mathrm{I}}^{\leq m}$ and T_∂ on $\mathcal{I}^{\partial\leq m}$ only, so (i)–(iii) hold for it as for the T it truncates; the pair lift of $\Phi^{(N)}_p$, which the code builds to chain degree 6, is such a truncation, Lemma 31.19's induction continuing it (v1.23.0-pre.36, the C-D8b referee's N2). Then:*

(i) *these comparison data extend to all degrees with every identity of (3)(b), (c); the extension is not unique, and nothing below depends on it;*

(ii) *in cone degree j, j_Φ, j_Ψ, $T'_{\mathcal{T}}$, T'_∂, $\eta_{\mathcal{T}}$ and $T^{\mathrm{c}}_{\mathcal{T}}$ use Φ, Ψ in chain degrees $\leq j+1$ and Y in chain degree j only — more precisely the bulk $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$ in chain degree $j+1$, and Φ_∂, Ψ_∂, Y_Φ, Y_Ψ in chain degree j —, so in cone degrees $j \leq m-1$ they are the maps of every extension; likewise $T'_{\mathcal{T}}$ on $C^q_{\mathcal{T},\ell}$ and T'_∂ on B^q_ℓ use the comparisons in chain degree q only, so H^q and $H^q(\partial)$, $q \leq m$, with the operators of (b) on them, are read from the data as well (v1.23.0-pre.36, the C-D8b referee's N4);*

(iii) *hence, for $j \leq m-1$, $T^{\mathrm{c}}_{\mathcal{T}}$ maps the cocycles of $C(\mathsf{a}^{(\ell)*})^j$ to cocycles and the coboundaries to coboundaries, and induces on $H^j(C(\mathsf{a}^{(\ell)*})) = H^{j+1}_c$ the operator of (b): **the read of H^q_c, $q \leq m$, needs the comparisons in chain degrees $\leq m$ only**;*

(iv) *the identity of (a), $d\eta_{\mathcal{T}} + \eta_{\mathcal{T}}d = T'_\partial\mathsf{a}^{(\ell)*} - \mathsf{a}^{(\ell)*}T'_{\mathcal{T}}$ on $C^{j+1}_{\mathcal{T},\ell}$, involves $\eta_{\mathcal{T}}$ in cone degree $j+1$, i.e. Φ, Ψ in chain degree $j+2$: it can be formed from the data iff $j \leq m-2$. The read of H^q_c does not use it: it holds for the maps of (iii) by (a).* As a certificate *of that read it bears on cone degree $q-2$, where (given $T'_{\mathcal{T}}$, T'_∂ cochain maps and the strict block) it is equivalent to $T^{\mathrm{c}}_{\mathcal{T}}$ commuting with D on cone degree $q-2$, which gives the coboundary half of (iii) (and is stronger than it; v1.23.0-pre.36, the C-D8b referee's N1); in cone degree $q-1$ the read needs only the cocycle half of (iii), which can be checked directly. At $\ell = 3$ it is empty in cone degrees $j \geq 1$ ($\mathsf{B}^j_3 = 0$ for $j \geq 2$).*

With $m = 5$ (the comparisons of the code, built to degree 5): H^q_c for $q \leq 5$, so $q_{\max} = 5$ is read with no comparison in degree 6, and the identity of (a) can be formed in cone degrees $j \leq 3$.

5. ***(The precision rule.)** Over $\mathbb{Z}_{(\ell)}$, $\ell \in \{2,3\}$:*

(a) *$C(\mathsf{a}^{(\ell)*})$ on $\mathcal{I}^{(\ell)}_{\mathcal{T},\mathsf{O}_\ell}$, $\mathsf{B}^{(\ell)}_{\mathcal{T},\mathsf{O}_\ell}$ is a bounded below complex of finitely generated free O_ℓ-modules, one integral form for every O_ℓ-algebra; Proposition 31.20(6) holds for it **verbatim** — the Smith-valuation read-out, the stopping rule, the a-priori exponents, the truncation —, with the rank sources read on **this** complex: the Betti recursion with the dimensions of Proposition 31.20(5)(b), or the reduction modulo a qualifying $\ell' \nmid \mathsf{q}_\ell$ of this tree cone ((6)(c), (d) there), of $\mathsf{B}^{(\ell)}_\ell(M)$ for the boundary ((6)(a)); (6)(b)'s "*$\operatorname{rank}\mathsf{P}(M)^0$*" reads $\operatorname{rank}C^0_{\mathcal{T},\ell}(M)$ here. The rank of a differential of $C(r_{\mathcal{I}})$, of $C(r_{\mathcal{I}^{(\ell)}})$, of the cylinder's $C(r_{\mathrm{Cyl}})$ (its differentials' rational ranks differ from those of $C(\mathsf{a}^{(\ell)*})$ by the contractible summand $\mathsf{B}\oplus\mathsf{B}[-1]$; v1.23.0-pre.34, referee N4) or of the bulk tree alone is **not** admissible for it, nor conversely.*

(b) *The maps of* (79) *are read by Proposition 31.10(2)(g); the certifying ranks of (g)(iii) for them are invariants of the maps: over* $\mathbb{Q}$, $\operatorname{rank}\operatorname{res}^0 = b_0(M)$, $\operatorname{rank}\operatorname{res}^1 = b_1(\partial;M) - \dim(M_{\mathbb{Q}}^{\vee})^{\Gamma}$, $\operatorname{rank}\operatorname{res}^q = 0$ *(*$q \geq 2$*)*, $\operatorname{rank}\operatorname{inc}_{c*}^q = b_q(\partial;M) - \operatorname{rank}\operatorname{res}^q$, $\operatorname{rank}\operatorname{pr}_{c*}^q = b_q(M) - \operatorname{rank}\operatorname{res}^q$ *(Proposition 31.20(5)(b)).*

(c) *Every matrix of* $\mathsf{a}^{(\ell)*}$, $\Phi^*_{\mathcal{T}}$, $\Psi^*_{\mathcal{T}}$, Φ^*_{∂}, Ψ^*_{∂}, Y^*_{Φ}, Y^*_{Ψ}, j_{Φ}, j_{Ψ}, $T^{\mathrm{c}}_{\mathcal{T}}$ *has its entries in* O_{ℓ} *and is* ***exact*** *modulo every* ℓ^K *(exact data, no Smith form: no digit is lost). A torsion operator is read on the generators of the tree's presentation by Proposition 31.15(3)* ($2\sigma_{\max} \leq K$)*, and its free part by (3′) there.*

(d) ***(Glueing.)*** $(\mathrm{Cyl}(\mathsf{a}^{(\ell)}), \mathsf{B}^{(\ell)}_{\mathcal{T}}) \otimes \mathbb{Z}_{(\ell)}$ *is a* $\mathbb{Z}_{(\ell)}$*-pair model, so Proposition 31.20(2)(b) admits it for the* ℓ*-primary parts and the* ℓ*-local maps of* (79)*, read on* $C(\mathsf{a}^{(\ell)*})$ *through* ϖ*; the glueing datum of the* $\mathbb{Z}[1/6]$*-piece* (cmp) *is unchanged. A map between the tree's* ℓ*-local piece and anything computed on* $\mathcal{I}$ *(a Hecke operator, an explicit comparison) passes through* j_{Φ}, j_{Ψ}.

6. ***(Refusal and scope.)*** *A run on the tree pair* ***refuses****, never truncates: when its predicted RSS exceeds the ceiling (rule 19, before building); when a read exhausts the precision (the stopping rule of (5)(a), (c)); for Hecke, on every condition of Proposition 31.21(5) (the pair lift's codes), and when* $q_{\max} \notin \{4,5\}$ *(*5 *is the top chain degree* m *of the comparisons, (4)(e); v1.23.0-pre.35); when an exact product of the (L)-, (L′)-built maps leaves* 64*-bit signed integers (the cap).* ***A failed certificate of a constructed map*** *— a chain-map identity of* $\mathsf{a}^{(\ell)}$, $\Phi^{\mathbb{Z}}_{\partial}$, $\Psi^{\mathbb{Z}}_{\partial}$*, the identities of (3)(a), (3)(c), the strict blocks of (3)(d), (4)(a) — is a* ***failure of the code*** *(`hecke::HeckeInternalError`), not a refusal: by (1)–(4) it cannot occur for a correct implementation.* ***Scope****: the groups and maps for every* ℓ*, in every degree; the production pieces are* $\ell \in \{2,3\}$ *(*$\ell \geq 5$*: the Bredon tree pair (2)(c) is a check; D4's pairs stay the route);* T_p, $p \nmid N$*, by transport only; not* U_p*, not nebentypus Hecke operators, not* $\Gamma_1(N)$*; no native* τ*-type datum on the tree.*

Proof. (1)(a) The column $F^{\mathfrak{e}} \to R[\Gamma/\langle\mathbf{c}\rangle]$ is a projective resolution (Proposition 31.12(1)(b)); $r_{(1-T)e_{\mathfrak{e}}}$ is a chain endomorphism (T commutes with $\mathbf{c}$ and $e_{\mathfrak{e}}$), lying over $y\langle\mathbf{c}\rangle \mapsto y\langle\mathbf{c}\rangle - yT\langle\mathbf{c}\rangle$ on $R[\Gamma/\langle\mathbf{c}\rangle]$. $\Gamma_{\infty}/\langle\mathbf{c}\rangle = \langle\bar{T}\rangle \cong \mathbb{Z}$ acts freely on the right of $\Gamma/\langle\mathbf{c}\rangle$ with orbit set $\Gamma/\Gamma_{\infty} = \mathbb{P}^1$ ((B2)), so on each orbit $\cong \mathbb{Z}$ the map $1-T$ is injective on $R[\mathbb{Z}]$ with cokernel R: it is injective with cokernel $R[\mathbb{P}^1]$. The cone of a map of resolutions has homology the cone of the map on H_0 [standard: [Wei94, §1.5]]: $H_1 = \ker = 0$, $H_0 = \mathrm{coker} = R[\mathbb{P}^1]$. $\varepsilon^{(\ell)}_{\mathsf{B}}$ is the column augmentation, a map of cones to the cone $C_{\bullet}(\mathcal{L}) \otimes R$ of $1-T$ on $R[\Gamma/\langle\mathbf{c}\rangle]$, a quasi-isomorphism by the five lemma, surjective. (1)(b) $e_{\mathfrak{e}}$ is central in $R[\langle S\rangle]$ and $R[\langle U\rangle]$ and $e_{\mathfrak{e}}e_S = e_S$, $e_{\mathfrak{e}}e_U = e_U$ (Proposition 31.12(1)(a)), and $e_{\mathfrak{e}}$ commutes with S^{-1}, T. *Chain map*, on e_t: $a_t\delta^S_t = \varepsilon_t a_{t-1}$, the S-half of the chain-map identity of the canonical lift (loc. cit.). On z_t, write $\gamma := (S^{-1}-T)e_{\mathfrak{e}} = S^{-1}(1-U)e_{\mathfrak{e}}$ ($S^{-1}U = T$) and α_t, β_t for the v^S_{t+1}-, v^U_{t+1}-coefficients; $\partial\mathsf{a}(\mathsf{z}_t) = \mathsf{a}(\partial\mathsf{z}_t)$ reads, on $F^{\mathfrak{e}}_{t-1}$: $-\gamma\varepsilon_t = -\varepsilon_t\gamma$ (central ε_t); on F^S_t: $\alpha_t\delta^S_{t+1} = (1-S^{-1})a_t - \varepsilon_t\alpha_{t-1}$ ($1-T-\gamma = 1-S^{-1}$ on e_S); on F^U_t: $\beta_t\delta^U_{t+1} = -\gamma b_t - \varepsilon_t\beta_{t-1}$. For even t ($\alpha_t = -S^{-1}e_S$, $\beta_t = S^{-1}e_U$, $\alpha_{t-1} = \beta_{t-1} = 0$, $\delta^S_{t+1} = (1-S)e_S$, $\delta^U_{t+1} = (1-U)e_U$): $-S^{-1}(1-S) = 1-S^{-1}$ and $a_t = 1$ (resp. e_S), and $S^{-1}(1-U)e_U = -\gamma b_t$ as $b_t = -e_U$ (-1 on the free model). For odd t ($\alpha_t = \beta_t = 0$, $\alpha_{t-1} = -S^{-1}e_S$, $\beta_{t-1} = S^{-1}e_U$, $\varepsilon_t = 1-\mathbf{c}$): $(1-S^{-1})(1+S) = S - S^{-1} = -(1-\mathbf{c})S^{-1}$ ($\mathbf{c}S^{-1} = S$), so $(1-S^{-1})a_t - \varepsilon_t\alpha_{t-1} = 0$; and $-\gamma b_t = S^{-1}(1-U)(1+U+U^2) = S^{-1}(1-\mathbf{c}) = \varepsilon_t\beta_{t-1}$ on the free model, $S^{-1}(1-U)e_U = S^{-1}(1-\mathbf{c})e_U$ at $\ell = 2$ ($Ue_U = \mathbf{c}e_U$), so $-\gamma b_t - \varepsilon_t\beta_{t-1} = 0$. Where a target column vanishes the corresponding equation is empty or reads $(1-S^{-1})e_S = 0$ ($S \in Q_S$ at $\ell \geq 3$), resp. $\gamma e_U = S^{-1}(1-U)e_U = 0$ ($U \in Q_U$ at $\ell \geq 5$). *Over* ε *and* f_{a}: in degree 0, $\mathsf{a}(\mathsf{e}_0) = a_0 s_0$ augments to $\mathfrak{v}_S$, $\varepsilon(a_0) = 1$ ($\varepsilon^{(\ell)}_{\mathcal{T}}(a_0\mathsf{v}^S_0) = a_0\Gamma_S = \mathfrak{v}_S = \mathsf{f}_{\mathsf{a}}\varepsilon^{(\ell)}_{\mathsf{B}}(\mathsf{e}_0)$, as $a_0 \in \{1, e_S\}$); in degree 1 only the edge component $\gamma\mathsf{v}^{\mathfrak{e}}_0$ survives $\varepsilon^{(\ell)}_{\mathcal{T}}$, giving $(S^{-1}-T)\mathfrak{e} = \mathsf{f}_{\mathsf{a}}(\mathfrak{e}_{\infty}) = \mathsf{f}_{\mathsf{a}}\varepsilon^{(\ell)}_{\mathsf{B}}(\mathsf{z}_0)$; both sides vanish on column degrees ≥ 1. (1)(c) Lemma 31.24; $\mathcal{I}^{(\ell)}_{\mathcal{T}}$ is an R-model (Proposition 31.12(1)(b)), $\mathsf{B}^{(\ell)}_{\mathcal{T}}$ resolves $R[\mathbb{P}^1]$ by (a), $\mathsf{a}^{(\ell)}$ lies over ε by (b).

(2) $\mathrm{Hom}(r_a, M)$ is $m \mapsto am$, and $\mathrm{Hom}(R[\Gamma]e_x, M) = M^{Q_x}$ (Proposition 31.12(1)(c), (2)'s

proof): $d^q(u,w)$ is $\varepsilon_{q+1}u$ on e_{q+1} and $(1-T)u-\varepsilon_q w$ on z_q; $(\mathsf{a}^*c)(\mathsf{e}_q)=a_q x$, $(\mathsf{a}^*c)(\mathsf{z}_{q-1})=-[q\text{ odd}]S^{-1}x+[q\text{ odd}]S^{-1}y+(S^{-1}-T)z$ ($e_U y=y$, $e_S x=x$). At $\ell\geq 3$ only degrees ≤ 1 remain. At $\ell\geq 5$, $(1-S^{-1})M^{-I}=(1-S)M^{-I}$ (S^{-1} is invertible and commutes with $1-S$), and $(1-T)M^{-I}+(1-S^{-1})M^{-I}$ is $\sum_g(g-1)M^{-I}$ (S, T generate Γ); with 2 invertible $M^{-I}\cong M_{\{\pm I\}}$, whose Γ-coinvariants are M_Γ ($\sum_g(g-1)M=(S-1)M+(T-1)M$ by $(gh-1)m=(g-1)hm+(h-1)m$). (d): Proposition 31.12(2)(e), (1)(c).

(3)(a) Inspection on the representatives, as for Proposition 31.12(4)(a): $\partial\mathsf{i}_\partial(\mathfrak{e}_\infty)=-(Tv_0-v_0)=\mathsf{i}_\partial((1-T)\mathfrak{v}_\infty)$; $\mathsf{r}_\partial(\partial v_2v_3)=T\mathfrak{v}_\infty-\mathfrak{v}_\infty=\partial(-\mathfrak{e}_\infty)$; $\partial\mathsf{G}_\partial(v_2)=v_2-v_0$; on v_2v_3, $\mathsf{G}_\partial(Tv_0-v_2)=-(v_0v_1+v_1v_2)=v_2v_3-\mathsf{i}_\partial\mathsf{r}_\partial(v_2v_3)$; $\mathsf{r}_\mathcal{T}(v_0)-\mathsf{f}_\mathsf{a}\mathsf{r}_\partial(v_0)=\mathfrak{v}_U-\mathfrak{v}_S=\partial\mathsf{g}_\square(v_0)$; on v_2v_3, $-S\mathfrak{e}+(S^{-1}-T)\mathfrak{e}=-T\mathfrak{e}=\mathsf{g}_\square(Tv_0-v_2)$ ($S^{-1}\mathfrak{e}=S\mathfrak{e}$, $\mathbf{c}$ fixing $\mathfrak{e}$); g': the identity $\mathsf{i}_\mathcal{T}\mathsf{r}_\mathcal{T}=\mathrm{id}-\partial\mathsf{G}_\mathcal{T}-\mathsf{G}_\mathcal{T}\partial$ of Proposition 31.12(4)(a) composed with $j_X\mathsf{i}_\partial$, and the $\mathsf{g}_\square$-identity composed with i_∂ (using $\mathsf{r}_\partial\mathsf{i}_\partial=\mathrm{id}$), give $j_X\mathsf{i}_\partial-\mathsf{i}_\mathcal{T}\mathsf{f}_\mathsf{a}=\partial(-\mathsf{g}')+(-\mathsf{g}')\partial$; the displayed values are $\mathsf{G}_\mathcal{T}$'s table. (b) The columns of $\mathsf{B}_\mathcal{T}$ are the tree's edge column, contracted by Proposition 31.12(4)(b)'s h with the same sign rules, and the descent through the two columns terminates as there. *The generalised solver* (v1.23.0-pre.34, referee m2; loc. cit. states it for $p\in\{\mathrm{cmp},\varepsilon_\mathcal{T}\}$ only): the proof of Proposition 31.12(4)(b) uses of p only that it is the column augmentation of finitely many free periodic columns. On one column, h (with h_{-1}) is a contracting homotopy of the augmented column, $\partial\mathsf{h}+\mathsf{h}\partial=\mathrm{id}$ in degrees ≥ 1 and $\partial\mathsf{h}=\mathrm{id}-\mathsf{h}_{-1}\varepsilon$ in degree 0, a $\mathbb{Z}$-linear identity on the basis ρg^j; the global sign ϵ_s of the column in the total complex is compensated by $\epsilon_s\mathsf{h}$. If $z\in\ker p$ is a cycle with top component z_s in column s, then z_s is a cycle of column s in column degree ≥ 1, or in column degree 0 with augmentation 0 (as $pz=0$), so $\partial(\epsilon_s\mathsf{h}(z_s))$ has column-s component z_s and its other components lie in columns $<s$; $z-\partial k_s$ is a cycle of $\ker p$ with no column-s component, and induction on s ends after $s_{\max}+1$ steps with $\partial\mathsf{h}_p(z)=z$; $\mathsf{h}_p(z)$ lies in column degrees ≥ 1, so $p\,\mathsf{h}_p(z)=0$. (L) and (L′) then follow from these two properties alone, by the induction on the degree of loc. cit. Every identity is $\mathbb{Z}$-linear on bases, so it survives $\otimes R$; the $R[\Gamma]$-linear extensions of (L), (L′) need only that $P'\otimes R$ be free over $R[\Gamma]$ on the given basis, and nothing is asked of the coefficients of A, B, f, g beyond $R[\Gamma]$-linearity, so denominators in O_ℓ and the idempotents e_x are allowed. For $p=\varepsilon_\mathsf{B}$ the columns are e ($s=0$) and z ($s=1$, $\epsilon=-1$, the cone's shifted column); for $p=\mathrm{cmp}|_{\mathcal{I}^\partial}$ the columns $G_{s,\bullet}$ over Σ^∂_s, $s\leq 1$, and $\epsilon_s\mathsf{h}$ maps each into itself, so the descent stays in $\mathcal{I}^\partial$ (as in the boundary step of the proof of Proposition 31.21(1)). Φ^∂_ℓ, Ψ^∂_ℓ: the column computation of Proposition 31.12(3) for $x=\mathfrak{e}$, and $r_{(1-T)e}$ commutes with r_{ce} for central scalars c; the corrections k, k' of loc. cit. arise only between columns of different stabilisers, of which $\mathsf{B}_\mathcal{T}$ has none. (c) (L′)'s hypotheses: $\varepsilon_\mathcal{T}\Phi^\mathbb{Z}_\mathcal{T}j=\mathsf{r}_\mathcal{T}\mathrm{cmp}j=\mathsf{r}_\mathcal{T}j_X\mathrm{cmp}|_{\mathcal{I}^\partial}$; $\varepsilon_\mathcal{T}\Psi^\mathcal{T}_\ell\mathsf{a}^{(\ell)}\Phi_\partial=\varepsilon^{(\ell)}_\mathcal{T}\mathsf{a}^{(\ell)}\Phi^\partial_\ell\Phi^\mathbb{Z}_\partial=\mathsf{f}_\mathsf{a}\varepsilon^{(\ell)}_\mathsf{B}\Phi^\partial_\ell\Phi^\mathbb{Z}_\partial=\mathsf{f}_\mathsf{a}\varepsilon_\mathsf{B}\Phi^\mathbb{Z}_\partial=\mathsf{f}_\mathsf{a}\mathsf{r}_\partial\mathrm{cmp}|_{\mathcal{I}^\partial}$ ($\varepsilon_\mathcal{T}\Psi^\mathcal{T}_\ell=\varepsilon^{(\ell)}_\mathcal{T}$, $\varepsilon^{(\ell)}_\mathsf{B}\Phi^\partial_\ell=\varepsilon_\mathsf{B}$, (1)(b) with the ℓ-local augmentations; tensored with R throughout; v1.23.0-pre.34, referee m4); and $\mathrm{cmp}\Psi_\mathcal{T}\mathsf{a}^{(\ell)}\Phi^\partial_\ell=\mathsf{i}_\mathcal{T}\mathsf{f}_\mathsf{a}\varepsilon_\mathsf{B}$, $\mathrm{cmp}j\Psi^\mathbb{Z}_\partial=j_X\mathsf{i}_\partial\varepsilon_\mathsf{B}$, with (a)'s homotopies. Then $\Phi_\mathcal{T}j-\mathsf{a}\Phi_\partial=\Phi^\mathcal{T}_\ell(\Phi^\mathbb{Z}_\mathcal{T}j-\Psi^\mathcal{T}_\ell\mathsf{a}\Phi_\partial)$ by $\Phi^\mathcal{T}_\ell\Psi^\mathcal{T}_\ell=\mathrm{id}$, and $\Psi_\mathcal{T}\mathsf{a}-j\Psi_\partial=(\Psi_\mathcal{T}\mathsf{a}\Phi^\partial_\ell-j\Psi^\mathbb{Z}_\partial)\Psi^\partial_\ell$ by $\Phi^\partial_\ell\Psi^\partial_\ell=\mathrm{id}$. (d) *Cochain maps*: the cone lemma ($*$) of Proposition 31.12's proof, dualised: $(\alpha,c)\mapsto(F^*\alpha,G^*c+K\alpha)$ commutes with the two D iff $dK+Kd=G^*\mathsf{a}^*-j^*F^*$, and (c) dualises to $j^*\Phi^*_\mathcal{T}-\Phi^*_\partial\mathsf{a}^*=\mathsf{Y}^*_\Phi d+d\mathsf{Y}^*_\Phi$ ($(\partial Y)^*=Y^*d$, $(Y\partial)^*=dY^*$), so $K=-\mathsf{Y}^*_\Phi$; likewise for j_Ψ. The strict identities are read off. *The comparison of pairs.* Put $\mathrm{pr}_\partial\colon\mathcal{I}\to\mathcal{I}^\partial$ the projection along the columns over $\Sigma\setminus\Sigma^\partial$, $\widehat{\mathsf{Y}}:=\mathsf{Y}_\Phi\mathrm{pr}_\partial$, and $\Theta_\Phi:=i_\mathsf{P}\Phi_\mathcal{T}-\partial K_\Phi-K_\Phi\partial\colon\mathcal{I}\otimes R\to\mathrm{Cyl}(\mathsf{a}^{(\ell)})$ with $K_\Phi(y):=(\mathsf{Y}_\Phi y,0,-\Phi_\partial y)$ on $\mathcal{I}^\partial$, 0 on the complement. Since $i_\mathsf{P}\mathsf{a}-i_\mathsf{B}=-(\partial s+s\partial)$ on B (s of Lemma 31.24's proof), $i_\mathsf{P}\Phi_\mathcal{T}j-i_\mathsf{B}\Phi_\partial=\partial K_\Phi j+K_\Phi\partial j$, so $\Theta_\Phi|_{\mathcal{I}^\partial}=i_\mathsf{B}\Phi_\partial$: Θ_Φ is a comparison of pairs $(\mathcal{I},\mathcal{I}^\partial)\to(\mathrm{Cyl},\mathsf{B})$ (over id_R: ∂K_Φ has augmentation 0). As $\pi_\mathrm{Cyl}\Theta_\Phi=\Phi_\mathcal{T}-\partial\widehat{\mathsf{Y}}-\widehat{\mathsf{Y}}\partial$,

$$\mathsf{j}_\Phi-(\Theta^*_\Phi[1],\Phi^*_\partial)\circ\varsigma'=D\Upsilon+\Upsilon D,\qquad\Upsilon(\alpha,c):=(-\widehat{\mathsf{Y}}^*\alpha,0)$$

(both sides send (α,c) to $(d\widehat{\mathsf{Y}}^*\alpha+\widehat{\mathsf{Y}}^*d\alpha,-\mathsf{Y}^*_\Phi\alpha)$, as $r\widehat{\mathsf{Y}}^*=\mathsf{Y}^*_\Phi$). And $\Theta_\Psi(x,b,u):=\Psi_\mathcal{T}x+j\Psi_\partial b-\mathsf{Y}_\Psi u$ is a comparison of pairs $(\mathrm{Cyl},\mathsf{B})\to(\mathcal{I},\mathcal{I}^\partial)\otimes R$ ($\partial\Theta_\Psi-\Theta_\Psi\partial=(\Psi_\mathcal{T}\mathsf{a}-j\Psi_\partial-\partial\mathsf{Y}_\Psi-\mathsf{Y}_\Psi\partial)u=0$) with $\mathsf{j}_\Psi=\varpi\circ(\Theta^*_\Psi[1],\Psi^*_\partial)$ **exactly**. Hence, by Lemma 31.19(2), (3) for the self-comparisons $\Theta_\Phi\Theta_\Psi$ of $(\mathrm{Cyl},\mathsf{B})$ and $\Theta_\Psi\Theta_\Phi$ of

$(\mathcal{I},\mathcal{I}^{\partial})$, and Lemma 31.24(2): $\mathsf{j}_\Psi\mathsf{j}_\Phi \simeq \varpi\big((\Theta_\Phi\Theta_\Psi)^*[1],(\Phi_\partial\Psi_\partial)^*\big)\varsigma' \simeq \varpi\varsigma' = \mathrm{id}$ and $\mathsf{j}_\Phi\mathsf{j}_\Psi \simeq (\Theta_\Phi^*[1],\Phi_\partial^*)\varsigma'\varpi(\Theta_\Psi^*[1],\Psi_\partial^*) \simeq \big((\Theta_\Psi\Theta_\Phi)^*[1],(\Psi_\partial\Phi_\partial)^*\big) \simeq \mathrm{id}$. (The five lemma alone would not do: a map of short exact sequences of cones whose outer maps induce the identity need not induce the identity on the cone, Proposition 15.16.) Independence: two choices give two comparisons of pairs, homotopic as pairs (Lemma 31.19(2)).

(4)(a) Expand $\mathsf{j}_\Psi T\mathsf{j}_\Phi(\alpha,c) = \mathsf{j}_\Psi(T_\mathrm{b}\Phi_\mathcal{T}^*\alpha,\ T_\partial\Phi_\partial^*c - T_\partial\mathsf{Y}_\Phi^*\alpha)$. A triangular endomorphism $(\alpha,c)\mapsto(A\alpha,Bc+E\alpha)$ of $C(\mathsf{a}^*)$ commutes with D iff A, B are cochain maps and $dE+Ed = B\mathsf{a}^*-\mathsf{a}^*A$ (compare the second components of $D(A\alpha,Bc+E\alpha)$ and of the image of $(-d\alpha,dc+\mathsf{a}^*\alpha)$); $T_\mathcal{T}^\mathrm{c}$ is a composite of cochain maps. *The pair model* (v1.23.0-pre.34, referee m1): Θ_Φ, Θ_Ψ are comparisons of pairs, so $r_\mathrm{Cyl}\Theta_\Psi^* = \Psi_\partial^* r$ and $r\,\Theta_\Phi^* = \Phi_\partial^* r_\mathrm{Cyl}$ strictly, and with $r\,T_\mathrm{b} = T_\partial r$ every factor of T^Cyl is diagonal; by the proof of (3)(d), $\mathsf{j}_\Psi = \varpi(\Theta_\Psi^*[1],\Psi_\partial^*)$ exactly and $\mathsf{j}_\Phi \simeq (\Theta_\Phi^*[1],\Phi_\partial^*)\varsigma'$ (by Υ), so $T_\mathcal{T}^\mathrm{c} - \varpi T^\mathrm{Cyl}\varsigma' = D\Upsilon'+\Upsilon' D$ with $\Upsilon' := \mathsf{j}_\Psi T\Upsilon$. That $\eta_\mathcal{T}$ need not vanish and the diagonal operator need not be a cochain map is an *observation* of the checks (arm H below, for $T=\theta^*$, $\theta=\mathrm{id}+\partial K+K\partial$, $K(\mathcal{I}^\partial)\subseteq\mathcal{I}^\partial$; the referee's arm H for T_ε), not a proved assertion; nothing else in (4) depends on it. (b) The blocks are read off; $H(\mathsf{j}_\Psi) = H(\mathsf{j}_\Phi)^{-1}$ by (3)(d), and Proposition 31.21(3) for $\Phi_p^{(N)}$. (c) By (3)(d), $H(T_1^\mathrm{c})H(T_2^\mathrm{c}) = H(\mathsf{j}_\Psi)H(T_1)H(\mathsf{j}_\Phi)H(\mathsf{j}_\Psi)H(T_2)H(\mathsf{j}_\Phi)$ and $H(\mathsf{j}_\Phi\mathsf{j}_\Psi)=\mathrm{id}$. (d) u acts by $\mathrm{Hom}(-,u)$, which commutes with precomposition by every chain map; Proposition 31.21(6)'s proof then applies. (e) (v1.23.0-pre.35.) *(i)* The targets $\mathcal{I}\otimes R$, $\mathcal{I}^\partial\otimes R$, $\mathcal{I}_\mathcal{T}^{(\ell)}$, $\mathsf{B}_\mathcal{T}^{(\ell)}$, $\mathsf{B}_\mathcal{T}\otimes R$ are acyclic in positive degrees ($\mathcal{I}$, $\mathcal{I}^\partial$ are resolutions by free abelian groups, hence $\mathbb{Z}$-split, and stay exact after $\otimes R$; the ℓ-local models by Proposition 31.12(1)(b) and (1)(a) here), and the sources are projective over $R[\Gamma]$. The comparison theorem's induction [standard: [Wei94, Thm. 2.2.6]] extends a family f_q, $q\le m$, with $\partial f_q = f_{q-1}\partial$: for a summand $R[\Gamma]e$ of P_{m+1} (e idempotent; $e=1$ where the source is free; v1.23.0-pre.36, the C-D8b referee's N3), $f_m\partial e$ is a cycle ($\partial f_m\partial = f_{m-1}\partial\partial = 0$) of positive degree m, hence a boundary ∂y, and $f_{m+1}(x) := xy$ for $x\in R[\Gamma]e$ (Γ-linear, and $\partial f_{m+1}(x) = x\,\partial y = xf_m\partial e = f_m\partial x$ since $x=xe$; equivalently, by projectivity). A homotopy Y_q, $q\le m-1$, with $f_q-g_q = \partial Y_q + Y_{q-1}\partial$ extends likewise: $z := f_m - g_m - Y_{m-1}\partial$ satisfies $\partial z = (f_{m-1}-g_{m-1}-\partial Y_{m-1})\partial = Y_{m-2}\partial\partial = 0$, so $z(e)=\partial y'$ and $Y_m(x) := xy'$ on each summand $R[\Gamma]e$, as before; then the maps in degree $m+1$, and so on. The conditions "over" the augmentations and the tables live in degrees 0, $1\le m$, and the proofs of (3)(d) and (4)(a)–(c) use of Φ, Ψ, Y only the identities (3)(b), (c) (not that they were built by (L), (L′)), so they apply to the extension. *(ii)* By the formulas of (3)(d) and (4)(a): on $(\alpha,c)\in C_{\mathcal{T},\ell}^{j+1}\oplus\mathsf{B}_\ell^j$, j_Φ uses $\Phi_\mathcal{T}^*$ in cochain degree $j+1$, Φ_∂^* in degree j and $\mathsf{Y}_\Phi^*\colon C^{j+1}\to\mathcal{I}^{\partial j}$, the dual of Y_Φ in chain degree j; likewise j_Ψ and T in cone degree j. *(iii)* Let $\widetilde{T}^\mathrm{c}$ be built from an extension; it is a cochain map by (4)(a), (b), and agrees with $T_\mathcal{T}^\mathrm{c}$ in cone degrees $\le m-1$ by (ii). For a cocycle g of cone degree $j\le m-1$, $DT_\mathcal{T}^\mathrm{c}g = D\widetilde{T}^\mathrm{c}g = \widetilde{T}^\mathrm{c}Dg = 0$ (the extension enters only through $\widetilde{T}^\mathrm{c}(0)=0$); for $g=Dx$, x of cone degree $j-1$, $T_\mathcal{T}^\mathrm{c}g = \widetilde{T}^\mathrm{c}Dx = D\widetilde{T}^\mathrm{c}x = DT_\mathcal{T}^\mathrm{c}x$; and the induced map is $H(\widetilde{T}^\mathrm{c})$, which is (b)'s and, by (c), does not depend on the extension. *(iv)* $\eta_\mathcal{T}^{j+1}$ is the off-diagonal block of $T_\mathcal{T}^\mathrm{c}$ in cone degree $j+1$, so by (ii) it uses Φ, Ψ in chain degree $j+2$; formable iff $j+2\le m$. By the triangular criterion of the proof of (a), $T_\mathcal{T}^\mathrm{c}$ commutes with D on cone degree j iff $T_\mathcal{T}'$ on C^{j+1}, T_∂' on B^j commute with d, the first component of $T_\mathcal{T}^\mathrm{c}(0,c)$ vanishes, and the identity of (a) holds on $C_{\mathcal{T},\ell}^{j+1}$; with $j=q-2$ this gives the coboundary half of (iii) for H_c^q (commuting with D maps coboundaries to coboundaries; the converse need not hold). At $\ell=3$, $\eta_\mathcal{T}^j$ has target $\mathsf{B}_3^j=0$ for $j\ge 2$.

(5)(a) The integral forms are Proposition 31.12(1)(d) and (1)(a) here (every coefficient of $\mathsf{B}_\mathcal{T}^{(\ell)}$ and $\mathsf{a}^{(\ell)}$ in $\mathsf{O}_\ell[\Gamma]$). Proposition 31.20(6)'s proof uses of the cone only that it is a bounded below complex of finitely generated free $\mathbb{Z}_{(\ell)}$-modules computing $H_c^\bullet$ and the boundary term (here through ϖ, Lemma 31.24), since the elementary divisors of a differential of any such complex are the torsion exponents of the next cohomology group, and the rank sources' criteria are properties of the groups; the exclusion of other complexes' ranks is (6)'s own argument

(the rank of a differential is a property of that differential). (b) The rational long sequence of Proposition 31.20(5)(b) is exact with $H^0_c = 0$, res^0 injective and $H^q(\Gamma; M_{\mathbb{Q}}) = H^q(\Gamma_\infty; M_{\mathbb{Q}}) = 0$ for $q \geq 2$. (c) The comparisons are built over $\mathbb{Z}$ (resp. O_ℓ: $\Phi^{\mathcal{T}}_\ell$, $\Psi^{\mathcal{T}}_\ell$, Φ^{∂}_ℓ, Ψ^{∂}_ℓ have coefficients in $\mathsf{O}_\ell[\Gamma]$, and (L), (L′) preserve integrality); products of exact matrices are exact. (d) Proposition 31.20(2)(b) and Lemma 31.24. (6) collects (1)–(5). □

Remark 31.26 (What the plan's §5 stated, and what holds). [proved here] (v1.23.0-pre.33.) (i) **The construction.** The owner's candidate — the cylinder or cone of $\mathrm{Ind}^{\Gamma}_{\Gamma_\infty} C(\text{axis of } T) \to C(\mathcal{T})$ — holds in the algebraic form of the plan's §5.1, with the *minimal* cusp resolution $(1, 2, 2, \dots$; the plan's "F^∞"), not the cellular axis $(2, 4, 4, \dots)$: the axis map a needs no geometric axis, only the path from $T\mathfrak{v}_S$ through $S^{-1}\mathfrak{v}_U$ $(= T\mathfrak{v}_U)$ to $\mathfrak{v}_S$, i.e. $\mathfrak{e}_\infty \mapsto (S^{-1} - T)\mathfrak{e}$, in the orientation of $\mathfrak{e}_\infty$ ($\partial\mathfrak{e}_\infty = (1 - T)\mathfrak{v}_\infty$, $\partial\mathsf{f}_{\mathsf{a}}(\mathfrak{e}_\infty) = \mathfrak{v}_S - T\mathfrak{v}_S$; v1.23.0-pre.34, referee N2). (ii) **The computed object is the cone of a^*, not the cylinder pair's cone**: the cylinder carries the boundary twice and is only the pair model that lets §31.4 apply (Lemma 31.24). (iii) **"No η appears on the tree side" (plan §5.4(c)1) holds for $C(r_{\mathrm{Cyl}})$ and fails for $C(\mathsf{a}^*)$**, the object the code computes (v1.23.0-pre.34, referee m1): transported along the strict comparisons of pairs Θ_Φ, Θ_Ψ, the operator on the pair model's cone is diagonal, $\eta = 0$, as the plan says; on $C(\mathsf{a}^*)$, which is not the cone of a restriction to a subcomplex, $T^{\mathrm{c}}_{\mathcal{T}} \simeq \varpi T^{\mathrm{Cyl}} \varsigma'$ carries the off-diagonal $\eta_{\mathcal{T}}$ of Proposition 31.25(4)(a), observed non-zero in the checks (not proved in general); it is determined, so the ambiguity of Proposition 15.16 does not return. (iv) The plan's "strict maps of pairs $(\mathcal{I}, \mathcal{I}^\partial) \rightleftarrows (\mathrm{Cyl}, \mathsf{B})$" exist ($\Theta_\Phi$, Θ_Ψ of the proof of (3)(d)); the code uses the cone maps j_Φ, j_Ψ of the homotopy-commutative squares instead, which are smaller and need no cylinder; that they are mutually inverse on cohomology is proved through Θ_Φ, Θ_Ψ, not by the five lemma. (v) The comparison is over $\mathbb{Z}$ and $\mathbb{Z}[\mathbb{P}^1]$ through $C_\bullet(\mathcal{T})$, $C_\bullet(\mathcal{L})$ and $C_\bullet(\partial X)$, as the plan said; the geometric statement through Ash's retraction stays **not needed**; "every edge lies on exactly two axes" (plan §5.1) is not used and not verified here. (vi) **Cost.** The groups need the tree cone only (ranks (2)(d)); the comparisons and the transport touch $\mathcal{I}$-sized cochains, as a product, never an elimination; the plan's proxy ($\approx \frac{1}{170}$ of the $\mathcal{I}^{(2)}$ pair in time) is a prediction, not a statement of this document.

A check outside the build, not a tag (v1.23.0-pre.33): `docs/implementation/slot24/cd8_check.py` (`python3 -I`, pure integers and fractions, no computer algebra system), output `cd8_check.out`. It imports the group ring, the tree models and $\Phi^{\mathcal{T}}_\ell$, $\Psi^{\mathcal{T}}_\ell$ of `cd6_tree_check.py` and the model $\mathcal{I}$, the solvers and (L), (L′) of `cd6_iso_check.py`, reads `d4_pair_oracle.tsv` as data, and builds everything of this subsection from its text: $\mathsf{B}^{(\ell)}_{\mathcal{T}}$, $\mathsf{a}^{(\ell)}$ at $\ell \in \{0, 2, 3, 5\}$, the tables of Definition 31.23(4), $\Phi^{\mathbb{Z}}_\partial$, $\Psi^{\mathbb{Z}}_\partial$, $\mathsf{H}^{\mathsf{B}}_\partial$, H^{X}_∂, y_ℓ, y'_ℓ, j_Φ, j_Ψ, the cylinder maps and $T^{\mathrm{c}}_{\mathcal{T}}$. Exactly in $\mathbb{Q}[\Gamma]$ to degree 6: $\partial^2 = 0$ on the four cusp models, $\mathsf{a}^{(\ell)}$ a chain map over f_{a}, the identities of (3)(a)–(c); on the 32 cells of the D4 oracle (level 1, $n \leq 12$; $\Gamma_0(N)$, $N \leq 13$), over $\mathbb{Z}/\ell^K$: (2)'s closed forms entry by entry, $H^q(\mathsf{B}^{(\ell)})$ against the oracle's boundary groups and H^q_c, $q \leq 4$, of the tree cone against its H^q_c (ℓ-parts at 2, 3, 5; ranks from the Betti recursion, certified pivots, a-priori exponents), the lengths of H^q of the tree cone against $(\mathcal{I}, \mathcal{I}^\partial)$'s cone, j_Φ, j_Ψ cochain maps with the strict inc_c, pr_c blocks and mutually inverse on cohomology modulo ℓ, Lemma 31.24(2)'s identities; and (4) for $T = \theta^*$ ($\theta = \mathrm{id} + \partial K + K\partial$, $K(\mathcal{I}^\partial) \subseteq \mathcal{I}^\partial$, not a fibre map) and for fibre maps $u \in \{1, 1 + 2J\}$ (J the all-ones endomorphism of $\mathbb{Z}[\mathbb{P}^1(\mathbb{F}_N)]$, $N \in \{5, 7\}$): $T^{\mathrm{c}}_{\mathcal{T}}$ a cochain map with the stated blocks, $\eta_{\mathcal{T}} \neq 0$, inducing $H(T)$. Planted faults, each detected: S for S^{-1} in a (at chain level, exactly in $\mathbb{Q}[\Gamma]$, at $\ell \in \{0, 2\}$ only: at $\ell \geq 3$ it is the same map, Definition 31.23(2); the C-D8 referee's check detects instead the dropped term $S^{-1}e_U\mathsf{v}^U_1$ at $\ell = 0, 2, 3$; v1.23.0-pre.34, referee m3); a sign of r_∂; $+\mathsf{Y}^*_\Phi$ in j_Φ; $1 - T^2$ for $1 - T$ in B; $\eta_{\mathcal{T}}$ dropped (for $T = \theta^*$ the diagonal operator fails to be a cochain map on all 32 cells at $\ell \in \{0, 2\}$ and on 24 of the 32 at $\ell = 3$ — at $\ell = 3$ the 7 cells with n odd have $M^{-I} = 0$, so no 3-local boundary and $\eta_{\mathcal{T}} = 0$, and at $(1, 0)$ it happens to commute; for fibre maps $\eta_{\mathcal{T}}$ is a cocycle and the diagonal operator is a cochain map). `CD8_CHECK degree=6`

`cells=32 checks=67252 mismatch=0 ... verdict=OK` (about 15 minutes). Not exercised: $T = \Phi_p^{(N)}$ (the pair lift is in the code only); the code's own $\mathcal{I}$ (the check's $\mathcal{I}$ is built by the same procedure, not read from `cohomology_app`). *Extended at v1.23.0-pre.35* by two arms: (M3) on every cell at $\ell \in \{0, 2\}$, $\rho_{M_N}(\mathbf{c}) = (-1)^n$, and the closed form of a^* with S for S^{-1} equals the true one modulo 2^{20} in every degree $q \le 4$ iff n is even (Definition 31.23(2)); (Q) the read at $q_{\max} = 5$ (Proposition 31.25(4)(e)): in cone degree 4, $T^{\mathrm{c}}_{\mathcal{T}}$ built from the maps of chain degree ≤ 5 maps every cocycle modulo p to a cocycle and induces $H(T)$ on H^5_c (non-zero modulo p on cells at $\ell = 0$ and $\ell = 2$), and with the degree-6 extension (the check builds its comparisons to degree 6) the identity of (4)(a) holds in cone degree 4 as well. `CD8_CHECK degree=6 cells=32 checks=68168 mismatch=0 ... verdict=OK` (about 17 minutes; (M3) $256 + 64$ checks, (Q) $3 \times 198 + 2$, $H^5_c \neq 0$ modulo p on all 32 cells at $\ell = 0$ and at $\ell = 2$).

Code map (v1.23.0-pre.33, **stated before D5c's code**, contract first, rule 21; the status of v1.23.0-pre.35). **Built, audited and merged** into `hecke-extensions` at `4064e164` (v1.23.0-pre.36; built at v1.23.0-pre.35 on branch `s24-d5cimpl`, then at `e720f11c`; the audit `docs/implementation/slot24/D5c_code_audit.md` at `f7af1744`, ACCEPT after fixes, 0 M / 4 m / 7 N, its m1–m4, N1, N2, N4, N6 fixed at `e75c3937`, N3, N7 this document's, N5 deferred (below); the C-D8b referee's Arm (G) at `3e4bc339`; `docs/implementation/slot24/DESCENT5c_report.md`, §9 for the fixes). The chain-level half is `Integral/Model/TreePair.{h,cpp}`: the cusp models (`CuspModelChain`), the axis map (`AxisMapChain`), the tables of Definition 31.23(4), and `TreePairComparison` ($\Phi^{\mathbb{Z}}_{\partial}$, $\Psi^{\mathbb{Z}}_{\partial}$, Φ^{∂}_{ℓ}, Ψ^{∂}_{ℓ}, y_{ℓ}, y'_{ℓ}, Y_{Φ}, Y_{Ψ}), whose `certify()` checks every identity of (1)(b) and (3)(a)–(c) exactly in $\mathbb{Q}[\Gamma]$, once per ℓ and process, in chain degrees ≤ 5, on D5's comparison now built to degree 5 (`kTreeComparisonTop`). The production half is `Integral/Descent/DescentTreePair.{h,cpp}`: `BuildTreePairComplex` (the closed forms of (2), read by D4's `RunPairLes` verbatim, with $d_{\mathsf{B}} d_{\mathsf{B}} = 0$ and $d_{\mathsf{B}} \mathsf{a}^* = \mathsf{a}^* d_{\mathcal{T}}$ checked entrywise on every call), `TreeConeRanksModPrime` (the rank source on **this** cone, (5)(a)), `TreePairCochains` (every map of (3)(d), (4) applied to vectors, never as a dense $\mathcal{I}$-sized matrix) and `ReadTreePairHecke`. **The tree cone is the default route** of the 2- and 3-adic pair pieces of `AssembleDescentPair`; the read-only rollbacks `HECKE_DESCENT_TREE_PAIR=0` and `HECKE_DESCENT_TREE=0` select D4's dense pair (`EFFECTIVE HeckeDescentTreePair route=...`). Validations: `IntegralDescentTreePairValidation` (standard; 2934 checks, 0 failing at the merge) and `IntegralDescentTreePairSlowValidation` (slow), registered in `Registry/Slot24/DESCENT5c.cpp`, sources `cmake/slot24/DESCENT5c.cmake`; D0–D5's and D3's standard names pass on the default and on both rollbacks, and `IntegralDescentPairValidation` prints its route (`EFFECTIVE IntegralDescentPairValidation`, `route=tree` or `route=dense`; the audit's N4). **Reach** (the report's §4): the 2-adic pair wall moves from $D \approx 1060$–1240 (D4: refusal, resp. 40 GiB) to the refusal at $D \approx 6340$ (rule 19, before building), ≈ 8000 at 40 GiB; the next wall is set by D2's dense Smith form with transforms, not by the model; largest measured cell $D = 6000$. (Re-timed nearly idle at the merge, the report's §9.4: $D = 852$ in 11.3 s, $D = 4500$ in 1325 s at a 12.8 GiB peak; the filed time model, $\propto D^{2.97}$, is an upper estimate at $D \ge 4500$, the two idle points giving exponent 2.86; the memory model is unchanged.) **Deferred** (the audit's N5): the validation-only fault hooks (`ActiveTreePairFault()`, a process-wide atomic as D3's `ScopedD3Fault`, and the planted error of `ScopedPlantedCocycleEta`) are read inside production paths (`BuildTreePairComplex`, the cusp model of `TreePair.cpp`, `j_phi`, `apply_hecke`, `y_psi_star`); safe while the validations run serially, as they do; making them thread-local or injected changes production signatures, and is left. The ℓ-local column augmentations $\varepsilon^{(\ell)}_{\mathcal{T}}$, $\varepsilon^{(\ell)}_{\mathsf{B}}$ of Definition 31.23(4) are the code's (`TreeModelChain::augment`: the cosets of the column groups in column degree 0, 0 above). `D3Refusal::HeldOutMismatch` is removed, which closes the open code item of §31.3's code map (the C-D8 referee's N7). Below, the obligations as stated before the code, with (e) and (f) amended at v1.23.0-pre.35 and (c), (e), (f), (g) at

v1.23.0-pre.36; on D5's `Integral/Model/TreeCochains` and `IsotropyModelTree` and D4's `Integral/Descent/DescentPair`: **Obligations of the code**: (a) $\mathsf{B}_{\mathcal{T}}^{(\ell)}(M)$ and $\mathsf{a}^{(\ell)*}$ in the closed form of (2), block by block for a module given by its block-monomial action, and the tree cone to cochain degree $q_{\max}+1$, $4 \le q_{\max} \le 5$ when Hecke is asked; (b) certification by (5)(a) on **this** cone, never with a rank of $(\mathcal{I}, \mathcal{I}^{\partial})$'s cone; the boundary by its closed form (Proposition 31.20(4), `BoundaryClosedForm`); (c) the tree cone as the route of the 2- and 3-adic pair pieces, **every** N (amended at v1.23.0-pre.36, the C-D8b referee's N7, the audit's N3: the code routes every 3-adic pair piece to the tree, $\nu_3(N) = 0$ included, and, since the audit's m3, $3 \in$ `extra_primes` with n odd, where the 3-local cusp cochains have rank $\operatorname{rank} e_{\mathfrak{e}} M = D \cdot [n \text{ even}] = 0 = \operatorname{rank} M^{-I}$, as in (2)(b); D4's 3-adic piece was dense everywhere, so nothing is lost), with a **read-only, value-preserving rollback** switch to D4's dense pair, read once per call — the preserved values are **the record's invariants** (the group types and the choice-free Hecke invariants of Proposition 31.15), not Smith generators or coordinates, which the two cones do not share; "invariants only", as ruled for the descent (v1.23.0-pre.34, referee N6); (d) $\Phi_{\partial}^{\mathbb{Z}}$, $\Psi_{\partial}^{\mathbb{Z}}$, y_{ℓ}, y'_{ℓ} by (L), (L′) on the code's $\mathcal{I}$, built once per run to the degree in use, with their certificates (chain maps, over r_{∂}, i_{∂}, the two square identities) checked when built, a failure thrown as `hecke::HeckeInternalError`; (e) $T_{\mathcal{T}}^{\mathrm{c}} = \mathsf{j}_{\Psi}\Phi_p^{(N)}\mathsf{j}_{\Phi}$ for every $p \nmid N$ with a certified pair lift, applied to vectors (the presentation's generators), never as a dense $\mathcal{I}$-sized matrix. **Certificates** (v1.23.0-pre.35; **restated at v1.23.0-pre.36** for the merged code, the D5c audit's m1 at `e75c3937` and the C-D8b referee's Arm (G) at `3e4bc339`, the C-D8b referee's m1; Proposition 31.25(4)(e) with $m = 5$): the read of H_c^q, $q \le q_{\max} \le 5$, rests on (1)–(4), the chain-level certificates of (d) (exact, independent of M and of D) and the pair lift's own; the cochain-level checks (`certify_transport`) are tests of the module-level assembly, a failure being a `HeckeInternalError`. In every cone degree $j \le q_{\max} - 1$ of the read they run on **the operator as applied** (`apply_hecke`, whose values on the presentation's generators are the values read): (i) its blocks equal the separately assembled $T'_{\mathcal{T}}$, $\eta_{\mathcal{T}}$, T'_{∂} (`bulk_hecke`, `eta`, `bdry_hecke`) — $T_{\mathcal{T}}^{\mathrm{c}}(\alpha, 0) = (T'_{\mathcal{T}}\alpha, \eta_{\mathcal{T}}\alpha)$ and $T_{\mathcal{T}}^{\mathrm{c}}(0, c) = (0, T'_{\partial}c)$, the strict block included — exactly modulo ℓ^K; (ii) $DT_{\mathcal{T}}^{\mathrm{c}} g = 0$ on the presentation's generators g modulo $\ell^{K-\sigma_{\max}}$; (iii) the identity of (4)(a) on the same α, with the values $\eta_{\mathcal{T}}\alpha$, $T'_{\mathcal{T}}\alpha$ just compared in (i), in every cone degree $j \le \min(q_{\max} - 1, 3)$ — all of them at $q_{\max} = 4$; at $q_{\max} = 5$ the top cone degree 4, which would need $\Phi_{\mathcal{T}}$ in degree 6 and which the read does not use ((4)(e)(iv)), is **counted as skipped** (`eta_skipped`, printed; (i) and (ii) still run there), and an arm (Q5) of the standard validation compares T_p on H_c^q, H^q, $H^q(\partial)$, $q \le 5$, with D4's `BuildConeHecke(...,6)` through $H(\mathsf{j}_{\Phi})$, $H(\Phi_{\mathcal{T}}^*)$, $H(\Phi_{\partial}^*)$. **The vectors** of (i) and (iii): for $D \le 1300$ (`kEtaEntrywiseMaxD`) **every basis vector** of $C_{\mathcal{T},\ell}^{j+1}$ and of B_{ℓ}^{j} (entrywise); above, the α- and c-parts of the generators **and** 64 random vectors (`kFreivaldsVectors`) of $C_{\mathcal{T},\ell}^{j+1}$ and of B_{ℓ}^{j}, uniform in $\mathbb{Z}/\ell^K$ (rejection sampling from splitmix64), seeded deterministically by (ℓ, N, n, j). *What is deterministic, what is probabilistic.* **Deterministic**: (1)–(4) and (4)(e); the chain-level certificates; (ii); and (i), (iii) on the basis vectors and on the generators. Entrywise ($D \le 1300$), (iii) is the identity of (4)(a) on all of $C_{\mathcal{T},\ell}^{j+1}$, which with (i) (the applied operator *is* the block operator) and $T'_{\mathcal{T}}$, T'_{∂} cochain maps (composites of certified ones) is equivalent to the applied operator commuting with D on cone degree j: both halves of (4)(e)(iii) are checked at cochain level. **Probabilistic**: (i), (iii) on the random vectors above $D = 1300$, a Freivalds test. For a non-zero error matrix E over $\mathbb{Z}/\ell^K$ write $E = \ell^s E'$, $E' \not\equiv 0 \bmod \ell$, $s < K$; for v uniform, $Ev \equiv 0$ needs $E'v \equiv 0 \bmod \ell$, so $\Pr[Ev = 0] \le 1/\ell$, and one check on 64 independent vectors misses with probability $\le \ell^{-64} \le 2^{-64}$. The bound is over the seed, for an error not depending on it (the stream taken as uniform); the seed is fixed per (ℓ, N, n, j), so a re-run repeats the same vectors. *What "on the generators" certifies* (the C-D8b referee's m1, N1, N5). (i) on the generators compares the values read with an independent assembly: it catches an error of the applied operator not shared by the separately assembled maps, whether or not its values are cocycles — such an error changes the read only by being non-zero on a generator. It cannot see

an error in a primitive shared by both sides. (iii) on the generators checks the identity, both terms, on their α-parts exactly modulo ℓ^K; these are d-cocycles modulo $\ell^{K-\sigma_{\max}}$ only, and to that precision, given (i) and T'_∂ a cochain map, it is the second component of (ii) on g: nothing beyond (i) and (ii). So an off-diagonal error shared by the applied operator and $\eta_{\mathcal{T}}$, with d_{B}-cocycle values, passes (i), (ii) and (iii) on the generators and can change the read (the C-D8b referee's Arm (G)); the identity on the random vectors sees it, through its term $\eta_{\mathcal{T}}^{j+1} d\alpha$. The validation pins one such plant (`ScopedPlantedCocycleEta`, $\ell = 2$, $(N, n) = (11, 0)$, T_3, $q = 2$): clean on the generators alone, with the read of H_c^3 changed; caught with the random vectors; caught entrywise. **Above $D = 1300$ the coboundary half of (4)(e)(iii) is not checked deterministically at cochain level**: the read takes $T^{\mathrm{c}}_{\mathcal{T}}$ on the generators only, its proof is (4)(e)(iii) with the chain-level certificates, and the identity on the random vectors tests it with the bound above. *Why this is admissible.* No value is read from a check, from the separately assembled maps or from the random vectors: the values read are the images of the generators under the applied operator, products of exact matrices ((5)(c)), computed deterministically; the precision rule (5) governs them — the Smith read-out, the rank sources, the exactness modulo ℓ^K —; and their correctness for every D is a deterministic statement, (4)(e)(iii) with the chain-level certificates (exact, not weakening with D). The checks, deterministic or random, can only add a failure (`HeckeInternalError`); they never change, select or refuse a value. A miss of the random vectors therefore leaves the read where the deterministic checks leave it. What is *tested and not proved* at cochain level above $D = 1300$ is the module-level assembly of the off-diagonal block outside the generators' span; that is the part checked with probability of a miss $\leq \ell^{-64}$. (Forming $\eta_{\mathcal{T}}$ as a matrix $C^{q+1}_{\mathcal{T},\ell} \to \mathsf{B}^q_\ell$ costs about $\operatorname{rank} C^{q+1}_{\mathcal{T},\ell}$ ($\approx (2 + \frac{1}{3})D$ at $\ell = 2$) applications of T_{b} to $\mathcal{I}$-sized cochains per degree and per p, of the order $p \cdot D^3$, that of the tree cone's own elimination; v1.23.0-pre.34, referee N5; measured at D5c: $\approx D^2$, the applications being block-monomial, ≈ 15 s per degree and per p at D4's wall; v1.23.0-pre.36, measured at the merge (the report's §9.6): (i) about doubles the entrywise transport, 24 s for T_3 at $D = 1152$ against the groups' ≈ 29 s; the random-vector mode grows as $\approx D^{1.1}$, 2.5 s at $D = 1152$, extrapolated — not measured — to under 2 % of the groups at $D \approx 3000$–6000.) Refusals exactly those of (6); (f) a validation with arms: groups against D4's pair on every cell where both run and against `d4_pair_oracle.tsv`; $\mathsf{j}_\Psi \mathsf{j}_\Phi = \mathrm{id}$, $\mathsf{j}_\Phi \mathsf{j}_\Psi = \mathrm{id}$ on cohomology at a few cells; T_p on the three terms through the tree equal to D4's through $H(\mathsf{j}_\Phi)$, and the characteristic polynomial of T_p on H_c^1 against the oracle's `HECKE` rows; negative controls, each restricted to where it is visible (v1.23.0-pre.34, referee m3): $+\mathsf{Y}^*_\Phi$ in j_Φ; $1 - T^2$ for $1 - T$ in B; S for S^{-1} in a **only where it can bite** (v1.23.0-pre.35): at chain level over $\mathbb{Q}[\Gamma]$ at $\ell \in \{0, 2\}$ (the certificate of (1)(b)), or on the cochains at $\ell = 2$ of a module on which $\mathbf{c}$ acts non-trivially — for M_N, n odd, modulo 2^K, $K \geq 2$ — and **asserted clean** (the same cochains) for n even at every ℓ and at $\ell \geq 3$ (Definition 31.23(2)); the code pins it caught at $(N, n) = (1, 5)$, $(3, 1)$ and at chain level, clean at $(1, 4)$ and at $\ell = 3$, $(2, 2)$; and at $\ell = 3$ instead the term $S^{-1} e_U \mathsf{v}_1^U$ of $\mathsf{a}^{(3)}(\mathsf{z}_0)$ dropped; $\eta_{\mathcal{T}}$ dropped **for an operator that is not a fibre map** (for a fibre map the diagonal is a cochain map), detected on at least one cell per $\ell \in \{2, 3\}$ for an operator for which $\eta_{\mathcal{T}}$ is not a cocycle — θ^* is one; $\Phi_p^{(N)}$ at the cells where this has been observed — and never asked at $\ell = 3$ with n odd ($M^{-I} = 0$, no 3-local boundary, $\eta_{\mathcal{T}} = 0$); each control prints its detection count, in the manner of the `EFFECTIVE` convention (v1.23.0-pre.36: met by the production certificate (i) itself, the audit's m1 — each fault line names the certificate that caught it, every operator fault caught at $\ell = 2$ and at $\ell = 3$; `TREEPAIR_FAULTS` caught= 19 clean= 3, the clean ones the two asserted-clean S-for-S^{-1} controls and the generators-only mode of the Arm (G) plant); every refusal code once; and $\mathsf{j}_\Psi \mathsf{j}_\Phi = \mathrm{id}$ on cohomology at a cell with non-trivial 2-torsion in $H_c^{\geq 3}$, where the five-lemma caveat bites (the referee's §5); (g) RSS predicted per cell (rule 19) and the cost lines at D4's wall ($D \approx 1060$–1240) and beyond, **including the cost of forming** $\eta_{\mathcal{T}}$ ((e); its dominant term, part of the RSS prediction), in a report `DESCENT5c_report.md` (v1.23.0-pre.36: met — `TreePairCochains::transport_bytes`

adds the transport's columns on the built layout to the prediction before any operator is formed, refusing `[RssCeiling]` above 0.9× the ceiling, the audit's N2).

Formalization note. Definition 31.23, Lemma 31.24 and Proposition 31.25 are homological algebra (cones, cylinders, the comparison theorem relative to a subcomplex) and identities in the group rings of $\mathbb{Z}/4$, $\mathbb{Z}/6$ and $\langle T, \mathbf{c}\rangle$; no `\LEAN` marker.

31.6 The Hecke algebra over $\mathbb{Z}$: structure, certificates, congruence modules

(v1.25.0-pre.0 on the branch `s24-wip-heckez-z0`, refereed A7 / A7b; **in the main line from v1.25.0-pre.1** (2026-11-06), a pre-release, not owner-released; the HeckeZ plan `docs/implementation/slot24/heckez/`, signed by the owner on 2026-11-04 with every question at its recommendation — in particular Q1(a), *residue-only* big integers, Q6(a), *the anemic algebra first at level* $N > 1$, and Q8(a), this text on its own branch; stages `01-math-core.md` v0.6, `03-bockstein-tor.md` v0.3, `04-record-census.md` v0.4, package Z0 of `06-work-packages.md` v0.4; the adversarial audits `A1-audit.md`, `A3-audit.md`, `P2a-final-math-audit.md`, `P3-reaudit.md`.) §31.1–§31.5 certify the *groups* $H^q(\Gamma_0(N); V)$, the characteristic polynomials $\chi_{p,1}^{(N)}$ over $\mathbb{Z}$ and the torsion Hecke modules (Theorem 31.16). This subsection fixes what these data, together with reads modulo ℓ^K on the free part, say about the *ring* generated by the Hecke operators over $\mathbb{Z}$: (i) the algebras and their coarse structure — the prime spectrum does not see the square-zero glue between the free part and the torsion (Definition 31.27, Propositions 31.28, 31.29, Lemma 31.30); (ii) how the local structure at a prime ℓ is read from mod-ℓ^K data, and exactly when it can and cannot be (Propositions 31.31–31.36); (iii) the universal coefficient sequences with their Hecke action, their splitting, the level-one picture at 2 and 3, and the glue J, which computed data bound but do not determine (Propositions 31.37–31.41, three named conjectures in Remarks 31.40–31.43); (iv) the descent of the structure to $\mathbb{Z}[1/6]$ and to 2, 3 (Theorem 31.44); (v) the global certificate and what is reachable with the shipped tools (Theorem 31.45, Remark 31.46); (vi) congruence modules and their certificates, Ramanujan's congruence modulo 691 for every m from one finite computation (Definition 31.47, Propositions 31.48, 31.49). The record that carries the structure is item 8 of §22.4. **Not here**: U_p, $p \mid N$, over $\mathbb{Z}$ (Theorem 31.16(5), (7)(c) assert nothing of it; its lift is Proposition 31.13(9), and whether a binary serves it is item 8 (d′)(v); v1.26.0-pre.11, confirm-referee C1); non-rational Eisenstein components and cusp–cusp congruence modules beyond bounds (the plan's Prop. 5.7, a sketch); Hida families ([U]); p-adic L-functions (postponed by the owner).

What is used. Theorem 31.16 and the statements it cites; Definition 30.1, Propositions 30.2, 30.3, Corollary 30.6, Proposition 30.7; Definition 23.20(3); Proposition 16.15, Lemmas 16.14, 16.10, Proposition 24.4, Claim 16.7; Proposition 31.15; [U, `lem:u-fibre`], [U, `def:u-congmod`]; [E, `cl:sturm`] (as corrected at [E] 1.24.0-pre.1). *Standing assumptions.* (H4), (B1)–(B3); $N \geq 1$; $n \geq 0$ **even** (for n odd, 2 kills $H^\bullet(\Gamma_0(N); V)$, the proof of Proposition 31.14(2), and everything below is empty); $k := n + 2$; $\Gamma := \Gamma_0(N)$; $H^q := H^q(\Gamma; V)$, group cohomology (Definition 23.20(3)). *Notation.* $\mathbb{T}$, $\mathbb{T}^{(q)}$, $\mathbb{T}^{\mathrm{an}}$, $\mathbb{T}^{\mathrm{fr}}$, $\mathbb{T}^{\mathrm{an,fr}}$, $\mathbb{T}^{\mathrm{tors}}$, $\mathbb{T}^{\mathrm{cusp}}$, $\mathbb{T}'$, $\mathbb{T}_\ell$, $\mathbb{T}_{\mathbb{Z}_\ell}$, $\mathcal{H}_N$, X_p, L, L_ℓ, I_{fr}, I_{tors}, J, $A_K(M)$, $A'_K(M)$, d, d^{an}, e_k, $e_k^{\mathrm{an}}(N)$, ϑ, f_ϑ, f_i, g_i, P_i, $\mathcal{O}_i$, disc_ϑ, S, $\mathsf{S}_{\mathrm{cand}}$, c, $\mathcal{S}$, $\mathcal{S}'$, $I_{\mathcal{S}}$, $\mathbb{T}_{\mathcal{S}}$, $C_{\mathbb{T}}$, η_L, $L_{\mathcal{S}}$, $c_{\mathcal{S}',p}$, $g_{\mathcal{P}}$, C_{ft}, and in Proposition 31.49(3) ν, η_q, $e_{\mathfrak{m}}^{\mathrm{cusp}}$ (the rank of the cuspidal block at $\mathfrak{m}$, not an exponent), h_q, F_q, $f_{q,\mathfrak{m}}$ (v1.25.0-pre.0b, A7-m7) are new symbols of this subsection and mean nothing else (rule (A5)). In particular ϑ is a primitive element, **not** the operator θ of §24; η_L (always with its subscript) is a congruence module, not the pair homotopy η; d is a rank, not the $d = \mathrm{rank}_{\mathbb{Z}} M$ of §31.4; $\mathcal{P}$ is a finite set of primes, as in Theorem 31.16; $\mathbb{T}$ here is the Hecke algebra of Definition 31.27, and Definition 30.1's general pair is written $(M, \mathbb{T}_\ell)$ when it is applied. Statement numbers of the plan (Prop. 2.2, B8, ...) are those of `01-math-core.md` and `03-bockstein-tor.md`.

Definition 31.27 (The Hecke algebras over $\mathbb{Z}$)**.** Let $\mathcal{H}_N := \mathbb{Z}[X_p : p \text{ prime}]$ act on H^q by $X_p \mapsto T_p$ for $p \nmid N$ (Propositions 31.13, 28.9; $p = \ell$ included) and $X_p \mapsto U_p$ for $p \mid N$ (on cohomology, Proposition 31.10(3)(b′); over $\mathbb{Z}$ the lift of Proposition 31.13(9), v1.26.0-pre.1, served by a binary or not according to item 8 (d′)(v) (v1.26.0-pre.11, confirm-referee C1); Theorem 31.16(5), (7)(c) assert nothing of it); $\mathcal{H}_N^{\mathrm{an}} := \mathbb{Z}[X_p : p \nmid N]$.

1. $\mathbb{T}^{(q)} := \mathrm{im}(\mathcal{H}_N \to \mathrm{End}_{\mathbb{Z}} H^q)$, and $\mathbb{T} := \mathbb{T}^{(1)}$, the **full** Hecke algebra; $\mathbb{T}^{\mathrm{an}}$, the **anemic** algebra, is the image of $\mathcal{H}_N^{\mathrm{an}}$. At $N = 1$ they agree.
2. $L := H^1/\mathrm{tors} = H^1_{\mathrm{fr}}(N)$; $\mathbb{T}^{\mathrm{fr}} := \mathrm{im}(\mathbb{T} \to \mathrm{End} L)$, $\mathbb{T}^{\mathrm{tors}} := \mathrm{im}(\mathbb{T} \to \mathrm{End}(\mathrm{tors}\, H^1))$, with kernels I_{fr}, I_{tors}, and $J := I_{\mathrm{fr}} \cap I_{\mathrm{tors}}$ (**the glue**); $\mathbb{T}^{\mathrm{an,fr}} := \mathrm{im}(\mathbb{T}^{\mathrm{an}} \to \mathrm{End} L) \subseteq \mathbb{T}^{\mathrm{fr}}$. Everything stated below for $\mathbb{T}^{\mathrm{fr}}$ and d holds verbatim for $\mathbb{T}^{\mathrm{an,fr}}$ and d^{an} (Proposition 31.29(3′)), except where Propositions 31.48(5) and 31.49 say otherwise.
3. $\mathbb{T}^{\mathrm{cusp}}$: the image of $\mathbb{T}^{\mathrm{fr}}$ on the cuspidal part $L \cap (L \otimes \mathbb{Q})_{\mathrm{cusp}}$ (equivalently on H^1_c's image $H^1_!$ modulo torsion, §31.4; the two agree by Proposition 31.29(1)).
4. For a prime ℓ: $L_\ell := L \otimes \mathbb{Z}_\ell$, $\mathbb{T}_\ell := \mathbb{T}^{\mathrm{fr}} \otimes \mathbb{Z}_\ell$ (acting on L_ℓ only), $\mathbb{T}_{\mathbb{Z}_\ell} := \mathbb{T} \otimes \mathbb{Z}_\ell$ (acting on $H^1_\ell := H^1 \otimes \mathbb{Z}_\ell$, torsion included); the two are never identified. For a finitely generated $\mathbb{T}$-module M and $K \geq 1$, $A_K(M) := \mathrm{im}(\mathbb{T} \to \mathrm{End}(M/\ell^K M))$; for the subalgebra $\mathbb{T}' \subseteq \mathbb{T}^{\mathrm{fr}}$ generated by the operators actually computed — X_p, $p \in \mathcal{P}$, and integer combinations of them — $A'_K(M)$ is the image of $\mathbb{T}'$. Without an argument, $A_K := A_K(L_\ell)$; it is Definition 30.1's A_K for the pair $(L_\ell, \mathbb{T}_\ell)$.

Definition 30.1 and Propositions 30.2, 30.3 are stated for a general pair (a free $\mathbb{Z}_\ell$-module of finite rank and a commutative subalgebra of its endomorphisms), and their proofs use nothing else; the level-one setting of §30 is that section's *application*. They are applied below at level N, at $\ell \in \{2, 3\}$ and to subalgebras $\mathbb{T}'_\ell$ (the plan's A1-m10). The census algebra of Proposition 30.7(4) is $A'_1(H^1)$, not $A_1(L)$. *Other degrees*: for $n > 0$, $H^0 = 0$; for $n = 0$, $H^0 = \mathbb{Z}$ with $T_p = p + 1$, $U_p = p$; for $q \geq 2$, H^q is finite, supported at 2 and 3 and periodic (Theorem 31.16(1), (6)), so $\mathbb{T}^{(q)}$, $q \neq 1$, is a finite algebra at 2 and 3, and nothing beyond Proposition 31.28(1) is asserted about it.

Proposition 31.28 (Coarse structure)**.** *[proved here] (the plan's Prop. 1.2.)*

(0) The operators X_p commute on H^q: at level 1 by Proposition 23.23; at level N [standard: [Shi71, Ch. 3]] (double cosets of $\Gamma_0(N)$ in its Hecke semigroup). So $\mathbb{T}^{(q)}$ is commutative.

1. *$\mathbb{T}$ is a finite $\mathbb{Z}$-algebra; $\mathbb{T}^{\mathrm{fr}}$ is free of finite rank $d := \mathrm{rank}_{\mathbb{Z}} \mathbb{T}^{\mathrm{fr}}$; $\mathbb{T}^{\mathrm{tors}}$ is finite, supported at the torsion primes $\{2, 3\} \cup \mathcal{B}(M_N)$ (Theorem 31.16(1)).*
2. *$I_{\mathrm{fr}} I_{\mathrm{tors}} = 0$; hence $J^2 = 0$, $\mathbb{T}/J \hookrightarrow \mathbb{T}^{\mathrm{fr}} \times \mathbb{T}^{\mathrm{tors}}$, and $\mathrm{Spec}\, \mathbb{T} = V(I_{\mathrm{fr}}) \cup V(I_{\mathrm{tors}}) = \mathrm{Spec}\, \mathbb{T}^{\mathrm{fr}} \cup \mathrm{Spec}\, \mathbb{T}^{\mathrm{tors}}$. $\mathbb{T} \to \mathbb{T}/J$ **induces a homeomorphism of spectra with equal residue fields**: the glue plays no role for prime ideals and components.*
3. *Every prime P with $P \cap \mathbb{Z} = 0$ contains I_{fr}. The minimal primes of $\mathbb{T}$ are (a) the primes of characteristic 0 — the minimal primes of $\mathbb{T}^{\mathrm{fr}}$, i.e. the primes of $\mathbb{T} \otimes \mathbb{Q}$, the **components** — and (b) the **torsion-only points**: the maximal ideals $\mathfrak{m} \supseteq I_{\mathrm{tors}}$ with $\mathfrak{m} \not\supseteq I_{\mathrm{fr}}$. Each of the latter is minimal and maximal, an isolated point, i.e. a finite local factor of $\mathbb{T}$.*

Proof. (0) is cited. (1) $\mathrm{End}_{\mathbb{Z}}$ of a finitely generated abelian group is a finitely generated $\mathbb{Z}$-module; a submodule of $\mathrm{End}(L)$ is free; $\mathbb{T}^{\mathrm{tors}} \subseteq \mathrm{End}(\mathrm{tors}\, H^1)$ is finite. (2) For $a \in I_{\mathrm{fr}}$, $aH^1 \subseteq \mathrm{tors}$, and $b \in I_{\mathrm{tors}}$ kills tors, so $ba = 0$. Then $V(I_{\mathrm{fr}}) \cup V(I_{\mathrm{tors}}) = V(I_{\mathrm{fr}} I_{\mathrm{tors}}) = V(0)$; J is nilpotent, so it lies in every prime, and $\mathbb{T} \to \mathbb{T}/J$ is a homeomorphism on spectra with equal residue fields. (3) $m := |\mathbb{T}^{\mathrm{tors}}|$ has $m \cdot 1 \in I_{\mathrm{tors}}$; if $P \cap \mathbb{Z} = 0$ then $m \notin P$, so $P \not\supseteq I_{\mathrm{tors}}$ and by (2) $P \supseteq I_{\mathrm{fr}}$. Let $\mathfrak{m}$ be as in (b) and $P \subseteq \mathfrak{m}$ prime. $P \supseteq I_{\mathrm{fr}}$ would give $\mathfrak{m} \supseteq I_{\mathrm{fr}}$; so $P \supseteq I_{\mathrm{tors}}$, and P is a prime of the finite ring $\mathbb{T}^{\mathrm{tors}}$, hence maximal: $P = \mathfrak{m}$. The primes of (a) are minimal: a prime below one of them has characteristic 0, contains I_{fr}, and the primes of the finite $\mathbb{Q}$-algebra $\mathbb{T}^{\mathrm{fr}} \otimes \mathbb{Q}$ are pairwise incomparable. Conversely (A7-m2), let P be minimal of characteristic ℓ. If $P \supseteq I_{\mathrm{fr}}$, then P is a prime of $\mathbb{T}^{\mathrm{fr}}$ over ℓ; $\mathbb{T}^{\mathrm{fr}}$ is $\mathbb{Z}$-free, hence flat, so going-down gives a prime

of $\mathbb{T}^{\mathrm{fr}}$ over 0 strictly inside P, against minimality. So $P \not\supseteq I_{\mathrm{fr}}$, hence $P \supseteq I_{\mathrm{tors}}$ by (2), and P is a maximal ideal as in (b). □

Proposition 31.29 (Rational structure; reducedness; the generation inputs)**.** *[proved here] given the cited standard statements (the plan's Props. 1.3, 1.4, 1.5(2)–(4)); (4)(c) [conjectural] for $k > 2$.*

1. $\mathbb{T}^{\mathrm{fr}}$ *is the subring* $\mathbb{Z}[X_p]$ *of* $\mathbb{T}_{\mathbb{Q}} := \mathbb{T} \otimes \mathbb{Q} \subseteq \mathrm{End}(L \otimes \mathbb{Q})$ *generated by the operators. It does not depend on the lattice: on every* $\mathbb{T}$*-stable lattice* $M \subset L \otimes \mathbb{Q}$ *the image of* $\mathcal{H}_N$ *is the same subring of* $\mathbb{T}_{\mathbb{Q}}$*. (Saturation, Proposition 31.32, does depend on the lattice.)*
2. *[standard: [DI95, Thm. 12.2.2, Prop. 12.4.10], as in Lemma 30.4; the compatibility with* U_p*,* $p \mid N$*, is the verified [GT] row* `lit:diamond-im-es` *]* $H^1 \otimes \mathbb{C} \cong M_k(\Gamma) \oplus \overline{S_k(\Gamma)}$ *as* $\mathcal{H}_N$*-modules. Hence* $\mathbb{T}_{\mathbb{Q}} \cong \mathbb{T}(M_k(\Gamma))_{\mathbb{Q}}$*; its systems are those of* $M_k(\Gamma)$*, Eisenstein ones included; a cuspidal system occurs in* $H^1 \otimes \overline{\mathbb{Q}}$ *twice, an Eisenstein one once (for the full algebra, when* $\mathbb{T}_{\mathbb{Q}}$ *is reduced).*
3. $d = \dim M_k(\Gamma) = (b_1 + e_k)/2$*,* $b_1 = \mathrm{rank}\, H^1$ *(the descent record),* $e_k = \dim \mathrm{Eis}_k(\Gamma)$*: the number* $c(N)$ *of cusps for* $k \geq 4$*,* $c(N) - 1$ *for* $k = 2$*.*

(3′) *[standard: [DI95, Thm. 6.2.3, Rmks. 6.3.3–6.3.4] (strong multiplicity one and the old-space decomposition for the cuspidal part), [Ste07, Thms. 5.8–5.10] (the basis* $E_k^{\psi,\bar{\psi}}(q^t)$ *of* $\mathrm{Eis}_k(\Gamma)$ *for the rest); quoted in the verified [GT] rows* `lit:hz-es-anemic`*,* `lit:hz-eis-dim` *(v1.25.0-pre.0b, A7-m5: in place of [DS05], not in the library, and [AL70], unquoted)] The anemic algebra has rank*

$$d^{\mathrm{an}} := \mathrm{rank}_{\mathbb{Z}}\, \mathbb{T}^{\mathrm{an,fr}} = \sum_{M \mid N} \dim S_k^{\mathrm{new}}(\Gamma_0(M)) + e_k^{\mathrm{an}}(N),$$

$$e_k^{\mathrm{an}}(N) := \#\{\psi \text{ primitive} : \mathrm{cond}(\psi)^2 \mid N\} - [k = 2,\, N = 1].$$

For $(N, k) \neq (1, 2)$ *(where* $e_2^{\mathrm{an}}(1) = 0$*, A7-N9),* $e_k^{\mathrm{an}}(N) = 1$ *iff* $16 \nmid N$ *and* $p^2 \nmid N$ *for every odd* p*; then the anemic Eisenstein part is* ***one rational system*** $T_p \mapsto 1 + p^{k-1}$*, carried by several Eisenstein series at once.* $d^{\mathrm{an}} = d$ *iff there are no old forms and every anemic Eisenstein system is carried by one series (always at* $N = 1$*).*

4. Reducedness *(a predictor only: every cell checks it, Theorem 31.45(C1)). (a) [standard: Petersson normality and the explicit Eisenstein basis: [DI95, §6], [Ste07, Thms. 5.9, 5.10], the verified [GT] row* `lit:hz-semisimple-an` *]* $\mathbb{T}^{\mathrm{an}}_{\mathbb{Q}}$ *is reduced. (b) [standard: the old-space decomposition [DI95, Rmk. 6.3.4]; [AL70]VERIFY] If* $p^3 \mid N$ *for some* p*, the full* $\mathbb{T}_{\mathbb{Q}}$ *is* ***not*** *reduced. (c) If* N *is cube-free and* $\alpha_p \neq \beta_p$ *(the roots of* $x^2 - a_p x + p^{k-1}$*) for every* $p \mid N$ *and every newform of level* $M \mid N$*,* $p \nmid M$*, then* $\mathbb{T}_{\mathbb{Q}}$ *is reduced; the hypothesis* $\alpha_p \neq \beta_p$ *is known for* $k = 2$ *[standard: [CE98]VERIFY] and is [conjectural] for* $k > 2$*.*
5. Generation inputs. *(a) [standard: [Stu87]; [Ste07, Thm. 9.23]; the integral duality* $\mathbb{T}^{\mathrm{cusp}} \times S_k(\mathbb{Z}) \to \mathbb{Z}$*,* $(t, f) \mapsto a_1(tf)$*, [DI95, Prop. 12.4.13]; quoted in the verified [GT] row* `lit:hz-sturm-integral` *(A7-N2)]* $\mathbb{T}^{\mathrm{cusp}}$ *is spanned* as a $\mathbb{Z}$-module *by the* T_n*,* $n \leq B_N := k\mu_N/12$*,* $\mu_N := [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$ *(the* T_n *with* $p \mid \gcd(n, N)$ *are words in* U_p*). (b) For* M_k *the* $\mathbb{Z}$*-module statement is false at primes* ≥ 5 *([E,* `cl:sturm`*](b)); that* $\mathbb{T}(M_k)_{\mathbb{Q}}$ *is spanned by* T_m*,* $m \leq \lfloor k/12 \rfloor + 1$*, is [conjectural] ([E,* `cl:sturm`*](a) as corrected at [E] 1.24.0-pre.1) and* ***is used nowhere here****. (c) Let* $\mathbb{T}' \subseteq \mathbb{T}''$ *be commutative rings and* M *a faithful* $\mathbb{T}''$*-module.* Cyclic-faithful*: if* $M = \mathbb{T}'v$*, then* $\mathbb{T}' = \mathbb{T}''$ *as images in* $\mathrm{End}M$*.* Commutant*: if* $\mathrm{End}_{\mathbb{T}'}(M)$ *is the image of* $\mathbb{T}'$*, then* $\mathbb{T}'' = \mathbb{T}'$*.*

Proof. (1) $\mathcal{H}_N \to \mathrm{End}(L) \subset \mathrm{End}(L \otimes \mathbb{Q})$ and $\mathcal{H}_N \to \mathrm{End}(M) \subset \mathrm{End}(L \otimes \mathbb{Q})$ have the same image as $\mathcal{H}_N \to \mathrm{End}(L \otimes \mathbb{Q})$. (2) is cited. (3) By (2), $\mathbb{T}_{\mathbb{Q}}$ acts faithfully on $M_k(\Gamma)$, and the action on $\overline{S_k}$ factors through it. The pairing $\mathbb{T}(M_k)_{\mathbb{Q}} \times M_k(\mathbb{Q}) \to \mathbb{Q}$, $(t, f) \mapsto a_1(tf)$, is perfect: if $a_1(tf) = 0$ for all t then $a_m(f) = a_1(T_m f) = 0$ for $m \geq 1$, so f is constant, and $f = 0$ as $k > 0$; if $a_1(tf) = 0$ for all f then $a_m(tf) = a_1(tT_m f) = 0$ for all f and m, so $t = 0$. So

$d = \dim M_k$, and $b_1 = \dim M_k + \dim S_k$, $\dim M_k = \dim S_k + e_k$. No reducedness is used. (3′) $\mathbb{T}^{\mathrm{an}}_{\mathbb{Q}}$ is reduced ((4)(a)) and acts faithfully on $M_k(\Gamma)$, so its dimension is the number of distinct anemic systems of $M_k(\Gamma)$ over $\overline{\mathbb{Q}}$. Cuspidal: every eigenform lies in the old space of a unique newform of level $M \mid N$, the old space shares its anemic system, and distinct newforms of levels dividing N have distinct anemic systems (strong multiplicity one). Eisenstein: $\mathrm{Eis}_k(\Gamma_0(N))$ has the basis $E_k^{\psi,\bar\psi}(q^t)$, $\mathrm{cond}(\psi)^2 t \mid N$ (for $k = 2$, $\psi = 1$: the combinations $E_2(q) - tE_2(q^t)$, $t > 1$), with anemic system $T_p \mapsto \psi(p) + \bar\psi(p)p^{k-1}$, independent of t; distinct ψ give distinct systems (the semisimple representations $\psi \oplus \bar\psi\varepsilon^{k-1}$, ε the cyclotomic character, are distinct, and are determined by their traces at almost all primes), and no Eisenstein system is cuspidal ([Del74]: at any prime $p \geq 7$, $p \nmid N$, $|\psi(p) + \bar\psi(p)p^{k-1}| \geq p^{k-1} - 1 > 2p^{(k-1)/2} \geq |a_p|$; A7-m1: the bound $1 + p^{k-1}$ is the trivial-ψ eigenvalue only, and at $k = 2$, $p \leq 5$ the inequality fails). (4) $\mathbb{T}_{\mathbb{Q}}$ is reduced iff each generator is semisimple (commuting semisimple operators generate a reduced algebra; conversely a reduced finite $\mathbb{Q}$-algebra is a product of number fields, whose elements have separable minimal polynomials). (a) The T_p, $p \nmid N$, are normal for the Petersson product on S_k and diagonal on the Eisenstein basis. (b), (c) On the old space of a newform f of level M, $N = Mp^r \cdot$ (prime to p), U_p acts on $v_j := f(q^{p^j})$, $0 \leq j \leq r$, by $U_p v_j = v_{j-1}$ ($j \geq 1$), $U_p v_0 = a_p v_0 - p^{k-1} v_1$ ($p \nmid M$), $U_p v_0 = a_p v_0$ ($p \mid M$). v_r is cyclic, so the minimal polynomial is the characteristic polynomial: $x^{r-1}(x^2 - a_p x + p^{k-1})$ ($p \nmid M$), $x^r(x - a_p)$ with $a_p \neq 0$ ($p \,\|\, M$), x^{r+1} ($p^2 \mid M$, $a_p = 0$). So U_p fails to be semisimple iff $v_p(N) \geq 3$ in each case, or $r \geq 1$ and $\alpha_p = \beta_p$. The Eisenstein old spaces behave the same way: for $k \geq 4$, $\alpha_p = \psi(p) \neq \bar\psi(p)p^{k-1} = \beta_p$ (different absolute values); for $k = 2$ the functional $v_j \mapsto p^{-j}$ on $\mathrm{span}\{E_2(q^{p^j})\}$ is a U_p-eigenvector of eigenvalue p cutting out the modular subspace, so the generalized 0-eigenspace, one Jordan block of size $r - 1$, lies in that subspace. At $p^3 \mid N$ the level-1 series E_k ($k \geq 4$) or the E_2-family ($k = 2$) contributes a nilpotent block of size ≥ 2, which is (b). (5)(a) By the duality, $\{T_n : n \leq B_N\}$ spans $\mathbb{T}^{\mathrm{cusp}}/\ell$ iff every $f \in S_k(\mathbb{Z})$ with $a_n(f) \equiv 0 \pmod{\ell}$ for $n \leq B_N$ is $\equiv 0 \pmod{\ell}$, which is Sturm's theorem modulo ℓ (the form he proved; the characteristic-0 vanishing statement would give only full rank). So $\Sigma + \ell\mathbb{T}^{\mathrm{cusp}} = \mathbb{T}^{\mathrm{cusp}}$ for $\Sigma := \sum_{n \leq B_N} \mathbb{Z}T_n$ and every ℓ; Σ has finite index prime to every prime, hence $\Sigma = \mathbb{T}^{\mathrm{cusp}}$. (b) is cited. (c) Cyclic-faithful: for $t \in \mathbb{T}''$ write $tv = t'v$, $t' \in \mathbb{T}'$; then $t(sv) = s(tv) = s(t'v) = t'(sv)$ for $s \in \mathbb{T}'$, so $t - t'$ kills M. Commutant: $\mathbb{T}''$ commutes with $\mathbb{T}'$. □

Checks (the plan's `s01r/dan.py`): $(N, n) = (1, 10)$: $b_1 = 3$, $c(1) = 1$, $d = d^{\mathrm{an}} = 2$; $(2, 6)$: $d = 3$, $d^{\mathrm{an}} = 1+1 = 2$; $(8, 10)$: $d = 13 = \dim M_{12}(\Gamma_0(8))$, $d^{\mathrm{an}} = (1+0+1+3)+1 = 6$. **The rank is an a-priori input of every certificate**: d for the full algebra, d^{an} for the anemic one, never the computed $\deg f_\vartheta$ — an operator that fails to separate two systems is otherwise indistinguishable from one that separates all (the plan's A1-M3). The cyclic-faithful and commutant certificates exist in code over $\mathbb{F}_\ell$ (`HeckeExt/HeckeGenerationCertificates`); on a direct summand of L at odd ℓ cyclic-faithful implies saturation (Proposition 30.2(2), sufficient condition), so for $\mathbb{T}^{\mathrm{fr}}$ Proposition 31.32 subsumes it; it matters for $\mathbb{T}^{\mathrm{tors}}$ and at $\ell = 2$.

Lemma 31.30 (The integral/ℓ-adic bridge)**.** *[proved here] (the plan's Lemma 1.6, HZ-0.) Let L be a free $\mathbb{Z}$-module of finite rank, $\mathbb{T}^{\mathrm{fr}} \subseteq \mathrm{End}_{\mathbb{Z}} L$ the subalgebra generated by a family of operators, $\mathbb{T}' \subseteq \mathbb{T}^{\mathrm{fr}}$ a subalgebra, ℓ a prime, L_ℓ, $\mathbb{T}_\ell$ as in Definition 31.27(4), $\mathbb{T}'_\ell := \mathbb{T}' \otimes \mathbb{Z}_\ell$.*

1. *$\mathbb{T}_\ell \to \mathrm{End}_{\mathbb{Z}_\ell}(L_\ell)$ is injective, with image the $\mathbb{Z}_\ell$-subalgebra generated by the base-changed operators; likewise for $\mathbb{T}'$.*
2. *$\mathrm{rank}_{\mathbb{Z}_\ell} \mathbb{T}_\ell = \mathrm{rank}_{\mathbb{Z}} \mathbb{T}^{\mathrm{fr}}$; so the d of Proposition 31.29(3) is the rank the local statements compare with.*
3. *If $\mathrm{rank}\, \mathbb{T}' = \mathrm{rank}\, \mathbb{T}^{\mathrm{fr}}$, then $\ell \nmid [\mathbb{T}^{\mathrm{fr}} : \mathbb{T}']$ iff $\mathbb{T}'_\ell = \mathbb{T}_\ell$. Hence $\mathbb{T}' = \mathbb{T}^{\mathrm{fr}}$ iff $\mathbb{T}'_\ell = \mathbb{T}_\ell$ for every ℓ.*

Proof. $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$, so $\mathbb{T}^{\mathrm{fr}} \hookrightarrow \mathrm{End}_{\mathbb{Z}} L$ stays injective after $\otimes\mathbb{Z}_\ell$, and $\mathrm{End}_{\mathbb{Z}}(L) \otimes \mathbb{Z}_\ell = \mathrm{End}_{\mathbb{Z}_\ell}(L_\ell)$ (L finite free); the image is the $\mathbb{Z}_\ell$-span of the words in the operators, which is (1). (2) is base change of a free module. (3) Tensoring $0 \to \mathbb{T}' \to \mathbb{T}^{\mathrm{fr}} \to \mathbb{T}^{\mathrm{fr}}/\mathbb{T}' \to 0$ with $\mathbb{Z}_\ell$ gives $\mathbb{T}_\ell/\mathbb{T}'_\ell \cong (\mathbb{T}^{\mathrm{fr}}/\mathbb{T}') \otimes \mathbb{Z}_\ell$, the ℓ-primary part of a finite group. □

Proposition 31.31 (Maximal ideals are detected modulo ℓ)**.** *[proved here] [standard: Cayley–Hamilton for finitely generated modules, [AM69, Prop. 2.4]] (the plan's Prop. 2.1.) Let R be a commutative ring, M a finitely generated faithful R-module, $\ell \in R$, $K \geq 1$ and $I_K := \ker(R \to \mathrm{End}(M/\ell^K M))$. Then $\ell^K R \subseteq I_K \subseteq \sqrt{\ell R}$. Hence $\mathrm{Spec}(R/I_K) \to \mathrm{Spec}\, R$ is a homeomorphism onto $V(\ell)$ with equal residue fields, for every $K \geq 1$.*

Proof. For $x \in I_K$, $xM \subseteq \ell^K M \subseteq (\ell R)M$, so Cayley–Hamilton gives $x^m + a_1 x^{m-1} + \cdots + a_m \in \mathrm{Ann}(M) = 0$ with $a_i \in (\ell R)^i$; hence $x^m \in \ell R$. □

How it is applied, and its limits. With $R = \mathbb{T} \otimes \mathbb{Z}_{(\ell)}$, $M = H^1 \otimes \mathbb{Z}_{(\ell)}$ (faithful by flat base change; $H^1 \otimes \mathbb{Z}_{(\ell)}$ is *not* faithful over $\mathbb{T}$ itself, an operator mapping H^1 into its prime-to-ℓ torsion acting as 0), and with $R = \mathbb{T}^{\mathrm{fr}} \otimes \mathbb{Z}_{(\ell)}$, $M = L \otimes \mathbb{Z}_{(\ell)}$, which detects $V(\ell) \subset \mathrm{Spec}\, \mathbb{T}^{\mathrm{fr}}$ and misses the torsion-only points. **A computation produces $A'_K(M)$, not $A_K(M)$**: $\mathrm{Spec}\, A'_K$ is the ℓ-fibre of $\mathrm{Spec}\, \mathbb{T}'$, onto which $V(\ell) \subset \mathrm{Spec}\, \mathbb{T}$ maps surjectively, not necessarily injectively. The two agree under *generation modulo ℓ*, $A'_1(M) = A_1(M)$, which Proposition 31.32 certifies. So "the census lists exactly the maximal ideals of $\mathbb{T}$ over ℓ" holds under the census's generation certificate (Corollary 30.6(2)), not unconditionally; two maximal ideals that differ on a computed operator are certainly distinct. At $\ell \geq 5$, $H^1(\Gamma; V/\ell) = H^1/\ell$ (Proposition 31.37(4)). At $\ell \in \{2, 3\}$, $H^2[\ell]$ is a *quotient* of $H^1(\Gamma; V/\ell)$, not a submodule, and the census algebra there can carry maximal ideals of $\mathbb{T}^{(2)}$ that are not maximal ideals of $\mathbb{T}^{(1)}$; the filter is applied on the integral model (Proposition 31.37(4)).

Proposition 31.32 (Generation by saturation)**.** *[proved here] (the plan's Prop. 2.2; it strengthens Corollary 30.6(2).) Let M be a $\mathbb{T}^{\mathrm{fr}}$-stable lattice in a $\mathbb{T}$-stable subspace of $L \otimes \mathbb{Q}_\ell$ on which $\mathbb{T}_\ell$ acts faithfully (e.g. $M = L_\ell$, or one ι-part when every system occurs in it; A7-N6), and $d = \mathrm{rank}\, \mathbb{T}^{\mathrm{fr}}$ from Proposition 31.29(3) — **an a-priori input**; for the anemic algebra read $\mathbb{T}^{\mathrm{an,fr}}$ and d^{an} of Proposition 31.29(3′); a rank read off a computed $\deg f_\vartheta$ is not admissible here.*

1. *$\dim_{\mathbb{F}_\ell} A'_1(M) \leq \mathrm{rank}\, \mathbb{T}'_\ell \leq d$. If $\dim_{\mathbb{F}_\ell} A'_1(M) = d$, then $\mathbb{T}'_\ell = \mathbb{T}_\ell$, $\mathbb{T}_\ell$ is saturated in $\mathrm{End}(M)$, and $\mathbb{T}_\ell/\ell = A'_1(M)$.*
2. *Conversely, if $\mathbb{T}_\ell$ is saturated in $\mathrm{End}M$ and $\mathbb{T}' + \ell\mathbb{T}_\ell = \mathbb{T}_\ell$ (true for a large enough finite set of operators), then $\dim A'_1(M) = d$. So the test succeeds for some finite operator set **iff** $\mathbb{T}_\ell$ is saturated on M.*
3. *If $\mathbb{T}_\ell$ is not saturated on M, no operator set passes. If $\mathrm{rank}\, \mathbb{T}' = d$, mod-$\ell^K$ data on M then give only $\mathbb{T}'_\ell \subseteq \mathbb{T}_\ell \subseteq \mathbb{T}'^{\mathrm{sat}} := (\mathbb{T}'_{\mathbb{Q}} \otimes \mathbb{Q}_\ell) \cap \mathrm{End}(M)$, with $\mathrm{length}(\mathbb{T}'^{\mathrm{sat}}/\mathbb{T}'_\ell)$ read by Proposition 30.3.*

Proof. (1) The first inequality is Proposition 30.2(3) at $K = 1$ for the pair $(M, \mathbb{T}'_\ell)$; the second holds as $\mathbb{T}' \subseteq \mathbb{T}^{\mathrm{fr}}$ (Lemma 31.30(2)). Equality forces $\mathrm{rank}\, \mathbb{T}' = d$, so $\mathbb{T}'_{\mathbb{Q}} = \mathbb{T}_{\mathbb{Q}}$, and it forces $\mathbb{T}'_\ell$ saturated (Proposition 30.2(2), (a)⇔(c)), i.e. $\mathbb{T}'_\ell = (\mathbb{T}_{\mathbb{Q}} \otimes \mathbb{Q}_\ell) \cap \mathrm{End}M \supseteq \mathbb{T}_\ell \supseteq \mathbb{T}'_\ell$ ($\mathbb{T}_\ell \subseteq \mathrm{End}M$ as M is $\mathbb{T}^{\mathrm{fr}}$-stable). (2) Nakayama over $\mathbb{Z}_\ell$ gives $\mathbb{T}'_\ell = \mathbb{T}_\ell$, and saturation gives $\dim A_1(M) = d$ (Proposition 30.2(2)(a)). $\mathbb{T}^{\mathrm{fr}}$ is a finitely generated $\mathbb{Z}$-module, spanned by finitely many words in finitely many X_p. (3) A pass would make $\mathbb{T}_\ell = \mathbb{T}'_\ell$ saturated, by (1). With $\mathrm{rank}\, \mathbb{T}' = d$, $\mathbb{T}'_{\mathbb{Q}} \otimes \mathbb{Q}_\ell = \mathbb{T}_{\mathbb{Q}} \otimes \mathbb{Q}_\ell$, so $\mathbb{T}_\ell \subseteq \mathbb{T}'^{\mathrm{sat}}$, the saturation of the pair $(M, \mathbb{T}'_\ell)$. □

The hypothesis is $\dim A'_1 = d$ with d from Proposition 31.29, not $\dim A'_1 = \mathrm{rank}\, \mathbb{T}'$: the latter makes $\mathbb{T}'$ saturated and says nothing about $\mathbb{T}'_{\mathbb{Q}} = \mathbb{T}_{\mathbb{Q}}$. Saturation holds at every multiplicity-one maximal ideal at odd ℓ (Proposition 30.7(5)); census rows with positive `sat_defect_dim` show that it fails in practice. *What saturation means* (not used by any certificate). $(\mathbb{T}_{\mathbb{Q}} \otimes \mathbb{Q}_\ell) \cap \mathrm{End}M$ is the multiplier ring $\{x : xM \subseteq M\}$, so $\mathbb{T}_\ell$ is saturated on M iff M is a module over no larger order; $\mathbb{T}_\ell$ is always saturated on its dual $\mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{T}_\ell, \mathbb{Z}_\ell)$ (if $x\mathbb{T}^\vee \subseteq \mathbb{T}^\vee$ then $x \cdot 1 \in \mathbb{T}^{\vee\vee} = \mathbb{T}_\ell$), which on the forms side is the lattice of $f \in M_k(\mathbb{Q}_\ell)$ with $a_m(f) \in \mathbb{Z}_\ell$ for $m \geq 1$ — not computable without the congruence modules. Unsaturation is a property of the *cohomology lattice*, not of $\mathbb{T}$.

Proposition 31.33 (Étale primes, and the candidate bad set)**.** *[proved here] (the plan's Prop. 2.3.) Let $\mathcal{O}$ be a finite free $\mathbb{Z}_\ell$-algebra with $\mathcal{O}\otimes\mathbb{Q}_\ell$ reduced, and $\mathcal{R}\subseteq\mathcal{O}$ a subalgebra of full rank (e.g. $\mathcal{R}=\mathbb{Z}_\ell[\vartheta]$ or $\mathbb{T}'_\ell$, $\mathcal{O}=\mathbb{T}_\ell$);* disc *is the Gram determinant of the trace form of the regular representation.*

1. $\operatorname{disc}(\mathcal{R}) = [\mathcal{O}:\mathcal{R}]^2\operatorname{disc}(\mathcal{O})$ *up to a unit. In particular, for $\vartheta\in\mathbb{T}^{\mathrm{fr}}$ with $\mathbb{Q}[\vartheta]=\mathbb{T}_\mathbb{Q}$, $\operatorname{disc}_\vartheta := \operatorname{disc}\mathbb{Z}[\vartheta] = \operatorname{disc}(f_\vartheta) = [\mathbb{T}^{\mathrm{fr}}:\mathbb{Z}[\vartheta]]^2\operatorname{disc}\mathbb{T}^{\mathrm{fr}}$, f_ϑ the minimal polynomial (the trace is that of $\mathbb{T}_\mathbb{Q}$, not of $L\otimes\mathbb{Q}$, which counts cuspidal systems twice).*
2. *If $v_\ell(\operatorname{disc}\mathcal{R})\le 1$, then $\mathcal{R}=\mathcal{O}$. If $v_\ell(\operatorname{disc}\mathcal{R})=0$, $\mathcal{O}$ is étale over $\mathbb{Z}_\ell$, every faithful lattice is saturated, and each maximal ideal lies on exactly one component. If $v_\ell(\operatorname{disc}\mathcal{R})=1$, then ℓ is odd, exactly one maximal ideal $\mathfrak{m}$ is ramified, it lies on one component, and $\mathcal{O}_\mathfrak{m}$ is a DVR with ramification index 2.*
3. ***Hence**, for ℓ with $\ell^2\nmid\operatorname{disc}_\vartheta$ (resp. $\ell^2\nmid\operatorname{disc}\mathbb{T}'$), $\mathbb{T}_\ell=\mathbb{Z}_\ell[\vartheta]\cong\mathbb{Z}_\ell[x]/(f_\vartheta)$ (resp. $=\mathbb{T}'_\ell$) **with no generation hypothesis**; its maximal ideals are $(\ell,h(\vartheta))$ for the irreducible factors $\bar h$ of $\bar f_\vartheta$, and component i passes through $(\ell,h(\vartheta))$ iff $\bar h\mid\bar f_i$ ($f_\vartheta=\prod_i f_i$ over $\mathbb{Q}$, one f_i per component).*
4. *Components $i\neq j$ meet above ℓ only if $\ell\mid\operatorname{Res}(f_i,f_j)$ for **every** such ϑ. If no two meet above ℓ, then $\mathbb{T}_\ell=\prod_i\mathcal{O}_i\otimes\mathbb{Z}_\ell$, $\mathcal{O}_i:=\mathbb{T}^{\mathrm{fr}}/P_i$ the Hecke order of component i (P_i its minimal prime).*
5. *If components meet at ℓ, then $v_\ell(\operatorname{disc}\mathbb{T}^{\mathrm{fr}})\ge 2\,v_\ell|C_\mathbb{T}(\mathcal{S},\mathcal{S}')|$ for the congruence module of Definition 31.47 ($\mathcal{S}\sqcup\mathcal{S}'$ all components).*

Proof. (1) A $\mathbb{Z}_\ell$-basis of $\mathcal{R}$ is the image of one of $\mathcal{O}$ under a matrix of determinant $[\mathcal{O}:\mathcal{R}]$ up to a unit, and the Gram matrices transform accordingly. $\mathbb{Z}[\vartheta]\cong\mathbb{Z}[x]/(f_\vartheta)$ has Gram determinant $\operatorname{disc}(f_\vartheta)$ in the basis $1,\vartheta,\dots,\vartheta^{d-1}$. (2) By (1), $2\operatorname{length}(\mathcal{O}/\mathcal{R})\le v_\ell\operatorname{disc}\mathcal{R}$, so $\mathcal{R}=\mathcal{O}$ when $v\le 1$. For $v=0$ the trace form of $\mathcal{O}/\ell$ is non-degenerate, so $\mathcal{O}/\ell$ is a separable $\mathbb{F}_\ell$-algebra and $\mathcal{O}$ is étale, a product of unramified DVRs, hence a maximal order; the multiplier ring of a faithful lattice is an order containing $\mathcal{O}$, hence equal to it. For $v=1$: the nilradical $\mathfrak{n}$ of $\mathcal{O}/\ell$ lies in the kernel of the reduced trace form ($\operatorname{Tr}(nx)=0$ for nilpotent nx), and $v_\ell\det$ of the Gram matrix is at least the dimension of that kernel, so $\dim_{\mathbb{F}_\ell}\mathfrak{n}\le 1$; and $\mathfrak{n}\neq 0$, since a reduced finite algebra over the perfect field $\mathbb{F}_\ell$ is separable, with non-degenerate trace form, which would give $v=0$. So $\mathfrak{n}$ is one-dimensional and sits in one local factor $\mathcal{O}_\mathfrak{m}/\ell$; its radical $\mathfrak{n}$ is a non-zero vector space over the residue field modulo $\mathfrak{n}^2$, of $\mathbb{F}_\ell$-dimension 1, so the residue field is $\mathbb{F}_\ell$, $\mathfrak{n}^2=0$, $\mathcal{O}_\mathfrak{m}/\ell\cong\mathbb{F}_\ell[\varepsilon]/(\varepsilon^2)$, and (Nakayama) $\mathcal{O}_\mathfrak{m}\cong\mathbb{Z}_\ell[x]/(F)$ with F monic of degree 2, $\bar F=(x-a)^2$. If $\mathcal{O}_\mathfrak{m}\otimes\mathbb{Q}$ were not a field, $\mathcal{O}_\mathfrak{m}$ would be of index $\ge\ell$ in $\mathbb{Z}_\ell\times\mathbb{Z}_\ell$ and $v\ge 2$ by (1); so it is a quadratic order with $v_\ell(\operatorname{disc})=1$, hence maximal (by (1)), ramified (an unramified one has unit discriminant), a DVR with ramification index 2, and it lies on one component. At $\ell=2$: $F=x^2+bx+c$ with $\bar F=(x-a)^2$ has b even, so $v_2(b^2-4c)\ge 2$; $v=1$ does not occur. (3) is (2) with $\mathcal{O}/\ell=\mathbb{F}_\ell[x]/(\bar f_\vartheta)$; the kernel of $\mathbb{T}_\ell\to\mathbb{Z}_\ell[x]/(f_i)$ is $P_i\mathbb{T}_\ell$, generated by $f_i(\vartheta)$, and $f_i(\vartheta)\in(\ell,h(\vartheta))$ iff $\bar h\mid\bar f_i$. (4) Write $\operatorname{Res}(f_i,f_j)=af_i+bf_j$, $a,b\in\mathbb{Z}[x]$, and evaluate at ϑ: $f_i(\vartheta)\in P_i$, $f_j(\vartheta)\in P_j$, so $\operatorname{Res}\in(P_i+P_j)\cap\mathbb{Z}\subseteq\mathfrak{m}\cap\mathbb{Z}=\ell\mathbb{Z}$ if $\mathfrak{m}\supseteq P_i+P_j$. If no two meet, every block $\mathbb{T}_\mathfrak{m}=e_\mathfrak{m}\mathbb{T}_\ell$ of the complete semilocal ring $\mathbb{T}_\ell$ lies on exactly one component, and the component idempotent $\varepsilon_i\in\mathbb{T}_\mathbb{Q}\otimes\mathbb{Q}_\ell$ is $\sum_{\mathfrak{m}\supseteq P_i}e_\mathfrak{m}\in\mathbb{T}_\ell$ ($\varepsilon_i e_\mathfrak{m}\neq 0$ iff $(\mathcal{O}_i)_\mathfrak{m}\neq 0$ iff $P_i\subseteq\mathfrak{m}$); then $\mathbb{T}_\ell=\prod_i\varepsilon_i\mathbb{T}_\ell$ and $\varepsilon_i\mathbb{T}_\ell$ is the image of $\mathbb{T}_\ell$ in $\mathcal{O}_i\otimes\mathbb{Q}_\ell$. (5) [U, `lem:u-fibre`] gives $\mathbb{T}^{\mathrm{fr}}=\mathbb{T}_\mathcal{S}\times_C\mathbb{T}_{\mathcal{S}'}$ with both projections onto and $(\mathbb{T}_\mathcal{S}\times\mathbb{T}_{\mathcal{S}'})/\mathbb{T}^{\mathrm{fr}}\cong C$; apply (1) to $\mathbb{T}^{\mathrm{fr}}\subseteq\mathbb{T}_\mathcal{S}\times\mathbb{T}_{\mathcal{S}'}$. □

So the **computable candidate bad set** is $\mathsf{S}_{\mathrm{cand}} := \{\ell:\ell^2\mid\operatorname{disc}\mathbb{T}'\}\cup\{\text{torsion primes}\}$, contained in $\{\ell:\ell^2\mid\operatorname{disc}_\vartheta\}\cup\{\text{torsion primes}\}$ and shrinking as the operator set grows; the set $\{\ell\mid\operatorname{disc}\mathbb{T}^{\mathrm{fr}}\}$ is not computable before $\mathbb{T}^{\mathrm{fr}}$ is known, and primes with $v_\ell=1$ need nothing.

Proposition 31.34 (Monogenic blocks: $K=1$ suffices)**.** *[proved here] [standard: a finite flat local $\mathbb{Z}_\ell$-algebra is Gorenstein iff its reduction modulo ℓ is, [BH93, Prop. 3.1.19(b)]VERIFY] (the plan's Prop. 2.4, its hypothesis as corrected by A1-M2.) Assume generation at ℓ certified ($\mathbb{T}'_\ell=$*

$\mathbb{T}_\ell$, *by Proposition 31.32 or by a theorem such as Proposition 31.49), $\vartheta \in \mathbb{T}'$ with $\mathbb{Q}[\vartheta] = \mathbb{T}_\mathbb{Q}$, and that $\bar\vartheta$ generates $\mathbb{T}_\ell/\ell\mathbb{T}_\ell$ as an $\mathbb{F}_\ell$-algebra. Then:*

1. $\mathbb{T}_\ell = \mathbb{Z}_\ell[\vartheta] \cong \mathbb{Z}_\ell[x]/(f_\vartheta)$.
2. *The maximal ideals are the irreducible factors $\bar h$ of $\bar f_\vartheta$; component i passes through $(\bar h)$ iff $\bar h \mid \bar f_i$.*
3. *For disjoint sets $\mathcal{S}, \mathcal{S}'$ of components, $C_\mathbb{T}(\mathcal{S}, \mathcal{S}') \otimes \mathbb{Z}_\ell \cong \mathbb{Z}_\ell[x]/(f_\mathcal{S}, f_{\mathcal{S}'})$, of length $v_\ell \operatorname{Res}(f_\mathcal{S}, f_{\mathcal{S}'})$, $f_\mathcal{S} := \prod_{i\in\mathcal{S}} f_i$.*
4. *Every $\mathbb{T}_\mathfrak{m}$ is a complete intersection and Gorenstein, and $\mathbb{T}_\mathfrak{m}/\ell$ has embedding dimension ≤ 1 over its residue field.*

Conversely, if some $\mathbb{T}_\mathfrak{m}/\ell$ has embedding dimension ≥ 2 — in particular if $\mathbb{T}_\mathfrak{m}$ is not Gorenstein — no ϑ generates. Computable form of the hypothesis. *By Proposition 30.2(3) for $(L_\ell, \mathbb{T}_\ell)$, $\mathbb{T}_\ell/\ell\mathbb{T}_\ell \to A_K/\ell A_K$ is an isomorphism of $\mathbb{F}_\ell$-algebras exactly when $\dim_{\mathbb{F}_\ell} A_K/\ell A_K = d$, and for every $K \ge c^{\mathrm{sat}} + 1$. So raise K until $\dim A_K/\ell A_K = d$ ($K = c^{\mathrm{sat}} + 1$ at the latest; c^{sat} need not be known in advance), then test whether $\bar\vartheta$ generates $A_K/\ell A_K$; under generation $A_K = A'_K$. On the Proposition 31.32 path ($\mathbb{T}_\ell$ saturated) $K = 1$. On an unsaturated lattice the test at $K = 1$ is* ***wrong****: A_1 is then a proper quotient of $\mathbb{T}_\ell/\ell$ (the plan's A1-M2: $\mathbb{T}_\ell = \{(a,b) \in \mathbb{Z}_\ell^2 : a \equiv b\ (\ell^2)\}$ on $\mathbb{Z}_\ell^2$, $A_1 = \mathbb{F}_\ell$, $\vartheta = (1+\ell^3, 1)$ generates A_1 but $\mathbb{Z}_\ell[\vartheta] = \{a \equiv b\ (\ell^3)\} \subsetneq \mathbb{T}_\ell$; there $c^{\mathrm{sat}} = 2$ and $A_3/\ell A_3 \cong \mathbb{F}_\ell[\varepsilon]/(\varepsilon^2)$, which $\bar\vartheta = 1$ does not generate).*

Proof. (1) $\mathbb{Z}_\ell[\vartheta] + \ell\mathbb{T}_\ell = \mathbb{T}_\ell$, so Nakayama gives equality, and $\mathbb{Z}_\ell[\vartheta] \cong \mathbb{Z}_\ell[x]/(f_\vartheta)$ as f_ϑ is the minimal polynomial and is monic. (2) $\mathbb{T}_\ell/\ell = \mathbb{F}_\ell[x]/(\bar f_\vartheta)$, and Proposition 31.33(3)'s argument. (3) The kernel of $\mathbb{Z}_\ell[x]/(f_\vartheta) \to \mathbb{Z}_\ell[x]/(f_\mathcal{S})$ is generated by $f_\mathcal{S}$ ($f_\mathcal{S} \mid f_\vartheta$), so $I_\mathcal{S} \otimes \mathbb{Z}_\ell = f_\mathcal{S}(\vartheta)\mathbb{T}_\ell$, likewise for $\mathcal{S}'$, and $C \otimes \mathbb{Z}_\ell = \mathbb{Z}_\ell[x]/(f_\vartheta, f_\mathcal{S}, f_{\mathcal{S}'}) = \mathbb{Z}_\ell[x]/(f_\mathcal{S}, f_{\mathcal{S}'})$. This is the cokernel of multiplication by $f_{\mathcal{S}'}(x)$ on the free $\mathbb{Z}_\ell$-module $\mathbb{Z}_\ell[x]/(f_\mathcal{S})$ of rank $\deg f_\mathcal{S}$, whose determinant is the norm $\prod_{f_\mathcal{S}(\alpha)=0} f_{\mathcal{S}'}(\alpha) = \operatorname{Res}(f_\mathcal{S}, f_{\mathcal{S}'})$; its length is the valuation. (4) $\mathbb{T}_\mathfrak{m} \cong \mathbb{Z}_\ell[x]/(F_\mathfrak{m})$ with $F_\mathfrak{m}$ the Hensel factor of f_ϑ at $\mathfrak{m}$, a quotient of a two-dimensional regular local ring by one element; $\mathbb{T}_\mathfrak{m}/\ell \cong \mathbb{F}_\ell[x]/(\bar h^a)$, whose maximal ideal is generated by $\bar h(x)$ and whose socle, $\bar h^{a-1}\mathbb{F}_\ell[x]/(\bar h^a)$, is one-dimensional over the residue field; so $\mathbb{T}_\mathfrak{m}/\ell$ is Gorenstein, and so is $\mathbb{T}_\mathfrak{m}$ (cited). Converse: a local $\mathbb{F}_\ell$-algebra generated by one element has a principal maximal ideal. □

The kind of a block. For $\mathbb{T}_\mathfrak{m} = \mathbb{Z}_\ell[x]/(F_\mathfrak{m})$ the kind — DVR, non-maximal order of one component, or a meeting of several components — is decided in two steps at finite precision: the ℓ-adic factorisation of $F_\mathfrak{m}$ (Hensel, from f_ϑ modulo a finite power of ℓ) gives the components through $\mathfrak{m}$ and their residue fields, which separates "meeting" from "one component"; for one component, whether $\mathbb{Z}_\ell[x]/(F_\mathfrak{m})$ is maximal is its index in the maximal order (Dedekind's criterion, or one Ore–Montes step when that is inconclusive). *Per block* (not asserted as a statement: the bookkeeping when the Hensel factors of distinct blocks are not coprime modulo ℓ is not written out): if for each block some computed $\vartheta_\mathfrak{m}$ generates $\mathbb{T}_\mathfrak{m}/\ell$ (tested in $e_\mathfrak{m} A_K/\ell A_K$, $K \ge c^{\mathrm{sat}}_\mathfrak{m} + 1$, or at $K = 1$ on a saturated block), then $\mathbb{T}_\mathfrak{m} = \mathbb{Z}_\ell[e_\mathfrak{m}\vartheta_\mathfrak{m}]$. *Example* (691). At $(1, 10)$, $\mathbb{T}^{\mathrm{fr}} = \{(a,b) \in \mathbb{Z}^2 : a \equiv b\ (691)\}$ (Eisenstein, Δ) has the $\mathbb{Z}$-basis $(1,1)$, $(691, 0)$, and $T_2 = (2049, -24) = -24\cdot(1,1) + 3\cdot(691, 0)$. Modulo 691, $T_2 \equiv -24 + 3\varepsilon$, $\varepsilon := (691, 0)$, $\varepsilon^2 \equiv 0$, so $\bar T_2$ generates $\mathbb{T}_{691}/691 = \mathbb{F}_{691}[\varepsilon]/(\varepsilon^2)$ (which is A_1 there: the lattice is saturated at 691). Hence $\mathbb{T}_{691} = \mathbb{Z}_{691}[T_2]$, and the depth of the congruence is $v_{691} \operatorname{Res}(x - 2049, x + 24) = v_{691}(2073) = 1$: **no** $K > 1$ **is needed at** 691. At 3, $\mathbb{Z}[T_2]$ has index 3 in $\mathbb{T}^{\mathrm{fr}}$ ($2073 = 3 \cdot 691$) and $\bar T_2$ does not generate. (Sign: [E, `cl:ramanujan`] uses $(0, 691) \equiv -(691, 0)$, hence its -3 against $+3$ here.)

Proposition 31.35 (Reconstruction from A_K)**.** *[proved here] (the plan's Prop. 2.5.) Let $\vartheta \in \mathbb{T}'$ with $\mathbb{Q}[\vartheta] = \mathbb{T}_\mathbb{Q}$, $\mathcal{R} := \mathbb{Z}_\ell[\vartheta] \subseteq \mathbb{T}_\ell$, ℓ^e the exponent of $\mathbb{T}_\ell/\mathcal{R}$, and assume A_K is the image of $\mathbb{T}_\ell$ itself (generation, certified or by a theorem). Let c^{sat} be that of Definition 30.1 for $(L_\ell, \mathbb{T}_\ell)$ (0 if saturated) and $\pi\colon \mathcal{R} \to A_K$ the restriction of the reduction.*

1. *For every $E \ge e$ and $K \ge E + c^{\mathrm{sat}}$: $\ell^E\mathbb{T}_\ell = \pi^{-1}(\ell^E A_K)$. So $\mathbb{T}_\ell = \ell^{-E}\pi^{-1}(\ell^E A_K) \subset \mathbb{Q}_\ell[x]/(f_\vartheta)$ is an explicit order.*

2. $e \le \lfloor v_\ell(\mathrm{disc}_\vartheta)/2 \rfloor$. ***Stopping rule***, *saturated case:* $Q_K := A_K/\pi(\mathcal{R}) \cong (\mathbb{T}_\ell/\mathcal{R})/\ell^K$ *has exponent* $\ell^{\min(e,K)}$, *so the first* K *with* $\exp Q_K < \ell^K$ *gives* $e = \log_\ell \exp Q_K$, *without* disc_ϑ. *Unsaturated case: the stopping rule does* ***not*** *apply* (Q_K *also carries the defect); use* $E := \lfloor v_\ell(\mathrm{disc}_\vartheta)/2 \rfloor$ *and* $K \ge E + c^{\mathrm{sat}}$, c^{sat} *off the plateau of Proposition 30.3(2).*

Proof. (1) "$\subseteq$": $\ell^E t \in \mathcal{R}$ for $t \in \mathbb{T}_\ell$, and $\pi(\ell^E t) = \ell^E \bar{t}$. "$\supseteq$": if $\pi(r) = \ell^E \bar{a}$ with $a \in \mathbb{T}_\ell$, then $r - \ell^E a \in \ker(\mathbb{T}_\ell \to A_K) = \mathbb{T}_\ell \cap \ell^K \mathbb{T}_\ell^{\mathrm{sat}} \subseteq \ell^{K-c^{\mathrm{sat}}} \mathbb{T}_\ell \subseteq \ell^E \mathbb{T}_\ell$ (Proposition 30.2(1)). (2) $e \le \mathrm{length}(\mathbb{T}_\ell/\mathcal{R}) \le v_\ell(\mathrm{disc}_\vartheta)/2$ by Proposition 31.33(1); in the saturated case $A_K = \mathbb{T}_\ell/\ell^K\mathbb{T}_\ell$, so $Q_K = \mathbb{T}_\ell/(\mathcal{R} + \ell^K\mathbb{T}_\ell)$, and a finite ℓ-group G of exponent ℓ^e has $G/\ell^K G$ of exponent $\ell^{\min(e,K)}$. □

Only *generation* is needed for reconstruction; saturation is one way to certify it. With generation certified there is also a **rational route** that needs no $K > 1$: $\mathbb{T}'$ is the $\mathbb{Z}$-algebra generated by the $g_p(\vartheta) \in \mathbb{Q}[x]/(f_\vartheta)$ with $X_p = g_p(\vartheta)$, computed by a Hermite normal form over $\mathbb{Z}$; the two routes cross-check each other.

Proposition 31.36 (Which components pass through a block)**.** *[proved here] (the plan's Prop. 2.6.) Let* ϑ *be as in Proposition 31.35,* $f_\vartheta = \prod_i f_i$, $g_i := f_\vartheta/f_i$, $\varepsilon_i \in \mathbb{T}_\mathbb{Q}$ *the idempotent of component* i, *and* $e \in \mathbb{T}_\ell$ *an idempotent (e.g. of a block of* $\mathbb{T}'_\ell$*). Then* $g_i(\vartheta)e \ne 0$ *iff* $\varepsilon_i e \ne 0$ *iff component* i *passes through some maximal ideal in the support of* e. *If* $g_i(\vartheta)e \ne 0$, *then* $g_i(\vartheta)e \not\equiv 0 \pmod{\ell^K}$ *on* L_ℓ *for every* $K > v_\ell \mathrm{Res}(f_i, g_i)$*; more generally for every* $K > v_\ell(D)$, $D \in (f_i, g_i) + \ell^K\mathbb{Z}[x]$ *an integer. No generation and no saturation is needed.*

Proof. $g_i(\vartheta)$ kills the components $j \ne i$ ($f_j \mid g_i$), and $g_i(\vartheta)\varepsilon_i$ is a unit of $\varepsilon_i\mathbb{T}_\mathbb{Q}$ ($\gcd(f_i, g_i) = 1$); so $g_i(\vartheta)e = 0$ iff $\varepsilon_i e = 0$. For a block idempotent $e_\mathfrak{m}$, $\varepsilon_i e_\mathfrak{m} \ne 0$ iff $(\mathcal{O}_i)_\mathfrak{m} \ne 0$ iff $P_i \subseteq \mathfrak{m}$. Precision: $X := eL_\ell \cap (L \otimes \mathbb{Q}_\ell)_i$ is non-zero when $\varepsilon_i e \ne 0$ and saturated in L_ℓ (eL_ℓ is a direct summand). Write $D = uf_i + vg_i + \ell^K h$; on X, $f_i(\vartheta) = 0$, so $Dx = v(\vartheta)g_i(\vartheta)x + \ell^K h(\vartheta)x$. If $g_i(\vartheta)eL_\ell \subseteq \ell^K L_\ell$, then $DX \subseteq \ell^K L_\ell \cap X = \ell^K X$, i.e. $v_\ell(D) \ge K$. $D = \mathrm{Res}(f_i, g_i)$, $h = 0$, is a case. □

The idempotent (the plan's A1-N7, F07-5). In practice e is computed in A'_K and lifted. $\ker(\mathbb{T}'_\ell \to A'_K)$ consists of elements $\ell^K x$ with x in the saturation of $\mathbb{T}'_\ell$, a finite $\mathbb{Z}_\ell$-algebra, so it lies in the Jacobson radical; idempotents lift uniquely modulo it, and exist because $\mathbb{T}'_\ell$ is ℓ-adically complete (lift at each finite level $A'_K \to A'_1$, a nilpotent kernel, and take the limit).

The coefficient sequences, their splitting, and the glue. The next four statements are the plan's stage 03 (`03-bockstein-tor.md` v0.3: B1, B7, B8, B8′, B9, B11–B15). Notation: $H^q_\ell := H^q \otimes \mathbb{Z}_\ell$, $\mathrm{tors}_\ell := \mathrm{tors}\, H^1_\ell$, e_ℓ its exponent (ℓ^{e_ℓ}), $L_q := H^q/\mathrm{tors}$.

Proposition 31.37 (Universal coefficients with the Hecke action)**.** *[proved here] [standard: the universal coefficient theorem for a complex of free abelian groups, [Wei94, §3.6]VERIFY; naturality of connecting maps, [Wei94, Thm. 1.3.1]] (the plan's B1, with B4(1) and the consequences of 01 §2.) Let* ℓ *be a prime and* $K \ge 1$.

1. *For every* $q \ge 0$ *there is a natural exact sequence*

$$0 \to H^q \otimes \mathbb{Z}/\ell^K \xrightarrow{\mathrm{red}_K} H^q(\Gamma; V/\ell^K) \xrightarrow{\beta_K} H^{q+1}[\ell^K] = \mathrm{Tor}_1^{\mathbb{Z}}(H^{q+1}, \mathbb{Z}/\ell^K) \to 0,$$

   β_K *the connecting map of* $0 \to V \xrightarrow{\ell^K} V \to V/\ell^K \to 0$*; it splits as a sequence of abelian groups, not naturally.*
2. *Every map in it commutes with every double-coset operator* $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$, g *in the Hecke semigroup of* $\Gamma_0(N)$*:* T_p *(*$p \nmid N$, $p = \ell$ *included),* U_p *(*$p \mid N$*) and their words.*

3. *The same holds for any complex C of finitely generated free $\mathbb{Z}_{(\ell)}$-modules quasi-isomorphic in degrees $\le q_{\max}$ to $\mathrm{Hom}_\Gamma(P_\bullet, V) \otimes \mathbb{Z}_{(\ell)}$, $P_\bullet$ a resolution by finitely generated free modules: $H^q(C/\ell^K) = H^q(\Gamma; V/\ell^K)$ for $q < q_{\max}$, with the sequence of (1), and a cochain endomorphism F of C induces compatible endomorphisms of the three terms.* ***That endomorphism is T_p on $H^q(\Gamma; V/\ell^K)$ only if F is a cochain-level lift of T_p over $\mathbb{Z}_{(\ell)}$*** *in the sense of Proposition 31.13(2) — not merely an F inducing T_p on $H^\bullet(C)$: the ambiguity is the component of $\mathrm{End}_{D(\mathbb{Z}_{(\ell)})}(C)$ in $\mathrm{Ext}^1(H^{q+1}(C), H^q(C))$, which acts as $H^{q+1}[\ell^K] \to H^q/\ell^K$, exactly on the* Tor *term (example: $C^0 = \mathbb{Z}a \oplus \mathbb{Z}b$, $C^1 = \mathbb{Z}c$, $da = 0$, $db = \ell c$; $F(a) = a$, $F(b) = b + a$, $F(c) = c$ is the identity on $H^\bullet(C)$ but not on $H^0(C/\ell)$). The shipped operators are such lifts (Propositions 31.13(2), 31.21(1)); an export built from integral-cohomology data alone would be wrong on the* Tor *part.*
4. Consequences. *$0 \subseteq \mathrm{tors}_q/\ell^K \subseteq H^q/\ell^K \subseteq H^q(\Gamma; V/\ell^K)$ is a filtration by Hecke submodules with graded pieces tors_q/ℓ^K, L_q/ℓ^K, $H^{q+1}[\ell^K]$; the middle step is the reduction of $0 \to \mathrm{tors}_q \to H^q \to L_q \to 0$, not a universal-coefficient sequence; $H^q/\ell^K = \ker \beta_K$ (on a free model: x lies in it iff $\ell^{-K} d\tilde{x}$ is a coboundary for a lift $\tilde{x}$), and there is no Hecke-equivariant complement in general. At $\ell \ge 5$, $H^1(\Gamma; V/\ell^K) = H^1/\ell^K$ (H^2 has no ℓ-torsion, Theorem 31.16(1)), which is L/ℓ^K exactly when $\ell \nmid |\mathrm{tors}\, H^1|$. At $\ell \in \{2, 3\}$, $H^1/\ell \subset H^1(\Gamma; V/\ell) \twoheadrightarrow H^2[\ell]$: $H^2[\ell]$ is a* quotient*, there is no canonical Hecke-equivariant copy of it inside, one exists iff the sequence splits $\mathbb{T}$-equivariantly (Proposition 31.38 guarantees it under disjoint supports and does not characterise it), and a mod-ℓ maximal ideal must be attributed to its degree through β_1 before it is reported as a prime of $\mathbb{T}$.*

Proof. (1) V is $\mathbb{Z}$-free, so the coefficient sequence is an exact sequence of $\mathbb{Z}[\Delta]$-modules, and $H^\bullet(\Gamma; -)$ is a δ-functor; the long exact sequence breaks into the displayed short ones because $H^q \xrightarrow{\ell^K} H^q$ has cokernel H^q/ℓ^K and kernel $H^q[\ell^K]$ (the case $K = 1$ is Lemma 16.10 and, at $\ell \ge 5$, Claim 16.7(2)). $\Gamma_0(N)$ is of type FP_∞, so $\mathrm{Hom}_\Gamma(P_\bullet, V)$ is a complex of free abelian groups and the universal coefficient theorem gives the splitting. (2) res, $(\mathrm{conj}_g)_*$ and cor are morphisms of δ-functors in the coefficient module (Claim 16.7(3); at level N the operator of Lemma 14.22(3), realised on the models by Proposition 31.13; U_p on cohomology, Proposition 31.10(3)(b′)). (3) A quasi-isomorphism of bounded-below complexes of free $\mathbb{Z}_{(\ell)}$-modules stays one after $\otimes \mathbb{Z}/\ell^K$ (its cone is acyclic, bounded below and free over a principal ideal domain, hence split exact); for a comparison in degrees $\le q_{\max}$ apply this to the brutal truncations, or compare the universal coefficient sequences by the five lemma in degrees $< q_{\max}$. C free gives the sequence for C, and F commutes with red_K and β_K by naturality, on the nose. The example: $H^0(C) = \mathbb{Z}a$, $H^1(C) = \mathbb{Z}/\ell$, F is the identity on both, and on $H^0(C/\ell) = \mathbb{F}_\ell a \oplus \mathbb{F}_\ell b$ it sends $b \mapsto b + a$. (4) $\mathrm{Tor}_1(L_q, \mathbb{Z}/\ell^K) = 0$ makes the first inclusion injective; the rest is (1), (2) and the definition of β_K. □

Nothing here depends on U_p over $\mathbb{Z}$ being built: the statement is about the operators on cohomology. The record of Theorem 31.16 predicts every mod-ℓ^K type at no cost (the plan's B2: $H^q(\Gamma; V/\ell^K) \cong (\mathbb{Z}/\ell^K)^{b_q} \oplus \bigoplus_i \mathbb{Z}/\ell^{\min(a_{q,i}, K)} \oplus \bigoplus_i \mathbb{Z}/\ell^{\min(a_{q+1,i}, K)}$ for $H^q_{(\ell)} \cong \mathbb{Z}_{(\ell)}^{b_q} \oplus \bigoplus_i \mathbb{Z}/\ell^{a_{q,i}}$; checked against the Lean-kernel dimensions of Remark 16.20 on 16 of 16 cases, the plan's §6.1).

Proposition 31.38 (Splitting by support)**.** *[proved here] (the plan's B7.) Let $0 \to S \to M \to Q \to 0$ be an exact sequence of finite modules over a commutative ring R (here $\mathbb{T}$, or the image of a subalgebra $\mathbb{T}'$). If $\mathrm{supp}_R(S) \cap \mathrm{supp}_R(Q) = \emptyset$, it splits uniquely as R-modules, $M = M_{\mathrm{supp}\, S} \oplus M_{\mathrm{supp}\, Q}$, cut out by the idempotents of the Artinian image of R in $\mathrm{End}(M)$; and $\mathrm{Hom}_R(Q, S) = \mathrm{Hom}_R(S, Q) = \mathrm{Ext}^1_R(Q, S) = 0$. For a subring $R' \subseteq R$, disjointness for R' implies disjointness for R (maximal ideals of R contract to those of R'), so* ***a separation computed from the operators in $\mathcal{P}$ is a certificate for all of $\mathbb{T}$.*** *The same holds for modules finitely generated over $\mathbb{Z}_\ell$ (e.g. L_ℓ), with the complete semilocal ring $\mathbb{T}_{\mathbb{Z}_\ell}$ (resp. $\mathbb{T}_\ell$) in place of the Artinian image.*

Proof. The image A of R in $\mathrm{End}M$ is finite, hence Artinian, $A = \prod_{\mathfrak{m}} A_{\mathfrak{m}}$; its idempotents act on the whole sequence, and on each factor one of the outer terms vanishes. Homomorphisms and extensions between modules supported on disjoint sets of idempotents vanish. A maximal ideal of R in both supports would contract to one of R' in both supports. In the $\mathbb{Z}_\ell$-finite case the same argument runs with the block idempotents of the complete semilocal ring. □

The separation certificate (the plan's K6). The torsion residues of T_p are the roots of $\prod_j P_j(x)$, the layer polynomials of Proposition 31.15(1)(a) — possibly several, possibly in an extension of $\mathbb{F}_\ell$ — so separation at T_p is $\gcd\big(\chi^{(N)}_{p,1} \bmod \ell,\ \prod_j P_j\big) = 1$ in $\mathbb{F}_\ell[x]$, certified by a Bézout pair or by $\mathrm{Res} \not\equiv 0 \pmod{\ell}$; a certificate of $O(b_1)$ small integers, not "a few". The per-operator test is weaker than the joint one (at $(1,18)$, $\ell = 5$, every T_p alone agrees with some free eigenvalue, but no free joint system agrees with the torsion's); the joint test needs the matrices. *When the sequences of Proposition 31.37(1) split.* At $\ell \geq 5$ the only non-trivial one is $q = 0$, an isomorphism; the question is the torsion/free extension inside H^1/ℓ^K (Proposition 31.41). At $\ell \in \{2,3\}$, level 1: never by support for the prime-to-ℓ algebra (Proposition 31.39(1)); T_ℓ may separate. At $\ell \in \{2,3\}$, level N: Proposition 16.15 is a level-one statement and fails at level N ($\Gamma_0(11)$, $n = 0$, $\ell = 2$: T_3 acts on the newform's line by $-1 \not\equiv 1 + 3$); supports must be computed.

Proposition 31.39 (Level one at $\ell \in \{2,3\}$)**.** *[proved here] given Proposition 16.15 and Lemma 16.14 (the plan's B8, B8′, B9, with A3-M1's correction). Let $N = 1$, n even, $\ell \in \{2,3\}$.*

1. *For every prime $p \neq \ell$ and every q: $T_p - (1+p) \in \ell\mathrm{End}(L_q \otimes \mathbb{Z}_{(\ell)})$, and $T_p - (1+p)$ is nilpotent on $\mathrm{tors}\, H^q_{(\ell)}$, on L_q/ℓ^K, on $H^{q+1}[\ell^K]$ and on $H^q(\Gamma; V/\ell^K)$. It is **not** nilpotent on $H^q_{(\ell)}$ in general (at $(1,10)$, $\ell = 3$, $T_2 - 3$ has the non-zero eigenvalues 2046 and -27 on $L \otimes \mathbb{Q}$); on $L_{(\ell)}$ it is only topologically nilpotent. Equivalently: $T_p - (1+p)$ lies in every maximal ideal of $\mathbb{T}$ above ℓ.*
2. *The maximal ideals of $\mathbb{T}$ above ℓ are $\mathfrak{m}_\psi := (\ell,\ T_p - (1+p)\ (p \neq \ell),\ \psi(T_\ell))$, ψ running over the irreducible factors modulo ℓ of the characteristic polynomial of T_ℓ on H^1/ℓ, which is $\chi^{(1)}_{\ell,1} \bmod \ell$ (free part) times $\prod_j P_j$ (the torsion layers of T_ℓ, Proposition 31.15(1)(a)) — both in the descent record. $\mathfrak{m}_\psi \in \mathrm{Spec}\,\mathbb{T}^{\mathrm{fr}}$ iff $\psi \mid \chi^{(1)}_{\ell,1} \bmod \ell$; $\mathfrak{m}_\psi \in \mathrm{Spec}\,\mathbb{T}^{\mathrm{tors}}$ iff $\psi \mid \prod_j P_j$. No generation hypothesis is needed.*
3. *T_ℓ is nilpotent on tors_ℓ; so $\mathrm{supp}(\mathrm{tors}_\ell) \subseteq \{\mathfrak{m}_x\}$.*
4. *(a) The torsion point $\mathfrak{m}_x$ coincides with a free point iff $\chi^{(1)}_{\ell,1}$ has the root 0 modulo ℓ, i.e. iff some cusp eigenform is non-ordinary at ℓ (the Eisenstein series has $a_\ell = 1 + \ell^{k-1} \equiv 1$); a certificate for all p. (b) $\mathbb{T}$ has a torsion-only point above ℓ iff $\mathrm{tors}_\ell \neq 0$ and every cusp eigenform is ℓ-ordinary. (c) The torsion never meets the Eisenstein point $\mathfrak{m}_{x-1}$.*
5. ***No cusp eigenform of level 1 is ℓ-ordinary.** Hence if $S_k \neq 0$ the torsion point $\mathfrak{m}_x$ is common with the cuspidal part, and a torsion-only point above ℓ exists iff $\mathrm{tors}_\ell \neq 0$ and $S_k = 0$ ($k \in \{4,6,8,10,14\}$).*
6. *$\dim_{\mathbb{F}_\ell} A_1(L/\ell) \leq 1 + \mathrm{rank}(T_\ell \mid L/\ell) \leq 3$. Hence $\mathbb{T}_\ell$ is **not saturated on L_ℓ whenever** $d = \dim M_k \geq 4$ ($k = 36$ or $k \geq 40$), at $\ell = 2$ and at $\ell = 3$, and Theorem 31.45(C3) cannot pass at 2 or 3 for any operator set. This is a statement about the cohomology lattice L (and its direct summands), not about every $\mathbb{T}^{\mathrm{fr}}$-stable lattice.*

Proof. (1) Proposition 16.15 gives $T_p = 1 + p$ on $H^q(\Gamma; V/\ell)$, and H^q/ℓ (a sub), tors/ℓ (a sub of that), L_q/ℓ (a subquotient) and $H^{q+1}[\ell]$ (a quotient) are subquotients of it (Proposition 31.37(4)); 0 on L_q/ℓ is the statement $T_p - (1+p) \in \ell\mathrm{End}(L_q \otimes \mathbb{Z}_{(\ell)})$. On a finite ℓ-group, or on a lattice modulo ℓ^K, an endomorphism that is 0 on every ℓ-adic layer is nilpotent. (2) Proposition 31.31 with $M = H^1 \otimes \mathbb{Z}_{(\ell)}$, faithful over $\mathbb{T} \otimes \mathbb{Z}_{(\ell)}$: $V(\ell) \cong \mathrm{Spec}\, A_1(H^1/\ell)$, and $\mathbb{T}$ acts on H^1/ℓ through $\mathbb{F}_\ell[T_\ell]$ by (1). In the Grothendieck group of finite $\mathbb{T}$-modules $[G/\ell G] = [G[\ell]]$ ($0 \to G[\ell] \to G \xrightarrow{\ell} G \to G/\ell G \to 0$), and $G[\ell]$ is filtered by the layers, whence the torsion factor.

(3) $\beta\colon H^0(\Gamma;V/\ell)\to H^1[\ell]$ is a T_ℓ-equivariant isomorphism (Lemma 16.10), and $T_\ell=U_\ell$ has square 0 on $H^0(\Gamma;M_n)$ (Proposition 24.4(2)); by the Grothendieck-group identity of (2), every composition factor of tors_ℓ is one of $H^1[\ell]$, on which T_ℓ acts by 0, so T_ℓ is nilpotent. (4) From (2), (3). (5) On L/ℓ, a quotient of the submodule H^1/ℓ of $H^1(\Gamma;M_n)$, $\operatorname{rank}T_\ell\le 2$ (Lemma 16.14: $\ell=2$ every $q\ge 1$; $\ell=3$, n even, $q=1$; rank does not grow on subquotients). The invertible Fitting part of T_ℓ on L/ℓ has dimension at most that rank, and it counts the eigenvalues of T_ℓ on $L\otimes\mathbb{Q}$ that are ℓ-adic units, with multiplicity. The Eisenstein system contributes 1; each cuspidal system occurs twice in L (Proposition 31.29(2)), so an ℓ-ordinary cusp eigenform would add ≥ 2, total $\ge 3>2$. $S_k=0$ exactly for $k\in\{4,6,8,10,14\}$. (6) $A_1(L/\ell)=\mathbb{F}_\ell[\bar T_\ell]$ by (1). If Y has rank r and g is the characteristic polynomial of Y on $\operatorname{im}Y$, then $g(Y)Y=0$, so $\deg m_Y\le r+1$. Saturation is $\dim A_1=d$ (Proposition 30.2(2)); and $\dim M_k\ge 4$ iff $k=36$ or $k\ge 40$. □

Checks (the plan's §6.2): at the eight level-one cells $n\in\{8,\dots,30\}$ the torsion's T_ℓ residues are $\{0\}$ at 2 and 3; the free ones are $\{1\}$ at $n=8,12$ ($S_k=0$) and $\{0,1\}$ otherwise — torsion-only points exactly at $n=8,12$, and at $(1,10)$ the torsion points over 2 and 3 **are** Δ's maximal ideals. The 2- and 3-adic structure of $\mathbb{T}^{\mathrm{fr}}$ at level 1 for $d\ge 4$ must therefore come from Proposition 31.49 or from $A_K/\ell A_K$ at $K\ge c^{\mathrm{sat}}+1$ under a generation theorem (Proposition 31.34, computable form).

Remark 31.40 (CB-1: $T_p=1+p$ on the 2-primary torsion at level one). [conjectural] (the plan's CB-1; recorded, **used by nothing**, no hypothesis structure.) For $N=1$, n even and every odd prime p, $T_p=1+p$ exactly on $\operatorname{tors}H^1(\mathrm{SL}_2(\mathbb{Z});V)_{(2)}$. Evidence: all eight checked cells, $n\le 30$, exponents up to 2^4 (e.g. $(1,16)$: $\mathbb{Z}/2\oplus\mathbb{Z}/4\oplus\mathbb{Z}/16$, $\ker(T_3-4)=\ker(T_5-6)=\ker(T_7-8)$ everything); consistent with the Frobenius classes h_j of Remark 16.8(2), on which $T_p\equiv(1+p)p^{a2^j}$ modulo 2^{j+1} (the plan's B10(2), from Lemma 24.2 and Proposition 24.4(1)), since $p^{2^j}\equiv 1 \pmod{2^{j+1}}$. The analogue at $\ell=3$ is **false**: the $\mathbb{Z}/9$ at $n=12$ and $n=30$ carries $(1+p)p^3 \bmod 9$, not $1+p$.

Proposition 31.41 (The glue J: what it is, the extension class, what computed data give, coincidence)**.** *[proved here] [standard: the Artin–Rees lemma, [AM69, Cor. 10.10]] for (5)(b)'s limit;* ***no stopping rule for that limit is asserted*** *(the plan's B11–B15, with A3-M2's restriction).*

1. (What J is.) $I_{\mathrm{fr}}=\mathbb{T}_{\mathrm{tors}}$, *the additive torsion subgroup of the ring* $\mathbb{T}$*; so* $J=\{t\in\mathbb{T}_{\mathrm{tors}}: t\cdot\operatorname{tors}H^1=0\}$ *and* $|\mathbb{T}_{\mathrm{tors}}|=|J|\cdot|\operatorname{im}(\mathbb{T}_{\mathrm{tors}}\to\mathbb{T}^{\mathrm{tors}})|$. *Every* $t\in J$ *induces* $\varphi_t\colon L\to\operatorname{tors}H^1$, *and* $t\mapsto\varphi_t$ *is an injective* $\mathbb{T}$*-linear map* $J\hookrightarrow\operatorname{Hom}_{\mathbb{T}}(L,\operatorname{tors}H^1)=\bigoplus_\ell\operatorname{Hom}_{\mathbb{T}}(L/\ell^{e_\ell},\mathrm{tors}_\ell)$. $J_{\mathfrak{m}}\ne 0$ *only at maximal ideals in* $\operatorname{supp}(L/\ell)\cap\operatorname{supp}(\mathrm{tors}_\ell)$.
2. (The extension class.) *Let* ε *be the class of* $0\to\mathrm{tors}\to H^1\to L\to 0$ *in* $\operatorname{Ext}^1_{\mathbb{T}}(L,\mathrm{tors})$ *and, for* $K\ge e_\ell$, ε_K *that of* $0\to\mathrm{tors}_\ell\to H^1_\ell/\ell^K\to L_\ell/\ell^K\to 0$. *(a) The* ℓ*-part of* ε *is the pull-back of* ε_K *along* $L_\ell\to L_\ell/\ell^K$, $H^1_\ell\cong(H^1_\ell/\ell^K)\times_{L_\ell/\ell^K}L_\ell$*; so* ε_ℓ *splits iff* ε_K *does, for every* $K\ge e_\ell$. *(b)* Split test*: in a basis adapted to* $\mathrm{tors}_\ell\subset H^1_\ell/\ell^K$ *each operator is* $\begin{pmatrix}B_p & 0\\ \beta_p & c_p\end{pmatrix}$ *(B_p on L/ℓ^K, c_p on tors_ℓ, $\beta_p\in\operatorname{Hom}(L/\ell^{e_\ell},\mathrm{tors}_\ell)$ the derivation part); ε_K splits for $\mathbb{T}'$ iff $hB_p-c_ph=\beta_p$ ($p\in\mathcal{P}$) has a solution $h\in\operatorname{Hom}(L/\ell^{e_\ell},\mathrm{tors}_\ell)$. Unsolvable for $\mathbb{T}'$ implies non-split for $\mathbb{T}$ (a certificate); solvable for $\mathbb{T}'$ implies split for $\mathbb{T}$ only if $\mathcal{P}$ generates the image of $\mathbb{T}$ on H^1_ℓ/ℓ^K. (c) Split implies $J_\ell=0$;* ***the converse fails****: on $H=\mathbb{Z}_\ell^2\oplus\mathbb{F}_\ell$ let $x(a,b,c)=(0,\ell b,\bar a)$; $\mathbb{Z}_\ell[x]\cong\mathbb{Z}_\ell[x]/(x^2-\ell x)$ acts faithfully on $L=\mathbb{Z}_\ell^2$ by $\operatorname{diag}(0,\ell)$, so $I_{\mathrm{fr}}=0=J$, but no complement is x-stable. (d) If $\mathbb{T}^{\mathrm{fr}}_{\mathfrak{m}}$ acts on $L_{\mathfrak{m}}$ by scalars and $\mathrm{tors}_{\mathfrak{m}}$ is one-dimensional with the same residual character, then $t-\lambda(t)\in J_{\mathfrak{m}}$ for every t, $J_{\mathfrak{m}}$ is spanned by the $\beta(T_p)$, and* ***non-split*** $\Leftrightarrow J_{\mathfrak{m}}\ne 0$ *there.*
3. (Reading J from mod-ℓ^K data.) *Let* $c:=c^{\mathrm{sat}}$ *for* $(L_\ell,\mathbb{T}_\ell)$, $K\ge e_\ell+c$, $A_K:=$ *the image of* $\mathbb{T}_{\mathbb{Z}_\ell}$ *in* $\operatorname{End}(H^1_\ell/\ell^K)$ *and* $J_K:=\ker\big(A_K\to\operatorname{End}(L_\ell/\ell^K)\times\operatorname{End}(\mathrm{tors}_\ell)\big)$. *Then* $J_\ell:=J\otimes\mathbb{Z}_\ell\to A_K$ *is injective with image* J_K. *For a subalgebra* $\mathbb{T}'$: $J\cap\mathbb{T}'\subseteq J\hookrightarrow$

$\mathrm{Hom}_{\mathbb{T}'}(L/\ell^{e_\ell}, \mathrm{tors}_\ell)$. ***What computed data give:*** *(i) this* sandwich *(a non-zero element of $J \cap \mathbb{T}'$ certifies $J_\ell \neq 0$; record status* `bounds`*); (ii) $J_\ell = 0$* exactly *where the supports separate ((1), Proposition 31.38) or where the upper bound has order* 1 *(status* `zero`*); (iii) $J_\ell = \ker(A'_K \to \cdots)$* ***only under*** *the mixed generation statement of Remark 31.43, which is [conjectural]: so J is never recorded as exact.*

4. (Coincidence certificates.) *If ε_K is non-split for some $\mathbb{T}'$, or $J \cap \mathbb{T}' \neq 0$, or $\mathrm{Hom}_{\mathbb{T}}(L, \mathrm{tors}_\ell) \neq 0$, then some maximal ideal of $\mathbb{T}$ contains I_{fr} and I_{tors}: a torsion system* equal for all p *to a free one. If moreover $\mathrm{tors}_{\ell,\mathfrak{m}'}$ is one-dimensional over $\mathbb{F}_\ell$ for the $\mathbb{T}'$-block $\mathfrak{m}'$ on which the non-split extension or the witness was found, then $\mathbb{T}$ acts on it by one character, a residual system of $\mathbb{T}^{\mathrm{fr}}$. (Coincidence is a* Spec *statement; this does not compute J.)*
5. (The fibre product and the local rings.) *(a) With $C_{\mathrm{ft}} := \mathbb{T}/(I_{\mathrm{fr}} + I_{\mathrm{tors}})$, the torsion–free congruence ring, finite and supported exactly on the common points: $\mathbb{T}/J \cong \mathbb{T}^{\mathrm{fr}} \times_{C_{\mathrm{ft}}} \mathbb{T}^{\mathrm{tors}}$. So $\mathbb{T}$ is determined by $\mathbb{T}^{\mathrm{fr}}$, $\mathbb{T}^{\mathrm{tors}}$, the two maps $\mathbb{T}^{\mathrm{fr}} \to C_{\mathrm{ft}} \leftarrow \mathbb{T}^{\mathrm{tors}}$ (not the ring C_{ft} alone), the $\mathbb{T}/J$-module J and the class of the square-zero extension $0 \to J \to \mathbb{T} \to \mathbb{T}/J \to 0$ (the trivial extension $\mathbb{T}/J \oplus J$ is in general a different ring). (b) $\mathbb{T}_{\mathbb{Z}_\ell} = \varprojlim_K A_K(H^1_\ell/\ell^K)$, and there is c' with $\mathbb{T}_{\mathbb{Z}_\ell} \cap \ell^K \mathrm{End}(H^1_\ell) \subseteq \ell^{K-c'}\mathbb{T}_{\mathbb{Z}_\ell}$; for $K \geq e_\ell$, $E_K := \{f : f(H^1_\ell) \subseteq \ell^K H^1_\ell\} \subseteq \ell^{K-e_\ell}\mathrm{End}(H^1_\ell)$; so $\mathbb{T}_{\mathbb{Z}_\ell}/\ell^{K-c'-e_\ell}$ is a quotient of A_K: the local rings at torsion-touching ideals, glue included, are a finite computation once c' is known* and *A_K itself, the image of all of $\mathbb{T}$, is known — i.e. under Remark 31.43 when only $\mathbb{T}'$ is computed.*

Proof. (1) $\mathbb{T}^{\mathrm{fr}}$ is torsion-free of rank $\mathrm{rank}\,\mathbb{T}$ ($\mathrm{tors} \otimes \mathbb{Q} = 0$), so $I_{\mathrm{fr}} = \ker(\mathbb{T} \to \mathbb{T}^{\mathrm{fr}})$ is finite, hence $\subseteq \mathbb{T}_{\mathrm{tors}}$, and $\mathbb{T}_{\mathrm{tors}} \subseteq I_{\mathrm{fr}}$ as $\mathbb{T}^{\mathrm{fr}}$ is torsion-free. $t \in I_{\mathrm{fr}}$ maps H^1 into tors, and $t \in I_{\mathrm{tors}}$ kills tors, so $t \in J$ factors through L; $\varphi_t = 0$ means $t = 0$. For $s \in \mathbb{T}$, $\varphi_{st} = s|_{\mathrm{tors}} \circ \varphi_t$, $\varphi_{ts} = \varphi_t \circ s|_L$ and $st = ts$, so φ_t is $\mathbb{T}$-linear and so is $t \mapsto \varphi_t$. The support statement is Proposition 31.38 (Hom between disjoint supports vanishes). (2)(a) $x \mapsto (\bar{x}, \pi x)$ is a map of extensions of L_ℓ by tors_ℓ (identity on $\mathrm{tors}_\ell = \mathrm{tors}_\ell/\ell^K$ for $K \geq e_\ell$, and on L_ℓ), hence an isomorphism (five lemma); a section of ε_K pulls back to one of ε_ℓ, and a section of ε_ℓ reduces to one of ε_K. (b) A complement is the graph of an h, and its T_p-stability is the displayed equation. (c) For $t \in J$ and a $\mathbb{T}$-stable complement L': $t(L') \subseteq L' \cap \mathrm{tors} = 0$ and $t(\mathrm{tors}) = 0$. The example: $x^2(a,b,c) = (0, \ell^2 b, 0)$, so $x(x-\ell) = 0$ and there is no relation of degree 1 (evaluate on $(1,0,0)$ and $(0,1,0)$); x acts on L by $\mathrm{diag}(0,\ell)$, whose minimal polynomial is $x(x-\ell)$, so $\mathbb{Z}_\ell[x] \to \mathrm{End}L$ is injective; a complement $\{(a,b,h(a,b))\}$ is x-stable only if $\bar{a} = h(0,\ell b) = 0$ for all a. (d) On the block the inner derivations are $(c_p - \lambda_p)h \equiv 0$; for $t \in \mathbb{T}_{\mathfrak{m}}$, $t - \lambda(t)$ is 0 on $L_{\mathfrak{m}}$ and on $\mathrm{tors}_{\mathfrak{m}}$, so lies in $J_{\mathfrak{m}}$, with $\varphi_{t-\lambda(t)} = \beta(t)$; every $j \in J_{\mathfrak{m}}$ has $\lambda(j) = 0$; β is a derivation, so its values are spanned by those on the generators T_p; and the split equation reads $\beta_p = 0$ for all p. (3) Injective: $t \in J_\ell$ zero on H^1_ℓ/ℓ^K has $t(H^1_\ell) \subseteq \ell^K H^1_\ell \cap \mathrm{tors}_\ell = 0$ ($K \geq e_\ell$). Onto J_K: lift $a \in J_K$ to $t \in \mathbb{T}_{\mathbb{Z}_\ell}$; $t|_{\mathrm{tors}} = 0$ and $t|_L \in \mathbb{T}_\ell \cap \ell^K \mathrm{End}(L_\ell) \subseteq \ell^{K-c}\mathbb{T}_\ell \subseteq \ell^{e_\ell}\mathbb{T}_\ell$ (Proposition 30.2(1)); choose $s \in \mathbb{T}_{\mathbb{Z}_\ell}$ with $s|_L = \ell^{-e_\ell} t|_L$ ($\mathbb{T}_{\mathbb{Z}_\ell} \to \mathbb{T}_\ell$ is onto). Then $t - \ell^{e_\ell} s \in J_\ell$, and $\ell^{e_\ell} s$ is 0 on H^1_ℓ/ℓ^K: for a free complement L', $\ell^{e_\ell} H^1_\ell = \ell^{e_\ell} L'$, and for $y \in L'$, $s(y) = w + \tau$ ($w \in L'$, τ torsion) gives $\ell^{e_\ell} s(y) = \ell^{e_\ell} w$, whose image in L is $t|_L(\bar{y}) \in \ell^K L$, so $\ell^{e_\ell} w \in \ell^K L'$. So $t - \ell^{e_\ell} s$ has image a. The sandwich is (1) with $\mathrm{Hom}_{\mathbb{T}} \subseteq \mathrm{Hom}_{\mathbb{T}'}$; (ii) is (1) and the order bound; (iii) is the case $A'_K = A_K$. (4) Non-split for $\mathbb{T}'$ implies non-split for $\mathbb{T}$, which by Proposition 31.38 (contrapositive) implies a common point; $J \neq 0$ or $\mathrm{Hom} \neq 0$ implies one by (1). A one-dimensional module has one maximal ideal. (5)(a) $\mathbb{T}/(I \cap I') \to \mathbb{T}/I \times_{\mathbb{T}/(I+I')} \mathbb{T}/I'$ is injective, and onto: given $(x + I, y + I')$ with $x - y = i + i'$, $z := x - i = y + i'$ maps to the pair; $I_{\mathrm{fr}} \cap I_{\mathrm{tors}} = J$. (b) The constant c' is Artin–Rees for $\mathbb{T}_{\mathbb{Z}_\ell} \subseteq \mathrm{End}(H^1_\ell)$ over $\mathbb{Z}_\ell$. With the bound on E_K below, $\ell^K \mathbb{T}_{\mathbb{Z}_\ell} \subseteq \ker(\mathbb{T}_{\mathbb{Z}_\ell} \to A_K) = \mathbb{T}_{\mathbb{Z}_\ell} \cap E_K \subseteq \ell^{K-c'-e_\ell}\mathbb{T}_{\mathbb{Z}_\ell}$, so the kernels define the ℓ-adic topology and the inverse limit is the completion, $\mathbb{T}_{\mathbb{Z}_\ell}$ itself. For $f \in E_K$, $K \geq e_\ell$: $f(H^1_\ell) \subseteq \ell^K H^1_\ell = \ell^K L'$, and $g := \ell^{-(K-e_\ell)} f$ is a well-defined endomorphism with values in $\ell^{e_\ell} L'$ (L' free), so $f \in \ell^{K-e_\ell}\mathrm{End}(H^1_\ell)$; the inclusion is strict in general (on $\mathbb{Z}/\ell \oplus \mathbb{Z}/\ell^2$, $e_1 \mapsto \ell e_2$ lies in E_1, not in $\ell\mathrm{End}$). □

Checks (the plan's §6.4, level 1, $\ell = 5$): at $(1,18)$ and $(1,20)$ T_5 separates the torsion from the free part — split, $J_5 = 0$, certified; at $(1,26)$ and $(1,30)$ the torsion meets a non-ordinary cuspidal point, the extension splits for the computed $\mathbb{T}'$ (every one of the 25 candidate torsion lines), $J_5 \cap \mathbb{T}' = 0$ and $|J_5| \le 25$ (`bounds`); the exact J_5 needs Remark 31.43. The torsion image $\mathrm{tors}_\ell \subset H^1/\ell^K$ is canonical but **not determined by the Hecke module H^1/ℓ^K alone** when $\mathrm{Hom}_{\mathbb{T}}(\mathrm{tors}, L/\ell^{e_\ell}) \neq 0$: either the verdict is checked for every candidate, or the torsion cocycles of Proposition 31.15(3) are exported with the matrices. Theorem 31.16(7)(a) leaves the extension class open; by (2)(a) it is determined by finite data, ε_K for any $K \ge e_\ell$, and from computed operators (2)(b) certifies a non-split verdict, a split one only under Remark 31.43 (A7-m3).

Remark 31.42 (CB-2: the ι-refinement of the bound on J). [conjectural] (the plan's CB-2, a sketch; recorded, used by nothing.) At odd ℓ the involution ι commutes with $\mathbb{T}$ (Obligation 9.10), φ_t respects the ι-signs, and at a cuspidal common point $L_{\mathfrak{m}} = L_{\mathfrak{m}}^+ \oplus L_{\mathfrak{m}}^-$ with one sign matching the torsion's; so $|J_{\mathfrak{m}}| \le |\mathrm{Hom}_{\mathbb{T},\iota}(L_{\mathfrak{m}}, \mathrm{tors}_{\mathfrak{m}})|$ (e.g. $|J_5| \le 5$ at $(1,26)$). It needs ι on the $\mathbb{Z}/\ell^K$ handle ([E, `sec:lvdepth2`] has it over $\mathbb{Z}/\ell^k$) and the sign of the torsion class; not run.

Remark 31.43 (CB-3: mixed generation). [conjectural] (the plan's CB-3, A3-M2; **nothing is certified under it**, and no hypothesis structure of the Lean lane may carry it silently.) For the computed set $\mathcal{P}$ (with an explicit bound), the image A'_K of $\mathbb{T}' = \mathbb{Z}[X_p : p \in \mathcal{P}]$ on the whole H^1_ℓ/ℓ^K — free part, torsion and glue — equals the image A_K of $\mathbb{T}_{\mathbb{Z}_\ell}$; equivalently, the cotangent space of $\mathbb{T}_{\mathfrak{m}}/\ell$, glue included, is spanned by the $T_p - \lambda(T_p)$, $p \in \mathcal{P}$. It is a Sturm-type statement for the mixed module; by Proposition 31.41(2)(d), at a scalar block the derivation β vanishes once it vanishes on a set of generators of that cotangent space. Under it, J_ℓ would be the kernel J_K of the computed algebra (Proposition 31.41(3)(iii)) and the local rings at torsion-touching ideals would be computed (Proposition 31.41(5)(b)). A non-split verdict does not need it (Proposition 31.41(4)).

Theorem 31.44 (Descent of the Hecke structure)**.** *(The plan's Thm. 3.1.) Write $\mathbb{Z}_{(6)}$ for the semilocalisation at 2, 3.*

1. (Fracture.) *[proved here] For every finitely generated abelian group X, $X \to X[1/6] \times X_{(6)}$ is injective with image $X[1/6] \times_{X \otimes \mathbb{Q}} X_{(6)}$ ($\mathbb{Z}_{(6)}[1/6] = \mathbb{Q}$); for $X = \mathbb{T}$ this is a fibre product of rings. Images commute with flat base change, so $\mathbb{T} \otimes R = \mathrm{im}(\mathcal{H}_N \otimes R \to \mathrm{End}_R(H^1 \otimes R))$ for $R \in \{\mathbb{Z}[1/6], \mathbb{Z}_{(2)}, \mathbb{Z}_{(3)}, \mathbb{Z}_\ell\}$: the algebra form of Propositions 23.7 and 23.8.*
2. (Spectra.) *[proved here] As sets, $\mathrm{Spec}\,\mathbb{T} = \mathrm{Spec}\,\mathbb{T}_{\mathbb{Q}} \sqcup \bigsqcup_\ell V(\ell)$; every prime of characteristic ℓ is maximal; $\mathrm{Spec}\,\mathbb{T}[1/6]$ is the open complement of $V(2) \sqcup V(3)$; a component P_i specialises to $\mathfrak{m}$ iff $P_i \subseteq \mathfrak{m}$ (Proposition 31.36).*
3. (Local–global.) *[proved here] $\mathbb{T}^{\mathrm{fr}} = \{x \in \mathbb{T}_{\mathbb{Q}} : x \in \mathbb{T}_\ell \ \forall \ell\}$, and for $\vartheta \in \mathbb{T}^{\mathrm{fr}}$ with $\mathbb{Q}[\vartheta] = \mathbb{T}_{\mathbb{Q}}$, $\mathbb{T}_\ell = \mathbb{Z}_\ell[\vartheta]$ at every ℓ with $\ell^2 \nmid \mathrm{disc}_\vartheta$ (Proposition 31.33(3)). So $\mathbb{T}^{\mathrm{fr}}$ is determined by f_ϑ and the finitely many $\mathbb{T}_\ell$, $\ell^2 \mid \mathrm{disc}_\vartheta$; and $\mathbb{T}' = \mathbb{T}^{\mathrm{fr}}$ iff $\mathbb{T}'_\ell = \mathbb{T}_\ell$ for every ℓ (Lemma 31.30(3)).*
4. (Above 2 and 3: what the truncated rings see.) *[proved here] given the model statements of Theorem 31.16(3), (4) and Propositions 31.12, 31.6. For $\ell \in \{2,3\}$ the inputs — $L \otimes \mathbb{Z}_{(\ell)}$ and $\mathrm{tors}\, H^1_{(\ell)}$ with the operators — are computed on the ℓ-local models, and:*
   (a) *the maximal ideals over ℓ and their residue fields are given by A'_K for **every** $K \ge 1$, once generation modulo ℓ holds (Propositions 31.31, 31.32); without it a computation sees the primes of $\mathbb{T}'$, onto which $\mathrm{Spec}\,\mathbb{T}$ maps;*
   (b) *which components pass through each maximal ideal: at $K > v_\ell \mathrm{Res}(f_i, g_i)$ (Proposition 31.36), with no generation needed;*
   (c) *$\mathbb{T}_\ell$, hence the local rings and the ℓ-parts of the congruence modules, always under generation: in the monogenic case at $K = 1$ when $\mathbb{T}_\ell$ is saturated on L_ℓ and at $K = c^{\mathrm{sat}} + 1$ when it is not (Proposition 31.34); at $K \ge e + c^{\mathrm{sat}}$ otherwise (Proposition 31.35);*
   (d) *$\mathbb{T}^{\mathrm{tors}} \otimes \mathbb{Z}_\ell$ exactly, as the image on the finite module (any K at least its exponent),*

under a generation certificate for the torsion (Proposition 31.29(5)(c)) over $\mathbb{Z}/\ell^e$, $e \geq 2$ in general ($\mathbb{Z}/4$ at $(1,10)$, $(\mathbb{Z}/4)^2$ at $(2,6)$); its input is the torsion module *with its operators (Theorem 31.16(3)), not the layer polynomials (at $(1,10)$ the layers of $\mathrm{tors}_2 = \mathbb{Z}/2 \oplus \mathbb{Z}/4$ read "x" for every T_p, while T_5 acts by 2 and $\ker T_2 \cong \mathbb{Z}/4$);*

(e) ***at level*** *1, $\ell \in \{2,3\}$: the maximal ideals above ℓ are the $\mathfrak{m}_\psi$ of Proposition 31.39(2),* ***known without any generation hypothesis****, and $\dim A_1(L/\ell) \leq 3$, so Theorem 31.45(C3) fails at 2, 3 for every $d \geq 4$; the 2- and 3-adic structure of $\mathbb{T}^{\mathrm{fr}}$ comes from Proposition 31.49 or from $A_K/\ell A_K$ at $K \geq c^{\mathrm{sat}}+1$ under a generation theorem. The same holds verbatim at $\ell \geq 5$, with inputs on the* O*-piece (Propositions 27.5(2), 28.9; the free part modulo $\ell^{K'}$ of Theorem 31.16(4)).* ***So "everything above p survives in the truncated rings" holds with the precise meaning (a)–(d); its only extra hypothesis is generation at p, and the same hypothesis is needed at every $\ell \geq 5$.***

5. (Torsion and glue.) *[proved here]* $\operatorname{Spec}\mathbb{T}$*, its components and its isolated torsion points do not depend on J (Proposition 31.28(2)). Whether a maximal ideal of $\mathbb{T}^{\mathrm{tors}}$* equals *one of $\mathbb{T}^{\mathrm{fr}}$ is decided by agreement on* ***all*** *T_p: a disagreement on a computed operator certifies that they differ ("differs from system X at T_p" is a certificate, "agrees on $\mathcal{P}$" is not). Equality is certified (i) at level 1, $\ell \in \{2,3\}$, for every cell, by T_ℓ (Proposition 31.39(4), (5): the torsion point is $\mathfrak{m}_x$, common with the cuspidal part exactly when $S_k \neq 0$); (ii) everywhere by Proposition 31.41(4). The residual case "split for $\mathbb{T}'$, common on $\mathcal{P}$" is not decided. The local rings at maximal ideals touching both parts are $\varprojlim_K A_K(H^1_{\mathfrak{m}}/\ell^K)$ (Proposition 31.41(5)(b)), computable from computed data only under Remark 31.43; without it J is* `bounds` *or* `zero` *(Proposition 31.41(3)), never exact.*

Proof. (1) Mayer–Vietoris for $\mathbb{Z} = \mathbb{Z}[1/6] \times_{\mathbb{Q}} \mathbb{Z}_{(6)}$, tensored with X (flat): the kernel consists of elements killed by a power of 6 and by an integer prime to 6; and $\mathbb{T} \otimes R \to \mathrm{End}(H^1) \otimes R = \mathrm{End}_R(H^1 \otimes R)$ stays injective with the base-changed image. (2) $\mathbb{T}/P$ is a finite domain when $P \cap \mathbb{Z} = \ell\mathbb{Z}$, hence a field; the rest is Proposition 31.28(3). (3) A lattice in a finite-dimensional $\mathbb{Q}$-vector space is the intersection of its ℓ-adic closures; then Proposition 31.33(3) and Lemma 31.30. (4) Propositions 31.31–31.36 with $M = L_\ell$ from the local model; the model statements say that the operators there induce $T_p \otimes \mathbb{Z}_{(\ell)}$; (e) is Proposition 31.39. (5) Proposition 31.28(2); a maximal ideal of $\mathbb{T}$ is determined by the residues of all its generators; the certificates are Propositions 31.39, 31.41. □

What is not available at 2 and 3 today (as written at v1.25.0-pre.0, before EXPORT and UP-Z; what a record reads at 2 and 3 is item 8, and for a `full` record at $N > 1$ item 8 (d′)(v), by the binary that wrote it; v1.26.0-pre.11, confirm-referee C1). U_p, $p \mid N$, on the local models over $\mathbb{Z}_{(\ell)}$; the ι-splitting at $\ell = 2$ (so multiplicity one cannot be read per sign there); and **any export of the operators on** $H^1_{(\ell)}/\ell^K$, at every level — the console's $\mathbb{Z}/\ell^k$ mode needs $\ell \geq 5$, and the descent's 2- and 3-local reads are internal. So (4)(a)–(d) at 2 and 3 are reachable today only through (4)(e) (level 1) and Proposition 31.49; the full algebra at level $N > 1$ stays partial at 2 and 3 until U_p exists there.

Theorem 31.45 (The global certificate)**.** *[proved here] (the plan's Thm. 4.1 and Prop. 4.2.) Let $\mathcal{P}$ be a finite set of primes, $\mathbb{T}' := \mathbb{Z}[X_p : p \in \mathcal{P}] \subseteq \mathbb{T}^{\mathrm{fr}}$, and $\vartheta \in \mathbb{T}'$. Let d be the a-priori rank of Proposition 31.29(3) (resp. d^{an} of (3′), with $\mathcal{P} \not\ni p$ for $p \mid N$ and $\mathbb{T}^{\mathrm{an,fr}}$ for $\mathbb{T}^{\mathrm{fr}}$ throughout). Suppose:*

(C1) rational*: $f_\vartheta :=$ the squarefree part of the characteristic polynomial χ_ϑ of ϑ on L (assembled over $\mathbb{Z}$ by CRT) has $\deg f_\vartheta = d$;*

(C2) bad set*: for one or more $\vartheta_j \in \mathbb{T}'$,* ***each passing (C1) itself****, with $D_j := \mathrm{disc}(f_{\vartheta_j})$, a factorisation $D_{j_0} = \pm\prod_i q_i^{e_i} \cdot \mathsf{c}$ of* ***one*** *of them whose factors q_i are* ***certified primes****, and the integer $\mathsf{c} \geq 1$ the* cofactor*; S is the set of* ***all*** *the q_i, except that a q_i with $v_{q_i}(D_j) \leq 1$ for some j ($v_{q_i}(D_j)$ read by residues)* ***may*** *be left out of S (v1.25.0-pre.0b, A7-m4: no gcd*

of the big integers D_j is formed; v1.25.0-pre.0c, A7b-m1: this drop rule is the only way the other ϑ_j enter);

(C3) local*: for each $\ell \in \mathsf{S}$, $\dim_{\mathbb{F}_\ell} A_1'(L/\ell) = d$.*

Then:

1. *$f_\vartheta(\vartheta) = 0$ on L, f_ϑ is the minimal polynomial of ϑ, $\mathbb{T}_\mathbb{Q} = \mathbb{Q}[\vartheta]$, and $\mathbb{T}_\mathbb{Q}$ is **reduced** — for a full ϑ (with U_p) as for an anemic one; no semisimplicity hypothesis is used.*
2. *$\mathbb{T}'_\ell = \mathbb{T}_\ell$ for every $\ell \nmid \mathsf{c}$, and for every ℓ when c is squarefree (in particular $\mathsf{c} = 1$). Then $\mathbb{T}' = \mathbb{T}^{\mathrm{fr}}$, and every prime ideal, component, local ring and congruence module of $\mathbb{T}^{\mathrm{fr}}$ is a finite computation from the structure constants of $\mathbb{T}'$ alone.*
3. (Termination of the rational step.) *(a) If the systems of $\mathbb{T}_\mathbb{Q}$ are separated by $\mathcal{P}$, then $\vartheta = \sum_{p\in\mathcal{P}} c_p X_p$ fails to be primitive only for c on at most $\binom{d}{2}$ proper hyperplanes, so some $c \in \{0,\dots,\binom{d}{2}\}^{\mathcal{P}}$ works; a finite separating $\mathcal{P}$ exists (the d systems are distinct ring maps out of $\mathcal{H}_N$, resp. $\mathcal{H}_N^{\mathrm{an}}$; one p per pair), and no a-priori bound on p is needed, since the search stops on $\deg f_\vartheta = d$, which it can test. (b) [standard: [Del74]] The roots of χ_ϑ are bounded by $R := \sum_p |c_p|(1+p^{k-1})$ and its coefficients by $\binom{b_1}{j}R^j$, so CRT over word primes stops after finitely many primes. (c) At a fixed $\ell \in \mathsf{S}$, enlarging $\mathcal{P}$ reaches (C3) after finitely many steps **iff** $\mathbb{T}_\ell$ is saturated on L_ℓ (Proposition 31.32(2), (3)); the algorithm cannot know which case holds and uses a cap.*

Proof. (1) The minimal polynomial m_ϑ of ϑ on $L\otimes\mathbb{Q}$ has the irreducible factors of χ_ϑ, so $f_\vartheta \mid m_\vartheta$ and $\deg m_\vartheta \geq d$; and $\deg m_\vartheta = \dim \mathbb{Q}[\vartheta] \leq \dim \mathbb{T}_\mathbb{Q} = d$ (Proposition 31.29(3), the perfect pairing, no reducedness used; resp. (3′)). Hence $m_\vartheta = f_\vartheta$, squarefree, and $\mathbb{Q}[\vartheta] = \mathbb{T}_\mathbb{Q}$ is reduced. (Taking f_ϑ to be the squarefree part of χ_ϑ matters: "f squarefree, $f(\vartheta) = 0$, $\deg f = d$" for an arbitrary f would admit $f = m_\vartheta h$.) (2) By (1) for ϑ_j, $\operatorname{rank}\mathbb{T}' = d = \operatorname{rank}\mathbb{T}^{\mathrm{fr}}$ and $\operatorname{disc}\mathbb{T}' \mid D_j$ (Proposition 31.33(1) for $\mathbb{Z}[\vartheta_j] \subseteq \mathbb{T}'$). Let $\ell \notin \mathsf{S}$. If $\ell = q_i$ was left out of S, then $v_\ell(D_j) \leq 1$ for some j by the drop rule of (C2). Otherwise $\ell \neq q_i$ for every i; if $\ell^2 \mid D_j$ for all j, then $\ell^2 \mid D_{j_0} = \pm\prod q_i^{e_i}\mathsf{c}$, so $\ell^2 \mid \mathsf{c}$: excluded when $\ell \nmid \mathsf{c}$ or c is squarefree. So $v_\ell(\operatorname{disc}\mathbb{T}'_\ell) \leq v_\ell(D_j) \leq 1$ for some j, and $\mathbb{T}'_\ell = \mathbb{T}_\ell$ by Proposition 31.33(2). At $\ell \in \mathsf{S}$ it is Proposition 31.32(1) through (C3). $\mathbb{T}' = \mathbb{T}^{\mathrm{fr}}$ is Lemma 31.30(3). (3)(a) Over $\overline{\mathbb{Q}}$ the reduced $\mathbb{T}_\mathbb{Q}$ has d characters; ϑ is primitive iff it separates every pair, and for a pair separated by some X_p the failing c form a proper hyperplane; a non-zero product of $\binom{d}{2}$ linear forms does not vanish on the whole grid $\{0,\dots,\binom{d}{2}\}^{\mathcal{P}}$. Distinct characters of $\mathcal{H}_N\otimes\overline{\mathbb{Q}}$ differ on a generator. (b) Deligne's bound for the cusp eigenvalues, $|a_p| \leq 2p^{(k-1)/2} \leq 1+p^{k-1}$; Eisenstein eigenvalues $\psi(p)+\bar\psi(p)p^{k-1}$; at $p \mid N$, $|a_p| \leq p^{k-1}$; then Vieta. (c) Proposition 31.32(2), (3). □

The cofactor test and the big-integer rule (owner ruling Q1(a) of 2026-11-04, *residue-only*). Only the square part of $\operatorname{disc}\mathbb{T}'$ matters. A cofactor $\mathsf{c} < B^3$ with no prime factor $\leq B$ that is neither prime nor a perfect square is a product of two distinct primes, hence squarefree and harmless; each step is single-word arithmetic for $B = 10^6$ ($B^3 < 2^{64}$): "no factor $\leq B$" by the trial division itself, "not prime" by a Fermat witness $a^{\mathsf{c}-1} \not\equiv 1 \pmod{\mathsf{c}}$, "not a square" by an exact integer square root. A cofactor that cannot be cleared leaves $\mathbb{T}^{\mathrm{fr}}$ certified at every $\ell \nmid \mathsf{c}$ (record status `partial(unfactored)`); a factor whose primality is not proved is part of the cofactor. Under Q1(a) the C++ build certifies what residues and single-word arithmetic decide: χ_ϑ, discriminants and resultants as exact values (residues modulo word primes, a certified bound, the limb output of §22.4 item 7 (l)); $v_q(D)$ for word-size q — from D mod q^{e+1} while $q^{e+1} < 2^{64}$, and in general by the cofactor route: reconstruct X from the residues Dq^{-e} mod p_i, check $q^eX \equiv D$ modulo word primes whose product exceeds the certified bound, and read X mod $q \neq 0$ from its mixed-radix digits; the cofactor exactly (its residues are $D\prod q_i^{-e_i}$); Pratt certificates for $q < 2^{64}$; the cofactor test. A factor above 2^{64} is `probable` in the record and part of the cofactor; the Lean kernel may check its Pratt certificate, but that result lives beside the record. Exact division, gcds, modular powers and square roots of big integers are multi-precision arithmetic and stay out of the certified build ([R-Q2] unchanged). Two consequences (the plan's `01-math-core.md`

§4 Q1(a), restored at A7-m4): **the gcd over several ϑ_j is used only through its word-size primes** ((C2) factors one D_{j_0}; $\ell^2 \mid \gcd_j D_j$ implies $\ell^2 \mid D_{j_0}$, so the proof of (2) needs nothing more), and $g_{\mathcal{P}}$ **is never formed as a big gcd**: its ℓ-parts for word-size ℓ are read by residues modulo ℓ^K ($v_\ell(g_{\mathcal{P}}) = \min_{p\in\mathcal{P}} v_\ell c_{\mathcal{S}',p}(\lambda(X_p))$), and as an upper bound one $c_{\mathcal{S}',p_0}(\lambda(X_{p_0}))$ with its cofactor serves. *As built* (v1.25.0-pre.2, Z0 referee F5; `Integral/HeckeZRational.cpp`, `GcdOf`): HZ-RAT takes (C2) on the gcd of the D_j over *every* primitive ϑ_j it found, and $g_{\mathcal{P}}$ on the gcd of the $c_{\mathcal{S}',p}(\lambda(X_p))$ over the primitive unit $\vartheta = T_p$, in both cases through word-size primes only (v_q of the gcd is $\min_j v_q(\cdot)$, read by residues); its cofactor is the smallest base cofactor, kept only if the residue route above shows that it divides every term and no certified prime divides it, otherwise `cofactor_test = upper_bound` (a bound, never `complete`). This is the same certificate: $v_\ell(\gcd_j D_j) = \min_j v_\ell(D_j)$, so `complete` on the gcd discharges (C2)'s drop rule for every j at once, and nothing but residue arithmetic enters. The mathematics and the soundness are the same under every option of Q1: a wrong C++ big integer can only make a Lean check fail.

Two forms, and what is reachable today. The theorem is used in two forms, both proved, differing only in the a-priori rank: **full** ($\mathcal{P}$ contains the $p \mid N$, $X_p = U_p$; $d = \dim M_k(\Gamma_0(N))$; conclusion $\mathbb{T}' = \mathbb{T}^{\mathrm{fr}}$) and **anemic** ($p \nmid N$ only; d^{an}; conclusion $\mathbb{T}' = \mathbb{T}^{\mathrm{an,fr}}$). "Certifiable" means certifiable by a proved statement once its inputs exist; what the shipped tools reach **today** is narrower, because every step that reads a module — (C3), Propositions 31.34–31.36, the at-root read of Proposition 31.48(5) — is stated for L_ℓ and needs the matrices of the operators on L/ℓ^K (for the glue, on the whole $H^1_{(\ell)}/\ell^K$). The console's $\mathbb{Z}/\ell^k$ mode at level 1, $\ell \geq 5$, delivers $H^1(\Gamma; V \otimes \mathbb{Z}/\ell^K) = H^1/\ell^K$, **not** L/ℓ^K; the two differ exactly at the cell's torsion primes ($0 \to \mathrm{tors}_\ell/\ell \to H^1/\ell \to L/\ell \to 0$; at level 1 every $\ell \geq 5$ with $\ell \leq n+1$ is one, $H^1[\ell] \cong H^0(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell^2) \neq 0$), where $\dim A_1'(H^1/\ell) \geq \dim A_1'(L/\ell)$, possibly strictly, and L/ℓ^K is not recoverable from the Hecke module H^1/ℓ^K alone. So **a certified status for (C3), Propositions 31.34–31.36 or the at-root read is never read from $H^1(\Gamma; V/\ell^K)$ at a torsion prime of the cell**. With U_p over $\mathbb{Z}$ not built (the state when this list was written, v1.25.0-pre.0, before EXPORT and UP-Z: for the `anemic` line (C3) and the lower bound are read since EXPORT, e.g. the `anemic` $(11, 0)$, item 8; for the `full` line see that line; v1.26.0-pre.11, confirm-referee C1):

- *level* 1: the full algebra ($= \mathbb{T}^{\mathrm{an}}$) is certifiable by this theorem and Propositions 31.48, 31.49, and **reachable today** at every $\ell \geq 5$ not dividing $|\mathrm{tors}\, H^1|$; at a torsion prime $\ell \geq 5$ only **blockwise**, at the blocks outside $\mathrm{supp}(\mathrm{tors}_\ell)$ (the torsion support is non-ordinary, so every ordinary block qualifies; the per-block (C3) compares with the per-block rank from the Hensel factors attributed by Proposition 31.36); at $\ell \in \{2, 3\}$ through Proposition 31.49 only;
- *level* $N > 1$, *anemic*: certifiable by (C1)–(C3) with d^{an}, and the congruence module between the cuspidal systems and the anemic Eisenstein system by Proposition 31.48(4) when $e_k^{\mathrm{an}}(N) = 1$; Proposition 31.49(1) is not available (Sturm needs every T_n, the U_p-words included). **Reachable today: (C1), (C2) and the upper bound $g_{\mathcal{P}}$ only**; (C3) and the lower bound after the export of the whole module $H^1_{(\ell)}/\ell^K$ with the matrices of every computed T_p by cochain-level lifts (Proposition 31.37(3)) — the free quotient alone is not enough;
- *level* $N > 1$, *full* (at v1.25.0 this line read "nothing is certified but two tags"; v1.26.0-pre.11, confirm-referee C1): what a record writes is item 8 (d′)(ii), (v), by the binary that wrote it. (C1) for the full algebra needs no U_p over $\mathbb{Z}$: the ϑ search reads U_p at the word primes by Proposition 28.9, so `G1.C1.status` can be `certified` (67 of the 85 records of Z1's recompute, a count at this version), and the words of G2 at $\ell \geq 5$ are read on that operator; the fields at $\ell \in \{2, 3\}$ follow item 8 (d′)(v), according to whether the binary carries the route over $\mathbb{Z}$. Two tags are computable without any lattice from $\chi_{\mathbb{Q}}(U_p)$

assembled by CRT from the level-N route over word primes: `upper_bound_certified` (Proposition 31.48(2) for $\vartheta = U_p \in \mathbb{T}^{\mathrm{fr}}$; e.g. $(2,6)$: $C(f, E^{(1)}) = 0$, $C(f, E^{(2^7)}) \mid 2^3 \cdot 17$, $C(E^{(1)}, E^{(2^7)}) \mid 127$) and, **for N a power of p**, `non_reduced_certified` (the counting criterion: e.g. $(8,10)$, $\chi_{\mathbb{Q}}(U_2) = x^{16}(x-1)(x-2048)(x^2+24x+2048)^2$, $d = 13$, $d^{\mathrm{an}} = 6$; a reduced full $\mathbb{T}_{\mathbb{Q}}$ would need ≥ 7 distinct systems with $U_2 \neq 0$, but $H^1 \otimes \mathbb{C} \supseteq M_{12}$ carries only 6 non-zero U_2-roots with multiplicity; the attribution of U_p-roots to systems uses Proposition 31.29(2) for U_p; for other N the criterion is **not proved** (v1.25.0-pre.2, Z0 referee F2; HZ-UTAG audit F1: with a second prime $q \mid N$ several systems over one anemic class may share $U_p = 0$, e.g. $N = 12$, $k = 4$: $U_2 \in \{1, 8, 0\}$, $U_3 \in \{1, 27\}$, so "reduced $\Rightarrow b_1 - m_0 \geq d - d^{\mathrm{an}}$" is unproved, with margin 0 at $(12,2)$); the code implements exactly this scope (at $(24,2)$ the raw inequality holds and the tag is withheld). $d^{\mathrm{an}} = d$ alone does not make the full algebra certified: the index $[\mathbb{T}^{\mathrm{fr}} : \mathbb{T}^{\mathrm{an,fr}}]$ needs the integral U_p, unless each U_p is *known* to lie in $\mathbb{T}^{\mathrm{an,fr}}$ (at $(11,0)$ $U_{11} = \mathrm{id}$ on L, a literature input).

Lean form (not a change of the theorem). The Lean certificate uses Proposition 31.33(2) only in its $v_\ell = 0$ clause (a Bézout witness $uf + vf' = D$); a prime with $v_\ell(D_j) = 1$ for every j is put into S (record reason `v1_pending_LC3b`) and closed by (C3) until the $v_\ell = 1$ clause is formalised. *The measured (C2) frontier* (v1.25.0-pre.0c, A7b-m1: re-measured on what (C2) forms, **one** D_{j_0}, with $\vartheta_{j_0} = T_2$; level 1, the discriminant of $\chi(T_2 \mid M_k)$, trial division to 10^5 and Pollard rho at 8 s per number, `docs/implementation/slot24/heckez/a7b_frontier_single.py`): the factorisation is complete for every $k \leq 64$ except $k = 52$, 60; at $k = 36$, 42, 44, 48, 50, 54, 56, 58, 62, 64 it contains a prime of 65–204 bits, so under Q1(a) these records read `partial(unfactored)` while the Lean lane can close them; $k = 52$, 60 keep cusp cofactors of 109 and 263 bits, and $k = 84$, 98 the composite cofactors of $\mathrm{num}(B_k/k)$ of 189 and 206 bits. This is the list of the earlier measurement on $\gcd_{p \leq 7}$, weight by weight: the index factor $[\mathbb{T}^{\mathrm{fr}} : \mathbb{Z}[T_2]]^2$ of D_{j_0} splits into primes below 2^{64} except for the square of a 65-, 72- resp. 67-bit prime at $k = 60$, 62, 64, weights already in the list. For $\vartheta_{j_0} = T_3$, T_5, T_7 the index factor is larger and often unsplit at this budget (`partial(unfactored)` for T_5, T_7 at most weights $k \geq 36$, for T_3 at $k \geq 52$; a new prime above 2^{64} at $k = 32$, 38, 40 or 46), so T_2 is the measured choice of ϑ_{j_0}. More operators barely help: $\mathrm{disc}\, \mathbb{T}^{\mathrm{fr}}$ divides every disc_ϑ and contains the discriminant of the cuspidal Hecke field. The congruence part (Propositions 31.48, 31.49) certifies beyond the frontier. *As built* (v1.25.0-pre.1, 2026-11-06): the C++ factor finder has no multi-limb rho or ECM yet, so the frontier the build reaches is lower than this measured one; see record item 8 (j) of §22.4. Every status it writes is one this subsection proves; the difference is coverage.

Remark 31.46 (The 691 caveat, revised). [proved here] for (5) and (3) (the plan's Rem. 4.5, Props. 4.3, 4.4). **Scope**: Propositions 31.32–31.36, Theorem 31.45(2), (3) and Definition 31.47 onwards assume $\mathbb{T}_{\mathbb{Q}}$ reduced — always for the anemic algebra, for the full one when (C1) passes; the full algebra at $p^3 \mid N$ is refused (Proposition 31.29(4)(b); $\mathbb{T}_{\mathbb{Q}}$ may then be non-monogenic, e.g. $\mathbb{Q}[x,y]/(x^2, y^2)$ when two primes have cube level).

1. *One operator overstates.* T_2 at $(1,10)$ gives $2073 = 3 \cdot 691$; the 3 is an index prime of $\mathbb{Z}[T_2]$, not a congruence. A congruence prime divides $\mathrm{Res}(f_{\vartheta,\mathcal{S}}, f_{\vartheta,\mathcal{S}'})$ for *every* ϑ, and the depth is $v_\ell \mathrm{Res}$ for a ϑ that generates modulo ℓ (Proposition 31.34(3)).
2. *Bad primes are not small.* Level-one Eisenstein congruence primes divide $\mathrm{num}(B_k/k)$: 691 ($k = 12$), 3617 (16), 43867 (18), $283 \cdot 617$ (20), $131 \cdot 593$ (22), $103 \cdot 2294797$ (24), 657931 (26), $9349 \cdot 362903$ (28), $1721 \cdot 1001259881$ (30). They come from rational data, not from a mod-ℓ census range.
3. *Existence is not depth; when $K > 1$ is needed.* $K = 1$ suffices for the maximal ideals and residue fields (Proposition 31.31), the saturation test (Proposition 31.32), the étale verdict (Proposition 31.33: no computation), and the whole local structure at a saturated block where a computed ϑ generates $\mathbb{T}_{\mathfrak{m}}/\ell$ (Proposition 31.34; on an unsaturated block under a generation theorem the test runs at $K = c^{\mathrm{sat}} + 1$, and the rational route needs no $K > 1$). $K > 1$ is needed when no computed ϑ generates (embedding dimension ≥ 2, every non-

Gorenstein $\mathbb{T}_{\mathfrak{m}}$, unless the rational route is used: Proposition 31.35), for Proposition 31.36 ($K > v_\ell \operatorname{Res}$), and for the at-root read (Proposition 31.48(5), $K > v_\ell c(\lambda)$). $K = 1$ alone cannot separate the kinds: $\mathbb{Z}_\ell \times_{\mathbb{F}_\ell} \mathbb{Z}_\ell$ (meeting, depth 1), $\mathbb{Z}_\ell \times_{\mathbb{Z}/\ell^2} \mathbb{Z}_\ell$ (depth 2), $\mathbb{Z}_\ell[\sqrt{\ell u}]$ (ramified DVR) and $\mathbb{Z}_\ell[\ell\sqrt{u}]$ (for u a non-square unit a non-maximal order of one component; for u a square and ℓ odd it is $\mathbb{Z}_\ell \times_{\mathbb{Z}/\ell} \mathbb{Z}_\ell$ up to isomorphism, a meeting of two components; A7-N7) all reduce to $\mathbb{F}_\ell[\varepsilon]/(\varepsilon^2)$.

4. *The census algebra is not $\mathbb{T}/\ell$.* It is $A_1'(H^1)$, which differs by the saturation defect (Remark 30.10), mixes in torsion (Proposition 30.7(1), (4)) and needs a generation certificate (Proposition 31.31); a census "non-Gorenstein" row is a statement about A_1', and about $\mathbb{T}_{\mathfrak{m}}$ only where $\mathbb{T}$ is saturated *and* generation is certified — both hold at $(1, 10)$, $\ell = 691$ (`length_A`$= 2$, `hilbert`$= [1, 1]$); at $(1, 22)$, $\ell = 103$, the census's second maximal ideal is the non-Eisenstein conjugate.
5. *Other primes say nothing at ℓ.* There are orders $\mathcal{O}_1 \neq \mathcal{O}_2$ in the same $\mathbb{T}_{\mathbb{Q}}$, with faithful lattices, whose mod-ℓ'^K data ($\ell' \neq \ell$, all K) and rational data all coincide: $\{(a, b) \in \mathbb{Z}^2 : a \equiv b\ (\ell)\}$ and $\{a \equiv b\ (\ell^2)\}$ on $\mathbb{Z}^2$ (both $\mathbb{Z}^2_{\ell'}$ at every $\ell' \neq \ell$, characteristic polynomial $(x - a)(x - b)$ for every element of both). **Adding primes ℓ' helps only to find the bad set and to assemble rational data, never to certify structure at a bad ℓ**; there one adds operators or raises K.
6. *Anemic versus full.* The anemic algebra merges systems that differ only at U_p: at $(2, 6)$ the two Eisenstein series ($U_2 = 1$, $U_2 = 2^7$) are anemically one; the Eisenstein–Eisenstein congruence at 127 is invisible anemically, and the anemic 2-part 2^7 of the cuspidal–Eisenstein congruence overstates the full one ($\leq 2^3$, from U_2: $128 - (-8) = 2^3 \cdot 17$).
7. *Factoring.* Only the square part of $\operatorname{disc} \mathbb{T}'$ must be factored (Theorem 31.45(C2)).
8. *Saturation fails outside multiplicity one*; then mod-ℓ^K data on the cohomology lattice cannot certify generation (Proposition 31.32(3)), but Ramanujan-type congruences are still certified by Propositions 31.48(4) and 31.49, which need no generation.
9. *Torsion systems.* Coincidence of a torsion maximal ideal with a free one is not certified by agreement on finitely many operators; it is certified at level 1 at 2, 3 by T_ℓ, and anywhere by a non-split extension (Theorem 31.44(5)).
10. Tor *terms.* 691 is not a torsion prime: at $(1, 10)$ no group H^q, $H^q(\partial)$, H^q_c has 691-torsion (integral census edition 1). It is invisible to every Bockstein and is the lattice index η_L of Definition 31.47: the congruence module is the *cokernel* of the lattice sequence $0 \to L_{\mathcal{S}} \oplus L_{\mathcal{S}'} \to L \to \eta_L \to 0$, not a term of a coefficient sequence; reduced modulo ℓ^K it appears as $\operatorname{Tor}_1(\eta_L, \mathbb{Z}/\ell^K) = \eta_L[\ell^K]$, the ℓ-part of η_L once $K \geq v_\ell(\exp \eta_L)$ (A7-N7). In degree ≥ 1, Tor terms of $H^q(\Gamma; V)$ occur only at 2 and 3, but H^2_c has $\ell \geq 5$ torsion (78 of the 210 pair cells of integral census edition 1, always $\ell \leq n$ or $\ell \mid N$) and no route computes a Hecke action on it.

Proof. (5): every element (a, b) of either order acts on $\mathbb{Z}^2$ diagonally with characteristic polynomial $(x - a)(x - b)$, and both orders become $\mathbb{Z}^2_{\ell'}$ after $\otimes\mathbb{Z}_{\ell'}$. (3): the four rings are free of rank 2 over $\mathbb{Z}_\ell$ with one maximal ideal and a square-zero cotangent line modulo ℓ; their kinds differ by Proposition 31.33(4), (5) and the index in the maximal order. The other items cite the statements named in them. □

Definition 31.47 (Congruence modules)**.** For disjoint sets $\mathcal{S}, \mathcal{S}'$ of components, $I_{\mathcal{S}} := \ker(\mathbb{T}^{\mathrm{fr}} \to \mathbb{T}_{\mathcal{S}})$, $\mathbb{T}_{\mathcal{S}}$ the image in $\prod_{i \in \mathcal{S}} K_i$ (K_i the field of component i). The **algebra congruence module** is $C_{\mathbb{T}}(\mathcal{S}, \mathcal{S}') := \mathbb{T}^{\mathrm{fr}}/(I_{\mathcal{S}} + I_{\mathcal{S}'})$, the finite-level integral form of [U, `def:u-congmod`]. When $\mathcal{S} \sqcup \mathcal{S}'$ is all components and L is a faithful lattice, the **lattice congruence module** is $\eta_L(\mathcal{S}, \mathcal{S}') := L/(L_{\mathcal{S}} \oplus L_{\mathcal{S}'})$, $L_{\mathcal{S}} := L \cap (L \otimes \mathbb{Q})_{\mathcal{S}}$; equivalently $\eta_L \cong \pi_{\mathcal{S}}(L)/L_{\mathcal{S}}$, $\pi_{\mathcal{S}}$ the rational projection. For $p \in \mathcal{P}$, $c_{\mathcal{S}',p}$ is the **minimal** polynomial of X_p on $(L \otimes \mathbb{Q})_{\mathcal{S}'}$ (in $\mathbb{Z}[x]$, a monic factor of χ), and when $\mathbb{T}_{\mathcal{S}} = \mathbb{Z}$ with character λ, $g_{\mathcal{P}} := \gcd_{p \in \mathcal{P}} c_{\mathcal{S}',p}(\lambda(X_p))$.

The characteristic polynomial on $(L\otimes\mathbb{Q})_{\mathcal{S}'}$ counts each cuspidal system twice ($(x+24)^2$ at $(1,10)$, giving 2073^2): a valid but not sharp bound, and the certificates below close only with the minimal polynomial. For non-rational Eisenstein components ($\mathbb{T}_{\mathcal{S}}=\mathcal{O}$, characters at level N) the bounds become ideals of $\mathcal{O}$ and the analogue of the at-root read has not been written; for two cuspidal sets neither side is cyclic, and $C_{\mathbb{T}}$ is computed from a certified $\mathbb{T}^{\mathrm{fr}}$ (Theorem 31.45) or bounded by Proposition 31.48(2) — **nothing more is asserted** (the plan's Prop. 5.7, a sketch; prototype [E, `cl:258alg`]).

Proposition 31.48 (Bounds, the Eisenstein certificate, the read at a root)**.** *[proved here] [standard: [Del74]] for (5)'s existence clause (the plan's Props. 5.2–5.5, with A1-m2, m3, m4, m6 and P2a-5). Let $\mathcal{S}\sqcup\mathcal{S}'$ be all components.*

1. *$I_{\mathcal{S}}L\subseteq L_{\mathcal{S}'}$ and $I_{\mathcal{S}'}L\subseteq L_{\mathcal{S}}$; so η_L is a $C_{\mathbb{T}}$-module and $\exp\eta_L\mid\exp C_{\mathbb{T}}$. This holds for every $\mathbb{T}'\subseteq\mathbb{T}^{\mathrm{fr}}$ as well.* ***The lower bound uses no generation.***
2. *Every element of $(I_{\mathcal{S}}+I_{\mathcal{S}'})\cap\mathbb{Z}$ kills $C_{\mathbb{T}}$, in particular $\mathrm{Res}(f_{\vartheta,\mathcal{S}},f_{\vartheta,\mathcal{S}'})$ for every $\vartheta\in\mathbb{T}^{\mathrm{fr}}$ with $\mathbb{Q}[\vartheta]=\mathbb{T}_{\mathbb{Q}}$; upper bounds from computed $\vartheta\in\mathbb{T}'$ are valid for $\mathbb{T}^{\mathrm{fr}}$.*
3. *$|C_{\mathbb{T}}|=[\mathbb{T}_{\mathcal{S}}\times\mathbb{T}_{\mathcal{S}'}:\mathbb{T}^{\mathrm{fr}}]$ and $\mathbb{T}^{\mathrm{fr}}=\mathbb{T}_{\mathcal{S}}\times_{C_{\mathbb{T}}}\mathbb{T}_{\mathcal{S}'}$.*
4. (The Eisenstein certificate.) *Let $\mathbb{T}_{\mathcal{S}}=\mathbb{Z}$ (one rational Eisenstein component, character λ). Then $C_{\mathbb{T}}\cong\mathbb{Z}/g$ is cyclic, $g\mathbb{Z}=\{a:(a,0)\in\mathbb{T}^{\mathrm{fr}}\}$, and*

$$\exp\eta_L\;\Big|\;g\;\Big|\;g_{\mathcal{P}}.$$

   ***If the outer terms agree,** g **equals them and** $\mathbb{T}^{\mathrm{fr}}=\mathbb{Z}\times_{\mathbb{Z}/g}\mathbb{T}_{\mathcal{S}'}$: $\lambda(t)\equiv\phi(t|_{\mathcal{S}'})\pmod g$ for every $t\in\mathbb{T}^{\mathrm{fr}}$, $\phi\colon\mathbb{T}_{\mathcal{S}'}\to\mathbb{Z}/g$ the induced ring map — every Hecke operator satisfies the congruence, not only the computed ones. ℓ-wise form: since $\exp\eta_L\mid g\mid g_{\mathcal{P}}$, "$\exp\eta_L=g_{\mathcal{P}}$" is equivalent to $v_\ell(\exp\eta_L)\geq v_\ell(g_{\mathcal{P}})$ for every prime $\ell\mid g_{\mathcal{P}}$; each is one read (5).*
5. (Reading η_L at a root.) *In the setting of (4) let $c:=c_{\mathcal{S}',p}$ with $c(\lambda(X_p))\neq0$. Such a p exists when $\mathcal{S}'$ is cuspidal and $k\geq3$ ($|a_p|\leq2p^{(k-1)/2}<1+p^{k-1}$; at $k=2$ choose p with $a_p\neq1+p$ for every cusp form). Let $r:=\operatorname{rank}L_{\mathcal{S}}$ ($r=1$ for the full algebra; $r\geq1$ for the merged anemic Eisenstein system at level $N>1$, e.g. $r=2$ at $(2,6)$). Then $c(\lambda)L_{\mathcal{S}}\subseteq c(X_p)L\subseteq L_{\mathcal{S}}$, and with $a_1\mid\cdots\mid a_r$ the elementary divisors of $c(X_p)L$ in $L_{\mathcal{S}}$,*

$$\eta_L\cong\textstyle\bigoplus_{i\leq r}\mathbb{Z}/\big(c(\lambda(X_p))/a_i\big),$$

   independently of p; for $r=1$, $|\eta_L|=c(\lambda(X_p))/a$. Locally: one Smith form of $c(X_p)$ mod ℓ^K on L, $K>v_\ell c(\lambda(X_p))$. At level $N>1$ for the full *algebra with several Eisenstein systems the old Eisenstein partner of λ lies in $\mathcal{S}'$ and agrees with λ at every T_p, $p\nmid N$, so only a U_p separates ($(2,6)$: $127=2^7-1$ from U_2), so the read needs U_p over $\mathbb{Z}$, and no route of this version performs it (item 8 (d′)(ii),* Why `lower` is `not_run`*; v1.26.0-pre.11, confirm-referee C1); for the anemic algebra with $e_k^{\mathrm{an}}(N)=1$, $\mathcal{S}'$ is cuspidal and the read works.*
6. (Free lattices.) *If $L_{\mathfrak{m}}\cong\mathbb{T}_{\mathfrak{m}}^a$, then $\eta_{L,\mathfrak{m}}\cong C_{\mathbb{T},\mathfrak{m}}^a$. At odd ℓ, an ι-part $L_{\mathfrak{m}}^{\pm}$ through which every component of $\mathfrak{m}$ passes and which is* cyclic *over $\mathbb{T}_{\mathfrak{m}}$ is free of rank one and gives $\eta_{L^{\pm},\mathfrak{m}}=C_{\mathbb{T},\mathfrak{m}}$. (At level 1 the Eisenstein class lies in L^+ only, so $L_{\mathfrak{m}}^-$ is not faithful at the Eisenstein ideal: [E,* `cl:ramanujan`*], $H^1_+\cong\mathbb{T}_{\mathfrak{m}}$, $H^1_-\cong\mathbb{F}_{691}$.)*

Proof. (1) $t\in I_{\mathcal{S}}$ kills $(L\otimes\mathbb{Q})_{\mathcal{S}}$, so $tL\subseteq L\cap(L\otimes\mathbb{Q})_{\mathcal{S}'}$; an integer $m\in I_{\mathcal{S}}+I_{\mathcal{S}'}$ therefore kills η_L, and $C_{\mathbb{T}}$ acts on $\eta_L=L/(L_{\mathcal{S}}\oplus L_{\mathcal{S}'})$ through $(I_{\mathcal{S}}+I_{\mathcal{S}'})L\subseteq L_{\mathcal{S}}\oplus L_{\mathcal{S}'}$. (2) Bézout for resultants, $\mathrm{Res}=af_{\mathcal{S}}+bf_{\mathcal{S}'}$, evaluated at ϑ: $f_{\mathcal{S}}(\vartheta)\in I_{\mathcal{S}}$, $f_{\mathcal{S}'}(\vartheta)\in I_{\mathcal{S}'}$. (3) [U, `lem:u-fibre`]. (4) [U, `lem:u-fibre`] with $A=\mathbb{Z}$ gives $C\cong\mathbb{Z}/\pi_A(I_{\mathcal{S}'})$ and $\pi_A(I_{\mathcal{S}'})=\{a:(a,0)\in\mathbb{T}^{\mathrm{fr}}\}$. Cayley–Hamilton on the $\mathcal{S}'$-part gives $c_{\mathcal{S}',p}(X_p)\in I_{\mathcal{S}'}$, with π_A-image $c_{\mathcal{S}',p}(\lambda(X_p))$, so $g\mid g_{\mathcal{P}}$; and C cyclic of order g has exponent g, so (1) gives $\exp\eta_L\mid g$. Equal outer terms give $g=g_{\mathcal{P}}$, and the fibre product is (3). The ℓ-wise form is the divisibility chain. (5) $c(X_p)$ is 0 on $(L\otimes\mathbb{Q})_{\mathcal{S}'}$

and $c(\lambda)$ on $(L\otimes\mathbb{Q})_{\mathcal{S}}$, so $c(X_p)L\subseteq L\cap(L\otimes\mathbb{Q})_{\mathcal{S}}=L_{\mathcal{S}}$, $c(\lambda)L_{\mathcal{S}}=c(X_p)L_{\mathcal{S}}\subseteq c(X_p)L$, and $\pi_{\mathcal{S}}=c(\lambda)^{-1}c(X_p)$; hence $\eta_L\cong\pi_{\mathcal{S}}(L)/L_{\mathcal{S}}\cong c(X_p)L/c(\lambda)L_{\mathcal{S}}$, read in a basis of $L_{\mathcal{S}}$ adapted to $c(X_p)L$ (each $a_i\mid c(\lambda)$). $L_{\mathcal{S}}$ is saturated in L, so the non-zero elementary divisors of $c(X_p)$ on L are the a_i, each of valuation $\le v_\ell c(\lambda)$, and r is known; a Smith form modulo ℓ^K with $K>v_\ell c(\lambda)$ reads them. The existence clause is Deligne's bound. (6) $L_{\mathcal{S}}=(\mathbb{T}\cap\varepsilon_{\mathcal{S}}\mathbb{T}_{\mathbb{Q}})^a=I_{\mathcal{S}'}^a$ locally, likewise $L_{\mathcal{S}'}=I_{\mathcal{S}}^a$, and $I_{\mathcal{S}}\cap I_{\mathcal{S}'}=0$, so $\eta=(\mathbb{T}/(I_{\mathcal{S}}+I_{\mathcal{S}'}))^a$; a cyclic faithful module is free of rank one. □

At $(1,10)$: $c_{\Delta,2}(2049)=2073=3\cdot691$, $c_{\Delta,3}(177148)=176896=2^8\cdot691$, $g_{\mathcal{P}}=691$; one read at $\ell=691$, $K=2$, with $v_{691}(\exp\eta_L)\ge1$ certifies $\mathbb{T}^{\mathrm{fr}}=\mathbb{Z}\times_{\mathbb{Z}/691}\mathbb{Z}$ and $\tau(m)\equiv\sigma_{11}(m)$ (mod 691) for **every** m ([E, `cl:ramanujan`] cites this for all m; here it is derived from one lattice computation). No 2- or 3-adic read is needed, and at level 1 none ever is once $2,3\in\mathcal{P}$: every level-one cusp eigenform is non-ordinary at 2 and 3 (Proposition 31.39(5)), so $c_{\mathcal{S}',2}(1+2^{k-1})$ and $c_{\mathcal{S}',3}(1+3^{k-1})$ are norms of elements $\equiv1$ modulo the primes above 2, resp. 3, and $g_{\mathcal{P}}$ is prime to 6 (A7-N4) — which is why Proposition 31.49(1) reaches $\ell\in\{2,3\}$. $|\eta_L|$ from T_2 ($a=3$) is a cross-check, not part of the certificate. The pair complex (Proposition 31.20; the denominator of the Eisenstein class) gives an independent reading, kept as a cross-check. *Factored* $g_{\mathcal{P}}$ (P2a-5). The ℓ-wise form needs the prime factorisation of $g_{\mathcal{P}}$, obtained without a big gcd: the word-size primes of one $c_{\mathcal{S}',p_0}(\lambda(X_{p_0}))$, each $v_\ell(g_{\mathcal{P}})$ read by residues modulo ℓ^K (Q1(a), A7-m4); the certificate closes **at every certified prime** $\ell\mid g_{\mathcal{P}}$, and when $g_{\mathcal{P}}$ keeps a cofactor (unfactored, or a factor only `probable`) the global fibre-product statement is `partial(unfactored)`, as is Proposition 31.49. *The word modulus* (the plan's FX-1, R-1). A read modulo ℓ^K needs ℓ^K below the engine's largest word modulus ($<2^{63}$); (C3) has $K=1$ (limit $\ell<2^{63}$), the at-root read $K>v_\ell c(\lambda)\ge1$, i.e. $K\ge2$ (limit $\ell<2^{31.5}$; $K=2$ at 691 and at every Eisenstein prime of a level-one cell with $k\le46$). A certified $\ell\mid g_{\mathcal{P}}$ whose at-root read needs $\ell^K\ge2^{63}$ cannot be read on any route of the plan; its ℓ-part stays `upper_bound_certified` (read reason `refused(word_modulus)`), final. At level 1, $k\le64$: no read at all at $k\in\{36,42,48,56,60,62,64\}$ ($\mathrm{num}(B_k/k)$ has a prime factor of 65–107 bits); (C3) yes and the at-root read refused at $k\in\{34,38,40,46,50,52,54,58\}$; the at-root read at $K=2$ on the 64-bit regimes for 2294797 ($k=24$) and the Eisenstein primes of $k=26,28,30,32,44$. *As built* (v1.25.0-pre.1, 2026-11-06; `docs/implementation/slot25/HZLOC_report.md`). At level 1 the at-root reads run on the 64-bit regimes of the console's module route: the seven named primes (2294797; 657931, 362903, 1001259881, 305065927; 2947939, 1798482437) are read at $K=2$ and every one closes. That route needs $K\ge2$, so a (C3) read at $K=1$ for $\ell\in[2^{31.5},2^{63})$ at level 1 is not served today; the code writes `refused(word_modulus)` although its modulus is below 2^{63}, and the record carries the **pending** `refused(k1_route)` (v1.25.0-pre.2, lead decision on O1: no new $K=1$ route now, an EXPORT-Z / slot-26 item; none of these weights is in the first golden set). At level $N>1$ the lattice comes from EXPORT, whose word ring has $\ell^K<2^{31}$: for $\ell\in[46341,2^{31})$ it serves $K=1$ only, so (C3) runs and the at-root read is `refused(export_word)`; for $\ell\ge2^{31}$ it serves no K and (C3) is `refused(export_word)` too; `word_modulus` ($\ell^K\ge2^{63}$) is tested first. `export_word` is pending: EXPORT-Z step 2 moves it (record item 8 (c)). The read set at ℓ contains T_ℓ for $\ell\nmid N$, $\ell\le23170$ (item 8 (j)).

Proposition 31.49 (The single-Eisenstein-component certificate; two further routes)**.** *(The plan's Prop. 5.6 and Rem. 5.6′.)*

1. *[proved here] given Proposition 31.29(5)(a). Suppose the Eisenstein part is one rational component (level* 1, $k\ge4$*; prime level* N, $k=2$*), let* $\mathcal{P}\supseteq\{p\le B_N\}$ *(with* U_N *if* $N>1$*),* $\mathcal{S}'$ *the cuspidal components and* $g_{\mathcal{P}}$ *as in Definition 31.47. If* $v_\ell(\exp\eta_L)\ge v_\ell(g_{\mathcal{P}})$ *for every prime* $\ell\mid g_{\mathcal{P}}$*, then*

$$\mathbb{T}^{\mathrm{fr}}=\mathbb{Z}\times_{\mathbb{Z}/g_{\mathcal{P}}}\mathbb{T}^{\mathrm{cusp}},\qquad \mathbb{T}^{\mathrm{cusp}}=\textstyle\sum_{n\le B_N}\mathbb{Z}\,T_n|_{\mathrm{cusp}},$$

glued by $T_n\mapsto\lambda(T_n)\bmod g_{\mathcal{P}}$*. This certifies* $\mathbb{T}^{\mathrm{fr}}$ *completely, at every prime,* 2 *and* 3

included, ***without saturation.*** ***Available today at level*** 1 ***only:*** *at prime level the $\mathbb{Z}$-span needs the T_n with $N \mid n$, i.e. U_N over $\mathbb{Z}$; for the anemic algebra Sturm gives no spanning set. If $\exp \eta_L < g_{\mathcal{P}}$ the certificate does not close (L not free at the Eisenstein ideal); $C_{\mathbb{T}}$ is then sandwiched, and Proposition 31.48(6) on the Eisenstein ι-part, or Propositions 31.32, 31.34 at that ℓ, may close it.*

2. (Bernoulli route, level 1.) *[proved here] given [U,* `pr:u-U1`*] (proved in [U] from Emerton's theorem) and [U,* `rmk:u-hidaM`*]. At level 1, $\ell \geq 5$, $(\ell-1) \nmid k$, the ℓ-part of $C_{\mathbb{T}}(\mathrm{Eis}, \mathrm{cusp})$ is $\mathbb{Z}/\ell^{v_\ell(B_k/k)}$: the lower bound is [U,* `thm:u-sandwich`*], the equality [U,* `pr:u-U1`*], and $\mathbb{T}^{\mathrm{fr}} \otimes \mathbb{Z}_\ell$ localised at the Eisenstein ideal is [U]'s $\mathbb{T}_k(a)_{\mathfrak{m}}$ ([U,* `rmk:u-hidaM`*] with Proposition 31.29). So where $v_\ell(g_{\mathcal{P}}) = v_\ell(B_k/k)$ the ℓ-part of (1) closes from $B_k/k \bmod \ell^K$ alone.* ***The at-root read stays the primary certificate****; this is an independent second one, a disagreement is a defect, and where the at-root read is* `refused(word_modulus)` *this route does* ***not*** *lift the ℓ-part above* `upper_bound_certified` *(a future option, owner-gated, with a Lean form still to be planned). At $(1, 10)$: $v_{691}(B_{12}/12) = 1 = v_{691}(g_{\mathcal{P}})$.*

3. (Mazur route, prime level, weight 2.) *[proved here] given [standard: [Maz77, II (9.7), Cor. (16.2), (16.6)]; the page images are quoted in the verified [GT] row* `lit:hz-mazur-eis`*] (v1.25.0-pre.0b: the valuation check, A7-M1). Let N be prime, $n = 0$, $\ell \geq 5$, $\ell \mid \nu := \mathrm{num}((N-1)/12)$, $\mathfrak{m}$ the Eisenstein maximal ideal of $\mathbb{T}^{\mathrm{cusp}}$ over ℓ, $\mathbb{T}^{\mathrm{cusp}}_{\mathfrak{m}}$ Mazur's cuspidal local ring and $e^{\mathrm{cusp}}_{\mathfrak{m}}$ its rank over $\mathbb{Z}_\ell$ (a block rank,* ***not*** *known a priori). For a prime $q \neq N$ let $\mathsf{h}_q \in \mathbb{Z}[x]$ be the characteristic polynomial of T_q on $S_2(\Gamma_0(N))$ — the monic square root of the cuspidal factor $\chi^{(N)}_{q,1}/(x-1-q)$ (each cuspidal system occurs twice in L, Proposition 31.29(2)) — and $\mathsf{F}_q \in \mathbb{Z}_\ell[x]$ its Hensel factor at $1+q$: the monic factor whose roots are exactly the roots of h_q congruent to $1+q$ modulo the maximal ideal of $\overline{\mathbb{Z}}_\ell$, with multiplicity ($\mathsf{h}_q$ need not be squarefree; A7b-N1). Let $f_{q,\mathfrak{m}}$ be the characteristic polynomial of T_q on $\mathbb{T}^{\mathrm{cusp}}_{\mathfrak{m}}$.* ***The check:***

$$v_\ell\big(\mathsf{F}_q(1+q)\big) = v_\ell(\nu).$$

If it holds, then $\mathsf{F}_q = f_{q,\mathfrak{m}}$, $e^{\mathrm{cusp}}_{\mathfrak{m}} = \deg \mathsf{F}_q$, $\mathbb{T}^{\mathrm{cusp}}_{\mathfrak{m}} = \mathbb{Z}_\ell[\eta_q] \cong \mathbb{Z}_\ell[x]/(\mathsf{F}_q)$, $\eta_q := T_q - 1 - q$, the Eisenstein ideal of $\mathbb{T}^{\mathrm{cusp}}_{\mathfrak{m}}$ is (η_q), and the $\mathfrak{m}$-block $\mathbb{T}^{\mathrm{fr}}_{\mathfrak{m}}$ of $\mathbb{T}^{\mathrm{fr}} \otimes \mathbb{Z}_\ell$ (the Eisenstein component included) has rank $e^{\mathrm{cusp}}_{\mathfrak{m}} + 1$:

$$\mathbb{T}^{\mathrm{fr}}_{\mathfrak{m}} = \mathbb{Z}_\ell[T_q]_{\mathfrak{m}} \cong \mathbb{Z}_\ell[x]/\big((x-1-q)\mathsf{F}_q\big) \cong \mathbb{Z}_\ell \times_{\mathbb{Z}/\ell^{v_\ell(\nu)}} \mathbb{Z}_\ell[x]/(\mathsf{F}_q),$$

$\mathbb{Z}_\ell[T_q]_{\mathfrak{m}}$ the image of $\mathbb{Z}_\ell[T_q]$ in that block; so $U_N \in \mathbb{T}^{\mathrm{an,fr}}_{\mathfrak{m}} = \mathbb{T}^{\mathrm{fr}}_{\mathfrak{m}}$. If it fails, nothing is certified by this q; a computed $v_\ell(\mathsf{F}_q(1+q)) < v_\ell(\nu)$ is impossible by the proof and is reported as a defect (A7b-N4). For a ***good*** *q in Mazur's sense — (i) q is not an ℓ-th power modulo N, i.e. $q^{(N-1)/\ell} \not\equiv 1 \pmod N$, and (ii) $\frac{q-1}{2} \not\equiv 0 \pmod \ell$, i.e. $q \not\equiv 1 \pmod \ell$, one modular power and one residue (A7b-N4) — the check fails* ***exactly*** *when a non-Eisenstein maximal ideal $\mathfrak{m}' \ni \ell$ of $\mathbb{T}^{\mathrm{cusp}}$ has a T_q-eigenvalue $\equiv 1+q$, and this happens for good q: at $(N, \ell, q) = (61, 5, 37)$, q is good, $\mathsf{h}_{37} = (x-8)(x^3+6x^2-36x-108)$, the rational newform of level 61 ($a_{37} = 8 \equiv 38$, $a_2 = -1 \not\equiv 3 \pmod 5$) enters F_{37}, $\deg \mathsf{F}_{37} = 2 > 1 = e^{\mathrm{cusp}}_{\mathfrak{m}}$ and $v_5(\mathsf{F}_{37}(38)) = 2 > 1 = v_5(\nu)$; at the same cell $q = 2$ passes. So goodness is how q is chosen and the check is the certificate: the record carries q, checks that it is good, and certifies only when the check holds, else tries the next good q (or splits h_q by the block attribution of Proposition 31.36). Everything is residue arithmetic modulo ℓ^K, $K > v_\ell(\nu)$: $\chi^{(N)}_{q,1}$ by residues, the exact division by $x-1-q$, the square root by the triangular recursion on the coefficients (ℓ odd), the Hensel factor by lifting the coprime factorisation modulo ℓ — no squarefree part and no gcd of big integers (owner ruling Q1(a)). For $\ell \,\|\, \nu$, $\mathsf{F}_q(x+1+q)$ is Eisenstein and $\mathbb{T}^{\mathrm{cusp}}_{\mathfrak{m}}$ a totally ramified DVR; for $\ell^2 \mid \nu$ (e.g. $N = 101$, $\nu = 25$) only $\mathbb{T}^{\mathrm{cusp}}_{\mathfrak{m}}/(\eta_q) = \mathbb{Z}/\ell^{v_\ell(\nu)}$ and F_q are certified, and the kind is read by Proposition 31.34(4). This needs only integral characteristic polynomials:* ***no*** *U_N* ***and no export****; it is local at $\mathfrak{m}$, and the global $\mathbb{Z}$-span still needs U_N.*

Proof. (1) Proposition 31.48(4) gives $C = \mathbb{Z}/g_{\mathcal{P}}$ and the fibre product; Proposition 31.29(5)(a) gives $\mathbb{T}^{\mathrm{cusp}}$ (the T_n, $n \le B_N$, are words in the X_p, $p \le B_N$); the gluing map is the ring map $\mathbb{T}^{\mathrm{cusp}} \to C$ of [U, `lem:u-fibre`], determined on the $\mathbb{Z}$-spanning set. (2) is the cited statements. (3) Write $\mathbb{T} := \mathbb{T}^{\mathrm{cusp}}_{\mathfrak{m}}$ (Mazur's $\mathbf{T}_{\mathfrak{P}}$, a completion and a direct factor of $\mathbb{T}^{\mathrm{cusp}} \otimes \mathbb{Z}_\ell$, containing $w = -U_N$; [GT] `lit:hz-mazur-eis`), $e := e^{\mathrm{cusp}}_{\mathfrak{m}}$ and $I_{\mathfrak{m}}$ its Eisenstein ideal. Mazur (9.7) gives $\mathbb{T}/I_{\mathfrak{m}} \cong \mathbb{Z}_\ell/\nu = \mathbb{Z}/\ell^{v_\ell(\nu)}$, and $\eta_q \in I_{\mathfrak{m}}$. $\mathbb{T}$ is free of rank e over $\mathbb{Z}_\ell$, and h_q is the product over the maximal ideals of $\mathbb{T}^{\mathrm{cusp}}$ over ℓ of the characteristic polynomials of T_q on their blocks of $S_2(\Gamma_0(N);\mathbb{Q}_\ell)$, each free of rank one over its block of $\mathbb{T}^{\mathrm{cusp}} \otimes \mathbb{Q}_\ell$ (the perfect pairing, Proposition 31.29(5)(a)); so $f_{q,\mathfrak{m}} \mid \mathsf{h}_q$, and as $\mathbb{T}$ is local with $\eta_q \in \mathfrak{m}$, every root of $f_{q,\mathfrak{m}}$ is $\equiv 1+q$: $\mathsf{F}_q = f_{q,\mathfrak{m}} G$, G the Hensel factor at $1+q$ of $\mathsf{h}_q/f_{q,\mathfrak{m}}$. $f_{q,\mathfrak{m}}(1+q) = \pm \mathrm{N}(\eta_q) \neq 0$ ($|a_q| \le 2\sqrt{q} < 1+q$), so $v_\ell(f_{q,\mathfrak{m}}(1+q)) = \mathrm{length}(\mathbb{T}/\eta_q\mathbb{T}) \ge \mathrm{length}(\mathbb{T}/I_{\mathfrak{m}}) = v_\ell(\nu)$, with equality iff $\eta_q\mathbb{T} = I_{\mathfrak{m}}$; and every root α of G has $v(1+q-\alpha) > 0$, so $v_\ell(G(1+q)) > 0$ unless $G = 1$. Hence the check holds iff $G = 1$ and $\eta_q\mathbb{T} = I_{\mathfrak{m}}$. Then $\mathbb{T} = \mathbb{Z}_\ell + I_{\mathfrak{m}} = \mathbb{Z}_\ell + \eta_q\mathbb{T}$, hence $\mathbb{T} = \mathbb{Z}_\ell[\eta_q] + \eta_q^j\mathbb{T}$ for every j, and $\mathbb{T} = \mathbb{Z}_\ell[\eta_q]$ ($\mathfrak{m}$-adic completeness; $\mathbb{Z}_\ell[\eta_q]$ is closed); its rank is e, so the minimal polynomial of T_q on $\mathbb{T}$ is $f_{q,\mathfrak{m}} = \mathsf{F}_q$ and $\mathbb{T} \cong \mathbb{Z}_\ell[x]/(\mathsf{F}_q)$. For a good q, (16.1), (16.2) and (16.6) give $\eta_q\mathbb{T} = I_{\mathfrak{m}}$, so the check reduces to $G = 1$; at $(61, 5, 37)$ the exact computation of the A7 fix round (`docs/implementation/slot24/heckez/Z0-report.md` §5) gives $G = x - 8$. The image of $\mathbb{Z}_\ell[T_q]$ in the $\mathfrak{m}$-block is $\mathbb{Z}_\ell[x]/((x-1-q)\mathsf{F}_q)$ (the minimal polynomial there; $\mathsf{F}_q(1+q) \neq 0$), and $\mathbb{Z}_\ell[T_q]_{\mathfrak{m}} \subseteq \mathbb{T}^{\mathrm{fr}}_{\mathfrak{m}} \subseteq \mathbb{Z}_\ell \times \mathbb{T}$, both inclusions of index $\ell^{v_\ell(\nu)}$ in the outer ring — the first by Proposition 31.34(3) for $\mathbb{Z}_\ell[x]/((x-1-q)\mathsf{F}_q)$, the second by Proposition 31.48(3) with $C_{\mathfrak{m}} = \mathbb{T}/I_{\mathfrak{m}} = \mathbb{Z}/\ell^{v_\ell(\nu)}$ ($I_{\mathcal{S}}$ maps onto the Eisenstein ideal, $U_N - 1 = -(1+w)$) — so they are equal, and U_N lies in the anemic algebra at $\mathfrak{m}$. For $\ell \,\|\, \nu$: $\bar{\mathsf{F}}_q(x+1+q) = x^e$ ($\mathbb{T}$ local, $\eta_q \in \mathfrak{m}$) and its constant term has valuation 1. □

Code map and reach (v1.25.0-pre.0). **Nothing of this subsection is built yet**; the C++ programme is the plan's slot-25 packages (`06-work-packages.md` §2.3): EXPORT (the whole module $H^1_{(\ell)}/\ell^K$ with the cochain-level operators, every level, $\ell \in \{2,3\}$ included; it lifts every `blocked(export)` of item 8), HZ-RAT (combinations $\vartheta = \sum c_p X_p$, χ_ϑ, discriminants, resultants, the cofactor route and the Pratt verifier below 2^{64}, under Q1(a)), HZ-UTAG (the two tags at $N > 1$), HZ-LOC (the local reads, the at-root read with the 64-bit regimes), HZ-TOR (separation, split test, the sandwich for J), HZ-REC (item 8), HZ-EMIT (the Lean certificates); UP-Z (U_p over $\mathbb{Z}$) in slot 26. Built today and used: Theorem 31.16's record (`integral_descent_record/v1`, item 7); `HeckeExt/HeckeRingHeckeAlgebra` and `HeckeSaturationDefect` (A'_K at level 1, $\ell \ge 5$); `HeckeExt/HeckeLocalAlgebra`, `HeckePresentation`, `HeckeGenerationCertificates` (over $\mathbb{F}_\ell$). The first golden objects of item 8 are planned for the slot-25 close (item 8 (k)).

Formalization note. No `\LEAN` marker is placed in this subsection. The plan's Lean lane (`07-lean-plan.md` v0.4) formalises Lemma 31.30 (HZ-0), Propositions 31.28, 31.31, 31.32, 31.33 ($v_\ell = 0$ clause), 31.34, 31.35, 31.36, Theorems 31.44(1)–(3), 31.45, Propositions 31.48, 31.49(1), 31.29(5)(c), and of the stage-03 block Propositions 31.37, 31.38, 31.39, 31.41(1)–(4), (5)(a), as [GT] entries, with per-cell certificates (K1–K10) re-checked by a verified checker; a marker enters only after the census line of §1.1's rule, and only after the owner has released [C] 1.24.0 (owner ruling Q8). Remarks 31.40–31.43 are never hypotheses of a Lean theorem.

32 The Moerdijk–Pronk model: the simplicial set in all degrees, refinements, morphisms

What this chapter fixes. Moerdijk and Pronk attach a simplicial set to a triangulated orbifold whose simplicial cohomology is the orbifold cohomology in *every* degree, from a *finite*

triangulation [MP99]; Remark 17.4 records why their construction could not be made normative here (effective charts, μ-maps whose independence of choices is stated without proof). This chapter states the global-quotient form of their object for $[\mathrm{SL}_2(\mathbb{Z})\backslash X]$ *without* those two obstacles, proves its comparison theorem independently of their argument, and fixes the maps that the code builds. (i) **Orbifold triangulations as finite data** and Moerdijk–Pronk's rules as decidable predicates (Definitions 32.1, 32.2, 32.3); the triangulations in use, M_7 and the refinement $\mathsf{R}_{\mathbb{Z}}$ (Definition 32.4, Table 9, Remark 32.5). (ii) **The choice-free MP complex** (Definition 32.6) and what it is (Remark 32.7). (iii) **The comparison theorem** (Theorem 32.12): its cohomology is $H^q(\Gamma; M)$ for every q, every ring and every module, $-I$ included, through the free bisimplicial Γ-set $\mathfrak{F}$ (Definition 32.8) and one assembly lemma used twice (Lemmas 32.9, 32.10, 32.11). (iv) **The ℓ-local hybrid** (Remark 32.13). (v) **All degrees**: the comparison lift Φ^{MP} (Definition 32.16) and the exact scope of the all-degrees statement (Proposition 32.17). (vi) **Morphisms** with the label factor (Definition 32.18), naturality and refinement invariance (Theorem 32.19), the Bredon comparison (Proposition 32.20), the boundary (Proposition 32.21), restriction and corestriction (Proposition 32.22), Hecke operators (Remark 32.23). (vii) Moerdijk–Pronk's teardrop (Example 32.24) and what is checked (Remark 32.25).

> **CONTRACT. No shipped value or default changes** (owner ruling Q4 of 2026-10-09). What $\mathbb{F}_\ell$ and $\mathbb{Z}/\ell^e$ *name* at $\ell \in \{2,3\}$ stays Definition 23.20: the Bredon object of Proposition 23.3, not group cohomology; in particular $\mathbb{F}_2$ and $\mathbb{F}_3$ keep their Bredon semantics. The MP model is a new object with its own validation names; its output lines carry `model=mp`; it adds no golden object, record, setting, console verb or env hook that changes a value. Whether $\mathbb{F}_2$ / $\mathbb{Z}/2^e$ should ever *mean* group cohomology is a separate, golden-gated owner question (the implementation plan's D1), not decided here.

Normative object. As in §17: "the orbifold cohomology of $[\Gamma\backslash X]$" is the cohomology of the Borel construction, i.e. group cohomology $H^q(\Gamma; M)$ in every degree (Definition 17.1(2)); every complex of this chapter is a computational model of it. Moerdijk–Pronk's sheaf-side identification of orbifold sheaf cohomology with the Borel construction is cited in Remark 17.4(1)(i) and is **not** re-proved; what is proved is the computational half, the analogue of their comparison theorem, for the global quotient and without their effectivity assumption. **Notation.** $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ unless a statement says "a group Γ"; Y a simplicial Γ-complex as in (G1); R a commutative ring, M an $R[\Gamma]$-module ($M = V_R$ with ρ_n in the numbers). $\mathfrak{T}$, Ar, S, $\underline{M}$, $\mathfrak{F}$, $\mathfrak{F}'$, $O(x)$, h, Φ^{MP}, $\varepsilon^{\mathsf{MP}}$, M_7, $\mathsf{R}_{\mathbb{Z}}$, $\mathsf{R}^+_{\mathbb{Z}}$, R_ζ, $\mathsf{R}_{\mathrm{good}}$, Tot, $C^\bullet_{\mathrm{un}}$, D, and (v1.22.0-pre.11) Q_0, $\mathcal{C}$, $\mathsf{p}(x)$, $\mathfrak{D}$, $\mathfrak{F}^Q$, $\mathsf{S}^{(\ell)}$, and the hypothesis tags (SC), (AC), (IO), (PD), (QD) are new symbols of this chapter and mean nothing else (rule (A5)): D denotes the degree from which periodicity transports every higher degree, $D = 3$ (Proposition 32.17(2); neither the matrix D of Remark 27.10(3) nor $D_{M,N,d}$), S is not the matrix S, $\mathfrak{F}'$ is not the column F'^{σ} of Definition 31.1, Φ^{MP} is not the cone map Φ_g nor Φ_ℓ, and $\varepsilon^{\mathsf{MP}}$ is not the augmentation ε (the implementation's reports write S_k, F', Φ, ε^* and $h_{\tilde\sigma}$). P_σ, Q_σ are those of Definition 31.1. The elliptic vertices are written $v_4 = \zeta^2$, $v_5 = \zeta$, $v_6 = i$ as in Definition 7.1; [MP] is `docs/implementation/slot22/MP_math_plan.md` v0.2.1, cited by section for the full derivations.

32.1 Orbifold triangulations as finite data

Definition 32.1 (Orbifold triangulation; descriptor; the four hypotheses)**.** Let Γ be a group and Y a finite-dimensional simplicial Γ-complex satisfying (G1) (simplicial action without inversions, finite stabilisers) and (G3) (finitely many orbits of simplices in each degree).

1. **(Descriptor.)** A *descriptor* of Y is the finite datum: for each degree q a set $\Sigma_q(Y)$ of orbit representatives $\iota(\sigma)$, the stabilisers Γ_σ, the transporters $\mathrm{Tr}(\sigma,\sigma') = \{\gamma \in \Gamma : \gamma\,\iota(\sigma') = \iota(\sigma)\}$ (finite unions of cosets of stabilisers), and the faces $\partial_j\iota(\sigma) = h \cdot \iota(\tau)$,

$h \in \Gamma$. The descriptors of this document are `integral_model::ModelData` (§23.2, §31's code map) and, in Lean, [GT]'s `XBulkDatum`. $\Sigma_{\max}(Y)$ denotes the representatives of the maximal simplices.

2. **(Isotropy datum; orbifold triangulation.)** The *isotropy datum* is $\sigma \mapsto \Gamma_\sigma$ (the pointwise stabiliser, which is the setwise one by (G1)) with, for a face τ of σ realised as $h\iota(\tau) \subseteq \iota(\sigma)$, the monomorphism $\Gamma_\sigma \to \Gamma_\tau$, $\gamma \mapsto h^{-1}\gamma h$ (well defined up to conjugation in Γ_τ). An **orbifold triangulation** $\mathfrak{T}$ of $[\Gamma\backslash Y]$ is a descriptor of Y together with its isotropy datum. *Effectivity is not assumed*: a normal subgroup acting trivially on Y — for $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ on X the centre $\{\pm I\}$, (H4)(iv) — lies in every Γ_σ and is kept.
3. **(The four hypotheses.)** (SC) *simplicial-complex property*: two closed simplices of Y meet in the empty set or a common face; (AC) *acyclicity*: the augmented chain complex $C_\bullet(Y) \to \mathbb{Z}$ is exact; (IO) *invariant order*: no simplex of Y has two vertices in one Γ-orbit, so ordering the vertices of each simplex by the index of their orbit is a Γ-invariant total order on every simplex; (PD) *purity*: every maximal simplex of Y has dimension $\dim Y$, so $\Sigma_{\max}$ is the set of top-dimensional representatives and every simplex lies in one. On a descriptor (SC), (IO) and (PD) are finite, decidable statements.

Definition 32.2 (Moerdijk–Pronk's rules as predicates)**.** Let $\mathfrak{T}$ be an orbifold triangulation of $[\Gamma\backslash Y]$. Following the rules (i), (ii) of an adapted triangulation in [MP99] (quoted in Remark 17.4(1)(ii)), read on the isotropy datum: [formally verified: [GT, `def:mp-datum`]] ((1)–(4) as decidable predicates on a finite descriptor, [GT]'s `MPDatum`, not on `XBulkDatum`; (3) in its finite-group form and in the strong form below)

1. **Adapted** (their rule (ii), *subgroup form*): every simplex σ of Y has a vertex $v(\sigma)$ with $\Gamma_\tau \subseteq \Gamma_{v(\sigma)}$ for every face τ of σ (point stabilisers, as subgroups of Γ after transport — *not* as orders).
2. **Adapted for $\mathfrak{T}^{(\ell)}$**: the same with $P_\tau \subseteq P_{v(\sigma)}$, the ℓ-Sylow datum of Definition 32.3. This is rule (ii) applied to the ℓ-relevant datum, *not* Moerdijk–Pronk-conformance of $[\Gamma\backslash Y]$, which concerns the actual isotropy.
3. **Good** (their chart condition): for every vertex v, the set $\mathsf{T}(v) := \{\gamma \in \Gamma : \gamma\,\mathrm{St}(v) \cap \mathrm{St}(v) \neq \emptyset\}$, $\mathrm{St}(v)$ the closed star, generates a finite subgroup of Γ. The *strong* form is $\mathsf{T}(v) \subseteq \Gamma_p$ for one vertex p of $\mathrm{St}(v)$ ($\mathsf{T}(v)$ in the stabiliser of one point of $\mathrm{St}(v)$; the stabiliser of any point lies, under (IO), in that of a vertex of its carrier). It is *not* $\mathsf{T}(v) \subseteq \Gamma_v$, which is strictly stronger (v1.22.0-pre.6; pre.5 wrote this form, which no code or table uses). [formally verified: [GT, `thm:mp-goodvtx`]] (the strong form as stated here, `goodAtB_iff_onePt`; and that $\mathsf{T}(v) \subseteq \Gamma_v$ fails at every vertex orbit of M_7 and at v_0, v_1, v_2, m on $\mathsf{R}_{\mathbb{Z}}$)
4. **Collar**: no simplex meets both the boundary subcomplex and a vertex whose stabiliser is not the kernel of the action.

Definition 32.3 (The ℓ-relevant structure)**.** Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, so every Γ_σ is cyclic, $\Gamma_\sigma = P_\sigma \times Q_\sigma$ with P_σ its ℓ-Sylow subgroup and $|Q_\sigma|$ prime to ℓ (Definition 31.1). The **ℓ-relevant orbifold structure** of $\mathfrak{T}$ is the pair $(\mathfrak{T}^{(\ell)}, M^Q)$: $\mathfrak{T}^{(\ell)}$ the isotropy datum with Γ_σ replaced by P_σ, and M^Q the *coefficient system* $\sigma \mapsto M^{Q_\sigma}$, contravariant along faces ($\Gamma_\sigma \subseteq \Gamma_\tau$ for $\tau \leq \sigma$ gives $Q_\sigma = Q_\tau \cap \Gamma_\sigma \subseteq Q_\tau$, hence $M^{Q_\tau} \subseteq M^{Q_\sigma}$), with P_σ acting on M^{Q_σ}. [formally verified: [GT, `def:mp-datum`]] (the ℓ-Sylow datum $\sigma \mapsto P_\sigma$ on a finite descriptor, as the input of Definition 32.2(2); not the coefficient system M^Q, not the table below) On X (owner ruling Q3 of 2026-10-09):

	ℓ-singular simplices, P_σ	elsewhere	Q-invariants kept
$\ell = 3$	the orbit of ζ^2: $P = \langle (ST)^2 \rangle \cong \mathbb{Z}/3$	$P = 1$	M^{-I}, except $M^{\langle S \rangle}$ at i; at ζ^2 M^{-I} with its $\mathbb{Z}/3$-action
$\ell = 2$	every simplex has $P \supseteq \{\pm I\}$; $P = \langle S \rangle \cong \mathbb{Z}/4$ at i	$P = \{\pm I\}$	M, except $M^{\langle (ST)^2 \rangle}$ at ζ^2
over $\mathbb{Z}$	$\langle ST \rangle \cong \mathbb{Z}/6$ at ζ^2, $\langle S \rangle$ at i	$\{\pm I\}$	none (no order is a unit)

"Relevant at ℓ" is thus a statement about which columns of a model may be truncated to degree 0, not about a different orbifold.

Definition 32.4 (The triangulations in use)**.** All are Γ-equivariant subdivisions of X ((H4)); they differ only in the descriptor. [formally verified: [GT, `def:mp-models`]] ((1) and (2) as MP descriptors: M_7 computed from the dump that defines [GT]'s `model7`, $\mathsf{R}_{\mathbb{Z}}$ from the C++ dump `ORBIFOLD_MP_DUMP`; the identification of the dumps with the geometric complexes is the trust boundary, as for (H4)(i)) [formally verified: [GT, `def:mp-rz-tau`]] ((2) as a Γ-complex descriptor with the X axioms: 9 vertices, 15 edges, 7 triangles of the lifted domain, all transporters; not (3))

1. M_7: X with the triangulation of Definition 7.1 (vertex orbits $\{v_0, v_3\}, \{v_1\}, \{v_2\}, \{v_4, v_5\}, \{v_6\}$; quotient counts $(V, E, F) = (5, 9, 5)$, the representatives Σ_s of Definition 23.4).
2. $\mathsf{R}_{\mathbb{Z}}$: the subdivision of the edge orbit of v_4v_6 at one interior point m (its stabiliser is $\{\pm I\}$, which fixes the edge pointwise), so that $v_5v_6 = S \cdot v_4v_6$ is cut at $m' := S\,m$. Lifted domain: vertices $v_0, \dots, v_6, m, m'$; triangles $v_0v_1v_4$, $v_1v_2v_6$, $v_2v_3v_5$, v_1v_4m, v_1mv_6, v_2v_5m', $v_2m'v_6$; relation rows those of Definition 7.2 with the row of $\{v_4, v_6\} \leftarrow \{v_5, v_6\}$ replaced by $v_5m' \leftarrow v_4m$, $m'v_6 \leftarrow mv_6$ and the vertex row $m' \leftarrow m$, all by S, and the central rows on every simplex. Stabilisers: $\langle ST\rangle$, $\langle TS\rangle$ at v_4, v_5, $\langle S\rangle$ at v_6, $\{\pm I\}$ elsewhere, m, m' included. Quotient counts $(V, E, F, \chi) = (6, 12, 7, 1)$. In code `SubdivideEdgeOrbit` applied to M_7 at the edge orbit whose ends have stabilisers of orders 6 and 4, found from the inventory, not by name.
3. Predicate rows only (no model is built on them): R_ζ, slot 17's subdivision of the three edge orbits at ζ^2 (Remark 31.4(3)); $\mathsf{R}^+_{\mathbb{Z}}$, $\mathsf{R}_{\mathbb{Z}}$ with the edges v_0v_4 (and v_3v_5), v_1v_4, v_2v_5 cut once; R_{good}, v_4v_6 cut twice (and its S-image), v_1v_4, v_1v_6, v_2v_5, v_2v_6 cut once and the two triangles at i coned off at their centres ([MP] §3.1; the table of `docs/implementation/slot22/mp/mp_rules.py`).

triangulation	(V, E, F, χ)	adapted for $\mathfrak{T}^{(3)}$	adapted for $\mathfrak{T}^{(2)}$	adapted (MP's rule)	good	fails at
M_7 (shipped)	$(5, 9, 5, 1)$	yes	yes	**no**	no	v_1, v_2, ζ^2, i
$\mathsf{R}_{\mathbb{Z}}$	$(6, 12, 7, 1)$	yes	yes	**yes**	no	v_1, v_2, m
R_ζ	$(8, 18, 11, 1)$	yes	yes	yes	no	three vertices
$\mathsf{R}^+_{\mathbb{Z}}$	$(9, 21, 13, 1)$	yes	yes	yes	no	m
R_{good}	$(13, 33, 21, 1)$	yes	yes	yes	**yes**	—

Table 9: Moerdijk–Pronk's rules on the candidate triangulations of X (Definitions 32.2, 32.4); quotient counts. The collar predicate is false on every row. "Fails at" lists the vertices whose closed star is not good (finite-group and strong form of Definition 32.2(3) alike). `OrbifoldMPRulesValidation`. Rows M_7 and $\mathsf{R}_{\mathbb{Z}}$ only, every column: [formally verified: [GT, `thm:mp-model7-rules`]] [formally verified: [GT, `thm:mp-rz-rules`]] [formally verified: [GT, `thm:mp-rz-dump`]] (the C++ predicate rows of these two rows are the Lean predicates).

Remark 32.5 (Reading the rules on X). [proved here] [verified computationally]^{r}

1. M_7 **is not adapted, at every** ℓ **and over** $\mathbb{Z}$**, for** $\mathrm{SL}_2(\mathbb{Z})$ **and for the effective** $\mathrm{PSL}_2(\mathbb{Z})$ **alike.** The edge v_4v_6 has stabiliser $\{\pm I\}$ (resp. 1) inside, $\langle ST\rangle \cong \mathbb{Z}/6$ and $\langle S\rangle \cong \mathbb{Z}/4$ (resp. $\mathbb{Z}/3$, $\mathbb{Z}/2$) at its ends (Lemma 7.3(1)), neither containing the other, so neither $v_1v_4v_6$ nor $v_2v_5v_6$ has a dominating vertex. Rule (ii) knows no prime.
2. $\mathsf{R}_{\mathbb{Z}}$ **is adapted**, and is the smallest listed refinement that is: each of its seven triangles has at most one vertex with stabiliser $\neq \{\pm I\}$, which dominates. So for Moerdijk–Pronk's construction *as they define it*, a refinement is forced, and it is the same $\mathsf{R}_{\mathbb{Z}}$ at $\ell = 2$, $\ell = 3$ and over $\mathbb{Z}$ (owner ruling Q2 of 2026-10-09). The Sylow data $\mathfrak{T}^{(3)}$, $\mathfrak{T}^{(2)}$ of M_7 are adapted ($\mathbb{Z}/3$ at ζ^2 only, resp. $\{\pm I\}$ everywhere and $\mathbb{Z}/4$ at i only); this enters no theorem below and is not conformance.

3. **Goodness** fails on $\mathsf{R}_{\mathbb{Z}}$ and $\mathsf{R}^+_{\mathbb{Z}}$, at different vertices (`good_fails_at`): on $\mathsf{R}^+_{\mathbb{Z}}$ at m only, on $\mathsf{R}_{\mathbb{Z}}$ at v_1, v_2 and m (the labels $[1,2,7]$ of the oracle); there the closed star meets both the orbit of ζ^2 and that of i, so $\mathsf{T}(v)$ contains generators of $\langle ST\rangle$ and $\langle S\rangle$, which generate Γ. Decidability: the finite subgroups of $\mathrm{SL}_2(\mathbb{Z})$ are cyclic of order ≤ 6 (a finite subgroup fixes a point of $\mathcal{H}$, so lies in that point's stabiliser, conjugate to a subgroup of $\langle S\rangle$ or $\langle ST\rangle$, (H4)(iv); the orders $1,2,3,4,6$ as listed in Proposition 29.32's proof), so the closure of $\mathsf{T}(v)$ under products may be declared infinite once it exceeds 6 elements; on the four oracle rows the finite-group form and the strong form of Definition 32.2(3) (some $p \in \mathrm{St}(v)$ with $\mathsf{T}(v) \subseteq \Gamma_p$) agree (`good_forms_agree`). The stronger reading $\mathsf{T}(v) \subseteq \Gamma_v$ would fail at every vertex orbit of M_7 and at v_0, v_1, v_2, m on $\mathsf{R}_{\mathbb{Z}}$ (Lean, `goodVtx_discrepancy`), which is why it is not the definition. Goodness is Moerdijk–Pronk's hypothesis for *their* sheaf-side proof only; nothing in this chapter uses it, nor the collar, nor adaptedness (Theorem 32.12, Remark 32.13, Theorem 32.19).
4. **The subgroup form is necessary**: comparing stabiliser orders (as `mp_rules.py` does) agrees with it on these rows only because every non-maximal stabiliser is the central $\{\pm I\}$; with a planted non-central sub-stabiliser the order form accepts what the subgroup form rejects.

[verified computationally]^{r}: `OrbifoldMPRulesValidation` (`fast`) reproduces Table 9 on its four oracle rows against `mp_rules.py` ($4 \times 3 = 12$ gated cells, `ORACLE mp_rules.py cells=12 agree=12`); the R_ζ row is reported only (an ungated `OUTPUT P1.12` line). (SC) and (IO) are checked on all five, the descriptor's quotient counts against `ModelData`'s representatives, and the negative controls of (4) and of a refused two-vertices-in-one-orbit order.

32.2 The Moerdijk–Pronk set, choice-free

Definition 32.6 (The MP set $\mathsf{S}(Y;\mathfrak{T})$, its local system and its cochains)**.** Let $\mathfrak{T}$ be an orbifold triangulation of $[\Gamma\backslash Y]$ with (SC) and (PD). For top representatives $\sigma, \sigma' \in \Sigma_{\max}$ put $\mathrm{Ar}(\sigma,\sigma') := \{\gamma \in \Gamma : \iota(\sigma) \cap \gamma\,\iota(\sigma') \neq \emptyset\}$ (the arrows $\sigma \xleftarrow{\gamma} \sigma'$). For $k \geq 0$,

$$\mathsf{S}_k := \big\{(\sigma_0,\dots,\sigma_k;\gamma_1,\dots,\gamma_k) :\ \sigma_i \in \Sigma_{\max},\ \gamma_i \in \Gamma,\ \iota(\sigma_0)\cap\gamma_1\iota(\sigma_1)\cap\gamma_1\gamma_2\iota(\sigma_2)\cap\cdots\cap\gamma_1\cdots\gamma_k\iota(\sigma_k) \neq \emptyset\big\}.$$

Faces: d_0 drops (σ_0,γ_1); d_i $(0<i<k)$ drops σ_i and replaces (γ_i,γ_{i+1}) by $\gamma_i\gamma_{i+1}$; d_k drops (σ_k,γ_k). Degeneracies insert $(\sigma_i,1)$. The **local system** $\underline{M}$ has fibre M at every $\sigma \in \Sigma_{\max}$, the arrow $\sigma_0 \xleftarrow{\gamma} \sigma_1$ acting by $m \mapsto \gamma\cdot m$. The **MP complex** is the normalised cochain complex

$$C^k(\mathsf{S};\underline{M}) := \{f\colon \mathsf{S}_k^{\mathrm{nd}} \to M\}, \qquad (df)(s) := \gamma_1\cdot f(d_0 s) + \sum_{i=1}^{k+1}(-1)^i f(d_i s) \quad (s \in \mathsf{S}_{k+1}^{\mathrm{nd}}),$$

with $f := 0$ on degenerate strings; $\mathsf{S}_k^{\mathrm{nd}}$ is the set of non-degenerate strings (no i with $\sigma_i = \sigma_{i+1}$ and $\gamma_{i+1} = 1$). The same formula on all functions $\mathsf{S}_k \to M$ defines the **unnormalised MP complex** $C^\bullet_{\mathrm{un}}(\mathsf{S};\underline{M})$, which contains $C^\bullet(\mathsf{S};\underline{M})$ as the subcomplex of functions vanishing on degenerate strings (v1.22.0-pre.6: named, so that the Lean markers can be clause-scoped). On the triangulations of Definition 32.4, $\mathsf{S}(\mathsf{M}_7)$ and $\mathsf{S}(\mathsf{R}_{\mathbb{Z}})$, with $M = V_R$, $\gamma\cdot m = \rho_n(\gamma)m$. [formally verified: [GT, `def:mp-cover`]] [formally verified: [GT, `lem:mp-set-gamma`]] [formally verified: [GT, `lem:mp-face-face`]] (S_k as the normal forms $g_0 = 1$ of $\mathfrak{F}'_k$ with the coordinates $(\sigma_i;\gamma_i)$ as an explicit bijection, the intersection read as a common vertex, which is the same under (SC), Remark 32.7(1); its faces and degeneracies and the simplicial identities, the middle faces as deletions) [formally verified: [GT, `thm:mp-dd`]] (the twisted differential and $d\circ d = 0$ on $C^\bullet_{\mathrm{un}}$) [formally verified: [GT, `thm:mp-norm-dgn`]] (the normalised complex: the functions vanishing on the degenerate strings $s_i(s)$ form a subcomplex of $C^\bullet_{\mathrm{un}}$ for the twisted differential, isomorphic to the cochains of the simplicial object of Definition 32.8's marker that vanish on its degeneracies, and its inclusion is a quasi-isomorphism; v1.22.0-pre.7)

Remark 32.7 (What the MP set is, and when it is Moerdijk–Pronk's). [proved here] [verified computationally]^{r}

1. **Finite and decidable.** $\mathrm{Ar}(\sigma,\sigma')$ is the finite union, over vertex pairs v of $\iota(\sigma)$, w of $\iota(\sigma')$ in one orbit, of the cosets $\mathrm{Tr}(v,w)$ of Γ_w; under (SC) a string is in S_k iff some vertex p of $\iota(\sigma_0)$ has $(\gamma_1\cdots\gamma_j)^{-1}p\in\iota(\sigma_j)$ for all j (a finite intersection of closed simplices is a common face, hence non-empty iff it has a vertex). The faces of a string are strings; S is a simplicial set. No chart, lift, μ-map, effectivity or adaptedness enters the definition. [formally verified: [GT, `thm:mp-datum-arrows`]] [formally verified: [GT, `thm:mp-models-cover`]] ($\mathrm{Ar}(\sigma,\sigma')$ the finite union of the cosets $\mathrm{Tr}(v,w)$ on a descriptor; on M_7 and $\mathsf{R}_{\mathbb{Z}}$ the counts of all strings $|\mathsf{S}_0|,|\mathsf{S}_1|=5,132$ and $7,172$)
2. **What it is.** Under (SC), $\mathsf{S}=\pi_0$ of the nerve of the action groupoid $\Gamma\ltimes Y$ restricted to the transversal $\bigsqcup_{\sigma\in\Sigma_{\max}}\iota(\sigma)$: the component of a k-simplex of that nerve is the closed simplex $\iota(\sigma_0)\cap\gamma_1\iota(\sigma_1)\cap\cdots$, which is contractible.
3. **When it is Moerdijk–Pronk's $S(\mathcal{T})$.** If Γ acts effectively, $\mathfrak{T}$ is adapted and the quotient is a simplicial complex, the intersection is a face τ with a maximal vertex $v(\tau)$ and the γ_i range over a coset of $\Gamma_{v(\tau)}$ — under adaptedness the stabilisers of the vertices shared by $\iota(\sigma_{i-1})$ and $\gamma_i\iota(\sigma_i)$ are nested in $\Gamma_{v(\tau)}$, so the union of cosets in (1) is a single coset $h\Gamma_{v(\tau)}$, and effectivity and the simplicial quotient make the shared vertex set the unique lift of $\bar\sigma_{i-1}\cap\bar\sigma_i$: $\mathsf{S}_k\cong\bigsqcup\Gamma^k_{v(\tau)}$, their formula (Remark 17.4(1)(ii)). For $\Gamma=\mathrm{SL}_2(\mathbb{Z})$, **not** effective, their own construction gives the $\mathrm{PSL}_2(\mathbb{Z})$-set (Remark 17.4(2)); S keeps $-I$ as an arrow $\sigma\xleftarrow{-I}\sigma$ of every object and is a **non-effective generalisation** of their object, not their object (owner ruling Q1 of 2026-10-09: this choice-free form is normative for the MP model). On a non-adapted $\mathfrak{T}$ $\mathrm{Ar}(\sigma,\sigma)$ need not be a group (on M_7, $\mathrm{Ar}(\sigma,\sigma)$ at $v_1v_4v_6$ is $\langle ST\rangle\cup\langle S\rangle$, eight elements), and their formula no longer describes S, which is still correct (Theorem 32.12).

[verified computationally]^{r} (`OrbifoldMPRulesValidation`, `OrbifoldMPNerveValidation`): $|\mathsf{S}_0^{\mathrm{nd}}|,\dots,|\mathsf{S}_3^{\mathrm{nd}}|=5,127,2541,53143$ on M_7 and $7,165,2927,56349$ on $\mathsf{R}_{\mathbb{Z}}$; all strings $|\mathsf{S}_1|,|\mathsf{S}_2|=172,3264$ on $\mathsf{R}_{\mathbb{Z}}$ by both S and the μ-form count $\sum_\tau|\Gamma_{v(\tau)}|^k$, against, on the non-adapted M_7, $132,2800$ by S versus $124,2704$ by the μ-form count (`ORACLE adapted_form_check.py cells=6 agree=6`); faces closed, simplicial identities exhaustive to $k=4$, every string once.

32.3 The comparison theorem

Definition 32.8 (The free bisimplicial Γ-set $\mathfrak{F}$ and the labelled Čech nerve $\mathfrak{F}'$)**.** Let Y, $\mathfrak{T}$ satisfy (SC), (IO), (PD). Let Y_s be the set of non-degenerate ordered s-simplices of Y (one increasing vertex tuple per geometric simplex, for the order of (IO)), a Δ-set with $0\le s\le\dim Y$. For $k\ge0$ put

$$\mathfrak{F}_{k,s}:=\big\{\big((g_0,\sigma_0),\dots,(g_k,\sigma_k);x\big):\ g_i\in\Gamma,\ \sigma_i\in\Sigma_{\max},\ x\in Y_s,\ x\subseteq g_0\iota(\sigma_0)\cap\cdots\cap g_k\iota(\sigma_k)\big\},$$

$\mathfrak{F}'_k:=\{((g_0,\sigma_0),\dots,(g_k,\sigma_k)):\bigcap_i g_i\iota(\sigma_i)\neq\emptyset\}$, with Γ acting by left multiplication on every g_i and on x. Faces in k drop a pair, degeneracies repeat one (simplicial in k); faces in s are those of $Y_\bullet$ (semi-simplicial in s). $\mathbb{Z}[\mathfrak{F}_{\bullet,\bullet}]$ is a double complex, bounded in s; Tot is its total complex; $\mathbb{Z}[\mathfrak{F}'_\bullet]$ is the chain complex of the simplicial set $\mathfrak{F}'$, augmented by $(g_0,\sigma_0)\mapsto1$, and $N\mathbb{Z}[\mathfrak{F}'_\bullet]$ its normalised quotient. For $x\in Y_s$, $O(x):=\{(g,\sigma):x\subseteq g\iota(\sigma)\}$. [formally verified: [GT, `def:mp-bisimp`]] ((IO), Y_s, $\mathfrak{F}_{k,s}$ in the bounded form $\dim Y\le2$ with both kinds of faces, and the augmented complexes $R[\mathfrak{F}_{\bullet,s}]\to R[Y_s]$, $R[\mathfrak{F}'_\bullet]\to R$, $R[Y_\bullet]\to R$ over every commutative ring; not the degeneracies in k, not $N\mathbb{Z}[\mathfrak{F}'_\bullet]$) [formally verified: [GT, `def:mp-simpK`]] [formally verified: [GT, `thm:mp-simpK-alt`]] (v1.22.0-pre.7: $[k]\mapsto R[\mathfrak{F}'_k]$ as a simplicial object in $R[\Gamma]$-modules, faces the deletions and degeneracies the repetitions, whose alternating face map complex is $R[\mathfrak{F}'_\bullet]$, over every commutative ring; not $N\mathbb{Z}[\mathfrak{F}'_\bullet]$ as a quotient complex)

Lemma 32.9 (Rows). *[proved here] [formally verified: [GT, `lem:mp-rows`]] (the augmented row contracted by the extra degeneracy, non-equivariantly, over every commutative ring) For each s, the augmentation $\mathbb{Z}[\mathfrak{F}_{\bullet,s}] \to \mathbb{Z}[Y_s]$, $((g_i,\sigma_i);x) \mapsto x$, is a quasi-isomorphism of complexes of $\mathbb{Z}[\Gamma]$-modules.*

Proof. It is Γ-equivariant, and a quasi-isomorphism is checked non-equivariantly. $\mathfrak{F}_{\bullet,s} = \bigsqcup_{x\in Y_s} O(x)^{\bullet+1}$, and $(g_i,\sigma_i) \in O(x)$ for each i is exactly the condition $x \subseteq \bigcap_i g_i\iota(\sigma_i)$. $O(x) \neq \emptyset$ by (PD). $O(x)^{\bullet+1}$ is the nerve of the chaotic groupoid on $O(x)$ (the Čech nerve of $O(x) \to \mathrm{pt}$), and for a fixed $o \in O(x)$ the extra degeneracy $(o_0,\dots,o_k) \mapsto (o,o_0,\dots,o_k)$ contracts the augmented complex $\mathbb{Z}[O(x)^{\bullet+1}] \to \mathbb{Z}$. The augmentation kills boundaries: $((a,b);x) \mapsto x - x$. □

Lemma 32.10 (Columns). *[proved here] [formally verified: [GT, `lem:mp-cols`]] (the cone contraction on the first vertex v^*, non-equivariant, for $\dim Y \leq 2$) For each k, the augmentation $\mathbb{Z}[\mathfrak{F}_{k,\bullet}] \to \mathbb{Z}[\mathfrak{F}'_k]$ (concentrated in $s=0$), $((g_i,\sigma_i);x) \mapsto ((g_i,\sigma_i))$, is a quasi-isomorphism of complexes of $\mathbb{Z}[\Gamma]$-modules.*

Proof. Again checked non-equivariantly, fibre by fibre. Over a string $t = ((g_i,\sigma_i)) \in \mathfrak{F}'_k$ the s-fibre is the set of non-degenerate ordered faces of $\tau_t := \bigcap_i g_i\iota(\sigma_i)$, which by (SC) (and induction on k) is a closed simplex, non-empty by the definition of $\mathfrak{F}'_k$; with the order of (IO) its augmented ordered chain complex is that of a standard simplex, contracted by the cone on its first vertex v^* ($w_0\cdots w_j \mapsto v^*w_0\cdots w_j$ if $w_0 \neq v^*$, else 0). *No equivariant homotopy is claimed*: $v^*(\tau_t)$ is not compatible with the k-faces (a k-face enlarges τ_t and may change its first vertex). The augmentation kills boundaries: an edge (w_0w_1) maps to $t-t$. □

Lemma 32.11 (Bounded assembly). *[proved here] Let $f\colon A \to B$ be a map of double complexes of modules over a ring, $A_{k,s} = B_{k,s} = 0$ unless $k \geq 0$ and $0 \leq s \leq N$. If f is a quasi-isomorphism on every row (fixed s), or on every column (fixed k), then $\mathrm{Tot}\, f$ is a quasi-isomorphism. [formally verified: [GT, `thm:mp-assembly-grid`]] (only the acyclic form used in the proof of Theorem 32.12(2): in a four-row bounded double complex with exact columns and acyclic upper rows the bottom row is acyclic; not the quasi-isomorphism form)*

Proof. Replace f by its mapping cone formed in the direction in which f is a quasi-isomorphism: a double complex C, bounded in the same way up to one step, with exact rows, resp. exact columns, and $\mathrm{Tot}\, C = \mathrm{Cone}(\mathrm{Tot}\, f)$ up to sign; it suffices that $\mathrm{Tot}\, C$ is acyclic. Exact rows: filter $\mathrm{Tot}\, C$ by s; the filtration has finitely many steps, each graded piece is a shifted row, hence acyclic, and induction over the long exact sequences gives the claim. Exact columns: filter by $k \leq p$; the filtration is exhaustive with $F_{-1} = 0$, its graded pieces are shifted columns, so every F_p is acyclic by induction, and F_{q+1} contains $\mathrm{Tot}\, C$ in every degree $\leq q+1$, so $H_q(\mathrm{Tot}\, C) = H_q(F_{q+1}) = 0$. (Equivalently the acyclic assembly lemma, [standard: [Wei94, Lemma 2.7.3], locator not checked against the text, VERIFY], the two totalisations coinciding because s is bounded.) □

Theorem 32.12 (Comparison: the MP complex computes group cohomology in every degree). *[proved here] Let Γ be a group and Y a simplicial Γ-complex with (G1), (G3), (SC), (AC), (IO), (PD) (Definition 32.1), and $\mathfrak{T}$ its orbifold triangulation. Then:*

1. *Γ acts freely on $\mathfrak{F}'_k$ (already on g_0); the normal form $g_0 = 1$ identifies $\Gamma\backslash\mathfrak{F}'_k$ with S_k, by $((1,\sigma_0),(\gamma_1,\sigma_1),(\gamma_1\gamma_2,\sigma_2),\dots) \leftrightarrow (\sigma_i;\gamma_i)$, $\gamma_i = g_{i-1}^{-1}g_i$; and S_k is finite. [formally verified: [GT, `lem:mp-free`]] (the free action and the normal form as a transversal) [formally verified: [GT, `lem:mp-set-finite`]] (finiteness, given finitely many maximal representatives and finite arrow sets)*
2. *$\mathbb{Z}[\mathfrak{F}'_\bullet] \to \mathbb{Z}$ and $N\mathbb{Z}[\mathfrak{F}'_\bullet] \to \mathbb{Z}$ are resolutions of $\mathbb{Z}$ by free $\mathbb{Z}[\Gamma]$-modules, the latter of rank $\left|\mathsf{S}_k^{\mathrm{nd}}\right|$ in degree k — the labelled Čech nerve of the cover of Y by its closed maximal simplices. [formally verified: [GT, `thm:mp-resolution`]] [formally verified:*

[GT, thm:mp-projres]] (the first: $R[\mathfrak{F}'_\bullet] \to R$ a resolution of R by projective $R[\Gamma]$-modules, for every commutative ring R with (AC) over R, (IO), finite vertex sets and $\dim Y \le 2$; not $N\mathbb{Z}[\mathfrak{F}'_\bullet]$) [formally verified: [GT, thm:mp-norm-heq]] (v1.22.0-pre.7: the normalised clause dually, at the level of cochains — the cochains vanishing on the degeneracies are a homotopy equivalent subcomplex of $\mathrm{Hom}_{R[\Gamma]}(R[\mathfrak{F}'_\bullet], M)$, every ring, every module; not $N\mathbb{Z}[\mathfrak{F}'_\bullet] \to \mathbb{Z}$ as a resolution, not its rank $\left|\mathsf{S}^{\mathrm{nd}}_k\right|$)

3. $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(N\mathbb{Z}[\mathfrak{F}'_\bullet], M) = C^\bullet(\mathsf{S}; \underline{M})$ ***on the nose***, $F \mapsto (f(s) := F(x_s))$, x_s *the normal form of s; the twist γ_1 of the differential is the re-normalisation of $d_0 x_s = \gamma_1 \cdot x_{d_0 s}$. Likewise* $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\mathfrak{F}'_\bullet], M) = C^\bullet_{\mathrm{un}}(\mathsf{S}; \underline{M})$. *[formally verified: [GT, thm:mp-homequiv]] (the unnormalised identity, over every commutative ring)*
4. *Hence, for every commutative ring R and every $R[\Gamma]$-module M, naturally in M,*

$$H^k\big(C^\bullet(\mathsf{S}(Y;\mathfrak{T}); \underline{M})\big) \;\cong\; H^k(\Gamma; M) \qquad \text{for every } k \ge 0,$$

and the same for $C^\bullet_{\mathrm{un}}(\mathsf{S}(Y;\mathfrak{T}); \underline{M})$. No hypothesis on $|\Gamma_\sigma|$ being a unit, no effectivity, no adaptedness and no choice of charts is used; the central $-I$ of $\mathrm{SL}_2(\mathbb{Z})$ is included. [formally verified: [GT, thm:mp-comparison]] (for $C^\bullet_{\mathrm{un}}$: every group, every commutative ring R, every $R[\Gamma]$-module, every k, with (AC) over R — implied by (AC) over $\mathbb{Z}$ —, (IO), finite vertex sets and $\dim Y \le 2$ as hypotheses) [formally verified: [GT, thm:mp-comparison-natural]] (naturality in M, for the isomorphism built through a homotopy equivalence with the bar resolution) [formally verified: [GT, thm:mp-comparison-normalised]] (v1.22.0-pre.7: the displayed isomorphism for the normalised $C^\bullet$, under the same hypotheses; on M_7 and $\mathsf{R}_\mathbb{Z}$ with no hypothesis left)
5. *For $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, both triangulations M_7 and $\mathsf{R}_\mathbb{Z}$ of Definition 32.4 satisfy the hypotheses: (G1), (AC) and (SC) by (H4) (X is a contractible simplicial Γ-complex, and a Γ-equivariant subdivision is again one, Remark 17.5(1)), (G3) by the finite quotient, (PD) since X is a 2-dimensional pseudo-manifold with boundary built from triangles, and (IO) by the vertex orbits listed in Definition 32.4, each triangle meeting each orbit at most once. [formally verified: [GT, thm:mp-models-cover]] ((IO), finite vertex sets and $\dim \le 2$ for both, on the descriptors of Definition 32.4's marker) [formally verified: [GT, thm:mp-ac-bridge]] [formally verified: [GT, thm:mp-model7-ac]] [formally verified: [GT, thm:mp-retr]] [formally verified: [GT, thm:mp-rz-ac]] ((AC) for both over every commutative ring: for M_7 from (H4)(ii) through the identification of the ordered with the oriented chains; for $\mathsf{R}_\mathbb{Z}$ (H4)(ii) of the subdivision, by a deformation retraction onto M_7 given by finite tables; so, with (4), the comparison for $C^\bullet_{\mathrm{un}}$ on M_7 and on $\mathsf{R}_\mathbb{Z}$ with no hypothesis left. (G1), (G3) are not restated)*

Proof. (1) Γ acts freely on the first label. A string is in $\mathfrak{F}'_k$ iff $\bigcap g_i \iota(\sigma_i) \neq \emptyset$, iff (multiplying by g_0^{-1}) its normal form satisfies the condition of S_k; distinct normal forms lie in distinct orbits. Finiteness: $\gamma_i \in \mathrm{Ar}(\sigma_{i-1}, \sigma_i)$, a finite set (Remark 32.7(1), (G1), (G3)). (2) By Lemma 32.9 the row augmentations form a map of double complexes $\mathbb{Z}[\mathfrak{F}_{\bullet,\bullet}] \to \mathbb{Z}[Y_\bullet]$ (the target concentrated in $k = 0$; the s-faces of both are those of $Y_\bullet$) that is a quasi-isomorphism on every row, so $\mathrm{Tot} \to C^{\mathrm{ord}}_\bullet(Y)$ is a quasi-isomorphism by Lemma 32.11; under (IO) the ordered chain complex of $Y_\bullet$ is the oriented chain complex $C_\bullet(Y)$ (one generator per simplex, the orientation induced by the order; a stored orientation differs by signs only), and $C_\bullet(Y) \to \mathbb{Z}$ is a quasi-isomorphism by (AC). By Lemma 32.10 the column augmentations form a map $\mathbb{Z}[\mathfrak{F}_{\bullet,\bullet}] \to \mathbb{Z}[\mathfrak{F}'_\bullet]$ (concentrated in $s = 0$; the k-faces of both drop a pair) that is a quasi-isomorphism on every column, so $\mathrm{Tot} \to \mathbb{Z}[\mathfrak{F}'_\bullet]$ is one by Lemma 32.11. The two composites $\mathrm{Tot} \to \mathbb{Z}$ agree (both send $((g,\sigma);v) \in \mathfrak{F}_{0,0}$ to 1 and vanish elsewhere), so $\mathbb{Z}[\mathfrak{F}'_\bullet] \to \mathbb{Z}$ is a quasi-isomorphism. It is free over $\mathbb{Z}[\Gamma]$ by (1). The projection onto the normalised quotient is a quasi-isomorphism [standard: [Wei94, §8.3], the normalised chain complex of a simplicial abelian group; section-level pointer, VERIFY], Γ preserves degeneracy, and the quotient is free on the non-degenerate normal forms. (3)

$\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma\cdot x_s], M) = M$ by $F \mapsto F(x_s)$. For $i \geq 1$, $d_i x_s$ is already in normal form and is $x_{d_i s}$; $d_0 x_s = ((\gamma_1, \sigma_1), (\gamma_1\gamma_2, \sigma_2), \dots) = \gamma_1 \cdot x_{d_0 s}$, so $F(d_0 x_s) = \gamma_1 \cdot f(d_0 s)$; degenerate strings are killed in the normalised quotient. Without the quotient the same computation gives the unnormalised identity. (4) By (2), $H^k(\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(N\mathbb{Z}[\mathfrak{F}'], M)) = \mathrm{Ext}^k_{\mathbb{Z}[\Gamma]}(\mathbb{Z}, M) = H^k(\Gamma; M)$, the definition of group cohomology, independent of the free resolution [standard: [Wei94, Thm. 2.2.6]], natural in M; over R, $R \otimes N\mathbb{Z}[\mathfrak{F}'] \to R$ is a free $R[\Gamma]$-resolution ($N\mathbb{Z}[\mathfrak{F}'] \to \mathbb{Z}$ is a bounded-below exact complex of free abelian groups, hence split over $\mathbb{Z}$), and $\mathrm{Hom}_{R[\Gamma]}(R \otimes N\mathbb{Z}[\mathfrak{F}'], M) = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(N\mathbb{Z}[\mathfrak{F}'], M)$. The same argument on the free resolution $\mathbb{Z}[\mathfrak{F}']$ of (2) gives the statement for $C^\bullet_{\mathrm{un}}$. Nothing in (1)–(3) used an order being invertible, effectivity or adaptedness. (5) (H4)(i),(ii) and Remark 17.5(1); the orbits are read off Definition 7.2 (Lemma 7.3(2)) and, for $\mathsf{R}_\mathbb{Z}$, the rows of Definition 32.4(2): $v_0v_1v_4$, $v_1v_2v_6$, $v_2v_3v_5$, $v_1v_4v_6$, $v_2v_5v_6$, v_1v_4m, v_1mv_6, v_2v_5m', $v_2m'v_6$ each meet the orbits $\{v_0, v_3\}$, $\{v_4, v_5\}$, $\{m, m'\}$ at most once. □

Reading. The resolution direction is the nerve direction k; the triangulation direction s stays finite ($0 \leq s \leq 2$). That is the precise sense in which a finite triangulation accounts for all degrees: $\mathsf{S}_k \neq \emptyset$ for every k (strings $\sigma \xleftarrow{w} \sigma \xleftarrow{w} \cdots$, $w \in \Gamma_\sigma$), and $H^k(\Gamma; \mathbb{F}_2) = \mathbb{F}_2$ for every k is seen by $\mathsf{S}(\mathsf{M}_7)$. Read the other way, row-wise the Γ-cochains of $\mathbb{Z}[\mathfrak{F}]$ compute $\bigoplus_{x \in \Gamma\backslash Y_s} H^k(\Gamma_x; M)$, the E_1 page of the isotropy spectral sequence of Theorem 17.3's proof; the MP set is that spectral sequence's total complex made finite in each bidegree by passing to π_0. Refinement is *not* needed for correctness: the theorem holds on M_7 at every ℓ and over $\mathbb{Z}$; what adaptedness buys is the description $\mathsf{S}_k \cong \bigsqcup \Gamma^k_{v(\tau)}$, i.e. being Moerdijk–Pronk's set in their form (Remark 32.7(3)).

Remark 32.13 (The ℓ-local hybrid is Proposition 31.2). [proved here] [verified computationally]^{r}

1. **What it is.** Over a $\mathbb{Z}_{(\ell)}$-algebra R, the hybrid complex of the ℓ-relevant structure (Definition 32.3) — the column of σ the inflation to Γ_σ of a resolution of R over $R[P_\sigma]$, with Hom into M equal to cochains of P_σ with values in M^{Q_σ} — is an ℓ-local isotropy model in the sense of Definition 31.1 with the column of σ any projective $R[\Gamma_\sigma]$-resolution of R through $R[\Gamma_\sigma/Q_\sigma]$: **the shipped $\mathcal{I}^{(\ell)}$ (`integral_model::LocalModel`) is its periodic-column instance**, and Proposition 31.2(2),(3) is its theorem (a projective resolution; $H^q(\Gamma; M)$ in every degree). No adaptedness is assumed. The bar-column form ([MP] §2.3: the inflated bar resolutions of P_σ, projective because $R[\Gamma_\sigma/Q_\sigma] = R[\Gamma_\sigma]e_\sigma$) satisfies the hypotheses of the same proof, which uses only projective columns with R-linear contracting homotopies; so it is again a projective resolution of R, homotopy equivalent to the periodic-column one over $R[\Gamma]$ [standard: [Wei94, Thm. 2.2.6]], and both compare with S through Theorem 32.12. They are compared *on cohomology*, never on cochains: the cochain-level models are choice-dependent.
2. **The three regimes** (Proposition 31.3, Remark 31.4(2)). $\ell = 3$: one resolved column, at ζ^2 ($P = \mathbb{Z}/3$), with row 0 $\bigoplus_\sigma M^{Q_\sigma}$ — *not* the Bredon value $M^{\langle ST\rangle}$ at the corner — and $H^q(\Gamma; M) = H^q(\mathbb{Z}/3; M^{-I})$ for $q \geq 2$. $\ell = 2$: every simplex keeps the central column $C^\bullet(\{\pm I\}; M^{Q_\sigma})$ and i carries $C^\bullet(\mathbb{Z}/4; M)$, a $\mathbb{Z}/2$-gerbe that no 2-local reduction removes (the class e of Lemma 14.22(1) has order 12). Over $\mathbb{Z}$: every column is the full one; nothing is a unit, and $H^2(\Gamma; \mathbb{Z}) \cong \mathbb{Z}/12$ (Lemma 14.22(1)) takes its 3-part from $\langle ST\rangle$ and its 2-part from $\{\pm I\}$ and $\langle S\rangle$.
3. **Gap G1, closed** (v1.22.0-pre.11; recorded as open until pre.10). In nerve form the ℓ-reduced object is a simplicial set with a *coefficient system* $\sigma \mapsto M^{Q_\sigma}$, not a local system. Pre.10 recorded that its description needs adaptedness for $\mathfrak{T}^{(\ell)}$ and a choice of lifts making the twists ℓ-elements. Neither is needed: Proposition 32.15 shows that the *choice-free* quotient $\mathfrak{D} = \mathfrak{F}'/\!\sim$ of the labelled nerve by the prime-to-ℓ parts at a dominant point is a projective resolution under the domination hypothesis (QD) of Definition 32.14, which bounds the Q's, not the Sylow parts, and holds on M_7 (not adapted) as well as on $\mathsf{R}_\mathbb{Z}$ at $\ell \in \{2, 3\}$. The subgroup form with labels in $P_{v(\tau)}$ does not exist for any choice of lifts (the cone-point holonomy, [MPT] `docs/latex/mp_triangulations.tex` §2.5). The

hybrid double complex stays the normative definition of the ℓ-relevant MP model; the Q-reduced nerve is a second projective resolution, compared with it *on cohomology* (Proposition 32.15(4)).

[verified computationally]^{r} (`OrbifoldMPComparisonValidation`, `OrbifoldMPAllDegreesValidation`): $\mathcal{I}^{(\ell)}$ at $\ell \in \{2,3\}$ has the dimensions of $C^{\bullet}_{\mathcal{I}}$ and of the oracle on $H^0,\dots,H^7$ for $n \le 12$, the strict comparison of Proposition 31.2(4) is an isomorphism on $H^{\le 2}$, and composed with Φ^{MP} (Definition 32.16) on to the nerve for $n \le 3$.

Definition 32.14 (Q-domination)**.** Let Γ, Y satisfy the hypotheses of Theorem 32.12 with every vertex stabiliser Γ_v finite abelian, and let ℓ be a prime. For a simplex x of Y let $Q_x \subseteq \Gamma_x = \bigcap_{w \in x} \Gamma_w$ be the subgroup of elements of order prime to ℓ (a subgroup, Γ_x being abelian; for $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ it is the Q_σ of Definition 31.1). Then $Q_x \subseteq Q_w$ for every vertex w of x, $\gamma Q_x \gamma^{-1} = Q_{\gamma x}$, and $Q_{x \cup x'} = Q_x \cap Q_{x'}$ whenever $x \cup x'$ is a simplex. The triangulation is **Q-dominated at ℓ, (QD)**, if there are a subgroup $Q_0 \subseteq \Gamma$ acting trivially on Y and a Γ-stable set $\mathcal{C}$ of vertices (the *Q-cone vertices*) such that (a) no simplex of Y has two vertices in $\mathcal{C}$; (b) $Q_w = Q_0$ for every vertex $w \notin \mathcal{C}$; (c) $Q_0 \subseteq Q_v$ for every $v \in \mathcal{C}$. The *dominant vertex* $\mathsf{p}(x)$ of a simplex x is its vertex in $\mathcal{C}$ if it has one, otherwise its first vertex for the order of (IO). Under (QD):

1. $Q_w \subseteq Q_{\mathsf{p}(x)}$ for every vertex w of x;
2. $\mathsf{p}(x') = \mathsf{p}(x)$ whenever $\mathsf{p}(x) \in x' \subseteq x$;
3. $\mathsf{p}(\gamma x) = \gamma\,\mathsf{p}(x)$ for $\gamma \in \Gamma$.

For $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ on M_7 and on $\mathsf{R}_{\mathbb{Z}}$: at $\ell = 3$, $Q_0 = \{\pm I\}$ and $\mathcal{C}$ the orbit of i ($Q_i = \langle S\rangle$, $Q = \{\pm I\}$ at every other vertex, ζ^2 included); at $\ell = 2$, $Q_0 = 1$ and $\mathcal{C}$ the orbit of ζ^2 ($Q_{\zeta^2} = \langle (ST)^2\rangle$, $Q = 1$ elsewhere, i included). (QD) is a finite, decidable statement on a descriptor; it is *not* adaptedness (Definition 32.2(1),(2)): M_7 is not adapted, yet (QD) holds on it at $\ell \in \{2,3\}$, since a triangle of M_7 containing both ζ^2 and i contains only one of them in $\mathcal{C}$. At $\ell \ge 5$, (QD) holds on $\mathsf{R}_{\mathbb{Z}}$ ($Q_0 = \{\pm I\}$, $\mathcal{C}$ the two elliptic orbits, which no triangle of $\mathsf{R}_{\mathbb{Z}}$ joins) but not on M_7.

Proof of (1)–(3) and of the $\mathrm{SL}_2(\mathbb{Z})$ *clause.* $Q_x \subseteq Q_w$ because $\Gamma_x \subseteq \Gamma_w$; conjugation carries Γ_x to $\Gamma_{\gamma x}$ and preserves orders; $\Gamma_{x\cup x'} = \Gamma_x \cap \Gamma_{x'}$. (1) If $\mathsf{p}(x) \in \mathcal{C}$, every other vertex w of x is outside $\mathcal{C}$ by (a), so $Q_w = Q_0 \subseteq Q_{\mathsf{p}(x)}$ by (b), (c); otherwise every $Q_w = Q_0$. (2) If $\mathsf{p}(x) \in \mathcal{C}$, it is the unique vertex of x' in $\mathcal{C}$; otherwise x, hence x', has no vertex in $\mathcal{C}$, and the first vertex of x lies in x', so it is the first vertex of x'. (3) $\mathcal{C}$ is Γ-stable and the order of (IO) is Γ-invariant. For $\mathrm{SL}_2(\mathbb{Z})$: the stabilisers are $\langle S\rangle \cong \mathbb{Z}/4$ at the orbit of i, $\langle ST\rangle \cong \mathbb{Z}/6$ at the orbit of ζ^2 and $\{\pm I\}$ at every other vertex (m included), and $\{\pm I\}$ acts trivially ((H4)(iv)); (a) holds by (IO), each triangle meeting each orbit at most once (Theorem 32.12(5)). At $\ell \ge 5$, $Q_x = \Gamma_x$, so every vertex with stabiliser $\ne Q_0$ must lie in $\mathcal{C}$, and $Q_0 \subseteq \{\pm I\}$; on M_7 the triangle $v_1v_4v_6$ then has two vertices in $\mathcal{C}$, on $\mathsf{R}_{\mathbb{Z}}$ no triangle has two non-central vertices (Remark 32.5(2)). □

Proposition 32.15 (The Q-reduced nerve: a choice-free projective resolution)**.** *[proved here] Let Γ, Y, $\mathfrak{T}$ satisfy the hypotheses of Theorem 32.12 with finite abelian vertex stabilisers, let ℓ be a prime with (QD) (Definition 32.14), and let R be a commutative ring with $|Q_v| \in R^\times$ for every vertex v. For $t = ((g_i,\sigma_i))_{i\le k} \in \mathfrak{F}'_k$ (Definition 32.8) put $\tau_t := \bigcap_i g_i\iota(\sigma_i)$, a closed simplex by (SC), and let $t \sim t'$ iff $t' = ((q_ig_i,\sigma_i))_i$ with $q_i \in Q_{\mathsf{p}(\tau_t)}$ ($k+1$ independent elements).*

1. *$\sim$ is an equivalence relation, compatible with the faces, the degeneracies and the Γ-action; $\mathfrak{D} := \mathfrak{F}'/\!\sim$ is a simplicial Γ-set, the stabiliser of the class $[t]$ is $Q_{\mathsf{p}(\tau_t)}$, and $\mathsf{p}(\tau_{t'}) = \mathsf{p}(\tau_t)$ for $t' \sim t$. In the normal form $g_0 = 1$ of Theorem 32.12(1), $\mathsf{S}^{(\ell)}_k := \Gamma\backslash\mathfrak{D}_k$ is the quotient of S_k by $(\sigma_i;\gamma_i) \mapsto (\sigma_i; c_i\gamma_i)$, $c_i \in Q_{\mathsf{p}_{i-1}}$ independent, $\mathsf{p}_{i-1} := (\gamma_1\cdots\gamma_{i-1})^{-1}\mathsf{p}(\tau_s)$; a class is degenerate iff some representative has $\sigma_i = \sigma_{i+1}$ and $\gamma_{i+1} \in Q_{\mathsf{p}_i}$.*
2. *$R[\mathfrak{D}_k] \cong \bigoplus_{[s]\in\mathsf{S}^{(\ell)}_k} R[\Gamma/Q_{\mathsf{p}(\tau_s)}]$ is a projective $R[\Gamma]$-module, and $R[\mathfrak{D}_\bullet] \to R$ and its normalised quotient $NR[\mathfrak{D}_\bullet] \to R$ are resolutions of R by projective $R[\Gamma]$-modules, the latter projective on the non-degenerate classes.*

3. $\mathrm{Hom}_{R[\Gamma]}(NR[\mathfrak{D}_\bullet], M)$ *is, on the nose, the complex* $C^\bullet(\mathsf{S}^{(\ell)}; M^Q)$ *of functions* f *on the non-degenerate classes with* $f(s) \in M^{Q_{\mathsf{p}(\tau_s)}}$ *(*$\mathsf{p}(\tau_s)$ *a vertex of* $\iota(\sigma_0)$*) and the differential of Definition 32.6,* $(df)(s) = \gamma_1 \cdot f(d_0 s) + \sum_{i\geq 1}(-1)^i f(d_i s)$*. Hence, for every* $R[\Gamma]$*-module* M*, naturally in* M*,* $H^k(C^\bullet(\mathsf{S}^{(\ell)}; M^Q)) \cong H^k(\Gamma; M)$ *for every* $k \geq 0$*; likewise for the unnormalised complex.*
4. *The projection* $\mathfrak{F}' \to \mathfrak{D}$ *is a map of projective resolutions over* id_R*, so it induces the identity of* $H^\bullet(\Gamma; M)$ *under (3) and Theorem 32.12(4). The* Q*-reduced nerve, the full MP complex and the hybrid of Remark 32.13 are compared* on cohomology *by the comparison theorem [standard: [Wei94, Thm. 2.2.6]], never on cochains.*
5. *For* $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ *on* M_7 *and on* $\mathsf{R}_{\mathbb{Z}}$*, at* $\ell \in \{2,3\}$*, the hypotheses hold for every* $\mathbb{Z}_{(\ell)}$*-algebra* R *(*$|Q_v| \in \{1,2,4\}$ *at* $\ell = 3$*,* $\in \{1,3\}$ *at* $\ell = 2$*). No adaptedness and no choice of lifts is used.*

Proof. (1) Let $t' = ((q_i g_i, \sigma_i))$ with $q_i \in Q_{\mathsf{p}}$, $\mathsf{p} := \mathsf{p}(\tau_t)$. Each q_i fixes p, so $\mathsf{p} \in \tau_{t'}$. If $\mathsf{p} \in \mathcal{C}$, it is the vertex of $\tau_{t'}$ in $\mathcal{C}$, so $\mathsf{p}(\tau_{t'}) = \mathsf{p}$; otherwise $Q_{\mathsf{p}} = Q_0$ acts trivially on Y and $\tau_{t'} = \tau_t$. Hence the relation is symmetric and transitive (the group Q_{p}^{k+1} acting). A face d_j enlarges τ: $\mathsf{p}(\tau_t)$ is a vertex of $\tau_{d_j t}$, so $Q_{\mathsf{p}(\tau_t)} \subseteq Q_{\mathsf{p}(\tau_{d_j t})}$ by Definition 32.14(1), and $t \sim t'$ implies $d_j t \sim d_j t'$; degeneracies keep τ. For $\gamma \in \Gamma$, $\gamma(q_i g_i) = (\gamma q_i \gamma^{-1})(\gamma g_i)$ with $\gamma Q_{\mathsf{p}} \gamma^{-1} = Q_{\gamma\mathsf{p}} = Q_{\mathsf{p}(\tau_{\gamma t})}$ (Definition 32.14(3)). As Γ acts freely on the first label of $\mathfrak{F}'$, $\gamma t \sim t$ iff $\gamma g_i = q_i g_i$ for all i, i.e. iff $\gamma \in Q_{\mathsf{p}(\tau_t)}$. In normal form, t' re-normalised by q_0^{-1} has $\gamma_i' = g_{i-1}^{-1}(q_{i-1}^{-1} q_i) g_{i-1}\, \gamma_i$, and $g_{i-1}^{-1} Q_{\mathsf{p}} g_{i-1} = Q_{\mathsf{p}_{i-1}}$ (Q_{p} abelian, the $q_{i-1}^{-1} q_i$ independent). A class is degenerate iff it is $[s_i u]$, i.e. iff some representative repeats a pair up to Q_{p}, which in normal form is the stated condition.

(2) The Γ-orbit of $[t]$ is $\Gamma/Q_{\mathsf{p}(\tau_t)}$ by (1), and $R[\Gamma/Q] = R[\Gamma] e_Q$, $e_Q := |Q|^{-1} \sum_{q\in Q} q$, is a direct summand of $R[\Gamma]$. The degenerate classes form a Γ-stable subset, so $NR[\mathfrak{D}_k]$ is the permutation module on the non-degenerate classes, projective likewise, and the projection $R[\mathfrak{D}_\bullet] \to NR[\mathfrak{D}_\bullet]$ is a quasi-isomorphism [standard: [Wei94, §8.3], as in the proof of Theorem 32.12(2), VERIFY]. Exactness of $R[\mathfrak{D}_\bullet] \to R$ is the argument of Theorem 32.12(2) with one quotient in each lemma. Let $\mathfrak{F}^Q_{k,s} := \mathfrak{F}_{k,s}/\!\equiv$, where $((g_i, \sigma_i); x) \equiv ((q_i g_i, \sigma_i); x)$ for $q_i \in Q_x$ — the relation by the *simplex* x, not by the dominant point. It is well defined ($Q_x \subseteq \Gamma_x$ fixes x, so $x \subseteq q_i g_i \iota(\sigma_i)$); the k-faces and degeneracies keep x, the s-faces $x \mapsto \partial_j x$ coarsen it ($Q_x \subseteq Q_{\partial_j x}$), and Γ acts as in (1); so $\mathfrak{F}^Q$ is a bisimplicial Γ-set (semi-simplicial in s) with $\mathfrak{F}^Q_{k,s} = \bigsqcup_{x\in Y_s} (O(x)/Q_x)^{k+1}$, Q_x acting on $O(x)$ by $(g,\sigma) \mapsto (qg, \sigma)$. *Rows* (fixed s): $O(x)/Q_x \neq \emptyset$ by (PD), and $R[(O(x)/Q_x)^{\bullet+1}] \to R$ is contracted by the extra degeneracy with a fixed class, as in Lemma 32.9; so $\mathrm{Tot}\, R[\mathfrak{F}^Q] \to R[Y_\bullet]$ is a quasi-isomorphism by Lemma 32.11, and $R[Y_\bullet] \to R$ is one by (IO) and (AC) (over $\mathbb{Z}$, hence over R: a bounded-below exact complex of free abelian groups is split). *Columns* (fixed k): the augmentation $\mathfrak{F}^Q_{k,s} \to \mathfrak{D}_k$ (in $s = 0$), $((g_i, \sigma_i); x) \mapsto [((g_i, \sigma_i))]$, is well defined, since for a vertex w of $x \subseteq \tau_t$, $Q_x \subseteq Q_w \subseteq Q_{\mathsf{p}(\tau_t)}$ (Definition 32.14(1)), and it commutes with the k-faces. The column splits over $[t] \in \mathfrak{D}_k$; the summand over $[t]$ is the augmented ordered chain complex of the Δ-set $Z_{[t]} := \{(t'; x) : t' \sim t,\ x \subseteq \tau_{t'}\}/\!\equiv$. Put $\mathsf{p} := \mathsf{p}(\tau_t)$, which is $\mathsf{p}(\tau_{t'})$ for every $t' \sim t$ by (1). Define $h([t]) := [(t; \mathsf{p})]$ and $h[(t'; x)] := (-1)^j [(t'; \mathsf{p} \cup x)]$ if $\mathsf{p} \notin x$, j the position of p in the ordered simplex $\mathsf{p} \cup x$, and 0 if $\mathsf{p} \in x$. This is well defined: $\mathsf{p} \cup x \subseteq \tau_{t'}$ is a simplex, its gluing group is $Q_{\mathsf{p}\cup x} = Q_{\mathsf{p}} \cap Q_x = Q_x$ (as $Q_x \subseteq Q_{\mathsf{p}}$), and the classes $[(t'; \mathsf{p})]$, $t' \sim t$, coincide because t' and t differ by elements of Q_{p}. As h changes only the simplex coordinate, $dh + hd = \mathrm{id}$ is the identity of the cone on a vertex of an ordered simplex, checked term by term (the faces of $\mathsf{p} \cup x$ other than x pair off with those of $h(\partial x)$, with signs $(-1)^j(-1)^i$ and $(-1)^i(-1)^{j-1}$ for $i < j$, $(-1)^j(-1)^{i+1}$ and $(-1)^i(-1)^j$ for $i \geq j$), including the augmentation degree; this is Lemma 32.10 with the cone point p in place of the first vertex. So every column is acyclic in positive degrees with $H_0 = R[\mathfrak{D}_k]$, and $\mathrm{Tot}\, R[\mathfrak{F}^Q] \to R[\mathfrak{D}_\bullet]$ is a quasi-isomorphism by Lemma 32.11. The two composites $\mathrm{Tot}\, R[\mathfrak{F}^Q] \to R$ agree (both send $((g,\sigma); v) \in \mathfrak{F}^Q_{0,0}$ to 1), so $R[\mathfrak{D}_\bullet] \to R$ is a quasi-isomorphism. (Adaptedness was not used; (QD) entered twice: $\mathsf{p}(\tau_{t'}) = \mathsf{p}(\tau_t)$ on a class, and

$Q_x \subseteq Q_{\mathsf{p}(\tau_t)}$.)

(3) $\mathrm{Hom}_{R[\Gamma]}(R[\Gamma/Q_{\mathsf{p}}]\cdot[x_s], M) = M^{Q_{\mathsf{p}}}$ by $F \mapsto F([x_s])$, x_s a normal form of s; two normal forms of one class differ by an element of Q_{p}, which fixes $F([x_s])$, so $f(s) := F([x_s])$ is well defined. As in Theorem 32.12(3), $d_i x_s = x_{d_i s}$ for $i \geq 1$ and $d_0 x_s = \gamma_1 \cdot x_{d_0 s}$, which gives the differential; $\mathsf{p}(\tau_s) \in \tau_s \subseteq \iota(\sigma_0)$. By (2) the cohomology is $\mathrm{Ext}^k_{R[\Gamma]}(R, M)$ [standard: [Wei94, Thm. 2.2.6]], which the free resolution $R \otimes \mathbb{Z}[\mathfrak{F}'_\bullet]$ identifies with $H^k(\Gamma; M)$ (Theorem 32.12(4)), naturally in M. Without the normalisation the same computation gives the unnormalised statement. (4) $R \otimes \mathbb{Z}[\mathfrak{F}'_\bullet] \to R[\mathfrak{D}_\bullet]$ is Γ-equivariant, commutes with faces and degeneracies (by (1)) and with the augmentations; two maps of projective resolutions over id_R are homotopic, so the induced map on Ext is the canonical identification. (5) Definition 32.14 and Theorem 32.12(5); the orders of the Q_v are read off its $\mathrm{SL}_2(\mathbb{Z})$ clause. □

[verified computationally][a] (not a registered validation): `docs/implementation/slot22/fmptri/fmptri_spotch` (the F-MPT referee's spot-check, check D) builds the choice-free $\mathfrak{D}$ on M_7 and on $\mathsf{R}_{\mathbb{Z}}$ and computes $\dim H^0, H^1, H^2$ of $C^\bullet(\mathsf{S}^{(\ell)}; M^Q)$, $M = \mathrm{Sym}^n \mathbb{F}_\ell^2$: equal to the tree oracle of `mpcore.py` on every cell, $\ell = 3$, $n \leq 4$ and $\ell = 2$, $n \leq 3$, on both triangulations, with $d \circ d = 0$, the d_0-twist representative-independent on all of degree 1 and every value in its fibre asserted; $\left|\mathsf{S}^{(\ell),\mathrm{nd}}_k\right|$, $k \leq 3$, is $5, 52, 510, 5386$ ($\ell = 3$) and $5, 63, 621, 5847$ ($\ell = 2$) on M_7, $7, 70, 588, 5656$ and $7, 101, 1007, 9053$ on $\mathsf{R}_{\mathbb{Z}}$. On $\mathsf{R}_{\mathbb{Z}}$ these classes are the quotient form of [MPT] (`docs/latex/mp_triangulations.tex` §2.5, in the μ-coordinates of its Proposition `prop:muform`), whose tables agree. Re-run of 2026-10-10: `FMPTRI_SPOTCHECK failing=0 verdict=OK`.

32.4 All degrees

Definition 32.16 (The comparison lift Φ^{MP})**.** Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $\mathfrak{T} \in \{\mathsf{M}_7, \mathsf{R}_{\mathbb{Z}}\}$, and $\mathcal{I}_\bullet$ the isotropy-resolution model of Definition 23.4 on X with the representatives of M_7 (the target for both triangulations). $\Phi^{\mathsf{MP}} \colon N\mathbb{Z}[\mathfrak{F}'_\bullet] \to \mathcal{I}_\bullet$ is the $\mathbb{Z}[\Gamma]$-linear map defined on the free generators e_s ($s \in \mathsf{S}^{\mathrm{nd}}_k$, the normal form x_s) by

$$\Phi_0(e_s) := h \cdot e_{0,v,0}, \qquad \Phi_k(e_s) := c\big(\Phi_{k-1}(d\, e_s)\big) \quad (k \geq 1),$$

where $h \cdot v$ is the first vertex of $\iota(\sigma_0)$ lying in the orbit of a representative $v \in \Sigma_0$ of M_7 (h any element with that property; the construction depends on the choices of the h and of c, and Proposition 32.17(1) shows that the homotopy class does not; the code's choice is recorded by `PhiLift`), $e_{0,v,0}$ the generator $1 \otimes 1$ of $\mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_v]} F_0^v \subseteq G_{0,0}$, $d\, e_s = \gamma_1 \cdot e_{d_0 s} + \sum_{i \geq 1} (-1)^i e_{d_i s}$, and c a $\mathbb{Z}$-linear contracting homotopy of the augmented $\mathcal{I}_\bullet \to \mathbb{Z}$ (it exists: Proposition 23.5(2), split over $\mathbb{Z}$). The code builds it to degree 3 (`orbifold_mp::PhiLift`, $c =$ `PeriodicityClass::fill`).

Proposition 32.17 (All degrees from a finite triangulation: what holds, and its exact scope)**.** *Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $\mathfrak{T} \in \{\mathsf{M}_7, \mathsf{R}_{\mathbb{Z}}\}$, R a commutative ring and V an $R[\Gamma]$-module.*

1. *[proved here] Φ^{MP} is a $\mathbb{Z}[\Gamma]$-chain map over the augmentations, unique up to homotopy, and $(\Phi^{\mathsf{MP}})^* \colon C^\bullet_{\mathcal{I}}(V) \to C^\bullet(\mathsf{S}; \underline{V})$ is a quasi-isomorphism in **every** degree over every R; both sides are $H^\bullet(\Gamma; V)$ (Theorem 32.12, Proposition 23.5(3)).*
2. *[proved here] **($D = 3$, the degree bound of the notation paragraph.)** For $q \geq 2 = \mathrm{vcd}(\Gamma) + 1$, cup product with the class $e \in H^2(\Gamma; \mathbb{Z})$ of Lemma 14.22(1) is an isomorphism $H^q(\Gamma; V) \to H^{q+2}(\Gamma; V)$ (Lemma 14.22(2)). Hence $H^{\leq 3}(\mathsf{S}; \underline{V})$ determines every degree: $H^{2j} \cong H^2$, $H^{2j+1} \cong H^3$ for $j \geq 1$, transported through (1).*
3. *[proved here] **(The torsion bound; why $\mathbb{Z}/2^k$, $k \leq 3$, and $\mathbb{F}_3$ suffice in degrees ≥ 2.)** $24 \cdot H^q(\Gamma; V_R) = 0$ for $q \geq 2$, every R and n (Claim 16.2(4)), and for $R = \mathbb{Z}$ there is no ℓ-torsion for $\ell \geq 5$ in degrees ≥ 2 (Proposition 23.16). So for $q \geq 2$ and $k \geq v_\ell(24)$*

($k \geq 3$ at $\ell = 2$, $k \geq 1$ at $\ell = 3$), $H^q(\Gamma; \mathrm{Sym}^n(\mathbb{Z}/\ell^k)^2) \cong H^q(\Gamma; V)_{(\ell)} \oplus H^{q+1}(\Gamma; V)_{(\ell)}$ (the universal coefficient sequence of the finite free complex $C^\bullet_{\mathcal{I}}(\mathbb{Z})$, both terms finite of exponent dividing $\ell^{v_\ell(24)}$), independently of k: the cells $k \leq 3$ at $\ell = 2$ and $k = 1$ at $\ell = 3$ are complete in degrees ≥ 2, not a choice.

4. *[verified computationally]^r* ***(What is certified.)*** *Φ^{MP} is built to degree 3 on both triangulations (55 816 generators on M_7, 59 448 on $\mathsf{R}_{\mathbb{Z}}$; 20 distinct group elements, coefficients ≤ 2 in absolute value) and certified exactly: $d\Phi = \Phi d$ over $\mathbb{Z}[\Gamma]$ and $\varepsilon \circ \Phi_0 = \varepsilon$, a* coefficient-free *certificate serving every n and every ring. On $C^\bullet_{\mathcal{I}}$, the cup product with e is certified (`PeriodicityClass` on $q_{\max} = 7$) and is an isomorphism $H^q \to H^{q+2}$, $q = 2, \dots, 5$, over $\mathbb{F}_\ell$, $\ell \in \{2, 3, 5, 7\}$, $n = 0, \dots, 12$ (52 rows). $(\Phi^{\mathsf{MP}})^*$ is measured an isomorphism on $H^{\leq 1}$ for $n \leq 12$, on H^2 for $n \leq 3$, and on H^3 for $\ell \in \{2, 3\}$, $n \in \{0, 2\}$. The certificate line `MP_ALL_DEGREES model=<M7|R_Z> ring=F_ℓ nerve_to=2 periodic_on=C_I from=2 via=thm:mpcomparison certified=4/4 scope=bulk verdict=OK` records these, and `certified=4/4` counts $n = 0, \dots, 3$.*
5. ***(Scope.)*** *The statement is about $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ and the* ***bulk*** *only: not the boundary (where 2-periodicity starts in degree 1 and is a separate statement, Remark 14.23(3)(a)), not the pair, and not a finite-index subgroup $B \leq \Gamma$ (which would need its own periodicity clause). On the MP complex the degrees above `nerve_to` are* transported *by (1) and (2), not computed.*

Proof. (1) The construction is well posed: $\varepsilon(\Phi_0(e_s)) = 1$, and by induction $\Phi_{k-1}(d\, e_s)$ is a cycle of $\mathcal{I}_{k-1}$ (augmentation $1 - 1 = 0$ at $k = 1$), so c fills it; extended Γ-linearly from the free generators it is a chain map over $\mathrm{id}_{\mathbb{Z}}$. Both $N\mathbb{Z}[\mathfrak{F}']$ (Theorem 32.12(2)) and $\mathcal{I}$ (Proposition 23.5(2)) are free resolutions of $\mathbb{Z}$, so a chain map over $\mathrm{id}_{\mathbb{Z}}$ is a homotopy equivalence, unique up to homotopy [standard: [Wei94, Thm. 2.2.6]], and $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, V)$ of it is a quasi-isomorphism. (2) is Lemma 14.22(2), which holds for every $\mathbb{Z}[\Gamma]$-module V, read through (1). (3) $C^\bullet_{\mathcal{I}}(\mathbb{Z})$ is a complex of free abelian groups of finite rank (Proposition 23.5(2),(3)) with $C^\bullet_{\mathcal{I}}(\mathbb{Z}) \otimes \mathbb{Z}/\ell^k = C^\bullet_{\mathcal{I}}(\mathbb{Z}/\ell^k)$, so the universal coefficient sequence $0 \to H^q(\mathbb{Z}) \otimes \mathbb{Z}/\ell^k \to H^q(\mathbb{Z}/\ell^k) \to \mathrm{Tor}(H^{q+1}(\mathbb{Z}), \mathbb{Z}/\ell^k) \to 0$ holds and splits. For $q \geq 2$ both $H^q(\mathbb{Z})$ and $H^{q+1}(\mathbb{Z})$ are finite and killed by 24; for a finite abelian group G with $\ell^a G_{(\ell)} = 0$ and $k \geq a$, $G \otimes \mathbb{Z}/\ell^k = G_{(\ell)} = \mathrm{Tor}(G, \mathbb{Z}/\ell^k)$. (4) is the record of the validations named in Remark 32.25. (5) is a statement of scope. □

Which part is load-bearing (the F-MP2 audit's finding 6). The all-degrees statement (1)–(2) rests on Theorem 32.12, the existence of Φ^{MP} and Lemma 14.22(2); the certificate of (4) shows that the *built* Φ^{MP} is such a lift. The measured isomorphisms on $H^{\leq 3}$ are consistency checks of the implementation (ρ_n, layouts, the length machinery) against the frozen oracle; they are not what makes the statement true.

32.5 Morphisms, naturality, and the maps that are needed

Definition 32.18 (Morphisms of MP data; the label factor)**.** Let $(\Gamma, Y, \mathfrak{T})$ and $(\Gamma', Y', \mathfrak{T}')$ satisfy the hypotheses of Theorem 32.12.

1. **(Simplicial morphism.)** A morphism (φ, f) is a homomorphism $\varphi\colon \Gamma \to \Gamma'$ and a φ-equivariant simplicial map $f\colon Y \to Y'$, recorded on the descriptors by a **label table** $\sigma \mapsto (\mathsf{h}_\sigma, \sigma')$, $\sigma \in \Sigma_{\max}$, $\mathsf{h}_\sigma \in \Gamma'$, $\sigma' \in \Sigma'_{\max}$, with $f(\iota(\sigma)) \subseteq \mathsf{h}_\sigma\, \iota'(\sigma')$, which is all that the induced maps of (3) use.
2. **(Refinement.)** For a Γ-equivariant subdivision $\tilde{Y}$ of Y (not a simplicial map $\tilde{Y} \to Y$): $\varphi = \mathrm{id}$ and a label table $\tilde{\sigma} \mapsto (\mathsf{h}_{\tilde{\sigma}}, \sigma)$ with $\tilde{\iota}(\tilde{\sigma}) \subseteq \mathsf{h}_{\tilde{\sigma}}\, \iota(\sigma)$ — the geometric simplex is unique, the label only up to right multiplication by Γ_σ ($= \{\pm I\}$ for a triangle of X under (IO), the setwise stabiliser being the pointwise one). The table need not come from a simplicial map; it is certified equally by a Γ-equivariant map of vertex sets sending the vertices of $\tilde{\iota}(\tilde{\sigma})$ into $\mathsf{h}_{\tilde{\sigma}}\, \iota(\sigma)$ (a common vertex then goes to a common vertex), which is how the Lean

marker below certifies $\mathsf{R}_{\mathbb{Z}} \to \mathsf{M}_7$ (v1.22.0-pre.7). [formally verified: [GT, `def:mp-morph`]] [formally verified: [GT, `def:mp-orbmap`]] (for (1) and (2): a morphism as a label table carrying non-empty intersections to non-empty ones, over any group homomorphism φ, given in particular by an equivariant vertex map with the containments; on descriptors, from decided orbit and label tables; the simplicial map f and the geometric containment in $|Y|$ are not formalised, only what the induced maps use)

3. **(The induced maps.)** In both cases

$$u\colon \mathfrak{F}'_k \to \mathfrak{F}''_k, \qquad ((g_i,\sigma_i)) \mapsto ((\varphi(g_i)\,\mathsf{h}_{\sigma_i},\sigma'_i)),$$

and on the quotients $u\colon (\sigma_i;\gamma_i) \mapsto (\sigma'_i;\ \mathsf{h}^{-1}_{\sigma_{i-1}}\varphi(\gamma_i)\,\mathsf{h}_{\sigma_i})$, a map of simplicial sets $\mathsf{S}(Y) \to \mathsf{S}(Y')$, covered by the morphism of local systems $\mathsf{h}_\sigma\colon \underline{M'}_{u(\sigma)} = M' \to M' = \underline{\varphi^*M'}_\sigma$. The **induced cochain map** is

$$(u^*f)(s) := \mathsf{h}_{\sigma_0}\cdot f(u(s)) \qquad (s=(\sigma_i;\gamma_i)\in \mathsf{S}^{\mathrm{nd}}_k;\ f:=0 \text{ on degenerate images}),$$

$C^\bullet(\mathsf{S}(Y');\underline{M'}) \to C^\bullet(\mathsf{S}(Y);\underline{\varphi^*M'})$. The factor h_{σ_0} — the **label factor** — is part of the map for every morphism: refinement, the forget-coset map of Proposition 32.22, the boundary inclusion of Proposition 32.21 and the g-twist alike (for the g-twist the label factor is $\rho_n(g)$ on a conj_g-twisted chain map of one $\mathfrak{F}'$, Remark 32.23, not a morphism of MP data; v1.22.0-pre.8). The same formula on all strings $s\in\mathsf{S}_k$ defines $u^*\colon C^\bullet_{\mathrm{un}}(\mathsf{S}(Y');\underline{M'}) \to C^\bullet_{\mathrm{un}}(\mathsf{S}(Y);\underline{\varphi^*M'})$, of which the map above is the restriction (u commutes with the degeneracies; v1.22.0-pre.7). [formally verified: [GT, `lem:mp-morph-simplicial`]] [formally verified: [GT, `def:mp-pull`]] (u on $\mathfrak{F}'$, equivariant and commuting with the faces, $u(x_s)=\mathsf{h}_{\sigma_0}\cdot x_{u(s)}$, the γ-coordinates of $u(s)$; u^* with the label factor on $C^\bullet_{\mathrm{un}}$, equal to $F\mapsto F\circ u$ on the equivariant cochains; not the normalised restriction)

Theorem 32.19 (Naturality and refinement invariance)**.** *[proved here] In the setting of Definition 32.18:*

1. *u is a φ-equivariant map of simplicial sets commuting with the augmentations to $\mathbb{Z}$; u^* is the map $F\mapsto F\circ\mathbb{Z}[u]$ of $\mathrm{Hom}_{\mathbb{Z}[\Gamma']}(N\mathbb{Z}[\mathfrak{F}''],M') \to \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(N\mathbb{Z}[\mathfrak{F}'],\varphi^*M')$ read on the normal forms, hence a cochain map; the assignment is compatible with composition when labels compose by $\mathsf{h}''_\sigma = \psi(\mathsf{h}_\sigma)\,\mathsf{h}'_{\sigma'}$. The same holds for the unnormalised complexes: u^* of Definition 32.18(3) on $C^\bullet_{\mathrm{un}}$ is $F\mapsto F\circ\mathbb{Z}[u]$ on $\mathrm{Hom}_{\mathbb{Z}[\Gamma']}(\mathbb{Z}[\mathfrak{F}''],M')$, a cochain map preserving the normalised subcomplexes (v1.22.0-pre.7). [formally verified: [GT, `thm:mp-pull-cochain`]] [formally verified: [GT, `lem:mp-pull-comp`]] [formally verified: [GT, `def:mp-zu`]] (the unnormalised clause: $\mathbb{Z}[u]$, over every commutative ring, a Γ-chain map over the augmentations, u^* precomposition with it, a cochain map for every representation, and composition; the normalised u^* is not stated in Lean)*
2. *The isomorphism of Theorem 32.12(4) is natural: under it, $H^\bullet(u^*)$ is the restriction $\varphi^*\colon H^\bullet(\Gamma';M') \to H^\bullet(\Gamma;\varphi^*M')$; likewise for $C^\bullet_{\mathrm{un}}$, and the inclusions $C^\bullet \subseteq C^\bullet_{\mathrm{un}}$ are quasi-isomorphisms commuting with u^* and inducing the identity of $H^\bullet(\Gamma;M)$, resp. $H^\bullet(\Gamma';M')$ (v1.22.0-pre.7). [formally verified: [GT, `thm:mp-pull-natural`]] [formally verified: [GT, `def:mp-barmap`]] [formally verified: [GT, `thm:mp-homotopy-exact`]] (for $C^\bullet_{\mathrm{un}}$, every group homomorphism φ, every commutative ring, every representation M', every degree, with (AC) over the ring, (IO), finiteness and $\dim\le 2$ for both data: under the isomorphisms of Theorem 32.12(4)'s naturality marker, $H^\bullet(u^*)$ is `mathlib`'s `groupCohomology.map` $\varphi(\mathrm{id})$; the transfer to $C^\bullet$ is by the normalised markers of Definition 32.6 and Theorem 32.12(4) on cohomology, not stated for the normalised u^*)*
3. ***(Refinement invariance.)** For a refinement, $u^*\colon C^\bullet(\mathsf{S}(Y);\underline{M}) \to C^\bullet(\mathsf{S}(\tilde{Y});\underline{M})$ is a quasi-isomorphism in every degree, inducing the identity of $H^\bullet(\Gamma;M)$, and independent of the label choices on cohomology (different labels give chain-homotopic maps); likewise on $C^\bullet_{\mathrm{un}}$. [formally verified: [GT, `thm:mp-refine-invariance`]] [formally verified:*

[GT, `thm:mp-label-indep`]] (for $C^\bullet_{\mathrm{un}}$ and $\varphi = \mathrm{id}$: a quasi-isomorphism in every degree, the identity under the comparison isomorphisms; label independence on cohomology *only — the chain homotopy between the two label choices is not formalised)*

4. ***(The label factor cannot be dropped.)*** *$f \mapsto f \circ u$ is a cochain map when all labels act on M' by one and the same central automorphism (all by $+1$, all by -1, or all in $\ker \rho'$, ρ' the action of Γ' on M'), but not in general: with mixed labels $\pm I$ it fails already at odd n, and with a constant non-central label h it would need γ and $h^{-1}\gamma h$ to act alike on M' for every arrow γ (v1.22.0-pre.6, referee F-MP2 F1: pre.5's sufficient condition "every label acts centrally, e.g. by ± 1" was false). On $\mathsf{R}_\mathbb{Z} \to \mathsf{M}_7$ every label is $\pm I$, the factor is $(\pm 1)^n$ per simplex, and it is visible for mixed labels at odd n.* [formally verified: *[GT, `thm:mp-untwisted`]] (the defect formula $d(f\circ u)(s) - (df)(u(s)) = (\varphi(\gamma_1) - \mathsf{h}_{\sigma_0}^{-1}\varphi(\gamma_1)\mathsf{h}_{\sigma_1}) \cdot f(u(d_0 s))$, the sufficient condition of one central automorphism, and the failure as soon as one string has a vector on which the two act differently)* [formally verified: *[GT, `thm:mp-rz-untwisted`]] (on $\mathsf{R}_\mathbb{Z} \to \mathsf{M}_7$: labels all I give a cochain map for every representation; mixed labels I, $-I$ do not, at $n = 1$ over $\mathbb{F}_3$)*

Proof. (1) $f(g\iota(\sigma)) = \varphi(g) f(\iota(\sigma)) \subseteq \varphi(g)\mathsf{h}_\sigma \iota'(\sigma')$ (for a refinement, $g\tilde{\iota}(\tilde{\sigma}) \subseteq g\mathsf{h}_{\tilde{\sigma}}\iota(\sigma)$ as subsets of $\left|\tilde{Y}\right| = |Y|$), so a non-empty intersection maps into a non-empty one — and this is all that the rest of the proof uses of f, resp. of the subdivision: any label table carrying non-empty intersections to non-empty ones gives the same u and u^* (v1.22.0-pre.7); dropping and repeating pairs commute with u; $u(\gamma \cdot x) = \varphi(\gamma) \cdot u(x)$. On the normal form, $u(x_s) = ((\mathsf{h}_{\sigma_0}, \sigma'_0), (\varphi(\gamma_1)\mathsf{h}_{\sigma_1}, \sigma'_1), \dots) = \mathsf{h}_{\sigma_0} \cdot x_{u(s)}$, since $\mathsf{h}_{\sigma_0}^{-1}\varphi(\gamma_1\gamma_2)\mathsf{h}_{\sigma_2} = (\mathsf{h}_{\sigma_0}^{-1}\varphi(\gamma_1)\mathsf{h}_{\sigma_1})(\mathsf{h}_{\sigma_1}^{-1}\varphi(\gamma_2)\mathsf{h}_{\sigma_2})$; so $(F \circ u)(x_s) = \mathsf{h}_{\sigma_0} \cdot F(x_{u(s)})$, which is the displayed formula (Theorem 32.12(3)); it is a cochain map because $\mathbb{Z}[u]$ is a chain map. The local systems are compatible with the arrows: $\mathsf{h}_{\sigma_0}(\mathsf{h}_{\sigma_0}^{-1}\varphi(\gamma_1)\mathsf{h}_{\sigma_1}) = \varphi(\gamma_1)\mathsf{h}_{\sigma_1}$. Composition: substitute. Without the normalised quotient the same computation gives the unnormalised clause; u commutes with the degeneracies, so $\mathbb{Z}[u]$ descends to $N\mathbb{Z}[\mathfrak{F}']$ and u^* preserves the functions vanishing on degenerate strings. (2) $N\mathbb{Z}[\mathfrak{F}'] \to \mathbb{Z}$ is a free $\mathbb{Z}[\Gamma]$-resolution and $\varphi^* N\mathbb{Z}[\mathfrak{F}''] \to \mathbb{Z}$ is acyclic, so $\mathbb{Z}[u]$ is *the* lift of $\mathrm{id}_\mathbb{Z}$ up to homotopy [standard: [Wei94, Thm. 2.2.6]], and $F \mapsto F \circ \mathbb{Z}[u]$ computes the restriction along φ on Ext [standard: [Bro82, Ch. III.8], the map (φ, id) of pairs (group, module); chapter-level pointer, VERIFY]. The same argument on the free resolutions $\mathbb{Z}[\mathfrak{F}']$, $\varphi^*\mathbb{Z}[\mathfrak{F}'']$ gives the unnormalised statement; the projection $\pi\colon \mathbb{Z}[\mathfrak{F}'] \to N\mathbb{Z}[\mathfrak{F}']$ is a map of free resolutions over $\mathrm{id}_\mathbb{Z}$, hence a homotopy equivalence (the same theorem), so the inclusion $C^\bullet \subseteq C^\bullet_{\mathrm{un}}$, which is $\mathrm{Hom}(\pi, M)$, is one inducing the identity of $H^\bullet(\Gamma; M)$, and $\pi \circ \mathbb{Z}[u] = N\mathbb{Z}[u] \circ \pi$ makes it commute with u^*. (3) With $\varphi = \mathrm{id}$, (2) gives $\mathrm{id}^* = \mathrm{id}$ on $H^\bullet(\Gamma; M)$, both sides being $H^\bullet(\Gamma; M)$ by Theorem 32.12. Two label tables give two Γ-maps $N\mathbb{Z}[\mathfrak{F}'_{\tilde{Y}}] \to N\mathbb{Z}[\mathfrak{F}'_Y]$ over $\mathrm{id}_\mathbb{Z}$, chain-homotopic by the same comparison theorem; likewise $\mathbb{Z}[\mathfrak{F}'_{\tilde{Y}}] \to \mathbb{Z}[\mathfrak{F}'_Y]$ for $C^\bullet_{\mathrm{un}}$. (4) Without the factor, $d(f\circ u)(s) - (df)(u(s)) = \gamma_1 \cdot f(u(d_0 s)) - (\mathsf{h}_{\sigma_0}^{-1}\gamma_1\mathsf{h}_{\sigma_1}) \cdot f(d_0 u(s))$, which vanishes for all f iff γ_1 and $\mathsf{h}_{\sigma_0}^{-1}\gamma_1\mathsf{h}_{\sigma_1}$ act alike on M' — for labels $\pm I$ iff $\rho'(\mathsf{h}_{\sigma_0}) = \rho'(\mathsf{h}_{\sigma_1})$ on every arrow, i.e. (Y being connected by (AC)) iff the labels are constant or n is even; with the factor, $\mathsf{h}_{\sigma_0}\mathsf{h}_{\sigma_0}^{-1}\gamma_1\mathsf{h}_{\sigma_1} = \gamma_1\mathsf{h}_{\sigma_1}$, which is the factor of $d_0 s$. For labels $\pm I$, $\rho_n(\pm I) = (\pm 1)^n$. □

[verified computationally]^{r} for (3), (4) on $\mathsf{R}_\mathbb{Z} \to \mathsf{M}_7$ (`OrbifoldMPRefinementValidation`): the seven labels (each $\mathsf{R}_\mathbb{Z}$ triangle in one M_7 triangle; the label coset $\{h, h\cdot(-I)\}$ of each simplex, both members passing containment; $h = I$ by the subdivision record); every string of $\mathsf{S}_{\le 3}(\mathsf{R}_\mathbb{Z})$ maps into $\mathsf{S}(\mathsf{M}_7)$ (1044 non-degenerate strings have a degenerate image); u^* a cochain map on 27 products over $\mathbb{F}_2$, $\mathbb{F}_3$, $\mathbb{Z}/4$; with $h\cdot(-I)$ on one simplex only, at $\mathbb{F}_3$, $n = 1$, the untwisted $f\circ u$ is *not* a cochain map (`REFINE_TWIST`); u^* an isomorphism on $H^{\le 2}$ at $\ell \in \{2, 3, 5\}$, $n \in \{0, 2, 4\}$, label-independent; a wrong label $h \cdot T$ fails containment and sends 7146 strings outside $\mathsf{S}(\mathsf{M}_7)$. [formally verified: [GT, `def:mp-rz-refine`]] (the seven labels, all $\mathsf{h} = I$, with the orbit table of the collapse of the new vertices onto vertices of M_7 recorded by `RZTau.cTab`, and the coset member $\mathsf{h}\cdot(-I)$ on one triangle; kernel-decided) [formally verified: [GT, `thm:mp-rz-refine`]]

(u^* a quasi-isomorphism on $C^\bullet_{\mathrm{un}}$ for every commutative ring and every representation, in every degree, the identity under the comparisons; both label choices alike on cohomology) [formally verified: [GT, `thm:mp-rz-untwisted`]] (the untwisted failure at $\mathbb{F}_3$, $n=1$, on one string)

Proposition 32.20 (The comparison with the shipped Bredon complex)**.** *Let* $\Gamma=\mathrm{SL}_2(\mathbb{Z})$, $\mathfrak{T}\in\{\mathsf{M}_7,\mathsf{R}_{\mathbb{Z}}\}$ *and put* $\varepsilon^{\mathsf{MP}}:=(\Phi^{\mathsf{MP}})^*\circ\mathrm{cmp}^*\colon C^\bullet_{\mathrm{eq}}(R)\to C^\bullet(\mathsf{S};\underline{V_R})$, *with* cmp^* *the map of Proposition 23.5(4) and* $C^\bullet_{\mathrm{eq}}(R)=\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(C_\bullet(X),V_R)$ *(Theorem 5.10).*

1. *[proved here]* $\varepsilon^{\mathsf{MP}}$ *is a cochain map, compatible with base change, and its homotopy class is canonical: any two* $\mathbb{Z}[\Gamma]$*-maps* $N\mathbb{Z}[\mathfrak{F}']\to C_\bullet(X)$ *over* $\mathrm{id}_{\mathbb{Z}}$ *are homotopic; in particular it is, up to homotopy, the map induced by the augmentation* $\mathrm{Tot}\to C_\bullet(X)$ *composed with an inverse of* $\mathrm{Tot}\to\mathbb{Z}[\mathfrak{F}']$*. Likewise, for any* Γ*-map* $\Phi\colon R[\mathfrak{F}'_\bullet]\to C_\bullet(X)\otimes R$ *over* id_R*, the map* $\Phi^*\colon C^\bullet_{\mathrm{eq}}(R)\to C^\bullet_{\mathrm{un}}(\mathsf{S};\underline{V_R})$ *is homotopic to* $\varepsilon^{\mathsf{MP}}$ *followed by the inclusion* $C^\bullet\subseteq C^\bullet_{\mathrm{un}}$ *(v1.22.0-pre.7). [formally verified: [GT, `def:mp-epsmp`]] [formally verified: [GT, `thm:mp-epsmp-canonical`]] (*$\varepsilon^{\mathsf{MP}}$ *built as* Φ^**,* Φ *the lift of* id_R *from* $R[\mathfrak{F}'_\bullet]$ *into* $C_\bullet(X)\otimes R$*, into the unnormalised MP complex, a cochain map at every ring; any two such lifts homotopic, so the class on cohomology is canonical;* Φ^{MP} *and* cmp *themselves and the compatibility with base change are not formalised)*
2. *[proved here] It is a quasi-isomorphism* ***if*** $6\in R^\times$ *(the cone of* cmp^* *is killed by* 48*, Proposition 23.5(4)): this is Theorem 17.3 re-proved on the MP model.* ***Not only if:*** *for* $R=\mathbb{Z}[1/2]$ *and* n *odd both sides are acyclic (Remark 7.7, second bullet, as corrected at v1.22.0-pre.4;* $H^\bullet(\Gamma;V_R)=0$ *since* $2\in R^\times$*), while* $3\notin R^\times$*. [formally verified: [GT, `thm:mp-epsmp-quasiiso`]] (the "if" only, with the stabiliser orders dividing* 12 *of (H4)(iv):* Φ *a homotopy equivalence and* Φ^* *a quasi-isomorphism for every representation, the Bredon side on* M_7 *and the MP side on* M_7 *or* $\mathsf{R}_{\mathbb{Z}}$*; not the "not only if" witness)*
3. *[proved here] At every* R*,* $\varepsilon^{\mathsf{MP}}$ *is injective on* H^0 *and* H^1*: the Bredon groups are the row* $t=0$ *of the isotropy spectral sequence of* $C^\bullet_{\mathcal{I}}$*, and the edge maps* $E_2^{p,0}\to H^p$ *are injective for* $p\le 1$ *(no differential reaches* $E^{p,0}$ *from a column* $p-r<0$*), as in the proof of Proposition 31.3(3). So at* $\ell\in\{2,3\}$ *the Bredon defect lives in the cokernel, never in the kernel, in degrees* ≤ 1*, and in degree* ≥ 2 *the Bredon side is* 0*.*
4. *[verified computationally]*^{r} *(`OrbifoldMPComparisonValidation`, from the shipped Bredon pipeline `EquivariantCohomologyOnK`): quasi-isomorphism at* $\ell\in\{5,7\}$*,* $n\le 12$ *(non-trivially on* H^0,H^1*; at* $q=2$ *both sides are* 0*); at* $(\ell,n)=(2,0),(2,2),(3,0),(3,2)$*,* $q=0,1,2$*: rank of* $\varepsilon^{\mathsf{MP}}$ *equal to the Bredon dimension, defects* $[0,1,1]$*,* $[0,1,2]$*,* $[0,1,1]$*,* $[0,0,0]$*; the "if, not iff" witness at* $\ell=3$*,* $n\in\{1,3,5\}$*.*

Proof. (1) Composite of cochain maps; base change from Proposition 23.5(4) and the definition of Φ^{MP} over $\mathbb{Z}$. $\mathrm{cmp}\circ\Phi^{\mathsf{MP}}$ is a Γ-map from a free complex to the acyclic augmented $C_\bullet(X)\to\mathbb{Z}$ over $\mathrm{id}_{\mathbb{Z}}$, unique up to homotopy [standard: [Wei94, Thm. 2.2.6]]. The same theorem, applied over R to the projective resolution $R[\mathfrak{F}'_\bullet]$ (Theorem 32.12(2)) and the exact $C_\bullet(X)\otimes R\to R$ ((H4)(ii)), makes any Φ over id_R homotopic to $(\mathrm{cmp}\circ\Phi^{\mathsf{MP}}\circ\pi)\otimes R$, $\pi\colon\mathbb{Z}[\mathfrak{F}']\to N\mathbb{Z}[\mathfrak{F}']$ the projection; dualised, this is the second sentence of (1), and with the inclusion $C^\bullet\subseteq C^\bullet_{\mathrm{un}}$ a quasi-isomorphism (Theorem 32.19(2)), clause (2) below holds for Φ^* as well. (2) $(\Phi^{\mathsf{MP}})^*$ is always a quasi-isomorphism (Proposition 32.17(1)) and cmp^* is one for $6\in R^\times$. The witness: the Bredon complex vanishes by (7) with 2 invertible, and $2H^\bullet(\Gamma;V_R)=0$ by centrality. (3) Filter $C^\bullet_{\mathcal{I}}(V_R)$ by s as in the proof of Proposition 23.5(4): $E_1^{s,t}=\bigoplus_{\sigma\in\Sigma_s}H^t(\Gamma_\sigma;V_R)$, row $t=0$ is $C^\bullet_{\mathrm{eq}}$, cmp^* is the inclusion of that row, so on cohomology it is the edge map $E_2^{p,0}\to E_\infty^{p,0}=F^pH^p\subseteq H^p$; $E_2^{p,0}\to E_\infty^{p,0}$ is the quotient by the images of $d_r\colon E_r^{p-r,r-1}\to E_r^{p,0}$, which vanish for $p\le 1<r$; then compose with the isomorphism $(\Phi^{\mathsf{MP}})^*$. Bredon $H^{\ge 2}=0$: Proposition 23.3(2). □

Proposition 32.21 (The boundary and the pair)**.** *[proved here] [verified computationally]*^{r} *Let* $\Gamma=\mathrm{SL}_2(\mathbb{Z})$ *and* ∂X *the horocycle subcomplex ((B1)–(B3)),* $\Gamma_\infty=\langle T\rangle\times\langle -I\rangle$*, with the top simplices of the boundary descriptor the three edges* v_0v_1*,* v_1v_2*,* v_2v_3*.*

1. *(AC) fails for ∂X ($H_0 = \mathbb{Z}[\Gamma/\Gamma_\infty]$), and the proof of Theorem 32.12, parts (1)–(2), gives a free resolution $N\mathbb{Z}[\mathfrak{F}'(\partial X)] \to \mathbb{Z}[\Gamma/\Gamma_\infty]$ of the* permutation module*; hence $H^\bullet(\mathsf{S}(\partial X);\underline{M}) \cong \mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma/\Gamma_\infty], M) \cong H^\bullet(\Gamma_\infty; M) = H^\bullet(\partial; M)$ of Definition 14.2, in every degree, by Shapiro's lemma.*
2. *The inclusion is the labelled morphism $j\colon \mathfrak{F}'(\partial X) \to \mathfrak{F}'(X)$, a boundary edge to the unique triangle of X containing it, with its label (Definition 32.18); j^* carries the label factor, its cone computes the relative object of Definition 17.1 (the MP-model form of Definition 15.1) — in the relative indexing $\mathrm{Cone}^q := C^q(\mathsf{S}(X);\underline{M}) \oplus C^{q-1}(\mathsf{S}(\partial X);\underline{M})$, $d(c,x) := (dc,\ j^*c - dx)$, so that $H^q(\mathrm{Cone}) = H^q(X, \partial X; M)$; $(c,x) \mapsto (c,-x)$ identifies it with $C(r)[-1]$ in the convention of Definition 15.1 (the shift negating the differential), so $H^q(\mathrm{Cone}) = H^{q-1}(C(r)) = H^q_c$ (v1.22.0-pre.7, the indexing of the code) —, and the long exact sequence of the pair holds unconditionally, at every ring, in every degree; under Theorem 32.12 and (1) j^* is the restriction $H^\bullet(\Gamma; M) \to H^\bullet(\Gamma_\infty; M)$.*

Proof. (1) Lemmas 32.9, 32.10, 32.11 hold for ∂X verbatim ((SC), (IO), (PD) are inherited by the subcomplex), so $\mathbb{Z}[\mathfrak{F}'(\partial X)] \simeq C_\bullet(\partial X)$, and $C_\bullet(\partial X) = \mathbb{Z}[\Gamma] \otimes_{\mathbb{Z}[\Gamma_\infty]} C_\bullet(\partial X_\infty)$ with the distinguished component contractible, so $C_\bullet(\partial X) \to H_0 = \mathbb{Z}[\Gamma/\Gamma_\infty]$ is a quasi-isomorphism; the source is free. Shapiro: [standard: [Bro82, Ch. III.6.2]], as in Definition 17.1(2) and Proposition 14.4(1). (2) Theorem 32.19(1) gives the cochain map; $\mathbb{Z}[j]$ is a Γ-map from the free resolution of (1) to the free resolution of $\mathbb{Z}$ over the augmentation $\varepsilon\colon \mathbb{Z}[\Gamma/\Gamma_\infty] \to \mathbb{Z}$, unique up to homotopy [standard: [Wei94, Thm. 2.2.6]], so $H^\bullet(j^*)$ is $\mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(\varepsilon, M)\colon \mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(\mathbb{Z}, M) \to \mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(\mathbb{Z}[\Gamma/\Gamma_\infty], M)$, which under Shapiro is $\mathrm{res}^\Gamma_{\Gamma_\infty}$ [standard: [Bro82, Ch. III.9], restriction as the map induced by $\mathbb{Z}[\Gamma/H] \to \mathbb{Z}$; VERIFY]; the cone and its sequence as in Lemma 15.2(1),(2), which use of the restriction only that it is a cochain map. □

[verified computationally]^{r} for (1), (2) (`OrbifoldMPBoundaryValidation`, `standard`, 332 checks; slot-22 MP-3, audited F-MP3, v1.22.0-pre.7; since MP-3b, v1.22.0-pre.8, also the digest of the oracle file actually read, gated against the frozen one): $H^0, \dots, H^3(\mathsf{S}(\partial X); \mathrm{Sym}^n R) = H^\bullet(\Gamma_\infty; \mathrm{Sym}^n R)$ on 134 cells ($\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, $n \le 12$; $\mathbb{Z}/4, \mathbb{Z}/8, \mathbb{Z}/9$ by lengths, $n \le 4$; the boundary descriptors of M_7 and $\mathsf{R}_\mathbb{Z}$) against an explicit resolution of Γ_∞ and the shipped circle model of Proposition 14.4; j^* a cochain map, its rank on $H^{\le 3}$ that of the restriction of $C^\bullet_\mathcal{I}$; the cone exact, with $H^\bullet(X, \partial X; \mathbb{F}_2) = [0,0,1,1]$ for $q = 0, \dots, 3$ at $n = 0$ on $\mathsf{S}_{\le 4}(\mathsf{M}_7)$, in particular $H^1(X, \partial X; \mathbb{F}_2) = 0$.

Proposition 32.22 (Restriction, corestriction and the sequence of a pair of groups)**.** *[proved here] [verified computationally]^{r} Let $B \le A \le \Gamma = \mathrm{SL}_2(\mathbb{Z})$ have finite index, acting on the same X with the triangulation $\mathfrak{T}$, and $\mathfrak{F}'_A$, $\mathfrak{F}'_B$ the labelled Čech nerves built from the A-, resp. B-orbit representatives.*

1. ***(Shapiro on the nose, on one $\mathfrak{F}'$.)*** *$\mathfrak{F}'_A$ is a free B-set, and $\mathrm{Hom}_{\mathbb{Z}[B]}(N\mathbb{Z}[\mathfrak{F}'_A], M) = \mathrm{Hom}_{\mathbb{Z}[A]}(N\mathbb{Z}[\mathfrak{F}'_A], \mathrm{Coind}^A_B M)$ is an identity of cochain complexes; on it res^A_B is the restriction of equivariance and cor^A_B the transfer $F \mapsto \sum_t r_t^{-1} \cdot F(r_t\,\cdot)$ over a transversal r_t of $B\backslash A$, both cochain maps, with $\mathrm{cor} \circ \mathrm{res} = [A : B]$. The same holds with $\mathfrak{F}'_A$ replaced by any free Γ-set resolving $\mathbb{Z}$ as in Theorem 32.12(2) (free over every subgroup), in particular by $\mathfrak{F}'_\Gamma$, on which $\mathrm{Hom}_{\mathbb{Z}[B]}(N\mathbb{Z}[\mathfrak{F}'_\Gamma], M) = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(N\mathbb{Z}[\mathfrak{F}'_\Gamma], \mathrm{Coind}^\Gamma_B M)$, res^A_B is induced by the Γ-map $\mathrm{Coind}^\Gamma_A M \to \mathrm{Coind}^\Gamma_B M$ (the inclusion of A-equivariant into B-equivariant functions $\Gamma \to M$) and cor^A_B by $\phi \mapsto (\gamma \mapsto \sum_t r_t^{-1}\phi(r_t\gamma))$; either nerve serves, the two realisations agreeing on* Ext *by transitivity of coinduction and the independence of* res, cor *from the projective resolution (v1.22.0-pre.7; the code works on $\mathfrak{F}'_\Gamma$).*
2. ***(Forget the coset.)*** *$\mathsf{S}(\mathfrak{F}'_B) = B\backslash\mathfrak{F}'_B$ and $A\backslash\mathfrak{F}'_A$ are different simplicial sets; the forget-coset map is a labelled morphism $\mathfrak{F}'_B \to \mathfrak{F}'_A$ of Definition 32.18 (labels the coset representatives,* not *central, so the label factor is load-bearing), a comparison of two free B-resolutions, unique up to homotopy; under Theorem 32.12 it induces res^A_B.*

3. ***(The pair of groups.)*** *The cone of* res *gives the long exact sequence of the* pair of groups (A, B) *in every degree and at every* ℓ *— the group-cohomology form of §29.8's level cone, which is the model's (Bredon) sequence at* $\ell \in \{2, 3\}$ *(Proposition 29.31(3)). It is* not *inflation–restriction (Proposition 29.36), a different statement with different terms.*

Proof. (1) B acts freely on the first label; $\mathrm{Coind}_B^A = \mathrm{Ind}_B^A$ for finite index, and the identity is the adjunction, termwise and compatible with the differentials; res, cor on Ext are computed on any projective resolution by these formulas [standard: [Bro82, Ch. III.9–10]]. (2) Theorem 32.19(1),(2) with φ the inclusion. (3) The cone of a cochain map has a long exact sequence (Lemma 15.2(2)'s argument); with (1), (2) its terms are $H^\bullet(A; M)$, $H^\bullet(B; M)$ and the relative groups of the pair. For $\mathfrak{F}'_\Gamma$ in (1): it is free over A and B (already on the first label), so the same adjunction gives $\mathrm{Hom}_{\mathbb{Z}[B]}(N\mathbb{Z}[\mathfrak{F}'_\Gamma], M) = \mathrm{Hom}_{\mathbb{Z}[\Gamma]}(N\mathbb{Z}[\mathfrak{F}'_\Gamma], \mathrm{Coind}_B^\Gamma M)$ with $\mathrm{Coind}_A^\Gamma \mathrm{Coind}_B^A = \mathrm{Coind}_B^\Gamma$, and the displayed maps are the transported res, cor [standard: [Bro82, Ch. III.9–10]]. □

[verified computationally]^{r} for (1)–(3) (`OrbifoldMPLevelConeValidation`, `standard`, 723 checks; slot-22 MP-3, audited F-MP3, v1.22.0-pre.7; MP-3b, audited F-MP3b, v1.22.0-pre.8): the five pairs $\Gamma_0(4) < \Gamma_0(2)$, $\Gamma_0(9) < \Gamma_0(3)$, $\Gamma_1(4) < \Gamma_0(4)$, $\Gamma^2 < \Gamma$, $\Gamma^3 < \Gamma$ over $\mathbb{F}_\ell$, $\ell \in \{2, 3\}$, $n \le 2$, on M_7: the B-descriptor with $\left|\mathsf{S}_k^{\mathrm{nd}}(B)\right| = [\Gamma : B] \left|\mathsf{S}_k^{\mathrm{nd}}(\mathsf{M}_7)\right|$ when $-I \in B$; res, cor on $\mathfrak{F}'_\Gamma$ with $\mathrm{cor} \circ \mathrm{res} = [A : B]$ on the cochains; the forget-coset map with its label factor (non-central labels, e.g. 10 of 15 for $\Gamma_0(2) \to \Gamma$; for $\Gamma_1(4) \to \Gamma_0(4)$ all labels are $+I$) equal to Shapiro $\circ$ res on $H^{\le 2}$, and dropping the factor breaks the cochain map; $H^q(A, B; M)$ from the cone equal to the long exact sequence of $0 \to \mathrm{Coind}_A M \to \mathrm{Coind}_B M \to Q \to 0$ on 21 cells ($q \le 2$; $q = 3$ for $\Gamma^2, \Gamma^3 < \Gamma$ at $n = 0$), e.g. $\Gamma^2 < \Gamma$, $\mathbb{F}_2$, $n = 0$: res $= 0$ on H^1, $H^\bullet(A, B) = [0, 1, 1, 1]$. Since MP-3b (v1.22.0-pre.8): the pair sequence over $\mathbb{Z}/\ell^k$ — $\mathbb{Z}/4$ ($\Gamma_0(4) < \Gamma_0(2)$, $n \le 1$ on M_7, $n = 1$ on $\mathsf{R}_\mathbb{Z}$), $\mathbb{Z}/9$ ($\Gamma_0(9) < \Gamma_0(3)$) and $\mathbb{Z}/25$ ($\Gamma_0(25) < \Gamma_0(5)$, a step of the Hida 5-tower), $n = 0$ unless stated: the lengths of $H^q(A)$, $H^q(B)$, of the image of res and of $H^q(A, B)$ (MP $q \le 2$, $C_\mathcal{I}^\bullet$ $q \le 3$) against an integral Mayer–Vietoris oracle (Smith normal form, universal coefficients), with torsion visible (over $\mathbb{Z}/4$ the lengths of $H^\bullet(\Gamma_0(2))$ are $[2, 4, 3, 3] \ne 2 \cdot [1, 2, 2, 2]$); the pairs $\Gamma^2 < \Gamma$ ($\mathbb{F}_2$) and $\Gamma_0(4) < \Gamma_0(2)$ ($\mathbb{F}_3$, $\mathbb{Z}/4$) on $\mathsf{R}_\mathbb{Z}$, with their B-descriptors; and $\Gamma_1(8) < \Gamma_1(4)$, $-I \notin A$ ($\mathbb{F}_2$; $\mathbb{F}_3$, $n = 1$; MP $q \le 1$, $C_\mathcal{I}^\bullet$ $q \le 3$): labels with $u = -I$ occur only when $-I \notin A$, and do occur, $\Gamma_1(8) \to \Gamma_1(4)$, 60 of 120 (representative-dependent); replacing them by their $u = I$ candidates moves strings out of S_A — a negative control that is structurally guaranteed ($-h \notin A$ is automatic since $-I \notin A$), which shows that the containment gate detects a $u = I$-only labelling. **Not gated at this version**: $q = 3$ on the nerve for the Γ_0 pairs, $\mathbb{Z}/\ell^k$ at $n \ge 2$, $\mathbb{Z}/49$ ($\Gamma_0(49)$ has 280 top orbits on M_7, beyond the nerve's bound), inflation–restriction.

Remark 32.23 (Hecke operators on the MP model). (Not a statement.) For $g \in \Delta$ the operator $T_{\Gamma g \Gamma} = \mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ needs a g-twisted chain datum (Definition 11.1) on a projective resolution. On the MP model this is built *on the MP resolution itself*: $N\mathbb{Z}[\mathfrak{F}'_\bullet]$ is free over $\mathbb{Z}[\Gamma'']$, $\Gamma'' = \Gamma \cap g\Gamma g^{-1}$, on $\{s_j e_s\}$ (s_j the representatives of $\Gamma''\backslash\Gamma$), and acyclic, so a conj_g-twisted chain map τ lifting $\mathrm{id}_\mathbb{Z}$ exists and is unique up to conj_g-twisted homotopy *over every commutative ring* (the comparison theorem; unlike Lemma 11.3(2) no hypothesis $6 \in R^\times$ is needed, because the source is free). It is built generator by generator with the MP model's own $\mathbb{Z}$-linear filling of $N\mathbb{Z}[\mathfrak{F}'_\bullet]$ (the double complex of the proof of Theorem 32.12 made explicit), $\tau_0(s_j e_s) = e_s$, $\tau_k(s_j e_s) = \mathrm{fill}(\tau_{k-1}(\partial\, s_j e_s))$; no datum of $C_\bullet(X)$ enters. The three maps are, in the Shapiro coordinates of Proposition 32.22(1), res the inclusion of invariants, $(\mathrm{conj}_g)_* \colon F \mapsto \rho_n(g)\, F \circ \tau$ — the factor $\rho_n(g)$ is the label factor of the g-twist, it makes F' Γ''-equivariant and cannot be dropped (Theorem 32.19(4)) — and cor the transfer; fused, $(Tf)(e_s) = \sum_j \sum c\, \rho_n(s_j^{-1} g \gamma) f(e_{s'})$ over the terms $c\gamma e_{s'}$ of $\tau(s_j e_s)$, the form of (41). Since $\Phi \circ \tau^{\mathrm{MP}}$ and $\tau_{\mathrm{Iso}} \circ \Phi$ are two such twisted maps into the acyclic $\mathrm{Iso}_\bullet$, $T_p\Phi^* = \Phi^* \hat{\Theta}_g$ on $H^\bullet$ over every ring. The g-twist is *not* a labelled morphism of MP data (Definition 32.18): X and gX are different truncations of $\mathcal{H}$, and a *canonical* geometric datum through a common truncation is gap G3, off the critical path — this remark shows

it is not needed to compute. [verified computationally]^{r} `OrbifoldMPHeckeValidation` (slot 22, MP-4): τ certified exactly over $\mathbb{Z}[\Gamma]$ on M_7 ($p \in \{2,3,5,7\}$, degrees ≤ 2) and $\mathsf{R}_{\mathbb{Z}}$ ($p \in \{2,3\}$, degrees ≤ 1); T_p a cochain map; $T_p\Phi^* = \Phi^*\hat{\Theta}_g$ on $H^{\le 1}$ over $\mathbb{F}_\ell$, $\ell \in \{2,3,5,7\}$, $n \le 12$, on $H^{\le 2}$ over $\mathbb{Z}/4, \mathbb{Z}/8, \mathbb{Z}/9, \mathbb{Z}/27$ ($n \in \{0,2\}$) and over $\mathbb{F}_2, \mathbb{F}_3$ ($n \le 3$); the characteristic polynomials on $H^0, H^1(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell)$ equal an independent amalgam-cocycle oracle, with $(x-\tau(p))^2(x-1-p^{11})$ at $n = 10$, $\ell \in \{5,7\}$; the Bredon operator of Section 11 is recovered on $\operatorname{im}\varepsilon^*$. Not built: H^3, T_{p^ν}, level N, the ι-sign. Notation (v1.22.0-pre.8; the text above is the audit F-MP4 §3's verbatim, except that its [verified computationally] is written [verified computationally]^{r} and its undefined "Section sec:hecke" is §11): $g = \mathrm{diag}(1,p)$, so $\Gamma'' = \Gamma_0(p)$ and $T_p = T = T_{\Gamma g \Gamma}$; $\Phi = \Phi^{\mathsf{MP}}$ of Definition 32.16; $\mathrm{Iso}_\bullet = \mathcal{I}_\bullet$, τ_{Iso} and $\hat{\Theta}_g = \Theta_{\mathcal{I},g}$ as in Proposition 23.13(1),(3); $\tau(p)$ in the last formula is Ramanujan's τ, not the chain map.

[verified computationally]^{r} for Remark 32.23 (`OrbifoldMPHeckeValidation`, `standard`, 2059 checks; slot-22 MP-4, audited F-MP4 (ACCEPT WITH REPAIRS); predictions `MP4_predictions.md` and the independent oracle `mp/mp4/mp4_oracle.py` committed before the first run; v1.22.0-pre.8). The scope of the record in the remark, with the *non-vacuous* counts (cells on which the operator acts on a non-zero space) printed and gated: $T_p\Phi^* = \Phi^*\hat{\Theta}_g$ on 354 of 592 cells (240 of the 448 cells on M_7 over $\mathbb{F}_\ell$; 114 of the 144 cells over $\mathbb{Z}/\ell^k$ and on $\mathsf{R}_{\mathbb{Z}}$); the characteristic polynomials against the oracle on 254 of 464 cells (220 of 416 on M_7, 34 of 48 on $\mathsf{R}_{\mathbb{Z}}$); the shipped Bredon T_p on 66 of 108 cells (15 of 24 over $\mathbb{F}_2, \mathbb{F}_3$, on $\operatorname{im}\varepsilon^*$, $n \in \{0,2\}$; 51 of 84 over $\mathbb{F}_5, \mathbb{F}_7$, ε^* an isomorphism, n even ≤ 12, $p \ne \ell$). The cochain-map identity $T\,d^q = d^q\,T$ is gated for $q \le 1$ on M_7 and $q = 0$ on $\mathsf{R}_{\mathbb{Z}}$. The uniqueness over every ring is what makes the cells over $\mathbb{Z}/4, \mathbb{Z}/8, \mathbb{Z}/9, \mathbb{Z}/27$ and $\mathbb{F}_2, \mathbb{F}_3$ meaningful, where Lemma 11.3(2) does not apply. In degree 2 the datum stops at τ_2: $d\tau_2 = \tau_1 d$ is certified, so T preserves 2-coboundaries and the class of $T_p\Phi^*z$ is defined, but $T\,d^2 = d^2\,T$ needs τ_3 and is checked only on the tested classes ($T_p\Phi^*z$ a cocycle, cohomologous to $\Phi^*\hat{\Theta}_g z$) — all of H^2, Φ^* being onto ((MPC4)); at $\ell \in \{2,3\}$, H^2 has no independent Hecke oracle (only $\hat{\Theta}_g$ through Φ^*). Negative controls: the label factor $\rho_n(g)$ dropped, or mismatched with the datum (gT for g), breaks the cochain map and the identity on H^1; a perturbed τ_1 fails the certificate. **Not gated at this version**: H^3 (τ_3, d^3); T_{p^ν}, $T(p,p)$ and the full operators (19); characteristic polynomials over $\mathbb{Z}/\ell^k$; $\mathsf{R}_{\mathbb{Z}}$ beyond $p \in \{2,3\}$, degrees ≤ 1; a second filling (the independence of the filling is argued, not tested); level N (T_p, U_p on the subgroup MP sets of Proposition 32.22); compatibility with j^* of Proposition 32.21; the ι-sign.

Example 32.24 (Moerdijk–Pronk's teardrop in the G^k-form). [verified computationally]^{r} *Tear-n* is the 2-sphere with one cone point of order n, not a global quotient, so it tests the G^k-form of their $S(\mathcal{T})$ (not Definition 32.6): six top simplices around and below the cone point, $\mathsf{S}_k = \bigsqcup C_n^k$ over the strings whose intersection contains the cone point, with their μ-maps on the edge ac ([MP] §6.1–6.2, read from the author source quoted in Remark 17.4). `OrbifoldMPTeardropValidation` (`fast`), against `docs/implementation/slot22/mp/teardrop.py`: for $n \in \{2,3,4\}$, $\ell \in \{2,3,5\}$, $\dim H^0, \dots, H^3 = 1,0,1,[\ell \mid n]$ — Moerdijk–Pronk's values $A, 0, A, H^3(C_n; A)$ — and $|\mathsf{S}_0|, \dots, |\mathsf{S}_4| = 6, 39, 177, 705, 2877$; $6, 48, 294, 1866, 13290$; $6, 57, 465, 4323, 44925$; $d \circ d = 0$ on every non-degenerate string to $k = 4$; negative controls ($\mu_{ac} = \mu_{ca}$, a second non-inverse μ pair, which does not model a C_{n+1} vertex group, trivial μ) fail (the first two break $d \circ d = 0$; trivial μ keeps it and fails the dimension gate). "Infinitely many degrees from one cone point" in its simplest form; the $\ell = 3$ structure of Definition 32.3 is the disk with one cone point of order 3 and a boundary circle, i.e. Tear-3 without its lower hemisphere ([MP] §6.5).

32.6 What is checked

Remark 32.25 (What is checkable: (MPC1)–(MPC12)). [proved here] [verified computationally]^{r} Predictions committed before each first run (rule 20) in `docs/implementation/slot22/MP_predictions.md`

(P1–P4 and the MP-2 addendum). Every validation below is registered in `Registry/Slot22/MP.cpp`, adds 0 golden objects, tags its output `model=mp`, ends with `ORBIFOLD_MP <name> checks=<k> failing=0`, and where a Python oracle is reproduced prints `ORACLE <script> cells=<k> agree=<k>`. The oracles are independent of the code under test: the frozen `audit/mv_check.out` (an independent Mayer–Vietoris on $\mathbb{Z}/4 *_{\mathbb{Z}/2} \mathbb{Z}/6$, own Sym^n, own ranks), the shipped Bredon pipeline, and the shipped $C^\bullet_{\mathcal{I}}$ (`Integral/Model/IsotropyModel`).

(MPC1) [proved here] (Definitions 32.2, 32.4, Remark 32.5) **Rules**: Table 9, (SC), (IO), the quotient counts, the subgroup-form and negative controls. `OrbifoldMPRulesValidation` (`fast`, 77 checks).

(MPC2) [proved here] (Definition 32.6, Remark 32.7) **The set**: $\left|\mathsf{S}^{\mathrm{nd}}_k\right|$, $k \le 3$, the μ-form counts, faces and simplicial identities, $d \circ d = 0$ formally in $\mathbb{Z}[\Gamma] \otimes \mathbb{Z}[\mathsf{S}^{\mathrm{nd}}_q]$ through degree 4. `OrbifoldMPRulesValidation`, `OrbifoldMPNerveValidation`.

(MPC3) [proved here] (Theorem 32.12) **The nerve against the oracle**: H^0, H^1 over $\mathbb{F}_\ell$, $\ell \in \{2,3,5,7\}$, $n = 0, \dots, 12$; H^2, $\ell \in \{2,3,5\}$, $n \le 3$ ($n \in \{4,6\}$ under the read-only hook `ORBIFOLD_MP_CELLS`); $\mathbb{Z}/\ell^k$ lengths, $\ell \in \{2,3\}$, $k \le 3$, against $C^\bullet_{\mathcal{I}}$'s integral groups through the universal coefficient theorem; odd n with 2 a unit: non-zero acyclic complexes; on M_7 and $\mathsf{R}_{\mathbb{Z}}$. `OrbifoldMPNerveValidation` (`standard`, 584 checks); H^3 at $\ell \in \{2,3\}$, $n \in \{0,2\}$ in `OrbifoldMPNerveH3Slow` (`slow`). Negative controls: the untwisted d_0, the inverse twist $\rho_n(\gamma_1^{-1})$, and a planted stabiliser $\Gamma_{\zeta^2} = \langle (ST)^2 \rangle$ without $-I$, which changes $H^2(\mathbb{F}_2)$ from 1 to 2 (the gated effect); that H^0, H^1 and the $\mathbb{Z}/\ell^k$ cells are unchanged is an ungated observation of the F-MP1 probe. **Hence the default H^2 cells are load-bearing for the central gerbe and never move behind a hook or into the slow tier; (AC), the one hypothesis of Theorem 32.12 not decided by a predicate, is guarded by this gate** (and, since v1.22.0-pre.6, proved in Lean for M_7 and $\mathsf{R}_{\mathbb{Z}}$ over every ring: the markers of Theorem 32.12(5)).

(MPC4) [proved here] (Definition 32.16, Proposition 32.17(1)) Φ^{MP}: certified to degree 3; $(\Phi^{\mathsf{MP}})^* \delta_I = d_{\mathsf{MP}} (\Phi^{\mathsf{MP}})^*$ exactly (54 products); an isomorphism $H^q(C^\bullet_{\mathcal{I}}) \to H^q(\mathsf{S})$ as in Proposition 32.17(4), and over $\mathbb{Z}/\ell^k$, $\ell \in \{2,3\}$, $k \in \{2,3\}$, $n \in \{0,2\}$; a perturbed Φ_1 fails the certificate. `OrbifoldMPComparisonValidation` (`standard`, 901 checks).

(MPC5) [proved here] (Proposition 32.20) $\varepsilon^{\mathsf{MP}}$: Proposition 32.20(4). `OrbifoldMPComparisonValidation`.

(MPC6) [proved here] (Remark 32.13) **The hybrid**: as stated there. `OrbifoldMPComparisonValidation`, `OrbifoldMPAllDegreesValidation`.

(MPC7) [proved here] (Theorem 32.19(3),(4)) **Refinement** $\mathsf{R}_{\mathbb{Z}} \to \mathsf{M}_7$: as recorded after the theorem. `OrbifoldMPRefinementValidation` (`standard`, 115 checks).

(MPC8) [proved here] (Proposition 32.17(2)–(4)) **All degrees**: $C^\bullet_{\mathcal{I}}$ and $\mathcal{I}^{(\ell)}$ $H^0, \dots, H^7$ over $\mathbb{F}_\ell$, $\ell \in \{2,3,5,7\}$, $n \le 12$, against the oracle (832 values); the cup product with e an isomorphism $H^q \to H^{q+2}$, $q = 2, \dots, 5$; the eight `MP_ALL_DEGREES` lines; negative control: $\ell \cdot E^*$ is not an isomorphism wherever $H^2 \neq 0$. `OrbifoldMPAllDegreesValidation` (`standard`, 1086 checks).

(MPC9) [verified computationally]^{r} (Example 32.24) **The teardrop**. `OrbifoldMPTeardropValidation` (`fast`, 66 checks).

(MPC10) [proved here] (Proposition 32.21; v1.22.0-pre.7) **The boundary and the pair**: as recorded after the proposition. `OrbifoldMPBoundaryValidation` (`standard`, 332 checks; 331 at v1.22.0-pre.7, MP-3b adds the oracle-file digest gate).

(MPC11) [proved here] (Proposition 32.22; v1.22.0-pre.7) **Restriction, corestriction, the pair of groups**: as recorded after the proposition; the LES-0 arm (iv) on the MP model; since MP-3b (v1.22.0-pre.8) also the pair sequence over $\mathbb{Z}/4$ ($\Gamma_0(4) < \Gamma_0(2)$, $n \le 1$), $\mathbb{Z}/9$ ($\Gamma_0(9) < \Gamma_0(3)$) and $\mathbb{Z}/25$ ($\Gamma_0(25) < \Gamma_0(5)$) against an integral Mayer–Vietoris oracle (Smith normal form, UCT), the pairs $\Gamma^2 < \Gamma$, $\Gamma_0(4) < \Gamma_0(2)$ on $\mathsf{R}_{\mathbb{Z}}$, and $\Gamma_1(8) < \Gamma_1(4)$ with $-I \notin A$ (forget-coset labels with $u = -I$ reached and load-bearing). `OrbifoldMPLevelConeValidation` (`standard`, 723 checks; 537 at v1.22.0-pre.7).

(MPC12) [verified computationally]^{r} (Remark 32.23; v1.22.0-pre.8) **Hecke operators on the MP model**: T_p, $p \in \{2,3,5,7\}$, on $H^{\leq 2}$ from the twisted chain map on the MP resolution, as recorded after the remark (non-vacuous counts included). `OrbifoldMPHeckeValidation` (`standard`, 2059 checks).

(v1.22.0-pre.7: (MPC10), (MPC11) added — slot-22 MP-3, audited by F-MP3 (ACCEPT WITH REPAIRS, the audit file `F-MP3code_audit.md`), with the predictions `MP3_predictions.md`, both in `docs/implementation/slot22/`. v1.22.0-pre.8: (MPC10), (MPC11) widened by MP-3b, audited F-MP3b, and (MPC12) added — slot-22 MP-4, audited F-MP4, predictions `MP4_predictions.md`; both audits ACCEPT WITH REPAIRS.) **Not checked at this version**: for (MPC11), $q = 3$ on the nerve for the Γ_0 pairs and $\mathbb{Z}/\ell^k$ at $n \geq 2$; for (MPC12), the list after Remark 32.23 (H^3, T_{p^ν}, characteristic polynomials over $\mathbb{Z}/\ell^k$, level N, the ι-sign).

Scope: what this chapter does not do. It does not re-prove Moerdijk–Pronk's sheaf-side theorem, and does not prove their μ-independence (it avoids it); it does not make any triangulation of this chapter, or the MP model, the shipped object or a default (the contract box above); it asserts the all-degrees statement for $\mathrm{SL}_2(\mathbb{Z})$ in the bulk only (Proposition 32.17(5)); it does not need goodness, a collar, or adaptedness for any theorem; it builds no *canonical geometric* Hecke datum on the MP model (gap G3; the computational datum of Remark 32.23 is a twisted chain map on the MP resolution, built to degree 2; v1.22.0-pre.8). The non-developable case (complexes of groups that are not global quotients, the teardrop) is recorded by Example 32.24 only.

Code map. **Built** (slot-22 MP-1a, MP-1b, MP-T, MP-2; audited: F-MP1, the MP-T audit, F-MP2 ACCEPT WITH REPAIRS, `docs/implementation/slot22/F-MP2_code_audit.md`), in `src/hacs/Orbifold/MP/`, compiled into `cohomology_app`: `Descriptor` (Definition 32.1, the predicates of Definition 32.2, read off `integral_model::ModelData`; $\mathsf{R}_\mathbb{Z}$ by `SubdivideEdgeOrbit`), `MPSet` and `Nerve` (Definition 32.6: the strings, faces, the twisted normalised coboundaries over $\mathbb{Z}/m$), `SparseLength` (lengths over $\mathbb{Z}/\ell^e$), `AmalgamOracle` (the C++ port of `mv_check.py`, and its frozen output as `MvCheckOracle.inc`), `Comparison` (`PhiLift`, Definition 32.16), `Refinement` (Definition 32.18(2), `RefinementPullback` with the label factor), `CochainMaps`, `Models`, `MP2Common`; the seven validations of Remark 32.25; registration `Registry/Slot22/MP.cpp`, `cmake/slot22/MP.cmake`. The scratch emitter `ORBIFOLD_MP_DUMP=<dir>` (no-op unless set, refused inside the tree) writes the $\mathsf{R}_\mathbb{Z}$ descriptor for the Lean cross-checks. **Built at slot 22** (MP-3, audited F-MP3; v1.22.0-pre.7): the boundary descriptor, j^* and the cone (Proposition 32.21), the subgroup descriptors, res, cor on $\mathfrak{F}'_\Gamma$, the forget-coset map and the Shapiro comparison with the label factor, the cone of a pair of groups (Proposition 32.22), with `OrbifoldMPBoundaryValidation`, `OrbifoldMPLevelConeValidation`; MP-3b (audited F-MP3b; v1.22.0-pre.8): `MP3bLevelArms`, `MP3bOracle` (the $\mathbb{Z}/\ell^k$, $\mathsf{R}_\mathbb{Z}$ and $-I \notin A$ pairs). **Built at slot 22** (MP-4, audited F-MP4; v1.22.0-pre.8): `MPFill` (the MP model's $\mathbb{Z}$-linear filling), `MPHecke` (`MPHeckeDatum`: the twisted datum, its certificate, the operator with the label factor, the factorisation $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$), `MP4Oracle.inc`, with `OrbifoldMPHeckeValidation` (Remark 32.23). No console verb, setting, record or golden object (owner ruling Q4).

Formalization note. The chapter is homological algebra over $\mathbb{Z}[\Gamma]$ on finite data, written to be Lean-friendly: every object is a `Finset` given a descriptor, $d \circ d = 0$ is a finite identity per degree, Theorem 32.12 is uniform in the coefficient ring with no invertibility hypothesis, and the per-fibre contractions are non-equivariant. The one homological input not in `mathlib` at the lane's pin is Lemma 32.11 (row-/column-wise quasi-isomorphism of a bounded double complex implies one on totals). The Lean rows L0–L10 of [MP] §8.2 (descriptor predicates, the set, the cochains, $\mathfrak{F}$, rows, columns, assembly, the resolution,

Theorem 32.12, Theorem 32.19, refinement $\mathsf{M}_7 \leftarrow \mathsf{R}_{\mathbb{Z}}$, Proposition 32.20(2)) are the milestone M6 "orbifold" item (owner ruling Q5), the teardrop is not formalised (Q7). **Markers** (v1.22.0-pre.6; pre.5 carried none): the slot-22 Lean packages MP-L1 / MP-L2 ([GT] Part G12, `docs/implementation/slot22/MPL1_report.md` v1.1) prove the rows L0–L7 and L0b, and 30 clause-scoped `\LEAN` markers on 11 labels record them: Definitions 32.2, 32.3, 32.4, Table 9 (rows M_7, $\mathsf{R}_{\mathbb{Z}}$), Definition 32.6, Remark 32.7(1), Definition 32.8, Lemmas 32.9, 32.10, 32.11 (acyclic form) and Theorem 32.12(1)–(5), the comparison for the *unnormalised* complex $C^{\bullet}_{\mathrm{un}}$, natural in M, unconditional on M_7 and $\mathsf{R}_{\mathbb{Z}}$. **v1.22.0-pre.7**: the slot-22 Lean package MP-L3 ([GT] Part G12, `docs/implementation/slot22/MPL3_report.md` v1.0) proves the rows L8–L10 and the normalised complex, and 25 further clause-scoped markers record them: Definition 32.6 (the normalised $C^{\bullet}$ as a subcomplex, its inclusion a quasi-isomorphism), Definition 32.8 ($\mathfrak{F}'$ as a simplicial object), Theorem 32.12(2) (normalised $\simeq$ unnormalised on cochains) and (4) (the comparison for $C^{\bullet}$), Definition 32.18(1)–(3), Theorem 32.19(1)–(4) and its $[\mathsf{verified\ computationally}]^{\mathrm{r}}$ record on $\mathsf{R}_{\mathbb{Z}} \to \mathsf{M}_7$, Proposition 32.20(1) and the "if" of (2). Lean states Theorem 32.19(1)–(3) and Proposition 32.20(1),(2) for $C^{\bullet}_{\mathrm{un}}$ (whence the unnormalised clauses written into them at pre.7), the label independence of Theorem 32.19(3) on cohomology only, $\varepsilon^{\mathsf{MP}}$ as Φ^* for a lift Φ of the identity (homotopic to the written one by Proposition 32.20(1)). Not formalised, so not marked: the normalised u^* as a cochain map in its own right, the chain homotopy between label choices, the ranks $\left|\mathsf{S}^{\mathrm{nd}}_k\right|$, the quasi-isomorphism form of Lemma 32.11, Proposition 32.20(2)'s "not only if" witness, (3), (4) and the base change, Proposition 32.17, Remark 32.13, Propositions 32.21, 32.22 (Lean rows L11–L16, MP-L4).

A Errata to the superseded documents

What this appendix fixes. One entry per defect of the superseded corpus that this document corrects, retires or inherits the duty to record. Each entry gives, in this order: **the defective statement** with its `path:line`; **the correct statement**; **the evidence**; and **where in this document** the correct statement is now made. The entries are the authority every later chapter cites for its corrections: a chapter that re-homes text from a superseded document is not finished until every entry touching its source has been applied.

Three of these are *contradictions* — two statements that cannot both be true (A.1, A.6, A.12); the rest are loose arguments, stale statements, dangling citations, or notation that cannot survive the merge. None of them is a defect of the shipped implementation: where a document and the code disagreed, the code was right in every case (A.1 most pointedly).

#	defect	corrected in
A.1	primitive vs. full T_{p^ν}: two numbered prescriptions off by $p^n I$	§10, §20 (20.4), §21
A.2	"$H^0(B) = Z^0(B) = B^0$"	§15
A.3	a realisation theorem proved over a field, applied over $\mathbb{Z}/\ell^e$	§15 (hypothesis at point of use)
A.4	τ_∂-uniqueness argued from a theorem whose hypothesis it avoids	§14 (14.10)
A.5	"$\tau(jx)$ … both are vertices"	§15 (15.6)
A.6	$T_p^2 = T_{p^2} + (p+1)p^n$ — wrong coefficient, false attribution	§10 (10.8)
A.7	`[C, thm:explicit_complex_equiv]` cited from five source files, label not in [C]	§5
A.8	"issues with torsion, which we ignore"	§7 (7.6, 7.7), §17
A.9	corestriction agrees only after $\gamma \mapsto \gamma^{-1}$, unmarked	§12 (12.7)

#	defect	corrected in
A.10	garbled "Simplicial interpretation" display	not re-homed; recorded only
A.11	basis vectors named η, δ, β	§2.4, §6, §16
A.12	$H^\bullet(\partial)$ defined as Γ_∞, constructed as $\langle T\rangle$	§14 (14.2, 14.4)
A.13	the cone-homotopy sign	§15 (derived, not asserted)
A.14	five uncorrected statements in a working note	§14 (14.16), §16 (16.26); rest in [E]
A.15	"cone by functoriality"	§15 is the definition
A.16	T_n vs. T_N for the composite index	§2.2, §10
A.17	no version marker on either superseded document	§1.4
A.18	the scope cut: nine labels move from [E] to [C]	§16
A.19	[v5] §12's map β is not a morphism of local systems	§12 (12.9)
A.20	[v5] §12.3's bar-cochain formula omits the coset permutation	§13 (13.3, 13.4)
A.21	"$\chi_1 \equiv \chi_2$ at every Dickson locus" — false as stated	§16 (16.24, 16.25)
A.22	the FP32 totality threshold $B \le 2^{10}$ does not follow from the strict criterion it is derived from	§19 (19.4, 19.5)
A.23	the bad-prime oracle asserts $U_4 = U_2^2$ at $\ell = 2$ citing a corollary that carries $\ell \ge 5$	§21 (oracle 7)

A.1 Primitive versus full T_{p^ν}: two numbered prescriptions off by $p^n I$

Survey finding D1; plan §11.2. The most serious defect of the superseded contract: it contradicts itself, and the two defective sites are a numbered obligation and a numbered oracle.

The defective statements.

- `hecke_action.tex`:1443–1444, inside Obligation 20.4: "compute $T_{\Gamma\,\mathrm{diag}(1,p^2)\,\Gamma}$ by the same pipeline with its p^2+p cosets and compare with $T_p^2 - p^{n+1}$".
- `hecke_action.tex`:1595–1596, oracle 6 of the validation plan: "$A^{(q)}(T_4) = A^{(q)}(T_2)^2 - 2^{n+1}I$ with T_4 computed independently from $\Gamma\,\mathrm{diag}(1,4)\,\Gamma$ (6 cosets)".

In both, p^2+p cosets (at $p=2$: 6) is the **primitive** double coset $\Delta^{\mathrm{prim}}_{p^2}$, while $T_p^2 - p^{n+1}I$ is the **full** operator. The two sides differ by exactly p^nI, so each prescription, read literally, *fails* against a correct implementation.

The correct statements. Both, with the entry point that emits each named:

> **CONTRACT.** For every degree q and every prime p, on $H^q(K,\mathcal{R};\mathrm{Sym}^n(\mathbb{F}_\ell^2))$:
>
> $$A^{(q)}\big(T^{\mathrm{full}}_{p^2}\big) = A^{(q)}(T_p)^2 - p^{n+1}I, \qquad A^{(q)}\big(T_{\Delta^{\mathrm{prim}}_{p^2}}\big) = A^{(q)}(T_p)^2 - p^{n+1}I - p^nI,$$
>
> the two differing by the diamond scalar $T(p,p) = p^nI$ of Proposition 10.5. At $p=2$: $A^{(q)}(T_4^{\mathrm{full}}) = A^{(q)}(T_2)^2 - 2^{n+1}I$ and $A^{(q)}(T_4^{\mathrm{prim}}) = A^{(q)}(T_2)^2 - 2^{n+1}I - 2^nI$.

Which the pipeline emits, from which entry point (this is the part the superseded text never said, and the part a reader must have):

entry point	emits
`HeckeOperatorFamily::T_p_pow`(p,ν)	the **full** T_{p^ν}, by the recursion of Thm. 10.8
`HeckeOperatorFamily::T_p_pow_direct`(p,ν)	the **full** T_{p^ν}, by the double-coset assembly (19)
`HeckeOperatorFamily::T_p_pow_primitive_direct`(p,ν)	the **primitive** $T_{\Delta^{\mathrm{prim}}_{p^\nu}}$, the raw pipeline

The raw pipeline `HeckeCosetTables`(p,ν)→`HeckeTauTable`→`HeckeOperatorOnK` computes the *primitive* operator — which is what made the defective wording plausible — but the entry point named `T_p_pow_direct` does not return it: it assembles the full operator from primitive terms by $T^{\mathrm{full}}_{p^\nu} = \sum_{0\le 2j\le \nu} p^{jn}\, T_{\Delta^{\mathrm{prim}}_{p^{\nu-2j}}}$, with $T_{\Delta^{\mathrm{prim}}_{p^0}} := I$. The three coincide at $\nu = 1$, so T_p is unambiguous.

Evidence. Four independent confirmations, obtained from the code and from a run rather than from prose.

1. **The classical decomposition** of the determinant-p^ν Hecke operator into primitive double cosets, [standard: [Shi71, Thm. 3.24(1), (2), p. 63], the $T(p^\nu)$ decomposition; v1.19.0-pre.5: notation corrected; v1.21.0-pre.1, R20-L: locator corrected — the earlier "Thm. 3.34" is the structure of the level-N ring $R(\Gamma',\Delta')$, not this decomposition], is what makes the full operator — and not the primitive one — satisfy the recursion of Theorem 10.8.
2. **The degree count**, which settles it with no convention at all: Δ^{prim}_p has $p+1$ cosets, the full Δ_{p^2} has $1+p+p^2$, and $\Delta^{\mathrm{prim}}_{p^2}$ has p^2+p. The missing coset is the central one, $T(p,p)$, contributing p^n.
3. **The implementation**, whose header states the distinction normatively and whose `T_p_pow_direct` implements the sum above (`Hecke/HeckeOperatorFamily.h`:136–173, `Hecke/HeckeOperatorFamily.cpp`:234–268).
4. **Two runs** on the tree of 2026-09-15 (`RelWithDebInfo`, `make -j16`), each exit code 0:
   - `./cohomology_app HeckeOperatorFamilySmokeValidation` → "PASSED (for $p\in\{2,3\}$, `ell` $\in\{5,13,101\}$, $n\in\{0,2,10\}$: …`T_p_pow(p,2)==T_p_pow_direct(p,2)` [FULL recursion == FULL double-coset], and `T_p_pow_direct(p,2)==T'_{p^2}+p^n*I` [primitive-vs-full decomposition witness])".
   - `./cohomology_app HeckeRelationsValidation` → "O9 PASSED at `(ell,n)` $= (13,0),(13,2),(13,10),(101,0),(101,2),(101,10)$ for all $q = 0..2$ [commutativity, `T_4=T_2^2-2^{n+1}I` (full, independent), …, decomposition witness]". Oracle O9.2 *is* the assertion $A^{(q)}(T_4^{\mathrm{full}}) = A^{(q)}(T_2)^2 - 2^{n+1}I$ and O9.5 *is* $A^{(q)}(T_4^{\mathrm{full}}) = A^{(q)}(T_4^{\mathrm{prim}}) + 2^n I$.

   A hand-checkable instance, from `HeckeOperatorSmokeValidation` (also exit 0) at $(p,\ell,n) = (2,5,0)$, where $A^{(0)}(T_2) = (3) = (1+p)$: the full operator gives $3^2 - 2^{0+1} = 7 \equiv 2 \pmod 5$, which is $\sigma_1(4) = 1+2+4$ (degree $1+p+p^2 = 7$); the primitive gives $7 - 2^0 = 6 \equiv 1$ (degree $p+p^2 = 6$).

Where the correct statement is now made. §10 (Definition 10.4, (18), (19) and the caveat to Theorem 10.8) — which the superseded document had already reworded correctly; §20, Obligation 20.4, with the corrected prescription; and §21, oracle 6, likewise. The half-applied state was itself recorded at the time (`docs/implementation/hecke_action_implementation_plan.md`:45 directed the rewording of §`sec:heckedatum` only; §`sec:obligations` and §`sec:example` were never in its scope, and `:81` carries the same stale phrasing).

A.2 "$H^0(B) = Z^0(B) = B^0$"

Plan §11.1 item 1; a loose step in the cone realisation argument.

Defective statement. `hecke_extensions_OQ2_verdict.md` §2.2: "$H^0(B) = Z^0(B) = B^0$ (no coboundaries in degree 0)". **Correct statement.** $Z^0(B) = \ker\big(d\colon B^0 \to B^1\big) \subseteq B^0$, with equality only if $d^0 = 0$. The first equality is right (there are indeed no coboundaries in degree 0); the second is not. **Evidence.** Immediate. The construction is unaffected: it only ever needs values in $Z^0(B)$. **Corrected in** §15, which states the realisation argument with $Z^0(B)$ and never with B^0.

A.3 A realisation theorem proved over a field, applied over $\mathbb{Z}/\ell^e$

Plan §11.1 item 2. The rule (A1) failure mode, in its purest form.

Defective statement. The realisation theorem of `hecke_extensions_OQ2_verdict.md` §2.2 — "every element of $\mathrm{Hom}(\ker r^1, \operatorname{coker} r^0)$ is realised" — uses a splitting of the cochain groups and is stated for *R a field*. The hypothesis is not repeated in the statement, and the surrounding architecture has since put $R = \mathbb{Z}/\ell^e$ and $\mathbb{Z}/m$ in scope. **Correct statement.** The theorem holds as stated for $R = \mathbb{F}_\ell$. It is **not** claimed over $\mathbb{Z}/\ell^e$ or $\mathbb{Z}/m$, where the splitting is unavailable. **Evidence.** The proof's use of a splitting of R-modules. **Corrected in** §15, where the hypothesis $R = \mathbb{F}_\ell$ is stated in the statement that uses it, and the $\mathbb{Z}/\ell^e$ case is explicitly not claimed.

A.4 τ_∂-uniqueness argued from a theorem whose hypothesis it avoids

Plan §11.1 item 3. Circular as written.

Defective statement. `hecke_extensions_OQ2_verdict.md` §2.6.1 derives the homotopy-uniqueness of τ_∂ from "the comparison theorem". **Why it is defective.** That theorem's relative-projectivity hypothesis is precisely what forces $\ell \geq 5$ in Lemma 11.3(2). Citing it makes the conclusion conditional on exactly the hypothesis the boundary construction is supposed to avoid. **Correct statement.** The conclusion is right over $\mathbb{Z}$ *for every* ℓ, by an explicit argument: $N_0(x)$ is the unique 1-chain of the contractible, one-dimensional component with $\partial N_0(x) = (\tau_\partial - \tau'_\partial)(x)$; $N_1 = 0$ because $Z_1(\partial X) = 0$; and the (T1)-extension is well defined because the boundary stabilisers are central and act trivially. **Corrected in** §14, Lemma 14.10, which gives that explicit argument and never the abstract citation.

A.5 "$\tau(jx)$ ... both are vertices"

Plan §11.1 item 4; loose wording, construction unaffected.

Defective statement. `hecke_extensions_OQ2_verdict.md` §2.3.1: "$\eta_0(x) :=$ an edge path in X from $\tau(jx)$ to $j\tau_\partial(x)$ (both are vertices)". **Correct statement.** $\tau(jx)$ is a 0-chain of augmentation 1, not necessarily a single vertex. The construction survives because X is connected and $\varepsilon\big(j\tau_\partial(x) - \tau(jx)\big) = 0$ (both (T3)s give $\varepsilon = 1$), so the difference is a boundary. **Evidence.** The augmentation computation above. **Corrected in** §15, Lemma 15.6, which states the identity and the augmentation argument rather than the recipe.

A.6 $T_p^2 = T_{p^2} + (p+1)p^n$ — wrong coefficient, false attribution

Survey finding D2; plan §11.1 item 5. A wrong formula, attributed to a document that does not contain it.

Defective statements. `hecke_extensions_OQ2_verdict.md`:278 — "$T_p^2 = T_{p^2} + (p+1)p^n$ with $T(p,p) = p^n$ on Sym^n, [C] Corollary on T_{p^2}". The same error stood at `hecke_extensions_mathematics.md`:63, as "$T(1,q)^2 = T(1,q^2) + (q+1)T(q,q)$" — in Shimura's notation (adopted since v1.19.0-pre.5) this is the *true* identity for the primitive $T(1,q^2) = T_{\Delta^{\mathrm{prim}}_{q^2}}$ $(T(q^2) = T(1,q^2) + T(q,q))$; the defect was its reading with $T(1,q^2)$ as the

full T_{q^2}, the inversion corrected at v1.19.0-pre.5, so the erratum concerns the naming, not the identity. **Correct statement.**

$$T_p^2 \;=\; T_{p^2} + p\cdot T(p,p) \;=\; T_{p^2} + p^{n+1},$$

the coefficient being p, not $p+1$. **Evidence**, three ways.

1. The superseded contract has it right — `hecke_action.tex`:507–516, Theorem 10.8(1): $T_{p^{\nu+1}} = T_p T_{p^\nu} - p\,T(p,p)\,T_{p^{\nu-1}} = T_p T_{p^\nu} - p^{n+1} T_{p^{\nu-1}}$, which at $\nu = 1$ with $T_{p^0} = I$ is the display above. So the attribution "[C] Corollary" is false: [C] says the opposite.
2. The classical statement, [standard: [Shi71, Ch. 3]]: $T_p^2 = T_{p^2} + p^{k-1}\langle p\rangle$ in weight k; here $k = n+2$, so $p^{k-1} = p^{n+1}$, and $\langle p\rangle = T(p,p)$ acts on Sym^n by p^n.
3. The degree count, which needs no convention: $\deg T_p^2 = (p+1)^2 = p^2+2p+1$, $\deg T_{p^2} = 1+p+p^2$ (the *full* operator), $\deg T(p,p) = 1$, so the coefficient is $(p+1)^2 - (1+p+p^2) = p$. The value $p+1$ is the *number of cosets* of Δ_p^{prim} — a different quantity, and presumably how it got in.

Corrected in §10, Theorem 10.8, which already carried the correct relation and carries it here unchanged. The defective instance is recorded because the review document that contains it stands as published and is not edited; `hecke_extensions_mathematics.md`:63 was corrected in place on 2026-09-15. The *conclusion* of the passage — that the Hecke relations hold on the nose on $H_c^\bullet$ — is unaffected.

A.7 A `[C, ...]` citation to a label that was not in [C]

Found during the architecture pass of 2026-09-15; not in the corpus survey.

Defective state. Five C++ source files cite `[C, thm:explicit_complex_equiv]` — the load-bearing isomorphism Ψ — namely `LocalSystemSL2Z.h`:60, `LocalSystemSL2Z.cpp`:190, `InducedLocalSystem.cpp`:150, `HeckeTessellationValidation.h`:36 and `.cpp`:396, and `HeckeTessellationSmokeValidation.cpp`:64. That label was in `monodromy_v5.tex`:587, **not** in [C]. `LocalSystemSL2Z.cpp`:190 hedges by naming both files ("in `docs/latex/hecke_action.tex` / `monodromy_v5.tex`"). **Correct state.** The citations were correct in intent and are now correct in fact: the framework of [v5] is re-homed here in full and the label comes with it. **Evidence.** The label census of the architecture pass; the label is absent from `hecke_action.tex` and present at `monodromy_v5.tex`:587. **Corrected in** §5, which carries Theorem 5.10 verbatim under that label. No source file changes. A label census run against this document (§22.5) would have caught this defect at the time it was introduced, and is registered for exactly that reason.

A.8 "issues with torsion, which we ignore"

Survey finding S1, first bullet. The [v5] §12 erratum block does not cover it.

Defective statement. `monodromy_v5.tex`:1490: "Since $\pi_1(X) \cong \Gamma$ (up to issues with torsion, which we ignore for this theoretical overview)". **Why it is defective.** That torsion is exactly what the model makes load-bearing. The central relation $-I$ is the difference between $H^\bullet = 0$ and a spuriously non-zero $H^\bullet$ for odd n; "ignore" is the wrong instruction, and the sentence appears in a passage a reader reaches while looking for the framework. **Correct statement.** $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ has torsion, π_1 of the quotient *orbifold* is Γ only in the orbifold sense, and the elliptic and central stabilisers enter the equivariance constraints explicitly. **Corrected in** §7: Remark 7.6 and Remark 7.7 state it as a hypothesis with its consequences, and §17 treats the $6 \notin R^\times$ regime properly rather than by exclusion.

A.9 The corestriction convention, unmarked at the source

Survey finding S1, second bullet.

Defective statement. `monodromy_v5.tex`:1536 gives the corestriction as $\omega_*(\psi) = \sum_{\gamma\in\Gamma''\backslash\Gamma}\rho_L(\gamma)\psi(\gamma^{-1})$, which agrees with the convention used here only after the substitution $\gamma \mapsto \gamma^{-1}$ in the summation — a fact recorded at `hecke_action.tex`:906 but *not* marked at [v5], where a transcriber will meet it first. **Correct statement.** True as stated, under its own convention; not interchangeable with the one used here without the substitution. **Corrected in** §12, Proposition 12.7: one convention, fixed once, with the substitution shown explicitly.

A.10 A garbled display in "Simplicial interpretation"

Survey finding S1, third bullet.

Defective statement. `monodromy_v5.tex`:1545–1553 contains `C_*(K^u; Z), W) -> C_eq^*(K,R;W)`: an unbalanced parenthesis and a map with no source. **Disposition. Not re-homed.** [v5] §12 is superseded in full (A.19, A.20); this passage is recorded here so that the supersession is complete and no reader mistakes the omission for an oversight. The mathematics the passage was reaching for is Theorem 5.10 (§5), stated correctly.

A.11 Basis vectors named η, δ, β

Survey collisions C6–C8. Three reserved symbols used for basis vectors, inside the cone section that also needs all three for something else.

Defective usage. `monodromy_v5.tex` §7–§8 writes $\eta_p^{(q)}$, $\delta_s^{(q)}$ and $\beta_j^{(q)}$ for the stored basis vectors of B^q_{eq}, Z^q_{eq} and C^q_{eq} (e.g. `:1098--1109`, `:1182`, `:1208--1211`, `:1325--1328`) — colliding with the pair homotopy η, the coset element δ_j and the right-coset representative β_j respectively, and additionally indexing the first by p, which is the Hecke prime. **Correct usage.** The basis-name scheme $c_j^{(q)}$ / $z_j^{(q)}$ / $b_j^{(q)}$ for C^q_{eq} / Z^q_{eq} / B^q_{eq}, indexed by j. **Corrected in** §2.4 (the scheme), §6 and §16 (the text that uses it). Typographic only: no label and no code identifier moves.

A.12 $H^\bullet(\partial)$ defined as one group and constructed as another

Survey finding D4; verdict §5.5. A definition and its own normative construction disagree inside one document.

Defective state. `hecke_extensions.tex`:189 *defines* $H^\bullet(\partial; M) := H^\bullet(\Gamma_\infty; M)$ with $\Gamma_\infty = \langle T\rangle \times \langle -I\rangle$, while its normative construction (A1), `:755--768`, is the circle model, which computes $H^\bullet(\langle T\rangle; M)$. The two agree only when $2 \in R^\times$. **Correct statement.** The definition is $H^\bullet(\Gamma_\infty; V_R)$. The circle model computes it *under the hypothesis* $2 \in R^\times$, and that hypothesis is part of the statement, not of a preamble. **Corrected in** §14: Definition 14.2 is the definition, Proposition 14.4 is the specialisation statement carrying $2 \in R^\times$ at the point of use, and §14.5 states what is *not* settled at $\ell = 2$ and the policy decided for it (14.24) rather than assuming it away.

A.13 The cone-homotopy sign

Survey finding D5; historical and annotated, but the wrong line is in the body.

Defective statement. `hecke_extensions_plan_audit.md`:271–274 asserts $dh + hd = r\,\Theta_g - \Theta_g^\partial\, r$. A dated correction box three lines later (`:276--288`) gives the right sign, but a reader who stops at the bullet does not reach it. **Correct statement.** $dh + hd = \Theta_g^\partial\, r - r\,\Theta_g$, which this document does not assert but *derives*, from the chain-level identity $\partial\eta + \eta\partial = j\tau_\partial - \tau j$ together with $h := \widetilde{\eta}$. **Evidence.** Re-derived independently twice in the corpus; the audit's own sweep found $17 + 14$ consistent occurrences elsewhere and **zero** silent uses of the wrong orientation, so nothing downstream was contaminated. **Corrected in** §15, where the sign is

fixed once at chain level and the cochain identity is a consequence — so there is no second place for it to be asserted differently.

A.14 Five uncorrected statements in a working note

Survey finding S2. The note is what two other documents cite as their source.

Defective state. `hecke_extensions_mathematics.md` carries five statements corrected only in [E] and in the [E] audit, never in the note itself: `:32` ("the exact '+1' should be double-checked" — the $\mathbb{Z}$-module Sturm statement for M_k is false at $\ell \geq 5$, counterexample $(n, \ell) = (32, 131)$); `:35` (the duality consequence stated for the *Loewy* series); `:39` (the Bellaïche–Khare recall, and "no single T_p ever suffices"); `:63` (the $(p+1)$ coefficient — A.6, since fixed in place); `:67` (the boundary characters presented in the β_j, i.e. right-coset, convention as though it were *the* answer, with only a "Convention warning" as a hedge). **Correct statements.** It is the *socle* series that reverses under duality, not the Loewy series; the boundary characters must be stated in the normative (left-coset δ_j) convention, the β_j convention giving the det-twisted operator and *swapping* the two families; the Sturm and Bellaïche–Khare items are as [E] carries them. **Corrected in** §14, Proposition 14.16 (the characters, in the normative convention, with the twist stated) and §16, Claim 16.26 (socle, not Loewy). The Sturm bound and the Bellaïche–Khare item are programme material and stay in [E], which already carries them correctly. **Status of the note.** `hecke_extensions_mathematics.md` is a working note. It is **not** a source of record, and nothing in this document or in [E] may cite it as one.

A.15 "cone by functoriality"

Survey finding S3.

Defective statement. `hecke_extensions_computational_strategy.md`:15 lists, as a Stage 0 deliverable, "a routine `hecke(q)` returning the matrices of T_p on all X_i (... cone by functoriality)". **Why it is defective.** "By functoriality" is not a definition. The restriction r is Hecke-equivariant only *up to homotopy*, and with a freely chosen cochain homotopy the induced operator on $X_3 = H^1_c$ is genuinely undetermined. Pinning it down is the entire content of §15. **Corrected in** §15, which *is* the definition: h is derived from η and never freely chosen (rule (A6)), and Proposition 15.12 is what makes the result independent of the remaining freedom.

A.16 T_n versus T_N for the composite index

Survey collision C2. The superseded contract contradicted itself on one symbol.

Defective usage. `hecke_action.tex`:80 (the abstract) and `:1439` write T_n for the composite-index operator while n is, throughout the same document, the symmetric power; `:518` correctly writes T_N. **Correct usage.** T_N with $N \in \mathbb{Z}_{>0}$, everywhere; n is the symmetric power and nothing else. **Corrected in** §2.2 (the rule) and §10 (the text). Typographic; no label moves.

A.17 No version marker on either superseded document

Survey finding S6.

Defective state. `hecke_action.tex`:72 and `monodromy_v5.tex`:41 both carry `\date{\today}`, so neither has a citable version and every rebuild silently re-dates the PDF. The committed `hecke_action.pdf` is dated 2026-07-03 and the `.tex` has not changed since, so nothing is out of sync today — but one rebuild would make the PDF and every "as of" statement about it unverifiable, with no diff to show for it. **Correct state.** An explicit version line and a changelog. **Corrected in** §1.4, which fixes the convention for this document, and in the frozen banners both superseded documents received at v1.0.0 ((V4)). `hecke_action.tex`

now carries *Version 1.0 — frozen 2026-07-02* and `monodromy_v5.tex` *Version 5.0 — frozen 2026-07-03*, each date being that of the last commit that changed the file's content; neither file uses `\today` any more, so a rebuild no longer re-dates either PDF and "as of `hecke_action.tex` v1.0" is a verifiable statement.

A.18 The scope cut: nine labels move from [E] into [C]

Not a defect: a recorded relocation, entered here because rule (L1) makes the move of a label an event that must be minuted.

What moved. `sec:six`, `cl:amalgam`, `cl:dimH1`, `cl:bockstein`, `cl:boundary`, `cl:les`, `eq:les`, `cl:chi` and `cl:pd` were defined in `hecke_extensions.tex` and are defined here, in §16. **Why.** They are facts about the objects, not a programme (§1), and they are the oracles Obligation 20.7 reads. A normative obligation may not depend on a versioned, churning companion. **What it costs.** Eleven citation sites in project documents, written `[E, ...]`, become `[C, ...]` — `cl:les` at five, `cl:pd` at two, the remaining six at one each. **No C++ source file cites any of the nine**, so no code changes. The labels themselves are unchanged, per rule (L1). **Completed at v1.0.0.** Until then the move was one-sided: the nine were defined here *and* still defined in the live `hecke_extensions.tex`, which is why (V5)(ii) held the leading version field at 0. `hecke_extensions.tex` v1.3 removes its nine copies, replaces them with a pointer section citing this document, and names as superseded the four of the nine whose content this document *corrected* rather than transported: `cl:chi`'s Dickson-locus sentence (entry A.21), `cl:boundary`'s unhypothesised $H^i(\partial) = 0$ for $i \geq 2$ (Remark 16.22), `cl:les`'s hypothesis pair (Remark 16.1) and `cl:pd`'s self-adjointness wording (Remark 16.27). One `[E, cl:les]` citation survives in the corpus, in the quoted body of `docs/implementation/hecke_extensions_OQ2_verdict.md`; it is a verbatim quotation inside a dated review record and is left as written.

A.19 [v5] §12: the map β is not a morphism of local systems

Recorded at `monodromy_v5.tex`:1345–1368 and superseded there; carried here because [v5] §12 is dropped and this document is now the only record.

Defective statement. `monodromy_v5.tex`:1532–1534 defines a "conjugation" map $\beta_*(\phi)(\gamma) = \rho_L(g)\phi(g^{-1}\gamma g)$ intended as a morphism $\mathrm{Ind}_{\Gamma'}^{\Gamma} V \to \mathrm{Ind}_{\Gamma''}^{\Gamma} V$, and builds the Hecke operator as $\omega \circ \beta \circ \alpha$ (§§12.5–12.7). **Correct statement — a no-go, not a repair.** Two independent reasons. *(i)* The formula does not define a function on Γ at all: $g^{-1}\gamma g \in \Gamma$ only for $\gamma \in \Gamma''$. *(ii)* More decisively, *no* morphism of Γ-modules $\mathrm{Ind}_{\Gamma'}^{\Gamma} V \to \mathrm{Ind}_{\Gamma''}^{\Gamma} V$ whatsoever can make $\omega \circ \mu \circ \alpha$ realise T_p: any such composite $V \to \mathrm{Ind}_{\Gamma'}^{\Gamma} V \to \mathrm{Ind}_{\Gamma''}^{\Gamma} V \to V$ lies in $\mathrm{End}_{\Gamma}(V)$, a scalar ring for irreducible $V = \mathrm{Sym}^{10}\mathbb{Q}^2$, whereas T_2 on $H^1(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^{10}\mathbb{Q}^2)$ has eigenvalues $-24, -24, 2049$ ([standard: Eichler–Shimura in weight 12, [Shi71, Ch. 8]]) and is therefore not scalar. **Corrected in** §12, Remark 12.9, which keeps that label and states the no-go; the middle map is written μ throughout (it is β in [v5] §12, and β_j is a coset representative here). The construction that replaces the sketch is the fused cochain map $\Theta_g = \omega_\bullet \circ \Xi_g$ of Definition 12.10, assembled by Algorithm 12.13, which needs neither the intermediate group Γ' nor any μ. **Consequence for the code.** The fix is implemented: the generalisation of a single constant fibre matrix to per-simplex sparse block rows (`EquivariantCochainComplexMorphismOnK`, `FiberBlockTable`); the legacy single-matrix constructor is unchanged and still produces byte-identical results.

A.20 [v5] §12.3: the bar-cochain formula omits the coset permutation

Recorded at `monodromy_v5.tex`:1369–1379; carried here for the same reason as A.19.

Defective statement. The "Algebraic Hecke Operator" formula of [v5] §12.3, $(T_g\phi)(\gamma_1,\dots,\gamma_s) = \sum_j \rho_L(g_j)\phi(g_j^{-1}\gamma_1 g_j,\dots,g_j^{-1}\gamma_s g_j)$, silently assumes that every γ_i stabilises every coset Γg_j. **Correct statement.** The classical formula must carry the coset permutation $j \mapsto \gamma * j$ explicitly; without it the formula does not commute with the bar differential. **Corrected in** §13: Remark 13.3 states the deviation and Proposition 13.4 gives the corrected degree-1 formula, both keeping their labels. The corrected formula is what the independent bar-cocycle route implements, and that route is cross-checked against the fused pipeline (Obligation 20.5).

A.21 "$\chi_1 \equiv \chi_2$ at every Dickson locus"

Found while Chapter 16 was written (2026-09-16), in a statement being re-homed from [E] by the scope cut of entry A.18. It is a defect in a claim*, not in a working note, and it is corrected at the label that carries it.*

Defective statement. `hecke_extensions.tex`:323–324, inside `cl:chi`: "$\chi_1 \equiv \chi_2$ at every Dickson locus, so Dickson classes cannot discriminate the two indexings ($T_2 = 1$ on L at $(6,5)$: $\chi_1(2) = 2+2^6 \equiv 1$)". **Why it is defective.** By that same claim's own coincidence criterion — $\chi_a \equiv \chi_{a'}$ iff $a \equiv a'$ or $a \equiv n+1-a' \pmod{\ell-1}$, which is correct and is proved at Claim 16.24(3) — one has $\chi_1 \equiv \chi_2$ iff $(\ell-1) \mid (n-2)$. On the Dickson locus $n = a(\ell+1)+b(\ell^2-\ell)$ one has $n \equiv 2a \pmod{\ell-1}$, so the condition reads $(\ell-1) \mid 2(a-1)$, which is *not* automatic. First counterexample: $(n,\ell) = (12,5)$, a Dickson locus ($a=2$, $b=0$, $\dim H^0 = 1$), where $\chi_1(2) = 2+2^{12} \equiv 3$ and $\chi_2(2) = 2^2+2^{11} \equiv 2$ in $\mathbb{F}_5$ — different. **Correct statement.** $\chi_1 \equiv \chi_2$ exactly when $(\ell-1) \mid (n-2)$; in particular at the *first* Dickson class $n = \ell+1$, which is [E]'s own example $(n,\ell) = (6,5)$, and there always, since then $n-2 = \ell-1$. The conclusion [E] drew the sentence for survives in that form: the first Dickson class never separates the left-coset from the right-coset indexing, so the separation must come from the Eisenstein eigenvalue $1+p^{n+1}$ instead. **Corrected in** §16: Claim 16.24(3) states the criterion and proves it, and Remark 16.25 states what is and is not true on the Dickson locus, with the counterexample and the computational check. `cl:chi` keeps its label (rule (L2)); what changes is one sentence of its commentary, and the change is recorded here and in Appendix E. **What it does not affect.** Nothing in this document or in [E] branches on the sentence. Proposition 14.16(2) and Obligation 14.19(vii) already settle the coset convention by the Eisenstein eigenvalue and not by a Dickson class.

A.22 The FP32 totality threshold does not follow from the criterion it is derived from

Found while Chapter 19 was written (2026-09-16), in a statement being re-homed. The threshold is correct and the shipped constant is right; the derivation *of it from the neighbouring corollary is off by the strictness of one inequality, and the two statements therefore do not compose as written.*

Defective statements. `hecke_action.tex`:1385–1391, inside `cor:fptotal`: "If $E(B) < \frac{1}{2}$ … the FP64 filter is *total*"; and :1394–1398, inside `cor:fp32`: "the filter is total iff $2^{-21}B^2 < \frac{1}{2}$, i.e. $B \le 2^{10}$". **Why it is defective.** At $B = 2^{10}$ the binary32 threshold is $E_{32}(2^{10}) = 2^{-21}\cdot 2^{20} = \frac{1}{2}$ exactly, which does *not* satisfy the strict criterion $E < \frac{1}{2}$ that `cor:fptotal` supplies; read literally, `cor:fp32`'s own criterion yields $B \le 1023$, not $B \le 2^{10}$. Separately, "iff" overstates: the criterion is sufficient for totality, not necessary. **Correct statement.** The criterion is the non-strict $E(B) \le \frac{1}{2}$ — the proof needs $E(B)+|\hat{x}-x| < 1$, and $|\hat{x}-x| \le \frac{1}{2}E(B)(1+2^{-53})$, so $E(B) \le \frac{1}{2}$ gives $\frac{3}{4}+2^{-55} < 1$ with room to spare. Under it, $B \le 2^{10}$ is certified in binary32 and $B \le 2^{24}$ in binary64, both as stated. **Evidence, and a second and stronger route.** Independently of any error estimate, both totality regimes are *exact*-arithmetic regimes: $2B^2 \le 2^{\mathrm{prec}}$ makes every product and the sum an exactly rep-

resentable integer, so $\hat{x} = x$ and $|\hat{x}| \le E(B)$ decides $x = 0$ whenever $E(B) < 1$. At $B = 2^{10}$ in binary32, $2B^2 = 2^{21} \le 2^{24}$; at $B = 2^{24}$ in binary64, $2B^2 = 2^{49} \le 2^{53}$. That is the justification the implementation's own validation records at its binary32 totality-boundary case (`Hecke/HeckeFpFilterValidation.cpp`, Phase E: "$E_{32}(2^{10}) = 0.5$ exactly — not < 0.5, so the totality guarantee here rests on the products/sums at this B staying exactly representable in binary32"), which is the same observation, made in the code and not in the contract. **Corrected in** §19: Corollary 19.4 states the criterion as $E(B) + |\hat{x} - x| < 1$ with the two sufficient conditions (i) $E(B) \le \frac{1}{2}$ and (ii) exact representability, and Corollary 19.5 derives $B \le 2^{10}$ from both. Both labels are preserved and both thresholds are unchanged (rule (L2): what changes is the criterion, not the constant). **What it does not affect.** Nothing computed. The constants $2^{-50}B^2$, $2^{-21}B^2$, 2^{24} and 2^{10} are exactly what `Hecke/Predicates.cpp` uses and what the contract box of §19 keeps binding; the 18 source sites citing `cor:fp32` and the 12 citing `cor:fptotal` continue to assert what they asserted.

A.23 The bad-prime oracle cites a corollary that excludes the bad prime

Found while Chapter 21 was written (2026-09-16), in the oracle list being re-homed. The shipped implementation is right; the contract text asserted, at $\ell \in \{2, 3\}$, an identity whose stated justification does not reach there.

Defective statement. `hecke_action.tex`:1604–1607, oracle 7 of the validation plan: "**Bad prime** $p = \ell = 2$, $n \in \{0, 2, 10\}$: pipeline must run and produce well-defined U_2 …; U_2-eigenvalue assertions are *not* made (hypothesis (H3): $\ell = 2$), only $U_4 = U_2^2$ (Corollary `cor:ulrecursion`) and idempotent-stability checks."

Why it is defective. The oracle is careful to withhold eigenvalue assertions at $\ell = 2$ and then asserts one identity anyway, on the strength of a corollary that does not cover it. Corollary 10.10(1) — $T_{\ell^\nu} = U_\ell^\nu$ for the *full* operators — is derived from the recursion of Theorem 10.8(1), which carries $6 \in R^\times$; at $\ell = 2$ that hypothesis fails and the recursion is not available, so the collapse $p^{n+1} = \ell^{n+1} \equiv 0$ has nothing to collapse. Clause (2) of the same corollary carries no hypothesis on ℓ but says something different — that the full and the primitive prime-power operators coincide on $\mathbb{F}_\ell$-coefficients for $n \ge 1$ — and it too fails to give $U_4 = U_2^2$. The statement may well be true at $\ell = 2$; what is wrong is that a numbered oracle asserts it with a citation that excludes its own setting, which is rule (A1)'s failure mode exactly.

Correct statement. Split the oracle by what each clause needs, which is what oracle 7 of §21 now does. For *every* ℓ and n: the pipeline runs to completion and certifies, and $U_\ell := T_\ell$ is well defined (Obligations 20.1, 20.3, 20.11, none of which carries a hypothesis on ℓ). For $\ell \ge 5$ only: $T_{\ell^2} = U_\ell^2$ and the rank-stabilisation check, by Corollary 10.10(1). At $\ell \in \{2, 3\}$ both are **recorded**, not asserted.

Evidence. The implementation already draws this line and records the reason. `HeckeRelationsValidation` hard-asserts $U_{\ell^2} = U_\ell^2$ and the rank $\operatorname{rank} U_\ell^r$ stabilisation at $p = \ell = 5$, and marks the same two quantities "`[A-F1 RECORD ONLY, NOT asserted]`" at $p = \ell \in \{2, 3\}$, where it asserts only that the pipeline completes and the objects are well typed (`Hecke/HeckeRelationsValidation.cpp`:273–330, 389–408); amendment A-F1 of the implementation plan is where that gating was decided. So the gap was noticed in the code and never fixed in the contract — the same shape as A.1, and the reason §21 is now written with the hypothesis on every clause.

Corrected in §21, oracle 7, which carries the split above and an erratum box naming this entry. **What it does not affect.** Nothing computed, and no label moves. Corollary 10.10 itself is unchanged: it was already apportioned in v0.5.0 (Appendix E), and this entry is about the oracle that cited it, not about the corollary.

B Index of statements with STATUS

What this appendix fixes. §1.1 declares that every assertion that is not a definition carries exactly one status tag and that this appendix indexes them. It does: one line per numbered statement of this document, giving its number, its kind, its label, the status tags it actually carries, and — as rule (CL4) directs — its provenance. It is **mechanically derived**, not curated: the status column reports the tags that occur in the statement and in its proof, and nothing is rounded up. Where the document's own rules are not met, the line says so rather than supplying the tag the rule would require.

How to read a line.

- **Status.** One or more of [proved here], [standard], [conjectural], [verified computationally], [open], as §1.1 defines them. **A statement with several clauses may carry different tags on different clauses**, and many do (Claim 16.24 carries all four of [proved here], [conjectural], [verified computationally] and a [standard] pointer, one per clause); this index does not resolve clause structure, and a multi-tag line means "these tags occur in this statement", never "this statement is all of these". The tags on a clause are at the clause.
- **[standard] carries its citation**, in the compact form `\cite[locator]{key}`: the same pointer the statement makes, with the sentence of justification left at the statement (rule (A2) binds the statement, and the terseness licence of (A4) applies here).
- **[verified computationally] is split, because the two kinds are not the same evidence.** [verified computationally]^{r} means a *registered* validation re-runs the check, so the claim is re-executable by name in the shipped tree; [verified computationally]^{a} means an *ad hoc* run recorded in the statement that nothing re-runs — a sweep done once while the chapter was written. Fifteen of the twenty-one are [verified computationally]^{r} (fourteen of nineteen until v1.8.0, which added 23.21 and the ad hoc 16.8) (eight until v1.3.3, which added 14.6, 14.16 and 14.19; eleven until v1.4.0, which added 15.23; twelve until v1.5.0, which added 15.17, whose dimensions are measured and re-runnable since the rewrite of its arithmetic; thirteen until v1.6.0, which added 9.8, whose dimension half is measured by `HeckeInvolutionValidation`) and six are [verified computationally]^{a}; the six are Remarks 15.15, 16.25, 21.3, 22.1, 16.8(3) (since v1.8.0) and Claim 16.24(2), and each of them says so in its own text. Neither kind upgrades to [proved here] (§1.1).
- **A dash** is not a missing tag: it is a statement that makes no assertion of its own. *definition* and *hypothesis* assert nothing; *procedure* (an `alg`) and *predicate* (an `obl`) are specifications, whose correctness is the statement they discharge and whose *status* is in §20.3; *commentary* is a remark that derives from, restates or points at tagged statements it cites. **[untagged]** marks the one place where a dash is a defect.
- **Provenance** ((CL4)) is [new], [re-homed] or [repaired], collapsed to the statement's *current* content: a statement re-homed in one version and repaired in a later one is [repaired] here, and one whose re-homed form was strictly generalised is [new]. [re-homed]/[repaired] marks the one statement whose clauses split (Assumption 3.17: (FD1)–(FD3) re-homed then repaired, (FD4) new). A star marks a statement for which Appendix E carries *no* provenance line at all.

CONTRACT. The column totals of this appendix and of Appendix E's provenance table do not agree, and neither is wrong. The changelog counts *tag instances* — a statement written in one version and revised in another is counted in both, which is what (CL3) requires of a per-version record — and counts container labels (`sec:`, `eq:`) named on a tagged line. This appendix counts *statements*, once each, at their current content. The 226 statements here split 146 [new] (one of them starred) / 45 [re-homed] / 34 [repaired], plus the one whose clauses split both ways (225 and 145 until v1.9.0 added one [new] line, 205 and 125 until v1.8.0 added

twenty), against the changelog's 142/45/38 tag instances over a larger label set (141/45/38 until v1.9.0, 121/45/38 until v1.8.0). The two [repaired] figures are the informative case and they differ for a stated reason: 38 statements were [repaired] in *some* version, and 34 of those are [repaired] at current content — three were later generalised and are now [new], and one is the split case. Quoting either total as the other is the misreading (CL4) exists to stop.

The counts. 434 numbered statements across Chapters 3–22, 23, 24, 25, 26, 27, 28, 29, 30, 31 and 32 (Chapters 1 and 2 carry rules and tables, not numbered statements). By kind: 119 `rmk`, 115 `prop`, 71 `def`, 64 `lem`, 15 `obl`, 22 `thm`, 7 `cl`, 5 `ass`, 7 `cor`, 4 `alg`, 3 `ex`, 2 `open` (the first decided at v1.3.0, the kind kept, rule (L1); the second 25.6, answered at v1.12.0, the kind kept likewise). By status, counting tag *occurrences* (so a multi-tag statement is counted once per tag): [proved here] 261, [standard] 116, [verified computationally] 58 (29 registered, 29 ad hoc), [conjectural] 19, [open] 6. 299 statements carry at least one tag and 135 carry none — 68 definitions, 11 obligations, 5 assumptions, 4 algorithms, 1 example, 45 remarks, and **one corollary**. (v1.25.0-pre.0, a branch pre-release: twenty-three lines added, [new] — §31.6's 31.27 and 31.47 (untagged), 31.28, 31.29 (with [conjectural]), 31.31, 31.34, 31.37, 31.41, 31.45, 31.48, 31.49 ([proved here], [standard]), 31.30, 31.32, 31.33, 31.35, 31.36, 31.38, 31.39, 31.44, 31.46 ([proved here]) and 31.40, 31.42, 31.43 ([conjectural]); so 411 → 434 lines, `def` 69 → 71, `lem` 63 → 64, `prop` 101 → 115, `thm` 20 → 22, `rmk` 115 → 119; [proved here] 243 → 261, [standard] 107 → 116, [conjectural] 15 → 19; tagged 278 → 299, untagged 133 → 135 (definitions 66 → 68). No tag set of an existing line moves; the statements are the last of their section, so no number moves.) (v1.23.0-pre.33: four lines added, [new] — §31.5's 31.23 (untagged), 31.24 ([proved here], [standard]), 31.25 ([proved here], [standard]) and 31.26 ([proved here]); so 407 → 411 lines, `def` 68 → 69, `lem` 62 → 63, `prop` 100 → 101, `rmk` 114 → 115; [proved here] 240 → 243, [standard] 105 → 107; tagged 275 → 278, untagged 132 → 133 (definitions 65 → 66). No tag set of an existing line moves; the statements are the last of their section, so no number moves.) (v1.23.0-pre.24: five lines added, [new] — §31.3's 31.13 ([proved here], [standard]), 31.14 ([proved here], [standard]), 31.15 ([proved here]), 31.16 ([proved here]) and 31.17 ([proved here]); so 397 → 402 lines, `prop` 95 → 98, `thm` 19 → 20, `rmk` 112 → 113; [proved here] 231 → 236, [standard] 100 → 102; tagged 266 → 271. No tag set of an existing line moves; the statements are the last of their section, so no number moves.) (v1.23.0-pre.26: five lines added, [new] — §31.4's 31.18 (untagged), 31.19 ([proved here], [standard]), 31.20 ([proved here], [standard]), 31.21 ([proved here], [standard]) and 31.22 ([proved here]); so 402 → 407 lines, `def` 67 → 68, `lem` 61 → 62, `prop` 98 → 100, `rmk` 113 → 114; [proved here] 236 → 240, [standard] 102 → 105; tagged 271 → 275, untagged 131 → 132 (definitions 64 → 65). No tag set of an existing line moves; the statements are the last of their section, so no number moves.) (v1.23.0-pre.29: no line added, no tag set moves — the C-D7p referee's fixes restate 31.21(1), (5)'s refusals and (1)'s proof inside the existing tags; the counts stand.) (v1.23.0-pre.28: no line added, no tag set moves — 31.21(1), (5) restated for every prime $p \nmid N$ inside its existing tags; the counts stand.) (v1.23.0-pre.27: no line added, no tag set moves — the C-D7 referee's fixes sharpen §31.4's statements inside their existing tags (31.20's locator list corrected, VII.6 withdrawn); the counts stand.) (v1.23.0-pre.25: no line added, no tag set moves — the C-D4 referee's fixes sharpen §31.3's five statements inside their existing tags; the counts stand.) (v1.23.0-pre.23: no line added, no tag set moves — the C-D3g referee's fixes sharpen 31.10(2)(g) and (2)(b) inside their existing tags; the counts stand.) (v1.23.0-pre.22: no line added, no tag set moves — 31.10 gains item (2)(g), proved by elementary row and column operations over $\mathbb{Z}_{(\ell)}$ inside its existing tags; the counts stand.) (v1.23.0-pre.21: no line added, no tag set moves — 31.12(4) is made effective inside its existing tags; the counts stand.) (v1.23.0-pre.20: two lines added, [new] — §31.2's 31.11 (untagged) and 31.12 ([proved here], [standard]); so 395 → 397 lines, `def` 66 → 67,

prop 94 → 95; [proved here] 230 → 231, [standard] 99 → 100; tagged 265 → 266, untagged 130 → 131 (definitions 63 → 64). No tag set of an existing line moves; the statements are the last of their section, so no number moves.) (v1.23.0-pre.19: no line added, no tag set moves — 31.10 gains item (3)(b′), proved by the double coset formula of [Bro82, Ch. III.9], already in its tag; the counts stand.) (v1.23.0-pre.18: one line added, [new] — §31.1's 31.10 ([proved here], [standard]); so 394 → 395 lines, prop 93 → 94; [proved here] 229 → 230, [standard] 98 → 99; tagged 264 → 265. No tag set of an existing line moves; the statement is the last of its section, so no number moves.) (v1.23.0-pre.17: one line added, [new] — §31.1's 31.9 ([proved here], [standard]); so 393 → 394 lines, prop 92 → 93; [proved here] 228 → 229, [standard] 97 → 98; tagged 263 → 264. No tag set of an existing line moves; the statement is the last of its section, so no number moves.) (v1.23.0-pre.16: one line added, [new] — 16.6 ([proved here], no [standard:)]; so 392 → 393 lines, lem 60 → 61; [proved here] 227 → 228; tagged 262 → 263. The tag of 16.12(2) loses its [standard:] on [Wil83] (the proposition keeps [standard] via [Wei94], 97 unchanged); 24.3's statements keep their citations as provenance. **Statement numbers after 16.5 in its section shift by one.**) (v1.23.0-pre.15: no line added; 16.5's tag set gains [proved here] (clause (2), proved in place) and drops [Wil83]; so [proved here] 226 → 227, tagged 262 unchanged. 31.7's citation list gains [Bro82, Ch. V.3].) (v1.23.0-pre.14: two lines added, [new] — §31.1's 31.7 ([proved here], [standard]) and 31.8 ([proved here]); so 390 → 392 lines, prop 90 → 92; [proved here] 224 → 226, [standard] 96 → 97; tagged 260 → 262. No tag set of an existing line moves.) (v1.23.0-pre.12: one line added, [new] — §31's 31.6 ([proved here], [standard]); so 389 → 390 lines, prop 89 → 90; [proved here] 223 → 224, [standard] 95 → 96; tagged 259 → 260. No tag set of an existing line moves. **Not counted, a gap of v1.23.0-pre.3 left to the 1.23.0 cut**: 30.11 and 30.12 have no line in this appendix yet.) (v1.22.0-pre.14: one line added, [new] — §16.4's 16.20 ([proved here]); so 388 → 389 lines, rmk 111 → 112; [proved here] 222 → 223; tagged 258 → 259. No tag set of an existing line moves; the twelve \LEAN markers are not counted.) (v1.22.0-pre.12: one line added, [new] — §16's 16.3 ([proved here]); so 387 → 388 lines, rmk 110 → 111; [proved here] 221 → 222; tagged 257 → 258. 16.2 gains a [proved here] at its clause (1) and 14.22 none, so no tag set of an existing line moves; the seven \LEAN markers are not counted.) (v1.22.0-pre.11: two lines added, [new] — §32's 32.14 (definition, untagged) and 32.15 ([proved here], [standard], [verified computationally][a]); so 385 → 387 lines, prop 88 → 89, def 65 → 66; [proved here] 220 → 221, [standard] 94 → 95, [verified computationally] 57 → 58 (ad hoc 28 → 29); tagged 256 → 257, untagged 129 → 130 (definitions 62 → 63). The F-L23 repairs change no tag set.) (v1.22.0-pre.10: eleven lines added, [new] — §16.4's 16.9, 16.10, 16.11, 16.12, 16.13, 16.14, 16.15 ([proved here], [standard]), 16.16 ([proved here], [conjectural]), 16.17 ([proved here], [standard]), 16.18 ([proved here]) and 16.19 (commentary); so 374 → 385 lines, lem 54 → 60, prop 85 → 88, rmk 108 → 110; [proved here] 210 → 220, [standard] 86 → 94 (eight lines; 16.15 counted once), [conjectural] 14 → 15; tagged 246 → 256, untagged 128 → 129 (remarks 44 → 45). No tag set of an existing line moves. v1.22.0-pre.9 added no line.) (v1.22.0-pre.8: no line added; one tag set grows — 32.23, commentary until pre.7, gains [verified computationally][r] ((MPC12), slot-22 MP-4), so [verified computationally] 56 → 57, registered 28 → 29, tagged 245 → 246, untagged 129 → 128 (remarks 45 → 44); no other count moves.) (v1.22.0-pre.7: no line added; two tag sets grow — 32.21 and 32.22 gain [verified computationally][r] ((MPC10), (MPC11), slot-22 MP-3), so [verified computationally] 54 → 56, registered 26 → 28; no other count moves.) (v1.22.0-pre.6: no line added and no tag moves; the line of 32.21 gains the locator [Bro82, Ch. III.9] of its repaired proof of (2), referee F-MP2 R2.) (v1.22.0-pre.5: twenty-three lines added, [new] — §32's 32.1, 32.2, 32.3, 32.4, 32.5 ([proved here], [verified computationally][r]), 32.6, 32.7 ([proved here], [verified computationally][r]), 32.8, 32.9, 32.10 ([proved here]), 32.11 ([proved here], [standard]), 32.12 ([proved here], [standard]), 32.13 ([proved here], [standard], [verified computationally][r]), 32.16, 32.17 ([proved here], [standard], [verified computationally][r]), 32.18, 32.19 ([proved here], [standard], [verified computationally][r]), 32.20 ([proved here], [standard], [verified computationally][r]), 32.21 and 32.22 ([proved here], [standard]),

32.23 (commentary), 32.24 ([verified computationally]^{r}) and 32.25 ([proved here], [verified computationally]^{r}); so 351 → 374 lines, `def` 57 → 65, `lem` 51 → 54, `thm` 17 → 19, `prop` 81 → 85, `rmk` 103 → 108, `ex` 2 → 3; [proved here] 197 → 210, [standard] 78 → 86, [verified computationally] 46 → 54 (registered 18 → 26); tagged 231 → 245, untagged 120 → 129 (definitions 54 → 62, remarks 44 → 45). No tag set of an existing line moves (the addenda to 17.4, 17.5 and 31.4 add no tag).) (v1.22.0-pre.3: twenty-one lines added, [new] — §29.8's 29.25, 29.26, 29.27, 29.28 ([proved here], [standard]), 29.29, 29.30, 29.31, 29.32, 29.33, 29.34 ([proved here], [standard], [verified computationally]^{a}), 29.35, 29.36 ([proved here], [standard]), 29.37 ([proved here], [standard]), 29.38, 29.39, 29.40 and 29.41 ([proved here], [verified computationally]^{a}), 29.42 ([proved here]), 29.43 (predicate), 29.44 ([proved here], [verified computationally]^{a}) and 29.45 (commentary); so 330 → 351 lines, `def` 53 → 57, `lem` 46 → 51, `prop` 73 → 81, `obl` 14 → 15, `rmk` 100 → 103; [proved here] 182 → 197, [standard] 74 → 78, [verified computationally] 42 → 46 (ad hoc 24 → 28), [open] 6 unchanged (after the F-LES repair R1, which made 29.34(3) and (LES9) [proved here]); tagged 216 → 231, untagged 114 → 120 (definitions 50 → 54, obligations 10 → 11, remarks 43 → 44). No tag set of an existing line moves.) (v1.17.0-pre.1: one line added, [new] — §30's 30.10 ([proved here]); two tag sets grow — 29.14 gains [verified computationally]^{r} ((NB4)(a), `HeckeNebentypusValidation` arm (X4)), 28.16 moves from commentary to [verified computationally]^{a} (slot-16 WP-13's measured fill ratios in its item (4)); so 317 → 318 lines, `rmk` 96 → 97; [proved here] 173 → 174, [verified computationally] 37 → 39 (registered 17 → 18, ad hoc 20 → 21); tagged 205 → 207, untagged 112 → 111 (remarks 43 → 42). The `\LEAN` marker of §1.1 is not counted.) (v1.17.0-pre.4: (NB4)(c)'s sharp form at $n = 0$ tagged [proved here] (given one [standard:] input, locator [AS86, §1.1] not verified; owner ruling of 2026-10-02). 29.14's tag set already holds [proved here], [standard], [conjectural] and [verified computationally]^{r} (the general (NB4) and (NB5) stay [conjectural]), so it is unchanged and **no count moves**: 318 lines; [proved here] 174, [standard] 70, [verified computationally] 39, [conjectural] 13, [open] 6; tagged 207, untagged 111.) (v1.17.0-pre.5: locators only — (NB4)(c)'s input verified ([AS86, Lemmas 1.1.1, 1.1.4]), 30.4's eigenbasis from [DI95], (LN5) from [Mer96, Théorème 2]; the code-map sentences of §31; **no tag or count moves**.) (v1.17.0: the first `\LEAN` / `\LEANlit` markers, 42 on 20 labels; markers are not counted, **no tag or count moves**.) (v1.21.0-pre.4: two lines added, [new] — §28's 28.18 ([proved here], one [standard] step with its locator not verified, [verified computationally]^{a}) and 28.19 ([standard], [conjectural], [verified computationally]^{a}); so 318 → 320 lines, `prop` 70 → 71, `rmk` 97 → 98; [proved here] 174 → 175, [standard] 70 → 72, [verified computationally] 39 → 41 (ad hoc 21 → 23), [conjectural] 13 → 14; tagged 207 → 209, untagged 111. No tag set of an existing line moves (28.12's addendum adds no tag; 28.3 stays a definition).) (v1.22.0-pre.1: ten lines added, [new] — §29.7's 29.15, 29.16 ([proved here], [standard]), 29.17, 29.18, 29.19, 29.20, 29.21 ([proved here], [standard]), 29.22 (each [proved here]), 29.23 ([proved here]; its (NB8) names [U]'s orientation, [verified computationally] at $r = 1$, counted here as [verified computationally]^{a}) and 29.24 (commentary: the code contract); so 320 → 330 lines, `def` 51 → 53, `lem` 42 → 46, `prop` 71 → 73, `rmk` 98 → 100; [proved here] 175 → 182, [standard] 72 → 74, [verified computationally] 41 → 42 (ad hoc 23 → 24); tagged 209 → 216, untagged 111 → 114 (definitions 48 → 50, remarks 42 → 43). No tag set of an existing line moves (29.4's sentence and the refusal box add no tag; the addendum after 27.11 is unnumbered prose and adds none).) (v1.16.0-pre.1: ten lines added, [new] — §28's 28.13, 28.14, 28.15, 28.16, 28.17 and §31's 31.1, 31.2, 31.3, 31.4, 31.5; so 307 → 317 lines, `rmk` 91 → 96, `prop` 66 → 70, `def` 50 → 51; [proved here] 165 → 173, [standard] 68 → 70, [verified computationally] 34 → 37 (ad hoc 17 → 20: slot-16 WP-12's `degen.py` and `negctl.py` in 28.15 and 28.17, slot-14 WP-Q1's `z3check.py` in 31.5), [conjectural] 12 → 13; tagged 197 → 205, untagged 110 → 112 (definitions 47 → 48, remarks 42 → 43).) (v1.15.0-pre.1: twenty-three lines added, [new] — §29's 29.1, 29.2, 29.3, 29.4, 29.5, 29.6, 29.7, 29.8, 29.9, 29.10, 29.11, 29.12, 29.13, 29.14 and §30's 30.1, 30.2, 30.3, 30.4, 30.5, 30.6, 30.7, 30.8, 30.9; so 284 → 307 lines, `rmk` 86 → 91, `prop` 59 → 66, `def` 47 → 50, `lem` 36 → 42, `thm` 16 → 17, `cor` 6 → 7; [proved here] 148 → 165, [standard] 62 → 68,

[verified computationally] 33 → 34 (ad hoc 16 → 17: slot-14 WP-C9's `theta_check.py` in 30.9), [conjectural] 11 → 12; tagged 179 → 197, untagged 105 → 110 (definitions 44 → 47, remarks 40 → 42). 27.12 gains a pointer sentence (P18) and moves no tag.) (v1.14.1: one line added, [new] — §27's 27.15 (the U_ℓ lever's rank claim, owner ruling [S14] Q-H); so 283 → 284 lines, `lem` 35 → 36; [proved here] 147 → 148; tagged 178 → 179, untagged unchanged. The [verified computationally][a] sentences added to (Hi11) and (Hi12) move no tag set.) (v1.14.0-pre.1: twelve lines added, [new] — §28's 28.1, 28.2, 28.3, 28.4, 28.5, 28.6, 28.7, 28.8, 28.9, 28.10, 28.11 and 28.12; so 271 → 283 lines, `rmk` 83 → 86, `prop` 58 → 59, `def` 44 → 47, `lem` 30 → 35; [proved here] 138 → 147, [standard] 57 → 62, [verified computationally] 30 → 33 (ad hoc 13 → 16: WP-4's scripts in 28.10, 28.11, 28.12), [conjectural] 10 → 11; tagged 168 → 178, untagged 103 → 105 (definitions 42 → 44). The [verified computationally][a] sentences added to 27.14 ((Hi11), (Hi12)) and 16.8 move no tag set: both already carry the tag.) (v1.13.0-pre.2: one line added, [new] — 27.7, [standard] (WP-3 referee M2); 27.14 gains [standard] ((Hi10)'s commutativity, referee m9); 27.9 moves to per-item tags (referee m1) with its tag set unchanged; so 270 → 271 lines, `rmk` 82 → 83, [standard] 55 → 57, tagged 167 → 168, untagged unchanged.) (v1.13.0-pre.1: thirteen lines added, [new] — §27's 27.1, 27.2, 27.3, 27.4, 27.5, 27.6, 27.8, 27.9, 27.10, 27.11, 27.12, 27.13 and 27.14; so 257 → 270 lines, `rmk` 79 → 82, `prop` 53 → 58, `def` 42 → 44, `lem` 29 → 30, `thm` 14 → 16; [proved here] 130 → 138, [standard] 49 → 55, [verified computationally] 26 → 30 (ad hoc 9 → 13: WP-2's `levelell_check.py` in 27.4, 27.5, 27.9 and 27.14), [conjectural] 9 → 10; tagged 157 → 167, untagged 100 → 103 (definitions 40 → 42, remarks 39 → 40: 27.6 is commentary). 26.6's (Hi3), (Hi4) gain a locator note and keep their tags.) (v1.12.0: no line added; 25.6 changes tag [open] → [proved here] (owner-signed release, 2026-09-26; 26.2 discharges it, as 14.24's tag moved at v1.3.0); so [proved here] 129 → 130, [open] 7 → 6; tagged/untagged unchanged, kind `open` unchanged.) (v1.12.0-pre.3: two lines added, [new] — 14.17 and 14.18, both [proved here]; so 255 → 257 lines, `prop` 52 → 53, `lem` 28 → 29, [proved here] 127 → 129, tagged 155 → 157. 14.16, 14.19, 16.24 and 14.7 gain status notes and keep their tags: 16.24(2)'s [conjectural] is now its multiset half's alone.) (v1.12.0-pre.1: six lines added, [new] — §26's 26.1, 26.2, 26.3, 26.4, 26.5 and 26.6; so 249 → 255 lines, `rmk` 77 → 79, `prop` 51 → 52, `def` 41 → 42, `lem` 27 → 28, `cor` 5 → 6; [proved here] 122 → 127, [standard] 47 → 49, [verified computationally] 24 → 25 (ad hoc 7 → 8: 26.6's $\tau(2411)$), [conjectural] 8 → 9; tagged 150 → 155, untagged 99 → 100 (definitions 39 → 40). The revised 25.6 and 25.4 keep their tags.) (v1.12.0-pre.2: no line added; 26.4 gains [verified computationally][a] (the WP-3 referee's R5), so [verified computationally] 25 → 26 (ad hoc 8 → 9); 26.6's new classical evidence (R8) is under its existing [verified computationally][a]; tagged and untagged unchanged.) (v1.11.0-pre.1: twenty lines added, [new] — 23.23 and Part VII's nineteen; so 228 → 248 lines, `rmk` 71 → 76, `prop` 46 → 51, `def` 37 → 41, `lem` 24 → 27, `thm` 13 → 14, `cor` 4 → 5, `open` 1 → 2; [proved here] 109 → 121, [standard] 41 → 47, [verified computationally] 22 → 23 (ad hoc 6 → 7: 24.4's script, which its own proof says is ad hoc), [conjectural] 7 → 8, [open] 6 → 7; tagged 135 → 149, untagged 93 → 99 (definitions 35 → 39, remarks 37 → 39). No existing line changes.) (v1.11.1: one line added, [new] — 25.4, [proved here] (the slot-11 WP-15 referee's ruling on F-4); so 248 → 249 lines, `rmk` 76 → 77, [proved here] 121 → 122, tagged 149 → 150; the revised 24.14 and 25.6 keep their tags.) (v1.11.0: no line added; 24.14 gains [verified computationally][r] — slot-11 WP-5 and WP-6's registered validations, its last paragraph — so [verified computationally] 23 → 24 (registered 16 → 17); tagged and untagged unchanged. The revised 20.10, 23.21, 25.6 and 23.23 keep their tags.) (v1.10.0: two lines added, [new] — 23.14, [verified computationally][r], and 23.15, untagged commentary; so 226 → 228 lines, `rmk` 69 → 71, [verified computationally] 21 → 22 (registered 15 → 16), tagged 134 → 135, untagged 92 → 93 (remarks 36 → 37); the revised 14.22, 23.13, 15.22, 23.20, 23.21, 23.22 and 17.4 keep their tags.) (v1.9.0: one line added, [new] — 23.18, [proved here]; so 225 → 226 lines, `rmk` 68 → 69, [proved here] 108 → 109, tagged 133 → 134; the revised 16.8, 23.20, 23.21 and 15.22 keep their tags.) (v1.8.0, re-counted mechanically from the table: 20 lines added, all [new] — §23's nineteen statements and 16.8; [proved here] 93 → 108,

[standard] 34 → 41, [verified computationally] 19 → 21 with 23.21 registered and 16.8 ad hoc, tagged/untagged 117/88 → 133/92; the figures before this sentence were 205 lines and the v1.6.0 counts below.) (v1.3.0: 14.22 added with [proved here] and a [standard] pointer; 14.21 lost its [conjectural] and 14.24 its [open], both by the owner decision and the lemma that discharge them; 14.23 is commentary. v1.3.3 added three [verified computationally]^{r} tags — 14.6, 14.16, 14.19 — and left this paragraph at "13 (8 registered)", which is corrected here to the table's own count. v1.4.0: 14.12 (procedure, its properties [proved here]) and 14.7 (commentary) untagged; 14.13 [proved here]; 15.23 [verified computationally]^{r}. v1.4.1: the split stood at 115/90 with 11 untagged obligations, which counted 14.19 — [verified computationally]^{r} since v1.3.3, and counted as such in the 17 — among the untagged; recounted from the table, 116/89 with 10. v1.5.0: 15.17 gains [verified computationally]^{r} beside the [proved here] and [open] it already carried, its arithmetic having been rewritten from measured values, so [verified computationally] goes 17 → 18 and its registered half 12 → 13; the tagged/untagged split 116/89 is unchanged, the remark having been tagged before. No other status column moved and no line was added or removed: v1.5.0 adds no numbered statement. v1.6.0: 9.8 gains [verified computationally]^{r}, its contingent mirror-split prediction having become a measured record, so [verified computationally] goes 18 → 19 and its registered half 13 → 14; the remark was untagged commentary before, so the tagged/untagged split goes 116/89 → 117/88 and the untagged remarks 36 → 35. No other status column moved and no line was added or removed: v1.6.0 adds no numbered statement either.)

> **CONTRACT. Two gaps in this document, found by compiling this index.** Neither is an erratum ((CL2): Appendix A records the superseded corpus), and neither is repaired here, because repairing another chapter's statement from an index is exactly the silent edit rule (V3) exists to prevent.
>
> 1. **Corollary 10.10 carries no status tag**, although §1.1 requires one of every assertion that is not a definition. Its two clauses are proved inline — (1) from Theorem 10.8(1) under $\ell \geq 5$, (2) from (19) with no hypothesis on ℓ — so the tag it should carry is [proved here]; the index records that it does not carry one rather than supplying it. It is the only untagged statement in the document that is not a definition, a hypothesis, a procedure, a predicate or commentary.
> 2. **Remark 11.2 has no line in Appendix E**: it was introduced in v0.5.0 alongside Definition 11.1's widening, which *is* recorded and which names it, but it never got a line of its own and appears in no "re-homed unchanged" roster. Its provenance is [new] on the evidence of the text; the star on its line records that the changelog does not say so. (CL1) requires one line per statement, so this is a gap in the record and not in the statement.

What this index is not. It is not a summary of what the document proves, and a reader looking for "what should be attacked first" should use Appendix E's roster of the 67 statements that no superseded document and no project note states — those are the ones an error in which cannot be caught by comparing documents. Appendix C is built on that roster for the same reason. Nor is it a coverage report: whether an obligation is *discharged* is §20.3, which is a different question from whether the statement it discharges is proved, and the two must not be read off one another.

no.	kind	label	status	provenance
§3 Simplicial complexes with relations, and the universal cover				
3.1	ass	`ass:K_contractible`	— *hypothesis*	[re-homed]

no.	kind	label	status	provenance
3.2	def	def:quotientcomplex	— *definition*	[new]
3.3	ass	ass:edge_non_ident	— *hypothesis*	[re-homed]
3.4	lem	lemma:pR_simplicial	[proved here]	[re-homed]
3.5	def	def:bdryint	— *definition*	[new]
3.6	ass	ass:open_interior	— *hypothesis*	[re-homed]
3.7	ex	ex:S1	**[untagged]**	[re-homed]
3.8	prop	prop:injectivity	[proved here]	[repaired]
3.9	def	def:saturated	— *definition*	[new]
3.10	prop	prop:saturated_complexes	[proved here]	[repaired]
3.11	def	def:universal_cover	— *definition*	[re-homed]
3.12	prop	prop:universalcover	[standard] [Hat02, §1.3, Props. 1.33–1.34], [Hat02, §1.3], [Hat02, Props. 1.33–1.34]	[re-homed]
3.13	rmk	rmk:injectivityofi	— *commentary*	[re-homed]
3.14	def	def:orderedrelation	— *definition*	[new]
3.15	def	def:relationdatum	— *definition*	[new]
3.16	def	def:kernelN	— *definition*	[new]
3.17	ass	ass:fund_domain	— *hypothesis*	[re-homed]/[repaired]
3.18	rmk	rmk:fd13centre	— *commentary*	[new]
3.19	rmk	rmk:fdconsequences	[proved here]	[new]
3.20	rmk	rmk:twosettings	— *commentary*	[new]
3.21	prop	prop:fund_group_action	[proved here][standard] [Hat02, §1.3]	[re-homed]
3.22	lem	lem:fd4criterion	[proved here]	[new]
3.23	rmk	rmk:fd4criterion	— *commentary*	[new]
3.24	thm	thm:fund_group_generators	[proved here]	[repaired]
§4 Matrix local systems and the three equivalent categories				
4.1	def	def:matrix_ls	— *definition*	[re-homed]
4.2	rmk	rmk:derivedmatrices	— *commentary*	[new]
4.3	prop	prop:matrix_rep_equiv	[proved here]	[re-homed]
4.4	thm	thm:lcs_rep_equiv	[standard] [Hat02, §1.3, Thm. 1.38, and §3.H]	[re-homed]
4.5	thm	thm:rep_ls_equiv	[proved here]	[re-homed]
4.6	cor	cor:threecategories	[proved here]	[new]
4.7	rmk	rmk:lcsnotfree	— *commentary*	[new]
§5 The explicit equivariant cochain complex				
5.1	rmk	rmk:universalcovercohomology	[proved here]	[re-homed]
5.2	thm	thm:cohom_equiv	[proved here][standard] [Bro82, Ch. I.4], [Bro82, Ch. I.7.4], [Bro82, Ch. III.1], [Hat02, §3.H]	[re-homed]
5.3	def	def:naivecochains	— *definition*	[new]
5.4	def	def:eq_cochains	— *definition*	[re-homed]
5.5	lem	lem:eqgroupoid	[proved here]	[new]
5.6	ass	ass:consistent_face_id	— *hypothesis*	[re-homed]
5.7	rmk	rmk:facesautomatic	[proved here]	[new]
5.8	lem	lemma:d_maps_C_eq	[proved here]	[re-homed]
5.9	def	def:eqcohomology	— *definition*	[new]
5.10	thm	thm:explicit_complex_equiv	[proved here]	[repaired]
5.11	rmk	rmk:psiuse	— *commentary*	[new]
5.12	rmk	rmk:psiinjectivelift	— *commentary*	[new]
§6 Systematic computation: bases, projectors, and the stored objects				
6.1	prop	prop:ACq	[proved here]	[new]
6.2	prop	prop:AZq	[proved here]	[new]
6.3	def	def:storedbases	— *definition*	[new]
6.4	prop	prop:Bqimage	[proved here]	[new]
6.5	prop	prop:projector	[proved here]	[new]
6.6	alg	alg:cohomology	— *procedure*	[new]
6.7	rmk	rmk:cost	— *commentary*	[new]
§7 Conventions, the two primes, and the 7-vertex model				
7.1	def	def:model7	— *definition*	[new]
7.2	def	def:model7rel	— *definition*	[new]
7.3	lem	lem:model7coincidences	[proved here][standard] [Ser73, Ch. VII, §1.2, Thms. 1–2]	[new]
7.4	prop	prop:model7h4	[proved here]	[new]
7.5	rmk	rmk:fd4computed	[verified computationally]^{r}	[new]
7.6	rmk	rmk:orbifold	[standard] [Ben91, §3.6], [Bro82, Ch. III.6], [Bro82, Ch. VII, §7]	[repaired]
7.7	rmk	rmk:negI	[proved here]	[repaired]
§8 The Sym^n convention, its semigroup extension, and the contragredient				

no.	kind	label	status	provenance
8.1	def	`def:rhon`	— *definition*	[re-homed]
8.2	prop	`prop:symconv`	[proved here]	[re-homed]
8.3	def	`def:semigroup`	— *definition*	[re-homed]
8.4	lem	`lem:detsym`	[proved here]	[re-homed]
8.5	def	`def:contragredient`	— *definition*	[new]
8.6	prop	`prop:pairing`	[proved here]	[new]
8.7	rmk	`rmk:selfadjoint`	— *commentary*	[new]
8.8	lem	`lem:symdual`	[proved here]	[new]
8.9	rmk	`rmk:divided`	— *commentary*	[new]
§9 The involution ι				
9.1	def	`def:iota`	— *definition*	[new]
9.2	lem	`lem:iotamodel`	[proved here]	[new]
9.3	rmk	`rmk:iotasigns`	— *commentary*	[new]
9.4	prop	`prop:iotacommutes`	[proved here]	[new]
9.5	rmk	`rmk:iotatausym`	[verified computationally]^{r}	[new]
9.6	prop	`prop:iotasplit`	[proved here]	[new]
9.7	rmk	`rmk:iotaelltwo`	— *commentary*	[new]
9.8	rmk	`rmk:iotapairing`	[verified computationally]^{r}	[new]
9.9	rmk	`rmk:iotasign`	— *commentary*	[new]
9.10	obl	`obl:iota`	[conjectural][verified computationally]^{r} [standard] [Ser73, Ch. VII, §3], [Shi71, Ch. 8], [Cre97, Ch. 2]	[new]
§10 The integral Hecke datum for $\mathrm{GL}(2)/\mathrm{SL}_2(\mathbb{Z})$				
10.1	prop	`prop:cosets`	[proved here][standard] [Shi71, §3.2, Lemmas 3.11–3.13]	[new]
10.2	def	`def:cosetreps`	— *definition*	[new]
10.3	lem	`lem:cocycle`	[proved here]	[new]
10.4	def	`def:operators`	[standard] [Shi71, §3.2, Thm. 3.24]	[re-homed]
10.5	prop	`prop:diamond`	[proved here]	[re-homed]
10.6	prop	`prop:fullassembly`	[proved here]	[new]
10.7	rmk	`rmk:rawprimitive`	— *commentary*	[new]
10.8	thm	`thm:relations`	[standard] [Shi71, Thm. 3.24], [AS86, §1], [Bro82, Ch. III.9–10], [Shi71, Thm. 3.24 and §3.2], [Shi71, Prop. 3.16]	[repaired]
10.9	rmk	`rmk:relcoeff`	— *commentary*	[new]
10.10	cor	`cor:ulrecursion`	**[untagged]**	[repaired]
10.11	rmk	`rmk:badprimesingular`	— *commentary*	[new]
§11 The primary definition: $T_{\Gamma g \Gamma}$ in all degrees via a twisted chain map				
11.1	def	`def:tau`	— *definition*	[repaired]
11.2	rmk	`rmk:taurings`	— *commentary*	[new]*
11.3	lem	`lem:tauexists`	[proved here][standard] [Ser73, Ch. VII, §1.2, Thm. 1(1),(3)]	[repaired]
11.4	rmk	`rmk:tauexistsscope`	[proved here][conjectural]	[new]
11.5	rmk	`rmk:tauintegral`	[verified computationally]^{r}	[new]
11.6	def	`def:hecke`	— *definition*	[re-homed]
11.7	thm	`thm:welldef`	[proved here]	[repaired]
11.8	rmk	`rmk:T3`	— *commentary*	[new]
11.9	rmk	`rmk:indeplevel`	— *commentary*	[repaired]
11.10	thm	`thm:classical`	[standard] [Bro82, Ch. III.9–10], [AS86, §1], [Har22, Ch. 2–3], [Shi71, Ch. 8], [Den23], [Bro82, Ch. III.9]	[re-homed]
11.11	rmk	`rmk:whichoperator`	— *commentary*	[new]
§12 Induced local systems and the maps α, μ, ω				
12.1	def	`def:inducedmodule`	— *definition*	[re-homed]
12.2	prop	`prop:blockmatrices`	[proved here]	[re-homed]
12.3	def	`def:inducedls`	— *definition*	[re-homed]
12.4	prop	`prop:lhcompliant`	[proved here][standard] [Bro82, Ch. III.6.2 and Exercise III.8.2]	[repaired]
12.5	rmk	`rmk:shapirouse`	— *commentary*	[new]
12.6	prop	`prop:alpha`	[proved here][standard] [Bro82, Ch. III.6.2 and Exercise III.8.2]	[re-homed]
12.7	prop	`prop:omega`	[proved here][standard] [Bro82, Ch. III.9–10]	[re-homed]
12.8	rmk	`rmk:omegaconvention`	— *commentary*	[new]
12.9	rmk	`rmk:nogo`	[proved here][standard] [Ser73, Ch. VII, §1], [Shi71, Ch. 8]	[repaired]
12.10	def	`def:xi`	— *definition*	[re-homed]
12.11	thm	`thm:xi`	[proved here]	[repaired]
12.12	rmk	`rmk:thetachoices`	— *commentary*	[new]

no.	kind	label	status	provenance
12.13	alg	`alg:assembly`	— *procedure*	[repaired]
§13 Equivalent formulations and cross-checks				
13.1	prop	`prop:tworoutes`	[proved here]	[new]
13.2	prop	`prop:mu`	[proved here]	[new]
13.3	rmk	`rmk:barformula`	[proved here]	[repaired]
13.4	prop	`prop:barformula`	[proved here][standard] [Bro82, Ch. III.9–10], [AS86, §1.1, p. 194], [Bro82, Ch. I.7.4]	[repaired]
13.5	rmk	`rmk:kappaexact`	— *commentary*	[new]
§14 The boundary datum				
14.1	rmk	`rmk:bdrymodel`	[proved here]	[new]
14.2	def	`def:bdrygroup`	— *definition*	[re-homed]
14.3	prop	`prop:bdrycohom`	[proved here][standard] [Bro82, Ch. VII.6]	[new]
14.4	prop	`prop:circlemodel`	[proved here][standard] [Bro82, Ch. III.6]	[repaired]
14.5	rmk	`rmk:bdryoddn`	[proved here]	[new]
14.6	prop	`prop:bdrycosets`	[proved here][verified computationally]^{r}	[new]
14.7	rmk	`rmk:bdrynugeneral`	— *commentary*	[new]
14.8	def	`def:taubdry`	— *definition*	[new]
14.9	lem	`lem:bdrycomponent`	[proved here]	[new]
14.10	lem	`lem:taubdryexists`	[proved here]	[repaired]
14.11	rmk	`rmk:taubdryintegral`	— *commentary*	[new]
14.12	alg	`alg:taubdry`	— *procedure*	[new]
14.13	rmk	`rmk:taubdrynoiota`	[proved here]	[new]
14.14	def	`def:bdryop`	— *definition*	[new]
14.15	prop	`prop:bdryop`	[proved here]	[new]
14.16	prop	`prop:bdrychar`	[proved here][verified computationally]^{r}	[repaired]
14.17	prop	`prop:bdrytranscription`	[proved here]	[new]
14.18	lem	`lem:bdrytorus`	[proved here]	[new]
14.19	obl	`obl:taubdry`	— *predicate*[verified computationally]^{r}	[new]
14.20	lem	`lem:ell2unbounded`	[proved here]	[new]
14.21	rmk	`rmk:ell2window`	— *commentary*	[new]
14.22	lem	`lem:bulkperiodic`	[proved here][standard] [Bro82, Ch. V, §3], [Bro82, Ch. III, §§5–6, 8–9, 10.2], [Ser80, Ch. I, §4.3]	[new]
14.23	rmk	`rmk:bulkperiodicscope`	— *commentary*	[new]
14.24	open	`open:ell2`	[proved here]	[new]
§15 The pair homotopy and the Hecke action on the mapping cone				
15.1	def	`def:pair`	— *definition*	[new]
15.2	lem	`lem:conerelative`	[proved here][standard] [Wei94, Thm. 1.3.1], [Wei94, §1.5]	[new]
15.3	def	`def:cosettransport`	— *definition*	[new]
15.4	lem	`lem:cosettransport`	[proved here]	[new]
15.5	def	`def:pairhomotopy`	— *definition*	[new]
15.6	lem	`lem:etaexists`	[proved here]	[repaired]
15.7	lem	`lem:etaunique`	[proved here]	[new]
15.8	lem	`lem:conesign`	[proved here]	[repaired]
15.9	def	`def:coneaction`	— *definition*	[new]
15.10	lem	`lem:derivedhomotopy`	[proved here]	[new]
15.11	thm	`thm:conewelldef`	[proved here][open][standard] [Har87, §§1–2], [BS73], [Wei94, Thm. 1.3.1]	[new]
15.12	prop	`prop:coneindep`	[proved here]	[new]
15.13	prop	`prop:strictpair`	[proved here][standard] [Ser73, Ch. VII, §1.2, Thm. 1(1),(3)]	[new]
15.14	rmk	`rmk:conesteinberg`	[proved here][conjectural][open][standard] [Bro82, Ch. VII.7], [Wei94, §§10.7, 5.7], [Shi71, Thm. 3.24], [AS86, §1], [Har87, §§1–2], [BS73]	[new]
15.15	rmk	`rmk:conestrict`	[proved here][conjectural][verified computationally]^{a}	[new]
15.16	prop	`prop:coneambiguity`	[proved here]	[repaired]
15.17	rmk	`rmk:coneambiguity691`	[proved here][verified computationally]^{r} [open]	[new]
15.18	prop	`prop:coneintegral`	[proved here][standard] [Wei94, §5.4], [Wei94, Thm. 1.3.1]	[new]
15.19	rmk	`rmk:conetorsion`	— *commentary*	[new]
15.20	prop	`prop:coneroutes`	[proved here][open]	[new]
15.21	obl	`obl:eta`	— *predicate*	[new]
15.22	obl	`obl:coneexact`	— *predicate*	[new]
15.23	rmk	`rmk:conerestrictsmallell`	[verified computationally]^{r}	[new]
§16 The six groups and the long exact sequence				
16.1	rmk	`rmk:sixscope`	— *commentary*	[new]

no.	kind	label	status	provenance
16.2	cl	`cl:amalgam`	[proved here][standard] [Ser80, Ch. I, §4], [Bro82, Ch. III.6], [Bro82, Ch. III.10.2], [Bro82, Ch. III.9–10]	[new]
16.3	rmk	`rmk:treeconeres`	[proved here]	[new]
16.4	lem	`lem:h2torsion`	[proved here][standard] [Bro82, Ch. VII.6]	[new]
16.5	cl	`cl:dimH1`	[proved here][standard] [Ser73, Ch. VII, §3], [Shi71, Ch. 8], [Cre97, Ch. 2], [Den23]	[re-homed]
16.6	lem	`lem:dicksonSL2`	[proved here]	[new]
16.7	cl	`cl:bockstein`	[proved here][standard] [Bro82, Ch. III, §§0–1]	[re-homed]
16.8	rmk	`rmk:bocksteinexp`	[proved here][verified computationally][a] [standard] [Wei94, Thm. 1.3.1], [CCS13, Thm. 4.5] (v1.24.0-rc.1)	[new]
16.9	lem	`lem:centralminusone`	[proved here][standard] [Bro82, Ch. III, §8]	[new]
16.10	lem	`lem:bocksteinallell`	[proved here][standard] [Bro82, Ch. III, §§0–1]	[new]
16.11	lem	`lem:cyclicres`	[proved here][standard] [Bro82, Ch. I.6, Ch. III.1]	[new]
16.12	prop	`prop:oddnltwo`	[proved here][standard] [Wei94, Thm. 1.3.1]	[new]
16.13	lem	`lem:oddnsplit`	[proved here][standard] [Wei94, Thm. 1.3.1]	[new]
16.14	lem	`lem:tellrank`	[proved here][standard] [Bro82, Ch. III.6]	[new]
16.15	prop	`prop:ltwothreescalar`	[proved here][standard] [Ser80, Ch. I, §4], [Shi71, §1.6], [Bro82, Ch. III, §9], [Bro82, Prop. III.9.5(ii)], [Bro82, Ch. VII.6]	[new]
16.16	prop	`prop:oddnhecke`	[proved here][conjectural]	[new]
16.17	lem	`lem:cprule`	[proved here][standard] [Har22, Ch. 2–3]	[new]
16.18	rmk	`rmk:standardtwist`	[proved here]	[new]
16.19	rmk	`rmk:oddnltwoerr`	— *commentary*	[new]
16.20	rmk	`rmk:lvdimsltwothree`	[proved here]	[new]
16.21	cl	`cl:boundary`	[proved here]	[new]
16.22	rmk	`rmk:bdryrepairs`	— *commentary*	[new]
16.23	cl	`cl:les`	[proved here][conjectural]	[new]
16.24	cl	`cl:chi`	[proved here][conjectural][verified computationally][a] [standard] [Ser73, Ch. VI, §4]	[new]
16.25	rmk	`rmk:chicoincide`	[verified computationally][a]	[new]
16.26	cl	`cl:pd`	[open]	[repaired]
16.27	rmk	`rmk:pdscope`	[open]	[new]
§17 The general orbifold model, and the torsion primes				
17.1	def	`def:borelmodel`	[proved here]	[new]
17.2	rmk	`rmk:borelfree`	[proved here][standard] [Wei94, Thm. 2.2.6], [Bro82, Ch. I.7.4]	[new]
17.3	thm	`thm:specialisation`	[proved here][standard] [Bro82, Ch. VII, §7, (7.10)], [Bro82, Ch. III.10.2]	[new]
17.4	rmk	`rmk:mpcaveat`	[conjectural]	[new]
17.5	rmk	`rmk:orbifoldrefine`	— *commentary*	[new]
§18 The combinatorial reduction primitive and integer bounds				
18.1	def	`def:primitives`	— *definition*	[re-homed]
18.2	rmk	`rmk:primitiveclosure`	— *commentary*	[new]
18.3	alg	`alg:tau`	— *procedure*	[repaired]
18.4	rmk	`rmk:certoverGeometry`	— *commentary*	[new]
18.5	lem	`lem:bounds`	[proved here]	[repaired]
18.6	rmk	`rmk:btabsmall`	[verified computationally][r]	[new]
18.7	thm	`thm:backend`	[proved here]	[repaired]
§19 The certified floating-point filter				
19.1	lem	`lem:predshapes`	[proved here]	[new]
19.2	thm	`thm:fp64`	[proved here]	[repaired]
19.3	rmk	`rmk:thresholdrounding`	[proved here]	[new]
19.4	cor	`cor:fptotal`	[proved here]	[repaired]
19.5	cor	`cor:fp32`	[proved here]	[repaired]
§20 Correctness obligations				
20.1	obl	`obl:welldef`	— *predicate*	[re-homed]
20.2	obl	`obl:indep`	— *predicate*	[repaired]
20.3	obl	`obl:integral`	— *predicate*	[re-homed]
20.4	obl	`obl:algebra`	[standard] [Shi71, Thm. 3.24, §3.2–3.3], [AS86, §1], [Bro82, Ch. III.9–10]	[repaired]
20.5	obl	`obl:agreement`	— *predicate*	[re-homed]
20.6	obl	`obl:shapiro`	— *predicate*	[re-homed]
20.7	obl	`obl:oracle`	[standard] [Shi71, Ch. 8], [Har22, Ch. 2–3], [Den23]	[repaired]
20.8	rmk	`rmk:dickson`	[proved here][verified computationally][r]	[re-homed]
20.9	obl	`obl:fp`	— *predicate*	[re-homed]

no.	kind	label	status	provenance
20.10	obl	obl:backend	— *predicate*	[re-homed]
20.11	obl	obl:tau	— *predicate*	[re-homed]
§21 Worked micro-example ($p = 2$) and the numerical oracles				
21.1	prop	prop:t2cocycle	[proved here][verified computationally][r]	[re-homed]
21.2	ex	ex:t2tau	[verified computationally][r]	[new]
21.3	rmk	rmk:dicksontwo	[verified computationally][a]	[new]
§22 Implementation notes: mapping to the code base				
22.1	rmk	rmk:labelcensus	[verified computationally][a]	[new]
§23 The integral line (v1.8.0)				
23.1	def	def:ringinforce	— *definition*	[new]
23.2	prop	prop:ringinforce	[proved here]	[new]
23.3	prop	prop:bredon	[proved here][standard] [Wei94, Thm. 2.2.6]	[new]
23.4	def	def:isomodel	— *definition*	[new]
23.5	prop	prop:isomodel	[proved here][standard] [Wei94, Thm. 2.2.6], [Bro82, Ch. III.6], [Bro82, Ch. III.10.2]	[new]
23.6	rmk	rmk:cafive	[proved here]	[new]
23.7	prop	prop:fracture	[proved here][standard] [Wei94, §1.5]	[new]
23.8	prop	prop:cohglue	[proved here]	[new]
23.9	prop	prop:descent	[proved here]	[new]
23.10	prop	prop:homvanish	[proved here][standard] [Bro82, Ch. III, §8], [Wei94, §10.7]	[new]
23.11	rmk	rmk:truncdevice	[proved here]	[new]
23.12	lem	lem:strictlifting	[proved here]	[new]
23.13	prop	prop:strictlift	[proved here][standard] [Wei94, Thm. 2.2.6]	[new]
23.14	rmk	rmk:strictliftbuilt	[verified computationally][r]	[new]
23.15	rmk	rmk:bredonhecke	— *commentary*	[new]
23.16	prop	prop:badprimes	[proved here]	[new]
23.17	prop	prop:certifiede	[proved here][standard] [Wei94, §3.6]	[new]
23.18	rmk	rmk:certifiedeperdiff	[proved here]	[new]
23.19	prop	prop:crtcharpoly	[proved here]	[new]
23.20	def	def:ringsemantics	— *definition*	[new]
23.21	rmk	rmk:integralevidence	[verified computationally][r]	[new]
23.22	rmk	rmk:integralscope	— *commentary*	[new]
23.23	prop	prop:relationsmodel	[proved here][standard] [Shi71, Thm. 3.24], [RW70, Thm. 1], [AS86, §1], [Bro82, Ch. III.9–10]	[new]
§24 The Dickson invariant and θ_{L} (v1.11.0-pre.1)				
24.1	def	def:dicksoninv	— *definition*	[new]
24.2	lem	lem:dicksontwist	[proved here]	[new]
24.3	prop	prop:dicksonH0	[proved here][standard] [Dic11, §§2–3], [Wil83, §1]	[new]
24.4	prop	prop:heckeH0	[proved here][standard] [Luc78, §XXI] [verified computationally][a]	[new]
24.5	rmk	rmk:heckeH0plan	— *commentary*	[new]
24.6	lem	lem:heckefibre	[proved here]	[new]
24.7	def	def:Qn	— *definition*	[new]
24.8	lem	lem:Qnfunctions	[proved here]	[new]
24.9	def	def:theta	— *definition*	[new]
24.10	prop	prop:thetatwist	[proved here]	[new]
24.11	thm	thm:thetales	[proved here][standard] [Wei94, Thm. 1.3.1], [Shi71, Ch. 8], [Den23]	[new]
24.12	rmk	rmk:thetasmallell	— *commentary*	[new]
24.13	rmk	rmk:thetaclassical	[standard] [Ser72], [SD73, §3], [Kat77], [Joc82a, Thm. 4.1], [AS86, Lemma 2.1], [Edi92, §3], [Joc82b], [Gro90], [CV92]	[new]
24.14	rmk	rmk:thetacheckable	[proved here][standard] [SD73, §1] [conjectural][verified computationally][r]	[new]
§25 The ordinary projector (v1.11.0-pre.1)				
25.1	def	def:ordinary	— *definition*	[new]
25.2	prop	prop:ordinary	[proved here]	[new]
25.3	cor	cor:thetanonordinary	[proved here]	[new]
25.4	rmk	rmk:ordtorsionfree	[proved here]	[new]
25.5	rmk	rmk:ordinaryeis	[proved here]	[new]
25.6	open	open:ordprecision	[proved here]	[new]
§26 Hida theory at level one: foundation (v1.12.0-pre.1)				
26.1	prop	prop:torsionprepass	[proved here][standard] [Bro82, Ch. III, §§0–1]	[new]
26.2	cor	cor:ordprecision	[proved here]	[new]

no.	kind	label	status	provenance
26.3	lem	`lem:ordexponent`	[proved here]	[new]
26.4	rmk	`rmk:ordexponentuse`	[proved here][verified computationally][a]	[new]
26.5	def	`def:hidatower`	— *definition*	[new]
26.6	rmk	`rmk:hidacheckable`	[proved here][standard] [Har22, Ch. 2–3], [Shi71, Ch. 8] [conjectural][verified computationally][a]	[new]
§27 Hida theory at level one: control, level ℓ, the lattice (v1.13.0-pre.1; 27.7 pre.2; 27.15 v1.14.1)				
27.1	def	`def:levelellmodel`	— *definition*	[new]
27.2	prop	`prop:levelcorrespondence`	[proved here][standard] [Bro82, Ch. III.9–10]	[new]
27.3	prop	`prop:levelellU`	[proved here]	[new]
27.4	prop	`prop:oldspace`	[proved here][verified computationally][a]	[new]
27.5	prop	`prop:levelorbit`	[proved here][standard] [Bro82, Ch. III.6.2] [verified computationally][a]	[new]
27.6	rmk	`rmk:levelcost`	— *commentary*	[new]
27.7	rmk	`rmk:levelscaling`	[standard] [Man72], [Mer94, §2], [Ste07, Ch. 3, 8], [Cre97, Ch. 2]	[new]
27.8	lem	`lem:levelelltorsion`	[proved here][standard] [Bro82, Ch. III.6.2]	[new]
27.9	prop	`prop:stabilisation`	[proved here][standard] [Hid86a, Prop. 4.7, Lemma 4.6], [Shi71, AL70] [verified computationally][a]	[new]
27.10	rmk	`rmk:stabconvention`	[proved here]	[new]
27.11	thm	`thm:hidacontrol`	[standard] [Hid86a, Thm. 3.1, Thm. 6.6], [Hid86b, Thm. 3.1, Cor. 3.2]	[new]
27.12	thm	`thm:hidaweight`	[standard] [Hid86b, Thm. 3.1], [Hid86a, Lemma 4.6, Prop. 4.7]	[new]
27.13	def	`def:hidalattice`	— *definition*	[new]
27.14	rmk	`rmk:hidacheckable2`	[proved here][standard] [Shi71, Ch. 3] [conjectural][verified computationally][a]	[new]
27.15	lem	`lem:rhorank`	[proved here]	[new]
§28 Level N: the composite coset table, the cokernel presentation and its Hecke action (v1.14.0-pre.1)				
28.1	def	`def:cosetrepsN`	— *definition*	[new]
28.2	lem	`lem:cosetrepsN`	[proved here]	[new]
28.3	def	`def:levelNmodel`	— *definition*	[new]
28.4	lem	`lem:treeres`	[proved here]	[new]
28.5	lem	`lem:manin`	[proved here][standard] [Man72]	[new]
28.6	def	`def:kappaN`	[proved here]	[new]
28.7	lem	`lem:levelduality`	[proved here][standard] [Bro82, Ch. III.8]	[new]
28.8	lem	`lem:merelpath`	[proved here][standard] [Mer94, §2]	[new]
28.9	prop	`prop:heilbronn`	[proved here]	[new]
28.10	rmk	`rmk:heilbronnconv`	[proved here][verified computationally][a]	[new]
28.11	rmk	`rmk:heilbronncost`	[standard] [Cre97, Ch. 2] [verified computationally][a]	[new]
28.12	rmk	`rmk:levelNcheckable`	[proved here][standard] [Shi71, Maz77, WWE20], [Mer96, Théorème 2] [conjectural][verified computationally][a]	[new]
28.13	prop	`prop:kappamodel`	[proved here]	[new]
28.14	prop	`prop:degenpres`	[proved here]	[new]
28.15	rmk	`rmk:degenpresconv`	[proved here][verified computationally][a]	[new]
28.16	rmk	`rmk:levelNscale`	[verified computationally][a]	[new]
28.17	rmk	`rmk:levelNcheckabletwo`	[proved here][standard] [AL70, BM16] [conjectural][verified computationally][a]	[new]
28.18	prop	`prop:levelNcurve`	[proved here][standard] [Ser80, Ch. I, §4.2] [verified computationally][a]	[new]
28.19	rmk	`rmk:levelNcurvefamily`	[standard] [Maz77, CE05], [Stu87, Thm. 1] [conjectural][verified computationally][a]	[new]
§29 Nebentypus: $\Gamma_1(N)$ and $(\Gamma_0(N), \chi)$ on the fixed model by Shapiro (v1.15.0-pre.1)				
29.1	def	`def:nebdata`	— *definition*	[new]
29.2	lem	`lem:nebdata`	[proved here]	[new]
29.3	lem	`lem:teichmuller`	[proved here]	[new]
29.4	rmk	`rmk:nebfield`	— *commentary*	[new]
29.5	lem	`lem:inducedchar`	[proved here]	[new]
29.6	lem	`lem:gammaonetable`	[proved here]	[new]
29.7	def	`def:nebmodel`	— *definition*	[new]
29.8	prop	`prop:nebdecomp`	[proved here]	[new]
29.9	prop	`prop:nebcorr`	[proved here][standard] [Bro82, Ch. III.9–10], [Shi71, §3.3]	[new]
29.10	thm	`thm:nebrelations`	[standard] [Shi71, §3.3], [Bro82, Ch. III.9–10]	[new]
29.11	prop	`prop:neborbit`	[proved here]	[new]
29.12	rmk	`rmk:nebheilbronn`	[proved here]	[new]
29.13	rmk	`rmk:nebcost`	— *commentary*	[new]

no.	kind	label	status	provenance
29.14	rmk	rmk:nebcheckable	[proved here][standard] [Shi71, CO77, Rib77], [AS86, Lemmas 1.1.1, 1.1.4] [conjectural][verified computationally][r]	[new]
29.15	def	def:nebwild	— *definition*	[new]
29.16	lem	lem:nebwild	[proved here][standard] [Was97, Lemma 1.4, Prop. 2.1]	[new]
29.17	def	def:nebgammaH	— *definition*	[new]
29.18	prop	prop:nebgammaH	[proved here]	[new]
29.19	prop	prop:nebgammaHheilbronn	[proved here]	[new]
29.20	lem	lem:nebshapiro	[proved here]	[new]
29.21	lem	lem:nebwildtors	[proved here][standard] [AS86, Lemma 1.1.1, p. 194]	[new]
29.22	lem	lem:nebchainrref	[proved here]	[new]
29.23	rmk	rmk:nebwildcheckable	[proved here][verified computationally][a]	[new]
29.24	rmk	rmk:nebcosetlevel	— *commentary*	[new]
29.25	def	def:levelpair	— *definition*	[new]
29.26	lem	lem:nebcompat	[proved here]	[new]
29.27	def	def:levelres	— *definition*	[new]
29.28	lem	lem:levelresmorph	[proved here][standard] [Bro82, Ch. III.9–10]	[new]
29.29	def	def:levelcone	— *definition*	[new]
29.30	lem	lem:levelconeexact	[proved here]	[new]
29.31	prop	prop:levelconeshapiro	[proved here]	[new]
29.32	prop	prop:modeltorsion	[proved here]	[new]
29.33	prop	prop:levelsplit	[proved here]	[new]
29.34	prop	prop:levelconehecke	[proved here][standard] [Bro82, Ch. III.9–10] [verified computationally][a]	[new]
29.35	def	def:normalpair	— *definition*	[new]
29.36	prop	prop:levelinfres	[proved here][standard] [Bro82, Ch. VII, §6; Ch. III, Cor. 10.2]	[new]
29.37	prop	prop:cyclicconе	[proved here][standard] [Bro82, Ch. III, §1, Example 2]	[new]
29.38	lem	lem:levelfreeeuler	[proved here]	[new]
29.39	lem	lem:unipotentinv	[proved here]	[new]
29.40	prop	prop:gamma0normal	[proved here][verified computationally][a]	[new]
29.41	prop	prop:hidanormalsteps	[proved here][verified computationally][a]	[new]
29.42	rmk	rmk:normalN1	[proved here]	[new]
29.43	obl	obl:levelconeexact	— *predicate*	[new]
29.44	rmk	rmk:lescheckable	[proved here][verified computationally][a]	[new]
29.45	rmk	rmk:oldspacecone	— *commentary*	[new]
§30 The $\mathbb{Z}/\ell^K$ Hecke algebra: saturation, the defect, the comparison with forms (v1.15.0-pre.1)				
30.1	def	def:satdata	— *definition*	[new]
30.2	prop	prop:saturation	[proved here]	[new]
30.3	prop	prop:satdefect	[proved here]	[new]
30.4	lem	lem:heckerankone	[proved here][standard] [DI95, §§3.5–3.6, Thm. 12.2.2, Prop. 12.4.10], [Shi71, Thm. 3.41, Ch. 8], [DS05, Thm. 5.5.4], [Ser73, Ch. VII]	[new]
30.5	lem	lem:freepart	[proved here]	[new]
30.6	cor	cor:stablecert	[proved here]	[new]
30.7	prop	prop:cohforms	[proved here][standard] [Shi71, Ch. 8]	[new]
30.8	prop	prop:hidaalgconst	[proved here][standard] [Hid86b, Thm. 3.1, Cor. 3.2]	[new]
30.9	rmk	rmk:thetaforms	[proved here][verified computationally][a]	[new]
30.10	rmk	rmk:sattor	[proved here]	[new]
§31 The ℓ-local isotropy model, and the closed form at $\ell = 3$ (v1.16.0-pre.1)				
31.1	def	def:isomodelell	— *definition*	[new]
31.2	prop	prop:isomodelell	[proved here][standard] [Bro82, Ch. III.6.2]	[new]
31.3	prop	prop:ellthreeclosed	[proved here]	[new]
31.4	rmk	rmk:isomodellmp	[proved here]	[new]
31.5	rmk	rmk:isomodellcheckable	[proved here][verified computationally][a]	[new]
31.6	prop	prop:certifiedeell	[proved here][standard] [Wei94, §3.6], [Bro82, Ch. III.10]	[new]
31.7	prop	prop:descentN	[proved here][standard] [Bro82, Ch. III.6.2, III.10.2, V.3], [Wei94, §10.7]	[new]
31.8	prop	prop:badprimesN	[proved here]	[new]
31.9	prop	prop:descentNthree	[proved here][standard] [Bro82, Ch. III.6.2], [Wei94, Thm. 2.2.6]	[new]
31.10	prop	prop:descentmaps	[proved here][standard] [Bro82, Ch. III.6.2, III.8–10, VII.6], [Wei94, Thm. 2.2.6]	[new]
31.11	def	def:isomodeltree	— *definition*	[new]
31.12	prop	prop:isomodeltree	[proved here][standard] [Wei94, Thm. 2.2.6], [Bro82, Ch. III.6.2]	[new]

no.	kind	label	status	provenance
31.13	prop	`prop:strictliftN`	[proved here][standard] [Bro82, Ch. III.8–10], [Wei94, Thm. 2.2.6]	[new]
31.14	prop	`prop:crtcharpolyN`	[proved here][standard] [DI95, Thm. 12.2.2, Prop. 12.4.10], [Del74], [DS05]	[new]
31.15	prop	`prop:torsionhecke`	[proved here]	[new]
31.16	thm	`thm:descentcert`	[proved here]	[new]
31.17	rmk	`rmk:descentcertscope`	[proved here]	[new]
31.18	def	`def:descentpair`	— *definition*	[new]
31.19	lem	`lem:pairlifting`	[proved here][standard] [Wei94, Thm. 2.2.6]	[new]
31.20	prop	`prop:descentpair`	[proved here][standard] [Bro82, Ch. III.2, III.5–6, III.8–9, V.1–3, VI.9], [Wei94, Thm. 1.3.1, Thm. 2.2.6]	[new]
31.21	prop	`prop:descentpairhecke`	[proved here][standard] [Bro82, Ch. III.2, III.8–10], [Wei94, Thm. 2.2.6]	[new]
31.22	rmk	`rmk:descentpairscope`	[proved here]	[new]
31.23	def	`def:treepair`	— *definition*	[new]
31.24	lem	`lem:treepaircyl`	[proved here][standard] [Wei94, §1.5]	[new]
31.25	prop	`prop:treepair`	[proved here][standard] [Wei94, Thm. 2.2.6, §1.5], [Bro82, Ch. III.6, III.8–9]	[new]
31.26	rmk	`rmk:treepairscope`	[proved here]	[new]
31.27	def	`def:heckeZ`	— *definition*	[new]
31.28	prop	`prop:heckeZspec`	[proved here][standard] [Shi71, Ch. 3]	[new]
31.29	prop	`prop:heckeZrat`	[proved here][standard] [DI95, Thm. 12.2.2, Props. 12.4.10, 12.4.13, §6], [Ste07, Thms. 5.8–5.10, 9.23], [AL70], [CE98], [Stu87] [conjectural]	[new]
31.30	lem	`lem:heckeZbridge`	[proved here]	[new]
31.31	prop	`prop:heckeZmax`	[proved here][standard] [AM69, Prop. 2.4]	[new]
31.32	prop	`prop:heckeZgen`	[proved here]	[new]
31.33	prop	`prop:heckeZetale`	[proved here]	[new]
31.34	prop	`prop:heckeZmono`	[proved here][standard] [BH93, Prop. 3.1.19]	[new]
31.35	prop	`prop:heckeZlocal`	[proved here]	[new]
31.36	prop	`prop:heckeZthrough`	[proved here]	[new]
31.37	prop	`prop:heckeZuct`	[proved here][standard] [Wei94, §3.6, Thm. 1.3.1]	[new]
31.38	prop	`prop:heckeZsplit`	[proved here]	[new]
31.39	prop	`prop:heckeZlevelone23`	[proved here]	[new]
31.40	rmk	`rmk:heckeZcbone`	[conjectural]	[new]
31.41	prop	`prop:heckeZglue`	[proved here][standard] [AM69, Cor. 10.10]	[new]
31.42	rmk	`rmk:heckeZcbtwo`	[conjectural]	[new]
31.43	rmk	`rmk:heckeZcbthree`	[conjectural]	[new]
31.44	thm	`thm:heckeZdescent`	[proved here]	[new]
31.45	thm	`thm:heckeZcert`	[proved here][standard] [Del74]	[new]
31.46	rmk	`rmk:heckeZcaveat`	[proved here]	[new]
31.47	def	`def:heckeZcong`	— *definition*	[new]
31.48	prop	`prop:heckeZcong`	[proved here][standard] [Del74]	[new]
31.49	prop	`prop:heckeZeiscert`	[proved here][standard] [Maz77, II (9.7), (16.1), Cor. (16.2), (16.6)]	[new]
§32 The Moerdijk–Pronk model (v1.22.0-pre.5)				
32.1	def	`def:mptri`	— *definition*	[new]
32.2	def	`def:mprules`	— *definition*	[new]
32.3	def	`def:mpell`	— *definition*	[new]
32.4	def	`def:mpRZ`	— *definition*	[new]
32.5	rmk	`rmk:mprulesread`	[proved here][verified computationally]^{r}	[new]
32.6	def	`def:mpset`	— *definition*	[new]
32.7	rmk	`rmk:mpwhat`	[proved here][verified computationally]^{r}	[new]
32.8	def	`def:mpfree`	— *definition*	[new]
32.9	lem	`lem:mprows`	[proved here]	[new]
32.10	lem	`lem:mpcols`	[proved here]	[new]
32.11	lem	`lem:mpassembly`	[proved here][standard] [Wei94, Lemma 2.7.3]	[new]
32.12	thm	`thm:mpcomparison`	[proved here][standard] [Wei94, §8.3, Thm. 2.2.6]	[new]
32.13	rmk	`rmk:mphybrid`	[proved here][standard] [Wei94, Thm. 2.2.6] [verified computationally]^{r}	[new]
32.14	def	`def:mpqd`	— *definition*	[new]
32.15	prop	`prop:mpqreduced`	[proved here][standard] [Wei94, Thm. 2.2.6, §8.3] [verified computationally]^{a}	[new]
32.16	def	`def:mpphi`	— *definition*	[new]
32.17	prop	`prop:mpalldegrees`	[proved here][standard] [Wei94, Thm. 2.2.6] [verified computationally]^{r}	[new]
32.18	def	`def:mpmorphism`	— *definition*	[new]
32.19	thm	`thm:mpnatural`	[proved here][standard] [Wei94, Thm. 2.2.6], [Bro82, Ch. III.8] [verified computationally]^{r}	[new]

no.	kind	label	status	provenance
32.20	prop	`prop:mpbredon`	[proved here][standard] [Wei94, Thm. 2.2.6] [verified computationally]^{r}	[new]
32.21	prop	`prop:mpboundary`	[proved here][standard] [Bro82, Ch. III.6.2, III.9] [verified computationally]^{r}	[new]
32.22	prop	`prop:mprescor`	[proved here][standard] [Bro82, Ch. III.9–10] [verified computationally]^{r}	[new]
32.23	rmk	`rmk:mphecke`	[verified computationally]^{r}	[new]
32.24	ex	`ex:mpteardrop`	[verified computationally]^{r}	[new]
32.25	rmk	`rmk:mpcheckable`	[proved here][verified computationally]^{r}	[new]

C Formalization notes

What this appendix is, and what it is not. The owner intends to commission an assessment of a Lean 4 / `mathlib4` formalization of this document. **This appendix is the input to that commission, not the commission.** It says what the document's chapters have each recorded about their own formalizability, reconciled into one place: which objects a formalization would have to build, which statements it should import rather than re-prove, which hypotheses it must not lose, and which statements are the cheapest and the most valuable to attack. It deliberately **does not** estimate effort, propose a schedule, or rank the work by cost: that is the assessment, and an appendix that pre-empted it with guesses would be supplying the answer it is meant to supply the data for.

> **CONTRACT. Every claim below about what `mathlib4` contains is a claim made by the chapter that made it, and was not re-verified against `mathlib4` in this pass.** No version of the library was consulted while this document was written; the per-chapter notes were written by authors reasoning about what a library of that scope plausibly carries. The commission must check each one. They are recorded here because a wrong belief about the library is cheap to correct and expensive to discover late — but a reader must not treat "`mathlib4` has this" in this document as a fact about `mathlib4`.

C.1 What a formalization would have to build

These have no counterpart the chapters expect to find, and are the substance of the work.

1. **The $(K, \mathcal{R})$ framework** (Chapter 3): Definition 3.15, the quotient complex 3.2, saturation 3.9, and the fundamental-domain hypotheses (FD1)–(FD4). The chapter's own note is that this is formalization-*friendly* rather than hard: for finite K with decidable equality on $o_\bullet(K^u)$ the coincidence groupoid is a decidable relation and (FD4) is reachability in a finite graph. It is nonetheless entirely new construction.
2. **The isomorphism** Ψ of Theorem 5.10(1), with Φ given by (5). It is finite, constructive, needs no topology, no field and no freeness, and is the statement six source files cite. It is the single most load-bearing object of Part I.
3. **Sym^n as a monoid representation over an arbitrary** R (Definition 8.1, Proposition 8.2). What is needed is not Sym^n of a module — the chapters expect that, via `MvPolynomial` — but the extension of the action from GL_2 to the monoid Δ of Definition 8.3, in the explicit basis $u_j = x^{n-j}y^j$ with the closed form (9), over $R = \mathbb{Z}/m$ as well as over a field. The contragredient D_n and the pairing (Definition 8.5, Proposition 8.6) come with it and are finite linear algebra over a commutative ring.

4. **Twisted chain data**: Definition 11.1's (T1)–(T3), the boundary analogue 14.8, and the pair homotopy 15.5. These are equational conditions on maps of complexes and are directly expressible; the *constructions* (Algorithm 18.3, Lemmas 14.10 and 15.6) are terminating procedures on finite data whose correctness reduces to $Z_1 = 0$ on a tree and $Z_2 = 0$ on a contractible 2-complex — simplicial chains, not homological algebra.
5. **The truncated tessellation and its boundary** ((H4), (B1)–(B3)), the pair $(X, \partial X)$ of 15.1 and the mapping cone with the derived homotopy $h = \widetilde{\eta}$ (15.9). The cone of a cochain map and the long exact sequence of a short exact sequence of complexes are expected from the library; the *pair homotopy* and its coset transport are not.
6. **The Hecke ring** $\mathcal{H}(\Delta, \Gamma)$ and double cosets of a group in a monoid (10.8, 20.4). Chapter 10's note offers the cheaper alternative explicitly: verify the $\nu = 2$ instance directly as a finite identity of cochain maps and induct, which is what `HeckeRelationsValidation` does numerically, rather than import a structure theorem.
7. **The free Γ-model** $F_\bullet \otimes C_\bullet(Y)$ of 17.1, whose clause (1) — $F \otimes_{\mathbb{Z}} M$ is free over $\mathbb{Z}[\Gamma]$ when F is free and M is $\mathbb{Z}$-free — the chapter calls easy, useful and worth doing early, because Remark 17.2, the removal of every invertibility hypothesis, is one line from it plus the fundamental lemma.
8. **An arithmetic model** for Part V. Theorem 18.7 is about fixed-width integer types and Theorem 19.2 about IEEE-754; both chapters recommend carrying these as *hypotheses on an abstract arithmetic model* rather than as facts about hardware, with (FP2) plus exact representability being all the proofs use.
9. **The integral line** (Chapter 23, since v1.8.0). Its note ranks the fracture square and cohomology-level glueing (Propositions 23.7, 23.8) first — elementary facts about flat abelian groups, and what makes the descent exact — then the strict lifting lemma 23.12, a finite induction; the Bredon identification (Proposition 23.3) needs a functor category and the comparison theorem in it, and the isotropy-resolution model (Proposition 23.5) the assembly of resolutions over a complex.

C.2 What to import rather than prove

The chapters are unanimous that these should be axioms or library imports, and say why in each case: Theorem 4.4 (the only genuinely topological input of Chapter 4, and used nowhere downstream — Remark 4.7); Theorem 5.2 and Theorem 5.10(2) ([Bro82, Ch. I.4], [Hat02, §3.H]); Proposition 3.12; the fundamental lemma of relative homological algebra, which is what Lemma 11.3(2) and, through it, Theorem 11.7(b),(c), Theorem 11.10, Theorem 15.11(c) and Proposition 9.4(3c) all rest on; Eichler–Shimura and the Dickson invariants, which are what Claim 16.5 imports and which make it and Claim 16.26 *not* targets at this stage; the Bass–Serre sequence (50), which a formalization may take as the definition of the amalgam datum; the isotropy spectral sequence behind Theorem 17.3(1); and the $H^2_{\mathrm{eq}} = 0$ vanishing that §13.4 carries by citation — which a formalization must import rather than read off an emitted 0×0 matrix.

Two further items are flagged by their chapters as things to *avoid* formalising in the obvious way. Remark 12.9(ii) is a non-constructive argument (irreducibility plus a numerical non-scalarity) and should be a statement about $\mathrm{End}_\Gamma(\mathrm{Sym}^{10}\mathbb{Q}^2)$ with an imported eigenvalue fact, not re-derived. Proposition 14.3 needs the Lyndon–Hochschild–Serre spectral sequence of a product with a cyclic factor; Chapter 14 recommends replacing it by the explicit $\langle T \rangle$-equivariant periodic resolution tensored with the two-term resolution over $\mathbb{Z}[\mathbb{Z}]$, which is finite and avoids the spectral sequence entirely.

C.3 Hypotheses a formalization must not lose

Rule (A1) exists because of exactly these, and the OQ2 audit's finding F5 — an unwritten load-bearing hypothesis — is the cautionary case. Five recur:

- (H4)(ii), $|K^u|$ *contractible and two-dimensional*: used in Lemma 15.7 (via $Z_2 = 0$), in the fills of Algorithm 18.3 step 3 and Lemma 15.6, and everywhere in Chapter 15.
- (B3), *boundary stabilisers are central*: three uses in Lemma 14.10 and three in Lemma 15.6, and it is what makes the (T1∂)-extension well defined.
- *Relative projectivity* in Lemma 11.3(2), whose entire content here is that 4 and 6 — the elliptic stabiliser orders of (H4)(iv) — are units. Stating it as $6 \in R^\times$ for a general commutative R, rather than as $\ell \geq 5$ for $\mathbb{F}_\ell$, is what puts it in the statement instead of the proof.
- Proposition 9.6(1)'s $\frac{1}{2}$, which must be carried as an Invertible $(2 : R)$ instance argument and **not** as a field hypothesis, since $R = \mathbb{Z}/\ell^e$ is in scope.
- The (T3) normalisation of Definition 11.1: without it the degree-0 operator is still well defined but is not normalised, and every H^0 eigenvalue oracle of §21 is off by the augmentation defect (Remark 11.8).

C.4 Shapes that recur, and should be factored once

Chapter 20 identifies three predicate shapes that every obligation reduces to — entrywise equality of two matrices over R; equality of two integer chains after canonicalisation; a magnitude recount against a stored bound — and the one obligation that is not of that shape, 20.9(iv), which quantifies over *builds* and which a formalization should replace by the statement that the pipeline's output is a function of the abstract sign predicate alone (Lemma 19.1 plus Theorem 19.2 give exactly that, and it is a better target than the two-build diff). Two further shapes recur across chapters and are worth one shared development: the **1-cocycle chain rule** (Lemma 10.3, reused verbatim by Proposition 12.2, by the $(*, \kappa)$ rule of Proposition 13.4 and by Lemma 15.4), and the **shared (T1)-extension helper** that Algorithm 18.3, 14.10 and 15.6 each use for both construction and certification — in the implementation it is deliberately one function, and a formalization that duplicates it loses the property that makes the certification independent of the construction only where it should be.

C.5 The targets: cheapest, and most valuable

These are two different lists, and the chapters say so. The cheapest statements are cheap because they are finite, constructive and self-contained; the most valuable are those an error in which could not be caught any other way. The second list is *not* constructed here: it is Appendix E's roster of the 67 statements that no superseded document and no project note states — the ones a referee should attack first, for the reason given there, namely that every other statement of this contract can be checked against a source and these cannot. A formalization's priority order should be read off that roster, and this appendix adds only the per-chapter judgement of which of them are also *tractable*.

part	cheapest first target	why, and whether it is in the roster
I, the framework	5.10(1); then 6.1, 6.2, 6.4, 6.5	Ψ needs no topology, no field and no freeness, and is cited by six source files; the four propositions are kernels, images and quotients of explicit matrices — "the most directly formalizable material in the document". All five are in the roster. The contract box of §6 must come with them: the *basis* is part of the specification, and a proof that fixes only the spans would not pin $A^{(q)}$
II, model and coefficients	7.7; (12)	the odd-n vanishing is two lines from centrality and $2 \in R^\times$, and it is oracle 2 of §21; (12) is what everything about the duality rests on. 8.6 is in the roster, 7.7 is not (it is re-homed)
III, the Hecke datum	10.3; then the equivariance half of 11.7 and 12.11(1),(2)	one line from uniqueness of coset representatives; the two halves are finite manipulations using only 10.3 and 8.2(2), need **no** hypothesis on ℓ, and are the chapters' own "natural first target of Part III". 10.3 is in the roster
III, the two bit-identities	13.1, 13.2(3)	the only two claims in the document where a *bit*-level agreement is mathematically forced rather than empirically observed — an integral identity in $\mathrm{M}_2(\mathbb{Z})$ pushed through ρ_n. Both are in the roster, and both are cheap
IV, the boundary	14.6; then 14.9, 14.10	a finite matrix-arithmetic statement, decidable for each N; the two that carry the content are constructive and need H_1 of a tree, **neither** the fundamental lemma **nor** any invertibility hypothesis. All three are in the roster
IV, the cone	15.4, 15.8, 15.10, 15.7, then 15.12(1)	**the chapter to attack first.** It is the one whose statements have *no* external corroboration at all: its only sources are project notes, which are not sources of record. In that order the work is four one-line identities, pure algebra on a direct sum, a substitution, two uses of $Z_2 = 0$, and then one displayed computation — and 15.12(1) is the theorem that matters
IV, six groups, orbifold model	16.2(2); 17.1(1)	the first is $\mathrm{Ext}^\bullet_{\mathbb{Z}[\Gamma]}(-, V_R)$ on one short exact sequence and is the most reusable thing in its chapter; the second is easy and unlocks 17.2. Both are in the roster. 16.5 and 16.26 are *not* targets: the first imports Eichler–Shimura and Dickson, the second is open
V, realisation	19.1; 19.4(ii)	19.1 is a finite case analysis over $\mathbb{Z}$ needing no floating-point model at all, and it is the *only* assertion in Part V that no source states — the whole of the rest of the part is re-homable. 19.4(ii) replaces an error estimate by a decidable integer-magnitude statement
the involution	9.1(i)–(ii), 9.2, 9.4(1),(2)	decidable predicates on finite data. 9.4(3c) is the only statement of that chapter invoking an abstract input, and inherits $6 \in R^\times$ verbatim
the obligations	all of §20	each is already a decidable predicate on a finite emitted table ((A4)), so the predicate needs no proof, only evaluation; what a formalization proves is the implication "predicate holds on the emitted data $\Rightarrow$ the stored $A^{(q)}$ is the matrix of $T_{\Gamma g \Gamma}$"

C.6 What this appendix does not supply

1. **Any cost.** No effort estimate, no ordering by expense, no schedule.
2. **Any verified statement about `mathlib4`.** See the box at the head of this appendix.
3. **An executable oracle for one chapter and one clause.** A formalization of Chapter 17 cannot be cross-checked against a run, and one of Chapter 15 in everything but Obligation 15.22(vii)'s cross-routes and PD adjointness (computed since v1.7.3, slot-6 WP-3 / WP-4 / WP-5), because nothing in the tree computed the rest of their objects (§22.2; Chapter 14 was a third until v1.3.3, Chapter 15 wholly uncomputed until v1.4.2 and computed in its η-half only until v1.5.2). For the rest of the document the implementation is a second, independent witness — 89 registered validations, and Appendix B's eleven [verified computationally]^{r} claims — and a formalization can be developed against it. For those two chapters it cannot, which is a fact about the input to the commission and not a judgement about the work.
4. **A decision on 14.24 — taken at v1.3.0, in the weakest form a formalisation could want.** The owner decided (OC-1a) on 2026-09-16: the decision is that $q_{\max}$ *is a parameter*, so a development takes it as a section variable and proves nothing about it, which is what 17.1 and 17.3 already do. What a formalisation of the truncated regime would still be formalising is a policy, not a theorem — unless (OC-1c) is proved, whose bulk half now is one: 14.22 is a natural target (a five lemma over 16.2(2), the cup product with a class in $H^2(\Gamma;\mathbb{Z})$, and the transfer's projection formula; `mathlib4` has group cohomology in low degrees but, as of this writing, neither cup products on it nor the transfer), and 14.23(3) lists the three statements that would complete it. The degree-generic statements of Chapters 15 and 17 are written so that none of this blocks the rest.

D Index of preserved labels

This appendix exists to make the label namespace machine-checkable: every label that a C++ source file or a project document cites is dereferenced here, so a missing, renamed or duplicated label is a build error rather than a silent breakage.

The 67 labels of `hecke_action.tex`

§1, §7, 7.6, §7.1, 7.7, §8, 8.1, 8.2, 8.3, 8.4, §10, 10.1, 10.4, (18), (19), 10.5, 10.8, 10.10, §11, (20), 11.1, 11.3, 11.4, 11.6, (21), 11.7, 11.9, 11.10, §12, 12.1, 12.2, (22), 12.3, 12.4, 12.6, 12.7, 12.9, 12.10, (24), (25), 12.11, 12.13, §13, 13.3, 13.4, §18, 18.1, 18.3, 18.5, 18.7, §19, 19.2, 19.4, 19.5, §20, 20.1, 20.2, 20.3, 20.4, 20.5, 20.6, 20.7, 20.8, 20.9, 20.10, 20.11, §21, §22.

The label of `monodromy_v5.tex` cited from C++ source

5.10.

Further `monodromy_v5.tex` labels carried with the re-homed text

3.1, 3.3, 3.4, 3.6, 3.8, 3.10, 3.11, 3.12, 3.13, 3.17, 3.21, 3.24, 4.1, 4.3, 4.4, 4.5, 5.1, 5.2, §5, 5.4, 5.6, 5.8, §6.

Labels re-homed from `hecke_extensions.tex` (cited from project documents as [E])

§16, 16.2, 16.5, 16.7, 16.21, 16.23, (52), 16.24, 16.26.

Labels new in this document

§2, §1.1, §1.2, §1.3, §1.4, 7.1, 7.2, 7.4, (7), (8), (9), §8.1, (10), 8.5, 8.6, (11), (12), 8.7, 8.8, (13), 8.9, §9, 9.1, (14), (15), (16), 9.2, 9.3, 9.4, 9.5, 9.6, 9.7, 9.8, 9.9, 9.10, §10.1, §10.2, 10.2, (17), 10.3, §10.3, 10.6, 10.7, §10.4, 10.9, 10.11, §11.1, §11.2, 11.2, 11.5, §11.3, 11.8, §11.4, 11.11, §14, (B1), (B2), (B3), 14.1, §14.1, 14.2, (28), 14.3, (29), 14.4, 14.5, §14.2, 14.6, 14.7, §14.3, 14.8, 14.9, 14.10, 14.11, 14.12, 14.13, §14.4, 14.14, (30), 14.15, 14.16, (31), 14.17, (32), 14.18, 14.19, §14.5, 14.20, 14.21, 14.24, (OC-1a), (OC-1b), (OC-1c), (OC-1d), §14.5, §15, 15.1, 15.5, 15.6, 15.7, 15.9, 15.11, 15.12, 15.18, 15.15, 15.20, 15.21, 15.22, 15.23, §15.1, (37), (38), 15.2, (39), (40), 15.3, (41), 15.4, §15.2, (42), §15.3, 15.8, (43), (44), (45), 15.10, (46), (47), 15.13, 15.14, 15.16, 15.17, §15.4, 15.19, §15.5, (48), §15.6, (49), 16.1, (50), (51), 16.4, 16.22, (53), (54), 16.25, 16.27, §17, (G1), (G2), (G3), (G4), 17.1, (55), 17.2, 17.3, 17.4, 17.5, §18.1, 18.2, §18.2, (56), 18.4, §18.3, 18.6, §18.4, §19.1, 19.1, (57), §19.2, (58), 19.3, (59), §20.1, (60), §20.2, §20.3, §22.5, §2.10, §23, §23.1, 23.1, 23.2, 23.3, (65), §23.2, 23.4, (IM1), (IM2), 23.5, 23.6, §23.3, 23.7, 23.8, 23.9, 23.10, 23.11, §23.4, 23.12, 23.13, 23.14, 23.15, §23.5, 23.16, 23.17, 23.18, 23.19, §23.6, 23.20, 23.21, 23.22, 16.8, §23.7, 23.23, §2.11, §24, 24.1, 24.2, (66), 24.3, 24.4, 24.5, 24.6, 24.7, 24.8, 24.9, 24.10, 24.11, (67), 24.12, §24.5, 24.13, 24.14, §25, 25.1, 25.2, 25.3, 25.4, 25.5, 25.6, §26, 26.1, 26.2, 26.3, 26.4, 26.5, 26.6, §27, §27.1, 27.1, 27.2, (68), 27.3, (69), (70), 27.4, 27.5, §27.2, 27.6, 27.7, §27.3, 27.8, 27.9, 27.10, 27.11, 27.12, §27.4, 27.13, 27.14, 27.15, §28, §28.1, 28.1, 28.2, §28.2, 28.3, 28.4, 28.5, 28.6, 28.7, §28.3, 28.8, 28.9, (71), 28.10, §28.4, 28.11, §28.5, 28.12, §28.6, 28.13, 28.14, (72), 28.15, 28.16, 28.17, (LN7), (LN8), (LN9), §28.7, 28.18, 28.19, §29, §29.1, 29.1, 29.2, 29.3, 29.4, §29.2, 29.5, (73), 29.6, 29.7, 29.8, §29.3, 29.9, (74), 29.10, §29.4, 29.11, 29.12, §29.5, 29.13, §29.6, 29.14, §29.7, 29.15, 29.16, 29.17, 29.18, 29.19, 29.20, 29.21, 29.22, 29.23, 29.24, §29.8, 29.25, 29.26, 29.27, (75), 29.28, 29.29, 29.30, (76), 29.31, 29.32, 29.33, 29.34, 29.35, 29.36, (77), 29.37, 29.38, 29.39, 29.40, 29.41, 29.42, 29.43, 29.44, 29.45, §30, §30.1, 30.1, 30.2, 30.3, §30.2, 30.4, 30.5, 30.6, §30.3, 30.7, §30.4, 30.8, §30.5, 30.9, §30.6, 30.10, §31, 31.1, 31.2, 31.3, 31.4, 31.5, 31.6, §31.1, 31.7, 31.8, 31.9, 31.10, (78), §31.2, 31.11, 31.12, §31.3, 31.13, 31.14, 31.15, 31.16, 31.17, §31.4, 31.18, 31.19, 31.20, (79), 31.21, 31.22, §31.5, 31.23, 31.24, 31.25, 31.26, (IL1), (IL2), (IL3), (IL4), (IL5), §32, §32.1, 32.1, 32.2, 32.3, 32.4, 9, 32.5, §32.2, 32.6, 32.7, §32.3, 32.8, 32.9, 32.10, 32.11, 32.12, 32.13, §32.4, 32.16, 32.17, §32.5, 32.18, 32.19, 32.20, 32.21, 32.22, 32.23, 32.24, §32.6, 32.25, (MPC1), (MPC2), (MPC3), (MPC4), (MPC5), (MPC6), (MPC7), (MPC8), (MPC9), (MPC10), (MPC11), (MPC12), §A, §B, §C, §D, §E.

Errata entries of Appendix A

§A.1, §A.2, §A.3, §A.4, §A.5, §A.6, §A.7, §A.8, §A.9, §A.10, §A.11, §A.12, §A.13, §A.14, §A.15, §A.16, §A.17, §A.18, §A.19, §A.20, §A.21, §A.22.

E Changelog

What this appendix fixes. The record that rule (V3) requires and that this document did not have for its first ten versions: one entry per version, and within it one line per statement whose content changed, naming the label. It also fixes what a version *is* for a document still under construction ((V5)), and marks which statements are new mathematics rather than transported from a superseded document.

> **CONTRACT. Versions 0.1.0–0.10.0 are assigned retrospectively.** The convention of §1.4 was written in v0.2.0 and then honoured by nobody, this appendix included, until v0.11.0. The first eleven versions below are therefore a numbering imposed *now* on the ten commits that built the document, reconstructed from those commits and checked against the document, not a series of releases that were cut at

the time. Where a change is not evidenced by a commit or visible in the text, it is not recorded here, and §E.1 says what was left out.

Release record

This is the supersession record of rule (V1): one entry per version, newest first. Up to v1.25.0-pre.7 it stood in the date line of the title page; the entries from v1.25.0-pre.7 down to v0.1.0, the sentence naming the superseded documents and the companion pointer that follow them are that date line's text, verbatim. The changelog proper, one entry per version with one line per changed statement (rule (V3)), follows the rules below.

Version 1.26.3 — 2026-11-09 (hecke-extensions slot 26, **signed by the owner 2026-11-09**: the owner-signed release of v1.26.3-pre.1; header, date and changelog only — no statement's mathematics, label, number, tag, status, written word or marker count (229 on 88 labels) changes; see the changelog)

Version 1.26.3-pre.1 — 2026-11-09 (hecke-extensions slot 26, a **pre-release, not signed**: text-only: private path removed, no change of mathematical content.)

Version 1.26.2 — 2026-11-09 (hecke-extensions slot 26, **signed by the owner 2026-11-09**: the owner-signed release of v1.26.2-pre.1; header, date and changelog only — no statement's mathematics, label, number, tag, status, written word or marker count (229 on 88 labels) changes; see the changelog)

Version 1.26.2-pre.1 — 2026-11-09 (hecke-extensions slot 26, a **pre-release, not signed**: paths only, no change of content (repository layout, owner ruling 2026-11-09).)

Version 1.26.1 — 2026-11-09 (hecke-extensions slot 26, **signed by the owner 2026-11-09**: the owner-signed release of v1.26.1-pre.1; header, date and changelog only — no statement's mathematics, label, number, tag, status, written word or marker count (229 on 88 labels) changes; see the changelog)

Version 1.26.1-pre.1 — 2026-11-09 (hecke-extensions slot 26, a **pre-release, not signed**: no change of mathematical content: machine names and private paths removed for the public release (owner ruling 2026-11-09).)

Version 1.26.0 — 2026-11-09 (hecke-extensions slot 26, **signed by the owner 2026-11-09**: the owner-signed release of v1.26.0-pre.11; header, date and changelog only — no statement's mathematics, label, number, tag, status, written word or marker count (229 on 88 labels) changes; see the changelog)

Version 1.26.0-pre.11 — 2026-11-09 (hecke-extensions slot 26, UP-Z (b), the condition C1 and should-fix W-A, W-B of the confirm-referee of v1.26.0-pre.10; **a pre-release, not owner-released**: status sentences of the descent and HeckeZ chapters and item 8 (d′)(ii), (vii) of §22.4, text only (binding changes **not yet seen by the owner**); no written word, golden entry, level-1 or `anemic` byte or label moves; no marker (229 on 88) moves; see the changelog);
Version 1.26.0-pre.10 — 2026-11-08 (hecke-extensions slot 26, UP-Z (b), the condition E1 of the referee of v1.26.0-pre.8 / pre.9; **a pre-release, not owner-released**: item 8 (d′)(ii), (vii) of §22.4, text only (binding changes **not yet seen by the owner**); no written word, golden entry, level-1 or `anemic` byte or label moves; no marker (229 on 88) moves; see the changelog);
Version 1.26.0-pre.9 — 2026-11-08 (hecke-extensions slot 26, UP-Z (b), Z2's words, contract first; **a pre-release, not owner-released, not refereed**: item 8 (d′)(ii), (v), (vii) of §22.4, text only (binding changes **not yet seen by the owner**); no golden entry, level-1 or `anemic` byte or label moves; no marker (229 on 88) moves; see the changelog)

Version 1.26.0-pre.8 — 2026-11-08 (hecke-extensions slot 26, UP-Z; **a pre-release, not owner-released**; the conditions of the confirm-referee of v1.26.0-pre.7, `docs/implementation/slot26/C126pre7_confirm_referee.md`: item 8 (c), (d′)(vii) of §22.4,

text only (binding changes **not yet seen by the owner**); no record word, hash input, golden entry or label moves; no marker (229 on 88) moves; see the changelog)

Version 1.26.0-pre.7 — 2026-11-08 (hecke-extensions slot 26, UP-Z; **a pre-release, not owner-released**; the conditions of the confirm-referee of v1.26.0-pre.6, `docs/implementation/slot26/C126pre6_confirm_referee.md`: item 8 (c), (d), (d′)(ii), (vii), (ix) of §22.4, text only (binding changes **not yet seen by the owner**); no record word, hash input, golden entry or label moves; no marker (229 on 88) moves; see the changelog)

Version 1.26.0-pre.6 — 2026-11-08 (hecke-extensions slot 26, UP-Z; **a pre-release, not owner-released**; the conditions of the referee-light of v1.26.0-pre.5, `docs/implementation/slot26/C126pre5_referee_light.md`: item 8 (c), (d), (d′)(ii), (v), (vii), (ix) of §22.4, text only (binding changes **not yet seen by the owner**); no record word, hash input, golden entry or label moves; no marker (229 on 88) moves; see the changelog)

Version 1.26.0-pre.5 — 2026-11-08 (hecke-extensions slot 26, UP-Z; **a pre-release, not owner-released**; the text aligned with the audited Z1 code, should-fix S1 (and S2's key) of `docs/implementation/slot26/Z1_code_audit.md`: item 8 (d′)(ii), (v), (vii) of §22.4 (binding changes **not yet seen by the owner**); no label is added, renamed or removed; no marker (229 on 88) moves; see the changelog)

Version 1.26.0-pre.4 — 2026-11-08 (hecke-extensions slot 26, UP-Z; **a pre-release, not owner-released**; the should-fix items of the confirm-referee of v1.26.0-pre.3 (ACCEPT): item 8 (d), (d′)(ii) of §22.4 (W3, W4, W5, N1; binding changes **not yet seen by the owner**); no label is added, renamed or removed; no marker (229 on 88) moves; see the changelog)

Version 1.26.0-pre.3 — 2026-11-08 (hecke-extensions slot 26, UP-Z; **a pre-release, not owner-released**; the conditions of the adversarial re-referee of v1.26.0-pre.2 (ACCEPT-with-conditions): item 8 (c), (d), (d′)(ii) of §22.4 (R1, R2, W1, W2; binding changes **not yet seen by the owner**); no label is added, renamed or removed; no marker (229 on 88) moves; see the changelog)

Version 1.26.0-pre.2 — 2026-11-08 (hecke-extensions slot 26, UP-Z; **a pre-release, not owner-released**; the fixes of the adversarial referee of v1.26.0-pre.1 (ACCEPT-with-conditions): `prop:strictliftN`(9) (S1, E1–E4, E7–E9) and item 8 (d′) of §22.4 (S3, S4, E5, E6; S4 a binding change **not yet seen by the owner**); no label is added, renamed or removed; no marker (229 on 88) moves; see the changelog)

Version 1.26.0-pre.1 — 2026-11-08 (hecke-extensions slot 26, UP-Z, contract first; **a pre-release, not owner-released, not yet refereed**; binding changes **not yet seen by the owner**: `prop:strictliftN`(9) (U_p, $p \mid N$, on the free model) and item 8 (d′) of §22.4 (the full-N reading of the caps text, frozen with v1); no label is added, renamed or removed; no marker (229 on 88) moves; see the changelog)

Version 1.25.0 — 2026-11-07 (hecke-extensions slot 25, **signed by the owner 2026-11-07**: the owner-signed release of v1.25.0-pre.8; header, date and changelog only — no statement's mathematics, label, number, tag, status or marker count (229 on 88 labels) changes; see the changelog)

Version 1.25.0-pre.8 — 2026-11-07 (hecke-extensions slot 25, LaTeX polish stage 5c; **a pre-release, not owner-released**; one binding change, **not yet seen by the owner**, approved by the owner in advance (ruling Q1 of 2026-11-06): rule (V1) amended, so that this record is the first subsection of Appendix E and no longer the date line; otherwise no change of content: a one-line date, the uniform author block, a generic abstract, the old date line and abstract moved here verbatim, two part headings; no label, number, statement, tag set or `\LEAN` marker (229 on 88) moves; see the changelog)

Version 1.25.0-pre.7 — 2026-11-06 (hecke-extensions slot 25, the HZ-RAT-PERF fix round after its audit; **a pre-release, not owner-released**; item 8 only, binding changes **not yet**

seen by the owner: the pending word `refused(word_prime_cap)` (the word-prime count cap 2^{21}), the closed list of `C2_status` as built, the cap in the unhashed provenance key; no label, tag set, marker, record or golden line moves; see the changelog);

Version 1.25.0-pre.6 — 2026-11-06 (hecke-extensions slot 25, the H1 merge after its audit; **a pre-release, not owner-released**; item 8 only, binding changes **not yet seen by the owner**: the G8 paragraph (the Hida block under `heckez.G8 = hida`: paths, closed lists, caps), golden only under the default settings; no label, tag set or marker moves; see the changelog);

Version 1.25.0-pre.5 — 2026-11-06 (hecke-extensions slot 25, the HZ-REC schema re-audit fix round; **a pre-release, not owner-released**; item 8 only, binding changes **not yet seen by the owner**: the Eisenstein multiplicity e_k, the cuspidal block iff $d \geq 2$, the ϑ search over $\mathcal{P} \cup \{T_\ell\}$, the second route's caps and `not_read`, the irreducibility cap 256, `[Lift]` by its certificate, the `not_run` words of v1; no label, tag set or marker moves; see the changelog);

Version 1.25.0-pre.4 — 2026-11-06 (hecke-extensions slot 25, the HZ-REC schema-audit fix round; **a pre-release, not owner-released**; item 8 only, binding changes **not yet seen by the owner**: the new final value `split_on_P`, G4–G6 per torsion point, no G8 in default records, the components, maximal ideals and caps text as written; no label, tag set or marker moves; see the changelog);

Version 1.25.0-pre.3 — 2026-11-06 (hecke-extensions slot 25, the Z0 re-check; **a pre-release, not owner-released**; text only: `export_word` also for K above EXPORT's $K_{\max}$ below the word bound; no label, tag set, marker or value moves; see the changelog);

Version 1.25.0-pre.2 — 2026-11-06 (hecke-extensions slot 25, the Z0 merge after its referee (REVISE); **a pre-release, not owner-released**; text only: the counting criterion of the full-algebra tag narrowed to N a power of p, `export_word` pending, the as-built refusal rule, the cofactor route as built, the closed `t_ell` list; no label, tag set, marker or value moves; see the changelog);

Version 1.25.0-pre.1 — 2026-11-06 (hecke-extensions slot 25, the Z0 merge; **a pre-release, not owner-released**, on top of the signed v1.24.0: the refereed §31.6 of v1.25.0-pre.0c (referees A7, A7b: ACCEPT) enters the main line with record item 8 of §22.4, reconciled with the code of slot 25 by dated notes; no marker (229 on 88 labels) or value moves; see the changelog);

Version 1.24.0 — 2026-11-05 (hecke-extensions slot 25, **signed by the owner 2026-11-05**: the owner-signed release of v1.24.0-rc.2; header, date and changelog only — no statement's mathematics, label, number, tag, status or marker count (229 on 88 labels) changes; see the changelog);

Version 1.24.0-rc.2 — 2026-11-05 (hecke-extensions slot 25, **the release candidate of 1.24.0: not owner-released**; rc.1 refereed ACCEPT_WITH_MINORS, `docs/implementation/slot25/RC124_referee.md`, its findings applied as text: the marker rule of §1.1 amended by one sentence, Remarks 16.8, 28.19 and Proposition 16.15 corrected; no label, tag set, marker or value moves; see the changelog);

Version 1.24.0-rc.1 — 2026-11-05 (hecke-extensions slot 25, **the release candidate of 1.24.0: not owner-released, not yet refereed**, on top of v1.24.0-pre.1: the Lean markers on Theorem 11.10 and (20) (229 on 88 labels), the phenomenology citations in Remarks 16.8, 28.19, Proposition 16.15 and (Hi11); no label or tag set moves; see the changelog);

Version 1.25.0-pre.0c — 2026-11-04 (hecke-extensions slot 24, HeckeZ Z0 after the A7b re-audit; **still a branch pre-release** on `s24-wip-heckez-z0`, **not owner-released**: the drop rule of Theorem 31.45(C2) and the (C2) frontier measured on one discriminant (A7b-m1); no label, number, tag set or marker moves; see the changelog);

Version 1.25.0-pre.0b — 2026-11-04 (hecke-extensions slot 24, HeckeZ Z0 fix round after the adversarial referee A7; **still a branch pre-release** on `s24-wip-heckez-z0`, **not**

owner-released: the Mazur route of Proposition 31.49(3) is certified by a valuation check, goodness of q alone being insufficient (A7-M1); the referee's seven MINOR findings and the cheap NOTEs; no label, number, tag set or marker moves; see the changelog);

Version 1.25.0-pre.0 — 2026-11-04 (hecke-extensions slot 24, HeckeZ package Z0; **a branch pre-release** on `s24-wip-heckez-z0`, **not owner-released, not refereed**; it enters the main line as v1.25.0-pre.1 only after the owner has released v1.24.0; on top of v1.24.0-pre.1: the new §31.6, the Hecke algebra over $\mathbb{Z}$ (23 statements appended at the end of their section, no number moves), item 8 of §22.4, and a pointer in Theorem 31.16(7)(a); no marker moves; see the changelog);

Version 1.24.0-pre.1 — 2026-11-03 (hecke-extensions slot 24, **a pre-release, not owner-released, not refereed**, on top of v1.23.0: FMTMP, §22.4 item 7 (l), *The faithful coefficients*; text only, no label moves; see the changelog);

Version 1.23.0 — 2026-11-03 (hecke-extensions slot 24, **signed by the owner 2026-11-03**: the owner-signed release of v1.23.0-pre.40; header, date and changelog only — no statement's mathematics, label, number, tag, status or marker count (222 on 86 labels) changes; see the changelog);

Version 1.23.0-pre.40 — 2026-11-02 (slot 24, **a pre-release, not owner-released, not refereed**, on top of v1.23.0-pre.39: ICENSUS-P2, §22.4 item 7 (k), *Parts not run*; text only, no label moves; see the changelog);

Version 1.23.0-pre.39 — 2026-11-02 (hecke-extensions slot 24, **a pre-release, not owner-released, not refereed**, on top of v1.23.0-pre.38: ICENSUS-P1, §22.4 item 7 (e), (j), the ker / coker read's width ladder; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.38 — 2026-10-28 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.23.0-pre.37: the pre.37 referee's fixes M1, M2, m1–m10, N1–N4, T1, T2 (`docs/implementation/slot24/CON_pre37_referee.md`, PASS-with-fixes), text only — item 7 of §22.4: the coefficient order (leading coefficient first, an exception to item 2), the digest's byte layout, the hash partition, and the code map as merged (`2aa61397`) with the golden event landed (`6958a5d2`); no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.37 — 2026-10-26 (hecke-extensions slot 24, **a pre-release, not owner-released, not refereed**, on top of v1.23.0-pre.36: the integral descent record `integral_descent_record/v1` gets its contract home, item 7 of §22.4 (D6, built on branch `s24-d6impl`, pending audit and merge; its golden event pending the owner's go); $\mathbb{Z}$ un-reserved for the descent only (Definition 23.20(3)); the ker / coker precision refusal renamed `[KerCokerPrecision]`; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.36 — 2026-10-26 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.23.0-pre.35: the pre.35 referee's fixes m1, N1–N7 (`docs/implementation/slot24/C-D8b_referee.md`, ACCEPT after fixes) and D5c **merged** (`4064e164`) — §31.5's certificates restated for the merged code (the applied operator's blocks, entrywise or on the generators and 64 random vectors), Proposition 31.25(4)(e) sharpened, the code map's status; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.35 — 2026-10-26 (hecke-extensions slot 24, **a pre-release, not owner-released, not refereed**, on top of v1.23.0-pre.34: §31.5 brought in line with the D5c code (branch `s24-d5cimpl`, pending audit) — S versus S^{-1} on cochains, the same for n even; a truncation statement, Proposition 31.25(4)(e), by which $q_{\max} = 5$ is read with the comparisons to degree 5; the code map's status, certificates and controls; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.34 — 2026-10-26 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.23.0-pre.33: the pre.33 referee's fixes m1–m4, N1–N8 (`docs/implementation/slot24/C-D8_referee.md`, ACCEPT after fixes) — in §31.5 the off-diagonal $\eta_{\mathcal{T}}$ arises on $C(\mathtt{a}^*)$, not on the pair model, and its non-vanishing is observed, not proved; the solver and (L), (L′) for any column augmentation; the code's negative controls restricted where they are vacuous; the ℓ-local column augmentations defined; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.33 — 2026-10-26 (hecke-extensions slot 24, **a pre-release draft, not owner-released, not refereed**, on top of v1.23.0-pre.32: §31.5, the tree cusp pair (plan milestone D5c, clause C-D8), contract first — the cusp model and the axis map in closed form, the tree cone and the cylinder lemma, effective comparisons with $(\mathcal{I}, \mathcal{I}^{\partial})$, T_p by transport with its off-diagonal term; the D3 merge recorded in §31.3's code map; four statements added at the end of their section, no number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.32 — 2026-10-26 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.23.0-pre.31: the pre.31 referee's fixes m1–m3, N1–N6 (`docs/implementation/slot24/C-D3h_referee.md`, ACCEPT after fixes) — in §31.3 the homotopy of two lifts is $\Theta_{h_{\mathcal{I}}}$ with no sign, the precision of Theorem 31.16(4) at $\ell \geq 5$ is $K > \sigma_{\max} + v_\ell(\chi(c))$, any exact lifts of the free basis are a basis; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.31 — 2026-10-26 (hecke-extensions slot 24, **a pre-release, not owner-released, not refereed**, on top of v1.23.0-pre.30: three clauses of §31.3 close the contract notes of D3 (Hecke over $\mathbb{Z}$, built on branch `s24-d3impl`) — the torsion read on a cokernel presentation (Proposition 31.15(3′)), admitted at $\ell \geq 5$ and at the residue primes; obligation (b) met through the lift; any admissible bulk lift, the read independent of it (Proposition 31.13(2)); the code map records D3 as built on its branch; no number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.30 — 2026-10-26 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.23.0-pre.29: a non-mathematical code-map update — D5 (the tree models) is built and merged (§31.2), and `hecke::HeckeInternalError` separates internal errors, which fail, from a real overflow, which refuses as `IntegerCap` (§31.4); no statement, label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.29 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.23.0-pre.28: the pre.28 referee's fixes M1, m1–m4, N1 (`docs/implementation/slot24/C-D7p_referee.md`, ACCEPT-with-fixes) — in §31.4 the refusals of T_p on the pair are exactly the code's eleven refusal codes, the forced component map proved in (1)'s proof, $q_{\max} \leq 5$; the code map records D4 as merged; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.28 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, refereed ACCEPT-with-fixes, on top of v1.23.0-pre.27 (C-D7 fixes applied): the owner's scope addition of 2026-10-25 — in §31.4 T_p on the boundary and the cone for every prime $p \nmid N$, at level 1 and level N, from the pair lift built on demand and certified at run time; the code map records D4's build; see the changelog);

Version 1.23.0-pre.27 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: the C-D7 referee's fixes M1–M3, m1–m5 (`docs/implementation/slot24/C-D7_referee.md`, ACCEPT-with-fixes) — in §31.4 H^2_c at $\ell \geq 5$ is H_0, a prime rank source for D^0, T_p on the cone certified only where a pair lift is built, the signs of the Ext identification, elementary periodicity proofs; see the changelog);

Version 1.23.0-pre.26 — 2026-10-25 (hecke-extensions slot 24, **a pre-release draft, not owner-released, not refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**:

milestone D4 of the descent plan, clause C-D7 — §31.4 (new, after §31.3, so that no statement number moves): the boundary pair and its cone over $\mathbb{Z}$ on $\mathcal{I}^{\partial} \subseteq \mathcal{I}$ and $\mathcal{I}^{(\ell)}$, glued, the boundary in closed form, compact support and the certified exponent, the cone with the strict pair datum and T_p, $p \nmid N$; see the changelog);

Version 1.23.0-pre.25 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: the C-D4 referee's fixes M1, M2, m1–m5, n1, t1–t3 (`docs/implementation/slot24/C-D4_referee.md`, ACCEPT-with-fixes) — in §31.3 the layer blocks choice-free modulo $\ell^{\min(j,d_j)}$, one route at $\ell = 3$ (the transport on $\mathcal{I}^{(3)}$, no Bredon operator), the necessity of $2\sigma_{\max} \leq K$, the word conditions of the recovery; see the changelog);

Version 1.23.0-pre.24 — 2026-10-25 (hecke-extensions slot 24, **a pre-release draft, not owner-released, not refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: milestone D3 of the descent plan, clauses C-D4 and C-D5 — §31.3 (new, after §31.2, so that no statement number moves): the level-N operator T_p, $p \nmid N$, on the free model and its transports, the characteristic polynomial over $\mathbb{Z}$ by residues with the output rule, the choice-free Hecke invariants of finite ℓ-groups, and the certified record; see the changelog);

Version 1.23.0-pre.23 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released, not re-refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: the C-D3g referee's fixes M1, m1–m5, t1, t2 (`docs/implementation/slot24/C-D3g_referee.md`, ACCEPT-with-fixes) — in Proposition 31.10(2)(g) the blocks of the transforms that are exact modulo ℓ^K, the lattice coordinates, the conservative refusal policy, the further operations and the certifying ranks of derived forms; (2)(b)'s sign convention; editorial fixes in Proposition 31.12; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.22 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, refereed ACCEPT-with-fixes, on top of v1.22.0, **signed by the owner 2026-10-16**: the D2 code audit's F1 and F8 (`docs/implementation/slot24/D2_code_audit.md`) — Proposition 31.10 gains item (2)(g), the precision rule for generators, coordinates and maps read from a certified Smith form, and (2)(b) states the signs of the cone's maps; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.21 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, the C-D6 referee's fixes M1, m1–m5 (`docs/implementation/slot24/C-D6_referee.md`, ACCEPT-with-fixes; not re-refereed), on top of v1.22.0, **signed by the owner 2026-10-16**: in Proposition 31.12, (4)'s comparisons with $\mathcal{I}$ over $\mathbb{Z}$ and every homotopy between their composites are constructed (the tree inside X, an explicit retraction and homotopy, the lifting formula (L)), and five minor repairs; no label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.20 — 2026-10-25 (hecke-extensions slot 24, **a pre-release draft, not owner-released, not refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: milestone D5 of the descent plan, clause C-D6, brought forward by the owner's ruling at gate G-T — §31.2 (new, after the last statement of §31.1, so that no statement number moves): Definition 31.11, the free and the ℓ-local tree models, and Proposition 31.12: they compute group cohomology in every degree for every coefficient module (level N by Shapiro), one integral form over O_ℓ with the certified-exponent rule, cochains and comparisons with the free tree in closed form, comparisons with $\mathcal{I}$ over $\mathbb{Z}$, the fibre maps and the level cone, the Hecke action by transport; bulk only; no existing label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.19 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, the C-D3 referee's fixes m1–m5

(`docs/implementation/slot24/C-D3_referee.md`, ACCEPT-with-fixes; not re-refereed), on top of v1.22.0, **signed by the owner 2026-10-16**: in Proposition 31.10, (3)(b)'s failure of the square with restriction is restricted to this document's U_p (the adjoint of the classical one), and the new item (3)(b′): corestriction commutes with U_p for $p \mid N$ and every M, over $\mathbb{Z}$ in every degree (the Hida tower); four wording and citation fixes; no number, label or marker moves; see the changelog);

Version 1.23.0-pre.18 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D3_referee.md`), on top of v1.22.0, **signed by the owner 2026-10-16**: milestone D2 of the descent plan, clause C-D3 — Proposition 31.10: fibre maps (restriction, trace, conjugation) commute strictly with every comparison and glue to maps over $\mathbb{Z}$; the level cone on every model and its long exact sequence over $\mathbb{Z}$ in every degree, with the certified exponent; the Hecke squares for $p \nmid M$; inflation–restriction over $\mathbb{Z}$ and the two-row case of a torsion-free B (the Hida p-tower); what a tree model must supply; one new statement, the last of its section, so no number moves; no marker moves; see the changelog);

Version 1.23.0-pre.17 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, refereed ACCEPT (`docs/implementation/slot24/C-D1_three_referee.md`), on top of v1.22.0, **signed by the owner 2026-10-16**: the D1 code audit's finding M1 — Proposition 31.9, the 3-adic shortcut at level N when $\Gamma_0(N)$ has no element of order 3: the tree complex computes the 3-primary part, $H^{\geq 2}$ has no 3-torsion, and the sparse cokernel route holds at $\ell = 3$; one new statement, the last of its section, so no number moves; no marker moves; see the changelog);

Version 1.23.0-pre.16 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, on top of v1.22.0, **signed by the owner 2026-10-16**: the Dickson-proof referee's fixes D-1–D-5 (ACCEPT-with-fixes, not re-refereed) — the proof of Claim 16.5(2) is now Lemma 16.6, valid at every prime ℓ and for GL_2 as well, cited by its three users; scope, tag and exposition nits; one new statement, no marker moves, statement numbers after Claim 16.5 in its section shift by one; see the changelog);

Version 1.23.0-pre.15 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, on top of v1.22.0, **signed by the owner 2026-10-16**: the C-D1 / C-D2 referee's fixes F1–F7 to §31.1 (ACCEPT-with-fixes, not re-refereed) — Claim 16.5(2), the Dickson invariants of $\mathrm{SL}_2(\mathbb{F}_\ell)$, proved in place, so no count of Proposition 31.8 rests on an unverified pointer, and the good-prime rule $\ell > n + 1$ proved without it; $b_1 \geq 1$ for every $N > 1$; the sparse cokernel's generators, their certificate, and periodicity at chain level; no label, number or marker moves; see the changelog);

Version 1.23.0-pre.14 — 2026-10-25 (hecke-extensions slot 24, **a pre-release draft, not owner-released, not refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: milestone D1 of the descent plan, clauses C-D1 and C-D2 — §31.1 (new, after the last statement of §31, so that no statement number moves): Proposition 31.7, the descent for every finite free coefficient module (level N, nebentypus), Hecke determined locally, the coherence term only when $b_0 b_1 = 0$, periodicity by $\mathrm{res}(e)\cup$, the good prime and the certified exponent on the direct Smith and sparse cokernel routes; Proposition 31.8, the closed-form bad-prime list $\mathcal{B}(n, N)$ with the torsion at $\ell \geq 5$ in H^1 only; no existing label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.13 — 2026-10-25 (hecke-extensions slot 24, **a pre-release, not owner-released**, on top of v1.22.0, **signed by the owner 2026-10-16**: the C-D0 referee's fixes F1–F8 to Proposition 31.6 (ACCEPT-with-fixes, not re-refereed) — the good prime certified for every M ($24 \cdot H^{\geq 2}(\Gamma; M) = 0$; at $q = 0$ the H^0 test against b_0), the two admissible sources of $\mathrm{rank}_{\mathbb{Q}}$, the valuations counted with multiplicity, the a-priori exponents for $q \geq 1$, and the code map's D0 obligations; no label, number, tag set or marker moves; see the

changelog);

Version 1.23.0-pre.12 — 2026-10-25 (hecke-extensions slot 24, **a pre-release draft, not owner-released, not refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: milestone D0 of the descent plan, clause C-D0 — Proposition 31.6 (new, [proved here], at the end of §31, so that no statement number moves): $\mathcal{I}^{(\ell)}$ built once over $\mathbb{Z}[1/\mathsf{q}_\ell]$, its base changes the complexes the code builds, the certified-exponent rule over $\mathbb{Z}_{(\ell)}$ with $\mathrm{rank}_{\mathbb{Q}}$ read at a good prime, and the Smith-valuation read-out; its code map; no existing statement, label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.11 — 2026-10-21 (hecke-extensions slot 24, **a pre-release, not owner-released, not refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: code-map text only, from two audited packages — G2-H (Obligation 20.1(iii) certified without forming $\Theta_{\mathrm{eq}}^{(q)}$ on the sat / Loewy runner's lean route, route 2 compared in place, lever `HECKE_MODULE_LEAN_PEAK`) in §20's code map, G6 (the cone certificate's span inclusions by the unit-row route and the field route's memory diet, lever `HECKE_FIELD_DIET`) in §22.1's row for §15; every lever value-neutral; no statement, label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.10 — 2026-10-20 (hecke-extensions slot 24, **a pre-release, not owner-released, not refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: code-map text only, from two audited packages — FNTT (the forms cross-route's transform-domain route, lever `HECKE_FORMS_NTT_CACHE`; the forms ℓ-range past 2^{16} restated, `forms::FormsSpaceSupported`) in §30's code map, MT-1 (the module route's pivot-search exit and further pool splits, lever `HECKE_MODULE_MT`, values independent of the thread count) in §20's; every lever value-neutral; no statement, label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.9 — 2026-10-20 (hecke-extensions slot 24, **a pre-release draft, not owner-released, not refereed**, on top of v1.22.0, **signed by the owner 2026-10-16**: package ELL-W — Lemma 18.5 gains clause (7), the wide tier (products of two stored-class matrices formed in 128-bit integers under $2NB \le 2^{63}-1$), Theorem 18.7 gains consequence (4), Obligation 20.10(ii) names its assertion; the reach of the τ-certification moves from $\ell \le 23170$ to $\ell \le 1048573$; no label, existing clause number, tag set or marker moves, and no value at $\ell \le 23170$; see the changelog);

Version 1.23.0-pre.8 — 2026-10-18 (hecke-extensions slot 23, **a pre-release, not owner-released**, on top of v1.22.0, **signed by the owner 2026-10-16**: two audited code-map sentences (G2-T2, the SIMD ρ_n fill and route 2's shared memo, levers `HECKE_RHO_SIMD`, `HECKE_ROUTE2_SHARED_MEMO`; G2-P, the module route on the thread pool, lever `HECKE_MODULE_PARALLEL`, values independent of the thread count) in §20's code map; code-map text only, every lever value-neutral, no statement, label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.7 — 2026-10-18 (hecke-extensions slot 23, **a pre-release, not owner-released**, on top of v1.22.0, **signed by the owner 2026-10-16**: one audited code-map sentence (G2-T, the module route's term evaluation and its value-neutral rollback lever `HECKE_MODULE_EVAL=0`) in §20's code map; code-map text only, no statement, label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.6 — 2026-10-18 (hecke-extensions slot 23, **a repair-only pre-release, not owner-released**, on top of v1.22.0, **signed by the owner 2026-10-16**: the fixes F1–F6 of the referee report `C1230_pre3_referee.md` (ACCEPT-WITH-FIXES, MINOR 2, NIT 4) to pre.1–pre.3 — the proof of Lemma 30.11 repaired (the τ-homotopy over $C^{\bullet}_{\mathrm{eq}}(\mathbb{Z}_\ell)$), its use of Theorem 10.8(1) named, K_{cls}, KMAX, K_{stab} attributed or defined, readings across levels added to (3) as a consequence of (1); otherwise no statement's mathematics, and no label, number, tag set or marker, moves; see the changelog);

Version 1.23.0-pre.5 — 2026-10-18 (hecke-extensions slot 23, **a pre-release, not owner-released**; pre.4 refereed (ACCEPT-WITH-FIXES, fixes applied), pre.5 adds one audited code-map sentence (G2-F, `ChainsOnly` τ tables): code-map text only — §30.7's code map built (G1), the module route's G2 re-implementation in §20's code map, the G2-X forms product in §30's, each rollback lever value-neutral; no statement, label, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.3 — 2026-10-16 (hecke-extensions slot 23, **a pre-release, not owner-released**, audited (ACCEPT-WITH-FIXES, fixes applied), on top of v1.22.0, signed by the owner 2026-10-16 (its date line below is left as released): new §30.7 — Lemma 30.11 (the precision levels of the module route are reductions of one run) and Obligation 30.12 (one module run per cell); two new statements, no existing statement, number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.2 — 2026-10-12 (hecke-extensions slot 23, **a repair-only pre-release, not owner-released**: the repairs of the referee report F-C123pre1 (ACCEPT; MINOR 2, NIT 5) to Version 1.23.0-pre.1; no label, statement number, tag set or marker moves; see the changelog);

Version 1.23.0-pre.1 — 2026-10-12 (hecke-extensions slot 23, **a pre-release, not owner-released; refereed by F-C123pre1**: Proposition 29.34(4) replaced — the level-correspondence operator is independent of the transversal — and the negative control of Obligation 29.43(ix) replaced by the criterion-fails control; no label, statement number, tag set or marker moves; see the changelog);

Version 1.22.0 — 2026-10-12 (hecke-extensions slot 22, the CLOSE-T release candidate from v1.22.0-pre.17; **owner sign-off pending**, not yet owner-released; header, date and changelog only — no statement's mathematics, label, number, tag, status or marker count (222 on 86 labels) changes; see the changelog);

Version 1.22.0-pre.17 — 2026-10-12 (hecke-extensions slot 22, **a pre-release, not owner-released**: CLOSE-T T1 repairs — the F-T1 release referee's MINOR-1 (the changelog of pre.9) and seven wording NITs; no marker, label, statement or value moves; see the changelog);

Version 1.22.0-pre.16 — 2026-10-12 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed**: LR-Y phase 5 — four `\LEAN` markers on Claim 16.2(1) (the tree sequence as one short exact sequence of Γ-representations) and (2) (the first map is $(\mathrm{res}_S, \mathrm{res}_U)$, the whole Mayer–Vietoris sequence in group-cohomology terms, naturality in the coefficients); two existing marker scopes shortened; no new statement, no existing statement moves; see the changelog);

Version 1.22.0-pre.15 — 2026-10-11 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed**: F-pre14 referee repairs — wording only; no marker, label or statement change; see the changelog);

Version 1.22.0-pre.14 — 2026-10-11 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed**: the Lean dimension results at $\ell \in \{2, 3\}$ — twelve `\LEAN` markers on Lemma 16.9 (its last clause, for fields), Proposition 16.12(2) (the value of h^0 at certified cells), Proposition 31.3(4) (for fields) and the new Remark 16.20 ($\dim H^q(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell^2)$, $\ell = 2, 3$, $n \leq 700$); one new statement, no existing statement moves; see the changelog);

Version 1.22.0-pre.13 — 2026-10-11 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed**: LR-Y phase 4 — four `\LEAN` markers on Claim 16.2(2) (the Mayer–Vietoris terms as group cohomology of the cyclic subgroups, the middle map $\mathrm{res}_S - \mathrm{res}_U$, the beginning) and Lemma 14.22(2) (a natural periodicity isomorphism); two existing marker scopes rewritten; no new statement, no existing statement moves; see the changelog);

Version 1.22.0-pre.12 — 2026-10-11 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed**: LR-Y phases 1–3 — seven \LEAN markers on Claim 16.2(1),(2), Lemma 14.22(2) and the new Remark 16.3 (the tree resolution over every ring); the Serre pointer of Claim 16.2(1) no longer offered as a hypothesis; one new statement, no existing statement moves; see the changelog);

Version 1.22.0-pre.11 — 2026-10-10 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed**: the F-L23 referee's repairs in §16.4; §32 — Definition 32.14 and Proposition 32.15, the Q-reduced nerve as a projective resolution, closing Gap G1; two new statements, no existing statement moves; see the changelog);

Version 1.22.0-pre.10 — 2026-10-10 (hecke-extensions slot 22, L23-0 extended, **a pre-release, not owner-released, not refereed**: §16.4 — odd n at $\ell = 2$ in closed form, the $\mathbb{Z}/2^k$ splitting, $T_p = 1 + p$ at $\ell \in \{2, 3\}$, $\operatorname{rank} T_\ell \leq 2$, the c_p rule; eleven new statements, no existing statement moves; see the changelog);

Version 1.22.0-pre.9 — 2026-10-10 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed**: §32 — code-vs-chapter wording repairs in Remark 32.5, (MPC3) and Example 32.24; no statement number or value moves; see the changelog);

Version 1.22.0-pre.8 — 2026-10-10 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed**: §32 — the pre.7 referee's four minor repairs; Remark 32.23 rewritten for the built MP-4 route (a twisted chain map on the MP resolution itself, unique over every ring) with its [verified computationally]^{r} record and (MPC12); the MP-3b coverage of (MPC10), (MPC11); no statement number or value moves; see the changelog);

Version 1.22.0-pre.7 — 2026-10-10 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed at pre.7** (refereed at pre.8, ACCEPT): §32 — 25 further \LEAN markers (Lean MP-L3: morphisms, naturality, refinement, $\varepsilon^{\mathsf{MP}}$, the normalised complex), the unnormalised clauses they need in Definition 32.18, Theorem 32.19 and Proposition 32.20, and the [verified computationally]^{r} records of Propositions 32.21, 32.22 (MP-3); no statement number or value moves; see the changelog);

Version 1.22.0-pre.6 — 2026-10-10 (hecke-extensions slot 22, **a pre-release, not owner-released**: §32 refereed by F-MP2 (contract part, ACCEPT WITH REPAIRS), repairs R1–R8 applied — Theorem 32.19(4)'s first sentence corrected; the strong form of goodness in Definition 32.2(3) is the one the code and Lean check; the first \LEAN markers of §32 (30 on 11 labels); no statement number, label, tag or value moves; see the changelog);

Version 1.22.0-pre.5 — 2026-10-09 (hecke-extensions slot 22, **a pre-release, not owner-released, not refereed at pre.5**: new chapter §32, the Moerdijk–Pronk model — the choice-free MP set, its comparison theorem in every degree over every ring, all degrees with its scope, morphisms with the label factor, the Bredon comparison; no shipped value or default changes, no existing number, label, tag or marker moves; see the changelog);

Version 1.22.0-pre.4 — 2026-10-09 (hecke-extensions slot 22, **a pre-release, not owner-released**: Remark 7.7, second bullet, the superfluous hypothesis $6 \in R^\times$ deleted — erratum, wording only; see the changelog);

Version 1.22.0-pre.3 — 2026-10-09 (hecke-extensions slot 22, **a pre-release, not owner-released**, refereed by F-LES with repairs R1–R6 applied: new last subsection §29.8 of §29 — the level-change cone for $B \leq A$ with compatible characters, its six-term sequence, inflation–restriction in the normal case, which level steps are normal, level-raising U_p settled negatively; no existing number, label, tag or marker moves; see the changelog);

Version 1.22.0-pre.2 — 2026-10-08 (hecke-extensions slot 22, **a pre-release, not owner-released**: Remark 28.19(1) reworded, Mazur's good-prime condition sufficient, not necessary, at $p = 2$; nothing else changes; see the changelog);

Version 1.22.0-pre.1 — 2026-10-06 (hecke-extensions slot 22, **a pre-release, not**

owner-released: new last subsection §29.7 of §29 — wild characters over the chain rings $\mathcal{O}_{j,e}$, the groups $\Gamma_{\mathsf{H}}(N)$, specialisation without division, (NB7)–(NB10) — as refereed by F-U5; an addendum after Theorem 27.11; no existing number, label, tag or marker moves; see the changelog);

Version 1.21.0 — 2026-10-06 (hecke-extensions slot 21, the close: **the release** from v1.21.0-pre.4 under the owner's decision D16 (accepted as recommended); header, date and changelog only — no statement's mathematics, label, number, tag, status or marker count (140 on 65 labels) changes; see the changelog);

Version 1.21.0-pre.4 — 2026-10-06 (hecke-extensions slot 21, **a pre-release, not owner-released**: new Proposition 28.18 — the weight-two presentation at every prime, $p \in \{2, 3\}$ included, is the cohomology of the open modular curve — with Remark 28.19, in a new last subsection of §28; no existing number, label, tag or marker moves; see the changelog);

Version 1.21.0-pre.3 — 2026-10-06 (hecke-extensions slot 21, **a pre-release, not owner-released**: one addendum sentence in (Hi5) — [U]'s (U17)(a) equals [E, `cl:lveis`](2) on the ordinary Eisenstein block for $n < \ell - 1$; no clause's mathematics, label, tag or marker count changes; see the changelog);

Version 1.21.0-pre.2 — 2026-10-05 (hecke-extensions slot 21, **a pre-release, not owner-released**: addenda only — [U]'s lattice-independence of ordinary parts recorded in Remark 27.10(3), the normalisation of Hida's P_n in Theorem 27.11(2), and literature locators read in the slot-21 literature pass; no clause's mathematics, label, tag or marker count changes; see the changelog);

Version 1.21.0-pre.1 — 2026-10-05 (hecke-extensions slot 21, step R20-L, **a pre-release, not owner-released**: the Shimura locators of the nine Shimura `\LEANlit` markers and of the Hecke-ring [standard] notes verified on the owner's scan and restated where the book numbers differ; no clause's mathematics, label, tag or marker count changes; see the changelog);

Version 1.20.0 — 2026-10-05 (hecke-extensions slot 20, the close: **the release** from v1.20.0-pre.2 under the owner's ruling O6, on the slot-20 referees F1 / F1′ (ACCEPT) and the Lean audit F3 of milestone M4 (ACCEPT WITH REPAIRS, repaired); the milestone-M4 `\LEAN` markers (140 on 65 labels); no statement's mathematics, label or count change; see the changelog);

Version 1.20.0-pre.2 — 2026-10-04 (hecke-extensions slot 20, the close delta for the Fable referee F1′, **a pre-release, not owner-released**: code-map text only — WP-MEAS, NB-4, WP-LV3 and S1 / S2 *built* in §22.2, ι^* over $\mathbb{Z}/\ell^k$ built in §22.1; no statement, tag, label or count change; see the changelog);

Version 1.20.0-pre.1 — 2026-10-04 (hecke-extensions slot 20, WP-2, **a pre-release, not owner-released**, for the slot-20 referee F1: code-map text only — the slot-20 packages WP-LV3, WP-MEAS and NB-4 planned in §22.1 and §22.2, their mathematics in [E] 1.20.0-pre.1; the milestone-M4 `\LEAN` markers follow the Lean audit F3 of slot 20; no statement, tag, label or count change; see the changelog);

Version 1.19.0 — 2026-10-04 (hecke-extensions slot 19, the close: **the release** from v1.19.0-pre.6 under the owner's ruling O3, on the slot-19 referees F1 / F1-delta's ACCEPT WITH REPAIRS; the milestone-M3 `\LEAN` markers (98 on 48 labels); no statement's mathematics, label or count change; see the changelog);

Version 1.19.0-pre.6 — 2026-10-04 (F1-delta m-C1: one sentence in err:e6, textual only; before that: hecke-extensions slot 19, WP-2 continued, **a pre-release, not owner-released**: Shimura's notation corrected (the full operator is his $T(p^\nu)$, his $T(1, p^\nu)$ the primitive coset); Proposition 7.4's (H4)(ii) clause records [GT]'s proof of acyclicity (slot-19 L19-3); a v1.19.0-pre.5 box in §22.2 (WP-BIL, WP-LIN built); no tag, label or count change; see the changelog);

Version 1.19.0-pre.4 — 2026-10-04 (hecke-extensions slot 19, WP-2 continued, **a pre-release, not owner-released**: code-map text only — WP-LV2 (the twisted Eisenstein route on §29's character model) and WP-HL (the truncated Λ-adic arm) built, in §22.1 and §22.2; no statement, tag, label or count change; see the changelog);

Version 1.19.0-pre.3 — 2026-10-04 (hecke-extensions slot 19, WP-2 continued, **a pre-release, not owner-released**: code-map text only — the slot-19 packages as built (WP-LC, WP-LV untwisted, WP-NF, NB-2) and as not built (WP-HL, WP-LV twisted) in §22.1 and §22.2; no statement, tag, label or count change; see the changelog);

Version 1.19.0-pre.2 — 2026-10-03 (hecke-extensions slot 19, WP-3 repairs, **a pre-release, not owner-released**: the slot-19 referee F1's ACCEPT, its typographic note m-C1 applied in three §22.1 rows, and the WP-LV pointer follows [E] 1.19.0-pre.2's saturation hypothesis; no statement, tag, label or count change; see the changelog);

Version 1.19.0-pre.1 — 2026-10-03 (hecke-extensions slot 19, WP-2, **a pre-release, not owner-released**, for the slot-19 referee F1: code-map text only — the slot-19 packages WP-NF, WP-LC, WP-LV, WP-HL and NB-2 planned in §22.1 and §22.2, their mathematics in [E] 1.19.0-pre.1 and [U] 0.3.0-pre.1; no statement, tag, label or count change; see the changelog);

Version 1.18.0 — 2026-10-03 (hecke-extensions slot 18, the close: **the release** from v1.18.0-pre.2 under the owner's ruling O3 of 2026-10-03, on the slot-18 referee F1's ACCEPT WITH REPAIRS; the milestone-M2 `\LEAN` markers (81 on 42 labels); the **erratum** in the proof of Theorem 11.7(c), confirmed by the Lean audit F3 and by the owner; no statement, tag, label or count change; see the changelog);

Version 1.18.0-pre.2 — 2026-10-03 (hecke-extensions slot 18, WP-3 repairs, **a pre-release, not owner-released**: the slot-18 referee F1's M-1, m-C1, m-C2 applied ((IL5)'s proof route and hypothesis, Remark 28.16(4)); the code maps of WP-5, WP-6, WP-7 as built and run; (IL3), (IL4) moved to [verified computationally][a]; an **erratum** in the proof of Theorem 11.7(c), for the owner's confirmation; see the changelog);

Version 1.18.0-pre.1 — 2026-10-03 (hecke-extensions slot 18, WP-2, **a pre-release, not owner-released**, for the slot-18 referee F1: contract-first code maps for slot-18 WP-5 (§28, Remark 28.16(4)), WP-6 (§29) and WP-7 (§31, the new checkable item (IL5)); Lemma 30.4's Eichler–Shimura input located in [DI95], verified; no statement's content, tag, label or count changes; see the changelog);

Version 1.17.0 — 2026-10-02 (hecke-extensions slot 17, the close: **the release** from v1.17.0-pre.5, under the owner's ruling O3 of 2026-10-01; the referee F1 (slot-17 WP-3) was ACCEPT WITH REPAIRS, 0 blocking, its repairs applied at pre.3 and pre.4; over pre.5 the first `\LEAN` / `\LEANlit` markers (Lean milestone M1, clause-scoped, 42 markers on 20 labels) — no statement, tag, label or count changes; see the changelog);

Version 1.17.0-pre.5 — 2026-10-02 (hecke-extensions slot 17, WP-2, **a pre-release, not owner-released**: pointers only — slot-17 R17-4's verified locators ((NB4)(c)'s input, [AS86, Lemmas 1.1.1, 1.1.4], now verified; Lemma 30.4's eigenbasis from [DI95]; (LN5) from [Mer96, Théorème 2]), two narrowed attributions to [AS86], and §31's status sentences and code map after slot-17 WP-7; no statement, tag, label or count changes; see the changelog);

Version 1.17.0-pre.4 — 2026-10-02 (hecke-extensions slot 17, WP-2, **a pre-release, not owner-released**: two owner rulings of 2026-10-02 on the slot-17 referee F1 — (NB4)(c)'s sharp form at $n = 0$ tagged [proved here] given one [standard:] input, its locator [AS86, §1.1] not verified; and, following [E]'s restatement of (R) on congruence-free classes, the final wording of (LN6)'s and §30's dependent sentences; no label or count changes; see the changelog);

Version 1.17.0-pre.3 — 2026-10-02 (hecke-extensions slot 17, WP-2 resumed, **a pre-release,**

not owner-released: the slot-17 referee F1's repairs — (NB4)(c)'s orientation proved, its last [standard:] input located but not verified, so the sharp form at $n = 0$ is [verified computationally]^{r}; the code maps of §28 (slot-17 WP-5) and §29 (slot-17 WP-6) built; Remark 28.16(4)'s $N = 99330$ point; five minors; no label, tag set or count changes; see the changelog);

Version 1.17.0-pre.2 — 2026-10-02 (hecke-extensions slot 17, WP-2, second pass, **a pre-release, not owner-released**: one citation repair in Lemma 30.4's proof — the eigenbasis from [Shi71, Thm. 3.41] or [DS05, §5.5], both not verified, multiplicity one from [Ser73, Ch. VII, §5]; no statement, tag or label changes; see the changelog);

v1.17.0-pre.1 — 2026-10-01 (hecke-extensions slot 17, WP-2, drafted during the slot-16 close, **a pre-release, not owner-released**: the deferred items of slot 16 for one referee (slot-17 WP-3, Fable) — Remark 30.10 (new), (NB4)'s sharp form at $n = 0$ as data with a proposed proof, the composite-N caveat of Remark 28.16(4), the code maps of slot 17's packages (contract first), and the `\LEAN` tag defined, not yet used; one numbered statement added, none removed or renamed; see the changelog);

v1.16.0 — 2026-10-01 (hecke-extensions slot 16, the close: **the release** from v1.16.0-pre.3, under the owner's ruling O2 of 2026-10-01; the referee (slot-16 WP-14) was ACCEPT WITH REPAIRS, 0 blocking, every repair applied at pre.3 and confirmed; over pre.3 verified locators, the reason the e_σ average is a no-op on d, and code-map and status sentences only — no statement added, no tag moved; see the changelog);

v1.16.0-pre.3 — 2026-10-01 (hecke-extensions slot 16, WP-12, **a pre-release, not owner-released**: the repairs answering the WP-14 Fable referee (ACCEPT WITH REPAIRS, 0 blocking, 1 major C-1, 8 minors C-2–C-9), all applied; no label added, renamed or removed, no count moves; see the changelog);

v1.16.0-pre.2 — 2026-10-01 (hecke-extensions slot 16, WP-12 follow-up, **a pre-release, not owner-released**: (NB5)'s product form squared, with its proof and the reduction rule (slot-16 WP-9 audit); no label added, renamed or removed; see the changelog);

v1.16.0-pre.1 — 2026-10-01 (hecke-extensions slot 16, WP-12, **a pre-release, not owner-released**: new §28.6 — the model-to-presentation map, the degeneracy maps α_d on the cokernel presentation (replacing v1.15.0-pre.3's unrefereed sentence), higher weight at composite N, (LN7)–(LN9) — and the new chapter §31, the ℓ-local isotropy model and the closed form at $\ell = 3$; ten numbered statements added, none removed or renamed, no tag moved; referee slot-16 WP-14; see the changelog);

v1.15.0 — 2026-10-01 (hecke-extensions slot 15, WP-18: **the release at the slot-15 close** from v1.15.0-pre.3, under the owner's ruling Q2 of 2026-09-30; the referee (slot-15 WP-4) was ACCEPT WITH REPAIRS, 0 blocking, all applied at pre.2; §29 released unbuilt; pre.3's one scope sentence released unrefereed; no statement added, no tag moved; see the changelog);

v1.15.0-pre.3 — 2026-09-30 (hecke-extensions slot 15, WP-17, **a pre-release, not owner-released**: one scope sentence in §28 on the degeneracy maps and the cokernel presentation; the sparse elimination built; no statement added, no tag moved; see the changelog);

v1.15.0-pre.2 — 2026-09-30 (hecke-extensions slot 15, WP-3 resumed, **a pre-release, not owner-released**: the repairs answering the WP-4 Fable referee, 0 blocking, all applied; $H_{N,p}$ and `HeckeHidaComponents` built; no statement added, no tag moved; see the changelog);

v1.15.0-pre.1 — 2026-09-30 (hecke-extensions slot 15, package WP-3, **a pre-release, not owner-released**: the new chapters §29 (nebentypus for every N over $\mathbb{F}_q$, contract first, not built) and §30 (the saturation statements behind the $\mathbb{Z}/\ell^K$ Hecke-algebra route, and the torsion-aware comparison with modular forms); no existing statement changed, no tag moved; see the changelog);

v1.14.2 — 2026-09-29 (hecke-extensions slot 14, owner ruling Q-F: §28's code maps now say *built* (slot-14 WP-12, `7a5158d`); no statement changed, no tag moved; see the changelog);

v1.14.1 — 2026-09-29 (hecke-extensions slot 14, owner rulings of 2026-09-29 on the contract-currency audit: the second cluster run's [verified computationally][a] sentences in (Hi11), (Hi12) and §27's code map ([S14] Q-B), and Lemma 27.15, the U_ℓ lever's rank claim (Q-H); no existing statement changed, no tag moved; see the changelog);

v1.14.0 — 2026-09-28 (hecke-extensions slot 14, package WP-13: **the owner-signed release** of §28 from v1.14.0-pre.2, released unbuilt. The referee was ACCEPT WITH REPAIRS with 0 blocking. The data sentences of the first cluster run are restored in (Hi11), (Hi12) and Remark 16.8, citing the extract `docs/matrix-engine/results/noctua2_run1/`, published on the owner's sign-off ([S14] Q2, Q3). Release note: `docs/implementation/slot14/WP-13_release_note.md`);

v1.14.0-pre.2: 2026-09-28 (hecke-extensions slot 14, package WP-5, **a pre-release, not owner-released**: the repairs answering the Fable referee of §28, ACCEPT WITH REPAIRS, 0 blocking, all eleven findings applied — a Hecke matrix on H^1 costs quadratically in m_N through the normal form of Rel_N (Remark 28.11), what the Eisenstein scan computed is stated ((LN4), (LN6)); the data remarks of the first cluster run taken back out of (Hi11), (Hi12) and Remark 16.8 ([S14] Q2's default), the provenance line of §27's code map kept. Referee: `docs/implementation/slot14/WP-5_referee.md`);

v1.14.0-pre.1: 2026-09-28 (hecke-extensions slot 14, package WP-4, **a pre-release, not owner-released**: a new chapter §28, **the contract first** — the binding coset table for every N (obstacle O3), the cokernel presentation pinned with Manin's sequence and a duality natural in restriction, corestriction and conjugation, and the Hecke action of T_p and U_p on it by Merel's matrices, transcribed and **proved** (obstacle O2), its cost and the predictions (LN1)–(LN6), among them mod-p Loewy length $e+1$ at Mazur's Eisenstein ideal; data remarks of the first cluster run added to (Hi11), (Hi12) and Remark 16.8, their publication the owner's (removed at pre.2). Report: `docs/implementation/slot14/WP-4_report.md`);

v1.13.0: 2026-09-27 (hecke-extensions slot 13, the close, package WP-15: **the owner-signed release** of §27 from v1.13.0-pre.2 — the referee ACCEPT with 0 blocking, slot-13 WP-6's arms held; one sentence of Definition 27.13 amended (the module route's Nakayama basis on Hida's lattice), the code maps read "built". Report: `docs/implementation/slot13/WP-15_report.md`);

v1.13.0-pre.2: 2026-09-26 (hecke-extensions slot 13, package WP-3: the repairs answering the Fable referee of §27, ACCEPT WITH REPAIRS, all twelve findings applied — Hida's lattice L_n is ρ_n conjugated by the binomial diagonal, not V (Remark 27.10(3), the variant (H) of Definition 27.13, (Hi8)'s arms), and the cost model's regimes and obstacles, with the Manin–Merel Hecke action on the cokernel named (Remark 27.7). Response: `docs/implementation/slot13/WP-3_response.md`);

v1.13.0-pre.1: 2026-09-26 (hecke-extensions slot 13, package WP-2, the pre-release of an X bump, **the contract first**: a new chapter §27 — level structure by Shapiro on the fixed model (the level-N correspondences of Proposition 27.2, the level-ℓ operator of Proposition 27.3, the old space, the orbit-wise cokernel and a cost model), the integral stabilisation (Proposition 27.9), Hida's control and weight independence as literature with verified locators, the lattice datum and the predictions (Hi7)–(Hi12), committed before their code. Report: `docs/implementation/slot13/WP-2_report.md`);

v1.12.1: 2026-09-26 (hecke-extensions slot 12, post-close, package S12-census2: a **non-normative** bump, code maps only — §26's code map records what is built, `Hecke/HeckeOrdinaryPart` and `HeckeExt/HeckeHidaTower`, and §22.1 names `HeckeExt/HeckeFinalityRoutes`; companion pointer [E] v1.13.1. Report: `docs/implementation/slot12/census2_report.md`);

v1.12.0: 2026-09-26 (hecke-extensions slot 12, post-close, **the owner-signed release**, from v1.12.0-pre.3: publishes §26 (the foundation of Hida theory at level 1) and **the Shapiro transcription in degree 1, proved** — Proposition 14.17 and Lemma 14.18 — both accepted by their Fable referees after repairs, all applied, none outstanding. **The one status change this version makes**: Question 25.6 is **answered** at $\ell \geq 5$, $n \geq 1$ by Corollary 26.2 (with Proposition 26.1 and Remark 25.4): the tag moves [open] → [proved here], and the item is retitled "— ANSWERED 2026-09-26", exactly as 14.24 lost its `[open]` tag at v1.3.0. No statement added, no label renamed or removed, one tag moves; the question text and the boxed statement of 25.6 are unchanged (rules (L1), (L2)). Report: this release note (owner sign-off 2026-09-26; `docs/implementation/slot12/WP-2_report.md`, `WP-3_referee.md`, `WP-16_report.md` and `WP-16_referee.md` record the refereed content));

v1.12.0-pre.3: 2026-09-26 (hecke-extensions slot 12, package WP-16, post-close, the third pre-release: **the Shapiro transcription in degree 1, proved** — Proposition 14.17 (the coordinate Φ an isomorphism over every ring at $\rho_n(-I) = \mathrm{id}$; an H^1 formula at every ν and every p; (31) at $\nu = 1$, with no hypothesis $p \in R^\times$) and Lemma 14.18 (boundary operators away from ℓ factor through the torus); status notes on Proposition 14.16, Obligation 14.19(v.b) and Claim 16.24(2). Two statements added, no label renamed or removed, no tag moved. Report: `docs/implementation/slot12/WP-16_report.md`);

v1.12.0-pre.2: 2026-09-26 (hecke-extensions slot 12, package WP-3, the second pre-release: **the Fable referee's repairs** (ten minor findings, all applied; none blocking, none major) — the scope clause of Proposition 26.1, the definition of the ℓ-adic projector and $e = e'$ in Corollary 26.2, the free-block sentence of Remark 26.4 with a [verified computationally] record, and (Hi2), (Hi3), (Hi5), (Hi6) made explicit. No label renamed or removed, no statement added. Report: `docs/implementation/slot12/WP-2_report.md`);

v1.12.0-pre.1: 2026-09-26 (hecke-extensions slot 12, package WP-2, the pre-release of an X bump, **the contract first**: a new chapter §26, the foundation of Hida theory at level 1 — the torsion pre-pass (Proposition 26.1: the ℓ-primary torsion of H^1 from $\rho_n(S)$, $\rho_n(T)$ alone), the certified precision before the run (Corollary 26.2, which answers Question 25.6 at $\ell \geq 5$, $n \geq 1$; the item's status macro waits for the release), the ordinary projector as one power of U (Lemma 26.3), the weight towers and the predictions (Hi1)–(Hi6), committed before their code. Six statements added, no label renamed or removed, no existing tag moved. Report: `docs/implementation/slot12/WP-2_report.md`);

v1.11.1: 2026-09-25 (hecke-extensions slot 11, package WP-17, the slot close, a Y bump: **the WP-15 referee's repairs** — Remark 24.14's provenance sentence restated to the reports, a new Remark 25.4 (the ordinary part is torsion-free at level 1, at $\ell \geq 5$, $n \geq 1$), Question 25.6 narrowed to the per-cell bound (still open), and wording. No normative statement moves, no label renamed or removed; one [proved here] remark added. Report: `docs/implementation/slot11/WP-17_report.md`);

v1.11.0: 2026-09-25 (hecke-extensions slot 11, package WP-15: **the slot's X bump published** from v1.11.0-pre.2 with what slot 11 built — Part VII's objects and every checkable item of Remark 24.14 (all held; (C9)'s indexing said), the lifts Proposition 23.23 needs and its recursion on the free model, the ℓ-adic digit route and the ordinary idempotent on it; Obligation 20.10(iv)'s scope sentence worded to the digit route as built; Question 25.6 still open. No statement added, no label renamed or removed; one [verified computationally] tag added. Report: `docs/implementation/slot11/WP-15_report.md`);

v1.11.0-pre.2: 2026-09-25 (hecke-extensions slot 11, package WP-2, the second pre-release: **the Fable referee's repairs** (one major, nine minor; Proposition 24.4 confirmed independently) — Obligation 20.10(iv)'s owner-ruled scope sentence for the fixed-width digit-plane route, Proposition 24.4 at $\ell = 2$, a multiplicity in (C5), the model's answer to (C7) recorded, and wording. No label renamed or removed, no tag moved. Report:

`docs/implementation/slot11/WP-2_report.md`);

v1.11.0-pre.1: 2026-09-25 (hecke-extensions slot 11, package WP-2, the pre-release of an X bump, **the contract first**: a new Part VII states multiplication by the Dickson invariant L and the operator θ_{L} it induces (§24) — the determinant twist on all of $\mathrm{M}_2(\mathbb{F}_\ell)$, $T_p\theta_{\mathsf{L}} = p\,\theta_{\mathsf{L}}T_p$ and $U_\ell\theta_{\mathsf{L}} = 0$ at cochain level, the quotient Q_n with a fixed basis, the long exact sequence with $\ker\theta_{\mathsf{L}} = \operatorname{im}\delta$ and its closed forms, $H^0 = \mathbb{F}_\ell[\mathsf{L},\mathsf{D}]$ with T_p and U_ℓ on it (Proposition 24.4: U_ℓ has rank ≤ 1 and square zero on H^0, and is *not* zero at $n = \ell^3 - \ell$), the literature on θ and what of it is checkable on the model — and the ordinary projector (§25), with the precision question open; and §23.7 states 10.8's recursion on the free model for every ring. Twenty statements added, no label renamed or removed, no existing tag moved. It carries its own identifier so that no version number is published twice under different text ((V1), (V2)); the referee's repairs will be v1.11.0-pre.2. Report: `docs/implementation/slot11/WP-2_report.md`);

v1.10.1: 2026-09-25 (hecke-extensions slot 10, package WP-15, phase 2, a Y bump: **the phase-1 referee's repairs — no normative predicate, no hypothesis, no displayed formula, no tag and no label moves**. The Fable referee of v1.10.0 accepted it after two major findings, each a one-line repair, and four minor ones on this document: Remark 23.15 dates the canonical τ to v1.1.0 and cites the one comparison this document has at $\ell \in \{2,3\}$, the legacy against the canonical τ, as an observation; its "compares nothing there" names the operators meant; Remark 23.14(3) counts 359 cells on eight rings; Remark 23.21 attributes its 224-element check correctly. Report: `docs/implementation/slot10/WP-15_report.md`);

v1.10.0: 2026-09-25 (hecke-extensions slot 10, package WP-15, an X bump on the owner's rulings of 2026-09-25, all at recommendation (A): **the strict lift of Proposition 23.13 is built** for $g = \operatorname{diag}(1,p)$, $p \in \{2,3,5\}$, in degrees $\leq q_{\max}+1$, certified exactly over $\mathbb{Z}$, and recorded with its measured integer sizes in a new Remark 23.14; a new Remark 23.15 beside Proposition 23.13(4) states that the operator on cohomology of the Bredon object at $\ell \in \{2,3\}$ is **not shown** independent of τ and that this document compares nothing there;
Lemma 14.22's proof fixes the sign of its Bockstein, so its class e is pinned;
Obligation 15.22(vii) over $\mathbb{Z}/\ell^e$, $\ell \geq 5$, is **discharged in its first two sentences at** $\ell^e < 2^{32}$, every E8 point, its third (the Poincaré-duality gate) recorded unavailable over the ring; §23 gains the lifted datum, the class e with its periodicity map, and the integral Hecke action read basis-free. Two statements added, no label renamed or removed, no existing tag moved. Report: `docs/implementation/slot10/WP-15_report.md`);

v1.9.1: 2026-09-25 (hecke-extensions slot 9, package WP-17, the slot close, a Y bump: **the referee's repairs and status — no normative predicate, no hypothesis, no displayed formula, no tag and no label moves**. The Fable referee of v1.9.0 accepted it after repairs (two major, both one-sentence, and four minor findings on this document): the slot's independent review is now in the tree and cited (`docs/implementation/slot9/WP-14_review.md`); Obligation 15.22(iv)'s as-built sentence follows the slot's trimming of the item-4b checks; two commit hashes are back-filled; one citation, one sentence's grammar and one code-map lead are repaired; the `Hecke*` count is re-derived. Report: `docs/implementation/slot9/WP-17_report.md`);

v1.9.0: 2026-09-24 (hecke-extensions slot 9, package WP-15, an X bump on the owner's rulings of 2026-09-24, all at recommendation (A): **Remark 16.8(2) is stated at every prime**, with its instances at $\ell \in \{2,3\}$; a new Remark 23.18 reads Proposition 23.17(3) **differential by differential**; Definition 23.20(2)'s stale refusal clause is replaced by what this build computes at $\ell \in \{2,3\}$ — the Bredon object's lengths and invariant factors — and what it refuses; Obligation 15.22(iv)'s non-field half is **checked** over $\mathbb{Z}/\ell^e$, $\ell \geq 5$ (phase E8 item 4b), while (vi) is partial there and (vii) holds by route 1 only, its composite half open; §23 is partly built (its free model, comparison map and integral groups for $n \leq 20$). No label renamed or removed, no existing tag moved; one statement added. Report:

`docs/implementation/slot9/WP-15_report.md`);

v1.8.1: 2026-09-24 (hecke-extensions slot 8, package WP-14, the slot close, a Y bump: **the referee's wording repairs and status — no normative predicate, no hypothesis, no displayed formula, no tag and no label moves**. The five minor findings of the referee of v1.8.0 are applied as wording (Lemma 23.12's proof, the notation row of $F^{\sigma}_{\bullet}$, Theorem 17.3's v1.8.0 sentence, Proposition 23.3's family, Remark 23.11's citation); the abstract's stale sentence is repaired; Appendix E gains the entries of v1.7.2–v1.7.4 and its resolvable commit hashes. Phase E8 items 2, 3 and 4a are implemented — the complex over the ring in force at $\mathbb{Z}/\ell^k$, $\ell \geq 5$, its reductions and Bockstein — and item 4b (the boundary and the cone over $\mathbb{Z}/\ell^k$) is the next slot's; §22.1 and §22.2 record it, and the registered-validation counts are re-derived (114; 65 `Hecke*`). Close report: `docs/implementation/slot8/WP-14_report.md`);

v1.8.0: 2026-09-24 (hecke-extensions slot 8, package WP-9, an X bump: **the integral line**, a new Part VI, §23 — what the shipped complex is over the ring in force (Bredon cohomology, Proposition 23.3), the free model on the *unrefined* tessellation with one integral comparison map (Definition 23.4, Proposition 23.5), the extension plan's contract item C-A5 re-scoped onto it and owner direction OD-T1 answered "not needed" (Remark 23.6, owner ruling 6c of 2026-09-24), the descent over $\mathbb{Z}$ (Propositions 23.7–23.10), the strict lift of the Hecke datum (existence only, Proposition 23.13), the bad primes, the certified-exponent rule and the characteristic polynomial over $\mathbb{Z}$ by residues, and what each coefficient ring names (Definition 23.20); and the F-A remark beside Claim 16.7 (Remark 16.8, owner ruling 10). Twenty statements added, (H3) and Theorem 17.3's closing paragraph amended, no label renamed or removed, no existing tag moved; the arbitrary-precision ban is unchanged. Report: `docs/implementation/slot8/WP-9_report.md`);

v1.7.4: 2026-09-23 (hecke-extensions slot 7, package WP-14, the slot close: **status only — no normative predicate, no hypothesis, no displayed formula, no tag and no label moves**. Phase E7 is **accepted**: Claim 16.26(1)–(2) are *tested* on all 54 cells of the extension stage's case list at $\ell \geq 5$ and fail on none, and the claim stays open (slot-7 WP-1); the extension record of §22.4 item 6 is `hecke_ext/v3` (slot-7 WP-3, stage 3), its new keys being [E]'s field inventory; §22.1 and §22.2 record it, and the registered-validation counts are re-derived (108; 61 `Hecke*`). Close report: `docs/implementation/slot7/WP-14_report.md`);

v1.7.3: 2026-09-22 (hecke-extensions slot 6, package WP-14, the slot close: **status only — no normative predicate, no hypothesis, no displayed formula, no tag and no label moves**. §8.1 is **implemented** (slot-6 WP-1), and Obligation 15.22(vii) is **implemented in all three sentences** — the strict-pair route (WP-4), route 3 (WP-5) and the Poincaré-duality gate for X_3 at $\ell \geq 5$ (WP-3), which Claim 16.26 still only *tests*; §22.1 and §22.2 record it, and the registered-validation counts are re-derived by `--list` (106; 60 `Hecke*`). Close report: `docs/implementation/slot6/WP-14_report.md`);

v1.7.2: 2026-09-22 (hecke-extensions slot 5, package WP-13, the slot close: **status only — no normative predicate, no hypothesis, no displayed formula, no tag and no label moves**. §22.1 records what slot 5 shipped, the registered-validation counts are re-derived by `--list` (100; 55 `Hecke*`), and the one point v1.7.1 left open for the owner — whether Obligation 14.19(vii) is a distinct weaker clause or a pointer to Obligation 15.22(iii) — is **decided: Reading A, a distinct weaker clause**, so v1.7.1's text stands as drafted. Close report: `docs/implementation/slot5/WP-13_report.md`);

v1.7.1: 2026-09-22 (hecke-extensions slot 5, package WP-12, the independent referee pass on v1.7.0, a Y bump: **pointers and self-contradictions only — no normative predicate, no hypothesis, no displayed formula, no tag and no label moves**. Five of the six repairs are of text v1.7.0 itself introduced: §22.4 item 6(f) bound "(a)–(e) and (g)–(h)" where the item's preamble binds "the nine below", leaving rule (i) bound and unbound at once; Remark 15.23's opening said "load-bearing for no clause" two lines before saying it still

discharges Obligation 14.19(vii); that clause's new asymmetry paragraph rested on "the data *this* chapter stores are τ_∂ and Θ_g^∂ alone", which the clause's own statement falsifies, the real missing datum being η and h; its opening sentence still called it a clause that "belongs to the pair and is stated in full at" Obligation 15.22, which v1.7.0 falsified by strengthening (iii) alone; and Obligation 15.22(iii)'s proof now names $z \in Z^q_{\mathrm{eq}}$, the "one more word" its author's own residual finding asked for. The sixth records that slot-5 WP-11a landed §22.4 item 5's refusal. **Open for the owner**: whether Obligation 14.19(vii) is a distinct weaker clause or a pointer to Obligation 15.22(iii) — this version takes the first reading, which is the one v1.7.0's own "the pair's form of" commits to, and flags the choice. Report: `docs/implementation/referee_report_2026-09-22_contract_v1_7_0.md`);

v1.7.0: 2026-09-22 (hecke-extensions slot 5, package WP-11, the slot's one X revision. **Obligation 15.22(iii)'s $\ell \in \{2,3\}$ carve-out on the r-half is dropped and the identity asserted at *every* ℓ, with the proof written in**: clause (i) carries no hypothesis on ℓ and aborts on failure, so on a stored cocycle z the difference $\Theta_g^\partial rz - r\Theta_g z$ is $d(hz)$, a coboundary, and $r^q A^{(q)} = A_\partial^{(q)} r^q$ on cohomology follows — the implication that was re-derived in three successive packages and was never contract text (owner ruling `[OD]` slot-4 close-out 3 of 2026-09-22; the hypothesis it uses re-checked against the shipped code by slot-5 WP-4). The clause states, in as many words, *what it does not say*: at $\ell \in \{2,3\}$ it is about the complexes the implementation computes and not about $H^\bullet(\Gamma; M_n)$, which Remark 15.19 keeps separate. And it states the **asymmetry**: Obligation 14.19(vii), in the boundary chapter, has neither η nor h and therefore no proof, so it keeps its carve-out and gains a pointer; Remark 15.23 is demoted to the measurement that preceded the proof, load-bearing for no clause. Obligation 15.22(v)'s closing sentence names *both* the stored object's certificate and the validation arm, the shipped `ConeCertificate` now carrying the unconditional first identity of (40) under its own guard. §22.4: item 5 makes the derived-complex *names* the record's own derived flag, so that `q_max_source = "set"` on a derived complex is refusable by a verifier and not only by the builder; item 6 becomes `hecke_ext/v2` with nine binding rules, (b), (c), (g), (h) and (i) new or amended, and (a) and (d) answer the two halves of v1.6.0's deferred item 3. Obligation 9.10(ii) gains the rider for a complex without exact group data. *Deferred again*: the ι-equivariance of h, which no accepted package measured. No label added, renamed or removed; no tag moves. Status sentences stay with the slot close except where this version's own text falsifies them. Report `docs/implementation/slot5/WP-11_report.md`);

v1.6.2: 2026-09-22 (hecke-extensions slot 4, package WP-11, the slot close, a Y bump: **status only — no normative statement, no hypothesis, no displayed formula, no asserted predicate, no tag and no label moves**. What is recorded: the *production* involution exists — §9's 2026-09-16 census, its specification paragraph, §22.1's row and §22.2 are rewritten from the pointers v1.6.1 left, naming `Hecke/HeckeInvolution`, `HeckeExt/HeckeSignSplit` and `HeckeInvolutionValidation` (slot-4 WP-3, phase E3); §22.4 item 6's status becomes "written, bulk complex" with `HeckeExt/HeckeExtDriver` named (WP-1, WP-2, phase E2); §22.1 gains `HeckeExt/HeckeExtRecord`, `HeckeExtReport`, `HeckeExtFinal`, `HeckePredictions` and `GloverFactors`, the two new settings keys `commutant_budget` and `final_mode`, and — in the Chapter 15 row — WP-4's three *value-preserving* certificate re-implementations, with the statement that no clause was sampled, relaxed or moved out of `compute()`; Remark 22.1 records the content-keyed allow-list (WP-6) and the slot-4 census run; and the validation counts go 89 → 94 and 45 → 49 `Hecke*`, both counted by `--list` and not asserted. Three referee questions of v1.6.1 (`ConeCertificate` C11 versus the strengthened clause (v); the $\ell \in \{2,3\}$ carve-out in Obligation 15.22(iii); the scope of the v1.6.1 pointer repair) are *not* decided here — they are the owner's, and the first two are slot-5 code. Report `docs/implementation/slot4/WP-11_report.md`);

v1.6.1: 2026-09-22 (hecke-extensions slot 4, package WP-9, the independent referee pass on

v1.6.0, a Z bump: **no hypothesis, displayed formula, asserted predicate, tag or label moves**. Two recorded measurements are corrected — Remark 9.8's "thirteen cells" counts $(10,691)$ twice and folds in four $n=0$ cells at which Obligation 9.10(v) makes no prediction (it is *eight*), and §15.3's closing paragraph said the commutator $[\iota_c, A_c^{(q)}]$ was recorded on the 15-cell grid where it is recorded at $(10,691)$ alone. Three status sentences that v1.6.0's own new text falsifies — §9's census ("no `HeckeInvolution` translation unit"), its specification paragraph and §22.1's row, plus §22.2, all saying that no production ι^* exists — carry a pointer, minimally, the rewrite itself remaining the slot close's; and the title page's companion pointer, three revisions stale, moves to [E] v1.5.1. Companion repairs at [E] v1.5.1: two false statements ([E]'s `cl:harder`, its new predicate paragraph at $\ell \in \{2,3\}$, and the Glover recursion's factor bound) and two proof defects. Report `docs/implementation/referee_report_2026-09-22_contract_v1_6_0.md`);

v1.6.0: 2026-09-21 (hecke-extensions slot 4, package WP-8, an X bump: the eight items Appendix E deferred at v1.5.2 are resolved — Obligation 15.22(i) "as matrices" named as the stored $C^\bullet_{\mathrm{eq}}$ basis with the flat identities *recorded*, (iii) given the $\ell \in \{2,3\}$ carve-out on the r-half, (iv)'s explicit im = ker check read degree-generically, (v) and (40)'s first identity freed of the hypothesis $H^2_{\mathrm{eq}} = 0$ (with Lemma 15.2(5) and its proof restructured), (vi) told where the *full* T_{p^ν} comes from, the preamble's "six maps" replaced by the three map families, Proposition 15.16(2) read inside $C^\bullet_{\mathrm{eq}}$ and Lemma 15.6(V)'s "single edge path" repaired — together with the normative findings of the slot-4 packages accepted when this revision started: §22.4 gains item 6, the extension record `hecke_ext/v1` and its six binding rules — two of them from the E2b driver, on what a *resumed* run's settings block contains and on what the incomplete-item count counts — and Remark 9.8's mirror-split prediction becomes a measured record while §15.3 gains a paragraph saying that $[\iota_c, A_c^{(q)}] = 0$ follows from nothing here. Owner confirmations of 2026-09-21: (a) yes, (b) yes, (c) per mode. No label added, renamed or removed; one tag moves. Status sentences are the slot close's. Report `docs/implementation/slot4/WP-8_report.md`);

v1.5.2: 2026-09-17 (hecke-extensions slot 3, package WP-11, the slot close, a Y bump: status only — strict locality mode implemented (the one false status sentence of v1.5.1, at §22.1); Chapter 15 implemented except Obligation 15.22(vii), Obligation 15.21 discharged in full; the U_ℓ repair, the Sturm sweep and the citation census in the code maps; the validation count $80 \to 89$. No normative statement, tag or label moves; the normative findings that arrived after v1.5.0 started are listed for the next X. Report `docs/implementation/slot3/WP-11_report.md`);

v1.5.1: 2026-09-17 (hecke-extensions slot 3, package WP-9, a Y bump: the referee pass on v1.5.0 — all ten resolutions accepted, (L1) and Appendix B's counts re-verified mechanically, and seven defects repaired, six of them in the U_ℓ item: Theorem 18.7(1)'s "$N \le 2^{15}$" is $N \le 23170$, the reach §10 already recorded; its $N \lesssim 4580$ is a *measurement*, not a consequence of its criterion; item (iv)'s "$B_{\mathrm{red}} \le 7C^2N^2$ by Lemma 18.5(6)" is proved for the entry point only and the rest is measured; Lemma 18.5(6)'s proof gains the two shapes the reduction applies and names the coordinate that attains the order; and Obligation 20.7's "off the Kummer loci E is a line" is measured, not proved; the seventh is rule (A4)'s example, which still quoted Obligation 15.21(ii) in the form v1.5.0 corrected. No normative statement, tag or label moves. Report `docs/implementation/slot3/WP-9_referee_report.md`);

v1.5.0: 2026-09-17 (hecke-extensions slot 3, package WP-8, an X bump: the seven items v1.4.2 deferred are resolved — Remark 15.17's arithmetic rewritten from the measured rank $r^1 = 1$ (its conclusion unchanged) and "the Eisenstein class" mod ℓ read as the joint eigenspace, here and at Obligation 20.7; Obligation 15.21(ii), (iv) and its preamble corrected; Lemma 15.6(V) names the constructive choice; §22.4 item 5 gains the rule for a map and for a derived complex; Obligation 15.22(iv) the per-complex meaning of "reported"; and Lemma 18.5 gains

clause (6), with Theorem 18.7(1) corrected, which is what the U_ℓ reach repair found — together with the normative findings of the slot-3 packages accepted at the time (WP-1, WP-5); no label added, renamed or removed; status sentences are the slot close's; report `docs/implementation/slot3/WP-8_report.md`);

v1.4.2: 2026-09-16 (hecke-extensions slot 2, package WP-13, the slot close, a Y bump: status only — §22.4 item 5 implemented at `hecke_report/v2`, the code maps and the obligation roster brought up to the slot-2 units, Obligation 15.21(i)–(iv) discharged, no normative statement changed; the findings the slot's packages handed to the next X revision are listed in Appendix E; report `docs/implementation/slot2/WP-13_report.md`);

v1.4.1: 2026-09-16 (hecke-extensions slot 2, package WP-10, a Y bump: referee pass on v1.4.0 — each of the nine resolutions re-derived and accepted, no normative statement changed; the prose and proof repairs are listed in Appendix E, the largest being Remark 14.13(3)'s $(2,1)$ exhibit, which v1.4.0 had placed on slots where no tie occurs; report `docs/implementation/slot2/WP-10_referee_report.md`);

v1.4.0: 2026-09-16 (hecke-extensions slot 2, package WP-9, an X bump: the nine open points that v1.3.3 recorded against chapter 14 are resolved, as the slot-1 reports proposed and the owner accepted — Obligation 14.19(v) is split into the three comparisons the chapter offers, (v.a)–(v.c), and is degree 0 alone at $\nu \geq 2$; its clause (vii) is qualified by $\chi_0(p) \neq \chi_1(p)$; Proposition 14.4(4) identifies the circle model with $\mathcal{R}_\partial$ *including* the $-I$ self-relations for every n; Proposition 14.16(2) says what the right-coset sum is and (3) states the $H^1 \cong (V_R)_T$ coordinate; the canonical τ_∂ is Algorithm 14.12, and Remark 14.13 proves that no vertex rule makes it ι-symmetric; the record fields of §22.4 item 5 are per complex; the $\ell \in \{2,3\}$ restriction equivariance is recorded in §15 as Remark 15.23; Remark 14.7 records the general-ν shape as not stated);

v1.3.3: 2026-09-16 (hecke-extensions slot 1 close, a Y bump: chapter 14 is now *implemented* — the certified Γ-level τ_∂ and the boundary operator Θ_g^∂ exist in the tree and Obligation 14.19 is discharged in full, (i)–(vii), by two registered validations — so every status site that said "three chapters state objects nothing implements" now says two; the code maps, the roster row, the oracle-family (A) run record, the validation counts ($68 \to 77$) and the build-surface target list are updated; no normative statement changed);

v1.3.2: 2026-09-16 (coherence pass after OQ3 and OC-1: ten stale or over-broad sites repaired, among them the direction of Remark 9.8's contingent prediction — duality pairs each ι^*-sign with the *opposite* one, the orientation-reversal factor having been omitted — the "for n even" qualification of Definition 9.1(iv)'s "not a pullback along any lift", and the provenance counts of Appendix E; no normative statement changed);

v1.3.1: 2026-09-16 (referee pass on Lemma 14.22, the audit the OC-1 decision asked for: every clause and step re-derived, tag [proved here] confirmed, no normative statement changed; four prose repairs listed in Appendix E; report `docs/implementation/referee_report_2026-09-16_bulkperiodic.md`);

v1.3.0: 2026-09-16 (**OC-1 decided**: owner decision of 2026-09-16 adopting (OC-1a) with three riders — 14.24 loses its `[open]` tag and becomes the binding truncation policy, Obligation 15.22(iv) and the record schema of §22 are narrowed to it; and the bulk periodicity above the virtual cohomological dimension, named in Remark 14.21(2) as an unverified citation, is *proved* as Lemma 14.22 together with its Hecke equivariance, so that half of (OC-1c)'s hypothesis is a theorem);

v1.2.0: 2026-09-16 (owner decision OQ3 resolved by **negating Definition 9.1(iv)**: $\iota^* := -\iota^*_{\mathrm{geo}}$, the geometric pullback twisted by the sign character, so that "+" is *by definition* the ι^*-eigenspace containing the Eisenstein classes; Obligation 9.10(v) becomes a gating definition-consistency oracle and holds on every cell, the split is $(d_{n+2}, d_{n+2}-1)$ as predicted, $\iota^* = -\mathrm{id}$ on $H^0(\Gamma;\mathbb{F}_\ell)$; every $\pm$ label switches against the geometric reading; the cost is

`docs/implementation/oq3_resolution.md`);

v1.1.1: 2026-09-16 (status only, no normative statement changed: Obligation 9.10(ii),(iii),(v) implemented and run by `HeckeIotaSignValidation` — the aborting content passes; the *recorded* sign prediction of (v) is false for the then-current involution, the Eisenstein class lying in its $H^1_{\mathrm{eq},-}$ with the mirror split $(d_{n+2}-1, d_{n+2})$; resolution deferred to the owner, `[OQ3-PENDING]`);

v1.1.0: 2026-09-16 (the canonical τ of Algorithm 18.3 re-baselined by owner decision to be ι-symmetric — the ι_K-equivariant tie-break of step (1b) and the tree-routed path templates of step 2; Obligation 9.10(iv) discharged *positively* on the canonical τ and gating there; Proposition 9.4(3b) available at $\ell \in \{2,3\}$; every frozen cochain-level reference value re-baselined, nothing on H^q_{eq} moved);

v1.0.0: 2026-09-16 (**the leading version field becomes** 1: rule (V5)(ii) discharged — (V4)'s two freezes carried out (`hecke_action.tex` frozen at v1.0, `monodromy_v5.tex` at v5.0, each with a superseded-by banner) and `hecke_extensions.tex` v1.3 revised to cite the nine labels of §16 rather than define them, so the label namespace is unambiguous; no normative statement of this document changed);

v0.19.0: 2026-09-16 (chapters 21–22 and Appendices B–C: the worked T_2 datum and the numerical oracles, with erratum E1 applied to oracle 6 and the new erratum E23 to oracle 7; the code map, the configuration surface and the record schema; the STATUS index; the formalization notes. **The last chapter stub is written, but the leading version field is held at** 0: rule (V5) is amended in this version to require an unambiguous label namespace as well, and nine labels of §16 are still defined both here and in the live `hecke_extensions.tex`; erratum E23);

v0.18.0: 2026-09-16 (chapters 18–20: the reduction primitive, the τ-table algorithm and the integer bounds; the certified floating-point filter; the correctness obligations, with erratum E1 applied to 20.4 and the whole obligation roster integrated; erratum E22);

v0.17.0: 2026-09-16 (chapters 16–17: the six groups, the long exact sequence and Poincaré duality, re-homed from [E] by the scope cut; the general orbifold model as definitions, with the construction deferred);

v0.16.0: 2026-09-16 (the τ-existence lemma apportioned: existence over $\mathbb{Z}$ for every ℓ, homotopy uniqueness still under $6 \in R^\times$; notation collisions C18, C19; a corrected bibliography entry);

v0.15.0: 2026-09-16 (chapter 15: the pair homotopy η, the derived h, the cone action Φ_g and its well-definedness);

v0.14.0: 2026-09-16 (chapter 14: the boundary datum, and the $\ell = 2$ truncation policy **OC-1** as a tagged working assumption);

v0.13.0: 2026-09-16 ($\iota_X \tau \iota_X \neq \tau$ computed and recorded; Obligation 9.10(i),(iv) discharged);

v0.12.0: 2026-09-15 (chapter 9: the involution ι);

v0.11.0: 2026-09-15 (the Changelog, Appendix E, back-filled; rule (V5));

v0.10.0: 2026-09-15 (FD4 proved for the model; FD1/FD3 repaired for a non-effective action);

v0.9.0: 2026-09-15 (Part I re-homed from `monodromy_v5.tex`; erratum E7 closed);

v0.8.0: 2026-09-15 (notation C17; two stale uniqueness claims);

v0.7.0: 2026-09-15 (chapters 12–13: induced systems, Ξ_g, Θ_g, the equivalent formulations);

v0.6.0: 2026-09-15 (notation C16, the lift macro);

v0.5.0: 2026-09-15 (chapters 10–11: the Hecke datum, τ, the definition);

v0.4.0: 2026-09-15 (notation C15);

v0.3.0: 2026-09-15 (chapters 7–8: conventions, the model, Sym^n and the contragredient);

v0.2.0: 2026-09-15 (chapter 1 and Appendix A, the errata);

v0.1.0: 2026-09-15 (skeleton).

Supersedes `hecke_action.tex` (frozen at v1.0, 2026-07-02) and `monodromy_v5.tex` (frozen at v5.0, 2026-07-03).

Companion specification: `hecke_extensions.tex` v1.16.3 (its 1.17.0-pre.4, in the tree beside this version's pre.4, restates [E, `pr:latticesplit`] (R) on congruence-free classes, owner ruling on B-1 of 2026-10-02; its 1.17.0-pre.3, in the tree beside this version's pre.3, carries the referee F1's repairs to [E, `pr:kfonset`] and leaves [E, `pr:latticesplit`] (R) for the owner's ruling on B-1; its 1.17.0-pre.2, beside this version's pre.2, restates [E, `pr:latticesplit`] (R) again; neither is released; v1.17.0-pre.1 moves the pointer there: the companion's patch v1.16.3, owner ruling O4 of 2026-10-01, adds a [proved here] remark and a [conjectural] prediction; its 1.17.0-pre.1, drafted with this version for the same referee, is not released; until v1.17.0-pre.1 the pointer read v1.16.2: v1.16.0 moves the pointer there: the companion's v1.16.0, owner-signed on 2026-10-01 after the slot-15 close (ruling S15-O2; [E, `pr:latticesplit`] restated), its patch v1.16.1 (census edition 3, evidence and pointers only) and its patch v1.16.2 (one verified locator, released with this document's v1.16.0); until v1.16.0 the pointer read v1.15.0: v1.14.0 moves the pointer there: the companion's v1.15.0, owner-released on 2026-09-28 together with this document's v1.14.0, carries evidence and pointer updates only; until v1.14.0 the pointer read v1.14.0: v1.14.0-pre.1 moves the pointer there: the companion's v1.14.0, owner-released on 2026-09-27 at the slot-13 close after this document's v1.13.0, touched only the companion; until v1.14.0-pre.1 the pointer read v1.13.1: v1.13.0 leaves it there, v1.13.0-pre.1 and pre.2 leave it there, slot-13 WP-2 and WP-3 touching only this document; v1.12.1 moves the pointer there: the companion's v1.13.0 and v1.13.1, slot-12 S12-FIN, the finality certificates, touched only the companion; until v1.12.1 the pointer read v1.12.1: v1.12.0-pre.1 leaves it there, slot-12 WP-2 touching only this document; v1.12.0, the owner-signed release, leaves it there as well, the release touching only this document; v1.11.1, published in the same package, slot-11 WP-17, as the companion's v1.12.1; v1.11.0 had it at v1.12.0, published in the same package, slot-11 WP-15; v1.11.0-pre.1 and v1.11.0-pre.2 had it at v1.11.0, the companion being at its pre-releases v1.12.0-pre.1 (slot-11 WP-4) and v1.12.0-pre.2 (slot-11 WP-18c) meanwhile — and, before either of them, at v1.11.1 (`4999a21`, slot-10 WP-4's W2 documents), which the pointer did not follow; the companion's v1.11.1 and v1.12.0-pre.1 both preceded this document's v1.11.0-pre.1 (v1.11.1: the WP-15 referee's m-4); v1.10.1, at whose commit the companion reads v1.11.0; v1.10.0; the companion's v1.11.0-pre.1 of slot-10 WP-2 and v1.11.0 of slot-10 WP-15, the same package as this version, which publishes it in its second phase — at this version's commit the companion reads v1.11.0-pre.2; v1.9.1 had it at v1.10.1, the companion's referee-repair Y bump at the slot-9 close; v1.9.0 had it at v1.10.0, the companion's v1.10.0-pre.1 of slot-9 WP-2 and v1.10.0 of slot-9 WP-15, the same package as this version, which publishes it in its second phase — at this version's commit the companion reads v1.10.0-pre.2; v1.8.1 had it at v1.9.1, the companion's v1.9.0-pre.1 and v1.9.0 X bumps of slot 8 ([E, `sec:ring`]) and its v1.9.1 referee pass; v1.8.0 and v1.7.4 had it at v1.8.2, the companion's own status bump at the slot-7 close, after its v1.8.0-pre.1 and v1.8.0 X bumps and the v1.8.1 referee pass; v1.7.3 had it at v1.7.2, the companion's status bump at the slot-6 close, after its v1.7.0 X bump and v1.7.1 referee pass; v1.7.2 had it at v1.6.2, the companion's slot-5 status bump; v1.7.1 had it at v1.6.1, repaired in the same referee pass, and v1.7.0 had moved it to v1.6.0 from v1.5.1, the pointer having stood at v1.3.3 since that companion's v1.4.0, three revisions).

The title page of v1.25.0-pre.7

The author line and the abstract of v1.0.0 to v1.25.0-pre.7, verbatim (the abstract as of v1.25.0-pre.7).

Complete as a document, incomplete as a programme. Every chapter 1–22 and every appendix is written, no stub remains, and at v1.0.0 the label namespace is unambiguous as well: (V4)'s two supersession freezes are done, and the nine labels of §16 that the live companion `hecke_extensions.tex` also defined are cited there rather than restated, so a reference `[C, cl:pd]` resolves to exactly one statement. Both criteria of (V5) are therefore met and the leading version field is 1. **That is a statement about this document and not about the programme.** Two chapters — 17 (the general orbifold model) and, since v1.8.0, 23 (the integral line: its free model, comparison map and descent) — state objects that *nothing in the implementation builds*, and say so in their own code maps and in §22.2 (v1.9.0: 23's free model, its comparison map and its integral groups for $n \leq 20$ are built, its lifted Hecke datum and descent are not; v1.10.0: the lifted Hecke datum is built for T_2, T_3, T_5, with the class e of Lemma 14.22 and the integral Hecke action read basis-free at small n — the descent assembly is still not; v1.11.0-pre.1: the two chapters of the new Part VII, 24 and 25, are written *before* their implementation, and their code maps say so; v1.11.0: 24 is built and tested, and the ordinary projector of 25 is built at $p = \ell$ on the ℓ-adic digit route, both recorded in their code map; 25.6 stays open; v1.11.1: its torsion half is answered by Remark 25.4, the per-cell bound stays open; v1.12.0-pre.1: a third chapter of Part VII, 26, is written *before* its implementation and its code map says so, and Corollary 26.2 answers the per-cell bound, the item's status pending the release; v1.12.0: released, and 25.6's tag moves [open] → [proved here]; v1.13.0-pre.1: a fourth chapter of Part VII, 27, the level-ℓ model, the stabilisation and the lattice datum, is written *before* its implementation and its code map says so; v1.14.0-pre.1: a fifth, 28, level N for every N — the composite coset table, the cokernel presentation and its Hecke action — likewise); 15 (the pair homotopy and the cone), which did so in whole or in part until v1.7.2, is implemented in full since v1.7.3, and 14 (the boundary datum), which did so until v1.3.2, since v1.3.3, its obligation discharged (§20.3) (v1.8.1: this sentence named 15 as unbuilt until then, stale since v1.7.3); the one construction item that was open, 14.24, was **decided by the owner on 2026-09-16** ((OC-1a), v1.3.0) and its dependent sites are enumerated in §14.5. Every numbered statement is indexed with its status and its provenance in Appendix B. What each version of this document changed, statement by statement, is Appendix E. It is cited as **[C]** — the same short key the 53 citing sites in the C++ source already use — and it carries verbatim every `\label` those sites cite.

The rules this appendix follows

(CL1) **One line per statement whose content changed, naming the label** ((V3)). A statement written into a stub that this document had already declared counts as changed: before, the label named `[to be written]`; after, it names an assertion. Statements transported verbatim modulo the notation crosswalk from a named source, with nothing added and nothing corrected, are not given a line each; they are named together in that version's *re-homed unchanged* roster, which lists every one of them, so the test "the contract must say which statements moved" is still met and "editorial pass" is still not an entry.

(CL2) **This document's history only.** The changelog says what changed between two versions of *this* document — which for this purpose is `gl2_contract.tex` together with the files it is built from: `gl2_notation.tex` (chapter 2), `gl2_preamble.tex` (the macros (A5) enforces) and `gl2_refs.bib` (the citations (A2) demands). The defects of the superseded

corpus are Appendix A's subject and are not restated here: an entry that exists because an erratum was applied names the erratum number (`err:e1`–`err:e21`) and stops. Adding or amending an erratum entry is itself a change to this document and does get a line, under that entry's label.

(CL3) **A statement written and later revised gets one line per version in which it changed**, not one summary line. The changelog is a per-version record, and "as of [C] v$X.Y$" must be answerable: `prop:tworoutes` was written in v0.7.0 asserting a uniqueness that the same version's `prop:mu`(3) had already made false, and was corrected in v0.8.0. It therefore takes a line in each, and collapsing the two would hide precisely what v0.7.0 said.

(CL4) **Provenance is marked, and only on statements.** A line whose subject is a numbered statement — an environment of kind `thm`, `lem`, `prop`, `cor`, `def`, `rmk`, `alg`, `obl`, `ass`, `ex`, `cl` or `open` (the last added to this list at v1.3.0, as the v0.14.0 entry asked) — carries exactly one of three tags. [new]: the statement appears in no superseded document and in no project note; it is first asserted and first proved here. [re-homed]: transported from a named superseded document, content unchanged beyond notation and any named erratum. [repaired]: transported, but the source statement was false, circular, ill-typed, or missing a hypothesis its own proof uses; the line names the erratum, or says that the defect was found in the pass that repaired it. The tag attaches to the *content*, not to the label, so a preserved label whose statement is strictly generalised here carries [new] for the generalisation and the line says so.

Section, display, rule, erratum and appendix labels carry no tag — `sec:`, `eq:`, the (L1)-style item labels of an enumeration, `err:`, `app:`. A subsection is a container and a numbered display is a formula; neither is a statement, and a *new container for re-homed mathematics* tagged [new] would read as new mathematics, which is exactly the misreading this rule exists to prevent. Where such a label shares a line with a statement, the tag on that line is the statement's and says nothing about the container; where it stands alone, the line's prose says what happened to it.

This is the only provenance marking the document currently carries; when Appendix B is written it should carry the same three values as a column, so that the marking reaches the statement itself and not only its changelog line.

(CL5) **Normative and non-normative entries are separated**, because (V2)'s bump rule reads off the split: a version with any entry in its *Normative* list increments X. "Change to a normative statement" includes *adding* one — otherwise writing a chapter would be a Y bump. Chapter 2 is declared normative and exhaustive and (A5) binds every later statement to it, so a change to its rules, its collision table or its crosswalk is normative. Recording an erratum, a formalization note, a code map, and a design direction that nothing in the document depends on are not.

(CL6) **A version is one commit.** The commits below each did one coherent thing, so no version splits or merges them (the first ten versions were assigned to their commits retrospectively, see the box above). The commit hash is given with each version where it is known, and is the evidence; `[hash]` marks an entry written before its commit existed and not yet back-filled.

Under (CL5) every version so far increments X: ten writing slots and one repair pass each added or changed a normative statement. The exceptions are v1.0.0, which moved the leading field under (V5), and the Y bumps v1.1.1 (evidence only), v1.3.1 (a referee pass), v1.3.2 (a coherence pass), v1.3.3 (evidence and status only), v1.4.1 (a referee pass), v1.4.2 (status only), v1.5.1 (a referee pass), v1.5.2 (status only), v1.6.1 (a referee pass), v1.6.2 (status only), v1.7.1 (a referee pass), v1.7.2, v1.7.3 and v1.7.4 (status only), v1.8.1 (a referee's wording and status; the list completed at v1.8.1, having stopped at v1.5.2) and v1.9.1 (a referee's repairs and status), which changed prose and no statement — which is itself information about how this document

has been built.

Provenance: the counts, and what they do not measure

Over all versions to date, with the (CL1) rosters folded in, and counting *statement-kind labels* as (CL4) defines them:

	labels	of which
[new]	236	148 assertions or definitions, 88 remarks (v1.4.0: `alg:taubdry`; `rmk:taubdrynoiota`, `rmk`
[re-homed]	45	4 `thm`, 3 `lem`, 9 `prop`, 4 `ass`, 12 `def`, 3 `rmk`, 1 `ex`, 2 `cl`, 7 `obl`
[repaired]	38	7 `thm`, 5 `lem`, 9 `prop`, 3 `cor`, 1 `ass`, 1 `def`, 2 `alg`, 6 `rmk`, 1 `cl`, 3 `obl`

Of the 236 [new] statements (226 until v1.16.0-pre.1, 203 until v1.15.0-pre.1, 191 until v1.14.0-pre.1, 190 until v1.13.0-pre.2, 177 until v1.13.0-pre.1, 175 until v1.12.0-pre.3, 169 until v1.12.0-pre.1, 168 until v1.11.1, 148 until v1.11.0-pre.1, 146 until v1.10.0, 145 until v1.9.0, 125 until v1.8.0) the 88 remarks are new *commentary* on re-homed mathematics — what a theorem does not say, which choices an object depends on, what a diagnostic may not be branched on — and are new writing rather than new mathematics (v1.4.0's `rmk:taubdrynoiota` carries a proof and `rmk:conerestrictsmallell` a [verified computationally] record, and both are listed with the boundary roster's remarks below; v1.11.1's `rmk:ordtorsionfree` carries a [proved here] corollary and is listed with Part VII below). The 130 assertions and definitions (125 until v1.16.0-pre.1, 116 until v1.14.0-pre.1, 106 until v1.13.0-pre.1 — the figure pre.1 left unchanged here, corrected at pre.2 —, 104 until v1.12.0-pre.3, 100 until v1.12.0-pre.1, 85 until v1.11.0-pre.1, 70 until v1.8.0) are the new mathematics, and they are listed below. Counting every label named on a tagged line, containers included, 194 sit on [new] lines and 80 on [re-homed] lines or in a roster (no such figure is carried for [repaired] lines: the versions before v0.18.0 did not record one). Those raw figures are *not* a provenance split — the excess over 125 and 45 is containers named alongside a statement — and quoting them as one is the misreading (CL4) exists to stop. Before that rule was imposed the tags reached 108 labels, which would have read as 108 new theorems.

> **CONTRACT. Pages and labels are different measurements, and neither refutes the other.** The architecture costed this document at roughly 62 % re-homing, 11 % re-casting and 27 % new *by page*, and writing slots 5–6 measured the re-homing rate at 1.0× over 14 pp of `monodromy_v5.tex` source. Both figures stand. The table above counts *labels*, and the two disagree for a structural reason: re-homed theorem-proof prose is label-sparse — one `\label` can carry a page of proof — while the apparatus this consolidation added is label-dense, since chapter 2's tables, the rules (L1)–(L6), (A1)–(A6) and (V1)–(V5), the twenty-two errata entries and the per-statement remarks each earn a label within a few lines. The conclusion a reader should draw from the two together is that **re-homing dominates by page and by theorem, while the new apparatus dominates by label** — not that either measurement is wrong.

The 85 statements that are new here, and why they are the ones to scrutinise. Each asserts something that no superseded document and no project note states anywhere. Every other statement of this contract can be checked against a source — that is what [re-homed] and [repaired] mean, and it is how the defects of Appendix A were found. These cannot be: an error in one of them cannot be caught by comparing documents, only by reading the proof or

by running the implementation. They are therefore the first thing a referee should attack and the first thing a formalisation should cover, and they are exactly the lines this appendix tags [new] whose subject is not a `rmk`.

- *Part I, the framework* (19): `def:relationdatum`, `def:orderedrelation`, `def:quotientcomplex`, `def:saturated`, `def:bdryint`, `def:kernelN`, `ass:fund_domain` (FD4), `lem:eqgroupoid`, `lem:fd4criterion`, `lem:model7coincidences`, `def:naivecochains`, `def:eqcohomology`, `cor:threecategories`, `prop:Bqimage`, `def:storedbases`, `prop:ACq`, `prop:AZq`, `prop:projector`, `alg:cohomology`.
- *Part II, the model and its coefficients* (11): `def:model7`, `def:model7rel`, `prop:model7h4`, `def:contragredient`, `prop:pairing`, `lem:symdual`, `def:iota`, `lem:iotamodel`, `prop:iotacommutes`, `prop:iotasplit`, `obl:iota`.
- *Part III, the Hecke datum and the maps* (6): `prop:cosets` — the extension to every ν; the $\nu = 1$ case is re-homed — `def:cosetreps`, `lem:cocycle`, `prop:fullassembly`, `prop:mu`, `prop:tworoutes`.
- *Part IV, the boundary* (13; 11 until v1.12.0-pre.3): `prop:bdrycohom`, `prop:bdrycosets`, `def:taubdry`, `lem:bdrycomponent`, `def:bdryop`, `prop:bdryop`, `obl:taubdry`, `lem:ell2unbounded`, `lem:bulkperiodic` (v1.3.0: the bulk periodicity above the vcd and its Hecke equivariance, a five-lemma argument on `cl:amalgam`(2) with an explicitly constructed class $e \in H^2(\Gamma;\mathbb{Z})$ — the statement no source of this corpus makes), `open:ell2`, `alg:taubdry` (v1.4.0: the canonical τ_∂ as a procedure, with its four proved properties), `prop:bdrytranscription` (v1.12.0-pre.3: the Shapiro transcription in degree 1, at every n, R and p, with an H^1 formula at every ν — the statement `prop:bdrychar`(3) had left unproved), `lem:bdrytorus` (v1.12.0-pre.3: the boundary operators away from ℓ factor through the torus; a referee should read `prop:bdrytranscription`'s steps 1–4 first). (`def:bdrygroup` is [re-homed] and `prop:circlemodel`, `lem:taubdryexists`, `prop:bdrychar` are [repaired]; they are checkable against a source and are therefore not in this roster. Three v1.4.0 remarks — `rmk:taubdrynoiota`, which carries a proof, `rmk:bdrynugeneral` and, in §15, `rmk:conerestrictsmallell` — are [new] commentary.)
- *Part IV, the pair homotopy and the cone* (15): `def:pair`, `lem:conerelative`, `def:cosettransport`, `lem:cosettransport`, `def:pairhomotopy`, `lem:etaunique`, `def:coneaction`, `lem:derivedhomotopy`, `thm:conewelldef`, `prop:coneindep`, `prop:strictpair`, `prop:coneintegral`, `prop:coneroutes`, `obl:eta`, `obl:coneexact`. **This is the chapter with no external corroboration at all**: its only sources are project notes (the OQ2 verdict and the plan), which are not sources of record, so every statement here is either first proved in this document or repaired from one (`lem:etaexists`, `lem:conesign`, `prop:coneambiguity` are [repaired], for `err:e5`, `err:e13`, `err:e2`/`err:e3`). Four remarks — `rmk:conesteinberg`, `rmk:conestrict`, `rmk:coneambiguity691`, `rmk:conetorsion` — are [new] as well and carry mathematics a referee should read (`rmk:conesteinberg`(ii) in particular: it is where the Hecke relations on $H_c^\bullet$ become a theorem rather than a citation).
- *Part IV, the six groups and the general orbifold model* (7): `cl:amalgam` — the Mayer–Vietoris sequence in *every* degree over *every* coefficient ring, and the torsion bound, where [E] stated only the $\ell \geq 5$ consequence — `lem:h2torsion`, `cl:boundary`, `cl:les` (the hypothesis-apportioned form, with $H_c^0 = 0$ unconditional), `cl:chi` (clause (3) proved), `def:borelmodel`, `thm:specialisation`. (`cl:dimH1` and `cl:bockstein` are [re-homed] and `cl:pd` is [repaired], so they are not in this roster. Seven further remarks — `rmk:sixscope`, `rmk:bdryrepairs`, `rmk:chicoincide`, `rmk:pdscope`, `rmk:borelfree`, `rmk:mpcaveat`, `rmk:orbifoldrefine` — are [new] commentary.)
- *Part V, realisation and certification* (1): `lem:predshapes` — every branch decision of `alg:tau` and `def:primitives` is one of exactly four two-term integer dot products, the fourth of which the superseded contract did not list. Part V is the most re-homable

part of the document and this is the only assertion in it that no source states; its four other [new] labels (rmk:primitiveclosure, rmk:certoverGeometry, rmk:btabsmall, rmk:thresholdrounding) are remarks, two of them carrying a [proved here] or a [verified computationally] claim a referee should still read — rmk:thresholdrounding, which is why the binding filter constant is $2^{-50}B^2$ and not the sharper bound the proof gives, and rmk:btabsmall.

- *Part VI, the integral line* (15, v1.8.0): def:ringinforce, prop:ringinforce, prop:bredon, def:isomodel, prop:isomodel, prop:fracture, prop:cohglue, prop:descent, prop:homvanish, lem:strictlifting, prop:strictlift, prop:badprimes, prop:certifiede, prop:crtcharpoly, def:ringsemantics. Like the cone chapter, its sources are project notes only (the integral-line design brief and its audit, the ℓ-adic investigation), which are not sources of record; two registered validations test prop:bredon(2) and prop:crtcharpoly, nothing tests the rest (v1.8.0; since v1.9.0 and v1.10.0 more do, rmk:integralevidence, which lists them). Five remarks — rmk:cafive, rmk:truncdevice, rmk:integralevidence, rmk:integralscope and, in §16, rmk:bocksteinexp — are [new]; three carry a [proved here] clause a referee should read (rmk:bocksteinexp(1),(2) above all). v1.9.0 adds a sixth, rmk:certifiedeperdiff ([proved here]), and states rmk:bocksteinexp(2) at every prime. v1.10.0 adds a seventh and an eighth: rmk:strictliftbuilt ([verified computationally], registered — the lifted datum as built, with its certificates and measured sizes) and rmk:bredonhecke (commentary — what this document does *not* compare at $\ell \in \{2,3\}$); and three registered validations now test prop:strictlift(1),(3),(4), lem:bulkperiodic(1)–(3) and the Hecke action on the free model (rmk:integralevidence). v1.11.0-pre.1 adds prop:relationsmodel: thm:relations's recursion on the free model over every ring.
- *Part VII, the Dickson invariant, θ_{L} and the ordinary projector* (14 assertions or definitions and 5 remarks, v1.11.0-pre.1): def:dicksoninv, lem:dicksontwist, prop:dicksonH0, prop:heckeH0, lem:heckefibre, def:Qn, lem:Qnfunctions, def:theta, prop:thetatwist, thm:thetales, def:ordinary, prop:ordinary, cor:thetanonordinary, open:ordprecision; remarks rmk:heckeH0plan, rmk:thetasmallell, rmk:thetaclassical, rmk:thetacheckable, rmk:ordinaryeis; v1.11.1 adds a sixth remark, rmk:ordtorsionfree ([proved here]: the torsion of $H^1 \otimes \mathbb{Z}_\ell$ is non-ordinary at $\ell \geq 5$, $n \geq 1$ — a corollary of cor:thetanonordinary(3) and prop:ordinary). Its sources are the slot-11 plan and its audit (project notes, not sources of record) and the literature of rmk:thetaclassical, none of it in docs/literature/; a referee should read prop:heckeH0(2)–(4) first, since it corrects the plan.
- *Part VII, §26, the foundation of Hida theory at level* 1 (4 assertions or definitions and 2 remarks, v1.12.0-pre.1): prop:torsionprepass, cor:ordprecision, lem:ordexponent, def:hidatower; remarks rmk:ordexponentuse ([proved here]: sharpness) and rmk:hidacheckable (the prediction list). Its sources are the slot-12 plan (a project note, not a source of record) and Hida's papers and book, none of them in docs/literature/; a referee should read prop:torsionprepass(1),(2),(6) and cor:ordprecision(1),(2) first, since they depart from the plan's statements.
- *Part VII, §27, control, level ℓ and the lattice* (10 assertions or definitions and 3 remarks, v1.13.0-pre.1; a fourth remark, rmk:levelscaling ([standard]: the regimes, the obstacles (O1)–(O6), the Manin–Merel action by pointer), v1.13.0-pre.2): def:levelellmodel, prop:levelcorrespondence, prop:levelellU, prop:oldspace, prop:levelorbit, lem:levelelltorsion, prop:stabilisation, thm:hidacontrol, thm:hidaweight, def:hidalattice; remarks rmk:levelcost (commentary: the cost model), rmk:stabconvention ([proved here]) and rmk:hidacheckable2 (the prediction list). Its sources are the slot-13 plan and its owner decisions (project notes) and Hida's two 1986 papers, read from the owner's read-only copies with every locator

checked against the page; a referee should read `prop:levelellU`, `prop:oldspace`(3) and `rmk:stabconvention` first, since the convention they fix makes $e \circ \mathrm{dg}_\ell$, not Hida's $e \circ \mathrm{res}$, the integral stabilisation. v1.14.1 adds `lem:rhorank` (the number of non-zero invariant factors of ρ_n at a matrix singular modulo ℓ, and its rank-one closed form over $\mathbb{F}_\ell$ — the U_ℓ lever's rank claim, checked at every block by the implementation).

- *Part VII, §28, level N: the composite coset table, the cokernel presentation and its Hecke action* (9 assertions or definitions and 3 remarks, v1.14.0-pre.1): `def:cosetrepsN`, `lem:cosetrepsN`, `def:levelNmodel`, `lem:treeres`, `lem:manin`, `def:kappaN`, `lem:levelduality`, `lem:merelpath`, `prop:heilbronn`; remarks `rmk:heilbronnconv` ([proved here]: the transcription and its trap), `rmk:heilbronncost` (the cost model) and `rmk:levelNcheckable` (the prediction list). Its sources are the slot-14 plan and the owner's standing direction on level N (project notes), and Manin, Merel, Stein, Cremona, Mazur, Merel 1996 and Wake–Wang-Erickson, none of them verified here ([S14] Q6's default, rule (A2)) — every proof in the chapter is complete without them; a referee should read `lem:levelduality`(5) and Step 3 of `prop:heilbronn`'s proof first, since the transcription (the adjugate, the row action, the admissible pairs at $p \mid N$) lives there.
- *Part VII, §29, nebentypus* (10 assertions or definitions and 4 remarks, v1.15.0-pre.1): `def:nebdata`, `lem:nebdata`, `lem:teichmuller`, `lem:inducedchar`, `lem:gammaonetable`, `def:nebmodel`, `prop:nebdecomp`, `prop:nebcorr`, `thm:nebrelations`, `prop:neborbit`; remarks `rmk:nebfield`, `rmk:nebheilbronn` ([proved here]), `rmk:nebcost` (the cost model) and `rmk:nebcheckable` (the prediction list). Its sources are the slot-15 plan and the owner's ruling Q2 of 2026-09-30 (project notes); the literature (Shimura §3.3, Cohen–Oesterlé, Ribet 1977, Ash–Stevens) is cited at chapter level and **not** verified; a referee should read `prop:nebcorr`(2),(4), `prop:neborbit`(4) and `rmk:nebheilbronn` first.
- *Part VII, §30, the $\mathbb{Z}/\ell^K$ Hecke algebra* (8 assertions or definitions and 1 remark, v1.15.0-pre.1): `def:satdata`, `prop:saturation`, `prop:satdefect`, `lem:heckerankone`, `lem:freepart`, `cor:stablecert`, `prop:cohforms`, `prop:hidaalgconst`; remark `rmk:thetaforms`. Its sources are the slot-14 Loewy note `docs/implementation/slot14/WP-M1_loewy_no` (Prop. 1, §2, §5.0), the proposals P15, P17, P18, P20 of slot-14 WP-6 and WP-D1, and the slot-15 plan's WP-19 (c) (commit `c1b45cd`); a referee should read `prop:saturation`(3) and `cor:stablecert` first, since the certified verdict of slot-15 WP-5 rests on them.
- *Part VII, §28.6, degeneracy maps on the presentation and level N at scale* (2 assertions and 3 remarks, v1.16.0-pre.1): `prop:kappamodel`, `prop:degenpres`; remarks `rmk:degenpresconv` ([proved here], [verified computationally][a]), `rmk:levelNscale` (the cost and memory model) and `rmk:levelNcheckabletwo` ((LN7)–(LN9)). Its sources are the slot-16 plan (WP-12 items (a), (c)) and v1.15.0-pre.3's unrefereed sentence, which it supersedes; a referee should read Step 3 of `prop:degenpres`'s proof (the re-indexing and the admissibility condition) and `prop:kappamodel`(2) first.
- *Part VII, §31, the ℓ-local isotropy model* (3 assertions or definitions and 2 remarks, v1.16.0-pre.1): `def:isomodelell`, `prop:isomodelell`, `prop:ellthreeclosed`; remarks `rmk:isomodellmp` (the MP constraint) and `rmk:isomodellcheckable` ((IL1)–(IL4)). Its source is `docs/implementation/slot14/WP-Q1_small_ell_orbifold_note.md` §§2–4; a referee should read `prop:ellthreeclosed`(3)'s spectral-sequence argument and `prop:isomodelell`(4) first.
- *Part VII, §29.8, the level-change cone and inflation–restriction* (18 assertions or definitions, 1 obligation and 3 remarks, v1.22.0-pre.3): `def:levelpair`, `lem:nebcompat`, `def:levelres`, `lem:levelresmorph`, `def:levelcone`, `lem:levelconeexact`, `prop:levelconeshapiro`, `prop:modeltorsion`, `prop:levelsplit`, `prop:levelconehecke`, `def:normalpair`, `prop:levelinfres`, `prop:cycliccone`, `lem:levelfreeeuler`, `lem:unipotentinv`, `prop:gamma0normal`, `prop:hidanormalsteps`; `obl:levelconeexact`; remarks `rmk:normalN1` ([proved here]), `rmk:lescheckable` ((LES1)–(LES9)) and `rmk:oldspacecone`

(deferred). Its source is `docs/implementation/slot22/LES0_spec.md` v0.1; a referee should read `prop:levelconehecke`(1)–(3) (the convention of $A'' = A \cap gAg^{-1}$; (3) is the level-raising U_p, [proved here] with negative content after the F-LES repair R1: the strict identity $U_p^B \circ \mathrm{res} = p \cdot \mathrm{dg}_g$ and the failing cell), `prop:cycliccone`(3)(c) and `prop:gamma0normal`'s local count first.

- *Part VII, §32, the Moerdijk–Pronk model* (17 assertions or definitions, 5 remarks and 1 example, v1.22.0-pre.5): `def:mptri`, `def:mprules`, `def:mpell`, `def:mpRZ` (with `tab:mprules`), `def:mpset`, `def:mpfree`, `lem:mprows`, `lem:mpcols`, `lem:mpassembly`, `thm:mpcomparison`, `def:mpphi`, `prop:mpalldegrees`, `def:mpmorphism`, `thm:mpnatural`, `prop:mpbredon`, `prop:mpboundary`, `prop:mprescor`; remarks `rmk:mprulesread`, `rmk:mpwhat`, `rmk:mphybrid`, `rmk:mphecke` (commentary), `rmk:mpcheckable` ((MPC1)–(MPC9)); example `ex:mpteardrop`. Its sources are `docs/implementation/slot22/MP_math_plan.md` v0.2.1 (audited), `MP_impl_plan.md` v0.2.1, the F-MP2 code audit and the built code; a referee should read `thm:mpcomparison`'s Step (2) with `lem:mpcols` (the non-equivariant fibre contraction and the bounded assembly), `def:mpmorphism`(3) with `thm:mpnatural`(1),(4) (the label factor), and `prop:mpalldegrees`(3),(5) (the torsion bound and the scope) first.
- *§16, §16.4, odd n at $\ell = 2$ and the prime-to-ℓ Hecke action at $\ell \in \{2,3\}$* (9 assertions and 2 remarks, v1.22.0-pre.10): `lem:centralminusone`, `lem:bocksteinallell`, `lem:cyclicres`, `prop:oddnltwo`, `lem:oddnsplit`, `lem:tellrank`, `prop:ltwothreescalar`, `prop:oddnhecke` (its clause (4) [conjectural]), `lem:cprule`; remarks `rmk:standardtwist` ([proved here]) and `rmk:oddnltwoerr` (commentary). Its sources are `docs/implementation/slot22/L23_` v0.2 (Theorem O) as repaired by its Fable review `L23_master_plan_review.md`, and the census reviews `census_l23_classification_review.md` (Theorems A, B(1),(2), D(1),(2)) and `census_l23_plan_v02_review.md` §1.2; a referee should read `prop:ltwothreescalar`'s Step 0(c) (the level-$\Gamma(\ell)$ operator, its twisted equivariance and the cusp value) and its $\ell = 2$ St_2 case, then `prop:oddnltwo`(2) ($c_U = 2h^0$) and `lem:tellrank`'s orbit count, first.
- *Part VII, §32, the Q-reduced nerve* (1 assertion and 1 definition, v1.22.0-pre.11): `def:mpqd`, `prop:mpqreduced`. Its source is the F-MPT referee `docs/implementation/slot22/mp_triang` §3, whose proof the editor checked and wrote out; a referee should read `prop:mpqreduced`(2)'s column step (the cone on the dominant vertex, well defined on the Q_x-classes) and (1)'s invariance of $\mathsf{p}(\tau)$ on a class first.
- *§16, the tree resolution over every ring* (1 remark, v1.22.0-pre.12): `rmk:treeconeres` ([proved here], three `\LEAN` markers). Its sources are the LR-Y phase reports `docs/implementation/slot22/LRY_phase1_report.md`, `LRY_phase2_report.md`, `LRY_phase3_report.md` §C and the Fable audits `LRY_audit.md`, `LRY3_audit.md`; a referee should read its orientation sentence against §28's tree $\mathcal{Y}$ and the periodicity indexing ($P_{n+2} = P_n$ for $n \geq 1$, $\partial_{n+2} = \partial_n$ for $n \geq 2$) first.
- *§16.4, the dimensions at $\ell \in \{2,3\}$ formalised* (1 remark, v1.22.0-pre.14): `rmk:lvdimsltwothree` ([proved here], nine `\LEAN` markers). Its sources are `docs/implementation/slot22/LV1_report.md`, `LV2_report.md`, the Fable audits `LV1_audit.md`, `LV2_audit.md` (§8 sets what may be said) and `LVprov_audit.md`; a referee should read its $\ell = 3$ formula against Proposition 31.3(2) and the sentence on the `L23Cells` cells (data, not markers) first.
- *Part VII, §31, the certified-exponent rule on $\mathcal{I}^{(\ell)}$* (1 assertion, v1.23.0-pre.12): `prop:certifiedeell`. Its sources are `docs/implementation/slot24/DESCENT_plan.md` v0.2 §2.7 (row C-D0), §2.1 and D0, and the audit finding M3 of `DESCENT_plan_audit.md`; a referee should read (1)'s replacement of "$\mathbb{Z}_{(\ell)}$-algebra" by "q_ℓ a unit" in Proposition 31.2's proof, and (5)'s reduction of the code's full layout to d^q, first. Refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D0_referee.md`); the fixes of v1.23.0-pre.13 add (4)(a)–(c) and (6)'s a-priori bound, which a re-referee should read first.

- *Part VII, §31.1, the descent at level N and its bad primes* (2 assertions, v1.23.0-pre.14): `prop:descentN`, `prop:badprimesN`. Their sources are `docs/implementation/slot24/DESCENT_plan.md` v0.3 §2.7 (rows C-D1, C-D2), §2.1, §2.4 D-δ4 and D1, the audit findings M8, m3, m4, m7 of `DESCENT_plan_audit.md`, and `DESCENT0_report.md` §1; a referee should read `prop:badprimesN`(2)'s image of $\Gamma_0(N)$ and Claim A (the Borel invariants), (3)(b)'s twist, `prop:descentN`(4) (the iff, and the values of b_0) and (7)(b) (generators, not a basis) first. Refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D1_referee.md`); the fixes of v1.23.0-pre.15 add the proof of `cl:dimH1`(2) and of $b_1 \geq 1$, which a re-referee should read first.
- *Part VII, §31.1, maps and the level cone over $\mathbb{Z}$* (1 assertion, v1.23.0-pre.18): `prop:descentmaps`. Its sources are `docs/implementation/slot24/DESCENT_plan.md` v0.3 §2.2 (the rows "Maps" and "Level-change LES"), §2.7 (row C-D3) and D2, §29.8, and the slot-22 LES-0 request; a referee should read (2)(f) at $\ell = 3$ (the $S[A]$-models), (3)(a)'s group computation (A'', A', the indices, $p \nmid M$), (4)(c)'s vanishing of $V(\chi')^B_S$ and (5)'s sufficiency claim first. Refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D3_referee.md`); the fixes of v1.23.0-pre.19 add (3)(b′) (the square with cor for $p \mid N$, every M) and restrict (3)(b) to this document's U_p, which a re-referee should read first. v1.23.0-pre.22 adds (2)(g) (the precision rule, from the D2 code audit's F1) and (2)(b)'s signs; refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D3g_referee.md`); the fixes of v1.23.0-pre.23 add (2)(g)(i)'s "moreover" (the blocks exact modulo ℓ^K) with (ii)'s lattice coordinates, and (iii)'s list of certifying ranks, which a re-referee should read first.
- *Part VII, §31.2, the tree models* (1 assertion, v1.23.0-pre.20): `prop:isomodeltree` (with `def:isomodeltree`). Its sources are `docs/implementation/slot24/DESCENT_plan.md` v0.4 §2.6, §2.7 (row C-D6), gate G-T and D5, `rmk:treeconeres`, `prop:descentmaps`(5), and the evidence of `DESCENT1_report.md`; a referee should read (1)(a) (well-definedness and the chain-map identities of the canonical lifts at $\ell = 2$), (3)'s corrections k, k' (above all κ'_t and w at $\ell = 2$) and the strict identity $\Phi^{\mathcal{T}}_\ell \Psi^{\mathcal{T}}_\ell = \mathrm{id}$, (1)(d)'s transfer of `prop:certifiedeell`(2)–(6), and (4)'s "over $\mathbb{Z}$, not over $C_\bullet(X)$" first. Refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D6_referee.md`); the fixes of v1.23.0-pre.21 make (4) effective — (4)(a)'s tables of $\mathsf{r}_{\mathcal{T}}$, $\mathsf{G}_{\mathcal{T}}$, (4)(b)'s solvers and the formulas (L), (L′), (4)(d)'s homotopies K, K' —, which a re-referee should read first.
- *Part VII, §31.3, Hecke T_p, $p \nmid N$, on the descent* (5 assertions, v1.23.0-pre.24): `prop:strictliftN`, `prop:crtcharpolyN`, `prop:torsionhecke`, `thm:descentcert`, `rmk:descentcertscope`. Their sources are `docs/implementation/slot24/DESCENT_plan.md` v0.4 §2.2 (the row "Hecke action"), §2.3, §2.4 D-δ1–D-δ3, D-δ7, §2.7 (rows C-D4, C-D5) and D3, §5 of `DESCENT_plan_audit.md` and its finding M5, and the owner's rulings [R-Q2]–[R-Q4], [R-GT]; a referee should read `prop:strictliftN`(3)(c) (the identification with $[AgA]$ without 6 inverted: the free source), (4) and (7) (the strict identities), `prop:crtcharpolyN`(3) (the bound is an input; its citations are not all verified) and (5) (the output rule), `prop:torsionhecke`(3) (the read needs $2\sigma_{\max} \leq K$, stronger than the coordinate rule of `prop:descentmaps`(2)(g)) and (4), and `thm:descentcert`(3), (7) against the plan's M5 first. Refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D4_referee.md`); the fixes of v1.23.0-pre.25 — `prop:torsionhecke`(1)(e), (2) (the layer blocks modulo ℓ^{e_j}, its proof and sharpness), `prop:strictliftN`(6)(b) with `thm:descentcert`(3) (the one route at $\ell = 3$) and the code map's obligations — are what a re-referee should read first. v1.23.0-pre.31 (D3's contract notes: `prop:torsionhecke`(3′), `prop:strictliftN`(2)) refereed ACCEPT after fixes (`docs/implementation/slot24/C-D3h_referee.md`); the fixes of v1.23.0-pre.32 — the unsigned $\mathsf{H} = \Theta_{h_{\mathcal{I}}}$ of `prop:strictliftN`(2), the precision of `thm:descentcert`(4) at $\ell \geq 5$, and the any-lifts argument of `prop:torsionhecke`(3′) — are what a re-referee

should read first.

- *Part VII, §31.4, the boundary pair and the cone over* $\mathbb{Z}$ (5 assertions, v1.23.0-pre.26): `def:descentpair`, `lem:pairlifting`, `prop:descentpair`, `prop:descentpairhecke`, `rmk:descentpairscope`. Their sources are `docs/implementation/slot24/DESCENT_plan.md` v0.4 §2.2 (the row "Boundary LES"), §2.4 D-δ6, §2.7 (row C-D7), D4 and §5.5, `DESCENT_plan_audit.md` on D-δ6, and the owner's rulings [R-Q2]–[R-Q4]; a referee should read `prop:descentpair`(1)(c) (the identification with $\mathrm{Ext}(\mathrm{St}, -)$ on a pair model, and the integral torsion-freeness of H^1_c), (1)(d) (the boundary of $\mathcal{I}^{(\ell)}$ at 2 and at $\ell \geq 3$), (2) (the pair map cmp as the only glueing datum), (4) (the closed form and its periodicity at $\varepsilon = \pm 1$), (5)(b) (the rational H^2_c by the Manin resolution and the adjoint computation) and (6) (the rank sources for D^0), and `prop:descentpairhecke`(1) (the pair lift for every p, and its homotopy uniqueness), (3) (the identification of the cone operator with the double-coset operator on $\mathrm{Ext}(\mathrm{St}, -)$ without 6 inverted) and (5) (what is certified at which p) first. Refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D7_referee.md`); the fixes of v1.23.0-pre.27 — `prop:descentpair`(5)(e) (H^2_c at $\ell \geq 5$ is H_0) with (6)(c), (1)(c)'s signs and their proof, the periodicity proofs of (4) and (5)(d), and `prop:descentpairhecke`(5) (weakened) — are what a re-referee should read first. v1.23.0-pre.28 (*not refereed*) restates `prop:descentpairhecke`(1), (5) and the code map (f) for every prime $p \nmid N$ (the pair lift built on demand, certified at run time); refereed ACCEPT-with-fixes (`docs/implementation/slot24/C-D7p_referee.md`); the fixes of v1.23.0-pre.29 — the refusal list of `prop:descentpairhecke`(1), (5) and the code map's (f) (the code's eleven refusal codes) and the forced-component-map paragraph of the proof of (1) — are what a re-referee should read first.
- *Part VII, §31.5, the tree cusp pair* (4 assertions, v1.23.0-pre.33, **not refereed**): `def:treepair`, `lem:treepaircyl`, `prop:treepair`, `rmk:treepairscope`. Their sources are `docs/implementation/slot24/DESCENT_plan.md` v0.4 §5 (the owner's candidate, the cusp model, §5.3's costs, §5.4(c)), §2.7 (row C-D8) and D5c, and the evidence of `DESCENT4_report.md` §4 and `DESCENT5_report.md`; a referee should read `def:treepair`(2) with `prop:treepair`(1)(b) (the axis map: well defined and a chain map at $\ell = 2$ in odd degrees), `lem:treepaircyl`(2), (3) (the deformation of the cylinder's cone onto the tree cone and the signs), `prop:treepair`(3)(c), (d) (the two squares by (L′), and $H(\mathsf{j}_\Psi) = H(\mathsf{j}_\Phi)^{-1}$ through the comparisons of pairs Θ_Φ, Θ_Ψ, not the five lemma) and (4)(a) (the off-diagonal $\eta_{\mathcal{T}}$, which corrects the plan) first. Refereed ACCEPT after fixes (`docs/implementation/slot24/C-D8_referee.md`, 0 M / 4 m / 8 N; independent check `cd8referee/`); the fixes of v1.23.0-pre.34 — `prop:treepair`(4)(a) and `rmk:treepairscope`(iii) (where $\eta_{\mathcal{T}}$ arises: on $C(\mathsf{a}^*)$, not on the pair model; its non-vanishing observed, not proved), the generalised solver of `prop:treepair`(3)(b), the ℓ-local column augmentations of `def:treepair`(4), and the code map's restricted negative controls (f) — are what a re-referee should read first. v1.23.0-pre.35 (*not refereed*): `def:treepair`(2)'s cochain statement ($\rho_{M_N}(\mathbf{c}) = (-1)^n$), the new `prop:treepair`(4)(e) (truncation) with its proof, and the code map's certificates paragraph (what "on the generators" certifies) are what a referee should read first. Refereed ACCEPT after fixes (`docs/implementation/slot24/C-D8b_referee.md`, 0 M / 1 m / 7 N; independent check `pre35referee/`); the fixes of v1.23.0-pre.36 — the code map's certificates (the applied operator's blocks; deterministic and probabilistic parts; admissibility) and `prop:treepair`(4)(e)'s hypothesis on T, (ii) and (iv) — are what a re-referee should read first.
- *Part VII, §31.6, the Hecke algebra over* $\mathbb{Z}$ (17 assertions, 2 definitions and 4 remarks, three of them [conjectural]; v1.25.0-pre.0, a branch pre-release, **not refereed**): `def:heckeZ`, `prop:heckeZspec`, `prop:heckeZrat`, `lem:heckeZbridge`, `prop:heckeZmax`,

prop:heckeZgen, prop:heckeZetale, prop:heckeZmono, prop:heckeZlocal, prop:heckeZthrough, prop:heckeZuct, prop:heckeZsplit, prop:heckeZlevelone23, prop:heckeZglue, thm:heckeZdescent, thm:heckeZcert, def:heckeZcong, prop:heckeZcong, prop:heckeZeiscert; remarks rmk:heckeZcbone, rmk:heckeZcbtwo, rmk:heckeZcbthree ([conjectural]) and rmk:heckeZcaveat ([proved here]). Their sources are the HeckeZ plan docs/implementation/slot24/hec (signed by the owner 2026-11-04): 01-math-core.md v0.6 §§1–5, 03-bockstein-tor.md v0.3 §§2–4, 04-record-census.md v0.4 §1 (item 8), and the audits A1-audit.md, A3-audit.md, P2a-final-math-audit.md, P3-reaudit.md, whose findings the stage files had applied; a referee should read thm:heckeZcert(C1)'s dimension count and (C2)'s set S with the cofactor, prop:heckeZgen(1) (the a-priori d), prop:heckeZmono's computable form (the test at $K \geq c^{\mathrm{sat}} + 1$ on an unsaturated lattice), prop:heckeZetale(2)'s $v_\ell = 1$ clause, prop:heckeZglue(3) (the sandwich; J never exact) and prop:heckeZeiscert(3) (Mazur's good prime and the index count) first. Refereed: A7 REVISE → A7b ACCEPT (docs/implementation/slot24/heckez/A7-Z0-referee.md, A7b-Z0-reaudit.md); the main-line merge v1.25.0-pre.1 refereed REVISE (docs/implementation/slot25/Z0_referee.md), its findings applied in v1.25.0-pre.2. The reconciliation notes dated 2026-11-06 (R1–R9 of docs/implementation/slot25/Z0_merge_report.md: item 8 (c), (d), (f), (j) and the code map; §31.6's two *As built* paragraphs, the cofactor route as built, the counting criterion's scope) are what a re-referee should read first.

v1.26.3 — 2026-11-09 — hecke-extensions slot 26: the owner-signed release of v1.26.3-pre.1

This version. **Signed by the owner 2026-11-09** (the owner's pre-signature, effective on the referee's ACCEPT, RELFIX_signed_referee.md): the owner-signed release of v1.26.3-pre.1; no change of mathematical content. v1.26.3-pre.1 (a pre-release on top of v1.26.2) is text-only: a private path removed. **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 229 **on** 88 **labels.**

- Header and \date; the Release record's first entry; this entry; the companion pointer of the v1.26.3-pre.1 entry.

Companion pointers: [E] 1.25.2, [U] 0.7.2, [X] v1.28; [C] and [U] are signed together.

v1.26.3-pre.1 — 2026-11-09 — hecke-extensions slot 26: text only, a private path removed

This version. A **pre-release, not signed**: text-only: private path removed, no change of mathematical content (release audit of 2026-11-09, finding S1). **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 229 **on** 88 **labels.**

- Header and \date; the Release record's first entry; this entry.
- The *Corroboration* paragraph after the proof of Proposition 14.17 names its ad hoc script by its file name prop_t_check.py instead of a path in the author's scratch directory.
- The two [verified computationally][a] data remarks of v1.14.1 (the second-cluster run at $n = 4150$ and at $(4830, 2411)$) cite "the WP-C10 report of the September run, §1" instead of that report's file name, which carried the cluster's name.

Companion pointers: [E] 1.25.2, [U] 0.7.2 (the same scrub; signed with v1.26.3), [X] v1.28.

v1.26.2 — 2026-11-09 — hecke-extensions slot 26: the owner-signed release of v1.26.2-pre.1

This version. **Signed by the owner 2026-11-09**: the owner-signed release of v1.26.2-pre.1; no change of content. v1.26.2-pre.1 (a pre-release on top of v1.26.1) is a paths-only update

for the repository layout (referee `LAYOUT_signed_referee.md`: ACCEPT). **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 229 **on** 88 **labels.**

- Header and `\date`; the Release record's first entry; this entry.

Companion pointers: [E] 1.25.2, [U] 0.7.1, [X] v1.28; [C] and [E] are signed together.

v1.26.2-pre.1 — 2026-11-09 — hecke-extensions slot 26: paths only

This version. A **pre-release, not signed**: paths only, no change of content (repository layout, owner ruling 2026-11-09). The program's sources moved under `src/` (`src/hacs/`, `src/ModArithmetic/`, `src/Engine/`, `src/Cuda/`; the old-to-new map of every file is `tools/layout/mapping.tsv`). **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 229 **on** 88 **labels.**

- Header and `\date`; the Release record's first entry; this entry.
- Twelve lines name a file by its new path: eleven long-form paths (`src/ModArithmetic/...`, `src/hacs/...`) in `sec:codemap`'s table and the paragraphs citing `ENGINE_API.md`, `DenseElim`, `FlatMatrix`, `Binary/Ring2k.h`, `Orbifold/MP/` and `HeckeIotaSignValidation`; and the 2026-09-16 grep census of the cone chapter, which now names `src/` (the same files).

Companion pointers: [E] 1.25.2 (paths only; signed with v1.26.2), [U] 0.7.1, [X] v1.28 (keeps its paths: history).

v1.26.1 — 2026-11-09 — hecke-extensions slot 26: the owner-signed release of v1.26.1-pre.1

This version. **Signed by the owner 2026-11-09**: the owner-signed release of v1.26.1-pre.1; no change of content. v1.26.1-pre.1 (a pre-release on top of v1.26.0) changed no mathematical content: machine names and private paths removed for the public release (owner ruling 2026-11-09); machine and cluster names were replaced by neutral words and private directory paths by prose or placeholders, in the text and in source comments. **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 229 **on** 88 **labels.**

- Header and `\date`; the Release record's first entry; this entry.

Companion pointers: [E] 1.25.1, [U] 0.7.1, [X] v1.28; all three binding documents are signed together.

v1.26.0 — 2026-11-09 — hecke-extensions slot 26: the owner-signed release of v1.26.0-pre.11

This version. **Signed by the owner 2026-11-09**: the owner-signed release of v1.26.0-pre.11; no change of content. It releases the content of v1.26.0-pre.1 to pre.11 as described in their entries below, including the binding changes there marked "not yet seen by the owner" (item 8 (c), (d), (d′) of §22.4 and `prop:strictliftN`(9)); the owner signs them with this release. **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 229 **on** 88 **labels.** By the test of item 8 (d′)(vii) a signature as such moves no written word: the values `gl2_contract 1.26.0-pre.5, item 8 (d')` and `gl2_contract 1.26.0-pre.9, item 8 (d')` of `G10.contract` stay, and neither becomes `1.26.0`.

- Header and `\date`; the Release record's first entry; this entry.

Companion pointers: [E] 1.25.0, [U] 0.7.0, [X] v1.28; [E] and [U] are not re-signed.

v1.26.0-pre.11 — 2026-11-09 — hecke-extensions slot 26, UP-Z (b): the conditions of the final confirm-referee of v1.26.0-pre.10

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.10. It discharges the condition C1 and the should-fix items W-A, W-B of the confirm-referee `docs/implementation/slot26/C126pre10_confirm_referee.md` (ACCEPT-with-conditions); disposition `docs/implementation/slot26/C126pre11_disposition.md`. Text only. Its binding changes are **not yet seen by the owner**. It states nothing of Z2 as done: each edited sentence holds before and after Z2's merge, or is scoped to the binary that wrote the record.

Normative.

- Item 8 (d′)(ii), *Why `lower` is `not_run`* (W-A): where no tag certifies there is no row; `G3.C_T.lower` is `not_run` there too, on the `open(cap)` records, outside the scope of `def:heckeZcong`, and also where η_L is defined and `prop:heckeZcong`(5) could read it with a T_p ($(11,0)$), which a `full` record of this version does not do. A future row: $X_p = T_p$ where a T_p separates, U_p where only a U_p does. (W-B): "the partition `U_p_tags` has no field". The word `not_run` and its final class do not move.
- Item 8 (d′)(vii): v1.26.0-pre.11 moves no written word by the test of that item, so the values of `G10.contract` stay `gl2_contract 1.26.0-pre.5, item 8 (d')` and `gl2_contract 1.26.0-pre.9, item 8 (d')`.

Status sentences (C1; the edits promised in the entry of v1.26.0-pre.1, audit F7). `thm:heckeZcert`, "what is reachable today": the list dated (v1.25.0-pre.0, before EXPORT and UP-Z); the `full` line at $N > 1$ no longer says "nothing is certified but two tags" ((C1) is certified without U_p over $\mathbb{Z}$ on 67 of Z1's 85 records), and points to item 8 (d′)(ii), (v). `thm:descentcert`(5), `prop:descentpairhecke` ("U_p is deferred"): nothing is asserted there of U_p, with a pointer to `prop:strictliftN`(9). `def:heckeZ` and the "Not here" of §31.6 ("not built"): the lift of `prop:strictliftN`(9), served by a binary or not according to (d′)(v). `prop:heckeZcong`(5): the read needs U_p over $\mathbb{Z}$ and no route of this version performs it. "What is not available at 2 and 3 today": dated. `prop:descentN`(5), "its cochain operator is deferred": lifted for the free model by `prop:strictliftN`(9). The "Not here" of §31.6, "What is not available" and `prop:descentN`(5) were found by a search for the same phrases, beyond the referee's list.

Unchanged. No written word or hash input moves; no label is added, renamed or removed; no `\LEAN` marker (229 on 88) moves; no golden entry, level-1 or `anemic` byte, settings text or fingerprint moves; no code change and no rerun.

v1.26.0-pre.10 — 2026-11-08 — hecke-extensions slot 26, UP-Z (b): the condition E1 of the referee of v1.26.0-pre.8 / pre.9

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.9. It discharges the condition E1 of the referee `docs/implementation/slot26/C126pre9_referee.md` (ACCEPT-with-conditions), its should-fix W-1 and nits N-1–N-3; disposition `docs/implementation/slot26/C126pre10_d` Text only. Its binding changes are **not yet seen by the owner**.

Normative.

- Item 8 (d′)(ii), *The G3 words from Z2 on* (E1, N-2): the reason for `G3.C_T.lower` = `not_run` is corrected. For $d \geq 3$, η_L of a pair is undefined; for $d = 2$ it is defined and no route of this version performs the read of Proposition 31.48(5) on a `U_p_tags` row; on the non-reduced branch, the scope of Definition 31.47. "No statement defines a lower bound" becomes "no statement prescribes a read of a lower bound"; the same in the entry of

v1.26.0-pre.9 below (`not_run` where no bound, resp. no lower bound, is read on the pairs). The word `not_run` and its final class do not move.
- Item 8 (d′)(ii) (W-1): those words presuppose that the U_p tag step ran at every $p \mid N$; where it did not (no record of this version), `G3.C_T[U_p].status` of that p, `G3.C_T.status` and `G3.C_T.lower` keep `blocked(U_p)`.
- Item 8 (d′)(ii) (N-1, N-3): the retained rule's "until Z2 lands" points at the per-record sense; a binary carrying the route writes these words, which ties (ii) to (vii).
- Item 8 (d′)(vii): v1.26.0-pre.10 moves no written word by the test of that item, so the values of `G10.contract` stay `gl2_contract 1.26.0-pre.5, item 8 (d')` and `gl2_contract 1.26.0-pre.9, item 8 (d')`.

Unchanged. No written word or hash input moves; no label is added, renamed or removed; no `\LEAN` marker (229 on 88) moves; no golden entry, level-1 or `anemic` byte, settings text or fingerprint moves; no code change and no rerun.

v1.26.0-pre.9 — 2026-11-08 — hecke-extensions slot 26, UP-Z (b): Z2's G3 words, contract first

This version is a pre-release, **not owner-released, not refereed**, on top of v1.26.0-pre.8; disposition `docs/implementation/slot26/C126pre9_disposition.md`. Text only. Its binding changes are **not yet seen by the owner**.

Normative.
- Item 8 (d′)(ii), new paragraph *The G3 words from Z2 on*: `G3.C_T[U_p].status`, `G3.C_T.status`, `G3.C_T.lower` (final; `not_run` where no bound, resp. no lower bound, is read on the `U_p_tags` pairs; this wording corrected at v1.26.0-pre.10), `open(sandwich)` at $\ell \in \{2, 3\}$ on an `open(cap)` record, and amendment A3 (`not_run` in the closed list `utag`). It names the words left open at v1.26.0-pre.4. The rules before Z2 stay in force for a binary without Z2's route.
- Item 8 (d′)(v): pointer to that paragraph.
- Item 8 (d′)(vii): the value of `G10.contract` is `gl2_contract 1.26.0-pre.9, item 8 (d')` for a binary that writes those words, and stays `gl2_contract 1.26.0-pre.5, item 8 (d')` for one that does not.

Unchanged. No label is added, renamed or removed; no `\LEAN` marker (229 on 88) moves; no golden entry, level-1 or `anemic` byte, settings text or fingerprint moves; records already written keep their value of `G10.contract` and stay valid, without a code change or a rerun.

v1.26.0-pre.8 — 2026-11-08 — hecke-extensions slot 26, UP-Z: the conditions of the confirm-referee of v1.26.0-pre.7

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.7. It discharges the conditions D1, D2 of the confirm-referee `docs/implementation/slot26/C126pre7_confirm_referee.md` (ACCEPT-with-conditions); disposition `docs/implementation/slot26/C126pre8_disposition.md`. Text only. Its binding changes are **not yet seen by the owner**.

Normative.
- Item 8 (c) (D1): "one reading apart" becomes two: beside the derived `maximal_ideals = not_run` of (d′)(ii), the aggregate `G3.C_T.status` while a `C_T` read or `G3.C_T.lower` it depends on is pending; it is rewritten when those move. No byte of a final record moves.

- Item 8 (d′)(vii) (D2): naming for the first time the word of a field that (d′) had left open moves a written word for a binary that writes it; the manifest asserts that no later version moved a written word *of a record carrying that value*, so the assertion stays true after a partial regeneration.

Unchanged. No label is added, renamed or removed; no `\LEAN` marker (229 on 88) moves; no record word, hash input (`record_fnv1a64`, `cert_sha256`), settings text or fingerprint, golden entry or level-1 / `anemic` byte moves; the value `gl2_contract 1.26.0-pre.5, item 8 (d')` that the code writes stays correct, without a code change or a rerun.

v1.26.0-pre.7 — 2026-11-08 — hecke-extensions slot 26, UP-Z: the confirm-referee's conditions

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.6. It discharges the conditions C1–C3, the should-fix items W-A, W-B, W-D and the nits N-a, N-b of the confirm-referee `docs/implementation/slot26/C126pre6_confirm_referee.md` (ACCEPT-with-conditions); disposition `docs/implementation/slot26/C126pre7_disposition.md`. Text only. Its binding changes are **not yet seen by the owner**.

Normative.
- Item 8 (d′)(ii) (C1): "not golden-eligible after Z2 alone" is stated only of a record carrying a field that waits on `HZ-EISUP` (the records with (C1) `certified`); the other 18 of Z1's 85 depend on the words Z2 fixes.
- Item 8 (d′)(ii), (c), (d) (C3): `maximal_ideals` = `not_run` beside a pending `generation.status` is a derived word on any record, not only on a `full` record at $N > 1$; no byte moves.
- Item 8 (d′)(vii) (C2): `<version>` is a decidable test (a change of the prescribed word or value of a written key or field other than `G10.contract`, or of the key's presence; a reason, a class, a pointer or a conflict resolved in favour of an already prescribed word moves nothing), "the code" is the binary that wrote the record, and the manifest applies the test.
- Item 8 (c) (W-A): `HZ-EISUP` named as the mover of `G1.components` and of the `maximal_ideals` of (d′)(ii).
- Item 8 (d′)(ii) (W-D): `HZ-EISUP` has its card in plan v0.6 (§2.2), sequenced right after Z2 by the owner's ruling of 2026-11-08; Z2 stays necessary and not sufficient.
- Item 8 (d′)(ix) (W-B): "exactly the status fields"; the non-status moves of `G0.operators` and `G1.S_cand.ell=<ℓ>.reason`, `.v_disc` are named.

Editorial. (d′)(vii) "A written value of this key" (N-a); (d′)(ii) the antecedent of "pending in any case" (N-b, also required by C3).

Unchanged. No label is added, renamed or removed; no `\LEAN` marker (229 on 88) moves; no record word, hash input (`record_fnv1a64`, `cert_sha256`), settings text or fingerprint, golden entry or level-1 / `anemic` byte moves; the value `gl2_contract 1.26.0-pre.5, item 8 (d')` that the code writes stays correct, without a code change or a rerun.

v1.26.0-pre.6 — 2026-11-08 — hecke-extensions slot 26, UP-Z: the referee-light's conditions

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.5. It discharges the conditions R1–R3, the should-fix items W1, W2 and the nits N1–N3 of the referee-light

docs/implementation/slot26/C126pre5_referee_light.md (ACCEPT-with-conditions); disposition docs/implementation/slot26/C126pre6_disposition.md. Text only. Its binding changes are **not yet seen by the owner**.

Normative.

- Item 8 (d′)(ii) (R1), with pointers in (c), (d), (d′)(v), (d′)(ix): on a `full` record at $N > 1$, `maximal_ideals` = `not_run` beside a pending `generation.status` is a derived word, rewritten when `generation.status` moves; `not_run` beside a final `generation.status` remains final. (ix) covers edition 1's `not_run` → `blocked(U_p)`.
- Item 8 (d′)(ii) (W1): the Eisenstein U_p systems are not built by Z2, and no card of plan v0.5 builds them; working name `HZ-EISUP`. Z2 is necessary and not sufficient.
- Item 8 (d′)(ii) (W2): `G1.components` = `not_run` at (C1) `open(cap)` stated explicitly (final; the first normative statement of a word the code already wrote).
- Item 8 (d′)(v) (R2): `G9.Up.*` is written by the package that serves the U_p matrix over $\mathbb{Z}$ inside the record build; no record of Z1 carries it; a dump run does not create it.
- Item 8 (d′)(vii) (R3): `<version>` is the last version that moved a written word (at this version `1.26.0-pre.5`); a written value is never rewritten; the manifest states the issuing version and the `G10.contract` value(s), and asserts that no written word moved against them.

Editorial. (d′)(ii) "the other four" (N1); one phrase for the source of v1.26.0-pre.5 (N2); the referee-light check recorded where it stood pending (N3).

Unchanged. No label is added, renamed or removed; no `\LEAN` marker (229 on 88) moves; no record word, hash input (`record_fnv1a64`, `cert_sha256`), settings text or fingerprint, golden entry or level-1 / `anemic` byte moves; the value `gl2_contract 1.26.0-pre.5, item 8 (d')` that the code writes stays correct, without a code change or a rerun.

v1.26.0-pre.5 — 2026-11-08 — hecke-extensions slot 26, UP-Z: the text aligned with the audited Z1 code

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.4. It takes should-fix S1 (and S2's key) of the Z1 code audit docs/implementation/slot26/Z1_code_audit.md (PASS-with-fixes) on the contract side: the text moves, not the code (disposition docs/implementation/slot26/C126pre5_disposition.md; a referee-light check was pending at this version, since done: docs/implementation/slot26/C126pre5_referee_light.md, discharged at v1.26.0-pre.6). Its binding changes are **not yet seen by the owner**.

Normative.

- Item 8 (d′)(ii) (S1(a)): where the search ran and (C1) is `certified`, `G1.components` = `blocked(U_p)`, and `G2.ell=<ℓ>.maximal_ideals` = `blocked(U_p)` where `generation.status` is `certified` — pending, until the Eisenstein U_p systems are built (Z2 or a named successor); replaces pre.4's inclusion of `G1.components` in "`blocked(U_p)` is no longer written in …". Elsewhere the existing `not_run` words of (d′)(ii) and (d). No final word is introduced.
- Item 8 (d′)(v) (S1(b)): `G9.Up.p=<p>.mat_sha256` unhashed (outside `record_fnv1a64`; inside `cert_sha256` as every G9 key), present only on `full` records at $N > 1$ where the U_p matrix over $\mathbb{Z}$ was computed, absent elsewhere.
- Item 8 (d′)(vii) (S2): the key is exactly `G10.contract` with the value `gl2_contract <version>, item 8 (d')`, `<version>` the implemented version.

Unchanged. No label is added, renamed or removed; no `\LEAN` marker (229 on 88) moves; only words written under `full_N`; no `record_fnv1a64`, no golden line and no byte of a level-1 or `anemic` record moves.

v1.26.0-pre.4 — 2026-11-08 — hecke-extensions slot 26, UP-Z: the confirm-referee's should-fix items

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.3. It takes the contract-side should-fix items W3, W4, W5 and the note N1 of the confirm-referee `docs/implementation/slot26/C126pre3_confirm.md` (ACCEPT: Z1 may merge on the contract side; disposition `docs/implementation/slot26/C126pre4_disposition.md`); W6 and N2–N4 are Z1 / K3 items. Its binding changes are **not yet seen by the owner**.

Normative.

- Item 8 (d′)(ii), the rule "no full record at $N > 1$ is final before Z2" (W3): the fields enumerated, exhaustively — `G3.C_T.lower = blocked(U_p)`; `G3.C_T.status = upper_bound_certified` where every U_p tag certifies a bound, else `blocked(U_p)`, sharpened to `certified` only together with `.lower`, by Z2; `G3.C_T.exact` absent; the per-ℓ words at $\ell \in \{2, 3\}$ by (v). Replaces "every other aggregate field whose value may need U_p".
- Item 8 (d′)(ii), the "otherwise" branch (W4): `G1.components` is the fifth ℓ-independent field kept `blocked(U_p)` / `blocked(export)` where the word-prime U_p route did not run.
- Item 8 (d′)(ii), G3 on the non-reduced branch (N1): `G3.C_T.status = blocked(U_p)` there (the tag's bound needs (C1) for $\vartheta = U_p$); the final words of `G3.C_T.status` and `.lower` on that branch are **left open for Z2** (`prop:heckeZcong`(1) uses no generation, so they do not follow from the present text), and Z2's contract text must name them before any such record is final.
- Item 8 (d), G1 (W5): the reading sentence of `partial(unfactored)` beside a non-certified (C1) — "the words the `anemic` records with (C1) `open(cap)` already carry" corrected: no record of edition 1 reaches that code path (every `anemic` record of edition 1 has (C1) `certified`).

Non-normative. The status note of (d′) (confirm-refereed); the header comment, `\date`, the Release record's first entry; this entry. No label is added, renamed or removed; no status tag or marker moves (229 on 88); no record, golden line, fingerprint (`0x8ffc29730c4f9672`) or digest moves; every changed word is written only under `full_N` (`algebra = full`, $N > 1$), and no record of edition 1 carries a word this version fixes that it did not carry before.

v1.26.0-pre.3 — 2026-11-08 — hecke-extensions slot 26, UP-Z: the re-referee's conditions

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.2. It discharges the conditions R1, R2 and the should-fix items W1, W2 of the adversarial re-referee `docs/implementation/slot26/C126pre2_rereferee.md` (ACCEPT-with-conditions; disposition `docs/implementation/slot26/C126pre3_disposition.md`). Its binding changes are **not yet seen by the owner**.

Normative.

- Item 8 (d′)(ii), the non-reduced branch (W1, R1(a); lead decision): `G1.C1.status = refused(non_reduced)` (final; replaces pre.2's `open(cap)`), `G1.theta = not_run` (replaces `open(cap)`), `G1.C2_status = not_run` (replaces `partial(unfactored)`); G2 `generation.status = open(sandwich)` at $\ell \in \mathsf{S}$, $\ell \geq 5$, and `blocked(U_p)` at

$\ell \in \mathsf{S} \cap \{2,3\}$ until Z2 (then `open(sandwich)`; `refused(no_lift)` on Z2-D's audited negative). Amendment A2 after the freeze (hash-neutral; no record carries the words yet); the verifier's `FINAL` set gains `refused(non_reduced)`.

- Item 8 (d′)(ii), the "otherwise" branch (R2): the final `C1` words only where the ϑ search ran with U_p at the word primes; else the four ℓ-independent fields keep `blocked(U_p)` / `blocked(export)` (pending), never `open(cap)`.
- Item 8 (d′)(ii) (R1(b)): `G3.C_T.lower`, and every aggregate field that may need U_p at $\ell \in \{2,3\}$, is `blocked(U_p)` on every `full` record at $N > 1$ until Z2 lands or Z2-D's negative is recorded and audited: no `full` record at $N > 1$ is final before Z2.
- Item 8 (c): the refusal reason `non_reduced` added to the closed list and defined (final).
- Item 8 (d), G1 (W2, W1): `C2_status`'s as-built `blocked(U_p)` "until UP-Z's (d′)(ii)", needing U_p "at the word primes" (was "over $\mathbb{Z}$"); `not_run` added (closed list of `C2`: five words); the reading of `partial(unfactored)` beside a non-certified (C1) (certifies nothing; no `anemic` byte moves).

Non-normative. The status note of (d′) (re-refereed); the header comment, `\date`, the Release record's first entry; this entry. No label is added, renamed or removed; no status tag or marker moves (229 on 88); no record, golden line, fingerprint (`0x8ffc29730c4f9672`) or digest moves.

v1.26.0-pre.2 — 2026-11-08 — hecke-extensions slot 26, UP-Z: the referee's conditions and fixes

This version is a pre-release, **not owner-released**, on top of v1.26.0-pre.1. It discharges every condition and finding of the adversarial referee `docs/implementation/slot26/C126pre1_referee.md` (ACCEPT-with-conditions; disposition `docs/implementation/slot26/C126pre2_disposition.md`). The one new binding rule (S4) is **not yet seen by the owner**.

Normative.

- Item 8 (d′)(ii) of §22.4 (S4): at a `full` record, $N > 1$, with `G1.reduced_Q = non_reduced_certified` (e.g. $(8,10)$) no ϑ search is run and `G1.C1.status = open(cap)` (final), `G1.theta = open(cap)`, `G1.components = not_run`, `G1.C2_status = partial(unfactored)`, `G3.partition = U_p_tags`; otherwise `G1.C1.status` $\in$ {`certified`, `open(cap)`} and `G1.reduced_Q` $\in$ {`true`, `undecided`} as stated; `blocked(U_p)` no longer in these ℓ-independent fields. Also (E5) $\mathcal{P}$ in the record's sense; (E6) the three re-read phrases enumerated, $g_{\mathcal{P}}$ of `S_rule` unchanged.
- Item 8 (d′)(iv) (S3): cites `rmk:stabconvention`(1), (2) and [U]'s orientation remark, and states that [U]'s "classical U" (Hecke systems, $\Gamma_1(\ell^r)$, χ) and `prop:levelconehecke`(3)(d) (the operator against dg_1, dg_p at $\Gamma_0(N)$, $\chi = 1$) concern different objects; one operator-level convention is not asserted.
- `prop:strictliftN`(9): (b) the classical operator written U_p^{cl} (E2), $A \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} A \neq AgA$ at $p \mid N$ (E9); (c) "their references to (7), (8) dropped" (E7); (d) the commutativity of U_p with the T_p is `prop:heckeZspec`(0) (E3); the proof (9)(c) discharges `prop:descentpairhecke`(1) at $p \mid N$ (S1: its item (1) reads nothing of N).

Non-normative. §31.3: the introduction points at (9) (E4); "What is used" names `prop:levelconehecke`(3)(d), `prop:descentmaps`(3)(b′) and says `def:heckeZ`, `prop:heckeZspec` are cited for identification only (S2); the notation sentence (E1, U_p^{cl}). Status notes of (9) and (d′): refereed ACCEPT-with-conditions (E8). E10: nothing to do (no `\LEAN` macro added). The header comment, `\date`, the Release record's first entry; this entry. No label is added,

renamed or removed; no status tag or marker moves (229 on 88); no record, golden line, fingerprint (`0x8ffc29730c4f9672`) or digest moves.

v1.26.0-pre.1 — 2026-11-08 — hecke-extensions slot 26, UP-Z: U_p at $p \mid N$, contract first

This version is a pre-release, **not owner-released** and **not yet refereed**, on top of the signed v1.25.0; its changes are **binding changes not yet seen by the owner**. They are condition C5 of the critical audit `docs/implementation/slot26/UPZ_design_audit.md` of the UP-Z design `docs/implementation/slot26/UPZ_design.md` (ACCEPT route (i), the invariant route, with conditions C1–C7): the text gates the merge of the code package Z1 and is refereed (adversarially) first. Its wording carries C1 (the freeze), C2 (the self-describing operator list), C4 (a)–(d) and C6 (the contract version in the provenance).

Normative.

- `prop:strictliftN`: new item (9), $p \mid N$: (a) $A'' = \Gamma_0(pN)$ of index p, the transversal $r_t = \begin{pmatrix} 1 & 0 \\ tN & 1 \end{pmatrix}$, $t < p$; (b) $U_p := [A \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} A]$ is the $U_p^{(N)}$ of `prop:levelcorrespondence`(4)(d), of `prop:heilbronn`(2) and of `def:heckeZ`, the adjoint U_p^* on p-old spaces (`prop:levelconehecke`(3)(d)); (c) items (2)–(6) and the independence of the lift hold verbatim with the same certified level-1 $\tau_{\mathcal{I}}$, so the induced-fibre operator, the glueing map and the transports to $\ell \in \{2, 3\}$ exist at $p \mid N$; (d) items (1), (7), (8) are not asserted. The proof (9) checks, item by item, that $p \nmid N$ enters only the hypothesis line and items (1), (7), (8) — the audit's §2.3 list, which also corrects the design's "(6)–(8)" to "(7), (8)". Title, hypothesis line and the subsection's notation sentence ("U_p does not occur") amended accordingly; the label and the status tag are unchanged.
- Item 8 of §22.4, new paragraph (d′) after (d): at $N > 1$, `algebra = full`, the operator list is `G1.P`'s T_p then U_p, $p \mid N$ (not `G0.operators`); $\mathcal{P}$ and its (C1) are the `anemic` ones; `theta_coeff_max` $= \binom{d}{2}$ with the full d; the operator is the adjoint-convention U_p, named; the routes (Heilbronn at $\ell \geq 5$, the free model at $\ell \in \{2, 3\}$; `blocked(U_p)` until built, never `refused(no_lift)` for want of a lift); the required hashed field `G1.theta_ops` and unhashed `G10.contract` on `full` records at $N > 1$; the freeze: any later change is a caps line or `/v2`, never an edit in place.

Non-normative. The header comment, `\date`, the Release record's first entry; this entry. Not changed (status edits after the Z1 merge, audit F7): `thm:descentcert`(5) "U_p is deferred" and (7)(c), `def:heckeZ`'s "not built over $\mathbb{Z}$". No label is added, renamed or removed; no status tag set or `\LEAN` marker moves (229 on 88); no record, golden line, fingerprint (`0x8ffc29730c4f9672`) or digest moves. Companion pointers: [E] 1.25.0, [U] 0.7.0, [X] v1.28.

v1.25.0 — 2026-11-07 — hecke-extensions slot 25: the owner-signed release of v1.25.0-pre.8

This version. **Signed by the owner 2026-11-07**: the owner-signed release of v1.25.0-pre.8; no change of content. It releases the content of v1.25.0-pre.1 to pre.8 as described in their entries below, including the binding changes there marked "not yet seen by the owner" (the refereed §31.6 and record item 8 of §22.4, rule (V1) as amended, Parts VIII and IX); the owner signs them with this release. **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 229 **on** 88 **labels.**

- Header and `\date`; the Release record's first entry; this entry.

Companion pointers: [E] 1.25.0, [U] 0.7.0, [X] v1.28.

v1.25.0-pre.8 — 2026-11-07 — hecke-extensions slot 25, LaTeX polish stage 5c: the title page

This version is a pre-release, **not owner-released**, on the branch `s25-wip-latexpolish-s5c`. Its one normative change, to rule (V1), is a **binding change not yet seen by the owner** in this form; the owner approved it in advance (ruling Q1 of 2026-11-06, `docs/hacs/OWNER_RULINGS_1106.md`) and signs it at the 1.25.0 release. The plan is stage 5c of `docs/hacs/LATEX_polish_plan.md`; the report is `docs/hacs/LATEX_polish_s5c_report.md`.

Normative.

- `V1`: the supersession record leaves the date line. Old: *The date line is* `Version X.Y --- YYYY-MM-DD`*, never* `\today`*, with the supersession record appended.* New: *The date line is* `Version X.Y --- YYYY-MM-DD` *and a status clause, never* `\today`*. The supersession record — one entry per version — is the first subsection of Appendix E.*

Non-normative. No statement, label, number or `\LEAN` marker moves.

- The date line is one line under the amended (V1). Its text up to v1.25.0-pre.7 — 167 version entries (v1.25.0-pre.7 down to v0.1.0), the sentence "Supersedes ..." and the companion pointer — is the Release record, the first subsection of this appendix, verbatim. Form only: each entry and the two closing sentences became a paragraph; the 53 line breaks \\ that ended the first 53 entries were dropped, and the run-on line of v0.10.0 to v0.2.0 was split before each "v0." (whitespace only).
- The author line "Orbifold Cohomology of GL(2)" is replaced by the uniform author block (owner ruling Q3 of 2026-11-06).
- A generic abstract replaces the old one, which named versions. The old author line and abstract are quoted verbatim in the second subsection of this appendix. The old abstract's one label-citation site (of `cl:pd`) is therefore still in this file, and the label census row of this file is unchanged.
- Two unnumbered part headings with their table-of-contents lines (owner ruling Q5 of 2026-11-06): "Part VIII — Level and character" before §28, and "Part IX — Integral and local structure" before §30. No section number or label moves; Part VII now ends after §27.
- Consequential edits of the part headings (owner ruling Q5), fix round 1 of the stage's audit; prose only, no statement changes. (a) the code map, §22.1: the rows of §28 and §29, "(Part VII)" → "(Part VIII)"; the rows of §30, §31, §31.1, §31.2, §31.3, §31.4, §31.5 and §32, "(Part VII)" → "(Part IX)". (b) §1, "How to read this document". Old: *Part VII (..., §27, since v1.13.0-pre.1, and §28, since v1.14.0-pre.1) is ... control and the lattice datum; then level N for every N — the composite coset table, the cokernel presentation and its Hecke action by Merel's matrices; it is placed after Part VI for the same reason. Chapter 2 ... binds all five parts;* New: *Part VII (..., and §27, since v1.13.0-pre.1) is ... control and the lattice datum; it is placed after Part VI for the same reason. Part VIII (§28, since v1.14.0-pre.1, and §29) is level N for every N — the composite coset table, the cokernel presentation and its Hecke action by Merel's matrices — and nebentypus; Part IX (§§30–32) is the $\mathbb{Z}/\ell^K$ Hecke algebra, the ℓ-local isotropy model with the descent at level N and the Hecke algebra over $\mathbb{Z}$, and the Moerdijk–Pronk model. Chapter 2 ... binds all nine parts;* ("five" had been stale since Part VI was added; the document has nine parts.)
- A consequence of the amended (V1): in §14.5's list, "the historical version lines of the date line" → "the historical version lines of the Release record (the first subsection of Appendix E; the date line until v1.25.0-pre.7)".
- The abstract is the stage audit's version: the Hecke algebra over $\mathbb{Z}$ is listed with the ℓ-local isotropy model (it is §31.6), and the status sentence follows Appendix B's wording.

- The appendix keeps its title "Changelog", which rule (V3) names; the plan's retitling ("Revision history") would have changed (V3)'s text.
- The header comment, `\date`; this entry. Companion pointers unchanged.

Not changed, listed for the owner: the counts in §1 ("53 C++ source files ... at 519 sites across 71 files") are stale; the label census (`docs/latex/label_census.tsv`) is the live record. Updating them is a prose change for a later version.

v1.25.0-pre.7 — 2026-11-06 — hecke-extensions slot 25, the HZ-RAT-PERF fix round

This version is a pre-release, **not owner-released**, on the branch `s25-wip-ratperffix`; its changes to item 8 of §22.4 are **binding changes not yet seen by the owner**, who signs them at the 1.25.0 release. HZ-RAT-PERF (value-neutral speed and memory of the rational step) added the named refusal `refused(word_prime_cap)`; its audit (`docs/implementation/slot25/HZRATPERF_audit.md`: PASS_WITH_MINORS for the code, F2 MAJOR: the word was code ahead of the binding text; F6: the draft sentence incomplete) asked for the text with or before the merge:

- (c): `refused(word_prime_cap)` in the list of pending statuses and beside "No other pending word exists"; a paragraph: what the list is (the `CrtPrimes` order without the divisors of N, prefix-stable, the U_p tag's list from min(2048, cap) primes), why the word is pending (only a larger cap lifts it), what a refused record contains, and that the word enters the frozen schema as amendment A1 (hash-neutral).
- G1: the closed list of `C2_status` as built (`complete`, `partial(unfactored)`, `blocked(U_p)`, `refused(word_prime_cap)`).
- (i): the count cap in the unhashed `word_primes=` provenance key.
- Header, `\date`; this entry.

No label is added, renamed or removed; no status tag set or `\LEAN` marker moves (229 on 88); no record, golden line, fingerprint (`0x8ffc29730c4f9672`) or digest moves.

v1.25.0-pre.6 — 2026-11-06 — hecke-extensions slot 25, the H1 merge

This version is a pre-release, **not owner-released**; its changes to item 8 of §22.4 are **binding changes not yet seen by the owner**, who signs them at the 1.25.0 release. The package H1 (the Eisenstein component over Λ, [U] part 6) writes G8 under `heckez.G8 = hida`; its audit (`docs/implementation/slot25/H1_audit.md`: PASS_WITH_MINORS; F1 MEDIUM, F2 MAJOR (report; an upstream HZ-RAT defect), F3–F5 MINOR, F6–F9 NOTE) recommended the text:

- (h): a G8 paragraph beside G5–G7 — only under `heckez.G8 = hida`; the paths `G8.scope`, `G8.ell=<l>.blocks`, `G8.ell=<l>.m[i].{ordinary, chain_key, g, L_k, v_Bk, structure, I0_generator}`; their closed lists and meanings, citing [U, `prop:u-H11`], [U, `prop:u-H12`] and the new [U, `rmk:u-H12general`] ([U] 0.7.0-pre.3); the caps as count-based record content, the q-expansion cap naming ϑ's operators (F4); G9 `K9.ell=<l>` (with `absent(not_applicable)` under `hida`, F3) and G10 `hida.ell=<l>`.
- (i): a pointer to it. (k): golden only under the default settings (F1; the code's `golden_eligible` now requires the default `heckez.G8 = off` line).
- Header, `\date`; this entry.

No label is added, renamed or removed; no status tag set or `\LEAN` marker moves (229 on 88); no default record, golden line or digest moves (F4 moves only the `hida` settings fingerprint and the three `hida` regression pins, which are never golden).

v1.25.0-pre.5 — 2026-11-06 — hecke-extensions slot 25, the HZ-REC schema re-audit fix round

This version is a pre-release, **not owner-released**, on the branch `s25-wip-hzrec`; its changes to item 8 of §22.4 are **binding changes not yet seen by the owner**, who signs them at the 1.25.0 release. The re-audit of pre.4's record (`docs/implementation/slot25/HZREC_schema_reaudit.md`: FAIL, narrow; R1 MAJOR, R2 MEDIUM, R3–R7 MINOR, R8–R12 NOTE) and the lead's decisions on it, each marked in the item:

- R1 (MAJOR): `components[0].mult` is e_k, the multiplicity of the Eisenstein system on H^1 (pre.4 said 1; wrong at $N > 1$ with $e_k \geq 2$); (d).
- R2: the second route's word caps in the settings text (`heckez.second_route`) and `not_read` also where `v_Bk = not_run`; (f), (j). Every record hash moves.
- R3, R4, R10, R11: the cuspidal block iff $d \geq 2$; the `not_run` of `components_through`, `pair_kinds.kind`, `G7.pair` and the cusp multiplicity at $N > 1$; the (C1) probe primes; where the `theory` tag is written; (d)–(i).
- R6: `[Lift]` is `refused(no_lift)` only for a certificate code, a cap of the lift is `blocked(export)`; (c).
- R8: the irreducibility cap 256 (was 64); R9: the ϑ search over $\mathcal{P} \cup \{T_\ell\}$ where $\mathcal{P}$ fails; both in the settings text; (d), (e), (j).
- Header, `\date`; this entry. R5, R7, R12 are code and documents (`docs/implementation/slot25/HZREC_r` "Re-audit fixes").

No label is added, renamed or removed; no status tag set or `\LEAN` marker moves (229 on 88); no golden line or digest moves (the record's golden objects are not yet emitted).

v1.25.0-pre.4 — 2026-11-06 — hecke-extensions slot 25, the HZ-REC schema-audit fix round

This version is a pre-release, **not owner-released**, on the branch `s25-wip-hzrec`; its changes to item 8 of §22.4 are **binding changes not yet seen by the owner**, who signs them at the 1.25.0 release. The schema audit of the built record (`docs/implementation/slot25/HZREC_schema_audit.md`: FAIL as a short fix round, 2 MAJOR, 4 MEDIUM, 8 MINOR, 2 NOTE) and the lead's decisions on it:

- F1 (MAJOR): the new final G6 value `split_on_P` for every $\mathbb{T}'$-split point; `split` (for $\mathbb{T}$) needs the mixed generation of Remark 31.43 ([conjectural]) and is not written in v1; (h), (k).
- F2 (MAJOR): no G8 field in default v1 records; H1 writes G8 under the non-default settings line `heckez.G8`; (i), (j).
- F3: G4–G6 per torsion point (a joint $\mathbb{T}'$-block with torsion) with its system, no per-ℓ aggregate; (g), (h).
- F5, F10: the components as written (the cuspidal part as one block, `irreducible` certified by a prime or `open(cap)`, p_0, no block at $d = e_k$), the closed lists of `type`, p_0 and `partition`; (d), (f).
- F6, F12: the maximal-ideal rule (a searched ϑ generating A_1'), `degree_attr`, `generation_tors`, $q = 2$ and G4's `lower` as v1 writes them; (e)–(h).
- F7, F8: `pending(HZ-TOR)` retired, the export's refusals mapped by name; (c). F11, F16: the caps text before it is frozen, defaults never change; (i), (j). O5, O6: `agrees = not_read`, `eta_L.sign`; (f).
- Header, `\date`, the item's status sentence; this entry. F4, F9, F13–F15 are code (`docs/implementation/slot25/HZREC_report.md`, "Schema-audit fixes").

No label is added, renamed or removed; no status tag set or `\LEAN` marker moves (229 on 88); no golden line or digest moves.

v1.25.0-pre.3 — 2026-11-06 — hecke-extensions slot 25, the Z0 re-check

This version is a pre-release, **not owner-released**; text only (`docs/implementation/slot25/Z0_recheck.md`). Item 8 (c): pre.2's binding rule (`export_word` iff $2^{31} \le \ell^K < 2^{63}$) left an EXPORT-served read at $\ell^K < 2^{31}$ with K above EXPORT's $K_{\max} = K_{\mathrm{run}} - \sigma_{\max} - 2\sigma^{(1)}_{\max}$ (the precision lost to the ℓ-torsion; level $N > 1$ at $\ell = 2, 3$ among others) without a record word, since `open(precision)` was defined as the settings' step cap. Now `export_word` iff $\ell^K < 2^{63}$ and $K > K_{\max}$ of EXPORT (the word bound included), pending as before (EXPORT-Z step 2 serves every K); `open(precision)` is the step cap only. No label is added, renamed or removed; no status tag set or `\LEAN` marker moves (229 on 88); no computed value, record, golden line or digest moves.

v1.25.0-pre.2 — 2026-11-06 — hecke-extensions slot 25, the Z0 merge after its referee

This version is a pre-release, **not owner-released**. v1.25.0-pre.1 was refereed REVISE (`docs/implementation/slot25/Z0_referee.md`: 0 BLOCKER, 3 MAJOR, 4 MINOR, 4 NOTE); pre.2 applies its findings as text and the lead's decision on its open item O1:

- F2 (**a statement narrowed**): §31.6, "Two forms", level $N > 1$, full — `non_reduced_certified` by the counting criterion is asserted only for N a power of p; for other N it is not proved (HZ-UTAG audit F1). This replaces pre.1's "no statement's mathematics changes" for that sentence; the code was already restricted.
- F3, F4: item 8 (c) — `refused(export_word)` is **pending** (EXPORT-Z step 2 widens the ring), not final; the binding rule `export_word` iff $\ell^K < 2^{63}$ and K exceeds EXPORT's $K_{\max}$ (the word bound $\ell^K \ge 2^{31}$ or its torsion precision; pre.3) at an EXPORT-served read, `word_modulus` tested first; the as-built rule ($\ell \ge 2^{31}$ refused at every K, $K = 1$ included; `open(precision)` at $\ell < 46341$ above EXPORT's $K_{\max}$); `open(precision)` added to the closed list (final: the step cap). Same in §31.6's *As built* paragraph.
- O1 (lead decision): no new $K = 1$ route now; (C3) at $K = 1$, level 1, $\ell \in [2^{31.5}, 2^{63})$ is the pending `refused(k1_route)`, an EXPORT-Z / slot-26 item, not a final narrowing of `word_modulus`.
- F5: the cofactor of (C2) and of $g_{\mathcal{P}}$ as built (the gcd over the primitive ϑ's, residue-checked, else `upper_bound`) in §31.6's cofactor paragraph and item 8 (d), (f).
- F6: item 8 (j) — `t_ell` $\in$ {`in_P`, `added`, `omitted(cap:ell>23170)`, `omitted(ell|N:U_ell_not_exported)`} the refusal name only; the cap 23170 in `P_rule`'s default text. (k): a pending refusal reason keeps a record out of the golden set.
- F7: Appendix B, the referee item "Part VII, §31.6" records A7 → A7b ACCEPT and this referee.
- Header, `\date`; this entry. F1, F8–F11 concern the merge report, the slot history and HZ-REC (`docs/implementation/slot25/Z0_merge_report.md`).

No label is added, renamed or removed; no status tag set or `\LEAN` marker moves (229 on 88); no computed value, record, golden line or digest moves. Companion pointers: [E] 1.24.0, [U] 0.6.0, [X] v1.28.

v1.25.0-pre.1 — 2026-11-06 — hecke-extensions slot 25, the Z0 merge: the Hecke algebra over $\mathbb{Z}$ in the main line

This version is a pre-release, **not owner-released**, on top of the signed v1.24.0. By the owner's ruling Q8 (2026-11-04) the HeckeZ text drafted and refereed on the branch `s24-wip-heckez-z0` (v1.25.0-pre.0, pre.0b, pre.0c below; referee A7 REVISE, re-audit A7b ACCEPT: `docs/implementation/slot24/heckez/A7-Z0-referee.md`, `A7b-Z0-reaudit.md`) enters the main line now that v1.24.0 is released. Taken exactly as Z0 defines it: §31.6 (23 statements), item 8 of §22.4, the pointer in `thm:descentcert`(7)(a), the notation list, 23 lines

of Appendix B, three bib entries and the pre.0–0c entries below; everything outside these keeps v1.24.0's text (the hunk list: `docs/implementation/slot25/Z0_merge_report.md`). Reconciled, each by a note dated 2026-11-06, with the plan's `04-record-census.md` (its slot-25 amendment of §1.5) and the code of slot 25 (EXPORT, HZ-RAT, HZ-UTAG, HZ-LOC):

- Item 8 (c): the refusal reason `export_word` (level $N > 1$, $\ell \geq 46341$, EXPORT's ring $\ell^K < 2^{31}$). (d): `cofactor_test` gains `upper_bound`. (j): the read set at ℓ is $\mathcal{P} \cup \{T_\ell\}$ for $\ell \nmid N$, $\ell \leq 23170$, with the hashed field `t_ell`; the factor budget as built. The code map.
- §31.6: the word-modulus paragraph as built (the seven 64-bit reads; (C3) at $K = 1$ for $\ell \in [2^{31.5}, 2^{63})$ at level 1 not served, an open item; `export_word`); the counting criterion of `thm:heckeZcert`'s full-algebra tag set only for N a power of p; the (C2) frontier as built.
- Header, `\date`, the body's two version pointers; this entry.

No label is added, renamed or removed beyond Z0's; no status tag set or `\LEAN` marker moves (229 on 88); no computed value, record, golden line or digest moves. Companion pointers: [E] 1.24.0 (the [E, `cl:sturm`](a) erratum Z0 cites is in it), [U] 0.6.0, [X] v1.28.

v1.24.0 — 2026-11-05 — hecke-extensions slot 25: the owner-signed release of v1.24.0-rc.2

This version. **Signed by the owner 2026-11-05**: the owner-signed release of v1.24.0-rc.2; no change of content. It releases the content of v1.24.0-pre.1, rc.1 and rc.2 as described in their entries below, each with the referee status stated there (rc.1 refereed ACCEPT_WITH_MINORS, all findings applied in rc.2). **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 229 **on** 88 **labels.**

- Header and `\date`; this entry.

Companion pointers: [E] 1.24.0, [U] 0.6.0, [X] v1.28.

v1.24.0-rc.2 — 2026-11-05 — hecke-extensions slot 25, RC124: the release candidate of 1.24.0, after its referee

This version is the release candidate of 1.24.0, **not owner-released**. v1.24.0-rc.1 was refereed ACCEPT_WITH_MINORS (`docs/implementation/slot25/RC124_referee.md`); rc.2 applies its findings as text (the map finding → fix: `docs/implementation/slot25/RC124_report.md`, "rc.1 → rc.2"):

- F1: Remark 16.8(4), the convention — [CCS13] grade with $\deg x = \deg y = 2$; their degree-$2n$ part is V and their condition is the one stated.
- F2, F6: §1.1 — the marker rule amended by one sentence (beside [proved here] or [standard:], and clause-scoped on definitions and displayed equations, as already applied); the rc.1 sentence shortened to a pointer, "the full form" → "the group-cohomological form". No marker or tag moves.
- F3, F4: Remark 28.19 — the printed `layers=` are the dimensions $\dim \mathfrak{m}^i W = 255 - 2i$; the curve-object confirmation and the 1297 sentence are reworded as reported records this document does not assert (nothing uses them; no tag added).
- F7: Remark 16.8, the scope sentence's two parentheses merged. F8: Proposition 16.15, the Hatada attribution; `gl2_refs.bib`: `calegari_emerton_largeindex`'s note gains Lemma 3.1's hypothesis and "journal version not checked".
- Header, `\date`, this entry.

No label is added, renamed or removed; no status tag set or marker moves (229 on 88); no computed value, record, golden line, digest, validation or code path moves. Companion pointers: [E] 1.24.0-rc.2, [U] 0.6.0, [X] v1.28.

v1.24.0-rc.1 — 2026-11-05 — hecke-extensions slot 25, RC124: the release candidate of 1.24.0

This version is the release candidate of 1.24.0: **not owner-released and not yet refereed** (its adversarial referee is `docs/implementation/slot25/RC124_referee.md`; the change list with each change's status is `docs/implementation/slot25/RC124_report.md`). It contains v1.24.0-pre.1 unchanged (FMTMP, §22.4 item 7 (l), owner request of 2026-11-03) and adds:

- **The Lean markers of question Q4** (approved in principle by the owner on 2026-10-18; `docs/implementation/slot24/LEAN_M6_report.md` §4 as corrected by `LEAN_M6_audit.md` F1, F3): 3 on Theorem 11.10 (`thm:m6-classical-theta`, `thm:m6-Tcls-res`, `thm:m6-hecke-res`) and 4 on (20) (`thm:m6-cor-res`, `thm:m6-corH-indep`, `thm:m6-cor-res-index`, `thm:m6-Tcls-res`), each clause-scoped by its parenthesis; the marker list of §1.1 brought up to date. 222 markers on 86 labels become 229 on 88 (215 `\LEAN`, 14 `\LEANlit`). The proposal's third marker (the witness of Proposition 32.20(2)) is not applied.
- **The phenomenology citations** (`docs/phenomenology/FINDINGS_SUMMARY.md` §3, `phenomena_catalogue.md` §11 items 2, 3, 5, 6; **not yet seen by the owner**): Remark 16.8 — clause (4) is now the theorem of Callegaro–Cohen–Salvetti [CCS13, Thm. 4.5] ([standard: ,] locator VERIFY), the full multiset of exponents at every prime, so the largest exponent equals the lower bound of (2) and is 1 off its loci; with the module convention checked against Proposition 8.2; clause (3) is a confirmation of it; "Not asserted" keeps only [L]'s growth estimate. Proposition 16.15 (F1) — its origin, [CE, Lemma 3.1] after Hatada. (Hi11) — a route to a proof through Hida's control modulo ℓ, not checked; the tag stays [conjectural]. Remark 28.19 — the locator Thm. 1.1; the heavy oracle $(21929, 2)$, $e_2 = 127$, is Calegari–Emerton's own example; their §6 sentence on 2143 is false (1297) and not used. `gl2_refs.bib`: `ccs13`, `calegari_emerton_largeindex`, `rustom_primepowers`.
- Header and `\date`; Appendix B's row of Remark 16.8 gains the citation (its tag set does not change); this entry.

No label is added, renamed or removed; no status tag set moves; no computed value, record, golden line, digest, validation or code path moves. Companion pointers: [E] 1.24.0-rc.1, [U] 0.6.0, [X] v1.28.

v1.25.0-pre.0c — 2026-11-04 — hecke-extensions slot 24, HeckeZ Z0: the A7b re-audit's MINOR (branch pre-release)

This version is still a **branch pre-release** on `s24-wip-heckez-z0`, not owner-released, on top of v1.25.0-pre.0b. The adversarial re-audit A7b (`docs/implementation/slot24/heckez/A7b-Z0-reaudit.md`) returned ACCEPT with 0 BLOCKER, 0 MAJOR, 1 MINOR and 5 NOTE findings. No label, number, tag set or marker moves (222 on 86 labels); no golden line, no code.

- `thm:heckeZcert`(C2) (A7b-m1). The clause on the other ϑ_j is now a statement: a certified prime q_i of D_{j_0} with $v_{q_i}(D_j) \leq 1$ for some j may be left out of S; the proof of (2) covers it (Proposition 31.33(2)). The measured (C2) frontier is re-measured on one D_{j_0}, $\vartheta_{j_0} = T_2$ (`a7b_frontier_single.py`): the same weights as on $\gcd_{p \leq 7}$; Q1(a)'s loss list $k \in \{36, 42, 44, 48, 50, 54, 56, 58, 62, 64\}$ is unchanged.
- NOTEs: `prop:heckeZeiscert`(3) "with multiplicity" (N1); "one modular power and one residue", a read below $v_\ell(\nu)$ is a defect (N4). N2, N3, N5 need no change.
- Header, `\date`; this entry.

v1.25.0-pre.0b — 2026-11-04 — hecke-extensions slot 24, HeckeZ Z0 fix round: the A7 referee's findings (branch pre-release)

This version is still a **branch pre-release** on `s24-wip-heckez-z0`, not owner-released, on top of v1.25.0-pre.0. The adversarial Opus referee A7 (`docs/implementation/slot24/heckez/A7-Z0-referee.md`) returned REVISE with 0 BLOCKER, 1 MAJOR, 7 MINOR and 10 NOTE findings; all MAJOR and MINOR findings are applied, and the NOTEs N1, N2, N4, N6–N9 (N3, N5, N10 need no change); the table is in `Z0-report.md` §5. No label, number, tag set or marker moves (222 on 86 labels); no golden line, no code.

- `prop:heckeZeiscert`(3) (A7-M1, MAJOR). A good prime q does not keep a non-Eisenstein block out of the Hensel factor at $1+q$: at $(N,\ell,q)=(61,5,37)$, q good, the rational newform of level 61 ($a_{37}=8\equiv 38 \bmod 5$) enters it, its degree is $2>1$ and its value at 38 has 5-adic valuation $2>1=v_5(\nu)$ (exact modular-symbol computation, `docs/implementation/slot24/heckez/a7fix_mazur_cx.py`; some good $q<60$ is contaminated at 27 of the 46 pairs (N,ℓ) with $N\le 260$). The statement now rests on the **valuation check** $v_\ell(\mathsf{F}_q(1+q))=v_\ell(\nu)$, which is necessary and sufficient for $\mathsf{F}_q=f_{q,\mathfrak{m}}$ and $(\eta_q)=I_{\mathfrak{m}}$; goodness is how q is chosen. F_q is the Hensel factor of h_q, the square root of the cuspidal factor — not of a squarefree part — so the route is residue-only (Q1(a)). The rank is $e^{\mathrm{cusp}}_{\mathfrak{m}}$ (A7-m7). Record item 8 (e): `mazur` carries both valuations and certifies only on agreement.
- `prop:heckeZrat`: (3′)'s proof, "no Eisenstein system is cuspidal", at $p\ge 7$ for every ψ (A7-m1); the citations of (3′), (4)(a), (4)(b), (5)(a) by the verified [GT] rows `lit:hz-es-anemic`, `lit:hz-eis-dim`, `lit:hz-semisimple-an`, `lit:hz-sturm-integral` (Diamond–Im Thm. 6.2.3, Rmks. 6.3.3–6.3.4, Prop. 12.4.13; Stein Thms. 5.8–5.10, 9.23) in place of Diamond–Shurman, and the Appendix B row (A7-m5, N2); (3′)'s "iff" excludes $(N,k)=(1,2)$ (N9).
- `prop:heckeZspec`(3): the converse (no other minimal primes) by going-down for the $\mathbb{Z}$-free $\mathbb{T}^{\mathrm{fr}}$ (A7-m2).
- `thm:descentcert`(7)(a)'s pointer and the closing sentence after `prop:heckeZglue`: the extension class is determined by finite data; a split verdict needs `rmk:heckeZcbthree`; "the one invariant … needs" becomes "the invariant … uses" (A7-m3). The record of item 7 still asserts neither.
- `thm:heckeZcert`(C2) factors one D_{j_0}, no gcd of big integers; the cofactor paragraph restores the plan's two Q1(a) sentences; item 8 (d), (f) and the "factored $g_{\mathcal{P}}$" paragraph accordingly (A7-m4).
- NOTEs: `prop:heckeZeiscert`(3) cites the verified row `lit:hz-mazur-eis` without `\VERIFYloc` (N1); $g_{\mathcal{P}}$ prime to 6 at level 1 (N4); `prop:heckeZgen`'s lattice in a stable subspace (N6); `rmk:heckeZcaveat`(3) the square-u case, (10) cokernel versus Tor (N7).
- Header, `\date`, the notation list, Appendix B's row of `prop:heckeZeiscert`; this entry.

Companion pointers: [E] 1.24.0-pre.2 ([E, `cl:sturm`](a): the condition for $k\equiv 2 \bmod 12$, A7-m6; range $4\le k\le 70$, N8), [U] 0.6.0.

v1.25.0-pre.0 — 2026-11-04 — hecke-extensions slot 24, HeckeZ package Z0: the Hecke algebra over $\mathbb{Z}$ (branch pre-release)

This version is a **branch pre-release** on `s24-wip-heckez-z0`, not owner-released, not refereed (an adversarial Opus referee follows, the plan's step A7), on top of v1.24.0-pre.1. By the owner's ruling Q8 of 2026-11-04 it enters the main line as **v1.25.0-pre.1** only after the owner has released v1.24.0; neither release waits for the other. Source: the HeckeZ plan `docs/implementation/slot24/heckez/`, signed 2026-11-04 with every question at its recommendation (`06-work-packages.md` v0.4 §5; package Z0, §2.3). No marker moves (222 on 86 labels); no golden line, no code.

- [new] §31.6, appended at the end of §31 (after §31.4 and §31.5), so no number moves. From `01-math-core.md` v0.6: `def:heckeZ` (untagged; Def. 1.1), `prop:heckeZspec` ([proved here], [standard]; Prop. 1.2), `prop:heckeZrat` ([proved here], [standard], [conjectural]; Props. 1.3, 1.4, 1.5(2)–(4)), `lem:heckeZbridge` ([proved here]; Lemma 1.6), `prop:heckeZmax` ([proved here], [standard]; Prop. 2.1), `prop:heckeZgen` ([proved here]; Prop. 2.2, Rem. 2.2′), `prop:heckeZetale` ([proved here]; Prop. 2.3), `prop:heckeZmono` ([proved here], [standard]; Prop. 2.4), `prop:heckeZlocal` ([proved here]; Prop. 2.5), `prop:heckeZthrough` ([proved here]; Prop. 2.6), `thm:heckeZdescent` ([proved here]; Thm. 3.1), `thm:heckeZcert` ([proved here], [standard]; Thm. 4.1, Prop. 4.2), `rmk:heckeZcaveat` ([proved here]; Rem. 4.5, Props. 4.3, 4.4), `def:heckeZcong` (untagged; Def. 5.1), `prop:heckeZcong` ([proved here], [standard]; Props. 5.2–5.5), `prop:heckeZeiscert` ([proved here], [standard]; Prop. 5.6, Rem. 5.6′). From `03-bockstein-tor.md` v0.3: `prop:heckeZuct` ([proved here], [standard]; B1 with B4(1)), `prop:heckeZsplit` ([proved here]; B7), `prop:heckeZlevelone23` ([proved here]; B8, B8′, B9), `prop:heckeZglue` ([proved here], [standard]; B11–B15), and the conjectures CB-1, CB-2, CB-3 as `rmk:heckeZcbone`, `rmk:heckeZcbtwo`, `rmk:heckeZcbthree` ([conjectural]; used by nothing). Not written as statements: the plan's Prop. 1.5(1) ([conjectural], used nowhere; [E, `cl:sturm`] as corrected at [E] 1.24.0-pre.1 is cited instead), Prop. 5.7 (a sketch), the per-block remark of Prop. 2.4 (a sketch) and the stopping rule of B15(2) (a sketch) — each is named in the text as not asserted.
- §22.4 gains item 8, `integral_hecke_structure/v1`, clauses (a)–(l) (`04-record-census.md` v0.4 §1.6; (l), the big-integer rule, at Q1(a)).
- `thm:descentcert`(7)(a): a sentence pointing to `prop:heckeZcong`(5) for the read at a root and to `prop:heckeZglue`(2)(a) for the extension class; the record of item 7 still asserts neither.
- Appendix B: 23 lines; `gl2_refs.bib`: `atiyah_macdonald`, `bruns_herzog`, `coleman_edixhoven`; header and `\date`; this entry.

Companion pointers: [E] 1.24.0-pre.1 (the [E, `cl:sturm`](a) erratum, owner ruling Q9), [U] 0.6.0.

v1.24.0-pre.1 — 2026-11-03 — hecke-extensions slot 24, FMTMP: the faithful coefficients of χ

This version is a pre-release, not owner-released, not refereed, on top of the signed v1.23.0. §22.4 item 7 gains the lettered item (l), *The faithful coefficients*: every coefficient of $\chi_{p,1}^{(N)}$ beyond 2^{63} is recorded also by its exact value, the canonical signed hexadecimal text of the integer, in the entry's list `big`, and hashed by one body line each; items (d) and (h) point to it. A record without such a coefficient keeps its bytes and its `record_fnv1a64`; the digest of (d) and the golden reference do not move; cell ids and resume keys are unchanged. Text only: no label, number, tag set or marker moves.

- §22.4 item 7: (l) new; one clause each in (d) and (h).
- Header and `\date`; this entry.

v1.23.0 — 2026-11-03 — hecke-extensions slot 24: the owner-signed release of v1.23.0-pre.40

This version. **Signed by the owner 2026-11-03** ("I sign off 1.23.0"): the owner-signed release of v1.23.0-pre.40; no change of content. It releases the content of v1.23.0-pre.1 … pre.40 as described in their entries below, each with the referee status stated there. **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 222 **on** 86 **labels.** The Lean markers on Theorem 11.10 and Equation 20 (Q4) are not part of this release.

- Header and `\date`; this entry.

Companion pointers: [E] 1.23.0, [U] 0.6.0, [X] v1.28.

v1.23.0-pre.40 — 2026-11-02 — hecke-extensions slot 24, ICENSUS-P2: the campaign's `parts` key

This version is a pre-release, not owner-released, not refereed, on top of v1.23.0-pre.39. §22.4 item 7 gains the lettered item (k), *Parts not run*: a descent campaign cell may compute only the groups (`parts = groups`) or the groups and the Hecke data (`parts = groups,hecke`); a part not run is marked `"ran":false`, a `groups` record's hashed body says `hecke=not_run`, a non-default value is part of the cell id, and the default `all` leaves every id, record and hash byte for byte as before. Text only: no label, number, tag set or marker moves, and no golden line.

v1.23.0-pre.39 — 2026-11-02 — hecke-extensions slot 24, the ker / coker width ladder (ICENSUS-P1)

This version. **A pre-release, not owner-released, not refereed.** On top of v1.23.0-pre.38, item 7 of §22.4:

- (e): the ker / coker read's working width is a ladder, $k_0 = \min(K_{\max}(\ell), 2a_{\max} + 3)$, then $K_{\max}(\ell)$ on a precision shortfall; the width is not in the canonical text and is not one of (e)'s caps.
- (j): the paragraph on the refusal at $(1, 200)$ kept as pre.38's text, as history, with its two misstatements marked (the width there is $k_0 = 11$, not 2^{30}; $W = 61$ is not what lifts the cell); the new paragraph "The width ladder": escalation on a precision shortfall only, at every ℓ; a failed safeguard refuses `[NotEndomorphism]`; why the read stays certified and no hash of an `ok` entry and no golden line moves; the six level-1 cells that now read; arm (L) and the oracle check at $(1, 48)$ (`docs/implementation/slot24/ICP1_acceptance.md`).

The code change it describes: `Integral/Descent/DescentHeckeTorsion.cpp` (`Invariants`), `Integral/Descent/DescentMaps` (`MapRes::precision`). No label, number, tag set or marker moves (222 on 86).

v1.23.0-pre.38 — 2026-10-28 — hecke-extensions slot 24, the pre.37 referee's fixes (CON)

This version. **A pre-release, not owner-released, not re-refereed.** It carries v1.23.0-pre.37, refereed PASS-with-fixes (package CON, P7 (a), `docs/implementation/slot24/CON_pre37_referee.m` 2 M / 10 m / 5 N / 2 T, the pre.30 code map included), and applies the fixes, each checked against the code; `docs/implementation/slot24/CON_pre38_fixes.md` maps finding to fix. Text only: no code, hash, record or golden line moves. No statement is weakened.

- M1, item 7 (d): `coefficients` is $c_0, \ldots, c_r$, c_i the coefficient of x^{r-i}, the leading 1 first — an explicit exception to item 2's low-to-high rule; the in-band key is an open code item.
- M2: item 7's code map restated as built, audited (`D6_code_audit.md`), fixed (`19a969de`), merged (`2aa61397`), golden R1 landed (`6958a5d2`); the console's prime bound (≤ 6 primes, each ≤ 97, $\sum p \leq 100$); arm (B); the four code items of v1.23.0-pre.37 closed; (f) met; (j) in the present tense, its interim reading rule vestigial; the status-table row of §31.3, the pointer in `thm:descentcert`(7)(c) and §31.3's code map updated.
- m1 (a): the two refusal mechanisms (`CoefficientRingK` at construction; `GateVerbsOnField`'s opt-in list, kept by arm (F)) and the residual risk. m2 (b): the record-level fields and the pair's `bulk` object, one complex reported twice. m3 (e): indexed by `torsion_primes` (n odd: $\ell = 2$ only). m4 (e): the three conditions of the ker / coker read, record content. m5 (g): the groups carry no refusal of their own; a non-refused entry may carry a refusal text. m6 (h): the hashed / unhashed partition exhaustive, `DescentRecordZ::body()`

normative for the order, `commute` unhashed. m7 (d): the digest's byte layout (every field one 64-bit word, least significant byte first; the sign as the word $1 + [c_i < 0]$). m8 (h): the $(1, 12)$ hashes illustrative, not pinned.

- m9: the status-table row of §31.2 brought up to its code map; "the tree bulk does not move D4's wall" put at v1.23.0-pre.30. m10: `def:ringsemantics`(3) at level N, and (4)'s "waits for the descent assembly" marked built.
- N1 (d): the undetermined case is not reachable from the record. N2 (g): a record built without its Hecke or pair part is not a conforming `v1` record. N3 (d): $q = 0$ omitted by choice. N4: four oracle tables. T1: the JSON key is `digest`. T2: [IB]'s OD-9 (a), OD-10 (a), recorded by [X] S8-6d. N5 (a source comment) is outside [C].
- One referee claim not taken: m7's "fixed widths" (u32 for r, i and the residues, u64 for the count) — `engine::FnvHasher::feed` takes one `uint64_t`, so every field enters as an 8-byte word.
- No label is removed, renamed or added; no number, tag set or marker moves (222 on 86 labels).

v1.23.0-pre.37 — 2026-10-26 — hecke-extensions slot 24, the integral descent record (D6)

This version. **A pre-release, not owner-released, not refereed.** It carries v1.23.0-pre.36 and gives the record of D6 (branch `s24-d6impl`, tip `538cf881`, `docs/implementation/slot24/DESCENT6_repor` pending audit and merge) its contract home. No statement is weakened.

- New item 7 of §22.4, `integral_descent_record/v1`: (a) ring $\mathbb{Z}$, object group, $\mathbb{Z}$ refused by name elsewhere, `Z/<m>` reserved; (b) $q_{\max} = 3$ derived from `thm:descentcert`(6) (checked: degrees 0, 1 and one period 2, 3 determine every degree, and 3 is the least such), `truncated = true`, `periodic_from = 2`; (c) groups, bad primes, K per prime as provenance; (d) the `char_Z` digest (FNV over $c_i \bmod 2^{64}$, $\bmod 3^{40}$, or the residues when undetermined): choice-free when determined, a fingerprint not a certificate; (e) the torsion Hecke invariants, `module_certified = false` not a refusal; (f) the pair, each complex $q_{\max} = 4$, the compact object's list of names required by item 5 and missing in `v1` (code item); (g) refusals: a refused record is never a valid empty answer; (h) the hashed and unhashed fields, justified by `thm:descentcert`'s choice-independence, `rmk:descentcertscope` and the read-only, value-preserving rollbacks `HECKE_DESCENT_TREE=0`, `HECKE_DESCENT_TREE_PAIR=0`; (i) `thm:descentcert`(4), (5) not recorded and not built; (j) the refusal at $(1, 200)$, $\ell = 2$.
- The refusal name: `[NotEndomorphism] ... (30 < 34)` at $(1, 200)$ is a precision shortfall of the ker / coker read at the ring width 2^{30} ($30 < 34$ counts certified pivots); the taxonomy misnamed it; the name is `[KerCokerPrecision]`, `[NotEndomorphism]` kept for a failed endomorphism safeguard. A code item; moves no hash and no golden line.
- `def:ringsemantics`(3): $\mathbb{Z}$ "(reserved)" becomes un-reserved for the integral descent only. `thm:descentcert`(7)(c), the status table's row for §31.3 and §31.3's code map point to item 7.
- No label is removed, renamed or added; no number, tag set or marker moves (222 on 86 labels).

v1.23.0-pre.36 — 2026-10-26 — hecke-extensions slot 24, the C-D8b referee's fixes; D5c merged

This version. **A pre-release, not owner-released, not re-refereed.** It carries v1.23.0-pre.35, refereed ACCEPT after fixes (0 M / 1 m / 7 N, `docs/implementation/slot24/C-D8b_referee.md`, with the author's response), and records D5c merged into `hecke-extensions` at `4064e164`. No statement is weakened.

- m1, code map of §31.5: the certificates restated for the merged code — `certify_transport` compares the blocks of the **applied** operator ($T'_{\mathcal{T}}$, $\eta_{\mathcal{T}}$, T'_{∂}, the strict block) with separately assembled ones and checks the identity of (4)(a) on those values; entrywise for $D \le 1300$; above, on the generators and 64 random vectors uniform in $\mathbb{Z}/\ell^K$, seeded by (ℓ, N, n, j), a miss $\le \ell^{-64}$ per check. What is deterministic and what probabilistic; admissible because no value is read from a check and the read's correctness is the deterministic (4)(e)(iii) with the chain-level certificates. "Not new coverage" removed.
- N1: `prop:treepair`(4)(e)(iv) and its proof: commuting with D *gives* the coboundary half of (iii); the code map says that above $D = 1300$ it is not checked deterministically at cochain level.
- N2: (4)(e)'s hypothesis admits the truncation of T to cochain degrees $\le m$ (the pair lift, built to degree 6, is one). N3: the proof of (e)(i) for projective, not free, sources. N4: (e)(ii) refined (bulk in chain degree $j+1$, boundary and homotopies in j) and extended to H^q, $H^q(\partial)$, $q \le m$. N5: the identity on the generators' α-parts is the second component of (ii) modulo $\ell^{K-\sigma_{\max}}$. N6: `def:treepair`(2), "computed from these cochains"; the chain-level constructions use $\mathtt{a}^{(\ell)}$.
- N7, code map: merged at `4064e164` (audit `D5c_code_audit.md`, its N5 deferred and recorded); obligation (c) every 2- and 3-adic pair piece, every N, including $3 \in$ `extra_primes` with n odd; the skipped top degree at $q_{\max} = 5$ counted (`eta_skipped`) with the (Q5) arm; (f) met by the production certificate; (g) met by `transport_bytes`. The status table and the remarks of §31.2 and §31.3 that said "pending" updated; D5c's two additions to D3's code (degree-5 comparison, `IsoHeckeN::apply_boundary`) recorded there.
- Checks re-run: `cd8_check.py` and the referee's `pre35referee/pre35_check.py`, both OK (lines in the author's response).
- No label is removed or renamed; no number, tag set or marker moves (222 on 86 labels).

v1.23.0-pre.35 — 2026-10-26 — hecke-extensions slot 24, §31.5 in line with the D5c code

This version. **A pre-release, not owner-released, not refereed.** It carries v1.23.0-pre.34 and changes §31.5 (and one code-map note of §31.3) after the D5c implementation (branch `s24-d5cimpl`, tip `e720f11c`, `docs/implementation/slot24/DESCENT5c_report.md` §6; pending audit and merge). No statement is weakened.

- `def:treepair`(2): at $\ell \in \{0, 2\}$ S and S^{-1} give different chain maps but, the difference being $(1-\mathbf{c})Se_x$, **the same cochains** on every module on which $\mathbf{c}$ acts trivially; $\rho_{M_N}(\mathbf{c}) = (-1)^n$, so for n even at every ℓ, with proof; for n odd the difference $2Se_x$ vanishes modulo 2. The code map's (f) and the check paragraph restate the negative control: at chain level, or on cochains at $\ell = 2$ with n odd (modulo 2^K, $K \ge 2$); asserted clean for n even and at $\ell \ge 3$ — as the code pins it ($(1,5)$, $(3,1)$ caught; $(1,4)$ clean).
- New `prop:treepair`(4)(e), with proof: comparisons given in chain degrees $\le m$ (homotopies $\le m-1$) extend (the comparison theorem; the targets are acyclic in positive degrees), and the maps in cone degrees $\le m-1$ are those of every extension, so the read of H^q_c, $q \le m$, needs nothing in degree $m+1$; the identity of (4)(a) in cone degree j needs chain degree $j+2$, is formable iff $j \le m-2$, and is not used by the read. With the code's $m = 5$, $q_{\max} = 5$ **stays admissible**; (6)'s refusal "$q_{\max} \notin \{4, 5\}$" is unchanged and points to (4)(e). The alternative, restricting the transport to $q_{\max} = 4$, is not needed.
- Code map: built on `s24-d5cimpl`, pending audit; files, the default route of the 2- and 3-adic pair pieces, the rollbacks `HECKE_DESCENT_TREE_PAIR=0`, `HECKE_DESCENT_TREE=0`, the two validations, the reach ($D \approx 1060$–1240 to the refusal at $D \approx 6340$, ≈ 8000 at 40 GiB; the next wall D2's dense Smith form with transforms). (e)'s certificates restated per cone degree ((i) strict block, (ii) cocycles on the generators in every read degree, (iii) the identity

of (4)(a) in cone degrees $\le \min(q_{\max}-1,3)$, entrywise for $D \le 1300$, on the generators above); "on the generators" is deterministic, partial and not probabilistic, and admissible under (5) because no value is read from it. The ℓ-local augmentations of `def:treepair`(4) are the code's. §31.3's open code item N7 (`D3Refusal::HeldOutMismatch`) closed on the branch.
- Check: `docs/implementation/slot24/cd8_check.py` gains arms (M3) and (Q); its line is recorded in the check paragraph.
- No label is removed or renamed; no number, tag set or marker moves (222 on 86 labels).

v1.23.0-pre.34 — 2026-10-26 — hecke-extensions slot 24, the C-D8 referee's fixes

This version. **A pre-release, not owner-released, not re-refereed.** It carries v1.23.0-pre.33, refereed ACCEPT after fixes (0 M / 4 m / 8 N, `docs/implementation/slot24/C-D8_referee.md`, independent check `cd8referee/`), and changes §31.5 (and one code-map note of §31.3). No statement is weakened except that "$\eta_{\mathcal{T}} \neq 0$ in general" is now an observation of the checks, not a proved assertion.

- **m1** `prop:treepair`(4)(a), its proof, `rmk:treepairscope`(iii) and the subsection's introduction: on the pair model $(\mathrm{Cyl}, \mathsf{B})$ the transported datum T^{Cyl} is strict ($\eta = 0$; the plan's sentence holds there); on $C(\mathsf{a}^*)$, not the cone of a restriction to a subcomplex, $T^{\mathrm{c}}_{\mathcal{T}} \simeq \varpi T^{\mathrm{Cyl}} \varsigma'$ acquires $\eta_{\mathcal{T}}$ (homotopy proved); "$\eta_{\mathcal{T}} \neq 0$, the diagonal not a cochain map" is observed (θ^*; the referee's T_ε), not proved; "forced" and "no strict pair datum exists on the tree" are gone.
- **m2** `prop:treepair`(3)(b): h_p, (L), (L′) for any column augmentation of a complex filtered by finitely many free periodic columns, with the sign rule per column, over every R after $\otimes R$ (coefficients in $\mathsf{O}_\ell[\Gamma]$ allowed), with proof; applied to ε_{B}, $\mathrm{cmp}|_{\mathcal{I}^\partial}$, $\varepsilon_{\mathcal{T}} \otimes R$, $\mathrm{cmp} \otimes R$. The citation "stated for any such p" is replaced.
- **m3** Code map (f) and the check paragraph: S for S^{-1} only at $\ell \in \{0,2\}$; at $\ell = 3$ the dropped term $S^{-1} e_U \mathsf{v}_1^U$; $\eta_{\mathcal{T}}$ dropped only for non-fibre operators, detected on a cell per $\ell \in \{2,3\}$, never at $\ell = 3$ with n odd; detection counts printed; an arm for $\mathsf{j}_\Psi \mathsf{j}_\Phi = \mathrm{id}$ at a cell with 2-torsion in $H^{\ge 3}_c$.
- **m4** `def:treepair`(4): $\varepsilon^{(\ell)}_{\mathcal{T}}$, $\varepsilon^{(\ell)}_{\mathsf{B}}$ defined; `prop:treepair`(1)(b) stated with them; $\Phi^{\mathcal{T}}_\ell$, $\Psi^{\mathcal{T}}_\ell$, Φ^{∂}_ℓ, Ψ^{∂}_ℓ lie over the identity of $C_\bullet(\mathcal{T}) \otimes R$, resp. $C_\bullet(\mathcal{L}) \otimes R$; the proof of (3)(c) uses them.
- **N1–N8** `lem:treepaircyl`: B finitely generated projective, zero in negative degrees; `rmk:treepairscope`(i): the path runs from $T\mathfrak{v}_S$ through $S^{-1}\mathfrak{v}_U$ to $\mathfrak{v}_S$; (3)(b): the parenthesis on Φ^{∂}_ℓ, scalars 1; (5)(a): $C(r_{\mathrm{Cyl}})$ not admissible, $\mathrm{rank}\, C^0_{\mathcal{T},\ell}(M)$; code map (e), (g): the cost of forming $\eta_{\mathcal{T}}$; (c): the rollback preserves the record's invariants; §31.3's code map: `D3Refusal::HeldOutMismatch` an open code item; `def:treepair`(2): S versus S^{-1} immaterial at $\ell \ge 3$.
- No label is removed or renamed; no number, tag set or marker moves (222 on 86 labels).

v1.23.0-pre.33 — 2026-10-26 — hecke-extensions slot 24, C-D8: the tree cusp pair (D5c), contract first

This version. **A pre-release draft, not owner-released, not refereed** (an adversarial referee follows). It carries v1.23.0-pre.32 and adds §31.5, the plan's clause C-D8 for milestone D5c, **stated before its code** (rule 21); the D3 code-map status of §31.3 records the merge. Source: `docs/implementation/slot24/DESCENT_plan.md` v0.4 §5, §2.7 (row C-D8), D5c; `DESCENT4_report.md` §4 (the 2-adic pair wall), `DESCENT5_report.md`; the referee reports `C-D6_referee.md`, `C-D7_referee.md`. Check `docs/implementation/slot24/cd8_check.py` (output `cd8_check.out`).

- [new] §31.5. `def:treepair` (untagged): the cusp model $\mathsf{B}_{\mathcal{T}}^{(\ell)}$ (the cone of $r_{(1-T)e_{\mathfrak{e}}}$ on the tree's edge column; ranks $1, 2, 2, \dots$ at $\ell = 2$, $1, 1$ at $\ell \geq 3$), the axis map $\mathsf{a}^{(\ell)} \colon \mathsf{B}_{\mathcal{T}}^{(\ell)} \to \mathcal{I}_{\mathcal{T}}^{(\ell)}$ in closed form ($\mathfrak{e}_\infty \mapsto (S^{-1} - T)\mathfrak{e}$ plus periodic column terms), the tree cone $C(\mathsf{a}^{(\ell)*})$, and the geometric tables i_∂, r_∂, G_∂, f_a, $\mathsf{g}_\square$, g' on the line complex $C_\bullet(\mathcal{L})$.
- [new] `lem:treepaircyl` ([proved here], [standard]): the mapping cylinder $(\mathrm{Cyl}(\mathsf{a}), \mathsf{B})$ is a pair model in the sense of `def:descentpair`; its cone deforms onto $C(\mathsf{a}^*)$ by explicit ϖ, ς, ς', $\mathsf{h}_{\mathrm{Cyl}}$, strictly compatible with inc_c, pr_c; so `prop:descentpair`(1) holds on the tree cone with its signs.
- [new] `prop:treepair` ([proved here], [standard]): what it computes in every degree over every O_ℓ-algebra; the cochains in closed form at $\ell = 2$, 3, ≥ 5 (at $\ell \geq 5$ the modular-symbol complex, H_c^2 the coinvariants); effective comparisons with $(\mathcal{I}, \mathcal{I}^\partial)$ — $\Phi_\partial^{\mathbb{Z}}$, $\Psi_\partial^{\mathbb{Z}}$ by (L), the square homotopies y_ℓ, y'_ℓ by (L′), the cone maps j_Φ, j_Ψ, mutually inverse on cohomology through comparisons of pairs; T_p by transport, $T_{\mathcal{T}}^{\mathrm{c}} = \mathsf{j}_\Psi \Phi_p^{(N)} \mathsf{j}_\Phi$, with a **non-zero, determined off-diagonal** $\eta_{\mathcal{T}}$; the precision rule; refusal (RSS, precision, the pair lift's codes, $q_{\max}$, the cap) and failure (a failed certificate of a constructed map) and scope.
- [new] `rmk:treepairscope` ([proved here]): where the plan's §5 is corrected — the minimal cusp resolution, the tree cone in place of the cylinder's, $\eta_{\mathcal{T}} \neq 0$ ("no η on the tree side" is withdrawn), the comparisons of pairs as proof devices, cost a prediction.
- §31.3's code map: D3 **audited and merged** at `dc8261a3`; the $\ell \geq 5$ cokernel torsion read refuses `[RssCeiling]` before building; `HeldOutMismatch` and `[ResidueRank]` throw `hecke::HeckeInternalError` (failures, not refusals). Pointers to §31.5 in §31.2 (scope, code map), §31.4 (introduction) and `rmk:integralscope`(5); the code-map table gains a row and records the D3 merge.
- Four statements added, the last of their section; no label is removed or renamed; no existing number or tag set moves; no `\LEAN` marker is added (222 on 86 labels).

v1.23.0-pre.32 — 2026-10-26 — hecke-extensions slot 24, the C-D3h referee's fixes

This version. **A pre-release, not owner-released, not re-refereed.** It carries v1.23.0-pre.31, refereed ACCEPT after fixes (0 M / 3 m / 6 N, `docs/implementation/slot24/C-D3h_referee.md`, check `pre31referee/pre31_check.py`), and changes §31.3 only. No statement is weakened.

- **m1, N4** Proposition 31.13(2), independence of the lift: the coboundary of $C_{\mathcal{I}}^\bullet$ is unsigned (Lemma 15.4(2)), so $\mathsf{H} = \Theta_{h_{\mathcal{I}}}$ in every degree; "±" and "re-signed degree by degree" are gone; H is "constructible", not "explicit".
- **m2** Theorem 31.16(4) at $\ell \geq 5$: with $k = K'$, $K > \sigma_{\max} + v_\ell(\chi_{p,1}^{(N)}(c))$ suffices ($\sigma_{\max}$ of Δ's certified form, 0 without ℓ-torsion), with its proof; a short read is refused, never wrong.
- **m3** Proposition 31.15(3′): any exact lifts $x_j \equiv \tilde{x}_j \pmod{\ell^K}$ form a basis and give the same X modulo ℓ^K ($P = I + \ell^K M$), proved.
- **N1–N3, N5, N6** (3′)'s independence per certifying read; $U = ST$ in Theorem 31.16's proof; the free part's K' is sharp; the code map notes that `certify()` at $q \leq 2$ is all the clause requires and that Rel_N-stability is a theorem; "What is used" completed.
- No label is removed or renamed; no number, tag set or marker moves (222 on 86 labels).

v1.23.0-pre.31 — 2026-10-26 — hecke-extensions slot 24, D3: the cokernel presentation, obligation (b) through the lift, any admissible lift

This version. **A pre-release, not owner-released, not refereed** (an adversarial referee follows). It carries v1.23.0-pre.30 and changes §31.3 only. Source: §6 of `docs/implementation/slot24/DESCENT3_report.md` (D3, branch `s24-d3impl`, tip `a1375159`, not merged), and the D3 code (`Integral/Descent/DescentHecke*`). No statement is

weakened; three are widened to what the code reads, each with its proof.

- **The cokernel presentation** (D3's note 1). New sub-item Proposition 31.15(3′): the read of (3) on $Q = \operatorname{coker}\Delta$ with an endomorphism F preserving $\operatorname{im}\Delta$, verbatim (proof: the two-term complex, with F^{q-1} chosen by freeness and never read); the free part Q/tors is read as well, modulo $\ell^{K'}$ (the "moreover" of Proposition 31.10(2)(g)(i)), as the input of (4); the invariants do not depend on the presentation. Proposition 31.14(2) admits $[H_{N,p}]$ on $M_{N,\mathbb{F}_{\ell'}}/\mathrm{Rel}_N$ (Proposition 28.9(1), (2) through $\varkappa_N$), Theorem 31.16(3), (4) admit it on $M_{N,\mathbb{Z}_{(\ell)}}/\mathrm{Rel}_N$ at $\ell \geq 5$, with the certifying rank $m_N(n+1) - b_1(M_N)$; the proofs say why (6 a unit, $\varkappa_N$ an isomorphism intertwining $[H_{N,p}]$ with T_p, flat base change).
- **Obligation (b) through the lift** (D3's note 2). Proposition 31.13(2): an integral table and a $\tau_{\mathcal{I}}$ certified by (T1)–(T3) and the strict identity on *every* basis element ($\mathrm{cmp}\tau_{\mathcal{I}}(s_j e_c) = 0$ at $t \geq 1$ included) make the table the table of a g-twisted chain datum over $\mathbb{Z}$; proof: $\mathrm{cmp}\tau_{\mathcal{I}}$ kills $\ker\mathrm{cmp}$ (its $t = 0$ part is $d^{(0)}$ of $t = 1$, by flatness of $\mathbb{Z}[\Gamma]$ over $\mathbb{Z}[\Gamma_\sigma]$), so it factors through cmp, and the factor inherits (T1)–(T3). A separate τ^s check on X is therefore not needed; the code obligation's "before the lift" reads "before the lift is used".
- **Any admissible lift** (D3's note 3). Proposition 31.13(2): "It is produced … with no boundary step" describes one construction; *every* additive $\tau_{\mathcal{I}}$ with (T1)–(T3) and the strict identity (fixed independently of r_t, N where (3)(c), (7) need it) is admissible, the bulk component of D4's pair lift among them; (3)(c) records the pair lift as fixed. *Independence of the lift*: two admissible lifts are homotopic through (T1)-maps (the fundamental lemma over $\mathbb{Z}[\Gamma''_g]$), and the operators differ by $d_{\mathrm{I}}\mathsf{H} + \mathsf{H}d_{\mathrm{I}}$ with H the formula (68) applied to the homotopy; so the record's reads coincide, and for a fixed Smith form even $\tilde{\Phi}$ where it is exact.
- **Code map**: a status paragraph — D3 **built on branch `s24-d3impl`, pending merge and audit**: the files, the two validations, the transport at 2 and 3 for every N, the presentation route at $\ell \geq 5$, obligations (a)–(d); **not built**: Theorem 31.16(4) at $\ell \in \{2, 3\}$, and (5). The code-map table row and the "What is used" list follow.
- No label is removed or renamed; the new sub-item (3′) has no label of its own. No number, tag set or marker moves (222 on 86 labels).

v1.23.0-pre.30 — 2026-10-26 — hecke-extensions slot 24, D5 merged: the code map of the tree models; `HeckeInternalError`

This version. **A pre-release, not owner-released, not re-refereed**; non-mathematical. It carries v1.23.0-pre.29 and changes only code-map text; no statement, label, number, tag set or marker moves (222 on 86 labels). Sources: `docs/implementation/slot24/DESCENT5_report.md` (with its §8) and the code; D5 merged at `5eb7e9de`.

- §31.2's code map (and the code-map table row): "Not built" replaced by **built and merged** — the files, the two validations, the tree as the default route for the dense ℓ-local bulk pieces of D0, D1, D2 with rollback `HECKE_DESCENT_TREE=0`, Hecke by transport as a hook whose real arm waits for D3, the tree cusp pair (D5c) not built. The other descent code maps were checked: D0–D2 and D4 already record what is built; the remaining "not built" items (D3's level-N operators, (7)(a)–(b), $\ell \geq 5$ "raise K") are still true.
- Code map (f) of §31.4 and the pre.29 changelog's "known width": `hecke::HeckeInternalError` (`bc128cb7`) — genuine internal errors propagate as failures; only a real overflow refuses as `IntegerCap`.

v1.23.0-pre.29 — 2026-10-25 — hecke-extensions slot 24, D4: the C-D7p referee's fixes M1, m1–m4, N1; the code map for the merged D4

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.28 (refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D7p_referee.md`: one binding clause wrong, four minors, three notes; the referee's independent check `pre28referee/pre28_pairlift_check.py`, 132 checks, 0 failing, rerun at this version) with every finding applied; **not re-refereed**. Only §31.4 and two historical parentheses change; no label, number, tag set or marker moves (222 on 86 labels). The code side of the report (`6daf1e03`) and the D4 code audit's fixes (`46fc0c73`) are merged with D4 at `3991a2f1`; source of the code's refusal list: `docs/implementation/slot24/DESCENT4_report.md` §11, checked against `Integral/Descent/DescentPair.h`, `DescentPairHecke.cpp`.

- **M1** ("refused exactly when" did not match the code): `prop:descentpairhecke`(1) now defines the cap as the code's checked bounds — 64-bit signed exact products, 32-bit signed τ-table coefficients, the coset tables' 32-bit unsigned domain, a failed self-certification of an input table — plus the RSS ceiling (4 GiB, rule 19, before building), and says that every other exception is a failure, not a refusal; (5) lists the eleven refusal conditions (a)–(k), the code's `PairRefusal` codes `NotPrime` … `StrictIdentity`, the strict identity and $\mathcal{I}^{\partial}$-closure on the layout included, and the cap during the operator's assembly as well; "in the last two cases" became "in cases (d)–(k)". The code map's (f) names the codes and the classification (`ClassifyPairException`); `rmk:descentpairscope`(iii) and the two historical parentheses (`rmk:strictliftbuilt`, `rmk:integralscope`(2)) point to (5). Recorded as a known width, not hidden: the exact-integer layer's overflow type is also thrown by some internal guards of the integer geometry on the lift's path, which the code therefore refuses as `IntegerCap` instead of failing.
- **m1**: the proof of (1) carries the sufficiency argument — (i) a twisted map on $H_0(\mathcal{I}^{\partial})$ by (T1), (T2) and $\mathcal{I}^{\partial}$-preservation, (ii) the parabolic $g^{-1}P_c^k g$ forces a multiple of $[g^{-1}c]$, (iii) `def:descentpair`'s augmentation compatibility and (T3) give 1; "Homotopy of pairs" cites it; the route through cmp and `lem:bdrycomponent` (which uses (B-ε)) kept as a second proof.
- **m2**: the pair is served for $q_{\max} \le 5$; $q_{\mathrm{top}} > 6$ (and $q_{\max} > 5$) is refused; higher degrees come from (7), never from extending the lift; the code map's (g) reads $4 \le q_{\max} \le 5$. **m3**: an overflow during the operator's assembly refuses (now true in the code). **m4**: the code map's new (i) lists the negative controls, the homotopy faults $\tau + dK + Kd$ (`StrictLift::with_homotopy`) refused by (S) and the boundary items alone among them.
- **N1**: (1) says the measured sizes belong to the shipped model (on the referee's model, $\approx \times 3$ per two degrees), which is why (7), not a longer lift, serves the degrees above 6. N2, N3: no change needed (N3's separate assembly of $\widetilde{T}_{\mathcal{I}^{\partial}}$ is merged and named in (f)).
- The code map: D4 **merged** at `3991a2f1` (no longer "not merged"), `Integral/Model/StrictLift.{h,cpp}` validation hook, `cmake/slot24/DESCENT4.cmake`, `Registry/Slot24/DESCENT4.cpp` added; the code-map table row likewise.

v1.23.0-pre.28 — 2026-10-25 — hecke-extensions slot 24, D4: T_p on the boundary and the cone for every prime $p \nmid N$ (owner scope addition)

This version. **A pre-release, not owner-released, not refereed** (C-D7 refereed v1.23.0-pre.26, its fixes at pre.27); it carries v1.23.0-pre.27 and amends §31.4 and two historical remarks to the owner's scope addition of 2026-10-25: D4 computes T_p on the boundary and the cone for *general* $p \nmid N$, at level 1 and level N, from the strict pair lift built on demand for each requested p, certified and cached. No label, number or marker moves (222 on 86 labels); no tag set moves. Source: `docs/implementation/slot24/DESCENT4_report.md` §§4–6.

- `prop:descentpairhecke`(1): "Built and certified: $p \in \{2, 3, 5\}$ …; for every other p

... not built" replaced: built on demand for every prime p, degrees ≤ 6, certified by `rmk:strictliftbuilt`(2)'s checks with the boundary items before first use, cached; refused on a failed certificate or at the cap (64-bit signed products); why the certificate suffices on its own (the component map on $H_0(\mathcal{I}^\partial)$ forced by (T1), (T3)); measured sizes and costs to $p = 127$ as evidence, no bound. Proof of (1): the boundary step on $\mathcal{I}^\partial$ spelled out; termination and integrality carry no restriction on p — no step was verified only at $p \in \{2, 3, 5\}$; only the implementation was, and the run-time certificate covers it.

- `prop:descentpairhecke`(5): certified at every ℓ and every N for every prime $p \nmid N$ for which a pair lift has been built and its certificate passed (at run time), the strict identity $r\widetilde{T}_{\mathcal{I}} = \widetilde{T}_{\mathcal{I}^\partial} r$ checked entrywise in every degree used; refused exactly for p not prime, $p \mid N$, a failed certificate or the cap (*incomplete*: corrected at v1.23.0-pre.29, referee M1, to the code's eleven refusal codes); pre.27's "in every other case (level N at $p \notin \{2, 3, 5\}$) ... refused" restricted to the last two.
- `rmk:descentpairscope`(iii) updated; the code map: **built** on branch `s24-d4impl` (`d82c7761`, not merged) — `Integral/Descent/DescentPair.{h,cpp}`, `DescentPairHecke.cpp`, `IntegralDescentPairValidation` (+`Slow`) — and (f) restated for every $p \nmid N$ (no longer "not a D4 obligation"); the code-map table row likewise.
- Historical, kept: `rmk:strictliftbuilt` (slot 10's $p \in \{2, 3, 5\}$) gains a "later builds" sentence; `rmk:integralscope`(2) a v1.23.0-pre.28 parenthesis.

v1.23.0-pre.27 — 2026-10-25 — hecke-extensions slot 24, D4: the C-D7 referee's fixes M1–M3, m1–m5

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.26 (refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D7_referee.md`: three fixes that matter, five minors, two notes) with every finding applied; **not re-refereed**. Only §31.4 changes; no label, number or marker moves (222 on 86 labels); one item, `prop:descentpair`(5)(e), is added inside an existing statement. No code exists yet for D4; the code map's obligations (c) and (f) are amended before it.

- **M1** the check paragraph: "on these cells H^2_c has no torsion at $\ell \ge 5$" was false by the script's own output ($\mathbb{Z}/4 \oplus \mathbb{Z}/60$ at $(1, 8)$, $\mathbb{Z}/2 \oplus \mathbb{Z}/12 \oplus \mathbb{Z}/1260$ at $(1, 12)$); replaced by (5)(e) and the cells' $\ell \ge 5$ parts; the Eisenstein reading withdrawn.
- **M2** `prop:descentpair` new (5)(e): $H^2_c \otimes \mathbb{Z}_{(\ell)} \cong H_0(\Gamma; M) \otimes \mathbb{Z}_{(\ell)}$ for $\ell \ge 5$, with proof (Manin's resolution, the norms of the stabilisers, generation by S, U); for M_N it is V_A, zero for $\ell > n + 1$ (Borel coinvariants, Nakayama). (6)(c): D^0 by the Betti recursion *or* a prime $\ell' \ge 5$ with $\ell' \nmid \operatorname{tors} H_0$ — for M_N every $\ell' \ge 5$, $\ell' > n + 1$, the bulk's rule; the a-priori cap on D^0 at $\ell \ge 5$; `rmk:descentpairscope`(ii) and the code map (c).
- **M3** `prop:descentpairhecke`(5) *weakened* (the referee's first option): T_p on the boundary and the cone is certified, at every ℓ, exactly at the p with a built pair lift ($p \in \{2, 3, 5\}$), and at $\ell \ge 5$, level 1, where the shipped cone operator is built; at level N and $p \notin \{2, 3, 5\}$, $\ell \ge 5$ included, it exists and is determined but is not computed, and the code refuses it; the code map (f) says it is not a D4 obligation; `rmk:descentpairscope`(iii).
- **m1** locators: Brown VII.6 withdrawn; (4)'s periodicity by the tensor resolution $F \otimes F'$ and the shift of the periodic resolution; (5)(d) by the coefficient sequence $0 \to M \to \operatorname{Hom}(\mathbb{Z}[\mathbb{P}^1], M) \to \operatorname{Hom}(\mathrm{St}, M) \to 0$ (note n1), with the Shapiro–cup compatibility for Coind stated once; pre.26's dangling "`lem:bulkperiodic`(4)" removed; III.9 read III.8–9.
- **m2** the cone is unbounded at 3 exactly when $H^q(A; V)_{(3)} \ne 0$ for some $q \ge 2$ (not in general). **m3** (4): order $n!$ for n even; $2^{\lfloor n/2 \rfloor + 1}$ for n odd. **m4** (1)(c)'s signs pinned: $\mathrm{inc}_{c*} = j^*_{\mathrm{St}}$, $\mathrm{pr}_{c*} = -\delta$ under the stated identification, with proof. **m5** (5)(c): exponent ≤ 12 observed, not asserted.
- `cd7_pair_check.py` gains arm R (M2; 308 checks): 2528 checks, 0 failing; the referee's

`cd7referee/cd7referee_check.py` rerun: 2457 checks, 0 failing.

v1.23.0-pre.26 — 2026-10-25 — hecke-extensions slot 24, D4: the boundary pair and the cone over $\mathbb{Z}$ (clause C-D7)

This version. **A pre-release draft, not owner-released, not refereed**; it carries v1.23.0-pre.25 plus one subsection, stated *before* D4's code (contract first, rule 21): clause C-D7 of `docs/implementation/slot24/DESCENT_plan.md` v0.4 (its D4, the §2.2 row "Boundary LES", D-δ6), under the owner's rulings [R-Q2] (no big integer), [R-Q3] (invariants only) and [R-Q4] (T_p, $p \nmid N$; U_p deferred). The tree cusp pair (D5c, C-D8) is not part of it. No marker moves (222 on 86 labels); the new statements are the last of their section, so no statement number moves.

- [new] §31.4, with a paragraph naming what it uses.
- [new] `def:descentpair`: S-pair models — $(\mathcal{I}, \mathcal{I}^{\partial})$ over every ring, $(\mathcal{I}^{(\ell)}, \mathcal{I}^{\partial^{(\ell)}})$ over O_ℓ, the cellular pair with 6 inverted — their cochains, cones and $H_c^\bullet$; comparisons and homotopies of pairs.
- [new] `lem:pairlifting` ([proved here], [standard:] [Wei94, Thm. 2.2.6]): comparisons of pairs exist, boundary first, are unique up to homotopy of pairs, and induce quasi-isomorphisms of cones with the cone homotopy Υ_k.
- [new] `prop:descentpair` ([proved here], [standard:] Brown, Weibel). (1) What a pair model computes over every ring; $H_c^q = \mathrm{Ext}^{q-1}(\mathrm{St}, M)$, the sequence (79) the Ext-sequence of $0 \to \mathrm{St} \to \mathbb{Z}[\mathbb{P}^1] \to \mathbb{Z} \to 0$; H_c^1 torsion-free over $\mathbb{Z}$; $\mathcal{I}^{\partial^{(2)}}$ free, $\mathcal{I}^{\partial^{(\ell)}}$ for $\ell \geq 3$ the circle model; the shipped E8 pair as the $\mathbb{Z}[1/6]$-piece at level 1. (2) The glued sequence over $\mathbb{Z}$: the pair map cmp the only glueing datum, exactness local, maps by the precision rule. (3) Level N by Shapiro, block-diagonal by cusps, widths from $\mathbb{P}^1(\mathbb{Z}/N)$. (4) The boundary in closed form $(X^t,\ X_t,\ (X/2)^t,\ (X/2)_t)$, $|F_{n,w}| = n!\, w^n$, bounded away from 2, periodic. (5) $H_c^0 = 0$, $\dim H_c^2 = \dim(M^\vee)^\Gamma$, $\dim H_c^1 = b_1$ (for $M_N(\chi)$), $48 H_c^q = 0$ ($q \geq 3$), the shapes at $\ell \geq 5,\ 3,\ 2$; periodicity, $q_{\max} \geq 4$ for the pair. (6) The certified exponent on boundary and cone: admissible rank sources (D^0 by the Betti recursion only), a-priori exponents, 14.24's truncation — the plan's D-δ6.
- [new] `prop:descentpairhecke` ([proved here], [standard:] Brown, Weibel). (1) The pair lift for every p: produced, certified by the existing checks, homotopy-unique as pairs; built for $p \in \{2, 3, 5\}$. (2) The level-N operator on $\mathcal{I}^{\partial}$, strict with r, $h = 0$. (3) What it induces — T_p of §31.3 on the bulk, the double-coset operator on $\mathrm{Ext}(\mathrm{St}, V_R)$ and on the boundary — choice-free over $\mathbb{Z}$; the sequence T_p-equivariant over $\mathbb{Z}$, $\mathbb{Z}_{(2)}$, $\mathbb{Z}_{(3)}$. (4) The pieces and the glue. (5) What is certified at which p: boundary and cone at $\ell \geq 5$ for every p, at $\ell \in \{2, 3\}$ only with a built pair lift; never the bulk-only lift on the cone. (6) D2's maps on the pair. (7) Periodicity, Hecke-equivariantly.
- [new] `rmk:descentpairscope` ([proved here]): where the plan's D4 wording is corrected — C-D7 a clause, D-δ6 sharpened (boundary unbounded only at 2, cone at 2 and 3, $q_{\max} \geq 4$), "built for $p \in \{2, 3, 5\}$" versus produced for every p, arm (Q) at level 1 only, coherence for the Hecke operator only, level N by cusps.
- `rmk:integralscope`(4), (5) and `thm:descentcert`(7)(c): pointers to §31.4 (no statement changes).
- A check outside the build, `docs/implementation/slot24/cd7_pair_check.py` (output `cd7_pair_check.out`): 24 cells, 2220 checks, 0 failing; three independent complexes (the tree pair with a computed comparison, the Manin resolution of St, the Bredon pair); planted faults each detected.
- The status appendix (402 → 407 lines), the label index, the code-map row and the referee pointer.

v1.23.0-pre.25 — 2026-10-25 — hecke-extensions slot 24, D3: the C-D4 referee's fixes M1, M2, m1–m5

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.24 (refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D4_referee.md`: two fixes that matter, five minors, one note, three trivial) with every finding applied; **not re-refereed**. Only §31.3 changes, with one sentence each in `prop:descentmaps`(2)(g)(ii) and `rmk:bredonhecke`; no statement, label, number or marker moves (222 on 86 labels). No code exists yet for D3; the code map's obligations are stated before it.

- **M1** `prop:torsionhecke`: new (1)(e) — the layer-j block's characteristic polynomial is choice-free modulo ℓ^{e_j}, $e_j = \min(j, d_j)$, d_j the distance from j to the nearest other layer (a closed path through a layer $a \neq j$ is divisible by $\ell^{|a-j|}$); (2) no longer says "only modulo ℓ": sharpness pairs on $\mathbb{Z}/\ell^{j\mp d} \oplus \mathbb{Z}/\ell^{j}$ for every $1 \leq d < j$ (e.g. $\mathbb{Z}/2 \oplus \mathbb{Z}/8$: modulo 4, not 8); `rmk:descentcertscope`(iii) and `thm:descentcert`(3) follow.
- **M2**, resolution (a): the 3-local piece of `thm:descentcert`(3) is the transport $T_p'^{(3)}$ (or $T'_{\mathcal{T},p}$) for *every* N; `prop:strictliftN`(6)(b) is restated for an integral datum certified by (T1)–(T3) over $\mathbb{Z}$ and is read by no statement; so the subsection's "Not used" list and `rmk:bredonhecke` (one sentence added) hold as written. Code obligation: no branch on $\nu_3(N)$, the shipped operator read only with 6 inverted.
- **m1** `prop:torsionhecke`(3): $2\sigma_{\max} \leq K$ is necessary for the module (a minimal certified example; 33 real cases in the referee's check), and is not `prop:descentmaps`(2)(g)(ii)'s conservative policy (that one reads exact vectors; a sentence there says so); the P_j, the Fitting types and the block polynomials modulo $\ell^{\min(e_j,K')}$ need only a certified form; `rmk:descentcertscope`(ii).
- **m2** `prop:crtcharpolyN`(4): residue primes $< 2^{32}$; a word mulmod for Horner modulo $Q > 2^{32}$; the negative case c^+ mod $Q - P$ mod Q.
- **m3** `prop:strictliftN`(2): τ_X is any g-twisted datum over $\mathbb{Z}$ — the canonical τ once certified, with no boundary datum at all; the procedure is uniform in p, its input is not. **m4** (3)(c): $\tau_{\mathcal{I}}$ is built once, independent of r_t and N. **m5** `thm:descentcert`(7)(a), `rmk:descentcertscope`(i): "not asserted" (finite determinacy expected for a squarefree χ, no bound; repeated roots at old forms) in place of "open mathematics".
- **n1** (2): $\ell' = p$ admissible. **t1** (3): attained iff $r_N > 0$. **t2** proof of (1): `prop:symconv`(2). **t3** the "What is used" list: `prop:levelconehecke`(4) with $\tau_{\mathcal{I}}$.
- The code map's four obligations (M2, m2, m3, m4); the check paragraph names the referee's check and its new arm `M`; the provenance lines, the Appendix B note (no line, no count moves) and its referee pointer.

v1.23.0-pre.24 — 2026-10-25 — hecke-extensions slot 24, D3: Hecke T_p, $p \nmid N$, on the descent (clauses C-D4, C-D5)

This version. **A pre-release draft, not owner-released, not refereed**; it carries v1.23.0-pre.23 plus one subsection, stated *before* D3's code (contract first, rule 21): clauses C-D4 and C-D5 of `docs/implementation/slot24/DESCENT_plan.md` v0.4, under the owner's rulings [R-Q2] (residues and a certified bound), [R-Q3] (invariants only; integer Hecke matrices are D8), [R-Q4] (T_p, $p \nmid N$; every U_p deferred) and [R-GT] (the tree is the 2-adic route). No marker moves (222 on 86 labels); the new statements are the last of their section, so no statement number moves.

- [new] §31.3, with a paragraph naming exactly the Hecke clauses it uses (and those it does not: the Bredon operator at $\ell \in \{2, 3\}$, `cor:stablecert`, any τ-table on the tree).
- [new] `prop:strictliftN` ([proved here], [standard:] [Bro82, Ch. III.8–10], [Wei94, Thm. 2.2.6]). (1) $AgA = A \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} A$ for $p \nmid N$ (explicit γ_1, γ_2): no adjoint question for

T_p. (2) A bulk strict lift $\tau_{\mathcal{I}}$ for every p, *constructed* by (L) of `prop:isomodeltree`(4)(b) without a boundary step; no size bound. (3) The induced-fibre form (68) on $\mathcal{I}_\bullet$: integral, a cochain map over every ring, inducing T_p in every degree with no hypothesis on 6. (4) cmp* intertwines it with the shipped operator on the nose; the glue. (5) Transports to $\mathcal{I}^{(\ell)}$ and to the tree models (the route at 2). (6) The cheap routes at $\ell \geq 5$ and at 3 when $\nu_3(N) = 0$. (7) The strict square with α for $p \nmid MN$: the level sequence (78) is T_p-equivariant over $\mathbb{Z}$ and over $\mathbb{Z}_{(2)}$, $\mathbb{Z}_{(3)}$ (what `prop:descentmaps`(3)(c) had left open). (8) T_p, $T_{p'}$ commute.

- [new] `prop:crtcharpolyN` ([proved here], [standard:] Diamond–Im, Deligne, Diamond–Shurman for the bound, which is the input). $\chi^{(N)}_{p,q}$ is 1 for $q \geq 2$, $x-(p+1)$ or 1 for $q=0$; $\chi^{(N)}_{p,1}$ from residues on the O-piece alone (no 2-, 3-adic input); $|\lambda| \leq 1+p^{n+1}$, coefficient bits $\mathsf{b}_{p,i} = r_N + i(n+1)\mathrm{bitlen}(p)$; recovery in 64-bit words (Garner, sign by digits, residues modulo any word); the output rule [R-Q2]: integer if certified and below 2^{63}, digits and bound if certified and larger, residues and bound if not certified — never a lifted integer.
- [new] `prop:torsionhecke` ([proved here]). For an endomorphism of a finite ℓ-group: the layer polynomials $P_j \in \mathbb{F}_\ell[x]$, the Fitting components (whose types are read off the P_j), ker/coker of $\phi - c$ and the joint module are choice-free; the characteristic polynomial over $\mathbb{Z}/\ell^j$ on Smith summands is not (an example). The exact read from a certified local Smith form needs $2\sigma_{\max} \leq K$. On a free quotient, the invariant factors of $\psi - c$ are certified by the count $r_k = r$, reached for $k > v_\ell(\chi(c))$, never at a root.
- [new] `thm:descentcert` ([proved here]): the record of C-D5 — groups, $\chi^{(N)}_{p,q}$, the torsion Hecke modules, the free-quotient invariants at non-roots, D2's maps with $\chi^{(N)}_{p,1} \mid \chi^{(MN)}_{p,1}$, every degree by periodicity — and what is not asserted (the lattice at the roots and the extension class: the plan's open D-δ7; tier B; U_p, nebentypus, the pair).
- [new] `rmk:descentcertscope` ([proved here]): where the plan's wording is corrected — torsion is not open; the bound is $2\sigma_{\max} \leq K$ on torsion and $v_\ell(\chi(c))$ on the lattice; "char poly on Smith summands" is not defined; χ needs no 2-, 3-adic input; C-D4′ not drafted.
- `rmk:integralscope`(2), (4) and `prop:descentmaps`(3)(c): pointers to §31.3 (no statement changes). Bibliography: [Del74] (locator not verified).
- A check outside the build, `docs/implementation/slot24/cd4_hecke_check.py` (output `cd4_hecke_check.out`): 21 cells, 931 checks, 0 failing; independent oracles (q-expansions, point counts on $X_0(11)$, genus-0 Eisenstein); planted faults each detected.
- The status appendix (397 → 402 lines), the label index, the code-map row and the referee pointer.

v1.23.0-pre.23 — 2026-10-25 — hecke-extensions slot 24, D2: the C-D3g referee's fixes M1, m1–m5, t1, t2

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.22 (refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D3g_referee.md`: no blocker, one medium, five minors, two trivial) with all eight findings applied; **not re-refereed**. Only `prop:descentmaps` and, editorially, `prop:isomodeltree` change; no statement, label, number or marker moves (222 on 86 labels). No code value changes: the code was already doing what the sharpened statement allows (comments in `Integral/Descent/DescentMaps` now cite it).

- **M1** (2)(g)(i) "moreover": the first s rows of $\tilde{L}$, the last $r-s$ columns of $\tilde{L}^{-1}$ and the last $c-s$ rows of $\tilde{R}^{-1}$ agree with L, L^{-1}, R^{-1} modulo ℓ^K (read off the factorisations of the Schur-complement proof); (ii) lattice coordinates: for $y \in A\mathbb{Z}^c_{(\ell)}$ the coordinates in the basis $(\tilde{L}^{-1}De_i)$ are $D^{-1}(Ly)_{\leq s}$, known modulo $\ell^{K-\sigma_i}$, and $(Ly)_{>s} \equiv 0 \pmod{\ell^K}$ — what `LatticeQuotient` reads.
- **m1** (ii): torsion coordinates are exact for every $\sigma_i < K$ (not "iff $\sigma_i \leq K'$"); `BuildPres`'s refusal at $K \leq 2\sigma_{\max}$ is a conservative policy, sufficient and not necessary.

- **m2** (iii): exact division by ℓ^{σ_i} (costs σ_i digits), reduction of torsion rows (free), mixing precisions (the minimum).
- **m3** (iii): the admissible certifying ranks of derived forms ($[F \mid \Delta_Y]$, L_f, the sublattice coordinates C, and $\mathrm{rank}_{\mathbb{Q}}$ of res, cor, j, the connecting map, c_σ, the invariants map), with the rational reason ($\mathrm{cor} \circ \mathrm{res} = [A : B]$).
- **m4** (g): the pivots are $u_i \ell^{\sigma_i}$, u_i units; the proof is verbatim.
- **m5** (2)(b): $\delta[x] = [d\tilde{x}]$ stated; the sign does not affect Hecke equivariance, only entrywise comparisons with an independent δ and compositions through δ.
- **t1** `prop:isomodeltree`(6)(b): "Φ_ℓ of `prop:isomodelell`(4), $\Psi_{\mathcal{T}}$ of (4)(d)". **t2** (4)(b): the five files calling `decompose()`.
- Pointers: the provenance line, (2)(g)'s heading, the Appendix B note (no line, no count moves) and its referee pointer.

v1.23.0-pre.22 — 2026-10-25 — hecke-extensions slot 24, D2: the code audit's precision rule and signs

This version. **A pre-release, not owner-released, not refereed**; it carries v1.23.0-pre.21 (re-refereed ACCEPT) plus two findings of the D2 code audit (`docs/implementation/slot24/D2_code_audit.m` ACCEPT-with-fixes, no value-changing defect) and the C-D6 re-referee's four editorial findings. Only `prop:descentmaps` and, editorially, `prop:isomodeltree` change; no statement, label, number or marker moves (222 on 86 labels).

- **F1** New item (2)(g): for a *certified* Smith form $LAR = \Sigma$ over $\mathbb{Z}/\ell^K$ ($s = \mathrm{rank}_{\mathbb{Q}} A$ pivots, all of valuation $< K$), an exact Smith form over $\mathbb{Z}_{(\ell)}$ exists whose transforms and their inverses agree with L, R, L^{-1}, R^{-1} modulo $\ell^{K-\sigma_{\max}}$; so kernel bases, generators, coordinates ($\sigma_i \le K - \sigma_{\max}$) and induced maps read from the transforms are exact to that precision; precisions decrease additively along a chain of certified forms, and an exhausted precision is a refusal. Proof: the audit's Appendix B (a Schur complement, forced to vanish by the certification), made integral over $\mathbb{Z}_{(\ell)}$. Without the certification the rule fails ($A = (\ell^K)$). Until now (d) and `prop:certifiedeell`(6) certified only group types; D2's maps rest on (2)(g).
- **F8** (2)(b): the maps $H^q(B) \to H^{q+1}(A, B)$, $H^q(A, B) \to H^q(A)$ are induced by $c' \mapsto (0, c')$, $(c, c') \mapsto c$; the second is $-\delta$ through π.
- **The C-D6 re-referee of pre.21** (`C-D6_referee.md`, "Re-referee of pre.21 (M1)": **ACCEPT**, four editorial findings), in `prop:isomodeltree`: **N1** (4)(b) and the code map cite `decompose()` of `PeriodicColumn.h` (signed lexicographic order, j relative to the fixed generator) as the representative convention; **N2** (4)(b): (L), (L′) finite per degree, the comparisons not periodic (coefficients grow linearly in q), built to the degree in use; **N3** (L′): p' any chain map, its surjectivity unused; **N4** (6)(b) names the effective comparison $\vartheta = \Phi_\ell \circ \Psi_{\mathcal{T}}$.
- Pointers: the provenance line, the code map (D2 built, `DescentMaps`), the Appendix B note (no line, no count moves) and its referee pointer.

v1.23.0-pre.21 — 2026-10-25 — hecke-extensions slot 24, D5: the C-D6 referee's fixes M1, m1–m5

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.20 (refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D6_referee.md`: no blocker, one major, five minors) with all six findings applied; **not re-refereed**. Only `prop:isomodeltree` changes; no statement, label, number or marker moves (222 on 86 labels).

- **M1** (4) is rewritten as (a)–(d) and made **effective**. (a) The tree is the Γ-subcomplex of X on the orbits of v_6, v_4, $\{v_4, v_6\}$ ($\mathsf{i}_{\mathcal{T}}$); an explicit retraction $\mathsf{r}_{\mathcal{T}}$ and homotopy $\mathsf{G}_{\mathcal{T}}$ on the 19 representatives, $\mathsf{r}_{\mathcal{T}}\mathsf{i}_{\mathcal{T}} = \mathrm{id}$, $\mathrm{id} - \mathsf{i}_{\mathcal{T}}\mathsf{r}_{\mathcal{T}} = \partial\mathsf{G}_{\mathcal{T}} + \mathsf{G}_{\mathcal{T}}\partial$ (`prop:bredon`(2) in

closed form). (b) The columns' $\mathbb{Z}$-linear contractions with fixed coset representatives, the solvers h_p of $\ker \mathrm{cmp}$ and $\ker \varepsilon_{\mathcal{T}}$ by descending column index, the lifting formula (L) (`lem:strictlifting` made effective) and its homotopy form (L′). (c) $\Phi^{\mathbb{Z}}_{\mathcal{T}}$ is (L) of $\mathsf{r}_{\mathcal{T}} \circ \mathrm{cmp}$, $\Psi^{\mathbb{Z}}_{\mathcal{T}}$ is (L) of $\mathsf{i}_{\mathcal{T}} \circ \varepsilon_{\mathcal{T}}$; explicit homotopies $\mathsf{H}_{\mathcal{T}}$, H_X, H_ℓ. (d) $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$ with explicit homotopies K, K'. (5)'s (T3) and (6) now rest on constructed maps; the code map names the route on the code's $\mathcal{I}$.

- **m1** `prop:isomodelell`(4), (5) are used over every O_ℓ-algebra; (4)(d) and (6)(b) record that their proofs need only $|Q_\sigma| \in R^\times$.
- **m2** (3): "strict deformation retract" replaced by "a retract with $\Psi^*\Phi^* = \mathrm{id}$ strictly and $\Phi^*\Psi^* \simeq \mathrm{id}$", the homotopy H^*_ℓ of (4)(c); no side conditions asserted.
- **m3** The proof of (1)(d)(iv): $a_{yx} = e_y a_{yx} e_x$ for the column differentials and the edge's $-\varepsilon_t$ by centrality of e_x in $R[\Gamma_x]$.
- **m4** (5): its last sentence is about groups and fibre-map-induced maps; other maps between the glued groups enter through $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$.
- **m5** (6) is stated for every cochain endomorphism T, instantiated at level N by T_p, $p \nmid N$ ([R-Q4]); pre.20's changelog line "(6) … stated now for $p \nmid N$" is to be read so. The proof of (1)(b): the sign reversal is relative to `def:isomodelell`'s $\bar{g} - 1$, not to `rmk:treeconeres`.
- A companion check, `docs/implementation/slot24/cd6_iso_check.py`, builds an $\mathcal{I}_\bullet$ and checks (4) entirely: 12 870 checks, 0 mismatches; the pre.20 check and the referee's scripts rerun unchanged, 0 mismatches.
- Pointers: the provenance line, the §22.1 row, the Appendix B note (no line, no count moves) and its referee pointer.

v1.23.0-pre.20 — 2026-10-25 — hecke-extensions slot 24, D5: the tree models, bulk only (clause C-D6)

This version. **A pre-release draft, not owner-released, not refereed**; it carries v1.23.0-pre.19 plus one new subsection with two statements, stated *before* D5's code (contract first, rule 21): clause C-D6 of `docs/implementation/slot24/DESCENT_plan.md` v0.4, whose milestone D5 the owner brought forward at gate G-T on 2026-10-25, ahead of D3 and D4 (the evidence: `DESCENT1_report.md`, the dense $\mathcal{I}^{(2)}$ memory wall at $D \approx 3\,080$, the tree at 2 at $D = 11\,110$). No marker moves (222 on 86 labels); the new statements are the last of their section, so no statement number moves.

- [new] §31.2 and `def:isomodeltree`: for x the two vertices and the edge of the Bass–Serre tree, ℓ-local columns $R[\Gamma]e_x$ (periodic where the ℓ-Sylow $P_x \neq 1$, degree 0 otherwise) over O_ℓ-algebras, and the model as the cone of a lift φ' of the tree boundary; canonical lifts at $\ell = 2$ ($(1, -e_U)$, $(1 + S, -e_U)$, $e_U = \frac{1}{3}(1 + U^2 + U^4)$), $\ell = 3$ and $\ell \geq 5$; the free model is `rmk:treeconeres`'s.
- [new] `prop:isomodeltree` ([proved here], [standard:] [Wei94, Thm. 2.2.6], [Bro82, Ch. III.6.2]). (1) Projective resolutions computing $H^\bullet(\Gamma; M)$ in every degree for every M ($\mathrm{Ind}^{\Gamma}_{\Gamma_0(N)}$ included), cochains $\bigoplus M^{Q_x}$ with ranks by traces, one integral form over O_ℓ, and `prop:certifiedeell`(2)–(6) on it with its own rank sources. (2) The cochains in closed form at $\ell = 2$, 3, ≥ 5; periodicity from degree 1, resp. 2; `prop:ellthreeclosed`(2) read off the cochains. (3) Closed-form comparisons $\Phi^{\mathcal{T}}_\ell$, $\Psi^{\mathcal{T}}_\ell$ with the free tree (column maps $|Q_x|^{\pm\lfloor t/2 \rfloor}$ and explicit corrections), $\Phi^{\mathcal{T}}_\ell \Psi^{\mathcal{T}}_\ell = \mathrm{id}$ strictly: the ℓ-local cochains are a strict deformation retract of the free tree's. (4) Comparisons $\Phi_{\mathcal{T}}$, $\Psi_{\mathcal{T}}$ with $\mathcal{I}$ over the augmentation to $\mathbb{Z}$ (`lem:strictlifting`), not over $C_\bullet(X)$, mutually inverse on cohomology. (5) (T1)–(T3) of `prop:descentmaps`(5) hold, so its fibre maps, level cone, long exact sequence and certified rule hold on the tree at $\ell \in \{2, 3\}$. (6) Hecke by transport, $T'_{\mathcal{T}} = \Psi^*_{\mathcal{T}} \circ T \circ \Phi^*_{\mathcal{T}}$: it induces the operator, equals the transport on $\mathcal{I}^{(\ell)}$ on

cohomology, respects relations up to homotopy and fibre maps strictly; stated now for $p \nmid N$ (the plan's [R-Q4]), implemented with D3. Bulk only: nothing on the boundary, the pair or the cusp (the plan's C-D8). A code-map paragraph (not built) and a formalization note (no marker).
- A check outside the build, `docs/implementation/slot24/cd6_tree_check.py` (own group ring, own induced module, the models built as chain data): the chain identities exactly in $\mathbb{Q}[\Gamma]$, and on 145 cells and 18 level pairs at $\ell = 2$, 3 (and 5, 7 on the Bredon tree) the closed forms, the groups against the oracle tables, the comparisons and the level cone: 49 303 checks, 0 mismatches.
- Pointers: the §22.1 row; Appendix B: two lines, 395 → 397, `def` 66 → 67, `prop` 94 → 95, [proved here] 230 → 231, [standard] 99 → 100, tagged 265 → 266; its referee pointer; Appendix D: three labels.

v1.23.0-pre.19 — 2026-10-25 — hecke-extensions slot 24, D2: the C-D3 referee's fixes m1–m5

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.18 (refereed ACCEPT-with-fixes, `docs/implementation/slot24/C-D3_referee.md`: no blocker, no major, five minors) with the five minors applied; **not re-refereed**. No statement, label, number or marker moves (222 on 86 labels).

- **m1** `prop:descentmaps`(3): (a) and (b) say that $[AgA]$, $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$, is *this document's* U_p, the adjoint U_p^* on p-old spaces (`prop:levelconehecke`(3)(d)); (b)'s failure of the square with res is that convention's, for $p \mid N$ — the classical $U_p = [A \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} A]$ has a strict square at $\chi = 1$ — and for $p \nmid N$, $p \mid M$ both squares fail. [new] item **(3)(b′)**: if $p \mid N$, $\mathrm{cor} \circ [BgB] = [AgA] \circ \mathrm{cor}$ on H^q for every q, every M, over every ring and over $\mathbb{Z}$, with its proof (the double coset formula with $A' = A \cap g^{-1}Ag$: $B \cap A' = B'$ and $A = A'B$ by the transversal $T^j \in B$); on the Hida tower $\Gamma_0(p^{r+1}) \subset \Gamma_0(p^r)$ corestriction is U_p-equivariant over $\mathbb{Z}$ in every degree. The proof's check paragraph records the referee's 32 cells plus 10 more run at this version (the square with cor: 15 cells with $p \mid N$ commute, 5 with $p \nmid N$ fail).
- **m2** (3)(a) cites the criterion `prop:levelconehecke`(1) alone; its instance (2) for U_p needs equal Γ_0-levels.
- **m3** (3)(c) assumes $p \nmid M$.
- **m4** (2)(d)(i): the rank is read on the reduction mod ℓ' of the model's integral form ($\mathcal{I}$ over $\mathbb{Z}$, $\mathcal{I}^{(\ell)}$ over O_ℓ with $\ell' \nmid \mathfrak{q}_\ell$), as in `prop:certifiedeell`(4)(c).
- **m5** (4)(c): $H^q(\Gamma_0(p^r); \mathrm{Sym}^n \mathbb{Z}^2) \cong H^{q-1}((\mathbb{Z}/p^r)^\times; H^1(\Gamma_1(p^r)))$ for $q \geq 2$, all of it 2- and 3-primary; no ℓ-torsion for $\ell \geq 5$ even when $\ell \mid |\mathsf{G}|$.
- The statement's provenance line, the §B note (no line, no count moves) and the referee pointer of the status appendix.

v1.23.0-pre.18 — 2026-10-25 — hecke-extensions slot 24, D2: maps and the level cone over $\mathbb{Z}$ (clause C-D3)

This version. **A pre-release, not owner-released, not refereed**; it carries v1.23.0-pre.17 (refereed ACCEPT, `docs/implementation/slot24/C-D1_three_referee.md`, four minor remarks, folded in below) plus one statement, stated *before* D2's code (contract first, rule 21): clause C-D3 of `docs/implementation/slot24/DESCENT_plan.md` v0.3, and the part of the owner's slot-22 LES-0 request (the level-change sequence for $\Gamma_0(MN) \subset \Gamma_0(N)$, inflation–restriction in the normal case, the Hida p-tower first) that §29.8 had left at $6 \in R^\times$. No marker

moves (222 on 86 labels); the new statement is the last of its section, so no statement number moves.

- [new] `prop:descentmaps` ([proved here], [standard:] [Bro82, Ch. III.6.2, III.8–10, VII.6], [Wei94, Thm. 2.2.6]). *Models*: an S-model is any finitely generated projective resolution of S over $S[\Gamma]$; comparisons are chain maps over id_S. (1) A $\mathbb{Z}[\Gamma]$-map of fibres commutes strictly with every comparison (cmp^*, Φ_ℓ^*, Ψ_ℓ^*, a tree model's), acts block by block on permutation-type models, and glues to the map over $\mathbb{Z}$; restriction, trace and conjugation are induced in every degree over every ring (no 6 inverted); the split case in all degrees. (2) The level cone on every model is exact and computes $H^{\bullet-1}(A; Q_{A/B})$; the long exact sequence (78) over $\mathbb{Z}$ in every degree (not six-term at 2, 3: $H^3(\Gamma_0(4), \Gamma_0(8)) = \mathbb{Z}/2$ at $n = 0$), periodic from degree 3, glued by `prop:cohglue`; the certified-K rule on the cone, $\mathrm{rank}_{\mathbb{Q}}$ from the Betti recursion $\beta_j = b_j(M_B) - b_j(M_A)$ or from the cone's own reduction at a qualifying prime; bad primes of the cone (degree 2 only at $\ell \geq 5$); the tree and sparse-cokernel routes at $\ell \geq 5$ and, through $S[A]$-models, at $\ell = 3$ when $\nu_3(N) = 0$. (3) res and cor commute with $[\,\cdot \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \cdot\,]$ in every degree when $p \nmid M$ (double coset formula): T_p for $p \nmid MN$, U_p for $p \mid N$; level raising fails (`prop:levelconehecke`(3)(c), lifted to $\mathbb{Z}$); the Hecke action on the relative term at 2, 3 is deferred to C-D4, with what C-D4 must supply. (4) Normal pairs: the G-action, the norm identities and the five-term sequence over $\mathbb{Z}$; for torsion-free B two rows over every ring, and for $n > 0$ $H^q(A) \cong H^{q-1}(\mathsf{G}; H^1(B))$ ($\Gamma_1(N) \trianglelefteq \Gamma_0(N)$, $N \geq 4$: the bottom of the Hida tower; $|\mathsf{G}|$ kills $H^{\geq 2}(A)$). (5) What an ℓ-local tree model (D5, C-D6) must supply: (T1)–(T3). A closing list of what is deferred and why. A check outside the build, `docs/implementation/slot24/cd3_descentmaps_check.py` (own cosets, modules, free resolution and exact Smith forms): 74 cone cells, 12 two-row cells, 888 oracle rows, 0 mismatches.
- The C-D1 three-shortcut referee's minor remarks: **m1** `prop:descentNthree`(1), why "no element of order 3" is "no U-fixed point"; **m2** `prop:modeltorsion`'s last line points to the full characterisation; **m3** the header of `cd1b3_three_shortcut_check.out` records one command per figure and the proof says $84 = 80 + 4$ in two runs, and "22 of 32"; **m4** the proof records the shared construction and the referee's independent closure.
- Pointers: §29.8's scope sentence, `rmk:integralscope`(4). §31.1's code-map paragraph (D2 not built; D1 merged at `1cc9a6d3`), Formalization note, the §22.1 row.
- Appendix B: one line, 394 → 395, `prop` 93 → 94, [proved here] 229 → 230, [standard] 98 → 99, tagged 264 → 265; Appendix D: two labels.

v1.23.0-pre.17 — 2026-10-25 — hecke-extensions slot 24, D1: the 3-adic shortcut at $\nu_3(N) = 0$

This version. **A pre-release, not owner-released, not refereed**; it carries v1.23.0-pre.16 plus one statement, written *after* D1's code (branch `s24-d1impl`, `726c3a67`, `87e8cf32`) to answer finding M1 of `docs/implementation/slot24/D1_code_audit.md` (commit `ffb59130`): the code's $\ell = 3$ route at $\nu_3(N) = 0$ was sound and agreed with $\mathcal{I}^{(3)}$ on every test cell, but `prop:descentN`(7) stated the route for $\ell \geq 5$ only. No marker moves (222 on 86 labels); the new statement is the last of its section, so no statement number moves.

- [new] `prop:descentNthree` ([proved here], [standard:] [Bro82, Ch. III.6.2], [Wei94, Thm. 2.2.6]). (1) $\nu_3(N) = 0$ iff $\Gamma_0(N)$ has no element of order 3 iff $9 \mid N$ or a prime $p \equiv 2 \pmod 3$ divides N; then every finite subgroup of $\Gamma_0(N)$ has order $1, 2, 4$, and M_N is induced over $\langle U^2 \rangle$. (2) $\mathrm{T}(M_N \otimes \mathbb{Z}_{(3)})$ is free, base-changes to every $\mathbb{Z}_{(3)}$-algebra, and computes $H^\bullet(\Gamma_0(N); V) \otimes \mathbb{Z}_{(3)}$, as does $C_{\mathrm{eq}}^\bullet$ ($\mathrm{cmp}^* \otimes \mathbb{Z}_{(3)}$ a quasi-isomorphism): **no 3-torsion in $H^{\geq 2}$**; checked against `prop:modeltorsion` and `prop:ellthreeclosed`(2),(3). (3) `prop:descentN`(7)(b) verbatim at $\ell = 3$: the certified-K rule $s = r = \mathrm{rank}\,\mathrm{T}^0 - b_0$ with closed forms, the stopping rule of `prop:certifiedeell`(6), the generators (orbit sums;

idempotent columns at the S-fixed points, order 4 prime to 3) and the F4 certificate; there are no U-fixed points. (4) n odd: nothing 3-primary. (5) The hypothesis is needed (oracle cells $(1,0)$, $(7,0)$, $(3,4)$ have 3-torsion in $H^{\geq 2}$); ν_2 plays no role; $9 \mid N$ is covered. A check outside the build, `docs/implementation/slot24/cd1b3_three_shortcut_check.py`: 84 cells, 0 mismatches.

- `prop:descentN`(7): a pointer to the new statement. §31.1's code-map paragraph and Formalization note; a §22.1 row for §31.1.
- Appendix B: one line, 393 → 394, `prop` 92 → 93, [proved here] 228 → 229, [standard] 97 → 98, tagged 263 → 264; Appendix D: one label.

v1.23.0-pre.16 — 2026-10-25 — hecke-extensions slot 24, D1: the Dickson-proof referee's fixes

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.15 plus the fixes of the Dickson-proof referee (`docs/implementation/slot24/C-D1_dickson_referee.md`, commit `509d67b5`: ACCEPT-with-fixes, 1 Minor, 4 nits; the mathematics was found correct at every prime); **not re-refereed**. No marker moves (222 on 86 labels); one statement is added, so the numbers after Claim 16.5 in its section shift by one (labels do not).

- **D-1** (Minor). The proof of Claim 16.5(2) is a standalone [new] `lem:dicksonSL2` ([proved here]), stated for every prime ℓ and every n, cited by `cl:dimH1`(2), `prop:oddnltwo`(2) (at $\ell = 2$) and `prop:dicksonH0`. "First non-zero at $n = \ell + 1$" is restricted to $\ell \geq 3$ ($\delta_2(2) = 1$, $Q = d_2$).
- **D-2**. `prop:oddnltwo`(2): the [standard:] tag on [Wil83] removed; a plain pointer to the lemma.
- **D-3**. `prop:dicksonH0`(2) rests on the SL_2 half of (1) only; the GL_2 half, $\mathbb{F}_\ell[L^{\ell-1}, Q]$, proved in the lemma ($g_A L = \det A \cdot L$, $g_A Q = Q$).
- **D-4**. Step 3: θ has a minimal polynomial over $\mathcal{K}^G$ because the upper bound made $\mathcal{K}$ algebraic over k; the five coefficients compared are named.
- **D-5**. `prop:descentN`(4): $b_1 = 1$ *exactly* at $N \in \{2, 3, 5, 7, 13\}$.
- Appendix B: one line, 392 → 393, `lem` 60 → 61, [proved here] 227 → 228, tagged 262 → 263; Appendix D: one label.

v1.23.0-pre.15 — 2026-10-25 — hecke-extensions slot 24, D1: the C-D1 / C-D2 referee's fixes

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.14 plus the fixes of the C-D1 / C-D2 referee (`docs/implementation/slot24/C-D1_referee.md`, commit `e28dcb09`: ACCEPT-with-fixes, 0 Blocker, 1 Major, 4 minor, 2 nits), applied to §31.1 and to Claim 16.5(2); **not re-refereed**. Every new claim is proved in place. No label, statement number or marker moves (222 markers on 86 labels); one tag set moves (Claim 16.5, below).

- **F1** (Major). (a) `prop:badprimesN`(2)'s proof: for $\ell > n > 0$, $\ell \nmid N$, Claim A and the lower unipotent give $s_\ell = 0$, so the good-prime rule $\ell > n + 1$ of `prop:descentN`(6), D1's production path, rests on Claims A and B alone. (b) `cl:dimH1`(2), $\mathbb{F}_\ell[x, y]^{\mathrm{SL}_2(\mathbb{F}_\ell)} = \mathbb{F}_\ell[L, Q]$, is [proved here] in place, for every prime ℓ, along the referee's sketch with two simplifications: the Moore determinant gives $\prod_v (T - v) = T^{\ell^2} - QT^\ell + L^{\ell-1}T$ (integrality); the degree $[\mathbb{F}_\ell(x, y) : \mathbb{F}_\ell(L, Q)] \leq |\mathrm{SL}_2(\mathbb{F}_\ell)|$ by a tower $k \subseteq k(x) \subseteq k(x, y)$, against $[\mathbb{F}_\ell(x, y) : \mathbb{F}_\ell(x, y)^G] \geq |G|$ from the trivial stabiliser of $x + y^2$, in place of freeness and Artin's theorem; a factorial ring is integrally closed. So the count $\delta_\ell(n)$ at $\ell \nmid N$, $5 \leq \ell \leq n$ (and at level 1, `prop:badprimes`(1)) no longer rests on the unverified [Wil83], and "not by omission" holds as stated. Pointers updated: `prop:oddnltwo`(2) (the $\ell = 2$ case, $L = d_3$, $Q = d_2$), the proof of `prop:badprimes`, `prop:dicksonH0`(1) (its SL_2 half; the GL_2 half keeps its citation).

- **F2**. `prop:descentN`(4): $6b_1 = 6 + m_N - 3\nu_2 - 4\nu_3 \geq 6$ for every $N > 1$ at $n = 0$, $\chi = 1$, from `prop:levelorbit`(3) and the local counts of the fixed points of S, ST (CRT, Hensel), replacing "by the genus formula, which is not proved here"; equality at $N = 2, 3, 5, 7, 13$ (the referee's "only at $N = 2$" is corrected; the minimum 6 stands). Nothing else uses it.
- **F3**. The code map: `SparseCokernel`'s generators are the averaging-idempotent columns and transported blocks, admissible for (7)(b); the blocker is the prime-field gate and the RREF elimination.
- **F4**. `prop:descentN`(7)(b): the certificate that a family generates $(V_{\mathbb{Z}/\ell^K})^{F_j}$ (fixed by a generator, spanning mod ℓ; Nakayama), and that the comparison value r is the closed form, never the number of rows.
- **F5**. `prop:descentN`(5): $\mathrm{Hom}(E, M)$ induces $e \cup -$ for every M (dimension shifting), and U_p's defining label (`prop:levelcorrespondence`, `prop:nebcorr` at $A = B = \Gamma_0(N)$).
- **F6, F7** (nits). (7)(a): the source by any generating family, the lifts taken in $e_\sigma M_{\mathbb{Z}_{(\ell)}}$. The subsection's preamble says why it sits under §31.
- Appendix B: `cl:dimH1`'s line gains [proved here] and drops [Wil83]; `prop:descentN`'s gains [Bro82, Ch. V, §3]; 392 lines, [proved here] 226 → 227.

v1.23.0-pre.14 — 2026-10-25 — hecke-extensions slot 24, D1: the descent at level N and its bad primes

This version. **A pre-release draft, not owner-released, not refereed**; it carries v1.23.0-pre.13 plus clauses C-D1 and C-D2 of `docs/implementation/slot24/DESCENT_plan.md` v0.3 (milestone D1; audit findings M8, m3, m4, m7 of `DESCENT_plan_audit.md`). Contract first (rule 21). **Additive**: one subsection after the last statement of §31, so that no statement number moves; no existing statement, tag set, marker or value moves (222 markers on 86 labels; the new statements carry none).

- [new] `prop:descentN` ([proved here]), for every $\mathbb{Z}[\Gamma]$-module M finitely generated and free over $\mathbb{Z}$, in particular $M_N = \mathrm{Ind}_{\Gamma_0(N)}^{\Gamma} V$ and $M_N(\chi)$ over $\mathsf{O}_\chi = \mathbb{Z}[\zeta_{m_\chi}]$: (1) $C_{\mathcal{I}}^\bullet(M)$, $C_{\mathrm{eq}}^\bullet(M)$, $\mathrm{T}(M)$ and their base changes; (2) Proposition 23.9 for M; (3) maps and Hecke operators determined locally, by Proposition 23.8's morphism clause alone; (4) the coherence term $\mathrm{Hom}^{-1} \cong \widehat{\mathbb{Q}}_6^{b_0 b_1}$, so Proposition 23.10(2) extends iff $b_0 b_1 = 0$ — for $M_N(\chi)$ iff $n > 0$ or $\chi \neq 1$; it fails at $n = 0$, $\chi = 1$, $N = 2$ (finding M8, the restriction the plan offered, made exact); (5) Remark 23.11 for M, and periodicity above the vcd by $\mathrm{res}_A^\Gamma e \cup -$, equivariant for every $[AgA]$, T_p ($p \nmid N$) included (finding m4); (6) the good prime: for M_N every $\ell' \geq 5$, $\ell' > n + 1$ ($\ell' \nmid N$ not needed); for $\chi \neq 1$, $\ell' \nmid m_\chi N$ needed (finding m3); (7) the certified exponent at $\ell \geq 5$ on the direct Smith form of $C_{\mathrm{eq}}^\bullet(M_{\mathbb{Z}/\ell^K})$ (the D0 report's recommendation) and on the level-N sparse cokernel over $\mathbb{Z}/\ell^K$ (finding m7), with $\mathrm{rank}_{\mathbb{Q}}\, d^0 = \mathrm{rank}\, C^0 - b_0$, and Proposition 31.6(4) at level N; (8) $M_N(\chi) \otimes \mathbb{Z}/\ell^K$ as a sum of §29's character models over $W_K(\mathbb{F}_q)$.
- [new] `prop:badprimesN` ([proved here]): (1) for every such M and $\ell \geq 5$ the torsion lies in H^1, with $\dim H^0(\Gamma; M/\ell) - b_0$ cyclic summands; (2) at level N the count is the level-1 Dickson count $\delta_\ell(n)$ for $\ell \nmid N$ and the Borel count $\beta_\ell(n)$ for $\ell \mid N$, which reaches $\ell = n+1$; the closed-form list $\mathcal{B}(n, N)$, exact at $\ell \geq 5$, with every $\ell > n + 1$ torsion-free by proof; $N = 1$ is Proposition 23.16; (3) nebentypus at $\ell \nmid 6m_\chi$: torsion only at $\ell \mid N$ with χ of ℓ-power conductor, count $\beta_\ell(n, t_\lambda)$; the primes $\ell \mid m_\chi$ not treated.
- §31.1's code map (not built: the direct Smith route, the sparse cokernel over $\mathbb{Z}/\ell^K$, the level-N assembly, E on $C_{\mathcal{I}}^\bullet(M_N)$, $M_N(\chi)$ over O_χ) and formalization note; an exact check outside the build (1376 cells, 0 mismatches), not a tag.
- Remark 23.22(4): one parenthesis (the level half of the plan's C-D7). §22.1's row for §31: one sentence.
- Appendix B: two lines, counts 390 → 392; Appendix D: three labels.

v1.23.0-pre.13 — 2026-10-25 — hecke-extensions slot 24, D0: the C-D0 referee's fixes

This version. **A pre-release, not owner-released**; it carries v1.23.0-pre.12 plus the required fixes of the C-D0 referee (`docs/implementation/slot24/C-D0_referee.md`, commit `4089cdf8`: ACCEPT-with-fixes, 0 Blocker, 2 Major, 6 minor), applied to Proposition 31.6; **not re-refereed**. Every new claim is proved in place. No label, statement number, tag set, marker or value moves (222 markers on 86 labels).

- **F1** (Major). (4) no longer leaves the qualifying primes an input for $M \neq V$: (4)(a) $24 \cdot H^j(\Gamma; M) = 0$ for $j \geq 2$ and every $\mathbb{Z}[\Gamma]$-module M (Mayer–Vietoris from (50) with Shapiro, the argument of Claim 16.2(4)), so every $\ell' \geq 5$ qualifies for d^q, $q \geq 1$ ($q = 1$, H^2, checked); (4)(b) at $q = 0$, ℓ' qualifies iff $\dim H^0(\Gamma; M \otimes \mathbb{F}_{\ell'}) = b_0$, measured on the same complex, and $\operatorname{rank}_{\mathbb{Q}} d^0 = \operatorname{rank} C^0_\ell - b_0$; $b_0 = 0$ ($n > 0$), 1 ($n = 0$) for V and $\operatorname{Ind}_{\Gamma_0(N)}^{\Gamma} V$, proved.
- **F2** (Major). (4)(c): the two admissible sources of $\operatorname{rank}_{\mathbb{Q}} d^q$ (this O_ℓ-complex mod ℓ'; certified Betti numbers through $\operatorname{rank} C^q_\ell - b_q - \operatorname{rank}_{\mathbb{Q}} d^{q-1}$); census, shipped-$C_7$ and $\mathcal{I}^{(\ell')}$ ranks stated inadmissible, with the reason. Code map: the `ell()` guard as a D0 requirement (audit note F-5).
- **F3**. (6): the Smith valuations a list with multiplicity, their number s.
- **F4**. (6): $a_t \leq v_\ell(24)$ for $q \geq 1$ (3 at 2, 1 at 3, 0 at $\ell \geq 5$), the stopping rule (raise K to the ring's cap, then refuse); the caps bind only at d^0.
- **F5**. Code map: the denominator check (every denominator divides a power of q_ℓ) as a D0 obligation.
- **F6**. Code map (ii): $W \leq 61$ reachable now through `ModCtx<uint64_t>` and `Ring2k`'s masking; $W \in [62, 64]$ needs the new reduction; §22.1's row, which said "the reduction to $\mathbb{Z}/2^{64}$", reworded to agree.
- **F7**. Code map (i′): the `DenseBuf` → `Ring2k` / `Ring3k` hand-off; $q_{\max} \geq 5$ for the (P) arm at degree 6.
- **F8**. `docs/implementation/slot24/DESCENT_plan.md` v0.3: row C-D0 ($|Q_\sigma| \in \{2, 4\}$ at $\ell = 3$, $\mathsf{q}_3 = 4$; "(1)–(3), (4) restated as 31.6(4)") and the good-prime wording of §2.1 and D0.
- Appendix B: the line of 31.6 names the proof's new [standard:] input [Bro82, Ch. III.10]; no count moves.

v1.23.0-pre.12 — 2026-10-25 — hecke-extensions slot 24, D0: the certified-exponent rule on $\mathcal{I}^{(\ell)}$

This version. **A pre-release draft, not owner-released, not refereed**; it carries v1.23.0-pre.11 plus clause C-D0 of `docs/implementation/slot24/DESCENT_plan.md` v0.2 (milestone D0; audit finding M3 of `DESCENT_plan_audit.md`: D0 certifies K on $\mathcal{I}^{(2)}$ and $\mathcal{I}^{(3)}$, and Proposition 23.17 is stated for complexes of free $\mathbb{Z}$-modules only). Contract first (rule 21): the plan requires a referee **before** D0's validation lands. **Additive**: one statement at the end of §31's statements, so that no statement number moves; no existing statement, tag set, marker or value moves (222 markers on 86 labels; the new statement carries none).

- [new] `prop:certifiedeell` ([proved here]): (1) Definition 31.1 and Proposition 31.2(1)–(3) over any ring in which $\mathsf{q}_\ell = \operatorname{lcm} |Q_\sigma|$ is a unit ($\mathsf{q}_2 = 3$, $\mathsf{q}_3 = 4$, $\mathsf{q}_{\ell \geq 5} = 12$); one model over $\mathsf{O}_\ell = \mathbb{Z}[1/\mathsf{q}_\ell]$ whose base changes to $\mathbb{Z}/\ell^K$ and $\mathbb{F}_{\ell'}$ ($\ell' \nmid \mathsf{q}_\ell$) are the complexes built; (2) its cochains free over O_ℓ, compatible with base change, computing $H^q(\Gamma; M) \otimes \mathsf{O}_\ell$; (3) Proposition 23.17(1)–(3) and Remark 23.18 over $\mathbb{Z}_{(\ell)}$; (4) $\operatorname{rank}_{\mathbb{Q}}$ at a prime $\ell' \nmid \mathsf{q}_\ell$ without ℓ'-torsion in H^{q+1}; (5) the code's full layout $\delta^q = P_{q+1} \delta^q P_q$ is $\operatorname{diag}(d^q, 0)$ over O_ℓ; (6) the Smith valuations over $\mathbb{Z}/\ell^K$ are the exponents $a_t < K$, their number r_K.
- Its code map (built: d once over $\mathbb{Q}[\Gamma]$, denominators $|Q_\sigma|$ only, reduced entrywise; not

built: the `Ring2k` / `Ring3k` read-out on $\mathcal{I}^{(\ell)}$, a reduction to $\mathbb{Z}/2^W$ for $W \geq 62$; `Ring3k`'s $K \leq 19$; the plan's "$|Q| \in \{1,2\}$ at $\ell = 3$" corrected to $\{2,4\}$).
- §31: "What this chapter fixes" item (v); the notation sentence names the new symbols; the scope sentence ("at best over $\mathbb{Z}[1/\mathfrak{q}_\ell]$"); the formalization note (no `\LEAN` marker). §22.1's row for §31: one sentence.
- Appendix B: one line, counts 389 → 390, with the note that the two pre.3 statements still lack lines; Appendix D: one label.

v1.23.0-pre.11 — 2026-10-21 — hecke-extensions slot 24: the G2-H and G6 code-map sentences

This version. **A pre-release, not owner-released, not refereed**; it carries v1.23.0-pre.10 plus **code-map text only**:

- §20, its code map: one sentence on G2-H (`docs/implementation/slot24/G2H_report.md` §9; audit `G2H_audit.md`, MERGE after fixes F1, F2; its note F6 asks for this sentence) — on the stream route under `RunModuleT`'s lean retention, Obligation 20.1(iii) certified by `unit_pivot_residual_product` without forming $\Theta_{\mathrm{eq}}^{(q)}$, the predicate exactly the read-off's residual, so the obligation is discharged as written; route 2 compared in place for 20.5(i); the early release of X_q, `Zc`; `howell_rows`' adopted store; rollback `HECKE_MODULE_LEAN_PEAK=0`. The report's draft is corrected in two places against `DenseElim.h`: the store is adopted only when also every row is a pivot row, and the adoption runs on the stream route in every caller, not only under the lean retention.
- §22.1, the row for §15: one sentence on G6 (`G6_report.md`; audit `G6_audit.md`, MERGE after fixes F1–F3) — the span inclusions C2 / C3 / C5 by the unit-row route, the shared Z, the census cone release, rollback `HECKE_FIELD_DIET=0`, and the audit's note N1: `tape::factor_basis`'s two re-blockings have no lever.
- Audit (`docs/implementation/slot24/C1230_pre11_audit.md`): the G6 sentence now names the tape route's operand threshold `kSpanTapeMinArea` (below it C2 / C3 / C5 take the dense route of before, unit rows or not) and the heap trim among the levers `HECKE_FIELD_DIET=0` rolls back.

No statement, label, number, tag set or `\LEAN` marker moves (222 on 86 labels); every lever selects a route, never a value; the golden reference and the digests are unchanged.

v1.23.0-pre.10 — 2026-10-20 — hecke-extensions slot 24: the FNTT and MT-1 code-map sentences

This version. **A pre-release, not owner-released, not refereed**; it carries v1.23.0-pre.9 plus **code-map text only**:

- §30, its code map: one sentence on FNTT (`docs/implementation/slot24/FNTT_report.md`; audit `FNTT_audit.md`, fixes applied) — `BuildCuspSpace` / `BuildModularSpace` in the NTT's transform domain and `HeckeOnCusp`'s delayed reduction, rollback `HECKE_FORMS_NTT_CACHE=0`; and the forms ℓ-range restated: the G2-X sentence's $\ell \notin [2, 2^{16})$ stays a statement about `MulSeriesFast`; a forms space is built for every $5 \leq \ell < 2^{16}$ and, for $2^{16} \leq \ell < 2^{27}$, only at $N < 1024$ or where the two-prime product is exact ($N(\ell-1)^2 < (5 \cdot 2^{25}+1)(7 \cdot 2^{26}+1)$, $2N - 1 \leq 2^{25}$), every other cell refused before the module run (the admitted range at the sat route's precision from `FNTT_audit.md` F5).
- §20, its code map: one sentence on MT-1 (`MT1_report.md`; audit `MT1_audit.md`, MERGE after fixes F1–F3) — the pivot-search exit of `cyclic_decomposition` at every thread count and the further splits over the consumer pool, Obligations 20.1 and 20.5(i) discharged as written, rollback `HECKE_MODULE_MT=0`.
- The check `docs/implementation/slot23/C1230_pre10_check.md` amended these two sentences in place (wording only, against the code): the word range is $5 \leq \ell < 2^{16}$;

`HeckeOnCusp`'s delayed reduction is gated by the two levers and $(m+1)\ell^2 < 2^{64}$ alone, not by the transform-domain route's conditions; the range figures 114868, 77702 are the largest admitted integers (primes 114859, 77699); under `ENGINE_HOST_ELIM_HOOK=0` only MT-1's assembly splits; the gate runs the read-off and the solves at pools 1, 2, 7, 16 (only `cyclic_decomposition` also at 3, 13).

The slot-24 record-format event L23-H1 (`orbifold_cell/v2`) concerns no text of this document. No statement, label, number, tag set or `\LEAN` marker moves (222 on 86 labels); every lever selects a route, never a value; the golden reference and the digests are unchanged.

v1.23.0-pre.9 — 2026-10-20 — hecke-extensions slot 24: ELL-W, the wide tier of the τ-certification (draft)

This version. **A pre-release draft, not owner-released, not refereed** (an adversarial audit of the clause follows; plan question Q5: a PROVED lemma is amended, release at v1.24); it carries v1.23.0-pre.8 plus:

- [repaired] `lem:bounds`: clause (7) appended (the wide tier: under $2NB \le 2^{63}-1$, storage in 64 bits, products of two stored-class operands and determinants exact in 128 bits, a fixed datum times a stored matrix in 64 bits, the reach $N \le 2^{20}-1$ when $B = 4N^2$), with its proof; clauses (1)–(6) and their proofs unchanged.
- [repaired] `thm:backend`: consequence (4) (the stored wide tier; why a stored 128-bit type would not extend it) and one proof sentence (the order of evaluation); (i)–(iv), (1)–(3) unchanged.
- Audit repairs (`docs/implementation/slot24/ELLW_audit.md`, F1): (4) states a sufficient condition ("under", not "iff"), says that (iii) is bounded through (iv), and names the checked, measured-only sites — the (P2) input $\gamma_a^{-1}\gamma_b$ ($\le 2B^2$) and the edge prefix, the (T1) conjugate ($\le 2N^2B$), the level-structure consumers; Lemma 18.5(7)(c) is restricted to the level-one consumers and (7) gains the same list. The statement stays [proved here] with an inline [verified computationally], as Theorem 18.7(iv) already does.
- `obl:backend`(ii): one sentence naming the wide tier's assertion (code 52) and the rollback (code 51 as before).
- `rmk:btabsmall`: the measured $B_{\mathrm{tab}} \approx 0.88\,N$ past 23170 and the zero escalations; §10's reach paragraph and the backend code map: the lift, the helpers, the lever `HECKE_TAU_WIDE=0`, the gates `HeckeWideTauValidation`, `HeckeWideTauSlowValidation`.

No label is added, renamed or removed; no existing clause number moves; no tag set or `\LEAN` marker moves (222 on 86 labels); the golden reference and the digests are unchanged (every value at $\ell \le 23170$ is computed by the same 64-bit operations).

v1.23.0-pre.8 — 2026-10-18 — hecke-extensions slot 23: the G2-T2 and G2-P code-map sentences

This version. **A pre-release, not owner-released, not refereed**; it carries v1.23.0-pre.7 plus **code-map text only**:

- §20, its code map: one sentence on G2-T2 (the SIMD ρ_n fill, rollback `HECKE_RHO_SIMD=0`; route 2's memo shared across chunks, rollback `HECKE_ROUTE2_SHARED_MEMO=0`; audit `docs/implementation/slot23/G2T2_audit.md`, PASS) and one on G2-P (the module route on the consumer pool, rollback `HECKE_MODULE_PARALLEL=0`, values independent of the thread count; audit `G2P_audit.md`, PASS, fixes F1–F4 applied).

No statement, label, number, tag set or `\LEAN` marker moves (222 on 86 labels); every lever selects a route, never a value; the golden reference and the digests are unchanged.

v1.23.0-pre.6 — 2026-10-18 — hecke-extensions slot 23: the repairs of C1230-pre3

This version. **A repair-only pre-release, not owner-released**; it carries v1.23.0-pre.5 (below) plus the fixes F1–F6 of `docs/implementation/slot23/C1230_pre3_referee.md` (Opus, adversarial; the referee of pre.1–pre.3: ACCEPT-WITH-FIXES, BLOCKER 0, MAJOR 0, MINOR 2, NIT 4). It sits on top of v1.22.0, **signed by the owner 2026-10-16** (fix F6: v1.22.0's own date line and changelog entry are left as released, "owner sign-off pending"; this entry and the `\date` line record the signature, so that the chain reads consistently). **No label is added, renamed or removed; no statement number, status tag or value moves; the marker count stays** 222 **on** 86 **labels** (none in §30.7).

- F1 (MINOR), the proof of Lemma 30.11, paragraph "The integral complex": the homotopy between the cochain maps of two integral τ-tables is now stated over $C = C^{\bullet}_{\mathrm{eq}}(\mathbb{Z}_\ell)$ with $\Upsilon = \widetilde{H}$ $\mathbb{Z}_\ell$-linear — Proposition 15.12(2) with Lemma 11.3(2) over $R = \mathbb{Z}_\ell$, $6 \in \mathbb{Z}_\ell^\times$, and Proposition 15.12(3),(4) when the transporters s_j or g are rechosen — in place of a homotopy "of $C^{\bullet}_{\mathrm{eq}}(\mathbb{Z})$ with Υ integral" (Proposition 15.18(4)), which the cited results do not supply over $\mathbb{Z}$. The conclusion only needs the homotopy over C and its reduction modulo ℓ^K, so the lemma is unchanged. Proposition 15.18(4)'s own wording of the ring of Υ in its τ-half (older text) is not edited here.
- F2 (MINOR), Lemma 30.11(1), its last sentence, and the proof of (2): (2), (3) and the module structure over $\mathbb{Z}[X_p : p \in S]$ in (1) use that the T_p commute on M_K, Theorem 10.8(1), whose proof uses Theorem 11.10; the sentence "(2)–(5) use neither" now says so. The formalization note of §30 adds Theorem 10.8(1) (its marker a `\LEANlit` with hypothesis `hcomm`) and Proposition 15.12(2)–(4) with Lemma 11.3(2) in place of Proposition 15.18(4).
- F3 (NIT), "Why this section exists": $K_{\mathrm{cls}} := \max(2, e_{\max} + 1)$ with $e_{\max}$ of Proposition 26.1 (which defines $e_{\max}$ only); KMAX defined as the route's level cap, a run setting (`LoewyOptions::kmax`, `SatOptions::kmax`, default 6), and K_{stab} as the first level at which a block reads stable in the sense of Corollary 30.6(1) (`LoewyBlock::K_stab`).
- F4 (NIT), Lemma 30.11(3) and its proof: readings of the tower $(M_K, r_{K',K})$ that use the maps between levels are covered, by (1)'s naturality; Obligation 30.12(iv) names the `LOEWY_RING` fields computed from the listed readings (`D`, `rank_T`, `free_level`, `free_hilbert`, `hidden`, `cf9`, `cert`).
- F5 (NIT, code, not [C]): `HeckeLevelConeValidation`'s two-transversal cell is retagged `(hecke_two_transversals)` and its two checks `(gated)`, no longer "NEG", matching Obligation 29.43(ix) (a message change only; no golden, record or transcript).
- Header and `\date`; this entry.
- The NITs N1–N3 of the check `docs/implementation/slot23/C1230_pre6_check.md` (ACCEPT-WITH-FIXES, NIT 3), applied within pre.6 (no version bump): N1, in F1's proof, H of Lemma 11.3(2) (additive) is replaced by the $\mathbb{Z}_\ell$-linear extension of its restriction to $C_\bullet(K^u) \otimes 1$, which keeps (T1) and the homotopy identity; N2, K_{stab} is described as the code computes it (the stopping rule read from $K = 2$, on a torsion block from K_{cls}; the field holds the last level read on a not-stable block and 0 on `reduced_rank1`); N3 (catalogue, not [C]) the two-transversal cell of `HeckeLevelConeValidation` is no longer listed under NEG. No statement, label, number, tag or marker moves.

v1.23.0-pre.5 — 2026-10-18 — hecke-extensions slot 23: the G2-F code-map sentence

This version. **A pre-release, not owner-released, not refereed**; it carries v1.23.0-pre.4 (below) plus **code-map text only**, fix F1 of `docs/implementation/slot23/G2F_audit.md` (Opus, adversarial; PASS-WITH-FIXES). **No statement's mathematics changes; no label**

is added, renamed or removed; no statement number, status tag or value moves; the marker count stays 222 **on** 86 **labels.**

- §20, its code map: one sentence on Obligation 20.11 on a `TauTableContents::ChainsOnly` table (slot-23 G2-F, the module route's table): it carries no consumer table, so (vi) has no object there and its certification is vacuous; (i)–(v), (vii) run unchanged in the constructor; the module route evaluates ρ_n mod ℓ^k from each term's exact γ_t.
- The status row of Obligation 20.11 in the roster: the same, in a few words, and `HeckeModuleStreamValidation` (its (T) arm) as a gate.
- Header and `\date`; this entry.

v1.23.0-pre.4 — 2026-10-18 — hecke-extensions slot 23: the G1 / G2 code maps

This version. **A pre-release, not owner-released**, in the 1.23.0 series (mid-slot practice: the X.Y.0 release candidate is cut at the slot close); refereed by `docs/implementation/slot23/C1230_codemap_ref` (Opus, adversarial; ACCEPT-WITH-FIXES, MAJOR 1, MINOR 3, NIT 6, every fix applied). It carries the content of v1.23.0-pre.1 … pre.3 as described in their entries below, each with the referee status stated there (pre.1 refereed by F-C123pre1; pre.2 its repairs; pre.3 audited, not refereed), plus **code-map text only** (the follow-up "[C] v1.23.0 code-map sentences for G1 / G2" of `docs/implementation/slot23/HANDOVER_1018.md`; refereed as above). A release referee's coverage record, as `docs/implementation/slot22/release_referee_coverage.md` was for v1.22.0, is due with the release candidate at the slot close. **No statement's mathematics changes; no label is added, renamed or removed; no statement number, status tag or value moves; the marker count stays** 222 **on** 86 **labels.**

- §30.7, its code map: **planned** → **built** (G1; `docs/implementation/slot23/G1_impl_audit.md` finding F8), with the clause-by-clause map of Obligation 30.12 onto `ringalg::LevelCache`, `ReduceToLevel`, `CheckDerivedModule`, `CheckRunModule`, `RunLevelPolicy` and the rollback `HECKE_PRECISION_REUSE=0` (value-neutral).
- §20, its code map: one paragraph, the slot-23 G2 re-implementation of the module route over $\mathbb{Z}/\ell^k$ (G2-A, -B, -C, -D, -E, -G), which discharges Obligations 20.1 and 20.5(i) as written, with the rollback `HECKE_MODULE_STREAM=0` (value-neutral); `G2_plan.md` §2.7's sentence, widened as `G2AB_report.md` §6 item 5 and `G2CDE_report.md` §6 item 5 asked.
- §30, its code map: one sentence on G1 / G2 and one on the G2-X forms product (NTT and CRT) with the rollback `HECKE_FORMS_FASTMUL=0` (value-neutral).
- §22.1, the row of §30: one pointer sentence.
- Header and `\date`; this entry.

v1.23.0-pre.3 — 2026-10-16 — hecke-extensions slot 23, G1: one module run per cell

This version. **A pre-release, not owner-released**; audited by `docs/implementation/slot23/G1_clause_au` (Opus, adversarial; ACCEPT-WITH-FIXES, minor 7, nit 5, every fix applied in this entry's text), in the 1.23.0 series on top of v1.22.0 (signed by the owner 2026-10-16; that entry's text is left as released). Licenses lever G1 of `docs/implementation/slot23/ELL_scout.md` §3 (owner decision 2026-10-16; note `docs/implementation/slot23/G1_clause_note.md`). **Additive**: one subsection and two statements at the end of §30, so that no statement number moves; no existing statement, tag set, marker or value moves (222 markers on 86 labels; the new statements carry none).

- [new] `sec:satprecisionsec`.
- [new] `lem:precisionreduction` ([proved here]): at $\ell \geq 5$, level 1, $M_K = H^1_{\mathrm{eq}}(C^\bullet_{\mathrm{eq}}(\mathbb{Z}/\ell^K))$ is $M_{K'}/\ell^K$ through the reduction map, Hecke-equivariantly, for $K \leq K'$ (because $H^2(C^\bullet_{\mathrm{eq}}(\mathbb{Z}_\ell)) = 0$); the presentation with exponents $\min(a_i, K)$; blocks and every

isomorphism-invariant reading of the route agree; downward only; the hypothesis is needed (the long exact sequence with $H^{q+1}[\ell^K]$; H^0_{eq} on the Dickson locus is not a reduction).

- [new] `obl:precisionreuse`: scope, the restart rule, the derived module exactly, invariance of what is printed (recorded), the per-K route as oracle and rollback arm with a byte-identity gate and two negative controls, record honesty for `K_run` (recorded), certificates on derived modules.
- The chapter's "What this chapter fixes" gains item (ix); its formalization note one sentence.
- The audit's fixes (F1–F12, resolution in the audit file): (1)'s second sentence rests on the naturality of Remark 7.6 in the ring and, for the classical T_p on $\mathbb{L}$, on Theorem 11.10 (F1); the hypothesis is "at each level an integral τ-table", through Proposition 15.18(4) applied to the bulk complex alone (F2, F11); (i) sends every failure of a run (refusal, exception, cap) to the per-K route (F3); (v) gains a digit-route cell (F4), negative controls with named checks (F5) and a tier (F7); (vi) the meaning of `K_run` and the places to update (F6); (3), (iv) list every printed field (F8); the cochain map of (5) is φ (F9); the pre.3 date line (F10); "a new run" per event (F12).
- **Referee request**: a referee should read Lemma 30.11(5)'s converse and its proof's identification of Θ_g at level K with the base change of one integral cochain map first, then Obligation 30.12(iv) against `ReadAlgebra` and `ReadDefect`.

v1.23.0-pre.2 — 2026-10-12 — hecke-extensions slot 23, LES-2: repairs of the referee report F-C123pre1

This version. **A repair-only pre-release, not owner-released.** Applies `docs/implementation/slot23/F-C` (Fable; verdict ACCEPT; BLOCKER 0, MAJOR 0, MINOR 2, NIT 5) to v1.23.0-pre.1. **No label, statement number, tag set, marker or value moves** (222 markers on 86 labels).

- MINOR-1: the "Not edited here" bullet of v1.23.0-pre.1 no longer quotes the (LES4) sentence as existing text; it is the audit's proposed addition.
- MINOR-2: Obligation 29.43(ix) now states what `HeckeLevelConeValidation` does: the two-transversal comparison on $C^\bullet_{\mathrm{eq}}$ is *gated* (cell tag `(NEG hecke_fixed_tables)`), the flat-cochain comparison is recorded, and neither is a negative control. (The tag's "NEG" is a naming matter for the LES-3 code pass; no code changed.)
- NIT-1 (scalar step of the proof of (4) cites $\Delta_0(N_A)$), NIT-2 (class maps named by the indices of (68); both `BuildLevelCorrespondence` and `BuildLevelCorrespondenceChi`), NIT-3 (rule cross-reference; "stored equivariant bases, $q = 0, 1, 2$"), NIT-4 (chapter comment before §29.8), NIT-5 (referee pointer): wording only.

v1.23.0-pre.1 — 2026-10-12 — hecke-extensions slot 23, LES-2: independence of the transversal

This version. **A pre-release, not owner-released; refereed by F-C123pre1 (ACCEPT), repaired at pre.2.** Applies the proposed edit of `docs/implementation/slot23/LES2_audit.md` §3.2 (Fable, MERGE after repairs) with one correction: the audit's sentence "on the flat cochains $C^q(K; L^\chi_A)$ the operator of (68) does depend on the transversal" is **not** carried over — repair R1 of the package refuted it (`LES2_predictions.md` §7, post-run note; `LES2_report.md` §3 item 2): the implemented operator uses r_t only through its coset $A''r_t$, so it is transversal-independent on the whole flat cochain space. **No label is added, renamed or removed; no statement number, existing tag set, marker or value moves** (222 markers on 86 labels).

- Proposition 29.34(4) *replaced* (still [proved here]): "Independently fixed tables" becomes "Independence of the transversal", by (T1) of Definition 11.1; the strict square holds for

every pair of transversals under (1); proof of (4) written out. The old text (the square commutes on cohomology only and a homotopy h is needed) is withdrawn, together with its rule "never (4)".

- Obligation 29.43(ix): the control "operators on independently fixed transversals" (it can never fire) is replaced by the criterion-fails control at ($\Gamma_0(25) \subset \Gamma_0(5)$, $\ell = 7$, $n = 0$); the equal-on-$C^{\bullet}_{\mathrm{eq}}$ record at $m_A > 1$ is stated as a record, not a control.
- **Not edited here:** the audit's *proposed* addition to `rmk:lescheckable` (LES4) ("... with any transversals under the criterion, (4)"; not present text of (LES4)), and the erratum pointers in `LES0_spec.md` §3.3 and `LES_predictions.md` §2, are left to the LES-3 text pass.
- **Refereed** (at pre.2): `docs/implementation/slot23/F-C123pre1_referee.md`, ACCEPT, MINOR-1 / MINOR-2 applied. (Original request: a referee should read Proposition 29.34(4) and its proof against the audit §3.2 and `BuildLevelCorrespondence`, and check that nothing else in the document still relies on the withdrawn statement.)

v1.22.0 — 2026-10-12 — hecke-extensions slot 22, CLOSE-T: the slot-22 release candidate, from v1.22.0-pre.17

This version. **RELEASE CANDIDATE from v1.22.0-pre.17; the owner's sign-off is pending (CLOSE-T step T8) — until it is recorded this text is not owner-released.** It releases the content of v1.22.0-pre.1 ... pre.17 as described in their entries below, each with the referee status stated there; the gaps in referee coverage were read by the Fable release referee `docs/implementation/slot22/F-T1_release_referee.md` (ACCEPT on every gap; its repairs are v1.22.0-pre.17). **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 222 **on** 86 **labels (**208 `\LEAN`**,** 14 `\LEANlit`**).**

- Header and `\date`; this entry.

Companion pointers: [E] 1.22.0, [U] 0.6.0, [L37] 0.5.2, [RC] 0.2.0, [MPT] 0.3.1, [CLI] 1.4.0, [P] 3.12.0, [PG] 1.1.0.

v1.22.0-pre.17 — 2026-10-12 — hecke-extensions slot 22: CLOSE-T T1 repairs (the F-T1 release referee)

This version. **A pre-release, not owner-released.** Repairs of `docs/implementation/slot22/F-T1_release` (Fable, ACCEPT on every gap; BLOCKER 0, MAJOR 0, MINOR 2, NIT 15), the [C] items only. **No statement, number, label, marker, tag set or value moves.**

- **MINOR-1:** the subsection of v1.22.0-pre.9 (below), whose `\date` line pointed at a missing changelog entry.
- **Wording / punctuation (NIT-1, -2, -3, -4, -8, -9, -10):** the pre.2 changelog bullet on LN23F §6 item 3; "membership in $\mathfrak{m}^2$ is read modulo p" and the reason for excluding $\ell = 2$ in Remark 28.19(1); the grammar after the (7) deletion; the order of two header comments of §32; the negative controls of Example 32.24 (which break $d \circ d = 0$, which the dimension gate); the full stop before "(This generalises ..." in Lemma 16.9.
- **Not applied:** NIT-5 (evidence wording at $\ell = 5 = p$), NIT-6 ("$M > 1$ prime to N" versus $M \nmid N$ in Proposition 29.41) and NIT-7 (the preamble of Proposition 29.34) change the scope of a statement and are left to slot 23; NIT-11 to NIT-13, NIT-15 concern [E] / [U].

v1.22.0-pre.16 — 2026-10-12 — hecke-extensions slot 22: the LR-Y phase-5 markers (the tree sequence in $\mathrm{Rep}\,k\,\Gamma$, the Mayer–Vietoris sequence in group cohomology, naturality)

This version. **A pre-release, not owner-released, not refereed** as [C] text; the text applied is §D of `docs/implementation/slot22/LRY_phase5_report.md` as repaired by the Fable audit `LRY5_audit.md` (ACCEPT, BLOCKER 0, MAJOR 0, MINOR 1, NIT 6; MINOR-1 names the identifications of [GT] `thm:mp-mv-terms` in two scopes), applied on the owner's approval of step LR-3 (2026-10-12). Lean side: the LR-Y package, phase 5 (`lean/GL2/MP/TreeSES.lean`, `lean/GL2/MP/TreeMV.lean`, [GT] rows `thm:mp-treeses-rep`, `thm:mp-mv-res`, `thm:mp-mv-les`, `thm:mp-mv-natural`; census `decls=2776 ... verdict=OK`). **No new label; no statement, number, existing tag set or value moves.** No shipped value, default, golden object, setting or console verb changes. Report: `docs/implementation/slot22/cpre16_report.md`.

- **The LR-Y phase-5 markers**, 4 `\LEAN` on 1 label that already carried markers, each with its own scope parenthesis (every [GT] row `proved`): `cl:amalgam` (1 on (1): `thm:mp-treeses-rep`, (50) as one short exact sequence in $\mathrm{Rep}\,k\,\Gamma$ over every commutative ring; 3 on (2): `thm:mp-mv-res`, the first map is $(\mathrm{res}_S, \mathrm{res}_U)$ through the bar-built comparison; `thm:mp-mv-les`, the whole sequence exact at every term, $q = 0$ included, with a connecting map defined by transport and characterised on cocycles, not identified with mathlib's connecting map of a coefficient sequence; `thm:mp-mv-natural`, each of the three maps natural in V). So 222 markers on 86 labels in all (208 `\LEAN`, 14 `\LEANlit`).
- **Two existing marker texts shortened** (no statement text moves): the scope of `thm:mp-model7-tree-orbits` in `cl:amalgam`(1) loses "the sequence is not assembled as one object of $\mathrm{Rep}\,k\,\Gamma$" (now `thm:mp-treeses-rep`), and the "Not in Lean" sentence of `cl:amalgam`(2) is removed (its three items — restriction, the connecting map in `groupCohomology` terms, naturality in V — are the three new markers). The `thm:mp-tree-mv` scope (built on Ext) and the `thm:mp-mv-low` scope (proved directly) stay as they are.
- The marker-history paragraph before §1.2 is brought to pre.16.

Other. The header comment, the date line. Appendix B and the new-statement list are unchanged (no new label, no tag set moves).

v1.22.0-pre.14 — 2026-10-11 — hecke-extensions slot 22: the Lean dimension results at $\ell \in \{2, 3\}$ (LV)

This version. **A pre-release, not owner-released, not refereed** as [C] text; worded within §8 of the Fable audit `docs/implementation/slot22/LV2_audit.md` (`3d7b868d`, ACCEPT-WITH-REPAIRS) and the provenance audit `LVprov_audit.md` (`a2ea84b5`). Lean side: the LV-0–LV-2 packages ([GT] Part MP, sections LV-1 and LV-2, rows `thm:lv-f1` ... `chk:lv3-cells`; census `decls=2683 ... verdict=OK`). **One new label; no existing statement text, number before it, tag set or value moves**; statement numbers after Remark 16.20 within §16 shift (labels do not). No shipped value, default, golden object, setting or console verb changes. Report: `docs/implementation/slot22/cpre14_report.md`.

- **The LV markers**, 12 `\LEAN` on 4 labels that carried none, each with its own scope parenthesis (every [GT] row `proved`, every one in the `GL2` library under [GT]'s census): `lem:centralminusone` (1, at its last clause: `thm:lv3-tree-negone`, *for a field* K with $2 \neq 0$ and a finite-dimensional module, not for a ring with $2 \in R^\times$), `prop:oddnltwo` (1, at (2): `chk:lv-cells`, the value of h^0 at the odd $n \leq 63$ and $n = 101, 129, 255$; not the formula, not the integral groups), `prop:ellthreeclosed` (1, at (4): `thm:lv3-tree-negone`, for fields) and the new `rmk:lvdimsltwothree` (9). So 218 markers on 86 labels in all

(204 \LEAN, 14 \LEANlit). The cells $n > 64$ of the library `L23Cells` are outside [GT]'s census and are cited as data (`docs/implementation/slot22/lv/`), never by a marker. `prop:ltwothreescalar` carries none.

- New `rmk:lvdimsltwothree` ([proved here]): $\dim_{\mathbb{F}_\ell} H^q(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell^2)$, $\ell \in \{2,3\}$, as ranks of explicit matrices (at $\ell = 3$, n even: $h^q = d - r_3 - r_U^-$, $q \geq 2$, and $h^1 = 2d - s - r_S^+ - r_3$), formalised on mathlib's `groupCohomology` with the module of (9), for every $n \leq 700$ and every q; a different model from the census's $C_{\mathcal{I}}^\bullet$, agreeing wherever both exist. It does not say "Lean-verified record".
- The marker-history paragraph before §1.2, which stopped at pre.8, is brought to pre.14.

Other. Appendix B: one line, counts 388 → 389; the header comment, the date line; the new-statement list of this appendix.

v1.22.0-pre.13 — 2026-10-11 — hecke-extensions slot 22: the LR-Y phase-4 markers (the Mayer–Vietoris terms, res − res, the natural periodicity)

This version. **A pre-release, not owner-released, not refereed** as [C] text; the text applied is §6 of the Fable audit `docs/implementation/slot22/LRY4_audit.md` (`9feb57e2`, ACCEPT-WITH-REPAIRS, BLOCKER 0, MAJOR 1 (process), MINOR 5, NIT 6) verbatim, which corrects the proposal of `LRY_phase4_report.md` §D by MINOR-1, -2, -3 and NIT-1 (MINOR-4, NIT-2–4 concern [GT] and Lean, MINOR-5, NIT-6 the report, MAJOR-1 the process). Lean side: the LR-Y package, phase 4 (`lean/GL2/MP/TreeCyclic.lean`, [GT] rows `thm:mp-cyc-coh` ... `thm:mp-periodic-natural`; census `decls=2477` ... `verdict=OK`). **No new label; no statement, number, existing tag set or value moves.** No shipped value, default, golden object, setting or console verb changes. Report: `docs/implementation/slot22/cpre13_report.md`.

- **The LR-Y phase-4 markers**, 4 \LEAN on 2 labels that already carried markers, each with its own scope parenthesis (every [GT] row `proved`): `cl:amalgam` (3, on (2): `thm:mp-mv-terms`, the periodic complexes compute mathlib's `groupCohomology` of $\langle S\rangle$, $\langle U\rangle$, $\langle -I\rangle$; `thm:mp-mv-resres`, under those identifications the middle map is $\mathrm{res}_S - \mathrm{res}_U$ with the sign as displayed; `thm:mp-mv-low`, the beginning $0 \to V^\Gamma \to V^S \oplus V^U \to V^{-I}$, proved directly), `lem:bulkperiodic` (1, on (2): `thm:mp-periodic-natural`, *a* periodicity isomorphism natural on `groupCohomology`, not shown to coincide with that of `thm:mp-tree-periodic` — not $e \cup -$, not restriction-compatibility, not the $q = 1$ clause). So 206 markers on 82 labels in all (192 \LEAN, 14 \LEANlit).
- **Two existing marker scopes rewritten** (no statement text moves): `cl:amalgam`(2)'s `thm:mp-tree-mv` (the cyclic-group terms are no longer "replaced by" the periodic complexes but the sequence is "built on" them; the comparison is named, mathlib's `groupCohomologyIso` through Ext; the "Not in Lean" list now holds only the map $H^q(\Gamma; V) \to H^q(\langle S\rangle; V) \oplus H^q(\langle U\rangle; V)$ as restriction, the connecting map in `groupCohomology` terms and naturality in V), and `lem:bulkperiodic`(2)'s `thm:mp-tree-periodic` (its "not naturality on $H^q(\Gamma; V)$ itself" is dropped, since the new marker covers naturality — for a periodicity isomorphism not shown to be the same one; LRY4 audit MINOR-1, -2).

Other. The header comment, the date line. Appendix B and the new-statement list are unchanged (no new label, no tag set moves).

v1.22.0-pre.12 — 2026-10-11 — hecke-extensions slot 22: the LR-Y markers on the amalgam, the bulk periodicity and the tree resolution

This version. **A pre-release, not owner-released, not refereed** as [C] text; the text applied is the proposal of `docs/implementation/slot22/LRY_phase3_report.md` §C as audited by Fable (`LRY3_audit.md`, `c3bbe552`, ACCEPT-WITH-REPAIRS, BLOCKER 0, MAJOR 0, MINOR 6, NIT 6), with MINOR-1–6 and the [C] NITs 1–3, 5 applied (NIT-4, NIT-6 and MINOR-6 concern [GT] and plan files, not [C]). Lean side: the LR-Y package (phases 1–3, [GT] Part G12, `lean/GL2/MP/TreeRetract.lean` ... `TreeCochains.lean`; census `decls=2344` ... `verdict=OK`). **One new label; no statement, number of an earlier statement, existing tag set or value moves**; statement numbers after Remark 16.3 within §16 shift (labels do not). No shipped value, default, golden object, setting or console verb changes. Report: `docs/implementation/slot22/cpre12_report.md`.

- **The LR-Y markers**, 7 `\LEAN` on 3 labels, each with its own scope parenthesis (every [GT] row `proved`): `cl:amalgam` (3: (1) `thm:mp-treeses-exact`, `thm:mp-model7-tree-orbits`; (2) `thm:mp-tree-mv`, the long exact sequence with the cyclic-group terms replaced by their periodic cochain complexes and the middle map induced by φ^* — not res − res for $q \geq 1$, not the Shapiro identifications, not naturality), `lem:bulkperiodic` (1: (2) `thm:mp-tree-periodic`, an isomorphism $H^q \cong H^{q+2}$, $q \geq 2$, natural on the explicit complex — not $e \cup -$, not the $q = 1$ clause, not restriction-compatibility; clause (1) carries none), `rmk:treeconeres` (3: `thm:mp-model7-cone-res`, `thm:mp-tree-mp-comparison`, `thm:mp-tree-cochains`). So 202 markers on 82 labels in all (188 `\LEAN`, 14 `\LEANlit`).
- New `rmk:treeconeres` ([proved here]): the mapping cone of the lift φ of the tree boundary between the 2-periodic resolutions is a free resolution $P_\bullet\colon \mathbb{Z}[\Gamma]^2 \leftarrow \mathbb{Z}[\Gamma]^3 \leftarrow \mathbb{Z}[\Gamma]^3 \leftarrow \cdots$ over every ring, its terms 2-periodic from degree 1 and its differentials from degree 2; $\mathrm{Hom}_\Gamma(P_\bullet, V)$ written out as $V^2 \to V^3 \to V^3 \to \cdots$; the orientation $\mathfrak{e} \mapsto \mathfrak{v}_S - \mathfrak{v}_U$, the negative of §28's convention for $\mathcal{Y}$ (which moves no statement); its relation to Lemma 28.4.
- `cl:amalgam`**(1), the one change to an existing clause's wording (owner-visible; LRY3 audit MINOR-4).** The last sentence of its standard pointer ([Ser80, Ch. I, §4]), "A reader who prefers not to rely on it may take (50) itself as the hypothesis; everything below uses the amalgam only through it", is replaced by "(50) is proved independently of this pointer (the markers below), so nothing in this document rests on it; the amalgam decomposition itself is used nowhere else": the trust basis of (50) moves from the pointer to the Lean proof. The clause also gains, after (50), the description of the tree inside Y (vertex set $\Gamma/\langle S\rangle \sqcup \Gamma/\langle U\rangle$, edge set $\Gamma/\langle -I\rangle$, as Lemma 14.22 cites it).

  Other. Appendix B: one line, counts 387 → 388; the header comment, the date line; the new-statement list of this appendix.

v1.22.0-pre.11 — 2026-10-10 — hecke-extensions slot 22: the F-L23 and F-MPT repairs, and the Q-reduced nerve (Gap G1)

This version. **A pre-release, not owner-released, not refereed.** Inputs: the Fable referee F-L23 of §16.4 (`docs/implementation/slot22/F-L23_referee.md`, `d3ef8b2d`, ACCEPT with two required and three recommended repairs) and the Fable referee F-MPT of [MPT] (`docs/implementation/slot22/mp_triangulations_referee.md`, `54a55ec0`), whose §3 proves the comparison theorem of the Q-reduced nerve. **Two new labels; no existing tag set, `\LEAN` marker or value moves**; statement numbers after Remark 32.13 within §32 shift (labels do not). No shipped value, default, golden object, setting or console verb changes. Report: `docs/implementation/slot22/pre11_repairs_report.md`.

- F-L23 R1 (required): `prop:oddnhecke`(4) — "equivalently" replaced by the one-way implication: the [conjectural] is the induced rank-one idempotent on $H^2(\Gamma;V)$; the stronger form on $H^1(\Gamma;M_n)$ implies it and is not asserted equivalent. F-L23 R2 (required): `prop:ltwothreescalar` — "the whole prime-to-ℓ Hecke algebra" replaced by "the algebra generated by all T_p".
- F-L23 R3: (I1)'s independence of the representatives proved in place as a cochain identity, the pointers aligned with `prop:barformula`'s; R4: (F1)'s module identification $H^1(\Gamma(3);\mathbb{F}_3) \cong \mathrm{Sym}^2\mathbb{F}_3^2$ derived from the cusp count; R5: Step 0(b) says $d_p = 1$ and the pair is the identity.
- New `def:mpqd` (Q-domination (QD), the dominant vertex $\mathsf{p}(x)$; holds on M_7 and $\mathsf{R}_{\mathbb{Z}}$ at $\ell \in \{2,3\}$) and `prop:mpqreduced` ([proved here], [verified computationally][a]): the choice-free quotient $\mathfrak{D} = \mathfrak{F}'/\!\sim$ is a projective resolution computing $H^\bullet(\Gamma;M)$ with the coefficient system M^Q and the twisted differential, without adaptedness and without lifts; `rmk:mphybrid`(3) rewritten from "Gap G1, recorded" to "closed".

Other. Appendix B: two lines, counts 385 → 387; the header comment, the date line; the new-statement list of this appendix.

v1.22.0-pre.10 — 2026-10-10 — hecke-extensions slot 22: odd n at $\ell = 2$, and $T_p = 1 + p$ at $\ell \in \{2,3\}$ (L23-0 extended)

This version. **A pre-release, not owner-released, not refereed** (the Fable referee F-L23 is pending). Inputs: the L23 master plan `docs/implementation/slot22/L23_master_plan.md` v0.2 and its Fable review `L23_master_plan_review.md` (`dd3254f3`; only what it marks PROVED enters as a theorem); the census reviews `census_l23_classification_review.md` (with its erratum §7) and `census_l23_plan_v02_review.md`; the owner's sign-off of the plan's seven decisions (2026-10-10). **New subsection §16.4, eleven new labels; no existing statement, tag, `\LEAN` marker or value moves**; statement numbers after §16.4 within §16 shift (labels do not). No shipped value, default, golden object, setting or console verb changes. Report: `docs/implementation/slot22/L23_0_report.md`.

- `lem:centralminusone` ([proved here]): $-I \mapsto -1$ forces $2H^q = 0$ for every module and degree, $\mathbb{Z}/2^k$ coefficients included (generalises `prop:badprimes`(2)); `lem:bocksteinallell` ([proved here]): `cl:bockstein`(2)'s Bockstein isomorphism at every prime, and its first isomorphism exactly where $H^2(\Gamma;V)[\ell] = 0$ (false at $\ell = 2$, n odd); `lem:cyclicres` ([proved here]): cyclic cohomology and restriction.
- `prop:oddnltwo` ([proved here]; the review's Theorem O(2)–(3) with its self-contained proof): Mayer–Vietoris collapse, $c_U = 2h^0$, $h^0 = \lfloor(\lfloor n/3\rfloor + 1)/2\rfloor$, $H^{2j+1} \cong \mathbb{F}_2^{h^0}$, $H^{2j+2} \cong \mathbb{F}_2^{(n+1)/2-h^0}$, periodicity from degree 1, the odd-degree classes as $e^j \cup \beta$ of Dickson invariants; `lem:oddnsplit` ([proved here]): $H^q(\Gamma;V/2^k) \cong H^q \oplus H^{q+1}$ Hecke-equivariantly for $k \geq 2$.
- `lem:tellrank` ([proved here]): $\operatorname{rank} T_\ell \leq 2$ through $H^q(\Gamma_0(\ell);\mathbb{F}_\ell)$, of dimensions $1,2,2,\ldots$ ($\ell = 2$) and $1,2,1,\ldots$ ($\ell = 3$); `prop:ltwothreescalar` ([proved here] given (F1) and (I1), both [standard] with section-level pointers **not** verified): $T_p = (1+p)\mathrm{id}$ on $H^q(\Gamma;\mathrm{Sym}^n\mathbb{F}_\ell^2)$, every q, $\ell \in \{2,3\}$, $p \neq \ell$ — the review's input (F2) (a Shapiro–Hecke compatibility) is not needed: the proof uses the level-$\Gamma(\ell)$ coset operator directly; `prop:oddnhecke`: $T_p = 0$ (p odd) on $H^q(V)$ and $H^q(V/2^k)$, $T_2 = 0$ in odd degrees ([proved here]), T_2 a rank-one idempotent in even degrees ([conjectural]).
- `lem:cprule` ([proved here], the census review's $c_p = \min(v_\ell(p^n - 1), \min \mathrm{ed}_{\mathrm{eis},p})$); `rmk:standardtwist` ([proved here], with the $p = 2$ clause); `rmk:oddnltwoerr` (the [R] B(6) erratum and the list of what is deliberately not stated: the Gouvêa bound, the $\ell = 3 \pm$ law, $e_1 = v_\ell(n) + 1$, the T_ℓ idempotent for even n, Hodge–Newton).

Other. Appendix B: eleven lines, counts 374 → 385; the header comment, the date line; the new-statement list of this appendix.

v1.22.0-pre.9 — 2026-10-10 — hecke-extensions slot 22: §32, code-vs-chapter wording (a)–(e)

This version. **A pre-release, not owner-released.** Wording only, five sites: Remark 32.5 (the goodness bullet: `good_fails_at` $\mathrm{R}_{\mathbb{Z}^+}$ at m, $\mathrm{R}_{\mathbb{Z}}$ at v_1, v_2, m; the finite-group and strong forms agree on the four oracle rows, `good_forms_agree`; the [verified computationally]^r sentence: $4 \times 3 = 12$ gated cells, the R_ζ row reported by the ungated `OUTPUT P1.12` line), Example 32.24 (the three negative controls named as built), (MPC3) (only the H^2 change is gated; the H^0 / H^1 / $\mathbb{Z}/\ell^k$ invariance is an ungated observation). No statement, number, label, tag, marker or value moves; not refereed at pre.9 (read by F-T1 at the close, `docs/implementation/slot22/F-T1_release_referee.md`, §3).

v1.22.0-pre.8 — 2026-10-10 — hecke-extensions slot 22: the pre.7 referee's repairs, the MP-4 Hecke remark, the MP-3b records of §32

This version. **A pre-release, not owner-released, not refereed.** Inputs: the Fable referee of pre.7 (`docs/implementation/slot22/C_pre7_referee.md`, ACCEPT, four minor findings F1–F4); the slot-22 code package MP-4 (`MP4_report.md`, merged at `eac3f3b3`) and its audit F-MP4 (`F-MP4_audit.md`, ACCEPT WITH REPAIRS, §3 the contract sentences); the package MP-3b (`MP3_report.md` §6, merged at `ec982f93`) and its audit F-MP3b (`F-MP3b_audit.md`, ACCEPT WITH REPAIRS, R1–R3). **No statement number, label of an existing statement, `\LEAN` marker or value moves**; one tag set grows ([verified computationally]^r); one item label is added ((MPC12)); no shipped value, default, golden object, setting or console verb changes. Map of every edit: `docs/implementation/slot22/C_pre8_report.md`.

- **The pre.7 referee, F1–F4** (all minor, the referee's exact text): the proof of Proposition 32.20, "clause (2) below holds for Φ^* as well" (F1, three consecutive "(2)"); Proposition 32.22(1), "any free Γ-set resolving $\mathbb{Z}$ as in Theorem 32.12(2) (free over every subgroup)" in place of the empty qualifier "free A-set that is also free over B" (F2); Definition 32.2(3), the stabiliser of a point lies in that of a vertex of its carrier "under (IO)" (F3); the proof of Theorem 32.19(4), "(Y being connected by (AC))" (F4, optional, applied). The referee's optional sentence of its §6 is not adopted (the counts carry it).
- **MP-4: Remark 32.23 rewritten** (F-MP4 §3 verbatim, F-MP4 F11 and R3): the route described at pre.7 (lifting the shipped τ-tables through Tot → $C_\bullet(X)$) was not taken, and its plan (slot 23, T_2, T_3, T_5 on $H^{\leq 3}$ against `IntegralHecke`) is superseded. Now: a conj_g-twisted chain map on the MP resolution $N\mathbb{Z}[\mathfrak{F}'_\bullet]$ itself, built with the MP model's own filling, *unique up to twisted homotopy over every commutative ring because* $N\mathbb{Z}[\mathfrak{F}']$ *is free over* $\mathbb{Z}[\Gamma'']$ — no hypothesis $6 \in R^\times$, unlike Lemma 11.3(2) (F-MP4 F1, "the new feature, unstated"); the label factor $\rho_n(g)$; the comparison $T_p\Phi^* = \Phi^*\hat{\Theta}_g$ over every ring; the g-twist not a labelled morphism, G3 not needed to compute. A notation sentence ties the audit's symbols to [C]'s (Φ^{MP}, $\Theta_{\mathcal{I},g}$, $\Gamma'' = \Gamma_0(p)$) and replaces its undefined "Section sec:hecke" by §11. A [verified computationally]^r record follows the remark (`OrbifoldMPHeckeValidation`, 2059 checks) with the non-vacuous counts of F-MP4 R1 (354/592, 254/464, 66/108), the degree-2 caveat of F-MP4 F7 and the not-gated list (F9: H^3, T_{p^ν}, $\mathbb{Z}/\ell^k$ charpolys, level N, j^*, the ι-sign, a second filling); the remark's tag: commentary → [verified computationally]^r; (MPC12) added. Definition 32.18(3): the g-twist's label factor is $\rho_n(g)$ on a twisted chain map of one $\mathfrak{F}'$, not a morphism of MP data. The scope paragraph, the code map and the status table of §22: MP-4 built ("builds no Hecke datum" → no *canonical geometric* one).

- **MP-3b** (`MP3_report.md` §6.4 with F-MP3b R1, R2): (MPC11) 537 → 723 checks, with the pair sequence over $\mathbb{Z}/4$, $\mathbb{Z}/9$, $\mathbb{Z}/25$ against an integral oracle, the pairs on $\mathsf{R}_{\mathbb{Z}}$ and $\Gamma_1(8) < \Gamma_1(4)$ with $-I \notin A$; (MPC10) 331 → 332 (the oracle-file digest); the "not gated" clauses for $\mathbb{Z}/\ell^k$, $\mathsf{R}_{\mathbb{Z}}$ and $-I \notin A$ removed from (MPC11), from the [verified computationally]^{r} record of Proposition 32.22 ("no label $-I$ occurs" → "labels with $u = -I$ occur *only* when $-I \notin A$, and do occur, 60 of 120", R2) and from the sentence after the list (the stale "Remark 32.23 (MP-4)" remainder dropped, R2); the negative control for the $u = -I$ branch called structurally guaranteed (R1). R3 is a docstring of `mp3b_oracle.py` (output unchanged), not a [C] matter.

Other. Appendix B: 32.23 gains [verified computationally]^{r}, [verified computationally] 56 → 57 (registered 28 → 29), tagged 245 → 246. §1.1 (no marker added), the header comment, the date line, the chapter's header comment, the label index ((MPC12)); pre.7 is recorded as refereed at pre.8.

v1.22.0-pre.7 — 2026-10-10 — hecke-extensions slot 22: the MP-L3 Lean markers, the unnormalised clauses, the MP-3 records of §32

This version. **A pre-release, not owner-released, not refereed at pre.7** (refereed at pre.8 by Fable, `C_pre7_referee.md`, ACCEPT; the four minor repairs F1–F4 applied there). Inputs: the slot-22 Lean package MP-L3 (`docs/implementation/slot22/MPL3_report.md` v1.0, merged at `c5617bc7`; [GT] Part G12), the slot-22 code package MP-3 (`MP3_report.md`) and its audit F-MP3 (`F-MP3code_audit.md`, ACCEPT WITH REPAIRS, §10 the contract sentences). **No statement number, label of an existing statement, or value moves**; two tag sets grow ([verified computationally]^{r}); no shipped value, default, golden object, setting or console verb changes. Map of every edit: `docs/implementation/slot22/C_pre7_report.md`.

- **The MP-L3 markers**, per MPL3 report §5 (every [GT] row `proved`): 25 `\LEAN` — `def:mpmorphism` (4: (1),(2) `def:mp-morph`, `def:mp-orbmap`; (3) `lem:mp-morph-simplicial`, `def:mp-pull`), `thm:mpnatural` (10: (1) `thm:mp-pull-cochain`, `lem:mp-pull-comp`, `def:mp-zu`; (2) `thm:mp-pull-natural`, `def:mp-barmap`, `thm:mp-homotopy-exact`; (3) `thm:mp-refine-invariance`, `thm:mp-label-indep`; (4) `thm:mp-untwisted`, `thm:mp-rz-untwisted`) and its [verified computationally]^{r} record (3: `def:mp-rz-refine`, `thm:mp-rz-refine`, `thm:mp-rz-untwisted`), `prop:mpbredon` (3: (1) `def:mp-epsmp`, `thm:mp-epsmp-canonical`; (2), the "if" only, `thm:mp-epsmp-quasiiso`), `def:mpset` (1: `thm:mp-norm-dgn`), `def:mpfree` (2: `def:mp-simpK`, `thm:mp-simpK-alt`), `thm:mpcomparison` (2: (2) `thm:mp-norm-heq`, (4) `thm:mp-comparison-normalised`). `thm:mp-dd`'s scope note "not the normalised complex" is dropped. So 195 markers on 79 labels in all (181 `\LEAN`, 14 `\LEANlit`).
- **The unnormalised clauses** (MPL3 report §4.1). Lean proves Theorem 32.19(1)–(3) for $C^\bullet_{\mathrm{un}}$; [C] now states them there too: Definition 32.18(3) defines u^* on all strings, the normalised map being its restriction; Theorem 32.19(1), (2), (3) gain "likewise for $C^\bullet_{\mathrm{un}}$", (2) also that the inclusions $C^\bullet \subseteq C^\bullet_{\mathrm{un}}$ are quasi-isomorphisms commuting with u^*; the proofs gain one sentence each (u commutes with the degeneracies; the projection $\mathbb{Z}[\mathfrak{F}'] \to N\mathbb{Z}[\mathfrak{F}']$ is a map of free resolutions over $\mathrm{id}_{\mathbb{Z}}$). The markers are scoped to the unnormalised clauses; the normalised u^* as a cochain map is not marked.
- **Refinement certified by a vertex map** (§4.2). Definition 32.18(2): the table need not come from a simplicial map; an equivariant map of vertex sets with the containments certifies it (how Lean certifies $\mathsf{R}_{\mathbb{Z}} \to \mathsf{M}_7$). Proof of Theorem 32.19(1): "non-empty to non-empty" is all that is used.
- **Label independence** (§4.3): [C]'s chain-homotopy claim in Theorem 32.19(3) stays (its proof, the comparison theorem, covers it); the marker is scoped to cohomology.

- $\varepsilon^{\mathsf{MP}}$ (§4.4). Proposition 32.20(1) gains: for any lift Φ of id_R from $R[\mathfrak{F}'_\bullet]$ into $C_\bullet(X) \otimes R$, Φ^* is homotopic to $\varepsilon^{\mathsf{MP}}$ followed by $C^\bullet \subseteq C^\bullet_{\mathrm{un}}$ (the proof: the same uniqueness up to homotopy, over R); so the Lean $\varepsilon^{\mathsf{MP}} = \Phi^*$ and the written $(\Phi^{\mathsf{MP}})^* \circ \mathrm{cmp}^*$ agree up to homotopy, and (2) holds for both.
- **MP-3** (F-MP3 §10). Proposition 32.22(1): the identity holds on any free A-set free over B, in particular on $\mathfrak{F}'_\Gamma$ (the code's choice), with res, cor written there and one proof sentence; Proposition 32.21(2): the relative indexing of the cone and its identification with $C(r)[-1]$; [verified computationally]^{r} records after both propositions (`OrbifoldMPBoundaryValidation`, `OrbifoldMPLevelConeValidation`), their tags [proved here] → [proved here] [verified computationally]^{r}; Remark 32.25 gains (MPC10), (MPC11) (two new item labels) and says what is not gated: the level pairs over $\mathbb{Z}/\ell^k$, on $\mathsf{R}_\mathbb{Z}$, and with $-I \notin A$. The audit's optional sentence for Theorem 32.19(4) ("non-central for every pair with $[\Gamma : B] > 2$") is not adopted: $\Gamma^2 < \Gamma$ has index 2 and non-central labels; the record of (MPC11) states the counts instead.

Other. Appendix B: two lines gain [verified computationally]^{r}, [verified computationally] 54 → 56 (registered 26 → 28). The code map and the status table of §32: MP-3 built. The formalization note of §32, §1.1, the header comment, the date line, the chapter's header comment, the label index ((MPC10), (MPC11)).

v1.22.0-pre.6 — 2026-10-10 — hecke-extensions slot 22: the F-MP2 referee repairs and the first Lean markers of §32

This version. **A pre-release, not owner-released.** §32 is refereed by F-MP2 (contract part, Fable, `docs/implementation/slot22/F-MP2_contract_referee.md`: ACCEPT WITH REPAIRS, one blocking finding), and its repairs R1–R8 are applied as given there; the slot-22 Lean packages MP-L1 / MP-L2 (`docs/implementation/slot22/MPL1_report.md` v1.1) supply one wording repair and the markers. **No statement number, label, tag set, status or value moves**; no shipped value, default, golden object, setting or console verb changes. Map of every edit: `docs/implementation/slot22/C_pre6_report.md`.

- **R1 (blocking), Theorem 32.19(4).** The first sentence was false: labels acting by ± 1 do *not* make $f \mapsto f \circ u$ a cochain map (mixed labels $\pm I$ fail at odd n, the `REFINE_TWIST` control); the sufficient condition is that all labels act by one and the same central automorphism. Proof: one clause (for labels $\pm I$, iff the labels are constant or n is even). The clause's last sentence and the theorem's tag are unchanged.
- **R2, Proposition 32.21(2).** The proof cited Theorem 32.19(2), which compares resolutions of $\mathbb{Z}$ over $\mathrm{id}_\mathbb{Z}$; it now lifts the augmentation $\mathbb{Z}[\Gamma/\Gamma_\infty] \to \mathbb{Z}$, takes $\mathrm{Ext}_{\mathbb{Z}[\Gamma]}(\varepsilon, M)$ and identifies it with $\mathrm{res}^\Gamma_{\Gamma_\infty}$ by Shapiro, [standard:] Weibel Thm. 2.2.6 and Brown Ch. III.9 (VERIFY, added to the list of the pre.5 item); its line in Appendix B reads [Bro82, Ch. III.6.2, III.9].
- **R3** Definition 32.1(3): (PD) is now pure dimension (as written it held in every finite-dimensional complex). **R4** Definition 32.16: the choices of h and c are named, the homotopy class is choice-free, the code's choice is `PhiLift`'s. **R5** Remark 32.13(1): the hybrid "is an ℓ-local isotropy model in the sense of" Definition 31.1 with any projective column (was "is the"). **R6** Remark 32.7(3): the coset sentence (adaptedness nests the shared stabilisers in $\Gamma_{v(\tau)}$); its [verified computationally]^{r} paragraph says which count is S's $(132, 2800)$ and which the μ-form's $(124, 2704)$. **R7** the symbol D of Proposition 32.17(2) declared in the notation paragraph.
- **R8, the eight hygiene items.** Proposition 32.21(1) "the proof of Theorem 32.12, parts (1)–(2)" and (2) "computes the relative object of Definition 17.1"; "Lemma " in the proof of Theorem 32.12(5); the cyclicity of the finite subgroups of $\mathrm{SL}_2(\mathbb{Z})$ argued in

Remark 32.5(3); the label coset $\{h, h \cdot (-I)\}$ in the [verified computationally]^{r} paragraph after Theorem 32.19; Definition 32.18(1) "recorded on the descriptors by a label table ..., which is all that the induced maps of (3) use"; the bound group element u renamed w in the Reading paragraph after Theorem 32.12 and in Definition 31.1's e_σ (a bound variable; no statement changes); the header comment and the pre.5 item now say "refereed at pre.6".

- **Definition 32.2(3), the strong form of goodness** (MPL1 report §4.1). Pre.5 wrote $\mathsf{T}(v) \subseteq \Gamma_v$; the C++ (`Good`, `good_strong`), `mp_rules.py`, Table 9's "fails at" column and Lean all use "$\mathsf{T}(v)$ in the stabiliser Γ_p of one vertex p of $\mathrm{St}(v)$"; the written form fails at every vertex orbit of M_7 and at v_0, v_1, v_2, m on $\mathsf{R}_{\mathbb{Z}}$ (Lean: `goodVtx_discrepancy`). The definition now states the form in use; Remark 32.5(3)'s "the finite-group and the strong form agree" is true for it and now says so. A wording repair of a definition: no value, table entry, tag or validation changes.
- **The unnormalised MP complex** $C^\bullet_{\mathrm{un}}$ (MPL1 report §4.2) is named in Definition 32.6, and Theorem 32.12(3), (4) state it beside the normalised one (the proof's argument verbatim on the resolution $\mathbb{Z}[\mathfrak{F}']$ of (2)), so that the markers are clause-scoped.
- **The first `\LEAN` markers of §32**, per MPL1 report §5 ([GT] Part G12, every row `proved`): 30 on 11 labels — `def:mprules` (2: `def:mp-datum`, `thm:mp-goodvtx`), `def:mpell` (1), `def:mpRZ` (2: `def:mp-models`, `def:mp-rz-tau`), `tab:mprules` (3, rows M_7, $\mathsf{R}_{\mathbb{Z}}$), `def:mpset` (4), `rmk:mpwhat` (2, item (1)), `def:mpfree` (1), `lem:mprows`, `lem:mpcols`, `lem:mpassembly` (1 each; the last in its acyclic form) and `thm:mpcomparison` (12: (1) `lem:mp-free`, `lem:mp-set-finite`; (2) `thm:mp-resolution`, `thm:mp-projres`; (3) `thm:mp-homequiv`; (4) `thm:mp-comparison`, `thm:mp-comparison-natural`; (5) `thm:mp-models-cover`, `thm:mp-ac-bridge`, `thm:mp-model7-ac`, `thm:mp-retr`, `thm:mp-rz-ac`, i.e. (AC) for M_7 and $\mathsf{R}_{\mathbb{Z}}$, (H4)(ii) of the subdivision included, and the comparison on both with no hypothesis left). So 170 markers on 76 labels in all (156 `\LEAN`, 14 `\LEANlit`); §1.1 names the list, the formalization note of §32 says what is not marked, (MPC3) that (AC) is now proved in Lean.

Other. Appendix B: no line or count moves (one locator, R2). Non-normative: the header comment, the date line, the chapter's header comment.

v1.22.0-pre.5 — 2026-10-09 — hecke-extensions slot 22: the Moerdijk–Pronk model (MP-CT)

This version. **A pre-release, not owner-released, not refereed at pre.5** (refereed at pre.6 by F-MP2, contract part, ACCEPT WITH REPAIRS; repairs R1–R8 applied there). Sources: `docs/implementation/slot22/MP_math_plan.md` v0.2.1 (audited, `MP_math_plan_audit.md`, repairs R1–R8), `MP_impl_plan.md` v0.2.1 §4 and its review (R1, R2, R5, R6), the owner rulings Q1–Q7 of 2026-10-09 (at the recommendations: Q1 the choice-free set is normative, Q2 $\mathsf{R}_{\mathbb{Z}}$ MP-conformant and M_7 economical, Q3 the ℓ-relevant structure, Q4 no shipped value or default changes, Q5 Lean L0–L10 as the M6 "orbifold" item, Q6 the 7.7 erratum of pre.4, Q7 no teardrop in Lean), the F-MP2 code audit `F-MP2_code_audit.md` §1.3 (the label factor $\rho(h_{\tilde{\sigma}_0})$) and the built code and validations (`MP1a_report.md`, `MP1b_report.md`, `MPT_report.md`, `MP2_report.md`). **No existing statement number, label, tag set or status moves; the marker count stays** 140 **on** 65 **labels** (the new labels carry none; no Lean module of the lane exists yet). **No shipped value, default, golden object, setting or console verb changes**; $\mathbb{F}_2$ and $\mathbb{F}_3$ keep the Bredon semantics of Definition 23.20.

- **New: §32**, a chapter after §31, before the appendices (the implementation plan named the place after §17; placed last, as §31 was, so that no chapter number moves): Definitions 32.1, 32.2, 32.3, 32.4 and Table 9 — orbifold triangulations as descriptors, Moerdijk–Pronk's rules as decidable predicates (subgroup form), the ℓ-relevant structure, M_7 and

$\mathsf{R}_{\mathbb{Z}}$; Remark 32.5 ([proved here], [verified computationally]^{r}); Definition 32.6, the choice-free MP set, and Remark 32.7 ([proved here], [verified computationally]^{r}: π_0 of the restricted action groupoid; Moerdijk–Pronk's own set exactly in the effective, adapted, simplicial case; a non-effective generalisation for $\mathrm{SL}_2(\mathbb{Z})$); Definition 32.8 and Lemmas 32.9, 32.10, 32.11 ([proved here], the last with a [standard:] alternative); **Theorem 32.12** ([proved here], [standard:] Weibel) — the MP complex computes $H^q(\Gamma; M)$ for every q, ring and module, $-I$ included, via the free resolution $\mathbb{Z}[\mathfrak{F}']$; Remark 32.13 (the hybrid is Proposition 31.2, its periodic-column instance; gap G1); Definition 32.16 and Proposition 32.17 ([proved here], [standard: ,] [verified computationally]^{r}: $D = 3$, the $24 \cdot H^{\geq 2} = 0$ completeness of $k \leq 3$, what is certified with its n-range, scope bulk and $\Gamma = \mathrm{SL}_2(\mathbb{Z})$); Definition 32.18 and Theorem 32.19 ([proved here], [standard: ,] [verified computationally]^{r}: naturality, refinement invariance, the label factor h_{σ_0}, which [MP] v0.2 had dropped at the quotient level and the code carried — F-MP2 §1.3); Proposition 32.20 ([proved here], [standard: ,] [verified computationally]^{r}: quasi-isomorphism if $6 \in R^\times$, the $\mathbb{Z}[1/2]$ odd-n witness against "only if", injective on $H^{\leq 1}$); Propositions 32.21, 32.22 ([proved here], [standard:] Brown; code planned, slot-23 MP-3); Remark 32.23 (commentary, MP-4; G3 open); Example 32.24 ([verified computationally]^{r}); Remark 32.25 ((MPC1)–(MPC9), mapping each clause to its registered validation); scope, code map, formalization note. The new [standard:] locators Weibel §8.3, Weibel Lemma 2.7.3 (an alternative to the written proof of Lemma 32.11) and Brown Ch. III.8 are VERIFY; Weibel Thm. 2.2.6 and Brown Ch. III.6.2, III.9–10 are the document's existing pointers (pre.6: Brown Ch. III.9 as used in the repaired proof of Proposition 32.21(2), restriction as the map induced by $\mathbb{Z}[\Gamma/H] \to \mathbb{Z}$, is added to the VERIFY list).

- **The forward-dated labels resolved.** `thm:mpcomparison` (the `via=` of the `MP_ALL_DEGREES` lines), `def:mpmorphism`, `rmk:mphybrid`, `prop:mpalldegrees`, `sec:mpmodel` and `def:mpset`, cited ahead of this version by the catalogue, the MP reports and a C++ output string, now exist.
- **Addenda** (no tag moves): Remark 17.4 (items (2), (3) resolved by §32, not withdrawn), Remark 17.5 (owner ruling Q2), Remark 31.4(3) (R_ζ against $\mathsf{R}_{\mathbb{Z}}$); a row of §22.1.

Other. Appendix B: twenty-three lines added (its counts paragraph). Appendix D: `sec:mpmodel` and the thirty-nine further new labels. The referee reading list gains one item. Non-normative: the header comment, the date line.

v1.22.0-pre.4 — 2026-10-09 — hecke-extensions slot 22: Remark 7.7, the superfluous hypothesis $6 \in R^\times$ (MP-0, erratum)

A pre-release, not owner-released; the erratum is owner-approved (question Q6 of the Moerdijk–Pronk mathematical plan). In Remark 7.7, second bullet (n odd, $\ell \neq 2$), the words "under the additional hypothesis $6 \in R^\times$" are deleted: the group-side argument ($-I$ is central, so it acts trivially on $H^\bullet(\Gamma; -)$ and by $(-1)^n = -1$ on the coefficients, whence $2H^\bullet = 0$ and, 2 being invertible, $H^\bullet = 0$) uses $2 \in R^\times$ only, which is the bullet's own hypothesis. Wording only: no statement, label, tag set, marker (140 on 65) or value moves. Predictions for the Moerdijk–Pronk validations: `docs/implementation/slot22/MP_predictions.md`.

v1.22.0-pre.3 — 2026-10-09 — hecke-extensions slot 22: the level-change cone and inflation–restriction (LES-C)

This version. **A pre-release, not owner-released**; refereed by F-LES (Fable, `docs/implementation/slot22` ACCEPT WITH REPAIRS), repairs R1–R6 applied (below). Source: `docs/implementation/slot22/LES0_sp` v0.1 §§1–2, written out with proofs; its owner questions Q1–Q7 taken at the spec's recommendations by the lead pending owner objection (the corrected normality picture; $\ell \in \{2, 3\}$

labelled Bredon, the group-cohomology sequence deferred; $\mathbb{F}_\ell$ and $\mathbb{Z}/\ell^K$ only; the old-space U_p cone deferred; no console verb; the text in [C] as §29.8). **No existing statement number, label, tag set or status moves; the marker count stays** 140 **on** 65 **labels** (the new labels carry none).

- **New:** **§29.8**, the last subsection of §29: Definitions 29.25, 29.27, 29.29, 29.35; Lemma 29.26 ([proved here]); Lemma 29.28 ([proved here], [standard:] Brown) — restriction and trace as fibre maps; Lemma 29.30 ([proved here]) — the six-term sequence at every ring and every ℓ, unconditionally; Propositions 29.31, 29.32, 29.33 ([proved here]); Proposition 29.34 ([proved here], [standard:] Brown, [verified computationally][a]; level-raising U_p [proved here] negative after R1); Propositions 29.36 ([proved here], [standard:] Brown, Hochschild–Serre) and 29.37 ([proved here], [standard:] Brown); Lemmas 29.38, 29.39 ([proved here]); Propositions 29.40, 29.41 ([proved here], [verified computationally][a]); Remark 29.42 ([proved here]); Obligation 29.43; Remark 29.44 ((LES1)–(LES9)); Remark 29.45; the code map (planned, slot-22 LES-1..3). All new Brown locators are VERIFY or marked as read at an earlier R20-L pass.
- **Departures from the spec**, each argued in the text: the same-level U_p square is [proved here] strict (the spec had U_p for $p \mid N_A$ in general); the level-raising U_p square does *not* commute in this document's convention (the spec asserted it on cohomology from the classical right cosets, which belong to the classical $U_p = [A\,\mathrm{diag}(p,1)A]$, not to this document's $U_p^{(N)} = U_p^*$); the Adamson / Bieri–Eckmann identification is a parenthesis, not a claim; χ_4 modulo 5 is odd (its model vanishes for even n) and has no values in $\mathbb{F}_7$; every cell of the spec's arm (i) is split ($\ell \nmid p+1$).
- **F-LES repairs** (2026-10-09, same pre-release). **R1**: Proposition 29.34(3) [open] → [proved here] [verified computationally][a] with negative content — (a) $U_p^B \circ (\alpha_{B/A})_* = p \cdot [A\,g\,B]$ as cochain maps (via (T1) of Definition 11.1, as in Proposition 27.4(1); $= 0$ at $\ell = p$), (b) $U_p^A = \mathrm{cor} \circ [A\,g\,A'']$ at cochain level, (c) for $B = A''$, $6, p \in R^\times$ the square commutes on H^q iff $\mathrm{dg}_g(H^q(A)) \subseteq \mathrm{res}\, H^q(A)$, and fails at ($\Gamma_0(25) \subset \Gamma_0(5)$, $\ell = 7$, $n = 0$) by an explicit Eisenstein computation, (d) this document's $U_p^{(N)}$ is U_p^* on old spaces, the classical $U_p = [A\,\mathrm{diag}(p,1)A]$ with $\tau' = S\tau S^{-1}$ is strict by (1) (deferred); the "not settled here" parenthesis and the double-coset count with its Brown III.9.5(ii) locator are gone (so R2 is void). Propagated to Obligation 29.43(vi) (the strict identity gated, the square recorded), (LES9) ([proved here]), Remark 29.45, Appendix B (two [open] tags go, 8 → 6; [verified computationally] 45 → 46), the reading list; evidence `les0/levelraise_up_check.py`, `les0/out_levelraise_up.txt`. **R3**: Proposition 29.34: $N_A \mid N_A'$, $N_B \mid N_B'$ stated, the coset invariant is $a_\gamma c_\gamma / N'$ mod p. **R4**: Proposition 29.41: the non-normality witness inside $\Gamma_0(N)$ is $\begin{pmatrix} 1 & 0 \\ NN' & 1 \end{pmatrix}$. **R5**: Lemma 29.28(3)'s [standard:] marked as read at R20-L; [Ada54], [BE78] added to the bibliography (VERIFY). **R6**: the T_p instance of Proposition 29.34(2) extended to $\Gamma(M)$, Γ^2, Γ^3; $\Gamma(3)$ added to Proposition 29.32's torsion-free list (Remark 29.42: $M \geq 3$). No label, number or marker moves.

Other. Appendix B: twenty-one lines added (its counts paragraph). Appendix D: `sec:nebles` and the twenty-four new labels. The referee reading list of Appendix E's provenance section gains one item. Two bibliography entries (R5). Non-normative: the header comment, the date line.

v1.22.0-pre.2 — 2026-10-08 — hecke-extensions slot 22: Remark 28.19(1), sufficient, not necessary (DOC22)

This version. **A pre-release, not owner-released.** One rewording, from `docs/implementation/slot22/LN2` §6 item 1 (a finding of LN23-F against the F4 referee's "iff"): in Remark 28.19(1), "By Mazur

this holds iff $\mathcal{F}$ contains a good prime" becomes "if", with the p-odd, $\ell = 2$ sentence and the $p = 2$ counter-example $(2473, 2)$ ([verified computationally]), and the next sentence reads "may read" for "reads". The remark's tags ([standard], [conjectural], [verified computationally][a]) are unchanged. Not applied from the same report: its §6 items 2 and 3 (the family-certificate figures; item 3 not added as a data line: $e_3(1747) = 3$ appears only through "Loewy $4 = e_3 + 1$" in item 1's sentence). No statement number, label, tag set or marker moves.

v1.22.0-pre.1 — 2026-10-06 — hecke-extensions slot 22: wild characters, $\Gamma_{\mathsf{H}}(N)$ and the tower at ℓ (P10-C)

This version. **A pre-release, not owner-released**; decisions taken as recommended. The text of `docs/implementation/slot22/P10C_contract_text.tex.txt` v0.2, part (A), inserted verbatim ("v1.22.0-pre.N" numbered 1), as refereed by the slot-22 Fable referee F-U5 (`docs/implementation/slot22/P10C_F4_referee_report.md`, ACCEPT WITH REPAIRS, 0 blocking / 3 major / 13 minor, all applied in that text; the annotations "v0.6.0-pre.2, F-U5 ..." inside it name the referee's revision v0.2 of the text, released here and in [U] 0.6.0-pre.1). Implementation spec `docs/implementation/slot22/P10_spec.md` v0.2. **No existing statement number, label, tag set or status moves; the marker count stays** 140 **on** 65 **labels** (the new labels carry none; no marked label is touched).

- **New: §29.7**, the last subsection of §29, so that no statement number of v1.21.0 moves: Definition 29.15 and Lemma 29.16 ([proved here], [standard:] Washington, VERIFY) — the chain rings $\mathcal{O}_{j,e}$ and the wild characters $\chi_{i,t}$; Definition 29.17, Propositions 29.18, 29.19 ([proved here]; F-U5 m6, m7) — $\Gamma_{\mathsf{H}}(N)$, its table, diamonds, operators and Heilbronn action; Lemma 29.20 ([proved here]; F-U5 M1: Γ' acts on P through the inverse of the quotient map) — specialisation and coinvariants without division; Lemmas 29.21 ([proved here], [standard:] Ash–Stevens) and 29.22 ([proved here]); Remark 29.23 ((NB7)–(NB10), [proved here]; (NB8)'s rank clause [proved here] *given* [U, `prop:u-towerorient`], [verified computationally] at $r = 1$, gated; F-U5 M3); Remark 29.24 (the code contract); the code map (planned, slot-22 P10).
- `rmk:nebfield`(2): one sentence (wild characters admitted over $\mathcal{O}_{j,e}$; over $\mathbb{F}_q$ and $W_K(\mathbb{F}_q)$ the refusal stands).
- The refusal box after `rmk:nebcost`: item (ii) narrowed, not reversed (`character_wild` over $\mathbb{F}_q$, $W_K(\mathbb{F}_q)$; new names `wild_ring_order`, `wild_precision`, `wild_tame_level`).
- After `thm:hidacontrol`: an unnumbered addendum to its (2) (`docs/implementation/slot22/P10C_adde` drafted by F-U5 and verified on Hida pp. 589, 593) — "no twist intervenes" is relative to Hida's own Λ-structure; the index a of $\mathcal{V}^0(a)$ is the natural one. An addendum, not an erratum; the theorem's statement is unchanged.

Other. Appendix B: ten lines added (its counts paragraph). Appendix D: `sec:nebwildsec` and the ten new labels. Macros `\Zl`, `\Ql`, `\VERIFYloc` provided (as in [U]). The one departure from the verbatim text: in (NB8) the two cross-document references to [U]'s `thm:u-towercontrol` and `prop:u-towerorient` are printed as label names (`\texttt`), not `\ref`, as [C] cites [U] elsewhere (a `\ref` into another document is undefined here). Non-normative: the header comment, the date line. Companion: [U] 0.6.0-pre.1 (part 5, [U, `sec:u-tower`]).

v1.21.0 — 2026-10-06 — hecke-extensions slot 21, the close: the slot-21 release, from v1.21.0-pre.4

This version. **RELEASE from v1.21.0-pre.4 under the owner's decision D16 of 2026-10-06** (release of the pre-release documents at the slot-21 close, accepted as recommended;

`docs/implementation/slot21/slot21_status_and_close_plan.md` §5). It releases the content of v1.21.0-pre.1 … pre.4 as described in their entries below, each with the referee status stated there. **No statement's content changes; no label is added, renamed or removed; no status tag moves; the marker count stays** 140 **on** 65 **labels.**

- Header and `\date`; this entry.

Companion pointers: [E] 1.21.0, [U] 0.5.0, [L37] 0.5.1, [RC] 0.1.0, [LG] 0.1.5, [EX] 1.4.1, [CLI] 1.3.0.

v1.21.0-pre.4 — 2026-10-06 — hecke-extensions slot 21: the weight-two curve object (LN23)

This version. **A pre-release, not owner-released**; the owner's decision O6 on slot-22 LN23, taken as the F4 referee recommended (`docs/implementation/slot22/LN23_F4_referee_report.md` §11, ACCEPT WITH REPAIRS, 0 B / 2 M / 9 m, the repairs in the delivered text). **No existing statement number, label, tag set or status moves; the marker count stays** 140 **on** 65 **labels** (the new labels carry none; no marked label is touched).

- **New: Proposition 28.18**, in a new last subsection §28.7 of §28, so that no statement number of v1.21.0-pre.3 moves ([proved here]; text as delivered in `docs/implementation/slot22/LN23_C_proposition.tex.txt`): at $n = 0$ the cokernel presentation with orbit-sum fixed spaces is the cellular H^1 of the quotient graph $\Gamma_0(N)\backslash\mathcal{T}$, i.e. $H^1(Y_0(N);\mathbb{Z}) = \mathrm{Hom}(\Gamma_0(N),\mathbb{Z})$, free of rank $2g + c - 1$, and its reduction $H^1(Y_0(N);\mathbb{F}_p)$ for *every* prime p — at $p \in \{2,3\}$ not group cohomology but the Bredon object of (H3); $H_{N,q}$ induces T_q over $\mathbb{Z}$ (Proposition 28.9 over $\mathbb{Z}[1/6]$ and torsion-freeness). One [standard] step (the spine, [Ser80, Ch. I, §4.2], locator not verified) with Proposition 23.3(2) as the alternative; [verified computationally][a] for the torsion of the two-/three-term lattice and the unimodular elimination (`docs/implementation/slot22/ln23ref/`).
- **New: Remark 28.19** ([standard:] Mazur II §§9, 16–17 read on the scan, Calegari–Emerton held, Sturm held; [conjectural]; [verified computationally][a]): Mazur's Eisenstein ideal at $p \in \{2,3\}$ on the curve object, and the two ways a finite Hecke family fails — generation (no Mazur good prime) and separation (another maximal ideal).
- `def:levelNmodel`(c): admits $p \in \{2,3\}$ at $n = 0$ on the curve object (a definition; no tag).
- (LN6) (in `rmk:levelNcheckable`): an addendum carrying the F2 caveat to the extension of (LN4), (LN6) to $p \in \{2,3\}$; no tag of the remark moves.

Other. §28's scope paragraph names the exception. Appendix B: two lines added (its counts paragraph). Appendix D: `sec:levelNcurve`, `prop:levelNcurve`, `rmk:levelNcurvefamily`. `gl2_refs.bib`: `calegari_emerton`. Non-normative: the header comment, the date line. Companions unchanged.

v1.21.0-pre.3 — 2026-10-06 — hecke-extensions slot 21: the (U17)(a) addendum

This version. **A pre-release, not owner-released.** One addendum sentence. **No clause's mathematics changes; no label, tag or status moves; the marker count stays** 140 **on** 65 **labels** (no marked label is touched).

- (Hi5) (in `rmk:hidacheckable`): for $n < \ell - 1$, [U]'s (U17)(a) (the ordinary $\mathfrak{m}_{\mathrm{Eis}}$-block of H^+ free of rank one over its Hecke algebra) is equivalent to [E, `cl:lveis`](2) on that block, as read in [E] after the proof of [E, `cl:lveis`] (`docs/implementation/slot21/P8_F4_referee_report.md` M3 and §4 item 3; [RC] §9.4); for $n \geq \ell - 1$ both stay conjectural. This is the row v1.21.0-pre.2 listed as "not done here".

Companions: [E] 1.21.0-pre.5, [U] 0.5.0-pre.6, [L37] 0.5.1-pre.6, [RC] 0.1.0-pre.8.

v1.21.0-pre.2 — 2026-10-05 — hecke-extensions slot 21: addenda from [U] part 4 and the literature pass

This version. **A pre-release, not owner-released.** Addenda and locator repairs only. **No clause's mathematics changes; no label, tag or status moves; the marker count stays** 140 **on** 65 **labels** (no marked label is touched).

- `rmk:stabconvention`(3): an *addendum*, not an erratum — since [U] v0.5.0-pre.2 (`lem:u-latticeord`, [proved here] there; the slot-21 referee F4 of [U] part 4, `docs/implementation/slot21/P8_F4_referee_report.md`, §4 item 1) the ordinary part does not see the lattice: D_* is an isomorphism on $e_1 H^q$ for every q, on ∂ and at level ℓ; (Hi8)'s arms (a) and (b) coincide and its report (c) (45/45) is a theorem. (Hi8)'s own text and tag are unchanged.
- `thm:hidacontrol`(2): "its normalisation is not restated here" now continues with the normalisation read on the page (Hida, Invent. p. 545, F4 M2): Hida's $P_n = (1+X) - (1+\ell)^{n+2}$, the P_k of item (3) with $k = n+2$ (F4 §4 item 2).
- Literature pass of 2026-10-05 (`docs/implementation/slot21/missing_literature.md` §7; LIT-15, LIT-20, LIT-23 received): `rmk:thetaclassical`(1) (Katz, Theorem (2), p. 55, for $N \geq 3$; the "iff" from Jochnowitz's local-components paper, Lemma 4.1(b), p. 257), (4) (that paper's §7, pp. 261–264); `rmk:nebcheckable` (NB1) (Cohen–Oesterlé, Théorème 1, pp. 71–72; Shimura Thm. 8.4 cuspidal only, R20-L). The [standard] tags stay; the bib notes of `katz1977`, `jochnowitz_local`, `cohen_oesterle` record the reading.
- Not done here (owner's call, listed): a [C] row for [U]'s (U17)(a) = [E, `cl:lveis`](2) on the ordinary block is in [E] 1.21.0-pre.1, not in [C].

Companions: [E] 1.21.0-pre.1, [U] 0.5.0-pre.3, [L37] 0.5.1-pre.2, [EX] 1.4.1-pre.1.

v1.21.0-pre.1 — 2026-10-05 — hecke-extensions slot 21, R20-L: the Shimura and Brown locators verified

This version. **A pre-release, not owner-released.** Step R20-L (slot-21 plan §0 item 7; condition (C1) of the owner's Q5 ruling, `docs/implementation/slot21/owner_rulings_Q5_1005.md`) read every locator of the nine Shimura `\LEANlit` markers on the owner's scan of [Shi71] Ch. 3 and Ch. 8 (page images), [Bro82] III.8–10 and [AS86] §1.1 (GDZ page images); report `docs/implementation/slot21/R20-L_report.md`. Every marked clause is **verified**; the locators, not the clauses, are restated. **No clause's mathematics changes; no label, tag or status moves; the marker count stays** 140 **on** 65 **labels.**

- The nine markers: "locator unverified (R20-L)" replaced by the verified locators — `hcomm`: Prop. 3.8 (p. 54) as applied in §3.2 (p. 56), for $\det < 0$ the same argument with a diagonal representative (Thm. 3.24 dropped: it does not state commutativity); `hTp`: Thm. 3.24(5) with Prop. 3.17 (Prop. 3.30 dropped: it is the level-N-to-level-1 homomorphism); `hcop`: Prop. 3.16 (Thm. 3.24(3) is the full-operator form); `hTpchi0`, `hTpchi`: Thm. 3.24(5) and Prop. 3.17 carried to $R(\Gamma_0(N), \Delta_0(N))$ by Prop. 3.31 (Shimura's ring carries no character; χ enters through $T(p,p)$); `hUchi`: Thm. 3.34(3) with Prop. 3.33. The transport is [AS86, §1.1, p. 194], where the action of the Hecke algebra on cohomology is stated as well known (with a reference to Kuga–Parry–Sah), not proved.
- [standard] notes: `prop:cosets` (§3.2, Lemmas 3.11–3.13, not §3.1; also its status row); (18) (Thm. 3.24(1), (2), (7): the notation confirmed); `thm:relations` (its note; the proof's "$T(p,p)^{\pm 1}$" corrected to Thm. 3.20's polynomial ring on $T(1,p)$, $T(p,p)$; the coprime clause Prop. 3.16 / Thm. 3.24(3), not "§3.3"; also its status row); `obl:algebra`'s note; `def:nebdata`(2) ((3.3.3), p. 68); the $\Gamma_1(N)$-double-coset statement in the proof of `prop:nebcorr` (printed Lemma 3.29 and Prop. 3.32–3.33,

not "Prop. 3.29"); `thm:nebrelations`'s note; A.1's evidence (Thm. 3.24(1), (2), not Thm. 3.34); `lem:heckerankone`'s secondary pointers (Thm. 3.41 / 3.42 and Prop. 8.5 verified; Diamond–Shurman still not); the two Thm. 8.4 sites (the parabolic part); one [Bro82] note (Cor. III.10.2 with Prop. III.9.5(ii)).
- §1.1's paragraph on the slot-20 `\LEANlit` markers; header and `\date`.

Left to the [GT] owner (not edited here, `lean/` untouched): the `\VERIFYloc` texts of `lit:hecke-Tp-prim`, `lit:hecke-coprime`, `lit:hecke-commute`, the nebentypus paragraph, `conv:hecke`, and `lit:hecke-classical`, whose Ash–Stevens half is **not as cited** (Ash–Stevens §1.1 gives the cocycle formula of T_g, not $\mathrm{cor} \circ c_g \circ \mathrm{res}$; no [C] marker rests on it). Companion pointers unchanged.

v1.20.0 — 2026-10-05 — hecke-extensions slot 20, the close: the release, from v1.20.0-pre.2, with the milestone-M4 \LEAN markers

This version. **RELEASE from v1.20.0-pre.2 under the owner's ruling O6 of 2026-10-04 (releases after the Fable windows F1 and F1′); the milestone-M4 markers clause by clause per L20-6 §8 as amended by L20-6_repairs** (`docs/implementation/slot20/L20-6_F3_audit.md`, `docs/implementation/slot20/L20-6_repairs.md`). Referees: F1 (`docs/implementation/slot20/WP-3_F1_review.md`) and F1′ (`docs/implementation/slot20/F` ACCEPT for [C] (code-map text, no finding); the Lean audit F3 ACCEPT WITH REPAIRS on M4 (one MAJOR, `thm:classical` formalised in a weak form only; repaired), Lean census `decls=1303 proved=1238 proved_mod_lit=37 stated=0 planned=0 sorries=0 global_axioms=0 ... verdict=OK`. **No statement's content changes; no label is added, renamed or removed; no status tag moves.**

- **The milestone-M4 markers** (42 new: 33 `\LEAN`, 9 `\LEANlit`; 140 markers on 65 labels in all, 126 `\LEAN` and 14 `\LEANlit`), clause by clause per L20-6 §8 as amended by L20-6_repairs: `lem:tauexists`(2) last sentence — `lem:m4-tauR-exists`; `def:hecke`, `prop:cosets`(5) — `lem:m4-cosetdata`; `thm:welldef`(a) — `prop:fullassembly-model` (τ over R); `prop:diamond` — `prop:diamond-model`; `prop:fullassembly` — the summand only; `def:operators`(1), (2), (3) — `def:m4-hecke`, (3) only in the assembled form; `thm:relations`(1), its caveat, (2) and `obl:algebra`'s commutativity, (ii), (iii) — `\LEANlit` modulo `hTp` / `hcomm` / `hcop` ([Shi71] Prop. 3.8, Thm. 3.24, Prop. 3.30, locators unverified, R20-L); `prop:lhcompliant`(4), `def:inducedmodule`, `prop:blockmatrices`, `def:levelellmodel` — `def:levelellmodel-model`; `prop:levelcorrespondence`(2), (3), (4)(b)(c) — `prop:levelcorrespondence-model`; `def:nebdata`(2)(3)(5), `lem:nebdata`(1)(2)(4) — `def:nebdata-model`; `prop:nebcorr` — `prop:nebcorr-model`; `thm:nebrelations`(2), (3) — `\LEANlit` modulo `hTpchi0`, `hTpchi`, `hUchi` (Thm. 3.34, 3.35, §3.3, unverified); `lem:merelpath` — `lem:merelpath-model`; `prop:heilbronn` — `prop:heilbronn-model`, `chk:merel-n5`; `lem:cosetrepsN`(4) — `prop:heilbronn-model`. Not tagged: `thm:classical`, (20) (not even its maps, although `def:m4-classical` formalises them), (18), the ring homomorphism of `obl:algebra`, choice-freeness of the level-N / nebentypus operators. §1.1 names the v1.20.0 list.
- Header and `\date`.

Companion pointers: [E] 1.20.0, [U] 0.4.0, [L37] 0.5.0, [EX] 1.4.0 (unchanged).

v1.20.0-pre.2 — 2026-10-04 — hecke-extensions slot 20, the close delta: code-map text

This version. A pre-release, **not owner-released**, for the slot-20 Fable referee F1′, made with [E] 1.20.0-pre.6, [U] 0.4.0-pre.3 and [L37] 0.5.0-pre.5 (list: `docs/implementation/slot20/close_delta.md`). **No statement's content changes; no label is added, renamed or removed; no status tag moves.**

- §22.2: the v1.20.0-pre.1 box gains a "built" paragraph — WP-MEAS, S1 / S2, NB-4, WP-LV3 merged, with their commits.
- §22.1, the §9 row: ι^* over $\mathbb{Z}/\ell^k$ built (WP-LV3(d)).
- Header and `\date`.

Companion pointers: [E] 1.20.0-pre.6, [U] 0.4.0-pre.3, [L37] 0.5.0-pre.5, [EX] 1.4.0 (unchanged).

v1.20.0-pre.1 — 2026-10-04 — hecke-extensions slot 20, WP-2: code-map text for slot 20

This version. A pre-release, **not owner-released**, for the slot-20 Fable referee F1 (WP-3), made with [E] 1.20.0-pre.1, [U] 0.4.0-pre.1 and [L37] 0.5.0-pre.1 (list of new and changed labels: `docs/implementation/slot20/WP-2_drafts.md`). **No statement's content changes; no label is added, renamed or removed; no status tag moves**; Appendices B and D are unchanged.

- §22.2: a v1.20.0-pre.1 box — WP-LV3, WP-MEAS, NB-4 planned, WP-JET value-neutral; their mathematics in [E] 1.20.0-pre.1 ([E, `sec:lvmeasure`], [E, `sec:lvdepth2`]).
- §22.1, the §9 row: ι^* over $\mathbb{Z}/\ell^k$ planned (WP-LV3(d)); Propositions 9.6, 9.4 cover it.
- Not in this pre-release: the milestone-M4 `\LEAN` markers (slot-20 L20-6 and its repairs); the Shimura locators stay marked unverified until R20-L reads the scan.
- Header and `\date`.

Companion pointers: [E] 1.20.0-pre.1, [U] 0.4.0-pre.1, [L37] 0.5.0-pre.1, [EX] 1.4.0 (unchanged).

v1.19.0 — 2026-10-04 — hecke-extensions slot 19, the close: the release, from v1.19.0-pre.6, with the milestone-M3 `\LEAN` markers

This version. **RELEASE from v1.19.0-pre.6 under the owner's ruling O3, on F1 / F1-delta ACCEPT WITH REPAIRS; the milestone-M3 markers clause by clause per L19-6 §8 as amended by L19-6_repairs §3 and L19-4 §3** (`docs/implementation/slot19/L19-6_F3_audit.md`, `docs/implementation/slot19/L19-6_repairs.md`, `docs/implementation/slot19/L19-4_report.md`). Referees: F1 (`docs/implementation/slot19/WP-3_F1_re`
and F1-delta (`docs/implementation/slot19/F1delta_referee_report.md`) ACCEPT WITH REPAIRS, repairs at pre.2 and pre.6; the Lean audit F3 ACCEPT WITH REPAIRS on M3, its eight MINORs repaired in Lean (Lean census `decls=1136 proved=1077 proved_mod_lit=19 stated=0 planned=16 sorries=0 global_axioms=0 verdict=OK`). **No statement's mathematics changes; no label is added, renamed or removed; no status tag moves**; Appendices B and D are unchanged (markers are not counted). The Shimura locators of v1.19.0-pre.5 remain marked unverified (from memory; a scan is requested).

- **The milestone-M3 markers** (17 new, all `\LEAN`; one `\LEANlit` becomes `\LEAN`; 98 markers on 48 labels in all, 93 `\LEAN` and 5 `\LEANlit`): `prop:model7h4`(ii) — the marker `prop:model7h4-ii-dim` ("the dimension only") replaced by `prop:model7h4-ii`: the acyclicity in the three forms of (37) and $\dim X = 2$; topological contractibility not formalised; (37) — `prop:model7h4-ii` (over $\mathbb{Z}$, on the model), `thm:xbulk-repaug` and `lem:xbulk-contractz` (after $\otimes R$); `rmk:orbifold` — `lem:xbulk-orient` (the direct-sum decomposition), `thm:xbulk-repaug` (the resolution), `thm:model7-rmkorbifold` (the isomorphism under $6 \in R^\times$, every acyclic descriptor with (H4)(iv), L in the $(G, \Gamma, \Delta, e, \rho)$ form; the statement, not the argument; K outside the descriptor framework not claimed); `lem:tauexists`(2) — `lem:tauexists-2-model` ((1) not); `thm:welldef`(b) — `thm:welldef-b` (τ over R, on H^n_{eq} itself); (c) — `thm:welldef-c` now at the whole of (c); (a) — any two choices of representatives; `def:hecke` — the caveat names τ over R on the model; `cor:stablecert`(1) — `\LEANlit` modulo Eichler–Shimura at level one (`hES`) in place of "modulo Lemma 30.4"; (2)'s first sentence — `cor:stablecert-uncond` (`\LEAN`); the cone chapter on the model (L19-4) — §15's standing hypotheses `thm:model7-pairhyp`,

`def:pair` `def:xbdry` and `thm:xbdry-pair`, (B1)–(B3) `def:model7bd` (the component map of (B2) not formalised), `lem:etaexists`, `lem:etaunique` and `thm:conewelldef`(b) `cor:model7-etaexists`, `obl:eta`'s link to Definition 15.5 `thm:etaspec-pairhomotopy` and `cor:etap2-pairhomotopy` (that realisations of the tables exist is not formalised). Unchanged: `lem:conerelative`(5) "given (4)"; `lem:heckerankone`, `prop:cohforms`. Not tagged: (H4)(ii) as contractibility of a space, `lem:tauexists`(1), `rmk:orbifold`'s general K and its proof route.

- `prop:model7h4`: the v1.19.0-pre.5 parenthesis at the (H4)(ii) marker ("this marker's scope moves only at the slot-19 close's tagging pass") is replaced by the marker's new scope.
- §1.1: the v1.19.0 list named. Header and `\date`.

Companion pointers: [E] 1.19.0, [U] 0.3.0, [L37] 0.4.0, [EX] 1.4.0 (unchanged).

v1.19.0-pre.6 — 2026-10-04 — F1-delta m-C1

A pre-release, **not owner-released**: the slot-19 F1-delta's m-C1 — one sentence in Appendix entry A.6 (Shimura's naming of $T(1,q^2)$), textual only. No statement, tag, label or count change.

v1.19.0-pre.5 — 2026-10-04 — hecke-extensions slot 19, WP-2 continued: Shimura's notation, H4(ii), WP-BIL and WP-LIN

A pre-release, **not owner-released**, made with [E] 1.19.0-pre.6, [U] 0.3.0-pre.4 and [L37] 0.4.0-pre.3.

- **Shimura's notation** (`docs/implementation/slot19/shimura_convention_check.md`): §10.1, Definition 10.4(3), Theorem 10.8's caveat and proof, the proof in §23.7 and Appendix entry A.1's evidence called the full operator (18) "Shimura's $T(1,p^\nu)$". In Shimura's notation $T(1,p^\nu)$ is the single double coset $\Delta_{p^\nu}^{\mathrm{prim}}$ and $T(p^\nu)$ the sum of all double cosets of determinant p^ν; the text now says $T(p^\nu)$ for the full operator (and $T(p)T(p^\nu) = T(p^{\nu+1}) + pT(p,p)T(p^{\nu-1})$, $T(M)T(N) = T(MN)$). The mathematics — the definition, the recursion, $T_p^2 = T_{p^2} + p^{n+1}$ — is unchanged; the later uses of $T(1,p^2)$ for the primitive coset (§28) were already in Shimura's notation. The locators (§3.1–3.3, Thm. 3.24, Thm. 3.34) are from memory: the book is not in the library and is to be checked when a scan is available.
- Proposition 7.4, the (H4)(ii) clause of its marker: "contractibility is stated in [GT], not proved" was superseded by slot-19 L19-3 (`model7_acyclic`: the acyclicity used through (37), proved combinatorially). A statement of fact; the marker itself moves only at the close.
- §22.2: a v1.19.0-pre.5 box — built: WP-BIL, WP-LIN.
- Header and `\date`.

No statement's mathematics changes; no label is added, renamed or removed; no tag moves. Companion pointers: [E] 1.19.0-pre.6, [U] 0.3.0-pre.4, [L37] 0.4.0-pre.3, [EX] 1.4.0 (unchanged).

v1.19.0-pre.4 — 2026-10-04 — hecke-extensions slot 19, WP-2 continued: WP-LV2 and WP-HL built

A pre-release, **not owner-released**, made with [E] 1.19.0-pre.5, [U] 0.3.0-pre.4 and [L37] 0.4.0-pre.2 (change table `docs/implementation/slot19/WP-2_pre5_changes.md`). **Code-map text only.**

- §22.1: "At v1.19.0-pre.4" in the rows of §29 (WP-LV2: the twisted route built), §27 (WP-HL built, `HeckeLambdaValidation`) and §30 (WP-LV2's free part and deep rungs).
- §22.2: a v1.19.0-pre.4 box — built: WP-LV twisted (WP-LV2), WP-HL.

- Header and `\date`.

No statement's content changes; no label is added, renamed or removed; no tag moves; Appendices B and D are unchanged; the milestone-M3 `\LEAN` markers still wait for the slot-19 close (F3's list). Companion pointers: [E] 1.19.0-pre.5, [U] 0.3.0-pre.4, [L37] 0.4.0-pre.2, [EX] 1.4.0 (unchanged).

v1.19.0-pre.3 — 2026-10-04 — hecke-extensions slot 19, WP-2 continued: the slot-19 code maps as built

A pre-release, **not owner-released**, made with [E] 1.19.0-pre.3, [U] 0.3.0-pre.3 and [L37] 0.4.0-pre.1 after the slot-19 Fable referee F4 (`docs/implementation/slot19/R19-6_F4_referee_report.md`; change table `docs/implementation/slot19/WP-2_pre3_changes.md`). **Code-map text only.**

- §22.1: "At v1.19.0-pre.3" in the rows of §30 (WP-LC and WP-LV untwisted: built and merged), §29 (WP-LV twisted: not built) and §27 (WP-HL: not built; R19-2's prototype).
- §22.2: a v1.19.0-pre.3 box — built: WP-LC, WP-LV untwisted, WP-NF, NB-2; not built: WP-HL, WP-LV twisted.
- Header and `\date`.

No statement's content changes; no label is added, renamed or removed; no tag moves; Appendices B and D are unchanged; the milestone-M3 `\LEAN` markers still wait for the slot-19 close (F3's list). Companion pointers: [E] 1.19.0-pre.3, [U] 0.3.0-pre.3, [L37] 0.4.0-pre.1, [EX] 1.4.0 (unchanged).

v1.19.0-pre.2 — 2026-10-03 — hecke-extensions slot 19, WP-3 repairs: typography

A pre-release, **not owner-released**. The slot-19 referee F1 (`docs/implementation/slot19/WP-3_F1_referee`
ACCEPTED v1.19.0-pre.1 with one typographic note (m-C1): in the §22.1 rows of §27, §29 and §30 the space before the full stop preceding "Planned at v1.19.0-pre.1" is removed and each planned sentence ends with a period. The §30 row's WP-LV pointer now reads "on a saturated block (`stable` at $K = 1$)", following [E] 1.19.0-pre.2's repair M-1 of [E, `cl:lveis`](3). **No statement, tag, label or count changes.** Companion pointers: [E] 1.19.0-pre.2, [U] 0.3.0-pre.2, [EX] 1.4.0 (unchanged).

v1.19.0-pre.1 — 2026-10-03 — hecke-extensions slot 19, WP-2: code-map text for slot 19

This version. A pre-release, **not owner-released**, for the slot-19 Fable referee F1 (WP-3). **No statement's content changes; no label is added, renamed or removed; no tag moves**; Appendices B and D are unchanged.

- §22.2: a v1.19.0-pre.1 box — WP-LC, WP-LV, WP-HL, WP-NF and NB-2, what each reads of this document and that none is built.
- §22.1: "Planned at v1.19.0-pre.1" in the rows of §27 (WP-HL), §29 (WP-LV, twisted) and §30 (WP-LC; WP-LV, untwisted).
- Not in this pre-release: the milestone-M3 `\LEAN` markers (they follow slot-19 L19-6 and the Lean audit F3), and the measurements of Remark 28.16(4)'s window (the open repair of the slot-18 F1, pending slot-19 WP-RC's runs).
- Header and `\date`.

Companion pointers: [E] 1.19.0-pre.1, [U] 0.3.0-pre.1, [EX] 1.4.0 (unchanged).

v1.18.0 — 2026-10-03 — hecke-extensions slot 18, the close: the release, from v1.18.0-pre.2, with the milestone-M2 \LEAN markers and an erratum

This version. Released at the slot-18 close (WP-18) under the owner's ruling O3 of 2026-10-03 ("release [C] 1.18.0, [E] 1.18.0, [EX] 1.4.0"), on the slot-18 referee F1's ACCEPT WITH REPAIRS (repairs at pre.2). **No statement's content changes; no label is added, renamed or removed; no tag moves** relative to pre.2; Appendices B and D are unchanged (markers are not counted).

- **Erratum** (**confirmed by the owner**, 2026-10-03): the proof of `thm:welldef`(c) — representatives $\{\gamma_1 s_j\}$ of $\Gamma_1''\backslash\Gamma$, not $\{s_j\gamma_1^{-1}\}$ — recorded at pre.2 and **confirmed** by the slot-18 Lean audit F3 with a counterexample ($\Gamma'' = \Gamma_0(p)$, $\gamma_1 = S$: the p elements $\gamma_1^{-1}s_j$, $j < p$, lie in one coset), now written into the proof's parenthesis. The statement, its tag and its conclusion are unchanged.
- **The milestone-M2 markers**, clause by clause per F3 §8 and its repairs (39 new markers, 36 \LEAN and 3 \LEANlit; 22 newly tagged labels; Lean census `decls=823 proved=764 proved_mod_lit=34 stated=1 sorries=1 global_axioms=0 hyp_bad=0 verdict=OK`): G2 — `def:model7` (the simplex list), `def:model7rel` (the eight rows), `lem:model7coincidences`(1),(2), `prop:model7h4` ((i) without $K^u \cong X$; (ii) the dimension only; (iii) (FD1), (FD2); (iv)), `def:eq_cochains`, `lem:eqgroupoid`, `lemma:d_maps_C_eq`, `thm:explicit_complex_equiv`(1), `thm:welldef` (the first bullet, (a) at cochain level, (c)'s transported half), `def:hecke`, `prop:cosets`(3) now with "group isomorphism" and (5) for any transversal, `prop:mu`(1), `prop:bdrycosets`(2) with the cusp characterisation, `prop:bdrychar`(1)'s first sentence; G3 — `obl:taubdry`(i)–(iv) and `obl:eta`(i)–(iv) (verified at the listed (p,ν)), `def:pairhomotopy`, `lem:conerelative`(2),(3),(5) (given (4)), `cl:les`(2)'s injectivity and Euler identity, `prop:coneambiguity`(2) over a field, `lem:etaexists`, `lem:etaunique`, `thm:conewelldef`(a),(b),(d), `prop:coneindep`(1); G10 — `lem:heckerankone` and `prop:cohforms`(3), (4)'s second surjection (\LEANlit, modulo `hES` / `hF`), `lem:freepart`(2) in full. Not tagged: (H4)(ii)'s contractibility, (FD3), (FD4), `thm:welldef`(b) and (c)'s general case, `thm:explicit_complex_equiv`(2), `obl:taubdry`(v), `obl:eta`(v),(vi), `lem:tauexists`, `rmk:orbifold` (G4: only the abstract projective reduction is in [GT], slot-18 L18-7).
- §1.1: the v1.18.0 list named. Header and `\date`.

Companion pointers: [E] 1.18.0, [EX] 1.4.0, [U] 0.2.1 (unchanged).

v1.18.0-pre.2 — 2026-10-03 — hecke-extensions slot 18, WP-3 repairs: the referee F1's repairs, the slot's code maps, and an erratum

This version. A pre-release, **not owner-released**; the release at the slot-18 close on the owner's yes (ruling O3). The slot-18 Fable referee F1 (`docs/implementation/slot18/WP-3_F1_referee_report.md`) read v1.18.0-pre.1 as **ACCEPT WITH REPAIRS**, 0 blocking, 1 major (M-1), 2 minor; all applied here (`docs/implementation/slot18/WP-3_repairs.md`). No statement's content changes, no label is added, renamed or removed; two tags move; Appendix B's counts are unchanged (the `IL3`, `IL4` items are not rows of it).

- **M-1** `IL5`: the proof route is `prop:strictlift`(3)'s argument with `prop:isomodelell`(4),(5) in place of `rmk:borelfree` (the model over Y' is projective, not free); the hypothesis "$\mathcal{I}^{(\ell)}(Y')$ is an ℓ-local isotropy model over Y'" is stated, (T1) is over Γ'', and the strict identity is named a certificate of the built map, not a hypothesis. The [proved here] tag is unchanged; the item gains its [verified computationally][a] line (slot-18 WP-7, arm (IL5)). **m-C2**: at $\ell = 2$ the item's checks are a code-path check.
- **m-C1** `rmk:levelNscale`(4): the figures given per cell (99330 at $k = 2$: $m_N = D_N =$ 304 128, 10 991 MiB; 90090 at $k = 4$: $m_N =$ 290 304, $D_N =$ 870 912, 9622 MiB). And the *outcome* (slot-18 WP-5): the structure-aware fill model **failed** its factor-1.5 rule, so

`hook_rss/v1` and `CokernelCostModel`'s law stay (a finding); W5-F1's root cause is the generator order (the fill at 99330, $k = 2$, is 54.45 M); the sorted order is built, value-free, gated on 21 rungs; the two window measurements are pending.

- **Code maps as built** (slot-18 WP-5, WP-6, WP-7, merged): §28 ("Built at v1.18.0-pre.2"), §29 (items (iv)–(vi) built; `rmk:nebcost`(4)), §31 ("Run at v1.18.0-pre.2": (v), (vi); the Scope paragraph), §22.1's three rows, §22.2's box; each pre.1 plan kept for the record.
- **Tag move** (the referee F1's C-4: "move both at the close on WP-7's `IL5` + (vi) pass"; WP-7 passed, `--all` 156/156): `IL3`, `IL4` from [verified computationally][a] *partial* to [verified computationally][a].
- **Erratum, for the owner's confirmation** (slot-18 L18-2, the Lean proof `GL2.Model.thetaG_transport` the proof of `thm:welldef`(c) took $\{s_j\gamma_1^{-1}\}$ as representatives of $\Gamma_1''\backslash\Gamma$ for $g_1 = \gamma_1 g \gamma_2$; that family is not a transversal in general. The correct representatives are $\{\gamma_1 s_j\}$, with $(\gamma_1 s_j)^{-1} g_1 = \delta_j \gamma_2$; with them the operator is reproduced verbatim. The statement (c), its tag and its conclusion are unchanged; only the proof text is corrected (and the sentence on which factors cancel).
- **Made explicit** (slot-18 L18-4, `GL2.Cone.eta_exists`'s hypothesis `hjaug`): `def:pair` states $\varepsilon \circ j = \varepsilon_\partial$, true because j is the inclusion of a subcomplex, and `lem:etaexists`(V) cites it; no statement changes.

Companion pointers: [E] 1.18.0-pre.2 (the referee F1's M-2 and minors; Prediction `pr:depth`'s $(2, 3)$ datum), [EX] 1.4.0-pre.2, [U] 0.2.1 (unchanged).

v1.18.0-pre.1 — 2026-10-03 — hecke-extensions slot 18, WP-2: contract first for slot 18, and the Eichler–Shimura locator

This version. A pre-release, **not owner-released**, for the slot-18 Fable referee F1 (WP-3); the release at the slot-18 close only on its acceptance and the owner's yes (ruling O3 of 2026-10-03; the `\LEAN` tags of slot 18's F3 list enter only at that release, WP-18). **No statement's content changes, no label is renamed or removed, no tag moves**; one checkable item is added; Appendix B's counts are unchanged.

- **New item** `IL5` in `rmk:isomodellcheckable` ([proved here] given (T1)–(T3) of the built map; committed before slot-18 WP-7's code, rule 20): the strict action on a synthetic refined descriptor agrees on cohomology with the transported one. `app:labelindex` gains it.
- **Contract first** (rule 21; nothing built): §28's code map (slot-18 WP-5) and `rmk:levelNscale`(4)'s paragraph *the structure-aware fill model* (a cost statement, with its replacement rule; the two window measurements; the order change gated by bit-identity); §29's code map (slot-18 WP-6: items (iv) $W_K(\mathbb{F}_q)$ at $d(\chi) \geq 2$, (v) the model-route T_{p^2} through $[\mathrm{diag}(1, p^2)]_\chi + \chi(p)p^n$ against `thm:nebrelations`(2), (vi) the WP-6 audit minors) and `rmk:nebcost`(4)'s pointer; §31's code map (slot-18 WP-7: (v) the strict lift on a refined descriptor and its MP requirements, (vi) `IL3` over $\mathbb{F}_9$ and $\mathbb{Z}/9$); §22.1's three rows; a new box in §22.2.
- **Locator** (the slot-17 referee F1's m-C4): `lem:heckerankone`'s Eichler–Shimura input is [DI95, Thm. 12.2.2, Prop. 12.4.10], verified on page images (slot-17 L17-5 repairs); [Shi71, Ch. 8] kept as an unverified secondary; `app:status`'s row follows.

Companion pointers: [E] 1.18.0-pre.1 (new: Claims `cl:seedtwist`, `cl:selfmirror`, `cl:dimformula`, Prediction `pr:depth`; `pr:kfonset` generalised), [EX] 1.4.0-pre.1, [U] 0.2.1 (unchanged).

v1.17.0 — 2026-10-02 — hecke-extensions slot 17, the close: the release, from v1.17.0-pre.5, with the first \LEAN markers

This version. The slot's X bump, whose text v1.17.0-pre.1 … pre.5 carried, is released at the slot-17 close (WP-18) under the owner's ruling O3 of 2026-10-01 ("release [C] v1.17.0, [E] 1.17.0, [U] 0.2.1 and [EX] 1.3.0 on F1's accept, with the \LEAN tags"). Its referee (slot-17 WP-3, Fable, `docs/implementation/slot17/WP-3_referee_F1.md`) was **ACCEPT WITH REPAIRS** for this document, 0 blocking, 1 major (M-1), 5 minor; the repairs were applied at pre.3, and M-1 was settled by the owner's ruling of 2026-10-02 at pre.4. **No statement is added, no label is renamed or removed, and no tag moves** relative to pre.5; Appendices B and D are unchanged.

Over pre.5: the Lean milestone M1.

- **The first markers**, exactly per `docs/implementation/slot17/lean_taggable_labels.md` (the F3 audit's clause-scoped list after its repairs; Lean census `decls=440 proved=383 proved_mod_lit=20 stated=13 sorries=13 global_axioms=0 hyp_bad=0 verdict=OK`): 39 \LEAN and 3 \LEANlit on 20 labels — Part 0: `prop:symconv`(2),(3), `lem:detsym`, `def:contragredient` ((11)), `prop:pairing`, `lem:symdual`; G1: `def:cosetreps`, `prop:cosets`(1)–(5) with the list's restrictions, `lem:cocycle`, `lem:bounds`(1)–(4) restricted, `prop:tworoutes`, `prop:bdrycosets`(1)–(4), `prop:bdrychar`(2); G10: `def:satdata`, `prop:saturation`(1)–(4) and (5)'s first sentence, `prop:satdefect`(1)–(3), `lem:freepart`(1)–(3), `cor:stablecert`(1) and (2)'s first sentence (\LEANlit, modulo Lemma 30.4), `prop:cohforms`(1),(2),(4) restricted, `prop:hidaalgconst`(1) (\LEANlit, modulo Theorem 27.11(3)), `rmk:thetaforms`(1)'s two power-series facts. Not tagged: `lem:heckerankone`, `prop:cohforms`(3),(5),(6), `prop:mu`, `prop:bdrychar`(1),(3), `cor:stablecert`(3), `lem:bounds`(5),(6), and the G3 cone statements (milestone M3).
- §1.1: the marker is clause-scoped, a definition may carry it, and the v1.17.0 list is named.

Companion pointers: [E] 1.17.0, [U] 0.2.1, [EX] 1.3.0.

v1.17.0-pre.5 — 2026-10-02 — hecke-extensions slot 17, WP-2: verified locators and the WP-7 code map

This version. A pre-release, **not owner-released**. Pointers proposed for WP-2 by other slot-17 packages; no statement, tag, label or count changes (`docs/implementation/slot17/WP-2_report.md` §8).

- **R17-4** (`docs/implementation/slot17/R17-4_report.md`, locators read on page images). (NB4)(c): the [standard:] input is [AS86, Lemma 1.1.1, p. 194] (the long exact sequence; the referee's recollection "Lemma 1.1.3" is restriction / corestriction) and [AS86, Def. 1.1.2, p. 194; Lemma 1.1.4, p. 195] (Shapiro for a compatible pair), **verified**; the "not verified" flag is dropped, the [proved here] tag of v1.17.0-pre.4 stands and the [verified computationally]^{r} fallback is not triggered; compatibility (b) is stated in Ash–Stevens' orientation ($\Gamma\Delta_0(\ell)$), with (c). Proposition 13.4 and (20): "in exactly this form" / "in this exact form" dropped — Ash–Stevens give the right-module form on homogeneous cochains (p. 194). Remark 24.13: item (3) cites [AS86, Lemma 2.1, p. 204] for the inclusion only, the equality and the twist bookkeeping being [E, `cl:glover`]'s and Proposition 24.10's; item (2) cites [Joc82a, Thm. 4.1, p. 276]. Lemma 30.4: the eigenbasis from [DI95, §§3.5–3.6, pp. 54–55] (in the library, verified; new bibliography key), Shimura and Diamond–Shurman kept as unverified secondary pointers. (LN5): [Mer96, Théorème 2, p. 72], verified, with its hypotheses ($p > 3$, $p \mid (N-1)/\gcd(N-1, 12)$, in this document's letters). The [AS86] sites of Proposition 23.23's [standard:] gain the page and lose "not verified". `app:status`: the rows of `prop:barformula`, `rmk:thetaclassical`, `rmk:levelNcheckable`, `rmk:nebcheckable`, `lem:heckerankone` name the locators; no count moves.

- **WP-7** (`docs/implementation/slot17/WP-7_report.md` §5, checked against `docs/implementation/slo
(IL3)'s and (IL4)'s status sentences record the runs ((IL3) at level N against the pipeline over $\mathbb{F}_3$ only, MINOR-3; (IL4) over $\mathbb{Z}/3^e$ and the pair and cone); the code map records the descriptor as data, the synthetic refined descriptor, and arm (REF) as a descriptor round trip (MINOR-1); the §22.1 row and the §22.2 box follow. The tags stay [verified computationally]^{a} partial; the move is for the next referee.
- **Lean** (`docs/implementation/slot17/lean_taggable_labels.md`): §1.1 notes that Corollary 30.6 will take `\LEANlit` (its [GT] proof carries Lemma 30.4 as a hypothesis); no marker is used at this version.
- The [L37] referee F4's pointers: [E, `pr:kfonset`] (c) records the window's last cell off 37 in [E] 1.17.0-pre.5; no sentence of this document states "$e = 2$ at the eleven cells".

Companion pointer: [E] 1.17.0-pre.5; [EX] 1.3.0-pre.4 and [U] 0.2.1-pre.2 unchanged.

v1.17.0-pre.4 — 2026-10-02 — hecke-extensions slot 17, WP-2: the owner's rulings on the referee F1's M-1 and B-1

This version. A pre-release, **not owner-released**. Two owner rulings of 2026-10-02 on the slot-17 Fable referee F1 (`docs/implementation/slot17/WP-3_referee_F1.md`):

- **M-1** (NB4)(c): the sharp form at $n = 0$ is tagged [proved here] (given the one [standard:] input named), as the referee proposed, with the locator [AS86, §1.1] flagged **not verified** — the pattern of `lem:heckerankone`. At pre.3 the referee's alternative ([verified computationally]^{r} only, [proved here] on verification) had been applied; the owner chose the proposed tag. Reading the locator stays on the owner's scan list (`docs/implementation/slot17/L17-5_report.md` §3). `app:status`: `rmk:nebcheckable`'s tag set already holds [proved here], so it is unchanged and no count moves; the counts paragraph records it. The §29 scope paragraph follows.
- **B-1** (the companion's [E, `pr:latticesplit`] (R), restated on *congruence-free* classes in [E] 1.17.0-pre.4): the two dependent sentences repaired at pre.3 (m-C3) take their final wording — (LN6)'s route-R3 sentence (level 1 shows Loewy length ≥ 3 only on classes that are not congruence-free) and §30's scope paragraph (the pre.2 Kummer-free form contradicted on the forms side, the regular $\ell = 139$ among the primes; (R) now on congruence-free classes).

No label added, renamed or removed; no count moves. Companion pointer: [E] 1.17.0-pre.4, [EX] 1.3.0-pre.4 in the tree.

v1.17.0-pre.3 — 2026-10-02 — hecke-extensions slot 17, WP-2 resumed: the referee F1's repairs

This version. A pre-release, **not owner-released**. The slot-17 Fable referee F1 (`docs/implementation/slot17/WP-3_referee_F1.md`) read v1.17.0-pre.2 as **ACCEPT WITH REPAIRS**, 0 blocking, 1 major (M-1), 5 minor, and WP-6's proposed code-map text as acceptable with repairs (M-2, m-W1–m-W3). Every repair is applied as worded except where stated; the referee file's "Repairs" table lists each.

- **M-1** (NB4)(c): the orientation step is proved in this document's conventions (Proposition 8.2, Definition 12.1, Definition 29.1(5)): $0 \to \mathrm{Sym}^{\ell-1-i} \otimes \det^{i} \to I \to \mathrm{Sym}^{i} \to 0$ is exact, so Sym^{i} is the untwisted constituent, and the second [standard:] (the composition factors) is no longer needed. The remaining [standard:] input carries the locator [AS86, §1.1], **not verified** (no copy in the library); the referee's alternative therefore applies — the sharp form at $n = 0$ is re-tagged [conjectural] → [verified computationally]^{r}, and [conjectural] → [proved here] is made when the locator is verified. `app:status`: the `ash_stevens` cite gains [§1.1]; the tag set already holds [verified computationally]^{r}, so no count moves.

- **M-2, m-W1–m-W3** §29's code map: slot-17 WP-6 recorded as built at this version (merged, audited ACCEPT): the sparse cokernel with a character over $\mathbb{F}_q$, and $H_{N,p^\nu,\chi}$ from Mer_{p^ν} — *tested, not proved* to be T_{p^ν} (Merel at composite determinant is not a statement of this document); the model-route double coset of Proposition 29.9 not built. The §22.1 row and §22.2's v1.17.0-pre.1 box follow.
- **WP-5** §28's code map: slot-17 WP-5 recorded as built (merged, audited ACCEPT WITH REPAIRS): the coset products in `int64` then `__int128`, bit-identical below the old reach; the ladder to $N = 90090$ at $k = 2$, 60060 at $k = 4$; the predictions' verdicts held and their numbers did not all hold (WP-5 audit MINOR-1). `rmk:levelNscale`(4): the $N = 99330$ point (an open measurement and a failed prediction, WP-5 audit MINOR-3).
- **m-C1** `rmk:sattor`(4): "every order is saturated on its $\mathbb{Z}_\ell$-dual". **m-C3** two sites cited [E] [E, `pr:latticesplit`] as "the census at level 1 never shows Loewy length ≥ 3": (LN6)'s route-R3 sentence and §30's scope paragraph, both repaired; pre.2's changelog sentence "nothing in this document cites (R)" is corrected in place. **m-C4** `lem:heckerankone`: [DS05, Thm. 5.5.4] named, still not verified. **m-C5** §31's code map, item (iv): a control cell with $-I \mapsto -1$. **m-C2** (gate `n12_held == 240` in arm (X4)) is a carry for a code package; no text change.

No label added, renamed or removed; no tag set of a numbered statement changes; no count moves. Companion pointer: [E] 1.17.0-pre.3 in the tree, whose [E, `pr:latticesplit`] (R) awaits the owner's ruling on the referee's B-1 (`LN6`'s and §30's repaired sentences follow its final form).

v1.17.0-pre.2 — 2026-10-02 — hecke-extensions slot 17, WP-2, second pass: the eigenbasis citation of Lemma 30.4

This version. A pre-release, **not owner-released**, for the same referee as pre.1 (slot-17 WP-3, Fable, window F1). One citation repair, the one proposed in the addendum of `docs/implementation/slot17/WP-3_referee_brief.md` (from slot-17 L17-5's report §7, finding 2, and its audit's (a)):

- `lem:heckerankone`'s proof cites its two [standard:] inputs separately. *The eigenbasis* of $S_k(\mathbb{C})$ (commuting Hecke operators, self-adjoint for the Petersson product): [Shi71, Thm. 3.41] or [DS05, §5.5], in place of pre.1's Shimura Thms. 3.24, 3.29, 3.34, 3.35. **Neither locator is verified** (rule (A2)): neither book is in the project's library, and both pointers were relayed by L17-5 unread. *Multiplicity one* (the distinctness of the systems): [Ser73, Ch. VII, §5, Thm. 7 and its Cor. 1, p. 102; Prop. 11, p. 101], verified on page images by L17-5. Serre's §5 gives only this half.
- `app:status`'s row for `lem:heckerankone` names [Shi71, Thm. 3.41] and [DS05, §5.5]; the bibliography gains `diamond_shurman`.

No statement, tag, label or count changes; no number moves. The companion's 1.17.0-pre.2 (in the tree) restates [E, `pr:latticesplit`] (R) again (owner ruling O-R17-2a of 2026-10-02); nothing in this document cites (R). *(Corrected at v1.17.0-pre.3: two sites do — §30's scope paragraph and (LN6)'s route-R3 sentence; both repaired there.)*

v1.17.0-pre.1 — 2026-10-01 — hecke-extensions slot 17, WP-2 (drafted during the slot-16 close): the deferred items, for one referee

This version. A pre-release, **not owner-released**. It drafts, for a single referee (slot-17 WP-3, Fable, window F1), the items that the slot-16 release left out because they are new mathematics (`docs/implementation/slot16/contract_release_report.md` §4), and writes slot 17's code-map texts before the code (rule 21). Release at the slot-17 close only on the referee's acceptance and the owner's yes (ruling O3 of 2026-10-01). Report `docs/implementation/slot17/WP-2_report.md`; referee brief `docs/implementation/slot17/WP-3_referee_`

Statements.

- **New: Remark 30.10**, in a new last subsection §30.6 of §30, so that no statement number of v1.16.0 moves ([proved here]; slot-16 WP-5, `WP-5_tor_note.md` §1, §5): the saturation defect's Tor term $I = \mathrm{Tor}_1^{\mathbb{Z}_\ell}(\mathbb{T}^{\mathrm{sat}}/\mathbb{T}, \mathbb{F}_\ell) = (\mathbb{T} \cap \ell\mathbb{T}^{\mathrm{sat}})/\ell\mathbb{T} = \ker(\mathbb{T}/\ell \to \mathrm{End}(L/\ell))$, its dimension $\varepsilon_1^{\mathrm{sat}}$, its reading at a stable K, and its distinction from $\mathrm{Tor}_1^{\mathbb{T}}(L, \mathbb{T}/\mathfrak{m})$. Over the note: item (4)'s "the converse fails" is proved here by an example ($\mathbb{T}^\vee$ over the non-Gorenstein order of Corollary 30.6(2)), which the note asserted without one.
- **(NB4)** (slot-16 WP-11, `WP-11_report.md` §1, §4): (a) the sharp form at $n = 0$ recorded as [verified computationally]^{r} (132/132, $\ell \le 43$), with the orientation it pins; (b) the $n > 0$ reading (Clebsch–Gordan, the H^0 correction at $i = n$; 240/240, reported); (c) a proof of (a) by the Jordan–Hölder long exact sequence, **proposed**: the referee accepts it ([conjectural] → [proved here] for the sharp form at $n = 0$) or downgrades the re-tag to [conjectural] → [verified computationally]^{r}. The general statement stays [conjectural].
- `rmk:levelNscale`(4): slot-16 WP-13's F-3 — the fitted fill law underestimates `final_nnz` at composite N, $k > 2$, by 1.47–6.80 ([verified computationally]^{a}); a statement about the cost model only.
- `lem:heckerankone`'s proof: the eigenbasis of $M_k(\mathbb{C})$ is cited from Petersson's adjointness ([Shi71] Ch. 3, **VERIFY**), the multiplicity-one part from Serre's *Cours* Ch. VII §5 with verified page locators (slot-17 L17-5, relayed by the lead): Serre's §5 alone gives the distinctness of the systems, not the existence of the basis. No statement or tag changes. (Its locators were replaced at v1.17.0-pre.2, below.)

Code maps, contract first (no statement changes). §28: the table's reach beyond composite $N \approx 5.5 \cdot 10^4$ (slot-17 WP-5, F-1). §29: the sparse cokernel with a character and $T_{p^\nu}^{(N,\chi)}$, $\nu \ge 2$ (WP-6). §31: (IL3) at level N, (IL4) over $\mathbb{Z}/3^e$, the pair and cone half of `prop:isomodelell`(5), and what the MP constraint requires of a synthetic refined descriptor (WP-7). §30: slot-16 WP-5's `HeckeExt/HeckeSaturationDefect` recorded as built. §22.1's row and a §22.2 box follow.

Other. §1.1: the `\LEAN` / `\LEANlit` marker (the Lean programme's Q9, owner-ruled 2026-10-01), defined and **unused**; it is not a status tag and is not counted. Appendix B: one line added, two tag sets grown (its counts paragraph). Appendix D: `sec:sattorsec`, `rmk:sattor`. Not in this version: slot-17 R17-1's double-divisibility observation (not merged when this draft was written; [E]'s owner may add it in a later patch). Non-normative: the header comment, the date line; the companion pointer stays at [E] v1.16.3 (its 1.17.0-pre.1 is in the tree, unreleased).

v1.16.0 — 2026-10-01 — hecke-extensions slot 16, the close: the release, from v1.16.0-pre.3

This version. The slot's X bump, whose text v1.16.0-pre.1, pre.2 and pre.3 carried, is released at the slot-16 close under the owner's ruling O2 of 2026-10-01 ("release [C] v1.16.0 and [U] 0.2.0 since WP-14 accepted and the repairs are confirmed"). Its referee (slot-16 WP-14, Fable, `docs/implementation/slot16/WP-14_referee.md`) was **ACCEPT WITH REPAIRS**, 0 blocking, 1 major (C-1), 8 minor; every repair was applied at pre.3 and **confirmed**, 16 of 16 findings with [U]'s, by `docs/implementation/slot16/WP-14_repair_confirmation.md`. **No statement is added, no label is renamed or removed, and no tag moves** relative to pre.3; Appendices B and D are unchanged. The edits over pre.3 are locators, one sentence of proof commentary, and code-map and status sentences; each is listed with its source in `docs/implementation/slot16/contract_release_report.md`.

Over pre.3.

- **Locators** (slot-16 WP-8, `docs/implementation/slot16/WP-8_citations.md` §7): (LN5)'s source restated as [WWE20, Thm. 1.5.1], verified on the arXiv text (Merel's own text not read); (LN6)'s [WWE20, p. 8, Cor. 7.2.2], verified on the arXiv text. The [conjectural] tags stay: the chain to $H^1(\Gamma_0(N);\mathbb{F}_p)$ is not carried out. `gl2_refs.bib`: `wwe`'s note; `haberland` and `kiming_verrill` added for [EX].
- **§31** (the slot-16 WP-16 report §3 and its audit's F-1, F-7): the proof of `prop:isomodelell`(1) keeps the average and says why it is a no-op on d (the lifts start at simplices of dimension ≥ 1, stabiliser $\{\pm I\}$, central) and that it is load-bearing in Ψ_ℓ; (IL3) and (IL4) carry [verified computationally][a] *partial* sentences ((IL3)'s level-N clause, (IL4) over $\mathbb{Z}/3^e$ and the pair and cone half of `prop:isomodelell`(5), not run); the code map names Ψ_ℓ's well-definedness check as where the average is tested and records the model as built.
- **Code maps** (§22.1, the chapters' own, §22.2's boxes): §28.6 built (slot-16 WP-13, `Hecke/HeckeLevelNDegen`); §29 built (slot-16 WP-10, WP-11, `Hecke/HeckeNebentypus`: (NB4)'s sharp form at $n = 0$ gated; $m = 2$ and $H^1_{\mathfrak{m}} \cong A^2_W$ on the dihedral blocks; $H^1_{\mathfrak{m}} \ncong \mathbb{T}^2_{\mathfrak{m}}$ on the F1 hidden blocks over the full Hecke algebra); §31 built at $\ell = 3$ (slot-16 WP-16, `Integral/Model/IsotropyModelEll`).
- **Verified, no edit**: the L16-2 audit's note 5 (referee C-9) is applied at pre.3 — (NB5)'s dictionary is [proved here] modulo (L1)–(L5) of [GT, `bridge:nebentypus-dictionary`], [conjectural] only for the literature inputs; the note's conditional second clause (an eigenbasis in place of "the newforms" if a cell with oldforms is added) has no cell to act on.
- **Not in this release** (new mathematics, for a referee): slot-16 WP-5's proposed remark on saturation and torsion, WP-11's proposed statements on (NB4) and the hidden blocks, the [L37] report's proposed window bound for [E, `pr:kfonset`]; and, outside this release's brief, WP-13's composite-N caveat (its F-3) to the cost figure of `rmk:levelNscale`(4).
- Non-normative: the header comment, the date line, the companion pointer (now [E] v1.16.2).

v1.16.0-pre.3 — 2026-10-01 — hecke-extensions slot 16, WP-12 resumed: the WP-14 referee's repairs

This version. The repairs answering the Fable referee `docs/implementation/slot16/WP-14_referee.md`: [C] **ACCEPT WITH REPAIRS**, 0 blocking, 1 major, 8 minor; every finding applied as worded. No label added, renamed or removed; no tag set of a numbered statement changes (`rmk:nebcheckable` already carries [proved here], [standard], [conjectural]), so Appendix B's counts are unchanged.

Repairs.

- **C-1** (major) `prop:isomodelell`(1),(4): every lift produced by the contracting homotopies is averaged, $y \mapsto e_\sigma y$, since $1 \otimes e_\sigma$ is Q_σ-fixed; constructively $y_0 := e_\sigma y_0'$, $k := e_\sigma k'$; §31's code map and (IL1) test $d \circ d = 0$, (IM1), (IM2) on the built complex.
- **C-2** `prop:kappamodel`: on a refinement, $\widetilde{e_{46}}$ is replaced by path$(\widetilde{v_4} \to \widetilde{v_6})$. **C-3** `prop:kappamodel`(2) and `prop:lhcompliant`(4): the identification is comparison of the projective resolution $C_\bullet(X) \otimes R$, then Shapiro.
- **C-4** (LN8): "the analogue of `prop:oldspace`(1),(3) with $\Gamma_0(M) \to \Gamma_0(Mp)$" ([standard], [verified computationally]), and one cell with $d = p \mid M$ for WP-13; (LN9): the torsion-freeness of $H^1(\Gamma_0(N);\mathrm{Sym}^{k-2}\mathbb{Z}_p^2)$ behind $p > k+1$. Same wording in `rmk:degenpresconv`(3).
- **C-5** `prop:degenpres`(7), `rmk:levelNscale`(5): at most $2m_N/m_d$ admissible i per ϵ; cost $O(|\mathrm{Mer}_d|\, m_M d_V^2)$ with the classes enumerated per ϵ. **C-6** `rmk:levelNscale`(4): `SparseCokernel`'s `final_nnz`. **C-7** `rmk:isomodellmp`(3): the simplices with $|P_\sigma| > 2$.
- **C-8** (NB5): $H \oplus H^{\mathrm{anti}}$ (not $H \oplus \bar{H}$ for non-real χ); the characteristic-0-to-$\mathbb{F}_q$ clause ($H^0 = 0$ on the cells); the reduction rule presumes $d(\chi) = 1$. **C-9** (the owner's

call, ruled): (NB5)'s eigenvalue dictionary is [proved here] modulo (L1)–(L5) of [GT, `bridge:nebentypus-dictionary`], [conjectural] only for those literature inputs.

v1.16.0-pre.2 — 2026-10-01 — hecke-extensions slot 16, WP-12 follow-up: (NB5)'s product form squared

This version. A repair of (NB5) in `rmk:nebcheckable`, ruled by the slot-16 WP-9 / WP-9b audit (`docs/implementation/slot16/WP-9_audit.md` §3): on the χ-part the cuspidal factor of the characteristic polynomial of T_p, U_p is $\prod_{f\in S_k(\Gamma_0(N),\chi)}(X-a_p(f))^2$, not $\prod_f (X-a_p(f))(X-\overline{a_p(f)})$ (the two agree for real χ; the written form failed in 20 of 24 non-real cases, the squared one held in 24 of 24). Added: the proof of the squaring by the involution of $\varepsilon = \mathrm{diag}(1,-1)$, which swaps the Eichler–Shimura halves ([proved here], with [standard:] Eichler–Shimura), and the reduction rule $\psi(g) = s$ (least primitive root) that makes the mod-ℓ reading unambiguous. The eigenvalue dictionary (χ versus $\bar\chi$) stays [conjectural] and L16-2's. No label added, renamed or removed; the tag set of `rmk:nebcheckable` already contains [proved here], [standard] and [conjectural], so no count moves. To be read by the WP-14 referee with v1.16.0-pre.1.

v1.16.0-pre.1 — 2026-10-01 — hecke-extensions slot 16, WP-12: degeneracy maps on the presentation, level N at scale, the ℓ-local isotropy model — the pre-release of an X bump

This version. Slot-16 WP-12 (plan `docs/implementation/slot16/00-slot16-plan.md` §2 "WP-12", items (a)–(c); owner rulings of 2026-10-01). **A pre-release, not owner-released**; the Fable referee is slot-16 WP-14, and the release at the slot close needs its ACCEPT and the owner's sign-off (ruling O2). One subsection added at the end of §28 and one chapter after §30, so that no chapter or subsection number moves; no existing statement changes, no label is renamed or removed, no tag moves.

New statements.

- **§28.6** [new] (contract first for slot-16 WP-13): `prop:kappamodel` ($\kappa_N^{\mathrm{mod}}(f) = -f(e_{46})$, the pinned model-to-presentation map; naturality for fibre maps — v1.15.0-pre.3's unrefereed scope sentence, now proved; the entrywise comparison of the model's and the presentation's operators), `prop:degenpres` ((72): α_d on the cokernel presentation by Mer_d through adj, for every d with $dM \mid N$, proved equal to Proposition 27.2(4)(c) on H^1; $d = 1$ is α^0; composite d; the trace $\omega \circ \alpha_d = [A\, g\, A]$; with a character), `rmk:degenpresconv` (two mis-transcriptions caught; [verified computationally][a]), `rmk:levelNscale` (item (c): the chapter's scope at $k > 2$ and composite N, the memory model of the normal form), `rmk:levelNcheckabletwo` ((LN7)–(LN9)).
- **§31** [new] (item (b)): `def:isomodelell`, `prop:isomodelell`, `prop:ellthreeclosed`, `rmk:isomodellmp`, `rmk:isomodellcheckable` ((IL1)–(IL4)).

Non-normative.

- §28's scope paragraph and code map; §22.1: one row amended, one added; §22.2: one box. Appendix B: 307 → 317 lines; Appendix D: the new labels; the provenance counts: [new] 226 → 236.
- `gl2_refs.bib`: `billerey_menares` added (not verified). The header comment and the date line. The companion pointer stays at [E] v1.16.1.
- Evidence scripts (tracked, read-only, not registered validations): `docs/implementation/slot16/wp12/de` `negctl.py`. Report: `docs/implementation/slot16/WP-12_report.md`, with the referee brief for WP-14.

v1.15.0 — 2026-10-01 — hecke-extensions slot 15, WP-18: the release at the close, from v1.15.0-pre.3

This version. The slot's X bump, whose text v1.15.0-pre.1, pre.2 and pre.3 carried, is released at the slot-15 close under the owner's ruling Q2 of 2026-09-30 ("release [C] v1.15.0 at the close if WP-4 accepts"). Its referee (slot-15 WP-4, Fable, `docs/implementation/slot15/WP-4_referee.md`) was **ACCEPT WITH REPAIRS** for §29 and `cor:stablecert`, and **ACCEPT** for `prop:saturation`, `prop:satdefect`, `prop:cohforms`, `prop:hidaalgconst` and `rmk:thetaforms`, with 0 blocking findings; every repair of its §9 was applied at pre.2. §29 is released *unbuilt* (contract first; the code is slot 16's, owner ruling Q3). §30's `cor:stablecert` is cited by the certified `stable` verdict of slot-15 WP-5 (Fable audit ACCEPT WITH REPAIRS, repaired). **No statement is added, no label is renamed or removed, and no tag moves** relative to pre.3; Appendices B and D are unchanged.

Not refereed (listed for the slot-16 referee). pre.3's one sentence in §28's scope paragraph (slot-15 WP-17, proposed by WP-8): a Γ-module map between induced modules induces the map on the cokernel presentations of Proposition 27.5(2), so the presentation-level fibre maps α^0, ω rest on a statement of this document, while α_d, $d > 1$, is asserted on the model only. It adds no numbered statement and moves no tag; it was written after WP-4's referee and is released *unrefereed*.

Non-normative.

- The header comment and the date line. The companion pointer stays at [E] v1.15.0: [E]'s tree text 1.15.1-pre.6 and [EX]'s 1.2.0-pre.13 are *not* released at this close; their releases (Q13: [E] 1.16.0, [EX] 1.2.0) wait for the owner's ruling on the [E, `pr:latticesplit`] edit (refuted as stated at $\ell = 37$; §30's scope sentence, pre.2's B4).
- Release note: `docs/implementation/slot15/WP-18_close_report.md`.

v1.15.0-pre.2 — 2026-09-30 — hecke-extensions slot 15, WP-3 resumed: the referee's repairs

This version. The repairs answering the Fable referee `docs/implementation/slot15/WP-4_referee.md` (commit `6215ccd`): §29 ACCEPT WITH REPAIRS; `prop:saturation`, `prop:satdefect`, `prop:hidaalgconst`, `rmk:thetaforms`, `prop:cohforms` ACCEPT; `cor:stablecert` ACCEPT WITH REPAIRS; **0 blocking**, every repair of its §9 applied. **No statement added or removed, no label renamed, no tag moved**; Appendices B and D unchanged.

Repairs.

- **A1** `prop:nebcorr`(4): the $\Gamma_1(N)/\Gamma_0(N)$ compatibility proved in-document (the coset bijection $A_1''\backslash\Gamma_1(N) \to A_0''\backslash\Gamma_0(N)$, $r_t \in \Gamma_1(N)$ for both models); [standard:] only for the general $\langle d\rangle$-commutation.
- **A2** `rmk:nebheilbronn`: the parity clause. **A4** (NB4): the sharp form with the two constituents Sym^i and $\mathrm{Sym}^{\ell-1-i} \otimes \det^i$ (still [conjectural]). **A5** `lem:gammaonetable`(3): $L_{\Gamma_1(N)} := L^{\rho_n}_{\Gamma_1(N)}$. **A9** `lem:teichmuller`: the Galois ring does not depend on the lift. **N1** `rmk:nebcost`(1): one solve per distinct scalar.
- **B1** `lem:heckerankone` for every idempotent e of $\mathbb{T}^{\mathbb{L}}$ (at most one Eisenstein system in any block); `cor:stablecert` for a maximal ideal $\mathfrak{m}'$ of $\mathbb{T}' = \mathbb{Z}_\ell[T_p : p \in S]$, the code's joint system — what slot-15 WP-5's certificate cites. **B4** §30's scope: faithfulness on $M_k(\mathbb{F}_\ell)$ is [E, `cl:mkfaithful`]; [E, `pr:latticesplit`] refuted as stated at $\ell = 37$.

Non-normative.

- Code maps: $H_{N,p}$ of (71) *built* (slot-15 WP-7, `Hecke/HeckeLevelNHeilbronn`, `HeckeLevelNHeilbronnValidation`; (LN3)'s Heilbronn half gains a [verified computationally][a] sentence); `HeckeExt/HeckeHidaComponents` *built* (slot-15 WP-6) in §22.1's §27 row. The header comment and the date line.

v1.15.0-pre.1 — 2026-09-30 — hecke-extensions slot 15, WP-3: nebentypus, and the $\mathbb{Z}/\ell^K$ Hecke algebra — the pre-release of an X bump

This version. Slot-15 WP-3 (plan `docs/implementation/slot15/00-slot15-plan.md` §2 "WP-3"; owner rulings of 2026-09-30, Q2: *general N, characters into $\mathbb{F}_q^\times$, $q = \ell^d$ general, no layering*; Q3: the nebentypus code is slot 16's; Q4: the certified `stable` verdict on the referee's ACCEPT). **A pre-release, not owner-released**; the Fable referee is slot-15 WP-4. Two chapters added after §28, before the appendices, so that no chapter number moves; no existing statement changes, no label is renamed or removed, no tag moves.

New statements.

- **Part VII, §29** [new] (contract first, rule 21; not built): `def:nebdata` and `lem:nebdata` ($\Gamma_1(N)$, $\Delta_0(N)$, characters $\chi\colon (\mathbb{Z}/N)^\times \to \mathbb{F}_q^\times$, the module $V_R(\chi)$ with $\rho_{n,\chi} = \chi(a)\rho_n$), `lem:teichmuller` (the Galois ring $W_K(\mathbb{F}_q)$ on $C_{\ell,d}$, the Teichmüller lift $\tilde{x}^{q^{K-1}}$), `rmk:nebfield`, `lem:inducedchar` (the character model is monomial: (73)), `lem:gammaonetable`, `def:nebmodel`, `prop:nebdecomp` (diamonds as fibre maps; the character model as the χ-eigenspace; (NB2), (NB3)), `prop:nebcorr` ((74): T_p, U_p, $T(p,p) = \chi(p)p^n$ on the level-1 τ-table), `thm:nebrelations`, `prop:neborbit` (orbit-wise, parity, cokernel, Euler characteristic), `rmk:nebheilbronn`, `rmk:nebcost`, the refusal box, `rmk:nebcheckable` ((NB1)–(NB6)).
- **Part VII, §30** [new]: `def:satdata`, `prop:saturation` (P15 = P-WP6-1: $\mathsf{A}_K = \mathbb{T}/(\mathbb{T} \cap \ell^K\mathbb{T}^{\mathrm{sat}})$, the stopping-rule equivalence), `prop:satdefect` (P17: $\varepsilon_K^{\mathrm{sat}} = \mathrm{length}((\mathbb{T}^{\mathrm{sat}}/\mathbb{T})[\ell^K])$), `lem:heckerankone`, `lem:freepart` (P-WP6-2), `cor:stablecert` (the certified reading and the generation caveat), `prop:cohforms` (the torsion-aware comparison, plan WP-19 (c) as corrected in `c1b45cd`), `prop:hidaalgconst` (P18), `rmk:thetaforms` (P20).

Non-normative.

- 27.12: one pointer sentence to `prop:hidaalgconst` (P18); no tag moves. §28's scope paragraph: a pointer to §29.
- §22.1: two rows; §22.2: one box. Appendix B: 284 → 307 lines; Appendix D: the new labels; the provenance counts: [new] 203 → 226.
- `gl2_refs.bib`: `cohen_oesterle`, `ribet77` added (not verified). The header comment and the date line. The companion pointer stays at [E] v1.15.0.

v1.14.2 — 2026-09-29 — hecke-extensions slot 14: §28 built (WP-12)

This version. A Z-bump by the owner's ruling **Q-F** (2026-09-29): slot-14 WP-12 (commit `7a5158d`, level N on the model by Shapiro, first build) is in the tree, refereed by the Fable code review of commit `64e6b9c` (`docs/implementation/slot14/WP-12_code_review.md` §4, whose texts are used verbatim; verdicts §5.1–§5.3 accept with change, §5.4 accept). **No statement changes, no label is added, renamed or removed, and no tag moves**; Appendix B and Appendix D are unchanged.

Non-normative.

- §28's code map, §22.1's row: "not built" → *built at v1.14.2*, naming `Hecke/HeckeLevelN` (`Gamma0LevelN`, `Gamma0SubTransversalN`, `InducedRelationsLevel`, `OrbitWiseFixedSpace`, `CokernelFromFixedSpaces`), *not* `Hecke/InducedLocalSystem` (review S-4). Not built: $H_{N,p}$, the sparse cokernel elimination, degeneracy maps at composite N.
- (O3) **closed**; (O2) stays open (its τ-route at composite N is built). The two consequential sites of §27 (code map and row; the scope paragraph) gain their v1.14.2 sentence.
- (LN3): the τ-route half's [verified computationally][a] sentence (20 cells). (LN6): the second-route [verified computationally][a] sentence with the $q \leq 7$ and e-read-off caveats and "the Loewy length is the subalgebra's".
- The r_t alternative (slot 13's `Gamma0SubTransversal`) is stated in the box after Lemma 28.2, not in Definition 28.1(6). Remark 27.6(3): ν_2, ν_3 checked against the table (one parenthesis).
- The header comment and the date line. The companion pointer stays at [E] v1.15.0.

v1.14.1 — 2026-09-29 — hecke-extensions slot 14: the second cluster run's data sentences, and the U_ℓ lever's rank lemma

This version. The owner's rulings of 2026-09-29 on that day's Fable audit, the file `AUDIT_contract_currency_2026-09-29.md` in the directory `docs/implementation/slot14/`: **Q-B** (the second-cluster-run data, job `3467349`, signed off; proposals L1–L4 of `WP-L1_binding_doc_proposals.md` in the same directory, with the audit's corrections: the probe's peaks and the sampler's stated apart, and the admission predictor's refit named) and **Q-H** (audit item C4, the lever lemma, into this document). A Y-level release by the owner's ruling: one new statement, proved, about the implementation's own matrices; **no existing statement changes, no label is renamed or removed, and no tag moves**. **Q-F**: slot-14 WP-12's "not built → built" flip of §28's three code-map sites is *not* made here (WP-12 is in review); it will be a later, separate bump.

New statement.

- `lem:rhorank` [new] — over $\mathbb{Z}/\ell^k$, $\rho_n(g)$ with $\ell \mid \det g$ has at most $\min(k, n+1)$ non-zero invariant factors (Smith form over $\mathbb{Z}_\ell$, ρ_n of a diagonal matrix); over $\mathbb{F}_\ell$ it is $w\,v^{\mathsf{T}}$ in closed form. The claim `Hecke/HeckeLowRank.h` `ENGINE_CHECK`s at every block (slot-13 WP-4; slot-14 G0's closed form). Placed after §27's code map; Appendix B gains its line (counts 283 → 284), Appendix D its label.

Non-normative (Q-B: L1–L4).

- `rmk:hidacheckable2`: (Hi11) gains a [verified computationally][a] sentence (L1: held at $n = 4150$, 711.8 s, 49.9 GiB, the probe's gated `HI11` line; $n = 3460$ refused by name); (Hi12) gains one (L2: held at $(4830, 2411)$, 68.4 GiB probe / 65.3 GiB sampler; $(2420, 2411)$ recomputed with the levers in 19.2 GiB; $(1210, 2411)$ reproduced) and its last clause is repaired (the build `e85e25a` prints a (Hi12) verdict). No item's tag moves.
- §27's code map (L3): the second run's provenance line, the refit of the campaign's admission predictor (`549200e`), and the warning that (O5)'s fit predates the slot-14 levers; the lever sentence points to `lem:rhorank`.
- §22.1's v1.13.0 scope box (L4): "at v1.14.1 a second cluster run added (Hi11) at $n = 4150$ and (Hi12) at $(4830, 2411)$, both held".
- The header comment and the date line. The companion pointer stays at [E] v1.15.0.

v1.14.0 — 2026-09-28 — hecke-extensions slot 14, WP-13: the owner-signed release, from v1.14.0-pre.2

This version. The owner releases the slot's X bump, whose text v1.14.0-pre.1 and pre.2 carried: §28 (slot-14 WP-4). Its referee (slot-14 WP-5, `docs/implementation/slot14/WP-5_referee.md`) was **ACCEPT WITH REPAIRS**, with 0 blocking, 3 should-fix and 8 minor findings, all applied at pre.2. The owner's decision of 2026-09-28, "Go for publication and release", answers [S14] Q2 and Q3 with the recommended options. §28 is released *unbuilt* (contract first, rule 21). Its code maps say "not built at v1.14.0". **No statement is added, no label is renamed or removed, and no tag moves**; Appendix B is unchanged. Release note: `docs/implementation/slot14/WP-13_release_note.md`.

Non-normative (the [S14] D1 routing, Q3).

- **P1** ((Hi12) run once at $(2420, 2411)$ and held), **P2** and **P8** ((Hi11) at $n = 1390$, 2080, 2770, and $(700, 691)$ at $m = 2$ on a second machine) and **P3** (Remark 16.8(3), the $\mathbb{Z}/\ell^2$ depth window, 17 cells, exactly the loci of (2)) are re-inserted *verbatim* from pre.1 (commit `cc97dd7`). pre.2 had taken them out under Q2's default, which no longer holds. Only their provenance clauses change: "staged, publication pending" becomes "published 2026-09-28 on the owner's sign-off", because the extract `docs/matrix-engine/results/noctua2_run1/` is now a published set. Each remains a [verified computationally][a] sentence. No item's tag moves, and the converse of Remark 16.8(2) is still not asserted.
- **P7** (the 493.3 GiB provenance) is restored to its pre.1 form in §27's code map. The §22.1 row and v1.13.0's scope box record that (Hi12) ran and held.
- *Deferred to slot 15*, as new statements that the rule for post-referee proposals excludes from a release: P14 (the ordinary part of section M as a [verified computationally] sentence; its cluster data are also unpublished), P15 (= slot-14 WP-6's P-WP6-1 and P-WP6-2, two propositions), P17 (the saturation identity for ε_K and the generation caveat) and P18 (an attribution of algebra-level consequences to Theorem 27.11 and Proposition 27.9). The release note gives the reasons.
- The header comment, the date line, the companion pointer (→ [E] v1.15.0), and the two code maps of §28.

v1.14.0-pre.2 — 2026-09-28 — hecke-extensions slot 14, WP-5: the referee's repairs

This version. The repairs answering the Fable referee of v1.14.0-pre.1 (`docs/implementation/slot14/WP-5_re`
ACCEPT WITH REPAIRS, 0 blocking, 3 should-fix, 8 minor; every proof of §28 correct as written). All eleven findings are applied (F10 is a check, no change). **No statement is added, no label is renamed or removed, and no tag moves**; no [proved here] statement's text changes. Still a pre-release ([S14] Q3). Report: `docs/implementation/slot14/WP-4_report.md`, section "pre.2 repairs".

Normative.

1. **F1** `rmk:heilbronncost`(3): a Hecke *matrix* on H^1 costs $O(|\mathrm{Mer}_p| \cdot d \cdot (\dim H^1)^2)$ through the $D_N \times \dim H^1$ normal-form table of Rel_N (or one sparse solve per column) — quadratic in m_N; obstacle (O1) loses the N_q^2 factor and the dense projector, not the quadratic order. (4) names the normal form as the route's one $D_N \times \dim H^1$ object, (2)'s "no $D_N \times D_N$ object" being about one application; (5) says "quadratic per matrix".
2. **F2** (LN4), (LN6): the scan's $W_{N,p}$ is joint for T_2, T_3, T_5, T_7 only and contains (LN4)'s, equality being part of the prediction; e is read off $\dim W_{N,p}$, so the cells are tabulated by $\dim W_{N,p} \in \{3, 5, 7\}$ and $e = 1, 2, 3$ only under (LN4).

3. **F3** `def:levelellmodel`: the composite-N table is fixed by `def:cosetrepsN`.
4. **F4** `def:cosetrepsN`(3): the case $N = 1$ ($(c_1, d_1) = (0, 1)$, $w_1 = I$).

Non-normative.

- **F5** the contractbox after `lem:cosetrepsN`: case (iii) is a bounded loop plus one inverse, table-building time. **F6** the notation paragraph lists Mer_n, W_A, $W_{N,p}$, μ_N, ζ_A, $H_{N,p}$, Div, D_0 as new symbols (rule (A5)), with $W_A \neq$ `def:rhon`'s W and $\mu_N \neq \mu_\bullet$. **F7** `lem:levelduality`(5): the Brown pointer is definitional. **F8** (LN3): the independent route tests `prop:heilbronn`(2), the τ-route half is WP-12's; the referee's seven re-run cells added. **F9** `rmk:heilbronncost`(1): $\sum_L x_0(L) = p^2 + 1$ spelled out, $|\mathrm{Mer}_p| / (p \log p)$ increasing. **F10** (`rmk:heilbronnconv`(2)) verified by the referee, no change.
- **F11**: [S14] Q2 is unanswered, so its default ("P1–P4, P8 left out of the binding drafts") stands. P1 ((Hi12)), P2 and P8 ((Hi11)) and P3 (`rmk:bocksteinexp`(3)) are taken back out, with their traces in §22.1's §27 row and §27's code map; P7 (the 493.3 GiB provenance line) is kept, reworded to stand alone. Re-insertion notes: `docs/implementation/slot14/WP-D1_binding_doc_proposals.md`. No label was cited by the removed text alone.
- The header comment, the date line, the two code maps of §28 ("not built through v1.14.0-pre.2"); v1.14.0-pre.1's commit hash filled in below.

v1.14.0-pre.1 — 2026-09-28 — hecke-extensions slot 14, WP-4: level N, part 2 — the pre-release of an X bump

This version. The slot's "contract first" package for level N (`docs/implementation/slot14/00-slot14-plan.`
§2 "WP-4", rules 20 and 21; the owner's questions of its §9 at their defaults, recorded as lead rulings in [X] v1.18.2 — in particular Q3: drafts only, no release; Q6: literature locators unverified). The Fable referee (slot-14 WP-5) follows and its repairs are v1.14.0-pre.2. **Twelve statements are added, [new]; no label is renamed or removed and no existing tag moves**; Appendix B gains twelve lines. Base `9269d25`. Report: `docs/implementation/slot14/WP-4_report.md`. Commit: `cc97dd7`.

Normative.

1. **Part VII, §28** [new]: `def:cosetrepsN`, `lem:cosetrepsN` (obstacle O3: the binding coset table for every N), `def:levelNmodel`, `lem:treeres`, `lem:manin`, `def:kappaN` (the isomorphism of `prop:levelorbit`(2), pinned), `lem:levelduality`, `lem:merelpath`, `prop:heilbronn` (obstacle O2: T_p, $p \nmid N$, and U_p, $p \mid N$, on the cokernel presentation, proved equal to `prop:levelcorrespondence`(4)(d) on H^1), `rmk:heilbronnconv`, `rmk:heilbronncost`, `rmk:levelNcheckable` (the predictions (LN1)–(LN6), committed before slot-14 WP-12's code).
2. **Where this departs from the plan's statements.** (a) The Heilbronn–Merel set enters *through the adjugate*: the value on block $j \mapsto j \cdot \epsilon$ is $\rho_n(\mathrm{adj}\, h_j(\epsilon))$, and Mer_p itself does not satisfy the right-coset condition the proof needs (`rmk:heilbronnconv`(2), with two mis-transcriptions shown to fail). (b) U_p, $p \mid N$, is not left open: the same formula with the non-primitive terms dropped. (c) The plan's "$|X_p| = O(p \log p)$" is not what Merel's set gives: $\leq p^2 + 1$ proved, about $0.4\, p \log^2 p$ counted; the $O(p \log p)$ Heilbronn set from continued fractions is named as an admissible replacement, not specified (`rmk:heilbronncost`(1)). (d) (N6) is sharpened to mod-p Loewy length $e + 1$ on the Eisenstein part of the full H^1 (the cuspidal part e, the Eisenstein class one more), from the scan ((LN6)). (e) The prediction labels are (LN1)–(LN6), not (N1)–(N6): `N3` is a notation rule of §2.

3. `rmk:levelscaling`: (O2), (O3) gain a status sentence (specified and proved, resp. pinned; not built).

Non-normative.

- **Data remarks** of `docs/implementation/slot14/WP-D1_binding_doc_proposals.md`, each marked as the first cluster run's, its publication pending the owner's sign-off ([S14] Q2): P1 ((Hi12) held at $(2420, 2411)$), P2 and P8 ((Hi11) at $n = 1390$, 2080, 2770; $(700, 691)$ at $m = 2$ on a second machine) as [verified computationally][a] sentences of the items; P3 (the $\mathbb{Z}/\ell^2$ depth window, 17 cells, exactly the loci of `rmk:bocksteinexp`(2)) in its clause (3); P7 (the 493.3 GiB of $(2420, 2411)$, 0.912 of the lever fit) in §27's code map rather than its Scope paragraph (a provenance line, not a scope statement). No tag set moves.
- Bibliography: [Mer96], [WWE20] added (not verified); the header comment, the date line, the abstract, "How to read this document" (Part VII), §22.1 (a row), §22.2 (a v1.14.0-pre.1 box), §27's Scope and code map, Appendix B (twelve lines, the counts), Appendix D, this appendix's provenance table and roster. The **companion pointer moves to [E] v1.14.0** (released at the slot-13 close after this document's v1.13.0, which left the pointer at v1.13.1).

Not done here, deliberately.

1. **No code, no build**: WP-12 (gated on WP-5's verdict and the owner's Q4) builds the table, the orbit-wise route and $H_{N,p}$; the sparse elimination at large N is slot 15's.
2. No literature locator is verified ([S14] Q6's default); copies of [Man72], [Mer94], [Maz77] and [Cre97] exist in the owner's read-only library (papers/) and were *not* read, the ruling being unverified.
3. No release: [C] stays at v1.13.0 as released; this text is a pre-release for the owner (Q3).

v1.13.0 — 2026-09-27 — hecke-extensions slot 13, the close (WP-15): the owner-signed release, from v1.13.0-pre.2

This version. The owner's release of the slot's X bump whose text v1.13.0-pre.1 and -pre.2 carried: §27 (slot-13 WP-2, refereed by WP-3, `docs/implementation/slot13/WP-3_referee.md`: **ACCEPT WITH REPAIRS**, 0 blocking, 2 major, 10 minor, all applied, none outstanding). Owner decision Q14 (2026-09-26) made the release conditional on the referee's verdict being ACCEPT with 0 blocking and on slot-13 WP-6's arms holding; both hold (`docs/implementation/slot13/WP-6_report.md`: (69) held against the independent group-cocycle route, Proposition 27.4(1), (2) as cochain maps, Proposition 27.9 modulo ℓ, ℓ^2, ℓ^3, (Hi10) on 92 cells). Proposition 27.9 is released with its per-item tags as at pre.2. **No statement is added, no label is renamed or removed, and no tag moves.**

Normative.

1. `def:hidalattice`, variant (H): the sentence "nothing else changes" is amended, as the slot-13 WP-13 code audit (A3) required: on the module route the (free) cochain module's Howell form may carry a non-unit pivot on Hida's lattice, and the route then takes a Nakayama basis, checked free by length (slot-13 WP-5); value-preserving wherever the route computed before.

Non-normative. The code maps of §27 and §22.1, and §22.2's box, read "built" (`Hecke/HeckeLevelEll`, `HeckeExt/HeckeHidaTower` part 2, `HeckeHidaHeavyCellProbe`); (Hi7)–(Hi11) held, (Hi12) not run. Companion pointer unchanged ([E] v1.13.1; its v1.14.0-pre.1 … pre.3 are slot-13 pre-releases). Report `docs/implementation/slot13/WP-15_report.md`.

v1.13.0-pre.2 — 2026-09-26 — hecke-extensions slot 13, WP-3: the referee's repairs

This version. The repairs answering the Fable referee of v1.13.0-pre.1 (`docs/implementation/slot13/WP-3_re`
ACCEPT WITH REPAIRS, 0 blocking, 2 major, 10 minor); **all twelve findings are applied**, none declined; the finding-by-finding table with locations is `docs/implementation/slot13/WP-3_respor`
One statement is added, [new] (`rmk:levelscaling`); no label is renamed or removed; two tag sets move (`rmk:hidacheckable2` gains [standard]; `prop:stabilisation` is tagged per item, its tag set unchanged); Appendix B gains one line. No code. Base `6be7502`. Commit: `[hash]`.

Normative.

1. **M1 — Hida's lattice.** `rmk:stabconvention`(3) replaces pre.1's false "up to the order of the basis": Hida's L_n carries $\rho_n^{\mathrm{H}}(A) = \rho_n(A^T)^T = D^{-1}\rho_n(A)D$, $D = \mathrm{diag}(\binom{n}{j})$, a different $\mathbb{Z}_\ell$-lattice from V unless $n+1 = c\ell^s$, $1 \le c \le \ell - 1$ (so always for $n < \ell$); what is lattice-independent (ranks, characteristic polynomials) and what is not (the finite-level module class). Glosses in `thm:hidacontrol`, `thm:hidaweight`; `def:hidalattice` names its lattice and gains the variant (H) on Hida's lattice (one `FibreRepresentation` variant); (Hi8) gains arms (a) V, (b) L_n, (c) the D_* comparison (reported), and its source is restated: (b) extends Hida's parabolic control to the full H^1, (a) is this document's extrapolation to a different lattice. (Hi8) stays [conjectural].
2. **M2 — the cost model.** `rmk:levelcost`(1), (5): a Hecke *matrix* on H^1 on the built dense route costs the $\Theta((N_q D_N)^2)$ projector per column, cubic in $m_N(k-1)$; the count of (5) is for the cochain map. New `rmk:levelscaling`: the two regimes (large N at small k; large k at level 1 or ℓ), the obstacles (O1)–(O6) (the owner's scaling requirement, tracked here), and the Hecke action on the cokernel presentation named as the Manin–Merel Heilbronn-matrix formula, **specified by pointer, not transcribed, not built** ([Man72], [Mer94, §2], [Ste07], [Cre97]; not verified).
3. **Minors.** m1: `prop:stabilisation` per-item tags, and its proof of (4) cites `rmk:stabconvention`(1). m2: `def:levelellmodel` (the count for composite N), `prop:levelcorrespondence`(4)(c) (prime-power d; composite d by composition). m3: `rmk:levelcost`(3), the exact ν_2, ν_3. m4: (Hi8)'s test is a decision procedure, no "inconclusive". m5: `thm:hidacontrol` records Hida's $p \ge 5$ (page image; the text layer says $p > 5$) and $N = 1$. m7: the Scope paragraph at $\ell \in \{2, 3\}$ (cochain level and the Bredon reading hold). m8: `rmk:stabconvention`(4), the U^* old-space matrix and $\omega\, e(U^*)\, \mathrm{dg}_1 \in \ell \cdot \mathbb{Z}_\ell^\times$. m9: (Hi10)'s commutativity [standard]. m6, m10: `prop:oldspace`'s [verified computationally][a] scope (cohomology only; the block formula untested until WP-6).

Non-normative.

- The chapter's code map, §22.1's row and §22.2's box (the variant (H), (O2)); the header comment, the date line, the companion pointer's wording (unchanged: [E] v1.13.1); Appendix B (the counts), Appendix D, the provenance table and roster (and the stale "106 assertions" left by pre.1, now 116); pre.1's commit back-filled. `prop:stabilisation`'s [verified computationally][a] gains the referee's $\ell = 13$ values.

What WP-5 and WP-6 must test differently. WP-5: (Hi8) arms (a) and (b) (the latter on the ρ_n^{H} fibre), (c) reported, by the decision procedure. WP-6: the pair $(e_\ell \circ \mathrm{dg}_\ell,\ c_{\mathrm{st}}^{-1}\omega)$ and the old-space relations, never a trace inverse in the U^* convention (`rmk:stabconvention`(4)); (69) is evaluated on the model for the first time there.

v1.13.0-pre.1 — 2026-09-26 — hecke-extensions slot 13, WP-2: control, level ℓ, the lattice — the pre-release of an X bump

This version. The slot's "contract first" package (`docs/implementation/slot13/00-slot13-plan.md` §2 "WP-2" and §9, the owner decisions of 2026-09-26, rules 20 and 21). It carries its own identifier so that no version number is published twice under different text ((V1), (V2)); the Fable referee (slot-13 WP-3) follows and its repairs are v1.13.0-pre.2. **Thirteen statements are added, [new]; no label is renamed or removed and no existing tag moves**; Appendix B gains thirteen lines. Base `26f3f45`. Report: `docs/implementation/slot13/WP-2_report.md`. Commit: `83376c5` (back-filled at pre.2, (CL6)).

Normative.

1. **Part VII, §27** [new]: `def:levelellmodel`, `prop:levelcorrespondence`, `prop:levelellU`, `prop:oldspace`, `prop:levelorbit`, `rmk:levelcost`, `lem:levelelltorsion`, `prop:stabilisation`, `rmk:stabconvention`, `thm:hidacontrol`, `thm:hidaweight`, `def:hidalattice`, `rmk:hidacheckable2` (the predictions (Hi7)–(Hi12), committed before slot-13 WP-5, WP-6 and WP-10's code).
2. **Where this departs from the plan's statements.** (a) The level-ℓ operator needs *no* new coset or τ-table: `prop:levelellU` reads the level-1 U_ℓ table, so the plan's fallback ("degeneracy-map comparison only") is not needed; the general form `prop:levelcorrespondence` covers every level-N Hecke operator, degeneracy and level-change map (the owner's level-$\Gamma_0(N)$ amendment). (b) The plan's "$e_{\mathrm{ord}}H(\Gamma_0(\ell)) \cong e_{\mathrm{ord}}H(\Gamma)$" is made precise: in this document's convention the integral isomorphism is $e_\ell \circ \mathrm{dg}_\ell$, and $e_\ell \circ \mathrm{dg}_1$ is divisible by ℓ (`prop:stabilisation`(1),(3)); Hida's "$e\circ$res" is the same statement for the Atkin–Lehner conjugate U^* (`rmk:stabconvention`). (c) The plan's old-space matrix is $\mathsf{M} = \left(\begin{smallmatrix} 0 & -\ell^n \\ \ell & T_\ell \end{smallmatrix}\right)$ in the basis $(\mathrm{dg}_1, \mathrm{dg}_\ell)$, with two of its relations exact at cochain level (`prop:oldspace`). (d) The plan's labels `thm:hidacontrol` / `thm:hidaweight` are literature ([standard]), with locators **verified** against the owner's read-only PDFs (the Q2 amendment); the book [Hid93] is not among them and stays unverified. (e) The plan's α_1, α_ℓ are written dg_1, dg_ℓ (rule (A5): α is `prop:alpha`'s). (f) (Hi10) is [proved here] as the plan says, but its $k = 2$ half rests on [standard] pointers not verified (Shimura, Atkin–Lehner).

Non-normative.

- `rmk:hidacheckable`'s (Hi3), (Hi4): a v1.13.0-pre.1 locator note each; tags unchanged.
- Bibliography: [AL70] added (not verified); the notes of [Hid86b] and [Hid86a] record the verified locators. Hida's papers are read from the owner's local library (papers/h/Hida/), read-only; nothing is copied into `docs/literature/` (the owner's Q2 amendment).
- The abstract, "How to read this document" (Part VII), §22.1 (a row), §22.2 (a v1.13.0-pre.1 box), Appendix B (thirteen lines, the counts), Appendix D, this appendix's provenance table and roster, the notation table of §2.11, the header comment and the date line. The companion pointer is unchanged ([E] v1.13.1).

Not done here, deliberately.

1. **No Hida theorem is proved**: (Hi7)–(Hi9), (Hi11), (Hi12) are [conjectural]; (Hi3), (Hi4) keep [conjectural] although their locators are now verified (a candidate for the release, the owner's call).
2. **No code**: the chapter's code map says what slot-13 WP-5, WP-6 and WP-10 are to build; the monomial and orbit-wise routes of §27.2 are an interface for general N.
3. No Hecke action on the cokernel presentation (`prop:levelorbit`(2)); no level ℓ^r, $r \geq 2$, beyond the general formula.

v1.12.1 — 2026-09-26 — hecke-extensions slot 12, post-close, S12-census2: the code maps catch up

This version. A **non-normative** Y bump ((V2)): code maps and the companion pointer only. **No statement, hypothesis, label or tag changes.** Base `bc96fae` (the finality landing S12-FIN, [E] v1.13.1). Report: `docs/implementation/slot12/census2_report.md`. Commit: `[hash]`.

Non-normative.

- §26's code map: "Not built at v1.12.0-pre.1 ... What is built will be recorded here" is replaced by the record of what slot-12 WP-4 and WP-5 built — `Hecke/HeckeOrdinaryPart` (`prop:torsionprepass`, `lem:ordexponent`; `HeckeOrdinaryPrecisionValidation`) and `HeckeExt/HeckeHidaTower` (`def:hidatower`, `Hi2`–`Hi6`; `HeckeHidaTowerValidation`, `HeckeHidaTowerSlowValidation`); §22.1's §26 row likewise.
- §22.1's §14 and §24 rows name `HeckeExt/HeckeFinalityRoutes` (slot-12 S12-FIN): (d1) "boundary" resting on `prop:bdrytranscription` and `lem:bdrytorus`, (d0) "dickson" on `prop:heckeH0`(1) — the certificates themselves are [E]'s (`cert:boundary`, `cert:dickson`, [E] v1.13.0–v1.13.1).
- The companion pointer: [E] v1.12.1 → v1.13.1. The header comment and the date line.

v1.12.0 — 2026-09-26 — hecke-extensions slot 12, post-close: the owner-signed release, from v1.12.0-pre.3

This version. The owner's release of the slot's X bump whose text v1.12.0-pre.1, -pre.2 and -pre.3 carried: §26, the foundation of Hida theory at level 1 (slot-12 WP-2, refereed by WP-3, `docs/implementation/slot12/WP-3_referee.md`: 0 blocking, 0 major, 10 minor, all applied), and the Shapiro transcription in degree 1 (Proposition 14.17, Lemma 14.18; slot-12 WP-16, refereed by `docs/implementation/slot12/WP-16_referee.md`: all three results ACCEPT, status notes ACCEPT WITH MINOR REPAIRS, eight minor findings, all applied) — both accepted after repairs, all applied, none outstanding. **The one status change this version makes**: Question 25.6 is answered at $\ell \geq 5$, $n \geq 1$ by Corollary 26.2 (with Proposition 26.1 and Remark 25.4), so its tag moves [open] → [proved here] and the item is retitled "— ANSWERED 2026-09-26", exactly as 14.24 lost its `[open]` tag at v1.3.0. **No statement is added, no label is renamed or removed, and one tag moves**; the question text and the boxed statement of 25.6 are unchanged (rules (L1), (L2)). Base `7d71fdc`: every slot-12 mathematics package (WP-2, WP-3, WP-16 and its referee repairs) merged. Report: this release note (no separate WP report; owner sign-off 2026-09-26, on `docs/implementation/slot12/WP-2_report.md`, `WP-3_referee.md`, `WP-16_report.md` and `WP-16_referee.md`). Commit: `[hash]`.

Normative.

1. `open:ordprecision`: retitled "Certified precision for the ordinary part — ANSWERED 2026-09-26"; the tag [open] is replaced by [proved here], discharged by `cor:ordprecision` at $\ell \geq 5$, $n \geq 1$ ($e = e'$ for the ordinary part, $K = e' + e_{\max}$ for the free quotient, $e_{\max}$ computable before the run from $\rho_n(S)$, $\rho_n(T)$ alone). At $\ell \in \{2, 3\}$ nothing is asserted; the question text and the boxed statement are unchanged.

Non-normative.

- Appendix B: the open row's status column ([proved here] in place of [open]) and the counts ([proved here] 129 → 130, [open] 7 → 6; tagged/untagged unchanged).
- The abstract's parenthesis on 25.6, the companion-pointer sentence (unchanged version, [E] v1.12.1), the header comment and the date line.

- **Outside this document**: `CLAUDE.md`'s and `Hecke/README.md`'s mentions of this document's version move to v1.12.0; `docs/implementation/hecke_extensions_implementation_plan.md` [X]'s deferred list marks "[C] v1.12.0 release" done (the release and the status change; the §26 code map's own "what is built will be recorded here" is untouched by this release — it is slot-12 WP-4/WP-5's code, not asked for here).

Not done here, deliberately.

1. **No mathematics changes**: every normative statement, hypothesis, displayed formula and label of v1.12.0-pre.1–pre.3 is carried verbatim; this version only lifts the pre-release status and answers 25.6.
2. **No finality certificate**: (d0)/(d1) are [E] matters and land with their code, as v1.12.0-pre.3 recorded.
3. **No code-map update**: §26's "Not built at v1.12.0-pre.1 ... What is built will be recorded here" is left for the package that lands that record.

v1.12.0-pre.3 — 2026-09-26 — hecke-extensions slot 12, WP-16 (post-close, owner-approved): the Shapiro transcription in degree 1, proved

This version. The one blocking finding (F1) of the slot-12 WP-13 referee (`docs/implementation/slot12/WP-13`) asked for a proof of "Proposition T", the identity `prop:bdrychar`(3) stated and left to Obligation 14.19(v.b); the finality proposal v2 (`docs/implementation/slot12/WP-12_finality_proposal.md` §3.3) sketched one. This version proves it, in a form **stronger than the one asked for**, and adds the lemma it was wanted for. **Two statements are added, [new]; no label is renamed or removed and no existing tag moves.** The finality certificates (d0)/(d1) are *not* added — they are [E] matters, owner-gated, and land with their code. Base `fb3fb84`. Report: `docs/implementation/slot12/WP-16_report.md`. Commit: `4ca3555` (the referee repairs: `[hash]`); pre.2's commit is `b73a292`. **Referee repairs P1–P8** (the Fable referee, `docs/implementation/slot12/WP-16_referee.md`: all three results ACCEPT, status notes ACCEPT WITH MINOR REPAIRS, eight minor findings, all applied within this version): P1 the OP1-box sentence scoped to (v.a)/(v.b), (v.c) a checked route; P2 "for n even and $p \neq \ell$" in `prop:bdrychar`(4)'s pointer; P3 `cl:chi`(2)'s "class of y^n carrying χ_0" recorded as proved for every even n; P4 `prop:bdrytranscription`(2)'s right-hand side well defined over every R for positive-diagonal representatives; P5 the corroboration paragraph: the orientation control is a sign control, the referee's extended grid and (T3∂)-breaking control cited; P6, P7 one clause each in the proof (why T is a shift; (B2) for the component of $s_i v_0$); P8 report only.

Normative.

1. `prop:bdrytranscription` [new] (with the display `eq:bdrygroupformnu`): (1) the coordinate Φ is well defined over every R, independent of the base vertex, and an isomorphism whenever $\rho_n(-I) = \mathrm{id}$ — which `prop:bdrychar`(3) had shown only inside a clause assuming $n < \ell$ and a field; (2) at every ν and every p, $\Phi \circ \Theta_g^{\partial}$ is an explicit sum over the left cosets B_N/Γ_∞; (3) at $\nu = 1$ it is `eq:bdrygroupform`. **Where this departs from the requested statement** (WP-13 F1, proposal §3.3): the hypothesis $p \in R^\times$ is *not needed* — the proof evaluates on one period and the correction terms telescope under $j \mapsto j \cdot T$, so no division by p occurs and $p = \ell$ is included; "n even" (or $2 = 0$) is needed only for Φ to be an isomorphism; (T3∂) is used only at the level of H_0; and the sketch's period argument (which needs $p \in R^\times$) is replaced. At $\nu \geq 2$ the proposition gives an H^1 formula without the double-coset structure `rmk:bdrynugeneral` leaves unstated.
2. `lem:bdrytorus` [new] (the proposal's "Lemma B′", WP-13 §2): over $\mathbb{F}_\ell$, n even, $p \neq \ell$, the boundary operator in degrees 0 and 1 lies in the image of $\mathbb{F}_\ell[\mathbf{T}(\mathbb{F}_\ell)]$, is diagonalisable, the operators commute and depend on $p \bmod \ell$ only; with the eigenvalue on a torus

weight vector — $\chi_k(p)$ in degree 0 and $\chi_{n-k}(p)$ in degree 1 (the proposal's "χ_k up to $b \leftrightarrow n+1-b$" in degree 1 is corrected). All parts [proved here]; part (1) rests on `prop:bdrytranscription`.

Non-normative.

- `prop:bdrychar`(3): the sentence "which this document does not prove" replaced by a pointer to `prop:bdrytranscription`(3), the old wording quoted; (4): a pointer to `lem:bdrytorus`.
- `obl:taubdry`(v): the sentence on what a disagreement is; (v.b) a status note — it is now corroboration of a theorem, testing the implementation; the $\nu \geq 2$ sentence notes that `eq:bdrygroupformnu` is not made a clause (owner confirmation 2 of the slot-2 plan stands). Predicates, rings and actions on failure unchanged.
- `cl:chi`(2): a status note — the semisimplicity half is proved (`lem:bdrytorus`), the multi-set half stays [conjectural]; the slot-12 evidence (WP-13's 2080 group-form cases, WP-14's 814/814 recorded operators) cited. The tag is unchanged.
- `rmk:bdrynugeneral`: one sentence (the new H^1 formula at every ν; the double-coset shape still not asserted). The decision box after `def:bdryop`: one sentence (OP1 unchanged).
- Appendix B (two lines, the counts), Appendix D (three labels), this appendix's provenance table and Part IV roster, the header comment and the date line.

Not done here, deliberately.

1. **No finality certificate**: (d0) "dickson" and (d1) "boundary" (proposal §3.1, §3.4) are [E] `sec:certgen` and schema matters and land with their code (owner-gated); this version only removes (d1)'s mathematical condition.
2. **No obligation clause added**: `eq:bdrygroupformnu` at $\nu \geq 2$ is not made a (v.b).
3. **No tag moves**: `cl:chi` keeps its tags; its (2) is not split into two labels.

v1.12.0-pre.2 — 2026-09-26 — hecke-extensions slot 12, WP-3: the Fable referee's repairs

This version. The Fable referee of v1.12.0-pre.1 (`docs/implementation/slot12/WP-3_referee.md`; **0 blocking, 0 major, 10 minor**, R1–R10, R2 and R5 optional) is disposed of: **every finding is applied**. No label is renamed or removed and no statement is added; one [verified computationally][a] tag is added (R5). Every change is marked "v1.12.0-pre.2" at its site. Base `dce4ab0`. Report: `docs/implementation/slot12/WP-2_report.md` §6. Commit: [hash]; pre.1's commit is `30e5b69`.

Normative.

1. `prop:torsionprepass`(6) (R1): $H^1(C^\bullet_{\mathrm{eq}}(\mathbb{Z}))$ agrees with $H^1(\Gamma; V)$ only after inverting 6 (`prop:isomodel`(4)); the conclusion (the $\mathbb{Z}_{(\ell)}$-types agree at $\ell \geq 5$) is unchanged.
2. `cor:ordprecision`(1) (R3): ϵ is Hida's projector $\lim_j U^{j!}$ on H, pinning "the ordinary part of $H^1 \otimes \mathbb{Z}_\ell$" as an object; the exponent is written $e = e'$ (was $\max(1, e')$, vacuous under $e' \geq 1$); `open:ordprecision`'s pre.1 note likewise.
3. `rmk:hidacheckable` (Hi2) (R7): the two equalities stated on named modules ($F'/\ell^{e'}F'$ and $H_{e'}$).

Non-normative.

- `prop:torsionprepass`(2) (R2): why $\ell \geq 5$ is needed for $e_{\max} = 0$ below $\ell + 1$ ($(2,2)$ has $\mathbb{Z}/2$).
- `cor:ordprecision` (R4): $h = 2d_k - 1$ (even n), 0 (odd n), known before the run.

- `rmk:ordexponentuse`(1) (R5): [verified computationally][a] — the exponent of $\mathrm{GL}_{\mathfrak{g}}(\mathbb{Z}/\ell^K)$ equals $N_{\mathrm{ord}}^{\mathrm{lcm}}$ in eleven tested cases (general equality not claimed); (2) (R6): the F'-block sentence reworded — $e_{\mathrm{ord}}(U_{F'})$ exists, and is canonical only modulo $\ell^{e'}$.
- `rmk:hidacheckable`: (Hi3) (R8) the weight-2 step explicit, $\Delta \equiv f_{11a} \pmod{11}$ named, the referee's classical check recorded under the item's [verified computationally][a]; (Hi5) (R9) the Eisenstein line a simultaneous eigenline; (Hi6) (R10) the Bezout step named and $\tau(2411)$'s independent re-derivation recorded.
- Appendix B: `rmk:ordexponentuse`'s row gains [verified computationally][a], the counts ([verified computationally] 25 → 26). Header comment, date line.

v1.12.0-pre.1 — 2026-09-26 — hecke-extensions slot 12, WP-2: the foundation of Hida theory at level one — the pre-release of an X bump

This version. The slot's "contract first" package (`docs/implementation/slot12/00-slot12-plan.md` §2 "WP-2", rules 20 and 21), written with the owner away for the slot: the plan's ruling L3 keeps `open:ordprecision`'s status macro until the owner-signed release. It carries its own identifier so that no version number is published twice under different text ((V1), (V2)); the Fable referee (slot-12 WP-3) follows and its repairs are v1.12.0-pre.2. **Six statements are added, [new]; no label is renamed or removed and no existing tag moves**; Appendix B gains six lines. Base `e2d137a`. Report: `docs/implementation/slot12/WP-2_report.md`. Commit: `[hash]`.

Normative.

1. **Part VII, §26** [new]: `prop:torsionprepass`, `cor:ordprecision`, `lem:ordexponent`, `rmk:ordexponentuse`, `def:hidatower`, `rmk:hidacheckable` (the predictions (Hi1)–(Hi6), committed before slot-12 WP-4 and WP-5's code).
2. **Where this departs from the plan's statements.** (a) The plan states the prepass at $\ell \geq 5$; `prop:torsionprepass`(1)–(5) hold at *every* prime for group cohomology (only $H^0(\Gamma; V) = 0$ is used), and what needs $\ell \geq 5$ is its use as a precision for the model (clause (6)). (b) The plan's working precision $K = k + e_{\max}$ for "the charpolys of every T_p on the ordinary part" is sufficient but not needed: `cor:ordprecision`(1) shows the ordinary part is read at the target precision itself, the torsion being non-ordinary; $K = e' + e_{\max}$ is what the *free quotient* needs (clauses (2)–(4), which also justify slot-11 WP-18c's free-block reading). (c) The plan's (H1)–(H6) and r are written (Hi1)–(Hi6) and $\mathfrak{g}$ (rule (A5): (H1)–(H4) and the restriction map r are taken); the exponent is N_{ord} (N is the Hecke level) and the target precision e' (k is the weight). (d) `lem:ordexponent` also gives the lcm variant and every multiple; `rmk:ordexponentuse` shows no term of the exponent can be dropped.
3. `open:ordprecision`: a v1.12.0-pre.1 status note — answered at $\ell \geq 5$, $n \geq 1$ by `cor:ordprecision`, pending release; [open] kept. The question text is kept.

Non-normative.

- `rmk:ordtorsionfree`: one parenthesis pointing at `cor:ordprecision`; its tag and statement are unchanged.
- Three bibliography entries ([Hid86b], [Hid86a], [Hid93]), none of them in `docs/literature/` and every locator into them flagged "not verified".
- The abstract, "How to read this document" (Part VII), §22.1 (a row), §22.2 (a v1.12.0-pre.1 box), Appendix B (six lines, the counts), Appendix D, this appendix's provenance table and rosters, the notation table of §2.11, the header comment and the date line. The companion pointer is unchanged ([E] v1.12.1).

Not done here, deliberately.

1. **No Hida theorem**: control and weight independence stay predictions ((Hi3), (Hi4), [conjectural]); they, the $\Gamma_0(\ell)$ model, the lattice and the large cells are slot 13's (§26's scope paragraph).
2. **No tag moves**; `open:ordprecision` keeps [open] (ruling L3).
3. **No code**: the chapter's code map says what slot-12 WP-4 and WP-5 are to build.

v1.11.1 — 2026-09-25 — hecke-extensions slot 11, WP-17 (the slot close): the WP-15 referee's repairs

This version. A Y bump ((V2)): the Fable referee of v1.11.0 (`docs/implementation/slot11/WP-15_referee.m` **accept with repairs, 0 B / 2 M / 6 m**) is disposed of. **No normative statement changes**; one [proved here] remark is added (new non-normative material, a corollary of statements already here); no label is renamed or removed. Every change is marked "v1.11.1" at its site. Base: `hecke-extensions` at `9833321`. Report: `docs/implementation/slot11/WP-17_report.md`. Commit: `[hash]`.

Non-normative.

- `rmk:thetacheckable` (M-1): the [verified computationally]^{r} paragraph's provenance parenthesis restated to the reports — WP-6's predictions committed before their runs; WP-5's one rule-20 deviation (two closed forms checked on their first run); WP-6 §1a's realignment of (C5) and (C7) after the first runs — with the v1.11.0 wording quoted. The item's evidence and tags are unchanged.
- `rmk:ordtorsionfree` **new**, [proved here] (m-2, the ruling on the author's F-4): at $\ell \geq 5$, $n \geq 1$ the torsion of $H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2) \otimes \mathbb{Z}_\ell$ is non-ordinary, so the ordinary part is torsion-free at level 1 and is the ordinary part of the free quotient; proof from `cor:thetanonordinary`(3) and `prop:ordinary`(1)–(4). Appendix B: one row, the counts (248 → 249).
- `open:ordprecision`: a v1.11.1 status note — the torsion half answered by `rmk:ordtorsionfree`; the per-cell bound computable before the run stays [open]. The question text is kept.
- Part VII's code map (m-3): the pre.1 sentence on `def:ordinary` marked "(through v1.11.0-pre.2)"; and the $\ell \in \{2, 3\}$ refusal recorded as [E, `sec:thetaobj`] (h)'s precondition (the ruling on F-2; the code changed at slot-11 WP-17).
- The companion pointer (m-4) names the companion's v1.11.1 (`4999a21`), which preceded this document's v1.11.0-pre.1 and which the pointer had skipped; the pointer reads [E] v1.12.1.
- §22.2: a v1.11.1 box with the counts re-derived from `--list` (141 registered, 82 `Hecke*`). The abstract's parenthesis. Header comment, date line.
- v1.11.0's commit (`a8e0afb`) back-filled ((CL6)).
- **Outside this document** (the same package): the companion's v1.12.1 (the referee's M-2, m-1, m-6); the console plan's m-5 (wording of a schedule row); the examples document [EX] v1.0.0 (owner sign-off); the implementation plan [X] v1.15.

Not done here, deliberately.

1. **No Hida section**: it is slot 12's (or slot 13's, by the owner's leave), with the per-cell bound of `open:ordprecision`.
2. **No tag moves**; `open:ordprecision` keeps [open].

v1.11.0 — 2026-09-25 — hecke-extensions slot 11, WP-15: the slot's X bump published, with what slot 11 built

This version. The slot's documents package (`docs/implementation/hecke_extensions_slot11_plan.md` §3 "WP-15", §4.3) publishes the X bump whose text v1.11.0-pre.1 and v1.11.0-pre.2 carried, with the status of what slot 11 built recorded at the sites that promised it ("recorded at v1.11.0"). Sources: the reports of slot-11 WP-5, WP-6, WP-11, WP-18a, WP-18b and WP-18c and the slot's independent review (`docs/implementation/slot11/WP-14_review.md`, `WP-14_review_pass2.md`), all in `docs/implementation/slot11/`. Base `1106200`: every slot-11 code package merged. One version is one text: pre.1 and pre.2 keep theirs. **No statement is added, no label is renamed or removed, and one tag is added** ([verified computationally]^{r} on `rmk:thetacheckable`); Appendix B changes in that row and in the counts. Every change is marked "v1.11.0" at its site. A short Fable referee of the delta since pre.2 follows (slot-11 WP-15). Report: `docs/implementation/slot11/WP-15_report.md`. Commit: `a8e0afb` (back-filled at v1.11.1, (CL6)).

Normative.

1. `obl:backend`(iv), the scope of the exclusion, worded to the digit route as built (`ENGINE_API.md` §13.11.2, §13.11.8): a digit primitive is a single-word operation or one product into a fixed double word (pre.2 said "a native w-bit operation of the mod-ℓ^m backend of (v)", and (v) is the mod-ℓ backend); K is fixed per cell before the digit run from the certified bound $K = k + e_{\max}$ and capped by [E]'s grammar; the integer-tier ladder governs `thm:backend`'s tables and is unchanged, and the digit route is not a rung of it (`ENGINE_API.md` §13.11.8 had called it "the fixed-width wide rung"; it now reads as here). The ban and its owner ruling are unchanged.
2. `rmk:thetacheckable`(C9): the indexing of rank δ is said — at the source $n' = n - 6$ ($n' \in \{30, 38\}$, the targets 36, 44), which is what the question needs, and at the cell itself (38, 42), which is what pre.2's parenthesis stated (slot-11 WP-6's finding 1, confirmed by the review). Both are true; the item's claim is unchanged.

Non-normative (status).

- `rmk:thetacheckable`: a last paragraph, [verified computationally]^{r}, with the registered checks (`HeckeDicksonThetaValidation`, `HeckeDicksonThetaHeckeValidation` and their `slow` companions) and their outcome: every checkable item held, (C7)'s open alternative decided at $n = 14$, (C9) decided cell by cell. The re-derivation note after `thm:thetales` points to it.
- §22.2: a v1.11.0 box. Part VII's code map: "Built at v1.11.0". §22.1: a Part VII row, and (g) in §23's row. The abstract's parenthesis.
- `prop:relationsmodel`: the note after its proof records slot-11 WP-11 — the lifts for $\mathrm{diag}(1, p^2)$, $\mathrm{diag}(1, p^3)$ and pI built and certified, clause (2) holding for $\nu \le 2$ on $H^q(C^\bullet_{\mathcal{I}}(R))$, $q \le 5$, at $\ell \in \{2, 3\}$ with $p = \ell$. `rmk:integralevidence`: a ninth bullet (`HeckeStrictLiftRelationsValidation`) and the title.
- `open:ordprecision`: a status note — slot-11 WP-18c's per-cell certified bound for the free part ($K = k + e_{\max}$) and the ordinary idempotent on it are built; neither answers the question, which stays [open]; its pointer into `ENGINE_API.md` corrected (§13.12 → §13.11.11).
- The commits of v1.11.0-pre.2 (`fb9dcbf`) and v1.10.1 (`bee8afb`) back-filled ((CL6)). Header comment, date line, companion pointer ([E] v1.12.0, published in the same package). The registered-validation counts of §22.1's obligations row stay left to the slot close.
- **Outside this document** (the same package): the companion's v1.12.0; the console plan's v3.9; the integral-line design brief's erratum items 17–19 (X7 in full, arbitrary k by

the digit induction, the ordinary projector as built); `ENGINE_API.md` §13.11.4, §13.11.6, §13.11.8 and §13.12.9 (the review's m-3 and m-4, and §13.11.8 worded to agree with `obl:backend`(iv)); the slot-11 plan's Hida-exponent line corrected.

Not done here, deliberately.

1. **No Hida section** (decision 2 (A)); `open:ordprecision` stays open.
2. **No tag upgrade**: the registered checks of Part VII and of `prop:relationsmodel` are evidence; no [proved here], [standard] or [conjectural] tag moves, and the literature locators of `rmk:thetaclassical` stay "not verified".
3. **Nothing of [EX]** (`docs/latex/hecke_examples.tex`): it cites this document and is not cited by it.

v1.11.0-pre.2 — 2026-09-25 — hecke-extensions slot 11, WP-2: the WP-3 referee's repairs

This version. The Fable referee of v1.11.0-pre.1 (slot-11 WP-3, `docs/implementation/slot11/WP-3_referee.`
accepted it after repairs: no blocking, one major (M-1), nine minor findings (m-1 … m-9), and **confirmed `prop:heckeH0` independently** (the plan's "$U_\ell = 0$ on all of H^0" is false; $U_5\mathsf{D}^6 = -\mathsf{L}^{20}$). One version is one text: pre.1 keeps its text under its number. **No label is renamed or removed and no tag moves**; Appendix B is unchanged except one added citation in `rmk:thetaclassical`'s row. Every change is marked "v1.11.0-pre.2" at its site. Base: `hecke-extensions` at `9c2ee40`. Report: `docs/implementation/slot11/WP-2_report.md`. Commit: `fb9dcbf` (back-filled at v1.11.0).

Normative.

1. **M-1** — `obl:backend`(iv) gains the *scope of the exclusion* (owner ruling (ii) of 2026-09-25, hecke-extensions slot 11): the ℓ-adic digit induction of `ENGINE_API.md` §13.11 with a digit count fixed per cell from the certified precision and native-word digit operations is not arbitrary precision; a run-time-growing digit count remains excluded. The referee's draft, verbatim. The ban itself is unchanged.
2. **m-1** — `prop:heckeH0`: the hypotheses are stated per clause ((3) for ℓ odd), and the proof of (4) gains the case $\ell = 2$ ($U_2\mathsf{D}^3 = \mathsf{L}^2$ at $m = 6 = \ell^3 - \ell$), so (4) holds at every prime as stated.

Non-normative.

- **m-2** — `rmk:thetacheckable`(C5): $H^1(M_{22})$ at $\ell = 11$ is $\theta_{\mathsf{L}} H^1(M_{10})$ plus the Eisenstein class plus a second $p + p^2$ class; that system has multiplicity 2 there.
- **m-3** — the paragraph after `thm:thetales` is a re-derivation note and no longer carries an ad hoc tag, so `app:status`'s row and counts are consistent as they stand.
- **m-4** — `rmk:thetasmallell`: which stabilisers fail at $\ell = 3$ ($\langle U \rangle$) and at $\ell = 2$ (all).
- **m-5** — `rmk:thetaclassical`(5): [CV92] beside [Gro90] (not verified), and the endpoint $k = \ell - 1$ named as the case (C8) uses.
- **m-7** — `prop:heckeH0`(3)'s proof: an unused clause deleted.
- **m-8** — pre.1's commit `3b223ef` back-filled.
- **m-9** — `rmk:thetacheckable`(C7): the pipeline's answer ($p^2\tau(p)$ first at $n = 14$, $\ell = 13$, multiplicity 2) recorded beside the unchanged disjunction.
- **m-6** concerns the WP-2 report only (the Ash–Stevens paper named).
- Header comment, date line, one bibliography entry.

v1.11.0-pre.1 — 2026-09-25 — hecke-extensions slot 11, WP-2: the Dickson invariant and θ_{L}, the ordinary projector, and the Hecke relations on the free model — the pre-release of an X bump

This version. The slot's "contract first" package (`docs/implementation/hecke_extensions_slot11_plan.md` §3 "WP-2", rule 21), written on the owner's rulings of 2026-09-25 (§10 there: every decision at recommendation; decision 2 (A), so no Hida section), with the plan's audit (`docs/implementation/hecke_extensions_slot11_plan_audit.md`) M-1, m-1, m-2, m-11, m-12, m-13 taken up. It carries its own identifier so that no version number is published twice under different text ((V1), (V2)); the Fable referee (slot-11 WP-3) follows and its repairs are v1.11.0-pre.2, the slot's documents package publishes v1.11.0. **Twenty statements are added, [new]; no label is renamed or removed and no existing tag moves**; Appendix B gains twenty lines. Base `54bc2b5` (R0). Report: `docs/implementation/slot11/WP-2_report.md`. Commit: `3b223ef` (back-filled at v1.11.0-pre.2).

Normative.

1. **Part VII, §24** [new]: `def:dicksoninv`, `lem:dicksontwist` ((66) for every $A \in \mathrm{M}_2(\mathbb{Z})$, singular included), `prop:dicksonH0`, `prop:heckeH0`, `rmk:heckeH0plan`, `lem:heckefibre`, `def:Qn` (the complement basis $(\bar{u}_0, \dots, \bar{u}_{\ell-1}, \bar{u}_{n^+})$ *fixed*), `lem:Qnfunctions`, `def:theta`, `prop:thetatwist` (the twist is p, derived from `def:rhon`'s covariance and (H2); the plan's audit m-1), `thm:thetales`, `rmk:thetasmallell` (audit m-12), `rmk:thetaclassical` (audit m-11), `rmk:thetacheckable` (WP-6's prediction list).
2. **The plan's "$U_\ell = 0$ on all of H^0" is not adopted**: `prop:heckeH0`(2)–(4) proves $U_\ell v = v(1,0)\, G_m$, rank ≤ 1 and $U_\ell^2 = 0$ on H^0, $U_\ell = 0$ there for $n < \ell^3 - \ell$, and $U_\ell \neq 0$ at $n = \ell^3 - \ell$; $U_5 \mathsf{D}^6 = -\mathsf{L}^{20}$ at $(120, 5)$, inside the slot's grid. The audit's M-1 had corrected an earlier formula and checked D, D^2 only. "Every Dickson class is non-ordinary" survives, as `cor:thetanonordinary`(2)–(3).
3. **Part VII, §25** [new]: `def:ordinary`, `prop:ordinary`, `cor:thetanonordinary`, `rmk:ordinaryeis` (audit m-13), `open:ordprecision` ([open]: the certified precision, slot 12's).
4. **§23.7** [new]: `prop:relationsmodel` — `thm:relations`'s recursion on $H^q(C^\bullet_{\mathcal{I}}(R))$ for every commutative R (slot-10 review finding 6; line 2's M-3). `thm:relations` itself is unchanged and keeps $6 \in R^\times$.

Non-normative.

- Notation §2.11 (sans-serif L, D and θ_{L}, since L, D_n and θ_g are taken, rule (A5); n^+, since N is the Hecke level only), five preamble macros, eight bibliography entries ([Dic11], [SD73], [Kat77], [Joc82b], [Edi92], [Gro90], [RW70], [Luc78]), none of them in `docs/literature/` and every locator into them flagged "not verified".
- The abstract, "How to read this document", §22.2 (a v1.11.0-pre.1 box), Appendix B (twenty lines, the counts), Appendix D, this appendix's provenance table and rosters, the header comment and the date line.
- Companion pointers unchanged ([E] v1.11.0); [E]'s `cl:Tlnil` is upgraded to a pointer at `prop:thetatwist` by slot-11 WP-4, not here.

Not done here, deliberately.

1. **No Hida section** (decision 2 (A)): the ordinary-part predictions at $(700, 691)$ and $(10, 2411)$, control and the lattice are slot 12's.
2. **No [verified computationally] for Part VII beyond the ad hoc script** of `prop:heckeH0`'s proof and the paragraph after `thm:thetales`: the registered checks are WP-5 and WP-6's, recorded at v1.11.0.

3. **No claim about θ_{L}'s long exact sequence at $\ell \in \{2,3\}$** (`rmk:thetasmallell`).

v1.10.1 — 2026-09-25 — hecke-extensions slot 10, WP-15, phase 2: the referee's repairs — a Y bump

This version. The second phase of the package that wrote v1.10.0 (`docs/implementation/hecke_extensions_s` §3 "WP-15"). The Fable referee of v1.10.0 and of the companion's v1.11.0-pre.2 (`docs/implementation/slot10/WP-15_referee_phase1.md`) accepted them after repairs: two major findings (M-1, M-2), each a one-line text repair, and six minor ones, of which m-1 … m-4 concern this document (m-5 the companion, m-6 the slot plan). One version is one text: v1.10.0 keeps its text under its number, and these repairs carry their own. **No normative predicate, no hypothesis, no displayed formula, no tag and no label moves**; Appendix B is unchanged. Every change is marked "v1.10.1" at its site. Base `47e3ff9`: every slot-10 code package merged. Report: `docs/implementation/slot10/WP-15_report.md`. Commit: `bee8afb` (back-filled at v1.11.0).

Non-normative.

- **M-1** — `rmk:bredonhecke` said that at $e = 1$ the shipped $\mathbb{F}_\ell$ pipeline computes (a) and (b) for the canonical τ "unchanged since v1.0.0". The canonical τ was re-defined at v1.1.0 and the golden file re-baselined (`rmk:iotatausym`, "Two consequences" (b)), so neither τ nor (a) is unchanged since v1.0.0. The sentence now reads: the canonical τ of `alg:tau` since v1.1.0; its operator (b) did not move in that re-baseline and equals the legacy τ's at every $\ell \in \{2,3\}$ golden point, `rmk:tauexistsscope`(1) — an observation, not a theorem. "Compares nothing there" becomes "compares none of these operators there" (Θ_g^s, the canonical Θ_g, $\Theta_{\mathcal{I},g}$), with a pointer to that observation.
- **M-2** — `rmk:strictliftbuilt`(3): 349 → 359 cells, an arithmetic slip inherited from a report's summary line (per ring $34, 34, 42, 55, 42, 42, 55, 55$).
- **m-1** — v1.10.0's header comment said `rmk:integralscope`(2) was "narrowed to what is not built"; it gains a pointer, and the comment now says so.
- **m-2** — `rmk:strictliftbuilt`(3): "on the rings where a Θ_g^s is shipped" becomes "on eight rings among those with a shipped Θ_g^s", since Θ_g^s is shipped over $\mathbb{F}_\ell$ at every ℓ.
- **m-3** — `rmk:bredonhecke`(a): why the canonical τ is integral — in practice, because since v1.1.0 it vanishes in degree 2 (`rmk:iotatausym`, consequence (a)), so `alg:tau`'s mod-ℓ triangle-fill fallback is never used.
- **m-4** — `rmk:integralevidence`, the `HeckePeriodicityClassValidation` bullet: $\kappa_C(\mathrm{res}_C\, e) = \chi_C$ is read on the stabilisers; the 224-element check is $\chi = \zeta_+$.
- **Status** — `rmk:integralevidence`'s check counts after slot-10 WP-8's validation carries: `HeckeFreeModelValidation` 803 → 801, `HeckePeriodicityClassValidation` 2482 → 2483. Slot-10 WP-7's levers (the ι sign split and route 2 of `HeckeOperatorOnK` as host tasks) and WP-8's ninth fingerprint bit move no value and touch no site of this document. The commit of v1.10.0 (`ca6c9b9`) is back-filled ((CL6)). Header comment, date line, companion pointer ([E] v1.11.0, published at this version's commit). The registered-validation counts stay left to the slot close.

v1.10.0 — 2026-09-25 — hecke-extensions slot 10, WP-15: the strict lift as built, the Bredon-object operator not compared, the sign of the Bockstein, and (vii) over the ring — an X bump

This version. The slot's one X revision of this document (`docs/implementation/hecke_extensions_slot10_pl` §3 "WP-15", §4.3), made on the owner's rulings of 2026-09-25 on the slot-10 decisions (§10 there, every one at recommendation (A)): 12 (A), this X with a Fable referee; 3 (A), `obl:coneexact`(vii) over $\mathbb{Z}/\ell^e$; 4 (A), the Hecke action on the Bredon object at

$\ell \in \{2,3\}$ declined, with a remark here. Sources: the reports of slot-10 WP-3, WP-10, WP-11 and WP-13 (`docs/implementation/slot10/`) and the slot's independent review (`docs/implementation/slot10/WP-14_review.md`, passes 1 and 2), whose findings M-1, M-3, m-1, m-6, m-7, m-15 and m-17 this version takes up. Base `f0640d0`: WP-2, WP-3, WP-6, WP-9, WP-10, WP-11, WP-12 and WP-13 merged; slot-10 WP-7 and WP-8 (lane B) touch no site of this document. **Two statements are added, [new]; one proof gains a sentence fixing a sign convention**; the rest is status. **No label is renamed or removed and no existing tag moves**; Appendix B gains two lines. Report: `docs/implementation/slot10/WP-15_report.md`. Commit: `ca6c9b9` (back-filled at v1.10.1).

Normative.

1. `lem:bulkperiodic`, proof, Step 0 — **the sign of δ is fixed**: the snake-lemma connecting map $\delta[c] = [d\tilde{c}]$, the cochain differential carrying no sign. Through v1.9.1 the step named the Bockstein without its sign, so clause (1)'s "unique class e" was determined only up to $\pm$; it is now the class the built one is (the slot's review, m-6). **Pins a convention**; no clause's truth moves, since (2) and (3) are insensitive to the sign.
2. `rmk:strictliftbuilt` [new] ([verified computationally]^{r}) — `prop:strictlift`(1), (3) and (4) as built for $g = \mathrm{diag}(1,p)$, $p \in \{2,3,5\}$, in degrees $\leq q_{\max}+1 = 6$: the construction (the preimage read off τ^s, $\ker \mathrm{cmp}$ contracted by the columns' closed-form homotopies, no solve), the five certificates, (3) and (4) in use, and the integer sizes measured (the first evidence on [IB]'s K1). Its last paragraph keeps "existence only" for every other g and degree. `prop:strictlift`'s closing paragraph and `rmk:integralscope`(2) gain pointers to it; neither statement changes.
3. `rmk:bredonhecke` [new] (commentary) — beside `prop:strictlift`(4): at $\ell \in \{2,3\}$ the operator on cohomology of the Bredon object is **not shown** independent of τ (homotopy uniqueness, `lem:tauexists`(2), needs $6 \in R^\times$), this document compares nothing there, and that is the absence of a proof, not a proof of dependence (owner ruling 4 (A); plan §9 audit finding M-2's wording). No line-2 package exhibited two τ with different operators, so the "not shown" reading stands.

Non-normative.

- `obl:coneexact`(vii) — a v1.10.0 status paragraph over $\mathbb{Z}/\ell^e$: **sentences 1–2 discharged at $\ell^e < 2^{32}$, every E8 point** (slot-10 WP-3), the cross-routes refusing above 2^{32} by the `uint32_t` regime rule; **sentence 3 recorded as unavailable over the ring** — the pairing and the contragredient refuse a composite ring. The slot plan's and WP-3's hand-over read "(vii) over the ring discharged"; the review's M-1 scoped it to the two sentences built, and this version follows the review. The v1.9.0 sentence is kept, in the past tense, with its staleness window (`c9684e9` to this version). The obligation roster's row says the same.
- `sec:codemap` — the §15 row gains the cross-routes over the ring; the §23 row gains (d) $\tau_{\mathcal{I}}$ and $\Theta_{\mathcal{I},g}$ (WP-10), (e) e and E_q (WP-13), (f) the basis-free Hecke action (WP-11). §23's closing "Code map" paragraph, stale from v1.9.0, is pointed at the row.
- `sec:notimplemented` — a v1.10.0 paragraph: (vii) over the ring as above; E8 complete; at $\ell \in \{2,3\}$ the refusals stand for `rmk:bredonhecke`'s reason; §23 with the lifted datum, e and the action built, $T_pT_{p'} = T_{p'}T_p$ tested and `thm:relations`(1)'s recursion not (a determinant-p^2 lift is not built; the review's M-3); the descent, the value $\mathbb{Z}$ and M1(b) not built.
- `rmk:integralevidence` — title and count ("slots 8, 9 and 10", eight validations) and three bullets: `HeckeStrictLiftValidation`, `HeckePeriodicityClassValidation` (with the identification of the realised connecting map, the review's m-7) and `HeckeIntegralActionValidation` (with the diagonalisability the congruence-module

readings assume, m-15, and the $(16, 3)$ row recorded without a T_3 oracle, m-17); the `HeckeFreeModelValidation` bullet names where its tests (3)–(5) are hosted. Tag unchanged.

- `def:ringsemantics`(3)(d), `rmk:mpcaveat`(4), §23's "What is not here", the abstract — pointers.
- Appendix B (two lines, the counts), Appendix D (two labels), this appendix's provenance table and Part VI roster (whose "nothing tests the rest", stale since v1.9.0, is pointed), the header comment, the date line and the companion pointer ([E] v1.11.0). The commit of v1.9.1 (`1f190e5`) is back-filled ((CL6)). Containers.
- **Outside this document** ((CL2)): the companion specification's v1.11.0 (whose pre-release v1.11.0-pre.2 is at this version's commit), the console plan's v3.8 and the design brief's slot-10 erratum items, by the same package.

Not done here, deliberately.

1. `obl:coneexact`**(vii) is not narrowed**: its third sentence stays as stated; over the ring it is recorded unavailable by its own "until then" rule, and the E7 pairing over a composite ring is code for a later slot.
2. **No effectivity statement for $\tau_{\mathcal{I}}$ is added to** `prop:strictlift`: the evidence is a remark, and "existence only" stays a statement about every g.
3. **The registered-validation counts** of §22.1's §20 row are left at v1.9.1's: slot-10 packages still running when this version was written may register validations, and the slot close re-derives them from `--list`.
4. **No statement about the Bredon-object operator at $\ell \in \{2, 3\}$ beyond** `rmk:bredonhecke`: in particular no claim that it depends on τ.

v1.9.1 — 2026-09-25 — hecke-extensions slot 9, WP-17, the slot close: the referee's repairs and status — a Y bump

This version. The slot close (`docs/implementation/hecke_extensions_slot9_plan.md` §3 "WP-17"). The Fable referee of v1.9.0 and of the companion's v1.10.0 (plan WP-15) accepted them after repairs: two major findings (M-1, M-2), each a one-sentence or one-file repair, and eleven minor ones, of which m-1 … m-4 concern this document (the others the companion, the console plan, the design brief and the slot close). **No normative predicate, no hypothesis, no displayed formula, no tag and no label moves**; Appendix B is unchanged. Report: `docs/implementation/slot9/WP-17_report.md`. Commit: `1f190e5` (back-filled at v1.10.0).

Non-normative.

- **M-1** — v1.9.0 cited "the slot's independent review (WP-14, pass 1)" as evidence, and no such file was in the tree: the reviewer's file was lost when the host ran out of disk on 2026-09-25. It is reconstructed from the reviewer's two returned summaries and committed as `docs/implementation/slot9/WP-14_review.md`, with a provenance note; `rmk:certifiedeperdiff` cites it. v1.9.0's own changelog text is left as written, and this entry resolves its citation.
- **M-2** — `obl:coneexact`(iv)'s as-built sentence named `HeckeReductionValidation` arm (4b-C) at the seven points. Since slot-9 WP-4t (owner-approved 2026-09-25, merged before v1.9.0 was written) the point $(60, 5)$ at both levels runs in `HeckeReductionCoverageSlowValidation`; arm (4b-C) covers the other five points and the two word-regime cells. The obligation roster's joint attribution was already true and is unchanged.
- **m-1** — the commits of v1.9.0 (`1d7599a`) and v1.8.1 (`b7391f7`) are back-filled, each found by `git log -S` on its changelog header ((CL6)).

- **m-2** — `def:ringsemantics`(2) and the §23 code-map row cited `prop:ringinforce`(2),(3) for the non-freeness of $C^{\bullet}_{\mathrm{eq}}(\mathbb{Z}/\ell^e)$ at $\ell \in \{2,3\}$; neither clause states it. The citation is now the invariants computation of (2)'s proof read over $\mathbb{Z}/2^e$ ($(V_R)^{\langle S\rangle} \cong \mathbb{Z}/2^e \oplus \mathbb{Z}/2^{e-1}$ at $n = 2$), which the companion's [E, `sec:ring`] (c1′) carries out.
- **m-3** — the grammar of `obl:coneexact`(vii)'s as-built sentence ("… and the relative sub-complex constructor that route 3 needs refuse …").
- **m-4** — the §23 code-map row opened "Partly tested, not built (since v1.8.0)" and later said "Since v1.9.0: partly built"; its lead now reads "Partly built since v1.9.0 (partly tested, not built, v1.8.0–v1.8.1)".
- **Status** — the §20 code-map row's count is re-derived from `--list`: 71 `Hecke*` entries of 125 registered (52 `fast`, 63 `standard`, 10 `slow`), the slot-9 names registered through the per-package fragments `Registry/Slot9/`; §22.2 gains a v1.9.1 sentence; (CL5)'s list of Y bumps gains v1.9.1; the header comment, the date line and the companion pointer ([E] v1.10.1).
- **Outside this document** ((CL2)): the companion's v1.10.1, the console plan's v3.7.1, the design brief's erratum item 8 and the implementation plan's v1.13, by the same package.

v1.9.0 — 2026-09-24 — hecke-extensions slot 9, WP-15: the F-A remark at every prime, the certified-exponent rule per differential, the coefficient ring at 2 and 3 as built, and phase E8 item 4b — an X bump

This version. The slot's one X revision of this document (`docs/implementation/hecke_extensions_slot9_pla` §3 "WP-15", scope 3, §4.3), made on the owner's rulings of 2026-09-24 on the slot-9 decisions (§10 there, every one at recommendation (A)): 10 (A), this X with a Fable referee; 3 (iii) (A), `obl:coneexact`(vii) over $\mathbb{Z}/\ell^e$ by route 1 only, its composite half recorded as open; 4 (A), the lift at $\ell \in \{2,3\}$ for the cohomology object only. Sources: the reports of slot-9 WP-2, WP-3, WP-4, WP-10, WP-11, WP-12 and WP-13 (`docs/implementation/slot9/`) and the slot's independent review (WP-14, pass 1), whose findings on this document are the per-differential remark and the two stale sentences. **One statement is added, [new]; one remark's clause is stated at every prime; one definition's clause is replaced by the as-built statement**; the rest is status. **No label is renamed or removed and no existing tag moves**; Appendix B gains one line. Report: `docs/implementation/slot9/WP-15_report.md`. Commit: `1d7599a`.

Normative.

1. `rmk:bocksteinexp`(2) — **stated at every prime** ℓ, 2 and 3 included; clauses (1), (3), (4) keep $\ell \geq 5$. The proof is unchanged except for one parenthesis: its only use of `cl:bockstein` is $H^0(\Gamma; V) = 0$, whose proof is about $\mathbb{Z}$-coefficients. The clause names the design brief's instances (F-4): $\mathbb{Z}/2^j$ at $n = 2^j$ from $m = 2$ at $\ell = 2$, and $\mathbb{Z}/9$ at $n = 12, 18$ from $m = 4, 6$ at $\ell = 3$; clause (3) records, outside its own scope, that the probe's table shows exactly these exponents at $n \leq 20$. **Widens a proved statement to what its proof already gave** (slot-8 WP-9 finding 6; owner ruling 10 (A)). Tags unchanged ([proved here], [verified computationally][a] and a standard-citation pointer).
2. `rmk:certifiedeperdiff` [new] ([proved here]) — `prop:certifiede`(3)'s hypothesis may be checked differential by differential, each d^q at its own level e_q; which levels the type of H^q and the universal-coefficient display need; clause (3) is the uniform case. A clarification, not a correction: no statement or proof of `prop:certifiede` changes. It is the reading line 2's milestone M0 uses (`HeckeFreeModelValidation`); slot-9 WP-11 finding 2, confirmed by the slot's review.
3. `def:ringsemantics`(2) — **the clause "and the implementation of this build refuses it there" is replaced** by the as-built statement: at $\ell \in \{2,3\}$ this build computes

the Bredon object's lengths and invariant factors through a flat-coordinate module route, and refuses the K-mode (undefined there, `prop:certifiede`(5)), the Hecke operator and the reductions by name. The clause was stale from slot-9 WP-3's merge (`b53836f`) until this version, a window the slot-9 plan recorded (§4.3) rather than closed by a second editor. **A status repair in a definition**: what the ring names is unchanged. Measured by `HeckeBredonTreeRingValidation`. (3)(a) and (3)(b) gain pointers: the ranks may come from $C^\bullet_{\mathcal{I}}(\mathbb{Z})$ as `prop:certifiede`(4) already allows, and "per degree" is read per differential.

4. `obl:coneexact`(iv) — **the parenthesis "until it exists the non-field case is recorded as unchecked" is restated**: a ring over which no module solve is built records the case unchecked; over $\mathbb{Z}/\ell^e$, $\ell \geq 5$, the check is made (the as-built paragraph names `ConeModuleCertificate` item 14 and `HeckeReductionValidation` arm (4b-C)), and the length identity, with the composite-zero check, implies exactness independently of the witnesses. The same change in the clause's last sentence. **No predicate moves**: the clause always required the solve over a non-field ring; what changes is that the conditional "until it exists" is discharged.

Non-normative.

- `obl:coneexact`(vi) and (vii) — one status paragraph each, over $\mathbb{Z}/\ell^e$: (vi) is partial there (the commutator asserted, arm (4b-H); the $\nu = 2$ recursion not computed over the ring); (vii) is route 1 only, its composite half open (owner ruling 3 (iii) (A)). The obligation roster's `obl:coneexact` row says the same.
- `sec:codemap` — the §15 row gains the dense module route (slot-9 WP-4: `ModuleConeT`, `ConeModuleCertificate`, `ModuleConeActionT`, `HeckeBoundaryOperatorT`; the `uint32_t` classes below 2^{32}, the templated route in every word regime); the §23 row is **partly built**: the scoped lift and the flat-coordinate route at $\ell \in \{2, 3\}$ (WP-3), $\mathcal{I}_\bullet$, $\mathcal{I}^\partial_\bullet$ and cmp (WP-10, L2-1a) and the integral groups for $n \leq 20$ (WP-11, M0).
- `sec:notimplemented` — a v1.9.0 paragraph: phase E8 complete as specified (items 1–4b; 4b over $\mathbb{Z}/\ell^e$ by route 1; `obl:coneexact`(vii)'s composite half open, (vi) partial over the ring); §23 partly built. The registered-validation counts are the slot close's and are not re-derived here.
- `rmk:integralevidence` — its title and count ("slots 8 and 9", five validations) and three bullets: `HeckeBredonTreeRingValidation`, `HeckeIsotropyModelValidation`, `HeckeFreeModelValidation`. Its tag is unchanged.
- §23's opening "What is not here", `rmk:mpcaveat`'s last parenthesis and the abstract — pointers: the free model, its comparison map and the integral groups at small n are built; the lifted datum and the descent are not.
- Appendix B (one line, the counts), Appendix D (one label), this appendix's provenance table and Part VI roster, the header comment, the date line and the companion pointer ([E] v1.10.0). Containers.
- **Outside this document** ((CL2)): the companion specification's v1.10.0 and the design brief's erratum block, by the same package.

Not done here, deliberately.

1. `obl:coneexact`**(vi) and (vii) are not narrowed.** Both stay as stated, over every ring with 6 invertible; this build discharges them over $\mathbb{Z}/\ell^e$ in part, and says so. The recursion over the ring and the two cross-routes over the ring are code, for a later slot.
2. **The registered-validation counts** of §22.1's §20 row are left at v1.8.1's: slot-9 packages still running when this version was written register or re-count validations, and the slot close re-derives them from `--list`.

3. **No obligation form of `prop:certifiede`**, and no record-schema rule for the integral line: line 2's objects carry no golden entry and no record key in slot 9.

v1.8.1 — 2026-09-24 — hecke-extensions slot 8, WP-14, the slot close: the referee's wording repairs, the changelog's gaps, and status — a Y bump

This version. The slot close (`docs/implementation/hecke_extensions_slot8_plan.md` §3 "WP-14", scope 4). The Fable referee of v1.8.0 (plan WP-9) accepted it with six minor findings, all wording, "none needs a Y bump before merge, all can go into the slot close's status pass"; m-1 to m-5 are applied here and m-6 concerns the plan's wording, not this document. **No normative statement, hypothesis, displayed formula, asserted predicate, tag or label moves**, so Appendix B's counts are unchanged. Close report: `docs/implementation/slot8/WP-14_report.md`. Commit: `b7391f7`.

Repairs (non-normative; the referee's m-1 to m-5).

1. `lem:strictlifting`, proof — "(for $q = 0$, $z = 0$)" assumed $E_{-1} = 0$, which the lemma does not; now "for $q = 0$ the same step applies; in the applications below $E_{-1} = 0$ and $z = 0$". The proof's general step already covered $q = 0$ (m-1).
2. `sec:notation-integral`, the row of $F^{\sigma}_{\bullet}$ — "an instance of the $F_{\bullet}$ of `def:borelmodel`" becomes "the analogue, for the stabiliser": the two are resolutions over $\mathbb{Z}[\Gamma_{\sigma}]$ and over $\mathbb{Z}[\Gamma]$, the same kind of object, not one object ((N2); m-2). A description, not a rule: the symbol and its meaning are unchanged.
3. `thm:specialisation`, the v1.8.0 sentence of its closing paragraph — now says what v1.8.0 adds for $(X, \partial X)$ (a finite-rank free model with its comparison map and a Hecke datum of the pair, as existence) instead of reading as if the unrefined case had been deferred; clauses (1)–(4) and their tags untouched (m-3).
4. `prop:bredon` — one parenthesis after the definition of $\mathcal{D}$: it is the family of all finite subgroups of Γ, which this document does not use (m-4).
5. `rmk:truncdevice` — the Hecke-equivariance clause cites `lem:bulkperiodic`(2),(3), not (2) alone (m-5).

Pre-existing gaps closed (non-normative).

- The abstract — "Two chapters, §15 and §17, state objects that nothing builds" had been stale since v1.7.3 (§15 implemented in full); it now names §17 and §23.
- This appendix — subsections for v1.7.2, v1.7.3 and v1.7.4 (the three status-only slot closes, which v1.8.0's entry found missing, "Not done here" item 4), reconstructed from their date lines, header comments and commits and adding nothing those do not say; the commit hashes of fifteen entries back-filled ((CL6); v0.13.0's commit could not be identified from the history and keeps its `[hash]`); (CL5)'s list of Y bumps completed from v1.6.1 on; §E.1's sentence that the citation census is "specified and not written", stale since hecke-extensions slot 3, pointed at `label_census.sh`.

Status (non-normative).

- `sec:codemap` — the §8 row gains ρ_n over $\mathbb{Z}/\ell^k$ (`ComputeSymmetricPowerMatrixRing`, slot-8 WP-3); the §16 row gains the reductions and the Bockstein over the ring in force (`Hecke/HeckeReduction`, WP-4) and the count identity of `rmk:bocksteinexp`(1) as it is asserted; the §23 row says the third complex of `def:ringinforce` at $\mathbb{Z}/\ell^k$, $\ell \geq 5$, is built (phase E8 items 2 and 3) and refused at $\ell \in \{2, 3\}$, as `prop:certifiede`'s scope clause requires; the §20 row's count is re-derived (65 `Hecke*`; 114 registered).
- `sec:notimplemented` — a v1.8.1 paragraph: the free models of §17 and §23 are unbuilt; the boundary and the cone over $\mathbb{Z}/\ell^k$ (phase E8 item 4b) are the next slot's, so the

non-field half of `obl:coneexact`(iv) stays unchecked.
- The header comment, the date line and the companion pointer ([E] v1.9.1). Containers.
- **Outside this document** ((CL2)): the extension plan's v1.12 and the engine contract's §13 pointer repairs, by the same package.

Not done here, deliberately.

1. `rmk:bocksteinexp`**(2) at every** ℓ**.** The referee recommends stating clause (2) for every ℓ at the next X (its proof uses no hypothesis on ℓ and at $\ell \in \{2,3\}$ proves the observed $\mathbb{Z}/2^j$ at $n = 2^j$ and $\mathbb{Z}/9$ at $n = 12, 18$); clause (1) and the summand count stay at $\ell \geq 5$. It changes a statement, so it is an X bump's, and it is recorded for the next one (owner default of 2026-09-24).
2. **Part VI's placement** is kept, as the referee recommends: moving it would renumber chapters 18–22 for no gain.

v1.8.0 — 2026-09-24 — hecke-extensions slot 8, WP-9: the integral line (L2-C), C-A5 re-scoped onto the unrefined tessellation, and the F-A remark — an X bump

This version. The slot's one X revision of this document (`docs/implementation/hecke_extensions_slot8_pla` §3 "WP-9", §4.3), made on the owner's rulings of 2026-09-24 (§10 there): 6b (A), the chapter must-accept; 6c (A), the extension plan's contract item C-A5 folded into it on the *unrefined* tessellation, OD-T1 answered "not needed"; 6d, the integral-line decisions OD-3 (c), OD-4 (a), OD-6 (a), OD-7 (a), OD-9 (a), OD-10 (a) (record keys only under the value naming $\mathbb{Z}$) and OD-11 (b)/(a) confirmed; 10 (A), the F-A remark beside `cl:bockstein`. Sources: the integral-line design brief, revision 2 (`docs/implementation/integral_line_design_brief.md`) and its audit, and the ℓ-adic investigation (`docs/implementation/e8_ladic_investigation.md`) — project notes, not sources of record; evidence: slot-8 WP-7 and WP-8. **Twenty numbered statements are added, all [new]**; one hypothesis ((H3)) and one theorem's closing paragraph (`thm:specialisation`) are amended; the notation chapter gains a subsection; **no label is renamed or removed and no existing tag moves**. Appendix B's counts are re-derived mechanically from its table. Report: `docs/implementation/slot8/WP-9_report.md`. Commit: `1760a4a`.

Normative.

1. `sec:integral` — **new chapter**, Part VI, placed after Part V so that no chapter number of v1.7.4 moves; containers `sec:integralobjects`, `sec:isomodel`, `sec:descent`, `sec:strictlift`, `sec:badprimes`, `sec:ringsemantics`, the display `eq:treemodel` and the rule tags `IM1`, `IM2`.
2. `def:ringinforce` [new] — the three complexes on the model (integral, its base change, the complex over the ring in force) and the notation rule that $C^\bullet_{\mathrm{eq}}(R)$ is always the third.
3. `prop:ringinforce` [new] — the reduction map injective, an isomorphism at $6 \in R^\times$ (from `prop:coneintegral`(2)); *not* a quasi-isomorphism at $(n, R) = (2, \mathbb{F}_2)$, with a proof (the design brief's audit had shown it on the golden $\dim H^0$ values); the odd-n divergence of the four objects, including $H^2(\Gamma; \mathrm{Sym}^1\mathbb{Z}^2) \neq 0$.
4. `prop:bredon` [new] — the shipped complex is Bredon cohomology over the ring in force: the fixed-set check on X and on the Bass–Serre tree, the $\mathbb{Z}[\Gamma]$-chain homotopy equivalence $C_\bullet(X) \simeq C_\bullet(\mathcal{T})$ through the orbit category of the family $\mathcal{D}$, the tree complex (65), independence of the triangulation. The design brief rested this on the classifying-space theory of proper actions with a pointer it had to drop as unverified; the proof here uses only the comparison theorem in a functor category and `cl:amalgam`(1).

5. `def:isomodel`, `prop:isomodel` [new] — the isotropy-resolution model on the unrefined $(X, \partial X)$, its existence by a terminating procedure (boundary summands first), what it computes over every ring in every degree, the one integral comparison map cmp^* (cone annihilated by 48, a quasi-isomorphism at $6 \in R^\times$), and its comparison with `def:borelmodel`. **The ranks are 5, 14, 19, 19, ...**: the design brief (§2.5(b), risk K10) and its audit (m-9) counted the 7, 11, 5 simplices of K instead of the 5, 9, 5 Γ-orbits of `lem:model7coincidences`(2), and so gave $7(n+1)$, $18(n+1)$, $23(n+1)$ where the cochain ranks are $5(n+1)$, $14(n+1)$, $19(n+1)$.
6. `rmk:cafive` [new] — C-A5 discharged item by item on the unrefined tessellation; OD-T1 answered "not needed", with the reasons already in this document; a refinement, if ever wanted, compares integrally.
7. `prop:fracture`, `prop:cohglue`, `prop:descent`, `prop:homvanish` [new] — the arithmetic fracture square as an exact sequence of complexes; glueing at the cohomology level exact over $\mathbb{Z}$, **with the design brief's finite-generation hypothesis removed** (the proof uses flatness only); the owner's descent realised by the one map cmp^*; and Hom^{-1} over $\mathbb{Q}_2 \times \mathbb{Q}_3$ zero for the bulk, the coherence datum real only for the pair (`prop:coneambiguity`).
8. `rmk:truncdevice` [new] — the untruncated complex is what is glued, the brutal truncation is `open:ell2`'s box on the free model, and the object is not perfect ($n = 0$).
9. `lem:strictlifting`, `prop:strictlift` [new] — the strict lift of the strict pair datum of `prop:strictpair` along cmp, boundary first, with vanishing pair homotopy; its coset transport intertwined with that of τ^s on the nose over every ring; agreement with the shipped operator on cohomology at $6 \in R^\times$ only. **Existence only**: effectivity and certification are code (risk K1).
10. `prop:badprimes` [new] — the bad primes as a closed-form list, and 2-torsion only at odd n.
11. `prop:certifiede` [new] — the certified-exponent rule for any complex of finitely generated free $\mathbb{Z}$-modules, hence at 2 and 3 on the free model; where the rank over $\mathbb{Q}$ comes from; and the **scope clause**: not the model complex over $\mathbb{Z}/\ell^e$ at $\ell \in \{2, 3\}$, which is no reduction of an integral complex.
12. `prop:crtcharpoly` [new] — the residue identity behind χ_p over $\mathbb{Z}$ and its determination by residues against a bound, with no integer beyond a word; the bound is an input.
13. `def:ringsemantics` [new] — what each coefficient ring names; the value naming $\mathbb{Z}$ is group cohomology, never $C^\bullet_{\mathrm{eq}}(\mathbb{Z})$, basis-free (OD-11 (b)); the setting itself is [E, `sec:ring`]'s and is not restated (single-editor rule).
14. `rmk:integralevidence` [new] ([verified computationally]^{r}) and `rmk:integralscope` [new] (commentary) — what the two slot-8 validations test, and what the chapter does not say (no arbitrary precision: OD-3 (c), `thm:backend`(3), `obl:backend`(iv) unchanged).
15. `rmk:bocksteinexp` [new] — **the F-A remark** beside `cl:bockstein`: the Bockstein counts the exponent-one summands, and the Frobenius lower bound (an element of order ℓ^{j+1} at $n = \ell^j m$, m a Dickson degree), both [proved here]; the probe's exact exponents at $(30, 5)$, $(60, 5)$, $(150, 5)$ and the rest of its sweep, [verified computationally]^{a}. **A remark, not a claim change**: `cl:bockstein` is untouched and nothing beyond "$\geq$" is asserted.
16. (H3) — **amended**: "no comparison map between the two is asserted here, in either direction" becomes "no isomorphism ... is asserted there", with the pointer to `prop:bredon` and cmp^*. **Narrows a negative statement to what stays true**; every hypothesis the comparison statements carry, $6 \in R^\times$, is unchanged.
17. `thm:specialisation` — its closing "Not asserted" paragraph gains the pointer that the deferred construction is no longer deferred for $(X, \partial X)$ itself. Clauses (1)–(4) and their tags are unchanged.
18. `sec:notation-integral` — new subsection of the notation chapter (normative

under (CL5)): the symbols of §23 and how the design notes' symbols read here; `gl2_preamble.tex` gains the seven macros behind them.

Non-normative.

- `rmk:orbifold` — its design-direction paragraph records the answer (a direction nothing depends on, (CL5)); `rmk:mpcaveat`(4), `rmk:orbifoldrefine` and `sec:orbifoldmodel`'s opening paragraph gain pointers.
- §1's reading guide names Part VI.
- `sec:codemap` — a row for `sec:integral`, and the `sec:six` / `sec:orbifoldmodel` row's last sentence, which this version falsifies (C-A5 no longer fixes a refinement); `sec:notimplemented` — a v1.8.0 paragraph, for the same reason. The fuller status is the slot close's.
- Appendix B (twenty lines, the counts), Appendix C (one item), Appendix D (the new labels), this appendix's provenance table and roster, the header comment and the date line. Containers.
- **Outside this document** ((CL2)): none. The extension plan records the re-scoping of C-A5, E9 and R20 at its next revision (the slot close); the companion specification is not touched.

Not done here, deliberately.

1. **The arbitrary-precision ban is not re-scoped** — `thm:backend`(3), the box of §19, `obl:backend`(iv): owner decision OD-3 (c).
2. **No record-schema rule for the value naming** $\mathbb{Z}$ — the keys are the companion specification's; `sec:recordschema` is not amended.
3. **The pair and the cone over** $\mathbb{Z}$ beyond `prop:strictlift`(1)'s datum, and any obligation form of `prop:certifiede` — a later revision's, with the code that needs them.
4. **Versions 1.7.2–1.7.4** (the three status-only slot closes) carry no subsection of this appendix; their record is the header comment and the date line. Found while this entry was written; not repaired here, since reconstructing another version's entry is the slot close's call.

v1.7.4 — 2026-09-23 — hecke-extensions slot 7, WP-14, the slot close: status only — a Y bump

This version (entry reconstructed at v1.8.1 from the date line, the header comment and the commit; nothing is added that those do not say). **No normative predicate, no hypothesis, no displayed formula, no tag and no label moves.** Close report: `docs/implementation/slot7/WP-14_report.md`. Commit: `cc166d4`.

Status (non-normative).

- `sec:codemap` and `sec:notimplemented` — phase E7 is **accepted**: `cl:pd`(1)–(2) are *tested* on all 54 cells of the extension stage's case list at $\ell \geq 5$ (slot-7 WP-1) and fail on none; `cl:pd` stays open. The extension record of `sec:recordschema` item 6 is `hecke_ext/v3` (slot-7 stage 3), its new keys being [E]'s field inventory, so no rule of item 6 moves. Ten mathematical keys are registered.
- The registered-validation counts re-derived: 106 → 108 registered, 60 → 61 `Hecke*`. Companion pointer [E] v1.8.2. The header comment and the date line.

v1.7.3 — 2026-09-22 — hecke-extensions slot 6, WP-14, the slot close: status only — a Y bump

This version (entry reconstructed at v1.8.1, as above). **No normative predicate, no hypothesis, no displayed formula, no tag and no label moves.** Close report: `docs/implementation/slot6/WP-14_report.md`. Commit: `41792ac`.

Status (non-normative).

- `sec:codemap` — `sec:contragredient` is **implemented** (slot-6 WP-1: the contragredient, Φ_n, the coefficient pairing, $H^*(D_n)$ and $A(D_n)(T_p)$); `obl:coneexact`(vii) is **implemented in all three sentences**: the strict-pair route (WP-4), route 3 (WP-5) and the Poincaré-duality gate for X_3 at $\ell \geq 5$ (WP-3), which `cl:pd` still only *tests*. The presentation and atlas units and identity-certificate items 6 and 7 are named.
- `sec:notimplemented` — one chapter unimplemented (§17), none partial except `obl:indep`'s clause (iii).
- The registered-validation counts re-derived by `--list`: 100 → 106 registered, 55 → 60 `Hecke*`. Companion pointer [E] v1.7.2. The header comment and the date line.

v1.7.2 — 2026-09-22 — hecke-extensions slot 5, WP-13, the slot close: status only — a Y bump

This version (entry reconstructed at v1.8.1, as above). **No normative predicate, no hypothesis, no displayed formula, no tag and no label moves.** Close report: `docs/implementation/slot5/WP-13_report.md`. Commit: `53563c3`.

Status (non-normative).

- `sec:codemap` records what slot 5 shipped: the record is `hecke_ext/v2`, and the boundary, pair and cone complexes are emitted objects (WP-1); the cone's $Z/B/H/P$ and the cone certificate's span inclusions run on the tape, and its item 11 evaluates (40)(1) on its own guard (WP-4, WP-5), both value-preserving; `obl:coneexact`(iii)'s r-half is asserted at every ℓ (WP-4), and the cone action's certificate gains item 12, record honesty (WP-12a, owner ruling Q2); `obl:iota`(ii)'s rider gets an emitted record (WP-12a, owner ruling Q3); `sec:recordschema` item 5's refusal is a `verify()` code (WP-11a); the ρ_n memo, the Dickson multiplication and the canonical field are named.
- **Decided**: the owner's reading of `obl:taubdry`(vii) left open by v1.7.1 — Reading A, a distinct weaker clause, so v1.7.1's text stands as drafted.
- The registered-validation counts re-derived by `--list`: 94 → 100 registered, 49 → 55 `Hecke*`. The header comment and the date line.

v1.7.1 — 2026-09-22 — hecke-extensions slot 5, WP-12: the independent referee pass on v1.7.0 — pointers and self-contradictions only, a Y bump

This version. The slot's independent referee pass on v1.7.0 and on [E] v1.6.0 (`docs/implementation/hecke_ext` §3 WP-12), written by an agent that did not write v1.7.0, under the operating rule of `[OD]` slot-4 close-out 4: a referee may repair a self-contradiction the version under review itself introduced, including in a status sentence, provided the repair is pointers-only; anything wider is a finding for the close. **No normative statement, hypothesis, displayed formula, asserted predicate, tag or label moves**, so Appendix B's counts are unchanged. Report: `docs/implementation/referee_report_2026-09-22_contract_v1_7_0.md`, with the full numbered finding list and the questions routed to the owner; package report `docs/implementation/slot5/WP-12_report.md`. Commit: `5d1c083`.

Repairs (non-normative; five of the six are of v1.7.0's own text).

1. `sec:recordschema` item 6(f) — **rule (i) was bound and unbound at once**. The item's preamble binds "the nine below — (a)–(f) since v1.6.0, (g)–(i) with `v2`", and [E] §7 says item 6(i) "now makes" the matching rule a rule; (f), which scopes what this document binds, still read "(a)–(e) and (g)–(h)". It now reads "(a)–(e) and (g)–(i)". (At v1.6.2 the list ended at (f) and the sentence was correct; v1.7.0 appended (g), (h) and (i) and updated (f) for two of the three.)
2. `rmk:conerestrictsmallell` — **"load-bearing for no clause" two lines before the clause it is load-bearing for**. The remark's new opening said it is load-bearing for no clause and then, in the same sentence, that it is "still the record that discharges Obligation 14.19(vii) at $\ell \in \{2, 3\}$". The v1.7.0 changelog entry has the qualifier the body lost; the body now reads "load-bearing for no clause of §15".
3. `obl:taubdry`(vii), the asymmetry paragraph — **a premise this clause's own statement falsifies**. The paragraph rested on "the data *this* chapter stores are τ_∂ and Θ^∂_g alone"; but (vii) reads the bulk $T_{\Gamma g \Gamma}$ and the restriction r, and says so itself ("the only assertion that ties (30)'s choice of coset system to the bulk"). The distinguishing datum is η and h, not the bulk operator, and the paragraph now says that. **The conclusion is unchanged** — the proof of `obl:coneexact`(iii) is unavailable without h — so no predicate moves.
4. `obl:taubdry`(vii), the opening sentence — **"belongs to the pair and is stated in full at `obl:coneexact`"**, which was exactly true while the two clauses had the same content and which v1.7.0 falsified by strengthening (iii) alone. Aligned with v1.7.0's own new wording in (iii) ("the pair's form of Obligation 14.19(vii)"): (vii) is the boundary datum's form of a statement expounded in full at `obl:coneexact`(iii). **Open for the owner**, below.
5. `obl:coneexact`(iii), the proof — z **is named**. The proof read "let z be a stored equivariant cocycle"; *this document uses "stored" of the flat matrices as well* (clause (i)'s own v1.6.0 paragraph: "the stored matrices are honest linear maps of the flat cochain spaces"), where clause (i) is recorded and false in 395 of 424 degrees, so the adjective carrying the argument was "equivariant" alone. It now reads $z \in Z^q_{\mathrm{eq}}$, "an element of the span of the stored $C^\bullet_{\mathrm{eq}}$ basis". This is WP-11's own residual finding 5, which asked for exactly one more word if a referee found the reading.
6. `sec:recordschema` item 5, the implementation note — **the refusal landed**. v1.7.0 wrote the rule and recorded that the tree did not yet enforce it; hecke-extensions slot-5 WP-11a added `HeckeExtRecord::verify()` code 22 and inverted `HeckeExtRecordValidation`'s arm 13 to assert it, with a positive control on a source complex. The note records that, and the clause is no longer ahead of the code. (The v1.7.0 changelog entry above is left as written: it was true when written.)

Open for the owner — not repaired here.

1. **Is `obl:taubdry`(vii) a distinct clause or a pointer to `obl:coneexact`(iii)?** Repair 4 takes the first reading, because v1.7.0's own (iii) calls itself "the pair's form of" (vii) and its asymmetry paragraph is built on the two binding different implementations. The second reading — that (vii) always was (iii), named in the boundary chapter for signposting — is equally available from the pre-v1.7.0 text, and under it v1.7.0 strengthened (vii) too, so (vii)'s carve-out sentence and (iii)'s whole asymmetry paragraph come out. *Which reading is normative is an owner choice*; this version makes the document consistent under one of them and records the other.
2. `hecke_cone_action/v1`**'s** `restriction_equivariance_asserted` **is** `false` **at** $\ell \in \{2, 3\}$, being the stored `six_invertible`, although slot-5 WP-4 made the r-half assertion unconditional in ℓ and this document asserts it at every ℓ since v1.7.0. A record that says "not

asserted" where the run asserted is the mirror of what rule (A6) forbids. A code change, so not this pass's.

3. `obl:iota`**(ii)'s rider asks every implementation to "record which rows it ran the exact half on"**; the shipped `IotaCertificate` carries it (per-row `has_exact`, plus `exact_relation_data`), but no *emitted* record does — ι is in neither `hecke_report/v2` nor `hecke_ext/v2`. Either the rider means the certificate, or it is an open implementation obligation of the same kind item 5's was.

v1.7.0 — 2026-09-22 — hecke-extensions slot 5, WP-11: `obl:coneexact`(iii) at every ℓ with its proof, the `hecke_ext/v2` schema rules, and four deferred items — an X bump

This version. The slot's one X revision (`docs/implementation/hecke_extensions_slot5_plan.md` §3 WP-11), written on the findings of the packages accepted when it started — slot-5 WP-4 (the cone contract tightenings), WP-1 (the `hecke_ext/v2` schema) and WP-10 (the console) — and on the owner's rulings of 2026-09-22. **The one clause that moves an asserted predicate is `obl:coneexact`(iii)**: its $\ell \in \{2,3\}$ carve-out on the r-half is dropped and the identity is asserted at every ℓ. **No label is added, renamed or removed, and no tag moves**, so Appendix B's counts are unchanged. Status sentences — code maps, the validation counts, `sec:notimplemented` — are **not** this version's and stay with the slot close, the exceptions being the five places this version's own text falsifies: `rmk:conerestrictsmallell`'s opening and closing sentences, `obl:taubdry`(vii)'s last sentence, `obl:coneexact`(v)'s division of labour between certificate and validation arm, the obligation roster's `obl:coneexact` row, and `sec:recordschema` item 6's "implemented at" paragraph. Report: `docs/implementation/slot5/WP-11_report.md`. Commit: `2390850`.

Normative.

1. `obl:coneexact`(iii) — **the r-half is asserted at every ℓ, and the proof is written in**. $r^q A^{(q)} = A_\partial^{(q)} r^q$ aborts on failure with no hypothesis on ℓ; clause (i), itself unconditional in ℓ and aborting, gives $\Theta_g^\partial rz - r\Theta_g z = (dh + hd)z = d(hz)$ on a stored equivariant cocycle, a coboundary, whence the identity on cohomology. **Widens an asserted predicate** — owner ruling `[OD]` slot-4 close-out 3, Option A, which also asks that the implication stop being re-derived and become contract text. Two further paragraphs are part of the clause and not commentary: *what it does not say* — at $\ell \in \{2,3\}$ the assertion is about $H^\bullet_{\mathrm{eq}}(K, \mathcal{R}; L)$, $H^\bullet(C^\bullet_\partial)$ and the stored maps, and `rmk:conetorsion` keeps those separate from $H^\bullet(\Gamma; M_n)$, so the clause neither asserts nor licenses the group-cohomological reading there — and *the asymmetry* with `obl:taubdry`(vii), below. Discharged as written by `HeckeConeActionValidation`, whose A6 arm re-derives the identity independently of the certificate and, since slot-5 WP-4, *asserts* at every ℓ what it previously only measured: 424/424 (cell, q) pairs, the 74 operators at $\ell \in \{2,3\}$ included; and by `ConeActionCertificate`, whose r-half guard lost its $6 \in R^\times$ conjunct in the same package.
2. `obl:taubdry`(vii) — **keeps its carve-out, and gains the reason**. The boundary chapter's stored data are τ_∂ and Θ_g^∂ alone: there is no η and no h, so item 1's proof is unavailable to an implementation of the boundary datum, which is what that obligation binds. The two clauses bind different implementations and therefore do not contradict each other; for one that discharges both, the identity is asserted at every ℓ under (iii) and this carve-out is vacuous. **No predicate moves here.** Discharged as before by `HeckeBoundaryOperatorValidation` (216/216 recorded at $\ell \in \{2,3\}$).
3. `rmk:conerestrictsmallell` — **demoted from the clause's authority to the measurement that preceded the proof**. It is [verified computationally]^{r}, which never up-

grades and never disappears ((CL4), §1.1), so the remark stays, its title now saying what it is; it remains the record that discharges `obl:taubdry`(vii) at $\ell \in \{2, 3\}$, and is load-bearing for no clause of §15. Its closing sentence, which said the 424/424 were "recorded there and not asserted", is corrected: they are asserted, on the strength of clause (i) and not of this record, which licenses nothing (rule (A6)). **Every recorded pair is unchanged.**

4. `obl:coneexact`(v) — **the unconditional first identity of** (40) **is the certificate's too**. The clause's closing sentence assigned the second identity to the stored object's certificate and the first to "the validation arm that carries it"; it now names *both*, each identity under its own minimal guard. **No predicate moves** — the clause already required the first identity unconditionally — but the division of labour it states is now the shipped one: `ConeCertificate`'s item C11 is split in place, keeping the item code, and the split is shown non-vacuous on a constructed cell (owner ruling `[OD]` slot-4 close-out 2, Option A; slot-5 WP-4 §3). Discharged by `ConeCertificate` item C11 and by `HeckeConeValidation`'s G5 arm and its new negative control.
5. `sec:recordschema` item 5 — **the names of the complexes a complex is derived from are the record's own derived flag**. Non-empty exactly on a derived complex; consequently `q_max_source = "set"` on a complex with a non-empty list is *non-conforming* and a verifier that checks internal consistency refuses it. **Adds a refusal, adds no field** (v1.5.0 already required the names). This closes slot-5 WP-1 finding 3, which that package deliberately left open because the choice changes what a conforming record may say; the alternative — that the rule binds only the builder — is rejected because it leaves the violation undetectable from the record, which is what this section exists to prevent. `HeckeExtRecordValidation` pins the gap today, in its negative form, and is where the refusal becomes an assertion.
6. `sec:recordschema` item 6 — `hecke_ext/v2`, with nine binding rules in place of six. (b) is amended: in the extension family a verdict that is a property of the whole run is carried **at record level, once, with a single** `resumable`, and the incomplete-item count is then a count of **decisions** — v1.6.0's rule stands for a schema that does replicate such a verdict, and the change of an existing key's *value* is why this is a version bump and not an amendment in place, made before the first emission (owner ruling `[OD]` slot-4 close-out 5). (c) gains the rule one level down: a key the schema requires but this record has not computed is `null`, never a default and never dropped — dropping makes a closed list a function of the cell, and a `false` on an unevaluated diagnostic destroys rule (A6)'s distinction (slot-5 WP-1 findings 2 and 4). (g): a record about several complexes reports the *maps* between them, per spot, with `obl:coneexact`(iv)'s notion of "reported" governing which spots exist and (c)'s `null` governing the rest. (h): a ±-split record is two further records of the same cell, each ideal carrying its record's sign, and the per-sign dimensions **sum** to the unsplit ones — *asserted*, being `prop:iotasplit`(1) itself. (i): a matching a record performed is a matching it reports, with its witness; a count is not a report. Discharged by `HeckeExtRecordValidation` and `HeckeExtRamanujanValidation`.
7. `sec:recordschema` item 6(a) and (d) — **the two halves of v1.6.0's deferred item 3, answered**. (a): the settings block does *not* name the engine settings it excludes; the criterion is that a setting belongs iff it may change a published value (owner decision R1) and the `ENGINE_*` family is excluded *by that rule*, since enumerating it would make a record a function of the build. (d): the closed list of certificate names lives in [E, §7] with the rest of the field inventory; this document binds that it is closed, that the two payload forms are distinguishable by a reader holding only the record, and that a payload outside it is non-conforming.
8. `obl:iota`(ii) — **the rider for a complex without exact group data**. The *fibre* half is asserted unconditionally, always. The *exact* half is asserted on every row whose γ the

complex stores, an implementation whose complex stores them may not skip it, and every implementation **records** which rows it ran it on, so that a complex carrying no exact rows discharges the fibre half and records the exact half as not evaluated rather than reporting the clause passed. **Names the data the clause reads; excuses nothing.** Deferred item 2 of the v1.6.0 entry (slot-4 WP-3 finding 3). Discharged by `HeckeInvolution`'s `IotaCertificate` (the recorded flag, and the `verify()` clause that re-derives the exact comparison where the rows are present) and by `HeckeInvolutionValidation`.

Non-normative.

- The obligation roster's `obl:coneexact` row — the (iii) cell rewritten to the new clause, with the assertion status of the validation's own re-derivation. Falsified by item 1.
- `sec:recordschema` item 6's "implemented at" paragraph and its status sentence — `hecke_ext/v2`, slot-5 WP-1, `HeckeExt/HeckeExtComplexRecords` named. Falsified by item 6. The fuller status is the slot close's.
- The title page's companion pointer — `hecke_extensions.tex` v1.5.1 → **v1.6.0**, that companion having been revised in this same package. Container.
- The header comment, the date line and this appendix. Containers.
- **Outside this document** ((CL2)): [E] v1.6.0 — the `det`-twist convention of `cl:glover`'s preamble recorded as **the gate on phase E7**, which [E] had not said; §7's two booked schema repairs marked taken at `hecke_ext/v2`; and the three schema findings of slot-5 WP-1 that are inventory rather than rule. [P] v3.3 — the two console verbs bound, the C5-early scripts, the counts. [X] and `CLAUDE.md` are the slot close's.

Deferred to the next X revision.

1. **The ι-equivariance of the derived homotopy** — a clause of `obl:eta` or a counterexample cell. Deferred at v1.6.0 because no shipped validation asserts it and one cell is not evidence, and deferred again here: of the packages accepted when this revision started, none widened the measurement. Slot-5 WP-1 widened the per-sign *dimension* assertion, which is `prop:iotasplit`(1) and not this; $[\iota_c, A_c^{(q)}] = 0$ is still recorded at $(10, 691)$ alone (§15.3's closing paragraph, corrected at v1.6.1), and $\iota h = h\iota$ is measured nowhere.
2. **Any normative finding of a package accepted after this revision started.** The slot rule routes it to the close's open-point list and to the next X, which is what keeps the document chain off the critical path.

v1.6.2 — 2026-09-22 — hecke-extensions slot 4, WP-11: the slot close — status only, a Y bump

This version. The slot close. **No normative statement, hypothesis, displayed formula, asserted predicate, tag or label moves**; every edit below reports what the tree now contains. The slot itself moved **no golden value and emitted no golden entry**: `git diff 9ae1d7a` is empty for `Golden/GoldenReference.inc`, for `docs/implementation/tau_table_p2.txt` and for `Hecke/HeckeReport.{h,cpp}`; `GOLDEN_STRICT=1` reports tier A 9446 failing 0 and tier B 4603 mismatching 0; `EngineDeterminismValidation` gives digest `0xe446b3bd2d28bd6e` on all eight arms.

Status recorded.

- **The production ι^* exists.** `sec:involution`'s 2026-09-16 census paragraph, its specification paragraph, `sec:codemap`'s row and `sec:notimplemented` are rewritten from the minimal pointers v1.6.1 left (the referee's B1), naming `Hecke/HeckeInvolution` (the morphism, with `compute()` discharging Obligation 9.10(i),(ii),(iii) blockwise and exactly over

$\mathbb{Z}$ and aborting on failure), `HeckeExt/HeckeSignSplit` (clause (v), $A_{\pm}^{(q)}$, the per-sign decomposition, the Eisenstein line) and `HeckeInvolutionValidation`, whose acceptance is *entrywise against the two validations' independent construction.* Slot-4 WP-3, phase E3.

- **The extension record is written, bulk complex.** `sec:recordschema` item 6 gains that status and names `HeckeExt/HeckeExtDriver`; `sec:codemap`'s level (b),(c) row gains `HeckeExt/HeckeExtRecord`, `HeckeExtReport`, `HeckeExtFinal`, the settings keys `commutant_budget` and `final_mode` (six keys, defaults and fingerprint 0 unchanged), and the first two E6 units `HeckeExt/HeckePredictions` and `HeckeExt/GloverFactors`, which realise no statement of this document. Slot-4 WP-1, WP-2, WP-7.
- **Chapter 15's cost work.** The row records WP-4's three re-implementations and states explicitly that *no clause was sampled, relaxed or moved out of* `compute()`: both certificates still run every item unconditionally, $A_c^{(q)}$ is bit-identical and both validations' printed output is byte-identical to the pre-slot-4 binary's. The cost *gate* is [X]'s E5 acceptance (d), not this document's — [C] states no cost bound anywhere — and WP-4's finding that (d) is not decidable as written is recorded there.
- **The citation census.** Remark 22.1 records the content-keyed allow-list (`relpath|<fragment>`, a stale entry failing, the retired `relpath:line` form rejected by name) and the slot-4 run: 429 labels defined, 1575 sites in 128 of 470 files, 4 allow-listed, 0 unresolved, 0 stale. Slot-4 WP-6.
- **Counts**, taken from `--list` and not asserted: 89 → 94 registered validations, 45 → 49 `Hecke*`. Appendix B's own row counts are unchanged, no statement having moved.

Left for the owner, decided nowhere here. The three questions v1.6.1's referee raised and did not answer stand open, verbatim: **(1)** `ConeCertificate` item C11 gates *both* identities of (40) on a stored $H^2_{\mathrm{eq}} = 0$ while clause (v) says the first may not be conditional — the clause is not false (it assigns the unconditional identity to the validation arm, and `HeckeConeValidation`'s G5 arm does assert it on all 74 cells), but splitting C11 so the certificate itself carries it is a *code* change and is slot 5's; **(2)** Obligation 15.22(iii)'s $\ell \in \{2, 3\}$ carve-out is weaker than clause (i) already proves, so asserting the r-half at every ℓ is a candidate for the next X; **(3)** whether the B1 pointer repair belonged to the referee pass at all — *decided by the slot lead: B1 stays*, minimal and pointers-only, and this version completes it.

v1.6.1 — 2026-09-22 — hecke-extensions slot 4, WP-9: referee pass on v1.6.0 — two recorded counts corrected, three self-contradicting status sentences pointed, a Z bump

This version. The independent referee pass the slot-4 plan puts after the document package (`hecke_extensions_slot4_plan.md` §3 WP-9), run on `sec:cone`, `sec:coneaction`, `sec:involution`, `sec:recordschema` and Appendix B from the diff, the source tree and the validations. **Changes no normative statement** ((V2), (CL5)): no hypothesis, displayed formula, asserted predicate, tag or label moves, and Appendix B's counts are unchanged (117/88, [verified computationally] 19) — they were recounted mechanically from the table and every one of v1.6.0's numbers is right. **The referee could not fault** the rewritten proof of `lem:conerelative`(5), the equivariant reading of `prop:coneambiguity`(2), `lem:etaexists`(V)'s repair, `obl:coneexact`(i)–(vi) as rewritten or `sec:recordschema` item 6, and confirmed each of them discharged as written by the validation that runs it. Report: `docs/implementation/referee_report_2026-09-22_contract_v1_6_0.md`. Commit: `8994c35`.

Normative. None. (CL5)'s test is met vacuously: the two items below correct *recorded measurements*, which rule (A6) makes records and not claims, and the third is a status pointer.

Corrections to recorded measurements and to status.

1. `rmk:iotapairing` — **"thirteen cells"** → ***eight***. The recorded grid is `HeckeInvolutionValidation`'s 15 cells $\ell \in \{2,3,5,7,691\} \times n \in \{0,2,10\}$; $(10,691)$ is one of the twelve with $\ell \neq 2$, so "$(10,691)$ and the twelve cells with $\ell \in \{3,5,7,691\}$" counts it twice, and the four at $n = 0$ — where $H^1_c = 0$ and `obl:iota`(v)'s d_k is not even defined, being stated for even $k \geq 4$ — were folded into the count while the sentence described only $n \in \{2,10\}$. The eight cells it describes are the eight it now claims, and the remark records that at $\ell = 3$ `obl:iota`(v)'s own hypothesis $\ell \geq 5$ does not hold, so the comparison there is measured against measured. [verified computationally]^{r} kept; every recorded pair is unchanged.
2. §15.3's closing paragraph — **the commutator is recorded at** $(10,691)$ **alone**, not on the 15-cell grid, whose arm builds ι_c, D and the $\pm$ dimensions and no cone Hecke action. The paragraph's own next sentence, "One cell is not evidence", already said so.
3. §9 and §22.2 — **three status sentences that v1.6.0's own new text falsifies** ("no `HeckeInvolution` translation unit"; the specification paragraph's "stands unchanged"; §22.1's "Both are validations: no *production* ι^* exists"; and §22.2's "what that chapter still lacks") each gain a **pointer** to `Hecke/HeckeInvolution` and `rmk:iotapairing`. This is the exception v1.6.0 made for `sec:recordschema` item 5 and owed here; the rewrite of the code maps and of the row itself stays with the slot close (`hecke_extensions_slot4_plan.md` §4.3), which is why the pointers say so.

Non-normative.

- The title page — **"Companion specification:** `hecke_extensions.tex` **v1.3.3"** → **v1.5.1**, three revisions stale (since [E] v1.4.0, slot 2) and not introduced by v1.6.0, but named because v1.6.0 moved [E]'s pointer to this document while leaving this document's pointer to [E]. Container.
- The header comment and the date line. Containers.
- **Outside this document** ((CL2)): [E] v1.5.1, the companion repairs — [E]'s `cl:harder`, its new predicate paragraph (*false* at its own edge cell $(n,\ell) = (0,2)$ and silent on every even k with $(\ell-1) \mid k$, i.e. on all of $\ell \in \{2,3\}$); `cl:glover`(4)'s raw-factor bound (*false*: two factors against one at $n = \ell$); and two proof defects in `cl:sharpbound`. [P] v3.2's one off-by-one ("thirteen keys" against "the twelve keys" at its §10 C2 (g)) is reported and left, [P] being outside this pass's editing scope; [X] and `CLAUDE.md` are the slot close's.

Left for the owner, decided nowhere here.

1. `ConeCertificate` item C11 gates *both* identities of (40) on a stored $H^2_{\mathrm{eq}} = 0$, while `obl:coneexact`(v) now says an implementation "may not make the assertion conditional" on it for the first. The clause is not false — it assigns the unconditional identity to the validation arm, and `HeckeConeValidation`'s does carry it on all 74 cells — but splitting C11 is a code change, and is not this pass's to make.
2. `obl:coneexact`(iii)'s new $\ell \in \{2,3\}$ carve-out on the r-half is weaker than clause (i) now proves: on an equivariant cocycle, (i) gives $r\Theta_g z - \Theta_g^{\partial} rz = -d(hz)$, a coboundary, at *every* ℓ. The carve-out is what v1.5.2's deferred list proposed and the owner pre-approved, so it stands; asserting the r-half at every ℓ is a candidate for the next X.
3. Whether item 3 above belongs to this pass at all or wholly to the slot close.

v1.6.0 — 2026-09-21 — hecke-extensions slot 4, WP-8: the eight items deferred at v1.5.2, and the normative findings of the accepted slot-4 packages — an X bump

This version. The X revision that v1.5.2's *Deferred to the next X* list asks for, run in slot 4 rather than slot 5 by owner confirmation (`docs/implementation/hecke_extensions_owner_decisions.md`,

"Slot 4 confirmations" of 2026-09-21, item 2; `hecke_extensions_slot4_plan.md` §3 WP-8). **All eight items are resolved**, the three that needed a real choice as the owner answered them — item 1(a) *yes*: (40)'s first identity is asserted on every computed cell, without consulting H^2_{eq}, so that a cell with $H^2_{\mathrm{eq}} \neq 0$ would be covered too (none occurs on the shipped grid); item 1(b) *yes*: the explicit im = ker check is required in every degree; item 1(c) *per mode* (an [E] item, not one of this document's). Together with them, the normative findings of the slot-4 packages accepted while this revision ran: WP-1's record-schema findings (`docs/implementation/slot4/WP-1_report.md` findings 1–3, 5), WP-3's involution findings (`WP-3_report.md` findings 1 and 5) and WP-2's two record-schema rules (`WP-2_report.md` findings 4 and 5). **WP-4's findings are not this document's**: E5 acceptance (d) is a clause of the implementation plan [X], not of this contract, which states no cost bound anywhere; its two defects — (d) pins neither the denominator's measurement protocol nor which E5 object the numerator is — go to the slot close with the measured numbers. **No label is added, renamed or removed**; one tag changes (`rmk:iotapairing` gains [verified computationally]^{r}) and Appendix B's counts move with it, to 117/88 and [verified computationally] 19 (14 registered). Status sentences — code maps, the obligation roster, `sec:notimplemented`, the validation counts — are **not** this version's and stay with the slot close (`hecke_extensions_slot4_plan.md` §4.3); the one exception is the half-sentence of `sec:recordschema` item 5 that said the extension record "is not yet written", which the new item 6 replaces and which would otherwise contradict this version's own text. Report: `docs/implementation/slot4/WP-8_report.md`. Commit: `ec92e3a` ((CL6), back-filled at v1.6.1).

Normative.

1. `obl:coneexact`(i) — **"as matrices" is named: on the stored $C^\bullet_{\mathrm{eq}}$ basis**. The cone identity $dh + hd = \Theta^\partial_g r - r\Theta_g$ and $D\Phi_g = \Phi_g D$ are asserted there, $D^2 = 0$ everywhere; the same two identities on the full flat cochain space are **recorded, never asserted**, the coset transport of `def:cosettransport` being defined on $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, V_R)$ (424/424 computed degrees on $C^\bullet_{\mathrm{eq}}$, 29 of 424 flat; $D\Phi_g = \Phi_g D$ flat on 1 of 212 operators). The qualifier is carried onto clause (viii)(b)'s tampered h'. Deferred item 1 (WP-6 finding 1); discharged as written by `HeckeConeActionValidation`.
2. `obl:coneexact`(iii) — **the $\ell \in \{2,3\}$ carve-out on the r-half**. $r^q A^{(q)} = A^{(q)}_\partial r^q$ is asserted for $6 \in R^\times$ and *recorded* otherwise, which is the carve-out `obl:taubdry`(vii) and `rmk:conerestrictsmallell` already carried and which (iii) contradicted by aborting with no exception; the other maps of (39) are asserted at every ℓ. Deferred item 2 (WP-6 finding 2); discharged as written by `HeckeConeActionValidation` (424/424 recorded).
3. `obl:coneexact`(iv) — **the explicit** im = ker **check is degree-generic**. "$\mathrm{im}(H^1_c \to H^1_{\mathrm{eq}}) = \ker r^1$ is checked explicitly" is replaced by $\mathrm{im}(\mathrm{pr}_{c*}^{q}) = \ker r^q$, both inclusions by solves, at *every* bulk spot all three of whose terms are reported. **Widens the asserted set** — owner item 1(b), answered *yes*. Deferred item 6 (WP-5 finding 5); discharged as written by `ConeCertificate` item C9, which already ran in every fully reported degree, and by `HeckeConeValidation`'s own inclusion checks.
4. `obl:coneexact`(v) and (40) — **the first identity is asserted without consulting H^2_{eq} at all**, the second where $H^2_{\mathrm{eq}} = 0$ holds in the stored data. **Weakens a hypothesis** — owner item 1(a), answered *yes*. Deferred item 5 (WP-5 finding 3); discharged as written by `HeckeConeValidation`, whose G5 arm asserts the first identity on all 74 cells of its grid, $\ell \in \{2,3\}$ included and with no reference to $\dim H^2_{\mathrm{eq}}$, and by `ConeCertificate` item C11 for the second. *Stated as a dependence, not as a counterexample*: on the shipped grid $H^2_{\mathrm{eq}} = 0$ everywhere, at $\ell \in \{2,3\}$ recorded rather than asserted, so no computed cell separates the two readings today; the clause says so in as many words rather than implying, as a first draft of it did, that the $\ell \in \{2,3\}$ cells are cells with $H^2_{\mathrm{eq}} \neq 0$.

5. `lem:conerelative`(5) — restructured to match: (40) is stated for a field, its **first identity with no further hypothesis** (exactness of (39) and $H^0_c = 0$ from clause (4) suffice) and its second under $H^2_{\mathrm{eq}} = 0$, which is also the hypothesis of the six-term window. **The proof is rewritten** in two halves accordingly, the first exhibiting the four-term exact sequence $0 \to H^0_{\mathrm{eq}} \to H^0(\partial) \to H^1_c \to \ker r^1 \to 0$ and the second saying where $H^2_{\mathrm{eq}} = 0$ is used. Joint with item 4 above; [proved here] kept.
6. `obl:coneexact`(vi) — **where the *full* T_{p^ν} comes from**: the cone pipeline at $g = \mathrm{diag}(1, p^\nu)$ emits the primitive operator, and the full one is assembled by (19), at $\nu = 2$ by adding $p^n I$ (`sec:primfull`, `err:e1`). Deferred item 4 (WP-6 finding 4); discharged as written by `ConeRelationsCertificate`.
7. `obl:coneexact`'s preamble — "the six maps of (39)" → **the three map families pr_{c*}^{q}, r^q, inc_{c*}^{q}, stored per degree**, with the eight non-trivial instances on the model complexes named and the two zero-dimensional ones marked. Deferred item 7 (WP-5 finding 11).
8. `prop:coneambiguity`(2) — **the realisation is a construction inside $C^\bullet_{\mathrm{eq}}$**: $C^1 := C^1_{\mathrm{eq}}(K, \mathcal{R}; L)$, $B^1 := d\,C^0_{\mathrm{eq}}$, $Z^1 := Z^1_{\mathrm{eq}}$ stated explicitly in the clause and in its proof, with the reason the flat reading is vacuous (on the contractible X the flat complex is acyclic in degree 1, (37)). Deferred item 3 (WP-6 finding 3, joint with item 1). [proved here] kept.
9. `lem:etaexists`(V) — "reduces to the single edge path" → "reduces to *the one path joining the two vertices*", with the reason (a stored path template may be several edges long). Deferred item 8 (WP-9 §7 item 4). The statement of (V) is untouched, as it was at v1.5.0.
10. `sec:recordschema` — **new item 6, the extension record `hecke_ext/v1`**: a second writer and not an extension of the per-degree one (whose text is a golden object); and its six binding rules — (a) the settings block carries value, `source` and the fingerprint of the non-default entries, which enters the record's hashes and the writing job's identity, so that *how* a value was set is itself value-bearing; (b) a budgeted item's `resumable` object, and the sidecar, whose name must be **injective** across several states of one kind and whose reference is **a path relative to the directory holding the record**; (c) an unavailable complex is the placeholder {`"status": "unavailable"`} and its degree fields are not emitted; (d) a non-final verdict's reason list, with `certificate_failed:<code or certificate name>` — a *closed name list* beside the clause index, because a certificate route may answer "no" with no `verify()` having failed — and `phase_reached` per item beside the run's high-water mark under separate keys; (e) the two recorded cone flags `phi_flat_identity` / `residual_flat_zero` of item 1 above, named here so that they have one spelling from the start; (f) what this document binds and what is [E, §7]'s. WP-1 findings 1–3 and 5 (`docs/implementation/slot4/WP-1_report.md`); discharged as written by `HeckeExtRecordValidation`, whose structural `verify()` runs (a)–(e). **Two further rules from the E2b driver** (slot-4 WP-2, `docs/implementation/slot4/WP-2_report.md` findings 4 and 5), each normative because each decides a published number: in (a), *a resumed run writes the settings block a fresh run at the same effective values would write*, so that a resume to the default budget is byte-identical to a fresh default run, fingerprint included — without the rule two conforming implementations differ in a published hash (discharged as written by `HeckeExtRamanujanValidation` arm 4, which asserts that byte identity); and in (b), *the count of incomplete resumable items counts record entries*, which may exceed the number of distinct incomplete decisions where a schema attaches a run-level verdict to each of several sub-objects — legal, and what the shipped record does with its per-ideal U_ℓ verdict, but a schema that does it must say so (discharged as written by `HeckeExtRecordValidation`, whose `verify()` asserts the count equals the number of incomplete entries).
11. `rmk:iotapairing` — **the mirror-split prediction becomes a record.** The remark

gains [verified computationally]^{r} and a closing paragraph: ι_∂^* and ι_c now exist, $r^q\iota^* = \iota_\partial^* r^q$ and $D\iota_c = \iota_c D$ are asserted, and on thirteen cells H_c^1 carries exactly the mirror of the split `obl:iota`(v) puts on H_{eq}^1. The *pairing* statement stays contingent on `cl:pd`; $\ell = 2$'s (k, k) pairs are correctly not summed. WP-3 finding 5; discharged as written by `HeckeInvolutionValidation`.

12. §15.3, a new closing paragraph — **what this document does *not* say about ι and the cone action.** $[\iota_c, A_c^{(q)}] = 0$ follows from nothing here: beyond `prop:iotacommutes` and `obl:taubdry` it needs $\iota_\partial^* h = h\,\iota^*$ on the stored $C_{\mathrm{eq}}^\bullet$ basis, and neither `lem:etaexists`'s construction nor `alg:taubdry`'s vertex policy is built to respect ι. The commutator is **recorded** zero at $(10, 691)$ for T_2 and is **asserted nowhere**; the statement, or a counterexample, is left for the next X. WP-3 finding 1, which proposed the recorded form rather than a proved one.

Non-normative.

- Appendix B — `rmk:iotapairing`'s row — *commentary* → [verified computationally]^{r}; the [verified computationally] split paragraph "thirteen of the eighteen" → "fourteen of the nineteen", with the v1.6.0 clause; the counts paragraph [verified computationally] 18 → 19 (13 → 14 registered), tagged/untagged 116/89 → 117/88, untagged remarks 36 → 35. Recounted mechanically from the table: 205 rows, [proved here] 93, [standard] 34, [verified computationally] 19, [conjectural] 7, [open] 6. No row added or removed — v1.6.0 adds no numbered statement.
- `sec:recordschema` item 5 — "the extension record of E2 … is not yet written" → a pointer to item 6. Container; the item's rule is unchanged.
- This appendix — the v1.5.2 entry's `[hash]` back-filled with `cfc6f0b` ((CL6)); the list of Y bumps under (CL5) is unchanged, v1.6.0 being an X. The header comment and the date line. Containers.
- **Outside this document** ((CL2)): [E] v1.5.0 (its own items (a)–(e), and the companion pointer → v1.6.0) and the console plan [P] v3.2, both revised by the same package; [X] and `CLAUDE.md` are the slot close's.

Deferred to the next X revision (slot 5, with E5 acceptance — findings no status bump may act on).

1. ι-equivariance of the derived homotopy: either a clause of `obl:eta` requiring $\iota_\partial^* h = h\,\iota^*$ where an ι-equivariant η has been constructed, or a counterexample cell. One measured cell is not evidence (WP-3 finding 1; the recorded form is item 12 above).
2. `obl:iota`(ii)'s exact half asks for the image triple "as exact $\mathrm{M}_2(\mathbb{Z})$ data together with stored simplex indices and their signs", which a local system carrying only $\rho_n(g_r)$ cannot supply; a rider saying what an implementation on a complex without exact group data must do instead — the fibre half $\rho_n(\iota) = M_r \rho_n(\iota) M_{r'}$, which is what makes ι_C preserve C_{eq}^q — would help a re-implementer (WP-3 finding 3).
3. Whether `sec:recordschema` item 6(a)'s settings block should also name the *engine* settings it deliberately excludes, and whether (d)'s closed name list belongs in this document or in [E, §7] alone. Raised by WP-1 finding 5's observation that [E, §7]'s `groups` object mixes groups with scalars indistinguishably — an [E] schema-hygiene point that item 6(f) leaves with [E].
4. Any normative finding of a slot-4 package accepted after this revision started (`hecke_extensions_slot4_plan.md` §4.3).

v1.5.2 — 2026-09-17 — hecke-extensions slot 3, WP-11: the slot close — status only: strict locality implemented, the cone chapter implemented except `obl:coneexact`(vii), the U_ℓ repair and the census in the code maps

This version. The close of hecke-extensions slot 3 (`docs/implementation/hecke_extensions_slot3_plan.md` §3 WP-11, §4.3: "WP-11 acts only on status"). Every sentence changed here reports what the tree does on 2026-09-17 at the slot's close (commit `d65e96f` before this version) and nothing else: the U_ℓ reach repair (WP-1), the Sturm sweep on real blocks (WP-2), strict locality mode (WP-3), the equivariant cone and its long exact sequence (WP-5), the cone Hecke action (WP-6), console C1a (WP-7, which realises no statement of this document) and the citation census (WP-10); the two document packages WP-8 / WP-9 are v1.5.0 / v1.5.1. The slot made **no golden re-baseline and no golden emission**: `GoldenReference.inc` and `tau_table_p2.txt` are byte-identical to the slot's base, and the determinism digest is unchanged. **Changes no normative statement** ((V2), (CL5)): no hypothesis, displayed formula, asserted predicate, tag or label moves; Appendix B's counts are unchanged (116/89). Report: `docs/implementation/slot3/WP-11_report.md`. Commit: `cfc6f0b`.

Normative. None. (CL5)'s test is met vacuously.

Non-normative.

- §22.1, the level (b), (c) row — "`strict` accepted and throwing a named not-implemented error until slot 3", **false since WP-3 merged**, the one outright-false status sentence v1.5.0 and v1.5.1 carried (WP-8 §6 item 1, WP-9 §7 item 3): strict mode implemented (the Frobenius radical, the Berlekamp subalgebra, the residue witness over $\bar{A}$, terminating, byte-identical to budget mode wherever budget mode certifies), `NotImplementedInThisBuild` surviving as a named error no settings value raises; the keys `u_ell`, `sturm`, `sturm_unavailable`; the extension layer's `HeckeExt/HeckeSturmSweep`. Container.
- `sec:cone`'s code map — "**Not implemented**: the cone … — slot 3" → **implemented except `obl:coneexact`(vii)**: `EquivariantMappingCone`, `Hecke/SL2ZBoundaryPair`, `Hecke/HeckeConeAction`, `EtaVertexPolicy` / `TauBoundaryVertexPolicy` and `table_fnv64()`, the two validations, the measured flat-versus-$C^\bullet_{\mathrm{eq}}$ reading of clause (i), the collectors defined and never emitted; the v1.4.2 paragraph kept below it as the record, its census sentence in the past tense, and a note that the class its item (c) anticipated now exists. Container.
- §20.3 — `obl:eta`: **partial → discharged** ((v) by `HeckeConeActionValidation`, the tampered h' failing exactly (v) at $(10, 691)$); `obl:coneexact`: **unimplemented → partial** (every clause but (vii), with the measured figures of WP-5 / WP-6); the preamble's count 80 → 89; "Reading the table": eleven → twelve rows discharged, two unimplemented → one partial. Container; the obligations' text unchanged.
- §22.1, the rows `sec:heckedatum` (`QuadPointT<I>`, `escalate_on_overflow`, `HeckeExt/HeckeUEllBudget`, the three U_ℓ validations), `sec:cone` (as above), `sec:six` / `sec:orbifoldmodel` (the restriction lifted as `Hecke/SL2ZBoundaryPair`) and `sec:obligations` (45 `Hecke*` entries); `lem:bounds` / `thm:backend`'s code-map paragraph in §18 (WP-8 §6 item 4). Containers.
- §22.2 — `sec:cone` "partial since v1.4.2" → implemented except clause (vii), PD adjointness and the record integration; and Appendix C's oracle item ("two chapters" → one chapter and one clause; the count 80 → 89). Containers.
- §22.5 item 2 and `rmk:labelcensus` — `label_census.sh` is written and registered as `ContractCensusValidation` (WP-10); the remark gains the registered run of the slot close (429 labels, 1408 sites in 121 files, 0 unresolved). Rule container; remark, tag [verified computationally] kept.

- `rmk:conerestrictsmallell` — the record gains the cone's own run (424/424, WP-6 finding 2). Remark, tag [verified computationally] kept; nothing is asserted that was recorded.
- `rmk:coneambiguity691` — the closing status sentence: [X]'s stale 4 is corrected at [X] v1.7, and the measured dim $H^1_c = 3$ is `HeckeConeValidation`'s. Remark, tags unchanged.
- `sec:involution`'s census sentence — the count 80 → 89; there is still no `HeckeInvolution` (phase E3 moved to slot 4 by owner confirmation). Container.
- This appendix — the v1.5.1 entry's `[hash]` back-filled with `de697e6` ((CL6); the commit as the referee made it, cherry-picked onto `hecke-extensions` as `ef8d1b4`, exactly as v1.5.0's `1f19cf5` landed as `821c621`); the list of Y bumps under (CL5) gains v1.5.2. The header comment and the date line. Containers.
- **Outside this document** ((CL2)): `CLAUDE.md` (the suite table re-measured, the slot-3 roster and architecture), `docs/KNOWN_ISSUES.md` §3 (the RSS figure reconciled with the frozen model), the implementation plan [X] v1.7, [E] v1.4.2 (status: strict mode, E1b, the cone; the [C] pointer → v1.5.2) and the console plan [P] v3.1 (status: C1a implemented).

Deferred to the next X revision (slot 5, with E5 acceptance — findings no status bump may act on).

1. `obl:coneexact`(i): "$dh + hd = \Theta_g^{\partial} r - r\Theta_g$ as matrices" and "$D\Phi_g = \Phi_g D$" read as statements on the stored $C^{\bullet}_{\mathrm{eq}}$ basis; the same identities on the full flat cochain space **recorded**, not asserted (they hold in 29 of 424 degrees and on 1 of 212 operators), the coset transport being defined on $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}(-, V)$; §22.4 to carry the two recorded flags (`phi_flat_identity`, `residual_flat_zero`). Normative (WP-6 finding 1).
2. `obl:coneexact`(iii) against `rmk:conerestrictsmallell`: (iii) aborts with no ℓ carve-out while the remark records the r-half at $\ell \in \{2, 3\}$; add the carve-out to (iii). Normative (WP-6 finding 2).
3. `prop:coneambiguity`(2): state $C^1 := C^1_{\mathrm{eq}}(K, \mathcal{R}; L)$ and $B^1 := d\,C^0_{\mathrm{eq}}$ explicitly (on the contractible X the flat reading is vacuous), and in `obl:coneexact`(viii)(b) that h' satisfies the cone identity on $C^{\bullet}_{\mathrm{eq}}$. Normative (WP-6 finding 3, joint with item 1).
4. `obl:coneexact`(vi) / §10.1: one sentence on where the *full* T_{p^2} comes from when the cone pipeline runs at $g = \mathrm{diag}(1, p^{\nu})$ ((19); at $\nu = 2$ adding $p^n I$). Wording (WP-6 finding 4).
5. (40): the first identity needs no $H^2_{\mathrm{eq}} = 0$ (exactness and $H^0_c = 0$ suffice), while `obl:coneexact`(v) gates both on it — a weakened hypothesis, i.e. a strengthening, for the owner (WP-5 finding 3; WP-8 §4 item 1).
6. `obl:coneexact`(iv): "$\mathrm{im}(H^1_c \to H^1_{\mathrm{eq}}) = \ker r^1$ is checked explicitly" read degree-generically, as implemented — widens the asserted set (WP-5 finding 5; WP-8 §4 item 2).
7. `obl:coneexact`'s preamble, "the six maps of (39)": on the model eight map instances, two zero-dimensional; a wording nit (WP-5 finding 11).
8. `lem:etaexists`(V): "the single edge path" → "the one path joining the two vertices" (WP-9 §7 item 4; not touched by a status bump).

Not for [C] and listed here only so that none is lost: [E] `sec:localcert`'s hypothesis on a *reported* matrix witness ($w \in A_S$, or commuting with the family), the sharpness identity $\dim_{\mathbb{F}_\ell} \bar{A} = \mathrm{lcm}_p \deg \pi_p$ on a local block, the mode-dependence of the sub-block order after a split and `phase_reached` per block versus high-water mark (WP-3 findings 2, 3, 4, 6 — the next [E] revision); [P]'s phase column in §3.5, `golden()`'s home, the extension layer's console bindings and C1b's obsolete `strict expect_error` line (WP-7 findings 1–4, WP-3 finding 7 — the next [P] revision, with C1b).

v1.5.1 — 2026-09-17 — hecke-extensions slot 3, WP-9: referee pass on v1.5.0 — the ten resolutions accepted, four arithmetic and two proof defects in the U_ℓ item repaired

This version. The independent read of v1.5.0 that `docs/implementation/hecke_extensions_slot3_plan.md` §3 WP-9 asks for, on the model of v1.4.1 (`9c54b91`) and v1.3.1 (`f69cda0`). Every one of the ten resolutions was re-derived: `rmk:coneambiguity691`'s arithmetic line by line against the measured row for $(10, 691)$ of `docs/implementation/slot3/WP-5_report.md` §3 ($\dim H_c^\bullet = [0,3,0]$, $\dim H_{\mathrm{eq}}^\bullet = [0,3,0]$, $\dim H^\bullet(\partial) = [1,1]$, $\operatorname{rank} r^0, r^1 = 0, 1$, $\dim H_!^1 = 2$) and against `eq:conedims`, `cl:dimH1`(1)–(2) and `docs/implementation/slot2/WP-7_report.md` §3.2–3.3; `obl:coneexact`(iv)'s per-complex "reported" against 14.24's box, its rider (iii) and the slot-2 owner confirmation 1 ("per complex, never as a maximum") — it is the box's own per-term sentence applied literally, it never asserts a spot with an unreported term, and where the three $q_{\max}$ agree (the only case the spelled-out clause defines) it reproduces that clause verbatim on a truncated model and adds, on an untruncated one, only spots all of whose terms are 0; the three (T1) helpers, the one conjugation primitive and the three `table_fnv64()` accessors against the tree (`Hecke/HeckeTauTable`, `Hecke/HeckeBoundaryDatum`, `Hecke/HeckePairHomotopy.h:247,252,253`); the "MAY omit" branch of `obl:eta`(iv) against the shipped solve regime (OP2's abort becomes *unconditional* there, hence is not weakened); `lem:etaexists`(V)'s routing and the η-record's four fields against `Hecke/HeckePairHomotopy.h`; and Appendix B's counts by a mechanical recount of its table (205 rows; [verified computationally] $= 18 = 13 + 5$, [proved here] 93, [standard] 34, [conjectural] 7, [open] 6, tagged/untagged 116/89 with 10 untagged obligations — every figure as v1.5.0 states it). (L1): no label was added, renamed or removed by v1.5.0 or by this version. **Changes no normative statement** ((V2), (CL5)): no hypothesis, displayed formula, asserted predicate, tag or label moves; the tier criteria of `thm:backend` are the same criteria in the same quantities. What was wrong lay in the U_ℓ item — two arithmetic slips of `thm:backend`(1), one unproved inference in its item (iv), two steps of `lem:bounds`(6)'s proof — with one over-stated sentence at `obl:oracle` and one stale quotation at (A4) beside them; each is repaired under its own statement's label ((L3)), and the seven repairs are listed below. Report: `docs/implementation/slot3/WP-9_referee_report.md`. Commit: `de697e6`.

Normative. None. (CL5)'s test is met vacuously.

Non-normative.

- `thm:backend` — clause (1), **two arithmetic corrections**, tag [proved here] kept, both criteria unchanged. *(a)* "for every $N \le 2^{15}$ with such a table" does not follow from $B \le 2^{31}$ and $B = \max(B_{\mathrm{tab}}, 4N^2)$: at $N = 2^{15}$ already $4N^2 = 2^{32}$. The condition is $4N^2 \le 2^{31} - 1$, i.e. $N \le 23170$ — which is exactly the reach §10 records in the same version, clause (e) of `obl:tau`'s certification being this criterion, so the two sites now agree. *(b)* "$7B_{\mathrm{red}}^2 < 2^{63}$ … means $N \lesssim 4580$" mixed a criterion with a measurement: the criterion gives $N \lesssim 2700$, and 4580 is the *measured* first overflow, which `rmk:btabsmall` derives from the intermediate actually formed, $\approx B_{\mathrm{red}}^2$. Both figures are kept, each with its own provenance; clause (2)'s $1.8 \cdot 10^8$, which *is* derived from the criterion, was what exposed the mismatch.
- `thm:backend` — item (iv), **an unproved inference removed**. "$B_{\mathrm{red}} \le 7C^2N^2$ by `lem:bounds`(6)" is not what that lemma proves: (6b) bounds the *entry* point z_0, and the reduction's own steps are not bounded by their entry norm by any argument in this document (a single S step on a point with $|z| < 1$ can multiply $\|z\|$ by a factor). The item now says so and cites `rmk:btabsmall`, where the constancy of $\max\|z\|$ along the reduction is [verified computationally] *measured* over $\ell \in \{11, \dots, 3617\}$ — which is all the clause

needs, its criteria being stated in B_{red} itself and clause (3) requiring overflow-checked arithmetic regardless. `lem:bounds`(6c)'s "with $\max_z\|z\|$ of order C^2N^2" is struck for the same reason and the measurement left where it is recorded. Nothing an implementation must satisfy changes.

- `lem:bounds` — proof of (6), **two steps repaired**, statement and tag unchanged. *(a)* (6a) bounds one Möbius step by $7\|\gamma\|^2\|z\|^2$, but `thm:backend`(iv) quotes $7\|z\|^2$, with no $\|\gamma\|$ — and $\|T^{-q}\| = |q|$ is not 1 for a shift by a large quotient, so the step needs the two shapes `alg:tau` step 1 actually applies: S, where $\|\gamma\| = 1$; and the *nearest*-integer shift, where $c = 0$, $\Delta = 1$ and $|u - qw| \le w/2$ collapse the formulas to magnitudes $\le 2\|z\|^2$. The proof now carries that specialisation. *(b)* the attainment argument of (6b) cited $v' = vw\Delta$, which for $\mathrm{adj}(g)s_j$ is of order C^2N, not C^2N^2; the coordinate that attains the order is w', through its summand $D\,c^2v^2$ with c of order N. Also $144.8\,N^2 = C^2N^2$ (here and at `rmk:btabsmall`) is an equality between 144.8 and 144; it is now $\approx$ / $\asymp$, and `rmk:btabsmall`'s int64 threshold is derived from the measured 144.8, which is where its 4580 comes from.
- (A4) — its worked example quoted `obl:eta`(ii) in the form v1.5.0 corrected ("both sides evaluated through the shared (T1) helper"), which was the one site of that wording left over: the other five — §22.1, `obl:tau`(iii), `alg:tau`(0), §10 and the v0.18.0 entry — are about the τ-table alone, where there *is* one helper (`eval_tau_on_translate`), and are correct as they stand. Rule, no tag.
- `obl:oracle` — one sentence of the v1.5.0 paragraph. "Off the Kummer loci E is a line" was asserted flatly; it is measured, on the 43 off-Dickson non-Kummer cells with $k \le 26$ of slot-2 WP-7, and is [E]'s `cl:harder` in general. The sentence says so, and the reason no assertion of the obligation moves is now stated positively — (i)–(iii) assert dimensions, `eq:esoracle` and the full/primitive distinction, none of which names a class inside E — instead of resting on the line case. Obligation, no tag.
- Appendix E — the v1.5.0 `lem:bounds` line gains [new] under (CL4)'s rule for a preserved label whose statement is strictly generalised (clause (6) is new content with its own proof), with the line saying that Appendix B's row stays [repaired] and no provenance count moves. That is the referee's answer to the open question WP-8's report raised under that head. (CL5)'s list of Y bumps gains v1.5.1. The v1.5.0 entry's `[hash]` back-filled with `1f19cf5` ((CL6)). The header comment and the date line. Containers, no tag.
- **Outside this document** ((CL2)): nothing. [E] v1.4.1 was read and accepted as it stands (the canonical generator exists by Nakayama, is a function of the reported data, and no claim of [E] depends on it, so the Y bump is right); `CLAUDE.md`, `docs/KNOWN_ISSUES.md`, [X] and [P] are the slot close's.

The ten resolutions of v1.5.0: accepted. All ten stand, eight of them unaltered. The two that moved are `thm:backend`/`lem:bounds` — item 7 of the deferred list, whose *mathematics* is accepted (the gap is real, clause (6) is the right place to close it, and the site, the bracket $3617 < N < 5003$ and the new reach $\ell \le 23170$ are re-derived here) and whose arithmetic and proof carried the six defects above. `obl:coneexact`(iv), which v1.5.0's entry names as "the one entry a referee should attack", is accepted as written: it is 14.24's own per-term formulation ("exact at every spot all three of whose terms are reported"), it is what the owner's slot-2 confirmation 1 decided ("per complex, never as a maximum"), and where the three $q_{\max}$ differ the single-$q_{\max}$ sentence has no reading at all — which is the defect the rule repairs, not a set it widens.

v1.5.0 — 2026-09-17 — hecke-extensions slot 3, WP-8: the seven deferred items of v1.4.2, and the normative findings of the accepted slot-3 packages — an X bump

This version. The one X revision of hecke-extensions slot 3 (`docs/implementation/hecke_extensions_slot3_`§3 WP-8, §2.2; the owner pre-approved the seven "as the slot-2 reports propose" on 2026-09-17, `docs/implementation/hecke_extensions_owner_decisions.md`, "Slot 3 confirmations" item 1). It does two things and nothing else. **(1)** It resolves the deferred list the v1.4.2 entry below hands it — all seven, none carried forward. **(2)** It applies the *normative* findings of the two slot-3 packages accepted when it started: WP-1, the U_ℓ exact-integer reach repair (`docs/implementation/slot3/WP-1_report.md`), and WP-5, the equivariant cone complex and its long exact sequence (`docs/implementation/slot3/WP-5_report.md`). **No label is added, renamed or removed** ((L1), (L6)): every change is a clause, a paragraph or a sentence inside a statement this document already had, so B's line count is unchanged and no provenance tag is created. **Status sentences are not this version's** (§3 WP-8 scope): the obligation roster, the code maps, §22.2 and §22.1 are v1.5.2's, so that a status never precedes the merge it describes — with the single exception the deferred list itself demands, item 7's U_ℓ reach record, which *is* the finding. The normative findings of WP-6 (the cone Hecke action), and any finding that arrives after this version started, go to the next X revision (slot 5), exactly as slot 2 handed its seven here. Report: `docs/implementation/slot3/WP-8_report.md`. Commit: `1f19cf5`.

Normative.

- `rmk:coneambiguity691` — **deferred item 1**, the rewrite. The displayed arithmetic was computed under $r^1 = 0$ at $(10, 691)$, which is false. Rewritten from the measured values: $\operatorname{rank} r^0 = 0$ and $\operatorname{coker} r^0$ one-dimensional; $\operatorname{rank} r^1 = 1$, $\dim \ker r^1 = \dim H^1_! = 2$; hence $\dim X_3 = 1 - 0 + 2 = \mathbf{3}$ by `eq:conedims`, confirmed by a direct $\dim H^1_c = 3$ (slot-2 WP-7 `HeckeBoundaryParabolicValidation`; slot-3 WP-5 `HeckeConeValidation`, which finds the same on all ten Kummer cells), and $\dim H^2_c = 0$; the ambiguity $\operatorname{Hom}(\ker r^1, \operatorname{coker} r^0) \cong \mathbb{F}^2_{691} \neq 0$. **The remark's conclusion is unchanged** — a free cochain-level h leaves the T_p-action on X_3 undetermined and the derived one does not — and it is why the remark exists; only two dimensions moved, and the v1.4.2 numbers are quoted at the end of the remark so that a reader of the old PDF is not left guessing. The new paragraph "Why $r^1 \neq 0$" states the mod-ℓ reading of "the Eisenstein class" as the joint $1 + p^{n+1}$-eigenspace E ([E] v1.4.0 `cl:harder`) and records that it is the one input this document does not prove. Remark; [open] kept, with its reason corrected (it had said "§16 not yet written at this version", which has not been true since v0.19.0), [proved here] kept, and [verified computationally]^{r} **added** — the dimensions are now measured and re-runnable by name. The one tag change of this version.
- `obl:oracle` — **deferred item 1**, second half. A new paragraph after `eq:esoracle`: the *factor* $X - (1 + p^{n+1})$ is well defined wherever the obligation applies; the *class* the second standard pointer names ([Har22, Ch. 2–3], a characteristic-0 statement) need not survive reduction as a line, and at a Kummer cell does not — read "the Eisenstein class modulo ℓ" as the joint eigenspace E throughout this document, `obl:iota`(v) giving the two conditions under which a class rather than a space is locatable. **No assertion of the obligation moves**: off the Kummer loci E is a line and the two readings coincide. Obligation; no tag change (the measurement is cited to `rmk:coneambiguity691`, which carries it).
- `obl:eta`(ii) — **deferred item 2**. "Both sides evaluated only through the shared (T1) helper — the same helper the construction uses" named a helper that does not exist. Corrected to "through each table's own (T1) helper, all built on the one (P1) split", with the three helpers and the one conjugation primitive

(`HeckeCosetTables::twisted_conjugate`) named, and what the clause is really about — construction and certification sharing the derivation — said in place of what it had said. The asserted predicate is unchanged. Obligation clause, no tag.

- `obl:eta`(iv) — **deferred item 3**. "Behind the explicit setting that permits it" presupposed a setting no implementation need have. Now: a mod-m path, *where one exists*, is behind an explicit setting (unchanged), **and an implementation MAY omit the mod-m regime altogether**, in which case a tier failure aborts unconditionally and the recorded regime is $\mathbb{Z}$ by construction. The shipped implementation is of that kind, and the absence of the setting is conformance rather than a gap. **OP2's abort is not weakened** — it is made unconditional on the omitting branch. Obligation clause, no tag.
- `lem:etaexists`(V) — **deferred item 4**. A sentence after the procedure naming the constructive choice the implementation makes for the free vertex chain: the pairwise path routing of `alg:tau` step 2, which needs no linear solve and reduces to the single edge path when both values are single vertices. **The statement is unchanged** — (V) still admits *any* 1-chain with the stated boundary, which `lem:etaunique` is what makes the naming harmless. Lemma, [proved here] kept.
- `sec:recordschema` item 5 — **deferred item 5**, and WP-5 finding 2. A new block, "Maps, and complexes built from complexes": a record about a *map* (r, η, h) carries the four fields of its **source** complex and the degree shift is a schema constant fixed by the map's definition, never a field; on a truncated free model the map is built and certified in the source degrees the box of 14.24 builds its source complex in, and reported in the source degrees $\le q_{\max}$(source). A record about a complex *derived* from others (the cone) carries `q_max_source = "derived"` unconditionally with the names of its sources, and its $q_{\max}$ / `truncated` are the stated functions of theirs; `"set"` is legal on a source complex only, which is rider (i) of 14.24 one level up. Container ((CL4): no tag), but a normative schema rule, hence this list.
- `obl:eta` preamble — **deferred item 6**. "Together with the stored τ- and τ_∂-tables it was built against" is now "together with the *table hashes* of" them, with a paragraph fixing what a table hash is (one function, `table_fnv64()`, on each of the three tables, the FNV-1a-64 of `ModArithmetic/Fnv.h`) and what it is for: *identifying and never certifying*. Two admissible η differ in the first hash while $A_c^{(q)}$ does not (`lem:etaunique`, `prop:coneindep`(1)), so a hash may be recorded and compared but not branched on ((A6)), and clause (i) — not a hash — is what ties the table to its certified inputs. The three accessors exist in the tree. Obligation preamble, no tag.
- `lem:bounds` [new] — **deferred item 7**, and the substance of WP-1's finding: a **new clause (6)**, with proof. Clauses (1)–(5) bound the matrices the pipeline *stores*; `alg:tau` step 1 also carries a quadratic point z on cleared denominators that none of them mentions. (6a) one Möbius step forms integers of magnitude $\le 7\|\gamma\|^2\|z\|^2$; (6b) the point step 1 starts from satisfies $\|z_0\| \le 7C^2N^2$ with $C = \max_v\|\mathrm{pos}(v)\|$ ($C = 12$ here), and the order C^2N^2 is attained; (6c) so the tier must be chosen against $\max(B, 7C^2N^2)$ and *its* square, not against B. Lemma, [proved here]; the tag is unchanged and the proof carries the new clause. **The bound was not wrong — it did not exist**, which is why this is a new clause and not a repair. ([new] added at v1.5.1 under (CL4)'s rule for a preserved label whose statement is strictly generalised: the tag on this *line* is the new clause's. Appendix B's row for `lem:bounds` stays [repaired] — the statement as a whole is still a transported-and-repaired one — so no provenance count moves.)
- `thm:backend` — **deferred item 7** continued. A new item (iv) in the closure list (step 1's Möbius steps, of magnitude $\le 7B_{\mathrm{red}}^2$ with $B_{\mathrm{red}} \le 7C^2N^2$ by `lem:bounds`(6), and *not* bounded by B); and clause (1) **corrected**: it had said that $2B^2 < 2^{63}$ "covers every $N < 2^{15}$ outright", which is false — at $N = 5003 < 2^{15}$ the certified B leaves $2B^2$ four orders of magnitude inside the tier while step 1 leaves it. Clause (1) now states the two

criteria separately, and clause (2) likewise for the 128-bit tier. The wording "the ladder is entered per operation" is item (iv)'s consequence and is what the implementation does. Theorem, [proved here] kept; clause (3) (arbitrary precision never required) is unaffected — the 192- and 320-bit widths bound $7B_{\mathrm{red}}^2$ a priori as well.

- `obl:coneexact`(iv) — WP-5 finding 1, the per-complex reading of "reported". A term $H^q(F)$ is reported iff $q < 0$, or $q \le q_{\max}(F)$, or F is not truncated; the clause's existing sentence is this rule in the case $q_{\max}(\text{bulk}) = q_{\max}(\text{boundary})$, which is what 14.24's spelled-out clause states. **Nothing on the shipped models changes** — all three complexes are finite and untruncated there — and on a truncated model the rule is the one `HeckeConeValidation` implements and exercises. Obligation clause, no tag. **This is the one entry a referee should attack**: on a hypothetical model with $q_{\max}(\text{bulk}) \neq q_{\max}(\text{boundary})$ it asserts more spots than the single-$q_{\max}$ sentence did.

Non-normative.

- `rmk:btabsmall` — a second [verified computationally] record beside the $B \le 196$ one: the measurement that found `lem:bounds`(6)'s gap ($\|z\| = 144.8\,N^2$ over $\ell \in \{11, \dots, 3617\}$, constant along the reduction, independent of n), the int64 threshold $N \ge 4580$ it predicts — exactly the slot-1 bracket $3617 < N < 5003$ — and the fact that the slot-1 record's attribution of that bracket to the coset construction was wrong. Remark, [verified computationally]^{r} kept.
- §10's U_ℓ reach record — **deferred item 7**'s status half, which the item explicitly asks this version for. The v1.4.2 sentence ("the repair did not run") is replaced by the measured outcome: the site was step 1, not the coset construction; `lem:bounds` did not fail; and after the per-operation escalation the binding constraint is again `lem:bounds`(5), giving $\ell \le 23170$, past which the table is *refused by* ***`obl:tau`****'s certification* rather than lost to an anonymous overflow. With the measured $(16, 5003)$ and $(16, 10007)$ points. Container.
- Appendix B — `rmk:coneambiguity691`'s status column and the two counts that follow from it: [verified computationally] 17 → 18, its registered half 12 → 13. The tagged/untagged split 116/89 is unchanged (the remark was already tagged), and no line was added or removed. Container.
- This appendix — the v1.4.2 entry's `[hash]` back-filled with `833c98f` ((CL6)); the deferred list of that entry marked resolved. The header comment and the date line. Containers.
- **Outside this document** ((CL2)): `hecke_extensions.tex` goes to [E] v1.4.1, stating `cert:cyclic`'s canonical generator as part of the certificate rather than only as an implementation note, and re-pointing its companion citation at [C] v1.5.0. `CLAUDE.md`, `docs/KNOWN_ISSUES.md` §3, the implementation plan [X] and the console plan [P] are slot 3's close (WP-11), not this version's.

The v1.4.2 deferred list: resolved. All seven items below are closed by this version — items 1–6 as the slot-2 reports proposed and the owner pre-approved, item 7 by WP-1's report, which found more than the item expected: not a status sentence alone but a missing clause of `lem:bounds` and a false clause of `thm:backend`(1). **Nothing is carried forward.** What is *not* here, and why: WP-5's findings 3 (`eq:conedims`'s first identity needs no $H^2_{\mathrm{eq}} = 0$) and 5 (`obl:coneexact`(iv)'s "$\mathrm{im}(H^1_c \to H^1_{\mathrm{eq}}) = \ker r^1$ is checked explicitly" read degree-generically) would each *strengthen* a statement or widen an obligation's asserted set, which §3 WP-8 of the slot plan reserves for the owner; they are listed in WP-8's report for the next X. So are the findings of slot-3 WP-3 (strict locality) that bear on [E] `sec:localcert` rather than on this document.

v1.4.2 — 2026-09-16 — hecke-extensions slot 2, WP-13: the slot close — status only: `sec:recordschema` item 5 implemented, the code maps for the slot-2 units, `obl:eta`(i)–(iv) discharged

This version. The close of hecke-extensions slot 2 (`docs/implementation/hecke_extensions_slot2_plan.md` §3 WP-13, §4.3: "status only"). Every sentence changed here reports what the tree does on 2026-09-16 at the slot's close (commit `45f7c4e` before this version), and nothing else: the slot's one golden re-baseline (WP-2, `hecke_report/v2`, 384 report lines, 0 elsewhere), the locality certificate inside the refinement loop and the first mathematical setting (WP-1), the local algebra and both generation certificates (WP-4, WP-5), $H^1_!$, Harder and the two recorded diagnostics (WP-7), and the certified η-table (WP-8). The slot's planned U_ℓ reach repair (WP-3) **did not run**, and the record of `sec:heckedatum` says so. **Changes no normative statement** ((V2), (CL5)): no hypothesis, displayed formula, asserted predicate, tag or label moves; Appendix B's counts are unchanged (116/89, WP-10's recount). Report: `docs/implementation/slot2/WP-13_report.md`. Commit: `833c98f`.

Normative. None. (CL5)'s test is met vacuously.

Non-normative.

- `sec:recordschema` — the writer's schema string (`v2` since v1.4.2), and item 5's status: "scheduled" → **implemented at** `hecke_report/v2`, per complex, derived from `HeckeReportInput::num_degrees`, which the writer requires; `"set"` reserved for E9; the campaign writer's three un-hashed fields; the re-baseline's count and note. Container; site (6) of `sec:oc1sites` unchanged.
- `rmk:conestrict`(1) — the [verified computationally] record's counts re-read on the canonical τ (v1.1.0): 10 of 12 vertex slots and 5 of 9 edge slots leave ∂X (`hecke::cusp_support_walk`, WP-7, confirmed by hand on `tau_table_p2.txt`); the "$j = 1$ slots all stay" sentence was the pre-v1.1.0 table's and is withdrawn with the old counts kept in a parenthesis; the second statistic, 11 of 12 and 6 of 9 slots with a non-zero η (WP-8), added and distinguished from the first. The conclusion of (1) is unchanged and holds a fortiori. Remark, tag [proved here] kept (it tags (2), not the record).
- `rmk:coneambiguity691` — a status sentence: r^1 is now computed and has rank 1 at $(10, 691)$ (WP-7), so $\dim \ker r^1 = 2$ and $\dim X_3 = 3$; the stale arithmetic is left standing under its [open] tag and listed below for the next X.
- `obl:eta` — the roster row of §20.3: "unimplemented" → **partial**, (i)–(iv) discharged and (vi) recorded by `HeckePairHomotopyValidation`, (v) awaiting the cone. Obligation text unchanged.
- `sec:cone`'s code map — "nothing in this chapter is implemented" → implemented in part: `Hecke/HeckePairHomotopy`, `Hecke/HeckeIntegerSolve`, the validation, what is still missing; the census sentence past tense. Container.
- `sec:heckedatum`'s U_ℓ reach record — a status sentence: the repair (slot-2 WP-3) did not run; slot 3. Container.
- `sec:codemap` — rows `sec:heckedatum` (`HeckeIntegerSolve`), `sec:boundarydatum` (`HeckeExt/HeckeBoundary`, `Hecke/HeckeBoundaryDiagnostics`), `sec:cone` split from `sec:six` / `sec:orbifoldmodel` (the former in part), `sec:obligations` (39 `Hecke*` entries), the level (b),(c) row (`HeckeModuleDecomposition`'s certified loop, `HeckeSettings`, `HeckeExt/HeckeLocalAlgebra`, `HeckeExt/HeckeGenerationCertificates`), and the plan pointer v1.5 → v1.6 in the `sec:symconv` and `sec:orbifoldmodel` rows. Container.
- `sec:notimplemented` — "two chapters have no implementation" → one (`sec:orbifoldmodel`), `sec:cone` partial; chapter 14's sentence gains $H^1_!$ and Harder. Container.
- `sec:oraclesnew` and `sec:example`'s closing paragraph — the family (A)/(B) status sen-

tences (the second had said "no boundary Hecke operator" since before v1.3.3). Containers.

- `sec:involution`'s census sentence, §20.3's preamble and Appendix C's oracle item — the count 77 → 80 registered validations, and the oracle item's two chapters → one and a half. Containers.
- This appendix — the v1.4.1 entry's `[hash]` back-filled with `9c54b91` ((CL6)); the list of Y bumps under (CL5) gains v1.4.2. The header comment and the date line. Containers.
- **Outside this document** ((CL2)): `CLAUDE.md` (the "[C] v1.3.3" pointer, the suite table, the slot-2 roster), `docs/KNOWN_ISSUES.md` §3 (the E1b booking, WP-3 not run) and the implementation plan [X] v1.6 — all WP-13's; `hecke_extensions.tex` stays at [E] v1.4.0 citing [C] v1.4.1, which is unaffected by a status bump (no label moved).

Deferred to the next X revision (findings the slot-2 packages handed to this version, none of which a status bump may act on).

1. `rmk:coneambiguity691`: the displayed arithmetic assumes $r^1 = 0$ at $(10, 691)$; measured $\operatorname{rank} r^1 = 1$, $\dim \ker r^1 = 2$, $\dim X_3 = 3$ (WP-7 §3.3). Rewrite from the measured values, with the mod-ℓ reading of "the Eisenstein class" at a Kummer cell as the joint eigenspace E ([E] v1.4.0 `cl:harder`); the same reading applies wherever `obl:oracle`(b)'s [standard:] citation speaks of the Eisenstein class mod ℓ (WP-7 finding 3, WP-11 item 4).
2. `obl:eta`(ii), "the shared (T1) helper — the same helper the construction uses": there is no one helper shared by three tables; each table has its own (T1) helper, all on the one (P1)/conjugation primitive `HeckeCosetTables::twisted_conjugate`. Proposed: "through each table's (T1) helper, all built on the one (P1) split" (WP-8 finding 1).
3. `obl:eta`(iv), "the explicit setting that permits" a mod-m fill: the shipped implementation has no mod-m path and no such setting; a tier failure aborts. Proposed: an implementation MAY omit the mod-m regime, and then the record's regime is $\mathbb{Z}$ by construction (WP-8 finding 2).
4. `lem:etaexists`(V): a sentence naming the constructive choice actually made — the pairwise path routing of `alg:tau` step 2, which needs no linear solve and reduces to the edge path when both values are single vertices (WP-8 finding 3).
5. `sec:recordschema` item 5 / `open:ell2` for a chain *map*: the η record carries the *source* (boundary) degrees $[0, 1]$ with $q_{\max} = 1$, $\eta(q)$ landing in bulk degree $q + 1$; the rule for a truncated free model (E9) is to be stated (WP-8 finding 4).
6. `obl:eta`'s preamble, "together with the stored τ- and τ_∂-tables it was built against": neither base table exposes a table hash; the η record computes both (`built_against.tau_fnv64` / `tau_bd_fnv64`). Proposed: the base tables expose `table_fnv64()` and E2's record carries the three hashes as tier-B objects (WP-8 finding 5).
7. `sec:heckedatum`'s U_ℓ record and `lem:bounds` / `thm:backend`: no finding — WP-3 did not run, so the reach limit stands as recorded and nothing is flagged; the repair's report (slot 3) may flag `lem:bounds` then.

v1.4.1 — 2026-09-16 — hecke-extensions slot 2, WP-10: referee pass on v1.4.0 — all nine accepted, tags kept, `rmk:taubdrynoiota`(3)'s exhibit moved to the right slots, prose repaired

This version. The independent read of v1.4.0 that `docs/implementation/hecke_extensions_slot2_plan.md` §3 WP-10 asks for, on the model of v1.3.1 (`f69cda0`): every one of the nine resolutions was re-derived — the arm map of `obl:taubdry`(v.a)–(v.c) against `Hecke/HeckeBoundaryOperator.cpp`; the general-ν coset shape; (FD2), (FD4) and the coincidence groupoid of $(K_\partial, \partial X)$ for `prop:circlemodel`(4); the bijectivity of Φ and the orientation sign; $\chi_0 - \chi_1 = (1-p)(1-p^n)$

and the Dickson-power congruence $b \equiv n - b \pmod{\ell - 1}$; $V_R^{Tp} = V_R^T$; `alg:taubdry` step by step against `Hecke/HeckeBoundaryDatum.cpp` and `boundary_line_nearest`; the 216/216 record against `run_restriction_arm`; the per-complex schema rule against owner confirmation 1 and `open:ell2` rider (i). **Changes no normative statement** ((V2), (CL5)): no hypothesis, displayed formula, asserted predicate or tag moves. One proof was wrong as written and is repaired under its own label ((L3)): `rmk:taubdrynoiota`(3) computed the $(2, 1)$ exhibit with g in place of g^{-1} (g^{-1} acts on $\mathcal{H}$ as $z \mapsto Nz$, `alg:tau` step 1), placing it on the slots Sv_1, Sv_2, where the transported real parts are $\mp\frac{1}{12}$ and no tie occurs; the asymmetry is real and sits on v_1, v_2 and v_0, v_3 of K_∂ — confirmed on the shipped table. (L1): the label set is unchanged (v1.3.3 → v1.4.0 added exactly the four labels the v1.4.0 entry names, none removed). Appendix B was recounted mechanically from its table and agrees with "The counts" in every figure except the tagged/untagged split, corrected below. Report: `docs/implementation/slot2/WP-10_referee_report.md`. Commit: `9c54b91` (the report, committed before any repair, is `2fed619`).

Normative. None. (CL5)'s test is met vacuously.

Non-normative.

- `rmk:taubdrynoiota` — proof and prose, tag [proved here] kept. (2): the sentence "consequently ... cannot be arranged by the choice of tie rule for all (p, ν) at once" was asserted without the step that the mirror point is ever a transported real part; it now says that an integer transported real part occurs at every $(2, \nu)$ ($x(v_0) = -N/2$) and obstructs every T-equivariant vertex rule there, and that for odd p it obstructs nothing; the proof gains the reduction of the identity at an integer to the identity at 0. (3): the exhibit moved from Sv_1, Sv_2 to v_1, v_2 (transported real parts $N \operatorname{Re} v = \mp\frac{1}{3}$, ties to v_1 and v_3) and v_0, v_3 ($x = \mp 1$, ties to $T^{-1}v_2$ and Tv_2, $\iota_X(T^{-1}v_2) = Tv_1$), with the general criterion for an integer transported real part ($x = (p^a \operatorname{Re} v + B)/p^{\nu-a}$; integral only for $p = 2$, $v \in \{v_0, v_3\}$) and a parenthetical recording what v1.4.0 had computed. (4): made consistent with (2)+(3) — the table-level identity fails at every $(2, \nu)$ under every T-equivariant rule; at odd p the mirror-consistent-tie-rule question stays not settled and not claimed. What is asserted after the repair is what (2) already asserted; it is now proved.
- `obl:taubdry` — prose, two clauses. (v.c): "at cochain level the two data are homotopic, not equal (`lem:taubdryexists`(3))" was unproved — the Γ_∞-level datum of route A is not a boundary g-twisted chain datum, and the decision box after `def:bdryop` calls the transcription a validation route only; the clause now says the two operators are not asserted equal and no chain homotopy is claimed, the asserted predicate (cohomology-level, entrywise, aborting) unchanged. (vii): "with the right-coset sum M_β of `prop:bdrychar`(2) in place of $A_\partial^{(1)}$" was a type error, M_β being an endomorphism of $V_R^T = H^0$; the H^1 control is the scalar $\chi_1(p)$ in place of $A_\partial^{(1)} = \chi_0(p)$ on the one-dimensional H^1 ($n < \ell$), which is what `HeckeBoundaryOperatorValidation` substitutes; the condition $\chi_0(p) \neq \chi_1(p)$, $r^1 \neq 0$ is unchanged. Obligation, no tag.
- `prop:circlemodel` — proof of (4), one parenthetical: "(FD1) fails, ∂X being disconnected" was false ((FD1) concerns $|K|$ and $\widetilde{D}$, both fine for the path); what fails is `ass:fund_domain`'s standing hypothesis that $|K^u|$ be connected, and the sentence now says so and names the face closure `ass:consistent_face_id` of $\mathcal{R}_\partial$, which the theorem's hypotheses require and the proof had not mentioned. Statement, hypotheses and tag unchanged.
- `alg:taubdry` — step 3, wording: the two vertex values are "already stored (a derived face's value being the one the step-0 helper produced)", which is what `construct_edge` does, in place of "fetched through the step-0 helper where a face is derived". Procedure, no tag.

- Appendix B — "The counts": 115/90 tagged/untagged with 11 untagged obligations → 116/89 with 10, since `obl:taubdry` has carried [verified computationally]^{r} since v1.3.3 and is counted in the 17; every other figure confirmed by a mechanical recount of the table (205 rows; 63 `rmk`, 4 `alg`; [proved here] 93, [verified computationally] 17 = 12 + 5; [new] 125, [re-homed] 45, [repaired] 34 + 1). This appendix — the v1.4.0 entry's `[hash]` backfilled with `75c0c27` ((CL6)); the list of Y bumps under (CL5) gains v1.4.1. The header comment and the date line. Containers, no tag.
- **Outside this document** ((CL2)): nothing — [E]'s pointer remains WP-11's, CLAUDE.md the close package's.

v1.4.0 — 2026-09-16 — hecke-extensions slot 2, WP-9: the nine open points of v1.3.3 resolved — `obl:taubdry`(v) split and (vii) qualified, `prop:circlemodel`(4), `alg:taubdry`, the per-complex record fields

This version. The slot's normative revision of the boundary chapter (`docs/implementation/hecke_extensions` §2.2 and §3 WP-9; report `docs/implementation/slot2/WP-9_report.md`). It resolves **exactly** the nine points that v1.3.3's entry below recorded as "open points recorded, not resolved", each as the slot-1 work-package report that found it proposed (`docs/implementation/slot1/WP-4a_report.md` findings 2, 3, 5; `WP-4b_report.md` findings 9–15, 17), under the owner's close-out decision (a) of slot 1 ("as the work-package reports propose, in the next normative (X) revision") and the two slot-2 confirmations that the reports had left open (confirmation 1: the record fields are per complex, never a maximum; confirmation 2: clause (v) is narrowed to degree 0 at $\nu \geq 2$, the general-ν lemma recorded as not stated). **This is an X bump** ((V2), (CL5)): an obligation's clauses change, a proposition gains a clause, a proposition's statement is made precise, an algorithm and a proved remark are added, and the record schema's rule changes. Nothing outside chapter 14, the two schema sites, one remark in §15 and the indices is touched; chapter 9's ± statements, `lem:bulkperiodic` and everything the concurrent E4b package may discharge are deliberately left for later bumps. No erratum is filed ((CL2)); (L2) is not engaged — every label keeps its subject, and the one statement that was false as written, `prop:circlemodel`'s "single relation", is corrected under its own label, as (L3) directs. Every validation that discharges `obl:taubdry` evaluates the rewritten clauses *as written*, with no arm changed (the roster row says which arm is which sub-clause). Commit: `75c0c27`.

Normative.

- `obl:taubdry` [new] — **clause (v)** (open point 1; [WP-4b] 10): *before*, "the operator obtained from `eq:bdrychain` and the operator $\sum_\delta \rho_n(\delta)(\cdot)$ written in the coordinates of `prop:circlemodel`(1) agree . . . in every computed degree" — one named comparison, where the chapter offers three and the implementation runs all three; *after*, three subclauses, each naming its degrees and ν: (v.a) degree 0, every ν, against `prop:bdryop`(2) through `prop:circlemodel`(1); (v.b) degree 1, $\nu = 1$, against `eq:bdrygroupform` in the coordinate Φ of `prop:bdrychar`(3); (v.c) every computed degree, $\nu = 1$, against the Γ_∞-level datum of the decision box after `def:bdryop`, asserted on cohomology only. **Clause (v) at $\nu \geq 2$** (open point 2; [WP-4b] 11; owner confirmation 2): *before*, silent — (v.b)'s and (v.c)'s inputs are $\nu = 1$ statements, so at $\nu \geq 2$ only the H^0 half was available; *after*, "at $\nu \geq 2$ clause (v) is (v.a) alone", H^1 there being covered by (vii) and `prop:bdryop`(1) as run. A narrowing to what is proved, not an extension. **Clause (vii)** (open point 5; [WP-4b] 13): *before*, "it holds for $1 + p^{n+1}$ with the left-coset operator and fails with the right-coset one" — over-broad, the failure being invisible where $\chi_0(p) = \chi_1(p)$ (62 of 232 $n < \ell$ cells) and never on H^0 of the model; *after*, "fails whenever $\chi_0(p) \neq \chi_1(p)$ in R and $r^1 \neq 0$", with the H^0 statement made exact: the control cannot bite on any power of the

first Dickson class ($b \equiv n - b \pmod{\ell - 1}$, `cl:chi`(3)), hence not on the model complexes below degree $\ell(\ell - 1)$; plus the pointer to `rmk:conerestrictsmallell`. Underspecified made precise; over-broad corrected.

- `prop:circlemodel` [repaired] — (open point 3; [WP-4b] 9) *before*, the preamble called $C^\bullet_{\mathrm{eq}}(K_\partial, \mathcal{R}_\partial; L)$ "the circle model with its single relation $f(v_3) = \rho_n(T) f(v_0)$" and identified it with $C^\bullet_\partial$, which is **false for n odd,** $2 \in R^\times$: $\mathrm{Hom}_{\mathbb{Z}[\Gamma]}$ carries the central $-I$ constraint and the single-relation model does not (the defect was found by the phase that implemented the chapter, and the implementation had already added the self-relations; a second repair of a [repaired] statement, (CL3)); *after*, $\mathcal{R}_\partial$ is *defined* as the cusp relation together with the $-I$ self-relations $(\sigma, \sigma, -I)$ on every simplex, and a new **clause (4)** proves $C^\bullet_{\mathrm{eq}}(K_\partial, \mathcal{R}_\partial; L) = C^\bullet_\partial$ for every n and every R — via (FD2) and (FD4) for $(K_\partial, \mathcal{R}_\partial)$ and the proof of `thm:explicit_complex_equiv`(1), which uses exactly those two — with the single-relation form kept as the corollary "literally when $\rho_n(-I) = \mathrm{id}$, i.e. n even or $2 = 0$ in R", the zero complex for n odd at $2 \in R^\times$, and the explicit statement that the single-relation model on V_R is not $C^\bullet_\partial$ there. Clauses (1)–(3) unchanged, numbering kept. Tag [proved here] kept, the new clause proved.
- `prop:bdrychar` [repaired] — **clause (3)** (open point 4; [WP-4b] 12): *before*, the identification $H^1(C^\bullet_\partial) \cong (V_R)_T$ that `eq:bdrygroupform` depends on was used silently in the proof; *after*, the statement carries it: $\Phi\colon [\varphi] \mapsto [\varphi(\langle v_0, Tv_0\rangle)]$, the signed sum over the oriented fundamental path, equivalently $[f(T)]$ for the group cocycle $f(\gamma) = \varphi(\langle v_0, \gamma v_0\rangle)$, with its inverse; the proof gains the verification that Φ is a well-defined bijection. **Resolved with one correction to the report's remark**: reversing the orientation gives $-\Phi$, which leaves the transported operator *unchanged*; what must agree is the orientation used by a representative map and by a lift, and with the two on opposite orientations the transport is off by -1 — which is the disagreement [WP-4b] observed. The statement says so. **Clause (2)** (open point 6; [WP-4b] 14): *before*, nothing said what object the right-coset substitution is; *after*, a clarifying clause: $M_\beta = \sum_b \rho_n(\beta_b) + \rho_n(\beta_\infty)$ is a well-defined endomorphism of V_R^T (proved: $T\beta_b = \beta_b T^p$, $T^p\beta_\infty = \beta_\infty T$, $V_R^{T^p} = V_R^T$ for $p \neq \ell$), the monomials are its eigenbasis, it is asserted as an operator on V_R^T only, it is not asserted to be induced by any cochain map, and the implementation builds it as a fibre-level negative control. Underspecified made precise, both clauses.
- `alg:taubdry` [new] — **new** (open point 7, first half; [WP-4a] 2, 5): the canonical τ_∂ as a terminating procedure beside `alg:tau`, per rule (A3): the slot/derivation step through the single (T1∂) helper, the transported real part $x = \mathrm{Re}(\gamma_{\mathrm{red}}\, g^{-1} s_j \mathrm{pos}(v))$, the vertex rule "nearest line vertex, ties to the larger real part", the edge walk, constructor-time certification, and the normalisation $\gamma_j = \mathrm{red}(g^{-1} s_j \infty)^{-1}$ of `prop:bdrycosets`(2). Four properties [proved here]: well defined and independent of the derivation order and of the cusp reduction (the rule is a Γ''-twisted-equivariant function of the vertex), a boundary datum over $\mathbb{Z}$ with $\{0, \pm 1\}$ coefficients, homotopic to every other admissible datum, and free of ℓ and n. It matches `Hecke/HeckeBoundaryDatum`'s `construct_vertex` / `construct_edge` / `try_derive` and `Hecke/SL2ZTessellationTables`' `boundary_line_nearest` step by step. `lem:taubdryexists`(1) gains a pointer to it (prose).
- `rmk:taubdrynoiota` [new] — **new**, [proved here] (open point 7, second half; [WP-4a] 3): (1) $\iota_X \tau_\partial \iota_X$ is an admissible boundary datum, homotopic to τ_∂; (2) no T-equivariant vertex rule — any tie rule, or none — is ι-equivariant, because at the mirror point $x = 0$ it would need an ι_X-fixed vertex of ∂X_∞ and there is none (the Γ_∞-orbit structure of `rmk:bdrymodel`); (3) the canonical table at $(p, \nu) = (2, 1)$ is not ι-symmetric, by an explicit computation on the slots Sv_1, Sv_2 (transported real parts $\mp\frac{1}{3}$, ties resolved to v_1 and v_3, $\iota_K v_3 = v_0 \neq v_1$); (4) what is and is not asserted — the table-level identity is asserted nowhere and recorded nowhere, and whether a mirror-consistent tie rule would give it at every (p, ν) is *not settled*. **Resolved with a scope correction to the report**: [WP-4a]

3's argument ("the tied set at the mirror is a full ι_K-orbit") proves the impossibility for the *rule*, not for the emitted table, since the mirror point need not be a transported real part; the remark says exactly what each argument proves. `obl:iota` unchanged.

- `rmk:bdrynugeneral` [new] — **new**, commentary, no tag (open point 2's companion; owner confirmation 2): records that no double-coset decomposition of B_N for $\nu \geq 2$, no general-ν `eq:bdrygroupform` and no Γ_∞-level datum for (v.c) at $\nu \geq 2$ is stated, and records the shape a later addition would have to prove (indexing by u and $b \bmod p^{\min(u,\nu-u)}$, $p^{\max(0,2u-\nu)}$ left cosets per double coset), explicitly as not asserted.
- `sec:recordschema` item 5 — (open point 9; [WP-4b] 17; owner confirmation 1) *before*, "the record carries three fields" — one $q_{\max}$ per record, the reported-degree list as the third field, and the instruction addressed to E5 "not implemented now"; *after*, the four fields `q_max`, `truncated`, `q_max_source`, `degrees` are **per complex**, never a maximum over a record; a single-complex record carries them at record level, a multi-complex record inside each complex's object; the campaign's reported-degree selection is a distinct job-level field; the job identity includes each complex's tuple (the campaign job-id fix is done, `CampaignSmokeValidation`); the model values are bulk 2/`false`/`"derived"`/$[0,1,2]$ and boundary 1/`false`/`"derived"`/$[0,1]$; the instruction is re-addressed to E0b (slot-2 WP-2) as *scheduled*, with the "implemented" sentence left to the version after the re-baseline. `open:ell2`'s boxed record sentence gains "per complex it is about" (its "verbatim the v0.14.0 working assumption" note says so); `sec:oc1sites` items (3) (the split clause) and (6) (the four fields) follow. Schema rule; container labels, no tag.

Non-normative.

- `rmk:conerestrictsmallell` [new] — **new** in §15, [verified computationally]^{r} (open point 8; [WP-4b] 15): `obl:taubdry`(vii) = `obl:coneexact`(iii) holds on all 216/216 (cell, q) pairs of `HeckeBoundaryOperatorValidation`'s grid at $\ell \in \{2,3\}$ (run 2026-09-16), recorded as a computational fact with the validation named, consistent with a pair homotopy over $\mathbb{Z}$, claimed nowhere, and licensing nothing ((A6)). No statement moves.
- `rmk:bdryoddn` — one sentence: since `prop:circlemodel`(4) the odd-n zero complex is literally what the model computes. Remark unchanged in claim; prose.
- `sec:boundarydatum` — code map: the (v) paragraph names the three sub-clauses and which arm discharges each; the "choices the implementation fixed" bullets point at `alg:taubdry`, `rmk:taubdrynoiota` and `prop:circlemodel`(4); the "what it records" paragraph's per-complex question is closed by `sec:recordschema` item 5; the (vii) bullet points at `rmk:conerestrictsmallell` and at the qualified clause. Container.
- `obl:taubdry` — roster row (`sec:obligationroster`): (v) per sub-clause with its arm, (vii)'s control failures as the qualified clause predicts, the $\ell \in \{2,3\}$ pointer. Evidence.
- Appendix B — four new rows (`rmk:bdrynugeneral` commentary, `alg:taubdry` procedure, `rmk:taubdrynoiota` [proved here], `rmk:conerestrictsmallell` [verified computationally]^{r}); the [verified computationally] split sentence (eleven of sixteen → twelve of seventeen); the counts (201 → 205 statements, 60 → 63 remarks, 3 → 4 algorithms, [proved here] 92 → 93, tagged 113 → 115 / untagged 88 → 90, [new] 121 → 125) — and the [verified computationally] figure of "The counts", which v1.3.3 had left at 13 (8 registered) against a table of 16 (11), corrected to 17 (12) with a note. Appendix D — the four labels dereferenced. This appendix — the provenance table ([new] 121 → 125: 70 assertions or definitions, 55 remarks; 190 → 194 labels on [new] lines) and the Part IV boundary roster (10 → 11, `alg:taubdry`). The header comment and the date line. Containers, no tag.
- **Open points recorded at v1.3.3: resolved, all nine** (the v1.3.3 entry below keeps its wording, as a published version's text does). (1) → `obl:taubdry`(v.a)–(v.c); (2) → `obl:taubdry`(v)'s $\nu \geq 2$ sentence and `rmk:bdrynugeneral`; (3) → `prop:circlemodel`(4); (4) → `prop:bdrychar`(3); (5) → `obl:taubdry`(vii); (6) → `prop:bdrychar`(2); (7) →

`alg:taubdry` and `rmk:taubdrynoiota`; (8) → `rmk:conerestrictsmallell`; (9) → `sec:recordschema` item 5. None remains open; the referee pass on this version is the slot's next package (WP-10, a Y bump), and its status sentence for `hecke_report/v2` the one after (WP-13).

- **Outside this document** ((CL2)): nothing — the companion `hecke_extensions.tex` pointer is left at v1.3.3 for the package that owns it (WP-11); CLAUDE.md is the close package's.

v1.3.3 — 2026-09-16 — hecke-extensions slot 1 close: chapter `sec:boundarydatum` implemented, `obl:taubdry` discharged, status sites and counts

This version. The close of slot 1 of the hecke-extensions stage (`docs/implementation/hecke_extensions_slot` §3 WP-7; `docs/implementation/slot1/WP-7_report.md`). **Changes no normative statement** ((V2), (CL5)): no definition, obligation, hypothesis, displayed formula or tag on a statement moves; what it records is *evidence and status*, exactly as v1.1.1 did for `sec:involution`. The seven clauses of `obl:taubdry`, written at v1.0.0 as a specification for the phase that would build the boundary datum, were implemented and run by that phase — slot-1 packages WP-4a (`Hecke/HeckeBoundaryDatum`, the boundary subcomplex of `Hecke/SL2ZTessellationTables`, `HeckeCosetTables::twisted_conjugate`) and WP-4b (`Hecke/HeckeBoundaryOperator`, the circle model as a library call in `LocalSystemSL2Z`) — and two registered validations, `HeckeBoundaryTauValidation` ((i)–(iv), 138 tables, 52 686 checks, tier `standard`) and `HeckeBoundaryOperatorValidation` ((v)–(vii), 546 cells, 19 669 checks, `standard`), evaluate every aborting predicate and record what the clauses mark as recorded; nothing failed. The registered-validation count moves from 68 to 77 (slot 1 added `HeckeLocalityValidation`, `HeckeLocalityProbe`, `HeckeIdentityValidation`, `HeckeSturmBoundValidation`, `HeckeExtArchitectureValidation`, `ConsoleC0Validation` and the two above, besides `GoldenDump`); the `Hecke*` count from 29 to 36. No golden value changed in the slot (`GOLDEN_STRICT=1`: tier A 9446 checked, 0 failing; tier B 4603 checked, 0 mismatching, against the unchanged reference). Commit: `13760f7`.

Normative. None. (CL5)'s test is met vacuously.

Non-normative.

- `sec:boundarydatum` — the "What this chapter fixes" paragraph gains one status sentence, and the chapter's code map is rewritten from a census that found nothing to the record of what was built: the four units, the two validations with their grids, the clause-by-clause [verified computationally] record of `obl:taubdry`(v)–(vii) (both routes of (v) asserted in `compute()`; (vi)'s asserted parts and the recorded 330/330 agreement of `cl:chi`(2); (vii) asserted on 864 pairs at $6 \in R^\times$ and recorded, 216/216, at $\ell \in \{2,3\}$; the right-coset negative control of `prop:bdrycosets`(4) returning χ_{b+1} and breaking (vii) on 170 cells), the choices the implementation fixed where the statements leave them open (the nearest-line-vertex rule with ties to the larger real part in `lem:taubdryexists`(1); the normalisation of γ_j in `prop:bdrycosets`(2); the reading of clause (iii)'s transporter set; $\mathcal{R}_\partial$ including the $-I$ self-relations, so that the circle model is $\mathrm{Hom}_{\mathbb{Z}[\Gamma_\infty]}$ literally for every n), and the recorded fields (`boundary_model`, `boundary_is_group_cohomology`, `q_max = 1` derived, `truncated = false`). The census's "simplices not those of K^u" is corrected to "slot set not that of K^u". Container and code map, no tag.
- `obl:taubdry` — roster row (`sec:obligationroster`): unimplemented → **discharged**, clause by clause, with the two validation names; the roster's count sentence ("ten of the fifteen rows" → eleven) and its prose ("three are unimplemented" → two, naming `obl:eta` and `obl:coneexact`). Obligation unchanged; evidence.

- `sec:oraclesnew` — opening paragraph: family (A) is a run, not a specification; family (A) gains its [verified computationally] run record (the numbers above, the $(2,5,2)$ instance reading 4/4/4). Container.
- `sec:codemap` — the chapter-14 row now names the four units and the statements each realises; the row for `sec:cone`, `sec:six`, `sec:orbifoldmodel` stays "—" and points at phase E9 / C-A5; the `sec:symconv` row records that D_n and the pairing are *scheduled* (phase E7); the `sec:obligations` row's `Hecke*` count; the `rmk:dickson` roster row names `hecke::IsDicksonExceptionalCount` (`Hecke/DicksonCount.h`), the one Dickson count that replaced three file-local copies (slot-1 WP-2; the golden `dickson` entry unchanged). Container.
- `sec:notimplemented` — "three chapters" → two, with a sentence on `sec:boundarydatum` in the style of the v1.1.1 sentence on `sec:involution`. Container.
- `sec:buildsurface` — five targets → six (`ConsoleCore`, the console's STL-only language core, slot-1 WP-5; it realises no statement of this document). Container.
- `sec:cone`'s code map — the census sentence no longer lists a missing τ_∂-table; the pair homotopy is what is still missing. Container.
- `thm:backend`'s consumer paragraph in `sec:heckedatum` (the U_ℓ entry-point paragraph) — gains a [verified computationally] record of a reach limit found by slot-1 WP-2: for U_ℓ at $\nu = 1$ the `int64` tier is exhausted between $\ell = 3617$ (runs) and $\ell = 5003$ (`HeckeOverflow` in the coset construction), independently of n, while $\ell = 43867$ with a good prime runs; recorded, not repaired (`docs/KNOWN_ISSUES.md` §3). Container.
- Appendix B — `prop:bdrycosets`, `prop:bdrychar` and `obl:taubdry` gain [verified computationally]^{r}; the count sentence (eight of thirteen → eleven of sixteen). Appendix C — "three chapters" → two, the validation count, "eight" → "eleven" [verified computationally]^{r} claims. §E.1 — "three chapters" → two. The header comment, the abstract, (V5)'s closing sentence, the involution chapter's census count, the date line and the companion pointer (`hecke_extensions.tex` v1.3.3). Containers, no tag.
- **Open points recorded, not resolved** (each would be an X bump by its own editor, or an owner decision; none blocks the discharge above, which holds under the statements as written): (1) `obl:taubdry`(v) names one comparison where the chapter offers three (`prop:bdryop`(2) on H^0, `eq:bdrygroupform` on H^1, and the Γ_∞-level datum of the decision box after `def:bdryop`); the implementation runs all of them — the clause should say which are meant. (2) At $\nu \geq 2$ only the H^0 half of (v) is available as stated: `prop:bdrycosets`(3) and `eq:bdrygroupform` are $\nu = 1$ statements, and the double-coset decomposition of B_N for general ν is not in the chapter. (3) `prop:circlemodel`'s identification with the circle model "with its single relation" holds for n even (or $\ell = 2$) only; $\mathrm{Hom}_{\mathbb{Z}[\Gamma_\infty]}$ carries the $-I$ constraint, and the implementation adds it (code map). (4) The $H^1 \cong (V_R)_T$ coordinate that `eq:bdrygroupform` depends on $(\varphi \mapsto \varphi([v_0, Tv_0]))$ is used silently in the proof of `prop:bdrychar`(3) and should be stated. (5) `obl:taubdry`(vii)'s "fails with the right-coset one" holds only where $\chi_0(p) \neq \chi_1(p)$ (`cl:chi`(3)); on 62 of the 232 $n < \ell$ cells the two coincide and the control cannot bite, and on H^0 it cannot bite at all on this model. (6) `prop:bdrychar`(2)'s right-coset substitution is an operator on V_R^T only because the monomials are an eigenbasis; it has no cochain-level meaning, which a reader may try to give it. (7) Whether the nearest-line-vertex rule should become an `alg:taubdry` beside `alg:tau`; and τ_∂ is not ι-symmetric, which no tie rule can repair (code map) — if §15 or the involution ever need $\iota_K \tau_\partial \iota_K = \tau_\partial$, that is a contract addition. (8) `obl:taubdry`(vii) holds at $\ell \in \{2, 3\}$ on the model complexes (216/216), consistent with a pair homotopy over $\mathbb{Z}$; recorded for §15, claimed nowhere. (9) `sec:recordschema` item 5 speaks of one $q_{\max}$; the boundary object derives 1 and the bulk 2 — E0b's record integration decides.
- **Outside this document** ((CL2)): `hecke_extensions.tex` v1.3.2 → v1.3.3 (the

(160, 691) probe verdict, the Sturm-bound values and implementation status, the reduced-character wording, the stale OQ2-table status sentence; PDF rebuilt); `hecke_extensions_implementation_plan.md` v1.5 → v1.5.1 (§9.3's verdicts and constants, §1.3.1's resume-invariance wording); `CLAUDE.md`; `docs/KNOWN_ISSUES.md`; the P8 comment of `HeckeIotaSignValidation.cpp` that v1.3.2 left.

v1.3.2 — 2026-09-16 — coherence pass after OQ3 and OC-1: stale status sites, an over-broad ± rule, the direction of `rmk:iotapairing`'s prediction, and the provenance counts

This version. Applies the required fixes of the read-only coherence review `docs/implementation/review_2026-` (89 sites checked after v1.2.0 and v1.3.0, 10 inconsistent), less the one that v1.3.1 had already made. **Changes no normative statement** ((V2), (CL5)): no definition, obligation, hypothesis, displayed formula or tag moves. One remark's *prediction* changes direction — `rmk:iotapairing`, indexed by Appendix B as commentary, tagged nothing, consumed by no obligation, oracle or code map, and stating of itself that it is "asserted here as nothing more" — and that is recorded with its own line below ((CL1), (CL3)) rather than by an X bump, which (V2) reserves for normative statements. Commit: `c2c6b44`.

Normative. None. (CL5)'s test is met vacuously.

Non-normative.

- `def:iota` — clause (iv), prose: "not the pullback along any lift of $z \mapsto -\bar{z}$" was false for n odd, where $\rho_n(-\iota) = -\rho_n(\iota)$ and ι^* *is* the pullback along $-\iota$; now qualified "for n even", with the odd case and why it is harmless ($H^\bullet_{\mathrm{eq}} = 0$ for $\ell \neq 2$, the sign invisible at $\ell = 2$) in a parenthesis (review R2). Two polish items: the sentence "with $\det|_\Gamma = 1$" now says what that buys (the relations of $\mathcal{R}$ are preserved, P2), and the flat-coordinate description is $-\varepsilon_\sigma \rho_n(\iota)$ at block position $(\sigma, |\iota_K \sigma|)$, the form of the code map and of the validation (P3); the matrix is unchanged, being its own transpose. Definition; the formula (16) untouched.
- The ± subsection's opening paragraph (container) — "a ± read from a source written before v1.2.0 … must be negated" was over-broad: only the *computed* signs of the v1.1.1 run record are geometric; the ± *predicted* from OQ3 (`obl:iota`(v)'s split, family (C) of `sec:oraclesnew`, the $H^1_\pm$ statements of `hecke_extensions.tex`) were written in the convention v1.2.0 adopted and hold as printed, as the v1.2.0 entry already said (R3).
- `rmk:iotapairing` — **the contingent prediction changes direction.** Until v1.3.2 the remark predicted that duality would pair each ι^*-sign with itself, from an expected equivariance $(-1)^n$ read off the coefficient level alone. That omitted the orientation factor: $z \mapsto -\bar{z}$ reverses the orientation of $\mathcal{H}$, so any pairing through a fundamental class of the pair — which is what `cl:pd`(1) asserts — picks up a further -1, the expected equivariance is $-(-1)^n$, equal signs annihilate for n even, and duality pairs $+$ with $-$ (the modular-symbol picture, [Cre97, Ch. 2]). The remark now says so, and adds the dimension cross-check from this document's own predictions: $(d_{n+2}, d_{n+2} - 1)$ on $H^1_{\mathrm{eq},\pm}$ (`obl:iota`(v)) against the mirror split on H^1_c forced by the odd one-dimensional $H^0(\partial; M_n)$, which self-pairing could not match. Its status is unchanged: contingent, "asserted here as nothing more", no tag. (Review R4, adjudicated by the referee: the reviewer's sign is right, and the contract's own dimension predictions independently exclude the old direction.)
- `rmk:iotaelltwo` — closing sentence: "open construction item … a tagged working assumption" (stale since v1.3.0) → "decided by the owner on 2026-09-16 (`oc1a`)". Remark, commentary (R5).

- The remark before `obl:coneexact` that names §15's two reading sites (container) — "open construction item" → decided, with the date and `oc1a` (R6).
- `sec:oc1sites` — item (7) gains the two status sites above, marked as missed at v1.3.0 (R5, R6). Container.
- `sec:oraclesnew` — family (C): "on every cell" → "on every cell on which it can be located (the Dickson cells carry no sign)" (P4). Container.
- `err:e12` — "what is not settled at $\ell = 2$" now also names the policy decided for it (P9). Erratum entry, no tag.
- Appendix E, the provenance table and prose — **recounted from Appendix B** (121 [new] rows: 52 `rmk` and 69 others, namely 21 `def`, 19 `prop`, 15 `lem`, 4 `cl`, 4 `obl`, 2 `thm`, 1 each of `alg`, `cor`, `ex`, `open`): the table's "70 assertions or definitions, 51 remarks" becomes 69 / 52, agreeing with the roster below it (which lists 69 and was right); the prose "118 / 49 / excess over 111 and 35" and the heading "The 68 statements" — stale since before v1.3.0 — become 121 / 52 / 121 and 45 / 69 (R10). (CL6) no longer says "ten commits" and says what `[hash]` means; the `[hash]` placeholders of v1.1.1, v1.2.0, v1.3.0 and v1.3.1 are back-filled (`a047147`, `c823f3a`, `b1c0aaa/4f84a32`, `f69cda0`; P11); the "Y field" sentence under the rules lists v1.0.0, v1.1.1, v1.3.1 and v1.3.2 (R9, completed).
- The date line (companion pointer `hecke_extensions.tex` v1.3.2, P1) and the header comment. Container, no tag.
- **Outside this document** ((CL2)): `hecke_extensions.tex` v1.3.1 → v1.3.2 — its (A2) closing sentence no longer calls the truncation policy "open … not decided here" but points at [C] `open:ell2` / `oc1a` (R8), its item `add:iota`'s "not a pullback along any lift" gains "for n even" (P5), and "ι is not implemented" becomes "no production implementation; built and `obl:iota` discharged by two validations" (P6); PDF rebuilt. `hecke_extensions_implementation_plan.md` risk R4(ii) and the Q2 row (R7, P10); `hecke_extensions_owner_decisions.md` summary row 3 (P7). Not applied: P8, a comment in `HeckeIotaSignValidation.cpp` — no C++ change in a documentation pass.

v1.3.1 — 2026-09-16 — referee pass on `lem:bulkperiodic`: verified, tag kept, prose repaired

This version. The audit the owner decision of 2026-09-16 asked for ("write the periodicity argument and then audit / referee the result, including Hecke actions"), executed on `lem:bulkperiodic` as written at v1.3.0. **Changes no normative statement** ((V2), (CL5)): every clause of the lemma and every step of its proof was re-derived independently — the five lemma and its lower edge, the cyclic periodicity with its degree-0 kernel $N\,M$, the well-posedness and conjugation-invariance of ζ_+ (in particular that S and S^{-1}, with $\zeta_+ = \pm i$, are not conjugate in $\mathrm{GL}_2(\mathbb{Q})^+$, so the definition is consistent), $H^2(\Gamma;\mathbb{Z}) \cong \mathbb{Z}/12$, the detection of $H^2(\Gamma'';\mathbb{Z})$ on finite cyclic subgroups, the projection formula and the compatible-pair bookkeeping of (20) — and checked against $H^\bullet(\Gamma;\mathbb{F}_2)$, $H^\bullet(\Gamma;\mathbb{F}_3)$ and the $6 \in R^\times$ case. No error was found; the tag [proved here] stands, the [standard] inputs being cited at the level §1.1 permits. Four prose defects were repaired, none touching a hypothesis, a displayed formula or a conclusion. Report: `docs/implementation/referee_report_2026-09-16_bulkperiodic.md`. Commit: `f69cda0`.

Normative. None. (CL5)'s test is met vacuously.

Non-normative.

- `lem:bulkperiodic` — prose only, three places. (i) The "External inputs" preamble claimed that every unverified pointer is "a direct check on the cochain formula"; that is true of (E1) and the first two items of (E2) and false of (E2)'s last item, (E3) and (E4),

which are now said to be used as stated, (E4) as the meaning of cor in (20); the preamble also now names [Bro82, Ch. III.10.2] (Steps 0, 5) and the in-place proof of the fixed-vertex statement (Step 3), which the count "four standard facts" had left out. (ii) Clause (3): the operator is now said to be (20) on ordinary group cohomology, defined for every R, n and ℓ with no model, the pointers to Theorem 11.10 and Remark 17.2 being identifications the clause does not depend on — where v1.3.0 called it "the Hecke operator of the free Γ-model" and cited Theorem 17.3(3), a $6 \in R^\times$ statement (rule (A1): the ring and n are now in the clause). (iii) The closing sentence of (3): $A^{(q)}(T_{\Gamma g \Gamma})$ is glossed as the matrix on $H^q(\Gamma; V_R)$, not the shipped model's at $\ell \in \{2, 3\}$. Lemma, tag unchanged.

- `rmk:ell2window` — clause (2): "proved from Claim 16.2(2)" now says from the tree sequence (50) of `cl:amalgam`(1), the Mayer–Vietoris sequence re-derived for an arbitrary coefficient module (`cl:amalgam`(2) is stated for V_R only). Remark, commentary.
- `rmk:bulkperiodicscope` — (3)(b): the identification $H_c^\bullet = H_\Gamma^\bullet(Y, \partial Y; V_R)$ now points at Definition 17.1, Theorem 17.3(2) being a $6 \in R^\times$ statement; the closing sentence of (3) said a run with $q_{\max} < 4$ "cannot exhibit one full period of the cone", which misstates what $q_{\max} = 4$ shows — it exhibits the first bulk isomorphism $H^2 \cong H^4$, while the cone's first, $H_c^3 \cong H_c^5$, needs $q_{\max} \geq 5$; the sentence now says so. Remark, commentary.
- `open:ell2` — the *non-normative recommendation* paragraph: "one full period of the bulk needs degrees 2 to 5 built, i.e. $q_{\max} \geq 4$" was internally inconsistent (degree 5 is built as scaffolding and never reported, so $H^3 \cong H^5$ is not observed at $q_{\max} = 4$); it now says which isomorphism $q_{\max} \geq 4$ exhibits and that the second and the cone's first need $q_{\max} \geq 5$. The recommended value $q_{\max} \geq 4$, the box, the riders and the decision are untouched. Policy, no tag.
- Appendix E — the sentence "the Y field has never yet been used" under the rules, stale since v1.1.1, corrected. Container, no tag.
- The date line and the header comment (version 1.3.1). Container, no tag.
- **Outside this document** ((CL2)): the referee report `docs/implementation/referee_report_2026-09-` — verdict per claim, the sanity computations $H^\bullet(\Gamma; \mathbb{F}_2)$, $H^\bullet(\Gamma; \mathbb{F}_3)$, $H^2(\Gamma; \mathbb{Z})$ written out, the external facts actually needed with precise statements, and what a phase-E5 computation on the free model could test.

v1.3.0 — 2026-09-16 — OC-1 decided: `open:ell2` becomes the binding truncation policy, and the bulk periodicity above the vcd is proved, Hecke-equivariantly

This version. Executes the owner decision of 2026-09-16 on the $\ell = 2$ truncation policy — "OC-1: adoption per brief, write the periodicity argument and then audit / referee the result, including Hecke actions" — on the decision brief `docs/implementation/oc1_decision_brief.md`: `oc1a` is adopted with the brief's three riders, and the argument the brief sketched in its §4.2 is written into the contract as `lem:bulkperiodic`, in a stronger form than sketched (a canonical class, and the Hecke equivariance the brief listed as the open half). The audit/referee pass the decision asks for is the *next* slot, not this one; the lemma is written to be refereed and says at each step which standard fact it uses. **This is an X bump** ((V2), (CL5)): `open:ell2` loses its `[open]` tag, and §1.1 makes the tag part of the statement; its boxed content becomes binding rather than assumed; `obl:coneexact`(iv), an obligation, changes wording; the record schema gains its adopted fields; and a new normative statement, `lem:bulkperiodic`, is added. **Ordering note**: v1.2.0 — OQ3, the ι sign and the $\pm$ decomposition of Chapter 9 — was prepared *in parallel* with this version, from the same v1.1.1 base, and integrated first; neither version edits a statement of the other, the two entries are independent, and the date line and this appendix list them in integration order (v1.3.0 above v1.2.0). No erratum is filed: (CL2) reserves Appendix A for the superseded corpus, and (L2) is not engaged, every label keeping its subject. Commit: `b1c0aaa` (integrated with v1.2.0, PDFs rebuilt, at `4f84a32`).

Normative.

- `open:ell2` — retitled "OC-1: the $\ell = 2$ truncation policy — DECIDED 2026-09-16"; the tag `[open]` and the header "reserved by the owner; a tagged working assumption, not a decision" are replaced by the decision sentence and its date; the boxed statement is adopted **verbatim** and followed by the three riders ($q_{\max}$ derived on a finite model and settable only where `truncated = true`, distinct from the reported-degree list; the periodicity is a theorem; `obl:coneexact`(iv) carries the box's qualification) and by the non-normative recommendation $q_{\max} \geq 4$ at $\ell \in \{2, 3\}$; "The alternatives, kept reachable" becomes "`oc1a` taken, `oc1c` kept reachable"; `oc1c`'s "the Farrell–Tate route named, and deliberately not asserted" and "Neither hypothesis is proved anywhere in this document" become a pointer to `lem:bulkperiodic` and the precise residue of `rmk:bulkperiodicscope`(3); "Recommendation, and why" becomes "Decision, and why". The labels `oc1a`–`oc1d` are kept ((L1)). Open construction item; status tag `[open]` removed, the truncation lemma's [proved here] kept; [new] content unchanged in its box ((CL4)).
- `lem:bulkperiodic` [new] — **new**: for $\Gamma = \langle S \rangle *_{\langle -I \rangle} \langle U \rangle$ and *any* $\mathbb{Z}[\Gamma]$-module V, cup product with an explicitly constructed generator $e \in H^2(\Gamma; \mathbb{Z}) \cong \mathbb{Z}/12$ — the unique class whose restriction to each finite cyclic subgroup is the eigenvalue character ζ_+ on the fixed point in $\mathcal{H}$ — is an isomorphism $H^q(\Gamma; V) \to H^{q+2}(\Gamma; V)$ for every $q \geq 2$, a surjection with kernel $\partial(N_{-I}V)$ for $q = 1$; e is Δ-invariant, and every $T_{\Gamma g \Gamma} = \mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ commutes with $e \cup -$, so the Hecke module $H^q(\Gamma; V_R)$ is 2-periodic in $q \geq 2$ for every R and n. Proof by the five lemma on `cl:amalgam`(2), a dimension-shifting proof of the periodicity of finite cyclic groups, detection of H^2 of a finite-index subgroup on its finite cyclic subgroups via the Bass–Serre tree, and the projection formula $\mathrm{cor}(\mathrm{res}\, u \cup y) = u \cup \mathrm{cor}\, y$ derived from Shapiro's lemma. Four standard inputs are cited at section level (Brown Ch. V.3, III.5–6, III.8–9; Serre, *Trees* I.4.3) with rule (A2) caveats where the pointer is unverified. Ordinary cohomology throughout; Farrell cohomology is neither used nor asserted about.
- `rmk:ell2window` — clause (2): the parenthesis "the standard route would be Farrell–Tate cohomology ... `[conjectural]` as a citation: this document has not verified the precise form" is replaced by the statement of what `lem:bulkperiodic` proves and by the note that Farrell theory is not used. The remark loses its only tag and becomes commentary; (L2) is not engaged, the clause now asserting *more*, not something different. Remark, tag `[conjectural]` removed.
- `obl:coneexact` — clause (iv): "Each composite of consecutive maps of (39) is zero in every computed degree" becomes "each composite both of whose end terms are reported is zero", with the truncated-model reading spelled out (only the incoming composite at $H^{q_{\max}}(\partial)$), and two **recorded** quantities at that spot: $\mathrm{rank}\, r^{q_{\max}}$ and the injection $\mathrm{coker}\, r^{q_{\max}} \hookrightarrow H_c^{q_{\max}+1}$, the latter at $\ell = 2$ a recorded lower bound and never a reported group (the brief's §4.3 under-claim). Obligation, predicate narrowed ((CL5): normative).
- §22, record schema item 5 (container, no tag) — "on a truncated free model they are the parameter `open:ell2` fixes ... which is why the shipped writer does not yet carry them" becomes the adopted fields (`q_max`, `truncated`, the reported-degree list, the derived/set marker) and a concrete instruction for phase E5 naming `Hecke/HeckeReport.{h,cpp}` and `Campaign/CampaignDriver.cpp`, and requiring the degree list and `q_max` in a record's identity.

Non-normative.

- `rmk:bulkperiodicscope` [new] — new commentary: ordinary versus Farrell cohomology; the (unverified, unused) identification of e with the Euler class of $B_3 \to \mathrm{SL}_2(\mathbb{Z})$; what `lem:bulkperiodic` makes of `oc1c` and the three things that remain open in it (a boundary periodicity isomorphism — candidate $\mathrm{res}^{\Gamma}_{\Gamma_\infty}\, e \cup -$, automatically compatible with r —, the $H^\bullet(\Gamma; \mathbb{Z})$-module structure of (39) on the cone, and the Hecke equivariance of both);

why $q_{\max} \geq 4$ is the observing bound.
- `sec:oc1sites` — item (1) reworded for a decided item; a seventh item added listing the sites that assert the item's *status* (the brief's finding: the list enumerated content sites only). Container, no tag.
- The status-asserting sites themselves (item (7) of `sec:oc1sites`): the abstract; `sec:scope`'s closing paragraph; (V5); `sec:model`'s $\ell = 2$ paragraph; `sec:boundarydatum`'s "What this chapter fixes"; the opening of `sec:ell2`; Appendix C's "A decision on `open:ell2`"; §E.1. Each now says "decided 2026-09-16" where it said "open". Containers, no tag.
- Appendix B — rows for `lem:bulkperiodic` and `rmk:bulkperiodicscope` added; `rmk:ell2window`'s row `[conjectural]` → commentary; `open:ell2`'s row loses `[open]`; the counts paragraph (201 statements, 23 `lem`, 60 `rmk`; [proved here] 92, [standard] 34, [conjectural] 7, [open] 6; 88 untagged) and the provenance box (121 [new]).
- (CL4) — the list of statement kinds gains `open`, as the v0.14.0 entry asked; the provenance table (121 [new], 70 assertions or definitions, 51 remarks) and the roster of new statements (Part IV, the boundary: 10) follow.
- The date line and the header comment (version 1.3.0, with the concurrency note on v1.2.0). Container, no tag.
- **Outside this document** ((CL2); recorded where they happened): the OC-1 entry in `docs/implementation/hecke_extensions_owner_decisions.md`, the "DECIDED" note at the head of the brief, and the pointers in `hecke_extensions_implementation_plan.md`, `hecke_extensions_open_questions.md`, `docs/console/00-console-interface-plan.md` and `lean_formalization_assessment.md`.

v1.2.0 — 2026-09-16 — OQ3 resolved: `def:iota`(iv) negated, "+" is by definition the Eisenstein side

This version. Executes the owner decision of 2026-09-16 on OQ3, taken on the v1.1.1 evidence: "negate `def:iota`(iv) in the following sense: we *define* '+' to be the sign where Eisenstein classes live; this means that we have to switch +/− notationally everywhere." Of the three realizations weighed in `docs/implementation/oq3_resolution.md` — global negation, notation-only, and a degree-parity twist — the owner's own (global negation) is the cheapest and is taken: $\iota^* := -\iota^*_{\mathrm{geo}}$ in every degree, the geometric pullback kept under its own name. Its honest cost, stated in the definition: ι^* is the geometric involution twisted by the sign character det of $\mathrm{GL}_2(\mathbb{Z})/\mathrm{SL}_2(\mathbb{Z})$, not the pullback along any lift of $z \mapsto -\bar{z}$; constants are odd ($\iota^* = -\mathrm{id}$ on $H^0(\Gamma;\mathbb{F}_\ell)$); on the boundary $H^0(\partial)$ is odd and $H^1(\partial)$ ($\ell > n$) even; at $\ell = 2$ nothing changes. What it buys: every $\pm$ prediction the document had written from OQ3 — $(d_{n+2}, d_{n+2}-1)$, Eisenstein in $H^1_{\mathrm{eq},+}$ — holds as printed, the former recorded half of `obl:iota`(v) becomes a gating definition-consistency oracle, and no PROVED statement of chapter 9 needs more than "the sign cancels". No pipeline value moved: ι^* occurs in no golden object, and `GoldenRegressionValidation` was run in the same process as the changed validation. The validation is `HeckeIotaSignValidation` (tier `standard`, 1.4 s, 4 104 267 exact assertions, 0 gating failures; it now builds both involutions and prints `IOTA_SIGN eisenstein=plus (geometric=minus) ... h0_anchor=-1 (geometric=+1)`). X increments ((V2), (CL5)): a definition, an obligation and a remark change. Commit: `c823f3a`.

Normative.
- `def:iota` [new] — clause (iv) restated: `eq:iotacochain` now defines ι^*_{geo} (the former ι^*) *and* $\iota^* := -\iota^*_{\mathrm{geo}}$, for every degree and every coefficient module (D_n included); ι^* is declared the involution of the document; the sign's nature ($(\det \otimes \rho_n)(\iota)$; an automorphism of $(K, \mathcal{R}, L)$ but no lift of $z \mapsto -\bar{z}$), its consequences ($-\mathrm{id}$ on $H^0(\Gamma;\mathbb{F}_\ell)$; invisible at $\ell = 2$) and the flat-coordinate block $\mp\rho_n(\iota)$ are stated. Clauses (i)–(iii) unchanged.

- `obl:iota` [new] — clause (v) rewritten: the sign statement ("the Eisenstein class lies in $H^1_{\mathrm{eq},+}$", by both routes, on every signed cell, with $\iota^* = -\iota^*_{\mathrm{geo}}$ entrywise) moves from *recorded* to *aborting* as a definition-consistency oracle, and what a failure means is stated (a non-uniform geometric sign, i.e. an ill-posed definition); the degree-0 sentence becomes $-\rho_n(\iota)$ on the invariants, $-\mathrm{id}$ at $n = 0$; the $\pm$ dimension statement stays recorded, with the reason (it is the even cuspidal split, the [conjectural] Eichler–Shimura prediction). The [verified computationally] record is rewritten for the v1.2.0 run, the geometric numbers kept as the recorded reading; `[OQ3-PENDING]` removed.
- `rmk:iotasign` [new] ((CL3)) — rewritten: the sign was an assertion about the geometric involution and was checked (the mechanism named); the resolution negated the definition and not the record; `iota_sign` now *names the normalisation* $\iota^* = -\iota^*_{\mathrm{geo}}$, by which a third party reproduces a record; what is not chosen (uniformity of the geometric sign, now gated by `obl:iota`(v)). Its prohibition on call-site flips is unchanged.

Non-normative.

- `prop:iotacommutes` — statement (2) names ι^*_{geo} alongside ι^* and transports through $\iota^{\#} := -\iota^{\#}_{\mathrm{geo}}$; the proofs of (2) and (3a) are written for the geometric pair with the one-line reduction (the sign cancels; $(-\mathrm{id})^2 = \mathrm{id}$). Content of the proposition unchanged, [proved here] tag unchanged, so no normative line (the statement's predicates are the same).
- `rmk:iotaelltwo` — one sentence: at $\ell = 2$, $-1 = +1$, so $\iota^* = \iota^*_{\mathrm{geo}}$ and the normalisation is invisible. Remark, no tag change.
- `rmk:iotapairing` — one sentence: the sign applies to both coefficient modules, the two factors -1 cancel in the pairing, and every $\pm$ consequence is invariant under negating both involutions at once. Remark, no tag change.
- `sec:involution` — the "What this chapter fixes" paragraph (the sign fixed, its cost named); the $\pm$ subsection's opening paragraph (the eigenspaces are the geometric ones with labels exchanged; how to read a pre-v1.2.0 $\pm$); the Eichler–Shimura paragraph after `obl:iota`; the code map (both involutions built, the sign gating, the production block $\mp\rho_n(\iota)$). Container, no tag ((CL4)).
- `sec:lateroblig`, `sec:obligationroster` (the `obl:iota` row and the "Reading the table" paragraph), `sec:oraclesnew` (opening and family (C), now carrying both runs), `sec:codemap` (the `sec:involution` row), `sec:notimplemented` (chapter 9 has no open item) — each updated from "pending" to the resolution. Containers, no tag.
- The date line and Appendix E's closing paragraph. Container, no tag.
- Appendix B is **unchanged**: `def:iota` stays a definition, `obl:iota`'s line already reads [conjectural] [verified computationally]^{r} [standard] (the [conjectural] is the dimension prediction, still recorded), and `rmk:iotasign` stays commentary.
- Outside this document, in the same commit: `hecke_extensions.tex` Stage 0 item (A3) and the `iota_sign` paragraph of its §7; `hecke_extensions_owner_decisions.md` §OQ3 (the decision of 2026-09-16 with the verdict that motivated it), `hecke_extensions_open_questions.md` items 3 and 18, the implementation plan's C-A1 and Q3 rows; `HeckeIotaSignValidation.{h,cpp}`, its registry comment and `Hecke/README.md`; and the banner on `docs/implementation/oq3_iota_sign_ve` saying its $\pm$ are geometric.

v1.1.1 — 2026-09-16 — `obl:iota`(ii),(iii),(v) run: status only, and the OQ3 sign comes out −

This version. **Changes no normative statement of this document**, which is why Y and not X increments ((V2), (CL5)): no clause was added, removed or amended, no predicate, hypothesis, ring or action-on-failure moved, and no displayed formula changed. What it records is *evidence*. The three clauses of `obl:iota` that Chapter 9's code map had listed as

unimplemented — (ii), (iii) and (v) — were implemented and run by the new registered validation `HeckeIotaSignValidation` (`src/hacs/Hecke/HeckeIotaSignValidation.{h,cpp}`, tier `standard`, 1.4 s, 4 103 488 exact assertions, 0 gating failures), which builds ι^* from `def:iota` alone. **Every aborting predicate of the obligation passes.** The half of clause (v) that the clause itself marks *recorded rather than aborting* — owner decision OQ3's "the Eisenstein class lies in $H^1_{\mathrm{eq},+}$" — is **false**: on all 74 cells of a 77-cell grid ($n \in \{2,\dots,30\}$ even, $\ell \in \{5,7,11,13,101,134217757,2^{31}-1\}$) on which a sign is defined, and by two independent routes that agree wherever both are defined, the class lies in $H^1_{\mathrm{eq},-}$, and on all 66 off-Dickson cells the $\pm$ split is the exact mirror $(d_{n+2}-1,\, d_{n+2})$ of the printed prediction. Per `rmk:iotasign` that is resolved by an owner decision on a normative statement — negate `def:iota`(iv), or amend OQ3 — and **the decision is pending**: this version takes neither, tags the place `[OQ3-PENDING]`, and leaves the predicted numbers standing next to the observed ones so that the two can be compared. Nothing was flipped at a call site, which is the failure mode `rmk:iotasign` exists to forbid. The evidence, the full 77-cell table and both resolutions in the owner's own terms: `docs/implementation/oq3_iota_sign_verification.md`. Commit: `a047147`.

Normative. None. (CL5)'s test is met vacuously: no entry below is a change to a statement's content.

Non-normative.

- `obl:iota` — a [verified computationally] evidence record added to clause (v), reporting the run above: what passes, what is recorded false, the two dimensions as observed, and the `[OQ3-PENDING]` tag. The clause's own text — its predicate, its $6 \in R^\times$ hypothesis, its displayed prediction, its action on failure and its degree-0 sentence — is untouched, so this is evidence about a statement and not a change to one. Obligation, no tag change ((CL4)).
- `rmk:iotasign` — one bullet added: the check it specifies has been run and fails, so the choice between its two named resolutions is now live. The remark's content is unchanged, because it is precisely the statement that says what such a failure means and what may not be done about it. Remark, no tag change.
- `sec:involution` — the chapter's "What this chapter fixes" paragraph (all five clauses now implemented and run, with the one recorded failure named) and its code map (the second validation, what it builds and which clauses it discharges; the census re-run; the `EquivariantCochainComplexMorphismOnK` paragraph re-labelled as the specification for a *production* ι^*, which still does not exist). Container, no tag ((CL4)).
- `sec:lateroblig` — the `obl:iota` bullet gains the (ii),(iii),(v) paragraph. Container, no tag.
- `sec:obligationroster` — the `obl:iota` row's status and validation columns, and the "Reading the table" paragraph, which now describes that row as evaluated in full with two negative records rather than as partly unimplemented. Container, no tag.
- `sec:oraclesnew` — the section's opening (family (C) is a run, not a specification) and family (C) itself, which gains a [verified computationally] paragraph with the observed split and loses the stale sentence calling the `obl:iota`(iv) diagnostic false without saying that v1.1.0 repaired it. Container, no tag.
- `sec:codemap` — the `sec:involution` row of the chapter-to-file table names both validations; the `sec:obligations` row's `Hecke*Validation` count moves from 27 to 29. Container, no tag.
- `sec:notimplemented` — Chapter 9 is removed from the list of chapters whose obligations are unimplemented; what it still lacks is named precisely (a production ι^*, and an owner decision). Container, no tag.
- The count of **registered validations** moves from 67 to 68 at its six occurrences

(`sec:involution`'s code map, `sec:boundarydatum`'s and `sec:cone`'s census paragraphs, `sec:obligationroster`, `sec:notimplemented`, and Appendix C's "second, independent witness"). Counts, no tag.

- The date line and Appendix E's closing paragraph ("not yet written"), which now names `obl:iota`(v)'s pending decision alongside `obl:iota`(iv)'s two-version record. Container, no tag.
- Appendix B is **unchanged**: it is mechanically derived from the tags a statement carries, `obl:iota`'s line already reads [conjectural] [verified computationally]^{r} [standard], and this version adds no tag to any statement — the new [verified computationally] record is a second [verified computationally]^{r} on a line that already has one, and `rmk:iotasign` stays commentary. The "eight of the thirteen" [verified computationally] count is therefore also unchanged.

v1.1.0 — 2026-09-16 — the canonical τ made ι-symmetric: owner decision, re-baseline

This version. Executes the owner decision of 2026-09-16 that `rmk:iotatausym` had recorded as available and not taken: the canonical τ of `alg:tau` is rebuilt so that $\iota_X \tau \iota_X = \tau$, which makes `prop:iotacommutes`(3b) — the only route to $[\iota^*, T_{\Gamma g \Gamma}] = 0$ at $\ell \in \{2,3\}$ — available for it. The feasibility pass found that the remark's proposed repair could not work as stated (no choice *within* an ι_K-swapped tied pair commutes with ι_K; the witness point $4i$ lies on the mirror), and that step 2 of `alg:tau` was a second obstruction the degree-0 witness could not show; both are repaired and both are recorded. Every frozen cochain-level reference value (`Golden/GoldenReference.inc`: 2097 of 14049 lines — every $\Theta^{(q)}_{\mathrm{flat}}$, its rank, every $\Theta^{(q)}_{\mathrm{eq}}$, the JSON records) was re-baselined and *no* cohomology-level value moved, $\ell \in \{2,3\}$ included; the object-by-object record is `docs/matrix-engine/rebaseline-2026-09-16-iota-symmetric-tau.md`. The pre-v1.1.0 τ is kept in the tree as `TauVertexPolicy::GeometricNearestLegacy` and re-run by the same validation, so the v0.13.0 numbers stay reproducible. X increments ((V2)): `alg:tau` and `obl:iota` change. Commit: `a047147`.

Normative.

- `alg:tau` [new] — step (1b): ties among nearest vertices are broken ι_K-equivariantly (a tied set containing a full ι_K-orbit goes to v_6, the unique ι_K-fixed vertex; otherwise to its member in the orbit of smallest index), in place of ascending vertex index. Step 2: the path templates are tree-routed — spoke to the Bass–Serre tree Y (the Γ-orbit of $\{v_4 v_6\}, \{v_5 v_6\}$), the unique Y-path, spoke back — so that concatenation along any word telescopes to a chain of the two endpoints alone, Γ- and ι_X-equivariant and antisymmetric, in place of breadth-first paths in ascending neighbour order. Neither change touches `def:tau`, `obl:tau` or step 4; the rule is new to this document and to every source.
- `obl:iota` [new] — clause (iv) amended: still recorded and still licensing nothing beyond (3b), but on the canonical τ of `alg:tau` a false value is now a *failure* (a regression gate on the construction, not a branch on the diagnostic); on any other admissible τ it is not. Its [verified computationally] record moves from "discharged, negatively" to "negatively at v0.13.0 on the then-canonical τ, positively since v1.1.0 on the canonical τ": 0 of 21528 slots over 144 tables, exact over $\mathbb{Z}$; the legacy τ reproduces its 3600 of 10764.

Non-normative.

- `rmk:iotatausym` [new] ((CL3)) — rewritten a second time. The v0.13.0 finding is kept verbatim as history (grid, counts by degree, the witness, the mechanism); added: why no tie-break of the tied set can repair it (the mirror argument, with i the only vertex of

X on the mirror), the three tied configurations that actually occur, the repair taken, the measurement that step 1 alone left 1848/5148 and 1068/2340 differing slots in degrees 1 and 2 (the second obstruction, in step 2), the verdict since v1.1.0 with its two stated limits (verified per table by a run, not proved for every (p, ν); ι^* on cochains still not implemented), the consequence $[\iota^*, T_{\Gamma g \Gamma}] = 0$ at $\ell \in \{2, 3\}$ for the canonical τ on the grid, the vanishing of the canonical τ in degree 2, and the re-baseline's counts. The sentence "a repair exists and is not taken here" is withdrawn: the repair as proposed there did not exist, and the one that does is taken.

- `rmk:certoverGeometry` [new] — item (1) revised: the (1b) tie-break and the step-2 templates are now ι_K-equivariant and the canonical τ is ι-symmetric; the pre-v1.1.0 choices were not, each on its own.
- `rmk:tauexistsscope` — one sentence added to the [conjectural] paragraph: the two certified τ the validation holds induce the *same* $A^{(q)}$ at every $\ell \in \{2, 3\}$ point of the golden grid, so no counterexample to homotopy uniqueness at $\ell \in \{2, 3\}$ is known there. Remark, no tag change.
- `sec:involution` — the chapter's "What this chapter fixes" paragraph (clause (iv) now holds on the canonical τ and gates it) and its code map (three policies, 432 tables, 64699 assertions; the recorded verdict positive on the canonical τ, negative on the legacy one). Container, no tag ((CL4)).
- `sec:lateroblig` — the clause-(iv) paragraph of the obligations summary, from "discharged negatively" to the two-version record and the route now open at $\ell \in \{2, 3\}$. Container, no tag.
- `sec:codemap` — the `alg:tau` line names both template families; the `sec:involution` row of the chapter-to-file table records that (iv) gates. Container, no tag.
- Appendix E's closing paragraph ("not yet written"): (iv) is no longer listed as discharged negatively. Appendix, no tag.

v1.0.0 — 2026-09-16 — the leading version field becomes 1: (V4)'s freezes, and the companion's half of the scope cut

This version. **Changes no normative statement of this document.** It records the discharge of (V5)(ii) — the criterion v0.19.0 added and did not meet — by the two acts that criterion names, both of which are edits to *other* files:

(1) **(V4)'s two supersession freezes, carried out.** `hecke_action.tex` is frozen at *Version 1.0 — 2026-07-02* and `monodromy_v5.tex` at *Version 5.0 — 2026-07-03*, each date being that of the last commit that changed the file's content, each replacing a `\date{\today}` (entry A.17), and each followed by a banner naming this document as the successor and Appendix A as required reading. Neither document's content was otherwise touched. [v5]'s banner says where `thm:explicit_complex_equiv` now lives — its five C++ citing sites use the `[C]` key and resolve here (entry A.7) — and that §12's β is not re-homed (entry A.19).

(2) `hecke_extensions.tex` **v1.3 cites the nine labels instead of defining them.** Its former §1.1–§1.2 and the boundary-eigenvalue claim of §1.4 are replaced by a pointer section; the four of the nine this document *corrected* are named there as superseded, with the correction and its site: `cl:chi`'s Dickson-locus sentence (entry A.21, `rmk:chicoincide`), `cl:boundary`'s $H^i(\partial) = 0$ for $i \geq 2$ without $2 \in R^\times$ (`rmk:bdryrepairs`), `cl:les`'s hypothesis pair (`rmk:sixscope`), `cl:pd`'s self-adjointness wording (`rmk:pdscope`). Its Appendix A loses the seven STATUS rows for the moved claims, and its `[C]` key now means this document.

The evidence that the criterion is met is `rmk:labelcensus`'s second run: 0 dangling `[C, ...]` citations, and an *empty* intersection between this document's label set and that of the only live companion. Commit: `b456448`. Page count unchanged apart from the paragraphs below.

Normative.

- `V4` [repaired] — from "each receive a frozen version marker and a superseded-by banner when this document ships. Freezing them is a separate, later step" to a record that the step is taken, with both frozen version numbers and what the banners say. The rule is unchanged; what changed is that it is now discharged, and (V4)'s last sentence no longer defers.
- `V5` [repaired] — (ii) recorded as **met**. The rule itself is untouched; the sentence "(ii) is not [met]" and the list of what blocked it are replaced by the discharge and a pointer to its evidence, and the paragraph now says in terms what a leading 1 does *not* assert.

Non-normative.

- `err:e17` — "Corrected in . . . the frozen banners that both superseded documents receive when this document ships" becomes a record of the two markers as written, with their dates and the reason those dates were chosen.
- `err:e18` — gains a *Completed at v1.0.0* paragraph: what the companion's v1.3 removed, which four of the nine it names as superseded, and the one surviving `[E, cl:les]` citation in the corpus — a verbatim quotation inside the dated review record `docs/implementation/hecke_extensions_OQ2_verdict.md`, left as written because rewriting a quotation would falsify the record.
- `rmk:labelcensus` — gains the v1.0.0 re-run (63 labels, 778 sites, 79 files, 0 dangling) *and* the two label-set checks the census does not perform. Those found one collision outside the nine, which the companion's v1.3 removed: it used `A1`–`A3` for its three required additions while this document uses them for authoring rules (A1)–(A3). No citation used either reading, so no citation census could have found it — which is now the stated reason for §22.5's guard to compare label sets and not only citations.
- The abstract and the header comment are re-written for the same reason, and §E.1's "Not yet written" and "two supersession steps" paragraphs are updated. **No chapter of Parts I–V was touched.**
- **What this version does not claim.** Not that the three unimplemented chapters are implemented, not that 14.24 is settled, not that 20.2 is complete, and not that §22.5's census guard is written. §E.1 keeps all four.

v0.19.0 — 2026-09-16 — chapters 21–22 and Appendices B–C: the worked example, the oracles, the machinery, the STATUS index and the formalization notes

This version. Writes §21, §22 with §22.5, and Appendices B and C, which were the document's last five declared stubs, adds one errata entry, and **amends (V5)**. (V5) as written would have taken the leading version field to 1 on the strength of the last stub alone. It now requires two criteria, and the second — an unambiguous label namespace — is **not** met: the nine labels that entry A.18's scope cut moved into §16 are still defined in the live `hecke_extensions.tex` as well, so a citation `[C, cl:pd]` does not resolve to a unique statement and (L6) cannot be honoured across the corpus. The version is therefore **0.19.0** and the leading 0 now carries that information as well as the stub count. Criterion (i) *is* met, and (V5) says so. None of this asserts that the three unimplemented chapters have been implemented (they have not) or that 14.24 is settled (it is not); §E.1 says so in those words. Commit: `0a3358b`. Pages 142–147, 148–151, 161–166 and 167–170, i.e. $6 + 4 + 6 + 4 = 20$ pp against an architecture budget of $4 + 2 + 5 + 2.5 = 13.5$ (the overrun is 1.5× overall, and is worst in §22 at 4 pp against 2 — its code map is a whole-document table, and the configuration surface and record schema are new material the architecture budgeted as "file→section map"). Sources: `hecke_action.tex`:1512–1612 (§21), :1613–1690 (§22); this document itself, read mechanically, for Appendix B; the

eighteen per-chapter *Formalization note* paragraphs for Appendix C; and the shipped tree, read and run on 2026-09-16, for every recorded number in all four.

Normative.

- `V5` [repaired] — **amended**. Its criterion for the leading version field was "the last stub is written"; it is now that criterion, (i), *and* (ii), an unambiguous label namespace: (V4)'s two freezes done, and no label of this document also defined by a live companion. The rule as it stood would have certified $1.X.Y$ with nine labels of §16 duplicated in `hecke_extensions.tex` — which (L1)–(L4) bind and (L6) is meant to machine-check, and which entry A.18 recorded as a relocation rather than a duplication. The nine are named in (V5) so that the criterion can be discharged by inspection. Owner decision of 2026-09-16.
- §21, (61), §21.1 — the T_2 datum. `prop:t2cocycle` [re-homed], (62), (63) — the twelve cocycles and the three permutations, transported unchanged from the source table (which is correct: it was recomputed from p alone in this pass and agrees entry for entry) but promoted from a display to a proposition with a proof, so that §11's forward reference has a numbered target.
- `ex:t2tau` [new], [verified computationally] — the τ-table at $p = 2$ as *emitted*: 69 chains, 15 triangle fills all over $\mathbb{Z}$, $B_{\mathrm{tab}} = 2$, $B = 16$, coefficients in $\{0, \pm 1\}$. The superseded text gave the count as "at most 69" and derived it from a bijection that `alg:tau`'s repair showed to be false; this example gives the count as a measurement and names step 0 as the reason the 69 slots are not 69 constructions. *Not* added to the roster of statements nothing else states: its content is a recorded run, not new mathematics.
- §21.2 [new] — a fully worked $(n, \ell, p) = (2, 5, 2)$, with ρ_n of the six blocks of α and ω and of the three δ_j printed over $\mathbb{Z}$, and the emitted $A^{(0)}(T_2) = (3)$, $A^{(1)}(T_2) = (4)$. The superseded chapter worked the datum symbolically in n and never instantiated it.
- §21.3 — the nine oracles, re-homed with the hypotheses written onto each clause ((A1)) and the deciding of abort-versus-record handed to `obl:oracle`(i),(ii) rather than restated. Two carry errata. **Oracle 6** carries `err:e1`: it now states both identities — full $T_4 = T_2^2 - 2^{n+1}I$ and primitive $= T_2^2 - 2^{n+1}I - 2^nI$ — with the emitting entry point of each, which is what `obl:oracle`(iii) binds it to. **Oracle 7** carries `err:e23`, new in this version.
- `err:e23` [new] — *found in this pass*, in the oracle list being re-homed: the superseded oracle 7 asserts $U_4 = U_2^2$ at $p = \ell = 2$ citing `cor:ulrecursion`, whose clause (1) carries $\ell \geq 5$ and whose clause (2) says something else. The implementation already records rather than asserts it at $\ell \in \{2, 3\}$ (amendment A-F1), so the defect is in the contract text alone. Oracle 7 now splits by hypothesis. `cor:ulrecursion` itself is unchanged.
- §21.4 [new] — the boundary, cone and involution oracle families, as *specifications*: each names the obligation it serves, and each is marked as not runnable, because the objects do not exist. Two numerical corrections were made while writing them and are the reason they are numbers and not prose: the boundary H^0 has rank $\lfloor n/\ell \rfloor + 1$ and not 1 (`prop:bdrychar`(1)), so the cone dimension book is instantiated at $(n, \ell, p) = (2, 5, 2)$, where $n < \ell$ and every term is independently known, and not at $n = 10$, $\ell = 5$, where it is not.
- `rmk:dicksontwo` [new], [verified computationally] — two implementations of the gating predicate of `rmk:dickson` exist in the tree under one name; the weaker one (first Dickson generator only) is sound on the grid it is used on, checked exhaustively, and unsound in general.
- §22, §22.1, §22.2, §22.3, §22.4 [new] — the whole-document code map; the boxed rule that **no switch may change a computed value**, with the two rollbacks, the two shadow builds and the execution switches classified by which of those they are; and the record schema, whose fifth item is site (6) of §14.5 and carries 14.24's $q_{\max}$, `truncated` and

reported-degree list. §22.2 is a contract box and states in one place what four chapters had each stated in their own code maps.

- §22.5 [new] — this document's four guards, of which three exist. `rmk:labelcensus` [new], [verified computationally]: the fourth, the citation census that would have caught `err:e7`, is *not written*, and was run by hand instead — 64 distinct labels cited from 78 files, 0 cited labels this document does not define.
- Appendix B [new] — 199 statements indexed with status and provenance, mechanically derived and build-gated: the number column is a real `\ref`, so an index line naming an undefined label fails the build, which is what (L6) requires and which was verified by deliberately mistyping one label. [verified computationally] is split into *registered* (8) and *ad hoc* (5), because a check the repository re-runs and a sweep done once are not the same evidence.
- Appendix C [new] — the eighteen per-chapter formalization notes reconciled into: what must be built, what must be imported, the five hypotheses that must not be lost, the shapes to factor once, and a cheapest-target table read against Appendix E's roster of the statements nothing else states. It performs *no* cost assessment — that is the commission — and it boxes the warning that every claim in this document about what `mathlib4` contains is unverified.

Non-normative.

- **Two gaps in this document were found by compiling Appendix B and are recorded, not repaired**, because repairing another chapter's statement from an index is the silent edit (V3) exists to prevent, and because (CL2) reserves Appendix A for the superseded corpus. **(i)** `cor:ulrecursion` carries no status tag although §1.1 requires one; it is the only untagged statement in the document that is not a definition, a hypothesis, a procedure, a predicate or commentary, and the tag it should carry is [proved here]. **(ii)** `rmk:taurings` has no line in this appendix: it entered in v0.5.0 alongside `def:tau`'s widening, which names it, but got no line of its own and is in no roster. Its line in Appendix B is starred for that reason.
- The provenance counts above are updated for this version: [new] 119 (69 assertions or definitions, 50 remarks), [re-homed] 45, [repaired] 38. Appendix B's column totals are *statement* counts at current content and are deliberately different; the contract box at the head of that appendix says why neither refutes the other.
- The roster of statements that no superseded document and no project note states is **unchanged** by this version. Of its four new statements one is [re-homed], two are remarks, and `ex:t2tau` is a recorded run.

v0.18.0 — 2026-09-16 — chapters 18–20: reduction, bounds, the floating-point filter, and the obligations

This version. Writes §18, §19 and §20, which between them were eighteen declared stubs, and adds one errata entry. All three are re-homings of `hecke_action.tex`:1162–1511, and almost all of the new writing is in the two places where re-homing was not enough: (A3) (a definition is the terminating procedure where the code's is), which turned `alg:tau` from a four-step sketch into a five-step algorithm whose emitted table is well defined, and (A4) (an obligation is a checkable predicate on emitted data), which turned eleven prose restatements of theorems into clause lists naming data, predicate, ring and failure action. Commit: `3d67bf5`. Pages 123–128, 129–131 and 132–140 (architecture budget $4 + 2 + 3$ pp; the overrun is 2.0× and is concentrated in §20 at 9 pp against 3 — the roster of §20.3 has fifteen rows, not eleven, because the obligations of §§9, 14 and 15 are integrated here, and (A4)'s four fields are written out for every clause of every one of them). Sources: `hecke_action.tex`:1162–1338 (§18), :1339–1407 (§19), :1408–

1511 (§20); and, for the constructive content of `alg:tau` and for every status claim in §20.3, the shipped tree read on 2026-09-16.

Normative.

- `def:primitives` [re-homed] — (P1)–(P4) with their output specifications. Two changes of content, both recorded in the statement: (P2)'s parenthetical "all intermediate matrices have entries bounded by the entries of the input" is dropped, because it contradicts `lem:bounds`(4) of the same source, which says *twice* the maximum and is what the implementation checks — the true statement about the input's own entries is that $|c|$ strictly decreases, and that is what is written; and (P3) is recorded as **specified with no live call site**, the closed forms of `def:cosetreps` producing δ_j directly. (P4) now specifies the canonical key (lexicographic minimum over the transporter set, sign from the vertex-slot permutation, quotient by $-I$) where the source said "a fixed choice of orbit section".
- `rmk:primitiveclosure` [new] — the closure statement: $\mathbb{Z}$ is touched in exactly four places, which is what makes `thm:backend` a statement about a finite workload.
- `alg:tau` [repaired], (56) — the τ-table construction. **The source's step list is not enough to determine its own output.** It stores "one representative per orbit, since $\{(i,\sigma)\} \leftrightarrow$ simplices of $\Gamma''\backslash K^u$" — and that bijection is false: $o_q(K)$ is not a section of the orbits, already because $\widetilde{v_3} = T \cdot \widetilde{v_0}$. The repaired algorithm has a **step 0**: one slot per Γ''-orbit is constructed and every other is derived through a single shared (T1) helper used by *both* construction and certification. Step 1 additionally carries the two branches that later chapters depend on — the exact-position hit, and the ascending-index tie-break that `rmk:iotatausym` shows is structurally incompatible with ι_K — and the adj(g) projective device that keeps the geometry rational-free. Step 3 carries owner decision OP2's default-forbidden mod-ℓ fallback. Step 4 is delegated to `obl:tau` rather than restated ((A4): one home).
- `rmk:certoverGeometry` [new] — certification over geometry, with the two consequences the rest of the document uses: the choices of steps 1–3 are free and changing one changes the table but not $A^{(q)}$; elliptic slots are forced by (T1), which explains the construction and proves nothing.
- `lem:bounds` [repaired] — the entry bounds. Clause (4) was asserted ("never exceed twice the maximum of their input entries") and is now **proved**, in the sharper form $\max(\|\gamma\|, 2)$, from $|a'| \le |c|/2$ and $b' = (a'd-1)/c$; the implementation checks the weaker constant at every step. Clauses (2) and (3) are stated in the level $N = p^\nu$, not in p: the source wrote $\|s_j\| \le p$ and $4p^2$, which are the $\nu = 1$ values of bounds that are really about $\mathbb{P}^1(\mathbb{Z}/N)$, and the implementation's certified B is $\max(B_{\text{tab}}, 4N^2)$.
- `rmk:btabsmall` [new] — [verified computationally]: C_K is never evaluated and $B \le 196$ over the 66 shipped points, so the first tier of `thm:backend` is never approached. (Re-reads the run already recorded at `rmk:tauintegral`; nothing new is executed.)
- `thm:backend` [repaired] — the backend. The source's clause (iii) bounds the predicates by $2B^2$ with B the bound on the *stored matrices*, which does not cover the one predicate whose inputs are not stored matrices: the nearest-vertex comparison of `alg:tau`(1b) forms a difference of squared distances on cleared denominators, needs the 128-bit tier and a gcd reduction before the 64-bit predicate can take it, and aborts if that is not enough. That clause is written out. The early-reduction scheme and the mod-ℓ box are transported.
- `lem:predshapes` [new], (57) — every branch decision is a two-term integer dot product, and there are exactly four: (D), (Q1), (Q2) and **(Q3)**, the cross-D nearest-vertex comparison $\operatorname{sign}(A + B\sqrt{3})$, which the source did not list at all. Proved, including the closure of the exact quadratic points under the Möbius action of a positive-determinant matrix. This is the lemma that makes the rest of §19 a self-contained statement about IEEE-754 rather than a claim about the pipeline.

- `thm:fp64` [repaired], (58) — the filter. Its floating-point hypotheses were unstated and its proof used all three: (FP1) exact input conversion, (FP2) the round-to-nearest standard model, and **(FP3) the evaluation order** $\mathrm{fl}(\mathrm{fl}(a_1b_1) \pm \mathrm{fl}(a_2b_2))$ **with no fused multiply-add**, which the implementation enforces with `volatile` intermediates and which (A1) requires in the statement. The source's displayed error chain does not parse as written; the bound is re-derived, and the sharper $\frac{1}{2}E(B)(1+2^{-53})$ is recorded because `rmk:thresholdrounding` needs it.
- `rmk:thresholdrounding` [new] — the threshold is itself computed in floating point, and the factor 2 of slack between the proved bound and the normative constant $E(B) = 2^{-50}B^2$ is what makes that sound. This is why the binding constant is not the sharper $2^{-51}B^2$.
- `cor:fptotal` [repaired], (59) — totality. The criterion is now the non-strict $E(B) \le \frac{1}{2}$, with a second and independent sufficient condition (exact representability, $2B^2 \le 2^{53}$), and "total *iff*" is corrected to a sufficient condition. Erratum `err:e22`.
- `cor:fp32` [repaired] — the binary32 analogue. Threshold $B \le 2^{10}$ *unchanged*, and now proved: it fails the source's own strict criterion with equality. Erratum `err:e22`.
- `obl:welldef` [re-homed] — recast as the three predicates that `alg:assembly`'s contract box delegates here: y_j satisfies every row of $\mathcal{R}$; $d\,y_j = 0$; the $\Theta_{\mathrm{eq}}^{(q)}$ solve succeeds with identically zero residual. All over R, all aborting.
- `obl:indep` [repaired] — independence. **The source asserted as a runtime test something the construction cannot test.** Clause (iii), independence of the double-coset representative g, is not reachable through an interface that fixes $g = \mathrm{diag}(1, N)$ and $\Gamma'' = \Gamma_0(N)$; what a run can do is the constant-h_j case of clause (i), and the obligation now says so and says what would make (iii) checkable. Clause (i) also records, rather than asserts, the cochain-level agreement under a non-trivial relabelling, because rebuilding the chain rebuilds τ (`rmk:indeplevel`); clause (ii) is gated on $6 \in R^\times$, which is where `lem:tauexists`(2) is used.
- `obl:integral` [re-homed] — four clauses: integrality of every δ, no inversion at determinant N, the integral normalisation $A^{(0)}(T_p) = 1 + p$ with coefficient p^{n+1}, and the bad prime through the same pipeline.
- `obl:algebra` [repaired] — **erratum E1, the second of its two landing sites** (`err:e1`). The source's runtime test compared the *primitive* $p^2 + p$-coset operator with $T_p^2 - p^{n+1}$, which is the *full* operator's identity, and is therefore wrong by $p^n I$. The obligation now asserts *both* identities, names which entry point emits which operator, and requires the full T_{p^2} of clause (ii) to be obtained by the double-coset assembly (19) rather than by the recursion being tested. Clauses (iv) and (v), the diamond and coprime multiplicativity, are new as clauses; (iv) carries no hypothesis on ℓ, by `prop:diamond`.
- `obl:agreement` [re-homed] — four clauses, each with its level of comparison fixed by `rmk:indeplevel`: cochain-level bit-identity only between constructions on the same datum, cohomology-level for the bar-cochain route of `prop:barformula`. The clause requiring agreement through the public evaluation interface is written out, because the internal assertion can be discharged against private state.
- `obl:shapiro` [re-homed] — the dimension comparison, with the observation that the *equality of the two computed dimensions* carries no hypothesis on ℓ while identifying either side with $H^\bullet(\Gamma_0(N); V_R)$ does.
- `obl:oracle` [repaired], (60) — the Eichler–Shimura oracle. The source wrote the cuspidal factor as $(X - a_p(f))(X - a_p(f))$, a doubled factor whose multiplicity is the point; it is now $(X - a_p(f))^2$ with the multiplicity attributed to the two summands of `cl:dimH1`(1). What is *asserted* and what is *recorded* is now part of the obligation — only independently verified $a_p(f)$ constants gate — and clause (iii) binds §21's oracle 6 to the full-versus-primitive distinction of `err:e1`.
- `rmk:dickson` [re-homed] — the exceptional loci. The invariant-theoretic input and its

citation are *deferred* to `cl:dimH1`(2) rather than restated with a second pointer; what is added is that both semigroup generators matter, with the witness $(n, \ell) = (20, 5)$ where $\ell + 1 \nmid n$, and [verified computationally] that the implemented predicate is semigroup membership and that $(20, 5)$ is in the shipped grid.

- `obl:fp` [re-homed] — four clauses. The differential test must not share its exact routine with the production fallback; the counters witnessing dead code are a diagnostic ((A6)); and clause (iv), the pure-integer bit-identity, is recorded as a **two-build protocol rather than a predicate a single run can evaluate**.
- `obl:backend` [re-homed] — five clauses: the certified bound recounted, the tier inequality asserted, overflow aborting rather than wrapping, the absence of arbitrary precision recorded rather than asserted at runtime, and the mod-ℓ backend exact for the ℓ in force.
- `obl:tau` [re-homed] — seven clauses, each named at the point `alg:tau`(4) defers to them, with the ring per clause ($\mathbb{Z}$, or $\mathbb{F}_\ell$ for an identity involving a mod-ℓ-solved chain) and the (T2) right-hand side required to go through the *same* (T1) helper the construction uses.
- §20.3 — the roster. Fifteen obligations with, for each, the ring, the failure action, the status and the registered validations behind it; ten are discharged by a registered validation, one (`obl:indep`) is partial for a stated structural reason, one clause of `obl:iota` is discharged *negatively*, and three (`obl:taubdry`, `obl:eta`, `obl:coneexact`) are unimplemented. [verified computationally] The one clause that is a protocol rather than a predicate, `obl:fp`(iv), was executed on 2026-09-16: two out-of-source builds, `HECKE_PURE_INTEGER` off and on, both reporting tier A 9446 failing 0 and tier B 4603 mismatching 0 against the same frozen reference under `GOLDEN_STRICT=1`, hence bit-identical on all 14049 hashed objects, with five named Hecke validations exiting 0 in both (`docs/matrix-engine/tools/cross_config.sh`). That is a recorded run of an existing script, not a registered validation, and the roster says so.

Non-normative.

- `err:e22` — a new errata entry: `cor:fp32`'s totality threshold $B \le 2^{10}$ does not follow from `cor:fptotal`'s strict criterion $E(B) < \frac{1}{2}$, which it meets with equality. The threshold and every shipped constant are unchanged; what was missing is the proof, which the implementation's own validation had already written down in a comment and the contract had not. Adding an erratum entry is itself a change to this document ((CL2)), and the summary table of `app:errata` gains its row.
- **One symbol was deliberately not introduced.** §19 needs the unit roundoff, whose customary letter is ε — which is `app:labelindex`'s ε, the augmentation, four pages earlier in `alg:tau`(4). Rather than add a twentieth collision row, the two values are written as the explicit constants 2^{-53} and 2^{-24} throughout and the quantity is given no letter; the chapter header records the decision. `gl2_notation.tex` is unchanged by this version. The four symbols §18 does need ($\|\cdot\|$, $\mathcal{F}$, pos, $\mathcal{N}$) are declared chapter-locally, in the manner of the proof of `prop:fullassembly`, and occur nowhere else.
- **No new statement of these three chapters carries a [conjectural] or an [open] tag**, and no [standard:] pointer is newly asserted: §18 and §19 are self-contained, and §20 cites [Shi71], [AS86], [Bro82], [Har22] and [Den23] only at pointers `thm:relations`, `thm:classical` and `cl:dimH1` already carry, and `obl:oracle` says so in its own statement rather than implying the texts were re-checked here. `gl2_refs.bib` is unchanged by this version.
- `app:changelog` — provenance counts updated from 111/35/29 to 116/44/38 ([new]/[re-homed]/[repaired]) and the [new] split from 67/44 to 68/48; this version contributes 5 [new], 9 [re-homed] and 9 [repaired] statements. The tagged-line totals go from 182/64 to 190/80, this version contributing 8 labels on [new] lines and 16 on [re-homed] lines, counted as every `\cl` on an item whose *subject* carries that tag. A third such figure is

not introduced: this version's [repaired] items name 21 labels, but earlier versions did not record theirs, and a column that starts at v0.18.0 would read as a total.

- `app:changelog-gaps` — §18, §19 and §20 removed from the stub roster, with their eighteen stub labels. `app:labelindex` — the nineteen new labels of the three chapters dereferenced, and `err:e22` added to the errata list, per (L6).
- The abstract, the file header and the date line: twenty of the twenty-two chapters are written.

v0.17.0 — 2026-09-16 — chapters 16–17: the six groups, the long exact sequence, and the general orbifold model

This version. Writes §16 and §17, which between them were eleven declared stubs. §16 is the **move out of [E]** that the scope cut requires (`err:e18`): nine labels that were defined in `hecke_extensions.tex` are defined here, each now carrying its own hypotheses ((A1)) and its own citation ((A2)) instead of a blanket "$\ell \geq 5$, n even" declared once. §17 is **definitions now, construction deferred** (owner decision OP4), written refinement-parametrically per owner direction OD-T1. Commit: `44c355e`. Pages 111–118 and 119–122 (architecture budget 4+3 pp; the overrun is 1.7×, and it is the apparatus (A1) and (A2) force — [E] proved none of these nine statements and cited three of them to a working note). Sources: `hecke_extensions.tex`:186–338 for §16; the OQ2 verdict §5, the implementation plan's C-A3(ii.7) and the Moerdijk–Pronk source in `docs/literature/orbifolds/` for §17; owner decisions OP4, OP5 and direction OD-T1. **One erratum lands here**, `err:e21`, found in this pass.

The three things a reader should check first. **(i)** `cl:pd` is stated in the *adjointness* form and in no other. [E]'s "T_q self-adjoint once D_n carries the adjugate Δ-structure" is true of $\langle \rho_n(g)v, \varphi\rangle = \langle v, \rho_n^\vee(\operatorname{adj} g)\varphi\rangle$ (`prop:pairing`(3), twist-free) and *false* of $\langle \rho_n(g)v, \rho_n^\vee(g)\varphi\rangle = (\det g)^n\langle v, \varphi\rangle$ (`prop:pairing`(4), which carries p^n). The v0.3.0 entry below recorded that it "must be re-homed that way"; it now is. **(ii)** `cl:les` drops the hypothesis $H^0_c = 0$, which `lem:conerelative`(4) makes unconditional, and drops "n even", which exactness never used; it *keeps* $H^2(\Gamma; V_R) = 0$, and `lem:h2torsion`(2) shows why — at $(\ell, n) = (2, 0)$, $H^q(\Gamma; \mathbb{F}_2) \neq 0$ for every $q \geq 0$. **(iii)** `err:e21`: one sentence of [E]'s `cl:chi` is false.

Normative.

- §16, `eq:sixgroups` — the chapter and the contract box fixing $X_1, \dots, X_6$, H_{tot} and d in the notation of §2.6, with the three recurring hypotheses named once and repeated in every statement.
- `cl:amalgam` [new] — **strictly generalised** ((CL4): the tag is on the content). [E] asserted the Mayer–Vietoris sequence and the vanishing $H^i(\Gamma; M) = 0$, $i \geq 2$, for $\ell \geq 5$, as [standard:]. Here: (1) the amalgam and its Bass–Serre tree sequence `eq:treeses`, the only external input; (2) the long exact sequence `eq:mvamalgam` in *every* degree over *every* commutative ring, proved from (1) by Shapiro; (3) the $6 \in R^\times$ consequences, with $V_R^S = (1 + \rho_n(S))V_R$ and $V_R^U = (1 + \rho_n(U) + \rho_n(U)^2)V_R$ derived rather than asserted; (4) a torsion bound of 24 proved from (2) with *no* citation at all, beside the sharp classical 12. The Euler-characteristic clause no longer names the quantity χ, which is the boundary character.
- `lem:h2torsion` [new] — (1) the vanishing for $6 \in R^\times$; **(2) the proof that it is not removable**: $H^q(\mathrm{SL}_2(\mathbb{Z}); \mathbb{F}_2) \neq 0$ for every $q \geq 0$, by the dimension count in `eq:mvamalgam` with $\mathbb{F}_2$-coefficients. This also settles, for $n = 0$, the half of `rmk:ell2window`(2) that was left open there; that remark is **not** edited (it is the v0.14.0 slot's record), and the new statement says so at its own site.

- `cl:dimH1` [re-homed] — content unchanged; restated in the closed form $2d_k - 1 + \dim H^0$ with the *same* d_k that `obl:iota`(v) defines, rather than in the symbols M_k, S_k, which would collide with $M_n = \mathrm{Sym}^n \mathbb{F}_\ell^2$ and with the operator set of [E] §7 (§2.7, C12, C14). No modular-form notation was added: see the non-normative note below.
- `cl:bockstein` [re-homed] — content unchanged; its *status* moves from [standard:] to [proved here], the $2,3$-torsion input now being `cl:amalgam`(4) and the rest the long exact cohomology sequence of $0 \to V \xrightarrow{\ell} V \to M_n \to 0$, whose connecting map *is* the Bockstein. Clause (3), the Hecke-equivariance, is given its reason: that sequence is one of Δ-modules, and res, $(\mathrm{conj}_g)_*$, cor are morphisms of δ-functors. The vanishing $H^0(\Gamma; V) = 0$ is proved in two lines rather than cited.
- `cl:boundary` [new] — **hypothesis repaired**. [E]'s "$H^i(\partial; M) = 0$ for $i \geq 2$" is false at $\ell = 2$ (`lem:ell2unbounded`); the statement now carries $2 \in R^\times$, and is proved from `prop:circlemodel` and `prop:bdrychar`(1) rather than asserted. `rmk:bdryrepairs` [new] records that repair and the C18 spelling Nm.
- `cl:les`, `eq:les`, `eq:dimbook`, `eq:r1formula` — [new] for the generalisation. Four clauses: (1) the sequence in all degrees with *no* hypothesis on ℓ or n, Hecke-equivariant by `thm:conewelldef`(a) — where [E] made equivariance conditional on "Φ_g a cochain map", which is now a theorem; (2) the collapse, from `lem:conerelative`(4),(5); (3) exactly when `eq:les` is a statement about `eq:sixgroups`; (4) the formula for r^1, proved *well defined* here and tagged [conjectural] as a citation for the identification, its only source being the working note `err:e14` declares not a source of record.
- `cl:chi` — [new] for clause (3), which is now [proved here] (Vandermonde on a generator of $\mathbb{F}_\ell^\times$, plus Dirichlet only for the passage from primes to residues). Clause (1) cites `prop:bdrychar`; clause (2), the general-n H^1 multiset, stays [conjectural] — it is what `obl:taubdry`(vi) records rather than gates — with a [verified computationally] corroboration on 468 cases.
- `cl:pd` [repaired] — restated in the adjointness form (see (i) above), with $\mathrm{adj}(\Gamma g \Gamma) = \Gamma g \Gamma$ from `prop:pairing`(5) supplying the *self*-adjointness and no determinant factor; `err:e14`'s socle-versus-Loewy correction carried; status [open], as `prop:coneroutes`(3) and `thm:conewelldef`(c′) already read it. `rmk:pdscope` [new] states the p^n twist, what the claim depends on, and what has no certificate yet.
- `rmk:sixscope` [new] — which two of [E]'s hypotheses are dropped and which one is not.
- §17, `def:borelmodel`, `eq:borelfree`, (G1)–(G4) [new] — the free Γ-model $F_\bullet \otimes C_\bullet(Y)$ of an *arbitrary* pair satisfying (G1)–(G4), its orbifold cochain complexes in a degree range $\leq q_{\max} + 1$, and twisted chain data on it, with (T1)–(T3) and the pair identity carried over verbatim. Stated so that a Γ-equivariant refinement of `def:model7` is admissible input, per OD-T1.
- `rmk:borelfree` [new] — why the free model is wanted: both clauses of `lem:tauexists` hold on it over $\mathbb{Z}$ for every ℓ, so `thm:conewelldef`(c) holds at every ℓ there, where on the 7-vertex model it is asserted only for $6 \in R^\times$ (`thm:conewelldef`(e)).
- `thm:specialisation` [new] — "the 7-vertex complex is *a* specialisation at $6 \in R^\times$", in a degree range and for an arbitrary input: (1) the isotropy spectral sequence collapses to $C^\bullet_{\mathrm{eq}}(K_Y, \mathcal{R}_Y; L)$; (2) the relative version is $H^\bullet_c$; (3) the two Hecke actions agree, both being $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$; (4) the shipped model is one admissible input, a refinement another.
- `rmk:orbifoldrefine` [new] — what a refinement preserves (`prop:model7h4`(i)–(iv), `def:cosetreps`), what must be recomputed (N_0, N_1, N_2, `def:model7rel`, the stored simplex order, `alg:tau`, `lem:bounds`, the golden references), that OD-T1 revises nothing, and that a refinement does not by itself make the complexes finite.
- `err:e21` — a new errata entry: [E]'s "$\chi_1 \equiv \chi_2$ at every Dickson locus" is false; the criterion is $(\ell - 1) \mid (n - 2)$, first counterexample $(n, \ell) = (12, 5)$. Adding an erratum entry is itself a change to this document ((CL2)), and the summary table of `app:errata` gains

its row.

Non-normative.

- `rmk:mpcaveat` [new] — Moerdijk–Pronk, cited for the Borel-construction identification and for $S(\mathcal{T})$ as one computational model, and *not* for the normative definition: their charts are effective and $[\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}]$ is not, and they twice omit the independence proof for μ. **Every quotation and line pointer in that remark was checked against `docs/literature/orbifolds/moerdijk_pronk_.../simplicial.tex` on 2026-09-16**; the published theorem *number* was not checked against the journal text and the remark says so.
- `gl2_refs.bib` — two entries added, `serre_trees` (the amalgam, its tree, and free subgroups of free products) and `wilkerson` (the Dickson invariants). **Neither is in `docs/literature/` and neither section pointer is verified**; both notes say so, and both statements carry either an independent proof from `eq:treeses` or a [verified computationally] corroboration. `moerdijk_pronk`'s existing note already said what `rmk:mpcaveat` now spells out.
- **No modular-form notation was added**, and `gl2_notation.tex` is unchanged by this version. Slot 7 recorded that the contract has no symbol for M_k or S_k and that chapter 16 might need them. It does not: the only statement that would use them, `cl:dimH1`, states the same numbers as d_k and $d_k - 1$, which `obl:iota`(v) already defines with its citation. Adding $\mathcal{M}_k$ and $\mathcal{S}_k$ would have cost a twentieth collision row — $\mathcal{S}$ is [E]'s operator set (C14) and M already has four live readings (C12) — for no statement this document makes.
- [verified computationally] **corroborations, both *ad hoc* and neither a registered validation.** `cl:dimH1`: the shipped campaign table `docs/matrix-engine/results/results_table.tsv` re-read on 2026-09-16 — 2118 rows, 710 distinct (n, ℓ) with $\ell \geq 5$, $n > 0$, 203 on the Dickson locus; 0 disagreements on $\dim H^0_{\mathrm{eq}}$, on $\dim H^1_{\mathrm{eq}} = 2d_k - 1 + \dim H^0_{\mathrm{eq}}$, and on $\dim H^2_{\mathrm{eq}} = 0$. `cl:chi`(2),(3) and `rmk:chicoincide`: direct computation over $\mathbb{F}_\ell$ — 468 operator spectra and 190 560 instances of the coincidence criterion, 0 disagreements. There is nothing in the tree to re-run for either; `obl:oracle` and `obl:taubdry`(vi) are where they would become registered.
- `app:changelog` — provenance counts updated from 97/33/28 to 111/35/29 ([new]/[re-homed]/[repaired]), the [new] split from 60/37 to 67/44, the tagged-line totals from 144/60 to 182/64 — this version contributes 38 labels on [new] lines and 4 on [re-homed] lines, counted as every `\cl` on an item whose *subject* carries that tag, which is the convention the entries above use and which is stated here because it is not self-evident from them — and a Part IV roster bullet of 7 new statements added.
- `sec:oc1sites` — site (4) updated: `def:borelmodel` and `thm:specialisation` now *are* stated in a degree range rather than being scheduled to be, and `rmk:orbifoldrefine`(4) is added as that chapter's third 14.24-dependent site. This is the only edit this version makes to §14, and it is the one `sec:oc1sites` asks for by existing.
- `app:changelog-gaps` — §16 and §17 removed from the stub roster. `app:labelindex` — the seventeen new labels of the two chapters dereferenced, and `err:e21` added to the errata list, per (L6).
- The abstract, the file header and the date line: seventeen of the twenty-two chapters are written.

v0.16.0 — 2026-09-16 — 11.3 apportioned; notation C18, C19; a wrong bibliography entry

This version. Three things, none of them a new chapter. (i) The hypothesis $6 \in R^\times$ on Lemma 11.3 is apportioned per clause: **existence holds over $\mathbb{Z}$ for every ℓ**, by the construc-

tion Proposition 15.13 already contained; **homotopy uniqueness keeps the hypothesis**. (ii) The two notation collisions that Chapters 14 and 15 flagged and could not close are closed, as C18 and C19. (iii) `gl2_refs.bib`'s `harder_eisenstein` carried the wrong title. Commit: `3c3366c`. Sources: this document only — Proposition 15.13 (v0.15.0), the two flagged notation items in the v0.15.0 and v0.14.0 entries below, and [Ser73] read against the text. No superseded document and no project note is an input, so nothing here is [re-homed].

The verdict, stated plainly. The strict-pair construction *does* establish existence of a datum satisfying (T1)–(T3) of `def:tau` over $\mathbb{Z}$ for every ℓ: steps (V), (E), (F) of `prop:strictpair` never invert a stabiliser order — (V) chooses the vertex value *inside* the fixed-point set of the twisted stabiliser, which exists by Serre's classification of the elliptic points, and (E), (F) fill freely because edge and triangle stabilisers are central — and step (B), the only part that uses (B1)–(B3) and τ_∂, is what makes the result a map of *pairs* and is not needed for bare existence. It does **not** establish uniqueness up to twisted homotopy: a homotopy H must have a *prescribed* boundary *and* be stabiliser-invariant, and only the averaging idempotent supplies that. **No downstream hypothesis is therefore weakened.** The statements checked, one by one, against their own proofs, and all unchanged: `thm:welldef`(b) and, through it, (c) in the general case; `prop:iotacommutes`(3c); `thm:classical` (both of its two uses); `prop:barformula`'s agreement clause; `prop:coneindep`(2); `thm:conewelldef`(c), (c′) and (e); `rmk:conesteinberg`(i),(ii). **Not** re-examined, and therefore not weakened either: everything whose $6 \in R^\times$ comes from `rmk:orbifold`/(H3) rather than from `lem:tauexists` — `prop:bdrycohom`, `thm:relations`, `obl:oracle`, `prop:iotasplit`, the $H^2_{\mathrm{eq}} = 0$ citation of §13.4 — none of which cites this lemma at all.

Normative.

- `lem:tauexists` — restated as two clauses with their hypotheses apportioned ((A1)). **(1) existence over $\mathbb{Z}$, no hypothesis on ℓ**, new here, with the terminating construction (V), (E), (F) written out and the missing step supplied (a finite subgroup of order 4 or 6 is cyclic and a generator is elliptic, hence has a fixed point); **(2) homotopy uniqueness**, the v0.5.0 statement verbatim with its hypothesis and its proof unchanged. The contract box after it is restated accordingly: at $\ell \in \{2, 3\}$ it is *(2) and not the lemma* that is unavailable. The label already stands in the [repaired] column from v0.5.0 and is not re-counted; clause (1) is new mathematics and is what this line records.
- `rmk:tauexistsscope` [new] — the apportionment: why the construction does not give uniqueness and what would settle it ([conjectural] either way); the list of downstream statements checked and left alone; and the warning that existence of *an* integral datum is not integrality of the *canonical* table, which is `rmk:tauintegral`'s subject and owner decision OP2's.
- `rmk:tauintegral` — the "what is *not* claimed" bullet corrected: it said Lemma 11.3 gives existence "*not* over $\mathbb{Z}$", which clause (1) makes false. What survives, and is now the bullet's point, is the distinction between existence of some integral datum and integrality of Algorithm 18.3's fills.
- `rmk:taubdryintegral` — the parallel bullet corrected: "that Lemma 11.3 holds over $\mathbb{Z}$ … it does not" narrowed to the uniqueness half, which is what that sentence was about.
- `prop:strictpair` — statement unchanged; its [standard:] pointer corrected from Serre's Thm. 2 (the generation by S and T) to Thm. 1(1),(3) (the fundamental domain and the stabilisers), the target list extended from $\{i, \zeta\}$ to $\{i, \zeta^2, \zeta\}$ to match the source exactly, and the fixed-point step made explicit. Verified against [Ser73] Ch. VII §1.2 itself on 2026-09-16.
- §2.7, row **C18** — N the Hecke level versus N the degree-ℓ boundary invariant $y^\ell - x^{\ell-1}y$; resolved to Nm, with its crosswalk row. Flagged by the v0.14.0 slot and resolved locally

in `prop:bdrychar`; that local note now names C18 instead of promising a revision.
- §2.7, row **C19** — the tilde, with its four live readings (the lift `\lift`; coset transport $\widetilde{F}$; the cochain-level $\widetilde{T}$; Part I's $\widetilde{K}$, $\widetilde{X}$). Resolved by a reading rule and *no* rename — the decoration is read from what it decorates — plus the standing rule that no new tilde may be introduced without a row. Flagged by the v0.15.0 slot; `def:cosettransport`'s notation note now names C19 and lists all four readings. The collision count is nineteen; §2.7's heading, its opening sentence and §2.3's `\lift` row are updated.

Non-normative.
- `gl2_refs.bib`, `harder_eisenstein` — the title was "Eisenstein cohomology for $\mathrm{SL}_2(\mathbb{Z})$ and special values of L-functions", which is not a paper of Harder's in Invent. Math. 89. The correct title is *Eisenstein cohomology of arithmetic groups. The case* GL_2; author, journal, volume, year and pages 37–118 were already right and are now verified. Under (CL2) the bibliography is part of this document, so this is an entry.
- `gl2_refs.bib`, `deng` — page range 87–156 added from the article. `serre_cours` — the note records what Thms. 1 and 2 of Ch. VII §1.2 actually say, so the next pointer cannot repeat the slip fixed above. The remaining 29 entries were spot-checked for author, title, journal, volume and year; no further error was found, and no section- or theorem-level pointer other than `serre_cours`'s was verified against a text.
- `rmk:conesteinberg`(iii) and `thm:conewelldef`(c′) — the (A2) caveat on unverified section numbers extended from [BS73] to [Har87], which it should always have covered; and [Har87]'s subject stated as GL_2 over a number field rather than $\mathrm{SL}_2(\mathbb{Z})$, matching the corrected title.
- Formalization notes of §11 and §14, and the standing-hypotheses paragraph of §11 — all rewritten to name the clause, since "the comparison theorem is what a formalisation must not lose" is now true of (2) and false of (1); (1) is the cheaper target and should be done first. Every reference to `lem:tauexists` in the document now names the clause it means, except the handful that mean the lemma as a whole or its contract box.
- The paragraph after `prop:strictpair` — rewritten: that proposition no longer makes a bulk-existence statement "`lem:tauexists` does not make", because it now does; what it adds is strictness. Neither proof uses the other.
- `app:changelog` — provenance counts updated from 96/33/28 to 97/33/28 ([new]/[rehomed]/[repaired]), the [new] split from 60/36 to 60/37 (the one addition is a remark), and the tagged-line totals from 143/60 to 144/60. `lem:tauexists`, `prop:strictpair`, `rmk:tauintegral` and `rmk:taubdryintegral` are labels already in the table and are not re-counted ((CL3), (CL4): the table counts labels, this appendix counts revisions).
- `app:labelindex` — `rmk:tauexistsscope` dereferenced, per (L6).
- The date line. The v0.14.0 and v0.15.0 entries below are **not** edited: they are the append-only record of what those slots found and flagged, and this entry is where the flags are discharged.

v0.15.0 — 2026-09-16 — the pair homotopy η and the cone action Φ_g

This version. Writes §15, which was a stub declaring twelve labels, and with it the Hecke action on compactly supported cohomology as a *definition*: the pair homotopy η at chain level, the derived h, the cone map Φ_g, and the theorem that the induced operator is independent of η over $\mathbb{Z}$ for every ℓ. Commit: `0ae11d3`. Sources: the OQ2 verdict §§0, 2.1–2.10, 4, 6 (`hecke_extensions_OQ2_verdict.md`); the implementation plan's amendment C-A3(ii)–(iii) and its decision record and errata (`hecke_extensions_implementation_plan.md`:412–566, 1994–2098); the OQ2 revision audit (F5, B.1–B.4); the specification's (A2) (`hecke_extensions.tex`:789–880); `monodromy_v5.tex`:1148–1265 for the cone convention, the only superseded-document input;

owner decisions OP2, OP3, OP6. **No superseded document states any of this**: the sources are project notes, so under (CL4) every statement below is [new] or — where a note's argument was defective — [repaired], and the [re-homed] roster for this version is **empty**. **Five errata land here**: `err:e2` and `err:e3` (`prop:coneambiguity`), `err:e5` (`lem:etaexists`), `err:e13` (`lem:conesign`, `lem:derivedhomotopy`) and `err:e15` (the chapter as a whole); `err:e14`'s cone half is applied by citing `thm:relations` and never restating the T_p^2 relation. Pages 91–106 (architecture budget: 8 pp; the overrun is 2.0×, the same rate as chapter 14's mathematics, and it is proofs: every statement is proved in full rather than by reference to the verdict, because the verdict is not a source of record). The sign of the cone identity was **re-derived for this version** from the cone differential alone and confirms the plan v1.2 / verdict §2.1 / revision audit B.1 orientation (`lem:conesign`); the claim that the shipped τ is not a map of pairs was **re-verified** on `docs/implementation/tau_table_p2.txt` (`rmk:conestrict`(1): 10 of the 21 stored boundary slots leave ∂X).

Normative.

- `eq:contractibleuse` — the three consequences of (H4)(ii) (ε an isomorphism on H_0, $Z_1 = B_1$, $Z_2 = 0$, over $\mathbb{Z}$) written once and cited at every use; the OQ2 revision audit's F5 found them load-bearing and unwritten in every source. Display label, no tag ((CL4)).
- `def:pair` [new] (`eq:conediff`) — the pair, $r = j^*$, the relative complex $\ker r$, the cone in `monodromy_v5.tex` §8's convention with the notation of §2.6 (q, (c, c_∂), $\mathrm{inc}_c/\mathrm{pr}_c$), and $H_c^q := H^{q-1}(C(r))$. The convention is re-homed; the pair and $H_c^\bullet$ as objects of the contract are new.
- `lem:conerelative` [new] (`eq:coneles`, `eq:conedims`) — r is onto (extension by zero, equivariant because ∂X is Γ-stable, and *not* a cochain map); the cone is a complex with the long exact sequence `eq:coneles` whose connecting map is $+r$; $(\ker r)[1] \to C(r)$ is a quasi-isomorphism, so $H_c^\bullet$ *is* relative equivariant cohomology; $H_c^q = 0$ for $q \le 0$ and $q \ge 3$ — in particular $H_c^0 = 0$ **with no hypothesis on** ℓ **or** n, which the verdict and the plan stated only for $6 \in R^\times$, n even; and the six-term window closes under the single hypothesis $H^2_{\mathrm{eq}} = 0$. New: no source proves any of these.
- `def:cosettransport` [new] (`eq:cosettransport`), `lem:cosettransport` [new] — the device of `thm:welldef`(b) named and generalised to any additive (T1)-map of any degree between the chain complexes of the pair, with its equivariance and *four identities* ($\widetilde{\partial F} = \widetilde{F}d$, $\widetilde{F\partial} = d\widetilde{F}$, $\widetilde{jG} = \widetilde{G}r$, $\widetilde{Gj} = r\widetilde{G}$) that make every later proof a two-line computation. The tilde on a map (coset transport) is distinguished from the tilde on a simplex (`\lift`, C16) at the point of definition.
- `def:pairhomotopy` [new] (`eq:pairhomotopy`) — the pair homotopy and the Hecke datum of the pair $(\tau, \tau_\partial, \eta)$; with the owner-decision box recording OP6, the sign $\partial\eta + \eta\partial = j\tau_\partial - \tau j$ and $h = \widetilde{\eta}$ with no sign, and stating what that decision does *not* move.
- `lem:etaexists` [repaired] (erratum `err:e5`) — existence over $\mathbb{Z}$ for every ℓ by a terminating construction ((A3)): the vertex step is the augmentation argument ($\varepsilon(j\tau_\partial x - \tau j x) = 1 - 1 = 0$), not "a path between two vertices"; the edge fills exist by $Z_1 = B_1$ and are *unique* by $Z_2 = 0$; (B3) makes the (T1)-extension well defined without inverting a stabiliser order, and the proof says at each step which hypothesis is doing the work.
- `lem:etaunique` [new] — two pair homotopies differ by $\partial\Lambda - \Lambda\partial$ for a **unique** $\Lambda\colon C_0(\partial X) \to C_2(X)$, with (T1) *automatic* from uniqueness; the degree-1 identity is forced. Stated with (H4)(ii) as the hypothesis (audit F5) and with the observation, new here, that (B3) is *not* needed for this lemma.
- `lem:conesign` [repaired] (erratum `err:e13`; `eq:conesign`) — for an *arbitrary* degree-(-1) map h, $D\Phi - \Phi D = (0, (dh + hd + r\Theta - \Theta^\partial r)c)$, so Φ is a cochain map iff $dh + hd = \Theta^\partial r - r\Theta$; and $\Phi\mathrm{inc}_c = \mathrm{inc}_c\Theta^\partial$, $\mathrm{pr}_c\Phi = \Theta\mathrm{pr}_c$ for every h. The sign is asserted here and nowhere else.
- `def:coneaction` [new] (`eq:derivedh`, `eq:conemap`) — the derived $h := \widetilde{\eta}$, the cone map

Φ_g, $T^c_{\Gamma g\Gamma}$ and $A^{(q)}_c$ by `alg:assembly` on $C(r)$; the degree bookkeeping (h^1 from vertices, h^2 from edges, $h^0 = h^3 = 0$); and the rule that h is never chosen at cochain level.

- `lem:derivedhomotopy` [new] — $dh+hd = \Theta^{\partial}_g r - r\Theta_g$ for the derived h, with no hypothesis on ℓ, as a substitution into `lem:cosettransport`: the cochain identity is *derived* from the chain-level convention (`err:e13`), which is why it has only one home.
- `thm:conewelldef` [new] — (a) Φ_g is a cochain map and every map of `eq:coneles` intertwines, no hypothesis on ℓ or n; (b) independence of η over $\mathbb{Z}$ for every ℓ; (c) for $6 \in R^\times$, independence of $\tau, \tau_\partial, s_j, g$, identification with $\mathrm{cor} \circ (\mathrm{conj}_g)_* \circ \mathrm{res}$ on the relative complex; (c′) the Hecke relations on the nose ([proved here] via the Steinberg form), the geometric identification ([standard:], section numbers of Borel–Serre unverified) and the PD adjointness ([open]), each with its own tag; (d) $\eta \equiv 0$ iff $\tau j = j\tau_\partial$, and the entrywise identity does not license $h = 0$; (e) the $6 \notin R^\times$ reading, as for $T_{\Gamma g\Gamma}$.
- `prop:coneindep` [new] (`eq:conehomotopyeta`, `eq:conehomotopytau`) — the four independence proofs by explicit cone homotopies: $\Upsilon = (0, -\widetilde{\Lambda}c)$ for η (over $\mathbb{Z}$, integral, filtration-preserving, non-zero only on $C(r)^1$); $\Upsilon = (-\widetilde{H}c, \widetilde{N_\partial} c_\partial)$ for (τ, τ_∂), with $6 \in R^\times$ entering only through the existence of H and the τ_∂-half unconditional; the s_j summand by summand with the tables fixed; g by transport.
- `prop:strictpair` [new] — strict pairs exist over $\mathbb{Z}$ for every ℓ, by a construction that pre-assigns the boundary slots and needs, beyond `eq:contractibleuse`, only that a finite subgroup of Γ of order 4 or 6 fixes a vertex of X (Serre, cited). The verdict's Prop. 2.6.2 argued through `alg:tau`; this is self-contained and, as a by-product, an *existence* statement for the bulk τ over $\mathbb{Z}$ with no hypothesis on ℓ, which `lem:tauexists` does not make and which does not repair that lemma's uniqueness half.
- `prop:coneambiguity` [repaired] (errata `err:e2`, `err:e3`) — the $\mathrm{Hom}(\ker r^{q+1}, \mathrm{coker}\, r^q)$ ambiguity of a free cochain-level h, over any R; its *realisation in full*, stated **over a field** with $Z^0(C^\bullet_\partial)$ and never B^0, and with $\mathbb{Z}/\ell^e$, $\mathbb{Z}/m$ explicitly not claimed; and the consequence that for the derived h the ambiguity is a cone homotopy and vanishes.
- `prop:coneintegral` [new] — tables over $\mathbb{Z}$ once, matrices per R; the cone over $\mathbb{Z}$ is a complex of free $\mathbb{Z}$-modules (saturated sublattices) and $C(r)_{\mathbb{Z}} \otimes R = C(r)_R$ for $6 \in R^\times$ (averaging); reductions and Bockstein connecting maps are $\Phi_{g,\mathbb{Z}}$-equivariant *on the nose*; two data give the same maps on every page of the ℓ-adic spectral sequence, by an integral filtration-preserving homotopy; and the observation, new here, that the η fills are unique over every $\mathbb{Z}/m$, so a mod-m regime on the η-table can only ever be a tier failure (OP2 applied).
- `prop:coneroutes` [new] (`eq:route3`) — the strict-pair route (needs $6 \in R^\times$, via `prop:coneindep`(2)) and route 3, $\Theta'_g = \Theta_g + dE + Ed$ with $E = \mathrm{ext}_0 \circ h$, which satisfies $r\Theta'_g = \Theta^{\partial}_g r$ on the nose and is cone-homotopic to Φ_g via $(Ec, 0)$ — *for every* ℓ, a distinction the sources do not draw; and the PD gate, stated conditionally on `cl:pd` (a stub) and tagged [open]. Owner decision OP3 (both routes committed; PD the gate) recorded.
- `obl:eta` [new] — six clauses on the η-table and the derived h per (A4): structural; the homotopy identity through the *shared* (T1) helper; (T1) bookkeeping and the `lem:bounds` recount; the solve regime under OP2 (a mod-m fill aborts by default, is a recorded non-integral datum behind an explicit setting, and is never mathematically necessary); **clause (v), that the stored h is the transport of the stored η, which is what pins $A^{(q)}_c$**; and the diagnostics `eta_zero`, `cochain_commutes`, recorded and never branched on ((A6)), with `eta_zero` ⇒ `cochain_commutes` the only asserted relation between them.
- `obl:coneexact` [new] — eight clauses over R: the cone identity and $D\Phi_g = \Phi_g D$; assembly; intertwining; exactness *where asserted*, with the non-field case recorded until a non-field elimination exists and with `open:ell2` narrowing the asserted spots; bookkeeping with $H^0_c = 0$ unconditional and $H^2_{\mathrm{eq}} = 0$ asserted only for $6 \in R^\times$; the Hecke relations

re-checked as theorems for $6 \in R^\times$, recorded otherwise; the two cross-routes and the PD gate; and the two mandatory validation arms (independence; the tampered-h negative control), the second being the executable form of the statement that clauses (i)–(vi) certify the sequence and *not* the module — any Φ of `prop:coneambiguity` passes them.

Non-normative.

- `rmk:conesteinberg` [new] — the status of the two sentences of `thm:conewelldef`(c) that reach outside the document. (i) The operator on the relative complex is canonical ([proved here]: two strict pairs give homotopic τ_{rel}, by extending the Λ of `lem:etaunique` by zero to a homotopy of pairs). (ii) **The Steinberg form**: for $6 \in R^\times$, $H_c^q \cong H^{q-1}(\Gamma; \mathrm{Hom}_{\mathbb{Z}}(\mathrm{St}, V_R))$ with St the Steinberg module, the cone operator is the double-coset operator on that Δ-module, and the Hecke relations therefore hold on $H_c^\bullet$ *on the same external inputs as* `thm:relations` — a theorem, not a citation of geometry; [proved here] as a sketch whose derived-category bookkeeping is named as the part a formalisation must supply. This route appears in no source. (iii) The identification with the Borel–Serre pair and $H_c^\bullet$ of the open orbifold: [standard:] on Harder §§1–2 and Borel–Serre, with the caveat, per (A2), that the section numbers of the latter are not verified here and the half is to be read as [conjectural] *as a citation*; nothing in the document depends on it. (iv) PD is `cl:pd`'s business and [open] here.
- `rmk:conestrict` [new] — (1) [verified computationally] the shipped τ is not a map of pairs: on `tau_table_p2.txt`, 8 of 12 stored boundary-vertex slots and 2 of 9 boundary-edge slots leave ∂X, with the specific slots named; a property of `alg:tau`'s nearest-vertex step, not of the model. (2) The entrywise identity $r\Theta_g = \Theta_g^\partial r$ does not license $h = 0$; whether the residual class vanishes for some actual (τ, τ_∂) is [conjectural] either way; the diagnostics gate nothing.
- `rmk:coneambiguity691` [new] — the $(10, 691)$ witness: $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0) \cong \mathbb{F}_{691}^3 \neq 0$ and $\dim X_3 = 4$, tagged [open] *in this document* because two of its three inputs are stubs of `sec:six` and the third is [E]'s; $X_2 \cong \mathbb{F}_{691}$ is [proved here] here via `prop:bdrychar`(1).
- `rmk:conetorsion` [new] — what holds at $6 \notin R^\times$ on the model complexes (everything that used no hypothesis on ℓ), what is only *defined* there, and the two sites at which this chapter reads `open:ell2`.
- §`sec:pair`, §`sec:pairhomotopy`, §`sec:coneaction`, §`sec:coneintegral`, §`sec:coneroutes`, §`sec:coneobligations` — the chapter's subsections. Containers, no tag ((CL4)).
- §`sec:oc1sites` — item 3 gains the two sites in §15 that read `open:ell2` (`def:coneaction`, `rmk:conetorsion`). Container, no tag.
- The code map records that **nothing in this chapter is implemented** — a census of the tree finds no η-table, no equivariant cone, no Φ_g, no $A_c^{(q)}$ and no cone validation — and that the one existing cone, `CochainComplexMorphism::ConeComplex`, is the non-equivariant model of `monodromy_v5.tex` §8; and it fixes the shape of the phase that builds the rest (E5), including the golden tiers of owner decision OP7.
- A notation point found while writing and *not* resolvable here: the tilde denotes both the lift of a simplex (`\lift`, C16) and the coset transport of a map (`thm:welldef`(b), and now `def:cosettransport`). They never meet in one formula and are typographically distinct (a decorated symbol versus a decorated operator), but §2.7 should record the pair at its next revision. `gl2_notation.tex` was not edited.
- `app:changelog` — provenance counts updated from 77/33/25 to 96/33/28 ([new]/[re-homed]/[repaired]), the [new] split from 45/32 to 60/36, the tagged-line totals from 114/60 to 143/60, and a Part IV cone roster of 15 new statements added; the sentence "over all eleven versions" corrected to "over all versions to date".
- `app:labelindex` — the forty-one labels of §15 dereferenced, per (L6) (thirteen declared by the stub, §15 included, twenty-eight new).

- `app:changelog-gaps` — §15 removed from the stub roster.
- The abstract, the file header and the date line: fifteen of the twenty-two chapters are written.

v0.14.0 — 2026-09-16 — the boundary datum, and 14.24

This version. Writes §14 and §14.5, which were a stub declaring nine labels, and states the $\ell = 2$ truncation policy 14.24 (**OC-1**) as a *tagged working assumption*, the owner having reserved the decision. Commit: `01b4b75`. Sources: the implementation plan's amendment C-A2 (`hecke_extensions_implementation_plan.md`:312–382) and C-A3(ii.2) (:412–429); the OQ2 verdict §§2.6, 5.1–5.5; the specification's Stage 0 item (A1) (`hecke_extensions.tex`:761–788); owner decisions OP1, OP2, OP5. All four sources are prose or Markdown — no superseded document states the Γ-level τ_∂, its existence proof, its integrality or the $\ell = 2$ degree behaviour — so all but one statement below is written rather than transported. **Three errata land here**: `err:e12` (a definition and its own normative construction disagreeing inside one document), `err:e4` (τ_∂-uniqueness derived from a theorem whose hypothesis the construction exists to avoid) and the boundary half of `err:e14` (the characters in the right-coset convention). The [re-homed] roster for this version is the single entry `def:bdrygroup`, transported unchanged from `sec:six`'s definition of X_2 and X_5. Pages 77–90 (architecture budget: 5 pp). **The overrun is** 2.8× **and it is 14.24**: the boxed assumption, the four enumerated alternatives, the spelled-out exactness clause, the site list of §14.5 and the recommendation are about 2.5 pp that the architecture costed as one label. The mathematics — §§14.1–14.4 — came in at about 8.5 pp against the same 5 pp, i.e. 1.7×, which is the ordinary *RC+NEW* rate.

Normative.

- `B1`, `B2`, `B3` — the chapter's own standing hypotheses: the boundary subcomplex, its components (each a contractible 1-complex), and the centrality of its stabilisers. Stated for *the boundary triangulation in force* and not for the four horocycle vertices of `def:model7`, per owner direction OD-T1 as recorded in `rmk:orbifold`; `rmk:bdrymodel` proves that the shipped triangulation and every equivariant refinement of it satisfy them. Container labels, no tag ((CL4)).
- `def:bdrygroup` [re-homed] — $\Gamma_\infty = \langle T\rangle \times \langle -I\rangle$ `eq:bdrysplit` and the *definition* $H^\bullet(\partial; V_R) := H^\bullet(\Gamma_\infty; V_R)$ (owner decision OP5, the Borel option), transported unchanged from `sec:six`. The label previously named `[to be written]`. It is the half of `err:e12` that was already right; the other half is `prop:circlemodel`.
- `prop:bdrycohom` [new] — the cohomology of Γ_∞ in *every* degree, as the two-column Lyndon–Hochschild–Serre sequence `eq:bdrylhs`. New: no source computes $H^q(\Gamma_\infty; V_R)$ for $q \geq 2$ at all.
- `prop:circlemodel` [repaired] (erratum `err:e12`) — what the circle model computes ($H^\bullet(\langle T\rangle; V_R)$, unconditionally, by Shapiro), and that it is $H^\bullet(\partial; V_R)$ **exactly when** $2 \in R^\times$, with the failure at $R = \mathbb{Z}/2^e$ exhibited in degree 2. [E] *defines* the boundary cohomology as $H^\bullet(\Gamma_\infty; M)$ at `:189` and then makes the circle model its normative construction at `:755--768`; the hypothesis separating the two was in neither statement and is now in this one.
- `prop:bdrycosets` [new] — B_N, its m left Γ_∞-cosets **for every** ν, the bijection $B_N/\Gamma_\infty \to \Gamma g \Gamma/\Gamma$, the decomposition of B_p into *two* Γ_∞-double cosets, and the right cosets β_b as the wrong system. [E] (A1) states the $\nu = 1$ table and the bijection without proof; the general-ν count, the two-double-coset structure and the proofs are new.
- `def:taubdry` [new] — the boundary g-twisted chain datum, with (T3∂) written as "the induced map on $H_0(\partial X)$ is π_* for a map of sets", which is the correct analogue of (T3) on a *disconnected* complex; the sources say "(T3) on every component" without saying what that means.

- `lem:bdrycomponent` [new] — (T1∂) and (T3∂) force the component map, and it is $c \mapsto g^{-1}c$ on $\mathbb{P}^1(\mathbb{Q})$. The plan asserts only the two cases $\infty \mapsto \infty$ and $0 \mapsto 0$, parenthetically and without proof; the closed form and the proof are new.
- `lem:taubdryexists` [repaired] (erratum `err:e4`) — existence over $\mathbb{Z}$ *for every* ℓ by a terminating construction ((A3)), coefficients in $\{0, \pm 1\}$, and homotopy uniqueness over $\mathbb{Z}$ for every ℓ with $N_\partial = 0$ in degree 1. The verdict states the conclusion at §2.6.1 but derives the uniqueness half from "the comparison theorem", whose relative-projectivity hypothesis is precisely what forces $\ell \geq 5$ in `lem:tauexists` — circular, and `err:e4`. The proof here is explicit and never cites that theorem; it also separates the two hypotheses it uses ((B3) for the (T1∂)-extension, (B2) for the fills) and says at each step which one is doing the work.
- `def:bdryop` [new] — Θ_g^∂ by `eq:bdrychain`, i.e. `eq:heckechain` with τ_∂ and **the same** s_j**,** δ_j **as the bulk**; with the owner-decision box recording OP1, that the Γ-level datum is normative and the left-coset $(\delta_\infty^\partial, \delta_t^\partial)$ operator is its Shapiro transcription and a validation route only.
- `prop:bdryop` [new] — Θ_g^∂ is well defined with no hypothesis on ℓ, and its independence of τ_∂ holds *over* $\mathbb{Z}$ *for every* ℓ, which the bulk statement `thm:welldef`(b) does not; and in degree 0 it is the classical double-coset action $\sum_{B_N/\Gamma_\infty} \rho_n(\delta)$. The degree-0 identification is new and is what licenses `prop:bdrychar`(1) for the normative operator.
- `prop:bdrychar` [repaired] (erratum `err:e14`, boundary half) — $V_R^T = R[x, \mathrm{Nm}]_n$ with a proof (no citation: the invariant-ring argument is four lines and a formalisation target), the characters χ_b on H^0, the χ_{b+1} of the right-coset system, and the $n < \ell$ case of H^1 with the group-cohomology form `eq:bdrygroupform`. [E]'s `cl:chi` is *not* superseded: its general-n H^1 multiset is left to §16 and named as such. What is repaired is `hecke_extensions_mathematics.md`:67, which presents the characters in the β_b convention as though it were *the* answer with only a "convention warning" as a hedge: here the normative convention is fixed, the det-twist is part of the statement, and `prop:bdrycosets`(4) says why the two conventions split B_p in opposite ways.
- `obl:taubdry` [new] — seven clauses, each naming its data, predicate, ring and failure action per (A4). Clause (iv) is where OP2 lands: on the boundary the mod-ℓ fallback *cannot arise*, so a τ_∂-table with `integral = false` is an implementation defect and **aborts**, where the bulk clause of `rmk:tauintegral` records a weaker datum instead.
- `lem:ell2unbounded` [new] — at $\ell = 2$, $H^q(\partial; V_R) \neq 0$ for *every* $q \geq 0$, with the exact sequence and, over a field, the dimensions $\dim V_R^T$ and $2 \dim V_R^T$. The plan states the non-vanishing; the sequence, the dimensions and the proof are new.
- `open:ell2` [new] — **OC-1**, stated as a tagged working assumption and *not* as a decision: $q_{\max}$ an explicit recorded run parameter defaulting to 2, complexes built to $q_{\max}+1$, cohomology reported only to $q_{\max}$, exactness asserted only where all three terms are reported. The four alternatives `oc1a`–`oc1d` are enumerated and kept reachable, the recommendation is `oc1a` with its reason (it is the only one that keeps `oc1c`, a stabilisation theorem, free), and §`sec:oc1sites` lists every site that reads the tag so the owner can flip it mechanically. (CL4)'s list of statement kinds does not name `open`; this is a numbered statement and is tagged as one, and that list should gain the kind at its next revision.

Non-normative.

- `rmk:bdrymodel` [new] — the shipped triangulation satisfies (B1)–(B3), and so does every Γ-equivariant refinement of it; what is *not* inherited by a refinement is named (orbit counts, table size, `lem:bounds`). This is the chapter's discharge of OD-T1.
- `rmk:bdryoddn` [new] — for n odd and $2 \in R^\times$ the boundary complex is zero, for the same reason as `rmk:negI` in the bulk, and both fail at $\ell = 2$.
- `rmk:taubdryintegral` [new] — the integrality claim, its two independent reasons, and

what it is not: it does not repair `lem:tauexists`, and it does not use the comparison theorem. Written in the four-field shape because this is the area in which an over-claim would be hardest to see (cf. `err:e3`).

- `rmk:ell2window` [new] — the three complexes that coincide for $6 \in R^\times$ and come apart at $\ell = 2$ (model / group-cohomological / free Γ-model), which of them the "unbounded" statement is about, and that `open:ell2` governs only the third. It also states explicitly that whether $H^q(\Gamma; V_R)$ is non-zero in every degree is **not** asserted, and that nothing here needs it — the Farrell–Tate route is named with a [conjectural] *as a citation*, per (A2).
- §`sec:bdrygroup`, §`sec:bdrycosets`, §`sec:taubdry`, §`sec:bdryop`, §`sec:ell2`, §`sec:oc1sites` — the chapter's subsections. Containers, no tag ((CL4)).
- A notation collision found while writing and *not* resolvable here: `cl:boundary` and `cl:chi` write the degree-ℓ boundary invariant $y^\ell - x^{\ell-1}y$ as N, while $N = p^\nu$ is this document's Hecke level. It is written Nm in `prop:bdrychar`, with the clash stated at the point of use; §2.7 lists seventeen collisions and should gain this as an eighteenth at its next revision. `gl2_notation.tex` was not edited.
- `prop:bdrychar`'s corroboration paragraph — an *ad hoc* polynomial check of all three parts over $\ell \in \{2,3,5,7,11,13\}$, $n \le 24$, $p \in \{2,3,5,7,11,13\}$, $p \neq \ell$, 0 disagreements, carrying **no** [verified computationally] tag because nothing in the tree re-runs it. `obl:taubdry`(vi) is where it becomes a registered check.
- `app:changelog` — provenance counts updated from 64/32/22 to 77/33/25 ([new]/[re-homed]/[repaired]), the [new] split from 36/28 to 45/32, the tagged-line totals from 94/58 to 114/60, and a Part IV roster of 9 new statements added.
- `app:labelindex` — the thirty-five labels of §14 and §14.5 dereferenced, per (L6).
- `app:changelog-gaps` — §14 and §14.5 removed from the stub roster.
- The abstract, the file header and the date line: fourteen of the twenty-two chapters are written.

v0.13.0 — 2026-09-16 — $\iota_X\tau\iota_X \neq \tau$, computed

This version. Records a computation that §9 asked for and that had never been performed. No statement is generalised, restricted or withdrawn: an obligation is *discharged*, and a remark that recorded a gap now records the gap's cause. Commit: `[hash]`; the evidence is the run of `HeckeIotaTauSymmetryValidation`, added to the code base as `e0b64c4` and re-run on 2026-09-16 for this entry. Pages 43–50 (was 43–48).

Normative.

- `obl:iota` [new] ((CL3): the statement was written in v0.12.0 and changes here) — clause (iv), the τ-symmetry diagnostic, gains its [verified computationally] discharge: the recorded value is *false* on the canonical τ at every $\ell \in \{2,3\}$ of the shipped grid. The predicate, the ring and the failure action are unchanged, and the clause explicitly stays in force for any other τ, vertex policy or refined triangulation. Clause (i) is recorded as passing. The verdict remains unbranchable ((A6)).

Non-normative.

- `rmk:iotatausym` [new] ((CL3)) — rewritten. v0.12.0 said the hypothesis of `prop:iotacommutes`(3b) was *not known*, and attributed the doubt to the stored edge-path templates of `alg:tau` step 2. Both halves are superseded by measurement: the hypothesis is **false**, with 3600 of 10764 slots differing, every comparison exact over $\mathbb{Z}$, and a minimal sign-free witness printed; and the mechanism is one level earlier — step 1's nearest-vertex tie-break, which is by ascending vertex index and is therefore structurally incompatible with ι-symmetry, since ι_K exchanges v_1 and v_2. The remark also records

that the negative is *certified*: τ^ι satisfies (T1)–(T3) of `def:tau` on every stored slot, so this is two admissible data for one g, exactly as `prop:iotacommutes`(3a) predicts. Consequence for the document: at $\ell \in \{2,3\}$ there is **still** no route to $[\iota^*, T_{\Gamma g \Gamma}] = 0$, and the gap is now open for a known reason. An ι-symmetric tie-break is recorded as an available repair that changes no normative statement but re-baselines every cochain-level reference value, and is *not* taken — an owner decision.

- `sec:involution` — the chapter's "What this chapter fixes" paragraph and its code map: the chapter is no longer wholly unimplemented. The code map now names the validation, says which clauses of `obl:iota` it discharges and which three it does not, and records that it re-verifies `lem:iotamodel`'s unstated hypothesis $\operatorname{Re} v_1 = -\operatorname{Re} v_2$ and certifies `prop:iotacommutes`(3a) on the shipped tables. The validation count in that paragraph moves from 66 to 67. Container, no tag ((CL4)).
- The date line.

v0.12.0 — 2026-09-15 — the involution ι

This version. Writes §9, which was a stub declaring three labels. Commit: `7beceb2`. Sources: the implementation plan's amendment C-A1 (`hecke_extensions_implementation_plan.md`:287–305), the specification's Stage 0 item (A3) (`hecke_extensions.tex`:882–889) and owner decision OQ3 — all three prose, none of them LaTeX, so every statement below is written rather than transported and carries [new]. There is no [re-homed] roster for this version and no [repaired] line: no superseded document states anything about ι, and no erratum lands here. Pages 43–48 (architecture budget: 3 pp).

Normative.

- `I1` — the chapter's own standing hypothesis, that the tessellation X and the fundamental domain K are stable under $z \mapsto -\bar{z}$. Stated for an arbitrary Γ-tessellation, not for the seven vertices, per owner direction OD-T1 as recorded in `rmk:orbifold`. Container label, no tag ((CL4)).
- `def:iota` [new] — $\iota = \operatorname{diag}(-1, 1)$, the conjugation table `eq:iotaconj`, the symmetric power `eq:rhoniota`, the realisations ι_K and ι_X on chains, and the cochain involution `eq:iotacochain`. The label previously named `[to be written]`. The definition is given on all of $\mathrm{M}_2(\mathbb{Z})$ rather than on Δ, which is available only because `def:rhon` was stated there: $\det \iota = -1$, so $\iota \notin \Delta$.
- `lem:iotamodel` [new] — the seven-vertex model satisfies (I1), with the vertex permutation and the orientation sign of every simplex. **It carries a hypothesis that no source states**: $\operatorname{Re} v_1 = -\operatorname{Re} v_2$. `def:model7` fixes $\operatorname{Re} v_0 = -\frac{1}{2}$ and $\operatorname{Re} v_3 = +\frac{1}{2}$ and leaves the two middle horocycle vertices free, so "the model is symmetric under $z \mapsto -\bar{z}$" is true of the model only under that placement. Nothing else in the document uses it, and it is compatible with `lem:model7coincidences`, whose proof needs only $|\mathrm{Re}| < \frac{1}{2}$ there.
- `prop:iotacommutes` [new] — (1) $(\iota_K, \rho_n(\iota))$ carries every relation of $\mathcal{R}$ to the inverse of a relation, hence is an automorphism of $\langle \mathcal{R} \rangle$ and not of the list; (2) ι^* is an involution of $C^\bullet_{\mathrm{eq}}$ commuting with d, and corresponds to $\iota^\#$ under Ψ of `thm:explicit_complex_equiv`; (3) the commutation with $T_{\Gamma g \Gamma}$, **split into three statements with three different hypotheses** — (3a) unconditional, at cochain level, against the ι-transported datum; (3b) unconditional and on the nose *if* $\iota_X \tau \iota_X = \tau$; (3c) on H^q_{eq} under $6 \in R^\times$, via `thm:welldef`(a),(b). The stub's title claimed only the unqualified "ι commutes with $T_{\Gamma g \Gamma}$", which is what (A1) forbids; the label keeps its subject and gains its hypotheses.
- `prop:iotasplit` [new] — the $\pm$ decomposition of H^q_{eq} holds when $2 \in R^\times$ and is *false in general* otherwise; at $\ell = 2$ what holds is the Hecke-stable filtration $0 \subseteq \operatorname{im} \mathfrak{n} \subseteq \ker \mathfrak{n} \subseteq H^q_{\mathrm{eq}}$ with $\mathfrak{n} = \iota^* - \mathrm{id}$ square-zero, and H^q_{eq} is a sum of trivial and free $\mathbb{F}_2[x]/(x^2)$-modules. The sources assert the splitting with no hypothesis at

all (`hecke_extensions_mathematics.md`:14, "it splits every cohomology group into ± eigenspaces"; `hecke_extensions.tex`:888, "$H_{\mathrm{tot}} = H^{+} \oplus H^{-}$ halves every matrix"), which is exactly the class of unstated invertibility hypothesis (A1) exists for.
- `obl:iota` [new] — five clauses, each naming its data, predicate, ring and failure action per (A4): (i) simplicial automorphism, (ii) local-system automorphism, (iii) cochain involution, (iv) the τ-symmetry *diagnostic* — recorded, never branched on ((A6)) — and (v) commutation plus the dim $H^1_{\mathrm{eq},\pm}$ oracle, which records rather than aborts because its mod-ℓ form is a prediction. The stub's title promised a "sign convention"; see `rmk:iotasign` for why there is none to fix.

Non-normative.
- `rmk:iotasigns` [new] — the six minus signs are forced by the stored orientation and are not a convention.
- `rmk:iotatausym` [new] — at $\ell \in \{2, 3\}$, `prop:iotacommutes`(3b) is the only available route to the commutation, and whether the canonical τ of `alg:tau` satisfies its hypothesis is **not known**. Recorded as a gap.
- `rmk:iotaelltwo` [new] — the three degeneracies at $\ell = 2$ ($\rho_n(\iota) = I$, no splitting, and the failure of `rmk:negI`'s odd-n vanishing) are distinct and coincide only in that regime.
- `rmk:iotapairing` [new] — $\rho_n^{\vee}(\iota) = (-1)^n \rho_n(\iota)$ and $\langle \rho_n(\iota)v, \rho_n^{\vee}(\iota)\varphi\rangle = (-1)^n\langle v, \varphi\rangle$ from `prop:pairing`(4), with an explicit statement that the cohomological half is `cl:pd`'s and is not available here, and that the actions of ι on $H^{\bullet}(\partial)$ and $H^{\bullet}_c$ belong to `sec:boundarydatum` and `sec:cone`. The duality-restricts-to-each-sign consequence is marked a prediction.
- `rmk:iotasign` [new] — `iota_sign`: the definition leaves no sign free (the two lifts ι, $-\iota$ of $z \mapsto -\bar{z}$ differ by $(-1)^n$ and agree wherever $H^{\bullet}_{\mathrm{eq}} \neq 0$ with $6 \in R^{\times}$), so owner decision OQ3 is an *assertion to be checked*, not a normalisation, and the record field says which convention a record was produced under.
- `app:changelog` — the provenance counts of this appendix updated from 54/31/23 to 64/36/28 [new], the Part II roster of new statements extended by five, the retrospective-numbering box scoped to the first eleven versions, and the (CL5) tally to ten writing slots.
- `app:labelindex` — the fourteen labels of §9 dereferenced, per (L6).
- `app:changelog-gaps` — §9 removed from the stub roster.
- The abstract and the date line: thirteen of the twenty-two chapters are written.

v0.11.0 — 2026-09-15 — the changelog

This version. Adds the record and the rule that makes the record's own version number well-formed. No mathematical statement changes.

Normative.
- `V5` — the pre-1.0 numbering rule: while any chapter is still a stub the version is $0.X.Y$, the leading 0 marking incompleteness and $X.Y$ carrying (V2)'s bump rules unchanged; the leading field becomes 1 when the last stub is written. Without it the date line below would violate (V1).

Non-normative.
- `app:changelog` — this appendix; versions 0.1.0–0.10.0 back-filled.
- `V3` — amended to name where the record lives.
- `sec:machinery` — the stub now points at this appendix for the record and at §1.4 for the convention, so the chapter that will describe the machinery does not have to re-derive

where the changelog went.
- `app:labelindex` — `app:changelog` dereferenced, per (L6).
- The abstract and the date line stop calling the document a skeleton: twelve of the twenty-two chapters and two of the four appendices are written.

v0.10.0 — 2026-09-15 — (FD4) proved; (FD1) and (FD3) repaired for a non-effective action

Commit `3e5ffbc`. 78 → 82 pages. Two connected repairs to Part I, arising from an audit of how `prop:model7h4` discharges (H4)(iii). No label altered; no source file touched.

Normative.

- `lem:fd4criterion`, `FD4a`, `FD4b` [new] — $\mathcal{R}$ generates the coincidence groupoid *iff* (a) at every simplex the $\mathcal{R}$-loops reach all of $\mathrm{Stab}_G(\widetilde{\sigma})$ and (b) every pair of distinct simplices with non-empty transporter is joined by $\mathcal{R}$. Completeness is split into local isotropy and global orbit content, each finitely decidable, with no enumeration of K^u or of G.
- `rmk:fd4criterion` [new] — why the criterion has that shape.
- `lem:model7coincidences`, `serreA`, `serreB` [new] — exactly four pairs of distinct simplices of the model are Γ-equivalent, derived from Serre's fundamental-domain theorems alone: a moved simplex's relative interior lies in $\mathrm{Re}\, z = \pm\frac{1}{2}$ or $|z| = 1$, hence has dimension ≤ 1.
- `prop:model7h4` [repaired] — (FD4) is now discharged by `lem:fd4criterion` rather than "by inspection" of the relation table: the three elliptic rows give (a) at v_4, v_5, v_6, the 23 central rows give (a) at the other 20 simplices, and four seed rows give (b); the eighth seed row is shown redundant for (FD4). Also corrected: the 31 rows are 8 seeds + 23 central, not $8 + 3 + 23$ — the three elliptic stabiliser rows are inside the eight.
- `def:kernelN` [new] — the ineffective kernel $N := \ker(G \to \mathrm{Aut}(K^u))$, normal, contained in every stabiliser, with every transporter a union of N-cosets; $N = 1$ in the free instance and $N = \{\pm I\}$ for the model. A non-effective action becomes an explicit supported case of Part I instead of an unstated exception.
- `ass:fund_domain`, (FD1) [repaired] — false as written. $-I$ acts trivially on the tessellation while being a non-identity group element, so every simplex admits the coincidence $(\sigma, \sigma, -I)$; the old K_{cl} was all of K, $\widetilde{D}$ was empty, and (FD1) failed outright. K_{cl} is now generated by the simplices a coincidence *moves*, and (FD1) additionally asserts that $\widetilde{D}$ is dense in $|K|$ — which `thm:fund_group_generators`'s proof already used and (FD3) only pretended to supply.
- `ass:fund_domain`, (FD3) [repaired] — disjointness is now asserted over G/N: $\gamma\widetilde{D} \cap \widetilde{D} = \emptyset$ for every $\gamma \notin N$, and nothing else. As written it failed at $\gamma = -I$ once $\widetilde{D}$ was repaired.
- `rmk:fdconsequences`, `eq:fddecomp` [new] — the former conclusion "hence $|K^u| = \bigsqcup_\gamma \gamma\widetilde{D}$" was a non-sequitur, wrong even for an effective action, since the union omits the translates of $|K_{\mathrm{cl}}|$; it holds only for a half-open domain, which this document neither needs nor builds. Removed and replaced by a proved decomposition, disjoint in its first half and not in its second.
- `rmk:fd13centre` [new] — why N is quantified over and never divided out, stated once and in Part I: the coefficient action does not factor through G/N — $\rho_n(-I) = (-1)^n\,\mathrm{id}$ is the odd-n vanishing mechanism — so quotienting would delete exactly the constraint (FD4) imposes. Owner decision, 2026-09-15, taken against the alternative of restating Part I over the effective quotient.
- `rmk:twosettings` [repaired] — both bullets updated: the free instance has $N = 1$, the model is non-effective with $N = \{\pm I\}$.
- `rmk:fd4computed` [new], [verified computationally] — (FD4) for the model re-verified as a finite computation: `HeckeTessellationValidation` computes $\langle\mathcal{R}\rangle$ and asserts exactly

conditions (a) and (b); re-run, PASS. Recorded with the caveat that the computation supplies $\langle\mathcal{R}\rangle$ but not the true stabilisers and orbits it is compared against.

v0.9.0 — 2026-09-15 — Part I re-homed from [v5]; erratum E7 closed

Commit `d5a2b2d`*, writing slots 5–6.* Chapters 3–6, 14 pp; 65 → 78 pages. All 26 assigned labels carried verbatim, each defined exactly once, 0 removed, 32 new under (L5).

Normative.

- `thm:explicit_complex_equiv` [re-homed] — `err:e7` closed: the label is now defined in this document under exactly the string six source files cite, and its part (1) is what those sites assert. The erratum entry records five files; there are six sites in six files (it lists all six paths but counts a validation `.h/.cpp` pair as one).
- `thm:explicit_complex_equiv` [repaired] — freeness of the action is a hypothesis of part (2) *only*, not of part (1). [v5] never marked the use, and the marking is what licenses this pipeline — whose action is not free — to cite part (1).
- `def:relationdatum` [new] — relations, transporters and the coincidence groupoid. [v5]'s relation datum is indexed by ordered pairs (u, v) with $u < v$ and therefore structurally cannot express a self-relation (σ, σ, γ) with $\gamma \neq 1$, while the model's $\mathcal{R}$ consists of seed rows *and* 23 central $-I$ rows. Transcribed as written, the re-homed theorem would not have covered the model it is cited for.
- `ass:fund_domain`, (FD1)–(FD3) [re-homed] — [v5]'s fundamental-domain hypothesis, written out as three separately citable and separately repairable conditions. All three are repaired in v0.10.0.
- `ass:fund_domain`, (FD4) [new] — the completeness condition that makes Ψ surjective. (H4)(iii)'s forward reference "presents the quotient in the sense of the framework" had, until this version, resolved to nothing.
- `rmk:twosettings` [new] — the two instances of the framework this document uses: the free one, and the model. Repaired in v0.10.0, once the model's action was recognised as non-effective.
- `def:orderedrelation`, `def:quotientcomplex`, `def:saturated`, `def:bdryint` [new] — the framework's definitional layer, stated rather than assumed.
- `lem:eqgroupoid` [new] — one groupoid argument replacing [v5]'s well-definedness case analysis, which offered "interior $\Rightarrow h = e$" or "boundary $\Rightarrow h = g_r^{-1}$" and had no case for $\gamma_1^{-1}\gamma_2 \in \mathrm{Stab}(\widetilde{\sigma})$ — the stabiliser case the model needs. It also makes visible that dropping the central rows breaks surjectivity of Ψ, which is the mathematical content of the `FIX-negI` change and until now lived only in an implementation-plan bullet.
- `prop:Bqimage` [new] — B^q_{eq} computed as the image basis of $d\,C^{q-1}_{\mathrm{eq}}$ equals [v5]'s $d\,W^{q-1}$, proved column for column. The shipped code takes the former and [v5] §7.2 computes the latter; the equality was justified only in a source comment.
- `prop:saturated_complexes` [repaired] — [v5] proved it "via explicit construction" and gave no construction; supplied.
- `prop:injectivity` [repaired] — its decomposition was asserted; now derived.
- `thm:fund_group_generators` [repaired] — [v5]'s 70-line path induction replaced by a 15-line simplicial argument, which also yields the non-free case.
- `prop:fund_group_action` [re-homed] — relation elements. Its part (1), that a non-empty transporter is a left coset of the stabiliser, is what `lem:fd4criterion` runs on in v0.10.0.
- `def:storedbases`, `prop:ACq`, `prop:AZq`, `prop:projector`, `eq:projcert`, `alg:cohomology` [new] — chapter 6. [v5] §7.2 is an informal recipe; the stored objects, the two cochain systems, the projector and its certificate are stated and proved here for the first time.
- `eq:psiinverse`, `eq:equivariance`, `eq:naivecoboundary`, `eq:quotientcoboundary`, `def:naivecochains`, `def:eqcohomology`, `cor:threecategories` [new] — the displayed

forms the later chapters transport formulas through.
- Nine hypotheses [v5] left implicit are now stated at the point of use ((A1)): action without inversions, so that $\mathrm{Tr}(\sigma,\sigma) = \mathrm{Stab}$; completeness (FD4); injectivity of $\sigma \mapsto \widetilde{\sigma}$; contractibility of $|K^u|$; and ring versus field — chapters 4–5 hold over any commutative ring, chapter 6 over a field. [v5] wrote F once and never marked its use, and $R = \mathbb{Z}/\ell^e$ is in scope.
- *Re-homed unchanged* (content as in [v5], notation crosswalk applied): `sec:framework`, `ass:K_contractible`, `ass:edge_non_ident`, `ass:open_interior`, `lemma:pR_simplicial`, `def:universal_cover`, `prop:universalcover`, `rmk:injectivityofi`, `ex:S1`, `sec:localsystems`, `def:matrix_ls`, `prop:matrix_rep_equiv`, `thm:lcs_rep_equiv`, `thm:rep_ls_equiv`, `sec:explicitcochaincomplex`, `rmk:universalcovercohomology`, `thm:cohom_equiv`, `def:eq_cochains`, `ass:consistent_face_id`, `lemma:d_maps_C_eq`, `sec:systematiccomputation`.

Non-normative.
- `err:e11` applied — the $c/z/b$ basis-name scheme, $w^{(q)}$ for W^q, and [v5]'s $\beta_j^{(q)}$ renamed $z_{1,j}^{(q)}$, since C8 reserves β for the right-coset representative.
- `rmk:derivedmatrices`, `rmk:lcsnotfree`, `rmk:psiuse`, `rmk:psiinjectivelift`, `rmk:facesautomatic`, `rmk:cost` [new] — what the equivalence does not say, what the theorem is used for, and the cost of the algorithm.
- `err:e8` and `err:e10` referenced, not restated; `err:e19` and `err:e20` deliberately not duplicated, chapters 3–6 touching neither μ nor the bar formula.
- Recorded, and repaired in v0.10.0: `prop:model7h4`'s "(H4)(iii) follows by inspection" now discharges a stronger condition than before — which it does, but by inspection of a 31-row table rather than by an argument.

v0.8.0 — 2026-09-15 — C17, and two stale uniqueness claims

Commit `24c235e`, slot 4 follow-ups. 65 pages, unchanged. Three corrections, all of them claims that were true when written in v0.7.0 and stopped being true later in v0.7.0 when `prop:mu`(3) was added ((CL3)).

Normative.
- `sec:notation-collisions`, row C17, with its `sec:notation-crosswalk` row — the relation element of $\mathcal{R}$ versus the Hecke element $g = \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}$; the two meet in one formula, `thm:xi`(1). Relation elements are $\gamma \in \Gamma$; g is the Hecke element only. Checked against what the sources write (subscripted g_r, not bare g).
- `rmk:indeplevel` [repaired] — the enumeration of licensed cochain-level bit-identities omitted $\omega_\bullet \circ \mu_\bullet \circ \alpha_\bullet$ against Θ_g. Added with its hypothesis: both sides consume the same g, the same s_j and the same τ-table, per `obl:agreement`. The bar-cocycle route is correctly excluded and was left unchanged.
- `prop:tworoutes` [repaired] — its contract box called the proposition "the one place in this document where bit-identity of cochain-level matrices is asserted". It is one of two; the other, `prop:mu`(3), is now named.

Non-normative.
- `def:model7rel`, `def:contragredient`, `sec:induced` — C17 conformance, $g \to \gamma$. The collision was wider than slot 4 reported: `def:model7rel` still wrote g in four places (table column, prose, contract box, and the contragredient local-system definition), and chapter 13 carried a paragraph describing the discrepancy as open.
- `sec:degreewise`, Formalization note — it called `prop:tworoutes` "the only claim in the document that a bit-level agreement is mathematically forced rather than empirically

observed". Corrected to two, with the other named and its proof route given. The forced-versus-observed distinction is intact; only the count was wrong, and a false uniqueness claim in a formalization note misleads precisely the reader it is addressed to.

v0.7.0 — 2026-09-15 — induced systems, Ξ_g, Θ_g, and the equivalent formulations

Commit `592c6e8`, writing slot 4. Chapters 12 and 13, 13 pp; 53 → 65 pages. All 17 assigned labels carried verbatim, 0 unplaceable, 18 new.

Normative.

- `prop:mu`, `sec:threemap` [new] — the middle cochain map, in closed form, derived and proved. The source's "Recovering the three-map factorization" remark is *circular*: $\beta^\bullet$ is "determined by $\beta^\bullet\alpha^\bullet = \Xi_g$ on $\operatorname{im}\alpha^\bullet$ and by the analogous formula in general", which is not a definition. A map the contract claimed to specify was in fact specified only by the code, and `HeckeCrossChecks` implements what is now proved.
- `eq:evalone` — the same remark's composite is *ill-typed*: $\rho_n(g)$ maps $V \to V$ while $\phi^{L'}$ lands in $\mathrm{Ind}_{\Gamma'}V$. Fixed by ordering the operations — evaluate at 1, then apply $\rho_n(g)$.
- `prop:lhcompliant` [repaired] — part (3) added: $M_H(-I) = (-1)^n I$, the odd-n vanishing mechanism. `def:inducedls` and `prop:lhcompliant` predate the central relations and gave no $M_H(-I)$ at all.
- `rmk:nogo`, `sec:nogo` [repaired] — `err:e19` applied: the map β of [v5] §12 is a no-go, not a repair; the middle map is μ.
- `rmk:barformula`, `prop:barformula`, `eq:barcorrected` [repaired] — `err:e20` applied: the corrected degree-1 group-cochain formula carries the coset permutation $j \mapsto \gamma * j$ explicitly. Both keep their labels ((L2)).
- `rmk:kappaexact` [new] — the corrected formula would otherwise invert a matrix of determinant N, which `obl:integral` forbids; the implementation uses the exact division $N\kappa = \mathrm{adj}(g)\, s_k \gamma\, s_j^{-1} g$.
- `prop:tworoutes`, `sec:chainpicture` [new] — the two bracketings agree bit for bit, proved term by term rather than observed.
- `eq:psiinv` — well-definedness of the inverse transport, from the strict fundamental domain *plus* the stabiliser and central relations.
- `thm:xi` [repaired] — $\widetilde{\partial\sigma} = \partial\widetilde{\sigma}$, from (H4)(iii), is stated as the hypothesis part (2) uses; and the proof's own collision (an oriented simplex called τ, while τ is the chain datum everywhere else) is resolved by C4.
- `alg:assembly` [repaired] — $P_q H_{\mathrm{eq}} = I$ and $P_q B_{\mathrm{eq}} = 0$ named as the defining property the algorithm consumes, rather than left to the reader.
- `rmk:omegaconvention` [new] — `err:e9`: the corestriction convention stated once, with the $\gamma \to \gamma^{-1}$ substitution shown.
- `rmk:thetachoices`, `rmk:shapirouse`, `sec:xitheta`, `sec:alphaomega`, `sec:assembly`, `sec:inducedmodule`, `sec:barformula`, `sec:degreewise` [new] — which choices Θ_g depends on, what the Shapiro statement is and is not for, and the degree-wise specialisations.
- $6 \in R^\times$ is isolated to exactly two statements of the two chapters; every other statement is over an arbitrary commutative R, and every chapter-12 statement is parametric in $(K, \mathcal{R})$, per owner direction OD-T1.
- *Re-homed unchanged*: `sec:induced`, `def:inducedmodule`, `prop:blockmatrices`, `eq:MH`, `def:inducedls`, `prop:alpha`, `prop:omega`, `def:xi`, `eq:xidef`, `eq:xiexplicit`, `sec:equivalence`.

Non-normative.

- Three route comparisons recorded at the level each actually holds, none overclaimed: fused versus route 2 bit-identical (`prop:tworoutes`); $\omega_\bullet\mu_\bullet\alpha_\bullet$ versus Θ_g bit-identical on the same data (`prop:mu`(3)); and the bar-cocycle route versus $A^{(1)}(T_p)$ *spectral only* — characteristic and minimal polynomial, never entrywise — because that route produces its matrix in an unrelated basis of H^1.

v0.6.0 — 2026-09-15 — C16 and the lift macro

Commit `3560525`, slot 3 follow-ups. 52 → 53 pages. Two notation gaps closed before slot 4, which carries the heaviest notation load, and one owner direction recorded.

Normative.

- `sec:notation-framework`, row C16 — the contract had *no* symbol for the inclusion $K \to K^u$: both sources write $i(\sigma)$, which C15 retired. The macro `\lift` is registered, with a collision row and two crosswalk rows ($i(\sigma) \to \widetilde{\sigma}$; $s_i, \delta_i, n_k^{(i,\sigma)} \to s_j, \delta_j, n_t^{(j,\sigma)}$), so the mechanical crosswalk later slots run cannot reintroduce either collision.
- `sec:notation-hecke` — the notation chapter's own Hecke table was stale against the rule it declares. It now writes $\Gamma = \bigsqcup_j \Gamma'' s_j$ and $\delta_j = s_j^{-1} g$, matching chapter 11, with C7, C10, C14 and the k-bullet re-indexed. Historical quotations of the superseded documents keep their i.

Non-normative.

- `sec:operatordef` — the distinguished lift is now written $\widetilde{\sigma}$ via the macro (C15, C16).
- `rmk:tauintegral` — $\tau(\partial\tilde{t}) \to \tau(\partial\widetilde{\sigma})$: t was naming a triangle while (N4) reserves it as a running index.
- `err:e11` — $\delta_i \to \delta_j$. Checked to be the contract's own voice and not a quotation of the defective text: an erratum's quoted statement must keep the old spelling.
- `rmk:orbifold` — owner direction OD-T1 (2026-09-15) recorded: a correct orbifold treatment at $\ell \in \{2, 3\}$ and over $\mathbb{Z}$ is expected to require a *refinement* of the 7-vertex triangulation. It is marked explicitly as a design direction and not a theorem; nothing stated in this document depends on it, and its force is on the authors of §17 and §14, who must phrase their definitions to admit a refinement.

v0.5.0 — 2026-09-15 — the Hecke datum, τ, and the definition of the operator

Commit `ef87af6`, writing slot 3. Chapters 10 and 11, 13 pp; 39 → 52 pages. All 17 assigned labels carried verbatim — including `alg:tau` (36 code citation sites) and `obl:tau` (28) — 0 unplaceable, 19 new.

Normative.

- `prop:cosets` [new] (extension of a re-homed statement) — stated and proved for every $\nu \geq 1$, via the bijection with $\mathbf{P}^1(\mathbb{Z}/N)$ and the count $p^\nu + p^{\nu-1}$. The source states it only for $g = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$, yet the pipeline runs at $g = \begin{pmatrix} 1 & 0 \\ 0 & p^\nu \end{pmatrix}$, and the code said so outright, calling its own $\nu \geq 2$ case an undocumented generalisation. The Hermite-normal-form citation the source leaned on is replaced by a proof.
- `def:cosetreps`, `eq:cocycle`, `lem:cocycle` [new] — the closed-form lifts are pinned *binding as written*, with the class map and its cocycle property. A generic extended-Euclid Bézout solve returns a different, equally valid lift, and the cochain-level matrices depend on the choice, so an implementation that "reimplements the obvious thing" would produce different matrices with no test to catch it.

- `prop:fullassembly`, `eq:fullassembly` [new] — the primitive-to-full assembly, promoted from a parenthetical to a proposition proved at cochain level with no hypothesis on ℓ. `err:e1`.
- `sec:primfull`, `rmk:whichoperator`, `rmk:rawprimitive` [new] — `err:e1` made unrepeatable: the two objects are named in a contract box before anything can refer to them, unqualified T_{p^ν} *always* means the full operator, and the entry-point table sits in the code map.
- `thm:relations` [repaired] — now carries $6 \in R^\times$, the hypothesis its own proof needs, since it routes through `obl:algebra` and `thm:classical`. The source omitted it, and so did a source-code comment, which was corrected separately in v0.6.0's commit.
- `cor:ulrecursion` [repaired] — split: part 1 needs $\ell \geq 5$, part 2 does not.
- `thm:welldef` [repaired] — split into what holds unconditionally (equivariance, part (a)), what needs $6 \in R^\times$ (part (b)), and what is unconditional for the transported choices but conditional in general (part (c)). The source blurred the three.
- `lem:tauexists` [repaired] — $6 \in R^\times$ made explicit *and* derived, from the elliptic stabiliser orders 4 and 6 of `prop:model7h4`(iv) rather than re-assumed; acyclicity from `prop:model7h4`(ii).
- `def:tau` [repaired] — widened from $\mathbb{F}_\ell$ to a general commutative R, recorded in `rmk:taurings` as a widening and not a retargeting ((L2)), because the integrality requirement and "tables once, matrices per R" both need it.
- `prop:diamond` [re-homed] — proved with no hypothesis on ℓ.
- `rmk:tauintegral` [new], [verified computationally] — τ's integrality stated narrowly: steps 1–2 of `alg:tau` are integral unconditionally, only the step-3 triangle fill can lose it, and there is *no* theorem that the fills solve over $\mathbb{Z}$ — `lem:tauexists` gives existence over R with 6 invertible. Verified per table, with the measured grid in the statement: 0 of 66 grid points used the mod-ℓ triangle solve, maximum certified bound $B = 196$, 69/69 passed.
- `rmk:relcoeff` [new] — the coefficient of $T_{p^{\nu-1}}$ is p^{n+1}, not $(p+1)p^n$; `err:e6`, settled by the degree count.
- `rmk:T3`, `rmk:badprimesingular` [new] — what (T3) is for, and why the singularity at $p = \ell$ is not a defect of the construction.
- *Re-homed unchanged*: `sec:heckedatum`, `sec:correspondencedatum`, `sec:operatordefs`, `def:operators`, `eq:fullop`, `sec:relations`, `sec:chainlevel`, `sec:correspondence`, `eq:corconjres`, `sec:taudatum`, `sec:operatordef`, `def:hecke`, `eq:heckechain`, `sec:classicalcomparison`, `thm:classical`, `rmk:indeplevel`.

Non-normative.

- Two citations that could not be checked were removed under (A2): the Hermite normal form is now proved inside `prop:cosets`, and Shimura Ch. 3 replaces a pair of unverifiable theorem numbers.

v0.4.0 — 2026-09-15 — C15: i is $\sqrt{-1}$ only

Commit `6d8abae`. 39 pages, unchanged; the change is entirely in chapter 2. Slot 2 had found a fifteenth symbol collision the notation chapter did not list and resolved it locally, but nothing told the remaining chapter authors to do the same.

Normative.

- `N4` — amended from "a running index is i, j, t or s" to "a running index is t, j or s; i is never an index". Owner decision, 2026-09-15; the rejected alternative, renaming the elliptic vertex value, churns a normative object for no gain.

- `sec:notation-collisions`, row C15 — i as a running index versus $i = \sqrt{-1} = v_6$, the elliptic point of the model; collision count updated.
- `sec:notation-crosswalk` — crosswalk row (i as an index $\to t/j/s$). It earns its place: all three source documents use i heavily as a running index, so the mechanical crosswalk later slots run over re-homed text would otherwise reintroduce the collision chapter by chapter.

Non-normative.

- `sec:notation-open` — C15 recorded as decided, not as an open item.

v0.3.0 — 2026-09-15 — conventions, the 7-vertex model, Sym^n and the contragredient

Commit `1c84f7c`, writing slot 2. Chapters 7 and 8, 9 pp against a 7 pp budget; 30 → 39 pages. All 12 assigned labels carried verbatim, 17 new. This is the first re-homing slot, and it established the notation-crosswalk workflow for the rest.

Normative.

- `H4`, `prop:model7h4` [new] — that K^u is the Γ-tessellation of the truncated half-plane is used by the source's `rmk:orbifold` but the source never states that it is contractible and two-dimensional, nor the stabiliser inventory. Both are load-bearing here and in chapter 15. Written as (H4), with `prop:model7h4` proving (i)–(iv). Contractibility goes by the horoball retraction, which needs $Y > 1$ — a condition the sources do not state either, and which is now in the proof.
- `H1`, `H2`, `H3` — the standing hypotheses, kept under those exact strings because they are cited by name from four source files and four plan documents.
- `sec:contragredient`, `def:contragredient`, `eq:adjrules`, `eq:dualaction`, `prop:pairing`, `eq:pairadjoint`, `lem:symdual`, `eq:BnGram` [new] — the contragredient D_n, the perfect pairing, and the integral invariant form, per owner decision OQ4. The adjointness identity $\langle \rho_n(g)v, \phi\rangle = \langle v, \rho_n^\vee(\mathrm{adj}\, g)\phi\rangle$ is exact, integral and carries no determinant factor; the *diagonal* identity does carry a p^n twist, so [E]'s "T_q self-adjoint once D_n carries the adjugate structure" is true of the adjointness form only and must be re-homed that way.
- `rmk:selfadjoint` [new] — states that `prop:pairing` asserts nothing about cohomology. The architecture's assigned caption for it, "self-adjointness of T_p", over-promises: the operators are not defined until chapters 10–12. Only the coefficient-level content is proved here, and chapter 16 must not assume more.
- `rmk:divided` [new] — divided powers, and what is not claimed.
- `rmk:orbifold` [repaired] — `err:e8` at its first landing site: the torsion is stated as a hypothesis and the place where ignoring it changes an answer is named, replacing "issues with torsion, which we ignore".
- `rmk:negI` [repaired] — `err:e8` at its second: the odd-n vanishing argument replaces the same phrase.
- `sec:conventions` — `err:e16` applied: the composite-index operator is T_N, never T_n, n being the Sym index.
- `def:model7`, `def:model7rel`, `eq:negI`, `eq:flatindex`, `eq:rhondef` [new] — the model, its relation table with a code tag per row, the central relation, the flat index, and the displayed ρ_n.
- *Re-homed unchanged*: `sec:model`, `sec:symconv`, `def:rhon`, `prop:symconv`, `def:semigroup`, `lem:detsym`.

Non-normative.

- The model was verified against the code rather than the prose: 7 vertices, 11 edges, 5 faces ($\chi = 1$), hence $|\mathcal{R}| = 8 + 23 = 31$.
- `gl2_refs.bib` gains `serre_cours`, the fundamental-domain, generation and stabiliser-order citation (A2) requires.
- A fifteenth symbol collision (i) was found and resolved locally only; closing it needed an owner call and became C15 in v0.4.0.

v0.2.0 — 2026-09-15 — scope, the errata appendix, and the T_{p^ν} semantics

Commit `7f1f5f2`, writing slot 1. Chapter 1 (3.5 pp) and Appendix A (7.5 pp, 20 entries); 18 → 30 pages. 38 labels added, 0 removed.

Normative.

- `sec:scope` — the [C]/[E] cut, moved: predictions and plain facts about the objects had been on the same side of it. Nine statement groups — the six groups $X_1, \dots, X_6$ and their long exact sequence, the boundary groups and their characters, Poincaré duality, the amalgam presentation and $\dim H^1$, and the Dickson/Bockstein statement — move into this document, recorded as `err:e18`.
- `sec:status` — the STATUS vocabulary. `\STD` takes a *mandatory* precise citation and will not compile without one, so "[standard]" with no pointer is a build error; [verified computationally] never upgrades to [proved here], and carries the range it covers and the validation that produces it.
- `sec:labelpolicy`, `L1`–`L6` — a label is an API: never renamed, never deleted, never retargeted to a statement asserting something different; a retired statement keeps its label, attached to the erratum that retires it.
- `sec:authoring`, `A1`–`A6` — every hypothesis at the point of use; no claim rests on "standard"; definitions constructive where the code is; obligations as checkable predicates on emitted data; one symbol one meaning; diagnostics never branched on.
- `sec:versioning`, `V1`–`V4` — this document's version convention, prompted by `err:e17`. Honoured for the first time in v0.11.0.

Non-normative.

- `app:errata` — the errata appendix, 20 entries `err:e1`–`err:e20`, each giving the defective statement with its `path:line`, the correct statement, the evidence, and the landing site.
- `err:e18`, `err:e19`, `err:e20` — three entries the architecture's list did not have: the [E]→[C] label move (a consequence of the scope cut, not a defect), [v5] §12's β no-go, and [v5] §12.3's bar-cochain formula omitting the coset permutation.
- `err:e1` — the primitive-versus-full T_{p^ν} semantics settled empirically rather than assumed, by running the pipeline and not by reading it: the entry point returns the *full* operator, the raw route the *primitive* double coset, and both entry points are named. The hand-checkable instance at $(p, \ell, n) = (2, 5, 0)$ is recorded with the entry.
- Erratum numbering unified to E1–E20 across the document and the architecture, which had been written from two different drafts; later chapters cite E-numbers in their template lines, so a chapter author reading the wrong list would have applied the wrong correction.

v0.1.0 — 2026-09-15 — the skeleton

Commit `91cf200`. 18 pages. The architecture pass: every chapter and appendix of the plan present, 144 labels declared, and every statement other than chapter 2 a stub reading `[to be written in step 3]`. The purpose was to prove that the merged label namespace compiles and is collision-free.

Normative.

- `sec:notation` — the notation chapter, written in full because it binds every later chapter, and declared normative and exhaustive. No document of the superseded corpus had one. `N1`–`N5`; `sec:notation-primes`, `sec:notation-framework`, `sec:notation-cochains`, `sec:notation-hecke`, `sec:notation-cone`; `sec:notation-collisions` with 14 resolved collisions C1–C14 (including `coll:p`, and the three the corpus survey missed: M with five meanings, $\mathcal{R}$ versus the coefficient ring, and s/S); `sec:notation-crosswalk`; `sec:notation-open`, `O1`–`O7`.

Non-normative.

- `app:labelindex` — every externally cited label dereferenced, so a missing, renamed or duplicated label is a build error and not a silent breakage ((L6)). The 68 preserved labels — 67 from `hecke_action.tex` plus `thm:explicit_complex_equiv` from `monodromy_v5.tex` — are placed, 0 unplaceable, 0 collisions; the three namespaces are pairwise disjoint.
- Every other label of the document is declared here as a stub; each is written by the version recorded above, and the ones still unwritten are listed in §E.1.

E.1 What is not recorded here

Not recorded, for want of evidence. Nothing. Every entry above is carried by a commit that names the statement it changed, by the label census of that commit's revision, or by the text itself. Two classes of change were deliberately *excluded* rather than lost:

- **Changes outside this document** (in the sense of (CL2)). Three of the ten commits also changed something else: v0.1.0's added the corpus survey, the architecture note and the build script; v0.2.0's reconciled the architecture note against what slot 1 had actually written; and v0.6.0's corrected a `HeckeOperatorFamily.h` comment that claimed the T_{p^ν} recursion is contract-backed unconditionally, while `thm:relations` requires $6 \in R^\times$, and filed owner direction OD-T1 in the owner-decision record. Those are not versions of this document and are recorded where they happened; the contract-side half of each is above.
- **Page counts and build numbers** are given per version as context, not as entries.

Not yet written. Nothing. v0.19.0 wrote the last five stubs — §21, §22, §22.5 and Appendices B and C — so (V5)(i) is met; v1.0.0 discharged (V5)(ii) (`rmk:labelcensus`), so the leading version field is 1. **A written document with an unambiguous namespace is still not a finished programme.** Two chapters state objects no code builds (§22.2; three until v1.3.3, when §14 was implemented); the construction item 14.24 is *decided* (v1.3.0, (OC-1a)) but governs a truncated free model that nothing builds, and the stabilisation theorem (OC-1c) that would supersede it is half proved (14.22) and half open (14.23(3)); 20.2 is partial; 9.10(iv), discharged negatively at v0.13.0, is discharged positively on the canonical τ since v1.1.0, and 9.10(v)'s sign prediction, discharged *negatively* for the geometric involution at v1.1.1, holds by definition and gates since the owner's v1.2.0 negation of `def:iota`(iv); and the citation-census guard of §22.5 was specified and not written until the slot-3 tooling (v1.8.1: `docs/matrix-engine/tools/label_census.sh`, run by `ContractCensusValidation` since hecke-extensions slot 3; this sentence had not been updated) — v1.0.0's census was run by hand, as `rmk:labelcensus` says, and found one collision that a citation census could not have found. (V5) does not claim otherwise: its leading field speaks about this document, not about the implementation.

The two supersession steps (V4) requires — freezing `hecke_action.tex` and `monodromy_v5.tex` with a version marker and a superseded-by banner — **were taken at v1.0.0**: the first is frozen at Version 1.0 (2026-07-02), the second at Version 5.0 (2026-07-03),

and each banner directs its reader to Appendix A and says that a `[C, <label>]` citation resolves in this document. Their content is otherwise unchanged; they remain the historical record, and a reader of either must still apply Appendix A.

References

[Ada54] Iain T. Adamson. Cohomology theory for non-normal subgroups and non-normal fields. *Proc. Glasgow Math. Assoc.*, 2:66–76, 1954. Relative cohomology of a group pair. Not held; bibliographic data not checked against the page (VERIFY at R20-L), no locator used. Added at [C] v1.22.0-pre.3 (F-LES repair R5) for prop:levelconeshapiro's orientation remark only.

[AL70] A. O. L. Atkin and J. Lehner. Hecke operators on $\Gamma_0(m)$. *Math. Ann.*, 185:134–160, 1970. Newforms; for an ℓ-new eigenform of weight k and trivial character, $a_\ell^2 = \ell^{k-2}$ (added at [C] v1.13.0-pre.1, slot-13 WP-2). NOT verified against the text (rule A2): no copy is available to this project.

[AM69] Michael F. Atiyah and Ian G. Macdonald. *Introduction to Commutative Algebra*. Addison-Wesley, 1969. Prop. 2.4 (Cayley–Hamilton for finitely generated modules), Cor. 10.10 (Artin–Rees). Not checked against the page (VERIFY); added at [C] v1.25.0-pre.0 (sec:heckeZ: prop:heckeZmax, prop:heckeZglue).

[AS86] Avner Ash and Glenn Stevens. Cohomology of arithmetic groups and congruences between systems of Hecke eigenvalues. *J. reine angew. Math.*, 365:192–220, 1986. §1.1 (p. 194: the Hecke action, Lemma 1.1.1 the long exact sequence, Def. 1.1.2 compatible pairs), Lemma 1.1.4 (p. 195, Shapiro), Lemma 2.1 and Thm. 2.2 (p. 204) verified on the GDZ page images, slot-17 R17-4; §1.1 (pp. 193–196, incl. Lemma 1.1.5, the character twist) re-read, slot-21 R20-L: the Hecke action is stated as well known (ref. Kuga–Parry–Sah), not proved.

[BC09] Joël Bellaïche and Gaëtan Chenevier. *Families of Galois representations and Selmer groups*, volume 324 of *Astérisque*. 2009.

[BE78] Robert Bieri and Beno Eckmann. Relative homology and Poincaré duality for group pairs. *J. Pure Appl. Algebra*, 13:277–319, 1978. $\Delta = \ker(\mathbb{Z}[G/S] \to \mathbb{Z})$ and the relative (co)homology of a group pair. Not held; bibliographic data not checked against the page (VERIFY at R20-L), no locator used. Added at [C] v1.22.0-pre.3 (F-LES repair R5) for prop:levelconeshapiro's orientation remark only.

[Bel21] Joël Bellaïche. *The Eigenbook: Eigenvarieties, families of Galois representations, p-adic L-functions*. Birkhäuser, 2021.

[Ben91] David J. Benson. *Representations and Cohomology I*, volume 30 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 1991.

[BH93] Winfried Bruns and Jürgen Herzog. *Cohen–Macaulay Rings*, volume 39 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 1993. Prop. 3.1.19(b): a Noetherian local ring is Gorenstein iff its quotient by a regular sequence is. Not checked against the page (VERIFY); added at [C] v1.25.0-pre.0 (sec:heckeZ: prop:heckeZmono).

[BK15] Joël Bellaïche and Chandrashekhar Khare. Level 1 Hecke algebras of modular forms modulo p. *Compositio Math.*, 151:397–415, 2015.

[BM16] Nicolas Billerey and Ricardo Menares. On the modularity of reducible mod l Galois representations. *Math. Res. Lett.*, 23(1):15–41, 2016. Cited for Eisenstein congruences of newforms of prime level at weight $k \geq 4$ (added at [C] v1.16.0-pre.1, slot-16 WP-12, (LN9)). Bibliographic data and content NOT verified against the text (rule A2).

[Bro82] Kenneth S. Brown. *Cohomology of Groups*, volume 87 of *Graduate Texts in Mathematics*. Springer-Verlag, 1982. Ch. III end of §8 and §§9–10 (pp. 80–85) read on the library's copy, slot-21 R20-L: Prop. III.9.5 (p. 82: (ii) cor $\circ$ res $= (G : H)$, (iii) the double-coset formula), Cor. III.10.2 (p. 84).

[BS73] Armand Borel and Jean-Pierre Serre. Corners and arithmetic groups. *Comment. Math. Helv.*, 48:436–491, 1973. The bordification $\bar{\mathcal{H}}$ and the pair $(\Gamma\backslash\bar{\mathcal{H}}, \partial)$ on which the cone chapter's geometric identification rests.

[CCS13] F. Callegaro, F. R. Cohen, and M. Salvetti. The cohomology of the braid group B_3 and of $\mathrm{SL}_2(\mathbb{Z})$ with coefficients in a geometric representation. *Q. J. Math.*, 64(3):847–889, 2013. arXiv:1204.5390. Full arXiv text read (literature search B, 2026-10-19): Thm. 4.5 (= Thm. 3.7(2)), the p-torsion of $H^1(\mathrm{SL}_2(\mathbb{Z}); M_n)$ for every prime p; theorem numbers are the arXiv version's. Journal data from the first author's CV. Added at [C] v1.24.0-rc.1 for rmk:bocksteinexp(4).

[CE] F. Calegari and M. Emerton. The Hecke algebra T_k has large index. arXiv:math/0311367. Full arXiv text read (literature search B, 2026-10-19): Lemma 3.1 ($a_\ell \equiv 1+\ell \bmod p$ at level $\Gamma(p)$, $p = 2, 3, 5$, for primes $\ell \equiv \pm 1 \bmod p$; credited to Hatada, their ref. [11]). Journal version not checked. Added at [C] v1.24.0-rc.1 for prop:ltwothreescalar (F1); note amended at v1.24.0-rc.2 (RC124 referee F8).

[CE98] Robert F. Coleman and Bas Edixhoven. On the semi-simplicity of the U_p-operator on modular forms. *Math. Ann.*, 310:119–127, 1998. $\alpha_p \neq \beta_p$ in weight 2. Not held; bibliographic data not checked (VERIFY); added at [C] v1.25.0-pre.0 (sec:heckeZ: prop:heckeZrat(4)(c)), a predictor only.

[CE05] Frank Calegari and Matthew Emerton. On the ramification of Hecke algebras at Eisenstein primes. *Invent. Math.*, 160:97–144, 2005. Prime level N, weight 2, $p \in \{2, 3\}$: the rank of the Eisenstein-localised cuspidal Hecke algebra. Held (LIT-30); §1 (Thms. 1.1–1.3) and §6 read by the slot-22 F4 referee of LN23. Added at [C] v1.21.0-pre.4 for rmk:levelNcurvefamily (entry as in loewy_and_galois_refs.bib); owner's personal communication: possible error or gap in the last technical steps, not independently verified.

[CO77] Henri Cohen and Joseph Oesterlé. Dimensions des espaces de formes modulaires. In *Modular Functions of One Variable VI*, volume 627 of *Lecture Notes in Math.*, pages 69–78. Springer, 1977. Cited for $\dim S_k(\Gamma_0(N), \chi)$ (added at [C] v1.15.0-pre.1, slot-15 WP-3). NOT verified against the text (rule A2). Since [C] v1.21.0-pre.2: Théorème 1, pp. 71–72 ($\dim S_k - \dim M_{2-k}$) read on the library copy (slot-21); no Eisenstein dimension for integral weight in the paper.

[Cre97] John E. Cremona. *Algorithms for Modular Elliptic Curves*. Cambridge University Press, 2 edition, 1997.

[CV92] Robert F. Coleman and José Felipe Voloch. Companion forms and Kodaira–Spencer theory. *Invent. Math.*, 110:263–281, 1992. Re-proves Gross's companion-form theorem (added at [C] v1.11.0-pre.2, the slot-11 WP-3 referee's m-5). Whether its range includes $k = \ell - 1$ is NOT verified against the text (rule A2).

[Del74] Pierre Deligne. La conjecture de Weil. I. *Publ. Math. Inst. Hautes Études Sci.*, 43:273–307, 1974. §8, the Ramanujan–Petersson bound $|a_p| \leq 2p^{(k-1)/2}$ for normalised eigenforms of weight $k \geq 2$, p prime to the level. Added at [C] v1.23.0-pre.24 (prop:crtcharpolyN(3), the bound that is an input). The locator Thm. 8.2 is NOT verified against the text (rule A2).

[Den23] Taiwang Deng. Torsions in cohomology of $\mathrm{SL}_2(\mathbb{Z})$ and congruence of modular forms. *J. Number Theory*, 246:87–156, 2023. Page range added 2026-09-16 from the article itself, `docs/literature/1-s2.0-S0022314X22002529-main.pdf`.

[Deo17] Shaunak V. Deo. Structure of Hecke algebras of modular forms modulo p. *Algebra & Number Theory*, 11:1–38, 2017.

[DI95] Fred Diamond and John Im. Modular forms and modular curves. In *Seminar on Fermat's Last Theorem (Toronto, 1993–1994)*, volume 17 of *CMS Conf. Proc.*, pages 39–133. Amer. Math. Soc., 1995. Library: papers/d/Diamond/ (a camera scan). §3.5, p. 54 (every eigenspace at most one-dimensional) and §3.6, p. 55 (self-adjointness, orthogonal decomposition into eigenspaces) verified on the page images (slot-17 L17-5 audit, R17-4). Added at [C] v1.17.0-pre.5 for lem:heckerankone's eigenbasis.

[Dic11] Leonard Eugene Dickson. A fundamental system of invariants of the general modular linear group with a solution of the form problem. *Trans. Amer. Math. Soc.*, 12:75–98, 1911. The invariants of $\mathrm{GL}_n(\mathbb{F}_q)$ and $\mathrm{SL}_n(\mathbb{F}_q)$; for $n = 2$ the ring of §24. NOT verified against the text (rule A2).

[Dro72] Ju. A. Drozd. Representations of commutative algebras. *Funct. Anal. Appl.*, 6:286–288, 1972.

[DS74] Pierre Deligne and Jean-Pierre Serre. Formes modulaires de poids 1. *Ann. Sci. École Norm. Sup.*, 7:507–530, 1974.

[DS05] Fred Diamond and Jerry Shurman. *A First Course in Modular Forms*, volume 228 of *Graduate Texts in Mathematics*. Springer-Verlag, 2005. §5.5 (the Petersson inner product, the adjoints of the Hecke operators, a basis of simultaneous eigenforms). Cited by lem:heckerankone (v1.17.0-pre.2) for the eigenbasis; a secondary pointer since v1.17.0-pre.5 (primary: diamond_im). NOT verified against the text: the book is not in the project's library (slot-17 L17-5).

[Edi92] Bas Edixhoven. The weight in Serre's conjectures on modular forms. *Invent. Math.*, 109:563–594, 1992. §3: Tate's θ-cycles. NOT verified against the text (rule A2).

[Gro90] Benedict H. Gross. A tameness criterion for Galois representations associated to modular forms (mod p). *Duke Math. J.*, 61:445–517, 1990. Companion forms. NOT verified against the text (rule A2).

[Hab83] K. Haberland. Perioden von Modulformen einer Variabler und Gruppencohomologie, III. *Math. Nachr.*, 112:297–315, 1983. Proves the projective-S_4 image of $E_4\Delta$ modulo 59 (added at [EX] 1.2.2, slot-16 WP-8). Bibliographic data verified (Kiming–Verrill [22], the publisher's record, doi 10.1002/mana.19831120115); the text itself NOT read (rule A2).

[Har87] Günter Harder. Eisenstein cohomology of arithmetic groups. The case GL_2. *Invent. Math.*, 89:37–118, 1987. STANDARD input to the cone chapter's identification of T_p on $H_c^\bullet$ with the correspondence on the Borel–Serre pair. Title, journal, volume, year and pages verified 2026-09-16 against the published record; the entry previously

carried the WRONG title ("Eisenstein cohomology for $\mathrm{SL}_2(\mathbb{Z})$ and special values of L-functions"), which is not a paper of this author in this volume. The SECTION pointer §§1–2 used in the text is NOT verified against the article (rule A2), exactly as for borel_serre.

[Har22] Günter Harder. Cohomology of arithmetic groups, vol. iii. manuscript; `docs/literature/Volume-III-07.-08.-2022.pdf`, 2022.

[Hat02] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002.

[Hid86a] Haruzo Hida. Galois representations into $\mathrm{GL}_2(\mathbf{Z}_p[[x]])$ attached to ordinary cusp forms. *Invent. Math.*, 85:545–613, 1986. The Λ-adic ordinary Hecke algebra and control (added at [C] v1.12.0-pre.1). Locators Thm. 3.1 (p. 562), (4.6a) and pp. 563–566, Lemma 4.6 (p. 571), Prop. 4.7 (p. 572), Thm. 6.6 (p. 590) VERIFIED against the PDF at [C] v1.13.0-pre.1, slot-13 WP-2; the source is read-only and is not copied into this repository.

[Hid86b] Haruzo Hida. Iwasawa modules attached to congruences of cusp forms. *Ann. Sci. École Norm. Sup. (4)*, 19:231–273, 1986. The ordinary projector; the rank of the ordinary part depends only on the weight modulo $p-1$ (added at [C] v1.12.0-pre.1, slot-12 WP-2). Locators Thm. 3.1 and Cor. 3.2, p. 250, VERIFIED against the PDF (numdam scan) at [C] v1.13.0-pre.1, slot-13 WP-2; the source is read-only and is not copied into this repository.

[Hid93] Haruzo Hida. *Elementary Theory of L-functions and Eisenstein Series*, volume 26 of *London Math. Soc. Student Texts*. Cambridge University Press, 1993. Ch. 7: p-adic Hecke algebras, the ordinary part, control and weight independence (added at [C] v1.12.0-pre.1). NOT verified against the text (rule A2).

[HM97] George Havas and Bohdan S. Majewski. Integer matrix diagonalization. *J. Symbolic Comput.*, 24:399–408, 1997.

[Joc82a] Naomi Jochnowitz. Congruences between systems of eigenvalues of modular forms. *Trans. Amer. Math. Soc.*, 270:269–285, 1982. Thms. 2.2, 2.4 (p. 272), Thm. 4.1 (p. 276: every system of level $N \leq 17$, 19 or 23 is a twist of a system of weight $\leq l+1$) and §6 (Thms. 6.3, 6.6, p. 280) verified on the AMS open-archive page images, slot-17 R17-4.

[Joc82b] Naomi Jochnowitz. A study of the local components of the Hecke algebra mod l. *Trans. Amer. Math. Soc.*, 270:253–267, 1982. θ-cycles. NOT verified against the text (rule A2). Since [C] v1.21.0-pre.2: Lemma 4.1(b), p. 257 (the filtration law, with equality iff $\ell \nmid w(f)$) and §7, pp. 261–264 (θ-cycles) read on the library copy (slot-21).

[Kat77] Nicholas M. Katz. A result on modular forms in characteristic p. In *Modular Functions of One Variable V (Bonn 1976)*, volume 601 of *Lecture Notes in Math.*, pages 53–61. Springer, 1977. The filtration of θf. NOT verified against the text (rule A2). Since [C] v1.21.0-pre.2: Theorem (2), p. 55, read on the library copy (slot-21 literature pass): exact filtration $k+p+1$ when $p \nmid k$, for $N \geq 3$.

[KV05] I. Kiming and H. A. Verrill. On modular mod ℓ Galois representations with exceptional images. *J. Number Theory*, 110:236–266, 2005. arXiv math/0311082. Cor. 1 (§3.1, p. 6) and §3.1.1 (pp. 6–7) verified on the arXiv v1 text (slot-16 WP-8); the journal data are from the publisher's record, the journal numbering not compared with arXiv v1. Added at [EX] 1.2.2.

[L̈] Frank Lübeck. Conway polynomials for finite fields. online tables; definition in Holt, Eick, O'Brien, *Handbook of Computational Group Theory*, §7.5.

[Luc78] Édouard Lucas. Théorie des fonctions numériques simplement périodiques. *Amer. J. Math.*, 1:184–240, 289–321, 1878. Lucas's theorem on binomial coefficients modulo a prime. NOT verified against the text (rule A2).

[Man72] Yuri I. Manin. Parabolic points and zeta-functions of modular curves. *Izv. Akad. Nauk SSSR Ser. Mat.*, 36:19–66, 1972.

[Maz77] Barry Mazur. Modular curves and the Eisenstein ideal. *Publ. Math. IHÉS*, 47:33–186, 1977.

[Mer94] Loïc Merel. Universal Fourier expansions of modular forms. In *On Artin's conjecture for odd 2-dimensional representations*, volume 1585 of *Lecture Notes in Math.*, pages 59–94. Springer, 1994.

[Mer96] Loïc Merel. L'accouplement de Weil entre le sous-groupe de Shimura et le sous-groupe cuspidal de $J_0(p)$. *J. Reine Angew. Math.*, 477:71–115, 1996. The criterion for the rank of Mazur's Eisenstein ideal to be at least 2 (added at [C] v1.14.0-pre.1, slot-14 WP-4, for rmk:levelNcheckable (LN5)). Théorème 2, p. 72 verified on the GDZ page images (slot-17 R17-4): for a prime $q \mid n$, the q-adic completion of the Hecke algebra is $\mathbf{Z}_q$ iff, when $q > 3$, $\prod_{l=1}^{(p-1)/2} l^l$ is not a q-th power in $\mathbf{Z}/p\mathbf{Z}$.

[MP99] Ieke Moerdijk and Dorette A. Pronk. Simplicial cohomology of orbifolds. *Indag. Math. (N.S.)*, 10:269–293, 1999. `docs/literature/orbifolds/`. Cited in §17 as one computational model, NOT as the normative definition: their charts are effective and $[\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}]$ is not.

[Rib76] Kenneth A. Ribet. A modular construction of unramified p-extensions of $\mathbb{Q}(\mu_p)$. *Invent. Math.*, 34:151–162, 1976.

[Rib77] Kenneth A. Ribet. Galois representations attached to eigenforms with Nebentypus. In *Modular Functions of One Variable V*, volume 601 of *Lecture Notes in Math.*, pages 17–51. Springer, 1977. Cited for CM forms with nebentypus (added at [C] v1.15.0-pre.1, slot-15 WP-3). NOT verified against the text (rule A2).

[Rin75] Claus Michael Ringel. The representation type of local algebras. In *Representations of Algebras (Ottawa 1974)*, volume 488 of *Lecture Notes in Math.*, pages 282–305. Springer, 1975.

[Rus] N. Rustom. Congruences modulo prime powers of Hecke eigenvalues in level 1. arXiv:1710.00928v4. Full arXiv text read (literature search B, 2026-10-19): quotes Hatada, Math. Ann. 239 (1979) 75–96 (Thms. 1.4, 1.5), which was not seen. Added at [C] v1.24.0-rc.1 for prop:ltwothreescalar (F1).

[RW70] Yoon Ho Rhie and George Whaples. Hecke operators in cohomology of groups. *J. Math. Soc. Japan*, 22:431–442, 1970. The Hecke ring acts on $H^q(\Gamma; M)$ for every Δ-module M. NOT verified against the text (rule A2).

[SD73] H. P. F. Swinnerton-Dyer. On ℓ-adic representations and congruences for coefficients of modular forms. In *Modular Functions of One Variable III (Antwerp 1972)*, volume 350 of *Lecture Notes in Math.*, pages 1–55. Springer, 1973. Mod-ℓ level-1 forms, θ and the filtration, the Ramanujan congruences. NOT verified against the text (rule A2).

[Ser72] Jean-Pierre Serre. Congruences et formes modulaires (d'après H. P. F. Swinnerton-Dyer). In *Séminaire Bourbaki 416 (1971/72)*. 1972. and *Formes modulaires et fonctions zêta p-adiques*, LNM 350 (1973).

[Ser73] Jean-Pierre Serre. *A Course in Arithmetic*, volume 7 of *Graduate Texts in Mathematics*. Springer-Verlag, 1973. Ch. VII, §1.2, Thms. 1–2: the fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathcal{H}$, the generation by T and S, and the elliptic points with their stabiliser orders. Cited by §7. Checked against the text 2026-09-16: Thm. 1 is the fundamental domain, with (1) surjectivity of $D \to \mathcal{H}/G$ and (3) the stabilisers $I(z)$, trivial except at i, $\rho = e^{2\pi i/3}$ and $-\bar{\rho} = e^{\pi i/3}$; Thm. 2 is the generation by S and T. Pointers to the elliptic points must name Thm. 1, not Thm. 2.

[Ser80] Jean-Pierre Serre. *Trees*. Springer-Verlag, 1980. Translated by J. Stillwell from *Arbres, amalgames,* SL_2, Astérisque 46 (1977). Cited in §16 for the amalgam decomposition $\mathrm{SL}_2(\mathbb{Z}) = \mathbb{Z}/4 *_{\mathbb{Z}/2} \mathbb{Z}/6$, its Bass–Serre tree, and the existence of a free subgroup of a free product of finite groups. The pointers used in the text are given at chapter-and-section level (Ch. I, §4) and are NOT verified against the book (rule A2): no copy is in `docs/literature/`. The statements themselves are used only through the short exact sequence (50), which a reader may take as the datum.

[She97] Jonathan R. Shewchuk. Adaptive precision floating-point arithmetic and fast robust geometric predicates. *Discrete & Comput. Geom.*, 18:305–363, 1997.

[Shi71] Goro Shimura. *Introduction to the Arithmetic Theory of Automorphic Functions*. Princeton University Press, 1971. Ch. 3 (pp. 51–95) and Ch. 8 (pp. 223–241) read on the owner's scan, slot-21 R20-L: Prop. 3.8 (p. 54), Prop. 3.16, 3.17 (pp. 57–58), Thm. 3.20 (p. 59), Thm. 3.24 (p. 63), Lemma 3.29 (p. 66), Prop. 3.30–3.33, Thm. 3.34, 3.35 (pp. 67–71), Prop. 3.36 (p. 72), Thm. 3.41, 3.42 (p. 77), Thm. 8.4 (p. 234), Prop. 8.5 (pp. 237–238).

[Ste07] William Stein. *Modular Forms, a Computational Approach*, volume 79 of *Graduate Studies in Mathematics*. American Mathematical Society, 2007.

[Stu87] Jacob Sturm. On the congruence of modular forms. In *Number Theory (New York 1984–85)*, volume 1240 of *Lecture Notes in Math.*, pages 275–280. Springer, 1987.

[Was97] Lawrence C. Washington. *Introduction to Cyclotomic Fields*, volume 83 of *Graduate Texts in Mathematics*. Springer, 2 edition, 1997.

[Wei94] Charles A. Weibel. *An Introduction to Homological Algebra*, volume 38 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 1994. 2.2.6 (the comparison theorem) is cited by the cone chapter — with its relative-projectivity hypothesis made explicit, since that hypothesis is exactly what forces $\ell \geq 5$ elsewhere.

[Wil83] Clarence Wilkerson. A primer on the Dickson invariants. In *Proceedings of the Northwestern Homotopy Theory Conference (Evanston, Ill., 1982)*, volume 19 of *Contemp. Math.*, pages 421–434. American Mathematical Society, 1983. Cited in §16 for the invariants of $\mathrm{SL}_2(\mathbb{F}_\ell)$ on $\mathbb{F}_\ell[x, y]$: a polynomial ring on generators of degrees $\ell + 1$ and $\ell^2 - \ell$. NOT verified against the text (rule A2); the consequence used — the count of monomials in those two degrees — is corroborated computationally on 710 pairs (n, ℓ) in §16.

[WWE20] Preston Wake and Carl Wang-Erickson. The rank of Mazur's Eisenstein ideal. *Duke Math. J.*, 169(1):31–115, 2020. The cuspidal Hecke algebra at Mazur's Eisenstein ideal modulo p as $\mathbb{F}_p[x]/(x^e)$ (added at [C] v1.14.0-pre.1, slot-14 WP-4, for

rmk:levelNcheckable (LN6)). Bibliographic data and statement NOT verified against the text (rule A2; [S14] Q6 default). Since [C] v1.16.0: Thm. 1.5.1 (p. 5), the display on p. 8 and Cor. 7.2.2 (p. 30) verified on the arXiv text 1707.01894v3 (slot-16 WP-8); the journal pagination is not checked.